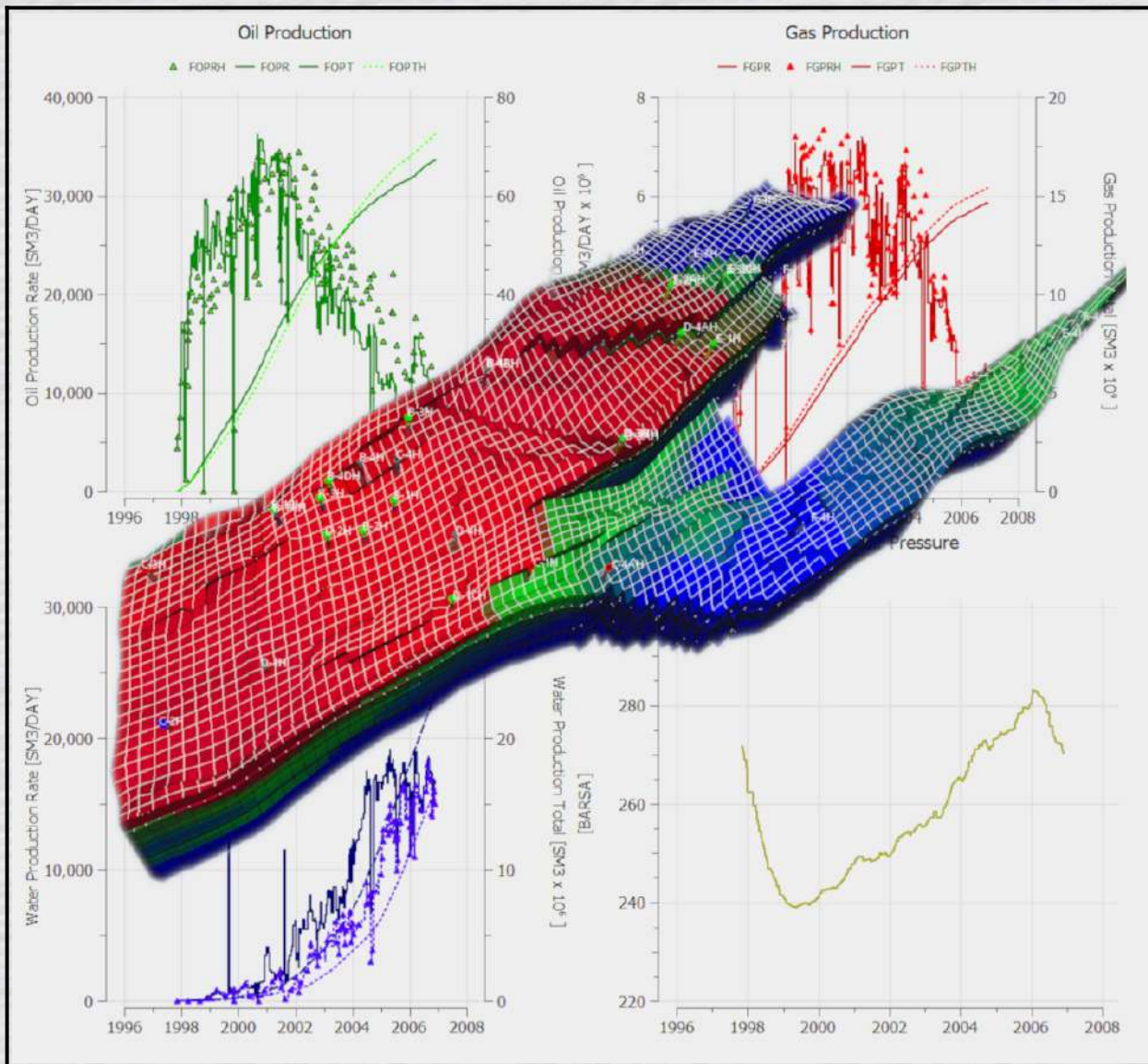


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OPM Flow Reference Manual

(2025-10)

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TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION.....	47
1.1 Overview.....	49
1.2 Black-Oil Model Equations.....	54
1.3 Standard Well Model.....	57
1.4 Multi-Segment Well Model.....	59
1.5 Salt Precipitation Model.....	61
1.6 Numerical Solution of Equations.....	63
1.6.1 Newton-Raphson Solver (Outer Iteration).....	64
1.6.2 Linear Solver (Inner Iteration).....	66
1.7 Formation Damage Models.....	68
1.7.1 Incompressible External Filter Cake Model - Linear Geometry.....	68
1.7.2 Incompressible External Filter Cake Model - Radial Geometry.....	68
CHAPTER 2: INSTALLING AND RUNNING FLOW.....	71
2.1 Installing OPM Flow.....	73
2.1.1 Ubuntu Linux 22.04 or 24.04 (64-bit version only).....	73
2.1.2 Red Hat Enterprise or CentOS (version 7 or 8).....	74
2.1.3 Windows 7 and 10 Using VirtualBox.....	74
2.1.4 Windows 10 and 11 - Windows Subsystem for Linux (WSL).....	80
2.1.5 Using a Docker Container.....	83
2.1.6 Installing from Source.....	83
2.2 Running OPM Flow From The Command Line.....	84
2.3 Running OPM Flow Using OPMRUN.....	107
2.4 Improving Simulator Convergence and Numerical Performance.....	115
2.4.1 Data Quality and Assurance.....	116
2.4.2 Convergence and Numerical Performance.....	136
2.4.3 Time Step Control Guidance.....	142
2.4.4 Newton-Raphson Solver Iteration Controls (Non-Linear Iterations).....	148
2.4.5 Linear Solver Controls and Options.....	153
CHAPTER 3: KEYWORD DOCUMENTATION STRUCTURE.....	159
3.1 Introduction.....	161
3.2 Keyword Definitions.....	162
3.3 Multi-Section Keywords.....	162
3.4 Keyword Formats.....	163
3.4.1 Keyword Format Type - Comment.....	163
3.4.2 Keyword Format Type - Activation.....	163
3.4.3 Keyword Format Type - Vector (Row Vector).....	164
3.4.4 Keyword Format Type - Vector (Columnar Vector).....	166
3.4.5 Keyword Format Type - Array.....	167
3.5 Input File Structure.....	168
CHAPTER 4: GLOBAL SECTION KEYWORDS.....	171
4.1 Introduction.....	173
4.2 Data Requirements.....	173
4.3 Keyword Definitions.....	175
4.3.1 COLUMNS - Define Input File Column Margins.....	176
4.3.2 DEBUG - Define the Debug Data to be Printed to File.....	177
4.3.3 ECHO - Activate Echoing of User Input Files to the Print File.....	178
4.3.4 END - Define the End of the Input File.....	179
4.3.5 ENDINC - Define the End of an Include File.....	180
4.3.6 ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	181
4.3.7 EXTRAPMS - Activate Extrapolation Warning Messages.....	182

4.3.8 FORMFEED - Defined the Print File Form-Feed Character.....	183
4.3.9 INCLUDE - Load Another Data File at the Current Position.....	184
4.3.10 MESSAGE - Output User Message.....	185
4.3.11 MESSAGES - Define Message Print Limits and Stop Limits.....	186
4.3.12 NOECHO - Deactivate Echoing of User Input Files to the Print File.....	188
4.3.13 NOWARN - Deactivate Warning Messages.....	189
4.3.14 SKIP - Activate Skipping of All Keywords and Input Data.....	190
4.3.15 SKIP100 - Activate Skipping of Keywords by Black-Oil Simulator.....	191
4.3.16 SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	193
4.3.17 WARN - Activate Warning Messages.....	195

CHAPTER 5: RUNSPEC SECTION..... 197

5.1 Introduction.....	199
5.2 Data Requirements.....	199
5.3 Keyword Definitions.....	214
5.3.1 ACTDIMS - ACTION Keyword Dimensions.....	215
5.3.2 ACTPARAM - Define Action Facility Target and Tolerance Parameters.....	216
5.3.3 AITS - Activate Intelligent Time Stepping.....	217
5.3.4 AITSOFF - Deactivate Intelligent Time Stepping.....	218
5.3.5 ALKALINE - Activate the Alkaline Phase and Model.....	219
5.3.6 API - Activate API Tracking.....	220
5.3.7 AQUODIMS - Define Aquifer Dimensions.....	221
5.3.8 AUTOREF - Define Auto Refinement Options.....	222
5.3.9 BIGMODEL - Activate Big Model Option (Retired).....	223
5.3.10 BIOFILM - Activate the Biofilm Model.....	224
5.3.11 BLACKOIL - Activate Black-Oil Phases.....	225
5.3.12 BPARA - Activate Block Parallel License.....	226
5.3.13 BPIDIMS - Define the Dimensions of the Interpolated Block Quantities.....	227
5.3.14 BRINE - Activate Brine Tracking Option.....	228
5.3.15 CART - Activate Cartesian Geometry.....	230
5.3.16 CBMOPTS - Define Coal Bed Methane Options.....	231
5.3.17 CO2SOL - Activate Dissolved CO2 in the Water Phase.....	232
5.3.18 CO2STORE - Activate the CO2 Storage Model.....	235
5.3.19 COAL - Activate the Coal Phase (CBM Model).....	241
5.3.20 COLUMNS - Define Input File Column Margins.....	242
5.3.21 COMPS - Activate Compositional Modeling Formulation.....	243
5.3.22 CPR - Activate Constrained Pressure Residual ("CPR") Linear Solver.....	244
5.3.23 DEBUG - Define the Debug Data to be Printed to File.....	246
5.3.24 DIFFUSE - Activate Molecular Diffusion Option.....	247
5.3.25 DIMENS - Define the Dimensions of the Model.....	248
5.3.26 DISGAS - Activate the Dissolved Gas Phase in the Model.....	249
5.3.27 DISGASW - Activate Dissolved Gas in the Water Phase in the Model.....	250
5.3.28 DISPDIMS - Define the Maximum Number of Dispersion Tables.....	251
5.3.29 DUALPERM - Activate Dual Permeability Model.....	252
5.3.30 DUALPORO - Activate Dual Porosity Model.....	253
5.3.31 DYNRDIMS - Define Dynamic Region Dimensions.....	254
5.3.32 ECHO - Activate Echoing of User Input Files to the Print File.....	255
5.3.33 ECLMC - Activate Multi-Component Brine Model.....	256
5.3.34 END - Define the End of the Input File.....	257
5.3.35 ENDINC - Define the End of an Include File.....	257
5.3.36 ENDSCALE - Activate Relative Permeability End-Point Scaling Option.....	258
5.3.37 ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	260
5.3.38 EOS - Specify Equation of State.....	261
5.3.39 EQLDIMS - Define the Equilibration Data Dimensions.....	262
5.3.40 EQLOPTS - Activate the Equilibration Options.....	263
5.3.41 EXTRAPMS - Activate Extrapolation Warning Messages.....	265
5.3.42 FAULTDIM - Define the Number of Fault Segments.....	266
5.3.43 FIELD - Activate the Oil Field System of Units for the Model.....	267
5.3.44 FMTHMD - Activate The Format History Match Gradient File Option.....	268
5.3.45 FMTIN - Activate The Format Input File Option.....	269
5.3.46 FMTOU - Activate The Format Output File Option.....	271
5.3.47 FOAM - Activate the Foam Phase and Model.....	273

5.3.48 FORMFEED - Defined the Print File Form-Feed Character.....	274
5.3.49 FRICTION - Activate Wellbore Friction Option.....	275
5.3.50 FULLIMP - Activate Fully Implicit Solution Option.....	276
5.3.51 GAS - Activate the Gas Phase in the Model.....	277
5.3.52 GASFIELD - Define Gas Field Operations Options.....	278
5.3.53 GASWAT - Activate the Gas-Water Model Formulation.....	279
5.3.54 GDIMS - Activate Instantaneous Gradient Option and Define Dimensions.....	280
5.3.55 GIMODEL - Activate Gi Pseudo Compositional Option.....	281
5.3.56 GRAVDR - Activate Gravity Drainage and Imbibition for Dual Porosity Model.....	282
5.3.57 GRAVDRB - Activate Vertical Discretized Gravity Drainage and Imbibition for Dual Porosity Model.....	283
5.3.58 GRAVDRM - Activate Alternative Gravity Drainage and Imbibition for Dual Porosity Model.....	284
5.3.59 GRIDOPTS - Grid Processing Options.....	285
5.3.60 H2SOL - Activate Dissolved H2 in the Water Phase.....	287
5.3.61 H2STORE - Activate the H2 Storage Model.....	291
5.3.62 HMDIMS - Define History Match Gradient Parameter Dimensions.....	298
5.3.63 HYST - Activate the Hysteresis Option (Retired).....	299
5.3.64 IMPES - Activate Implicit Pressure Explicit Saturation Solution Option.....	300
5.3.65 IMPLICIT - Activate Fully Implicit Solution Option.....	301
5.3.66 INCLUDE - Load Another Data File at the Current Position.....	302
5.3.67 INSPEC - Activate the INSPEC File Option.....	303
5.3.68 LAB - Activate the Laboratory System of Units for the Model.....	304
5.3.69 LGR - Define Local Grid Refinement Dimensions and Parameters.....	305
5.3.70 LGRCOPY - Activate Local Grid Refinement Inheritance.....	307
5.3.71 LICENSES - Define Required Licenses for Run.....	308
5.3.72 LIVEOIL - Activate the Live Oil Phase (Oil with Free and Dissolved Gas).....	309
5.3.73 LOAD - Load a SAVE File for a Fast Restart.....	310
5.3.74 LOWSALT - Activate the Low Salt Brine Phase in the Brine Model.....	311
5.3.75 MEMORY - Define Allocated Memory (Retired).....	312
5.3.76 MESSAGE - Output User Message.....	313
5.3.77 MESSAGES - Define Message Print Limits and Stop Limits.....	313
5.3.78 MESSSRVC - Activate or Deactivate Database Message File Output.....	314
5.3.79 METRIC - Activate the Metric System of Units for the Model.....	315
5.3.80 MICP - Activate the Microbially Induced Calcite Precipitation Model.....	316
5.3.81 MINNPCOL - Define the Minimum Number of Newton Iterations Used to Update Well Targets.....	317
5.3.82 MISCIBLE - Define Miscibility Todd-Longstaff Parameters.....	318
5.3.83 MONITOR - Activate Output of the Monitoring Data and File.....	319
5.3.84 MSGFILE - Activate or Deactivate Message File Output.....	320
5.3.85 MULTIN - Activate the Non-Unified Multiple Input File Option.....	321
5.3.86 MULTOUT - Activate the Non-Unified Multiple Output File Option.....	323
5.3.87 MULTOUTS - Activate Non-Unified Multiple Summary Output File Option.....	325
5.3.88 MULTREAL - Activate Commercial Simulator's Multi-Realization License.....	326
5.3.89 NETWORK - Activate the Extended Network Option and Define Parameters.....	327
5.3.90 NINEPOIN - Activate the Nine-Point Discretization Option.....	328
5.3.91 NMATRIX - Activate the Discretized Matrix Dual Porosity Option.....	329
5.3.92 NNEWTF - Activate the Non-Newtonian Fluid Model.....	330
5.3.93 NOCASC - Activate Linear Solver Tracer Algorithm.....	331
5.3.94 NODPPM - Deactivate Fracture Porosity-Permeability Calculation.....	332
5.3.95 NOECHO - Deactivate Echoing of User Input Files to the Print File.....	333
5.3.96 NOHYST - Deactivate the Hysteresis Option.....	334
5.3.97 NOINSPEC - Deactivate Output of the INIT Index File.....	335
5.3.98 NOMONITO - Deactivate Output of the Monitoring Data and File.....	336
5.3.99 NONNC - Deactivate Non-Neighbor Connections.....	337
5.3.100 NORSSPEC - Deactivate Output of the RESTART Index File.....	338
5.3.101 NOSIM - Activate the No Simulation Mode for Data File Checking.....	339
5.3.102 NOWARN - Deactivate Warning Messages.....	340
5.3.103 NRSOUT - Defined Maximum Number of RESTART Elements.....	341
5.3.104 NSTACK - Define the Stack Length for the Iterative Linear Solver.....	342
5.3.105 NUMRES - Define the Number of Reservoir Grids.....	343
5.3.106 NUPCOL - Define the Maximum Number of Newton Iterations Used to Update Well Targets.....	344

5.3.107 OIL - Activate the Oil Phase in the Model.....	345
5.3.108 OPTIONS - Activate Various Program Options.....	346
5.3.109 PARALLEL - Define Parallel Run Configuration.....	347
5.3.110 PARTTRAC - Activate and Define Partitioned Tracer Option.....	348
5.3.111 PATHS - Define Filename Directory Path Aliases.....	349
5.3.112 PEDIMS - Define Petro-Elastic Model Regions and Table Dimensions.....	350
5.3.113 PETOPTS - Define Petrel and Generic Simulation File Options.....	351
5.3.114 PIMTDIMS - Define Well Productivity Scaling Table Dimensions.....	352
5.3.115 PINTDIMS - Define Polymer Molecular Weight Model Table Dimensions.....	353
5.3.116 POLYMER - Activate the Polymer Phase in the Model.....	355
5.3.117 POLYMW - Activate the Polymer Molecular Weight Transport Option.....	356
5.3.118 PRECSALT - Activate the OPM Flow Salt Precipitation Model.....	357
5.3.119 PSTEADY - Activate Pseudo Steady State Flow Calculation Option.....	361
5.3.120 RADIAL - Radial Grid Activation Option.....	362
5.3.121 REGDIMS - Define the Maximum Number of Regions for a Region Array.....	363
5.3.122 RIVRDIMS - Define the River Dimensions and Associated Data.....	365
5.3.123 ROCKCOMP - Activate Rock Compaction.....	366
5.3.124 RPTCPL - Activate Couple Simulation Reporting.....	368
5.3.125 RPTHMD - Define Well History Match Gradient Reporting Options.....	369
5.3.126 RPTRUNSP - Activate RUNSPEC Reporting.....	370
5.3.127 RSSPEC - Activate Output of the RESTART Index File.....	371
5.3.128 RUNSPEC - Define the Start of the RUNSPEC Section of Keywords.....	372
5.3.129 SAMG - Activate Algebraic Multi-Grid Linear Solver.....	373
5.3.130 SATOPTS - Activate Relative Permeability Assignment Options.....	374
5.3.131 SAVE - Activate Output of a SAVE File for Fast Restarts.....	378
5.3.132 SCDPDIMS - Define Scale Deposition and Damage Table Dimensions.....	379
5.3.133 SKIP - Activate Skipping of All Keywords and Input Data.....	380
5.3.134 SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	380
5.3.135 SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	380
5.3.136 SMRYDIMS - Define Maximum Number of Summary Vectors to be Written.....	381
5.3.137 SOLVDIMS - Define PEBI Grid Nested Factorization Solver Dimensions.....	382
5.3.138 SOLVENT - Activate the SOLVENT Phase in the Model.....	383
5.3.139 SPIDER - Spider Grid Activation Option.....	384
5.3.140 START - Simulation Start Date.....	385
5.3.141 SURFACT - Activate the Surfactant Phase in the Model.....	386
5.3.142 SURFACTW - Activate the Surfactant Phase with Wettability Changes in the Model.....	387
5.3.143 TABDIMS - Define the Number of Tables and the Table Dimensions.....	388
5.3.144 TEMP - Activate the Temperature Modeling Option.....	391
5.3.145 THERMAL - Activate the Thermal Modeling Option.....	392
5.3.146 TITLE - Define the Title for the Input Deck.....	396
5.3.147 TRACERS - Activate Tracer Options and Set Tracer Array Dimensions.....	397
5.3.148 TRPLPORO - Activate the Triple Porosity Model Option.....	399
5.3.149 UDADIMS - Define the Dimensions of the User Defined Arguments.....	400
5.3.150 UDQDIMS - Define the Dimensions of the User Defined UDQ Feature.....	401
5.3.151 UDQPARAM - Define Parameters for the User Defined Quantity Feature.....	403
5.3.152 UDTDIMS - Define the Dimensions of the User Defined Tables.....	405
5.3.153 UNCODHMD - Activate History Match Gradient Unencoded Output.....	406
5.3.154 UNIFIN - Activate The Unified Input File Option.....	407
5.3.155 UNIFOUT - Activate The Unified Output File Option.....	409
5.3.156 UNIFOUTS - Activate The Unified Output Summary File Option.....	411
5.3.157 UNIFSAVE - Activate The Unified Output Save File Option.....	412
5.3.158 VAPOIL - Activate the Vaporized Oil in Wet Gas Phase in the Model.....	413
5.3.159 VAPWAT - Activate Vaporized Water in the Dry and Wet Gas Phases.....	414
5.3.160 VE - Activate Vertical Equilibrium Model (Global).....	415
5.3.161 VFPIDIMS - Injection Vertical Flow Performance Table Dimensions.....	416
5.3.162 VFPPDIMS - Production Vertical Flow Performance Table Dimensions.....	417
5.3.163 VISAGE - Activate External Reservoir Geo-Mechanics VISAGE Option.....	418
5.3.164 VISCD - Activate Dual Porosity Viscous Displacement Option.....	419
5.3.165 WARN - Activate Warning Messages.....	420
5.3.166 WATER - Activate the Water Phase in the Model.....	421
5.3.167 WELLDIMS - Define the Wells and Group Dimensions.....	422
5.3.168 WPOTCALC - Well Potential Calculation Options.....	424
5.3.169 WSEGDIMS - Define Multi-Segment Well Dimensions.....	425

CHAPTER 6: GRID SECTION.....427

6.1 Introduction.....	429
6.2 Data Requirements.....	429
6.2.1 Cartesian Regular Grids.....	429
6.2.2 Irregular Corner-Point Grids.....	432
6.2.3 Radial Grids.....	435
6.2.4 Spider Grids.....	439
6.2.5 Rock Properties.....	442
6.3 Keyword Definitions.....	445
6.3.1 ACTNUM - Set the Status of a Grid Block To Active or Inactive.....	446
6.3.2 ADD - Add a Constant to a Specified Array.....	447
6.3.3 ADDREG - Add a Constant to an Array based on a Region Number.....	449
6.3.4 ADDZCORN - Add a Constant to the ZCORN Depth Array.....	452
6.3.5 AMALGAM - Define LGR Amalgamations.....	453
6.3.6 AQUANCON - Define Analytical Connections to the Grid.....	454
6.3.7 AQUCON - Define Numerical Aquifer Connections to the Grid.....	457
6.3.8 AQUCT - Define Carter-Tracy Analytical Aquifers.....	460
6.3.9 AQUNNC - Define Numerical Aquifer Non-Neighbor Connections.....	464
6.3.10 AQUNUM - Define Numerical Aquifer Properties.....	465
6.3.11 AUTOCOAR - Define Auto Refinement Grid Coarsen Area.....	469
6.3.12 BCCON - Define Boundary Conditions Connections.....	470
6.3.13 BOUNDARY - Define a Boundary Box for Printing.....	473
6.3.14 BOX - Define a Range of Grid Blocks to Enter Property Data.....	474
6.3.15 BTOBALFA - Dual Porosity Matrix to Fracture Multiplier (All Cells).....	476
6.3.16 BTOBALFV - Dual Porosity Matrix to Fracture Multiplier (Individual Cells).....	477
6.3.17 CARFIN - Define a Cartesian Local Grid Refinement.....	478
6.3.18 COALNUM - Define the Coal Region Numbers.....	480
6.3.19 COARSEN - Define Grid Coarsening Cells.....	481
6.3.20 COLLAPSE - Define Compressed Vertical Equilibrium Cells.....	482
6.3.21 COLUMNS - Define Input File Column Margins.....	483
6.3.22 COORD - Define a Set of Coordinates Lines for a Reservoir Grid.....	484
6.3.23 COORDSYS - Define Coordinate Grid Options.....	486
6.3.24 COPY - Copy Array Data to Another Array.....	487
6.3.25 COPYBOX - Copy Array Data Defined by a Box.....	489
6.3.26 COPYREG - Copy an Array to Another Array based on a Region Number.....	492
6.3.27 CRITPERM - Define Minimum Permeability for Vertical Equilibrium Grid Cell Compression.....	494
6.3.28 DEBUG - Define the Debug Data to be Printed to File.....	495
6.3.29 DIFFMR - Define Grid Block Radial Direction Diffusivity Multipliers.....	496
6.3.30 DIFFMR- - Define Grid Block Negative Radial Direction Diffusivity Multipliers.....	497
6.3.31 DIFFMTH- - Define Grid Block Negative Theta Direction Diffusivity Multipliers.....	498
6.3.32 DIFFMTH- - Define Grid Block Theta Direction Diffusivity Multipliers.....	499
6.3.33 DIFFMX - Define Grid Block X-Direction Diffusivity Multipliers.....	500
6.3.34 DIFFMX- - Define Grid Block Negative X-Direction Diffusivity Multipliers.....	501
6.3.35 DIFFMY - Define Grid Block Y-Direction Diffusivity Multipliers.....	502
6.3.36 DIFFMY- - Define Grid Block Negative Y-Direction Diffusivity Multipliers.....	503
6.3.37 DIFFMZ - Define Grid Block Z-Direction Diffusivity Multipliers.....	504
6.3.38 DIFFMZ- - Define Grid Block Negative Z-Direction Diffusivity Multipliers.....	505
6.3.39 DISPERC - Define the Mechanical Dispersivity for All Cells.....	506
6.3.40 DOMAINS - Define the Parallel Domain Properties.....	507
6.3.41 DPGRID - Activate The Matrix Cell to Fracture Cell Option.....	508
6.3.42 DPNUM - Define Dual and Single Porosity Grid Block Array.....	509
6.3.43 DR - Define the Size of Grid Blocks in the R Direction for All Cells.....	510
6.3.44 DRV - Define the Size of Grid Blocks in the R Direction via a Vector.....	512
6.3.45 DTHETA - Define the Size of Grid Blocks in the THETA Direction for All Cells.....	513
6.3.46 DTHETAV - Sets the Size of Grid Blocks in THETA Direction via a Vector.....	514
6.3.47 DUMPFLUX - Activate Writing Out of a Flux File.....	515
6.3.48 DX - Define the Size of Grid Blocks in the X Direction for All Cells.....	516
6.3.49 DXV - Define the Size of Grid Blocks in the X Direction via a Vector.....	517
6.3.50 DY - Define the Size of Grid Blocks in the Y Direction for All Cells.....	518
6.3.51 DYV - Define the Size of Grid Blocks in the Y Direction via a Vector.....	519

6.3.52 DZ - Define the Size of Grid Blocks in the Z Direction for All Cells.....	520
6.3.53 DZMATRIX - Matrix Block Height for Gravity Drainage Model For All Cells.....	521
6.3.54 DZMTRX - Matrix Block Height for Gravity Drainage Model for the Grid.....	522
6.3.55 DZMTRXV - Matrix Block Height for Gravity Drainage Model For All Cells.....	523
6.3.56 DZNET - Define Grid Block Net Thickness for All Cells.....	524
6.3.57 DZV - Define the Size of Grid Blocks in the Z Direction via a Vector.....	525
6.3.58 ECHO - Activate Echoing of User Input Files to the Print File.....	526
6.3.59 END - Define the End of the Input File.....	526
6.3.60 ENDBOX - Define the End of the BOX Defined Grid.....	527
6.3.61 ENDFIN - End the Definition of a Local Grid Refinement.....	528
6.3.62 ENDINC - Define the End of an Include File.....	529
6.3.63 ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	529
6.3.64 EQLZCORN - Modify the Depth of the Corner-Point Depth Array.....	530
6.3.65 EQUALREG - Sets an Array to a Constant by Region Number.....	531
6.3.66 EQUALS - Sets a Specified Array to a Constant.....	533
6.3.67 EXTFIN - Define an External Unstructured Local Grid Refinement.....	536
6.3.68 EXTHOST - Define Host Cells for External LGRs.....	537
6.3.69 EXTRAPMS - Activate Extrapolation Warning Messages.....	538
6.3.70 EXTREPGL - Define Host Cells for External Unstructured LGRs.....	539
6.3.71 FAULTS - Define Faults in the Grid Geometry.....	540
6.3.72 FILEUNIT - Activate Unit Consistency Checking.....	542
6.3.73 FLUXNUM - Define the Flux Regions.....	543
6.3.74 FLUXREG - Define Active Flux Regions.....	544
6.3.75 FLUXTYPE - Defines the Flux Boundary Type.....	545
6.3.76 FORMFEED - Defined the Print File Form-Feed Character.....	546
6.3.77 GDFILE - Load a Grid File.....	547
6.3.78 GDORIENT - Define Grid Orientation Parameters.....	548
6.3.79 GETDATA - Load and Assign Data Array from INIT or RESTART Files.....	549
6.3.80 GRID - Define the Start of the GRID Section of Keywords.....	550
6.3.81 GRIDFILE - Set the Grid File Output Options.....	551
6.3.82 GRIDUNIT - Define the Grid Units.....	552
6.3.83 HALFTRAN - Activate Half Block Transmissibility Calculations.....	553
6.3.84 HEATCR - Define Reservoir Rock Heat Capacity for All Cells.....	554
6.3.85 HEATCRT - Define Reservoir Rock Heat Capacity Temperature Dependence for All Cells.....	555
6.3.86 HMAQUNUM - History Match Numerical Aquifer Gradient Parameters.....	556
6.3.87 HMFAULTS - History Match Fault Gradient Parameters.....	557
6.3.88 HMMLAQUN - History Match Numerical Aquifer Gradient Multipliers.....	558
6.3.89 HMMLT - History Match Grid Permeability Gradient Cumulative Multipliers.....	559
6.3.90 HMMMREGT - History Match Region Transmissibility Gradient Cumulative Multipliers.....	560
6.3.91 HMMULRGT - History Match Region Transmissibility Parameters.....	561
6.3.92 HMMULTFT - History Match Fault Transmissibility Gradient Cumulative Multipliers.....	562
6.3.93 HMMULTSG - History Match Dual porosity Sigma Gradient Cumulative Multipliers.....	563
6.3.94 HRFIN - Define the Ratio of LGR Grid Blocks in the R-Direction.....	564
6.3.95 HXFIN - Define the Ratio of LGR Grid Blocks in the X-Direction.....	565
6.3.96 HYFIN - Define the Ratio of LGR Grid Blocks in the Y-Direction.....	566
6.3.97 HZFIN - Define the Ratio of LGR Grid Blocks in the Z-Direction.....	567
6.3.98 IHOST - Assign LGRs to Parallel Process Number.....	568
6.3.99 IMPORT - Import Grid File Data at the Current Position.....	569
6.3.100 INCLUDE - Load Another Data File at the Current Position.....	570
6.3.101 INIT - Activate the INIT File Option.....	571
6.3.102 INRAD - Define the Inner Radius of a Radial Grid.....	572
6.3.103 IONROCK - Define the Ion Exchange Capacity for All the Cells.....	573
6.3.104 ISOLNUM - Define the Independent Reservoir Regions.....	574
6.3.105 JFUNC - Activate the Leverett J-function Option.....	575
6.3.106 JFUNCRCR - Activate the Leverett J-function Saturation Table Option.....	578
6.3.107 LGRCOPY - Activate Local Grid Refinement Inheritance.....	581
6.3.108 LINKPERM - Assign Cell Permeabilities to Cell Faces.....	582
6.3.109 LTOSIGMA - Dual Porosity Viscous Displacement Sigma Parameters.....	583
6.3.110 LX - Dual Porosity Viscous Displacement X Direction Matrix Size for All Cells.....	584
6.3.111 LXFIN - Define Logarithmic LGR Grid Block Spacing in the X-Direction.....	586
6.3.112 LY - Dual Porosity Viscous Displacement Y Direction Matrix Size for All Cells.....	587
6.3.113 LYFIN - Define Logarithmic LGR Grid Block Spacing in the Y-Direction.....	589

6.3.114	LZ - Dual Porosity Viscous Displacement Z Direction Matrix Size for All Cells.....	590
6.3.115	LZFIN - Define Logarithmic LGR Grid Block Spacing in the Z-Direction.....	592
6.3.116	MAPAXES - Define the Map Origin Input Data.....	593
6.3.117	MAPUNITS - Define the Map Axes Units.....	594
6.3.118	MAXVALUE - Sets a Maximum Value for an Array Element.....	595
6.3.119	MESSAGE - Output User Message.....	597
6.3.120	MESSAGES - Define Message Print Limits and Stop Limits.....	597
6.3.121	MINNNCT - Set a Minimum Non-Nighbor Connection Transmissibility.....	598
6.3.122	MINPORV - Set a Minimum Grid Block Pore Volume Threshold for All Cells.....	599
6.3.123	MINPV - Set a Minimum Grid Block Pore Volume Threshold for All Cells.....	600
6.3.124	MINPVV - Set a Minimum Grid Block Pore Volume Threshold for Individual Cells.....	601
6.3.125	MINVALUE - Set a Minimum Value for an Array Element.....	603
6.3.126	MPFANUM - Define Multi-Point Flux Discretization Regions.....	605
6.3.127	MPFNNC - Define Multi-Point Flux Non-Nighbor Connections.....	606
6.3.128	MULTFLT - Multiply the Transmissibility of a Defined Fault by a Constant.....	607
6.3.129	MULTIPLY - Multiply a Specified Array by a Constant.....	608
6.3.130	MULTIREG - Multiply an Array by a Constant based on a Region Number.....	610
6.3.131	MULTNUM - Define the Multiple Transmissibility Regions.....	613
6.3.132	MULTPV - Multiply Cell Pore Volumes by a Constant.....	614
6.3.133	MULTR - Multiply Cell Transmissibility in the +R Direction.....	615
6.3.134	MULTR- - Multiply Cell Transmissibility in the -R Direction.....	616
6.3.135	MULTREGD - Multiply Diffusivities Between Regions.....	617
6.3.136	MULTREGH - Multiply Thermal Conductivities Between Regions.....	620
6.3.137	MULTREGP - Multiply Pore Volumes Based On Region Number.....	623
6.3.138	MULTREGT - Multiply Transmissibilities Between Regions.....	624
6.3.139	MULTTHT - Multiply Cell Transmissibility in the +Theta Direction.....	626
6.3.140	MULTTHT- - Multiply Cell Transmissibility in the -Theta Direction.....	627
6.3.141	MULTX - Multiply Cell Transmissibility in the +X Direction.....	629
6.3.142	MULTX- - Multiply Cell Transmissibility in the -X Direction.....	630
6.3.143	MULTY - Multiply Cell Transmissibility in the +Y Direction.....	631
6.3.144	MULTY- - Multiply Cell Transmissibility in the -Y Direction.....	632
6.3.145	MULTZ - Multiply Cell Transmissibility in the +Z Direction.....	633
6.3.146	MULTZ- - Multiply Cell Transmissibility in the -Z Direction.....	634
6.3.147	NEWTRAN - Activate Irregular Corner-Point Grid Transmissibilities.....	635
6.3.148	NINENUM - Define the Nine-Point Discretization Region.....	636
6.3.149	NMATOPTS - Define the Discretized Matrix Dual Porosity Parameters.....	637
6.3.150	NNC - Define Non-Nighbor Connections Between Cells Manually.....	638
6.3.151	NODPPM - Deactivate Fracture Porosity-Permeability Calculation.....	641
6.3.152	NOECHO - Deactivate Echoing of User Input Files to the Print File.....	641
6.3.153	NOGGF - Deactivate Output of Grid Geometry File.....	642
6.3.154	NOWARN - Deactivate Warning Messages.....	643
6.3.155	NTG - Define the Net-to-Gross Ratio for All the Cells.....	644
6.3.156	NXFIN - Define the Number of LGR Grid Blocks in the X-Direction.....	645
6.3.157	NYFIN - Define the Number of LGR Grid Blocks in the Y-Direction.....	646
6.3.158	NZFIN - Define the Number of LGR Grid Blocks in the Z-Direction.....	647
6.3.159	OLDTRAN - Activate Cartesian Regular Grid Transmissibilities.....	648
6.3.160	OLDTRANR - Activate Radial Regular Grid Transmissibilities.....	649
6.3.161	OPERATE - Define Mathematical Operations on Arrays.....	650
6.3.162	OPERATER - Define Mathematical Operations on Arrays by Region.....	653
6.3.163	OPERNUM - Define Regions for Mathematical Operations on Arrays.....	657
6.3.164	OUTRAD - Define the Outer Radius of a Radial Grid.....	658
6.3.165	PARAOPTS - Define Parallel Run Options.....	660
6.3.166	PEBI - Activate and Defined PEBI Grid Options.....	661
6.3.167	PERMAVE - Define Average Transmissibility Coefficients.....	662
6.3.168	PERMJFUN - Define Leverett J-Function Permeability for All Cells.....	663
6.3.169	PERMR - Define the Permeability for Each Cell in the R Direction.....	664
6.3.170	PERMTHT - Define the Permeability for Each Cell in the THETA Direction.....	665
6.3.171	PERMX - Define the Permeability in the X Direction for All the Cells.....	666
6.3.172	PERMY - Define the Permeability in the Y Direction for All the Cells.....	667
6.3.173	PERMZ - Define the Permeability in the Z Direction for All the Cells.....	668
6.3.174	PETGRID - Load a Generic Simulation Grid File.....	670
6.3.175	PINCH - Define Pinch-Out Layer Options.....	671
6.3.176	PINCHNUM - Define Pinch-Out Regions for the PINCHREG Keyword.....	673

6.3.177 PINCHOUT - Define Pinch-Out Layers Option (Fixed).....	674
6.3.178 PINCHREG - Define Pinch-Out Region Options.....	675
6.3.179 PINCHXY - Define Pinch-Out Areal Options.....	677
6.3.180 PORO - Define the Porosity Values for All the Cells.....	678
6.3.181 PYEND - End the Definition of a PYINPUT Section.....	679
6.3.182 PYINPUT - Define the Start of a PYINPUT Section.....	681
6.3.183 QMOBIL - Activate or Deactivate LGR End-Point Mobility Correction.....	683
6.3.184 RADFIN - Define a Radial Local Grid Refinement with One Column.....	684
6.3.185 RADFIN4 - Define a Radial Local Grid Refinement with Four Columns.....	685
6.3.186 REFINE - Start the Definition of a Local Grid Refinement.....	686
6.3.187 RESVNUM - Define Reservoir Coordinate Data Set.....	687
6.3.188 ROCKFRAC - Define the Rock Volume to Bulk Volume Fraction for All the Cells.....	689
6.3.189 RPTGRID - Define GRID Section Reporting.....	690
6.3.190 RPTGRIDL - Define GRID Section Reporting for LGRs.....	692
6.3.191 RPTINIT - Define Output to the INIT File.....	694
6.3.192 RPTISOL - Activate Isolated Reservoir Number Reporting.....	695
6.3.193 SIGMA - Dual Porosity Matrix to Fracture Sigma (All Cells).....	696
6.3.194 SIGMAGD - Dual Porosity Matrix to Fracture Sigma for Gravity Drainage (All Cells)....	697
6.3.195 SIGMAGDV - Dual Porosity Matrix to Fracture Sigma Gravity Drainage (Individual Cells)	698
6.3.196 SIGMAV - Dual Porosity Matrix to Fracture Sigma (Individual Cells).....	699
6.3.197 SKIP - Activate Skipping of All Keywords and Input Data.....	700
6.3.198 SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	700
6.3.199 SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	700
6.3.200 SMULTX - Multiply Cell Transmissibility in the +X Direction (Auto-Refinement).....	701
6.3.201 SMULTY - Multiply Cell Transmissibility in the +Y Direction (Auto-Refinement).....	702
6.3.202 SMULTZ - Multiply Cell Transmissibility in the +Z Direction (Auto-Refinement).....	703
6.3.203 SOLVDIRS - Define Linear Solver Principal Directions.....	704
6.3.204 SOLVNUM - Define PEBI Grid Correspondence to Solver Order.....	705
6.3.205 SPECGRID - Define the Dimensions of a Corner-Point Grid.....	706
6.3.206 THCGAS - Define Gas Phase Thermal Conductivity for All Cells.....	707
6.3.207 THCOIL - Define Oil Phase Thermal Conductivity for All Cells.....	709
6.3.208 THCONR - Define Rock and Fluid Thermal Conductivity for All Cells.....	711
6.3.209 THCONSF - Define Gas Saturation Dependent Thermal Conductivity Scaling Factor for	
All Cells.....	712
6.3.210 THCROCK - Define Reservoir Rock Thermal Conductivity for All Cells.....	714
6.3.211 THCWATER - Define Water Phase Thermal Conductivity for All Cells.....	716
6.3.212 THPRESFT - Define Fault Threshold Pressures.....	718
6.3.213 TOPS - Define the Depth at the Center of the Top Face for Each Cell.....	720
6.3.214 TRANGL - Define Non-Neighbor Connections Between Global and LGR Cells Manually	722
6.3.215 USEFLUX - Activate Flux Boundary Model and Define Flux File.....	724
6.3.216 USENOFLO - Activate Flux Boundary Model Without a Flux File.....	725
6.3.217 VEDEBUG - Vertical Equilibrium Debug Data Output.....	726
6.3.218 VEFIN - Activate Vertical Equilibrium Model (LGR).....	727
6.3.219 WARN - Activate Warning Messages.....	728
6.3.220 ZCORN - Define the Depth of Each Corner-Point of a Grid Block.....	729

CHAPTER 7: EDIT SECTION.....731

7.1 Introduction.....	733
7.2 Data Requirements.....	733
7.3 Keyword Definitions.....	735
7.3.1 ADD - Add a Constant to a Specified Array.....	735
7.3.2 ADDREG - Add a Constant to an Array based on a Region Number.....	735
7.3.3 BOUNDARY - Define a Boundary Box for Printing.....	735
7.3.4 BOX - Define a Range of Grid Blocks to Enter Property Data.....	735
7.3.5 COLUMNS - Define Input File Column Margins.....	735
7.3.6 COPY - Copy Array Data to Another Array.....	735
7.3.7 COPYREG - Copy an Array to Another Array based on a Region Number.....	736
7.3.8 DEBUG - Define the Debug Data to be Printed to File.....	736
7.3.9 DEPTH - Edits the Depth at the Center of Each Cell.....	737
7.3.10 DIFFR - Define Grid Block Radial Direction Diffusivity Values.....	738
7.3.11 DIFFTHT - Define Grid Block Theta Direction Diffusivity Values.....	739

7.3.12 DIFFX - Define Grid Block X-Direction Diffusivity Values.....	740
7.3.13 DIFFY - Define Grid Block Y-Direction Diffusivity Values.....	741
7.3.14 DIFFZ - Define Grid Block Z-Direction Diffusivity Values.....	742
7.3.15 ECHO - Activate Echoing of User Input Files to the Print File.....	743
7.3.16 EDIT - Define the Start of the EDIT Section of Keywords.....	744
7.3.17 EDITNNC - Scale Non-Neighbor Connections Between Cells Manually.....	745
7.3.18 EDITNNCR - Reset Non-Neighbor Connections Between Cells Manually.....	748
7.3.19 END - Define the End of the Input File.....	751
7.3.20 ENDBOX - Define the End of the BOX Defined Grid.....	751
7.3.21 ENDFIN - End the Definition of a Local Grid Refinement.....	751
7.3.22 ENDINC - Define the End of an Include File.....	751
7.3.23 ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	751
7.3.24 EQUALREG - Sets an Array to a Constant by Region Number.....	751
7.3.25 EQUALS - Sets a Specified Array to a Constant.....	751
7.3.26 EXTRAPMS - Activate Extrapolation Warning Messages.....	752
7.3.27 FILEUNIT - Activate Unit Consistency Verification.....	752
7.3.28 FORMFEED - Defined the Print File Form-Feed Character.....	752
7.3.29 GETDATA - Load and Assign Data Array from INIT or RESTART File.....	752
7.3.30 HMMULT - History Match Grid Transmissibility & Pore Volume Gradient Cumulative Multipliers.....	753
7.3.31 IMPORT - Import Grid File Data at the Current Position.....	754
7.3.32 INCLUDE - Load Another Data File at the Current Position.....	754
7.3.33 LGRCOPY - Activate Local Grid Refinement Inheritance.....	754
7.3.34 MAXVALUE - Sets a Maximum Value for an Array Element.....	754
7.3.35 MESSAGE - Output User Message.....	754
7.3.36 MESSAGES - Define Message Print Limits and Stop Limits.....	754
7.3.37 MINVALUE - Set a Minimum Value for an Array Element.....	755
7.3.38 MULTFLT - Multiply the Transmissibility of a Defined Fault by a Constant.....	755
7.3.39 MULTIPLY - Multiply a Specified Array by a Constant.....	755
7.3.40 MULTIREG - Multiply an Array by a Constant based on a Region Number.....	755
7.3.41 MULTPV - Multiply Cell Pore Volumes by a Constant.....	755
7.3.42 MULTR - Multiply Cell Transmissibility in the +R Direction.....	755
7.3.43 MULTR- - Multiply Cell Transmissibility in the -R Direction.....	756
7.3.44 MULTREGD - Multiply Diffusivities Between Regions.....	756
7.3.45 MULTREGH - Multiply Thermal Conductivities Between Regions.....	756
7.3.46 MULTREGP - Multiply Pore Volumes Based On Region Number.....	756
7.3.47 MULTREGT - Multiply Transmissibilities Between Regions.....	756
7.3.48 MULTTHT - Multiply Cell Transmissibility in the +Theta Direction.....	756
7.3.49 MULTTHT- - Multiply Cell Transmissibility in the -Theta Direction.....	757
7.3.50 MULTX - Multiply Cell Transmissibility in the +X Direction.....	757
7.3.51 MULTX- - Multiply Cell Transmissibility in the -X Direction.....	757
7.3.52 MULTY - Multiply Cell Transmissibility in the +Y Direction.....	757
7.3.53 MULTY- - Multiply Cell Transmissibility in the -Y Direction.....	757
7.3.54 MULTZ - Multiply Cell Transmissibility in the +Z Direction.....	757
7.3.55 MULTZ- - Multiply Cell Transmissibility in the -Z Direction.....	757
7.3.56 NOECHO - Deactivate Echoing of User Input Files to the Print File.....	758
7.3.57 NOWARN - Deactivate Warning Messages.....	758
7.3.58 OPERATE - Define Mathematical Operations on Arrays.....	758
7.3.59 OPERATER - Define Mathematical Operations on Arrays by Region.....	758
7.3.60 PORV - Define the Pore Volumes for All the Cells.....	759
7.3.61 PYEND - End the Definition of a PYINPUT Section.....	760
7.3.62 PYINPUT - Define the Start of a PYINPUT Section.....	760
7.3.63 REFIN - Start the Definition of a Local Grid Refinement.....	760
7.3.64 SKIP - Activate Skipping of All Keywords and Input Data.....	760
7.3.65 SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	760
7.3.66 SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	760
7.3.67 TRANR - Define the Transmissibility in the +R Direction for All the Cells.....	761
7.3.68 TRANTH - Define the Transmissibility in the +Theta Direction for All the Cells.....	762
7.3.69 TRANX - Define the Transmissibility in the X Direction for All the Cells.....	763
7.3.70 TRANY - Define the Transmissibility in the Y Direction for All the Cells.....	764
7.3.71 TRANZ - Define the Transmissibility in the Z Direction for All the Cells.....	765
7.3.72 WARN - Activate Warning Messages.....	766

CHAPTER 8: PROPS SECTION.....767

8.1 Introduction.....	769
8.2 Data Requirements.....	769
8.2.1 Fluid Properties.....	769
8.2.2 Classification of Reservoirs by Fluid Type.....	776
8.2.3 Fluid Property Tables.....	783
8.2.4 Saturation Tables (Relative Permeability and Capillary Pressure Tables).....	788
8.2.5 Saturation Table Generation - Corey Curves.....	795
8.2.6 Saturation Table Generation - LET Functions.....	797
8.2.7 Saturation Table Allocation.....	808
8.3 Keyword Definitions.....	809
8.3.1 ACF - Define Acentric Factors.....	810
8.3.2 ACTCO2S - Activity Model for CO2.....	811
8.3.3 ADD - Add a Constant to a Specified Array.....	812
8.3.4 ADDREG - Add a Constant to an Array based on a Region Number.....	812
8.3.5 ADSALNOD - Salt Concentration Based on SATNUM Array.....	813
8.3.6 ADSORP - Define Generalized Langmuir Adsorption Function.....	815
8.3.7 ALKADS - Define Alkaline Adsorption Functions.....	816
8.3.8 ALKROCK - Define Rock Alkaline Properties.....	817
8.3.9 ALPOLADS - Polymer Adsorption versus Alkaline Concentration Multipliers.....	818
8.3.10 ALSURFAD - Surfactant Adsorption versus Alkaline Concentration Multipliers.....	819
8.3.11 ALSURFST - Water-Oil Surface Tension versus Alkaline Concentration Multipliers.....	820
8.3.12 APIGROUP - Define API Tracking Number of Grouped Oil PVT Tables.....	821
8.3.13 AQUCT - Define Carter-Tracy Analytical Aquifers.....	822
8.3.14 AQU TAB - Define Carter-Tracy Aquifer Influence Functions.....	823
8.3.15 BDENSITY - Define the Surface Brine Density for the Fluid.....	831
8.3.16 BGGI - Define Gi Gas Formation Volume Factor Pressure Tables.....	833
8.3.17 BIC - Binary Interaction Coefficients.....	834
8.3.18 BIOF PARA - Define Properties for Models with Biofilms.....	835
8.3.19 BOGI - Define Gi Oil Formation Volume Factor Pressure Tables.....	837
8.3.20 BOX - Define a Range of Grid Blocks to Enter Property Data.....	838
8.3.21 C NAMES - Define Compositional Component Names.....	839
8.3.22 COALADS - Define Gas and Solvent Relative Adsorption Tables.....	840
8.3.23 COALPP - Define Gas and Solvent Partial Pressure Adsorption Tables.....	841
8.3.24 COLUMNS - Define Input File Column Margins.....	842
8.3.25 COPY - Copy Array Data to Another Array.....	842
8.3.26 COPYBOX - Copy Array Data Defined by a Box.....	842
8.3.27 COPYREG - Copy an Array to Another Array based on a Region Number.....	842
8.3.28 DEBUG - Define the Debug Data to be Printed to File.....	842
8.3.29 DENAQA - Specify Ezrokhi Coefficients for Aqueous Density.....	843
8.3.30 DENSITY - Define the Surface Oil, Water Gas Densities for the Fluids.....	845
8.3.31 DEPTH TAB - River Time and Depth Tables.....	847
8.3.32 DIAGDISP - Activate Alternate Form of Tracer Dispersion.....	848
8.3.33 DIFFAGAS - Define PVT Region Gas Component Diffusion Coefficients (Mass Fraction Formulation).....	849
8.3.34 DIFFAWAT - Define PVT Region Water Component Diffusion Coefficients (Mass Fraction Formulation).....	851
8.3.35 DIFFC - Define PVT Region Molecular Diffusion Tables.....	853
8.3.36 DIFFCGAS - Define PVT Region Gas Component Diffusion Coefficients.....	855
8.3.37 DIFFCOAL - Define Coal Bed Methane Gas Diffusion Data.....	857
8.3.38 DIFFCWAT - Define PVT Region Water Component Diffusion Coefficients.....	858
8.3.39 DIFFDP - Activate Dual Porosity Molecular Diffusion for Matrix-Fracture Flow Only.....	860
8.3.40 DIFFMICP - Define PVT Region MICP Diffusion Coefficients (Mass Concentration Formulation).....	861
8.3.41 DIFFMMF - Define Diffusivity Multipliers for Matrix-Fractures.....	863
8.3.42 DISPERSE - Define Dispersion Tables.....	864
8.3.43 DPKRMOD - Modify Matrix Oil Relative Permeability Data.....	865
8.3.44 DSPDEINT - Activate Brine Tracer Dispersion Interpolation by Water Density.....	866
8.3.45 ECHO - Activate Echoing of User Input Files to the Print File.....	867
8.3.46 EHYSYSTR - Define Hysteresis Model and Parameters.....	868
8.3.47 EHYSYTRR - Define Hysteresis Model and Parameters via SATNUM.....	871

8.3.48	END - Define the End of the Input File.....	872
8.3.49	ENDBOX - Define the End of the BOX Defined Grid.....	872
8.3.50	ENDFIN - End the Definition of a Local Grid Refinement.....	872
8.3.51	ENDINC - Define the End of an Include File.....	872
8.3.52	ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	872
8.3.53	ENKRVD - Define Relative Permeability End-Points versus Depth Functions.....	873
8.3.54	ENPCVD - Define Maximum Capillary Pressure versus Depth Functions.....	874
8.3.55	ENPTVD - Define Relative Permeability Saturation End-Points versus Depth.....	875
8.3.56	ENSPCVD - Define Capillary Pressure End-Points versus Depth.....	876
8.3.57	EOS - Specify Equation of State.....	877
8.3.58	EPSDBGS - Write End-Point Debug Data to the DEBUG File (Multiple).....	878
8.3.59	EPSDEBUG - Write End-Point Debug Data to the DEBUG File (Individual).....	879
8.3.60	EQUALREG - Sets an Array to a Constant by Region Number.....	880
8.3.61	EQUALS - Sets a Specified Array to a Constant.....	880
8.3.62	ESSNODE - Define Salt Concentration Data for Water-Oil Surface Tension.....	881
8.3.63	EXTRAPMS - Activate Extrapolation Warning Messages.....	882
8.3.64	FHERCHBL - Define Herschel-Bulkley Data versus Polymer Concentration.....	883
8.3.65	FILEUNIT - Activate Unit Consistency Verification.....	884
8.3.66	FILLEPS - Activate Saturation End-Point Export to the INIT File.....	885
8.3.67	FOAMADS - Define Foam Rock Adsorption Tables.....	886
8.3.68	FOAMDCYO - Define Foam Decay versus Oil Saturation Tables.....	888
8.3.69	FOAMDCYW - Define Foam Decay versus Water Saturation Tables.....	889
8.3.70	FOAMFCN - Define Foam Gas Mobility Reduction versus Capillary Number.....	890
8.3.71	FOAMFRM - Define Foam Gas Mobility Reduction versus Reference Mobility.....	891
8.3.72	FOAMFSC - Define Foam Gas Mobility versus Surfactant Concentration Functions.....	892
8.3.73	FOAMFSO - Define Foam Gas Mobility Reduction versus Oil Saturation.....	895
8.3.74	FOAMFST - Define Foam Gas-Water Surface Tension versus Surfactant Concentration.....	896
8.3.75	FOAMFSW - Define Foam Gas Mobility Reduction versus Water Saturation.....	897
8.3.76	FOAMMOB - Define Foam Gas Mobility versus Foam Concentration Tables.....	898
8.3.77	FOAMMOBP - Define Foam Mobility Reduction versus Oil Pressure.....	900
8.3.78	FOAMMOBS - Define Foam Mobility Reduction versus Shear.....	901
8.3.79	FOAMOPTS - Define Foam Model Options.....	902
8.3.80	FOAMROCK - Define Foam Rock Properties.....	903
8.3.81	FORMFEED - Defined the Print File Form-Feed Character.....	905
8.3.82	GASDENT - Define Gas Density Temperature Coefficients.....	906
8.3.83	GASJT - Define Gas Joule-Thomson Coefficient.....	908
8.3.84	GASVISCT - Define Gas Viscosity versus Temperature Functions.....	910
8.3.85	GETDATA - Load and Assign Data Array from INIT or RESTART File.....	912
8.3.86	GIALL - Define Gi Values and PVT Properties versus Pressure.....	913
8.3.87	GINODE - Define Gi Node Values.....	914
8.3.88	GRAVCONS - Re-Define Gravity Constant.....	915
8.3.89	GRAVITY - Define the Surface Oil, Water Gas Gravities for the Fluids.....	916
8.3.90	GSF - Gas Saturation Function Tables (Gas-Water Systems).....	918
8.3.91	HA - History Match End-Point Gradient Additive Modifier.....	920
8.3.92	HDISP - Define Tracer Mechanical Dispersivity Parameters.....	922
8.3.93	HM - History Match End-Point Gradient Multiplicative Modifier.....	923
8.3.94	HMMROCK - History Match Rock Compressibility Gradient Cumulative Multipliers.....	925
8.3.95	HMMROCKT - History Match Rock Compaction Gradient Cumulative Multipliers.....	926
8.3.96	HMPROPS - History Match End-Point Section Start.....	927
8.3.97	HMROCK - History Match Rock Compressibility Gradient Parameters.....	928
8.3.98	HMROCKT - History Match Rock Compaction Gradient Parameters.....	929
8.3.99	HMRREF - History Match Rock Table Reference Pressure Values.....	930
8.3.100	HWKRO - End-Point Scaling of Grid Cell Kro(Swl) (High Salinity and Water Wet).....	931
8.3.101	HWKRORG - End-Point Scaling of Grid Cell Kro(Sgcr) (High Salinity and Water Wet).....	932
8.3.102	HWKRORW - End-Point Scaling of Grid Cell Kro(Swcr) (High Salinity and Water Wet).....	933
8.3.103	HWKRW - End-Point Scaling of Grid Cell Krw(Sw = 1.0) (High Salinity and Water Wet).....	934
8.3.104	HWKRWR - End-Point Scaling of Grid Cell KRWR(Sw=1.0) (High Salinity and Water Wet).....	935
8.3.105	HWPWCW - End-Point Scaling of Grid Cell Water Capillary Pressure (High Salinity and Water Wet).....	936
8.3.106	HWSOGCR - End-Point Scaling Grid Cell SOGCR (High Salinity and Water Wet).....	937
8.3.107	HWSOWCR - End-Point Scaling Grid Cell SOWCR (High Salinity and Water Wet).....	938
8.3.108	HWSWCR - End-Point Scaling Grid Cell SWCR (High Salinity and Water Wet).....	939

8.3.109 HWSWL - End-Point Scaling Grid Cell SWL (High Salinity and Water Wet).....	940
8.3.110 HWSWLPC - End-Point Scaling Grid Cell SWLPC (High Salinity and Water Wet).....	941
8.3.111 HWSWU - End-Point Scaling Grid Cell SWU (High Salinity and Water Wet).....	942
8.3.112 HYDRHEAD - Define Hydraulic Head Output Reference Data.....	943
8.3.113 HYMOBGDR - Activate Carlson and Killough Alternative Drainage Hysteresis Option.	944
8.3.114 HYSTCHCK - Activate Hysteresis Imbibition and Drainage End-Point Validation.....	945
8.3.115 IKRG - End-Point Scaling of Grid Cell Krg(Sgu) (Imbibition).....	946
8.3.116 IKRGR - End-Point Scaling of Grid Cell Krgr(1-Sogcr) (Imbibition).....	948
8.3.117 IKRO - End-Point Scaling of Grid Cell Kro(Swl) (Imbibition).....	950
8.3.118 IKRORG - End-Point Scaling of Grid Cell Kro(Sgcr) (Imbibition).....	952
8.3.119 IKRORW - End-Point Scaling of Grid Cell Kro(Swcr) (Imbibition).....	954
8.3.120 IKRW - End-Point Scaling of Grid Cell Krw(Sw =1.0) (Imbibition).....	956
8.3.121 IKRWR - End-Point Scaling of Grid Cell KRWR(Sowcr) (Imbibition).....	958
8.3.122 IMKRVD - Imbibition Relative Permeability End-Points versus Depth Functions.....	960
8.3.123 IMPCVD - Imbibition Maximum Capillary Pressure versus Depth Functions.....	961
8.3.124 IMPORT - Import Grid File Data at the Current Position.....	962
8.3.125 IMPTVD - Imbibition Relative Permeability Saturation End-Points versus Depth.....	963
8.3.126 IMSPCVD - Imbibition Capillary Pressure Connate Saturations versus Depth.....	964
8.3.127 INCLUDE - Load Another Data File at the Current Position.....	965
8.3.128 INTPC - Activate Dual Porosity Integrated Capillary Pressure Option.....	966
8.3.129 IONXROCK - Define Ion Exchange Constant by Saturation Table Regions.....	967
8.3.130 IONXSURF - Define Surfactant Ion Exchange Constant by Saturation Table Regions....	968
8.3.131 IPCG - End-Point Scaling of Grid Cell Gas Capillary Pressure (Imbibition).....	969
8.3.132 IPCW - End-Point Scaling of Grid Cell Water Capillary Pressure (Imbibition).....	970
8.3.133 ISGCR - End-Point Scaling of Grid Cell Critical Gas Saturation (Imbibition).....	971
8.3.134 ISGL - End-Point Scaling of Grid Cell Connate Gas Saturation (Imbibition).....	972
8.3.135 ISGLPC - End-Point Scaling of Grid Cell Capillary Pressure Connate Gas Saturation (Imbibition).....	973
8.3.136 ISGU - End-Point Scaling of Grid Cell Maximum Gas Saturation (Imbibition).....	975
8.3.137 ISOGCR - End-Point Scaling of Grid Cell Critical Oil Saturation to Gas (Imbibition)....	976
8.3.138 ISOWCR - End-Point Scaling of Grid Cell Critical Oil Saturation to Water (Imbib.).....	978
8.3.139 ISWCR - End-Point Scaling of Grid Cell Critical Water Saturation (Imbibition).....	980
8.3.140 ISWL - End-Point Scaling of Grid Cell Connate Water Saturation (Imbibition).....	981
8.3.141 ISWLPC - End-Point Scaling Grid Cell Capillary Pressure Connate Water Saturations (Imbibition).....	982
8.3.142 ISWU - End-Point Scaling of Grid Cell Maximum Water Saturation (Imbibition).....	984
8.3.143 KRG - End-Point Scaling of Grid Cell Krg(Sgu) (Drainage).....	985
8.3.144 KRGR - End-Point Scaling of Grid Cell Krgr(1-Sogcr) (Drainage).....	988
8.3.145 KRO - End-Point Scaling of Grid Cell Kro(Swl) (Drainage).....	990
8.3.146 KRORG - End-Point Scaling of Grid Cell Kro(Sgcr) (Drainage).....	993
8.3.147 KRORW - End-Point Scaling of Grid Cell Kro(Swcr) (Drainage).....	995
8.3.148 KRW - End-Point Scaling of Grid Cell Krw(Sw =1.0) (Drainage).....	997
8.3.149 KRWR - End-Point Scaling of Grid Cell KRWR(Sowcr) (Drainage).....	1000
8.3.150 LANGMPL - Define Langmuir Pressure Grid Cell Multiplier.....	1002
8.3.151 LANGMUIR - Langmuir Adsorption Isotherm Tables.....	1003
8.3.152 LANGSOLV - Langmuir Adsorption Isotherm Solvent Tables.....	1004
8.3.153 LCUNIT - Define Linear Combination Rate and Volume Units.....	1005
8.3.154 LKRO - End-Point Scaling of Grid Cell Kro(Swl) (Low Salinity and Oil Wet).....	1006
8.3.155 LKRORG - End-Point Scaling of Grid Cell Kro(Sgcr) (Low Salinity and Oil Wet).....	1007
8.3.156 LKRORW - End-Point Scaling of Grid Cell Kro(Swcr) (Low Salinity and Oil Wet).....	1008
8.3.157 LKRW - End-Point Scaling of Grid Cell Krw(Sw =1.0) (Low Salinity and Oil Wet).....	1009
8.3.158 LKRWR - End-Point Scaling of Grid Cell KRWR(Sowcr) (Low Salinity and Oil Wet).....	1010
8.3.159 LPCW - End-Point Scaling of Grid Cell Water Capillary Pressure (Low Salinity and Oil Wet).....	1011
8.3.160 LSALTFNC - Define Low Salt Weighting Factors versus Salt Concentration Functions	1012
8.3.161 LSOGCR - End-Point Scaling Grid Cell SOGCR (Low Salinity and Oil Wet).....	1013
8.3.162 LSOWCR - End-Point Scaling Grid Cell SOWCR (Low Salinity and Oil Wet).....	1014
8.3.163 LSWCR - End-Point Scaling Grid Cell SWCR (Low Salinity and Oil Wet).....	1015
8.3.164 LSWL - End-Point Scaling Grid Cell SWL (Low Salinity and Oil Wet).....	1016
8.3.165 LSWLPC - End-Point Scaling Grid Cell SWLPC (Low Salinity and Oil Wet).....	1017
8.3.166 LSWU - End-Point Scaling Grid Cell SWU (Low Salinity and Oil Wet).....	1018
8.3.167 LWKRO - End-Point Scaling of Grid Cell Kro(Swl) (Low Salinity and Water Wet).....	1019
8.3.168 LWKRORG - End-Point Scaling of Grid Cell Kro(Sgcr) (Low Salinity and Water Wet)...	1020

8.3.169 LWKRORW - End-Point Scaling of Grid Cell Kro(Swcr) (Low Salinity and Water Wet)...	1021
8.3.170 LWKRW - End-Point Scaling of Grid Cell Krw(Sw =1.0) (Low Salinity and Water Wet)	1022
8.3.171 LWKRWR - End-Point Scaling of Grid Cell KRWR(Sw=1.0) (Low Salinity and Water Wet)	1023
8.3.172 LWPCW - End-Point Scaling of Grid Cell Water Capillary Pressure (Low Salinity and Water Wet).....	1024
8.3.173 LWSOGR - End-Point Scaling Grid Cell SOGR (Low Salinity and Water Wet).....	1025
8.3.174 LWSOWCR - End-Point Scaling Grid Cell SOWCR (Low Salinity and Water Wet).....	1026
8.3.175 LWSWCR - End-Point Scaling Grid Cell SWCR (Low Salinity and Water Wet).....	1027
8.3.176 LWSWL - End-Point Scaling Grid Cell SWL (Low Salinity and Water Wet).....	1028
8.3.177 LWSWLPC - End-Point Scaling Grid Cell SWLPC (Low Salinity and Water Wet).....	1029
8.3.178 LWSWU - End-Point Scaling Grid Cell SWU (Low Salinity and Water Wet).....	1030
8.3.179 MASSFLOW - Define River Mass Flow versus Time Tables.....	1031
8.3.180 MAXVALUE - Sets a Maximum Value for an Array Element.....	1032
8.3.181 MESSAGE - Output User Message.....	1032
8.3.182 MESSAGES - Define Message Print Limits and Stop Limits.....	1032
8.3.183 MICPPARA - Define Microbially Induced Calcite Precipitation Parameters (Retired)...	1033
8.3.184 MINVALUE - Set a Minimum Value for an Array Element.....	1034
8.3.185 MISC - Define Solvent Miscibility-Immiscibility Transform Functions.....	1035
8.3.186 MLANG - Define Langmuir Maximum Gas Concentration for All Grid Cells.....	1037
8.3.187 MLANGSLV - Define Langmuir Maximum Solvent Concentration for All Grid Cells....	1038
8.3.188 MSFN - Miscible Normalized Relative Permeability Tables.....	1039
8.3.189 MULTIPLY - Multiply a Specified Array by a Constant.....	1040
8.3.190 MULTIREG - Multiply an Array by a Constant based on a Region Number.....	1040
8.3.191 MW - Molecular Weights.....	1041
8.3.192 NCOMPS - Confirm Number of Compositional Components.....	1042
8.3.193 NOECHO - Deactivate Echoing of User Input Files to the Print File.....	1043
8.3.194 NOWARN - Deactivate Warning Messages.....	1043
8.3.195 NOWARNP - Deactivate End-Point Scaling Warning Messages.....	1044
8.3.196 OILDENT - Define Oil Density Temperature Coefficients.....	1045
8.3.197 OILJT - Define Oil Joule-Thomson Coefficient.....	1047
8.3.198 OILVISCT - Define Oil Viscosity versus Temperature Functions.....	1049
8.3.199 OPERATE - Define Mathematical Operations on Arrays.....	1051
8.3.200 OPERATER - Define Mathematical Operations on Arrays by Region.....	1051
8.3.201 OVERBURD - Define Rock Overburden Pressure versus Depth Tables.....	1052
8.3.202 PCFACT - Capillary Pressure Multiplication Factor as a Function of Porosity Change.	1054
8.3.203 PCG - End-Point Scaling of Grid Cell Maximum Gas Capillary Pressure (Drainage)....	1056
8.3.204 PCG32D - Gas-Oil Capillary Pressure versus Oil and Water Saturation Tables.....	1057
8.3.205 PCRT - Critical Pressures.....	1058
8.3.206 PCW - End-Point Scaling of Grid Cell Water Capillary Pressure (Drainage).....	1059
8.3.207 PCW32D - Water-Oil Capillary Pressure versus Oil and Gas Saturation Tables.....	1061
8.3.208 PECOEFS - Define Petro-Elastic Model Coefficients.....	1062
8.3.209 PEGTAB - Petro-Elastic Pressure Shear Modulus Table.....	1063
8.3.210 PEKTAB - Petro-Elastic Pressure Bulk Modulus Table.....	1064
8.3.211 PERMFAC - Permeability Multiplication Factor as a Function of Porosity Change....	1065
8.3.212 PLMIXPAR - Define the Polymer Todd-Longstaff Mixing Parameters.....	1067
8.3.213 PLYADS - Define Polymer Rock Adsorption Tables.....	1068
8.3.214 PLYADSS - Define Polymer Rock Adsorption with Salt Dependence Tables.....	1070
8.3.215 PLYATEMP - Define Polymer Adsorption Table Temperature.....	1072
8.3.216 PLYCAMAX - Define Polymer-Rock Maximum Adsorption by Cell.....	1074
8.3.217 PLYDHFLF - Define Polymer Thermal Degradation Half-Life Tables.....	1075
8.3.218 PLYESAL - Define Polymer Effective Salinity Coefficient.....	1077
8.3.219 PLYKRRF - Define Polymer Rock Permeability Reduction by Cell.....	1078
8.3.220 PLYMAX - Define Polymer-Salt Viscosity Mixing Concentrations.....	1079
8.3.221 PLYMWIJ - Polymer Molecular Weight Model Throughput and Velocity Table.....	1081
8.3.222 PLYRMDEN - Define Polymer Model In Situ Rock Density.....	1084
8.3.223 PLYROCK - Define Polymer-Rock Properties.....	1085
8.3.224 PLYSHEAR - Activate and Define Polymer Shearing Parameters.....	1087
8.3.225 PLYSHLOG - Activate and Define the Polymer Shearing Logarithmic Parameters.....	1089
8.3.226 PLYTRRF - Define Polymer Rock Permeability Reduction versus Temperature.....	1092
8.3.227 PLYTRRFA - Define Polymer Rock Permeability Reduction versus Temperature Option	1093
8.3.228 PLYVISC - Define Polymer Viscosity Scaling Factors.....	1094

8.3.229	PLYVISCS - Define Polymer-Salt Viscosity Scaling Factors.....	1096
8.3.230	PLYVISCT - Define Polymer-Temperature Viscosity Scaling Factors.....	1097
8.3.231	PLYVMH - Polymer Molecular Weight Model Polymer Viscosity Constants.....	1098
8.3.232	PLYVSCST - Define Polymer-Salt-Temperature Viscosity Scaling Factors.....	1102
8.3.233	PMAX - Maximum and Minimum Pressure for Total Compressibility Check.....	1103
8.3.234	PMISC - Define Miscibility versus Pressure Tables.....	1104
8.3.235	PPCWMAX - Define SWATINIT Calculated Capillary Pressure Constraints.....	1106
8.3.236	PROPS - Define the Start of the PROPS Section of Keywords.....	1108
8.3.237	PVCO - Oil PVT Properties for Dead Oil (Constant Compressibility).....	1109
8.3.238	PVCO - Oil PVT Properties for Live Oil.....	1111
8.3.239	PVDG - Gas PVT Properties for Dry Gas.....	1114
8.3.240	PVDO - Oil PVT Properties for Dead Oil.....	1117
8.3.241	PVDS - Solvent PVT Properties for the Solvent Model.....	1119
8.3.242	PVTG - Gas PVT Properties for Wet Gas with Vaporized Oil.....	1121
8.3.243	PVTGW - Gas PVT Properties for Dry Gas with Vaporized Water.....	1125
8.3.244	PVTGWO - Gas PVT Properties for Wet Gas with Vaporized Water and Oil.....	1128
8.3.245	PVTO - Oil PVT Properties for Live Oil.....	1131
8.3.246	PVTSOL - Oil PVT Properties for Live Oil versus CO2 Mass Fraction.....	1135
8.3.247	PVTW - Define Water Fluid Properties for Various Regions.....	1140
8.3.248	PVTWSALT - Define Brine Water Fluid Properties for Various Regions.....	1142
8.3.249	PVZG - Gas PVT Properties for Dry Gas (Z-Factor).....	1145
8.3.250	PYEND - End the Definition of a PYINPUT Section.....	1148
8.3.251	PYINPUT - Define the Start of a PYINPUT Section.....	1148
8.3.252	QHRATING - Define River Mass Flow versus Depth Tables.....	1149
8.3.253	REFINE - Start the Definition of a Local Grid Refinement.....	1150
8.3.254	RIVRSEC - Define River Cross-Section versus Depth Parameters.....	1151
8.3.255	RKTRMDIR - Activate ROCKTAB Keyword Directional Transmissibility Multipliers.....	1152
8.3.256	ROCK - Define the Rock Compressibility for Various Regions.....	1153
8.3.257	ROCK2D - Pore Volume Compaction versus Pressure and Sw Tables.....	1155
8.3.258	ROCK2DTR - Transmissibility Compaction versus Pressure and Sw Tables.....	1157
8.3.259	ROCKOPTS - Define Rock Compaction and Compressibility Options.....	1159
8.3.260	ROCKPAMA - Define Coal Palmer-Mansorri Rock Model Parameters.....	1161
8.3.261	ROCKTAB - Rock Compaction Tables.....	1162
8.3.262	ROCKTABH - Rock Compaction Hysteresis Tables.....	1165
8.3.263	ROCKTABW - Rock Compaction Tables (Water Induced).....	1168
8.3.264	ROCKTHSG - Rock Compaction Hysteresis Tables (Dual Porosity).....	1169
8.3.265	ROCKTSIG - Rock Compaction Tables (Dual Porosity).....	1170
8.3.266	ROCKWNOD - Water Saturation Values for Compaction Pressure-Sw Tables.....	1171
8.3.267	RPTPROPS - Define PROPS Section Reporting.....	1173
8.3.268	RSCONST - Define Constant GOR (Rs) for All Dead Oil PVT Fluids.....	1175
8.3.269	RSCONSTT - Define Constant GOR (Rs) for Each Dead Oil PVT Fluid.....	1176
8.3.270	RSGI - Define Gas-Oil Ratio versus Pressure and Gi Tables.....	1178
8.3.271	RTEMP - Define the Initial Reservoir Temperature for the Model.....	1179
8.3.272	RTEMPA - Define the Initial Reservoir Temperature for the Model.....	1180
8.3.273	RTEMPVD - Define the Initial Reservoir Temperature versus Depth Tables.....	1181
8.3.274	RVCONST - Define Constant CGR (Rv) for All Dry Gas PVT Fluids.....	1182
8.3.275	RVCONSTT - Define Constant CGR (Rv) for Each Dry Gas PVT Fluid.....	1183
8.3.276	RVGI - Define Condensate-Gas Ratio versus Pressure and Gi Tables.....	1185
8.3.277	RWGSALT - Water Vaporization versus Pressure and Salt Concentration.....	1186
8.3.278	SALINITY - Define the Reservoir Salinity for All Cells.....	1188
8.3.279	SALTMF - Define the Salt Liquid-Phase Mole Fraction for All Cells.....	1189
8.3.280	SALTNODE - Salt Concentration Based PVTNUM Array.....	1190
8.3.281	SALTSOL - Define the Salt Solubility Limit by Region.....	1191
8.3.282	SCALECRS - Define End-Point Scaling Option.....	1193
8.3.283	SCALELIM - End-Point Scaling versus Depth Maximum Water Saturation.....	1195
8.3.284	SDENSITY - Define the Miscible or Solvent Surface Gas Density.....	1196
8.3.285	SGCR - End-Point Scaling Grid Cell Critical Gas Saturations.....	1197
8.3.286	SGCWMIS - Miscible Critical Gas versus Water Saturation Functions.....	1198
8.3.287	SGF32D - Gas Saturation Tables versus Oil and Water Saturations.....	1199
8.3.288	SGFN - Gas Saturation Tables (Format Type 2).....	1200
8.3.289	SGL - End-Point Scaling Grid Cell Connate Gas Saturations.....	1202
8.3.290	SGLPC - End-Point Scaling Grid Cell Capillary Pressure Connate Gas Saturations.....	1203
8.3.291	SGOF - Gas-Oil Saturation Tables versus Gas (Format Type 1).....	1205

8.3.292	SGOFLET - Gas-Oil LET Relative Permeability Functions.....	1207
8.3.293	SGU - End-Point Scaling Grid Cell Gas Saturation.....	1211
8.3.294	SGWFLET - Gas-Water LET Relative Permeability Functions.....	1212
8.3.295	SGWFN - Gas-Water Saturation Tables (Format Type 2).....	1216
8.3.296	SHRATE - Activate Log-based Polymer Shearing and Define the Shear Rate Constant	1218
8.3.297	SKIP - Activate Skipping of All Keywords and Input Data.....	1219
8.3.298	SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	1219
8.3.299	SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	1219
8.3.300	SKPRPOLY - Polymer Molecular Weight Model Polymer Injection Skin Table.....	1220
8.3.301	SKPRWAT - Polymer Molecular Weight Model Water Injection Skin Table.....	1223
8.3.302	SKRO - End-Point Scaling of Grid Cell Kro(Swl) (Surfactant).....	1226
8.3.303	SKRORG - End-Point Scaling of Grid Cell Kro(Sgr) (Surfactant).....	1228
8.3.304	SKROW - End-Point Scaling of Grid Cell Kro(Swcr) (Surfactant).....	1229
8.3.305	SKRW - End-Point Scaling of Grid Cell Krw(Sw =1.0) (Surfactant).....	1230
8.3.306	SKRWR - End-Point Scaling of Grid Cell KRWR(Sowcr) (Surfactant).....	1231
8.3.307	SLGOF - Gas-Oil Saturation Tables versus Gas (Format Type 1).....	1233
8.3.308	SOCRS - End-Point Scaling Grid Cell Miscible Critical Oil Saturation with Respect to Water.....	1235
8.3.309	SOF2 - Oil Saturation Tables with Respect to Gas or Water (Format Type 2).....	1236
8.3.310	SOF3 - Oil Saturation Tables with Respect to Gas and Water (Format Type 2).....	1238
8.3.311	SOF32D - Oil Saturation Tables with Respect to Water and Gas (Three Phase).....	1240
8.3.312	SOGCR - End-Point Scaling Grid Cell Critical Oil Saturation with Respect to Gas.....	1241
8.3.313	SOMGAS - STONE1 Model Minimum Oil Saturation versus Gas Saturation.....	1242
8.3.314	SOMWAT - STONE1 Model Minimum Oil Saturation versus Water Saturation.....	1243
8.3.315	SORWMIS - Miscible Residual Oil versus Water Saturation Functions.....	1244
8.3.316	SOWCR - End-Point Scaling Grid Cell Critical Oil Saturation with Respect to Water.....	1245
8.3.317	SPECHEAT - Define the Specific Heat of Oil, Water and Gas.....	1246
8.3.318	SPECROCK - Define the Specific Heat of the Reservoir Rock.....	1248
8.3.319	SSFN - Solvent and Gas Relative Permeability Tables.....	1250
8.3.320	SSGCR - End-Point Scaling Grid Cell Surfactant Critical Gas Saturations.....	1252
8.3.321	SSGL - End-Point Scaling Grid Cell Surfactant Connate Gas Saturations.....	1253
8.3.322	SSOGR - End-Point Scaling Grid Cell Surfactant Critical Oil Saturation with Respect to Gas.....	1254
8.3.323	SSOWCR - End-Point Scaling Grid Cell Surfactant Critical Oil Saturation with Respect to Water.....	1255
8.3.324	SSWCR - End-Point Scaling Grid Cell Critical Water Saturation.....	1256
8.3.325	SSWL - End-Point Scaling Grid Cell Surfactant Connate Water Saturation.....	1257
8.3.326	SSWU - End-Point Scaling Grid Cell Surfactant Maximum Water Saturation.....	1258
8.3.327	STOG - Define Capillary Pressure Oil-Gas Surface Tension versus Pressure.....	1259
8.3.328	STONE - Activate Stone's Second Three Phase Oil Relative Permeability Model (Alias for STONE2).....	1260
8.3.329	STONE1 - Activate Stone's First Three Phase Oil Relative Permeability Model.....	1261
8.3.330	STONE1EX - Define Stone's First Three Phase Oil Relative Permeability Parameter.....	1262
8.3.331	STONE2 - Activate Stone's Second Three Phase Oil Relative Permeability Model.....	1263
8.3.332	STOW - Define Capillary Pressure Oil-Water Surface Tension versus Pressure.....	1264
8.3.333	STWG - Define Capillary Pressure Water-Gas Surface Tension versus Pressure.....	1265
8.3.334	SURFADDW - Defined Surfactant Adsorbed Concentration versus Wettability Fraction.....	1266
8.3.335	SURFADS - Define Surfactant Rock Adsorption Tables.....	1267
8.3.336	SURFCAPD - Capillary Number versus Miscibility Tables.....	1269
8.3.337	SURFESAL - Define Surfactant Effective Salinity Coefficient.....	1270
8.3.338	SURFROCK - Define Surfactant-Rock Properties.....	1271
8.3.339	SURFST - Surfactant Water-Oil Surface Tension versus Surfactant Concentration.....	1272
8.3.340	SURFSTES - Surfactant Water-Oil Surface Tension versus Surfactant and Salt Concentrations.....	1273
8.3.341	SURFVISC - Surfactant Solution Viscosity versus Concentration.....	1274
8.3.342	SWATINIT - Define the Initial Water Saturation Array for Capillary Pressure Scaling.....	1276
8.3.343	SWCR - End-Point Scaling Grid Cell Critical Water Saturation.....	1277
8.3.344	SWF32D - Water Saturation Tables with Respect to Oil and Gas (Three Phase).....	1278
8.3.345	SWFN - Water Saturation Tables (Format Type 2).....	1279
8.3.346	SWL - End-Point Scaling Grid Cell Connate Water Saturation.....	1281
8.3.347	SWLPC - End-Point Scaling Grid Cell Capillary Pressure Connate Water Saturations.....	1282
8.3.348	SWOF - Water-Oil Saturation Tables (Format Type 1).....	1284

8.3.349 SWOFLET - Water-Oil LET Relative Permeability Functions.....	1286
8.3.350 SWU - End-Point Scaling Grid Cell Maximum Water Saturation.....	1290
8.3.351 TCRIT - Critical Temperatures.....	1291
8.3.352 TEMPNODE - Temperature Table for Polymer Solution Viscosity.....	1292
8.3.353 TEMPTVD - Activate Temperature Flux Limited Transport Option.....	1293
8.3.354 TEMPVD - Define the Initial Reservoir Temperature versus Depth Table.....	1294
8.3.355 THCO2MIX - Specify Thermal Mixing Models.....	1295
8.3.356 TLMIXPAR - Define the Miscible Todd-Longstaff Mixing Parameters.....	1296
8.3.357 TOLCRIT - Define The Critical Saturation Tolerance.....	1297
8.3.358 TPAMEPS - Volumetric Strain versus Coal Gas Concentration Tables.....	1298
8.3.359 TPAMEPSS - Volumetric Strain versus Coal Solvent Concentration Tables.....	1299
8.3.360 TRACER - Define Passive Tracer Variables.....	1300
8.3.361 TRACERKM - Multi-Partitioned Tracer Option K(P) Tables.....	1302
8.3.362 TRACERKP - Standard Partitioned Tracer Option K(P) Tables.....	1303
8.3.363 TRACITVD - Activate and Define Tracer Implicit Flux Limited Transport Option.....	1304
8.3.364 TRACTVD - Activate Tracer Explicit Flux Limited Transport Option.....	1305
8.3.365 TRADS - Environmental Tracer Adsorption Tables.....	1306
8.3.366 TRDCY - Environmental Tracer Decay Tables.....	1307
8.3.367 TRDIF - Tracer Diffusion Tables.....	1308
8.3.368 TRDIS - Tracer Dispersion Table Number Allocation.....	1309
8.3.369 TRNHD - Activate Dispersion Non-Homogeneous Diffusion Option.....	1310
8.3.370 TRROCK - Environmental Tracer-Rock Property Data.....	1311
8.3.371 TZONE - End-Point Scaling Transition Zone Options.....	1312
8.3.372 VCRIT - Critical Volumes.....	1314
8.3.373 VDFLOW - Velocity Dependent Flow Coefficient for Grid Block Flow (Grid).....	1315
8.3.374 VDFLOWR - Velocity Dependent Flow Coefficient for Grid Block Flow (Region).....	1316
8.3.375 VEFRACTION - Vertical Equilibrium Relative Permeability Fraction (Grid).....	1317
8.3.376 VEFRACTION - Vertical Equilibrium Capillary Pressure Fraction (Grid).....	1318
8.3.377 VEFRACTION - Vertical Equilibrium Capillary Pressure Fraction (Cell).....	1319
8.3.378 VEFRACTION - Vertical Equilibrium Relative Permeability Fraction (Cell).....	1320
8.3.379 VISCAQA - Specify Ezrokhi Coefficients for Aqueous Viscosity.....	1321
8.3.380 VISCREF - Define Viscosity-Temperature Reference Conditions.....	1323
8.3.381 WAGHYSTR - Define Water-Alternating-Gas Hysteresis Parameters.....	1325
8.3.382 WARN - Activate Warning Messages.....	1328
8.3.383 WATDENT - Define Water Density Temperature Coefficients.....	1329
8.3.384 WATJT - Define Water Joule-Thomson Coefficient.....	1331
8.3.385 WATVISCT - Define Water Viscosity versus Temperature Functions.....	1333
8.3.386 WSF - Water Saturation Function Tables (Gas-Water Systems).....	1335
8.3.387 ZMFVD - Define Compositional Components versus Depth.....	1337

CHAPTER 9: REGIONS SECTION..... 1339

9.1 Introduction.....	1341
9.2 Data Requirements.....	1341
9.3 Keyword Definitions.....	1346
9.3.1 ADD - Add a Constant to a Specified Array.....	1346
9.3.2 ADDREG - Add a Constant to an Array based on a Region Number.....	1346
9.3.3 BOUNDARY - Define a Boundary Box for Printing.....	1346
9.3.4 BOX - Define a Range of Grid Blocks to Enter Property Data.....	1346
9.3.5 COLUMNS - Define Input File Column Margins.....	1346
9.3.6 COPY - Copy Array Data to Another Array.....	1346
9.3.7 COPYBOX - Copy Array Data Defined by a Box.....	1346
9.3.8 COPYREG - Copy an Array to Another Array based on a Region Number.....	1347
9.3.9 DEBUG - Define the Debug Data to be Printed to File.....	1347
9.3.10 ECHO - Activate Echoing of User Input Files to the Print File.....	1347
9.3.11 END - Define the End of the Input File.....	1347
9.3.12 ENDBOX - Define the End of the BOX Defined Grid.....	1347
9.3.13 ENDFIN - End the Definition of a Local Grid Refinement.....	1347
9.3.14 ENDINC - Define the End of an Include File.....	1348
9.3.15 ENDNUM - Define the End-Point Scaling Depth Region Numbers.....	1349
9.3.16 ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	1350
9.3.17 EOSNUM - Define the Equation of State Region Numbers.....	1351
9.3.18 EQLNUM - Define the Equilibration Region Numbers.....	1352

9.3.19	EQUALREG - Sets an Array to a Constant by Region Number.....	1353
9.3.20	EQUALS - Sets a Specified Array to a Constant.....	1353
9.3.21	EXTRAPMS - Activate Extrapolation Warning Messages.....	1353
9.3.22	FILEUNIT - Activate Unit Consistency Verification.....	1353
9.3.23	FIP - Define the Fluid In-Place Names and Region and Numbers.....	1354
9.3.24	FIPNUM - Define the Fluid In-Place Region Numbers.....	1357
9.3.25	FIPOWG - Activate Oil, Gas, and Water FIP Zone Reporting.....	1359
9.3.26	FORMFEED - Defined the Print File Form-Feed Character.....	1360
9.3.27	GETDATA - Load and Assign Data Array from INIT or RESTART File.....	1360
9.3.28	HA - History Match End-Point Gradient Additive Modifier.....	1360
9.3.29	HBNUM - Define Herschel-Bulkley Region Numbers.....	1361
9.3.30	HM - History Match Region Gradient Parameters.....	1362
9.3.31	HMPROPS - History Match End-Point Section Start.....	1364
9.3.32	HWSNUM - Define the Saturation Table Region Numbers (High Salinity and Water Wet).....	1365
9.3.33	IMBNUM - Define the Imbibition Saturation Table Region Numbers.....	1366
9.3.34	IMBNUMMF - Define the Imbibition Saturation Table Region Numbers (Matrix-Fracture).....	1367
9.3.35	IMPORT - Import Grid File Data at the Current Position.....	1368
9.3.36	INCLUDE - Load Another Data File at the Current Position.....	1368
9.3.37	KRNUM - Define the Directional Saturation Table Region Numbers.....	1369
9.3.38	KRNUMMF - Define the Saturation Table Region Numbers (Matrix-Fracture).....	1371
9.3.39	LSLTWNUM - Define the Low Salt Water Wet Saturation Table Region Numbers.....	1372
9.3.40	LSNUM - Define the Low Salt Oil Wet Saturation Table Region Numbers.....	1373
9.3.41	LWSLTNUM - Define the Low Salt Oil Wet Saturation Table Region Numbers.....	1374
9.3.42	LWSNUM - Define the Low Salt Water Wet Saturation Table Region Numbers.....	1375
9.3.43	MESSAGE - Output User Message.....	1376
9.3.44	MESSAGES - Define Message Print Limits and Stop Limits.....	1376
9.3.45	MISCNUM - Define the Miscibility Region Numbers.....	1377
9.3.46	MULTIPLY - Multiply a Specified Array by a Constant.....	1378
9.3.47	MULTIREG - Multiply an Array by a Constant based on a Region Number.....	1378
9.3.48	NOECHO - Deactivate Echoing of User Input Files to the Print File.....	1378
9.3.49	NOWARN - Deactivate Warning Messages.....	1378
9.3.50	OPERATE - Define Mathematical Operations on Arrays.....	1378
9.3.51	OPERATER - Define Mathematical Operations on Arrays by Region.....	1378
9.3.52	OPERNUM - Define Regions for Mathematical Operations on Arrays.....	1379
9.3.53	PENUM - Define the Petro-Elastic Region Numbers.....	1380
9.3.54	PLMIXNUM - Define the Polymer Region Numbers.....	1381
9.3.55	PVTNUM - Define the PVT Regions.....	1382
9.3.56	PYEND - End the Definition of a PYINPUT Section.....	1384
9.3.57	PYINPUT - Define the Start of a PYINPUT Section.....	1384
9.3.58	REFINE - Start the Definition of a Local Grid Refinement.....	1384
9.3.59	REGIONS - Define the Start of the REGIONS Section of Keywords.....	1385
9.3.60	RESIDNUM - Define Vertical Equilibrium Residual Flow Region Numbers.....	1386
9.3.61	ROCKNUM - Define Rock Compaction Table Region Numbers.....	1387
9.3.62	RPTREGS - Define REGIONS Section Reporting.....	1388
9.3.63	SATNUM - Define the Saturation Table Region Numbers.....	1390
9.3.64	SKIP - Activate Skipping of All Keywords and Input Data.....	1391
9.3.65	SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	1391
9.3.66	SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	1391
9.3.67	SURFNUM - Define the Surfactant Miscible Saturation Table Region Numbers.....	1392
9.3.68	SURFWNUM - Define the Saturation Table Region Numbers (High Salinity and Water Wet).....	1393
9.3.69	TNUM - Define Passive Tracer Concentration Regions.....	1394
9.3.70	TRKPF - Define Partitioned Tracer Regions.....	1396
9.3.71	WARN - Activate Warning Messages.....	1398
9.3.72	WH2NUM - Define WAG Hysteresis Saturation Table Region Numbers (Two Phase).....	1399
9.3.73	WH3NUM - Define WAG Hysteresis Saturation Table Region Numbers (Three Phase).....	1400

CHAPTER 10: SOLUTION SECTION..... 1401

10.1	Introduction.....	1403
10.2	Data Requirements.....	1407

10.2.1	Equilibrium Initialization.....	1407
10.2.2	Enumeration Initialization.....	1408
10.2.3	Restart Initialization.....	1410
10.2.4	Section Keywords.....	1413
10.3	Keyword Definitions.....	1422
10.3.1	ADD - Add a Constant to a Specified Array.....	1422
10.3.2	ADDREG - Add a Constant to an Array based on a Region Number.....	1422
10.3.3	APIVD - Equilibration Oil API Gravity versus Depth Tables.....	1423
10.3.4	AQANCONL - Define Analytical Connections to a LGR Grid.....	1425
10.3.5	AQANNC - Define Analytic Aquifer Non-Neighbor Connections.....	1428
10.3.6	AQANTRC - Define Analytic Aquifer Initial Tracer Concentrations.....	1429
10.3.7	AQUALIST - Define An Analytic Aquifer Name to Aquifer Numbers.....	1430
10.3.8	AQUANCON - Define Analytical Connections to the Grid.....	1431
10.3.9	AQUCHGAS - Define Constant Pressure Gas Analytical Aquifer Properties.....	1432
10.3.10	AQUCHWAT - Define Constant Pressure Water Analytical Aquifer Properties.....	1433
10.3.11	AQUCON - Define Numerical Aquifer Connections to the Grid.....	1434
10.3.12	AQUCT - Define Carter-Tracy Analytical Aquifers.....	1434
10.3.13	AQUFET - Define Fetkovich Analytical Aquifer and Connections.....	1435
10.3.14	AQUFETP - Define Fetkovich Analytical Aquifers.....	1438
10.3.15	AQUFLUX - Define Constant Flux Analytical Aquifer.....	1441
10.3.16	BC - Define Boundary Conditions.....	1443
10.3.17	BOUNDARY - Define a Boundary Box for Printing.....	1447
10.3.18	BOX - Define a Range of Grid Blocks to Enter Property Data.....	1447
10.3.19	COLUMNS - Define Input File Column Margins.....	1447
10.3.20	COPY - Copy Array Data to Another Array.....	1447
10.3.21	COPYREG - Copy an Array to Another Array based on a Region Number.....	1447
10.3.22	DATUM - Define the Datum Depth for the Model.....	1448
10.3.23	DATUMR - Define Datum Depths for the FIPNUM Regions.....	1449
10.3.24	DATUMRX - Define Datum Depths for the FIP Allocated Regions.....	1450
10.3.25	DEBUG - Define the Debug Data to be Printed to File.....	1451
10.3.26	DYNAMICR - Start of Dynamic Region Parameter Definition.....	1452
10.3.27	ECHO - Activate Echoing of User Input Files to the Print File.....	1453
10.3.28	END - Define the End of the Input File.....	1453
10.3.29	ENDBOX - Define the End of the BOX Defined Grid.....	1453
10.3.30	ENDDYN - End of Dynamic Region Parameter Definition.....	1454
10.3.31	ENDFIN - End the Definition of a Local Grid Refinement.....	1455
10.3.32	ENDINC - Define the End of an Include File.....	1455
10.3.33	ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	1455
10.3.34	EQUALREG - Sets an Array to a Constant by Region Number.....	1455
10.3.35	EQUALS - Sets a Specified Array to a Constant.....	1455
10.3.36	EQUIL - Define the Equilibration Initialization Data.....	1456
10.3.37	EXTRAPMS - Activate Extrapolation Warning Messages.....	1461
10.3.38	FILEUNIT - Activate Unit Consistency Verification.....	1461
10.3.39	FORMFEED - Defined the Print File Form-Feed Character.....	1461
10.3.40	GASCONC - Define the Initial Equilibration Coal Gas Concentration for All Grid Blocks.....	1462
10.3.41	GASSATC - Define the Initial Equilibration Saturated Coal Gas Concentration for All Grid Blocks.....	1463
10.3.42	GCVD - Define Equilibration Coal Gas Concentration versus Depth Tables.....	1464
10.3.43	GETDATA - Load and Assign Data Array from INIT or RESTART File.....	1466
10.3.44	GETGLOB - Activate Loading of Global Grid Restart Data Option.....	1467
10.3.45	GI - Define the Initial Equilibration Gi Values for All Grid Blocks.....	1468
10.3.46	HMAQUCT - History Match Carter-Tracy Aquifer Gradient Parameters.....	1469
10.3.47	HMAQUFET - History Match Fetkovich Aquifer Gradient Parameters.....	1470
10.3.48	HMMLCTAQ - History Match Carter-Tracy Aquifer Gradient Multipliers.....	1471
10.3.49	HMMLFTAQ - History Match Fetkovich Aquifer Gradient Multipliers.....	1472
10.3.50	HMMLTWCN - History Match Well Connection and Skin Multipliers.....	1473
10.3.51	HMWELCON - History Match Well Connection and Skin Parameters.....	1474
10.3.52	IMPORT - Import Grid File Data at the Current Position.....	1475
10.3.53	INCLUDE - Load Another Data File at the Current Position.....	1475
10.3.54	MESSAGE - Output User Message.....	1475
10.3.55	MESSAGES - Define Message Print Limits and Stop Limits.....	1475

10.3.56 MULTIPLY - Multiply a Specified Array by a Constant.....	1475
10.3.57 MULTIREG - Multiply an Array by a Constant based on a Region Number.....	1475
10.3.58 NOECHO - Deactivate Echoing of User Input Files to the Print File.....	1476
10.3.59 NOHMD - Deactivate History Match Gradient Derivative Calculations.....	1477
10.3.60 NOHMO - Deactivate History Match Gradient Derivative Calculations (Alias).....	1478
10.3.61 NOWARN - Deactivate Warning Messages.....	1479
10.3.62 OILAPI - Define the Initial Equilibration Oil API for All Grid Blocks.....	1480
10.3.63 OPERATE - Define Mathematical Operations on Arrays.....	1481
10.3.64 OPERATER - Define Mathematical Operations on Arrays by Region.....	1481
10.3.65 OUTSOL - Define Data to be Written to the RESTART File (Retired).....	1482
10.3.66 PBUB - Define the Initial Equilibration Bubble-Point Pressure for All Grid Blocks.....	1483
10.3.67 PBVD - Equilibration Bubble-Point versus Depth Tables.....	1484
10.3.68 PDEW - Define the Initial Equilibration Dew-Point Pressure for All Grid Blocks.....	1484
10.3.69 PDVD - Define Equilibration Dew-Point versus Depth Tables.....	1486
10.3.70 PRESSURE - Define the Initial Equilibration Pressures for All Grid Blocks.....	1487
10.3.71 PRVD - Define the Initial Equilibration Pressures versus Depth.....	1488
10.3.72 PYEND - End the Definition of a PYINPUT Section.....	1490
10.3.73 PYINPUT - Define the Start of a PYINPUT Section.....	1490
10.3.74 RAINFALL - Constant Flux Aquifer Rainfall Flux by Month.....	1491
10.3.75 RBEDCONT - Define River Grid Block Contact Area versus Depth.....	1492
10.3.76 REFIN - Start the Definition of a Local Grid Refinement.....	1493
10.3.77 RESTART - Restart Run From an Existing Restart File.....	1494
10.3.78 RIVERSYS - Define River System (Branch Structure and Boundary Conditions).....	1496
10.3.79 RPTRST - Define Data to be Written to the RESTART File.....	1497
10.3.80 RPTSOL - Define SOLUTION Section Reporting.....	1503
10.3.81 RS - Define the Initial Equilibration GOR (Rs) for All Grid Blocks.....	1506
10.3.82 RSVD - Equilibration Dissolved Gas-Oil Ratio (Rs) versus Depth Tables.....	1507
10.3.83 RSW - Define the Initial Equilibrium Solution Gas in Water Ratio for All Grid Blocks.....	1509
10.3.84 RTEMP - Define the Initial Reservoir Temperature for the Model.....	1511
10.3.85 RTEMPA - Define the Initial Reservoir Temperature for the Model.....	1511
10.3.86 RTEMPVD - Define the Initial Reservoir Temperature versus Depth Tables.....	1512
10.3.87 RV - Define the Initial Equilibration CGR (Rv) for All Grid Blocks.....	1514
10.3.88 RVVD - Equilibration Vaporized Oil-Gas Ratio (Rv) versus Depth Tables.....	1515
10.3.89 RVW - Define the Initial Equilibration Vaporized Water in Gas Ratio for All Grid Blocks.....	1517
10.3.90 RVWVD - Equilibration Vaporized Water-Gas Ratio (Rvw) versus Depth Tables.....	1519
10.3.91 SALT - Define the Initial Equilibration Salt Concentration for All Grid Blocks.....	1521
10.3.92 SALT - Define the Initial Precipitated Salt Volume Fraction for All Grid Blocks.....	1522
10.3.93 SALT - Define the Initial Precipitated Salt Volume Fraction versus Depth Tables.....	1524
10.3.94 SALTREST - Define the Restart Salt Concentration for All Grid Blocks.....	1526
10.3.95 SALTVD - Equilibration Salt Concentration versus Depth Tables.....	1527
10.3.96 SBIOF - Define The Initial Equilibration Biofilm Volume Fraction For All Grid Blocks.....	1530
10.3.97 SCALC - Define The Initial Equilibration Calcite Volume Fraction For All Grid Blocks.....	1531
10.3.98 SCVD - Define Equilibration Coal Solvent Concentration versus Depth Tables.....	1532
10.3.99 SFOAM - Define the Initial Equilibration Foam Concentration for All Grid Blocks.....	1534
10.3.100 SGAS - Define the Initial Equilibration Gas Saturation for All Grid Blocks.....	1535
10.3.101 SKIP - Activate Skipping of All Keywords and Input Data.....	1536
10.3.102 SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	1536
10.3.103 SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	1536
10.3.104 SMICR - Define The Initial Equilibration Microbial Concentration For All Grid Blocks.....	1537
10.3.105 SOIL - Define the Initial Equilibration Oil Saturation for All Grid Blocks.....	1538
10.3.106 SOLUTION - Define the Start of the SOLUTION Section of Keywords.....	1539
10.3.107 SOLVCONC - Define the Initial Coal Solvent Concentration for All Grid Blocks.....	1540
10.3.108 SOLVFRAC - Define the Initial Gas Solvent Fraction for All Grid Blocks.....	1541
10.3.109 SOXYG - Define The Initial Equilibration Oxygen Concentration For All Grid Blocks.....	1542
10.3.110 SPOLY - Define the Initial Equilibration Polymer Concentration for All Grid Blocks.....	1543
10.3.111 SPOLYMW - Define The Initial Equilibration Polymer Molecular Weights For All Grid Blocks.....	1544
10.3.112 SSOL - Define the Initial Equilibration Solvent Saturation for All Grid Blocks.....	1545
10.3.113 SUREA - Define The Initial Equilibration Urea Concentration For All Grid Blocks.....	1546
10.3.114 SURF - Define the Initial Equilibration Surfactant Concentration for All Grid Blocks.....	1547
10.3.115 SWAT - Define the Initial Equilibration Water Saturation for All Grid Blocks.....	1548

10.3.116 TBLK - Define Tracer Initial Grid Block Concentrations.....	1549
10.3.117 TEMPI - Define the Initial Temperature Values for All Cells.....	1551
10.3.118 TEMPVD - Define the Initial Reservoir Temperature versus Depth Tables.....	1552
10.3.119 THPRES - Define Equilibration Region Threshold Pressures.....	1554
10.3.120 TVDP - Define the Initial Equilibration Tracer Saturation versus Depth Functions....	1556
10.3.121 VAPPARS - Oil Vaporization Parameters.....	1559
10.3.122 VISDATES - Define External Reservoir Geo-Mechanics VISAGE Stress Dates.....	1561
10.3.123 VISOPTS - Define External Reservoir Geo-Mechanics VISAGE Options.....	1562
10.3.124 WARN - Activate Warning Messages.....	1563

CHAPTER 11: SUMMARY SECTION.....1565

11.1 Introduction.....	1567
11.2 Data Requirements.....	1570
11.2.1 Summary Variable Mnemonic Syntax.....	1570
11.2.2 Summary Variable Format.....	1573
11.2.3 Aquifer Variables.....	1577
11.2.4 Field, Group, Well, Well Connection, and Completion Variables.....	1579
11.2.5 Field, Group and Well Control Mode Variables.....	1587
11.2.6 Field, Region, and Block Variables.....	1590
11.2.7 Field and Region Recovery Variables.....	1596
11.2.8 Option Specific Variables - Alkaline Model.....	1598
11.2.9 Option Specific Variables - API and Tracer Tracking.....	1599
11.2.10 Option Specific Variables - Asphaltene Model.....	1602
11.2.11 Option Specific Variables - Brine Model.....	1603
11.2.12 Option Specific Variables - CO2STORE/H2STORE Model.....	1605
11.2.13 Option Specific Variables - Coal Bed Methane Model.....	1607
11.2.14 Option Specific Variables - Conductive Faults Model.....	1607
11.2.15 Option Specific Variables - Environmental Tracers.....	1607
11.2.16 Option Specific Variables - Formation Damage Model.....	1608
11.2.17 Option Specific Variables - Foam Model Tracers.....	1609
11.2.18 Option Specific Variables - Gas Field Operations Model.....	1610
11.2.19 Option Specific Variables - Gas Lift Optimization Model.....	1611
11.2.20 Option Specific Variables - Gas Calorific Value Reporting.....	1612
11.2.21 Option Specific Variables - Gi Psuedo-Compositional Model.....	1612
11.2.22 Option Specific Variables - LGR and Coarsening.....	1612
11.2.23 Option Specific Variables - Biofilm And MICP Model.....	1613
11.2.24 Option Specific Variables - Multi-Segment Wells.....	1614
11.2.25 Option Specific Variables - Network Model.....	1617
11.2.26 Option Specific Variables - OPM Flow Simulation Performance.....	1619
11.2.27 Option Specific Variables - Polymer Model.....	1624
11.2.28 Option Specific Variables - Pseudo Steady State.....	1627
11.2.29 Option Specific Variables - Reservoir Coupling.....	1627
11.2.30 Option Specific Variables - Solvent Model.....	1628
11.2.31 Option Specific Variables - Surfactant Model.....	1629
11.2.32 Option Specific Variables - Thermal Model.....	1630
11.2.33 Option Specific Variables - User Define Quantities.....	1631
11.2.34 Option Specific Variables - Wellbore Friction model.....	1633
11.3 Keyword Definitions.....	1634
11.3.1 ALL - Export Standard Summary Variable Vectors to File.....	1635
11.3.2 COLUMNS - Define Input File Column Margins.....	1638
11.3.3 DATE - Activate the DATE Option for the SUMMARY File.....	1639
11.3.4 DEBUG - Define the Debug Data to be Printed to File.....	1640
11.3.5 ECHO - Activate Echoing of User Input Files to the Print File.....	1640
11.3.6 END - Define the End of the Input File.....	1640
11.3.7 ENDINC - Define the End of an Include File.....	1640
11.3.8 ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	1640
11.3.9 EXCEL - Activate the EXCEL Option for the SUMMARY File.....	1641
11.3.10 EXTRAPMS - Activate Extrapolation Warning Messages.....	1642
11.3.11 FMWSET - Export Well Status Vectors for the Field to File.....	1643
11.3.12 FORMFEED - Defined the Print File Form-Feed Character.....	1645
11.3.13 GMWSET - Export Well Status Vectors by Group to File.....	1646

11.3.14 INCLUDE - Load Another Data File at the Current Position.....	1648
11.3.15 MESSAGE - Output User Message.....	1648
11.3.16 MESSAGES - Define Message Print Limits and Stop Limits.....	1648
11.3.17 MONITOR - Activate Output of the Monitoring Data and File.....	1648
11.3.18 NARROW - Activate Run Summary Narrow Column Output Option.....	1649
11.3.19 NEWTON - Activate Newton Iteration SUMMARY Output.....	1650
11.3.20 NMESSAGE - Export Cumulative Message Summary Variables to File.....	1651
11.3.21 NOECHO - Deactivate Echoing of User Input Files to the Print File.....	1653
11.3.22 NOMONITO - Deactivate Output of the Monitoring Data and File.....	1653
11.3.23 NOWARN - Deactivate Warning Messages.....	1653
11.3.24 OFM - Activate OFM File Output of the SUMMARY Data.....	1654
11.3.25 PERFORMA - Export Standard Simulator Performance Summary Variables to File....	1655
11.3.26 PYEND - End the Definition of a PYINPUT Section.....	1657
11.3.27 PYINPUT - Define the Start of a PYINPUT Section.....	1657
11.3.28 RPTONLY - Activate the Report Time Steps Only Option for the SUMMARY File.....	1658
11.3.29 RPTONLYO - Deactivate the Report Time Steps Only Option for the SUMMARY File. .	1659
11.3.30 RPTSMRY - Activate or Deactivate Summary List Report.....	1660
11.3.31 RUNSUM - Activate RSM File Output of the SUMMARY Data.....	1661
11.3.32 SEPARATE - Activate the Separate RSM File Output Option.....	1662
11.3.33 SKIP - Activate Skipping of All Keywords and Input Data.....	1663
11.3.34 SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	1663
11.3.35 SKIP300 - Activate Skipping of Compositional Keywords and Input Data.....	1663
11.3.36 SUMMARY - Define the Start of the SUMMARY Section of Keywords.....	1664
11.3.37 SUMTHIN - Define SUMMARY Data Reporting Time Steps.....	1665
11.3.38 WARN - Activate Warning Messages.....	1666

CHAPTER 12: SCHEDULE SECTION.....1667

12.1 Introduction.....	1669
12.2 Data Requirements.....	1670
12.2.1 History Matching Runs.....	1680
12.2.2 Prediction Runs.....	1682
12.2.3 Well Productivity.....	1687
12.3 Keyword Definitions.....	1690
12.3.1 ACTION - Define Action Conditions and Command Processing (Field).....	1691
12.3.2 ACTIONG - Define Action Conditions and Command Processing (Groups).....	1692
12.3.3 ACTIONR - Define Action Conditions and Command Processing (Regions).....	1693
12.3.4 ACTIONS - Define Action Conditions and Command Processing (Well Segments).....	1694
12.3.5 ACTIONW - Define Action Conditions and Command Processing (Wells).....	1695
12.3.6 ACTIONX - Define Action Conditions and Command Processing.....	1702
12.3.7 AITS - Activate Intelligent Time Stepping.....	1712
12.3.8 AITSOFF - Deactivate Intelligent Time Stepping.....	1712
12.3.9 APILIM - Define API Tracking Grid Block Limits.....	1713
12.3.10 AQUCHGAS - Define Constant Pressure Gas Analytical Aquifer Properties.....	1714
12.3.11 AQUCHWAT - Define Constant Pressure Water Analytical Aquifer Properties.....	1714
12.3.12 AQUCT - Define Carter-Tracy Analytical Aquifers.....	1714
12.3.13 AQUCWFAC - Modify Constant Pressure Water Analytical Aquifer Properties.....	1715
12.3.14 AQUFETP - Define Fetkovich Analytical Aquifers.....	1716
12.3.15 AQUFLUX - Define Constant Flux Analytical Aquifer.....	1716
12.3.16 BCPROP - Define Boundary Conditions Properties.....	1717
12.3.17 BOUNDARY - Define a Boundary Box for Printing.....	1722
12.3.18 BOX - Define a Range of Grid Blocks to Enter Property Data.....	1722
12.3.19 BRANPROP - Define Network Branch Properties for Extended Network Option.....	1723
12.3.20 CALTRAC - Define a Gas Calorific Value Tracer.....	1727
12.3.21 CECON - Define Well Connections Economic Limit Criteria.....	1728
12.3.22 CECONT - Define Well Connections Tracer Economic Limit Criteria.....	1729
12.3.23 COLUMNS - Define Input File Column Margins.....	1730
12.3.24 COMPDAT - Define Well Connections to the Grid.....	1731
12.3.25 COMPDATL - Define Well Connections to a LGR Grid.....	1735
12.3.26 COMPDATM - Define Well Connections to an Amalgamated LGR Grid.....	1739
12.3.27 COMPFLSH - Assign COMPDAT Differential-Flash Liberation Ratios.....	1740
12.3.28 COMPIMB - Assign Imbibition Saturation Tables to Well Connections.....	1741

12.3.29	COMPINJK - Assign Injection well Relative Permeability Values.....	1743
12.3.30	COMPLMPL - Assign Well LGR Connections to Completions.....	1744
12.3.31	COMPLUMP - Assign Well Connections to Completions.....	1747
12.3.32	COMPOFF - Deactivate Network Automatic Compressors.....	1749
12.3.33	COMPORD - Define Well Connection Ordering.....	1750
12.3.34	COMPRIV - Define Grid Cell Connections to a River.....	1752
12.3.35	COMPRP - Re-Scale Fluid Saturations of Well Connections.....	1753
12.3.36	COMPRPL - Re-Scale Fluid Saturations of Well LGR Connections.....	1754
12.3.37	COMPSEGL - Define Well Connections for Multi-Segment Wells in a LGR.....	1755
12.3.38	COMPSEGS - Define Well Connections for Multi-Segment Wells.....	1758
12.3.39	COMPTRAJ - Define Well Trajectory Connections to the Grid.....	1761
12.3.40	COMPVE - Re-Define Well Connection Depths.....	1765
12.3.41	COMPVEL - Re-Define Well LGR Connection Depths.....	1766
12.3.42	CPIFACT - Define Well Connection Transmissibility Multipliers.....	1767
12.3.43	CPIFACTL - Define Well Connection Transmissibility Multipliers in a LGR.....	1768
12.3.44	CSKIN - Re-Define Well Connection Skin Factors.....	1769
12.3.45	DATES - Advance Simulation by Reporting Date.....	1771
12.3.46	DCQDEFN - Define Gas DCQ Units as Rate or Energy.....	1774
12.3.47	DEBUG - Define the Debug Data to be Printed to File.....	1775
12.3.48	DELAYACT - Define Delayed Action Keywords.....	1776
12.3.49	DIFFMMF - Define Diffusivity Multipliers for Matrix-Fractures.....	1777
12.3.50	DIMPES - Define IMPES Dynamic Solution Parameters.....	1778
12.3.51	DIMPLICIT - Activate Fully Implicit Dynamic Solution Formulation.....	1779
12.3.52	DRILPRI - Define Prioritized Drilling Queue Priority Parameters.....	1780
12.3.53	DRSDT - Solution Gas (Rs) Maximum Rate of Increase Parameters.....	1781
12.3.54	DRSDTCON - CO2 Convective Dissolution Parameters.....	1783
12.3.55	DRSDTR - Solution Gas (Rs) Maximum Rate of Increase Parameters by Region.....	1787
12.3.56	DRVDT - Vaporized Oil (Rv) Maximum Rate of Increase Parameters.....	1789
12.3.57	DRVDT - Vaporized Oil (Rv) Maximum Rate of Increase Parameters by Region.....	1790
12.3.58	DUMPCUPL - Activate Output to the Reservoir Coupling File.....	1792
12.3.59	DYNAMICR - Define Dynamic Region Parameters.....	1793
12.3.60	ECHO - Activate Echoing of User Input Files to the Print File.....	1793
12.3.61	END - Define the End of the Input File.....	1793
12.3.62	ENDACTIO - End the Definition of ACTION Commands.....	1794
12.3.63	ENDBOX - Define the End of the BOX Defined Grid.....	1795
12.3.64	ENDDYN - End of Dynamic Region Parameter Definition.....	1795
12.3.65	ENDFIN - End the Definition of a Local Grid Refinement.....	1795
12.3.66	ENDINC - Define the End of an Include File.....	1795
12.3.67	ENDSKIP - Deactivate Skipping of Keywords and Input Data.....	1795
12.3.68	EPSDBGS - Write End-Point Debug Data to the DEBUG File (Multiple).....	1795
12.3.69	EPSDEBUG - Write End-Point Debug Data to the DEBUG File (Individual).....	1795
12.3.70	EXCAVATE - Set the Status of a Grid Block To Active or Excavate.....	1796
12.3.71	EXIT - Exit Simulation from within an Action Section.....	1797
12.3.72	EXTRAPMS - Activate Extrapolation Warning Messages.....	1799
12.3.73	FBHPDEF - Define Well Default BHP Target and Constraints.....	1800
12.3.74	FILEUNIT - Activate Unit Consistency Verification.....	1801
12.3.75	FORMFEED - Defined the Print File Form-Feed Character.....	1801
12.3.76	GASBEGIN - Define Start of Annual Scheduling Section.....	1802
12.3.77	GASEND - Define End of Annual Scheduling Section.....	1803
12.3.78	GASFCOMP - Define Automatic Gas Compressors.....	1804
12.3.79	GASFDECR - Define Field Gas Sales Contract Monthly Reduction.....	1805
12.3.80	GASFDELC - Define Gas Deliverability Calculation.....	1806
12.3.81	GASFTARG - Define Field Gas Sales Contract Monthly Target.....	1807
12.3.82	GASMONT - Define Start of Annual Scheduling Event.....	1808
12.3.83	GASPERIO - Advance Simulation by Gas Contract Period.....	1809
12.3.84	GASYEAR - Advance Simulation by Gas Contract Year.....	1810
12.3.85	GCALECON - Group Economic Criteria for Production Calorific Groups.....	1811
12.3.86	GCONCAL - Group Production Calorific Targets.....	1813
12.3.87	GCONENG - Group Production Energy Targets.....	1815
12.3.88	GCONINJE - Group Injection Targets and Constraints.....	1816
12.3.89	GCONPRI - Group Production Priority Targets and Constraints.....	1820
12.3.90	GCONPROD - Group Production Targets and Constraints.....	1824
12.3.91	GCONSALE - Define Group Sales Gas Production Targets and Constraints.....	1830

12.3.92	GCONSUMP - Define Group Gas Consumption and Gas Import Targets.....	1833
12.3.93	GCONTOL - Define Group Constraint Tolerance.....	1836
12.3.94	GCUTBACK - Define Group Cutback Limits and Parameters.....	1838
12.3.95	GCUTBACT - Define Group Tracer Cutback Limits and Parameters.....	1839
12.3.96	GDCQ - Define Group Multiple Daily Contract Quantities.....	1840
12.3.97	GDCQECON - Group Economic Criteria for DCQ Production Groups.....	1841
12.3.98	GDRILPOT - Define Group Potential Rates for Automatic Drilling.....	1843
12.3.99	GECON - Group Economic Criteria for Production Groups.....	1844
12.3.100	GECONT - Group Tracer Economic Criteria for Production Groups.....	1847
12.3.101	GEFAC - Define Group Efficiency.....	1850
12.3.102	GLIFTLIM - Group Artificial Lift Constraints.....	1852
12.3.103	GLIFTOPT - Define Group Gas Optimization Limits.....	1854
12.3.104	GNETDP - Group Network Pressure and Rate Controls.....	1856
12.3.105	GNETINJE - Define Group Injection Network Configuration.....	1857
12.3.106	GNETPUMP - Standard Network Automatic Compressor and Pumps.....	1858
12.3.107	GPMaint - Define Group Pressure Maintenance Targets and Controls.....	1859
12.3.108	GRADGRUP - Define Group History Match Gradient File Output.....	1863
12.3.109	GRADRESV - Define Solution Derivative History Match Gradient Output.....	1864
12.3.110	GRADRFT - Define RFT Derivative History Match Gradient Output.....	1865
12.3.111	GRADWELL - Define Well History Match Gradient File Output.....	1866
12.3.112	GRDREACH - Define River and Grid Block Connections.....	1867
12.3.113	GRUPMAST - Define Master and Slave Groups.....	1868
12.3.114	GRUPNET - Define Group Standard Network Parameters.....	1869
12.3.115	GRUPRIG - Group Drilling and Workover Rig Specifications.....	1873
12.3.116	GRUPLAV - Define Slave Groups in Slave Reservoirs.....	1874
12.3.117	GRUPTARG - Modify Group Targets and Constraints Values.....	1875
12.3.118	GRUPTREE - Define Group Tree Hierarchy.....	1877
12.3.119	GSATINJE - Define Group Satellite Injection Rates.....	1879
12.3.120	GSATPROD - Define Group Satellite Production Rates.....	1883
12.3.121	GSEPCOND - Assign Group Separators.....	1887
12.3.122	GSSCPTST - Perform Sustainable Capacity Test.....	1888
12.3.123	GSWINGF - Define Group Multiple Gas Contract Parameters.....	1889
12.3.124	GTADD - Add a Constant to a Group Target or Constraint.....	1890
12.3.125	GTMULT - Multiply Group Target or Constraint by a Constant.....	1891
12.3.126	GUIDECAL - Scale Guide Rates Based on Gas Calorific Value.....	1892
12.3.127	GUIDERAT - Define Group Guide Rate Formula.....	1893
12.3.128	GUPFREQ - Instantaneous Gradient Option Update Frequency.....	1898
12.3.129	GWRTWCV - Instantaneous Gradient Option Well Variables.....	1899
12.3.130	HMWPIMLT - History Match Well Productivity Index Parameters.....	1900
12.3.131	INCLUDE - Load Another Data File at the Current Position.....	1901
12.3.132	LGRFREE - Activate Local Grid Refinement Independent Time Steps.....	1902
12.3.133	LGRLOCK - Deactivate Local Grid Refinement Independent Time Steps.....	1903
12.3.134	LGROFF - Deactivate a Local Grid Refinement.....	1904
12.3.135	LGRON - Activate a Local Grid Refinement.....	1905
12.3.136	LIFTOPT - Activate Gas Lift Optimization.....	1906
12.3.137	LINCOM - Define Linear Combination Coefficients.....	1908
12.3.138	MATCORR - Activate the Material Balance Correction Option.....	1909
12.3.139	MESSAGE - Output User Message.....	1910
12.3.140	MESSAGES - Define Message Print Limits and Stop Limits.....	1910
12.3.141	MESSOPTS - Reset Severity Level for Forced Time Steps.....	1911
12.3.142	MULSGGD - Multiply Matrix-Fracture Coupling for Oil-Gas Gravity Drainage for All Cells.....	1912
12.3.143	MULSGGDV - Multiply Matrix-Fracture Coupling for Oil-Gas Gravity Drainage for Individual Cells.....	1913
12.3.144	MULTFLT - Multiply the Transmissibility of a Defined Fault by a Constant.....	1914
12.3.145	MULTPV - Multiply Cell Pore Volumes by a Constant.....	1914
12.3.146	MULTR - Multiply Cell Transmissibility in the +R Direction.....	1914
12.3.147	MULTR- - Multiply Cell Transmissibility in the -R Direction.....	1914
12.3.148	MULTREGT - Multiply Transmissibilities Between Regions.....	1914
12.3.149	MULTSIG - Multiply Matrix-Fracture Coupling for All Cells.....	1915
12.3.150	MULTSIGV - Multiply Matrix-Fracture Coupling for Individual Cells.....	1916
12.3.151	MULTTHT - Multiply Cell Transmissibility in the +Theta Direction.....	1917
12.3.152	MULTTHT- - Multiply Cell Transmissibility in the -Theta Direction.....	1917

12.3.153	MULTX - Multiply Cell Transmissibility in the +X Direction.....	1917
12.3.154	MULTX- - Multiply Cell Transmissibility in the -X Direction.....	1917
12.3.155	MULTY - Multiply Cell Transmissibility in the +Y Direction.....	1917
12.3.156	MULTY- - Multiply Cell Transmissibility in the -Y Direction.....	1917
12.3.157	MULTZ - Multiply Cell Transmissibility in the +Z Direction.....	1917
12.3.158	MULTZ- - Multiply Cell Transmissibility in the -Z Direction.....	1918
12.3.159	NCONSUMP - Node Gas Consumption (Extended Network).....	1919
12.3.160	NEFAC - Node Efficiency Factors (Extended Network).....	1920
12.3.161	NETBALAN - Network Balancing Parameters.....	1921
12.3.162	NETCOMPA - Define Automatic Compressors (Extended Network).....	1925
12.3.163	NEXT - Maximum Next Time Step Size (Alias for NEXTSTEP).....	1926
12.3.164	NEXTSTEP - Maximum Next Time Step Size.....	1927
12.3.165	NEXTSTPL - Maximum Next Time Step Size (LGR).....	1930
12.3.166	NODEPROP - Define Network Node Properties for Extended Network.....	1931
12.3.167	NOECHO - Deactivate Echoing of User Input Files to the Print File.....	1934
12.3.168	NOHMD - Deactivate History Match Gradient Derivative Calculations.....	1934
12.3.169	NOHMO - Deactivate History Match Gradient Derivative Calculations (Alias).....	1934
12.3.170	NOSIM - Activate the No Simulation Mode for Data File Checking.....	1934
12.3.171	NOWARN - Deactivate Warning Messages.....	1934
12.3.172	NUPCOL - Define the Number of Newton Iterations Used to Update Well Targets....	1935
12.3.173	NWATREM - Node Water Removal (Extended Network).....	1936
12.3.174	OUTSOL - Define Data to be Written to the RESTART File (Retired).....	1937
12.3.175	PICOND - Define the Generalized Pseudo Pressure Parameters.....	1938
12.3.176	PIMULTAB - Define Well Productivity Index versus Water Cut Tables.....	1939
12.3.177	PLYADS - Define Polymer Rock Adsorption Tables.....	1941
12.3.178	PLYDHFLF - Define Polymer Thermal Degradation Half-Life Tables.....	1941
12.3.179	PLYMAX - Define Polymer-Salt Viscosity Mixing Concentrations.....	1941
12.3.180	PLYROCKM - Modify Polymer-Rock Properties.....	1942
12.3.181	PLYSHEAR - Activate and Define Polymer Shearing Parameters.....	1943
12.3.182	PLYSHLOG - Activate and Define the Polymer Shearing Logarithmic Parameters....	1943
12.3.183	PLYVISC - Define Polymer Viscosity Scaling Fact.....	1943
12.3.184	PLYVISCS - Define Polymer-Salt Viscosity Scaling Factors.....	1943
12.3.185	PLYVISCT - Define Polymer-Temperature Viscosity Scaling Factors.....	1943
12.3.186	PLYVSCST - Define Polymer-Salt-Temperature Viscosity Scaling Factors.....	1944
12.3.187	PRIORITY - Activate and Define Well Prioritization Coefficients.....	1945
12.3.188	PRORDER - Define a Group Production Rules Sequence.....	1947
12.3.189	PYACTION - Define Python Based Action Conditions and Command Processing.....	1948
12.3.190	PYEND - End the Definition of a PYINPUT Section.....	1955
12.3.191	PYINPUT - Define the Start of a PYINPUT Section.....	1955
12.3.192	QDRILL - Define Sequential Drilling Queue Wells.....	1956
12.3.193	RAINFALL - Constant Flux Aquifer Rainfall Flux by Month.....	1957
12.3.194	RCMASTS - Reservoir Coupling Group Minimum Time Step for Flow Restriction.....	1958
12.3.195	REACHES - Define River Reaches Structure.....	1959
12.3.196	READDATA - Read Schedule Data Based on Current Time Step.....	1960
12.3.197	REFINE - Start the Definition of a Local Grid Refinement.....	1961
12.3.198	RIVDEBUG - Define the Debug Data to be Printed to File (Rivers).....	1962
12.3.199	RIVERSYS - Define River System (Branch Structure and Boundary Conditions).....	1963
12.3.200	RIVRPROP - Modify River Reaches Properties.....	1964
12.3.201	RIVSALT - Define River Upstream Flow Salt Concentrations.....	1965
12.3.202	RIVTRACE - Define River Upstream Flow Tracer Concentrations.....	1966
12.3.203	RPTHMG - Define Well History Match Gradient Reporting (Groups).....	1967
12.3.204	RPTHMW - Define Well History Match Gradient Reporting (Wells).....	1968
12.3.205	RPTONLY - Activate the Report Time Steps Only Option for the SUMMARY File.....	1969
12.3.206	RPTONLYO - Deactivate the Report Time Steps Only Option for the SUMMARY File	1969
12.3.207	RPTRST - Define Data to be Written to the RESTART File.....	1969
12.3.208	RPTSCHED - Define SCHEDULE Section Reporting.....	1970
12.3.209	SAVE - Activate Output of a SAVE File for Fast Restarts.....	1978
12.3.210	SCDATAB - Well Connection PI Multipliers versus Scale Deposit.....	1979
12.3.211	SCDETAB - Well Connection Karst Aquifer Properties for Scale Deposit.....	1980
12.3.212	SCDPTAB - Well Connection Scale Deposition Tables.....	1981
12.3.213	SCDPTRAC - Allocate Sea Water Tracer for Scale Deposition.....	1982
12.3.214	SCHEDULE - Define the Start of the SCHEDULE Section of Keywords.....	1983
12.3.215	SEPVALS - Define Separator Oil Formation Volume Factor and GOR.....	1984

12.3.216	SHRATE - Activate and Define the Polymer Shearing Logarithmic Parameters.....	1985
12.3.217	SIMULATE - Activate the Simulation Mode.....	1986
12.3.218	SKIP - Activate Skipping of All Keywords and Input Data.....	1987
12.3.219	SKIP100 - Activate Skipping of Black-Oil Keywords and Input Data.....	1987
12.3.220	SKIP300 - Activate Skipping of Keywords by Compositional Simulator.....	1987
12.3.221	SKIPREST - Activate Skipping of Restart Schedule Data.....	1988
12.3.222	SLAVES - Define Slave Reservoir Simulation Parameters.....	1989
12.3.223	SOURCE - Define Source Term.....	1990
12.3.224	SUMTHIN - Define SUMMARY DATA Reporting Time Steps.....	1992
12.3.225	SURFVISC - Surfactant Solution Viscosity versus Concentration.....	1992
12.3.226	SWINGFAC - Define Field Gas Contract Parameters.....	1993
12.3.227	TIGHTEN - Tighten and Relax Numerical Controls.....	1994
12.3.228	TIGHTENP - Tighten and Relax Numerical Controls Individually.....	1995
12.3.229	TIME - Advance Simulation by Cumulative Reporting Time.....	1996
12.3.230	TSTEP - Advance Simulation by Reporting Time.....	1998
12.3.231	TUNING - Numerical Tuning Control.....	2000
12.3.232	TUNINGDP - Numerical Tuning Control for High Throughput Cases.....	2005
12.3.233	TUNINGH - Numerical Tuning Control for History Match Gradient Calculations.....	2007
12.3.234	TUNINGL - Numerical Tuning Control for All LGRs.....	2008
12.3.235	TUNINGS - Numerical Tuning Control for Individual LGRs.....	2013
12.3.236	UDQ - Declare User Define Quantities ("UDQ").....	2018
12.3.237	UDT - Declare User Define Tables ("UDT").....	2026
12.3.238	USECUPL - Load a Reservoir Coupling File.....	2031
12.3.239	VAPPARS - Oil Vaporization Parameters.....	2032
12.3.240	VFPCHK - Define Production Vertical Flow Performance BHP Check.....	2033
12.3.241	VFPINJ - Define Injection Vertical Flow Performance Tables.....	2035
12.3.242	VFPPROD - Define Production Vertical Flow Performance Tables.....	2039
12.3.243	VFPTABL - Define Production Vertical Flow Performance ALQ Interpolation.....	2047
12.3.244	WAITBAL - Wait On Network Balance Before Allowing Further Actions.....	2048
12.3.245	WALKALIN - Define Water Injection Alkaline Concentration.....	2049
12.3.246	WALQCALC - Define Well VFP Surface ALQ Phase Density.....	2050
12.3.247	WAPI - Define Oil Well Injection API Gravity.....	2051
12.3.248	WARN - Activate Warning Messages.....	2052
12.3.249	WBHGLR - Define Well Bottom-Hole GLR Constraint.....	2053
12.3.250	WBOREVOL - Define Effective Wellbore Storage Volume.....	2054
12.3.251	WCALCVAL - Define Gas Well Calorific Value.....	2055
12.3.252	WCONHIST - Define Well Historical Production Rates and Pressures.....	2056
12.3.253	WCONINJ - Well Injection Targets and Constraints.....	2061
12.3.254	WCONINJE - Well Injection Targets and Constraints.....	2062
12.3.255	WCONINJH - Well Historical Observed Injection Rates and Pressures.....	2066
12.3.256	WCONINJP - Define Well Injection Targets and Constraints for Pattern Flood Wells.....	2070
12.3.257	WCONPROD - Define Well Production Targets and Constraints.....	2071
12.3.258	WCUTBACK - Define Well Cutback Limits and Parameters.....	2078
12.3.259	WCUTBACT - Define Well Tracer Cutback Limits and Parameters.....	2079
12.3.260	WCYCLE - Define Automatic Well Opening and Closing Cycling Parameters.....	2080
12.3.261	WDFAC - Define Gas Flow Dependent Skin Factor.....	2083
12.3.262	WDFACCOR - Gas Flow Dependent Skin Factor (Correlation).....	2084
12.3.263	WDRILPRI - Add Wells to the Drilling Priority Drilling Queue.....	2086
12.3.264	WDRILRES - Activate Prevention of Multi-Completions in the Same Cell for Queued Wells.....	2087
12.3.265	WDRILTIM - Define Drilling Parameters for Automatic Drilling of New Wells.....	2088
12.3.266	WECON - Well Economic Criteria for Production Wells.....	2089
12.3.267	WECONINJ - Well Economic Criteria for Injection Wells.....	2094
12.3.268	WECONT - Well Economic Tracer Criteria for Production Wells.....	2096
12.3.269	WEFAC - Define Well Efficiency.....	2099
12.3.270	WELCNTL - Modify Well Control and Targets.....	2101
12.3.271	WELDEBUG - Define the Well Debug Data to be Printed to File.....	2103
12.3.272	WELDRAW - Define Maximum Draw Down for Production Wells.....	2104
12.3.273	WELEVNT - Define Well WPWEM Summary Mnemonic Output Value.....	2105
12.3.274	WELLSTRE - Define Injection Stream Composition.....	2106
12.3.275	WELMOVE - Move Global Well Into an LGR.....	2108
12.3.276	WELOPEN - Define Well and Well Connection Flowing Status.....	2109
12.3.277	WELOPENL - Define Well and Well Connections Flowing Status (LGR).....	2113

12.3.278 WELPI - Define Well Productivity and Injectivity Indices.....	2116
12.3.279 WELPRI - Assign Well Priority.....	2118
12.3.280 WELSEGS - Define Multi-Segment Wells and Their Segment Structure.....	2119
12.3.281 WELSOMIN - Define Well Connection Minimum Oil Saturation for Opening.....	2128
12.3.282 WELSPECL - Define Well Specifications for Local Grid Refinements.....	2129
12.3.283 WELSPECS - Define Well Specifications.....	2134
12.3.284 WELTARG - Modify Well Target and Constraint Values.....	2139
12.3.285 WELTRAJ - Define Well Trajectory Data.....	2142
12.3.286 WFOAM - Define Well Foam Injection Concentrations.....	2145
12.3.287 WFRICSEG - Convert Friction Well to Multi-Segment Well.....	2146
12.3.288 WFRICSG - Convert Friction Well to Multi-Segment Well (LGR).....	2147
12.3.289 WFRICTN - Define Well as a Friction Well.....	2148
12.3.290 WFRICTNL - Define Well as a Friction Well (LGR).....	2149
12.3.291 WGASPROD - Define Sale Gas Well Production Targets.....	2150
12.3.292 WGORPEN - Define Well GOR Penalty Parameters.....	2151
12.3.293 WGRUPCON - Define Well Guide Rates for Group Control.....	2152
12.3.294 WHEDREFD - Define Well Hydraulic Head Reference Depth.....	2154
12.3.295 WHISTCTL - Define Well Historical Target Phase.....	2155
12.3.296 WHTEMP - Define Well Tubing Head Temperature Parameters.....	2157
12.3.297 WINJCLN - Clean a Fraction of a Deposited Filter Cake.....	2159
12.3.298 WINJDAM - Define Well Filter Cake Properties.....	2161
12.3.299 WINJFCNC - Define Injection Well Filtrate Concentration.....	2164
12.3.300 WINJGAS - Define Gas Injection Properties for a Well.....	2165
12.3.301 WINJMULT - Define Well Pressure Dependent Injectivity Multipliers.....	2168
12.3.302 WINJTEMP - Define Injection Fluid Thermal Properties.....	2171
12.3.303 WLIFT - Define Well Re-Tubing, THP and Lift Switching Workover Operations.....	2173
12.3.304 WLIFTOPT - Define Well Gas Lift Optimization Parameters.....	2174
12.3.305 WLIMTOL - Define Well Constraint Tolerance.....	2178
12.3.306 WLIST - Define Well Lists (Static).....	2179
12.3.307 WLISTARG - Modify Well List Target and Constraint Values (Static).....	2181
12.3.308 WLISTNAM - Define Well Lists (WLISTARG).....	2184
12.3.309 WMICP - Define Water Injection Well's Microbial, Oxygen, And Urea Concentrations.....	2186
12.3.310 WNETCTRL - Define Well Control for Network Control Option.....	2189
12.3.311 WNETDP - Define Well THP to Network Pressure Drop.....	2190
12.3.312 WORKLIM - Define Well Workover Time.....	2191
12.3.313 WORKTHP - Define Well Workover Options for THP Killed Wells.....	2192
12.3.314 WPAVE - Well Block Average Pressure Calculation Parameters for All Wells.....	2193
12.3.315 WPAVEDEP - Define Well Reference Depth for Pressure Calculations.....	2195
12.3.316 WPIMULT - Define Well Connection Multipliers.....	2197
12.3.317 WPIMULTL - Define Well Connection Multipliers (LGR).....	2200
12.3.318 WPITAB - Assign Well Productivity Index versus Water Cut Tables.....	2203
12.3.319 WPLUG - Define Well Plug Back Length.....	2204
12.3.320 WPMITAB - Assign Well Polymer Molecular Model Injection Tables.....	2205
12.3.321 WPOLYMER - Define Water Injection Well Polymer and Salt Concentrations.....	2207
12.3.322 WPOLYRED - Define Well Polymer-Water Viscosity Reduction Factor.....	2210
12.3.323 WREGROUP - Automatic Re-Assignment of Wells to Groups.....	2211
12.3.324 WRFT - Activate Well RFT Reporting to the RFT File.....	2212
12.3.325 WRFTPLT - Activate Well RFT and PLT Reporting to the RFT File.....	2214
12.3.326 WSALT - Define Water Injection Well Salt Concentrations.....	2217
12.3.327 WSCCLEAN - Well Deposited Scale Adjustment.....	2219
12.3.328 WSCCLENL - Well Deposited Scale Adjustment (LGR).....	2220
12.3.329 WSCTAB - Assign Well Scale Deposition and Scale Damage Tables.....	2221
12.3.330 WSEGAICD - Define Multi-Segment Well Autonomous ICD Connections.....	2222
12.3.331 WSEGDFIN - Define Multi-Segment Well Drift Flux Slip Model Input Data.....	2230
12.3.332 WSEGDFMD - Define Multi-Segment Well Drift Flux Slip Model.....	2231
12.3.333 WSEGDFPA - Define Multi-Segment Well Drift Flux Slip Model Parameters.....	2232
12.3.334 WSEGEXSS - Define Multi-Segment Well Import-Export Segment Volumes.....	2233
12.3.335 WSEGFLIM - Define Multi-Segment Well Artificial Choke Connections.....	2234
12.3.336 WSEGFMOD - Define Multi-Segment Well Model.....	2235
12.3.337 WSEGINIT - Define Multi-Segment Well Initial Conditions.....	2236
12.3.338 WSEGITER - Define Multi-Segment Wells Iteration Parameters.....	2237
12.3.339 WSEGLABY - Define Multi-Segment Well Labyrinth ICD Connections.....	2238

12.3.340 WSEGLINK - Define Multi-Segment Well Looped Flow Paths.....	2239
12.3.341 WSEGMULT - Define Multi-Segment Well Frictional Pressure Loss Multipliers.....	2240
12.3.342 WSEGPROP - Modify Multi-Segment Wells and Their Segment Structure.....	2241
12.3.343 WSEGPULL - Define a Multi-Segment Well Down-Hole Separator Pump.....	2243
12.3.344 WSEGSEP - Define a Multi-Segment Well Down-Hole Separator.....	2244
12.3.345 WSEGSICD - Define Multi-Segment Well Spiral ICD Connections.....	2245
12.3.346 WSEGSOLV - Define Multi-Segment Well Iterative Linear Solver Parameters.....	2252
12.3.347 WSEGTABL - Assign Multi-Segment Well VLP Tables to Segments.....	2253
12.3.348 WSEGVALV - Define Multi-Segment Well Sub-Critical Valve.....	2254
12.3.349 WSKPTAB - Assign Well Polymer Molecular Model Skin Tables.....	2259
12.3.350 WSOLVENT - Define Gas Injection Well Solvent Fraction.....	2261
12.3.351 WSURFACT - Define Water Injection Well Surfactant Concentration.....	2263
12.3.352 WTADD - Add a Constant to a Well Target or Constraint.....	2264
12.3.353 WTEMP - Define An Injection Well's Fluid Temperature.....	2265
12.3.354 WTEMPQ - Output Well Names and Well Lists to the Print File.....	2266
12.3.355 WTEST - Well Testing Criteria for Re-Opening Closed Wells.....	2267
12.3.356 WTHPMAX - Define a Well's Maximum Flowing THP for Shut-In.....	2269
12.3.357 WTMULT - Multiply a Well Target or Constraint by a Constant.....	2270
12.3.358 WTRACER - Define An Injection Well's Tracer Concentration.....	2272
12.3.359 WVFDP - Modify Well BHP Obtained from VFP Tables.....	2275
12.3.360 WVFPEXP - Define Well VFP Interpolation Options.....	2277
12.3.361 WWPAVE - Well Block Average Pressure Calculation Parameters for Individual Wells.....	2280
12.3.362 ZIPP2OFF - Deactivate Automatic Time Step Control.....	2283
12.3.363 ZIPPY2 - Activate Automatic Time Step Control.....	2284

APPENDIX A: KEYWORD INDEX - ALPHABETIC LISTING.....2285

A.....	2287
B.....	2289
C.....	2290
D.....	2292
E.....	2295
F.....	2297
G.....	2299
H.....	2303
I.....	2308
J.....	2314
K.....	2315
L.....	2318
M.....	2321
N.....	2324
O.....	2326
P.....	2327
Q.....	2332
R.....	2333
S.....	2336
T.....	2343
U.....	2346
V.....	2347
W.....	2348
X.....	2353
Y.....	2354
Z.....	2355

APPENDIX B: OPM FLOW RELEASE HISTORY.....2357

B.1 Release 2025-10.....	2360
B.1.1 New and Deprecated Command Line Options.....	2361
B.1.2 New Features.....	2363
B.1.3 Improvements.....	2365

B.1.4 Bug Fixes.....	2368
B.1.5 Known Issues.....	2370
B.2 Release 2025-04.....	2372
B.2.1 New and Deprecated Command Line Options.....	2373
B.2.2 New Features.....	2375
B.2.3 Improvements.....	2377
B.2.4 Bug Fixes.....	2382
B.2.5 Known Issues.....	2384
B.3 Release 2024-10.....	2387
B.3.1 New and Deprecated Command Line Options.....	2388
B.3.2 New Features.....	2391
B.3.3 Improvements.....	2392
B.3.4 Bug Fixes.....	2395
B.3.5 Known Issues.....	2399
B.4 Release 2024-04.....	2401
B.4.1 New and Deprecated Command Line Options.....	2402
B.4.2 New Features.....	2404
B.4.3 Improvements.....	2406
B.4.4 Bug Fixes.....	2412
B.4.5 Known Issues.....	2415
B.5 Release 2023-10.....	2418
B.5.1 New and Deprecated Command Line Options.....	2418
B.5.2 New Features.....	2421
B.5.3 Improvements.....	2422
B.5.4 Bug Fixes.....	2426
B.5.5 Known Issues.....	2428
APPENDIX C: OPMRUN - FLOW JOB SCHEDULER.....	2431
C.1 Introduction.....	2435
C.2 OPMRUN Functionality.....	2437
C.2.1 Add Job and Select Run Type.....	2438
C.2.2 Edit Job Data and Parameter File.....	2440
C.2.3 Load Previously Saved Queue.....	2442
C.2.4 Reset Job Queue Parameters.....	2443
C.2.5 Run Jobs in Queue with Various Options.....	2444
C.2.6 Menu Options.....	2447
C.3 OPMRUN Tools.....	2451
C.3.1 OPMRUN Tools: Job File Compression Utility for Saving Space and Archiving.....	2451
C.3.2 OPMRUN Tools: Simulator Input/Keywords.....	2452
C.3.3 OPMRUN Tools: Simulator Input/Production Schedule.....	2463
C.3.4 OPMRUN Tools: Simulator Input/Sensitivities.....	2466
C.3.5 OPMRUN Tools: Simulator Input/Well Specification.....	2472
C.3.6 OPMRUN Tools: ResInsight.....	2475
C.3.7 OPMRUN Tools: Well Trajectory Conversion.....	2475
APPENDIX D: PYTHON INTERFACE TO OPM FLOW.....	2479
D.1 Introduction.....	2482
D.2 Working with the Input Deck.....	2482
D.2.1 The Parser Object.....	2482
D.2.2 Loading an Input Deck.....	2483
D.2.3 Units.....	2483
D.2.4 Accessing the Input Deck.....	2484
D.2.5 Special Parsing.....	2485
D.2.6 Creating EclipseState and Grid Objects.....	2486
D.2.7 Creating a Schedule.....	2487
D.3 Working with the Result Files.....	2487
D.3.1 Loading a Grid File.....	2487

D.3.2 Loading a Summary Case of Results.....	2487
D.3.3 Loading a Restart File.....	2488

APPENDIX E: RUNNING PREVIOUS RELEASES OF OPM FLOW.....2489

E.1 Running OPM Flow 2025-04.....	2492
E.2 Running OPM Flow 2024-10.....	2514
E.3 Running OPM Flow 2024-04.....	2535
E.4 Running OPM Flow 2023-10.....	2555

APPENDIX F: OPM FLOW OUTPUT FILE FORMATS.....2575

F.1 Overview.....	2580
F.1.1 File Types.....	2580
F.1.2 File Naming Conventions.....	2585
F.1.3 Unformatted File Format Considerations.....	2587
F.1.4 File Output Format.....	2588
F.2 EGRID - Model Structural Data for Irregular Corner-Point Grids File.....	2591
F.2.1 EGRID Header Keywords.....	2593
F.2.2 EGRID Global Irregular Corner Point Grid Keywords.....	2597
F.2.3 EGRID LGR Irregular Corner Point Grid Keywords.....	2601
F.2.4 EGRID Non-Neighbor Connections for Irregular Corner Point Grid Keywords.....	2604
F.3 EGRID - Model Structural Data for Unstructured Grids File.....	2607
F.4 EGRID - Model Structural Data for Mixed Grids File.....	2607
F.5 GRID - Model Structural Data File.....	2607
F.5.1 RESTART Data - LGR Grid Header Keywords.....	2607
F.6 INIT - Model Initialization and Static Data File.....	2610
F.6.1 INIT Index File.....	2610
F.6.2 INIT Data File Specification.....	2611
F.6.3 INIT Data - Static Data Header Keywords.....	2614
F.6.4 INIT Data - Static Grid Array Data Keywords.....	2619
F.6.5 INIT Data - LGR Static Data Header Keywords.....	2622
F.6.6 INIT Data - LGR Grid Array Data Keywords.....	2625
F.6.7 INIT Data - Static Tabular Data Header Keyword.....	2627
F.6.8 INIT Data - Static Tabular and Constant Data Array Keywords.....	2630
F.7 LOG File - OPM Flow Terminal Output.....	2632
F.7.1 LOG File - Header.....	2632
F.7.2 LOG File - Reading Deck.....	2633
F.7.3 LOG File - Processing Data.....	2633
F.7.4 LOG File - Saturation Function Diagnostics.....	2634
F.7.5 LOG File - Main Simulation Loop.....	2635
F.7.6 LOG File - End of Simulation.....	2636
F.8 RESTART Files - Dynamic Data (Solution, Groups, Wells, Connections, Aquifers, etc.).....	2637
F.8.1 RESTART Index Files.....	2638
F.8.2 RESTART Data File Specification.....	2639
F.8.3 RESTART Data - Header Keywords.....	2641
F.8.4 RESTART Data - Group, Well, and Connection Data Keywords.....	2653
F.8.5 RESTART Data - UDQ and ACTIONX Keywords.....	2673
F.8.6 RESTART Data - Aquifer Data Keywords.....	2680
F.8.7 RESTART Data - Hidden Keyword.....	2685
F.8.8 RESTART Data - Solution Data keyword.....	2686
F.8.9 RESTART Data - LGR Solution Data Keywords.....	2694
F.9 RFT Files - Pressure, Saturation and Production Log Data.....	2697
F.9.1 RFT Data File Specification.....	2697
F.9.2 RFT Data File - Time and Date keywords.....	2698
F.9.3 RFT Data File - Well and Connection Data Keyword.....	2699
F.9.4 RFT Data File - RFT Solution Data Keyword.....	2701

F.9.5 RFT Data File - Solution PLT Data Keywords.....	2703
F.9.6 RFT Data File - Solution Multi-Segment Well Keywords.....	2703
F.9.7 RFT Data File - Solution River Keywords.....	2703
F.10 SUMMARY Files - Time Based Vector Data.....	2704
F.10.1 Enhanced SUMMARY Data File Keywords.....	2704
F.10.2 SUMMARY Index File keywords.....	2707
F.10.3 SUMMARY Data File Keywords.....	2712
F.11 SAVE Files - Initialization and Solution Data.....	2714

INDEX OF TABLES

Table 2.1: OPM Flow Command Line Options.....	105
Table 2.2: Oil-Water Saturation Function and Derivative Table.....	122
Table 2.3: Oil-Water Saturation Function and Derivative Table (Modified).....	125
Table 2.4: Major Solver Time Stepping Controls.....	147
Table 2.5: Newton-Raphson Solver Controls.....	153
Table 2.6: Linear Solver Property Tree versus Command Line Parameters.....	154
Table 2.7: Linear Solver Controls.....	158
Table 3.1: Example Keyword Table Section.....	162
Table 3.2: ADD Keyword Table Section.....	162
Table 3.3: PORO Keyword Description.....	167
Table 3.4: OPM Flow Input Deck Sections.....	169
Table 4.1: Alphabetic List of GLOBAL Section Keywords.....	174
Table 4.2: EXTRAPMS Keyword Description.....	182
Table 4.3: FORMFEED Keyword Description.....	183
Table 4.4: INCLUDE Keyword Description.....	184
Table 4.5: MESSAGES Keyword Description.....	187
Table 5.1: Minimum Set of RUNSPEC Keywords Required by OPM Flow.....	200
Table 5.2: Alphabetic List of RUNSPEC Keywords.....	209
Table 5.3: ACTDIMS Keyword Description.....	215
Table 5.4: ACTPARAM Keyword Description.....	216
Table 5.5: AQUDIMS Keyword Description.....	221
Table 5.6: BRINE Keyword Description.....	228
Table 5.7: COMPS Keyword Description.....	243
Table 5.8: CPR Keyword Description.....	244
Table 5.9: DIMENS Keyword Description.....	248
Table 5.10: ENDSCALE Keyword Description.....	259
Table 5.11: EQLDIMS Keyword Description.....	262
Table 5.12: EQLOPTS Keyword Description.....	263
Table 5.13: FAULTDIM Keyword Description.....	266
Table 5.14: FMTIN Keyword Description.....	270
Table 5.15: FMOUT Keyword Description.....	272
Table 5.16: FRICTION Keyword Description.....	275
Table 5.17: GRAVDRM Keyword Description.....	284
Table 5.18: GRIDOPTS Keyword Description.....	285
Table 5.19: LGR Keyword Description.....	305
Table 5.20: MINNPCOL Keyword Description.....	317
Table 5.21: MISCIBLE Keyword Description.....	318
Table 5.22: MSGFILE Keyword Description.....	320
Table 5.23: MULTIN Keyword Description.....	322
Table 5.24: MULTOUT Keyword Description.....	324
Table 5.25: NETWORK Keyword Description.....	327
Table 5.26: NMATRIX Keyword Description.....	329
Table 5.27: NNEWTF Keyword Description.....	330
Table 5.28: NRSOUT Keyword Description.....	341
Table 5.29: NSTACK Keyword Description.....	342
Table 5.30: NUMRES Keyword Description.....	343
Table 5.31: NUPCOL Keyword Description.....	344
Table 5.32: OPTIONS Keyword Description.....	346
Table 5.33: PARALLEL Keyword Description.....	347
Table 5.34: PATHS Keyword Description.....	349
Table 5.35: PIMTDIMS Keyword Description.....	352
Table 5.36: PINTDIMS Keyword Description.....	353
Table 5.37: REGDIMS Keyword Description.....	364
Table 5.38: ROCKCOMP Keyword Description.....	367
Table 5.39: SATOPTS Keyword Description.....	376
Table 5.40: SATOPTS Relative Permeability Function Allocation Keywords.....	377
Table 5.41: SCDPDIMS Keyword Description.....	379
Table 5.42: SMRYDIMS Keyword Description.....	381
Table 5.43: START Keyword Description.....	385
Table 5.44: TABDIMS Keyword Description.....	390
Table 5.45: OPM Flow's THERMAL Option Associated Keywords.....	394

Table 5.46: TITLE Keyword Description.....	396
Table 5.47: TRACERS Keyword Description.....	398
Table 5.48: TRPLPORO Keyword Description.....	399
Table 5.49: UDADIMS Keyword Description.....	400
Table 5.50: UDQDIMS Keyword Description.....	402
Table 5.51: UDQPARAM Keyword Description.....	403
Table 5.52: UDTDIMS Keyword Description.....	405
Table 5.53: UNIFIN Keyword Description.....	408
Table 5.54: UNIFOUT Keyword Description.....	410
Table 5.55: VFPPIDIMS Keyword Description.....	416
Table 5.56: VFPPDIMS Keyword Description.....	417
Table 5.57: WELLDIMS Keyword Description.....	423
Table 5.58: WSEGDIMS Keyword Description.....	425
Table 6.1: Key Static Grid Properties.....	442
Table 6.2: ACTNUM Keyword Description.....	446
Table 6.3: ADD Keyword Description.....	447
Table 6.4: ADD Keyword Applicable Arrays by Section.....	448
Table 6.5: ADDREG Keyword Description.....	449
Table 6.6: ADDREG Keyword Applicable Arrays by Section.....	450
Table 6.7: AQUANCON Keyword Description.....	455
Table 6.8: AQUCON Keyword Description.....	458
Table 6.9: AQUCT Keyword Description.....	461
Table 6.10: AQUNUM Keyword Description.....	466
Table 6.11: BCCON Keyword Description.....	470
Table 6.12: BOX Keyword Description.....	474
Table 6.13: CARFIN Keyword Description.....	479
Table 6.14: COALNUM Keyword Description.....	480
Table 6.15: COORD Keyword Description.....	484
Table 6.16: COORDSYS Keyword Description.....	486
Table 6.17: COPY Keyword Description.....	487
Table 6.18: COPY Keyword Applicable Arrays by Section.....	488
Table 6.19: COPYBOX Keyword Description.....	490
Table 6.20: COPYBOX Keyword Applicable Arrays by Section.....	490
Table 6.21: COPYREG Keyword Description.....	492
Table 6.22: COPYREG Keyword Applicable Arrays by Section.....	493
Table 6.23: DISPERC Keyword Description.....	506
Table 6.24: DR Keyword Description.....	510
Table 6.25: DRV Keyword Description.....	512
Table 6.26: DTHETA Keyword Description.....	513
Table 6.27: DTHETA Keyword Description.....	514
Table 6.28: DX Keyword Description.....	516
Table 6.29: DXV Keyword Description.....	517
Table 6.30: DY Keyword Description.....	518
Table 6.31: DYV Keyword Description.....	519
Table 6.32: DZ Keyword Description.....	520
Table 6.33: DZNET Keyword Description.....	524
Table 6.34: DZV Keyword Description.....	525
Table 6.35: EQUALREG Keyword Description.....	531
Table 6.36: EQUALREG Keyword Applicable Arrays by Section.....	532
Table 6.37: EQUALS Keyword Description.....	533
Table 6.38: EQUALS Keyword Applicable Arrays by Section.....	534
Table 6.39: FAULTS Keyword Description.....	540
Table 6.40: FILEUNIT Keyword Description.....	542
Table 6.41: FLUXNUM Keyword Description.....	543
Table 6.42: GDFILE Keyword Description.....	547
Table 6.43: GRIDFILE Keyword Description.....	551
Table 6.44: GRIDUNIT Keyword Description.....	552
Table 6.45: HEATCR Keyword Description.....	554
Table 6.46: HEATCRT Keyword Description.....	555
Table 6.47: HMMLT Keyword List.....	559
Table 6.48: HRFIN Keyword Description.....	564
Table 6.49: HXFIN Keyword Description.....	565
Table 6.50: HYFIN Keyword Description.....	566

Table 6.51: HZFIN Keyword Description.....	567
Table 6.52: IMPORT Keyword Description.....	569
Table 6.53: INRAD Keyword Description.....	572
Table 6.54: ISOLNUM Keyword Description.....	574
Table 6.55: JFUNC Keyword Description.....	576
Table 6.56: JFUNCR Keyword Description.....	579
Table 6.57: LX Keyword Description.....	584
Table 6.58: LY Keyword Description.....	587
Table 6.59: LZ Keyword Description.....	590
Table 6.60: MAPAXES Keyword Description.....	593
Table 6.61: MAPUNITS Keyword Description.....	594
Table 6.62: MAXVALUE Keyword Description.....	595
Table 6.63: MAXVALUE Keyword Applicable Arrays by Section.....	596
Table 6.64: MINPORV Keyword Description.....	599
Table 6.65: MINPV Keyword Description.....	600
Table 6.66: MINPVV Keyword Description.....	601
Table 6.67: MINVALUE Keyword Description.....	603
Table 6.68: MINVALUE Keyword Applicable Arrays by Section.....	604
Table 6.69: MULTFLT Keyword Description.....	607
Table 6.70: MULTIPLY Keyword Description.....	608
Table 6.71: MULTIPLY Keyword Applicable Arrays by Section.....	609
Table 6.72: MULTIREG Keyword Description.....	610
Table 6.73: MULTIREG Keyword Applicable Arrays by Section.....	611
Table 6.74: MULTNUM Keyword Description.....	613
Table 6.75: MULTPV Keyword Description.....	614
Table 6.76: MULTR Keyword Description.....	615
Table 6.77: MULTR- Keyword Description.....	616
Table 6.78: MULTREGD Keyword Description.....	618
Table 6.79: MULTREGH Keyword Description.....	621
Table 6.80: MULTREGP Keyword Description.....	623
Table 6.81: MULTREGT Keyword Description.....	625
Table 6.82: MULTTHT Keyword Description.....	626
Table 6.83: MULTTHT- Keyword Description.....	627
Table 6.84: MULTX Keyword Description.....	629
Table 6.85: MULTX- Keyword Description.....	630
Table 6.86: MULTY Keyword Description.....	631
Table 6.87: MULTY- Keyword Description.....	632
Table 6.88: MULTZ Keyword Description.....	633
Table 6.89: MULTZ- Keyword Description.....	634
Table 6.90: NINENUM Keyword Description.....	636
Table 6.91: NNC Keyword Description.....	639
Table 6.92: NTG Keyword Description.....	644
Table 6.93: NXFIN Keyword Description.....	645
Table 6.94: NYFIN Keyword Description.....	646
Table 6.95: NZFIN Keyword Description.....	647
Table 6.96: OPERATE Keyword Description.....	651
Table 6.97: OPERATE Keyword Applicable Arrays by Section.....	651
Table 6.98: OPERATER Keyword Description.....	654
Table 6.99: OPERATER Keyword Applicable Arrays by Section.....	654
Table 6.100: OPERNUM Keyword Description.....	657
Table 6.101: OUTRAD Keyword Description.....	658
Table 6.102: OUTRAD Radial Grid Example.....	659
Table 6.103: PEBI Keyword Description.....	661
Table 6.104: PERMJFUN Keyword Description.....	663
Table 6.105: PERMR Keyword Description.....	664
Table 6.106: PERMTHT Keyword Description.....	665
Table 6.107: PERMX Keyword Description.....	666
Table 6.108: PERMY Keyword Description.....	667
Table 6.109: PERMZ Keyword Description.....	668
Table 6.110: PINCH Keyword Description.....	672
Table 6.111: PINCHNUM Keyword Description.....	673
Table 6.112: PINCHREG Keyword Description.....	676
Table 6.113: PINCHXY Keyword Description.....	677

Table 6.114: PORO Keyword Description.....	678
Table 6.115: PYINPUT Keyword Description.....	681
Table 6.116: RESVNUM Keyword Description.....	687
Table 6.117: NTG Keyword Description.....	689
Table 6.118: RPTGRID Keyword Description.....	690
Table 6.119: RPTGRIDL Keyword Description.....	693
Table 6.120: SMULTX Keyword Description.....	701
Table 6.121: SMULTY Keyword Description.....	702
Table 6.122: SMULTX Keyword Description.....	703
Table 6.123: SPECGRID Keyword Description.....	706
Table 6.124: THCGAS Keyword Description.....	707
Table 6.125: THCOIL Keyword Description.....	709
Table 6.126: THCONR Keyword Description.....	711
Table 6.127: THCROCK Keyword Description.....	712
Table 6.128: THCROCK Keyword Description.....	714
Table 6.129: THCWATER Keyword Description.....	716
Table 6.130: THPRESFT Keyword Description.....	718
Table 6.131: TOPS Keyword Description.....	720
Table 6.132: TRANGL Keyword Description.....	722
Table 6.133: ZCORN Keyword Description.....	729
Table 7.1: EDIT Section Arrays Available for Modification.....	733
Table 7.2: Depth Keyword Description.....	737
Table 7.3: EDITNNC Keyword Description.....	746
Table 7.4: EDITNNCR Keyword Description.....	749
Table 7.5: HMMULT Keyword List.....	753
Table 7.6: PORV Keyword Description.....	759
Table 7.7: TRANR Keyword Description.....	761
Table 7.8: TRANR Keyword Description.....	762
Table 7.9: TRANX Keyword Description.....	763
Table 7.10: TRANY Keyword Description.....	764
Table 7.11: TRANZ Keyword Description.....	765
Table 8.1: General Reservoir Fluid Classification.....	782
Table 8.2: Reservoir Fluids Composition Indicators.....	782
Table 8.3: Oil Sub-Classification (after Wilmon).....	782
Table 8.4: Fluid Property Keywords versus Oil, Gas and Water Fluid Type.....	783
Table 8.5: Fluid Property Keywords versus CO ₂ (Storage), Foam, Polymer, Solvent, and MICP Fluid Types.....	785
Table 8.6: Oil-Water Relative Permeability End-Point Data Definitions.....	789
Table 8.7: Gas-Oil Relative Permeability End-Point Data Definitions.....	790
Table 8.8: Gas-Water Relative Permeability End-Point Data Definitions.....	792
Table 8.9: Saturation Table Formats and Phases.....	793
Table 8.10: Corey Exponent Values.....	796
Table 8.11: LET Modeling of Flooding Processes.....	797
Table 8.12: LET Functions (DIC1) versus SWOFLET Keyword.....	801
Table 8.13: LET Functions (IDC2) versus SWOFLET Keyword.....	802
Table 8.14: LET Functions (CDI2) versus SGOFLET Keyword.....	804
Table 8.15: LET Functions (DCI1) versus SGWFLET Keyword.....	806
Table 8.16: LET Functions (DCI1) versus SGWFLET Keyword.....	808
Table 8.17: ACF Keyword Description.....	810
Table 8.18: ACTCO2S Keyword Description.....	811
Table 8.19: ADSALNOD Keyword Description.....	813
Table 8.20: AQUATAB Keyword Description.....	823
Table 8.21: Carter-Tracy Aquifer Influence Functions (RD =1.5, 2.0, 2.5 and 3.0).....	824
Table 8.22: Carter-Tracy Aquifer Influence Functions (RD =3.5, 4.0, 4.5 and 5.0).....	825
Table 8.23: Carter-Tracy Aquifer Influence Functions (RD =6.0, 7.0, 8.0 and 9.0).....	826
Table 8.24: Carter-Tracy Aquifer Influence Functions (RD =10 and Finite Linear).....	827
Table 8.25: Carter-Tracy Infinite Radial Aquifer Influence Function (Default).....	829
Table 8.26: BDENSITY Keyword Description.....	831
Table 8.27: BIC Keyword Description.....	834
Table 8.28: BIOFPARA Keyword Description.....	836
Table 8.29: CNAMEs Keyword Description.....	839
Table 8.30: DENAQA Keyword Description.....	843
Table 8.31: DENSITY Keyword Description.....	845

Table 8.32: DIFFAGAS Keyword Description.....	849
Table 8.33: Diffusivity Conversion Factors.....	850
Table 8.34: DIFFAWAT Keyword Description.....	851
Table 8.35: Diffusivity Conversion Factors.....	852
Table 8.36: DIFFC Keyword Description.....	854
Table 8.37: DIFFCGAS Keyword Description.....	855
Table 8.38: Diffusivity Conversion Factors.....	856
Table 8.39: DIFFCWAT Keyword Description.....	858
Table 8.40: Diffusivity Conversion Factors.....	859
Table 8.41: DIFFMICP Keyword Description.....	861
Table 8.42: EHYSTR Keyword Description.....	870
Table 8.43: EHYSTRR Keyword Description.....	871
Table 8.44: EOS Keyword Description.....	877
Table 8.45: FOAMADS Keyword Description.....	886
Table 8.46: FOAMFSC Keyword Description.....	893
Table 8.47: FOAMMOB Keyword Description.....	898
Table 8.48: FOAMOPTS Keyword Description.....	902
Table 8.49: FOAMROCK Keyword Description.....	903
Table 8.50: GASDENT Keyword Description.....	906
Table 8.51: GASJT Keyword Description.....	908
Table 8.52: GASVISCT Keyword Description.....	910
Table 8.53: GRAVCONS Keyword Description.....	915
Table 8.54: GRAVITY Keyword Description.....	916
Table 8.55: GSF Keyword Description.....	918
Table 8.56: HA Keyword List.....	921
Table 8.57: HM Keyword List.....	924
Table 8.58: IKRG Keyword Description.....	946
Table 8.59: Critical Displacement Relationships.....	947
Table 8.60: IKRGR Keyword Description.....	948
Table 8.61: Critical Displacement Relationships.....	948
Table 8.62: IKRO Keyword Description.....	950
Table 8.63: Critical Displacement Relationships.....	951
Table 8.64: IKRORG Keyword Description.....	952
Table 8.65: Critical Displacement Relationships.....	952
Table 8.66: IKRORW Keyword Description.....	954
Table 8.67: Critical Displacement Relationships.....	954
Table 8.68: IKRW Keyword Description.....	956
Table 8.69: Critical Displacement Relationships.....	957
Table 8.70: IKRWR Keyword Description.....	958
Table 8.71: Critical Displacement Relationships.....	958
Table 8.72: IPCG Keyword Description.....	969
Table 8.73: IPCW Keyword Description.....	970
Table 8.74: ISGCR Keyword Description.....	971
Table 8.75: ISGL Keyword Description.....	972
Table 8.76: ISGLPC Keyword Description.....	973
Table 8.77: ISGU Keyword Description.....	975
Table 8.78: ISOGCR Keyword Description.....	976
Table 8.79: ISOWCR Keyword Description.....	978
Table 8.80: ISWCR Keyword Description.....	980
Table 8.81: ISWL Keyword Description.....	981
Table 8.82: ISWLPC Keyword Description.....	982
Table 8.83: ISWU Keyword Description.....	984
Table 8.84: KRG Keyword Description.....	985
Table 8.85: Critical Displacement Relationships.....	986
Table 8.86: KRGR Keyword Description.....	988
Table 8.87: Critical Displacement Relationships.....	988
Table 8.88: KRO Keyword Description.....	990
Table 8.89: Critical Displacement Relationships.....	991
Table 8.90: KRORG Keyword Description.....	993
Table 8.91: Critical Displacement Relationships.....	993
Table 8.92: KRORW Keyword Description.....	995
Table 8.93: Critical Displacement Relationships.....	995
Table 8.94: KRW Keyword Description.....	997

Table 8.95: Critical Displacement Relationships.....	998
Table 8.96: KRWR Keyword Description.....	1000
Table 8.97: Critical Displacement Relationships.....	1000
Table 8.98: MISC Keyword Description.....	1035
Table 8.99: MSFN Keyword Description.....	1039
Table 8.100: MW Keyword Description.....	1041
Table 8.101: NCOMPS Keyword Description.....	1042
Table 8.102: OILDENT Keyword Description.....	1045
Table 8.103: OILJT Keyword Description.....	1047
Table 8.104: OILVISCT Keyword Description.....	1049
Table 8.105: OVERBURD Keyword Description.....	1052
Table 8.106: PCFACT Keyword Description.....	1054
Table 8.107: PCG Keyword Description.....	1056
Table 8.108: PCRIT Keyword Description.....	1058
Table 8.109: PCW Keyword Description.....	1059
Table 8.110: PERMFACT Keyword Description.....	1065
Table 8.111: PLMIXPAR Keyword Description.....	1067
Table 8.112: PLYADS Keyword Description.....	1068
Table 8.113: PLYADSS Keyword Description.....	1070
Table 8.114: PLYATEMP Keyword Description.....	1072
Table 8.115: PLYDHFLF Keyword Description.....	1075
Table 8.116: PLYMAX Keyword Description.....	1079
Table 8.117: PLYMWINJ Keyword Description.....	1082
Table 8.118: PLYROCK Keyword Description.....	1085
Table 8.119: PLYSHEAR Keyword Description.....	1087
Table 8.120: PLYSHLOG Keyword Description.....	1090
Table 8.121: PLYVISC Keyword Description.....	1094
Table 8.122: PLYVMH Keyword Description.....	1098
Table 8.123: PMISC Keyword Description.....	1104
Table 8.124: PPCWMAX Keyword Description.....	1106
Table 8.125: PVCDO Keyword Description.....	1109
Table 8.126: PVCO Keyword Description.....	1112
Table 8.127: PVDG Keyword Description.....	1114
Table 8.128: PVDO Keyword Description.....	1117
Table 8.129: PVDS Keyword Description.....	1119
Table 8.130: PVTG Keyword Description.....	1122
Table 8.131: PVTGW Keyword Description.....	1126
Table 8.132: PVTGWO Keyword Description.....	1129
Table 8.133: PVTO Keyword Description.....	1132
Table 8.134: PVTSQL Keyword Description.....	1137
Table 8.135: PVTW Keyword Description.....	1140
Table 8.136: PVTWSALT Keyword Description.....	1143
Table 8.137: PVZG Keyword Description.....	1145
Table 8.138: ROCK Keyword Description.....	1153
Table 8.139: ROCK2D Keyword Description.....	1155
Table 8.140: ROCK2DTR Keyword Description.....	1157
Table 8.141: ROCKOPTS Keyword Description.....	1160
Table 8.142: ROCKTAB Keyword Description.....	1163
Table 8.143: ROCKTABH Keyword Description.....	1166
Table 8.144: ROCKWNOD Keyword Description.....	1171
Table 8.145: RPTPROPS Keyword Description.....	1173
Table 8.146: RSCONST Keyword Description.....	1175
Table 8.147: RSCONSTT Keyword Description.....	1176
Table 8.148: RTEMP Keyword Description.....	1179
Table 8.149: RTEMPA Keyword Description.....	1180
Table 8.150: RVCONST Keyword Description.....	1182
Table 8.151: RVCONSTT Keyword Description.....	1183
Table 8.152: RWGSALT Keyword Description.....	1186
Table 8.153: SALINITY Keyword Description.....	1188
Table 8.154: SALTMF Keyword Description.....	1189
Table 8.155: SALTNODE Keyword Description.....	1190
Table 8.156: SALTSOL Keyword Description.....	1191
Table 8.157: SCALECRS Keyword Description.....	1193

Table 8.158: End-Point Arrays Used in the End-Point Scaling Options.....	1193
Table 8.159: SDENSITY Keyword Description.....	1196
Table 8.160: SGCR Keyword Description.....	1197
Table 8.161: SGCWMIS Keyword Description.....	1198
Table 8.162: SGFN Keyword Description.....	1200
Table 8.163: SGL Keyword Description.....	1202
Table 8.164: SGLPC Keyword Description.....	1203
Table 8.165: SGOF Keyword Description.....	1205
Table 8.166: SGOFLET Keyword Description.....	1209
Table 8.167: SGU Keyword Description.....	1211
Table 8.168: SGWFLET Keyword Description.....	1214
Table 8.169: SGWFN Keyword Description.....	1216
Table 8.170: SHRATE Keyword Description.....	1218
Table 8.171: SKPRPOLY Keyword Description.....	1221
Table 8.172: SKPRWAT Keyword Description.....	1224
Table 8.173: SKRO Keyword Description.....	1226
Table 8.174: SKRORG Keyword Description.....	1228
Table 8.175: SKRORW Keyword Description.....	1229
Table 8.176: SKRW Keyword Description.....	1230
Table 8.177: SKRWR Keyword Description.....	1231
Table 8.178: SLGOF Keyword Description.....	1233
Table 8.179: SOGCR Keyword Description.....	1235
Table 8.180: SOF2 Keyword Description.....	1236
Table 8.181: SOF3 Keyword Description.....	1238
Table 8.182: SOGCR Keyword Description.....	1241
Table 8.183: SORWMIS Keyword Description.....	1244
Table 8.184: SOWCR Keyword Description.....	1245
Table 8.185: SPECHEAT Keyword Description.....	1246
Table 8.186: SPECROCK Keyword Description.....	1248
Table 8.187: SSFN Keyword Description.....	1250
Table 8.188: SSGCR Keyword Description.....	1252
Table 8.189: SSGL Keyword Description.....	1253
Table 8.190: SSOGCR Keyword Description.....	1254
Table 8.191: SSOWCR Keyword Description.....	1255
Table 8.192: SSWCR Keyword Description.....	1256
Table 8.193: SSWL Keyword Description.....	1257
Table 8.194: SSWU Keyword Description.....	1258
Table 8.195: STONE1EX Keyword Description.....	1262
Table 8.196: SURFADS Keyword Description.....	1267
Table 8.197: SURFROCK Keyword Description.....	1271
Table 8.198: SURFVISC Keyword Description.....	1274
Table 8.199: SWATINIT Keyword Description.....	1276
Table 8.200: SWCR Keyword Description.....	1277
Table 8.201: SWFN Keyword Description.....	1279
Table 8.202: SWL Keyword Description.....	1281
Table 8.203: SWLPC Keyword Description.....	1282
Table 8.204: SWOF Keyword Description.....	1284
Table 8.205: SWOFLET Keyword Description.....	1288
Table 8.206: SWU Keyword Description.....	1290
Table 8.207: TCRIT Keyword Description.....	1291
Table 8.208: THCO2MIX Keyword Description.....	1295
Table 8.209: TLMIXPAR Keyword Description.....	1296
Table 8.210: TOLCRIT Keyword Description.....	1297
Table 8.211: TRACER Keyword Description.....	1301
Table 8.212: TZONE Keyword Description.....	1312
Table 8.213: VCRIT Keyword Description.....	1314
Table 8.214: VISCAQA Keyword Description.....	1321
Table 8.215: VISCREF Keyword Description.....	1323
Table 8.216: WAGHYSTR Keyword Description.....	1327
Table 8.217: WATDENT Keyword Description.....	1329
Table 8.218: WATJT Keyword Description.....	1331
Table 8.219: WATVISCT Keyword Description.....	1333
Table 8.220: WSF Keyword Description.....	1335

Table 8.221: ZMFVD Keyword Description.....	1337
Table 9.1: REGION Section Allocation Array Summary.....	1343
Table 9.2: ENDNUM Keyword Description.....	1349
Table 9.3: EOSNUM Keyword Description.....	1351
Table 9.4: EQLNUM Keyword Description.....	1352
Table 9.5: FIP Keyword Description.....	1354
Table 9.6: FIPNUM Keyword Description.....	1357
Table 9.7: HM Region Grid Gradient Parameter Keyword List.....	1362
Table 9.8: HM Region End-Point Gradient Parameter Keyword List.....	1363
Table 9.9: HWSNUM Keyword Description.....	1365
Table 9.10: IMBNUM Keyword Description.....	1366
Table 9.11: KRNUM Keyword Description.....	1369
Table 9.12: LSLTWNUM Keyword Description.....	1372
Table 9.13: LSNUM Keyword Description.....	1373
Table 9.14: LWSLTNUM Keyword Description.....	1374
Table 9.15: LWSNUM Keyword Description.....	1375
Table 9.16: MISCNUM Keyword Description.....	1377
Table 9.17: PLMIXNUM Keyword Description.....	1381
Table 9.18: PVTNUM Keyword Description.....	1382
Table 9.19: RESIDNUM Keyword Description.....	1386
Table 9.20: ROCKNUM Keyword Description.....	1387
Table 9.21: RPTREGS Keyword Description.....	1388
Table 9.22: SATNUM Keyword Description.....	1390
Table 9.23: SURFNUM Keyword Description.....	1392
Table 9.24: SURFWNUM Keyword Description.....	1393
Table 9.25: TNUM Keyword Name Format.....	1394
Table 9.26: TNUM Keyword Data Description.....	1394
Table 9.27: TRKPF Keyword Name Format.....	1396
Table 9.28: TRKPF Keyword Data Description.....	1396
Table 9.29: WH2NUM Keyword Description.....	1399
Table 9.30: WH3NUM Keyword Description.....	1400
Table 10.1: Minimum Set of SOLUTION Equilibrium Keywords Required by OPM Flow.....	1408
Table 10.2: Minimum Set of SOLUTION Enumeration Keywords Required by OPM Flow.....	1410
Table 10.3: SOLUTION Restart Initialization Keywords Required by OPM Flow.....	1411
Table 10.4: Alphabetic List of SOLUTION Keywords.....	1419
Table 10.5: APIVD Keyword Description.....	1423
Table 10.6: AQANCONL Keyword Description.....	1426
Table 10.7: AQUFET Keyword Description.....	1436
Table 10.8: AQUFETP Keyword Description.....	1439
Table 10.9: AQUFLUX Keyword Description.....	1441
Table 10.10: BC Keyword Description.....	1445
Table 10.11: DATUM Keyword Description.....	1448
Table 10.12: DATUMR Keyword Description.....	1449
Table 10.13: DATUMR Keyword Description.....	1450
Table 10.14: EQUIL Keyword Description.....	1459
Table 10.15: GASCONC Keyword Description.....	1462
Table 10.16: GASSATC Keyword Description.....	1463
Table 10.17: GCVD Keyword Description.....	1464
Table 10.18: OILAPI Keyword Description.....	1480
Table 10.19: PBUB Keyword Description.....	1483
Table 10.20: PBVD Keyword Description.....	1484
Table 10.21: PDEW Keyword Description.....	1485
Table 10.22: PDVD Keyword Description.....	1486
Table 10.23: PRESSURE Keyword Description.....	1487
Table 10.24: PRVD Keyword Description.....	1488
Table 10.25: RESTART Keyword Description.....	1494
Table 10.26: RPTRST Keyword Description.....	1499
Table 10.27: Data Sets Automatically Written to the RESTART File.....	1501
Table 10.28: Data Sets Automatically Written to the RESTART File for Visualization Only.....	1502
Table 10.29: RPTSOL Keyword Description.....	1504
Table 10.30: RS Keyword Description.....	1506
Table 10.31: RSVD Keyword Description.....	1507
Table 10.32: RSW Keyword Description.....	1509

Table 10.33: RTEMPVD Keyword Description.....	1512
Table 10.34: RV Keyword Description.....	1514
Table 10.35: RVVD Keyword Description.....	1515
Table 10.36: RVW Keyword Description.....	1517
Table 10.37: RVWVD Keyword Description.....	1519
Table 10.38: SALT Keyword Description.....	1521
Table 10.39: SALTP Keyword Description.....	1522
Table 10.40: SALTPVD Keyword Description.....	1524
Table 10.41: SALTREST Keyword Description.....	1526
Table 10.42: SALTVD Keyword Description.....	1527
Table 10.43: SBIOF Keyword Description.....	1530
Table 10.44: SCALC Keyword Description.....	1531
Table 10.45: SCVD Keyword Description.....	1532
Table 10.46: SFOAM Keyword Description.....	1534
Table 10.47: SGAS Keyword Description.....	1535
Table 10.48: SMICR Keyword Description.....	1537
Table 10.49: SOIL Keyword Description.....	1538
Table 10.50: SOLVCONC Keyword Description.....	1540
Table 10.51: SOLVFRAC Keyword Description.....	1541
Table 10.52: SOXYG Keyword Description.....	1542
Table 10.53: SPOLY Keyword Description.....	1543
Table 10.54: SPOLYMW Keyword Description.....	1544
Table 10.55: SSOL Keyword Description.....	1545
Table 10.56: SUREA Keyword Description.....	1546
Table 10.57: SURF Keyword Description.....	1547
Table 10.58: SWAT Keyword Description.....	1548
Table 10.59: TBLK Keyword Description.....	1549
Table 10.60: TEMPI Keyword Description.....	1551
Table 10.61: TEMPVD Keyword Description.....	1552
Table 10.62: THPRES Keyword Description.....	1554
Table 10.63: TVDP Keyword Name Format.....	1556
Table 10.64: TVDP Keyword Description.....	1557
Table 10.65: VAPPARS Keyword Description.....	1559
Table 11.1: OPM Flow Summary Output File Summary.....	1568
Table 11.2: RUNSPEC Input and Output File Format Keywords.....	1568
Table 11.3: OPM Flow File Naming Conventions.....	1569
Table 11.4: Summary Variable Mnemonic Syntax.....	1571
Table 11.5: Summary Variable Mnemonics Examples.....	1571
Table 11.6: Summary Variable Mnemonics Examples Not Following the General Syntax.....	1572
Table 11.7: Summary Variable Format.....	1576
Table 11.8: Aquifer Summary Variables.....	1577
Table 11.9: Field, Group, Well, Well Connection, and Completion Summary Variables.....	1586
Table 11.10: Field and Group Control Mode Reporting.....	1587
Table 11.11: Well Control Mode Reporting.....	1589
Table 11.12: Block, Field, and Region Summary Variables.....	1595
Table 11.13: Field and Region Summary Recovery Variables.....	1596
Table 11.14: API and Tracer Tracking Summary Variables.....	1600
Table 11.15: Brine and Multi-Component Brine Model Summary Variables.....	1604
Table 11.16: CO2STORE/H2STORE Field, Group and Well Summary Variables.....	1605
Table 11.17: CO2STORE/H2STORE Field, Region, and Block Summary Variables.....	1606
Table 11.18: Formation Damage Model Summary Variables.....	1608
Table 11.19: Foam Model Tracers Summary Variables.....	1609
Table 11.20: Gas Lift Optimization Model Summary Variables.....	1611
Table 11.21: Biofilm and MICP Model Summary Variables.....	1613
Table 11.22: Multi-Segment Wells Summary Variables.....	1616
Table 11.23: Network Model Summary Variables.....	1618
Table 11.24: OPM Flow Simulator Performance Summary Variables.....	1623
Table 11.25: Polymer Model Summary Variables.....	1626
Table 11.26: Solvent Model Summary Variables.....	1628
Table 11.27: Thermal Model Summary Variables.....	1630
Table 11.28: User Define Quantities Summary Variables.....	1631
Table 11.29: Standard Production, Injection, and Pressure Summary Variables.....	1636
Table 11.30: Standard Aquifer Summary Variables.....	1636

Table 11.31: FMWSET - Standard Field and Group Well Status Summary Variables.....	1644
Table 11.32: GMWSET - Standard Field and Group Well Status Summary Variables.....	1647
Table 11.33: Simulator Performance Summary Variables (Cumulative Messages).....	1651
Table 11.34: Simulator Performance Summary Variables (Numerical Performance).....	1656
Table 11.35: RPTSMRY Keyword Description.....	1660
Table 11.36: SUMTHIN Keyword Description.....	1665
Table 12.1: Major Well Specification Keywords.....	1672
Table 12.2: Major Group Specification Keywords.....	1673
Table 12.3: Schedule Advancement, Control, and Reporting Keywords.....	1675
Table 12.4: ACTIONW Keyword Description.....	1697
Table 12.5: ACTIONW Supported Well SUMMARY Variables.....	1699
Table 12.6: ACTIONX Keyword Description.....	1704
Table 12.7: ACTIONX Variable Definitions.....	1708
Table 12.8: ACTIONX Schedule Section Keyword Status.....	1709
Table 12.9: BCPROP Keyword Description.....	1720
Table 12.10: BRANPROP Keyword Description.....	1724
Table 12.11: COMPDAT Keyword Description.....	1733
Table 12.12: COMPDATL Keyword Description.....	1737
Table 12.13: COMPIMB Keyword Description.....	1742
Table 12.14: COMPLUPL Keyword Description.....	1745
Table 12.15: COMPLUMP Keyword Description.....	1748
Table 12.16: COMPORD Keyword Description.....	1750
Table 12.17: COMPSEGL Keyword Description.....	1757
Table 12.18: COMPSEGS Keyword Description.....	1759
Table 12.19: COMPTRAJ Keyword Description.....	1763
Table 12.20: CSKIN Keyword Description.....	1769
Table 12.21: DATES Keyword Description.....	1771
Table 12.22: DRSDT Keyword Description.....	1781
Table 12.23: DRSDTCON Keyword Description.....	1784
Table 12.24: DRSDTR Keyword Description.....	1787
Table 12.25: DRVDT Keyword Description.....	1789
Table 12.26: DRVDTR Keyword Description.....	1790
Table 12.27: EXIT Keyword Description.....	1797
Table 12.28: FBHPDEF Keyword Description.....	1800
Table 12.29: GCALECON Keyword Description.....	1812
Table 12.30: GCONCAL Keyword Description.....	1813
Table 12.31: GCONENG Keyword Description.....	1815
Table 12.32: GCONINJE Keyword Description.....	1819
Table 12.33: GCONPRI Keyword Description.....	1822
Table 12.34: GCONPROD Keyword Description.....	1828
Table 12.35: GCONSALE Keyword Description.....	1832
Table 12.36: GCONSUMP Keyword Description.....	1834
Table 12.37: GCONTOL Keyword Description.....	1837
Table 12.38: GDCQECON Keyword Description.....	1841
Table 12.39: GECON Keyword Description.....	1846
Table 12.40: GECONT Keyword Description.....	1848
Table 12.41: GEFAC Keyword Description.....	1850
Table 12.42: GLIFTLIM Keyword Description.....	1852
Table 12.43: GLIFTOPT Keyword Description.....	1854
Table 12.44: GPMAINT Keyword Description.....	1861
Table 12.45: GRUPNET Keyword Description.....	1871
Table 12.46: GRUPTARG Keyword Description.....	1876
Table 12.47: GRUPTREE Keyword Description.....	1877
Table 12.48: GSATINJE Keyword Description.....	1880
Table 12.49: GSATPROD Keyword Description.....	1884
Table 12.50: GUIDERAT Keyword Description.....	1895
Table 12.51: LGRFREE Keyword Description.....	1902
Table 12.52: LGRLOCK Keyword Description.....	1903
Table 12.53: LGROFF Keyword Description.....	1904
Table 12.54: LGRON Keyword Description.....	1905
Table 12.55: LIFTOPT Keyword Description.....	1907
Table 12.56: NEFAC Keyword Description.....	1920
Table 12.57: NETBALAN Keyword Description.....	1923

Table 12.58: NEXT Keyword Description.....	1926
Table 12.59: NEXTSTEP Keyword Description.....	1927
Table 12.60: NEXTSTPL Keyword Description.....	1930
Table 12.61: NODEPROP Keyword Description.....	1932
Table 12.62: PIMULTAB Keyword Description.....	1939
Table 12.63: PRIORITY Keyword Description.....	1946
Table 12.64: PYACTION Keyword Description.....	1949
Table 12.65: PYACTION Module Script Definition.....	1952
Table 12.66: RPTSCHED Keyword Description.....	1971
Table 12.67: SCDPTRAC Keyword Description.....	1982
Table 12.68: SOURCE Keyword Description.....	1991
Table 12.69: TIME Keyword Description.....	1996
Table 12.70: TSTEP Keyword Description.....	1998
Table 12.71: TUNING Keyword Description.....	2003
Table 12.72: TUNINGDP Keyword Description.....	2005
Table 12.73: TUNINGH Keyword Description.....	2007
Table 12.74: TUNINGL Keyword Description.....	2011
Table 12.75: TUNINGS Keyword Description.....	2017
Table 12.76: UDQ Keyword Description.....	2019
Table 12.77: UDQ - Description of Functions.....	2022
Table 12.78: UDQ - User Defined Argument Supported Keywords.....	2023
Table 12.79: UDT Keyword Description.....	2027
Table 12.80: VFCHK Keyword Description.....	2033
Table 12.81: VFPINJ Keyword Description.....	2036
Table 12.82: VFPPROD Keyword Description.....	2044
Table 12.83: VFPTABL Keyword Description.....	2047
Table 12.84: WCONHIST Keyword Description.....	2059
Table 12.85: WCONINJE Keyword Description.....	2064
Table 12.86: WCONINJH Keyword Description.....	2068
Table 12.87: WCONPROD Keyword Description.....	2075
Table 12.88: WCYCLE Keyword Description.....	2081
Table 12.89: WDFAC Keyword Description.....	2083
Table 12.90: WDFACCOR Keyword Description.....	2084
Table 12.91: WECON Keyword Description.....	2092
Table 12.92: WECONINJ Keyword Description.....	2094
Table 12.93: WECONT Keyword Description.....	2097
Table 12.94: WEFAC Keyword Description.....	2099
Table 12.95: WELCNTL Keyword Description.....	2102
Table 12.96: WELLSTRE Keyword Description.....	2106
Table 12.97: WELOPEN Keyword Description.....	2109
Table 12.98: WELOPENL Keyword Description.....	2114
Table 12.99: WELPI Keyword Description.....	2116
Table 12.100: WELSEGS Keyword Description.....	2123
Table 12.101: WELSPECL Keyword Description.....	2133
Table 12.102: WELSPECS Keyword Description.....	2137
Table 12.103: WELTARG Keyword Description.....	2140
Table 12.104: WELTRAJ Keyword Description.....	2143
Table 12.105: WFOAM Keyword Description.....	2145
Table 12.106: WGRUPCON Keyword Description.....	2153
Table 12.107: WHEDREFD Keyword Description.....	2154
Table 12.108: WHISTCTL Keyword Description.....	2155
Table 12.109: WHTEMP Keyword Description.....	2157
Table 12.110: WINJCLN Keyword Description.....	2159
Table 12.111: WINJDAM Keyword Description.....	2162
Table 12.112: WINJFCNC Keyword Description.....	2164
Table 12.113: WINJGAS Keyword Description.....	2166
Table 12.114: WINJMULT Keyword Description.....	2169
Table 12.115: WINJTEMP Keyword Description.....	2171
Table 12.116: WLIFTOPT Keyword Description.....	2176
Table 12.117: WLIST Keyword Description.....	2179
Table 12.118: WLISTARG Keyword Description.....	2182
Table 12.119: WLISTNAM Keyword Description.....	2184
Table 12.120: WMICP Keyword Description.....	2186

Table 12.121: WPAVE Keyword Description.....	2194
Table 12.122: WPAVDEP Keyword Description.....	2195
Table 12.123: WPIMULT Keyword Description.....	2197
Table 12.124: WPIMULT Keyword Description.....	2201
Table 12.125: WPITAB Keyword Description.....	2203
Table 12.126: WPMITAB Keyword Description.....	2205
Table 12.127: WPOLYMER Keyword Description.....	2208
Table 12.128: WRFT Keyword Description.....	2212
Table 12.129: WRFTPLT Keyword Description.....	2215
Table 12.130: WSALT Keyword Description.....	2217
Table 12.131: WSEGAICD Keyword Description.....	2225
Table 12.132: WSEGPROP Keyword Description.....	2242
Table 12.133: WSEGSICD Keyword Description.....	2247
Table 12.134: WSEGVALV Keyword Description.....	2256
Table 12.135: WELVALV Conversion Factor Constants and Units.....	2257
Table 12.136: WSKPTAB Keyword Description.....	2259
Table 12.137: WSOLVENT Keyword Description.....	2261
Table 12.138: WSURFACT Keyword Description.....	2263
Table 12.139: WTEMP Keyword Description.....	2265
Table 12.140: WTEST Keyword Description.....	2268
Table 12.141: WTMULT Keyword Description.....	2271
Table 12.142: WTRACER Keyword Description.....	2273
Table 12.143: WVFDPD Keyword Description.....	2275
Table 12.144: WVFPEXP Keyword Description.....	2278
Table 12.145: WWPAVE Keyword Description.....	2281
Table B.1: OPM Flow 2025-10 New and Deprecated Command Line Options.....	2363
Table B.2: OPM Flow 2025-04 New and Deprecated Command Line Options.....	2375
Table B.3: OPM Flow 2024-10 New and Deprecated Command Line Options.....	2391
Table B.4: OPM Flow 2024-04 New and Deprecated Command Line Options.....	2404
Table B.5: New SUMMARY Keywords for the 2024-04 Release.....	2411
Table B.6: New Initialization Output for the 2024-04 Release.....	2411
Table B.7: New Restart Output for the 2024-04 Release.....	2412
Table B.8: OPM Flow 2023-10 New and Deprecated Command Line Options.....	2420
Table B.9: New SUMMARY Keywords for the 2023-10 Release.....	2425
Table E.1: OPM Flow 2025-04 Command Line Options.....	2513
Table E.2: OPM Flow 2024-10 Command Line Options.....	2534
Table E.3: OPM Flow 2024-04 Command Line Options.....	2554
Table E.4: OPM Flow 2023-10 Command Line Options.....	2573
Table F.1: OPM Flow Output File Types Summary.....	2583
Table F.2: RUNSPEC Input and Output File Format Keywords.....	2584
Table F.3: OPM Flow File Naming Conventions.....	2586
Table F.4: Big-Endian and Little-Endian Representation.....	2587
Table F.5: Output File Keyword Format.....	2588
Table F.6: INTEHEAD Keyword - IWEL Example.....	2590
Table F.7: EGRID - Model Structural Data for Irregular Corner-Point Grids.....	2593
Table F.8: EGRID Header Keywords.....	2595
Table F.9: EGRID Global Irregular Corner Point Grid Keywords.....	2599
Table F.10: EGRID LGR Irregular Corner Point Grid Keywords.....	2602
Table F.11: EGRID NNC Keywords for Irregular Corner Point Grids.....	2605
Table F.12: RESTART Data - LGR Grid Header Keywords.....	2608
Table F.13: INIT Data File Structure.....	2613
Table F.14: INIT Data - Static Data Header Keywords.....	2617
Table F.15: INIT Data - Static Grid Array Data Keywords.....	2621
Table F.16: INIT Data - LGR Static Data Header Keywords.....	2623
Table F.17: INIT Data - LGR Grid Array Data Keywords.....	2625
Table F.18: INIT Data - Static Tabular Data Header Keyword.....	2629
Table F.19: INIT Data - Static Tabular and Constant Data Array Keywords.....	2630
Table F.20: RESTART Data File Structure.....	2640
Table F.21: RESTART Data - Header Keywords.....	2651
Table F.22: RESTART Data - Group, Well, and Connection Keywords (Groups).....	2658
Table F.23: RESTART Data - Group, Well, and Connection Keywords (Multi-Segment Wells).....	2661
Table F.24: RESTART Data - Group, Well, and Connection Keywords (Wells).....	2667
Table F.25: RESTART Data - Group, Well, and Connection Keywords (Connections).....	2669

Table F.26: RESTART Data – Group, Well, and Connection Keywords (Tracers).....	2670
Table F.27: RESTART Data – Group, Well, and Connection Keywords (Network).....	2671
Table F.28: RESTART Data – UDQ Keywords.....	2675
Table F.29: RESTART Data – ACTIONX Keywords.....	2678
Table F.30: RESTART Data – Analytical Aquifer Data Keywords.....	2682
Table F.31: RESTART Data – Numerical Aquifer Data Keywords.....	2683
Table F.32: RESTART Data – Hidden Keyword.....	2685
Table F.33: RESTART Data – Solution Data keyword.....	2686
Table F.34: RESTART Data – Solution Data Keyword for Tracer Concentration Name.....	2687
Table F.35: RESTART Data – Solution Data Keywords.....	2692
Table F.36: RESTART Data – LGR Solution Data Keywords.....	2694
Table F.37: RFT Data File Format.....	2698
Table F.38: RFT Data File – Time and Date Keywords.....	2698
Table F.39: RFT Data File – Well and Connection Data Keyword.....	2700
Table F.40: RFT Data File – RFT Solution Data Keywords.....	2701
Table F.41: Enhanced SUMMARY File Keywords.....	2706
Table F.42: SUMMARY Index File Keywords (Global).....	2709
Table F.43: SUMMARY Index File Keywords (LGR).....	2710
Table F.44: SUMMARY Index File Keywords (Commercial Simulator).....	2710
Table F.45: SUMMARY Data File Keywords.....	2712
Table F.46: SUMMARY Data File Keywords (LGR).....	2713

INDEX OF FIGURES

Figure 1.1: Enhanced Oil Recovery Processes.....	50
Figure 1.2: Multi-Segment Well Model - Segment Structure.....	59
Figure 1.3: Multi-Segment Well Model - Well Example with Main Wellbore Branch and Four Offset Branches.....	60
Figure 1.4: Numerical Solution of Equations Schematic.....	63
Figure 1.5: Numerical Solution of Equations Work Flow.....	66
Figure 2.1: VirtualBox Main Screen Showing Various Guest Virtual Machines.....	76
Figure 2.2: Ubuntu Live CD Initial Display.....	77
Figure 2.3: Ubuntu Guest Virtual Machine Desktop in VirtualBox Example.....	78
Figure 2.4: Ubuntu Guest Virtual Machine- Showing a Command Terminal.....	78
Figure 2.5: Ubuntu Guest Virtual Machine - Running OPM Flow Help.....	79
Figure 2.6: Windows 10 - Activate WSL.....	80
Figure 2.7: Windows 10 - Application Store.....	81
Figure 2.8: Windows 10 - WSL Installation of <i>Ubuntu</i> 18.04 LTS.....	81
Figure 2.9: Windows 10 WSL - Running OPM Flow Help.....	82
Figure 2.10: OPMRUN - Initial Display.....	107
Figure 2.11: OPMRUN - Display Elements.....	108
Figure 2.12: OPMRUN - Add Job Dialog.....	108
Figure 2.13: OPMRUN - Load Job Queue Dialog Box and Queue Display.....	109
Figure 2.14: OPMRUN - Edit Job Parameter File.....	110
Figure 2.15: OPMRUN - Select Run Option Dialog.....	110
Figure 2.16: OPMRUN - Running Jobs in the Queue.....	111
Figure 2.17: OPMRUN - Session Log File.....	111
Figure 2.18: Non-Orthogonal Grid Cells Along the Fault Plane Example.....	116
Figure 2.19: Norne Field Orthogonal Grid Example.....	117
Figure 2.20: Isolated Cells Example.....	119
Figure 2.21: Corey Oil-Water Curves and Derivative Plot.....	121
Figure 2.22: Corey Oil-Water Curves and Derivative Plot (Extended).....	123
Figure 2.23: Compressibility Pressure Effect on Gas and Liquids.....	124
Figure 2.24: Example Rock Oil-Water Relative Permeability Curves.....	126
Figure 2.25: Example Normalize Oil-Water Relative Permeability Curves.....	126
Figure 2.26: Example Normalize Oil-Water Relative Permeability Curves.....	127
Figure 2.27: RFT Measured and Model Pressure Data versus Depth Example.....	129
Figure 2.28: Norne Scaled Capillary Pressure Example.....	130
Figure 2.29: Norne Model Initialization Stability Example.....	131
Figure 2.30: PVT and Equilibration Data Validation Example.....	132
Figure 2.31: VFP Table Header Example.....	133
Figure 2.32: VFPPROD BHP versus THP Example.....	134
Figure 2.33: Non-Linear and Linear Iterations Example No. 1 (Commercial Simulator).....	136
Figure 2.34: Non-Linear and Linear Iterations Example No. 2 (OPM Flow).....	137
Figure 2.35: INFOSTEP Report Example.....	139
Figure 2.36: INFOITER Report Example.....	140
Figure 2.37: Time Step Advancement in OPM Flow.....	142
Figure 2.38: Report Step and Time Step Report.....	144
Figure 2.39: Report Step and Time Step Summary Reporting Example.....	148
Figure 2.40: Time Step Summary Showing Non-Convergence Example.....	150
Figure 2.41: Linear Solver Property Tree.....	153
Figure 2.42: Time Step Summary Reporting Linear Solver Example.....	155
Figure 2.43: Number of Linear Iterations Per Newton Iteration Report Example.....	156
Figure 3.1: Table of Contents Footer.....	161
Figure 3.2: Keyword Index Footer.....	161
Figure 3.3: Alphabetic Listing of Keywords.....	161
Figure 6.1: Numerical Model Dimensions.....	429
Figure 6.2: SPE Simulation Case #01 Cartesian Regular Grid.....	430
Figure 6.3: Norne Field Grid Skeleton.....	432
Figure 6.4: Norne Field Corner-Point Geometry Example.....	433
Figure 6.5: Norne Grid Ternary Solution Display.....	434
Figure 6.6: Cartesian and Radial Grids Comparison.....	435
Figure 6.7: 3D Radial Grid (Odeh Example).....	435
Figure 6.8: Odeh Radial Grid in OPM ResInsight (Commercial Simulator Run).....	437
Figure 6.9: Odeh Radial Grid in OPM ResInsight (OPM Flow Run).....	438

Figure 6.10: 3-D Spider Grid (Odeh Example).....	439
Figure 7.1: Volve Full Field Model PORV Array.....	734
Figure 7.2: Volve Full Field Model TRANX Array.....	734
Figure 8.1: Reservoir Fluid Phase Diagram.....	769
Figure 8.2: Low Shrinkage Oil Phase Diagram.....	777
Figure 8.3: High Shrinkage Oil Phase Diagram.....	778
Figure 8.4: Dry Gas Phase Diagram.....	779
Figure 8.5: Wet Gas Phase Diagram.....	780
Figure 8.6: Retrograde Condensate Gas Phase Diagram.....	781
Figure 8.7: Volve Field Live Oil PVT Data.....	786
Figure 8.8: Volve Field Gas Dry PVT Data.....	787
Figure 8.9: Example Oil-Water Relative Permeability Curves.....	789
Figure 8.10: Example Gas-Oil Relative Permeability Curves.....	790
Figure 8.11: Example Gas-Water Relative Permeability Curves.....	791
Figure 8.12: Volve Oil-Water Relative Permeability - Unscaled.....	794
Figure 8.13: Volve Oil-Water Relative Permeability - Scaled.....	794
Figure 8.14: Volve Gas-Oil Relative Permeability - Unscaled.....	794
Figure 8.15: Volve Gas-Oil Relative Permeability - Scaled.....	794
Figure 8.16: Oil-Water Corey Example (no=0.5, nw=0.75).....	796
Figure 8.17: Oil-Water Corey Example (no=1.0, nw=1.0).....	796
Figure 8.18: : Oil-Water Corey Example (no=3.0, nw=4.0).....	796
Figure 8.19: Oil-Water Relative Permeability LET Empirical Parameters Example.....	799
Figure 8.20: Oil-Water Primary Drainage (DIC1) Capillary Pressure LET Example.....	800
Figure 8.21: Oil Water Primary Imbibition (IDC2) Capillary Pressure LET Example.....	800
Figure 9.1: Volve Full Field Model SATNUM Array.....	1344
Figure 9.2: Volve Full Field Model EQUIL Array.....	1345
Figure 12.1: Calibration of Well Productivity Index.....	1683
Figure 12.2: Well Calibration Example.....	1684
Figure 12.3: Extend Network Example.....	1725
Figure 12.4: Norne Group Tree Hierarchy Example.....	1878
Figure 12.5: Extend Network Example.....	1932
Figure 12.6: RPTSCHED: FIP (Balance) Report.....	1972
Figure 12.7: RPTSCHED: FIPRESV (Reservoir Volumes) Report.....	1972
Figure 12.8: RPTSCHED: WELLS - Injection Sub-Report.....	1973
Figure 12.9: RPTSCHED: WELLS - Production Sub-Report.....	1973
Figure 12.10: RPTSCHED: WELLS - Cumulative Production and Injection Sub-Report.....	1974
Figure 12.11: RPTSCHED: WELSPECS - Production Well Control Sub-Report.....	1974
Figure 12.12: RPTSCHED: WELSPECS - Well Connection Data Sub-Report.....	1975
Figure 12.13: RPTSCHED: WELSPECS - Multi Segment Well Structure Sub-Report.....	1976
Figure 12.14: RPTSCHED: WELSPECS - Multi Segment Well Connection Sub-Report.....	1976
Figure 12.15: Multi-Segment Well OP01 Completion 3D View.....	2127
Figure 12.16: Multi-Segment Well Example (Wells OP01 and OP02).....	2229
Figure C.1: OPMRUN - Initial Display.....	2437
Figure C.2: OPMRUN - Display Elements.....	2438
Figure C.3: OPMRUN - Add Job Dialog Box.....	2438
Figure C.4: OPMRUN - Add Job and Select Run Type.....	2439
Figure C.5: OPMRUN - Edit Job Data and Parameter File.....	2440
Figure C.6: OPMRUN - Edit Job Parameter File.....	2441
Figure C.7: OPMRUN - Load Job Queue Dialog Box and Queue Display.....	2442
Figure C.8: OPMRUN - Reset Job Queue Parameters.....	2443
Figure C.9: OPMRUN - Reset Job Queue Parameters Dialog.....	2443
Figure C.10: OPMRUN - Run Jobs in the Queue.....	2444
Figure C.11: OPMRUN - Select Run Option Dialog.....	2444
Figure C.12: OPMRUN - Log File Creation.....	2445
Figure C.13: OPMRUN - Session Log.....	2446
Figure C.14: OPMRUN - Edit Data File.....	2447
Figure C.15: OPMRUN - Set Default Parameter Set.....	2448
Figure C.16: OPMRUN - Set Application Options.....	2449
Figure C.17: OPMRUN - Set Project Directories.....	2449
Figure C.18: OPMRUN Tools -Compression Utility.....	2451
Figure C.19: OPMRUN Tools - Compression Utility Application.....	2451
Figure C.20: OPMRUN Tools - Keywords Application.....	2452
Figure C.21: OPMRUN Tools - Keywords HEADER Comment Example.....	2453

Figure C.22: OPMRUN Tools - Keywords File Menu.....	2454
Figure C.23: OPMRUN Tools - Keywords Edit Menu.....	2454
Figure C.24: OPMRUN Tools - Keywords Generate Menu.....	2454
Figure C.25: OPMRUN Tools - Keywords Help.....	2454
Figure C.26: OPMRUN Tools - Keywords Generator Help.....	2455
Figure C.27: OPMRUN Tools - Keywords Velocity Template Help.....	2455
Figure C.28: OPMRUN Tools - Keywords File Imports.....	2456
Figure C.29: OPMRUN Tools - Keywords Section Standard Set of Keywords Example.....	2457
Figure C.30: OPMRUN Tools - Keywords Summary Section Keywords.....	2458
Figure C.31: OPMRUN Tools - Keywords Schedule Section and Date Schedule.....	2459
Figure C.32: OPMRUN Tools - Keywords Data (Sets) Option.....	2460
Figure C.33: OPMRUN Tools - Keywords Model Option.....	2461
Figure C.34: OPMRUN Tools - Keywords User Templates Option.....	2462
Figure C.35: OPMRUN Tools - Production Schedule Input File Example.....	2463
Figure C.36: OPMRUN Tools - Production Schedule Application.....	2464
Figure C.37: OPMRUN Tools - Production Schedule Output Example.....	2465
Figure C.38: OPMRUN Tools - Sensitivity Application.....	2466
Figure C.39: OPMRUN Tools - Sensitivity Base Case Data File Example.....	2467
Figure C.40: OPMRUN Tools - Sensitivity Load Factors Example.....	2468
Figure C.41: OPMRUN Tools - Sensitivity Factor Editing Example.....	2469
Figure C.42: OPMRUN Tools - Sensitivity Scenario Option Selection Example.....	2470
Figure C.43: OPMRUN Tools - Sensitivity Case Generation and OPMRUN Queue File.....	2471
Figure C.44: OPMRUN Tools - Well Specification OPM ResInsight Well Completion File Format.....	2472
Figure C.45: OPMRUN Tools - Well Specification OPM ResInsight Reservoir Layer File.....	2473
Figure C.46: OPMRUN Tools - Well Specification Application.....	2474
Figure C.47: OPMRUN Tools - Well Specification WELSPECS and COMPDAT Output Example.....	2474
Figure C.48: OPMRUN Tools - Well Specification OPM ResInsight Perforation File Output.....	2475
Figure C.49: OPMRUN Tools - Well Trajectory Conversion Input File.....	2475
Figure C.50: OPMRUN Tools - Well Trajectory Application.....	2476
Figure C.51: OPMRUN Tools - Well Trajectory Output Trajectory File.....	2477
Figure F.1: IWEL Property Keyword Example.....	2590
Figure F.2: ICON Property Keyword Example.....	2591

CHAPTER 1: INTRODUCTION

1.1 OVERVIEW

The Open Porous Media (“OPM”) initiative was started in 2009 to encourage open innovation and reproducible research on modeling and simulation of porous media processes. OPM was initially founded as a collaboration between groups at Equinor (formerly Statoil), SINTEF, the University of Stuttgart, and the University of Bergen, but over time, several other groups and individuals have joined and contributed. What today forms the OPM suite of software, has mainly been developed by SINTEF, NORCE (formerly IRIS), Equinor, Ceetron Solutions (OPM ResInsight), Poware Software Solutions, Dr. Blatt HPC-Simulation-Software & Services, and OPM-OP. The initial vision was to create long-lasting, efficient, and well-maintained, open-source software for simulating flow and transport in porous media. The scope has later been extended to also provide open data sets, thus making it easier to benchmark, compare, and test different mathematical models, computational methods, and software implementations.

All software in OPM is distributed under the GNU General Public License (“GPL”), [version 3](#), with the option of using newer versions of the license. Datasets are distributed under the Open Data Commons Open Database License (“ODbL”), [version 1.0](#), with individual content under the Database Contents License (“DbCL”), [version 1.0](#).

The project’s numerical simulator, OPM Flow, is a fully-implicit, black-oil simulator capable of running industry-standard simulation models¹. The simulator’s input file is compatible with the commercial simulator, although OPM Flow has extra keywords for supporting additional functionality not found in the commercial simulator. The simulator’s output files, except for the print and debug files, are compatible with the commercial simulator, and are readable by commercial post-processing software as well as by OPM ResInsight. OPM ResInsight is the Open Porous Media’s post-processing software.

The simulator is implemented using automatic differentiation to enable rapid development of new fluid models. In most simulators the coefficients of the linearized systems of equations (Jacobian matrix) required by the non-linear Newton Solver, have traditionally been computed by evaluating the closed-form expressions obtained by differentiating the discretized flow equations analytically. Differentiating these flow equations manually and programming the resulting formulas is time-consuming and error-prone. Thus, even simple extensions in functional dependencies can be difficult to accommodate into existing simulators. Automatic differentiation eliminates these drawbacks.

OPM Flow runs under Linux and Windows Subsystem for Linux, and the program is in active development with new features added in each bi-annual release. Currently the program has the following functionality:

Model Formulation

- Black-oil with dissolved gas and vaporized oil.
- Rock-dependent capillary and relative-permeability curves.
- End-point scaling and hysteresis.
- Oil vaporization controls ([VAPPARS](#)).
- Rock compaction.

Enhance Oil Recovery Models

Enhanced Oil Recovery (“EOR”) is a term applied to methods used for recovering oil from a petroleum reservoir beyond that recoverable by primary and secondary methods. Primary recovery normally refers to production using the energy inherent in the reservoir from gas under pressure or a natural water drive. Secondary recovery usually refers to injection of water or water flooding. Thus, Enhanced Oil Recovery is often synonymous with tertiary recovery. Improved Oil Recovery (“IOR”) and Advanced Oil Recovery (“AOR”) have similar meaning, except they also apply to primary and secondary methods, and sometimes

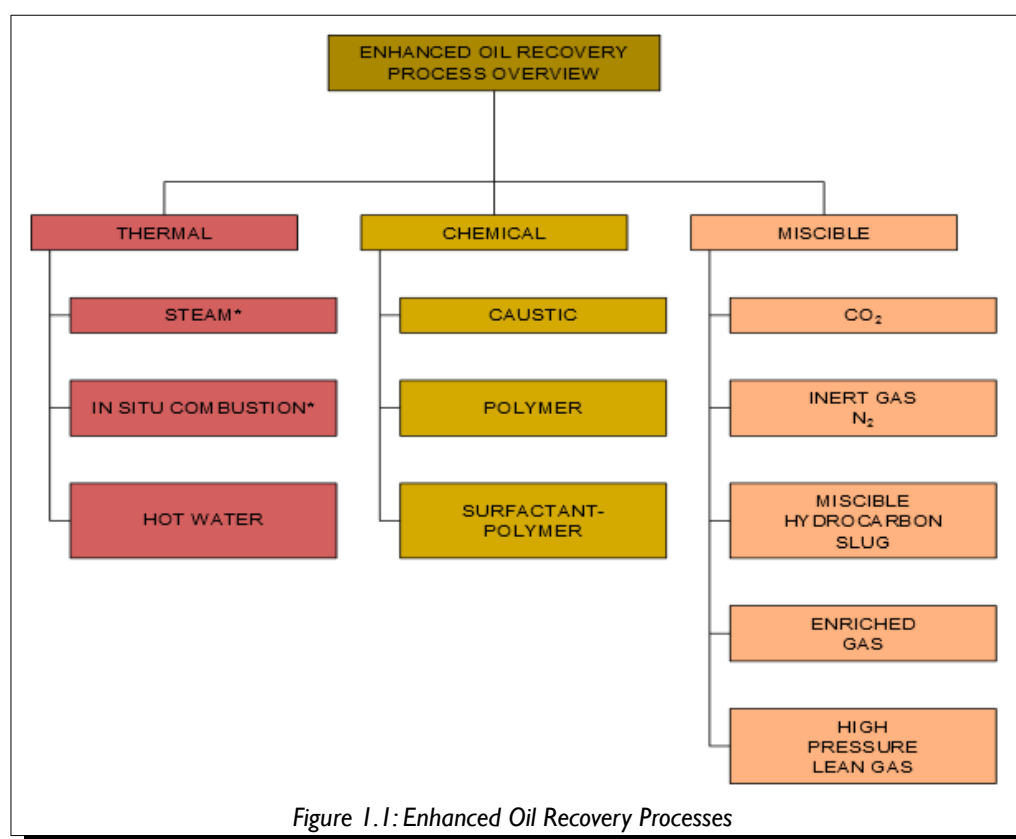
¹ Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkor, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.

EOR methods can be used earlier in the sequence. Originally water flooding was considered as Enhanced Oil Recovery, but now EOR is generally considered to follow water flooding.

There are four major EOR groups of methods:

- thermal recovery,
- gas miscible recovery,
- chemical flooding, and
- microbial flooding.

These main groups are further subdivided into individual processes as illustrated in Figure 1.1. For example the thermal recovery methods consist of steam flooding and cyclic steam stimulation, as well as in-situ combustion. Figure 1.1 also includes hot water as a process for thermal EOR, but this process is rarely used in the industry as a drive mechanism, as steam contains significantly more heat than hot water and therefore steam is more effective in heating up the in situ oil. Note that microbial EOR methods, are not shown in Figure 1.1, due to there limited use in the industry.



Chemical flooding methods include polymer flooding (including polymer gels), micellar-polymer flooding, and alkaline flooding. Quite often these type of floods are combined to form a more efficient displacement process, for example in an Alkaline Surfactant Polymer (“ASP”) process.

Conventional gas flooding is an immiscible process, that is the oil and gas phases remain separate during the displacement drive. For the EOR gas drive process, the displacement is miscible, here the gas “mixes” with the oil to form a single phase fluid in the reservoir. Gas miscible processes are subdivided into carbon dioxide flooding, cyclic carbon dioxide stimulation, nitrogen flooding, and nitrogen-CO₂ flooding.

OPM Flow has several EOR models based on extending the black-oil model with a fourth component in the simulator:

- **CO₂ Standard EOR Model:** OPM Flow's CO₂ Standard EOR model is based on the simulators Solvent model by extending the black-oil formulation with a fourth component by adding a CO₂ component to the gas phase. This is the standard formulation used by most black-oil simulators.
- **CO₂ Dynamic EOR Model:** Although the CO₂ Standard EOR model is commonly used, the formulation cannot account for the mass transfer of the various components and phases. The extended black-oil formulations often poorly represent the PVT properties of the oil-CO₂ mixtures, resulting in poor agreement with the compositional formulation. OPM Flow's CO₂ Dynamic EOR Model black-oil formulation attempts to overcome this limitation by making the black-oil properties dependent on the fraction of CO₂ in the cell². This approach models the oil-CO₂ mixture more accurately, and thus give results closer to the compositional simulator. Note the model is not restricted to CO₂ utilization, as long as the appropriate oil-injection-gas dependent properties are entered in the model.
- **Foam Model:** An experimental foam module has been added to OPM Flow 2019-10 release. With this it is possible to simulate certain types of surfactant injection. Such injection stimulates formation of foam to change mobility ratios, and give better reservoir sweep. The implemented foam model treats surfactant transported in the gas or water phase, and reduces the mobility of that phase depending on the surfactant concentration. In addition to mobility reduction, foam adsorption to the reservoir rock is also included in the model.
- **Polymer Model (Standard):** OPM Flow's polymer model is based on a black-oil polymer formulation, which is developed by extending the black-oil model with a polymer component. The effects of the polymer mixing are simulated based on the Todd-Longstaff³ mixing model, and adsorption, dead pore space, and permeability reduction effects are also considered. A logarithmic shear thinning/thickening model has also been incorporated since the 2015-10 Release (see [Flow-polymer](#)). Note that the Polymer model has now been incorporated into the main OPM Flow simulator and is no longer a separate simulator.
- **Polymer Model (Molecular Weight Transport):** In addition to the standard polymer model, the simulator also has the Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity. Here, the *Mark-Howink equation*, also known as the Mark-Howink-Staudinger equation⁴, ⁵, and ⁶ is used to calculate the intrinsic viscosity as a function of the molecular weight of the polymer.
- **Solvent Model:** Similar to the polymer model, OPM Flow's solvent model again uses an extra component by extending the black-oil formulation with a fourth component in the simulator by adding a solvent component to the gas phase (see [Flow-solvent](#)). Note that Solvent model has now been incorporated into the main OPM Flow simulator and is no longer a separate simulator.

CO₂ Storage:

- Brine-CO₂ PVT module that generates brine and CO₂ PVT properties directly eliminating user input for PVT.
- Thermal properties generated by OPM Flow include the CO₂ enthalpy represented as a table internally in the simulator. The liquid enthalpy depends on the dissolved CO₂ and salinity as well as

² T. H. Sandve, O. Sæviareid and I. Aavatsmark: "Improved Extended Blackoil Formulation – for CO₂ EOR Simulations." in *ECMOR XVII – The 17th European Conference on the – Mathematics of Oil Recovery*, September 2020.

³ M. R. Todd and W. J. Longstaff, *The Development, Testing, and Application Of a Numerical Simulator for Predicting Miscible Flood Performance*. In: *J. Petrol. Tech.* 24.7 (1972), pages 874-882.

⁴ H. Mark, in R. Saenger, *Der feste Koerper*, Hirzel, Leipzig, 1938.

⁵ R. Houwink, *J. Prakt. Chem.*, Vol. 157, Issue 1-3, p. 15 (1940).

⁶ H. Staudinger, *Die Hochmolekulare Organischen Verbindungen*, Julius Springer, Berlin 1932.

pressure and temperature. The water enthalpy is automatically calculated and modified to account for salinity and for CO₂ following.

- Diffusion for fine grid modeling is implemented via coefficients that depend on temperature, pressure, and salinity. The diffusion coefficient is computed internally for pure water and modified to account for salinity. The effect of the porous media on the diffusion is modeled using an industry standard relationship. In addition, the coefficient can also be given as an input parameter using a keyword.
- Convective dissolution of CO₂ into in situ brine within a grid cell is implemented together with a convective dissolution parameter for controlling the rate of dissolution.
- Microbially Induced Calcite Precipitation (“MICP”) model for the modeling of biochemical processes in creating barriers by calcium carbonate cementation, the technology has the potential to be used for sealing leakage zones in geological formations⁷ and ⁸.

Other OPM Flow Specific Enhancements:

- The Salt Precipitation and Water Evaporation Model: The model accounts for water evaporation of the in situ brine and the resulting increasing dissolved salt concentration. Once the salt solubility limit is exceeded, salt precipitation occurs, that results in a decrease in porosity and permeability, and consequently a decrease in gas production. This feature is primary used in modeling gas fields under depletion.
- Thermal Model Enhancements: The energy black-oil implementation in OPM Flow is a mixture of the commercial simulators black-oil temperature and compositional thermal options. These commercial simulator options are two separate modeling facilities in the commercial simulator. OPM Flow’s thermal implementation is based on solving the energy equation fully coupled with the black-oil equations so the results are not directly equivalent to the commercial simulator’s black-oil temperature or compositional thermal formulations. In addition, OPM Flow can take into account the Joule–Thomson effect for oil, gas, and water. This behavior is especially important in gas wells, when a real gas expands, resulting in the temperature of the gas dropping.
- Biofilm Effects in Underground Applications: Biofilm-related effects in subsurface applications such as hydrogen storage include reduced injectivity and hydrogen loss. The conceptual model includes the following mechanisms: (1) Biofilm is present in the storage site prior to injection and (2) the biofilm consumes injected H₂/CO₂, leading to clogging effects.

Description of Geology:

- Rectilinear and fully-unstructured grid.
- Corner-point grids from the commercial simulator input, including fault and region multipliers, minimum pore volume, pinch outs, etc.
- Spider grids that are similar to radial grids for modeling near well bore effects that enable visualization in OPM ResInsight.
- Analytical and numerical aquifers.

⁷ Landa-Marbán, D., Tveit, S., Kumar, K., Gasda, S.E., 2021. Practical approaches to study microbially induced calcite precipitation at the eld scale. *Int. J. Greenh. Gas Control* 106, 103256. <https://doi.org/10.1016/j.ijggc.2021.103256>.

⁸ Landa-Marbán, D., Kumar, K., Tveit, S., Gasda, S.E., 2021. Numerical studies of CO₂ leakage remediation by micp-based plugging technology. In: Røkke, N.A. and Knuutila, H.K. (Eds) *Short Papers from the 11th International Trondheim CCS conference*, ISBN: 978-82-536-1714-5, 284-290.

Well and Group Controls:

- Bottom-hole pressure and surface/reservoir rate.
- Group controls, including voidage replacement, network model, gas lift optimization, guide rate control for producers and injection groups and well, etc.
- Shut/stop/open individual completions.
- History-matching wells.
- Multi-segment wells together with various inflow control devices ("ICDs").
- Well lists.
- Action conditions and command processing via the [ACTIONX](#) and [UDQ](#) keywords employed in the commercial simulator.
- Python interface to control and process simulator data.

Input and Output:

- General reader/parser for Eclipse input decks.
- XML-based or simple text-format input of additional parameters.
- Flexible output of summary and restart files in the commercial simulator's format.
- Logging to terminal and print file.
- Front-end graphical interface for running the simulator with various tools, including: case compression and uncompression, keyword generation etc.

Simulation Technology:

- Fully-implicit in time.
- Automatic differentiation.
- Two-point flux approximation in space with upstream-mobility weighting.
- Flexible assembly through the use of automatic differentiation.
- Bi-conjugate Gradient method (BiCG-stab) and a restarted Generalized Minimal Residual ("GMRes") iterative linear solvers with ILU0 and Constraint Pressure Residual ("CPR") preconditioners.
- Adaptive step-size controls.
- Parallel processing.

The following section briefly describe the simulator's formulation, equation, and the various numerical procedures available to solve the equations.

Note

The material in the following sections is taken from Rasmussen et al.⁹, and the reader is encouraged to read the full paper to obtain a fuller understanding of the simulator's formulation and the resulting equations.

1.2 BLACK-OIL MODEL EQUATIONS

The black-oil equations constitute the most widely used flow model in the simulation of hydrocarbon reservoirs. The model is based on the premise that we have three different fluid phases (aqueous, oleic, and gaseous), and three (pseudo) components: water, oil, and gas. Oil and gas are not single hydrocarbon species, but represent all hydrocarbon species that exist in liquid and vapor form at surface conditions. Mixing is allowed, in the sense that both oil and gas can be found in the oleic phase, in the gaseous phase, or in both. The amount of dissolved gas in the oleic phase and vaporized oil in the gaseous phase must be kept track of on a grid cell basis. In the following discussions the subscripts w represents the water, o the oil, and g the gas quantities related to each of the phases or components.

The black-oil equations can be deduced from conservation of mass for each component with suitable closure relationships such as Darcy's law and initial and boundary conditions. The equations are discretized in space with an upstream finite-volume scheme using a two-point flux approximation, and in time using an implicit (backward) Euler scheme. The resulting equations are solved simultaneously in a fully implicit formulation by a Newton-type linearization with a properly preconditioned, iterative linear solver (see Rasmussen et al.¹⁰).

The conservation laws form a system of partial differential equations, one for each pseudo component α :

$$\phi_{ref} \frac{\partial A_\alpha}{\partial t} + \nabla \cdot \mathbf{u}_\alpha + q_\alpha = 0 \quad (1.1)$$

where the accumulation terms and fluxes are given by:

$$A_w = m_\phi b_w s_w \quad \mathbf{u}_w = b_w \mathbf{v}_w \quad (1.2)$$

$$A_o = m_\phi (b_o s_o + r_{og} b_g s_g) \quad \mathbf{u}_o = b_o \mathbf{v}_o + r_{og} b_g \mathbf{v}_g \quad (1.3)$$

$$A_g = m_\phi (b_g s_g + r_{go} b_o s_o) \quad \mathbf{u}_g = b_g \mathbf{v}_g + r_{go} b_o \mathbf{v}_o \quad (1.4)$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v}_\alpha = -\lambda_\alpha \mathbf{K} (\nabla p_\alpha - \rho_\alpha \mathbf{g}) \quad (1.5)$$

In addition, the following closure relations should hold:

$$s_w + s_o + s_g = 1 \quad (1.6)$$

$$p_{c,ow} = p_o - p_w \quad (1.7)$$

$$p_{c,og} = p_o - p_g \quad (1.8)$$

⁹ Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.

¹⁰ Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.

where:

ϕ	= porosity, can be a function of (oil) pressure: $\phi = m_\phi(p_o)\phi_{ref}$.
m_ϕ	= pore volume multiplier as function of pressure.
ϕ_{ref}	= reference porosity, a constant in time but varying in space.
p_α	= pressure for phase α .
b_α	= shrinkage/expansion factor for phase α defined as the ratio of the surface volume at standard conditions to the reservoir volume for a given amount of fluid: $b_\alpha = V_{surface;\alpha} / V_{reservoir;\alpha}$. For the oil phase, b_o is called the shrinkage factor, whereas for gas b_g is called the expansion factor. The reciprocal quantity is called the formation volume factor, and is usually denoted with a capital B: $B_\alpha = 1 / b_\alpha$. The variable is usually a function of phase pressure and composition: $b_\alpha = b_\alpha(p_\alpha, r_{go}, r_{og})$.
r_{go}	= gas-oil ratio ("GOR"), ratio of dissolved gas to oil in the oleic phase, commonly called r_s or R_s in other literature.
r_{og}	= oil-gas ratio or condensate-gas ratio ("CGR"), ratio of vaporized oil to gas in the gaseous phase, commonly called r_v or R_v in other literature.
S_α	= saturation of phase α , the pore volume fraction occupied by the phase. The saturations of all phases sum to one.
$p_{c,\alpha\beta}$	= capillary pressure between phases α and β . Typically a function of saturation: $p_{c,\alpha\beta} = p_{c,\alpha\beta}(S_\alpha)$.
$\mathbf{K}$	= permeability of the porous medium.
$k_{r;\alpha}$	= relative permeability for phase α modeling the reduction in effective permeability for a fluid phase in the presence of other phases. Typically a function of saturation.
μ_α	= viscosity of phase α , typically a function of phase pressure and composition.
λ_α	= mobility of phase α , given by $\lambda_\alpha = k_{r,\alpha} / \mu_\alpha$.
$\mathbf{v}_\alpha$	= velocity of phase α .
$\mathbf{u}_\alpha$	= velocity of component α .
$\rho_{S,\alpha}$	= surface density of phase α at one atmosphere, a given constant.
ρ_α	= density of phase α in the reservoir. For water, $\rho_w = b_w \rho_{S,w}$. For the oleic phase the relationship is more complex since it must include dissolved gas: $\rho_o = b_o(\rho_{S,o} + r_{go} \rho_{S,g})$, and similar for the gaseous phase we have: $\rho_g = b_g(\rho_{S,g} + r_{og} \rho_{S,o})$.
$\mathbf{g}$	= gravitational acceleration vector.
q_α	= well out flux density of pseudo component α , (negative for well in-flow). The form of this term depends on the well model used, either the standard well model or the multi-segment model.

Equations (1.1) to (1.8) are then transformed to the discrete form using a forward finite difference scheme. In the following, let subscript $\bar{i}$ denote a discrete quantity defined in cell $\bar{i}$ and subscript $\bar{i}\bar{j}$ denote a discrete quantity defined at the connection between two cells $\bar{i}$ and $\bar{j}$. Two cells share a connection if they are adjacent geometrically in the computational grid or share an explicit Non-Neighbor Connection ("NNC")¹¹. For example $v_{o,\bar{i}\bar{j}}$ is the oil flux from cell $\bar{i}$ to cell $\bar{j}$. For oriented quantities such as fluxes, the orientation is taken to be from cell $\bar{i}$ to cell $\bar{j}$, and the quantity is skew-symmetric ($v_{o,\bar{i}\bar{j}} = -v_{o,\bar{j}\bar{i}}$), whereas non-oriented quantities such as the transmissibility $T_{\bar{i}\bar{j}}$ are symmetric ($T_{\bar{i}\bar{j}} = T_{\bar{j}\bar{i}}$). Quantities with superscript 0 are taken at the start of the discrete time step, other quantities are at the end of the time step. Superscripts or subscripts applied to an expression in parenthesis apply to each element in the expression.

¹¹ Connections between cells due to geometric adjacency are automatically calculated by the simulator, including across faults in the grid. In addition, the user can specify extra connections between arbitrary cells using the NNC keyword in the GRID section.

The discretized equations and residuals are, for each pseudo-component α and cell $\bar{i}$:

$$R_{\alpha,i} = \frac{\phi_{ref,i} V_i}{\Delta t} (A_{\alpha,i} - A_{\alpha,i}^0) + \sum_{j \in C(i)} u_{\alpha,ij} + q_{\alpha,i} = 0 \quad (1.9)$$

where the accumulation terms are the same as equations (1.2) to (1.4) and fluxes u_{α} are of similar form to the velocities, $\mathbf{u}_{\alpha}$ in the aforementioned equations; thus we have

$$A_{wi} = (m_{\phi} b_w s_w)_i \quad u_{wi} = (b_w v_w)_i \quad (1.10)$$

$$A_{oi} = (m_{\phi} (b_o s_o + r_{og} b_g s_g))_i \quad u_{oi} = (b_o v_o + r_{og} b_g v_g)_i \quad (1.11)$$

$$A_{gi} = (m_{\phi} (b_g s_g + r_{go} b_o s_o))_i \quad u_{gi} = (b_g v_g + r_{go} b_o v_o)_i \quad (1.12)$$

In addition, equations (1.6) to (1.8) also hold for each cell i , that is:

$$(s_w + s_o + s_g)_i = 1 \quad (1.13)$$

$$(p_{c,ow})_i = (p_o - p_w)_i \quad (1.14)$$

$$(p_{c,og})_i = (p_o - p_g)_i \quad (1.15)$$

Finally, the fluxes are given for each connection $\bar{i}\bar{j}$ by the following relationships:

$$(b_{\alpha} v_{\alpha})_{ij} = (b_{\alpha} \lambda_{\alpha} m_T)_{U(\alpha,ij)} T_{ij} \Delta \Phi_{\alpha,ij} \quad (1.16)$$

$$(r_{\beta\alpha} b_{\alpha} v_{\alpha})_{ij} = (r_{\beta\alpha} b_{\alpha} \lambda_{\alpha} m_T)_{U(\alpha,ij)} T_{ij} \Delta \Phi_{\alpha,ij} \quad (1.17)$$

$$\Delta \Phi_{\alpha,ij} = p_{\alpha,i} - p_{\alpha,j} - g \rho_{\alpha,ij} (z_i - z_j) \quad (1.18)$$

$$\rho_{\alpha,ij} = \frac{(\rho_{\alpha,i} + \rho_{\alpha,j})}{2} \quad (1.19)$$

$$U(\alpha, ij) = \begin{cases} i, & \Delta \Phi_{\alpha,ij} \geq 0 \\ j, & \Delta \Phi_{\alpha,ij} < 0 \end{cases} \quad (1.20)$$

where:

V	= cell volume.
Δt	= time step length for the current Euler step.
v_{α}	= volume flux of phase α (oriented quantity).
u_{α}	= surface volume flux of pseudo component α (oriented quantity).
T_{ij}	= transmissibility factor for a connection, derived from permeability.
m_T	= transmissibility multiplier as function of pressure.
$C(i)$	= connections from cell $\bar{i}$, i.e. the set of cells connected to it.
$U(\alpha, ij)$	= upwind cell for phase α for the connection between cells $\bar{i}$ and $\bar{j}$.
$\Delta \Phi_{(\alpha,ij)}$	= potential difference for phase α for the connection between cells $\bar{i}$ and $\bar{j}$ (oriented quantity).
g	= gravitational acceleration in the z-direction.
z_i	= depth of center of cell $\bar{i}$.

For a grid containing a total of n active cells, the material balance relationship expressed by equation (1.9) results in a system of $3n$ equations for the three phase formulation (oil, water, and gas). Similarly, for the two phase model, oil-water or gas-water, there are $2n$ equations to be solved. In addition, there are also the sink and source terms associated with the wells. The solution variables for the system of equations is dependent on the state of a cell, basically, for:

- 1) **Non-miscible:** Here pressure of one of the phases (which we take to be oil pressure P_o), S_w , and S_g as our primary variables. The pressure P_o will then behave mostly in an elliptic fashion, whereas the saturations S_w and S_g are more hyperbolic.
- 2) **Miscible flow:** Since the gaseous phase may disappear if all the gas dissolves into the oleic phase, and similarly the oleic phase may disappear if all the oil vaporizes into the gaseous phase, one cannot always use S_g for our third variable. Instead we use the third variable to track the composition of the phase that has not disappeared. We therefore use either S_g , r_{go} , or r_{og} as our third primary variable, depending on the fluid state, and we call that variable x :

$$x = \begin{cases} S_g, & \text{all three phases present,} \\ r_{go}, & \text{no gaseous phase,} \\ r_{og}, & \text{no oleic phase.} \end{cases} \quad (1.21)$$

This choice is made separately for each cell i depending on its state.

The next two sections describe the formulation of the two well models implemented in OPM flow: the Standard Well Model and the Multi-Segment Well Model.

1.3 STANDARD WELL MODEL

For the standard well model, in a three phase black-oil system, there are four primary variables, three fluid variables: Q_t , F_w , and F_g , which describe the fluid composition within the wellbore¹², as depicted in the following equation:

$$Q_t = \sum_{\alpha \in \{o, g, w\}} g_{\alpha} Q_{\alpha}, \quad F_w = \frac{g_w Q_w}{Q_t}, \quad F_g = \frac{g_g Q_g}{Q_t} \quad (1.22)$$

where:

$$\begin{aligned} Q_t &= \text{the weighted total flow rate,} \\ F_w &= \text{the weighted fraction of water, and} \\ F_g &= \text{the weighted fraction of gas.} \end{aligned}$$

Here, Q_{α} is the flow rate of component α under surface conditions and g_{α} is a weighting factor. For gas, this factor is typically chosen to be a small value, for example, 0.01 to avoid gas fractions close to unity¹².

The fourth variable is the bottom-hole pressure (P_{bhp}), that is the pressure in the wellbore at the datum depth.

Volumetric inflow rates at reservoir conditions are calculated as:

$$q_{\alpha, j}^r = T_{w, j} M_{\alpha, j} [p_j - (p_{bhp, w} + h_{w, j})] \quad (1.23)$$

where:

$$q_{\alpha, j}^r = \text{is the flow rate of phase } \alpha \text{ through connection } j. \text{ The rate is negative for flow from the wellbore to reservoir, and positive for flow into the opposite direction.}$$

¹² J. Holmes, *Enhancements to the strongly coupled, fully implicit well model: wellbore crossflow modeling and collective well control*, in: SPE Reservoir Simulation Symposium, Society of Petroleum Engineers, 1983.

$T_{w,j}$	= connection transmissibility factor.
$M_{\alpha,j}$	= the mobility for phase α at the connection j . For injection wells, the total mobility of the injecting phase is used, as per Holmes12.
P_j	= pressure of the grid block that contains the connection j .
$p_{bhp,w}$	= the bottom-hole pressure of well w .
$h_{w,j}$	= pressure difference within the wellbore between connection j and the well's bottom-hole datum depth. The pressure differences $h_{w,j}$ between connection j and the datum point are computed explicitly based on the fluid composition in the wellbore at the start of the time step.

To keep the system closed, we introduce conservation equations for each component using:

$$R_{\alpha,w} = \frac{A_{\alpha,w} - A_{\alpha,w}^0}{\Delta t} + Q_{\alpha} - \sum_{j \in C(w)} q_{\alpha,j} = 0 \quad (1.24)$$

Equation (1.24) is solved in a fully implicit and coupled way with the black-oil equations, as per equation (1.9)). Here, $C(w)$ is the set of connections of the well w , $q_{\alpha,j}$ is the flow rate of phase α through connection j under surface condition, which can be calculated from $q_{\alpha,j}^r$ defined in equation (1.23); the relationship is similar to the one between component and mass fluxes, as per equations (1.2) through (1.4). The storage term $A_{\alpha,w}$ describes the amount of component α in the wellbore (a small volume, that is 0.1 ft³ as per Holmes12), and it is introduced for better stability of the well solution.

In addition, we need equations that model how the wells are controlled. For wells controlled by a prescribed bottom-hole pressure (*bhp*) target, we have:

$$R_{c,w} = p_{bhp,w} - p_{bhp,w}^{target} = 0 \quad (1.25)$$

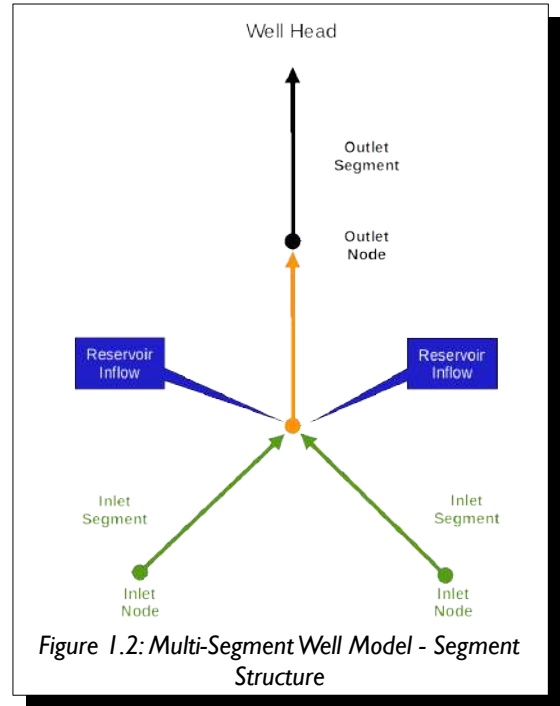
whereas for a rate-controlled well we have:

$$R_{c,w} = Q_{\alpha} - Q_{\alpha}^{target} = 0 \quad (1.26)$$

where Q_{α}^{target} is the desired surface-volume rate of the controlled component α , typically oil or gas rate for a production well. The Standard Well model handles cross-flow, that is flow exiting from one well connection and entering another connection; however, this model assumes that the fluid composition is uniform throughout the wellbore (given by F_g and F_w), and thus the same composition will be injected through all injecting connections. When this is not a reasonable assumption, then the Multi-Segment model should be used instead, which is the topic of the next section.

1.4 MULTI-SEGMENT WELL MODEL

Multi-segment wells were introduced to handle more complex well configurations, including for example horizontal, multi-lateral and fish bone well types, as well Inflow Control Devices (“ICD”)¹³. This model divides the wellbore into a number of arbitrary segments. Each segment has a segment node and a flow path to the neighboring segment in the direction towards the well head, that defines as the outlet segment (Figure 1.2). Most segments also have inlet segment neighbors in the direction away from the well head. For multi-lateral wells, a segment node must be placed at the branch junction. Segments at branch junction points have additional inlet segments and segments at the end of branches do not have inlet segments. The top segment of each well does not have an outlet segment. The pressure at this segment is the bottom-hole pressure of the well, whereas the component rates equal the component rates for the well as a whole. For example, Figure 1.3 on the following page, shows the main wellbore branch together with four offsetting branches.



Each segment has the node pressure p as a primary variable in addition to the primary variables Q_t , F_w , and F_g introduced for the standard well model, as per equation (1.22). Similarly, the component equations, (1.24), are extended to include the additional $Q_{\alpha,i}$ terms that represent the flow rate from the inlet segments $I(n)$ of segment n , as well as the inflow from the reservoir through the segment's connections $C(n)$. Thus, the equivalent standard well equation (1.24) for multi-segment wells is:

$$R_{\alpha,n} = \frac{A_{\alpha,n} - A_{\alpha,n}^0}{\Delta t} + \sum_{i \in I(n)} Q_{\alpha,i} - \sum_{j \in C(w)} q_{\alpha,j} + Q_{\alpha,n} = 0 \quad (1.27)$$

Each segment can thus receive inflow from more than one connection or no connection through the segment node, whereas each connection can only contribute to one segment. In the standard well model, the connection is located at the centroid of the grid block. In the multi-segment well model, the connection and segment node can be located at any place within the grid block. The inflow calculation of the connection in the multi-segment well model is as follows:

$$q_{\alpha,j}^r = T_{w,j} M_{\alpha,j} (p_j + H_{c,j} - p_n - H_{nc}) \quad (1.28)$$

where:

- p_j = the pressure at the cell center of the grid block that contains connection j ,
- $H_{c,j}$ = the hydrostatic pressure difference between the cell center and the connection,
- p_n = the pressure of segment n , and
- H_{nc} = the hydrostatic pressure difference between the connection and the segment.

¹³ J. Holmes, T. Barkve, O. Lund, Application of a multisegment well model to simulate flow in advanced wells, in: European Petroleum Conference, Society of Petroleum Engineers, 1998.

The last equation for each segment describes the pressure relationships between the segment n and its outlet segment m :

$$R_{p,n} = p_n - p_m - H_h - H_f - H_a = 0 \quad (1.29)$$

Here, the H terms represent the hydrostatic, frictional, and acceleration pressure drops between the segments.

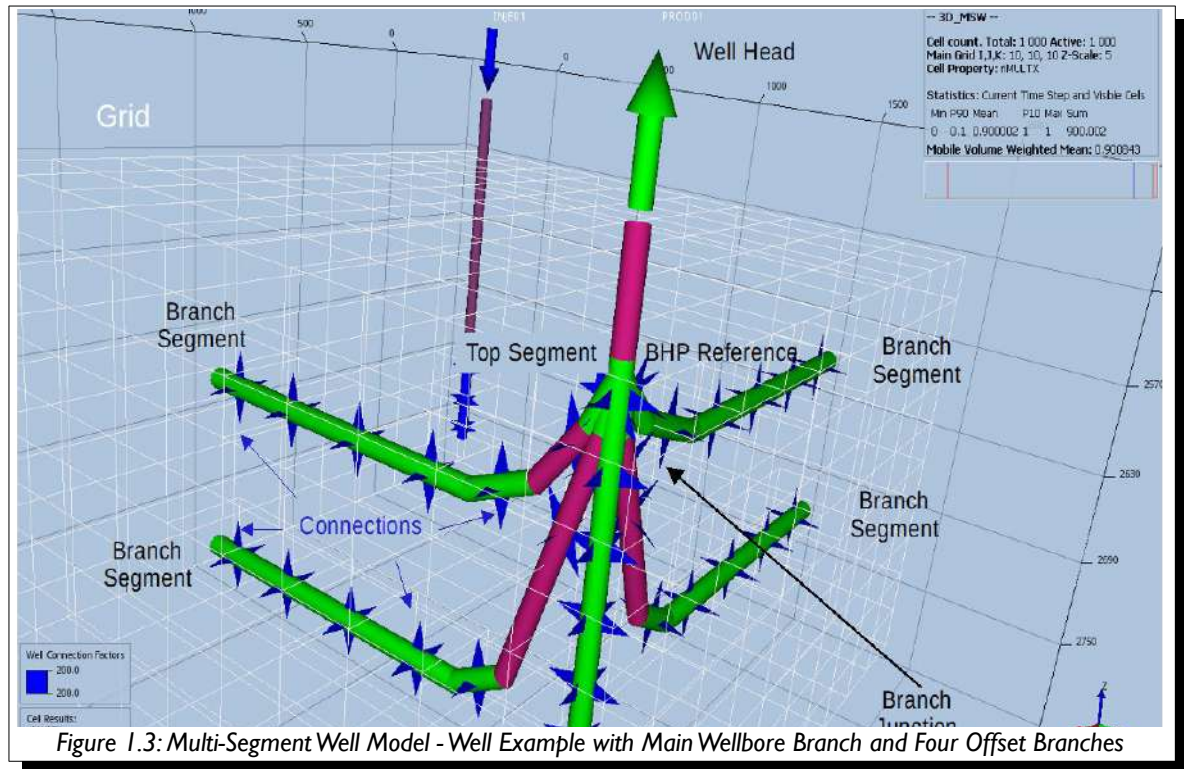
The top segment does not have an outlet segment, and hence the pressure equation is replaced by a well control equation, which is the same as for the standard well model, that is either equation (1.25) or (1.26).

As with the standard well model, the well equations are solved with the reservoir mass conservation equations in a coupled, fully implicit way. The well equations are added to the global system using a Schur¹⁴ complement approach:

$$(A - \sum_w C_w D_w^{-1} B_w) x_r = R_r - \sum_w C_w D_w^{-1} R_w \quad (1.30)$$

$$x_w = D_w^{-1} (R_w - B_w x_r)$$

Figure 1.3 depicts a multi-segment well with the main wellbore branch as the vertical segment, together with four offsetting horizontal branches.



¹⁴ Zhang, Fuzhen (2005). Zhang, Fuzhen (ed.). *The Schur Complement and Its Applications. Numerical Methods and Algorithms. Vol. 4.* Springer. doi:10.1007/b105056. ISBN 0-387-24271-6.

1.5 SALT PRECIPITATION MODEL

OPM Flow's salt precipitation and water evaporation model is an extension of the black-oil model with a water in gas component, plus addition of a solid salt phase component to the salt transport equation. The effects of porosity and permeability reduction effects are also considered. The oil, gas, and water and salt conservation equations including these extensions are given by equations (1.31) to (1.34).

Oil:

$$\frac{\partial}{\partial t} [\phi (b_o s_o + r_{og} b_g s_g)] + \nabla \cdot (b_o v_o + r_{og} b_g v_g) + q_o = 0 \quad (1.31)$$

Gas:

$$\frac{\partial}{\partial t} [\phi (b_g s_g + r_{go} b_o s_o)] + \nabla \cdot (b_g v_g + r_{go} b_o v_o) + q_g = 0 \quad (1.32)$$

Water:

$$\frac{\partial}{\partial t} [\phi (b_w s_w + r_{wg} b_g s_g)] + \nabla \cdot (b_w v_w + r_{wg} b_g v_g) + q_w = 0 \quad (1.33)$$

Salt:

$$\frac{\partial}{\partial t} [\phi b_w s_w c_w^{salt} + m_\phi \phi_0 s_s \rho^{salt}] + \nabla \cdot (c_w^{salt} b_w v_w) + c_w^{salt} q_w = 0 \quad (1.34)$$

where:

- ϕ = porosity.
- m_ϕ = pore volume multiplier as function of pressure.
- ϕ_0 = reference porosity, a constant in time but varying in space.
- p_α = pressure for phase α .
- b_α = shrinkage/expansion factor for phase α defined as the ratio of the surface volume at standard conditions to the reservoir volume for a given amount of fluid: $b_\alpha = V_{surface; \alpha} / V_{reservoir; \alpha}$. For the oil phase, b_o is called the shrinkage factor, whereas for gas b_g is called the expansion factor. The reciprocal quantity is called the formation volume factor, and is usually denoted with a capital B: $B_\alpha = 1 / b_\alpha$. The variable is usually a function of phase pressure and composition.
- r_{go} = gas-oil ratio ("GOR"), ratio of dissolved gas to oil in the oleic phase, commonly called r_s or R_s in other literature.
- r_{og} = oil-gas ratio or condensate-gas ratio ("CGR"), ratio of vaporized oil to gas in the gaseous phase, commonly called r_v or R_v in other literature.
- r_{wg} = evaporated water in gas ratio.
- s_α = saturation of phase α , the pore volume fraction occupied by the phase.
- s_s = saturation of precipitated salt (volume fraction).
- ρ_α = surface density of phase α at one atmosphere, a given constant.
- ρ^{salt} = density of the solid salt.

q_α	= well out flow of component α , (negative for well in-flow). The form of this term depends on the well model used, either the standard well model or the multi-segment model.
C_w^{salt}	= salt concentration in water (mass per unit volume).

The extension has two extra primary variable switching mechanisms. One between S_w and r_{wg} , when water disappears and reappears, and one between C_w^{salt} and S_s when the salt concentration reaches the (user defined) solubility limit and salt precipitates or when solid salt dissolves. Furthermore, in the new formulation porosity becomes:

$$\phi = (1 - s_s) m_\phi \phi_0 \quad (1.35)$$

The resulting (reduced) permeability, k , is based on a user specified permeability-porosity relation $k=k(\phi)$.

Both for dry and humid gas and wet and humid gas the PVT data can be provided using dedicated PVT tables. For the capillary pressure data it is recommended to extend its values towards zero water saturation to permit water saturations below the irreducible water saturation due to water evaporation.

For further details see Machado et al.¹⁵

¹⁵ I.C. G. Machado, P. Egberts, J. Alvestad, O. S. Hustad. Salt Precipitation and Water Evaporation Modelling in a Black-Oil Reservoir Simulator. SPE Reservoir Simulation Conference 2023 (SPE-212257).

1.6 NUMERICAL SOLUTION OF EQUATIONS

Efficiently solving the aforementioned equations is non-trivial, and involves a step iterative process that consists of an outer, Newton-Raphson type method, of a series of iterations, and an inner set of linear iterations, as depicted in Figure 1.4, which shows a simplified schematic of the iterative scheme.

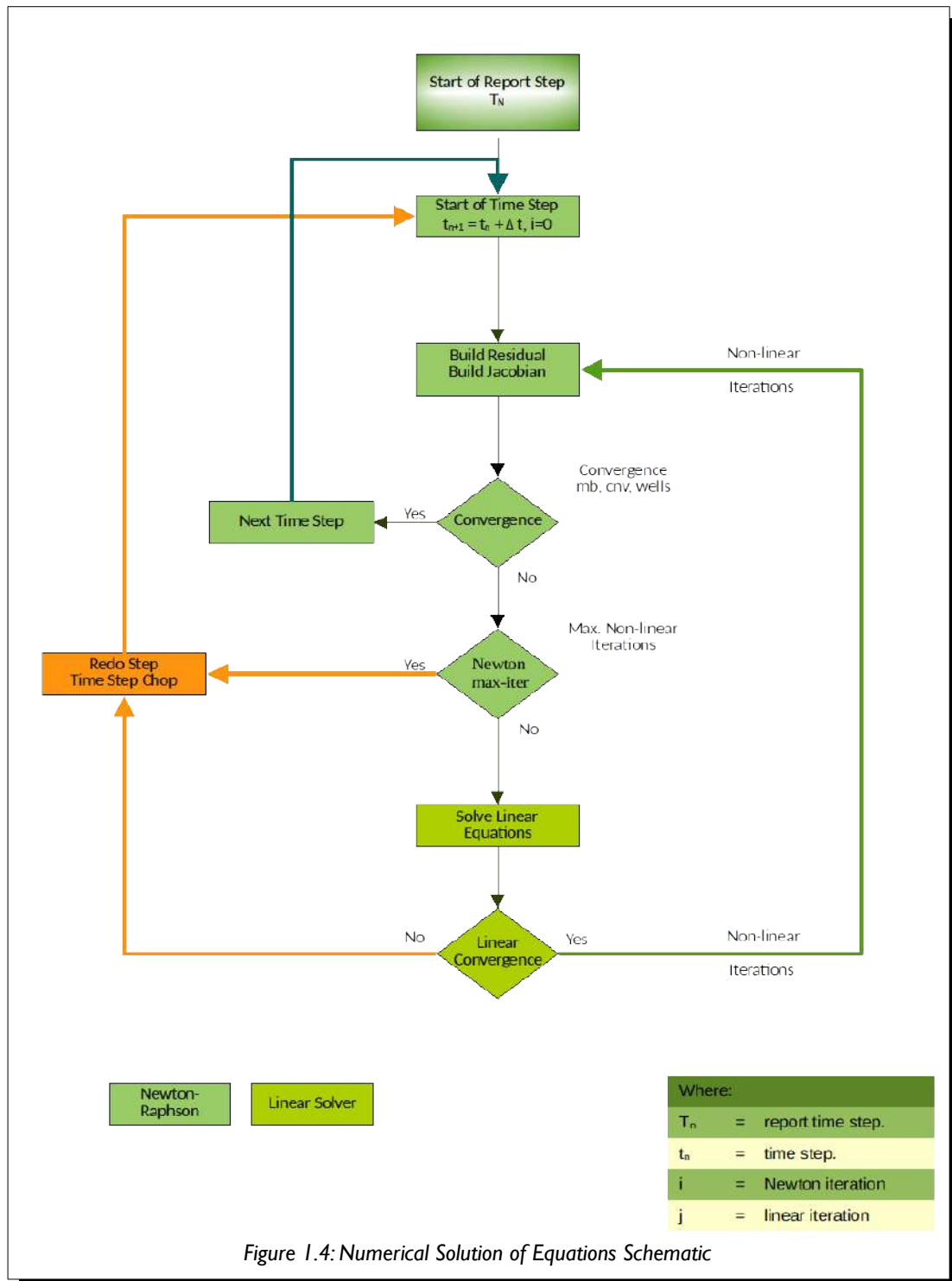


Figure 1.4: Numerical Solution of Equations Schematic

The next two sections attempt to give an overview of the two numerical schemes.

1.6.1 NEWTON-RAPHSON SOLVER (OUTER ITERATION)

The aforementioned reservoir and well equations define a large system of (fully implicit) non-linear equations, which one can write in compact residual form as $R(y) = 0$, where y is the vector of primary variables, p_o , s_w , x . This system is solved using a Newton-Raphson type method.

Let $y_n = (p_o, s_w, x)$ be the primary variables after n Newton iterations, and given the initial state y_0 , the solution of $R(y) = 0$ can be found by iteratively solving:

$$J(y_n)(y_{n+1} - y_n) = -R(y_n) \quad (1.36)$$

until $R(y_n)$ satisfies a given tolerance. Creating the Jacobian matrix $J(y_n)$ and solving this linearized problem is the core computational task of the simulator, and its performance depends largely on how this part of the code is programmed.

The coefficients of the linearized systems required by the non-linear Newton solver, $J(y_n)$ in equation (1.36), have traditionally been computed by evaluating closed-form expressions obtained by differentiating the discretized flow equations analytically. Differentiating these flow equations manually and programming the resulting formulas is both time-consuming and error-prone. The problem is particularly pronounced when extending an existing simulator with new functional relationships. If a quantity is set to depend on a different or new (primary) variable, the changes in derivatives will propagate through the chain rule to the Jacobians of all quantities that depend on the modified quantity. Thus, even very simple extensions in functional dependencies may cause large modifications throughout the simulator code.

Automatic Differentiation (“AD”) is a way to mitigate this problem¹⁶ and ¹⁷, and there are many examples of AD tools used for scientific computing, such as Sacado¹⁸, ADOL-C¹⁹ or ADET²⁰; with the latter used for the AD-GRPS²¹ reservoir simulator at Stanford University.

The OPM Flow’s implementation of AD introduces a class that mimics the behavior of the built-in floating point types of C++ (double and float) as closely as possible. The AD object contains a value and a fixed number of derivatives and defines all basic arithmetic operators as well as common mathematical functions. Also, to allow easy comparisons with code that uses standard floating-point objects, all functions that work on AD-objects can also be used with objects of built-in floating-point types. For further detailed information on OPM Flow’s implementation of AD, again see Rasmussen et al.²².

When solving the Newton-Raphson equation (1.36), the first iteration at the new time step uses the solution at the end of the previous time step (y_n), as an initial estimate for the new time step (y_{n+1}). Thus, the Newton-Raphson iteration is expected to converge rapidly for sufficiently small time steps. For large time steps and/or large changes in the solution, typically caused by changing well controls, the Newton-Raphson method may converge poorly or not at all. There are several means in which one can increase the probability of convergence, namely:

- ¹⁶ R. D. Neidinger, *Introduction to automatic differentiation and matlab object-oriented programming*, SIAM Rev. 52 (3) (2010) 545{563. doi:10.1137/080743627.
- ¹⁷ A. Griewank, A. Walther, *Evaluating Derivatives, Principles and Techniques of Algorithmic Differentiation*, second edition, SIAM, Philadelphia, 2008.
- ¹⁸ Eric T. Phipps, Roscoe A. Bartlett, David M. Gay, Robert J. Hoekstra, *Large-Scale Transient Sensitivity Analysis of a Radiation-Damaged Bipolar Junction Transistor via Automatic Differentiation Advances in Automatic Differentiation*, Springer, 2008.
- ¹⁹ A. Walther, A. Griewank, *Getting started with ADOL-C*, in: U. Naumann, O. Schenk (Eds.), *Combinatorial Scientific Computing*, Chapman-Hall CRC Computational Science, 2012, Ch. 7, pp. 181-202.
- ²⁰ Rami M. Younis, *Modern Advances In Software And Solution Algorithms For Reservoir Simulation*, PhD thesis, Stanford University, (August 2011)
- ²¹ AD-GRPS, Stanford Earth, Stanford University, <https://suetri-b.stanford.edu/research/ad-grps>.
- ²² Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.

- 1) Restrict the change allowed per iteration and prevent the solution from jumping far across the boundary between saturated and undersaturated states in a single iteration. Thus, like the commercial simulator, OPM Flow implements the Appleyard²³ chop technique, which is often sufficient for the method to converge. However, time step chops²⁴ may still occur in order to obtain convergence for some difficult cases, particularly for models with very fine grids.
- 2) Secondly, since the third primary variable, α , may change solution state from one iteration to the next, this may disturb the convergence, and therefore a small threshold is applied (ϵ) to the variable-switching logic for equation (1.21), in order to prevent oscillation between states.
- 3) Thirdly, and frequently the main cause of non-convergence and the lack of smoothness of the residuals, is the discontinuity or inconsistent input data, particularly the relative permeability and capillary pressure functions. Thus, using Corey²⁵ or Lomeland et al.²⁶ type of curves for these data may well improve convergence performance, see [SGOFLET – Gas-Oil LET Relative Permeability Functions](#), [SGWFLET – Gas-Water LET Relative Permeability Functions](#), and [SWOFLET – Water-Oil LET Relative Permeability Functions](#), for OPM Flow's implementation of the Lomeland type of curves.
- 4) The final point that influences the convergence of the non-linear solver is the accuracy of the linear solvers, as discussed in the next section.

For more information about run time options and numerical controls, see section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

²³ J. Appleyard, I. M. Cheshire, Nested factorization, in: 7th SPE Symposium on Reservoir Simulation, San Francisco, USA, Society of Petroleum Engineers, 1983. doi:10.2118/12264-MS.

²⁴ A time step chop occurs when convergence is not achieved within the allowable number of iterations. For example, if the simulator was attempting to take a 30 day time step and failed, then the time step would be cutback, to say ten days, and the iterations restarted. If this 10 day time step also failed to converge, then a second time step chop would occur reducing the new time step, to say three days. This process is repeated until either convergence is achieved, or the time step is too small to chop and the simulation stops.

²⁵ Corey, A.T.: "The Interrelation Between gas and Oil Relative Permeabilities", *Production Mon.*, 19. 38. (1954).

²⁶ Lomeland F., 2018. Overview Of The Let Family Of Versatile Correlations For Flow Functions. Paper SCA2018-056 presented at the International Symposium of the Society of Core Analysts held in Trondheim, Norway, 27-30 August 2018.

1.6.2 LINEAR SOLVER (INNER ITERATION)

As mentioned previously, efficiently solving the aforementioned equations is non-trivial, and involves a step iterative process that consists of an outer, Newton-Raphson type method, series of iterations, and an inner set of linear iterations, as depicted in the detailed work flow of the iterative process in Figure I.5.

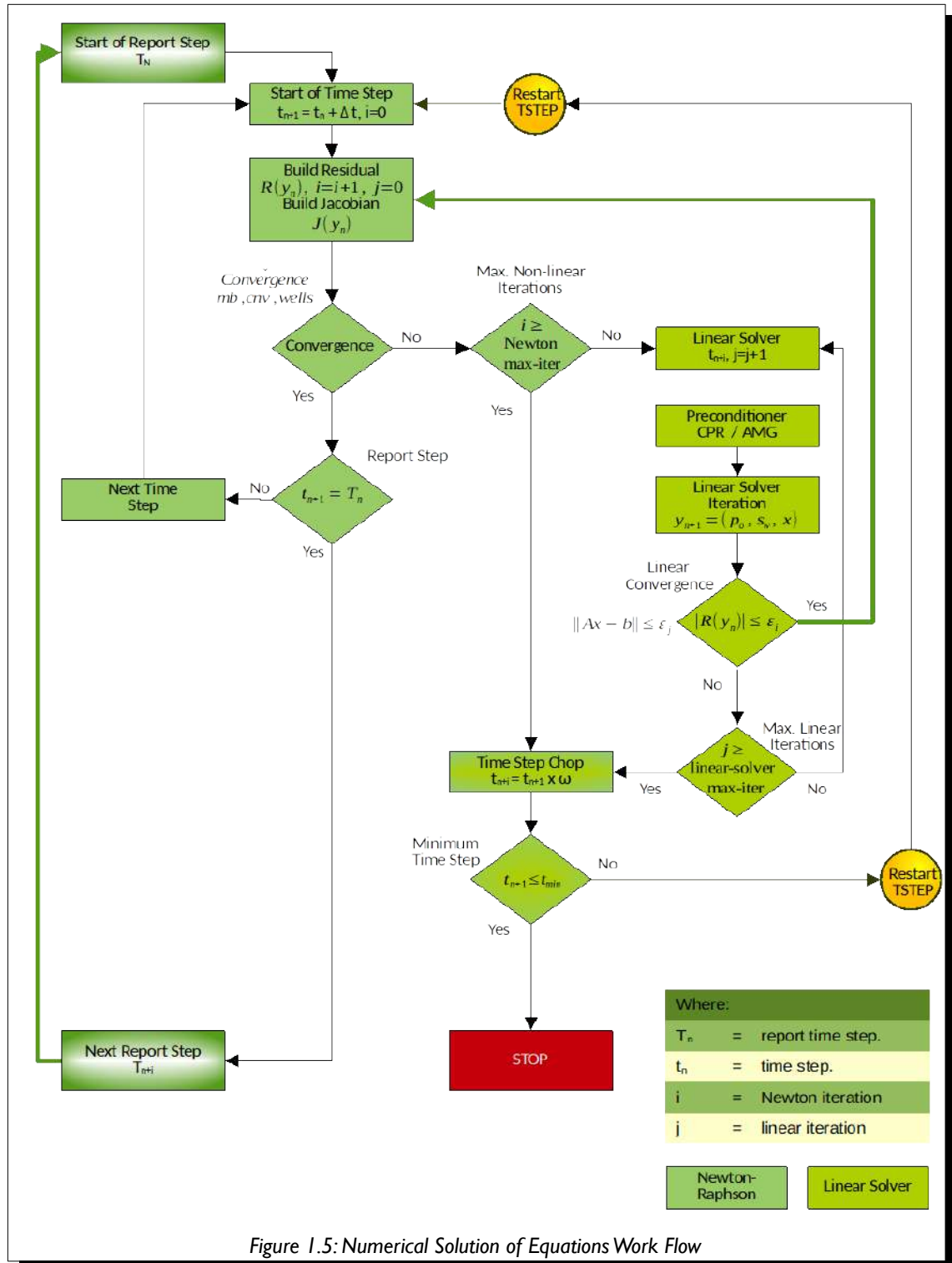


Figure I.5: Numerical Solution of Equations Work Flow

Depending on the fluid properties and other reservoir specific parameters, the system of equations as described by equation (1.9), that is:

$$R_{\alpha,i} = \frac{\phi_{ref,i} V_i}{\Delta t} (A_{\alpha,i} - A_{\alpha,i}^0) + \sum_{j \in C(i)} u_{\alpha,ij} + q_{\alpha,i} = 0 \quad (1.9)$$

could exhibit elliptic or degenerate parabolic behavior, or could even be hyperbolic for specific setups. Therefore, solving the linearized system stemming from these equations is challenging, since it is likely non-symmetric and ill-conditioned. Thus, instead of solving equation (1.36), that is:

$$J(y_n)(y_{n+1} - y_n) = -R(y_n) \quad ((1.36))$$

directly, the solution is approximated by an iterative linear solver. Two variants are available in OPM Flow:

- 1) a stabilized Bi-conjugate Gradient method (BiCG-stab), and
- 2) a restarted Generalized Minimal Residual (GMRes) solver.

By default, BiCG-stab is used. In addition, there are several options available for preconditioning the linear system, such as the incomplete lower triangular-upper triangular factorization with level of fill in 0 (ILU0). The default option is based on an algebraic multi-grid (AMG) method for preconditioning, named the Constraint Pressure Residual (CPR) preconditioner (originally introduced by Wallis²⁷ using the approach of Scheichl et al.²⁸). The solvers and preconditioners are implemented in the Dune module dune-istl and described by Blatt et al.^{29, 30}, and ³¹. The interface between OPM Flow and the linear solver packages has been designed in a flexible way so that besides dune-istl other linear algebra packages could be used.

For more information about run time options and numerical controls, see section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface. In addition, section [2.4 Improving Simulator Convergence and Numerical Performance](#) provides some practical advice on how to address numerical performance issues.

²⁷ J. Wallis, *Incomplete Gaussian Elimination as a Preconditioning for Generalized Conjugate Gradient Acceleration*, in: *7th SPE Symposium on Reservoir Simulation*, San Francisco, USA, Society of Petroleum Engineers, 1983. doi:10.2118/12265-MS.

²⁸ R. Scheichl, M. Roland, J. Wendebourg, *Decoupling and block preconditioning for sedimentary basin simulations*, *Computational Geosciences* 7 (2003) 295-318.

²⁹ M. Blatt, *A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients*, Ph.D. thesis, Ruprecht-Karls-Universität at Heidelberg (2010).

³⁰ M. Blatt, P. Bastian, *The Iterative Solver Template Library*, in: B. Kågström, E. Elmroth, J. Dongarra, J. Waśniewski (Eds.), *Applied Parallel Computing. State of the Art in Scientific Computing*, Vol. 4699 of *Lecture Notes in Computer Science*, Springer, 2007, pp. 666-675.

³¹ M. Blatt, P. Bastian, *On the generic parallelisation of iterative solvers for the finite element method*, *Int. J. Comput. Sci. Engrg.* 4 (1) (2008) 56-69. doi:10.1504/IJCSE.2008.021112.

1.7 FORMATION DAMAGE MODELS

1.7.1 INCOMPRESSIBLE EXTERNAL FILTER CAKE MODEL – LINEAR GEOMETRY

This simple model assumes that all of the filtrate is captured on the formation surface to form a filter cake of constant porosity and permeability. Hence, the filter cake will simply grow in thickness as a function of the volume of deposited material, and a simple volume balance assuming linear flow yields,

$$A_f l_c(t)(1-\phi_c) = f_s \int_0^t q(\tau) d\tau = f_s Q_c(t) \quad (1.36)$$

Here A_f is the area of flow, l_c is the filter cake thickness, ϕ_c is the filter cake porosity, f_s is the volume concentration of suspended material, q is the injection rate, Q_c is the cumulative injected volume, and t is time. The filter cake thickness is hence,

$$l_c(t) = \frac{f_s Q_c(t)}{A_f(1-\phi_c)}. \quad (1.36)$$

Assuming a constant filter cake permeability k_c , and viscosity μ , Darcy's law for the pressure drop across the filtrate gives the damage pressure drop ΔP_d as,

$$\Delta P_d(t) = \frac{q \mu f_s Q_c(t)}{k_c A_f^2 (1-\phi_c)} \quad (1.36)$$

and the skin factor as,

$$S = \frac{f_s}{1-\phi_c} \frac{k}{k_c} \frac{Q_c(t)}{A_f r_w} = \frac{l_c}{r_w} \frac{k}{k_c} \quad (1.36)$$

Here k is the rock permeability. The linear flow assumption is valid for flow near a fracture surface, and it is also a decent approximation for radial flow assuming that the filter cake is thin compared to the well radius.

1.7.2 INCOMPRESSIBLE EXTERNAL FILTER CAKE MODEL – RADIAL GEOMETRY

Deposition Inside the Wellbore

Simple volume balance on deposited filtrate inside a well in radial geometry yields, for a 2D cross section perpendicular to the well,

$$\pi r_w^2 - \pi r_c^2 = \frac{f_s}{1-\phi_c} \int_0^t q_h(t) dt = \frac{f_s}{1-\phi_c} Q_h(t) \quad (1.36)$$

Here q_h and Q_h are rate and cumulative injected volume per length, respectively. This may be solved for the filtrate inner radius r_c to compute the cake thickness $r_w - r_c$ as,

$$\Delta r_c(t) = r_w - r_c = r_w - \sqrt{r_w^2 - \frac{f_s Q_h(t)}{\pi(1-\phi_c)}} \quad (1.36)$$

The pressure drop across the filter cake is then given as,

$$\Delta P_c = \frac{q\mu}{2\pi k_c} \ln \frac{r_c}{r_w} \quad (1.36)$$

The skin is simply,

$$S = \frac{k}{k_c} \ln \frac{r_w}{r_c} \quad (1.36)$$

Deposition in the Reservoir

If the filtrate is instead deposited in the near-well reservoir (i.e., outside the well), we will have $r_c > r_w$, and the skin factor will be,

$$S = \frac{k}{k_c} \ln \frac{r_c}{r_w} \quad (1.36)$$

For implementation convenience and consistency with the linear geometry case the filter cake thickness may be expressed in terms of a connection area, $A_f = 2\pi r_w h$, so that with Q being the cumulative injected volume, the filter cake radius follows from,

$$r_c^2 = r_w^2 + \frac{2r_w f_s Q(t)}{A_f(1-\phi_c)} \quad (1.36)$$

CHAPTER 2: INSTALLING AND RUNNING FLOW

2.1 INSTALLING OPM FLOW

OPM Flow can be installed and used in a variety of ways. Perhaps the simplest way is to use the operating system's package manager to install the OPM modules as binary packages, including OPM Flow and other OPM programs. That way all prerequisite libraries will automatically be installed on the system. This is supported on Ubuntu Linux version 22.04 or 24.04 and Red Hat Enterprise (or CentOS) version 7 or 8.

OPM Flow can also be used via a Docker container. This allows OPM Flow to run on any system that supports Docker, including Microsoft Windows, without installing prerequisite libraries or affecting any other software on your system.

Finally, OPM Flow can be installed by compiling from source on Linux or macOS systems.

2.1.1 UBUNTU LINUX 22.04 OR 24.04 (64-BIT VERSION ONLY)

The easiest way to install the Ubuntu packages is to first add the OPM personal package archive (ppa). In order to do that we need to install the apt-add-repository command.

```
sudo apt-get update  
sudo apt-get install software-properties-common
```

Then we add the repository, and run update again:

```
sudo apt-add-repository ppa:opm/ppa  
sudo apt-get update
```

At this point, all the OPM modules should be available to install. To see a list of (for example) the opm-simulators packages:

```
apt-cache search opm-simulators
```

Then to install the opm-simulators programs (including OPM Flow) and their dependencies:

```
sudo apt-get install mpi-default-bin  
sudo apt-get install libopm-simulators-bin python3-opm-common
```

To optionally install the OPM Python bindings use:

```
sudo apt-get install python3-opm-simulators
```

Notes

- 1) The mpi install above is required for the mpi libraries which the OPM executables are linked to, even those that are not intended to be run with mpirun. The reason the OPM packages do not depend on that package is that by Debian policy the user should be allowed to choose which MPI implementation to use (above we picked the default set by Ubuntu).
- 2) If an old versions of prerequisite libraries have already been installed (for example from installing a previous release of OPM Flow) it might be necessary to upgrade them via the following commands:

```
sudo apt-get update
```

```
sudo apt-get upgrade
```

- 3) Some users have experienced trouble with OPM Flow simply aborting with no error message. In some cases that can be related to locale issues. Check by running the “locale” command. OPM Flow requires the “C” or an English locale to run correctly. One way to fix this can be to put the following in your “.bash_profile” (note the initial period or full stop in the filename), and open a new terminal (command) window:

```
LANG="en_US.UTF-8"

export LANG

LC_ALL="en_US.UTF-8"

export LC_ALL
```

2.1.2 RED HAT ENTERPRISE OR CENTOS (VERSION 7 OR 8)

First add the OPM package repository:

```
sudo yum-config-manager --add-repo \
http://www.opm-project.org/package/opm.repo
```

The OPM software is split in several packages. To list all available OPM packages one can use:

```
sudo yum search opm-
```

For example you can install the opm-simulators binary package to get access to the OPM Flow reservoir simulator:

```
sudo yum install opm-simulators-bin python3-opm-common
```

Development packages are available using the -devel suffix. For example, to install the opm-upscaling development package use:

```
sudo yum install opm-upscaling-devel
```

Notes

- 1) If you wish to install the MPI parallel version of Flow, you must choose an MPI implementation, either mpich or openmpi, and instead use the following command:

```
sudo yum install opm-simulators-openmpi-bin
```

- 2) For mpich, just replace openmpi with mpich in the above command. In order to run Flow in parallel using mpirun you must also load the required module first:

```
module load mpi/openmpi-x86_64
```

To use mpich instead, replace openmpi with mpich in the above command.

2.1.3 WINDOWS 7 AND 10 USING VIRTUALBOX

Oracle VM VirtualBox's (formerly Sun VirtualBox, Sun xVM VirtualBox and Innotek VirtualBox) is a free and open source hosted hypervisor for x86 virtualization, developed by Oracle Corporation. The software was

originally created by Innotek, who were acquired by Sun Microsystems in 2008, which was in turn acquired by Oracle in 2010. VirtualBox can be installed on Windows, macOS, Linux, Solaris and Open Solaris operating systems. The software supports the creation and management of guest virtual machines running Windows, Linux, BSD, OS/2, Solaris, Haiku, and OSx86, as well as limited virtualization of macOS guests on Apple hardware. For some guest operating systems, a "Guest Additions" package of device drivers and system applications is available, which typically improves performance, especially that of graphics³².

The OPM web site outlines the install process using VirtualBox, Vagrant³³ and a Vagrant file to automatically set up a virtual environment – see https://opm-project.org/?page_id=294&page=2 for details on this approach.

Alternatively, one can use the more standard approach of installing virtual machines using VirtualBox. The basic outline for this method to use VirtualBox to run a guest Linux distribution and install OPM Flow in the guest Linux virtual machine. Note only the Linux distributions mentioned previously should be used as a guest virtual machine, although any of the Ubuntu "flavors" (Kubuntu, Lubuntu, Ubuntu Mate, etc.) will work as well. The steps are:

1) Installing VirtualBox³⁴

- First download the latest VirtualBox software from Oracle's web site at <https://www.virtualbox.org/> for Windows host machines.
- Install the software via double clicking on the executable and following the on screen instructions.

2) Create a Guest Linux Virtual Machine.

- Download a suitable Linux distribution in ISO format that supports running OPM Flow, for example Ubuntu 24.04 LTS (64-bit version only) from <https://ubuntu.com/download/desktop> or CentOS Linux (version 7 or 8) from <https://www.centos.org/download/>.
- Start VirtualBox, which should display a screen similar to Figure 2.1 without all the guest virtual machines.

³² After <https://en.wikipedia.org/wiki/VirtualBox>.

³³ See <https://www.vagrantup.com/> for details

³⁴ See also <https://www.virtualbox.org/wiki/Documentation> for further and more detailed information.

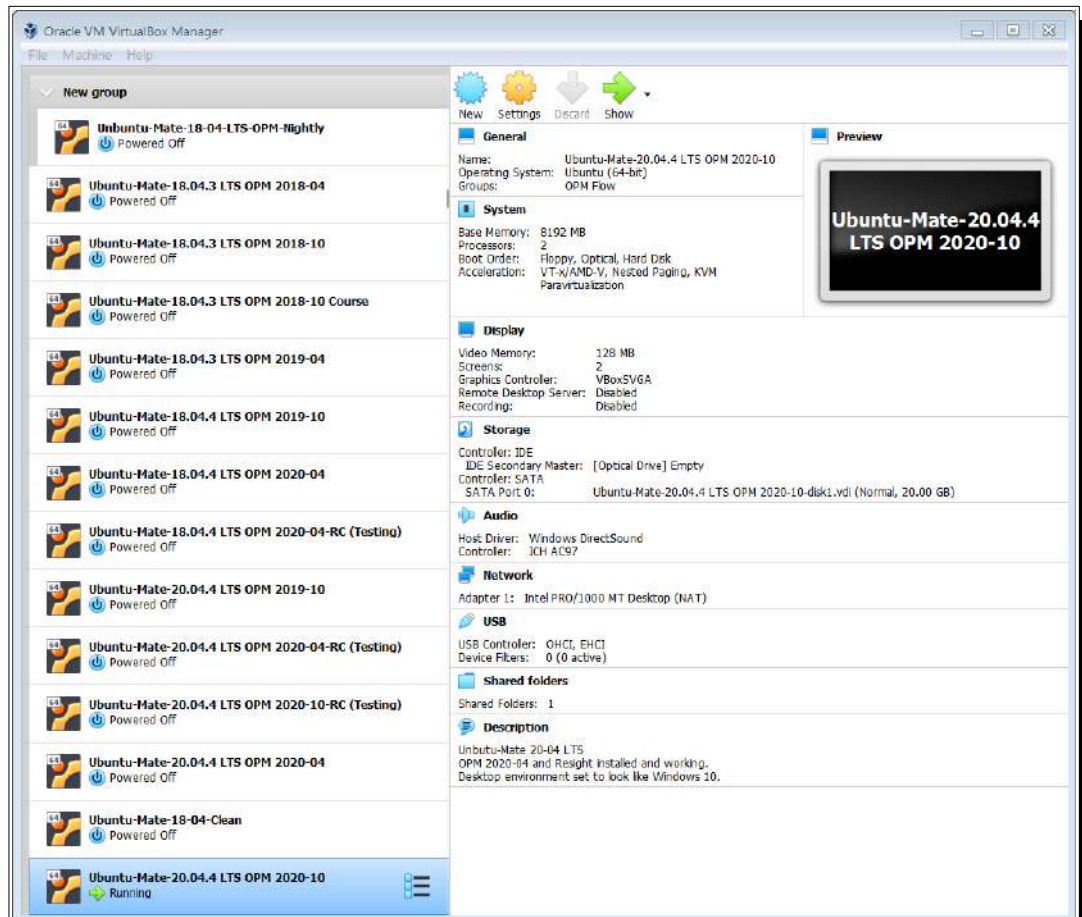


Figure 2.1: VirtualBox Main Screen Showing Various Guest Virtual Machines

- Then click the "New" icon button to install a new guest virtual machine.
- Identify the operating system type (Linux) and the Linux distribution that one wishes to install (Ubuntu 64-bit or RedHat for one of the CentOS distributions).
- Set the amount of RAM, should be 4GB or more; note the guest machine properties can subsequently be changed after the guest machine has been created.
- Create a virtual hard drive, the default virtual disk of VDI is normally used, and select either the dynamic or the fixed option. Next select the size of the virtual disk, the size should be sufficient to allow for the operating system and any other software, including OPM Flow, to be installed. For Ubuntu 24.04 LTS the operating system needs at least 10 GB. VirtualBox will then create the guest virtual machine.
- The next step is to start the guest virtual machine and installing the Linux operating system. This can be done by double clicking on the guest virtual machine and then selecting the previously downloaded IOS file. For Ubuntu 20.04 LTS the machine will boot into a live distribution, that is the guest virtual machine is running from a "live CD", shown below.

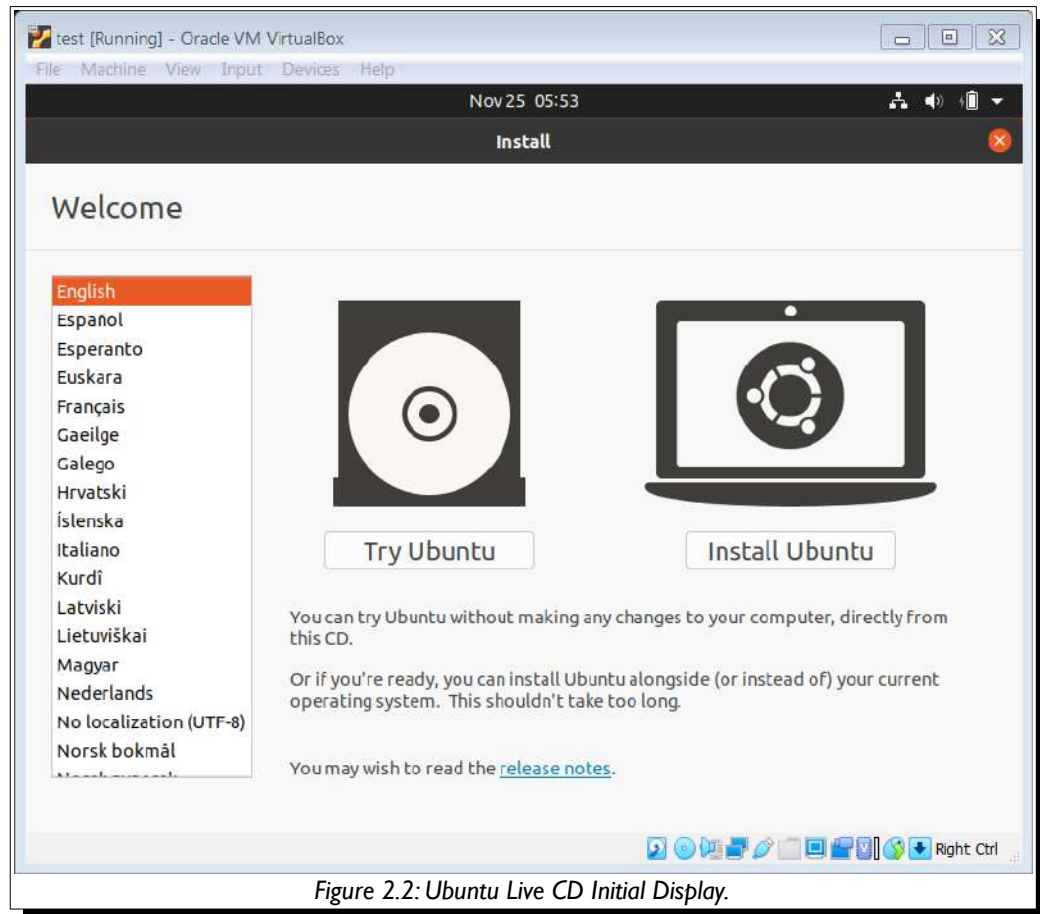


Figure 2.2: Ubuntu Live CD Initial Display.

- Select the “Install Ubuntu” option to install the operating system. The “Try Ubuntu” option means the guest machine will run from a “Live CD”.
- Follow the on screen instructions to install the operating system, together with any updates suggested by the installer. Once the installation is complete the guest machine will re-boot as a normal guest Linux machine.

3) Installing OPM Flow in the Guest Virtual Machine.

- Boot up your virtual machine by selecting the machine and selecting the “Start” icon on the main VirtualBox screen (Figure 2.1). One should then be presented with the newly created guest virtual machine, similar to one shown in Figure 2.3, that displays the standard Ubuntu desktop.

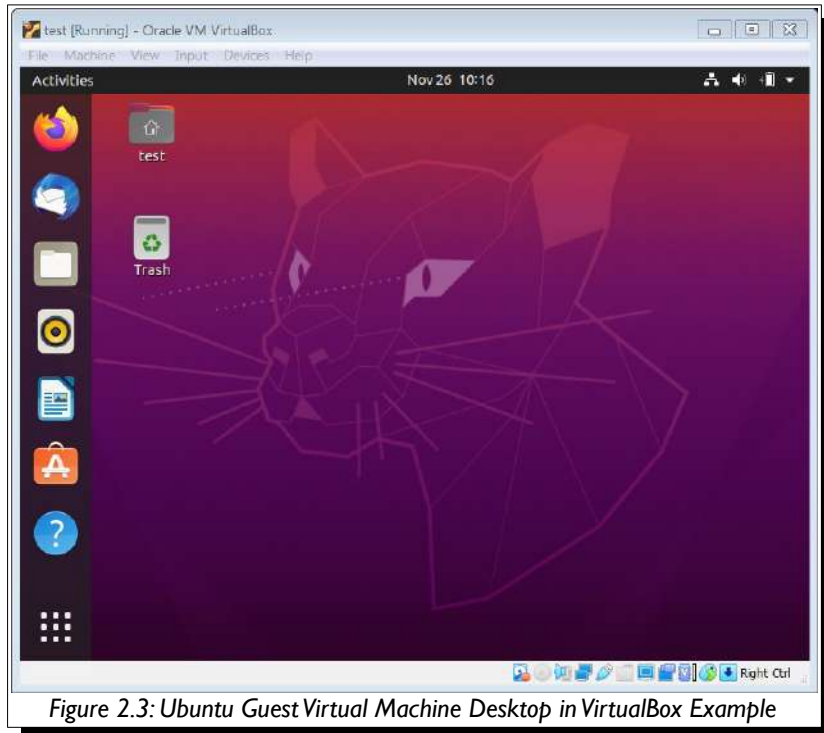


Figure 2.3: Ubuntu Guest Virtual Machine Desktop in VirtualBox Example

- In order to install OPM Flow in the guest virtual machine one first needs to open a command terminal using ALT+CNTRL+T key sequence (Figure 2.4).

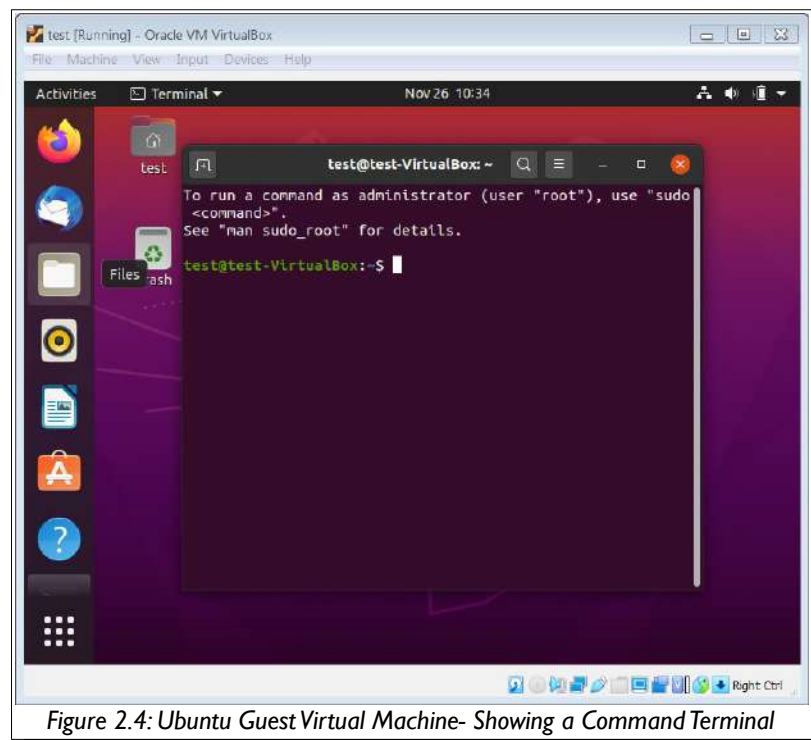


Figure 2.4: Ubuntu Guest Virtual Machine- Showing a Command Terminal

- One can then use the same commands as for a host machine, as described earlier in Sections [2.1.1 Ubuntu Linux 22.04 or 24.04 \(64-bit version only\)](#) and [2.1.2 Red Hat Enterprise or CentOS \(version 7 or 8\)](#) to install the software. To check if OPM Flow is installed, execute the following command in the terminal window:

```
flow --help
```

which should then display the command line parameters for the installed version of OPM Flow, as depicted in Figure 2.5

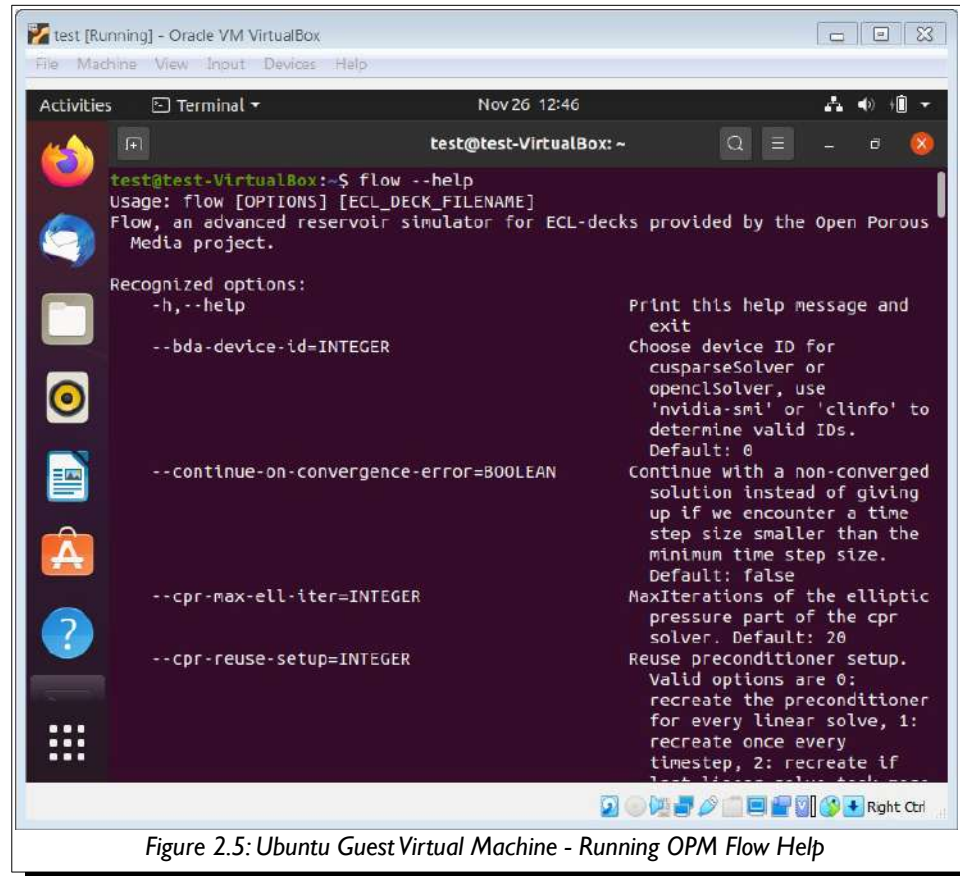


Figure 2.5: Ubuntu Guest Virtual Machine - Running OPM Flow Help

4) Using OPM Flow in a Linux Virtual Machine.

- The guest virtual machine only has access to files within the guest machine. To enable the guest to have access to the host machine files; on the main VirtualBox screen (Figure 2.1), select the “Setting” icon and then the “Shared Folders” option to set a share directory between the guest and the host machines. One can then have all the input and output files on the host machine and run OPM Flow in the guest virtual machine.
- Note one should shut down the guest virtual machine in the normal way to avoid potential file corruptions.
- One may also take snapshots of the current state of the virtual machine, as a form of backup if one changes the configuration of the virtual machine, or if one updates to the latest release of OPM Flow.

Historically this has been one of the preferred methods to run OPM Flow on Windows machines and still has several advantages, including being able to export and import various guest virtual machines for testing various versions of the simulator. However, Microsoft, with the introduction of Windows Subsystem for Linux (“WSL”) has create a more seamless method for running OPM Flow on Windows 10 and higher machines. This option is outlined in the next section.

2.1.4 WINDOWS 10 AND 11 – WINDOWS SUBSYSTEM FOR LINUX (WSL)

The first release of WSL was on August 2nd, 2016 and there has been various updates to this version which is known as WSL 1. On May 6th, 2019 WSL 2 was announced, that was a complete re-write of the Windows Subsystem for Linux, and offers significant performance improvements over WSL 1. Note that WSL 2 requires Windows 10 version 1903 or higher, with Build 18362 or higher, for x64 systems. WSL 2 is based on a virtual machine with an actual Linux Kernel that can immediately react to system calls. Such an architecture enables Full System call capabilities.

To run OPM Flow under WSL 1 or WSL 2 one has to first activate WSL, then install a Linux distribution from the Application store, and then finally install OPM Flow. The detail instruction are as follows:

- 1) Activating Windows Subsystem for Linux.
 - On the Windows machine, Click Start and navigate to Control Panel, then Select the Programs Category. Under Programs and Features, click on Turn Windows Features on or off and click the check box next to the Windows Subsystem for Linux option, to enable WSL.

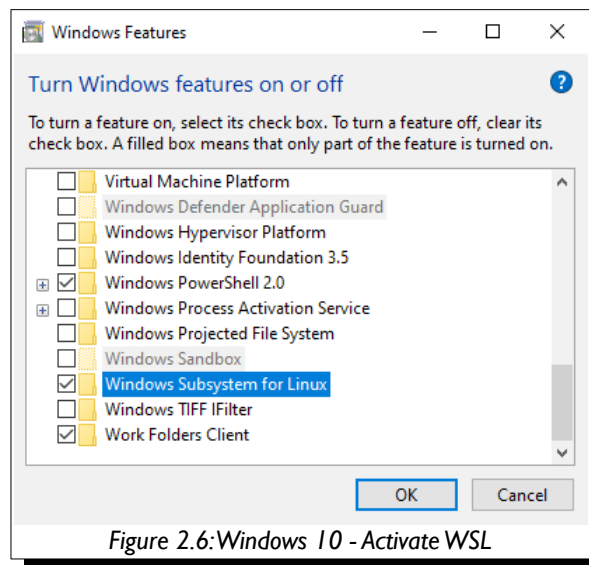


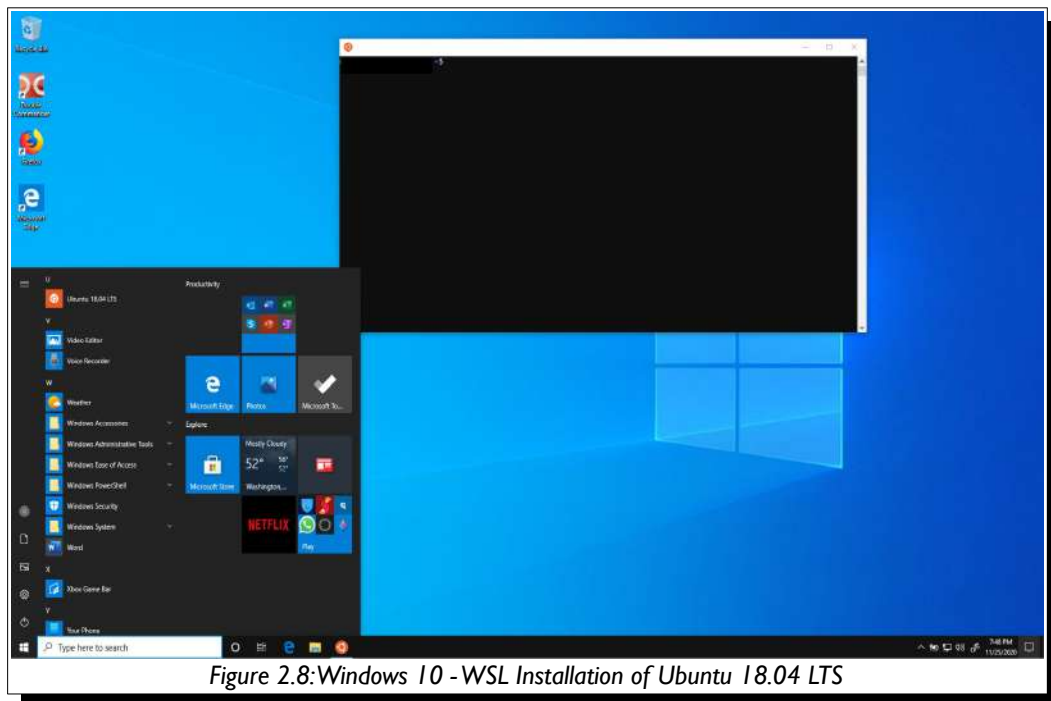
Figure 2.6: Windows 10 - Activate WSL

You may also have to ensure that CPU virtualization has been activated in your computer's BIOS.

- 2) Installing a Linux Distribution.
 - To install the Linux distribution in Windows, click Start and then select the Microsoft Store.
 - Using the search box available in the Microsoft Store, pick one of the distributions that is compatible with OPM Flow (Figure 2.7). Note that for Ubuntu these are usually the two latest LTS versions before the release, i.e. version "20XX.04 LTS" where XX is a multiple of 4.



- After selecting the distribution, click Install or Get to install the Linux Distribution. Note the installer will ask you to sign into your account, but you can close the pop-up window if you don't want to sign in at this time.
- On completion of the installation process the Linux distribution will be installed on Windows and ready for use. Furthermore, the distribution will be available directly in the Start Menu itself in the form of a standard everyday application (Figure 2.8).



- When starting the Linux Distribution for the first time, a brief setup process will take place. Basically, it will ask for the username and password you want to set for the Linux environment, a similar process as when you install a Linux distribution via VirtualBox.
- To check the version of WSL that will be used to run the Linux distribution, in PowerShell use the following command:

```
wsl -l -v
```

- To set the WSL for a given distribution, for example Ubuntu-24-04 to WSL 2, in PowerShell use the following command:

```
wsl --set-version Ubuntu-24-04 2
```

- And finally to set the default version for WSL for all Linux distributions, use the following PowerShell command:

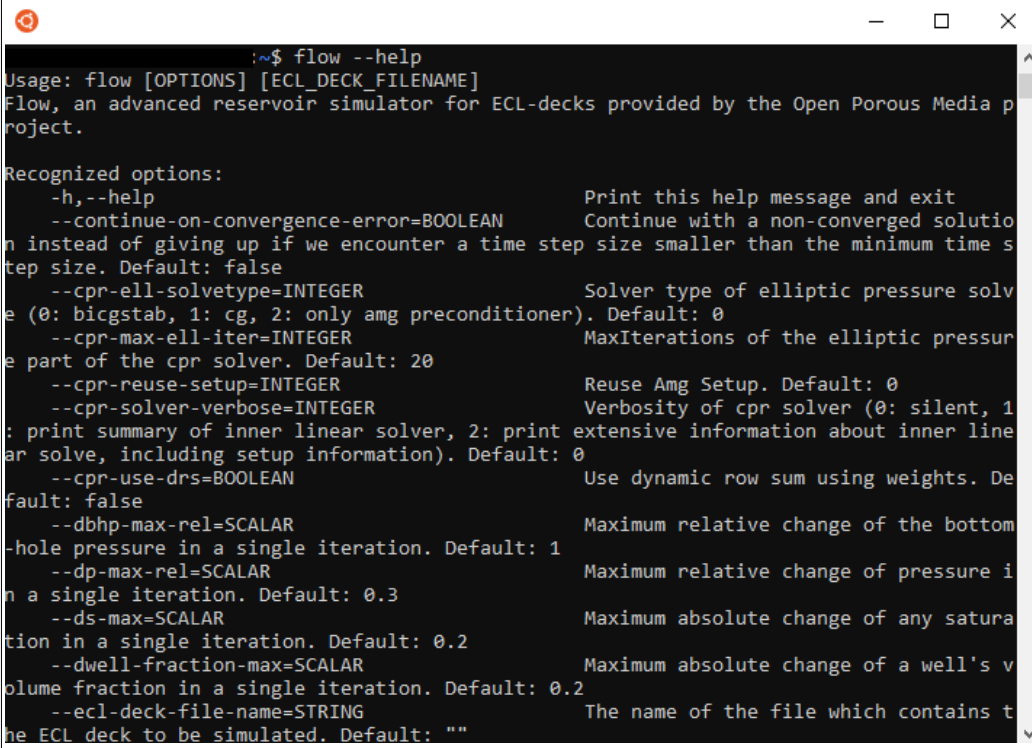
```
wsl --set-default-version 2
```

3) Installing OPM Flow in a WSL Linux Distribution:

- Click on installed Linux distribution in the Start Menu to open a Linux terminal session, and then execute the Linux commands as described earlier in Sections [2.1.1 Ubuntu Linux 22.04 or 24.04 \(64-bit version only\)](#) and [2.1.2 Red Hat Enterprise or CentOS \(version 7 or 8\)](#) to install the software.
- To check if OPM Flow is installed, execute the following command in the Linux terminal window:

```
flow --help
```

which should then display the command line parameters for the installed version of OPM Flow, as depicted in Figure 2.9.



```

~$ flow --help
Usage: flow [OPTIONS] [ECL_DECK_FILENAME]
Flow, an advanced reservoir simulator for ECL-decks provided by the Open Porous Media project.

Recognized options:
  -h, --help                                Print this help message and exit
  --continue-on-convergence-error=BOOLEAN   Continue with a non-converged solution instead of giving up if we encounter a time step size smaller than the minimum time step size. Default: false
  --cpr-ell-solver-type=INTEGER              Solver type of elliptic pressure solver (0: bicgstab, 1: cg, 2: only amg preconditioner). Default: 0
  --cpr-max-ell-iter=INTEGER                 MaxIterations of the elliptic pressure part of the cpr solver. Default: 20
  --cpr-reuse-setup=INTEGER                  Reuse Amg Setup. Default: 0
  --cpr-solver-verbose=INTEGER               Verbosity of cpr solver (0: silent, 1: print summary of inner linear solver, 2: print extensive information about inner linear solve, including setup information). Default: 0
  --cpr-use-drs=BOOLEAN                     Use dynamic row sum using weights. Default: false
  --dbhp-max-rel=SCALAR                     Maximum relative change of the bottom-hole pressure in a single iteration. Default: 1
  --dp-max-rel=SCALAR                       Maximum relative change of pressure in a single iteration. Default: 0.3
  --ds-max=SCALAR                           Maximum absolute change of any saturation in a single iteration. Default: 0.2
  --dwell-fraction-max=SCALAR               Maximum absolute change of a well's volume fraction in a single iteration. Default: 0.2
  --ecl-deck-file-name=STRING               The name of the file which contains the ECL deck to be simulated. Default: ""

```

Figure 2.9: Windows 10 WSL - Running OPM Flow Help

One can also use the Windows command terminal, instead of the WSL Linux Terminal, to do the same thing by adding WSL character string before the command, that is:

```
wsl flow --help
```

4) Using OPM Flow via the Windows WSL Linux Virtual Machine.

- Similar to running OPM Flow using a Linux distribution installed in VirtualBox, the WSL Linux virtual machine only has access to files within WSL virtual machine. To access the Windows file

system, one has to execute the Linux CD command in the WSL Linux Terminal, the command has the form:

```
cd /mnt/<drive letter>
```

For example, to access the Windows system drive C one would use the following command:

```
cd /mnt/c
```

One can then run OPM Flow from the desired directory in the WSL Linux virtual machine and use OPM ResInsight under Windows to view the results of the simulation run.

User experience indicates this approach of running OPM Flow under Windows is the most integrated and convenient method, compared to the VirtualBox approach, especially for the less technically advance users.

2.1.5 USING A DOCKER CONTAINER

See the tutorial on the OPM website [Running Flow in Docker](#).

2.1.6 INSTALLING FROM SOURCE

See instructions on the OPM website [Building from Source](#).

2.2 RUNNING OPM FLOW FROM THE COMMAND LINE

This section describes how to run the OPM Flow simulator and the various command line options associated with this release, the command line syntax for running OPM Flow is:

```
flow [OPTIONS] CASENAME
```

and typing the following command from your terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA. Alternatively, you can type the path to the binary and the *.DATA file, as per the example below:

```
path_to_binary/flow path_to_data/CASENAME
```

The simulator can also be run using command line parameters for example to run the previous CASENAME one could use:

```
flow --ecl-deck-file-name=path_to_data/CASENAME
```

Note that there must be no spaces around the equals sign and there should be no spaces in CASENAME. If there are spaces in CASENAME, which can typically occur on Windows based systems, then the CASENAME should be enclosed in quotes; thus instead of CASE NAME use "CASE NAME".

By default output files are generated in the same folder as the *.DATA file. To override this one can use the --output-dir command line parameter, for instance:

```
flow --output-dir=foo CASENAME
```

will send the output files to the foo directory.

It is also possible to put multiple command line options together in a parameter file. To do so, put one option on each line of the file with ".param" as the extension and pass that filename as a command line parameter to OPM Flow using:

```
flow --parameter-file=CASENAME.param
```

to instruct OPM Flow to read the parameter file. Note, it is good practice to name both the *.DATA and parameter file with the same CASENAME, to avoid confusion. However, if one wishes to use the same parameter file for various cases one can specify the parameter file followed by the DATA file, for example:

```
flow --parameter-file=path_to_params/ALLCASES.param CASENAME.DATA
```

If OPM Flow is installed with parallel capabilities then:

```
mpirun -np 4 flow --parameter-file=path_to_data/CASENAME.param
```

will run OPM Flow on four nodes etc.

Note

On Red Hat based distributions, including CentOS distributions, one needs to install the openmpi version of the simulator, opm-simulators-openmpi-bin, via:

```
sudo yum install opm-simulators-openmpi-bin
```

and then set the MPI version to use. The way this is done is by first making the module command available, by running the following command:

```
./usr/share/Modules/init/bash
```

Then query for the installed modules with:

```
module avail
```

And lastly, you tell the system it to use openmpi with:

```
module add mpi/openmpi-x86_64
```

On Ubuntu based distributions, there is no need for this as openmpi is installed and mpirun just works.

A complete list of the active user facing command line options and their function is given in the Table 2.1 for the current release. These commands are the options listed with command line --help option, there is also a --help-all option that lists all the command line options included in the release, including experimental, obsolete, hidden and deprecated options.

Additional tutorials for running OPM Flow are available on the OPM website in the [Tutorials](#) section.

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
1	-h or --help	A character string. Print the help message that gives a brief description of the available command line parameters.	N/A
2	--help-all	Prints all the command line options included in the release, including obsolete, hidden and deprecated options.	N/A
3	--accelerator-mode	A character string. Choose a linear solver, usage: '--accelerator-mode=[none cusparse opencl amgcl rocalution rocsparse]'.	"none"
4	--action-parsing-strictness	A character string. Set strictness of parsing process for ActionX and PyAction. Available options are normal (do not apply keywords that have not been tested for ActionX or PyAction) and low (try to apply all keywords, beware: the simulation outcome might be incorrect).	"normal"
5	--add-corners	A boolean value. Add corners to partition.	false
6	--allow-distributed-wells	A boolean value. Allow the perforations of a well to be distributed to interior of multiple processes.	false
7	--allow-splitting-inactive-wells	A boolean value. Allow inactive (never non-shut) wells to be split across multiple domains.	true

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
8	--alternative-well-rate-init	A boolean value. Use alternative well rate initialization procedure.	true
9	--check-group-constraints-inner-well-iterations	A boolean value. Allow checking of group constraints during inner well iterations.	true
10	--check-satfunc-consistency	Saturation function consistency checking is now enabled by default. A boolean value. Whether or not to check saturation function consistency requirements.	true
11	--continue-on-convergence-error	A boolean value. Continue with a non-converged solution instead of giving up if we encounter a time step size smaller than the minimum time step size.	false
12	--convergence-monitoring	A boolean value. Enable convergence monitoring.	false
13	--convergence-monitoring-cut-off	An integer value. Cut off limit for convergence monitoring.	6
14	--convergence-monitoring-decay-factor	A positive real value. Decay factor for convergence monitoring.	0.75
15	--cpr-reuse-interval	An integer value. Reuse preconditioner interval. Used when CprReuseSetup is set to 4, then the preconditioner will be fully recreated instead of reused every N linear solve, where N is this parameter.	30
16	--cpr-reuse-setup	A positive integer. Specify if the CPR solver should re-use the AMG setup. Valid options are: 0 : Recreate the preconditioner for every linear solve, 1 : Recreate once every time step, 2 : Recreate if last linear solve took more than 10 iterations, 3 : Never recreate, or 4 : Recreate every N linear solves, where N is the parameter defined by --cpr-reuse-interval.	4
17	--cpr-weights-thread-parallel	A boolean value. Enable OpenMP thread parallelization of CPR weight calculation. This can improve performance for large models but is disabled by default. Usage: --cpr-weights-thread-parallel=[true false].	false
18	--dbhp-max-rel	A positive real value. Maximum relative change of the bottom-hole pressure in a single iteration.	1
19	--debug-emit-cell-partition	A boolean value. Whether or not to emit cell partitions as a debugging aid.	false

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
20	--debug-verbosity-level	An integer value. Set debug verbosity level globally. Default is 1, increasing values give additional output and 0 disables most messages to the .DBG file.	1
21	--dp-max-rel	A positive real value. Maximum relative change of pressure in a single iteration.	0.3
22	--ds-max	A positive real value. Maximum absolute change of any saturation in a single iteration.	0.2
23	--dwell-fraction-max	A positive real value. Maximum absolute change of a well's volume fraction in a single iteration.	0.2
24	--ecl-deck-file-name	A character string. The name of the file which contains the ECL deck to be simulated.	""
25	--ecl-output-double-precision	A boolean value. Tell the output writer to use double precision. Useful for 'perfect' restarts.	false
26	--ecl-output-interval	An integer value. The number of report steps that ought to be skipped between two writes of ECL results.	-1
27	--edge-conformal	A boolean value. Edge conformal cornerpoint processing.	false
28	--edge-weights-method	This option is now specified by a string rather than an integer. A defined character string. Specify the edge-weighting strategy: 1) "uniform", 2) "transmissibility", or 3) "logtrans" for the logarithm of transmissibility.	"transmissibility"
29	--enable-adaptive-time-stepping	A boolean value. Use adaptive time stepping between report steps.	true
30	--enable-async-ecl-output	A boolean value. Write the ECL-formatted results in a non-blocking way (i.e., using a separate thread).	true
31	--enable-async-vtk-output	A boolean value. Dispatch a separate thread to write the VTK output.	true
32	--enable-drift-compensation	A boolean value. Enable partial compensation of systematic mass losses via the source term of the next time step.	false
33	--enable-dry-run	A boolean value. Specify if the simulation should actually run (true), or just check the input deck (false).	"auto"

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		This option is equivalent to activating the NOSIM keyword in the RUNSPEC section of the input deck. The default value of "auto" will use whatever is stipulated in the input deck via the NOSIM keyword.	
34	--enable-ecl-output	A boolean value. Write binary output which is compatible with the commercial Eclipse simulator.	true
35	--enable-esmry	A boolean value. Write ESMRY file for fast loading of summary data.	false
36	--enable-gravity	A boolean value. Use the gravity correction for the pressure gradients.	true
37	--enable-grid-adaptation	A boolean value. Enable adaptive grid refinement/coarsening.	false
38	--enable-intensive-quantity-cache	A boolean value. Turn on caching of intensive quantities.	true
39	--enable-logging-fallout-warning	A boolean value. Developer option to see whether logging was on non-root processors. In that case it will be appended to the *.DBG or *.PRT files.	false
40	--enable-opm-rst-file	A boolean value. Include OPM-specific keywords in the ECL restart file to enable restart of OPM simulators from these files.	false
41	--enable-storage-cache	A boolean value. Store previous storage terms and avoid re-calculating them.	true
42	--enable-terminal-output	A boolean value. Print high-level information about the simulation's progress to the terminal.	true
43	--enable-thermodynamic-hints	A boolean value. Enable thermodynamic hints.	false
44	--enable-tuning	A boolean value. Read the time stepping and convergence parameters from the TUNING keyword in the input deck, if set to true. Note that only some of the TUNING keyword parameters are processed.	false
45	--enable-vtk-output	A boolean value. Global switch for turning on writing VTK files.	false
46	--enable-well-operability-check	A boolean value. Enable the well operability checking.	true
47	--enable-well-operability-check-iter	A boolean value. Enable the well operability checking during iterations.	false

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
48	--enable-write-all-solutions	A boolean value. Write all solutions to disk instead of only the ones for the report steps.	false
49	--end-time	A positive real value. The simulation time at which the simulation is finished [s].	1e+100
50	--explicit-rock-compaction	A boolean value. Use pressure from end of the last time step when evaluating rock compaction.	false
51	--external-partition	A character string. Name of file from which to load an externally generated partitioning of the model's active cells for MPI distribution purposes. If empty, the built-in partitioning method will be employed.	""
52	--force-disable-fluid-in-place-output	A boolean value. Do not print fluid-in-place values after each report step even if requested by the deck. Note this parameter will override the print request in the input deck.	false
53	--force-disable-resv-fluid-in-place-output	A boolean value. Do not print reservoir volumes values after each report step even if requested by the deck. Note this parameter will override the print request in the input deck.	false
54	--full-time-step-initially	A boolean value. Always attempt to finish a report step using a single substep.	false
55	--gpu-aware-mpi	A boolean value. MPI communication use GPU aware MPI in the sense that it will use GPU direct communication. Setting this to true will require that the MPI implementation supports GPU direct communication. If you are unsure, set this to false. Usage: --gpu-aware-mpi=[true false].	false
56	--gpu-device-id	An integer value. Choose device ID for cusparseSolver or openclSolver, use 'nvidia-smi' or 'clinfo' to determine valid IDs.	0
57	--hy-ne-config-file	A character string. Use config files for Hybrid Newton.	"hybridNewtonConfig.json"
58	--ignore-keywords	A character string. List of Eclipse keywords which should be ignored. As a ':' separated string.	""
59	--ilu-fillin-level	An integer value. The fill-in level of the linear solver's ILU preconditioner.	0
60	--ilu-redblack	A boolean value. Use red-black partitioning for the ILU preconditioner.	false

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
61	--ilu-relaxation	A real value ranging from zero to one. The relaxation factor of the linear solver's ILU preconditioner.	0.9
62	--ilu-reorder-spheres	A boolean value. Whether to reorder the entries of the matrix in the red-black ILU preconditioner in spheres starting at an edge. If false the original ordering is preserved in each color. Otherwise why try to ensure D4 ordering (in a 2D structured grid, the diagonal elements are consecutive).	false
63	--imbalance-tol	A positive real value. Tolerable imbalance of the loadbalancing.	1.1
64	--initial-time-step-in-days	A positive real value. The size of the initial time step in days.	1
65	--initial-time-step-size	A positive real value. The size of the initial time step [s].	86400
66	--inj-mult-damp-mult	A real value ranging from zero to one. Injection multiplier dampening factor (dampening multiplied by this each time oscillation is detected).	0.9
67	--inj-mult-min-damp-factor	A real value ranging from zero to one. Minimum injection multiplier dampening factor (maximum dampening level).	0.05
68	--inj-mult-osc-threshold	A positive real value. Injection multiplier oscillation threshold (used for multiplier dampening).	0.1
69	--input-skip-mode	A defined character string. Set compatibility mode for the <i>SKIP100</i> and <i>SKIP300</i> keywords. Options are: 1) 100: skip <i>SKIP100</i>...<i>ENDSKIP</i> and keep <i>SKIP300</i>..<i>ENDSKIP</i>, 2) 300: skip <i>SKIP300</i>...<i>ENDSKIP</i> and keep <i>SKIP100</i>..<i>ENDSKIP</i>, or 3) all: skip both <i>SKIP100</i>...<i>ENDSKIP</i> and <i>SKIP300</i>...<i>ENDSKIP</i>.	"100"
70	--linear-solver	A defined character string. Specify the configuration of the linear solver, valid values are: 1) cprw^{35 36 37} (default),	"cprw"

³⁵ Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

³⁶ R. Scheichl, M. Roland, J. Wendebourg, Decoupling and block preconditioning for sedimentary basin simulations, *Computational Geosciences* 7 (2003) 295-318.

³⁷ Klemetsdal, Ø.S., Møyner, O. & Lie, K.A. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. *Comput Geosci* 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		2) ilu0, 3) dilu, 4) cpr (an alias for cprw), 5) cpr_quasiimpes, 6) cpr_trueimpes, 7) cpr_trueimpesanalytic, 8) amg³⁸, 9) hybrid (experimental), or 10) alternatively the name of a JSON file that contains the linear solver configuration parameters. Option (1) extends the existing Constrained Pressure Residual ("CPR") preconditioner to include wells. This option can also be invoked via the <i>CPR</i> keyword in the <i>RUNSPEC</i> section; however, the command line parameter takes precedence. For option (10) one enters a character string enclosed in quotes that defines the filename of a JSON configuration file for the flexible linear solver system. In this case the file extension should be '.json'. Note that the Print (PRT) file contains the "Property tree for the linear solver" listing, which is the JSON specification of the current case, and can be used to configure a user specific linear solver JSON file.	
71	--linear-solver-accelerator	A character string. Choose the backend for the linear solver, usage: '--linear-solver-accelerator=[cpu gpu]'.	"cpu"
72	--linear-solver-ignore-convergence-failure	A boolean value. Continue with the simulation like nothing happened after the linear solver did not converge. The default value for "CPR" solvers is 20.	false
73	--linear-solver-max-iter	An integer value. The maximum number of iterations of the linear solver.	200
74	--linear-solver-print-json-definition	A boolean value. Write the JSON definition of the linear solver setup to the DBG file. The default value for "CPR" solvers is 0.005.	true
75	--linear-solver-reduction	A positive real value. The minimum reduction of the residual which the linear solver must achieve.	0.01
76	--linear-solver-restart	An integer value. The number of iterations after which GMRES is restarted.	40

³⁸ M. Blatt, A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients, Ph.D. thesis, Ruprecht-Karls-Universität Heidelberg (2010).

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
77	--linear-solver-verbosity	An integer value. Specify the output from the linear solver: 0 : No extra output, 1 : output per solution iteration, or 2 : output per iteration.	0
78	--load-file	A character string. FileName for .OPMRST file used to load serialized state. If empty, CASENAME.OPMRST is used.	""
79	--load-step	An integer value. Specify if the simulator should load the serialized state from OPM Flow's version of the restart file, and should be set to either a specific report step, or 0 to load the last stored report step. The default value of -1 does not load the data from the OPM Flow specific restart file. OPM Flow's version of the restart file, is written using the --save-step=N option.	-1
80	--local-domains-ordering-measure	A character string. Subdomain ordering measure. Allowed values are 'maxpressure', 'averagepressure' and 'residual'.	"maxpressure"
81	--local-domains-partition-well-neighbor-levels	An integer value. Number of neighbor levels around wells to include in the same domain during NLDD partitioning.	1
82	--local-domains-partitioning-imbalance	A real value greater than one. Subdomain partitioning imbalance tolerance. 1.03 is 3 percent imbalance.	1.03
83	--local-domains-partitioning-method	A character string. Subdomain partitioning method. Allowed values are 'zoltan', 'simple', and the name of a partition file ending with '.partition'.	"zoltan"
84	--local-solve-approach	A character string. Choose local solve approach. Valid choices are jacobi and gauss-seidel.	"gauss-seidel"
85	--local-tolerance-scaling-cnv	A positive real value. Set lower than 1.0 to use stricter convergence tolerance for local solves.	0.1
86	--local-tolerance-scaling-mb	A positive real value. Set lower than 1.0 to use stricter convergence tolerance for local solves.	1
87	--local-well-solve-control-switching	A boolean value. Allow control switching during local well solutions.	true
88	--matrix-add-well-contributions	A boolean value. Explicitly specify the influences of wells between cells in the Jacobian and preconditioner matrices.	false

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
89	--max-inner-iter-ms-wells	An integer value. Maximum number of inner iterations for multi-segment wells.	100
90	--max-inner-iter-wells	An integer value. Maximum number of inner iterations for standard wells.	50
91	--max-local-solve-iterations	An integer value. Max iterations for local solves with NLDD nonlinear solver.	20
92	--max-newton-iterations-with-inner-well-iterations	An integer value. Maximum newton iterations with inner well iterations.	8
93	--max-pressure-change-ms-wells	A positive real value. Maximum relative pressure change for a single iteration of the multi-segment well model.	1e+06
94	--max-residual-allowed	A positive real value. Absolute maximum tolerated for residuals without cutting the time step size.	1e+07
95	--max-single-precision-days	A positive real value. Maximum time step size where single precision floating point arithmetic can be used solving for the linear systems of equations.	20
96	--max-temperature-change	A positive real value. Maximum absolute change of temperature in a single iteration.	5
97	--max-time-step-divisions	An integer value. The maximum number of divisions by two of the timestep size before the simulation bails out.	10
98	--max-time-step-size	A positive real value. The maximum size to which all time steps are limited to [s].	inf
99	--max-well-status-switch-for-wells	An integer value. Maximum number of status switching (stop<->open) for a well during a time-step.	99
100	--max-well-status-switch-in-inner-iter-wells	An integer value. Maximum number of status switching (stop<->open) for a well during inner iterations.	99
101	--max-welleq-iter	An integer value. Maximum number of iterations to determine solution the well equations.	30
102	--maximum-number-of-group-switches	An integer value. Maximum number of times a group can switch to the same control.	3
103	--maximum-number-of-well-switches	An integer value. Maximum number of times a well can switch to the same control.	3
104	--maximum-water-saturation	A positive real value. Maximum water saturation.	1
105	--metis-params	A character string. Configuration of Metis	"default"

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		partitioner. You can request a configuration to be read from a JSON file by giving the filename here, with extension '.json'. See http://glaros.dtc.umn.edu/gkhome/fetch/sw/metis/manual.pdf for available METIS options.	
106	--milu-variant	Option strings have been converted to lower case. A defined character string. Specify which variant of the modified ILU preconditioner ought to be used. Possible variants are: 1) ilu: plain ILU (default), 2) milu_1: lump diagonal with dropped row entries, 3) milu_2: lump diagonal with the sum of the absolute values of the dropped row entries, 4) milu_3: if diagonal is positive add sum of dropped row entries. Otherwise subtract them, 5) milu_4: if diagonal is positive add sum of dropped row entries. Otherwise do nothing.	"ilu"
107	--min-strict-cnv-iter	An integer value. Minimum number of Newton iterations before relaxed tolerances can be used for the CNV convergence criterion.	-1
108	--min-strict-mb-iter	An integer value. Minimum number of Newton iterations before relaxed tolerances can be used for the MB convergence criterion. Default -1 means that the relaxed tolerance is used when maximum number of Newton iterations are reached.	-1
109	--min-time-step-based-on-newton-iterations	A real value greater than or equal to zero. The minimum time step size (in days for field and metric unit and hours for lab unit) can be reduced to based on newton iteration counts.	0
110	--min-time-step-before-shutting-problematic-wells-in-days	A real value greater than or equal to zero. The minimum time step size in days for which problematic wells are not shut.	0.01
111	--min-time-step-size	A real value greater than or equal to zero. The minimum size to which all time steps are limited to [s].	0
112	--network-max-outer-iterations	An integer value. Maximum outer number of iterations in the network solver before giving up.	10

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
I 13	--network-max-pressure-update-in-bars	A positive real value. Maximum pressure update in the inner network pressure update iterations.	5
I 14	--network-max-strict-outer-iterations	An integer value. Maximum outer iterations in network solver before relaxing tolerance.	10
I 15	--network-max-sub-iterations	An integer value. Maximum number of sub-iterations to update network pressures (within a single well/group control update).	20
I 16	--network-pressure-update-damping-factor	A real value ranging from zero to one. Damping factor in the inner network pressure update iterations.	0.1
I 17	--newton-max-error	A positive real value. The maximum error tolerated by the Newton method to which does not cause an abort.	1e+100
I 18	--newton-max-iterations	An integer value. The maximum number of Newton iterations per time step.	20
I 19	--newton-max-relax	A positive real value. The maximum relaxation factor of a Newton iteration.	0.5
I 20	--newton-min-iterations	An integer value. The minimum number of Newton iterations per time step.	2
I 21	--newton-relaxation-type	A character string. The type of relaxation used by the Newton method. Valid options are: dampen or sor.	"dampen"
I 22	--newton-target-iterations	An integer value. The 'optimum' number of Newton iterations per time step.	10
I 23	--newton-tolerance	A positive real value. The maximum raw error tolerated by the Newton method for considering a solution to be converged.	0.01
I 24	--newton-verbose	A boolean value. Specify whether the Newton method should inform the user about its progress or not.	true
I 25	--newton-write-convergence	A boolean value. Write the convergence behaviour of the Newton method to a VTK file.	false
I 26	--nlld-local-linear-solver	A character string. Configuration of NLDD local linear solver. Valid options are: ilu0 (default), dilu, cpr_quasiimpes and amg. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json'.	"ilu0"
I 27	--nlld-local-linear-solver-max-iter	An integer value. The maximum number of iterations of the NLDD local linear solver.	200
I 28	--nlld-local-linear-solver-reduction	A positive real value. The minimum	0.01

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		reduction of the residual which the NLDD local linear solver must achieve.	
129	--nldd-num-initial-newton-iter	An integer value. Number of initial global Newton iterations when running the NLDD nonlinear solver.	1
130	--nldd-relative-mobility-change-tol	A positive real value. Threshold for single cell relative mobility change in the NLDD solver.	0.1
131	--nonlinear-solver	A character string. Choose nonlinear solver. Valid choices are newton or nldd.	"newton"
132	--num-local-domains	An integer value. Number of local domains for NLDD nonlinear solver.	0
133	--num-overlap	An integer value. Numbers of layers overlap in parallel partition.	1
134	--num-pressure-points-equil	An integer value. Number of pressure points (in each direction) in tables used for equilibration.	2000
135	--num-satfunc-consistency-sample-points	An integer value. Maximum number of reported failures for each individual saturation function consistency check.	5
136	--nupcol-group-rate-tolerance	A positive real value. Tolerance for acceptable changes in VREP/REIN group rates.	0.001
137	--opencl-ilu-parallel	A boolean value. Parallelize ILU decomposition and application on GPU.	true
138	--opencl-platform-id	An integer value. Choose platform ID for openclSolver, use 'clinfo' to determine valid platform IDs.	0
139	--output-dir	A character string. The directory to which result files are written.	""
140	--output-extra-convergence-info	A defined character string. Specify whether to provide additional convergence output to separate files for diagnostic purposes. The available options are: 1) "none" results in no extra output and overrides all other options. 2) "steps" writes out convergence information per time step, to a CASENAME.INFOSTEP file. The file is useful for identifying numerical issues. 3) "iterations" writes out non-linear convergence metrics, i.e., the MB and CNV values, per phase, for each non-linear iteration in each time step, to a CASENAME.INFOITER file. Options can be combined with commas,	"none"

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		e.g."steps,iterations" for multiple outputs. The default value of "none" prevents the two files from being written out, for better compatibility with the commercial simulator.	
141	--output-interval	An integer value. Specify the number of report steps between two consecutive writes of restart data.	1
142	--output-mode	A defined character string. Specify the output to the *.PRT and *.DBG files: 1) "none": No output to the files. 2) "log" or false: Output logging information only. 3) "all" or true: Output everything. For example to just output logging information use: <code>--output-mode="log"</code> or <code>--output-mode=false</code>	"all"
143	--owner-cells-first	A boolean value. Order cells owned by rank before ghost/overlap cells.	true
144	--parameter-file	A character string. An .ini file which contains a set of run-time parameters.	""
145	--parsing-strictness	A character string. Set strictness of parsing process. Available options are: 1) "normal": stop for critical errors, and for unsupported keywords that would change the simulator results if supported. 2) "high": stop for all errors, that is even for unsupported keywords that do not effect the results, for example <i>ECHO</i> and <i>NOECHO</i> . 3) "low": same as normal, except do not stop due to unsupported keywords that would change the simulator results if supported, and even if marked critical.	"normal"
146	--partition-method	Options are now specified by strings rather than integers. A character string. Choose partitioning strategy: 1) "simple", 2) "zoltan", 3) "metis", 4) "zoltanwell" Zoltan with all cells	"zoltanwell"

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		perforated by a well represented by a single vertex.	
147	--pre-solve-network	A boolean value. Pre solve and iterate the network model at start-up.	true
148	--predetermined-time-steps-file	A character string. A file with a list of predetermined time step sizes (one time step per line).	""
149	--pressure-max	A positive real value. Maximum absolute pressure.	1e+99
150	--pressure-min	A real value. Minimum absolute pressure.	-1e+99
151	--pressure-scale	A positive real value. Scaling of pressure primary variable.	1
152	--pri-var-oscilation-threshold	A positive real value. The threshold value for the primary variable switching conditions after its meaning has switched to hinder oscillations.	1e-05
153	--print-parameters	An integer value. Specify the run time parameters to be printed at the start of the run: 0 : No output to the files. 1 : Output to *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
154	--project-saturations	A boolean value. Option for doing saturation projection.	false
155	--regularization-factor-wells	A positive real value. Regularization factor for wells.	100
156	--relaxed-linear-solver-reduction	A positive real value. The minimum reduction of the residual which the linear solver need to achieve for the solution to be accepted.	0.01
157	--relaxed-max-pv-fraction	A positive real value. The fraction of the pore volume of the reservoir where the volumetric error (CNV) may be violated during strict Newton iterations.	0.03
158	--relaxed-pressure-tol-msw	A positive real value. Relaxed tolerance for the MSW pressure solution.	10000
159	--relaxed-well-flow-tol	A positive real value. Relaxed tolerance for the well flow residual.	0.001
160	--restart-time	A real value. The simulation time at which a restart should be attempted [s].	-1e+35
161	--restart-writing-interval	An integer value. The frequencies of which time steps are serialized to disk.	16777215

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
162	--save-file	A character string. FileName for .OPMRST file used for saving serialized state. If empty, CASENAME.OPMRST is used.	""
163	--save-step	A character string. Specify if the simulator should save the serialized state of the OPM Flow simulator at one or more report steps to a special *.OPMRST file. This is in addition to the normal restart files written, and consumes significantly more space than the normal restart files, but restarting OPM Flow from this file using the <code>--load-step</code> option deviates less from the original run, compared to restarting using the normal restart features. The files produced are not compatible with other simulators, and also will not be compatible between different releases of OPM Flow. The parameter should be set to one of the following: 1) "all" to save all report steps, 2) "last" to save every step, keeping only the last. 3) "x" to save every x'th step, or 4) "x" to save a specific time step. Use negative values of "x" to keep only the last written step. The default value of "" does not write anything to the OPM Flow specific restart file.	""
164	--scale-linear-system	A boolean value. Scale linear system according to equation scale and primary variable types.	false
165	--sched-restart	A boolean value. Determine for a restart run if the case should initialize wells and groups from the historical SCHEDULE section (true), or from the well and group data on the restart file (false). Note that the commercial simulator always uses data from the restart file (false).	false
166	--serial-partitioning	A boolean value. Perform partitioning for parallel runs on a single process.	false
167	--shut-unsolvable-wells	A boolean value. Shut unsolvable wells.	true
168	--slave	A boolean value. Specify if the simulation is a slave simulation in a master-slave simulation. Note that reservoir coupling is not currently supported. This is a developer option that should not be used.	false
169	--solve-welleq-initially	A boolean value. Fully solve the well equations before each iteration of the	true

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		reservoir model. Note that the well equations are always added to the full system and solved until convergence.	
170	--solver-continue-on-convergence-failure	A boolean value. Continue instead of stop when minimum solver time step is reached.	false
171	--solver-growth-factor	A real value greater than one. Specify the growth factor a time step can be increased by when recovering from one or more time step chops, subject to the maximum allowable time step size set by the <code>--solver-max-time-step-in-days</code> parameter. For example, if the current time step has converged at 5 days after a time step chop, and <code>--solver-growth-factor</code> is set to the default value of 2.0, then the next time step will be $2.0 * 5$ days, that is at 10 days.	2
172	--solver-max-growth	A real value greater than one. Specify the maximum growth factor a time step can be increased by after a report time step, subject to the maximum allowable time step size set by the <code>--solver-max-time-step-in-days</code> parameter. Thus, if the current time step has converged at 5 days after at a report step, and <code>--solver-max-growth</code> is set to the default value of 3.0, then the next time step will be $3.0 * 5$ days, that is at 15 days.	3
173	--solver-max-restarts	An integer value. The maximum number of breakdowns before a substep is given up and the simulator is terminated.	10
174	--solver-max-time-step-in-days	A positive real value. The maximum size of a time step in days.	365
175	--solver-min-time-step	A positive real value. The minimum size of a time step in days for field and metric and hours for lab. If a step cannot converge without getting cut below this step size the simulator will stop.	1e-12
176	--solver-restart-factor	A real value between zero and one. Specify the time step chop factor of the time step after a convergence failure. For example, if the current non-convergent time step is 30 days and <code>--solver-restart-factor</code> is set to the default value, then the time step will be repeated using $0.33 * 30$ days as the time step, that is 9.9 days.	0.33
177	--solver-verbosity	An integer value. Specify the "chattiness" of the non-linear solver itself.	1
178	--strict-inner-iter-wells	An integer value. Number of inner well iterations with strict tolerance.	40

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
179	--strict-outer-iter-wells	An integer value. Number of newton iterations for which wells are checked with strict tolerance.	6
180	--temperature-max	A positive real value. Maximum absolute temperature.	1e+09
181	--temperature-min	A real value. Minimum absolute temperature.	0
182	--threads-per-process	An integer value. The maximum number of threads to be instantiated per process ('-1' means 'automatic').	2
183	--time-step-after-event-in-days	A real value. Specify the maximum allowed time step after an event; for example, when a well is open or closed etc. The default value of -1 means that events do not affect the time stepping.	-1
184	--time-step-control	A character string. The algorithm used to determine time-step sizes. Valid options are: 1) 'pid': PID controller based adaptive time step control as suggested by Turek and Kuzmin³⁹. 2) 'pid+iteration': Use PID and linear iteration numbers to guide the time step. 3) 'pid+newtoniteration': Use PID and non-linear iterations numbers to guide the time step. 4) 'iterationcount', 5) 'newtoniterationcount', 6) 'general3rdorder': Use general 3rd order controller, or 7) 'hardcoded': Use time steps supplied by user via the time-step-control-file-name parameter.	"pid+newtoniteration"
185	--time-step-control-decay-damping-factor	A positive real value less than or equal to one. The decay rate of the time step decrease when the target iterations is exceeded.	1
186	--time-step-control-decay-rate	A positive real value less than one. The decay rate of the time step size of the number of target iterations is exceeded.	0.75
187	--time-step-control-file-name	A character string. Specify the name of a file where time steps are specified. For instance time steps can be generated by the summary application as per the following UNIX command line:	"timesteps"

³⁹ Algebraic Flux Correction III. Incompressible Flow Problems. Uni Dortmund, Turek and Kuzmin, January, 2006 (DOI: 10.1007/3-540-27206-2_8)

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		<code>path_to_application/summary DECK</code> <code>TIME > timesteps</code> Where: DECK is the name of the data deck you want to get the time steps from, TIME tells the application to return the timing for the run, and "filename" is the name of the file the times are piped to.	
188	--time-step-control-growth-damping-factor	A real value greater than one. The growth rate of the time step increase when the target iterations is undercut.	3.2
189	--time-step-control-growth-rate	A real value greater than one. The growth rate of the time step size of the number of target iterations is undercut.	1.25
190	--time-step-control-max-reduction-time-step	A positive real value (only applicable for the general 3rd order controller, using 'control-error-filtering' as time-step-control-tolerance-test-version). If the (proposed) relative change in time step size is larger than this parameter, the time step will be rejected.	0.1
191	--time-step-control-parameters	A character string (only applicable for the general 3rd order controller). Parameters for the general 3rd order controller. Should be given as 'beta_1;beta_2;beta_3;alpha_2;alpha_3'.	"0.125;0.25;0.125;0.75;0.25"
192	--time-step-control-reject-completed-step	A boolean value (only applicable for the general 3rd order controller). Include rejection of completed time steps if the relative change is larger than the time step control tolerance.	false
193	--time-step-control-safety-factor	A positive real value less than or equal to one. Value to be multiplied with the time step control tolerance to ensure that the target relative change is lower than the tolerance.	0.8
194	--time-step-control-target-iterations	An integer value. The number of linear iterations which the time step control scheme should aim for (if applicable).	30
195	--time-step-control-target-newton-iterations	An integer value. The number of Newton iterations which the time step control scheme should aim for (if applicable).	8
196	--time-step-control-tolerance	A positive real value. The tolerance used by the time step size control algorithm.	0.1
197	--time-step-control-tolerance-test-version	A character string (only applicable for the general 3rd order controller). Ways to decide if the time step should be rejected.	"standard"

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		Options: 'standard' and 'control-error-filtering'. The standard version compares relative change to tolerance directly to decide if the time step should be rejected, while the control-error-filtering version compares the relative change in time step size to the max-reduction-time-step parameter.	
198	--time-step-verbosity	An integer value. Specify the "chattiness" during the time integration.	1
199	--tolerance-cnv	A positive real value. Local convergence tolerance (maximum of local saturation errors).	0.01
200	--tolerance-cnv-energy	A positive real value. Local energy convergence tolerance (maximum of local energy errors).	0.01
201	--tolerance-cnv-energy-relaxed	A positive real value. Relaxed local energy convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1
202	--tolerance-cnv-relaxed	A positive real value. Relaxed local convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1
203	--tolerance-energy-balance	A positive real value. Tolerated energy balance error relative to (scaled) total energy present.	1e-07
204	--tolerance-energy-balance-relaxed	A positive real value. Relaxed tolerated energy balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
205	--tolerance-mb	A positive real value. Tolerated mass balance error relative to total mass present.	1e-07
206	--tolerance-mb-relaxed	A positive real value. Relaxed tolerated mass balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
207	--tolerance-pressure-ms-wells	A positive real value. Tolerance for the pressure equations for multi-segment wells.	1000
208	--tolerance-well-control	A positive real value. Tolerance for the well control equations.	1e-07
209	--tolerance-wells	A positive real value. Well convergence tolerance.	0.0001
210	--update-equations-scaling	A boolean value. Update scaling factors for mass balance equations during the run.	false

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
211	--use-average-density-ms-wells	A boolean value. Approximate segment densities by averaging over segment and its outlet.	false
212	--use-gmres	A boolean value. Use the Generalized Minimal Residual ("GMRES") ⁴⁰ and ⁴¹ solver instead of Biconjugate Gradient Stabilized ("BiCGSTAB") ⁴² and ⁴³ as the linear solver within the Newton iterations.	false
213	--use-hy-ne	A boolean value. Whether or not to use Hybrid Newton.	false
214	--use-implicit-ipr	A boolean value. Compute implicit IPR for stability checks and stable solution search.	true
215	--use-multisegment-well	A boolean value. Use the well model for multi-segment wells instead of the one for single-segment wells.	true
216	--use-update-stabilization	A boolean value. Try to detect and correct oscillations or stagnation during the Newton method.	true
217	--verify-gpu-aware-mpi	A boolean value. Verify that the MPI implementation supports GPU aware MPI. If this is set to true *and* --gpu-aware-mpi=true, the simulation will fail if the MPI implementation does not support GPU aware MPI. Note that the verification is not exhaustive, and some configurations will not verify, but will work in practice. Usage: --verify-gpu-aware-mpi=[true false].	false
218	--water-only-threshold	A positive real value. Cells with water saturation above or equal is considered one-phase water only.	1
219	--well-group-constraints-max-iterations	An integer value. Maximum number of iterations in the well/group switching algorithm.	1
220	--zoltan-imbalance-tol	A real value. Tolerable imbalance of the loadbalancing provided by Zoltan. DEPRECATED: Use --imbalance-tol instead.	1.1
221	--zoltan-params	A character string. Configuration of Zoltan partitioner. Valid options are: graph, hypergraph or scotch. Alternatively, you can	"graph"

⁴⁰ Y. Saad, A flexible inner-outer preconditioned GMRES algorithm, *SIAM J. Sci. Statist. Comput.*, 14, (1993).

⁴¹ Y. Saad and M.H. Schultz, GMRES: A generalized minimum residual algorithm for solving nonsymmetric linear systems, *SIAM J. Sci. Statist. Comput.*, 7 (1986), pp. 856-869.

⁴² Van der Vorst, H. A. (1992). "Bi-CGSTAB: A Fast and Smoothly Converging Variant of Bi-CG for the Solution of Nonsymmetric Linear Systems". *SIAM J. Sci. Stat. Comput.* 13 (2): 631–644. doi:10.1137/0913035. hdl:10338.dmlcz/104566

⁴³ Sleijpen, G. L. G.; Fokkema, D. R. (November 1993). "BiCGstab(l) for linear equations involving unsymmetric matrices with complex spectrum" (PDF). *Electronic Transactions on Numerical Analysis*. Kent, OH: Kent State University. 1: 11–32. ISSN 1068-9613.

OPM Flow Command Line Options			
No.	Variable Name	Description	Default
		request a configuration to be read from a JSON file by giving the filename here, with extension '.json'. See https://sandialabs.github.io/Zoltan/ug_html/ug.html for available Zoltan options.	
222	--zoltan-phg-edge-size-threshold	A real value ranging from zero to one. Low-level threshold fraction in the range [0,1] controlling which hypergraph edge to omit. Used if --zoltan-params="graph" or if --zoltan-params="hypergraph".	0.35
Notes: - Cells colored green in the No. column indicate new command line parameters for this release. Similarly for the Default column, cells colored green indicate the default value has changed from the previous release. - Items shaded in gray are considered to be developer options that should be used with caution as the values associated with these command line parameters are subject to change. Cells colored orange in the No. column indicate new developer options for this release. - Cells colored red in the No. column indicate command line parameters that have either been deprecated or removed from this release. - VTK Graphics Command Line Parameters⁴⁴ set of commands are no longer listed via the basic help command line command: <pre>flow --help</pre> instead use: <pre>flow --help-all</pre> to get a list of supported command line Parameters. - The --enable-vtk-output option above, if set to "true" will write out the data as stipulated by the compile options. One may need to compile the source code to obtain the desired output. - As per all UNIX and LINUX based systems the input is case sensitive. - If a parameter file is used to set one of the above parameters, the leading "--" should not be incorporated in the parameter file.			

Table 2.1: OPM Flow Command Line Options

As per all UNIX and LINUX based systems the input is case sensitive. The command line parameters must either be entered as lower case or CamelCase, for example:

```
flow --enable-dry-run=false CASENAME.DATA
```

or:

```
flow --EnableDryRun=false CASENAME.DATA
```

If a parameter file is used to set one of the above parameters, the leading "--" should not be incorporated in the parameter file, for example on the command line use:

```
flow --enable-dry-run=false CASENAME.DATA
```

but in the parameter file use:

```
ec1-deck-file-name=CASENAME.DATA
```

```
enable-dry-run=false
```

⁴⁴ Virtualization ToolKit ("VTK") files used to visualize the 3D results from the model using ParaView, an open-source, multi-platform general purpose data analysis and visualization application (<https://www.paraview.org/overview/>). For the commercial simulator's compatible output files OPM's ResInsight 3D visualization package can be used instead and is specifically tailored for OPM Flow and the commercial simulator.

or:

```
EclDeckFileName=CASENAME.DATA
```

```
EnableDryRun=false
```

OPM Flow prints out the command line run time and compile time parameters in CamelCase in both the *.PRT and *.DBG files for reference, one can therefore use this as basis for creating additional run specific parameter files.

Note

It is worth considering having a parameter file for each run of the form CASENAME.param, in order to re-run a case using the same parameters or to use the same parameters for other cases.

Example

The following example shows the parameter file NORNE_ATW2013.PARAM taken from the Norne project, where the comments below (preceded with “#”) explain the parameter settings used in the file.

```
# -----  
# INPUT AND OUTPUT OPTIONS  
# -----  
#  
# Input File  
#  
ecl-deck-file-name=NORNE_ATW2013.DATA  
#  
# Output and Output Directory  
#  
output-dir='$HOME\OPM\NORNE\  
# -----  
# NEWTON SOLVER PARAMETER  
# -----  
#  
# Define Numerical Tolerances  
#  
tolerance-cnv=1e-2  
tolerance-mb=1e-5  
tolerance-wells=1e-2  
#  
# Set Min Newton Solver iterations to 1 and Max to 15  
#  
newton-min-iterations=1  
newton-max-iterations=15  
#  
# -----
```

Notice that the leading “--” have not be incorporated in the parameter file, as per the notes in Table 2.1.

In order to run the deck file NORNE_ATW2013.DATA in OPM Flow using the above parameter file one could use either of the following commands:

```
flow --parameter-file=NORNE_ATW2013.PARAM
```

or:

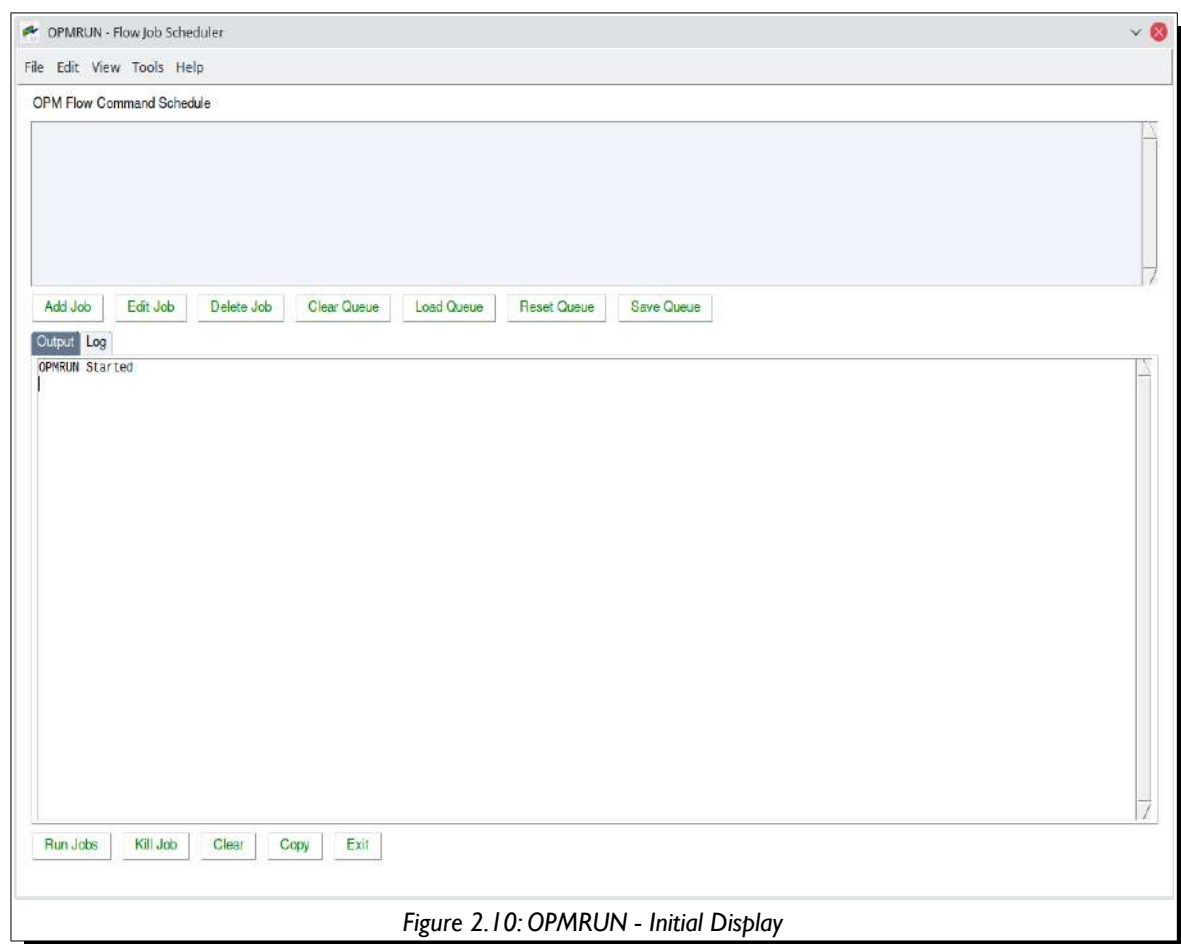
```
flow --ParameterFile=NORNE_ATW2013.PARAM
```

2.3 RUNNING OPM FLOW USING OPMRUN

This section briefly describes how to run the OPM Flow simulator using the Graphical User Interface (“GUI”) program called OPMRUN, a more detailed description is given in [APPENDIX C: OPMRUN – FLOW JOB SCHEDULER](#), that also describes the various tools included with OPMRUN. The program is written in Python 3 and has been tested under Ubuntu-Mate 24.04 TLS. The software can be downloaded from the following link:

<https://github.com/OPM/opm-utilities/tree/master/opmruntime>

The intent is to develop a graphical user interface to OPM Flow that has similar functionality to the commercial simulator’s program, with the targeted audience being Reservoir Engineers in a production environment. Developers and experienced Linux users will already have compatible work flows. OPMRUN enables the editing and management of OPM Flow’s run time parameters, setting up job queues to run a series of simulation jobs sequentially, as well as the management of the job queues. Figure 2.10 shows the initial display.



Upon launch the program runs OPM Flow to get a list of command line parameters from the current version of OPM Flow. These default parameters can be edited for each case, or alternative default parameter sets can be loaded from an existing parameter file from another job, or a *.PRT file from a completed simulation.

As can be seen in Figure 2.11 the program has upper and lower display elements. The upper element shows a list of simulation jobs that are in the job queue and the lower element consists of two elements, one for the OPM Flow Output (the terminal output from OPM Flow) and a second element (OPM Run Log) that is a session log of the jobs run by OPMRUN. Clicking the OPM Flow Output and OPM Run Log tabs switches the display on the lowered element between two display types.

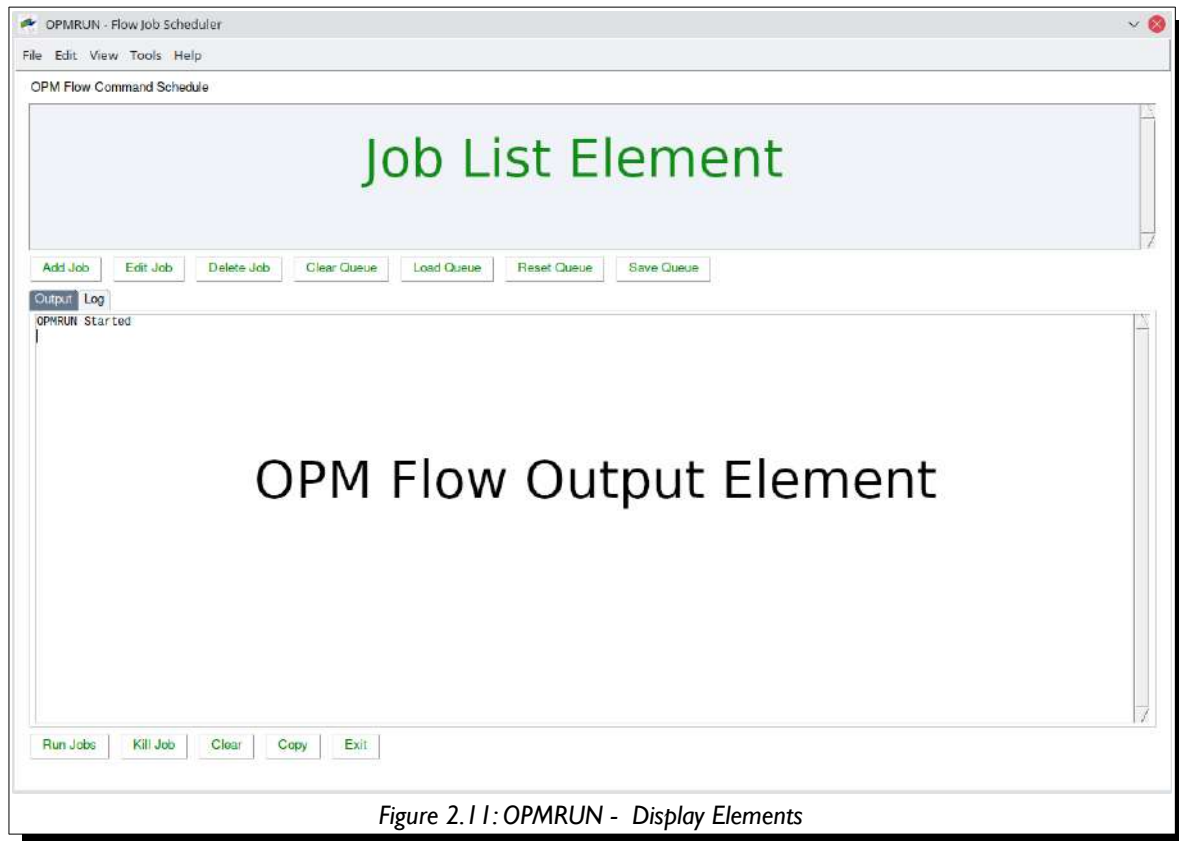


Figure 2.11: OPMRUN - Display Elements

To add jobs to the queue use the Add Job button or load an existing job queue using the Load Queue button. Jobs can be edited or deleted from the queue using the Edit Job and Delete Job buttons, and a series of jobs can be saved as a job queue by using the Save Queue button. The Clear Queue button deletes all jobs from the queue.

Pressing the Add Job button will display the following dialogue box:

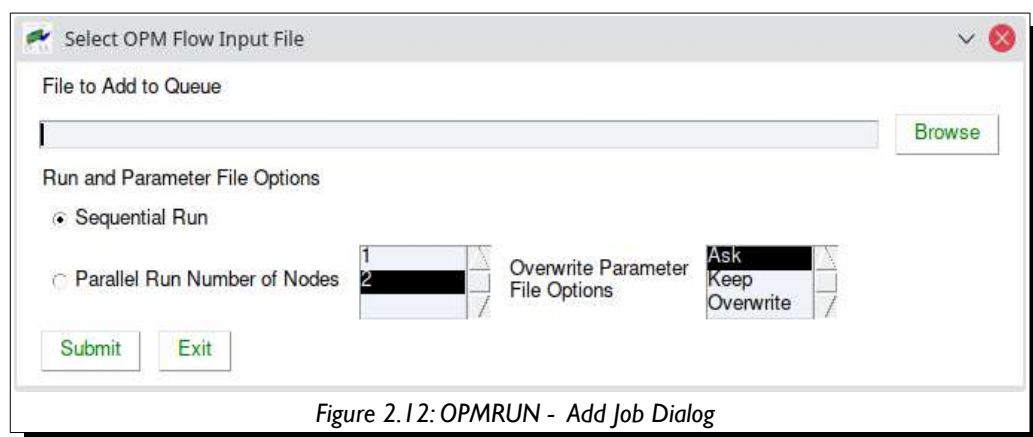


Figure 2.12: OPMRUN - Add Job Dialog

Use the *Browse* button to select the input file to add to the queue, then select the Run Parameters for this input file, then press the Submit button to add the input file to the job queue.

The Overwrite Parameter File Options allow for different default treatments of existing *.PARAM files, which is particularly useful when adding multiple jobs at the same time. The reason for this is because different versions of OPM Flow have different parameter sets and if a newer version of OPM Flow runs with a previous version's *.PARAM file then the simulator will stop with an error message for the various invalid

parameters for the current version of the simulator. To load a previously saved job queue, press the *Load Queue* button this will display a dialog box allowing the user to select a queue file (*.que), after pressing the OK button the jobs will be displayed in the Job List Element as illustrated in Figure 2.13.

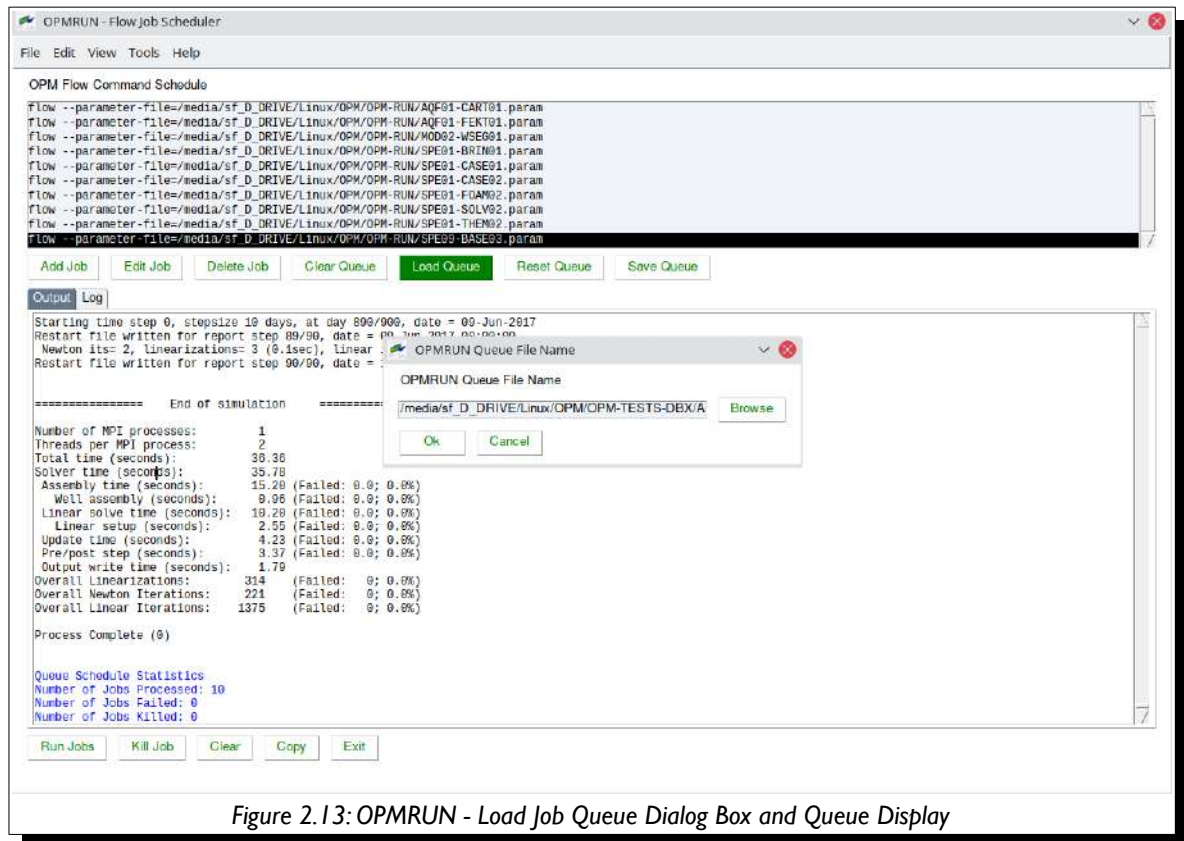


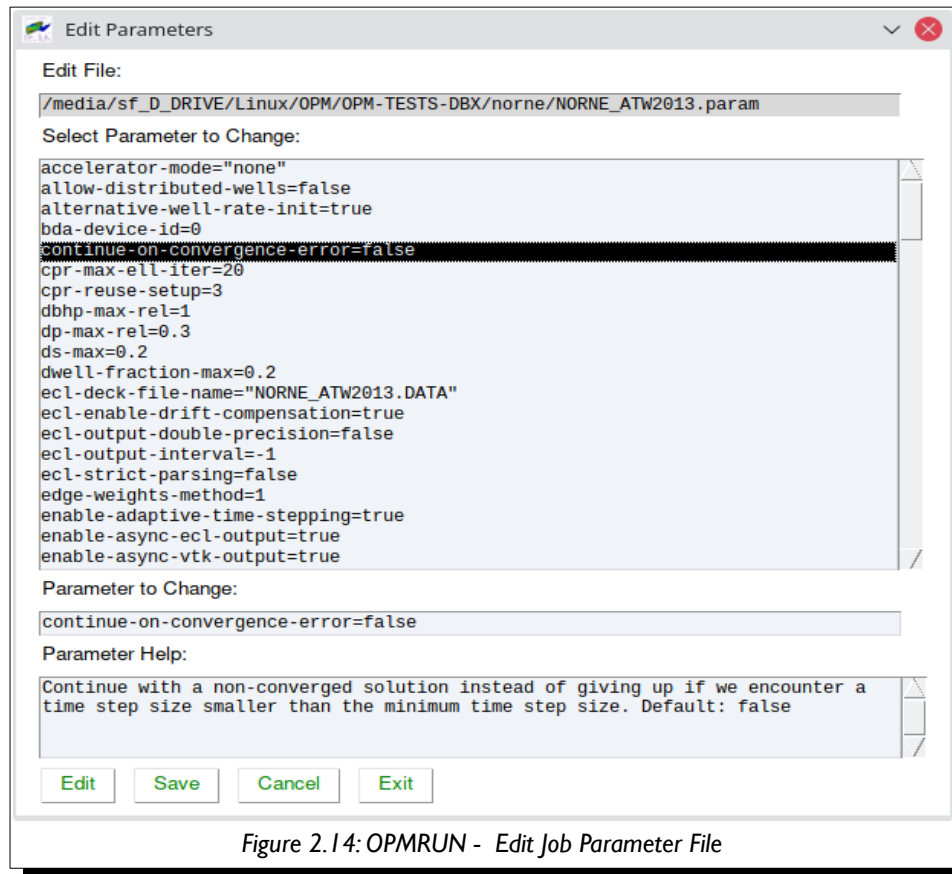
Figure 2.13: OPMRUN - Load Job Queue Dialog Box and Queue Display

Jobs in the queue can be edited by selecting the *Edit Job* button that will display two options: one to edit the input file using the defined editor and the second to edit the OPM Flow Parameter File. If the second option is selected OPMRUN will display a dialog box that shows a list of the OPM Flow command line parameters together with the parameter help information (Figure 2.14). Selecting a parameter from the list and selecting the Edit button will display the setting for the selected parameter (alternatively one can double click the required entry). One can then edit the parameter as required. Use the Save button to save the change and use the Exit but to save all the changes to the parameter file. The Cancel button will cancel all changes to the parameter file.

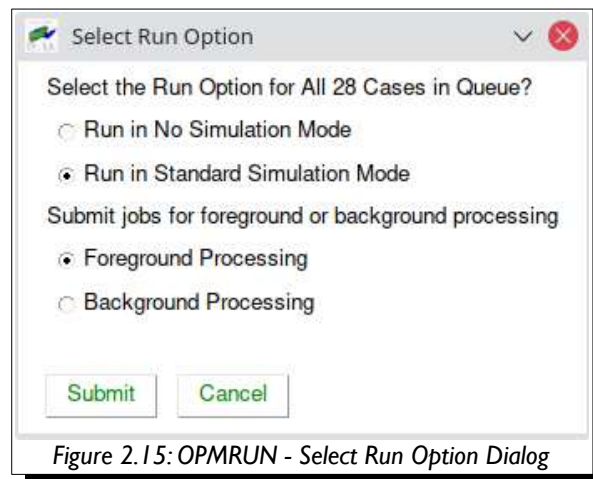
Alternatively one can use the:

- 1) Edit OPM Flow Parameter menu option to edit the parameter file for a job.
- 2) List OPM Flow Parameters menu option to list the commands in the parameter file for a job.
- 3) Set OPM Flow Default Parameters to set the default parameters for all subsequent jobs added to the queue. This option allows the user to load a default set of parameters from (1) OPM Flow, (2) an OPM Flow Parameter File, or (3) an OPM Flow print file (*.PRT).

One can also right-click on a job in the *Job List Element* and select one of the available options.



Selecting the *Run Jobs* button displays the Select Run Option dialog box shown in Figure 2.15.



The Run in No Simulation Mode option is equivalent to setting the *NOSIM* option in the input deck for all jobs in the queue (see section 5.3.101 *NOSIM – Activate the No Simulation Mode for Data File Checking* and the *--enable-dry-run* command line parameter in Table 2.1 in section 2.2 *Running OPM Flow From The Command Line*). This allows for checking all the jobs at once.

Selecting Run in Standard Simulation Mode will run all the jobs in the queue sequentially, with the OPM Flow terminal output directed to OPM Flow Output Element, as shown in Figure 2.16. The terminal output is also directed to a *.LOG file as well, similar to what the commercial simulator does.

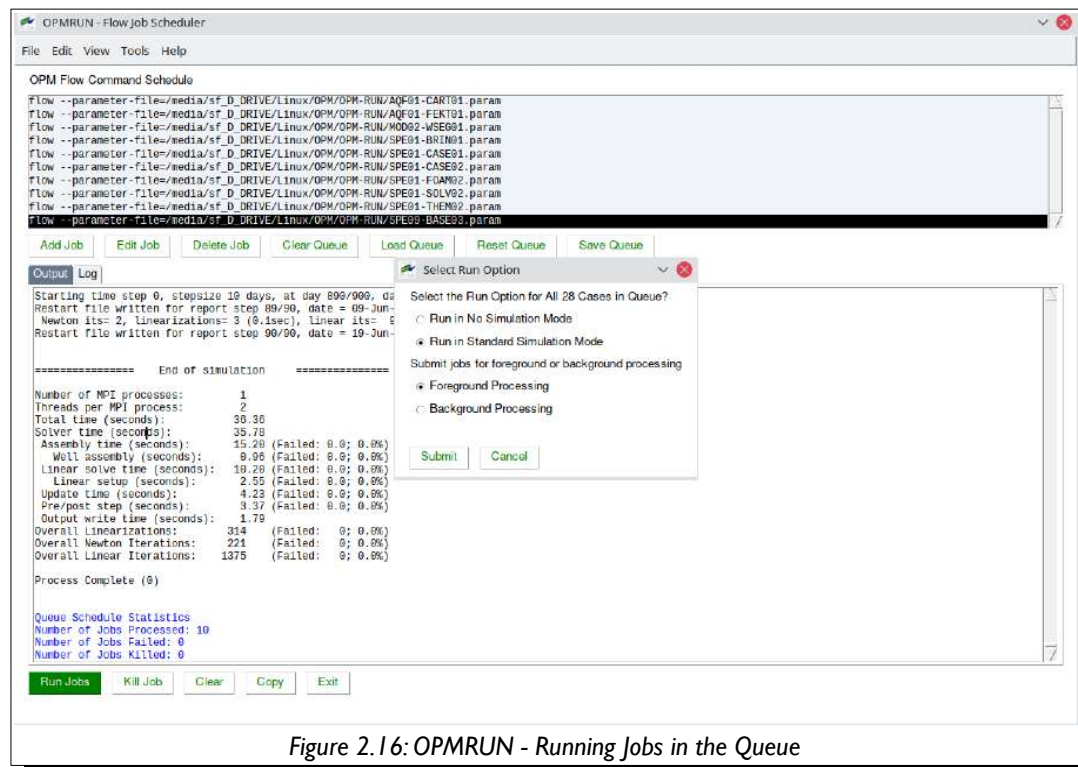
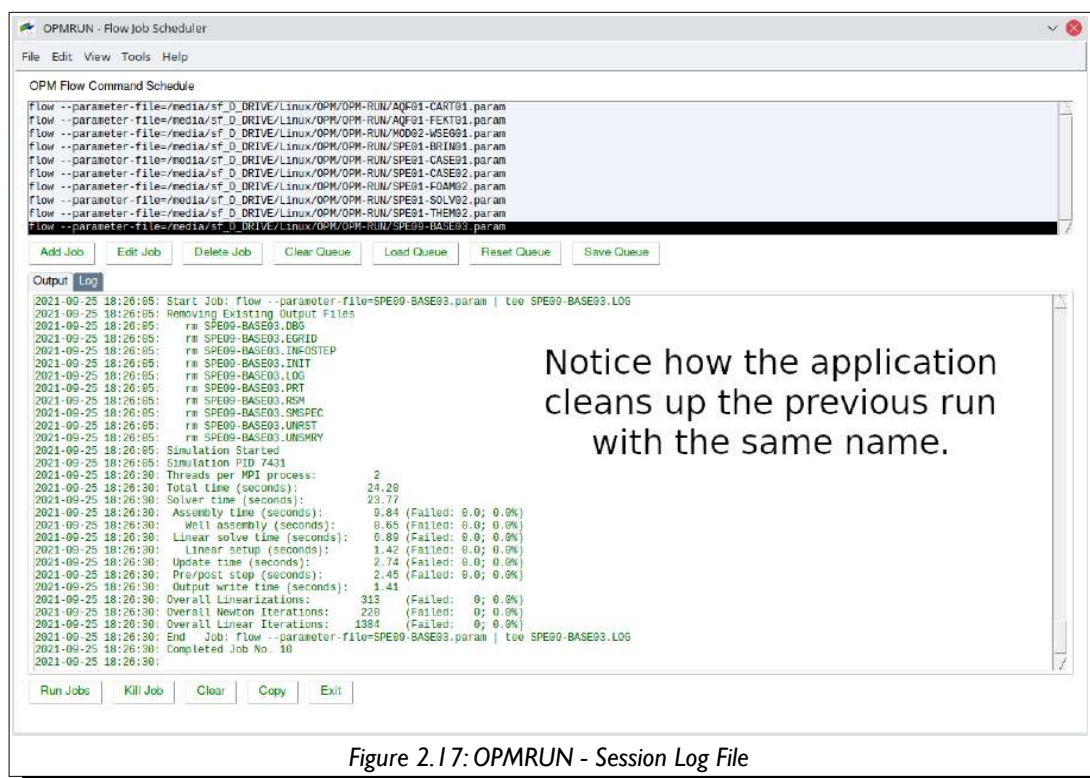


Figure 2.16: OPMRUN - Running Jobs in the Queue

Clicking the OPM Run Log tab displays the OPMRUN's session log file that records the time and date of the major events that have occurred, including the start and end times of each run. Notice also how OPMRUN deletes all the existing output files for a given job, if they exists, before running OPM Flow.



Notice how the application cleans up the previous run with the same name.

Figure 2.17: OPMRUN - Session Log File

The Kill button will ask the user if the current running job should be killed, and if the job is to be killed, the application will prompt as whether or not all the jobs in the queue should be killed.

The *Clear* clears the OPM Flow Output Element from the currently displayed tab (*Output* or *Log*) and the *Copy* button copies the data to the clipboard.

Finally, an example *.param file generated by OPMRUN for the 2022-10 release is shown below and on the following pages for reference:

```
#
# OPMRUN Parameter File
#
# File Name   : "/media/Fields/Norne/Model/NOR-BASE-A00.param"
# Created By  : opmuser
# Date Created: 2022-11-01 13:12:44
#
accelerator-mode="none"
allow-distributed-wells=false
alternative-well-rate-init=true
bda-device-id=0
continue-on-convergence-error=false
cpr-max-ell-iter=20
cpr-reuse-interval=10
cpr-reuse-setup=3
dbhp-max-rel=1
dp-max-rel=0.3
ds-max=0.2
dwell-fraction-max=0.2
ecl-deck-file-name="NOR-BASE-A00.DATA"
ecl-enable-drift-compensation=true
ecl-output-double-precision=false
ecl-output-interval=-1
ecl-strict-parsing=false
edge-weights-method=1
enable-adaptive-time-stepping=true
enable-async-ecl-output=true
enable-async-vtk-output=true
enable-dry-run="auto"
enable-ecl-output=true
enable-esmry=false
enable-logging-fallout-warning=false
enable-opm-rst-file=false
enable-storage-cache=true
enable-terminal-output=true
enable-tuning=false
enable-vtk-output=false
enable-well-operability-check=true
enable-well-operability-check-iter=false
enable-write-all-solutions=false
force-disable-fluid-in-place-output=false
force-disable-resv-fluid-in-place-output=false
fpga-bitstream=""
full-time-step-initially=false
ignore-keywords=""
ilu-fillin-level=0
ilu-redblack=false
ilu-relaxation=0.9
ilu-reorder-spheres=false
initial-time-step-in-days=1
linear-solver="ilu0"
linear-solver-ignore-convergence-failure=false
linear-solver-max-iter=200
linear-solver-reduction=0.01
linear-solver-require-full-sparsity-pattern=false
linear-solver-restart=40
```

```
linear-solver-verbosity=0
matrix-add-well-contributions=false
max-inner-iter-ms-wells=100
max-inner-iter-wells=50
max-newton-iterations-with-inner-well-iterations=8
max-pressure-change-ms-wells=1e+06
max-residual-allowed=1e+07
max-single-precision-days=20
max-temperature-change=5
max-welleq-iter=30
maximum-number-of-well-switches=3
milu-variant="ILU"
min-strict-cnv-iter=0
min-time-step-based-on-newton-iterations=0
min-time-step-before-shutting-problematic-wells-in-days=0.01
newton-max-iterations=20
newton-max-relax=0.5
newton-min-iterations=1
newton-relaxation-type="dampen"
opencl-ilu-parallel=true
opencl-platform-id=0
output-dir=""
output-interval=1
output-mode="all"
owner-cells-first=true
parameter-file=""
pri-var-oscillation-threshold=1e-05
print-parameters=2
print-properties=2
project-saturations=false
regularization-factor-wells=100
relaxed-max-pv-fraction=0.03
relaxed-pressure-tol-msw=10000
relaxed-well-flow-tol=0.001
scale-linear-system=false
sched-restart=false
serial-partitioning=false
shut-unsolvable-wells=true
solve-welleq-initially=true
solver-continue-on-convergence-failure=false
solver-growth-factor=2
solver-max-growth=3
solver-max-restarts=10
solver-max-time-step-in-days=365
solver-min-time-step=1e-12
solver-restart-factor=0.33
solver-verbosity=1
strict-inner-iter-wells=40
strict-outer-iter-wells=6
temperature-max=1e+09
temperature-min=0
threads-per-process=-1
time-step-after-event-in-days=-1
time-step-control="pid+newtoniteration"
time-step-control-decay-damping-factor=1
time-step-control-decay-rate=0.75
time-step-control-file-name="timesteps"
time-step-control-growth-damping-factor=3.2
time-step-control-growth-rate=1.25
time-step-control-target-iterations=30
time-step-control-target-newton-iterations=8
time-step-control-tolerance=0.1
time-step-verbosity=1
tolerance-cnv=0.01
tolerance-cnv-relaxed=1
```

```
tolerance-mb=1e-06
tolerance-pressure-ms-wells=1000
tolerance-well-control=1e-07
tolerance-wells=0.0001
update-equations-scaling=false
use-gmres=false
use-multisegment-well=true
use-update-stabilization=true
zoltan-imbalance-tol=1.1
#
# End of Parameter File
```

In the above example, all the default parameters have been used.

In addition to the functionality described in this section, the application has various additional features accessible through the application's menus including:

- 1) Configuring OPMRUN various options, default editor, default terminal console for background jobs, setting default project directories, etc.
- 2) Various additional simulation input generation and conversion utilities are available including:
 - Compressing a job to save space (DATA, and all OPM Flow output files) and uncompressing previously compressed jobs.
 - Keywords, a keyword generator based on the Apache Velocity Template Language ("VTL"). The templates can therefore also be used with any editor that supports VTL, jEdit for example. There is one template per keyword, with the formatting the same as the OPM Flow manual. Over 450 templates are currently implemented. One can also customize the existing templates as well as creating User defined templates. The keywords are examples, one still has to edit the resulting deck with the actual required data, but the format with comments should make this a straight forward process.
 - A Production Schedule application that takes a comma delimited CSV file containing historical production and injection data and converts the data to an OPM Flow [SCHEDULE](#) file using the [WCONHIST](#) series of keywords. Currently only production data is supported.
 - Sensitivities application that generates sensitivity cases based on a "Base" case file. The Base file contains "Factors" (variable names), \$X01, \$X02, etc., that are substituted with user defined values using the data entered and the type of Sensitivity Scenario selected.
 - A Well Specification application that uses the standard well export files from OPM ResInsight to reformat the data in a more user-friendly manner for the [WELSPECS](#) and [COMPDAT](#) keywords. Optionally, the application can generate the [COMPLUMP](#) keyword based on the OPM ResInsight layers file, with one completion per defined reservoir layer.
 - Calling OPM ResInsight and loading the currently selected job into OPM ResInsight for viewing.
 - A Well Trajectory Conversion application that converts a Schlumberger Petrel exported well trajectory file into an OPM ResInsight file, containing all the wells.

All the aforementioned options and tools are described in detail in APPENDIX C:OPMRUN – FLOW JOB SCHEDULER,

Note

OPMRUN can be run as either a Linux or a Windows application. If the software is installed and run under Windows, then the program automatically handles the file naming conventions of the two operating systems. That is under Windows, the application uses the Windows file name to select the files, and then automatically converts the file name to the equivalent Linux file name to run under WWSL.

2.4 IMPROVING SIMULATOR CONVERGENCE AND NUMERICAL PERFORMANCE

In general, most simulator performance issues are caused by the data used by the simulator, rather than the numerical algorithms used to solve the equations presented in sections [1.2 Black-Oil Model Equations](#) through [1.4 Multi-Segment Well Model](#). Thus, it is important to verify all warning messages and to address the issues raised by these warnings. Naturally, some warnings can be safely ignored, while others should not. Just because the simulator runs with warning messages does not mean that the results are reasonable, or that numerical performance has not been impacted.

This section is therefore broken into two parts, the first part discusses potential data errors and how to address them, and the second section outlines the numerical controls available to improve the simulator's performance. The latter should only be done after the data quality issues have been rectified, although, it is not uncommon for users to skip the data quality step altogether, and jump to "tuning" the model's numerical controls - to be clear, skipping the data quality step is strongly not recommended.

2.4.1 DATA QUALITY AND ASSURANCE

GRID Section

Grid Configuration

In full field models grid distortions may occur due to faulting and the structural complexity of the geological model, and ideally one wishes to have all the cells to be orthogonal. Grid cells are orthogonal if all grid lines intersect at a right angles, that is at 90° . This is because orthogonal grids offer significant advantages in solving a system of partial differential equations used computational fluid dynamics. In addition, they ensure that the fluid flow is not being unduly influenced by the grid orientation effects, as cell to cell flow only occurs across connected cell faces.

Most earth modeling software can produce a histogram of the cell angles as an aide to checking the orthogonality of the grid. A common cause for cells not being orthogonal, as well as creating other grid artifacts, is the grid cells following actual the faulting traces, as oppose to using "zig-zag" fault traces that follow the grid. A typical fault trace grid is illustrated in Figure 2.18.

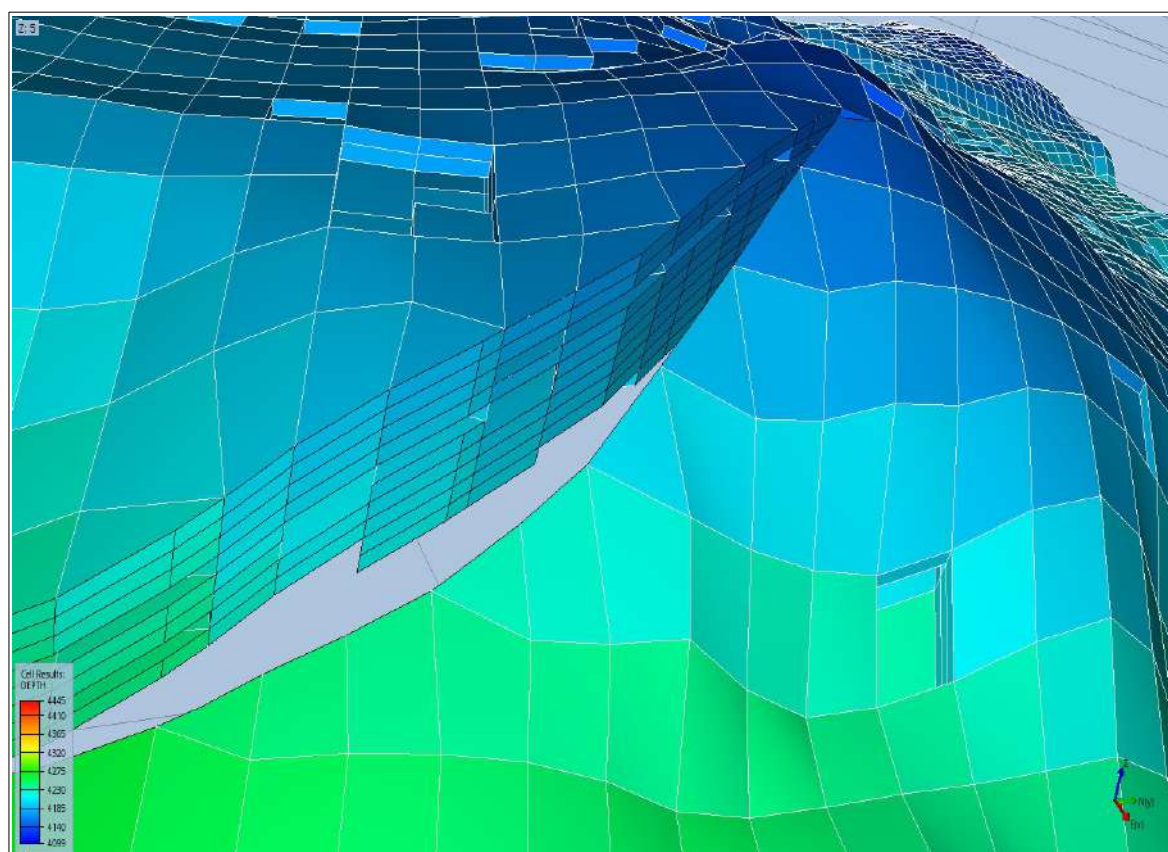


Figure 2.18: Non-Orthogonal Grid Cells Along the Fault Plane Example

Notice in Figure 2.18 how the cells along the fault plane are severely distorted, with some cells having only three faces instead of four in the x-y plane. Now compare this to the Norne model shown in Figure 2.19, which has more complex faulting, and yet the grid along the fault planes is mainly orthogonal.

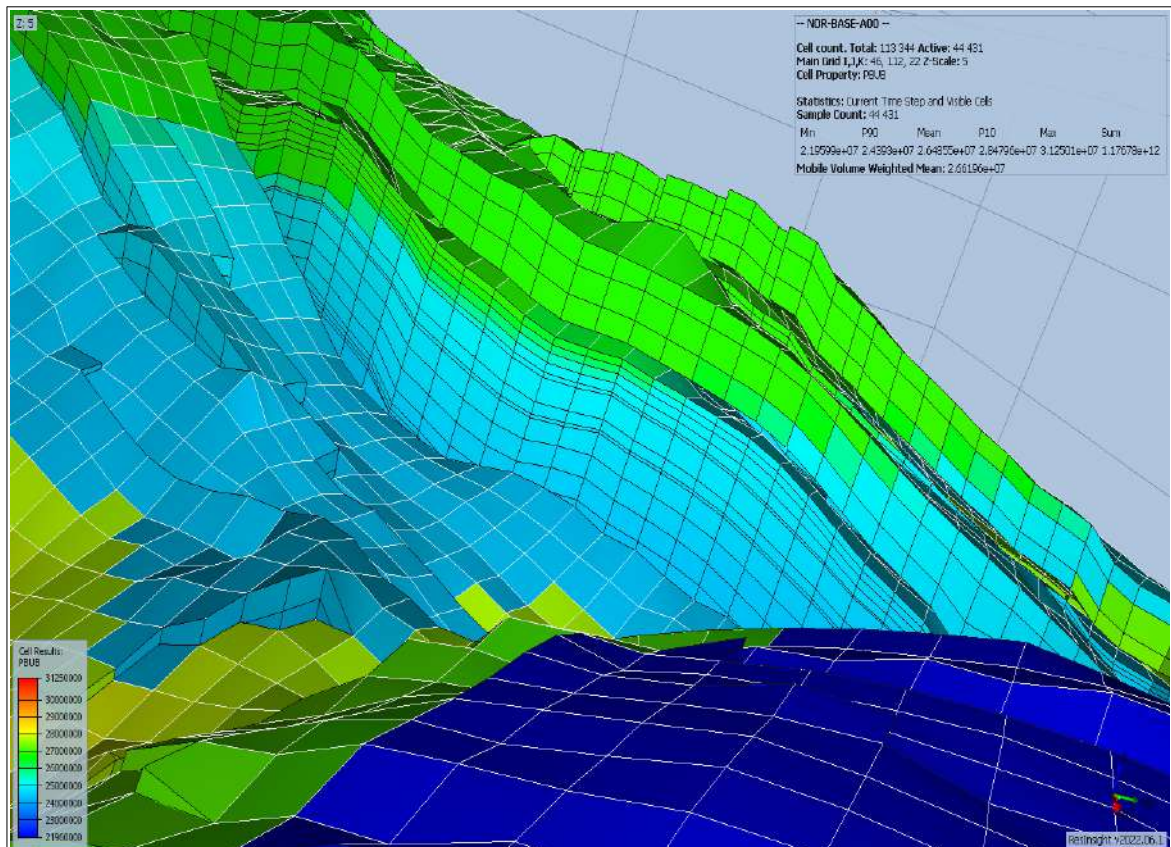


Figure 2.19: Norne Field Orthogonal Grid Example

Grid Property Data and Active Cells

There is a current trend in the industry for not applying cut-offs to the petrophysical parameters used to generate the grid property data (porosity, net-to-gross, etc.). The philosophy behind this approach is that the simulator should determine what is reservoir and is capable of contributing to flow within the reservoir, and what is non-reservoir. At first glance this would appear reasonable, but unfortunately the inclusion of low pore volume and low permeability grid cells will have a dramatic negative impact on the model's numerical convergence and run times. This is because one ends up with many more active cells than otherwise, plus the fact that low pore volumes grid blocks will be connected to high pore volume cells. Because of this interconnectedness, a change in a large pore volume grid block will have disproportional change on a low pore volume cell, which will result in smaller time step sizes in order to satisfy the numerical controls.⁴⁵ That is, the low pore volume cell throughput will influence the simulator's selected time step size disproportionately.

Nevertheless, even if the more conventional approach to the model build is undertaken by applying some form of petrophysical cut-offs, there may well still be grid blocks containing small pore volumes. In this case the **MINPV** keyword in the **GRID** section can be used to set a minimum pore volume threshold in which cells below the threshold value will be automatically made inactive. It is not possible to give absolute guidance for a suitable pore volume value to be used with **MINPV**, the best approach is to run some sensitivities to investigate the impact of different pore volume thresholds on the reduction in the number of cells and the consequent change in the fluid in-place volumes. One should also be mindful not to remove thin high permeability cells that have small pore volumes due to their size, as this will effect the model's flow behavior.

Finally, if the **PINCH** keyword in the **GRID** section is also in the input deck, then one must be cognizant of the interaction between the two keywords.

⁴⁵ The large pore volume differences make the solution of the linear system of equations more challenging, due to the residual errors in certain grid cells or areas within the grid, that are much harder to reduce with iterative solvers. This results in more convergence failures, smaller time steps, etc. Similarly, for highly varying cell sizes.

Grid Coarsening

As the number of active grid blocks directly effect a model's run time, reducing the number of active cells by coarsening areas of model that are of secondary importance, can be beneficial to the overall run time of a model. An example of where this approach should be considered is the modeling of aquifers. It is not necessary to fully model an aquifer, by having the model cells cover the whole area of the aquifer. Instead, incorporating either analytical aquifers via the [AQUCT – Define Carter-Tracy Analytical Aquifers](#) and [AQUFETP – Define Fetkovich Analytical Aquifers](#) keywords, or numerical aquifers using the [AQUNUM – Define Numerical Aquifer Properties](#) keyword, can effectively re-produce the aquifer response. Both types of aquifers have advantages and disadvantages.

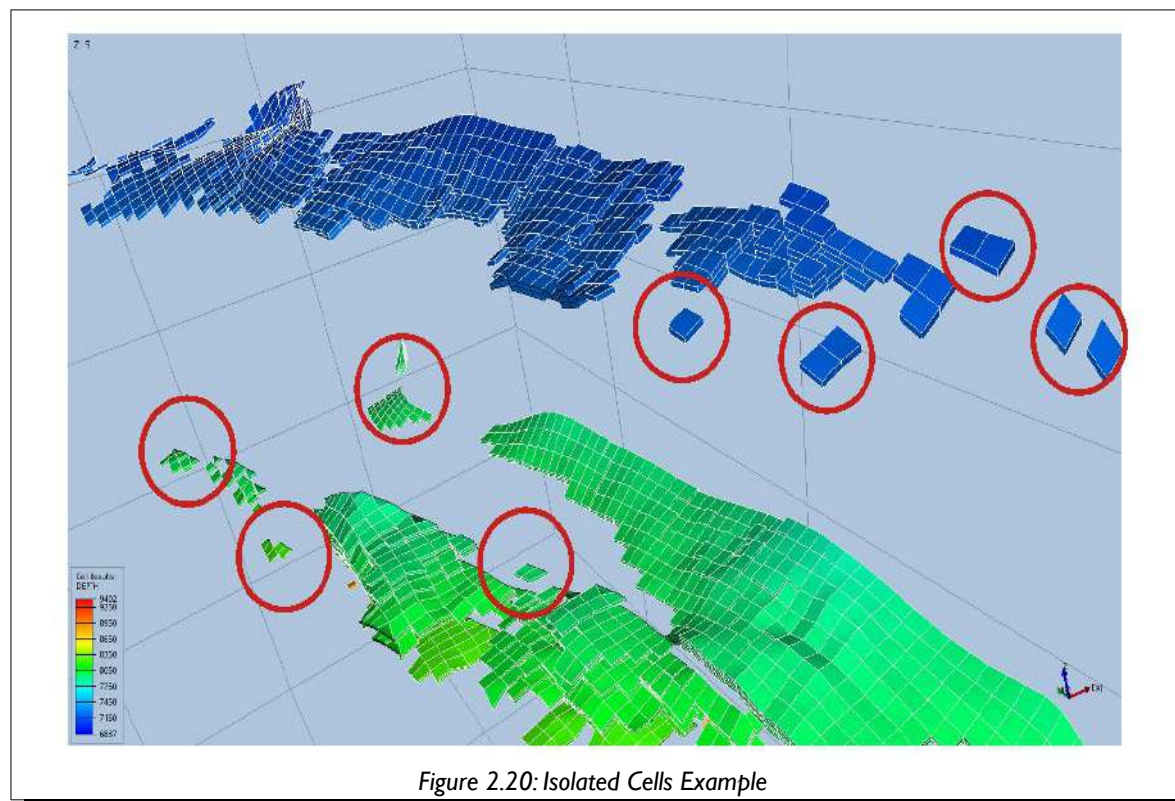
Analytical aquifers have the advantage that their response and behavior can be matched by using material balance techniques to quantify the aquifer properties, by history matching the measured pressure response in the hydrocarbon zones. This is an efficient analytical modeling work flow to identify the aquifer response, the properties derived can therefore be more or less directly incorporated into the dynamic model. The disadvantage is that analytical aquifers are not directly comparable with geological interpretations, which may, or may not, be important, depending upon the circumstances.

The opposite is true for numerical aquifers in that an approximate aquifer volume and property data can be matched with current geological interpretation, but cannot be history matched in most material balance software, as numerical aquifers in general are not supported.

Nevertheless, no matter which type of aquifer is employed, they should not be directly connected to hydrocarbon bearing cells, ideally there should be between three to five water bearing cells between the aquifer connected cells and the hydrocarbon cells. However, the water cells may vary in size, such that water cells connected to the hydrocarbon zone are of similar size, and then increase in size towards the edge of the model. The reason for this is that we wish to limit "small" hydrocarbon cells being directly coupled to "large" water cells, that will hinder the length of time steps selected by the simulator due to the convergence criteria.

Grid Isolated Cells

Simulation models should only include grid blocks that influence the flow and recovery in the model. One common issue with some of the geo-statistical formulations used to generate the grid and the associated grid property data, is the tendency to create isolated grid blocks. These isolated grid blocks are not connected to the rest of the model and therefore do not influence the behavior of the model. Figure 2.20 shows a typical example, with the cells circled in red being the isolated cells.



Unfortunately, these cells will be included in all the calculations, and may cause numerical issues due to their unchanging state, both pressure and saturation. They will also unnecessarily add to the active cell count, and therefore effect the simulation run time. In addition, they will give a false impression of the average reservoir pressure, as the isolated cell pressures will remain at initial conditions throughout the simulation run. If model convergence is proving difficult, then this may be a potential cause of the problem. In any case, it is recommended that the identified isolated cells be removed by making them inactive via the [ACTNUM](#) keyword in the [GRID](#) section.

If there are only a few isolated cells then they can manually be made inactive; however, in many cases this not possible due to the number of isolated grid blocks. In this case, it is better to revert back to the static model and fix the issue in the earth modeling software. Indeed, if there are numerous isolated grid blocks this may be indicative that an inappropriate approach has been used to generate the static model, although it may be also due to the depositional environment as well.

PROPS Section

This section describes potential issues associated with the property data and is split into two separate sub-sections:

- 1) The first sub-section covers the rock properties, and in particular the saturation functions (relative permeability curves), that can have a dramatic impact on reservoir performance, especially for the non-linear solver. As will be shown later, it is important that the derivatives of the relative permeability curves are smooth and monotonically increasing or decreasing depending on the phase of the relative permeability fluid. Particular care must be taken if end-point scaling is being used, as it is not uncommon for the resulting de-normalized curves to be inappropriate, even though the normalized curves are reasonable and correct.
- 2) The second sub-section addresses issues that may arise with the fluid properties, and inconsistent PVT data that may result in the simulation struggling to converge in the linear solver.

PROPS Section: Saturation Function (Derivative)

In general, one should not enter laboratory derived relative permeability data directly into the simulator. This is because it is dubious that laboratory measurements on 1 1/2 inch core plugs can represent the flow in 100 x 100 x 1.0 m grid cell in the model. There are various upscaling techniques that can account for the scaling issue, using various forms of relative permeability curves, including the laboratory curves, piston-like curves and thickness average curves etc. However, the resulting pseudo relative permeability curves tend to not have smooth derivatives, which will have a detrimental impact on convergence efficiency. Thus, it is common practice to use the laboratory derived curves, either in an upscaling workflow or as part of the input workflow, and to fit a Corey⁴⁷ type curve to the "rock" curves in order to smooth out any discontinuities.

Unfortunately, neither OPM Flow nor the commercial simulator, unlike some other simulators, support the direct entry of Corey type curves; however, OPM Flow does support the more advanced and flexible LET family of models instead. See section [8.2.6 Saturation Table Generation - LET Functions](#) and Lomeland et al⁴⁶ for further information on the model, as well as the [SGOFLET – Gas-Oil LET Relative Permeability Functions](#), [SGWFLET – Gas-Water LET Relative Permeability Functions](#), and the [SWOFLET – Water-Oil LET Relative Permeability Functions](#) keywords in the [PROPS](#) section.

Corey⁴⁷ combined the work of Purcell⁴⁸ and Burdine⁴⁹ that was widely accepted for its simplicity. His original equations were developed for the drainage cycle in water-wet sandstones, but have also been used in carbonate formations. Corey's original water-oil equations were as follows:

$$k_{ro}(S_w) = \left[\frac{1 - S_w}{1 - S_{wcr}} \right]^{n_o} \quad (2.1)$$

$$k_{rw}(S_w) = \left[\frac{S_w - S_{wcr}}{1 - S_{wcr}} \right]^{n_w} \quad (2.2)$$

where:

$k_{ro}(S_w)$	= relative permeability to oil,
$k_{rw}(S_w)$	= relative permeability to water,
n_o	= Corey oil exponent, set to four in the original paper,
n_w	= Corey water exponent, set to four in the original paper,

⁴⁶ Lomeland F., Ebeltoft E. and Thomas W.H., 2005. A New Versatile Relative Permeability Correlation. Paper SCA2005-32 presented at the International Symposium of the Society of Core Analysts held in Toronto, Canada, 21-25 August, 2005.

⁴⁷ Corey, A.T.: "The Interrelation Between gas and Oil Relative Permeabilities", *Production Mon.*, 19. 38. (1954).

⁴⁸ Purcell, W. R., "Capillary Pressures- Their Measurement Using Mercury and the Calculation of Permeability Therefrom", *Transactions AIME*, 186, 39 (1949).

⁴⁹ Burdine, N.T., "Relative Permeability Calculations from Pore Size Distribution Data", *Transactions AIME*, 198, 71 (1953).

S_w = water saturation, and
 S_{wcr} = critical water saturation.

The denominator in equations (2.1) and (2.2) scales the water saturation to the mobile water phase. There are several forms of these equations, with the most common normalizing the saturation over the mobile hydrocarbon phase, as depicted in equations (2.3) and (2.4).

$$k_{ro}(S_w) = k_{row} \left[\frac{1 - S_w - S_{orw}}{1 - S_{orw} - S_{wcr}} \right]^{n_o} \quad (2.3)$$

$$k_{rw}(S_w) = k_{rww} \left[\frac{S_w - S_{wcr}}{1 - S_{orw} - S_{wcr}} \right]^{n_w} \quad (2.4)$$

where:

k_{row} = maximum oil relative permeability at S_{wcr} ,
 k_{rw} = maximum water relative permeability at S_{orw} ,
 S_{wcr} = critical water saturation, and
 S_{orw} = residual oil saturation under a water flood ([SOWCR](#)).

Figure 2.21 shows a typical Corey plot, with the plot at the top of the figure showing the Corey plot and the bottom plot showing the derivative plot, the data is also tabulated in Table 2.2. Notice in Figure 2.21 that the water curve does not extend to 100% water saturation, this is an error as the water below the oil-water contact would have restrictive water flow.

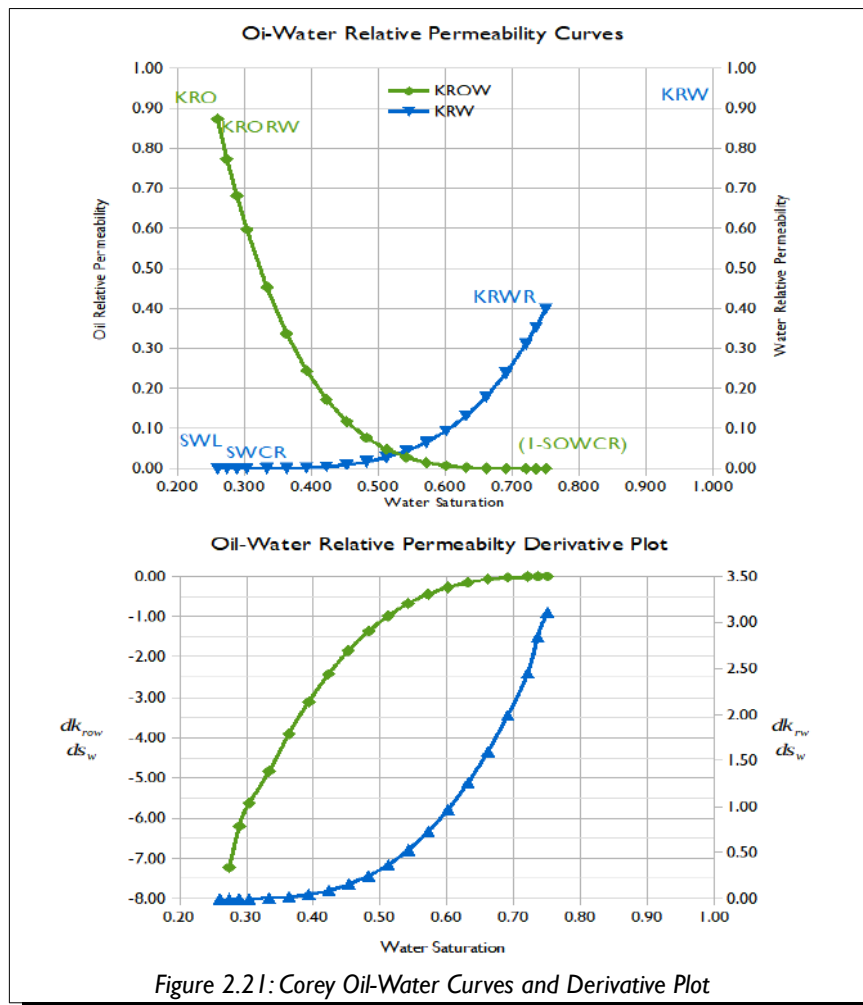


Figure 2.21: Corey Oil-Water Curves and Derivative Plot

Oil-Water Saturation Function and Derivative Table						
Sw	Krw	Krow	Pcow (psia)	$\frac{dk_{rw}}{dS_w}$	$\frac{dk_{row}}{dS_w}$	$\frac{dP_{cow}}{dS_w}$
0.2600	0.0000	0.8750	16.4040	0.00	0.00	0.00
0.2740	0.0000	0.7737	13.4170	0.00	-7.23	-213.13
0.2889	0.0000	0.6814	11.0910	0.00	-6.20	-156.36
0.3038	0.0000	0.5976	9.2570	0.00	-5.63	-123.34
0.3335	0.0002	0.4535	6.6120	0.01	-4.84	-88.90
0.3633	0.0008	0.3372	4.8610	0.02	-3.91	-58.87
0.3930	0.0022	0.2448	3.6610	0.05	-3.11	-40.33
0.4228	0.0049	0.1728	2.8150	0.09	-2.42	-28.44
0.4525	0.0096	0.1181	2.2030	0.16	-1.84	-20.55
0.4823	0.0171	0.0775	1.7520	0.25	-1.36	-15.18
0.5120	0.0282	0.0484	1.4120	0.37	-0.98	-11.42
0.5418	0.0440	0.0283	1.1520	0.53	-0.67	-8.74
0.5715	0.0656	0.0153	0.9500	0.73	-0.44	-6.78
0.6013	0.0944	0.0074	0.7920	0.97	-0.27	-5.34
0.6310	0.1318	0.0030	0.6650	1.26	-0.15	-4.25
0.6608	0.1793	0.0010	0.5640	1.60	-0.07	-3.42
0.6905	0.2386	0.0002	0.4810	1.99	-0.03	-2.78
0.7203	0.3115	0.0000	0.4130	2.45	-0.01	-2.28
0.7351	0.3537	0.0000	0.3840	2.84	0.00	-1.97
0.7500	0.4000	0.0000	0.3570	3.11	0.00	-1.80
1.0000	0.9000	0.0000	0.1270	2.00	0.00	-0.92

Table 2.2: Oil-Water Saturation Function and Derivative Table

If we correct the data shown in Figure 2.21 by extending the water curve to *KRW*, then we obtain the result shown in Figure 2.22, but notice what has happened to the water derivative plot, which now has a discontinuity because of the extension. Thus, small changes to the relative permeability curves can have an undue influence on the simulator's performance. In this particular case, the effect is minor as it is only one point that has the issue, but it is not uncommon to have multiple discontinuities in numerous curves that will have an impact on convergence.

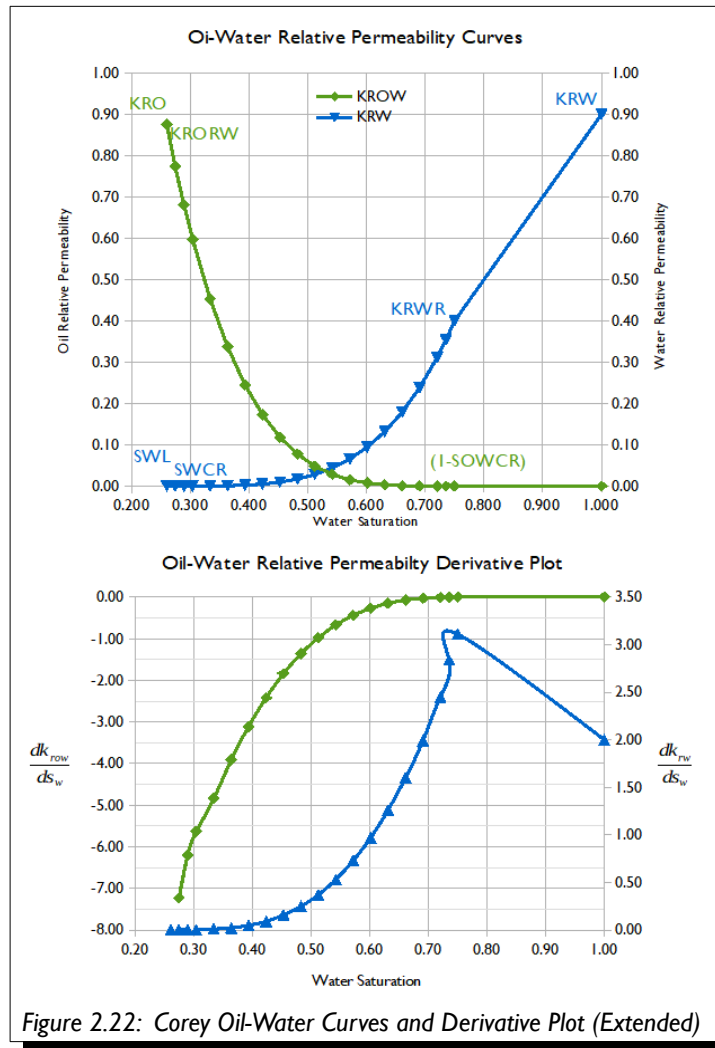


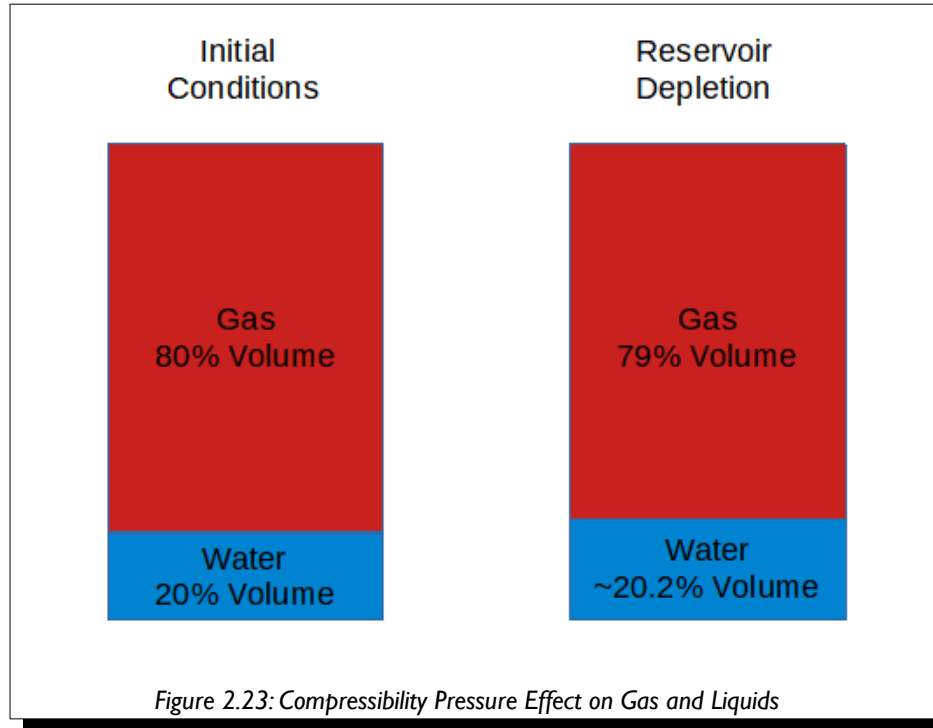
Figure 2.22: Corey Oil-Water Curves and Derivative Plot (Extended)

To summarize, if convergence is an issue verify that both the entered relative permeability data and capillary pressure do not have numerous discontinuities in the derivative plots.

The next section discusses the impact and relationship of lowest saturation and critical saturation, in terms of pressure depletion. However, again, care should be taken to avoid any potential discontinuities in the derivatives between these to data points.

PROPS Section: Saturation Function (Lowest Saturation and Critical Saturation)

In most cases, time step chops caused by saturation changes may be reduced by avoiding having the lowest and critical saturations be the same. For example for the water phase, this would mean having different values for [SWL](#) and [SWCR](#). The reason for this due to the compressibility difference between the gas and liquids (oil and water), as illustrated Figure 2.23. Here we assume that initially 80% of cell's volume of 100 m^3 is occupied by gas and the remaining 20% by water.



Assuming there is no flow in or out of the grid block and the cell pressure decreases due to production elsewhere in the model, that results in a small reduction (1.0%) in the [cell's pore volume](#). And since gas is much more compressible than oil and water, then the gas volume will decrease by approximately 1.0%; whereas the water will increase slightly to approximately 20.2%⁵⁰. Thus, if [SWL](#) was set to 20%, then the 0.2% increase in water saturation would become mobile and start to flow to the other connect grid blocks. This can be a cause of instability.

The solution is to enter values for [SWL](#) and [SWCR](#), such that there is a small difference between them, around 2% to 5% for example, instead of this:

```
--
--      WATER RELATIVE PERMEABILITY TABLES (SWFN)
--
SWFN
--      SWAT      KRW      PCOW
--      FRAC      FRAC      PSIA
--      -----
--      0.30      0.0000      0.6190
--      0.35      0.0005      0.3891
--      0.40      0.0028      0.2279
--      0.50      0.0237      0.0602
--      0.60      0.0931      0.0077
--      0.80      0.5000      0.0000
--      0.90      0.7000      0.0000
--      1.00      0.9000      0.0000
--
-- / TABLE NO. 1
```

⁵⁰ Assuming that the initial pore volume of the grid block was 100 m^3 , then a 1.0% reduction would result in a pore volume 99 m^3 , and 20 m^3 of water in 99 m^3 pore volume results in 20.2% of the grid block being occupied by water.

Amend the table as follows:

```
--
--      WATER RELATIVE PERMEABILITY TABLES (SWFN)
--
SWFN
--      SWAT      KRW      PCOW
--      FRAC      FRAC      PSIA
--      -----
--      0.30      0.0000      0.6190
--      0.31      0.0000      1*
--      0.35      0.0005      0.3891
--      0.40      0.0028      0.2279
--      0.50      0.0237      0.0602
--      0.60      0.0931      0.0077
--      0.80      0.5000      0.0000
--      0.90      0.7000      0.0000
--      1.00      0.9000      0.0000
```

/ TABLE NO. 1

The incorporation of the row for Sw equal to 0.31 should assist convergence and will not influence the oil in-place, although it will slightly reduce the water mobility between the water saturations of 0.30 and 0.31. Secondly, as one can see from Table 2.3 the additional water saturation value did not cause any derivative discontinuities.

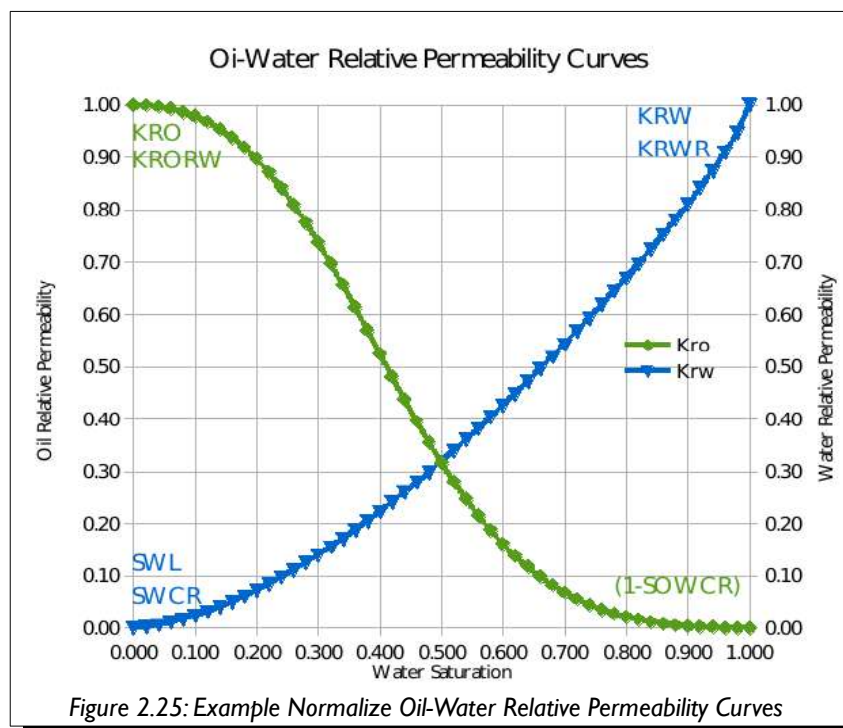
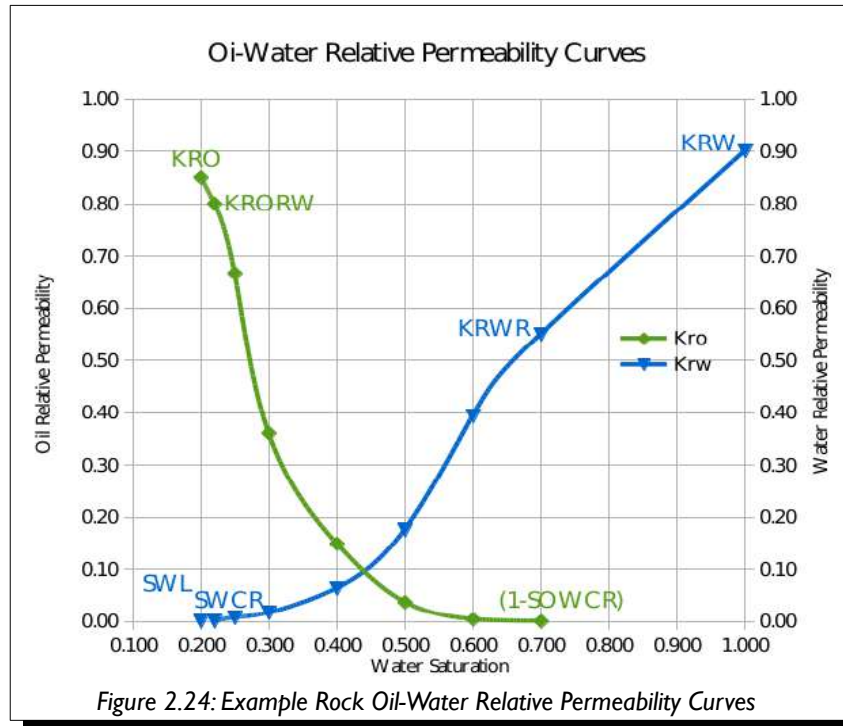
Oil-Water Saturation Function and Derivative Table						
Sw	Krw	Krow	Pcow (psia)	$\frac{dk_{rw}}{dS_w}$	$\frac{dk_{row}}{dS_w}$	$\frac{dP_{cow}}{dS_w}$
0.30	0.0000	0.6190	9.7247			
0.31	0.0000	0.5918	9.1501	0.00	-5.44	-114.92
0.35	0.0005	0.3891	5.6425	0.01	-4.50	-77.95
0.40	0.0028	0.2279	3.4623	0.05	-3.22	-43.60
0.50	0.0237	0.0602	1.5494	0.21	-1.68	-19.13
0.60	0.0931	0.0077	0.7989	0.69	-0.52	-7.50
0.80	0.5000	0.0000	0.3110	2.03	-0.04	-2.44
0.90	0.7000	0.0000	0.2190	2.00	0.00	-0.92
1.00	0.9000	0.0000	0.1270	2.00	0.00	-0.92

Table 2.3: Oil-Water Saturation Function and Derivative Table (Modified)

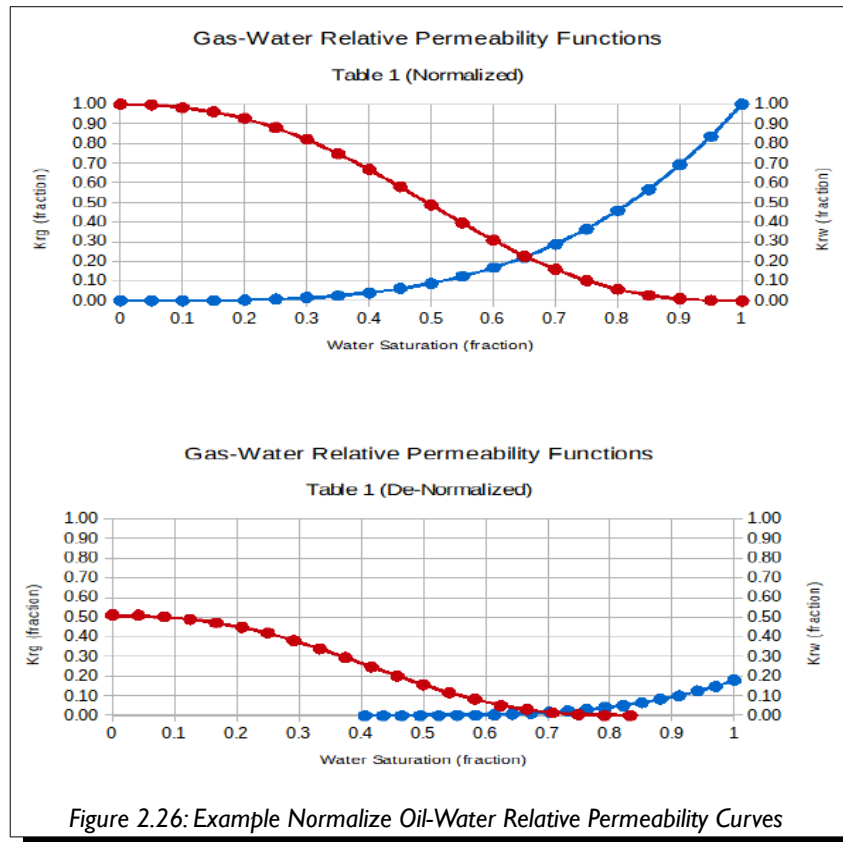
Note also that same approach can also be applied to the oil phase.

PROPS Section: Saturation Table End-point Scaling

The end-point scaling functionality in OPM Flow gives tremendous flexibility, allowing one to use a limited set of relative permeability functions by accounting for rock property variation, such that each cell in the model has its own relative permeability function. However, both the original (normalized) input functions and the grid cell's scaled (or de-normalized) functions should be checked for consistency. In general, it is best to input the original curves as standardized curves that have distinct end-points, rather than fully normalized curves, in the classical sense. The reason for this is shown in Figure 2.24 and Figure 2.25.



Notice how the distinction between *KRO* and *KRORW* is no longer present in the normalized curve in Figure 2.25, and similarly for *KRW* and *KRWR*, and *SWL* and *SWCR*. Thus, the simulator will ignore one of each pair of end-points - this can lead to unexpected results for the de-normalized grid cell based saturation functions. An example is shown in Figure 2.26 for a gas-water saturation table. Notice how the de-normalized end-point water relative permeability, *KRW*, is only 0.20 versus 1.00 for the normalized value of *KRW*.



Consequently, water influx will be inhibited leading to unreliable results in terms of water production, and potentially gas production if aquifers are present in the model.

As mentioned previously, entering a correct original curve that is not normalized will automatically resolve these type of issues. If a normalized curve is entered, then ensure that simulator can distinguish between *SWL* and *SWCR* etc., based on the variable definitions provided in Table 8.6, Table 8.7 and Table 8.8.

Finally, see the SCALCERS keyword in the *PROPS* section, that defines the options used in the re-scaling process, including the options for two point and three point scaling. Whatever option is selected, always verify that the de-normalized curves that will be used in the model are correct and reasonable.

PROPS Section: Fluid Properties

It is important that the fluid properties are consistent to avoid convergence issues, and thus the simulator performs a series of validation checks to ensure that data is consistent. The following example indicates the oil formation volume factor is not monotonically increasing for increasing GOR.

--					
--		OIL PVT TABLE FOR LIVE OIL			
--					
PVTO					
--	RS	PSAT	BO	VISC	
--	MSCF/STB	PSIA	RB/STB	CPoise	
--	-----	-----	-----	-----	
.....					
					/ TABLE NO. 43
	0.000	1.010	1.0388000	120924.000	
		2510.0	0.9801748	529773.445	/
	3.909	10.0	1.0563000	45850.000	
		2510.0	0.9969008	200315.440	/
	5.725	20.0	1.0617000	34198.000	
		2510.0	1.0022360	148947.825	/
	6.787	30.0	1.0648000	28808.000	
		2510.0	1.0054019	125083.718	/
	7.540	40.0	1.0671000	25508.000	
		2510.0	1.0078136	110411.450	/
	8.125	50.0	1.0688000	23210.000	
		2510.0	1.0096595	100151.779	/
	9.940	100.0	1.0742000	17312.000	
		2510.0	1.0159689	73535.291	/
	11.002	150.0	1.0774000	14583.000	
		2510.0	1.0202071	60960.876	/
	11.756	200.0	1.0797000	12912.000	
		2510.0	1.0235993	53105.652	/
	12.340	250.0	1.0814000	11749.000	
		2510.0	1.0264272	47530.728	/
	12.818	300.0	1.0829000	10877.000	
		2510.0	1.0290688	43270.164	/
	13.221	350.0	1.0841000	10191.000	
		2510.0	1.0314284	39854.509	/
	13.571	400.0	1.0851000	9631.000	
		2510.0	1.0336002	37015.563	/
	13.880	450.0	1.0861000	9163.000	
		2510.0	1.0357743	34599.472	/
	14.156	500.0	1.0869000	8763.000	
		2510.0	1.0377596	32498.634	/
	14.405	550.0	1.0877000	8417.000	
		2510.0	1.0397467	30648.326	/
	14.633	600.0	1.0884000	8113.000	
		2510.0	1.0416399	28994.746	/
	14.843	650.0	1.0890000	7843.000	
		2510.0	1.0434389	27501.353	/
	15.037	700.0	1.0896000	7601.000	
		2510.0	1.0452393	26140.639	/
	17.800	2500.0	1.0222000	4436.000	
		2510.0	1.0219701	4495.778	/
					/ TABLE NO. 44

The data is displayed above and the offending entries are shown in bold. These type of PVT warning messages are important and should be addressed before continuing with the simulation run.

SOLUTION Section: Initialization

The first quality control check that should be performed once the model has been initialized, is to check that the pressure distribution matches the observed data, this is especially important when there are multiple stacked reservoirs in different hydraulic pressures zones, as per the likes of the oil and gas fields in the Malay Basin and the deep water fields in Indonesia and the Gulf of Mexico. One effective way to do this is to plot the initial pressure data versus depth from exploration wells, and compared this to the equivalent model pressures at the same depth, as shown in Figure 2.27. Here the measured data is from the RFT⁵¹ ("Repeat Formation Tester") tool that measures the observed pressure profile in the open-hole well bore.

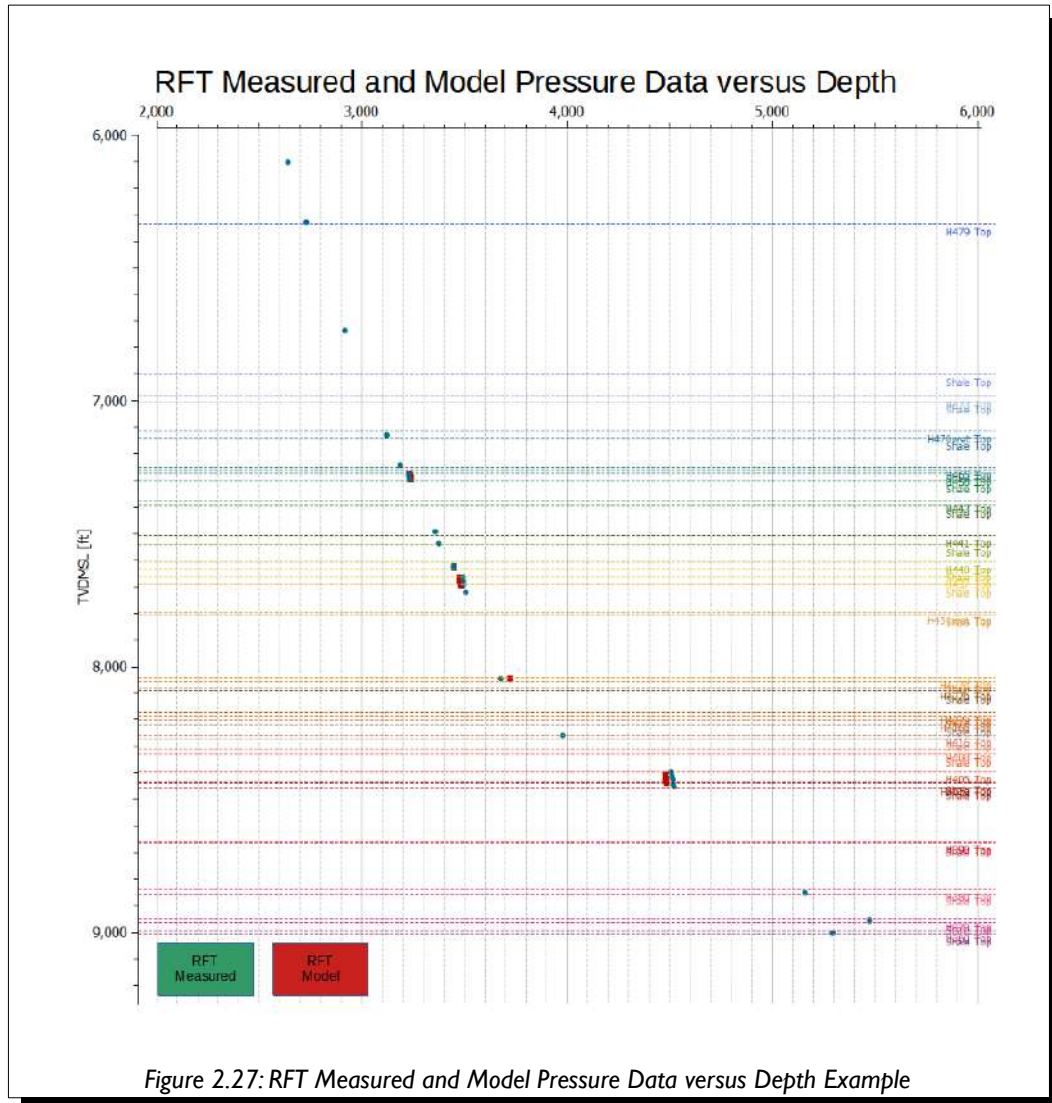


Figure 2.27 was created in OPM ResInsight, and shows good agreement between the two data sets. This type of plot is commonly constructed when assessing the history matching field performance, but is also useful in verifying the initial pressure distribution. Note that for simple models with only a few reservoir zones, then this plot can be replaced by simply checking the three dimensional pressure distribution.

⁵¹ Repeat Formation Tester (RFT) tool is an open hole device which is an updated version of the Formation Interval Tester (FIT), both of which are run on wire line. Both tools take multiple pressure readings (at various depths) thus enabling a pressure depth profile to be obtained from the formation, and, in addition, they can also take fluid samples from the formation. The latest tool available is the Modular Formation Dynamics Tester (MDT), which, as its name suggests, is a modular tool that can be assembled in different configurations depending on what are the objectives for running the tool. Thus, the term RFT measurements applies to all tools that measure a pressure profiles versus depth (RFT/FIT/MDT etc.).

If the [SWATINIT](#) keyword in the [PROPS](#) section is used to initialize the model by setting each grid block's initial water saturation ("Sw"), then OPM Flow will re-scale the water-oil capillary pressure curves entered via the [SWFN](#) saturation functions in the [PROPS](#) section, so that the resulting initialized Sw matches the values in the [SWATINIT](#) array.

Normally the [SWATINIT](#) array is generated in the static earth model when calculating the hydrocarbons in-place volumes using a Saturation Height Functions ("SHF") derived from capillary pressure functions. Static earth models do not directly use capillary pressure in these type of calculations as individual cell pressures are not available. There is therefore some potential for inconsistencies to arise between the two sets of formulations. This is normally manifested by extreme scaling in the scaled capillary pressure values calculated by the simulator. These large scaled capillary pressures values can result in numerical issues, and should be addressed, by verifying that the SHF used in the earth model to derive the [SWATINIT](#) array and the unscaled capillary pressure values on the [SWFN](#) keyword are consistent. Note in general, the scaled capillary pressure values calculated by the simulator, should be roughly in the same range as the entered unscaled data, as shown in Figure 2.28 for the Norne field.

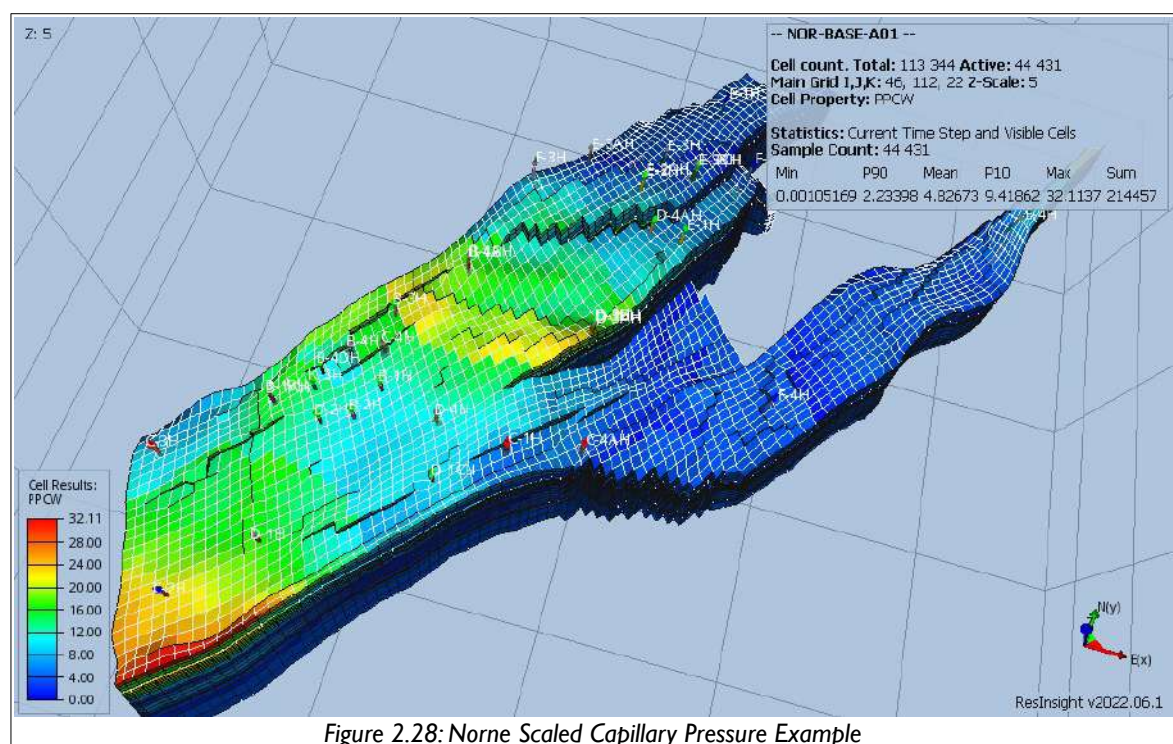
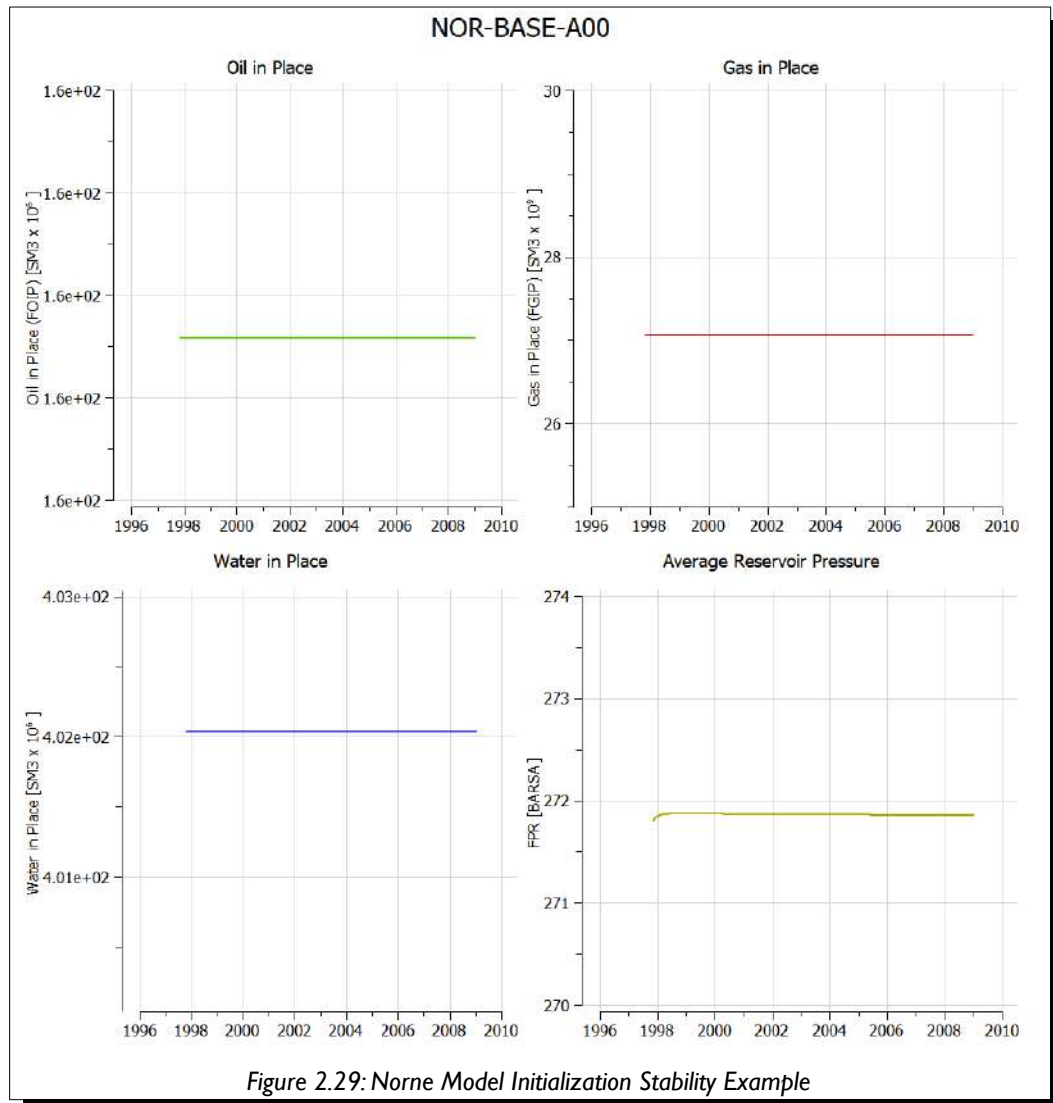


Figure 2.28: Norne Scaled Capillary Pressure Example

Note that although the [PPCWMAX](#) keyword can be used to set a maximum scaled capillary pressure value, a more technically sound approach is to resolve these inconsistencies before continuing with the model build.

Once a model is built and can be initialized, it is good practice to run the model with no production for several years to ensure that the initialization is stable. Under these circumstances, there should be no fluid movement, or only very limited movement, and the pressure should remain stable, that is constant, although a small drift may be observable. An example for the Norne model is shown in Figure 2.29 on the field level, although it can, and should be performed on a region by region basis if the model is not stable.



Potential causes for the instability are:

- 1) Aquifer Not in Equilibrium with the Connected Hydrocarbon Zones. One can check for this by switching off all the aquifer connections and see if the instability is still present. If this corrects the issue then verify the aquifer connections by reviewing the AQUIFERA array in the *.INIT file in OPM ResInsight. OPM Flow writes out the AQUIFERA value as $2^{(AQUID-1)}$ for cells connected to aquifer AQUID. If a cell is connected to multiple analytical aquifers then AQUIFERA is summed for all aquifers connected to a cell. Note that connecting cells to multiple aquifers should be avoided.

Note aquifers should only be connected to grid blocks containing 100% water, or at the very least any hydrocarbons in the connected cells should be immobile.

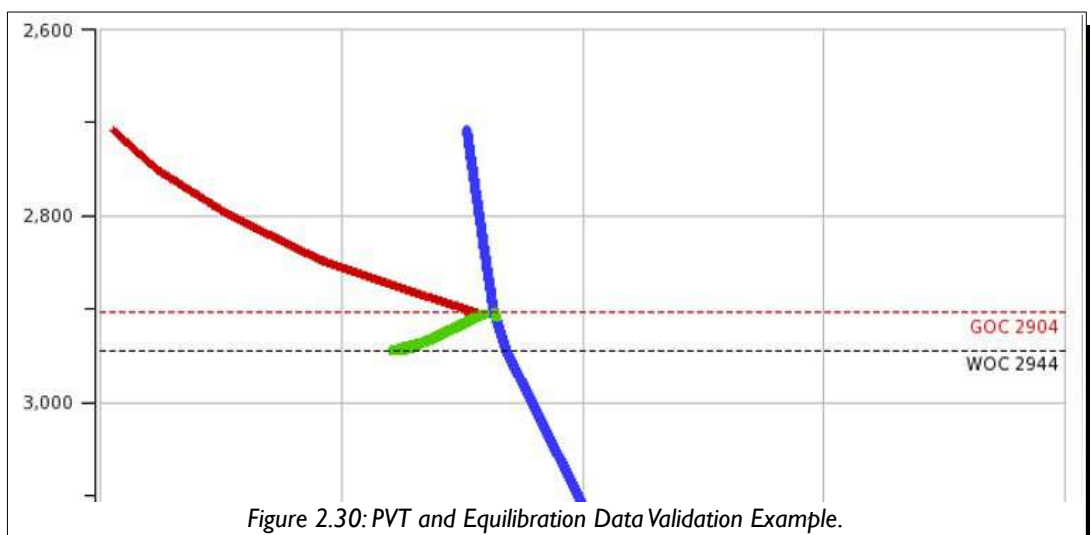
- 2) Different PVT Regions in Communication. Recall that fluid properties are associated with a grid cell, so if two different PVT regions are in communication, then a m^3 of fluid from PVT region one flowing into PVT region two, will automatically assume the properties of PVT region two. This creates a material balance error that will result in numerical difficulties. One may be able to identify this by reviewing the inter-region flows to identify the source of the problem, either via the *.PRT file or the summary file out. It typically occurs across fault planes, when one PVT region butts against a different PVT region along the fault face.

There are several solutions to this, firstly, one can set the transmissibility along the fault plane to zero using the [MULTFLT](#) keyword in the [GRID](#) or [EDIT](#) sections.

Secondly, create a [MULTNUM](#) array based on the [PVTNUM](#) array, as the [PVTNUM](#) keyword is not available in the [GRID](#) and [EDIT](#) sections. Then use the [MULTREGT](#) keyword with [MULTNUM](#) array in the [GRID](#) or [EDIT](#) sections to set the transmissibility between the offending regions to zero.

Finally, one can use the [THPRES](#) keyword in the [SOLUTION](#) section combined with the [EQLNUM](#) array to isolate the offending regions. Normally, but not always, the [PVTNUM](#) and [EQLNUM](#) arrays are identical, if not, then it is relatively easy to copy the aforementioned [MULTNUM](#) array to the [EQLNUM](#) array, then add the additional equilibration data to the [EQUIL](#) keyword. One advantage of this option is that it will cater for the scenario when the PVT regions are initially isolated, but after a certain production period, the faults break down and there is communication between the two regions. In this scenario, one still has the material balance error as different fluids inter-mix. If this is the case then the [API](#) Tracking option should be used to account for this behavior. See the [API](#) keyword in the [RUNSPEC](#) on how to activate this option.

- 3) Different Equilibration Regions in Communication. This is similar to (2) above and the remedies are also similar, so the solution is not repeated here.
- 4) PVT Data Not Consistent with Equilibration Data. In an oil-gas system the saturation pressure, that is the bubble-point pressure should be the pressure at the GOC, and similarly for the dew point. Conveniently, OPM ResInsight can produce a plot, as depicted in Figure 2.30. The figure illustrates that gas dew point pressure gradient (shown in red), and the oil bubble-point pressure gradient (shown in green) from the PVT data intersect at the GOC, which matches the initial pressure profile (shown in blue) from the grid cells. Thus, confirming the consistency of the two data sets.

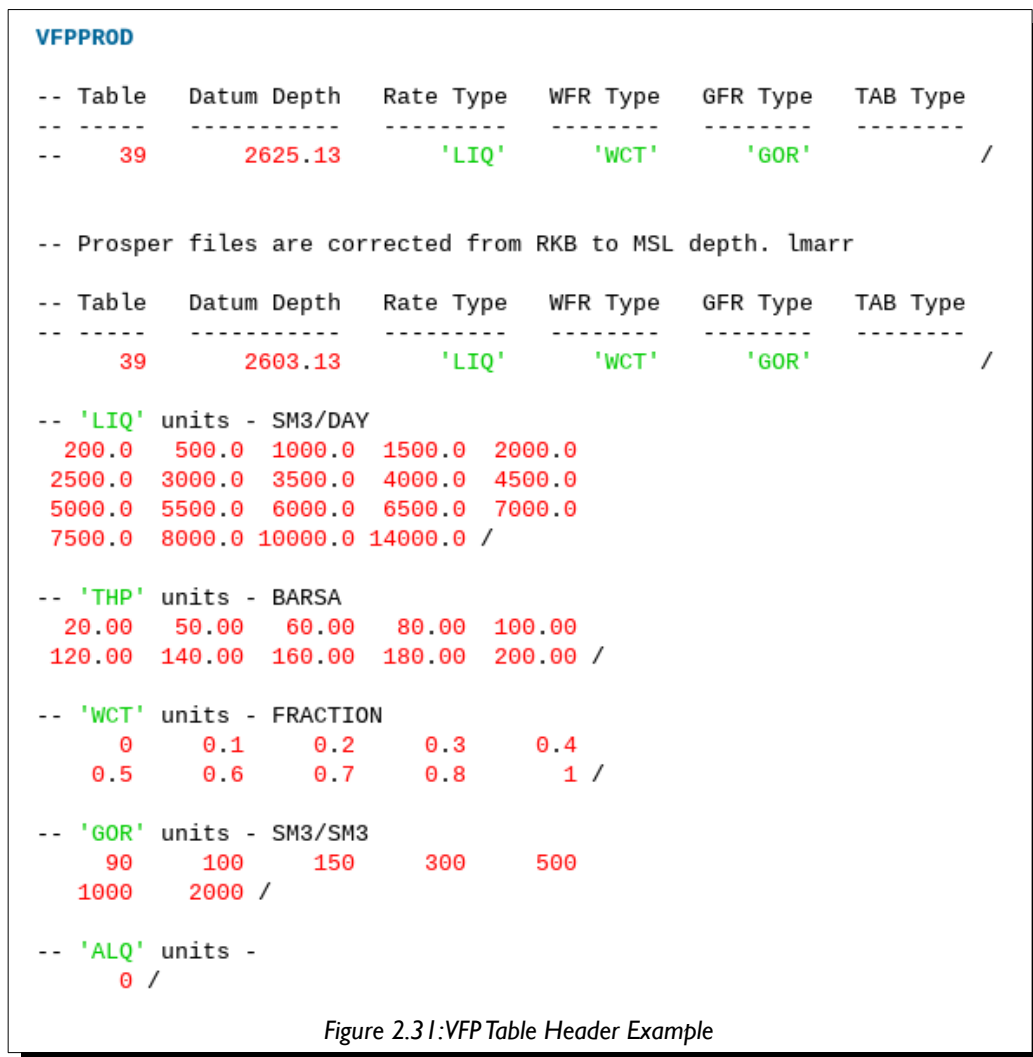


SCHEDULE Section

A common issue in most simulation models that employ Vertical Lift Performance ("VLP") tables to relate a well's Tubing Head Pressure ("THP") to its Bottom-Hole Pressure ("BHP"), is the consistency of the data. Note that this is not just limited to wells, but also the network pipelines, that describe the pressure drop between the inlet node and the outlet node.

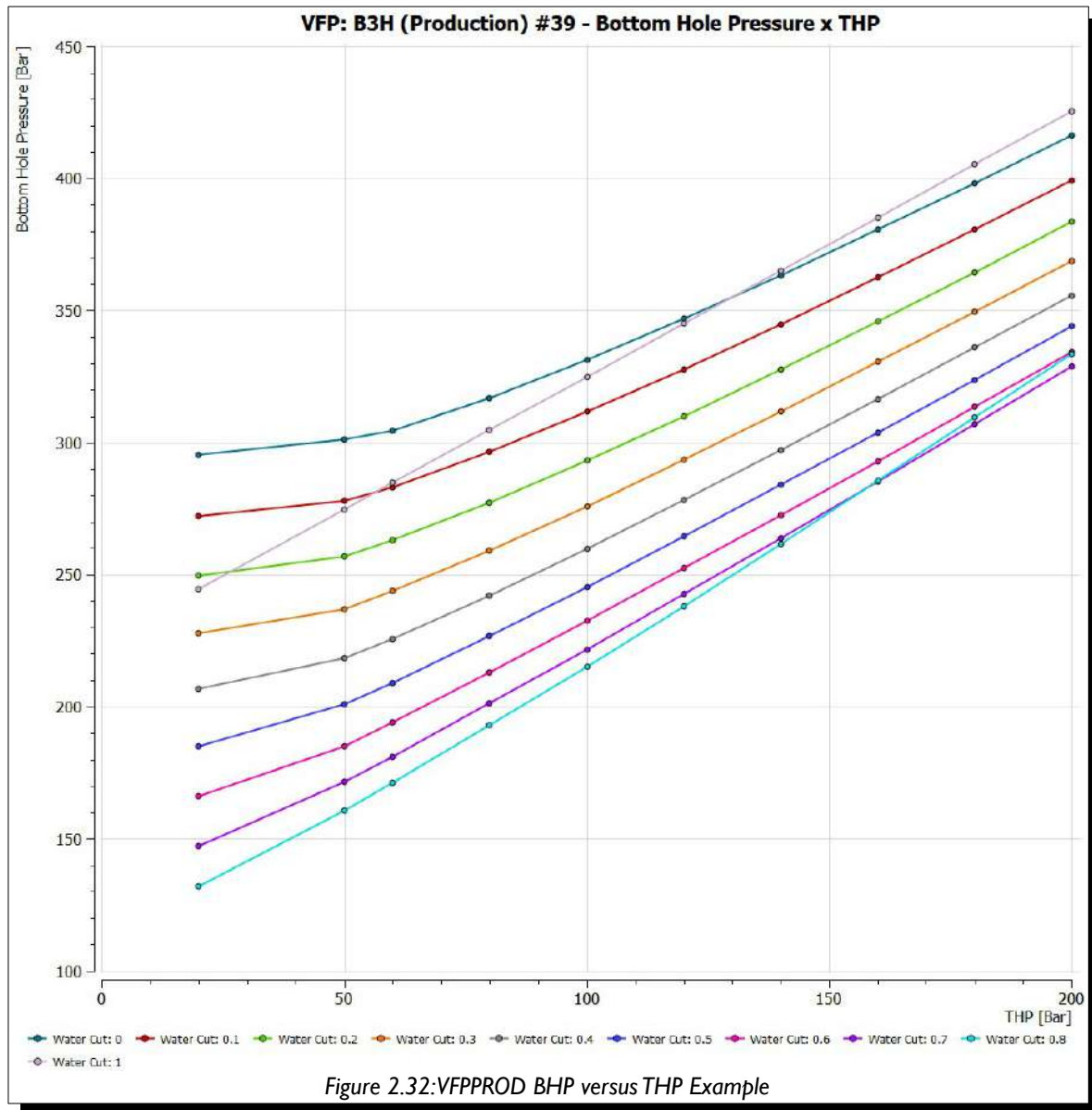
Warning: VFPPROD table 39 has 5 non-monotonic points of BHP(THP)
 In /media/OPM/Fields/Norne/Model/INCLUDE/VFP/B3H.Ecl line 68
 This may cause convergence issues due to switching between BHP and THP control.

Generally, the cause of this warning is because the VFP tables have been generated with too much fidelity or resolution. Users often consider that having more detailed and higher resolution VFP tables will give more accurate results. This is not always the case and in this particular example, may very well end up causing the wells using this particular VFP table to oscillate between control modes. Figure 2.31 shows the VFP header section of the offending table, that declares the values used to generate the table.



At first glance this looks reasonable, but the problem is that the variables (LIQ, THP, WCT and GOR) are at a too high resolution, which causes some of the resulting curves to cross, due to the numerical accuracy when generating the curves. In addition, note that the maximum water cut is defined as 100%, which is unlikely to occur in practice, perhaps 98% would be more appropriate as the maximum value. OPM

ResInsight is able to load [VFPPROD](#) and [VFPINJ](#) tables and to plot the data. Here, Figure 2.32 shows a plot of BHP versus THP with the various water cut values represented by the individual curves. The figure clearly illustrates the crossing of the water cut line of 80% over the 70% water cut curve, as well as the issue with the 100% water cut curve.



The solution is to re-generate the offending VFP tables using a coarser set of variable values. The Petroleum Experts' PROSPER⁵² program, as well as other nodal analysis software programs used to generate VFP tables for simulators, have options to generate the parameters based on the number of entries, and the minimum and maximum values for a variable, combined with various interval spacing options. Keeping the same range and selecting four or five entries and using the geometric spacing option normally resolves this type of issue. The simulator will then interpolate the coarser curves and avoid any issues with crossing curves. Note that it is important to ensure that the variable ranges cover the full range of values expected to occur during the simulation, as extrapolation, unlike interpolation of the curves, is more prone to be problematic.

⁵² IPM Prosper Version 16.5 User Manual - November 2020, Petroleum Experts, Edinburgh, Scotland (www.petex.com).

Note that this warning message can be safely ignored:

```
Warning: LIQ preferred phase not supported for well OP-01, using OIL instead
```

as it refers to the reference phase in a well's productivity index calculation. This is because OPM Flow only currently supports [OIL](#), [GAS](#) and [WATER](#) well type fluids, as defined by the [WELSPECS](#)(TYPE) parameter in the [SCHEDULE](#) section, i.e., liquid (LIQ) is not supported. For producing wells this mostly matters if one plots the WPI summary vector (productivity index for well's preferred phase). In the current treatment WPI will not have contributions from the water phase if the declared preferred phase is LIQ. Thus, if history matching a well's actual productivity index, this should be done on one of the recognized fluid well types. For injection wells the preferred phase defined by the [WELSPECS](#) keyword does not matter, since the preferred phase is (typically) reset to the injected phase via the [WCONINJE](#) or [WCONINJH](#) keywords.

2.4.2 CONVERGENCE AND NUMERICAL PERFORMANCE

Given that all the potential issues with the input data have been successfully resolved, there may still be problems with convergence and numerical performance, due to the nature of the problem the simulator is trying to solve. In Figure 2.33, at first glance, the non-linear iterations would appear to be reasonable, ranging from one to four, combined with linear iterations averaging around ten per time step. However, the time steps are very small resulting in nearly 12,500 Newton iterations and over 60,000 linear iterations in the commercial simulator; for this relatively simple problem. In fact, at the time of the study⁵³, the commercial simulator was unable to complete this simulation. Re-running the model with the latest version of the commercial simulator (Figure 2.33), still gives incorrect results compared to the published data.

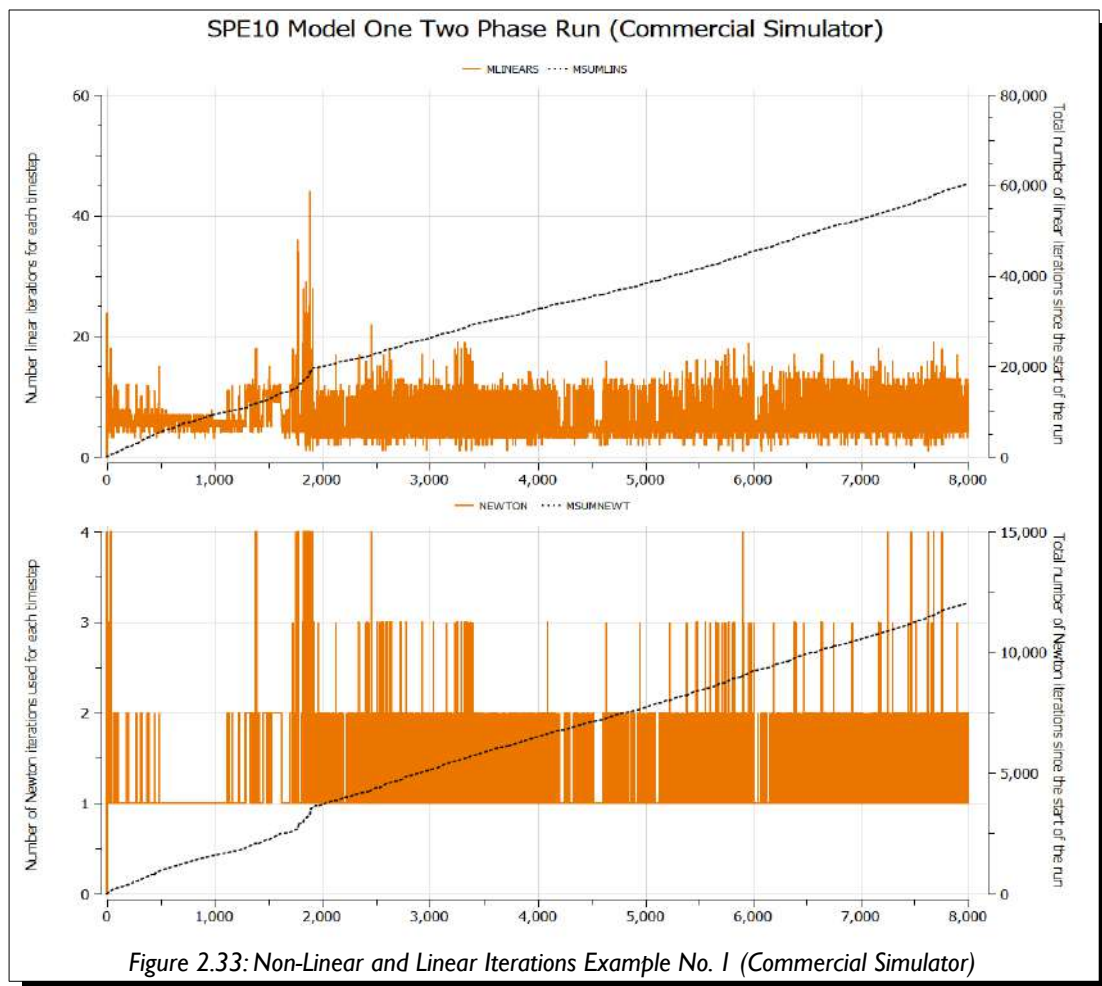


Figure 2.33: Non-Linear and Linear Iterations Example No. 1 (Commercial Simulator)

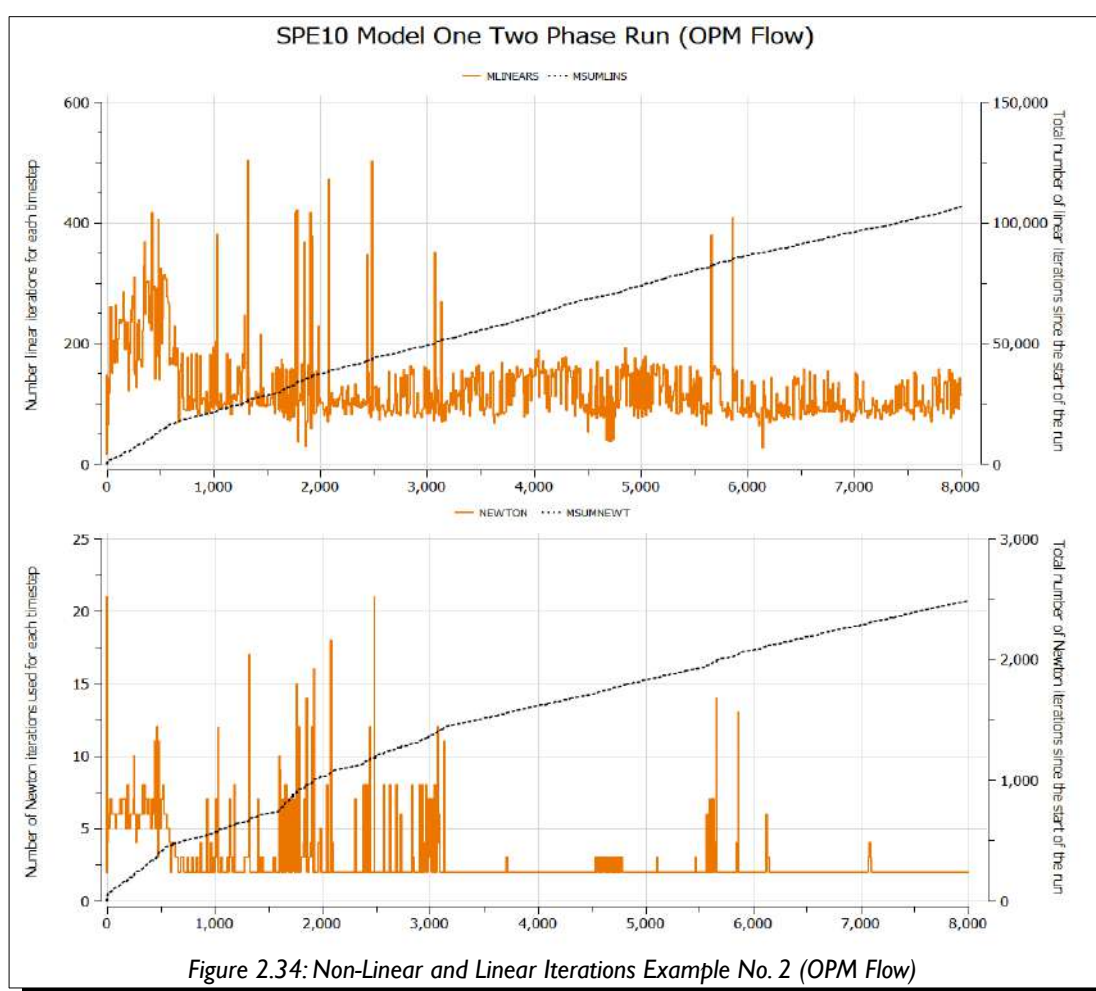
Fortunately, there are several ways to remedy poor performance, including:

- 1) The simplest and common approach is to manually restrict the time step size to reduce the number of time steps chops. Time step chops occurs when the simulator fails to converge at a given time step (t_{n+1}), and therefore restarts at the last successful time step (t_n) and advances to the next time step using a smaller time step size (Δt_{n+1}) than before (Δt_n). This was the approach used for the example in Figure 2.33, which although the model now runs, the results are incorrect.
- 2) Alternatively, one can change the numerical controls for the Newton-Raphson ("Newton" or non-linear solver) iteration, by changing the minimum and/or maximum number of Newton iterations. Although in general this may resolve some time step chops it can also result in a significant increase in run times, and should therefore be used with caution.

⁵³ Christie, M. A., and M. J. Blunt. "Tenth SPE Comparative Solution Project: A Comparison of Upscaling Techniques." *SPE Res Eval & Eng* 4 (2001): 308–317. doi: <https://doi.org/10.2118/72469-PA>.

- 3) The last option is to change the numerical parameters for the linear solver, which is generally preferable to modifying the Newton iteration parameters. OPM Flow contains various options in terms of the type of linear solver and the linear solver preconditioner to be employed, in addition to various standard linear solver parameters employed by the commercial simulator (minimum, maximum number of iterations etc.).

For example, Figure 2.34 illustrates the impact of selecting the more appropriate numerical controls. Here, the number of Newton iterations are greatly reduced to around 2,500 at the expense of increasing the number of linear iterations from ~60,000 to over 100,000. More importantly, the simulator results are correct, in that they are close to the published oil, gas, and pressure values⁵³.



Note that the simulator has various other numerical performance parameters than can use used to identify the cause for under performance.

Note

To obtain the numerical performance data, the easiest way is to add the [PERFORMA](#) keyword in the [SUMMARY](#) section, that enables the writing out of a series of performance keywords to the summary file, which can then be plotted in OPM ResInsight as per Figure 2.33 and Figure 2.34.

In addition to the aforementioned [PERFORMA](#) keyword in the [SUMMARY](#) section, which is available for both OPM Flow and the commercial simulator, OPM Flow has two additional numerical performance reports that may help in identifying numerical issues. Both of these OPM Flow specific reports tabulate the numerical performance data available in the print file, in an easily readable format.

The first is the INFOSTEP report illustrated in Figure 2.35, which summarizes the numerical performance for each time step. The Assembly, LSetup, LSolve, LocSol, Update and Output header columns represent the time in seconds taken to complete the particular work item.

Time(day)	TStep(day)	Assembly	LSetup	LSolve	LocSol	Update	Output	WellIt	Lins	NewtIt	LinIt	Conv
0	1	0.0117	0.0217	0.0345	0.0000	0.0057	0.0000	0	16	15	19	1
1	0.204970966	0.0025	0.0037	0.0061	0.0000	0.0012	0.0000	0	4	3	4	1
1.204970966	0.446033467	0.0025	0.0036	0.0065	0.0000	0.0013	0.0000	0	4	3	4	1
1.651004434	0.617940361	0.0026	0.0037	0.0062	0.0000	0.0012	0.0000	0	4	3	4	1
2.268944796	1.150119293	0.0034	0.0050	0.0082	0.0000	0.0017	0.0000	0	5	4	4	1
3.419064089	2.527926281	0.0051	0.0100	0.0173	0.0000	0.0028	0.0000	0	8	7	11	1
5.94699037	2.026504815	0.0039	0.0063	0.0107	0.0000	0.0021	0.0000	0	6	5	7	1
7.973495185	2.026504815	0.0031	0.0055	0.0089	0.0000	0.0017	0.0000	0	5	4	5	1
10	5.268912519	0.0039	0.0068	0.0123	0.0000	0.0021	0.0000	0	6	5	8	1
15.26891252	4.731087481	0.0031	0.0052	0.0094	0.0000	0.0016	0.0000	0	5	4	6	1
20	10	0.0049	0.0085	0.0189	0.0000	0.0027	0.0000	0	8	7	18	1
30	10	0.0042	0.0085	0.0158	0.0000	0.0024	0.0000	0	7	6	13	1
40	10	0.0044	0.0075	0.0142	0.0000	0.0024	0.0000	0	7	6	11	1
50	10	0.0031	0.0048	0.0091	0.0000	0.0017	0.0000	0	5	4	6	1
60	10	0.0034	0.0050	0.0116	0.0000	0.0017	0.0000	0	5	4	11	1
[...]												
420	10	0.0025	0.0037	0.0078	0.0000	0.0013	0.0000	0	4	3	7	1
430	10	0.0031	0.0064	0.0115	0.0000	0.0017	0.0000	0	5	4	8	1
440	10	0.0060	0.0120	0.0213	0.0000	0.0036	0.0000	0	10	9	14	1
450	5	0.0046	0.0083	0.0163	0.0000	0.0026	0.0000	0	7	6	11	1
455	5	0.0043	0.0080	0.0145	0.0000	0.0024	0.0000	0	7	6	11	1
460	5	0.0026	0.0041	0.0073	0.0000	0.0014	0.0000	0	4	3	5	1
465	5	0.0037	0.0074	0.0125	0.0000	0.0020	0.0000	0	6	5	9	1
470	10	0.0074	0.0144	0.0254	0.0000	0.0043	0.0000	0	10	9	14	1
480	5	0.0038	0.0061	0.0111	0.0000	0.0019	0.0000	0	6	5	8	1
485	5	0.0043	0.0078	0.0129	0.0000	0.0024	0.0000	0	7	6	8	1
490	5	0.0042	0.0079	0.0137	0.0000	0.0024	0.0000	0	7	6	10	1
495	5	0.0026	0.0051	0.0086	0.0000	0.0013	0.0000	0	4	3	6	1
500	10	0.0050	0.0090	0.0155	0.0000	0.0028	0.0000	0	8	7	11	1
510	10	0.0048	0.0090	0.0179	0.0000	0.0028	0.0000	0	8	7	14	1
[...]												

Figure 2.35: INFOSTEP Report Example

Here, the *Assembly* column represents the time taken to assemble the equations, *LSetup* is the time taken to setup the linear solver equations, *LSolve* is the linear solver time, *LocSol* is the local solver time, *Update* is the time taken in updating the results, and *Output* is the time used for writing out the results, that is the output to the print, debug, summary, and restart files.

The *WellIt*, *Lins*, *NewtIt*, and *LinIt* column headers represent the number of iterations used to complete the time step for a given work item. Thus for time step one, the number of well iterations was zero (*WellIt*), the number of times the linear equations were setup was 16 (*Lins*), the number of Newton iterations used to complete the time step was 15 (*NewtIt*), and the total number of linear iterations to completed the time step was 19 (*LinIt*). The last column, *Conv*, indicates if the time step had successfully converged given the current convergence criteria, with a one indicating success and a zero indicating a failure.

By default the report is not generated to maintain compatibility with commercial simulator. To generate the report one can use the `--output-extra-convergence-info` command line parameter; for instance:

```
flow --output-extra-convergence-info="steps" CASENAME
```

will write the report to the CASENAME.INFOSTEP file in the same directory as the other output files.

The second report, INFOITER, contains more detailed information by reporting the numerical performance per iteration, as can be seen in Figure 2.36.

Figure 2.36: INFOITER Report Example

Here, the *Iteration* column contains the number of Newton iterations, same as the *NewIt* column in the INFOSTEP report. Note that adding one to the values in the *Iteration* column gives the number of times the linear equations were setup. Columns starting *CnvPvFrac* give the fractions of the pore volume that have converged under the standard convergence criteria, have converged under the relaxed convergence criteria, and have not converged. Similarly, columns starting *CnvCellCnt* give the number of cells that have converged under the standard convergence criteria, have converged under the relaxed convergence criteria, and have not converged. Columns starting *Penalty* indicate the penalty values for non-convergence, distance decay, and large well residuals. Columns containing *MB* indicate a phase's normalized material balance error, and columns containing *CNV* show a phase's normalized residual error. The last column, *WellStatus*, indicates if the iteration converged (CONV) or failed (FAIL).

Like the INFOSTEP report, the INFOITER report by default is not generated in order to maintain compatibility with commercial simulator. To generate the report one can use the `--output-extra-convergence-info` command line parameter, for instance:

```
flow --output-extra-convergence-info="iterations" CASENAME
```

will write the report to the CASENAME.INFOITER file in the same directory as the other output files. Finally to generate both reports use the following command line option.

```
flow -output-extra-convergence-info="steps,iterations" CASENAME
```

The following sections now attempt to give some guidance on how to use the numerical controls to resolve the simulator's performance, assuming that all the issues with the input data have been successfully resolved.

2.4.3 TIME STEP CONTROL GUIDANCE

Figure 2.37 illustrates how the simulator advances from the start of the simulation to the end, and shows two user defined report steps, each containing two simulator selected time steps. Each time step within a report time step contains one to three Newton iterations, and each Newton iteration contains a various number of linear iterations.

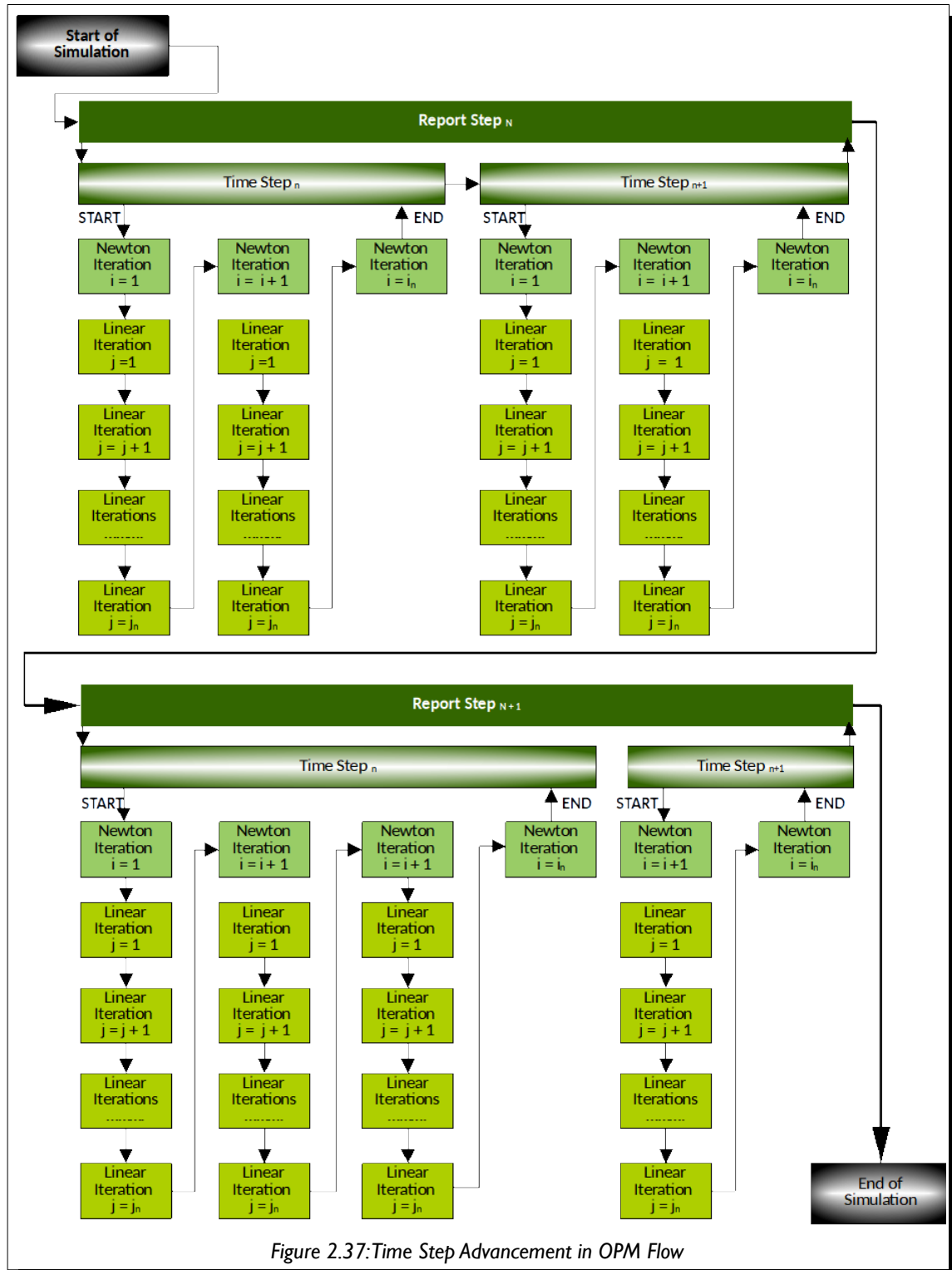


Figure 2.37: Time Step Advancement in OPM Flow

In an actual model the number of time steps are determined by the simulator's internal adaptive time stepping algorithm, combined with any user settings overriding the internal stepping algorithm and parameters. The default initial time step of one day can be overwritten via either the command line parameter:

`--initial-time-step-in-days`

or via the *TUNING – Numerical Tuning Control* keyword in the *SCHEDULE* section, that offers more flexibility, as the initial time step size can be reset at various report time steps, by repeating the keyword and setting *TUNING*(TSINIT) to a given value. Alternatively, *TUNING*(TSMINZ) may be used to define the minimum length of all time steps. Note, however, the simulator only reads some of the parameters from the *TUNING* keyword if, and only if, the following command line parameter has been set equal to true:

`--enable-tuning=true`

One can also set the maximum time step size via both the command line parameter:

`--solver-max-time-step-in-days`

or via the *TUNING*(TSMAXZ) parameter, that defines the maximum length of the next time step following *TUNING*(TSINIT).

There are many other time stepping command line parameters, including those that are equivalent to all the first record parameters on the *TUNING* keyword; however, provided there are no data inconsistencies within the input deck, resetting the maximum time step size from the default value of 365 days may improve numerical performance. Typically the default value of 365 days is not binding, as the report time steps are more restrictive. So for example, in a history matching model, the production data is normally entered on a monthly basis, so the maximum report time step size would be 31 days. Even so, time step chops may still occur even in this scenario, and therefore reducing the maximum time step to half a month, that is 16 days may avoid time step chops. Sixteen days is preferred over 15 days as the maximum time step, because it enables only two successful time steps to complete a 31 day month (a 16 day plus a 15 day time step). If 15 days was used for the maximum time step size, then one would need three successful time steps to complete a 31 day month. In some cases using 11, or even a five day maximum time step may increase computational efficiency by avoiding expensive computational time step chops.

Basically, one has to balance the maximum length of time steps with the required number of time steps to reach the end of the run, versus the total elapsed time⁵⁴. Time steps chops are computationally expensive and thus should therefore be minimized in order to reduce run times.

Figure 2.38 shows a typical report step and time step report from the simulator that contains two time step chops shown in the red outline boxes. At the start of the report:

- 1) The simulation has advanced to 00:00:00 on October 9th, 1967 at report step 15 (Current Step) out of 33 steps (End Step) and after an elapsed time of 250 days out of a total of 2000 days needed to complete the run.
- 2) As requested, the simulator writes out a restart record at report step 15, that is October 9th, 1967.
- 3) The simulator attempts to advance the simulation from October 9th, 1967 by taking a time step of 50 days, as determined by the internal adaptive time stepping algorithm. If successful, this would advance the simulator from the current report step 15 at an elapsed time of 250 days to the next report step 16 at an elapsed time of 300 days.
- 4) However, the time step fails due to a *Numerical problem encountered*, that is either the non-linear or linear solver failed to convergence within the maximum number of iterations. Thus, the time step was chopped from 50 days to 16.5 days.

⁵⁴ Elapsed real time, real time, wall-clock time, or wall time is the actual time taken from the start of a simulation to the end. In other words, it is the difference between the time at which the simulation finishes and the time at which the simulation started.

- 5) The simulator restarts again from report step 15, that is October 9th, 1967 and attempts to advance 16.5 days to October 25th, 1967. If successful, this would advance the simulator by 16.5 days from the current elapsed time of 250 days towards the next report step at 300 days.
- 6) The 16.5 day time step converges in three Newton iterations using a total of 342 linear iterations, and the simulator has successfully advanced to October 25th, 1967.
- 7) The simulator then attempts to advance 33.5 days, but again fails due to a *Numerical problem encountered*, that is either the non-linear or the linear solver again failed to convergence within the maximum number of iterations. Thus, the time step was chopped from 33.5 days to 11.055 days.
- 8) The simulator restarts again from the last successful time step, that is October 25th, 1967 and attempts to advance to November 5th, 1967. The 11.055 day time step converges in three Newton iterations using a total of 317 linear iteration, and the simulator has advanced to November 5th, 1967.
- 9) The simulator's internal adaptive time stepping algorithm has determined that the next time step should be 22.445 days to coincide with the next report time step (report step 16) at November 28th, 1967. Again, the 22.445 day time step is successful and converges in three Newton iteration using a total of 318 linear iterations.
- 10) The simulation has now advanced to 00:00:00 on November 28th, 1967 at report step 16 (Current Step) out of 33 steps (End Step) and after an elapsed time of 300 days out of a total of 2000 days needed to complete the run.

If the maximum time step was set to 16 days then advancing 50 days to November 28th, 1967 would take four time steps with a good chance of avoiding time step chops and thus reducing run times.

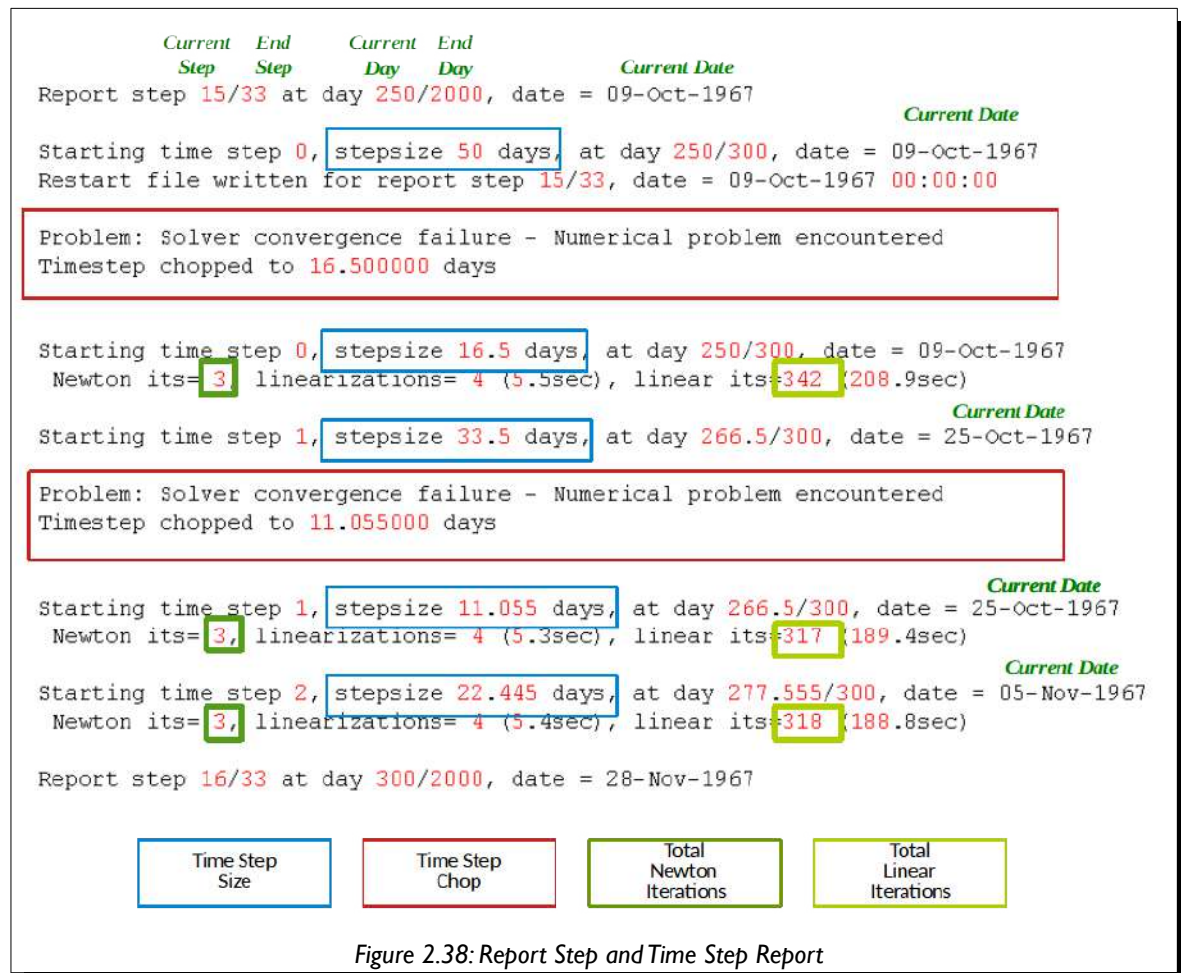


Figure 2.38: Report Step and Time Step Report

Various command line parameters can be used to control the behavior of the time stepping algorithm, as summarized in the Table 2.4, that lists the important time stepping control parameters for the simulator, including the [TUNING](#) equivalent parameters where applicable.

Major Solver Time Stepping Controls			
No.	Command Line Parameter	TUNING Keyword Equivalent	Description
1	--initial-time-step-in-days	TSINIT	A real positive value. Set the size of the initial time step in days. The default value is one day. Note that as the TUNING keyword can be entered multiple times, then TUNING(TSINIT) can be reset multiple times, whereas as --initial-time-step-in-days, is only set once at the beginning of the run.
2	--solver-growth-factor	TSDIFF	A real positive value. Set the growth factor a time step can be increased by when recovering from one or more time step chops, subject to the maximum allowable time step size set by the --solver-max-time-step-in-days parameter. For example, if the current time step has converged at 5 days after a time step chop, and --solver-growth-factor is set to the default value of 2.0, then the next time step will be 2.0 * 5 days, that is at 10 days.
3	--solver-max-growth	TSFMAX	A real positive value. Set the maximum growth factor a time step can be increased by after a report time step, subject to the maximum allowable time step size set by the --solver-max-time-step-in-days parameter. Thus, if the current time step has converged at 5 days after at a report step, and --solver-max-growth is set to the default value of 3.0, then the next time step will be 3.0 * 5 days, that is at 15 days.
4	--solver-max-restarts		A positive integer. Set the number of allowed consecutive restarts (or time step chops) before the simulation is terminated. Here the default value is 10. TUNING (TSMCHP) sets the minimum length of all chopped time steps.
5	--solver-max-time-step-in-days	TSMAXZ	A real positive value. Set the maximum allowed time step size in days. The default value of 365 days is rarely used in practise due to the report time steps being more restrictive, although in some specialized models (CO2 long term prediction cases) this may indeed be the upper limit. As mentioned previously, if the time step chops are mostly occuring when the time step size is greater than Δt, then setting this parameter to a value less than Δt may reduce the number of time step chops. Provided of cause the value of Δt is reasonable.

6	--solver-min-time-step	TSMINZ	A real positive value. Set the minimum size of a time step in days for field and metric units, and hours for laboratory units If a time step cannot converge without getting cut below this time step size the simulator will stop, unless overwritten by the --solver-continue-on-convergence-failure command line parameter (see Table 2.1 for further details). Note if the current time step is smaller than this minimum and has not been subject to a time step chop, then in some circumstances it may still be accepted due to other numeric controls.
7	--solver-restart-factor	TSFCNV	A real positive value. Set the time step chop factor of the time step after a convergence failure. For example, if the current non-convergent time step is 30 days and this parameter is set to the default value of 0.33, then the time step will be repeated using $0.33 * 30$ days as the time step, that is 9.9 days.
Notes: 1) Cells colored orange under the TUNING Equivalent Keyword column with no TUNING keyword parameter do not have an equivalent to the OPM Flow command line parameter.			

Table 2.4: Major Solver Time Stepping Controls

Note

OPM Flow also has a series of command line parameters of the form `--time-step*` that are subordinate to the `--solver*` series of command line parameters. The former, `--time-step*`, are used to select the time step control algorithm and associated parameters, and in general should only be used by users with appropriate experience.

As mentioned above, the `--time-step*` series of command line parameters should in general not be used; however, the following parameter may be useful if group and well events are resulting in time step chops:

`--time-step-after-event-in-days`

If the simulation is running smoothly with monthly historical report time steps, except when a new well is bought on stream in a given month, or a group control is reset resulting in a significant change in production or injection, and the event causes time step chops, then using `--time-step-after-event-in-days` may improve the situation. The parameter can be used to automatically enforce a small time step when an event occurs, thus reducing the probability, or even eliminating a time step chop. Typically, a one or five day time step is used, as per:

`--time-step-after-event-in-days=5.0`

The drawback is that the control is applied to the complete run, which again needs to be taken into consideration.

Alternatively, one could use the [NEXT](#) or [NEXTSTEP](#) keywords in the [SCHEDULE](#) section, to incorporate similar behavior at specific times in the [SCHEDULE](#) section. See [NEXT – Maximum Next Time Step Size \(Alias for NEXTSTEP\)](#) or [NEXTSTEP – Maximum Next Time Step Size](#) for further information.

2.4.4 NEWTON-RAPHSON SOLVER ITERATION CONTROLS (NON-LINEAR ITERATIONS)

Figure 2.39 shows typical time step output for two report time steps, March 1st and April 1st, 2022; with both solving the equations with two Newton iterations per time step, and between four to six linear iterations per time step. The numerical count for the number of linearizations is the number of times the Jacobian and the residuals have been constructed, and is generally one more than the number of Newton iterations. Figure 2.39, thus indicates very good performance, especially as the first iteration is taken from the previous successful time step. Note however, that the model is only using five day time steps in this case, which significantly contributes to the convergence stability of both the non-linear and linear solvers.

```
Report step 3/61 at day 59/1825, date = 01-Mar-2022

Starting time step 0, stepsize 5 days, at day 59/90, date = 01-Mar-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 5 (0.0sec)

Starting time step 1, stepsize 5 days, at day 64/90, date = 06-Mar-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 4 (0.0sec)

Starting time step 2, stepsize 5 days, at day 69/90, date = 11-Mar-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 4 (0.0sec)

Starting time step 3, stepsize 5 days, at day 74/90, date = 16-Mar-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 4 (0.0sec)

Starting time step 4, stepsize 5 days, at day 79/90, date = 21-Mar-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 4 (0.0sec)

Starting time step 5, stepsize 3 days, at day 84/90, date = 26-Mar-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 5 (0.0sec)

Starting time step 6, stepsize 3 days, at day 87/90, date = 29-Mar-2022

Report step 4/61 at day 90/1825, date = 01-Apr-2022

Starting time step 0, stepsize 5 days, at day 90/120, date = 01-Apr-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 5 (0.0sec)

Starting time step 1, stepsize 5 days, at day 95/120, date = 06-Apr-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 5 (0.0sec)

Starting time step 2, stepsize 5 days, at day 100/120, date = 11-Apr-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 4 (0.0sec)

Starting time step 3, stepsize 5 days, at day 105/120, date = 16-Apr-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 5 (0.0sec)

Starting time step 4, stepsize 5 days, at day 110/120, date = 21-Apr-2022
Newton its= 2, linearizations= 3 (0.0sec), linear its= 6 (0.0sec)

Starting time step 5, stepsize 5 days, at day 115/120, date = 26-Apr-2022
```

Figure 2.39: Report Step and Time Step Summary Reporting Example

The number of Newton iterations taken to solve a time step is indicative of the degree of ease the simulator has in converging to a solution, with higher numbers of iterations indicating an increasingly difficult problem.

Note, unlike Figure 2.39, the number of non-linear iterations will vary as the simulation progresses, with periods of relatively efficient convergence and periods where the simulator is struggling to reach non-linear convergence. The latter is normally associated with operational events, for example new wells being brought on stream, major changes to group controls etc.

Note

As mentioned previously, in general the number of Newton iterations is indicative of the ease of convergence, with one to three iterations indicating that converge was relatively easy and efficient, four to nine iterations indicating an increasingly difficult problem, and greater than ten would tend to suggest that there may be an issue with the model.

It is important to stress that the convergence of the non-linear solver is very dependent on the accuracy of the linear solver. If the accuracy of the linear solver is too relaxed or is not fully converged, the non-linear solver may fail to converge. For example, if the linear solver solution is not sufficiently converged, then the non-linear solution may no longer be well-posed⁵⁵. That is the solution starts digressing from the true solution, and additional iterations only exasperate the problem. Normally, under these circumstances OPM Flow will chop the time step, which should correct the divergence by reverting to previous converged time step. However, if the time step is too small to chop and the run continues with the minimum time step, then the divergence from the true solution will continue. This may manifest itself with various error messages and NaNs ("Not a Number"), or eventually with the simulator aborting. Figure 2.40 shows an example of this behavior.

⁵⁵ A problem in differential equations is said to be well-posed if: (1) a solution exists; (2) that the solution is unique; (3) and the solution changes continuously with the changes in the data.

```

Starting time step 0, stepsize 0.1089 days, at day 0/31, date = 01-Jan-2020
Problem: solver convergence failure - Numerical problem encountered
Timestep chopped to 0.035937 days

Starting time step 0, stepsize 0.035937 days, at day 0/31, date = 01-Jan-2020
Problem: Solver convergence failure - Iteration limit reached
Timestep chopped to 0.011859 days

Starting time step 0, stepsize 0.0118592 days, at day 0/31, date = 01-Jan-2020
Well OP01 will be shut because it cannot get converged.
Problem:
Problematic well(s) were shut: OP01 (retrying timestep)

Starting time step 0, stepsize 0.0118592 days, at day 0/31, date = 01-Jan-2020
Newton its= 2, linearizations= 3 (0.2sec), linear its= 0 (0.0sec)

Starting time step 1, stepsize 0.0237184 days, at day 0.0118592/31, date = 01-Jan-2020
flow: /home/david/OPM/GitHub/opm-simulators/opm-simulators/timestepping/TimeStepControl.cpp:140: virtual double
[david-VirtualBox:252949] *** Process received signal ***
[david-VirtualBox:252949] Signal: Aborted (6)
[david-VirtualBox:252949] Signal code: (-6)
[david-VirtualBox:252949] [ 0] /lib/x86_64-linux-gnu/libc.so.6(+0x42520) [0x7f4c57af1520]
[david-VirtualBox:252949] [ 1] /lib/x86_64-linux-gnu/libc.so.6(pthread_kill+0x12c) [0x7f4c57b45a7c]
[david-VirtualBox:252949] [ 2] /lib/x86_64-linux-gnu/libc.so.6(raise+0x16) [0x7f4c57af1476]
[david-VirtualBox:252949] [ 3] /lib/x86_64-linux-gnu/libc.so.6(abort+0xd3) [0x7f4c57ad77f3]
[david-VirtualBox:252949] [ 4] /lib/x86_64-linux-gnu/libc.so.6(+0x2871b) [0x7f4c57ad771b]
[david-VirtualBox:252949] [ 5] /lib/x86_64-linux-gnu/libc.so.6(+0x39e96) [0x7f4c57ae8e96]
[david-VirtualBox:252949] [ 6] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x194c2f4) [0x556648c6f2f4]
[david-VirtualBox:252949] [ 7] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x194c33a) [0x556648c6f33a]
[david-VirtualBox:252949] [ 8] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x91c686) [0x556647c3f686]
[david-VirtualBox:252949] [ 9] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x91e9cd) [0x556647c419cd]
[david-VirtualBox:252949] [10] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x91fa3b) [0x556647c42a3b]
[david-VirtualBox:252949] [11] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x90dc2e) [0x556647c30c2e]
[david-VirtualBox:252949] [12] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x87c7f1) [0x556647b9f7f1]
[david-VirtualBox:252949] [13] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x25c6a5) [0x55664758f6a5]
[david-VirtualBox:252949] [14] /lib/x86_64-linux-gnu/libc.so.6(+0x29d90) [0x7f4c57ad8d90]
[david-VirtualBox:252949] [15] /lib/x86_64-linux-gnu/libc.so.6(__libc_start_main+0x80) [0x7f4c57ad8e40]
[david-VirtualBox:252949] [16] /home/david/OPM/GitHub/opm-simulators/build/bin/flow(+0x3902a5) [0x5566476b32a5]
[david-VirtualBox:252949] *** End of error message ***
Aborted (core dumped)
Process Complete (0)

```

Figure 2.40: Time Step Summary Showing Non-Convergence Example.

Fortunately, OPM Flow has various controls that can resolve these type of issues similar to the commercial simulator, and in addition, and unlike the commercial simulator, OPM Flow has several linear solvers and preconditioners that increase the likelihood a solution may be found for various non-convergence issues.

As mentioned previously in section 1.6 Numerical Solution of Equations, convergence is achieved when $R(y_n)$, the residual, in equation (2.5), satisfies a given tolerance.

$$J(y_n)(y_{n+1} - y_n) = -R(y_n) \quad (2.5)$$

where:

- $J(y_n)$ = the Jacobian matrix,
- y_n = the solution of the non-linear equations at the nth non-linear iteration, and
- y_{n+1} = the solution of the non-linear equations iteration at the n+1 iteration.

However, as the iterations proceed during a time step the residual convergence criteria is reduced in an effort to avoid a time step chop. Thus, by relaxing the convergence criteria as the non-linear iterations progress, the simulator is attempting to avoid throwing away all the previous work performed in trying to reach a solution for the time step.

From the user prospective, there are several command-line options that can be used to control the Newton-Raphson solver, in particular:

- 1) To modify the maximum number of non-linear iterations per time step, the following command line parameter can be used:

`--newton-max-iterations`

Although, the default value of 20 is sufficiently large that it should not be necessary to increase this value.

- 2) There is also a command that sets the minimum number of Newton iterations per time step used by the simulator.

`--newton-min-iterations`

The default value of two ensures that at least two Newton iterations are performed after the previous time step. Increasing this parameter, may help convergence in difficult time steps, by ensuring slightly more accurate results in preceding "easy" time steps.

- 3) Thirdly, the mass balance error as a reservoir average saturation error, can be use to slackened or tighten the material balance tolerance.

`--tolerance-mb`

In general, if the non-linear solver is having difficulty in converging, then tightening the linear mass balance error as a reservoir average saturation error, may help the non-linear solver converge more rapidly.

- 4) Fourthly, maximal local residual error can be slackened or tighten using:

`--tolerance-cnv`

command line parameter. And may be used to set the maximal local residual error.

For completeness, Table 2.5 lists the major non-linear command line parameters used to control the behavior of the non-linear solver, including those that relate to relaxed convergence criteria. In general, it is recommended not to modify these parameters, unless one has the sufficient expertise in this area.

Major Newton-Raphson Solver Controls			
No.	Command Line Parameter	TUNING Keyword Equivalent	Description
1	<code>--newton-max-iterations</code>	NEWTMX	A positive integer. Set the maximum number of Newton iterations per time step used by the simulator. The default value is 20 which compares to the commercial simulator's value of 12, so there should not really be a requirement to increase this parameter.
2	<code>--newton-min-iterations</code>	NEWTMN	A positive integer. Set the minimum number of Newton iterations per time step used by the simulator. The default values for <code>--newton-min-iterations</code> and <code>NEWTMN</code> are 2 and 1, respectively. The value of one ensures that at least one Newton iteration is performed after the previous time step. Increasing this parameter, to say three for example, may help convergence in difficult time steps, by ensuring slightly more accurate results in preceding "easy" time steps.

Major Newton-Raphson Solver Controls			
No.	Command Line Parameter	TUNING Keyword Equivalent	Description
3	--newton-max-relax		A real value ranging from zero to one. Set the maximum relaxation factor of a Newton iteration used by the simulator. The default is 0.5. <u>In general, it is not recommended to modify this parameter.</u>
4	--newton-relaxation-type		A character string. The type of relaxation used by the Newton method. Valid options are: dampen (default) or sor. <u>In general, it is not recommended to modify this parameter.</u>
5	--max-newton-iterations-with-inner-welliterations	MXWSIT	A positive integer. Set the maximum Newton iterations with inner well iterations. The default value is eight, the same as the commercial simulator. If there are issues with well convergence, then sometimes increasing this value, to say 12, may improve the situation.
6	--min-strict-cnv-iter		An integer. Set the minimum number of Newton iterations before relaxed tolerances are used for the CNV convergence criterion. The default value -1 means that relaxed convergence criterion is applied after the initial iteration, that is iteration zero in Figure 2.42 in section 2.4.5 Linear Solver Controls and Options. Increasing this parameter may improve convergence by applying stricter convergence controls over additional Newton iterations, but may also increase the number of Newton iterations over a time step before the time step is accepted.
7	--tolerance-cnv	TRGCNV	A real positive value. Set the maximum non-linear tolerance error. This is the local convergence tolerance (maximum of local saturation errors). The default value for --tolerance-cnv is 0.01 and for TRGCNV is 0.001.
8	--tolerance-cnv-relaxed	XXXCNV	A real positive value. Set the maximum non-linear relaxed tolerance error. The default value for --tolerance-cnv-relaxed is 1 and for XXXCNV is 0.01.
9	--tolerance-mb	TRGMBE	A real positive value. Set the maximum mass balance error, that is the tolerated mass balance error relative to total mass present. The default value is 1.0×10^{-7} .
10	--tolerance-mb-relaxed	XXXMBE	A real positive value. Set the maximum mass balance relaxed error. The default value is 1.0×10^{-6} .

Major Newton-Raphson Solver Controls			
No.	Command Line Parameter	TUNING Keyword Equivalent	Description
Notes: - Cells colored gray under the No. column, are not recommend to be modified. - Cells colored orange under the <i>TUNING</i> Equivalent Keyword column with no <i>TUNING</i> keyword parameter do not have an equivalent to the OPM Flow command line parameter. - Cells colored orange under the <i>TUNING</i> Equivalent Keyword column with a <i>TUNING</i> keyword parameter, means OPM Flow does not support reading this parameter from the <i>TUNING</i> keyword and therefore the OPM Flow command line option should be used.			

Table 2.5: Newton-Raphson Solver Controls

Detailed descriptions of all of the command line parameters are given in Table 2.1 in section [2.2 Running OPM Flow From The Command Line](#).

2.4.5 LINEAR SOLVER CONTROLS AND OPTIONS

There are numerous options for the linear solver, which can be overwhelming even for the most experienced users, given the rich feature set enabled by the various command line parameters. Fortunately, the simulator summarizes the options in use in the *Linear Solver Property Tree* printed in both the *.PRT and *.DBG files, right after processing the input data and just before the simulation starts.

```
{
  "maxiter": "20",
  "tol": "0.005000000000000001",
  "verbosity": "0",
  "solver": "bicgstab",
  "preconditioner": {
    "type": "cprw",
    "use_well_weights": "false",
    "add_wells": "true",
    "weight_type": "trueimpes",
    "pre_smooth": "0",
    "post_smooth": "1",
    "finesmooother": {
      "type": "paroverilu0",
      "relaxation": "1"
    }
  },
  "verbosity": "0",
  "coarsesolver": {
    "maxiter": "1",
    "tol": "0.10000000000000001",
    "solver": "loopsolver",
    "verbosity": "0",
    "preconditioner": {
      "type": "amg",
      "alpha": "0.3333333333330003",
      "relaxation": "1",
      "iterations": "1",
      "coarsenTarget": "1200",
      "pre_smooth": "1",
      "post_smooth": "1",
      "beta": "0",
      "smoother": "ilu0",
      "verbosity": "0",
      "maxlevel": "15",
      "skip_isolated": "0",
      "accumulate": "1",
      "prolongationdamping": "1",
      "maxdistance": "2",
      "maxconnectivity": "15",
      "maxaggsz": "6",
      "minaggsz": "4"
    }
  }
}
```

Figure 2.41: Linear Solver Property Tree

Figure 2.41 above shows an example of the Linear Property Tree, based on using mostly the linear solver default parameters. Here the tol value of 0.005 is equivalent to the --tolerance-cnv command line parameter,

and maxiter is equivalent to --linear-solver-max-iter. The key parameters are summarized in Table 2.6. See also [Customizing the Linear Solver in OPM Flow](#).

No.	PropertyTree Parameter	Equivalent Command Line Parameter
1	tol	--tolerance-cnv
2	maxiter	--linear-solver-max-iter
3	verbosity	--linear-solver-verbosity
4	solver	--use-gmres=false
5	preconditioner	
6	type	--linear-solver
Notes: 1) BiCGSTAB stands for Biconjugate Gradient Stabilized linear solver. 2) GMRES is for the Generalized Minimal Residual linear solver.		

Table 2.6: Linear Solver PropertyTree versus Command Line Parameters

Table 2.7 lists the major linear solver command line parameters discussed in this section.

Figure 2.42 shows typical output for the linear solver for the March 1st report step advancing to the next report step. Naturally, the Newton and linear iteration counts are the same as for Figure 2.39 for the common periods; however, oil, gas, and water material balance values are printed for each non-linear iteration, together with the respective convergence values.

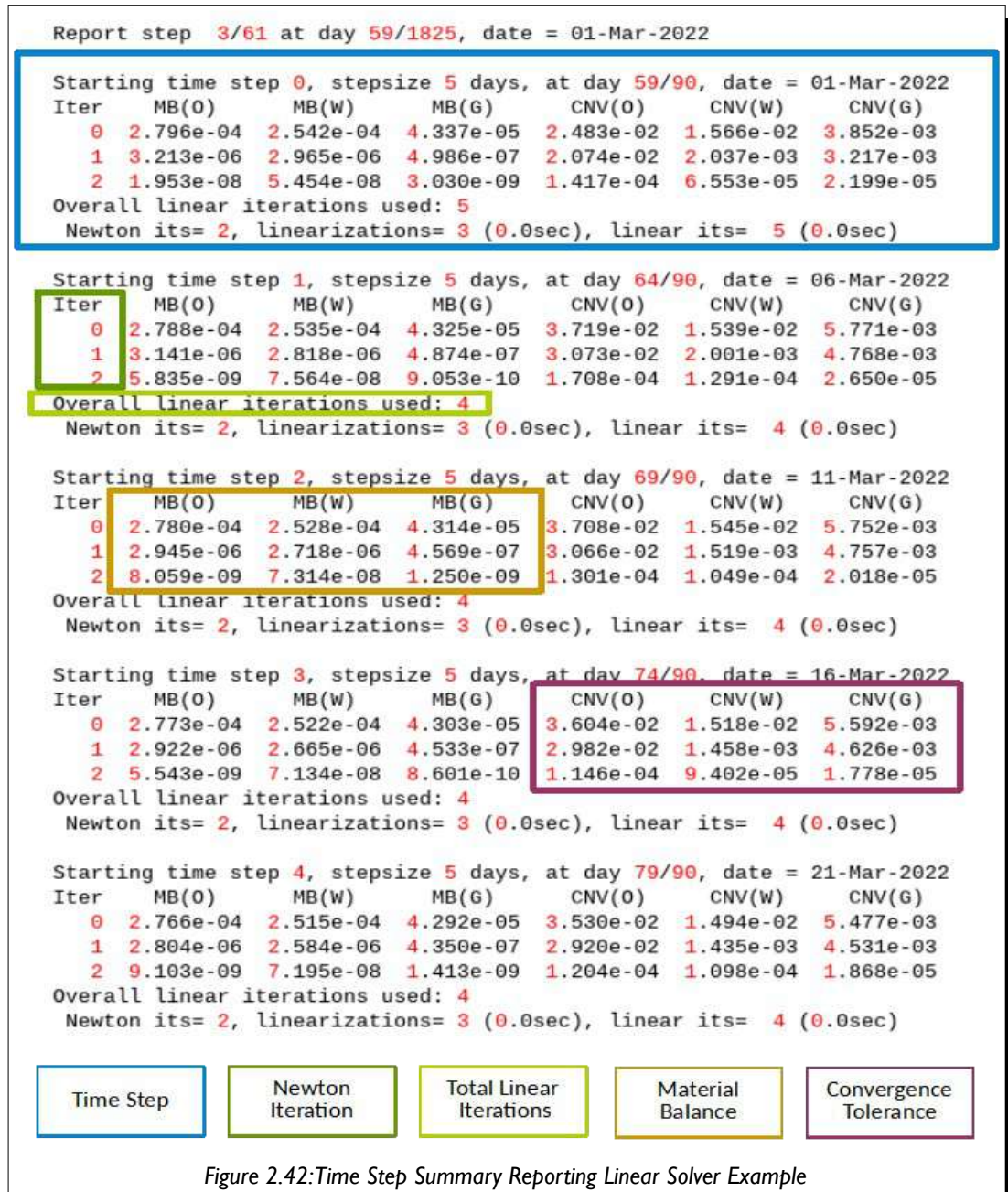


Figure 2.42: Time Step Summary Reporting Linear Solver Example

The total number of linear iterations displayed in Figure 2.42 is the total number of iterations for all the non-linear iterations, not for a particular non-linear iteration. To get the number of linear iterations per non-linear iteration use the following command line parameter:

```
--linear-solver-verbosity=1
```

which will produce a report similar to that shown in Figure 2.43, with the IT parameter indicating the number of linear iterations for the given Newton iteration.

```
Starting time step 0, stepsize 0.0118592 days, at day 0/10, date = 01-Jan-1967
=== Dune::BiCGSTABSolver
=== rate=3.08338e-09, T=0.000472239, TIT=0.000472239, IT=1
=== Dune::BiCGSTABSolver
=== rate=0.35949, T=0.00317358, TIT=0.000634715, IT=5
=== Dune::BiCGSTABSolver
=== rate=0.51016, T=0.00358736, TIT=0.00051248, IT=7
=== Dune::BiCGSTABSolver
=== rate=0.0874993, T=0.0032023, TIT=0.00160115, IT=2
=== Dune::BiCGSTABSolver
=== rate=0.0976473, T=0.00200675, TIT=0.00100338, IT=2
=== Dune::BiCGSTABSolver
=== rate=0.260942, T=0.000812008, TIT=0.000270669, IT=3
=== Dune::BiCGSTABSolver
=== rate=0.00482599, T=0.00161764, TIT=0.00161764, IT=1
=== Dune::BiCGSTABSolver
=== rate=0.0938774, T=0.00202222, TIT=0.00101111, IT=2
=== Dune::BiCGSTABSolver
=== rate=0.20818, T=0.000666642, TIT=0.000222214, IT=3
=== Dune::BiCGSTABSolver
=== rate=0.155462, T=0.000522632, TIT=0.000261316, IT=2
Newton its=10, linearizations=11 (0.0sec), linear its= 28 (0.0sec)
```

Figure 2.43: Number of Linear Iterations Per Newton Iteration Report Example

As mentioned previously, the convergence of the non-linear solver is dependent on the accuracy of the linear solvers. Solving the linear system to a loose accuracy may lead to poor convergence of the non-linear solver. The tuning of the time stepping procedure, and the tolerance used in the linear and non-linear solvers are highly related, and optimal choices are case and problem dependent.

From the user prospective, there are several command-line options that can be used to configure and control the linear solver, in particular:

- 1) Using an alternative linear solver by changing

```
--use-gmres
```

The default value of false sets the linear solver to the Biconjugate Gradient Stabilized ("BiCGSTAB") linear solver, whereas a value of true will set the linear solver to use the Generalized Minimal Residual linear solver ("GMRES"). If the default linear solver is not converging then changing to the GMRES solver may help. Note, that the GMRES linear solver has a slightly higher memory consumption, as it stores some of the previous search iterations as part of the optimization process. The default number of search iterations stored is 15, which means 14 more vectors than BiCGSTAB solver.

- 2) Changing the linear solver preconditioner can dramatically improve the performance of the linear solver, which is accomplished by the following command line parameter:

```
--linear-solver
```

In many cases using the default Constrained Pressure Residual ("CPR") preconditioner, instead of the widely used Incomplete Lower-triangular Upper-triangular factorization with level of fill in 0 ("ILU0"), results in difficult problems being solved at the expense of additional work. For example, model two in the Tenth SPE Comparative Solution Project⁵⁶, fails to run using the ILU0 preconditioner, but runs satisfactory with the CPR preconditioner. Note that the commercial simulator also has a CPR option

⁵⁶ Christie, M. A., and M. J. Blunt. "Tenth SPE Comparative Solution Project: A Comparison of Upscaling Techniques." *SPE Res Eval & Eng* 4 (2001): 308–317. doi: <https://doi.org/10.2118/72469-PA>.

that is implemented via the [CPR](#) keyword in the [RUNSPEC](#), but the implementation is different to that of OPM Flow.

- 3) And finally increasing the number of linear iterations may be appropriate, via:

```
--linear-solver-max-iter
```

that defines the maximum number of linear iterations. Sometimes increasing this value may improve performance. Note that for the default linear solver CPRW the flag is ignored, but one could set this and additional values using a JSON file, see [Customizing the Linear Solver in OPM Flow](#) for details.

Major Linear Solver Controls			
No.	Command Line Parameter	TUNING Keyword Equivalent	Description
I	--linear-solver		A defined string. Specify the configuration of the linear solver; valid values are: 1) cprw^{57 58 59} (default), 2) ilu0, 3) dilu, 4) cpr (an alias for cprw), 5) cpr_quasiimpes, 6) cpr_trueimpes, 7) cpr_trueimpesanalytic, 8) amg⁶⁰, 9) hybrid (experimental), or 10) a file name that has the extension “.json”, that contains the linear solver configuration parameters. Option (1) extends the existing Constrained Pressure Residual (“CPR”) preconditioner to include wells. This option can also be invoked via the CPR keyword in the RUNSPEC section; however, the command line parameter takes precedence. For option (10) one enters a character string enclosed in quotes that defines the filename of a JSON configuration file for the flexible linear solver system. In this case the file extension should be ‘.json’. Note that the *.PRT file contains the “Property tree for the linear solver” listing, which is the JSON specification of the current case, and can be used to configure a user specific linear solver JSON file.

⁵⁷ Wallis, J. R., Little, T. E., and Nolen, J. S. "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

⁵⁸ Scheichl, R., Roland, M., and Wendebourg, J. Decoupling and block preconditioning for sedimentary basin simulations, *Computational Geosciences* 7 (2003) 295–318.

⁵⁹ Klemetsdal, Ø.S., Møyner, O., and Lie, K.A. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. *Comput Geosci* 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

Major Linear Solver Controls			
No.	Command Line Parameter	TUNING Keyword Equivalent	Description
2	--linear-solver-ignore-convergence-failure		A boolean value set to true or false. If set to true convergences failures in the linear solver are ignored. <u>This option should be used with care, as the results may be unreliable.</u>
3	--linear-solver-max-iter	LITMAX	A positive integer value. Set the maximum number of linear iterations.
4	--linear-solver-reduction		A real positive value. Set the minimum reduction of the residual for the linear solver. The linear solver convergences when the residual is reduced sufficiently. <u>In general, it is not recommended to modify this parameter.</u>
5	--linear-solver-restart		A positive integer value. Set the number of iterations after which GMRES is restarted. <u>In general, it is not recommended to modify this parameter.</u>
6	--use-gmres		A boolean value. Use Generalized Minimal Residual ("GMRES") solver instead of Biconjugate Gradient Stabilized ("BiCGSTAB") as the linear solver within the Newton iterations.
Notes: - Cells colored gray under the No. column, are not recommend to be modified. - Cells colored orange under the <i>TUNING</i> Equivalent Keyword column with no <i>TUNING</i> keyword parameter do not have an equivalent OPM Flow command line parameter. - Cells colored orange under the <i>TUNING</i> Equivalent Keyword column with a <i>TUNING</i> keyword parameter, means OPM Flow does not support reading this parameter from the <i>TUNING</i> keyword and therefore the OPM Flow command line option should be used.			

Table 2.7: Linear Solver Controls

Again detailed descriptions of all of the command line parameters are given in Table 2.1 in section 2.2 Running OPM Flow From The Command Line.

⁶⁰ Blatt, M. A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients, Ph.D. thesis, Ruprecht-Karls-Universit t Heidelberg (2010).

CHAPTER 3: KEYWORD DOCUMENTATION STRUCTURE

3.1 INTRODUCTION

The OPM Flow manual is constructed in a manner to enable the reader to reference various parts of the document by using the table contents or by simply pressing on a link embedded in the text. This automatic cross referencing has been extensively employed to ensure effective documentation of the keywords used by the simulator.

There are several key features that can be used to navigate the manual in an efficient manner. The first is the “footer” which can be used to move to various sections in the manual. For example, the “Table of Contents” footer shown below:

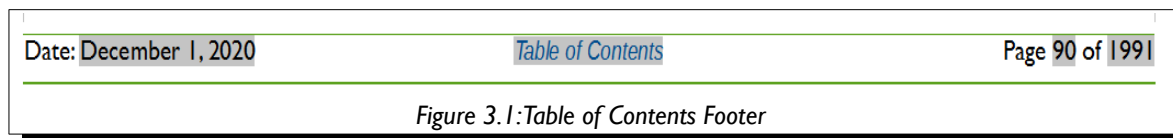


Figure 3.1: Table of Contents Footer

allows the reader to move to the Table of Contents by clicking on the link (highlighted in italic blue text on a gray background in Figure 3.1). Note also that the entries in the table of contents are also “clickable” enabling the reader to move to the desired entry.

The second type of footer is the “Keyword Index” footer that also contains the Table of Contents link mentioned above. This footer is illustrated in Figure 3.2 and is displayed on a keyword definition page.

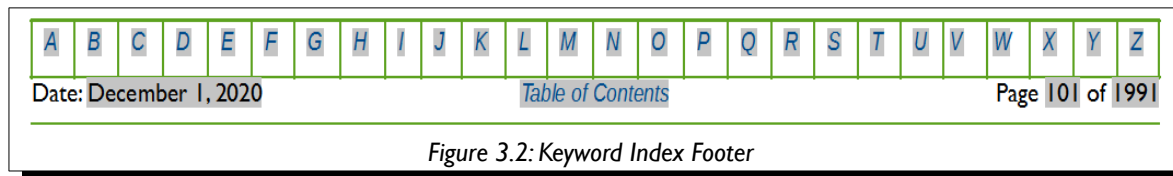


Figure 3.2: Keyword Index Footer

Clicking on a letter (highlighted in italic blue text on a gray background in) takes the reader to an alphabetic listing of all the keywords beginning with the selected letter (Figure 3.3).

A							
RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
Alphabetic Listing Of Keywords Starting With The Letter A							Status
ACTDIMS – ACTION Keyword Dimensions							
ACTION – Define Action Conditions and Command Processing (Field)							
ACTIONG – Define Action Conditions and Command Processing (Groups)							

Figure 3.3: Alphabetic Listing of Keywords

The list is color coded, so one instantly knows what keywords are implemented, with green colored cells indicating the keyword is available and is mostly or fully implemented. Cells colored orange show that keyword is recognized but not implemented in OPM Flow. Finally, cells colored red mean that keyword is available in the commercial simulator but has not been implemented in OPM Flow, and may cause an error if used in the input deck. Clicking on a keyword in the list will move the reader to the keyword definition. Note also that clicking one of the section names ([RUNSPEC](#), [GRID](#), etc.) in Figure 3.3, will take the reader to the beginning of the selected section.

Finally, in the PDF version of the manual if one displays the “bookmarks” in the PDF reading software one can jump to a particular keyword or section without having to scroll up or down.

3.2 KEYWORD DEFINITIONS

Each keyword is defined in it's own section that contains a section header, that contains the keyword name in capital letters followed by a brief description of the keyword's function. This is then followed by the Keyword Table Section which defines the status of the keyword and in which sections of the input deck the keyword can be utilized. Table 3.1 illustrates a typical Keyword Table Section defining the keyword status with the various OPM Flow sections.

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Table 3.1: Example Keyword Table Section

The cells are colored in such a manner as to quickly indicate to the reader the keyword's section availability and function availability, with green colored cells indicating the keyword is available for this section and is mostly or fully implemented. Cells colored gray indicate that keyword cannot be used in that particular section, and cells colored orange show that the keyword is only partially implemented within OPM Flow, for example OPM Flow may just recognize the keyword and ignore the keyword's function. Finally, cells colored red mean that the keyword is available in the commercial simulator but has not been implemented in OPM Flow. In this scenario the keyword should not be used in OPM Flow as it will result in unpredictable results, including causing the simulator to abort or throw an exception.

3.3 MULTI-SECTION KEYWORDS

As there are numerous keywords that can be used within multiple OPM Flow sections of the input file, for example the [ADD](#) and [EQUALS](#) keywords, there is a need to avoid duplication of the keyword definitions but at the same time attempt to define only those keywords for a given section. Thus for multi-section keywords, the keyword is defined in the first available section that the keyword can be found. The Keyword Table Section as shown below for the [ADD](#) keyword below, indicates in which sections the keyword can be utilized.

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Table 3.2: ADD Keyword Table Section

Here the [ADD](#) keyword can be used in the [GRID](#), [EDIT](#), [PROPS](#), [REGIONS](#), and [SOLUTION](#) sections as indicated by those cells colored green and not for the cells colored in light gray. A full description of the [ADD](#) keyword is in the [GRID](#) section. In the [EDIT](#), [PROPS](#), [REGIONS](#), and [SOLUTION](#) sections there is a brief description of the [ADD](#) keyword with a link to the full description, as shown below.

The [ADD](#) keyword adds a constant to a specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [ADD](#) keyword is being applied. See [ADD – Add a Constant to a Specified Array](#) in the [GRID](#) section for a full description.

A complete list of keywords in alphabetic order is given in [APPENDIX A: KEYWORD INDEX - ALPHABETIC LISTING](#), and clicking on a specific keyword in the appendix will take the reader to the keyword definition in a particular section.

3.4 KEYWORD FORMATS

All keywords in OPM Flow should be entered in capital case and start in column one, lowercase entry of keywords will produce errors, and keywords not starting in column one will not be recognized. There are five types of keyword format types used by OPM Flow for data input. The description of the five types is given in the next three sections together with some examples.

3.4.1 KEYWORD FORMAT TYPE – COMMENT

Comments in the input deck can occur anywhere in the file are preceded by “--” in columns one and two, for example for the [EQLDIMS](#) keyword:

```
--
--      MAX      MAX      RSVD      TVDP      TVDP
--      EQLNUM    DEPTH    NODES    TABLE    NODES
EQLDIMS
      9          1*       20        1*        1*
/
```

In addition, comments can be placed after “/” that terminates a record entry as shown below;

```
--
-- -- ARRAY      CONSTANT -- ----- BOX -----
--                               I1  I2   J1  J2   K1  K2
MULTIPLY
  'PERMZ'      0.50000      1*  1*   1*  1*   1*  1* / PERMZ * 0.5
/
```

3.4.2 KEYWORD FORMAT TYPE – ACTIVATION

This type of keyword format only consists of the keyword itself and is usually used to invoke a feature or to switch on or off a processing feature. The keyword is documented by describing the functionality or action the keyword performs, followed by an example. Examples of this type of keyword include [API](#) (to switch on [API](#) tracking), [GAS](#) (to activate the gas phase in the model), [ECHO](#) (to switching echoing of the input file to the output file), and [SKIP](#) (for skipping parts of the input deck). For example the [GAS](#) keyword in the [RUNSPEC](#) section would be described as:

Description

This keyword indicates that the gas phase is present in the model and must be used for oil-gas, gas-water, oil-water-gas input decks that contain the gas phase. The keyword will also invoke the data input file checking to ensure that all the required gas phase input parameters are defined in the input deck.

There is no data required for this keyword.

Example

```
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
```

The above example declares that the gas phase is active in the model.

3.4.3 KEYWORD FORMAT TYPE - VECTOR (ROW VECTOR)

Vector based keywords consist of the keyword followed by a vector of parameters on a separate line and may consists of multiple lines of vectors with each line representing a data set (see the second example for this type of vector keyword). The vector may contain integer, real, and character parameters depending on the keywords requirements. This type of keyword is documented by describing the functionality or action the keyword performs, a table describing the parameters associated with the keyword, followed by one or two examples on how to use the keyword. For example the [DIMENS](#) keyword in the [RUNSPEC](#) section would be described as:

Description

[DIMENS](#) defines the dimensions of the model entered as integer vector. The keyword can be used with all grid types.

No.	Name	Description	Default
1	NX	The number of grid blocks in the x direction for Cartesian grids or the number of grid blocks in the r direction for radial grids	None
2	NY	The number of grid blocks in the y direction for Cartesian grids or the number of grid blocks in the theta direction for radial grids.	None
3	NZ	The number of grid blocks in the z direction for both Cartesian and radial grids.	None

Notes:

- 1) The keyword is terminated by a "/".

Table 3.3: DIMENS Keyword Description

Note that NX, NY, and NZ are not maximum values but the actual size of the grid. OPM Flow applies these parameters when reading in particular data sets. For example if NX, NY, and NZ are set to 10, 10, and 10 respectively, then for the grid property data like [PORO](#); OPM Flow expects to read in 10 x 10 x 10 or 1,000 porosity values for the [PORO](#) array. If the number of porosity values is not equal to 1,000 then OPM Flow will produce an error.

Example

```
--
--      MAX      MAX      MAX
--      NDIVIX  NDIVIY  NDIVIZ
DIMENS      46      112      22      /
```

The above example defines the dimensions for the Norne model of 36 cells in the x direction, 122 cells in the y direction, and 22 cells in the z direction.

For vector keywords that have parameters associated with units, then there is a slightly different table format from the one used above to take into account documenting the default values for the three sets of units supported by OPM Flow, for example for the [ROCK](#) keyword is describe as follows:

Description

ROCK defines the rock compressibility for various regions in the model. The number of **ROCK** vector data sets is defined by the NTPVT parameter on the **TABDIMS** keyword in the **RUNSPEC** section and the allocation of the **ROCK** tables to different grid blocks in the model is done via the **PVTNUM** keyword in the **REGIONS** section. One data set consists of one record or line which is terminated by a “/”. This keyword must be defined in the OPM Flow input deck.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	Pref	Pref is a real number defining the reference pressure for the other parameters for this data set.			Default
		psia 1.032	barsa 1.032	atma 1.032	
2	Cf	Cf is a real number defining the rock compressibility at the reference pressure, Cf(Pref) and is defined as: $C_f = -\frac{1}{V}\left(\frac{dV}{dP}\right)$			Defined
		l/psia 0.0	l/barsa 0.0	l/atma 0.0	

Notes:

1) The each data set terminated by terminated by a “/” at the end of the line, there is no terminator for the keyword.

Table 3.4: ROCK Keyword Description

Example

The following shows the **PVTW** keyword for when NTPVT on the **TABDIMS** keyword in the **RUNSPEC** section is set to one.

```
--
-- ROCK COMPRESSIBILITY
--
-- (1) REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--     AS THE PORV IS ALREADY AT RESERVOIR CONDITIONS (FLOW USES THE REFERENCE
--     PRESSURE) TO CONVERT THE GIVEN PORV TO RESERVOIR CONDITIONS USING THE DATA
--     ON THE ROCK KEYWORD)
--
ROCK
  3566.9    5.0E-06                / ROCK COMPRESSIBILITY REGION 1
  3966.9    5.5E-06                / ROCK COMPRESSIBILITY REGION 2
  4566.9    6.0E-06                / ROCK COMPRESSIBILITY REGION 3
```

There is no terminating “/” for this keyword.

In this case the example shows a multiple data set entry of the vector format keyword, with three **ROCK** data sets being defined by the keyword.

3.4.4 KEYWORD FORMAT TYPE – VECTOR (COLUMNAR VECTOR)

Columnar vector based keywords consist of the keyword followed by a columnar vector of parameters in a separate column for each parameter. The vector may contain integer, real, and character parameters depending on the keywords requirements. This type of keyword is documented by describing the functionality or action the keyword performs, a table describing the parameters associated with the keyword, followed by one or two examples on how to use the keyword. For example the [SWFN](#) keyword in the [PROPS](#) section would be described as:

Description

The [SWFN](#) keyword defines the water relative permeability and water-oil capillary pressure data versus water saturation tables for when water is present in the input deck. This keyword should only be used if water is present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the water saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRW	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the water relative permeability with respect to gas saturation. The first value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
3	PCWO	A columnar vector of real values that are either equal or increasing down the column that defines the water-oil relative capillary pressure. If the SWATINIT keyword has been used to initialize the model then columnar vector has to be strictly monotonically increasing.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 3.3: SWFN Keyword Description

Example

```
--
--      WATER RELATIVE PERMEABILITY TABLES (SWFN)
--
SWFN
--      SWAT      KRW      PCOW
--      FRAC      FRAC      PSIA
--      -----
--      0.15      0.00000      1*
--      0.30      0.00050      1*
--      0.40      0.00390      1*
--      0.50      0.01500      1*
--      0.60      0.04100      1*
--      0.65      0.06250      1*
--      0.70      0.09150      1*
--      0.80      0.17850      1*
```

```

0.90      0.31640      1*
0.95      0.40960      1*
1.00      0.52200      1*
/ TABLE NO. 1

```

The example defines two [SWFN](#) tables for use when water is present in the run. In the tables the water-oil capillary pressure data has been defaulted with “1*” and will be set to zero as there are no other values for the water-oil capillary pressure columns.

3.4.5 KEYWORD FORMAT TYPE – ARRAY

This type of keyword defines a property for the grid or an area of the grid using a previously entered [BOX](#) keyword to define the area where the property will be defined. For array data a full set of values for each element in the array is required. For example, the documentation for the [PORO](#) array would be:

Description

[PORO](#) defines the porosity for all the cells in the model via an array. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PORO	PORO is an array of real numbers assigning the porosity values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*100.0.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- I) The keyword is terminated by a “/”.

Table 3.3: [PORO](#) Keyword Description

See also the [DX](#), [DY](#), and [TOPS](#) keywords to fully define a Cartesian Regular Grid.

Example

```

--
--  DEFINE GRID BLOCK POROSITY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PORO
  300*0.300
/

```

3.5 INPUT FILE STRUCTURE

OPM Flow input files are similar to commercial simulators that are used in the oil and gas industry, that is the input file is separated into sections in an effort to avoid errors in an engineer's input data and a computer programmer's code to interpret the data. OPM Flow has been designed from an engineer's prospective with an input structure similar to Schlumberger's industry wide ECLIPSE 100⁶¹ simulator. Table 3.4 lists the various sections together with a brief description of purpose of each section, as well as declaring if a section is mandatory or not for a run to form a valid input deck.

Section Name	Description	Required Optional
RUNSPEC	This is the first section in the OPM Flow input file and defines the key parameters for the simulator including the dimensions of the model, phases present in the model (oil, gas, and water for example), number of tables for a given property, and the maximum number of rows for each table, the maximum number of groups, wells and well completions, as well as various options to be invoked by OPM Flow.	Required
GRID	Defines the basic grid properties, including structure, faults, and various static rock properties (porosity, permeability, etc.). Upon completion of reading this section, the software calculates the pore volume (PORV) for each cell and the transmissibilities (TRANX , TRANX , and TRANZ) between all neighbouring cells, as well as calculating the transmissibilities of the Non-Neighbor Connections ("NNC").	Required
EDIT	The properties calculated by OPM Flow in the GRID section are available for editing in this section (PORV , TRANX , etc.).	Optional
PROPS	This section defines the fluid properties for all the phases present in the run, for example oil viscosity, oil formation volume factor, etc. The section also defines the rock flow properties such as the relative permeabilities and the capillary pressure functions.	Required
REGIONS	This section allows the engineer to define various regions in the model for reporting purposes and to define how the fluid and rock properties defined in the PROPS section are allocated throughout the model.	Optional
SOLUTION	Defines the parameters to initialize the model, fluid contacts, reservoir pressures, etc., together with the data from the previous sections. This section, if requested, reports the initial in-place volumes for phases present in the model, as well as the average pressure for the various defined regions.	Required
SUMMARY ¹	Defines the vectors of summary results versus time to be written to various output files for reviewing the results of the simulation. This includes field, group, well and well completion production, and injection results, for example field oil rate versus time. Grid block property results can also be reported versus time, for example grid block pressure versus time ² .	Optional
SCHEDULE	This is the final section that defines the time dependent data such as the field, group and well parameters, targets and constraints, the numerical controls, the operating schedule, and the reporting requirements.	Required

⁶¹ ECLIPSE Industry-Reference Reservoir Simulator – Reference Manual 2019.1, Schlumberger.

Section Name	Description	Required Optional
Notes: 1) Although the SUMMARY section is optional, it is nearly always included in order to obtain summary results (for example, production and injection profiles) from the simulation run. 2) The OPM ResInsight three-dimensional visualization software has a feature to create a cross-plot of a grid block property versus time, so it should not generally be necessary to export the grid block data to the SUMMARY report output files.		

Table 3.4: OPM Flow Input Deck Sections

CHAPTER 4: GLOBAL SECTION KEYWORDS

4.1 INTRODUCTION

Keywords used in this section can be used in all input file sections, that is the [RUNSPEC](#), [GRID](#), [EDIT PROPS](#), [REGIONS](#), [SOLUTION](#), [SUMMARY](#), and [SCHEDULE](#) sections.

4.2 DATA REQUIREMENTS

There are no data requirements for the GLOBAL section keywords, as all the keywords are optional and considered to be utility keywords that can be used throughout the input deck. A complete list of these global keywords is given in Table 4.1.

No.	GLOBAL Keyword	Description	Section Applied
1	COLUMNS	COLUMNS – Define Input File Column Margins . The COLUMNS keyword defines the input file column margins; characters outside the margins are ignored by the input parser.	ALL
2	DEBUG	DEBUG – Define the Debug Data to be Printed to File . This keyword defines the debug data to be written to the debug file (*.DBG), it is ignored by OPM Flow.	ALL
3	ECHO	ECHO – Activate Echoing of User Input Files to the Print File . Turns on echoing of all the input files to the print file.	ALL
4	END	END – Define the End of the Input File . This keyword marks the end of the input file and can occur in any section. Any keywords and data after the END keyword are ignored.	ALL
5	ENDINC	ENDINC – Define the End of an Include File . This keyword marks the end of an include file specified on the INCLUDE keyword. When the ENDINC keyword is encountered in the INCLUDE file, input data is read from the next keyword in the current file.	ALL
6	ENDSKIP	ENDSKIP – Deactivate Skipping of Keywords and Input Data . Turns off skipping of keywords that was activated by the SKIP, SKIP100, or SKIP300 keywords.	ALL
7	EXTRAPMS	EXTRAPMS – Activate Extrapolation Warning Messages . The EXTRAPMS keyword activates extrapolation warning messages for when simulator extrapolates the PVT or VFP tables.	ALL
8	FORMFEED	FORMFEED – Defined the Print File Form-Feed Character . The keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files.	ALL
9	INCLUDE	INCLUDE – Load Another Data File at the Current Position . The INCLUDE keyword informs OPM Flow to continue reading input data from the specified INCLUDE file.	ALL
10	MESSAGE	MESSAGE – Output User Message . The MESSAGE keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files.	ALL
11	MESSAGES	MESSAGES – Define Message Print Limits and Stop Limits . The MESSAGES keyword defines the print and stops levels for various messages.	ALL

No.	GLOBAL Keyword	Description	Section Applied
12	NOECHO	<i>NOECHO – Deactivate Echoing of User Input Files to the Print File.</i> Turns off echoing of all the input files to the print file.	ALL
13	NOWARN	<i>NOWARN – Deactivate Warning Messages.</i> Turns off warning messages to be printed to the print file.	ALL
14	SKIP	<i>SKIP – Activate Skipping of All Keywords and Input Data.</i> The keyword turns on skipping of keywords until the ENDSKIP activation keyword is encountered.	ALL
15	SKIP100	<i>SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator.</i> This keyword turns on skipping of all keywords and input data by the commercial black-oil simulator until the ENDSKIP keyword is encountered.	ALL
16	SKIP300	<i>SKIP300 – Activate Skipping of Keywords by Compositional Simulator.</i> This keyword turns on skipping of all keywords and input data by the commercial compositional simulator until the ENDSKIP keyword is encountered.	ALL
17	WARN	<i>WARN – Activate Warning Messages.</i> Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default.	ALL
Notes: 1) Note that not all keywords and features listed above are implemented in OPM Flow. Cells not colored in the No. column indicate the keyword is supported or partially supported by OPM Flow, cells colored gray indicate that the keyword is not applicable, and finally, cells colored in orange indicate keywords that are not supported by OPM Flow and will be ignored by the simulator.			

Table 4.1: Alphabetic List of GLOBAL Section Keywords

A more detailed description of the keywords is presented in the following sections.

4.3 KEYWORD DEFINITIONS

4.3.1 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

The *COLUMNS* keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

4.3.2 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

4.3.3 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      SWITCH OFF ECHOING OF INPUT FILES
--
NOECHO
--
--      INCLUDE SIMULATION GRID WITH SLOPING FAULTS
--
INCLUDE
      './INCLUDE/GRID/IRAP_1005.GRDECL' /
--
--      SWITCH ON ECHOING OF INPUT FILES
--
ECHO
```

The examples deactivates the echoing of the input files, reads in the grid geometry data using the [INCLUDE](#) keyword, and then activates the echoing of the input files again.

Note

Especially for the large voluminous data sets in the [GRID](#) section, it is good practice to deactivate the echoing of the input files when loading this data to avoid the print output file becoming too large to view in a text editor.

4.3.4 END – DEFINE THE END OF THE INPUT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

There is no data required for this keyword.

Example

```

-----
-- SCHEDULE SECTION - 2006-01-01
-----
RPTSCHED
'WELLS=2'      'WELSPECS'      'CPU=2'      'FIP=2'      /

DATES
 1 JAN  2006  /
/

RPTSCHED
'NOTHING'      /

DATES
 1 APR  2006  /
 1 JUL  2006  /
 1 OCT  2006  /
/
ECHO
--
-- *****
-- END OF FILE
-- *****
END
-----
-- SCHEDULE SECTION - 2007-01-01
-----
RPTSCHED
'WELLS=2'      'WELSPECS'      'CPU=2'      'FIP=2'      /

DATES
 1 JAN  2007  /
/

```

In the above example OPM Flow will process the data up to October 1, 2006 only, and then start to run the simulation. All keywords after the [END](#) file keyword will not be read or processed.

4.3.5 ENDINC – DEFINE THE END OF AN INCLUDE FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

There is no data required for this keyword.

Example

```

-----
-- SCHEDULE SECTION - 2006-01-01
-----
RPTSCHED
'WELLS=2'      'WELSPECS'      'CPU=2'      'FIP=2'      /

DATES
 1 JAN  2006  /
/

RPTSCHED
'NOTHING'      /

DATES
 1 APR  2006  /
 1 JUL  2006  /
 1 OCT  2006  /
/
ECHO
--
-- *****
-- END OF INCLUDE FILE PROCESSING
-- *****
ENDINC
-----
-- SCHEDULE SECTION - 2007-01-01
-----
RPTSCHED
'WELLS=2'      'WELSPECS'      'CPU=2'      'FIP=2'      /

DATES
 1 JAN  2007  /
/

```

In the above example OPM Flow will process the data up to October 1, 2006 only, and return control to the file that called the [INCLUDE](#) keyword, and then continue processing the input files. All keywords after the [ENDINC](#) keyword in the [INCLUDE](#) FILE will not be read or processed.

4.3.6 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#) or [SKIP300](#) keywords. Each [SKIP](#), [SKIP100](#) or [SKIP300](#) keyword should be paired with an [ENDSKIP](#) keyword.

There is no data required for this keyword.

The nesting of pairs of either the [SKIP](#), [SKIP100](#), or [SKIP300](#) keyword and the [ENDSKIP](#) keyword is not supported by OPM Flow.

Example

```
--
--      SWITCH ON SKIPPING OF KEYWORDS AND DATA
--
SKIP
--
--      INCLUDE SIMULATION GRID WITH SLOPING FAULTS
--
INCLUDE
      './INCLUDE/GRID/IRAP_1005.GRDECL' /
--
--      SWITCH ON READING OF KEYWORDS AND DATA
--
ENDSKIP
```

The example skips reading of the grid geometry data input using the [INCLUDE](#) keyword, and then reverts back to reading the remainder of the input file.

4.3.7 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **EXTRAPMS** keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

No.	Name	Description	Default
I	EXTRAP	Defines a single integer that activates the extrapolation warning message options for PVT and VFP tables. EXTRAP can have the following values: 0 – No warning messages are give (the default). 1 – PVT table extrapolation warnings are printed. 2 – VFP table extrapolation warnings are printed. 3 – PVT and VFP table extrapolation warnings are printed. 4 - PVT and VFP table extrapolation warnings are printed with additional information.	0
Notes: - In addition extrapolation warnings will also be given for Rs and Rv if options (1), (3), and (4) are requested. - The keyword is terminated by a “/”.			

Table 4.2: EXTRAPMS Keyword Description

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      ACTIVATE EXTRAPOLATION MESSAGES
--
EXTRAPMS
    2
/
```

The above example activates the default the VFP table extrapolation warnings option.

4.3.8 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *FORMFEED* keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be place at the very top of the input file.

No.	Name	Description	Default
1	<i>FORMFEED</i>	Defines a single integer that defines the carriage control character activates, and should be set to: 1) 1 – Standard FORTRAN (the default). 2) 2 – Form-feed character ASCII(12) 3) 3 – None.	0
Notes: 1) The keyword is terminated by a “/”.			

Table 4.3: FORMFEED Keyword Description

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE EXTRAPOLATION MESSAGES
--
FORMFEED      3                                     /
```

The above example sets the carriage return character to no form-feed character.

4.3.9 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *INCLUDE* keyword informs OPM Flow to continue reading input data from the specified *INCLUDE* file. When the end of the *INCLUDE* file is reached, or the *ENDINC* keyword is encountered in the included file, input data is read from the next keyword in the current file. Although *INCLUDE* files can be nested, that is *INCLUDE* files within *INCLUDE* files etc., in practice this should be avoided due to the complexity of tracking the files.

No.	Name	Description	Default
I	FILENAME	A character string enclosed in quotes that defines a file to read in and be processed by OPM Flow.	None
Notes: I) The keyword is terminated by a “/”.			

Table 4.4: INCLUDE Keyword Description

Examples

The first example shown below loads the grid file from the same directory as the data file.

```
--
--      LOAD INCLUDE FILE
--
INCLUDE      'NOR-OPM-A00-GRID.inc'  /
```

The next example loads the same file one directory above from where the data file is located.

```
--
--      LOAD INCLUDE FILE
--
INCLUDE      '../NOR-OPM-A00-FAULTS.inc'  /
```

The final example loads the same file from a separate include directory found in the parent directory relative to where the data file is located.

```
--
--      LOAD INCLUDE FILE
--
INCLUDE      '../INCLUDE/NOR-OPM-A00-FAULTS.inc'  /
```


4.3.10 MESSAGE – OUTPUT USER MESSAGE

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

4.3.11 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MESSAGES** keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

No.	Name	Description	Default
1	PRTSMESG	An integer defining the maximum number of MESSAGE type messages to be printed after which this type of message stops printing.	1,000,000
2	PRTSCOMT	An integer defining the maximum number of COMMENT type messages be printed after which this type of message stops printing.	1,000,000
3	PRTSWARN	An integer defining the maximum number of WARNING type messages be printed after which this type of message stops printing.	10,000
4	PRTSPROB	An integer defining the maximum number of PROBLEM type messages to be printed after which this type of message stops printing.	100
5	PRTSERRS	An integer defining the maximum number of ERROR type messages to be printed after which this type of message stops printing.	100
6	PRTSBUGS	An integer defining the maximum number of BUG type messages to be printed after which this type of message stops printing.	100
7	STOPMESG	An integer defining the maximum number of MESSAGE type messages to be printed after which OPM Flow terminates the run. Not used by OPM Flow.	1,000,000
8	STOPCOMT	An integer defining the maximum number of COMMENT type messages to be printed after which OPM Flow terminates the run. Not used by OPM Flow.	1,000,000
9	STOPWARN	An integer defining the maximum number of WARNING type messages to be printed after which OPM Flow terminates the run. Not used by OPM Flow.	10,000
10	STOPPROB	An integer defining the maximum number of PROBLEM type messages to be printed after which OPM Flow terminates the run. Not used by OPM Flow.	100
11	STOPERRS	An integer defining the maximum number of ERROR type messages to be printed after which OPM Flow terminates the run. Not used by OPM Flow.	10
12	STOPBUGS	An integer defining the maximum number of BUG type messages to be printed after which OPM Flow terminates the run. Not used by OPM Flow.	1

No.	Name	Description	Default
13	PRTGRPMS	An integer defining the maximum number of GROUP MESSAGE type messages to be printed after which this type of message stops printing. Not used by OPM Flow.	10
Notes: 1) MESSAGE type messages are informative messages. 2) COMMENT type messages are probably not data errors. 3) WARNING type message are possible data errors and should be investigated. 4) PROBLEM type errors messages are associated with numerical problems. 5) ERROR type messages are errors that need to be fixed before the simulator can run the input deck. 6) BUG type of messages are potential programming errors. 7) The keyword is terminated by a “/”.			

Table 4.5: MESSAGES Keyword Description

Example

```
--
--      MESS  COMMT WARN  PROBL ERROR BUG   MESS COMMT WARN  PROBL ERROR BUG
--      LIMIT LIMIT LIMIT LIMIT LIMIT LIMIT STOP STOP  STOP  STOP  STOP  STOP
MESSAGES
      1*    1*    1*    1500 1*    1*    1*    1*    1*    1000 1*    1*  /
```

The above example sets the PROBLEM type message print limit to 1,500 and the stop limit to 1,000.

4.3.12 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active, but can subsequently be switched off by the [NOECHO](#) activation keyword.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      SWITCH OFF ECHOING OF INPUT FILES
--
NOECHO
--
--      INCLUDE SIMULATION GRID WITH SLOPING FAULTS
--
INCLUDE      './INCLUDE/GRID/IRAP_1005.GRDECL' /
--
--      SWITCH ON ECHOING OF INPUT FILES
--
ECHO
```

The example deactivates the echoing of the input files, reads in the grid geometry data using the [INCLUDE](#) keyword, and then activates the echoing of the input files again.

Note

Especially for the large voluminous data sets in the [GRID](#) section, it is good practice to deactivate the echoing of the input files when loading this data to avoid the print output file becoming too large to view in a text editor.

4.3.13 NOWARN – DEACTIVATE WARNING MESSAGES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

It is recommended that [WARN](#) should always be used and action taken if necessary. For subsequent runs, the warning messages can be turned off.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      SWITCH OFF WARNING MESSAGES
NOWARN
--
--      INCLUDE SIMULATION GRID WITH SLOPING FAULTS
--
INCLUDE
'./INCLUDE/GRID/IRAP_1005.GRDECL' /
--
--      SWITCH ON WARNING MESSAGES
--
WARN
```

The example deactivates the warning messages before reading the grid geometry data using the [INCLUDE](#) keyword, and then activates the warning messages after reading the [INCLUDE](#) file.

4.3.14 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword. See also the [SKIP100](#) and [SKIP300](#) keywords.

There is no data required for this keyword.

The nesting of pairs of the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords and the [ENDSKIP](#) keyword is not supported by OPM Flow.

Example

```
--
--      SWITCH ON SKIPPING OF ALL KEYWORDS AND DATA
--
SKIP
--
--      INCLUDE SIMULATION GRID WITH SLOPING FAULTS
--
INCLUDE
      './INCLUDE/GRID/IRAP_1005.GRDECL' /
--
--      SWITCH ON READING OF ALL KEYWORDS AND DATA
--
ENDSKIP
```

The example skips reading of the grid geometry data input using the [INCLUDE](#) keyword, and then reverts back to reading the remainder of the input file.

4.3.15 SKIP100 – ACTIVATE SKIPPING OF KEYWORDS BY BLACK-OIL SIMULATOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator. Each [SKIP100](#) keyword should be paired with an [ENDSKIP](#) keyword. See also the [SKIP](#) and [SKIP300](#) keywords.

There is no data required for this keyword.

By default OPM Flow behaves like the commercial black-oil simulator. However, this behaviour can be changed using the --input-skip-mode command line parameter.

The nesting of pairs of either the [SKIP](#), [SKIP100](#), or [SKIP300](#) keyword and the [ENDSKIP](#) keyword is not supported by OPM Flow.

Example

If the following example is read using the commercial black-oil simulator then the [SKIP300](#) keyword will be ignored and the black-oil PVT data will be read, the [SKIP100](#) keyword will cause the compositional PVT data to be ignored, and then the remainder of the input will be read after the next [ENDSKIP](#) keyword is encountered.

```
--
--      ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR
--
SKIP300
--
--      INCLUDE BLACK-OIL PVT DATA
--
INCLUDE
    './INCLUDE/'BLACK-OIL-PVT'      /
--
--      DEACTIVATE SKIPPING OF KEYWORDS
--
ENDSKIP
--
--      ACTIVATE SKIPPING OF KEYWORDS BY BLACK-OIL SIMULATOR
--
SKIP100
--
--      INCLUDE COMPOSITIONAL PVT DATA
--
INCLUDE
    './INCLUDE/'COMPOSITION-PVT-EOS'  /
--
--      SWITCH ON READING OF ALL KEYWORDS AND DATA
--
ENDSKIP
--
--      WATER PVT TABLE
--
PVTW
--      REF PRES  BW      CW      VISC      VISC
--      PSIA      RB/STB  1/PSIA  CPOISE    GRAD
```

--	-----	-----	-----	-----	-----	
--	4840.0	1.019	2.7E-6	0.370	1*	/ WATER DATA REGION 1
--						
--	OIL	WAT	GAS			
--	DENSITY	DENSITY	DENSITY			
--	-----	-----	-----			
DENSITY						
--	39.0	62.37	0.04520			/ PVT DATA REGION 1
--	ROCK COMPRESSIBILITY					
--						
--	REF PRES	CF				
--	PSIA	1/PSIA				
--	-----	-----				
ROCK						
--	3966.9	5.0E-06				/ ROCK COMPRESSIBILITY

4.3.16
SKIP300
– ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator. Each [SKIP300](#) keyword should be paired with an [ENDSKIP](#) keyword. See also the [SKIP](#) and [SKIP100](#) keywords.

There is no data required for this keyword.

By default OPM Flow behaves like the commercial black-oil simulator. However, this behaviour can be changed using the --input-skip-mode command line parameter.

The nesting of pairs of either the [SKIP](#), [SKIP100](#), or [SKIP300](#) keyword and the [ENDSKIP](#) keyword is not supported by OPM Flow.

Example

If the following example is read using the commercial compositional simulator then the [SKIP300](#) keyword will cause the black-oil PVT data to be ignored, the input will continue to be read after the next [ENDSKIP](#) keyword is encountered, and the [SKIP100](#) keyword will be ignored and the compositional PVT data will be read along with the remainder of the input.

```
--
--      ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR
--
SKIP300
--
--      INCLUDE BLACK-OIL PVT DATA
--
INCLUDE
    './INCLUDE/'BLACK-OIL-PVT'      /
--
--      DEACTIVATE SKIPPING OF KEYWORDS
--
ENDSKIP
--
--      ACTIVATE SKIPPING OF KEYWORDS BY BLACK-OIL SIMULATOR
--
SKIP100
--
--      INCLUDE COMPOSITIONAL PVT DATA
--
INCLUDE
    './INCLUDE/'COMPOSITION-PVT-EOS'  /
--
--      SWITCH ON READING OF ALL KEYWORDS AND DATA
--
ENDSKIP
--
--      WATER PVT TABLE
--
PVTW
--      REF PRES  BW      CW      VISC      VISC
--      PSIA      RB/STB  1/PSIA  CPOISE   GRAD
```

--	-----	-----	-----	-----	-----	
	4840.0	1.019	2.7E-6	0.370	1*	/ WATER DATA REGION 1
--						
--	OIL	WAT	GAS			
--	DENSITY	DENSITY	DENSITY			
--	-----	-----	-----			
DENSITY						
	39.0	62.37	0.04520			/ PVT DATA REGION 1
--	ROCK COMPRESSIBILITY					
--						
--	REF PRES	CF				
--	PSIA	1/PSIA				
--	-----	-----				
ROCK						
	3966.9	5.0E-06				/ ROCK COMPRESSIBILITY

4.3.17 WARN – ACTIVATE WARNING MESSAGES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

It is recommended that [WARN](#) should always be used and action taken if necessary for the initial runs, once the run has been “cleaned up” the warning messages can be turned off.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      SWITCH OFF WARNING MESSAGES
--
NOWARN

--
--      INCLUDE SIMULATION GRID WITH SLOPING FAULTS
--
INCLUDE
    ' ./INCLUDE/GRID/IRAP_1005.GRDECL ' /

--
--      SWITCH ON WARNING MESSAGES
--
WARN
```

The example deactivates the warning messages before reading the grid geometry data using the [INCLUDE](#) keyword, and then activates the warning messages after reading the [INCLUDE](#) file.

CHAPTER 5: RUNSPEC SECTION

5.1 INTRODUCTION

This is the first section in the OPM Flow input file and defines the key parameters for the simulator including the dimensions of the model, phases present in the model (oil, gas, and water for example), number of tables for a given property and the maximum number of rows for each table, the maximum number of groups, wells and well completions, as well as various options to be invoked by the simulator.

The section is defined by the [RUNSPEC](#) keyword and is terminated by the [GRID](#) keyword.

Keywords associated with the [RUNSPEC](#) section are either Activation type keywords to switch on a feature or declare a phase, or a Vector (Row Vector) type of keyword that declares various options or array dimensions. Examples of Vector type keywords are the [SATOPTS](#) keyword that activates relative permeability assignment options, and the [WELLDIMS](#) keyword that defines the maximum number of wells, completions per well, maximum number of groups, etc. for the run. The keywords can be in any order, but the values entered impact the data the simulator expects in subsequent sections. So for example if the [VAPOIL](#) keyword has been invoked to declare that the vaporized oil phase (condensate) in wet gas⁶² is present in the run, then the simulator expects the [PVTG](#) keyword in the [PROPS](#) section to define the wet gas PVT properties. If the [PVDG](#) keyword (dry gas⁶³ PVT data set) has been entered instead in the [PROPS](#) section, then the simulator will report this as an error when it reads the [PVDG](#) keyword or after processing the [PROPS](#) section.

Note that in some instances the numerical values entered need to be the actual values, as per the [DIMENS](#) keyword, or maximum values, for example the previously mentioned [WELLDIMS](#) keyword. If the requirement for the keyword is for the actual value and an incorrect value has been entered then this may or may not be directly trapped by the simulator, and may result in numerous error message associated with the data instead. Thus, if the error reporting messages in a given section are associated with data that is required to be declared via a [RUNSPEC](#) keyword, one should also verify that the [RUNSPEC](#) keyword has been correctly defined with the associated problematic data error messages. This is particularly true for the [TABDIMS](#) keyword in the [RUNSPEC](#) section where some of the variables are defined as the actual number of tables, whereas the length of the tables are a maximum value. To be clear, if NTSFUN on the [TABDIMS](#) keyword has been set to five, then there must be exactly five sets of relative permeability data entered in the [PROPS](#) section, otherwise data input errors will occur.

5.2 DATA REQUIREMENTS

Table 5.1 outlines the minimum set of keywords required by OPM Flow in order for the simulator to successfully parse the RUNSPEC section.

No.	RUNSPEC Keyword	Description	Section Applied
1	TITLE	TITLE – Define the Title for the Input Deck - this is printed on most reports routed to the print file.	RUNSPEC
2	DIMENS	DIMENS – Define the Dimensions of the Model - the values entered must be the exact dimensions of the grid, otherwise the simulator will report errors.	GRID

⁶² Natural gas that contains significant heavy hydrocarbons such as propane, butane, and other liquid hydrocarbons is known as wet gas or rich gas.

⁶³ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates

No.	RUNSPEC Keyword	Description	Section Applied
3	OIL, DISGAS, GAS, VAPOIL, and/or WATER	<i>OIL – Activate the Oil Phase in the Model</i> <i>DISGAS – Activate the Dissolved Gas Phase in the Model</i> <i>GAS – Activate the Gas Phase in the Model</i> <i>VAPOIL – Activate the Vaporized Oil in Wet Gas Phase in the Model</i> <i>WATER – Activate the Water Phase in the Model</i> Only the phases present in the run should be declared, so for a dry gas run then only GAS and WATER keywords should be declared, since water is always present in hydrocarbon accumulations.	PROPS
4	FIELD, LAB, or METRIC	<i>FIELD – Activate the Oil Field System of Units for the Model</i> <i>LAB – Activate the Laboratory System of Units for the Model</i> <i>METRIC – Activate the Metric System of Units for the Model</i> States the system of units to be used for the input deck.	ALL
5	WELLDIMS	<i>WELLDIMS – Define the Wells and Group Dimensions</i> Sets the well and group dimensions etc.	SCHEDULE
Notes: 1) This limited input for the RUNSPEC section is never used in practice, as real full field models are more complicated.			

Table 5.1: Minimum Set of RUNSPEC Keywords Required by OPM Flow

If the minimum set of RUNSPEC keywords is only entered, then the default values of the missing keywords will apply. This means for example only one set of PVT, relative permeability tables, and one equilibrium region will be assumed by the simulator.

A complete list of RUNSPEC keywords in alphabetic order is shown in Table 5.2 together with a generalized Topic column that classifies the functionality of the keyword. Note that not all keywords and features listed in Table 5.2 are implemented in OPM Flow. Cells not colored in the No. column indicate the keyword is supported, cells colored gray indicate that the keyword is not applicable, and finally, cells colored in orange indicate keywords that are not supported by OPM Flow.

No.	RUNSPEC Keyword	Description	Topic
1	ACTDIMS	<i>ACTDIMS – ACTION Keyword Dimensions.</i>	Action
2	ACTPARAM	<i>ACTPARAM – Define Action Facility Target and Tolerance Parameters.</i>	Action
3	AITS	<i>AITS – Activate Intelligent Time Stepping.</i>	Numeric
4	AITSOFF	<i>AITSOFF – Deactivate Intelligent Time Stepping.</i>	Numeric
5	ALKALINE	<i>ALKALINE – Activate the Alkaline Phase and Model.</i>	Alkaline
6	API	<i>API – Activate API Tracking.</i>	API
7	AQUDIMS	<i>AQUDIMS – Define Aquifer Dimensions.</i>	Aquifer
8	AUTOREF	<i>AUTOREF – Define Auto Refinement Options.</i>	LGR

No.	RUNSPEC Keyword	Description	Topic
9	BIGMODEL	<i>BIGMODEL – Activate Big Model Option (Retired).</i>	N/A
10	BIOFILM	<i>BIOFILM – Activate the Biofilm Model.</i> This is an OPM Flow specific keyword used to investigate biofilm effects in underground applications. The module requires that either the <i>CO2STORE</i> or <i>H2STORE</i> keywords to be active.	Biofilm
11	BLACKOIL	<i>BLACKOIL – Activate Black-Oil Phases.</i>	Phases
12	BPARA	<i>BPARA – Activate Block Parallel License.</i>	N/A
13	BPIDIMS	<i>BPIDIMS – Define the Dimensions of the Interpolated Block Quantities.</i>	Output
14	BRINE	<i>BRINE – Activate Brine Tracking Option.</i>	Brine
15	CART	<i>CART – Activate Cartesian Geometry.</i> This the default grid type that is normally not required to be entered.	Input
16	CBMOPTS	<i>CBMOPTS – Define Coal Bed Methane Options.</i>	CBM
17	CO2SOL	<i>CO2SOL – Activate Dissolved CO2 in the Water Phase.</i> This is an OPM Flow implementation that is different to the commercial simulator. The keyword is a compositional keyword in the commercial simulator but has been implemented in OPM Flow's black-oil model. The keyword activates the carbon dioxide storage model for the run to account for both carbon dioxide and water phase solubility the simulators CO2-Brine PVT model.	Phases
18	CO2STORE	<i>CO2STORE – Activate the CO2 Storage Model.</i> This is an OPM Flow implementation that is different to the commercial simulator, the keyword is a compositional keyword in the commercial simulator but has been implemented in OPM Flow's black-oil model. The keyword activates the carbon dioxide storage model for the run to account for both carbon dioxide and water phase solubility the simulators CO2-Brine PVT model.	Phases
19	COAL	<i>COAL – Activate the Coal Phase (CBM Model).</i>	CBM
20	COLUMNS	<i>COLUMNS – Define Input File Column Margins.</i>	Input
21	COMPS	<i>COMPS – Activate Compositional Modeling Formulation.</i> OPM Flow does not currently support the general compositional modelling formulation. This keyword is only supported by OPM Flow when the two component gas-water CO2 storage model has been activated using the <i>CO2STORE</i> keyword.	Phases
22	CPR	<i>CPR – Activate Constrained Pressure Residual ("CPR") Linear Solver.</i> See section <i>2.2 Running OPM Flow From The Command Line</i> on how to invoke various numerical schemes via the OPM Flow command line interface.	Numeric

No.	RUNSPEC Keyword	Description	Topic
23	DEBUG	<i>DEBUG – Define the Debug Data to be Printed to File.</i>	Output
24	DIFFUSE	<i>DIFFUSE – Activate Molecular Diffusion Option.</i>	Diffusion
25	DIMENS	<i>DIMENS – Define the Dimensions of the Model.</i> The values entered must be the exact dimensions of the grid, otherwise the simulator will report errors.	Input
26	DISGAS	<i>DISGAS – Activate the Dissolved Gas Phase in the Model.</i>	Phases
27	DISGASW	<i>DISGASW – Activate Dissolved Gas in the Water Phase in the Model</i> This is an OPM Flow specific keyword that activates dissolved gas in the water phase in the model.	Phases
28	DISPDIMS	<i>DISPDIMS – Define the Maximum Number of Dispersion Tables.</i>	Dispersion
29	DUALPERM	<i>DUALPERM – Activate Dual Permeability Model.</i>	Dual-Porosity
30	DUALPORO	<i>DUALPORO – Activate Dual Porosity Model.</i>	Dual-Porosity
31	DYNRDIMS	<i>DYNRDIMS – Define Dynamic Region Dimensions.</i>	Input
32	ECHO	<i>ECHO – Activate Echoing of User Input Files to the Print File.</i>	Output
33	ECLMC	<i>ECLMC – Activate Multi-Component Brine Model.</i>	Brine
34	END	<i>END – Define the End of the Input File.</i>	Input
35	ENDINC	<i>ENDINC – Define the End of an Include File.</i>	Input
36	ENDSCALE	<i>ENDSCALE – Activate Relative Permeability End-Point Scaling Option.</i>	End-Point
37	ENDSKIP	<i>ENDSKIP – Deactivate Skipping of Keywords and Input Data.</i>	Input
38	EOS	<i>EOS – Specify Equation of State.</i> OPM Flow does not currently support the general compositional modelling formulation.	Phases
39	EQLDIMS	<i>EQLDIMS – Define the Equilibration Data Dimensions.</i>	Equilibration
40	EQLOPTS	<i>EQLOPTS – Activate the Equilibration Options.</i>	Equilibration
41	EXTRAPMS	<i>EXTRAPMS – Activate Extrapolation Warning Messages.</i>	Output
42	FAULTDIM	<i>FAULTDIM – Define the Number of Fault Segments.</i>	Input
43	FIELD	<i>FIELD – Activate the Oil Field System of Units for the Model.</i>	Input
44	FMTHMD	<i>FMTHMD – Activate The Format History Match Gradient File Option.</i>	Output
45	FMTIN	<i>FMTIN – Activate The Format Input File Option.</i>	Input
46	FMTOUT	<i>FMTOUT – Activate The Format Output File Option.</i>	Output
47	FOAM	<i>FOAM – Activate the Foam Phase and Model.</i>	Foam

No.	RUNSPEC Keyword	Description	Topic
48	FORMFEED	<i>FORMFEED – Defined the Print File Form-Feed Character.</i>	Output
49	FRICTION	<i>FRICTION – Activate Wellbore Friction Option.</i>	Well
50	FULLIMP	<i>FULLIMP – Activate Fully Implicit Solution Option.</i> See section 2.2 Running OPM Flow From The Command Line on how to invoke various numerical schemes via the OPM Flow command line interface.	Numeric
51	GAS	<i>GAS – Activate the Gas Phase in the Model.</i>	Phases
52	GASFIELD	<i>GASFIELD – Define Gas Field Operations Options.</i>	Gas Field
53	GASWAT	<i>GASWAT – Activate the Gas-Water Model Formulation.</i>	Phases
54	GDIMS	<i>GDIMS – Activate Instantaneous Gradient Option and Define Dimensions .</i>	Gradient
55	GIMODEL	<i>GI – Define the Initial Equilibration Gi Values for All Grid Blocks.</i>	Phases
56	GRAVDR	<i>GRAVDR – Activate Gravity Drainage and Imbibition for Dual Porosity Model.</i>	Dual-Porosity
57	GRAVDRB	<i>GRAVDRB – Activate Vertical Discretized Gravity Drainage and Imbibition for Dual Porosity Model .</i>	Dual-Porosity
58	GRAVDRM	<i>GRAVDRM – Activate Alternative Gravity Drainage and Imbibition for Dual Porosity Model.</i>	Dual-Porosity
59	GRIDOPTS	<i>GRIDOPTS – Grid Processing Options.</i>	Input
60	H2SOL	<i>H2SOL – Activate Dissolved H2 in the Water Phase.</i> The keyword activates the dissolved hydrogen in the water phase, where hydrogen is represented by the SOLVENT pseudo component.	Phases
61	H2STORE	<i>H2STORE – Activate the H2 Storage Model.</i> The keyword activates the hydrogen storage model for the run to account for both hydrogen and water phase solubility using the simulator's H2-Brine PVT model.	Phases
62	HMDIMS	<i>HMDIMS – Define History Match Gradient Parameter Dimensions.</i>	Gradient
63	HYST	<i>HYST – Activate the Hysteresis Option (Retired).</i> The HYST keyword activates the hysteresis option, the keyword should be avoided and the hysteresis option should be enabled by the HYSTER parameter on the SATOTPS keyword see - SATOPTS – Activate Relative Permeability Assignment Options .	Hysteresis
64	IMPES	<i>IMPES – Activate Implicit Pressure Explicit Saturation Solution Option.</i> See section 2.2 Running OPM Flow From The Command Line on how to invoke various numerical schemes via the OPM Flow command line interface.	Numeric

No.	RUNSPEC Keyword	Description	Topic
65	IMPLICIT	<i>IMPLICIT – Activate Fully Implicit Solution Option.</i> See section 2.2 Running OPM Flow From The Command Line on how to invoke various numerical schemes via the OPM Flow command line interface.	Numeric
66	INCLUDE	<i>INCLUDE – Load Another Data File at the Current Position.</i>	Input
67	INSPEC	<i>INSPEC – Activate the INSPEC File Option.</i>	Output
68	LAB	<i>LAB – Activate the Laboratory System of Units for the Model.</i>	Input
69	LGR	<i>LGR – Define Local Grid Refinement Dimensions and Parameters.</i>	LGR
70	LGRCOPY	<i>LGRCOPY – Activate Local Grid Refinement Inheritance.</i>	LGR
71	LICENSES	<i>LICENSES – Define Required Licenses for Run.</i>	N/A
72	LIVEOIL	<i>LIVEOIL – Activate the Live Oil Phase (Oil with Free and Dissolved Gas).</i>	Phases
73	LOAD	<i>LOAD – Load a SAVE File for a Fast Restart.</i>	Input
74	LOWSALT	<i>LOWSALT – Activate the Low Salt Brine Phase in the Brine Model.</i>	Brine
75	MEMORY	<i>MEMORY – Define Allocated Memory (Retired).</i>	N/A
76	MESSAGE	<i>MESSAGE – Output User Message.</i> The MESSAGE keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files.	Output
77	MESSAGES	<i>MESSAGES – Define Message Print Limits and Stop Limits.</i> The MESSAGES keyword defines the print and stops levels for various messages.	Output
78	MESSRVC	<i>MESSRVC – Activate or Deactivate Database Message File Output.</i>	Output
79	METRIC	<i>METRIC – Activate the Metric System of Units for the Model.</i>	Input
80	MICP	<i>MICP – Activate the Microbially Induced Calcite Precipitation Model.</i> This is an OPM Flow specific keyword that activates the Microbial Induced Calcite Precipitation ("MICP") Model used to investigate leakage remediation. The module requires that both the <i>MICP</i> and <i>WATER</i> keywords in the <i>RUNSPEC</i> to be active.	MICP
81	MINNPCOL	<i>MINNPCOL – Define the Minimum Number of Newton Iterations Used to Update Well Targets.</i> This is an OPM Flow specific keyword that sets the minimum number of Newton iterations, as oppose to the commercial simulator's NUPCOL keyword that defines the maximum number of Newton iterations within a time step, after which well targets are frozen.	Numeric
82	MISCIBLE	<i>MISCIBLE – Define Miscibility Todd-Longstaff Parameters.</i>	Miscible
83	MONITOR	<i>MONITOR – Activate Output of the Monitoring Data and File.</i>	Output
84	MSGFILE	<i>MSGFILE – Activate or Deactivate Message File Output.</i>	Output

No.	RUNSPEC Keyword	Description	Topic
85	MULTIN	<i>MULTIN – Activate the Non-Unified Multiple Input File Option.</i>	Input
86	MULTOUT	<i>MULTOUT – Activate the Non-Unified Multiple Output File Option.</i>	Output
87	MULTOUTS	<i>MULTOUTS – Activate Non-Unified Multiple Summary Output File Option.</i>	Output
88	MULTREAL	<i>MULTREAL – Activate Commercial Simulator's Multi-Realization License.</i>	N/A
89	NETWORK	<i>NETWORK – Activate the Extended Network Option and Define Parameters.</i>	Network
90	NINEPOIN	<i>NINEPOIN – Activate the Nine-Point Discretization Option.</i>	Numeric
91	NMATRIX	<i>NMATRIX – Activate the Discretized Matrix Dual Porosity Option.</i>	Dual-Porosity
92	NNEWTF	<i>NNEWTF – Activate the Non-Newtonian Fluid Model.</i>	Numeric
93	NOCASC	<i>NOCASC – Activate Linear Solver Tracer Algorithm.</i> See section 2.2 <i>Running OPM Flow From The Command Line</i> on how to invoke various numerical schemes via the OPM Flow command line interface.	Tracers
94	NODPPM	<i>NODPPM – Deactivate Fracture Porosity-Permeability Calculation</i>	Dual-Porosity
95	NOECHO	<i>NOECHO – Deactivate Echoing of User Input Files to the Print File.</i>	
96	NOHYST	<i>NOHYST – Deactivate the Hysteresis Option.</i>	Hysteresis
97	NOINSPEC	<i>NOINSPEC – Deactivate Output of the INIT Index File.</i>	Output
98	NOMONITO	<i>NOMONITO – Deactivate Output of the Monitoring Data and File.</i>	Output
99	NONNC	<i>NONNC – Deactivate Non-Neighbor Connections.</i>	Input
100	NORSSPEC	<i>NORSSPEC – Deactivate Output of the RESTART Index File.</i>	Output
101	NOSIM	<i>NOSIM – Activate the No Simulation Mode for Data File Checking.</i>	Numeric
102	NOWARN	<i>NOWARN – Deactivate Warning Messages.</i>	Output
103	NRSOUT	<i>NRSOUT – Defined Maximum Number of RESTART Elements .</i>	Output
104	NSTACK	<i>NSTACK – Define the Stack Length for the Iterative Linear Solver.</i>	Numeric
105	NUMRES	<i>NUMRES – Define the Number of Reservoir Grids.</i> Currently, this should only be set to one in OPM Flow.	Input
106	NUPCOL	<i>NUPCOL – Define the Maximum Number of Newton Iterations Used to Update Well Targets.</i>	Numeric
107	OIL	<i>OIL – Activate the Oil Phase in the Model.</i>	Phases
108	OPTIONS	<i>OPTIONS – Activate Various Program Options.</i>	Options

No.	RUNSPEC Keyword	Description	Topic
109	PARALLEL	<i>PARALLEL – Define Parallel Run Configuration.</i> The keyword defines the run to use parallel processing and sets the domain decomposition options. See section 2.2 Running OPM Flow From The Command Line on how to run OPM Flow in parallel mode.	Numeric
110	PARTTRAC	<i>PARTTRAC – Activate and Define Partitioned Tracer Option.</i>	Tracers
111	PATHS	<i>PATHS – Define Filename Directory Path Aliases.</i>	Input
112	PEDIMS	<i>PEDIMS – Define Petro-Elastic Model Regions and Table Dimensions.</i>	Petro-Elastic
113	PETOPTS	<i>PETOPTS – Define Petrel and Generic Simulation File Options.</i>	Input
114	PIMTDIMS	<i>PIMTDIMS – Define Well Productivity Scaling Table Dimensions.</i>	Wells
115	PINTDIMS	<i>PINTDIMS – Define Polymer Molecular Weight Model Table Dimensions.</i> This is an OPM Flow specific keyword for the simulator's Polymer Molecular Weight Transport option that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure.	Polymer
116	POLYMER	<i>POLYMER – Activate the Polymer Phase in the Model.</i>	Polymer
117	POLYMW	<i>POLYMW – Activate the Polymer Molecular Weight Transport Option</i> This is an OPM Flow specific keyword for the simulator's Polymer Molecular Weight Transport option that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure.	Polymer
118	PRECSALT	<i>PRECSALT – Activate the OPM Flow Salt Precipitation Model.</i> This is an OPM Flow specific keyword for the simulator's Salt Precipitation Model, note that this is an extension to the commercial simulator's Brine Model.	Water Vaporization
119	PSTEADY	<i>PSTEADY – Activate Pseudo Steady State Flow Calculation Option.</i>	Numeric
120	RADIAL	<i>RADIAL – Radial Grid Activation Option.</i>	Radial
121	REGDIMS	<i>REGDIMS – Define the Maximum Number of Regions for a Region Array.</i>	Input
122	RIVRDIMS	<i>RIVRDIMS – Define the River Dimensions and Associated Data.</i>	Rivers
123	ROCKCOMP	<i>ROCKCOMP – Activate Rock Compaction .</i>	Compaction
124	RPTCPL	<i>RPTCPL – Activate Couple Simulation Reporting.</i>	Output
125	RPTHMD	<i>RPTHMD – Define Well History Match Gradient Reporting Options.</i>	Gradient
126	RPTRUNSP	<i>RPTRUNSP – Activate RUNSPEC Reporting.</i>	Output
127	RSSPEC	<i>RSSPEC – Activate Output of the RESTART Index File.</i>	Output

No.	RUNSPEC Keyword	Description	Topic
128	RUNSPEC	<i>RUNSPEC – Define the Start of the RUNSPEC Section of Keywords</i>	
129	SAMG	<i>SAMG – Activate Algebraic Multi-Grid Linear Solver.</i> See section 2.2 Running OPM Flow From The Command Line on how to invoke various numerical schemes via the OPM Flow command line interface.	Numeric
130	SATOPTS	<i>SATOPTS – Activate Relative Permeability Assignment Options .</i>	Input
131	SAVE	<i>SAVE – Activate Output of a SAVE File for Fast Restarts.</i>	Output
132	SCDPDIMS	<i>SCDPDIMS – Define Scale Deposition and Damage Table Dimensions.</i>	Scale Deposition
133	SKIP	<i>SKIP – Activate Skipping of All Keywords and Input Data.</i>	Input
134	SKIP100	<i>SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator.</i>	Input
135	SKIP300	<i>SKIP300 – Activate Skipping of Keywords by Compositional Simulator.</i>	Input
136	SMRYDIMS	<i>SMRYDIMS – Define Maximum Number of Summary Vectors to be Written .</i>	Output
137	SOLVDIMS	<i>SOLVDIMS – Define PEBI Grid Nested Factorization Solver Dimensions .</i> See section 2.2 Running OPM Flow From The Command Line on how to invoke various numerical schemes via the OPM Flow command line interface.	Numeric
138	SOLVENT	<i>SOLVFRAC – Define the Initial Gas Solvent Fraction for All Grid Blocks</i>	Solvent
139	SPIDER	<i>SPIDER – Spider Grid Activation Option.</i> This is an OPM Flow specific keyword for the simulator's Spider Grid option, that emulates a Radial Grid via corner-point geometry. The option employs the standard GRID section radial grid keywords to construct the grid - see the example in the section on Spider Grids (<i>Spider Grids</i>).	Radial
140	START	<i>START – Simulation Start Date.</i>	Numeric
141	SURFACT	<i>SURFACT – Activate the Surfactant Phase in the Model.</i>	Surfactant
142	SURFACTW	<i>SURFACTW – Activate the Surfactant Phase with Wettability Changes in the Model.</i>	Surfactant
143	TABDIMS	<i>TABDIMS – Define the Number of Tables and the Table Dimensions.</i>	Input
144	TEMP	<i>TEMP – Activate the Temperature Modeling Option.</i> The temperature option (TEMP keyword) and the thermal option (THERMAL keyword) are two separate modeling facilities in the commercial simulator. See THERMAL – Activate the Thermal Modeling Option for additional information.	Thermal

No.	RUNSPEC Keyword	Description	Topic
145	THERMAL	<i>THERMAL – Activate the Thermal Modeling Option.</i> This keyword activates OPM Flow's thermal modeling option. The energy black-oil implementation in OPM Flow is a mixture of the commercial simulators black-oil and the commercial simulator's "compositional thermal" keywords, as well as some OPM Flow specific keywords. See the keyword for further information.	Thermal
146	TITLE	<i>TITLE – Define the Title for the Input Deck.</i>	Input
147	TRACERS	<i>TRACERS – Activate Tracer Options and Set Tracer Array Dimensions.</i>	Tracers
148	TRPLPORO	<i>TRPLPORO – Activate the Triple Porosity Model Option.</i>	Triple-Porosity
149	UDADIMS	<i>UDADIMS – Define the Dimensions of the User Defined Arguments.</i>	Action
150	UDQDIMS	<i>UDQDIMS – Define the Dimensions of the User Defined UDQ Feature.</i>	Action
151	UDQPARAM	<i>UDQPARAM – Define Parameters for the User Defined Quantity Feature.</i>	Action
152	UDTDIMS	<i>UDTDIMS – Define the Dimensions of the User Defined Tables.</i>	Action
153	UNCODHMD	<i>UNCODHMD – Activate History Match Gradient Unencoded Output.</i>	Gradient
154	UNIFIN	<i>UNIFIN – Activate The Unified Input File Option.</i>	Input
155	UNIFOUT	<i>UNIFOUT – Activate The Unified Output File Option.</i>	Output
156	UNIFOUTS	<i>UNIFOUTS – Activate The Unified Output Summary File Option.</i>	Output
157	UNIFSAVE	<i>UNIFSAVE – Activate The Unified Output Save File Option.</i>	Output
158	VAPOIL	<i>VAPOIL – Activate the Vaporized Oil in Wet Gas Phase in the Model</i>	Phases
159	VAPWAT	<i>VAPWAT – Activate Vaporized Water in the Dry and Wet Gas Phases.</i> This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by declaring that vaporized water is present in the run.	Water Vaporization
160	VE	<i>VE – Activate Vertical Equilibrium Model (Global).</i>	Input
161	VFPIDIMS	<i>VFPIDIMS – Injection Vertical Flow Performance Table Dimensions.</i>	Well
162	VFPPDIMS	<i>VFPPDIMS – Production Vertical Flow Performance Table Dimensions.</i>	Well
163	VISAGE	<i>VISAGE – Activate External Reservoir Geo-Mechanics VISAGE Option</i> .	N/A
164	VISCD	<i>VISCD – Activate Dual Porosity Viscous Displacement Option .</i>	Dual-Porosity
165	WARN	<i>WARN – Activate Warning Messages.</i>	Output
166	WATER	<i>WATER – Activate the Water Phase in the Model.</i>	Phases
167	WELLDIMS	<i>WELLDIMS – Define the Wells and Group Dimensions.</i>	Well
168	WPOTCALC	<i>WPOTCALC – Well Potential Calculation Options.</i>	Well

No.	RUNSPEC Keyword	Description	Topic
169	WSEGDIMS	WSEGDIMS – Define Multi-Segment Well Dimensions.	Multi-Segment Well

Notes:

- 1) Note that not all keywords and features listed above are implemented in OPM Flow. Cells not colored in the No. column indicate the keyword is supported or partially supported by OPM Flow, cells colored gray indicate that the keyword is not applicable, and finally, cells colored in orange indicate keywords that are not supported by OPM Flow and may cause the simulation to be stopped.
- 2) For the Topic column the definition of the entries is defined as follows:

1) Action: Keywords associated with the ACTIONX keyword or OPM Flow's Python facility. 2) API: Keywords associated with the API option. 3) Biofilm: BIOFILM model activation. 4) Brine: Keywords associated with the BRINE option. 5) CBM: COAL BED METHANE activation and options (not implemented in OPM Flow). 6) Diffusion: DIFFUSION model activation and options. 7) Dispersion: DISPERSION model activation and options. 8) Dual-Porosity: DUAL POROSITY and DUAL PERMEABILITY model activation and parameters. 9) End-Point: END-POINT SCALING activation and options. 10) Equilibration EQUIL section control and options. 11) Gas Field: GAS FIELD model activation, control and options. 12) Gradient: Keywords associated with the GRADIENT option, used in history matching. 13) Foam: FOAM model activation. 14) Hysteresis: HYSTERESIS model.	15) Input: Input control and options. 16) LGR: LOCAL GRID REFINEMENT dimensions and options. 17) MICP: MICROBIAL INDUCED CALCITE PRECIPITATION model activation. 18) Multi-Segment Well: Multi-segment well dimensions and options. 19) Network: Activation and options for the NETWORK model. 20) Numeric: Keywords related to the various numeric control and options for the run. 21) Output: Output control and options. 22) Phases: Declarations related to the active phases. 23) Scale Deposition: Scale and BRINE options. 24) Solvent: SOLVENT phase options 25) Surfactant: SURFACTANT phase activation and options. 26) Thermal: THERMAL model activation and options. 27) Tracers: TRACER model options. 28) Triple-Porosity: TRIPLE POROSITY model option activation. 29) Water Vaporization: WATER VAPORIZATION activation.
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Table 5.2: Alphabetic List of RUNSPEC Keywords

Note that a number of keywords can be classified under several topics, so the Topic column should only be used as a general reference guide.

Example

A typical RUNSPEC section using various keywords is shown below and on the next few pages for reference.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      DEFINE THE TITLE FOR THE RUN
--
TITLE
CASE CASENAME - SIMULATION RUN TITLE
--
--      DEFINE THE START DATE FOR THE RUN
--
START
      01 'JAN' 2021 /
--
--      SWITCH NO SIMULATION MODE FOR DATA CHECKING COMMENT OUT TO RUN THE MODEL
--
NOSIM
--
-- FLUID TYPES AND TRACER OPTIONS
--
--
--      OIL PHASE IS PRESENT IN THE RUN
--
OIL
--
--      WATER PHASE IS PRESENT IN THE RUN
--
WATER
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
--
--      DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN
--
DISGAS
--
--      VAPORIZED OIL IN WET GAS IS PRESENT IN THE RUN
--
VAPOIL
--
--      NUMBER AND TYPE OF TRACERS
--
--      NO OIL  NO WAT  NO GAS  NO ENV  DIFF    MAX    MIN    TRACER
--      TRACERS TRACERS TRACERS TRACERS CONTL  NONLIN NONLIN NONLIN
TRACERS
      0          0          1          0      'NODIFF' 1*      1*      1* /
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-- GRID AND EQUILBRATION DIMENSIONS AND OPTIONS										

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-- GROUP, WELL AND VFP TABLE DIMENSIONS											

--											
-- WELL WELL GRUPS GRUPS											
-- MXWELS MXCONS MXGRPS MXGRPW											
WELLDIMS											
60 110 18 40 /											
--											
-- DEFINE WELL PRODUCTIVITY SCALING TABLE DIMENSIONS											
-- MAX MAX											
-- TABLES ENTRIES											
PIMTDIMS											
1 51 /											
--											
-- INJECTING VFP TABLES											
-- VFP VFP VFP											
-- MXMFLO MXMTHP NMMVFT											
VFPIDIMS											
10 10 12 /											
--											
-- PRODUCING VFP TABLES											
-- VFP VFP VFP VFP VFP VFP											
-- MXMFLO MXMTHP MXMWFR MXMGFR MXMALQ NMMVFT											
VFPPDIMS											
20 10 10 10 6 9 /											

-- MISCELLANEOUS OPTIONS											

--											
-- USER DEFINED ARGUMENT DIMENSIONS FACILITY											
-- MAX MAX MAX MAX MAX MAX MAX MAX MAX MAX RAND											
-- FUNCS ITEMS CONNS FIELD GROUP REGS SEGTM WELL AQUF BLCKS OPT											
UDQDIMS											
16 16 0 0 0 0 0 0 0 0 N /											
--											
-- USER DEFINED ARGUMENT DIMENSIONS											
-- NO. NOT TOTAL											
-- ARGS USED UDQ											
UDADIMS											
2 1* 1* /											
--											
-- USER DEFINED DEFAULT VALUES											
-- SEED RANGE UNDEFINED COMPARISON											
-- INTG -AND+ VALUE TOLERANCE											
UDQPARAM											
1 1* 0.0 1.0E-4 /											

-- NUMERICAL AND RUN CONTROL OPTIONS											

--											
-- SET STACK SIZE FOR LINEAR SOLVER											
--											
NSTACK											
30 /											
--											
-- DEFINE THE NUMBER OF ITERATIONS TO UPDATE WELL FLOW TARGETS											
--											
NUPCOL											
3 /											

```
--
--      SKIP      ACTIVATE
--      OPTIONS   OPTION
OPTIONS
      77*0      1
--
-- -----
-- INPUT AND OUTPUT OPTIONS
-- -----
--
--      METRIC SYSTEM OF UNITS FOR BOTH INPUT AND OUTPUT
--
METRIC
--
--      SWITCH ON THE UNIFIED INPUT FILES OPTION
--
UNIFIN
--
--      SWITCH ON THE UNIFIED OUTPUT FILES OPTION
--
UNIFOUT
--
--      MESS  COMMT WARN  PROBL ERROR BUG  MESS COMMT WARN  PROBL ERROR BUG
--      LIMIT LIMIT LIMIT LIMIT LIMIT LIMIT STOP STOP  STOP  STOP  STOP  STOP
MESSAGES
      3000  1*    1000  1000  1*    1*    1*    1*    9000  1*    9000  1*    /
--
--      DEBUG PRINTING OPTIONS
--
DEBUG
      8*0    1      11*0  1      30*0
--
-- =====
```

The above example is rather extensive but is not unrealistic in terms of the features being activated and the array size declarations.

On 32-bit systems, array sizes had to be carefully managed to fit within limited addressable memory. On modern 64-bit systems such as Linux, this limitation is no longer a concern.

5.3 KEYWORD DEFINITIONS

5.3.1 ACTDIMS – ACTION KEYWORD DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [ACTDIMS](#) keyword defines the maximum number of properties associated with the [ACTION](#) keyword. The [ACTION](#) keyword allows the user to enter computational logic and calculations to the simulation run based on the how the simulation run is proceeding.

No.	Name	Description	Default
1	MXACTNS	A positive integer value that defines the maximum number of ACTION keywords defined in the input deck.	2
2	MXLINES	A positive integer value that defines the maximum number of lines in an ACTION statement.	50
3	MXCHARS	A positive integer value that defines the maximum characters in an ACTION statement.	80
4	MXSTATMS	A positive integer value that defines the maximum number of conditional statements in the ACTION statement.	3
Notes: 1) The keyword is terminated by a “/”.			

Table 5.3:ACTDIMS Keyword Description

Only the [ACTIONX](#) keyword and computational logic have been implemented in OPM Flow.

Example

```
--      ACTION      ACTION      ACTION      ACTION
--      MXACTNS      MXLINES      MXCHARS      MXSTATMS
ACTDIMS      2          50          80          3          /
```

The above example defines the default values for the [ACTDIMS](#) keyword.

5.3.2 ACTPARAM – DEFINE ACTION FACILITY TARGET AND TOLERANCE PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **ACTPARAM** keyword defines the maximum target percent value for the **ACTION** series of keywords and the fractional equality tolerance for determining if two numbers are numerically equal when comparing values using the **ACTION** series of keywords. The **ACTION** keyword allows the user to enter computational logic and calculations to the simulation run based on the how the simulation run is proceeding.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	MXTOLS	A positive real value that defines the maximum target percent number for the <i>ACTION</i> series of keywords. The default value of 100 means the target is not applied.			Defined
		percent 100.0	percent 100.0	percent 100.0	
2	MXEQLS	MXEQLS a real positive number greater than zero and less than one that defines the tolerance used to determine if two real values are equal for comparing values in the <i>ACTION</i> series of keywords. Floating-point numbers (as implemented in computers) are never exact, one cannot compare floating point numbers for exact equality. Thus, MXEQLS defines a tolerance. For example, the default value of 1×10^{-4} means that if the difference between two real values is less than 1×10^{-4} then the values are considered equal.			Defined
		fraction 1×10^{-4}	fraction 1×10^{-4}	fraction 1×10^{-4}	
Notes: 1) The keyword is terminated by a “/”.					

Table 5.4: ACTPARAM Keyword Description

Although this keyword is read by OPM Flow, the **ACTION** and **UDQ** computational logic and calculations have not been fully implemented and therefore this keyword should not be used as it may result in OPM Flow terminating.

Example

```
--
-- ACTION ACTION
-- MXTOLS MXEQLS
ACTPARAM 5.0 1.0E-4 /
```

The above example defines the maximum tolerance to be 5% and the equality tolerance to be the default value of 1.0×10^{-4} .

5.3.3 AITS – ACTIVATE INTELLIGENT TIME STEPPING

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

Turns on the commercial simulator's intelligent time stepping.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

5.3.4 AITSOFF – DEACTIVATE INTELLIGENT TIME STEPPING

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

Turns off the commercial simulator's intelligent time stepping.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

5.3.5 ALKALINE – ACTIVATE THE ALKALINE PHASE AND MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that an alkaline phase is present in the model and to activate the Alkaline Model in the run. The keyword will also invoke data input file checking to ensure that all the required Alkaline Model input parameters are defined in the input deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ALKALINE PHASE IS PRESENT IN THE RUN
--
ALKALINE
```

The above example declares that the alkaline phase is active in the model to activate Alkaline Model.

5.3.6 API – ACTIVATE API TRACKING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on API tracking so that the various “oil types” are tracked in the model.

In many reservoirs the initial API gravity of oil varies with depth due to the heavy viscous fractions occupying the deepest part of the reservoir whilst the lighter more mobile fractions will occupy the upper part of the reservoir. As a reservoir is depleted the API gravity of oil in a cell will gradually change as the different fluids mix.

In OPM Flow it is possible to define different PVT regions in a reservoir, as in all finite difference formulated simulators, oil moving from one region to another will suddenly assume the properties of that region it has moved to. The fluid type tracking option allows the smooth change of PVT properties in a cell to be simulated by correlating PVT properties against the API gravity of oil.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE THE API TRACKING OPTION
--
API
```

The above example switches on the API tracking facility.

5.3.7 AQUDIMS – DEFINE AQUIFER DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **AQUDIMS** keyword defines the dimensions of the various aquifer property data. The data is normally entered on a single line and is terminated by a “/”.

No.	Name	Description	Default
1	MXNAQN	A positive integer value that defines the AQUNUM keyword maximum number of lines associated with this keyword, that is the maximum number of numerical aquifers	1
2	MXNAQC	A positive integer value that defines the AQUCON keyword maximum number of lines of connection data associated with this keyword, that is the maximum number of lines of connection data for numerical aquifers.	1
3	NIFTBL	A positive integer value that defines the AQUTAB keyword maximum number of Carter-Tracy aquifer tables associated with this keyword.	1
4	NRIFTB	A positive integer value that defines the AQUTAB keyword maximum number of rows in the Carter-Tracy aquifer tables associated with this keyword. NRIFTB must not be less than 36 in order to accommodate the default infinite acting Carter-Tracy aquifer influence function.	36
5	NANAQ	A positive integer value that defines the AQUFETP , AQUFLUX and AQUCT maximum number of analytical aquifers defined by these three keywords.	1
6	NCAMAX	A positive integer value that defines the maximum number of cells connected to an <u>analytical aquifer</u> .	1
7	MXNALI	A positive integer value that defines the maximum number of aquifer lists.	0
8	MXAAQL	A positive integer value that defines the maximum number of <u>analytical aquifers</u> in any single aquifer list as defined by (7).	0

Notes:

- 1) The keyword is terminated by a “/”.

Table 5.5: AQUDIMS Keyword Description

Example

```
--
--      AQF      AQF      AQF      AQF      AQF      AQF      AQF      AQF
--      MXAQN    MXNAQC   NIFTBL  NRIFTB   NANAQU   NCAMAX  MXNALI  MXAAQL
AQUDIMS 1*      1*      1*      1*      1*      1*      1*      1*      /
```

The above example defines the default values for the **AQUDIMS** keyword.

5.3.8 AUTOREF – DEFINE AUTO REFINEMENT OPTIONS

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

The *AUTOREF* keyword activates the Auto Refinement option and defines the parameters for this feature.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.9 BIGMODEL – ACTIVATE BIG MODEL OPTION (RETIRED)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The original intention in the commercial simulator was to define an optimized memory allocation method to handle large models; this has since become redundant and has been retired in the commercial simulator.

This keyword is ignored by both OPM Flow and the commercial simulator and has no effect on the simulation but is documented here for completeness.

5.3.10 BIOFILM – ACTIVATE THE BIOFILM MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [BIOFILM](#) keyword activates the biofilm model for the run. Biofilm-related effects in subsurface applications such as hydrogen storage include reduced injectivity and hydrogen loss. The conceptual model includes the following mechanisms: (1) Biofilm is present in the storage site prior to injection and (2) the biofilm consumes injected H₂/CO₂, leading to clogging effects.

Note

This is an OPM Flow specific keyword used to investigate biofilm effects in underground applications. The module requires that either the [CO2STORE](#) or [H2STORE](#) keywords in the [RUNSPEC](#) to be active.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE BIOFILM MODEL
--
BIOFILM
--
--      ACTIVATE THE H2STORE MODEL
--
WATER
GAS
H2STORE
```

The above example declares that the [BIOFILM](#) model is active for the run and activates the [H2STORE](#) model. For a complete example of this model, see [H2STORE_BIOFILM.DATA](#).

5.3.11 BLACKOIL – ACTIVATE BLACK-OIL PHASES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword switches on the black-oil formulation, and is equivalent to setting the phases present in the model to be oil, vaporized oil, gas, and dissolved gas. Note if water is present in the model this needs to be explicitly stated via the [WATER](#) keyword in the [RUNSPEC](#) section (see also the [LIVEOIL](#) keywords in the [RUNSPEC](#) section). The keyword is used by the commercial simulator’s compositional [THERMAL](#) option to set the phases present in the model.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

The following example activates the black-oil phases in the model.

```
--
--      ACTIVATE BLACK-OIL PHASES
--
BLACKOIL
```

Alternatively one could explicitly declare the phases using the following keywords in the [RUNSPEC](#) section.

```
--
--      OIL PHASE IS PRESENT IN THE RUN
--
OIL
--
--      VAPORIZED OIL IN WET GAS IS PRESENT IN THE RUN
--
VAPOIL
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
--
--      DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN
--
DISGAS
```

The above example switches on the black-oil phases in the model.

5.3.12 BPARA – ACTIVATE BLOCK PARALLEL LICENSE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [BPARA](#) keyword activates the block parallel license in the commercial simulator. There is no data required for this keyword; however the keyword should be followed by the [PARALLEL](#) keyword in the [RUNSPEC](#) section, as illustrated in the example below.

There is no data required for this keyword and there is no terminating “/” for this keyword.

OPM Flow is an open source project and therefore there is no license management of the various implemented options or the number of cores/threads that can be utilized; hence this keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

See section [2.2 Running OPM Flow From The Command Line](#) on how to run OPM Flow in parallel mode.

Example

```
--
--      ACTIVATE BLOCK PARALLEL LICENSE
--
BPARA
--
--      PARALLEL MULTI-CORE OPTIONS
--      NDMAIN      MACHINE TYPE
PARALLEL
      8      DISTRIBUTED
/
```

The above example sets the number of domains (or processors) to eight and for the simulation to run in block parallel mode. This has no effect in OPM Flow input decks.

5.3.13

BPIDIMS – DEFINE THE DIMENSIONS OF THE INTERPOLATED BLOCK QUANTITIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The BPDIMS keyword defines the dimensions of the interpolated grid block quantities for the BPR_X, BHD_X,BHDF_X, BSCN_X and BCTRA_X, etc. variables declared in the SUMMARY section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.14 BRINE – ACTIVATE BRINE TRACKING OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [BRINE](#) keyword activates the standard Brine Tracking model and optionally defines the water phase to have various salinities if the [ECLMC](#) keyword in the [RUNSPEC](#) section has been used to activate the Multi-Component Brine model, that allows for the water phase to have multiple water salinities. Note that the Multi-Component Brine model is not supported by OPM Flow.

No.	Name	Description	Default																										
I	SALTS	An optional character vector string that defines the salts to be tracked for when the Multi-Component Brine model has been activated by the <i>ECLMC</i> keyword in the <i>RUNSPEC</i> section. SALTS should be set to one or more of the following salt chemical formulae: <table><thead><tr><th>Salt Name</th><th>Salt Chemical Formulae</th></tr></thead><tbody><tr><td>Sodium Chloride</td><td>NaCl</td></tr><tr><td>Potassium Chloride</td><td>KCl</td></tr><tr><td>Calcium Chloride</td><td>CaCl₂</td></tr><tr><td>Magnesium Chloride</td><td>MgCl₂</td></tr><tr><td>Sodium Carbonate</td><td>Na₂CO₃</td></tr><tr><td>Potassium Carbonate</td><td>K₂CO₃</td></tr><tr><td>Calcium Carbonate</td><td>CaCO₃</td></tr><tr><td>Magnesium Carbonate</td><td>MgCO₃</td></tr><tr><td>Sodium Sulfate</td><td>Na₂SO₄</td></tr><tr><td>Potassium Sulfate</td><td>K₂SO₄</td></tr><tr><td>Calcium Sulfate</td><td>CaSO₄</td></tr><tr><td>Magnesium Sulfate</td><td>MgSO₄</td></tr></tbody></table> Note that the <i>ECLMC</i> option is currently not available in OPM Flow, so only the <i>BRINE</i> keyword without the optional <i>SALT</i> variables should be declared in the input deck.	Salt Name	Salt Chemical Formulae	Sodium Chloride	NaCl	Potassium Chloride	KCl	Calcium Chloride	CaCl ₂	Magnesium Chloride	MgCl ₂	Sodium Carbonate	Na ₂ CO ₃	Potassium Carbonate	K ₂ CO ₃	Calcium Carbonate	CaCO ₃	Magnesium Carbonate	MgCO ₃	Sodium Sulfate	Na ₂ SO ₄	Potassium Sulfate	K ₂ SO ₄	Calcium Sulfate	CaSO ₄	Magnesium Sulfate	MgSO ₄	None
Salt Name	Salt Chemical Formulae																												
Sodium Chloride	NaCl																												
Potassium Chloride	KCl																												
Calcium Chloride	CaCl ₂																												
Magnesium Chloride	MgCl ₂																												
Sodium Carbonate	Na ₂ CO ₃																												
Potassium Carbonate	K ₂ CO ₃																												
Calcium Carbonate	CaCO ₃																												
Magnesium Carbonate	MgCO ₃																												
Sodium Sulfate	Na ₂ SO ₄																												
Potassium Sulfate	K ₂ SO ₄																												
Calcium Sulfate	CaSO ₄																												
Magnesium Sulfate	MgSO ₄																												

Notes:

1)

There is no data required for this keyword if the standard Brine Tracking option is being activated and there should be no terminating “/” in this case. However, if the Multi-Component Brine Tracking option has been invoked by the *ECLMC* keyword, a list of SALTS must be supplied and in this case.

2)

The keyword is terminated by a “/”.

Table 5.6: BRINE Keyword Description

See also the [PRECSALT](#) and [VAPWAT](#) keywords in the [RUNSPEC](#) section that activates OPM Flow's Salt Precipitation model, and the [PVTWSALT](#) keyword in the [PROPS](#) section to define the water properties with respect to salt concentration.

Example

The first example activates the standard Brine model and has no terminating “/”.

```
--  
--      ACTIVATE STANDARD BRINE MODEL IN THE RUN  
--  
BRINE
```

The second example illustrates how to activate OPM Flow's Salt Precipitation model.

```
--  
--      ACTIVATE STANDARD BRINE MODEL IN THE RUN  
--  
BRINE  
--  
--      ACTIVATE THE OPM FLOW SALT PRECIPITATION MODEL (OPM FLOW KEYWORD)  
--  
PRECSALT  
--  
--      VAPORIZED WATER IN DRY/WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)  
--  
VAPWAT
```

The third and final example activates the Multi-Component brine model with four different salts.

```
--  
--      ACTIVATE MULTI-COMPONENT BRINE MODEL  
--  
ECLMC  
--  
--      DEFINE WATER PHASE MULTI-COMPONENT BRINE COMPONENTS  
--  
--      SALT1    SALT2    SALT3    SALT4    SALT5  
BRINE          NACL     CACL2    MGC03    K2C03  
/
```

This option is currently not available in OPM Flow.

5.3.15
CART – ACTIVATE CARTESIAN GEOMETRY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

CART activates the Cartesian grid geometry for the main model, as oppose to a radial geometry. This is the default geometry and therefore the keyword does have to be used to activate this type of geometry.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.16 CBMOPTS – DEFINE COAL BED METHANE OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword sets the options for the Coal Bed Methane model which is activated via the [COAL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.17 CO2SOL – ACTIVATE DISSOLVED CO2 IN THE WATER PHASE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **CO2SOL** keyword activates dissolved carbon dioxide (CO₂) in the water phase, where CO₂ is represented by the **SOLVENT** pseudo component, using the simulator's CO₂-Brine PVT model. This keyword is a compositional keyword in the commercial simulator but has been implemented in OPM Flow's black-oil model.

The **CO2SOL** keyword can be used when modelling CO₂ injection in depleted hydrocarbon reservoirs. See also the **CO2STORE** keyword in the **RUNSPEC** section which can be used when modelling CO₂ injection in saline aquifers.

The CO₂-Brine PVT model computes the PVT properties such as density, viscosity, and enthalpy internally as functions of pressure, temperature, and composition by using analytic correlations and models from the literature rather than by interpolation of tabulated values in the input deck. These values are transformed to the standard black-oil equivalent PVT tables internally by the simulator. A full description of the underlying PVT models is described by Sandve et al.⁶⁴. Note that the CO₂-Brine PVT properties depend on the temperature and salinity and these must therefore be entered in the **PROPS** section. The reservoir temperature can be defined using, e.g., the **RTEMP** keyword. Region based salinity can be provided using the **SALINITY** keyword.

The **CO2SOL** keyword must be used together with the **SOLVENT** keyword in the **RUNSPEC** section. The **SOLVENT** keyword activates the four component black-oil model: oil, water and gas, plus a solvent (in this case CO₂).

The **DISGASW** keyword in the **RUNSPEC** section can be used to model dissolution of CO₂ in the Brine. Note that hydrocarbon gas is not allowed to dissolve in the Brine (this is considered a reasonable assumption for most hydrocarbon gases).

The hydrocarbon reservoir fluid properties should be set up as normal apart from the water PVT properties, which will be defined by the simulators' CO₂-Brine PVT model, so the water PVT keywords (e.g. **PVTW**) are not required. The CO₂ PVT properties will also be defined by the CO₂-Brine PVT model so the solvent PVT keywords (e.g. **SDENSITY**, **PVTSOL**) are also not required.

The oil-water and gas-water saturation functions should be set up as normal. The CO₂ and gas miscible relative permeability tables should be defined using the **SSFN** keyword in the **PROPS** section.

The initial CO₂ saturations in the model can be specified by enumeration using the **SSOL** keyword in the **SOLUTION** section.

The CO₂ fraction in a well's injection stream can be defined using the **WSOLVENT** keyword in the **SCHEDULE** section.

There is no data required for this keyword and there is no terminating "/" for this keyword.

Example

The example shows the usage of **CO2SOL** to model CO₂ injection in a live-oil reservoir. The **RUNSPEC** section includes the **CO2SOL** keyword, which activates dissolved carbon dioxide (CO₂) in the water phase. The **SOLVENT** keyword activates the solvent pseudo component, which represents CO₂. The **DISGASW** keyword has been used to activate dissolved CO₂ in the brine.

⁶⁴ Tor Harald Sandve I, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.


```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
-- FLUID TYPES AND TRACER OPTIONS
--
--
--      ACTIVATE DISSOLVED CO2 IN WATER (OPM FLOW KEYWORD)
--
CO2SOL
--
--      SOLVENT PHASE IS PRESENT IN THE RUN
--
SOLVENT
--
--      DISSOLVED CO2 IN WATER IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
DISGASW

```

The second part of the example covers the additional data required in the [PROPS](#) section, in which the CO₂-gas miscible relative permeabilities are defined using the [SSFN](#) keyword. The initial temperature and salinity are defined using the [RTEMP](#) and [SALINITY](#) keywords. The oil and gas PVT properties should be defined in the normal way. No water PVT data is required.

```

-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP
      80.0                                / RESERVOIR TEMP
--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALINITY
      0.7                                /
--
-- SOLVENT RELATIVE PERMEABILITY TABLES
--
SSFN
-- SGAS KRGT KRST
-- FRAC
--
--      -----
0.0000 0.0000 0.0000
1.0000 1.0000 1.0000 /

```

The third part of the example covers initialising the model in the [SOLUTION](#) section. Here the [EQUIL](#) keyword is used as normal to equilibrate the oil, gas and water phases, and the [SSOL](#) keyword to specify that there is initially no CO₂ present in the gas phase.

```

-- =====
--
-- SOLUTION SECTION
--
-- =====

```

```

SOLUTION
--
--          DATUM    DATUM    OWC    PCOW    GOC    PCGO    RS    RV    N    E300    RVW
--          DEPTH    PRESS    DEPTH    ----    DEPTH    ----    OPT    OPT    OPT    OPT    OPT
EQUIL
          2000.0    200.0    1900.0    0.00    2000.0    0.00    1*    1*    1*    2*    1*  /
--
-- DEFINE INITIAL EQUILIBRATION SOLVENT SAT VALUES FOR ALL CELLS IN THE MODEL
-- BASED ON NX = 100, NY = 100 AND NZ = 3
--
SSOL
          30000*0.0  /

```

For the summary section, the CO₂ storage quantity can be tracked using the solvent injection and production volumes as shown below.

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--          EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--          FIELD SOLVENT INJECTION RATE
FNIR
--
--          FIELD SOLVENT PRODUCTION RATE
FNPR
--
--          FIELD SOLVENT INJECTION TOTAL
FNIT
--
--          FIELD SOLVENT PRODUCTION TOTAL
FNPT

```

The final part of the example covers the [SCHEDULE](#) section. The [WSOLVENT](#) keyword is used to specify the injected solvent fraction (in this case pure CO₂).

```

-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--          RESTART CONTROL BASIC = 1 (LAST REPORT TIME ONLY)
--
RPTRST
          'BASIC=1'  'ALLPROPS'  /
--
-- DEFINE GAS INJECTION WELL SOLVENT FRACTION
--
-- WELL SOLVENT
-- NAME FRACTION
-- -----
WSOLVENT
          GI01 1.0000  /
/

```

5.3.18 CO2STORE – ACTIVATE THE CO2 STORAGE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **CO2STORE** keyword activates the carbon dioxide (CO₂) storage model for the run to account for both carbon dioxide and water phase solubility, via the simulator's CO₂-Brine PVT model. This keyword is a compositional keyword in the commercial simulator but has been implemented in OPM Flow's black-oil model.

The CO₂-Brine PVT model computes the PVT properties such as density, viscosity, and enthalpy internally as functions of pressure, temperature, and composition by using analytic correlations and models from the literature rather than by interpolation from tabulated values. These values are transformed to the standard black-oil equivalent PVT tables internally by the simulator. A full description of the underlying PVT models is described by Sandve et al.⁶⁵. This means that the normal PVT keywords like **DENSITY**, **PVTO**, **PVDG** etc. are not required by OPM Flow when this model is activated, and if entered will be ignored by the simulator. Note that the CO₂-Brine PVT properties depend on the temperature and salinity and these must therefore be entered in the **PROPS** or **SOLUTION** sections. The reservoir temperature can be defined using, e.g., the **RTEMP** keyword. Region based salinity can be provided using the **SALINITY** keyword.

The **CO2STORE** keyword must be used together with either: (1) the **GAS** and **WATER** keywords (or alternatively the **GASWAT** keyword), or (2) the **GAS** and **OIL** keywords in the **RUNSPEC** section. It is recommended that the standard method option (1) is used.

The **DISGASW** keyword in the **RUNSPEC** section can be used with option (1) to model dissolution of CO₂ in the Brine.

Option (1) has the advantage that it can be used with the **VAPWAT** and **PRECSALT** keywords in the **RUNSPEC** section to model the impact of both vaporization of residual water and salt precipitation in the near wellbore region on injectivity of CO₂ injection wells.

In option (1), the **GAS** and **WATER** (or **GASWAT**) keywords declare that the gas and water phases are present in the model. When the **CO2STORE** option is used the water phase represents the brine and the gas phase represents CO₂. Note that the input and output keywords need to be consistent with this assumption, e.g., **GSF** (gas saturation function) and **WSF** (water saturation function) should be used for the CO₂-Brine relative permeability, etc.

Although, the **DISGAS** and **VAPWAT** keywords can be used with option (2) to model CO₂ dissolution and water vaporization, salt precipitation is not supported with option (2).

In option (2), the **GAS** and **OIL** keywords declare that the gas and oil phases are present in the model. Internally when **CO2STORE** is used the oil phase refers to the brine and the gas phase to CO₂. Again, the input and output keywords need to be consistent with this assumption, e.g., **SGOF** (gas-oil relative permeability) is used for the CO₂-Brine relative permeability, **FOIP** (Field Oil-In-Place) shows the total amount of brine in the reservoir, etc.

There is no data required for this keyword and there is no terminating "/" for this keyword.

Examples

The first example shows the standard usage of **CO2STORE** with Option (1) the Gas-Water model (**GASWAT**). Here we also activate the dissolved gas in water (**DISGASW**) and vaporized water in gas (**VAPWAT**) options in the **RUNSPEC** section.

⁶⁵ Tor Harald Sandve I, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
-- FLUID TYPES AND TRACER OPTIONS
--
--
--      ACTIVATE CO2 STORAGE IN THE MODEL (OPM FLOW CO2 STORAGE KEYWORD)
--
CO2STORE
--
--      ACTIVATE GAS-WATER THE MODEL (OPM FLOW KEYWORD)
--
GASWAT
--
--      DISSOLVED GAS IN WATER IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
DISGASW
--
--      VAPORIZED WATER IN DRY/WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
VAPWAT
```

The second part of the example covers the data required for the [PROPS](#) section, in which the two-phase relative permeability functions are set using [GSF](#) and [WSF](#) keywords.

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP
--      80.0 / RESERVOIR TEMP
--
--      ROCK COMPRESSIBILITY
--
--      REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--
--      REF PRES  CF
--      BARSA     1/BARSA
--      -----
ROCK
--      200.0      5.0E-05 / ROCK COMPRESSIBILITY
--
--      GAS RELATIVE PERMEABILITY TABLES (OPM FLOW KEYWORD)
--
GSF
--      SGAS      KRG      PCGW
--      FRAC      PSIA
--      -----
--      0.000      0.000      0.0
--      0.080      0.001      0.0
--      0.170      0.010      0.0
--      0.350      0.050      0.0
--      0.530      0.200      0.0
```

```

0.620    0.350    0.0
0.650    0.390    0.0
0.710    0.560    0.0
0.800    1.000    0.0  / TABLE NO. 01

--
--      WATER RELATIVE PERMEABILITY TABLES (OPM FLOW KEYWORD)
--
WSF
--      SWAT      KRW
--      FRAC
--      -----
--      0.200    0.0000
--      0.400    0.1000
--      0.800    0.5000
--      1.000    1.0000          / TABLE NO. 01

--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALINITY
0.7
```

No other data is required to define the fluid and rock properties in the [PROPS](#) section as the data is generated from internal analytic correlations and models by the simulator. Finally, note that units for salinity are to the 10⁻³, thus for metric units we have 10⁻³ x kg-M/kg.

The third part of the example covers initializing the model in the [SOLUTION](#) section. Here we set the [EQUIL](#)(EQLOPT6) parameter equal to one, to use table number one of the [RVWVD](#) keyword, in order to set the vaporized water versus depth distribution for the model.

```

-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      SYSTEM IS SATURATED WITH WATER
--
--      DATUM    DATUM    OWC    PCOW    GOC    PCGO    RS    RV    N    E300    RVW
--      DEPTH    PRESS    DEPTH    ----    DEPTH    ----    OPT    OPT    OPT    OPT    OPT
EQUIL
2000.0  200.0    1800.0  0.00    1800.0  0.00    1*    1*    1*    2*    1  /
--
--      WATER VAPOR RATIO VS DEPTH (OPM FLOW KEYWORD)
--
--      DEPTH    RVW
--      -----
--      STB/MSCF
RVWVD
1000.0    0.000
3000.0    0.000          / RVW VS DEPTH EQUIL REGN 01
```

In the [SUMMARY](#) section, the simulator supports summary vectors specific to CO₂ storage (see Section [11.1.12 Option Specific Variables - CO2STORE/H2STORE Model](#)) including those shown below.

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
```

```
-- EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
-- FIELD C02 DISSOLVED IN WATER PHASE
FWCD
--
-- FIELD C02 TRAPPED IN GAS PHASE
FGCDI
--
-- FIELD C02 MOBILE IN GAS PHASE
FGCDM
```

The final part of the example covers the [SCHEDULE](#) section. The standard [WCONINJE](#) keyword is used to set the gas injection rate, in this case 100,000 sm³/day of CO₂.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
-- RESTART CONTROL BASIC (LAST=1, ALL=2, YEARLY=4, MONTHLY=5, TSTEP=6)
--
RPTRST
' BASIC=2' ALLPROPS TEMP RWSAT RVWSAT /
--
-- WELL SPECIFICATION DATA
--
-- WELL GROUP LOCATION BHP PHASE DRAIN INFLOW OPEN CROSS PVT
-- NAME NAME I J DEPTH FLUID AREA EQUANS SHUT FLOW TABLE
WELSPEDS
INJ1 'G1' 25 25 1* GAS /
/
--
-- WELL CONNECTION DATA
--
-- WELL --- LOCATION --- OPEN SAT CONN WELL KH SKIN D DIR
-- NAME II JJ K1 K2 SHUT TAB FACT DIA FACT FACT FACT PEN
COMPDAT
INJ1 25 25 1 10 OPEN 1* 1* 0.3 /
/
--
-- WELL INJECTION CONTROLS
--
-- WELL FLUID OPEN/ CNTL SURF RESV BHP THP VFP
-- NAME TYPE SHUT MODE RATE RATE PRES PRES TABLE
WCONINJE
INJ1 GAS OPEN RATE 100000 1* 300 /
/
--
-- ADVANCE SIMULATION BY REPORTING TIME
--
TSTEP
36*30.4375
/
```

Note in order to get the liquid phase mole fractions of CO₂, that is, the mole fractions of CO₂ in the water phase (XMFCO2), and the vapor phase mole fractions (YMFWAT) to the restart file, one must use the command line parameter enable-opm-rst-file set equal to true.

The second example shows how to use [CO2STORE](#) with the alternative option (2). The example below declares that the carbon dioxide storage model is active for the run to account for both carbon dioxide and water phase solubility using OPM Flow’s CO₂-Brine PVT model. Option (2) is used where the [OIL](#) phase refers to the brine and the [GAS](#) phase to CO₂.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
-- FLUID TYPES AND TRACER OPTIONS
--
--
--      ACTIVATE CO2 STORAGE IN THE MODEL (OPM FLOW CO2 STORAGE KEYWORD)
--
CO2STORE
--
--      OIL PHASE IS PRESENT IN THE RUN BUT IS THE BRINE PHASE FOR CO2STORE
--
OIL
--
--      GAS PHASE IS PRESENT IN THE RUN BUT IS THE CO2 PHASE FOR CO2STORE
--
GAS
--
--      DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN (CO2 IN BRINE)
--
DISGAS
--
--      VAPORIZED OIL IN WET GAS IS PRESENT IN THE RUN (WATER IN CO2)
--
VAPOIL
```

The second part of the example covers the data required for the [PROPS](#) section, in which the input keywords need to be consistent with the [OIL](#) phase referring to the Brine and the [GAS](#) to CO₂; that is [SGOF](#) (gas-oil relative permeability) is used to define the CO₂-Brine relative permeability table.

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP
90.0                                / RESERVOIR TEMPERATURE
--
--      ROCK COMPRESSIBILITY
--
--      REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--
--      REF PRES   CF
--      BARSA      1/BARSA
--      -----
ROCK
200.0      5.0E-05                    / ROCK COMPRESSIBILITY
--
--      GAS-OIL RELATIVE PERMEABILITY TABLES (SGOF) – CO2STORE PHASES
```



```
SGOF
--      SG      KRG      KROG      PCOG
--      FRAC
--      -----
--      0.00000  0.000000  1.00000  0.0000
--      1.00000  1.000000  0.00000  0.0000      /  TABLE No. 01
--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
--      SALINITY
--      0.7      /
```

The third part of the example covers initializing the model in the [SOLUTION](#) section. Here we set the [EQUIL](#)(EQLOPT1 and EQLOPT2) parameters equal to one, to use table number one of the [RSVD](#) and [RVVD](#) keywords, in order to set the initial dissolved CO₂ and vaporized water versus depth distribution for the model.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      SYSTEM IS SATURATED WITH OIL (BRINE)
--
--      DATUM  DATUM  OWC  PCOW  GOC  PCGO  RS  RV  N  E300  RVW
--      DEPTH  PRESS  DEPTH  ----  DEPTH  ----  OPT OPT OPT  OPT  OPT
EQUIL
--      2000.0  200.0  2200.0  0.00  1800.0  0.00  1  1  1*  2*  1*  /
--
--      DEPTH  RS
--      -----  MSCF/STB
--
--      RSVD
--      1000.0  0.000
--      3000.0  0.000      /
--
--      DEPTH  RVW
--      -----  STB/MSCF
--
--      RVVD
--      1000.0  0.000
--      3000.0  0.000      /
```

See also the [DRSDTCN](#) keyword in the [SCHEDULE](#) section for details of how to control convective dissolution of carbon dioxide into the in-situ brine.

5.3.19 COAL – ACTIVATE THE COAL PHASE (CBM MODEL)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *COAL* keyword activates the coal phase and the Coal Bed Methane (“CBM”) model for the run.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE COAL PHASE (CBM MODEL) IN THE MODEL
--
COAL
```

The above example declares that the Coal phase is active in the run and activates the CBM model option.

5.3.20 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The [COLUMNS](#) keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the GLOBAL section for a full description.

5.3.21 COMPS – ACTIVATE COMPOSITIONAL MODELING FORMULATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COMPS](#) keyword activates the Compositional Modeling Formulation, and declares the number of components active in the model.

OPM Flow does not currently support the general compositional modeling formulation.

Note

This keyword is only supported by OPM Flow when the two component gas-water CO2 storage model has been activated using the [CO2STORE](#) keyword and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section.

Secondly, although OPM Flow parses the keyword, the simulator currently ignores the data for this keyword.

No.	Name	Description	Default
I	COMPS	A positive integer defining the number of compositional components active in the model. Only the default value of two is currently supported by OPM Flow.	2
Notes: I) The keyword is terminated by a “/”.			

Table 5.7: COMPS Keyword Description

Example

The following example shows how to request a two component compositional modeling formulation to be used with the [CO2STORE](#) and [GASWAT](#) options.

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC

--
--      ACTIVATE CO2 STORAGE IN THE MODEL (OPM FLOW CO2 STORAGE KEYWORD)
--
CO2STORE
--
--      ACTIVATE COMPOSITIONAL MODELING FORMULATION (OPM FLOW KEYWORD)
--
COMPS
--      2
--
--      ACTIVATE GAS-WATER THE MODEL (OPM FLOW KEYWORD)
--
GASWAT

```

5.3.22 CPR – ACTIVATE CONSTRAINED PRESSURE RESIDUAL (“CPR”) LINEAR SOLVER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

Turns on the Constrained Pressure Residual (“CPR”)⁶⁶, ⁶⁷ and ⁶⁸ preconditioner linear solver option, and declares how the solver should be applied. The keyword is equivalent to using the OPM Flow command line parameter `--linear-solver= “cprw”`. Note that if the command line has been used, then this will take precedence over the [CPR](#) keyword.

No.	Name	Description	Default
I	CPROPTN	A defined character string that determines how the CPR linear solver should be applied, and should be set to one of the following: 1) ORIGINAL: Here the solver is applied for the whole of the simulation. 2) ADAPTIVE: This option applies the more computational demanding CPR linear solver, only in parts of the run that would benefit from its use, for example when linear convergence is challenging. Note that OPM Flow only supports the ORIGINAL option, which is the default value in OPM Flow, whereas the default value in the commercial simulator is ADAPTIVE..	ORIGINAL
Notes: 1) The keyword is terminated by a “/”.			

Table 5.8: CPR Keyword Description

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

Example

```
--
--      ACTIVATE CONSTRAINED PRESSURE RESIDUAL LINEAR SOLVER FOR THE RUN
--
CPR
/
```

The above example activates the Constrained Pressure Residual preconditioner linear solver using the default option, that is the ORIGINAL option for OPM Flow.

To enable the CPR preconditioner linear solver option using the command line parameter, use:

```
flow --linear-solver="cpr" CASENAME.DATA
```

⁶⁶ Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

⁶⁷ R. Scheichl, M. Roland, J. Wendebourg. Decoupling and block preconditioning for sedimentary basin simulations, *Computational Geosciences* 7 (2003) 295-318.

⁶⁸ Klemetsdal, Ø.S., Møyner, O. & Lie, KA. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. *Comput Geosci* 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

which should usually be combined with the `-matrix-add-well-contributions=true` option, that is:

```
flow --linear-solver=cpr --matrix-add-well-contributions=true CASENAME.DATA
```

However, the new CPRW preconditioner, that includes wells does not need the latter option, so in this instance use:

```
flow --linear-solver=cprw CASENAME.DATA
```

As of this release the CPRW preconditioner, should be considered experimental, and the recommended configuration for running this is:

```
flow --linear-solver=cprw --linear-solver-reduction=0.005 --cpr-reuse-setup=4
--cpr-reuse-interval=10 CASENAME.DATA
```

See new Rasmussen et al.⁶⁹ for further information on the available numerical algorithms available in OPM Flow.

⁶⁹ Atgeirr Flø Rasmussen, Tor Harald Sandve, Kai Bao, Andreas Lauser, Joakim Hove, Bård Skaflestad, Robert Klöfkorn, Markus Blatt, Alf Birger Rustad, Ove Sævareid, Knut-Andreas Lie, Andreas Thune, The Open Porous Media Flow reservoir simulator, *Computers & Mathematics with Applications*, Volume 81, 2021, Pages 159-185, ISSN 0898-1221, <https://doi.org/10.1016/j.camwa.2020.05.014>. (<https://www.sciencedirect.com/science/article/pii/S0898122120302182>).

5.3.23 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the GLOBAL section for a full description.

5.3.24 DIFFUSE – ACTIVATE MOLECULAR DIFFUSION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFUSE](#) keyword activates OPM Flow’s Molecular Diffusion option based on fluid and grid data (Sandve et al.⁷⁰), similar to the commercial simulator. For field-scale simulations diffusion is a sub-grid phenomenon and is typically not explicitly represented in the flow equations. However, for simulations on the laboratory scale diffusion plays a direct role and therefore needs to be explicitly represented in the flow equations. Diffusion coefficients that control the diffusion depend on temperature, pressure, and salinity. In OPM Flow, the diffusion coefficients are computed internally for pure water using McLachlan and Danckwerts⁷¹ and modified to account for salinity using Ratcliff and Holdcroft⁷². The effect of the porous media on the diffusion is modeled using the relation suggested by Millington and Quirk⁷³. The coefficients can also be given for each PVT region as an input parameter using the [DIFFC](#) keyword in the [PROPS](#) section.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Note

The option has been tested in combination with the [CO2STORE](#) keyword, but not for the general case at this point.

Example

```
--
--      ACTIVATE MOLECULAR DIFFUSION OPTION
--
DIFFUSE
```

The above example switches on the molecular diffusion facility.

⁷⁰ Tor Harald Sandve I, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

⁷¹ McLachlan, C. N. S., & Danckwerts, P.V. (1972). Desorption of carbon dioxide from aqueous potash solutions with and without the addition of arsenite as a catalyst. *Trans. Inst. Chem. Eng.* 50, 300-309.

⁷² Ratcliff, G.A., & Holdcroft, J. G. (1963). Diffusivities of gases in aqueous electrolyte solutions. *Trans. Inst. Chem. Eng.* 41(10), 315-319.

⁷³ Millington, R. J., & Quirk, J. P. (1961). Permeability of porous solids. *Transactions of the Faraday Society*, 57, 1200-1207

5.3.25 DIMENS – DEFINE THE DIMENSIONS OF THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

DIMENS defines the dimensions of the model entered as integer vector. The keyword can be used with all grid types.

No.	Name	Description	Default
1	NX	A positive integer value that defines the number of grid blocks in the x direction for Cartesian grids or the number of grid blocks in the r direction for radial grids	None
2	NY	A positive integer value that defines the number of grid blocks in the y direction for Cartesian grids or the number of grid blocks in the theta direction for radial grids.	None
3	NZ	A positive integer value that defines the number of grid blocks in the z direction for both Cartesian and radial grids.	None
Notes: 1) The keyword is terminated by a "/".			

Table 5.9: DIMENS Keyword Description

Note that NX, NY and NZ are not maximum values but the actual size of the grid. OPM Flow applies these parameters when reading in particular data sets. For example if NX, NY, and NZ are set to 10, 10 and 10 respectively, then for the grid property data like **PORO**. OPM Flow expects to read in 10 x 10 x 10 or 1,000 porosity values for the **PORO** array. If the number of porosity values is not equal to 1,000 then OPM Flow will produce an error.

Example

```
--
--      MAX      MAX      MAX
--      NDIVIX  NDIVIY  NDIVIZ
DIMENS
      46      112      22      /
```

The above example defines the dimensions for the Norne model of 46 cells in the x direction, 112 cells in the y direction and 22 cells in the z direction.

5.3.26 DISGAS – ACTIVATE THE DISSOLVED GAS PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that dissolved gas is present in live⁷⁴ oil in the model and the keyword should only be used if there is both oil and gas phases in the model. The keyword may be used for oil-water and oil-water-gas input decks that contain the oil and gas phases. The keyword will also invoke data input file checking to ensure that all the required oil and gas phase input parameters are defined in the input deck.

If the oil has a constant and uniform dissolved gas concentration, Gas-Oil Ratio (“GOR”), and if the reservoir pressure never drops below the saturation pressure (bubble point pressure), then the model can be run more efficiently by omitting the **GAS** and **DISGAS** keywords from the **RUNSPEC** section, treating the oil as a dead oil⁷⁵, and defining a constant Rs (GOR) value with keyword **RSCONST** or **RSCONSTT** in the **PROPS** section. This results in the model being run with as a dead oil problem with no active gas phase. However, OPM Flow takes into account the constant Rs in the calculations and reporting.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN
--
DISGAS
```

The above example declares that the dissolved gas in the oil phase is active in the model.

⁷⁴ “Live” oil is oil that contains gas in solution, which is normally the case for most conventional oil reservoirs. However, for oil reservoirs classified as heavy oil reservoirs, the in situ dissolved gas may be negligible and oil would then be classified as gas-free oil which is commonly referred to as “dead” oil.

⁷⁵ “Dead” oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

5.3.27 DISGASW – ACTIVATE DISSOLVED GAS IN THE WATER PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword indicates that dissolved gas is present in the water phase in the model. The keyword may only be used for gas-water input decks that contain just the gas and the water phases, which must also be declared. The keyword will also invoke data input file checking to ensure that all the required gas and water phase input parameters are defined in the input deck.

Note

This is an OPM Flow specific keyword for the simulator's Dissolved Gas in Water Model that is activated by declaring that this phase is present in the run.

The activation of this phase may be used for modeling the production of gas from the water in gas fields, where gas has dissolved into the in situ water phase. Although this is always the case, the volumes are usually insignificant compared with the volumes in the hydrocarbon zones, and therefore in nearly all cases the dissolved gas volumes are usually ignored in terms of resource volumes. For gas fields, the interest is normally the amount of water in the gas phase, as this influences the liquid loading rate as the water is liberated from the gas as the fluid transverses up the wellbore.

The second, and primary, application of this black-oil formulation is for CO₂ storage, where injection of CO₂ injection is being conducted in either depleted gas fields or in water filled aquifer zones. Activating the dissolved gas in the water phase under these circumstances will allow the CO₂ to dissolve into the water phase based on the entered PVT properties in the [PROPS](#) section. In this case the [CO2STORE](#) keyword in the [RUNSPEC](#) section should also be present in the deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
--
--      WATER PHASE IS PRESENT IN THE RUN
--
WATER
--
--      DISSOLVED GAS IN WATER IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
DISGASW
--
--      ACTIVATE CO2 STORAGE IN THE MODEL (OPM FLOW CO2 STORAGE KEYWORD)
--
CO2STORE
```

The above example declares that the gas and water phases are present, and that the dissolved gas in the water phase is also active in the model, for modeling CO₂ storage.

5.3.28 DISPDIMS – DEFINE THE MAXIMUM NUMBER OF DISPERSION TABLES

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Description

The *DISPDIMS* key defines the maximum number of dispersion tables, and the maximum number of velocity and concentration elements per table.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.29 DUALPERM – ACTIVATE DUAL PERMEABILITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **DUALPERM** keyword activates the Dual Permeability option and the Dual Porosity option for the run. In a dual porosity system flow occurs between the matrix and the fracture only, whereas in a dual permeability system flow also occurs between the matrix grid blocks⁷⁶, ⁷⁷, ⁷⁸, ⁷⁹ and ⁸⁰.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE DUAL PERMEABILITY MODEL
--
DUALPERM
```

The above example declares that the Dual Permeability option is active for the run.

⁷⁶ Warren, J.E. and Root, P.J. 1963. The Behavior of Naturally Fractured Reservoirs. SPE J. 3 (3): 245–255. SPE-426-PA. <http://dx.doi.org/10.2118/426-PA>.

⁷⁷ Gringarten, A.C. 1984. Interpretation of Tests in Fissured and Multilayered Reservoirs With Double-Porosity Behavior: Theory and Practice. J Pet Technol 36 (4): 549-564. SPE-10044-PA. <http://dx.doi.org/10.2118/10044-PA>.

⁷⁸ Serra, K., Reynolds, A.C., and Raghavan, R. 1983. New Pressure Transient Analysis Methods for Naturally Fractured Reservoirs (includes associated papers 12940 and 13014). J Pet Technol 35 (12): 2271-2283. SPE-10780-PA. <http://dx.doi.org/10.2118/10780-PA>.

⁷⁹ Barenblatt, G.E., Zheltov, I.P., and Kochina, I.N. 1960. Basic Concepts in the Theory of Homogeneous Liquids in Fissured Rocks. J. Appl. Math. Mech. 24: 1286-1303.

⁸⁰ Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. “Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs,” paper SPE 5719, Society of Petroleum Engineers Journal (1976) 16, No. 6, 317-326.

5.3.30 DUALPORO – ACTIVATE DUAL POROSITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DUALPORO* keyword activates the Dual Porosity option for the run. In a dual porosity system flow occurs between the matrix and the fracture only, whereas in a dual permeability system flow also occurs between the matrix grid blocks⁸¹, ⁸², ⁸³, ⁸⁴ and ⁸⁵.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE DUAL POROSITY MODEL
--
DUALPORO
```

The above example declares that the Dual Porosity option is active for the run.

⁸¹ Warren, J.E. and Root, P.J. 1963. The Behavior of Naturally Fractured Reservoirs. SPE J. 3 (3): 245–255. SPE-426-PA. <http://dx.doi.org/10.2118/426-PA>.

⁸² Gringarten, A.C. 1984. Interpretation of Tests in Fissured and Multilayered Reservoirs With Double-Porosity Behavior: Theory and Practice. J Pet Technol 36 (4): 549-564. SPE-10044-PA. <http://dx.doi.org/10.2118/10044-PA>.

⁸³ Serra, K., Reynolds, A.C., and Raghavan, R. 1983. New Pressure Transient Analysis Methods for Naturally Fractured Reservoirs (includes associated papers 12940 and 13014). J Pet Technol 35 (12): 2271-2283. SPE-10780-PA. <http://dx.doi.org/10.2118/10780-PA>.

⁸⁴ Barenblatt, G.E., Zheltov, I.P., and Kochina, I.N. 1960. Basic Concepts in the Theory of Homogeneous Liquids in Fissured Rocks. J. Appl. Math. Mech. 24: 1286-1303.

⁸⁵ Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. “Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs,” paper SPE 5719, Society of Petroleum Engineers Journal (1976) 16, No. 6, 317-326.

5.3.31 DYNRDIMS – DEFINE DYNAMIC REGION DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DYNRDIMS* keyword defines the dimensions for the parameters used by the Dynamic Regions facility, including the maximum number of dynamic regions. The Dynamic Regions facility allows for property and reporting regions to vary as the run progresses, based on the parameters and logic defined by the *DYNAMICR* keyword in the *SOLUTION* and *PROPS* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.32 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

5.3.33 ECLMC – ACTIVATE MULTI-COMPONENT BRINE MODEL

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

The **ECLMC** keyword activates the Multi-Component Brine model that allows for the water phase to have multiple water salinities. The keyword should be used in conjunction with the **BRINE** keyword in the **RUNSPEC**. Both keywords must be specified to activate the Multi-Component Brine model, whereas the **BRINE** keyword only is required to activate the standard brine tracking model.

There is no data required for this keyword and there is no terminating "/" for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

The first example activates the standard Brine Tracking model.

```
--
--      ACTIVATE STANDARD BRINE MODEL
--
BRINE
```

The next example shows the *ECLMC* and *BRINE* keywords for when the Multi-Component Brine model is required.

```
--
--      ACTIVATE MULTI-COMPONENT BRINE MODEL
--
ECLMC
--
--      DEFINE WATER PHASE MULTI-COMPONENT BRINE COMPONENTS
--
--      SALT1      SALT2      SALT3      SALT4      SALT5
BRINE      NACL      CACL2      MGC03
```

The above example activates the Multi-Component Brine model with three different water salinities.

5.3.34 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the GLOBAL section for a full description.

5.3.35 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the GLOBAL section for a full description.

5.3.36 ENDSCALE – ACTIVATE RELATIVE PERMEABILITY END-POINT SCALING OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **ENDSCALE** keyword activates OPM Flow's relative permeability end-point scaling option. The relative permeability functions are defined using the either the:

- 1) **SWOF**, **SGOF**, **SLGOF** series of saturation functions, or the
- 2) **SWFN**, **SGFN**, **SGWFN**, **SOF2**, **SOF3**, **SOF32D** series of functions.

And are allocated to the grid cells via the **SATNUM** keyword.

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **SWL**, **SWCR**, **SWU**, **SGL**, **SGCR**, **SGU**, **SOWCR**, and **SOGCR** saturation grid arrays for the saturation end-points, and the **KRG**, **KRORG**, **KRORW** and **KRW** relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **SWLX**, **SWLY** and **SWLZ** instead of **SWL** etc. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is **SWLX**, **SWLX-**, **SWLY**, **SWLY-**, **SWLZ** and **SWLZ-**, instead of **SWL** or the **SWLX**, **SWLY** and **SWLZ** set of keywords.

The keyword also defines the number of saturation end-point tables that allows for the re-scaling of the saturation functions to be a function of depth as opposed to being a grid property array. This is accomplished via the **ENKRVD** and **ENPTVD** keywords in the **PROPS** section.

No.	Name	Description	Default
I	DIRECT	A character string that activates or deactivates directional end-point scaling option. If DIRECT is set to NODIR then directional end-point scaling is switched off and the same saturation function is used in the x, y and z directions (unless activated otherwise by the SATOPTS keyword in the RUNSPEC section). In this case the SWL, SWCR, SWU, SGL, SGCR, SGU, SOWCR and SOGCR saturation grid arrays and the KRG, KRORG, KRORW and KRW relative permeability grid cell arrays should be used to enter the grid block end-point data. If DIRECT is set to DIRECT then directional end-point scaling is switched on and different saturation functions are used in the x, y and z directions (unless activated otherwise by the SATOPTS keyword in the RUNSPEC section). Here the directional form of the SWL, SWCR, SWU, SGL, SGCR, SGU, SOWCR and SOGCR saturation grid arrays and the KRG, KRORG, KRORW and KRW relative permeability grid cell arrays should be use to enter the grid block end-point data. For example SWLX, SWLY and SWLZ for SWL. Only the default option is supported by OPM Flow.	NODIR

No.	Name	Description	Default
2	IRREVERS	A character string that activates or deactivates non-reversible end-point scaling option. If IRREVERS is set to REVERS then the end-point scaling is set to reversible and results in the same set of end-point arrays being used for flow from the x_i to x_{i+1} direction as for the flow from the x_i to the x_{i-1} for all directions (x, y and z). Here the SWLX, SWLY and SWLZ series of keywords should be used instead of SWL type of keywords. Alternatively, if IRREVERS is set to IRREVERS then the end-point scaling is set to non-reversible and results in different sets of end-point arrays being applied for flow from the x_i to x_{i+1} direction and the x_i to the x_{i-1} direction, for all directions (x, y, z). in this case the SWLX+, SWLX-, SWLY+, SWLY-, SWLZ+ and SWLZ- series of keywords should be utilized instead of SWL or the SWLX, SWLY and SWLZ set of keywords. Only the default option is supported by OPM Flow.	REVERS
3	NTENDP	A positive integer that defines the maximum number of saturation end-point depth tables. The end-point depth tables are used to re-scale the saturation tables as a function of depth as opposed to being a grid block property. NTENDP may also be specified on the TABDIMS keyword, and if specified on both here and on the TABDIMS keyword the maximum value of the two is used. Only the default option is supported by OPM Flow.	1
4	NNODES	A positive integer the defines the maximum number entries for saturation end-point depth tables.	20
5	MODE	A positive integer that activates the options for temperature dependent saturation end-point scaling in the commercial compositional simulator. MODE should be defaulted with either 1* or zero, which means that scaling can only be performed by grid block end-point scaling properties or via saturation end-point depth tables.	0
Notes: - Note that the IRREVERS option can only be set to IRREVERS if the DIRECT parameter is set equal to DIRECT. - The keyword is terminated by a “/”.			

Table 5.10: ENDSCALE Keyword Description

Example

```
--      DIRC   REVERSE   MAX      MAX
--      SCALE  SCALE     TABLES  NODES
ENDSCALE
      NODIR  REVERS    1*        1*                               /
```

The above example invokes the end-point scaling option with end-point scaling being non-directional and reversible with the default number of saturation end-point depth tables (one) with 20 entries per table.

5.3.37 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the GLOBAL section for a full description.

5.3.38 EOS – SPECIFY EQUATION OF STATE

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Description

The *EOS* keyword specifies the Equation Of State (EOS) to be used for each EOS region. The keyword should only be used if the compositional mode has been requested using the *COMPS* keyword in the *RUNSPEC* section.

See *EOS – Specify Equation of State* in the *PROPS* section for a full description.

5.3.39 EQLDIMS – DEFINE THE EQUILIBRATION DATA DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **EQLDIMS** keyword defines the maximum number of properties associated with equilibrating the model, that is initializing the model. A reservoir grid can be separated into separate regions in order to apply different pressure regimes and/or fluid contacts. Care should be taken that the different regions are not in communication if the pressures or fluid contacts are different for the various regions, as this would lead to an unstable initialization and would also imply errors in the model description as implemented.

No.	Name	Description	Default
1	NTEQUL	A positive integer value that defines the number of equilibration regions entered using the EQLNUM keyword in the REGIONS section and the number of entries associated with the EQUIL keyword in the SOLUTION section.	1
2	NPRSVD	A positive integer value setting the number of pressure versus depth entries used by OPM Flow in determining equilibration parameters. Unless there is a requirement for a very fine equilibration this parameter should be defaulted.	100
3	NDRXVD	A positive integer value that defines the maximum number of depth entries in equilibration property versus depth tables (RSVD , RVVD , PBVD or PDVD etc.) as defined in the SOLUTION section.	20
4	NTTRVD	A positive integer that defines the maximum number of TVDP tables that describe the initial tracer concentration versus depth.	1
5	NSTRVD	A positive integer that defines the maximum number of depth entries in the TVDP tables as described in (4)	20
Notes: 1) NTEQUL is the exact number of entries must be entered on the EQUIL keyword, otherwise OPM Flow will report an error. It is not a maximum number as for the other region arrays. 2) The keyword is terminated by a “/”.			

Table 5.11: EQLDIMS Keyword Description

It is common that the **EQLNUM** and **FIPNUM** arrays are identical so that the fluid in-place reporting matches the equilibration regions. Thus, in order to avoid errors in this case, one should just use one array (say the **FIPNUM** property array) and use the **COPY** keyword to generate the **EQLNUM** array.

Example

```
--
--      MAX      MAX      RSVD      TVDP      TVDP
--      EQLNUM  DEPTH  NODES  TABLE  NODES
EQLDIMS
      9        1*       20        1*       1*
/
```

The above example defines nine equilibration regions the default values for the remaining parameters on the **EQLDIMS** keyword.

5.3.40 EQLOPTS – ACTIVATE THE EQUILIBRATION OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [EQLOPTS](#) keyword defines the equilibration options by stating the character command to activate an option to be used for initializing the model. Multiple commands may be utilized to activate several equilibration options following the keyword.

No.	Name	Description	Default
1	MOBILE	A character string that activates the mobile fluid critical saturation end point correction. If the MOBILE command is stated then this option is activated. This option is not supported by OPM Flow but would change the results if supported so the simulation will be stopped. Therefore this character string should be omitted.	None
2	QUIESC	A character string that activates the initial quiescence option that modifies the equilibrium calculated phase pressures to ensure that a steady state solution is obtained. This options ensures that there is no flow potential between the grid blocks in a given region, which is the normal case when block-centered equilibration is used by setting BOINIT on the EQUIL keyword to zero in the SOLUTION section. If the QUIESC command is stated then this option is activated. This option is not supported by OPM Flow but would change the results if supported so the simulation will be stopped. Therefore this character string should be omitted.	None
3	THPRES	A character string that activates the inter-region equilibration flow option. This option allows for a threshold pressure variable entered via the THPRES keyword to define a pressure which prevents flow between regions until the THPRES value between regions is exceeded. If the THPRES command is stated then this option is activated.	None
4	IRREVERS	A character string that activates the irreversible inter-region equilibration flow option. This option can only be invoked if the THPRES command has been stated. The option allows for different THPRES values for different directions. If the IRREVERS command is stated then this option is activated. This option is not supported by OPM Flow but would change the results if supported so the simulation will be stopped. Therefore this character string should be omitted.	None
Notes: 1) The keyword is terminated by a “/”.			

Table 5.12: EQLOPTS Keyword Description

Example

```
--
--      ACTIVATE EQUILIBRATION OPTIONS
--      MOBILE END-POINT(MOBILE) STEADY STATE(QUIESC) THRESHOLD(THPRES)
--      IRREVERSIBLE THRESHOLD(IRREVERS)
EQLOPTS
      'THPRES'  'IRREVERS'
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

The above example activates the threshold pressure option with different threshold pressure for different directions.

5.3.41 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

5.3.42 FAULTDIM – DEFINE THE NUMBER OF FAULT SEGMENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *FAULTDIM* keyword defines the maximum number of records (or segments) that can be entered with the *FAULTS* keyword. The *FAULTS* keyword defines the faults in the grid than can be used for setting (or re-setting) transmissibility barriers across the fault planes.

No.	Name	Description	Default
I	MFSEGS	A positive integer value that defines the maximum number of records (segments) for the <i>FAULTS</i> keyword.	0
Notes: I) The keyword is terminated by a “/”.			

Table 5.13: FAULTDIM Keyword Description

Example

```
--
--      FAULT
--      SEGMS
--
FAULTDIM
      10000
/
```

The above example defines the maximum number of records that can be entered using the *FAULT* keyword to be 10,000 segments.

5.3.43 FIELD – ACTIVATE THE OIL FIELD SYSTEM OF UNITS FOR THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the oil *FIELD* system of units for the model.

OPM Flow has three sets of units, namely: *METRIC*, *FIELD* and *LAB* and one of these keyword should be invoked in the *RUNSPEC* section to avoid any ambiguity. Both the simulator input and output units are controlled by including one of the *METRIC*, *FIELD* or *LAB* keywords in the *RUNSPEC* section of the input file.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      SWITCH ON THE FIELD SYSTEM OF UNITS FOR BOTH INPUT AND OUTPUT
--
FIELD
```

The above example switches on the *FIELD* system of units for the model.

5.3.44 FMTHMD – ACTIVATE THE FORMAT HISTORY MATCH GRADIENT FILE OPTION

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Description

This keyword switches on formatted output for the History Match Gradient output file, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.45 FMTIN – ACTIVATE THE FORMAT INPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword switches on the Format Input Files option for all input files. Similar to the commercial simulator, OPM Flow has various options for reading various input files and writing the resulting OPM Flow output files as described in Table 5.14.

Process	Keyword	Description	Files
Input	FMTIN	A character string that defines the input files to be formatted as ASCII i.e. text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that used as input, for for example when restarting from another case. <u>If the keyword is omitted then the default is for binary file input.</u>	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	MULTIN	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I
	UNIFIN	A character string that defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	FMTOUT	A character string that sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.PRT, *.LOG and *.DBG files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that binary files are outputted. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	MULTOUT	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I

Process	Keyword	Description	Files
	UNIFOUT	A character string that defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. If the keyword is omitted then the default is for one file per reporting time step.	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. Their main advantage is that if a number of simulations reside in one directory, their output is organized. There is no limit on the number of reporting steps that a unified file can store.			

Table 5.14: FMTIN Keyword Description

There is no data required for this keyword.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--          SWITCH ON THE FORMAT INPUT FILES OPTION
--
FMTIN
```

The above example switches on the format input file option.

5.3.46 FMTOUT – ACTIVATE THE FORMAT OUTPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the Format Output Files option for all output files. Similar to the commercial simulator, OPM Flow has various options for reading various input files and writing the resulting OPM Flow output files as described in Table 5.15.

Process	Keyword	Description	Files
Input	<i>FMTIN</i>	A character string that defines the input files to be formatted as ASCII i.e. text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that used as input, for for example when restarting from another case. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTIN</i>	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I
	<i>UNIFIN</i>	A character string that defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	<i>FMTOUT</i>	A character string that sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.PRT, *.LOG and *.DBG files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that binary files are outputted. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTOUT</i>	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I

Process	Keyword	Description	Files
	UNIFOUT	A character string that defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. Their main advantage is that if a number of simulations reside in one directory, their output is organized. There is no limit on the number of reporting steps that a unified file can store.			

Table 5.15: FMOUT Keyword Description

There is no data required for this keyword.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--          SWITCH ON THE FORMAT OUTPUT FILES OPTION
--
FMTOUT
```

The above example switches on the format output file option.

5.3.47 FOAM – ACTIVATE THE FOAM PHASE AND MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the foam phase and modeling option. The keyword will also invoke data input file checking to ensure that all the required foam phase input parameters are defined in the input deck. Note in the commercial simulator the [FOAM](#) phase and model can be used in conjunction with the [POLYMER](#) and [SURFACT](#) phases; this is not the case for OPM Flow. OPM Flow's [FOAM](#) phase and model is a standalone implementation and cannot be used in conjunction with either the [POLYMER](#) or [SURFACT](#) phases.

Foam flooding is an enhanced oil recovery flood process that attempts to control injected gas breakthrough in an oil reservoir by changing the mobility of the injected fluid.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE FOAM PHASE IN THE MODEL
--
FOAM
```

The above example declares that the foam phase is active in the model.

5.3.48 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be place at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a dull description.

5.3.49 FRICTION – ACTIVATE WELLBORE FRICTION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **FRICTION** keyword activates the Wellbore Friction option and defines the maximum number of wellbore friction wells together with the maximum number of well branches.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	MXWELS	A positive integer defining the maximum number of wellbore friction wells for this model.	0
3	MXBRAN	A positive integer defining the maximum number of branches per well. The default value of one implies a standard well with no branches.	1
Notes: 1) The keyword is terminated by a "/".			

Table 5.16: FRICTION Keyword Description

Example

```
--
--      WELL      BRANCH
--      MXWELS    MXBRAN
FRICTION
      5          1
/
```

The above example defines the maximum number of wellbore friction wells to be five and the maximum number of branches set to one, for standard wells.

5.3.50 FULLIMP – ACTIVATE FULLY IMPLICIT SOLUTION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *FULLIMP* keyword activates the Fully Implicit Solution formulation and solution options. OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness. The keyword as the same function as the *IMPLICIT* keyword in the *RUNSPEC* section.

There is no data required for this keyword and there is no terminating “/” for this keyword.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

Example

```
--
--      ACTIVATES THE FULLY IMPLICIT SOLUTION OPTION
--
FULLIMP
```

The above example switches on the fully implicit solution option; however, this has no effect in OPM Flow input decks.

5.3.51 GAS – ACTIVATE THE GAS PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicate that the gas phase is present in the model and must be used for oil-gas, gas-water, oil-water-gas input decks that contain the gas phase. The keyword will also invoke data input file checking to ensure that all the required gas phase input parameters are defined in the input deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
```

The above example declares that the gas phase is active in the model.

5.3.52 GASFIELD – DEFINE GAS FIELD OPERATIONS OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *GASFIELD* keyword activates and specifies the Gas Field Operations options and determines if extended compressors are present in the run and if the expedited first pass DCQ calculation should be used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.53 GASWAT – ACTIVATE THE GAS-WATER MODEL FORMULATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the two-phase Gas-Water model, as such it is equivalent to using both the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section..

Note

This is an OPM Flow keyword, and should not be confused with the more general version of the [GASWAT](#) keyword used in the commercial compositional simulator.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      ACTIVATE CO2 STORAGE IN THE MODEL (OPM FLOW CO2 STORAGE KEYWORD)
--
CO2STORE
--
--      ACTIVATE COMPOSITIONAL MODELING FORMULATION (OPM FLOW KEYWORD)
--
COMPS
--      2
--
--      ACTIVATE GAS-WATER THE MODEL (OPM FLOW KEYWORD)
--
GASWAT

```

The above example declares that the run should use the Gas-Water model, together with the [CO2STORE](#) option, and with two components.

5.3.54 GDIMS – ACTIVATE INSTANTANEOUS GRADIENT OPTION AND DEFINE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [GDIMS](#) keyword activates the Instantaneous Gradient option and defines the maximum dimensions as used by the [GWRTWCV](#) keyword in the [SCHEDULE](#) section. The Instantaneous Gradient option calculates derivatives of solution quantities at the current time step with respect to variations in the variables at the current time step. This is different to Gradient option that calculates the derivatives of solution quantities at the current time step with respect to variations in the variables at the initial time step, that is a time equal to zero. Consequently, the Instantaneous Gradient option can be switched on and off by the [GUPFREQ](#) keyword in the [SCHEDULE](#) section, whereas the Gradient option cannot.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.55 GIMODEL – ACTIVATE GI PSEUDO COMPOSITIONAL OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, *GIMODEL*, activates the Gi Pseudo Compositional option for gas condensate and volatile oil fluids.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

The accuracy of gas condensate and volatile oil modeling using a black-oil reservoir simulator is now firmly accepted in the industry; both depletion and gas cycling above the saturation point can be modeled and yield an acceptable match compared with compositional simulators. The main deficiency with the existing black-oil formulation is the treatment of gas injection below the saturation point, where the compositional effect of the stripping of liquid components is inversely proportional to their molecular weights, is not considered. This is because the black-oil model assumes that the saturated hydrocarbon fluid properties are only functions of pressure. Thus, when dry gas is injected into a condensate reservoir below the saturation pressure the injected gas continues to re-vaporize liquid at a rate governed only by the cell pressures. Compositional modeling indicate that this not the case.

The GI option attempts to overcome the limitation of the standard black-oil approach by extending the black-oil model using the method of Cook et al.⁸⁶ to take into account the fluid property changes occurring during gas injection. This is done by extending the fluid property treatment so that the saturated fluid properties depend on pressure (as per the standard black-oil formulation) and also on an additional parameter which characterizes the compositional changes in the reservoir liquid and vapor phases at constant pressure.

There is no data required for this keyword.

Example

```
--
--          ACTIVATE THE GI PSEUDO COMPOSITIONAL OPTION
--
GIMODEL
```

The above example switches on the Gi Pseudo Compositional option.

⁸⁶ Cook, R. E., Jacoby, R. H., and Ramesh, A. B. "A Beta-type Reservoir Simulator for Approximating Compositional Effects During Gas Injection" paper SPE 4272, Society of Petroleum Engineers Journal (1974) 14, No. 5, 471-481.

5.3.56 GRAVDR – ACTIVATE GRAVITY DRAINAGE AND IMBIBITION FOR DUAL POROSITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on gravity drainage and imbibition modeling between the matrix and the fracture grid blocks in dual porosity and dual permeability runs. Note that either [DZMTRX](#) or [DZMTRXV](#) keywords in the [GRID](#) section should be used to set the matrix vertical dimensions if this option is activated.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE GRAVITY DRAINAGE AND IMBIBITION FOR DUAL POROSITY MODEL
--
GRAVDR
```

The above example switches on the gravity drainage and imbibition option for the run.

5.3.57 GRAVDRB – ACTIVATE VERTICAL DISCRETIZED GRAVITY DRAINAGE AND IMBIBITION FOR DUAL POROSITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on vertical discretized gravity drainage and imbibition modeling between the matrix and the fracture grid blocks in dual porosity and dual permeability runs. Note that the geometry of the matrix sub-cells should be set to VERTICAL on the NMATOPS keyword in the [GRID](#) section if this option is activated.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE VERTICAL DISCRETIZED GRAVITY DRAINAGE AND IMBIBITION
--
GRAVDRB
```

The above example switches on the vertical discretized gravity drainage and imbibition option for the run.

5.3.58 GRAVDRM – ACTIVATE ALTERNATIVE GRAVITY DRAINAGE AND IMBIBITION FOR DUAL POROSITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the alternative gravity drainage and imbibition modeling between the matrix and the fracture grid blocks in dual porosity and dual permeability runs. Either the [GRAVDRM](#) or [GRAVDR](#) keywords should be used to activate this standard or alternative type of formulation.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	OPTIONI	A defined character string that sets the matrix flow in and out of the matrix block option, and should be set to one of the following: 1) YES: oil flow is bi-directional, that is oil can flow into and out of the matrix block. 2) NO: oil flow is uni-directional, that is oil can flow out of the matrix block.	YES
Notes: 1) The keyword is terminated by a “/”.			

Table 5.17: GRAVDRM Keyword Description

Example

```
--
--      ACTIVATE ALTERNATIVE GRAVITY DRAINAGE AND IMBIBITION MODEL
--
--      MATRIX
--      OPTION
GRAVDRM
      YES
```

The above example switches on the alternative gravity drainage and imbibition option for the run and sets oil flow to be bi-directional, that is oil can flow into and out of the matrix block.

5.3.59 GRIDOPTS – GRID PROCESSING OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

GRIDOPTS activates the negative directional dependent transmissibility multipliers option, defines the maximum number of **MULTNUM** regions and the number of **PINCHNUM** regions for the model.

No.	Name	Description	Default
1	TRANMULT	A character string that activates the negative directional dependent transmissibility multipliers option by setting TRANMULT to YES. Setting the value to NO switches off this option. OPM Flow uses a positive directional dependent transmissibility formulation to describe the flow between two cells, that is for cell (I, J, K) OPM Flow calculates the x face transmissibility between (I, J, K) and (I + 1, J, K) cell face. Modification to the transmissibilities in this case is accomplished by the MULTX, MULTY and MULTZ keywords. Setting TRANMULT to YES invokes the option to use a negative directional dependent multiplier scheme using the MULTX-, MULTY- and MULTZ- keywords. In this case OPM Flow applies the x face transmissibility between (I - 1, J, K) and (I, J, K) cell face when using the MULTX-, MULTY- and MULTZ- keywords. Note that if TRANMULT is defaulted, and there are negative directional dependent multiplier keywords in the input deck, then OPM Flow will continue to process the MULTX-, MULTY- and MULTZ keywords correctly. Whereas, the commercial simulator will terminate with an error.	NO
2	NRMULT	A positive integer value that defines the maximum number of MULTNUM regions for the MULTNUM array. The MULTNUM array is used in the GRID section to define various inter-region transmissibility regions in the model and NRMULT sets the maximum number of regions which is the maximum value of an element in the MULTNUM array. Inter-region MULTNUM transmissibility multipliers can be defined using the MULTREGT and regional pore volumes multipliers can be set using the MULTREGP keyword.	0
3	NRPINC	A positive integer value that defines the maximum number of PINCHNUM regions for the PINCHNUM array. The PINCHNUM array is used in the GRID section to define various regions in the model and NRPINC sets the maximum of regions which is the maximum value of an element in the PINCHNUM array. Each regions criteria for setting the pinch out criteria is set by the PINCHREG keyword.	0
Notes: 1) The keyword is terminated by a “/”.			

Table 5.18: GRIDOPTS Keyword Description

See also the **MULTNUM**, **MULTREGP**, **MULTREGT**, **PINCHNUM**, and **PINCHREG** keywords.

Example

```
--
--      NEG      MAX      MAX
--      MULTS    MULTNUM PINCHNUM
GRIDOPTS
      NO      9      1*      /
```

The above example switches off the negative directional dependent transmissibility multipliers option and defines the maximum of *MULTNUM* regions to be nine,. The NRPINC parameter is defaulted which means there the maximum number of *PINCHREG* regions is zero.

5.3.60 H2SOL – ACTIVATE DISSOLVED H2 IN THE WATER PHASE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **H2SOL** keyword activates dissolved hydrogen (H_2) in the water phase, where H_2 is represented by the **SOLVENT** pseudo component, using the simulator's H_2 -Brine PVT model. This keyword is a compositional keyword in the commercial simulator but has been implemented in OPM Flow's black-oil model.

The **H2SOL** keyword can be used when modelling H_2 injection in depleted hydrocarbon reservoirs. See also the **H2STORE** keyword in the **RUNSPEC** section which can be used when modelling H_2 injection in saline aquifers.

The H_2 -Brine PVT model computes the PVT properties such as density, viscosity, and enthalpy internally as functions of pressure, temperature, and composition by using analytic correlations and models from the literature rather than by interpolation of tabulated values in the input deck. These values are transformed to the standard black-oil equivalent PVT tables internally by the simulator. Dissolved hydrogen in brine is modeled using Li et al.⁸⁷. Hydrogen gas density was modeled using Leachman et al.⁸⁸. Other properties are modeled the same way as in **CO2STORE** but modified for hydrogen-brine (a similar approach was recently reported by Raad et al.⁸⁹). A full description of the underlying PVT models used by **CO2STORE** is described by Sandve et al.⁹⁰. Note that the H_2 -Brine PVT properties depend on the temperature and salinity and these must therefore be entered in the **PROPS** section. The reservoir temperature can be defined using, e.g., the **RTEMP** keyword. Region based salinity can be provided using the **SALINITY** keyword.

The **H2SOL** keyword must be used together with the **SOLVENT** keyword in the **RUNSPEC** section. The **SOLVENT** keyword activates the four component black-oil model: oil, water and gas, plus a solvent (in this case H_2).

The **DISGASW** keyword in the **RUNSPEC** section can be used to model dissolution of H_2 in the Brine. Note that hydrocarbon gas is not allowed to dissolve in the Brine (this is considered a reasonable assumption for most hydrocarbon gases).

The hydrocarbon reservoir fluid properties should be set up as normal apart from the water PVT properties, which will be defined by the simulators' H_2 -Brine PVT model, so the water PVT keywords (e.g. **PVTW**) are not required. The H_2 PVT properties will also be defined by the H_2 -Brine PVT model so the solvent PVT keywords (e.g. **SDENSITY**, **PVTSOL**) are also not required.

The oil-water and gas-water saturation functions should be set up as normal. The H_2 and gas miscible relative permeability tables should be defined using the **SSFN** keyword in the **PROPS** section.

The initial H_2 saturations in the model can be specified by enumeration using the **SSOL** keyword in the **SOLUTION** section.

The H_2 fraction in a well's injection stream can be defined using the **WSOLVENT** keyword in the **SCHEDULE** section.

There is no data required for this keyword and there is no terminating "/" for this keyword.

⁸⁷ Dedong Li, Christof Beyer, Sebastian Bauer, A unified phase equilibrium model for hydrogen solubility and solution density, *International Journal of Hydrogen Energy*, Volume 43, Issue 1, 2018, Pages 512-529.

⁸⁸ J. W. Leachman, R. T. Jacobsen, S. G. Penoncello, E. W. Lemmon; *Fundamental Equations of State for Parahydrogen, Normal Hydrogen, and Orthohydrogen*. *J. Phys. Chem. Ref. Data* 1 September 2009; 38 (3): 721–748.

⁸⁹ Seyed Mostafa Jafari Raad, Ehsan Ranjbar, Hassan Hassanzadeh, Yuri Leonenko, *Hydrogen-brine mixture PVT data for reservoir simulation of hydrogen storage in deep saline aquifers*, *International Journal of Hydrogen Energy*, Volume 48, Issue 2, 2023, Pages 696-708.

⁹⁰ Tor Harald Sandve, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. *Convective dissolution in field scale CO_2 storage simulation using the OPM Flow simulator*. Submitted to TCCS 11 – Trondheim Conference on CO_2 Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

Example

The example shows the usage of [H2SOL](#) to model H₂ injection in a live-oil reservoir. The [RUNSPEC](#) section includes the [H2SOL](#) keyword, which activates dissolved hydrogen (H₂) in the water phase. The [SOLVENT](#) keyword activates the solvent pseudo component, which represents H₂. The [DISGASW](#) keyword has been used to activate dissolved H₂ in the brine.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
-- -----
-- FLUID TYPES AND TRACER OPTIONS
-- -----
--
--      ACTIVATE DISSOLVED H2 IN WATER (OPM FLOW KEYWORD)
--
H2SOL
--
--      SOLVENT PHASE IS PRESENT IN THE RUN
--
SOLVENT
--
--      DISSOLVED H2 IN WATER IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
DISGASW
```

The second part of the example covers the additional data required in the [PROPS](#) section, in which the H₂-gas miscible relative permeabilities are defined using the [SSFN](#) keyword. The initial temperature and salinity are defined using the [RTEMP](#) and [SALINITY](#) keywords. The oil and gas PVT properties should be defined in the normal way. No water PVT data is required.

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP
--      80.0 / RESERVOIR TEMP
--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALINITY
--      0.7 /
--
-- SOLVENT RELATIVE PERMEABILITY TABLES
--
SSFN
-- SGAS KRGT KRST
-- FRAC
-- -----
--      0.0000 0.0000 0.0000
--      1.0000 1.0000 1.0000 /
```

The third part of the example covers initialising the model in the [SOLUTION](#) section. Here the [EQUIL](#) keyword is used as normal to equilibrate the oil, gas and water phases, and the [SSOL](#) keyword to specify that there is initially no H₂ present in the gas phase.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      DATUM      DATUM      OWC      PCOW      GOC      PCGO      RS      RV      N      E300      RVW
--      DEPTH      PRESS      DEPTH      ----      DEPTH      ----      OPT      OPT      OPT      OPT      OPT
EQUIL
      2000.0      200.0      2100.0      0.00      2000.0      0.00      1*      1*      1*      2*      1* /
--
-- DEFINE INITIAL EQUILIBRATION SOLVENT SAT VALUES FOR ALL CELLS IN THE MODEL
-- BASED ON NX = 100, NY = 100 AND NZ = 3
--
SSOL
      30000*0.0 /
```

For the summary section, the H₂ storage quantity can be tracked using the solvent injection and production volumes as shown below.

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--      FIELD SOLVENT INJECTION RATE
FNIR
--
--      FIELD SOLVENT PRODUCTION RATE
FNPR
--
--      FIELD SOLVENT INJECTION TOTAL
FNIT
--
--      FIELD SOLVENT PRODUCTION TOTAL
FNPT
```

The final part of the example covers the [SCHEDULE](#) section. The [WSOLVENT](#) keyword is used to specify the injected solvent fraction (in this case pure H₂).

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      RESTART CONTROL BASIC = 4 (ALL=2, YEARLY=4, MONTHLY=5, TSTEP=6)
--
RPTRST
      'BASIC=1'  'ALLPROPS'  /
```



```
--  
-- DEFINE GAS INJECTION WELL SOLVENT FRACTION  
--  
-- WELL SOLVENT  
-- NAME FRACTION  
-- -----  
WSOLVENT  
  GI01 1.0000 /  
/
```

5.3.61 H2STORE – ACTIVATE THE H2 STORAGE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **H2STORE** keyword activates the hydrogen (H₂) storage model for the run to account for both hydrogen and water phase solubility. **H2STORE** is similar to **CO2STORE**, which activates the carbon dioxide (CO₂) storage model.

The H₂-Brine PVT model computes the PVT properties such as density, viscosity, and enthalpy internally as functions of pressure, temperature, and composition by using analytic correlations and models from the literature rather than by interpolation from tabulated values. These values are transformed to the standard black-oil equivalent PVT tables internally by the simulator. Dissolved hydrogen in brine is modeled using Li et al.⁹¹. Hydrogen gas density was modeled using Leachman et al.⁹². Other properties are modeled the same way as in **CO2STORE** but modified for hydrogen-brine (a similar approach was recently reported by Raad et al.⁹³). A full description of the underlying PVT models used by **CO2STORE** is described by Sandve et al.⁹⁴. This means that the normal PVT keywords like **DENSITY**, **PVTO**, **PVDG** etc. are not required by OPM Flow when this model is activated, and if entered will be ignored by the simulator. Note that the H₂-Brine PVT properties depend on the temperature and salinity and these must therefore be entered in the **PROPS** or **SOLUTION** sections. The reservoir temperature can be defined using, e.g., the **RTEMP** keyword. Region based salinity can be provided using the **SALINITY** keyword.

The **H2STORE** keyword must be used with either: (1) the **GAS** and **WATER** keywords (or alternatively the **GASWAT** keyword), or (2) the **GAS** and **OIL** keywords in the **RUNSPEC** section. It is recommended that the standard option (1) is used.

The **DISGASW** keyword can also be used with option (1) to model dissolution of H₂ in the Brine.

Option (1) has the advantage that it can be used with the **VAPWAT** and **PRECSALT** keywords to model the impact of both vaporization of residual water and salt precipitation in the near wellbore region on injectivity of H₂ injection wells.

In option (1), the **GAS** and **WATER** (or **GASWAT**) keywords declare that the gas and water phases are present in the model. When the **H2STORE** option is used the water phase represents the brine and the gas phase represents H₂. Note that the input and output keywords need to be consistent with this assumption, e.g., **GSF** (gas saturation function) and **WSF** (water saturation function) should be used for the H₂-Brine relative permeability, etc.

Although, the **DISGAS** and **VAPWAT** keywords can be used with option (2) to model water vaporization and H₂ dissolution, salt precipitation is not currently supported with option (2).

In option (2), the **GAS** and **OIL** keywords declare that the gas and oil phases are present in the model. Internally when **H2STORE** is used the oil phase refers to the brine and the gas phase to H₂. Again, the input and output keywords need to be consistent with this assumption, e.g., **SGOF** (gas-oil relative permeability) is used for the H₂-Brine relative permeability, FOIP (Field Oil-In-Place) shows the total amount of brine in the reservoir, etc.

⁹¹ Dedong Li, Christof Beyer, Sebastian Bauer, A unified phase equilibrium model for hydrogen solubility and solution density, *International Journal of Hydrogen Energy*, Volume 43, Issue 1, 2018, Pages 512-529.

⁹² J. W. Leachman, R. T. Jacobsen, S. G. Penoncello, E. W. Lemmon; *Fundamental Equations of State for Parahydrogen, Normal Hydrogen, and Orthohydrogen*. *J. Phys. Chem. Ref. Data* 1 September 2009; 38 (3): 721–748.

⁹³ Seyed Mostafa Jafari Raad, Ehsan Ranjbar, Hassan Hassanzadeh, Yuri Leonenko, *Hydrogen-brine mixture PVT data for reservoir simulation of hydrogen storage in deep saline aquifers*, *International Journal of Hydrogen Energy*, Volume 48, Issue 2, 2023, Pages 696-708.

⁹⁴ Tor Harald Sandve, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. *Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator*. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Examples

The first example shows the standard useage of [H2STORE](#) with Option (I) the Gas-Water model ([GASWAT](#)). Here we also activate the dissolved gas in water ([DISGASW](#)) and vaporized water in gas ([VAPWAT](#)) options.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
-- FLUID TYPES AND TRACER OPTIONS
--
--
--      ACTIVATE H2 STORAGE IN THE MODEL (OPM FLOW H2 STORAGE KEYWORD)
H2STORE
--
--      ACTIVATE GAS-WATER THE MODEL (OPM FLOW KEYWORD)
GASWAT
--
--      DISSOLVED GAS IN WATER IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
DISGASW
--
--      VAPORIZED WATER IN DRY/WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
VAPWAT
```

The second part of the example covers the data required for the [PROPS](#) section, in which the two-phase relative permeability functions are set using [GSF](#) and [WSF](#) keywords.

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP
--      80.0 / RESERVOIR TEMP
--
--      ROCK COMPRESSIBILITY
--
--      REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--
--      REF PRES  CF
--      BARSA    1/BARSA
--      -----
ROCK
--      200.0    5.0E-05 / ROCK COMPRESSIBILITY
--
--      GAS RELATIVE PERMEABILITY TABLES (OPM FLOW KEYWORD)
GSF
```

```
--      SGAS      KRG      PCGW
--      FRAC      PSIA
--      -----
--      0.000      0.000      0.0
--      0.080      0.001      0.0
--      0.170      0.010      0.0
--      0.350      0.050      0.0
--      0.530      0.200      0.0
--      0.620      0.350      0.0
--      0.650      0.390      0.0
--      0.710      0.560      0.0
--      0.800      1.000      0.0      / TABLE NO. 01
--
--      WATER RELATIVE PERMEABILITY TABLES (OPM FLOW KEYWORD)
--
WSF
--      SWAT      KRW
--      FRAC
--      -----
--      0.200      0.0000
--      0.400      0.1000
--      0.800      0.5000
--      1.000      1.0000      / TABLE NO. 01
--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALINITY
0.7      /
```

No other data is required to define the fluid and rock properties in the [PROPS](#) section as the data is generated from internal analytic correlations and models by the simulator. Finally, note that units for salinity are to the 10⁻³, thus for metric units we have 10⁻³ x kg-M/kg.

The third part of the example covers initializing the model in the [SOLUTION](#) section. Here we use the [EQUIL](#)(EQLOPT6) parameter equal to one, to use table number one of the [RVWVD](#) keyword, in order to set the vaporized water versus depth distribution for the model.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      SYSTEM IS SATURATED WITH WATER
--
--      DATUM      DATUM      OWC      PCOW      GOC      PCGO      RS      RV      N      E300      RVW
--      DEPTH      PRESS      DEPTH      ----      DEPTH      ----      OPT      OPT      OPT      OPT      OPT
EQUIL
      2000.0      200.0      1800.0      0.00      1800.0      0.00      1*      1*      1*      2*      1 /
--
--      WATER VAPOR RATIO VS DEPTH (OPM FLOW KEYWORD)
--
--      DEPTH      RVW
--                  STB/MSCF
--      -----
RVWVD
      1000.0      0.000
      3000.0      0.000
                                     / RVW VS DEPTH EQUIL REGN 01
```

In the [SUMMARY](#) section, the simulator supports summary vectors specific to CO₂ storage (see Section [11.1.12 Option Specific Variables - CO2STORE/H2STORE Model](#)) many of these can also be used for H₂ storage including those shown below.

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--      FIELD H2 DISOLVED IN WATER PHASE
FWCD
--
--      FIELD H2 TRAPPED IN GAS PHASE
FGCDI
--
--      FIELD H2 MOBILE IN GAS PHASE
FGCDM
```

The final part of the example covers the [SCHEDULE](#) section. The standard [WCONINJE](#) keyword is then used to set the gas injection rate, in this case 100,000 sm³/day of H₂.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      RESTART CONTROL BASIC (LAST=1, ALL=2, YEARLY=4, MONTHLY=5, TSTEP=6)
--
RPTRST
      'BASIC=1' ALLPROPS TEMP RWSAT RVWSAT
--
--      WELL SPECIFICATION DATA
```

```

--
-- WELL      GROUP      LOCATION  BHP      PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME      NAME              I    J  DEPTH  FLUID   AREA   EQUANS  SHUT  FLOW   TABLE
WELSPEDS
INJ1        'G1'              25    25  1*      GAS                                /
/
--
--          WELL CONNECTION DATA
--
-- WELL      --- LOCATION ---  OPEN  SAT  CONN  WELL  KH  SKIN  D  DIR
-- NAME      II  JJ  K1  K2  SHUT  TAB  FACT  DIA  FACT  FACT  FACT  PEN
COMPDAT
INJ1        25  25    1  10  OPEN   1*   1*   0.3                                /
/
--
--          WELL INJECTION CONTROLS
--
-- WELL      FLUID  OPEN/  CNTL  SURF  RESV  BHP  THP  VFP
-- NAME      TYPE   SHUT   MODE  RATE  RATE  PRES  PRES  TABLE
WCONINJE
INJ1        GAS    OPEN    RATE 100000  1*    300                                /
/
--
--          ADVANCE SIMULATION BY REPORTING TIME
--
TSTEP
          36*30.4375
/

```

Note in order to get the liquid phase mole fractions of H₂, that is, the mole fractions of H₂ in the water phase (XMFH2), and the vapor phase mole fractions (YMFVAT) to the restart file, one must use the command line parameter enable-opm-rst-file set equal to true.

The second example shows how to use [H2STORE](#) with the alternative option (2). The example below declares that the hydrogen storage model is active for the run to account for both hydrogen and water phase solubility using OPM Flow's H₂-Brine PVT model. Option (2) is used where the [OIL](#) phase refers to the brine and the [GAS](#) phase to H₂.

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
-- -----
-- FLUID TYPES AND TRACER OPTIONS
-- -----
--
--          ACTIVATE H2 STORAGE IN THE MODEL (OPM FLOW H2 STORAGE KEYWORD)
--
H2STORE
--
--          OIL PHASE IS PRESENT IN THE RUN BUT IS THE BRINE PHASE FOR H2STORE
--
OIL
--
--          GAS PHASE IS PRESENT IN THE RUN BUT IS THE H2 PHASE FOR H2STORE
--
GAS
--
--          DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN (H2 IN BRINE)

```

```
--
DISGAS
--
--          VAPORIZED OIL IN WET GAS IS PRESENT IN THE RUN (WATER IN H2)
--
VAPOIL

The second part of the example covers the data required for the PROPS section, in which the input
keywords need to be consistent with the OIL phase referring to the Brine and the GAS to H2; that is SGOF
(gas-oil relative permeability) is used to define the H2-Brine relative permeability table.

-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP
90.0                                / RESERVOIR TEMPERATURE
--
--      ROCK COMPRESSIBILITY
--
--      REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--
--      REF PRES   CF
--      BARSA      1/BARSA
--      -----
ROCK
200.0      5.0E-05                                / ROCK COMPRESSIBILITY
--
--      GAS-OIL RELATIVE PERMEABILITY TABLES (SGOF) - H2STORE PHASES
SGOF
--      SG          KRG          KROG          PCOG
--      FRAC                PSIA
--      -----
--      0.00000      0.000000      1.00000      0.0000
--      1.00000      1.000000      0.00000      0.0000                                / TABLE No. 01
--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALINITY
0.7                                /
```

The third part and final part of the example covers initializing the model in the *SOLUTION* section. Here we set the *EQUIL*(EQLOPT1 and EQLOPT2) parameters equal to one, to use table number one of the *RSVD* and *RVVD* keywords, in order to set the initial dissolved H₂ and vaporised water versus depth distribution for the model.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      SYSTEM IS SATURATED WITH OIL (BRINE)
--
--      DATUM      DATUM      OWC      PCOW      GOC      PCGO      RS      RV      N      E300      RVW
```

--	DEPTH	PRESS	DEPTH	----	DEPTH	----	OPT	OPT	OPT	OPT	OPT	
EQUIL	2000.0	200.0	2200.0	0.00	1800.0	0.00	1	1	1*	2*	1*	/
--	DEPTH	RS										
--		MSCF/STB										
--	-----	-----										
RSVD	1000.0	0.000										
	3000.0	0.000							/			
--	DEPTH	RV										
--		STB/MSCF										
--	-----	-----										
RVVD	1000.0	0.000										
	3000.0	0.000							/			

5.3.62 HMDIMS – DEFINE HISTORY MATCH GRADIENT PARAMETER DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, *HMDIMS*, defines the maximum parameter dimensions for the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

5.3.63 HYST – ACTIVATE THE HYSTERESIS OPTION (RETIRED)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *HYST* keyword activates the hysteresis option, the keyword should be avoided and the hysteresis option should be enabled by the *HYSER* parameter on the *SATOTPS* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.64
IMPES
– ACTIVATE IMPLICIT PRESSURE EXPLICIT SATURATION SOLUTION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *IMPES* keyword activates the Implicit Pressure Explicit Saturation formulation and solution options, commonly know as *IMPES*. OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

Example

```
--
--      ACTIVATE THE IMPES SOLUTION OPTION
--
IMPES
```

The above example switches on the *IMPES* solution option; however, this has no effect in OPM Flow input decks.

5.3.65 IMPLICIT – ACTIVATE FULLY IMPLICIT SOLUTION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *IMPLICIT* keyword activates the Fully Implicit Solution formulation and solution options. OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness. The keyword as the same function as the *FULLIMP* keyword in the *RUNSPEC* section.

There is no data required for this keyword and there is no terminating “/” for this keyword.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

Example

```
--
--      ACTIVATES THE FULLY IMPLICIT SOLUTION OPTION
--
IMPLICIT
```

The above example switches on the fully implicit solution option; however, this has no effect in OPM Flow input decks.

5.3.66 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDINC](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the GLOBAL section for a full description.

5.3.67 INSPEC – ACTIVATE THE INSPEC FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the writing of the *INIT* Index file that specifies and defines the format and data type written to the *.INIT data file. The *.INIT data file contains the static data specified in the *GRID*, *PROPS* and *REGIONS* sections. For example, the *PORO*, *PERM* and *NTG* arrays from the *GRID* section. The data is used in post-processing software, for example OPM ResInsight, to visualize the static grid properties.

The *INIT* Index file can either be written out in formatted form as ASCII i.e. text files, if the *FMTOUT* keyword has been activated (**.FINSPEC*), or binary format (**.INSPEC*) if the *FMTOUT* keyword has not been activated. If the *INIT* keyword in the *GRID* section has been used to switch on the writing of the *. *INIT* data file then a binary *INIT* Index file is automatically written out as well, unless the *NOINSPEC* keyword in the *RUNSPEC* section has been used to switch off the writing of the *INIT* Index file.

Note that most post-processing software require the *.INSPEC file to load the *.INIT data set, although OPM ResInsight does not require this file to be able to load the *.INIT data file.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE WRITING THE INIT INDEX FILE FOR POST-PROCESSING
--
INSPEC
```

The above example switches on the writing of the *INIT* Index file for post-processing in ResInsight.

5.3.68 LAB – ACTIVATE THE LABORATORY SYSTEM OF UNITS FOR THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the LABORATORY system of units for the model.

OPM Flow has three sets of units, namely: [METRIC](#), [FIELD](#) and [LAB](#) and one of these keyword should be invoked in the [RUNSPEC](#) section to avoid any ambiguity. Both the simulator input and output units are controlled by including one of the [METRIC](#), [FIELD](#) or [LAB](#) keywords in the [RUNSPEC](#) section of the input file.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      SWITCH ON THE LABORATORY SYSTEM OF UNITS FOR BOTH INPUT AND OUTPUT
--
LAB
```

The above example switches on the LABORATORY system of units for the model.

5.3.69 LGR – DEFINE LOCAL GRID REFINEMENT DIMENSIONS AND PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [LGR](#) keyword defines the maximum dimensions and parameters for the Local Grid Refinement (“LGR”) option.

Currently, OPM Flow does not support the local grid refinement feature and therefore this keyword is ignored by the simulator, but is documented here for completeness.

No.	Name	Description	Default
1	MAXLGR	A positive integer value that defines the maximum number of LGRs in the model.	0
2	MAXCLS	A positive integer value that defines the maximum number of grid blocks in all the LGRs.	0
3	MCOARS	A positive integer value that defines the maximum number of amalgamated coarse grid blocks in the model.	0
4	MAMALG	A positive integer value that defines the maximum number of LGR amalgamations in the model.	0
5	MXLALG	A positive integer value that defines the maximum number of LGRs in any amalgamation in the model.	0
6	LSTACK	A positive integer that defines the maximum number of previous search directions stored by the linear solver for the LGR. See the NSTACK keyword in the RUNSPEC section for a full description.	10
7	INTOPT	A character string set to either INTERP to activate the Quandalle ⁹⁵ pressure correction, or NOINTERP to deactivate this option. The option applies bi-linear interpolation to the global cells surrounding an LGR in order to improve the accuracy of the flow calculations between the LGR and the host cells.	NOINTERP
8	NCHCOR	A positive integer value that defines the maximum number of grid blocks within a coarsened grid that overlap parallel domain boundaries for when the Parallel option has been invoked by the PARALLEL keyword in the RUNSPEC section. OPM Flow uses a different numerical scheme which makes this parameter redundant, see section 2.2 Running OPM Flow From The Command Line on how to run OPM Flow in parallel mode.	0

Notes:

- 1) The keyword is terminated by a “/”.

Table 5.19: LGR Keyword Description

⁹⁵ Quandalle, Philippe & Besset, P. (1985). Reduction of Grid Effects Due to Local Sub-Gridding in Simulations Using a Composite Grid. 10.2118/13527-MS.

Example

```
--
--      LOCAL GRID REFINEMENT DIMENSIONS AND PARAMETERS
--
--      LGR      LGR      LGR      LGR      LGR      LGR      LGR      LGR
--      MAXLGR   MAXCLS   MCOARS   MAMALG   MXLALG   LSTACK  INTOPT  NCHCOR
LGR
      10        1000     1*       1*       1*       1*       INTERP  1*      /
```

The above example sets the maximum number of LGRs to 10 and the maximum number of grid blocks a LGR may contain to 1,000, and that the Quandalle pressure correction should be used to improve the flow calculation.

5.3.70 LGRCOPY – ACTIVATE LOCAL GRID REFINEMENT INHERITANCE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *LGRCOPY* keyword activates the Local Grid Refinement (“LGR”) Inheritance option that allows the LGR to inherit the properties of the global or host cell containing an LGR grid block when it is defined, as opposed to the normal process of applying this transform at the end of the *GRID* section. *LGRCOPY* can be used in the *RUNSPEC*, *GRID* and *EDIT* sections. If used in the *RUNSPEC* section then the option is applied to all LGRs defined in the input file, whereas if used in the *GRID* or *EDIT* sections the keyword must be placed inside a *LGR* definition section, that is between a *CARFIN* (Cartesian LGR grid) or *RADFIN/RADFIN4* (radial LGR grid) keyword and an *ENDFIN* keyword. In the latter case inheritance is applied on an individual LGR basis.

Currently, OPM Flow does not support the local grid refinement feature and therefore this keyword is ignored by the simulator.

Example

The following example activates the LGR Inheritance option for all LGRs in the model.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--         ACTIVATE LOCAL GRID REFINEMENT INHERITANCE
--
LGRCOPY
```

5.3.71 LICENSES – DEFINE REQUIRED LICENSES FOR RUN

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the additional software licenses that are required to invoke various licensed options in the commercial simulator at the start of the run. The commercial simulator requests a license when keywords associated with a licensed option is encountered in the input deck, this may result in the license being unavailable at the time of request and after the simulation has been initiated, resulting in the run terminating.This keyword avoids this scenario by reserving the license at the start of the run.

OPM Flow is an open source project and therefore there is no license management of the various implemented options; hence this keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.72
LIVEOIL – ACTIVATE THE LIVE OIL PHASE (OIL WITH FREE AND DISSOLVED GAS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates oil, free and dissolved gas in the model and therefore makes the oil phase live oil⁹⁶ in the black-oil formulation, and is equivalent to setting the phases present in the model to be oil, dissolved gas, gas and water phases. Note if water is present in the model this needs to be explicitly stated via the [WATER](#) keyword in the [RUNSPEC](#) section (see also the [BLACKOIL](#) and [DEADOIL](#) keywords in the [RUNSPEC](#) section). The keyword is used by the commercial simulator's compositional [THERMAL](#) option to set the phases present in the model.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

The following example activates the black-oil phases in the model.

```
--
--      ACTIVATE LIVE-OIL PHASE
--
LIVEOIL
```

Alternatively one could explicitly declare the phases using the following keywords in the [RUNSPEC](#) section.

```
--
--      OIL PHASE IS PRESENT IN THE RUN
--
OIL
--
--      DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN
--
DISGAS
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
--
--      WATER PHASE IS PRESENT IN THE RUN
--
WATER
```

The above example switches on the oil, dissolved gas, gas and water phases in the model.

⁹⁶
“Live” oil is oil that contains gas in solution, which is normally the case for most conventional oil reservoirs. However, for oil reservoirs classified as heavy oil reservoirs, the in situ dissolved gas may be negligible and oil would then be classified as gas-free oil which is commonly referred to as “dead” oil.

5.3.73 LOAD – LOAD A SAVE FILE FOR A FAST RESTART

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [LOAD](#) keyword loads a previously generated [SAVE](#) file to enable a fast restart. A [SAVE](#) file contains all the data from a previous run's [RUNSPEC](#), [GRID](#), [EDIT](#), [PROPS](#) and [REGIONS](#) sections, and thus there is no need for the simulator to calculate various parameters, including grid block transmissibilities etc. This allows for the current run to restart quicker than a conventional restart run using the [RESTART](#) keyword in the [SOLUTION](#) section via a [RESTART](#) file (*.UNRST or *.FUNRST etc.). The keyword should be the first keyword in the input deck and the [RUNSPEC](#), [GRID](#), [EDIT](#), [PROPS](#) and [REGIONS](#) sections should be deleted from the input deck.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.74 LOWSALT – ACTIVATE THE LOW SALT BRINE PHASE IN THE BRINE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [LOWSALT](#), activates the low salt brine phase for the Brine option and also activates the Brine option. See also the [BRINE](#) keyword in the [RUNSPEC](#) section.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE THE LOW SALT BRINE PHASE FOR THE BRINE OPTION
--
LOWSALT
```

The above example declares that the low salt brine phase is active in the model for the Brine option.

5.3.75 MEMORY – DEFINE ALLOCATED MEMORY (RETIRED)

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

This keyword defines the memory allocation for the run.

OPM Flow and now the commercial simulator uses dynamic memory allocation and therefore the keyword has no effect and is ignored by both simulators.

5.3.76 MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the GLOBAL section for a full description.

5.3.77 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the GLOBAL section for a full description.

5.3.78
MESSSRVC
– ACTIVATE OR DEACTIVATE DATABASE MESSAGE FILE OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [MESSSRVC](#) keyword activates or deactivates output to the database message file (*.DBPRTX). The file contains all the messages from run in binary format and is used in some post-processing software to annotate production line plots from the run.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.79 METRIC – ACTIVATE THE METRIC SYSTEM OF UNITS FOR THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the *METRIC* system of units for the model.

OPM Flow has three sets of units, namely: *METRIC*, *FIELD* and *LAB* and one of these keyword should be invoked in the *RUNSPEC* section to avoid any ambiguity. Both the simulator input and output units are controlled by including one of the *METRIC*, *FIELD* or *LAB* keywords in the *RUNSPEC* section of the input file.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      SWITCH ON THE METRIC SYSTEM OF UNITS FOR BOTH INPUT AND OUTPUT
--
METRIC
```

The above example switches on the *METRIC* system of units for the model.

5.3.80 MICP – ACTIVATE THE MICROBially INDUCED CALCITE PRECIPITATION MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [MICP](#) keyword activates the Microbially Induced Calcite Precipitation (“MICP”) model for the run. [MICP](#) is a new and sustainable technology which utilizes biochemical processes to create barriers by calcium carbonate cementation, the technology has the potential to be used for sealing leakage zones in geological formations. Further information on the mathematical model can be found in the open-access publications Landa-Marbán et al⁹⁷ and ⁹⁸.

Note

This is an OPM Flow specific keyword used to investigate leakage remediation. The module requires that both the [MICP](#) and [WATER](#) keywords in the [RUNSPEC](#) to be active.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE MICROBIAL INDUCED CALCITE PRECIPITATION MODEL
--
MICP
--
--      WATER PHASE IS PRESENT IN THE RUN
--
WATER
```

The above example declares that the [MICP](#) model is active for the run and activates the water phase for the [MICP](#) model. For a complete example of this model, see [MICP.DATA](#).

⁹⁷ Landa-Marbán, D., Tveit, S., Kumar, K., Gasda, S.E., 2021. Practical approaches to study microbially induced calcite precipitation at the eld scale. *Int. J. Greenh. Gas Control* 106, 103256. <https://doi.org/10.1016/j.ijggc.2021.103256>.

⁹⁸ Landa-Marbán, D., Kumar, K., Tveit, S., Gasda, S.E., 2021. Numerical studies of CO2 leakage remediation by micp-based plugging technology. In: Røkke, N.A. and Knuutila, H.K. (Eds) *Short Papers from the 11th International Trondheim CCS conference*, ISBN: 978-82-536-1714-5, 284-290.

5.3.81 MINNPCOL – DEFINE THE MINIMUM NUMBER OF NEWTON ITERATIONS USED TO UPDATE WELL TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [MINNPCOL](#) keyword defines the minimum number of Newton iterations within a time step where the well production and injection targets may be updated, after which the well targets will be frozen until the time step calculations have converged and the time step is complete.

Note

This is an OPM Flow specific keyword that sets the minimum number of Newton iterations, as opposed to the commercial simulator's [NUPCOL](#) keyword that defines the maximum number of Newton iterations within a time step, after which well targets are frozen.

Wells under group control may suffer from some dependency with other wells in the same group that are under group control. This may cause some oscillation in the production and injection well rates within the group. In order to avoid this, the [NUPCOL](#) keyword in the [RUNSPEC](#) section can be used to set the maximum number of Newton iterations within a time step, after which the group well rates are frozen until the time step has converged. Reducing the potential for well rate oscillations within the time step may result in the group targets and limits not being exactly met in this case. Increasing the value of [NUPCOL](#), will improve the accuracy of the group targets and limits at the expense of computational efficiency. Here, the [MINNPCOL](#) keyword sets the minimum number of Newton iterations within a timestep where the well production and injection targets may be updated.

See also section [2.2 Running OPM Flow From The Command Line](#) on how to set various other numerical control parameters for OPM Flow.

No.	Name	Description	Default
I	MINNPCOL	A positive integer that defines the minimum number of Newton iterations within a timestep where well targets may be updated. Note that the default value of six is higher than the commercial black-oil simulator's default maximum value of three used on the NUPCOL keyword in the RUNSPEC section.	6
Notes: I) The keyword is terminated by a "/".			

Table 5.20: MINNPCOL Keyword Description

Example

```
--
--      DEFINE THE MIN NUMBER OF ITERATIONS TO UPDATE WELL FLOW TARGETS
--
MINNPCOL      6      /
```

The above example sets the [MINNPCOL](#) value to its default value of six.

5.3.82 MISCIBLE – DEFINE MISCIBILITY TODD-LONGSTAFF PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [MISCIBLE](#) keyword defines the options associated with the Todd-Longstaff⁹⁹ mixing parameters used for when polymer flooding or CO2 EOR simulation cases are being run.

No.	Name	Description	Default
1	NTMISC	A positive integer value that declares the number miscible residual oil saturations versus water saturations tables for SORWMIS keyword and the number Todd-Longstaff mixing parameters entries on the TLMIXPAR keyword.	1
2	NSMISC	A positive integer value that sets the maximum number of entries (or rows) for each SORWMIS table defined by the SORWMIS keyword.	20
3	MISOPT	A character string that defines the numerical dispersion control options for the oil and gas relative permeability curves, set to either NONE or TWOPOINT: 1) NONE – standard single point up streaming, that is using the immediate neighbor 2) TWOPOINT – two-point up streaming, that is using the immediate neighbor plus one cell for better numerical dispersion control but with a higher computational cost. <u>Only the default value of NONE is supported.</u>	NONE
Notes: 1) The keyword is terminated by a “/”.			

Table 5.21: MISCIBLE Keyword Description

Example

```
--
--      NTAB      MAX      UPSTRM
--      NTMISC    NSMISC    MISOPT
MISCIBLE
      1          20          NONE
```

The above example defines the default values for the [MISCIBLE](#) keyword, that is one table with a maximum of 20 rows per table using the standard one cell upstream option.

⁹⁹ M. R. Todd and W. J Longstaff, *The Development, Testing, and Application Of a Numerical Simulator for Predicting Miscible Flood Performance*. In: J. Petrol.Tech. 24.7 (1972), pages 874-882.

5.3.83 MONITOR – ACTIVATE OUTPUT OF THE MONITORING DATA AND FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *MONITOR* keyword activates the writing out of the run time monitoring information used by post-processing graphics software to display run time information, for example the simulated production and injection rates and cumulative values. OPM Flow does not have this functionality.

Hence, OPM Flow ignores this keyword but it is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE MONITORING OUTPUT DATA AND FILES
--
MONITOR
```

The above example switches on the output required for run time monitoring required by post-processing graphics software to review the simulation results in real time as the run progresses; however, this has no effect in OPM Flow input decks.

5.3.84 MSGFILE – ACTIVATE OR DEACTIVATE MESSAGE FILE OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MSGFILE keyword activates or deactivates the message file output used by pre- and post-processing software. Note that message file processing is not available in OPM Flow.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
I	MSGOPT	A positive integer set to 0 for to deactivate message file output or 1 to activate message file output.	1
Notes: 1) The keyword is terminated by a “/”.			

Table 5.22: MSGFILE Keyword Description

Example

```
--
--      OUTPUT
--      OPTN
MSGFILE      0
/
```

The above example deactivates the message file output, but the keyword is ignored by OPM Flow.

5.3.85 MULTIN – ACTIVATE THE NON-UNIFIED MULTIPLE INPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the Multiple Input Files option for all input files. Similar to the commercial simulator, OPM Flow has various options for reading various input files and writing the resulting OPM Flow output files as described in Table 5.23.

Process	Keyword	Description	Files
Input	<i>FMTIN</i>	A character string that defines the input files to be formatted as ASCII i.e. text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that used as input, for for example when restarting from another case. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTIN</i>	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I
	<i>UNIFIN</i>	A character string that defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	<i>FMTOUT</i>	A character string that sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.PRT, *.LOG and *.DBG files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that binary files are outputted. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTOUT</i>	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I

Process	Keyword	Description	Files
	UNIFOUT	A character string that defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. Their main advantage is that if a number of simulations reside in one directory, their output is organized. There is no limit on the number of reporting steps that a unified file can store.			

Table 5.23: [MULTIN](#) Keyword Description

There is no data required for this keyword.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--      ACTIVATE THE MULTIPLE INPUT FILES OPTION
--
MULTIN
```

The above example switches on the multiple input file option.

5.3.86 MULTOUT – ACTIVATE THE NON-UNIFIED MULTIPLE OUTPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the Multiple Output Files option for all output files. Similar to the commercial simulator, OPM Flow has various options for reading various input files and writing the resulting OPM Flow output files as described in Table 5.24.

Process	Keyword	Description	Files
Input	FMTIN	A character string that defines the input files to be formatted as ASCII i.e. text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that used as input, for example when restarting from another case. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	MULTIN	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I
	UNIFIN	A character string that defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	FMTOUT	A character string that sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.PRT, *.LOG and *.DBG files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that binary files are outputted. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	MULTOUT	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I

Process	Keyword	Description	Files
	UNIFOUT	A character string that defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. If the keyword is omitted then the default is for one file per reporting time step.	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. Their main advantage is that if a number of simulations reside in one directory, their output is organized. There is no limit on the number of reporting steps that a unified file can store.			

Table 5.24: MULTOUT Keyword Description

There is no data required for this keyword.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--      ACTIVATE THE MULTIPLE OUTPUT FILES OPTION
--
MULTOUT
```

The above example switches on the multiple output file option.

5.3.87 MULTOUTS – ACTIVATE NON-UNIFIED MULTIPLE SUMMARY OUTPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the Multiple Output Files option for [SUMMARY](#) files only, and overwrites the [UNIFOUT](#) keyword in the [RUNSPEC](#) section that activates the Unified Output Files option for all output files.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--      ACTIVATE MULTIPLE OUTPUT SUMMARY FILES ONLY OPTION
--
MULTOUTS
```

The above example switches on the multiple output file option.

5.3.88 MULTREAL – ACTIVATE COMMERCIAL SIMULATOR’S MULTI-REALIZATION LICENSE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *MULTREAL* keyword activates the commercial simulator’s Multi-Realization License option.

OPM Flow is an open source project and therefore there is no license management of the various implemented options; hence this keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.89 NETWORK – ACTIVATE THE EXTENDED NETWORK OPTION AND DEFINE PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the Extended Network option and defines the maximum number of nodes and links (branches) in the network. The Extended Network option is a different facility to the Standard Network facility, as such, this keyword should only be used if the former network is required for the run.

No.	Name	Description	Default
1	NODMAX	NODMAX is a positive integer that defines the maximum number of nodes in the Extended Network model.	None
2	NBRMAX	NBRMAX is a positive integer that defines the maximum number of links in the Extended Network model.	None
3	NBCMAX	NBCMAX is a positive integer that defines the maximum number of branches that can be connected to a node in the Extended Network model, used in the commercial compositional simulator. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of 20.	20
Notes: 1) The keyword is terminated by a “/”.			

Table 5.25: NETWORK Keyword Description

Example

```
--
--      ACTIVATE THE EXTENDED NETWORK OPTION AND DEFINE PARAMETERS
--
--      MAX.      MAX      NOT
--      NODE      LINK      USED
NETWORK      10      12      1*
/
```

In the above example the maximum number of nodes is set equal to ten and the maximum number of links (or branches) is set equal to 12, for the Extended Network option.

5.3.90 NINEPOIN – ACTIVATE THE NINE-POINT DISCRETIZATION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [NINEPOIN](#) keyword activates the Nine-Point Discretization formulation for the whole grid. If the keyword is absent from the run then the conventional standard five-point discretization formulation is used for the model. The nine-point scheme is based on adding additional non-neighbor connections between the diagonal neighbors in the areal plane, in order to reduce grid orientation effects¹⁰⁰.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

In none Local Grid Refinement runs the [NINENUM](#) keyword in the [GRID](#) section may be use to optionally set parts of the grid to use nine-point discretization and the remaining regions to use the conventional standard five-point discretization formulation.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE NINE-POINT DISCRETIZATION OPTION
--
NINEPOIN
```

The above example switches on the Nine-Point Discretization option for the whole grid.

¹⁰⁰
Yanosik, J. L. and McCracken, T. A. “A Nine-Point, Finite-Difference Reservoir Simulator for Realistic Prediction of Adverse Mobility Ratio Displacements,” paper SPE 5734, Society of Petroleum Engineers Journal (1979) 19, No. 4, 253-262.

5.3.91 NMATRIX – ACTIVATE THE DISCRETIZED MATRIX DUAL POROSITY OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [NMATRIX](#) keyword activates the Discretized Matrix Dual Porosity option and specifies the number of sub-grid blocks in the actual matrix grid blocks. See also the NMATOPS keyword in the [GRID](#) section that defines various parameters for this option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	NMATRIX	A positive integer value that specifies the number of sub-grid blocks in the actual matrix grid blocks.	I
Notes: I) The keyword is terminated by a “/”.			

Table 5.26: NMATRIX Keyword Description

Note the keyword cannot be used in conjunction with the [TRPLPORO](#) keyword, which is also in the [RUNSPEC](#) section.

Example

```
--
--      SUB-GRIDS
--      NMATRIX
NMATRIX      4
/
```

The above example activates the Discretized Matrix Dual Porosity option and specifies the number of sub-grid blocks in the actual matrix grid block to be four.

5.3.92 NNEWTF – ACTIVATE THE NON-NEWTONIAN FLUID MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the Non-Newtonian Fluid phase and model for when the polymer phase is present in the model, as indicated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	NTHRBL	A positive integer that defines the maximum number of Herschel-Bulkley versus polymer concentration tables to be used with the polymer model, as entered via the FHERCHBL keyword in the PROPS section. The tables are allocated to different parts of the grid by the HBNUM keyword in the REGIONS section	NTPVT
2	NLNHBL	A positive integer that defines the maximum number of rows for each table entered by the FHERCHBL keyword in the PROPS section.	2
Notes: 1) The exact number of NTHRBL tables are required to completed the data set. For example, if NTHRBL is set equal to three, then there must be three tables entered for the FHERCHBL keyword. 2) The keyword is terminated by a “/”.			

Table 5.27: NNEWTF Keyword Description

Example

```

--
--      MAX      MAX
--      NTHRBL  NLNHBL
NNEWTF
      3      5
/

```

The above example defines maximum number of Herschel-Bulkley tables to be three with a maximum number of rows for each table set to five.

5.3.93 NOCASC – ACTIVATE LINEAR SOLVER TRACER ALGORITHM

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

NOCASC keyword activates the linear solver tracer algorithm for single phase tracers.

OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      TRACER SOLVER OPTION
--
NOCASC
```

The above example switches on the linear solver tracer algorithm; however, this has no effect in OPM Flow input decks.

5.3.94

NODPPM

–

DEACTIVATE

FRACTURE

POROSITY-PERMEABILITY

CALCULATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [NODPPM](#) keyword deactivates the default behavior of multiplying the fracture porosity by the fracture permeability to calculate the effective fracture permeability in dual porosity and dual permeability runs. Either the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section must be declared in the input file in order to use this keyword. If the default calculation is switched off by this keyword, then the effective fracture permeability is taken to be those entered for the fracture using the [PERMX](#), [PERMY](#) and [PERMZ](#) keywords in the [GRID](#) section. If the keyword is absent from the input deck, then the entered [PERMX](#), [PERMY](#) and [PERMZ](#) arrays for the fractures are multiplied by fracture [PORO](#) array values in order to obtain the effective fracture permeability.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE FRACTURE POROSITY-PERMEABILITY CALCULATION
--
NODPPM
```

The above example switches off the default behavior of multiplying the fracture porosity by the fracture permeability to calculate the effective fracture permeability.

5.3.95 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active, but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

5.3.96 NOHYST – DEACTIVATE THE HYSTERESIS OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NOHYST* keyword deactivates the Hysteresis option and informs the simulator to ignore the *IMBNUM* array in the *REGIONS* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE THE HYSTERESIS OPTION
--
NOHYST
```

The above example switches off the default behavior of multiplying the fracture porosity by the fracture permeability to calculate the effective fracture permeability.

5.3.97 NOINSPEC – DEACTIVATE OUTPUT OF THE INIT INDEX FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NOINSPEC* keyword deactivates the writing out of the INIT index file (*.INSPEC). The initialization data (or static data) is written out to two files one file contains the data, *.INIT, and the second file contains an index of the data (*.INSPEC) stored in the *.INIT file. This functionality is redundant as most post-processing software require the *.INSPEC file to load the *.INIT data set.

Hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE OUTPUT OF THE INIT INDEX FILE *.INSPEC
--
NOINSPEC
```

The above example switches off the writing of the INIT index file (*.INSPEC); however, this has no effect in OPM Flow input decks.

5.3.98 NOMONITO – DEACTIVATE OUTPUT OF THE MONITORING DATA AND FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NOMONITO* keyword deactivates the writing out of the run time monitoring information used by post-processing graphics software to display run time information, for example the simulated production and injection rates and cumulative values. OPM Flow does not have this functionality.

Hence, OPM Flow ignores this keyword but it is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE MONITORING OUTPUT DATA AND FILES
--
NOMONITO
```

The above example switches off the output required for run time monitoring required by post-processing graphics software to review the simulation results in real time as the run progresses; however, this has no effect in OPM Flow input decks.

5.3.99 NONNC – DEACTIVATE NON-NEIGHBOR CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NONNC* keyword deactivates non-neighbor connections (“NNCs”) in the current run. NNCs create off-diagonal elements in the Jacobi matrix that impact the numerical efficiency of the solution algorithms, and thus if the run does not contain NNCs then there is the potential for greater computation efficiency. Unfortunately, nearly all models, except for the most simple models, generate NNCs via for example:

- 1) aquifer connections,

2) faults, and

3) manually entered NNCs, including those automatically generate by pre-processing software.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE NON-NEIGHBOR CONNECTIONS
--
NONNC
```

The above example switches off the NNCs.

5.3.100 NORSSPEC – DEACTIVATE OUTPUT OF THE RESTART INDEX FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NORSSPEC* keyword deactivates the writing out of the *RESTART* index file (*.RSSPEC). The restart data (pressure, saturations etc. through time for each active cell) are written out to two files one file contains the data, *.UNRST for example, and the second file contains an index of the data (*.RSSPEC) stored in the *.UNRST file. This functionality is redundant as most post-processing software require the *.RSSPEC file to load the *.UNRST data set.

Hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE OUTPUT OF THE RESTART INDEX FILE *.RSSPEC
--
NORSSPEC
```

The above example switches off the writing of the restart index file (*.RSSPEC); however, this has no effect in OPM Flow input decks.

5.3.101
NOSIM – ACTIVATE THE NO SIMULATION MODE FOR DATA FILE CHECKING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

NOSIM switches the mode of OPM Flow to data input checking mode. In this mode the input file is read and all messages and print instructions are sent to the respective output files. The *SCHEDULE* section is read but the simulation is not performed.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

The example below switches OPM Flow to no simulation mode for data checking of the input deck.

```
--
--      SWITCH NO SIMULATION MODE FOR DATA CHECKING COMMENT OUT TO RUN THE MODEL
--
NOSIM
```

And the next example shows how to commented out the *NOSIM* activation keyword so that the simulation will proceed.

```
--
--      SWITCH NO SIMULATION MODE FOR DATA CHECKING COMMENT OUT TO RUN THE MODEL
--
-- NOSIM
```

Note

Simulation input decks are complex and are therefore prone to typing errors, thus before submitting a run that will take over 15 minutes or so, it is a good idea to run the model with the *NOSIM* option. If no errors are found then the *NOSIM* keyword should be commented out by placing “--” before the keyword, and then re-running the model.

Alternatively, one could use OPMRUN to run all the jobs in the queue in *NOSIM* mode and have the software re-run jobs in simulation mode if there are no errors.

5.3.102 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

5.3.103 NRSOUT – DEFINED MAXIMUM NUMBER OF RESTART ELEMENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NRSOUT* keyword specifies the maximum number of elements that can be written to the *RESTART* file at each reporting time step.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	NRSOUT	A positive integer value that specifies the maximum number of elements that can be written to the <i>RESTART</i> file at each reporting time step.	3600
Notes: I) The keyword is terminated by a “/”.			

Table 5.28: NRSOUT Keyword Description

Example

```
--
--      MAX
--      NRSOUT
NRSOUT      6000      /
```

The above example sets the maximum number of elements that can be written to the *RESTART* file at each reporting time step to 6000.

5.3.104 NSTACK – DEFINE THE STACK LENGTH FOR THE ITERATIVE LINEAR SOLVER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **NSTACK** keyword defines the maximum number of previous search directions stored by the linear solver. Increasing the value of **NSTACK** may improve the efficiency of the solver on difficult problems, but will increase the memory requirements of the simulator. The default value of 10 should be sufficient for most problems; however, if OPM Flow is having issues with the convergence of the linear questions then increasing **NSTACK** and the value of **LITMAX** on the **TUNING** keyword may improve performance.

OPM Flow uses a different numerical scheme which makes this keyword redundant; see section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
I	NSTACK	A positive integer that defines the maximum number of previous search directions stored by the linear solver.	10

Notes:

- 1) The value of **NSTACK** and the value of **LITMAX** on the **TUNING** keyword are related such that **NSTACK** should always be less than or equal to **LITMAX**.
- 2) The keyword is terminated by a "/".

Table 5.29: NSTACK Keyword Description

Example

```
--
--          SET STACK SIZE FOR LINEAR SOLVER
--
NSTACK
      30
/
```

The above example sets maximum number of previous search directions stored by the linear solver to 30, this has no effect in OPM Flow input decks.

Note

If the run is suffering from linear convergence problems, then check the data first for any data issues before manipulating the numerical control parameters. For example, if OPM Flow has written some **WARNING** messages with respect to end-point scaling, etc., then resolve these messages first before adjusting the numerical controls.

5.3.105 NUMRES – DEFINE THE NUMBER OF RESERVOIR GRIDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NUMRES* keyword defines the number of reservoir grids (*COORD* data sets) that the simulator should process. OPM Flow currently only supports a single reservoir grid.

No.	Name	Description	Default
I	NUMRES	A positive integer that defines the number of <i>COORD</i> data sets to be processed by OPM Flow. OPM Flow currently only supports a single reservoir grid and so this item should be defaulted (I*) or set to one.	I
Notes: I) The keyword is terminated by a “/”.			

Table 5.30: NUMRES Keyword Description

Example

```
--
--      DEFINE THE NUMBER OF RESERVOIR GRIDS (COORD DATA SETS)
--
NUMRES      1                                     /
```

The above example sets the maximum number of *COORD* data sets to be processed to one.

5.3.106 NUPCOL – DEFINE THE MAXIMUM NUMBER OF NEWTON ITERATIONS USED TO UPDATE WELL TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [NUPCOL](#) keyword defines the maximum number of Newton iterations within a time step that may be used to update the well production and injection targets, after which the well targets will be frozen until the time step calculations have converged and the time step is complete.

Wells under group control may suffer from some dependency with other wells in the same group that are under group control. This may cause some oscillation in the production and injection well rates within the group. In order to avoid this, after the number Newton iterations within a time step surpasses [NUPCOL](#), the group well rates are frozen until the time step has converged. Reducing the potential for well rate oscillations within the time step may result in the group targets and limits not being exactly met in this case. Increasing the value of [NUPCOL](#) to greater than the default value of 12, will improve the accuracy of the group targets and limits at the expense of computational efficiency.

See also section [2.2 Running OPM Flow From The Command Line](#) on how to set various other numerical control parameters for OPM Flow.

No.	Name	Description	Default
I	NUPCOL	A positive integer that defines the maximum number of Newton iterations used to update well targets within a time step. Note that default value of 12 is different to the commercial black-oil simulator's default value of three.	12
Notes: I) The keyword is terminated by a "/".			

Table 5.31: NUPCOL Keyword Description

See also the [MINNPCOL](#) keyword in the [RUNSPEC](#) section that sets the minimum number of Newton iterations within a time step that may be used to update the well production and injection targets, after which the well targets will be frozen until the time step calculations have converged and the time step is complete. Note that [MINNPCOL](#) is an OPM Flow specific keyword.

Example

```
--
--      DEFINE THE MAX NUMBER OF ITERATIONS TO UPDATE WELL FLOW TARGETS
--
NUPCOL
      12
/
```

The above example sets the [NUPCOL](#) value to its default value of 12.

5.3.107 OIL – ACTIVATE THE OIL PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that the oil phase is present in the model and must be used for oil-gas, oil-water, oil-water-gas input decks that contain the oil phase. The keyword will also invoke data input file checking to ensure that all the required oil phase input parameters are defined in the input deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      OIL PHASE IS PRESENT IN THE RUN
--
OIL
```

The above example declares that the oil phase is active in the model.

5.3.108 OPTIONS – ACTIVATE VARIOUS PROGRAM OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *OPTIONS* keyword activates various program options in the commercial simulator. Currently, none of the options available in the commercial simulator are implemented in OPM Flow, and it is unlikely that this keyword will be supported in the future releases of OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Description	Default
I - 273	Commercial simulator options keyword, used to to switch on or off specific features. The keyword is commonly used to revert the simulator's behavior to past functionality that has been depreciated in the current version of the commercial simulator, for backward compatibility with previous models.	0
Notes: I) The keyword is terminated by a "/".		

Table 5.32: OPTIONS Keyword Description

Examples

```
--
--      SKIP      ACTIVATE
--      OPTIONS   OPTION
OPTIONS
      77*0      1
/
```

The above example activates the use of scratch files for pre-processing grid geometry data for non-neighbor connections. Note if multiple options are required then one can just repeat the format of the example to activate multiple options as the keyword does not overwrite previous entries. So for example:

```
--
--      SKIP      ACTIVATE
--      OPTIONS   OPTION
OPTIONS
      7*0      1
/
--
--      SKIP      ACTIVATE
--      OPTIONS   OPTION
OPTIONS
      77*0      1
/
--
--      SKIP      ACTIVATE
--      OPTIONS   OPTION
OPTIONS
      177*0     1
/
```

Could be used to activate the 8, 78 and 178 options if they were available.

5.3.109 PARALLEL – DEFINE PARALLEL RUN CONFIGURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PARALLEL](#) keyword defines the run to use parallel processing and sets the domain decomposition options. See section [2.2 Running OPM Flow From The Command Line](#) on how to run OPM Flow in parallel mode.

No.	Name	Description	Default
1	NPROCS	A positive integer that defines the number of domains or parallel processors to use for this run.	1
2	RTYPE	A character string set to either SERIAL to run the parallel code in serial mode for testing the code, or DISTRIBUTED to full utilize parallel processing.	DISTRIBUTED
Notes: 1) The keyword is terminated by a "/".			

Table 5.33: PARALLEL Keyword Description

OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness.

Example

```
--
--          PARALLEL MULTI-CORE OPTIONS
--          NDMAIN      MACHINE TYPE
PARALLEL
          2          DISTRIBUTED          /
```

The above example sets the number of domains (or processors) to two and for the simulation to run in parallel mode. This has no effect in OPM Flow input decks.

5.3.110 PARTTRAC – ACTIVATE AND DEFINE PARTITIONED TRACER OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *PARTTRAC* keyword activates the Partitioned Tracer option and defines the maximum number of partitioned tracers, the number of *TRACERKP* or *TRACERKM* partitioning tables in the *PROPS* section, and the maximum number of number of rows in the *TRACERKP* or *TRACERKM* partitioning tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.111 PATHS – DEFINE FILENAME DIRECTORY PATH ALIASES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

PATHS allows the user to define alias directory filenames to avoid long filenames with the **INCLUDE**, **IMPORT**, **RESTART** or **GDFILE** keywords. To use the alias with the aforementioned keywords **PATHS** should be prefixed with the \$ symbol.

No.	Name	Description	Default
1	ALIAS	A character string enclosed in quotes defining the alias.	None
2	DIRC	A character string enclosed in quotes defining the directory filename.	None
Notes: 1) Multiple entries must be on separate lines (see the example). 2) The keyword is terminated by a “/”.			

Table 5.34: PATHS Keyword Description

Examples

```
--
--      PATH      PATH
--      ALIAS     DIRECTORY FILENAME
PATHS
      'GRID'      '/DISK1/NORNE/2017/GRID-INCLUDES' /
      'SCHD'      '/DISK1/NORNE/2017/SCHD-INCLUDES' /
/
```

The above example defines “GRID” and “SCHD” aliases in the **RUNSPEC** section than can be used in the **GRID** and **SCHEDULE** sections of the input deck. The next example shows how to use the “GRID” alias with the **INCLUDE** keyword in the **GRID** section.

```
--
--      LOAD INCLUDE FILES
--
INCLUDE      '$GRID/PORO.INC' /

INCLUDE      '$GRID/PERMX.INC' /

INCLUDE      '$GRID/NTG.INC' /
```

Here the porosity, permeability and net-to-gross arrays are loaded in the **GRID** section using the directory filename aliases declared in the **RUNSPEC** section.

5.3.112 PEDIMS – DEFINE PETRO-ELASTIC MODEL REGIONS AND TABLE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PEDIMS](#) keyword defines the number of petro-elastic regions to be used with the [PENUM](#) keyword in the [REGIONS](#) section and the number of rows in the PEGTAB0 to PEGTAB7 keywords, as well as the number of rows in the PEKTAB0 to PEKTAB7 keywords in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.113 PETOPTS – DEFINE PETREL AND GENERIC SIMULATION FILE OPTIONS

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

The *PETOPTS* keyword defines various Petrel and Generic Simulation (*.GSG) file options.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.114 PIMTDIMS – DEFINE WELL PRODUCTIVITY SCALING TABLE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PIMTDIMS keyword defines the maximum number of *PIMULTAB* tables and the maximum number of entries (or rows) per *PIMULTAB* table. The *PIMULTAB* keyword is used to define a well’s productivity index factor as a function of a well’s producing water cut.

No.	Name	Description	Default
1	NTPIMT	A positive integer value that defines the maximum number of <i>PIMULTAB</i> keywords defined in the input deck.	0
2	NRPIMT	A positive integer value defining the maximum number of entries (rows) in the <i>PIMULTAB</i> keyword.	0
Notes: 1) The keyword is terminated by a “/”.			

Table 5.35: PIMTDIMS Keyword Description

Example

```
--
--      MAX      MAX
--      TABLES  ENTRIES
PIMTDIMS
      1          51
/
```

The above example defines that there is one *PIMULTAB* table with a maximum number of 51 rows.

5.3.115 PINTDIMS – DEFINE POLYMER MOLECULAR WEIGHT MODEL TABLE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PINTDIMS](#) keyword defines the number of property tables used in the OPM Flow's Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure. This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated. The [PINTDIMS](#) keyword defines the maximum number of tables for the [SKPRWAT](#), [SKPRPOLY](#), and [PLYMWINJ](#) keywords, and the number of entries in the [PLYVMH](#) keyword. All the aforementioned keywords are in the [PROPS](#) section.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

No.	Name	Description	Default
1	NTSKWAT	NTSKWAT is a positive integer that defines the number of SKPRWAT tables in the PROPS section, used to describe the relationship of wellbore skin pressure as a function of water throughput and water velocity, for the simulator's Polymer Molecular Weight Transport option.	1
2	NTSKPOLY	NTSKPOLY is a positive integer that defines the number of SKPRPOLY tables in the PROPS section, used to describe the relationship of wellbore skin pressure as a function of polymer throughput and polymer velocity, for the simulator's Polymer Molecular Weight Transport option.	1
3	NTPMWINJ	NTPMWINJ is a positive integer that defines the number of PLYMWINJ tables in the PROPS section, used to describe the relationship of the injected polymer molecular weight as a function of polymer throughput and polymer velocity, for the simulator's Polymer Molecular Weight Transport option.	1
4	NPLYVMH	NPLYVMH is a positive integer that defines the maximum number of entries (rows) in the PLYVMH table in the PROPS section, used to describe the relationship of the injected polymer viscosity as a function of polymer molecular weight and polymer concentration, for the simulator's Polymer Molecular Weight Transport option.	1
Notes: 1) The keyword is terminated by a "/".			

Table 5.36: PINTDIMS Keyword Description

The [SKPRWAT](#), [SKPRPOLY](#), [PLYMWINJ](#), and [PLYVMH](#) keywords in the [PROPS](#) section, are the additional keywords required for the Polymer Molecular Weight Transport option. Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

```
--
--      POLYMER MOLECULAR WEIGHT TRANSPORT TABLES (OPM FLOW RUNSPEC KEYWORD)
--
--      NO.      NO.      NO.      NO.
--      NTSKWAT  NTSKPOLY  NTPMWINJ  NPLYVMH
PINTDIMS
      2          2          2          1          /
```

The above example declares two [SKPRWAT](#), [SKPRPOLY](#), and [PLYMWINJ](#) keywords in the [PROPS](#) section will be used, as well as the default value of one for the number of rows in the [PLYVMH](#) keyword.

5.3.116 POLYMER – ACTIVATE THE POLYMER PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that the polymer phase is present in the model and to activate the polymer flooding model. The keyword will also invoke data input file checking to ensure that all the required polymer phase input parameters are defined in the input deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE POLYMER PHASE IN THE MODEL
--
POLYMER
```

The above example declares that the polymer phase is active in the model.

5.3.117 POLYMW – ACTIVATE THE POLYMER MOLECULAR WEIGHT TRANSPORT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates OPM Flow's Polymer Molecular Weight Transport option that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure. This keyword should only be used if the [POLYMER](#) keyword in the [RUNSPEC](#) section is also activated. The keyword will also invoke data input file checking to ensure that all the required polymer phase input parameters are defined in the input deck.

In the [PROPS](#) section the [PLYVMH](#), [PLYMWINJ](#), [SKPRWAT](#) and [SKPRPOLY](#) keywords are used to define the additional polymer and water properties for the model. And the [SPOLYMW](#) keyword in the [SOLUTION](#) section may be used to set the initial molecular weight for each grid block in the model, in order to initialize the model. In the [SCHEDULE](#) section, the [WPMITAB](#) keyword should be used to associate a water injection well with the appropriate [PLYMWINJ](#) table. Finally, the [WSKPTAB](#) keyword, also in the [SCHEDULE](#) section, may be used to define the [SKPRPOLY](#) and [SKPRWAT](#) tables associated with the skin pressure for polymer and water injection. All the aforementioned keywords are OPM Flow specific keywords, that are not available in the commercial simulator.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE THE POLYMER PHASE IN THE MODEL
--
POLYMER
--
--      ACTIVATE THE POLYMER MOLECULAR WEIGHT TRANSPORT OPTION
--
POLYMW
```

The above example declares that the polymer phase is present, and that the simulator's Polymer Molecular Weight Transport option should be used instead of the standard polymer model.

5.3.118 PRECSALT – ACTIVATE THE OPM FLOW SALT PRECIPITATION MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the OPM Flow Salt Precipitation model that accounts for salt precipitating out of the water phase when the water is being vaporized into the gas phase and the dissolved salt reaches the solubility limit as the pressure in the reservoir is being depleted (see the [VAPWAT](#) keyword in the [RUNSPEC](#) section). This facility is an extension to the standard Brine model, and as such the [BRINE](#) keyword in the [RUNSPEC](#) must also be present in the input deck. In general, if the [PRECSALT](#) keyword has been activated in the input deck then the [VAPWAT](#) keyword should also be activated. The keyword should only be used if both water and gas phases are active in the model.

Note

This is an OPM Flow specific keyword for the simulator’s Salt Precipitation model, note that this is an extension to the commercial simulator’s Brine model.

If the keyword is present in the input deck then the [SALTSOL](#) keyword in the [PROPS](#) section also needs to be present in the input deck to define the salt solubility. In addition, either the [SALTPVD](#) or [SALTP](#) keywords in the [SOLUTION](#) section should be used to define the initial salt precipitated saturation.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

The first part of the example shows the keywords for the [RUNSPEC](#) section.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
-----
-- FLUID TYPES AND TRACER OPTIONS
-----
--
--      OIL PHASE IS PRESENT IN THE RUN
--
OIL
--
--      WATER PHASE IS PRESENT IN THE RUN
--
WATER
--
--      GAS PHASE IS PRESENT IN THE RUN
--
GAS
--
--      DISSOLVED GAS IN LIVE OIL IS PRESENT IN THE RUN
--
DISGAS
```

```
--
--      ACTIVATE STANDARD BRINE MODEL IN THE RUN
--
BRINE
--
--      ACTIVATE THE OPM FLOW SALT PRECIPITATION MODEL (OPM FLOW KEYWORD)
--
PRECSALT
```

The above example declares that the oil, water, gas, dissolved gas, and brine phases are present in the model, and activates the Brine ([BRINE](#) keyword) and Salt Precipitation ([PRECSALT](#) keyword) models.

The second part of the example shows the additional PVT fluid keywords require for the Salt Precipitation model in the [PROPS](#) section. Note that in addition to the Salt Precipitation model specific keywords, the standard [DENSITY](#) (or [GRAVITY](#)), [PVDG](#), and [PVTO](#) keywords are required.

```
-- =====
--
--  PROPS SECTION
--
-- =====
PROPS
--
--      WATER SALT PVT TABLE
--
PVTWSALT
--      REF PRES      REF SALT
--      PSIA          LB/STB
--      -----
--      3000.0        0.000                                / REFERENCE DATA
--
--      SALTCONC      BW          CW          VISC          VISC
--      LB/STB         RB/STB      1/PSIA      CPOISE      GRAD
--      -----
--      0.0           1.0          3.0E-6      1.0          1.0E-6
--      10.0           0.8          3.0E-6      1.0          1.0E-6      / SALT CONCENTRATION
--
--      SET SALT SOLUBILITY LIMIT FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALTSOL
--      MAX          SALT
--      SALTSOL      DENSITY
--      0.500        135.46867445023383                      /
```

In addition, to the above the [PERMFACT](#) keyword may be used to account for the reduction in porosity and permeability due to salt precipitating into the pore space. Again, in addition to the Salt Precipitation model specific keywords, the standard [ROCK](#), [SGOF](#), and [SWOF](#) keywords are required.

```
--
-- PERMEABILITY FACTOR REDUCTION DUE TO SALT (OPM FLOW KEYWORD)
--
PERMFACT
--      PORO      PERM
--      FACTOR    FACTOR
--      -----
--      0.0       0.000
--      0.4       0.005
--      0.6       0.010
--      0.9       0.100
--      1.0       1.000
--
/                                     / TABLE NO. 01
```

In order to initialize the model in the [SOLUTION](#) section, apart from the standard equilibration keywords, the following Salt Precipitation model specific keywords should be included.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
-- PRECIPITATED SALT VOLUME FRACTION VERSUS DEPTH (OPM FLOW KEYWORD)
--
--      DEPTH      SALTPSAT
--      -----
SALTPVD
--      8300.0     0.010
--      8450.0     0.010
--
/ SALT VS DEPTH EQUIL REGN 01
--
--      DEPTH      SALT-1      SALT-2      SALT-3
--      NAME      SALTCON     SALTCON     SALTCON
--      -----
SALTV D
--      8300.0     0.500
--      8450.0     0.500
--
/ SALT VS DEPTH EQUIL REGN 01
```

Note that OPM Flow does not support Multi-Component Brine model, thus there should only one column of salt concentrations.

Finally, in the [SCHEDULE](#) section, one may optionally one can add the injections well's injection salt concentrations, as shown below.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
-- DEFINE WATER INJECTION WELL SALT CONCENTRATIONS
--
-- WELL  SALT-1      SALT-2      SALT-3      SALT-4
-- NAME  SALTCON     SALTCON     SALTCON     SALTCON
--      -----
WSALT
'INJ'   0.0000
/
```

Normally, with this option one would also include vaporized water via the [VAPWAT](#) keyword in the [RUNSPEC](#) section, as well as the [PVTGW](#) keyword to define gas PVT properties for dry gas, or the [PVTGWO](#) keyword that declares the gas PVT properties for wet gas. Both keywords are in the [PROPS](#) section. The [RWGSALT](#)

keyword in the [PROPS](#) section, may be used to define the relationship of water vaporization versus pressure and salt concentration. Again, the keyword is in the [PROPS](#) section. Secondly, in the [SOLUTION](#) section, the [RVW](#) keyword can be used to set the initial equilibration vaporized water in gas ratio values for all grid cells in the model for Enumeration Initialization, or the [RVWVD](#) keyword that declares the vaporized water-gas ratio (Rvw) versus depth tables for each equilibration region, for Equilibrium Initialization.

5.3.119 PSTEADY – ACTIVATE PSEUDO STEADY STATE FLOW CALCULATION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *PSTEADY* keyword activates Pseudo Steady State Flow Calculation option by advancing the simulator until it reaches a pseudo steady state flow and then sets the date to the date defined on this keyword, that is written to the *RESTART* file. Keyword also includes parameters defining the conditions for pseudo steady flow state.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.120 RADIAL – RADIAL GRID ACTIVATION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

RADIAL activates the radial grid geometry option for the model, if this keyword and the *SPIDER* keyword are omitted then Cartesian geometry is assumed by OPM Flow.

See also the *SPIDER* keyword in the *RUNSPEC* section (*SPIDER – Spider Grid Activation Option*). This is an OPM Flow specific keyword for the simulator’s Spider Grid option, that emulates a Radial Grid via corner-point geometry. Both Spider and Radial grids will be displayed the same in OPM ResInsight, as they both use Irregular Corner-Point Grid geometry. The difference is that the radial model’s pore volumes are adjusted to match the radial geometry pore volumes, whereas, the Spider grid volumes are not adjusted. See the examples in the section on Radial Grids (*Radial Grids*) and Spider Grids (*Spider Grids*) for further information.

Example

```
--
--      DEFINE RADIAL GRID GEOMETRY
--
RADIAL
```

The example activates the radial grid geometry option.

5.3.121 REGDIMS – DEFINE THE MAXIMUM NUMBER OF REGIONS FOR A REGION ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [REGDIMS](#) keyword defines the maximum number of regions for various region arrays used in the model. Note that the maximum number of [FIPNUM](#) regions can be defined both on this keyword and the [TABDIMS](#) keyword, if it set in both locations the maximum value is used. The reason for this type of inconsistency is due to the commercial simulator evolving with time as new features were added, but at the same time having to maintain backward input deck compatibility.

No.	Name	Description	Default
1	NTFIP	A positive integer defining the maximum number of regions in the FIPNUM region array. If additional sets of fluid in-place regions have been defined, as per the FIPxxxxx series of fluid in-place region keywords, then NTFIP should be set to the maximum number of regions in either the FIPNUM or FIPxxxxx associated arrays. Thus, if the maximum number of regions in the FIPNUM array is 12, and the maximum value in the FIPxxxxx series of arrays is 20, then 20 should be entered for NTFIP. Note that this parameter may also be set on the TABDIMS keyword as well. If NTFIP is set in both places, then the maximum value is used.	1
2	NMFIPR	A positive integer defining the total maximum number of fluid in-place regions. The number of FIPNUM regions are defined by NTFIP. However, if additional sets of fluid in-place regions are required, as per the FIPxxxxx series of fluid in-place region keywords, then these are to be defined here by adding the number of FIPxxxxx arrays to the value NTFIP. So for example, if NTFIP equals five and the number of distinct FIPxxxxx regions is three, then the value to enter for NMFIPR is eight.	1
3	NRFREG	A positive integer defining the maximum number of independent reservoir regions in the ISOLNUM region array.	0
4	MXNFLX	A positive integer defining the maximum number of flux regions in the FLUXNUM region array. MXNFLX can also be defined on the TABDIMS keywords as well. If MXNFLX is defined both here and on the TABDIMS keyword then the maximum value of the two is used.	0
5	NUSREG	A positive integer defining the maximum user defined regions in a commercial simulator's compositional model. This parameter is included for compatibility and should be defaulted as it is not used in OPM Flow.	0
6	NTCREG	A positive integer defining the maximum number of regions in the COALNUM region array. This parameter is included for compatibility and should be defaulted as it is not used in OPM Flow.	1
7	NOPREG	A positive integer defining the maximum number of regions in the OPERNUM region array.	0

No.	Name	Description	Default
8	NWKDREG	A positive integer defining the maximum number of real double-precision work arrays for use with the OPERATE and OPERATER keywords. This parameter is included for compatibility and should be defaulted as it is not used in OPM Flow.	0
9	NWKIREG	A positive integer defining the maximum number of integer work arrays for use with the OPERATE and OPERATER keywords. This parameter is included for compatibility and should be defaulted as it is not used in OPM Flow.	0
10	NPLMIX	A positive integer defining the maximum number of regions in the PLMIXNUM region array.	1
Notes: 1) The keyword is terminated by a “/”.			

Table 5.37: REGDIMS Keyword Description

Example

```
--
--      MAX      TOTAL  INDEP  FLUX    TRACK  CBM    OPERN  WORK  WORK  POLY
--      FIPNUM  REGNS  REGNS  REGNS  REGNS  REGNS  REGNS  REAL  INTG  REGNS
REGDIMS
      9        12      1*      1*      1*      1*      1*      1*      1*      1*  /
```

The above example defines the number of [FIPNUM](#) regions to be nine and the number of FIPxxx type of regions to be three (12 – 9), the rest of the region sizes are set to the default values.

5.3.122 RIVRDIMS – DEFINE THE RIVER DIMENSIONS AND ASSOCIATED DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

RIVRDIMS defines the river system array dimensions used with the *REACHES* keyword and other river keywords in the *SOLUTION* and *SCHEDULE* sections. The keyword also enables the River option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.123 ROCKCOMP – ACTIVATE ROCK COMPACTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ROCKCOMP](#) keyword activates rock compaction and defines various rock compaction options for the run. By default OPM Flow models rock compaction via pore volume compressibility as entered on the [ROCK](#) keyword in the [PROPS](#) section. This keyword enables pressure dependent pore volume and transmissibility multipliers for rock compaction that are entered in the [PROPS](#) section using the [ROCKTAB](#) keyword. Currently OPM Flow only supports the default options for rock compaction.

No.	Name	Description	Default
1	ROCKOPT	A character string that defines the rock compaction option based on one of the following character strings: 1) REVERS: Rock compaction is reversible with increasing pressure. The rock compaction multipliers should be entered via the ROCKTAB keyword in the PROPS section. This is the default value. 2) IRREVERS: Rock compaction is irreversible, that is the rock expansion does not occur when the pressure subsequently increases. 3) HYSTER: Invokes the hysteresis rock compaction option. 4) BOBERG: Rock compaction hysteresis is modeled using the Boberg formulation¹⁰¹. 5) REVLIMIT: Activates the reversible hysteresis rock compaction option that limits the pore volume subject to reversibility based on the minimum pressure in a grid block and the initial water saturation. This option is only intended to be used with the water induced compaction model, neither of which are currently supported by OPM Flow.. 6) PALM-MAN: Rock compaction hysteresis is modeled using the Palmer-Mansoori¹⁰² and ¹⁰³ formulation for coal bed methane reservoirs, neither of which are supported by OPM Flow. 7) NONE: Deactivates rock compaction, unless the water induced compaction model has been invoked. Only the REVERS and IRREVERS options are supported by OPM Flow.	REVERS
2	NTROCC	A positive integer that defines the number of rock compaction tables, that is the number of ROCKTAB tables to be used by OPM Flow.	1
3	WATINOPT	A character string that states if the water induced rock compaction option should be used (YES) or not (NO). If set to YES then either the ROCKTABW or the ROCK2D and ROCKWNOD keywords should be entered in the PROPS section.	NO

¹⁰¹ Beattie, C.I., Boberg, T.C., and McNab, G.S. "Reservoir Simulation of Cyclic Steam Stimulation in the Cold Lake Oil Sands," paper SPE 18752, Society of Petroleum Engineers Journal, (1991) 6, No. 2, 200-206.

¹⁰² Palmer, I. and Mansoori, J. "How Permeability Depends on Stress and Pore Pressure in Coalbeds: A New Model," paper SPE 52607, SPE Reservoir Evaluation & Engineering (1998) 1, No. 6, 539-544.

¹⁰³ Clarkson, C.R., Pan, Z., Palmer, I. and Harpalani, S. "Predicting Sorption-Induced Strain and Permeability Increase With Depletion for Coalbed-Methane Reservoirs", SPE 114778-PA, SPE Journal (2010) 15, No. 1, 152–159.

No.	Name	Description	Default
4	PORTXROP	A character string that specifies the model to be used for when transmissibility is dependent on porosity, and should be set to either: 1) EXP: An exponential porosity-transmissibility relationship should be used. 2) CZ: The Carmen-Kozeny¹⁰⁴ ¹⁰⁵ and ¹⁰⁶ porosity-transmissibility relationship should be used. This option is used in the commercial compositional simulator and is therefore ignored by OPM Flow.	1*
5	CARKZEXP	The exponent constant in the Carmen-Kozeny porosity-transmissibility equation for when PORTXROP has been set to CZ. This option is used in the commercial compositional simulator and is therefore ignored by OPM Flow.	0.0
Notes: 1) The keyword is terminated by a “/”.			

Table 5.38: ROCKCOMP Keyword Description

Example

```
--
--      ROCK   NUMBER   WAT      POR-TRAN
--      OPTN   TABLES  INDUCE   OPTION
ROCKCOMP
      REVERS 5          NO        1*                               /
```

The above example defines the default values for the ROCKCOMP keyword with five rock compaction tables.

¹⁰⁴ J. Kozeny, "Ueber kapillare Leitung des Wassers im Boden." *Sitzungsber Akad.Wiss.,Wien*, 136(2a): 271-306, 1927.

¹⁰⁵ P.C. Carman, "Fluid flow through granular beds." *Transactions, Institution of Chemical Engineers, London*, 15: 150-166, 1937.

¹⁰⁶ P.C. Carman, "Flow of gases through porous media." *Butterworths, London*, 1956

5.3.124 RPTCPL – ACTIVATE COUPLE SIMULATION REPORTING

FRUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the couple simulation reporting, that results in the simulator writing out various initialization data and simulation data in order for external “controlling programs” to interactively manage the simulation. There is no data required for this keyword but the keyword should be terminated by a “/”.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      ACTIVATE COUPLE SIMULATION REPORTING
--
RPTCPL
/
```

The above example switches on couple simulation reporting; however, this has no effect in OPM Flow input decks.

5.3.125 RPTHMD – DEFINE WELL HISTORY MATCH GRADIENT REPORTING OPTIONS

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

This keyword, *RPTHMD*, defines the options and level of history match output that should be written to history match file (*.HMD), for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

5.3.126 RPTRUNSP – ACTIVATE RUNSPEC REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates reporting of all the [RUNSPEC](#) options utilized in the run. There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      ACTIVATE RUNSPEC SECTION REPORTING
--
RPTRUNSP
```

The above example switches on [RUNSPEC](#) reporting; however, this has no effect in OPM Flow input decks.

5.3.127 RSSPEC – ACTIVATE OUTPUT OF THE RESTART INDEX FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [RSSPEC](#) keyword activates the writing out of the [RESTART](#) index file (*.RSSPEC). The restart data (pressure, saturations etc. through time for each active cell) are written out to two files one file contains the data, *.UNRST for example, and the second file contains an index of the data (*.RSSPEC) stored in the *.UNRST file. This keyword is somewhat redundant as the [RESTART](#) index file is written out by default. See the [NORSSPEC](#) keyword in the [RUNSPEC](#) section that deactivates the writing out of the file.

Hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE OUTPUT OF THE RESTART INDEX FILE *.RSSPEC
--
RSSPEC
```

The above example switches on the writing of the restart index file (*.RSSPEC); however, this has no effect in OPM Flow input decks.

5.3.128 RUNSPEC – DEFINE THE START OF THE RUNSPEC SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *RUNSPEC* activation keyword marks the start of the *RUNSPEC* section that defines the key parameters for the simulator including the dimensions of the model, phases present in the model (oil, gas and water for example), number of tables for a given property and the maximum number of rows for each table, the maximum number of groups, wells and well completions, as well as various options to be invoked by OPM Flow.

Apart from COMMENTS entered by “--” in columns one and two, this keyword should be the first keyword in the input deck.

There is no data required for this keyword and there is no keyword terminating “/”

Example

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
```

The above example marks the start of the *RUNSPEC* section in the OPM Flow data input file.

5.3.129 SAMG – ACTIVATE ALGEBRAIC MULTI-GRID LINEAR SOLVER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword actives the algebraic multi-grid linear solve; note this solver is not available to the general public in the commercial simulator.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

5.3.130 SATOPTS – ACTIVATE RELATIVE PERMEABILITY ASSIGNMENT OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **SATOPTS** keyword activates OPM Flow's relative permeability assignment options. The relative permeability functions are defined using the either the:

- 1) **SWOF**, **SGOF**, **SLGOF** series of saturation functions, or the
- 2) **SWFN**, **SGFN**, **SGWFN**, **SOF2**, **SOF3**, **SOF32D** series of functions.

The allocation of the relative permeability tables to the grid cells is dependent on the options selected on this keyword (**SATOPTS**).

If the **DIRECT** option has been activated and the **IRREVERS** option has not been invoked on the **SATOPTS** keyword, then different relative permeability functions are used for the x, y, and z directions. Here the **KRNUMX**, **KRNUMY** and **KRNUMZ** keywords are used for Cartesian grids to allocate the relative permeability tables to the cells. For Radial grids the **KRNUMR**, **KRNUMT** and **KRNUMZ** keywords should be used. This results in the same relative permeability curves being used in both the x_i to $x_i - 1$ and the x_i to $x_i + 1$ flow directions. Similarly for the y direction the same curves are used for the y_i to $y_i - 1$ and the y_i to $y_i + 1$ flow directions. And again for the z direction, the same relative permeability function is used for flow in the z_i to $z_i - 1$ and the z_i to $z_i + 1$ flow directions.

If the **DIRECT** option has been activated and the **IRREVERS** option has also been invoked on the **SATOPTS** keyword, then **KRNUMX**, **KRNUMY** and **KRNUMZ** keywords are used for Cartesian grids to allocate the relative permeability tables in the x_i to $x_i + 1$, y_i to $y_i + 1$, z_i to $z_i + 1$, flow directions, respectively. For Radial grids the **KRNUMR**, **KRNUMT** and **KRNUMZ** keywords should be used. For flow in the x_i to $x_i - 1$ flow directions, etc., the **KRNUMX-**, **KRNUMY-** and **KRNUMZ-** keywords are used for Cartesian grids and the **KRNUMR-**, **KRNUMT-** and **KRNUMZ-** are used for radial grids.

The **HYSTER** option activates relative permeability hysteresis for the non-wetting phases (liquid and vapour) such that their values depend on whether the non-wetting phase saturation is increasing or decreasing. For this option the user specifies two sets of saturation functions, one for a drainage process (decreasing wetting phase saturation) and one for an imbibition process (increasing wetting phase saturation). The **SATNUM** and **IMBNUM** keywords should then be used to identify which saturation table applies to drainage and imbibition respectively. For a process starting at the maximum wetting phase saturation, on the drainage curve, with the wetting phase saturation decreasing, the drainage curve is followed. Similarly for a process starting at the minimum wetting phase saturation with the wetting phase saturation increasing, the imbibition curve is followed. If the drainage or imbibition process is reversed at some point, then the curve does not necessarily run back over its previous values. In OPM Flow the Carlson¹⁰⁷ model may be used to describe relative permeability hysteresis and the Killough¹⁰⁸ model may be used for relative permeability or capillary pressure. See also the **EHYSTR** keyword used to define the hysteresis model that is applied.

If the **DIRECT** option has been activated and the **IRREVERS** option has not been invoked on the **SATOPTS** keyword, then the same set of keywords as for the **DIRECT** only option are used to assign the drainage relative permeability curves, that is: **KRNUMX**, **KRNUMY** and **KRNUMZ**, plus the **IMBNUMX**, **IMBNUMY**, and **IMBNUMZ**, keywords for the imbibition curves. If the **DIRECT** option has been activated and the **IRREVERS** option has also been invoked on the **SATOPTS** keyword, then the same set are keywords as for the **DIRECT** and **IRREVERS** option are used to assign the drainage relative permeability curves, that is: **KRNUMX**, **KRNUMX-**, etc., plus the **IMBNUMX**, **IMBNUMY**, **IMBNUMZ**, **IMBNUMX-**, **IMBNUMY-**, **IMBNUMZ-** keywords for the imbibition curves. See Table 5.40 for the various relative permeability table allocation keywords required for different combinations of the **DIRECT**, **IRREVERS** and **HYSTER** options.

¹⁰⁷ Carlson, F.M. (1981) SPE 10157, presented at the 56th Annual SPE Fall Meeting, San Antonio, 1981

¹⁰⁸ Killough, J. E. "Reservoir Simulation with History-dependent Saturation Functions," paper SPE 5106, Society of Petroleum Engineers Journal (1976) 16, No. 1, 37-48.

The keyword should be followed by one or more of the following keyword options.

No.	Name	Description	Default
1	DIRECT	A character string that activates the directional relative permeability assignment option. If the DIRECT option is specified then directional relative permeability assignment is activated and different relative permeability functions are assigned to the x, y and z directions. In this case the KRXUMX, KRYUMY and KRNUMZ keywords are used for Cartesian grids to allocate the relative permeability tables. For Radial grids the KRXUMR, KRNUMT and KRNUMZ keywords should be used.	None
2	IRREVERS	A character string that activates the irreversible directional relative permeability assignment option. If the IRREVERS option is specified then the relative permeability assignment is set to irreversible and results in different sets of relative permeability tables being applied for flow in the xi to xi + 1 direction and the xi to the xi – 1 direction, for all directions (x, y, z). In this case the KRXUMX, KRYUMY and KRNUMZ keywords are used for Cartesian grids to allocate the relative permeability tables in the xi to xi + 1 flow directions etc. For Radial grids the KRXUMR, KRNUMT and KRNUMZ keywords should be used. For flow in the xi to xi – 1 flow directions, etc., the KRXUMX-, KRYUMY- and KRNUMZ- keywords are used for Cartesian grids and the KRXUMR-, KRNUMT- and KRNUMZ- are used for radial grids. Only the default option is supported by OPM Flow, that is irreversible permeability tables are not supported.	None
3	HYSTER	A character string that activates the hysteresis option. If the HYSTER and DIRECT options have been activated and the IRREVERS has not been invoked on the SATOPTS keyword, then different relative permeability functions are used for the x, y, and z directions and for the drainage and imbibition processes. Here the drainage relative permeability curves are allocated via the KRXUMX, KRYUMY and KRNUMZ keywords for Cartesian grids and the KRXUMR, KRNUMT and KRNUMZ keywords for radial grids. The imbibition relative permeability curves are allocated via the IMBNUMX, IMBNUMY and IMBNUMZ keywords for Cartesian grids and the IMBNUMR, IMBNUMT and IMBNUMZ keywords for radial grids. If the HYSTER, DIRECT and IRREVERS options have been activated, then different relative permeability functions are used for the x, y, and z directions, each flow direction and for the drainage and imbibition processes. In addition to the aforementioned relative permeability allocation keywords for the xi to xi + 1 flow direction etc., the xi to xi – 1 flow directions keywords, KRXUMX-, KRYUMY- and KRNUMZ- are used for Cartesian grids and the KRXUMR-, KRNUMT- and KRNUMZ- are used for radial grids. The imbibition relative permeability curves are allocated via the IMBNUMX-, IMBNUMY- and IMBNUMZ- keywords for Cartesian grids and the IMBNUMR, IMBNUMT and IMBNUMZ- keywords for radial grids.	None

No.	Name	Description	Default
4	SURFTENS	A character string that activates the capillary pressure surface tension pressure dependency option. Only the default option is supported by OPM Flow, that is capillary pressure surface tension pressure dependency option is not supported.	None
Notes: - Note that the IRREVERS option can only be activated if the DIRECT option is activated at the same time. See Table 5.40 for the various relative permeability table allocation keywords. - The keyword is terminated by a “/”.			

Table 5.39: SATOPTS Keyword Description

For clarity the following table outlines the keywords that should be used in allocating the relative permeability tables for the various SATOPTS options.

Option	Cartesian		Radial	
DIRECT Flow in all directions	KRNUMX KRNUMY KRNUMZ		KRNUMR KRNUMT KRNUMZ	
DIRECT and IRREVERS Flow in the i to i +1 directions.	KRNUMX, KRNUMY KRNUMZ		KRNUMR KRNUMT KRNUMZ	
Flow in the i to i -1 directions.	KRNUMX- KRNUMY- KRNUMZ-		KRNUMR- KRNUMT- KRNUMZ-	
DIRECT and HYSTER Flow in all directions.	Drainage KRNUMX KRNUMY KRNUMZ	Imbibition IMBNUMX IMBNUMY IMBNUMZ	Drainage KRNUMR KRNUMT KRNUMZ	Imbibition IMBNUMR IMBNUMT IMBNUMZ
DIRECT, IRREVERS and HYSTER Flow in the i to i +1 directions.	Drainage KRNUMX KRNUMY KRNUMZ	Imbibition IMBNUMX IMBNUMY IMBNUMZ	Drainage KRNUMR KRNUMT KRNUMZ	Imbibition IMBNUMR IMBNUMT IMBNUMZ
Flow in the i to i -1 directions.	KRNUMX- KRNUMY- KRNUMZ-	IMBNUMX- IMBNUMY- IMBNUMZ-	KRNUMR- KRNUMT- KRNUMZ-	IMBNUMR- IMBNUMT- IMBNUMZ-
Notes: Note the drainage and imbibition classification is related to the wetting phase, that may be either oil or water; however, water is normally assumed in most cases but there are exceptions to this, especially for heavy oils.				

2)
Keywords colored orange indicate keywords that are not currently supported in OPM Flow.

Table 5.40: SATOPTS Relative Permeability Function Allocation Keywords

Examples

The first example activates the directional relative permeability assignment option only and hence the following keywords are used to allocate the relative permeability arrays for Cartesian grids: [KRUNUMX](#), [KRNUMY](#), and [KRNUMZ](#).

```
--
--      ACTIVATE RELATIVE PERMEABILITY ASSIGNMENT HYSTERESIS OPTIONS
--      DIRECTTIONAL(DIRECT) IRREVERSIBLE(IRREVERS) HYSTERESIS(HYSTER)
SATOPTS
      'DIRECT' /
```

The next example activates the directional and irreversible relative permeability assignment options, and hence the following keywords are used to allocate the relative permeability arrays for Cartesian grids: [KRUNUMX](#), [KRNUMY](#), [KRNUMZ](#), [KRUNUMX-](#), [KRNUMY-](#) and [KRNUMZ-](#).

```
--
--      ACTIVATE RELATIVE PERMEABILITY ASSIGNMENT HYSTERESIS OPTIONS
--      DIRECTTIONAL(DIRECT) IRREVERSIBLE(IRREVERS) HYSTERESIS(HYSTER)
SATOPTS
      'DIRECT'   'IRREVERS' /
```

Finally, the last example invokes the directional, irreversible and hysteresis assignment options.

```
--
--      ACTIVATE RELATIVE PERMEABILITY ASSIGNMENT HYSTERESIS OPTIONS
--      DIRECTTIONAL(DIRECT) IRREVERSIBLE(IRREVERS) HYSTERESIS(HYSTER)
SATOPTS
      'DIRECT'   'IRREVERS'   'HYSTER' /
```

In this case the drainage relative permeability curves are allocated by the [KRUNUMX](#), [KRNUMY](#), [KRNUMZ](#), [KRUNUMX-](#), [KRNUMY-](#), [KRNUMZ-](#) keywords, and the imbibition relative permeability curves are allocated by the [IMBNUMX](#), [IMBNUMY](#), [IMBNUMZ](#), [IMBNUMX-](#), [IMBNUMY-](#), [IMBNUMZ-](#) keywords.

Note

This keyword activates how relative permeability curves are assigned in the model. The [ENDSCALE](#) keyword allows the end-point scaling also to vary with direction, flow direction and hysteresis process, resulting in a great deal of flexibility.

Whether or not all these features should be used though is another question.

5.3.131 SAVE – ACTIVATE OUTPUT OF A SAVE FILE FOR FAST RESTARTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates output of a SAVE file for fast restarts which are read in by the [LOAD](#) keyword in the [RUNSPEC](#) section. No data is required for this keyword.

OPM Flow does not support the SAVE file format for fast restarts like the commercial simulator, but instead writes a standard [RESTART](#) file at the requested time step in the [SCHEDULE](#) section, which can then be used to restart the simulation at a given point in time via the [RESTART](#) keyword in the [RUNSPEC](#) section.

Example

The first example requests that a SAVE file be written out in the [RUNSPEC](#) section; however, OPM Flow will not write a [RESTART](#) record if the [SAVE](#) keyword is encountered in the [RUNSPEC](#) section.

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      WRITE OUT SAVE FILE FOR FAST RESTARTS
--
SAVE

```

The second example shows how the keyword is used in the [SCHEDULE](#) section.

```

-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE

-- -----
-- SCHEDULE SECTION - 2020-01-01
-- -----
RPTSCHED      'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'      /

DATES
      1  APR   2021  /
/
--
--      WRITE OUT SAVE FILE FOR FAST RESTARTS
--
SAVE

DATES
      1  JULY   2021  /
      1  OCT    2021  /
/

```

Here OPM Flow will write out a [RESTART](#) file instead of a SAVE file at April 1st, 2021.

5.3.132 SCDPDIMS – DEFINE SCALE DEPOSITION AND DAMAGE TABLE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SCDPDIMS](#) keyword defines the number of tables used in the Scale Deposition option and the maximum number of entries for the various tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	NTSCDP	NTSCDP is a positive integer that defines the number of SCDPTAB scale deposition tables used in the Scale Deposition option.	0
2	NPSCDP	NPSCDP is a positive integer that defines the maximum number of entries (or rows) in any one SCDPTAB scale deposition table defined in the input deck.	0
3	NTSCDA	NTSCDA is a positive integer that defines the number of SCDATAB scale damage tables used in the Scale Deposition option.	0
4	NPSCDA	NPSCDA is a positive integer that defines the maximum number of entries (or rows) in any one SCDATAB scale damage table defined in the input deck.	0
5	Not Used		1*
6	Not Used		1*
7	NTSCDE	NTSCDE is a positive integer that defines the number of SCDETAB karst aquifer dissolution tables used in the Scale Deposition option.	0
Notes: 1) The keyword is terminated by a "/".			

Table 5.41: SCDPDIMS Keyword Description

Example

```
--
--      NO.      MAX      NO.      MAX      NOT      NOT      NO.
--      NTSCDP   NPSCDP   NTSCDA   NPSCDA   USED     USED     NTSCDE
SCDPDIMS      5        10        4        10        1*      1*      3
                                                    /
```

The above example defines the number of [SCDPTAB](#) scale deposition tables to be five with a maximum number of rows for each table set to 10, the maximum number of [SCDATAB](#) scale damage tables to be four with a maximum number of 10 rows per table, and the maximum number of [SCDETAB](#) karst aquifer dissolution tables to be three.

5.3.133 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

5.3.134 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

5.3.135 SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

5.3.136
SMRYDIMS – DEFINE MAXIMUM NUMBER OF SUMMARY VECTORS TO BE WRITTEN

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SMRYDIMS](#) keyword defines the maximum number of summary vectors to be written out to the [SUMMARY](#) file (*.SUMMARY).

OPM Flow uses dynamic memory allocation and therefore the keyword has no effect and is ignored by the simulator, but is documented here for completeness.

No.	Name	Description	Default
I	NSUMMX	A positive integer that defines the maximum number of summary vectors to be written out to the SUMMARY file (*.SUMMARY).	10000
Notes: I) The keyword is terminated by a “/”.			

Table 5.42: SMRYDIMS Keyword Description

Example

```
--
--      SET THE MAXIMUM NUMBER OF SUMMARY VECTORS THAT CAN BE WRITTEN OUT
--
SMRYDIMS
      10000
/
```

The above example sets maximum number of summary vectors that can be written out to the [SUMMARY](#) file to the default value of 10,000; however, this has no effect in OPM Flow input decks.

5.3.137 SOLVDIMS – DEFINE PEBI GRID NESTED FACTORIZATION SOLVER DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SOLVDIMS](#) defines the unstructured Perpendicular Bisector (“PEBI”)¹⁰⁹ and ¹¹⁰ grid nested factorization solver dimensions. This keyword is generated by an external pre-processing program for generating simulation grids.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

¹⁰⁹
Heinemann, Z.E. and Brand, C.W. 1988. Gridding Techniques in Reservoir Simulation. Proc., First Intl. Forum on Reservoir Simulation, Alpbach, Austria, 339.

¹¹⁰
Heinemann, Z.E., Brand, C.W., Munka, M. et al. 1991. Modeling Reservoir Geometry With Irregular Grids. SPE Res Eng 6 (2): 225–232. SPE-18412-PA. <http://dx.doi.org/10.2118/18412-PA>

5.3.138 SOLVENT – ACTIVATE THE SOLVENT PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that the solvent phase is present in the model and to activate the four component solvent model for this run. In addition to this keyword, the oil, water and gases phases should also be declared for the run using the [OIL](#), [WATER](#) and [GAS](#) keywords. The keyword will also invoke data input file checking to ensure that all the required Solvent phase input parameters are defined in the input deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      SOLVENT PHASE IS PRESENT IN THE RUN
--
SOLVENT
```

The above example declares that the solvent phase is active in the model.

5.3.139 SPIDER – SPIDER GRID ACTIVATION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SPIDER](#) activates the OPM Flow spider grid geometry option for the model, if this keyword and the [RADIAL](#) keyword are omitted then Cartesian geometry is assumed by OPM Flow. Note that is an OPM Flow specific keyword and option.

Spider Grids basically take the standard radial geometry keywords in the grid section and transform the grid specification to an Irregular Corner-Point Grid which can then be viewed in OPM ResInsight in a more intuitive form. The only difference is that in the [RUNSPEC](#) section the [SPIDER](#) keyword is used instead of the [RADIAL](#) keyword. One drawback of the approach that only “cake-slides” can be used for Spider Grids; that is theta must be less than 360 degrees and the dimension must be one, not the full radial geometry.

See also the [RADIAL](#) keyword in the [RUNSPEC](#) section ([RADIAL – Radial Grid Activation Option](#)). Both Spider and Radial grids will be displayed the same in OPM ResInsight, as they both use Irregular Corner-Point Grid geometry. The difference is that the radial model’s pore volumes are adjusted to match the radial geometry pore volumes, whereas, the Spider grid volumes are not adjusted. See the examples in the section on Radial Grids ([Radial Grids](#)) and Spider Grids ([Spider Grids](#)) for further information

Example

```
--
--      DEFINE SPIDER GRID GEOMETRY (OPM FLOW RADIAL GRID KEYWORD)
--
SPIDER
```

The above example actives OPM Flow’s spider grid geometry option.

5.3.140 START – SIMULATION START DATE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword sets the start date for the simulation switches. If the [DATES](#) keyword is to be used during the simulation, then a start date should be entered.

No.	Name	Description	Default
1	DAY	A positive integer that defines the day of the month, the value should be greater than or equal to one and less than or equal to 31.	None
2	MONTH	Character string for the month and should be one of the following 'JAN', 'FEB', 'MAR', 'APR', 'MAY', 'JUN', 'JUL' (or 'JLY'), 'AUG', 'SEP', 'OCT', 'NOV', or 'DEC'	None
3	YEAR	A positive four digit integer value of the start year, which must be specified fully by four digits, that is 1986.	None
Notes: 1) The keyword is terminated by a "/".			

Table 5.43: START Keyword Description

Example

```
--
--      DEFINE THE START DATE FOR THE RUN
--
START      01 'JAN' 2014      /
```

The above example sets the start date for the run to be January 1, 2014.

Note

Whenever possible it is a good idea to always set the start date to be at the beginning of the year as per the example. As like most simulators, OPM Flow reports are always stated at the number of days from the start date (and sometimes at a given date). If the start date is at the beginning of the year, then calculating the actual date is relatively straight forward and simple.

5.3.141 SURFACT – ACTIVATE THE SURFACTANT PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that the surfactant phase is present in the model and to activate the surfactant flooding model. The keyword will also invoke data input file checking to ensure that all the required surfactant phase input parameters are defined in the input deck. See also the [SURFACTW](#) keyword in the [RUNSPEC](#) section that activates the surfactant phase, but with the changes to the wettability option activated as well.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE THE SURFACTANT PHASE IN THE MODEL
--
SURFACT
```

The above example declares that the surfactant phase is active in the model.

5.3.142 SURFACTW – ACTIVATE THE SURFACTANT PHASE WITH WETTABILITY CHANGES IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that the surfactant phase is present in the model and to activate the surfactant flooding mode with Changes to Wettability option activated as well. The keyword will also invoke data input file checking to ensure that all the required surfactant phase input parameters are defined in the input deck. See also the [SURFACT](#) keyword in the [RUNSPEC](#) section that activates the surfactant phase only, that is without the Changes to the Wettability option.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE THE SURFACTANT PHASE WITH WETTABILITY CHANGES IN THE MODEL
--
SURFACTW
```

The above example declares that the surfactant phase is active in the model together with the wettability changes.

5.3.143 TABDIMS – DEFINE THE NUMBER OF TABLES AND THE TABLE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [TABDIMS](#) keyword defines the maximum number of tables for a given table type dataset and the maximum number of entries for the various tables. The commercial simulator combines both the black-oil and compositional simulator variables on this keyword; however, although all the parameters are explained below only the black-oil parameters are used by OPM Flow.

No.	Name	Description	Default
1	NTSFUN	A positive integer that defines the number of relative permeability table sets defined in the input deck. The tables are allocated to different parts of the grid by the SATNUM keyword.	1
2	NTPVT	A positive integer that defines the number of fluid property table sets defined in the input deck. The tables are allocated to different parts of the grid by the PVTNUM keyword.	1
3	NSSFUN	A positive integer that defines the maximum number of saturation entries in the relative permeability tables defined in the input deck.	20
4	NPPVT	A positive integer that defines the maximum number of pressure entries in the PVT tables.	20
5	NTFIP	A positive integer defining the maximum number of regions in the FIPNUM region array. Note that this parameter may also be set on the REGDIMS keyword as well. If NTFIP is set in both places then the maximum value is used.	1
6	NRPVT	A positive integer that defines the maximum number of Rs and Rv entries in the PVT tables. If the DISGAS and VAPOIL options have not been activated then this parameter is ignored.	20
7	NRVPVT	A positive integer that defines the maximum number of Rv entries in the PVT tables for the commercial compositional simulator.	20
8	NTENDP	A positive integer that defines the maximum number of saturation end-point depth tables. The end-point depth tables are used to re-scale the saturation tables as a function of depth as oppose to being a grid block property. NTENDP may also be specified on the ENDSCALE keyword, and if specified on both here and on the ENDSCALE keyword the maximum value of the two is used.	1
9	NMEOSR	A positive integer that defines the maximum number of reservoir equations of states for the commercial compositional simulator.	1
10	NMEOSS	A positive integer that defines the maximum number of separator or surface equations of states for the commercial compositional simulator.	1
11	MXNFLX	A positive integer defining the maximum number flux regions in the FLUXNUM region array. MXNFLX can also be defined on the REGDIMS keywords as well. If MXNFLX is defined both here and on the REGDIMS keyword then the maximum value of the two is used.	10

No.	Name	Description	Default
12	MXNTHR	A positive integer that defines the maximum number of thermal regions for the commercial compositional simulator.	1
13	NTROCC	A positive integer that defines the number of rock compressibility entries enter by the ROCK keyword defined in the input deck. The ROCK data is allocated to different parts of the grid by the either the PVTNUM or ROCKNUM keywords in the REGIONS section. If NTROCC is defaulted then the PVTNUM array will be used to allocate the ROCK keyword data to the grid blocks; whereas, if a value is entered then the ROCKNUM array will be used instead. See also the ROCKOPTS keyword in the PROPS section that can be used to redefine if the PVTNUM, ROCKNUM or SATNUM arrays should be employed to allocate the ROCK keyword data.	1*
14	MXNPMR	A positive integer that defines the maximum number of pressure maintenance regions for the commercial compositional simulator.	0
15	NTABKT	A positive integer that defines the maximum number of temperature dependent K-value tables for the when the thermal option is activated in the commercial compositional simulator.	0
16	NTALPHA	A positive integer that defines the maximum number of transport coefficient tables for the commercial compositional simulator.	0
17	NASPKA	A positive integer that defines the maximum number of maximum number of entries in the ASPKDAM keyword tables for the commercial compositional simulator.	10
18	MXRAWG	A positive integer that defines the maximum number of maximum number of entries in the ASPREWG keyword tables for the commercial compositional simulator.	10
19	MXRASO	A positive integer that defines the maximum number of pressure maintenance regions for the commercial compositional simulator.	10
20		Not Used	1*
21	MCASPP	A positive integer that defines the maximum number of column entries in the ASPPW2D keyword tables for the commercial compositional simulator.	5
22	MRASPP	A positive integer that defines the maximum number of row entries in the ASPPW2D keyword tables for the commercial compositional simulator.	5
23	MXRATF	A positive integer that defines the maximum number of entries in the ASPWETF table for the commercial compositional simulator.	5
24	MXNKVT	A positive integer that defines the maximum number of composition dependent K-value tables for the commercial compositional simulator.	0
25	RESVED	Not Used	1*

No.	Name	Description	Default
Notes: - Table sets are groups of keywords that need to be defined to complete a table set. For example if NTPVT is set to three, then there must be three <i>PVTO</i> tables and three <i>PVTG</i> tables to complete the three PVT data set. - The keyword is terminated by a “/”.			

Table 5.44: TABDIMS Keyword Description

Example

```
--
--      NO.      NO.      MAX      MAX      MAX      MAX      E300
--      NTSFUN   NTPVT    NSSFUN  NPPVT    NTFIP    NRPVT    BLANK  NTEND
TABDIMS      15       9       40       30       1*       1*       1*       1          /
```

The above example defines number of relative permeability tables to be 15 with a maximum number of rows for each table set to 40, and the number of PVT tables to be nine with a maximum number of 30 rows per table.

5.3.144 TEMP – ACTIVATE THE TEMPERATURE MODELING OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the temperature modeling option. There is no data required for this keyword.

The temperature option (*TEMP* keyword) and the thermal option (*THERMAL* keyword) are two separate modeling facilities in the commercial simulator, although some keywords can be used by both options, for example the *RTEMP* keyword. OPM Flow’s temperature model is based on solving the energy equation fully coupled with the black-oil equations so the results are not directly equivalent to the commercial simulator’s black-oil *TEMP* or compositional *THERMAL* formulations. To activate OPM Flow’s temperature model use the *TEMP* keyword in the *RUNSPEC* section.

OPM Flow’s temperature model is similar to its thermal model except that the work term is neglected and the internal energy is equal to the enthalpy.

Example

```
--
--      ACTIVATE THE TEMPERATURE MODELING OPTION (OPM FLOW TEMPERATURE OPTION
ONLY)
--
TEMP
```

The above example activates the temperature modeling option.

5.3.145 THERMAL – ACTIVATE THE THERMAL MODELING OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the thermal modeling option. There is no data required for this keyword. The energy black-oil implementation in OPM Flow is a mixture of the commercial simulators black-oil and the commercial simulators “compositional thermal” keywords, as well as some OPM Flow specific keywords.

The temperature option (*TEMP* keyword) and the thermal option (*THERMAL* keyword) are two separate modeling facilities in the commercial simulator, although some keywords can be used by both options, for example the *RTEMP* keyword. OPM Flow’s thermal model is based on solving the energy equation fully coupled with the black-oil equations so the results are not directly equivalent to the commercial simulator’s black-oil *TEMP* or compositional *THERMAL* formulations.

Keywords specifically associated with both OPM Flow’s *THERMAL* and the commercial simulators *TEMP* and *THERMAL* options are listed in Table 5.45 for ease of reference.

Section	Keyword	Function	OPM Flow	Commercial	
			<i>THERMAL</i> MODEL	<i>TEMP</i> MODEL	<i>THERMAL</i> MODEL
<i>RUNSPEC</i>	ROCKDIMS	Thermal Rock Dimensions of Over and Underburden Rock Types			
	<i>TEMP</i>	Activate the Temperature Modeling Option			Black-Oil
	<i>THERMAL</i>	Activate the Thermal Modeling Option			
<i>GRID</i>	ROCKCON	Thermal Rock Over and Underburden Connection Data			
	ROCKPROP	Thermal Rock Over and Underburden Property Data			
	<i>HEATCR</i>	Rock Heat Capacity.			
	<i>HEATCRT</i>	Rock Heat Capacity Temperature.			
	<i>THCGAS</i>	Gas Phase Thermal Conductivity.			
	<i>THCOIL</i>	Oil Phase Thermal Conductivity.			
	<i>THCONR</i>	Thermal Conductivity of liquids and reservoir rock.			
	<i>THCONSF</i>	Thermal Conductivity of liquids and reservoir rock scaling factor applied to <i>THCONR</i> to account for gas saturation.			
	<i>THCROCK</i>	Rock Thermal Conductivity.			
	THCSOLID	Solid Phase Thermal Conductivity.			

Section	Keyword	Function	OPM Flow	Commercial	
			THERMAL MODEL	TEMP MODEL	THERMAL MODEL
	THCWATER	Water Thermal Conductivity.			
PROPS	HEATVAP	Thermal Oil Component Heat of Vaporization			
	GASDENT	Gas Density Temperature Coefficients (OPM Flow keyword).			
	GASJT	Gas Joule-Thomson Coefficient (OPM Flow keyword).			
	GASVISCT	Gas Viscosity versus Temperature Functions (OPM Flow black-oil keyword).			
	OILDENT	Oil Density Temperature Coefficients (OPM Flow keyword).			
	OILJT	Oil Joule-Thomson Coefficient (OPM Flow keyword).			
	OILVISCT	Oil Viscosity versus Temperature Functions (OPM Flow black-oil keyword).			
	RTEMP	Constant Initial Reservoir Temperature.			
	RTEMPA	Constant Initial Reservoir Temperature.			
	RTEMPVD	Initial Reservoir Temperature versus Depth.			
	SPECHA	Thermal Specific Heat of Oil Component			
	SPECHEAT	Specific Heat of Oil,Water and Gas			
	SPECROCK	Specific Heat of the Reservoir Rock			
	TEMPVD	Initial Reservoir Temperature versus Depth.			
	THANALB	Activate Thermal Analytic Water Density Option			
	THERMEXI	Liquid Components Thermal Expansion Coefficient			
	WATDENT	Oil Density Temperature Coefficients.			
	WATJT	Water Joule-Thomson Coefficient (OPM Flow keyword).			
	WATVISCT	Oil Viscosity versus Temperature Function.			

Section	Keyword	Function	OPM Flow	Commercial	
			THERMAL MODEL	TEMP MODEL	THERMAL MODEL
REGIONS	THERMNUM	Thermal Region Numbers.			
SOLUTION	RTEMP	Constant Initial Reservoir Temperature.			
	RTEMPA	Constant Initial Reservoir Temperature.			
	RTEMPVD	Initial Reservoir Temperature versus Depth.			
	TEMPI	Initial Reservoir Temperature for All Cells.			
	TEMPVD	Initial Reservoir Temperature versus Depth.			
SCHEDULE	WTEMP	Set An Injection Well's Fluid Temperature			
	WINJTEMP	Define Injection Fluid Thermal Properties			
Notes: 1) Cells colored green implies the keyword can be used with this model formulation. 2) Cells colored orange means the keyword is recognized by OPM Flow's parser but is ignored and not used by OPM Flow. 3) Cells colored red should not be used with this model formulation. 4) The list is focused on the OPM Flow implementation of the energy and black-oil formulation and therefore does not necessary include all the commercial simulator's compositional keywords.					

Table 5.45: OPM Flow's THERMAL Option Associated Keywords

In thermal runs a producing well's bottom-hole temperature is calculated based on a weighted average of the temperature in the grid cell connections open to flow in the producing well, that is the reported bottom-hole temperature, TBHT, is calculated as:

$$T_{BHT} = \frac{\sum_{i=1}^M W_i T_i}{\sum_{i=1}^M W_i} \quad (5.1)$$

with

$$W_i = \sum_p^N (\rho_{pi}^r) (q_{pi}^r) c_p \quad (5.2)$$

The term $(\rho_{pi}^r) (q_{pi}^r) c_p$ is the energy rate density (J/(K s)) of phase p,

where:

N = number of phases,

M	=	number of open connections in the well,
i	=	the open connection index,
p	=	the phase index, oil, water and gas.
r	=	indicating that the parameter is evaluated at reservoir conditions,
T	=	temperature (K),
q	=	connection flow rate (m ³ /s),
ρ	=	fluid density (kg/m ³), and
c	=	specific heat capacity (J/(K kg)) (see the SPECHEAT keyword in the PROPS section).

The current implementation makes use of the specific internal energy:

$$e_{pi} = c_p \cdot T_i \quad (5.3)$$

derived from the specific enthalpy, h_{pi}

$$e_{pi} = h_{pi} - \frac{P_{pi}}{\rho_{pi}^r} \quad (5.4)$$

where P_{pi} is connection grid block pressure of phase p.

The phase rates at surface conditions (q_{p1}^s) are converted to reservoir in situ rates (q_{p1}^r) using the phase formation volume factor, B_{pi} , via:

$$q_{pi}^r = \frac{q_{pi}^s}{B_{pi}^{-1}} \quad (5.5)$$

And thus equation (5.2) can be simplified to:

$$W_i = \sum_p^N (\rho_{pi}^r) \left(\frac{q_{pi}^s e_{pi}}{B_{pi}^{-1} \cdot T_i} \right) \quad (5.6)$$

Example

```
--
--      ACTIVATE THE THERMAL MODELING OPTION (OPM FLOW THERMAL OPTION ONLY)
--
THERMAL
```

The above example activates the thermal modeling option.

5.3.146 TITLE – DEFINE THE TITLE FOR THE INPUT DECK

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *TITLE* keyword defines the title for the input deck. The title text will be printed on all reports so as to act as a reference for the run.

No.	Name	Description	Default
I	TITLE	A character string that defines the title for the input deck	None
Notes: 1) All the characters on the line following the <i>TITLE</i> keyword are processed as a string and therefore there is no need to enclose the title in quotes. 2) There is no terminator “/” for the keyword.			

Table 5.46: TITLE Keyword Description

Note

It is good practice to include the name of the input file in the tittle (without the extension) for when cross checking results from multiple cases.

Example

```
--
--      DEFINE THE TITLE FOR THE RUN
TITLE
SPE01-THEM01-OPM1810-R01 - OPM THERMAL OPTION RUN
```

The above example defines the title for the run to be “SPE01-THEM01-OPM1810-R01 - OPM THERMAL OPTION RUN”.

5.3.147 TRACERS – ACTIVATE TRACER OPTIONS AND SET TRACER ARRAY DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

The **TRACERS** keyword defines the number of tracers in the model and the various passive tracer tracking options.

No.	Name	Description	Default
1	MXOILTR	A positive integer defining the maximum number of passive oil tracers defined using the TRACER keyword.	0
2	MXWATTR	A positive integer defining the maximum number of passive water tracers defined using the TRACER keyword.	0
3	MXGASTR	A positive integer defining the maximum number of passive gas tracers defined using the TRACER keyword.	0
4	MXENVTR	A positive integer defining the maximum number of passive environmental tracers defined using the TRACER keyword. Environmental tracers are used with the Environment Tracer model that takes into account tracer adsorption and decay. Only the default value of zero is currently supported.	0
5	DIFFOPT	A character string defining the numerical diffusion option for tracer tracking runs that should be set to: 1) DIFF activates the numerical diffusion control options. 2) NODIFF deactivates the numerical diffusion control options. Only the default value of NODIFF is supported	NODIFF
6	MXITRTR	A positive integer defining the maximum number of non-linear iterations to be used when the tracer option is activated.	12
7	MNITRTR	A positive integer defining the minimum number of non-linear iterations to be used when the tracer option is activated.	1
8	NONLIN	A character string stating if passive tracers as should be linear (NO) or non-linear (YES). Only the default value of NO is supported.	No
9	LNCONFAC	A real value defining the initial linear convergence factor. The default value of 1* means the parameter will not be utilized.	1*
10	NLCONFAC	A real value defining the initial non-linear convergence factor. The default value of 1* means the parameter will not be utilized.	1*
11	CONFAC	A real value defining the LNCONFAC and NLCONFAC convergence factors to be used after the initial convergence factor has been applied.	1.0
12	NUMCONF	A positive integer defining the maximum number of times CONFAC can be used.	0

No.	Name	Description	Default
Notes: I) The keyword is terminated by a “/”.			

Table 5.47:TRACERS Keyword Description

See also the [TRACER](#) keyword in the [PROPS](#) section that defines the individual tracers.

Example

```
--
--      NO OIL  NO WAT  NO GAS  NO ENV  DIFF    MAX    MIN    TRACER
--      TRACERS TRACERS TRACERS TRACERS CONTL  NONLIN NONLIN NONLIN
TRACERS
      0        7        1        0      'NODIFF' 1*    1*    1*           /
```

The above example defines seven tracers in the water phase and one tracer in the gas phase.

5.3.148 TRPLPORO – ACTIVATE THE TRIPLE POROSITY MODEL OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [TRPLPORO](#) keyword activates the Triple Porosity Model option that models matrix, fractures and vuggy porosity for carbonate reservoirs, and specifies the number of matrix porosity systems

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	TRPLPORO	A positive integer value that specifies the number of matrix porosity systems in the model. TRPLPORO should be set to either: 1) TRPLPORO set equal to 2, if the vugs are only connected to the fractures, so that the porosity system is matrix and connected vugs, or, 2) TRPLPORO set equal to 3, if the vugs are connected to the fractures and the matrix, so that the porosity system is matrix, connected vugs, and isolated vugs.	I
Notes: I) The keyword is terminated by a "/".			

Table 5.48:TRPLPORO Keyword Description

Note the keyword cannot be used in conjunction with the [NMATRIX](#) keyword, which is also in the [RUNSPEC](#) section.

Example

```
--
--      TRPLPORO
--      OPTION
TRPLPORO
3
```

The above example activates the Triple Porosity Model option and specifies the porosity system is matrix, connected vugs, and isolated vugs.

5.3.149 UDADIMS – DEFINE THE DIMENSIONS OF THE USER DEFINED ARGUMENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the dimensions of the User Defined Arguments (“UDA”) used by OPM Flow that can be applied to various connection, group, and well keywords in the [SCHEDULE](#) section. UDAs are defined by the [UDQ](#) keyword that is used to specify values to be constants, [SUMMARY](#) variables, as defined in [SUMMARY](#) section, or a formula using various mathematical functions together with constants and [SUMMARY](#) variables.

Although this keyword is read by OPM Flow and the [ACTION](#) and [UDQ](#) computational logic and calculations have been implemented, one should used caution using this facility as it may result in OPM Flow aborting.

No.	Name	Description	Default
1	NMUDA	NMUDA is a positive integer that defines the number of arguments in a SCHEDULE section keyword, that are replaced by numeric UDQ values.	0
2	IGNORED	Not used and should be defaulted.	1*
3	MXUDA	MXUDA is a positive integer that defines the maximum number of unique arguments in a keyword that are replaced by numeric UDA values. Note that MXUDA differs from NMUDA, for example: 1) If only the oil rate argument of, say the WCONPROD keyword is specified by a UDA, then both NMUDA and MXUDA equal one. 2) However, if a second WCONPROD uses a different UDA, then NMUDA equals two, but MXUDA would still be one. 3) Finally, if the same two distinct UDAs are used separately in two lines of WCONPROD data, then both NMUDA and MXUDA must be set to two. As MXUDA's default value is 100 then this only needs to be increased where the same UDA is used more than 100 times.	100
Notes: 1) The keyword is terminated by a “/”.			

Table 5.49: UDADIMS Keyword Description

Note that OPM Flow has a more restricted [UDQ](#) feature set than the commercial simulator, so not all options and functions are currently available.

Example

```
--
--      USER DEFINED ARGUMENT DIMENSIONS
--      NO.      NOT      TOTAL
--      ARGS    USED    UDQ
UDADIMS
      10        1*      10
```

In the above example both NMUDA and MXUDA are set equal to ten.

5.3.150 UDQDIMS – DEFINE THE DIMENSIONS OF THE USER DEFINED UDQ FEATURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the dimensions associated with the [UDQ](#) keyword used in OPM Flow to calculate various user defined values in the [SCHEDULE](#) section. The [UDQ](#) keyword defined variables can be constants, [SUMMARY](#) variables, as defined in the [SUMMARY](#) section, or a formula using various mathematical functions together with constants and [SUMMARY](#) variables.

Although this keyword is read by OPM Flow and the [ACTION](#) and [UDQ](#) computational logic and calculations have been implemented, one should used caution using this facility as it may result in OPM Flow aborting.

No.	Name	Description	Default
1	MXFUNS	A positive integer that defines the maximum number of functions that can be included when defining a UDQ definition. This should also include any brackets that will be used in the UDQ definition.	16
2	MXITEMS	MXITEMS is a positive integer that defines the maximum number of ITEMS allowed in an UDQ definition.	16
3	MXUDC	MXUDC is a positive integer that defines the maximum number of user defined CONNECTION quantities allowed in an UDQ definition.	0
4	MXUDF	MXUDF is a positive integer that defines the maximum number of user defined FIELD quantities allowed in an UDQ definition.	0
5	MXUDG	MXUDG is a positive integer that defines the maximum number of user defined GROUP quantities allowed in an UDQ definition.	0
6	MXUDR	MXUDR is a positive integer that defines the maximum number of user defined REGION quantities allowed in an UDQ definition.	0
7	MXUDS	MXUDS is a positive integer that defines the maximum number of user defined SEGMENT quantities allowed in an UDQ definition.	0
8	MXUDW	MXUDW is a positive integer that defines the maximum number of user defined WELL quantities allowed in an UDQ definition.	0
9	MXUDA	MXUDA is a positive integer that defines the maximum number of user defined AQUIFER quantities allowed in an UDQ definition.	0
10	MXUDB	MXUDB is a positive integer that defines the maximum number of user defined BLOCK quantities allowed in an UDQ definition.	0
11	RSEED	RSEED is a character string that determines if a new random number seed should be generated for restart runs for use in the UDQ functions RANDN, RANDU RRNDN and RRNDU. If RSEED is set to Y than a new seed will be generated and if set to the default value of N or I* then the same seed of the “base” simulation will be employed. See also the RSEED integer variable on the UDQPARAM keyword in the RUNSPEC section to set the random number seed for the current run. This feature is not supported by OPM Flow.	N

No.	Name	Description	Default
Notes: 1) The keyword is terminated by a “/”.			

Table 5.50: UDQDIMS Keyword Description

Note that OPM Flow has a more restricted [UDQ](#) feature set than the commercial simulator, so not all options and functions are available.

Example

```
--
--      USER DEFINED ARGUMENT DIMENSIONS FACILITY
--      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      RAND
--      FUNCS    ITEMS    CONNS    FIELD    GROUP    REGS    SEGTM    WELL    AQUF    BLCKS    OPT
UDQDIMS
      50        25        0        50        50        0        0        0        0        0        0        N /
```

In this case the maximum number of functions that can be included when defining a [UDQ](#) definition is set to 50, maximum number of items allowed in an [UDQ](#) definition is 25, the maximum number of user defined field quantities allowed in an [UDQ](#) definition is 50, and the maximum number of user defined group quantities allowed in an [UDQ](#) definition is also 50. All other parameters are defaulted including the RSEED variable (the same seed of the “base” simulation will be employed).

5.3.151 UDQPARAM – DEFINE PARAMETERS FOR THE USER DEFINED QUANTITY FEATURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the dimensions of the User Defined Arguments (“UDA”) used by OPM Flow that can be applied to various connection, group, and well keywords in the [SCHEDULE](#) section. UDAs are defined by the [UDQ](#) keyword that is used to specify values to be constants, [SUMMARY](#) variables, as defined in [SUMMARY](#) section, or a formula using various mathematical functions together with constants and [SUMMARY](#) variables.

Although this keyword is read by OPM Flow and the [ACTION](#) and [UDQ](#) computational logic and calculations have been implemented, one should used caution using this facility as it may result in OPM Flow aborting.

No.	Name	Description	Default
1	RSEED	RSEED is a positive integer greater than zero that sets a new random number seed for use in the UDQ functions RANDN, RANDU RRNDN and RRNDU. See also the RSEED character variable on the UDQDIMS keyword in the RUNSPEC section to default the random number seed for a restart run. This feature is not supported by OPM Flow.	1
2	RANGE	RANGE is a real positive value greater than or equal to one and less than or equal to 1.0×10^{20} , that sets the absolute range for the user defined quantities. The default value of 1×10^{20} sets the range from -1×10^{20} to $+1 \times 10^{20}$.	1×10^{20}
3	DEFAULT	DEFAULT is real value that is the default numerical value given to undefined UDQ variables and should be in the same range as RANGE.	0.0
4	TOLUDQ	TOLUDQ a real positive number greater than zero and less than one that defines the tolerance used to determine if two real values are equal. Floating-point numbers (as implemented in computers) are never exact, one cannot compare floating point numbers for exact equality. Thus, TOLUDQ defines a tolerance. For example, the default value of 1×10^{-4} means that if the difference between two real values is less than 1×10^{-4} then the values are considered equal.	1×10^{-4}

Notes:

- The keyword is terminated by a “/”.

Table 5.51: UDQPARAM Keyword Description

Note that OPM Flow has a more restricted [UDQ](#) feature set than the commercial simulator, so not all options and functions are available.

Example

```
--
--      USER DEFINED DEFAULT VALUES
--      SEED      RANGE      UNDEFINED      COMPARISON
--      INTG      -AND+      VALUE          TOLERANCE
UDQPARAM
      1          1.0E20      0.0          1.0E-4          /
```

The example explicitly sets the default values for all four variables on the UDAPARAM keyword, namely the random seed to one, the range to 1×10^{20} , the undefined [UDQ](#) variables to zero, and the comparison tolerance to 1.0×10^{-4} .

5.3.152 UDTDIMS – DEFINE THE DIMENSIONS OF THE USER DEFINED TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the dimensions of the User Defined Tables (“UDT”) used by OPM Flow that can be used as lookup tables when assigning values to User Defined Quantities (“UDQ”) using the [UDQ](#) keyword in the [SCHEDULE](#) section. UDTs are defined by the [UDT](#) keyword in the [SCHEDULE](#) section.

No.	Name	Description	Default
1	MXUDT	MXUDT is a positive integer that defines the maximum number of User Defined Tables	0
2	NUDT	NUDT is a positive integer that defines the maximum number of rows in any given User Defined Table.	0
3	MXINTP	MXINTP is a positive integer that defines the maximum number of interpolation points allowed in any given dimension.	0
4	MXDIMS	MXDIMS is a positive integer that defines the maximum number of dimensions in any given User Defined Table. Only one dimensional tables are currently supported by OPM Flow.	0

Notes:

- 1) The keyword is terminated by a “/”.

Table 5.52: UDTDIMS Keyword Description

Example

```
--
--      USER DEFINED TABLE DIMENSIONS
--
--      MAX      MAX      MAX      MAX
--      TABLES  ROWS    INTPOL  DIMS
UDTDIMS
      3          20      3          2                               /
```

In the above example the maximum number of [UDT](#) tables is set to three and the maximum number of rows for each table is 20, the maximum number of interpolation points in any given dimension is set to three and the maximum number of dimensions is defined as two.

5.3.153
UNCODHMD
– ACTIVATE HISTORY MATCH GRADIENT UNENCODED OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

UNCODHMD
activates the history match gradient unencoded output for the history match gradient output file. Unencoded files allows external programs to read this file type.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```

--
--      ACTIVATE HISTORY MATCH GRADIENT UNENCODED OUTPUT
--
UNCODHMD

```

The above example switches on the unified output file option.

5.3.154 UNIFIN – ACTIVATE THE UNIFIED INPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the Unified Input Files option for all input files. Similar to the commercial simulator, OPM Flow has various options for reading various input files and writing the resulting OPM Flow output files as described in Table 5.53.

Process	Keyword	Description	Files
Input	<i>FMTIN</i>	A character string that defines the input files to be formatted as ASCII i.e. text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that used as input, for for example when restarting from another case. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTIN</i>	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I
	<i>UNIFIN</i>	A character string that defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	<i>FMTOUT</i>	A character string that sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.PRT, *.LOG and *.DBG files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that binary files are outputted. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTOUT</i>	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I

Process	Keyword	Description	Files
	UNIFOUT	A character string that defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. Their main advantage is that if a number of simulations reside in one directory, their output is organized. There is no limit on the number of reporting steps that a unified file can store.			

Table 5.53: UNIFIN Keyword Description

There is no data required for this keyword and there is no terminating “/” for this keyword.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--          SWITCH ON THE UNIFIED INPUT FILES OPTION
--
UNIFIN
```

The above example switches on the unified input file option.

5.3.155 UNIFOUT – ACTIVATE THE UNIFIED OUTPUT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on the Unified Output Files option for all output files. Similar to the commercial simulator, OPM Flow has various options for reading various input files and writing the resulting OPM Flow output files as described in Table 5.54.

Process	Keyword	Description	Files
Input	<i>FMTIN</i>	A character string that defines the input files to be formatted as ASCII i.e. text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that used as input, for for example when restarting from another case. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTIN</i>	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I
	<i>UNIFIN</i>	A character string that defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	<i>FMTOUT</i>	A character string that sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.PRT, *.LOG and *.DBG files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that binary files are outputted. If the keyword is omitted then the default is for binary file input.	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTOUT</i>	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X000I *.SMSPEC *.S000I

Process	Keyword	Description	Files
	UNIFOUT	A character string that defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. If the keyword is omitted then the default is for one file per reporting time step.	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. Their main advantage is that if a number of simulations reside in one directory, their output is organized. There is no limit on the number of reporting steps that a unified file can store.			

Table 5.54: UNIFOUT Keyword Description

There is no data required for this keyword and there is no terminating “/” for this keyword.

See also [OPM FLOW OUTPUT FILE FORMATS](#) for a more detailed description of the various file types (ASCII or binary) and file structure formats (unified or non-unified formats).

Example

```
--
--          SWITCH ON THE UNIFIED OUTPUT FILES OPTION
--
UNIFOUT
```

The above example switches on the unified output file option.

5.3.156 UNIFOUTS – ACTIVATE THE UNIFIED OUTPUT SUMMARY FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [UNIFOUTS](#) keyword causes the [SUMMARY](#) file output files to be a unified file, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single [SUMMARY](#) file will be generated, as opposed to one file per report time step. See also the [MULTOUT](#) keyword in the [RUNSPEC](#) section that sets both the [SUMMARY](#) and [RESTART](#) files to be non-unified multiple files, as opposed to unified files. Note also that [UNIFOUTS](#) keyword has precedence over the [MULTOUT](#) keyword for [SUMMARY](#) files.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      ACTIVATE THE UNIFIED OUTPUT SUMMARY FILE OPTION
--
UNIFOUTS
```

The above example switches on writing of unified [SUMMARY](#) output files.

5.3.157 UNIFSAVE – ACTIVATE THE UNIFIED OUTPUT SAVE FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [UNIFSAVE](#) keyword causes the [SAVE](#) file output file to be a unified file, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single [SAVE](#) file will be generated, as opposed to one file per report time step. See also the [MULTOUT](#) keyword in the [RUNSPEC](#) section that sets both the [SUMMARY](#) and [RESTART](#) files to be non-unified multiple files, as opposed to unified files.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      ACTIVATE THE UNIFIED OUTPUT SAVE FILE OPTION
--
UNIFSAVE
```

The above example switches on writing of unified [SAVE](#) output files.

5.3.158 VAPOIL – ACTIVATE THE VAPORIZED OIL IN WET GAS PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
----------------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicates that vaporized oil (more commonly referred to as condensate) is present in wet¹¹¹ gas in the model and the keyword should only be used if there is both oil and gas phases in the model. The keyword may be used for gas-water and oil-water-gas input decks that contain the oil and gas phases. The keyword will also invoke data input file checking to ensure that all the required oil and gas phase input parameters are defined in the input deck.

If the gas has a constant and uniform vaporized oil concentration, Condensate-Gas Ratio (“CGR”), and if the reservoir pressure never drops below the saturation pressure (dew point pressure), then the model can be run more efficiently by omitting the **OIL** and **VAPOIL** keywords from the **RUNSPEC** section, treating the gas as a dry gas¹¹², and defining a constant Rv (CGR) value with keyword **RVCONST** or **RVCONSTT** in the **PROPS** section. This results in the model being run with as a dry gas problem with no active oil (condensate) phase. However, OPM Flow takes into account the constant Rv in the calculations and reporting.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      VAPORIZED OIL IN WET GAS IS PRESENT IN THE RUN
--
VAPOIL
```

The above example declares that the vaporized oil, i.e. condensate, in the gas phase is active in the model.

¹¹¹ Natural gas that contains significant heavy hydrocarbons such as propane, butane and other liquid hydrocarbons is known as wet gas or rich gas. The general rule of thumb is if the gas contains less methane (typically less than 85% methane) and more ethane, and other more complex hydrocarbons, it is labeled as wet gas. Wet gas normally has GOR's less than 100,000 scf/stb or 18,000 Sm³/m³, with the condensate having a gravity greater than 50 °API.

¹¹² Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

5.3.159 VAPWAT – ACTIVATE VAPORIZED WATER IN THE DRY AND WET GAS PHASES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword indicates that vaporized water is present in the gas phase and the keyword should only be used if both water and gas phases are present in the model. [VAPWAT](#) should also be used in conjunction with the [PRECSALT](#) keyword in the [RUNSPEC](#) section in order to activate OPM Flow's Salt Precipitation model. [VAPWAT](#) may be used for gas-water and oil-water-gas input decks that contain the oil, gas and water phases.

Note

This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by declaring that vaporized water is present in the run.

Note that if the [VAPWAT](#) keyword is in the input deck then either the [PVTGW](#) or [PVTGWO](#) keywords in [PROPS](#) section should be used to defined the gas and water PVT properties.

Secondly, if both the [VAPWAT](#) keyword and the [PRECSALT](#) keyword (used to activate the OPM Flow's Salt Precipitation model) are present in the input deck, then the [RWGSALT](#) keyword in the [PROPS](#) section also needs to be present.

There is no data required for this keyword and there is no terminating "/" for this keyword.

Example

```
--
--      VAPORIZED WATER IN DRY/WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
VAPWAT
```

The above example declares that the vaporized water is present in the gas phase and is active in the model.

5.3.160
VE – ACTIVATE VERTICAL EQUILIBRIUM MODEL (GLOBAL)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword actives the Vertical Equilibrium (“VE”) model for the global grid and optionally specifies the type of VE model. The VE model type can either be compressed for merging columns of grid blocks into a single grid block, or uncompressed for the standard VE model.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.161

VFPIDIMS

–

INJECTION

VERTICAL

FLOW

PERFORMANCE

TABLE

DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

VFPIDIMS keyword defines the maximum dimensions of the injection well Vertical Lift Performance (“VFP”) tables defined by *VFPINJ* keyword. The VFP tables for the producing wells are defined by the *VFPPDIMS* keyword.

No.	Name	Description	Default
1	MXMFLO	A positive integer that defines the maximum number of injection rate entries for the <i>VFPINJ</i> keyword.	0
2	MXMTHP	A positive integer that defines the maximum number of THP entries for the <i>VFPINJ</i> keyword.	0
3	MXVFPTAB	A positive integer that defines the maximum number of <i>VFPINJ</i> tables entered through the <i>VFPINJ</i> keyword.	0
Notes: 1) The keyword is terminated by a “/”.			

Table 5.55:VFPIDIMS Keyword Description

Example

```
--      INJECTING VFP TABLES
--      VFP      VFP      VFP
--      MXMFLO  MXMTHP  NMMVFT
VFPIDIMS
      10      10      12
                                     /
```

The above example defines that the maximum number of injection rates and THP entries on the *VFPINJ* keyword is 10, and the number of *VFPINJ* tables is 12.

5.3.162 VFPPDIMS – PRODUCTION VERTICAL FLOW PERFORMANCE TABLE DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

VFPPDIMS keyword defines the maximum dimensions of the production well Vertical Lift Performance (“VFP”) tables defined by *VFPPROD* keyword. The VFP tables for the injection wells are defined by the *VFPIDIMS* keyword.

No.	Name	Description	Default
1	MXMFLO	A positive integer that defines the maximum number of production flow rate entries for the <i>VFPPROD</i> keyword.	0
2	MXMTHP	A positive integer that defines the maximum number of THP entries for the <i>VFPPROD</i> keyword.	0
3	MXMWFR	A positive integer that defines the maximum number of water fraction entries (WOR,WCUT, GWR etc.) for the <i>VFPPROD</i> keyword.	0
4	MXMGFR	A positive integer that defines the maximum number of gas fraction entries (GOR, GLR, OGR etc.) for the <i>VFPPROD</i> keyword.	0
5	MXMALQ	A positive integer that defines the maximum number of artificial lift quantity entries for the <i>VFPPROD</i> keyword.	0
6	MXVFPTAB	A positive integer that defines the maximum number of <i>VFPPROD</i> tables entered through the <i>VFPPROD</i> keyword.	0
Notes: 1) The keyword is terminated by a “/”.			

Table 5.56:VFPPDIMS Keyword Description

Example

```
--      PRODUCING VFP TABLES
--      VFP      VFP      VFP      VFP      VFP      VFP
--      MXMFLO  MXMTHP  MXMWFR  MXMGFR  MXMALQ  NMMVFT
VFPPDIMS
      20      10      10      10      6      9      /
```

Here the example shows that there are a maximum of 20 flow rates, 10 THP entries, 10 water and gas fraction entries, and six artificial lift entries for the nine *VFPPROD* VFP production tables.

5.3.163 VISAGE – ACTIVATE EXTERNAL RESERVOIR GEO-MECHANICS VISAGE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *VISAGE* keyword activates the External Reservoir Geo-Mechanics VISAGE option. The keyword should not be used in input decks as the associated data is generated by an external program.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

5.3.164 VISCD – ACTIVATE DUAL POROSITY VISCOUS DISPLACEMENT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [VISCD](#) keyword activates the Dual Porosity Viscous Displacement option for dual porosity and dual permeability models, and therefore requires either the [DUALPORO](#) or [DUALPERM](#) keyword to be entered in the [RUNSPEC](#) section to activate either one of these options. The [VISCD](#) option is used to model the viscous displacement of fluids from the matrix by the fracture pressure gradient, for when the fracture system has a more moderate permeability, and flow to and from the matrix caused by the fracture pressure gradient acts as an additional production mechanism¹¹³. Normally this mechanism is ignored as the pressure gradient in the fracture system is small due to the very high permeability of the fracture system. See the [LX](#), [LY](#) and [LZ](#) keywords in the [GRID](#) section that define representative matrix grid block sizes.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE DUAL POROSITY VISCOUS DISPLACEMENT OPTION
--
VISCD
```

The above example activates the dual porosity viscous displacement option.

¹¹³ Gilman, J. R. and Kazemi, H. “Improved Calculation for Viscous and Gravity Displacement in Matrix Blocks in Dual-Porosity Simulators,” paper SPE 16010 (includes a number of associated papers), Journal of Petroleum Technology (1988) 40, No. 1, 60-70.

5.3.165 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

5.3.166 WATER – ACTIVATE THE WATER PHASE IN THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword indicate that the water phase is present in the model and must be used for gas-water, oil-water, oil-water-gas input decks that contain the water phase. The keyword will also invoke data input file checking to ensure that all the required water phase input parameters are defined in the input deck.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      WATER PHASE IS PRESENT IN THE RUN
--
WATER
```

The above example declares that the water phase is active in the model.

5.3.167 WELLDIMS – DEFINE THE WELLS AND GROUP DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[WELLDIMS](#) defines various well and group dimensions for the run. The commercial simulator combines both the black-oil and compositional simulator variables on this keyword; however, although all the parameters are explained below only the black-oil parameters are used by OPM Flow.

No.	Name	Description	Default
1	MXWELS	A positive integer defining the maximum number of wells for this model.	0
2	MXCONS	A positive integer defining the maximum number of grid block connections per well for this model.	0
3	MXGRPS	A positive integer defining the maximum number of groups for this model.	0
4	MXGRPW	A positive integer defining the maximum number of wells that can belong to a group in the model and the maximum number of child groups in a group. Note that MXGRPW sets both the maximum number of wells in a group <u>and</u> the maximum number of child groups in a group. The former applies to groups that contain wells and the latter applies to groups that contain other groups. See also the GRUPTREE keyword in the SCHEDULE section to define group hierarchy.	0
5	MXSTAGE	A positive integer defining the maximum number of stages per separator for this model. This option is ignored by OPM Flow.	5
6	MXSTRMS	A positive integer defining the maximum number of well streams for this model. This option is ignored by OPM Flow.	10
7	MXMIXS	A positive integer defining the maximum number of mixtures for this model. This option is ignored by OPM Flow.	5
8	MXSEPS	A positive integer defining the maximum number of separators for this model. This option is ignored by OPM Flow.	4
9	MXCOMPS	A positive integer defining the maximum number of mixture components in a mixture for the model. This option is ignored by OPM Flow.	3
10	MXDOCOMP	A positive integer defining the maximum number of well completions that can cross a parallel run domain boundary when the PARALLEL option has been activated. This option is ignored by OPM Flow.	0
11	MXWSLIST	A positive integer defining the maximum number of well lists that a well may be concurrent belong to at one time for this model. This option is ignored by OPM Flow.	1

No.	Name	Description	Default
12	MXWLISTS	A positive integer defining the maximum number of dynamic well lists for this model. This option is ignored by OPM Flow.	1
13	MXWSECD	A positive integer defining the maximum number of secondary wells for this model. This option is ignored by OPM Flow.	10
14	MXNGPP	A positive integer defining the maximum number of row entries per completion in the generalized pseudo-pressure tables used for to calculate the blocking factor associated with condensate drop-out in gas condensate reservoirs. If the generalized pseudo-pressure option has not been activated then this is ignored. This option is ignored by OPM Flow.	201
Notes: 1) Only parameters (1) to (4) are used by OPM Flow. 2) The keyword is terminated by a "/".			

Table 5.57: WELLDIMS Keyword Description

Example

```
--
--      WELL      WELL      GRUPS   GRUPS
--      MXWELS   MXCONS   MXGRPS  MXGRPW
WELLDIMS
      60         110       18        40
/
```

The above example defines the maximum number of wells to be 60 with 110 completions per well, and maximum number of groups to be 18 with maximum number of wells per group of 40. All other parameters are defaulted.

5.3.168 WPOTCALC – WELL POTENTIAL CALCULATION OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

WPOTCALC defines how shut-in and stopped wells should have their well potentials calculated. Well potentials for wells under these conditions need to have their potentials calculated if they are in a Priority Drilling Queue via the WDRILPRI keyword in the SCHEDULE section, or the Prioritization option has been enabled by the PRIORITY keyword in the SCHEDULE section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

5.3.169 WSEGDIMS – DEFINE MULTI-SEGMENT WELL DIMENSIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The WSEGDIMS keyword defines the multi-segment well dimensions for the multi-segment well model and the keyword is obligatory if multi-segment wells are being employed in the model.

No.	Name	Description	Default
1	MXWELS	A positive integer defining the maximum number of multi-segment wells for this model.	0
2	MXSEGS	A positive integer defining the maximum number of segments per well for this model.	1
3	MXBRAN	A positive integer defining the maximum number of branches per multi-segment well, including the main branch groups for this model.	1
4	MXLINKS	A positive integer defining the maximum number of segment links per multi-segment well. The simulator does not currently support multi-segment chord links, and therefore this parameter is ignored.	0

Notes:

- 1) The keyword is terminated by a “/”.

Table 5.58: WSEGDIMS Keyword Description

Example

```
--
--      WELL      WELL      BRANCH  SEGMENT
--      MXWELS    MXSEGS    MXBRAN  MXLINKS
WSEGDIMS
      5           100       10       10                               /
```

The above example defines the maximum number of multi-segment wells to be five with up to 100 segments per multi-segment well, a maximum number of 10 branches per multi-segment well, and up to 10 segment links per multi-segment well.

CHAPTER 6: GRID SECTION

6.1 INTRODUCTION

The [GRID](#) section defines the basic grid properties, including structure, faults, and various static rock properties (porosity, permeability, etc.). The information in this section will be used by the software to calculate the pore volume ([PORV](#)) for each cell, the cell mid-point depths, and the regular transmissibilities ([TRANX](#), [TRANX](#), and [TRANZ](#)) between all the cells, as well as across faults. The OPM Flow calculated parameters can then be edited in the [EDIT](#) section.

All models can be classified by the number of dimensions as show in Figure 6.1 (after Mattax¹¹⁴). The zero and one dimension models are employed in analytical modeling, while the higher dimensions are used in numerical modeling. The term 4D modeling refers to a 3D model with the fourth dimension being the time domain derived from time-lapse seismic, that is the comparison of 3D seismic surveys at two or more points in time.

OPM Flow enables the user to define 1D, 2D and 3D models using three types of grids: Cartesian Regular Grid, Radial Grid, and Irregular Corner-Point Grids. The first two type of grids are rather limited in their ability to describe the structural complexity of oil and gas reservoirs; however, this simplicity allows the engineer to quickly build simple models to investigate reservoir performance. Indeed in the early days of numerical modeling back in the late 1970's two-dimensional cross-section and radial models were the main models used to predict reservoir performance due to limited computer resources at the time. That is not to say that full field models were not developed, but that these full field models were very coarse in comparison to what is designed and built today using static earth modeling software.

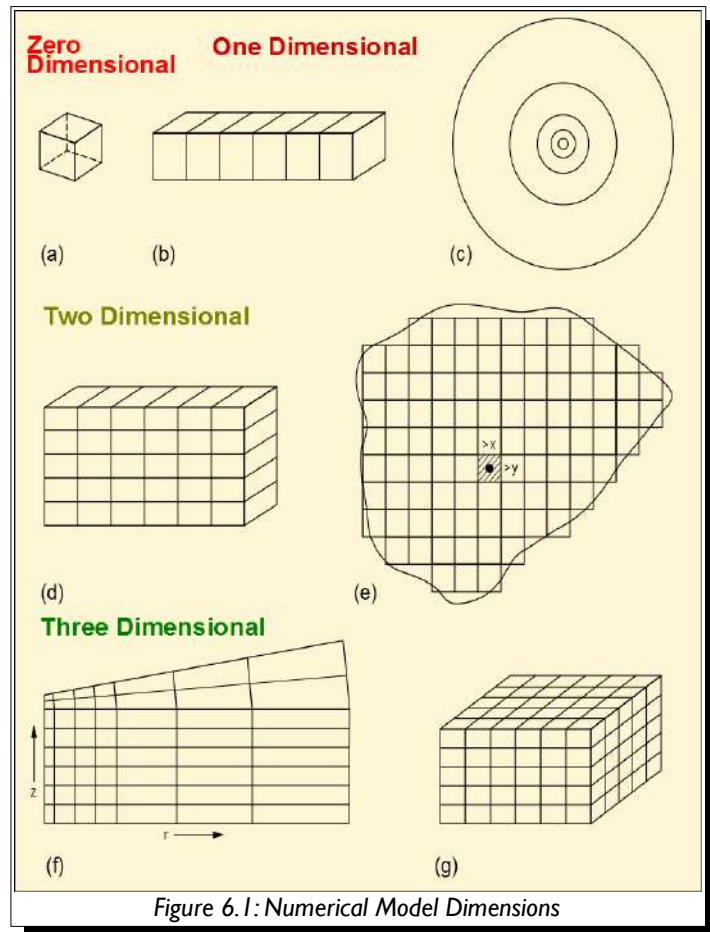


Figure 6.1: Numerical Model Dimensions

A brief introduction to the three types of grid and the data required to fully define the structural element of the grid together with the rock properties necessary to complete the [GRID](#) section data requirements is outlined in the following section. This is then followed by the keyword definitions applicable to this section.

6.2 DATA REQUIREMENTS

6.2.1 CARTESIAN REGULAR GRIDS

This type of grid defines a regular orthogonal grid based on defining the sizes of all the cells in the x, y, and z directions and is normally employed when a complex structural model is not required. Figure 6.2 shows the SPE Comparative Solution Project Number 1 (“SPE-CSP01”) as documented by Odeh¹¹⁵.

¹¹⁴ Mattax, C.C. and Dalton R.L. 1990. *Reservoir Simulation*. Society of Petroleum Engineers, Henry L. Doherty Series, Monograph Vol. 13

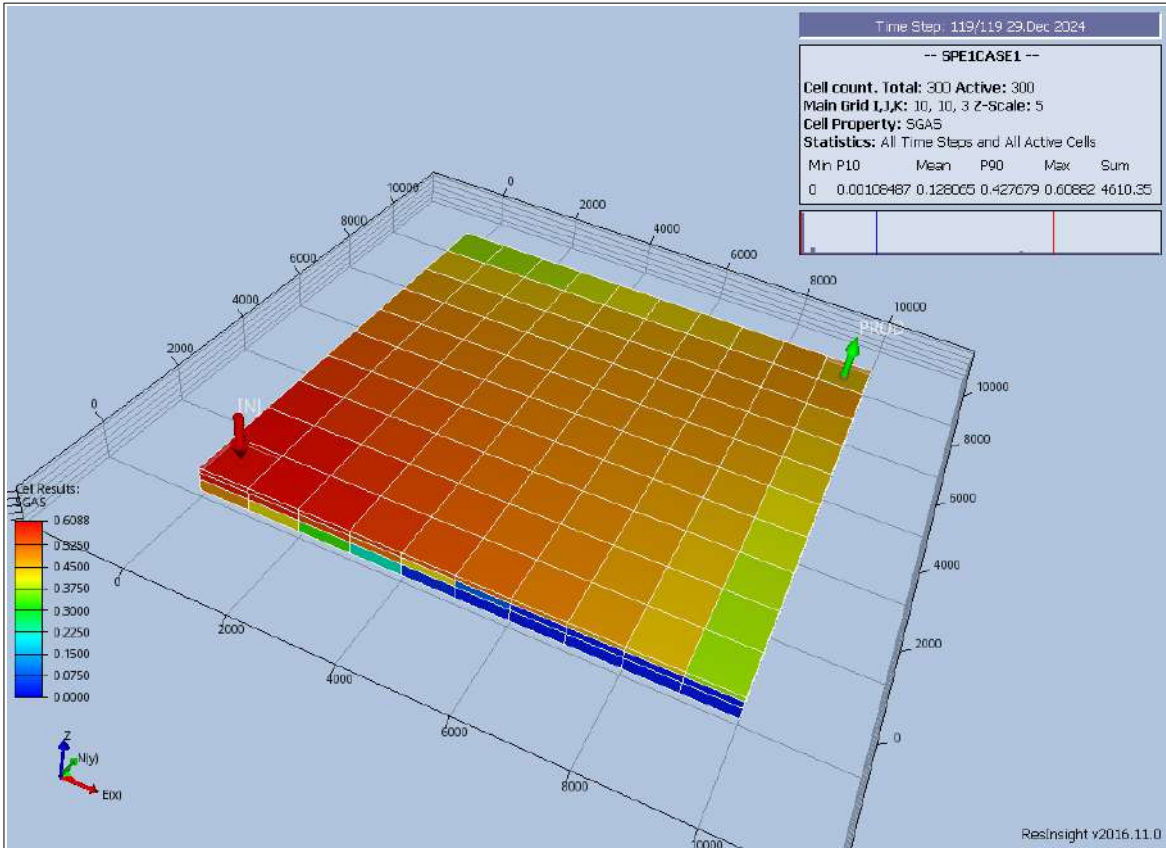


Figure 6.2: SPE Simulation Case #01 Cartesian Regular Grid

The model consists of a simple 10 × 10 × 3 (NX, NY, NZ) grid and is defined using the following [GRID](#) section keywords to define the grid geometry:

¹¹⁵ Odeh,A. “Comparison of Solutions to a Three Dimensional Black-Oil Reservoir Simulation Problem.” JPT 33 (1981):13-25.

```
-- =====
--
-- GRID SECTION
--
-- =====
GRID
--
--      DEFINE GRID BLOCK SIZES IN THE X DIRECTION BASED ON NX x NY x NZ = 300)
--      (There Are In Total 300 Cells With Length 1000ft In X-Direction)
DX
      300*1000
--
--      DEFINE GRID BLOCK SIZES IN THE Y DIRECTION (BASED ON NX x NY x NZ = 300)
--      (There Are In Total 300 Cells With Length 1000ft In Y-Direction)
DY
      300*1000
--
--      DEFINE GRID BLOCK SIZES IN THE Z DIRECTION (BASED ON NX x NY x NZ = 300)
--
DZ
      100*20.0    100*30.0    100*50.0
--
--      DEFINE GRID BLOCK TOPS FOR THE TOP LAYER (BASED ON NX = 100, NY = 100)
--      (Layer 2 and 3 TOPS Calculated by Simulator)
TOPS
      25*3100    25*3105    25*3110
```

The rock property data required to complete the GRID section is as follows:

```
--
--      DEFINE POROSITY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PORO
      300*0.300
--
--      DEFINE PERMY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PERMX
      100*500.0    100*50.0    100*200.0
--
--      DEFINE PERMY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PERMY
      100*500.0    100*50.0    100*200.0
--
--      DEFINE PERMZ DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--      (Not Defined in Original Paper So Assume That PERMX = PERMY = PERMZ)
PERMZ
      100*500.0    100*50.0    100*200.0
```

The above keywords define all the properties required for the GRID section for this type of grid geometry.

6.2.2 IRREGULAR CORNER-POINT GRIDS

This type of grid is an industry standard grid used to represent the structure of complex reservoirs¹¹⁶. Here a static modeling application is used to build the grid which is then exported and imported into a dynamic simulation model. Figure 6.3 illustrates the skeleton grid for the Norne Field which has dimensions of 46 x 112 x 22 in the x, y, and z dimensions respectively. This results in a total of 113,344 cells although not all of these cells will be active in the model.

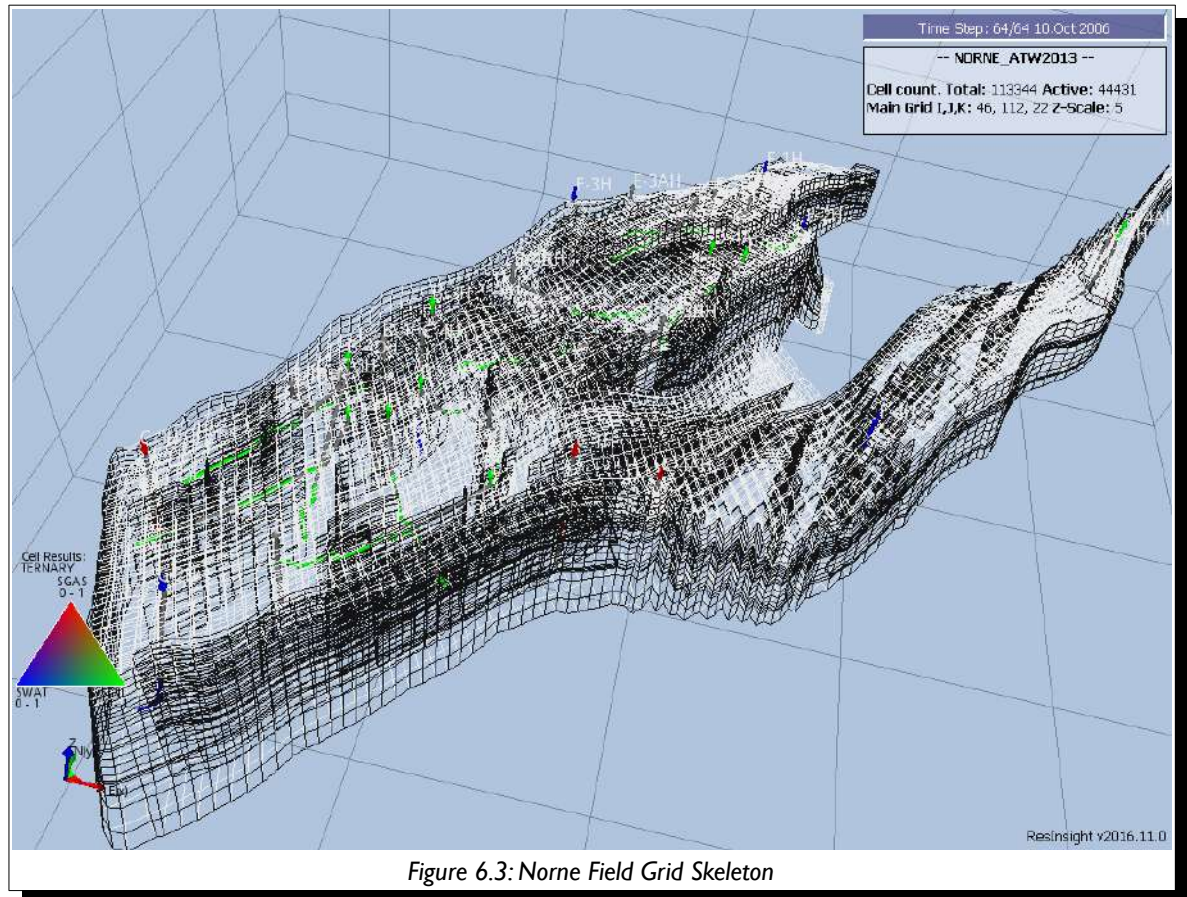


Figure 6.3: Norne Field Grid Skeleton

Similar to a Cartesian Regular Grid the grid geometry and properties must be defined for each cell. The formulation of the grid geometry is based on a corner-point geometry, where basically the coordinate lines or pillars are given, then the top and bottom surfaces for the cell are defined by specifying the depth (z-coordinates) of the cell's corner points along each of the four adjacent pillars. The cell then forms an irregular hexahedron as depicted in Figure 6.4. Note that the figure shows a corner-point cell which is more or less orthogonal, which is ideally what we want to minimize grid orientation effects.

The data required to define this type of grid consists of the *SPECGRID* to define the dimensions of the grid, that is:

--	MAX	MAX	MAX	MAX	GRID	
--	NDIVIX	NDIVIY	NDIVIZ	NUMRES	TYPE	
SPECGRID						
	46	112	22	1	F	/

¹¹⁶ K. Ponting, D. Corner Point Geometry in Reservoir Simulation, Conference Proceedings, ECMOR I - 1st European Conference on the Mathematics of Oil Recovery, Jul 1989, cp-234-00003 dDOI= "https://doi.org/10.3997/2214-4609.201411305", European Association of Geoscientists & Engineers.

A portion of the coordinate line data defined by the [COORD](#) keyword from the Norne model is shown on the next page.

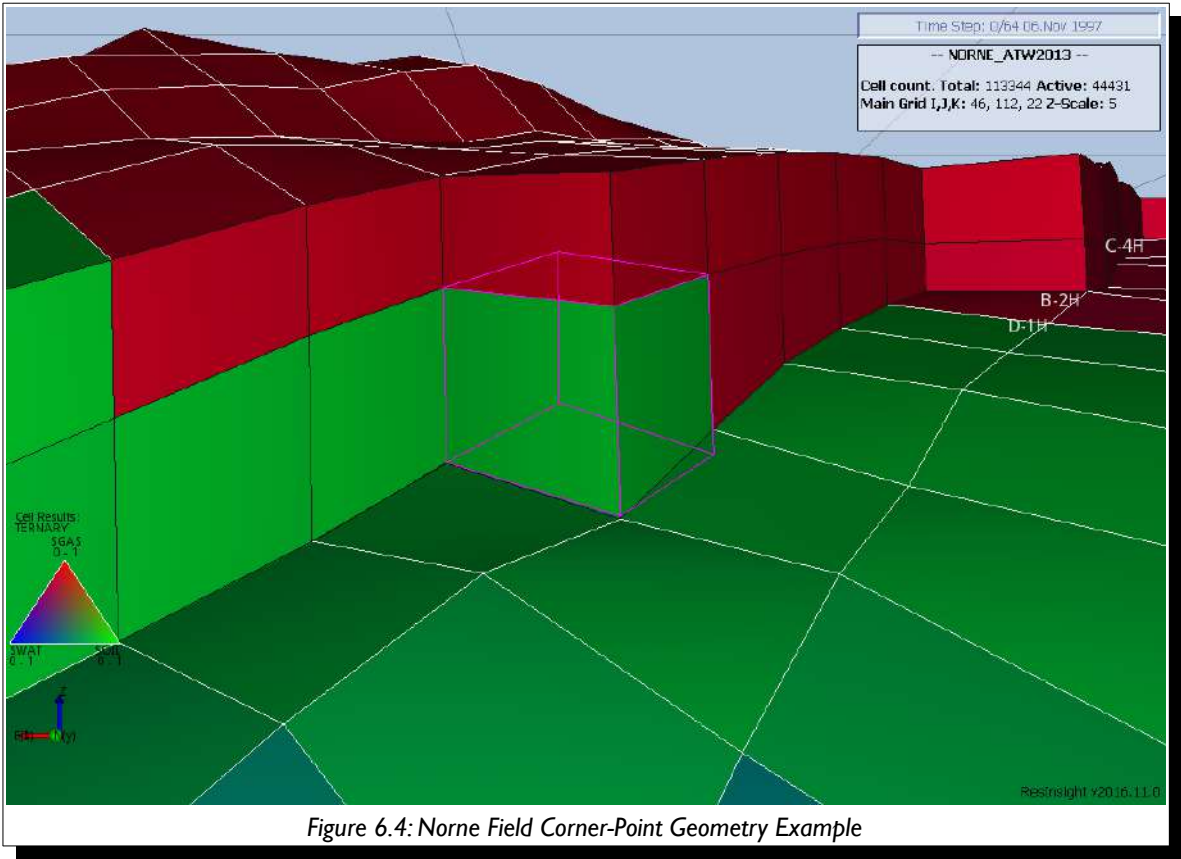


Figure 6.4: Norne Field Corner-Point Geometry Example

COORD					
-- X1	Y1	Z1	X2	Y2	Z2
----	----	----	----	----	----
453114.000	7319921.000	3037.473	453114.000	7319921.000	3132.831
453155.031	7319840.000	2983.933	453142.750	7319864.000	3173.572
453196.094	7319759.000	3005.969	453171.500	7319807.500	3215.836
453237.156	7319678.000	3000.265	453200.250	7319751.000	3217.250
453278.188	7319597.000	2989.348	453229.031	7319694.000	3213.951
453319.250	7319516.500	2995.680	453257.781	7319637.500	3215.323
453356.250	7319443.500	3000.855	453308.750	7319537.000	3220.549
453393.250	7319370.500	3005.252	453359.688	7319436.500	3210.393
453423.969	7319310.000	3030.862	453394.219	7319368.500	3203.438
453454.688	7319249.500	3036.870	453428.719	7319300.500	3190.770
453485.406	7319189.000	3038.017	453463.219	7319232.500	3190.660
453516.125	7319128.000	3045.027	453497.750	7319164.500	3188.813
453546.844	7319067.500	3055.410	453532.250	7319096.500	3185.966
453577.562	7319007.000	3066.541	453566.750	7319028.500	3184.325
453608.281	7318946.500	3076.624	453601.250	7318960.500	3183.584
453639.000	7318886.000	3086.938	453635.781	7318892.500	3184.057
453669.719	7318825.500	3096.153	453670.281	7318824.500	3185.988
453700.438	7318765.000	3104.703	453704.781	7318756.500	3188.598
453731.156	7318704.500	3097.016	453739.281	7318688.500	3180.484
453761.875	7318644.000	3088.539	453773.812	7318620.500	3177.091
453780.000	7318608.000	3098.118	453796.562	7318575.500	3176.401
453798.125	7318572.500	3096.691	453819.344	7318530.500	3172.299

/

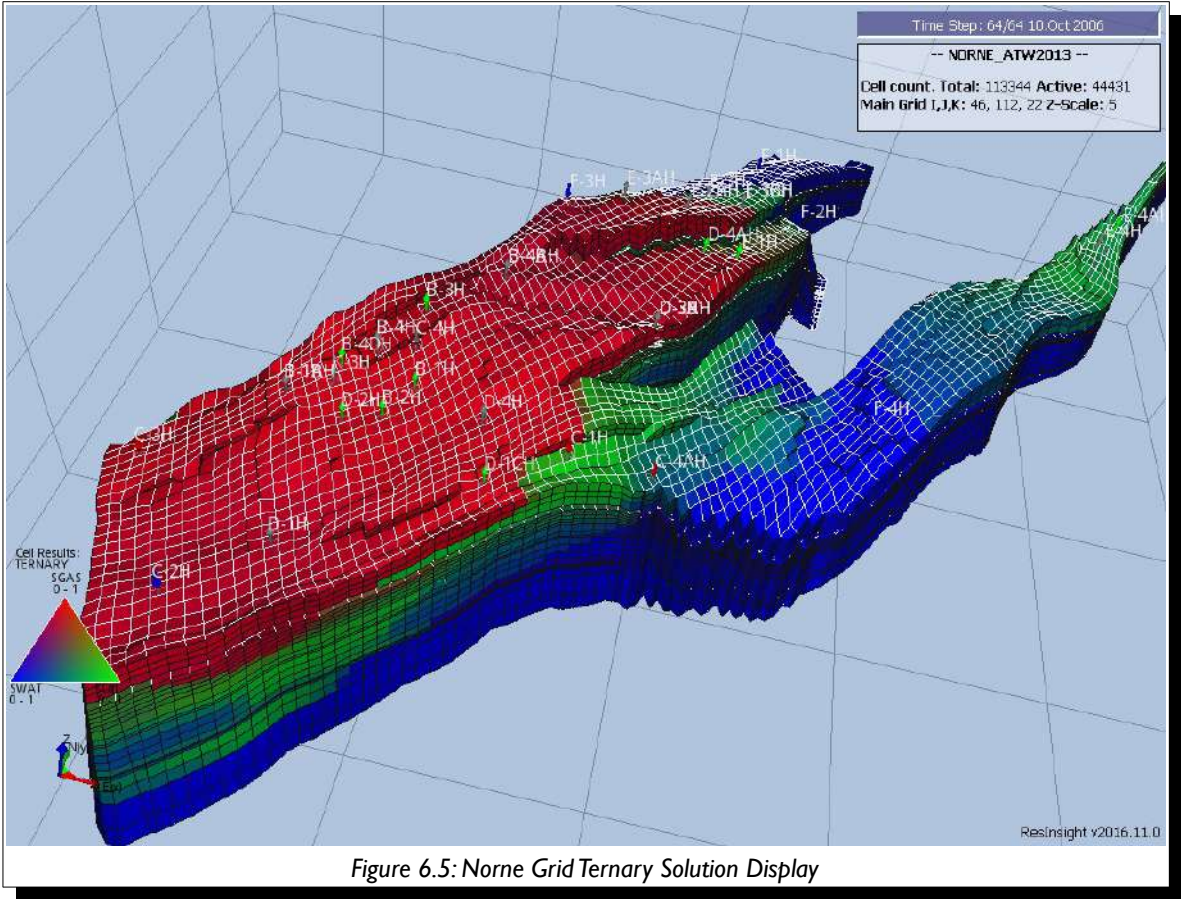
The final keyword to define an Irregular Corner-Point geometry grid is the [ZCORN](#) keyword that defines the depths of the cell corners. A portion of the ZCORN data from the Norne model is shown below.

ZCORN

3037.473	2983.933	2983.933	3005.969	3005.969	3000.265
3000.265	2989.348	2989.348	2995.680	2995.680	3000.855
3000.855	3005.252	3005.252	3030.862	3030.862	3036.870
3036.870	3038.017	3038.017	3045.027	3045.027	3055.410
3055.410	3066.541	3066.541	3076.624	3076.624	3086.938
3086.938	3096.153	3096.153	3104.703	3104.703	3097.016
3097.016	3088.539	3088.539	3098.118	3098.118	3096.691
3096.691	3093.886	3093.886	3085.393	3085.393	3081.957
3081.957	3080.645	3080.645	3115.021	3115.021	3130.474
3130.474	3204.674	3204.674	3193.187	3193.187	3169.512
3169.512	3101.928	3101.928	3044.277	3044.277	3023.930
3023.930	2964.244	2964.244	2900.178	2900.178	2875.715
2875.715	2864.913	2864.913	2855.256	2855.256	2841.119
2841.119	2826.261	2826.261	2806.556	2806.556	2781.052
2781.052	2791.720	2791.720	2817.940	2817.940	2813.308
2813.308	2788.492				

/

The rock property data required to complete the GRID section is the same as for a Cartesian Regular grid, as defined in section [6.2.1 Cartesian Regular Grids](#) and the data is defined using the same keywords. The resulting Norne model showing the ternary solution variable is displayed in Figure 6.5.



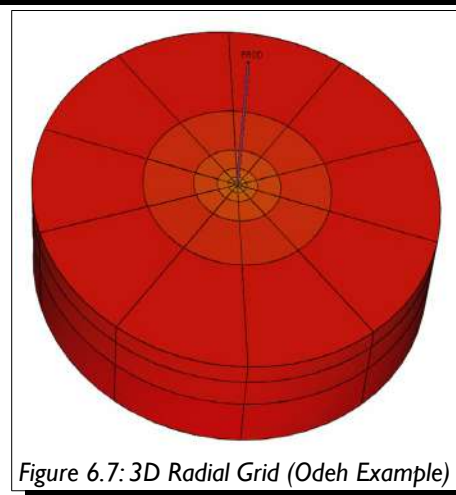
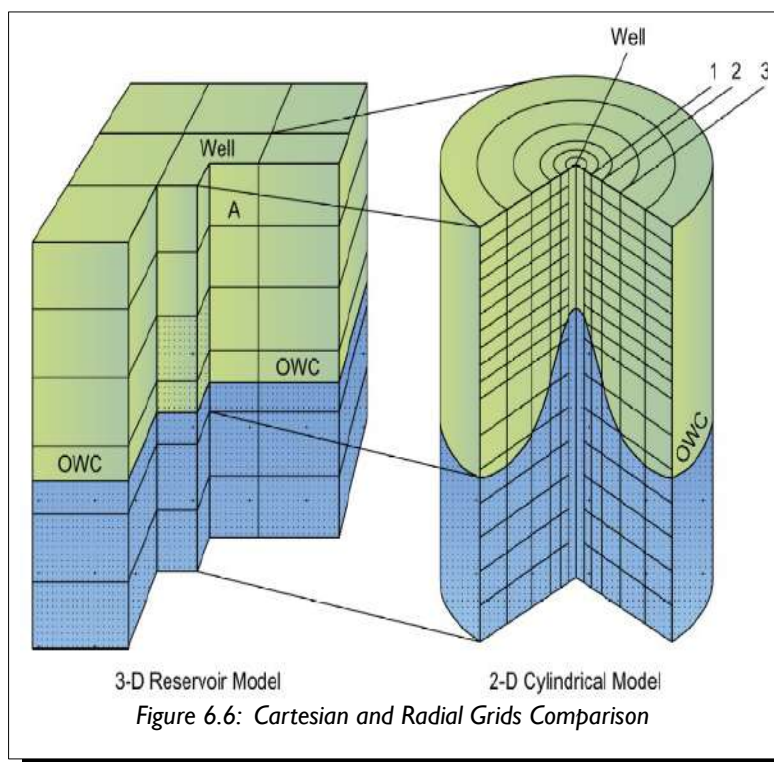
6.2.3 RADIAL GRIDS

Radial flow exists near the wellbore and linear or Cartesian flow occurs within the reservoir away from the wellbore. For field-wide studies the growth or decay of reservoir volumes activated by unsteady-state flow is small and hence linear flow is modeled via a linear grid, using either a Cartesian Regular Grid or an Irregular Corner-Point Grid, in these type of studies. On the other hand, for converging flow near wells the growth or decay of reservoir volumes activated by unsteady-state flow is large and radial flow should be modeled. Thus, radial flow is important in only very localized areas around the wells and linear flow is modeled in most reservoir studies, for example in full field or sector models. Consequently for investigating near well bore effects, radial models are used to investigate the behavior around the wellbore, for example in gas cusping and water coning studies, as one can see from Figure 6.6.

In both Cartesian Regular and Irregular Corner-Point grids the dimension nomenclature is (x, y, z) for the x , y , and z planes. The corresponding nomenclature for Radial Grids is (r, θ, z) for r or radius plane, θ for the angular plane measured in degrees, and z for the z plane.

In Figure 6.6 the radial grid has been labeled as a 2-D Cylindrical grid because there is no flow in the θ direction only in the r and z planes, whereas the 3-D Reservoir grid in the figure has flow in both the x and y planes as well as the z plane. Figure 6.7 shows a typical 3-D radial grid with dimensions $(10, 10, 3)$ in the (r, θ, z) dimensions taken from Odeh¹¹⁷.

Radial grids normally have a fine grid near the wellbore which then expands logarithmically away from the wellbore. This is illustrated in Figure 6.7 where one can clearly see the four most outer rings, but not the inner six. This type of grid makes solving the flow equations more challenging than conventional Cartesian and Irregular Corner-Point grids, where the cell pore volume distribution is relatively similar for neighboring cells, compared with Radial grids. For comparison, a typical conventional grid will have grid blocks size of 300, 300, 3 feet (or approximately 100m x 100m x 1m) spacing in the (x, y, z) plane; whereas a radial grid will utilize an inner most radius of between 0.25 to 1.0 foot in the R direction, and then logarithmically expand from the inner radius. The Odeh¹¹⁷ example employs 0.25 ft. for the inner radius, and 1.75, 2.32, 5.01, 10.84, 23.39, 50.55, 109.21, 235.92, 509.68, and 1101.08 ft. for the outer radii.



¹¹⁷ Comparison of Solutions to a Three-Dimensional Black-Oil Reservoir Simulation Problem by Aziz S. Odeh, – Journal of Petroleum Technology, January 1981.

Figure 6.7 is based on a 10 x 10 x 3 grid, where NX is the R dimensions, NY the THETA dimensions, and NZ the z dimensions on the DIMENS keyword in the RUNSPEC section. Thus, in order to fully define this radial grid the following RUNSPEC and GRID section keywords are required:

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--          MAX      MAX      MAX
--          NDIVIX   NDIVIY   NDIVIZ
DIMENS
--          10      10      3
--
--          DEFINE RADIAL GRID GEOMETRY
--
RADIAL
-- =====
--
-- GRID SECTION
--
-- =====
GRID
--
--          INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD
--          0.25
--
--          DEFINE GRID BLOCK SIZES IN THE R DIRECTION
--
DRV
--          1.75   2.32   5.01   10.84   23.39   50.55   109.21   235.92   509.68   1101.08 /
--
--          DEFINE GRID BLOCK SIZES IN THE THETA DIRECTION (BASED ON NY = 10)
--
DTHETAV
--          10*36
--
--          DEFINE GRID BLOCK Z DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 300)
--
DZ
--          100*20.0   100*30.0   100*50.0
--
--          DEFINE GRID BLOCK TOPS FOR THE TOP LAYER (NX=10,NY=10, and NZ=3)
--
TOPS
--          100*8325
--
The rock property data required to complete the GRID section is as follows:
--
--          DEFINE POROSITY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PORO
--          300*0.300
--
--          DEFINE GRID BLOCK PERMR DATA FOR ALL CELLS (BASED ON NR x NY x NZ = 300)
--
PERMR
--          100*500.0   100*50.0   100*200.0
--

```

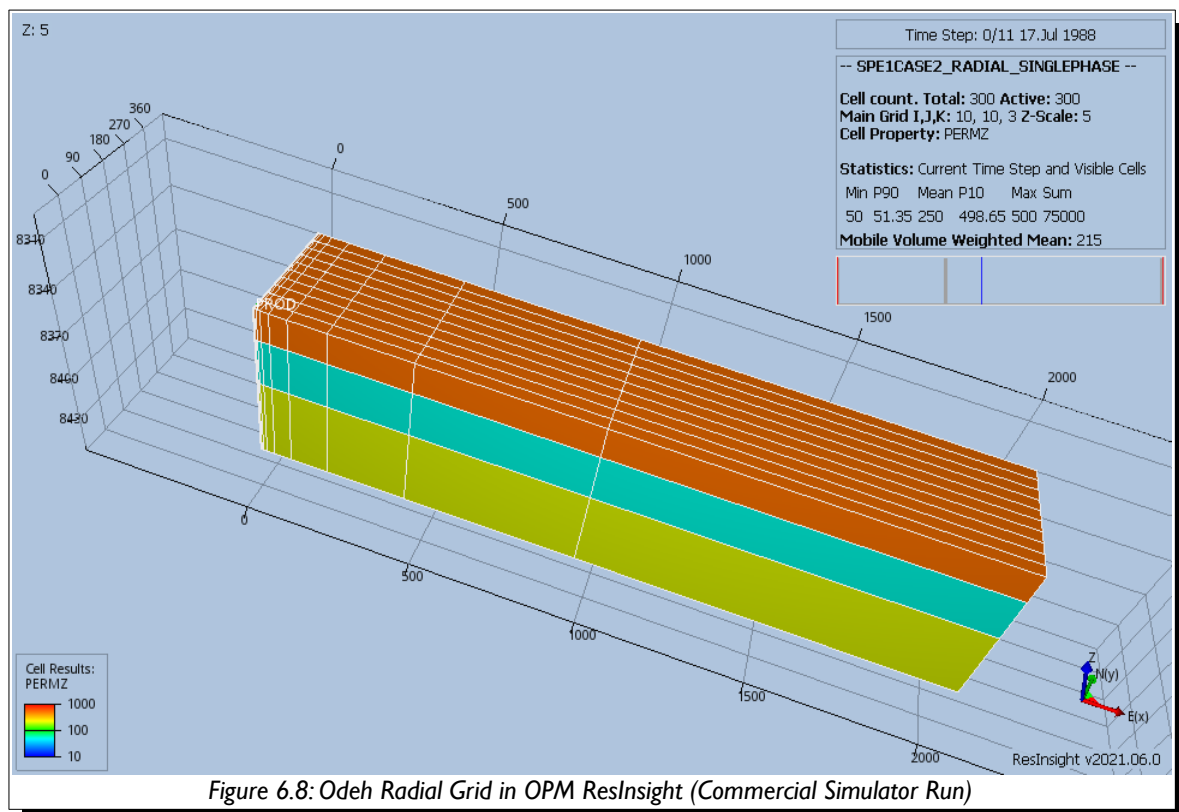
```
--      DEFINE GRID BLOCK PERMTHT DATA FOR ALL CELLS (NX x NY x NZ = 300)
--
PERMTHT
100*500.0   100*50.0   100*200.0   /
--
--      DEFINE GRID BLOCK Z DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 300)
--
PERMZ
100*500.0   100*50.0   100*200.0   /
```

The above keywords define all the properties required for the RUNSPEC and GRID sections for this type of grid geometry.

Note

Radial grids are currently not fully implemented in OPM Flow as the completion of the circle in the THETA direction is not supported. For reference only, this feature is activated via the COMPLETE parameter on the COORDSYS keyword in the GRID section.

OPM ResInsight is the 3D visualization software that is used to view the grid block property data and OPM Flow results (grid block pressures, fluid saturations, etc.); unfortunately, the software does not render radial grids correctly, and currently there is no plan to implement this feature. Figure 6.8 shows how the Odeh¹¹⁸ radial model is rendered in OPM ResInsight and this should be compared to Figure 6.7 which displays the correct rendering of the grid.



¹¹⁸ Comparison of Solutions to a Three-Dimensional Black-Oil Reservoir Simulation Problem by Aziz S. Odeh,– Journal of Petroleum Technology, January 1981.

The previous figure (Figure 6.8) shows how the model would be displayed if the commercial simulator run was loaded into OPM ResInsight. Previously, an OPM Flow run would also be displayed in the same manner.

To overcome this OPM ResInsight display limitation, OPM Flow now converts radial grids to Irregular Corner-Point Grids (Cartesian coordinates) and adjusts the model to reflect radial coordinates. Thus, the OPM Flow radial model has the same pore volumes as the commercial simulator radial model.

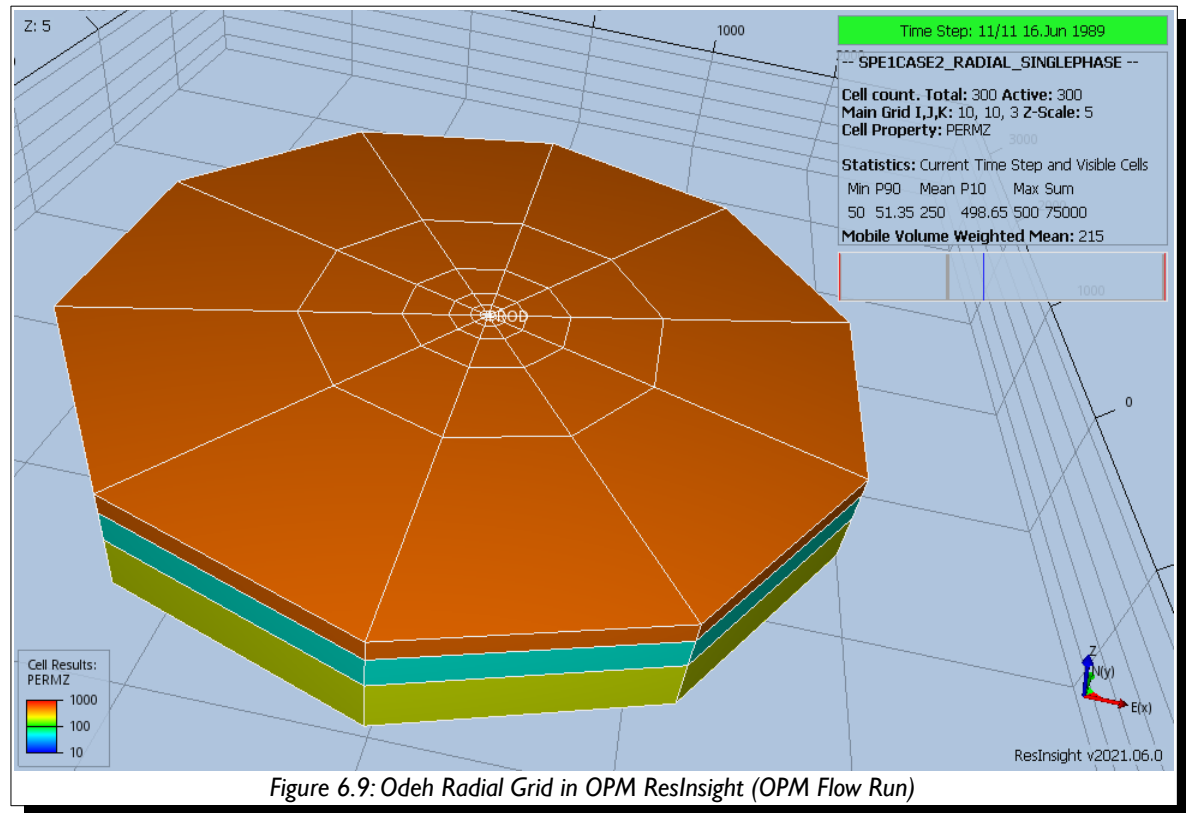


Figure 6.9 shows how the OPM Flow run is now displayed in OPM ResInsight. As will be shown in section 6.2.4 Spider Grids, both Spider and Radial grids will be displayed the same, as they both use Irregular Corner-Point Grid geometry. The difference is that the radial model's pore volumes are adjusted to match the radial geometry pore volumes, whereas, the Spider grid volumes are not adjusted.

6.2.4 SPIDER GRIDS

As mentioned in the previous section, OPM ResInsight does not render radial grids correctly and currently there is no plan to implement this feature. To overcome this limitation, Spider Grids have been implemented. They convert standard radial geometry keywords in the grid section into an Irregular Corner-Point Grid, which can then be visualized in OPM ResInsight in a more intuitive form. The only difference is that in the **RUNSPEC** section the **SPIDER** keyword is used instead of the **RADIAL** keyword.

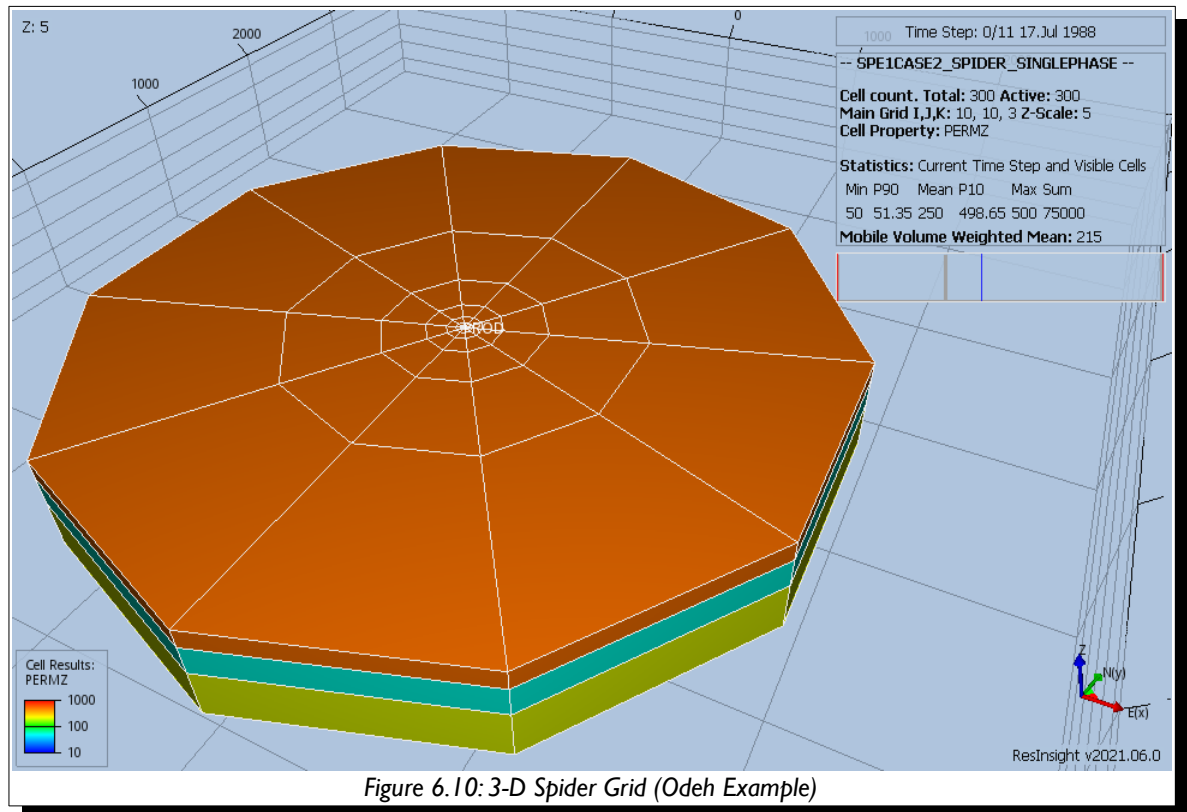


Figure 6.10 illustrates the spider grid for the Odeh¹¹⁸ example, which is not dissimilar to the conventional radial model shown in Figure 6.7 and is identical to OPM Flow's radial model implementation shown in Figure 6.9. Note that Figure 6.7 was not generated by OPM ResInsight,

Naturally, there will be differences in the results between radial and spider grid formulations, but the user should use their own judgement if the differences are relevant.

Again, Figure 6.10 is based on a 10 x 10 x 3 grid, where NX is the R dimensions, NY the THETA dimensions and NZ the z dimensions on the DIMENS keyword in the RUNSPEC section. The only difference between this example and the previous example for radial grids is that the **RADIAL** keyword has been replaced by the **SPIDER** keyword in the RUNSPEC section, all the other keywords are exactly the same.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--          MAX      MAX      MAX
--          NDIVIX   NDIVIY   NDIVIZ
DIMENS
10          1          3                                     /
--
--          DEFINE SPIDER GRID GEOMETRY (OPM FLOW RADIAL GRID KEYWORD)
--
SPIDER
-- =====
--
-- GRID SECTION
--
-- =====
GRID
--
--          INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD
0.25
/
--
--          DEFINE GRID BLOCK SIZES IN THE R DIRECTION
--
DRV
1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.08 /
--
--          DEFINE GRID BLOCK SIZES IN THE THETA DIRECTION (BASED ON NY = 10)
--
DTHETAV
1*45                                     /
--
--          DEFINE GRID BLOCK Z DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 30)
--
DZ
10*20.0   10*30.0   10*50.0                                     /
--
--          DEFINE GRID BLOCK TOPS FOR THE TOP LAYER (NX=10,NY=1,and NZ=3)
--
TOPS
10*8325                                     /
```

The rock property data required to complete the GRID section is as follows:

```
--
--          DEFINE POROSITY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 30)
--
PORO
30*0.300                                     /
--
--          DEFINE GRID BLOCK PERMR DATA FOR ALL CELLS (BASED ON NR x NY x NZ = 30)
--
PERMR
10*500.0   10*50.0   10*200.0                                     /
```

```
--
--      DEFINE GRID BLOCK PERMTHT DATA FOR ALL CELLS (NX x NY x NZ = 30)
--
PERMTHT      10*500.0      10*50.0      10*200.0      /
--
--      DEFINE GRID BLOCK Z DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 300)
--
PERMZ        10*500.0      10*50.0      10*200.0      /
```

The above keywords define all the properties required for the GRID section for this type of grid geometry.

6.2.5 ROCK PROPERTIES

Regardless of the grid type used to define the structural component of the model, certain static properties must be specified to complete the grid definition. These include identifying active and inactive grid blocks, porosity, permeability, and reservoir quality through the net-to-gross fraction ("NTG"). These parameters must be set for each cell in the model

Property	Description	Cartesian And Irregular Corner-Point Grids Keywords	Radial Grid Keywords
Active and Inactive cells	Defines if a cell in the model is active by setting the ACTNUM property for a cell to either one or inactive by setting the value to zero. Cells that are inactive in the model are ignored computationally and can act as barriers to flow. Thus, a shale in a conventional reservoir is normally treated as non-reservoir and is made inactive either by setting the ACTNUM , PORO , or NTG to zero for the cells representing the shale.	ACTNUM	
Porosity	Porosity is a measure of the space in a reservoir rock. It is defined as the fraction of the total bulk volume of the rock not occupied by solids, that is it is the fraction of the cell that is porous and contains the reservoir fluids.	PORO	
Reservoir Quality	Reservoir quality of the cell in terms of the gross volume derived from the structural grid and the net volume available for fluid flow in the model is expressed as a fraction from zero to one. A zero values means the cell does not contribute to flow and therefore is made inactive. A value of one means the gross and net volumes are identical for the cell	NTG	
Permeability	Permeability is a measure of the ease with which a fluid will flow through a porous medium. In numerical models permeability is dependent on the direction of flow, that is x, y, and z directions in Cartesian and Irregular Corner-Point Grids, and the radial, theta, and z directions in radial grids. There are various formulations for permeability, absolute permeability, effective permeability, gas permeability, liquid permeability, etc., and the values are saturation dependent. Thus, values entered should be consistent with the relative permeability entered in the PROPS section. Normally Kair (Sg=1.0) should be entered for the cell permeability and the values may or not be corrected for overburden or humidity drying effects. Correcting for liquid flow and saturation end points etc., is accomplished by the relative permeability curves. For example, if Kair (Sg=1.0) has been entered for the cell permeability when Krg (Sg=1-Swc) should be less than one.	PERMX PERMY PERMZ	PERMR PERMTHT PERMZ

Table 6.1: Key Static Grid Properties

Note

Static grid properties are frequently generated from a static earth model using petrophysical evaluation of the well logs and propagated through to the model based on a variety of geostatistical techniques.

Petrophysical evaluations are conducted in either the “Total” or the “Effective” porosity domain, and it is important that all the rock property data entered into the model is of the same basis. It is not important which porosity domain is used, as long as all the data is in the same domain.

Pore volume and transmissibility are common terms in the reservoir simulation vernacular. Pore volume is self-explanatory, that is, given the grid property data the pore volume for each cell is calculated using:

$$PV = \text{Cell Gross Volume} \times \text{PORO} \times \text{NTG} \times \text{ACTNUM} \quad (6.1)$$

Where

PV	= the pore volume of a cell,
Cell Gross Volume	= the gross volume (or bulk volume) calculated from the structural parameters of the cell,
PORO	= cell porosity,
NTG	= cell net-to-gross ratio, and
ACTNUM	= active and inactive cell indicator.

Any cell with a pore volume equal to zero is made inactive automatically in the model. However, there may be some cells that have small pore volumes that may negatively impact computational performance of the model. If this is the case then the MINPV and MINPORV keywords in the GRID section can be used to make these cells inactive.

There has been a trend in the industry in recent years to not apply petrophysical cut-offs to determine net volumes in static models. This results in large models with numerous cells with very low porosity values (less than 0.01 for example) and corresponding very low permeabilities. The theory behind this approach is that the numerical model will determine the effective (or net) reservoir. However, the approach is questionable as by not applying net pay cut-offs, the in-place and recoverable volumes may not satisfy Reserve reporting requirements and guidelines to various agencies. Secondly, although this may be appropriate in unconventional reservoirs, as all the cells in the model will have similar values of porosity and permeability, but in conventional reservoirs this methodology will lead to severe computational issues when attempting to run the model, due to very tight cells being next to relative high permeability cells. Again, the MINPV and MINPORV keywords can be used to resolve this issue.

Transmissibility on the other hand is more complex as it relates the flow from one cell face to another cell face and is a function of the area open to flow, the direction of flow, the permeability, saturation, and viscosity of the phases flowing between the cells. For single phase flow in a Cartesian grid the x-direction transmissibility is of the form:

$$T_{x_{i+1/2,j}} = \left[\frac{k_x h (\Delta y)}{\mu (\Delta x)} \right]_{i+1/2,j} \quad (6.2)$$

As transmissibility is a property of the flow between two cell faces, not a block centered grid cell property like porosity or permeability, then the nomenclature for transmissibility is different. In OPM Flow, the transmissible of cell face $T_x(i, j, k)$ is the transmissibility between cells (i, j, k) and $(i+1, j, k)$. In some simulators it would be between (i, j, k) and $(i-1, j, k)$. This is important to note if manual modifications to cell connections are to be made in the model.

Modifications to basic grid property data (porosity, permeability, etc.) can only be done in the [GRID](#) section, thereafter only the calculated pore volumes and transmissibilities are available for adjustment in the [EDIT](#) section. Some engineers use the [GRID](#) section for the basic model formulation, and the [EDIT](#) section to document the changes to the “base” model using the PORV,TRANX,TRANY,TRANZ, etc. keywords.

6.3 KEYWORD DEFINITIONS

6.3.1 ACTNUM – SET THE STATUS OF A GRID BLOCK TO ACTIVE OR INACTIVE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **ACTNUM** keyword specifies which grid blocks are either active or inactive. A grid block is automatically set to inactive if its pore volume is less than the value entered using the **MINPV** keyword. The **ACTNUM** keyword can be used to also make blocks with pore volumes greater than **MINPV** inactive.

No.	Name	Description	Default
I	ACTNUM	An array of integers equal to either 0 or 1 that define the activity of each cell in the model. A value of 0 indicates the cell is inactive. Grid blocks are ordered with the I index cycling fastest, followed by the J and K indices. Repeat counts may be used, for example 20*I.	I

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** keyword.
- 2) The keyword is terminated by a "/".

Table 6.2: ACTNUM Keyword Description

Examples

The example below sets several cells to be inactive for a 4 x 5 x 2 model.

```
ACTNUM
-- Layer 1
0 0 1 1
0 0 1 1
1 1 1 1
1 1 1 1
1 1 1 1
-- Layer 2
1 1 1 1
1 1 1 1
1 1 1 1
1 1 1 1
0 0 0 0
/
```

Alternatively the **EQUALS** keyword could be employed to accomplish the same task, that is:

```
--          -- ARRAY      CONSTANT --          ----- BOX -----
--          I1  I2    J1  J2    K1  K2
EQUALS
      'ACTNUM'      1.0000      1*  1*    1*  1*    1*  1* / SET ACTIVE CELLS
      'ACTNUM'      0.0000      1   2     1   2     1   1 / SET INACTIVE CELLS
      'ACTNUM'      0.0000      1   4     5   5     2   2 / SET INACTIVE CELLS
/
```

6.3.2 ADD – ADD A CONSTANT TO A SPECIFIED ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **ADD** keyword adds a constant to a specified array or part of an array. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **ADD** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the property to be modified.	None
2	CONSTANT	An integer or real value to be added to the ARRAY in the same units as the ARRAY property.	0
3	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal 1 and less than or equal to I2 and NX.	1
4	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX	NX
5	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal 1 and less than or equal to J2 and NY.	1
6	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
7	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1
8	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.3:ADD Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.4. Cells colored red indicate arrays that are not currently supported by OPM Flow operations.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

ADD Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANY	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TRLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	DILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.4: ADD Keyword Applicable Arrays by Section

Example

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
ADD
      PERMX      20.000      1*  1*   1*  1*   1*  1* / ADD 20 mD TO PERMX
/
```

The above example adds 20 md to the [PERMX](#) property array throughout the model.

6.3.3 ADDREG – ADD A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **ADDREG** keyword adds a constant to a specified array or part of an array based on cells with a specific region number. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **ADDREG** keyword is read by the simulator. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **ADDREG** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None
2	CONSTANT	An integer or real value to be added to the ARRAY in the same units as the ARRAY property for a given REGION.	0
3	REGION NUMBER	REGION NUMBER is a positive integer representing the region for which the CONSTANT in (2) should be applied.	None
4	REGION ARRAY	The REGION ARRAY to use for applying the CONSTANT in (2) based on the REGION NUMBER in (3). REGION ARRAY can have the following values: 1) F for the FLUXNUM array 2) M for the MULTNUM array 3) O for the OPERNUM array	M

Notes:

- Where the REGION NUMBER should be less than or equal to the maximum number of regions as defined on the **REGDIMS** keyword for the **FIPNUM** and **OPERNUM** arrays or the **GRIDOPTS** keyword for the **MULTNUM** array in the **RUNSPEC** section.
- If the **GRIDOPTS** keyword is not present in the **RUNSPEC** section or if the **GRIDOPTS** keyword is present but the maximum number of **MULTNUM** regions (NRMULT) equals zero then REGION ARRAY will default to F for the **FLUXNUM** array.
- Each record must be terminated by a "/" and the keyword is terminated by a "/".

Table 6.5: ADDREG Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.6. Cells colored red indicate arrays that are not supported by OPM Flow operations.

ADDREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANX	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFEX	SGU	PVTNUM	TBLK		

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

ADDREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	DILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.6: ADDREG Keyword Applicable Arrays by Section

Example

```
--
-- FIRST DEFINE THE PROPERTY ARRAYS AND MULTNUM ARRAYS FOR 10 X 10 X 20 MODEL
--
-- ARRAY      CONSTANT --  ----- BOX -----
--                                I1  I2   J1  J2   K1  K2
EQUALS
  PORO        0.2000      1*  1*   1*  1*   1*  1* / PORO TO 0.20 IN MODEL
  PERMX       100.00      1*  1*   1*  1*   1*  1* / PERMX TO 0.10 IN MODEL
  MULTNUM      1          1*  1*   1*  1*   1*  1* / MULTNUM IN MODEL
  MULTNUM      2          1*  5    1   5    6   6 / MULTNUM IN MODEL
  MULTNUM      3          1*  1*   1*  1*   10  10 / MULTNUM IN MODEL
/
--
-- NOW RESET PORO AND PERMX BASED ON THE MULTNUM REGION NUMBER
--
-- ADD A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER
--
-- ARRAY      CONSTANT  REGION  REGION ARRAY
--              VALUE    NUMBER   M / F / O
ADDREG
  PORO        0.050     1        M                /
  PORO        0.100     2        M                /
  PORO       -0.050     3        M                /
  PERMX       25.00     1        M                /
  PERMX      100.0      2        M                /
  PERMX     -50.00     3        M                /
/
```

The example first defines the [PORO](#) and [PERMX](#) property arrays for the model and then sets the [MULTNUM](#) array to 1 for all cells in the model, after which selected areas of model are assigned various [MULTNUM](#)

integer values. The [ADDREG](#) can then be invoked to add or subtract constant values from the [PORO](#) and [PERMX](#) arrays for the various [MULTNUM](#) regions.

6.3.4 ADDZCORN – Add a Constant to the ZCORN Depth Array

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [ADDZCORN](#) keyword adds a constant to the [ZCORN](#) array or part of the array based on cells defined in the specified input box. The constant can be real or integer and can be negative or positive.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.5 AMALGAM – DEFINE LGR AMALGAMATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [AMALGAM](#) keyword defines a Cartesian Local Grid Refinements (“LGR”) amalgamations, that is merging several LGRs into one amalgamated [LGR](#).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.6 AQUANCON – DEFINE ANALYTICAL CONNECTIONS TO THE GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **AQUANCON** keyword defines how analytical aquifers are connected to the simulation grid, this includes the Carter-Tracy, Fetkovich and Constant Flux analytical aquifers, all of which are implemented in OPM Flow. Carter-Tracy analytical aquifers are characterized by the **AQUCT** keyword in the **GRID** section and Fetkovich analytical aquifers are defined by the **AQUFETP** keyword in the **SOLUTION** section. Finally, the Constant Flux aquifer is defined by the **AQUFLUX** keyword in **SOLUTION** section.

Note that numerical aquifers are connected to the grid using the **AQUCON** keyword in the **GRID** section and that both aquifer types dimensions are declared by the **AQUDIMS** keyword in the **RUNSPEC** section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	AQUID is a positive integer greater than or equal to one and less than the maximum number of analytical aquifers as defined by the NANAQU variable on the AQUDIMS keyword in the RUNSPEC section, that defines the aquifer to be connected to the grid.			None
2	I1	A positive integer that defines the lower bound of the cells in the I-direction to be connected to the aquifer and must be greater than or equal to one and less than or equal to I2 and NX.			I
3	I2	A positive integer that defines the upper bound of the cells in the I-direction to be connected to the aquifer and must be greater than or equal to I1 and less than or equal to NX			NX
4	J1	A positive integer that defines the lower bound of the cells in the J-direction to be connected to the aquifer and must be greater than or equal to one and less than or equal to J2 and NY.			I
5	J2	A positive integer that defines the upper bound of the cells in the J-direction to be connected to the aquifer and must be greater than or equal to J1 and less than or equal to NY.			NY
6	K1	A positive integer that defines the lower bound of the cells in the K-direction to be to be connected to the aquifer and must be greater than or equal to one and less than or equal to K2 and NZ.			I
7	K2	A positive integer that defines the upper bound of the cells in the K-direction to be connected to the aquifer and must be greater than or equal to K1 and less than or equal to NZ.			NZ
8	AQUFACE	AQUFACE is a character string that sets the connection “face” of the cells declared by this record and should be set to one of the following: 1) X+,Y+, or Z+ for the positive direction, or X-,Y- or Z- for the negative direction transmissibilities. 2) I+, J+, or K+ for the positive direction, or I-, J- or K- for the negative direction transmissibilities.			None
9	AQUFLUX	AQUFLUX is a positive real value that sets the fraction of the total influx between the aquifer and the defined cells declared on this keyword. If defaulted the cell face area for each cell is applied and if a value is declared then this value is applied to all cells declared by this record.			I*

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

No.	Name	Description			Default
		Field	Metric	Laboratory	
		ft ²	m ²	cm ²	
10	AQUCOEF	AQUCOEF is a real positive values that scales the calculated connection between the aquifer and the cells declared on this record.			1.0
		dimensionless	dimensionless	dimensionless	
11	AQUOPT	AQUOPT is a character string that sets the cell face connection and should be set to one of the following: 1) YES: Aquifer connections <u>can adjoint</u> to active cells allowing for connections inside the reservoir grid. It is not recommended to use this option without thoroughly checking the connections in the model. 2) NO: Aquifer connections <u>cannot adjoint</u> to active cells preventing connections inside the reservoir grid. This is the recommended and the default value.			NO

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the [DIMENS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record must be terminated by a "/" and the keyword is terminated by a "/".

Table 6.7: AQUANCON Keyword Description

Note

If the [AQUANCON](#) keyword has been utilized in the run deck then OPM Flow will write the [AQUIFERA](#) array to the *.INIT file in order to visualize the aquifer connections in OPM ResInsight.

This is accomplished by setting the [AQUIFERA](#) value to 2^{^(AQUID-1)} for cells connected to aquifer [AQUID](#). If a cell is connected to multiple analytical aquifers then [AQUIFERA](#) is summed for all aquifers connected to a cell. Note that connecting cells to multiple aquifers is best avoided.

Example

The following example defines aquifer number one connected to the I+ face of various cells in the model.

```
--
--
--              ANALYTIC AQUIFER CONNECTION
--
--      ID      ----- BOX -----  CONNECT  AQF   AQF   ADJOIN
--      NUMBER I1  I2   J1  J2   K1  K2   FACE    INFLX MULTI CELLS
AQUANCON
1         57  57   28  36   46  58   'I+'    1*   1*   'NO'   /
1        111 111   38  41   22  31   'I+'    1*   1*   'NO'   /
1         96  96   44  49   22  31   'I+'    1*   1*   'NO'   /
1         43  43   28  35   54  58   'I+'    1*   1*   'NO'   /
1         98  98   38  42   32  40   'I+'    1*   1*   'NO'   /
1         79  79   41  67    5  11   'I+'    1*   1*   'NO'   /
1         61  61   48  72   12  17   'I+'    1*   1*   'NO'   /
/
```

See the [AQUCT](#) keyword in the [GRID](#) section for a complete example on defining and connecting a Carter-Tracy aquifer to a simulation grid.

6.3.7 AQUICON – DEFINE NUMERICAL AQUIFER CONNECTIONS TO THE GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

AQUICON keyword defines how numerical aquifers are connected to the simulation grid and these type of aquifers are characterized by the **AQUNUM** keyword in the **GRID** section. Analytical aquifers are connected to the simulation grid by the **AQUANCON** keyword in the **GRID** section, this includes the Carter-Tracy and Fetkovich analytical aquifers, both of which are implemented in OPM Flow. Both aquifer types dimensions are declared by the **AQUDIMS** keyword in the **RUNSPEC** section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	AQUID is a positive integer greater than or equal to one and less than or equal to the maximum number of numerical aquifers as defined by the MXNAQN variable on the AQUDIMS keyword in the RUNSPEC section, that defines the aquifer to be connected to the grid.			None
2	I1	A positive integer that defines the lower bound of the cells in the I-direction to be connected to the aquifer and must be greater than or equal to one and less than or equal to I2 and NX.			I
3	I2	A positive integer that defines the upper bound of the cells in the I-direction to be connected to the aquifer and must be greater than or equal to I1 and less than or equal to NX			NX
4	J1	A positive integer that defines the lower bound of the cells in the J-direction to be connected to the aquifer and must be greater than or equal to one and less than or equal to J2 and NY.			J
5	J2	A positive integer that defines the upper bound of the cells in the J-direction to be connected to the aquifer and must be greater than or equal to J1 and less than or equal to NY.			NY
6	K1	A positive integer that defines the lower bound of the cells in the K-direction to be to be connected to the aquifer and must be greater than or equal to one and less than or equal to K2 and NZ.			K
7	K2	A positive integer that defines the upper bound of the cells in the K-direction to be connected to the aquifer and must be greater than or equal to K1 and less than or equal to NZ.			NZ
8	AQUFACE	AQUFACE is a character string that sets the connection “face” of the cells declared by this record and should be set to one of the following: 1) X+,Y+, or Z+ for the positive direction, or X-,Y- or Z- for the negative direction transmissibilities. 2) I+, J+, or K+ for the positive direction, or I-, J- or K- for the negative direction transmissibilities.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	AQUMULT	AQUMULT is a positive real number greater than or equal to zero that scales the OPM Flow calculated transmissibility between the AQUID aquifer and the grid cell connections defined by this record. The default value of one sets the transmissibility between the aquifer and grid cells to the OPM Flow calculated value.			1.0
		dimensionless	dimensionless	dimensionless	
10	AQUOPT1	AQUOPT1 is a defined integer value set to either zero or one, that defines the area to be used in calculating the connection transmissibility between the aquifer and the grid cells: 1) A value of zero means the cross-sectional defined on the AQUNUM keyword will be used, whereas, 2) A value of one means the cross-sectional defined by the grid cell connections defined by this record will be used.			0
		dimensionless	dimensionless	dimensionless	
11	AQUOPT2	AQUOPT2 is a character string that sets the cell face connection and should be set to one of the following: 1) YES: Aquifer connections <u>can adjoin</u> to active cells allowing for connections inside the reservoir grid. It is not recommended to use this option without thoroughly checking the connections in the model. 2) NO: Aquifer connections <u>cannot adjoin</u> to active cells preventing connections inside the reservoir grid. This is the recommended and the default value.			NO
12	VEOPT1	Vertical Equilibrium Option Number 1– Not Used			1
13	VEOPT2	Vertical Equilibrium Option Number 2– Not Used			1
Notes: 1) Where NX, NY and NZ are the dimensions of the model as defined on the DIMENS keyword in the RUNSPEC section. 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.					

Table 6.8: AQUICON Keyword Description

Note

If the [AQUICON](#) keyword has been utilized in the run deck then OPM Flow will write the AQUIFERN array to the *.INIT file in order to visualize the aquifer connections in OPM ResInsight.

This is accomplished by setting the AQUIFERN value to $2^{(AQUID-1)}$ for cells connected to aquifer AQUID. If a cell is connected to multiple numerical aquifers then AQUIFERN is summed for all aquifers connected to a cell. Note that connecting cells to multiple aquifers is best avoided.

Finally, for cells representing the numerical aquifers themselves, AQUIFERN is set to minus AQUID.

Example

The following example defines numerical aquifer number one connected to the I+ face of various cells in the model.

```
--
--          NUMERICAL AQUIFER CONNECTIONS
--
--          ID          BOX          CONNECT  TRANS  TRANS  ADJOIN
--          NUMBER I1  I2    J1  J2    K1  K2    FACE    MULT  OPTN  CELLS
AQUICON
1          57  57    28  36    46  58    'I+'    1*    1*    'NO'    /
1          111 111    38  41    22  31    'I+'    1*    1*    'NO'    /
1          96  96    44  49    22  31    'I+'    1*    1*    'NO'    /
1          43  43    28  35    54  58    'I+'    1*    1*    'NO'    /
1          98  98    38  42    32  40    'I+'    1*    1*    'NO'    /
1          79  79    41  67     5  11    'I+'    1*    1*    'NO'    /
1          61  61    48  72    12  17    'I+'    1*    1*    'NO'    /
/
```

See the [AQUENUM](#) keyword in the [GRID](#) section for a complete example on defining and connecting a numerical aquifer to a simulation grid.

6.3.8 AQUCT – DEFINE CARTER-TRACY ANALYTICAL AQUIFERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [AQUCT](#) keyword defines Carter-Tracy^{119 120} analytical aquifers, the properties of the aquifer, including the Carter-Tracy aquifer influence function associated with the aquifer, defined by the [AQUCTAB](#) keyword in the [PROPS](#) section.

Each row entry in the [AQUCT](#) keyword defines one Carter-Tracy aquifer.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	A positive integer greater than or equal to one and less than or equal to NANAQU on the AQUIDIMS keyword in the RUNSPEC section, that defines the Carter-Tracy aquifer number.			I
2	DATUM	DATUM is a single positive value that defines the Carter-Tracy reference datum depth for PRESS.			None
		feet	m	cm	
3	PRESS	PRESS is a single positive value that defines the aquifer pressure at DATUM. If PRESS is defaulted then the simulator will set the aquifer's initial reservoir pressure to be in equilibrium with the cells the aquifer is contacted to.			I *
		psia	barsa	atma	
4	PERM	PERM is a real positive number that assigns the permeability to the aquifer.			None
		mD	mD	mD	
5	PORO	PORO is a real positive number greater than zero and less than or equal to one that assigns the porosity to the aquifer.			None
		dimensionless	dimensionless	dimensionless	
6	RCOMP	RCOMP is a real number defining the total (rock and water) compressibility (Ct) at the DATUM pressure.			None
		1/psia	1/barsa	1/atma	
7	RE	RE is a real positive number that defines the Carter-Tracy aquifer external radius.			None
		feet	m	cm	
8	DZ	DZ is a real positive number that defines the Carter-Tracy aquifer average net thickness.			None
		feet	m	cm	

¹¹⁹ Carter, R. D. and Tracy, G.W. "An Improved Method for Calculating Water Influx," *Transactions of AIME* (1960) 219, 215-417.

¹²⁰ Van Everdingen, A. & Hurst, W., *The Application of the Laplace Transformation to Flow Problems in Reservoirs. Petroleum Transactions, AIME* (December, 1949).

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	ANGLE	ANGLE is a real positive number that defines the angle of influence, that is the angular connection between the aquifer and the hydrocarbon reservoir. A value of 360° degrees, the default value, indicates that the aquifer completely surrounds the hydrocarbon reservoir.			360.0
		degrees	degrees	degrees	
10	PVTNUM	PVTNUM is positive integer greater than zero and less than the NTPVT variable on the TABDIMS keyword in the RUNSPEC section, that defines the PVTW table allocated to the Carter-Tracy aquifer.			1
11	AQUTAB	AQUTAB is positive integer greater than zero and less than the NIFTBL variable as declared on the AQUDIMS keyword in the RUNSPEC section, that defines the AQUTAB table allocated to this Carter-Tracy aquifer. The default value of one sets the internal infinite acting Carter-Tracy aquifer influence table not the first table in the AQUTAB keyword in the PROPS section The first table in the AQUTAB keyword is considered to be table number two. The internal table, AQUTAB equal to one, is based on the Radial Flow, Constant Pressure and Constant Terminal Rate Cases for Infinite Reservoirs (Table I) in Van Everdingen and Hurst's ¹²¹ paper.			1
12	SALTCON	SALTCON is a real positive number that defines the initial salt concentration in the aquifer, for when with simulator's Brine Model has been activated via the BRINE keyword in the RUNSPEC section.			0.0
		lb/stb	kg/sm ³	gm/scc	
13	TEMP	TEMP is a real positive number that defines the initial temperature of the aquifer at DATUM for use with OPM Flow's thermal option. The THERMAL keyword in the RUNSPEC section must be activated to use this option.			1*
		°F	°C	°C	
Notes:					
1) The keyword is followed by up to NANAQ records as defined on the AQUDIMS keyword in the RUNSPEC section					
2) Each record is terminated by a “/” and the keyword should be terminated by a “I”.					

Table 6.9: AQUCT Keyword Description

Note

OPM Flow includes the infinite acting Carter-Tracy aquifer influence table as a default for table number one; thus data entered on [AQUTAB](#) keyword starts from table number two.

In order to full define a Carter-Tracy aquifer one has to define the aquifer properties via the [AQUCT](#) keyword, the Carter-Tracy influence function via the [AQUTAB](#) keyword in the [PROPS](#) section ([AQUTAB – Define Carter-Tracy Aquifer Influence Functions](#)), if the default infinite acting table is not being employed, and

¹²¹ Van Everdingen, A. & Hurst, W., The Application of the Laplace Transformation to Flow Problems in Reservoirs. *Petroleum Transactions, AIME* (December, 1949).

finally, how the aquifer is connected to the reservoir using the [AQUANCON](#) keyword in the [GRID](#) or [SOLUTION](#) sections.

Example

Given the following grid and aquifer dimensions in the [RUNSPEC](#) section:

```
--
--      MAX      MAX      MAX
--      NDIVIX  NDIVIY  NDIVIZ
DIMENS
      20      1      5
/

--      AQF      AQF      AQF      AQF      AQF      AQF      AQF      AQF
--      MXAQN  MXNAQC  NIFTBL  NRIFTB  NANAQU  NCAMAX  MXNALI  MXAAQL
AQUDIMS
      1*      1*      5      100      1      1*      1*      1*
/
```

And the [AQUTAB](#) keyword in the [PROPS](#) section

```
--
--      CARTER-TRACY AQUIFER INFLUENCE TABLES
--      (STARTS FROM TABLE NO. 2, AS DEFAULT IS TABLE NO. 1)
--
AQUTAB
--      DIMLESS      DIMLESS
--      TIME          PRESSURE
--      -----
--      0.01          0.112
--      0.05          0.229
--      0.10          0.315
--      0.15          0.376
--      0.20          0.424
--      0.22          0.443
--      0.24          0.459
--      0.26          0.476
--      0.28          0.492
--      0.30          0.507
--      0.32          0.522
--      0.34          0.536
--      0.36          0.551
--      0.38          0.565
--      0.40          0.579
--      0.42          0.593
--      0.44          0.607
--      0.46          0.621
--      0.48          0.634
--      0.50          0.648
--      0.60          0.715
--      0.70          0.782
--      0.80          0.849
--      0.90          0.915
--      1.00          0.982
--      2.00          1.649
--      3.00          2.316
--      5.00          3.649
--      10.00         6.982
--      20.00        13.649
--      30.00        20.316
--      50.00        33.649
--      100.00       66.982
--      200.00      133.649
--      300.00      200.316
```

500.00333.649

1000.00666.982 /

The Carter-Tracy aquifer is defined in the [GRID](#) or [SOLUTION](#) sections as:

```
-- =====
--
-- SOLUTION SECTION
-- =====
SOLUTION
--
--              CARTER-TRACY AQUIFER DESCRIPTION
--
--      ID      DATUM      AQF      AQF      AQF      AQF      AQF      AQF      AQF      INFL      PVT      AQU
--      NUM     DEPTH     PRESS     PERM     PORO     RCOMP     RE      DZ      ANGLE     NUM     TAB
--
-- AQUICT
--      1      2000.0     269      100.0     0.30     3.0e-5     330     10.0     360.0     1      2      /
/
```

And the connection of the aquifer is set in the [GRID](#) or [SOLUTION](#) sections as:

```
--
--              ANALYTIC AQUIFER CONNECTION
--
--      ID      ----- BOX -----      CONNECT      AQF      AQF      ADJOIN
--      NUMBER  I1  I2   J1  J2   K1  K2   FACE      INFLX     MULTI     CELLS
--
-- AQUANCON
--      1        1   1    1   1    1   1    J-       1.0      1.0      'NO'
/
```

Here one Carter-Tracy aquifer is connected to a single cell (I, I, I) at the J- face (or X- face) of the cell.

6.3.9 AQUNNC – DEFINE NUMERICAL AQUIFER NON-NEIGHBOR CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [AQUNNC](#) keyword defines Numerical Aquifer Non-Neighbor Connections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.10 AQUNUM – DEFINE NUMERICAL AQUIFER PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [AQUNUM](#) keyword defines the properties of numerical aquifers, including which grid blocks in the model should be utilized as part of the numerical aquifer. Each row entry in the [AQUNUM](#) keyword defines one numerical aquifer. Note that a numerical aquifer may consists of more than one grid cell, in order to better describe the water influx from the aquifer to the grid.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	AQUID is a positive integer greater than or equal to one and less than or equal to the maximum number of numerical aquifers as defined by the MXNAQN variable on the AQUDIMS keyword in the RUNSPEC section, that defines the aquifer to be connected to the grid.			None
2	I	A positive integer that defines the cell in the I-direction that represents the AQUID aquifer, and which must be greater than or equal to one and less than or equal to NX.			None
3	J	A positive integer that defines the cell in the J-direction that represents the AQUID aquifer, and which must be greater than or equal to one and less than or equal to NY.			None
4	K	A positive integer that defines the cell in the K-direction that represents the AQUID aquifer, and which must be greater than or equal to one and less than or equal to NZ.			None
5	AREA	AREA is a real positive value that defines the cross-sectional area of the aquifer used in calculating the aquifer connection transmissibility. The actual transmissibility between the numerical aquifer and the connected grid cell is the harmonic average of the aquifer connection transmissibility and the calculated connected cell transmissibility. The value entered for AREA does not effect the visualization of the cell in OPM ResInsight, as it is only used in calculating the transmissibility. Note that the AQUOPT1 variable on the AQUCON keyword in the GRID section allows one to use the value entered for AREA or to use the grid cell cross-sectional area instead for the transmissibility calculation.			None
		ft ²	m ²	cm ²	
6	LENGTH	LENGTH is a real positive value that defines the length of the numerical aquifer. Similar to the AREA variable, LENGTH is not constrained by the original host cell size and the value entered does not effect the visualization of the cell in OPM ResInsight.			None
		feet	m	cm	
7	PORO	PORO is a real positive number greater than zero and less than or equal to one that assigns the porosity to the numerical aquifer.			None
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	PERM	PERM is a real positive number that assigns the permeability to the numerical aquifer.			None
		mD	mD	mD	
9	DATUM	DATUM is a real positive number that sets the reference datum depth of the numerical aquifer. Similar to the AREA variable, DATUM is not constrained by the original host cell depth and the value entered does not effect the visualization of the cell in OPM ResInsight. If defaulted then the depth of cell (I, J, K) will be used			Cell Depth
		feet	m	cm	
10	PRESS	PRESS is a single positive value that defines the numerical aquifer pressure at DATUM. If PRESS is defaulted then the simulator will set the aquifer's initial reservoir pressure to be in equilibrium with the cells the aquifer is contacted to. This is the preferred manner to initialize the numerical aquifer.			I*
		psia	barsa	atma	
11	PVTNUM	PVTNUM is positive integer greater than zero and less than the NTPVT variable on the TABDIMS keyword in the RUNSPEC section, that defines the PVTW table allocated to the numerical aquifer. If defaulted then the PVT tables allocated to cell (I, J, K) will be used.			Cell PVTNUM
12	SATNUM	SATNUM is positive integer greater than zero and less than the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section, that defines the saturation tables allocated to the numerical aquifer. If defaulted then the saturation tables allocated to cell (I, J, K) will be used.			Cell SATNUM

Notes:

- Where NX, NY and NZ are the dimensions of the model as defined on the [DIMENS](#) keyword in the [RUNSPEC](#) section.
- The keyword is followed by up to MXNAQN records as defined on the [AQUDIMS](#) keyword in the [RUNSPEC](#) section
- Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 6.10: AQUNUM Keyword Description

Numerical aquifers are modeled as one-dimensional, with aquifer flow assumed to be in the direction defined by LENGTH, and flux out of the aquifer is through the cross sectional area defined by AREA. Note that, all the aquifer cells must be isolated from the reservoir cells, with only the [AQUCON](#) connections connecting to the actual reservoir cells.

The values entered on the [AQUNUM](#) keyword are used to calculate the aquifer's pore volume and the transmissibility between the aquifer and the connected cell faces defined on the [AQUCON](#) keyword. Thus:

- The aquifer's pore volume is always calculated from the data entered on the [AQUNUM](#) keyword using $Pore\ Volume = AREA \times LENGTH \times PORO$ and any modifications to the host cell values performed in either the [GRID](#) or [EDIT](#) sections are always ignored for cells declared as numerical aquifers cells.

- 2) For the transmissibility calculation either the cross-sectional area (AREA) defined on the [AQUNUM](#) keyword may be used or the connecting cell cross-sectional area by setting the AQUOPT1 variable on the [AQUCON](#) keyword.

In order to fully define a numerical aquifer one has to define the aquifer properties via the [AQUNUM](#) keyword, and how the aquifer is connected to the reservoir using the [AQUCON](#) keyword in the [GRID](#) or [SOLUTION](#) sections.

Note

If the [AQUCON](#) keyword has been utilized in the run deck then OPM Flow will write the AQUIFERN array to the *.INIT file in order to visualize the aquifer connections in OPM ResInsight.

This is accomplished by setting the AQUIFERN value to $2^{(AQUID-1)}$ for cells connected to aquifer AQUID. If a cell is connected to multiple numerical aquifers then AQUIFERN is summed for all aquifers connected to a cell. Note that connecting cells to multiple aquifers is best avoided.

Finally for cells representing the numerical aquifers themselves, AQUIFERN is set to minus AQUID.

Using one aquifer cell should generally be sufficient, provided the aquifer properties are constant. However, employing multiple cells may be appropriate if the aquifer properties vary with distance or depth, which is not uncommon. Secondly, having multiple cells may help to minimize throughput-related convergence problems by increasing the pore volumes exponentially away from the reservoir cells. Typically, three to five aquifer cells are employed under these circumstances.

The aquifer pore volume can be used to define the amount of pressure support; whereas, the aquifer transmissibility will influence the responsiveness of the aquifer.

See also the [AQUDIMS](#) keyword in the [RUNSPEC](#) section that defines the numerical aquifer dimensions.

Example

Given the following grid and aquifer dimensions in the [RUNSPEC](#) section:

```
--
--      MAX      MAX      MAX
--      NDIYIX  NDIYIY  NDIYIZ
DIMENS
      96      58      28
--
--      AQF      AQF      AQF      AQF      AQF      AQF      AQF      AQF      AQF
--      MXAQN  MXNAQC NIFTBL NRIFTB NANAQU NCAMAX MXNALI MXAAQL
AQUDIMS
      3      3      1*      1*      1*      1*      1*      1*      1*
```

The following numerical aquifer definition:

```
--
--      NUMERICAL AQUIFER DESCRIPTION
--
--      ID      - LOCATION - AQF      AQF      AQF      AQF      AQF      AQF      AQF      PVT SATNUM
--      NUMBER  I1  J1  K1  AREA      LENGTH  PORO      PERM  DEPTH  PRES  TAB  TAB
--
--      AQUNUM
--      1      1  3  28  5.20E03  2.0E3    0.05    0.3  1*      1*      1*  1*
--      1      1  2  28  5.20E06  2.0E6    0.05    0.3  1*      1*      1*  1*
--      1      1  1  28  5.20E09  2.0E9    0.05    0.3  1*      1*      1*  1*
```

defines one numerical aquifer consisting of three cells. The connection to the grid cells would take the form of:

```
--
--      NUMERICAL AQUIFER CONNECTIONS
--
--      ID      ----- BOX -----  CONNECT  TRANS  TRANS  ADJOIN
--      NUMBER I1  I2   J1  J2   K1  K2   FACE    MULT  OPTN  CELLS
AQUCON
      1          1   1    4  58   28  28   'K+'    1.3   1*    1*      /
      1          2  96    1  58   28  28   'K+'    1.3   1*    1*      /
/
that creates a basal aquifer122.
```

¹²² Basal Aquifer: An aquifer located at the bottom of a geologic unit.

6.3.11
AUTOCOAR – DEFINE AUTO REFINEMENT GRID COARSEN AREA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [AUTOCOAR](#) keyword defines an area in the global grid that should be coarsen for when the [AUTOREF](#) keyword has been declared in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.12 BCCON – DEFINE BOUNDARY CONDITIONS CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **BCCON** keyword defines the grid connections for the boundary conditions.

Together the **BCCON** and **BCPROP** keywords define the boundary conditions for the model, and can be used to set boundary conditions for when external influx or efflux volumes are influencing the reservoir pressure and production history. For example, when the average reservoir pressure remains constant throughout the production period due to water influx, or gas migration from an external source.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	INDEX	A positive integer that identifies the boundary condition.			None
2	I1	A positive integer that defines the lower bound of the grid in the I-direction for which the boundary conditions are to be applied, must be greater than or equal to I1 and less than or equal to I2 and NX.			I
2	I2	A positive integer that defines the upper bound of the grid in the I-direction for which the boundary conditions are to be applied, must be greater than or equal to I1 and less than or equal to NX			NX
3	J1	A positive integer that defines the lower bound of the grid in the J-direction for which the boundary conditions are to be applied, must be greater than or equal to J1 and less than or equal to J2 and NY.			J
4	J2	A positive integer that defines the upper bound of the grid in the J-direction for which the boundary conditions are to be applied, must be greater than or equal to J1 and less than or equal to NY.			NY
5	K1	A positive integer that defines the lower bound of the grid in the K-direction for which the boundary conditions are to be applied, must be greater than or equal to one and less than or equal to K2 and NZ.			K
6	K2	A positive integer that defines the upper bound of the grid in the K-direction for which the boundary conditions are to be applied, must be greater than or equal to K1 and less than or equal to NZ.			NZ
7	DIRECT	A character string that defines the direction to apply the boundary conditions, and should be set to one of the following X, Y, or Z for the positive direction, or X-, Y- or Z- for the negative direction.			None

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.11: BCCON Keyword Description

See also the [AQUFLUX](#) keyword that is supported by OPM Flow in both the [SOLUTION](#) and [SCHEDULE](#) sections, to define a constant flux analytical aquifer.

If the [BCCON](#) and [BCPROP](#) keywords are not present in the input deck, then the boundary conditions for the model are set to be no flow, which is the normal behavior in both OPM Flow and the commercial simulator.

The [BC](#) keyword has been replaced by the [BCCON](#) and [BCPROP](#) keywords.

Examples

The first example shows how to set a constant pressure boundary using TYPE equal to FREE:

```
--
--      DEFINE BOUNDARY CONDITIONS CONNECTION (OPM FLOW GRID KEYWORD)
--
--INDEX  ----- BOX ----- BC
--      I1  I2   J1  J2   K1  K2  DIRC
BCCON
1         1   1    1   1*   1   1*  X-   /
2         1   1*   1   1    1   1*  Y    /
/

--
--      DEFINE BOUNDARY CONDITIONS PROPERTIES (OPM FLOW SCHEDULE KEYWORD)
--
--INDEX  BC      BC      BC      BC      BC
--      TYPE     PHASE   RATE   PRESS   TEMP
BCPROP
1        FREE     1*      1*      1*      1*   /
2        FREE
/
```

With this option it is only necessary to define the boundary cells and all the other parameters (COMPONENT, RATE, PRESS, and TEMP) can be defaulted, as they are ignored when TYPE equals FREE.

The next example is based on NX, NY and NZ equal to 20, 1, 10 respectively, on the [DIMENS](#) keyword in the [RUNSPEC](#) section, and shows how different boundary types can be assigned to different parts of the model.

```
--
--      DEFINE BOUNDARY CONDITIONS CONNECTION (OPM FLOW GRID KEYWORD)
--
--INDEX  ----- BOX ----- BC
--      I1  I2   J1  J2   K1  K2  DIRC
BCCON
1         1   1    1   1    1   10  X-   /
2        20  20    1   1    1   10  X    /
/

--
--      DEFINE BOUNDARY CONDITIONS PROPERTIES (OPM FLOW SCHEDULE KEYWORD)
--
--INDEX  BC      BC      BC      BC      BC
--      TYPE     PHASE   RATE   PRESS   TEMP
BCPROP
1  DIRICHLET      GAS      1*      256.0  100.0 /
2        FREE      4*
/
```

The last example shows how the DIRICHLET boundary condition option may be used:

```
--
--          DEFINE BOUNDARY CONDITIONS CONNECTION (OPM FLOW GRID KEYWORD)
--
--INDEX  ----- BOX ----- BC
--      I1 I2  J1 J2  K1 K2  DIRC
BCCON
1      1  1   1  1*  1  1*  X-  /
2      1  1*  1  1   1  1*  Y   /
/

--
--          DEFINE BOUNDARY CONDITIONS PROPERTIES (OPM FLOW SCHEDULE KEYWORD)
--
--INDEX  BC      BC      BC      BC      BC
--      TYPE      PHASE  RATE  PRESS  TEMP
BCPROP
1      DIRICHLET  WAT     1*     256.0  100.0 /
2      DIRICHLET  WAT     1*     1*      100.0 /
/
```

Here, the first line sets both the pressure and temperature at the boundary, and the second line defaults the pressure entry, so that the simulator calculated initial boundary pressure will be used.

6.3.13 BOUNDARY – DEFINE A BOUNDARY BOX FOR PRINTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *BOUNDARY* keyword defines a rectangular grid for printing various arrays to the output print file (*.PRT); thus, avoiding printing all the elements in the selected array.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.14 BOX – DEFINE A RANGE OF GRID BLOCKS TO ENTER PROPERTY DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

BOX defines a range of grid blocks for which subsequent data is assigned for all the cells in the defined **BOX**. Values are set for cells within the defined box grid using natural reading order, initially along the I-direction then J-direction and finally the K-direction. If fewer values are assigned than exist within the defined block space, then subsequent values are set starting from the next block that was not previously assigned for that property. This is the same behavior as applies to setting grid properties for an unboxed grid.

Note that the **BOX** grid is reset by the keyword **ENDBOX** by resetting the current defined **BOX** to be the whole grid. The keyword can be used for any array and for all grid types.

No.	Name	Description	Default
1	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal to 1 and less than or equal to I2 and NX.	1
2	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX.	NX
3	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal to 1 and less than or equal to J2 and NY.	1
4	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
5	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1
6	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) The keyword is terminated by a "/".

Table 6.12: BOX Keyword Description

See also the **ADD**, **COPY**, **ENDBOX**, **EQUALS**, and **MULTIPLY** keywords can also be used to enter data in a subset of the model.

Examples

```
--
--      DEFINE A BOX GRID FOR THE BOTTOM LAYER OF A 100 X 100 X 20 MODEL
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
      1*   1*   1*   1*   20  20 / SELECT THE BOTTOM LAYER
--
--      DEFINE THE POROSITY AND OTHER PROPERTIES ON THE BOX GRID
--
--      PORO
--      10000*0.300
--
--      /
--      PERMX
--      5000*100.0   5000*75.0
--
--      /
--      NTG
--      10000*0.500
--
--      /
--
--      RESET THE INPUT BOX TO BE THE FULL MODEL
--
--      ENDBOX
```

The above example set the [BOX](#) grid to be the last layer in the model which means that 100 x 100, that is 10,000 values need to entered for each property.

Alternatively, one could use the [EQUALS](#) keyword to accomplish the same thing.

```
-- -- ARRAY      CONSTANT -- ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
'PORO'          0.3000      1*   1*   1*   1*   20  20 / PORO  TO 0.30 IN LAYER 20
'PERMX'         100.00      1*   1*   1    50   20  20 / PERMX TO 100. IN LAYER 20
'PERMX'          75.00      1*   1*   51  100   20  20 / PERMX TO 75.0 IN LAYER 20
'NTG'           0.5000      1*   1*   1*   1*   20  20 / NTG   TO 0.50 IN LAYER 20
/
```

Note

It is good practice to always use the [ENDBOX](#) keyword to reset the input back to the full grid when all the modifications for a sub-grid have been completed.

6.3.15
BTOTALFA – DUAL POROSITY MATRIX TO FRACTURE MULTIPLIER (ALL CELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [BTOTALFA](#) keyword defines a dual porosity matrix to fracture multiplier that is applied to all cells, for when the Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section.

See also the [BTOTALFV](#) keyword in the [GRID](#) section that applies a multipliers to individual cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.16
BTOTALFV – DUAL POROSITY MATRIX TO FRACTURE MULTIPLIER (INDIVIDUAL CELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [BTOTALFV](#) keyword defines a dual porosity matrix to fracture multiplier that is applied to individual cells, for when the Dual Porosity model has been invoked by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section.

See also the BTOTALFAV keyword in the [GRID](#) section that applies a constant multiplier to all cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.17 CARFIN – DEFINE A CARTESIAN LOCAL GRID REFINEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[CARFIN](#) defines a Cartesian Local Grid Refinement (“LGR”) in a cell or a group of cells in the host grid, for when LGRs have been activated for the input deck using the [LGR](#) keyword in the [RUNSPEC](#) section. The keyword marks the start of an [LGR](#) description section and all subsequent keywords between the [CARFIN](#) and [ENDFIN](#) keywords are deemed to be associated with the current [LGR](#) and not the host grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the LGR is being defined.	None
2	I1	A positive integer that defines the lower index of the global or host grid in the I-direction to be refined; must be greater than or equal 1 and less than or equal to I2 and NX on the DIMENS keyword in the RUNSPEC section.	None
3	I2	A positive integer that defines the upper index of the global or host grid in the I-direction to be refined; must be greater than or equal 1 and I1, and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.	None
4	J1	A positive integer that defines the lower index of the global or host grid in the J-direction to be refined; must be greater than or equal 1 and less than or equal to J2 and NY on the DIMENS keyword in the RUNSPEC section.	None
5	J2	A positive integer that defines the upper index of the global or host grid in the J-direction to be refined; must be greater than or equal 1 and J1, and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.	None
6	K1	A positive integer that defines the lower index of the global or host grid in the K-direction to be refined; must be greater than or equal 1 and less than or equal to K2 and NZ on the DIMENS keyword in the RUNSPEC section.	None
7	K2	A positive integer that defines the upper index of the global or host grid in the K-direction to be refined; must be greater than or equal 1 and K1, and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.	None
8	NX	A positive integer value that defines the number of LGR grid blocks in the x direction for Cartesian grids or the number of grid blocks in the r direction for radial grids	None
9	NY	A positive integer value that defines the number of LGR grid blocks in the y direction for Cartesian grids or the number of grid blocks in the theta direction for radial grids.	None
10	NZ	A positive integer value that defines the number of LGR grid blocks in the z direction for both Cartesian and radial grids.	None
11	MXWELS	A positive integer defining the maximum number of wells contained in this LGR .	None

No.	Name	Description	Default
12	HOSTNAME	A character string of up to eight characters in length that defines the host grid name for nested refinements. The default value of "GLOBAL" sets the host name to the global grid, that is for a conventional LGR. A nested refinement is when the HOSTNAME is a previously declared LGR for which the current LGR is specifying a further LGR refinement.	GLOBAL
Notes: 1) The keyword is terminated by a "/".			

Table 6.13: CARFIN Keyword Description

Note that if the Dual Porosity option has been activated by either the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section, then the host grid definition (I1-I2, J1-J2, K1-K2) applies only to the matrix cells; however, the [LGR](#) NZ parameter in this case must include the fracture blocks, similar to the NZ parameter on the [DIMENS](#) keyword. This means that all property data should be entered for both the matrix and fracture cells in the [LGR](#) description.

Example

The example below defines an [LGR](#) in the global grid, named [LGR-OP01](#) with a maximum of one well allowed in the [LGR](#).

```
--
--      CARFIN LGR GRID COMMANDS
--
--      LGR      ----- HOST GRID -----  -- CARFIN GRID --  MAX  HOST
--      NAME      I1  I2  J1  J2  K1  K2      NX   NY   NZ   WELLS  NAME
CARFIN
      LGR-OP01    24  24  87  87   1   50      3    3   50    1    GLOBAL /

ENDFIN
```

Here the one global cell in the areal plane (24, 87) is divided into three [LGR](#) cells in the x-direction and three cells in the y-direction. Since no other property data is given, then the [LGR](#) cells take their properties from the host grid, that is the global grid.

6.3.18 COALNUM – DEFINE THE COAL REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COALNUM](#) keyword defines the coal region numbers for each grid block used with the Coal Bed Methane option (“CBM”). OPM Flow does not have a CBM option; however, the keyword is documented here for completeness.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	COALNUM	COALNUM defines an array of positive integers assigning a grid cell to a particular coal region. The maximum number of COALNUM regions is set by the NTCREG variable on REGDIMS keywords in the RUNSPEC section.	I

Notes:

- 1) A total of NX x NY x NZ integer values, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section, must be specified for the array.
- 2) If a cell is not assigned a [COALNUM](#) region then the default value of I will be used.
- 3) [COALNUM](#) value of 0 sets the cell be a non-coal region.
- 4) The keyword is terminated by a “/”.

Table 6.14: COALNUM Keyword Description

Example

The example below sets three [COALNUM](#) regions for a 4 x 5 x 2 model.

```
--
--      DEFINE COALNUM REGIONS FOR ALL CELLS
--
COALNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1  1
/
```

The above will no effect in an OPM Flow input deck.

6.3.19 COARSEN – DEFINE GRID COARSENING CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COARSEN](#) keyword defines how a set of cells should be coarsened for when the Local Grid Refinement (“LGR”) option has been activated by [LGR](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.20 COLLAPSE – DEFINE COMPRESSED VERTICAL EQUILIBRIUM CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COLLAPSE](#) keyword defines the which cells can be collapsed in a collapsed Vertical Equilibrium (“VE”) run when the [VE](#) option has been invoked via the [VE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

6.3.21 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The [COLUMNS](#) keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the GLOBAL section for a full description.

6.3.22 COORD – DEFINE A SET OF COORDINATES LINES FOR A RESERVOIR GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

COORD defines a set of coordinate lines or pillars for a reservoir grid via an array. A total of $6 \times (NX+1) \times (NY+1)$ values must be specified for each coordinate data set (or reservoir). For multiple reservoirs, where **NUMRES** is greater than one, there must be $6 \times (NX+1) \times (NY+1) \times \text{NUMRES}$ values. In OPM Flow **NUMRES** can only be set to one.

For Cartesian geometry, each line is defined by the (x, y, z) coordinates of two distinct points on the line. The lines are entered with I cycling fastest then J. For radial geometry, each line is defined by the (r, theta, z) coordinates of two distinct points on the line. The lines are entered with R cycling fastest then THETA.

The keyword can only be used with Irregular Corner-Point Grids.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	X1-Column	Top X coordinate			
2	Y1-Column	Top Y coordinate			
3	Z1-Column	Top Z coordinate			
4	X2-Column	Base X coordinate			
5	Y2-Column	Base Y coordinate			
6	Z2-Column	Base Z coordinate			
		feet	metres	cm	None

Notes:

- 1) THETA values are in degrees for all units.
- 2) Regular Cartesian and Radial grid keywords cannot be used with this keyword, that is: *DR*, *DRV*, *DTHETA*, *DTHETA*, *DX*, *DXV*, *DY*, *DYV*, *DZ*, *INRAD*, and *TOPS*.
- 3) The keyword is terminated by a “/”.

Table 6.15: COORD Keyword Description

See also the **SPECGRID**, **COORDSYS** and **ZCORN** keywords to fully define an Irregular Corner-Point Grid.

Example

```
--
--      SPECIFY VERTICAL COORDINATE LINES FOR A REGULAR 3 x 2 GRID
--      (DX = 100 and DY = 200)
--
--      X1      Y1      Z1      X2      Y2      Z2
--      ---      ---      ----      ---      ---      ----
COORD
      0          0      1000        0          0      5000
    100          0      1000       100          0      5000
    200          0      1000       200          0      5000
    300          0      1000       300          0      5000
      0      200      1000        0      200      5000
    100      200      1000       100      200      5000
    200      200      1000       200      200      5000
    300      200      1000       300      200      5000
      0      400      1000        0      400      5000
    100      400      1000       100      400      5000
    200      400      1000       200      400      5000
    300      400      1000       300      400      5000
/
```

The above example defines vertical coordinate lines for a regular 3 by 2 grid with *DX* equal to 100 and *DY* equal to 200.

6.3.23 COORDSYS – DEFINE COORDINATE GRID OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword sets various options for when multiple grid systems are being used, as declared by the [NUMRES](#) keyword in the [RUNSPEC](#) section. OPM Flow does not support multiple grid systems. The keyword is also used to stipulate for radial grids if the completion of the circle in the THETA direction should be implemented using non-neighbor connections.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
1	K1	A positive integer that defines the lower bound of the array in the K-direction for the given grid system.	None
2	K2	A positive integer that defines the upper bound of the array in the K-direction for the given grid system.	None
3	COMPLETE	COMPLETE is a defined character string that determines for radial grids if the circle should be completed in THETA direction, and should be set to COMP to complete the circle, or INCOMP for not completing the circle.	INCOMP
4	CONNECT	A defined character string that declares how the reservoir below should be connected to the given reservoir, and should be set to JOIN to connect the two reservoirs by calculating the inter-reservoir transmissibilities, or SEPARATE to isolate the reservoirs.	SEPARATE
5	R1	R1 is a positive integer defining the lower reservoir unit that is connected to the given reservoir unit.	Current Reservoir Record
6	R2	R2 is a positive integer defining the upper reservoir unit that is connected to the given reservoir unit.	

Notes:

- 1) The [DIMENS](#) keyword defines the dimensions of the model (NX, NY and NZ) and the [NUMRES](#) keyword the number of multiple grid systems (reservoirs), both keywords are in the [RUNSPEC](#) section.
- 2) Each record must be terminated by a "/" and there must be exactly [NUMRES](#) records entered for the keyword.
- 3) There is no "/" terminator for this keyword.

Table 6.16: COORDSYS Keyword Description

Example

```
--
--      DEFINE COORDINATE GRID OPTIONS
--
--      K1      K2      COMP      CONNECT      LOWER      UPPER
--      Layer   Layer   CIRCLE   RES           RES           RES
COORDSYS
--      1      1      COMP                               /
/
```

The above example connects the circle in the THETA direction for the [RADIAL](#) model, for when the number of grids have been set to one via the [NUMRES](#) keyword in the [SCHEDULE](#) section.

6.3.24 COPY – COPY ARRAY DATA TO ANOTHER ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COPY](#) keyword copies an array (or part of an array) to another array or part of an array. The arrays can be integer or real valued; however, the arrays that can be operated on are dependent on which section the [COPY](#) keyword is being applied in.

No.	Name	Description	Default
1	ARRAY-1	A character string of up to eight characters in length that defines the keyword identifying the array to be copied from.	None
2	ARRAY-2	A character string of up to eight characters in length that defines the keyword identifying the array to be copied to.	None
3	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal 1 and less than or equal to I2 and NX.	1
4	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX	NX
5	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal 1 and less than or equal to J2 and NY.	1
6	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
7	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1
8	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the [DIMENS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.17: COPY Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.18. Cells colored red indicate arrays that are not supported by OPM Flow operations.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

COPY Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANX	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TRBK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	DILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.18: COPY Keyword Applicable Arrays by Section

Example

```
--
--          SOURCE      DESTIN.      ----- BOX -----
--          I1  I2    J1  J2    K1  K2
COPY
          PERMX      PERMY      1*  1*    1*  1*    1*  1* / CREATE PERMY
          PERMX      PERMZ      1*  1*    1*  1*    1*  1* / CREATE PERMZ
/

--
--          ARRAY      CONSTANT      ----- BOX -----
--          I1  I2    J1  J2    K1  K2
MULTIPLY
          PERMZ      0.50000      1*  1*    1*  1*    1*  1* / PERMZ * 0.5
/
```

The above example copies *PERMX* array to the *PERMY* and *PERMZ* arrays in the *GRID* section for all grid blocks in the model. The *PERMZ* array is then multiplied by 0.5 for all grid blocks in the model.

6.3.25 COPYBOX – COPY ARRAY DATA DEFINED BY A BOX

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COPYBOX](#) keyword copies an array (or part of an array) to another part of the same array. The array can be real or integer depending on the array type; however, the array that can be operated on is dependent on which section the [COPYBOX](#) keyword is being used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	ARRAY-I	The name of the array to be copied This is the keyword name identifying the property and is up to eight characters in length and enclosed in quotes.	None
2	I1	A positive integer that defines the SOURCE lower bound of the array in the I-direction to be modified must be greater than or equal 1 and less than or equal to I2 and NX.	1
3	I2	A positive integer that defines the SOURCE upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX	NX
4	J1	A positive integer that defines the SOURCE lower bound of the array in the J-direction to be modified must be greater than or equal 1 and less than or equal to J2 and NY.	1
5	J2	A positive integer that defines the SOURCE upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
6	K1	A positive integer that defines the SOURCE lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1
7	K2	A positive integer that defines the SOURCE upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ
8	I3	A positive integer that defines the DESTINATION lower bound of the array in the I-direction to be modified must be greater than or equal 1 and less than or equal to I2 and NX.	1
9	I4	A positive integer that defines the DESTINATION upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX	NX
10	J3	A positive integer that defines the DESTINATION lower bound of the array in the J-direction to be modified must be greater than or equal 1 and less than or equal to J2 and NY.	1
11	J4	A positive integer that defines the DESTINATION upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY

No.	Name	Description	Default
12	K3	A positive integer that defines the DESTINATION lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	I
13	K4	A positive integer that defines the DESTINATION upper bound of the array in the K-direction to be modified must be greater than or equal to KI and less than or equal to NZ.	NZ
Notes: 1) Where NX, NY and NZ are the dimensions of the model as defined on the DIMENS keyword in the RUNSPEC section. 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.			

Table 6.19: COPYBOX Keyword Description

Note that the [SOURCE](#) and DESTINATION arrays must be of the same size in all dimensions and the applicable arrays for each section are defined in Table 6.20.

COPYBOX Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX		SWL	ENDNUM			
DY		SWCR	EQLNUM			
DZ		SWU	FIPNUM			
PERMX		SGL	IMBNUM			
PERMY		SGCR	MISCNUM			
PERMZ		SGU	PVTNUM			
MULTX		KRW	ROCKNUM			
MULTY		KRO	SATNUM			
MULTZ		KRG	WH2NUM			
DR		PCG				
DTHETA		PCW				
PERMR						
PERMTHT						
DZNET						
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFTHT						

Table 6.20: COPYBOX Keyword Applicable Arrays by Section

Example

```
--
--          SOURCE  ----- SOURCE BOX -----  ----- DESTINATION BOX -----
--          ARRAY   I1  I2   J1  J2   K1  K2   I1  I2   J1  J2   K1  K2
COPYBOX
          PORO     1*  1*   1*  1*   12  14   1*  1*   1*  1*   15  17 / PORO
          PERMX     1*  1*   1*  1*   12  14   1*  1*   1*  1*   15  17 / PERMX
/
```

The above example copies all the [PORO](#) and [PERMX](#) values in layers I2 to I4 to layers I5 and I7.

6.3.26 COPYREG – COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **COPYREG** keyword copies a specified array or part of an array based on cells with a specific region number to another array. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **COPYREG** keyword is read by the simulator. The property arrays can be real or integer valued depending on the property array type; however, the property arrays that can be operated on are dependent on which section the **COPYREG** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY-1	A character string of up to eight characters in length that defines the keyword identifying the array to be copied from.	None
2	ARRAY-2	A character string of up to eight characters in length that defines the keyword identifying the array to be copied to.	None
3	REGION NUMBER	Integer REGION NUMBER is the region for which the array data in (1) should be copied to array data in (2).	None
4	REGION ARRAY	The REGION ARRAY to use for selecting the REGION NUMBER in (3) for selecting the data to be copied. REGION ARRAY can have the following values: 1) F for the FLUXNUM array 2) M for the MULTNUM array 3) O for the OPERNUM array	M

Notes:

- 1) Where the REGION NUMBER should be less than or equal to the maximum number of regions as defined on the **REGDIMS** keyword for the **FIPNUM** and **OPERNUM** arrays or the **GRIDOPTS** keyword for the **MULTNUM** array in the **RUNSPEC** section.
- 2) If the **GRIDOPTS** keyword is not present in the **RUNSPEC** section or if the **GRIDOPTS** keyword is present but the maximum number of **MULTNUM** regions (NRMULT) equals zero then REGION ARRAY will default to F for the **FLUXNUM** array.
- 3) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.21: COPYREG Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.22. Cells colored red indicate arrays that are not supported by OPM Flow operations.

COPYREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANX	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

COPYREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
PERMZ	DIFFX	SGU	PVTNUM	TBLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	OILAPI		
MULTZ	TRANR	KRG	WPNUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPPANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.22: COPYREG Keyword Applicable Arrays by Section

Example

```
--
--      COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER
--
--      ARRAY      ARRAY      REGION      REGION ARRAY
--      FROM        TO        NUMBER      M / F / O
COPYREG
    PERMX      PERMY      1      M      / COPY PERMX TO PERMY
    PERMX      PERMZ      1      M      / COPY PERMX TO PERMZ
/
--
--      NOW RESET PERMZ BASED ON THE MULTNUM REGION NUMBER
--
--      MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER
--
--      ARRAY      CONSTANT  REGION      REGION ARRAY
--      VALUE      VALUE     NUMBER      M / F / O
MULTIREG
    PERMX      0.95      1      M      /
/
```

The above example first copies the [PERMX](#) property array for region number one to the [PERMY](#) and [PERMZ](#) property arrays for region one using the [MULTNUM](#) array to define the region numbers. After which [PERMZ](#) property array for region one is multiplied by 0.5 using the [MULTIREG](#) keyword.

6.3.27 CRITPERM – DEFINE MINIMUM PERMEABILITY FOR VERTICAL EQUILIBRIUM GRID CELL COMPRESSION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [CRITPERM](#) keyword is used to define the minimum permeability for Vertical Equilibrium(“VE”) grid cell compression, for when the Vertical Equilibrium formulation has been activated by the [VE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.28 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the GLOBAL section for a full description.

6.3.29
DIFFMR
– DEFINE GRID BLOCK RADIAL DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFMR* keyword defines the radial direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.30 DIFFMR- – DEFINE GRID BLOCK NEGATIVE RADIAL DIRECTION
DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFMR-* keyword defines the negative radial direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

6.3.31 DIFFMTH- – DEFINE GRID BLOCK NEGATIVE THETA DIRECTION
DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFMR-](#) keyword defines the negative theta direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.32 DIFFMTHT – DEFINE GRID BLOCK THETA DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFMTHT](#) keyword defines the theta direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.33 DIFFMX – DEFINE GRID BLOCK X-DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [DIFFMX](#) keyword defines the x-direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.34
DIFFMX- – DEFINE GRID BLOCK NEGATIVE X-DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFMX-* keyword defines the negative x-direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.35 DIFFMY – DEFINE GRID BLOCK Y-DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [DIFFMY](#) keyword defines the y-direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.36 DIFFMY- – DEFINE GRID BLOCK NEGATIVE Y-DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFMY](#)- keyword defines the negative y-direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.37 DIFFMZ – DEFINE GRID BLOCK Z-DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFMZ* keyword defines the z-direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.38
DIFFMZ- – DEFINE GRID BLOCK NEGATIVE Z-DIRECTION DIFFUSIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFMZ*- keyword defines the negative z-direction diffusivity multipliers for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.39 DISPERC – DEFINE THE MECHANICAL DISPERSIVITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DISPERC](#) keyword defines the mechanical dispersivity for all cells in the model via an array.

No.	Name	Description			Default
			Metric	Laboratory	
1	DISPERC	DISPERC is an array of real positive values that defines the mechanical dispersivity for each cell in the model. Repeat counts may be used, for example 20*1.0.			None
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 6.23: [DISPERC](#) Keyword Description

Note

The option has been tested in combination with the [CO2STORE](#), [H2STORE](#), [BIOFILM](#), or [MICP](#) keywords, but not for the general case at this point.

See also the [CO2STORE](#) and [H2STORE](#) keywords in the [RUNSPEC](#) section that active OPM Flow's CO₂ and H₂ storage models respectively.

Example

The example sets the mechanical dispersivity for all cells in the model to 1.0.

```
--
--      SET MECHANICAL DISPERSIVITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
DISPERC      1000*1.0      /
```


6.3.40 DOMAINS – DEFINE THE PARALLEL DOMAIN PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DOMAINS](#) keyword defines the parallel domain properties for when parallel processing has been invoked by the [PARALLEL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

See section [2.2 Running OPM Flow From The Command Line](#) on how to run OPM Flow in parallel mode.

6.3.41 DPGRID – ACTIVATE THE MATRIX CELL TO FRACTURE CELL OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DPGRID](#) keyword activates the matrix cell to fracture cell option for dual porosity runs for when a Dual Porosity model has been activated by either the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section. The keyword allows for only the matrix grid data to be entered and the missing fracture cells are set to the inputted matrix cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.42 DPNUM – DEFINE DUAL AND SINGLE POROSITY GRID BLOCK ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

In dual porosity runs only, that is not dual permeability runs, the [DPNUM](#) keyword defines which wells should be treated as single porosity cells and which cells should be treated as dual porosity cells, for when the Dual Porosity model has been activated by the [DUALPORO](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.43 DR – DEFINE THE SIZE OF GRID BLOCKS IN THE R DIRECTION FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DR](#) defines the size of all grid blocks in the R direction via an array for each cell in the model. The [RADIAL](#) or [SPIDER](#) keyword in the [RUNSPEC](#) section should be activated to indicate that radial or spider geometry is being used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DR	DR is an array of real numbers describing the cell size in the R direction for each cell in the model in a radial grid. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The first layer (K=1) must always be fully defined; however, subsequent layers may be default and if defaulted they will take the values from the layer above.
- 3) The keyword is terminated by a "/".

Table 6.24: DR Keyword Description

Note that the [SPIDER](#) keyword activates OPM Flow's radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: [INRAD](#), [DRV](#), [DTHETAV](#), [DZIDZV](#) etc. in the [GRID](#) section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model's pore volume to reflect radial coordinates; thus, overcoming the display limitation.

See also the [DRV](#), [DTHETAV](#), [DZ](#) and [TOPS](#) keywords in the [GRID](#) section to fully define a radial or spider grid model.

Example

Given the dimensions of the grid in the [RUNSPEC](#) section to be 10,1, 8 for NX, NY and NZ respectively, we first define the inner radius of the radial model,

```
--
--      INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD      0.25
/
```

and then [DR](#) should be defined as:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

```
--
--      DEFINE GRID BLOCK R DIRECTION CELL SIZE
--
DR
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
/
```

The above example defines the size of the cells in the R direction based on 80 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

Note that since the first layer (K=1) must be defined and subsequent layers default to the layer above then:

```
--
--      INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD
      0.25
/
--
--      DEFINE GRID BLOCK R DIRECTION CELL SIZE
--
DR
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.0
/
```

is equivalent to previous example.

6.3.44 DRV – DEFINE THE SIZE OF GRID BLOCKS IN THE R DIRECTION VIA A VECTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DRV](#) defines the size of grid blocks in the R direction via a vector as opposed to defining the property for each cell for a Radial or Spider Grid. The [RADIAL](#) or [SPIDER](#) keyword in the [RUNSPEC](#) section should be activated to indicate that radial or spider geometry is being used.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DRV	DRV is a vector of real numbers describing the cell size for the grid blocks in the R direction in a radial grid. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX parameter on the *DIMENS* keyword in the *RUNSPEC* section
- 2) The keyword is terminated by a “/”.

Table 6.25: DRV Keyword Description

Note that the [SPIDER](#) keyword activates OPM Flow's radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: [INRAD](#), [DRV](#), [DTHETAV](#), [DZIDZV](#) etc. in the [GRID](#) section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model's pore volume to reflect radial coordinates; thus, overcoming the display limitation.

See also the [DR](#), [DTHETAV](#), [DZ](#) and [TOPS](#) keywords in the [GRID](#) section to fully define a radial or spider grid model.

Example

```
--
--      INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD
      0.25
/
--
--      DEFINE GRID BLOCK SIZES IN THE R DIRECTION
--
DRV
      1.75  2.32  5.01  10.84  23.39  50.55  109.21  235.92  509.68  1101.08 /
```

The above example defines the size of the cells in the R direction based on NX equals 10 on the [DIMENS](#) keyword in the [RUNSPEC](#) section. Note the [INRAD](#) keyword to define the inner radius of the radial grid.

6.3.45 DTHETA – DEFINE THE SIZE OF GRID BLOCKS IN THE THETA DIRECTION FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DTHETA](#) defines the size of all grid blocks in the Theta direction via an array for each cell in model. The [RADIAL](#) or [SPIDER](#) keyword in the [RUNSPEC](#) section should be activated to indicate that radial or spider geometry is being used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DTHETA	DTHETA is an array of real numbers describing the cell size in the THETA direction in radial grids for each cell in the model. Repeat counts may be used, for example 10*25.0			None
		degrees	degrees	degrees	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.26: DTHETA Keyword Description

Note that the [SPIDER](#) keyword activates OPM Flow’s radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: [INRAD](#), [DRV](#), [DTHETAV](#), [DZ/DZV](#) etc. in the [GRID](#) the section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model’s pore volume to reflect radial coordinates; thus, overcoming the display limitation. See also the [DRV](#), [DTHETAV](#), [DZ](#) and [TOPS](#) keywords in the [GRID](#) section to fully define a radial or spider grid model.

Example

Given the dimensions of the grid in the [RUNSPEC](#) section to be 10, 6, 1 for NX, NY and NZ respectively, then [DTHETA](#) should be defined as:

```
--
--      DEFINE GRID BLOCK SIZES IN THE THETA DIRECTION
--
DTHETA
    10*60.0
    10*60.0
    10*60.0
    10*60.0
    10*60.0
    10*60.0
/
```

The above example defines the size of the cells in the R direction based on 60 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.46 DTHETAV – SETS THE SIZE OF GRID BLOCKS IN THETA DIRECTION VIA A VECTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

DTHETAV defines the size of grid blocks in the THETA direction via a vector as opposed to defining the property for each cell for a Radial Grid. The *RADIAL* or *SPIDER* keyword in the *RUNSPEC* should be activated to indicate that radial geometry is being used.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DTHETAV	DTHETAV is a vector of real numbers describing the cell size for the grid blocks in the THETA direction in a radial grid. Repeat counts may be used, for example 10*100.0.			None
		degrees	degrees	degrees	
Notes: 1) The number of entries should correspond to the NY parameter of the DIMENS keyword in the RUNSPEC section. 2) The keyword is terminated by a “/”.					

Table 6.27: *DTHETA* Keyword Description

Note that the *SPIDER* keyword activates OPM Flow's radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: *INRAD*, *DRV*, *DTHETAV*, *DZIDZV* etc. in the *GRID* the section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model's pore volume to reflect radial coordinates; thus, overcoming the display limitation.

See also the *DRV*, *DZV* and *TOPS* keywords to fully define a radial or spider grid model.

Example

```
--
--      DEFINE GRID BLOCK SIZES IN THE THETA DIRECTION (BASED ON NY = 6)
--
DTHETAV      60.0  60.0  60.0  60.0  60.0  60.0      /
```

The above example defines the size of the cells in the THETA direction based on NY equals six in the *DIMENS* keyword in the *RUNSPEC* section.

6.3.47 DUMPFLUX – ACTIVATE WRITING OUT OF A FLUX FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the writing out of a full field (the full grid) FLUX file for later processing in a Flux Boundary run. The Flux Boundary feature allows for the segmentation of the full grid into flux boundary areas which allow for a sub-area of the grid to be run and at the same time model the flux across the boundary derived from the main grid. The object of this feature is to be able to investigate the performance of various areas of the model without having to run the full field, thus improving computational efficiency and run times, but still obtain “reasonable” results due to the incorporation of the fluxes across the boundary.

This feature is not available in OPM Flow; however it is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE WRITING OUT OF A FLUX FILE
--
DUMPFLUX
```

The above example switches on the writing of the FLUX output file; the keyword has no effect and is ignored by the simulator.

6.3.48
DX – DEFINE THE SIZE OF GRID BLOCKS IN THE X DIRECTION FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

DX defines the size of all grid blocks in the X direction via an array for each cell in a Cartesian Regular Grid model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DX	DX is an array of real numbers describing the cell size in the X direction for each cell in the model. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2) The keyword is terminated by a “/”.

Table 6.28: DX Keyword Description

See also the *DY*, *DZ* and *TOPS* keywords to fully define a Cartesian Regular Grid.

Example

```
--
--      DEFINE GRID BLOCK X DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 300)
--
DX
      300*1000
/
```

The above example defines the size of the cells in the X direction based on 300 cells in the model as defined by the *DIMENS* keyword in the *RUNSPEC* section.

6.3.49 DXV – DEFINE THE SIZE OF GRID BLOCKS IN THE X DIRECTION VIA A VECTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DXV](#) defines the size of grid blocks in the X direction via a vector as opposed to defining the X direction cell size for each cell for a Cartesian Regular Grid.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DXV	DXV is a vector of real numbers describing the cell size for the grid blocks in the X direction. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

1)

The number of entries should correspond to the NX parameter on the *DIMENS* keyword in the *RUNSPEC* section.

2)

The keyword is terminated by a “/”.

Table 6.29: DXV Keyword Description

See also the [DYV](#), [DZV](#) and [TOPS](#) keywords to fully define a Cartesian Regular Grid.

Example

```
--
--      DEFINE GRID BLOCK X DIRECTION CELL SIZE (BASED ON NX = 5)
--
DXV      5*100      /
```

The above example defines the size of the cells in the X direction based on NX equals 5 on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.50 DY – DEFINE THE SIZE OF GRID BLOCKS IN THE Y DIRECTION FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

DY defines the size of all grid blocks in the Y direction via an array for each cell in a Cartesian Regular Grid model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DY	DY is an array of real numbers describing the cell size in the Y direction for each cell in the model. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2) The keyword is terminated by a “/”.

Table 6.30: DY Keyword Description

See also the DX, DZ and TOPS keywords to fully define a Cartesian Regular Grid.

Example

```
--
--      DEFINE GRID BLOCK Y DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 300)
--
DY      300*1000 /
```

The above example defines the size of the cells in the Y direction based on 300 cells in the model as defined by the DIMENS keyword in the RUNSPEC section.

6.3.51 DYV – DEFINE THE SIZE OF GRID BLOCKS IN THE Y DIRECTION VIA A VECTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DYV](#) defines the size of grid blocks in the Y direction via a vector as opposed to defining the Y direction cell size for each cell for a Cartesian Regular Grid.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DYV	DYV is a vector of real numbers describing the cell size for the grid blocks in the Y direction. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

1) The number of entries should correspond to the NY parameter on the *DIMENS* keyword in the *RUNSPEC* section.

2) The keyword is terminated by a “/”.

Table 6.31: DYV Keyword Description

See also the [DXV](#), [DZV](#) and [TOPS](#) keywords to fully define a Cartesian Regular Grid.

Example

```
--
--      DEFINE GRID BLOCK Y DIRECTION CELL SIZE (BASED ON NY = 5)
--
DYV      5*100      /
```

The above example defines the size of the cells in the Y direction based on NY equals 5 on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.52
DZ – DEFINE THE SIZE OF GRID BLOCKS IN THE Z DIRECTION FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

DZ defines the size of all grid blocks in the Z direction via an array for each cell in a Cartesian Regular Grid model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DZ	DZ is an array of real numbers describing the cell size in the Z direction for each cell in the model. Repeat counts may be used, for example 10*100.0.			None
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.32: DZ Keyword Description

See also the DX, DY and TOPS keywords to fully define a Cartesian Regular Grid.

Example

```
--
--      DEFINE GRID BLOCK Z DIRECTION CELL SIZE (BASED ON NX x NY x NZ = 300)
--
DZ      100*20.0   100*30.0   100*50.0      /
```

The above example defines the size of the cells in the Z direction based on 300 cells in the model as defined by the DIMENS keyword in the RUNSPEC section.

6.3.53 DZMATRIX – MATRIX BLOCK HEIGHT FOR GRAVITY DRAINAGE MODEL FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DZMATRIX](#) keyword defines the matrix block height for the gravity drainage model by grid block for when the Dual Permeability or Dual Porosity models are activated by the [DUALPERM](#) and [DUALPORO](#) keywords and the Gravity Drainage option is invoked via the [GRAVDR](#) and [GRAVDRM](#) keywords. All keywords are in the [RUNSPEC](#) section.

The keyword is identical to the [DZMTRXV](#) keyword in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

6.3.54 DZMTRX – MATRIX BLOCK HEIGHT FOR GRAVITY DRAINAGE MODEL FOR THE GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DZMTRX](#) keyword defines a constant matrix block height for the gravity drainage model for the entire grid for when the Dual Permeability or Dual Porosity models are activated by the [DUALPERM](#) and [DUALPORO](#) keywords and the Gravity Drainage option is invoked via the [GRAVDR](#) and [GRAVDRM](#) keywords. All keywords are in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.55
DZMTRXV – MATRIX BLOCK HEIGHT FOR GRAVITY DRAINAGE MODEL FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DZMATRIX](#) keyword defines the matrix block height for the gravity drainage model by grid block for when the Dual Permeability or Dual Porosity models are activated by the [DUALPERM](#) and [DUALPORO](#) keywords and the Gravity Drainage option is invoked via the [GRAVDR](#) and [GRAVDRM](#) keywords. All keywords are in the [RUNSPEC](#) section.

The keyword is identical to the [DZMATRIX](#) keyword in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.56 DZNET – DEFINE GRID BLOCK NET THICKNESS FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DZNET](#) defines the net thickness of all grid blocks in the Z direction via an array for each cell in a Cartesian Regular Grid model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DZNET	DZNET is an array of real numbers describing the net thickness in the Z direction for each cell in the model. Repeat counts may be used, for example 10*100.0. If the value for a grid block is not defined then the grid block size (DZ) is assigned to the missing values.			DZ
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.33: DZNET Keyword Description

See also the [DX](#), [DY](#), [DZ](#), [NTG](#) and [TOPS](#) keywords to fully define a Cartesian Regular Grid.

Example

```
--
--      DEFINE GRID BLOCK Z DIRECTION NET THICKNESS(BASED ON NX x NY x NZ = 300)
--
DZNET      100*15.0    100*25.0    00*45.0      /
```

The above example defines the net thickness of the cells in the Z direction based on 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.57 DZV – DEFINE THE SIZE OF GRID BLOCKS IN THE Z DIRECTION VIA A VECTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

DZV defines the size of grid blocks in the Z direction via a vector as opposed to defining the thickness property for each cell. The keyword is used for both Cartesian Regular Grids and Radial Grids.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DZV	DZV is a vector of real numbers describing the cell size for the grid blocks in the Z direction. Repeat counts may be used, for example 10*20.0.			None
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NZ parameter on the *DIMENS* keyword in the *RUNSPEC* section..
- 2) The keyword is terminated by a “/”.

Table 6.34: DZV Keyword Description

See also the DXV, DYV and TOPS keywords for a Cartesian Regular Grid and DRV, DTHETAV and TOPS keywords to fully define a Radial Grid model.

Example

```
--
--      DEFINE GRID BLOCK SIZES IN THE Z DIRECTION (BASED ON NZ = 20)
--
DZV
      3.0   5.0   3.0   2.0   5.0  15*3.0
                                                    /
```

The above example defines the size of the cells in the Z direction based on NZ equals 20 on the DIMENS keyword in the RUNSPEC section.

6.3.58 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

6.3.59 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the GLOBAL section for a full description.

6.3.60 ENDBOX – DEFINE THE END OF THE BOX DEFINED GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword marks the end of a previously defined [BOX](#) sub-grid as defined by a previously entered [BOX](#) keyword. The keyword resets the input grid to be the full grid as defined by the NX, NY, and NZ variables on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

There is no data required for this keyword.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1
--
--                                     /  DEFINE BOX AREA
--
--      DEFINE GRID BLOCK PERMZ DATA FOR THE INPUT BOX
--
--      PERMZ
--      6*0.01
--
--                                     /
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

The above example defines a subset of the grid and sets the cells [PERMZ](#) values to 0.01 for that area. After which the [ENDBOX](#) keyword resets the input to be the full grid.

Note

It is good practice to always use the [ENDBOX](#) keyword to reset the input back to the full grid when all the modifications for a sub-grid have been completed.

6.3.61 ENDFIN – END THE DEFINITION OF A LOCAL GRID REFINEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *ENDFIN* keyword defines the end of a Cartesian or radial local grid refinement (“LGR”) definition and a *LGR* property definition data set. In the *GRID* section the *CARFIN*, *RADFIN*, and *RADFIN4* keywords defines the start of an *LGR* description section, whereas the *REFINE* keyword in the *EDIT*, *PROPS*, *REGIONS*, *SOLUTION* and *SCHEDULE* section defines the start. The *REFINE* keyword can also be used in the *GRID* section provided the *LGR* has been previously specified by the *CARFIN*, *RADFIN*, or *RADFIN4* keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

The example below is based on using the *CARFIN* keyword in the *GRID* section to define an *LGR* in the global grid, named *LGR-OP01* with a maximum of one well allowed in the *LGR*.

```
--
--      CARFIN LGR GRID COMMANDS
--
--      LGR      ----- FINE GRID -----      -- CARFIN GRID --      MAX      HOST
--      NAME      I1  I2  J1  J2  K1  K2      NX      NY      NZ      WELLS      NAME
CARFIN
      LGR-OP01    24  24  87  87   1  50      3      3      50      1      GLOBAL /

ENDFIN
```

Here the one global cell in the areal plane (24, 87) is divided into three *LGR* cells in the x-direction and three cells in the y-direction. Since no other property data is given, then the *LGR* cells take their properties from the host grid, that is the global grid.

6.3.62 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the GLOBAL section for a full description.

6.3.63 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the GLOBAL section for a full description.

6.3.64 EQLZCORN – MODIFY THE DEPTH OF THE CORNER-POINT DEPTH ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [EQLZCORN](#) keyword modifies the depth of a corner point of a grid block on the pillars defining the reservoir grid. The keyword can be only used be used with Irregular Corner-Point Grids.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.65 EQUALREG – SETS AN ARRAY TO A CONSTANT BY REGION NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **EQUALREG** keyword sets a specified array to a constant for cells with a specific region number. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **EQUALREG** keyword is read by the simulator. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **EQUALREG** keyword is being applied in.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None
2	CONSTANT	An integer or real value to be assigned to the ARRAY in the same units as the ARRAY property for a given REGION	0
3	REGION NUMBER	REGION NUMBER is a positive integer representing the region for which the CONSTANT in (2) should be applied	None
4	REGION ARRAY	The REGION ARRAY to use for applying the CONSTANT in (2) based on the REGION NUMBER in (3). REGION ARRAY can have the following values: 1) F for the FLUXNUM array 2) M for the MULTNUM array 3) O for the OPERNUM array	M

Notes:

- Where the REGION NUMBER should be less than or equal to the maximum number of regions as defined on the **REGDIMS** keyword for the **FIPNUM** and **OPERNUM** arrays or the **GRIDOPTS** keyword for the **MULTNUM** array in the **RUNSPEC** section.
- If the **GRIDOPTS** keyword is not present in the **RUNSPEC** section or if the **GRIDOPTS** keyword is present but the maximum number of **MULTNUM** regions (NRMULT) equals zero then REGION ARRAY will default to F for the **FLUXNUM** array.
- Each record must be terminated by a "/" and the keyword is terminated by a "/".

Table 6.35: EQUALREG Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.36. Cells colored red indicate arrays that are not supported by OPM Flow operations.

EQUALREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANX	SGL	IMBNUM	RV		

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

EQUALREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TBLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	DILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.36: EQUALREG Keyword Applicable Arrays by Section

Example

```
-- FIRST DEFINE MULTNUM ARRAYS FOR 10 X 10 X 20 MODEL
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      MULTNUM      1          1* 1*  1* 1*  1* 1*      / MULTNUM IN MODEL
      MULTNUM      2          1* 1*  1* 1*  6  6      / MULTNUM IN MODEL
      MULTNUM      3          1* 1*  1* 1* 10 10      / MULTNUM IN MODEL
/
--      NOW SET PORO AND PERMX BASED ON THE MULTNUM REGION NUMBER
--
--      SETS A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER
--
--      ARRAY      CONSTANT  REGION  REGION ARRAY
--      VALUE      NUMBER    M / F / O
EQUALREG
      PORO        0.200      1        M
      PORO        0.150      2        M
      PORO        0.120      3        M
      PERMX       100.00      1        M
      PERMX       75.00      2        M
      PERMX       50.00      3        M
/
```

The example first defines the *MULTNUM* array to 1 for all cells in the model, after which selected areas of model are assigned various *MULTNUM* integer values. The *EQUALREG* can then be invoked to set constant values for the *PORO* and *PERMX* arrays for the various *MULTNUM* regions.

6.3.66 EQUALS – SETS A SPECIFIED ARRAY TO A CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **EQUALS** keyword sets a specified array or part of an array to a constant. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **EQUALS** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the property to be modified.	None
2	CONSTANT	An integer or real value that the ARRAY will be set to in the same units as the ARRAY property.	None
3	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal to one and less than or equal to I2 and NX.	1
4	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX	NX
5	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal to one and less than or equal to J2 and NY.	1
6	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
7	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1
8	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.37: EQUALS Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.38. Cells colored red indicate arrays that are not supported by OPM Flow operations.

EQUALS Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

EQUALS Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
PERMX	TRANY	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TBLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	OILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPPANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.38: EQUALS Keyword Applicable Arrays by Section

Note

Unlike the commercial simulator, using the **EQUALS** keyword to setup the structure of the grid using the **DX**, **DY**, **DZ** and **TOPS** keywords should be avoided as it may cause OPM Flow to prematurely fail during the initialization.

See the second example on the correct way to setup this type of grid.

Examples

The first example resets the **PERMX**, **PERMY** and **PERMZ**, arrays to 0.10, 0.10, and 0.01 for all cells in layer five, respectively.

```
--
--  ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
--
EQUALS
  PERMX      0.1000      1*  1*   1*  1*   5   5 / PERMX TO 0.10 IN LAYER 5
  PERMY      0.1000      1*  1*   1*  1*   5   5 / PERMY TO 0.10 IN LAYER 5
  PERMZ      0.0100      1*  1*   1*  1*   5   5 / PERMZ TO 0.01 IN LAYER 5
/
```

The second example illustrates how to correctly setup a Cartesian Regular Grid in OPM Flow, given the [DIMENS](#) keyword in the [RUNSPEC](#) section is set to:

```
--
--          MAX          MAX          MAX
--          NDIVIX      NDIVIY      NDIVIZ
DIMENS
          10           10           3
/
```

and the following keywords in the [GRID](#) section:

```
--
--          ACTIVATE IRREGULAR CORNER-POINT GRID TRANSMISSIBILITIES
--
OLDTRAN
--
--          DEFINE GRID BLOCK X DIRECTION CELL SIZE (BASED ON NX = 5)
--
DX
          300*1000
/
--
--          DEFINE GRID BLOCK Y DIRECTION CELL SIZE (BASED ON NY = 5)
--
DY
          300*1000
/
--
--          DEFINE GRID BLOCK SIZES IN THE Z DIRECTION
--
DZ
          100*20  100*30  100*50
/
--
--          DEFINE GRID BLOCK TOPS FOR THE TOP LAYER
--
TOPS
          100*8325
/
--
--          ARRAY          CONSTANT          ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
          PERMX          500.000          1*  1*   1*  1*   1  1 / Layer #01 Properties
          PERMY          500.000
          PERMZ          20.000
          PORO           0.300
          NTG            1.000
/
          PERMX          50.000          1*  1*   1*  1*   2  2 / Layer #02 Properties
          PERMY          50.000
          PERMZ          50.000
          PORO           0.300
          NTG            1.000
/
          PERMX          200.000          1*  1*   1*  1*   3  3 / Layer #03 Properties
          PERMY          200.000
          PERMZ          200.000
          PORO           0.300
          NTG            1.000
/
/
```

Notice that the [DX](#), [DY](#), [DZ](#) and [TOPS](#) keywords are defined separately, that is they are not included in the [EQUALS](#) keyword.

6.3.67 EXTFIN – DEFINE AN EXTERNAL UNSTRUCTURED LOCAL GRID REFINEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [EXTFIN](#) keyword defines an external Unstructured Local Grid Refinement (“LGR”) in a cell or a group of cells in the global grid, and for when LGRs have been activated for the model using the [LGR](#) keyword in the [RUNSPEC](#) section. Note the global grid can be either structured, see the [EXTREPGL](#) keyword in the [GRID](#) section for global structure grids, or unstructured, see the [EXTHOST](#) keyword in the [GRID](#) section for unstructured global grids.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.68 EXTHOST – DEFINE HOST CELLS FOR EXTERNAL LGRs

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [EXTHOST](#) keyword defines the host global grid blocks for an external Local Grid Refinement (“LGR”) for when LGRs have been activated for the model using the [LGR](#) keyword in the [RUNSPEC](#) section, and the global grid is an unstructured grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.69 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

6.3.70 EXTREPGL – DEFINE HOST CELLS FOR EXTERNAL UNSTRUCTURED LGRs

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [EXTREPGL](#) keyword defines the host global grid blocks for an external Unstructured Local Grid Refinement (“LGR”) for when LGRs have been activated for the model using the [LGR](#) keyword in the [RUNSPEC](#) section, and the global grid is a structured grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.71 FAULTS – DEFINE FAULTS IN THE GRID GEOMETRY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **FAULTS** keyword defines the faults in the grid geometry and the keyword is normally exported with the grid geometry **COORD** and **ZCORN** data sets from static earth modeling software. Note that the **FAULTS** keyword is not required to describe the structural geometry as this is already accounted for in the **COORD** and **ZCORN** data sets, but instead lists the fault traces with respect to the grid. Once the fault traces have been defined with the **FAULTS** keyword then the fault transmissibilities can be modified by the **MULTFLT** keyword. Note that without the **FAULTS** keyword one would still get proper cross-fault transmissibilities but they would not be modifiable using **MULTFLT** keyword.

No.	Name	Description	Default
1	FLTNAME	FLTNAME is a character string enclosed in quotes with a maximum length of eight characters, that defines the name of the fault.	None
2	I1	The lower bound of the fault's I-direction range must be greater than or equal to one and less than or equal to I2 and NX.	None
3	I2	The upper bound of the fault's I-direction range must be greater than or equal to I1 and less than or equal to NX	None
4	J1	The lower bound of the fault's J-direction range must be greater than or equal to one and less than or equal to J2 and NY.	None
5	J2	The upper bound of the fault's J-direction range must be greater than or equal to J1 and less than or equal to NY.	None
6	K1	The lower bound of the fault's K-direction range must be greater than or equal to one and less than or equal to K2 and NZ.	None
7	K2	The upper bound of the fault's K-direction range must be greater than or equal to K1 and less than or equal to NZ.	None
8	FLTFACE	FLTFACE is a character string enclosed in quotes with a maximum length of two characters, that classifies the fault face. 1) If TRANSMULT on the GRIDOPTS keyword in the RUNSPEC section is set to NO then FLTFACE can have values of X,Y, or Z. 2) Alternatively, if TRANSMULT on the GRIDOPTS keyword in the RUNSPEC section is set to YES then FLTFACE can have values of X,Y, or Z for the positive direction, or X-, Y- or Z- for the negative direction transmissibilities.	None

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) The **FAULTDIM** keyword in the **RUNSPEC** defines the maximum number of records (or segments) that can be entered with the **FAULTS** keyword.
- 3) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.39: FAULTS Keyword Description

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Example

The example below defines two fault traces, the first being the 'M_WEST' fault and the second the 'BC' fault trace.

```
--
--      DEFINE FAULTS IN THE GRID GEOMETRY
--
--      FAULT      ----- FAULT TRACE -----
--      NAME      I1   I2   J1   J2   K1   K2   FACE
FAULTS
    'M_WEST'      5     5     3     3     1    22   'X' /
    'M_WEST'      5     5     4     4     1    22   'X' /
    'M_WEST'      5     5     5     5     1    22   'X' /
    'M_WEST'      5     5     6     6     1    22   'X' /
    'M_WEST'      5     5     7     7     1    22   'X' /
    'M_WEST'      5     5     8     8     1    22   'X' /
    'M_WEST'      5     5     9     9     1    22   'X' /
    'M_WEST'      5     5    10    10     1    22   'X' /
    'M_WEST'      5     5    11    11     1    22   'X' /
.....
    'BC'          43    43     8     8     1    22   'Y' /
    'BC'          42    42     9     9     1    22   'X' /
    'BC'          44    44     8     8     1    22   'Y' /
    'BC'          45    45     8     8     1    22   'Y' /
    'BC'          46    46     8     8     1    22   'Y' /
    'BC'          31    31     9     9     1    22   'Y' /
    'BC'          30    30    10    10     1    22   'X' /
    'BC'          32    32     9     9     1    22   'Y' /
    'BC'          33    33     9     9     1    22   'Y' /
    'BC'          34    34     9     9     1    22   'Y' /
    'BC'          35    35     9     9     1    22   'Y' /
    'BC'          36    36     9     9     1    22   'Y' /
    'BC'          37    37     9     9     1    22   'Y' /
    'BC'          38    38     9     9     1    22   'Y' /
    'BC'          39    39     9     9     1    22   'Y' /
    'BC'          40    40     9     9     1    22   'Y' /
.....
/
```

6.3.72 FILEUNIT – ACTIVATE UNIT CONSISTENCY CHECKING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The `FILEUNIT` keyword defines the units of the data set, and is used to verify that the units in the input deck and any associated include files are consistent. The keyword does not provide for the conversion between different sets of units.

No.	Name	Description	Default
1	<code>FILEUNIT</code>	A character string that defines the units of the data set, and should be set to: 1) <code>FIELD</code> for field units, 2) <code>METRIC</code> for metric units, or 3) <code>LAB</code> for laboratory units	None
Notes: 1) No unit conversion is performed. 2) The keyword is terminated by a “/”.			

Table 6.40: FILEUNIT Keyword Description

OPM Flow's behavior is controllable through the "UNIT_SYSTEM_MISMATCH" environment variable. The default behavior if the check fails (i.e., if one of the `INCLUDE` files has a unit system different from the main run specification) is to terminate the simulation with an error.

Example

```
--
--      ACTIVATE UNIT CONSISTENCY CHECKING
--
FILEUNIT
      FIELD
/
```

The above example defines the data set units to be `FIELD` units.

6.3.73 FLUXNUM – DEFINE THE FLUX REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [FLUXNUM](#) keyword defines the flux region numbers for each grid block, as such there must be one entry for each cell in the model. The array is used with the Flux Boundary option to define the various flux regions; however, the Flux Boundary option has not been implemented in OPM Flow. In addition, the array can be used with the [EQUALREG](#), [ADDREG](#), [COPYREG](#), [MULTIREG](#), [MULTREGP](#) and [MULTREGT](#) keywords in calculating various grid properties in the [GRID](#) section. This facility has been implemented in OPM Flow.

No.	Name	Description	Default
I	FLUXNUM	FLUXNUM defines an array of positive integers assigning a grid cell to a particular flux region. The maximum number of flux regions is set by the MXNFLX variable on the REGDIMS keyword in the RUNSPEC section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.41: FLUXNUM Keyword Description

Examples

The example below sets three [FLUXNUM](#) regions for a 4 x 5 x 2 model.

```
--
--      DEFINE FLUXNUM REGIONS FOR ALL CELLS
--
FLUXNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the [EQUALS](#) keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
    FLUXNUM      1                1*  1*   1*  1*   1*  1* / SET REGION 1
    FLUXNUM      2                1   2    1   2    1   1 / SET REGION 2
    FLUXNUM      3                1   2    1   2    2   2 / SET REGION 3
/
```

6.3.74 FLUXREG – DEFINE ACTIVE FLUX REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *FLUXREG* is used in conjunction with the *USEFLUX* keyword in runs with have multiple flux regions, to reduce the number of flux regions, that is the keyword specifies which flux regions are active and which are not in the current run.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.75 FLUXTYPE – DEFINES THE FLUX BOUNDARY TYPE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FLUXTYPE](#) keyword defines the type of flux boundary to be used in the run.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.76 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be place at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a dull description.

6.3.77 GDFILE – LOAD A GRID FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GDFILE](#) keyword loads a GRID file that contains the structural data for the grid as a set of topological cuboidal cells, and EGRID files that contain structural and property data. Note OPM Flow only supports reading in EGRID files at this time.

No.	Name	Description	Default
1	GRIDFILE	A character string enclosed in quotes that defines the GRID or EGRID file to be read in and be processed by OPM Flow. Again, OPM Flow only supports reading in EGRID files.	None
2	FMTOPT	A defined character string that defines the format of the GRID or EGRID file to be read and should be set to one of the following: 1) FORMATTED: If the file is formatted as ASCII i.e. a text file, as oppose to a binary file. The option can be abbreviated to just the letter F. 2) UNFORMATTED: If the file is in binary format, note this option can be abbreviated to just the letter U. If the variable FMTOPT is omitted then the default is for binary file input for the commercial simulator; whereas, OPM Flow derives FMTOPT from the file extension (*.EGRID or *.FEGRID), making FMTOPT superfluous.	U
Notes: 1) The keyword is terminated by a “/”.			

Table 6.42: GDFILE Keyword Description

See also the [GRIDFILE](#) keyword in the [GRID](#) section for exporting the GRID and EGRID files from the current simulation run.

Examples

The first example shown below loads the NOR-OPM-A00-GRID.EGRID file in binary format from the same directory as the data file.

```
--
--      LOAD A GRID FILE
--
GDFILE      'NOR-OPM-A00-GRID.EGRID'      /
```

The next example loads the same EGRID file one directory above from where the data file is located.

```
--
--      LOAD a GRID FILE
--
GDFILE      '..\NOR-OPM-A00-GRID.EGRID'    /
```

6.3.78 GDORIENT – DEFINE GRID ORIENTATION PARAMETERS

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

This keyword defines the grid orientation parameters for post-processing applications.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

6.3.79
GETDATA – LOAD AND ASSIGN DATA ARRAY FROM INIT OR RESTART FILES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [GETDATA](#) keyword loads a data array from a previously generated [INIT](#) or [RESTART](#) file and assigns the loaded array to either same array in the run or another property array.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.80 GRID – DEFINE THE START OF THE GRID SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *GRID* activation keyword marks the end of the *RUNSPEC* section and the start of the *GRID* section that defines the key grid property data for the simulator including the grid structure, porosity, permeability and other relevant grid property data.

There is no data required for this keyword.

Example

```
-- =====  
--  
-- GRID SECTION  
--  
-- =====  
GRID
```

The above example marks the end of the *RUNSPEC* section and the start of the *GRID* section in the OPM Flow data input file.

6.3.81 GRIDFILE – SET THE GRID FILE OUTPUT OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword controls the output of a standard [GRID](#) or extended [GRID](#) file, as well as the extensible EGRID file for post-processing applications. The extended and extensible [GRID](#) formats are comparable; however, the extensible [GRID](#) format is more compact and is the only format supported by OPM Flow.

No.	Name	Description	Default
1	NGRID	A positive integer that controls the output of the GRID geometry file: 0) - for no GRID file to be written out. 1) - for the standard GRID file to be written out. 2) - for the extended GRID file to be written out. Only the default value of zero is supported.	0
2	NEGRID	A positive integer that controls the output of the EGRID geometry file: 0) - for no extensible GRID file to be written out. 1) - for the extensible GRID file to be written out. Only the default value of one is supported.	1
Notes: 1) The keyword is terminated by a “/”.			

Table 6.43: GRIDFILE Keyword Description

Example

```
--
--      GRID FILE OUTPUT OPTIONS
--      GRID      EGRID
--      OPTN      OPTN
GRIDFILE
      0          1
/
```

The above example defines that no [GRID](#) file will be written out and that the extensible [GRID](#) (that is the EGRID geometry format) file will be produced. This is the only configuration that OPM Flow supports

6.3.82 GRIDUNIT – DEFINE THE GRID UNITS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GRIDUNIT](#) keyword defines the units of the grid data. It is usually output by pre-processing software when exporting the grid geometry. The data is not used by OPM Flow intrinsically, but is merely written to the output EGRID file, as specified by the [GRIDFILE](#) keyword, for the use of post-processing software like OPM ResInsight.

No.	Name	Description	Default
1	GRIDUNIT	A character string that defines the units of the coordinates stated on the MAPAXES keyword, and should be set to: 1) FEET for field units, 2) METRES for metric units, or 3) CM for laboratory units.	METRES
2	MAPOPT	A character string that defines if the grid data are measured relative to the map, or relative to the origin as stated on the MAPAXES keyword. MAPOPT should either be left blank (the default) indicating the origin is relative to the origin on the MAPAXES keyword, or set equal to MAP measured relative to the map.	I*
Notes: 1) Note the alternative spelling METRES, that is METERS is not recognized. 2) The keyword is terminated by a "/".			

Table 6.44: GRIDUNIT Keyword Description

Example

```
--
--      SET THE GRID UNITS FOR THE GRID
--
GRIDUNIT
      METRES
```

The above example defines that the [GRID](#) units to be metric.

6.3.83 HALFTRAN – ACTIVATE HALF BLOCK TRANSMISSIBILITY CALCULATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [HALFTRAN](#) keyword activates the half block transmissibility calculation option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

6.3.84 HEATCR – DEFINE RESERVOIR ROCK HEAT CAPACITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [HEATCR](#) keyword defines the reservoir rock volumetric heat capacity for all cells for when OPM Flow's thermal calculation is activated by the [THERMAL](#) keywords in the [RUNSPEC](#) section.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	HEATCR	HEATCR is an array of real positive numbers that define reservoir rock volumetric heat capacity of a grid block. Repeat counts may be used, for example 3000*25.0			None
		Btu/ft3/°R	kJ/m3/K	J/cm3/K	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.45: HEATCR Keyword Description

Note this keyword is incompatible with [SPECROCK](#) keyword in the [PROPS](#) section.

Example

```
--
--      DEFINE GRID BLOCK RESERVOIR ROCK HEAT CAPACITY
--      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--      KEYWORD IS INCOMPATIBLE WITH THE SPECROCK KEYWORD
--      (OPM FLOW THERMAL OPTION ONLY)
--
```

```
HEATCR      300*32.0      /
```

The above example defines the reservoir rock volumetric heat capacity of 32.0 for each cell in the 300 grid block model.

6.3.85 HEATCRT – DEFINE RESERVOIR ROCK HEAT CAPACITY TEMPERATURE DEPENDENCE FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **HEATCRT** keyword defines the reservoir rock volumetric heat capacity temperature dependence for all cells for when OPM Flow's thermal calculation is activated by the **THERMAL** keywords in the **RUNSPEC** section.

This keyword can only be used if OPM Flow's thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	HEATCRT	HEATCRT is an array of real positive numbers that define reservoir rock volumetric heat capacity temperature dependence of a grid block. Repeat counts may be used, for example 3000*0.05			None
		Btu/ft3/°R2	kJ/m3/K2	J/cm3/K2	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.46: HEATCRT Keyword Description

Note this keyword is incompatible with **SPECROCK** keyword in the **PROPS** section.

The data for this keyword and the **HEATCR** keyword are used to calculate the reservoir rock volumetric heat capacity temperature dependence using the following relationship:

$$\text{Heat Capacity of Rock} = \text{HEATCR}(\text{Temp} - \text{Temp}_{\text{ref}}) + \frac{\text{HEATCRT}(\text{Temp} - \text{Temp}_{\text{ref}})^2}{2} \quad (6.3)$$

Example

```
--
--      DEFINE RESERVOIR ROCK HEAT CAPACITY TEMPERATURE DEPENDENCE
--      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--      KEYWORD IS INCOMPATIBLE WITH THE SPECROCK KEYWORD
--      (OPM FLOW THERMAL OPTION ONLY)
--
HEATCRT
      300*0.05
```

The above example defines the reservoir rock volumetric heat capacity temperature dependence of 0.05 for each cell in the 300 grid block model.

6.3.86 HMAQUNUM – HISTORY MATCH NUMERICAL AQUIFER GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [HMAQUNUM](#) keyword defines the history match numerical aquifer gradient parameters for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and numerical aquifers have been specified in the model via the [AQUNUM](#) keyword and connected to the grid using [AQUCON](#) keyword.All keywords are in the [GRID](#) section.

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of aquifers that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.87 HMFAULTS – HISTORY MATCH FAULT GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [HMFAULTS](#) keyword defines the history match faults gradient parameters for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and for when the [FAULTS](#) keyword in the [GRID](#) section has been used to define faults in the model.

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of faults that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.88
HMMLAQUN
– HISTORY MATCH NUMERICAL AQUIFER GRADIENT MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [HMMLAQUN](#) keyword defines the history match numerical aquifer gradient multipliers for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and numerical aquifers have been specified in the model via the [AQUNUM](#) keyword and connected to the grid using the [AQUCON](#) keyword. All keywords are in the [GRID](#) section.

Multipliers can be declared for numerical aquifers’ pore volume, permeability, and aquifer to grid connection factors.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.89

HMMLT – HISTORY MATCH GRID PERMEABILITY GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *HMMLT* series of keywords defines the history match gradient cumulative permeability multipliers, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section. The keyword consists of the first five characters of “HMMLT” followed by a two or three character string shown in Table 6.47, that determines the permeability direction, for example, HMMLTPX.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Mnemonic	Cartesian Grid		Radial Grid	
	Grid Keyword	<i>HMMULT</i> Keyword	Grid Keyword	<i>HMMULT</i> Keyword
PX/PR	<i>PERMX</i>	HMMLTPX	<i>PERMR</i>	HMMLTPR
PXY	PERMXY	HMMLTPXY		
PY/THT	<i>PERMY</i>	HMMLTPY	<i>PERMTHT</i>	HMMLTTH
PZ	<i>PERMZ</i>	HMMLTPZ	<i>PERMZ</i>	HMMLTPZ

Table 6.47: *HMMLT* Keyword List

See also the *HMMULT* keyword in the *EDIT* section

6.3.90
HMMMREGT – HISTORY MATCH REGION TRANSMISSIBILITY GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [HMMMREGT](#) keyword multiplies the transmissibility between two regions by a constant, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The constant should be a real number. Unlike the [MULTREGT](#) keyword in the [GRID](#) section, the [HMMMREGT](#) keyword modifications are cumulative.

Note that the [HMMMREGT](#) keyword only declares the two regions and the multiplier between those regions, the transmissibility direction (DIR on the [MULTREGT](#) keyword), type of transmissibility multiplier (TYPE on the [MULTREGT](#) keyword), and the region number array to use (ARRAY on the [MULTREGT](#) keyword), are all taken from the MULTREGY keyword. For example, the region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [MULTREGT](#) keyword is read by the simulator, and before the [HMMMREGT](#) keyword is used.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

6.3.91 HMMULRGT – HISTORY MATCH REGION TRANSMISSIBILITY PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

HMMULRGT defines the transmissibility between two regions gradient parameters, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section.

See also the *HMDIMS* keyword in the *RUNSPEC* section that specifies the dimensions for the gradient option, including the maximum number of gradient regions that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

6.3.92
HMMULTFT
– HISTORY MATCH FAULT TRANSMISSIBILITY GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

HMMULTFT defines the history match fault transmissibility gradient cumulative multipliers to be applied to the fault transmissibilities for faults declared by the **FAULT** keyword in the *GRID* section, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section. The keyword allows for the re-scaling of the existing fault transmissibilities calculated by OPM Flow, or if the *MULTFLT* keyword has been entered, then *HMMULTFT* is applied to the existing *MULTFLT* multipliers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.93
HMMULTSG – HISTORY MATCH DUAL POROSITY SIGMA GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[HMMULTSG](#) defines the history match dual porosity sigma parameter gradient cumulative multipliers applied to the dual porosity sigma value declared by the [SIGMAV](#) and [SIGMAGDV](#) keywords in the [PROPS](#) section, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. In addition to the [HMDIMS](#) keyword, either the [DUALPERM](#) keyword that activates the Dual Permeability option, or the [DUALPORO](#) keyword that activates the Dual Porosity option for the run, must be declared in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.94 HRFIN – DEFINE THE RATIO OF LGR GRID BLOCKS IN THE R-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

HRFIN¹²³ defines the ratio of grid blocks for the **DRV** keyword in the r-direction via a vector within a Local Grid Refinement (“LGR”) as opposed to defining the size for each cell for a Radial **LGR** Grid. The **LGR** keyword in the **RUNSPEC** section should be activated to indicate an **LGR** is being used, and the keyword **HRFIN** should be placed in between the **RADIN** (or **RAFDIN4**) and **ENDFIN** keywords in the **GRID** section. The **DRV** keyword in the **GRID** section defines the radial grid size in terms of the length, that is feet for field units, this keyword defines the length as the ratio of the previous cell size, starting with the inner radius (**INRAD**).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	HRFIN	HRFIN is a vector of real numbers describing the ratio of cell size for the grid blocks in the r-direction in a radial LGR for the DRV keyword. Repeat counts may be used, for example 2*1.5.			None
		dimensionless	dimensionless	dimensionless	

Notes:

1) The number of entries should correspond to the NR_parameter minus one on the RADFIN or RADFIN4 keywords in the GRID section

2) The keyword is terminated by a “/”.

Table 6.48: HRFIN Keyword Description

See also the **DR**, **DRV**, **DTHETAV**, and **DZ** keywords in the **GRID** section to fully define a radial **LGR** model.

Example

```
--
--      INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD    0.25                                     /
--
--      DEFINE GRID BLOCK DRV RATIOS IN THE R DIRECTION
--
HRFIN
      1.50  2.00  3.00  5.00  7.00  10.00                                     /
```

The above example defines the size of the cells in the R direction based on NR equals 7, resulting in NR-I values on the **RADFIN** keyword in the **GRID** section. Note the **INRAD** keyword to define the inner radius of the radial grid.

¹²³ Radial grids are not currently implemented in this version of OPM Flow, but is expected to be incorporated in a future release.

6.3.95 HXFIN – DEFINE THE RATIO OF LGR GRID BLOCKS IN THE X-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

HXFIN defines the split ratio of grid blocks for the *DXV* keyword in the x-direction via a vector within a Local Grid Refinement (“LGR”) as opposed to defining the size for each cell for a Cartesian *LGR* Grid. The *LGR* keyword in the *RUNSPEC* section should be activated to indicate an *LGR* is being used, and the keyword *HXFIN* should be placed in between the *CARFIN* and *ENDFIN* keywords in the *GRID* section. The *DXV* keyword in the *GRID* section defines the grid size in terms of the length, that is feet for field units, this keyword defines the length as the ratio of the coarse cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	HXFIN	HXFIN is a vector of real numbers describing the ratio of cell size for the grid blocks in the x-direction in a Cartesian LGR grid. Repeat counts may be used, for example 2*2.0.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX parameter on the CARFIN keyword in the GRID section.
- 2) The keyword is terminated by a “/”.

Table 6.49: HXFIN Keyword Description

See also the *CARFIN*, *ENDFIN*, *HYFIN*, and *HZFIN* keywords in the *GRID* section to fully define a Cartesian *LGR* grid model.

Example

```
--
-- CARFIN LGR GRID COMMANDS
--
-- LGR      ----- HOST GRID -----  -- CARFIN GRID --  MAX  HOST
-- NAME      I1  I2  J1  J2  K1  K2      NX   NY   NZ    WELLS  NAME
CARFIN
LGR-OP01    24  25  86  87   1  50       5    3   50     1    GLOBAL /
--
-- DEFINE LGR GRID BLOCK IN THE X-DIRECTION
NXFIN
3  2
/
--
-- DEFINE GRID BLOCK LGR RATIOS IN THE X-DIRECTION
--
HXFIN
1.00  2.00  3.00  2.00  1.00
/
ENDFIN
```

The above example defines the size of the cells in the x-direction based on NX equals five on the *CARFIN* keyword in the *GRID* section.

6.3.96 HYFIN – DEFINE THE RATIO OF LGR GRID BLOCKS IN THE Y-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

HYFIN defines the split ratio of grid blocks for the **DYV** keyword in the y-direction via a vector within a Local Grid Refinement (“LGR”) as opposed to defining the size for each cell for a Cartesian **LGR** Grid. The **LGR** keyword in the **RUNSPEC** section should be activated to indicate an **LGR** is being used, and the keyword **HYFIN** should be placed in between the **CARFIN** and **ENDFIN** keywords in the **GRID** section. The **DYV** keyword in the **GRID** section defines the grid size in terms of the length, that is feet for field units, this keyword defines the length as the ratio of the coarse cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	HYFIN	HYFIN is a vector of real numbers describing the ratio of cell size for the grid blocks in the y-direction in a Cartesian LGR grid. Repeat counts may be used, for example 2*2.0.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NY parameter on the CARFIN keyword in the GRID section.
- 2) The keyword is terminated by a “/”.

Table 6.50: HYFIN Keyword Description

See also the **CARFIN**, **ENDFIN**, **HXFIN**, and **HZFIN** keywords in the **GRID** section to fully define a Cartesian **LGR** grid model.

Example

```
--
-- CARFIN LGR GRID COMMANDS
--
-- LGR ----- HOST GRID ----- -- CARFIN GRID -- MAX HOST
-- NAME I1 I2 J1 J2 K1 K2 NX NY NZ WELLS NAME
CARFIN
LGR-OP01 24 25 86 87 1 50 3 5 50 1 GLOBAL /
--
-- DEFINE LGR GRID BLOCK IN THE Y-DIRECTION
NYFIN
3 2 /
--
-- DEFINE GRID BLOCK LGR RATIOS IN THE Y-DIRECTION
--
HYFIN
1.00 2.00 3.00 2.00 1.00 /
ENDFIN
```

The above example defines the size of the cells in the y-direction based on NY equals five on the **CARFIN** keyword in the **GRID** section.

6.3.97 HZFIN – DEFINE THE RATIO OF LGR GRID BLOCKS IN THE Z-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

HZFIN defines the split ratio of grid blocks for the *DZV* keyword in the z-direction via a vector within a Local Grid Refinement (“LGR”) as opposed to defining the size for each cell for a Cartesian *LGR* Grid. The *LGR* keyword in the *RUNSPEC* section should be activated to indicate an *LGR* is being used, and the keyword *HYFIN* should be placed in between the *CARFIN* and *ENDFIN* keywords in the *GRID* section. The *DZV* keyword in the *GRID* section defines the grid size in terms of the length, that is feet for field units, this keyword defines the length as the ratio of the coarse cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	HYZIN	<i>HZFIN</i> is a vector of real numbers describing the ratio of cell size for the grid blocks in the z-direction in a Cartesian <i>LGR</i> grid. Repeat counts may be used, for example 2*2.0.			None
		dimensionless	dimensionless	dimensionless	
Notes: 1) The number of entries should correspond to the NZ parameter on the <i>CARFIN</i> keyword in the <i>GRID</i> section. 2) The keyword is terminated by a “/”.					

Table 6.51: HZFIN Keyword Description

See also the *CARFIN*, *ENDFIN*, *HXFEN*, and *HYFIN* keywords in the *GRID* section to fully define a Cartesian *LGR* grid model.

Example

```
--
-- CARFIN LGR GRID COMMANDS
--
-- LGR      ----- HOST GRID -----  -- CARFIN GRID --  MAX  HOST
-- NAME      I1  I2  J1  J2  K1  K2      NX   NY   NZ   WELLS  NAME
CARFIN
LGR-OP01    24  25  86  87   1  50       5    3   100    1   GLOBAL /
--
-- DEFINE LGR GRID BLOCK IN THE Z-DIRECTION
NZFIN
50*2                                              /
--
-- DEFINE GRID BLOCK LGR RATIOS IN THE Z-DIRECTION
--
HZFIN
50*2.0                                           /
ENDFIN
```

The above example defines the size of the cells in the z-direction based on NZ equals 100 on the *CARFIN* keyword in the *GRID* section.

6.3.98 IHOST – Assign LGRs to PARALLEL PROCESS NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The IHOST keyword assigns Local Grid Refinements (“LGR”) to a parallel process number, for when the [PARALLEL](#) keyword has been invoked in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.99 IMPORT – IMPORT GRID FILE DATA AT THE CURRENT POSITION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **IMPORT** keyword informs the simulator to import data from the specified **IMPORT** file. When the end of the **IMPORT** file is reached, input data is read from the next keyword in the current file. Normally **IMPORT** files are generated by grid pre-processing software and the keyword allows for both formatted and unformatted (binary) files to be loaded. The keyword can be used to import any valid grid arrays within a section, for example the **EQLNUM** array in the **REGIONS** section

No.	Name	Description	Default
1	FILENAME	A character string enclosed in quotes that defines a file to be imported and to be processed by OPM Flow.	None
2	FMTOPT	A defined character string that defines the format of the file to be imported and should be set to one of the following: 1) FORMATTED : If the file is formatted as ASCII i.e. a text file, as oppose to a binary file. The option can be abbreviated to just the letter F. 2) UNFORMATTED : If the file is in binary format, note this option can be abbreviated to just the letter U. If the variable FMTOPT is omitted then the default is for binary file input.	U
Notes: 1) The keyword is terminated by a "/".			

Table 6.52: IMPORT Keyword Description

Examples

The first example shown below loads the grid file from the same directory as the data file.

```
--
--      LOAD A IMPORT FILE
--
IMPORT      'NOR-OPM-A00-GRID.EGRID'  /
```

The next example loads the same file one directory above from where the data file is located.

```
--
--      LOAD A IMPORT FILE
--
IMPORT      '..\NOR-OPM-A00-GRID.EGRID'  /
```

The final example loads the same file from a separate include subdirectory in the parent directory of the data file.

```
--
--      LOAD A IMPORT FILE
--
IMPORT      '..\INCLUDE\NOR-OPM-A00-GRID.EGRID'  /
```

6.3.100 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDING](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the GLOBAL section for a full description.

6.3.101 INIT – ACTIVATE THE INIT FILE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword switches on the writing of the [INIT](#) file that contains the static data specified in the [GRID](#), [PROPS](#) and [REGIONS](#) sections. For example, the [PORO](#), PERM and [NTG](#) arrays from the [GRID](#) section. The data is used in post-processing software, for example OPM ResInsight, to visualize the static grid properties.

The [INIT](#) file can either be written out in formatted form as ASCII i.e. text files, if the [FMTOUT](#) keyword has been activated, or binary format if the [FMTOUT](#) keyword has not been activated. Normally, this option is always activated by the user and when activated the binary form of the file is used.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE WRITING THE INIT FILE FOR POST-PROCESSING
INIT
```

The above example switches on the writing of the [INIT](#) file for post-processing in ResInsight.

6.3.102 INRAD – DEFINE THE INNER RADIUS OF A RADIAL GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[INRAD](#) defines the inner radius of the reservoir model for a radial grid geometry. The [RADIAL](#) or [SPIDER](#) keyword in the [RUNSPEC](#) should be activated to indicate that radial geometry is being used.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	INRAD	A single real positive number defining the inner radius of a radial grid.			None
		feet	m	cm	
Notes: I) The keyword is terminated by a “/”.					

Table 6.53: INRAD Keyword Description

Note that the [SPIDER](#) keyword activates OPM Flow’s radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: [INRAD](#), [DRV](#), [DTHETA](#), [DZ/DZV](#) etc. in the [GRID](#) the section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model’s pore volume to reflect radial coordinates; thus, overcoming the display limitation.

See also the [DR](#), [DRV](#), [DTHETA](#), [DTHETA](#) and [TOPS](#) keywords to fully define a radial or spider grid in the model.

Example

```
--
--      INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD      0.25      /
```

The above example defines the inner radius of a radial grid to be 0.25 feet.

6.3.103 IONROCK – DEFINE THE ION EXCHANGE CAPACITY FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *IONROCK* keyword defines the ion exchange capacity for all the cells in the model, for when the brine phase has been activated by the *BRINE* keyword and the Multi-Component Brine model, that allows for the water phase to have multiple water salinities, has been activated by the *ECLMC* keyword. Both keywords are in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.104 ISOLNUM – DEFINE THE INDEPENDENT RESERVOIR REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *ISOLNUM* keyword defines areas of the grid that consists of isolated reservoirs where the only form of communication between the reservoirs is via wellbore connections. This enables the reservoir flow equations to be solved independently for greater computational efficiency.

No.	Name	Description	Default
I	<i>ISOLNUM</i>	<i>ISOLNUM</i> defines an array of positive integers assigning a grid cell to a particular isolated reservoir region. The maximum number of <i>ISOLNUM</i> regions is set by the <i>NRFREG</i> variable on the <i>REGDIMS</i> keyword in the <i>RUNSPEC</i> section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section, unless the <i>BOX</i> keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the <i>BOX</i> statement. 2) Setting <i>ISOLNUM</i> for a cell to zero makes the cell inactive. 3) The keyword is terminated by a “/”.			

Table 6.54: ISOLNUM Keyword Description

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

The example below defines three separate independent reservoirs; the first reservoir covers the whole grid and layers 1 to 50, reservoir two cover the whole grid and layers 52 to 150, and finally the third reservoir again covers the whole grid but with layers 152 to 300. The layers 51 and 151 are shale layers made inactive by setting *ISOLNUM* to zero.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      ISOLNUM      1           1*  1*   1*  1*   1  50 / DEFINED RESERVOIR 1
      ISOLNUM      0           1*  1*   1*  1*   51 51 / DEFINED A SHALE
      ISOLNUM      2           1*  1*   1*  1*   52 150 / DEFINED RESERVOIR 2
      ISOLNUM      0           1*  1*   1*  1*  151 151 / DEFINED A SHALE
      ISOLNUM      3           1*  1*   1*  1*  152 300 / DEFINED RESERVOIR 3
/
```

Note the above example has no effect as the keyword is ignored by the simulator.

6.3.105 JFUNC – ACTIVATE THE LEVERETT J-FUNCTION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **JFUNC** keyword activates the Leverett-J-Function¹²⁴ option which is a commonly used technique to normalize capillary pressure based on laboratory measured core plugs porosity and permeability values and the resulting capillary pressure data. The keyword performs the calculation based on the parameters on the this keyword combined with a cells porosity and permeability to perform the scaling globally.

The keyword should only be used if end-point scaling is switched on using the **ENDSCALE** keyword in the **RUNSPEC** section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	JFOPT	A character string that defines which capillary data sets the J-Function option should be applied to, based on the following options: 1) WATER : apply the J-Function option to the water-oil capillary pressure data only. 2) GAS : apply the J-Function option to the gas-oil capillary pressure data only. 3) BOTH : apply the J-Function option to the water-oil and the gas-oil capillary pressure data.			BOTH
2	OWSTEN	A positive real number that defines oil-water surface tension used to de-normalized J-Function data entered in the PROPS section.			None
		dynes/cm	dynes/cm	dynes/cm	
3	OGSTEN	A positive real number that defines oil-gas surface tension used to de-normalized J-Function data entered in the PROPS section.			None
		dynes/cm	dynes/cm	dynes/cm	
4	ALPHA	A positive real value that defines an alternative power value for the porosity term in the J-Function equation, that is instead of $\sqrt{\frac{k}{\phi}}$ use $\frac{k^{0.5}}{\phi^\alpha}$ instead in the transformation.			0.5
5	BETA	A positive real number that defines an alternative power value for the permeability term in the J-Function equation, that is instead of $\sqrt{\frac{k}{\phi}}$ use $\frac{k^\beta}{\phi^{0.5}}$ instead in the transformation.			0.5

¹²⁴ Leverett, M. C.; "Capillary Behaviour in Porous Solids", Trans. AIME (1941) 142, 152-168.

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	PERM	PERM is a character string that sets the permeability array to be used in the transform, based on the following options: 1) X: use the <i>PERMX</i> array. 2) XY: use the average of the <i>PERMX</i> and <i>PERMY</i> arrays. 3) Y: use the <i>PERMY</i> array. 4) Z: use the <i>PERMZ</i> array. 5) U: use the <i>PERMJFUN</i> array			XY

Notes:

1) The keyword is terminated by a “/”.

Table 6.55: JFUNC Keyword Description

Just like the relative permeability data capillary pressure data are measured on core plugs with varying quality and perhaps from different reservoirs. It is therefore necessary to determine averaged data, before employing the data in engineering calculations. This is commonly done by using the Leverett J-function¹²⁵, which is defined as:

$$J(S_w) = \frac{P_{c,res}(S_w) \sqrt{\frac{k}{\phi}}}{\sigma} \quad (6.4)$$

Where:

$J(S_w)$	= dimensionless function of water saturation
$P_c(S_w)$	= capillary pressure (kPa)
k	= permeability, (m ²)
ϕ	= porosity (fraction)
σ	= interfacial tension (mN/m)
Θ	= contact angle

Sometimes the equation is stated with the $\cos \theta$ term included, that is:

$$J(S_w) = \frac{P_{c,res}(S_w) \sqrt{\frac{k}{\phi}}}{\sigma \cos \Theta} \quad (6.5)$$

Since the above function is just a normalizing function, then units are not important, as long as when we de-normalize the *average* curve we use the same unit set. Secondly, if all the capillary pressure data has been converted to reservoir conditions, we actually ignore the denominator as it is a constant, and we can therefore just use:

$$J(S_w) = P_{c,res}(S_w) \sqrt{\frac{k}{\phi}} \quad (6.6)$$

¹²⁵ Leverett, M. C.; "Capillary Behaviour in Porous Solids", Trans. AIME (1941) 142, 152-168.

However, in the simulator it is necessary to use the formal definition as outlined in equation (6.4). In addition to the standard the equation the keyword allows for de-normalizing the curve to use alternative power functions instead of the standard 0.5 used in equation (6.4), that is:

$$J(S_w) = \frac{P_{c,res}(S_w) \left(\frac{k^\beta}{\phi^\alpha} \right)}{\sigma} \quad (6.7)$$

Where:

$J(S_w)$	=	dimensionless function of water saturation
$P_c(S_w)$	=	capillary pressure (kPa)
k	=	permeability, (m ²)
ϕ	=	porosity (fraction)
σ	=	interfacial tension (mN/m)
Θ	=	contact angle
α	=	porosity power value
β	=	permeability value

The **JFUNC** keyword allows the data entered as capillary pressure in the saturation tables, for example, by using the **SGFN** and **SWFN** keywords in the **PROPS** section to be treated as J-functions instead, and to de-normalize these curves for each active cell in the model using the options and values defined with the **JFUNC** keyword combined with a cells porosity and permeability values.

Note

If either the **JFUNC** or **JFUNCR** keywords are used to activate J-Function scaling then the **ENDSCALE** keyword in the **RUNSPEC** section must also be present in the input deck, in order for the dimensionless J-function values entered on the **SWFN**, **SGFN** or the **SWOF**, **SGOF**, **SLGOF** keywords to be re-scaled to capillary pressure data.

Note if the **ENDSCALE** keyword is absent, then like the commercial simulator, J-Function scaling is not performed, and the values entered on the **SWFN**, **SGFN** or the **SWOF**, **SGOF**, **SLGOF** keywords are used as entered.

See also the **JFUNCR** keyword in the **GRID** section that performs similar calculations based on the J-Function parameters being declared by saturation table number.

Example

```
--
--      DEFINE LEVERETT J-FUNCTION PARAMETERS
--      JFUN      OILWAT  GASOIL  PORO    PERM    PERM
--      OPTN      SDENS   SDEN    ALPHA   BETA    OPTN
JFUNC
      WATER      22.5    1*      0.5     0.5     XY      /
```

The above example results in the oil-water capillary pressure data entered on the **SWFN** keyword in the **PROPS** section being treated as J-Functions, and that the J-Function should be de-normalized using an oil-water surface density of 22.5 dynes/cm, using the default power values and the average of the **PERMX** and **PERMY** values for each grid block.

6.3.106 JFUNC R – ACTIVATE THE LEVERETT J-FUNCTION SATURATION TABLE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

JFUNC R keyword activates Leverett-J-Function¹²⁶ Saturation Table option which is a commonly used technique to normalize capillary pressure base on laboratory measured core plugs porosity and permeability values and the resulting capillary pressure data. This keyword is an extension of the **JFUNC** keyword in the **GRID** section that uses the parameters on the **JFUNC** keyword combined with a cell's porosity and permeability to perform the scaling globally. In comparison, the **JFUNC R** allows for the J-Function parameters to be declared per saturation table number, resulting in greater flexibility.

The keyword should only be used if end-point scaling is switched on using the **ENDSCALE** keyword in the **RUNSPEC** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	JFOPT	A character string that defines which capillary data sets the J-Function option should be applied to, based on the following options: 1) WATER : apply the J-Function option to the water-oil capillary pressure data only. 2) GAS : apply the J-Function option to the gas-oil capillary pressure data only. 3) BOTH : apply the J-Function option to the water-oil and the gas-oil capillary pressure data.			BOTH
2	OWSTEN	A positive real number that defines oil-water surface tension used to de-normalized J-Function data entered in the PROPS section.			None
		dynes/cm	dynes/cm	dynes/cm	
3	OGSTEN	A positive real number that defines oil-gas surface tension used to de-normalized J-Function data entered in the PROPS section.			None
		dynes/cm	dynes/cm	dynes/cm	
4	ALPHA	A positive real value that defines an alternative power value for the porosity term in the J-Function equation, that is instead of $\sqrt{\frac{k}{\phi}}$ use $\frac{k^{0.5}}{\phi^\alpha}$ instead in the transformation.			0.5
5	BETA	A positive real number that defines an alternative power value for the permeability term in the J-Function equation, that is instead of $\sqrt{\frac{k}{\phi}}$ use $\frac{k^\beta}{\phi^{0.5}}$ instead in the transformation.			0.5

¹²⁶ Leverett, M. C.; "Capillary Behaviour in Porous Solids", Trans. AIME (1941) 142, 152-168.

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	PERM	PERM is a character string that sets the permeability array to be used in the transform, based on the following options: 1) X: use the PERMX array. 2) XY: use the average of the PERMX and PERMY arrays. 3) Y: use the PERMY array. 4) Z: use the PERMZ array. 5) U: use the PERMJFUN array			XY

Notes:

- 1) The keyword is followed by exactly NTSFUN rows of data, as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each row is terminated by a “/” and there is no “/” terminator for the keyword.

Table 6.56: JFUNCR Keyword Description

Just like the relative permeability data capillary pressure data are measured on core plugs with varying quality and perhaps from different reservoirs. It is therefore necessary to determine averaged data, before employing the data in engineering calculations. This is commonly done by using the Leverett J-function¹²⁷, which is defined as:

$$J(S_w) = \frac{P_{c,res}(S_w) \sqrt{\frac{k}{\phi}}}{\sigma} \quad (6.8)$$

Where:

$J(S_w)$	=	dimensionless function of water saturation
$P_c(S_w)$	=	capillary pressure (kPa)
k	=	permeability, (m ²)
ϕ	=	porosity (fraction)
σ	=	interfacial tension (mN/m)
Θ	=	contact angle

Sometimes the equation is stated with the $\cos \theta$ term included, that is:

$$J(S_w) = \frac{P_{c,res}(S_w) \sqrt{\frac{k}{\phi}}}{\sigma \cos \Theta} \quad (6.9)$$

Since the above function is just a normalizing function, then units are not important, as long as when we de-normalize the *average* curve we use the same unit set. Secondly, if all the capillary pressure data has been converted to reservoir conditions, we can actually ignore the denominator as it is a constant, and we can therefore just use:

$$J(S_w) = P_{c,res}(S_w) \sqrt{\frac{k}{\phi}} \quad (6.10)$$

¹²⁷ Leverett, M. C.; “Capillary Behaviour in Porous Solids”, *Trans. AIME* (1941) 142, 152-168.

However, in the simulator it is necessary to use the formal definition as outlined in equation (6.8). In addition to the standard the equation the keyword allows for de-normalizing the curve to use alternative power functions instead of the standard 0.5 used in equation (6.8), that is:

$$J(S_w) = \frac{P_{c,res}(S_w) \left(\frac{k^\beta}{\phi^\alpha} \right)}{\sigma} \quad (6.11)$$

Where:

$J(S_w)$	=	dimensionless function of water saturation
$P_c(S_w)$	=	capillary pressure (kPa)
k	=	permeability, (m ²)
ϕ	=	porosity (fraction)
σ	=	interfacial tension (mN/m)
Θ	=	contact angle
α	=	porosity power value
β	=	permeability value

The **JFUNC** keyword allows the data entered as capillary pressure in the saturation tables, for example, by using the **SGFN** and **SWFN** keywords in the **PROPS** section to be treated as J-functions instead, and to de-normalize these curves for each active cell in the model using the options and values defined with the **JFUNC** keyword combined with a cells porosity and permeability values.

Note

If either the **JFUNC** or **JFUNCR** keywords are used to activate J-Function scaling then the **ENDSCALE** keyword in the **RUNSPEC** section must also be present in the input deck, in order for the dimensionless J-function values entered on the **SWFN**, **SGFN** or the **SWOF**, **SGOF**, **SLGOF** keywords to be re-scaled to capillary pressure data.

Note if the **ENDSCALE** keyword is absent, then like the commercial simulator, J-Function scaling is not performed, and the values entered on the **SWFN**, **SGFN** or the **SWOF**, **SGOF**, **SLGOF** keywords are used as entered.

See also the **JFUNC** keyword in the **GRID** section that uses the parameters on the **JFUNC** keyword combined with a cell's porosity and permeability to perform the scaling globally.

Example

The example below assumes NTSFUN is equal to five on the **TABDIMS** keyword in the **RUNSPEC** section.

```
--
--      DEFINE LEVERETT J-FUNCTION PARAMETERS BY SATURATION TABLES
--      JFUN      OILWAT  GASOIL  PORO    PERM    PERM
--      OPTN      SDENS   SDEN    ALPHA   BETA     OPTN
JFUNCR
      WATER      22.5    1*      0.5     0.5     XY
      WATER      22.5    1*      0.5     0.5     XY
      WATER      22.5    1*      0.5     0.5     XY
      WATER      22.5    1*      0.5     0.5     XY
      WATER      22.5    1*      0.5     0.5     XY
```

Here the oil-water capillary pressure data entered on the **SWFN** keyword in the **PROPS** section are treated as J-Functions, and that the J-Function should be de-normalized using an oil-water surface density of 22.5 dynes/cm, using the default power values and the average of the **PERMX** and **PERMY** values for each grid block, for all five tables. Note that since all the **JFUNCR** parameters are the same for all saturation tables then the **JFUNC** keyword could be used instead in this instance.

6.3.107 LGRCOPY – ACTIVATE LOCAL GRID REFINEMENT INHERITANCE

The [LGRCOPY](#) keyword activates Local Grid Refinement (“LGR”) Inheritance option that allows the [LGR](#) to inherit the properties of the global or host cell containing a [LGR](#) grid block at the start of the [GRID](#) section, as oppose to the normal process of applying this transform at the end of the [GRID](#) section. [LGRCOPY](#) can be used in the [RUNSPEC](#), [GRID](#) and [EDIT](#) sections. If used in the [RUNSPEC](#) section then the option is applied to all LGRs defined in the input file, whereas if used in the [GRID](#) and [EDIT](#) sections the keyword must be placed inside a [LGR](#) definition section, that is between a [CARFIN](#) (Cartesian [LGR](#) grid) or [RADIN/RADIN4](#) (radial [LGR](#) grid) and the [ENDFIN](#) keyword. In the latter case inheritance is applied on an individual [LGR](#) basis.

See [LGRCOPY – Activate Local Grid Refinement Inheritance](#) in the [RUNSPEC](#) section for a full description.

6.3.108 LINKPERM – ASSIGN CELL PERMEABILITIES TO CELL FACES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LINKPERM](#) keyword assigns the grid cell permeabilities entered via the [PERMX](#), [PERMY](#) and [PERMZ](#) keywords to a cell face ($I\pm$, $J\pm$, or $K\pm$) and results in the simulator using these values directly in the calculating the transmissibility between grid blocks. This is different to the conventional way of entering permeability data that consists of entering the cell centered permeability and the simulator calculating a weighted average transmissibility based on the cell centered permeability of the up-stream and down-stream grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.109
LTOSIGMA
– DUAL POROSITY VISCOUS DISPLACEMENT SIGMA PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LTOSIGMA](#) keyword defines parameters to calculate the sigma factor in conjunction with the data entered via the [LX](#), [LY](#) and [LZ](#) keywords in the [GRID](#) section, for when the [VISCD](#) keyword has been used in the [RUNSPEC](#) section to activate the Dual Porosity Viscous Displacement option. In addition, either the [DUALPORO](#) or [DUALPERM](#) keyword should be entered in the [RUNSPEC](#) section to activate the dual porosity or dual permeability models.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.110 LX – DUAL POROSITY VISCOUS DISPLACEMENT X DIRECTION MATRIX SIZE FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LX](#) keyword defines the size of “representative” matrix grid blocks in the X direction via an array in dual porosity and dual permeability runs, for when the [VISCD](#) keyword has been used in the [RUNSPEC](#) section to activate the dual porosity viscous displacement option. In addition, either the [DUALPORO](#) or [DUALPERM](#) keyword should be entered in the [RUNSPEC](#) section to activate the dual porosity or dual permeability models. The [VISCD](#) option is used to model the viscous displacement of fluids from the matrix by the fracture pressure gradient, for when the fracture system has a more moderate permeability, and flow to and from the matrix caused by the fracture pressure gradient acts as an additional production mechanism.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	LX	LX is an array of real numbers describing the “representative” cell size in the X direction for each cell in the model. Repeat counts may be used, for example 10*100.0.			0
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x (NZ/2) parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.

Table 6.57: LX Keyword Description

If the [VISCD](#) keyword has been used to activate the Dual Porosity Viscous Displacement option and [LX](#) has not been specified then [LX](#) is set to zero in the calculation of the viscous displacement term. See also the [LY](#), [LZ](#) and [LTOSIGMA](#) keywords in the [GRID](#) section.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                      / DEFINE BOX AREA
--
--      DEFINE DUAL POROSITY VISCOUS DISPLACEMENT X DIRECTION MATRIX SIZE
--
LX      6*10.0                      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a subset of the grid and the size of the “representative” matrix cells in the X direction to 10.0 ft.; after which the [ENDBOX](#) keyword resets the input to be the full grid.

6.3.111 LXFIN – DEFINE LOGARITHMIC LGR GRID BLOCK SPACING IN THE X-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LXFIN](#) keyword defines the parameters for automatically generating a Local Grid Refinement (“LGR”) grid in the X direction based on logarithmic block spacing, for when the [LGR](#) option has been activated by the [LGR](#) keyword in the [RUNSPEC](#) section. [LXFIN](#) should be placed in between the [CARFIN](#) and [ENDFIN](#) keywords in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.112 LY – DUAL POROSITY VISCOUS DISPLACEMENT Y DIRECTION MATRIX SIZE FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LY](#) keyword defines the size of “representative” matrix grid blocks in the Y direction via an array in dual porosity and dual permeability runs, for when the [VISCD](#) keyword has been used in the [RUNSPEC](#) section to activate the dual porosity viscous displacement option. In addition, either the [DUALPORO](#) or [DUALPERM](#) keyword should be entered in the [RUNSPEC](#) section to activate the dual porosity or dual permeability models. The [VISCD](#) option is used to model the viscous displacement of fluids from the matrix by the fracture pressure gradient, for when the fracture system has a more moderate permeability, and flow to and from the matrix caused by the fracture pressure gradient acts as an additional production mechanism.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	LY	LY is an array of real numbers describing the “representative” cell size in the Y direction for each cell in the model. Repeat counts may be used, for example 10*100.0.			0
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x (NZ/2) parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 6.58: LY Keyword Description

If the [VISCD](#) keyword has been used to activate the Dual Porosity Viscous Displacement option and [LY](#) has not been specified then [LY](#) is set to zero in the calculation of the viscous displacement term.

See also the [LX](#), [LZ](#) and [LTOSIGMA](#) keywords in the [GRID](#) section.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                      / DEFINE BOX AREA
--
--      DEFINE DUAL POROSITY VISCOUS DISPLACEMENT Y DIRECTION MATRIX SIZE
--
LY      6*15.0                      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a subset of the grid and the size of the “representative” matrix cells in the Y direction to 15.0 ft.; after which the [ENDBOX](#) keyword resets the input to be the full grid.

6.3.113
LYFIN – DEFINE LOGARITHMIC LGR GRID BLOCK SPACING IN THE Y-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [LYFIN](#) keyword defines the parameters for automatically generating a Local Grid Refinement (“LGR”) grid in the Y direction based on logarithmic block spacing, for when the [LGR](#) option has been activated by the [LGR](#) keyword in the [RUNSPEC](#) section. [LYFIN](#) should be placed in between the [CARFIN](#) and [ENDFIN](#) keywords in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.114 LZ – DUAL POROSITY VISCOUS DISPLACEMENT Z DIRECTION MATRIX SIZE FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LZ](#) keyword defines the size of “representative” matrix grid blocks in the Z direction via an array in dual porosity and dual permeability runs, for when the [VISCD](#) keyword has been used in the [RUNSPEC](#) section to activate the dual porosity viscous displacement option. In addition, either the [DUALPORO](#) or [DUALPERM](#) keyword should be entered in the [RUNSPEC](#) section to activate the dual porosity or dual permeability models. The [VISCD](#) option is used to model the viscous displacement of fluids from the matrix by the fracture pressure gradient, for when the fracture system has a more moderate permeability, and flow to and from the matrix caused by the fracture pressure gradient acts as an additional production mechanism.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	LZ	LZ is an array of real numbers describing the “representative” cell size in the Z direction for each cell in the model. Repeat counts may be used, for example 10*100.0.			0
		feet	m	cm	

Notes:

- 1) The number of entries should correspond to the NX x NY x (NZ/2) parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 6.59: LZ Keyword Description

If the [VISCD](#) keyword has been used to activate the Dual Porosity Viscous Displacement option and [LZ](#) has not been specified then [LZ](#) is set to zero in the calculation of the viscous displacement term.

See also the [LX](#), [LY](#) and [LTOSIGMA](#) keywords in the [GRID](#) section.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                      / DEFINE BOX AREA
--
--      DEFINE DUAL POROSITY VISCOUS DISPLACEMENT Z DIRECTION MATRIX SIZE
--
LZ
      6*3.0                      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The example defines a subset of the grid and the size of the “representative” matrix cells in the Y direction to 15.0 ft.; after which the [ENDBOX](#) keyword resets the input to be the full grid.

6.3.115
LZFIN – DEFINE LOGARITHMIC LGR GRID BLOCK SPACING IN THE Z-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LZFIN](#) keyword defines the parameters for automatically generating a Local Grid Refinement (“LGR”) grid in the Z direction based on logarithmic block spacing, for when the [LGR](#) option has been activated by the [LGR](#) keyword in the [RUNSPEC](#) section. [LZFIN](#) should be placed in between the [CARFIN](#) and [ENDFIN](#) keywords in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.116 MAPAXES – DEFINE THE MAP ORIGIN INPUT DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

MAPAXES specifies the origin of the map used to create the grid. It is usually output by pre-processing software when exporting the grid geometry. The data is not used by OPM Flow intrinsically, but is merely written to the output EGRID file, as specified by the **GRIDFILE** keyword, for the use of post-processing software like OPM ResInsight.

No.	Name	Description			Default
		Field (feet)	Metric (metres)	Laboratory (metres)	
1	X1	X1 is a real number that defines the x co-ordinate of a point on the y-axis.			None
2	Y1	Y1 is a real number that defines the y co-ordinate of a point on the y-axis.			None
3	X2	X2 is a real number that defines the x co-ordinate of the origin.			None
4	Y2	Y2 is a real number that defines the y co-ordinate of the origin.			None
5	X3	X3 is a real number that defines the x co-ordinate of a point on the x-axis.			None
6	Y3	Y3 is a real number that defines the y co-ordinate of a point on the x-axis.			None
Notes: 1) The keyword is terminated by a “/”.					

Table 6.60: MAPAXES Keyword Description

Example

```
--
--      ----- MAPAXES -----
--      X1      Y1      X2      Y2      X3      Y3
MAPAXES  0.0      100.0    0.0      0.0      100.0    0.0  /
```

The above example defines the map axes to be exported to the grid file for use by post-processing software.

6.3.117 MAPUNITS – DEFINE THE MAP AXES UNITS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [MAPUNITS](#) keyword defines the units of the coordinates stated on the [MAPAXES](#) keyword. It is usually output by pre-processing software when exporting the grid geometry. The data is not used by OPM Flow intrinsically, but is merely written to the output EGRID file, as specified by the [GRIDFILE](#) keyword, for the use of post-processing software like OPM ResInsight.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
1	MAPUNITS	A character string that defines the units of the coordinates stated on the MAPAXES keyword, and should be set to: 1) FEET for field units 2) METRES for metric units, or 3) CM for laboratory units	METRES
Notes: 1) Note the alternative spelling of METRES, that is METERS is not recognized. 2) The keyword is terminated by a "/".			

Table 6.61: MAPUNITS Keyword Description

Example

```
--
--      SET THE MAP UNITS FOR THE MAPAXES KEYWORD
MAPUNITS      METRES      /
```

The above example specifies the units on the [MAPAXES](#) to be the default METRES.

6.3.118 MAXVALUE – SETS A MAXIMUM VALUE FOR AN ARRAY ELEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MAXVALUE** keyword sets a maximum value for the specified array or part of an array. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **MAXVALUE** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None
2	CONSTANT	CONSTANT is a positive integer or positive real value that an ARRAY element will be reset to if an element in the defined input BOX , as defined by items (3) to (8), is greater than CONSTANT. CONSTANT has in the same units as the ARRAY property.	None
3	I1	The lower bound of the array in the I-direction to be modified must be greater than or equal to one and less than or equal to I2 and NX.	I
4	I2	The upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX.	NX
5	J1	The lower bound of the array in the J-direction to be modified must be greater than or equal to one and less than or equal to J2 and NY.	I
6	J2	The upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
7	K1	The lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	I
8	K2	The upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ
Notes: 1) Where NX, NY and NZ are the dimensions of the model as defined on the DIMENS keyword in the RUNSPEC section. 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.			

Table 6.62: MAXVALUE Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.63. Cells colored red indicate arrays that are not supported by OPM Flow operations.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

MAXVALUE Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL				
DY	PORV	SWCR				
DZ	TRANX	SWU				
PERMX	TRANY	SGL				
PERMY	TRANZ	SGCR				
PERMZ	DIFFX	SGU				
MULTX	DIFFY	KRW				
MULTY	DIFFZ	KRO				
MULTZ	TRANR	KRG				
DR	TRANHT	PCG				
DTHETA	DIFFR	PCW				
PERMR	DIFFHT					
PERMHT						
DZNET						
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.63: MAXVALUE Keyword Applicable Arrays by Section

Example

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
MAXVALUE
      PERMX      1.0E2      1*  1*  1*  1*  1*  1* / MAX VALUE FOR PERMX
      PERMY      1.0E2      1*  1*  1*  1*  1*  1* / MAX VALUE FOR PERMY
      PERMZ      1.0E1      1*  1*  1*  1*  1*  1* / MAX VALUE FOR PERMZ
/
```

The above example resets the maximum values for the *PERMX*, *PERMY* and *PERMZ*, arrays to 100.0, 100.0 and 10.0, respectively, for all cells.

6.3.119 MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the GLOBAL section for a full description.

6.3.120 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the GLOBAL section for a full description.

6.3.121 MINNNCT – SET A MINIMUM NON-NEIGHBOR CONNECTION TRANSMISSIBILITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [MINNNCT](#) keyword defines a minimum non-neighbor connection transmissibility below which the non-neighbor connection is deleted. The keyword allows for three minimum values, one for the transmissibility, one for the diffusivity and one for the thermal transmissibility. If the keyword is absent from the input deck then no minimum cut-off is applied.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.122 MINPORV – SET A MINIMUM GRID BLOCK PORE VOLUME THRESHOLD FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[MINPORV](#) defines a minimum threshold pore volume that makes all grid blocks whose pore volume is below this value inactive in the model (inactive cells are not used in OPM Flow calculations). Note this keyword is different to the [MINPVV](#) keyword in the [GRID](#) section, that sets a minimum threshold pore volume for individual cells in the model.

This keyword is an alias for the [MINPV](#) keyword in the [GRID](#) section, and thus provides the same functionality.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	MINPORV	MINPORV is a real positive number that defines the minimum pore volume for a cell to be active in the model.			Defined
		rb 1.0e-6	rm ³ 1.0e-6	rcc 1.0e-6	
Notes:					
I) The keyword is terminated by a “/”.					

Table 6.64: MINPORV Keyword Description

The [MINPORV](#), [MINPV](#), and [MINPVV](#) keywords only apply their minimum threshold pore volume values to active cells. Thus, cells that have been made inactive via setting their [ACTNUM](#) values to zero, remain inactive, even if their pore volume exceeds the values set by the [MINPORV](#), [MINPV](#), and [MINPVV](#) keywords.

Secondly, although the [MINPORV](#) keyword allows one to set a minimum threshold pore volume below the default value, this is not recommended, as cells with small pore volumes can cause significant numerical convergence errors. Thus, in practice, values greater than the default values are normally applied to eliminate cells that have relatively small pore volumes. In addition, the simulator reports the total pore volume and the number of cells made inactive, as well as the pore volume reduction, when the keyword is invoked; allowing one to run some sensitivities to the minimum pore volume value.

See also the [PINCH](#) keyword for the treatment of inactive grid cells and pinch-outs.

Example

```
--
--      MINIMUM PORE VOLUME FOR ACTIVE CELLS
--
MINPORV      500.0      /
```

The above example defines 500 rb (or m3) as the minimum pore volume for a cell to be active in the model.

6.3.123 MINPV – SET A MINIMUM GRID BLOCK PORE VOLUME THRESHOLD FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[MINPV](#) defines a minimum threshold pore volume that makes all grid blocks whose pore volume is below this value inactive in the model (inactive cells are not used in OPM Flow calculations). Note this keyword is different to the [MINPVV](#) keyword in the [GRID](#) section, that sets a minimum threshold pore volume for individual cells in the model.

Note that the [MINPORV](#) keyword is an alias for the [MINPV](#) keyword in the [GRID](#) section, and thus provides the same functionality.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	MINPV	MINPV is a real positive number that defines the minimum pore volume for a cell to be active in the model.			Defined
		rb 1.0e-6	rm ³ 1.0e-6	rcc 1.0e-6	

Notes:

I) The keyword is terminated by a “/”.

Table 6.65: MINPV Keyword Description

The [MINPV](#), [MINPORV](#), and [MINPVV](#) keywords only apply their minimum threshold pore volume values to active cells. Thus, cells that have been made inactive via setting their [ACTNUM](#) values to zero, remain inactive, even if their pore volume exceeds the values set by the [MINPV](#), [MINPORV](#), and [MINPVV](#) keywords.

Secondly, although the [MINPV](#) keyword allows one to set a minimum threshold pore volume below the default value, this is not recommended, as cells with small pore volumes can cause significant numerical convergence errors. Thus, in practice, values greater than the default values are normally applied to eliminate cells that have relatively small pore volumes. In addition, the simulator reports the total pore volume and the number of cells made inactive, as well as the pore volume reduction, when the keyword is invoked; allowing one to run some sensitivities to the minimum pore volume value.

See also the [PINCH](#) keyword for the treatment of inactive grid cells and pinch-outs.

Example

```
--
--      MINIMUM PORE VOLUME FOR ACTIVE CELLS
--
MINPV      500.0 /
```

The above example defines 500 rb (or m3) as the minimum pore volume for a cell to be active in the model.

6.3.124 MINPVV – SET A MINIMUM GRID BLOCK PORE VOLUME THRESHOLD FOR INDIVIDUAL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

[MINPVV](#) is an array that defines the minimum threshold pore volume for each cell, that makes grid blocks whose pore volume is below this value inactive in the model (inactive cells are not used in OPM Flow calculations).

Note this keyword is different to the [MINPV](#) and [MINPORV](#) keywords in the [GRID](#) section, that set a constant minimum threshold pore volume for all cells in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	MINPVV	MINPVV is an array of real positive numbers that defines the minimum pore volumes for each cell in the model in order for the cells to be active.			Defined
		rb 1.0e-6	rm ³ 1.0e-6	rcc 1.0e-6	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.66: MINPVV Keyword Description

The [MINPVV](#), [MINPV](#), and [MINPORV](#) keywords only apply their minimum threshold pore volume values to active cells. Thus, cells that have been made inactive via setting their [ACTNUM](#) values to zero, remain inactive, even if their pore volume exceeds the values set by the [MINPVV](#), [MINPV](#), and [MINPORV](#) keywords.

Secondly, although the [MINPVV](#) keyword allows one to set a minimum threshold pore volume below the default value, this is not recommended, as cells with small pore volumes can cause significant numerical convergence errors. Thus, in practice, values greater than the default values are normally applied to eliminate cells that have relatively small pore volumes. In addition, the simulator reports the total pore volume and the number of cells made inactive, as well as the pore volume reduction, when the keyword is invoked; allowing one to run some sensitivities to the minimum pore volume value.

See also the [PINCH](#) keyword for the treatment of inactive grid cells and pinch-outs.

Example

The example below shows how to define 500 rb (or m3) as the minimum pore volume for all cells in layer 19 to be active in the model, and 750 rb (or m3) as the minimum pore volume for all cells in layer 20, by using the `BOX` keyword to set the portion of the grid of interest.

```
--
--      DEFINE A BOX GRID FOR THE BOTTOM TWO LAYERS OF A 100 X 100 X 20 MODEL
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX
      1*  1*    1*  1*    19  20 / SELECT THE BOTTOM LAYERS
--
--      MINIMUM PORE VOLUME FOR INDIVIDUAL CELLS TO BE ACTIVE
--
MINPVV
      10000*500.0    10000*750.0
/
--
--      RESET THE INPUT BOX TO BE THE FULL MODEL
--
ENDBOX
```

Although this will work in the commercial simulators, it does not currently work in OPM Flow, that is one cannot use the `MINPVV` keyword in conjunction with the `BOX` keyword, as shown in the aforementioned example.

Instead one can use:

```
--
--      MINIMUM PORE VOLUME FOR INDIVIDUAL CELLS TO BE ACTIVE
--
MINPVV
      10000*          10000*          10000*    10000*    10000*
      10000*          10000*          10000*    10000*    10000*
      10000*          10000*          10000*    10000*    10000*
      10000*          10000*          10000*
      10000*500.0    10000*750.0
/
```

To accomplish the same thing, where the `10000*` instructs the simulator to use the default value of 1.0×10^{-6} for 10,000 cells, which in this case is one layer.

6.3.125 MINVALUE – SET A MINIMUM VALUE FOR AN ARRAY ELEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MINVALUE** keyword sets a minimum value for the specified array or part of an array. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **MINVALUE** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None
2	CONSTANT	CONSTANT is a positive integer or positive real value that an ARRAY element will be reset to if an element in the defined input BOX , as defined by items (3) to (8), is less than CONSTANT. CONSTANT has in the same units as the ARRAY property.	None
3	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal to one and less than or equal to I2 and NX.	I
4	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX.	NX
5	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal to one and less than or equal to J2 and NY.	I
6	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
7	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	I
8	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.67: MINVALUE Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.68. Cells colored red indicate arrays that are not supported by OPM Flow operations.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

EQUALS Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL				
DY	PORV	SWCR				
DZ	TRANX	SWU				
PERMX	TRANX	SGL				
PERMY	TRANZ	SGCR				
PERMZ	DIFFX	SGU				
MULTX	DIFFY	KRW				
MULTY	DIFFZ	KRO				
MULTZ	TRANR	KRG				
DR	TRANHT	PCG				
DTHETA	DIFFR	PCW				
PERMR	DIFFHT					
PERMHT						
DZNET						
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.68: MINVALUE Keyword Applicable Arrays by Section

Example

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--
MINVALUE
      PERMX      1.0E0      1*  1*  1*  1*  1*  1* / MINIMUM PERMX
      PERMY      1.0E0      1*  1*  1*  1*  1*  1* / MINIMUM PERMY
      PERMZ      1.0E-1      1*  1*  1*  1*  1*  1* / MINIMUM PERMZ
/
```

The above example resets the minimum values for the *PERMX*, *PERMY* and *PERMZ*, arrays to 1.0, 1.0 and 0.1, respectively, for all cells.

6.3.126 MPFANUM – DEFINE MULTI-POINT FLUX DISCRETIZATION REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *MPFANUM* keyword defines regions in the model where the multi-point flux discretization should be applied.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.127 MPFNNC – DEFINE MULTI-POINT FLUX NON-NEIGHBOR CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [MPFNNC](#) keyword defines multi-point flux non-neighbor connections explicitly.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.128 MULTFLT – MULTIPLY THE TRANSMISSIBILITY OF A DEFINED FAULT BY A CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTFLT enables the transmissibilities across defined faults, as declared by the **FAULTS** keyword, to be modified. The keyword allows for the re-scaling of the existing fault transmissibilities calculated by OPM Flow, for example setting a fault to be completely sealing by setting the multiplier to zero.

No.	Name	Description	Default
1	FLTNAME	FLTNAME is a character string enclosed in quotes with a maximum length of eight characters, that defines the name of the fault that FLTMULT will be applied to. FLTNAME must have previously been defined using the FAULTS keyword in GRID section	None
2	FLT-TRS	A positive real number that sets the transmissible multiplier to be applied to the FLTNAME transmissibilities positive real number that sets the transmissible multiplier to be applied to the FLTNAME transmissibilities.	1.0
3	FLT-DIF	A positive real number that sets the diffusivity multiplier to be applied to the FLTNAME diffusivities. This option should only be used if the Diffusion option has been made activate by the DIFFUSE keyword in the RUNSPEC section. OPM Flow does not support this Diffusion option.	1.0

Notes:

- 1) Repeated entries of the same FLTNAME will result in all but the last entry being overwritten.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.69: MULTFLT Keyword Description

Example

```
--
--      MODIFY THE TRANSMISSIBILITES ACROSS DEFINED FAULTS
--
--      FAULT      TRANS      DIFUSS
--      NAME      MULTIPLIER  MULTIPLIER
MULTFLT
      'FAULT01'    0.0                / FAULT MULTIPLIERS
      'FAULT02'    0.0                / FAULT MULTIPLIERS
      'FAULT03'    0.0                / FAULT MULTIPLIERS
/
```

The above example sets the fault transmissibility multiplier for defined faults named FAULT01, FAULT02, and FAULT03 to zero making the faults sealing in the model.

6.3.129 MULTIPLY – MULTIPLY A SPECIFIED ARRAY BY A CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MULTIPLY** keyword multiplies a specified array or part of an array by a constant. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **MULTIPLY** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the property to be modified.	None
2	CONSTANT	An integer or real value to multiply the ARRAY by in the same units as the ARRAY property.	None
3	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal to one and less than or equal to I2 and NX.	1
4	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX.	NX
5	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal to one and less than or equal to J2 and NY.	1
6	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY
7	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1
8	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the **DIMENS** keyword in the **RUNSPEC** section.
- 2) Each record must be terminated by a "/" and the keyword is terminated by a "/".

Table 6.70: MULTIPLY Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.71. Cells colored red indicate arrays that are not supported by OPM Flow operations.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

MULTIPLY Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANY	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TRLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	DILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.71: MULTIPLY Keyword Applicable Arrays by Section

Example

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      MULTIPLY      I1  I2   J1  J2   K1  K2
--      PERMZ      0.50000      1*  1*   1*  1*   1*  1* / PERMZ * 0.5
/
```

The above example multiplies the [PERMZ](#) property array by 0.5 throughout the model.

6.3.130 MULTIREG – MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MULTIREG** keyword multiplies an array or part of an array by a constant for cells with a specific region number. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **MULTIREG** keyword is read by the simulator. The constant can be an integer or real value depending on the array type; however, the arrays that can be operated on are dependent on which section the **MULTIREG** keyword is being applied in.

No.	Name	Description	Default
1	ARRAY	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None
2	CONSTANT	An integer or real value to multiply the ARRAY by in the same units as the ARRAY property for a given REGION.	0
3	REGION NUMBER	REGION NUMBER is a positive integer representing the region for which the CONSTANT in (2) should be applied.	None
4	REGION ARRAY	The REGION ARRAY to use for applying the CONSTANT in (2) based on the REGION NUMBER in (3). REGION ARRAY can have the following values: 1) F for the FLUXNUM array 2) M for the MULTNUM array 3) O for the OPERNUM array	M

Notes:

- 1) Where the REGION NUMBER should be less than or equal to the maximum number of regions as defined on the **REGDIMS** keyword for the **FIPNUM** and **OPERNUM** arrays or the **GRIDOPTS** keyword for the **MULTNUM** array in the **RUNSPEC** section.
- 2) If the **GRIDOPTS** keyword is not present in the **RUNSPEC** section or if the **GRIDOPTS** keyword is present but the maximum number of **MULTNUM** regions (NRMULT) equals zero then REGION ARRAY will default to F for the **FLUXNUM** array.
- 3) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.72: MULTIREG Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.73. Cells colored red indicate arrays that are not supported by OPM Flow operations.

MULTIREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANX	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

MULTREG Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
PERMZ	DIFFX	SGU	PVTNUM	TBLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	OILAPI		
MULTZ	TRANR	KRG	WAZNUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPPANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.73: MULTIREG Keyword Applicable Arrays by Section

Example

```
--
-- FIRST DEFINE THE PROPERTY ARRAYS AND MULTNUM ARRAYS FOR 10 X 10 X 20 MODEL
--
--   ARRAY      CONSTANT      ----- BOX -----
--           I1  I2   J1  J2   K1  K2
EQUALS
  PORO      0.2000      1*  1*   1*  1*   1*  1* / PORO TO 0.20 IN MODEL
  PERMX     100.00      1*  1*   1*  1*   1*  1* / PERMX TO 0.10 IN MODEL
  MULTNUM    1          1*  1*   1*  1*   1*  1* / MULTNUM IN MODEL
  MULTNUM    2          1*  5    1   5    6   6 / MULTNUM IN MODEL
  MULTNUM    3          1*  1*   1*  1*   10  10 / MULTNUM IN MODEL
/
--
-- NOW RESET PORO AND PERMX BASED ON THE MULTNUM REGION NUMBER
--
-- MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER
--
--   ARRAY      CONSTANT  REGION  REGION ARRAY
--           VALUE      NUMBER    M / F / O
MULTIREG
  PORO      1.050      1         M           /
  PORO      1.100      2         M           /
  PORO      0.950      3         M           /
  PERMX     1.25       1         M           /
  PERMX     1.30       2         M           /
  PERMX     0.90       3         M           /
/
```

The example first defines the [PORO](#) and [PERMX](#) property arrays for the model and then sets the [MULTNUM](#) array to 1 for all cells in the model, after which selected areas of model are assigned various [MULTNUM](#) integer values. The [MULTIREG](#) can then be invoked to multiply the [PORO](#) and [PERMX](#) arrays by a constant for the various [MULTNUM](#) regions.

6.3.131 MULTNUM – DEFINE THE MULTIPLE TRANSMISSIBILITY REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *MULTNUM* keyword defines the inter-region transmissibility region numbers for each grid block, as such there must be one entry for each cell in the model. The array can be used with the *EQUALREG*, *ADDREG*, *COPYREG*, *MULTIREG*, *MULTREGP* and *MULTREGT* keywords in calculating various grid properties in the *GRID* section.

No.	Name	Description	Default
I	<i>MULTNUM</i>	<i>MULTNUM</i> defines an array of positive integers assigning a grid cell to a particular inter-region transmissibility region. The maximum number of <i>MULTNUM</i> regions is set by the NRMULT variable on the <i>GRIDOPTS</i> keyword in the <i>RUNSPEC</i> section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section, unless the <i>BOX</i> keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the <i>BOX</i> statement. 2) The keyword is terminated by a “/”.			

Table 6.74: MULTNUM Keyword Description

Examples

The example below sets three *MULTNUM* regions for a 4 x 5 x 2 model.

```
--
--      DEFINE MULTNUM REGIONS FOR ALL CELLS
--
MULTNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/

Alternatively the EQUALS keyword could be employed to accomplish the same task, that is:
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
    MULTNUM      1              1*  1*   1*  1*   1*  1* / SET REGION 1
    MULTNUM      2              1   2    1   2    1   1 / SET REGION 2
    MULTNUM      3              1   2    1   2    2   2 / SET REGION 3
/
```

One can then increase *PERMX* by 25% in region three only.

```
--
--      MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER
--
--      ARRAY      CONSTANT  REGION  REGION ARRAY
--      VALUE      NUMBER    M / F / O
MULTIREG
    PERMX      1.25      3          M
/
```

6.3.132 MULTPV – MULTIPLY CELL PORE VOLUMES BY A CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTPV multiplies the pore volumes of a cell by a real positive constant for all the cells in the model via an array. An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTPV** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
1	MULTPV	MULTPV is an array of real positive numbers assigning the pore volume multipliers for each cell in the model. Repeat counts may be used, for example 20*100.0.	1.0

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a "/".

Table 6.75: MULTPV Keyword Description

See also the **MULTREGP** keyword for scaling the cell pore volumes by region numbers.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   3                / DEFINE BOX AREA
--
--      SET PORE VOLUME MULTIPLIERS
--
MULTPV   18*0.0500                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.05 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.133 MULTR – MULTIPLY CELL TRANSMISSIBILITY IN THE +R DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

MULTR multiplies the transmissibility between two cell faces in the +R direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I+1, J, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTR** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword should only be used with radial and spider grids, as declared by the **RADIAL** or **SPIDER** keywords in the **RUNSPEC** section.

No.	Name	Description	Default
I	MULTR+	MULTR+ is an array of real positive numbers assigning the transmissibility multipliers in the +R direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.76: MULTR Keyword Description

See also the **MULTR-**, **MULTTHT**, **MULTTHT-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between between grid cells in the R direction.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      10  10   1   6   1   3
--                                     / DEFINE BOX AREA
--
--      SET MULTR+ TRANSMISSIBILITY MULTIPLIERS
--
--      MULTR
--      18*0.300
--                                     /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.134 MULTR- – MULTIPLY CELL TRANSMISSIBILITY IN THE -R DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTR- multiplies the transmissibility between two cell faces in the -R direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I-I, J, K) and (I, J, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTR-** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword should only be used with radial and spider grids, as declared by the **RADIAL** or **SPIDER** keywords in the **RUNSPEC** section.

No.	Name	Description	Default
I	MULTR-	MULTR- is an array of real positive numbers assigning the transmissibility multipliers in the -R direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.77: MULTR- Keyword Description

Note that OPM Flow does not require the **GRIDOPTS(TRANMULT)** parameter in the **RUNSPEC** section to be set to YES, in order to use this and other negative directional dependent multiplier keywords in the input deck. Whereas, the commercial simulator will terminate with an error if the keyword is present, and the **GRIDOPTS(TRANMULT)** parameter has not been set to YES.

See also the **MULTR**, **MULTTHT**, **MULTTHT-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between cells in the R direction.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                / DEFINE BOX AREA
--
--      SET MULTR- TRANSMISSIBILITY MULTIPLIERS
--
MULTR-   6*0.500                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.5 scaling multiplier for the six cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

6.3.135 MULTREGD – MULTIPLY DIFFUSIVITIES BETWEEN REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *MULTREGD* keyword multiplies the diffusivity between two regions by a constant. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *MULTREGD* keyword is read by the simulator. The constant should be a real number.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	REGION1	A positive integer value that defines the from REGION number for which the CONSTANT in (3) should be applied.	None
2	REGION2	A positive integer value that defines the to REGION number for which the CONSTANT in (3) should be applied.	None
3	CONSTANT	A real value to multiply the diffusivity between REGION1 and REGION2.	1
4	DIR	A character string that defines the direction to apply the diffusivity multiplier between the two regions, should be set to one of the following X,Y,Z,XY,XZ,YZ, or XYZ.	XYZ
5	TYPE	A character string that defines the type of connections the diffusivity multiplier should be applied to, should be one of the following: 1) <i>NNC</i> – Only apply the diffusivity multiplier between REGION1 and REGION2 to non-neighbor connections. 2) <i>NONNC</i> – Do not apply the diffusivity multiplier between REGION1 and REGION2 to non-neighbor connections. 3) <i>ALL</i> - Apply the diffusivity multiplier between REGION1 and REGION2 to all connections.	<i>ALL</i>
6	REGION ARRAY	The REGION ARRAY to use for applying the CONSTANT in (3) based on the regions REGION1 and REGION2 in (1 and 2). REGION ARRAY can have the following values: 1) F for the <i>FLUXNUM</i> array 2) M for the <i>MULTNUM</i> array 3) O for the <i>OPERNUM</i> array	M

Notes:

- Where REGION1 and REGION2 should be less than or equal to the maximum number of regions as defined on the *REGDIMS* keyword for the *FIPNUM* and *OPERNUM* arrays or the *GRIDOPTS* keyword for the *MULTNUM* array in the *RUNSPEC* section.
- If REGION1 is defaulted then the diffusivities between REGION2 and all other regions will be multiplied. Similarly, if REGION2 is defaulted then the diffusivities between REGION1 and all other regions will be multiplied. If both REGION1 and REGION2 are defaulted then the diffusivities between all pairs of regions will be multiplied.
- If REGION1 and REGION2 are equal and positive then all the diffusivities within that region as well as the diffusivities between that region and all other regions will be multiplied.
- If the same combination of REGION1 and REGION2 is repeated then only the latest multiplier will be used regardless of whether the REGION ARRAY specified is the the same or not.
- If the *GRIDOPTS* keyword is not present in the *RUNSPEC* section or if the *GRIDOPTS* keyword is present but the maximum number of *MULTNUM* regions (NRMULT) equals zero then REGION ARRAY will default to F for the *FLUXNUM* array.
- Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.78: MULTREGD Keyword Description

Example

```
--
--      MULTIPLY DIFFUSIVITIES BETWEEN RESERVOIRS
--
--      REGION  REGION  DIFFS  DIREC  NNC  REGION ARRAY
--      FROM    TO      MULT   OPT    OPTS  M / F / O
MULTREGD
      1*        1*        1.05   1*     'ALL '  M           / ALL REGIONS
/
```

The above example multiplies the diffusivities between all the *MULTNUM* regions by 1.05 in all directions and for all connections types.

6.3.136 MULTREGH – MULTIPLY THERMAL CONDUCTIVITIES BETWEEN REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *MULTREGH* keyword multiplies the thermal conductivity between two regions by a constant. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *MULTREGT* keyword is read by the simulator. The constant should be a real number.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	REGION1	A positive integer value that defines the from REGION number for which the CONSTANT in (3) should be applied.	None
2	REGION2	A positive integer value that defines the to REGION number for which the CONSTANT in (3) should be applied.	None
3	CONSTANT	A real value to multiply the thermal conductivity between REGION1 and REGION2.	1
4	DIR	A character string that defines the direction to apply the thermal conductivity multiplier between the two regions, should be set to one of the following X,Y,Z, XY, XZ, YZ, or XYZ.	XYZ
5	TYPE	A character string that defines the type of connections the thermal conductivity multiplier should be applied to, should be one of the following: 1) <i>NNC</i> – Only apply the thermal conductivity multiplier between REGION1 and REGION2 to non-neighbor connections. 2) <i>NONNC</i> – Do not apply the thermal conductivity multiplier between REGION1 and REGION2 to non-neighbor connections. 3) <i>ALL</i> - Apply the thermal conductivity multiplier between REGION1 and REGION2 to all connections.	<i>ALL</i>
6	REGION ARRAY	The REGION ARRAY to use for applying the CONSTANT in (3) based on the regions REGION1 and REGION2 in (1 and 2). REGION ARRAY can have the following values: 1) F for the <i>FLUXNUM</i> array. 2) M for the <i>MULTNUM</i> array. 3) O for the <i>OPERNUM</i> array.	M

Notes:

- Where REGION1 and REGION2 should be less than or equal to the maximum number of regions as defined on the *REGDIMS* keyword for the *FIPNUM* and *OPERNUM* arrays or the *GRIDOPTS* keyword for the *MULTNUM* array in the *RUNSPEC* section.
- If REGION1 is defaulted then the thermal conductivities between REGION2 and all other regions will be multiplied. Similarly, if REGION2 is defaulted then the thermal conductivities between REGION1 and all other regions will be multiplied. If both REGION1 and REGION2 are defaulted then the thermal conductivities between all pairs of regions will be multiplied.
- If REGION1 and REGION2 are equal and positive then all the thermal conductivities within that region as well as the thermal conductivities between that region and all other regions will be multiplied.
- If the same combination of REGION1 and REGION2 is repeated then only the latest multiplier is used regardless of whether the REGION ARRAY specified is the the same or not.
- If the *GRIDOPTS* keyword is not present in the *RUNSPEC* section or if the *GRIDOPTS* keyword is present but the maximum number of *MULTNUM* regions (NRMULT) equals zero then REGION ARRAY will default to F for the *FLUXNUM* array.
- Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.79: MULTREGH Keyword Description

Example

```
--
--      MULTIPLY THERMAL CONDUCTIVITIES BETWEEN RESERVOIRS
--
--      REGION  REGION  CONDS  DIREC  NNC  REGION ARRAY
--      FROM    TO      MULT   OPT   OPTS  M / F / O
MULTREGH
      1*      1*      1.05   1*    'ALL '  M          / ALL REGIONS
/
```

The above example multiplies the thermal conductivities between all the [MULTNUM](#) regions by 1.05 in all directions and for all connections types.

6.3.137 MULTREGP – MULTIPLY PORE VOLUMES BASED ON REGION NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MULTREGP** keyword multiplies the pore volume of a cell by a constant for all cells with a specific region number. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **MULTREGP** keyword is read by the simulator. The constant should be a real number.

No.	Name	Description	Default
1	REGION	REGION is a positive integer representing the region for which the CONSTANT in (2) should be applied.	None
2	CONSTANT	A real value to multiply the pore volume by for a given REGION.	1
3	REGION ARRAY	The REGION ARRAY to use for applying the CONSTANT in (2) based on the REGION in (1). ARRAY can have the following values: 1) F for the FLUXNUM array. 2) M for the MULTNUM array. 3) O for the OPERNUM array.	M

Notes:

- Where the REGION should be less than or equal to the maximum number of regions as defined on the **REGDIMS** keyword for the **FIPNUM** and **OPERNUM** arrays or the **GRIDOPTS** keyword for the **MULTNUM** array in the **RUNSPEC** section.
- If the **GRIDOPTS** keyword is not present in the **RUNSPEC** section or if the **GRIDOPTS** keyword is present but the maximum number of **MULTNUM** regions (NRMULT) equals zero then REGION ARRAY will default to F for the **FLUXNUM** array.
- Each record must be terminated by a "/" and the keyword is terminated by a "/".

Table 6.80: MULTREGP Keyword Description

Example

```
--
-- RESET PORE VOLUME FOR DIFFERENT REGIONS
--
--      REGION      PORV      REGION ARRAY
--      NUMBER      MULT      M / F / O
MULTREGP
      1      1.0456573      M      / Fault Block 1
      2      0      M      / Fault Block 2
      3      0.9756715      M      / Fault Block 3
      4      0      M      / Inactive Blocks
/
```

The above example re-scales the pore volumes for **MULTNUM** regions one and three and makes regions two and four inactive by setting their pore volumes to zero.

6.3.138 MULTREGT – MULTIPLY TRANSMISSIBILITIES BETWEEN REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **MULTREGT** keyword multiplies the transmissibility between two regions by a constant. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **MULTREGT** keyword is read by the simulator. The constant should be a real number.

No.	Name	Description	Default
1	REGION1	A positive integer value that defines the from REGION number for which the CONSTANT in (3) should be applied.	None
2	REGION2	A positive integer value that defines the to REGION number for which the CONSTANT in (3) should be applied.	None
3	CONSTANT	A real positive value to multiply the transmissibility between REGION1 and REGION2.	1
4	DIR	A character string that defines the direction to apply the transmissibility multiplier between the two regions, should be set to one of the following X,Y,Z, XY,XZ,YZ, or XYZ.	XYZ
5	TYPE	A character string that defines the type of connections the transmissibility multiplier should be applied to, should be one of the following: 1) NNC – Only apply the transmissibility multiplier between REGION1 and REGION2 to non-neighbor connections. 2) NONNC – Do not apply the transmissibility multiplier between REGION1 and REGION2 to non-neighbor connections. 3) ALL – Apply the transmissibility multiplier between REGION1 and REGION2 to all connections.	ALL
6	REGION ARRAY	A single character that defines the REGION ARRAY that is used to specify the regions identified by REGION1 and REGION2. REGION ARRAY can have the following values: 1) F for the FLUXNUM array 2) M for the MULTNUM array 3) O for the OPERNUM array	M

Notes:

- Where REGION1 and REGION2 should be less than or equal to the maximum number of regions as defined on the **REGDIMS** keyword for the **FLUXNUM** and **OPERNUM** arrays or the **GRIDOPTS** keyword for the **MULTNUM** array in the **RUNSPEC** section.
- If REGION1 is defaulted then the transmissibilities between REGION2 and all other regions will be multiplied. Similarly, if REGION2 is defaulted then the transmissibilities between REGION1 and all other regions will be multiplied. If both REGION1 and REGION2 are defaulted then the transmissibilities between all pairs of regions will be multiplied.
- If REGION1 and REGION2 are equal and positive then all the transmissibilities within that region as well as the transmissibilities between that region and all other regions will be multiplied.
- If the same combination of REGION1 and REGION2 is repeated then only the latest multiplier is used regardless of whether the REGION ARRAY specified is the the same or not.
- If the **GRIDOPTS** keyword is not present in the **RUNSPEC** section or if the **GRIDOPTS** keyword is present but the maximum number of **MULTNUM** regions (NRMULT) equals zero then REGION ARRAY will default

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

to F for the [FLUXNUM](#) array.

6) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.81: MULTREGT Keyword Description

Note

Note if the [MULTREGT](#) keyword is used in the [EDIT](#) section, OPM Flow will always apply the changes irrespective, of if the [TRANX](#), [TRANY](#) and [TRANZ](#) transmissibility arrays have been entered or not in the [EDIT](#) section.

This behavior is different to the commercial simulator that only applies the keyword if the transmissibility arrays have been entered in the [EDIT](#) section.

Example

```
--
--      SET TRANSMISSIBILITES ACROSS DIFFERENT RESERVOIRS TO ZERO
--
--      REGION      REGION      TRANS      DIREC      NNC      REGION ARRAY
--      FROM        TO          MULT        OPT        OPTS     M / F / O
MULTREGT
      1*           1*           0.0         1*         'ALL'    M           / ALL REGIONS SEALED
/
```

The above example isolates all regions from one another by setting the transmissibility for the [MULTNUM](#) regions to zero in all directions and for all connections types.

6.3.139 MULTTHT – MULTIPLY CELL TRANSMISSIBILITY IN THE +THETA DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTTHT multiples the transmissibility between two cell faces in the +Theta direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J+1, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTTHT** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword should only be used with radial and spider grids, as declared by the **RADIAL** or **SPIDER** keywords in the **RUNSPEC** section.

No.	Name	Description	Default
I	MULTTHT+	MULTTHT+ is an array of real positive numbers assigning the transmissibility multipliers in the +Theta direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.82: MULTTHT Keyword Description

See also the **MULTTHT-**, **MULTR**, **MULTR-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between I grid cells in the Theta direction.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      10  10    1   6    1   3
--
--                                     / DEFINE BOX AREA
--
--      SET MULTTHT+ TRANSMISSIBILITY MULTIPLIERS
--
--      MULTTHT
--      18*0.300
--
--                                     /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.140 MULTTHT- – MULTIPLY CELL TRANSMISSIBILITY IN THE -THETA DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTTHT- multiplies the transmissibility between two cell faces in the -Theta direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J-I, K) and (I, J, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTTHT-** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword should only be used with radial and spider grids, as declared by the **RADIAL** or **SPIDER** keywords in the **RUNSPEC** section.

No.	Name	Description	Default
I	MULTTHT-	MULTTHT- is an array of real positive numbers assigning the transmissibility multipliers in the -Theta direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 6.83: MULTTHT- Keyword Description

Note that OPM Flow does not require the **GRIDOPTS**(TRANMULT) parameter in the **RUNSPEC** section to be set to YES, in order to use this and other negative directional dependent multiplier keywords in the input deck. Whereas, the commercial simulator will terminate with an error if the keyword is present, and the **GRIDOPTS**(TRANMULT) parameter has not been set to YES.

See also the **MULTTHT**, **MULTR**, **MULTR-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between I grid cells in the Theta direction.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                / DEFINE BOX AREA
--
--      SET MULTTHT- TRANSMISSIBILITY MULTIPLIERS
--
MULTTHT- 6*0.500                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.5 scaling multiplier for the six cells defined by the preceding [BOX](#) statement. The [ENDBOX](#) keyword resets the input box to the full grid.

6.3.141 MULTX – MULTIPLY CELL TRANSMISSIBILITY IN THE +X DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTX multiplies the transmissibility between two cell faces in the +X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I+1, J, K).

An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTX** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
I	MULTX+	MULTX+ is an array of real positive numbers assigning the transmissibility multipliers in the +X direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.84: MULTX Keyword Description

See also the **MULTX-**, **MULTY**, **MULTY-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      10  10   1   6   1   3                               / DEFINE BOX AREA
--
--      SET MULTX+ TRANSMISSIBILITY MULTIPLIERS
--
MULTX
--      18*0.300                                              /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.142 MULTX- – MULTIPLY CELL TRANSMISSIBILITY IN THE -X DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTX- multiplies the transmissibility between two cell faces in the -X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I-I, J, K) and (I, J, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTX-** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
1	MULTX-	MULTX- is an array of real positive numbers assigning the transmissibility multipliers in the -X direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a "/".			

Table 6.85: MULTX- Keyword Description

Note that OPM Flow does not require the **GRIDOPTS**(TRANMULT) parameter in the **RUNSPEC** section to be set to YES, in order to use this and other negative directional dependent multiplier keywords in the input deck. Whereas, the commercial simulator will terminate with an error if the keyword is present, and the **GRIDOPTS**(TRANMULT) parameter has not been set to YES.

See also the **MULTX**, **MULTY**, **MULTY-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                / DEFINE BOX AREA
--
--      SET MULTX- TRANSMISSIBILITY MULTIPLIERS
--
MULTX-    6*0.500                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.5 scaling multiplier for the six cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.143 MULTY – MULTIPLY CELL TRANSMISSIBILITY IN THE +Y DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTY multiplies the transmissibility between two cell faces in the +Y direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J+1, K).

An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTY** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
I	MULTY+	MULTY+ is an array of real positive numbers assigning the transmissibility multipliers in the +Y direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.86: MULTY Keyword Description

See also the **MULTY-**, **MULTX**, **MULTX-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      10  10   1   6    1   3                               / DEFINE BOX AREA
--
--      SET MULTY+ TRANSMISSIBILITY MULTIPLIERS
--
MULTY
--      18*0.300                                              /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.144 MULTY- – MULTIPLY CELL TRANSMISSIBILITY IN THE -Y DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTY- multiplies the transmissibility between two cell faces in the -Y direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J-1, K) and (I, J, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTY-** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
1	MULTY-	MULTY- is an array of real positive numbers assigning the transmissibility multipliers in the -Y direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a "/".

Table 6.87: **MULTY-** Keyword Description

Note that OPM Flow does not require the **GRIDOPTS**(TRANMULT) parameter in the **RUNSPEC** section to be set to YES, in order to use this and other negative directional dependent multiplier keywords in the input deck. Whereas, the commercial simulator will terminate with an error if the keyword is present, and the **GRIDOPTS**(TRANMULT) parameter has not been set to YES.

See also the **MULTY**, **MULTX**, **MULTX-**, **MULTZ** and **MULTZ-** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   1                / DEFINE BOX AREA
--
--      SET MULTY- TRANSMISSIBILITY MULTIPLIERS
--
MULTY-    6*0.500                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.5 scaling multiplier for the six cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.145 MULTZ – MULTIPLY CELL TRANSMISSIBILITY IN THE +Z DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTZ multiplies the transmissibility between two cell faces in the +Z direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J, K+1).

An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTZ** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
1	MULTZ+	MULTZ+ is an array of real positive numbers assigning the transmissibility multipliers in the +Z direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a "/".			

Table 6.88: MULTZ Keyword Description

See also the **MULTZ-**, **MULTX**, **MULTX-**, **MULTY** and **MULTY-** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   18   1   1                / DEFINE BOX AREA
--
--      SET MULTZ+ TRANSMISSIBILITY MULTIPLIERS
--
MULTZ      18*0.300                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.146 MULTZ- – MULTIPLY CELL TRANSMISSIBILITY IN THE -Z DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

MULTZ- multiplies the transmissibility between two cell faces in the -Z direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K-1) and (I, J, K). An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTZ-** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description	Default
1	MULTZ-	MULTZ- is an array of real positive numbers assigning the transmissibility multipliers in the -X direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a "/".			

Table 6.89: MULTZ- Keyword Description

Note that OPM Flow does not require the **GRIDOPTS**(TRANMULT) parameter in the **RUNSPEC** section to be set to YES, in order to use this and other negative directional dependent multiplier keywords in the input deck. Whereas, the commercial simulator will terminate with an error if the keyword is present, and the **GRIDOPTS**(TRANMULT) parameter has not been set to YES.

See also the **MULTZ**, **MULTX**, **MULTX-**, **MULTY** and **MULTY-** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      10  10   1   6   1   1                                / DEFINE BOX AREA
--
--      SET MULTZ- TRANSMISSIBILITY MULTIPLIERS
--
MULTZ-
--      6*0.500                                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.5 scaling multiplier for the six cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.147 NEWTRAN

–

ACTIVATE

IRREGULAR

CORNER-POINT

GRID

TRANSMISSIBILITIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword switches on Irregular Corner-Point Grid geometry transmissibility calculation, which is the default option for this type of grid. Grids defined with the [COORD](#) and [ZCORN](#) keywords will always invoke this option by default.

For Cartesian Regular Grids defined by the [DX](#), [DY](#), and [DZ](#) series of keywords the block center geometry transmissibility calculations should be activated via the [OLDTRAN](#) keyword. Again this is automatically invoked if this type of grid is being employed.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE IRREGULAR CORNER-POINT GRID TRANSMISSIBILITIES
--
NEWTRAN
```

The above example manually activates Irregular Corner-Point Grid transmissibility calculations.

6.3.148 NINENUM – DEFINE THE NINE-POINT DISCRETIZATION REGION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **NINENUM** keyword defines areas in the grid that should use the Nine-Point Discretization formulation by setting a grid block's **NINENUM** value to one, or zero for the conventional standard five-point discretization formulation, for when the Nine-Point Discretization formulation has been activated by the **NINEPOIN** keyword in the **RUNSPEC** section. There should be a **NINENUM** value for each grid block in the model. Note that if the **NINEPOIN** keyword in the **RUNSPEC** section has been invoked and the **NINENUM** keyword has not been used in the input deck, then all the grid will use the nine-point scheme.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	NINENUM	NINENUM defines an integer array of zeros and ones assigning a grid cell to a particular discretization region, a value of zero for five-point or a value of one for nine-point discretization. Note that the default value of one implies a cell is included in the Nine-Point Discretization region; thus, if a cell is to use the conventional standard five-point finite difference discretization formulation, then NINENUM must be explicitly set to zero.	1
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a "/".			

Table 6.90: NINENUM Keyword Description

The **NINENUM** keyword cannot be used in models with Local Grid Refinements ("LGR") to set different discretization regions within the model, that is if LGRs are present in the model either all the grid uses nine-point discretization, if **NINEPOIN** is present in the **RUNSPEC** section, or five-point if **NINEPOIN** is absent.

Example

The example below sets a portion of the model to us the Nine-Point Discretization formulation.

```
--
--      DEFINE NINE-POINT DISCRETIZATION REGION FOR ALL CELLS
--
--      ARRAY          CONSTANT          ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      NINENUM'        0                  1*  1*   1*  1*   1*  1* / FIVE-POINT
      NINENUM'        1                  1*  1*   1*  1*   1   5  / NINE-POINT
/
```

Here the first line sets all the grid to us the five-point discretization formulation, all values set to zero, and then the second line sets all the cells in the layers one to five to use the nine-point discretization formulation.

6.3.149
NMATOPTS
– DEFINE THE DISCRETIZED MATRIX DUAL POROSITY PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [NMATOPTS](#) keyword defines the Discretized Matrix Dual Porosity parameters for when the Discretized Matrix Dual Porosity option has been activated by [NMATRIX](#) keyword in the [RUNSPEC](#) section. The option allows the matrix grid blocks to be subdivided into smaller cells for more accurate flow calculations, in particular the modeling of transient flow within the matrix grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.150 NNC – DEFINE NON-NEIGHBOR CONNECTIONS BETWEEN CELLS MANUALLY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

NNC enables Non-Neighbor Connections (“NNC”) to be manually defined. This keyword is normally generated by static modeling software as opposed to be manually entered in the OPM Flow input deck due to the verbosity and complexity of calculating the required parameters for this keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	I1	A positive integer that defines the first grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.			None
2	J1	A positive integer that defines the first grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.			None
3	K1	A positive integer that defines the first grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.			None
4	I2	A positive integer that defines the second grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.			None
5	J2	A positive integer that defines the second grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.			None
6	K2	A positive integer that defines the second grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.			None
7	TRANSNNC	TRANSNNC is a positive real number greater than or equal to zero that defines the transmissibility between the first grid block (I1, J1, K1) and the second grid block (I2, J2, K2). The default value of zero sets the transmissibility between the two cells to zero.			0.0
		cPrb/day/psia	cPrm ³ /day/bars	cPrcc/hr/atm	
8	ISATNUM1	ISATNUM1 is a positive integer defining which saturation table number (relative permeability table) to be used for flow from the first grid block to the second grid block. The default value of zero means the existing saturation table allocated to the upstream cell (I1, J1, K1).			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	ISATNUM2	ISATNUM2 is a positive integer defining which saturation table number (relative permeability table) to be used for flow from the second grid block to the first grid block. The default value of zero means the existing saturation table allocated to the downstream cell (I2,J2,K2).			0
10	IPRSNUM1	IPRSNUM1 is a positive integer defining which pressure table number (PVT table) to be used for flow from the first grid block to the second grid block. The default value of zero means the existing PVT table allocated to the upstream cell (I1,J1,K1).			0
11	IPRSNUM2	IPRSNUM2 is a positive integer defining which pressure table number (PVT table) to be used for flow from the second grid block to the first grid block. The default value of zero means the existing PVT table allocated to the downstream cell (I2,J2,K2).			0
12	FACE1	FACE1 is a character string that defines the face associated with flow from the first grid block to the second grid block, where FACE1 can have values of: X+, X-,Y+,Y-, Z+, or Z-.			None
13	FACE2	FACE2 is a character string that defines the face associated with flow from the second grid block to the first grid block, where FACE2 can have values of: X+, X-,Y+,Y-, Z+, or Z-.			None
14	DIFFNNC	DIFFNNC is a positive real number that defines the diffusivity between the first grid block (I1,J1,K1) and the second grid block (I2,J2,K2).			0.0
		feet	meters	cm	
15	DISPNNC	DISPNNC is a positive real number that defines the dispersion coefficient $\frac{1}{(Area \times Porosity)}$ between the first grid block (I1,J1,K1) and the second grid block (I2,J2,K2), used with the <i>DISPERSE</i> option.			0.0
		ft ²	m ²	cm ²	
16	AREANNC	AREANNC is a positive real number that defines the area associated with the connection between the first grid block (I1,J1,K1) and the second grid block (I2,J2,K2).			None
		ft ²	m ²	cm ²	
17	PERMNNC	AREANNC is a positive real number that defines the permeability associated with the connection between the first grid block (I1,J1,K1) and the second grid block (I2,J2,K2).This used by the non-Darcy option.			None
		mD	mD	mD	

Notes:

1) Only functionality defined by items (1) to (7) are activated in OPM Flow.

2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 6.91: NNC Keyword Description

Note that although items (8) to (17) for this keyword are not available in OPM Flow, even if they were, it is strongly recommended that these items are defaulted if the data is being entered manually, as opposed to being generated by a pre-processing software.

Care should be taken that cells in different [PVTNUM](#) regions (see the [PVTNUM](#) keyword in the [REGIONS](#) section) are not connected, since the fluid properties are associated with a cell. If for example, a rbb1 or a rm3 of oil flows from [PVTNUM](#) region 1 to [PVTNUM](#) region 2, then the oil properties of that oil will change from the PVT 1 data set to the PVT data set 2. This will result in material balance errors, that may or may not cause numerical issues.

If the transmissibility across a fault needs to be modified see the [FAULTS](#) and [MULTFLT](#) keywords in the [GRID](#) section for an alternative and less complicated method to modifying fault transmissibilities. Transmissibility between reservoir regions can be modified by using [MULTREGT](#) keyword, provided [MULTNUM](#) has been used to define the inter-region transmissibility region numbers for each grid block. Finally, the [MULTX](#), [MULTY](#) and [MULTZ](#) series of keywords can be used to modify transmissibility between various cells. All the aforementioned keywords are described in the [GRID](#) section.

Example

```
--
--      MANUALLY DEFINE NON-NEIGHBOR CONNECTIONS
--
--      ----- BOX -----      -- TRANSNNC --
--      I1   J1   K1      I2   J2   K2
NNC
      1     1     1       1     1     2      0.2500    / SET NNC FOR FAULT
      1     1     2       1     1     3      0.2500    / SET NNC FOR FAULT
      1     1     3       1     1     4      0.2500    / SET NNC FOR FAULT
/
```

The above example defines the transmissibility between cells (1, 1, 1) and (1, 1, 2), (1, 1, 2) and (1, 1, 3) and finally between (1, 1, 3) and (1, 1, 4) to be 0.2500.

6.3.151

NODPPM

–

DEACTIVATE

FRACTURE

POROSITY-PERMEABILITY

CALCULATION

The [NODPPM](#) keyword deactivates the default behavior of multiplying the fracture porosity by the fracture permeability to calculate the effective fracture permeability in dual porosity and dual permeability runs. Either the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section must be declared in the input file in order to use this keyword. If the default calculation is switched off by this keyword, then the effective fracture permeability is taken to be those entered for the fracture using the [PERMX](#), [PERMY](#) and [PERMZ](#) keywords in the [GRID](#) section. If the keyword is absent from the input deck, then the entered [PERMX](#), [PERMY](#) and [PERMZ](#) arrays for the fractures are multiplied by fracture [PORO](#) array values in order to obtain the effective fracture permeability.

See [NODPPM – Deactivate Fracture Porosity-Permeability Calculation](#) in the [RUNSPEC](#) section for a full description.

6.3.152

NOECHO

–

DEACTIVATE

ECHOING

OF

USER

INPUT

FILES

TO

THE

PRINT

FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active. but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the [GLOBAL](#) section for a full description.

6.3.153 NOGGF – DEACTIVATE OUTPUT OF GRID GEOMETRY FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword deactivates the output of a standard [GRID](#) or extended [GRID](#) file, as well as the extensible EGRID file for post-processing applications.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE GRID GEOMETRY OUTPUT
--
NOGGF
```

The above example switches off the default behavior of writing out the grid geometry files.

6.3.154 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

6.3.155 NTG – DEFINE THE NET-TO-GROSS RATIO FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[NTG](#) defines the Net-to-Gross Ratio (“NTG”) for all the cells in the model via an array. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NTG	NTG is an array of real numbers greater than or equal to zero and less than or equal to one, that are assigned the net-to-gross ratio values for each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 200*0.850.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Setting a cells [NTG](#) value to zero will make the cell inactive, similar to setting the cells [ACTNUM](#) property to zero.
- 2) The keyword is terminated by a “/”.

Table 6.92: NTG Keyword Description

See also the [PORO](#), [PERMX](#), [PERMY](#) and [PERMZ](#) keywords to fully define a grid’s properties.

Example

```
--
--      DEFINE GRID BLOCK NTG DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
NTG
      100*1.000    100*0.850    100*0.500      /
```

The above example defines a constant [NTG](#) of 1.00 for the first 100 cells, then 0.85 for the second 100 hundred cells, and finally 0.500 for the last 100 cell, for the 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.156 NXFIN – DEFINE THE NUMBER OF LGR GRID BLOCKS IN THE X-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

NXFIN defines the number of Local Grid Refinement (“LGR”) cells within a global or host cell in the x-direction via a vector, as opposed to defining the size for each cell for a Cartesian **LGR** Grid. The **LGR** keyword in the **RUNSPEC** section should be activated to indicate an **LGR** is being used, and the keyword **NXFIN** should be placed in between the **CARFIN** and **ENDFIN** keywords in the **GRID** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	NXFIN	NXFIN is a vector of integer numbers describing the number of LGR cells within each defined global or host grid block in the x-direction in a Cartesian LGR grid. Repeat counts may be used, for example 2*2.0.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX parameter on the **CARFIN** keyword in the **GRID** section.
- 2) The keyword is terminated by a “/”.

Table 6.93: NXFIN Keyword Description

See also the **CARFIN**, **ENDFIN**, **NYFIN**, and **NZFIN** keywords in the **GRID** section to fully define a Cartesian **LGR** grid model.

Example

```
--
--      CARFIN LGR GRID COMMANDS
--
--      LGR      ----- HOST GRID -----  -- CARFIN GRID --  MAX      HOST
--      NAME      I1  I2  J1  J2  K1  K2      NX      NY      NZ      WELLS      NAME
CARFIN
      LGR-OP01    24  25  87  87   1  50        8      1      50        1      GLOBAL /
--
--      DEFINE LGR GRID BLOCKS IN THE X-DIRECTION
--
NXFIN
      4    4
--
ENDFIN
```

The above example splits the global cells (24-25,87, 1-50) into four and four **LGR** grid blocks in the x-direction, and since the **HXFIN** keyword has not been supplied, then the host cells will split into equal proportions.

6.3.157 NYFIN – DEFINE THE NUMBER OF LGR GRID BLOCKS IN THE Y-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

NYFIN defines the number of Local Grid Refinement (“LGR”) cells within a global or host cell in the x-direction via a vector, as opposed to defining the size for each cell for a Cartesian **LGR** Grid. The **LGR** keyword in the **RUNSPEC** section should be activated to indicate an **LGR** is being used, and the keyword **NYFIN** should be placed in between the **CARFIN** and **ENDFIN** keywords in the **GRID** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	NYFIN	NYFIN is a vector of integer numbers describing the number of LGR cells within each defined global or host grid block in the y-direction in a Cartesian LGR grid. Repeat counts may be used, for example 2*2.0.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NY parameter on the CARFIN keyword in the GRID section.
- 2) The keyword is terminated by a “/”.

Table 6.94: NYFIN Keyword Description

See also the **CARFIN**, **ENDFIN**, **NXFIN**, and **NZFIN** keywords in the **GRID** section to fully define a Cartesian **LGR** grid model.

Example

```
--
--      CARFIN LGR GRID COMMANDS
--
--      LGR      ----- HOST GRID -----  -- CARFIN GRID --  MAX      HOST
--      NAME      I1  I2  J1  J2  K1  K2      NX      NY      NZ      WELLS  NAME
CARFIN
      LGR-OP01    24  24  86  87   1  50      1      8      50      1      GLOBAL /
--
--      DEFINE LGR GRID BLOCKS IN THE Y-DIRECTION
--
NYFIN
      4   4
--
--
--      ENDFIN
```

The above example splits the global cells (24, 86-87,1-50) into four and four **LGR** grid blocks in the y-direction and since the **HYFIN** keyword has not been supplied, then the host cells will split into equal proportions.

6.3.158 NZFIN – DEFINE THE NUMBER OF LGR GRID BLOCKS IN THE Z-DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[NZFIN](#) defines the number of Local Grid Refinement (“LGR”) cells within a global or host cell in the z-direction via a vector, as opposed to defining the size for each cell for a Cartesian [LGR](#) Grid. The [LGR](#) keyword in the [RUNSPEC](#) section should be activated to indicate an [LGR](#) is being used, and the keyword [NXFIN](#) should be placed in between the [CARFIN](#) and [ENDFIN](#) keywords in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	NZFIN	NZFIN is a vector of integer numbers describing the number of LGR cells within each defined global or host grid block in the x-direction in a Cartesian LGR grid. Repeat counts may be used, for example 2*2.0.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX parameter on the CARFIN keyword in the GRID section.
- 2) The keyword is terminated by a “/”.

Table 6.95: NZFIN Keyword Description

See also the [CARFIN](#), [ENDFIN](#), [NXFIN](#), and [NYFIN](#) keywords in the [GRID](#) section to fully define a Cartesian [LGR](#) grid model.

Example

```
--
--          CARFIN LGR GRID COMMANDS
--
--          LGR          ----- HOST GRID -----   -- CARFIN GRID --  MAX      HOST
--          NAME          I1  I2  J1  J2  K1  K2          NX    NY    NZ    WELLS    NAME
CARFIN
          LGR-OP01      24  24  86  86   1  50           8     1   100     1     GLOBAL /
--
--          DEFINE LGR GRID BLOCKS IN THE Z-DIRECTION
--
NZFIN
          50*2
/

ENDFIN
```

The above example splits the global cells (24, 86, 1-50) into two [LGR](#) grid blocks per host cell in the z-direction, and since the [HZFIN](#) keyword has not been supplied, then the host cells will split into equal proportions.

6.3.159 OLDTRAN – ACTIVATE CARTESIAN REGULAR GRID TRANSMISSIBILITIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword switches on Cartesian Regular Grids geometry transmissibility calculation (or block centered transmissibility calculations), which is the default option for this type of grid. Grids defined by the [DX](#), [DY](#), and [DZ](#) series of keywords will always invoke this option by default.

For Irregular Corner-Point Grids defined by the [COORD](#) and [ZCORN](#) keywords Irregular Corner-Point Grid geometry transmissibility calculations should be activated via the [NEWTRAN](#) keyword. Again this is automatically invoked if this type of grid is being employed.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE CARTESIAN REGULAR GRID TRANSMISSIBILITIES
--
OLDTRAN
```

The above example manually activates Cartesian Regular Grid transmissibility calculations.

6.3.160 OLDTRANR – ACTIVATE RADIAL REGULAR GRID TRANSMISSIBILITIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword switches on Radial Regular Grids geometry transmissibility calculation (or block centered transmissibility calculations), which is the default option for this type of grid. Grids defined by the [DR](#), [DTHETA](#), and [DZ](#) series of keywords will always invoke this option by default.

For Irregular Corner-Point Grids defined by the [COORD](#) and [ZCORN](#) keywords Irregular Corner-Point Grid geometry transmissibility calculations should be activated via the [NEWTRAN](#) keyword. Again this is automatically invoked if this type of grid is being employed.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE RADIAL REGULAR GRID TRANSMISSIBILITIES
--
OLDTRANR
```

The above example manually activates Radial Regular Grid transmissibility calculations.

6.3.161 OPERATE – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **OPERATE** keyword performs a mathematical operation on a specified array or part of an array, optionally using another specified array as input to the operation. The keyword allows for various mathematical functions and their associated variables to be defined and applied to the specified array. Input constants can be integer or real valued depending on the array type; however, the arrays that can be operated on are dependent on which section the **OPERATE** keyword is being applied in.

No.	Name	Description	Default														
1	Y	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None														
2	I1	A positive integer that defines the lower bound of the array in the I-direction to be modified must be greater than or equal to one and less than or equal to I2 and NX.	1														
3	I2	A positive integer that defines the upper bound of the array in the I-direction to be modified must be greater than or equal to I1 and less than or equal to NX	NX														
4	J1	A positive integer that defines the lower bound of the array in the J-direction to be modified must be greater than or equal to one and less than or equal to J2 and NY.	1														
5	J2	A positive integer that defines the upper bound of the array in the J-direction to be modified must be greater than or equal to J1 and less than or equal to NY.	NY														
6	K1	A positive integer that defines the lower bound of the array in the K-direction to be modified must be greater than or equal to one and less than or equal to K2 and NZ.	1														
7	K2	A positive integer that defines the upper bound of the array in the K-direction to be modified must be greater than or equal to K1 and less than or equal to NZ.	NZ														
8	EQUATION	A defined character string of up to eight characters that defines the mathematical function to be applied, using the X array and the ALPHA and BETA constants declared on this keyword. EQUATION should be set to one of the following character strings: <table><tr><td>‘MULTA’ - $Y = \alpha X + \beta$</td><td>‘ADDX’ - $Y = \alpha + X$</td></tr><tr><td>‘POLY’ - $Y = Y + \alpha X^\beta$</td><td>‘COPY’ - $Y = X$</td></tr><tr><td>‘SLOG’ - $Y = 10^{\alpha + \beta X}$</td><td>‘MAXLIM’ - $Y = \min(\alpha, X)$</td></tr><tr><td>‘LOG10’ - $Y = \log(X)$</td><td>‘MINLIM’ - $Y = \max(\alpha, X)$</td></tr><tr><td>‘LOGE’ - $Y = \ln(X)$</td><td>‘MULTP’ - $Y = \alpha X^\beta$</td></tr><tr><td>‘INV’ - $Y = \frac{1}{X}$</td><td>‘ABS’ - $Y = (X)$</td></tr><tr><td>‘MULTX’ - $Y = \alpha X$</td><td>‘MULTIPLY’ - $Y = XY$</td></tr></table>	‘MULTA’ - $Y = \alpha X + \beta$	‘ADDX’ - $Y = \alpha + X$	‘POLY’ - $Y = Y + \alpha X^\beta$	‘COPY’ - $Y = X$	‘SLOG’ - $Y = 10^{\alpha + \beta X}$	‘MAXLIM’ - $Y = \min(\alpha, X)$	‘LOG10’ - $Y = \log(X)$	‘MINLIM’ - $Y = \max(\alpha, X)$	‘LOGE’ - $Y = \ln(X)$	‘MULTP’ - $Y = \alpha X^\beta$	‘INV’ - $Y = \frac{1}{X}$	‘ABS’ - $Y = (X) $	‘MULTX’ - $Y = \alpha X$	‘MULTIPLY’ - $Y = XY$	None
‘MULTA’ - $Y = \alpha X + \beta$	‘ADDX’ - $Y = \alpha + X$																
‘POLY’ - $Y = Y + \alpha X^\beta$	‘COPY’ - $Y = X$																
‘SLOG’ - $Y = 10^{\alpha + \beta X}$	‘MAXLIM’ - $Y = \min(\alpha, X)$																
‘LOG10’ - $Y = \log(X)$	‘MINLIM’ - $Y = \max(\alpha, X)$																
‘LOGE’ - $Y = \ln(X)$	‘MULTP’ - $Y = \alpha X^\beta$																
‘INV’ - $Y = \frac{1}{X}$	‘ABS’ - $Y = (X) $																
‘MULTX’ - $Y = \alpha X$	‘MULTIPLY’ - $Y = XY$																

No.	Name	Description	Default
9	X	A character string of up to eight characters in length that defines the keyword identifying the array to be used as an input parameter.	None
10	ALPHA	An integer or real value that is the α variable in the EQUATION function.	None
11	BETA	An integer or real value that is the β variable in the EQUATION function.	None
Notes: - Where NX, NY and NZ are the dimensions of the model as defined on the DIMENS keyword in the RUNSPEC section. - Each record must be terminated by a “/” and the keyword is terminated by a “/”.			

Table 6.96: OPERATE Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.97 . Cells colored red indicate arrays that are not supported by OPM Flow operations.

OPERATE Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANX	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TBLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	OILAPI		
MULTZ	TRANR	KRG	WH2NUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPFANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.97: OPERATE Keyword Applicable Arrays by Section

Note that care should be exercised when performing operations on integer array data as all transforms are performed using floating point arithmetic operations. In addition, operations on any of the transmissibility arrays ([TRANX](#), [TRANX-](#), [TRANY](#), [TRANY-](#), [TRANZ](#), and [TRANZ-](#)) may result in untended consequences as these arrays have zero values on the boundary of the grid. The [OPERATE](#) keyword in OPM Flow uses SI units. Operations where [PERMX](#), [PERMY](#) and [PERMZ](#) keywords are the output array, and the system of units is set to [FIELD](#) or [LAB](#), will need to be corrected by multiplying by a conversion factor of 9.87E-16. This is not necessary with the commercial simulator. In this use OPM ResInsight to verify and visually inspect the results.

Example

The first example uses the MULTP function combined with the Net-to-Gross ([NTG](#)) array to re-scale the [MULTX](#), [MULTY](#) and [MULTZ](#) arrays to reduce the transmissibility in three separate reservoirs based on the reservoir quality ([NTG](#)).

```
--
--      MATHEMATICAL OPERATIONS ON ARRAYS BY CELL
--
--      OUTPUT  ----- BOX ----- OPERATION  INPUT  ALPHA  BETA
--      ARRAY   I1  I2   J1  J2    K1  K2  -----  ARRAY   -----
OPERATE
MULTX  1*  1*   1*  1*      1  32  'MULTP'   NTG    1.00  0.75 / RES1
MULTY  1*  1*   1*  1*      1  32  'MULTP'   NTG    1.00  0.75 / RES1
MULTZ  1*  1*   1*  1*      1  32  'MULTP'   NTG    1.00  0.75 / RES1

MULTX  1*  1*   1*  1*     34  64  'MULTP'   NTG    1.00  0.85 / RES2
MULTY  1*  1*   1*  1*     34  64  'MULTP'   NTG    1.00  0.85 / RES2
MULTZ  1*  1*   1*  1*     34  64  'MULTP'   NTG    1.00  0.85 / RES2

MULTX  1*  1*   1*  1*     67  96  'MULTP'   NTG    1.00  0.50 / RES3
MULTY  1*  1*   1*  1*     67  96  'MULTP'   NTG    1.00  0.50 / RES3
MULTZ  1*  1*   1*  1*     67  96  'MULTP'   NTG    1.00  0.50 / RES3
/
```

The next example shows how to set the maximum gas saturation ([SGU](#)) based on the minimum (lowest) water saturation ([SWL](#)) when using the End-Point Scaling option.

```
--
--      MATHEMATICAL OPERATIONS ON ARRAYS
--
--      OUTPUT  ----- BOX ----- OPERATION  INPUT  ALPHA  BETA
--      ARRAY   I1  I2   J1  J2    K1  K2  -----  ARRAY   -----
OPERATE
SGU      1*  1*   1*  1*      1*  1*  'MULTA'   SWL    -1.0  1.0    /
/
```

The above example sets the maximum gas saturation to be one minus the minimum water saturation.

6.3.162 OPERATER – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS BY REGION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **OPERATER** keyword is similar to the **OPERATE** keyword, except it applies the mathematical operation on specific regions, whereas, **OPERATE** applies the operations on a cell by cell basis. Here the **OPERATER** keyword performs a mathematical operation on a specified property array, optionally using another property array as input to the function. The keyword allows for various mathematical functions and their associated variables to be defined and applied to the specified region data. Input constants can be integer or real valued depending on the array type; however, the arrays that can be operated on are dependent on which section the **OPERATER** keyword is being applied in.

No.	Name	Description	Default
1	Y	A character string of up to eight characters in length that defines the keyword identifying the array to be modified.	None
2	REGION	REGION is a positive integer representing the region for which the EQUATION should be applied. The default is to use the region number from the OPERNUM keyword; however this can be reset to another region array via the ARRAY item on this keyword, provided the array exists at the time the keyword is declared in the input deck. Note also the OPERNUM keyword must precede the use of the OPERATER keyword.	0
3	EQUATION	A defined character string of up to eight characters that defines the mathematical function to be applied, using the X array and the ALPHA and BETA constants declared on this keyword. EQUATION should be set to one of the following character strings: 'MULTA' - $Y = \alpha X + \beta$ 'POLY' - $Y = Y + \alpha X^\beta$ 'SLOG' - $Y = 10^{\alpha + \beta X}$ 'LOG10' - $Y = \log(X)$ 'LOGE' - $Y = \ln(X)$ 'INV' - $Y = \frac{1}{X}$ 'MULTX' - $Y = \alpha X$ 'ADDX' - $Y = \alpha + X$ 'COPY' - $Y = X$ 'MAXLIM' - $Y = \min(\alpha, X)$ 'MINLIM' - $Y = \max(\alpha, X)$ 'MULTP' - $Y = \alpha X^\beta$ 'ABS' - $Y = (X)$ 'MULTIPLY' - $Y = XY$	None
4	X	A character string of up to eight characters in length that defines the keyword identifying the array to be used as an input parameter.	None
5	ALPHA	An integer or real value that is the α variable in the EQUATION function.	None
6	BETA	An integer or real value that is the β variable in the EQUATION function.	None

No.	Name	Description	Default
7	ARRAY	The name of the array for which the REGION variable references. This can be any standard region array as declared in the REGION section (<i>FIPNUM</i>, <i>PVTNUM</i>, etc.), provided the array exists at the time the <i>OPERATER</i> keyword is invoked. In addition, the <i>MULTNUM</i>, <i>FLUXNUM</i> and <i>OPERNUM</i> may be used. Only the default value of <i>OPERNUM</i> is supported by OPM Flow.	<i>OPERNUM</i>
Notes: - Where the REGION should be less than or equal to the maximum number of regions as defined on the <i>REGDIMS</i> keyword for the <i>FIPNUM</i> and <i>OPERNUM</i> arrays or the <i>GRIDOPTS</i> keyword for the <i>MULTNUM</i> array in the <i>RUNSPEC</i> section.. - Each record must be terminated by a “/” and the keyword is terminated by a “/”.			

Table 6.98: OPERATER Keyword Description

Examples of the arrays most commonly operated on in each section are given in Table 6.99. Cells colored red indicate arrays that are not supported by OPM Flow operations.

OPERATER Keyword And Variable Options By Section						
GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
DX	DEPTH	SWL	ENDNUM	PRESSURE		
DY	PORV	SWCR	EQLNUM	SWAT		
DZ	TRANX	SWU	FIPNUM	SGAS		
PERMX	TRANY	SGL	IMBNUM	RV		
PERMY	TRANZ	SGCR	MISCNUM	RS		
PERMZ	DIFFX	SGU	PVTNUM	TBLK		
MULTX	DIFFY	KRW	ROCKNUM	GI		
MULTY	DIFFZ	KRO	SATNUM	OILAPI		
MULTZ	TRANR	KRG	WAGNUM	SALT		
DR	TRANHT	PCG		GASCONC		
DTHETA	DIFFR	PCW		SOLVCONC		
PERMR	DIFFHT			SOLVFRAC		
PERMHT				SFOAM		
DZNET				SPOLY		
PORO						
NTG						
FLUXNUM						
MULTNUM						
MPPANUM						
DIFFX						
DIFFY						
DIFFZ						
DIFFR						
DIFFHT						

Table 6.99: OPERATER Keyword Applicable Arrays by Section

Note that care should be exercised when performing operations on integer array data as all transforms are performed using floating point arithmetic operations. In addition, operations on any of the transmissibility arrays (*TRANX*, *TRANX-*, *TRANY*, *TRANY-*, *TRANZ*, and *TRANZ-*) may result in unintended consequences as these arrays have zero values on the boundary of the grid. In this use OPM ResInsight to verify and visually inspect the results.

Note

The *OPERATER* and *OPERATE* keywords allow for great flexibility in generating or modifying the simulator’s input arrays. In addition, OPM Flow also has a Python facility to manipulate and calculate data that offers even greater flexibility, but note that this feature is not compatible with the commercial simulator.

Finally, OPM ResInsight, the post-processing plotting software, has both Octave and Python scripting facilities that enable both data generation and visual inspection of the results. The resulting calculated arrays can then be exported from OPM ResInsight and “included” back into OPM Flow, thus maintaining compatibility with the commercial simulator.

Example

The first example uses the *MULTP* function combined with the Net-to-Gross (*NTG*) array to re-scale the *MULTX*, *MULTY* and *MULTZ* arrays to reduce the transmissibility in three separate reservoirs based on the reservoir quality (*NTG*). This keyword sequence should be in the *GRID* section.

```
--
--      MATHEMATICAL OPERATIONS ON ARRAYS BY REGION
--
--      OUTPUT  REGN  OPERATION  SOURCE  ALPHA  BETA  REGN
--      ARRAY   NUM   TYPE        ARRAY   CONST  CONST ARRAY
OPERATER
MULTX    1      'MULTP'    NTG     1.00   0.75   / RES1
MULTY    1      'MULTP'    NTG     1.00   0.75   / RES1
MULTZ    1      'MULTP'    NTG     1.00   0.75   / RES1

MULTX    2      'MULTP'    NTG     1.00   0.85   / RES2
MULTY    2      'MULTP'    NTG     1.00   0.85   / RES2
MULTZ    2      'MULTP'    NTG     1.00   0.85   / RES2

MULTX    3      'MULTP'    NTG     1.00   0.50   / RES3
MULTY    3      'MULTP'    NTG     1.00   0.50   / RES3
MULTZ    3      'MULTP'    NTG     1.00   0.50   / RES3

/
```

Notice that the *ARRAY* variable has been defaulted, resulting in *OPERNUM* being the regional array for the *REGION* variable.

The next example shows how to set the maximum gas saturation (*SGU*) based on the minimum (lowest) water saturation (*SWL*) when using the End-Point Scaling option, in the *PROPS* section.

```
--
--      MATHEMATICAL OPERATIONS ON ARRAYS
--
--      OUTPUT  REGN  OPERATION  SOURCE  ALPHA  BETA  REGN
--      ARRAY   NUM   TYPE        ARRAY   CONST  CONST ARRAY
OPERATER
SGU         1      'MULTA'    SWL     -1.0    1.0    /
SGU         2      'MULTA'    SWL     -1.0    1.0    /
SGU         3      'MULTA'    SWL     -1.0    1.0    /

/
```

The above example sets the maximum gas saturation to be one minus the minimum water saturation for regions one to three.

The final example shows how to reset the `FIPNUM` array when the exported array from the earth model does not correspond to the simulator's desired numbering scheme.

```
--
--      MATHEMATICAL OPERATIONS ON ARRAYS BY REGION
--
--      RESET FIPNUM BASED ON MULTNUM AND OPERNUM
--
--      DESTIN   REGN   OPERATION   SOURCE   ALPHA   BETA   INPUT   SEGNUM   EQUIL
--      ARRAY    NUM    TYPE         ARRAY    CONST  CONST  ARRAY   NUMBER   NUMBER
OPERATER
FIPNUM    26      'MULTA'    'MULTNUM' 0.00     1       /    26      1
FIPNUM    44      'MULTA'    'MULTNUM' 0.00     2       /    44      2
FIPNUM    62      'MULTA'    'MULTNUM' 0.00     3       /    62      3
FIPNUM    98      'MULTA'    'MULTNUM' 0.00     4       /    98      4
FIPNUM   116      'MULTA'    'MULTNUM' 0.00     5       /   116     5
FIPNUM   134      'MULTA'    'MULTNUM' 0.00     6       /   134     6
FIPNUM    46      'MULTA'    'MULTNUM' 0.00     7       /    46     7
FIPNUM    64      'MULTA'    'MULTNUM' 0.00     8       /    64     8
FIPNUM    82      'MULTA'    'MULTNUM' 0.00     9       /    82     9
FIPNUM   226      'MULTA'    'MULTNUM' 0.00    10      /   226    10
FIPNUM   262      'MULTA'    'MULTNUM' 0.00    11      /   262    11
FIPNUM   280      'MULTA'    'MULTNUM' 0.00    12      /   280    12
FIPNUM   298      'MULTA'    'MULTNUM' 0.00    13      /   298    13
FIPNUM    33      'MULTA'    'MULTNUM' 0.00    14      /    33    14
FIPNUM    51      'MULTA'    'MULTNUM' 0.00    15      /    51    15
FIPNUM   105      'MULTA'    'MULTNUM' 0.00    16      /   105    16
FIPNUM   159      'MULTA'    'MULTNUM' 0.00    17      /   159    17
FIPNUM   195      'MULTA'    'MULTNUM' 0.00    18      /   195    18
FIPNUM   267      'MULTA'    'MULTNUM' 0.00    19      /   267    19
FIPNUM   303      'MULTA'    'MULTNUM' 0.00    20      /   303    20
FIPNUM   321      'MULTA'    'MULTNUM' 0.00    21      /   321    21
FIPNUM   339      'MULTA'    'MULTNUM' 0.00    22      /   339    22
FIPNUM    54      'MULTA'    'MULTNUM' 0.00    23      /    54    23
FIPNUM    72      'MULTA'    'MULTNUM' 0.00    24      /    72    24
FIPNUM   108      'MULTA'    'MULTNUM' 0.00    25      /   108    25
FIPNUM   144      'MULTA'    'MULTNUM' 0.00    26      /   144    26
FIPNUM   270      'MULTA'    'MULTNUM' 0.00    27      /   270    27
/
```

Note that operation can only be done in the `REGION` section as `FIPNUM` is only available for use in this section and that the `ARRAY` variable has been defaulted, resulting in `OPERNUM` being the regional array for the `REGION` variable.

6.3.163 OPERNUM – DEFINE REGIONS FOR MATHEMATICAL OPERATIONS ON ARRAYS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the *OPERATER* region numbers for each grid block. The *OPERNUM* keyword defines the region numbers for each grid block, as such there must be one entry for each cell in the model. The array can also be used with the *EQUALREG*, *ADDREG*, *COPYREG*, *MULTIREG*, *MULTREGP* and *MULTREGT* keywords, as well as the *OPERATER* keyword in calculating various grid properties in the *GRID* and *REGION* section.

No.	Name	Description	Default
I	<i>OPERNUM</i>	<i>OPERNUM</i> defines an array of positive integers greater than or equal to one that assigns a grid cell to a particular <i>OPERNUM</i> region. The maximum number of <i>OPERNUM</i> regions is set by the NOPREG variable on the <i>REGDIMS</i> keyword in the <i>RUNSPEC</i> section. Note that the default value of zero implies that the calculations requested by the <i>OPERATER</i> keyword will not be performed.	0

Notes: - The number of entries should correspond to the NX x NY x NZ parameters on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section, unless the <i>BOX</i> keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the <i>BOX</i> statement. - The keyword is terminated by a “/”.			
---	--	--	--

Table 6.100: OPERNUM Keyword Description

Examples

The example below sets three *OPERNUM* regions for a 4 x 5 x 2 model.

```
--
--          DEFINE OPERNUM REGIONS FOR ALL CELLS
--
OPERNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the *EQUALS* keyword could be employed to accomplish the same task, that is:

```
--
--          ARRAY          CONSTANT          ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
    OPERNUM          1          1*  1*   1*  1*   1*  1* / SET REGION 1
    OPERNUM          2          1   2    1   2    1   1 / SET REGION 2
    OPERNUM          3          1   2    1   2    2   2 / SET REGION 3
/
```

One can then increase *PERMX* by 25% in region three only.

```
--
--          MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER
--
--          ARRAY          CONSTANT  REGION  REGION ARRAY
--          VALUE          NUMBER    M / F / O
MULTIREG
    PERMX          1.25          3          0          /
/
```

6.3.164 OUTRAD – DEFINE THE OUTER RADIUS OF A RADIAL GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

OUTRAD defines the OUTER radius of the reservoir model for a radial or spider grid geometry. The **RADIAL** or **SPIDER** keyword in the **RUNSPEC** should be activated to indicate that radial or spider geometry is being used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	OUTRAD	A single real positive number greater than INRAD defining the outer radius of a radial grid.			None
		feet	m	cm	
Notes: I) The keyword is terminated by a “I”.					

Table 6.101: OUTRAD Keyword Description

Note that the **SPIDER** keyword activates OPM Flow’s radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: **INRAD**, **DRV**, **DTHETAV**, **DZ/DZV** etc. in the **GRID** section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model’s pore volume to reflect radial coordinates; thus, overcoming the display limitation.

The keyword allows for an alternative method of entering the size of the R direction grid cells instead of entering the data using the **DR** or **DRV** keywords in the **GRID** section. Given the internal radius set by the **INRAD** keyword, the external radius set by the **OUTRAD** keyword and the number of grid cells in the R direction set by the NX variable on the **DIMENS** keyword in the **RUNSPEC** section, the R direction cells sizes are computed automatically on a geometric spacing, as defined by:

$$\frac{R_i}{R_{i-1}} = \left(\frac{OUTRAD}{R_{i-1}} \right)^{\frac{1}{(NX - i_j + 1)}} \quad (6.12)$$

or

$$R_i = (R_{i-1}) \left(\frac{OUTRAD}{R_{i-1}} \right)^{\frac{(i - i_j + 1)}{(NX - i_j + 1)}} \quad (6.13)$$

and the **DR** value for the i^{th} cell, that is the value that can also be manually entered on the **DR** keyword in the **GRID** section, is given by:

$$DR_i = R_i - R_{i-1} \quad (6.14)$$

Where:

DR_i = **DR** value for the i^{th} cell
 R_i = current total radius for the i radii.
 R_{i-1} = total radius for the $i - 1$ radii.

NX (NR)

= number of radial grid cells excluding the inner radius

OUTRAD

= the outer radius of the radial grid, the value includes the inner radius.

For example, given an inner radius set to 0.25, an outer radius of 2,050 and the number of cells in the R direction set to ten, then Table 6.102 shows the grid size calculations.

OUTRAD Radial Grid Example			
INRAD	0.25		
OUTRAD	2050.0		
NX	10		
NX	Ri	DR	Ratio
0	0.250	0.250	
1	0.616	0.366	1.463
2	1.516	0.900	2.463
3	3.733	2.217	2.463
4	9.193	5.460	2.463
5	22.638	13.445	2.463
6	55.748	33.109	2.463
7	137.281	81.533	2.463
8	338.058	200.777	2.463
9	832.477	494.420	2.463
10	2050.000	1217.523	2.463
Total		2050.000	

Table 6.102: OUTRAD Radial Grid Example

See also the [DR](#), [DRV](#), [DTHETA](#), [DTHETAU](#) and [TOPS](#) keywords to fully define a Radial or Spider Grid.

Example

```

--
--      INNER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
INRAD
0.25
--
--      OUTER RADIUS OF FIRST GRID BLOCK IN THE RADIAL DIRECTION
--
OUTRAD
2050.0

```

The above example defines the inner radius to be 0.25 and the outer radius to be 2,050 feet. Note that the outer radius includes the inner radius.

6.3.165 PARAOPTS – DEFINE PARALLEL RUN OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PARAOPTS](#) keyword defines various options for parallel runs, for when the Parallel option has been invoked by the [PARALLEL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See section [2.2 Running OPM Flow From The Command Line](#) on how to run OPM Flow in parallel mode.

6.3.166 PEBI – ACTIVATE AND DEFINED PEBI GRID OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PEBI activates the unstructured Perpendicular Bisector (“PEBI”) ¹²⁸ and ¹²⁹ loading of grid data generated by an external pre-processing program for generating simulation grids.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	OPTION1	A defined character string that activates or deactivates the checking of negative transmissibility values. OPTION1 should be set to YES to check for negative values, or NO switches off this option.	NO
2	OPTION2	A defined character string that activates or deactivates the calculation of pore volumes and transmissibilities. OPTION2 should be set to YES if the pore volumes and transmissibilities are provided, or NO for the values to calculated by the simulator.	NO

Table 6.103: PEBI Keyword Description

Example

```
--
--      OPTION1  OPTION2
--      CHECK    CALCULATE
PEBI
      NO        YES
```

The above example switches off the negative transmissibility check and requests that the simulator calculates pore volumes and transmissibilities as they are not provided by the input data.

¹²⁸ Heinemann, Z.E. and Brand, C.W. 1988. Gridding Techniques in Reservoir Simulation. Proc., First Intl. Forum on Reservoir Simulation, Alpbach, Austria, 339.

¹²⁹ Heinemann, Z.E., Brand, C.W., Munka, M. et al. 1991. Modeling Reservoir Geometry With Irregular Grids. SPE Res Eng 6 (2): 225–232. SPE-18412-PA. <http://dx.doi.org/10.2118/18412-PA>

6.3.167 PERMAVE – DEFINE AVERAGE TRANSMISSIBILITY COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *PERMAVE* keyword defines the three directional exponent coefficients used to average the grid block permeabilities between two neighboring cells when calculating the transmissibility between the two blocks. The keyword can be used to change from the default weighted harmonic averaging (coefficient set equal to -1), to geometric (coefficient equal to zero), or to arithmetic averaging (coefficient equal to 1). The three coefficients represent the averaging in the x-, y- and z-directions.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.168 PERMJFUN – DEFINE LEVERETT J-FUNCTION PERMEABILITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PERMJFUN defines the permeability to be used in de-normalizing the Leverett J-Functions¹³⁰ for when the PERM variable on the **JFUNC** or the **JFUNC** keyword in the **GRID** section has been set to “U”, as oppose to using **PERMX**, **PERMY**, **PERMZ** arrays etc.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PERMJFUN	PERMJFUN is an array of real positive numbers assigning the permeability to be used in de-normalizing the Leverett J-Function to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		mD	mD	mD	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 6.104: PERMJFUN Keyword Description

For grid blocks that have not been assigned a **PERMJFUN** value the default directional permeability will be used, that is the average of **PERMX** and **PERMY**.

See also the **PERMX**, **PERMY** and **PERMZ** keywords to fully define the permeability for the model.

Example

```
--
--      DEFINE GRID BLOCK PERMJFUN FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PERMJFUN
    100*500.0    100*50.0    100*200.0    /
```

The above example defines the **PERMJFUN** to be 500.0, 50.0, and 200.0 for the first, second and third layers in the model for all 300 cells, as defined by the **DIMENS** keyword in the **RUNSPEC** section.

¹³⁰ Leverett, M. C.; “Capillary Behaviour in Porous Solids”, Trans. AIME (1941) 142, 152-168.

6.3.169 PERMR – DEFINE THE PERMEABILITY FOR EACH CELL IN THE R DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[PERMR](#) sets the permeability for each cell in the R direction in a radial geometry grid. The [RADIAL](#) or [SPIDER](#) keywords in the [RUNSPEC](#) should be activated to indicate that radial or spider geometry is being used.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PERMR	PERMR is an array of real positive numbers assigning the permeability in the R direction to each cell in the model.This equivalent to PERMX in a Cartesian grid. Repeat counts may be used,for example 20*100.0.			None
		mD	mD	mD	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.

2)

The keyword is terminated by a “/”.

Table 6.105: PERMR Keyword Description

Note that the [SPIDER](#) keyword activates OPM Flow's radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: [INRAD](#), [DRV](#), [DTHETAV](#), [DZIDZV](#) etc. in the [GRID](#) the section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model's pore volume to reflect radial coordinates; thus, overcoming the display limitation.

See also the [PERMTHT](#) and [PERMZ](#) keywords to fully define the permeability for a radial or spider grid model.

Example

```
--
--      DEFINE GRID BLOCK PERMR DATA FOR ALL CELLS (BASED ON NR x NY x NZ = 300)
--
PERMR      100*500.0    100*50.0    100*200.0      /
```

The above example defines the [PERMR](#) to be 500.0, 50.0, and 200.0 for the first, second and third layers in the model for all 300 cells, as defined by the [DIMENS](#) keyword (10, 10, 3) in the [RUNSPEC](#) section.

6.3.170 PERMTHT – DEFINE THE PERMEABILITY FOR EACH CELL IN THE THETA DIRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PERMTHT sets the permeability for each cell in the THETA direction in a radial geometry grid. The **RADIAL** or **SPIDER** keyword in the **RUNSPEC** should be activated to indicate that radial or spider geometry is being used.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PERMTHT	PERMTHT is an array of real positive numbers assigning the permeability in the THETA direction to each cell in the model. This equivalent to PERMY in a Cartesian grid. Repeat counts may be used, for example 20*100.0.			None
		mD	mD	mD	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.106: PERMTHT Keyword Description

Note that the **SPIDER** keyword activates OPM Flow's radial grid geometry option for the model. This keyword will create a spiderweb-shaped grid based on a corner-point grid using the standard radial grid keywords: **INRAD**, **DRV**, **DTHETAV**, **DZIDZV** etc. in the **GRID** the section. A spider grid can be viewed in 3D in OPM ResInsight unlike radial grids that cannot be viewed in the software. To overcome this, the simulator now converts radial grids to Irregular Corner-Point Grids and adjusts the model's pore volume to reflect radial coordinates; thus, overcoming the display limitation.

See also the **PERMR** and **PERMZ** keywords to fully define the permeability for a radial or spider grid model.

Example

```
--
--      DEFINE GRID BLOCK PERMTHT DATA FOR ALL CELLS (NX x NY x NZ = 300)
--
PERMTHT      100*500.0    100*50.0    100*200.0      /
```

The above example defines the **PERMTHT** to be 500.0, 50.0, and 200.0 for the first, second and third layers in the model for all 300 cells, as defined by the **DIMENS** keyword (10, 10, 3) in the **RUNSPEC** section.

6.3.171 PERMX – DEFINE THE PERMEABILITY IN THE X DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PERMX defines the permeability in the X direction for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PERMX	PERMX is an array of real positive numbers assigning the permeability in the X direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		mD	mD	mD	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.107: PERMX Keyword Description

See also the **PERMY** and **PERMZ** keywords to fully define the permeability for the model.

Example

```
--
--      DEFINE GRID BLOCK PERMX DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PERMX      100*500.0   100*50.0   100*200.0      /
```

The above example defines the **PERMX** to be 500.0, 50.0, and 200.0 for the first, second and third layers in the model for all 300 cells, as defined by the **DIMENS** keyword in the **RUNSPEC** section.

6.3.172 PERMY – DEFINE THE PERMEABILITY IN THE Y DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PERMY defines the permeability in the Y direction for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PERMY	PERMY is an array of real positive numbers assigning the permeability in the Y direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		mD	mD	mD	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.108: PERMY Keyword Description

See also the **PERMX** and **PERMZ** keywords to fully define the permeability for the model.

Example

```
--
--      DEFINE GRID BLOCK PERMY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PERMY      100*500.0   100*50.0   100*200.0      /
```

The above example defines the **PERMY** to be 500.0, 50.0, and 200.0 for the first, second and third layers in the model for all 300 cells, as defined by the **DIMENS** keyword in the **RUNSPEC** section.

6.3.173 PERMZ – DEFINE THE PERMEABILITY IN THE Z DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[PERMZ](#) defines the permeability in the Z direction for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PERMZ	PERMZ is an array of real positive numbers assigning the permeability in the Z direction to each cell in the model. Repeat counts may be used, for example 200*50.0.			None
		mD	mD	mD	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.109: PERMZ Keyword Description

Note

Although [PERMX](#) and [PERMY](#) are commonly set to be equal, [PERMZ](#) is typically not equal to either [PERMX](#) or [PERMY](#). Normally [PERMZ](#) is set as a fraction of [PERMX](#) with typical values ranging from 0.1 to 0.5 times [PERMX](#).

See also the [PERMX](#) and [PERMY](#) keywords to fully define the permeability for the model.

Example

The example below defines the [PERMZ](#) to be 50.0, 5.0, and 20.0 for the first, second and third layers in the model for all 300 cells, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

```
--
--      DEFINE GRID BLOCK PERMZ DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PERMZ      100*50.0      100*5.0      100*20.0      /
```

The next example sets [PERMX](#) to be 500.0, 50.0, and 200.0 for the first, second and third layers in the model for all 300 cells, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section. It then copies the [PERMX](#) values to the [PERMY](#) and [PERMZ](#) arrays, and finally multiplies [PERMZ](#) by 0.1 times to get the final values for [PERMZ](#).

```
--
--      DEFINE GRID BLOCK PERMX DATA FOR ALL CELLS
--
PERMX      100*500.0    100*50.0    100*200.0
--
--      SOURCE      DESTIN.      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
COPY
      PERMX      PERMY      1*  1*   1*  1*   1*  1* / CREATE PERMY
      PERMX      PERMZ      1*  1*   1*  1*   1*  1* / CREATE PERMZ
/
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
MULTIPLY
      PERMZ      0.10000      1*  1*   1*  1*   1*  1* / PERMZ * 0.1
/
```

The above sequence of keywords is quite common in input decks, that is copying the [PERMX](#) data to the [PERMY](#) and [PERMZ](#) arrays and then adjusting the [PERMY](#) and [PERMZ](#) arrays as required using the [MULTIPLY](#) keyword.

6.3.174 PETGRID – LOAD A GENERIC SIMULATION GRID FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *PETGRID* keyword instructs the simulator to load a Generic Simulation Grid (*.GSG) file that contains grid geometry data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.175 PINCH – DEFINE PINCH-OUT LAYER OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PINCH](#) keyword defines the parameters used to control the generation of Non-Neighbor Connections (“NNCs”) in the vertical (K) direction due to layers pinching out. This keyword is applied to all layers in the model as opposed to the [PINCHREG](#) keyword that offers more flexibility by applying the pinch-out controls to various regions in the model defined by the [PINCHNUM](#) keyword.

OPM Flow will automatically generate connections between non neighbor cells in the vertical direction based on the parameters on this keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PINCHTHK	A real number defining the pinch-out threshold thickness for any cell. NNCs are generated across inactive cells having a vertical thickness less than PINCHTHK.			Defined
		ft. 0.001	m 0.001	cm 0.001	
2	PINCHOPT	A character string controlling the generation of pinch-outs when the MINPV keyword has been used to deactivate cells with small pore volumes. PINCHOPT can either be set to: 1) GAP to allow the generation of NNCs across cells that have been made inactive with the MINPV keyword when the thickness is greater than PINCHTHK threshold. 2) NOGAP to enforce the strict adherence to the PINCHTHK threshold whether or not cells have been made inactive due to the MINPV keyword.			GAP
3	PINCHGAP	A real number defining the maximum “empty” thickness allowed between grid blocks in adjacent grid layers for a non-zero transmissibility to exist between them.			Defined
		ft. 1.0E20	m 1.0E20	cm 1.0E20	
4	PINCHCAL	A character string controlling the calculation of the pinch-out transmissibilities. PINCHCAL can either be set to: 1) TOPBOT results in the pinch-out transmissibility being calculated from the half-cell Z-direction transmissibilities of the active cells on either side of the pinched-out layers. 2) ALL results in the pinch-out transmissibility being calculated from the Z-direction transmissibilities harmonic average of all the cells between the active cells on either side of the pinched-out layers.			TOPBOT

Notes:

Table 6.110: PINCH Keyword Description

Examples

The first example below will create NNCs between the cells above and below any cell having vertical thickness less than 0.01 in either feet or metres.

```
--
--      SET PINCH-OUT PARAMETERS FOR CALCULATING PINCH-OUT PROPERTIES
--
PINCH
--      THRESHOLD      GAP      EMPTY      TRANS      MULTZ
--      THICKNESS      NO GAP    GAP      CALC      CALC
--      0.01           1*       1*       1*       1*
```

For the second example, the **MINPV** keyword is used to set the minimum pore volume to 500 m³ (metric units) and then the **PINCH** keyword is invoked with PINCHGAP set equal to GAP as follows:

```
--
--      MINIMUM PORE VOLUME FOR ACTIVE CELLS
--
MINPV      500.0
--
--      SET PINCH-OUT CRITERIA FOR THE MODEL
--
PINCH
--      THRESHOLD      GAP      EMPTY      TRANS      MULTZ
--      THICKNESS      NO GAP    GAP        CALC       CALC
--      0.1            GAP      1*        1*        1*
```

In the above example the `MINPV` keyword will deactivate all cells with pore volumes less than 500 m3. These deactivated cells are inactive in the model and therefore are not included in the flow calculations; however, by default they will result in no-flow barriers but may not be thin enough for `PINCH` to create NNCs across them. By setting `PINCHGAP` equal to `GAP` on the `PINCH` keyword (the default setting), then OPM Flow generates NNCs across the cells that have been deactivated by the `MINPV` keyword. However, in this case there may be grid blocks in the model with a pore volume greater than `MINPV` but a thickness less than the pinch-out threshold. These cells will not be deactivated by the `PINCH` keyword.

6.3.176 PINCHNUM – DEFINE PINCH-OUT REGIONS FOR THE PINCHREG KEYWORD

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **PINCHNUM** keyword defines the pinch-out region numbers for each grid block, as such there must be one entry for each cell in the model. The array is used with the **PINCHREG** keyword to set the pinch-out options and threshold thickness for each region.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	PINCHNUM	PINCHNUM defines an array of positive integers assigning a grid cell to a particular PINCHNUM region. The maximum number of PINCHNUM regions is set by the NRPIN variable on the GRIDOPTS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 6.111: PINCHNUM Keyword Description

Example

The example below sets defines three **PINCHNUM** regions for various layers in a model based on the model's layering.

```
--
--          ARRAY          CONSTANT          ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      MULTNUM      1          1*  1*   1*  1*   1*  1* / SET REGION 1
      MULTNUM      2          1   2    1   2    10  50 / SET REGION 2
      MULTNUM      3          1   2    1   2    51 100 / SET REGION 3
/
```

One can then set the pinch-out criteria for each region.

```
--
--          SET PINCH-OUT CRITERIA VIA THE PINCHNUM REGION
--
PINCHREG
--          THRESHOLD      GAP      EMPTY      TRANS
--          THICKNESS      NO GAP    GAP        CALC
--          0.1             1*       1*         1*                / PINCHNUM 01
--          1.0             1*       10          1*                / PINCHNUM 02
--          1.0             NOGAP    20         1*                / PINCHNUM 03
```

The above example sets the default pinch-out criteria for grid blocks defined as region one via the **PINCHNUM** array and different criteria for regions two and three.

6.3.177 PINCHOUT – DEFINE PINCH-OUT LAYERS OPTION (FIXED)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PINCHOUT](#) keyword activates the generation of Non-Neighbor Connections (“NNCs”) in the vertical (K) direction due to layers pinching out, using a constant threshold thickness of 0.001 for all unit systems. See also the [PINCH](#) keyword in the [GRID](#) section that allows for specifying the threshold thickness and other parameters on a layer basis, and the [PINCHREG](#) keyword that applies the pinch-out controls to various regions in the model defined by the [PINCHNUM](#) keyword.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

The example will create NNCs between the cells above and below any cell having vertical thickness less than 0.001 in either feet or metres.

```
--
--      SET PINCH-OUT CRITERA WITH CONSTANT THRESHOLD THICKNESS OF 0.001
--
PINCHOUT
```


6.3.178 PINCHREG – DEFINE PINCH-OUT REGION OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PINCHREG](#) keyword defines the parameters used to control the generation of Non-Neighbor Connections (“NNCs”) in the vertical (K) direction due to layers pinching out in combination with the [PINCHNUM](#) keyword. This allows different regions in the model to use different criteria in controlling the how pinch-outs are generated. The keyword should contain NRPINC records defining the criteria for each pinch-out region defined with the [PINCHNUM](#) keyword. NRPINC is the maximum number of [PINCHNUM](#) regions defined via the [GRIDOPTS](#) keyword in the [RUNSPEC](#) section.

An alternative method to set the pinch-out criteria is to use the [PINCH](#) keyword, that applies the criteria to the whole model.

OPM Flow will automatically generate connections between non neighbor cells in the vertical direction based on the parameters on this keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PINCHTHK	A real number defining the pinch-out threshold thickness for any cell. NNCs are generated across inactive cells having a vertical thickness less than PINCHTHK.			Defined
		ft. 0.001	m 0.001	cm 0.001	
2	PINCHOPT	A character string controlling the generation of pinch-outs when the MINPV keyword has been used to deactivate cells with small pore volumes. PINCHOPT can either be set to: 1) GAP to allow the generation of NNCs across cells that have been made inactive with the MINPV keyword when the thickness is greater than PINCHTHK threshold. 2) NOGAP to enforce the strict adherence to the PINCHTHK threshold whether or not cells have been made inactive due to the MINPV keyword.			GAP
3	PINCHGAP	A real number defining the maximum “empty” thickness allowed between grid blocks in adjacent grid layers for a non-zero transmissibility to exist between them.			Defined
		ft. 1.0E20	m 1.0E20	cm 1.0E20	
4	PINCHCAL	A character string controlling the calculation of the pinch-out transmissibilities. PINCHCAL can either be set to: 1) TOPBOT results in the pinch-out transmissibility being calculated from the half-cell Z-direction transmissibilities of the active cells on either side of the pinched-out layers. 2) ALL results in the pinch-out transmissibility being calculated from the Z-direction transmissibilities harmonic average of all the cells between the active cells on either side of the pinched-out layers.			TOPBOT

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	PINCHMUL	A character string controlling the calculation of the pinch-out transmissibilities when adjustments have been made by the <i>MULTZ</i> keyword. PINCHMUL can either be set to: 1) TOP results in the pinch-out transmissibility being calculated from the active cell at the top of the pinch-out. 2) <i>ALL</i> results in the pinch-out transmissibility being calculated from the minimum value of the <i>MULTZ</i> of the active cell at the top of the pinch-out and all the inactive cells in the pinch-out vertical column. Note if PINCHCAL has been set equal to <i>ALL</i> then PINCHMUL is reset to TOP, irrespective of the entered value for PINCHMUL.			TOP

Notes:

1) The keyword should contain NRPINC records defining the criteria for each pinch-out region defined with the *PINCHNUM* keyword. NRPINC is the maximum number of *PINCHNUM* regions defined via the *GRIDOPTS* keyword in the *RUNSPEC* section.

2) Each record must be terminated by a “/” there is no keyword terminating “/”.

Table 6.112: PINCHREG Keyword Description

Example

```

--
--          SET PINCH-OUT CRITERIA VIA THE PINCHNUM REGION
--
PINCHREG
--          THRESHOLD   GAP      EMPTY   TRANS
--          THICKNESS   NO GAP   GAP      CALC
--          0.1          1*       1*       1*           / PINCHNUM 01
--          1.0          1*       10       1*           / PINCHNUM 02
--          1.0          NOGAP    20       1*           / PINCHNUM 03

```

The above example sets the default pinch-out criteria for grid blocks defined as region one via the [PINCHNUM](#) array and different values for regions two and three.

6.3.179 PINCHXY – DEFINE PINCH-OUT AREAL OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PINCHXY](#) keyword defines the x-direction and y-direction threshold thickness used to control the generation of Non-Neighbor Connections (“NNCs”) in the x- and y- directions for missing cells in the areal plane.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PINCHTHX	A real number defining the pinch-out threshold width for any cell in the x-direction. NNCs are generated across inactive cells having a width less than PINCHTHX in the x-direction.			Defined
		ft. 0.001	m 0.001	cm 0.001	
2	PINCHTHY	A real number defining the pinch-out threshold width for any cell in the y-direction. NNCs are generated across inactive cells having a width less than PINCHTHY in the y-direction.			Defined
		ft. 0.001	m 0.001	cm 0.001	
Notes: 1) The keyword is terminated by a “/”.					

Table 6.113: PINCHXY Keyword Description

Example

The example below will create NNCs between the cells in the areal plane having cell widths less than 0.01 in either feet or metres in both the x- and y-directions.

```
--
--      SET PINCH-OUT PARAMETERS FOR AREAL PLANE
--
PINCHXY
--      X-DIRC      Y-DIRC
--      THRESHOLD   THRESHOLD
--
--      0.01        0.01
```

6.3.180 PORO – DEFINE THE POROSITY VALUES FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[PORO](#) defines the porosity for all the cells in the model via an array. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PORO	PORO is an array of real positive numbers that are greater than or equal to zero and less than or equal to one that are the porosity values for each cell in the model. Repeat counts may be used, for example 3000*0.15			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.114: PORO Keyword Description

See also the [NTG](#), [PERMX](#), [PERMY](#) and [PERMZ](#) keywords to fully define a grid's properties

Example

```
--
--   DEFINE GRID BLOCK POROSITY DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PORO
    300*0.300
/
```

The above example defines a constant porosity of 0.300 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.181 PYEND – END THE DEFINITION OF A PYINPUT SECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulator's input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords.

[PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword (this keyword) on a separate single line.

There is no data required for this keyword and there is no terminating "/" for this keyword.

Although this keyword is read by OPM Flow and the script processing has been implemented, one should use caution when using this facility as it may result in OPM Flow aborting. This is because the [PYINPUT](#) facility allows the user to implement complex functionality and the implementation is new for the 2020-04 release. Users should therefore use caution when using this facility.

Note

This is an OPM Flow specific keyword for the simulator's scripting facility using the standard Python interpreter; as such it gives more flexibility than the commercial simulator's data editing keywords ([ADD](#), [EQUAL](#), [MULTIPLY](#), etc.), although OPM Flow also supports these keywords as well.

The [PYINPUT](#) facility should be considered experimental as details of the OPM Flow - Python interface might change for future releases. In particular, the current implementation is quite minimal; however, future releases are expected to add more entry points in the simulator's deck class which can be used to manipulate the input deck as the data is loaded. As a user you are encouraged to come with wishes in this regard.

The [PYINPUT](#) facility is very powerful and allows for any piece of Python code to be included and run, including potentially malicious code. The important point is to scrutinize the Python code in between [PYINPUT](#) and [PYEND](#) in a deck you receive from other parties.

See also the [PYACTION](#) keyword in the [SCHEDULE](#) section which is also part of OPM Flow's Python scripting facility, that loads a standard Python script file that can be used to define a series of conditions and actions as the simulation proceeds through time.

Example

The example shows how to construct the [DX](#) variable in the [GRID](#) section and to add the resulting [DX](#) array as part of the input deck.

```
--
--      START OF PYINPUT SECTION
--
PYINPUT
#
# Import Numpy Model
#
import numpy as np
#
# Define DX and Get the Input Decks Unit Systems
#
dx = np.array([100.0, 100.0, 100.0, 100.0])
active_unit_system = context.deck.active_unit_system()
default_unit_system = context.deck.default_unit_system()
#
# Set DX in the Input Deck
#
kw = context.DeckKeyword( context.parser['DX'], dx, active_unit_system,
default_unit_system )
context.deck.add(kw)

PYEND
```

The active Parser objects are accessible as `context.parser` and the active Deck object is available as `context.deck`.

6.3.182 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **PYINPUT** and **PYEND** keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulator’s input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator.

PYINPUT declares the start of a **PYINPUT** Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line. A **PYINPUT** Definition Section is terminated by a **PYEND** keyword on a separate single line.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Although this keyword is read by OPM Flow and the script processing has been implemented, one should use caution when using this facility as it may result in OPM Flow aborting. This is because the **PYINPUT** facility allows the user to implement complex functionality and the implementation is new for the 2020-04 release. Users should therefore use caution when using this facility.

Note

This is an OPM Flow specific keyword for the simulator’s scripting facility using the standard Python interpreter; as such it gives more flexibility than the commercial simulator’s data editing keywords (**ADD**, **EQUALS**, **MULTIPLY**, etc.), although OPM Flow also supports these keywords as well.

The **PYINPUT** facility should be considered experimental as details of the OPM Flow - Python interface might change for future releases. In particular, the current implementation is quite minimal; however, future releases are expected to add more entry points in the simulator’s deck class which can be used to manipulate the input deck as the data is loaded. As a user you are encouraged to come with wishes in this regard.

The **PYINPUT** facility is very powerful and allows for any piece of Python code to be included and run, including potentially malicious code. The important point is to scrutinize the Python code in between **PYINPUT** and **PYEND** in a deck you receive from other parties.

No.	Name	Description	Default
PYINPUT		PYINPUT declares the start of a PYINPUT Definition Section. This is then followed by any number Python commands.	Not Applicable
I-I	PYTHON	A series of standard Python commands with one line per command. The active Parser objects are accessible as context.parser and the active Deck object is available as context.deck.	
PYEND		PYEND declares the end of a PYINPUT Definition Section. The Python code between PYINPUT and PYEND is read and executed, and the simulator then returns to reading the normal simulation input deck.	
Notes:			
I) There is no terminating “/” for this keyword, instead the PYEND keyword terminates the keyword.			

Table 6.115: PYINPUT Keyword Description

The `PYINPUT/PYEND` set of keywords is a result of combining two programming languages, the interactive Python interpreter and OPM Flow’s source code language C++. When combining two languages one extends and embeds one into the other. When extending Python with C++ the functionality implemented in C++ is made available to Python applications, when embedding Python in C++ one can call Python functions from within C++. The `PYINPUT/PYEND` set of keywords is based on embedding a Python interpreter in the C++ OPM Flow simulator, but the Python code actually runs as part of the `PYINPUT` keyword is based on wrapping C++ objects in Python, that is extending Python.

The Python code in between the `PYINPUT` and `PYEND` keywords are imported during processing of the input deck and as such this implies that basic Python syntax checking is performed during reading the Python script.

See also the `PYACTION` keyword in the `SCHEDULE` section which is also part of OPM Flow’s Python scripting facility, that loads a standard Python script file that can be used to define a series of conditions and actions as the simulation proceeds through time.

Example

The example shows how to construct the `DX` variable in the `GRID` section and to add the resulting `DX` array as part of the input deck.

```
--
--      START OF PYINPUT SECTION
--
PYINPUT
#
# Import Numpy Model
#
import numpy as np
#
# Define DX and Get the Input Decks Unit Systems
#
dx = np.array([100.0, 100.0, 100.0, 100.0])
active_unit_system = context.deck.active_unit_system()
default_unit_system = context.deck.default_unit_system()
#
# Set DX in the Input Deck
#
kw = context.DeckKeyword( context.parser['DX'], dx, active_unit_system,
default_unit_system )
context.deck.add(kw)

PYEND
```

The active Parser objects is accessible as `context.parser` and the active Deck object is available as `context.deck`.

6.3.183
QMOBIL
– ACTIVATE OR DEACTIVATE LGR END-POINT MOBILITY CORRECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [QMOBIL](#) keyword activates or deactivates the end-point mobility correction for Local Grid Refinements (“LGR”), for when LGRs have been activated for the input deck using the [LGR](#) keyword in the [RUNSPEC](#) section. [QMOBIL](#) should be placed in between the [LGR](#) definition keywords [CARFIN](#), or RADIN (or RAFDIN4) and the [ENDFIN](#) keyword in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.184 RADFIN – DEFINE A RADIAL LOCAL GRID REFINEMENT WITH ONE COLUMN

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines a radial local grid refinement using one columns Local grid refinement is currently not supported by OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.185 RADFIN4 – DEFINE A RADIAL LOCAL GRID REFINEMENT WITH FOUR COLUMNS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines a radial local grid refinement using four columns. Local grid refinement is currently not supported by OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.186 REFINE – START THE DEFINITION OF A LOCAL GRID REFINEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [REFINE](#) keyword defines the start of a Cartesian or radial Local Grid Refinement (“LGR”) definition that sets the properties of the selected [LGR](#). The keyword is then followed by the property keywords associated with the section where the keyword is being invoked. For example, if the [REFINE](#) keyword is used in the [GRID](#) section then most of the keywords in that section can be used to set the grid properties for the [LGR](#).

The [ENDFIN](#) keyword is used to terminate the [LGR](#) definition.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.187 RESVNUM – DEFINE RESERVOIR COORDINATE DATA SET

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [RESVNUM](#) keyword is used to define the start of a reservoir coordinate data set and stipulates the reservoir number for the data set. The keyword is used in conjunction with the [COORD](#) keyword in the [GRID](#) section, that specifies a set of coordinate lines or pillars for a reservoir grid via an array. Note that the [COORD](#) keyword should immediately follow the [RESVNUM](#) keyword.

Although the keyword is processed by OPM Flow, the keyword is effectively ignored as only the default value of one is supported.

No.	Name	Description	Default
1	RESVNUM	A positive integer values that defines the reservoir coordinate data set, or the independent reservoir, for which the subsequent COORD data is to be associated with. RESVNUM should be less than or equal to NUMRES on the NUMRES keyword in the RUNSPEC section. OPM Flow currently only accepts a single data set, that is the default value of one.	1

Notes:

- 1) The keyword is terminated by a “/”.

Table 6.116: RESVNUM Keyword Description

See the [NUMRES](#) keyword in the [RUNSPEC](#) section that defines the number of reservoir grids ([COORD](#) data sets) that the simulator should process.

The facility is useful to combine two separate reservoir grids into one model in the simulator.

Example

```
--
--      NUMRES
--      NUMBER
RESVNUM
    1
/

--
-- SPECIFY VERTICAL COORDINATE LINES FOR A REGULAR 3 x 2 GRID
--(DX = 100 and DY = 200)
--
-- X1      Y1      Z1      X2      Y2      Z2
-- ----      ---      ----      ---      ---      ----
COORD
    0          0      1000          0          0      5000
    100         0      1000         100         0      5000
    200         0      1000         200         0      5000
    300         0      1000         300         0      5000
    0        200      1000          0        200      5000
    100       200      1000         100       200      5000
    200       200      1000         200       200      5000
    300       200      1000         300       200      5000
    0        400      1000          0        400      5000
    100       400      1000         100       400      5000
    200       400      1000         200       400      5000
    300       400      1000         300       400      5000
/

--
--      NUMRES
--      NUMBER
RESVNUM
    2
/

--
-- SPECIFY VERTICAL COORDINATE LINES FOR A REGULAR 3 x 2 GRID
--(DX = 100 and DY = 200)
--
-- X1      Y1      Z1      X2      Y2      Z2
-- ----      ---      ----      ---      ---      ----
COORD
    0          0      1000          0          0      5000
    100         0      1000         100         0      5000
    200         0      1000         200         0      5000
    300         0      1000         300         0      5000
    0        200      1000          0        200      5000
    100       200      1000         100       200      5000
    200       200      1000         200       200      5000
    300       200      1000         300       200      5000
    0        400      1000          0        400      5000
    100       400      1000         100       400      5000
    200       400      1000         200       400      5000
    300       400      1000         300       400      5000
/
```

6.3.188 ROCKFRAC – DEFINE THE ROCK VOLUME TO BULK VOLUME FRACTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[ROCKFRAC](#) defines the rock volume to bulk volume fraction for all the cells. The keyword can be used with all grid types. Rock volume of a grid block is calculated by multiply a cell's bulk volume by it's [ROCKFRAC](#) volume. A cell's rock volume is used in the Coal option to calculate the adsorbed gas in the rock (coal), as well as the Thermal and Temp options to calculate the energy is stored in the rock.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	ROCKFRAC	ROCKFRAC is an array of real numbers greater than or equal to zero and less than or equal to one, that are assigned the rock volume to bulk volume fraction values for each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 200*0.850.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Setting a cells ROCKFRAC value to zero will make the cell inactive, similar to setting the cells ACTNUM property to zero.
- 2) The keyword is terminated by a “/”.

Table 6.117: [NTG](#) Keyword Description

See also the [PORO](#), [PERMX](#), [PERMY](#), [PERMZ](#) and [NTG](#) keywords to fully define a grid's properties.

Example

```
--
--      DEFINE GRID ROCKFRAC DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
ROCKFRAC
    100*1.000    100*0.850    100*0.500    /
```

The above example defines a constant [ROCKFRAC](#) of 1.00 for the first 100 cells, then 0.85 for the second 100 hundred cells, and finally 0.500 for the last 100 cell, for the 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.189 RPTGRID – DEFINE GRID SECTION REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the data in the [GRID](#) section that is to be printed to the output print file in human readable format. The keyword has two distinct forms, the first of which consists of the keyword followed by a series of integers on the next line indicating the data to be printed (see the first example). This is the original format in the commercial simulator and was subsequently superseded by the second format. The second format consists of the keyword followed by a series of character strings that indicate the data to be printed. In most cases the character string is the keyword used to load the data in the OPM Flow input deck, for example [PORO](#) for the porosity array. It is anticipated that OPM Flow will eventually support the functionality of the second format only, the first format although recognized will be completely ignored.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
1	ALLNCC	Print all the non-neighbor connections.	N/A
2	COORD	Print the coordinate lines.	N/A
3	COORDYS	Print the coordinate systems.	N/A
4	DEPTH	Print grid cells center depths.	N/A
....			N/A

Notes:

- 1) The keyword is terminated by a “/”.

Table 6.118: RPTGRID Keyword Description

Note

This keyword has the potential to produce very large print files that some text editors may have difficulty loading, coupled with the fact that reviewing the data in this format is very cumbersome.

A more efficient solution is to load the *.INIT file into OPM ResInsight to view the data graphically, this also has the benefit of being able to filter the grid based on I, J, K ranges and grid properties.

Examples

The first example shows the original format of this keyword; although the keyword and format are recognized by OPM Flow, the format is ignored and is unlikely to be implemented in the simulator.

```
--  
--      DEFINE GRID SECTION REPORT OPTION (ORIGINAL FORMAT)  
--  
RPTGRID      1      2*0      1      3*1      /
```

The next example shows the second format of the keyword which may be supported in a future release of OPM Flow.

```
--      DEFINE GRID SECTION REPORT OPTIONS  
--  
RPTGRID      DX      DY      DZ      DEPTH      PORO      PERMX      /
```

6.3.190 RPTGRIDL – DEFINE GRID SECTION REPORTING FOR LGRs

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the data in the [GRID](#) section that is to be printed to the output print file in human readable format for Local Grid Refinements (“LGRs”), for when LGRs have been activated for the input deck using the [LGR](#) keyword in the [RUNSPEC](#) section.

The keyword has two distinct forms, the first of which consists of the keyword followed by a series of integers on the next line indicating the data to be printed (see the first example). This is the original format in the commercial simulator and was subsequently superseded by the second format. The second format consists of the keyword followed by a series of character strings that indicate the data to be printed. In most cases the character string is the keyword used to load the data in the OPM Flow input deck, for example [PORO](#) for the porosity array. It is anticipated that OPM Flow will eventually support the functionality of the second format only, the first format although recognized will be completely ignored.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
1	ALLNCC	Print all the non-neighbor connections.	N/A
2	COORD	Print the coordinate lines.	N/A
3	COORDYS	Print the coordinate systems.	N/A
4	DEPTH	Print grid cells center depths.	N/A
....			N/A
24	ALLNNC	ALLNNC is a defined positive integer that specifies the type of Non-Neighbor Connections (“NNC”) to be printed, and should be set to one of the follow: 1) To print the NNCs within the LGRs, and the connections between the local and host cells to the print file (*.PRT). 2) To print the NNCs within the LGRs, and the connections between the local and host cells to the print (*.PRT) and debug files (*.DBG). 3) Same as (2) but the data in the debug file (*.DBG) is written out in an alternative format.	N/A
....			N/A
57	EXTHOST	EXTHOSTS outputs host cells for Perpendicular Bisector (“PEBI”) ¹³¹ and ¹³² LGRs.	
....			N/A
Notes:			

¹³¹ Heinemann, Z.E. and Brand, C.W. 1988. Gridding Techniques in Reservoir Simulation. Proc., First Intl. Forum on Reservoir Simulation, Alpbach, Austria, 339.

¹³² Heinemann, Z.E., Brand, C.W., Munka, M. et al. 1991. Modeling Reservoir Geometry With Irregular Grids. SPE Res Eng 6 (2): 225–232. SPE-18412-PA. <http://dx.doi.org/10.2118/18412-PA>

No.	Name	Description	Default
1)		The keyword is terminated by a "/".	

Table 6.119: RPTGRIDL Keyword Description

Note

This keyword has the potential to produce very large print files that some text editors may have difficulty loading, coupled with the fact that reviewing the data in this format is very cumbersome.

A more efficient solution is to load the *.INIT file into OPM ResInsight to view the data graphically, this also has the benefit of being able to filter the grid based on I,J, K ranges and grid properties.

Examples

The first example shows the original format of this keyword; although the keyword and format are recognized by OPM Flow, the format is ignored and is unlikely to be implemented in the simulator.

```
--
--      DEFINE LGR GRID SECTION REPORT OPTION (ORIGINAL FORMAT)
--
RPTGRIDL
      1          2*0      1          3*1          /
```

The next example shows the second format of the keyword which may be supported in a future release of OPM Flow.

```
--      DEFINE LGR GRID SECTION REPORT OPTIONS
--
RPTGRIDL
      DX          DY          DZ          DEPTH      PORO      PERMX          /
```

6.3.191 RPTINIT – DEFINE OUTPUT TO THE INIT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the data in the [GRID](#) and [EDIT](#) sections that is to be written out to the [INIT](#) file (*.INIT or *.FINIT). The format consists of the keyword followed by a series of character strings that indicate the data to be written. In most cases the character string is the keyword used to load the data into the OPM Flow input deck, for example [PORO](#) for the porosity array in the [GRID](#) section. In addition, values either read or calculated by the simulator in the [EDIT](#) section can also be written to the [INIT](#) file. Again the keyword or property name is used as the mnemonic for the character string, for example the [PORV](#), [TRANX](#) keywords etc. If the [RPTINIT](#) keyword is not used in the input deck then a default set of data array are written to the file, in this case the actual data written is dependent on the model's configuration and the options being used.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

6.3.192 RPTISOL – ACTIVATE ISOLATED RESERVOIR NUMBER REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *RPTISOL* keyword activates the isolated reservoir report that generates an array of isolated region numbers that is printed in the debug file (*.DBG). The main purpose of this facility is to use the generated array as input to the *ISOLNUM* keyword in the *GRID* section in conjunction with the Independent Reservoir Regions option. If the model can be divided into isolated reservoirs then the individual reservoirs may be solved independently, resulting in increased computational efficient, compared with solving the model as a whole.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      ACTIVATE ISOLATED RESERVOIR NUMBER REPORTING
--
RPTISOL
```

The above example activates the isolated reservoir report that generates an array of isolated region numbers to the debug file (*.DBG).

6.3.193 SIGMA – DUAL POROSITY MATRIX TO FRACTURE SIGMA (ALL CELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SIGMA](#) keyword defines the dual porosity matrix to fracture transmissibility multiplier, sigma, that is applied to all cells, for when the Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section. Sigma (σ) takes into account the matrix-fracture interface area per unit volume and was defined by Kazemi et al¹³³ to be:

$$\sigma = 4 \left(\frac{1}{l_x^2} + \frac{1}{l_y^2} + \frac{1}{l_z^2} \right) \quad (6.15)$$

Where l_x , l_y and l_z are not the grid block dimensions in the model in the respective directions, but the dimensions of the blocks of the matrix material. In practice, σ is used as a tuning parameter in dual porosity runs to match reservoir and well performance.

See also the [SIGMAV](#) keyword in the [GRID](#) section that supplies the sigma values on an individual cells basis.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

¹³³ Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. "Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs," paper SPE 5719, Society of Petroleum Engineers Journal (1976) 16, No. 6, 317-326.

6.3.194 SIGMAGD – DUAL POROSITY MATRIX TO FRACTURE SIGMA FOR GRAVITY DRAINAGE (ALL CELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SIGMAGD](#) keyword defines the dual porosity matrix to fracture transmissibility multiplier, sigma, that is applied to all cells, for when the Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section. In addition, the [GRAVDR](#) keyword in the [RUNSPEC](#) section should be used to enable the Gravity Drainage model for the run. Sigma (σ) takes into account the matrix-fracture interface area per unit volume and was defined by Kazemi et al¹³⁴ to be:

$$\sigma = 4 \left(\frac{1}{l_x^2} + \frac{1}{l_y^2} + \frac{1}{l_z^2} \right) \quad (6.16)$$

Where l_x , l_y and l_z are not the grid block dimensions in the model in the respective directions, but the dimensions of the blocks of the matrix material. In practice, σ is used as a tuning parameter in dual porosity runs to match reservoir and well performance.

Note that [SIGMAGD](#) keyword data is used for areas being swept by gas and the [SIGMA](#) keyword data is used when the area is being invaded by water. See also the [SIGMAGDV](#) keyword in the [GRID](#) section that supplies the sigma values on an individual cells basis

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

¹³⁴ Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. "Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs," paper SPE 5719, Society of Petroleum Engineers Journal (1976) 16, No. 6, 317-326.

6.3.195 SIGMAGDV – DUAL POROSITY MATRIX TO FRACTURE SIGMA GRAVITY DRAINAGE (INDIVIDUAL CELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SIGMAGD](#) keyword defines the dual porosity matrix to fracture transmissibility multiplier, sigma, that is applied to individual cells, for when the Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section. In addition, the [GRAVDR](#) keyword in the [RUNSPEC](#) section should be used to enable the Gravity Drainage model for the run. Sigma (σ) takes into account the matrix-fracture interface area per unit volume and was defined by Kazemi et al¹³⁵ to be:

$$\sigma = 4 \left(\frac{1}{l_x^2} + \frac{1}{l_y^2} + \frac{1}{l_z^2} \right) \quad (6.17)$$

Where l_x , l_y and l_z are not the grid block dimensions in the model in the respective directions, but the dimensions of the blocks of the matrix material. In practice, σ is used as a tuning parameter in dual porosity runs to match reservoir and well performance.

Note that SIGMAGDV keyword data is used for areas being swept by gas and the [SIGMA](#) keyword data is used when the area is being invaded by water. See also the [SIGMAGD](#) keyword in the [GRID](#) section that supplies a constant sigma value for all cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

¹³⁵ Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. "Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs," paper SPE 5719, Society of Petroleum Engineers Journal (1976) 16, No. 6, 317-326.

6.3.196 SIGMAV – DUAL POROSITY MATRIX TO FRACTURE SIGMA (INDIVIDUAL CELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SIGMAV](#) keyword defines a dual porosity matrix to fracture multiplier, sigma, that is applied to individual cells, for when the Dual Porosity model has been invoked by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section. Sigma (σ) takes into account the matrix-fracture interface area per unit volume and was defined by Kazemi et al¹³⁶ to be:

$$\sigma = 4 \left(\frac{1}{l_x^2} + \frac{1}{l_y^2} + \frac{1}{l_z^2} \right) \quad (6.18)$$

Where l_x , l_y and l_z are not the grid block dimensions in the model in the respective directions, but the dimensions of the blocks of the matrix material. In practice, σ is used as a tuning parameter in dual porosity runs to match reservoir and well performance.

See also the [SIGMA](#) keyword in the [GRID](#) section that supplies a constant sigma to all cells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

¹³⁶ Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. "Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs," paper SPE 5719, Society of Petroleum Engineers Journal (1976) 16, No. 6, 317-326.

6.3.197 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

6.3.198 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

6.3.199 SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

6.3.200 SMULTX – MULTIPLY CELL TRANSMISSIBILITY IN THE +X DIRECTION (AUTO-REFINEMENT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SMULTX multiplies the transmissibility between two cell faces in the +X direction between cells in a host base grid and the connecting auto-refined grid cells, via an array, that is the keyword sets the transmissibility multiplier of block (Ihost, Jhost, Khost) in the host base grid, multiplies the transmissibility all the cells (Iauto, Jauto, Kauto) and (I+Iauto, Jauto, Kauto) in the auto-refinement grid. The Auto Refinement option must be enabled to use this keyword via the **AUTOREF** keyword in the **RUNSPEC** section.

An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **SMULTX** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	SMULTX+	SMULTX+ is an array of real positive numbers assigning the transmissibility multipliers in the +X direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.120: SMULTX Keyword Description

See also the **MULTX**, **MULTY** and **MULTZ** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   3                / DEFINE BOX AREA
--
--      SET SMULTX+ TRANSMISSIBILITY MULTIPLIERS
--
SMULTX
      18*0.300                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.201 SMULTY – MULTIPLY CELL TRANSMISSIBILITY IN THE +Y DIRECTION (AUTO-REFINEMENT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SMULTY multiplies the transmissibility between two cell faces in the +Y direction between cells in a host base grid and the connecting auto-refined grid cells, via an array, that is the keyword sets the transmissibility multiplier of block (Ihost, Jhost, Khost) in the host base grid, multiplies the transmissibility all the cells (Iauto, Jauto, Kauto) and (Iauto, J+Iauto, Kauto) in the auto-refinement grid. The Auto Refinement option must be enabled to use this keyword via the **AUTOREF** keyword in the **RUNSPEC** section.

An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **SMULTY** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	SMULTY+	SMULTY+ is an array of real positive numbers assigning the transmissibility multipliers in the +X direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) The keyword is terminated by a “/”.			

Table 6.121: SMULTY Keyword Description

See also the **MULTX**, **MULTY** and **MULTZ** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   3                / DEFINE BOX AREA
--
--      SET SMULTY+ TRANSMISSIBILITY MULTIPLIERS
--
SMULTY      18*0.300                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.202 SMULTZ – MULTIPLY CELL TRANSMISSIBILITY IN THE +Z DIRECTION (AUTO-REFINEMENT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SMULTZ multiplies the transmissibility between two cell faces in the +Z direction between cells in a host base grid and the connecting auto-refined grid cells, via an array, that is the keyword sets the transmissibility multiplier of block (Ihost, Jhost, Khost) in the host base grid, multiplies the transmissibility all the cells (Iauto, Jauto, Kauto) and (Iauto, Jauto, K+Iauto) in the auto-refinement grid. The Auto Refinement option must be enabled to use this keyword via the **AUTOREF** keyword in the **RUNSPEC** section.

An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **SMULTZ** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	SMULTZ+	SMULTZ+ is an array of real positive numbers assigning the transmissibility multipliers in the +X direction to each cell face in the model. Repeat counts may be used, for example 20*100.0.	1.0

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 6.122: SMULTX Keyword Description

See also the **MULTX**, **MULTY** and **MULTZ** keywords for scaling transmissible between grid cells.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   3                / DEFINE BOX AREA
--
--      SET SMULTZ+ TRANSMISSIBILITY MULTIPLIERS
--
SMULTZ   18*0.300                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The above example defines a 0.3 scaling multiplier for the 18 cells defined by the preceding **BOX** statement. The **ENDBOX** keyword resets the input box to the full grid.

6.3.203 SOLVDIRS – DEFINE LINEAR SOLVER PRINCIPAL DIRECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SOLVDIRS](#) keyword defines the linear solver principal directions, which should be set to XY,XZ,YX,YZ, ZX, or ZY. The default direction is based on the direction of the highest transmissibility and [SOLVDIRS](#) allows for over writing the default direction for when linear convergence of the equations are problematic.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

6.3.204 SOLVNUM – DEFINE PEBI GRID CORRESPONDENCE TO SOLVER ORDER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SOLVNUM](#) defines the unstructured Perpendicular Bisector (“PEBI”) ¹³⁷ and ¹³⁸ grid correspondence to the nested factorization solver order, for when the grid has been entered as a [PEBI](#) list. This keyword is generated by an external pre-processing program for generating simulation grids.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

¹³⁷

Heinemann, Z.E. and Brand, C.W. 1988. Gridding Techniques in Reservoir Simulation. Proc., First Intl. Forum on Reservoir Simulation, Alpbach, Austria, 339.

¹³⁸

Heinemann, Z.E., Brand, C.W., Munka, M. et al. 1991. Modeling Reservoir Geometry With Irregular Grids. SPE Res Eng 6 (2): 225–232. SPE-18412-PA. <http://dx.doi.org/10.2118/18412-PA>

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

6.3.205 SPECGRID – DEFINE THE DIMENSIONS OF A CORNER-POINT GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SPECGRID](#) defines the dimensions of corner-point and radial grids in the x, y, and z directions as well as the number of reservoirs, where each reservoir has its own set of corner-point geometry data.

The keyword can only be used with Irregular Corner-Point Grids and Radial Grids.

No.	Name	Description	Default
1	NDIVIX	A positive integer value that defines the number of cells in the X or R direction	1
2	NDIVIY	A positive integer value that defines the number of cells in the Y or THETA direction	1
3	NDIVZ	A positive integer value that defines the number of cells in the Z direction	1
4	NUMRES	A positive integer values that defines number of coordinate data sets, or independent reservoirs in the model. OPM Flow currently only accepts a single data set, that is the default value of one.	1
5	TYPE	A character string set to either T or F that defines the type of grid to be defined by subsequent keywords: 1) T = Radial grid with radial coordinates 2) F = Cartesian grid Only the default option F is supported by OPM Flow.	F

Notes:

- 1) The dimensions are also entered on the [DIMENS](#) keyword in the [RUNSPEC](#) section and the two sets of numbers should be consistent.
- 2) The keyword is terminated by a “/”.

Table 6.123: SPECGRID Keyword Description

See also the [COORD](#), [COORDSYS](#) and [ZCORN](#) keywords to fully define an Irregular Corner-Point Grid.

Example

```
--
--      MAX      MAX      MAX      MAX      GRID
--      NDIVIX   NDIVIY   NDIVIZ   NUMRES   TYPE
SPECGRID
      46        112        22        1        F                                     /
```

The above example defines a 46 x 112 x 22 grid with one set of irregular corner-point data.

6.3.206 THCGAS – DEFINE GAS PHASE THERMAL CONDUCTIVITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **THCGAS** keyword defines the gas phase thermal conductivity for when the thermal calculation is activated by the **THERMAL** keyword in the **RUNSPEC** section, and should be used in conjunction with **THCROCK** keyword in the **GRID** section.

This keyword can only be used if the thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	THCGAS	THCGAS is an array of real positive numbers that define the thermal conductivity of the gas phase in each grid block. Repeat counts may be used, for example 3000*20.0			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 6.124: THCGAS Keyword Description

Note that there two ways to define the rock and in situ fluids thermal conductivity:

- 1) Either by using the **THCONR** keyword to define the combined rock and fluid conductivity, and optionally the **THCONSF** keyword in the **GRID** section, or
- 2) by specifying the rock and fluid conductivities individually using the **THCROCK**, **THCOIL**, **THCGAS**, and **THCWATER** keywords in the **GRID** section.

Hence, the **THCROCK** and **THCONR** keywords are mutually exclusive.

Here, the **THCGAS** keyword is used in conjunction with the other thermal conductivity arrays to calculate the porosity weighted thermal conductivity of a grid block using:

$$\text{Average Thermal Conductivity} = \frac{\text{PORO} \times (\text{THCOIL} + \text{THCGAS} + \text{THCWATER} + \text{THCSOLID})}{\text{NUMBER OF PHASES IN THE MODEL}} \times (1 - \text{PORO}) \times \text{THCROCK} \quad (6.19)$$

See also the **THCOIL**, and **THCWATER**, and **THCROCK** keywords in the **GRID** section. The commercial compositional simulator's THCSOLID keyword is not supported or required by OPM Flow.

Example

```
--
--      DEFINE GRID BLOCK GAS PHASE THERMAL CONDUCTIVITY
-      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
THCGAS      300*20.0      /
```

The above example defines the gas phase thermal conductivity of 20.0 for each cell in the 300 grid block model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.207 THCOIL – DEFINE OIL PHASE THERMAL CONDUCTIVITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [THCOIL](#) keyword defines the oil phase thermal conductivity for when the thermal calculation is activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section, and should be used in conjunction with [THCROCK](#) keyword in the [GRID](#) section.

This keyword can only be used if the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	THCOIL	THCOIL is an array of real positive numbers that define the thermal conductivity of the oil phase in each grid block. Repeat counts may be used, for example 3000*20.0			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.125: THCOIL Keyword Description

Note that there two ways to define the rock and in situ fluids thermal conductivity:

- 1) Either by using the [THCONR](#) keyword to define the combined rock and fluid conductivity, and optionally the [THCONSF](#) keyword in the [GRID](#) section, or
- 2) by specifying the rock and fluid conductivities individually using the [THCROCK](#), [THCOIL](#), [THCGAS](#), and [THCWATER](#) keywords in the [GRID](#) section.

Hence, the [THCROCK](#) and [THCONR](#) keywords are mutually exclusive.

Here, the [THCOIL](#) keyword is used in conjunction with the other thermal conductivity arrays to calculate the porosity weighted thermal conductivity of a grid block using:

$$\text{Average Thermal Conductivity} = \frac{\text{PORO} \times (\text{THCOIL} + \text{THCGAS} + \text{THCWATER} + \text{THCSOLID})}{\text{NUMBER OF PHASES IN THE MODEL}} \times (1 - \text{PORO}) \times \text{THCROCK} \quad (6.20)$$

See also the [THCGAS](#), and [THCWATER](#), and [THCROCK](#) keywords in the [GRID](#) section. The commercial compositional simulator's THCSOLID keyword is not supported or required by OPM Flow.

Example

```
--
--      DEFINE GRID BLOCK OIL PHASE THERMAL CONDUCTIVITY
-      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
THCOIL      300*20.0      /
```

The above example defines the oil phase thermal conductivity of 20.0 for each cell in the 300 grid block model, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.208 THCONR – DEFINE ROCK AND FLUID THERMAL CONDUCTIVITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **THCONR** keyword defines the reservoir rock plus fluid thermal conductivity for all cells for when the thermal calculation is activated by the **THERMAL** keywords in the **RUNSPEC** section.

This keyword can only be used if the thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	THCONR	THCONR is an array of real positive numbers that define the combined rock and fluid conductivity of a grid block. Repeat counts may be used, for example 3000*25.0			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.

Table 6.126: THCONR Keyword Description

Note that there two ways to define the rock and in situ fluids thermal conductivity:

- 1) Either by using the **THCONR** keyword to define the combined rock and fluid conductivity, and optionally the **THCONSF** keyword in the **GRID** section, or
- 2) by specifying the rock and fluid conductivities individually using the **THCROCK**, **THCOIL**, **THCGAS**, and **THCWATER** keywords in the **GRID** section.

Hence, the **THCROCK** and **THCONR** keywords are mutually exclusive.

Example

```
--
--      DEFINE GRID BLOCK ROCK-FLUID THERMAL CONDUCTIVITY
--      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
THCONR
      300*25.0
```

The above example defines the combined rock and fluid thermal conductivity of 25.0 for each cell in the 300 grid block model, as defined by the **DIMENS** keyword in the **RUNSPEC** section.

6.3.209 THCONSF – DEFINE GAS SATURATION DEPENDENT THERMAL CONDUCTIVITY SCALING FACTOR FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **THCONSF** keyword defines a gas saturation dependent scaling factor to the fluid and reservoir rock thermal conductivities entered via the **THCONR** keyword in the **GRID** section, for when the thermal calculation is activated by the **THERMAL** keyword in the **RUNSPEC**.

This keyword can only be used if the thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	THCONSF	THCONSF is an array of real positive numbers, greater than zero and less than or equal to one, that define the gas saturation dependent scaling factor that is applied to the THCONR data, entered via the THCONR keyword, to adjust the thermal conductivity of the reservoir cells in each grid block. Repeat counts may be used, for example 3000*0.15			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.127: **THCROCK** Keyword Description

Note that there two ways to define the rock and in situ fluids thermal conductivity:

- 1) Either by using the **THCONR** keyword to define the combined rock and fluid conductivity, and optionally the **THCONSF** keyword in the **GRID** section, or
- 2) by specifying the rock and fluid conductivities individually using the **THCROCK**, **THCOIL**, **THCGAS**, and **THCWATER** keywords in the **GRID** section.

Hence, the **THCROCK** and **THCONR** keywords are mutually exclusive.

Here, the **THCONSF** keyword defines a scaling factor which is a function of the gas saturation that scales a cells total thermal conductivity (reservoir fluids plus reservoir rock) entered via the **THCONR** keyword in the **GRID** section. This combination of keywords, **THCONSF** and **THCONR** implies that the oil and water phase thermal conductivities are saturation independent with respect to the liquid phase, and that only the gas saturation influences a cell's thermal conductivity as entered via the **THCONR** keyword.

Thus, **THCONSF** scales the **THCONR** values via a multiplier Ω , by:

$$\Omega_{i,j,k} = (1 - \text{THCONSF} \times \text{Gas Saturation})_{i,j,k} \quad (6.21)$$

See also the [THCGAS](#), [THCOIL](#), [THCWATER](#) and [THROCK](#) keywords in the [GRID](#) section, for an alternative way to enter the thermal conductivity properties. However, the [THCONSF](#) keyword cannot be used with the [THCGAS](#), [THCOIL](#), [THCWATER](#) and [THCROCK](#) keywords. Secondly, the commercial compositional simulator's [THCSOLID](#) keyword is not supported or required by OPM Flow.

Example

```
--  
--      DEFINE GRID SGAS DEPENDENT SCALING FACTOR FOR THE THCONR ARRAY  
--      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)  
--      (OPM FLOW THERMAL OPTION ONLY)  
--  
THCONSF      300*0.12      /
```

The above example defines the gas saturation thermal conductivity scaling factor to be applied to the [THCONR](#) to be 0.12 for all 300 cells in the model, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.210 THCROCK – DEFINE RESERVOIR ROCK THERMAL CONDUCTIVITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **THCROCK** keyword defines the reservoir rock thermal conductivity for when the thermal calculation is activated by the **THERMAL** keyword in the **RUNSPEC** section.

This keyword can only be used if the thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	THCROCK	THCROCK is an array of real positive numbers that define the thermal conductivity of the reservoir rock in each grid block. Repeat counts may be used, for example 3000*20.0			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 6.128: THCROCK Keyword Description

Note that there two ways to define the rock and in situ fluids thermal conductivity:

- 1) Either by using the **THCONR** keyword to define the combined rock and fluid conductivity, and optionally the **THCONSF** keyword in the **GRID** section, or
- 2) by specifying the rock and fluid conductivities individually using the **THCROCK**, **THCOIL**, **THCGAS**, and **THCWATER** keywords in the **GRID** section.

Hence, the **THCROCK** and **THCONR** keywords are mutually exclusive.

Here, the **THCROCK** keyword is used in conjunction with the other thermal conductivity arrays to calculate the porosity weighted thermal conductivity of a grid block using:

$$\text{Average Thermal Conductivity} = \frac{\text{PORO} \times (\text{THCOIL} + \text{THCGAS} + \text{THCWATER} + \text{THCSOLID})}{\text{NUMBER OF PHASES IN THE MODEL}} \times (1 - \text{PORO}) \times \text{THCROCK} \quad (6.22)$$

See also the **THCOIL**, and **THCGAS**, and **THCWATER** keywords in the **GRID** section. The commercial compositional simulator's THCSOLID keyword is not supported or required by OPM Flow.

Example

```
--
--      DEFINE GRID BLOCK RESERVOIR ROCK THERMAL CONDUCTIVITY
-      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
THCROCK      300*20.0      /
```

The above example defines the reservoir rock thermal conductivity of 20.0 for each cell in the 300 grid block model, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.211 THCWATER – DEFINE WATER PHASE THERMAL CONDUCTIVITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **THCWATER** keyword defines the water phase thermal conductivity for when the thermal calculation is activated by the **THERMAL** keyword in the **RUNSPEC** section, and should be used in conjunction with **THCROCK** keyword in the **GRID** section.

This keyword can only be used if the thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	THCWATER	THCWATER is an array of real positive numbers that define the thermal conductivity of the water phase in each grid block. Repeat counts may be used, for example 3000*20.0			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 6.129:THCWATER Keyword Description

Note that there two ways to define the rock and in situ fluids thermal conductivity:

- 1) Either by using the **THCONR** keyword to define the combined rock and fluid conductivity, and optionally the **THCONSF** keyword in the **GRID** section, or
- 2) by specifying the rock and fluid conductivities individually using the **THCROCK**, **THCOIL**, **THCGAS**, and **THCWATER** keywords in the **GRID** section.

Hence, the **THCROCK** and **THCONR** keywords are mutually exclusive.

Here, the **THCWATER** keyword is used in conjunction with the other thermal conductivity arrays to calculate the porosity weighted thermal conductivity of a grid block using:

$$\text{Average Thermal Conductivity} = \frac{\text{PORO} \times (\text{THCOIL} + \text{THCGAS} + \text{THCWATER} + \text{THCSOLID})}{\text{NUMBER OF PHASES IN THE MODEL}} \times (1 - \text{PORO}) \times \text{THCROCK} \quad (6.23)$$

See also the **THCGAS**, and **THCOIL**, and **THCROCK** keywords in the **GRID** section. The commercial compositional simulator's THCSOLID keyword is not supported or required by OPM Flow

Example

```
--
--      DEFINE GRID BLOCK WATER PHASE THERMAL CONDUCTIVITY
-      FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
THCWATER
      300*20.0                                     /
```

The above example defines the water phase thermal conductivity of 20.0 for each cell in the 300 grid block model, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

6.3.212 THPRESFT – DEFINE FAULT THRESHOLD PRESSURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [THPRESFT](#) keyword defines a fault threshold pressures that prevents fluid flow from occurring across the fault plane until the threshold pressure is exceeded, for when the threshold pressure option has been activated via the THRPRES variable on the [EQLOPTS](#) keyword in the [RUNSPEC](#) section.

Each row entry in the [THPRESFT](#) keyword defines a fault threshold pressure.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	FLTNAME	FLTNAME is a character string enclosed in quotes with a maximum length of eight characters, that defines the name of the fault. FLTNAME must have been previously defined using the <i>FAULTS</i> keyword in the GROD section, otherwise an error will occur.			None
2	PRESS	PRESS is a single positive real value that defines the threshold pressure for the fault (FLTNAME). If PRESS is defaulted then the simulator will set the threshold pressure to zero, that is the fault is open to flow along the fault plane.			0
		psia	barsa	atma	
Notes: 1) If there are multiple entries for FLTNAME only the last entry is applied. 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.					

Table 6.130: THPRESFT Keyword Description

See also the [FAULTS](#) keyword in the [GRID](#) section that is used to created named faults. Normally this fault data is automatically generated by pre-processing software that generates the static model.

Example

The example below defines two fault traces, 'M_WEST' and 'BC' fault having threshold pressures of 1000.0 and 2000 psis respectively.

```

--
--      DEFINE FAULTS IN THE GRID GEOMETRY
--
--      FAULT      ----- FAULT TRACE -----
--      NAME      I1   I2   J1   J2   K1   K2   FACE
FAULTS
      'M_WEST'    5     5     3     3     1    22   'X'   /
      'M_WEST'    5     5     4     4     1    22   'X'   /
      'M_WEST'    5     5     5     5     1    22   'X'   /
.....
      'BC'        43    43     8     8     1    22   'Y'   /
      'BC'        42    42     9     9     1    22   'X'   /
      'BC'        44    44     8     8     1    22   'Y'   /
.....
/
--
--      DEFINE FAULT THRESHOLD PRESSURES
--
--      FAULT      THRESHOLD
--      NAME      PRESSURE
THPRESFT
      'M_WEST'    1000.0
      'BC'        1200.0
/

```

6.3.213 TOPS – DEFINE THE DEPTH AT THE CENTER OF THE TOP FACE FOR EACH CELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[TOPS](#) defines the depth of the top face of each cell in the model.

It can only be used with the Cartesian Regular Grid or Radial Grid models.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	<i>TOPS</i>	<i>TOPS</i> is an array of real numbers defining the depth at the top face of each cell in the model. One can either just enter the <i>TOPS</i> for the first layer only based on NX x NY entries and OPM Flow will calculate the remaining <i>TOPS</i> based on either <i>DZ</i> or <i>DZV</i> . Alternatively NX x NY x NZ <i>TOPS</i> may be entered for each cell in the model. See the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section for the definition of NX, NY and NZ. Repeat counts may be used, for example 10*5201.0.			None
		feet	m	cm	

Notes:

I) The keyword is terminated by a “/”.

Table 6.131: TOPS Keyword Description

See also the DEPTHS keyword to define the structural depth for the cells.

Examples

The example below defines the [TOPS](#) of the cells for each cell for NX = 5, NY = 5 and NZ = 3 model, as well as the X and Y direction cells sizes.

```
--
--      DEFINE GRID BLOCK TOPS FOR THE TOP LAYER (NX=5, NY=5, and NZ=3)
--
TOPS
      25*3100  25*3105  25*3110
--
--      DEFINE GRID BLOCK X DIRECTION CELL SIZE (BASED ON NX = 5)
--
DXV
      5*100
--
--      DEFINE GRID BLOCK X DIRECTION CELL SIZE (BASED ON NY = 5)
--
Dyv
      5*100
```

A second example is shown on the following page.

This example defines the same grid as before but with the [TOPS](#) keyword only defining the top layer and [DZV](#) keyword defining the cells thickness.

```
--
--      DEFINE GRID BLOCK TOPS FOR THE TOP LAYER (NX = 5, NY = 5, NZ = 3)
--
TOPS
      25*3100
--
--      DEFINE GRID BLOCK Z DIRECTION CELL SIZE (BASED ON NZ = 3)
--
DZV
      3*5.0
--
--      DEFINE GRID BLOCK X DIRECTION CELL SIZE (BASED ON NX = 5)
--
DXV
      5*100
--
--      DEFINE GRID BLOCK Y DIRECTION CELL SIZE (BASED ON NY = 5)
--
DYV
      5*100
```

6.3.214 TRANGL – DEFINE NON-NEIGHBOR CONNECTIONS BETWEEN GLOBAL AND LGR CELLS MANUALLY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

TRANGL enables Non-Neighbor Connections (“NNC”) between the global cells and the Local Grid Refinement (“LGR”) cells to be manually specified, as oppose to the simulator calculating the transmissibilities. The **LGR** keyword in the **RUNSPEC** section should be utilized to define the presence of LGRs in the model and to define various **LGR** dimension parameters.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	I1	A positive integer that defines the <i>LGR</i> grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the <i>CARFIN</i> keyword in the <i>GRID</i> section.			None
2	J1	A positive integer that defines the <i>LGR</i> grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the <i>CARFIN</i> keyword in the <i>GRID</i> section.			None
3	K1	A positive integer that defines the <i>LGR</i> grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the <i>CARFIN</i> keyword in the <i>GRID</i> section.			None
4	I2	A positive integer that defines the GLOBAL grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section.			None
5	J2	A positive integer that defines the GLOBAL grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section.			None
6	K2	A positive integer that defines the GLOBAL grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section.			None
7	TRANSNNC	TRANSNNC is a positive real number greater than or equal to zero that defines the transmissibility between the GLOBAL grid block (I1, J1, K1) and the <i>LGR</i> grid block (I2, J2, K2). The default value of zero sets the transmissibility between the two cells to zero.			0.0
		cPrb/day/psia	cPrm ³ /day/bars	cPrcc/hr/atm	
Notes:					
1) Each record must be terminated by a “/” and the keyword is terminated by a “/”.					

Table 6.132:TRANGL Keyword Description

Example

```
--
--      MANUALLY DEFINE LGR-GLOBAL GRID NON-NEIGHBOR CONNECTIONS
--
--      ----LGR-----   ---GLOBAL-----   -- TRANSNCC --
--      I1   J1   K1     I2   J2   K2
TRANGL
      1     1     1      1     1     2          0.2500   /
      1     1     2      1     1     3          0.2500   /
      1     1     3      1     1     4          0.2500   /
/
```

The above example defines the transmissibility between *LGR* cell (1, 1, 1) and global cell (1, 1, 2), *LGR* cell (1, 1, 2) and global cell (1, 1, 3) and finally between *LGR* cell (1, 1, 3) and global cell (1, 1, 4) to be 0.2500.

6.3.215 USEFLUX – ACTIVATE FLUX BOUNDARY MODEL AND DEFINE FLUX FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [USEFLUX](#) keyword activates the Flux Boundary model and defines the name of the FLUX file. Only grid blocks that have been declared by the [FLUXREG](#) keyword in the [GRID](#) section to be in an active flux region, are active for the run.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.216 USENOFLO – ACTIVATE FLUX BOUNDARY MODEL WITHOUT A FLUX FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The USENOFLUX keyword activates the Flux Boundary model without a FLUX file. The [USEFLUX](#) keyword should still be in the input deck, but in this case the FLUX filename is ignored. The option is useful when the no-flow boundary condition is a reasonable assumption and avoids the pre-cursor run used to generate the FLUX file via the [DUMPFLUX](#) keyword in the [GRID](#) section. Only grid blocks that have been declared by the [FLUXREG](#) keyword in the [GRID](#) section to be in an active flux region, are active for the run.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```
--
--      ACTIVATE FLUX BOUNDARY MODEL WITHOUT A FLUX FILE
--
USEFLUX
/

USENOFLO
```

The above example activates the Flux Boundary model without a FLUX file.

6.3.217 VEDEBUG – VERTICAL EQUILIBRIUM DEBUG DATA OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the debug Vertical Equilibrium (“VE”) data to be written to the debug file (*.DBG), for when the [VE](#) model has been activated by the [VE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

6.3.218 VEFIN – ACTIVATE VERTICAL EQUILIBRIUM MODEL (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

If the [VE](#) keyword in the [RUNSPEC](#) section has been used to activate the Vertical Equilibrium (“VE”) model for the global grid, then the [VEFIN](#) keyword may used to set various options for the Local Grid Refinements (“LGR”). The [LGR](#) keyword in the [RUNSPEC](#) section should be activated to indicate the presence of LGRs and the keyword [VEFIN](#) should be placed in between the [CARFIN](#) and [ENDFIN](#) keywords in the [GRID](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

6.3.219 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

6.3.220 ZCORN – DEFINE THE DEPTH OF EACH CORNER-POINT OF A GRID BLOCK

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	-------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[ZCORN](#) defines the depth of each corner point of a grid block on the pillars defining the reservoir grid. A total of $8 \times NX \times NY \times NZ$ values are needed to fully define all the depths in the model. The depths specifying the top of the first layer are entered first with one point for each pillar for each grid block. The points are entered with the X axis cycling fastest. Next come the depths of the bottom of the first layer. The top of layer two follows etc.

The keyword can be only used be used with Irregular Corner-Point Grids.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	ZCORN	An array of depths with 8 depths for each cell, for a total of 8 x Nx x NY x NZ entries			None
		feet	metres	cm	

Notes:

1) Regular Cartesian and Radial grid keywords cannot be used with this keyword, that is: DR, DRV, DTHETA, DTHETA, DX, DXV, DY, DYV, DZ, DZV, INRAD, and TOPS.

2) The keyword is terminated by a “/”.

Table 6.133: ZCORN Keyword Description

See also the [SPECGRID](#), [COORD](#) and [COORDSYS](#) keywords to fully define an Irregular Corner-Point Grid.

Example

```
--
-- SPECIFY CORNER-POINT DEPTHS FOR A 3 x 2 x 2 GRID,
-- WITH CONSTANT SLOPE IN THE X AND Y DIRECTIONS
-- SUCH THAT ALL CORNER POINTS OF NEIGHBOURING BLOCKS ALIGN
ZCORN
--
-- top of layer 1
--
1450 1500 1500 1550 1550 1600
1500 1550 1550 1600 1600 1650
1500 1550 1550 1600 1600 1650
1550 1600 1600 1650 1650 1700
--
-- bottom of layer 1
--
1460 1510 1510 1560 1560 1610
1510 1560 1560 1610 1610 1660
1510 1560 1560 1610 1610 1660
1560 1610 1610 1660 1660 1710
--
-- top of layer 2
--
1460 1510 1510 1560 1560 1610
1510 1560 1560 1610 1610 1660
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

	1510	1560	1560	1610	1610	1660
	1560	1610	1610	1660	1660	1710
--						
--			bottom of layer 2			
--						
	1470	1520	1520	1570	1570	1620
	1520	1570	1570	1620	1620	1670
	1520	1570	1570	1620	1620	1670
/	1570	1620	1620	1670	1670	1720

The above example defines depths of the vertical coordinate lines for a regular 3 by 2 by 2 grid with a constant slope in the x and y directions such that all the corner points of neighboring blocks are aligned.

CHAPTER 7: EDIT SECTION

7.1 INTRODUCTION

This section allows users to define changes to be applied after OPM Flow has processed the data in the [GRID](#) section, that is the resulting pore volume ([PORV](#)) and transmissibility arrays ([TRANX](#), [TRANY](#), and [TRANZ](#)). The entered primary static arrays ([PORO](#), [PERMX](#), etc.) in the [GRID](#) section are no longer available and all modifications are applied to the pore volume and transmissibility arrays.

Historically the intention of this section was allow for the editing of the processed data; however, the features available in this section have, through time, migrated to the [GRID](#) section. For example the array operator keywords like [ADD](#), [COPY](#), [MULTIPLY](#), etc. are available in the [GRID](#) section and thus enable editing of the primary static arrays ([PORO](#), [PERMX](#), etc.). This increased capability in the [GRID](#) section has therefore made the [EDIT](#) section somewhat redundant.

Nevertheless the section is sometimes used by users to incorporate history matching parameter changes and by static earth modeling software packages to import the pore volumes and transmissibilities calculated in the static model directly into the numerical model via the [EDIT](#) section. Although the latter work flow is not very common.

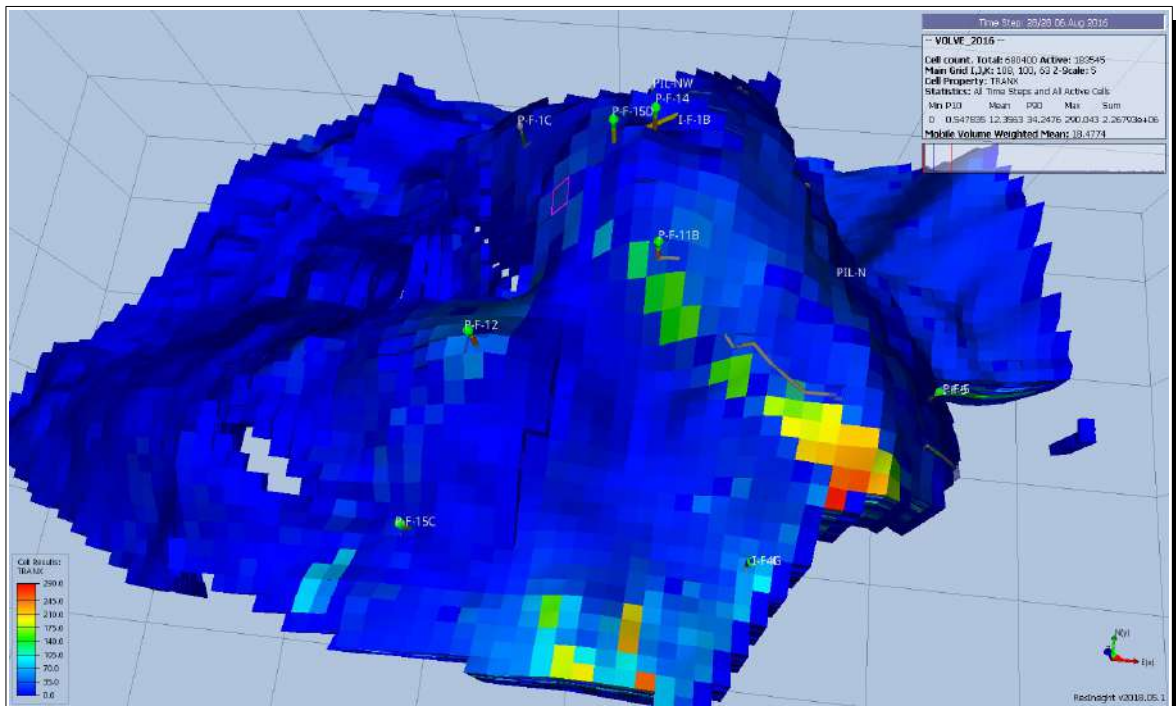
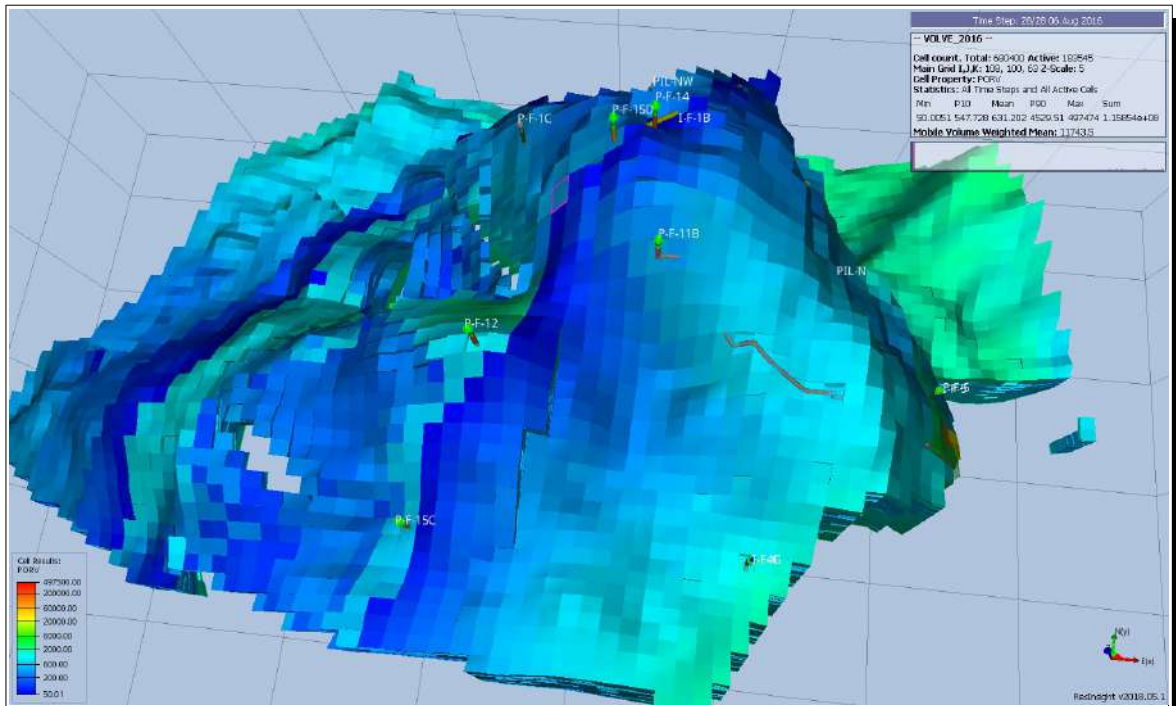
7.2 DATA REQUIREMENTS

As the primary purpose of this section is to modify the simulator’s calculated pore volumes and transmissibilities, then the properties used to define these arrays must have been fully defined in the [GRID](#) section. The arrays available for modification in the [EDIT](#) section are listed in Table 7.1 together with the associated [GRID](#) arrays used to generate the [EDIT](#) property array.

Cartesian And Irregular Corner-Point Grids		Radial Grid Keywords	
Keywords			
GRID	EDIT	GRID	EDIT
TOPS	DEPTH	TOPS	DEPTH
DX	PORV	DR	PORV
DY		DTHETA	
DZ		DZ	
DZNET		DZNET	
PORO		PORO	
NTG		NTG	
PERMX	TRANX	PERMR	TRANR
MULTX	TRANY	MULTR	TRANR
PERMY		PERMTHT	
MULTY		MULTTHT	
PERMZ	TRANZ	PERMZ	TRANZ
MULTZ		MULTZ	
Notes:			
1) Currently Radial Grids have not been implemented in OPM Flow.			
2) The GRID property association to the EDIT property is only indicative as several variables, DZNET and NTG for example, are also used in the transmissibility calculations.			

Table 7.1: EDIT Section Arrays Available for Modification

An example pore volume array (PORV property) from the Volve¹³⁹ field is shown in Figure 7.1 and Figure 7.2 illustrates the model's transmissibility in the x-direction (TRANX).



¹³⁹ The Volve Data was approved for data sharing in 2018 by the initiative of the last Operating company, Equinor and approved by the license partners ExxonMobil E&P Norway AS and Bayerngas Norge AS in the end of 2017.

7.3 KEYWORD DEFINITIONS

7.3.1 ADD – ADD A CONSTANT TO A SPECIFIED ARRAY

The **ADD** keyword adds a constant to a specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the **ADD** keyword is being applied.

See [ADD – Add a Constant to a Specified Array](#) in the [GRID](#) section for a full description.

7.3.2 ADDREG – ADD A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER

The **ADDREG** keyword adds a constant to a specified array or part of an array based on cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the **ADDREG** keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the **ADDREG** keyword is being used.

See [ADDREG – Add a Constant to an Array based on a Region Number](#) in the [GRID](#) section for a full description.

7.3.3 BOUNDARY – DEFINE A BOUNDARY BOX FOR PRINTING

The **BOUNDARY** keyword defines a rectangular grid for printing various arrays to the output print file (*.PRT); thus, avoiding printing all the elements in the selected array.

See [BOUNDARY – Define a Boundary Box for Printing](#) in the [GRID](#) section for a full description.

7.3.4 BOX – DEFINE A RANGE OF GRID BLOCKS TO ENTER PROPERTY DATA

BOX defines a range of grid blocks for which subsequent data is assigned for all the cells in the defined **BOX**. Note that the **BOX** grid is reset by the keyword **ENDBOX** by resetting the current defined **BOX** to be the whole grid. The keyword can be used for any array and for all grid types.

See [BOX – Define a Range of Grid Blocks to Enter Property Data](#) in the [GRID](#) section for a full description.

7.3.5 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The **COLUMNS** keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the [GLOBAL](#) section for a full description.

7.3.6 COPY – COPY ARRAY DATA TO ANOTHER ARRAY

The **COPY** keyword copies an array (or part of an array) to another array or part of an array. The arrays can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the **COPY** keyword is being used.

See [COPY – Copy Array Data to Another Array](#) in the [GRID](#) section for a full description.

7.3.7 COPYREG – COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER

The [COPYREG](#) keyword copies a specified array or part of an array based on cells with a specific region number to another array. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [COPYREG](#) keyword is read by the simulator. The property arrays can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [COPYREG](#) keyword is being used.

See [COPYREG – Copy an Array to Another Array based on a Region Number](#) in the [GRID](#) section for a full description.

7.3.8 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the [GLOBAL](#) section for a full description.

7.3.9 DEPTH – EDITS THE DEPTH AT THE CENTER OF EACH CELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **DEPTH** keywords modifies the depth at the center of selected cells in the model. The cells **DEPTH** are calculated by OPM Flow at the end of the **GRID** section and this keyword allows the user to adjust the calculated depths in the **EDIT** section. The area to be modified can be defined via the various grid selection keywords, **ADD**, **BOX**, **EQUALS**, etc., and areas that are not selected remain unchanged.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DEPTH	DEPTH is an array of real numbers defining the depth at the center of each cell in the model. Only the values in the currently defined input BOX needed be entered. Repeat counts may be used, for example 30*5201.0.			None
		feet	m	cm	
Notes: I) The keyword is terminated by a “/”.					

Table 7.2: Depth Keyword Description

See also the **TOPS** keyword to define the top structural depth for the cells.

Examples

The example below modifies the **DEPTH** of the cells for a selection of 10 cells from an NX = 10, NY = 11 and NZ = 20 model.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      1  10   11  11   20  20 / SET BOX AREA TO BE MODIFIED
/
--
--      SET GRID BLOCK CENTER DEPTH FOR THE GRID BLOCKS
--
DEPTH    10*3500.0
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

Alternatively the **EQUALS** keyword can be used to perform the same edit.

```
--
--      ARRAY          CONSTANT          ----- BOX -----
--
--      I1  I2   J1  J2   K1  K2
EQUALS   DEPTH        3500.0           1  10   11  11   20  20 / RESET DEPTH
/
```


7.3.10
DIFFR – DEFINE GRID BLOCK RADIAL DIRECTION DIFFUSIVITY VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFR](#) keyword defines the radial direction diffusivity values for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

7.3.11 DIFFTHT – DEFINE GRID BLOCK THETA DIRECTION DIFFUSIVITY VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFTHT* keyword defines the theta direction diffusivity values for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

7.3.12 DIFFX – DEFINE GRID BLOCK X-DIRECTION DIFFUSIVITY VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFX* keyword defines the x-direction diffusivity values for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

7.3.13 DIFFY – DEFINE GRID BLOCK Y-DIRECTION DIFFUSIVITY VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *DIFFY* keyword defines they y-direction diffusivity values for cells in the current input box for when the Diffusivity option has been activated by the *DIFFUSE* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

7.3.14 DIFFZ – DEFINE GRID BLOCK Z-DIRECTION DIFFUSIVITY VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFZ](#) keyword defines the z-direction diffusivity values for cells in the current input box for when the Diffusivity option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

7.3.15 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

7.3.16 EDIT – DEFINE THE START OF THE EDIT SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [EDIT](#) activation keyword marks the end of the [GRID](#) section and the start of the [EDIT](#) section that enables modifications to the OPM Flow calculated properties derived from the data entered in the [GRID](#) section, for example grid block pore volumes via the [PORV](#) array and the transmissibilities via the [TRANX](#), [TRANY](#) and [TRANZ](#) family of keywords.

There is no data required for this keyword.

Example

```
-- =====  
--  
-- EDIT SECTION  
--  
-- =====  
EDIT
```

The above example marks the end of the [GRID](#) section and the start of the [EDIT](#) section in the OPM Flow data input file.

7.3.17 EDITNNC – SCALE NON-NEIGHBOR CONNECTIONS BETWEEN CELLS MANUALLY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	-------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[EDITNNC](#) enables Non-Neighbor Connections (“NNC”), entered via the [NNC](#) keyword or calculated by the simulator, to be multiplied (re-scaled) by a constant. For example, if the existing transmissibility between non-neighbor connections is Told and the multiplier is C, then the resulting transmissibility, Tnew, will be $T_{new} = C \times T_{old}$. Only previously defined [NNC](#)s entered via the [NNC](#) keyword or calculated by the simulator can be edited, otherwise a warning message will be printed.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	I1	A positive integer that defines the first grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.			None
2	J1	A positive integer that defines the first grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.			None
3	K1	A positive integer that defines the first grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.			None
4	I2	A positive integer that defines the second grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.			None
5	J2	A positive integer that defines the second grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.			None
6	K2	A positive integer that defines the second grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.			None
7	TRANSMUL	TRANSMUL is a positive real number greater than or equal to zero that defines a constant that scales the transmissibility between the first grid block (I1,J1,K1) and the second grid block (I2,J2,K2). The default value of one means no scaling will be applied.			1
		dimensionless	dimensionless	dimensionless	
8	ISATNUM1	The default value of zero means the existing saturation table allocated to the upstream cell (I1,J1,K1).			0
9	ISATNUM2	ISATNUM2 is a positive integer defining which saturation table number (relative permeability table) to be used for flow from the second grid block to the first grid block. The default value of zero means the existing saturation table allocated to the downstream cell (I2,J2,K2).			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
10	IPRSNUM1	IPRSNUM1 is a positive integer defining which pressure table number (PVT table) to be used for flow from the first grid block to the second grid block. The default value of zero means the existing PVT table allocated to the upstream cell (I1,J1,K1).			0
11	IPRSNUM2	IPRSNUM2 is a positive integer defining which pressure table number (PVT table) to be used for flow from the second grid block to the first grid block. The default value of zero means the existing PVT table allocated to the downstream cell (I2,J2,K2).			0
12	FACE1	FACE1 is a character string that defines the face associated with flow from the first grid block to the second grid block, where FACE1 can have values of: X+, X-, Y+, Y-, Z+, or Z-. FACE1 is used with the commercial simulator's Vertical Equilibrium option which is not supported by OPM Flow.			None
13	FACE2	FACE2 is a character string that defines the face associated with flow from the second grid block to the first grid block, where FACE2 can have values of: X+, X-, Y+, Y-, Z+, or Z-. FACE2 is used with the commercial simulator's Vertical Equilibrium option which is not supported by OPM Flow.			None
14	DIFFNNC	DIFFNNC is a positive real number greater than or equal to zero that scales the diffusivity between the first grid block (I1, J1, K1) and the second grid block (I2, J2, K2).			0.0
		dimensionless	dimensionless	dimensionless	
Notes: 1) Each record must be terminated by a "/" and the keyword is terminated by a "/".					

Table 7.3: EDITNNC Keyword Description

Note that although items (8) to (14) for this keyword are not available in OPM Flow, even if they were, it is strongly recommended that these items are defaulted if the data is being entered manually, as opposed to being generated by pre-processing software.

Care should be taken that cells in different [PVTNUM](#) regions (see the [PVTNUM](#) keyword in the [REGIONS](#) section) are not connected, since the fluid properties are associated with a cell. If for example, a rbbl or a rm³ of oil flows from [PVTNUM](#) region 1 to [PVTNUM](#) region 2, then the oil properties of that oil will change from the PVT 1 data set to the PVT data set 2. This will result in material balance errors, that may or may not cause numerical issues.

If the transmissibility across a fault needs to be modified see the [FAULTS](#) and [MULTFLT](#) keywords in the [GRID](#) section for an alternative and less complicated method to modifying fault transmissibilities. Transmissibility between reservoir regions can be modified by using [MULTREGT](#) keyword, provided [MULTNUM](#) has been used to define the inter-region transmissibility region numbers for each grid block. Finally, the [MULTX](#), [MULTY](#) and [MULTZ](#) series of keywords can be used to modify transmissibility between various cells. All the aforementioned keywords are described in the [GRID](#) section.

See also the [EDITNNCR](#) keyword in the [EDIT](#) section that resets an existing [NNC](#) to a user defined value.

Example

```
--
--      MANUALLY RESCALE NON-NEIGHBOR CONNECTIONS
--
--      ----- BOX -----      -- TRANSMUL --
--      I1   J1   K1   I2   J2   K2
EDITNCC
      1     1     1     1     1     2      0.2000    / SET NNC FOR FAULT
      1     1     2     1     1     3      0.2000    / SET NNC FOR FAULT
      1     1     3     1     1     4      0.2000    / SET NNC FOR FAULT
/
```

The above example multiplies the transmissibility between cells (1, 1, 1) and (1, 1, 2), (1, 1, 2) and (1, 1, 3) and finally between (1, 1, 3) and (1, 1, 4) by 0.200.

7.3.18 EDITNNCR – RESET NON-NEIGHBOR CONNECTIONS BETWEEN CELLS MANUALLY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

EDITNNCR enables Non-Neighbor Connections (“NNC”), entered via the **NNC** keyword or calculated by the simulator, to be reset to a user defined value. Only previously defined **NNC**’s entered via the **NNC** keyword or calculated by the simulator can be edited, otherwise a warning message will be printed. See also the **EDITNNC** keyword in the **EDIT** section that scales an existing **NNC**.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	I1	A positive integer that defines the first grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.			None
2	J1	A positive integer that defines the first grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.			None
3	K1	A positive integer that defines the first grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.			None
4	I2	A positive integer that defines the second grid block in the I-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NX on the DIMENS keyword in the RUNSPEC section.			None
5	J2	A positive integer that defines the second grid block in the J-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NY on the DIMENS keyword in the RUNSPEC section.			None
6	K2	A positive integer that defines the second grid block in the K-direction in a non-neighbor connection, must be greater than or equal to one and less than or equal to NZ on the DIMENS keyword in the RUNSPEC section.			None
7	TRANSNNC	TRANSNNC is a positive real number greater than or equal to zero that defines the transmissibility between the first grid block (I1, J1, K1) and the second grid block (I2, J2, K2). This value cannot be defaulted and must be defined.			None
		cP.r/b/day/psia	cP.rm ³ /day/bars	cPrcc/hr/atm	
8	ISATNUM1	ISATNUM1 is a positive integer defining which saturation table number (relative permeability table) to be used for flow from the first grid block to the second grid block. The default value of zero means the existing saturation table allocated to the upstream cell (I1,J1,K1).			0
9	ISATNUM2	ISATNUM2 is a positive integer defining which saturation table number (relative permeability table) to be used for flow from the second grid block to the first grid block. The default value of zero means the existing saturation table allocated to the downstream cell (I2,J2,K2).			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
10	IPRSNUM1	IPRSNUM1 is a positive integer defining which pressure table number (PVT table) to be used for flow from the first grid block to the second grid block. The default value of zero means the existing PVT table allocated to the upstream cell (I1,J1,K1).			0
11	IPRSNUM2	IPRSNUM2 is a positive integer defining which pressure table number (PVT table) to be used for flow from the second grid block to the first grid block. The default value of zero means the existing PVT table allocated to the downstream cell (I2,J2,K2).			0
12	FACE1	FACE1 is a character string that defines the face associated with flow from the first grid block to the second grid block, where FACE1 can have vales of: X+, X-,Y+,Y-, Z+, or Z-.			None
13	FACE2	FACE2 is a character string that defines the face associated with flow from the second grid block to the first grid block, where FACE2 can have vales of: X+, X-,Y+,Y-, Z+, or Z-.			None
14	DIFFNNC	DIFFNNC is a positive real number greater than or equal to zero that scales the diffusivity between the first grid block (I1, J1, K1) and the second grid block (I2, J2, K2). The default value is the value calculated in the GRID section.			*
		feet	meters	cm	
Notes: 1) Each record must be terminated by a “/” and the keyword is terminated by a “/”.					

Table 7.4: EDITNNCR Keyword Description

Note that although items (8) to (14) for this keyword are not available in OPM Flow, even if they were, it is strongly recommended that these items are defaulted if the data is being entered manually, as opposed to being generated by pre-processing software.

If the transmissibility across a fault needs to be modified see the [FAULTS](#) and [MULTFLT](#) keywords in the [GRID](#) section for an alternative and less complicated method to modifying fault transmissibilities. Transmissibility between reservoir regions can be modified by using [MULTREGT](#) keyword, provided [MULTNUM](#) has been used to define the inter-region transmissibility region numbers for each grid block. Finally, the [MULTX](#), [MULTY](#) and [MULTZ](#) series of keywords can be used to modify transmissibility between various cells. All the aforementioned keywords are described in the [GRID](#) section.

Example

```
--
--      MANUALLY RESET NON-NEIGHBOR CONNECTIONS
--
--      ----- BOX -----      -- TRANSNNC --
--      I1   J1   K1   I2   J2   K2
EDITNCCR
      1     1     1     1     1     2      0.2500  / RESET NNC TRANS FOR FAULT
      1     1     2     1     1     3      0.2500  / RESET NNC TRANS FAULT
      1     1     3     1     1     4      0.2500  / RESET NNC TRANS FAULT
/
```

The above example res-sets the transmissibility between cells (1, 1, 1) and (1, 1, 2), (1, 1, 2) and (1, 1, 3) and (1, 1, 3) and (1, 1, 4) to be 0.2500.

7.3.19 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the *END* keyword are ignored.

See *END – Define the End of the Input File* in the GLOBAL section for a full description.

7.3.20 ENDBOX – DEFINE THE END OF THE BOX DEFINED GRID

This keyword marks the end of a previously defined *BOX* sub-grid as defined by a previously entered *BOX* keyword. The keyword resets the input grid to be the full grid as defined by the NX, NY, and NZ variables on the *DIMENS* keyword in the *RUNSPEC* section.

See *ENDBOX – Define the End of the BOX Defined Grid* in the *GRID* section for a full description.

7.3.21 ENDFIN – END THE DEFINITION OF A LOCAL GRID REFINEMENT

NDFIN defines the end of a Cartesian or radial local grid refinement (“LGR”) definition and a *LGR* property definition data set.

See *ENDFIN – End the Definition of a Local Grid Refinement* in the *GRID* section for a full description.

7.3.22 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the *INCLUDE* keyword. When the *ENDINC* keyword is encountered in the *INCLUDE* file, input data is read from the next keyword in the current file. Any keywords and data after the *ENDINC* keyword in the *INCLUDE* file are ignored.

See *ENDINC – Define the End of an Include File* in the GLOBAL section for a full description.

7.3.23 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The *ENDSKIP* keyword deactivates the skipping of keywords that was activated by the *SKIP*, *SKIP100*, or *SKIP300* keywords. Each *SKIP* keyword should be paired with an *ENDSKIP* keyword.

See *ENDSKIP – Deactivate Skipping of Keywords and Input Data* in the GLOBAL section for a full description.

7.3.24 EQUALREG – SETS AN ARRAY TO A CONSTANT BY REGION NUMBER

The *EQUALREG* keyword sets a specified array to a constant for cells with a specific region number. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *EQUALREG* keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the *EQUALREG* keyword is being used.

See *EQUALREG – Sets an Array to a Constant by Region Number* in the *GRID* section for a full description.

7.3.25 EQUALS – SETS A SPECIFIED ARRAY TO A CONSTANT

The *EQUALS* keyword sets a specified array or part of an array to a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the *EQUALS* keyword is being used.

See *EQUALS – Sets a Specified Array to a Constant* in the *GRID* section for a full description.

7.3.26 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

7.3.27 FILEUNIT – ACTIVATE UNIT CONSISTENCY VERIFICATION

The [FILEUNIT](#) keyword defines the units of the data set, and is used to verify that the units in the input deck and any associated include files are consistent. The keyword does not provide for the conversion between different sets of units.

See [FILEUNIT – Activate Unit Consistency Checking](#) in the [GRID](#) section for a full description.

7.3.28 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be place at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a dull description.

7.3.29 GETDATA – LOAD AND ASSIGN DATA ARRAY FROM INIT OR RESTART FILE

The [GETDATA](#) keyword loads a data array from a previously generated [INIT](#) or [RESTART](#) file and assigns the loaded array to either same array in the run or another array name.

See [GETDATA – Load and Assign Data Array from INIT or RESTART Files](#) in the [GRID](#) section for a full description

7.3.30
HMMULT – HISTORY MATCH GRID TRANSMISSIBILITY & PORE VOLUME GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *HMMULT* series of keywords defines the history match gradient cumulative permeability multipliers, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section. The keyword consists of the first six characters of “HMMULT” followed by a one or two character string shown in Table 7.5, that determines the transmissibility direction, for example, HMMULTX.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Mnemonic	Cartesian Grid		Radial Grid	
	Grid Keyword	<i>HMMULT</i> Keyword	Grid Keyword	<i>HMMULT</i> Keyword
X/R	<i>MULTX</i>	HMMULTX	<i>MULTR</i>	HMMULTR
XY		HMMULTXY		
Y/HT	<i>MULTY</i>	HMMULTY	<i>MULTTHT</i>	HMMULTTH
Z	<i>MULTZ</i>	HMMULTZ	<i>MULTZ</i>	HMMULTZ
PV	<i>MULTPV</i>	HMMULTPV	<i>MULTPV</i>	HMMULTPV

Table 7.5: *HMMULT* Keyword List

See also the *HMMLT* keyword in the *GRID* section.

7.3.31 IMPORT – IMPORT GRID FILE DATA AT THE CURRENT POSITION

The **IMPORT** keyword informs the simulator to continue reading input data from the specified **IMPORT** file. When the end of the **IMPORT** file is reached, input data is read from the next keyword in the current file. Normally **IMPORT** files are generated by grid pre-processing software and the keyword allows for both formatted and unformatted (binary) files to be loaded.

See [IMPORT – Import Grid File Data at the Current Position](#) in the **GRID** section for a full description.

7.3.32 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The **INCLUDE** keyword informs OPM Flow to continue reading input data from the specified **INCLUDE** file. When the end of the **INCLUDE** file is reached, or the **ENDINC** keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the **GLOBAL** section for a full description.

7.3.33 LGRCOPY – ACTIVATE LOCAL GRID REFINEMENT INHERITANCE

The **LGRCOPY** keyword activates Local Grid Refinement (“LGR”) Inheritance option that allows the **LGR** to inherit the properties of the global or host cell containing a **LGR** grid block at the start of the **GRID** section, as oppose to the normal process of applying this transform at the end of the **GRID** section. **LGRCOPY** can be used in the **RUNSPEC**, **GRID** and **EDIT** sections. If used in the **RUNSPEC** section then the option is applied to all LGRs defined in the input file, whereas if used in the **GRID** and **EDIT** sections the keyword must be placed inside a **LGR** definition section, that is between a **CARFIN** (Cartesian **LGR** grid) or **RADIN/RADIN4** (radial **LGR** grid) and the **ENDFIN** keyword. In the latter case inheritance is applied on an individual **LGR** basis.

See [LGRCOPY – Activate Local Grid Refinement Inheritance](#) in the **RUNSPEC** section for a full description.

7.3.34 MAXVALUE – SETS A MAXIMUM VALUE FOR AN ARRAY ELEMENT

The **MAXVALUE** keyword sets a maximum value for the specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the **MAXVALUE** keyword is being used.

See [MAXVALUE – Sets a Maximum Value for an Array Element](#) in the **GRID** section for a full description.

7.3.35 MESSAGE – OUTPUT USER MESSAGE

The **MESSAGE** keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the **MESSAGES** keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the **GLOBAL** section for a full description.

7.3.36 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The **MESSAGES** keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the **GLOBAL** section for a full description.

7.3.37 MINVALUE – SET A MINIMUM VALUE FOR AN ARRAY ELEMENT

The [MINVALUE](#) keyword sets a minimum value for the specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [MINVALUE](#) keyword is being used.

See [MINVALUE – Set a Minimum Value for an Array Element](#) in the [GRID](#) section for a full description.

7.3.38 MULTFLT – MULTIPLY THE TRANSMISSIBILITY OF A DEFINED FAULT BY A CONSTANT

[MULTFLT](#) enables the transmissibilities across defined faults, as declared by the [FAULTS](#) keyword, to be modified. They keyword allows for the re-scaling of the existing fault transmissibilities calculated by OPM Flow, for example setting a fault to be completely sealing by setting the multiplier to zero..

See [MULTFLT – Multiply the Transmissibility of a Defined Fault by a Constant](#) in the [GRID](#) section for a full description.

7.3.39 MULTIPLY – MULTIPLY A SPECIFIED ARRAY BY A CONSTANT

The [MULTIPLY](#) keyword multiplies a specified array or part of an array by a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [MULTIPLY – Multiply a Specified Array by a Constant](#) in the [GRID](#) section for a full description.

7.3.40 MULTIREG – MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER

The [MULTIREG](#) keyword multiplies an array or part of an array by a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [MULTIREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [MULTIREG](#) keyword is being used.

See [MULTIREG – Multiply an Array by a Constant based on a Region Number](#) in the [GRID](#) section for a full description.

7.3.41 MULTPV – MULTIPLY CELL PORE VOLUMES BY A CONSTANT

[MULTPV](#) multiplies the pore volumes of a cell by a real positive constant for all the cells in the model via an array. An alternative to defining the complete array is to use the [BOX](#) keyword to define an area of the grid and then use the [MULTPV](#) keyword to set the multipliers just for the area defined by the [BOX](#) keyword (see the example).

See [MULTPV – Multiply Cell Pore Volumes by a Constant](#) in the [GRID](#) section for a full description.

7.3.42 MULTR – MULTIPLY CELL TRANSMISSIBILITY IN THE +R DIRECTION

[MULTR](#) multiplies the transmissibility between two cell faces in the +R direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I+I, J, K). The keyword should only be used with radial grids, as declared by the [RADIAL](#) keyword in the [RUNSPEC](#) section.

See [MULTR – Multiply Cell Transmissibility in the +R Direction](#) in the [GRID](#) section for a full description.

7.3.43 MULTR- – MULTIPLY CELL TRANSMISSIBILITY IN THE -R DIRECTION

MULTR- multiplies the transmissibility between two cell faces in the -X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I-I, J, K) and (I, J, K). The keyword should only be used with radial grids, as declared by the *RADIAL* keyword in the *RUNSPEC* section.

See *MULTR- – Multiply Cell Transmissibility in the -R Direction* in the *GRID* section for a full description.

7.3.44 MULTREGD – MULTIPLY DIFFUSIVITIES BETWEEN REGIONS

The *MULTREGT* keyword multiplies the diffusivity between two regions by a constant. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *MULTREGT* keyword is read by the simulator. The constant should be a real number.

See *MULTREGD – Multiply Diffusivities Between Regions* in the *GRID* section for a full description.

7.3.45 MULTREGH – MULTIPLY THERMAL CONDUCTIVITIES BETWEEN REGIONS

The *MULTREGH* keyword multiplies the thermal conductivity between two regions by a constant. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *MULTREGT* keyword is read by the simulator. The constant should be a real number.

See *MULTREGH – Multiply Thermal Conductivities Between Regions* in the *GRID* section for a full description.

7.3.46 MULTREGP – MULTIPLY PORE VOLUMES BASED ON REGION NUMBER

The *MULTREGP* keyword multiplies the pore volume of a cell by a constant for all cells with a specific region number. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *MULTREGP* keyword is read by the simulator. The constant should be a real number.

See *MULTREGP – Multiply Pore Volumes Based On Region Number* in the *GRID* section for a full description.

7.3.47 MULTREGT – MULTIPLY TRANSMISSIBILITIES BETWEEN REGIONS

The *MULTREGT* keyword multiplies the transmissibility between two regions by a constant. The region number array can be *FLUXNUM*, *MULTNUM* or *OPERNUM* and these arrays must be defined and be available before the *MULTREGT* keyword is read by the simulator. The constant should be a real number.

See *MULTREGT – Multiply Transmissibilities Between Regions*in the *GRID* section for a full description.

7.3.48 MULTTHT – MULTIPLY CELL TRANSMISSIBILITY IN THE +THETA DIRECTION

MULTTHT multiplies the transmissibility between two cell faces in the +Theta direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J+I, K). The keyword should only be used with radial grids, as declared by the *RADIAL* keyword in the *RUNSPEC* section.

See *MULTTHT – Multiply Cell Transmissibility in the +Theta Direction* in the *GRID* section for a full description.

7.3.49

MULTTHT-

–

MULTIPLY CELL TRANSMISSIBILITY IN THE -THETA DIRECTION

MULTTHT- multiplies the transmissibility between two cell faces in the -Theta direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J-I, K) and (I, J, K). The keyword should only be used with radial grids, as declared by the *RADIAL* keyword in the *RUNSPEC* section.

See *MULTTHT- – Multiply Cell Transmissibility in the -Theta Direction* in the *GRID* section for a full description.

7.3.50

MULTX

–

MULTIPLY CELL TRANSMISSIBILITY IN THE +X DIRECTION

MULTX multiplies the transmissibility between two cell faces in the +X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I+I, J, K).

See *MULTX – Multiply Cell Transmissibility in the +X Direction* in the *GRID* section for a full description.

7.3.51

MULTX-

–

MULTIPLY CELL TRANSMISSIBILITY IN THE -X DIRECTION

MULTX- multiplies the transmissibility between two cell faces in the -X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I-I, J, K) and (I, J, K).

See *MULTX- – Multiply Cell Transmissibility in the -X Direction* in the *GRID* section for a full description.

7.3.52

MULTY

–

MULTIPLY CELL TRANSMISSIBILITY IN THE +Y DIRECTION

MULTY multiplies the transmissibility between two cell faces in the +Y direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J+I, K).

See *MULTY – Multiply Cell Transmissibility in the +Y Direction* in the *GRID* section for a full description.

7.3.53

MULTY-

–

MULTIPLY CELL TRANSMISSIBILITY IN THE -Y DIRECTION

MULTY- multiplies the transmissibility between two cell faces in the -Y direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J-I, K) and (I, J, K).

See *MULTY- – Multiply Cell Transmissibility in the -Y Direction* in the *GRID* section for a full description.

7.3.54

MULTZ

–

MULTIPLY CELL TRANSMISSIBILITY IN THE +Z DIRECTION

MULTZ multiplies the transmissibility between two cell faces in the +Z direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J, K+I).

See *MULTZ – Multiply Cell Transmissibility in the +Z Direction* in the *GRID* section for a full description.

7.3.55

MULTZ-

–

MULTIPLY CELL TRANSMISSIBILITY IN THE -Z DIRECTION

MULTZ- multiplies the transmissibility between two cell faces in the -Z direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K-I) and (I, J, K).

See *MULTZ- – Multiply Cell Transmissibility in the -Z Direction* in the *GRID* section for a full description.

7.3.56
NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active. but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

7.3.57
NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword.The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

7.3.58
OPERATE – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS

This keyword, [OPERATE](#), defines mathematical operations on property arrays ([NTG](#), [PORO](#) etc.) and optionally using another property array as input to the function. The keyword allows for various mathematical functions and their associated variables to be defined and applied to the selected array data. Input constants can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [OPERATE – Define Mathematical Operations on Arrays](#) in the [GRID](#) section for a full description.

7.3.59
OPERATER – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS BY REGION

This keyword defines the mathematical operations on arrays for specific regions in the commercial simulator and is currently not supported by OPM Flow. However, similar functionality is provided by the [ADD](#) and [MULTIPLY](#) keywords.

See [OPERATER – Define Mathematical Operations on Arrays by Region](#) in the [GRID](#) section for a full description.

7.3.60 PORV – DEFINE THE PORE VOLUMES FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[PORV](#) defines the pore volumes for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry. The keyword effectively overwrites previously entered and calculated data. The area to be modified can be defined via the various grid selection keywords, [ADD](#), [BOX](#), [EQUALS](#), etc., and areas that are not selected remain unchanged.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PORV	PORV is an array of real positive numbers assigning a pore volume to each cell in the model. Only the values in the currently defined input BOX needed be entered. Repeat counts may be used, for example 20*100.0.			None
		rb	rm ³	rcc	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) Values not reset by this keyword remain unaltered.
- 3) The keyword is terminated by a "/".

Table 7.6: PORV Keyword Description

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      1* 100    1* 100    20  20
--
--      SET PORV FOR THE GRID BLOCKS
--
--      PORV
--      1000*0.00
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

Here the [BOX](#) statement is used to define the input grid for the [PORV](#) keyword, which overwrites the pore volume previously calculated with pore volume values of zero, resulting in a no-flow boundary in that part of the field between layers 19 and 21, since layer 20 is deactivated. The [ENDBOX](#) keyword resets the input box to the full grid.

7.3.61 PYEND – END THE DEFINITION OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword on a separate single line.

See [PYEND – End the Definition of a PYINPUT Section](#) in the [GRID](#) section for a full description.

7.3.62 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator. [PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line.

See [PYINPUT – Define the Start of a PYINPUT Section](#) in the [GRID](#) section for a full description.

7.3.63 REFINES – START THE DEFINITION OF A LOCAL GRID REFINEMENT

The [REFINE](#) keyword defines the start of a Cartesian or radial local grid refinement (“LGR”) definition that sets the properties of the selected [LGR](#). The keyword is then followed by the property keywords associated with the section where the keyword is being invoked. For example, if the [REFINE](#) keyword is used in the [GRID](#) section then most of the keywords in that section can be used to set the grid properties for the [LGR](#).

See [REFINE – Start the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

7.3.64 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the [GLOBAL](#) section for a full description.

7.3.65 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the [GLOBAL](#) section for a full description.

7.3.66 SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the [GLOBAL](#) section for a full description.

7.3.67 TRANR – DEFINE THE TRANSMISSIBILITY IN THE +R DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

TRANR defines the transmissibility in the +R direction for all the cells in the model via an array. The keyword can only be used with Radial Grid geometry grids. The keyword effectively overwrites previously entered and calculated data. The transmissibility overwritten is the +R face transmissibility of each grid block, that is for cell (I,J,K) the transmissibility between cells (I,J,K) and (I+1,J,K).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TRANR	TRANR is an array of real positive numbers assigning the transmissibility in the R direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		cP.rb/day/psia	cP.rm ³ /day/bars	cP.rcc/hr/atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) Values not reset by this keyword remain unaltered.
- 3) The keyword is terminated by a "/".

Table 7.7:TRANR Keyword Description

See also the **TRANRTH** and **TRANRZ** keywords to modify the transmissibilities in the other directions.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      1   1   10  10   1   120
--
--      SET TRANR+ TRANSMISSIBILITY
--
TRANR    120*0.00
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

Here the **BOX** statement is used to define the input grid for the **TRANR** keyword, which overwrites the transmissibility previously calculated with transmissibility values of zero, resulting in a no-flow boundary in that part of the grid. The **ENDBOX** keyword resets the input box to the full grid.

7.3.68 TRANTHT – DEFINE THE TRANSMISSIBILITY IN THE +THETA DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

TRANTHT defines the transmissibility in the +Theta direction for all the cells in the model via an array. The keyword can only be used with Radial Grid geometry grids. The keyword effectively overwrites previously entered and calculated data. The transmissibility overwritten is the +Theta face transmissibility of each grid block, that is for cell (I,J,K) the transmissibility between cells (I,J,K) and (I,J+1,K).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TRANHT	TRANHT is an array of real positive numbers assigning the transmissibility in the +Theta direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		cP.rb/day/psia	cP.rm ³ /day/bars	cP.rcc/hr/atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) Values not reset by this keyword remain unaltered.
- 3) The keyword is terminated by a "/".

Table 7.8: **TRANR** Keyword Description

See also the **TRANR** and **TRANYZ** keywords to modify the transmissibilities in the other directions.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      10  10   1   6   1   3                / DEFINE BOX AREA
--
--      SET TRANHT TRANSMISSIBILITY
--
TRANHT
      18*0.00                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

Here the **BOX** statement is used to define the input grid for the **TRANHT** keyword, which overwrites the transmissibility previously calculated with transmissibility values of zero, resulting in a no-flow boundary in that part of the grid. The **ENDBOX** keyword resets the input box to the full grid.

7.3.69 TRANX – DEFINE THE TRANSMISSIBILITY IN THE X DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

TRANX defines the transmissibility in the X direction for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry. The keyword effectively overwrites previously entered and calculated data. The transmissibility overwritten is the +X face transmissibility of each grid block, that is for cell (I, J, K) the transmissibility between cells (I, J, K) and (I+1, J, K).

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TRANX	TRANX is an array of real positive numbers assigning the transmissibility in the X direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		cPrb/day/psia	cPrm ³ /day/bars	cPrcc/hr/atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) Values not reset by this keyword remain unaltered.
- 3) The keyword is terminated by a "/".

Table 7.9: TRANX Keyword Description

See also the **TRANX** and **TRANXZ** keywords to modify the transmissibilities in the other directions.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      1   1    10  10   1  120                / DEFINE BOX AREA
--
--      SET TRANX+ TRANSMISSIBILITY
--
TRANX    120*0.00                                /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

Here the **BOX** statement is used to define the input grid for the **TRANX** keyword, which overwrites the transmissibility previously calculated with transmissibility values of zero, resulting in a no-flow boundary in that part of the field. The **ENDBOX** keyword resets the input box to the full grid.

7.3.70 TRANY – DEFINE THE TRANSMISSIBILITY IN THE Y DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

TRANY defines the transmissibility in the Y direction for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry. The keyword effectively overwrites previously entered and calculated data. The transmissibility overwritten is the +Y face transmissibility of each grid block, that is for cell (I, J, K) the transmissibility between cells (I, J, K) and (I, J+1, K).

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TRANY	TRANY is an array of real positive numbers assigning the transmissibility in the Y direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		cPrb/day/psia	cPrm ³ /day/bars	cPrcc/hr/atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) Values not reset by this keyword remain unaltered.
- 3) The keyword is terminated by a "/".

Table 7.10: TRANY Keyword Description

See also the **TRANX** and **TRANZ** keywords to modify the transmissibilities in the other directions.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      1   1    10  10   1  120
--                                     / DEFINE BOX AREA
--
--      SET TRANY+ TRANSMISSIBILITY
--
--      TRANY
--      120*0.00
--                                     /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

Here the **BOX** statement is used to define the input grid for the **TRANY** keyword, which overwrites the transmissibility previously calculated with transmissibility values of zero, resulting in a no-flow boundary in that part of the field. The **ENDBOX** keyword resets the input box to the full grid.

7.3.71 TRANZ – DEFINE THE TRANSMISSIBILITY IN THE Z DIRECTION FOR ALL THE CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

TRANZ defines the transmissibility in the z direction for all the cells in the model via an array. The keyword can be used for all grid types, except for the Radial Grid geometry. The keyword effectively overwrites previously entered and calculated data. The transmissibility overwritten is the +Z face transmissibility of each grid block, that is for cell (I, J, K) the transmissibility between cells (I, J, K) and (I, J, K+1).

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TRANZ	TRANZ is an array of real positive numbers assigning the transmissibility in the Z direction to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		cPrb/day/psia	cPrm ³ /day/bars	cPrcc/hr/atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) Values not reset by this keyword remain unaltered.
- 3) The keyword is terminated by a "/".

Table 7.11: TRANZ Keyword Description

See also the **TRANX** and **TRANZ** keywords to modify the transmissibilities in the other directions.

Example

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      1* 100   1* 100   20  20                               / DEFINE BOX AREA
--
--      SET TRANZ+ TRANSMISSIBILITY
--
TRANZ
--      1000*0.00                                               /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

Here the **BOX** statement is used to define the input grid for the **TRANZ** keyword, which overwrites the transmissibility previously calculated with transmissibility values of zero, resulting in a no-flow boundary in that part of the field between layers 20 and 21. The **ENDBOX** keyword resets the input box to the full grid.

7.3.72 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

CHAPTER 8: PROPS SECTION

8.1 INTRODUCTION

The **PROPS** section is the section that contains the fluid property keywords used to define the PVT behavior of the fluids in the model and therefore the data is fluid type dependent. For example, if the oil phase has been activated in the **RUNSPEC** section via the **OIL** keyword then one of the oil PVT keywords needs to be defined in this section. The other main type of data required for the **PROPS** section is the saturation tables that govern how the various fluids flow in the model relative to the other phases. Again, the saturation tables are fluid type dependent.

Various other fluid and rock property data are also entered in this section including rock compressibility via the **ROCK** keyword for the standard rock compressibility model, and rather unusually, the Carter-Tracy influence function entered via the **AQUTAB** keyword.

8.2 DATA REQUIREMENTS

For all phases activated in the model a complete description of the PVT behavior (PVT tables) and fluid flow behavior (saturation tables) is required. As the data is fluid type dependent, the next two sections attempt to define the appropriate keywords for the various activated fluid types. The first section outlines the fluid PVT property requirements and the second section summarizes the keywords associated with fluid flow property data, namely the relative permeability and capillary pressure data.

8.2.1 FLUID PROPERTIES

A phase is a portion of a system which is (1) homogeneous in composition, (2) bounded by a physical surface, and (3) mechanically separable from other phases which may be present. The most important phases which occur in petroleum production are liquid phases (oil, distillate, condensate, and water) and a vapor or gas phase (natural gas) component. Separators are used to separate the gas, water, and hydrocarbon liquid phases produced from wells. Solid phases also occur in petroleum production, mainly as:

- (1) paraffin or other deposits within the producing formation, in the tubing, and in the surface equipment;
- (2) gas hydrates which may freeze and clog gas-flow lines and equipment; and
- (3) sand or reservoir rock¹⁴⁰.

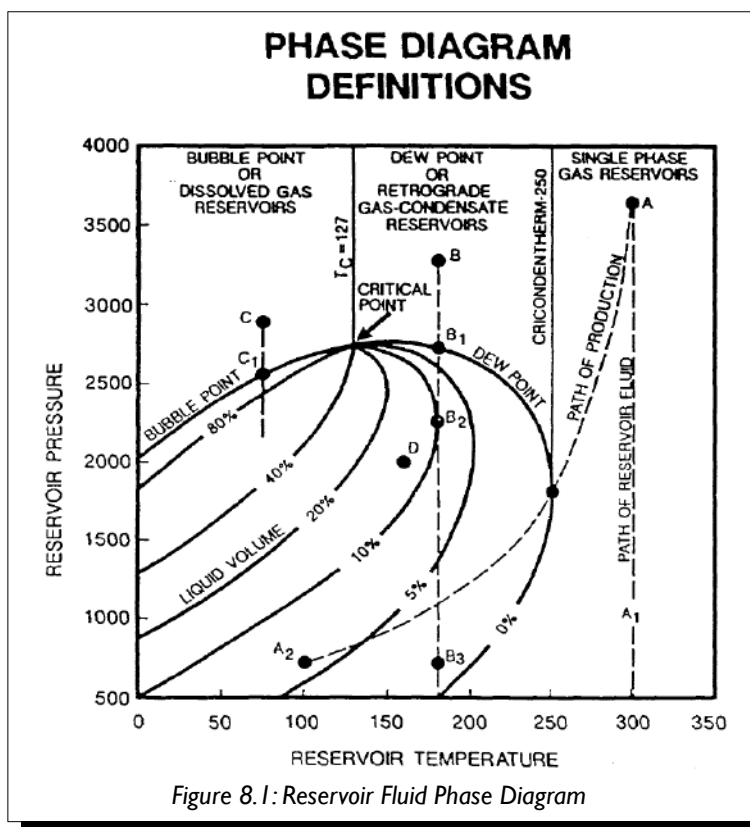


Figure 8.1: Reservoir Fluid Phase Diagram

The phase behavior of petroleum systems is represented by a Pressure-Volume-Temperature ("PVT") phase diagram, as illustrated in Figure 8.1, which shows a typical phase diagram after Clark¹⁴¹.

¹⁴⁰ Bradley, H. B.: "Petroleum Engineering Handbook", Society of Petroleum Engineers, Richardson, Texas, USA.

¹⁴¹ Clark, Norman, "It Pays to Know Your Petroleum", World Oil, March and April, 1953.

Several important definitions should be noted from Figure 8.1 that describes how a fluid behaves with respect to the composition, pressure, and temperature, namely:

- **Bubble Point:** Pressure at which the first bubble of gas is formed.
- **Dew Point:** Pressure at which the last drop of liquid turns to gas.
- **Critical Point:** Point at which all properties of liquid and gas are identical, the point where the dew point and bubble point curves join.
- **Cricondentherm:** Temperature above which two phases cannot co-exist regardless of pressure.
- **Cricondenbar:** Pressure above which two phases cannot co-exist regardless of the temperature

Multi-component hydrocarbon systems are most accurately defined by their compositions, which are usually expressed as mole fractions of the components methane, ethane, propane, isobutane, n-butane, iso-pentane, n-pentane, hexanes, and heptanes and heavier (heptanes plus). The heptanes-plus fraction is usually defined by its average molecular weight, its specific gravity, and/or the fractions which boil off in selected temperature ranges. In addition to hydrocarbons, many naturally occurring petroleum systems also contain hydrogen sulphide, nitrogen, carbon dioxide, and helium in quantities which are sufficiently large to appreciably affect the phase behavior of the system (see GPA Standard 2145¹⁴² and Standing (1952)¹⁴³ for additional information).

Gas Fluid Properties

The Ideal Gas Law is based on the theory of gases, a mathematical equation called an Equation-of-State that can be derived to express the relationship existing between pressure, volume, and temperature, for a given quantity of gas. This relationship is called the Ideal Gas Law, and is expressed mathematically as:

$$PV = nRT \quad (8.1)$$

where

- | | | |
|---|---|---|
| P | = | pressure (psia or Pa), |
| T | = | temperature (°R = 459.67 + °F or K = 273.15 + °C), |
| V | = | volume of the gas (ft ³ or m ³), |
| n | = | number of moles of gas (lb-mole or g-mol), and |
| R | = | universal gas constant which has the value of approximately 10.732 psi.ft ³ /lb-mole/°R or 8.314 Pa.m ³ /g-mol/K. |

However, the above equation of state is not applicable for **Real Gases** observed in petroleum reservoirs. In dealing with gases at a very low pressure, the ideal gas relationship (equation of state) is a convenient and accurate, within 2 - 3% at atmospheric conditions, while for pressures and temperatures found in petroleum reservoirs the physical properties calculated can lead to errors in excess of 500%. This is because real gases (natural gases) do not behave as an ideal gas. Basically the magnitude of deviations of real gases from the conditions of the ideal gas law increases with increasing pressure and temperature, and varies widely with the composition of the gas. The reason for this is that the perfect gas law was derived under the assumption that the volume of molecules is insignificant and no molecular attraction or repulsion takes place. This is not the case for real gases.

In order to express a more exact relationship between the variables P, V, and T, a correction factor called the gas compressibility factor (the gas deviation factor, or simply the Z-factor) must be introduced into the ideal gas law, that is:

¹⁴² Gas Processors Association, GPA, "Standard 2145 - Physical Constants of Paraffin Hydrocarbons and Other Components of Natural gas", 1986.

¹⁴³ Standing, M. B.: "Volumetric and Phase Behaviour of Oil Field Hydrocarbon Systems", Renihold Publishing Corp., New York City (1952).

$$PV = ZnRT \quad (8.2)$$

where

P	=	pressure (psia or Pa),
T	=	temperature ($^{\circ}\text{R} = 459.67 + ^{\circ}\text{F}$ or $\text{K} = 273.15 + ^{\circ}\text{C}$),
V	=	volume of the gas (ft^3 or m^3),
n	=	number of moles of gas (lb-mole or g-mol),
R	=	universal gas constant which has the value of approximately 10.732 psi.ft ³ /lb-mole/ $^{\circ}\text{R}$ or 8.314 Pa.m ³ /g-mol/K, and
Z	=	compressibility factor for the real gas.

For an ideal gas, the gas compressibility factor is equal to one, while for a real gas, the Z-factor is greater or less than one depending on the pressure, temperature, and composition of the gas. The Z-factor can be calculated from the composition of the gas or from various correlations including Standing¹⁴⁴ and Dranchuk, P.M., Purvis, R.A., and Robinson¹⁴⁵.

Gas Density (ρ_g): Depends on composition, temperature, and pressure at which the gas is measured, it is defined as the ratio of mass (m) to volume (V), that is:

$$\rho_g = \frac{m}{V} = \frac{P \text{ MW}}{ZRT} \quad (8.3)$$

where MW is the molecular weight of the gas and ρ_g has units of lbm/ft³ and kg/m³ for field and metric units, respectively in equation (8.3). Gas density is often reported as Relative Density, that is relative to air, as per 0.65 (air = 1).

Gas Formation Volume Factor (E): Is used to relate the volume of gas, as measured at reservoir conditions, to the volume of gas as measured at standard conditions (60 $^{\circ}\text{F}$ and 14.7 psia, or 15 $^{\circ}\text{C}$ and 101.325 kPa). This gas property is then defined as the actual volume occupied by a certain amount of gas at a specified pressure and temperature, divided by the same amount of gas at standard conditions. Thus the gas formation volume factor can be expressed as:

$$E = \frac{V_{sc}}{V_i} \quad (8.4)$$

and substituting equation (8.3) into (8.4) we can derive the gas formation volume factor for a given reservoir pressure and temperature:

$$E = \left(\frac{P_i}{P_{sc}} \right) \left(\frac{T_{sc}}{T_i} \right) \left(\frac{1}{Z_i} \right) \quad (8.5)$$

Note

In oilfield units when we are dealing with oil reservoirs the gas formation volume factor is more commonly denoted as B_g and has the units of rb/scf. Whereas for gas reservoirs, the gas formation volume factor is E_g and has the units scf/rcf and is often described as the Gas Expansion Factor or GEF.

¹⁴⁴ Standing, M.B. and Katz, D.L.: "Density of Natural Gases," Trans., AIME (1942) 146, 140-44.

¹⁴⁵ Dranchuk, P. M., Purvis, R. A., and Robinson, D. B., "Computer Calculations of Natural Gas Compressibility Factors Using the Standing and Katz Correlation", Institute of Petroleum Technical Series, No. IP 74-008, 1974.

Gas Isothermal Compressibility: A knowledge of the variability of a fluid compressibility with pressure and temperature is essential in performing many reservoir engineering calculations. For a liquid phase, the compressibility is small and is usually assumed to be constant. Whereas for a gas phase, the compressibility is neither small nor constant. By definition the isothermal gas compressibility, c_g , of a substance is the change in volume per unit volume, for a unit change in pressure, i.e.

$$c_g = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (8.6)$$

Most simulators calculate the isothermal gas compressibility based on the entered PVT data, so it is not necessary to enter this variable directly. If needed the correlation of Meehan, D. N., and Lyons, W. K. ¹⁴⁶ may be used to estimate the values.

Gas Viscosity (μ_g): The gas viscosity of a fluid is a measure of the internal fluid friction (resistance) to flow. If the friction between layers of the fluid is small, i.e., low viscosity, an applied shearing force will result in a large velocity gradient. As viscosity increases, each fluid layer exerts a larger frictional drag on the adjacent layers and velocity gradient increases. The viscosity of a fluid is generally defined as the ratio of the shear force per unit area to the local velocity gradient. Note that gas viscosity is not normally measured in the laboratory because it can be estimated precisely from empirical correlations. Gas viscosity increases with increasing temperature and pressure. Usually it is not measured but is obtained from the Carr, Kobayashi and Burrows¹⁴⁷ correlation, which include corrections for H₂S, CO₂, and N₂. For sour gases, this correlation is preferred to the Lee, Gonzalez and Eakin¹⁴⁸ formulation (which does not account for H₂S, CO₂ and N₂). Typically, gas viscosity is in the range of 0.015 to 0.03 cp.

Condensate-Gas Ratio (“CGR”): Defines the volume of hydrocarbon liquid that will dissolve in a unit volume of gas at a certain pressure and temperature and has units of stb/MMscf and sm³/sm³ for field and metric units, respectively. Field units values range from 1 or 2 stb/MMscf for dry gas reservoirs, up to 100 stb/MMscf for wet gas reservoirs. The empirical correlation of Meethan Vogel¹⁴⁹ can be used to estimate CGR (R_v).

Water Gas Ratio (“WGR”): Gas in the reservoir is usually saturated with water vapor, but as the gas moves up the wellbore, the temperature and pressure decrease, and water condenses out of the gas. Often, water that is produced at the surface is not formation water but is water of condensation. Confirmation of this is a low water-gas ratio (1 – 5 bbl/MMscf) and low water salinity (less than 100 ppm NaCl). The water vapor content of a gas depends on pressure, temperature, and composition of the gas. As pressure decreases, the water vapor content increases. Conversely, as temperature decreases, the water vapor content decreases. Units are stb/Mscf (although often quoted as stb/MMscf) and sm³/sm³ for field and metric units, respectively. Bubacek’s¹⁵⁰ empirical correlation for sweet natural gas (methane) may be used for gases without H₂S to calculate WGR or McKetta and Wehe’s¹⁵¹ correlation that takes into account water salinity and gas gravity. Finally Robinson et al¹⁵² and Wichert and Wichert’s¹⁵³ correlations may be used for sour natural gases (those containing H₂S).

¹⁴⁶ Meehan, D. N., and Lyons, W. K., “Programmable Calculations for Gas Compressibility”, *Oil and Gas Journal*, Oct. 8, 1979.

¹⁴⁷ Carr, N.L., Kobayshi, R., and Burrows, D.B., “Viscosity of Natural Cases Under Pressure”, *Trans. A.I.M.E.* (1954), 201 pp. 264-272.

¹⁴⁸ Lee, A.L., Gonzalez M.H. and Eakin B.E.: “The Viscosity of Natural Gases”, *J. Pet. Tech.* (1966) 18, pp. 997-1000.

¹⁴⁹ Meehan, D. N., and Vogel, E. L, *HP-41 Reservoir Engineering Manual*, PennWell Books, 1982.

¹⁵⁰ Bukacek, R.F.: “Equilibrium Moisture Content of Natural Gases,” *Bull., Inst. of Gas Technology Bulletin* (1955).

¹⁵¹ McKetta, J.J. and Wehe, A.H.: “Use This Chart for Water Vapor Content of Natural Gases,” *Petroleum Refiner* (Aug. 1958) 153-54.

¹⁵² Robinson, J. N., et al., “Estimation of the Water Content of Sour Natural Gases”, *Paper Number SPE 6098, 51st Annual Fall Technical Conference, SPE, New Orleans, Oct. 3–6, 1976.*

¹⁵³ Wichert, G. C. and Wichert, E., “Chart Estimates Water Content of Sour Natural Gas”, *O&GJ*, March 29, 1993, pp. 61-64.

Oil Fluid Properties

For oil and condensate fluids we have a similar set of properties; however, oil properties are nearly always measured experimentally in the laboratory, as although there are many correlations available, they generally have low predictability unless tuned with the laboratory measured data. Thus, the main oil properties are:

Oil Density (ρ_o): Like gas density, oil density depends on composition, temperature, and pressure at which the fluid is measured, it is defined as the ratio of mass (m) to volume (V), that is:

$$\rho_o = \frac{m}{V} \quad (8.7)$$

with units of lbm/ft^3 , kg/m^3 , and g/cm^3 . Note that density is quoted in various units even within a given unit system. In oilfield (English) units it is quite common to quote density as a gradient that is psi/ft , this is because fluid gradients are a common form of in-situ measurement that can be easily related to density by:

$$\rho_o = 144 \times \text{fluid gradient} \text{ (psi/ft)} \quad (8.8)$$

For example, for an oil density measured as 0.338 psi/ft , the actual in-situ density would simply be:

$$\rho_o = 144 \times 0.338 = 48.7 \frac{\text{lbm}}{\text{ft}^3} \quad (8.9)$$

Oil density can range from as low as about 0.270 psi/ft (0.62 g/cc) in the Malay Basin (offshore Malaysia) to as high as 0.429 psi/ft (0.99 g/cc) in the heavy oil deposits in Cold Lake (Canada). However, most common oil densities range from around $0.78 - 0.85 \text{ g/cc}$ ($0.338 - 0.369 \text{ psi/ft}$). When using oil field units most engineers drop the m in the lbm units term, and therefore quote density as lb/ft^3 (or lb/cu. ft.), as lbm is implied.

Oil Relative Density (γ_o) defines the ratio of the density of a known material to the density of reference material, at standard conditions of pressure and temperature. Standard conditions vary throughout the world, but for oil field units one normally uses 14.7 psia and 60°F , while for SI units some areas use 101.325 kPa and 15°C . The reference material for liquids is always water, and thus the **Oil Relative Density** is defined as:

$$\gamma_o = \frac{\rho_o}{\rho_w} \quad (8.10)$$

Since the density of water at 14.7 psia and 60°F is 62.4 lb/ft^3 , then we have:

$$\gamma_o = \frac{\rho_o}{62.4} \quad (8.11)$$

For SI units, with density measured in kg/m^3 , relative density is:

$$\gamma_o = \frac{\rho_o}{1000} \quad (8.12)$$

Commonly relative density is quoted with reference to the reference phase, for example, $\gamma_o = 0.780$ (water = 1).

The American Petroleum Institute ("API") classifies oils based on an API gravity (γ_{API}), or degrees API ($^\circ\text{API}$), the relationship between relative density (γ_o) of oil and API gravity (γ_{API}) is given by:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

$$\gamma_{API} = \frac{141.5}{\gamma_o} - 131.5 \quad (8.13)$$

where

γ_o = oil gravity (water = 1.0),
 ρ_o = oil density (lb/ft³ or kg/m³), and
 γ_{API} = API gravity (°API).

API density is commonly reported at the well site.

Bubble Point (Saturation Pressure): The bubble point pressure, or saturation pressure (P_b or P_{sat}), is defined as the pressure at which the first bubbles of gas appear from a liquid as the pressure declines. The correlations given in the oil formation volume section (later on), also have the equivalent bubble point pressure correlations. It should be noted that the comments addressing the application of these correlations in the oil formation volume factor section, are equally applicable here.

Gas-Oil Ratio (“GOR”): The GOR, (R_s) is defined as the number of standard cubic meters of gas that will dissolve in one stock tank meter cubed of crude oil at a certain pressure and temperature. The solubility of a natural gas in a crude oil is a strong function of the pressure, the temperature, the API gravity, and gas gravity. For a particular gas and crude oil to exist at a constant temperature, the solubility increases with pressure until the saturation pressure is reached. At the saturation pressure (bubble point pressure) all the available gases are dissolved in the oil, and the gas solubility reaches its maximum value. Rather than measuring the amount of gas that will dissolve in given stock tank crude oil as the pressure is increased, it is customary to determine the amount of gas that will come out of the sample of reservoir crude oil.

Oil Formation Volume Factor (B_o): The oil formation factor, B_o , is defined as the ratio of the volume of oil (plus the gas in solution) at the prevailing reservoir pressure and temperature, to the volume of oil at standard conditions. Oil formation volume factor is generally greater than or equal to one. Mathematically we can express the oil formation volume factor as:

$$B_o = \frac{V_{rc}}{V_{sc}} \quad (8.14)$$

where

B_o = oil formation volume factor (rb/stb or m³/Sm³),
 V_{rc} = volume of oil at reservoir conditions (rb or m³), and
 V_{sc} = volume of oil at standard conditions (stb or Sm³).

Values of oil FVF at reservoir temperature and various reservoir pressures can be obtained from a standard PVT analysis of a reservoir fluid sample. However, if this data is unavailable, the geologist/engineer must then resort to empirical correlations, such as Standing¹⁵⁴, Vasquez and Beggs¹⁵⁵, or Glasø¹⁵⁶. Generally B_o ranges from a low of 1.05 for heavy oils to a high of 2.5 rb/stb for volatile oils. **Care should be exercised when applying any of the aforementioned correlations, as there is significant variation between the various correlations. In general, a given correlation may be more appropriate for a geological basin based on comparing actual measured PVT data from other fields with the correlation predicted results.**

Oil Isothermal Compressibility (c_o): The oil isothermal compressibility is defined as the rate of change in volume with pressure increase per unit volume of liquid, all variables other than pressure being constant. Thus, by definition, isothermal compressibility of a substance is defined mathematically by the following expression:

¹⁵⁴ Standing, M. B.: “A Pressure-Volume-Temperature Correlation for Mixtures of California Oils and Gases”, *Drill, and Prod. Prac., API* (1947).

¹⁵⁵ Vasquez M. and Beggs H. D.: “Correlations for Fluid Physical Property Prediction”, *J. Pet. Tech.* (June 1980).

¹⁵⁶ Glasø, O.: “Generalized Pressure-Volume-Temperature Correlations”, *J. Pet. Tech.* (May 1980).

$$c_o = - \frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T \quad (8.15)$$

where

- c_o = oil isothermal compressibility (1/kPa),
- V = oil volume (m³), and
- P = pressure (kPa).

In terms of the oil formation volume factor, we can derive:

$$c_o = - \frac{1}{B_o} \frac{dB_o}{dP} \quad (8.16)$$

where

- c_o = oil compressibility (1/kPa), and
- B_o = oil formation volume factor (m³/Sm³).

Generally, the isothermal compressibility is determined from a laboratory PVT study. In cases where a study has not been performed, the correlations of Vasquez and Beggs¹⁵⁷ or Ramey¹⁵⁸ may be used.

Oil Viscosity (μ_o): Crude oil viscosity is an important physical property that controls and influences the flow of oil through porous media and pipes. The viscosity, in general, is defined as the internal resistance of the fluid to flow, and oil viscosity is a strong function of the temperature, pressure, oil gravity, gas gravity, and gas solubility. Oil viscosity can be classified into three main groups, based on the nature or state of the crude with respect to pressure, these are:

- **Dead Oil Viscosity** The dead oil viscosity is defined as the viscosity of the oil at atmospheric pressure and temperature conditions (no gas in solution).
- **Saturated Oil Viscosity** The saturated (bubble point) oil viscosity is defined as the viscosity of oil at the bubble point pressure and reservoir temperature.
- **Undersaturated Oil Viscosity** The undersaturated oil viscosity is defined as the viscosity of the oil at a pressure above the bubble point and at reservoir temperature.

Whenever possible, oil viscosity should be determined in the laboratory at reservoir conditions, and is normally included in a standard PVT analysis of the crude. However, if no data is available, the correlations of Beggs and Robinson¹⁵⁹ and Andrade-Guzman-Reynolds¹⁶⁰ may be employed.

¹⁵⁷ Vasquez M. and Beggs H. D.: "Correlations for Fluid Physical Property Prediction", J. Pet. Tech. (June 1980).

¹⁵⁸ Ramey, H. J.: "Rapid Methods for Estimating Reservoir Compressibilities", J. Pet. Tech. (April 1964).

¹⁵⁹ Beggs, H. D. and Robinson J. R.: "Estimating the Viscosity of Crude Oil Systems", J. Pet. Tech. (Sept., 1975).

¹⁶⁰ Andrade, E. N. da C., Nature, 125 (1930), 309.

Water Fluid Properties

Water fluid properties are similar to the oil fluid properties, and include:

Water Density (ρ_w): Water density has the same definition as oil density, equation (8.7), with units of lbm/ft³, kg/m³, and g/cm³. Fresh water density at 14.7 psia and 60 °F is 62.4 lb/ft³, and for SI units at 101.3 kPa and 15 °C, the density is 1000 kg/m³. Note that density is quoted in various units even within a given unit system. In oilfield (English) units it is quite common to quote density as a gradient that is psi/ft, when laboratory data or actual water samples are unavailable. The density of formation water at reservoir conditions can be estimated roughly (usually to within +/-10%) from correlations such as McCain's¹⁶¹ and ¹⁶² correlations.

Water Formation Volume Factor (B_w): The water formation volume factor has the same definition as for the oil formation volume factor. Normally this property is estimated from correlations based on salt content, for example the Numbere et al¹⁶³ correlation.

Water Viscosity (μ_w): Again, the definition is the same as for oil and is normally estimated from correlations based on salt content, for example by the Meehan¹⁶⁴.

8.2.2 CLASSIFICATION OF RESERVOIRS BY FLUID TYPE

Reservoir classification by fluid type includes the following reservoir fluids:

- Low Shrinkage Oil Reservoirs.
- High Shrinkage Oil Reservoirs.
- Volatile Oil Reservoirs.
- Dry Gas Reservoirs.
- Wet Gas Reservoirs.
- Retrograde Condensate Gas Reservoirs.

The following sections briefly outline the various reservoir types based on their fluid properties.

¹⁶¹ McCain, W.D. Jr.: McCain, W.D. Jr. 1990. *The Properties of Petroleum Fluids*, second edition. Tulsa, Oklahoma: PennWell Books.

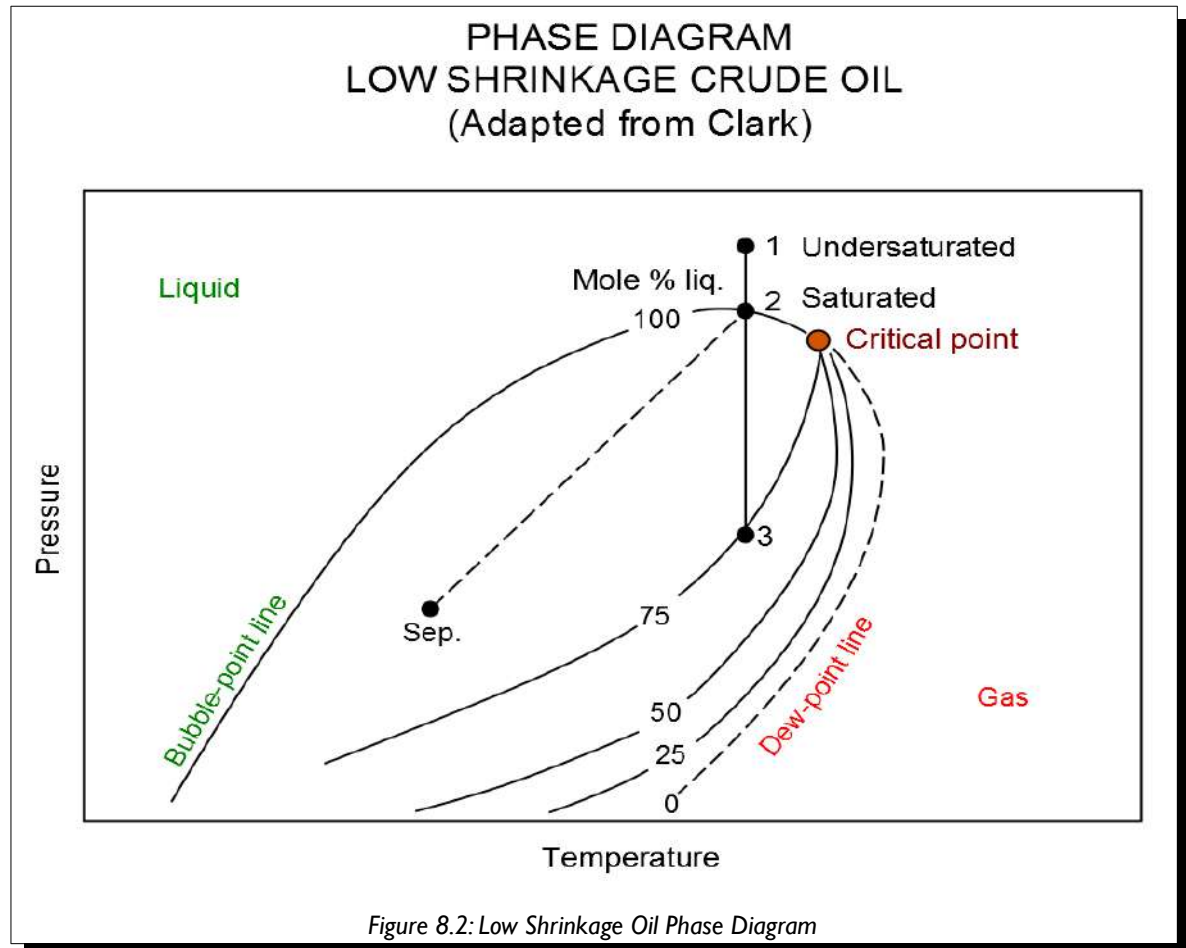
¹⁶² Cain Jr., W.D. 1991. *Reservoir-Fluid Property Correlations-State of the Art (includes associated papers 23583 and 23594)*. SPE Res Eng 6 (2): 266-272. SPE-18571-PA. <http://dx.doi.org/10.2118/18571-PA>.

¹⁶³ Numbere, D., Brigham, W. E., and Standing, M. B., *Correlations for Physical Properties of Petroleum Reservoir Brines*, Petroleum Research Institute, Stanford University, November, 1977).

¹⁶⁴ Meehan, D. N., "Estimating Water Viscosity at Reservoir Conditions", *Petroleum Engineer*, July 1980.

Low Shrinkage Oil Reservoirs

The phase diagram for a typical low shrinkage oil is shown in Figure 8.2 (after Clark¹⁶⁵), in this case the two-phase region covers a wide range of pressure and temperature (the critical temperature is above the reservoir temperature). The line marked 1 and 2 shows the effect of pressure reduction, with constant reservoir temperature, as the reservoir is produced. The dashed line shows the conditions encountered as the fluid travels up the tubing string and into the surface facilities.



If the reservoir pressure and temperature is at point 2 on the phase diagram, then the oil is at bubble point pressure and is said to be saturated, i.e., the oil contains the maximum amount of dissolved gas at the given conditions. However, if the reservoir conditions are at point 1 on the phase diagram, then the oil is said to be undersaturated. That is the oil is able to dissolve more gas at the current reservoir pressure and temperature.

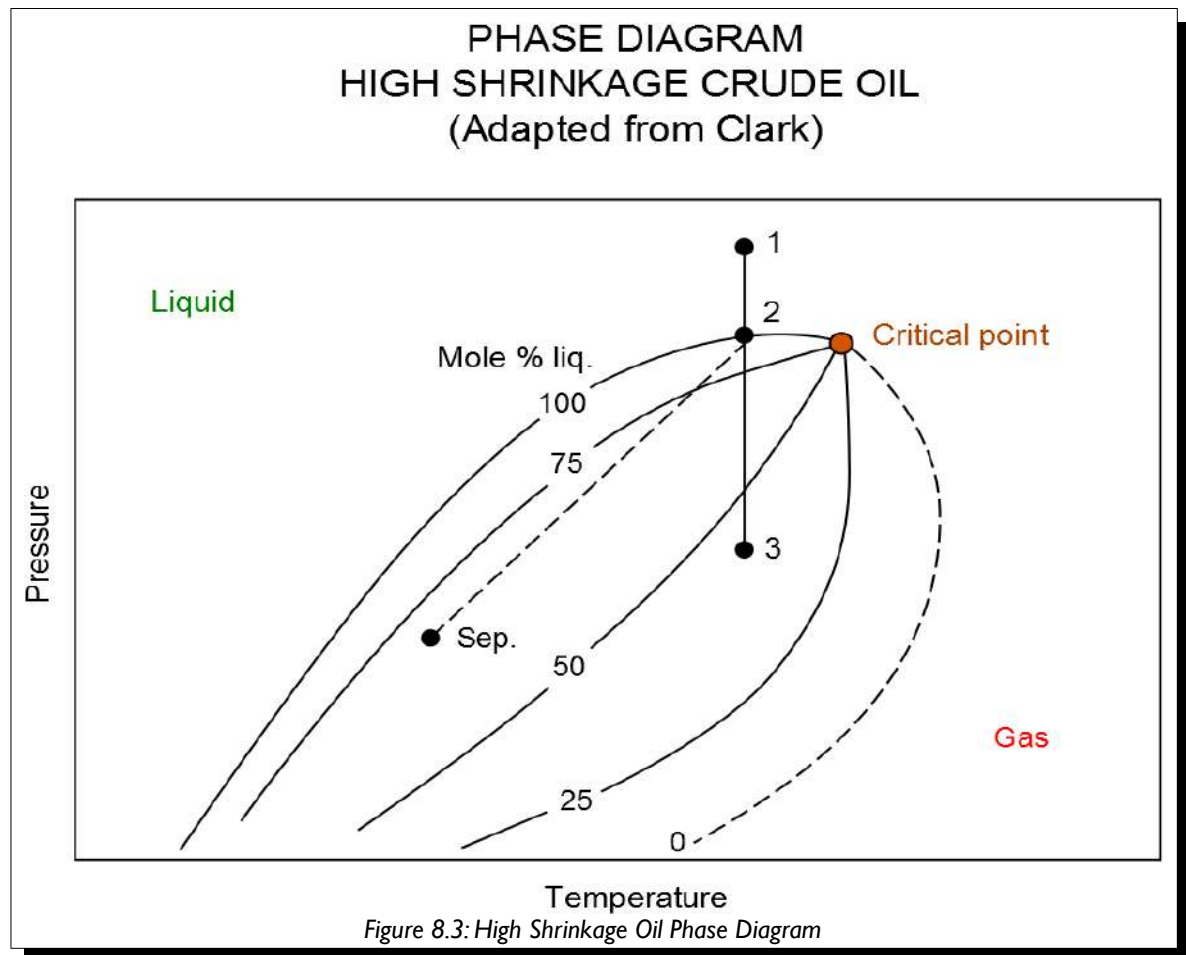
The term low shrinkage oil is derived from the fact that oil shrinkage is small, i.e., with an oil formation volume factors less than 1.5. This oil can be described as having a broad based phase envelope, high percentage of liquid, high proportion of heavier hydrocarbons, GOR's less than 500 scf/stb or 100 Sm³/m³, oil gravity 30 °API or heavier, and the stock tank liquid being black or a deep color. If the oil has a very low GOR and therefore has an oil formation volumes factor close to one, as observed in heavy oil reservoirs, then it is common to model this type of reservoir neglecting the gas phase. That is this type of reservoir fluid is typically modeled as a two-phase oil-water system using a black-oil formulation, which results in greater

¹⁶⁵ Clark, Norman, "It Pays to Know Your Petroleum", *World Oil*, March and April, 1953.

computational efficient due to having only two phases as opposed to the three phases in an oil-gas-water system.

High Shrinkage Oil Reservoirs

A high shrinkage oil's phase diagram is shown in Figure 8.3.

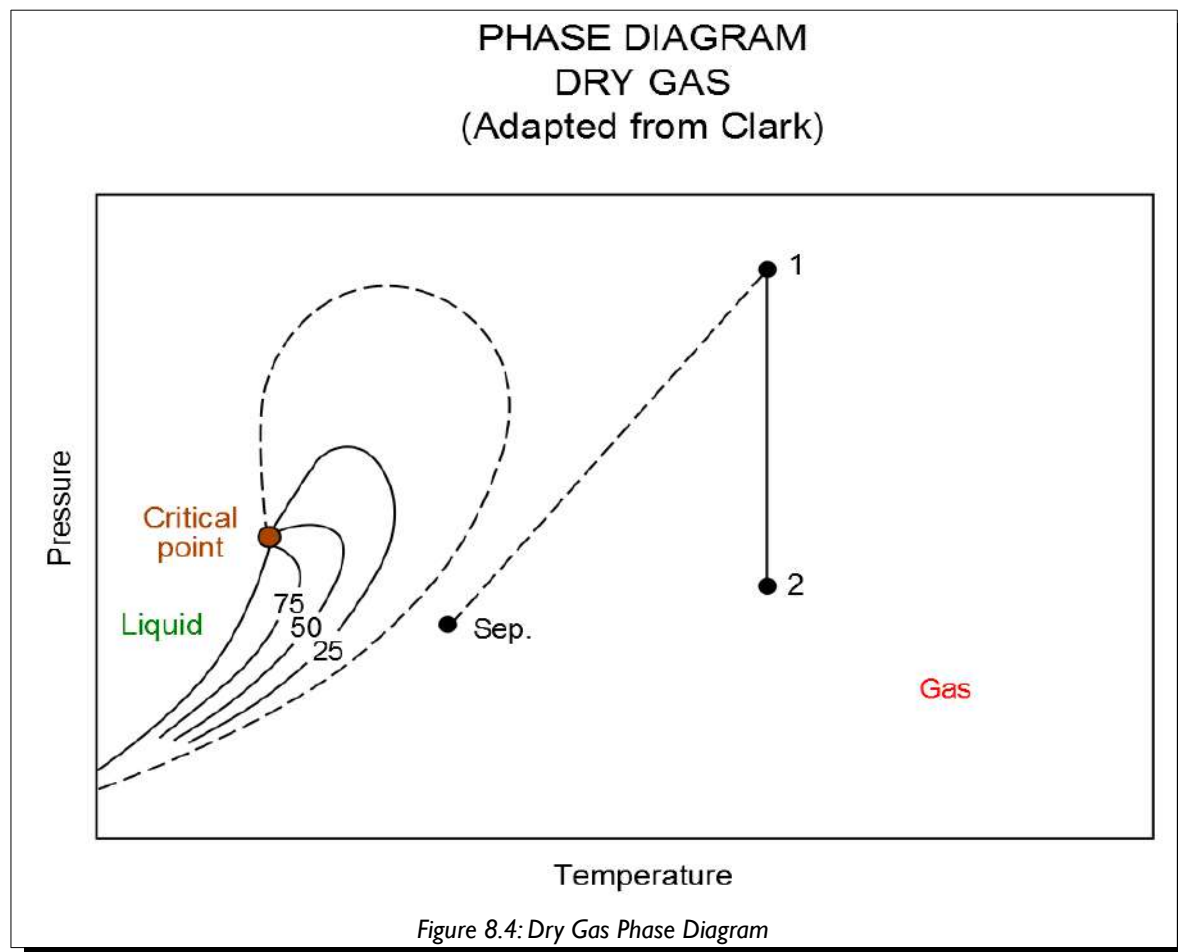


This type of reservoir fluid is characterized by not so broad a phase envelope, fewer heavier hydrocarbons, deep colored stock tank fluid, oil gravity lighter than 50 °API, GOR less than 8000 scf/stb or 1500 Sm³/m³, and oil formation volume factors greater than 1.5 Sm³/m³.

Dry Gas Reservoirs

Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface.

The phase diagram for a dry gas reservoir is illustrated in Figure 8.4.



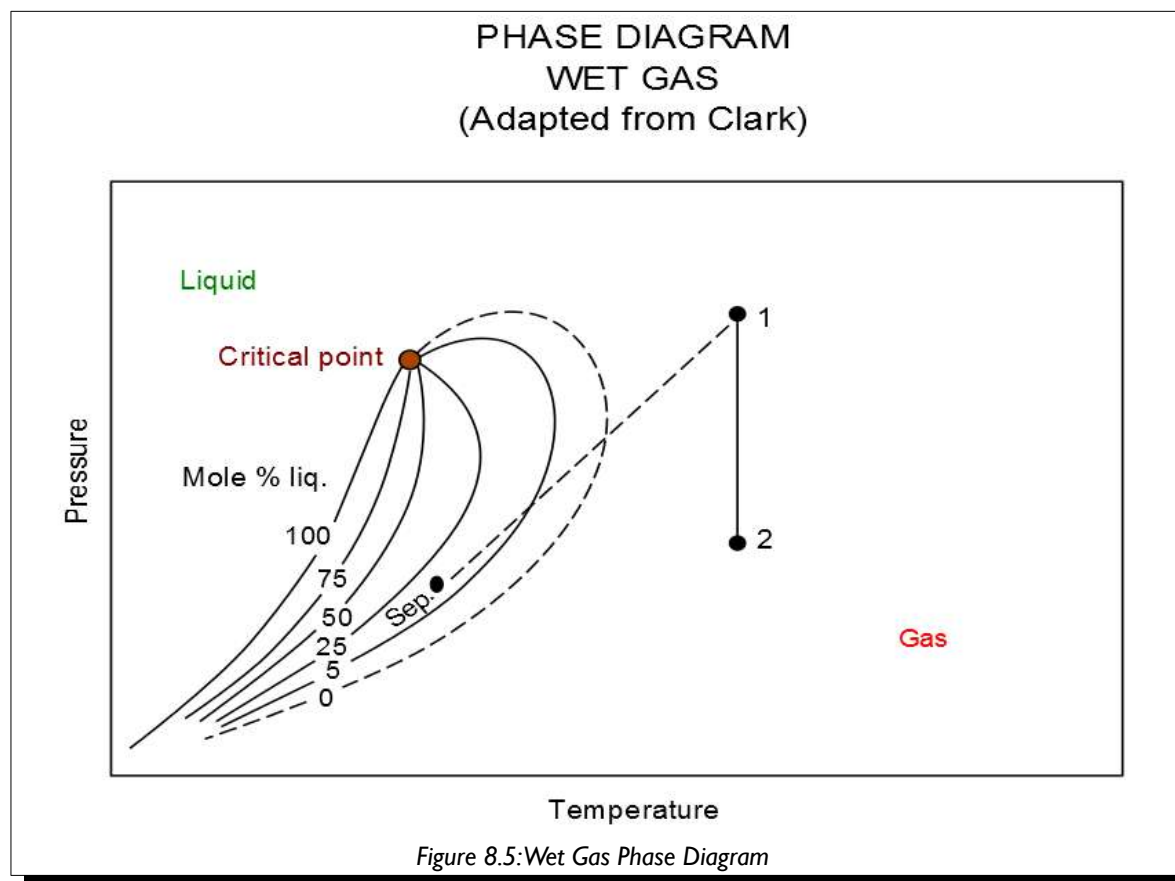
Notice that the separator conditions lie outside the phase envelope in the gas region. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has a GOR greater than 100,000 scf/stb or 18,000 Sm³/m³.

These type of reservoirs are typically modeled as two-phase gas-water systems in black-oil formulations, which results in greater computational efficient due to having only two phases as oppose to three phases in an oil-gas-water system. Also it is not uncommon to model dry gas reservoirs using analytical models, for example material balance, as often the reservoirs behave like “tanks” either with or without aquifer influx.

Wet Gas Reservoirs

Natural gas that contains significant heavy hydrocarbons such as propane, butane, and other liquid hydrocarbons is known as wet gas or rich gas. The general rule of thumb is if the gas contains less methane (typically less than 85% methane) and more ethane, and other more complex hydrocarbons, it is labelled as wet gas.

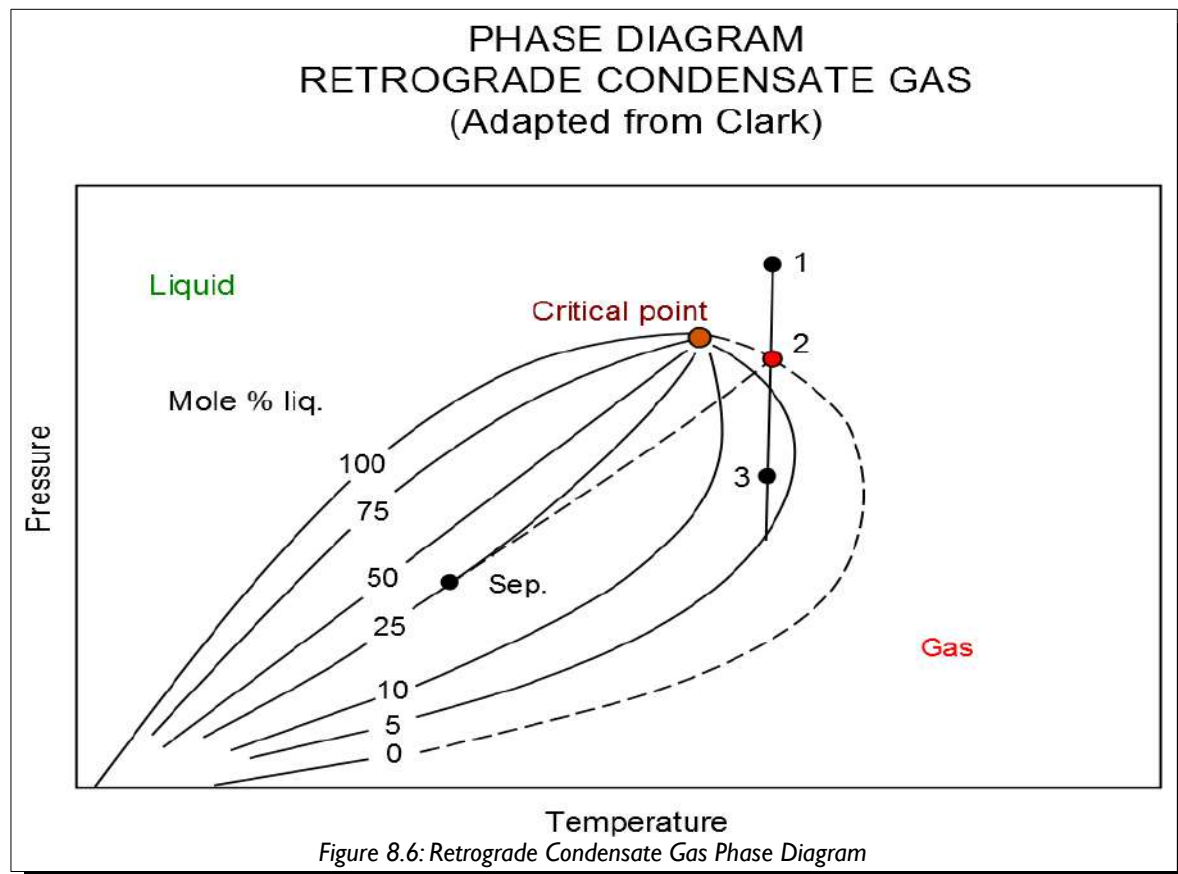
As can be seen from the figure below, wet gas exists in the gas phase in the reservoir throughout depletion of the reservoir pressure. However, the separator conditions lie in the two-phase region, resulting in liquid production (condensate).



Wet gas normally has a GOR less than 100,000 scf/stb or 18,000 Sm³/m³, with the condensate having a gravity greater than 50 °API.

Retrograde Condensate Gas Reservoirs

If the reservoir temperature lies between the critical point and the cricondentherm, a retrograde reservoir exists. As the pressure declines to point 2, which is the dew point, liquids begin to form in the reservoir. As the pressure further declines to point 3, the liquid yield increases. Further reduction in the pressure causes the liquid to vaporize back into the gas phase.



The liquid drop out in the reservoir can cause a well's productivity to be reduced, due to the system moving from a two-phase relative permeability effect, to a three phase effect. This is normally diagnosed by the well having a high skin factor.

This type of fluid has more lighter hydrocarbons than high shrinkage oils, and fewer heavier hydrocarbons than high shrinkage oils. The gravity can be as high as 60 °API, with GOR up to 70,000 scf/stb or 13,000 Sm³/m³. Generally the stock tank liquid is water-white, or slightly colored.

Reservoir Classification

The following tables outline some general parameters used to classify reservoir fluids. Table 8.1 outlines some general producing characteristics used to define reservoir fluids.

Item	Crude Oil	Volatile Oil	Gas Condensate	Gas
Initial Separator Gas-Liquid Ratio (scf/stb)	< 2,000	2,000 – 6,000	6,000 – 100,000 Generally 7,000 – 15,000	> 100,000 Usually below 300,000
Color of Produced Liquid	Black to Light Green	Dark Straw Colored	Colorless to Light Straw Colored	Colorless
Usual API Gravity Range of Produced Fluid	10 – 45+	40 – 50	45 – 65	–
Composition of Produced Fluid	C ₇₊ : usually > 40%	C ₇₊ : 10 – 40%	C ₇₊ : 2 – 10%	Primarily Methane
Oil Formation Volume Factor (rb/stb)	< 2.0 frequently < 1.5	> 2.0 often > 2.5		

Table 8.1: General Reservoir Fluid Classification

In terms of fluid composition reservoir fluids can be indicated by the values shown in Table 8.2 below.

Item	Gas	Gas Condensate	Volatile Oil	Crude Oil
Methane (mole%)	96	87	64	49
Ethane (mole %)	3	4	8	3
Heptanes Plus (mole %)	1	3	15	42
Solution GOR (scf/stb)	105,000	18,200	2,000	625
Stock Tank Oil °API	68	61	50	34

Table 8.2: Reservoir Fluids Composition Indicators

In addition reservoir oil is further divided into Light, Medium, Heavy, Extra Heavy, and Bituman (Wilmon¹⁶⁶) as shown in Table 8.3 below:

Temperature	STOCK TANK OIL °API	In Situ Viscosity (cp)
Light	> 31.1	
Medium	22.3 – 31.1	
Heavy	10.0 – 22.3	100 – 10,000
Extra Heavy	< 10.0	< 10,000
Bitumen	< 10.0	> 10,000

Table 8.3: Oil Sub-Classification (after Wilmon)

¹⁶⁶ Wilmon, G. J.: “Economic Outlook for Extra-heavy Oils, Natural Bitumens and Shale Oils”, World Petroleum Congress (1987).

8.2.3 FLUID PROPERTY TABLES

Table 8.4 outlines the oil, gas, and water fluid types that can be active in the model, together with the related RUNSPEC section keywords that activate the phases, versus the PVT keywords that can be used to define the PVT behavior. The table also includes several water types, the standard water phase used in most simulations models, the Brine water phase used in the Brine Tracking model to track the flow of brine through the simulation grid and the effect of brine on reservoir performance, and finally OPM Flow's Vaporized Water phase that is used in the simulator's Salt Precipitation model. The latter water phase is not available in the commercial simulator.

Fluid Property Keywords Versus Oil, Gas & Water Fluid Types								
Item	Oil		Gas				Water	
Fluid Type	Dead Oil	Live Oil	Dry Gas	Wet Gas	Dry / Wet Gas with Vaporized Water	CO ₂ (EOR)	Water	Brine
RUNSPEC Keywords	OIL	OIL	GAS	GAS	GAS	GAS	WATER	WATER
		DISGAS		VAPOIL	None / VAPOIL			
					VAPWAT ⁵			
					WATER			
					BRINE			BRINE
Pressure Dependent PVT	PVCDO	PVCO	PVDG	PVTG	PVTWSALT	PVTSOL	PVTW	PVTWSALT
	PVDO	PVTO	PVZG		PVTGW ⁵	PVDS		
					PVTGWO ⁵			
Miscellaneous	RSCONST		RVCONST		RWGSALT ⁵			
	RSCONSTT		RVCONSTT					
Surface Density	DENSITY				BDENSITY	SDENSITY	DENSITY	BDENSITY
	GRAVITY						GRAVITY	
Notes:								
1) Cells colored in gray with no mnemonic indicate that there is no keyword for this combination.								
2) Cells colored orange show keywords that have not been implemented in OPM Flow for the given fluid type.								
3) When two or more keywords are stated for the RUNSPEC keyword for a given fluid type, then all are required to define the given phase.								
4) When two keywords are stated for the Pressure Dependent PVT and Miscellaneous data for a given fluid type, then either one can be used to define the PVT behavior for the given phase.								
5) OPM Flow specific keywords.								

Table 8.4: Fluid Property Keywords versus Oil, Gas and Water Fluid Type

For the Dead Oil phases the RSCONST and RSCONSTT keywords are used to set a constant gas-oil ratio (R_s). In this case the R_s is independent of the reservoir pressure and R_s is also negligible, as in for example heavy oil type fluids.

Similarly for the Dry Gas phase, where the RVCONST and RVCONSTT keywords are used to set a condensate-gas ratio (R_v) which is independent of the reservoir pressure and is also negligible, as in for example dry gas type fluids.

For the Vaporized Water phase and model, both dry and wet gas can be incorporated. Note that the PVTGW, PVTGWO, RWGSALT, and VAPWAT keywords are OPM Flow specific keywords.

In addition for the Brine phase, then either the SALT or SALTVD keywords in the SOLUTION section should be used to define the initial equilibration salt concentration for the model.

CO₂ can either be used as Enhanced Oil Recovery (“EOR”) fluid by injecting the CO₂ into an oil reservoir, or injected for CO₂ storage. For the former, CO₂ is declared via the GAS keyword in the RUNSPEC section and the PVT data is entered via the standard gas fluid properties for the hydrocarbon gas and either the PVDS or the PVT SOL keywords are used to describe the interaction of the in situ oil and the injected CO₂.

In addition to the above the ROCK keyword should be used to define the rock compressibility.

Similarly, Table 8.5 outlines the fluid property data keywords for the CO₂ storage, foam, polymer, solvent, and MICP phases. Note that for these phases multiple keywords can be used to define the desired property behavior.

Fluid Property Keywords Versus CO ₂ (Storage), Foam, Polymer, Solvent, And MICP Fluid Types					
Fluid Type	CO ₂ (Storage)	Foam	Polymer	Solvent	MICP ⁷
RUNSPEC Keywords	CO2STORE	FOAM	POLYMER	SOLVENT	MICP
	GAS				WATER
	WATER				
Pressure Dependent PVT	Not required, calculated via correlations			PVDS	PVTW
				PVT SOL ⁶	
Surface Density				SDENSITY	DENSITY
Miscellaneous	SALINITY ⁵	FOAMADS	PLMIXPAR		BIOFARA
		FOAMDCYO	PLYADS		
		FOAMDCYW	PLYADSS		
		FOAMFCN	PLYATEMP		
		FOAMFRM	PLYCAMAX		
		FOAMFSC	PLYDHFLF		
		FOAMFSO	PLYESAL		
		FOAMFST	PLYKRRF		
		FOAMFSW	PLYMAX		
		FOAMMOB	PLYMWINJ8		
		FOAMMOBP	PLYRMDEN		
		FOAMMOBS	PLYROCK		
		FOAMOPTS	PLYSHEAR		
		FOAMROCK	PLYSHLOG		
			PLYTRRF		

Fluid Property Keywords Versus CO ₂ (Storage), Foam, Polymer, Solvent, And MICP Fluid Types					
Fluid Type	CO ₂ (Storage)	Foam	Polymer	Solvent	MICP ⁷
			PLYTRRFA		
			PLYVISC		
			PLYVISCs		
			PLYVISCT		
			PLYVMH8		
			PLYVSCST		
			SKPRPOLY ⁸		
Miscellaneous			SKPRWAT ⁸		
Notes: - Cells colored in gray with no mnemonic indicate that there is no keyword for this combination. - Cells colored orange show keywords that have not been implemented in OPM Flow. - When two or more keywords are stated for the RUNSPEC keyword for a given fluid type, then all are required to define the given phase. - When multiple keywords are stated for the Miscellaneous data for a given fluid type, then several keywords can be used to defined the fluid property behavior. - SALINITY is OPM Flow specific keyword, that defines the salinity for all cells in the model. - PVTSOL is used to model CO₂ interaction with in situ oil only and is an OPM Flow specific keyword. - OPM Flow's implementation of the Microbial Induced Calcite Precipitation ("MICP") model used to investigate leakage remediation. All the keywords are specific to OPM Flow. - OPM Flow has an additional formulation to the standard polymer flooding model, compared to the commercial simulator, known as the Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity. This model is activated via the POLYMER and POLYMW keywords in the RUNSPEC section. The model does not account for non-Newtonian flow; the apparent viscosity is simply set equal to the zero-shear viscosity. Secondly, the standard polymer property data keywords: PLYROCK, PLYADS, PLYMAX, etc., are still required to fully describe the polymer fluid.					

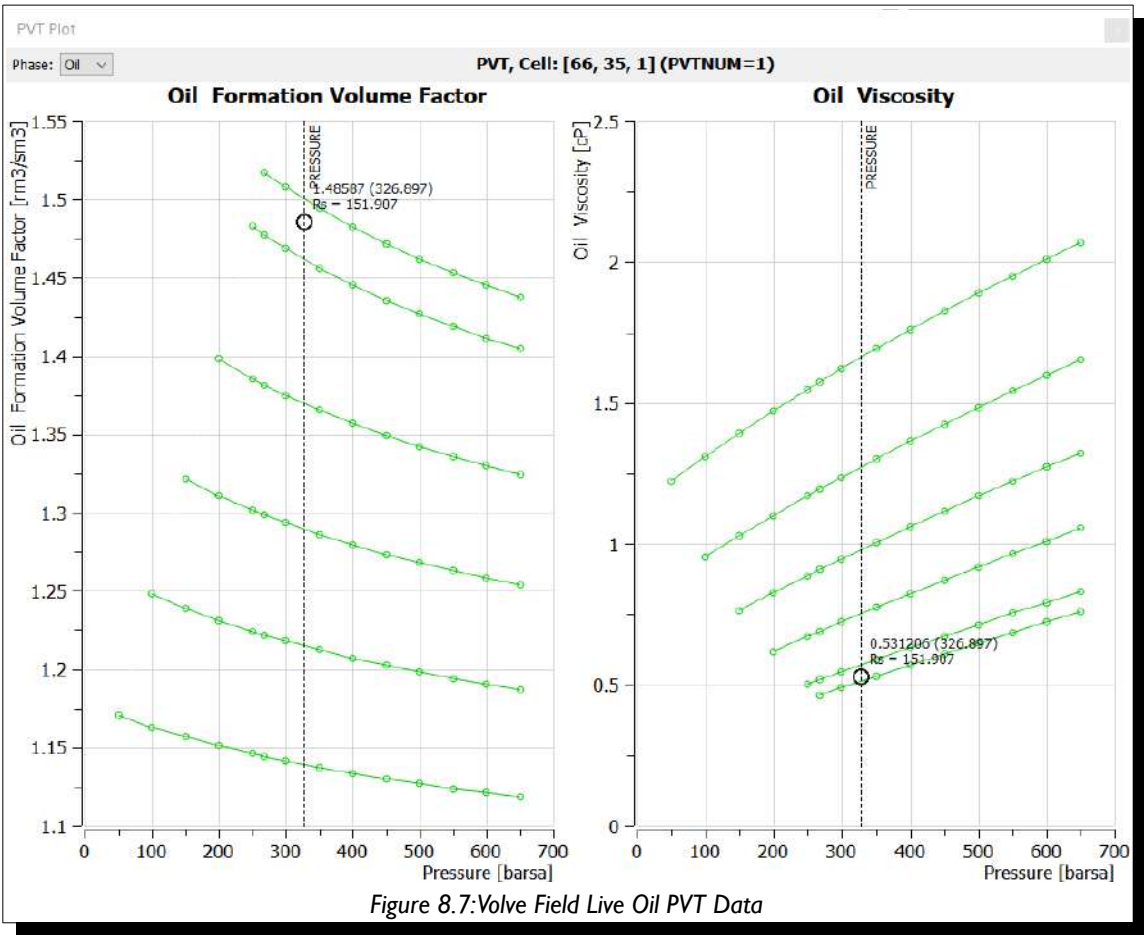
Table 8.5: Fluid Property Keywords versus CO₂ (Storage), Foam, Polymer, Solvent, and MICP Fluid Types

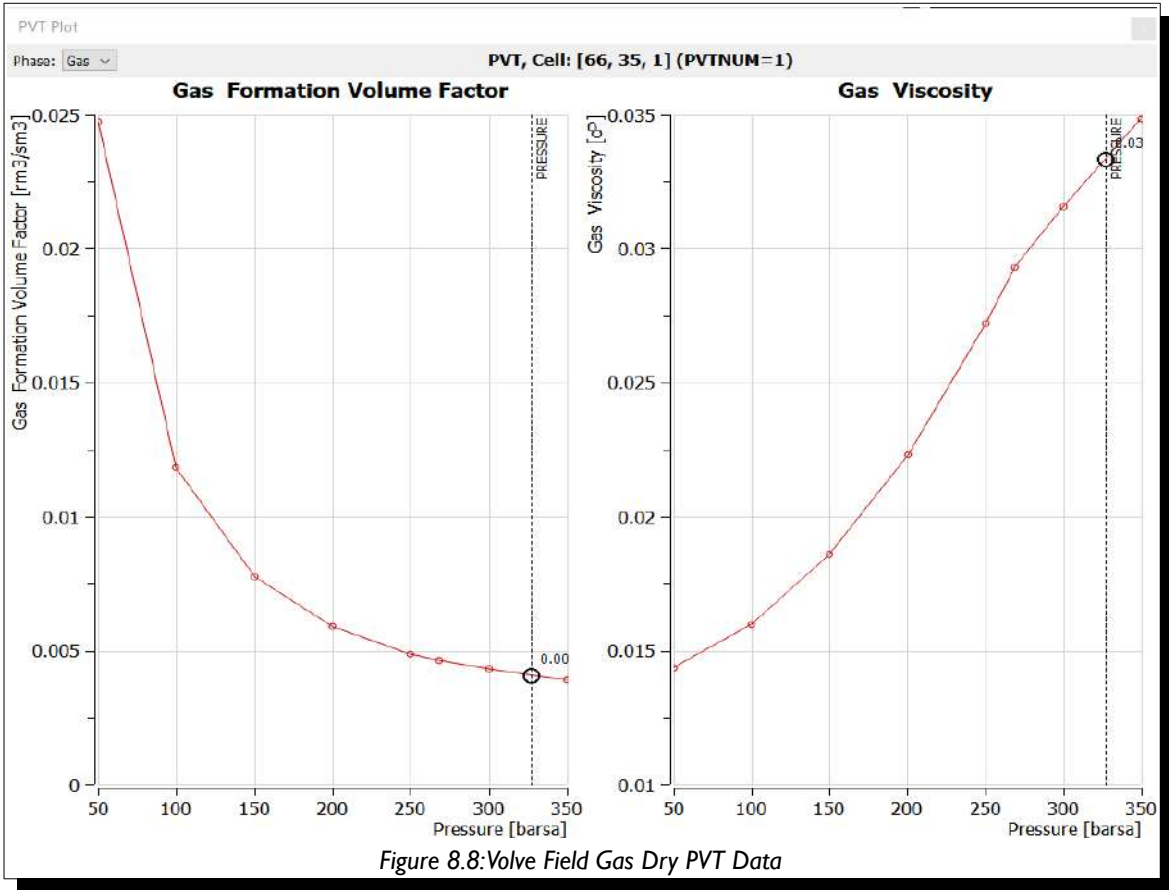
As mentioned previously, CO₂ can either be used as an EOR fluid by injecting the gas into an oil reservoir, or injected for CO₂ storage. For the latter, CO₂ is declared via the GAS and CO2STORE keywords in the RUNSPEC section, and, except for the SALINITY keyword, all the remaining PVT data is calculated automatically via correlations by the simulator. A full description of the underlying PVT models is described by Sandve et al.¹⁶⁷

Typical live oil and dry gas PVT data from the Volve¹⁶⁸ field is shown in Figure 8.7 and Figure 8.8, respectively. To fully define the live oil, dry gas, and water PVT properties for Volve the DENSITY, PVDG, PVTO, and PVTW keywords are employed.

¹⁶⁷ Sandve, T. H., Gasda, S. E., Rasmussen, A., and Rustad, A. B. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS II – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

¹⁶⁸ The Volve Data was approved for data sharing in 2018 by the initiative of the last Operating company, Equinor and approved by the license partners ExxonMobil E&P Norway AS and Bayerngas Norge AS in the end of 2017.





8.2.4 SATURATION TABLES (RELATIVE PERMEABILITY AND CAPILLARY PRESSURE TABLES)

Saturation tables contain the relative permeability and capillary pressure data as a function of fluid saturation and are used to both initialize the model and to describe multi-phase flow in the reservoir. Multiple saturation tables can be entered and allocated to various areas in the model, based on rock typing. Alternatively, a limited number of saturation tables may be entered and allocated by region and combined with end-point scaling option to enable a more robust reservoir rock characterization.

Note

Confusion abounds with the terms: critical water saturation, connate water saturation, and irreducible water saturation definitions, as the terms are extensively used interchangeably to define the maximum water saturation at which the water phase will remain immobile. For reference:

Connate Water Saturation: Water trapped in the interstices of the sediments at the time of deposition, as opposed to water that migrated into the formations after deposition¹⁶⁹. Normally the nomenclature of S_{wc} is used to describe this type of water saturation.

Irreducible Water Saturation: The fraction of the pore volume occupied by water in a reservoir at maximum hydrocarbon saturation. It represents water that has not been displaced by hydrocarbons because it is trapped by adhering to rock surfaces, trapped in small pore spaces and narrow interstices, etc. Irreducible water saturation is an equilibrium situation. It differs from residual water saturation, the value measured by core analysis, because of filtrate invasion and the gas expansion that occurs when a core is removed from the bottom of the hole to the surface. Also called immovable water¹⁷⁰. Here the nomenclature for irreducible water saturation is S_{wirr} .

Critical Water Saturation: Defines the largest water saturation for which the water relative permeability is zero.

In addition to the above definitions, the simulator uses the term SWL, which is lowest water saturation in a relative permeability table. Thus, SWL can represent either the connate water saturation or the irreducible water saturation in the relative permeability tables and functions.

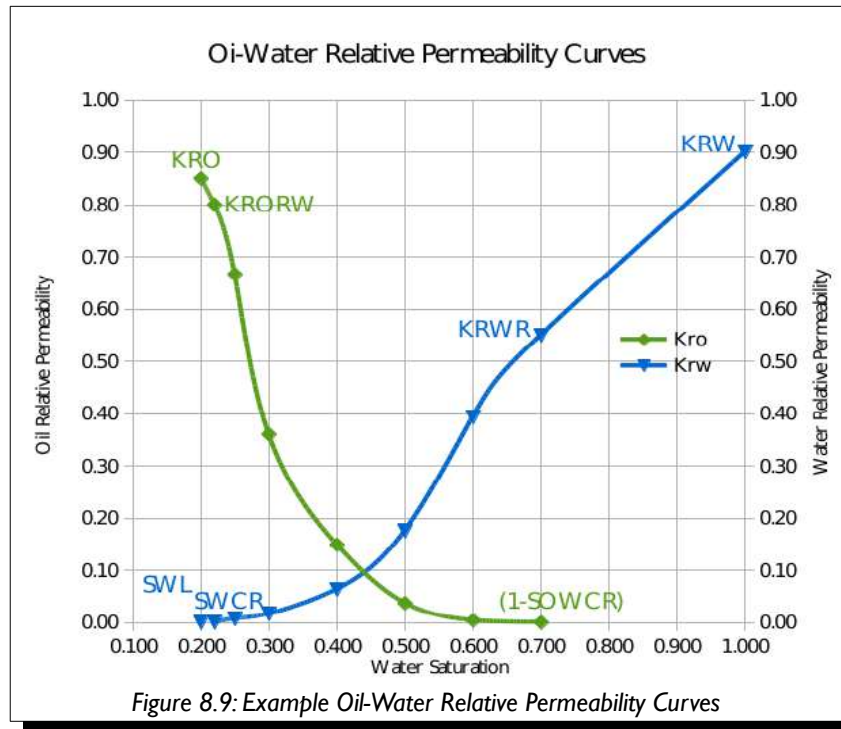
Secondly, the term irreducible water saturation is replaced by term critical water saturation (SWCR) in the relative permeability tables and function, as both terms are equivalent with regards to the mobility of the water phase. This terminology is also consistent with the critical values of the oil and gas phases, which are dependent on the displacing phase.

Thirdly, it is not uncommon for SWL to be set equal to the critical water saturation; however, care should be taken in this instance if end-point scaling is being used to ensure that the cell scaled relative permeability curves are as one might expect.

A typical set of oil-water relative permeability curves is shown in Figure 8.9 indicating the oil end-point data (KRO, KRORW, and $(1 - SOWCR)$) and the water end-point data (KRWR, KRW, SWL, and SWCR).

¹⁶⁹ https://wiki.seg.org/wiki/Dictionary:Connate_water

¹⁷⁰ https://wiki.seg.org/wiki/Dictionary:Irreducible_water_saturation

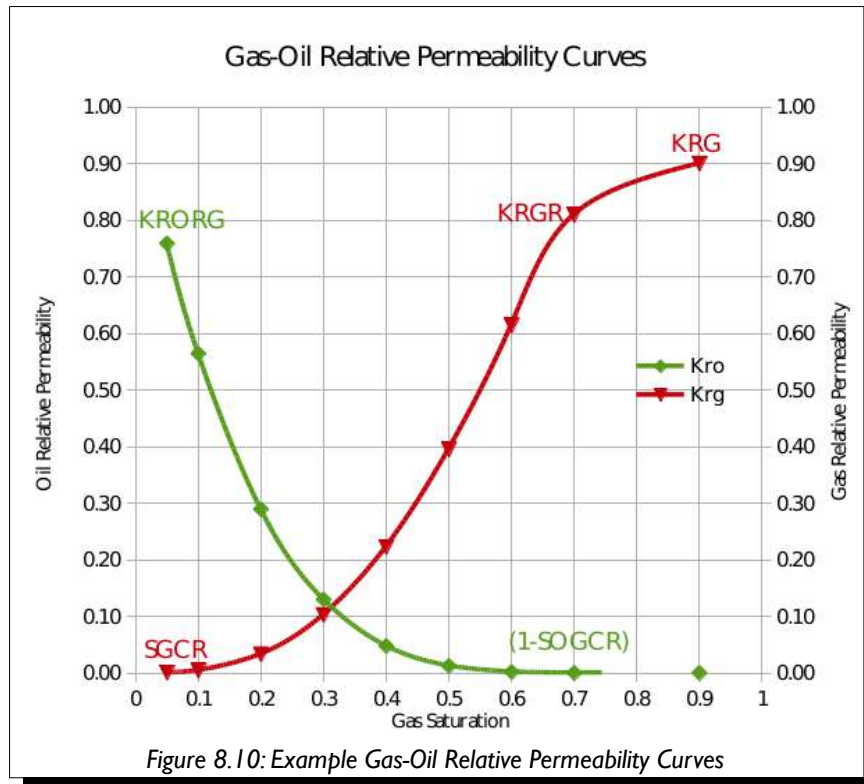


The associated oil-water end-point definitions are outlined in the following table:

Type	End-Point Keyword	Oil-Water End-Point Definitions
Saturation	SOWCR	Critical oil-in-water saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-water system.
	SWL	Connate water saturation, that is the lowest water saturation in a water saturation function table.
	SWCR	Critical water saturation, that is the largest water saturation for which the water relative permeability is zero.
	SWU	Maximum water saturation in a water saturation table. This is commonly set equal to one, unless there is a residual oil saturation below the defined OWC.
Relative Permeability	KRO	Relative permeability of oil at the maximum oil saturation.
	KRORW	Relative permeability of oil at the critical water saturation
	KRW	Relative permeability of water at the maximum water saturation (normally the maximum water saturation is one).
	KRWR	Relative permeability of water at the residual oil saturation (or at the residual gas saturation in a gas-water run).
Capillary Pressure	SWLPC	Capillary pressure connate water saturation, that is the lowest water saturation in a water saturation function table to be scaled independently from the SWL relative permeability end-point value.

Table 8.6: Oil-Water Relative Permeability End-Point Data Definitions

Similarly for gas-oil systems, Figure 8.10 illustrates a typical gas-oil relative permeability set of curves indicating the oil end-point data (KRORG and $(1 - \text{SOGCR})$) and the gas end-point data (KRGR, KRG, and SGCR).

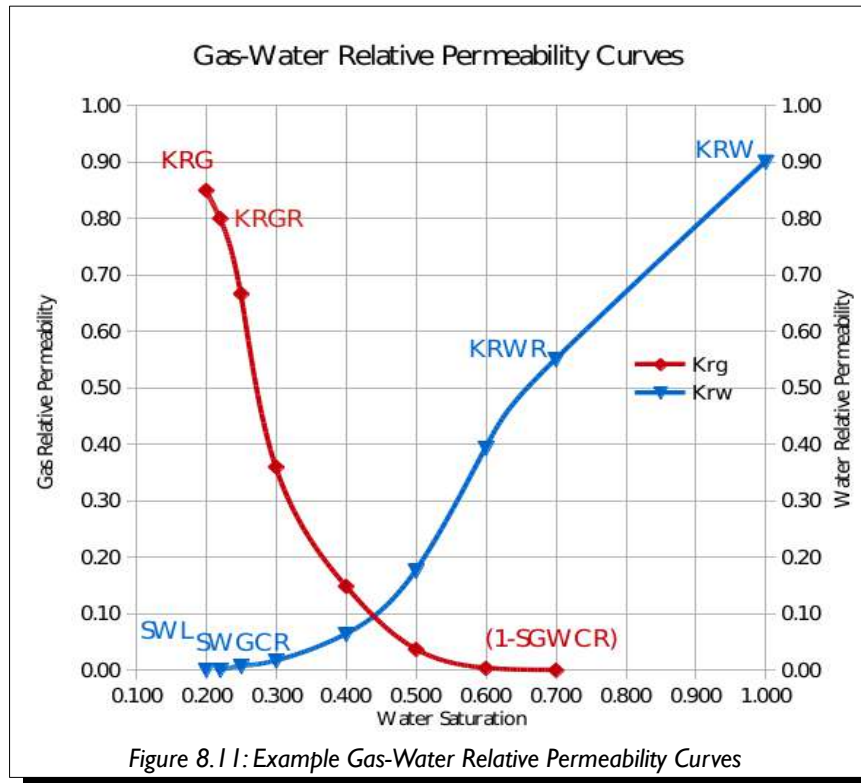


The gas-oil end-point definitions are outlined in the following table:

Type	End-Point Keyword	Gas-Oil End-Point Definitions
Saturation	SOGCR	Critical oil-in-gas saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-gas-connate-water system.
	SGL	Connate gas saturation, that is the lowest gas saturation in a gas saturation function table.
	SGCR	Critical gas saturation, that is the largest gas saturation for which the gas relative permeability is zero.
	SGU	Maximum gas saturation in a gas saturation table. This is generally set equal to one minus the connate water saturation.
Relative Permeability	KRG	Relative permeability of gas at the maximum gas saturation.
	KRGR	Relative permeability of gas at the residual oil saturation (or the critical water saturation in a gas-water run).
	KRORG	Relative permeability of oil at the critical gas saturation.
Capillary Pressure	SGLPC	Capillary pressure connate gas saturation, that is the smallest gas saturation in a gas saturation function table to be scaled independently from the SGL relative permeability end-point value.

Table 8.7: Gas-Oil Relative Permeability End-Point Data Definitions

Finally, for two phase gas-water systems, Figure 8.11 illustrates a typical set of gas-water relative permeability curves indicating the gas end-point data (KRG, KRGR, and $(1 - \text{SGWCR})$) and the water end-point data (KRW, KRWR, SWL, and SWGCR).



The gas-oil end-point definitions are outlined in the following table:

Type	End-Point Keyword	Gas-Water End-Point Definitions
Saturation	SGL	Connate gas saturation, that is the lowest gas saturation in a gas saturation function table.
	SGU	Maximum gas saturation in a gas saturation table.
	SGWCR	Critical gas-in-water saturation, that is the largest gas saturation for which the gas relative permeability is zero in a gas-water system.
	SWL	Connate water saturation, that is the lowest water saturation in a water saturation function table.
	SWGCR	Critical water saturation, that is the largest water saturation for which the water relative permeability is zero in a gas-water system.
	SWU	Maximum water saturation in a water saturation table.
Relative Permeability	KRG	Relative permeability of gas at the maximum gas saturation.
	KRGR	Relative permeability of gas at the residual oil saturation (or the critical water saturation in a gas-water run).

Type	End-Point Keyword	Gas-Water End-Point Definitions
	KRWR	Relative permeability of water at the residual oil saturation (or the residual gas saturation in a gas-water run).
Capillary Pressure	SWLPC	Capillary pressure connate water saturation, that is the smallest water saturation in a water saturation function table to be scaled independently from the SWL relative permeability end-point value.

Table 8.8: Gas-Water Relative Permeability End-Point Data Definitions

End-point scaling is activated in the [RUNSPEC](#) section with the **ENDSCALE** keyword and the data used to apply end-point scaling is entered in the [PROPS](#) section using the end-point keywords defined in Table 8.6, Table 8.7, and Table 8.8 to define each grid block's end-point data. There are also direction dependent versions of these keywords for when directional end-point scaling has been activated. For example for critical water saturation, **SWCR** is used with non-directional end-point scaling and the **SWCRX**, **SWCRX-**, **SWCRY**, **SWCRY-**, **SWCRZ**, and **SWCRZ-** series of keywords is used for when directional end-point scaling has been activated. In addition, there is also the facility to incorporate end-point scaling based on the drainage and/or imbibition process which again can be either non-directional or directional.

Note

If the hysteresis model option has been activated on the **SATOPTS** keyword in the **RUNSPEC** section, then the standard end-point scaling arrays, for example **KRORW**, can be used to scale the drainage tables, and the equivalent imbibition arrays suffixed with the letter **I**, for example **IKRORW**, can be used to scale the imbibition tables.

However, if the hysteresis model option has not been activated, then only the standard keywords can be used to define the end-point scaling parameters, that is **KRORW** should be used and not **IKRORW**, even though an imbibition process with the wetting phase (normally water) increasing is usually being modelled.

Saturation functions can be entered via several keywords consisting of two format types as depicted in the following table:

Format Type One				Format Type Two			
Keyword	Oil	Gas	Water	Keyword	Oil	Gas	Water
SGOF	Pcog			SGFN ³		Pcog	
SLGOF	Pcog			SGWFN		Pcgw	
SWOF	Pcwo		Pcwo	SOF2 ⁴	No Pc		
				SOF3 ⁵	No Pc		
				SOF32D	No Pc		
				SWFN			Pcwo
OPM Flow LET ⁶				Format Type Three ⁷			
Keyword	Oil	Gas	Water	Keyword	Oil	Gas	Water
SGOFLET	Pcog			GSF		Pcgw	

SWGFLET		Pc _{gw}	WSF			No Pc
SWOFLET	Pc _{wo}		Pc _{wo}			
Notes: 1) Cells colored in gray with no mnemonic indicate that there is no keyword for this combination. 2) Cells colored orange show keywords that have not been implemented in OPM Flow. 3) In gas-water systems, the gas-water Pc_{gw} data should be entered on the SWFN keyword and the Pc_{og} on the SGFN keyword should be set to zero. 4) The SOF2 keyword defines the relative permeability in oil-gas and oil-water runs only, and the miscible hydrocarbon in SOLVENT runs. This keyword should not be used to define the oil relative permeability when oil, gas and water are present. 5) Defines oil relative permeability with respect to water and oil relative permeability with respect to gas. 6) OPM Flow specific keywords that implement the LET family of saturation functions, see Lomeland et al.¹⁷¹ 7) Format Type Three keywords are available in the commercial compositional simulator only, and will cause an error if used in the commercial black-oil simulator.						

Table 8.9: Saturation Table Formats and Phases

Note that only one format type can be used in each run to define the required saturation functions, that is one cannot combine the keywords from the different format types in the same input deck.

Figure 8.12 shows an unscaled oil-water relative permeability curve from the Volve model, and Figure 8.13 displays the corresponding scaled relative permeability curves after applying the end-point scaling parameters. Similarly for the gas-oil curves, Figure 8.14 illustrates the unscaled curves and Figure 8.15 the scaled curves. The plots indicate that the end-point scaling has been applied correctly.

¹⁷¹ Lomeland F., Ebeltoft E., and Thomas W.H., 2005. A New Versatile Relative Permeability Correlation. Paper SCA2005-32 presented at the International Symposium of the Society of Core Analysts held in Toronto, Canada, 21-25 August, 2005.

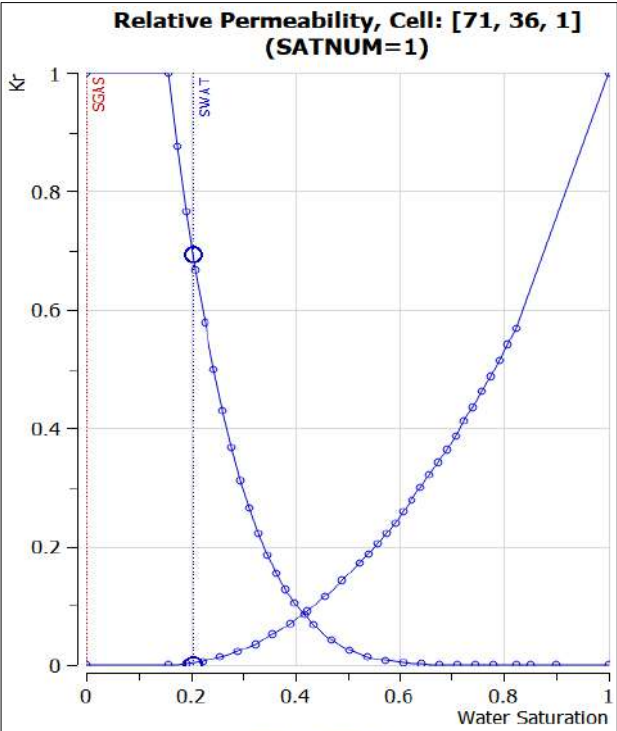


Figure 8.12:Volve Oil-Water Relative Permeability - Unscaled

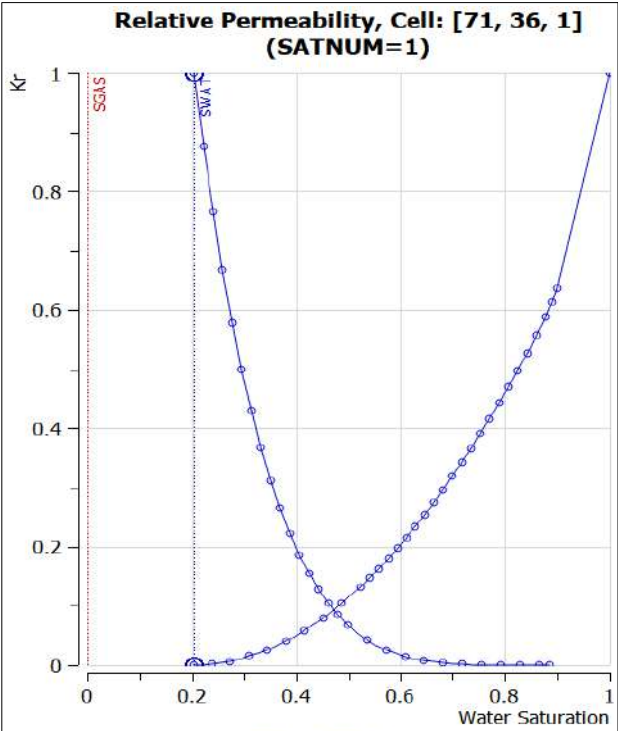


Figure 8.13:Volve Oil-Water Relative Permeability - Scaled

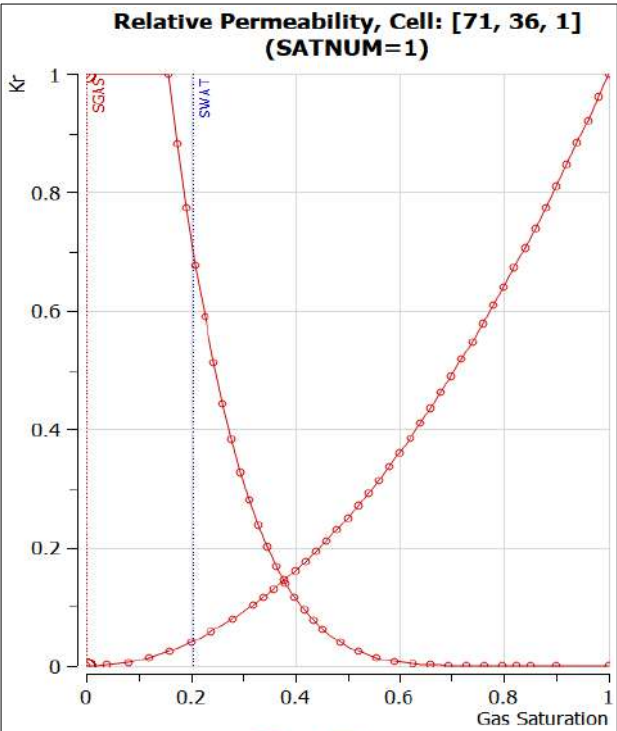


Figure 8.14: Volve Gas-Oil Relative Permeability - Unscaled

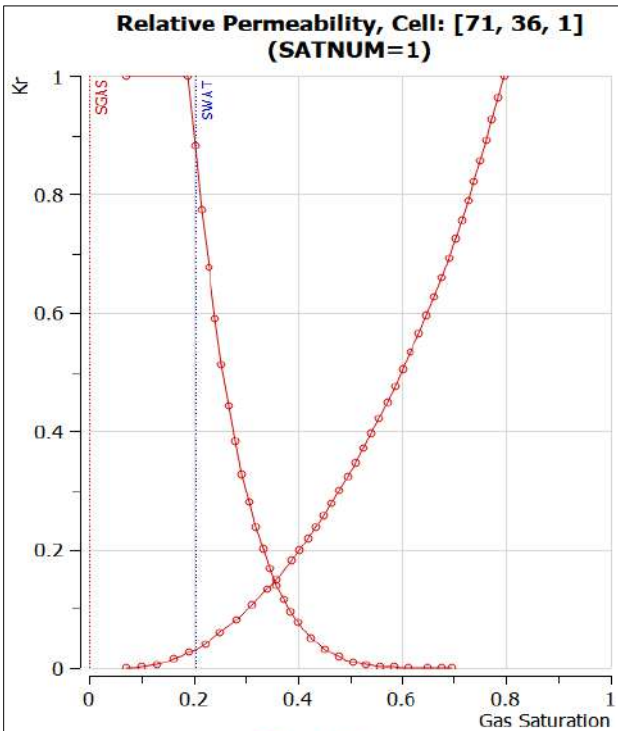


Figure 8.15: Volve Gas-Oil Relative Permeability - Scaled

8.2.5 SATURATION TABLE GENERATION - COREY CURVES

As mentioned in the [2.4.1 Data Quality and Assurance](#) section, Corey relative permeability curves are often used to "smooth" the intended relative permeability curves to be used in the simulator in order to improve convergence.

Corey¹⁷² combined the work of Purcell¹⁷³ and Burdine¹⁷⁴ that was widely accepted for its simplicity. His original equations were developed for the drainage cycle in water-wet sandstones, but have also been used in carbonate formations. Corey's original water-oil equations were as follows:

$$k_{ro}(S_w) = \left[\frac{1 - S_w}{1 - S_{wcr}} \right]^{n_o} \quad (8.17)$$

$$k_{rw}(S_w) = \left[\frac{S_w - S_{wcr}}{1 - S_{wcr}} \right]^{n_w} \quad (8.18)$$

where

$k_{ro}(S_w)$ = relative permeability to oil,
 $k_{rw}(S_w)$ = relative permeability to water,
 n_o = Corey oil exponent, set to four in the original paper,
 n_w = Corey water exponent, set to four in the original paper,
 S_w = water saturation, and
 S_{wcr} = critical water saturation.

The denominator in equations (8.17) and (8.18) scales the water saturation to the mobile water phase. There are several forms of these equations, with the most common normalizing the saturation over the mobile hydrocarbon phase, as depicted in equations (8.19) and (8.20).

$$k_{ro}(S_w) = k_{row} \left[\frac{1 - S_w - S_{orw}}{1 - S_{orw} - S_{wcr}} \right]^{n_o} \quad (8.19)$$

$$k_{rw}(S_w) = k_{rww} \left[\frac{S_w - S_{wcr}}{1 - S_{orw} - S_{wcr}} \right]^{n_w} \quad (8.20)$$

where

k_{row} = maximum oil relative permeability at S_{wcr} ,
 k_{rww} = maximum water relative permeability at S_{orw} ,
 S_{wcr} = critical water saturation, and
 S_{orw} = residual oil saturation under a water flood (SOWCR).

Similar equations exist for gas-oil and water-oil systems. As mentioned above, the denominator in the equations, normalizes the saturation to the mobile phase, as a consequence, it is still necessary to extend the resulting Corey water curve to 100% water saturation in order to correctly model the water leg. In terms of the values for the various Corey exponents, Table 8.10 offers some guidelines based on the rock's wettability.

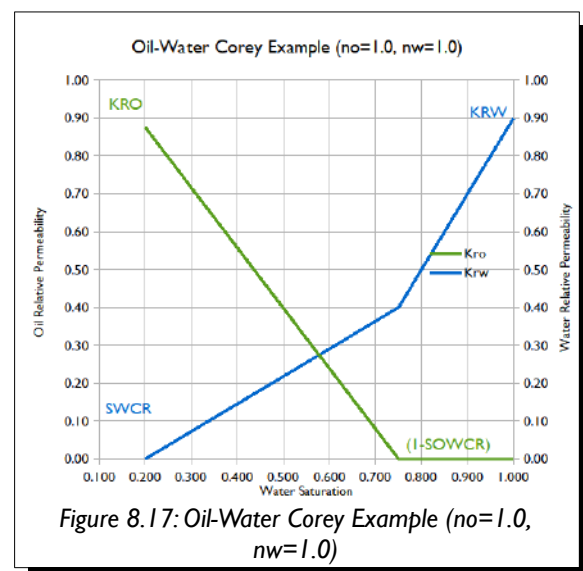
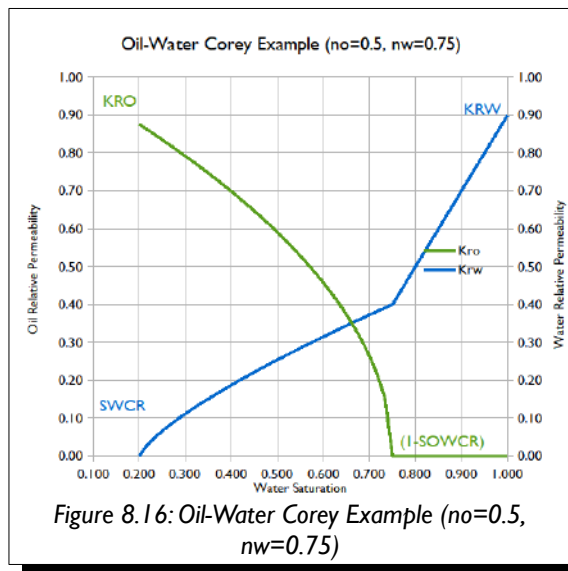
¹⁷² Corey, A.T.: "The Interrelation Between gas and Oil Relative Permeabilities", *Production Mon.*, 19. 38. (1954).

¹⁷³ Purcell, W. R., "Capillary Pressures- Their Measurement Using Mercury and the Calculation of Permeability Therefrom", *Transactions AIME*, 186, 39 (1949).

¹⁷⁴ Burdine, N.T., "Relative Permeability Calculations from Pore Size Distribution Data", *Transactions AIME*, 198, 71 (1953).

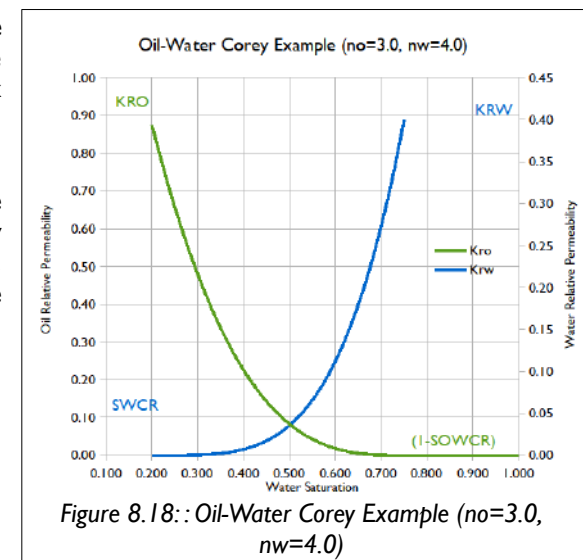
Corey Guide (Water-Oil)			Corey Guide (Gas-Oil)		
Wettability	Kro	Krw	Wettability	Kro	Krg
Water wet	2 - 4	5 - 8	Water wet		
Intermediate	3 - 6	3 - 5	Intermediate		
Oil wet	6 - 8	2 - 3	Oil wet	4 - 8	2 - 4

Table 8.10: Corey Exponent Values



As can be seen in Figure 8.16 to Figure 8.18, the Corey exponents less than one result in a concave curve, and values great than one result in a convex curve, where as a value of one gives a straight line.

Unfortunately, neither OPM Flow nor the commercial simulator, unlike some other simulators, support the direct entry of Corey type curves, and thus Corey curves have to be generated outside of the simulator. Fortunately, OPM Flow does support the more advanced and flexible LET family of models instead, which are described in the next section.



8.2.6 SATURATION TABLE GENERATION - LET FUNCTIONS

OPM Flow has implemented the LET family of saturation functions models, based on the work of Lomeland et al. ^{175-176 177 and 178} via the *SGOFLET* and *SWOFLET* keywords in this section. The keywords are used as replacements for the SGOF and SWOF keywords for three-phase oil-gas-water systems, and the SGWFN keyword for gas-water systems. Note there are two versions of the LET functions, LET¹⁷⁵ for two-phase flowing conditions and LETx¹⁷⁹ for three-phase flowing conditions. All three keywords implement the LET version for two phase systems.

The functions are dependent on the drainage and imbibition cycle of the wetting phase as well as drainage and inhibition cycle number, since a reservoir may undergo several flooding events. To account for this the system defines the flooding event using the three saturations: S_w , S_o , and S_g together with the state of the three saturations during the flooding event. The saturation state can be increasing, decreasing, or constant for a given flooding event cycle number (n). Thus, $S_w(D)$, $S_o(I)$, $S_g(C)I$, or $DICI$, means the water phase is decreasing, the oil phase is increasing, and the gas phase is constant for the primary or first cycle (n equals one). This is the case when oil is migrating into the reservoir rock and displacing the initial water contained with the reservoir. The various flooding processes are outlined in Table 8.11

Modeling Of Flooding Processes				
Oil Field		Gas Field		Comment
Oil-Water Oil Leg	Oil-Gas Oil-Leg	Gas-Water Gas Cap	Oil-Gas Gas Cap	Nomenclature LET(S_w, S_o, S_g, n)
DICI	CDI2	DCI1	CDI2	Primary starting point.
IDC2	CID3	ICD2	CID3	Second event
DIC3	CID4	DCI3	CID4	Third event

Table 8.11: LET Modeling of Flooding Processes

Thus, DICI is the well-known primary (first inflow number) drainage for an oil/water system, and IDC2 is the well-known (second general inflow number) imbibition for an oil/water system.

The following section describe the various flooding scenarios and the standard LET equations used in the literature.

¹⁷⁵ Lomeland F., Ebeltoft E. and Thomas W.H., 2005. A New Versatile Relative Permeability Correlation. Paper SCA2005-32 presented at the International Symposium of the Society of Core Analysts held in Toronto, Canada, 21-25 August, 2005.

¹⁷⁶ Lomeland F. and Ebeltoft E., 2008. A New Versatile Capillary Pressure Correlation. Paper SCA2008-08 presented at the International Symposium of the Society of Core Analysts held in Abu Dhabi, UAE, 29 Oct – 2 Nov., 2008.

¹⁷⁷ Lomeland F., Hasanov B., Ebeltoft E. and Berge M., 2012. A Versatile Representation of Up-scaled Relative Permeability for Field Applications. Paper SPE 154487-MS presented at the EAGE Annual Conference & Exhibition incorporating SPE Europec held in Copenhagen, Denmark, 4-7 June 2012.

¹⁷⁸ Lomeland F., 2018. Overview Of The Let Family Of Versatile Correlations For Flow Functions. Paper SCA2018-056 presented at the International Symposium of the Society of Core Analysts held in Trondheim, Norway, 27-30 August 2018.

¹⁷⁹ Lomeland F. and Ebeltoft E., 2013. Versatile Three-phase Correlations for Relative Permeability and Capillary Pressure. Paper SCA2013-034 presented at the International Symposium of the Society of Core Analysts held in Napa Valley, California, USA, 16-19 September, 2013.

Oil Displacing Water: Primary Drainage Cycle (DIC1)

For the Primary Drainage, assuming water is the wetting phase, then, $S_w(D)$, $S_o(I)$, $S_g(C)I$, or $DIC1$, means the water phase is decreasing, the oil phase is increasing and the gas phase is constant for the primary or first cycle (n equals one). This is the case for when oil is migrating into the reservoir rock and displacing the initial water contained within the reservoir. Under these circumstance the LET normalized water saturation and relative permeability functions are defined as:

$$K_{row} = \frac{K_{rot}(1 - S_{wp})^{L_o}}{(1 - S_{wp})^{L_o} + E_o S_{wp}^{T_o}} \quad (8.21)$$

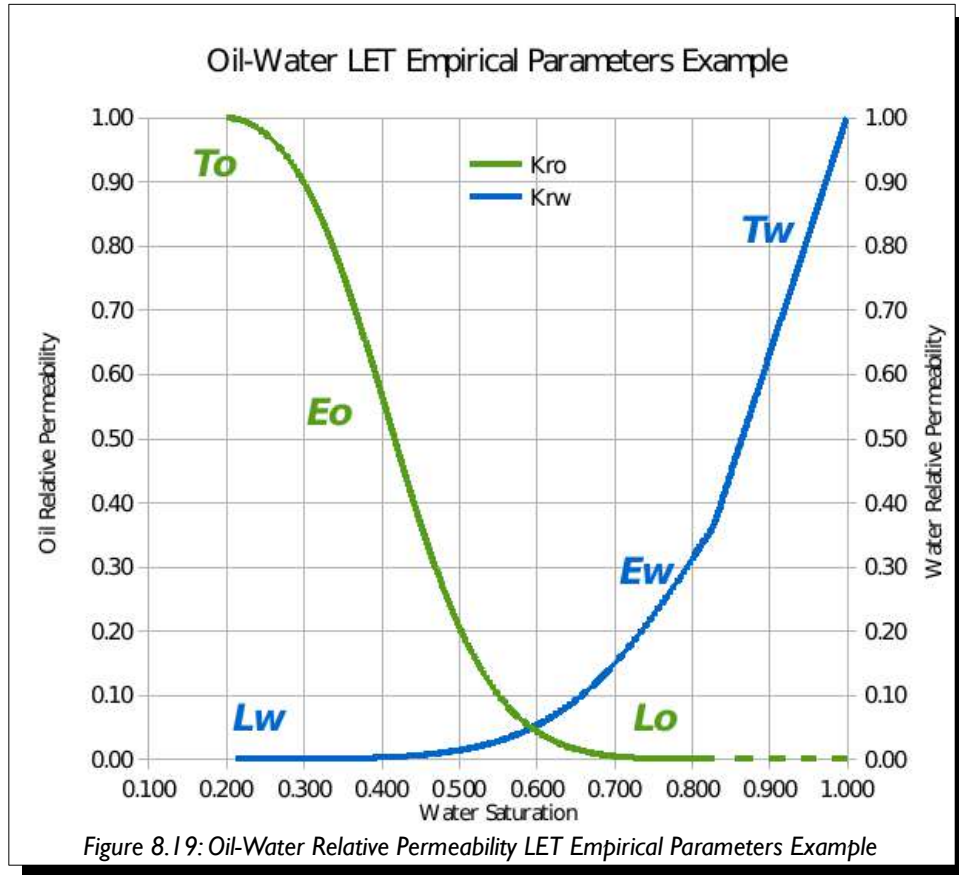
$$K_{rw} = \frac{K_{rwt} S_w^{L_w}}{S_w^{L_w} + E_w (1 - S_{wp})^{T_w}} \quad (8.22)$$

$$S_{wp} = \frac{S_w - S_{wirr}}{1 - S_{wirr}} \quad (8.23)$$

where

E_α	= empirical E parameter for the α phase,
K_{row}	= oil relative permeability with respect to water saturation,
K_{rot}	= base or top oil relative permeability, used to scale the maximum oil permeability,
K_{rw}	= water relative permeability,
K_{rwt}	= base or top water relative permeability, used to scale the maximum water permeability,
L_α	= empirical L parameter for the α phase,
T_α	= empirical T parameter for the α phase,
S_w	= water saturation,
S_{wirr}	= irreducible water saturation, and
S_{wp}	= normalize water saturation.

The LET empirical parameters E, L, and T influence the shape of the relative permeability curves, as depicted in Figure 8.19



For the primary drainage or first cycle, DIC1, with the water phase decreasing, the oil phase increasing and the gas phase constant, the LET capillary pressure functions are:

$$Y = \frac{P_{cow} - P_{ct}}{P_{cir} - P_{ct}} \text{ and } Y = F(S_{wp}) \text{ where } S_{wirr} \leq S_w \leq 1.0 \quad (8.24)$$

$$P_{cow}(S_w = S_{wirr}) = P_{cir} \text{ and } P_{cow}(S_w = 1.0) = P_{cit} \quad (8.25)$$

$$F = \frac{(1 - S_{wp})^L}{(1 - S_{wp})^L + ES_{wp}^T} \text{ and } S_{wp} = \frac{S_w - S_{wirr}}{1 - S_{wirr}} \quad (8.26)$$

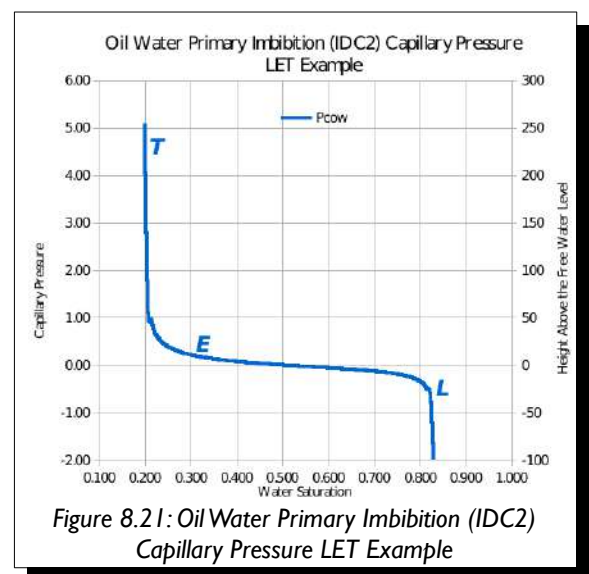
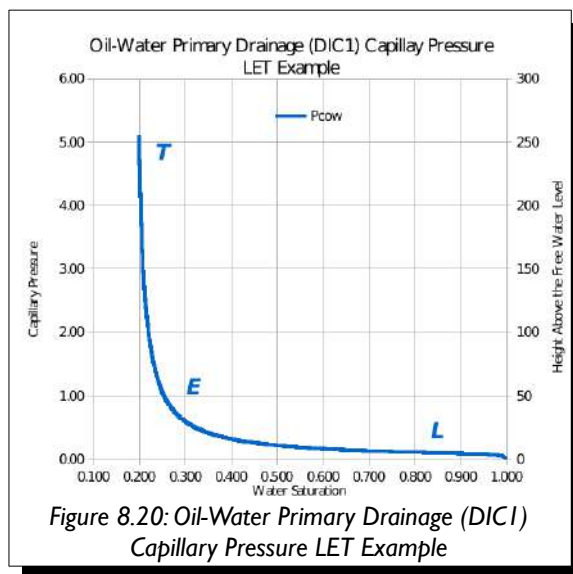
where

- E = empirical E parameter,
- L = empirical L parameter,
- P_{cir} = capillary pressure at irreducible water saturation,
- P_{ct} = capillary threshold / entry pressure,
- T = empirical T parameter,
- S_w = water saturation,
- S_{wirr} = irreducible water saturation, and
- S_{wp} = normalize water saturation.

The capillary pressure LET functions with either L or T equal to one are called a semi-simple LET functions, and LET functions with both L and T equal to one are called a simple LET function. The simple LET function is equal to the P_c correlation of Honapour et al.¹⁸⁰ except for the arrangement of the empirical coefficients.

The L parameter describes the lower part of the curve, whereas the T parameter describes the top part of the curve in a similar to how the L parameter describes the lower part of the curve. Finally, the E parameter describes the position of the slope of the curve, that is the elevation. An E value of one is neutral, and the position of the slope is governed by the L and T parameters in this case. Whereas, increasing the value of the E moves the slope towards the high end of the curve, and decreasing the value of the E moves the slope towards the lower end of the curve. Lombard et al.¹⁷⁶ noted that experience using the LET correlation indicate that L, E, and T values should be greater than or equal to 0.01, with no upper limit.

Figure 8.20 shows a typical LET derived drainage oil-water capillary pressure curve, and Figure 8.21 shows a typical imbibition curves, the later is discussed in the following section.



OPM has implemented a more restrictive form of the general LET functions, more akin to the original paper¹⁷⁵, as depicted in Figure 8.20. Table 8.12 relates the general LET equations presented previously with the parameters on the [SWOFLET](#) keyword.

LET Function (DIC1) versus SWOFLET Keyword					
SWOFLET	LET Function	SWOFLET	LET Function	SWOFLET	LET Function
SWL	S_{wirr}	SORW		PCIR	P_{cir}
SWCR	S_{wirr}	SOWCR		PCIT	P_{ct}
LWAT	L_w	LOIL	L_o	LPC	L
EWAT	E_w	EOIL	E_o	EPC	E
TWAT	T_w	TOIL	T_o	TPC	T
KRTWAT	K_{rwt}	KRTOIL	K_{rot}		

¹⁸⁰ Honarpour M.M., Djabbarah N.F. and Kralik J.G., 2004. Expert-Based Methodology for Primary Drainage Capillary Pressure Measurements and Modeling. Paper SPE-88709 presented at the 11th Abu Dhabi International Petroleum Exhibition and Conference held in Abu Dhabi, UAE, 10-13 Oct., 2004.

LET Function (DICI) versus SWOFLET Keyword					
SWOFLET	LET Function	SWOFLET	LET Function	SWOFLET	LET Function
Notes: 1) The current implementation of the LET functions is restricted to a single branch capillary pressure curves as shown in Figure 8.20, and not the multi-branch curve shown in Figure 8.21.					

Table 8.12: LET Functions (DICI) versus SWOFLET Keyword

Water Displacing Oil: Primary Imbibition Cycle (IDC2)

Similarly for the oil producing phase, where the water phase is increasing either by replacing the displaced oil or via water injection, that is $S_w(I)$, $S_o(D)$, $S_g(C)$, for the second flooding event, IDC2. Under this scenario the LET equations are:

$$K_{row} = \frac{K_{rot}(1 - S_{wn})^{L_o}}{(1 - S_{wn})^{L_o} + E_o S_{wn}^{T_o}} \quad (8.27)$$

$$K_{rw} = \frac{K_{rwr} S_{wn}^{L_w}}{S_{wn}^{L_w} + E_w (1 - S_{wn})^{T_w}} \quad (8.28)$$

$$S_{wn} = \frac{S_w - S_{wirr}}{1 - S_{orw} - S_{wirr}} \quad (8.29)$$

where

E_α	= empirical E parameter for the α phase,
K_{row}	= oil relative permeability with respect to water saturation,
K_{rot}	= base or top oil relative permeability, used to scale the maximum oil permeability,
K_{rw}	= water relative permeability,
K_{rwr}	= base or top water relative permeability, used to scale the maximum water permeability,
L_α	= empirical L parameter for the α phase,
T_α	= empirical T parameter for the α phase,
S_{orw}	= residual oil saturation under water displacement,
S_w	= water saturation,
S_{wirr}	= irreducible water saturation, and
S_{wn}	= normalize water saturation.

For the primary imbibition or second cycle, IDC2, with the water phase increasing, the oil phase decreasing and the gas phase constant, the LET capillary pressure functions are:

$$P_{cow} = (P_{cir} - P_{ct})G_n + (P_{cor} - P_{ct})F_n + P_{ct} \quad (8.30)$$

$$\text{where } S_{wir} \leq S_w \leq (1 - S_{orw}) \text{ and } P_{cow}(S_w = S_{wzo}) = 0 \Rightarrow E_{sn}$$

$$G_n = \frac{(1 - S_{wn})^{L_{sn}}}{(1 - S_{wn})^{L_{sn}} + E_{sn} S_{wn}^{T_{sn}}} \quad F_n = \frac{S_{wn}^{L_{fn}}}{S_{wn}^{L_{fn}} + E_{fn} (1 - S_{wn})^{T_{fn}}} \quad (8.31)$$

$$S_{wn} = \frac{S_w - S_{wirr}}{1 - S_{orw} - S_{wirr}} \quad (8.32)$$

where

E_{fn}	= forced imbibition empirical E parameter,
E_{sn}	= spontaneous imbibition empirical E parameter,
L_{fn}	= forced imbibition empirical L parameter,
L_{sn}	= spontaneous imbibition empirical L parameter,
P_{cir}	= capillary pressure at irreducible water saturation,
P_{ct}	= capillary threshold / entry pressure,
T_{fn}	= forced imbibition empirical T parameter,
T_{sn}	= spontaneous imbibition empirical T parameter,
S_{orw}	= residual oil saturation under water displacement,
S_w	= water saturation,
S_{wirr}	= irreducible water saturation, and
S_{wn}	= normalize water saturation.

Spontaneous imbibition refers to the wetting phase displacing the non-wetting phase, resulting in capillary pressure values greater than zero. For example, in a water wet oil reservoir under water drive, the water will easily imbibe into the pore space and displace the non-wetting oil phase. Forced imbibition is when the non-wetting phase displaces the wetting phase, and results in a negative capillary pressure values necessary to overcome the capillary force holding the wetting phase in-place. Thus, for a water wet oil reservoir, with the oil zone expanding into the water zone, an additional force is required to displace the water.

Table 8.13 relates the general LET equations presented previously with the parameters on the [SWOFLET](#) keyword.

LET Function (IDC2) versus SWOFLET Keyword					
SWOFLET	LET Function	SWOFLET	LET Function	SWOFLET	LET Function
SWL	S_{wirr}	SORW	S_{orw}	PCIR	P_{cir}
SWCR	S_{wirr}	SOWCR	S_{orw}	PCIT	P_{ct}
LWAT	L_w	LOIL	L_o	LPC	L_{sn}
EWAT	E_w	EOIL	E_o	EPC	E_{sn}
TWAT	T_w	TOIL	T_o	TPC	T_{sn}
KRTWAT	K_{rwr}	KRTOIL	K_{rot}		
Notes: 1) The current implementation of the LET functions is restricted to a single branch capillary pressure curves as shown in Figure 8.20, and not the multi-branch curve shown in Figure 8.21.					

Table 8.13: LET Functions (IDC2) versus SWOFLET Keyword

Gas Displacing Oil: Primary Imbibition (CDI2)

For the oil producing phase, where the gas phase is increasing either by replacing the displaced oil or via gas injection, CDI2, the LET equations are:

$$K_{rog} = \frac{K_{rot}(1 - S_{gn})^{L_o}}{(1 - S_{gn})^{L_o} + E_o S_{gn}^{T_o}} \quad (8.33)$$

$$K_{rg} = \frac{K_{rgt} S_{gn}^{L_g}}{S_{gn}^{L_g} + E_g (1 - S_{gn})^{T_g}} \quad (8.34)$$

$$S_{gn} = \frac{S_g}{1 - S_{org} - S_{wirr}} \quad (8.35)$$

where

- E_α = empirical E parameter for the α phase,
- K_{rog} = oil relative permeability with respect to gas saturation,
- K_{rot} = base or top oil relative permeability, used to scale the maximum oil permeability,
- K_{rg} = gas relative permeability,
- K_{rgt} = base or top gas relative permeability, used to scale the maximum gas permeability,
- L_α = empirical L parameter for the α phase,
- T_α = empirical T parameter for the α phase,
- S_g = gas saturation,
- S_{gn} = normalize gas saturation,
- S_{org} = residual oil saturation under gas displacement, and
- S_{wirr} = irreducible water saturation.

Similarly for the capillary pressure CDI2, with the water phase constant, the oil phase decreasing and the gas phase increasing, the LET capillary pressure functions are:

$$P_{cog} = (P_{cir} - P_{ct})G_n + (P_{cor} - P_{ct})F_n + P_{ct} \quad (8.36)$$

where $S_{wir} \leq S_w \leq (1 - S_{org} - S_{wirr})$ and $P_{cog}(S_w = S_{wzo}) = 0 \Rightarrow E_{sn}$

$$G_n = \frac{(1 - S_{gn})^{L_{sn}}}{(1 - S_{gn})^{L_{sn}} + E_{sn} S_{gn}^{T_{sn}}} \quad F_n = \frac{S_{gn}^{L_{fn}}}{S_{gn}^{L_{fn}} + E_{fn} (1 - S_{gn})^{T_{fn}}} \quad (8.37)$$

$$S_{gn} = \frac{S_g}{1 - S_{org} - S_{wirr}} \quad (8.38)$$

where

- E_{fn} = forced imbibition empirical E parameter,
- E_{sn} = spontaneous imbibition empirical E parameter,

L_{fn}	= forced imbibition empirical L parameter,
L_{sn}	= spontaneous imbibition empirical L parameter,
P_{cir}	= capillary pressure at irreducible water saturation,
P_{ct}	= capillary threshold/entry pressure,
T_{fn}	= forced imbibition empirical T parameter,
T_{sn}	= spontaneous imbibition empirical T parameter,
S_g	= gas saturation,
S_{gn}	= normalize gas saturation,
S_{org}	= residual oil saturation under gas displacement, and
S_{wirr}	= irreducible water saturation.

Table 8.14 relates the general LET equations presented previously with the parameters on the [SGOFLET](#) keyword.

LET Function (CDI2) versus SGOFLET Keyword					
SGOFLET	LET Function	SGOFLET	LET Function	SGOFLET	LET Function
SGL		SORG	S_{org}	PCIR	P_{cir}
SGCR		SOGCR	S_{org}	PCIT	P_{ct}
LGAS	L_g	LOIL	L_o	LPC	L_{sn}
EGAS	E_g	EOIL	E_o	EPC	E_{sn}
TGAS	T_g	TOIL	T_o	TPC	T_{sn}
KRTGAS	K_{rgt}	KRTOIL	K_{rot}		
Notes: 1) The current implementation of the LET functions is restricted to a single branch capillary pressure curves as shown in Figure 8.20, and not the multi-branch curve shown in Figure 8.21. 2) The S_{wirr} parameter value is taken from the SWOFFLET keyword.					

Table 8.14: LET Functions (CDI2) versus SGOFLET Keyword

Gas Displacing Water: Primary Drainage Cycle (DCI1)

For the Primary Drainage, assuming water is the wetting phase, then, $S_w(D)$, $S_o(C)$, $S_g(I)$ I, or DCI1, means the water phase is decreasing, the oil phase is constant, and the gas phase is increasing for the primary or first cycle (n equals one). This is the case for when gas is migrating into the reservoir rock and displacing the initial water contained within the reservoir. Under these circumstance the LET normalized water saturation and relative permeability functions are defined as:

$$K_{rgw} = \frac{K_{rgt}(1 - S_{wp})^{L_g}}{(1 - S_{wp})^{L_g} + E_g S_{wp}^{T_g}} \quad (8.39)$$

$$K_{rw} = \frac{K_{rwt} S_{wp}^{L_w}}{S_{wp}^{L_w} + E_w (1 - S_{wp})^{T_w}} \quad (8.40)$$

$$S_{wp} = \frac{S_w - S_{wirr}}{1 - S_{wirr}} \quad (8.41)$$

where

- E_α = empirical E parameter for the α phase,
- K_{rgw} = gas relative permeability with respect to water saturation,
- K_{rgt} = base or top gas relative permeability, used to scale the maximum gas permeability,
- K_{rw} = water relative permeability,
- K_{rwt} = base or top water relative permeability, used to scale the maximum water permeability,
- L_α = empirical L parameter for the α phase,
- T_α = empirical T parameter for the α phase,
- S_w = water saturation,
- S_{wirr} = irreducible water saturation, and
- S_{wp} = normalize water saturation.

For the primary drainage or first cycle, DCII, with the water phase decreasing, the oil phase constant and the gas phase increasing, the LET capillary pressure functions are:

$$Y = \frac{P_{cgw} - P_{ct}}{P_{cir} - P_{ct}} \text{ and } Y = F(S_{wp}) \text{ where } S_{wirr} \leq S_w \leq 1.0 \quad (8.42)$$

$$P_{cgw}(S_w = S_{wirr}) = P_{cir} \text{ and } P_{cgw}(S_w = 1.0) = P_{cit} \quad (8.43)$$

$$F = \frac{(1 - S_{wp})^L}{(1 - S_{wp})^L + E S_{wp}^T} \text{ and } S_{wp} = \frac{S_w - S_{wirr}}{1 - S_{wirr}} \quad (8.44)$$

where

- E = empirical E parameter,
- L = empirical L parameter,
- P_{cir} = capillary pressure at irreducible water saturation,
- P_{ct} = capillary threshold/entry pressure,
- T = empirical T parameter,
- S_w = water saturation,
- S_{wirr} = irreducible water saturation, and
- S_{wp} = normalize water saturation.

Table 8.15 relates the general LET equations presented previously with the parameters on the [SGWFLET](#) keyword.

LET Function (DCII) versus SGWFLET Keyword					
SGWFLET	LET Function	SGWFLET	LET Function	SGWFLET	LET Function
SGL		SWL	S_{wirr}	PCIR	P_{cir}
SGCR		SWCR	S_{wirr}	PCIT	P_{ct}

LET Function (DCII) versus SGWFLET Keyword					
SGWFLET	LET Function	SGWFLET	LET Function	SGWFLET	LET Function
LGAS	L_g	LWAT	L_w	LPC	L
EGAS	E_g	EWAT	E_w	EPC	E
TGAS	T_g	TWAT	T_w	TPC	T
KRTGAS	K_{rgt}	KRTWAT	K_{rwr}		
Notes: 1) The current implementation of the LET functions is restricted to a single branch capillary pressure curves as shown in Figure 8.20, and not the multi-branch curve shown in Figure 8.21.					

Table 8.15: LET Functions (DCII) versus SGWFLET Keyword

Note

The SGWFLET keyword has not been implemented, as of this current release.

Water Displacing Gas: Primary Imbibition Cycle (ICD2)

Similarly for the gas producing phase, where the water phase is increasing either by replacing the displaced gas or via water influx from the aquifer, that is $S_w(I)$, $S_o(C)$, $S_g(D)$, for the second flooding event, IDC2. Under this scenario the LET equations are:

$$K_{rgw} = \frac{K_{rgt}(1 - S_{wn})^{L_o}}{(1 - S_{wn})^{L_o} + E_o S_{wn}^{T_o}} \quad (8.45)$$

$$K_{rw} = \frac{K_{rwr} S_{wn}^{L_w}}{S_{wn}^{L_w} + E_w (1 - S_{wn})^{T_w}} \quad (8.46)$$

$$S_{wn} = \frac{S_w - S_{wirr}}{1 - S_{grw} - S_{wirr}} \quad (8.47)$$

where

E_α	= empirical E parameter for the α phase,
K_{rgw}	= gas relative permeability with respect to water saturation,
K_{rot}	= base or top gas relative permeability, used to scale the maximum gas permeability,
K_{rw}	= water relative permeability,
K_{rwr}	= base or top water relative permeability, used to scale the maximum water permeability,
L_α	= empirical L parameter for the α phase,
T_α	= empirical T parameter for the α phase,
S_{ogw}	= residual gas saturation under water displacement,
S_w	= water saturation,

S_{wirr} = irreducible water saturation, and
 S_{wn} = normalize water saturation.

For the primary imbibition or second cycle, IDC2, with the water phase increasing, the oil phase decreasing and the gas phase constant, the LET capillary pressure functions are:

$$P_{cgw} = (P_{cir} - P_{ct})G_n + (P_{cgr} - P_{ct})F_n + P_{ct} \quad (8.48)$$

where $S_{wir} \leq S_w \leq (1 - S_{gorw})$ and $P_{cgw}(S_w = S_{wzo}) = 0 \Rightarrow E_{sn}$

$$G_n = \frac{(1 - S_{wn})^{L_{sn}}}{(1 - S_{wn})^{L_{sn}} + E_{sn} S_{wn}^{T_{sn}}} \quad F_n = \frac{S_{wn}^{L_{fn}}}{S_{wn}^{L_{fn}} + E_{fn} (1 - S_{wn})^{T_{fn}}} \quad (8.49)$$

$$S_{wn} = \frac{S_w - S_{wirr}}{1 - S_{grw} - S_{wirr}} \quad (8.50)$$

where

E_{fn} = forced imbibition empirical E parameter,
 E_{sn} = spontaneous imbibition empirical E parameter,
 L_{fn} = forced imbibition empirical L parameter,
 L_{sn} = spontaneous imbibition empirical L parameter,
 P_{cir} = capillary pressure at irreducible water saturation,
 P_{ct} = capillary threshold / entry pressure,
 T_{fn} = forced imbibition empirical T parameter,
 T_{sn} = spontaneous imbibition empirical T parameter,
 S_{ogw} = residual gas saturation under water displacement,
 S_w = water saturation,
 S_{wirr} = irreducible water saturation, and
 S_{wn} = normalize water saturation.

Table 8.16 relates the general LET equations presented previously with the parameters on the [SGWFLET](#) keyword.

LET Function (DCII) versus SGWFLET Keyword					
SGWFLET	LET Function	SGWFLET	LET Function	SGWFLET	LET Function
SGL		SWL	S_{wirr}	PCIR	P_{cir}
SGCR		SWCR	S_{wirr}	PCIT	P_{ct}
LGAS	L_g	LWAT	L_w	LPC	L_{sn}
EGAS	E_g	EWAT	E_w	EPC	E_{sn}
TGAS	T_g	TWAT	T_w	TPC	T_{sn}
KRTGAS	K_{rgt}	KRTWAT	K_{rwr}		
Notes:					

LET Function (DCII) versus SGWFLET Keyword					
SGWFLET	LET Function	SGWFLET	LET Function	SGWFLET	LET Function
1) The current implementation of the LET functions is restricted to a single branch capillary pressure curves as shown in Figure 8.20, and not the multi-branch curve shown in Figure 8.21.					

Table 8.16: LET Functions (DCII) versus SGWFLET Keyword

Note

The SGWFLET keyword has not been implemented, as of this current release.

8.2.7 SATURATION TABLE ALLOCATION

If none of the relative permeability assignment options have been activated on the SATOPTS keyword in the RUNSPEC section then saturation tables may be allocated to each cell using the SATNUM keyword in the REGIONS section.

If the directional relative permeability option has been activated by the DIRECT parameter on the SATOPTS keyword then direction dependent relative permeability tables may be allocated to each cell using the KRNUMX, KRNUMY, and KRNUMZ series of keywords in the REGIONS section. If the irreversible directional relative permeability option has also been activated by the DIRECT and IRREVERS parameters on the SATOPTS keyword then relative permeability tables may be allocated to each face of each cell using the KRNUMX, KRNUMX-, KRNUMY, KRNUMY-, KRNUMZ, and KRNUMZ- series of keywords.

If the hysteresis option has been activated by the HYSTER parameter on the SATOPTS keyword then the "drainage" and imbibition saturation function tables may be allocated to each cell using the SATNUM and IMBNUM keywords respectively in the REGIONS section.

If both the hysteresis and directional relative permeability options have been activated then the "drainage" relative permeabilities can be allocated using the KRNUMX, KRNUMY, and KRNUMZ series of keywords, and the imbibition relative permeabilities can be allocated using the IMBNUMX, IMBNUMY, and IMBNUMZ series of keywords.

If the hysteresis, directional, and irreversible relative permeability options have been activated then the "drainage" relative permeabilities can be allocated using the KRNUMX, KRNUMX-, KRNUMY, KRNUMY-, KRNUMZ, and KRNUMZ- series of keywords, and the imbibition relative permeabilities can be allocated using the IMBNUMX, IMBNUMX-, IMBNUMY, IMBNUMY-, IMBNUMZ, and IMBNUMZ- series of keywords.

The above discussion assumes that the model has a Cartesian grid; however, if the model has a radial grid then the equivalent radial saturation table allocation keywords should be used. Thus, the equivalent KRNUM keywords would be KRNUMR, KRNUMR-, KRNUMT, KRNUMT-, KRNUMZ, and KRNUMZ-, and the equivalent IMBNUM keywords would be IMBNUMR, IMBNUMR-, IMBNUMT, IMBNUMT-, IMBNUMZ, and IMBNUMZ-. Note that the KRNUMZ, KRNUMZ-, IMBNUMZ, and IMBNUMZ- keywords are the same for both Cartesian and radial grids.

8.3 KEYWORD DEFINITIONS

8.3.1 ACF – DEFINE ACENTRIC FACTORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [ACF](#) keyword defines the acentric factors for each of the compositional components active in the model and for each equation of state. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description	Default
I	ACF	A series of real numbers that define the acentric factors for each of the compositional components active in the model.	None
Notes: 1) The keyword is followed by NMEOSR records as specified by the TABDIMS keyword in the RUNSPEC section. 2) Each record must contain COMPS values as specified by the COMPS keyword in the RUNSPEC section. 3) Each record is terminated by a "/" and there is no "/" terminator for the keyword.			

Table 8.17:ACF Keyword Description

Examples

The following example defines the acentric factors for each component in a single three-component equation of state model.

```
--
-- Acentric Factors
--
ACF
  0.0108  0.2273  0.3434 /
```

The following example defines the acentric factors for each component in two three-component equation of state models.

```
--
-- Acentric Factors
--
ACF
  0.0108  0.2273  0.3434 /
  0.0110  0.2281  0.3428 /
```


8.3.2 ACTCO2S – ACTIVITY MODEL FOR CO2

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The keyword **ACTCO2S** specifies the activity model for salting-out effects when calculating mutual solubility in the CO2 storage module which is activated by the **CO2STORE** keyword in the **RUNSPEC** section.

No.	Name	Description	Default
I	ACTMODEL	A positive integer value selecting the activity model for the salting-out effect. The choices are: 1) The Rumpf et al.¹⁸¹ model as detailed in the paper by Spycher and Pruess¹⁸². The model was calibrated in the P-T range 5 to 96 bar and 40 to 160°C, and was shown in Spycher and Pruess¹⁸² to be accurate in the P-T range 1 to 600 bar and 12 to 100°C. Note that this model requires a (fixed-point) iteration procedure to compute solubility, but usually converges within few iterations. 2) The Duan and Sun¹⁸³ model as detailed in the paper by Spycher and Pruess¹⁸⁴. The model was recalibrated to the P-T range 1 to 600 bar and 12 to 300°C, and was shown to be accurate within the same range. Note that above 99°C a (fixed-point) iteration procedure is required to compute solubility, while below 99°C the computations are done in a direct manner without needing iterations. 3) The Duan and Sun¹⁸³ model as detailed in the paper by Spycher and Pruess¹⁸². The model was calibrated to the P-T range 0 to 2000 bar and 0 to 260°C, and was shown in Spycher and Pruess¹⁸² to be accurate in the P-T range 1 to 600 bar and 12 to 100°C. Note that no iterations are required to compute the solubility.	3
Notes: I) The keyword is terminated by a “/”.			

Table 8.18: ACTCO2S Keyword Description

Example

The following example activates activity model number 1.

```
--
--          ACTIVITY MODEL FOR SALTING-OUT EFFECTS IN CO2
--
ACTCO2S
      1                                     /
```

¹⁸¹ Rumpf, B., Nicolaisen, H., Ocal, C., and Maurer, G., 1994. Solubility of carbon dioxide in aqueous solutions of sodium chloride: experimental results and correlation. *J. Sol. Chem.* 23, 431-448.

¹⁸² Spycher, N., and Pruess, K., 2005. CO2-H2O mixtures in the geological sequestration of CO2. II. Partitioning in chloride brines at 12–100°C and up to 600 bar. *Geochimica Et Cosmochimica Acta* 69, 3309–3320.

¹⁸³ Duan Z., and Sun R., 2003. An improved model calculating CO2 solubility in pure water and aqueous NaCl solutions from 257 to 533 K and from 0 to 2000 bar. *Chem. Geol.* 193, 257–271.

¹⁸⁴ Spycher, N., and Pruess, K., 2009. A Phase-Partitioning Model for CO2–Brine Mixtures at Elevated Temperatures and Pressures: Application to CO2-Enhanced Geothermal Systems. *Transp Porous Med* 82, 173–196.

8.3.3 ADD – ADD A CONSTANT TO A SPECIFIED ARRAY

The [ADD](#) keyword adds a constant to a specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [ADD](#) keyword is being applied.

See [ADD – Add a Constant to a Specified Array](#) in the [GRID](#) section for a full description.

8.3.4 ADDREG – ADD A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER

The [ADDREG](#) keyword adds a constant to a specified array or part of an array based on cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [ADDREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [ADDREG](#) keyword is being used.

See [ADDREG – Add a Constant to an Array based on a Region Number](#) in the [GRID](#) section for a full description.

8.3.5 ADSALNOD – SALT CONCENTRATION BASED ON SATNUM ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

ADSALNOD defines the salt concentration value based on a cells **SATNUM** number. The **ADSALNOD** property is used in the calculation of a polymer viscosity when the polymer and the salt options has been activated by the **POLYMER** and **BRINE** keywords in the **RUNSPEC** section. In the **RUNSPEC** section the number of **SATNUM** functions is declared by the **NTSFUN** variable on the **TABDIMS** keyword and allocated to individual cells by the **SATNUM** property array in the **REGIONS** section. **NSSFUN** on the **TABDIMS** keyword in the **RUNSPEC** section defines the maximum number of rows (or saturation values) in the relative permeability saturation tables and also sets the maximum number of entries for each **ADSALNOD** data set. The number of values for each data set must correspond to the number of polymer solution adsorption entries on the **PLYADSS** keyword. For example, if there are three sets of relative permeability tables and four values on the **PLYADSS** keyword, then three **ADSALNOD** data sets with four values of salt concentrations need to be entered.

The salt concentrations within each data set should be positive and monotonically increasing and each **ADSALNOD** data set is delimited by “/” including the last data set.

No.	Name	Description			Default
I	SALTCON	Field	Metric	Laboratory	None
		A real positive columnar vector that sets the salt concentrations for the given relative permeability saturation tables.			
		lb/stb	kg/sm3	gm/scc	
Notes: I) Each data set must be terminated by a “/” including the last data set.					

Table 8.19: ADSALNOD Keyword Description

An alternative manner of entering the salt concentrations is by utilizing the **PVTNUM** region array by using the **SALTNODE** keyword in the **PROPS** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

Given three sets of relative permeability tables and four values on the [PLYADSS](#) keyword, then the data salt concentration should be entered as follows:

```
--
--   SETS SALT CONCENTRATION FOR POLYMER SOLUTION ADSORPTION
--   VIA SATNUM ARRAY ALLOCATION
--
--   SALT
--
ADSALNOD
  1.0
  5.0
 10.5
 25.0      / SATNUM TABLE NO. 01
  1.0
  3.0
  7.5
 15.0      / SATNUM TABLE NO. 02
  1.0
  7.5
 20.5
 35.0      / SATNUM TABLE NO. 03
```

See also the [SALTNODE](#) keyword.

8.3.6 ADSORP – DEFINE GENERALIZED LANGMUIR ADSORPTION FUNCTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ADSORP](#) keyword defines the parameters for the generalized Langmuir Adsorption¹⁸⁵ function for when the polymer, surfact, alkaline, foam and tracers phases have been activated in the [RUNSPEC](#) section by the [POLYMER](#), [SURFACT](#), [ALKALINE](#), [FOAM](#) and [TRACER](#) keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

185

Langmuir, Irving (June 1918). "The Adsorption of Gases on Plane Surface of Glass, Mica and Platinum".*The Research Laboratory of the General Electric Company*. 40 (9): 1361–1402. doi:10.1021/ja02242a004

8.3.7 ALKADS – DEFINE ALKALINE ADSORPTION FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ALKADS](#) defines the alkaline adsorption functions for when the alkaline model has been activated via the [ALKALINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.8 ALKROCK – DEFINE ROCK ALKALINE PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ALKROCK](#) keyword defines the rock alkaline properties for when the alkaline model has been activated via the [ALKALINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.9 ALPOLADS – POLYMER ADSORPTION VERSUS ALKALINE CONCENTRATION MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

ALPOLDS defines the polymer adsorption versus alkaline concentration multipliers for when the alkaline model has been activated via the [ALKALINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.10 ALSURFAD – SURFACTANT ADSORPTION VERSUS ALKALINE CONCENTRATION MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

ALSURAD defines the surfactant adsorption versus alkaline concentration multipliers for when the alkaline model has been activated via the [ALKALINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.11

ALSURFST – WATER-OIL SURFACE TENSION VERSUS ALKALINE CONCENTRATION MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ALSURFST](#) keyword defines the water-oil surface tension versus alkaline concentration multipliers for when the alkaline model has been activated via the [ALKALINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.12 APIGROUP – DEFINE API TRACKING NUMBER OF GROUPED OIL PVT TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [APIGROUP](#) keyword defines the maximum number of groups of oil PVT tables when the [API](#) tracking option has been activated via the [API](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.13 AUCT – DEFINE CARTER-TRACY ANALYTICAL AQUIFERS

The [AUCT](#) keyword defines a Carter Tracy aquifer, the properties of the aquifer, including the Carter-Tracy aquifer influence function associated with the aquifer, defined by the [AUTAB](#) keyword in the [PROPS](#) section.

See [AUCT – Define Carter-Tracy Analytical Aquifers](#) in the [GRID](#) section for a full description

8.3.14 AQUTAB – DEFINE CARTER-TRACY AQUIFER INFLUENCE FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **AQUTAB** keyword defines additional Carter-Tracy¹⁸⁶ aquifer functions to be used in the model. Carter-Tracy representation of the aquifer influx is via a *qw* term in the non-linear aquifer influence function *Q(t)*. It allows the water influx from the aquifer to be represented in the simulator by assuming that there is a constant water influx rate over finite time periods. It is derived from the superposition methods of van Everdingen and Hurst¹⁸⁷, whose superposition methods are not suitable for implementation in reservoir simulation software, although they are very useful in interpreting aquifer response. The storage requirements and calculation complexity of handling the resulting superposition formulae can be largely eliminated by use of the Carter-Tracy approximate water influx method.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TD	Dimensionless Time			None
		dimensionless	dimensionless	dimensionless	
2	PD	Dimensionless Pressure			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NIFTBL tables as declared on the **AQUDIMS** keyword in the **RUNSPEC** section.
- 2) Each table must contain at least two complete rows with a maximum of NRIFTB rows as declared on the **AQUDIMS** keyword in the **RUNSPEC** section. Note that NRIFTB must not be less than 36 in order to accommodate the default infinite acting Carter-Tracy aquifer influence function.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.20: AQUTAB Keyword Description

Note

OPM Flow includes the infinite acting Carter-Tracy aquifer influence table as a default for table number one; thus data entered on this keyword starts from table number two.

Table 8.21 to Table 8.24 outline the standard finite tables derived by van Everdingen and Hurst that are used for the Carter-Tracy analytical aquifers and are taken from Bradely¹⁸⁸ table 38-6 on page 38-12. In the tables

*r*_D is defined as the ratio of the aquifer external radius divided by hydrocarbon radius, that is: $r_D = \frac{r_e}{r_o}$.

¹⁸⁶ Carter, R. D., and Tracy, G.W. “An Improved Method for Calculating Water Influx.” *Transactions of AIME*, Vol. 219 (1960), pp 415-417.

¹⁸⁷ Van Everdingen, A. F., and Hurst, W. “The Application of the Laplace Transform to Flow Problems in Reservoirs.” *Transactions of AIME*, Vol. 186 (1949), pp. 305-324.

¹⁸⁸ Bradley Howard B., et al., *Petroleum Engineering Handbook*, Society of Petroleum Engineers (June 1989), ISBN:9781555630102.

Carter-Tracy Aquifer Influence Functions								
No.	$R_D = 1.5$ Dimensionless		$R_D = 2.0$ Dimensionless		$R_D = 2.5$ Dimensionless		$R_D = 3.0$ Dimensionless	
	t_D	p_D	t_D	p_D	t_D	p_D	t_D	p_D
1	0.0600	0.2510	0.2200	0.4430	0.4000	0.5650	0.5200	0.6270
2	0.0800	0.2880	0.2400	0.4590	0.4200	0.5760	0.5400	0.6360
3	0.1000	0.3220	0.2600	0.4760	0.4400	0.5870	0.5600	0.6450
4	0.1200	0.3550	0.2800	0.4920	0.4600	0.5980	0.6000	0.6620
5	0.1400	0.3870	0.3000	0.5070	0.4800	0.6080	0.6500	0.6830
6	0.1600	0.4200	0.3200	0.5220	0.5000	0.6180	0.7000	0.7030
7	0.1800	0.4520	0.3400	0.5360	0.5200	0.6280	0.7500	0.7210
8	0.2000	0.4840	0.3600	0.5510	0.5400	0.6380	0.8000	0.7400
9	0.2200	0.5160	0.3800	0.5650	0.5600	0.6470	0.8500	0.7580
10	0.2400	0.5480	0.4000	0.5790	0.5800	0.6570	0.9000	0.7760
11	0.2600	0.5800	0.4200	0.5930	0.6000	0.6660	0.9500	0.7910
12	0.2800	0.6120	0.4400	0.6070	0.6500	0.6880	1.0000	0.8060
13	0.3000	0.6440	0.4600	0.6210	0.7000	0.7100	1.2000	0.8650
14	0.3500	0.7240	0.4800	0.6340	0.7500	0.7310	1.4000	0.9200
15	0.4000	0.8040	0.5000	0.6480	0.8000	0.7520	1.6000	0.9730
16	0.4500	0.8840	0.6000	0.7150	0.8500	0.7720	2.0000	1.0760
17	0.5000	0.9640	0.7000	0.7820	0.9000	0.7920	3.0000	1.3280
18	0.5500	1.0440	0.8000	0.8490	0.9500	0.8120	4.0000	1.5780
19	0.6000	1.1240	0.9000	0.9150	1.0000	0.8320	5.0000	1.8280
20			1.0000	0.9820	2.0000	1.2150		
21			2.0000	1.6490	3.0000	1.5960		
22			3.0000	2.3160	4.0000	1.9770		
23			5.0000	3.6490	5.0000	2.3580		

Table 8.21: Carter-Tracy Aquifer Influence Functions ($R_D = 1.5, 2.0, 2.5$ and 3.0)

Carter-Tracy Aquifer Influence Functions								
No.	R _D = 3.5		R _D = 4.0		R _D = 4.5		R _D = 5.0	
	Dimensionless		Dimensionless		Dimensionless		Dimensionless	
	t _D	p _D	t _D	p _D	t _D	p _D	t _D	p _D
1	1.0000	0.8020	1.5000	0.9270	2.0000	1.0230	3.0000	1.1670
2	1.1000	0.8300	1.6000	0.9480	2.1000	1.0400	3.1000	1.1800
3	1.2000	0.8570	1.7000	0.9680	2.2000	1.0560	3.2000	1.1920
4	1.3000	0.8820	1.8000	0.9880	2.3000	1.0720	3.3000	1.2040
5	1.4000	0.9060	1.9000	1.0070	2.4000	1.0870	3.4000	1.2150
6	1.5000	0.9290	2.0000	1.0250	2.5000	1.1020	3.5000	1.2270
7	1.6000	0.9510	2.2000	1.0590	2.6000	1.1160	3.6000	1.2380
8	1.7000	0.9730	2.4000	1.0920	2.7000	1.1300	3.7000	1.2490
9	1.8000	0.9940	2.6000	1.1230	2.8000	1.1440	3.8000	1.2590
10	1.9000	1.0140	2.8000	1.1540	2.9000	1.1580	3.9000	1.2700
11	2.0000	1.0340	3.0000	1.1840	3.0000	1.1710	4.0000	1.2810
12	2.2500	1.0830	3.5000	1.2550	3.2000	1.1970	4.2000	1.3010
13	2.5000	1.1300	4.0000	1.3240	3.4000	1.2220	4.4000	1.3210
14	2.7500	1.1760	4.5000	1.3920	3.6000	1.2460	4.6000	1.3400
15	3.0000	1.2210	5.0000	1.4600	3.8000	1.2690	4.8000	1.3600
16	4.0000	1.4010	5.5000	1.5270	4.0000	1.2920	5.0000	1.3780
17	5.0000	1.5790	6.0000	1.5940	4.5000	1.3490	5.5000	1.4240
18	6.0000	1.7570	6.5000	1.6600	5.0000	1.4030	6.0000	1.4690
19			7.0000	1.7270	5.5000	1.4570	6.5000	1.5130
20			8.0000	1.8610	6.0000	1.5100	7.0000	1.5560
21			9.0000	1.9940	7.0000	1.6150	7.5000	1.5980
22			10.0000	2.1270	8.0000	1.7190	8.0000	1.6410
23					9.0000	1.8230	9.0000	1.7250
24					10.0000	1.9270	10.0000	1.8080
25					11.0000	2.0310	11.0000	1.8920
26					12.0000	2.1350	12.0000	1.9750
27					13.0000	2.2390	13.0000	2.0590
28					14.0000	2.3430	14.0000	2.1420
29					15.0000	2.4470	15.0000	2.2250

Table 8.22: Carter-Tracy Aquifer Influence Functions (RD =3.5, 4.0, 4.5 and 5.0)

Carter-Tracy Aquifer Influence Functions								
No.	R _D = 6.0		R _D = 7.0		R _D = 8.0		R _D = 9.0	
	Dimensionless		Dimensionless		Dimensionless		Dimensionless	
	t _D	p _D	t _D	p _D	t _D	p _D	t _D	p _D
1	4.0000	1.2750	6.0000	1.4360	8.0000	1.5560	10.0000	1.6510
2	4.5000	1.3220	6.5000	1.4700	8.5000	1.5820	10.5000	1.6730
3	5.0000	1.3640	7.0000	1.5010	9.0000	1.6070	11.0000	1.6930
4	5.5000	1.4040	7.5000	1.5310	9.5000	1.6310	11.5000	1.7130
5	6.0000	1.4410	8.0000	1.5590	10.0000	1.6530	12.0000	1.7320
6	6.5000	1.4770	8.5000	1.5860	10.5000	1.6750	12.5000	1.7500
7	7.0000	1.5110	9.0000	1.6130	11.0000	1.6970	13.0000	1.7680
8	7.5000	1.5440	9.5000	1.6380	11.5000	1.7170	13.5000	1.7860
9	8.0000	1.5760	10.0000	1.6630	12.0000	1.7370	14.0000	1.8030
10	8.5000	1.6070	11.0000	1.7110	12.5000	1.7570	14.5000	1.8190
11	9.0000	1.6380	12.0000	1.7570	13.0000	1.7760	15.0000	1.8350
12	9.5000	1.6680	13.0000	1.8010	13.5000	1.7950	15.5000	1.8510
13	10.0000	1.6980	14.0000	1.8450	14.0000	1.8130	16.0000	1.8670
14	11.0000	1.7570	15.0000	1.8880	14.5000	1.8310	17.0000	1.8970
15	12.0000	1.8150	16.0000	1.9310	15.0000	1.8490	18.0000	1.9260
16	13.0000	1.8730	17.0000	1.9740	17.0000	1.9190	19.0000	1.9550
17	14.0000	1.9310	18.0000	2.0160	19.0000	1.9860	20.0000	1.9830
18	15.0000	1.9880	19.0000	2.0580	21.0000	2.0510	22.0000	2.0370
19	16.0000	2.0450	20.0000	2.1000	23.0000	2.1160	24.0000	2.0900
20	17.0000	2.1030	22.0000	2.1840	25.0000	2.1800	26.0000	2.1420
21	18.0000	2.1600	24.0000	2.2670	30.0000	2.3400	28.0000	2.1930
22	19.0000	2.2170	26.0000	2.3510	35.0000	2.4990	30.0000	2.2440
23	20.0000	2.2740	28.0000	2.4340	40.0000	2.6580	34.0000	2.3450
24	25.0000	2.5600	30.0000	2.5170	45.0000	2.8170	38.0000	2.4460
25	30.0000	2.8460					40.0000	2.4960
26							45.0000	2.6210
27							50.0000	2.7460

Table 8.23: Carter-Tracy Aquifer Influence Functions (RD =6.0, 7.0, 8.0 and 9.0)

Carter-Tracy Aquifer Influence Functions				
No.	$R_D = 10.0$ Dimensionless		Finite Linear Dimensionless	
	t_D	p_D	t_D	p_D
1	12.0000	1.7320	0.0050	0.0798
2	12.5000	1.7500	0.0100	0.1130
3	13.0000	1.7680	0.0200	0.1596
4	13.5000	1.7840	0.0600	0.2764
5	14.0000	1.8010	0.0800	0.3192
6	14.5000	1.8170	0.1000	0.3568
7	15.0000	1.8320	0.1200	0.3909
8	15.5000	1.8470	0.1600	0.4515
9	16.0000	1.8620	0.2000	0.5052
10	17.0000	1.8900	0.2400	0.5544
11	18.0000	1.9170	0.3000	0.6228
12	19.0000	1.9430	0.4000	0.7294
13	20.0000	1.9680	0.6000	0.9328
14	22.0000	2.0170	0.8000	1.1333
15	24.0000	2.0630	1.0000	1.3333
16	26.0000	2.1080		
17	28.0000	2.1510		
18	30.0000	2.1940		
19	32.0000	2.2360		
20	34.0000	2.2780		
21	36.0000	2.3190		
22	38.0000	2.3600		
23	40.0000	2.4010		
24	50.0000	2.6040		
25	60.0000	2.8060		
26	70.0000	3.0080		

Table 8.24: Carter-Tracy Aquifer Influence Functions ($R_D = 10$ and Finite Linear)

For the finite linear Carter-Tracy influence function in Table 8.24 set the inner radius of the aquifer to the length of linear aquifer and the angle of influence to $= \frac{360 \times \text{Width}}{(2 \times \pi \times \text{Length})}$ on the [AQUCT](#) keyword in the grid section.

For reference, Table 8.25 outlines the content terminal rate case for an infinite aquifer derived by van Everdingen and Hurst, that is the default table number one used for the Carter-Tracy analytical aquifers, as implemented in OPM Flow.

Carter-Tracy Infinite Radial Aquifer Influence Function (Default)					
No.	Infinite Dimensionless		No.	Infinite Dimensionless	
	t_D	p_D		t	p_D
1	1.0×10^{-2}	0.1120	27	20,00	1.9600
2	5.0×10^{-2}	0.2290	28	25,00	2.0670
3	1.0×10^{-1}	0.3150	29	30,00	2.1470
4	1.5×10^{-1}	0.3760	30	40,00	2.2820
5	2.0×10^{-1}	0.4240	31	50,00	2.3880
6	2.5×10^{-1}	0.4690	32	60,00	2.4760
7	3.0×10^{-1}	0.5030	33	70,00	2.5500
8	4.0×10^{-1}	0.5640	34	80,00	2.6150
9	5.0×10^{-1}	0.6160	35	90,00	2.6720
10	6.0×10^{-1}	0.6590	36	100,00	2.7230
11	7.0×10^{-1}	0.7020	37	150,00	2.9210
12	8.0×10^{-1}	0.7350	38	200,00	3.0640
13	9.0×10^{-1}	0.7720	39	250,00	3.1730
14	1.0	0.8020	40	300,00	3.2630
15	1.5	0.9270	41	400,00	3.4060
16	2.0	1.0200	42	500,00	3.5160
17	2.5	1.1010	43	600,00	3.6080
18	3.0	1.1690	44	700,00	3.6840
19	4.0	1.2750	45	800,00	3.7500
20	5.0	1.3620	46	900,00	3.8090
21	6.0	1.4360	47	1000,00	3.8600
22	7.0	1.5000	48	2000,00	4.1801
23	8.0	1.5560	49	5000,00	4.6350

Carter-Tracy Infinite Radial Aquifer Influence Function (Default)					
No.	Infinite Dimensionless		No.	Infinite Dimensionless	
24	9.0	1.6040	50	8000,00	4.8732
25	10.0	1.6510	51	10000,00	4.9815
26	15.0	1.8290			

Table 8.25: Carter-Tracy Infinite Radial Aquifer Influence Function (Default)

For an overview of analytical aquifers see [Dake¹⁸⁹](#).

Example

```
--
--      CARTER-TRACY AQUIFER INFLUENCE TABLES
--      (STARTS FROM TABLE NO. 2, AS DEFAULT IS TABLE NO. 1)
--
AQUITAB
--
--      TD      PD
--      -----
--      0.06      0.251
--      0.08      0.288
--      0.10      0.322
--      0.12      0.355
--      0.14      0.387
--      0.16      0.420
--      0.18      0.452
--      0.20      0.484
--      0.22      0.516
--      0.24      0.548
--      0.26      0.580
--      0.28      0.612
--      0.30      0.644
--      0.35      0.724
--      0.40      0.804
--      0.45      0.884
--      0.50      0.964
--      0.55      1.044
--      0.60      1.124
--
--
--      TD      PD
--      -----
--      0.22      0.443
--      0.24      0.459
--      0.26      0.476
--      0.28      0.492
--      0.30      0.507
--      0.32      0.522
--      0.34      0.536
--      0.36      0.551
--      0.38      0.565
--      0.40      0.579
--      0.42      0.593
--      0.44      0.607
--
--
--      / RD=1.5 TABLE NO. 02
```

¹⁸⁹ Dake, L.P. *Fundamentals of Reservoir Engineering*, Amsterdam, The Netherlands, Elsevier Science BV (1978) Chapter 9.

0.46	0.621	
0.48	0.634	
0.50	0.648	
0.6	0.715	
0.7	0.782	
0.8	0.849	
0.9	0.915	
1.0	0.982	
2.0	1.649	
3.0	2.316	
5.0	3.649	/ RD=2.0 TABLE NO. 03

The above example defines tables two and three Carter-Tracy aquifer influence tables.

8.3.15 BDENSITY – DEFINE THE SURFACE BRINE DENSITY FOR THE FLUID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

BDENSITY defines the brine surface density for when the brine phase has been activated in the model by the **BRINE** keyword in the **RUNSPEC** section. The number of **BDENSITY** vector data sets is defined by the NTPVT parameter on the **TABDIMS** keyword in the **RUNSPEC** section. Each record consists of a maximum of NPPVT values, as declared on the **TABDIMS** keyword in the **RUNSPEC** section, with each value representing a brine surface density.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

The keyword is used in conjunction with the **PVTWSALT** keyword in the **PROPS** section, with each brine density value matching with the salt concentration values in column I of each table in the **PVTWSALT** keyword. Note that the **BDENSITY** keyword is optional, and if absent from the input file, then the brine surface densities will be set to the water density values declared via the **DENSITY** keyword in the **PROPS** section. In this case there is no variation in brine surface density with respect to salt concentration.

No.	Name	Description			Default
I	WATDEN	Field	Metric	Laboratory	None
		WATDEN is a real monotonically increasing positive row vector that defines the brine density at surface conditions for the given salt concentrations on the corresponding <i>PVTWSALT</i> keyword in the <i>PROPS</i> section. There should be one row element for each salt concentration columnar element (SALTCON) on the <i>PVTWSALT</i> keyword.			
		lb/ft ³	kg/m	gm/cc	

Notes:

I) Each row vector data set must be terminated by a “/” including the last data set.

Table 8.26: BDENSITY Keyword Description

Note

In OPM Flow the tracer equations are solved decoupled from the reservoir equations at the end of a time step. For each tracer an implicit system is solved, however, the tracer equations are linear, resulting in converge in two iterations. However, the Brine phase is solved fully implicitly and is fully coupled with the other flow equations.

This is different to the commercial simulator, where the tracer equations are solved explicitly after the flow equations have converged at the end of a time step. This can lead to numerical instabilities if there are large variations in brine densities.

Example

The following shows the *BDENSITY* and *PVTWSALT* keywords for when NTPVT on the *TABDIMS* keyword in the *RUNSPEC* section is set equal to two and NPPVT is set to greater than four on the *TABDIMS* keyword.

```
--
--      BRINE WATER DENSITY DATA FOR PVTWSALT KEYWORD
--
--      SALTCON  SALTCON  SALTCON  SALTCON  SALTCON
--      DENSITY  DENSITY  DENSITY  DENSITY  DENSITY
--      -----  -----  -----  -----  -----
BDENSITY
      62.20      63.50      64.75      65.90
      64.00      65.50      67.00
                                     / FOR PVTWSALT TABLE 1
                                     / FOR PVTWSALT TABLE 2
--
--      WATER SALT PVT TABLE
--
PVTWSALT
--      REF PRES  REF SALT
--      PSIA      LB/STB
--      -----
--      4500.0    0.000
--                                     / TABLE NO. REF. DATA
--
--      SALTCONC  BW      CW      VISC      VISC
--      LB/STB    RB/STB  1/PSIA  CPOISE   GRAD
--      -----
--      0.0       1.020    2.7E-6  0.370    0.0
--      2.0       1.010    2.7E-6  0.370    0.0
--      4.0       1.000    2.7E-6  0.370    0.0
--      10.0      0.950    2.7E-6  0.370    0.0
--                                     / TABLE NO. 01 SALT DATA
--
--      REF PRES  REF SALT
--      PSIA      LB/STB
--      -----
--      4000.0    0.000
--                                     / TABLE NO. 02 REF. DATA
--
--      SALTCONC  BW      CW      VISC      VISC
--      LB/STB    RB/STB  1/PSIA  CPOISE   GRAD
--      -----
--      0.0       1.005    2.5E-6  0.320    0.0
--      6.0       0.985    2.5E-6  0.320    0.0
--      12.0      0.930    2.5E-6  0.320    0.0
--                                     / TABLE NO. 02 SALT DATA
```

8.3.16 BGGI – DEFINE GI GAS FORMATION VOLUME FACTOR PRESSURE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [BGGI](#) keyword defines Gi gas formation volume factor as a function of Gi and pressure for when the Gi option has been invoked via the [GIMODEL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

The accuracy of gas condensate and volatile oil modeling using a black-oil reservoir simulator is now firmly accepted in the industry; both depletion and gas cycling above the saturation point can be modeled and yield an acceptable match compared with compositional simulators. The main deficiency with the existing black-oil formulation is the treatment of gas injection below the saturation point, where the compositional effect of the stripping of liquid components is inversely proportional to their molecular weights, is not considered. This is because the black-oil model assumes that the saturated hydrocarbon fluid properties are only functions of pressure. Thus, when dry gas is injected into a condensate reservoir below the saturation pressure the injected gas continues to re-vaporize liquid at a rate governed only by the cell pressures. Compositional modeling indicate that this not the case.

The [GI](#) option attempts to overcome the limitation of the standard black-oil approach by extending the black-oil model using the method of Cook et al.¹⁹⁰ to take into account the fluid property changes occurring during gas injection. This is done by extending the fluid property treatment so that the saturated fluid properties depend on pressure (as per the standard black-oil formulation) and also on an additional parameter which characterizes the compositional changes in the reservoir liquid and vapor phases at constant pressure.

See also the [PVTG](#) and [GINODE](#) keywords in the [PROPS](#) section.

¹⁹⁰ Cook, R. E., Jacoby, R. H., and Ramesh, A. B. "A Beta-type Reservoir Simulator for Approximating Compositional Effects During Gas Injection" paper SPE 4272, Society of Petroleum Engineers Journal (1974) 14, No. 5, 471-481.

8.3.17 BIC – BINARY INTERACTION COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [BIC](#) keyword defines the binary interaction coefficients for each pair of compositional components active in the model and for each equation of state. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description	Default
I	BIC	A series of real numbers that define the binary interaction coefficients k_{ij} for each pair (i, j) of compositional components active in the model. The matrix k_{ij} is symmetrical with zeroes on the diagonal so only the portion of the matrix below the diagonal is required (i.e., $i = \{2, \dots, \text{COMPS}\}$ and $j < i$), where COMPS is specified by the COMPS keyword in the RUNSPEC section. The values are ordered with j cycling fastest (i.e. $k_{21}, k_{31}, k_{32}, k_{41}, k_{42}, k_{43}, \dots$).	0
Notes: 1) The keyword is followed by NMEOSR records as specified by the TABDIMS keyword in the RUNSPEC section. 2) Each record must contain $\text{COMPS} \times (\text{COMPS} - 1) / 2$ values as specified by the COMPS keyword in the RUNSPEC section. 3) Each record is terminated by a "/" and there is no "/" terminator for the keyword.			

Table 8.27: [BIC](#) Keyword Description

Examples

The following example defines the binary interaction coefficients for each pair of compositional components in a single three-component equation of state model.

```
--
-- Binary Interaction Coefficients
--
BIC
  0.1209
  0.1310  0.0339 /
```

The following example defines the binary interaction coefficients for each component in two three-component equation of state models.

```
--
-- Binary Interaction Coefficients
--
BIC
  0.1209
  0.1310  0.0339 /
  0.1209
  0.1310  0.0339 /
```


8.3.18 BIOFPARA – DEFINE PROPERTIES FOR MODELS WITH BIOFILMS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [BIOFPARA](#) keyword defines parameters for models including biofilms. For the parameters, biomass means both suspended microbes in the water phase (labeled as microbial) and biofilm. Currently the two available models with biofilm effects are the [BIOFILM](#) and the [MICP](#) model. See Landa-Marbán et al¹⁹¹ and ¹⁹² for further information on the [MICP](#) model parameters, which are also used in the [BIOFILM](#) model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DENSBIOF	A real positive value that defines the density of the biofilm.			None
		lb/rft ³	kg/rm ³	gm/rcc	
2	DEATRTE	A real value that defines the biomass death rate.			None
		l/days	l/days	l/hours	
3	GROWRATE	A real positive value that defines the maximum specific biomass growth rate.			None
		l/days	l/days	l/hours	
4	HALFGROW	A real positive value that defines the half-velocity coefficient in the Monod equation for the biomass growth rate.			None
		lb/rft ³	kg/rm ³	gm/rcc	
5	YIELGROW	A real value that defines the yield biomass growth coefficient.			None
		dimensionless	dimensionless	dimensionless	
6	OXYGSUBS	A real value that defines the mass ratio of oxygen consumed to substrate used for biomass growth.			None
		dimensionless	dimensionless	dimensionless	
7	ATTARATE	A real positive value that defines the microbial attachment rate.			None
		l/days	l/days	l/hours	
8	DETRATE	A real positive value that defines the biofilm detachment rate.			None
		l/days	l/days	l/hours	
9	DETAEXPO	A real value that defines the exponent in the norm for the water velocity in the detachment term (this value was set to 0.58 in the MICP publication).			None

¹⁹¹ Landa-Marbán, D., Tveit, S., Kumar, K., Gasda, S.E., 2021. Practical approaches to study microbially induced calcite precipitation at the field scale. *Int. J. Greenh. Gas Control* 106, 103256. <https://doi.org/10.1016/j.jggc.2021.103256>.

¹⁹² Landa-Marbán, D., Kumar, K., Tveit, S., Gasda, S.E., 2021. Numerical studies of CO₂ leakage remediation by micp-based plugging technology. In: Røkke, N.A. and Knuutila, H.K. (Eds) *Short Papers from the 11th International Trondheim CCS conference*, ISBN: 978-82-536-1714-5, 284-290.

No.	Name	Description			Default
		Field	Metric	Laboratory	
		dimensionless	dimensionless	dimensionless	
10	UREARATE	A real positive value that defines the maximum rate of urea utilization.			None
		l/days	l/days	l/hours	
11	HALFUREA	A real positive value that defines the half-velocity coefficient in the Monod equation for the urea utilization rate.			None
		lb/rft ³	kg/rm ³	gm/rcc	
12	DENSALC	A real positive value that defines the calcite density.			None
		lb/rft ³	kg/rm ³	gm/rcc	
13	YIELURCA	A real value that defines the yield coefficient in the calcite precipitation term (units of produced calcite over units of urea utilization).			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN vectors as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Items 10 to 13 are only relevant for the [MICP](#) model.
- 3) Each record is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.28: BIOFPARA Keyword Description

Note

This is an OPM Flow specific keyword.

Example

The example below is based on metric units, with NTSFUN equal to two on the [TABDIMS](#) keyword.

```
--
--      DEFINE BIOMASS PARAMETERS FOR THE MICP MODEL
--
--      DENS    DEAT    GROW    HALF    YIEL    OXYG
--      BIOF    RATE    RATE    GROW    GROW    SUBS
--      -----
--      ATTA    DETA    DETA    UREA    HALF    DENS    YIEL
--      RATE    RATE    EXPO    RATE    UREA    CALC    URCA
--      -----
BIOFPARA
35          0.0275  3.6      2E-05    0.5      0.5
0.0735     2.6E-13 0.58     1391     21.3     2710    1.67 /
10          0.0275  3.6      2E-05    0.5      0.5
0.0735     1.1E-05 0.58     1391     21.3     2710    1.67 /
```

8.3.19 BOGI – DEFINE GI OIL FORMATION VOLUME FACTOR PRESSURE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [BOGI](#) keyword defines Gi oil formation volume factor as a function of Gi and pressure for when the Gi option has been invoked via the [GIMODEL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

The accuracy of gas condensate and volatile oil modeling using a black-oil reservoir simulator is now firmly accepted in the industry; both depletion and gas cycling above the saturation point can be modeled and yield an acceptable match compared with compositional simulators. The main deficiency with the existing black-oil formulation is the treatment of gas injection below the saturation point, where the compositional effect of the stripping of liquid components is inversely proportional to their molecular weights, is not considered. This is because the black-oil model assumes that the saturated hydrocarbon fluid properties are only functions of pressure. Thus, when dry gas is injected into a condensate reservoir below the saturation pressure the injected gas continues to re-vaporize liquid at a rate governed only by the cell pressures. Compositional modeling indicate that this not the case.

The [GI](#) option attempts to overcome the limitation of the standard black-oil approach by extending the black-oil model using the method of Cook et al.¹⁹³ to take into account the fluid property changes occurring during gas injection. This is done by extending the fluid property treatment so that the saturated fluid properties depend on pressure (as per the standard black-oil formulation) and also on an additional parameter which characterizes the compositional changes in the reservoir liquid and vapor phases at constant pressure.

See also the [PVTG](#) and [GINODE](#) keywords in the [PROPS](#) section.

¹⁹³ Cook, R. E., Jacoby, R. H., and Ramesh, A. B. "A Beta-type Reservoir Simulator for Approximating Compositional Effects During Gas Injection" paper SPE 4272, Society of Petroleum Engineers Journal (1974) 14, No. 5, 471-481.

8.3.20 BOX – DEFINE A RANGE OF GRID BLOCKS TO ENTER PROPERTY DATA

[BOX](#) defines a range of grid blocks for which subsequent data is assigned for all the cells in the defined [BOX](#). Note that the [BOX](#) grid is reset by the keyword [ENDBOX](#) by resetting the current defined [BOX](#) to be the whole grid. The keyword can be used for any array and for all grid types.

See [BOX – Define a Range of Grid Blocks to Enter Property Data](#) in the [GRID](#) section for a full description.

8.3.21 C NAMES – DEFINE COMPOSITIONAL COMPONENT NAMES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [C NAMES](#) keyword defines the names for each of the compositional components active in the model. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

Note

This keyword is only supported by OPM Flow when the two phase gas-water CO₂ storage model has been activated using the [CO2STORE](#) keyword and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section. Only the component names "H₂O", "CO₂" and "NaCl" (water, CO₂ and salt respectively) are recognized when the [C NAMES](#) keyword is used with the [CO2STORE](#) keyword; any other component names are ignored.

No.	Name	Description	Default
I	C NAMES	A series of character strings of up to eight characters in length that define the names for each of the compositional components active in the model.	None
Notes: - The keyword must contain COMPS values as specified by the COMPS keyword in the RUNSPEC section. - The keyword is terminated by a "/".			

Table 8.29: C NAMES Keyword Description

Example

The following example defines how to confirm a three component formulation, together with defining the names of the compositional components, to be used with the [CO2STORE](#) and [GASWAT](#) options.

```
--
--      CONFIRM NUMBER OF COMPOSITIONAL COMPONENTS (OPM FLOW KEYWORD)
--
NCOMPS      3
--
--      DEFINE COMPOSITIONAL COMPONENTS NAMES (OPM FLOW KEYWORD)
--
C NAMES
      'H2O '
      'CO2 '
      'NaCl '
--
```

8.3.22 COALADS – DEFINE GAS AND SOLVENT RELATIVE ADSORPTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COALADS](#) keyword defines the gas and solvent relative adsorption tables for when the coal phase has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.23

COALPP – DEFINE GAS AND SOLVENT PARTIAL PRESSURE ADSORPTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COALPP](#) keyword defines the gas and solvent partial pressure adsorption tables for when the coal phase has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.24 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The [COLUMNS](#) keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the GLOBAL section for a full description.

8.3.25 COPY – COPY ARRAY DATA TO ANOTHER ARRAY

The [COPY](#) keyword copies an array (or part of an array) to another array or part of an array. The arrays can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [COPY](#) keyword is being used.

See [COPY – Copy Array Data to Another Array](#) in the [GRID](#) section for a full description.

8.3.26 COPYBOX – COPY ARRAY DATA DEFINED BY A BOX

The [COPYBOX](#) keyword copies an array (or part of an array) to part of the same array. The array can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [COPYBOX](#) keyword is being used.

See [COPYBOX – Copy Array Data Defined by a Box](#) in the [GRID](#) section for a full description.

8.3.27 COPYREG – COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER

The [COPYREG](#) keyword copies a specified array or part of an array based on cells with a specific region number to another array. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [COPYREG](#) keyword is read by the simulator. The property arrays can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [COPYREG](#) keyword is being used.

See [COPYREG – Copy an Array to Another Array based on a Region Number](#) in the [GRID](#) section for a full description.

8.3.28 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the GLOBAL section for a full description.

8.3.29 DENAQA – SPECIFY EZROKHI COEFFICIENTS FOR AQUEOUS DENSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DENAQA](#) keyword specifies the three Ezrokhi coefficients for each compositional component and for each equation of state that are used to calculate the aqueous phase density. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is only supported by OPM Flow when the two phase gas-water CO₂ storage model has been activated using the [CO2STORE](#) keyword and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section. The component names "H2O", "CO2" and "NACL" (water, CO₂ and salt respectively) must be specified using the [CNames](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	COEFS	A series of real numbers that define the three Ezrokhi coefficients for each of the compositional components active in the model.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NMEOSR records as specified by the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each record must contain three values for each compositional component, where the number of components is specified by the *COMPS* keyword in the *RUNSPEC* section.
- 3) Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.30: DENAQA Keyword Description

Examples

The following example defines the Ezrokhi coefficients for each component in a single three-component equation of state model.

```
--
-- Ezrokhi Coefficients for Aqueous Density Calculation
--
DENAQA
--COEFF0    COEFF1    COEFF2
  2.0E-6    -5.0E-7    2.0E-9
  2.0E-6    -5.0E-7    2.0E-9
  2.0E-6    -5.0E-7    2.0E-9    /
```

The following example defines the Ezrokhi coefficients for each component in two three-component equation of state models.

```
--
-- Ezrokhi Coefficients for Aqueous Density Calculation
--
DENAQA
--COEFF0    COEFF1    COEFF2
  2.0E-6    -5.0E-7    2.0E-9
  2.0E-6    -5.0E-7    2.0E-9
  2.0E-6    -5.0E-7    2.0E-9    /
```

--
2.0E-6 -5.0E-7 2.0E-9
2.0E-6 -5.0E-7 2.0E-9
2.0E-6 -5.0E-7 2.0E-9 /

8.3.30 DENSITY – DEFINE THE SURFACE OIL, WATER GAS DENSITIES FOR THE FLUIDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	------------------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

DENSITY defines the oil, water and gas surface densities for the fluids for various regions in the model. The number of **DENSITY** vector data sets is defined by the NTPVT parameter on the **TABDIMS** keyword in the **RUNSPEC** section and the allocation of the **DENSITY** data sets to different grid blocks in the model is done via the **PVTNUM** keyword in the **REGIONS** section. One data set consists of one record or line which is terminated by a “/”. The surface density or gravity must be entered using either the **DENSITY** or **GRAVITY** keywords irrespective of which phases are active in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	OILDEN	OILDEN is a real number defining the density of the oil phase at surface conditions.			Defined
		lb/ft ³ 37.457	kg/m ³ 600	gm/cc 0.6	
2	WATDEN	WATDEN is a real number defining the density of the water phase at surface conditions.			Defined
		lb/ft ³ 62.366	kg/m ³ 999.014	gm/cc 0.999014	
3	GASDEN	GASDEN is a real number defining the density of the gas phase at surface conditions.			Defined
		lb/ft ³ 0.062428	kg/m ³ 1.000	gm/cc 0.001	

Notes:

1)

The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.

2)

Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.31: DENSITY Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table. See the third example for an illustration on how to use this feature.

According to the SPE SI standard¹⁹⁴, **Relative Density** (γ) replaces **Specific Gravity** as the term used to define the ratio of the density of a known material to the density of reference material, at standard conditions of pressure and temperature. Standard conditions vary throughout the world, but for oil field units one normally uses 14.7 psia and 60 °F, whereas for SI units some areas use 101.325 kPa and 15 °C.

See also the **GRAVITY** keyword in the **PROPS** section, that can be used to enter the relative density values instead of the density numbers.

¹⁹⁴ The SI Metric System of Units and SPE Metric Standard, Adopted for Use as a Voluntary Standard by the SPE Board of Directors, June 1983, Society of Petroleum Engineers.

Examples

The following shows the [DENSITY](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to one.

```
--
--      OIL      WAT      GAS
--      DENSITY  DENSITY  DENSITY
--      -----  -----  -----
DENSITY
      39.0      62.37      0.04520                / PVT DATA REGION 1
```

The next example shows the [DENSITY](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to three.

```
--
--      OIL      WAT      GAS
--      DENSITY  DENSITY  DENSITY
--      -----  -----  -----
DENSITY
      38.0      62.30      0.04500                / PVT DATA REGION 1
      39.0      62.37      0.04520                / PVT DATA REGION 2
      40.0      62.40      0.04800                / PVT DATA REGION 3
```

The third, and final, example shows the [DENSITY](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to four. Here table two defaults to table one, and table four defaults to table three.

```
--
--      OIL      WAT      GAS
--      DENSITY  DENSITY  DENSITY
--      -----  -----  -----
DENSITY
      38.0      62.30      0.04500                / PVT DATA REGION 1
                                           / PVT DATA REGION 2
      39.0      62.37      0.04520                / PVT DATA REGION 3
                                           / PVT DATA REGION 3
```

Again, note that there is no terminating “/” for this keyword.

8.3.31 DEPTHTAB – RIVER TIME AND DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [DEPTHTAB](#), defines the river time and depth tables, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.32 DIAGDISP – ACTIVATE ALTERNATE FORM OF TRACER DISPERSION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [DIAGDISP](#), activates the alternate form of tracer dispersion matrix for when the Tracer facility has been activated by the [TRACERS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.33 DIFFAGAS – DEFINE PVT REGION GAS COMPONENT DIFFUSION COEFFICIENTS (MASS FRACTION FORMULATION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFAGAS](#) keyword defines the gas diffusion coefficients assuming a mass fraction formulation for each compositional component in the model and for each PVT region, for when the Molecular Diffusion option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is optional as OPM Flow will automatically calculate the coefficients assuming the mole fraction formulation, as described by Sandve et al.¹⁹⁵, if the [DIFFAGAS](#) and [DIFFCGAS](#) keywords are absent from the input deck. The keyword thus allows one to overwrite the automatically calculated values.

The keyword should only be used if the [CO2STORE](#) or [H2STORE](#) keywords and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section, have also been activated for the gas-water two component model.

See also the [DIFFCGAS](#) and [DIFFCWAT](#) keywords that assume the standard mole fraction formulation for diffusion rather than the mass fraction formulation assumed by the [DIFFAGAS](#) and [DIFFAWAT](#) keywords. The [DIFFAGAS](#) and [DIFFAWAT](#) keywords cannot be used in combination with the [DIFFCGAS](#) and [DIFFCWAT](#) keywords.

Note

This is an OPM Flow keyword used with OPM Flow's [CO2STORE](#) or [H2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section, and should not be confused with the version of the [DIFFAGAS](#) keyword used in the commercial compositional simulator that defines activity corrected diffusion coefficients.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	CO2DIFF	A real positive number that declares the CO ₂ or H ₂ in gas diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
2	WATDIFF	A real positive number that specifies the water in gas diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.32: DIFFAGAS Keyword Description

¹⁹⁵ Tor Harald Sandve, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

Note

The option has been tested in combination with the [CO2STORE](#) keyword, but not for the general case at this point.

See also the [DIFFUSE](#) keyword in the [RUNSPEC](#) section to activate the Molecular Diffusion option and the [DIFFAWAT](#) keyword in the [PROPS](#) section that defines the water diffusion coefficients for each compositional component in the model and for each PVT region. Finally, for gas-oil systems the [DIFFC](#) keyword in the [PROPS](#) section should be used.

Normally diffusion coefficients are measured in laboratory units, that is cm^2/s , for ease of use, Table 8.33 outlines the conversion factors for converting the laboratory measured diffusion coefficients to those used by the simulator.

Diffusivity Conversion Factors		
Laboratory Measured Units	Conversion Factor	Simulator Units
$1 \text{ cm}^2/\text{s}$	92.9979 ft^2/day	Field
	8.64 m^2/day	Metric
	3600 cm^2/hour	Laboratory

Table 8.33: Diffusivity Conversion Factors

Example

The example below is based on field units, with NTPVT equal to three on the [TABDIMS](#) keyword.

```
--
--          PVT REGION GAS COMPONENT DIFFUSION COEFFICIENTS (OPM FLOW KEYWORD)
--
DIFFAGAS
--          CO2 IN      WAT IN
--          GAS DF      GAS DF
--          -----
--          0.160      0.150      / PVT REGION NO. 01
--          0.165      0.155      / PVT REGION NO. 02
--                                   / PVT REGION NO. 03
```

Here the third PVT region has no values for the two component diffusion coefficients, and therefore the simulator will use correlations to define the diffusivity coefficients for this PVT region.

8.3.34 DIFFAWAT – DEFINE PVT REGION WATER COMPONENT DIFFUSION COEFFICIENTS (MASS FRACTION FORMULATION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFAWAT](#) keyword defines the water diffusion coefficients assuming a mass fraction formulation for each compositional component in the model and for each PVT region, for when the Molecular Diffusion option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section.

This keyword is optional as OPM Flow will automatically calculate the coefficients assuming the mole fraction formulation, as described by Sandve et al.¹⁹⁶, if the [DIFFAWAT](#) and [DIFFCWAT](#) keywords are absent from the input deck. The keyword thus allows one to overwrite the automatically calculated values.

The keyword should only be used if the [CO2STORE](#) or [H2STORE](#) keywords and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section, have also been activated for the gas-water two component model.

See also the [DIFFCGAS](#) and [DIFFCWAT](#) keywords that assume the standard mole fraction formulation for diffusion rather than the mass fraction formulation assumed by the [DIFFAGAS](#) and [DIFFAWAT](#) keywords. The [DIFFAGAS](#) and [DIFFAWAT](#) keywords cannot be used in combination with the [DIFFCGAS](#) and [DIFFCWAT](#) keywords.

Note

This is an OPM Flow specific keyword used with OPM Flow's [CO2STORE](#) or [H2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	CO2DIFF	A real positive number that declares the CO ₂ or H ₂ in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
2	WATDIFF	A real positive number that specifies the water in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	

Notes:

1)

The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.

2)

Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.34: DIFFAWAT Keyword Description

Note

The option has been tested in combination with the [CO2STORE](#) keyword, but not for the general case at this point.

¹⁹⁶ Tor Harald Sandve, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

See also the [DIFFUSE](#) keyword in the [RUNSPEC](#) section to activate the Molecular Diffusion option and the [DIFFAGAS](#) keyword in the [PROPS](#) section that defines the gas diffusion coefficients for each compositional component in the model and for each PVT region. Finally, for gas-oil systems the [DIFFC](#) keyword in the [PROPS](#) section should be used.

Normally diffusion coefficients are measured in laboratory units, that is cm^2/s , for ease of use, Table 8.35 outlines the conversion factors for converting the laboratory measured diffusion coefficients to those used by the simulator.

Diffusivity Conversion Factors		
Laboratory Measured Units	Conversion Factor	Simulator Units
1 cm^2/s	92.9979 ft^2/day	Field
	8.64 m^2/day	Metric
	3600 cm^2/hour	Laboratory

Table 8.35: Diffusivity Conversion Factors

Example

The example below is based on field units, with NTPVT equal to three on the [TABDIMS](#) keyword.

```
--
--      PVT REGION WATER COMPONENT DIFFUSION COEFFICIENTS (OPM FLOW KEYWORD)
--
DIFFAWAT
--      CO2 IN      WAT IN
--      WAT DF      WAT DF
--      -----
--      0.160      0.150      / PVT REGION NO. 01
--      0.165      0.155      / PVT REGION NO. 02
--                               / PVT REGION NO. 03
```

Here the third PVT region has no values for the two component diffusion coefficients, and therefore the simulator will use correlations to define the diffusivity coefficients for this PVT region.

8.3.35 DIFFC – DEFINE PVT REGION MOLECULAR DIFFUSION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DIFFC](#) keyword defines the molecular weight of the fluids and diffusion coefficients between phases for each PVT region, for when the Molecular Diffusion option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section. This keyword is optional as OPM Flow will automatically calculate the coefficients, as described by Sandve et al.¹⁹⁷, if the [DIFFC](#) keyword is absent from the input deck. The keyword thus allows one to overwrite the automatically calculated values.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	OILMW	OILMW is a real positive number that specifies the molecular weight of the oil in the given PVT region.			None
		lb/lb-M	kg/kg-M	gm/gm-M	
2	GASMW	GASMW is a real positive number that defines the molecular weight of the gas in the given PVT region.			None
		lb/lb-M	kg/kg-M	gm/gm-M	
3	GASGASDF	A real positive number that defines the gas in gas diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
4	OILGASDF	A real positive number that declares the oil in gas diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
5	GASOILDF	A real positive number that specifies the gas in oil diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
6	OILOILDF	A real positive number that defines the oil in oil diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
7	GASOILCD	A real positive number that defines the gas in oil cross phase diffusion coefficient in the given PVT region. This parameter is ignored by OPM Flow and should be defaulted or set equal to zero.			0.0
		ft ² /day	m ² /day	cm ² /hour	

¹⁹⁷ Tor Harald Sandve I, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	OILOILCD	A real positive number that defines the oil in oil cross phase diffusion coefficient in the given PVT region. This parameter is ignored by OPM Flow and should be defaulted or set equal to zero.			0.0
		ft ² /day	m ² /day	cm ² /hour	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.36: DIFFC Keyword Description

Note

The option has been tested in combination with the [CO2STORE](#) keyword, but not for the general case at this point.

See also the [DIFFUSE](#) keyword in the [RUNSPEC](#) section to activate the Molecular Diffusion option.

Example

The example below is based on field units, with NTPVT equal to three on the [TABDIMS](#) keyword.

```
--
--      PVT REGION MOLECULAR DIFFUSION TABLES
--
DIFFC
--      OIL      GAS      GAS IN   OIL IN   GAS IN   OIL IN   GAS IN   OIL IN
--      MW      MW      GAS DF   GAS DF   OIL DF   OIL DF   OIL CD   OIL CD
--      -----
--      103.20  1.120  1.35E-6  1.05E-7  4.50E-7  1.05E-8
--      102.00  1.130  1.25E-6  1.25E-7  4.80E-7  1.05E-8
--      100.00  1.250  1.22E-6
--                                     /TAB-1
--                                     /TAB-2
--                                     /TAB-3
```

Here the third PVT region has no values for the various oil related diffusion coefficients.

8.3.36 DIFFCGAS – DEFINE PVT REGION GAS COMPONENT DIFFUSION COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **DIFFCGAS** keyword defines the gas diffusion coefficients assuming the standard mole fraction formulation for each compositional component in the model and for each PVT region, for when the Molecular Diffusion option has been activated by the **DIFFUSE** keyword in the **RUNSPEC** section.

This keyword is optional as OPM Flow will automatically calculate the coefficients assuming the mole fraction formulation, as described by Sandve et al.¹⁹⁸, if the **DIFFAGAS** and **DIFFCGAS** keywords are absent from the input deck. The keyword thus allows one to overwrite the automatically calculated values.

The keyword should only be used if the **CO2STORE** or **H2STORE** keyword and either the **GASWAT** or the **GAS** and **WATER** keywords in the **RUNSPEC** section, have also been activated for the gas-water two component model.

See also the **DIFFAGAS** and **DIFFAWAT** keywords that assume a mass fraction formulation for diffusion rather than the standard mole fraction formulation assumed by the **DIFFCGAS** and **DIFFCWAT** keywords. The **DIFFAGAS** and **DIFFAWAT** keywords cannot be used in combination with the **DIFFCGAS** and **DIFFCWAT** keywords.

Note

This is an OPM Flow keyword used with OPM Flow's **CO2STORE** or **H2STORE** and **GASWAT** keywords in the **RUNSPEC** section, and should not be confused with the more general version of the **DIFFCGAS** keyword used in the commercial compositional simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	CO2DIFF	A real positive number that declares the CO ₂ or H ₂ in gas diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
2	WATDIFF	A real positive number that specifies the water in gas diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	

Notes:

1)

The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.

2)

Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.37: DIFFCGAS Keyword Description

¹⁹⁸ Tor Harald Sandve, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

Note

The option has been tested in combination with the [CO2STORE](#) keyword, but not for the general case at this point.

See also the [DIFFUSE](#) keyword in the [RUNSPEC](#) section to activate the Molecular Diffusion option and the [DIFFCWAT](#) keyword in the [PROPS](#) section that defines the water diffusion coefficients for each compositional component in the model and for each PVT region. Finally, for gas-oil systems the [DIFFC](#) keyword in the [PROPS](#) section should be used.

Normally diffusion coefficients are measured in laboratory units, that is cm^2/s , for ease of use, Table 8.38 outlines the conversion factors for converting the laboratory measured diffusion coefficients to those used by the simulator.

Diffusivity Conversion Factors		
Laboratory Measured Units	Conversion Factor	Simulator Units
$1 \text{ cm}^2/\text{s}$	92.9979 ft^2/day	Field
	8.64 m^2/day	Metric
	3600 cm^2/hour	Laboratory

Table 8.38: Diffusivity Conversion Factors

Example

The example below is based on field units, with NTPVT equal to three on the [TABDIMS](#) keyword.

```
--
--          PVT REGION GAS COMPONENT DIFFUSION COEFFICIENTS (OPM FLOW KEYWORD)
--
DIFFCGAS
--          CO2 IN      WAT IN
--          GAS DF      GAS DF
--          -----
--          0.160      0.150      / PVT REGION NO. 01
--          0.165      0.155      / PVT REGION NO. 02
--                                   / PVT REGION NO. 03
```

Here the third PVT region has no values for the two component diffusion coefficients, and therefore the simulator will use correlations to define the diffusivity coefficients for this PVT region.

8.3.37 DIFFCOAL – DEFINE COAL BED METHANE GAS DIFFUSION DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The DIFF keyword defines the coal bed methane diffusion data for when the Coal Bed Methane option has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.38 DIFFCWAT – DEFINE PVT REGION WATER COMPONENT DIFFUSION COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **DIFFCWAT** keyword defines the water diffusion coefficients assuming the standard mole fraction formulation for each compositional component in the model and for each PVT region, for when the Molecular Diffusion option has been activated by the **DIFFUSE** keyword in the **RUNSPEC** section.

This keyword is optional as OPM Flow will automatically calculate the coefficients assuming the mole fraction formulation, as described by Sandve et al.¹⁹⁹, if the **DIFFAWAT** and **DIFFCWAT** keywords are absent from the input deck. The keyword thus allows one to overwrite the automatically calculated values.

The keyword should only be used if the **CO2STORE** or **H2STORE** keyword and either the **GASWAT** or the **GAS** and **WATER** keywords in the **RUNSPEC** section, have also been activated for the gas-water two component model.

See also the **DIFFAGAS** and **DIFFAWAT** keywords that assume a mass fraction formulation for diffusion rather than the standard mole fraction formulation assumed by the **DIFFCGAS** and **DIFFCWAT** keywords. The **DIFFAGAS** and **DIFFAWAT** keywords cannot be used in combination with the **DIFFCGAS** and **DIFFCWAT** keywords.

Note

This is an OPM Flow keyword used with OPM Flow's **CO2STORE** or **H2STORE** and **GASWAT** keywords in the **RUNSPEC** section, and should not be confused with the more general version of the **DIFFCWAT** keyword used in the commercial compositional simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	CO2DIFF	A real positive number that declares the CO ₂ or H ₂ in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
2	WATDIFF	A real positive number that specifies the water in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	

Notes:

1)

The keyword is followed by NTPVT tables as declared on the **TABDIMS** keyword in the **RUNSPEC** section.

2)

Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.39: DIFFCWAT Keyword Description

¹⁹⁹ Tor Harald Sandve, Sarah E. Gasda, Atgeirr Rasmussen, and Alf Birger Rustad. Convective dissolution in field scale CO₂ storage simulation using the OPM Flow simulator. Submitted to TCCS 11 – Trondheim Conference on CO₂ Capture, Transport and Storage Trondheim, Norway – June 21-23, 2021.

Note

The option has been tested in combination with the [CO2STORE](#) keyword, but not for the general case at this point.

See also the [DIFFUSE](#) keyword in the [RUNSPEC](#) section to activate the Molecular Diffusion option and the [DIFFCGAS](#) keyword in the [PROPS](#) section that defines the gas diffusion coefficients for each compositional component in the model and for each PVT region. Finally, for gas-oil systems the [DIFFC](#) keyword in the [PROPS](#) section should be used.

Normally diffusion coefficients are measured in laboratory units, that is cm^2/s , for ease of use Table 8.40 outlines the conversion factors for converting the laboratory measured diffusion coefficients to those used by the simulator.

Diffusivity Conversion Factors		
Laboratory Measured Units	Conversion Factor	Simulator Units
$1 \text{ cm}^2/\text{s}$	92.9979 ft^2/day	Field
	8.64 m^2/day	Metric
	3600 cm^2/hour	Laboratory

Table 8.40: Diffusivity Conversion Factors

Example

The example below is based on field units, with NTPVT equal to three on the [TABDIMS](#) keyword.

```
--
--          PVT REGION WATER COMPONENT DIFFUSION COEFFICIENTS (OPM FLOW KEYWORD)
--
DIFFCWAT
--          CO2 IN      WAT IN
--          WAT DF      WAT DF
--          -----
--          0.160      0.150      / PVT REGION NO. 01
--          0.165      0.155      / PVT REGION NO. 02
--                                   / PVT REGION NO. 03
```

Here the third PVT region has no values for the two component diffusion coefficients, and therefore the simulator will use correlations to define the diffusivity coefficients for this PVT region.

8.3.39
DIFFDP
– ACTIVATE DUAL POROSITY MOLECULAR DIFFUSION FOR MATRIX-FRACTURE FLOW ONLY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, *DIFFDP*, activates the dual porosity molecular diffusion for matrix-fracture flow only option for when the Dual Porosity option has be activated by either the *DUALPORO* or *DUALPERM* keywords, and the Diffusivity option has been activated by the *DIFFUSE* keywords; three keywords are in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.40 DIFFMICP – DEFINE PVT REGION MICP DIFFUSION COEFFICIENTS (MASS CONCENTRATION FORMULATION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [DIFFMICP](#) keyword defines the diffusion coefficients assuming the mass concentration formulation for each component dissolved in water and for each PVT region, for when the molecular diffusion option has been activated by the [DIFFUSE](#) keyword in the [RUNSPEC](#) section. The keyword should only be used if either the [MICP](#) or [BIOFILM](#) model has been activated in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	MICRDIFF	A real positive number that declares the microbial concentration in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
2	OXYGDIFF	A real positive number that declares the oxygen concentration in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	
3	UREADIFF	A real positive number that declares the urea concentration in water diffusion coefficient in the given PVT region.			None
		ft ² /day	m ² /day	cm ² /hour	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Item 1 is only relevant for the *BIOFILM* model.
- 3) Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.41: DIFFMICP Keyword Description

Example

The example below is based on metric units, with NTPVT equal to one on the [TABDIMS](#) keyword.

```
--
--      PVT REGION MICP COMPONENT DIFFUSION COEFFICIENTS (OPM FLOW KEYWORD)
--
DIFFMICP
--      MICR IN      OXYG IN      UREA IN
--      WAT DF      WAT DF      WAT DF
--      -----
--      2E-04      2E-04      1E-04                / PVT REGION NO. 01
```


8.3.41 DIFFMMF – DEFINE DIFFUSIVITY MULTIPLIERS FOR MATRIX-FRACTURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [DIFFMMF](#), defines the diffusivity multipliers for matrix-fractures for when the Dual Porosity option has be activated by either the [DUALPORO](#) or [DUALPERM](#) keywords, or the Coal Bed Methane option is selected by the [COAL](#) keyword, and the Diffusivity option has been activated by the [DIFFUSE](#) keywords; all four keywords are in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.42 DISPERSE – DEFINE DISPERSION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [DISPERSE](#), defines the dispersion tables for when the Dispersion option has been activated via declaring the dimensions of the [DISPERSE](#) tables using the [DISPDIMS](#) keyword and activating the Tracer option via the [TRACERS](#) keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.43 DPKRMOD – MODIFY MATRIX OIL RELATIVE PERMEABILITY DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [DPKRMOD](#) keyword can be used to modify the matrix oil relative permeability data (oil-water; oil-gas) and the scaling of the fracture to matrix relative permeabilities, for dual porosity runs for when a Dual Porosity model has been activated by either the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.44 DSPDEINT – ACTIVATE BRINE TRACER DISPERSION INTERPOLATION BY WATER DENSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [DSPDEINT](#), activates the brine tracer dispersion interpolation by water density option for when the Brine phase is activated in the model by the [BRINE](#) keyword in the [RUNSPEC](#) section and the [DISPERSE](#) keyword in the [PROPS](#) section is in the input file. They keyword cause the lookup and interpolation of the [DISPERSE](#) tracer concentration to water density, that is the tracer concentration data on the [DISPERSE](#) keyword has been replaced by the water density data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.45 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

8.3.46 EHSTR – DEFINE HYSTERESIS MODEL AND PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [EHSTR](#) keyword defines the hysteresis model and associated parameters when the hysteresis option has been activated by the [HYSTER](#) variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. Both the [Carlson²⁰⁰](#) and [Killough²⁰¹](#) models are available.

No.	Name	Description	Default																																				
1	HYSTRCP	HYSTRCP is a positive real value that defines the Killough curvature parameter for capillary pressure hysteresis model. The value should range from 0.05 to 0.10. This item is not supported by OPM Flow and should be defaulted (1*).	0.1																																				
2	HYSTMOD	An integer value that determines the relative permeability hysteresis model to be used depending on the phase and the wettability of the system. HYSTMOD should be set to one of the following values: <table><tr><th colspan="3">Water Wet Hysteresis Models</th></tr><tr><th>HYSTMOD</th><th>Non-Wetting Phases</th><th>Wetting Phase</th></tr><tr><td>-1</td><td colspan="2">Equilibration option for equilibrating the model with the SATNUM (drainage curves) and running the model with imbibition curves (IMBNUM). This option implies no hysteresis.</td></tr><tr><td>0</td><td>Carlson Hysteresis Model</td><td>SATNUM</td></tr><tr><td>1</td><td>Carlson Hysteresis Model</td><td>IMBNUM</td></tr><tr><td>2</td><td>Killough Hysteresis Model</td><td>SATNUM</td></tr><tr><td>3</td><td>Killough Hysteresis Model</td><td>IMBNUM</td></tr><tr><td>4</td><td>Killough Hysteresis Model</td><td>Killough Hysteresis Model</td></tr><tr><th colspan="3">Oil Wet to Water Wet Water Wet</th></tr><tr><td>5</td><td>Carlson Non- Wetting Modeling for Gas and Water</td><td>SATNUM</td></tr><tr><td>6</td><td>Killough Non- Wetting Modeling for Gas and Water</td><td>SATNUM</td></tr><tr><td>7</td><td>Killough Non- Wetting Modeling for Gas and Water</td><td>Killough Non- Wetting Modeling for the Wetting Oil Phase</td></tr></table>	Water Wet Hysteresis Models			HYSTMOD	Non-Wetting Phases	Wetting Phase	-1	Equilibration option for equilibrating the model with the SATNUM (drainage curves) and running the model with imbibition curves (IMBNUM). This option implies no hysteresis.		0	Carlson Hysteresis Model	SATNUM	1	Carlson Hysteresis Model	IMBNUM	2	Killough Hysteresis Model	SATNUM	3	Killough Hysteresis Model	IMBNUM	4	Killough Hysteresis Model	Killough Hysteresis Model	Oil Wet to Water Wet Water Wet			5	Carlson Non- Wetting Modeling for Gas and Water	SATNUM	6	Killough Non- Wetting Modeling for Gas and Water	SATNUM	7	Killough Non- Wetting Modeling for Gas and Water	Killough Non- Wetting Modeling for the Wetting Oil Phase	0
Water Wet Hysteresis Models																																							
HYSTMOD	Non-Wetting Phases	Wetting Phase																																					
-1	Equilibration option for equilibrating the model with the SATNUM (drainage curves) and running the model with imbibition curves (IMBNUM). This option implies no hysteresis.																																						
0	Carlson Hysteresis Model	SATNUM																																					
1	Carlson Hysteresis Model	IMBNUM																																					
2	Killough Hysteresis Model	SATNUM																																					
3	Killough Hysteresis Model	IMBNUM																																					
4	Killough Hysteresis Model	Killough Hysteresis Model																																					
Oil Wet to Water Wet Water Wet																																							
5	Carlson Non- Wetting Modeling for Gas and Water	SATNUM																																					
6	Killough Non- Wetting Modeling for Gas and Water	SATNUM																																					
7	Killough Non- Wetting Modeling for Gas and Water	Killough Non- Wetting Modeling for the Wetting Oil Phase																																					

²⁰⁰ Carlson, F. M. "Simulation of Relative Permeability Hysteresis to the Non-Wetting Phase," paper SPE 10157, presented at the SPE Annual Technical Conference & Exhibition, San Antonio, Texas, USA (October 5-7, 1981).

²⁰¹ Killough, J. E. "Reservoir Simulation with History-dependent Saturation Functions," paper SPE 5106, Society of Petroleum Engineers Journal (1976) 16, No. 1, 37-48.

No.	Name	Description	Default
		Only the HYSTMOD options not colored orange are currently support by OPM Flow.	
3	HYSTREL	HYSTREL is a positive real number that defines the Killough's wetting phase relative permeability curvature parameter. This parameter is only applicable if HYSMOD is set to either 4 or 7. This item is not supported by OPM Flow and should be defaulted (I*).	1.0
4	HYSTSGR	HYSTSGR is a positive real number that sets a scaling parameter for the trapped non-wetting phase saturation in the Killough model. This item is not supported by OPM Flow and should be defaulted (I*).	0.1
5	HYSTOPT	A defined character string that determines if the hysteresis model should be activated for relative permeability, capillary pressure, or both, and should be set to one of the following: 1) BOTH: apply hysteresis modeling to both relative permeability, and capillary pressure curves. 2) PC: apply hysteresis modeling to capillary pressure curves only. 3) KR: apply hysteresis modeling to relative permeability curves only.	BOTH
6	HYSTSCAN	A defined character string that determines the shape of Killough capillary pressure scanning curves when secondary reversal occurs, that is for a drainage, imbibition, drainage cycle. 1) RETR: Secondary drainage curves re-traverses the same scanning curve. 2) NEW: Secondary drainage curves follows a new scanning curve and further reversals also generate a new scanning curve. Only the RETR option is supported by OPM Flow.	RETR
7	HYSTMOB	A defined character string that determines how to apply the mobility control correction invoked by the MOBILE variable on the EQLOPTS keyword in the RUNSPEC section. HYSTMOB should be set to one of the following: 1) DRAIN: Only the drainage curve end-points are modified. 2) BOTH: Both the drainage and imbibition curve end-points are modified. The Mobility Control option is not supported in OPM Flow so this parameter has no effect and will be ignored. This item should be defaulted.	DRAIN
8	HYSTWET	A defined character string that sets the wetting phase between oil and gas in three phase systems and should be set to one of the following: 1) OIL: Oil is set as the <u>wetting phase</u> between oil and gas, and the oil relative permeability with respect to gas is determined by HYSTMOD for the wetting phase. 2) GAS: Oil is set as the <u>non-wetting phase</u> between oil and gas, and the oil relative permeability with respect to gas is determined by HYSTMOD for the non-wetting phase. Note for all the above cases the gas relative permeability is always modelled as a non-wetting phase. Note oil is always the wetting phase to gas in two phase systems. This item is not supported by OPM Flow and should be defaulted (I*).	OIL

No.	Name	Description	Default
9	HYBAKOIL	Baker model one or model two relative permeability oil phase hysteresis option used in the commercial compositional simulator. This item is not supported by OPM Flow and should be defaulted (I*).	NO
10	HYBAKGAS	Baker model one or model two relative permeability gas phase hysteresis option used in the commercial compositional simulator. This item is not supported by OPM Flow and should be defaulted (I*).	NO
11	HYBAKWAT	Baker model one or model two relative permeability water phase hysteresis option used in the commercial compositional simulator. This item is not supported by OPM Flow and should be defaulted (I*).	NO
12	HYTHRESH	Killough's hysteresis threshold saturation value used in the commercial compositional simulator. This item is not supported by OPM Flow and should be defaulted (I*).	0.0
13	HYSWETRP	Killough's hysteresis wetting phase modification used in the commercial black-oil simulator. This item is not supported by OPM Flow and should be defaulted (I*).	0
14	HYSPCSCL	An integer value that determines if capillary pressure scaling should be enabled to correct the construction of the scanning capillary pressure curve. 1) 0: Capillary pressure scaling is disabled. 2) 1: Capillary pressure scaling is enabled. This option is currently disabled by default but this is likely to change as the correction is believed to be a bug fix. For more information see #6383 . This is an OPM Flow specific item that is not supported by the commercial simulator.	0
Notes: I) The keyword is terminated by a "/".			

Table 8.42: EHYSTR Keyword Description

Example

```
--
--      HYSTERESIS MODEL AND PARAMETERS
--
--      PC-CUR  MODEL  RELPERM  TRAPPED  OPTION  SHAPE  MOBILIT  WET
--      HYSTRCP HYSTMOD HYSTREL  HYSTSGR  HYSTOPT  HYSTSCAN HYSTMOB  HYSTWET
EHYSTR
      0.1      0      0.1      1*      KR      1*      1*      1*      /
```

The above example defines the hysteresis model and parameters used in the Norne model. Here the default value is used for the Killough curvature parameter for capillary pressure hysteresis mode, the Carlson hysteresis model is used for the non-wetting phase and [SATNUM](#) for the wetting phase, 0.1 is used for Killough's wetting phase relative permeability curvature parameter (this parameter is ignored because the Carlson model has been selected), the default values for the trapped non-wetting phase saturation in the Killough mode (again, this parameter is ignored because the Carlson model has been selected, and the hysteresis modeling is only applied to relative permeability curves).

8.3.47 EHYSTRR – DEFINE HYSTERESIS MODEL AND PARAMETERS VIA SATNUM

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **EHYSTRR** keyword defines the hysteresis model and associated parameters via the drainage **SATNUM** allocation region array, for when the hysteresis option has been activated by the **HYSTER** variable on the **SATOPTS** keyword in the **RUNSPEC** section. Only the Killough²⁰² model is available for this keyword and the keyword is optional.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped. See also the **EHYSTR** keyword in the **RUNSPEC** for an alternative keyword to enter the hysteresis model and associated parameters that is supported by OPM Flow

No.	Name	Description	Default
1	HYSTRCP	HYSTRCP is a positive real value that defines the Killough curvature parameter for capillary pressure hysteresis model. The value should range from 0.05 to 0.10.	0.1
2	HYSTREL	HYSTREL is a positive real number that defines the Killough's wetting phase relative permeability curvature parameter. This parameter is ignored if HYSMOD on the EHYSTR keyword is not set to 4.	1.0
3	HYSTSGR	HYSTSGR is a positive real number that sets a scaling parameter for the trapped non-wetting phase saturation in the Killough model.	0.1
Notes: 1) The keyword is followed by NTSFUN rows as declared on the TABDIMS keyword in the RUNSPEC section. 2) There is no "/" terminator for the keyword.			

Table 8.43: EHYSTRR Keyword Description

Example

```
--
--      HYSTERESIS MODEL AND PARAMETERS VIA SATNUM
--
--      PC-CUR  RELPERM  TRAPPED
--      HYSTRCP  HYSTREL  HYSTSGR
EHYSTRR
    0.04      1.0      1*          / SATNUM REGION 1
    0.06      1.0      1*          / SATNUM REGION 2
    0.08      1.0      1*          / SATNUM REGION 3
    0.10      1.0      1*          / SATNUM REGION 4
    0.10      1.0      1*          / SATNUM REGION 5
```

The above example defines the hysteresis model and parameters for when NTSFUN equals five on the **TABDIMS** keyword in the **RUNSPEC** section, that is for five **SATNUM** regions.

²⁰² Killough, J. E. "Reservoir Simulation with History-dependent Saturation Functions," paper SPE 5106, Society of Petroleum Engineers Journal (1976) 16, No. 1, 37-48.

8.3.48 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the GLOBAL section for a full description.

8.3.49 ENDBOX – DEFINE THE END OF THE BOX DEFINED GRID

This keyword marks the end of a previously defined [BOX](#) sub-grid as defined by a previously entered [BOX](#) keyword. The keyword resets the input grid to be the full grid as defined by the NX, NY, and NZ variables on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

See [ENDBOX – Define the End of the BOX Defined Grid](#) in the [GRID](#) section for a full description.

8.3.50 ENDFIN – END THE DEFINITION OF A LOCAL GRID REFINEMENT

[ENDFIN](#) defines the end of a Cartesian or radial local grid refinement (“LGR”) definition and a [LGR](#) property definition data set.

See [ENDFIN – End the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

8.3.51 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the GLOBAL section for a full description.

8.3.52 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the GLOBAL section for a full description.

8.3.53 ENKRVD – DEFINE RELATIVE PERMEABILITY END-POINTS VERSUS DEPTH FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the maximum oil, gas, and water relative permeability values versus depth for the three phases and for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. This functionality is not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.54 ENPCVD – DEFINE MAXIMUM CAPILLARY PRESSURE VERSUS DEPTH FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the maximum gas-oil and water-oil capillary pressure values versus depth for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.55
ENPTVD – DEFINE RELATIVE PERMEABILITY SATURATION END-POINTS VERSUS DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the variation of the relative permeability saturation end-points ([SWL](#), [SWCR](#), etc.) for all three phases versus depth for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. This functionality is not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.56 ENSPCVD – DEFINE CAPILLARY PRESSURE END-POINTS VERSUS DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the variation of the capillary pressure saturation end-points, connate gas ([SGL](#)) and connate water ([SWL](#)), versus depth for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. This functionality is not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.57 EOS – SPECIFY EQUATION OF STATE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *EOS* keyword specifies the Equation Of State (*EOS*) to be used for each *EOS* region. The keyword should only be used if the compositional mode has been requested using the *COMPS* keyword in the *RUNSPEC* section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description	Default
I	EQUATION	A defined character string that specifies the Equation of State to be used, and should be set to one of the following: 1) PR: Peng-Robinson, 2) RK: Redlich-Kwong, or 3) SRK: Soave-Redlich-Kwong, or 4) ZJ: Zudkevitch-Joffe-Redlich-Kwong.	PR
Notes: 1) The keyword is followed by NMEOSR records as specified by the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section. 2) Each record is terminated by a “/” and there is no “/” terminator for the keyword.			

Table 8.44: *EOS* Keyword Description

Example

The following example defines Peng-Robinson as the Equation of State for a model with a single *EOS* region.

```
--
-- SPECIFY EQUATION OF STATE
--
EOS
    PR
    /
```

The following example defines Peng-Robinson as the Equation of State for *EOS* region one, and Soave-Redlich-Kwong as the Equation of State for *EOS* region two in a model with two *EOS* regions.

```
--
-- SPECIFY EQUATION OF STATE
--
EOS
    PR
    RK
    /
    /
```

8.3.58 EPSDBGS – WRITE END-POINT DEBUG DATA TO THE DEBUG FILE (MULTIPLE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [EPSDBGS](#), defines the end-point debug data for multiple grid blocks that should be written to the [DEBUG](#) file (*.DBG) for when the End-Point Scaling option has been activated by the [ENDSCALE](#) keyword in the [RONSPEC](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

8.3.59 EPSDEBUG – WRITE END-POINT DEBUG DATA TO THE DEBUG FILE (INDIVIDUAL)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [EPSDEBUG](#), defines the end-point debug data for individual grid blocks that should be written to the Debug file (*.DBG) for when the End-Point Scaling option has been activated by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

8.3.60 EQUALREG – SETS AN ARRAY TO A CONSTANT BY REGION NUMBER

The [EQUALREG](#) keyword sets a specified array to a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [EQUALREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [EQUALREG](#) keyword is being used.

See [EQUALREG – Sets an Array to a Constant by Region Number](#) in the [GRID](#) section for a full description.

8.3.61 EQUALS – SETS A SPECIFIED ARRAY TO A CONSTANT

The [EQUALS](#) keyword sets a specified array or part of an array to a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [EQUALS](#) keyword is being used.

See [EQUALS – Sets a Specified Array to a Constant](#) in the [GRID](#) section for a full description.

8.3.62

ESSNODE – DEFINE SALT CONCENTRATION DATA FOR WATER-OIL SURFACE TENSION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [ESSNODE](#), defines the salt concentration data that is used in calculating the water-oil surface tension for when the Brine option has been activated by the [BRINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.63 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

8.3.64 FHERCHBL – DEFINE HERSCHEL-BULKLEY DATA VERSUS POLYMER CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FHERCHBL](#) keyword defines Herschel-Bulkley rheological property data for Non-Newtonian fluids versus polymer concentration, for when the Polymer option has been invoked via the [POLYMER](#) keyword in the [RUNSPEC](#) section and Non-Newtonian Fluid phase has been declared active by the [NNEWTF](#) keyword, also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.65 FILEUNIT – ACTIVATE UNIT CONSISTENCY VERIFICATION

The [FILEUNIT](#) keyword defines the units of the data set, and is used to verify that the units in the input deck and any associated include files are consistent. The keyword does not provide for the conversion between different sets of units.

See [FILEUNIT – Activate Unit Consistency Checking](#) in the [GRID](#) section for a full description.

8.3.66 FILLEPS – ACTIVATE SATURATION END-POINT EXPORT TO THE INIT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword switches on the export of the saturation end-point data ([SWL](#), [SWCR](#), [SOWCR](#) array etc.) to the *.INIT file so that the data can be viewed in post-processing software like OPM ResInsight.

There is no data required for this keyword.

Example

```
--
--      ACTIVATE SATURATION END-POINT EXPORT TO THE INIT FILE
--
FILLEPS
```

The above example switches on the export of the end-point saturation data to the *.INIT file.

8.3.67 FOAMADS – DEFINE FOAM ROCK ADSORPTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMADS](#) keyword defines the foam rock adsorption tables for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	FOAMCON	A columnar vector of real monotonically increasing down the column values that defines the foam concentration in the solution surrounding the rock. The first entry should be zero to define a no foam concentration data set. Units are dependent on the transport phase specified via the FOAMOPT I variable on the <i>FOAMOPTS</i> keyword in the <i>PROPS</i> section.			None
		Gas: lb/Mscf Water: lb/stb	Gas: kg/sm ³ Water: kg/sm ³	Gas: gm/scc Water: gm/scc	
2	FOAMRATI	A columnar vector of real increasing down the column values that defines the mass of adsorbed foam per unit mass of rock for a given FOAMCON. The first table data set entry should be zero to define a no foam concentration data set.			None
		lb/lb	kg/kg	gm/gm	
Notes: 1) The keyword is followed by NTSFUN tables as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section. 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section. 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.					

Table 8.45: FOAMADS Keyword Description

Example

```
--
--          FOAM ROCK ADSORPTION TABLE
--
FOAMADS
--          FOAM          FOAM
--          FOAMCON       FOAMRATI
--          -----
--              0.0        0.000000
--              2.0        0.000003
--              4.0        0.000005
--              6.0        0.000007
--              8.0        0.000009
--             10.0        0.000011
--             12.0        0.000012
--             14.0        0.000015
--                                / TABLE NO. 01
--          FOAM          FOAM
--          FOAMCON       FOAMRATI
--          -----
--              0.0        0.000000
--              3.0        0.000004
--              5.0        0.000006
--              7.0        0.000008
--              8.0        0.000009
--             10.0        0.000011
--                                / TABLE NO. 02
```

The above example defines two foam rock adsorption tables assuming NTSFUN equals two and NSSFUN is greater than or equal to eight on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

8.3.68
FOAMDCYO – DEFINE FOAM DECAY VERSUS OIL SATURATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMDCYO](#) keyword defines the foam decay half-life versus oil saturation for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.69
FOAMDCYW
– DEFINE FOAM DECAY VERSUS WATER SATURATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMDCYW](#) keyword defines the foam decay half-life versus water saturation for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.70
FOAMFCN
– DEFINE FOAM GAS MOBILITY REDUCTION VERSUS CAPILLARY NUMBER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMFCN](#) keyword defines the reduction in gas mobility versus capillary number, for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section, and the FOAMOPT2 parameter on the [FOAMOPTS](#) keyword in the [PROPS](#) section has been set to the character string FUNC.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.71
FOAMFRM
– DEFINE FOAM GAS MOBILITY REDUCTION VERSUS REFERENCE MOBILITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMFRM](#) keyword defines the reduction in gas mobility versus the reference mobility reduction factor, for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section, and the FOAMOPT2 parameter on the [FOAMOPTS](#) keyword in the [PROPS](#) section has been set to the character string FUNC.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.72 FOAMFSC – DEFINE FOAM GAS MOBILITY VERSUS SURFACTANT CONCENTRATION FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMFSC](#) keyword defines the reduction in gas mobility as a function of the foam surfactant concentration within a grid block. The Foam option must be activated by the [FOAM](#) keyword in the [RUNSPEC](#) section in order to use this keyword. In addition, the FOAMOPT2 parameter on the [FOAMOPTS](#) keyword in the [PROPS](#) section must be set to the character string FUNC, in order to activate the functional form of the gas mobility reduction calculations.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	FOAMCON	A real positive value that defines the foam surfactant concentration at which foam modeling becomes active in the model and a strong foam is formed. FOAMCON cannot be defaulted and must be specified for the first table. Subsequent tables can be defaulted and will in this case use the previous tables' entries as the default value.			None
		lb/stb	kg/sm ³	gm/scc	
2	FOAMEXP	A real positive value that defines an exponent that determines the gradient in the change of the reduction in gas mobility due to foam (es in equation (8.51). Note if e_s is less than one then the slope of F_s in equation (8.51) will be infinite at C_s equal to zero. In this case, small surfactant concentrations have a significant effect on the mobility, especially if the reference concentration C_{sr} is also small. If this is the case use MINSURF on this keyword to set a minimum surfactant concentration to avoid small-scale numerical errors from affecting the simulation.			Defined
		dimensionless 1.0	dimensionless 1.0	dimensionless 1.0	
3	MINSURF	MINSURF is a real positive value that defines the minimum surfactant concentration for which the reduction in gas mobility will be calculated. The default value of 1×10^{-20} implies that there is no minimum			Defined
		lb/stb 1×10^{-20}	kg/sm ³ 1×10^{-20}	gm/scc 1×10^{-20}	
4	MINSWAT	MINSWAT is a real positive value less than 1.0 that sets the minimum water saturation below which foam has no effect. The default value of 1×10^{-6} implies that there is no minimum. <u>Note that this parameter is only used in the commercial simulator's compositional simulator and is therefore not used by OPM Flow or the commercial simulators black-oil simulator.</u>			Defined
		dimensionless 1×10^{-6}	dimensionless 1×10^{-6}	dimensionless 1×10^{-6}	

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) The keyword is followed by NTSFUN tables as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section. 2) Each table must contain just one row and one row only. 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.					

Table 8.46: FOAMFSC Keyword Description

The gas mobility reduction as a function of surfactant concentration is of the form:

$$F_s = \left(\frac{C_s}{C_s^r} \right)^{e_s} \quad (8.51)$$

Where:

- Fs = the resulting gas mobility reduction factor as a function of surfactant concentration,
- Cs = surfactant concentration,
- Crs = reference surfactant concentration, that is $C_s < C_{sr}$ defines a weak foam and $C_s > C_{sr}$ defines a strong foam (FOAMCON), and
- es = exponent that determines the gradient in the change of the reduction in gas mobility due to foam (FAOAMEXP).

The functional form of the reduction in gas mobility factor (Mrf) is:

$$M_{rf} = \frac{1}{1 + (M_r \times F_s \times F_w \times F_o \times F_c)} \quad (8.52)$$

Where:

- Mr = the reference mobility reduction factor, see the [FOAMFRM](#) keyword in the [PROPS](#) section,
- Fs = gas mobility reduction factor as a function of surfactant concentration, see the [FOAMFSC](#) keyword in the [PROPS](#) section,
- Fw = gas mobility reduction factor as a function of water saturation, see the [FOAMFSW](#) keyword in the [PROPS](#) section,
- Fo = gas mobility reduction factor as a function of oil saturation, see the [FOAMFSO](#) keyword in the [PROPS](#) section, and
- Fc = gas mobility reduction factor as a function of capillary number, see the [FOAMFCN](#) keyword in the [PROPS](#) section.

See also the [FOAM](#) keyword in the [RUNSPEC](#) section, the [FOAMADS](#), [FOAMMOB](#), [FOAMOPTS](#) and [FOAMROCK](#) keywords in the [PROPS](#) section.

Example

```
--
--          FOAM GAS MOBILITY VERSUS SURFACTANT CONCENTRATION FUNCTIONS
--
FOAMFSC
--          FOAM          FOAM          FOAM          FOAM
--          FOAMCON      FOAMEXP      MINSURF      MINSWAT
--          -----      -
0.001          1.010
0.002          1.000
/ TABLE NO. 01
/ TABLE NO. 02
/ TABLE NO. 03 (DEFAULTED)
0.001          0.850  1.0E-10
0.002          1.030
0.002          1.000
/ TABLE NO. 05
/ TABLE NO. 06
```

Here, NTSFUN equals six on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and therefore six entries are required for the [FOAMFSC](#) keyword. Table number three is completed defaulted and will therefore use all the properties from the previous table, that is table number two.

8.3.73
FOAMFSO – DEFINE FOAM GAS MOBILITY REDUCTION VERSUS OIL SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMFSO](#) keyword defines the reduction in gas mobility versus oil saturation, for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section, and the FOAMOPT2 parameter on the [FOAMOPTS](#) keyword in the [PROPS](#) section has been set to the character string FUNC.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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8.3.74
FOAMFST – DEFINE FOAM GAS-WATER SURFACE TENSION VERSUS SURFACTANT CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMFST](#) keyword defines the gas-water surface tension versus the foam surfactant concentration, for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section, and the FOAMOPT2 parameter on the [FOAMOPTS](#) keyword in the [PROPS](#) section has been set to the character string FUNC.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.75
FOAMFSW – DEFINE FOAM GAS MOBILITY REDUCTION VERSUS WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMFRM](#) keyword defines the reduction in gas mobility versus water saturation, for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section, and the FOAMOPT2 parameter on the [FOAMOPTS](#) keyword in the [PROPS](#) section has been set to the character string FUNC.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.76 FOAMMOB – DEFINE FOAM GAS MOBILITY VERSUS FOAM CONCENTRATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [FOAMMOB](#) keyword defines the reduction in gas mobility as a function of the foam concentration within a grid block. The Foam option must be activated by the [FOAM](#) keyword in the [RUNSPEC](#) section in order to use this keyword. In addition, this keyword must be supplied if the foam model is activated.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	FOAMCON	A columnar vector of real monotonically increasing down the column values that defines the foam concentration for the corresponding gas mobility reduction factor (FOAMRATI). The first entry should be zero to define a no foam concentration data set. Units are dependent on the transport phase specified via the FOAMOPT I variable on the FOAMOPTS keyword in the PROPS section. FOAMOPT I should be set to either GAS or WATER .			None
		Gas: lb/Mscf Water: lb/stb	Gas: kg/sm ³ Water: kg/sm ³	Gas: gm/scc Water: gm/scc	
2	FOAMRATI	A columnar vector of real decreasing down the column values that defines the corresponding gas mobility reduction factor for a given FOAMCON. The first table data set entry should be one to define a no foam concentration data set. Each FOAMCON/FOAMRATI data set should be terminated by a “/”			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.47: FOAMMOB Keyword Description

See also the [FOAM](#) keyword in the [RUNSPEC](#) section, the [FOAMADS](#), [FOAMOPTS](#) and [FOAMROCK](#) keywords in the [PROPS](#) section.

8.3.77
FOAMMOBP
– DEFINE FOAM MOBILITY REDUCTION VERSUS OIL PRESSURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *FOAMMOBP* keyword defines the reduction in foam mobility reduction versus oil pressure, for when the Foam option has been activated by the *FOAM* keyword in the *RUNSPEC*.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.78 FOAMMOBS – DEFINE FOAM MOBILITY REDUCTION VERSUS SHEAR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMMOBS](#) keyword defines the reduction in foam mobility reduction versus shear, for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.79 FOAMOPTS – DEFINE FOAM MODEL OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [FOAMOPTS](#) keyword defines the transport phase for the foam (gas, water or solvent) and how gas mobility reduction should be calculated for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description	Default
1	FOAMOPT1	A defined character string that specifies the transport phase for the foam, and should be set to one of the following: 1) GAS : for the foam to be transport in the gas phase., 2) WATER : for the foam to be transported in the water phase, or 3) SOLVENT : for the foam to be transported in the solvent phase.	GAS
2	FOAMOPT2	A defined character string that specifies the method to be used to calculate the reduction in gas mobility, and should be set to one of the following: 1) TAB: Sets the reduction in gas mobility is to be calculated based on tables using the FOAMMOB keyword as a function of foam concentration, the FOAMMOBS keyword as a function of shear, or as a function of pressure using the FOAMMOBP keyword. All keywords are in the PROPS section. 2) FUNC: Sets the reduction in gas mobility to be calculated based on a function defined via the FOAMFRM , FOAMFSC , FOAMFSW , FOAMFSO , FOAMFCN , or FOAMFST keywords in the PROPS section. Only the default value of TAB is currently supported by OPM Flow.	TAB
Notes: 1) The keyword is terminated by a “/”.			

Table 8.48: FOAMOPTS Keyword Description

Example

```
--
--      FOAMOPT1  FOAMOPT2
--      PHASE      MOBILITY
--      -----
FOAMOPTS
      GAS          TAB                               / FOAM MODEL OPTIONS
```

The above example defines the transport phase is to be gas and the gas mobility reduction is to use a table as defined by the [FOAMMOB](#) keyword as a function of foam concentration, the [FOAMMOBS](#) keyword as a function of shear, or as a function of pressure using the [FOAMMOBP](#) keyword.

8.3.80 FOAMROCK – DEFINE FOAM ROCK PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FOAMROCK](#) keyword defines the foam rock properties for when the Foam option has been activated by the [FOAM](#) keyword in the [RUNSPEC](#) section.

The keyword is recognized by the input deck parser and simulator support is available in the experimental "ebos" simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ADINDEX	A positive integer of 1 or 2 that defines foam desorption option, as per: 1) then foam desorption may occur by retracing the foam adsorption isotherm when the local foam concentration in the solution decreases. 2) then no foam desorption may occur. Only the default value of 1 is supported by OPM Flow.			Defined
		dimensionless 1	dimensionless 1	dimensionless 1	
2	DENSITY	A real value that defines the rock in situ density, that is at reservoir conditions.			None
		lb/rb	kg/rm ³	gm/rcc	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain just one row and one row only.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.49: FOAMROCK Keyword Description

Note

In the commercial simulator if the [POLYMER](#) and [SURFACT](#) phases have been activated in conjunction with the [FOAM](#) phase then the mass density of rock will be set by the [PLYROCK](#), [SURFROCK](#), or the [FOAMROCK](#) keywords depending on the order entered in the run deck. This is not the case for OPM Flow.

OPM Flow's [FOAM](#) phase is a standalone implementation and cannot be used in conjunction with the either the [POLYMER](#) or [SURFACT](#) phases.

Example

```
--
--      FOAM-ROCK PROPERTIES
--
FOAMROCK
--      DESORP      INSITU
--      OPTN        DENSITY
--      -----
--      1           1800.0           / TABLE NO. 01
--      2           1980.0           / TABLE NO. 02
--      1           2005.0           / TABLE NO. 03
```

The above example defines three foam-rock tables, based on the NTSFUN variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to three.

There is no terminating “/” for this keyword.

8.3.81 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be place at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a dull description.

8.3.82 GASDENT – DEFINE GAS DENSITY TEMPERATURE COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[GASDENT](#) defines the gas density as a function of temperature coefficients for when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#). Note this is an OPM Flow keyword used with OPM Flow's black-oil thermal model that is not available in the commercial simulator's black-oil thermal formulation.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	TEMP is a real positive value greater than zero that defines the absolute reference temperature used with TEXP1 and TEXP2 to estimate the change in gas density with respect to temperature.			Defined
		°R 527.67	°K 293.15	°K 293.15	
2	TEXP1	TEXP1 is a real positive value greater than zero that defines the gas thermal expansion coefficient of the first order.			Defined
		1/°R 1.67 x 10 ⁻⁴	1/°K 3.0 x 10 ⁻⁴	1/°K 3.0 x 10 ⁻⁴	
3	TEXP2	TEXP2 is a real positive value greater than zero that defines the gas thermal expansion coefficient of the second order.			Defined
		1/°R2 9.26 x 10 ⁻⁷	1/°K2 3.0 x 10 ⁻⁶	1/°K2 3.0 x 10 ⁻⁶	

Notes:

- 1) The keyword is followed by NTPVT records as declared on the TABDIMS keyword in the RUNSPEC section.
- 2) Each data set is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.50: GASDENT Keyword Description

The gas density at a given pressure and temperature is calculated from its value at surface conditions and the gas expansion factor (the reciprocal of the gas formation volume factor) as shown in the following equation:

$$\rho_g(p, T) = \rho_g(p_s, T_s) b_g(p, T) \quad (8.53)$$

Where the temperature dependence of the gas expansion factor relative to its value at the reference temperature is calculated as shown in the following equation:

$$b_g(p,T)=\frac{b_g(p,T_{ref})}{1+c_1(T-T_{ref})+c_2(T-T_{ref})^2}$$

(8.54)

Where:

- ρ_g

= gas density
- b_g

= gas expansion factor
- p

= pressure
- T

= temperature
- c_1, c_2

= thermal expansion coefficients to first and second order
- s

= subscript indicating surface conditions
- ref

= subscript indicating reference conditions

Example

The following example shows the [GASDENT](#) keyword using the default values, for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to two.

```
--
--      GAS DENSITY TEMPERATURE COEFFICIENTS (OPM FLOW THERMAL KEYWORD)
--
--      GAS      DENSITY      DENSITY
--      TEMP      COEFF1      COEFF2
--      -----
GASDENT
      1*          1*          1*          / TABLE NO. 01
      1*          1*          1*          / TABLE NO. 02
```

There is no terminating “/” for this keyword.

8.3.83 GASJT – DEFINE GAS JOULE-THOMSON COEFFICIENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[GASJT](#) activates the gas Joule-Thomson²⁰³ effect in temperature calculations, and defines the gas Joule-Thomson Coefficient ("JTC") at a given reference pressure, for when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#).

Note

This is an OPM Flow keyword used with OPM Flow's black-oil thermal model, that is not available in the commercial simulator's black-oil thermal formulation.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model, and does not include the Joule-Thomson effect in temperature calculations.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A real positive value that defines the reference pressure for the corresponding Joule-Thomson Coefficient, GASJTC.			None
		psia	barsa	atma	
2	GASJTC	GASJTC is a real positive or negative value that defines the gas phase Joule-Thomson Coefficient. If the value is defaulted (I*) or set to 0, then GASJTC is internally calculated using the thermal gas density data on the GASDENT keyword in the PROPS section. If a non-zero value is specified, then the GASJTC is assumed to be constant and equal to that value.			0
		oF/psia	°C/barsa	°C/atma	

Notes:

- 1) The keyword is followed by NTPVT records as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each data set is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.51: GASJT Keyword Description

The Joule–Thomson effect is when a real gas, as oppose to an ideal gas, expands, resulting in the temperature of the gas dropping. During passage of a gas through a choke, the internal energy is transferred to kinetic energy with a corresponding reduction in temperature as the velocity increases. The effect for natural gas is approximately 7 °F for every 100 psi pressure reduction, or 0.5 oC per barsa²⁰⁴, is valid for "normal" pressures and temperatures at the surface.

²⁰³ *Natural Gas Engineering (McGraw-Hill chemical engineering series), Donald L Katz, Robert I Lee, McGraw-Hill Education, 1990 (ISBN 0071007776, 9780071007771).*

²⁰⁴ https://petrowiki.spe.org/Glossary:joule-thompson_effect.

Thermodynamically, the Joule–Thomson coefficient is defined as the isenthalpic²⁰⁵ change in temperature in a fluid caused by a unitary pressure drop, as shown in the following equation:

$$\eta = \left(\frac{\partial T}{\partial P} \right) \quad (8.55)$$

One can also express enthalpy changes in terms of pressure, temperature and volume changes:

$$\eta = \frac{RT^2}{Pc_p} \left(\frac{\partial Z}{\partial T} \right)_P \quad (8.56)$$

Where:

η	=	Joule–Thomson coefficient,
C_g	=	specific heat at constant pressure,
P	=	pressure,
R	=	gas constant,
T	=	temperature, and
Z	=	gas compressibility factor.

Example

The following example shows the [GASJT](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section, and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set equal to two.

```
--
--      GAS JOULE-THOMSON COEFFICIENT (OPM FLOW EXTENSION KEYWORD)
--
--      REF      GAS
--      PRESS    JTC
--      -----
GASJT
      20.0      1*          / TABLE NO. 01
      20.0      0.50       / TABLE NO. 02
```

Here the first entry is defaulted, and the simulator will therefore calculate the gas JTC internally using the data on the [GASDENT](#) keyword in the [PROPS](#) section.

There is no terminating “/” for this keyword.

²⁰⁵ An isenthalpic process or isoenthalpic process, is a process that proceeds without any change in enthalpy, H ; or specific enthalpy, h .

8.3.84 GASVISCT – DEFINE GAS VISCOSITY VERSUS TEMPERATURE FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[GASVISCT](#) defines the gas viscosity as a function of temperature for when OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#). The reference pressure for this table is given by the [VISCREF](#) keyword in the [PROPS](#) section. Note this is an OPM Flow keyword used with OPM Flow’s black-oil thermal model that is not available in the commercial simulator’s black-oil thermal formulation. However, the keyword and similar functionality is available in the commercial compositional simulator.

This keyword can only be used if OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	A columnar vector of real monotonically increasing down the column values that defines the temperature values.			None
		°F	°C	°C	
2	VIS	A columnar vector of real increasing down the column values that defines the gas viscosity for the corresponding temperature values (TEMP). VIS should be given at the reference pressure defined by the PRESS variable on the VISCREF keyword.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.52: GASVISCT Keyword Description

Example

The following example shows the [GASVISCT](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set equal to one.

```
--
--      GAS VISCOSITY VERSUS TEMPERATURE TABLES (OPM FLOW EXTENSION KEYWORD)
--
--      GAS      GAS
--      TEMP     VISC
--      -----
GASVISCT
      100.0      0.0500
      110.0      0.0550
      120.0      0.0580
      150.0      0.0620
      165.0      0.0625
```

/ TABLE NO. 01

There is no terminating “/” for this keyword.

8.3.85 GETDATA – LOAD AND ASSIGN DATA ARRAY FROM INIT OR RESTART FILE

The [GETDATA](#) keyword loads a data array from a previously generated [INIT](#) or [RESTART](#) file and assigns the loaded array to either same array in the run or another array name.

See [GETDATA – Load and Assign Data Array from INIT or RESTART Files](#) in the [GRID](#) section for a full description.

8.3.86 GIALL – DEFINE GI VALUES AND PVT PROPERTIES VERSUS PRESSURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *GIALL* keyword defines the *GI* values and the associated *RVGI*, *RSGI*, *BGGI* and *BOGI* values as a function of pressure, for when the *GI* Pseudo Compositional option has been activated in the model via the *GIMODEL* keyword in the *RUNSPEC* section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.87 GINODE – DEFINE GI NODE VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GINODE](#) keyword defines the Gi node values used when the [GIMODEL](#) keyword in the [RUNSPEC](#) section has been used to activate the [GI](#) Pseudo Compositional option for the run. The keyword is used in conjunction with the [RSGI](#), [RVGI](#), [BGGI](#) and [BOGI](#) keywords in the [PROPS](#) section to describe the fluid properties for the [GI](#) Pseudo Compositional option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.88 GRAVCONS – RE-DEFINE GRAVITY CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GRAVCONS](#) keyword re-defines the gravity constant used in various calculations from the default value used by the simulator. Normally this keyword should not be used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	GRAVCONS	GRAVCONS is a positive real number number that defines the gravity constant used in various calculations.			Defined
		ft²psi/lb 0.00694	m²bars/kg 0.0000981	cm²atm/gm 0.000968	
Notes: I) The keyword is terminated by a “/”.					

Table 8.53: GRAVCONS Keyword Description

Example

```
--
--      RE-DEFINE GRAVITY CONSTANT
--
GRAVITY      0.0000980665      /
```

The above example re-defines the gravity constant to be 0.0000980665 ft²psi/lb from the default value of 0.00694 ft²psi/lb.

8.3.89 GRAVITY – DEFINE THE SURFACE OIL, WATER GAS GRAVITIES FOR THE FLUIDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

GRAVITY defines the oil **API** gravity and water and gas surface specific gravities for the fluids for various regions in the model. The number of **GRAVITY** vector data sets is defined by the NTPVT parameter on the **TABDIMS** keyword in the **RUNSPEC** section and the allocation of the **GRAVITY** data sets to different grid blocks in the model is done via the **PVTNUM** keyword in the **REGION** section. One data set consists of one record or line which is terminated by a “/”.

This surface density or gravity must be entered using either the **DENSITY** or **GRAVITY** keywords irrespective of which phases are active in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	OILAPI	OILAPI is a real number defining the API gravity of the oil phase at surface conditions. The American Petroleum Institute (“API”) classifies oils based on an API gravity (γ_{API}), or degrees API ($^{\circ}API$), the relationship between relative density (γ_o) of oil and API gravity (γ_{API}) is given by: $\gamma_{API} = \frac{141.5}{\gamma_o} - 131.5$			None
		$^{\circ}API$	$^{\circ}API$	$^{\circ}API$	
2	WATGRAV	WATGRAV is a real number defining the specific gravity of the water phase relative to pure water at surface conditions.			Defined
		(water =1.0) 0.7773	(water =1.0) 0.7773	(water =1.0) 0.7773	
3	GASGRAV	GASGRAV is a real number defining the specific gravity of the gas phase relative to air at surface conditions.			Defined
		(air =1.0) 1.000	(air =1.0) 1.000	(air =1.0) 1.000	

Notes:
1) The keyword is followed by NTPVT tables as declared on the TABDIMS keyword in the RUNSPEC section.
2) Each data set is by terminated by a “/” at the end of the line and there is no “/” terminator for the keyword.

Table 8.54: GRAVITY Keyword Description

According to the SPE SI standard²⁰⁶, **Relative Density** (γ) replaces **Specific Gravity** as the term used to define the ratio of the density of a known material to the density of reference material, at standard conditions of pressure and temperature. Standard conditions vary throughout the world, but for oil field units one normally uses 14.7 psia and 60 °F, while for SI units some areas use 101.325 kPa and 15 °C.

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the

²⁰⁶ The SI Metric System of Units and SPE Metric Standard, Adopted for Use as a Voluntary Standard by the SPE Board of Directors, June 1983, Society of Petroleum Engineers.

prior table is copied to the current table. See the third example for an illustration on how to use this feature. See also the [DENSITY](#) keyword in the [PROPS](#) section.

Examples

The following shows the [GRAVITY](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to one.

```
--
--      OIL      WAT      GAS
--      GRAVITY  GRAVITY  GRAVITY
--      -----  -----  -----
GRAVITY
      39.0      1.012      0.650      / GRAVITY PVT DATA REGION 1
```

The next example shows the [GRAVITY](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to three.

```
--
--      OIL      WAT      GAS
--      GRAVITY  GRAVITY  GRAVITY
--      -----  -----  -----
GRAVITY
      37.0      1.012      0.650      / GRAVITY PVT DATA REGION 1
      38.0      1.012      0.646      / GRAVITY PVT DATA REGION 2
      39.0      1.012      0.640      / GRAVITY PVT DATA REGION 3
```

The third and final example shows the [GRAVITY](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to four. Here table two defaults to table one, and table four defaults to table three.

```
--
--      OIL      WAT      GAS
--      GRAVITY  GRAVITY  GRAVITY
--      -----  -----  -----
GRAVITY
      37.0      1.012      0.650      / GRAVITY PVT DATA REGION 1
                                   / GRAVITY PVT DATA REGION 2
      38.0      1.012      0.646      / GRAVITY PVT DATA REGION 3
                                   / GRAVITY PVT DATA REGION 4
```

Again, note that there is no terminating “/” for this keyword.

8.3.90 GSF – GAS SATURATION FUNCTION TABLES (GAS-WATER SYSTEMS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GSF](#) keyword defines the gas relative permeability and gas-water capillary pressure data versus gas saturation tables for when only gas and water are present in the input deck. This keyword should only be used if the gas and water phases are present in the run, and can therefore also be used with the [CO2STORE](#) and [H2STORE](#) models. In addition, the keyword must be used in conjunction with the [WSF](#) keyword in the [PROPS](#) section, that defines the water relative permeability versus water saturation for gas-water systems.

Note

GSF is a compositional keyword in the commercial compositional simulator, and will therefore cause an error in the commercial black-oil simulator.

Currently, both the [GSF](#) and [WSF](#) keywords can only be used with the [CO2STORE](#) and [H2STORE](#) models.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGAS	A columnar vector of real values monotonically increasing down the column starting from zero and terminating at one minus the connate water saturation, that defines the gas saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRG	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas relative permeability. Note that the first entry in the column must be zero.			None
		dimensionless	dimensionless	dimensionless	
3	PCWG	A columnar vector of real values that are either equal or increasing down the column that defines the gas-water capillary pressure.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.55: GSF Keyword Description

See also the [WSF – Water Saturation Function Tables \(Gas-Water Systems\)](#) keyword in the [PROPS](#) section, that defines the water saturation as a function of water saturation for when only the gas and water phases are present in the model.

Example

--	GAS RELATIVE PERMEABILITY TABLES (OPM FLOW KEYWORD)			
--				
--				
GSF				
--	SGAS	KRG	PCGW	
--	FRAC		PSIA	
--	-----	-----	-----	
	0.00	0.0000	0.0	
	0.05	0.0000	0.0	
	0.10	0.0000	0.0	
	0.15	0.0000	0.0	
	0.20	0.0002	0.0	
	0.25	0.0010	0.0	
	0.30	0.0062	0.0	
	0.35	0.0140	0.0	
	0.40	0.0273	0.0	
	0.45	0.0450	0.0	
	0.50	0.0707	0.0	
	0.55	0.1020	0.0	
	0.60	0.1412	0.0	
	0.65	0.1870	0.0	
	0.70	0.2412	0.0	
	0.77	0.3288	0.0	
	0.82	0.4000	0.0	
	0.85	0.4450	0.0	/ TABLE NO. 01
--	SGAS	KRG	PCGW	
--	FRAC		PSIA	
--	-----	-----	-----	
	0.00	0.0000	0.0	
	0.05	0.0000	0.0	
	0.10	0.0000	0.0	
	0.15	0.0000	0.0	
	0.20	0.0002	0.0	
	0.25	0.0010	0.0	
	0.30	0.0062	0.0	
	0.35	0.0140	0.0	
	0.40	0.0273	0.0	
	0.45	0.0450	0.0	
	0.50	0.0707	0.0	
	0.55	0.1020	0.0	
	0.60	0.1412	0.0	
	0.65	0.1870	0.0	
	0.70	0.2412	0.0	
	0.77	0.3288	0.0	
	0.82	0.4000	0.0	
	0.85	0.4450	0.0	/ TABLE NO. 02

The example defines two *GSF* tables for when gas and water are present in the input deck. In the tables the gas-water capillary pressure data has been set to zero.

8.3.91 HA – HISTORY MATCH END-POINT GRADIENT ADDITIVE MODIFIER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [HA](#) series of keywords defines the history match end-point gradient parameters used to set the additive cumulative end point data, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. In addition, the End-Point Scaling option must also be activated by the [ENDSCALE](#) keyword which is also in the [RUNSPEC](#) section. The keyword consists of the first two characters of “HA” followed by the end-point keyword shown in Table 8.56, for example, HASWL.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Type	End-Point Keyword	Oil-Water End-Point Definitions
Saturation	SWL	Connate water saturation, that is the smallest water saturation in a water saturation function table.
	SWCR	Critical water saturation, that is the largest water saturation for which the water relative permeability is zero.
	SOWCR	Critical oil-in-water saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-water system.
Relative Permeability	KRW	Relative permeability of water at the maximum water saturation (normally the maximum water saturation is one).
	KRO	Relative permeability of oil at the maximum oil saturation.
	KRWR	Relative permeability of water at the residual oil saturation or the residual gas saturation in a gas-water run.
	KRORW	Relative permeability of oil at the critical water saturation.
Capillary Pressure	SWLPC	Capillary pressure connate water saturation, that is the smallest water saturation in a water saturation function table.
Type	End-Point Keyword	Gas-Oil End-Point Definitions
Saturation	SGL	Connate gas saturation, that is the smallest gas saturation in a gas saturation function table.
	SGCR	Critical gas saturation, that is the largest gas saturation for which the gas relative permeability is zero.
	SOGCR	Critical oil-in-gas saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-gas-connate water system.
Relative Permeability	KRG	Relative permeability of gas at the maximum gas saturation.
	KRGR	Relative permeability of gas at the residual oil saturation or the critical water saturation in a gas-water run.
	KRORG	Relative permeability of oil at the critical gas saturation.

Type	End-Point Keyword	Oil-Water End-Point Definitions
Capillary Pressure	SGLPC	Capillary pressure connate gas saturation, that is the smallest gas saturation in a gas saturation function table.

Table 8.56: [HA](#) Keyword List

See also the [HMPROPS](#) keyword in the [PROPS](#) section that allows the use of the [ADD](#), [BOX](#), [EQUALS](#), [COPY](#), [MINVALUE](#), and [MAXVALUE](#) keywords to be used with the [HA](#) and [HM](#) series of keywords.

8.3.92 HDISP – DEFINE TRACER MECHANICAL DISPERSIVITY PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HDISP](#) keyword is combined with three character tracer name, specified by the [TRACER](#) keyword in the [PROPS](#) section, to define the tracer’s mechanical dispersivity parameters.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.93 HM – HISTORY MATCH END-POINT GRADIENT MULTIPLICATIVE MODIFIER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [HM](#) series of keywords defines the history match end-point gradient parameters used to set the multiplicative cumulative end point data, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. In addition, the End-Point Scaling option must also be activated by the [ENDSCALE](#) keyword which is also in the [RUNSPEC](#) section. The keyword consists of the first two characters of “HM” followed by the end-point keyword shown in Table 8.57, for example, HMSWL.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Type	End-Point Keyword	Oil-Water End-Point Definitions
Saturation	SWL	Connate water saturation, that is the smallest water saturation in a water saturation function table.
	SWCR	Critical water saturation, that is the largest water saturation for which the water relative permeability is zero.
	SOWCR	Critical oil-in-water saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-water system.
Relative Permeability	KRW	Relative permeability of water at the maximum water saturation (normally the maximum water saturation is one).
	KRO	Relative permeability of oil at the maximum oil saturation.
	KRWR	Relative permeability of water at the residual oil saturation or the residual gas saturation in a gas-water run.
	KRORW	Relative permeability of oil at the critical water saturation.
Capillary Pressure	SWLPC	Capillary pressure connate water saturation, that is the smallest water saturation in a water saturation function table.
Type	End-Point Keyword	Gas-Oil End-Point Definitions
Saturation	SGL	Connate gas saturation, that is the smallest gas saturation in a gas saturation function table.
	SGCR	Critical gas saturation, that is the largest gas saturation for which the gas relative permeability is zero.
	SOGCR	Critical oil-in-gas saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-gas-connate water system.
Relative Permeability	KRG	Relative permeability of gas at the maximum gas saturation.
	KRGR	Relative permeability of gas at the residual oil saturation or the critical water saturation in a gas-water run.
	KRORG	Relative permeability of oil at the critical gas saturation.

Type	End-Point Keyword	Oil-Water End-Point Definitions
Capillary Pressure	SGLPC	Capillary pressure connate gas saturation, that is the smallest gas saturation in a gas saturation function table.

Table 8.57: [HM](#) Keyword List

See also the [HMPROPS](#) keyword in the [PROPS](#) section that allows the use of the [ADD](#), [BOX](#), [EQUALS](#), [COPY](#), [MINVALUE](#), and [MAXVALUE](#) keywords to be used with the [HA](#) and [HM](#) series of keywords.

8.3.94
HMMROCK – HISTORY MATCH ROCK COMPRESSIBILITY GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HMMROCK](#) defines the rock compressibility gradient cumulative multipliers to be applied to the rock compressibility as defined by the [ROCK](#) keyword in the [PROPS](#) section, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The constant should be a real number.

The allocation of the [ROCK](#) tables to different grid blocks in the model is done via the [PVTNUM](#) or the [SATNUM](#) keywords in the REGION section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

8.3.95
HMMROCKT – HISTORY MATCH ROCK COMPACTION GRADIENT CUMULATIVE MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

HMMROCKT defines the rock compaction gradient cumulative multipliers to be applied to the compaction data entered by the ROCTAB or *ROCKTABH* keywords in the PRROPS section, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section

The *ROCKTAB* keyword defines the rock compaction attributes to be applied for when the rock compaction option has been invoked by the *ROCKCOMP* keyword in the *RUNSPEC* section. *ROCKTAB* defines pore volume and transmissibility multipliers versus pressure that are used in the compaction calculations. If the *RKTRMDIR* has been activated in the *PROPS* section, then the transmissibility multiplier is directional dependent

This keyword should only be used if compaction option has been enabled.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

8.3.96 HMPROPS – HISTORY MATCH END-POINT SECTION START

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

HMPROPS defines the start of a history match end-points section, for when the History Match Gradient option has been activated by the *HMDIMS* keyword in the *RUNSPEC* section. In addition, the End-Point Scaling option must also be activated by the *ENDSCALE* keyword which is also in the *RUNSPEC* section. The keyword allows for the *BOX*, *EQUALS*, *COPY*, *MINVALUE*, *MAXVALUE* and *ADD* keywords to be used with the *HA* and *HM* series of keywords that reference the end-point scaling arrays, that is: HMKRG, HMKRGR, HMKRO, HMKRORG, HMKRORW, HMKRW, HMKRWR, HMPCW, HMPCG, HMSGCR, HMSOWCR, HMSOGCR, HMSWCR, and HMSWL keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.97
HMROCK
– HISTORY MATCH ROCK COMPRESSIBILITY GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMROCK](#) keyword defines the history match rock compressibility gradient parameters for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of rock gradient parameters that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.98
HMROCKT
– HISTORY MATCH ROCK COMPACTION GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMROCKT](#) keyword defines the history match rock compaction gradient parameters for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and the history match rock compaction data has been entered via the [HMMROCKT](#), [ROCKTAB](#) and [ROCKTABH](#) keywords in the [PROPS](#) section.

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of rock gradient parameters that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.99
HMRREF
– HISTORY MATCH ROCK TABLE REFERENCE PRESSURE VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMRREF](#) keyword defines the history match rock compaction reference pressure gradient values to be used in conjunction with [HMMROCKT](#), [ROCKTAB](#) and [ROCKTABH](#) keywords in the [PROPS](#) section, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The history match rock compaction data is entered via the [HMMROCKT](#), [ROCKTAB](#) and [ROCKTABH](#) keywords in the [PROPS](#) section.

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of rock gradient parameters that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.100 HWKRO – END-POINT SCALING OF GRID CELL KRO(SWL) (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWKRO](#) defines the scaling parameter for the oil relative permeability value at the connate water saturation ([SWL](#)), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil relative permeability in the high salinity water wet oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.101 HWKRORG – END-POINT SCALING OF GRID CELL KRO(SGCR) (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWKRORG](#) defines the scaling parameter for the relative permeability of oil at the critical gas saturation ([SGCR](#)), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil relative permeability in the high salinity water wet oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.102
HWKRORW – END-POINT SCALING OF GRID CELL KRO(SWCR) (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWKRORW](#) defines the scaling parameter for the relative permeability of oil at the critical water saturation ([SWCR](#)), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil relative permeability in the high salinity water wet oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.103 HWKRW – END-POINT SCALING OF GRID CELL KRW(Sw =1.0) (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWKRW](#) defines the scaling parameter at the maximum oil relative permeability value ([SWU](#)), that is for Sw = 1.0, for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water relative permeability in the high salinity water wet water relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.104 HWKRWR – END-POINT SCALING OF GRID CELL KRWR(Sw=1.0)
(HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWKRWR](#) defines the scaling parameter at the maximum oil relative permeability value ([SWU](#)), that is for $S_w = 1.0$, for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water relative permeability in the high salinity water wet water relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.105
HWPCW – END-POINT SCALING OF GRID CELL WATER CAPILLARY PRESSURE (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

HWPCW defines the maximum water-oil pressure values for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The *ENDSCALE* keyword in the *RUNSPEC* section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the *LOWSALT* keyword in the *RUNSPEC* section and the Surfactant Wettability option activated by the *SURFACT* or *SURFACTW* keywords, which are also in the *RUNSPEC* section. The keyword re-scales the oil-water capillary pressure in the high salinity water wet capillary saturation tables from a cell's assigned saturation function by the grid block's *HWPCW* value.

The capillary pressure for a grid block is scaled by:

$$P_c = P_{c_{TABLE}} \left(\frac{HWPCW}{P_{c_{TABLE-MAX}}} \right)$$

(8.57)

Where:

- P_c

=

the resulting high salinity water wet water capillary pressure for a grid cell.

HWPCW

=

the maximum capillary pressure from the *HWPCW* array for a given cell.

P_{c_{TABLE}}

=

the capillary pressure in the high salinity water wet water capillary pressure table allocated to the grid block.

P_{c_{TABLE-MAX}}

=

the maximum capillary pressure in the high salinity water wet water capillary pressure table allocated to the grid block (that is at the connate water saturation).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.106 HWSOGR – END-POINT SCALING GRID CELL SOGR (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWSOGR](#) defines the critical oil saturation with respect to gas (“SOGR”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil saturation in the high salinity water wet oil-gas relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.107
HWSOWCR – END-POINT SCALING GRID CELL SOWCR (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWSOWCR](#) defines the critical oil saturation with respect to water (“SOWCR”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil saturation in the high salinity water wet water-oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.108
HWSWCR – END-POINT SCALING GRID CELL SWCR (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWSWCR](#) defines the critical water saturation (“SWCR”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the high salinity water wet water-oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.109
HWSWL – END-POINT SCALING GRID CELL SWL (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWSWL](#) defines the connate water saturation (“SWL”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the high salinity water wet water-oil relative permeability saturation tables, as well as the associated capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.110
HWSWLPC – END-POINT SCALING GRID CELL SWLPC (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWSWLPC](#) defines the capillary pressure connate water saturation (“SWLPC”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the high salinity water wet water-oil capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

Note the keyword only applies the scaling to the capillary pressures tables, unlike the [HWSWL](#) keyword that applies the scaling to both the capillary pressure and relative permeability tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.111
HWSWU – END-POINT SCALING GRID CELL SWU (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[HWSWU](#) defines the maximum water saturation(“SWU”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the high salinity water wet water-oil relative permeability saturation tables, as well as the associated capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.112 HYDRHEAD – DEFINE HYDRAULIC HEAD OUTPUT REFERENCE DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HYDRHEAD](#) keyword defines the hydraulic head reference data for when the hydraulic head information is requested to be written out via one on the [SUMMARY](#) keywords (BHD, BHDF, etc.) in the [SUMMARY](#) section, or to the [RESTART](#) file via the HYDH or HYDHFw variables on the [RESTART](#) keyword

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.113
HYMOBGDR – ACTIVATE CARLSON AND KILLOUGH ALTERNATIVE DRAINAGE HYSTERESIS OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [HYMOBGDR](#), activates the Carlson and Killough alternative secondary drainage hysteresis option for when the hysteresis option has been activated by the HYSTER variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, and either the Carlson²⁰⁷ or Killough²⁰⁸ models have been selected via the [EHYSTR](#) keyword in the [PROPS](#) section. Due to numerical accuracy, the gas saturation may fall below the critical gas saturation ([SGCR](#)), that is the largest gas saturation for which the gas relative permeability is zero, and gas would therefore be immobile until the gas saturation increases above [SGCR](#). This option overcomes this situation by letting the gas become mobile once it starts increasing, effectively setting the [SGCR](#) to the current gas saturation.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```

--
--      ACTIVATE CARLSON AND KILLOUGH ALTERNATIVE DRAINAGE HYSTERESIS OPTION
--
HYMOBGDR

```

²⁰⁷
Carlson, F. M. “Simulation of Relative Permeability Hysteresis to the Non-Wetting Phase,” paper SPE 10157, presented at the SPE Annual Technical Conference & Exhibition, San Antonio, Texas, USA (October 5-7, 1981).

²⁰⁸
Killough, J. E. “Reservoir Simulation with History-dependent Saturation Functions,” paper SPE 5106, Society of Petroleum Engineers Journal (1976) 16, No. 1, 37-48.

8.3.114 HYSTCHCK – ACTIVATE HYSTERESIS IMBIBITION AND DRAINAGE END-POINT VALIDATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HYSTCHCK](#) keyword activate the hysteresis imbibition and drainage end-point check to validate that the two sets of end-points are consistent, for when the Hysteresis option has been activated by the HYSTER variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, and the [ENDSCALE](#) keyword in the [RUNSPEC](#) section has been activated to enable end-point scaling.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.115 IKRG – END-POINT SCALING OF GRID CELL KRG(SGU) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

IKRG defines the imbibition scaling parameter at the maximum gas relative permeability value (*ISGU*), normally *ISGU* is equal to 1.0 - Swc, for all the cells in the model via an array. The *ENDSCALE* keyword in the *RUNSPEC* section should be activated to enable end-point scaling and the use of this keyword. In addition, the *HYSTER* option on the *SATOPTS* keyword in the *RUNSPEC* section has to be activated to invoke the Hysteresis option. The *SCALCERS* keyword in the *PROPS* section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	<i>IKRG</i>	<i>IKRG</i> is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling <i>IKRG</i> imbibition values for each cell in the model. Repeat counts may be used, for example 50*0.400. dimensionless			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the *IKRGX±*, *IKRGY±* and *IKRGZ±* series of the keywords should be used.

Table 8.58: *IKRG* Keyword Description

For the two point scaling option and for the *IKRGR* gas relative permeability array NOT present in the input deck the krg value for a grid block is scaled by:

$$k_{rg} = k_{rg_{TABLE}} \left(\frac{IKRG}{k_{rg_{TABLE-MAX}}} \right) \quad (8.58)$$

Where:

- krg* = the resulting krg value for a grid cell.
 - IKRG* = the scaling gas relative permeability value from the *IKRG* array for a given cell.
 - k_{rg_{TABLE}}*
 - k_{rg_{TABLE-MAX}}*
- = the gas relative permeability from a grid block's gas-oil table at the grid blocks gas saturation.
- = the maximum gas relative permeability from a grid block's gas-oil table, that is at the connate water saturation (Swc).

If the *IKRGR* keyword is present in the input deck then the scaling matches the imbibition relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the *SCALECRS* keyword in the *PROPS* section the critical displacing phase is defined as:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - ISOGCR - ISWL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - ISOGCR - ISWL$
3	Gas-Water	$S_{critical} = 1.0 - ISWCR$

Table 8.59: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the *ISWL*, *ISWCR*, *ISWU*, *ISGL*, *ISGCR*, *ISGU*, *ISOWCR*, and *ISOGCR* saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUY* and *ISWUZ* instead of *ISWU*. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the *IKRG*, *IKRGR*, *IKRO*, *IKRORG*, *IKRORW*, *IKRW* and *IKRWR* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGY* and *IKRGZ* instead of *IKRG*. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGX-*, *IKRGY*, *IKRGY-*, *IKRGZ* and *IKRGZ-*, instead of the *IKRG* keyword.

Example

The example below defines an input box for the whole grid and for layers one to three, for layer one *IKRG* is set equal to 0.550, for layer two *IKRG* equals 0.575, and for layer three *IKRG* equals 0.600.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      1*  1*    1*  1*    1    3
--
--      / DEFINE BOX AREA
--
--      SET IKRG VALUES FOR THREE LAYERS IN THE MODEL
--
--      IKRG
--      1000*0.550  1000*0.575  1000*0.600
--
--      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

8.3.116 IKRGR – END-POINT SCALING OF GRID CELL KRGR(1-SOGCR) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

IKRGR defines the imbibition scaling parameter at the relative permeability of gas at residual oil saturation ($1 - ISOGCR$), or critical water saturation in a gas-water run (Swc), for all the cells in the model via an array. The *ENDSCALE* keyword in the *RUNSPEC* section should be activated to enable end-point scaling and the use of this keyword. In addition, the *HYSTER* option on the *SATOPTS* keyword in the *RUNSPEC* section has to be activated to invoke the Hysteresis option. The *SCALCERS* keyword in the *PROPS* section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	<i>IKRGR</i>	<i>IKRGR</i> is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned imbibition scaling <i>IKRGR</i> values for each cell in the model. In addition, for a given grid block <i>IKRGRGT</i> should be less than <i>IKRG</i> . Repeat counts may be used, for example 50*0.400.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a "/".
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the *IKRGRX* $\pm$, *IKGRY* $\pm$ and *IKGRZ* $\pm$ series of the keywords should be used

Table 8.60: *IKRGR* Keyword Description

When the *IKRGR* keyword is present in the input deck then the scaling matches the imbibition relative permeability at the critical saturation of the displacing phase (oil or water).

If three point scaling option has been selected via the *SCALECRS* keyword in the *PROPS* section the critical displacing phase is defined as:

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - ISOGCR - ISWL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - ISOGCR - ISWL$
3	Gas-Water	$S_{critical} = 1.0 - ISWCR$

Table 8.61: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the *ISWL*, *ISWCR*, *ISWU*, *ISGL*, *ISGCR*, *ISGU*, *ISOWCR*, and *ISOGCR* saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used,

that is *ISWUX*, *ISWUY* and *ISWUZ* instead of *ISWU*, There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the *IKRG*, *IKRGR*, *IKRO*, *IKRORG*, *IKRORW*, *IKRW* and *IKRWR* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGY* and *IKRGZ* instead of *IKRG*, There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGX-*, *IKRGY*, *IKRGY-*, *IKRGZ* and *IKRGZ-*, instead of the *IKRG* keyword.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one *IKRRG* is set equal to 0.500, for layer two *IKRGR* equals 0.570, and for layer three *IKRGR* equals 0.580.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX
      1*  1*    1*  1*    1    3
--
--
--      SET IKRGR VALUES FOR THREE LAYERS IN THE MODEL
--
IKRGR
      10000*0.500  10000*0.570  10000*0.580
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the *EQUALS* keyword instead.

```
--
--      ARRAY          CONSTANT          ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      IKRGR'          0.5000          1*  1*    1*  1*    1    1  / IKRGR FOR LAYER 1
      IKRGR'          0.5700          1*  1*    1*  1*    2    2  / IKRGR FOR LAYER 2
      IKRGR'          0.5800          1*  1*    1*  1*    3    3  / IKRGR FOR LAYER 3
/
```

8.3.117 IKRO – END-POINT SCALING OF GRID CELL KRO(SWL) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

IKRO defines the scaling parameter for the imbibition oil relative permeability value at the connate water saturation (**ISWL**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the **HYSTER** option on the **SATOPTS** keyword in the **RUNSPEC** section has to be activated to invoke the Hysteresis option. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	IKRO	IKRO is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned imbibition scaling IKRO values for each cell in the model. Repeat counts may be used, for example 50*0.500.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **IKROX±**, **IKROY±** and **IKROZ±** series of the keywords should be used.

Table 8.62: IKRO Keyword Description

For the two point scaling option and for the **IKRORW** or **IKRORG** oil imbibition relative permeability arrays NOT being present in the input deck the kro value for a grid block is scaled by:

$$k_{ro} = k_{ro\text{ TABLE}} \left(\frac{IKRO}{k_{ro\text{ TABLE-MAX}}} \right) \quad (8.59)$$

Where:

- kro** = the resulting kro value for a grid cell.
- IKRO** = the scaling oil relative permeability value from the **IKRO** array for a given cell.
- $k_{ro\text{ TABLE}}$ = the oil relative permeability from a grid block's oil relative permeability table at the grid blocks oil saturation.
- $k_{ro\text{ TABLE-MAX}}$ = the maximum oil relative permeability from a grid block's oil relative table, that is at the critical water saturation (**Swcr**).

If the **IKRORW** or **IKRORG** keywords are present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

No	Keywords Present	Critical Saturation
1	<i>KRORW</i>	$S_{critical} = 1.0 - SWCR - SGL$
2	<i>KRORG</i>	$S_{critical} = 1.0 - SGCR - SWL$

Table 8.63: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the *ISWL*, *ISWCR*, *ISWU*, *ISGL*, *ISGCR*, *ISGU*, *ISOWCR*, and *ISOGCR* saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUY* and *ISWUZ* instead of *ISWU*. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the *IKRG*, *IKRGR*, *IKRO*, *IKRORG*, *IKRORW*, *IKRW* and *IKRWR* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGY* and *IKRGZ* instead of *IKRG*. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGX-*, *IKRGY*, *IKRGY-*, *IKRGZ* and *IKRGZ-*, instead of the *IKRG* keyword.

Example

The example below defines an input box for the whole grid and for layers one to three, for layer one *IKRO* is set equal to 0.850, for layer two *IKRO* equals 0.875, and for layer three *IKRO* equals 0.900.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      IKRO          0.8500        1*  1*   1*  1*   1   1  / IKRO FOR LAYER 1
      IKRO          0.8750        1*  1*   1*  1*   2   2  / IKRO FOR LAYER 2
      IKRO          0.9000        1*  1*   1*  1*   3   3  / IKRO FOR LAYER 3
/
```

8.3.118 IKRORG – END-POINT SCALING OF GRID CELL KRO(SGCR) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

IKRORG defines the scaling parameter for the imbibition relative permeability of oil at the critical gas saturation (**ISGCR**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the **HYSTER** option on the **SATOPTS** keyword in the **RUNSPEC** section has to be activated to invoke the Hysteresis option. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	IKRORG	IKRORG is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned imbibition scaling IKRORG values for each cell in the model. Repeat counts may be used, for example 50*0.850.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **IKRORGX±**, **IKRORGY±** and **IKRORGZ±** series of the keywords should be used.

Table 8.64: IKRORG Keyword Description

When the **IKRORG** keyword is present in the input deck then the scaling matches the imbibition relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

No	Keywords Present	Critical Saturation
1	IKRORW	$S_{critical} = 1.0 - ISWCR - ISGL$
2	IKRORG	$S_{critical} = 1.0 - ISGCR - SWL$

Table 8.65: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **ISWL**, **ISWCR**, **ISWU**, **ISGL**, **ISGCR**, **ISGU**, **ISOWCR**, and **ISOGCR** saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **ISWUX**, **ISWUY** and **ISWUZ** instead of **ISWU**. There is also the facility to make the directional end-point

scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the *IKRG*, *IKRGR*, *IKRO*, *IKROGR*, *IKRORW*, *IKRW* and *IKRWR* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGY* and *IKRGZ* instead of *IKRG*, There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGX-*, *IKRGY*, *IKRGY-*, *IKRGZ* and *IKRGZ-*, instead of the *IKRG* keyword.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one *IKROGR* is set equal to 0.755, for layer two *IKROGR* equals 0.775, and for layer three *IKROGR* equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX
      1*  1*    1*  1*    1    3
                                     / DEFINE BOX AREA
--
--      SET IKROGR VALUES FOR THREE LAYERS IN THE MODEL
--
IKROGR
      10000*0.755  10000*0.775  10000*0.800
                                     /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the *EQUALS* keyword instead.

```
--
--      ARRAY          CONSTANT          ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      IKROGR          0.7550          1*  1*    1*  1*    1    1 / IKROGR FOR LAYER 1
      IKROGR          0.7750          1*  1*    1*  1*    2    2 / IKROGR FOR LAYER 2
      IKROGR          0.8000          1*  1*    1*  1*    3    3 / IKROGR FOR LAYER 3
/
```

8.3.119 IKRORW – END-POINT SCALING OF GRID CELL KRO(SWCR) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

IKRORW defines the scaling parameter for the imbibition relative permeability of oil at the critical water saturation (**ISWCR**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the **HYSTER** option on the **SATOPTS** keyword in the **RUNSPEC** section has to be activated to invoke the Hysteresis option. The **SCALECRS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	IKRORW	IKRORW is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned imbibition scaling IKRORW values for each cell in the model. Repeat counts may be used, for example 50*0.850			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **IKRORWX±**, **IKRORWY±** and **IKRORWZ±** series of the keywords should be used.

Table 8.66: IKRORW Keyword Description

When the **IKRORW** keyword is present in the input deck then the scaling matches the imbibition relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

No	Keywords Present	Critical Saturation
1	IKRORW	$S_{critical} = 1.0 - ISWCR - ISGL$
2	IKRORG	$S_{critical} = 1.0 - ISGCR - ISWL$

Table 8.67: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **ISWL**, **ISWCR**, **ISWU**, **ISGL**, **ISGCR**, **ISGU**, **ISOWCR**, and **ISOGCR** saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **ISWUX**, **ISWUY** and **ISWUZ** instead of **ISWU**. There is also the facility to make the directional end-point

scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the *IKRG*, *IKRGR*, *IKRO*, *IKRORG*, *IKRORW*, *IKRW* and *IKRWR* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGY* and *IKRGZ* instead of *IKRG*, There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGX-*, *IKRGY*, *IKRGY-*, *IKRGZ* and *IKRGZ-*, instead of the *IKRG* keyword.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one *IKRORW* is set equal to 0.755, for layer two *IKRORW* equals 0.775, and for layer three *IKRORW* equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX
      1*  1*    1*  1*    1    3
--
--      SET IKRORW VALUES FOR THREE LAYERS IN THE MODEL
--
IKRORW
      10000*0.755  10000*0.775  10000*0.800
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the *EQUALS* keyword instead.

```
--
--      ARRAY          CONSTANT          ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      IKRORW          0.7550          1*  1*    1*  1*    1    1  / IKRORW FOR LAYER 1
      IKRORW          0.7750          1*  1*    1*  1*    2    2  / IKRORW FOR LAYER 2
      IKRORW          0.8000          1*  1*    1*  1*    3    3  / IKRORW FOR LAYER 3
/
```

8.3.120 IKRW – END-POINT SCALING OF GRID CELL KRW(Sw =1.0) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

IKRW defines the scaling parameter at the maximum imbibition water relative permeability value (*ISWU*), that is for $Sw = 1.0$, for all the cells in the model via an array. The *ENDSCALE* keyword in the *RUNSPEC* section should be activated to enable end-point scaling and the use of this keyword. In addition, the *HYSTER* option on the *SATOPTS* keyword in the *RUNSPEC* section has to be activated to invoke the Hysteresis option. The *SCALCERS* keyword in the *PROPS* section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	<i>IKRW</i>	<i>IKRW</i> is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned imbibition scaling <i>IKRW</i> values for each cell in the model. Repeat counts may be used, for example 50*1.000.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the *IKRWX*±, *IKRWY*± and *IKRWZ*± series of the keywords should be used.

Table 8.68: *IKRW* Keyword Description

For the two point scaling option and for the *IKRWR* water relative permeability array NOT present in the input deck the *krw* value for a grid block is scaled by:

$$k_{rw} = k_{rw\ TABLE} \left(\frac{IKRW}{k_{rw\ TABLE\ -\ MAX}} \right) \quad (8.60)$$

Where:

- krw = the resulting *IKRW* value for a grid cell.
- IKRW* = the scaling water relative permeability value from the *IKRW* array for a given cell.
- $k_{rw\ TABLE}$ = the water relative permeability from a grid block's oil relative permeability table at the grid blocks water saturation.
- $k_{rw\ TABLE\ -\ MAX}$ = the maximum water relative permeability from a grid block's water relative table, that is at the maximum water saturation.

If the *IKRWR* keyword is present in the input deck then the scaling matches the imbibition relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the [SCALECRS](#) keyword in the [PROPS](#) section the critical displacing phase is defined as:

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - ISOWCR - ISGL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - ISOWCR - ISGL$
3	Gas-Water	$S_{critical} = 1.0 - ISGCR$

Table 8.69: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISWUX](#), [ISWUY](#) and [ISWUZ](#) instead of [ISWU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISWUX](#), [ISWUX-](#), [ISWUY](#), [ISWUY-](#), [ISWUZ](#) and [ISWUZ-](#), instead of the [ISWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [IKRG](#), [IKRGR](#), [IKRO](#), [IKRORG](#), [IKRORW](#), [IKRW](#) and [IKRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [IKRGX](#), [IKRGY](#) and [IKRGZ](#) instead of [IKRG](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [IKRGX](#), [IKRGX-](#), [IKRGY](#), [IKRGY-](#), [IKRGZ](#) and [IKRGZ-](#), instead of the [IKRG](#) keyword.

Example

The example below defines an input box for the whole grid and for layers one to three, for layer one [IKRW](#) is set equal to 0.850, for layer two [IKRW](#) equals 0.875, and for layer three [IKRW](#) equals 0.900.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      1*   1*    1*  1*    1   3
--
--      SET IKRW VALUES FOR THREE LAYERS IN THE MODEL
--
IKRW      10000*0.850  10000*0.875  10000*0.900
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

8.3.121 IKRWR – END-POINT SCALING OF GRID CELL KRWR(SOWCR) (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

IKRWR defines the scaling parameter at the imbibition critical oil to water saturation value (*ISOWCR*), for the imbibition water relative permeability curve, for all the cells in the model via an array. The *ENDSCALE* keyword in the *RUNSPEC* section should be activated to enable end-point scaling and the use of this keyword. In addition, the *HYSTER* option on the *SATOPTS* keyword in the *RUNSPEC* section has to be activated to invoke the Hysteresis option. The *SCALCERS* keyword in the *PROPS* section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	<i>IKRWR</i>	<i>IKRWR</i> is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned imbibition scaling <i>IKRWR</i> values for each cell in the model. Repeat counts may be used, for example 50*1.000.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the *IKRWRX* $\pm$, *IKRWRY* $\pm$ and *IKRWRZ* $\pm$ series of the keywords should be used.

Table 8.70: IKRWR Keyword Description

When the *IKRWR* keyword is present in the input deck then the scaling matches the imbibition relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the *SCALECRS* keyword in the *PROPS* section the critical displacing phase is defined as:

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - ISOWCR - ISGL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - ISOWCR - ISGL$
3	Gas-Water	$S_{critical} = 1.0 - ISGCR$

Table 8.71: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the *ISWL*, *ISWCR*, *ISWU*, *ISGL*, *ISGCR*, *ISGU*, *ISOWCR*, and *ISOGCR* saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used,

that is *ISWUX*, *ISWUY* and *ISWUZ* instead of *ISWU*, There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the *IKRG*, *IKRGR*, *IKRO*, *IKRORG*, *IKRORW*, *IKRW* and *IKRWR* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGY* and *IKRGZ* instead of *IKRG*, There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *IKRGX*, *IKRGX-*, *IKRGY*, *IKRGY-*, *IKRGZ* and *IKRGZ-*, instead of the *IKRG* keyword.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one *IKRWR* is set equal to 0.755, for layer two *IKRWR* equals 0.775, and for layer three *IKRWR* equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX
      1*  1*    1*  1*    1    3
--
--
--      SET IKRWR VALUES FOR THREE LAYERS IN THE MODEL
--
IKRWR
      10000*0.755  10000*0.775  10000*0.800
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the *EQUALS* keyword instead.

```
--
--      ARRAY          CONSTANT          ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      IKRWR           0.7550           1*  1*    1*  1*    1    1  / IKRWR FOR LAYER 1
      IKRWR           0.7750           1*  1*    1*  1*    2    2  / IKRWR FOR LAYER 2
      IKRWR           0.8000           1*  1*    1*  1*    3    3  / IKRWR FOR LAYER 3
/
```

8.3.122
IMKRVD – IMBIBITION RELATIVE PERMEABILITY END-POINTS VERSUS DEPTH FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the maximum imbibition oil, gas, and water relative permeability versus depth for the three phases. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the HYSTER option on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has to be activated to invoke the Hysteresis option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.123 IMPCVD – IMBIBITION MAXIMUM CAPILLARY PRESSURE VERSUS DEPTH FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the maximum imbibition gas-oil and water-oil capillary pressure values versus depth for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section and the HYSTER option on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has been activated to invoke the Hysteresis option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.124 IMPORT – IMPORT GRID FILE DATA AT THE CURRENT POSITION

The *IMPORT* keyword informs the simulator to continue reading input data from the specified *IMPORT* file. When the end of the *IMPORT* file is reached, input data is read from the next keyword in the current file. Normally *IMPORT* files are generated by grid pre-processing software and the keyword allows for both formatted and unformatted (binary) files to be loaded.

See *IMPORT – Import Grid File Data at the Current Position* in the *GRID* section for a full description.

8.3.125
IMPTVD
–
IMBIBITION
RELATIVE
PERMEABILITY
SATURATION
END-POINTS
VERSUS
DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the variation of the imbibition relative permeability saturation end-points ([SWL](#), [SWCR](#), etc.) for all three phases versus depth., for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section, and the HYSTER option on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has been activated to invoke the Hysteresis option. This functionality is not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.126
IMSPCVD – IMBIBITION CAPILLARY PRESSURE CONNATE SATURATIONS VERSUS DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the imbibition capillary pressure gas and water connate saturations values versus depth for when the end-point scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section, and the HYSTER option on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has been activated to invoke the Hysteresis option. This functionality is not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.127 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDING](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the GLOBAL section for a full description.

8.3.128 INTPC – ACTIVATE DUAL POROSITY INTEGRATED CAPILLARY PRESSURE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *INTPC* keyword activates the integrated capillary pressure option for the oil, gas or both phases, for when a Dual Porosity model has been activated by either the *DUALPORO* or *DUALPERM* keywords in the *RUNSPEC* section. In addition, the keyword can only be used if the Gravity Drainage option has been specified by either the *GRAVDR* or *GRAVDRM* in the *RUNSPEC* section. Basically, activating this feature results in the simulator adjusting the capillary pressure curves by integrating the matrix capillary pressure curves over the matrix block height to calculate the average saturation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.129 IONXROCK – DEFINE ION EXCHANGE CONSTANT BY SATURATION
TABLE REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [IONXROCK](#) keyword activates ion exchange and defines the ion exchange constant by saturation table regions, for when the brine phase has been activated by the [BRINE](#) keyword and the Multi-Component Brine model, that allows for the water phase to have multiple water salinities, has been activated by the [ECLMC](#) keyword. Both keywords are in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.130 IONXSURF – DEFINE SURFACTANT ION EXCHANGE CONSTANT BY SATURATION TABLE REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [IONXROCK](#) keyword activates ion exchange on surfactant micellae²⁰⁹ and defines the ion exchange constant by surfactant equivalent molecular weight for saturation table regions, for when the brine and surfactant phases has been activated by the [BRINE](#) and [SURFACT](#) keywords, and the Multi-Component Brine model, that allows for the water phase to have multiple water salinities, has been activated by the [ECLMC](#) keyword. All three keywords are in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²⁰⁹ Particle of colloidal dimensions that exists in equilibrium with the molecules or ions in solution from which it is formed. A micella or micelle (plural micellae or micelles, respectively) is an aggregate (or supramolecular assembly) of surfactant molecules dispersed in a liquid colloid. A typical micella in aqueous solution forms an aggregate with the hydrophilic "head" regions in contact with surrounding solvent, sequestering the hydrophobic single-tail regions in the micella centre (<https://en.wikipedia.org/wiki/Micelle>).

8.3.131 IPCG – END-POINT SCALING OF GRID CELL GAS CAPILLARY PRESSURE (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[IPCG](#) defines the maximum imbibition gas-oil capillary pressure values for all the cells in the model via an array. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the [HYSTER](#) option on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has to be activated to invoke the hysteresis option. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	IPCG	IPCG is an array of positive real numbers assigning the maximum imbibition gas capillary pressure values for each cell in the model. Repeat counts may be used, for example 30*100.0.			None
		psia	bars	atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 8.72: IPCG Keyword Description

The capillary pressure for a grid block is scaled by:

$$P_c = P_{c_{TABLE}} \left(\frac{IPCG}{P_{c_{TABLE-MAX}}} \right) \quad (8.61)$$

Where:

- P_c = the resulting imbibition gas-oil capillary pressure for a grid cell.
- $IPCG$ = the maximum capillary pressure from the [IPCG](#) array for a given cell.
- $P_{c_{TABLE}}$ = the capillary pressure in the imbibition capillary pressure table allocated to the grid block.
- $P_{c_{TABLE-MAX}}$ = the maximum capillary pressure in the imbibition capillary pressure table allocated to the grid block at $S_g = 1 - S_{wco}$.

See also the [PCG](#) keyword for the equivalent drainage functionality.

Example

```
--
--      DEFINE GRID BLOCK IPCG DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
IPCG
      100*50.0  100*75.0  100*125.0 /
```

The above example defines the [IPCG](#) for 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.132 IPCW – END-POINT SCALING OF GRID CELL WATER CAPILLARY PRESSURE (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[IPCW](#) defines the maximum imbibition water-oil or water-gas capillary pressure values for all the cells in the model via an array. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the [HYSTER](#) option on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has to be activated to invoke the hysteresis option. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	IPCW	IPCW is an array of positive real numbers assigning the maximum imbibition water capillary pressure values for each cell in the model. Repeat counts may be used, for example 30*100.0.			None
		psia	bars	atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 8.73: IPCW Keyword Description

The capillary pressure for a grid block is scaled by:

$$P_c = P_{C_{TABLE}} \left(\frac{IPCW}{P_{C_{TABLE-MAX}}} \right) \quad (8.62)$$

Where:

- P_c = the resulting imbibition water capillary pressure for a grid cell.
- $IPCW$ = the maximum capillary pressure from the [IPCW](#) array for a given cell.
- $P_{C_{TABLE}}$ = the capillary pressure in the imbibition capillary pressure table allocated to the grid block.
- $P_{C_{TABLE-MAX}}$ = the maximum capillary pressure in the imbibition capillary pressure table allocated to the grid block (that is at the connate water saturation).

See also the [PCW](#) keyword for the equivalent drainage functionality.

Example

```
--
--      DEFINE GRID BLOCK IPCW DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
IPCW      100*50.0  100*75.0  100*125.0  /
```

The above example defines the [IPCW](#) for 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.133 ISGCR – END-POINT SCALING OF GRID CELL CRITICAL GAS SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISGCR](#) defines the imbibition critical gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The critical gas saturation is defined as the maximum gas saturation for which the gas relative permeability is zero in a two-phase relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISGCR	ISGCR is an array of real numbers assigning the critical gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.03			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the critical gas saturation used with the end-point scaling option.

Table 8.74: ISGCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISGCRX](#), [ISGCRY](#) and [ISGCRZ](#) instead of [ISGCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISGCRX](#), [ISGCRX-](#), [ISGCRY](#), [ISGCRY-](#), [ISGCRZ](#) and [ISGCRZ-](#), instead of the [ISGCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISGCR DATA FOR CELLS (NX x NY x NZ = 300)
ISGCR
      300*0.050 /
```

The above example defines a constant critical gas saturation of 0.05 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.134 ISGL – END-POINT SCALING OF GRID CELL CONNATE GAS SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISGL](#) defines the imbibition connate gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The connate gas saturation is defined as the minimum gas saturation in a two-phase gas relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISGL	ISGL is an array of real numbers assigning the connate gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.03 dimensionless			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the connate gas saturation array used with the end-point scaling option.

Table 8.75: ISGL Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISGLX](#), [ISGLY](#) and [ISGLZ](#) instead of [ISGL](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISGLX-](#), [ISGLY-](#), [ISGLZ-](#) and [ISGLZ-](#), instead of the [ISGL](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISGL DATA FOR ALL CELLS (NX x NY x NZ = 300)
ISGL      300*0.030 /
```

The above example defines a constant connate gas saturation of 0.03 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.135 ISGLPC – END-POINT SCALING OF GRID CELL CAPILLARY PRESSURE CONNATE GAS SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

ISGLPC defines the imbibition connate gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the *ENDSCALE* keyword in the *RUNSPEC* section and the hysteresis model option has been activated on the *SATOPTS* keyword in the *RUNSPEC* section. The connate gas saturation is defined as the minimum gas saturation in a two-phase gas relative permeability table. The keyword only applies the scaling to the imbibition capillary pressures tables, unlike the *ISGL* keyword that applies the scaling to both the capillary pressure and relative permeability tables. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	<i>ISGLPC</i>	<i>ISGLPC</i> is an array of real numbers assigning the connate gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the <i>DIMENS</i> keyword. If <i>ISGLPC</i> is omitted from the input deck the values will be defaulted to those on the <i>ISGL</i> series of keywords. If the <i>ISGL</i> series of keywords are missing from the input deck then the values are taken from the cell allocated capillary pressure table. Repeat counts may be used, for example 30*0.03			Taken from <i>SGL</i> or from the cell allocated capillary pressure table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a "/".
- 3) Note this is the directional independent version of the connate gas saturation array used with the end-point scaling option.

Table 8.76: ISGLPC Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the *ISWL*, *ISWCR*, *ISWU*, *ISGL*, *ISGCR*, *ISGU*, *ISOWCR*, and *ISOGCR* saturation grid arrays for the saturation end-points, and the *IKRG*, *IKRORG*, *IKRORW* and *IKRW* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *ISGLX*, *ISGLY* and *ISGLZ* instead of *ISGL* or *ISGLPC*. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISGLX*, *ISGLX-*, *ISGLY*, *ISGLY-*, *ISGLZ* and *ISGLZ-*, instead of the *ISGL* or *ISGLPC* keywords.

Missing Some Functionality - Use with Caution.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISGLPC DATA FOR ALL CELLS
-      (NX x NY x NZ = 300)
ISGLPC      300*0.030      /
```

The above example defines a constant connate gas saturation of 0.03 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.136 ISGU – END-POINT SCALING OF GRID CELL MAXIMUM GAS SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISGU](#) defines the imbibition maximum gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The maximum gas saturation is defined as the maximum gas saturation in a two-phase gas relative permeability table. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISGU	ISGU is an array of real numbers assigning the maximum gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.70			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the maximum gas saturation array used with the end-point scaling option.

Table 8.77: ISGU Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISGUX](#), [ISGUY](#) and [ISGUZ](#) instead of [ISGU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISGUX-](#), [ISGUY-](#), [ISGUZ-](#) and [ISGUZ-](#), instead of the [ISGU](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISGU DATA FOR ALL CELLS (NX x NY x NZ = 300)
ISGU      300*0.700 /
```

The above example defines a constant connate gas saturation of 0.70 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.137 ISOGCR – END-POINT SCALING OF GRID CELL CRITICAL OIL SATURATION TO GAS (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISOGCR](#) defines the imbibition critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to gas is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase gas-oil relative permeability table. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISOGCR	ISOGCR is an array of real numbers assigning the critical oil saturation with respect to gas values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the critical gas saturation used with the end-point scaling option.

Table 8.78: ISOGCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISOGCRX](#), [ISOGCRY](#) and [ISOGCRZ](#) instead of [ISOGCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISOGCRX-](#), [ISOGCRY-](#), [ISOGCRZ-](#), instead of the [ISOGCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISOGCR DATA FOR ALL CELLS
--      (NX x NY x NZ = 300)
--
ISOGCR
      300*0.200 /
```

The above example defines a constant critical gas saturation of 0.20 to all 300 cells in the model as defined by the *DIMENS* keyword in the *RUNSPEC* section.

8.3.138 ISOWCR – END-POINT SCALING OF GRID CELL CRITICAL OIL SATURATION TO WATER (IMBIB.)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISOWCR](#) defines the imbibition critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to water is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase oil-water relative permeability table. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISOWCR	ISOWCR is an array of real numbers assigning the critical oil saturation with respect to water values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the critical gas saturation used with the end-point scaling option.

Table 8.79: ISOWCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISOWCRX](#), [ISOWCRY](#) and [ISOWCRZ](#) instead of [ISOWCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISOWCRX-](#), [ISOWCRY-](#), [ISOWCRZ-](#), instead of the [ISOWCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISOWCR DATA FOR ALL CELLS
--      (NX x NY x NZ = 300)
--
ISOWCR
      300*0.200 /
```

The above example defines a constant critical gas saturation of 0.20 to all 300 cells in the model as defined by the *DIMENS* keyword in the *RUNSPEC* section

8.3.139 ISWCR – END-POINT SCALING OF GRID CELL CRITICAL WATER SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISWCR](#) defines the imbibition critical water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The critical water saturation is defined as the maximum water saturation for which the water relative permeability is zero in a two-phase relative permeability table. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISWCR	ISWCR is an array of real numbers assigning the critical water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.20			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the critical water saturation used with the end-point scaling option.

Table 8.80: ISWCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISWCRX](#), [ISWCRY](#) and [ISWCRZ](#) instead of [ISWCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISWCRX-](#), [ISWCRX-](#), [ISWCRY-](#), [ISWCRY-](#), [ISWCRZ-](#) and [ISWCRZ-](#), instead of the [ISWCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISWCR DATA FOR ALL CELLS
--      (NX x NY x NZ = 300)
--
ISWCR
  300*0.200 /
```

The above example defines a constant critical water saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.140 ISWL – END-POINT SCALING OF GRID CELL CONNATE WATER SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	------------------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[ISWL](#) defines the imbibition connate water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The connate water saturation is defined as the minimum water saturation in a two-phase water relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	ISWL	ISWL is an array of real numbers assigning the connate water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.15			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the connate gas saturation array used with the end-point scaling option.

Table 8.81: ISWL Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISWLX](#), [ISWLY](#) and [ISWLZ](#) instead of [ISWL](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISWLX](#), [ISWLX-](#), [ISWLY](#), [ISWLY-](#), [ISWLZ](#) and [ISWLZ-](#), instead of the [ISWL](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISWL DATA FOR ALL CELLS (NX x NY x NZ = 300)
--
ISWL      300*0.150 /
```

The above example defines a constant connate water saturation of 0.15 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.141 ISWLPC – END-POINT SCALING GRID CELL CAPILLARY PRESSURE CONNATE WATER SATURATIONS (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ISWLPC](#) defines the imbibition connate water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) in the [RUNSPEC](#) section and the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The connate water saturation is defined as the minimum water saturation in a two-phase water relative permeability table. The keyword only applies the scaling to the imbibition capillary pressures tables, unlike the [ISWL](#) keyword that applies the scaling to both the capillary pressure and relative permeability tables. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ISWLPC	ISWLPC is an array of real numbers assigning the connate water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. If ISWLPC is omitted from the input deck the values will be defaulted to those on the ISGL series of keywords. If the ISWL series of keywords are missing from the input deck then the values are taken from the cell allocated capillary pressure table. Repeat counts may be used, for example 30*0.15			Taken from SWL or from the cell allocated capillary pressure table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the connate gas saturation array used with the end-point scaling option.

Table 8.82: ISWLPC Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [ISWL](#), [ISWCR](#), [ISWU](#), [ISGL](#), [ISGCR](#), [ISGU](#), [ISOWCR](#), and [ISOGCR](#) saturation grid arrays for the saturation end-points, and the [IKRG](#), [IKRORG](#), [IKRORW](#) and [IKRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [ISWLX](#), [ISWLY](#) and [ISWLZ](#) instead of [ISWL](#) or [ISWLPC](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [ISWLX-](#), [ISWLY-](#), [ISWLZ-](#), instead of the [ISWL](#) or [ISWLPC](#) keywords.

Missing Some Functionality - Use with Caution.

Example

```
--  
--      DEFINE GRID BLOCK END-POINT ISWLPC DATA FOR ALL CELLS (  
--      NX x NY x NZ = 300)  
--  
ISWLPC      300*0.150      /
```

The above example defines a constant connate water saturation of 0.15 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.142 ISWU – END-POINT SCALING OF GRID CELL MAXIMUM WATER SATURATION (IMBIBITION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

ISWU defines the imbibition maximum water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the *ENDSCALE* in the *RUNSPEC* section and the hysteresis model option has been activated on the *SATOPTS* keyword in the *RUNSPEC* section. The maximum water saturation is defined as the maximum water saturation in a two-phase water relative permeability table. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	<i>ISWU</i>	<i>ISWU</i> is an array of real numbers assigning the maximum water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the <i>DIMENS</i> keyword. Repeat counts may be used, for example 30*0.70			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this is the directional independent version of the maximum water saturation array used with the end-point scaling option.

Table 8.83: *ISWU* Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the *ISWL*, *ISWCR*, *ISWU*, *ISGL*, *ISGCR*, *ISGU*, *ISOWCR*, and *ISOGCR* saturation grid arrays for the saturation end-points, and the *IKRG*, *IKRORG*, *IKRORW* and *IKRW* relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUY* and *ISWUZ* instead of *ISWU*. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is *ISWUX*, *ISWUX-*, *ISWUY*, *ISWUY-*, *ISWUZ* and *ISWUZ-*, instead of the *ISWU* keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT ISWU DATA FOR ALL CELLS (NX x NY x NZ = 300)
--
ISWU      300*0.700 /
```

The above example defines a constant connate gas saturation of 0.70 to all 300 cells in the model as defined by the *DIMENS* keyword in the *RUNSPEC* section.

8.3.143 KRG – END-POINT SCALING OF GRID CELL KRG(SGU) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

KRG defines the scaling parameter at the maximum drainage gas relative permeability value (**SGU**), normally **SGU** is equal to 1.0 - Swc, for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRG	KRG is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRG values for each cell in the model. Repeat counts may be used, for example 50*0.400. dimensionless			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KRGX±**, **KRGY±** and **KRGZ±** series of the keywords should be used.

Table 8.84: KRG Keyword Description

For the two point scaling option and for the **KRGR** gas relative permeability array NOT present in the input deck the krg value for a grid block is scaled by:

$$k_{rg} = k_{rg\,TABLE} \left(\frac{KRG}{k_{rg\,TABLE-MAX}} \right) \quad (8.63)$$

Where:

- krg** = the resulting krg value for a grid cell.
- KRG** = the scaling gas relative permeability value from the **KRG** array for a given cell.
- $k_{rg\,TABLE}$ = the gas relative permeability from a grid block's gas-oil table at the grid blocks gas saturation.
- $k_{rg\,TABLE-MAX}$ = the maximum gas relative permeability from a grid block's gas-oil table, that is at the connate water saturation (Swc).

If the **KRGR** keyword is present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the [SCALECRS](#) keyword in the [PROPS](#) section the critical displacing phase is defined as:

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - SOGCR - SWL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - SOGCR - SWL$
3	Gas-Water	$S_{critical} = 1.0 - SWCR$

Table 8.85: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUY](#) and [SWUZ](#) instead of [SWU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUX-](#), [SWUY](#), [SWUY-](#), [SWUZ](#) and [SWUZ-](#), instead of the [SWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KRGR](#), [KRO](#), [KROGR](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KRGX](#), [KRGY](#) and [KRGZ](#) instead of [KRG](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KRGX](#), [KRGX-](#), [KRGY](#), [KRGY-](#), [KRGZ](#) and [KRGZ-](#), instead of the [KRG](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays prefixed with the letter I, for example [IKRG](#), can be used to scale [KRG](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRG](#) is set equal to 0.555, for layer two [KRG](#) equals 0.575, and for layer three [KRG](#) equals 0.600.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      1*  1*    1*  1*    1   3
--
--      / DEFINE BOX AREA
--
--      SET KRG VALUES FOR THREE LAYERS IN THE MODEL
--
--      KRG
--      10000*0.555  10000*0.575  10000*0.600
--
--      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```


The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      KRG          0.5550        1*  1*   1*  1*   1   1  / KRG FOR LAYER 1
      KRG          0.5750        1*  1*   1*  1*   2   2  / KRG FOR LAYER 2
      KRG          0.6000        1*  1*   1*  1*   3   3  / KRG FOR LAYER 3
/
```

8.3.144 KRGR – END-POINT SCALING OF GRID CELL KRGR(1-SOGCR) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

KRGR defines the scaling parameter at the relative permeability of gas at residual oil saturation ($1 - \text{SOGCR}$), or critical water saturation in a gas-water run (Swc), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRGR	KRGR is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRGR values for each cell in the model. In addition, for a given grid block KRGR should be less than KRG . Repeat counts may be used, for example 50*0.400.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a "/".
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KRGRX**, **KRGRY** and **KRGRZ** series of the keywords should be used.

Table 8.86: KRGR Keyword Description

When the **KRGR** keyword is present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase (oil or water).

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{\text{critical}} = 1.0 - \text{SOGCR} - \text{SWL}$
2	Gas-Oil-Water	$S_{\text{critical}} = 1.0 - \text{SOGCR} - \text{SWL}$
3	Gas-Water	$S_{\text{critical}} = 1.0 - \text{SWCR}$

Table 8.87: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **SWL**, **SWCR**, **SWU**, **SGL**, **SGCR**, **SGU**, **SOWCR**, and **SOGCR** saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **SWUX**, **SWUY** and **SWUZ** instead of **SWU**. There is also the facility to make the directional end-point scaling reversible or

non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUX-](#), [SWUY](#), [SWUY-](#), [SWUZ](#) and [SWUZ-](#), instead of the [SWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KRGR](#), [KRO](#), [KROGR](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KRGRX](#), [KRGRY](#) and [KRGRZ](#) instead of [KRGR](#), There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KRGRX](#), [KRGRX-](#), [KRGRY](#), [KRGRY-](#), [KRGRZ](#) and [KRGRZ-](#), instead of the [KRGR](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays suffixed with the letter I, for example [IKRGR](#), can be used to define the [KRGR](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRGR](#) is set equal to 0.500, for layer two [KRGR](#) equals 0.570, and for layer three [KRGR](#) equals 0.580.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      1*  1*    1*  1*    1    3
--
--      SET KRGR VALUES FOR THREE LAYERS IN THE MODEL
--
KRGR      10000*0.500  10000*0.570  10000*0.580
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
KRGR      0.5000      1*  1*    1*  1*    1    1  / KRGR FOR LAYER 1
KRGR      0.5700      1*  1*    1*  1*    2    2  / KRGR FOR LAYER 2
KRGR      0.5800      1*  1*    1*  1*    3    3  / KRGR FOR LAYER 3
/
```

8.3.145 KRO – END-POINT SCALING OF GRID CELL KRO(SWL) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

KRO defines the scaling parameter for the drainage oil relative permeability value at the connate water saturation (**SWL**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRO	KRO is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRO values for each cell in the model. Repeat counts may be used, for example 50*0.500.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KROX±**, **KROY±** and **KROZ±** series of the keywords should be used.

Table 8.88: KRO Keyword Description

For the two point scaling option and for the **KRORW** or **KRORG** oil relative permeability arrays NOT being present in the input deck the kro value for a grid block is scaled by:

$$k_{ro} = k_{ro\ TABLE} \left(\frac{KRO}{k_{ro\ TABLE - MAX}} \right) \quad (8.64)$$

Where:

- k_{ro} = the resulting kro value for a grid cell.
- KRO = the scaling oil relative permeability value from the **KRO** array for a given cell.
- $k_{ro\ TABLE}$ = the oil relative permeability from a grid block's oil relative permeability table at the grid blocks oil saturation.
- $k_{ro\ TABLE - MAX}$ = the maximum oil relative permeability from a grid block's oil relative table, that is at the critical water saturation (**Swcr**).

If the **KRORW** or **KRORG** keywords are present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the [SCALECRS](#) keyword in the [PROPS](#) section the critical displacing phase is defined as:

No	Keywords Present	Critical Saturation
1	KRORW	$S_{critical} = 1.0 - SWCR - SGL$
2	KRORG	$S_{critical} = 1.0 - SGCR - SWL$

Table 8.89: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUY](#) and [SWUZ](#) instead of [SWU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUX-](#), [SWUY](#), [SWUY-](#), [SWUZ](#) and [SWUZ-](#), instead of the [SWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KRGR](#), [KRO](#), [KRORG](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KROX](#), [KROY](#) and [KROZ](#) instead of [KRO](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KROX](#), [KROX-](#), [KROY](#), [KROY-](#), [KROZ](#) and [KROZ-](#), instead of the [KRO](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays suffixed with the letter I, for example [IKRO](#), can be used to define the [KRO](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRO](#) is set equal to 0.855, for layer two [KRO](#) equals 0.875, and for layer three [KRO](#) equals 0.900.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      1*   1*   1*   1*   1   3
--
--      / DEFINE BOX AREA
--
--      SET KRO VALUES FOR THREE LAYERS IN THE MODEL
--
--      KRO
--      10000*0.855  10000*0.875  10000*0.900
--
--      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      KRO      0.8550      1*  1*  1*  1*    1  1  / KRO FOR LAYER 1
      KOG      0.8750      1*  1*  1*  1*    2  2  / KRO FOR LAYER 2
      KRO      0.9000      1*  1*  1*  1*    3  3  / KRO FOR LAYER 3
/
```

8.3.146 KRORG – END-POINT SCALING OF GRID CELL KRO(SGCR) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

KRORG defines the scaling parameter for the drainage relative permeability of oil at the critical gas saturation (**SGCR**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRORG	KRORG is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRORG values for each cell in the model. Repeat counts may be used, for example 50*0.850.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KRORGX±**, **KRORGY±** and **KRORGZ±** series of the keywords should be used.

Table 8.90: KRORG Keyword Description

When the **KRORG** keyword is present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

No	Keywords Present	Critical Saturation
1	KRORW	$S_{critical} = 1.0 - SWCR - SGL$
2	KRORG	$S_{critical} = 1.0 - SGCR - SWL$

Table 8.91: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **SWL**, **SWCR**, **SWU**, **SGL**, **SGCR**, **SGU**, **SOWCR**, and **SOGCR** saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **SWUX**, **SWUY** and **SWUZ** instead of **SWU**. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is **SWUX**, **SWUX-**, **SWUY**, **SWUY-**, **SWUZ** and **SWUZ-**, instead of the **SWU** keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KRGR](#), [KRORG](#), [KRORGRG](#), [KRORGRW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KRORGX](#), [KRORGY](#) and [KRORGZ](#) instead of [KRORG](#), There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KRORGX](#), [KRORGX-](#), [KRORGY](#), [KRORGY-](#), [KRORGZ](#) and [KRORGZ-](#), instead of the [KRORG](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays suffixed with the letter I, for example [IKRORG](#), can be used to define the [KRORG](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRORG](#) is set equal to 0.755, for layer two [KRORG](#) equals 0.775, and for layer three [KRORG](#) equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX
      1*  1*    1*  1*    1    3
--
--      SET KRORG VALUES FOR THREE LAYERS IN THE MODEL
--
KRORG
      10000*0.755  10000*0.775  10000*0.800
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--
--      ARRAY          CONSTANT          ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      KRORG           0.7550           1*  1*    1*  1*    1    1  / KRORG FOR LAYER 1
      KRORG           0.7750           1*  1*    1*  1*    2    2  / KRORG FOR LAYER 2
      KRORG           0.8000           1*  1*    1*  1*    3    3  / KRORG FOR LAYER 3
/
```


8.3.147 KRORW – END-POINT SCALING OF GRID CELL KRO(SWCR) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

KRORW defines the scaling parameter for the drainage relative permeability of oil at the critical water saturation (**SWCR**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALECRS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRORW	KRORW is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRORW values for each cell in the model. Repeat counts may be used, for example 50*0.850			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KRORWX±**, **KRORWY±** and **KRORWZ±** series of the keywords should be used.

Table 8.92: KRORW Keyword Description

When the **KRORW** keyword is present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

No	Keywords Present	Critical Saturation
1	KRORW	$S_{critical} = 1.0 - SWCR - SGL$
2	KRORG	$S_{critical} = 1.0 - SGCR - SWL$

Table 8.93: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **SWL**, **SWCR**, **SWU**, **SGL**, **SGCR**, **SGU**, **SOWCR**, and **SOGCR** saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **SWUX**, **SWUY** and **SWUZ** instead of **SWU**. There is also the facility to make the directional end-point scaling reversible or

non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUX-](#), [SWUY](#), [SWUY-](#), [SWUZ](#) and [SWUZ-](#), instead of the [SWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KGR](#), [KRORW](#), [KRORWRG](#), [KRORWRW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KRORWX](#), [KRORWY](#) and [KRORWZ](#) instead of [KRORW](#), There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KRORWX](#), [KRORWX-](#), [KRORWY](#), [KRORWY-](#), [KRORWZ](#) and [KRORWZ-](#), instead of the [KRORW](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays suffixed with the letter I, for example [IKRORW](#), can be used to define the [KRORW](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRORW](#) is set equal to 0.755, for layer two [KRORW](#) equals 0.775, and for layer three [KRORW](#) equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
BOX      1*  1*    1*  1*    1    3
--
--      SET KRORW VALUES FOR THREE LAYERS IN THE MODEL
--
KRORW
      10000*0.755  10000*0.775  10000*0.800
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      KRORW      0.7550      1*  1*    1*  1*    1    1  / KRORW FOR LAYER 1
      KRORW      0.7750      1*  1*    1*  1*    2    2  / KRORW FOR LAYER 2
      KRORW      0.8000      1*  1*    1*  1*    3    3  / KRORW FOR LAYER 3
/
```

8.3.148 KRW – END-POINT SCALING OF GRID CELL KRW(Sw =1.0) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

KRW defines the scaling parameter at the maximum drainage water relative permeability value (**SWU**), that is for $Sw = 1.0$, for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRW	KRW is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRW values for each cell in the model. Repeat counts may be used, for example 50*1.000.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KRWX**, **KRWY** and **KRWZ** series of the keywords should be used.

Table 8.94: KRW Keyword Description

For the two point scaling option and for the **KRW** water relative permeability array NOT present in the input deck the **krw** value for a grid block is scaled by:

$$k_{rw} = k_{rw_TABLE} \left(\frac{KRW}{k_{rw_TABLE-MAX}} \right) \quad (8.65)$$

Where:

- krw** = the resulting **KRW** value for a grid cell.
- KRW** = the scaling water relative permeability value from the **KRW** array for a given cell.
- k_{rw_TABLE} = the water relative permeability from a grid block's oil relative permeability table at the grid blocks water saturation.
- $k_{rw_TABLE-MAX}$ = the maximum water relative permeability from a grid block's water relative table, that is at the maximum water saturation.

If the **KRW** keyword is present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - SOWCR - SGL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - SOWCR - SGL$
3	Gas-Water	$S_{critical} = 1.0 - SGCR$

Table 8.95: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUY](#) and [SWUZ](#) instead of [SWU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUX-](#), [SWUY](#), [SWUY-](#), [SWUZ](#) and [SWUZ-](#), instead of the [SWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KRGR](#), [KRW](#), [KRORG](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KRWX](#), [KRWY](#) and [KRWZ](#) instead of [KRW](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KRWX](#), [KRWX-](#), [KRWY](#), [KRWY-](#), [KRWZ](#) and [KRWZ-](#), instead of the [KRW](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays suffixed with the letter I, for example [IKRW](#), can be used to define the [KRW](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRW](#) is set equal to 0.855, for layer two [KRW](#) equals 0.875, and for layer three [KRW](#) equals 0.900.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
--      1*  1*   1*  1*   1   3
--                                     / DEFINE BOX AREA
--
--      SET KRW VALUES FOR THREE LAYERS IN THE MODEL
--
--      KRW
--      10000*0.855  10000*0.875  10000*0.900
--                                     /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      KRW          0.8550      1*  1*   1*  1*   1   1  / KRW FOR LAYER 1
      KRW          0.8750      1*  1*   1*  1*   2   2  / KRW FOR LAYER 2
      KRW          0.9000      1*  1*   1*  1*   3   3  / KRW FOR LAYER 3
/
```

8.3.149 KRWR – END-POINT SCALING OF GRID CELL KRWR(SOWCR) (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

KRWR defines the scaling parameter at the drainage critical oil to water saturation value (**SOWCR**), for the drainage water relative permeability curve, for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The **SCALCERS** keyword in the **PROPS** section defines the options used in the re-scaling process, the options are two point scaling and three point scaling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	KRWR	KRWR is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling KRWR values for each cell in the model. Repeat counts may be used, for example 50*1.000.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) Note this the directional independent version of the keyword used with the end-point scaling option. If directional end-point scaling has been activated then the **KRWRX±**, **KRWRY±** and **KRWZR±** series of the keywords should be used.

Table 8.96: KRWR Keyword Description

When the **KRWR** keyword is present in the input deck then the scaling matches the relative permeability at the critical saturation of the displacing phase.

If three point scaling option has been selected via the **SCALECRS** keyword in the **PROPS** section the critical displacing phase is defined as:

No	Phases Present	Critical Saturation
1	Gas-Oil	$S_{critical} = 1.0 - SOWCR - SGL$
2	Gas-Oil-Water	$S_{critical} = 1.0 - SOWCR - SGL$
3	Gas-Water	$S_{critical} = 1.0 - SGCR$

Table 8.97: Critical Displacement Relationships

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **SWL**, **SWCR**, **SWU**, **SGL**, **SGCR**, **SGU**, **SOWCR**, and **SOGCR** saturation grid arrays for the saturation end-points. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **SWUX**, **SWUY** and **SWUZ** instead of **SWU**. There is also the facility to make the directional end-point scaling reversible or

non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUX-](#), [SWUY](#), [SWUY-](#), [SWUZ](#) and [SWUZ-](#), instead of the [SWU](#) keyword.

End-point scaling also allows the entered relative permeability functions to be scale on the relative permeability values using the [KRG](#), [KRGR](#), [KRWR](#), [KRORG](#), [KRORW](#), [KRWR](#) and [KRVRR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [KRWRX](#), [KRWRY](#) and [KRWRZ](#) instead of [KRWR](#), There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [KRWRX-](#), [KRWRX-](#), [KRWRX-](#), [KRWRX-](#), [KRWRX-](#) and [KRWRX-](#), instead of the [KRWR](#) keyword.

If the hysteresis model option has been activated on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the equivalent imbibition arrays suffixed with the letter I, for example [IKRWR](#), can be used to define the [KRWR](#) for the relative permeability imbibition tables.

Examples

The first example defines an input box for the whole grid and for layers one to three, for layer one [KRWR](#) is set equal to 0.755, for layer two [KRWR](#) equals 0.775, and for layer three [KRWR](#) equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX
      1*  1*   1*  1*   1   3                               / DEFINE BOX AREA
--
--      SET KRWR VALUES FOR THREE LAYERS IN THE MODEL
--
KRWR
      10000*0.755  10000*0.775  10000*0.800                               /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

The next example does exactly the same thing using the [EQUALS](#) keyword instead.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
EQUALS
      KRWR      0.7550      1*  1*   1*  1*   1   1  / KRWR FOR LAYER 1
      KRWR      0.7750      1*  1*   1*  1*   2   2  / KRWR FOR LAYER 2
      KRWR      0.8000      1*  1*   1*  1*   3   3  / KRWR FOR LAYER 3
/
```

8.3.150 LANGMPL – DEFINE LANGMUIR PRESSURE GRID CELL MULTIPLIER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [LANGMPL](#), defines the coal bed methane Langmuir Adsorption²¹⁰ pressure multiplier for each grid block, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section. The keyword applies the multiplier to the pressure values in a cell's allocated Langmuir table when calculating a cell's adsorption capacity. See the [LANGMUIR](#) keyword in the [PROPS](#) section for specifying the Langmuir tables for the model.

Note that if he Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section, then [LANGMPL](#) applies to only the matrix grid block.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²¹⁰
Langmuir, Irving (June 1918). "The Adsorption of Gases on Plane Surface of Glass, Mica and Platinum". *The Research Laboratory of the General Electric Company*. 40 (9): 1361–1402. doi:10.1021/ja02242a004

8.3.151 LANGMUIR – LANGMUIR ADSORPTION ISOTHERM TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LANGMUIR](#) keyword defines the coal bed methane Langmuir Adsorption Isotherms²¹¹ tables, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section. See the [COALNUM](#) keyword in the [GRID](#) section for allocating the Langmuir tables to the grid blocks and also the [LANGMPL](#) keyword in the [PROPS](#) section for re-scaling the pressure values in the tables that are allocated to a cell.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

211

Langmuir, Irving (June 1918). "The Adsorption of Gases on Plane Surface of Glass, Mica and Platinum".*The Research Laboratory of the General Electric Company*. 40 (9): 1361–1402. doi:10.1021/ja02242a004

8.3.152 LANGSOLV – LANGMUIR ADSORPTION ISOTHERM SOLVENT TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LANGMUIR](#) keyword defines the coal bed methane Langmuir Adsorption Isotherms²¹² Solvent tables, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword and the Solvent phase has been declared by the [SOLVENT](#) keyword in the [RUNSPEC](#) section. See the [COALNUM](#) keyword in the [GRID](#) section for allocating the Langmuir solvent tables to the grid blocks, and also the [LANGMUIR](#) keyword in the [PROPS](#) section for defining the Langmuir Adsorption Isotherm tables. Keywords [COALADS](#) and [COALPP](#), also in the [PROPS](#) section, are used to specify the relative adsorption data in runs containing the solvent phase.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

212

Langmuir, Irving (June 1918). "The Adsorption of Gases on Plane Surface of Glass, Mica and Platinum".*The Research Laboratory of the General Electric Company*. 40 (9): 1361–1402. doi:10.1021/ja02242a004

8.3.153 LCUNIT – DEFINE LINEAR COMBINATION RATE AND VOLUME UNITS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LCUNIT](#) keyword defines the units for the Linear Combination facility which allows for a linear combination of oil, gas and water rates and volumes to be used as combination targets and constraints in controlling group and well production and injection data. See also the [LINCOM](#) in the [SCHEDULE](#) section that defines the actual linear combination equation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.154 LKRO – END-POINT SCALING OF GRID CELL Kro(SWL) (LOW SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LKRO](#) defines the scaling parameter for the oil relative permeability value at the connate water saturation ([SWL](#)), for all the cells in the model via an array, for when the Low Salt option and the End-point Scaling options has been activated by the [LOWSALT](#) and the [ENDSCALE](#) keywords in the [RUNSPEC](#) section. The data is used to scale the oil relative permeability in the low salinity oil wet oil relative permeability saturation tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.155
LKRORG – END-POINT SCALING OF GRID CELL KRO(SGCR) (LOW SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LKRORG](#) defines the scaling parameter for the relative permeability of oil at the critical gas saturation ([SGCR](#)), for all the cells in the model via an array, for when the Low Salt option and the End-point Scaling options has been activated by the [LOWSALT](#) and the [ENDSCALE](#) keywords in the [RUNSPEC](#) section. The data is used to scale the oil relative permeability in the low salinity oil wet oil relative permeability saturation tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.156
LKRORW – END-POINT SCALING OF GRID CELL KRO(SWCR) (Low SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LKRORW](#) defines the scaling parameter for the relative permeability of oil at the critical water saturation ([SWCR](#)), for all the cells in the model via an array, for when the Low Salt option and the End-point Scaling options has been activated by the [LOWSALT](#) and the [ENDSCALE](#) keywords in the [RUNSPEC](#) section. The data is used to scale the oil relative permeability in the low salinity oil wet oil relative permeability saturation tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.157 LKRW – END-POINT SCALING OF GRID CELL KRW(Sw =1.0) (Low SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

LKRW defines the scaling parameter at the maximum oil relative permeability value (*SWU*), that is for $S_w = 1.0$, for all the cells in the model via an array, for when the Low Salt option and the End-point Scaling options has been activated by the *LOWSALT* and the *ENDSCALE* keywords in the *RUNSPEC* section. The data is used to scale the water relative permeability in the low salinity oil wet water relative permeability saturation tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.158
LKRWR – END-POINT SCALING OF GRID CELL KRWR(SOWCR) (Low SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LKRWR](#) defines the scaling parameter at the maximum oil relative permeability value ([SWU](#)), that is for $S_w = 1.0$, for all the cells in the model via an array, for when the Low Salt option and the End-point Scaling options has been activated by the [LOWSALT](#) and the [ENDSCALE](#) keywords in the [RUNSPEC](#) section. The data is used to scale the water relative permeability in the low salinity oil wet water relative permeability saturation tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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8.3.159
LPCW – END-POINT SCALING OF GRID CELL WATER CAPILLARY PRESSURE (LOW SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

LPCW defines the maximum oil-water pressure values for all the cells in the model via an array, for when the Low Salt option and the End-point Scaling options has been activated by the *LOWSALT* and the *ENDSCALE* keywords in the *RUNSPEC* section. The keyword re-scales the oil-water capillary pressure in the low salinity oil wet capillary saturation tables from a cell's assigned saturation function by the grid block's *LPCW* value.

The capillary pressure for a grid block is scaled by:

$$P_c = P_{c_{TABLE}} \left(\frac{LPCW}{P_{c_{TABLE-MAX}}} \right)$$

(8.66)

Where:

- P_c

=

the resulting low salt oil wet water capillary pressure for a grid cell.
- $LPCW$

=

the maximum capillary pressure from the *LPCW* array for a given cell.
- $P_{c_{TABLE}}$

=

the capillary pressure in the low salt oil wet capillary pressure table allocated to the grid block.
- $P_{c_{TABLE-MAX}}$

=

the maximum capillary pressure in the low salt oil capillary pressure table allocated to the grid block (that is at the connate water saturation).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.160
LSALTFNC – DEFINE LOW SALT WEIGHTING FACTORS VERSUS SALT CONCENTRATION FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LSALTFNC](#) keyword defines the low salt weighting factors versus salt concentration functions for when the Low Salt option has been activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section. The tables are used to modify the oil and water relative permeability saturation end-points, as well as the water-oil capillary pressure end-points, for different salt concentrations within a grid cell.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.161
LSOGCR – END-POINT SCALING GRID CELL SOGCR (Low SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LSOGCR](#) defines the critical oil saturation with respect to gas (“SOGCR”), for all the cells in the model via an array, for when the Low Salt option has been selected. The data is used to scale the oil saturation in the low salinity oil wet oil-gas relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.162
LSOWCR – END-POINT SCALING GRID CELL SOWCR (LOW SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LSOWCR](#) defines the critical oil saturation with respect to water (“SOWCR”), for all the cells in the model via an array, for when the Low Salt option has been selected. The data is used to scale the oil saturation in the low salinity oil wet water-oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.163 LSWCR – END-POINT SCALING GRID CELL SWCR (Low SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LSWCR](#) defines the critical water saturation (“SWCR”), for all the cells in the model via an array, for when the Low Salt option has been selected. The data is used to scale the water saturation in the low salinity oil wet water-oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.164
LSWL – END-POINT SCALING GRID CELL SWL (LOW SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LSWL](#) defines the connate water saturation (“SWL”), for all the cells in the model via an array, for when the Low Salt option has been selected. The data is used to scale the water saturation in the low salinity oil wet water-oil relative permeability saturation tables, as well as the associated capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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8.3.165 LSWLPC – END-POINT SCALING GRID CELL SWLPC (Low SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LSWLPC](#) defines the capillary pressure connate water saturation (“SWLPC”), for all the cells in the model via an array, for when the Low Salt option has been selected. The data is used to scale the water saturation in the low salinity oil wet water-oil capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

Note the keyword only applies the scaling to the capillary pressures tables, unlike the [LSWL](#) keyword that applies the scaling to both the capillary pressure and relative permeability tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.166 LSWU – END-POINT SCALING GRID CELL SWU (LOW SALINITY AND OIL WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LSWU](#) defines the maximum water saturation(“SWU”), for all the cells in the model via an array, for when the Low Salt option has been selected. The data is used to scale the water saturation in the low salinity oil wet water-oil relative permeability saturation tables, as well as the associated capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.167 LWKRO – END-POINT SCALING OF GRID CELL Kro(SWL) (Low SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWKRO](#) defines the scaling parameter for the oil relative permeability value at the connate water saturation ([SWL](#)), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil relative permeability in the low salinity water wet oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.168 LWKRORG – END-POINT SCALING OF GRID CELL KRO(SGCR) (LOW SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWKRORG](#) defines the scaling parameter for the relative permeability of oil at the critical gas saturation ([SGCR](#)), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil relative permeability in the low salinity water wet oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.169 LWKRORW – END-POINT SCALING OF GRID CELL KRO(SWCR) (LOW SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWKRORW](#) defines the scaling parameter for the relative permeability of oil at the critical water saturation ([SWCR](#)), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil relative permeability in the low salinity water wet oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.170
LWKRW – END-POINT SCALING OF GRID CELL KRW(Sw =1.0) (Low SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWKRW](#) defines the scaling parameter at the maximum water relative permeability value ([SWU](#)), that is for $S_w = 1.0$, for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water relative permeability in the low salinity water wet water relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.171
LWKRWR
– END-POINT SCALING OF GRID CELL KRWR(Sw=1.0)
(Low SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWKRWR](#) defines the scaling parameter at the critical oil to water saturation value ([SOWCR](#)), for the water relative permeability curve, for all the cells in the model via an array, and for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water relative permeability in the low salinity water wet water relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.172 LWPCW – END-POINT SCALING OF GRID CELL WATER CAPILLARY PRESSURE (LOW SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

LWPCW defines the maximum water-oil pressure values for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The *ENDSCALE* keyword in the *RUNSPEC* section should be activated to enable end-point scaling and the use of this keyword. In addition the Low Salt option should be activated by the *LOWSALT* keyword in the *RUNSPEC* section and the Surfactant Wettability option activated by the *SURFACT* or *SURFACTW* keywords, which are also in the *RUNSPEC* section. The keyword re-scales the oil-water capillary pressure in the low salinity water wet capillary saturation tables from the cell's assigned saturation function by the grid block's *LWPCW* value.

The capillary pressure for a grid block is scaled by:

$$P_c = P_{c_{TABLE}} \left(\frac{HWPCW}{P_{c_{TABLE-MAX}}} \right)$$

(8.67)

- Where:
- P_c

=

the resulting high salinity water wet water capillary pressure for a grid cell.
- $HWPCW$

=

the maximum capillary pressure from the *HWPCW* array for a given cell.
- $P_{c_{TABLE}}$

=

the capillary pressure in the high salinity water wet water capillary pressure table allocated to the grid block.
- $P_{c_{TABLE-MAX}}$

=

the maximum capillary pressure in the high salinity water wet water capillary pressure table allocated to the grid block (that is at the connate water saturation).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.173
LWSOGR – END-POINT SCALING GRID CELL SOGR (Low SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWSOGR](#) defines the critical oil saturation with respect to gas (“SOGR”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil saturation in the low salinity water wet oil-gas relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.174 LWSOWCR – END-POINT SCALING GRID CELL SOWCR (Low SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWSOWCR](#) defines the critical oil saturation with respect to water (“SOWCR”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the oil saturation in the low salinity water wet water-oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.175
LWSWCR – END-POINT SCALING GRID CELL SWCR (Low SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWSWCR](#) defines the critical water saturation (“SWCR”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the low salinity water wet water-oil relative permeability saturation tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.176 LWSWL – END-POINT SCALING GRID CELL SWL (LOW SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWSWL](#) defines the connate water saturation (“SWL”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the low salinity water wet water-oil relative permeability saturation tables, as well as the associated capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.177 LWSWLPC – END-POINT SCALING GRID CELL SWLPC (LOW SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWSWLPC](#) defines the capillary pressure connate water saturation (“SWLPC”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the low salinity water wet water-oil capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

Note the keyword only applies the scaling to the capillary pressures tables, unlike the [LWSWL](#) keyword that applies the scaling to both the capillary pressure and relative permeability tables.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.178
LWSWU – END-POINT SCALING GRID CELL SWU (LOW SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[LWSWU](#) defines the maximum water saturation(“SWU”), for all the cells in the model via an array, for when the Low Salt and Surfactant Wettability options have been selected. The data is used to scale the water saturation in the low salinity water wet water-oil relative permeability saturation tables, as well as the associated capillary pressure tables. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Low Salt option should be activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section and the Surfactant Wettability option activated by the [SURFACT](#) or [SURFACTW](#) keywords, which are also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.179 MASSFLOW – DEFINE RIVER MASS FLOW VERSUS TIME TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [MASSFLOW](#) keyword defines the upstream river mass flow versus time tables for rivers, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.180 MAXVALUE – SETS A MAXIMUM VALUE FOR AN ARRAY ELEMENT

The [MAXVALUE](#) keyword sets a maximum value for the specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [MAXVALUE](#) keyword is being used.

See [MAXVALUE – Sets a Maximum Value for an Array Element](#) in the [GRID](#) section for a full description.

8.3.181 MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the GLOBAL section for a full description.

8.3.182 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the GLOBAL section for a full description.

8.3.183 MICPPARA – DEFINE MICROBially INDUCED CALCITE PRECIPITATION PARAMETERS (RETIRED)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [MICPPARA](#) keyword used to define the model parameters for when the [MICP](#) model had been activated via the [MICP](#) keyword in the [RUNSPEC](#) section.

The parameters in this keyword are now defined by the [BIOFPARA](#) and [PERMFACT](#) keywords.

8.3.184 MINVALUE – SET A MINIMUM VALUE FOR AN ARRAY ELEMENT

The [MINVALUE](#) keyword sets a minimum value for the specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [MINVALUE](#) keyword is being used.

See [MINVALUE – Set a Minimum Value for an Array Element](#) in the [GRID](#) section for a full description.

8.3.185 MISC – DEFINE SOLVENT MISCIBILITY-IMMISCIBILITY TRANSFORM FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

MISC defines the transformation between the miscible and immiscible relative permeability models, for when the *MISCIBLE* and *SOLVENT* keywords in the *RUNSPEC* section have been activated. The keyword can only be used with the *MISCIBLE* option and for when the oil, water, gas and solvent phases are active in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SSOL	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the solvent fraction with respect to the solvent and gas saturation, and is defined by: $\frac{S_s}{(S_g + S_s)}$ Where Sg is the gas saturation and Ss is the solvent saturation. Note that the first entry in the columnar vector should be zero and the last entry should be one to fully define the solvent fraction range.			None
		dimensionless	dimensionless	dimensionless	
2	MISC	A columnar vector of real equal or increasing down the column values that are greater than or equal to zero and less than one, that define the corresponding miscibility for the corresponding solvent fraction SSOL. The first entry in the columnar vector should be zero and the last entry should be one to fully define the miscible-immiscible relationship.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTMISC tables as declared on the MISCIBLE keyword in the RUNSPEC section.
- 2) Each table must contain a minimum of two rows and a maximum of NSMISC rows as declared on the MISCIBLE keyword in the RUNSPEC section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.98: MISC Keyword Description

Example

```
--
-- SOLVENT MISCIBILITY-IMMISCIBILITY TRANSFORM TABLE
--
MISC
-- SSOL      MISC
-- FRAC      FRAC
-- -----
--    0.0000    0.0000
--    0.2000    0.2500
--    0.5000    0.7500
--    1.0000    1.0000
--                                     / TABLE NO. 01
--
-- SSOL      MISC
-- FRAC      FRAC
-- -----
--    0.0000    0.0000
--    0.3000    0.2500
--    0.6000    1.0000
--    1.0000    1.0000
--                                     / TABLE NO. 02
```

The above example defines two solvent miscible-immiscible transform tables assuming NTMISC equals two and NSMISC is greater than or equal to four on the *MISCIBLE* keyword in the *RUNSPEC* section.

8.3.186 MLANG – DEFINE LANGMUIR MAXIMUM GAS CONCENTRATION FOR ALL GRID CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [MLANG](#), defines the coal bed methane Langmuir Adsorption²¹³ maximum gas concentration for each grid cell used to scale the Langmuir isotherm table allocated to the cell, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section. See the [LANGMUIR](#) keyword in the [PROPS](#) section for specifying the Langmuir tables for the model.

Note that if the Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section, then [MLANG](#) applies to only the matrix grid block.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²¹³ Langmuir, Irving (June 1918). "The Adsorption of Gases on Plane Surface of Glass, Mica and Platinum". *The Research Laboratory of the General Electric Company*. 40 (9): 1361–1402. doi:10.1021/ja02242a004

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.187 MLANGSLV – DEFINE LANGMUIR MAXIMUM SOLVENT CONCENTRATION FOR ALL GRID CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [MLANGSLV](#), defines the coal bed methane Langmuir Adsorption²¹⁴ maximum solvent concentration for each grid cell used to scale the Langmuir isotherm solvent table allocated to the cell, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword in the [RUNSPEC](#) section. In addition, the Solvent phase must have been declared by the [SOLVENT](#) keyword in the [RUNSPEC](#) section. See the [COALNUM](#) keyword in the [GRID](#) section for allocating the Langmuir solvent tables to the grid blocks, and also the [LANGMUIR](#) keyword in the [PROPS](#) section for defining the Langmuir Adsorption Isotherm tables. Keywords [COALADS](#) and [COALPP](#), also in the [PROPS](#) section, are used to specify the relative adsorption data in runs containing the solvent phase.

Note that if he Dual Porosity model has been activated by either the [DUALPORO](#) or the [DUALPERM](#) keywords in the [RUNSPEC](#) section, then [MLANGSLV](#) applies to only the matrix grid block.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²¹⁴ Langmuir, Irving (June 1918). "The Adsorption of Gases on Plane Surface of Glass, Mica and Platinum". *The Research Laboratory of the General Electric Company*. 40 (9): 1361–1402. doi:10.1021/ja02242a004

8.3.188 MSFN – MISCIBLE NORMALIZED RELATIVE PERMEABILITY TABLES

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Description

The *MSFN* keyword defines the miscible normalized relative permeability tables for when the *MISCIBLE* and or *SOLVENT* options have been activated in the *RUNSPEC* section using the respective keyword. The *MISCIBLE* keyword invokes a three component formulation (oil, water and solvent gas or an oil, water and solvent oil). Whereas the *SOLVENT* keyword results in a four component model (oil, water and gas plus a solvent). This keyword should only be used if the *MISCIBLE* and or *SOLVENT* options have been activated.

No.	Name	Description	Default
1	SGAS	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the gas plus solvent saturation.	None
2	KRSG	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas plus solvent relative permeability multiplier.	None
3	KRO	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability multiplier.	None

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by "/" and there is no "/" terminator for the keyword.

Table 8.99: MSFN Keyword Description

Example

MISCIBLE NORMALIZED RELATIVE PERMEABILITY TABLES			
MSFN	SGAS	KRSG	KRO
	FRAC		
	0.0000	0.0000	1.0000
	1.0000	1.0000	0.0000
			/ TABLE NO. 01
	SGAS	KRSG	KRO
	FRAC		
	0.0000	0.0000	1.0000
	0.2000	0.2000	0.8000
	0.4000	0.3000	0.7000
	0.6000	0.4000	0.6000
	0.8000	0.5000	0.4000
	1.0000	1.0000	0.0000
			/ TABLE NO. 02

The above example defines two MSN tables for use the *MISCIBLE* and *SOLVENT* options.

8.3.189 MULTIPLY – MULTIPLY A SPECIFIED ARRAY BY A CONSTANT

The [MULTIPLY](#) keyword multiplies a specified array or part of an array by a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [MULTIPLY – Multiply a Specified Array by a Constant](#) in the [GRID](#) section for a full description.

8.3.190 MULTIREG – MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER

The [MULTIREG](#) keyword multiplies an array or part of an array by a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [MULTIREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [MULTIREG](#) keyword is being used.

See [MULTIREG – Multiply an Array by a Constant based on a Region Number](#) in the [GRID](#) section for a full description.

8.3.191 MW – MOLECULAR WEIGHTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [MW](#) keyword defines the molecular weights for each of the compositional components active in the model and for each equation of state. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	MW	A series of real numbers that define the molecular weights for each of the compositional components active in the model.			None
		lb/lb-M	kg/kg-M	gm/gm-M	
Notes: 1) The keyword is followed by NMEOSR records as specified by the TABDIMS keyword in the RUNSPEC section. 2) Each record must contain COMPS values as specified by the COMPS keyword in the RUNSPEC section. 3) Each record is terminated by a “/” and there is no “/” terminator for the keyword.					

Table 8.100: [MW](#) Keyword Description

Examples

The following example defines the molecular weights for each component in a single three-component equation of state model.

```
--
-- Molecular Weights
--
MW
  16.0425  44.009  18.01528 /
```

The following example defines the molecular weights for each component in two three-component equation of state models.

```
--
-- Molecular Weights
--
MW
  16.0425  44.009  18.01528 /
  16.0425  44.009  18.01528 /
```

8.3.192 NCOMPS – CONFIRM NUMBER OF COMPOSITIONAL COMPONENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword confirms the maximum number of compositional components in the model, as such it should have the same number of components as that declared via the [COMPS](#) keyword in the [RUNSPEC](#) section. The keyword should only be used if the [CO2STORE](#) keyword and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section, have also be activated for the gas-water two component model.

Note

This is an OPM Flow keyword used with OPM Flow's [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section, and should not be confused with the more general version of the [NCOMPS](#) keyword used in the commercial compositional simulator.

Secondly, although OPM Flow parses the keyword, the simulator currently ignores the data for this keyword.

No.	Name	Description	Default
1	NCOMPS	A positive integer defining the maximum number of compositional components in the model. Secondly the number of components must be the same as that enter via the COMPS keyword in the RUNSPEC section. Only the default value of two is currently supported by OPM Flow.	2
Notes: 1) The keyword is terminated by a "/".			

Table 8.101: NCOMPS Keyword Description

Example

The following example defines how to confirm a two component formulation to be used with the [CO2STORE](#) and [GASWAT](#) options.

```
--
--      CONFIRM NUMBER OF COMPOSITIONAL COMPONENTS (OPM FLOW KEYWORD)
--
NCOMPS      2      /
```


8.3.193 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active. but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

8.3.194 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

8.3.195 NOWARNEP – DEACTIVATE END-POINT SCALING WARNING MESSAGES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *NOWARNEP* keyword deactivates the writing out of warning messages associated with checking the consistency of saturation table end-points; however error messages are still reported by the simulator.

Hence, OPM Flow ignores this keyword but it is documented here for completeness.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
--
--      DEACTIVATE END-POINT SCALING WARNING MESSAGES
--
NOWARNEP
```

The above example switches off the writing out of warning messages associated with checking the consistency of saturation table end-points;

8.3.196 OILDENT – DEFINE OIL DENSITY TEMPERATURE COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

OILDENT defines the oil density as a function of temperature coefficients for when OPM Flow’s thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC**. Note this is an OPM Flow keyword used with OPM Flow’s black-oil thermal model that is not available in the commercial simulator’s black-oil thermal formulation.

This keyword can only be used if OPM Flow’s thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	TEMP is a real positive value greater than zero that defines the absolute reference temperature used with TEXP1 and TEXP2 to estimate the change in oil density with respect to temperature.			Defined
		°R 527.67	K 293.15	K 293.15	
2	TEXP1	TEXP1 is a real positive value greater than zero that defines the oil thermal expansion coefficient of the first order.			Defined
		1/°R 1.67 × 10 ⁻⁴	1/K 3.0 × 10 ⁻⁴	1/K 3.0 × 10 ⁻⁴	
3	TEXP2	TEXP2 is a real positive value greater than zero that defines the oil thermal expansion coefficient of the second order.			Defined
		1/°R ² 9.26 × 10 ⁻⁷	1/K ² 3.0 × 10 ⁻⁶	1/K ² 3.0 × 10 ⁻⁶	

Notes:

1)

The keyword is followed by NTPVT records as declared on the **TABDIMS** keyword in the **RUNSPEC** section.

2)

Each data set is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.102: OILDENT Keyword Description

The oil density at a given pressure and temperature is calculated from its value at surface conditions and the oil shrinkage factor (the reciprocal of the oil formation volume factor) as shown in the following equation:

$$\rho_o(p, T) = \rho_o(p_s, T_s) b_o(p, T) \quad (8.68)$$

Where the temperature dependence of the oil shrinkage factor relative to its value at the reference temperature is calculated as shown in the following equation:

$$b_o(p,T)=\frac{b_o(p,T_{ref})}{1+c_1(T-T_{ref})+c_2(T-T_{ref})^2}$$

(8.69)

Where:

- ρ_o

= oil density
- b_o

= oil shrinkage factor
- p

= pressure
- T

= temperature
- c_1, c_2

= thermal expansion coefficients to first and second order
- s

= subscript indicating surface conditions
- ref

= subscript indicating reference conditions

Example

The following example shows the [OILDENT](#) keyword using the default values, for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to two.

```
--
--      OIL DENSITY TEMPERATURE COEFFICIENTS (OPM FLOW THERMAL KEYWORD)
--
--      OIL      DENSITY  DENSITY
--      TEMP      COEFF1   COEFF2
--      -----
OILDENT
      1*          1*       1*          / TABLE NO. 01
      1*          1*       1*          / TABLE NO. 02
```

There is no terminating “/” for this keyword.

8.3.197 OILJT – DEFINE OIL JOULE-THOMSON COEFFICIENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[OILJT](#) activates the oil Joule-Thomson effect²¹⁵ in temperature calculations, and defines the oil Joule-Thomson Coefficient (“JTC”) at a given reference pressure, for when OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#).

Note

This is an OPM Flow keyword used with OPM Flow’s black-oil thermal model, that is not available in the commercial simulator’s black-oil thermal formulation.

This keyword can only be used if OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model, and does not include the Joule-Thomson effect in temperature calculations.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A real positive value that defines the reference pressure for the corresponding Joule-Thomson Coefficient, OILJTC.			None
		psia	barsa	atma	
2	OILJTC	OILJTC is a real positive or negative value that defines the oil phase Joule-Thomson Coefficient. If the value is defaulted (1*) or set to 0, then OILJTC is internally calculated using the thermal oil density data on the <i>OILDENT</i> keyword in the <i>PROPS</i> section. If a non-zero value is specified, then the OILJTC is assumed to be constant and equal to that value.			0
		oF/psia	°C/barsa	°C/atma	

Notes:

1) The keyword is followed by NTPVT records as declared on the *TABDIMS* keyword in the *RUNSPEC* section.

2) Each data set is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.103: OILJT Keyword Description

The Joule–Thomson effect is when a real gas, as oppose to an ideal gas, expands, resulting in the temperature of the gas dropping²¹⁶. For liquids the effect is the opposite, that is the internal energy is transferred to kinetic energy with a corresponding increase in temperature as velocity increases.

Thermodynamically, the Joule–Thomson coefficient is defined as the isenthalpic²¹⁷ change in temperature in a fluid caused by a unitary pressure drop, as shown in the following equation:

²¹⁵ The Joule–Thomson coefficient is defined as the change in temperature with respect to an increase in pressure at constant enthalpy.

²¹⁶ Natural Gas Engineering (McGraw-Hill chemical engineering series), Donald L Katz, Robert I Lee, McGraw-Hill Education, 1990 (ISBN 0071007776, 9780071007771).

$$\eta = \left(\frac{\partial T}{\partial P} \right) \quad (8.70)$$

Which can also express as²¹⁸:

$$\eta = (T\alpha - 1) \frac{1}{(\rho C_p)} - \left(\frac{g}{C_p} \frac{dp}{dz} \right)^{-1} \quad (8.71)$$

Setting the gravity term, g , to zero we have:

$$\eta = (T\alpha - 1) \frac{1}{(\rho C_b)} \quad (8.72)$$

Where:

η	= Joule–Thomson coefficient (oC/Pa),
α	= thermal expansivity at constant pressure (1/oC),
C_p	= specific heat at constant pressure (J/kg oC),
g	= gravitational acceleration (m/s ²)
P	= pressure (Pa),
T	= temperature (oC), and
z	= height (m).

Example

The following example shows the [OILJT](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section, and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set equal to two.

```
--
--      OIL JOULE-THOMSON COEFFICIENT (OPM FLOW EXTENSION KEYWORD)
--
--      REF      OIL
--      PRESS    JTC
--      -----
OILJT
      20.0      1*                / TABLE NO. 01
      20.0      -0.20            / TABLE NO. 02
```

Here the first entry is defaulted, and the simulator will therefore calculate the oil JTC internally using the data on the [OILDENT](#) keyword in the [PROPS](#) section.

There is no terminating “/” for this keyword.

²¹⁷ An isenthalpic process or isoenthalpic process, is a process that proceeds without any change in enthalpy, H ; or specific enthalpy, h .

²¹⁸ Pippard, A.B.: Elements of Classical Thermodynamics: For Advanced Students of Physics. Cambridge University Press, Cambridge, UK (1957)

8.3.198 OILVISCT – DEFINE OIL VISCOSITY VERSUS TEMPERATURE FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[OILVISCT](#) defines the oil viscosity as a function of temperature for when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. The reference pressure and solution gas-oil ratio of the oil for this table is given by the [VISCREF](#) keyword in the [PROPS](#) section. Note this is an OPM Flow keyword used with OPM Flow's black-oil thermal model that is not available in the commercial simulator's black-oil thermal formulation.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	A columnar vector of real monotonically increasing down the column values that defines the temperature values.			None
		°F	°C	°C	
2	VIS	A columnar vector of real increasing down the column values that defines the oil viscosity for the corresponding temperature values (TEMP). VIS should be given at the reference pressure and solution gas-oil ratio as defined by PRESS and RS variables on the VISCREF keyword.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.104: OILVISCT Keyword Description

There is no terminating "/" for this keyword.

Example

The following example shows the [OILVISCT](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set equal to one.

```
--
--      OIL VISCOSITY VERSUS TEMPERATURE TABLES (OPM FLOW EXTENSION KEYWORD)
--
--      OIL      OIL
--      TEMP     VISC
--      -----
OILVISCT
      100.0      0.600
      110.0      0.650
      120.0      0.680
      150.0      0.720
      165.0      0.725
                                     /  TABLE NO. 01
```


8.3.199 OPERATE – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS

This keyword, [OPERATE](#), defines mathematical operations on property arrays ([NTG](#), [PORO](#) etc.) and optionally using another property array as input to the function. The keyword allows for various mathematical functions and their associated variables to be defined and applied to the selected array data. Input constants can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [OPERATE – Define Mathematical Operations on Arrays](#) in the [GRID](#) section for a full description.

8.3.200 OPERATER – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS BY REGION

This keyword defines the mathematical operations on arrays for specific regions in the commercial simulator and is currently not supported by OPM Flow. However, similar functionality is provided by the [ADD](#) and [MULTIPLY](#) keywords.

See [OPERATER – Define Mathematical Operations on Arrays by Region](#) in the [GRID](#) section for a full description.

8.3.201 OVERBURD – DEFINE ROCK OVERBURDEN PRESSURE VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **OVERBURD** keyword defines the overburden pressures versus depth relationship to be applied for when the rock compaction option has been invoked by the **ROCKCOMP** keyword in the **RUNSPEC** section.

The rock compaction pore volume and transmissibility multipliers, entered via the **ROCKTAB**, **ROCK2D** and **ROCK2DTR** keywords, are applied to the pore pressure, unless the **OVERBURD** keyword is included in the input deck. When the **OVERBURD** keyword is present the multipliers are applied to the effective pore volume pressure, that is $P_{(effective)} = P_{(Pressure)} - P_{(overburden)}$. If the keyword is not present in the input deck then the overburden pressure is set to zero.

This keyword should only be used if compaction option has been enabled.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth for corresponding overburden pressure parameter PRESS.			None
		feet	m	cm	
2	PRESS	A columnar vector of real monotonically increasing down the column values that defines the corresponding overburden pressure for the given DEPTH.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTROCC tables as declared on the **ROCKCOMP** keyword in the **RUNSPEC** section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.105: OVERBURD Keyword Description

See also the **ROCKTAB**, **ROCK2D**, **ROCK2DTR**, and **ROCKWNOD** keywords in the **PROPS** section.

Examples

The example below defines three overburden tables, assuming NTROCC is equal to three on the [ROCKCOMP](#) keyword and NPPVT is greater than or equal to four on the [TABDIMS](#) keyword.

```
--
--      OVERBURDEN PRESSURE VERSUS DEPTH TABLES
--
OVERBURD
--      DEPTH      OVERBURDEN
--      FEET       PRESSURE
--      -----
--      1000.0     300.000
--      2000.0     600.000
--      3000.0     900.000
--      4000.0     1200.000
--                                     / TABLE NO. 01
--      DEPTH      OVERBURDEN
--      FEET       PRESSURE
--      -----
--      1000.0     200.000
--      2000.0     400.000
--      3000.0     800.000
--      4000.0     1000.000
--                                     / TABLE NO. 02
--      DEPTH      OVERBURDEN
--      FEET       PRESSURE
--      -----
--      1000.0     400.000
--      2000.0     800.000
--      3000.0     1100.000
--      4000.0     1500.000
--                                     / TABLE NO. 03
```

Note that there must be exactly NTROCC tables entered for this keyword, otherwise an error will occur.

```
--
--      ROCK COMPACTION TABLES
--
ROCKTAB
--      PRESS      PORV      TX(YZ)
--      MULT       MULT
--      -----
--      1000.0     0.9600    0.9650
--      1500.0     0.9800    0.9850
--      3000.0     0.9900    0.9950
--      4500.0     1.0000    1.0000
--      4750.0     1.0100    1.0100
--                                     / TABLE NO. 01
--      PRESS      PORV      TX(YZ)
--      MULT       MULT
--      -----
--      1000.0     0.9600    0.9650
--      1500.0     0.9800    0.9850
--      3000.0     0.9900    0.9950
--      4500.0     1.0000    1.0000
--      4750.0     1.0100    1.0100
--                                     / TABLE NO. 02
--      PRESS      PORV      TX(YZ)
--      MULT       MULT
--      -----
--      1000.0     0.9600    0.9650
--      2000.0     0.9800    0.9850
--      3000.0     0.9900    0.9950
--      4000.0     1.0100    1.0100
--                                     / TABLE NO. 03
```

Here [ROCKTAB](#) tables one and two are identical.

8.3.202 PCFACT – CAPILLARY PRESSURE MULTIPLICATION FACTOR AS A FUNCTION OF POROSITY CHANGE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	------------------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PCFACT defines the capillary pressure multiplication factor due to a change in porosity. Currently the keyword is used in conjunction with OPM Flow's Salt Precipitation model, in which the pore space is reduced due to salt precipitating in the pore space, causing a reduction in porosity and an associated increase in capillary pressure.

Note

This is an OPM Flow specific keyword for the simulator's Salt Precipitation model that is activated by the **BRINE** and **PRECSALT** keywords and declaring that vaporized water is present in the run via the **VAPWAT** keyword. All three keywords are in the **RUNSPEC** section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	POROFAC	A real monotonically increasing positive columnar vector that defines the porosity factor ($\frac{\phi}{\phi_0}$) for the corresponding PCFAC vector. In the simulator's Salt Precipitation model, the maximum value of ϕ is ϕ_0 , implying a maximum value of one for POROFAC.			None
		dimensionless	dimensionless	dimensionless	
2	PCFAC	A real positive monotonically decreasing columnar vector that defines the capillary pressure (p_c) multiplier associated with POROFAC and used to scale a grid block's capillary pressure due to the reduction in pore volume caused by salt precipitation. Where: $PCFAC = m \left(\frac{\phi}{\phi_0} \right)$ $\text{with } p_c = m \left(\frac{\phi}{\phi_0} \right) p_{c0}$			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 3) There must be same number of entries for each column.
- 4) Each table is terminated by a single "/" and there is no "/" terminator for the keyword.

Table 8.106: PCFACT Keyword Description

The porosity reduction is a function of the volume fraction of salt (S_s) precipitated out of the vaporized water phase, that is:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

$$\phi = (1 - s_s) \phi_0 \quad (8.73)$$

The capillary pressure factor data can be estimated from the permeability and porosity factors defined by the [PERMFACT](#) keyword in the [PROPS](#) section using the Leverett J-function, for example:

$$\frac{p_c}{p_{c0}} = \left(\frac{\phi}{\phi_0} \frac{k_0}{k} \right)^{\frac{1}{2}} \quad (8.74)$$

Where:

- ϕ_0, ϕ = the initial and scaled porosity,
- k_0, k = the initial and scaled permeability, and
- p_{c0}, p_c = the initial and scaled capillary pressure.

Example

The example below defines two [PCFACT](#) tables assuming NTSFUN equals two and NSSFUN is greater than or equal to five on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--          CAPILLARY PRESSURE FACTOR INCREASE DUE TO SALT PRECIPITATION
--          (OPM FLOW KEYWORD)
--
PCFACT
--          PORO          PC
--          FACTOR        FACTOR
--          -----
--          0.00          2.0000
--          0.25          2.0000
--          0.50          1.4142
--          0.75          1.1547
--          1.00          1.0000
--                                     / TABLE NO. 01
--
--          0.00          2.0000
--          0.25          2.0000
--          0.50          1.4142
--          0.75          1.1547
--          1.00          1.0000
--                                     / TABLE NO. 02
```

Note that the capillary pressure factor has been capped for POROFAC less than 0.25.

8.3.203 PCG – END-POINT SCALING OF GRID CELL MAXIMUM GAS CAPILLARY PRESSURE (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PCG defines the maximum drainage gas-oil capillary pressure values for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. The keyword can be used with all grid types.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped. See also the **IPCG** keyword for the equivalent imbibition functionality.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PCG	PCG is an array of positive real numbers assigning the maximum drainage gas-oil capillary pressure values for each cell in the model. Repeat counts may be used, for example 30*100.0.			None
		psia	bars	atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) If the **HYSTER** on the **SATOPTS** keyword in the **RUNSPEC** section has been activated to invoke hysteresis then **PCG** scales the drainage curve and **IPCG** scales the imbibition curve.
- 3) The keyword is terminated by a "/".

Table 8.107: PCG Keyword Description

The capillary pressure for a grid block is scaled by:

$$P_c = P_{C_{TABLE}} \left(\frac{PCG}{P_{C_{TABLE-MAX}}} \right) \quad (8.75)$$

Where:

- P_c = the resulting drainage gas-oil capillary pressure for a grid cell.
- PCG = the maximum capillary pressure from the **PCG** array for a given cell.
- $P_{C_{TABLE}}$ = the capillary pressure in the drainage capillary pressure table allocated to the grid block.
- $P_{C_{TABLE-MAX}}$ = the maximum capillary pressure in the drainage capillary pressure table allocated to the grid block at $S_g = 1 - S_{wco}$.

Example

```
--
--      DEFINE GRID BLOCK PCG DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PCG
      100*50.0  100*75.0  100*125.0 /
```

The above example defines the **PCW** for 300 cells in the model as defined by the **DIMENS** keyword in the **RUNSPEC** section.

8.3.204
PCG32D – GAS-OIL CAPILLARY PRESSURE VERSUS OIL AND WATER SATURATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [PCG32D](#), enables the gas-oil capillary pressure data to be entered as a function of both oil and water saturations. The keyword should be used in conjunction with the [SGF32D](#) keyword in the [PROPS](#) section. See also the [PCW32D](#) keyword in the [PROPS](#) section that provides similar functionality for the water-oil capillary pressure data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.205 PCRIT – CRITICAL PRESSURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PCRIT](#) keyword defines the critical pressures for each of the compositional components active in the model and for each equation of state. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PCRIT	A series of real numbers that define the critical pressures for each of the compositional components active in the model.			None
		psia	barsa	atma	

Notes:

1)

The keyword is followed by NMEOSR records as specified by the *TABDIMS* keyword in the *RUNSPEC* section.

2)

Each record must contain *COMPS* values as specified by the *COMPS* keyword in the *RUNSPEC* section.

3)

Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.108: [PCRIT](#) Keyword Description

Examples

The following example defines the critical pressures for each component in a single three-component equation of state model.

```
--
-- Critical Pressures
--
PCRIT
  1070.132 667.0576 616.5844 /
```

The following example defines the critical pressures for each component in two three-component equation of state models.

```
--
-- Critical Pressures
--
PCRIT
  1070.132 667.0576 616.5844 /
  1070.132 667.0576 616.5844 /
```


8.3.206 PCW – END-POINT SCALING OF GRID CELL WATER CAPILLARY PRESSURE (DRAINAGE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[PCW](#) defines the maximum drainage water-oil or water-gas capillary pressure values for all the cells in the model via an array. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. The keyword can be used with all grid types.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PCW	PCW is an array of positive real numbers assigning the maximum drainage water capillary pressure values for each cell in the model. Repeat counts may be used, for example 30*100.0.			None
		psia	bars	atm	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If the HYSTER on the [SATOPTS](#) keyword in the [RUNSPEC](#) section has been activated to invoke hysteresis then [PCW](#) scales the drainage curve and [IPCW](#) scales the imbibition curve.
- 3) The keyword is terminated by a "/".

Table 8.109: PCW Keyword Description

See also the [IPCW](#) keyword for the equivalent imbibition functionality.

The capillary pressure for a grid block is scaled by:

$$P_c = P_{c_{TABLE}} \left(\frac{PCW}{P_{c_{TABLE-MAX}}} \right) \quad (8.76)$$

Where:

- P_c = the resulting drainage water capillary pressure for a grid cell.
- PCW = the maximum capillary pressure from the [PCW](#) array for a given cell.
- $P_{c_{TABLE}}$ = the capillary pressure in the drainage capillary pressure table allocated to the grid block.
- $P_{c_{TABLE-MAX}}$ = the maximum capillary pressure in the drainage capillary pressure table allocated to the grid block (that is at the connate water saturation).

Example

```
--
--      DEFINE GRID BLOCK PCW DATA FOR ALL CELLS (BASED ON NX x NY x NZ = 300)
--
PCW      100*50.0  100*75.0  100*125.0      /
```

The above example defines the [PCW](#) for 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.207 PCW32D – WATER-OIL CAPILLARY PRESSURE VERSUS OIL AND GAS SATURATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [PCW32D](#), enables the water-oil capillary pressure data to be entered as a function of both oil and gas saturations. The keyword should be used in conjunction with the [SWF32D](#) keyword in the [PROPS](#) section. See also the [PCG32D](#) keyword in the [PROPS](#) section that provides similar functionality for the gas-oil capillary pressure data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.208
PECOEFS – DEFINE PETRO-ELASTIC MODEL COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PECOEFS](#) keyword defines the Petro-Elastic model coefficients.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.209 PEGTAB – PETRO-ELASTIC PRESSURE SHEAR MODULUS TABLE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PEGTAB](#) series of keywords define a series of coefficients of a polynomial function used in the calculation of the shear modulus in the petro-elastic model. The series of keywords consist of: PEGTAB0 PEGTAB1, PEGTAB2, PEGTAB3, PEGTAB4, PEGTAB5, PEGTAB6, and PEGTAB7.

This series of keywords are ignored by OPM Flow and have no effect on the simulation.

8.3.210 PEKTAB – PETRO-ELASTIC PRESSURE BULK MODULUS TABLE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PEKTAB](#) series of keywords define a series of coefficients of a polynomial function used in the calculation of the bulk modulus in the petro-elastic model. The series of keywords consist of: PEKTAB0 PEKTAB1, PEKTAB2, PEKTAB3, PEKTAB4, PEKTAB5, PEKTAB6, and PEKTAB7.

This series of keywords are ignored by OPM Flow and have no effect on the simulation.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.211 PERMFACT – PERMEABILITY MULTIPLICATION FACTOR AS A FUNCTION OF POROSITY CHANGE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PERMFACT defines the permeability multiplication factor due to a change in porosity. The keyword is used in conjunction with OPM Flow's Salt Precipitation model, in which the pore space is reduced due to salt precipitating in the pore space, causing a reduction in porosity and an associated reduction in permeability. This keyword is also used for the **BIOFILM** and **MICP** model, where the pore space is reduced due to biofilm formation, and for the **MICP** model, also due to calcite precipitation.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	POROFAC	A real monotonically increasing positive columnar vector that defines the porosity ($\frac{\phi}{\phi_0}$) factor for the corresponding PERMFAC vector. In the simulator's Salt Precipitation model, the maximum value of ϕ is ϕ_0 , implying a maximum value of one for POROFAC.			None
		dimensionless	dimensionless	dimensionless	
2	PERMFAC	A real positive monotonically increasing columnar vector that defines the permeability (k) multiplier associated with POROFAC and used to scale a grid block's permeability due to the reduction in pore volume caused by salt precipitation. Where: $PERMFAC = m(\phi)$ $\text{with } k = m(\phi)k_0$			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 3) There must be same number of entries for each column.
- 4) Each table is terminated by a single "/" and there is no "/" terminator for the keyword.

Table 8.110: PERMFACT Keyword Description

The porosity reduction can be written as a function proportional to the initial porosity, e.g., considering the volume fraction of salt (S_s) precipitated out of the vaporized water phase, that is:

$$\phi = (1 - s_s)\phi_0 \quad (8.77)$$

The porosity and associated permeability factor data can be calculated using a permeability-porosity

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

relationship, for example:

$$\frac{k}{k_o} = \left(\frac{\phi - \phi_c}{\phi_o - \phi_c} \right)^{\gamma} \quad (8.78)$$

Where:

- k_o and ϕ_o = the initial permeability and porosity,
- k and ϕ = the actual permeability and porosity that changes due to the model dynamics (e.g., precipitation or dissolution of salt, biofilm formation, calcite precipitation),
- ϕ_c = the residual porosity at which permeability is zero, and
- γ = a positive exponent.

See also Kozeny-Carment (extended)^{219, 220} and ²²¹ and Verma-Pruess²²² for additional functional forms that can be used to derive the tabulated data that can be entered via the [PERMFACT](#) keyword.

Example

The example below defines two [PERMFACT](#) tables assuming NTSFUN equals two and NSSFUN is greater than or equal to five on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--      PERMEABILITY FACTOR AS A FUNCTION OF POROSITY REDUCTION
--      (OPM FLOW KEYWORD)
--
PERMFACT
--      PORO      PERM
--      FACTOR    FACTOR
--      -----
--      0.00      0.0000
--      0.25      0.0625
--      0.50      0.2500
--      0.75      0.5625
--      1.00      1.0000
--                               / TABLE NO. 01
--
--      -----
--      0.00      0.0000
--      0.25      0.0625
--      0.50      0.2500
--      0.75      0.5625
--      1.00      1.0000
--                               / TABLE NO. 02
```

Both tables use equation (8.78) with Φ_c equal to zero and γ equal to 2.0. Note that [PERMFACT](#) changes the permeability in all three dimensions, that is the [PERMX](#), [PERMY](#) and [PERMZ](#) arrays are all modified.

²¹⁹ J. Kozeny, "Ueber kapillare Leitung des Wassers im Boden." *Sitzungsber Akad. Wiss., Wien*, 136(2a): 271-306, 1927.

²²⁰ P.C. Carman, "Fluid flow through granular beds." *Transactions, Institution of Chemical Engineers, London*, 15: 150-166, 1937.

²²¹ P.C. Carman, "Flow of gases through porous media." *Butterworths, London*, 1956.

²²² Verma, A., & Pruess, K. (1988). Thermohydrological conditions and silica redistribution near high-level nuclear wastes emplaced in saturated geological formations. *Journal of Geophysical Research*, 93, 1159-1173.

8.3.212 PLMIXPAR – DEFINE THE POLYMER TODD-LONGSTAFF MIXING PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PLMIXPAR](#) keyword defines the Todd-Longstaff²²³ mixing parameters for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. This keyword must be present in the input deck if the [POLYMER](#) keyword has been activated.

Note that this keyword is used only for the polymer option, if the [MISCIBLE](#) keyword in the [RUNSPEC](#) section has been invoked then in addition the [TLMIXPAR](#) keyword is also required to define the Todd-Longstaff mixing parameters for the [MISCIBLE](#) option.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PLMVIS	A real positive value that is greater than or equal to zero and less than or equal to one, that defines the viscosity Todd-Longstaff mixing parameter for each polymer region.			None
		dimensionless	dimensionless	dimensionless	

Notes:

1) The keyword is followed by NPLMIX tables as declared on the *REGDIMS* keyword in the *RUNSPEC* section for each rock region. There should be only one row per table.

2) Each entry is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.111: PLMIXPAR Keyword Description

Example

```
--
--      POLYMER TODD-LONGSTAFF MIXING PARAMETERS
--
PLMIXPAR
--      PLM
--      VISCOS
--      -----
--      0.3500                      / TABLE NO. 01
--      0.2500                      / TABLE NO. 02
--      0.6500                      / TABLE NO. 03
```

The above example defines three polymer Todd-Longstaff mixing parameter data sets, based on the NPLMIX variable on the [REGDIMS](#) keyword in the [RUNSPEC](#) section being equal to three.

²²³ Todd, M. and Longstaff, W. “The Development, Testing and Application of a Numerical Simulator for Predicting Miscible Flood Performance,” paper SPE 3484, Journal of Canadian Petroleum Technology (1972) 24, No. 7, 874-882.

8.3.213 PLYADS – DEFINE POLYMER ROCK ADSORPTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYADS](#) keyword defines the rock polymer adsorption tables for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. Alternatively, the functions can be entered via the [PLYADSS](#) keyword in the [PROPS](#) section for when salt sensitivity is to be considered.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	POLCON	A columnar vector of real monotonically increasing down the column values that defines the polymer concentration in the solution surrounding the rock. The first entry should be zero to define a no polymer concentration.			None
		lb/stb	kg/sm ³	gm/scc	
2	POLRATIO	A columnar vector of real increasing down the column values that defines the mass of adsorbed polymer per unit mass of rock. The first entry should be zero to define a zero ratio of polymer concentration.			None
		lb/lb	kg/kg	gm/gm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.112: PLYADS Keyword Description

See also the [PLYADSS](#) keyword in the [PROPS](#) section to also define rock polymer adsorption tables when the polymer concentration is a function of salinity.

8.3.214 PLYADSS – DEFINE POLYMER ROCK ADSORPTION WITH SALT DEPENDENCE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PLYADSS](#) keyword defines the rock polymer adsorption tables for when the polymer and the salt options has been activated by the [POLYMER](#) and [BRINE](#) keywords in the [RUNSPEC](#) section.

Note that the [BRINE](#) option is not currently supported by OPM Flow; however, the polymer rock adsorption functions without salt dependence may be entered via the [PLYADS](#) keyword in the [PROPS](#) section, for when salt sensitivity is not to be considered.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	POLCON	A columnar vector of real monotonically increasing down the column values that defines the polymer concentration in the solution surrounding the rock. The first entry should be zero to define a no polymer and no salt concentration data set. POLCON should only be given for the first entry of the POLCON/POLRATIO set and skipped until another POLCON/POLRATIO table is entered.			None
		lb/stb	kg/sm ³	gm/scc	
2	POLRATIO	A columnar vector of real increasing down the column values that defines the mass of adsorbed polymer per unit mass of rock of the saturated concentration of polymer adsorbed by the rock for a given POLCON and the salt concentration given by SALTCON on the ADSALNOD keyword in the PROPS section. The first table data set entry should be zero to define a no polymer and no salt concentration data set. Subsequent POLRATIO values define the POLCON/POLRATIO combinations for a given salt concentration as listed (and in the same order) by the SALTCON variable on the ADSALNOD keyword in the PROPS section. Each POLCON/POLRATIO/ SALT data sets should be terminated by a “/”			None
		lb/lb	kg/kg	gm/gm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.113: PLYADSS Keyword Description

See also the [PLYADS](#) keyword in the [PROPS](#) section to also define rock polymer adsorption tables when the polymer concentration is not a function of salinity.

Example

```
--
--      SETS SALT CONCENTRATION FOR POLYMER SOLUTION ADSORPTION
--      VIA SATNUM ARRAY ALLOCATION
--
--      SALT
--
ADSALNOD
      1.0
      5.0
      10.5
      25.0      / SATNUM TABLE NO. 01

--
--      POLYMER ROCK ADSORPTION WITH SALT DEPENDENCY TABLE
--
PLYADSS
--      POLYMER      POLYMER
--      POLCON      POLRATIO
--      -----
--              0.0      0.00000
--                      0.00000
--                      0.00000
--                      0.00000      / TABLE NO. 01
--      POLYMER      POLYMER
--      POLCON      POLRATIO
--      -----
--              1.0      0.00002
--                      0.00003
--                      0.00004
--                      0.00005      / TABLE NO. 02
--
--      POLYMER      POLYMER
--      POLCON      POLRATIO
--      -----
--              2.0      0.00003
--                      0.00004
--                      0.00005
--                      0.00006      / TABLE NO. 03
--      POLYMER      POLYMER
--      POLCON      POLRATIO
--      -----
--              3.0      0.00004
--                      0.00005
--                      0.00006
--                      0.00007      / TABLE NO. 04
```

The above example defines four polymer rock adsorption tables for four salt concentration on the [ADSALNOD](#) keyword, assuming NTSFUN equals one and NSSFUN is greater than or equal to four on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

There is no terminating “/” for this keyword.

8.3.215 PLYATEMP – DEFINE POLYMER ADSORPTION TABLE TEMPERATURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the polymer adsorption temperature for subsequent polymer adsorption tables entered via the [PLYADS](#) and [PLYADSS](#) keywords in the [PROPS](#) section. The Polymer option must have been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section and the Thermal option invoked by the [THERMAL](#) keyword, also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PLYATEMP	Single real positive value that defines polymer adsorption temperature for subsequent polymer adsorption tables.			None
		°F	°C	°C	
Notes: I) The keyword is terminated by a “/”.					

Table 8.114: PLYATEMP Keyword Description

Example

The example shows how to enter the polymer adsorption data using the [PLYADS](#) keyword for two different temperatures.

```
--
--      RESERVOIR
--      TEMPERATURE
--      -----
PLYATEMP
      60.0                               / TEMPERATURE
--
--      POLYMER ROCK ADSORPTION TABLE
--
PLYADS
--      POLYMER      POLYMER
--      POLCON       POLRATIO
--      -----
--      0.0          0.00000
--      2.0          0.00003
--      4.0          0.00005
--      6.0          0.00007
--      8.0          0.00009
--      10.0         0.00011
--      12.0         0.00012
--      14.0         0.00015                               / TABLE NO. 01
```

--			
--	POLYMER	POLYMER	
--	POLCON	POLRATIO	
--	-----	-----	
	0.0	0.00000	
	3.0	0.00004	
	5.0	0.00006	
	7.0	0.00008	
	8.0	0.00009	
	10.0	0.00011	/ TABLE NO. 02
--			
--	RESERVOIR		
--	TEMPERATURE		
--	-----		
PLYATEMP			
	120.0		/ TEMPERATURE
--			
--	POLYMER ROCK ADSORPTION TABLE		
--			
PLYADS			
--	POLYMER	POLYMER	
--	POLCON	POLRATIO	
--	-----	-----	
	0.0	0.00000	
	2.0	0.00003	
	4.0	0.00005	
	6.0	0.00007	
	8.0	0.00009	
	10.0	0.00011	
	12.0	0.00012	
	14.0	0.00015	/ TABLE NO. 01
--			
--	POLYMER	POLYMER	
--	POLCON	POLRATIO	
--	-----	-----	
	0.0	0.00000	
	3.0	0.00004	
	5.0	0.00006	
	7.0	0.00008	
	8.0	0.00009	
	10.0	0.00011	/ TABLE NO. 02

Here the first [PLYATEMP](#) keyword defines the temperature to be 60 oF for the subsequent two polymer rock adsorption tables, assuming NTSFUN equals four and NSSFUN is greater than or equal to eight on the [TABDIMS](#) keyword in the [RUNSPEC](#) section. The next [PLYATEMP](#) keyword defines the temperature to be 120 oF for the subsequent two polymer rock adsorption tables.

8.3.216
PLYCAMAX – DEFINE POLYMER-ROCK MAXIMUM ADSORPTION BY CELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYCAMAX](#) keyword defines the maximum polymer-rock adsorption value used in the calculation of the resistance factor for the water phase by individual grid block, for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. See the POLMAX parameter on the [PLYROCK](#) keyword in the [PROPS](#) section for setting the property for the whole grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.217 PLYDHFLF – DEFINE POLYMER THERMAL DEGRADATION HALF-LIFE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **PLYDHFLF** keyword defines the polymer thermal degradation half-life with respect to temperature functions for when the polymer option has been activated by the **POLYMER** keyword in the **RUNSPEC** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	A columnar vector of real monotonically increasing down the column values that defines the polymer thermal degradation temperature.			None
		°F	°C	°C	
2	POLHFLF	A columnar vector of real values that defines the corresponding polymer half-life.			None
		days	days	hours	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.115: PLYDHFLF Keyword Description

Example

```
--
--      POLYMER THERMAL DEGRADATION HALF-LIFE TABLE
--
PLYDHFLF
--      POLYMER      POLYMER
--      TEMP          HALF-LIFE
--      -----
--           0.0      365.000
--          40.0      200.000
--          80.0      150.000
--         120.0      100.000
--      POLYMER      POLYMER
--      TEMP          HALF-LIFE
--      -----
--           0.0      365.000
--          50.0      175.000
--          75.0      140.000
--         100.0      120.000
--         125.0      90.000
--         150.0      85.000
--
-- / TABLE NO. 01
--
-- / TABLE NO. 02
```

The example defines two polymer thermal degradation half-life tables, based on the NTPVT variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to two and NPPVT variable on the same keyword being greater than or equal to six.

8.3.218 PLYESAL – DEFINE POLYMER EFFECTIVE SALINITY COEFFICIENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [PLYESAL](#), defines the polymer effective salinity coefficient as well as enabling the effective salinity calculation for polymer adsorption. The keyword should only be used if the [BRINE](#) keyword has been declared to activate the brine phase, the [ECLMC](#) keyword to enable the Multi-Component Brine model, and the [POLYMER](#) keyword has been used to activate the polymer phase. All three keywords are in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.219
PLYKRRF – DEFINE POLYMER ROCK PERMEABILITY REDUCTION BY CELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYKRRF](#) keyword defines the polymer rock permeability reduction factor to the water phase by individual cell, for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. [PLYKRRF](#) should consist of an array of real positive values that are greater than or equal to one. See the PERMFAC parameter on the [PLYROCK](#) keyword in the [PROPS](#) section for setting the property for the whole grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.220 PLYMAX – DEFINE POLYMER-SALT VISCOSITY MIXING CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [PLYMAX](#) keyword defines maximum polymer and salt concentrations that are to be used in the mixing parameter calculation of the fluid component viscosities, for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

Note that If the [BRINE](#) option has not be activated by the [BRINE](#) keyword in the [RUNSPEC](#) section, then the salt concentrations in the second column are ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	POLCON	A real value that defines the polymer concentration in the solution which is used to calculate maximum polymer fluid component viscosity.			None
		lb/stb	kg/sm ³	gm/scc	
2	SALTCON	A real value that defines the salt concentration in the solution which is used to calculate maximum polymer fluid component viscosity. Note that If the BRINE option has not been activated by the BRINE keyword in the RUNSPEC section, then this variable is ignored; however, there should still be dummy entries in this case. This variable is ignored as the BRINE and POLYMER combination is not implemented in OPM Flow.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

- 1) The keyword is followed by NPLMIX tables as declared on the [REGDIMS](#) keyword in the [RUNSPEC](#) section, for each rock region. There should be only one row per table.
- 2) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.116: PLYMAX Keyword Description

Note

Currently, combining the [BRINE](#) and [POLYMER](#) models is not implemented in OPM Flow, and therefore SALTCON parameter on the [PLYMAX](#) keyword is ignored.

Example

```
--
--      POLYMER-SALT VISCOSITY MIXING CONCENTRATIONS
--
PLYMAX
--      POLYMER      SALT
--      POLCON      SALTCON
--      -----
--      0.0100      0.0500      / TABLE NO. 01
--      0.0075      0.0400      / TABLE NO. 02
--      0.0050      0.0300      / TABLE NO. 03
```

The above example defines three polymer-salt viscosity mixing concentrations, based on the NPLMIX variable on the [REGDIMS](#) keyword in the [RUNSPEC](#) section being equal to three.

8.3.221 PLYMWINJ – POLYMER MOLECULAR WEIGHT MODEL THROUGHPUT AND VELOCITY TABLE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [PLYMWINJ](#), describes the relationship of the injected polymer molecular weight as a function of polymer throughput and polymer velocity, for the simulator's Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity. The table is a two dimensional table that relates the polymer throughput values and velocity values to derive the resulting molecular weight of the injected polymer, which is then used via the [PLYVHM](#) keyword in the [PROPS](#) section, to derive the polymer molecular weight scaled viscosity.

This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

The model has been tested using metric units; however, using either field or laboratory units with the option should be considered experimental.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	PLYMWNUM	A positive integer value greater than zero and less than or equal to the NTPMWINJ variable, as defined on the PINTDIMS keyword in the RUNSPEC section, that defines the PLYMWINJ Polymer Molecular Weight Model throughput and velocity table number.			None
2-1	THRUPUT	A real positive monotonically increasing vector, that defines the polymer throughput values. The first entry should be zero to define a no throughput data set, and each vector record should be on a separate line terminated by a "/".			None
		feet ³ /feet ²	m ³ /m ²	cm ³ /cm ²	
3-1	VELOCITY	A real positive monotonically increasing vector, that defines the polymer velocity values. The first entry should be zero to define a no velocity data set, and each vector record should be on a separate line terminated by a "/".			None
		feet/day	m/day	cm/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4-1	<i>POLYMW</i>	A series of real positive vectors representing the polymer molecular weights for all combinations of the polymer throughput values (THRUPUT) and velocity values (VELOCITY), organized as a series of vectors <i>POLYMW</i>(THRUPUT,VELOCITY). Thus, the first vector represents the molecular weights of the first THRUPUT value and each entry in the vector is the corresponding molecular weight of the associated VELOCITY vector. Each vector should be on a separate line and should be terminated by a “/”. Thus, if THRUPUT has three entries and VELOCITY has four, then there should be three vectors, with each vector containing four elements representing molecular weight values, as a function of THRUPUT and VELOCITY.			None
		lb/lb-M	kg/kg-M	gm/gm-M	

Notes:

1) The keyword should be terminated by a “/”, but may be repeated up to NTPMWINJ times, as per the *PINTDIMS* keyword in the *RUNSPEC* section, to allow for the input of multiple tables.

Table 8.117: PLYMWINJ Keyword Description

Unlike other [PROPS](#) section table keywords, that enable multiple tables following the keyword to be entered, the [PLYMWINJ](#) keyword requires that the keyword itself must be repeated for each table, as is shown in the example on the following page.

See also the [WPMITAB](#) keyword in the [SCHEDULE](#) section, that assigns the [PLYMWINJ](#) tables to the water injection wells, as well as the [SKPRWAT](#), [SKPRPOLY](#), and [PLYVMH](#) keywords, in the [PROPS](#) section, that are the additional property keywords required for the Polymer Molecular Weight Transport option.

The [WSKPTAB](#) keyword in the [SCHEDULE](#) section may be used to assign the [SKPRPOLY](#) and [SKPRWAT](#) tables to water injections wells, that enable the calculation of the wellbore skin pressure based on the fluids being injected.

Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

Given NTPMWINJ equals two on the [PINTDIMS](#) keyword in the [RUNSPEC](#) section, then two PLYWINJ tables are required to be entered:

```
--
--      POLYMER MOLECULAR WEIGHT MODEL THROUGHPUT AND VELOCITY TABLE
--      (OPM FLOW PROPS KEYWORD)
--
PLYMWINJ
      1                                / TABLE NUMBER
--
--      THROUGHPUT VALUES
--
      0.0      200.0    400.0          /
--
--      VELOCITY VALUES
--
      0.0      50.0     80.0     100.0 /
--
--      POLYMW VALUES
--
      20.0     19.0     18.0     16.0 / POLYMW(OUTPUT=1, VELOCITY=1 TO N)
      20.0     16.0     14.0     12.0 / POLYMW(OUTPUT=2, VELOCITY=1 TO N)
      20.0     12.0      8.0      4.0 / POLYMW(OUTPUT=3, VELOCITY=1 TO N)
/
PLYMWINJ
      2                                / TABLE NUMBER
--
--      THROUGHPUT VALUES
--
      0.0      200.0    400.0          /
--
--      VELOCITY VALUES
--
      0.0      50.0     70.0     100.0 /
--
--      POLYMW VALUES
--
      20.0     19.0     18.0     16.0 / POLYMW(OUTPUT=1, VELOCITY=1 TO N)
      20.0     16.0     14.0     12.0 / POLYMW(OUTPUT=2, VELOCITY=1 TO N)
      20.0     12.0      8.0      4.0 / POLYMW(OUTPUT=3, VELOCITY=1 TO N)
/
```

As mentioned previously, the [PLYMWINJ](#) keyword requires that the keyword itself must be repeated for each table, as is shown in the above example.

8.3.222 PLYRMDEN – DEFINE POLYMER MODEL IN SITU ROCK DENSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [PLYRMDEN](#) keyword defines the in situ rock density at reservoir conditions by individual cell, for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. [PLYRMDEN](#) should consist of an array of real positive values. See the [DENSITY](#) parameter on the [PLYROCK](#) keyword in the [PROPS](#) section for setting the property for the whole grid.

8.3.223 PLYROCK – DEFINE POLYMER-ROCK PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYROCK](#) keyword defines rock properties for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PSPACE	A real positive value that is greater than or equal to zero and less the maximum water saturation and less than one, that defines available pore space for this rock type.			None
		dimensionless	dimensionless	dimensionless	
2	PERMFAC	A real positive value that is greater than or equal to one that defines decrease in the rock permeability to the water phase when the maximum amount of polymer has been adsorbed.			None
		dimensionless	dimensionless	dimensionless	
3	DENSITY	A real value that defines the rock in-situ density, that is at reservoir conditions.			None
		lb/rb	kg/rm³	gm/rcc	
4	ADINDEX	A positive integer of 1 or 2 that defines the polymer desorption option. 1) then polymer desorption may occur by retracing the polymer adsorption isotherm when the local polymer concentration in the solution decreases. 2) then no polymer desorption may occur.			Defined
		dimensionless 1	dimensionless 1	dimensionless 1	
5	POLMAX	A real positive non-zero value that defines the maximum polymer adsorption to be used in the calculation of the resistance factor for the water phase.			None
		lb/lb	kg/kg	gm/gm	

Notes:
1) The keyword is followed by NTSFUN tables as declared on the TABDIMS keyword in the RUNSPEC section for each polymer flooding region. There should be only one row per table.
2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.118: PLYROCK Keyword Description

Example

```
--
--      POLYMER-ROCK PROPERTIES
--
PLYROCK
--      PORE      PERM      INSITU      DESORP      MAX
--      SPACE     FACTOR    DENSITY    OPTN         POLY
--      -----
--      0.1200    1.7500    1800.0    1            0.00012      / TABLE NO. 01
--      0.1300    1.8500    1980.0    2            0.00015      / TABLE NO. 02
--      0.1500    1.9500    2005.0    1            0.00014      / TABLE NO. 03
```

The above example defines three polymer-rock tables, based on the NTSFUN variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to three.

There is no terminating “/” for this keyword.

The above example activates the polymer shear thinning-thickening option and defines two polymer shear thinning-thickening tables, based on the NTPVT variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to two and NPPVT variable on the same keyword being greater than or equal to six.

8.3.225 PLYSHLOG – ACTIVATE AND DEFINE THE POLYMER SHEARING LOGARITHMIC PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates and defines the parameters for the logarithm-based polymer shear thinning/thickening option.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	POLCON	A real positive value that defines the reference polymer concentration for the VELOCITY and VISFAC data for this keyword.			None
		lb/stb	kg/sm ³	gm/scc	
I-2	SALTCON	A real positive value that defines the reference salt concentration for the VELOCITY and VISFAC data for this keyword. Note that If the BRINE option has not been activated by the BRINE keyword in the RUNSPEC section, then this variable is ignored. This variable is ignored as the BRINE option is not implemented in OPM Flow.			None
		lb/stb	kg/sm ³	gm/scc	
I-3	TEMP	A real positive value defines the reference polymer temperature for the VELOCITY and VISFAC data for this keyword. Note that If the TEMP option has not been activated by the TEMP keyword in the RUNSPEC section, then this variable is ignored. This variable is ignored as the TEMP and POLYMER options combination is not implemented in OPM Flow.			None
		°F	°C	°C	
I-4	/	Record terminated by a “/”			Not Applicable
2-1	VELOCITY	A columnar vector of real monotonically increasing down the column values that defines the water-polymer flow velocity for the reference conditions of POLCON, SALTCON and TEMP . The VELOCITY value for the first row in the table should be a very small value that is greater than zero and less than 1 x 10 ⁻⁶ .			None
		feet/day	m/day	cm/hour	
2-2	VISFAC	A columnar vector of real positive values that define the dimensionless shear effect multiplier for the given VELOCITY entry for the reference conditions of POLCON, SALTCON and TEMP . Normally VISFAC value for the first row in the table should be one.			None
		dimensionless	dimensionless	dimensionless	
I-4	/	Record terminated by a “/”			Not Applicable

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes:					
1)	The keyword is followed by NTPVT tables as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section.				
2)	Each table must be entered with two records, with entries 1-1, 1-2, 1-3 and 1-4 representing record number one and 2-1, 2-2 and 2-3 representing record number two in the “No.” column in this table.				
3)	Each of the records are terminated by a “/” and is explicitly shown in the above rows.				
4)	For record number two a minimum of two rows and a maximum of NPPVT rows, as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section, are required.				
5)	There is no “/” terminator for the keyword.				

Table 8.120: PLYSHLOG Keyword Description

See the [PLYSHEAR](#) keyword for the alternative polymer shear thinning/thickening option that is also implemented in OPM Flow.

Example

The following example show how to enter two *PLYSHLOG* tables given that the NTPVT variable on the *TABDIMS* keyword in the *RUNSPEC* section is set equal to two.

```
--
-- POLYMER SHEARING LOGARITHMIC PARAMETERS
--
PLYSHLOG
-- REF          REF          REF
-- POLCON       SALTCON      TEMP
-- -----
0.5
/
--
-- VELOCITY     VISFAC
-- -----
0.0000001    1.00
0.000001     1.10
0.0001       1.30
0.001        1.47
0.01         1.67
0.1          2.00
1.0          2.20
10.0         2.30
100.0        2.40
1000.0       2.40
--
-- / TABLE NO. 01
-- REF          REF          REF
-- POLCON       SALTCON      TEMP
-- -----
0.5
/
```


--		
--	VELOCITY	VISFAC
--	-----	-----
	0.0000001	1.00
	0.000001	1.10
	0.0001	1.35
	0.001	1.57
	0.01	1.87
	0.1	2.20
	1.0	2.40
	10.0	2.60
	100.0	2.65
	1000.0	2.65

/ TABLE NO. 02

The example activates the polymer logarithmic shear thinning-thickening option and defines two polymer shear thinning-thickening tables, based on the NTPVT variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to two and NPPVT variable on the same keyword being greater than or equal to ten.

8.3.226 PLYTRRF – DEFINE POLYMER ROCK PERMEABILITY REDUCTION VERSUS TEMPERATURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYTRRF](#) keyword defines the polymer rock permeability reduction factor to the water phase as a function of temperature, for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. See the PLYTRRF keyword for the options on how this data is used in the polymer model and the PERMFAC parameter on the [PLYROCK](#) keyword for setting the property for the whole grid for a constant temperature. Both keywords are in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.227
PLYTRRFA
– DEFINE POLYMER ROCK PERMEABILITY REDUCTION VERSUS TEMPERATURE OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYTRRFA](#) keyword defines the how the polymer rock permeability reduction factor to the water phase as a function of temperature data, entered via the [PLYTRRA](#) keyword in the [PROPS](#) section, should be used. This keyword should only be used if the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. See the [PERMFAC](#) parameter on the [PLYROCK](#) keyword in the [PROPS](#) section for setting the property for the whole grid for a constant temperature.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.228 PLYVISC – DEFINE POLYMER VISCOSITY SCALING FACTORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PLYSVISC defines the polymer viscosity scaling factors used to determine the relationship of pure water viscosity with respect to increasing polymer concentration within a grid block. The polymer option must be activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section in order to use this keyword.

The [BRINE](#) option in the [RUNSPEC](#) should be deactivated if this keyword is to be used.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	POLCON	A columnar vector of real monotonically increasing down the column values that defines the polymer concentration in the solution surrounding the rock. The first entry should be zero to define a no polymer concentration.			None
		lb/stb	kg/sm ³	gm/scc	
2	VISFAC	A columnar vector of real increasing or equal values that defines a factor that scales the effective viscosity of the solution for the given POLCON entry. Normally VISFAC value for the first row in the table should be one.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.121: PLYVISC Keyword Description

Example

```
--
--      POLYMER VISCOSITY SCALING FACTOR TABLES
--
PLYVISC
--      POLYMER      VISCOSITY
--      POLCON      VISFAC
--      -----
--      0.0000      1.000
--      0.0002      10.000
--      0.0004      20.000
--      0.0008      40.000
--
-- / TABLE NO. 01
```

--	POLYMER	VISCOSITY
--	POLCON	VISFAC
--	-----	-----
	0.0000	1.000
	0.0003	10.000
	0.0005	20.000
	0.0007	40.000
	0.0009	45.000
	0.0011	55.000

/ TABLE NO. 02

The example defines two polymer viscosity scaling factor tables, based on the NTPVT variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to two and NPPVT variable on the same keyword being greater than or equal to six.

8.3.229 PLYVISCS – DEFINE POLYMER-SALT VISCOSITY SCALING FACTORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PLYSVISCS defines the polymer-salt viscosity scaling factor tables applied to pure water that are used to determine the viscosity of a polymer-salt mixture with respect to increasing polymer concentration within a grid block. The polymer option must be activated by the [POLYMER](#) keyword, as well as the brine phase declared by the [BRINE](#) keyword in the [RUNSPEC](#) section in order to use this keyword. However the ECLM keyword in the [RUNSPEC](#) must not be used with this keyword.

See also the [PLYVSCST](#) keyword in the [PROPS](#) section to enter polymer viscosity scaling factor data that is dependent on both salt and reservoir temperature.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.230
PLYVISCT
–
DEFINE POLYMER-TEMPERATURE VISCOSITY SCALING FACTORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PLYVISCT defines the polymer-temperature viscosity scaling factor tables applied to pure water that are used to determine the viscosity of the polymer at a given temperature with respect to increasing polymer concentration within a grid block. Both the polymer option must be activated by the [POLYMER](#) keyword and the temperature option invoked by the [TEMP](#) keyword in the [RUNSPEC](#) section in order to use this keyword. However the [BRINE](#) keyword in the [RUNSPEC](#) must not be used with this keyword, that is the salt sensitivity options should be deactivated.

See also the [PLYVSCST](#) keyword in the [PROPS](#) section to enter polymer viscosity scaling factor data that is dependent on both salt and reservoir temperature.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.231 PLYVMH – POLYMER MOLECULAR WEIGHT MODEL POLYMER VISCOSITY CONSTANTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [PLYVMH](#), defines the constants used to calculate viscosity of the polymer solution as a function of the polymer molecular weight and the polymer concentration, for the simulator's Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity. The keyword consists of a series of row vectors, which each vector having four elements, that define the constants used in calculating the polymer viscosity

This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator. The model has been tested using metric units; however, using either field or laboratory units with the option should be considered experimental.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	MHK	The Mark-Houwink K polymer specific constant in the Mark-Houwink equation, see equation (8.83).			None
			ml/g		
2	MHA	The Mark-Houwink exponent parameter a, in the Mark-Houwink equation, see equation (8.83).			None
		dimensionless	dimensionless	dimensionless	
3	GAMMA	The γ intrinsic water-polymer viscosity constant in the Huggins equation, see equation (8.85).			None
4	KAPPA	The κ intrinsic water-polymer viscosity constant in the Huggins equation, see equation (8.85).			None

Notes:

1) The keyword is followed by NPLYVHM records as declared on the [PINTDIMS](#) keyword in the [RUNSPEC](#) section, with each record terminated by a “/” and the keyword itself should be terminated by a “/”.

Table 8.122: PLYVMH Keyword Description

The high molecular weight of polymers greatly increase the viscosity of the injected water in which they are dissolved, thus adjusting the mobility ration of the displacing phase. The increase in viscosity is caused by strong internal friction between the randomly coiled and swollen macro molecules and the surrounding

water molecules. And is dependent on both the nature of the polymer and the injected water acting as the solvent. There are several formulations of viscosity associated with polymer rheology, namely:

Relative viscosity is defined as:

$$\eta_{rel} = \frac{\eta}{\eta_s} \quad (8.79)$$

where:

- η_{rel} = is the relative viscosity,
- η = is the viscosity of the solution, and
- η_s = is the viscosity of the solvent (injected water).

Specific viscosity is defined as:

$$\begin{aligned} \eta_{sp} &= \frac{(\eta - \eta_s)}{\eta_s} \\ &= \frac{\eta}{\eta_s} - 1 \\ &= \eta_{rel} - 1 \end{aligned} \quad (8.80)$$

where:

- η_{sp} = is the specific water-polymer viscosity, and
- η_s = is the viscosity of the solvent (injected water).

Reduced Specific Viscosity is given by:

$$\begin{aligned} \eta_{red} &= \frac{\eta_{sp}}{C_p} \\ &= \frac{(\eta_{rel} - 1)}{C_p} \end{aligned} \quad (8.81)$$

where:

- η_{red} = is the reduced specific water-polymer viscosity, and
- C_p = is the polymer concentration.

Finally, the Intrinsic Viscosity, which is defined as a measure for the internal friction in polymer solutions at the limit of zero polymer concentration. Thus, this quantity describes the effect of completely separated polymer chains on the solution viscosity, and is defined as:

$$\begin{aligned} [\eta] &= \lim_{C_p \rightarrow 0} \frac{\eta_{sp}}{C_p} \\ &= \lim_{C_p \rightarrow 0} \frac{\eta - \eta_0}{\eta_0 C_p} \end{aligned} \quad (8.82)$$

where:

- $[\eta]$ = is the intrinsic water-polymer viscosity and describes the increase in viscosity arising from an individual polymer chain and is a measure of the polymers' thickening power, and
- η_0 = zero-shear viscosity.

For any given solvent pair, the intrinsic viscosity increases as the molecular weight of the polymer increases; here, the *Mark-Houwink equation*, also known as the *Mark-Houwink-Staudinger equation* ²²⁴, ²²⁵, and ²²⁶ is used to calculate $[\eta]$, that is:

$$[\eta] = K \cdot M_w^a \quad (8.83)$$

where:

- K = is the polymer specific constant in the Mark-Houwink equation, the MHK parameter in Table 8.122,
- M_w = is the polymer molecular weight, and
- a = the exponent constant in the Mark-Houwink equation, the MHA parameter in Table 8.122.

The Mark-Houwink parameters can be determined from a double logarithmic plot of intrinsic viscosity versus molecular weight which yields straight lines, that is:

$$\ln([\eta]) = \ln(K) + a \times \ln(M_w) \quad (8.84)$$

OPM Flow uses a form of the Huggins²²⁷ equation to calculate the polymer apparent viscosity, as shown below:

$$\frac{\eta_0}{\eta_s} = 1 + \mathcal{J}'(X + \kappa X^2) \quad (8.85)$$

where $X = [\eta]C_p$

where:

- $\mathcal{J}'$ = a user defined constant, GAMMA in Table 8.122, and
- κ = a user defined constant, KAPPA in Table 8.122.

Which can be used to calculate the zero-shear viscosity, η_0 , based on the quadratic function in equation (8.85) time polymer concentration(C_p) and polymer intrinsic viscosity described in equation (8.83).

In terms of the keyword units, given the intrinsic viscosity in ml/g, polymer concentration in kg/m³, and the molecular weight as kg/kg-M, then equation (8.83) becomes:

$$[\eta] = K(M_w \cdot 1.0 \times 10^{-3})^a \cdot 1.0 \times 10^{-3} \quad (8.86)$$

and equation (8.85) becomes:

$$\frac{\eta_0}{\eta_s} = 1 + \mathcal{J}'(X + \kappa X^2) \quad (8.87)$$

where $X = 1.0 \times 10^{-6} [\eta]C_p$

Note that the model does not account for non-Newtonian flow; the apparent viscosity is simply set equal to the zero-shear viscosity, and that the model only considers the full mixing between the polymer and water.

²²⁴ H. Mark, in R. Saenger, *Der feste Koerper*, Hirzel, Leipzig, 1938.

²²⁵ R. Houwink, *J. Prakt. Chem.*, Vol. 157, Issue 1-3, p. 15 (1940).

²²⁶ H. Staudinger, *Die Hochmolekulare Organischen Verbindungen*, Julius Springer, Berlin 1932.

²²⁷ Huggins, M. L. 1942. The viscosity of dilute solutions of long-chain molecules. IV. Dependence on concentration.» *Journal of the American Chemical Society* 64 (11): 2716-2718.

See also the [PLYMWINJ](#) keyword in the [PROPS](#) section, that describes the relationship of the injected polymer molecular weight as a function of polymer throughput and polymer velocity. Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

Given NPLYVMH equals two on the [PINTDIMS](#) keyword in the [RUNSPEC](#) section, then:

```
--
--      POLYMER MOLECULAR WEIGHT MODEL POLYMER VISCOSITY CONSTANTS
--      (OPM FLOW PROPS KEYWORD)
--
--      MHK      MHA      VISC      VISC
--      CONST    EXPON    GAMMA    KAPPA
PLYVMH
      0.02      0.50      0.40      0.60      /
      0.03      0.51      0.42      0.63      /
/
```

Two sets of data should be entered with the [PLYVMH](#) keyword, as shown above.

8.3.232 PLYVSCST – DEFINE POLYMER-SALT-TEMPERATURE VISCOSITY SCALING FACTORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

PLYVSCST defines the polymer-salt-temperature viscosity scaling factor tables applied to pure water that are used to determine the viscosity of the polymer at a given salt concentration and for a given temperature, with respect to increasing polymer concentration within a grid block. Both the polymer option must be activated by the *POLYMER* keyword and the temperature option invoked by the *TEMP* keyword in the *RUNSPEC* section in order to use this keyword. In addition, the *BRINE* keyword in the *RUNSPEC* must also be invoked. The keyword is used in conjunction with the *SALTNODE* keyword to define the various salt concentrations and the *TEMPNODE* keyword to define the various reservoir temperatures. Both keywords are in the *PROPS* section.

See also the *PLYVISC* keyword in the *PROPS* section to enter polymer viscosity scaling factor data that is dependent just salt concentration and the *PLYVISC* keyword in the *PROPS* section to enter polymer viscosity scaling factor data that is dependent just on reservoir temperature.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.233 PMAX – MAXIMUM AND MINIMUM PRESSURE FOR TOTAL COMPRESSIBILITY CHECK

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [PMAX](#) keyword defines the maximum and minimum pressures expected to be encountered during the run. The data is used to perform the PVT total compressibility check that ensures that the total compressibility of a mixture of oil-gas, for when the gas-oil ratio is increasing for an oil, or the condensate gas ratio is increasing for a gas condensate, is positive respect to pressure. The total compressibility check is used to ensure that the entered oil and gas PVT data is consistent. If the check fails for given oil-gas mixture at a given pressure, resulting in a negative total compressibility, then this will result in numerical instabilities in the run causing this simulator difficulties in converging to a solution.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

The above example defines two miscibility versus pressure tables assuming NTMISC equals two and NSMISC is greater than or equal to six on the [MISCIBLE](#) keyword in the [RUNSPEC](#) section.

8.3.235 PPCWMAX – DEFINE SWATINIT CALCULATED CAPILLARY PRESSURE CONSTRAINTS

RUNSPEC	GRID	EDIT	PPCWMAX	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **PPCWMAX** keyword defines the maximum capillary pressure allowed when scaling the capillary pressure tables to match the inputted **SWATINIT** array. This is primarily used for when the **SWATINIT** array has values of water saturation above the connate water saturation significantly outside than capillary pressure transition zone, that is high on the structure. In this case OPM Flow may generate large values for the capillary pressure which may result in numerical converge problems. This keyword sets the maximum allowable calculated capillary pressure and how the water saturation should be treated when the limit is exceeded.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PCWO	A columnar vector of real values that defines the maximum allowable capillary pressure for each SATNUM region. The default value of infinity means there is no limit applied.			Infinity
		psia	barsa	atma	
2	OPTN	A columnar vector of character strings that should be set to: 1) NO: To ignore the SWATINIT value for the offending cell for when PCWO is exceeded. In this cases the capillary pressure for the block is set to the maximum (PCWO) and the water saturation is re-calculated based on PCWO. 2) YES: To set the SWATINIT value to the connate water saturation for the offending cell for when PCWO is exceeded. In this case the capillary pressure is set to the maximum value of the appropriate SATNUM table and the initial water saturation is calculated to be consistent with the tables maximum capillary pressure. This results in the capillary pressures not being re-scale for the offending cell.			No

Notes:

- 1) The keyword is followed by NTSFUN rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 2) Each row show contain two values representing PCWO and OPTN values.
- 3) Each row is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.124: PPCWMAX Keyword Description

Note

Using this keyword to limit the re-scaled grid block capillary pressure values will effect the fluids in-place when the simulator has to re-calculate values due to the capillary pressure limit being exceeded.

In addition, the high grid block capillary pressures may be indicative of an inconsistency between the tabular [SATNUM](#) capillary pressure values and the provided [SWATINIT](#) array water saturations. This inconsistency may be a result of the [SWATINIT](#) array being derived using a saturation height function, as is customary in static modeling software, and the numerical models tabulated capillary pressure. Rather than resetting the maximum calculated capillary pressure using the [PPCWMAX](#) keyword, it may be more appropriate to investigate the reason for the high capillary pressures values first, prior to applying the keyword.

Example

```
--
--      SET MAXIMUM PC FOR SWATINIT INITIALIZATION
--      MAX      MATCH
--      PC      SWATINIT
--      -----
PPCWMAX
      100.0      YES      / TABLE NO 01
      125.0      YES      / TABLE NO 02
      135.0      YES      / TABLE NO 03
```

The above example sets the maximum capillary pressure for three saturation regions to 100, 125 and 135 with [SWATINIT](#) reset to the connate water saturation for when the capillary pressure limit is exceeded.

8.3.236 PROPS – DEFINE THE START OF THE PROPS SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PROPS](#) activation keyword marks the end of the [EDIT](#) section and the start of the [PROPS](#) section that defines the key fluid and rock property data for the simulator

There is no data required for this keyword.

Example

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
```

The above example marks the end of the [EDIT](#) section and the start of the [PROPS](#) section in the OPM Flow data input file.

8.3.237 PVCDO – OIL PVT PROPERTIES FOR DEAD OIL (CONSTANT COMPRESSIBILITY)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[PVCDO](#) defines the oil PVT properties for dead oil²²⁸ with constant compressibility. If the oil has a constant and uniform dissolved gas concentration, Gas-Oil Ratio (“GOR”), and if the reservoir pressure never drops below the saturation pressure (bubble point pressure), then the model can be run more efficiently by omitting the [GAS](#) and [DISGAS](#) keywords from the [RUNSPEC](#) section, treating the oil as a dead oil, and defining a constant Rs (GOR) value with keyword [RSCONST](#) or [RSCONSTT](#) in the [PROPS](#) section. This results in the model being run as a dead oil problem with no active gas phase. However, OPM Flow takes into account the constant Rs in the calculations and reporting.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	PRESS is a real positive value defining the oil reference pressure for the other parameters for this data set.			None
		psia	barsa	atma	
2	OFVF	OFVF is a real positive value defining the oil formation volume factor (Bo) at the reference pressure.			None
		rb/stb	rm³/sm³	rcc/scc	
3	OCOMP	OCOMP is a real positive value defining the oil compressibility (Co) at the oil reference pressure and is defined as: $C_o = -\frac{1}{B_o} \left(\frac{dB_o}{dP} \right)$			None
		l/psia	l/barsa	l/atma	
4	OVISC	OVISC is a real positive value defining the oil viscosity (μo) at the oil reference pressure.			None
		cP	cP	cP	
5	OVISCOMP	OVISCOMP is a real positive value defining the oil viscosibility (μoc) at the oil reference pressure and is defined as: $\mu_{oc} = \frac{1}{\mu_o} \left(\frac{d\mu_o}{dP} \right)$			None
		l/psia	l/barsa	l/atma	

Notes:

1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.125: PVCDO Keyword Description

²²⁸ “Dead” oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

See also the [RSCONST](#) and [RSCONSTT](#) keywords to define the constant Rs for dead oil and [PVDO](#) as an alternative keyword to enter the dead oil properties.

Example

```
--
--      OIL PVT TABLE FOR DEAD WITH CONSTANT COMPRESSIBILITY
--
PVCDO
--      REF PRES      BO          CO          VISC          VISC
--      PSIA          RB/STB      1/PSIA      CPOISE      GRAD
--      -----
--      3840.0      1.080      1.5E-6      1.750      0.0      / TABLE NO. 01
--      3840.0      1.100      1.5E-6      1.050      0.0      / TABLE NO. 02
--      3840.0      1.120      1.6E-6      0.950      0.0      / TABLE NO. 03
--      3840.0      1.140      1.7E-6      0.850      0.0      / TABLE NO. 04
--      3840.0      1.160      1.7E-6      0.800      0.0      / TABLE NO. 05
```

The above example defines five dead oil PVT tables with constant compressibility and viscosity, and assumes that NTPVT equals five on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

There is no terminating “/” for this keyword.

8.3.238 PVCO – OIL PVT PROPERTIES FOR LIVE OIL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PVCO defines the oil PVT properties for live²²⁹ and the keyword should only be used if there is both oil and gas phases in the model. This keyword should be used when the **DISGAS** keyword has been declared in the **RUNSPEC** section indicating that dissolved gas (more commonly referred to as solution gas) is present in the oil. The keyword may be used for oil-water and oil-water-gas input decks. This is an alternative keyword to the **PVTO** keyword in the **PROPS** section that also enables entering live oil PVT data. Here, the **PVCO** keyword assumes that for the undersaturated oil with a given Gas-Oil Ratio (“GOR” or “Rs”), the oil compressibility is independent of the pressure. Hence, it is not necessary to enter the undersaturated oil formation volume factor versus pressure data. Similarly, the viscosity of the same type of oil is assumed to have a pressure independent “viscosity” derivative, and therefore it is not necessary to enter undersaturated viscosity versus pressure data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; however, the **PVTO** keyword in the **PROPS** section may be used to enter live oil PVT data instead.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	PRESS is a real columnar vector of real monotonically increasing down the column values that defines the oil phase saturation pressure (bubble-point pressure), that defines the <i>RS</i> , oil formation volume factor and the oil viscosity at PRESS.			None
		psia	barsa	atma	
2	<i>RS</i>	<i>RS</i> is a real monotonically increasing down the column values that defines the saturated gas-oil ratio (“GOR”) or Rs, for the given value of PRESS.			I*
		Mscf/stb	sm³/sm³	scc/scc	
3	OFVF	OFVF is a real positive value defining the oil saturated formation volume factor (Bo) at the saturation pressure PRESS.			None
		rb/stb	rm³/sm³	rcc/scc	
4	OVISC	OVISC is a real positive value defining the oil viscosity (μo) at the oil saturated reference pressure, PRESS.			I*
		cP	cP	cP	
5	OCOMP	OCOMP is a real positive value defining the oil compressibility (Co) at the saturated oil reference pressure and is defined as: $C_o = -\frac{1}{B_o} \left(\frac{dB_o}{dP} \right)$			I*
		l/psia	l/barsa	l/atma	

²²⁹ “Live” oil is oil that contains gas in solution, which is normally the case for most conventional oil reservoirs. However, for oil reservoirs classified as heavy oil reservoirs, the in situ dissolved gas may be negligible and oil would then be classified as gas-free oil which is commonly referred to as “dead” oil.

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	OVISCOMP	OVISCOMP is a real positive value defining the oil viscosity (μ_{oc}) at the saturated oil reference pressure with the given RS , where (μ_{oc}) is defined as: $\mu_{oc} = -\frac{1}{\mu_o} \left(\frac{d\mu_o}{dP} \right)$			1*
		l/psia	l/barsa	l/atma	

Notes:

1)

The keyword is followed by NTPVT tables as declared on the $TABDIMS$ keyword in the $RUNSPEC$ section.

2)

Items (2) to (6) may be defaulted, in which case linear interpolation will be used to calculate the missing values. In addition, OVISCOMP, item (6), may be completely defaulted, which sets this data to zero.

3)

Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.126: PVCO Keyword Description

Example

```
--
--      OIL PVT TABLE FOR LIVE OIL
--
PVCO
--      PSAT      RS      B0      VISC      OIL      OIL
--      PSIA      MSCF/STB  RB/STB  CPOISE  COMPRES  VISCOS
--      -----
--      14.7      0.0010  1.05340  1.7230  3.0E-5  1*
--      500.0      0.0890  1.08890  1.1670  1*      1*
--      1000.0     0.2060  1.13850  0.8570  1*      1*
--      1500.0     0.3360  1.19640  0.6840  1*      1*
--      2000.0     0.4750  1.26110  0.5750  1*      1*
--      2500.0     0.6220  1.33160  0.5000  1*      1*
--      3000.0     0.7750  1.40740  0.4450  1*      1*
--      3500.0     0.9330  1.48790  0.4020  1*      1*
--      4000.0     1.0960  1.57280  0.3680  1*      1*
--      4258.0     1.1800  1.61760  0.3530  1*      1*
--      4500.0     1.2630  1.66190  0.3400  1*      1*
--      5000.0     1.4340  1.75480  0.3170  1*      1*
--      5500.0     1.6060  1.85020  0.2980  1*      1*      / TABLE NO. 01
--
--      PSAT      RS      B0      VISC      OIL      OIL
--      PSIA      MSCF/STB  RB/STB  CPOISE  COMPRES  VISCOS
--      -----
--      14.7      0.0010  1.05340  1.7230  3.0E-5  1*
--      500.0      0.0890  1.08890  1.1670  1*      1*
--      1000.0     0.2060  1.13850  0.8570  1*      1*
--      1500.0     0.3360  1.19640  0.6840  1*      1*
--      2000.0     0.4750  1.26110  0.5750  1*      1*
--      2500.0     0.6220  1.33160  0.5000  1*      1*
--      3000.0     0.7750  1.40740  0.4450  1*      1*
--      3500.0     0.9330  1.48790  0.4020  1*      1*
--      4000.0     1.0960  1.57280  0.3680  1*      1*
--      4258.0     1.1800  1.61760  0.3530  1*      1*
--      4500.0     1.2630  1.66190  0.3400  1*      1*
--      5000.0     1.4340  1.75480  0.3170  1*      1*
--      5500.0     1.6060  1.85020  0.2980  1*      1*      / TABLE NO. 02
```

The example defines two live oil PVT tables with constant compressibility above the saturation pressure, and assumes that NTPVT equals two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

8.3.239 PVDG – GAS PVT PROPERTIES FOR DRY GAS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	------------------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PVDG defines the gas PVT properties for dry gas²³⁰. If the gas has a constant and uniform vaporized oil concentration, Condensate-Gas Ratio (“CGR”), and if the reservoir pressure never drops below the saturation pressure (dew point pressure), then the model can be run more efficiently by omitting the [OIL](#) and [VAPOIL](#) keywords from the [RUNSPEC](#) section, treating the gas as a dry gas, and defining a constant Rv (CGR) value with keyword [RVCONST](#) or [RVCONSTT](#) in the [PROPS](#) section. This results in the model being run with as a dry gas problem with no active oil (condensate) phase. However, OPM Flow takes into account the constant Rv in the calculations and reporting.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A columnar vector of real monotonically increasing down the column values that defines the gas phase pressure.			None
		psia	barsa	atma	
2	GFVF	A columnar vector of real decreasing down the column values that defines the corresponding gas phase formation volume factor.			None
		rb/Mscf	rm ³ /sm ³	rcc/scc	
3	GVISC	A columnar vector of real increasing down the column values that defines the corresponding gas phase viscosity.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.127: PVDG Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table. See the second example for an illustration on how to use this feature.

See also the [RVCONST](#) and [RVCONSTT](#) keywords to define the constant Rv for dry gas.

²³⁰ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

Example

The first example below defines two dry gas PVT tables assuming NTPVT equals two and NPPVT is greater than or equal to 22 on the *TABDIMS* keyword in the *RUNSPEC* section.

--	GAS PVT TABLE FOR DRY GAS			
--				
PVDG				
--	PRES	BG	VISC	
--	PSIA	RB/MSCF	CPOISE	
--	-----	-----	-----	
	14.7	197.8092	0.0129	
	50.0	65.9364	0.0130	
	100.0	31.6495	0.0130	
	230.0	13.8813	0.0131	
	460.0	6.8210	0.0132	
	690.0	4.4703	0.0135	
	920.0	3.2968	0.0138	
	1150.0	2.6113	0.0141	
	1380.0	2.1560	0.0145	
	1610.0	1.8316	0.0150	
	1840.0	1.5952	0.0155	
	2070.0	1.4129	0.0161	
	2300.0	1.2700	0.0167	
	2372.0	1.2305	0.0169	
	2530.0	1.1551	0.0174	
	2760.0	1.0621	0.0181	
	2990.0	0.9841	0.0189	
	3220.0	0.9190	0.0196	
	3450.0	0.8638	0.0204	
	4500.0	0.6910	0.0242	
	6000.0	0.5616	0.0293	/ TABLE NO. 01
--				
--	PRES	BG	VISC	
--	PSIA	RB/MSCF	CPOISE	
--	-----	-----	-----	
	14.7	265.0126	0.0133	
	50.0	66.2531	0.0133	
	100.0	33.1266	0.0133	
	230.0	14.4552	0.0134	
	460.0	7.0357	0.0136	
	690.0	4.6493	0.0138	
	920.0	3.4417	0.0140	
	1150.0	2.7227	0.0144	
	1380.0	2.2522	0.0147	
	1610.0	1.9158	0.0151	
	1840.0	1.6702	0.0156	
	2070.0	1.4805	0.0162	
	2300.0	1.3317	0.0167	
	2372.0	1.2927	0.0169	
	2530.0	1.2119	0.0173	
	2760.0	1.1135	0.0180	
	2990.0	1.0325	0.0187	
	3220.0	0.9637	0.0194	
	3450.0	0.9055	0.0201	
	4500.0	0.7228	0.0236	
	6000.0	0.5837	0.0285	/ TABLE NO. 02

The second example defines four dry gas PVT tables assuming NTPVT equals four and NPPVT is greater than or equal to 22 on the [TABDIMS](#) keyword in the [RUNSPEC](#) section. Here table two defaults to table one, and table four defaults to table three.

GAS PVT TABLE FOR DRY GAS			
PVDG			
PRES	BG	VISC	
PSIA	RB/MSCF	CPOISE	
-----	-----	-----	
14.7	197.8092	0.0129	
50.0	65.9364	0.0130	
100.0	31.6495	0.0130	
230.0	13.8813	0.0131	
460.0	6.8210	0.0132	
690.0	4.4703	0.0135	
920.0	3.2968	0.0138	
1150.0	2.6113	0.0141	
1380.0	2.1560	0.0145	
1610.0	1.8316	0.0150	
1840.0	1.5952	0.0155	
2070.0	1.4129	0.0161	
2300.0	1.2700	0.0167	
2372.0	1.2305	0.0169	
2530.0	1.1551	0.0174	
2760.0	1.0621	0.0181	
2990.0	0.9841	0.0189	
3220.0	0.9190	0.0196	
3450.0	0.8638	0.0204	
4500.0	0.6910	0.0242	
6000.0	0.5616	0.0293	/ TABLE NO. 01
			/ TABLE NO. 02
PRES	BG	VISC	
PSIA	RB/MSCF	CPOISE	
-----	-----	-----	
14.7	265.0126	0.0133	
50.0	66.2531	0.0133	
100.0	33.1266	0.0133	
230.0	14.4552	0.0134	
460.0	7.0357	0.0136	
690.0	4.6493	0.0138	
920.0	3.4417	0.0140	
1150.0	2.7227	0.0144	
1380.0	2.2522	0.0147	
1610.0	1.9158	0.0151	
1840.0	1.6702	0.0156	
2070.0	1.4805	0.0162	
2300.0	1.3317	0.0167	
2372.0	1.2927	0.0169	
2530.0	1.2119	0.0173	
2760.0	1.1135	0.0180	
2990.0	1.0325	0.0187	
3220.0	0.9637	0.0194	
3450.0	0.9055	0.0201	
4500.0	0.7228	0.0236	
6000.0	0.5837	0.0285	/ TABLE NO. 03
			/ TABLE NO. 04

There is no terminating “/” for this keyword.

8.3.240 PVDO – OIL PVT PROPERTIES FOR DEAD OIL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PVDO defines the oil PVT properties for dead oil²³¹. If the oil has a constant and uniform dissolved gas concentration, Gas-Oil Ratio (“GOR”), and if the reservoir pressure never drops below the saturation pressure (bubble point pressure), then the model can be run more efficiently by omitting the [GAS](#) and [DISGAS](#) keywords from the [RUNSPEC](#) section, treating the oil as a dead oil, and defining a constant Rs (GOR) value with keyword [RSCONST](#) or [RSCONSTT](#) in the [PROPS](#) section. This results in the model being run as a dead oil problem with no active gas phase. However, OPM Flow takes into account the constant Rs in the calculations and reporting.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A columnar vector of real monotonically increasing down the column values that defines the oil phase pressure.			None
		psia	barsa	atma	
2	OFVF	A columnar vector of real decreasing down the column values that defines the corresponding oil phase formation volume factor.			None
		rb/stb	rm ³ /sm ³	rcc/scc	
3	OVISC	A columnar vector of real increasing down the column values that defines the corresponding oil phase viscosity.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.128: PVDO Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table. See the second example for an illustration on how to use this feature.

See also the [RSCONST](#) and [RSCONSTT](#) keywords to define the constant Rs for dead oil and [PVDO](#) as an alternative keyword to enter the dead oil properties.

²³¹ “Dead” oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

Example

The example below defines two dead oil PVT tables with variable viscosity and compressibility with respect to pressure, and assumes that NTPVT equals two and NPPVT is greater than or equal to eight on the *TABDIMS* keyword in the *RUNSPEC* section.

--	OIL PVT TABLE FOR DEAD OIL		
--			
PVDO			
--	PSAT	BO	VISC
--	PSIA	RB/STB	CPOISE
--	-----	-----	-----
	400	1.0102	1.16
	1200	1.0040	1.164
	2000	0.9960	1.167
	2800	0.9880	1.172
	3600	0.9802	1.177
	4400	0.9724	1.181
	5200	0.9646	1.185
	5600	0.9607	1.19
			/ TABLE NO. 01
--	-----	-----	-----
	800	1.0255	1.14
	1600	1.0172	1.14
	2400	1.0091	1.14
	3200	1.0011	1.14
	4000	0.9931	1.14
	4800	0.9852	1.14
	5600	0.9774	1.14
			/ TABLE NO. 02

The second example defines four dead oil PVT tables with variable viscosity and compressibility with respect to pressure, and assumes that NTPVT equals four and NPPVT is greater than or equal to eight on the *TABDIMS* keyword in the *RUNSPEC* section. Here table two defaults to table one, and table four defaults to table three.

--	OIL PVT TABLE FOR DEAD OIL			
--				
--				
PVDO				
--	PSAT	BO	VISC	
--	PSIA	RB/STB	CPOISE	
--	-----	-----	-----	
	400	1.0102	1.16	
	1200	1.0040	1.164	
	2000	0.9960	1.167	
	2800	0.9880	1.172	
	3600	0.9802	1.177	
	4400	0.9724	1.181	
	5200	0.9646	1.185	
	5600	0.9607	1.19	/ TABLE NO. 01
				/ TABLE NO. 02
	800	1.0255	1.14	
	1600	1.0172	1.14	
	2400	1.0091	1.14	
	3200	1.0011	1.14	
	4000	0.9931	1.14	
	4800	0.9852	1.14	
	5600	0.9774	1.14	/ TABLE NO. 03
				/ TABLE NO. 04

There is no terminating “/” for this keyword.

8.3.241 PVDS – SOLVENT PVT PROPERTIES FOR THE SOLVENT MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PVDS defines the solvent PVT properties for use with **SOLVENT** option. The solvent is treated as an additional dry gas phase within the model. This keyword should only be used if the **SOLVENT** model has been invoked in the **RUNSPEC** section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A columnar vector of real monotonically increasing down the column values that defines the solvent phase pressure.			None
		psia	barsa	atma	
2	GVFV	A columnar vector of real decreasing down the column values that defines the corresponding solvent phase formation volume factor.			None
		rb/Mscf	rm ³ /sm ³	rcc/scc	
3	GVISC	A columnar vector of real increasing down the column values that defines the corresponding solvent phase viscosity.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.129: PVDS Keyword Description

Example

```
--
--      GAS SOLVENT PVT TABLE
--
PVDS
--      PRES      BG      VISC
--      PSIA      RB/MSCF  CPOISE
--      -----
--      700.0      4.4703   0.0135
--      920.0      3.2968   0.0138
--      1150.0     2.6113   0.0141
--      1380.0     2.1560   0.0145
--      1610.0     1.8316   0.0150
--      1840.0     1.5952   0.0155
--      2070.0     1.4129   0.0161
--      2300.0     1.2700   0.0167
--      2372.0     1.2305   0.0169
--      2530.0     1.1551   0.0174
--      2760.0     1.0621   0.0181
--      2990.0     0.9841   0.0189
--      3220.0     0.9190   0.0196
--      3450.0     0.8638   0.0204
--      4500.0     0.6910   0.0242
--      6000.0     0.5616   0.0293
--
--
--      PRES      BG      VISC
--      PSIA      RB/MSCF  CPOISE
--      -----
--      700.0      4.6493   0.0138
--      920.0      3.4417   0.0140
--      1150.0     2.7227   0.0144
--      1380.0     2.2522   0.0147
--      1610.0     1.9158   0.0151
--      1840.0     1.6702   0.0156
--      2070.0     1.4805   0.0162
--      2300.0     1.3317   0.0167
--      2372.0     1.2927   0.0169
--      2530.0     1.2119   0.0173
--      2760.0     1.1135   0.0180
--      2990.0     1.0325   0.0187
--      3220.0     0.9637   0.0194
--      3450.0     0.9055   0.0201
--      4500.0     0.7228   0.0236
--      6000.0     0.5837   0.0285
--
--
--      / TABLE N0. 01
--
--
--      PRES      BG      VISC
--      PSIA      RB/MSCF  CPOISE
--      -----
--      700.0      4.6493   0.0138
--      920.0      3.4417   0.0140
--      1150.0     2.7227   0.0144
--      1380.0     2.2522   0.0147
--      1610.0     1.9158   0.0151
--      1840.0     1.6702   0.0156
--      2070.0     1.4805   0.0162
--      2300.0     1.3317   0.0167
--      2372.0     1.2927   0.0169
--      2530.0     1.2119   0.0173
--      2760.0     1.1135   0.0180
--      2990.0     1.0325   0.0187
--      3220.0     0.9637   0.0194
--      3450.0     0.9055   0.0201
--      4500.0     0.7228   0.0236
--      6000.0     0.5837   0.0285
--
--
--      / TABLE N0. 02
```

The above example defines two solvent PVT tables assuming NTPVT equals two and NPPVT is greater than or equal to 16 on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

There is no terminating “/” for this keyword.

8.3.242 PVTG – GAS PVT PROPERTIES FOR WET GAS WITH VAPORIZED OIL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

PVTG defines the gas PVT properties for wet gas²³². This keyword should be used when the **VAPOL** keyword has been declared in the **RUNSPEC** section indicating that that vaporized oil (more commonly referred to as condensate) is present in the wet gas phase. The keyword may be used for gas-water and oil-water-gas input decks that contain the oil and gas phases.

No.	Name		Description			Default
			Field	Metric	Laboratory	
1	PRESS		A real monotonically increasing down the column vector that defines the gas phase pressure, associated with the saturated condensate-gas ratio ("CGR") or Rv, the gas formation volume factor and the gas viscosity for the corresponding pressure for the stated saturated RVS. For a given PRESS the variability of the gas formation volume factor and the gas viscosity with respect to the under-saturated Rv is optionally included as a sub table under RVU, FVFU and VISU columns, that is it is not necessary to repeat PRESS for each sub table entry. However, each sub table must be terminated by a "I". The under saturated Rv entries are optional, except for perhaps the last PRESS entry to define the PVT properties above the initial saturation pressure.			None
			psia	barsa	atma	
2	RVS	RVU	A columnar vector of real positive number for both the saturated (RVS) and under saturated (RVU) Rv sub table entries. The RVS entry on the main table is the saturated CGR at the pressure indicated by PRESS and may be increasing or decreasing in value as PRESS varies. Subsequent under-saturated Rv for a sub table at the given PRESS, as defined by RVU, are monotonically decreasing for entries in a given sub table.			None
			stb/Mscf	sm ³ /sm ³	scc/scc	
3	FVFS	FVFU	A columnar vector of real decreasing down the column values that defines the corresponding gas phase formation volume factor for a given pressure (PRESS) and for a given Rv (either RVS or RVU).			None
			rb/Mscf	rm ³ /sm ³	rcc/scc	
4	VISS	VISU	VISS a columnar vector of real increasing down the column values that defines the corresponding gas phase viscosity for a given pressure (PRESS) and for a given RVS. VISU a columnar vector of real decreasing from VISS down the column values that defines the corresponding gas phase viscosity for a given pressure (PRESS) and for a given RVU.			None

²³² Natural gas that contains significant heavy hydrocarbons such as propane, butane and other liquid hydrocarbons is known as wet gas or rich gas. The general rule of thumb is if the gas contains less methane (typically less than 85% methane) and more ethane, and other more complex hydrocarbons, it is labeled as wet gas. Wet gas normally has GOR's less than 100,000 scf/stb or 18,000 Sm³/m³, with the condensate having a gravity greater than 50 oAPI.

No.	Name		Description			Default
			Field	Metric	Laboratory	
			cP	cP	cP	
Notes: 1) The keyword is followed by NTPVT tables as declared on the TABDIMS keyword in the RUNSPEC section. 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows and NRPVT Rv entries as declared on the TABDIMS keyword in the RUNSPEC section. 3) Apart from the PRESS data there must be same number of entries for each column. 4) Each sub table defining the under saturated gas properties must be terminated by “/”. 5) Each main table is terminated by a single “/” on a separate line and there is no “/” terminator for the keyword.						

Table 8.130: PVTG Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table. See the second example for an illustration on how to use this feature.

Note

If the [VAPWAT](#) keyword in the [RUNSPEC](#) section is also present in the input deck, then the [PVTG](#) keyword in the [PROPS](#) section should be used to define the gas properties as function of pressure and [RV](#), assuming water-saturated gas. Also, in this case, the [PVTGW](#) keyword, also in the [PROPS](#) section, should also be in the input deck. In this case, [PVTGW](#) defines the gas properties as function of pressure and [RVW](#), assuming oil-saturated gas.

Example

The first example defines two wet gas PVT tables assuming NTPVT equals two, NPPVT is greater than or equal to eight, and NRPVT greater than or equal to two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```

--
--      GAS PVT TABLE FOR WET GAS WITH VAPORIZED OIL
--
PVTG
--      PRES      RV      BG      VISC
--      BARSA      SM^3/SM^3  RM^3/SM^3  CPOISE
--      -----
--      20      0.000132  0.042340  0.01344
--              0      0.042310  0.01389      /
--      40      0.000124  0.020460  0.01420
--              0      0.020430  0.01450      /
--      60      0.000126  0.013280  0.01526
--              0      0.013250  0.01532      /
--      80      0.000135  0.009770  0.01660
--              0      0.009730  0.01634      /
--      100     0.000149  0.007730  0.01818
--              0      0.007690  0.01752      /
--      120     0.000163  0.006426  0.01994
--              0      0.006405  0.01883      /
--      140     0.000191  0.005541  0.02181
--              0      0.005553  0.02021      /
--      160     0.000225  0.004919  0.02370
--              0      0.004952  0.02163      /
--                                     / TABLE NO. 1
--
--      PRES      RV      BG      VISC
--      BARSA      SM^3/SM^3  RM^3/SM^3  CPOISE
--      -----
--      20      0.000132  0.042340  0.01344      /
--      40      0.000124  0.020460  0.01420      /
--      60      0.000126  0.013280  0.01526      /
--      80      0.000135  0.009770  0.01660      /
--      100     0.000149  0.007730  0.01818      /
--      120     0.000163  0.006426  0.01994      /
--      140     0.000191  0.005541  0.02181      /
--      160     0.000225  0.004919  0.02370
--              0      0.004952  0.02163      /
--                                     / TABLE NO. 2

```

The second example defines four wet gas PVT tables assuming NTPVT equals four, NPPVT is greater than or equal to eight, and NRPVT greater than or equal to two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section. Here table two defaults to table one, and table four defaults to table three.

```
--
--      GAS PVT TABLE FOR WET GAS WITH VAPORIZED OIL
--
PVTG
--      PRES      RV      BG      VISC
--      BARSA      SM^3/SM^3      RM^3/SM^3      CPOISE
--      -----
--      20      0.000132      0.042340      0.01344
--              0      0.042310      0.01389      /
--      40      0.000124      0.020460      0.01420
--              0      0.020430      0.01450      /
--      60      0.000126      0.013280      0.01526
--              0      0.013250      0.01532      /
--      80      0.000135      0.009770      0.01660
--              0      0.009730      0.01634      /
--      100     0.000149      0.007730      0.01818
--              0      0.007690      0.01752      /
--      120     0.000163      0.006426      0.01994
--              0      0.006405      0.01883      /
--      140     0.000191      0.005541      0.02181
--              0      0.005553      0.02021      /
--      160     0.000225      0.004919      0.02370
--              0      0.004952      0.02163      /
--                                     / TABLE NO. 1
--                                     / TABLE NO. 2
--
--      PRES      RV      BG      VISC
--      BARSA      SM^3/SM^3      RM^3/SM^3      CPOISE
--      -----
--      20      0.000132      0.042340      0.01344      /
--      40      0.000124      0.020460      0.01420      /
--      60      0.000126      0.013280      0.01526      /
--      80      0.000135      0.009770      0.01660      /
--      100     0.000149      0.007730      0.01818      /
--      120     0.000163      0.006426      0.01994      /
--      140     0.000191      0.005541      0.02181      /
--      160     0.000225      0.004919      0.02370
--              0      0.004952      0.02163      /
--                                     / TABLE NO. 3
--                                     / TABLE NO. 4
```

Notice that in both examples there is no terminating “/” for this keyword only for a table and a sub table.

8.3.243 PVTGW – GAS PVT PROPERTIES FOR DRY GAS WITH VAPORIZED WATER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

PVTGW defines the gas PVT properties for dry gas²³³ with vaporized water. This keyword should be used when the **VAPWAT** keyword has been declared in the **RUNSPEC** section indicating that vaporized water is present in the dry gas phase. The keyword may be used for gas-water and oil-water-gas input decks that contain the dry gas and vaporized water phases.

Note

This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by declaring that vaporized water is present in the run.

No.	Name		Description			Default
			Field	Metric	Laboratory	
1	PRESS		A real monotonically increasing down the column vector that defines the gas phase pressure, associated with the corresponding saturated water-gas ratio ("WGR") or R_w, the gas formation volume factor, and the gas viscosity for the stated saturated RWS. For a given PRESS the variability of the gas formation volume factor and the gas viscosity with respect to the under-saturated R_w is optionally included as a sub table under RWU, FVFU and VISU columns, that is it is not necessary to repeat PRESS for each sub table entry. However, each sub table must be terminated by a "/". The under saturated R_w entries are optional, except for perhaps the last PRESS entry to define the PVT properties above the initial saturation pressure.			None
			psia	barsa	atma	
2	RWS	RWU	A columnar vector of real positive numbers for both the saturated (RWS) and under saturated (RWU) R_w sub table entries. The RWS entry on the main table is the saturated WGR at the pressure indicated by PRESS and may be increasing or decreasing in value as PRESS varies. Subsequent under-saturated R_w for a sub table at the given PRESS, as defined by RWU, are monotonically decreasing for entries in a given sub table.			None
			stb/Mscf	sm ³ /sm ³	scc/scc	
3	FVFS	FVFU	A columnar vector of real decreasing down the column values that defines the corresponding gas phase formation volume factor for a given pressure (PRESS) and for a given R_w (either RWS or RWU).			None

²³³ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

No.	Name		Description			Default
			Field	Metric	Laboratory	
			rb/Mscf	rm ³ /sm ³	rcc/scc	
4	VISS	VISU	VISS a columnar vector of real increasing down the column values that defines the corresponding gas phase viscosity for a given pressure (PRESS) and for a given RWS. VISU a columnar vector of real decreasing from VISS down the column values that defines the corresponding gas phase viscosity for a given pressure (PRESS) and for a given RWU.			None
			cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows and NRPVT Rw entries as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Apart from the PRES data there must be same number of entries for each column.
- 4) Each sub table defining the under saturated gas properties must be terminated by “/”.
- 5) Each main table is terminated by a single “/” on a separate line and there is no “/” terminator for the keyword.

Table 8.131: PVTGW Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table.

Note

If both the [VAPIWAT](#) and [VAPOIL](#) keywords have been declared in the [RUNSPEC](#) section indicating that both vaporized water and vaporized oil are present in the wet gas, then the [PVTGW](#) keyword should be used along with the [PVTG](#) keyword in the [PROPS](#) section to fully define the wet gas PVT properties. The [PVTGW](#) keyword should be used to define the gas properties as a function of pressure and water-gas ratio ([RVW](#)), assuming oil-saturated gas. The [PVTG](#) keyword should be used to define the gas properties as a function of pressure and oil-gas ratio ([RV](#)), assuming water-saturated gas.

Alternatively, the [PVTGWO](#) keyword in the [PROPS](#) section may be used instead of the [PVTGW](#) and [PVTG](#) keywords to fully define the wet gas PVT properties.

See also the [PVTG](#) keyword in the [PROPS](#) section that defines the wet gas PVT for when vaporized oil is present in the gas phase. Alternatively, the [PVTGWO](#) keyword in the [PROPS](#) section may be utilized instead of [PVTG](#) and [PVTGW](#) to fully define the wet gas PVT properties, for when both vaporized oil and water are present in the gas phase.

Example

```
--
--      GAS PVT TABLE FOR DRY GAS WITH VAPORIZED WATER (OPM FLOW KEYWORD)
--
PVTGW
--      PRES      RW      BG      VISC
--      PSIA      SM^3/SM^3  RM^3/SM^3  CPOISE
--      -----
--      300      0.000479  0.042340  0.01344
--              0          0.042310  0.01389      /
--      600      0.000469  0.020460  0.01420
--              0          0.020430  0.01450      /
--      900      0.000403  0.013280  0.01526
--              0          0.013250  0.01532      /
--      1200     0.000354  0.009770  0.01660
--              0          0.009730  0.01634      /
--      1500     0.000272  0.007730  0.01818
--              0          0.007690  0.01752      /
--      1800     0.000225  0.006426  0.01994
--              0          0.006405  0.01883      /
--      2100     0.000191  0.005541  0.02181
--              0          0.005553  0.02021      /
--      2400     0.000163  0.004919  0.02370
--              0          0.004952  0.02163      /
--                                     / TABLE NO. 1
--
--      PRES      RW      BG      VISC
--      PSIA      SM^3/SM^3  RM^3/SM^3  CPOISE
--      -----
--      300      0.000479  0.042340  0.01344      /
--      600      0.000469  0.020460  0.01420      /
--      900      0.000403  0.013280  0.01526      /
--      1200     0.000354  0.009770  0.01660      /
--      1500     0.000272  0.007730  0.01818      /
--      1800     0.000225  0.006426  0.01994      /
--      2100     0.000191  0.005541  0.02181      /
--      2400     0.000163  0.004919  0.02370      /
--                                     / TABLE NO. 2
```

The above example defines two dry gas PVT tables assuming NTPVT equals two and NPPVT is greater than or equal to eight on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

Notice that there is no terminating “/” for this keyword only for a table and a sub table.

8.3.244 PVTGWO – GAS PVT PROPERTIES FOR WET GAS WITH VAPORIZED WATER AND OIL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

PVTGWO defines the gas PVT properties for wet gas²³⁴ with vaporized water and oil. This keyword should be used when the **VAPOIL** and **VAPWAT** keywords have been declared in the **RUNSPEC** section indicating that vaporized oil and water are present in the wet gas phase. The keyword may be used for oil-water-gas input decks that contain the wet gas with vaporized oil and water phases.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Note

This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by declaring that vaporized water is present in the run.

No.	Name		Description			Default
			Field	Metric	Laboratory	
1	PRESS		A real monotonically increasing down the column vector that defines the gas phase pressure, associated with the saturated water-gas ratio (“WGR”) or R_w , the saturated condensate-gas ratio (“CGR”) or R_v , the gas formation volume factor, and the gas viscosity for the corresponding pressure for the stated saturated RWS. For a given PRESS the variability of the gas formation volume factor and the gas viscosity with respect to the under-saturated R_w and R_v is optionally included as a sub table under RWU, RVU, FVFU and VISU columns, that is it is not necessary to repeat PRESS for each sub table entry. However, each sub table must be terminated by a “/”. The under saturated R_w and R_v entries are optional, except for perhaps the last PRESS entry to define the PVT properties above the initial saturation pressure.			None
			psia	barsa	atma	
2	RWS	RWU	A columnar vector of real positive numbers for both the saturated (RWS) and under saturated (RWU) R_w sub table entries. The RWS entry on the main table is the saturated WGR at the pressure indicated by PRESS and may be increasing or decreasing in value as PRESS varies. Subsequent under-saturated R_w for a sub table at the given PRESS, as defined by RWU, are monotonically decreasing for entries in a given sub table.			None
			stb/Mscf	sm^3/sm^3	rcc/scc	

²³⁴ Natural gas that contains significant heavy hydrocarbons such as propane, butane and other liquid hydrocarbons is known as wet gas or rich gas. The general rule of thumb is if the gas contains less methane (typically less than 85% methane) and more ethane, and other more complex hydrocarbons, it is labeled as wet gas. Wet gas normally has GOR's less than 100,000 scf/stb or 18,000 Sm^3/m^3 , with the condensate having a gravity greater than 50 oAPI.

No.	Name		Description			Default
			Field	Metric	Laboratory	
3	RVS	RVU	A columnar vector of real positive numbers for both the saturated (RVS) and under saturated (RVU) Rv sub table entries. The RVS entry on the main table is the saturated CGR at the pressure indicated by PRESS and may be increasing or decreasing in value as PRESS varies. Subsequent under-saturated Rv for a sub table at the given PRESS, as defined by RVU, are monotonically decreasing for entries in a given sub table.			None
			stb/Mscf	sm ³ /sm ³	rcc/scc	
4	FVFS	FVFU	A columnar vector of real decreasing down the column values that defines the corresponding gas phase formation volume factor for a given pressure (PRESS) and for a given Rw (either RWS or RWU) and Rv (either RVS or RVU).			None
			rb/Mscf	rm ³ /sm ³	rcc/scc	
5	VISS	VISU	VISS a columnar vector of real increasing down the column values that defines the corresponding gas phase viscosity for a given pressure (PRESS) and for a given RWS and RVS. VISU a columnar vector of real decreasing from VISS down the column values that defines the corresponding gas phase viscosity for a given pressure (PRESS) and for a given RWU and RVU.			None
			cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows and NRPVT Rw entries as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 3) Apart from the PRES data there must be same number of entries for each column.
- 4) Each sub table defining the under saturated gas properties must be terminated by “/”.
- 5) Each main table is terminated by a single “/” on a separate line and there is no “/” terminator for the keyword.

Table 8.132: PVTGWO Keyword Description

See also the [PVTG](#) keyword in the [PROPS](#) section that defines the wet gas PVT properties when vaporized oil is present in the gas phase, and the [PVTGW](#) keyword in the [PROPS](#) section that defines the dry gas PVT properties when vaporized water is present in the gas phase. As an alternative to using the [PVTGWO](#) keyword, the [PVTGW](#) and [PVTG](#) keywords may be combined to fully define the wet gas PVT properties if both vaporized water and vaporized oil are present in the gas phase.

8.3.245 PVTO – OIL PVT PROPERTIES FOR LIVE OIL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

PVTO defines the oil PVT properties for live oil²³⁵ and the keyword should only be used if there is both oil and gas phases in the model. This keyword should be used when the **DISGAS** keyword has been declared in the **RUNSPEC** section indicating that dissolved gas (more commonly referred to as solution gas) is present in the oil. The keyword may be used for oil-water and oil-water-gas input decks.

No.	Name		Description			Default
			Field	Metric	Laboratory	
1	RS		A real monotonically increasing down the column values that defines the saturated gas-oil ratio (“GOR”) or Rs, that defines the oil formation volume factor and the oil viscosity for the tabulated corresponding pressure for stated saturated RS. For a given RS the variability of the oil formation volume factor and the oil viscosity with respect to the saturated RS and pressure is optionally included as a sub table under PRSU, FVFU and VISU columns, that is it is not necessary to repeat RS for each sub table entry. However, each sub table must be terminated by a “/”. The under-saturated PRSU entries are optional, except for perhaps the last RS entry to define the PVT properties above the initial saturation pressure. If there are no following under-saturated PRSU entries then the RS entry row should be terminated by a “/”, if there are under-saturated PRSU entries then the last PRSU entry row should be terminated by a “/”.			None
			Mscf/stb	sm ³ /sm ³	scc/scc	
2	PRSS	PRSU	PRSS is a real columnar vector of real monotonically increasing down the column values that defines the oil phase saturation pressure (bubble-point pressure), that defines the oil formation volume factor and the oil viscosity for the corresponding PRSS pressure for a given saturated RS. PRSU is a real columnar vector of real monotonically increasing down the column values that defines the oil phase under-saturated pressure that defines the oil formation volume factor and the oil viscosity for the corresponding PRSU pressure for a given saturated RS. Note that PRSU should be greater than PRSS.			None
			psia	barsa	atma	
3	FVFS	FVFU	FVFS is a columnar vector of real increasing down the column values that defines the corresponding oil phase saturated formation volume factor for a given pressure (PRSS) and for a given RS. FVFU is a columnar vector of real decreasing down the column values that defines the corresponding oil phase under-saturated formation volume factor for a given pressure (PRSU) and for a given RS.			None
			rb/stb	rm ³ /sm ³	rcc/scc	

²³⁵ “Live” oil is oil that contains gas in solution, which is normally the case for most conventional oil reservoirs. However, for oil reservoirs classified as heavy oil reservoirs, the in situ dissolved gas may be negligible and oil would then be classified as gas-free oil which is commonly referred to as “dead” oil.

No.	Name		Description			Default
			Field	Metric	Laboratory	
4	VISS	VISU	VISS a columnar vector of real increasing down the column values that defines the corresponding oil phase saturated viscosity for a given pressure (PRSS) and for a given <i>RS</i> . If this is the only entry for a given <i>RS</i> and PRSS then the record should be terminate by a “/”.			None
			VISU a columnar vector of real decreasing from VISS down the column values that defines the corresponding oil phase under-saturated viscosity for a given pressure (PRSU) and for a given <i>RS</i> . If this is the only entry for a given <i>RS</i> and PRSU then the record should be terminate by a “/”.			
			cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows and NRPVT Rs entries as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 3) Apart from the *RS* data there must be same number of entries for each column.
- 4) Each sub table defining the under saturated oil properties must be terminated by “/”.
- 5) Each main table is terminated by a single “/” on a separate line and there is no “/” terminator for the keyword.

Table 8.133: PVT0 Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table. See the second example for an illustration on how to use this feature.

Example

The first example defines live oil PVT tables assuming NTPVT equals two, NPPVT is greater than or equal to two, and NRPVT is greater than or equal to 18 on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--      OIL PVT TABLE FOR LIVE OIL
--
PVT0
--      RS          PSAT          BO          VISC
--      MSCF/STB    PSIA          RB/STB      CPOISE
--      -----
--      0.0010      14.7          1.05340     1.7230 /
--      0.0890      500.0         1.08890     1.1670 /
--      0.2060      1000.0        1.13850     0.8570 /
--      0.3360      1500.0        1.19640     0.6840 /
--      0.4750      2000.0        1.26110     0.5750 /
--      0.5480      2250.0        1.29570     0.5340 /
--      0.6220      2500.0        1.33160     0.5000 /
--      0.6980      2750.0        1.36890     0.4700 /
--      0.7750      3000.0        1.40740     0.4450 /
--      0.8530      3250.0        1.44710     0.4220 /
--      0.9330      3500.0        1.48790     0.4020 /
--      1.0140      3750.0        1.52980     0.3840 /
--      1.0960      4000.0        1.57280     0.3680 /
--      1.1800      4258.0        1.61760     0.3530 /
--      1.2630      4500.0        1.66190     0.3400 /
--      1.3480      4750.0        1.70780     0.3280 /
--      1.4340      5000.0        1.75480     0.3170 /
--      1.6060      5500.0        1.85020     0.2980 /
--                      6242.0        1.83040     0.3186 /
--                                     / TABLE NO. 1
--
--
--      RS          PSAT          BO          VISC
--      MSCF/STB    PSIA          RB/STB      CPOISE
--      -----
--      0.0010      14.7          1.05340     1.7230 /
--      0.0390      250.0         1.06830     1.4220 /
--      0.0890      500.0         1.08890     1.1670 /
--      0.1460      750.0         1.11250     0.9850 /
--      0.2060      1000.0        1.13850     0.8570 /
--      0.2700      1250.0        1.16660     0.7590 /
--      0.3360      1500.0        1.19640     0.6840 /
--      0.4050      1750.0        1.22800     0.6240 /
--      0.4750      2000.0        1.26110     0.5750 /
--      0.5480      2250.0        1.29570     0.5340 /
--      0.6220      2500.0        1.33160     0.5000 /
--      0.6980      2750.0        1.36890     0.4700 /
--      0.7750      3000.0        1.40740     0.4450 /
--      0.8530      3250.0        1.44710     0.4220 /
--      0.9330      3500.0        1.48790     0.4020 /
--      1.0140      3750.0        1.52980     0.3840 /
--      1.0960      4000.0        1.57280     0.3680 /
--      1.1800      4258.0        1.61760     0.3530 /
--      1.2630      4500.0        1.66190     0.3400 /
--      1.3480      4750.0        1.70780     0.3280 /
--      1.4340      5000.0        1.75480     0.3170 /
--      1.6060      5500.0        1.85020     0.2980 /
--                      6242.0        1.83040     0.3186 /
--                                     / TABLE NO. 2
```

Notice that there must be at least two entries for the last Rs value to enable the simulator to interpolate over the undersaturated pressure region.

The second example defines live oil PVT tables assuming NTPVT equals four, NPPVT is greater than or equal to two, and NRPVT is greater than or equal to 13 on the [TABDIMS](#) keyword in the [RUNSPEC](#) section. Here, tables two to four all default to table number one.

```
--
--      OIL PVT TABLE FOR LIVE OIL
--
--
PVT0
--      RS      PSAT      BO      VISC
--      MSCF/STB  PSIA      RB/STB  CPOISE
--      -----  -
--      0.0010      14.7      1.05340  1.7230  /
--      0.0890      500.0      1.08890  1.1670  /
--      0.2060     1000.0      1.13850  0.8570  /
--      0.3360     1500.0      1.19640  0.6840  /
--      0.4750     2000.0      1.26110  0.5750  /
--      0.5480     2250.0      1.29570  0.5340  /
--      0.6220     2500.0      1.33160  0.5000  /
--      0.7750     3000.0      1.40740  0.4450  /
--      0.9330     3500.0      1.48790  0.4020  /
--      1.0960     4000.0      1.57280  0.3680  /
--      1.2630     4500.0      1.66190  0.3400  /
--      1.4340     5000.0      1.75480  0.3170  /
--      1.6060     5500.0      1.85020  0.2980  /
--              6242.0      1.83040  0.3186  /
--
--      / TABLE NO. 1
--      / TABLE NO. 2
--      / TABLE NO. 3
--      / TABLE NO. 4
```

Again, note that there is no terminating “/” for this keyword only for a table and a sub table.

8.3.246 PVT SOL – OIL PVT PROPERTIES FOR LIVE OIL VERSUS CO₂ MASS FRACTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

PVT SOL defines the live oil PVT properties as a function of CO₂ mass fraction. The keyword automatically invokes the simulator's CO₂ Dynamic EOR Model²³⁶, that uses a fourth component to model the injected CO₂, for use in evaluating CO₂ Enhanced Oil Recovery ("EOR") projects. Normally CO₂ EOR projects are evaluated via compositional simulators to account for the mass transfer of the various components and phases. Unfortunately, compositional models are computationally expensive compared to the black-oil approach, which for field studies is challenging, especially if an ensemble approach is being used to capture the uncertainties. Previous extended black-oil formulations often poorly represent the PVT properties of the oil-CO₂ mixtures, resulting in poor agreement with the compositional formulation.

To overcome the limitations of the standard four component black-oil formulation, OPM Flow uses an improved extended black-oil formulation, the CO₂ Dynamic EOR Model, with the black-oil properties dependent on the fraction of CO₂ in the cell, as described by the **PVT SOL** keyword. This approach models the oil-CO₂ mixture more accurately and thus give results closer to the compositional simulator. Data for the keyword should normally be derived from laboratory or numerical slim-tube experiments based on one-dimensional compositional Equation Of State ("EOS") simulations.

Note, if this keyword is absent from the input deck then the CO₂ Standard EOR Model is used instead.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	CO2	A real monotonically increasing down the column values that stipulates the CO ₂ mass fraction, that defines the oil and gas properties, formation volume factor, viscosity etc., for the tabulated corresponding pressure for the stated CO ₂ mass fraction. CO ₂ should be greater than or equal to zero and less than or equal to one. Note it is not necessary to repeat the value of CO ₂ for the pressure column. However, for a given CO ₂ mass fraction, the last pressure entry (PRESS) of the sub table should be terminated by a "/".			None
		dimensionless	dimensionless	dimensionless	
2	PRESS	PRESS is a real columnar vector of real monotonically increasing down the column values that defines the oil phase saturation pressure (bubble-point pressure), that defines the oil formation volume factor and the oil viscosity for the corresponding PRESS pressure for a given saturated value of CO ₂ .			None
		psia	barsa	atma	
3	OFVF	OFVF is a columnar vector of real increasing down the column values that defines the corresponding oil phase saturated formation volume factor for a given pressure (PRESS) and for a given saturated value of CO ₂ .			None
		rb/stb	rm ³ /sm ³	rcc/scc	

²³⁶ T. H. Sandve, O. Sæviareid and I. Aavatsmark: "Improved Extended Blackoil Formulation – for CO₂ EOR Simulations." in ECMOR XVII – The 17th European Conference on the – Mathematics of Oil Recovery, September 2020.

No.	Name		Description			Default
			Field	Metric	Laboratory	
4		GFVF	GFVF is a columnar vector of real decreasing down the column values that defines the corresponding gas phase saturated formation volume factor for a given pressure (PRESS) and for a given saturated value of CO ₂ .			None
			rb/Mscf	rm ³ /sm ³	rcc/scc	
5		RS	RS is a real monotonically increasing down the column values that defines the saturated gas-oil ratio ("GOR") or Rs, for the given value of PRESS and for a given saturated value of CO ₂ .			None
			Mscf/stb	sm ³ /sm ³	scc/scc	
6		RV	RV is a real monotonically increasing down the column values that defines the saturated condensate-gas ratio ("CGR") or Rv, for the given value of PRESS and for a given saturated value of CO ₂ .			None
			stb/Mscf	sm ³ /sm ³	scc/scc	
7		XVOL	XVOL is a real positive value greater than or equal to zero and less than or equal to one, that stipulates the volumetric fraction of CO ₂ in the oil phase, that is: $XVOL = \frac{Volume_{oil}(CO_2)}{Volume_{oil}(CO_2) + Volume_{oil}(Gas) + Volume_{oil}(Oil)}$ where: Volume _{oil} (CO ₂) is the surface volume of CO ₂ in the oil phase, Volume _{oil} (Gas) is the surface volume of gas in the oil phase, and Volume _{oil} (Oil) is the surface volume of oil in the oil phase.			None
			dimensionless	dimensionless	dimensionless	
8		YVOL	YVOL is a real positive value greater than or equal to zero and less than or equal to one, that defines the volumetric fraction of CO ₂ in the gas phase, that is: $YVOL = \frac{Volume_{gas}(CO_2)}{Volume_{gas}(CO_2) + Volume_{gas}(Gas) + Volume_{gas}(Oil)}$ where: Volume _{gas} (CO ₂) is the surface volume of CO ₂ in the gas phase, Volume _{gas} (Gas) is the surface volume of gas in the gas phase, and Volume _{gas} (Oil) is the surface volume of oil in the gas phase.			None
			dimensionless	dimensionless	dimensionless	
9		OVISC	OVISC is a columnar vector of real decreasing down the column values that defines the corresponding oil phase saturated viscosity for a given pressure (PRESS) and for a given saturated value of CO ₂ .			None
			cP	cP	cP	

No.	Name		Description			Default
			Field	Metric	Laboratory	
10		GVISC	GVISC is a columnar vector of real increasing down the column values that defines the corresponding gas phase saturated viscosity for a given pressure (PRESS) and for a given saturated value of CO2.			None
			cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows and NRPVT Rs entries as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Except for the CO2 column, there must be same number of entries for each column.
- 4) Each sub table defining the properties for a given value of CO2 must be terminated by "/" and the keyword itself is terminated by "/".

Table 8.134: PVTSQL Keyword Description

See also the [PVDS](#) keyword in the [PROPS](#) section that can also be used to model CO2 injection, using the [SOLVENT](#) model (CO2 Standard EOR), this is the conventional approach, and does not take into account black-oil properties being dependent on the fraction of CO2.

Note

In the CO2 Dynamic EOR Model the oil properties within a cell are dependent on the CO2 saturation in the cell. However, the model is not restricted to just CO2 utilization, as long as the appropriate oil-injection-gas dependent properties are entered in the model via the [PVTSQL](#) keyword.

Thus, N2 or other miscible gases may be used instead of CO2, provided due care is taken in generating the required data for the [PVTSQL](#) keyword.

Another example would be a gas-condensate re-cycling project where the produced gas is stripped of the liquids and then re-injected back into the reservoir in order to maintain the reservoir pressure. Here, the CO2 fraction in the [PVTSQL](#) keyword would be replaced by the lean injected gas fraction. Again the other properties would be derived from laboratory or numerical slim-tube experiments based on one-dimensional compositional [EOS](#) simulations.

As an aside, the scope and accuracy of gas condensate modeling using black-oil reservoir simulators is well established in the industry; both depletion and gas cycling above the dew point can be modeled and yield an adequate match with the results from multi-component compositional simulators. The main inadequacy is with the treatment of gas injection below the dew point where the primary compositional effect, the stripping of liquid components is inversely proportional to their molecular weights, is completely ignored. The standard black-oil model assumes that the saturated hydrocarbon fluid properties are functions of pressure only and disregards any compositional dependence in the saturated fluid PVT properties. Thus, when dry gas is injected into a condensate below its dew point the gas continues to re-vaporize liquid at a rate governed only by the ambient pressure. The vapor saturates over a zone whose thickness is of the order of one grid block; in particular all the liquid in the vicinity of the injectors evaporates rapidly. Results obtained with fully compositional simulation models suggest that liquid saturation profiles would vary more slowly with increasing distance from the gas injectors. This compositional effect can thus be modeled via the CO2 Dynamic EOR Model, although there are no test cases at this stage.

Example

--	OIL PVT PROPERTIES FOR LIVE OIL VERSUS CO2 MASS FRACTION									
--										
PVTSOL										
-- CO2	PSAT	BO	BG	RS	RV	XVOL	YVOL	OIL VISC	GAS VISC	
-- MFRAC	PSIA	RB/STB	RB/MSCF	MSCF/STB	STB/MSCF			CPOISE	CPOISE	
0.000	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.0000	0.0000	2.0340e-01	1.3016e-02	
	725.18	1.0655	6.0855e+00	0.1302	1.4432e-04	0.0000	0.0000	2.0340e-01	1.3016e-02	
	870.22	1.0806	5.0340e+00	0.1591	1.9594e-04	0.0000	0.0000	1.9973e-01	1.3222e-02	
	1015.26	1.0961	4.2814e+00	0.1886	2.8389e-04	0.0000	0.0000	1.9609e-01	1.3452e-02	
	1160.30	1.1121	3.7174e+00	0.2189	4.1914e-04	0.0000	0.0000	1.9246e-01	1.3708e-02	
	1305.33	1.1285	3.2799e+00	0.2499	6.0779e-04	0.0000	0.0000	1.8886e-01	1.3994e-02	
	1450.37	1.1454	2.9314e+00	0.2819	8.5105e-04	0.0000	0.0000	1.8527e-01	1.4309e-02	
	1595.41	1.1629	2.6479e+00	0.3148	1.1490e-03	0.0000	0.0000	1.8171e-01	1.4657e-02	
	1740.45	1.1809	2.4133e+00	0.3487	1.5030e-03	0.0000	0.0000	1.7819e-01	1.5037e-02	
	1885.49	1.1996	2.2163e+00	0.3838	1.9155e-03	0.0000	0.0000	1.7469e-01	1.5450e-02	
	2030.52	1.2189	2.0490e+00	0.4200	2.3899e-03	0.0000	0.0000	1.7122e-01	1.5897e-02	
	2175.56	1.2389	1.9055e+00	0.4574	2.9302e-03	0.0000	0.0000	1.6779e-01	1.6377e-02	
	2320.60	1.2597	1.7812e+00	0.4962	3.5412e-03	0.0000	0.0000	1.6440e-01	1.6890e-02	
	2465.64	1.2812	1.6728e+00	0.5364	4.2273e-03	0.0000	0.0000	1.6104e-01	1.7435e-02	
	2610.67	1.3037	1.5775e+00	0.5782	4.9942e-03	0.0000	0.0000	1.5772e-01	1.8011e-02	
	2755.71	1.3270	1.4933e+00	0.6216	5.8471e-03	0.0000	0.0000	1.5443e-01	1.8617e-02	
	2900.75	1.3514	1.4185e+00	0.6668	6.7918e-03	0.0000	0.0000	1.5117e-01	1.9253e-02	
	3045.79	1.3768	1.3517e+00	0.7139	7.8347e-03	0.0000	0.0000	1.4795e-01	1.9917e-02	
	3137.32	1.3937	1.3132e+00	0.7450	8.5549e-03	0.0000	0.0000	1.4592e-01	2.0353e-02	
	4061.05	1.3705	1.3132e+00	0.7450	8.5549e-03	0.0000	0.0000	1.5939e-01	2.0353e-02	/
0.010	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.0464	0.0381	2.0301e-01	1.3203e-02	
	725.18	1.0670	6.0041e+00	0.1341	1.4845e-04	0.0464	0.0381	2.0301e-01	1.3203e-02	
0.100	3135.04	1.4082	1.2950e+00	0.7755	8.9113e-03	0.0298	0.0277	1.4422e-01	2.0743e-02	
	4061.05	1.3840	1.2950e+00	0.7755	8.9113e-03	0.0298	0.0277	1.5772e-01	2.0743e-02	/
0.200	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.3426	0.3023	1.9984e-01	1.4740e-02	
	725.18	1.0795	5.3329e+00	0.1666	1.8243e-04	0.3426	0.3023	1.9984e-01	1.4740e-02	
0.300	3116.25	1.5281	1.1444e+00	1.0268	1.1849e-02	0.2492	0.2431	1.3023e-01	2.3961e-02	
	4061.05	1.4955	1.1444e+00	1.0268	1.1849e-02	0.2492	0.2431	1.4400e-01	2.3961e-02	/
0.400	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.5303	0.4898	1.9699e-01	1.5838e-02	
	725.18	1.0909	4.8402e+00	0.1964	2.1919e-04	0.5303	0.4898	1.9699e-01	1.5838e-02	
0.500	4061.05	1.6501	1.0017e+00	1.3720	1.6728e-02	0.4220	0.4240	1.2963e-01	2.7757e-02	/
	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.6486	0.6159	1.9471e-01	1.6588e-02	
0.600	725.18	1.1003	4.4992e+00	0.2210	2.5440e-04	0.6486	0.6159	1.9471e-01	1.6588e-02	
	4061.05	1.8529	8.8189e-01	1.8200	2.4541e-02	0.5491	0.5597	1.1636e-01	3.2195e-02	/
0.700	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.7303	0.7059	1.9292e-01	1.7131e-02	
	725.18	1.1079	4.2506e+00	0.2413	2.8832e-04	0.7303	0.7059	1.9292e-01	1.7131e-02	
0.800	4061.05	2.1298	7.8515e-01	2.4251	3.7664e-02	0.6468	0.6616	1.0441e-01	3.7755e-02	/
	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.7905	0.7733	1.9156e-01	1.7543e-02	
0.900	725.18	1.1140	4.0628e+00	0.2581	3.2217e-04	0.7905	0.7733	1.9156e-01	1.7543e-02	
	4061.05	2.5282	7.1557e-01	3.2879	6.0741e-02	0.7245	0.7379	9.3885e-02	4.5038e-02	/
1.000	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.8372	0.8259	1.9061e-01	1.7868e-02	
	725.18	1.1188	3.9168e+00	0.2720	3.5813e-04	0.8372	0.8259	1.9061e-01	1.7868e-02	
1.000	4061.05	3.1455	6.8307e-01	4.6154	1.0176e-01	0.7880	0.7951	8.4761e-02	5.4583e-02	/
	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.8755	0.8686	1.9018e-01	1.8132e-02	
1.000	725.18	1.1223	3.8017e+00	0.2833	4.0084e-04	0.8755	0.8686	1.9018e-01	1.8132e-02	
	4061.05	3.3568	6.6803e-01	5.4784	1.3282e-01	0.9106	0.9124	8.9321e-02	6.2163e-02	/
1.000	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.9520	0.9591	1.8807e-01	1.8720e-02	
	725.18	1.1335	3.5254e+00	0.3153	5.0228e-04	0.9520	0.9591	1.8807e-01	1.8720e-02	
1.000	4061.05	2.7406	5.5115e-01	3.5079	8.5588e-02	0.8828	0.8905	8.1404e-02	7.0425e-02	/
	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	0.9846	0.9855	1.9003e-01	1.8852e-02	
1.000	725.18	1.1293	3.5076e+00	0.3007	4.6299e-04	0.9846	0.9855	1.9003e-01	1.8852e-02	
1.000	4061.05	1.9854	5.0045e-01	2.0535	3.9219e-02	0.9400	0.9448	1.0088e-01	6.5831e-02	/
	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	1.0000	1.0000	1.7716e-01	1.8887e-02	
1.000	725.18	1.1418	3.5421e+00	0.2487	1.7810e-10	1.0000	1.0000	1.7716e-01	1.8887e-02	
1.000	4061.05	1.2145	4.3773e-01	0.6633	4.2745e-09	1.0000	1.0000	1.4288e-01	6.2211e-02	/
	14.50	1.0000	1.7810e+02	0.0000	0.0000e+00	1.0000	1.0000	1.4288e-01	6.2211e-02	/

The above example defines one oil PVT Properties for Live Oil versus CO2 Mass Fraction table, and assumes that NTPVT equals one the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

8.3.247 PVTW – DEFINE WATER FLUID PROPERTIES FOR VARIOUS REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[PVTW](#) defines the water properties for various regions in the model. The number of [PVTW](#) vector data sets is defined by the NTPVT parameter on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and the allocation of the [PVTW](#) tables to different grid blocks in the model is done via the [PVTNUM](#) keyword in the [REGIONS](#) section. One data set consists of one record or line which is terminated by a “/”. If the water phase is active in the model, which is normally the case, then this keyword must be defined in the OPM Flow input deck.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRES	PRES is a real number defining the water reference pressure (P) for the other parameters for this data set.			None
		psia	barsa	atma	
2	WFVF	WFVF is a real number defining the water formation volume factor (Bw) at the water reference pressure.			Defined
		rb/stb 1.0	rm ³ /sm ³ 1.0	rcc/scc 1.0	
3	WCOMP	WCOMP is a real number defining the water compressibility (Cw) at the water reference pressure and is defined as: $C_w = -\frac{1}{B_w}\left(\frac{dB_w}{dP}\right)$			Defined
		l/psia 0.00004	l/barsa 0.00004	l/atma 0.00004	
4	WVISC	WVISC is a real number defining the water viscosity (μw) at the water reference pressure.			Defined
		cP 0.50	cP 0.50	cP 0.50	
5	WVISCMP	WVISCMP is a real number defining the water viscosibility (μwc) at the water reference pressure, μwc(Pref) and is defined as: $\mu_{wc} = -\frac{1}{\mu_w}\left(\frac{d\mu_w}{dP}\right)$			Defined
		l/psia 0.0	l/barsa 0.0	l/atma 0.0	
Notes: 1) The keyword is followed by NTPVT tables as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section. 2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.					

Table 8.135: PVTW Keyword Description

Note that provided the first table has been entered, subsequent tables may be defaulted, in this case the prior table is copied to the current table. See the third example for an illustration on how to use this feature.

Examples

The following shows the *PVTW* keyword for when NTPVT on the *TABDIMS* keyword in the *RUNSPEC* section is set to one.

```
--
--      WATER PVT TABLE
--
PVTW
--      REF PRES  BW      CW      VISC      VISC
--      PSIA      RB/STB   1/PSIA   CPOISE   GRAD
--      -----
--      4840.0    1.019    2.7E-6    0.370    1*          / TABLE NO. 01
```

The next example shows the *PVTW* keyword for when NTPVT on the *TABDIMS* keyword in the *RUNSPEC* section is set to three.

```
--
--      WATER PVT TABLE
--
PVTW
--      REF PRES  BW      CW      VISC      VISC
--      PSIA      RB/STB   1/PSIA   CPOISE   GRAD
--      -----
--      4640.0    1.008    2.5E-6    0.350    1*          / TABLE NO. 01
--      4840.0    1.019    2.7E-6    0.370    1*          / TABLE NO. 02
--      4940.0    1.030    2.8E-6    0.390    1*          / TABLE NO. 03
```

The above example defines three water PVT tables and assumes that NTPVT equals three on the *TABDIMS* keyword in the *RUNSPEC* section.

The third, and final example, shows the *PVTW* keyword for when NTPVT on the *TABDIMS* keyword in the *RUNSPEC* section is set to four. Here table two defaults to table one, and table four defaults to table three

```
--
--      WATER PVT TABLE
--
PVTW
--      REF PRES  BW      CW      VISC      VISC
--      PSIA      RB/STB   1/PSIA   CPOISE   GRAD
--      -----
--      4640.0    1.008    2.5E-6    0.350    1*          / TABLE NO. 01
--                                     / TABLE NO. 02
--      4940.0    1.030    2.8E-6    0.390    1*          / TABLE NO. 03
--                                     / TABLE NO. 04
```

Note that there is no terminating “/” for this keyword.

8.3.248 PVTWSALT – DEFINE BRINE WATER FLUID PROPERTIES FOR VARIOUS REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[PVTWSALT](#) defines the brine water properties for various regions in the model, for when the brine phase has been activated by the [BRINE](#) keyword in the [RUNSPEC](#) section. In this case [PVTWSALT](#) is used instead of [PVTW](#) in the input file. However, if the [ECLMC](#) keyword has been entered in the [RUNSPEC](#) section to invoke the Multi-Component Brine model, the [PVTW](#) keyword should be used instead of [PVTWSALT](#), as with this combination the salinity effect on the density is ignored.

The number of [PVTWSALT](#) table data sets is defined by the NTPVT parameter on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and the allocation of the [PVTWSALT](#) tables to different grid blocks in the model is done via the [PVTNUM](#) keyword in the [REGION](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	PRESS	Single real positive value that defines the reference pressure for the data in the following records (Pref). PRESS should be approximately equal to the average reservoir pressures in the model. The simulator uses the previous time step values to forecast the current time step water properties by linear interpolation. If PRESS is not representative of the average reservoir pressures in the model then the linear interpolation might result in nonphysical values of the water saturation and water viscosity.			None
		psia	barsa	atma	
1-2	SALTSURF	A real value that defines the reference salt concentration in the solution in the surface stock tank water (Cs, ref) If defaulted SALTSURF is taken as the minimum salt concentration entered in the SALTCON columnar vector in the second record for this keyword. This should be in most cases be zero.			Defined
		lb/stb	kg/sm ³	gm/scc	
1-3	/	Record terminated by a "/"			Not Applicable
2-1	SALTCON	A columnar vector of real monotonically increasing values that defines the salt concentration in the solution (Cs).			None
		lb/stb	kg/sm ³	gm/scc	
2-2	WVVF	WVVF is a real columnar vector defining the water formation volume factor (Bw) at the reference pressure PRESS, for the corresponding salt concentration SALTCON.			None
		rb/stb	rm ³ /sm ³	rcc/scc	

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-3	WCOMP	WCOMP is a real columnar vector defining the water compressibility (Cw) at the water reference pressure PRESS, for the corresponding salt concentration SALTCON.The water compressibility is defined as: $C_w = -\frac{1}{B_w}\left(\frac{dB_w}{dP}\right)$			None
		l/psia	l/bars	l/atma	
2.4	WVISC	WVISC is a real columnar vector defining the water viscosity (μw) at the water reference pressure PRESS, for the corresponding salt concentration SALTCON.			None
		cP	cP	cP	
2.5	WVISCMP	WVISCMP is a real columnar vector defining the water viscosibility (μwc) at the water reference pressure PRESS, for the corresponding salt concentration SALTCON. The water viscosibility is defined as: $\mu_{wc} = -\frac{1}{\mu_w}\left(\frac{d\mu_w}{dP}\right)$			None
		l/psia	l/barsa	l/atma	
2-6	/	Table and record terminated by a “/”			Not Applicable

Notes:

- 1) The keyword is followed by NTPVT data sets as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and each data set consists of two records, with items 1-1 to 1-3 representing record one items and 2-1 to 2-6 representing record number two items, etc., in the "No." column in this table.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.136: PVTWSALT Keyword Description

As mentioned above, the simulator first calculates the water properties as functions of the salt concentration at the previous time step by linear interpolation in salt concentration for water compressibility (C_w), water viscosibility (μ_{wc}), $\frac{1}{B_w}$ and $\frac{1}{B_w \mu_w}$. It then calculates the values of B_w and $B_w \mu_w$ at the current time step using the current pressure P, using the following equations:

$$B_w(P, C_s) = \frac{B_w(P_{ref}, C_{s,ref})}{1 + C_w(P - P_{ref}) + \frac{(C_w(P - P_{ref}))^2}{2}} \quad (8.88)$$

and

$$B_w(P, C_s) \mu_w(P, C_s) = \frac{B_w(P_{ref}, C_{s,ref}) \mu_w(P_{ref}, C_{s,ref})}{1 + (C_w - \mu_{wc})(P - P_{ref}) + \frac{((C_w - \mu_{wc})(P - P_{ref}))^2}{2}} \quad (8.89)$$

8.3.249 PVZG – GAS PVT PROPERTIES FOR DRY GAS (Z-FACTOR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

PVZG defines the gas PVT properties for dry gas²³⁷ via the gas compressibility factor (z-factor), instead of the gas formation volume factor. If the gas has a constant and uniform vaporized oil concentration, Condensate-Gas Ratio (“CGR”), and if the reservoir pressure never drops below the saturation pressure (dew point pressure), then the model can be run more efficiently by omitting the [OIL](#) and [VAPOIL](#) keywords from the [RUNSPEC](#) section, treating the gas as a dry gas, and defining a constant Rv (CGR) value with keyword [RVCONST](#) or [RVCONSTT](#) in the [PROPS](#) section. This results in the model being run with as a dry gas problem with no active oil (condensate) phase. However, OPM Flow takes into account the constant Rv in the calculations and reporting.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	RTEMP	Single real positive value that defines the reservoir temperature for the data in the following records.			
		°F	°C	°C	
1-2	/	Record terminated by a “/”			Not Applicable
2-1	PRESS	A columnar vector of real monotonically increasing down the column values that defines the gas phase pressure.			None
		psia	barsa	atma	
2-2	GZFAC	A columnar vector of real values that defines the corresponding gas phase z-factor at the given pressure, PRESS.			None
		dimensionless	dimensionless	dimensionless	
2-3	GVISC	A columnar vector of real increasing down the column values that defines the corresponding gas phase viscosity.			None
		cP	cP	cP	

Notes:

1) The keyword is followed by NTPVT data sets as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and each data set consists of two records, with items 1-1 to 1-2 representing record one items and 2-1 to 2-3 representing record number two items, etc., in the “No.” column in this table.

2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.137: PVZG Keyword Description

²³⁷ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

See also the [RVCONST](#) and [RVCONSTT](#) keywords to define the constant Rv for dry gas.

The ideal gas law provides a relationship between the pressure, the temperature and the specific volume of an ideal gas (pure component). This relationship is modified by use of a compressibility factor, [Z²³⁸](#), to account for deviations, from ideal, to the behavior of real gases. The PVT relation for a real gas can be defined by:

$$PV = ZnRT \quad (8.90)$$

As the gas formation volume factor is used to relate the volume of gas, as measured at reservoir conditions, to the volume of gas as measured at standard conditions (60 °F and 14.7 psia, or 15 °C and 101.325 kPa). This gas property is then defined as the actual volume occupied by a certain amount of gas at a specified pressure and temperature, divided by the same amount of gas at standard conditions. Thus, using the above equation one can obtain the gas volumes at reservoir and standard conditions, i.e.

$$V_{sc} = \frac{Z_{sc} nRT_{sc}}{P_{sc}} \quad (8.91)$$

$$V_i = \frac{Z_i nRT_i}{P_i} \quad (8.92)$$

Thus the gas formation volume factor can be expressed as:

$$E = \frac{V_{sc}}{V_i} \quad (8.93)$$

And substituting equation (8.91) and (8.92) into (8.93) we obtain

$$E = \left(\frac{P_i}{P_{sc}} \right) \left(\frac{T_{sc}}{T_i} \right) \left(\frac{1}{Z_i} \right) \quad (8.94)$$

Incorporating standard pressure and temperature values gives in SI units:

$$E = \left(\frac{P_i}{101.325} \right) \left(\frac{273.15 + 15}{T_i} \right) \left(\frac{1}{Z_i} \right) = 2.84 \left(\frac{P_i}{Z_i T_i} \right) \quad (8.95)$$

or in field units:

$$E = \left(\frac{P_i}{14.7} \right) \left(\frac{460 + 60}{T_i} \right) \left(\frac{1}{Z_i} \right) = 35.37 \left(\frac{P_i}{Z_i T_i} \right) \quad (8.96)$$

Where,

E	= gas formation volume factor (scf /rcft or Sm ³ /m ³)
P	= pressure (psia or kPa)
P_{sc}	= pressure standard conditions (psia or kPa)
P_i	= initial reservoir pressure (psia or kPa)
V	= volume (ft ³ or m ³)
T	= absolute temperature (°R or K)
T_{sc}	= temperature at standard conditions (°R or K)
T_i	= initial reservoir temperature (°R or K)
R	= gas constant (10.73 or 8.314)

[RTEMP](#) on this keyword is the T_i in the above equations and P_i is PRESS columnar vector.

²³⁸ Standing, M. B.: "Volumetric and Phase Behaviour of Oil Field Hydrocarbon Systems", Renihold Publishing Corp., New York City (1952).

Example

```

---
--      GAS PVT TABLE USING GAS Z-FACTOR
--
PVZG
--      RESERVOIR TEMPERATURE FOR Z TO BG CONVERSION
--
180.0                                     /
--
--      PRES      ZG      VISC
--      PSIA      DIMLESS  CPOISE
--      -----
14.7      0.998970  0.0130
250.0     0.976260  0.0131
500.0     0.954790  0.0134
750.0     0.932050  0.0137
1000.0    0.912990  0.0142
1250.0    0.896320  0.0147
1500.0    0.881610  0.0152
1750.0    0.870830  0.0159
2000.0    0.863130  0.0166
2250.0    0.858920  0.0173
2500.0    0.857800  0.0181
2750.0    0.860430  0.0189
3000.0    0.866440  0.0197
3250.0    0.874980  0.0206
3500.0    0.885470  0.0214
3750.0    0.898350  0.0223
4000.0    1.025120  0.0277
--
--
--      GAS PVT TABLE USING GAS Z-FACTOR
--
PVZG
--      RESERVOIR TEMPERATURE FOR Z TO BG CONVERSION
--
180.0                                     /
--
--      PRES      ZG      VISC
--      PSIA      DIMLESS  CPOISE
--      -----
14.7      0.998970  0.0130
250.0     0.976260  0.0131
500.0     0.954790  0.0134
750.0     0.932050  0.0137
1000.0    0.912990  0.0142
1250.0    0.896320  0.0147
1500.0    0.881610  0.0152
1750.0    0.870830  0.0159
2000.0    0.863130  0.0166
2250.0    0.858920  0.0173
2500.0    0.857800  0.0181
2750.0    0.860430  0.0189
3000.0    0.866440  0.0197
3250.0    0.874980  0.0206
3500.0    0.885470  0.0214
3750.0    0.898350  0.0223
4000.0    1.025120  0.0277
--
--
--      / TABLE NO 01
--
--      / TABLE NO 01

```

The above example defines two dry *PVZG* tables assuming NTPVT equals two and NPPVT is greater than or equal to 17 on the *TABDIMS* keyword in the *RUNSPEC* section. There is no terminating “/” for this keyword.

8.3.250 PYEND – END THE DEFINITION OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulator's input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword on a separate single line.

See [PYEND – End the Definition of a PYINPUT Section](#) in the [GRID](#) section for a full description.

8.3.251 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulator's input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator. [PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line.

See [PYINPUT – Define the Start of a PYINPUT Section](#) in the [GRID](#) section for a full description.

8.3.252 QHRATING – DEFINE RIVER MASS FLOW VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [QHRATING](#) keyword defines a river’s mass flow rate versus depth tables, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.253 REFINE – START THE DEFINITION OF A LOCAL GRID REFINEMENT

The [REFINE](#) keyword defines the start of a Cartesian or radial local grid refinement (“LGR”) definition that sets the properties of the selected [LGR](#). The keyword is then followed by the property keywords associated with the section where the keyword is being invoked. For example, if the [REFINE](#) keyword is used in the [GRID](#) section then most of the keywords in that section can be used to set the grid properties for the [LGR](#).

See [REFINE – Start the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

8.3.254 RIVRXSEC – DEFINE RIVER CROSS-SECTION VERSUS DEPTH PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [RIVRXSEC](#) keyword defines a river's cross-sectional area and perimeter versus depth parameters. The River option must be activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section in order to use this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.255 RKTRMDIR – ACTIVATE ROCKTAB KEYWORD DIRECTIONAL TRANSMISSIBILITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword activates the directional transmissibility multipliers for the [ROCKTAB](#) keyword. This results in two additional columns being inputted on the [ROCKTAB](#) keyword. This feature is currently not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.256 ROCK – DEFINE THE ROCK COMPRESSIBILITY FOR VARIOUS REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

ROCK defines the rock compressibility for various regions in the model. The number of **ROCK** vector data sets is defined by the NTPVT parameter on the **TABDIMS** keyword in the **RUNSPEC** section and the allocation of the **ROCK** tables to different grid blocks in the model is done via the **PVTNUM** keyword in the **REGIONS** section. One data set consists of one record or line which is terminated by a “/”.

This keyword must be defined in the OPM Flow input deck.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	PRESS is a real number defining the rock reference pressure for the other parameters for this data set.			Default
		psia 14.7	barsa 1.0132	atma 1.0	
2	RCOMP	RCOMP is a real number defining the rock compressibility (cf) at the rock reference pressure and is defined as: $c_f = -\frac{1}{V} \left(\frac{dV}{dP} \right)$			Defined
		l/psia 0.0	l/barsa 0.0	l/atma 0.0	

Notes:

- The keyword is followed by NTPVT vectors as declared by the NTPVT parameter on the **TABDIMS** keyword in the **RUNSPEC** section.
Note, however, that if **ROCKOPTS**(**ROCKOPT3**) parameter has been used to set the allocation of the **ROCK** data via the **ROCKNUM** array, then the number of **ROCK** vectors should correspond to the value entered on **TABDIMS**(**NTROCC**) in the **RUNSPEC** section. Similarly, if the **ROCKOPTS**(**ROCKOPT3**) has been used to set the assignment of the **ROCK** data via the **SATNUM** array, then the number of vectors should correspond to the value entered via the **TABDIMS**(**NTSFUN**) parameter, since the tables will be allocated via the **SATNUM** array.
- Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.138: ROCK Keyword Description

The simulator adjusts the pore volume based on the reference pressure (PRESS), that is:

$$V(P_i) = V(P_r) \left(1 + c_f(P_i - P_r) + \frac{(c_f(P_i - P_r))^2}{2} \right) \quad (8.97)$$

where:

- c_f = rock compressibility (RCOMP) at the reference pressure (PRESS),
- P_i = initial grid cell pressure,
- P_r = reference pressure (PRESS),
- $V(P_i)$ = pore volume at initial conditions, and
- $V(P_r)$ = pore volume at at the reference pressure.

Note

If the Rock Compaction option has been activated via the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section, then the [ROCKTAB](#) keyword in the [PROPS](#) section should be used instead of [ROCK](#) keyword.

See also the [ROCKOPTS](#) and [ROCKTAB](#) keywords in the [PROPS](#) section.

Examples

The following shows the [ROCK](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to one.

```
--
--      ROCK COMPRESSIBILITY
--
--      REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--      AS THE PORV IS ALREADY AT RESERVOIR CONDITIONS (OPM FLOW USES THE
--      REFERENCE PRESSURE) TO CONVERT THE GIVEN PORV TO RESERVOIR CONDITIONS
--      USING THE DATA ON THE ROCK KEYWORD)
--
--      REF PRES   CF
--      PSIA       1/PSIA
--      -----
--
ROCK      3966.9      5.0E-06                      / ROCK COMPRESSIBILITY
```

The next example shows the [ROCK](#) keyword for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to three.

```
--
--      ROCK COMPRESSIBILITY
--
--      REFERENCE PRESSURE IS TAKEN FROM THE HCPV WEIGHTED RESERVOIR PRESSURE
--      AS THE PORV IS ALREADY AT RESERVOIR CONDITIONS (OPM FLOW USES THE
--      REFERENCE PRESSURE) TO CONVERT THE GIVEN PORV TO RESERVOIR CONDITIONS
--      USING THE DATA ON THE ROCK KEYWORD)
--
--      REF PRES   CF
--      PSIA       1/PSIA
--      -----
--
ROCK      3566.9      5.0E-06                      / ROCK COMPRESSIBILITY REGION 1
          3966.9      5.5E-06                      / ROCK COMPRESSIBILITY REGION 2
          4566.9      6.0E-06                      / ROCK COMPRESSIBILITY REGION 3
```

The above example defines three [ROCK](#) tables and assumes that NTPVT equals three on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

There is no terminating “/” for this keyword, and thus the number entries must match the value entered via the [TABDIMS](#)(NTPVT), [TABDIMS](#)(NTROCC), or [TABDIMS](#)(NTSFUN) parameters, depending on the option selected via the [ROCKOPTS](#) keyword.

8.3.257 ROCK2D – PORE VOLUME COMPACTION VERSUS PRESSURE AND SW TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ROCK2D](#) keyword defines rock compressibility pore volume multipliers as a function of pressure and water saturation (“Sw”) for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section. The pressure values are defined on this keyword and the water saturations are declared on the associated [ROCKWNOD](#) keyword in the [PROPS](#) section

The rock compaction pore volume and transmissibility multipliers, entered via the [ROCKTAB](#), [ROCK2D](#) and [ROCK2DTR](#) keywords, are applied to the pore pressure, unless the [OVERBURD](#) keyword in [PROPS](#) section is included in the input deck. When the [OVERBURD](#) keyword is present the multipliers are applied to the effective pore volume pressure, that is $P_{(effective)} = P_{(Pressure)} - P_{(overburden)}$. If the keyword is not present in the input deck then the overburden pressure is set to zero.

This keyword should only be used if compaction option has been enabled.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A columnar vector of real monotonically increasing down the column values that defines the corresponding overburden pressure for the subsequent MULT columnar vector.			None
		psia	bars	atm	
2	MULT	A columnar vector of real equal or decreasing down the column values that are less than or equal to one, that defines the rock compressibility pore volume multipliers corresponding to PRESS and for each water saturation entry in the ROCKWNOD keyword.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTROCC tables as declared on the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.139: ROCK2D Keyword Description

See also the [OVERBURD](#), [ROCKTAB](#), [ROCK2DTR](#), and [ROCKWNOD](#) keywords in the [PROPS](#) section.

Example

The following example defines two pore volume compaction tables, assuming NTROCC is equal to two on the [ROCKCOMP](#) keyword and NSSFUN is greater than or equal to four on the [TABDIMS](#) keyword.

--			
--		ROCK COMPACTION VERSUS PRESSURE AND SW TABLES	
--			
ROCK2D			
--	PRESS	PORV	FIRST ROCK2D TABLE DATA
--	PSIA	MULTIPLIER	
--	-----	-----	
--	0.0	0.850	
		0.850	
		0.850	
		0.085	/ P-SW SET TABLE NO. 01
--	PRESS	PORV	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	1000.0	0.900	
		0.900	
		0.900	
		0.900	/ P-SW SET TABLE NO. 01
--	PRESS	PORV	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	2500.0	0.950	
		0.950	
		0.950	
		0.950	/ P-SW SET TABLE NO. 01
--	PRESS	PORV	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	5000.0	1.000	
		1.000	
		1.000	
		1.000	/ P-SW SET TABLE NO. 01
--			
--	PRESS	PORV	SECOND ROCK2D TABLE DATA
--	PSIA	MULTIPLIER	
--	-----	-----	
--	0.0	0.800	
		0.800	
		0.800	
		0.800	/ P-SW SET TABLE NO. 02
--	PRESS	PORV	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	1000.0	0.880	
		0.880	
		0.880	
		0.880	/ P-SW SET TABLE NO. 02
--	PRESS	PORV	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	2500.0	0.950	
		0.950	
		0.950	
		0.950	/ P-SW SET TABLE NO. 02
--	PRESS	PORV	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	5000.0	1.000	
		1.000	
		1.000	
		1.000	/ P-SW SET TABLE NO. 02

Note that there must be exactly NTROCC tables entered for this keyword, otherwise an error will occur.

8.3.258 ROCK2DTR – TRANSMISSIBILITY COMPACTION VERSUS PRESSURE AND SW TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ROCK2DTR](#) keyword defines rock compressibility transmissibility multipliers as a function of pressure and water saturation ("Sw") for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section. The pressure values are defined on this keyword and the water saturations are declared on the associated [ROCKWNOD](#) keyword in the [PROPS](#) section

The rock compaction pore volume and transmissibility multipliers, entered via the [ROCKTAB](#), [ROCK2D](#) and [ROCK2DTR](#) keywords, are applied to the pore pressure, unless the [OVERBURD](#) keyword in [PROPS](#) section is included in the input deck. When the [OVERBURD](#) keyword is present the multipliers are applied to the effective pore volume pressure, that is $P_{(effective)} = P_{(Pressure)} - P_{(overburden)}$. If the keyword is not present in the input deck then the overburden pressure is set to zero.

This keyword should only be used if compaction option has been enabled.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A columnar vector of real monotonically increasing down the column values that defines the corresponding overburden pressure for the subsequent MULT columnar vector.			None
		psia	bars	atm	
2	MULT	A columnar vector of real equal or decreasing down the column values that are less than or equal to one, that defines the rock compressibility transmissibility multipliers corresponding to PRESS and for each water saturation entry in the ROCKWNOD keyword.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTROCC tables as declared on the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.140: ROCK2DTR Keyword Description

See also the [OVERBURD](#), [ROCKTAB](#), [ROCK2D](#), and [ROCKWNOD](#) keywords in the [PROPS](#) section.

Example

The following example defines two rock compressibility transmissibility compaction tables, assuming NTROCC is equal to two on the [ROCKCOMP](#) keyword and NSSFUN is greater than or equal to four on the [TABDIMS](#) keyword.

--			
--		TRANSMISSIBILITY COMPACTION VERSUS PRESSURE AND SW TABLES	
--			
--	ROCK2DTR		
--	PRESS	TRAN	FIRST ROCK2DTR TABLE DATA
--	PSIA	MULTIPLIER	
--	-----	-----	
--	0.0	0.850	
		0.850	
		0.850	
		0.085	/ P-SW SET TABLE NO. 01
--	PRESS	TRAN	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	1000.0	0.900	
		0.900	
		0.900	
		0.900	/ P-SW SET TABLE NO. 01
--	PRESS	TRAN	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	2500.0	0.950	
		0.950	
		0.950	
		0.950	/ P-SW SET TABLE NO. 01
--	PRESS	TRAN	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	5000.0	1.000	
		1.000	
		1.000	
		1.000	/ P-SW SET TABLE NO. 01
--			
--	PRESS	TRAN	SECOND ROCK2DTR TABLE DATA
--	PSIA	MULTIPLIER	
--	-----	-----	
--	0.0	0.800	
		0.800	
		0.800	
		0.800	/ P-SW SET TABLE NO. 02
--	PRESS	TRAN	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	1000.0	0.880	
		0.880	
		0.880	
		0.880	/ P-SW SET TABLE NO. 02
--	PRESS	TRAN	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	2500.0	0.950	
		0.950	
		0.950	
		0.950	/ P-SW SET TABLE NO. 02
--	PRESS	TRAN	
--	PSIA	MULTIPLIER	
--	-----	-----	
--	5000.0	1.000	
		1.000	
		1.000	
		1.000	/ P-SW SET TABLE NO. 02

Note that there must be exactly NTROCC tables entered for this keyword, otherwise an error will occur.

8.3.259 ROCKOPTS – DEFINE ROCK COMPACTION AND COMPRESSIBILITY OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ROCKOPTS](#) keyword defines various options with respect to rock compaction and rock compressibility.

No.	Name	Description	Default
1	ROCKOPT1	A defined character string that specifies how the overburden pressures supplied by the OVERBURD keyword are applied to the tabulated pressures in the ROCKTAB keywords: 1) STRESS: Use this option if the overburden pressures on the OVERBURD keyword are greater than the fluid pressure which results in the effective fluid pressure being negative. To avoid the rock compaction tables being entered with negative pressure values use this option. In this case the pore volume and transmissibility multipliers will be tabulated against the effective overburden pressure. 2) PRESSURE: In this case the pore volume and transmissibility multipliers should be tabulated against the effective fluid pressure. This the default value. ROCKOPT1 should be set to PRESSURE if the OVERBURD is not used in the input deck. Only the default value of PRESSURE is supported.	PRESSURE
2	ROCKOPT2	A defined character string that sets the reference pressure option: 1) STORE: Copies the initial calculated grid block pressures into the overburden pressure array, resulting in the pore volumes being referenced at the initial pressures instead of the reference pressures as per the ROCKTAB keyword. 2) NOSTORE: This option results in the pore volumes being referenced as per the ROCKTAB keyword. This is the default value. Note that STORE option should not be used with the OVERBURD keywords as the OVERBURD data will be overwritten.	NOSTORE
3	ROCKOPT3	A defined character string that specifies which region array should be used to allocate the various ROCK and ROCKTAB property tables in the model: 1) ROCKOPT3, should be set to ROCKNUM, SATNUM or PVTNUM. 2) If the parameter is defaulted or the ROCKOPT keyword is not present in the deck, then the PVTNUM array is used. 3) Secondly, if ROCKOPT3 is set to ROCKNUM but the NTROCC parameter on the TABDIMS keyword in the RUNSPEC section is defaulted, then again the PVTNUM array will be utilized. Only the PVTNUM and ROCKNUM options are currently supported.	PVTNUM

No.	Name	Description	Default
4	ROCKOPT4	A defined character string that sets the initial conditions for the HYSTER and BOBERG options: 1) DEFLATION: This option defines the reservoir rock to be fully compacted and the deflation curve is used to calculated the initial pore volume and transmissibility multipliers.This is the default value. 2) ELASTIC: This option sets the pore volume and transmissibility multipliers to one, as the reservoir rock is set to lie on the elastic curve. This parameter is ignored by OPM Flow as the ROCKCOMP(ROCKOPT) options of HYSTER and BOBERG are not supported by the simulator.	DEFLATION
Notes: 1) The keyword is terminated by a “/”.			

Table 8.141: ROCKOPTS Keyword Description

Example

```
--
--      ROCKOPT1  ROCKOPT2  ROCKOPT3  ROCKOPT4
--      PRS/STRE  NO/STORE  ARRAY
--      -----  -----  -----  -----
ROCKOPTS
      PRESSURE  NOSTORE      PVTNUM                / ROCK COMP OPTIONS
```

The above example defines the default values for the [ROCKOPTS](#) keyword.

8.3.260 ROCKPAMA – DEFINE COAL PALMER-MANSORRI ROCK MODEL PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[ROCKPAMA](#) defines the Palmer-Mansoori²³⁹ and ²⁴⁰ parameters used for this rock model, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword, and PALM-MAN has been declared for the ROCKOPT variable on the [ROCKCOMP](#) keyword; both keywords are in the [RUNSPEC](#) section. The Palmer-Mansoori rock model is used to calculate the impact on pore volume and permeability due to rock compaction.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

239

Palmer, I. and Mansoori, J. "How Permeability Depends on Stress and Pore Pressure in Coalbeds: A New Model," paper SPE 52607, SPE Reservoir Evaluation & Engineering (1998) 1, No. 6, 539-544.

240

Clarkson, C.R., Pan, Z., Palmer, I. and Harpalani, S. "Predicting Sorption-Induced Strain and Permeability Increase With Depletion for Coalbed-Methane Reservoirs", SPE 114778-PA, SPE Journal (2010) 15, No. 1, 152–159.

8.3.261 ROCKTAB – Rock Compaction Tables

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ROCKTAB](#) keyword defines the rock compaction attributes to be applied when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section. [ROCKTAB](#) defines pore volume and transmissibility multipliers versus pressure that are used in the compaction calculations. If the [RKTRMDIR](#) has been activated in the [PROPS](#) section, then the transmissibility multiplier is directional dependent and two additional columns are used to define the y and z direction transmissibility multipliers.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	If the ROCKOPT I variable has been set to PRESSURE on the ROCKOPTS keyword in the PROPS section, then PRESS should be a columnar vector of real monotonically increasing down the column values, that define the reference pressure for which the other parameters correspond to.			None
		If ROCKOPT I has been set to STRESS, then PRESS should be a columnar vector of real monotonically decreasing down the column values.			
		psia	bars	atm	
2	PORV	A columnar vector of real positive values that are either equal or increasing down the column that define the rock pore volume multiplier for a given PRESS.			None
		dimensionless	dimensionless	dimensionless	
3	TRANS	If the RKTRMDIR is absent from the input deck, then TRANS is a columnar vector of real positive values that are either equal or increasing down the column that define the x, y, and z directional transmissibility multipliers for the corresponding PRESS.			None
		If the RKTRMDIR is present in the input deck, then TRANS is a columnar vector of real positive values that are either equal or increasing down the column that define only the x directional transmissibility multipliers for the corresponding PRESS.			
		dimensionless	dimensionless	dimensionless	
4	TRANSY	If the RKTRMDIR is absent from the input deck, then TRANSY is ignored.			None
		If the RKTRMDIR is present in the input deck, then TRANSY is a columnar vector of real positive values that are either equal or increasing down the column that define only the y directional transmissibility multipliers for the corresponding PRESS.			
		dimensionless	dimensionless	dimensionless	
5	TRANSZ	If the RKTRMDIR is absent from the input deck, then TRANSZ is ignored.			None
		If the RKTRMDIR is present in the input deck, then TRANSZ is a columnar vector of real positive values that are either equal or increasing down the column that define only the z directional transmissibility multipliers for the corresponding PRESS.			

No.	Name	Description			Default
		Field	Metric	Laboratory	
		dimensionless	dimensionless	dimensionless	
Notes: 1) The keyword is followed by NTROCC tables as declared on the ROCKCOMP keyword in the RUNSPEC section. 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the TABDIMS keyword in the RUNSPEC section. 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.					

Table 8.142: ROCKTAB Keyword Description

Examples

The example below defines two rock compaction tables, assuming NTROCC is equal to two on the [ROCKCOMP](#) keyword and NPPVT is greater than or equal to five on the [TABDIMS](#) keyword and that the [RKTRMDIR](#) keyword is present in the input deck.

```
--
--      ROCK COMPACTION TABLES
--
ROCKTAB
--      PRESS      PORV      TX(YZ)      TY      TZ
--              MULT      MULT      MULT      MULT
--      -----
--      1000.0      0.9600      0.9650      0.9650      0.9650
--      1500.0      0.9800      0.9850      0.9850      0.9500
--      3000.0      0.9900      0.9950      0.9950      0.9950
--      4500.0      1.0000      1.0000      1.0000      1.0000
--      4750.0      1.0100      1.0100      1.0100      1.0100      / TABLE NO. 01
--      PRESS      PORV      TX(YZ)      TY      TZ
--              MULT      MULT      MULT      MULT
--      -----
--      1000.0      0.9600      0.9650      0.9650      0.9650
--      1500.0      0.9800      0.9850      0.9850      0.9500
--      3000.0      0.9900      0.9950      0.9950      0.9950
--      4500.0      1.0000      1.0000      1.0000      1.0000
--      4750.0      1.0100      1.0100      1.0100      1.0100      / TABLE NO. 02
```

As the x, y and z directional transmissibility multipliers are identical in the above example, we could eliminate the [RKTRMDIR](#) keyword from the input deck and enter the data in the three column format, as shown on the next page.

--

--

--

ROCKTAB

--

--

--

1000.0

1500.0

3000.0

4500.0

4750.0

--

--

--

1000.0

1500.0

3000.0

4500.0

4750.0

--

--

--

ROCK COMPACTION TABLES

PRESS

PORV

TX(YZ)

MULT

MULT

0.9600

0.9800

0.9900

1.0000

1.0100

PORV

TX(YZ)

MULT

MULT

0.9650

0.9850

0.9950

1.0000

1.0100

0.9650

0.9850

0.9950

1.0000

1.0100

/ TABLE NO. 01

/ TABLE NO. 02

The net result of the two examples in this case is identical.

8.3.262 ROCKTABH – Rock Compaction Hysteresis Tables

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ROCKTABH](#) keyword defines the rock compaction hysteresis attributes to be applied for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section. [ROCKTABH](#) defines pore volume and transmissibility multipliers versus pressure that are used in the compaction calculations. If the [RKTRMDIR](#) has been activated in the [PROPS](#) section, then the transmissibility multiplier is directional dependent and two additional columns are used to define the y and z direction transmissibility multipliers. The keyword should only be used if the Rock Compaction Hysteresis option has been activated by either setting the ROCKOPT parameter on the [ROCKCOMP](#) keyword to HYSTER or BOBERG.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	If the ROCKOPTI variable has been set to <i>PRESSURE</i> on the <i>ROCKOPTS</i> keyword in the <i>PROPS</i> section, then PRESS should be a columnar vector of real monotonically <u>increasing</u> down the column values, that define the reference pressure for which the other parameters correspond to.			None
		If ROCKOPTI has been set to STRESS, then PRESS should be a columnar vector of real monotonically <u>decreasing</u> down the column values.			
		psia	bars	atm	
2	<i>PORV</i>	A columnar vector of real positive values that are either equal or increasing down the column that define the rock pore volume multiplier for a given PRESS.			None
		dimensionless	dimensionless	dimensionless	
3	TRANS	If the <i>RKTRMDIR</i> is absent from the input deck, then TRANS is a columnar vector of real positive values that are either equal or increasing down the column that define the x, y, and z directional transmissibility multipliers for the corresponding PRESS.			None
		If the <i>RKTRMDIR</i> is present in the input deck, then TRANS is a columnar vector of real positive values that are either equal or increasing down the column that define only the x directional transmissibility multipliers for the corresponding PRESS.			
		dimensionless	dimensionless	dimensionless	
4	TRANSY	If the <i>RKTRMDIR</i> is absent from the input deck, then TRANSY is ignored.			None
		If the <i>RKTRMDIR</i> is present in the input deck, then TRANSY is a columnar vector of real positive values that are either equal or increasing down the column that define only the y directional transmissibility multipliers for the corresponding PRESS.			
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	TRANSZ	If the RKTRMDIR is absent from the input deck, then TRANSZ is ignored. If the RKTRMDIR is present in the input deck, then TRANSZ is a columnar vector of real positive values that are either equal or increasing down the column that define only the z directional transmissibility multipliers for the corresponding PRESS.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTROCC tables as declared on the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.143: ROCKTABH Keyword Description

Each data set consists of columnar vectors of pore volume and transmissibility multipliers versus pressure that specify the elastic contraction and expansion and of the reservoir rock. The deflation curve is derived from the first data elements on each elastic curve. If the ROCKOPT parameter on the [ROCKCOMP](#) keyword has been set to HYSTER, then the dilation curves are extrapolated to infinite pressure, that is the curves are unbounded. However, if [ROCKCOMP](#) is set to BOBERG the last points of each elastic curve are used as the dilation curves.

Example

The example below defines two rock compaction tables, assuming NTROCC is equal to two on the [ROCKCOMP](#) keyword and NPPVT is greater than or equal to four on the [TABDIMS](#) keyword and that the [RKTRMDIR](#) keyword is not present in the input deck.

```
--
--          ROCK COMPACTION HYSTERESIS TABLES
--
ROCKTABH
--          PRESS      PORV      TX(YZ)      TY      TZ
--          MULT      MULT      MULT      MULT      MULT
--          -----
--          1500.0      0.9600      0.9800
--          2500.0      0.9700      0.9850
--          3500.0      0.9800      0.9900
--          4500.0      0.9900      0.9950
--                                     / NPPVT = 1
--          2500.0      0.9900      0.9900
--          3500.0      0.9950      0.9950
--          4750.0      0.9980      0.9980
--                                     / NPPVT = 2
--          3500.0      1.0000      1.0000
--          5500.0      1.0100      1.0100
--                                     / NPPVT = 3
--          4500.0      1.0100      1.0100
--          5750.0      1.0200      1.0200
--                                     / NPPVT = 4
--                                     / TABLE NO. 01
--
--          PRESS      PORV      TX(YZ)      TY      TZ
--          MULT      MULT      MULT      MULT      MULT
--          -----
--          1500.0      0.9400      0.9700
--          2750.0      0.9400      0.9700
--                                     / NPPVT = 1
--          2250.0      0.9800      0.9900
--          3250.0      0.9800      0.9900
--                                     / NPPVT = 2
--          3000.0      1.0000      1.0000
--          4250.0      1.0000      1.0000
--                                     / NPPVT = 3
--          4550.0      1.0200      1.0100
--          5750.0      1.0200      1.0100
--                                     / NPPVT = 4
--                                     / TABLE NO. 02
```

Here the deflation curve is define for table number one is:

1500.0	0.9600	0.9800
2500.0	0.9900	0.9900
3500.0	1.0000	1.0000
4500.0	1.0100	1.0100

and for table number 2:

1500.0	0.9400	0.9700
2250.0	0.9800	0.9900
3250.0	1.0000	1.0000
4250.0	1.0200	1.0100

And the dilation curve is define for table number one is:

4500.0	0.9900	0.9950
4750.0	0.9980	0.9980
5500.0	1.0100	1.0100
5750.0	1.0200	1.0200

and for table number 2:

2250.0	0.9400	0.9700
3250.0	0.9800	0.9900
4250.0	1.0000	1.0000
5250.0	1.0200	1.0100

8.3.263 ROCKTABW – Rock Compaction Tables (Water Induced)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ROCKTABW](#) keyword defines the rock compaction tables induced by increasing water saturation within a grid cell due to water invasion, for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section. [ROCKTABW](#) defines pore volume and transmissibility multipliers versus water saturation that are used in the compaction calculations. The keyword should be used together with the [ROCK](#), [ROCKTAB](#) or [ROCKTABH](#) keywords that specify the pore volume and transmissibility multipliers as functions of pressure. Alternatively the [ROCKWNOD](#), [ROCK2D](#) and [ROCK2DTR](#) keywords can be used to enter two dimensional tables of the data. All keywords are in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.264
ROCKTHSG
–
Rock Compaction Hysteresis Tables (Dual Porosity)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ROCKTHSG](#) keyword defines the rock compaction hysteresis attributes to be applied for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section and the either the Dual Permeability or Dual Porosity models are activated by the [DUALPERM](#) and [DUALPORO](#) keywords in the [RUNSPEC](#) section. [ROCKTHSG](#) specifies sigma multipliers versus pressure that are used in the dual porosity rock compaction calculations. The keyword should only be used if the Rock Compaction Hysteresis option has been activated by either setting the ROCKOPT parameter on the [ROCKCOMP](#) keyword to one of the available options.

Each data set consists of columnar vectors of sigma multipliers versus pressure that specify the elastic contraction and expansion and of the reservoir rock. The deflation curve is derived from the first data elements on each elastic curve. If the ROCKOPT parameter on the [ROCKCOMP](#) keyword has been set to HYSTER, then the dilation curves are extrapolated to infinite pressure, that is the curves are unbounded. However, if [ROCKCOMP](#) is set to BOBERG the last points of each elastic curve are used as the dilation curves.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.265 ROCKTSIG – Rock COMPACTION TABLES (DUAL POROSITY)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ROCKTSIG](#) keyword defines the rock compaction attributes to be applied for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section, and the either the Dual Permeability or Dual Porosity models are activated by the [DUALPERM](#) and [DUALPORO](#) keywords in the [RUNSPEC](#) section. [ROCKTSIG](#) specifies sigma multipliers versus pressure that are used in the dual porosity rock compaction calculations. The keyword should only be used if the Rock Compaction Hysteresis option has been activated by either setting the ROCKOPT parameter on the [ROCKCOMP](#) keyword to one of the available options.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.266 ROCKWNOD – WATER SATURATION VALUES FOR COMPACTION PRESSURE-SW TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ROCK2D](#) and the [ROCK2DTR](#) keywords in the [PROPS](#) section define rock compressibility pore volume and transmissibility multipliers as a function of pressure and water saturation (“Sw”), for when the rock compaction option has been invoked by the [ROCKCOMP](#) keyword in the [RUNSPEC](#) section. The pressure values are defined on [ROCK2D](#) and the [ROCK2DTR](#) keywords together with the multipliers. This keyword [ROCKWNOD](#), defines the water saturations that are used in conjunction with the [ROCK2D](#) and the [ROCK2DTR](#) keywords.

This keyword should only be used if compaction option has been enabled.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real monotonically increasing down the column values that defines the water saturations to be associated with the data on the ROCK2D and the ROCKTR keywords.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTROCC tables as declared on the ROCKCOMP keyword in the RUNSPEC section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the TABDIMS keyword in the RUNSPEC section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.144: ROCKWNOD Keyword Description

See also the [OVERBURD](#), [ROCKTAB](#), [ROCK2D](#) and [ROCK2DTR](#) keywords in the [PROPS](#) section.

Example

The following example defines two [ROCKWNOD](#) tables for the pore volume and transmissibility compaction tables, assuming NTROCC is equal to two on the [ROCKCOMP](#) keyword and NSSFUN is greater than or equal to four on the [TABDIMS](#) keyword.

```
--
--      WATER SATURATION VALUES FOR COMPACTION PRESSURE-SW TABLES
--
ROCKWNOD
--      COMPACT
--      SWAT
--      -----
--      0.000
--      0.200
--      0.400
--      1.000                                / P-SW SET TABLE NO. 01
--      COMPACT
--      SWAT
--      -----
--      0.000
--      0.250
--      0.750
--      1.000                                / P-SW SET TABLE NO. 02
```

Note that there must be exactly NTROCC tables entered for this keyword, otherwise an error will occur.

8.3.267 RPTPROPS – DEFINE PROPS SECTION REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the data in the [PROPS](#) section that is to be printed to the output print file in human readable format. The keyword has two distinct forms, the first of which consists of the keyword followed by a series of integers on the next line indicating the data to be printed (see the first example). This is the original format in the commercial simulator and was subsequently superseded by the second format. The second format consists of the keyword followed by a series of character strings that indicate the data to be printed. In most cases the character string is the keyword used to load the data in the OPM Flow input deck, for example [PVDG](#) for the dry gas PVT tables. It is anticipated that OPM Flow will eventually support the functionality of the second format only, the first format although recognized will be completely ignored.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
1	PVDG	Print dry gas PVT tables	N/A
2	PVTG	Print wet gas PVT tables	N/A
3	SGFN	Print gas relative permeability saturation function tables.	N/A
4	SGL	Print connate gas saturation array.	N/A
....			N/A

Notes:

- 1) The keyword is terminated by a “/”.

Table 8.145: RPTPROPS Keyword Description

Note

Except for tabular like data, [PVDG](#) etc., this keyword has the potential to produce very large print files that some text editors may have difficulty loading. A more efficient solution for array type data is to load the *.INIT file into OPM ResInsight to view the data graphically, this also has the benefit of being able to filter the grid based on I, J, K ranges and grid properties.

Examples

The first example shows the original format of this keyword; although the keyword and format are recognized by OPM Flow, the format is ignored and is unlikely to be implemented in the simulator.

```
--
--      DEFINE PROPS SECTION REPORT OPTION (ORIGINAL FORMAT)
--
RPTPROPS      1          2*0      1          3*1          /
```

The next example shows the second format of the keyword which may be supported in a future release of OPM Flow.

```
--
--      DEFINE PROPS SECTION REPORT OPTIONS
--
RPTPROPS      PVD0      S0F2      SGFN      SWFN          /
```

8.3.268 RCONST – DEFINE CONSTANT GOR (Rs) FOR ALL DEAD OIL PVT FLUIDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

RCONST defines a constant Gas-Oil Ratio (“GOR”), for all dead oil²⁴¹ PVT fluids. If the oil has a constant and uniform dissolved gas concentration, GOR, and if the reservoir pressure never drops below the saturation pressure (bubble point pressure), then the model can be run more efficiently by omitting the **GAS** and **DISGAS** keywords from the **RUNSPEC** section, treating the oil as a dead oil, and defining a constant Rs (GOR) value with keywords **RCONST** or **RCONSTT** in the **PROPS** section. This results in the model being run as a dead oil problem with no active gas phase. However, OPM Flow takes into account the constant Rs in the calculations and reporting.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RS	A real positive value that defines the dead oil GOR for all oil PVT tables in the model			None
		Mscf/stb	sm ³ /sm ³	scf/scf	
2	PRESS	A real positive value that defines that saturation pressure (bubble point pressure) for all the oil PVT tables in the model.			None
		psia	barsa	atma	

Notes:

1) The keyword is terminated by a “/”.

Table 8.146: RCONST Keyword Description

See also the **RCONSTT** keyword to define a different constant Rs to the various dead oil PVT tables and the **PVDO** and **PVCDO** keywords to enter the dead oil properties. All of the aforementioned keywords are in the **PROPS** section.

Example

The example sets the dead oil GOR to 5 scf/stb and the bubble point pressure to 14.7 psia.

```
--
--      DEAD OIL PVT CONSTANT GOR AND SATURATION PRESSURE
--
RCONST
--      RS      PSAT
--      MSCF/STB PSIA
--      -----
--      0.0050   14.7
```

²⁴¹ “Dead” oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

8.3.269 RCONSTT – DEFINE CONSTANT GOR (Rs) FOR EACH DEAD OIL PVT FLUID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

RCONSTT defines a constant Gas-Oil Ratio (“GOR”), for each dead oil²⁴² PVT fluid in the model. If the oil has a constant and uniform dissolved gas concentration, GOR, and if the reservoir pressure never drops below the saturation pressure (bubble point pressure), then the model can be run more efficiently by omitting the **GAS** and **DISGAS** keywords from the **RUNSPEC** section, treating the oil as a dead oil, and defining a constant Rs (GOR) value with keywords **RCONST** or **RCONSTT** in the **PROPS** section. This results in the model being run as a dead oil problem with no active gas phase. However, OPM Flow takes into account the constant Rs in the calculations and reporting.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RS	A real positive columnar vector that defines the dead oil GOR for each oil PVT table in the model			None
		Mscf/stb	sm ³ /sm ³	scc/scc	
2	PRESS	A real positive columnar vector that defines the saturation pressure (bubble point pressure) for each the oil PVT table in the model.			None
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by NTPVT rows as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each row must contain two values representing the *RS* and PRESS variables.
- 3) Each row is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.147: RCONSTT Keyword Description

See also the **RCONST** keyword to define a constant Rs to all the various dead oil PVT tables and the **PVDO** and **PVCDO** keywords to enter the dead oil properties. All of the aforementioned keywords are in the **PROPS** section.

²⁴² “Dead” oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

Example

The example sets the dead oil GOR to 5, 6.5 and 8.0 scf/stb for PVT tables one, two and three, respectively and the bubble point pressure to 14.7 psia for all three tables.

```
--
--      DEAD OIL PVT CONSTANT GOR AND SATURATION PRESSURE
--
RSCONSTT
--      RS      PSAT
--      MSCF/STB PSIA
--      -----
--      0.0050      14.7      / TABLE NO. 01
--      0.0065      14.7      / TABLE NO. 02
--      0.0080      14.7      / TABLE NO. 03
```

8.3.270
RSGI – DEFINE GAS-OIL RATIO VERSUS PRESSURE AND Gi TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [RSGI](#) keyword specifies the saturated oil Gas-Oil Ratio (“GOR”) factors used to specify the variation of the maximum possible GOR of oil with respect to pressure and Gi values, for when the [GIMODEL](#) keyword in the [RUNSPEC](#) section has been used to activate the [GI](#) Pseudo Compositional option for the run. See also the [GINODE](#), [RSGI](#), [RVGI](#), [BGGI](#) and [BOGI](#) keywords in the [PROPS](#) section to describe the fluid properties for the [GI](#) Pseudo Compositional option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.271 RTEMP – DEFINE THE INITIAL RESERVOIR TEMPERATURE FOR THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the initial reservoir temperature for the model. Note that the [RTEMP](#) keyword is an alias for [RTEMPA](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

The initial reservoir temperature must be defined when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil temperature model, and the [THERMAL](#) keyword to activate the compositional thermal model.

The initial reservoir temperature should be defined when OPM Flow's CO₂ or H₂ storage option has been activated by the [CO2STORE](#) or [H2STORE](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	RTEMP	Single real positive value that defines the reservoir temperature for the model.			None
		°F	°C	°C	
Notes: I) The keyword is terminated by a “/”.					

Table 8.148: RTEMP Keyword Description

See also the [RTEMPVD](#) keyword in [SOLUTION](#) section to define the reservoir temperature as a function of depth.

Example

```
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMP      190.0 / RESERVOIR TEMPERATURE
```

The above example defines the reservoir temperature to be 190 oF.

8.3.272 RTEMPA – DEFINE THE INITIAL RESERVOIR TEMPERATURE FOR THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the initial reservoir temperature for the model. Note that the [RTEMPA](#) keyword is an alias for [RTEMP](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

The initial reservoir temperature must be defined when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil temperature model, and the [THERMAL](#) keyword to activate the compositional thermal model.

The initial reservoir temperature should be defined when OPM Flow's CO₂ or H₂ storage option has been activated by the [CO2STORE](#) or [H2STORE](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	RTEMPA	Single real positive value that define the reservoir temperature for the model.			None
		°F	°C	°C	
Notes: I) The keyword is terminated by a “/”.					

Table 8.149: RTEMPA Keyword Description

See also the [RTEMPVD](#) keyword in [SOLUTION](#) section to define the reservoir temperature as a function of depth.

Example

```
--
--      RESERVOIR
--      TEMPERATURE
--      -----
RTEMPA      190.0 / RESERVOIR TEMPERATURE
```

The above example defines the reservoir temperature to be 190 oF.

8.3.273
RTEMPVD – DEFINE THE INITIAL RESERVOIR TEMPERATURE VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the initial reservoir temperature versus depth tables for each equilibration region. Note that the [RTEMPVD](#) keyword is an alias for [TEMPVD](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

See [RTEMPVD – Define the Initial Reservoir Temperature versus Depth Tables](#) in the [SOLUTION](#) section, as the keyword is documented in the [SOLUTION](#) section, the same as the commercial simulator, but it can also be used in the [PROPS](#) section by OPM Flow.

8.3.274 RVCONST – DEFINE CONSTANT CGR (Rv) FOR ALL DRY GAS PVT FLUIDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

RVCONST defines a constant Condensate-Gas Ratio (“CGR” or Rv), for all dry gas²⁴³ PVT fluids. If the gas has a constant and uniform dissolved condensate concentration, and if the reservoir pressure never drops below the saturation pressure (dew point pressure), then the model can be run more efficiently by omitting the **OIL** and **VAPOIL** keywords from the **RUNSPEC** section, treating the gas as a dry gas, and defining a constant Rv (CGR) value with keywords **RVCONST** or **RVCONSTT** in the **PROPS** section. This results in the model being run as a dry gas problem with no active oil phase. However, OPM Flow takes into account the constant Rv in the calculations and reporting.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RV	A real positive value that defines the dry gas CGR for all dry gas PVT tables in the model			None
		stb/Mscf	sm ³ /sm ³	scc/scc	
2	PRESS	A real positive value that defines that saturation pressure (dew point pressure) for all the dry gas PVT tables in the model.			0.0
		psia	barsa	atma	
Notes: 1) The keyword is terminated by a “/”.					

Table 8.150: RVCONST Keyword Description

See also the **RVCONSTT** keyword to define a different constant Rv to the various dry gas PVT tables and the **PVDG** keyword to enter the dry gas properties. All of the aforementioned keywords are in the **PROPS** section.

Example

The example sets the dry gas CGR to 5 stb/MMscf and the bubble point pressure to 14.7 psia.

```
--
--      DRY GAS PVT CONSTANT GCR AND SATURATION PRESSURE
--
RVCONST
--      RV      PSAT
--      STB/MSCF PSIA
--      -----
--      0.0050   14.7
```

²⁴³ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

8.3.275 RVCONSTT – DEFINE CONSTANT CGR (Rv) FOR EACH DRY GAS PVT FLUID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

RVCONSTT defines a constant Condensate-Gas Ratio (“CGR” or Rv), for each dry gas²⁴⁴ PVT fluid. If the gas has a constant and uniform dissolved condensate concentration, and if the reservoir pressure never drops below the saturation pressure (dew point pressure), then the model can be run more efficiently by omitting the **OIL** and **VAPGAS** keywords from the **RUNSPEC** section, treating the gas as a dry gas, and defining a constant Rv (CGR) value with keywords **RVCONST** or **RVCONSTT** in the **PROPS** section. This results in the model being run as a dry gas problem with no active oil phase. However, OPM Flow takes into account the constant Rv in the calculations and reporting.

This keyword is ignored by OPM Flow but is documented here for completeness.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RS	A real positive value that defines the dry gas CGR for each dry gas PVT table in the model			None
		stb/Mscf	sm ³ /sm ³	scc/scc	
2	PRESS	A real positive value that defines that saturation pressure (dew point pressure) for each dry gas PVT table in the model.			0.0
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by NTPVT rows as declared on the **TABDIMS** keyword in the **RUNSPEC** section.
- 2) Each row must contain two values representing the **RV** and **PRESS** variables.
- 3) Each row is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.151: RVCONSTT Keyword Description

See also the **RVCONST** keyword to define a constant Rv to all the various dry gas PVT tables and the **PVDG** keyword to enter the dry gas properties. All of the aforementioned keywords are in the **PROPS** section.

²⁴⁴ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

Example

The example sets the dry gas CGR to 5, 6.5 and 8.0 stb/MMscf for PVT tables one, two and three, respectively and the bubble point pressure to 14.7 psia for all three tables.

```
--
--      DRY GAS PVT CONSTANT GCR AND SATURATION PRESSURE
--
RVCONSTT
--      RV      PSAT
--      STB/MSCF PSIA
--      -----
--      0.0050      14.7      / TABLE NO. 01
--      0.0065      14.7      / TABLE NO. 02
--      0.0080      14.7      / TABLE NO. 03
```

8.3.276
RVGI – DEFINE CONDENSATE-GAS RATIO VERSUS PRESSURE AND GI TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [RVGI](#) keyword specifies the saturated gas Condensate-Gas Ratio (“CGR”) factors used to specify the variation of the maximum possible CGR of gas with respect to pressure and Gi values, for when the [GIMODEL](#) keyword in the [RUNSPEC](#) section has been used to activate the [GI](#) Pseudo Compositional option for the run. See also the [GINODE](#), [RSGI](#), [RVGI](#), [BGGI](#) and [BOGI](#) keywords in the [PROPS](#) section to describe the fluid properties for the [GI](#) Pseudo Compositional option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.277 RWGSALT – WATER VAPORIZATION VERSUS PRESSURE AND SALT CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[RWGSALT](#) defines the relationship of water vaporization versus pressure and salt concentration. This keyword should be used when the [VAPWAT](#) keyword has been declared in the [RUNSPEC](#) section indicating that vaporized water is present in the gas phase. In addition, if the Salt Precipitation model has been activated via the [BRINE](#) and [PRECSALT](#) keywords, also in the [RUNSPEC](#) section, then this keyword must be present. The keyword may be used for gas-water and oil-water-gas input decks that contain the either dry or wet gas and vaporized water phases.

Note

This is an OPM Flow specific keyword for the simulator's Water Vaporization and Salt Precipitation Models, note that these are extensions to the simulator's standard Brine model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A real monotonically increasing down the column values that define the gas phase pressure, that together with salt concentration, defines the vaporized water in gas ratio ("VWGR") or R_w			None
		psia	barsa	atma	
2	SALTCON	A real monotonically increasing positive columnar vector defining the salt concentration in water.			None
		lb/stb	kg/sm ³	gm/scc	
3	RW	A columnar vector of real positive number values defining the vaporized water in gas ratio (R_w) that for a given PRESS and SALTCON.			None
		stb/Mscf	sm ³ /sm ³	scc/scc	

Notes:

- Each table is terminated by a "/" including the last table; however, there is no "/" terminator for the keyword.

Table 8.152: RWGSALT Keyword Description

Since the water component is in both the water and the gas phases, [RWGSALT](#) controls the amount of water component evaporated from the water phase into the gas phase, which is a function of both the water phase salinity and a grid cells pressure. The keyword should be used in conjunction with the dry gas [PVTGW](#) keyword or the wet gas [PVTGWO](#) keyword, both of which are in the [PROPS](#) section.

Example

The example defines two [RWGSALT](#) tables assuming NTPVT equals two and NPPVT is greater than or equal to eight on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

--	WATER VAPORIZATION TABLE FOR BRINE (OPM FLOW KEYWORD)		
--			
--			
RWGSALT			
--	PRES	SALTCONC	RW
--	PSIA	LB/STB	STB/MSCF
--	-----	-----	-----
	300	0	0.000132
		0.5	0.000132
		1	0.000132
	600	0	0.000132
		0.5	0.000132
		1	0.000132
	900	0	0.000132
		0.5	0.000132
		1	0.000132
	1200	0	0.000132
		0.5	0.000132
		1	0.000132
	1500	0	0.000132
		0.5	0.000132
		1	0.000132
	1800	0	0.000132
		0.5	0.000132
		1	0.000132
	2100	0	0.000132
		0.5	0.000132
		1	0.000132
	2400	0	0.000132
		0.5	0.000132
		1	0.000132
			/
			/ Table NO. 1
--	PRES	SALTCONC	RW
--	PSIA	LB/STB	STB/MSCF
--	-----	-----	-----
	300	0	0.000132
		0.5	0.000132
		1	0.000132
	600	0	0.000132
		0.5	0.000132
		1	0.000132
	900	0	0.000132
		0.5	0.000132
		1	0.000132
	1200	0	0.000132
		0.5	0.000132
		1	0.000132
	1500	0	0.000132
		0.5	0.000132
		1	0.000132
	1800	0	0.000132
		0.5	0.000132
		1	0.000132
	2100	0	0.000132
		0.5	0.000132
		1	0.000132
	2400	0	0.000132
		0.5	0.000132
		1	0.000132
			/
			/ Table NO. 2

8.3.278 SALINITY – DEFINE THE RESERVOIR SALINITY FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SALINITY](#) keyword defines a uniform salinity for all cells in the model. The keyword should only be used with OPM Flow's CO₂-Brine model which is activated via the [CO2STORE](#) keyword in the [RUNSPEC](#) section. This keyword is a compositional keyword in the commercial simulator but has been implemented in OPM Flow's black-oil CO₂-Brine model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SALINITY	A real positive value that defines the salinity for all grid blocks in the model for when the CO ₂ -Brine model has been activated. Note that the units for salinity are molality, that is gm-M/Kg, and therefore the units are defined as given below with the 10 ⁻³ prefix.			0
		10 ⁻³ × lb-M/lb	10 ⁻³ × kg-M/kg	10 ⁻³ × gm-M/gm	
Notes: I) The keyword is terminated by a “/”.					

Table 8.153: SALINITY Keyword Description

See also the [CO2STORE](#) keyword in the [RUNSPEC](#) section.

Example

The example sets the salt salinity for all cells in the model to 0.001 lb-M/lb.

```
--
--      SET SALINITY FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALINITY
    1.0 /
```

Note that units for salinity are to the 10⁻³, that is a value of 0.001 lb-M/lb should be entered as 1.0 (10⁻³ × lb-M/lb), as per the example.

8.3.279 SALTMF – DEFINE THE SALT LIQUID-PHASE MOLE FRACTION FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SALTMF](#) keyword defines a uniform salt liquid-phase mole fraction for all cells in the model. The keyword should only be used with OPM Flow's CO₂-Brine model which is activated via the [CO2STORE](#) keyword in the [RUNSPEC](#) section.

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	0
I	SALTMF	A real positive value that defines the salt liquid-phase mole fraction for all grid blocks in the model for when the CO ₂ -Brine model has been activated.			
		mole fraction	mole fraction	mole fraction	
Notes: I) The keyword is terminated by a “/”.					

Table 8.154: SALTMF Keyword Description

See also the [CO2STORE](#) keyword in the [RUNSPEC](#) section.

Example

The example sets the salt liquid-phase mole fraction for all cells in the model to 0.018.

```
--
--      SET SALT LIQUID-PHASE MOLE FRACTION FOR ALL CELLS (OPM FLOW KEYWORD)
--
SALTMF      0.018      /
```

8.3.280 SALTNODE – SALT CONCENTRATION BASED PVTNUM ARRAY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

SALTNODE defines the salt concentration value based on a cells **PVTNUM** number. The **SALTNODE** property is used in the calculation of a polymer viscosity when the polymer and the salt options has been activated by the **POLYMER** and **BRINE** keywords in the **RUNSPEC** section. In the **RUNSPEC** section the number of **PVTNUM** functions is declared by NTPVT variable on the **TABDIMS** keyword and allocated to individual cells by the **PVTNUM** property array in the **REGIONS** section. NPPVT on the **TABDIMS** keyword in the **RUNSPEC** section defines the maximum number of rows (or pressure values) in the PVT tables and also sets the maximum number of entries for each **SALTNODE** data set. The number of values for each data set must correspond to the number of polymer solution adsorption entries on the **PLYADSS** keyword. For example if there are three sets of PVT tables and four values on the **PLYADSS** keyword, then three **SALTNODE** data sets with four values of salt concentrations need to be entered.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SALTNODE	A real monotonically increasing positive columnar vector defining the salt concentration for a given PVTNUM table.			None
		lb/stb	kg/sm3	gm/scc	
Notes: I) Each table is terminated by a “/” including the last table; however, there is no “/” terminator for the keyword.					

Table 8.155: SALTNODE Keyword Description

An alternative manner of entering the salt concentrations is by utilizing the **PVTNUM** region array by using the **ADSALNOD** keyword in the **PROPS** section.

Example

Given three sets of relative permeability tables and four values on the **PLYADSS** keyword and two **SALTNODE** data sets with four values of salt concentrations then the data should be entered as follows:

```
--
-- SETS SALT CONCENTRATION FOR POLYMER SOLUTION ADSORPTION
-- VIA PVTNUM ARRAY ALLOCATION
--
-- SALT
--
SALTNODE
  1.0
  5.0
 10.5
 25.0      / PVTNUM TABLE NO. 01
  1.0
  3.0
  7.5
 15.0      / PVTNUM TABLE NO. 02
```

See also the **ADSALNOD** keyword.

8.3.281 SALTSOL – DEFINE THE SALT SOLUBILITY LIMIT BY REGION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[SALTSOL](#) defines a grid block's maximum salt solubility for each [PVTNUM](#) region. The keyword should only be used with OPM Flow's Salt Precipitation model which is activated via the [PRECSALT](#) keyword in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword for the simulator's Salt Precipitation model that is activated by the [PRECSALT](#) keyword and declaring that vaporized water is present in the run via the [VAPWAT](#) in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SALTSOL	A real positive value that defines the maximum salt solubility for all grid blocks in a PVTNUM region.			None
		lb/stb	kg/sm3	gm/scc	
2	SALTDEN	SALTDEN is a real number defining the density of salt at surface conditions.			Defined
		lb/ft ³ 135.469	kg/m ³ 2170	gm/cc 2.170	

Notes:

1)

The keyword is followed by NTPVT tables as declared on the TABDIMS keyword in the RUNSPEC section.

2)

Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.156: SALTSOL Keyword Description

See also the [PRECSALT](#) and [VAPWAT](#) keywords in the [RUNSPEC](#) section and the [PVTGW](#) and [PVTGWO](#) keywords in the [PROPS](#) section.

Example

The first example sets the maximum salt solubility for all cells in the model to 134.6 lb/stb, assuming that there is only one PVT region, that is NTPVT is equal to one on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--      SET SALT SOLUBILITY LIMIT FOR EACH REGION (OPM FLOW KEYWORD)
--
SALTSOL
--      MAX          SALT
--      SALTSOL      DENSITY
--      134.6        1*                               /
```

The 134.6 lb/stb, (380 kg/sm³ or 0.384 gm/scc for metric and laboratory units, respectively) is based on the solubility of NACL at 212 °F (100 °C) and should be used with care.

The next example shows how to set the maximum salt solubility for when NTPVT is equal to three on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--      SET SALT SOLUBILITY LIMIT FOR EACH REGION (OPM FLOW KEYWORD)
--
SALTSOL
--      134.6                               / PVT REGION NO. 1
--      124.0                               / PVT REGION NO. 2
--                                           / PVT REGION NO. 3
```

Here the last entry, which is for region number three, is defaulted, and results in region's three maximum salt solubility to take the previous value, in this case 124.0 lb/stb.

8.3.282 SCALECRS – DEFINE END-POINT SCALING OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SCALECRS](#) keyword sets the end-point scaling option to be either two-point or three-point scaling, for when the End-Point Scaling option has been invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. This determines which end-points on the relative permeability curves are used for scaling based on the supplied end-point arrays ([SGCR](#), [SWCR](#), etc.).

No.	Name	Description	Default
I	SCALEOPT	SCALEOPT is a character string that sets the end-point scaling option and should be set to either NO or YES: 1) NO: Activates two-point end-point scaling. 2) YES: Activates three-point end-point	NO
Notes: I) The keyword is terminated by a “/”.			

Table 8.157: SCALECRS Keyword Description

The end-point scaled for each option and the arrays used in the end-point scaling are summarized in the following table:

Option	Phases	Relative Permeability End-Point	Minimum Saturation End-Point	Middle Saturation End-Point	Maximum Saturation End-Point
Two-Point	Water	KRW	SWCR		SWU
	Gas	KRG	SGCR		SGU
	Oil-Water	KRORW	SOWCR		$(1.0 - \text{SWL} - \text{SGL})$
	Oil-Gas	KRORG	SOGCR		$(1.0 - \text{SWL} - \text{SGL})$
Three-Point	Water	KRW	SWCR	$(1.0 - \text{SOWCR} - \text{SGL})$	SWU
	Gas	KRG	SGCR	$(1.0 - \text{SOGCR} - \text{SWL})$	SGU
	Oil-Water	KRORW	SOWCR	$(1.0 - \text{SWCR} - \text{SGL})$	$(1.0 - \text{SWL} - \text{SGL})$
	Oil-Gas	KRORG	SOGCR	$(1.0 - \text{SGCR} - \text{SGL})$	$(1.0 - \text{SWL} - \text{SGL})$
	Two Phase Gas-Water Simulations				
	Water	KRW	SWCR	$(1.0 - \text{SGCR})$	SWU
	Gas	KRG	SGCR	$(1.0 - \text{SWCR})$	SGU

Table 8.158: End-Point Arrays Used in the End-Point Scaling Options

See also the [TZONE](#) keyword in the [PROPS](#) section that sets the transition zone end-point scaling options for the oil, gas and water phases.

Example

```
--
--      TWO-POINT END-POINT SCALING IS NO THREE POINT IS YES
--
--      SCALEOPT
--      -----
SCALECRS      YES                               / SCALING OPTION
```

The above example activates three-point end-point scaling of the relative permeability curves.

8.3.283 SCALELIM – END-POINT SCALING VERSUS DEPTH MAXIMUM WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the maximum water saturation allowed in a cell for when the end-point versus depth tables are used in the End-Point Scaling option to calculate the water saturation for a grid block. The End-Point Scaling option must be invoked by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section to use this keyword, and the keyword may only be used in two phase runs containing water, or if the Miscible Flood option has been activated by the [MISCIBLE](#) keyword in the [RUNSPEC](#) section. This keyword functionality is not supported in OPM Flow.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.284 SDENSITY – DEFINE THE MISCIBLE OR SOLVENT SURFACE GAS DENSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *SDENSITY* keyword defines density at surface conditions of either the miscible injection gas for when the *MISCIBLE* option has been invoked in the *RUNSPEC* section, or the solvent for when the *SOLVENT* option has been invoked in the *RUNSPEC* section. This keyword must be invoked if either the *MISCIBLE* or *SOLVENT* options have been activated in the *RUNSPEC* section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SDENSITY	SDENSITY is a real positive number defining the density at surface conditions of either: 1) the miscible injection gas for when the MISCIBLE option has been invoked in the RUNSPEC section, or, 2) the solvent for when the SOLVENT option has been invoked in the RUNSPEC section.			None
		lb/ft ³	kg/m ³	gm/cc	

Notes:

1) The keyword is followed by NTPVT tables as declared on the TABDIMS keyword in the RUNSPEC section.

2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.159: SDENSITY Keyword Description

In addition to this keyword, the surface density or gravity of the in-place natural gas must be entered using either the *DENSITY* or *GRAVITY* keywords.

Examples

The following shows the *SDENSITY* keyword for when NTPVT on the *TABDIMS* keyword in the *RUNSPEC* section is set to one.

```
--
--      MIS-SOL
--      DENSITY
--      -----
SDENSITY 0.04520 / MIS-SOL DENSITY
```

The next example shows the *SDENSITY* keyword for when NTPVT on the *TABDIMS* keyword in the *RUNSPEC* section is set to three.

```
--
--      MIS-SOL
--      DENSITY
--      -----
SDENSITY 0.04520 / MIS-SOL DENSITY 1
          0.05520 / MIS-SOL DENSITY 2
          0.06420 / MIS-SOL DENSITY 3
```

There is no terminating “/” for this keyword.

8.3.285 SGCR – END-POINT SCALING GRID CELL CRITICAL GAS SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[SGCR](#) defines the critical gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The critical gas saturation is defined as the maximum gas saturation for which the gas relative permeability is zero in a two-phase relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGCR	SGCR is an array of real numbers assigning the critical gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.03			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the critical gas saturation used with the end-point scaling option. If directional end-point scaling has been activated then the [SGCRX±](#), [SGCRY±](#) and [SGCRZ±](#) series of keyword should be used.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.160: SGCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRGR](#), [KRO](#), [KROGR](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SGCRX](#), [SGCRY](#) and [SGCRZ](#) instead of [SGCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SGCRX-](#), [SGCRY-](#), [SGCRZ-](#), [SGCRX±](#), [SGCRY±](#), [SGCRZ±](#) instead of the [SGCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SGCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SGCR
      300*0.050
```

The above example defines a constant critical gas saturation of 0.05 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.286 SGCWMIS – MISCIBLE CRITICAL GAS VERSUS WATER SATURATION FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SGCWMIS](#) defines the dependency between the miscible critical gas saturation and the water saturation, for when the [MISCIBLE](#) keyword in the [RUNSPEC](#) section has been activated. The keyword can only be used with the [MISCIBLE](#) option and for when the oil, water and gas phases are active in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the water saturation.			None
		dimensionless	dimensionless	dimensionless	
2	SGCMIS	A columnar vector of real equal or increasing down the column values that are greater than or equal to zero and less than one, that define the corresponding miscible gas critical gas saturation for the corresponding water saturation SWAT .			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTMISC tables as declared on the [MISCIBLE](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSMISC rows as declared on the [MISCIBLE](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.161: SGCWMIS Keyword Description

Example

```
--
--      MISCIBLE CRITICAL GAS VERSUS WATER SATURATION TABLE
--
SGCWMIS
--      SWAT      SGRMIS
--      FRAC      FRAC
--      -----
--      0.0000      0.0000
--      0.2000      0.0300
--      1.0000      0.0300 / TABLE NO. 01
--
--      SWAT      SGRMIS
--      FRAC      FRAC
--      -----
--      0.0000      0.0000
--      0.3000      0.0500
--      1.0000      0.0500 / TABLE NO. 02
```

The above example defines two miscible critical gas saturation versus water saturation tables assuming NTMISC equals two and NSMISC is greater than or equal to three on the [MISCIBLE](#) keyword in the [RUNSPEC](#) section.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.287
SGF32D
– GAS SATURATION TABLES VERSUS OIL AND WATER SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SGF32D](#) keyword defines the gas relative permeability as a function of both oil and water saturations. This keyword should only be used if the gas is present in the run.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

See also the [SWOF](#), [SGOF](#), [SLGOF](#) series of keywords and the [SWFN](#), [SGFN](#), [SGWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#) series of keywords to enter relative permeability data.

8.3.288 SGFN – GAS SATURATION TABLES (FORMAT TYPE 2)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SGFN](#) keyword defines the gas relative permeability and oil-gas capillary pressure data versus gas saturation tables for when gas is present in the input deck. This keyword should only be used if the gas is present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGAS	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the gas saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRG	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas relative permeability.			None
		dimensionless	dimensionless	dimensionless	
3	PCOG	A columnar vector of real values that are either equal or increasing down the column that defines the oil-gas capillary pressure.			None
		psia	bars	atm	

Notes:

1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.162: SGFN Keyword Description

See also the [GSF – Gas Saturation Function Tables \(Gas-Water Systems\)](#) keyword in the [PROPS](#) section, that defines the gas relative permeability and gas-water capillary pressure data as a function of gas saturation, for when only the gas and water phases are present in the model. Note that the [GSF](#) keyword must be used in conjunction with the [WSF – Water Saturation Function Tables \(Gas-Water Systems\)](#) keyword, also in the [PROPS](#) section. [WSF](#) defines the water saturation as a function of water saturation for when only the gas and water phases are present in the model.

Example

--			
--	GAS RELATIVE PERMEABILITY TABLES (SGFN)		
--			
SGFN			
--	SGAS	KRG	PCGO
--	FRAC		PSIA
--	-----	-----	-----
	0.00	0.0000	0.0
	0.05	0.0000	0.0
	0.10	0.0000	0.0
	0.15	0.0000	0.0
	0.20	0.0002	0.0
	0.25	0.0010	0.0
	0.30	0.0062	0.0
	0.35	0.0140	0.0
	0.40	0.0273	0.0
	0.45	0.0450	0.0
	0.50	0.0707	0.0
	0.55	0.1020	0.0
	0.60	0.1412	0.0
	0.65	0.1870	0.0
	0.70	0.2412	0.0
	0.77	0.3288	0.0
	0.82	0.4000	0.0
	0.85	0.4450	0.0
			/ TABLE NO. 01
--	SGAS	KRG	PCGO
--	FRAC		PSIA
--	-----	-----	-----
	0.00	0.0000	0.0
	0.05	0.0000	0.0
	0.10	0.0000	0.0
	0.15	0.0000	0.0
	0.20	0.0002	0.0
	0.25	0.0010	0.0
	0.30	0.0062	0.0
	0.35	0.0140	0.0
	0.40	0.0273	0.0
	0.45	0.0450	0.0
	0.50	0.0707	0.0
	0.55	0.1020	0.0
	0.60	0.1412	0.0
	0.65	0.1870	0.0
	0.70	0.2412	0.0
	0.77	0.3288	0.0
	0.82	0.4000	0.0
	0.85	0.4450	0.0
			/ TABLE NO. 02

The example defines two *SGFN* tables for when gas is present in the input deck.

8.3.289 SGL – END-POINT SCALING GRID CELL CONNATE GAS SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SGL](#) defines the connate gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The connate gas saturation is defined as the minimum gas saturation in a two-phase gas relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGL	SGL is an array of real numbers assigning the connate gas saturation values to each cell in the model.The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.03			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the connate gas saturation array used with the end-point scaling option.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.163: SGL Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRORG](#), [KRORW](#) and [KRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SGLX](#), [SGLY](#) and [SGLZ](#) instead of [SGL](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SGLX](#), [SGLX-](#), [SGLY](#), [SGLY-](#), [SGLZ](#) and [SGLZ-](#), instead of the [SGL](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SGL DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SGL
      300*0.030
/
```

The above example defines a constant connate gas saturation of 0.03 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.290 SGLPC – END-POINT SCALING GRID CELL CAPILLARY PRESSURE CONNATE GAS SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SGLPC defines the connate gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the **ENDSCALE** keyword in the **RUNSPEC** section. The connate gas saturation is defined as the minimum gas saturation in a two-phase gas relative permeability table. The keyword only applies the scaling to the drainage capillary pressure tables, unlike the **SGL** keyword that applies the scaling to both the capillary pressure and relative permeability tables. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGLPC	SGLPC is an array of real numbers assigning the connate gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. If SGLPC is omitted from the input deck the values will be defaulted to those on the SGL series of keywords. If the SGL series of keywords are missing from the input deck then the values are taken from the cell allocated capillary pressure table. Repeat counts may be used, for example 30*0.03			Taken from SGL or from the cell allocated capillary pressure table.
		dimensionless	dimensionless	dimensionless	

Notes:

- Note this the directional independent version of the connate gas saturation array used with the end-point scaling option. If directional end-point scaling has been activated then the **SGLX±**, **SGLY±** and **SGZ±** series of keyword should be used.
- The keyword is terminated by a "/".

Table 8.164: SGLPC Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the **SWL**, **SWCR**, **SWU**, **SGL**, **SGCR**, **SGU**, **SOWCR**, and **SOGCR** saturation grid arrays for the saturation end-points, and the **KRG**, **KRGR**, **KRO**, **KROGR**, **KRORW**, **KRW** and **KRWR** relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is **SGLX**, **SGLY** and **SGLZ** instead of **SGL** or **SGLPC**. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is **SGLX-**, **SGLY-**, **SGLZ-** and **SGLZ-**, instead of the **SGL** or **SGLPC** keywords.

Missing Some Functionality - Use with Caution.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SGLPC DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SGLPC      300*0.030      /
```

The above example defines a constant connate gas saturation of 0.03 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.291 SGOF – GAS-OIL SATURATION TABLES VERSUS GAS (FORMAT TYPE 1)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SGOF](#) keyword defines the oil and gas relative permeability and oil-gas capillary versus gas saturation tables for when oil and gas are present in the input deck. This keyword should only be used if both oil and gas are present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGAS	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the gas saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRG	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas relative permeability.			None
		dimensionless	dimensionless	dimensionless	
3	KRO	A columnar vector of real values that are either equal or decreasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability with respect to gas and connate water saturation. When water is active in the run, the first entry the column, that is at $k_{rog}(S_g = 0)$, must be the same as the first entry in the corresponding SWOF table, that is at $k_{row}(S_o = 1 - S_{wco})$. The last value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
4	PCOG	A columnar vector of real values that are either equal or increasing down the column that defines the oil-gas relative capillary pressure.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.165: SGOF Keyword Description

8.3.292 SGOFLET – GAS-OIL LET RELATIVE PERMEABILITY FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SGOFLET](#) defines the relative permeability and capillary pressure parameters for the gas-oil LET family of models. Both the gas and oil phases should be made active in the model via the [GAS](#) and [OIL](#) keywords in the [RUNSPEC](#) section. See section [8.2.6 Saturation Table Generation - LET Functions](#) and Lomeland et al.^{245,246 247} and ²⁴⁸ for further information on the model.

The keyword is used as a replacement for the [SGOF](#) keyword for three-phase oil-gas-water systems, and the LET series of keywords cannot be combined with the standard set of relative permeability keywords.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGL	SGL is a real positive number less than one that defines the connate gas saturation, that is smallest gas saturation in the LET function.			0.0
		dimensionless	dimensionless	dimensionless	
2	SGCR	SGCR is a real positive number greater than or equal to SGL and less than one, that defines the critical gas saturation, that is the largest gas saturation for which the gas relative permeability is zero.			0.0
		dimensionless	dimensionless	dimensionless	
3	LGAS	LGAS is a real positive number that defines the LET Lower empirical parameter Lg for the gas phase with the associated oil phase in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
4	EGAS	EGAS is a real positive number that defines the LET Elevation empirical parameter Eg for the gas phase with the associated oil phase in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	

²⁴⁵ Lomeland F., Ebeltoft E. and Thomas W.H., 2005. A New Versatile Relative Permeability Correlation. Paper SCA2005-32 presented at the International Symposium of the Society of Core Analysts held in Toronto, Canada, 21-25 August, 2005.

²⁴⁶ Lomeland F. and Ebeltoft E., 2008. A New Versatile Capillary Pressure Correlation. Paper SCA2008-08 presented at the International Symposium of the Society of Core Analysts held in Abu Dhabi, UAE, 29 Oct. – 2 Nov., 2008.

²⁴⁷ Lomeland F., Hasanov B., Ebeltoft E. and Berge M., 2012. A Versatile Representation of Up-scaled Relative Permeability for Field Applications. Paper SPE 154487-MS presented at the EAGE Annual Conference & Exhibition incorporating SPE Europec held in Copenhagen, Denmark, 4-7 June 2012.

²⁴⁸ Lomeland F., 2018. Overview Of The Let Family Of Versatile Correlations For Flow Functions. Paper SCA2018-056 presented at the International Symposium of the Society of Core Analysts held in Trondheim, Norway, 27-30 August 2018.

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	TGAS	TGAS is a real positive number that defines the LET Top empirical parameter T_g for the gas phase with the associated oil phase in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
6	KRTGAS	KRTGAS is a real positive number less than one, that defines the relative permeability of gas at the maximum gas saturation K_{rgt} in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
7	SORG	SORG is a real positive number less than one that defines the residual oil saturation in a gas-oil system in the LET equations.			0.0
		dimensionless	dimensionless	dimensionless	
8	SOGCR	SOGCR is a real positive number less than one that defines critical oil-in-gas saturation, that is the largest oil saturation for which the oil relative permeability is zero in a gas-oil system.			0.0
		dimensionless	dimensionless	dimensionless	
9	LOIL	LOIL is a real positive number that defines the LET Lower empirical parameter L_o for the oil phase with the associated gas phase in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
10	EOIL	EOIL is a real positive number that defines the LET Elevation empirical parameter E_o for the oil phase with the associated gas phase in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
11	TOIL	TOIL is a real positive number that defines the LET Top empirical parameter T_o for the oil phase with the associated gas phase in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
12	KRTOIL	KRTOIL is a real positive number less than or equal to one, that defines the relative permeability of oil at the residual oil saturation, K_{rot} in the LET relative permeability equation.			1.0
		dimensionless	dimensionless	dimensionless	
13	LPC	LPC is a real positive number that defines the gas-oil LET Lower empirical parameter L in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	
14	EPC	EPC is a real positive number that defines the gas-oil LET Elevation empirical parameter E in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
15	TPC	TPC is a real positive number that defines the gas-oil LET Top empirical parameter T in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	
16	PCIR	PCIR is a real positive number that defines the gas-oil capillary pressure at connate water saturation (<i>SWL</i>) in the LET capillary pressure equations.			0.0
		psia	bars	atm	
17	PCIT	PCIT is a real positive number that defines the gas-oil threshold capillary pressure at the maximum water saturation in the LET capillary pressure equations.			0.0
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN definitions as declared on the *TABDIMS* keyword in the *RUNSPEC* section, with each definition consisting of the above 17 LET parameters.
- 2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.166: SGOFLET Keyword Description

Note there are two versions of the LET functions, LET²⁴⁵ for two-phase flowing conditions and LETx²⁴⁹ for three-phase flowing conditions. This keyword implements the LET version for a gas-oil system.

The functions are dependent on the drainage and imbibition cycle of the wetting phase as well as drainage and inhibition cycle number, since a reservoir may undergo several flooding events. To account for this the system defines the flooding event using the three saturations: *Sw*, *So*, and *Sg* together with the state of the three saturations during the flooding event. The saturation state can be Increasing, Decreasing, or Constant, for a given flooding event cycle number (*n*). Thus, *Sw(D)*, *So(I)*, *Sg(C)* or *DIC1*, means the water phase is decreasing, the oil phase is increasing and the gas phase is constant for the primary or first cycle (*n* equals one). This is case for when oil is migrating into the reservoir rock and displacing the initial water contained with the reservoir.

Note

All the LET parameters are dependent on the flooding event and flooding cycle, and thus are expected vary as such. To be clear, the values of [SGCR](#), *Lg*, *Lo* etc. should be different for each flooding cycle.

See also the [SGWFLET – Gas-Water LET Relative Permeability Functions](#), and [SWOFLET – Water-Oil LET Relative Permeability Functions](#) keywords in this section.

²⁴⁹ Lomeland F. and Ebeltoft E., 2013. Versatile Three-phase Correlations for Relative Permeability and Capillary Pressure. Paper SCA2013-034 presented at the International Symposium of the Society of Core Analysts held in Napa Valley, California, USA, 16-19 September, 2013.

Example

The following example uses the [SGOFLET](#) keyword to define two relative gas-oil relative permeability tables, based on NTSFUN equals two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--      SGOFLET - GAS-OIL LET REL. PERMEABILITY FUNCTIONS (OPM FLOW KEYWORD)
--
SGOFLET
--      SGL          SGCR          L-GAS          E-GAS          T-GAS          KRT-GAS
--      SOR          SOGCR         L-OIL           E-OIL           T-OIL          KRT-OIL
--      L-PC         E-PC          T-PC           PCIR          PCIT
--      -----
--      0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--      0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--      1*           1*           1*           1*           1*
--
--                                     / TABLE NO. 01
--
--      0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--      0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--      1*           1*           1*           1*           1*
--
--                                     / TABLE NO. 02
```

Here the [SGOFLET](#) keyword parameters are all set to their default values.

8.3.293 SGU – END-POINT SCALING GRID CELL GAS SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SGU](#) defines the maximum gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The maximum gas saturation is defined as the maximum gas saturation in a two-phase gas relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SGU	SGU is an array of real numbers assigning the maximum gas saturation values to each cell in the model. The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.70			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	
Notes: 1) Note this the directional independent version of the maximum gas saturation array used with the end-point scaling option. 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table. 3) The keyword is terminated by a "/".					

Table 8.167: SGU Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRORG](#), [KRORW](#) and [KRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SGUX](#), [SGUY](#) and [SGUZ](#) instead of [SGU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SGUX-](#), [SGUY-](#), [SGUZ-](#) instead of the [SGU](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SGU DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SGU
      300*0.700 /
```

The above example defines a constant connate gas saturation of 0.70 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.294 SGWFLET – GAS-WATER LET RELATIVE PERMEABILITY FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

SWGLET defines the relative permeability and capillary pressure parameters for the water-gas LET family of models. Both the gas and water phases should be made active in the model via the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section. This keyword should only be used in two-phase models containing the gas and water phases only, that is the oil phase must be absent. See section [8.2.6 Saturation Table Generation - LET Functions](#) and Lomeland et al.^{250,251 252} and ²⁵³ for further information on the model.

The keyword is used as a replacement for the [SGWFN](#) keyword for two-phase gas-water systems, and the LET series of keywords cannot be combined with the standard set of relative permeability keywords.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGL	SGL is a real positive number less than one that defines the connate gas saturation, that is smallest gas saturation in the LET function.			0.0
		dimensionless	dimensionless	dimensionless	
2	SGCR	SGCR is a real positive number greater than or equal to SGL and less than one, that defines the critical gas saturation, that is the largest gas saturation for which the gas relative permeability is zero.			0.0
		dimensionless	dimensionless	dimensionless	
3	LGAS	LGAS is a real positive number that defines the LET Lower empirical parameter Lg for the gas phase with the associated water phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
4	EGAS	EGAS is a real positive number that defines the LET Elevation empirical parameter Eg for the gas phase with the associated water phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	

²⁵⁰ Lomeland F., Ebeltoft E. and Thomas W.H., 2005. A New Versatile Relative Permeability Correlation. Paper SCA2005-32 presented at the International Symposium of the Society of Core Analysts held in Toronto, Canada, 21-25 August, 2005.

²⁵¹ Lomeland F. and Ebeltoft E., 2008. A New Versatile Capillary Pressure Correlation. Paper SCA2008-08 presented at the International Symposium of the Society of Core Analysts held in Abu Dhabi, UAE, 29 Oct. – 2 Nov., 2008.

²⁵² Lomeland F., Hasanov B., Ebeltoft E. and Berge M., 2012. A Versatile Representation of Up-scaled Relative Permeability for Field Applications. Paper SPE 154487-MS presented at the EAGE Annual Conference & Exhibition incorporating SPE Europec held in Copenhagen, Denmark, 4-7 June 2012.

²⁵³ Lomeland F., 2018. Overview Of The Let Family Of Versatile Correlations For Flow Functions. Paper SCA2018-056 presented at the International Symposium of the Society of Core Analysts held in Trondheim, Norway, 27-30 August 2018.

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	TGAS	TGAS is a real positive number that defines the LET Top empirical parameter Tg for the gas phase with the associated water phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
6	KRTGAS	KRTGAS is a real positive number less than one, that defines the relative permeability of gas at the maximum gas saturation Krgt in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
7	SWL	SWL is a real positive number less than one, that defines the connate water saturation, that is the smallest water saturation in the LET function.			0.0
		dimensionless	dimensionless	dimensionless	
8	SWCR	SWCR is a real positive number greater than or equal to SWL and less than one, that defines the critical water saturation, that is the largest water saturation for which the water relative permeability is zero, Swirr in equations.			0.0
		dimensionless	dimensionless	dimensionless	
9	LWAT	LWAT is a real positive number that defines the LET Lower empirical parameter Lw for the water phase with the associated gas phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
10	EWAT	EWAT is a real positive number that defines the LET Elevation empirical parameter Ew for the water phase with the associated gas phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
11	TWAT	TWAT is a real positive number that defines the LET Top empirical parameter Tw for the water phase with the associated gas phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
12	KRTWAT	KRTWAT is a real positive number less than one, that defines the relative permeability of water at the maximum water saturation (normally the maximum water saturation is one) Krwt in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
13	LPC	LPC is a real positive number that defines the gas-water LET Lower empirical parameter L in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
14	EPC	EPC is a real positive number that defines the gas-water LET Elevation empirical parameter E in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	
15	TPC	TPC is a real positive number that defines the gas-water LET empirical parameter T in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	
16	PCIR	PCIR is a real positive number that defines the gas-water capillary pressure at connate water saturation (<i>SWL</i>) in the LET capillary pressure equations.			0.0
		psia	bars	atm	
17	PCIT	PCIT is a real positive number that defines the gas-water threshold capillary pressure at the maximum water saturation in the LET capillary pressure equations.			0.0
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN definitions as declared on the *TABDIMS* keyword in the *RUNSPEC* section, with each definition consisting of the above 17 LET parameters.
- 1) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.168: SGWFLET Keyword Description

Note there are two versions of the LET functions, LET²⁵⁰ for two-phase flowing conditions and LETx²⁵⁴ for three-phase flowing conditions. This keyword implements the LET version for a gas-water system.

The functions are dependent on the drainage and imbibition cycle of the wetting phase as well as drainage and inhibition cycle number, since a reservoir may undergo several flooding events. To account for this the system defines the flooding event using the three saturations: Sw, So, and Sg together with the state of the three saturations during the flooding event. The saturation state can be Increasing, Decreasing, or Constant, for a given flooding event cycle number (n). Thus, Sw(D), So(C), Sg(I), or DCII, means the water phase is decreasing, the oil phase is constant and the gas phase is increasing for the primary or first cycle (n equals one). This is the case for when gas is migrating into the reservoir rock and displacing the initial water contained within the reservoir.

Note

All the LET parameters are dependent on the flooding event and flooding cycle, and thus are expected to vary as such. To be clear, the values of *SWCR*, Lg, Lw etc. should be different for each flooding cycle.

See also the *SGOFLET – Gas-Oil LET Relative Permeability Functions*, and *SWOFLET – Water-Oil LET Relative Permeability Functions* keywords in this section, that may be used for three-phase systems.

²⁵⁴ Lomeland F. and Ebeltoft E., 2013. Versatile Three-phase Correlations for Relative Permeability and Capillary Pressure. Paper SCA2013-034 presented at the International Symposium of the Society of Core Analysts held in Napa Valley, California, USA, 16-19 September, 2013.

Example

The following example uses the `SWGFL` keyword to define two relative gas-water relative permeability tables, based on `NTSFUN` equals two on the `TABDIMS` keyword in the `RUNSPEC` section.

```
--
--          SWGFL - GAS-WATER LET REL. PERMEABILITY FUNCTIONS (OPM FLOW KEYWORD)
--
SWGFL
--          SGL          SGR          L-GAS          E-GAS          T-GAS          KRT-GAS
--          SWL          SWR          L-WAT          E-WAT          T-WAT          KRT-WAT
--          L-PC          E-PC          T-PC          PCIR          PCIT
--
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          1*           1*           1*           1*           1*
--
--                                     / TABLE NO. 01
--
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          1*           1*           1*           1*           1*
--
--                                     / TABLE NO. 02
```

Here the `SWGFL` keyword parameters are all set to their default values.

8.3.295 SGWFN – GAS-WATER SATURATION TABLES (FORMAT TYPE 2)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SGWFN](#) keyword defines the gas and water relative permeability and gas-water capillary pressure data versus gas saturation tables for when gas and water are present in the input deck. This keyword should only be used if gas and water are present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGAS	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the gas saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRG	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas relative permeability. Note that the first entry in the column must be zero.			None
		dimensionless	dimensionless	dimensionless	
3	KRW	A columnar vector of real values that are either equal or decreasing down the column and that are greater than or equal to zero and less than or equal to one that defines the water relative permeability with respect to gas saturation. The last value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
4	PCGW	A columnar vector of real values that are either equal or increasing down the column that defines the gas-water relative capillary pressure.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.169: SGWFN Keyword Description

8.3.296
SHRATE – ACTIVATE LOG-BASED POLYMER SHEARING AND DEFINE THE SHEAR RATE CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the logarithm-based polymer shear thinning/thickening option and defines the shear rate constant. This keyword can only be used in conjunction with the [PLYSHLOG](#) in the [PROPS](#) section

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SHRATE	A positive real value that defines the shear rate constant.			4.8
		dimensionless	dimensionless	dimensionless	
Notes: 1) This keyword is followed by NTPVT values as declared on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section. 2) Each value is terminated by a "/" and there is no "/" terminator for the keyword.					

Table 8.170: SHRATE Keyword Description

Example

The following example activates the logarithm-based polymer shear thinning/thickening option and defines the shear rate constants for a run with two PVT regions.

```
--
--      ACTIVATE LOG-BASED POLYMER SHEAR THINNING-THICKENING OPTION
--      AND DEFINE THE SHEAR RATE CONSTANT
--
SHRATE
--      SHEAR RATE
--      CONSTANT
--      4.8      /
--      4.8      /
```


8.3.297 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

8.3.298 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

8.3.299 SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

8.3.300 SKPRPOLY – POLYMER MOLECULAR WEIGHT MODEL POLYMER INJECTION SKIN TABLE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [SKPRPOLY](#), describes the relationship of a water injection well's injected polymer skin pressure as a function of polymer throughput and water velocity, for the simulator's Polymer Molecular Weight Transport option. The table is a two dimensional table that relates the polymer throughput values and water velocity values to derive the resulting wellbore skin pressure of the injected polymer, which is then used to calculate the total wellbore skin pressure based on the polymer concentration.

This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator. The model has been tested using metric units; however, using either field or laboratory units with the option should be considered experimental.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	SKPRPNUM	A positive integer value greater than zero and less than or equal to the NTSKPOLY variable, as defined on the PINTDIMS keyword in the RUNSPEC section, that defines the SKPRPOLY Polymer Molecular Weight Model polymer injection skin pressure table number.			None
1-2	POLCON	A real positive value that the defines the reference polymer concentration for the table.			None
		lb/stb	kg/sm ³	gm/scc	
2-1	THRUPUT	A real positive monotonically increasing vector, that defines the polymer throughput values. The first entry should be zero to define a no throughput data set, and each vector record should be on a separate line terminated by a “/”.			None
		feet ³ /feet ²	m ³ /m ²	cm ³ /cm ²	
3-1	VELOCITY	A real positive monotonically increasing vector, that defines the polymer velocity values. The first entry should be zero to define a no velocity data set, and each vector record should be on a separate line terminated by a “/”.			None
		feet/day	m/day	cm/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4-1	PRESS	A series of real positive vectors representing the wellbore skin pressure, for the given reference polymer concentration (POLCON), for all combinations of the polymer throughput values (THRUPUT) and velocity values (VELOCITY), organized as a series of vectors PRESS(THRUPUT, VELOCITY). Thus, the first vector represents the wellbore skin pressure of the first THRUPUT value and each entry in the vector is the corresponding wellbore skin pressure of the associated VELOCITY vector. Each vector should be on a separate line and should be terminated by a “/”. Thus, if THRUPUT has three entries and VELOCITY has four, then there should be three vectors, with each vector containing four elements representing wellbore skin pressure values, as a function of THRUPUT and VELOCITY.			None
		psia	barsa	atma	

Notes:

1) The keyword should be terminated by a “/”, but may be repeated up to NTSKPOLY times, as per the *PINTDIMS* keyword in the *RUNSPEC* section, to allow for the input of multiple tables.

Table 8.171: SKPRPOLY Keyword Description

Unlike other [PROPS](#) section table keywords, that enable multiple tables following the keyword to be entered, the [SKPRPOLY](#) keyword requires that the keyword itself must be repeated for each table, as is shown in the example on the following page.

The [WSKPTAB](#) keyword in the [SCHEDULE](#) section may be used to assign the [SKPRPOLY](#) and [SKPRWAT](#) tables to water injections wells, that enable the calculation of the wellbore skin pressure based on the fluids being injected.

See also the [SKPRWAT](#), [PLYMWINJ](#), and [PLYVMH](#) keywords, in the [PROPS](#) section, that are the additional property keywords required for the Polymer Molecular Weight Transport option. In addition, see also the [WPMITAB](#) keyword in the [SCHEDULE](#) section, that assigns the [PLYMWINJ](#) tables to the water injection wells.

Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

Given NTSKPOLY equals two on the [PINTDIMS](#) keyword in the [RUNSPEC](#) section, then two [SKPRPOLY](#) tables are required to be entered:

```
--
--      POLYMER MOLECULAR WEIGHT MODEL POLYMER INJECTION SKIN TABLE
--      (OPM FLOW PROPS KEYWORD)
--
SKPRPOLY
--      TABLE      POLYMER REF
--      NO.          CONCENTRATION
--      1            2.00                      /
--
--      THROUGHPUT VALUES
--
--      0.0          200.0    400.0          /
--
--      VELOCITY VALUES
--
--      0.0          50.0     80.0     100.0 /
--
--      PRESS SKIN VALUES
--
--      0.0          10.0     20.0     40.0 / PRESS(THRUPUT=1, VELOCITY=1 TO N)
--      0.0          50.0     100.0    200.0 / PRESS(THRUPUT=2, VELOCITY=1 TO N)
--      0.0          100.0    200.0    400.0 / PRESS(THRUPUT=3, VELOCITY=1 TO N)
/
SKPRPOLY
--      TABLE      POLYMER REF
--      NO.          CONCENTRATION
--      2            2.0                      /
--
--      THROUGHPUT VALUES
--
--      0.0          200.0    400.0          /
--
--      VELOCITY VALUES
--
--      0.0          30.0     50.0     100.0 /
--
--      PRESS SKIN VALUES
--
--      0.0          10.0     20.0     40.0 / PRESS(THRUPUT=1, VELOCITY=1 TO N)
--      0.0          50.0     100.0    200.0 / PRESS(THRUPUT=2, VELOCITY=1 TO N)
--      0.0          100.0    200.0    400.0 / PRESS(THRUPUT=3, VELOCITY=1 TO N)
/
```

As mentioned previously, the [SKPRPOLY](#) keyword requires that the keyword itself must be repeated for each table, as is shown in the above example.

8.3.301 SKPRWAT – POLYMER MOLECULAR WEIGHT MODEL WATER INJECTION SKIN TABLE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [SKPRWAT](#), describes the relationship of a water injection well's injected water skin pressure as a function of water throughput and water velocity, for the simulator's Polymer Molecular Weight Transport option. The table is a two dimensional table that relates the water throughput values and water velocity values to derive the resulting wellbore skin pressure of the injected water, which is then used to calculate the total wellbore skin pressure based on the polymer concentration.

This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

The model has been tested using metric units; however, using either field or laboratory units with the option should be considered experimental.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	SKPRWNUM	A positive integer value greater than zero and less than or equal to the NTSKWAT variable, as defined on the PINTDIMS keyword in the RUNSPEC section, that defines the SKPRWAT Polymer Molecular Weight Model water injection skin pressure table number.			None
2-1	THRUPUT	A real positive monotonically increasing vector, that defines the water throughput values. The first entry should be zero to define a no throughput data set, and each vector record should be on a separate line terminated by a “/”.			None
		feet ³ /feet ²	m ³ /m ²	cm ³ /cm ²	
3-1	VELOCITY	A real positive monotonically increasing vector, that defines the water velocity values. The first entry should be zero to define a no velocity data set, and each vector record should be on a separate line terminated by a “/”.			None
		feet/day	m/day	cm/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4-1	PRESS	A series of real positive vectors representing the wellbore skin pressure, for all combinations of the water throughput values (THRUPUT) and velocity values (VELOCITY), organized as a series of vectors PRESS(THRUPUT,VELOCITY). Thus, the first vector represents the wellbore skin pressure of the first THRUPUT value and each entry in the vector is the corresponding wellbore skin pressure of the associated VELOCITY vector. Each vector should be on a separate line and should be terminated by a “/”. Thus, if THRUPUT has three entries and VELOCITY has four, then there should be three vectors, with each vector containing four elements representing wellbore skin pressure values, as a function of THRUPUT and VELOCITY.			None
		psia	barsa	atma	

Notes:

1) The keyword should be terminated by a “/”, but may be repeated up to NTSKWAT times, as per the *PINTDIMS* keyword in the *RUNSPEC* section, to allow for the input of multiple tables.

Table 8.172: SKPRWAT Keyword Description

Unlike other [PROPS](#) section table keywords, that enable multiple tables following the keyword to be entered, the [SKPRWAT](#) keyword requires that the keyword itself must be repeated for each table, as is shown in the example on the following page.

The [WSKPTAB](#) keyword in the [SCHEDULE](#) section may be used to assign the [SKPRPOLY](#) and [SKPRWAT](#) tables to water injections wells, that enable the calculation of the wellbore skin pressure based on the fluids being injected.

See also the [SKPRPOLY](#), [PLYMWINJ](#), and [PLYVMH](#) keywords, in the [PROPS](#) section, that are the additional property keywords required for the Polymer Molecular Weight Transport option. In addition, see also the [WPMITAB](#) keyword in the [SCHEDULE](#) section, that assigns the [PLYMWINJ](#) tables to the water injection wells.

Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

Given NTSKWAT equals two on the [PINTDIMS](#) keyword in the [RUNSPEC](#) section, then two [SKPRWAT](#) tables are required to be entered:

```

--
--      POLYMER MOLECULAR WEIGHT MODEL WATER INJECTION SKIN TABLE
--      (OPM FLOW PROPS KEYWORD)
--
SKPRWAT
1                                / TABLE NUMBER
--
--      THROUGHPUT VALUES
--
0.0      200.0    400.0      /
--
--      VELOCITY VALUES
--
0.0      50.0     70.0     100.0 /
--
--      PRESS SKIN VALUES
--
0.0      2.0      4.0      8.0  / PRESS(THRUPUT=1, VELOCITY=1 TO N)
0.0      20.0     40.0     80.0 / PRESS(THRUPUT=2, VELOCITY=1 TO N)
0.0      50.0     100.0    200.0 / PRESS(THRUPUT=3, VELOCITY=1 TO N)
/
SKPRWAT
2                                / TABLE NUMBER
--
--      THROUGHPUT VALUES
--
0.0      200.0    400.0      /
--
--      VELOCITY VALUES
--
0.0      30.0     50.0     100.0 /
--
--      PRESS SKIN VALUES
--
0.0      2.0      4.0      8.0  / PRESS(THRUPUT=1, VELOCITY=1 TO N)
0.0      20.0     40.0     80.0 / PRESS(THRUPUT=2, VELOCITY=1 TO N)
0.0      50.0     100.0    200.0 / PRESS(THRUPUT=3, VELOCITY=1 TO N)
/

```

As mentioned previously, the [SKPRWAT](#) keyword requires that the keyword itself must be repeated for each table, as is shown in the above example.

8.3.302 SKRO – END-POINT SCALING OF GRID CELL KRO(SWL) (SURFACTANT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SKRO](#) defines the scaling parameter for the surfactant oil relative permeability value at the connate water saturation ([SWL](#)), for all the cells in the model via an array. The [ENDSCALE](#) keyword in the [RUNSPEC](#) section should be activated to enable end-point scaling and the use of this keyword. In addition, the Surfactant option must be enabled by either the [SURFST](#) or [SURFSTES](#) keywords in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SKRO	SKRO is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling SKRO values for each cell in the model. Repeat counts may be used, for example 50*0.500.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 8.173: SKRO Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be scale on the relative permeability values using the [SKRO](#), [SKROG](#), [SKRORW](#), [SKRW](#) and [SKRWR](#) surfactant relative permeability grid cell arrays for the relative permeability end-point data.

Example

The example defines an input box for the whole grid and for layers one to three, for layer one *SKRO* is set equal to 0.850, for layer two *SKRO* equals 0.875, and for layer three *SKRO* equals 0.900.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      1*  1*   1*  1*   1   3                / DEFINE BOX AREA
--
--      SET SKRO VALUES FOR THREE LAYERS IN THE MODEL
--
SKRO     1000*0.855  1000*0.875  1000.0.900      /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

8.3.303 SKRORG – END-POINT SCALING OF GRID CELL KRO(SGCR) (SURFACTANT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SKRORG defines the scaling parameter for the surfactant relative permeability of oil at the critical gas saturation (**SGCR**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the Surfactant option must be enabled by either the **SURFST** or **SURFSTES** keywords in the **RUNSPEC** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SKRORG	SKRORG is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling SKRORG values for each cell in the model. Repeat counts may be used, for example 50*0.850.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 8.174: SKRORG Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be scale on the relative permeability values using the **SKRO**, **SKRORG**, **SKRORW**, **SKRW** and **SKRWR** surfactant relative permeability grid cell arrays for the relative permeability end-point data.

Example

The example uses the **EQUALS** keyword to set layer one **SKRORG** equal to 0.750, layer two **SKRORG** equals 0.775, and layer three **SKRORG** equals 0.800.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      SKRORG      0.7550      1*  1*   1*  1*   1   1  / SKRORG FOR LAYER 1
      SKRORG      0.7750      1*  1*   1*  1*   2   2  / SKRORG FOR LAYER 2
      SKRORG      0.8000      1*  1*   1*  1*   3   3  / SKRORG FOR LAYER 3
/
```

8.3.304 SKRORW – END-POINT SCALING OF GRID CELL KRO(SWCR) (SURFACTANT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SKRORW defines the scaling parameter for the surfactant relative permeability of oil at the critical water saturation (**SWCR**), for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the Surfactant option must be enabled by either the **SURFST** or **SURFSTES** keywords in the **RUNSPEC** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SKRORW	SKRORW is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling SKRORW values for each cell in the model. Repeat counts may be used, for example 50*0.850			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 8.175: SKRORW Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be scale on the relative permeability values using the **SKRO**, **SKRORG**, **SKRORW**, **SKRW** and **SKRWR** surfactant relative permeability grid cell arrays for the relative permeability end-point data.

Example

The example defines an input box for the whole grid and for layers one to three, for layer one **SKRORW** is set equal to 0.750, for layer two **SKRORW** equals 0.775, and for layer three **SKRORW** equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2  J1  J2  K1  K2
BOX      1*  1*  1*  1*  1   3                / DEFINE BOX AREA
--
--      SET SKRORW VALUES FOR THREE LAYERS IN THE MODEL
--
SKRORW   1000*0.755  1000*0.775  1000.0.800    /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
ENDBOX
```

8.3.305 SKRW – END-POINT SCALING OF GRID CELL KRW(Sw =1.0) (SURFACTANT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SKRW defines the scaling parameter at the maximum surfactant water relative permeability value (**SWU**), that is for $Sw = 1.0$, for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the Surfactant option must be enabled by either the **SURFST** or **SURFSTES** keywords in the **RUNSPEC** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SKRW	SKRW is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling SKRW values for each cell in the model. Repeat counts may be used, for example 50*1.000.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the $NX \times NY \times NZ$ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 8.176: SKRW Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be scale on the relative permeability values using the **SKRO**, **SKROG**, **SKRORW**, **SKRW** and **SKRWR** surfactant relative permeability grid cell arrays for the relative permeability end-point data.

Example

The example uses the **EQUALS** keyword to set **SKRW** for layer one equal to 0.850, layer two **SKRW** to 0.875, and layer three **KRW** to 0.900.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
EQUALS
      SKRW      0.8550      1*  1*   1*  1*   1   1  / SKRW FOR LAYER 1
      SKRW      0.8750      1*  1*   1*  1*   2   2  / SKRW FOR LAYER 2
      SKRW      0.9000      1*  1*   1*  1*   3   3  / SKRW FOR LAYER 3
/
```

8.3.306 SKRWR – END-POINT SCALING OF GRID CELL KRWR(SOWCR) (SURFACTANT)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SKRWR defines the scaling parameter at the critical oil to water saturation value (**SOWCR**), for the surfactant water relative permeability curve, for all the cells in the model via an array. The **ENDSCALE** keyword in the **RUNSPEC** section should be activated to enable end-point scaling and the use of this keyword. In addition, the Surfactant option must be enabled by either the **SURFST** or **SURFSTES** keywords in the **RUNSPEC** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SKRWR	SKRWR is an array of positive real numbers which are greater than zero and less than or equal to 1.0, that are the assigned scaling SKRWR values for each cell in the model. Repeat counts may be used, for example 50*1.000.			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.

Table 8.177: SKRWR Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be scale on the relative permeability values using the **SKRO**, **SKROG**, **SKRORW**, **SKRW** and **SKRWR** surfactant relative permeability grid cell arrays for the relative permeability end-point data.

Example

The first example defines an input box for the whole grid and for layers one to three, for layer one *SKRWR* is set equal to 0.750, for layer two *SKRWR* equals 0.775, and for layer three *SKRWR* equals 0.800.

```
--
--      DEFINE INPUT BOX FOR EDITING INPUT ARRAYS (NX=100, NY=100)
--
--      ----- BOX -----
--      I1  I2   J1  J2   K1  K2
BOX      1*  1*   1*  1*   1   3
--                                     / DEFINE BOX AREA
--
--      SET SKRWR VALUES FOR THREE LAYERS IN THE MODEL
--
--      SKRWR
--      1000*0.755  1000*0.775  1000.0.800
--                                     /
--
--      DEFINE END OF INPUT BOX EDITING OF INPUT ARRAYS
--
--      ENDBOX
```

8.3.307 SLGOF – GAS-OIL SATURATION TABLES VERSUS GAS (FORMAT TYPE 1)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SLGOF](#) keyword defines the oil and gas relative permeability and oil-gas capillary pressure versus liquid saturation tables for when oil and gas are present in the input deck. This keyword should only be used if both oil and gas are present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SLIQ	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the liquid saturation, that is the connate water saturation (SWL) plus the oil saturation. The first entry should correspond to residual liquid, that is $S_{wc} + S_{org}$ and the last entry should be 1.0 to correspond to a gas saturation of zero.			None
		dimensionless	dimensionless	dimensionless	
2	KRG	A columnar vector of real values that are either equal or decreasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas relative permeability..			None
		dimensionless	dimensionless	dimensionless	
3	KRO	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability with respect to gas and connate water saturation. When water is active in the run, the last entry the column, that is at $k_{rog}(S_g = 0)$, must be the same as the first entry in the corresponding SWOF table, that is at $k_{row}(S_o = 1 - S_{wco})$. The first value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
4	PCOG	A columnar vector of real values that are either equal or decreasing down the column that defines the oil-gas relative capillary pressure.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.178: SLGOF Keyword Description

Example

--	GAS-OIL RELATIVE PERMEABILITY TABLES (SLGOF)			
--	SLGOF			
--	SLIQ	KRG	KROG	PCOG
--	FRAC			PSIA
--	-----	-----	-----	-----
	0.30060	0.55000	0.0000	0.0000
	0.31480	0.42500	0.2848	0.0000
	0.32910	0.35000	0.3091	0.0000
	0.34330	0.27500	0.4333	0.0000
	0.35760	0.25000	0.5576	0.0000
	0.37180	0.22500	0.5818	0.0000
	0.38610	0.20000	0.6061	0.0000
	0.40030	0.17500	0.6303	0.0000
	0.41450	0.15000	0.6545	0.0000
	0.42880	0.12500	0.6788	0.0000
	0.44300	0.10000	0.7030	0.0000
	0.45730	0.07500	0.7273	0.0000
	0.47150	0.05000	0.7515	0.0000
	0.48580	0.02500	0.7758	0.0000
	0.50000	0.00000	0.8000	0.0000
	0.80000	0.00000	0.9000	0.0000
	/ TABLE No. 01			
--	-----	-----	-----	-----
	0.30060	0.55000	0.0000	0.0000
	0.31480	0.42500	0.2848	0.0000
	0.32910	0.35000	0.3091	0.0000
	0.34330	0.27500	0.4333	0.0000
	0.35760	0.25000	0.5576	0.0000
	0.37180	0.22500	0.5818	0.0000
	0.38610	0.20000	0.6061	0.0000
	0.40030	0.17500	0.6303	0.0000
	0.41450	0.15000	0.6545	0.0000
	0.42880	0.12500	0.6788	0.0000
	0.44300	0.10000	0.7030	0.0000
	0.45730	0.07500	0.7273	0.0000
	0.47150	0.05000	0.7515	0.0000
	0.48580	0.02500	0.7758	0.0000
	0.50000	0.00000	0.8000	0.0000
	0.80000	0.00000	0.9000	0.0000
	/ TABLE No. 02			

The example defines two *SLGOF* tables for use when oil, gas and water are present in the run.

8.3.308 SOCRS – END-POINT SCALING GRID CELL MISCIBLE CRITICAL OIL SATURATION WITH RESPECT TO WATER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[SOCRS](#) defines the miscible critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to water is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase oil-water relative permeability table. The keyword is used with the Surfactant model to re-scale the surfactant relative permeability saturation tables allocated to a grid block by the [SURFNUM](#) keyword in the [REGIONS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOCRS	SOCRS is an array of real numbers assigning the critical oil saturation with respect to water values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the critical gas saturation used with the end-point scaling option. If directional end-point scaling has been activated then the SOCRSX± , SOCRSY± and SOCRSZ± series of keyword should be used.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.179: [SOGCR](#) Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOCRS](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRORG](#), [KRORW](#) and [KRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SOCRS DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SOCRS
      300*0.200 /
```

The above example defines a constant critical oil saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.309 SOF2 – OIL SATURATION TABLES WITH RESPECT TO GAS OR WATER (FORMAT TYPE 2)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SOF2](#) keyword defines the oil relative permeability versus oil saturation tables for when oil and gas or oil and water are present in the input deck. The keyword is also used to define the relative permeability of the miscible hydrocarbon phase in [SOLVENT](#) runs. This keyword should only be used if oil is present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOIL	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the oil or the hydrocarbon solvent saturation. For two phase runs the oil saturation should be entered and for when the SOLVENT option has been activated in the RUNSPEC section the total hydrocarbon phase (including the solvent) should be entered, that is SOIL = So + Sg + Ss.			None
		dimensionless	dimensionless	dimensionless	
3	KRO	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability with respect to gas and connate water saturation. For two phase runs the oil relative permeability should be entered and for when the SOLVENT option has been activated in the RUNSPEC section the relative permeability of the miscible hydrocarbon phase with respect to water. The last value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.180: SOF2 Keyword Description

Note this keyword should only be used for when the [SOLVENT](#) keyword in the [RUNSPEC](#) section has been activated. It should not be use for two-phase oil-water runs.

8.3.310 SOF3 – OIL SATURATION TABLES WITH RESPECT TO GAS AND WATER (FORMAT TYPE 2)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SOF3](#) keyword defines the oil relative permeability versus oil saturation tables for when oil, gas and water are present in the input deck. The keyword should only be used if oil, gas and water are present in the input deck.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOIL	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the oil or the hydrocarbon solvent saturation. The final entry should be at the connate water saturation, that is 1 - Swc.			None
		dimensionless	dimensionless	dimensionless	
3	KROW	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability with respect to oil and water saturation. The first value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
4	KROG	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability with respect to oil, gas and connate water saturation. The first value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.181: SOF3 Keyword Description

Example

--				
--		OIL RELATIVE PERMEABILITY TABLES (SOF3)		
--				
SOF3				
--	SOIL	KRO	KROG	
--	FRAC	FRAC	FRAC	
--	-----	-----	-----	
	0.00	0.000000	0.00000	
	0.05	1.197e-5	0.00000	
	0.10	0.000191	0.00000	
	0.15	0.000969	0.00000	
	0.20	0.003065	0.00000	
	0.25	0.007483	0.00000	
	0.30	0.015517	0.05932	
	0.35	0.028747	0.13158	
	0.40	0.049041	0.21082	
	0.45	0.078555	0.29960	
	0.50	0.119730	0.40095	
	0.55	0.175297	0.51818	
	0.60	0.248272	0.65476	
	0.65	0.341961	0.81420	
	0.70	0.459956	1.00000	
	0.75	0.606134	1.00000	
	0.80	0.784664	1.00000	
	0.85	1.000000	1.00000	/ TABLE NO. 1
--	-----	-----	-----	
	0.00	0.000000	0.00000	
	0.05	1.197e-5	0.00000	
	0.10	0.000191	0.00000	
	0.15	0.000969	0.00000	
	0.20	0.003065	0.00000	
	0.25	0.007483	0.00000	
	0.30	0.015517	0.05932	
	0.35	0.028747	0.13158	
	0.40	0.049041	0.21082	
	0.45	0.078555	0.29960	
	0.50	0.119730	0.40095	
	0.55	0.175297	0.51818	
	0.60	0.248272	0.65476	
	0.65	0.341961	0.81420	
	0.70	0.459956	1.00000	
	0.75	0.606134	1.00000	
	0.80	0.784664	1.00000	
	0.85	1.000000	1.00000	/ TABLE NO. 2

The example defines two [SOF3](#) tables for when oil,gas and water are present in the input deck.

8.3.311 SOF32D – OIL SATURATION TABLES WITH RESPECT TO WATER AND GAS (THREE PHASE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SOF32D](#) keyword defines the three phase oil relative permeability versus water and gas saturation tables for when oil, gas and water are present in the input deck. The keyword should only be used if oil, gas and water are present in the input deck. Normally the simulator calculates the three-phase oil relative permeabilities based on the entered two phase tables of water-oil and gas-oil, combined with the [STONE1](#) and [STONE2](#) keywords in the [PROPS](#) section that determine the method used to generate the thee phase oil relative permeability curves. [SOF32D](#) allows for the direct input of the three phase tables, as such the [STONE1](#) and [STONE2](#) keywords should not be entered if [SOF32D](#) is used in the input deck.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.312 SOGCR – END-POINT SCALING GRID CELL CRITICAL OIL SATURATION WITH RESPECT TO GAS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[SOGCR](#) defines the critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to gas is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase gas-oil relative permeability table. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOGCR	SOGCR is an array of real numbers assigning the critical oil saturation with respect to gas values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30 dimensionless			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the critical gas saturation used with the end-point scaling option. If directional end-point scaling has been activated then the [SOGCRX_±](#), [SOGCRY_±](#) and [SOGCRZ_±](#) series of keyword should be used.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.182: SOGCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRGR](#), [KRO](#), [KRORG](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SOGCRX](#), [SOGCRY](#) and [SOGCRZ](#) instead of [SOGCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SOGCRX](#), [SOGCRX-](#), [SOGCRY](#), [SOGCRY-](#), [SOGCRZ](#) and [SOGCRZ-](#), instead of the [SOGCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SOGCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SOGCR
      300*0.200 /
```

The above example defines a constant critical gas saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.313 SOMGAS – STONE1 MODEL MINIMUM OIL SATURATION VERSUS GAS SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the minimum oil saturation as a function of gas saturation for Stone’s²⁵⁵ first three phase oil relative permeability model as modified by Aziz and Settari²⁵⁶. If the [STONE1](#) and [STONE2](#) keywords are not present in the input deck then the default three phase oil relative permeability model is employed. The [SOMGAS](#) and [STONE1](#) keywords should only be used in three phase runs containing the oil, gas and water phases. The keyword is optional.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

255

Stone, H. L. “Probability Model for Estimating Three-Phase Relative Permeability,” paper SPE 2116, Journal of Canadian Petroleum Technology (1973) 22, No. 2, 214-218.

256

Aziz, K. and Settari, A. Petroleum Reservoir Simulation, London, UK, Applied Science Publishers (1979), page 398.

8.3.314 SOMWAT – STONE1 MODEL MINIMUM OIL SATURATION VERSUS WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the minimum oil saturation as a function of water saturation for Stone’s²⁵⁷ first three phase oil relative permeability model as modified by Aziz and Settari²⁵⁸. If the [STONE1](#) and [STONE2](#) keywords are not present in the input deck then the default three phase oil relative permeability model is employed. The [SOMWAT](#) and [STONE1](#) keywords should only be used in three phase runs containing the oil, gas and water phases. The keyword is optional.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

257

Stone, H. L. “Probability Model for Estimating Three-Phase Relative Permeability,” paper SPE 2116, Journal of Canadian Petroleum Technology (1973) 22, No. 2, 214-218.

258

Aziz, K. and Settari, A. Petroleum Reservoir Simulation, London, UK, Applied Science Publishers (1979), page 398.

8.3.315 SORWMIS – MISCIBLE RESIDUAL OIL VERSUS WATER SATURATION FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

SORWMIS defines the dependency between the miscible residual oil saturation and the water saturation, for when the **MISCIBLE** keyword in the **RUNSPEC** section has been activated. The keyword can only be used with the **MISCIBLE** option and for when the oil, water and gas phases are active in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the water saturation.			None
		dimensionless	dimensionless	dimensionless	
2	SORMIS	A columnar vector of real equal or increasing down the column values that are greater than or equal to zero and less than one, that define the corresponding miscible residual oil saturation for the corresponding water saturation SWAT .			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTMISC tables as declared on the **MISCIBLE** keyword in the **RUNSPEC** section.
- 2) Each table must contain a minimum of two rows and a maximum of NSMISC rows as declared on the **MISCIBLE** keyword in the **RUNSPEC** section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.183: SORWMIS Keyword Description

Example

```
--
--      MISCIBLE RESIDUAL OIL VERSUS WATER SATURATION TABLE
--
SORWMIS
--      SWAT      SORWMIS
--      FRAC      FRAC
--      -----
--      0.0000      0.0000
--      0.2000      0.0000
--      1.0000      0.0000 / TABLE NO. 01
--
--      SWAT      SORWMIS
--      FRAC      FRAC
--      -----
--      0.0000      0.0000
--      0.3000      0.1000
--      0.7500      0.1500 / TABLE NO. 02
```

The above example defines two miscible residual oil versus water saturation tables assuming NTMISC equals two and NSMISC is greater than or equal to three on the **MISCIBLE** keyword in the **RUNSPEC** section.

8.3.316 SOWCR – END-POINT SCALING GRID CELL CRITICAL OIL SATURATION WITH RESPECT TO WATER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SOWCR](#) defines the critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to water is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase oil-water relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOWCR	SOWCR is an array of real numbers assigning the critical oil saturation with respect to water values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the critical gas saturation used with the end-point scaling option. If directional end-point scaling has been activated then the [SOWCRX±](#), [SOWCRY±](#) and [SOWCRZ±](#) series of keyword should be used.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.184: SOWCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRGR](#), [KRO](#), [KRORG](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SOWCRX](#), [SOWCRY](#) and [SOWCRZ](#) instead of [SOWCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SOWCRX-](#), [SOWCRY-](#), [SOWCRZ-](#), instead of the [SOWCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SOWCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SOWCR
```

300*0.200

/

The above example defines a constant critical oil saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.317 SPECHEAT – DEFINE THE SPECIFIC HEAT OF OIL, WATER AND GAS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SPECHEAT](#) defines the specific heat of the oil, water and gas phases for various PVT regions in the model for when the [THERMAL](#) option has been activated in the [RUNSPEC](#) section. The number of [SPECHEAT](#) vector data sets is defined by the NTPVT parameter on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and the allocation of the [SPECHEAT](#) data sets to different grid blocks in the model is done via the [PVTNUM](#) keyword in the [REGIONS](#) section.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	A columnar vector of real monotonically increasing down the column values that define the temperature for the corresponding oil, water and gas specific heat values.			None
		°F	°C	°C	
2	OILSHEAT	OILSHEAT is a columnar vector of positive real numbers defining the specific heat of oil at the corresponding temperature, TEMP .			None
		Btu/lb/oR	kJ/kg/K	J/gm/K	
3	WATSHEAT	WATSHEAT is a columnar vector of positive real numbers defining the specific heat of water at the corresponding temperature, TEMP .			None
		Btu/lb/oR	kJ/kg/K	J/gm/K	
4	GASSHEAT	GASHEAT is a columnar vector of positive real numbers defining the specific heat of gas at the corresponding temperature, TEMP .			None
		Btu/lb/oR	kJ/kg/K	J/gm/K	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.185: SPECHEAT Keyword Description

See also the [SPECROCK](#) keyword to define the reservoir rock specific heat.

Example

The example below defines three fluid phases specific heat versus temperature tables assuming NTPVT equals three and NPPVT is greater than or equal to two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--          SPECIFIC HEAT OF OIL, WATER AND GAS TABLE
--
SPECHEAT
--          TEMP          SPECHEAT    SPECHEAT    SPECHEAT
--                   OIL          WATER          GAS
--          -----
--                   0.000      0.5000      1.5000      0.5000
--                   250.000    0.5000      1.5000      0.5000      / TABLE NO. 01
--          TEMP          SPECHEAT    SPECHEAT    SPECHEAT
--                   OIL          WATER          GAS
--          -----
--                   0.000      0.5500      1.5000      0.5000
--                   260.000    0.5500      1.5000      0.5000      / TABLE NO. 02
--          TEMP          SPECHEAT    SPECHEAT    SPECHEAT
--                   OIL          WATER          GAS
--          -----
--                   0.000      0.5500      1.5500      0.5000
--                   270.000    0.6000      1.5500      0.5000      / TABLE NO. 03
```

There is no terminating “/” for this keyword.

8.3.318 SPECROCK – DEFINE THE SPECIFIC HEAT OF THE RESERVOIR ROCK

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SPECROCK](#) defines the specific heat of the reservoir rock for various PVT regions in the model for when the [THERMAL](#) option has been activated in the [RUNSPEC](#) section. The number of [SPECROCK](#) vector data sets is defined by the NTSFUN parameter on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and the allocation of the [SPECROCK](#) data sets to different grid blocks in the model is done via the [SATNUM](#) keyword in the [REGIONS](#) section.

This keyword can only be used if OPM's Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	A columnar vector of real monotonically increasing down the column values that define the temperature for the corresponding rock specific heat values.			None
		°F	°C	°C	
2	ROCKHEAT	ROCKHEAT is a columnar vector of positive real numbers defining the specific heat of the rock at the corresponding temperature, TEMP .			None
		Btu/ft3/oR	kJ/m3/K	J/cc/K	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.186: SPECROCK Keyword Description

See also the [SPECHEAT](#) keyword to define the specific heat relationships for the oil, water and gas phases.

Example

The example below defines three rock specific heat versus temperature tables assuming NTSFUN equals three and NSSFUN is greater than or equal to two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--          SPECIFIC HEAT OF ROCK
--
SPECROCK
--          TEMP          SPECHEAT
--                      ROCK
--          -----
--              0.000      20.000
--             250.000      20.000
--                               / TABLE NO. 01
--          -----
--              0.000      21.000
--             260.000      21.000
--                               / TABLE NO. 02
--          -----
--              0.000      23.000
--             270.000      23.000
--                               / TABLE NO. 03
```

There is no terminating “/” for this keyword.

8.3.319 SSFN – SOLVENT AND GAS RELATIVE PERMEABILITY TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SSFN](#) keyword defines the miscible normalized relative permeability tables for when the [SOLVENT](#) option has been activated in the [RUNSPEC](#) section using the respective keyword. The [SOLVENT](#) keyword results in a four component model (oil, water and gas, plus a solvent). This keyword should only be used if the [SOLVENT](#) option has been activated.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGAS	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the gas plus solvent saturation ration which is defined as either: $\frac{S_g}{(S_g + S_s)} \text{ or } \frac{S_s}{(S_g + S_s)}$ Where Sg is the gas saturation and Ss is the solvent saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRGt	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the gas relative permeability. The resulting gas relative permeability is calculated from: $k_{rg} = k_{rgt}(S_g + S_s)k_{rg}^t$ where krg^t is the data in this column and krgt is the gas relative permeability from the SGFN keyword.			None
		dimensionless	dimensionless	dimensionless	
3	KRSt	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the solvent relative permeability. The resulting solvent relative permeability is calculated from: $k_{rs} = k_{rgt}(S_g + S_s)k_{rs}^t$ where krS^t is the data in this column and krgt is the gas relative permeability from the SGFN keyword.			None
		dimensionless	dimensionless	dimensionless	

Notes:

1) The keyword is followed by NTSFUN tables as declared on the TABDIMS keyword in the RUNSPEC section

2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the TABDIMS keyword in the RUNSPEC section.

3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.187: SSFN Keyword Description

Example

```
--
--      SOLVENT RELATIVE PERMEABILITY TABLES
--
SSFN
--      SGAS      KRGD      KRST
--      FRAC
--      -----
--      0.0000    0.0000    0.0000
--      1.0000    1.0000    1.0000      / TABLE NO. 01
--
--      -----
--      0.0000    0.0000    0.0000
--      0.2000    0.2000    0.3000
--      0.4000    0.3000    0.5000
--      0.6000    0.4000    0.7000
--      0.8000    0.5000    0.7500
--      1.0000    1.0000    1.0000      / TABLE NO. 02
```

The above example defines two [SSFN](#) tables for use with the [SOLVENT](#) option.

8.3.320 SSGCR – END-POINT SCALING GRID CELL SURFACTANT CRITICAL GAS SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSGCR](#) defines the surfactant critical gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The critical gas saturation is defined as the maximum gas saturation for which the gas relative permeability is zero in a two-phase relative permeability table. [SSGCR](#) is used to scale the surfactant oil relative permeability to gas data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSGCR	SSGCR is an array of real numbers assigning the surfactant critical gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.03			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	
Notes: 1) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table. 2) The keyword is terminated by a “/”.					

Table 8.188: SSGCR Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the SKRG, SKROG, SKROW and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSGCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSGCR
      300*0.050 /
```

The above example defines a constant surfactant critical oil saturation with respect to gas of 0.05 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.321 SSGL – END-POINT SCALING GRID CELL SURFACTANT CONNATE GAS SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSGL](#) defines the surfactant connate gas saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The connate gas saturation is defined as the minimum gas saturation in a two-phase gas relative permeability table. [SSGL](#) is used to scale the surfactant oil and water relative permeability data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSGL	SSGL is an array of real numbers assigning the surfactant connate gas saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.03			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 2) The keyword is terminated by a "/".

Table 8.189: SSGL Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the [SKRG](#), [SKROG](#), [SKROW](#) and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSGL DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSGL      300*0.030 /
```

The above example defines a constant surfactant connate gas saturation of 0.03 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.322 SSOGCR – END-POINT SCALING GRID CELL SURFACTANT CRITICAL OIL SATURATION WITH RESPECT TO GAS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSOGCR](#) defines the surfactant critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to gas is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase gas-oil relative permeability table. [SSOGCR](#) scales the surfactant oil relative permeability to gas data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSOGR	SSOGR is an array of real numbers assigning the surfactant critical oil saturation with respect to gas values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30 dimensionless			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

1)

If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.

2)

The keyword is terminated by a "/".

Table 8.190: SSOGCR Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the [SKRG](#), [SKROG](#), [SKROW](#) and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSOGCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSOGCR
      300*0.200 /
```

The above example defines a surfactant constant critical oil saturation with respect to gas of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.323 SSOWCR – END-POINT SCALING GRID CELL SURFACTANT CRITICAL OIL SATURATION WITH RESPECT TO WATER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSOWCR](#) defines the surfactant critical oil saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The critical oil saturation with respect to water is defined as the maximum oil saturation for which the oil relative permeability is zero in a two-phase oil-water relative permeability table. [SSOWCR](#) scales the surfactant oil relative permeability to water data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSOWCR	SSOWCR is an array of real numbers assigning the surfactant critical oil saturation with respect to water values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.30			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

1)

If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.

2)

The keyword is terminated by a “/”.

Table 8.191: SSOWCR Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the [SKRG](#), [SKROG](#), [SKROW](#) and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSOWCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSOWCR
      300*0.200 /
```

The above example defines a constant critical oil saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.324 SSWCR – END-POINT SCALING GRID CELL CRITICAL WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSWCR](#) defines the surfactant critical water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The critical water saturation is defined as the maximum water saturation for which the water relative permeability is zero in a two-phase relative permeability table. [SSWCR](#) scales the surfactant water relative permeability data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSWCR	SSWCR is an array of real numbers assigning the surfactant critical water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.20			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

1)

If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.

2)

The keyword is terminated by a "/".

Table 8.192: SSWCR Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the SKRG, SKROG, SKROW and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSWCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSWCR
      300*0.200 /
```

The above example defines a constant surfactant critical water saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.325 SSWL – END-POINT SCALING GRID CELL SURFACTANT CONNATE WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSWL](#) defines the surfactant connate water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The connate water saturation is defined as the minimum water saturation in a two-phase water relative permeability table. [SSWL](#) scales the surfactant oil relative permeability to water and gas data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSWL	SSWL is an array of real numbers assigning the connate water saturation values to each cell in the model.The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.15			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 2) The keyword is terminated by a "/".

Table 8.193: SSWL Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the SKRG, SKROG, SKROW and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSWL DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSWL
      300*0.150
/
```

The above example defines a constant surfactant connate water saturation of 0.15 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.326 SSWU – END-POINT SCALING GRID CELL SURFACTANT MAXIMUM WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SSWU](#) defines the surfactant maximum water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword and the surfactant phase has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. The maximum water saturation is defined as the maximum water saturation in a two-phase water relative permeability table. [SSWU](#) scales the surfactant water relative permeability data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SSWU	SSWU is an array of real numbers assigning the surfactant maximum water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.70			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

1)

If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.

2)

The keyword is terminated by a “/”.

Table 8.194: SSWU Keyword Description

End-point scaling allows the entered surfactant relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SSWL](#), [SSWCR](#), [SSWU](#), [SSGL](#), [SSGCR](#), [SSGU](#), [SSOWCR](#), and [SSOGCR](#) saturation grid arrays for the saturation end-points, and the [SKRG](#), [SKROG](#), [SKROW](#) and [SKRW](#) relative permeability grid cell arrays for the relative permeability end-point data.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SSWU DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SSWU
      300*0.700 /
```

The above example defines a constant surfactant connate gas saturation of 0.70 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.327
STOG
– DEFINE CAPILLARY PRESSURE OIL-GAS SURFACE TENSION VERSUS PRESSURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [STOG](#) keyword defines capillary pressure oil-gas surface tension versus pressure tables used in adjusting the pressure independent capillary pressure vectors in the [SGFN](#), [SGOF](#) or [SLGOF](#) saturation tables, entered by their respective keywords in the [PROPS](#) section. The [SATOPTS](#) keyword in the [RUNSPEC](#) section should state the SURFTENS character string to activate the Capillary Pressure Surface Tension Pressure Dependency option. If the [STOG](#) keyword is not supplied then no capillary pressure surface tension pressure scaling will occur and the capillary pressure values on the [SGFN](#), [SGOF](#) or [SLGOF](#) saturation tables will be used directly.

Capillary pressure surface tension pressure scaling can also be used with multi-segment wells, but this facility has not been incorporated in OPM Flow’s multi-segment well implementation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.328 STONE – ACTIVATE STONE’S SECOND THREE PHASE OIL RELATIVE PERMEABILITY MODEL (ALIAS FOR STONE2)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	------------------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword is an alias for [STONE2](#) keyword that activates Stone’s²⁵⁹ second three phase oil relative permeability model as modified by Aziz and Settari²⁶⁰. If the [STONE](#), [STONE1](#) and [STONE2](#) keywords are not present in the input deck then the default three phase oil relative permeability model is employed. The [STONE](#) keyword should only be used in three phase runs containing the oil, gas and water phases.

There is no data required for this keyword.

Example

```
--
--      ACTIVATE STONE’S SECOND THREE PHASE RELATIVE PERMEABILITY MODEL
--
STONE
```

The above example switches on the Modified Stone three phase relative permeability model.

²⁵⁹ Stone, H. L. “Estimation of Three-Phase Relative Permeability and Residual Oil Data,” *Journal of Canadian Petroleum Technology* (1973) 12, No. 4, 53-61.

²⁶⁰ Aziz, K. and Settari, A. *Petroleum Reservoir Simulation*, London, UK, Applied Science Publishers (1979), page 398.

8.3.329 STONE1 – ACTIVATE STONE’S FIRST THREE PHASE OIL RELATIVE PERMEABILITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword activates Stone’s²⁶¹ first three phase oil relative permeability model as modified by Aziz and Settari²⁶². If the [STONE1](#) and [STONE2](#) keywords are not present in the input deck then the default three phase oil relative permeability model is employed. The [STONE1](#) keyword should only be used in three phase runs containing the oil, gas and water phases.

There is no data required for this keyword.

Example

```
--
--      ACTIVATE STONE’S FIRST THREE PHASE RELATIVE PERMEABILITY MODEL
--
STONE1
```

The above example switches on the Modified Stone three phase relative permeability model.

²⁶¹ Stone, H. L. “Probability Model for Estimating Three-Phase Relative Permeability,” paper SPE 2116, Journal of Canadian Petroleum Technology (1973) 22, No. 2, 214-218.

²⁶² Aziz, K. and Settari, A. Petroleum Reservoir Simulation, London, UK, Applied Science Publishers (1979), page 398.

8.3.330 STONE1EX – DEFINE STONE’S FIRST THREE PHASE OIL RELATIVE PERMEABILITY PARAMETER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the exponent used in Stone’s²⁶³ first three phase oil relative permeability model as modified by Aziz and Settari²⁶⁴. The [STONE1EX](#) keyword should only be used in three phase runs containing the oil, gas and water phases and when the [STONE1](#) keyword in the [PROPS](#) section has been used to activate Stone’s first three phase oil relative permeability model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	STONEPARI	A real positive value that defines the exponent to be used in the Modified Stone first three phase oil relative permeability model.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN records as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record must contain only one value and is terminated by a “/”
- 3) There is no “/” terminator for the keyword.

Table 8.195: STONE1EX Keyword Description

If the [STONE](#), [STONE1](#) and [STONE2](#) keywords are not present in the input deck then the default three phase oil relative permeability model is employed.

Example

Given NTSFUN equals five on the [TABDIMS](#) keyword in the [RUNSPEC](#) section, then:

```
--
--      DEFINE STONE'S FIRST THREE PHASE RELATIVE PERMEABILITY MODEL PARAMETER
--
STONE1EX
      1.000          / SATURATION TABLE NO. 01
      1.000          / SATURATION TABLE NO. 02
      2.000          / SATURATION TABLE NO. 03
      1.000          / SATURATION TABLE NO. 04
      3.000          / SATURATION TABLE NO. 05
```

Defines the exponents to be used in the Modified Stone first three phase oil relative permeability model, for each of the five saturation tables.

²⁶³ Stone, H. L. “Probability Model for Estimating Three-Phase Relative Permeability,” paper SPE 2116, Journal of Canadian Petroleum Technology (1973) 22, No. 2, 214-218.

²⁶⁴ Aziz, K. and Settari, A. Petroleum Reservoir Simulation, London, UK, Applied Science Publishers (1979), page 398.

8.3.331 STONE2 – ACTIVATE STONE’S SECOND THREE PHASE OIL RELATIVE PERMEABILITY MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword activates Stone’s²⁶⁵ second three phase oil relative permeability model as modified by Aziz and Settari²⁶⁶. If the [STONE](#), [STONE1](#) and [STONE2](#) keywords are not present in the input deck then the default three phase oil relative permeability model is employed. The [STONE2](#) keyword should only be used in three phase runs containing the oil, gas and water phases.

There is no data required for this keyword.

Example

```
--
--      ACTIVATE STONE’S SECOND THREE PHASE RELATIVE PERMEABILITY MODEL
--
STONE2
```

The above example switches on the Modified Stone three phase relative permeability model.

²⁶⁵ Stone, H. L. “Estimation of Three-Phase Relative Permeability and Residual Oil Data,” *Journal of Canadian Petroleum Technology* (1973) 12, No. 4, 53-61.

²⁶⁶ Aziz, K. and Settari, A. *Petroleum Reservoir Simulation*, London, UK, Applied Science Publishers (1979), page 398.

8.3.332 STOW – DEFINE CAPILLARY PRESSURE OIL-WATER SURFACE TENSION VERSUS PRESSURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [STOW](#) keyword defines capillary pressure oil-water surface tension versus pressure tables used in adjusting the pressure independent capillary pressure vectors in the [SWFN](#) or [SWOF](#) saturation tables, entered by their respective keywords in the [PROPS](#) section. The [SATOPTS](#) keyword in the [RUNSPEC](#) section should state the SURFTENS character string to activate the Capillary Pressure Surface Tension Pressure Dependency option. If the [STOW](#) keyword is not supplied then no capillary pressure surface tension pressure scaling will occur and the capillary pressure values on the [SWFN](#) or [SWOF](#) saturation tables will be used directly.

Capillary pressure surface tension pressure scaling can also be used with multi-segment wells, but this facility has not been incorporated in OPM Flow’s multi-segment well implementation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.333 STWG – DEFINE CAPILLARY PRESSURE WATER-GAS SURFACE TENSION VERSUS PRESSURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [STWG](#) keyword defines capillary pressure water-gas surface tension versus pressure tables for use with multi-segment wells. This facility has not been incorporated in OPM Flow’s multi-segment well implementation. Note that [STWG](#) is not required for Capillary Pressure Surface Tension Pressure Dependency option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.334 SURFADDW – DEFINED SURFACTANT ADSORBED CONCENTRATION VERSUS WETTABILITY FRACTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SURFADDW](#) defines tables of surfactant adsorbed concentration versus wettability fraction for when the [SURFACTW](#) keyword in the [RUNSPEC](#) section as been declared to activate the surfactant phase with changing wettability. The tables consists of columnar vectors of adsorbed surfactant concentration versus a wettability fraction that indicates the fraction of phase wettability. Here, a wettability fraction of zero indicates a 100% water wet rock resulting in the [SURFNUM](#) allocated saturation tables being used, and a value of one meaning 100% oil wet rock, with the [SATNUM](#) allocated saturations tables being employed. Both the [SURFNUM](#) and [SATNUM](#) keywords are in the [REGIONS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.335 SURFADS – DEFINE SURFACTANT ROCK ADSORPTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SURFADS](#) keyword defines the rock surfactant adsorption tables for when the surfactant option has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SURCON	A columnar vector of real monotonically increasing down the column values that defines the surfactant concentration in the solution surrounding the rock. The first entry should be zero to define a no surfactant concentration.			None
		lb/stb	kg/sm ³	gm/scc	
2	SURREATIO	A columnar vector of real increasing down the column values that defines the mass of adsorbed surfactant per unit mass of rock of the saturated concentration of surfactant adsorbed by the rock. The first entry should be zero to define a zero ratio of surfactant concentration.			None
		lb/lb	kg/kg	gm/gm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.196: SURFADS Keyword Description

See also the [ADSORP](#) keyword in the [PROPS](#) section that employs adsorption functions, as oppose to adsorption tables, to define rock surfactant adsorption behavior.

8.3.336 SURFCAPD – CAPILLARY NUMBER VERSUS MISCIBILITY TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SURFCAPD](#) keyword defines the relationship between the log of the capillary number and the level of miscibility, for when the Surfactant option has been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section. A value of zero for the level of miscibility means fully immiscible conditions and consequently a value of one implies fully miscible conditions.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.337 SURFESAL – DEFINE SURFACTANT EFFECTIVE SALINITY COEFFICIENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [SURFESAL](#), defines the surfactant effective salinity coefficient as well as enabling the effective salinity calculation for surfactant adsorption. The keyword should only be used if the [BRINE](#) keyword has been declared to activate the brine phase, the [ECLMC](#) keyword to enable the Multi-Component Brine model, and the [SURFACT](#) keyword has been used to activate the surfact phase. All three keywords are in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.338 SURFROCK – DEFINE SURFACTANT-ROCK PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SURFROCK](#) keyword defines rock properties for when the Surfactant option has been activated by the SURFACTANT keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	ADINDX	A positive integer of 1 or 2 that defines the surfactant desorption option. 1) then surfactant desorption may occur by retracing the surfactant adsorption isotherm when the local surfactant concentration in the solution decreases. 2) then no surfactant desorption may occur.			Defined
		dimensionless 1	dimensionless 1	dimensionless 1	
2	DENSITY	A real value that defines the rock in-situ density, that is at reservoir conditions.			None
		lb/rtb	kg/rm ³	gm/rcc	

Notes:
1) The keyword is followed by NTSFUN tables as declared on the TABDIMS keyword in the RUNSPEC section for each surfactant flooding region. There should be only one row per table.
2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.197: SURFROCK Keyword Description

Example

```
--
--      SURFACTANT-ROCK PROPERTIES
--
SURFROCK
--      DESORP      INSITU
--      OPTN        DENSITY
--      -----
--          1      1800.0      / TABLE NO. 01
--          2      1980.0      / TABLE NO. 02
--          1      2005.0      / TABLE NO. 03
```

The above example defines three surfactant-rock tables, based on the NTSFUN variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to three.

There is no terminating "/" for this keyword.

8.3.339 SURFST – SURFACTANT WATER-OIL SURFACE TENSION VERSUS SURFACTANT CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SURFST](#) keyword defines surfactant water-oil surface tension versus surfactant concentration in the water phase tables, used in adjusting the pressure independent capillary pressure vectors in the [SWFN](#) or [SWOF](#) saturation tables, entered by their respective keywords in the [PROPS](#) section. [SURFST](#) is also used to adjust the relative permeability curves on the aforementioned tables via the capillary number. The Surfactant option must have been activated by the SURFACTANT keyword in the [RUNSPEC](#) section to use this keyword and either this keyword or the [SURFSTES](#) keyword, also in the [PROPS](#) section, is obligatory in this case.

See also the [SURFSTES](#) that defines the surfactant water-oil surface tension as a function of surfactant concentration in the water phase and salt concentration or the effective salinity.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.340 SURFSTES – SURFACTANT WATER-OIL SURFACE TENSION VERSUS SURFACTANT AND SALT CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SURFSTES](#) keyword defines surfactant water-oil surface tension versus surfactant concentration in the water phase tables, used in adjusting the pressure independent capillary pressure vectors in the [SWFN](#) or [SWOF](#) saturation tables, entered by their respective keywords in the [PROPS](#) section. [SURFSTES](#) is also used to adjust the relative permeability curves on the aforementioned tables via the capillary number. The Surfactant option must have been activated by the SURFACTANT keyword in the [RUNSPEC](#) section to use this keyword and either this keyword or the [SURFST](#) keyword, also in the [PROPS](#) section, is obligatory in this case. In addition, the [BRINE](#) keyword in the [RUNSPEC](#) section must be activated and the [ESSNODE](#) keyword in the [PROPS](#) section must be used to define the salt concentration or the effective salinity.

See also the SURFSTS that defines the surfactant water-oil surface tension as a function of surfactant concentration in the water phase only.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.341 SURFVISC – SURFACTANT SOLUTION VISCOSITY VERSUS CONCENTRATION

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
----------------	-------------	-------------	--------------	----------------	-----------------	----------------	-----------------

Description

SURFSVISC defines the surfactant viscosity relationship of solution water viscosity with respect to increasing surfactant concentration within a grid block. The surfactant option must be activated by the *SURFACT* keyword in the *RUNSPEC* section in order to use this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SURFCON	A columnar vector of real monotonically increasing down the column values that defines the surfactant concentration in the solution surrounding the rock. The first entry should be zero to define a no surfactant concentration.			None
		lb/stb	kg/sm ³	gm/scc	
2	SURFVISC	A columnar vector of real positive values that defines the solution water viscosity of the solution for the given SURFCON entry at the reference pressure value, PRES, entered on the <i>PVTW</i> keyword in the <i>PROPS</i> section.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.198: SURFVISC Keyword Description

Example

```
--
--          SURFACTANT SOLUTION WATER VISCOSITY TABLES
--
SURFVISC
--          SURF          VISCOSITY
--          SURFCON      SURFVISC
--          -----
--          0.0000       0.3500
--          0.0100       0.3900
--          0.0200       0.4200
--          0.0300       0.4300
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

--	SURF	VISCOSITY
--	SURFCON	VISFAC
--	-----	-----
	0.0000	1.000
	0.0003	10.000
	0.0005	20.000
	0.0007	40.000
	0.0009	45.000
	0.0011	55.000

/ TABLE NO. 02

The example defines two surfactant viscosity scaling factor tables, based on the NTPVT variable on the [TABDIMS](#) keyword in the [RUNSPEC](#) section being equal to two and NPPVT variable on the same keyword being greater than or equal to six.

8.3.342 SWATINIT – DEFINE THE INITIAL WATER SATURATION ARRAY FOR CAPILLARY PRESSURE SCALING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[SWATINIT](#) defines the initial water saturation for all the cells in the model via an array. The keyword can be used with all grid types. [SWATINIT](#) is used to initialize the model by setting each grid block's initial water saturation ("Sw"). If the array is present in the input deck, then OPM Flow will re-scale the water-oil capillary pressure curves entered via the [SWFN](#) saturation functions in the [PROPS](#) section, so that the resulting initialized Sw matches the values in the [SWATINIT](#) array.

Normally the [SWATINIT](#) array is generated in the static earth model when calculating the hydrocarbons in-place volumes using Saturation Height Functions ("SHF") derived from capillary pressure functions. Static earth models do not directly use capillary pressure in these type of calculations as individual cell pressures are not required. There is therefore some potential for inconsistencies to arise between the two sets of formulations. This is normally manifested by extreme scaling in the scaled capillary pressure values calculated by the simulator. If this is the case then the [PPCWMAX](#) keyword can be used to set a maximum scaled capillary pressure value. Note that as large values of scaled capillary pressures can result in numerical issues, a more technically sound approach would be to resolve these inconsistencies before continuing with the model build.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWATINIT	SWATINIT is an array of real positive numbers that are greater than or equal to zero and less than or equal to one, that define the initial water saturation values to each cell in the model. Repeat counts may be used, for example 3000*0.15			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 8.199: SWATINIT Keyword Description

See also the [PPCWMAX](#) to control the re-scaling of the capillary pressure entries on the [SWFN](#) saturation function keyword in the [PROPS](#) section.

Example

```
--
--      DEFINE GRID BLOCK INITIAL SW DATA FOR ALL CELLS
--      (BASED ON NX x NY x NZ = 300)
--
SWATINIT
      300*0.300
/
```

The above example defines a constant initial water saturation of 0.300 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.343 SWCR – END-POINT SCALING GRID CELL CRITICAL WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SWCR](#) defines the critical water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The critical water saturation is defined as the maximum water saturation for which the water relative permeability is zero in a two-phase relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWCR	SWCR is an array of real numbers assigning the critical water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.20			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the critical water saturation used with the end-point scaling option. If directional end-point scaling has been activated then the [SWCRX±](#), [SWCRY±](#) and [SWCRZ±](#) series of keyword should be used.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.200: SWCR Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRGR](#), [KRO](#), [KRORG](#), [KRORW](#), [KRW](#) and [KRWR](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWCRX](#), [SWCRY](#) and [SWCRZ](#) instead of [SWCR](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWCRX-](#), [SWCRY-](#), [SWCRZ-](#), instead of the [SWCR](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SWCR DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SWCR
      300*0.200 /
```

The above example defines a constant critical water saturation of 0.20 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.344 SWF32D – WATER SATURATION TABLES WITH RESPECT TO OIL AND GAS (THREE PHASE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SWF32D](#) keyword defines the three phase water relative permeability versus oil and gas saturation tables for when oil, gas and water are present in the input deck. The keyword should only be used if oil, gas and water are present in the input deck. Normally the simulator calculates the three-phase oil relative permeabilities based on the entered two phase tables of water-oil and gas-oil, combined with the [STONE1](#) and [STONE2](#) keywords in the [PROPS](#) section that determine the method used to generate the three phase water relative permeability curves. [SWF32D](#) allows for the direct input of the three phase tables, as such the [STONE1](#) and [STONE2](#) keywords should not be entered if [SWF32D](#) is used in the input deck.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.345 SWFN – WATER SATURATION TABLES (FORMAT TYPE 2)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SWFN](#) keyword defines the water relative permeability and water-oil capillary pressure data versus water saturation tables for when water is present in the input deck. This keyword should only be used if water is present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the water saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRW	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the water relative permeability with respect to gas saturation. The first value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
3	PCWO	A columnar vector of real values that are either equal or increasing down the column that defines the water-oil relative capillary pressure. If the SWATINIT keyword has been used to initialize the model then columnar vector has to be strictly monotonically increasing.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Items (2) and (3) may be defaulted, in which case linear interpolation or constant extrapolation will be used to calculate the missing values. At least two values must be entered for item (2), and at least one value must be entered for item (3).
- 4) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.201: SWFN Keyword Description

Example

--			
--	WATER RELATIVE PERMEABILITY TABLES (SWFN)		
--			
SWFN			
--	SWAT	KRW	PCOW
--	FRAC	FRAC	PSIA
--	-----	-----	-----
	0.15	0.00000	0.0
	0.20	6.25e-6	0.0
	0.25	0.00010	0.0
	0.30	0.00050	0.0
	0.35	0.00160	0.0
	0.40	0.00390	0.0
	0.45	0.00810	0.0
	0.50	0.01500	0.0
	0.55	0.02560	0.0
	0.60	0.04100	0.0
	0.65	0.06250	0.0
	0.70	0.09150	0.0
	0.75	0.12960	0.0
	0.80	0.17850	0.0
	0.85	0.24010	0.0
	0.90	0.31640	0.0
	0.95	0.40960	0.0
	1.00	0.52200	0.0
	-----	-----	-----
--			/ TABLE NO. 1
	0.15	0.00000	0.0
	0.20	6.25e-6	0.0
	0.25	0.00010	0.0
	0.30	0.00050	0.0
	0.35	0.00160	0.0
	0.40	0.00390	0.0
	0.45	0.00810	0.0
	0.50	0.01500	0.0
	0.55	0.02560	0.0
	0.60	0.04100	0.0
	0.65	0.06250	0.0
	0.70	0.09150	0.0
	0.75	0.12960	0.0
	0.80	0.17850	0.0
	0.85	0.24010	0.0
	0.90	0.31640	0.0
	0.95	0.40960	0.0
	1.00	0.52200	0.0
			/ TABLE NO. 2

The example defines two *SWFN* tables for use when water is present in the run.

8.3.346 SWL – END-POINT SCALING GRID CELL CONNATE WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SWL](#) defines the connate water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The connate water saturation is defined as the minimum water saturation in a two-phase water relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWL	SWL is an array of real numbers assigning the connate water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.15			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this is the directional independent version of the connate gas saturation array used with the end-point scaling option.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.202: SWL Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRORG](#), [KRORW](#) and [KRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWLX](#), [SWLY](#) and [SWLZ](#) instead of [SWL](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWLX](#), [SWLX-](#), [SWLY](#), [SWLY-](#), [SWLZ](#) and [SWLZ-](#), instead of the [SWL](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SWL DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SWL
      300*0.150 /
```

The above example defines a constant connate water saturation of 0.15 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.347 SWLPC – END-POINT SCALING GRID CELL CAPILLARY PRESSURE CONNATE WATER SATURATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	------------------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SWLPC](#) defines the connate water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The connate water saturation is defined as the minimum water saturation in a two-phase water relative permeability table. The keyword only applies the scaling to the drainage capillary pressures tables, unlike the [SWL](#) keyword that applies the scaling to both the capillary pressure and relative permeability tables. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWLPC	SWLPC is an array of real numbers assigning the connate water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. If SWLPC is omitted from the input deck the values will be defaulted to those on the SWL series of keywords. If the SWL series of keywords are missing from the input deck then the values are taken from the cell allocated capillary pressure table. Repeat counts may be used, for example 30*0.03			Taken from SGL or from the cell allocated capillary pressure table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this the directional independent version of the connate gas saturation array used with the end-point scaling option. If directional end-point scaling has been activated then the [SWLX±](#), [SWLY±](#) and [SWZ±](#) series of keyword should be used.
- 2) The keyword is terminated by a “/”.

Table 8.203: SWLPC Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KROG](#), [KROW](#) and [KRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWLX](#), [SWLY](#) and [SWLZ](#) instead of [SWL](#) or [SWLPC](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWLX-](#), [SWLY-](#), [SWLZ-](#), instead of the [SWL](#) or [SWLPC](#) keywords.

Missing Some Functionality - Use with Caution.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SWLPC DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SWLPC      300*0.150
```

The above example defines a constant connate water saturation of 0.15 to all 300 cells in the model as defined by the *DIMENS* keyword in the *RUNSPEC* section.

8.3.348 SWOF – WATER-OIL SATURATION TABLES (FORMAT TYPE 1)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SWOF](#) keyword defines the water and oil relative permeability and water-oil capillary pressure data versus water saturation tables for when water and oil are present in the input deck. This keyword should only be used if water and oil are present in the run.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real monotonically increasing down the column values starting from zero and terminating at one, that defines the water saturation. The first entry is the connate water saturation S_{wc} and the last entry should be 1.0.			None
		dimensionless	dimensionless	dimensionless	
2	KRW	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the water relative permeability with respect to gas saturation. The first value in the column should be zero.			None
		dimensionless	dimensionless	dimensionless	
3	KRO	A columnar vector of real values that are either equal or decreasing down the column and that are greater than or equal to zero and less than or equal to one that defines the oil relative permeability with respect to oil and water saturation. When gas is active in the run, the first entry the column, that is at $k_{row}(S_o = 1 - S_{wc})$, must be the same as the first entry in the corresponding SGOF or SLGOF table, that is at $k_{rog}(S_g = 0)$. The first value in the column should be one.			None
		dimensionless	dimensionless	dimensionless	
4	PCWO	A columnar vector of real values that are either equal or decreasing down the column that defines the water-oil relative capillary pressure. If the SWATINIT keyword has been used to initialize the model then columnar vector has to be strictly monotonically increasing.			None
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.204: SWOF Keyword Description

Example

The following example is based on NTSFUN equals two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--      WATER-OIL RELATIVE PERMEABILITY TABLES (SWOF)
--
SWOF
--      SWAT      KRW      KROW      PCOW
--      FRAC
--      -----
--      0.200000    0.0000    0.9000    0.000000
--      0.238616    0.0002    0.7664    0.000000
--      0.245309    0.0004    0.7443    0.000000
--      0.261989    0.0010    0.6907    0.000000
--      0.303091    0.0044    0.5671    0.000000
--      0.368269    0.0191    0.3962    0.000000
--      0.435026    0.0519    0.2528    0.000000
--      0.486387    0.0940    0.1643    0.000000
--      0.550683    0.1725    0.0803    0.000000
--      0.575342    0.2115    0.0559    0.000000
--      0.599076    0.2542    0.0367    0.000000
--      0.621294    0.2991    0.0223    0.000000
--      0.642171    0.3458    0.0120    0.000000
--      0.658984    0.3868    0.0061    0.000000
--      0.671123    0.4183    0.0030    0.000000
--      0.679268    0.4403    0.0015    0.000000
--      0.684963    0.4562    0.0008    0.000000
--      0.688893    0.4674    0.0004    0.000000
--      0.692025    0.4765    0.0002    0.000000
--      0.694641    0.4841    0.0001    0.000000
--      0.696976    0.4910    0.0000    0.000000
--      0.699099    0.4973    0.0000    0.000000
--      0.700000    0.5000    0.0000    0.000000
--      1.000000    0.9000    0.0000    0.000000
--
--
--      / TABLE NO. 01
--
--      -----
--      0.200000    0.0000    0.9000    0.000000
--      0.238616    0.0002    0.7664    0.000000
--      0.245309    0.0004    0.7443    0.000000
--      0.261989    0.0010    0.6907    0.000000
--      0.303091    0.0044    0.5671    0.000000
--      0.368269    0.0191    0.3962    0.000000
--      0.435026    0.0519    0.2528    0.000000
--      0.486387    0.0940    0.1643    0.000000
--      0.550683    0.1725    0.0803    0.000000
--      0.575342    0.2115    0.0559    0.000000
--      0.599076    0.2542    0.0367    0.000000
--      0.621294    0.2991    0.0223    0.000000
--      0.642171    0.3458    0.0120    0.000000
--      0.658984    0.3868    0.0061    0.000000
--      0.671123    0.4183    0.0030    0.000000
--      0.679268    0.4403    0.0015    0.000000
--      0.684963    0.4562    0.0008    0.000000
--      0.688893    0.4674    0.0004    0.000000
--      0.692025    0.4765    0.0002    0.000000
--      0.694641    0.4841    0.0001    0.000000
--      0.696976    0.4910    0.0000    0.000000
--      0.699099    0.4973    0.0000    0.000000
--      0.700000    0.5000    0.0000    0.000000
--      1.000000    0.9000    0.0000    0.000000
--
--      / TABLE NO. 01
```

The example defines two [SWOF](#) tables for use when water and oil are present in the run. In the tables the water-oil capillary pressure data has been set to zero.

8.3.349 SWOFLET – WATER-OIL LET RELATIVE PERMEABILITY FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	--------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

SWOFLET defines the relative permeability and capillary pressure parameters for the water-oil LET family of models. Both the oil and water phases should be made active in the model via the [OIL](#) and [WATER](#) keywords in the [RUNSPEC](#) section. See section [8.2.6 Saturation Table Generation - LET Functions](#) and Lomeland et al. ^{267,268 269} and ²⁷⁰ for further information on the model.

The keyword is used as a replacement for the [SWOF](#) keyword for three-phase oil-gas-water systems, and the LET series of keywords cannot be combined with the standard set of relative permeability keywords.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWL	SWL is a real positive number less than one, that defines the connate water saturation, that is the smallest water saturation in the LET function.			0.0
		dimensionless	dimensionless	dimensionless	
2	SWCR	SWCR is a real positive number greater than or equal to SWL and less than one, that defines the critical water saturation, that is the largest water saturation for which the water relative permeability is zero, Swirr in equations.			0.0
		dimensionless	dimensionless	dimensionless	
3	LWAT	LWAT is a real positive number that defines the LET Lower empirical parameter Lw for the water phase with the associated oil phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
4	EWAT	EWAT is a real positive number that defines the LET Elevation empirical parameter Ew for the water phase with the associated oil phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	

²⁶⁷ Lomeland F., Ebeltoft E. and Thomas W.H., 2005. A New Versatile Relative Permeability Correlation. Paper SCA2005-32 presented at the International Symposium of the Society of Core Analysts held in Toronto, Canada, 21-25 August, 2005.

²⁶⁸ Lomeland F. and Ebeltoft E., 2008. A New Versatile Capillary Pressure Correlation. Paper SCA2008-08 presented at the International Symposium of the Society of Core Analysts held in Abu Dhabi, UAE, 29 Oct. – 2 Nov., 2008.

²⁶⁹ Lomeland F., Hasanov B., Ebeltoft E. and Berge M., 2012. A Versatile Representation of Up-scaled Relative Permeability for Field Applications. Paper SPE 154487-MS presented at the EAGE Annual Conference & Exhibition incorporating SPE Europec held in Copenhagen, Denmark, 4-7 June 2012.

²⁷⁰ Lomeland F., 2018. Overview Of The Let Family Of Versatile Correlations For Flow Functions. Paper SCA2018-056 presented at the International Symposium of the Society of Core Analysts held in Trondheim, Norway, 27-30 August 2018.

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	TWAT	TWAT is a real positive number that defines the LET Top empirical parameter T_w for the water phase with the associated oil phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
6	KRTWAT	KRTWAT is a real positive number less than one, that defines the relative permeability of water at the maximum water saturation (normally the maximum water saturation is one) K_{rwt} in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
7	SORW	SORW is a real positive number less than one that defines the residual oil saturation in an oil-water system in the LET equation.			0.0
		dimensionless	dimensionless	dimensionless	
8	SOWCR	SOWCR is a real positive number less than one that defines critical oil-in-water saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-water system.			0.0
		dimensionless	dimensionless	dimensionless	
9	LOIL	LOIL is a real positive number that defines the LET Lower empirical parameter L_o for the oil phase with the associated water phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
10	EOIL	EOIL is a real positive number that defines the LET Elevation empirical parameter E_o for the oil phase with the associated water phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
11	TOIL	TOIL is a real positive number that defines the LET Top empirical parameter T_o for the oil phase with the associated water phase in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
12	KRTOIL	KRTOIL is a real positive number less than or equal to one, that defines the relative permeability of oil at the residual oil saturation, K_{rot} in the LET relative permeability equations.			1.0
		dimensionless	dimensionless	dimensionless	
13	LPC	LPC is a real positive number that defines the water-oil LET Lower empirical parameter L in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	
14	EPC	EPC is a real positive number that defines the water-oil LET Elevation empirical parameter E in the LET capillary pressure equation.			1.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
15	TPC	TPC is a real positive number that defines the water-oil LET empirical parameter T in the LET capillary pressure equations.			1.0
		dimensionless	dimensionless	dimensionless	
16	PCIR	PCIR is a real positive number that defines the water-oil capillary pressure at connate water saturation (<i>SWL</i>) in the LET capillary pressure equations.			0.0
		psia	bars	atm	
17	PCIT	PCIT is a real positive number that defines the water-oil threshold capillary pressure at the maximum water saturation in the LET capillary pressure equations.			0.0
		psia	bars	atm	

Notes:

- 1) The keyword is followed by NTSFUN definitions as declared on the *TABDIMS* keyword in the *RUNSPEC* section, with each definition consisting of the above 17 LET parameters.
- 1) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.205: SWOFLET Keyword Description

Note there are two versions of the LET functions, LET²⁶⁸ for two-phase flowing conditions and LETx²⁷¹ for three-phase flowing conditions. This keyword implements the LET version for an oil-water system.

The functions are dependent on the drainage and imbibition cycle of the wetting phase as well as drainage and inhibition cycle number, since a reservoir may undergo several flooding events. To account for this the system defines the flooding event using the three saturations: *Sw*, *So*, and *Sg* together with the state of the three saturations during the flooding event. The saturation state can be Increasing, Decreasing, or Constant, for a given flooding event cycle number (*n*). Thus, *Sw(D)*, *So(I)*, *Sg(C)* or *DIC1*, means the water phase is decreasing, the oil phase is increasing and the gas phase is constant for the primary or first cycle (*n* equals one). This is case for when oil is migrating into the reservoir rock and displacing the initial water contained with the reservoir.

Note

All the LET parameters are dependent on the flooding event and flooding cycle, and thus are expected vary as such. To be clear, the values of [SWCR](#), *Lo*, *Lw* etc. should be different for each flooding cycle.

See also the [SGOFLET – Gas-Oil LET Relative Permeability Functions](#), and the [SGWFLET – Gas-Water LET Relative Permeability Functions](#) keywords in this section.

²⁷¹ Lomeland F. and Ebeltoft E., 2013. Versatile Three-phase Correlations for Relative Permeability and Capillary Pressure. Paper SCA2013-034 presented at the International Symposium of the Society of Core Analysts held in Napa Valley, California, USA, 16-19 September, 2013.

Example

The following example uses the [SWOFLET](#) keyword to define two relative oi-water relative permeability tables, based on NTSFUN equals two on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.

```
--
--          SWOFLET - WATER-OIL LET REL. PERMEABILITY FUNCTIONS (OPM FLOW KEYWORD)
--
SWOFLET
--          SWL          SWCR          L-WAT          E-WAT          T-WAT          KRT-WAT
--          SOR          SOWCR         L-OIL          E-OIL          T-OIL          KRT-OIL
--          L-PC          E-PC          T-PC          PCIR          PCIT
--          -----
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          1*          1*          1*          1*          1*
--
--
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          0.00000      0.0000      1.00000      1.00000      1.00000      1.00000
--          1*          1*          1*          1*          1*
--
--
--          / TABLE NO. 01
--
--
--          / TABLE NO. 02
```

Here the [SWOFLET](#) keyword parameters are all set to their default values.

8.3.350 SWU – END-POINT SCALING GRID CELL MAXIMUM WATER SATURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SWU](#) defines the maximum water saturation for all the cells in the model via an array when the end-point scaling option has been invoked via the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The maximum water saturation is defined as the maximum water saturation in a two-phase water relative permeability table.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWU	SWU is an array of real numbers assigning the maximum water saturation values to each cell in the model. The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword. Repeat counts may be used, for example 30*0.70			Taken from cell allocated relative permeability table.
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) Note this is the directional independent version of the maximum water saturation array used with the end-point scaling option.
- 2) If the value for a cell has been defaulted then OPM Flow uses the value from the cell's relative permeability table.
- 3) The keyword is terminated by a "/".

Table 8.206: SWU Keyword Description

End-point scaling allows the entered relative permeability functions to be re-scaled based on the saturation end-points allocated to each cell by the [SWL](#), [SWCR](#), [SWU](#), [SGL](#), [SGCR](#), [SGU](#), [SOWCR](#), and [SOGCR](#) saturation grid arrays for the saturation end-points, and the [KRG](#), [KRORG](#), [KRORW](#) and [KRW](#) relative permeability grid cell arrays for the relative permeability end-point data. In addition end-point scaling may be directional dependent in which case the directional dependent versions of the aforementioned arrays should be used, that is [SWUX](#), [SWUY](#) and [SWUZ](#) instead of [SWU](#). There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected the non-reversible versions of the aforementioned arrays should be used, that is [SWUX-](#), [SWUY-](#), [SWUZ-](#), instead of the [SWU](#) keyword.

Example

```
--
--      DEFINE GRID BLOCK END-POINT SWU DATA FOR ALL CELLS
--      (FOR NX x NY x NZ = 300)
--
SWU
      300*0.700 /
```

The above example defines a constant connate gas saturation of 0.70 to all 300 cells in the model as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

8.3.351 TCRIT – CRITICAL TEMPERATURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [TCRIT](#) keyword defines the critical temperatures for each of the compositional components active in the model and for each equation of state. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TCRIT	A series of real numbers that define the critical temperatures for each of the compositional components active in the model.			None
		°R	K	K	
Notes: 1) The keyword is followed by NMEOSR records as specified by the TABDIMS keyword in the RUNSPEC section. 2) Each record must contain COMPS values as specified by the COMPS keyword in the RUNSPEC section. 3) Each record is terminated by a “/” and there is no “/” terminator for the keyword.					

Table 8.207: [TCRIT](#) Keyword Description

Examples

The following example defines the critical temperatures for each component in a single three-component equation of state model.

```
--
-- Critical Temperatures
--
TCRIT
  547.4412 343.0152 665.802 /
```

The following example defines the critical temperatures for each component in two three-component equation of state models.

```
--
-- Critical Temperatures
--
TCRIT
  547.4412 343.0152 665.802 /
  547.4412 343.0152 665.802 /
```

8.3.352
TEMPNODE
–
TEMPERATURE TABLE FOR POLYMER SOLUTION VISCOSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the reservoir temperature table used to calculate the polymer solution viscosity when the temperature option has been activated by the [TEMP](#) keyword in the [RUNSPEC](#) section in the commercial simulator. Naturally, the polymer option must also be activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section in order to use this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.353 TEMPTVD – ACTIVATE TEMPERATURE FLUX LIMITED TRANSPORT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[TEMPTVD](#) activates the Temperature Flux Limited Transport option in the commercial simulator, to reduce numerical dispersion for when either the [TEMP](#) or [THERMAL](#) keywords in the [RUNSPEC](#) section have been declared.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.354
TEMPVD
– DEFINE THE INITIAL RESERVOIR TEMPERATURE VERSUS DEPTH TABLE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the initial reservoir temperature versus depth tables for each equilibration region. Note that the [TEMPVD](#) keyword is an alias for [RTEMPVD](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

See [TEMPVD – Define the Initial Reservoir Temperature versus Depth Tables](#) in the [SOLUTION](#) section, as the keyword is documented in the [SOLUTION](#) section in this instance, but it can also be used in the [PROPS](#) section by OPM Flow.

8.3.355 THCO2MIX – SPECIFY THERMAL MIXING MODELS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [THCO2MIX](#) keyword specifies the thermal mixing models for salt in the water phase, CO₂ in the liquid phase and vaporized water in gas phase.

This is an OPM Flow specific keyword that should only be used if the [CO2STORE](#) keyword has been specified in the [RUNSPEC](#) section.

No.	Name	Description	Default
1	SALTMOD	A defined character string that specifies the thermal mixing model for salt in the liquid phase, and should be set to one of the following: 1) NONE: pure water enthalpy will be used, or 2) MICHAELIDES: account for salt concentration according to Michaelides ²⁷² , 1981.	MICHAELIDES
2	LIQMOD	A defined character string that specifies the thermal mixing model for CO ₂ in the liquid phase, and should be set to one of the following: 1) NONE: do not account for CO ₂ in brine, 2) IDEAL: account for CO ₂ assuming an ideal mixture (based on mass fractions), or 3) DUANSUN: also add the heat of dissolution for CO ₂ according to Dun and Sun ²⁷³ , 2003 (Fig. 6).	DUANSUN
3	GASMOD	A defined character string that specifies the thermal mixing model for vaporized water in the gas phase, and should be set to one of the following: 1) NONE: pure CO ₂ enthalpy will be used, or 2) IDEAL: account for vaporized water assuming an ideal mixture (based on mass fractions).	NONE
Notes: 1) The keyword is terminated by a "/".			

Table 8.208: [THCO2MIX](#) Keyword Description

Example

The following example specifies the default thermal mixing models for salt in the liquid phase, CO₂ in the liquid phase, and vaporized water in the gas phase.

```
--
-- SPECIFY THERMAL MIXING MODELS
--
-- SALT      LIQUID      GAS
-- -----
THCO2MIX
MICHAELIDES DUANSUN  NONE  /
```

²⁷² Michaelides, E. E., "Thermodynamic Properties of Geothermal Fluids", Geothermal Resources Council, Trans. Vol. 5 (361-364), October 1981.

²⁷³ Duan, Zhenhao, and Rui Sun. "An improved model calculating CO₂ solubility in pure water and aqueous NaCl solutions from 273 to 533 K and from 0 to 2000 bar." Chemical geology 193.3-4 (2003): 257-271.

8.3.356 TLMIXPAR – DEFINE THE MISCIBLE TODD-LONGSTAFF MIXING PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **TLMIXPAR** keyword defines the Todd-Longstaff²⁷⁴ mixing parameters, for when either the miscible or solvent options have been activated by the **MISCIBLE** or **SOLVENT** keywords in the **RUNSPEC** section. This keyword must be present in the input deck if the **MISCIBLE** or **SOLVENT** keywords have been activated.

Note that If the **POLYMER** option has been activated by the **POLYMER** keyword in the **RUNSPEC** section, then this keyword is ignored and the mixing parameters are taken from the **PLMIXPAR** keyword instead.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TLMVIS	A real positive value that is greater than or equal to zero and less than or equal to one, that defines the viscosity Todd-Longstaff mixing parameter for each miscibility region.			None
		dimensionless	dimensionless	dimensionless	
2	TLMDEN	A real positive value that is greater than or equal to zero and less than or equal to one, that defines the density Todd-Longstaff mixing parameter for each miscibility region.			The same value as entered for TLMVIS
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTMISC tables as declared on the *MISCIBLE* keyword in the *RUNSPEC* section for each rock region. There should be only one row per table.
- 2) Each entry is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.209: TLMIXPAR Keyword Description

Example

```
--
--          TODD-LONGSTAFF MIXING PARAMETERS
--
TLMIXPAR
--          TLM          TLM
--          VISCOS       DENSITY
--          -----
--          0.3500       0.3500          / TABLE NO. 01
--          0.2500       1*              / TABLE NO. 02
--          0.6500       0.7500          / TABLE NO. 03
```

The above example defines three Todd-Longstaff mixing parameter data sets, based on the NTMISC variable on the **MISCIBLE** keyword in the **RUNSPEC** section being equal to three.

²⁷⁴ Todd, M. and Longstaff, W. “The Development, Testing and Application of a Numerical Simulator for Predicting Miscible Flood Performance,” paper SPE 3484, Journal of Canadian Petroleum Technology (1972) 24, No. 7, 874-882.

8.3.357 TOLCRIT – DEFINE THE CRITICAL SATURATION TOLERANCE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Critical fluid saturations are determined from the relative permeability tables, that is the last saturation in a relative permeability table where the relative permeability of a phase is set equal to zero. Since floating-point numbers (as implemented in computers) are never exact, one cannot compare floating point numbers for exact equality. Thus, this keyword defines a value below which is considered equivalent to zero in determining the critical saturation for a phase.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TOLCRIT	<i>TOLCRIT</i> is a real positive number greater than zero and less than one that defines the critical saturation tolerance used to determine the critical saturation of a fluid in the relative permeability tables. The default value of 1×10^{-6} means that relative permeability values less than this value will be treated as being equal to zero.			1×10^{-6}
		dimensionless	dimensionless	dimensionless	
Notes: I) The keyword is terminated by a “/”.					

Table 8.210: TOLCRIT Keyword Description

See also section [8.2.4 Saturation Tables \(Relative Permeability and Capillary Pressure Tables\)](#) for a description of the relative permeability tables and the various end-point definitions, including oil, water and gas critical saturations.

Example

```

---
--      SET THE CRITICAL SATURATION TOLERANCE
--
TOLCRIT
      1.0E-6
/

```

The above example defines the critical saturation tolerance to be the default value of 1×10^{-6} .

8.3.358 TPAMEPS – VOLUMETRIC STRAIN VERSUS COAL GAS CONCENTRATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

TPAMEPS defines the volumetric strain versus coal gas concentration tables, for when the Coal Bed Methane option has been activated via the COAL keyword, and PALM-MAN has been declared for the ROCKOPT variable on the ROCKCOMP keyword; both keywords are in the RUNSPEC section. The Palmer-Mansoorii²⁷⁵ and ²⁷⁶ rock model is used to calculate the impact on pore volume and permeability due to rock compaction.

See also the ROCKPARMA keyword in the PROPS section that defines the Palmer-Mansoori parameters.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²⁷⁵
Palmer, I. and Mansoori, J. "How Permeability Depends on Stress and Pore Pressure in Coalbeds: A New Model," paper SPE 52607, SPE Reservoir Evaluation & Engineering (1998) 1, No. 6, 539-544.

²⁷⁶
Clarkson, C.R., Pan, Z., Palmer, I. and Harpalani, S. "Predicting Sorption-Induced Strain and Permeability Increase With Depletion for Coalbed-Methane Reservoirs", SPE 114778-PA, SPE Journal (2010) 15, No. 1, 152–159.

8.3.359 TPAMEPSS – VOLUMETRIC STRAIN VERSUS COAL SOLVENT CONCENTRATION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[TPAMEPSS](#) defines the volumetric strain versus coal solvent concentration tables, for when the Coal Bed Methane option has been activated via the [COAL](#) keyword, and PALM-MAN has been declared for the ROCKOPT variable on the [ROCKCOMP](#) keyword; both keywords are in the [RUNSPEC](#) section. The Palmer-Mansoori²⁷⁷ and ²⁷⁸ rock model is used to calculate the impact on pore volume and permeability due to rock compaction.

See also the ROCKPARMA keyword in the [PROPS](#) section that defines the Palmer-Mansoori parameters.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

277

Palmer, I. and Mansoori, J. "How Permeability Depends on Stress and Pore Pressure in Coalbeds: A New Model," paper SPE 52607, SPE Reservoir Evaluation & Engineering (1998) 1, No. 6, 539-544.

278

Clarkson, C.R., Pan, Z., Palmer, I. and Harpalani, S. "Predicting Sorption-Induced Strain and Permeability Increase With Depletion for Coalbed-Methane Reservoirs", SPE 114778-PA, SPE Journal (2010) 15, No. 1, 152–159.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

8.3.360 TRACER – DEFINE PASSIVE TRACER VARIABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [TRACER](#) keyword defines a series of passive tracers that are associated with a phase (oil, water, or gas) in the model. The maximum number of tracers for each phase are declared on the [TRACERS](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NAME	A three letter character string defining the tracer's name. NAME is used by the TNUM keyword in the REGIONS section, and unlike other keywords, the TNUM keyword itself must be concatenated with the phase and the name of the tracer defined by NAME. Similarly for the TVDP keyword in the SOLUTION section, where the TVDP keyword itself must be concatenated with the either letter F (for free) or S (for solution), followed by the name of the tracer defined by NAME. Note it is best to void names beginning with the letters F, S, and T as these names may create naming issues in post-processing software.			None
2	PHASE	A three letter character string that defines the tracer given by NAME to a particular fluid phase. The character should be set to OIL , WAT or GAS .			None
3	UNITS	The units for the tracer. This can be any unit but should be consistent with values entered via the TBLK and TVDP keywords in the SOLUTION section. The default values are the same as the PHASE in the model.			Same as the phases in the model
		Liquid: stb Gas: Mscf	Liquid: sm ³ Gas: sm ³	Liquid: scc Gas: scc	
4	SOLPHASE	A three or four letter character string defining the partitioned tracer's solution phase. The character string should be set to OIL, WAT, GAS or MULT. Note that SOLPHASE only needs to be defined if the partitioned tracer option has been activate with the PARTTRAC keyword in the RUNSPEC section.			None
5	KPNUM	The table number to be used with the partitioned tracers defined by the PARTTRAC, TRACERKP and TRACERKM keywords. Note that KPNUM only needs to be defined if the partitioned tracer option has been activate with the PARTTRAC keyword in the RUNSPEC section.			None
6	ADSPHASE	A three letter character string defining the phase used for the adsorption calculation for when the MULT option has been for SOLPHASE. The character string should be set to OIL, WAT, GAS or ALL. Note that ADSPHASE only needs to be defined if the partitioned tracer option has been activate with the PARTTRAC keyword in the RUNSPEC section.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) Each record (or row) should be terminated by a “/” and the keyword is terminated by a “/”.					

Table 8.211:TRACER Keyword Description

See also the [TNUM](#) keyword in the REGION section that defines tracer regions, and the [TVDP](#) keyword in the [SOLUTION](#) section that sets the initial tracer saturation for all the cells as a function of depth.

Example

```
--
--      DEFINE TRACER NAMES
--
--      TRACER      TRACER
--      NAME        PHASE
--      -----
TRACER
      'IGS'         'GAS'           / GAS INJECTOR
      'DGS'         'GAS'           / DISOLVED GAS
      'IW1'         'WAT'           / WAT INJECTOR 1
      'IW2'         'WAT'           / WAT INJECTOR 2
/
```

The above example defines four passive tracers one for a gas injection well, one for tracking the dissolved gas, and two to track the injected water from two water injection wells.

8.3.361 TRACERKM – MULTI-PARTITIONED TRACER OPTION K(P) TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRACERKM](#), defines the Multi-Partitioned Tracer option K(P) tables, for when the Partitioned Tracer option has been activate with the [PARTTRAC](#) keyword in the [RUNSPEC](#) section, and the SOLPHASE parameter on the [TRACER](#) keyword in the [PROPS](#) section has been set to MULT to activate the Multi-Partitioned Tracer option. Multi-partitioned tracers can partition into any number of phases (oil, water, gas etc.) and have adsorption, decay and diffusion parameters specific to each phase; whereas the standard partitioned tracers only have a “free” and “solution” phases. For the [TRACERKM](#) keyword the K(P) tables relate the ratio of the reference phase to the other phases versus pressure. So for example, given a multi-partitioned tracer in oil, water and gas, with the water phase acting as the reference phase, then [TRACERKM](#) would consist of columnar vectors of:

$$K_{ow}(P) = \frac{C_{oil}}{C_{water}} \quad \text{and} \quad K_{gw}(P) = \frac{C_{gas}}{C_{water}} \quad (8.98)$$

Where:

$K_{ow}(P)$	= multi-partitioned oil-water K(P)
$K_{gw}(P)$	= multi-partitioned gas-water K(P)
C_{oil}	= oil concentration
C_{gas}	= gas concentration
C_{water}	= water concentration

See also the [TRACERKP](#) keyword in the [PROPS](#) section that provides similar data for standard partitioned tracers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.362 TRACERKP – STANDARD PARTITIONED TRACER OPTION K(P) TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRACERKP](#), defines the Standard Partitioned Tracer option K(P) tables, for when the Partitioned Tracer option has been activate with the [PARTTRAC](#) keyword in the [RUNSPEC](#) section. Standard partitioned tracers only have a “free” and “solution” phases; whereas, Multi-partitioned tracers can partition into any number of phases (oil, water, gas etc.) and have adsorption, decay and diffusion parameters specific to each phase. For the [TRACERKP](#) keyword the K(P) tables relate the ratio of the reference phase (the “free” phase) to the solution phase versus pressure. So for example, given a standard partitioned tracer in oil and gas, with the oil phase acting as the reference phase, then [TRACERKP](#) would consist of columnar vectors of:

$$K(P) = \frac{C_{gas}}{C_{oil}} \quad (8.99)$$

Where:

$K(P)$ = standard partitioned K(P)
 C_{oil} = oil concentration
 C_{gas} = gas concentration

See also the [TRACERKM](#) keyword in the [PROPS](#) section that provides similar data for tmulti-partitioned tracers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.363 TRACITVD – ACTIVATE AND DEFINE TRACER IMPLICIT FLUX LIMITED TRANSPORT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[TRACITVD](#) activates the Tracer Implicit Flux Limited Transport option and sets various parameters for this option. Basically the option is used to control numerical dispersion for tracers. Both the [TRACERS](#) keyword in the [RUNSPEC](#) section and the [TRACER](#) keyword in the [PROPS](#) section must be declared to activate tracers and to define the tracers.

See also the [TRACTVD](#) keyword in the [PROPS](#) section activates the Tracer Explicit Flux Limited Transport option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.364 TRACTVD – ACTIVATE TRACER EXPLICIT FLUX LIMITED TRANSPORT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[TRACTVD](#) activates the Tracer Explicit Flux Limited Transport option. Basically the option is used to control numerical dispersion for tracers. Both the [TRACERS](#) keyword in the [RUNSPEC](#) section and the [TRACER](#) keyword in the [PROPS](#) section must be declared to activate tracers and to define the tracers.

See also the [TRACITVD](#) keyword in the [PROPS](#) section activates the Tracer Implicit Flux Limited Transport option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.365 TRADS – ENVIRONMENTAL TRACER ADSORPTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRADS](#), specifies the environmental tracer adsorption tables that describe how a tracer is absorbed by the surrounding rock, for when the MXENVTR parameter has been set to greater than zero on the [TRACERS](#) keyword in the [RUNSPEC](#) section to activate environmental tracers. The keyword can only be used with environmental tracers.

Unlike other keywords, the [TRADS](#) keyword must be concatenated with the three character name of the tracer declared by [TRACER](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.366 TRDCY – ENVIRONMENTAL TRACER DECAY TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRDCY](#), specifies the environmental tracer decay tables that specifies the tracer decay half-life, for when the MXENVTR parameter has been set to greater than zero on the [TRACERS](#) keyword in the [RUNSPEC](#) section to activate environmental tracers. The keyword can only be used with environmental tracers.

Unlike other keywords, the TRDCY keyword must be concatenated with the three character name of the tracer declared by [TRACER](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.367 TRDIF – TRACER DIFFUSION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRDIF](#), specifies the tracer diffusion tables that specify the diffusion coefficient for a tracer. The keyword can be used with Environmental Tracers if the `MXENVTR` parameter has been set greater than zero on the [TRACERS](#) keyword in the [RUNSPEC](#) section. When used with a Standard Partitioned Tracer the diffusion coefficient applies to the solution phase, whereas as for a Multi-Partitioned Tracer the diffusion coefficient can be entered for each defined tracer phase. Unlike other keywords, the [TRADS](#) keyword must be concatenated with the three character name of the tracer declared by [TRACER](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.368 TRDIS – TRACER DISPERSION TABLE NUMBER ALLOCATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRDIS](#), specifies the tracer diffusion tables that should be allocated to a tracer; the actual dispersion tables are specified by the [DISPERSE](#) keyword in the [PROPS](#) section. The keyword can be used with Environmental Tracers if the [MXENVTR](#) parameter has been set greater than zero on the [TRACERS](#) keyword in the [RUNSPEC](#) section. The option does not work with two-phase Standard Partitioned Tracers and Multi-Partitioned Tracers. Unlike other keywords, the [TRADS](#) keyword must be concatenated with the three character name of the tracer declared by [TRACER](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.369 TRNHD – ACTIVATE DISPERSION NON-HOMOGENEOUS DIFFUSION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [TRNHD](#) keyword activates the Dispersion Non-Homogeneous Diffusion option for when tracer dispersion is independent of velocity or tracer concentration. Unlike other keywords, the [TRNHD](#) keyword must be concatenated with the name of the tracer declared by [TRACER](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

8.3.370 TRROCK – ENVIRONMENTAL TRACER-ROCK PROPERTY DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [TRROCK](#), specifies the environmental tracer rock data for the tracer adsorption model, for when the MXENVTR parameter has been set to greater than zero on the [TRACERS](#) keyword in the [RUNSPEC](#) section to activate environmental tracers. The keyword can only be used with environmental tracers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.371 TZONE – END-POINT SCALING TRANSITION ZONE OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [TZONE](#) keyword sets the transition end-point scaling options for the oil, water and gas phases, for when the End-Point Scaling option has been activated by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section. The keyword determines if the phase critical saturation should or should not be set to the initial immobile saturation in areas where the initial saturation is below the entered critical saturation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	OILZONE	OILZONE is a single character that sets the oil phase transition zone end-point scaling option and should be set to either T or F: 1) T: for true, results in the SOWCR values being adjusted to the initial immobile saturation for oil-water or oil-water-miscible gas simulations. For oil-gas simulations the SOGCR values are modified to be the initial immobile saturation. The modifications only occur in cells where the initial saturation is below the entered critical saturation. 2) F: for false, means the critical saturations are not modified.	F
2	WATZONE	WATZONE is a single character that sets the water phase transition zone end-point scaling option and should be set to either T or F: 1) T: for true, results in the SWCR values being adjusted to the initial immobile saturations. The modifications only occur in cells where the initial saturation is below the entered critical saturation values (SWCR). 2) F: for false, means the critical saturations are not modified.	F
3	GASZONE	GASZONE is a single character that sets the gas phase transition zone end-point scaling option and should be set to either T or F: 1) T: for true, results in the SGCR values being adjusted to the initial immobile saturation for oil-gas or gas-water simulations. The modifications only occur in cells where the initial saturation is below the entered critical saturation (SGCR). 2) F: for false, means the critical saturations are not modified.	F

Notes:

- 1) The keyword is terminated by a “/”.

Table 8.212: TZONE Keyword Description

See also the [SCALECRS](#) keyword in the [PROPS](#) section that sets the end-point scaling option to be either two-point or three-point scaling.

Example

```
--
--      END-POINT SCALING TRANSITION ZONE OPTIONS
--
--      OILZONE  WATZONE  GASZONE
--      -----  -----  -----
TZONE      F          T          F                      / SCALING OPTION
```

The above example results in the [SWCR](#) values being adjusted to the initial immobile saturations.

8.3.372 VCRIT – CRITICAL VOLUMES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [VCRIT](#) keyword defines the critical volumes for each of the compositional components active in the model and for each equation of state. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	VCRIT	A series of real numbers that define the critical volumes for each of the compositional components active in the model.			None
		ft ³ /lb-M	m ³ /kg-M	m ³ /kg-M	
Notes: 1) The keyword is followed by NMEOSR records as specified by the TABDIMS keyword in the RUNSPEC section. 2) Each record must contain COMPS values as specified by the COMPS keyword in the RUNSPEC section. 3) Each record is terminated by a “/” and there is no “/” terminator for the keyword.					

Table 8.213:VCRIT Keyword Description

Examples

The following example defines the critical volumes for each component in a single three-component equation of state model.

```
--
-- Critical Volumes
--
VCRIT
  1.505  1.585  3.250 /
```

The following example defines the critical volumes for each component in two three-component equation of state models.

```
--
-- Critical Volumes
--
VCRIT
  1.505  1.585  3.250 /
  1.505  1.585  3.250 /
```


8.3.373
VDFLOW – VELOCITY DEPENDENT FLOW COEFFICIENT FOR GRID BLOCK FLOW (GRID)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[VDFLOW](#) activates non-Darcy flow between grid blocks and defines a constant non-Darcy flow coefficient for the whole grid, the coefficient only applies to the gas phase. The coefficient is normally derived from well tests or calculated analytically based on the coefficient of inertial resistance, usually known as β , in Forchheimer's flow equation,²⁷⁹ ²⁸⁰ and ²⁸¹. Dake²⁸², in chapter eight, reports a typical value of β to be 10.07 cm-l.

See also the [VDFLOWR](#) keyword in the [PROPS](#) section that allows the non-Darcy coefficient to be entered for individual regions, and the [WDFAC](#) and [WDFACCOR](#) keywords in the [SCHEDULE](#) section that assigns the non-Darcy coefficient to well connections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²⁷⁹
Geertsma, J., 1974. Estimating the Coefficient of Inertial Resistance in Fluid Flow Through Porous Media. Soc.Pet.Eng.J., October: 445-450.

²⁸⁰
Gewers, C.W.W. and Nichol, L.R., 1969. Gas Turbulence Factor in a Microvugular Carbonate. J.Can.Pet.Tech., April.

²⁸¹
Wong, S.W., 1970. Effects of Liquid Saturation on Turbulence Factors for Gas Liquid Systems. J.Can.Pet.Tech., October

²⁸²
Dake, L.P. Fundamentals of Reservoir Engineering, Amsterdam, The Netherlands, Elsevier Science BV (1978) Chapter 8.6, pages 252-257.

8.3.374
VDFLOWR – VELOCITY DEPENDENT FLOW COEFFICIENT FOR GRID BLOCK FLOW (REGION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[VDFLOW](#) activates non-Darcy flow between grid blocks and defines a constant non-Darcy flow coefficient for individual regions allocated by the [SATNUM](#) keyword in the [REGIONS](#) section. Note that the coefficient only applies to the gas phase. The coefficient is normally derived from well tests or calculated analytically based on the coefficient of inertial resistance, usually known as β , in Forchheimer's flow equation,²⁸³ ²⁸⁴ and ²⁸⁵. Dake²⁸⁶, in chapter eight, reports a typical value of β to be 10.07 cm-I.

See also the [VDFLOW](#) keyword in the [PROPS](#) section that allows the non-Darcy coefficient to be entered for the whole grid, and the [WDFAC](#) and [WDFACCOR](#) keywords in the [SCHEDULE](#) section that assigns the non-Darcy coefficient to well connections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

283
Geertsma, J., 1974. Estimating the Coefficient of Inertial Resistance in Fluid Flow Through Porous Media. Soc.Pet.Eng.J., October: 445-450.

284
Gewers, C.W.W. and Nichol, L.R., 1969. Gas Turbulence Factor in a Microvugular Carbonate. J.Can.Pet.Tech., April.

285
Wong, S.W., 1970. Effects of Liquid Saturation on Turbulence Factors for Gas Liquid Systems. J.Can.Pet.Tech., October

286
Dake, L.P. Fundamentals of Reservoir Engineering, Amsterdam, The Netherlands, Elsevier Science BV (1978) Chapter 8.6, pages 252-257.

8.3.375
VEFRAC – VERTICAL EQUILIBRIUM RELATIVE PERMEABILITY FRACTION (GRID)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the Vertical Equilibrium (“VE”) relative permeability weighting factor (α) used to calculate the [VE](#) relative permeability curves to be used in the simulation, for when the [VE](#) option has been activated by the [VE](#) keyword in the [RUNSPEC](#) section. If $\alpha = 1.0$, then the [VE](#) model calculated relative permeability curves will be used, and if $\alpha = 0.0$, then the curves entered via the [SWOF](#), [SGOF](#), [SLGOF](#) series of keywords or the [SWFN](#), [SGFN](#), [SGWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#) series of keywords, will be used. A value of α between zero and one will result in weighted average relative permeability curves being employed, that is:

$$VE_{(average)} = (1.0 - \alpha) \times (SATNUM_{curves}) + \alpha \times (VE Model_{curves})$$

(8.100)

Note that [VEFRAC](#) sets α for the whole grid; whereas, the [VEFRACV](#) keyword in the [PROPS](#) section assigns α on a cell by cell basis, See also the [VEFRACP](#) and [VEFRACPV](#) keywords that apply the weighting factors to the capillary pressure data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.376
VEFRACP – VERTICAL EQUILIBRIUM CAPILLARY PRESSURE FRACTION (GRID)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the Vertical Equilibrium (“VE”) capillary pressure weighting factor (α) used to calculate the [VE](#) capillary pressure curves to be used in the simulation, for when the [VE](#) option has been activated by the [VE](#) keyword in the [RUNSPEC](#) section. If $\alpha = 1.0$, then the [VE](#) model calculated capillary pressure curves will be used, and if $\alpha = 0.0$, then the curves entered via the [SWOF](#), [SGOF](#), [SLGOF](#) series of keywords or the [SWFN](#), [SGFN](#), [SGWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#) series of keywords, will be used. A value of α between zero and one will result in weighted average capillary pressure curves being employed, that is:

$$VE_{(average)} = (1.0 - \alpha) \times (SATNUM_{curves}) + \alpha \times (VE Model_{curves})$$

(8.101)

Note that [VEFRACP](#) sets α for the whole grid; whereas, the [VEFRACPV](#) keyword in the [PROPS](#) section assigns α on a cell by cell basis, See also the [VEFRAC](#) and [VEFRACV](#) keywords that apply the weighting factors to the relative permeability data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.377
VEFRACPV – VERTICAL EQUILIBRIUM CAPILLARY PRESSURE FRACTION (CELL)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the Vertical Equilibrium (“VE”) capillary pressure weighting factor (α) used to calculate the [VE](#) capillary pressure curves to be used in the simulation, for when the [VE](#) option has been activated by the [VE](#) keyword in the [RUNSPEC](#) section. If $\alpha = 1.0$, then the [VE](#) model calculated capillary pressure curves will be used, and if $\alpha = 0.0$, then the curves entered via the [SWOF](#), [SGOF](#), [SLGOF](#) series of keywords or the [SWFN](#), [SGFN](#), [SGWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#) series of keywords, will be used. A value of α between zero and one will result in weighted average capillary pressure curves being employed, that is:

$$VE_{(average)} = (1.0 - \alpha) \times (SATNUM_{curves}) + \alpha \times (VE Model_{curves})$$

(8.102)

Note that [VEFRACPV](#) sets α on a cell by cell basis; whereas, the [VEFRACP](#) keyword in the [PROPS](#) section assigns α for the whole grid, See also the [VEFRAC](#) and [VEFRACV](#) keywords that apply the weighting factors to the relative permeability data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.378

VEFRACV

–

VERTICAL

EQUILIBRIUM

RELATIVE

PERMEABILITY

FRACTION

(CELL)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the Vertical Equilibrium (“VE”) relative permeability weighting factor (α) used to calculate the [VE](#) relative permeability curves to be used in the simulation, for when the [VE](#) option has been activated by the [VE](#) keyword in the [RUNSPEC](#) section. If $\alpha = 1.0$, then the [VE](#) model calculated relative permeability curves will be used, and if $\alpha = 0.0$, then the curves entered via the [SWOF](#), [SGOF](#), [SLGOF](#) series of keywords or the [SWFN](#), [SGFN](#), [SGWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#) series of keywords, will be used. A value of α between zero and one will result in weighted average relative permeability curves being employed, that is:

$$VE_{(average)} = (1.0 - \alpha) \times (SATNUM_{curves}) + \alpha \times (VE\ Model_{curves})$$

(8.103)

Note that [VEFRACV](#) sets α on a cell by cell basis; whereas, the [VEFRAC](#) keyword in the [PROPS](#) section assigns α for the whole grid, See also the [VEFRACP](#) and [VEFRACPV](#) keywords that apply the weighting factors to the capillary pressure data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

8.3.379 VISCAQA – SPECIFY EZROKHI COEFFICIENTS FOR AQUEOUS VISCOSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [VISCAQA](#) keyword specifies the three Ezrokhi coefficients for each compositional component and for each equation of state that are used to calculate the aqueous phase viscosity. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is only supported by OPM Flow when the two phase gas-water CO₂ storage model has been activated using the [CO2STORE](#) keyword and either the [GASWAT](#) or the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section. The component names "H2O", "CO2" and "NACL" (water, CO₂ and salt respectively) must be specified using the [CNames](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	COEFS	A series of real numbers that define the three Ezrokhi coefficients for each of the compositional components active in the model.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NMEOSR records as specified by the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each record must contain three values for each compositional component, where the number of components is specified by the *COMPS* keyword in the *RUNSPEC* section.
- 3) Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.214: VISCAQA Keyword Description

Examples

The following example defines the Ezrokhi coefficients for each component in a single three-component equation of state model.

```
--
-- Ezrokhi Coefficients for Aqueous Viscosity Calculation
--
VISCAQA
--COEFF0    COEFF1    COEFF2
  2.0E-7     0.0       1.0E-9
  2.0E-7     0.0       1.0E-9
  2.0E-7     0.0       1.0E-9   /
```

The following example defines the Ezrokhi coefficients for each component in two three-component equation of state models.

```
--
-- Ezrokhi Coefficients for Aqueous Viscosity Calculation
--
VISCAQA
--COEFF0    COEFF1    COEFF2
  2.0E-7     0.0       1.0E-9
  2.0E-7     0.0       1.0E-9
  2.0E-7     0.0       1.0E-9   /
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

--
2.0E-7 0.0 1.0E-9
2.0E-7 0.0 1.0E-9
2.0E-7 0.0 1.0E-9 /

8.3.380 VISCREF – DEFINE VISCOSITY-TEMPERATURE REFERENCE CONDITIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[VISCREF](#) defines the reference conditions for the viscosity-temperature tables, [GASVISCT](#), [OILVISCT](#) and [WATVISCT](#), for when the thermal option has been activated by [THERMAL](#) keyword in the [RUNSPEC](#) section. This keyword can only be used if the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRES	PRES is a real positive number defining the reference pressure for the viscosity and temperature tables			None
		psia	barsa	atma	
2	RS	RS is a real positive number defining the reference gas-oil ratio for when the model contains gas dissolved as activated by the DISGAS keyword in the RUNSPEC section			None
		Mscf/stb	sm ³ /sm ³	scc/scc	
3	API	API is a real number defining the oil API for when the API tracking option has been invoked by the API keyword in the RUNSPEC section. Note that OPM Flow does not support API tracking, and therefore this variable is ignored.			None
		°API	°API	°API	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the TABDIMS keyword in the RUNSPEC section.
- 2) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.215: VISCREF Keyword Description

OPM Flow currently does not support [API](#) tracking and therefore item (3) of this keyword is ignored. See also the [OILVISCT](#), [GASVISCT](#) and [WATVISCT](#) keywords in the [PROPS](#) section.

Example

The following example shows the [VISCREF](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to five.

```
--
--      REF      REF      REF
--      PRESSURE  GOR      API
--      -----
VISCREF
      3000.0      0.500      / TABLE NO. 01
      3200.0      0.550      / TABLE NO. 02
      3300.0      0.580      / TABLE NO. 03
      3400.0      0.620      / TABLE NO. 04
      3500.0      0.625      / TABLE NO. 05
```

There is no terminating “/” for this keyword.

8.3.381 WAGHYSTR – DEFINE WATER-ALTERNATING-GAS HYSTERESIS PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the parameters for the Water-Alternating-Gas (“WAG”) hysteresis option, for when the hysteresis option has been activated by the [WAGHYSTR](#) variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section.

The WAG recovery mechanism is an Enhanced Oil Recovery (“EOR”) process to optimize oil recovery by improving volumetric sweep efficiency. It was originally proposed as a method to improve the sweep efficiency of gas by using water to control the mobility ratio and to stabilize the front (Caudle and Dyes, 1958²⁸⁷; Christensen et al., 1998²⁸⁸; and Christensen et al., 2001²⁸⁹). WAG injection can lead to improved oil recovery by combining better mobility control and contacting upswept zones, and by leading to improved microscopic displacement. Although initially the injected gas was immiscible with respect to the oil (WAG Immiscible) the more common process is WAG Miscible, with alternating different types of hydrocarbon gases and non-hydrocarbon gases, such as N2 and CO2 Gases. WAG flooding has been successfully applied to various fields worldwide.

Only the gas phase relative permeability WAG hysteresis model has been implemented in the simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	LANDS_PAR AMETER	A real value greater than zero that defines Land’s parameter, C. The Land’s parameter controls how the trapped gas saturation depends on the maximum gas saturation attained and the critical gas saturation; and the shape of the imbibition curve. $S_{gtrap} = S_{gcr} + \frac{(S_{gm} - S_{gcr})}{(1 + C(S_{gm} - S_{gcr}))}$ where, S_{gtrap} is the trapped gas saturation, S_{gm} is the maximum gas saturation attained, and S_{gcr} is the critical gas saturation. Values of the Land’s parameter that are too small give a trapped gas saturation close to the maximum gas saturation attained. This results in an unphysical steep relative permeability curve giving potential convergence problems.			None
		dimensionless	dimensionless	dimensionless	
2	SECONDARY DRAINAGE_ REDUCTION	A real value greater than or equal to zero that defines the secondary drainage reduction factor, alpha. As alpha increases the reduction in gas mobility on secondary drainage increases.			0.0

²⁸⁷ Caudle, B. H., & Dyes, A. B. (1958, January 1). Improving Miscible Displacement by Gas-Water Injection. Society of Petroleum Engineers.

²⁸⁸ Christensen, J. R., Stenby, E. H., & Skauge, A. (1998, January 1). Review of WAG Field Experience. Society of Petroleum Engineers. doi:10.2118/39883-MS.

²⁸⁹ Christensen, J. R., Stenby, E. H., & Skauge, A. (2001, April 1). Review of WAG Field Experience. Society of Petroleum Engineers. doi:10.2118/71203-PA.

No.	Name	Description			Default
		Field	Metric	Laboratory	
		dimensionless	dimensionless	dimensionless	
3	GAS_MODEL	A defined character string that defines whether the gas hysteresis model should be used, and should be set to one of the following character strings: 1) YES: Use the WAG hysteresis model for the gas phase relative permeability. 2) NO: Turn off the WAG hysteresis model and use the drainage curve. Only the YES option is currently supported by the simulator.			YES
4	RES_OIL	A defined character string that defines whether the residual oil model should be used, and should be set to one of the following character strings: 1) YES: Use the trapped gas to modify the residual oil saturation (SOM) in the STONE I three-phase oil relative permeability model. No action is taken unless the STONE1 keyword has been entered. 2) NO: Do not modify the residual oil saturation. Only the NO option is currently supported by the simulator.			YES
5	WATER_MODEL	A defined character string that defines whether the water hysteresis model should be used, and should be set to one of the following character strings: 1) YES: Use the WAG hysteresis model for the water phase relative permeability 2) NO: Turn off the WAG hysteresis model. Note that the hysteresis model specified in EHYSTR keyword applies. Only the NO option is currently supported by the simulator.			YES
6	IMB_LINEAR_FRACTION	A real value greater than zero that defines the imbibition curve linear fraction. This is the fraction of the curve between Sgm and Sgtrap that uses a linear transformation. A non-zero value for the linear fraction prevents the potential infinite gradient in the imbibition curve when using the Carlson analytic model.			0.1
		dimensionless	dimensionless	dimensionless	
7	THREEPHASE_SAT_LIMIT	A real value between zero and one that defines the three-phase model threshold saturation. When the water saturation exceeds this threshold above the connate water saturation the gas (non-wetting) phase hysteresis switches from the two-phase model to the three-phase model. In the two-phase model a secondary drainage process follows the imbibition curve. However, if the water saturation exceeds the connate saturation by the given threshold, at the beginning of the secondary drainage process a three-phase secondary drainage curve is followed. This value also defines the minimum percentage change in gas saturation to allow switching from drainage to imbibition curve and vice-versa. This threshold allows better control of the numerical sensitivity of the system, preventing it from being too unstable.			0.001
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	RES_OIL_MODIFIED_FRACTION	A real value between zero and one that defines the residual oil modification fraction. This is the fraction of the trapped gas saturation subtracted from the residual oil (SOM) in the <i>STONE I</i> three-phase oil relative permeability model. This is not supported and will be ignored by the simulator.			1.0
		dimensionless	dimensionless	dimensionless	

Notes:

1)

The keyword is followed by one record for each saturation table region with each record terminated by a “/”.

2)

The number of saturation table regions NTSFUN is specified by the *TABDIMS* keyword in the *RUNSPEC* section.

Table 8.216: WAGHYSTR Keyword Description

Example

The following example defines the WAG hysteresis model parameters using the [WAGHYSTR](#) keyword for a case with three saturation table regions

```
--
--      WAG HYSTERESIS PARAMETERS
--
-- LAND  ALPHA  GAS    RES    WATER  LINEAR
-- C     FACTOR  MODEL  OIL    MODEL  FRAC
WAGHYSTR
2.0     1.0     YES    YES    YES    0.2  /
2.0     1.0     YES    NO     /
2.0     /
```

Here, saturation table region one uses the gas WAG hysteresis, residual oil and water WAG hysteresis models, with a Land's parameter of 2.0, an alpha factor of 1.0, and a imbibition curve linear factor of 0.2. The gas and water WAG hysteresis models are used in region two but the residual oil model is turned off, with a Land's parameter of 2.0, an alpha factor of 1.0, and a default imbibition curve linear factor of 0.1. A Land's parameter of 2.0 has been specified for region three with the remainder of the parameters defaulted.

Note that the above example uses some options that are not currently supported by OPM Flow.

8.3.382 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

8.3.383 WATDENT – DEFINE WATER DENSITY TEMPERATURE COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WATDENT defines the water density as a function of temperature coefficients for when the thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC**. Note this is an OPM Flow keyword used with OPM Flow's black-oil thermal model that is not available in the commercial simulator's black-oil thermal formulation.

This keyword can only be used if OPM Flow's thermal option has been activated by the **THERMAL** keyword in the **RUNSPEC** section. Note this is different to the commercial simulator that uses the **TEMP** keyword in the **RUNSPEC** section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	TEMP is a real positive value greater than zero that defines the absolute reference temperature used with TEXP1 and TEXP2 to estimate the change in water density with respect to temperature.			Defined
		°R 527.67	K 293.15	K 293.15	
2	TEXP1	TEXP1 is a real positive value greater than zero that defines the water thermal expansion coefficient of the first order.			Defined
		1/°R 1.67 x 10 ⁻⁴	1/K 3.0 x 10 ⁻⁴	1/K 3.0 x 10 ⁻⁴	
3	TEXP2	TEXP2 is a real positive value greater than zero that defines the water thermal expansion coefficient of the second order.			Defined
		1/°R2 9.26 x 10 ⁻⁷	1/K2 3.0 x 10 ⁻⁶	1/K2 3.0 x 10 ⁻⁶	

Notes:

1) The keyword is followed by NTPVT records as declared on the TABDIMS keyword in the RUNSPEC section.

2) Each data set is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.217: WATDENT Keyword Description

The water density at a given pressure and temperature is calculated from its value at surface conditions and the water shrinkage factor (the reciprocal of the water formation volume factor) as shown in the following equation:

$$\rho_w(p, T) = \rho_w(p_s, T_s) b_w(p, T) \quad (8.104)$$

Where the temperature dependence of the water shrinkage factor relative to its value at the reference temperature is calculated as shown in the following equation:

$$b_w(p,T)=\frac{b_w(p,T_{ref})}{1+c_1(T-T_{ref})+c_2(T-T_{ref})^2}$$

(8.105)

Where:

- ρ_w

= water density
- b_w

= water shrinkage factor
- p

= pressure
- T

= temperature
- c_1, c_2

= thermal expansion coefficients to first and second order
- s

= subscript indicating surface conditions
- ref

= subscript indicating reference conditions

Example

The following example shows the [WATDENT](#) keyword using the default values, for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to two.

```
--
--      WATER DENSITY TEMPERATURE COEFFICIENTS (OPM FLOW EXTENSION KEYWORD)
--
--      WATER      DENSITY      DENSITY
--      TEMP      COEFF1      COEFF2
--      -----
WATDENT
      1*          1*          1*          / TABLE NO. 01
      1*          1*          1*          / TABLE NO. 02
```

There is no terminating “/” for this keyword.

8.3.384 WATJT – DEFINE WATER JOULE-THOMSON COEFFICIENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WATJT](#) activates the water Joule-Thomson effect²⁹⁰ in temperature calculations, and defines the water Joule-Thomson Coefficient (“JTC”) at a given reference pressure, for when OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#).

Note

This is an OPM Flow keyword used with OPM Flow’s black-oil thermal model, that is not available in the commercial simulator’s black-oil thermal formulation.

This keyword can only be used if OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model, and does not include the Joule-Thomson effect in temperature calculations.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESS	A real positive value that defines the reference pressure for the corresponding Joule-Thomson Coefficient, WATJTC.			None
		psia	barsa	atma	
2	WATJTC	WATJTC is a real positive or negative value that defines the water phase Joule-Thomson Coefficient. If the value is defaulted (I*) or set to 0, then WATJTC is internally calculated using thermal water density data on the WATSDENT keyword in the PROPS section. If a non-zero value is specified, then the WATJTC is assumed to be constant and equal to that value.			0
		oF/psia	°C/barsa	°C/atma	

Notes:

- 1) The keyword is followed by NTPVT records as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each data set is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.218: WATJT Keyword Description

The Joule–Thomson effect is when a real gas, as oppose to an ideal gas, expands, resulting in the temperature of the gas dropping²⁹¹. For liquids the effect is the opposite, that is the internal energy is transferred to kinetic energy with a corresponding increase in temperature as velocity increases.

²⁹⁰ The Joule–Thomson coefficient is defined as the change in temperature with respect to an increase in pressure at constant enthalpy.

²⁹¹ Natural Gas Engineering (McGraw-Hill chemical engineering series), Donald L Katz, Robert I Lee, McGraw-Hill Education, 1990 (ISBN 0071007776, 9780071007771).

Thermodynamically, the Joule–Thomson coefficient is defined as the isenthalpic²⁹² change in temperature in a fluid caused by a unitary pressure drop, as shown in the following equation:

$$\eta = \left(\frac{\partial T}{\partial P} \right) \quad (8.106)$$

Which can also express as²⁹³:

$$\eta = (T\alpha - 1) \frac{1}{(\rho C_p)} - \left(\frac{g}{C_p} \frac{dp}{dz} \right)^{-1} \quad (8.107)$$

Setting the gravity term, g , to zero we have:

$$\eta = (T\alpha - 1) \frac{1}{(\rho C_b)} \quad (8.108)$$

Where:

η	= Joule–Thomson coefficient (oC/Pa),
α	= thermal expansivity at constant pressure (1/oC),
C_p	= specific heat at constant pressure (J/kg oC),
g	= gravitational acceleration (m/s ²)
P	= pressure (Pa),
T	= temperature (oC), and
z	= height (m).

Example

The following example shows the [WATJT](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section, and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set equal to two.

```
--
--      WATER JOULE-THOMSON COEFFICIENT (OPM FLOW EXTENSION KEYWORD)
--
--      REF      OIL
--      PRESS    JTC
--      -----
WATJT
      20.0      1*                / TABLE NO. 01
      20.0      -0.20            / TABLE NO. 02
```

Here the first entry is defaulted, and the simulator will therefore calculate the water JTC internally using the data on the [WATDENT](#) keyword in the [PROPS](#) section.

There is no terminating “/” for this keyword.

²⁹² An isenthalpic process or isoenthalpic process, is a process that proceeds without any change in enthalpy, H; or specific enthalpy, h.

²⁹³ Pippard, A.B.: Elements of Classical Thermodynamics: For Advanced Students of Physics. Cambridge University Press, Cambridge, UK (1957)

8.3.385 WATVISCT – DEFINE WATER VISCOSITY VERSUS TEMPERATURE FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WATVISCT](#) defines the water viscosity as a function of temperature for when thermal option has been activated by the [THERMAL](#) keywords in the [RUNSPEC](#). The reference pressure for this table is given by the [VISCREF](#) keyword in the [PROPS](#) section.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TEMP	A columnar vector of real monotonically increasing down the column values that defines the temperature values.			None
		°F	°C	°C	
2	VIS	A columnar vector of real decreasing down the column values that defines the water viscosity for the corresponding temperature values (TEMP). VIS should be given at the reference pressure defined by the PRESS variable on the VISCREF keyword.			None
		cP	cP	cP	

Notes:

- 1) The keyword is followed by NTPVT tables as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NPPVT rows as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 8.219: WATVISCT Keyword Description

Example

The following example shows the [WATVISCT](#) keyword for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section and for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set equal to one.

```
--
--      WATER VISCOSITY VERSUS TEMPERATURE TABLES
--
--      WATER      WATER
--      TEMP      VISC
--      -----
WATVISCT
      100.0      0.625
      110.0      0.620
      120.0      0.580
      150.0      0.550
      165.0      0.500
                                     / TABLE NO. 01
```

There is no terminating “/” for this keyword.

8.3.386 WSF – WATER SATURATION FUNCTION TABLES (GAS-WATER SYSTEMS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSF](#) keyword defines the water relative permeability data versus water saturation tables for when only gas and water are present in the input deck. This keyword should only be used if the gas and water phases are present in the run, and can therefore also be used with the [CO2STORE](#) and [H2STORE](#) models. In addition, the keyword must be used in conjunction with the [GSF](#) keyword in the [PROPS](#) section, that defines the gas relative permeability and gas-water capillary pressure data versus gas saturation for gas-water systems.

Note

[WSF](#) is a compositional keyword in the commercial compositional simulator, and will therefore cause an error in the commercial black-oil simulator.

Currently, both the [GSF](#) and [WSF](#) keywords can only be used with the [CO2STORE](#) and [H2STORE](#) models.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	A columnar vector of real values monotonically increasing down the column starting from the connate water saturation and terminating at one, that defines the water saturation.			None
		dimensionless	dimensionless	dimensionless	
2	KRW	A columnar vector of real values that are either equal or increasing down the column and that are greater than or equal to zero and less than or equal to one that defines the water relative permeability with respect to water saturation. Note that the first entry in the column must be zero.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTSFUN tables as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 2) Each table must contain a minimum of two rows and a maximum of NSSFUN rows as declared on the *TABDIMS* keyword in the *RUNSPEC* section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 8.220:WSF Keyword Description

See also the [GSF – Gas Saturation Function Tables \(Gas-Water Systems\)](#) keyword in the [PROPS](#) section, that defines the gas relative permeability and gas-water capillary pressure data as a function of gas saturation, for when only the gas and water water phases are present in the model.

8.3.387 ZMFVD – DEFINE COMPOSITIONAL COMPONENTS VERSUS DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ZMFVD](#) keyword defines the compositional component mole fractions, for each component, as a function of depth, as such the keyword should have the same number of component columnar vectors as that declared via the [COMPS](#) keyword in the [RUNSPEC](#) section, and the [NCOMPS](#) keyword in the [PROPS](#) section. The keyword should only be used if the [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section have also be activated for the gas-water two component model.

Note

This is an OPM Flow keyword used with OPM Flow's [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section, and should not be confused with the more general version of the [ZMFVD](#) keyword used in the commercial compositional simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding compositional component mole fractions. The default number of DEPTH values is 20, as defined by the NDRXVD parameter on the EQLDIMS keyword in the RUNSPEC section, and which may be used to reset the number of DEPTH values.			None
		feet	m	cm	
2	ZCOMP	A series of columnar vectors, with each columnar vector representing a compositional component mole fraction as a function of DEPTH. In addition, the sum of the compositional component mole fractions must sum to one for a given DEPTH value, otherwise an error will occur. Secondly, if the composition is independent of depth, then only one single row may be entered. Note that the number of columnar vectors, should be the same as that entered via the NCOMPS keyword in the PROPS section, and the COMPS keyword in the RUNSPEC section. Finally, only the default value of two components are currently supported by OPM Flow.			None
		mole fraction	mole fraction	mole fraction	

Notes:

1) The keyword is terminated by a “/”.

Table 8.221: ZMFVD Keyword Description

Example

The following example defines how to confirm a two component formulation, together with defining the names of the composition components, as well as the compositional gradient, to be used with the [CO2STORE](#) and [GASWAT](#) options.

```
--
--      CONFIRM NUMBER OF COMPOSITIONAL COMPONENTS (OPM FLOW KEYWORD)
--
NCOMPS      2                                     /
--
--      DEFINE COMPOSITIONAL COMPONENTS NAMES (OPM FLOW KEYWORD)
--
CNames      'CO2'
            'H2O'                                     /
--
--      COMPOSITIONAL COMPONENT MOLE FRACTIONS VS DEPTH (OPM FLOW KEYWORD)
--
--      DEPTH      CO2      H2O
ZMFVD
2000          0.0      1.0
2100          0.0      1.0
/
```


CHAPTER 9: REGIONS SECTION

9.1 INTRODUCTION

The [REGIONS](#) section defines how various properties in the [PROPS](#) and [SOLUTION](#) sections are allocated to individual cells within the model, as well as defining various fluid in-place reporting regions. This is accomplished by assigning an integer value to each cell that represents the data set of the property to be assigned to the grid block.

9.2 DATA REQUIREMENTS

OPM Flow, like most numerical modeling software, uses a default value of one for the various region arrays and thus if there is only one PVT data set for example, then there is no need to define the region array associated with allocating the PVT tables (via the [PVTNUM](#) keyword in the [REGIONS](#) section), as all cells will be allocated PVT table number one. However, if there are more than one PVT table entered in the [PROPS](#) section and [PVTNUM](#) is not defined in the model, then PVT tables greater than one will not be used and there will be no warning message indicating the fact.

Table 9.1 summarizes the various property region data and their associated keywords.

No.	REGIONS Keyword	Standard Property Regions	Property Section
1	EQLNUM	EQLNUM – Define the Equilibration Region Numbers. Equilibrium region allocation based on the EQUIL keyword in the SOLUTION section.	SOLUTION
2	FIPNUM	FIPNUM – Define the Fluid In-Place Region Numbers. Fluid In-Place reporting via the FIPNUM array that divides the model into different fluid in-place reporting regions.	REGION
3	PVTNUM	PVTNUM – Define the PVT Regions. PVT table allocation of the DENSITY, PVDG, PVDO, PVTG, PVTO, PVCO, PVTW, and ROCK tables.	PROPS
4	SATNUM	SATNUM – Define the Saturation Table Region Numbers. Saturation (relative permeability) table allocation of the SGFN, SWFN, SOF2, SOF3, SGOF, and SWOF tables. This array is for the standard drainage, as oppose to the imbibition curves, used in all models. There are separate region arrays for alternative relative permeability curves.	PROPS
Function Specific Regions			
5	ENDNUM	ENDNUM – Define the End-Point Scaling Depth Region Numbers. ENPTVD and ENKRVD versus depth table allocation for when ENDSIZE option has been activated in the RUNSPEC section.	PROPS
6	FIP	FIP – Define the Fluid In-Place Names and Region and Numbers. Fluid In-Place reporting via the FIP array that divides the model into different fluid in-place reporting regions.	REGION
7	FIPOWG	FIPOWG – Activate Oil, Gas, and Water FIP Zone Reporting. Fluid In-Place reporting for the total oil, gas, and water in the model. There is no region property array for this keyword, as the simulator calculates the total phase volumes based on the contacts entered via the EQUIL keyword in the SOLUTION section.	REGION
8	HBNUM	HBNUM – Define Herschel-Bulkley Region Numbers. Herschel-Bulkley rheological property table region numbers.	PROPS

No.	REGIONS Keyword	Standard Property Regions	Property Section
9	HWSNUM	<i>HWSNUM – Define the Saturation Table Region Numbers (High Salinity and Water Wet).</i> High salinity water wet saturation table allocation using the high salinity water wet saturation SWFN and SOFN tables. This is an alias for the SURFWNUM property region which is also not supported by OPM Flow.	PROPS
10	IBNUM	<i>IBNUM – Define the Imbibition Saturation Table Region Numbers.</i> Imbibition saturation table allocation of the SWFN, SOF2, SOF3, or SWOF imbibition tables.	PROPS
11	IBNUMMF	<i>IBNUMMF – Define the Imbibition Saturation Table Region Numbers (Matrix-Fracture).</i> Imbibition saturation tables region numbers for flow between the matrix and fracture blocks, for when the Hysteresis option has been activated in Dual Porosity or Dual Permeability runs.	PROPS
12	KRNUM	<i>KRNUM – Define the Directional Saturation Table Region Numbers.</i> Direction dependent saturation tables region numbers for when Directional Dependent Saturation Function option has been activated by the DIRECT parameter on the SATOPTS.	PROPS
13	KRNUMMF	<i>KRNUMMF – Define the Saturation Table Region Numbers (Matrix-Fracture).</i> Drainage saturation tables region numbers for flow between the matrix and fracture blocks in a Dual Porosity or Dual Permeability model.	PROPS
14	LSLTWNUM	<i>LSLTWNUM – Define the Low Salt Water Wet Saturation Table Region Numbers.</i> Low salt water wet saturation tables region numbers for when the Low Salinity option for the Brine model and the Surfactant Wettability option have been activated for the model.	PROPS
15	LSNUM	<i>LSNUM – Define the Low Salt Oil Wet Saturation Table Region Numbers.</i> Low salt oil wet saturation tables region numbers for the Low Salinity Brine model.	PROPS
16	LWSLTNUM	<i>LWSLTNUM – Define the Low Salt Oil Wet Saturation Table Region Numbers.</i> Low salt oil wet saturation table region numbers for the Low Salinity Brine model.	PROPS
17	LWSNUM	<i>LWSNUM – Define the Low Salt Water Wet Saturation Table Region Numbers.</i> Low salt water wet saturation table region numbers for when the Low Salinity option for the Brine model and the Surfactant Wettability option have been activated for the model.	PROPS
18	MISCNUM	<i>MISCNUM – Define the Miscibility Region Numbers.</i> Miscible regions based on the TLMIXPAR records when the MISCIBLE or SOLVENT keywords have been activated in the RUNSPEC section.	PROPS
19	OPERNUM	<i>OPERNUM – Define Regions for Mathematical Operations on Arrays.</i> OPERATER keyword region numbers that can also be used with the EQUALREG, ADDREG, COPYREG, MULTIREG, MULTREGP, and MULTREGT keywords, as well as the OPERATER keyword in calculating various grid properties in the GRID and REGION section.	GRID EDIT PROPS REGION SOLUTION

No.	REGIONS Keyword	Standard Property Regions	Property Section
20	PENUM	<i>PENUM – Define the Petro-Elastic Region Numbers.</i> Petro-elastic region numbers for petro-elastic coefficients, bulk modulus functions and shear modulus functions.	PROPS
21	PLMIXNUM	<i>PLMIXNUM – Define the Polymer Region Numbers.</i> Polymer region numbers for the mixing tables, and the maximum polymer and salt concentrations for the Polymer option.	PROPS
22	RESIDNUM	<i>RESIDNUM – Define Vertical Equilibrium Residual Flow Region Numbers.</i> Vertical Equilibrium residual flow calculation saturation tables or when region numbers for the Vertical Equilibrium option.	PROPS
23	ROCKNUM	<i>ROCKNUM – Define Rock Compaction Table Region Numbers.</i> Rock compaction table allocation for when the ROCKCOMP keyword has been activated in the RUNSPEC section, that allocates the ROCKTAB series of tables to a cell.	PROPS
24	SURFNUM	<i>SURFNUM – Define the Surfactant Miscible Saturation Table Region Numbers.</i> Surfactant saturation (relative permeability) tables allocation allocating the SWFN, SOF2, SOF3, or SWOF as miscible tables.	PROPS
25	SURFWNUM	<i>SURFWNUM – Define the Saturation Table Region Numbers (High Salinity and Water Wet).</i> High salinity water wet saturation table allocation using the high salinity water wet saturation SWFN and SOFN tables.	PROPS
26	TNUM	<i>TNUM – Define Passive Tracer Concentration Regions.</i> Partitioned Tracer option region numbers for this option.	PROPS
27	TRKPF	<i>TRKPF – Define Partitioned Tracer Regions.</i> Defines the regions associated with partition tracers and the partitioning tables allocated to grid blocks for when the Partitioned Tracer option has been enabled by the PARTTRAC keyword in the RUNSPEC section.	PROPS
28	WH2NUM	<i>WH2NUM – Define WAG Hysteresis Saturation Table Region Numbers (Two Phase).</i> Two phase Water-Alternating-Gas (“WAG”) hysteresis saturation table region numbers for when the hysteresis option has been activated by the WAGHYSTR variable on the SATOPTS keyword in the RUNSPEC.	PROPS
29	WH3NUM	<i>WH3NUM – Define WAG Hysteresis Saturation Table Region Numbers (Three Phase).</i> Three phase WAG hysteresis saturation table region numbers for when the hysteresis option has been activated by the WAGHYSTR variable on the SATOPTS keyword in the RUNSPEC.	PROPS
Notes: - Note that not all keywords and features listed above are implemented in OPM Flow. Cells not colored in the No. column indicate the keyword is supported or partially supported by OPM Flow, cells colored gray indicate that the keyword is not applicable and will be ignored by the simulator, and finally, cells colored in orange indicate keywords that are not supported by OPM Flow and will be ignored by the simulator. - Note that is common to set the FIPNUM array to be equal to the EQLNUM to have fluid in-place reporting for each equilibrium region, this can be done by using the COPY keyword to copy the EQLNUM array to the FIPNUM array.			

Table 9.1: REGION Section Allocation Array Summary

Note

Earth modeling software may export the region arrays to be loaded into the simulator with values starting from zero instead of one. This may result in an error message if these cells are active in the model, if the cells are inactive then the zero values will be ignored by the simulator.

A complete property data set associated with a region keyword is required for each region number allocated to the cells. For example, if the fluid properties for the model are the same (for example, PVTO and PVDG keyword data), but the rock compressibility is varying with depth resulting in, say three different ROCK keyword records, then there has to be three PVTNUM regions and three complete PVT data sets in order to allocate the three ROCK records. This would mean that the PVTO and PVDG records, in this instance, would have to be repeated three times to match the three ROCK keyword records.

Example SATNUM and EQUIL arrays from the Volve²⁹⁴ field are displayed in Figure 9.1 and Figure 9.2, respectively.

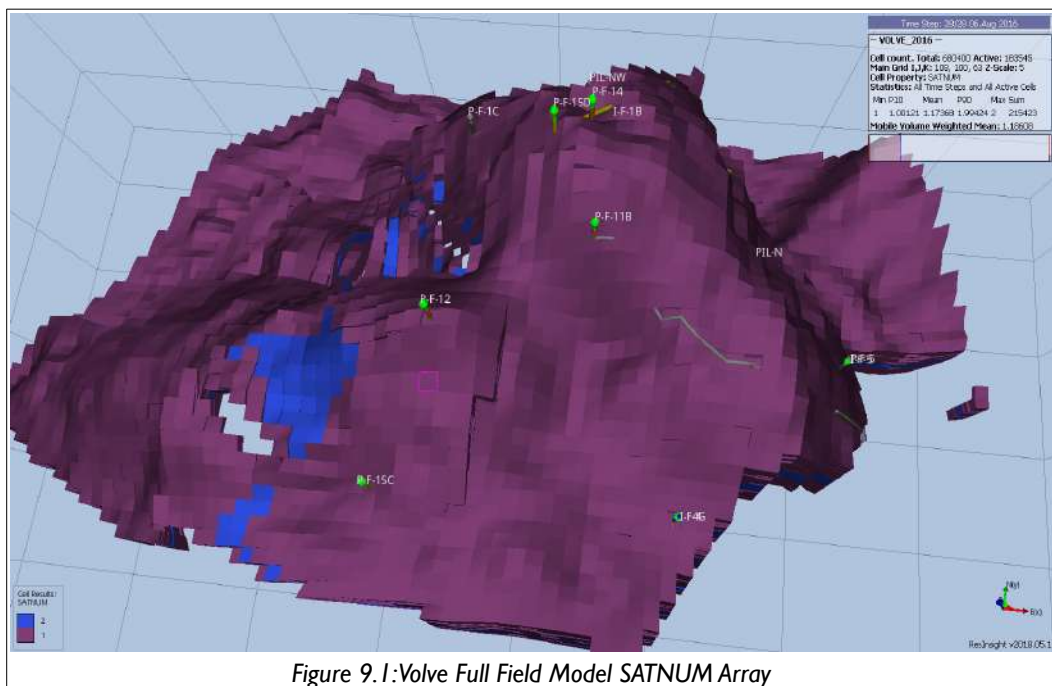
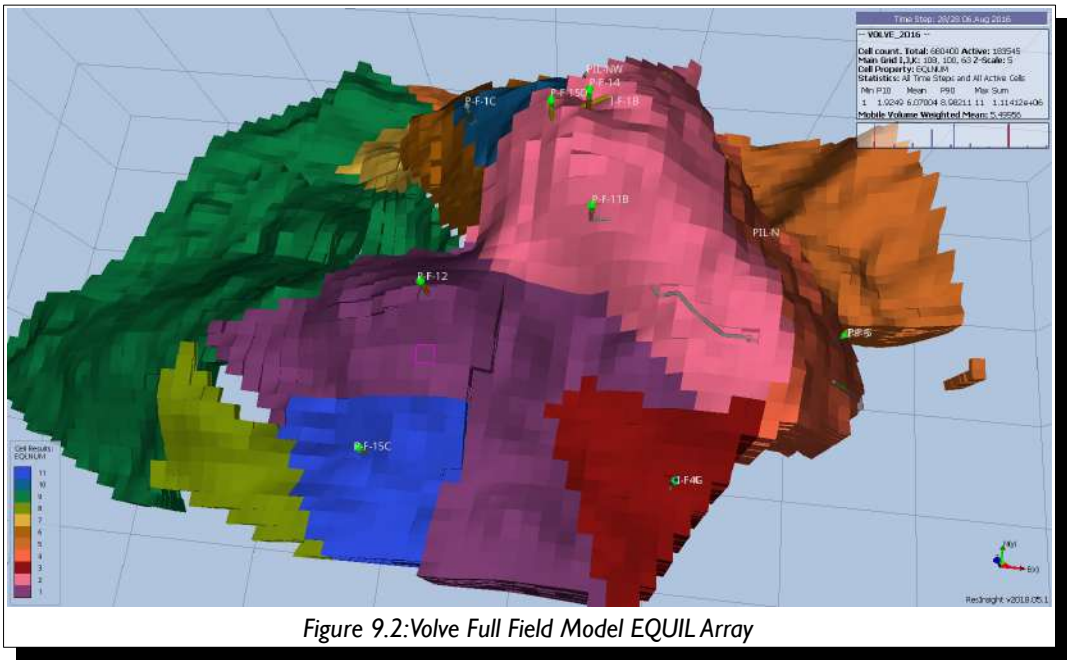


Figure 9.1: Volve Full Field Model SATNUM Array

²⁹⁴ The Volve Data was approved for data sharing in 2018 by the initiative of the last Operating company, Equinor and approved by the license partners ExxonMobil E&P Norway AS and Bayerngas Norge AS in the end of 2017.



9.3 KEYWORD DEFINITIONS

9.3.1 ADD – ADD A CONSTANT TO A SPECIFIED ARRAY

The **ADD** keyword adds a constant to a specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the **ADD** keyword is being applied.

See [ADD – Add a Constant to a Specified Array](#) in the **GRID** section for a full description.

9.3.2 ADDREG – ADD A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER

The **ADDREG** keyword adds a constant to a specified array or part of an array based on cells with a specific region number. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **ADDREG** keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the **ADDREG** keyword is being used.

See [ADDREG – Add a Constant to an Array based on a Region Number](#) in the **GRID** section for a full description.

9.3.3 BOUNDARY – DEFINE A BOUNDARY BOX FOR PRINTING

The **BOUNDARY** keyword defines a rectangular grid for printing various arrays to the output print file (*.PRT); thus, avoiding printing all the elements in the selected array.

See [BOUNDARY – Define a Boundary Box for Printing](#) in the **GRID** section for a full description.

9.3.4 BOX – DEFINE A RANGE OF GRID BLOCKS TO ENTER PROPERTY DATA

BOX defines a range of grid blocks for which subsequent data is assigned for all the cells in the defined **BOX**. Note that the **BOX** grid is reset by the keyword **ENDBOX** by resetting the current defined **BOX** to be the whole grid. The keyword can be used for any array and for all grid types.

See [BOX – Define a Range of Grid Blocks to Enter Property Data](#) in the **GRID** section for a full description.

9.3.5 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The **COLUMNS** keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the **GLOBAL** section for a full description.

9.3.6 COPY – COPY ARRAY DATA TO ANOTHER ARRAY

The **COPY** keyword copies an array (or part of an array) to another array or part of an array. The arrays can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the **COPY** keyword is being used.

See [COPY – Copy Array Data to Another Array](#) in the **GRID** section for a full description.

9.3.7 COPYBOX – COPY ARRAY DATA DEFINED BY A BOX

The **COPYBOX** keyword copies an array (or part of an array) to part of the same array. The array can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the **COPYBOX** keyword is being used.

See [COPYBOX – Copy Array Data Defined by a Box](#) in the **GRID** section for a full description.

9.3.8 COPYREG – COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER

The [COPYREG](#) keyword copies a specified array or part of an array based on cells with a specific region number to another array. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [COPYREG](#) keyword is read by the simulator. The property arrays can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [COPYREG](#) keyword is being used.

See [COPYREG – Copy an Array to Another Array based on a Region Number](#) in the [GRID](#) section for a full description.

9.3.9 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the [GLOBAL](#) section for a full description.

9.3.10 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the [GLOBAL](#) section for a full description.

9.3.11 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the [GLOBAL](#) section for a full description.

9.3.12 ENDBOX – DEFINE THE END OF THE BOX DEFINED GRID

This keyword marks the end of a previously defined [BOX](#) sub-grid as defined by a previously entered [BOX](#) keyword. The keyword resets the input grid to be the full grid as defined by the NX, NY, and NZ variables on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

See [ENDBOX – Define the End of the BOX Defined Grid](#) in the [GRID](#) section for a full description.

9.3.13 ENDFIN – END THE DEFINITION OF A LOCAL GRID REFINEMENT

[ENDFIN](#) defines the end of a Cartesian or radial local grid refinement (“LGR”) definition and a [LGR](#) property definition data set.

See [ENDFIN – End the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

9.3.14 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the GLOBAL section for a full description.

9.3.15 ENDNUM – DEFINE THE END-POINT SCALING DEPTH REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **ENDNUM** keyword defines the end-point scaling depth table region numbers for each grid block. The end-point scaling depth tables for various regions are defined by the **ENPVTD**²⁹⁵ and the **ENKRVD**²⁹⁶ keywords in the **PROPS** section. In the **RUNSPEC** section the **NTENDP** variable on the **ENDSCALE** keyword defines the maximum number of depth tables.

No.	Name	Description	Default
1	ENDNUM	ENDNUM defines an array of positive integers assigning a grid cell to a particular end-point scaling depth table region. The maximum number of ENDNUM regions is set by the NTENDP variable on the ENDSCALE keyword in the RUNSPEC section.	1
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a ENDNUM region number then the default value of one will be used. 3) The keyword is terminated by a “/”.			

Table 9.2: ENDNUM Keyword Description

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Examples

The example below sets three **ENDNUM** regions for a 4 x 5 x 2 model.

```
--
--      DEFINE ENDNUM REGIONS FOR ALL CELLS
--
ENDNUM
      2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
      3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the **EQUALS** keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      ENDNUM      1           1*  1*   1*  1*   1*  1* / SET REGION 1
      ENDNUM      2           1   2    1   2    1   1 / SET REGION 2
      ENDNUM      3           1   2    1   2    2   2 / SET REGION 3
/
```

²⁹⁵ This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

²⁹⁶ This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

9.3.16 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the GLOBAL section for a full description.

9.3.17 EOSNUM – DEFINE THE EQUATION OF STATE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [EOSNUM](#) keyword defines the Equation Of State ([EOS](#)) region number for all the cells in the model via an array. The region number specifies which Equation Of State is used in each grid block. The keyword should only be used if the compositional mode has been requested using the [COMPS](#) keyword in the [RUNSPEC](#) section.

OPM Flow does not currently support the general compositional modeling formulation.

This keyword is not supported by OPM Flow but it will be parsed and its data ignored.

No.	Name	Description	Default
1	EOSNUM	EOSNUM is an array of positive integers less than or equal to NMEOSR that define the EOS region number for each cell in the model. The NMEOSR variable on the TABDIMS keyword in the RUNSPEC section defines the number of EOS regions in the model.	1
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a region then the default value will be used. 3) The keyword is terminated by a "/".			

Table 9.3: [EOSNUM](#) Keyword Description

Examples

The example below sets two [EOSNUM](#) regions for a 4 x 5 x 2 model.

```
--
--      DEFINE EOSNUM REGIONS FOR ALL CELLS
--
EOSNUM
      20*1
      20*2
/
```

9.3.18 EQLNUM – DEFINE THE EQUILIBRATION REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **EQLNUM** keyword defines the equilibration region numbers for each grid block. The equilibration data for various regions are defined in the **SOLUTION** section. For example, the **EQUIL** keyword in the **SOLUTION** section defines the initial pressures and fluid contacts for each equilibration region identified by the **EQLNUM** region array.

No.	Name	Description	Default
I	EQLNUM	EQLNUM defines an array of positive integers assigning a grid cell to a particular equilibration region. The maximum number of EQLNUM regions is set by the NTEQL variable on the EQLDIMS keyword in the RUNSPEC section.	I

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The **EQLNUM** and **PVTNUM** arrays need to be consistent, that is the all cells with the same **EQLNUM** can only belong to one **PVTNUM** region.
- 3) If a cell is not assigned a **EQLNUM** region number then the default value will be used.
- 4) The keyword is terminated by a “/”.

Table 9.4: EQLNUM Keyword Description

Examples

The example below sets three **EQLNUM** regions for a 4 x 5 x 2 model.

```
--
--      DEFINE EQLNUM REGIONS FOR ALL CELLS
--
EQLNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the **EQUALS** keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
    EQLNUM'        1              1*  1*   1*  1*   1*  1* / SET REGION 1
    EQLNUM'        2              1   2    1   2    1   1 / SET REGION 2
    EQLNUM'        3              1   2    1   2    2   2 / SET REGION 3
/
```

9.3.19 EQUALREG – SETS AN ARRAY TO A CONSTANT BY REGION NUMBER

The [EQUALREG](#) keyword sets a specified array to a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [EQUALREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [EQUALREG](#) keyword is being used.

See [EQUALREG – Sets an Array to a Constant by Region Number](#) in the [GRID](#) section for a full description.

9.3.20 EQUALS – SETS A SPECIFIED ARRAY TO A CONSTANT

The [EQUALS](#) keyword sets a specified array or part of an array to a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [EQUALS](#) keyword is being used.

See [EQUALS – Sets a Specified Array to a Constant](#) in the [GRID](#) section for a full description.

9.3.21 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the [GLOBAL](#) section for a full description.

9.3.22 FILEUNIT – ACTIVATE UNIT CONSISTENCY VERIFICATION

The [FILEUNIT](#) keyword defines the units of the data set, and is used to verify that the units in the input deck and any associated include files are consistent. The keyword does not provide for the conversion between different sets of units.

See [FILEUNIT – Activate Unit Consistency Checking](#) in the [GRID](#) section for a full description.

9.3.23 FIP – DEFINE THE FLUID IN-PLACE NAMES AND REGION AND NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **FIP** keyword defines the fluid in-place name and the associated region numbers for each grid block. The simulator can print out summaries of the fluid in-place in each region, the current flow rates between regions, and the cumulative flows between regions. This keyword is not in the standard keyword format due to the fluid in-place name being concatenated to the keyword **FIP** to fully define the keyword.

Note that the total number of **FIP** and **FIPNUM** regions must be defined by the **NMFIPR** variable on the **REGDIMS** keyword, or the **NTFIP** variable on the **TABDIMS** keyword. Both keywords are in the **RUNSPEC** section.

No.	Name	Description	Default
1	FIPNAME	A character string of up to eight characters, consisting of FIP as the first three characters followed by up to a five letter character string defining the fluid in-place's name.	None
2	FIPNUM	FIPNUM defines an array of positive integers greater than or equal to one, that assigns a grid cell to a particular fluid in-place region named by FIPNAME . The maximum number of FIP and FIPNUM regions is set by the REGDIMS (NMFIPR) or the TABDIMS (NTFIP) keywords(variables) in the RUNSPEC section. If both REGDIMS (NMFIPR) and TABDIMS (NTFIP) have been defined then the maximum of the two is used.	1

Notes:

- 1) The number of entries should correspond to the **NX** x **NY** x **NZ** parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) If a cell is not assigned a **FIP** region number by the end of the **REGION** section then the fault value of one will be used.
- 3) The keyword is terminated by a "/".

Table 9.5: FIP Keyword Description

The keyword behaves the same as the **FIPNUM** keyword except the full name of the keyword, including the concatenated characters, are used as the property region name. For example, if we wish define a fluid in-place region name called **UNIT**, then the keyword would be **FIPUNIT**.

Note

The commercial simulator prints out a fluid in-place report if the **FIP** option on the **RPTSCHEG** keyword is set equal to three, that is: **FIP=3**. This option is currently not available in OPM Flow.

The region property data for **FIP** arrays can be written to the **SUMMARY** file, and the **RSM** file if requested, similar to the **FIPNUM** regions, with some caveats:

- 1) Only **SUMMARY** keywords for regions may be used, that is the **SUMMARY** variable name must begin with the letter **R**.
- 2) The **SUMMARY** variable name must consist of a character string length of exactly five characters, if less than five characters, then the "_" character should be used to fill out the **SUMMARY** variable name. For example, instead of **RPRUNIT**, use **RPR__UNI**, or instead of **ROIUNIT**, use **ROI__UNI**.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- 3) Only the first three characters of the [FIP](#) region name should be concatenated with the [SUMMARY](#) variable name. This means if the [FIP](#) region name is UNIT, the [FIP](#) keyword would be FIPUNIT; however, to access the regional pressures for FIPUNIT, one should use RPR__UNI, or to access the regional oil in-place one would use ROIP__UNI.

See also the [FIPOWG](#) keyword in the [REGIONS](#) section that automatically defines the fluid-in-place regions at the start of the run based the gas, oil and water zones at the time the model was initialized.

Example

The example below defines a region name of UNIT and sets three FIPUNIT regions for a 4 x 5 x 2 model.

```
--
--      DEFINE FIPUNIT FIP REGIONS FOR ALL CELLS
--
FIPUNIT
      2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1  1
      3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1  1  1
/
```

The second example is based on the Norne model that defines two [FIP](#) regions based on geological layers and numerical layers using the [EQUALS](#) keyword.

```
--      FIPGL BASED ON GEOLOGICAL LAYERS
--      ARRAY          CONSTANT          ----- BOX -----
--                                     I1  I2  J1  J2  K1  K2
EQUALS
      FIPGL          1          1  46      1 112      1  3  /  Garn
      FIPGL          2          1  46      1 112      4  4  /  Not
      FIPGL          3          1  46      1 112      5  5  /  Ile 2.2
      FIPGL          4          1  46      1 112      6  8  /  Ile 2.1
      FIPGL          5          1  46      1 112      9 11  /  Ile 1
      FIPGL          6          1  46      1 112     12 12  /  Tofte 2.2
      FIPGL          7          1  46      1 112     13 15  /  Tofte 2.1
      FIPGL          8          1  46      1 112     16 18  /  Tofte 1
      FIPGL          9          1  46      1 112     19 22  /  Tilje

--
--      FIPNL BASED ON NUMERICAL LAYERS
--
      FIPNL          1          1  46      1 112      1  1  /  Garn 3
      FIPNL          2          1  46      1 112      2  2  /  Garn 2
      FIPNL          3          1  46      1 112      3  3  /  Garn 1w
      FIPNL          4          1  46      1 112      4  4  /  Not
      FIPNL          5          1  46      1 112      5  5  /  Ile 2.2
      FIPNL          6          1  46      1 112      6  6  /  Ile 2.1.3
      FIPNL          7          1  46      1 112      7  7  /  Ile 2.1.2
      FIPNL          8          1  46      1 112      8  8  /  Ile 2.1.1
      FIPNL          9          1  46      1 112      9  9  /  Ile 1.3
      FIPNL         10          1  46      1 112     10 10  /  Ile 1.2
      FIPNL         11          1  46      1 112     11 11  /  Ile 1.1
      FIPNL         12          1  46      1 112     12 12  /  Tofte 2.2
      FIPNL         13          1  46      1 112     13 13  /  Tofte 2.1.3
      FIPNL         14          1  46      1 112     14 14  /  Tofte 2.1.2
      FIPNL         15          1  46      1 112     15 15  /  Tofte 2.1.1
      FIPNL         16          1  46      1 112     16 16  /  Tofte 1.2.2
      FIPNL         17          1  46      1 112     17 17  /  Tofte 1.2.1
      FIPNL         18          1  46      1 112     18 18  /  Tofte 1.1
      FIPNL         19          1  46      1 112     19 19  /  Tilje 4
      FIPNL         20          1  46      1 112     20 20  /  Tilje 3
      FIPNL         21          1  46      1 112     21 21  /  Tilje 2
      FIPNL         22          1  46      1 112     22 22  /  Tilje 1
/
```

Then in order to get the reservoir pressure for the regions and the in-place oil volumes [SUMMARY](#) variables written to the [SUMMARY](#) file, one would use the following [SUMMARY](#) variable names in the [SUMMARY](#) section

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--      FIP REGION REPORTING
--
ROIP_GL
/
ROIP_NL
/
RPR__GL
/
RPR__NL
/
```

Notice how the “_” character has been used to extend the [SUMMARY](#) variable name to five characters.

9.3.24 FIPNUM – DEFINE THE FLUID IN-PLACE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [FIPNUM](#) keyword defines the fluid in-place region numbers for each grid block. The simulator can print out summaries of the fluid in-place in each region, the current flow rates between regions, and the cumulative flows between regions.

Note that the total number of [FIPNUM](#) and [FIP](#) regions must be defined by the NMFIPR variable on the [REGDIMS](#) keyword, or the NTFIP variable on the [TABDIMS](#) keyword. Both keywords are in the [RUNSPEC](#) section.

No.	Name	Description	Default
1	FIPNUM	FIPNUM defines an array of positive integers greater than or equal to one, that assigns a grid cell to a particular fluid in-place region. The maximum number of FIPNUM and FIP regions is set by the REGDIMS(NMFIPR) or the TABDIMS(NTFIP) keywords(variables) in the RUNSPEC section. If both REGDIMS(NMFIPR) and TABDIMS(NTFIP) have been defined then the maximum of the two is used.	1
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If acell is not assigned a FIPNUM region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.6: FIPNUM Keyword Description

Note

In most simulation models the [FIPNUM](#) array is used to define various regions in the model for fluid in-place reporting and to identify (or report) the flow between the different regions. When calibrating a model's in-place volumes it would be useful to use the [FIPNUM](#) array combined with the [MULTREGP](#) keyword to accomplish this. However, the [FIPNUM](#) array cannot be used in the [GRID](#) section.

A work around is to:

- 1) Use the [FIPNUM](#) array but change the keyword to [MULTNUM](#) and incorporate this keyword or [INCLUDE](#) file in the [GRID](#) section.
- 2) Use the [MULTREGP](#) to calibrate the fluid in-place volumes for the various regions.
- 3) In the [REGIONS](#) section, use the [COPY](#) keyword to copy the [MULTNUM](#) array to the [FIPNUM](#) array.

The above work flow will ensure that both arrays and the reporting of fluid in-place regions are consistent.

Examples

The example below sets three *FIPNUM* regions for a 4 x 5 x 2 model.

```
--
--      DEFINE FIPNUM REGIONS FOR ALL CELLS
--
FIPNUM
      2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1  1
      3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the *EQUALS* keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      FIPNUM      1          1*  1*   1*  1*   1*  1* / SET REGION 1
      FIPNUM      2          1  2    1  2    1  1  / SET REGION 2
      FIPNUM      3          1  2    1  2    2  2  / SET REGION 3
/
```

9.3.25 FIPOWG – ACTIVATE OIL, GAS, AND WATER FIP ZONE REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [FIPOWG](#) keyword activates automatic fluid in-place reporting based on the initial oil, gas and water zones defined by the initial equilibration. The fluid contacts on the [EQUIL](#) keyword in the [SOLUTION](#) section determine the reporting fluid category a grid cell belongs to. For example all grid cells with depths above the gas-oil contact on the [EQUIL](#) keyword will be assigned to the gas zone and reported accordingly. Similarly, grid cells with depths between the gas-oil contact and the water-oil contact will be assigned to the oil zone. And finally, grid cells with depths below the oil-water contact will be assigned to the water zone. The simulator can print out summaries of the fluid in-place in each region, the current flow rates between regions, and the cumulative flows between regions.

Note that the total number of [FIP](#) and [FIPNUM](#) regions must be defined by the NMFIPR variable on the [REGDIMS](#) keyword in the [RUNSPEC](#) section.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Example

```
--
--      ACTIVATE OIL, GAS, AND WATER FIP ZONE REPORTING
--
FIPOWG
```

The above example switches on automatic fluid in-place reporting based on the initial oil, gas and water zones defined by the initial equilibration.

9.3.26 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be placed at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a full description.

9.3.27 GETDATA – LOAD AND ASSIGN DATA ARRAY FROM INIT OR RESTART FILE

The [GETDATA](#) keyword loads a data array from a previously generated [INIT](#) or [RESTART](#) file and assigns the loaded array to either same array in the run or another array name.

See [GETDATA – Load and Assign Data Array from INIT or RESTART Files](#) in the [GRID](#) section for a full description.

9.3.28 HA – HISTORY MATCH END-POINT GRADIENT ADDITIVE MODIFIER

The [HA](#) series of keywords defines the history match end-point gradient parameters used to set the additive cumulative end point data, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

See [HA – History Match End-Point Gradient Additive Modifier](#) in the [PROPS](#) section for a complete description.

9.3.29 HBNUM – DEFINE HERSCHEL-BULKLEY REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [HBNUM](#) keyword defines the Herschel-Bulkley rheological property table region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which table of Herschel-Bulkley rheological property data is assigned to a grid block, for when the Polymer option has been invoked via the [POLYMER](#) keyword in the [RUNSPEC](#) section and the Non-Newtonian Fluid phase has been declared active by the [NNEWTF](#) keyword, also in the [RUNSPEC](#) section. The [FHERCHBL](#) keyword in the [PROPS](#) section is used to specify the Herschel-Bulkley rheological property table data.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

9.3.30 HM – HISTORY MATCH REGION GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [HM](#) series of keywords in the REGION section defines the history match gradient regions and sub-regions, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

For grid properties, the region name (or region property array) is based on the property arrays defined in Table 9.7.

Property Array	Region Name	Grid Property Data Description
PERMX	HMPERMX	Permeability multipliers in the x-direction.
PERMXY	HMPRMXY	Permeability multipliers in the x-direction and y-direction
PERMY	HMPERMY	Permeability multipliers in the y-direction.
PERMZ	HMPERMZ	Permeability multipliers in the z-direction.
PORV	HMPORV	Pore volume multiplier
SIGMA	HMSIGMA	Dual porosity and/or dual permeability SIGMA multiplier
TRANX	HMTRANX	Transmissibility multipliers in the x-direction.
TRANXY	HMTRNXY	Transmissibility multipliers in the x-direction and y-direction
TRANZ	HMTRANZ	Transmissibility multipliers in the z-direction.

Table 9.7: [HM](#) Region Grid Gradient Parameter Keyword List

In addition, if the End-Point Scaling option has been activated by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section, then the history match gradient regions and sub-regions for the end-point data can be specified. In this the keyword consists of the first two characters of “HM” followed by the end-point keyword (Table 9.8), for example, HMSWL.

Type	End-Point Keyword	Oil-Water End-Point Definitions
Saturation	HMSWL	Connate water saturation, that is the smallest water saturation in a water saturation function table.
	HMSWCR	Critical water saturation, that is the largest water saturation for which the water relative permeability is zero.
	HMSOWCR	Critical oil-in-water saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-water system.

Type	End-Point Keyword	Oil-Water End-Point Definitions
Relative Permeability	HMKRW	Relative permeability of water at the maximum water saturation (normally the maximum water saturation is one).
	HMKRO	Relative permeability of oil at the maximum oil saturation.
	HMKRWR	Relative permeability of water at the residual oil saturation or the residual gas saturation in a gas-water run.
	HMKRORW	Relative permeability of oil at the critical water saturation.
Capillary Pressure	HMSWLPC	Capillary pressure connate water saturation, that is the smallest water saturation in a water saturation function table.
Type	End-Point Keyword	Gas-Oil End-Point Definitions
Saturation	HMSGL	Connate gas saturation, that is the smallest gas saturation in a gas saturation function table.
	HMSGCR	Critical gas saturation, that is the largest gas saturation for which the gas relative permeability is zero.
	HMSOGR	Critical oil-in-gas saturation, that is the largest oil saturation for which the oil relative permeability is zero in an oil-gas-connate water system.
Relative Permeability	HMKRG	Relative permeability of gas at the maximum gas saturation.
	HMKRGR	Relative permeability of gas at the residual oil saturation or the critical water saturation in a gas-water run.
	HMKRORG	Relative permeability of oil at the critical gas saturation.
Capillary Pressure	HMSGLPC	Capillary pressure connate gas saturation, that is the smallest gas saturation in a gas saturation function table.

Table 9.8: *HM Region End-Point Gradient Parameter Keyword List*

9.3.31 HMPROPS – HISTORY MATCH END-POINT SECTION START

[HMPROPS](#) defines the start of a history match end-points section, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. In addition, the End-Point Scaling option must also be activated by the [ENDSCALE](#) keyword which is also in the [RUNSPEC](#) section. The keyword allows for the [BOX](#), [EQUALS](#), [COPY](#), [MINVALUE](#), [MAXVALUE](#) and [ADD](#) keywords to be used with the [HM](#) series of keywords that reference the end-point scaling arrays, that is: [HMKRG](#), [HMKRGR](#), [HMKRO](#), [HMKRORG](#), [HMKRORW](#), [HMKRW](#), [HMKRWR](#), [HMPCW](#), [HMPCG](#), [HMSGCR](#), [HMSOWCR](#), [HMSOGCR](#), [HMSWCR](#), and [HMSWL](#).

See [HMPROPS – History Match End-Point Section Start](#) in the [PROPS](#) section for a full description.

9.3.32 HWSNUM – DEFINE THE SATURATION TABLE REGION NUMBERS (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **HWSNUM** keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the mode, for when the Surfactant Wettability option has been selected. The keyword may also be used with the Low Salt Brine option, in this case the water wet curves are calculated as a function of the low and high water wet salinity curves. The region number specifies which set of relative permeability tables are used to calculate the relative permeability and capillary pressure in a grid block. Note that the keyword is obligatory if the **SURFACTW** keyword in the **RUNSPEC** section has been used to invoke the Surfactant Wettability option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	HWSNUM	HWSNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of HWSNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) If a cell is not assigned a **SATNUM** region then the default value will be used.
- 3) The keyword is terminated by a “/”.

Table 9.9: HWSNUM Keyword Description

Examples

The example below sets three **HWSNUM** regions for a 4 x 5 x 2 model.

```
--
--      DEFINE HWSNUM REGIONS FOR ALL CELLS
--
HWSNUM
      2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
      3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the **EQUALS** keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      HWSNUM      1             1*  1*   1*  1*   1*  1* / SET REGION 1
      HWSNUM      2             1  2    1  2    1  1  / SET REGION 2
      HWSNUM      3             1  2    1  2    2  2  / SET REGION 3
/
```

9.3.33 IMBNUM – DEFINE THE IMBIBITION SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **IMBNUM** keyword defines the imbibition saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables (**SGFN**, **SWFN**, **SOF2**, **SOF3**, **SOF32D**, **SGOF**, **SLGOF** and **SWOF**) are used to calculate the relative permeability and capillary pressure in a grid block.

No.	Name	Description	Default
1	IMBNUM	IMBNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of IMBNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The keyword is terminated by a “/”.
- 3) If a cell is not assigned an **IMBNUM** region then the default value of one will be used.

Table 9.10: IMBNUM Keyword Description

In addition, saturation table assignment may be directional dependent in which case the directional dependent versions of the aforementioned array should be used, that is **IMBNUMX**, **IMBNUMY** and **IMBNUMZ** instead of **IMBNUM**. There is also the facility to make the directional end-point scaling reversible or non-reversible and if the non-reversible option is selected, the non-reversible versions of the aforementioned arrays should be used, that is **IMBNUMX-**, **IMBNUMY-**, **IMBNUMZ-**, instead of the **IMBNUM** keyword. For reference, see Table 5.40, that lists the various keywords that may be used with directional dependent relative permeability tables.

Note, currently **IMBNUMX-**, **IMBNUMY-**, and **IMBNUMZ-** are not supported by OPM Flow.

Example

The example below sets three **IMBNUM** regions for a 4 x 5 x 2 model using the **EQUALS** keyword.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      IMBNUM      1          1*  1*   1*  1*   1*  1* / SET REGION 1
      IMBNUM      2          1   2    1   2    1   1 / SET REGION 2
      IMBNUM      3          1   2    1   2    2   2 / SET REGION 3
/
```

9.3.34
IMBNUMMF – DEFINE THE IMBIBITION SATURATION TABLE REGION NUMBERS (MATRIX-FRACTURE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [IMBNUMMF](#) keyword defines the imbibition saturation tables (relative permeability and capillary pressure tables) region numbers for flow between the matrix and fracture blocks, for when the HYSTER option on the [SATOPTS](#) keyword has been invoked to activate the Hysteresis option, and the Dual Porosity or Dual Permeability models have been activated via the [DUALPORO](#) or [DUALPERM](#) keywords. All keywords are in the [RUNSPEC](#) section.

The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure between the matrix and fracture blocks. The keyword is optional and any cell not assigned a value will use the assignment from the [IMBNUM](#) array.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

9.3.35 IMPORT – IMPORT GRID FILE DATA AT THE CURRENT POSITION

The [IMPORT](#) keyword informs the simulator to continue reading input data from the specified [IMPORT](#) file. When the end of the [IMPORT](#) file is reached, input data is read from the next keyword in the current file. Normally [IMPORT](#) files are generated by grid pre-processing software and the keyword allows for both formatted and unformatted (binary) files to be loaded.

See [IMPORT – Import Grid File Data at the Current Position](#) in the [GRID](#) section for a full description.

9.3.36 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDINC](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the [GLOBAL](#) section for a full description.

9.3.37 KRNUM – DEFINE THE DIRECTIONAL SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [KRNUM](#) keyword defines the direction dependent saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block face, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure in a grid block. The keyword should only be used if Directional Dependent Saturation Function option has been activated by the [DIRECT](#) parameter on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. Otherwise the standard none directional relative permeability curves should be assigned by the [SATNUM](#) keyword in the [REGIONS](#) section.

This keyword is not in the standard keyword format due to the cell face (X, X-, Y, Y-, Z, and Z- for Cartesian grids and R, R-, T, T-, Z, and Z- for radial grids) being concatenated to the end of the keyword [KRNUM](#) to fully define the keyword.

No.	Name	Description	Default
1	KRNUM	KRNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of KRNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [KRNUM](#) region then the default value of one will be used.
- 3) The keyword is terminated by a “/”.

Table 9.11: [KRNUM](#) Keyword Description

If the Directional Dependent Saturation Function option has been activated by the [DIRECT](#) parameter on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the [KRNUMX](#), [KRNUMY](#) and [KRNUMZ](#) versions of the keyword should be used for cartesian grids. Secondly, if the Non-Reversible End-Point Scaling option has also been activated by the [IRREVERS](#) parameter on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the non-reversible versions of the [KRNUM](#) should be used, that is [KRNUMX](#), [KRNUMX-](#), [KRNUMY](#), [KRNUMY-](#), [KRNUMZ](#) and [KRNUMZ-](#). For reference, see Table 5.40, that lists the various keywords that may be used with directional dependent relative permeability tables.

The keywords [KRNUMX-](#), [KRNUMY-](#), and [KRNUMZ-](#) are not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

The example below sets the directional saturation tables in all three directions using the [EQUALS](#) keyword.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--
EQUALS      I1  I2    J1  J2    K1  K2
            KRXUMX      1      1*  1*    1*  1*    1*  1* / SET X-DIR TABLES
            KRXUMY      2      1*  1*    1*  1*    1*  1* / SET Y-DIR TABLES
            KRXUMZ      3      1*  1*    1*  1*    1*  1* / SET Z-DIR TABLES
/
```


9.3.38

KRNUMMF – DEFINE THE SATURATION TABLE REGION NUMBERS (MATRIX-FRACTURE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [KRNUMMF](#) keyword defines the drainage saturation tables (relative permeability and capillary pressure tables) region numbers for flow between the matrix and fracture blocks, for when the Dual Porosity or Dual Permeability models have been activated via the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section.

The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure between the matrix and fracture blocks. The keyword is optional and any cell not assigned a value will use the assignment from the [SATNUM](#) array.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

9.3.39 LSLTWNUM – DEFINE THE LOW SALT WATER WET SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **LSLTWNUM** keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables (**SWFN**, **SOF3** and related keywords) are used to calculate the relative permeability and capillary pressure in a grid block. The keyword should only be used if the Low Salinity option for the Brine model and the Surfactant Wettability option have been activated by the **LOWSALT** and **SURFACTW** keywords, respectively, in the **RUNSPEC** section.

The water wet curves are calculated as a weighted average of the low salinity saturation tables (allocated by this keyword) and the high salinity water wet saturation tables (allocated by the **SURFWNUM** keyword in the **REGIONS** section), using the weights provided by the **LSALTFNC** keyword in the **PROPS** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	LSLTWNUM	LSLTWNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of LSLTWNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a LSLTWNUM region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.12: LSLTWNUM Keyword Description

Example

The example below sets three **LSLTWNUM** regions for the model.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      LSLTWNUM      1          1*  1*    1*  1*    1*  1* / SET REGION 1
      LSLTWNUM      2          1   2     1   2     1   1 / SET REGION 2
      LSLTWNUM      3          1   2     1   2     2   2 / SET REGION 3
/
```

9.3.40 LSNUM – DEFINE THE LOW SALT OIL WET SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [LSNUM](#) keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables ([SWFN](#), [SOF3](#) and related keywords) are used to calculate the relative permeability and capillary pressure in a grid block. The keyword should only be used if the Low Salinity option for the Brine model has been activated in the [RUNSPEC](#) section using the [LOWSALT](#) keyword.

The oil wet curves are calculated as a weighted average of the low salinity saturation tables (allocated by this keyword) and the high salinity oil wet saturation tables (allocated by the [SATNUM](#) keyword), using the weights provided by the [LSALTFNC](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	LSNUM	LSNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of LSNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [LSNUM](#) region then the default value will be used.
- 3) The keyword is terminated by a “/”.

Table 9.13: LSNUM Keyword Description

If the Surfactant Wettability option have been activated by the [SURFACTW](#) keyword, the [LSNUM](#) tables correspond to the immiscible low salinity curves.

Example

The example below sets three [LSNUM](#) regions for the model.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      LSNUM      1          1*  1*    1*  1*    1*  1* / SET REGION 1
      LSNUM      2          1   2     1   2     1   1 / SET REGION 2
      LSNUM      3          1   2     1   2     2   2 / SET REGION 3
/
```

9.3.41 LWSLTNUM – DEFINE THE LOW SALT OIL WET SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [LWSLTNUM](#) keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables ([SWFN](#), [SOF3](#) and related keywords) are used to calculate the relative permeability and capillary pressure in a grid block. The keyword should only be used if the Low Salinity option for the Brine model has been activated by the [LOWSALT](#) keyword in the [RUNSPEC](#) section.

The oil wet curves are calculated as a weighted average of the low salinity saturation tables (allocated by this keyword) and the high salinity oil wet saturation tables (allocated by the [SATNUM](#) keyword in the [REGIONS](#) section), using the weights provided by the [LSALTFC](#) keyword in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	LWSLTNUM	LWSLTNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of LSLTNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a LWSLTNUM region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.14: LWSLTNUM Keyword Description

Example

The example below sets three [LWSLTNUM](#) regions for the model.

```

--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      LWSLTNUM      1          1*  1*    1*  1*    1*  1* / SET REGION 1
      LWSLTNUM      2          1   2     1   2     1   1 / SET REGION 2
      LWSLTNUM      3          1   2     1   2     2   2 / SET REGION 3
/

```

9.3.42 LWSNUM – DEFINE THE LOW SALT WATER WET SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **LWSNUM** keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables (**SWFN**, **SOF3** and related keywords) are used to calculate the relative permeability and capillary pressure in a grid block. The keyword should only be used if the Low Salinity option for the Brine model and the Surfactant Wettability option have been activated by the **LOWSALT** and **SURFACTW** keywords, respectively, in the **RUNSPEC** section.

The water wet curves are calculated as a weighted average of the low salinity saturation tables (allocated by this keyword) and the high salinity water wet saturation tables (allocated by the **HWSNUM** keyword), using the weights provided by the **LSALTFNC** keyword in the **PROPS** section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	LWSNUM	LWSNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of LWSNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a LWSNUM region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.15: LWSNUM Keyword Description

The **HWSNUM** allocated tables correspond to the immiscible high salinity water wet curves.

Example

The example below sets three **LWSNUM** regions for the model.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--      I1  I2    J1  J2    K1  K2
EQUALS
      LWSNUM      1          1*  1*    1*  1*    1*  1* / SET REGION 1
      LWSNUM      2          1   2     1   2     1   1 / SET REGION 2
      LWSNUM      3          1   2     1   2     2   2 / SET REGION 3
/
```

9.3.43 MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the GLOBAL section for a full description.

9.3.44 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the GLOBAL section for a full description.

9.3.45 MISCNUM – DEFINE THE MISCIBILITY REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [MISCNUM](#) keyword defines the miscibility region number mixing tables as defined by the [TLMIXPAR](#) keyword in the [PROPS](#) section, for when the miscibility option has been activated by the [MISCIBLE](#) keyword in the [RUNSPEC](#) section. [MISCNUM](#) also allocates miscible residual oil saturation versus water saturation tables ([SORWMIS](#) keyword in the [PROPS](#) section) used to calculate the relative permeability and PVT properties for a grid cell.

Note that although this keyword can only be used when the miscibility option is active, it is not necessary to use this keyword even if the [MISCIBLE](#) keyword in the [RUNSPEC](#) has been activated as the default value of one will be applied to all grid blocks. Secondly, a value of zero for a grid cell results in immiscible fluids in that grid cell.

No.	Name	Description	Default
1	MISCNUM	MISCNUM defines an array of positive integers greater than or equal to zero, that assign a grid cell to a particular table of mixing parameters as defined by the TLMIXPAR and SORWMIS keywords. A value of zero sets the fluids within a grid cell to be immiscible. The maximum number of MISCNUM regions is set by the NTMIS variable on the MISCIBLE keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [MISCNUM](#) region then the default value of one will be used.
- 3) The keyword is terminated by a “/”.

Table 9.16: MISCNUM Keyword Description

See also the [TLMIXPAR](#) and [SORWMIS](#) keyword in the [PROPS](#) section.

Example

The example below sets three [MISCNUM](#) regions in the model on a layer by layer basis, using the [EQUALS](#) keyword.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      MISCNUM      1           1*  1*   1*  1*   1  12 / SET REGION 1
      MISCNUM      2           1*  1*   1*  1*  13  55 / SET REGION 2
      MISCNUM      3           1*  1*   1*  1*  56 120 / SET REGION 3
/
```

9.3.46 MULTIPLY – MULTIPLY A SPECIFIED ARRAY BY A CONSTANT

The [MULTIPLY](#) keyword multiplies a specified array or part of an array by a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [MULTIPLY – Multiply a Specified Array by a Constant](#) in the [GRID](#) section for a full description.

9.3.47 MULTIREG – MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER

The [MULTIREG](#) keyword multiplies an array or part of an array by a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [MULTIREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [MULTIREG](#) keyword is being used.

See [MULTIREG – Multiply an Array by a Constant based on a Region Number](#) in the [GRID](#) section for a full description.

9.3.48 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active, but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the [GLOBAL](#) section for a full description.

9.3.49 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the [GLOBAL](#) section for a full description.

9.3.50 OPERATE – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS

This keyword, [OPERATE](#), defines mathematical operations on property arrays ([NTG](#), [PORO](#) etc.) and optionally using another property array as input to the function. The keyword allows for various mathematical functions and their associated variables to be defined and applied to the selected array data. Input constants can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [OPERATE – Define Mathematical Operations on Arrays](#) in the [GRID](#) section for a full description.

9.3.51 OPERATER – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS BY REGION

This keyword defines the mathematical operations on arrays for specific regions in the commercial simulator and is currently not supported by OPM Flow. However, similar functionality is provided by the [ADD](#) and [MULTIPLY](#) keywords.

See [OPERATER – Define Mathematical Operations on Arrays by Region](#) in the [GRID](#) section for a full description.

9.3.52 OPERNUM – DEFINE REGIONS FOR MATHEMATICAL OPERATIONS ON ARRAYS

This keyword defines the [OPERATE](#) region numbers for each grid block. The [OPERATE](#) keyword defines mathematical operations on arrays in the commercial simulator and is currently not supported by OPM Flow. However, similar functionality is provided by the [ADD](#) and [MULTIPLY](#) keywords combined with [MULTNUM](#) region array.

See [OPERNUM – Define Regions for Mathematical Operations on Arrays](#) in the [GRID](#) section for a full description.

9.3.53 PENUM – DEFINE THE PETRO-ELASTIC REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The PENNUM keyword defines the petro-elastic region number for each grid block that is used to assign the of petro-elastic coefficients, bulk modulus functions and shear modulus functions as defined by the [PECOEFS](#), [PEKTAB](#) and [PEGTAB](#) series of keywords in the [PROPS](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

9.3.54 PLMIXNUM – DEFINE THE POLYMER REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [PLMIXNUM](#) keyword defines the polymer region number for each grid block that is used to assign the mixing tables as well as the maximum polymer and salt concentrations, as defined by the [PLMIXPAR](#) and [PLYMAX](#) keywords in the [PROPS](#) section, for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

The maximum polymer concentration and the associated salt concentration are declared on the [PLYMAX](#) keyword.

No.	Name	Description	Default
1	PLMIXNUM	PLMIXNUM defines an array of positive integers greater than or equal to one, that assign a grid cell to a particular table of mixing parameters as defined by the PLMIXPAR and PLYMAX keywords. The maximum number of PLMIXNUM regions is set by the NPLMIX variable on the REGDIMS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [PLMIXNUM](#) region then the default value of one will be used.
- 3) The keyword is terminated by a "/".

Table 9.17: PLMIXNUM Keyword Description

See also the [PLYADS](#), [PLYADSS](#), [PLYDHLF](#), [PLYMAX](#), [PLYROCK](#), [PLYSHEAR](#), [PLYSHLOG](#) and [PLYVISC](#) keywords in the [PROPS](#) section.

Example

The example below sets three [PLMIXNUM](#) regions in the model on a layer by layer basis, using the [EQUALS](#) keyword.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      PLMIXNUM      1           1*  1*   1*  1*   1  12 / SET REGION 1
      PLMIXNUM      2           1*  1*   1*  1*  13  55 / SET REGION 2
      PLMIXNUM      3           1*  1*   1*  1*  56 120 / SET REGION 3
/
```

9.3.55 PVTNUM – DEFINE THE PVT REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **PVTNUM** keyword defines the PVT region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of PVT tables (**DENSITY**, **PVDG**, **PVDO**, **PVTG**, **PVTO**, **PVCO**, **PVTW** and **ROCK**) are used to calculate the PVT properties in a grid block.

No.	Name	Description	Default
1	PVTNUM	PVTNUM defines an array of positive integers assigning a grid cell to a particular PVT region. The maximum number of PVTNUM regions is set by the NTPVT variable on the TABDIMS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) The **EQLNUM** and **PVTNUM** arrays need to be consistent, that is the all cells with the same **PVTNUM** can only belong to one **EQLNUM** region.
- 3) If a cell is not assigned a **PVTNUM** region then the default value will be used.
- 4) The keyword is terminated by a "/".

Table 9.18: PVTNUM Keyword Description

Note

Care should be taken that cells in different **PVTNUM** regions are not in communication, since the fluid properties are associated with a cell. If for example, a rbbl or a rm3 of oil flows from **PVTNUM** region 1 to **PVTNUM** region 2, then the oil properties of that oil will change from the PVT 1 data set to the PVT data set 2. This will result in material balance errors, that may or may not cause numerical issues.

To avoid this one should use the **MULTNUM** (or **FLUXNUM**, or **OPERNUM**) array with the **MULTREGT** array to ensure that the various **PVTNUM** regions are not in communication.

Examples

The example below sets three **PVTNUM** regions for a 4 x 5 x 2 model.

```
--
--      DEFINE PVTNUM REGION FOR ALL CELLS
--
PVTNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the [EQUALS](#) keyword could be employed to accomplish the same task, that is:

```
--
--          ARRAY          CONSTANT          ----- BOX -----
--          I1  I2    J1  J2    K1  K2
EQUALS
          PVTNUM          1          1*  1*    1*  1*    1*  1* / SET REGION 1
          PVTNUM          2          1   2     1   2     1   1 / SET REGION 2
          PVTNUM          3          1   2     1   2     2   2 / SET REGION 3
/
```

There third example shows how to ensure the various PVT regions are isolated. First of all define the [MULTNUM](#) array in the [GRID](#) section and ensure all the regions are isolated.

```
-- =====
--
-- GRID SECTION
--
-- =====
GRID
--
--          ARRAY          CONSTANT          ----- BOX -----
--          I1  I2    J1  J2    K1  K2
EQUALS
          MULTNUM          1          1*  1*    1*  1*    1*  1* / SET REGION 1
          MULTNUM          2          1   2     1   2     1   1 / SET REGION 2
          MULTNUM          3          1   2     1   2     2   2 / SET REGION 3
/
--
--          SET TRANSMISSIBILITES ACROSS DIFFERENT RESERVOIRS TO ZERO TO ISOLATE
--          RESERVOIRS
--
--          REGION  REGION  TRANS  DIREC  NNC  REGION ARRAY
--          FROM    TO      MULT   OPT    OPTS  M / F / O
MULTREGT
          1*        1*        0.0    1*    'ALL'  M          / ALL REGIONS SEALED
/
```

Then in the [REGIONS](#) section copy the [MULTNUM](#) array to the [PVTNUM](#) array.

```
-- =====
--
-- REGIONS SECTION
--
-- =====
REGIONS
--
--          COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER
--
--          ARRAY  ARRAY  REGION  REGION ARRAY
--          FROM   TO     NUMBER  M / F / O
COPYREG
          MULTNUM  PVTNUM  1       M          / COPY MULT TO PVT 1
          MULTNUM  PVTNUM  2       M          / COPY MULT TO PVT 2
          MULTNUM  PVTNUM  3       M          / COPY MULT TO PVT 3
/
```

All the separate PVT regions are now isolated.

9.3.56 PYEND – END THE DEFINITION OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword on a separate single line.

See [PYEND – End the Definition of a PYINPUT Section](#) in the [GRID](#) section for a full description.

9.3.57 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator. [PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line.

See [PYINPUT – Define the Start of a PYINPUT Section](#) in the [GRID](#) section for a full description.

9.3.58 REFINE – START THE DEFINITION OF A LOCAL GRID REFINEMENT

The [REFINE](#) keyword defines the start of a Cartesian or radial local grid refinement (“LGR”) definition that sets the properties of the selected [LGR](#). The keyword is then followed by the property keywords associated with the section where the keyword is being invoked. For example, if the [REFINE](#) keyword is used in the [GRID](#) section then most of the keywords in that section can be used to set the grid properties for the [LGR](#).

See [REFINE – Start the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

9.3.59 REGIONS – DEFINE THE START OF THE REGIONS SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [REGIONS](#) activation keyword marks the end of the [PROPS](#) section and the start of the [REGIONS](#) section that defines how various fluid and rock property data defined in the [PROPS](#) section are allocated to the individual cells in the model.

There is no data required for this keyword.

Example

```
-- =====
--
-- REGIONS SECTION
--
-- =====
REGIONS
```

The above example marks the end of the [PROPS](#) section and the start of the [REGIONS](#) section in the OPM Flow data input file.

9.3.60 RESIDNUM – DEFINE VERTICAL EQUILIBRIUM RESIDUAL FLOW REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [RESIDNUM](#) keyword defines the Vertical Equilibrium (“VE”) residual flow calculation saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure in a grid block. The keyword should only be used if the Vertical Equilibrium option has been invoked via the [VE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	RESIDNUM	RESIDNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of RESIDNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [RESIDNUM](#) region then the default value will be used.
- 3) The keyword is terminated by a “/”.

Table 9.19: RESIDNUM Keyword Description

Note that any capillary pressure data on the relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) will be ignored by the [VE](#) option.

Example

The example below sets three [RESIDNUM](#) regions for the model, by first setting all values to one, then setting layers 2 to 10 to two, and finally setting layers 30 to 50 to three.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      SATNUM      1          1*  1*   1*  1*   1*  1* / SET REGION 1
      SATNUM      2          1   2    1   2    2  10 / SET REGION 2
      SATNUM      3          1   2    1   2    30  50 / SET REGION 3
/
```


9.3.61 ROCKNUM – DEFINE ROCK COMPACTION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [ROCKNUM](#) keyword defines the rock compaction table region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of rock compaction tables defined by the [ROCKTAB](#) keyword are used to calculate the rock compaction in a grid block.

No.	Name	Description	Default
I	ROCKNUM	ROCKNUM defines an array of positive integers assigning a grid cell to a particular rock compaction table region. The maximum number of ROCKNUM regions is set by the NTROCC variable on the ROCKCOMP keyword in the RUNSPEC section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a ROCKNUM region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.20: ROCKNUM Keyword Description

Examples

The example below sets three [ROCKNUM](#) regions for a 4 x 5 x 2 model.

```
--
--      DEFINE ROCKNUM REGION FOR ALL CELLS
--
ROCKNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the [EQUALS](#) keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
    ROCKNUM      1          1*  1*   1*  1*   1*  1* / SET REGION 1
    ROCKNUM      2          1   2    1   2    1   1 / SET REGION 2
    ROCKNUM      3          1   2    1   2    2   2 / SET REGION 3
/
```

9.3.62 RPTREGS – DEFINE REGIONS SECTION REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the data in the [REGIONS](#) section that is to be printed to the output print file in human readable format. The keyword has two distinct forms, the first of which consists of the keyword followed by a series of integers on the next line indicating the data to be printed (see the first example). This is the original format in the commercial simulator and was subsequently superseded by the second format. The second format consists of the keyword followed by a series of character strings that indicate the data to be printed. In most cases the character string is the keyword used to load the data in the OPM Flow input deck, for example [FIPNUM](#) for the fluid in-place array. It is anticipated that OPM Flow will eventually support the functionality of the second format only, the first format although recognized will be completely ignored.

No.	Name	Description	Default
1	EQLNUM	Print the equilibration region array.	N/A
2	FIPNUM	Print the fluid in-place array.	N/A
3	PVTNUM	Print the PVT table assignment array.	N/A
4	SATNUM	Print the saturation function (relative permeability) assignment array.	N/A
....			N/A
Notes: 1) The keyword is terminated by a "/".			

Table 9.21: RPTREGS Keyword Description

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Note

This keyword has the potential to produce very large print files that some text editors may have difficulty loading, coupled with the fact that reviewing the data in this format is very cumbersome. A more efficient solution is to load the *.INIT file into OPM ResInsight to view the data graphically, this also has the benefit of being able to filter the grid based on I, J, K ranges and grid properties.

Examples

The first example shows the original format of this keyword; although the keyword and format are recognized by OPM Flow, the format is ignored and is unlikely to be implemented in the simulator.

```
--
--      DEFINE REGIONS SECTION REPORT OPTION (ORIGINAL FORMAT)
--
RPTREGS
      1          2*0      1          3*1          /
```

The next example shows the second format of the keyword which may be supported in a future release of OPM Flow.

```
--          DEFINE REGIONS SECTION REPORT OPTIONS
--
RPTREGS          FIPMUM      EQLNUM      PVTNUM      SATNUM          /
```

9.3.63 SATNUM – DEFINE THE SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [SATNUM](#) keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure in a grid block.

No.	Name	Description	Default
I	SATNUM	SATNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of SATNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a SATNUM region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.22: SATNUM Keyword Description

Examples

The example below sets three [SATNUM](#) regions for a 4 x 5 x 2 model.

```
--
--      DEFINE SATNUM REGIONS FOR ALL CELLS
--
SATNUM
    2  2  1  1  2  2  1  1  1  1  1  1  1  1  1  1  1  1
    3  3  1  1  3  3  1  1  1  1  1  1  1  1  1  1  1  1
/
```

Alternatively the [EQUALS](#) keyword could be employed to accomplish the same task, that is:

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
    SATNUM        1              1*  1*   1*  1*   1*  1* / SET REGION 1
    SATNUM        2              1   2    1   2    1   1 / SET REGION 2
    SATNUM        3              1   2    1   2    2   2 / SET REGION 3
/
```

9.3.64
SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

9.3.65
SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

9.3.66
SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

9.3.67 SURFNUM – DEFINE THE SURFACTANT MISCIBLE SATURATION TABLE REGION NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [SURFNUM](#) keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the model. The region number specifies which set of oil-water relative permeability tables ([SWFN](#), [SOF2](#), [SOF3](#), and [SWOF](#)) are used to calculate the relative permeability and capillary pressure in a grid block. In this case the [SURFNUM](#) allocated tables assume that oil and water are miscible, whereas the [SATNUM](#) allocated tables are used to allocate the immiscible saturation tables. To use this keyword the Surfactant option must have been activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	SURFNUM	SURFNUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of SURFNUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	I

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [SURFNUM](#) region then the default value will be used.
- 3) The keyword is terminated by a “/”.

Table 9.23: SURFNUM Keyword Description

Example

The example below sets three [SURFNUM](#) for the model.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--
--      I1  I2    J1  J2    K1  K2
EQUALS
      SURFNUM      1          1*  1*    1*  1*    1*  1* / SET REGION 1
      SURFNUM      2          1*  1*    1*  1*    1   1 / SET REGION 2
      SURFNUM      3          1*  1*    1*  1*    2   2 / SET REGION 3
/
```

9.3.68
SURFNUM
– DEFINE THE SATURATION TABLE REGION NUMBERS (HIGH SALINITY AND WATER WET)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *SURFNUM* keyword defines the saturation tables (relative permeability and capillary pressure tables) region numbers for each grid block, as such there must be one entry for each cell in the mode, for when the Surfactant Wettability option has been selected. The keyword may also be used with the Low Salt Brine option, in this case the water wet curves are calculated as a function of the low and high water wet salinity curves. The region number specifies which set of relative permeability tables are used to calculate the relative permeability and capillary pressure in a grid block. Note that the keyword is obligatory if the *SURFACTW* keyword in the *RUNSPEC* section has been used to invoke the Surfactant Wettability option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
I	<i>SURFNUM</i>	<i>SURFNUM</i> defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of <i>SURFNUM</i> regions is set by the NTSFUN variable on the <i>TABDIMS</i> keyword in the <i>RUNSPEC</i> section.	I

Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the <i>DIMENS</i> keyword in the <i>RUNSPEC</i> section, unless the <i>BOX</i> keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the <i>BOX</i> statement. 2) If a cell is not assigned a <i>SATNUM</i> region then the default value will be used. 3) The keyword is terminated by a “/”.			
---	--	--	--

Table 9.24: SURFNUM Keyword Description

Example

The example below sets three *SURFNUM* regions for the model.

```

--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      SURFNUM      1           1*  1*   1*  1*   1*  1* / SET REGION 1
      SURFNUM      2           1   2    1   2    1   1 / SET REGION 2
      SURFNUM      3           1   2    1   2    2   2 / SET REGION 3
/

```

9.3.69 TNUM – DEFINE PASSIVE TRACER CONCENTRATION REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	----------------	--------------------------	-------------------------	--------------------------

Description

The [TNUM](#) keyword defines the regions associated with the series of tracers associated with a phase (oil, water, or gas) in the model. The maximum number of tracers for each phase are declared on the [TRACER](#) keyword in the [RUNSPEC](#) section. Unlike other keywords, the [TNUM](#) keyword must be concatenated with the phase and the name of the tracer declared by [TRACER](#) keyword in the [PROPS](#) section. Table 9.25 outlines the format of the [TNUM](#) keyword name.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	TNUM	A four letter character equal to TNUM that is the root keyword name for this data set array.	None
2	PHASE	A one letter character string that must be equal to F or S, that is concatenated to TNUM . The letter F states that the tracer is for the “free” phase, for example oil or water, as well as gas cap gas (free gas). The letter S signifies that the tracer is a “solution” phase tracer, for example gas dissolved in oil (as activated by the DISGAS keyword in the RUNSPEC section), or condensate (vaporized oil) in the gas (as per the VAPOIL keyword in the RUNSPEC section). Note tracers that are defined by the letter S to be in the “solution” phase, must also be initialized by the “free” phase as well.	None
3	NAME	A three letter character string defining the tracer’s name, which is concatenate to TNUM and PHASE to given the full name of the keyword Note it is best to void names beginning with the letters F, S, and T as these names may great naming issues in post-processing software.	None

Table 9.25: [TNUM](#) Keyword Name Format

Following the declaration of the full keyword name, TNUMPHASENAME, the keyword is followed by the data as outlined below.

No.	Name	Description	Default
1	TNUMREG	TUNREG defines an array of positive integers assigning a grid cell to a particular tracer table region. The maximum number of TNUMREG regions is set by the NTTRVD variable on the EQLDIMS keyword in the RUNSPEC section.	1
Notes: 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement. 2) If a cell is not assigned a TNUMREG region then the default value will be used. 3) The keyword is terminated by a “/”.			

Table 9.26: [TNUM](#) Keyword Data Description

See also the [TRACER](#) keyword in the [PROPS](#) section and the [TBLK](#) keyword in the [SOLUTION](#) section.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

Example

First define four passive tracers one for a free gas, one for dissolved gas, one for oil and one to track the water.

```
--
--      DEFINE TRACER NAMES
--
--      TRACER      TRACER
--      NAME        PHASE
--      -----
TRACER
      'GCG'         'GAS'           / GAS CAP GAS
      'DGS'         'GAS'           / DISOLVED GAS
      'OIL'         'OIL'           / OIL
      'WAT'         'WAT'           / WAT
/
```

Given a 100 x 100 x 5 grid with *DISGAS* activated in the *RUNSPEC* section, then the following *TNUM* keywords define the various tracer regions given that NTTRVD equals four on the *EQLDIMS* keyword in the *RUNSPEC* section.

```
--
--      DEFINE PASSIVE TRACER CONCENTRATION REGIONS
--
--      TNUMFGCG
--      1000*1
--      1000*2
--      1000*2
--      1000*2
--      1000*2
/
--      TNUMSDGS
--      1000*1
--      1000*1
--      1000*1
--      1000*1
--      1000*1
/
--      TNUMFOIL
--      1000*3
--      1000*3
--      1000*3
--      1000*3
--      1000*3
/
--      TNUMFWAT
--      1000*4
--      1000*4
--      1000*4
--      1000*4
--      1000*4
/
```

The keyword name is derived from the *TNUM* keyword, plus either F or S, plus the tracer name declared in the *TRACER* keyword. For example for the gas cap (free gas) this would be *TNUM*+F+*GAS* to give the TNUMFGAS keyword. And for the dissolved (solution) gas this would be *TNUM*+S+DGS resulting in the TNUMSDGS keyword.

9.3.70 TRKPF – DEFINE PARTITIONED TRACER REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **TRKPF** keyword defines the regions associated with the series of partition tracers and the partitioning tables allocated to grid blocks in the model, for when the Partitioned Tracer option has been enabled by the **PARTTRAC** keyword in the **RUNSPEC** section. The maximum number of tracers for each phase are declared on the **TRACERS** keyword in the **RUNSPEC** section. Unlike other keywords, the **TRKPF** keyword must be concatenated with the name of the tracer declared by **TRACER** keyword in the **PROPS** section. Table 9.27 outlines the format of the **TRKPF** keyword name.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	TRKPF	A five letter character string equal to TRKPF that is the root keyword name for this data set array.	None
2	NAME	A three letter character string defining the tracer's name, as declared by the TRACER keyword, which is concatenate to TRKPF to given the full name of the keyword Note it is best to void names beginning with the letters F, S, and T as these names may great naming issues in post-processing software.	None

Table 9.27: **TRKPF** Keyword Name Format

Following the declaration of the full keyword name, **TRKPFNAME**, the keyword is followed by the data as outlined below.

No.	Name	Description	Default
1	TRKPFREG	TRKPFREG defines an array of positive integers assigning a grid cell to a particular tracer table region. The maximum number of TRKPFREG regions is set by the NTTRVD variable on the EQLDIMS keyword in the RUNSPEC section.	1

Notes:

- 1) The number of entries should correspond to the **NX x NY x NZ** parameters on the **DIMENS** keyword in the **RUNSPEC** section, unless the **BOX** keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the **BOX** statement.
- 2) If a cell is not assigned a **TRKPFREG** region then the default value will be used.
- 3) The keyword is terminated by a "/".

Table 9.28: **TRKPF** Keyword Data Description

See also the **TRACER** and **TRACERKP** keywords in the **PROPS** section and the **TBLK** keyword in the **SOLUTION** section.

Example

First define one mult-partitioned tracer for the water phase.

```
--
--      DEFINE TRACER NAMES
--
--      TRACER      TRACER      TRACER      PARTITION      NUM      ADSOR
--      NAME        PHASE      VOLUME     PHASE            K(P)      PHASE
--      -----
TRACER      'WAT'      'WAT'      1*           MULT           2          ALL      / WAT
/
```

Then for a given a 100 x 100 x 5 grid assign the partitioned tracer regions and K(P) tables, based on two regions.

```
--
--      DEFINE PARTITIONED TRACER REGIONS
--
TRKPFWAT
    1000*1
    1000*1
    1000*2
    1000*2
    1000*2
/
```

The keyword name is derived from the [TRKPF](#) keyword, plus the tracer name declared in the [TRACER](#) keyword, in this case the keyword name is TRKPFWAT.

9.3.71 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

9.3.72 WH2NUM – DEFINE WAG HYSTERESIS SATURATION TABLE REGION NUMBERS (TWO PHASE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [WH2NUM](#) keyword defines the two phase Water-Alternating-Gas (“WAG”) hysteresis tables (relative permeability and capillary pressure tables) region numbers for each grid block, for when the hysteresis option has been activated by the [WAGHYSTR](#) variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure in a grid block. Note that this keyword if the two phase water relative permeabilities WAG option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	WH2NUM	WH2NUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of WH2NUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	Taken from cell allocated SATNUM

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [WH2NUM](#) region then the default value will be used.
- 3) The keyword is terminated by a “/”.

Table 9.29: [WH2NUM](#) Keyword Description

Example

The example below sets three [WH2NUM](#) regions for a model.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      WH2NUM      1           1*  1*   1*  1*   1*  1* / SET REGION 1
      WH2NUM      2           1   2    1   2    1   1 / SET REGION 2
      WH2NUM      3           1   2    1   2    2   2 / SET REGION 3
/
```

9.3.73 WH3NUM – DEFINE WAG HYSTERESIS SATURATION TABLE REGION NUMBERS (THREE PHASE)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [WH3NUM](#) keyword defines the three phase Water-Alternating-Gas (“WAG”) hysteresis tables (relative permeability and capillary pressure tables) region numbers for each grid block, for when the hysteresis option has been activated by the [WAGHYSTR](#) variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. The region number specifies which set of relative permeability tables ([SGFN](#), [SWFN](#), [SOF2](#), [SOF3](#), [SOF32D](#), [SGOF](#), [SLGOF](#) and [SWOF](#)) are used to calculate the relative permeability and capillary pressure in a grid block. Note that this keyword if the three phase water relative permeabilities WAG option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	WH3NUM	WH3NUM defines an array of positive integers assigning a grid cell to a particular saturation table region. The maximum number of WH3NUM regions is set by the NTSFUN variable on the TABDIMS keyword in the RUNSPEC section.	Taken from cell allocated SATNUM

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) If a cell is not assigned a [WH3NUM](#) region then the default value will be used.
- 3) The keyword is terminated by a “/”.

Table 9.30: [WH3NUM](#) Keyword Description

Example

The example below sets three [WH3NUM](#) regions for a model.

```
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      WH3NUM      1           1*  1*   1*  1*   1*  1* / SET REGION 1
      WH3NUM      2           1   2    1   2    1   1 / SET REGION 2
      WH3NUM      3           1   2    1   2    2   2 / SET REGION 3
/
```

CHAPTER 10: SOLUTION SECTION

10.1 INTRODUCTION

The [SOLUTION](#) section defines the initial conditions for the model, that is the conditions at time zero, before any flow occurs in the reservoir. As such, sufficient information from the previous sections is required for the simulator to calculate the initial saturation and pressure distributions through out the model, such that the fluids are in gravity/hydrostatic equilibrium. This means that if the simulator is run for a period of time without any production or injection then there should be no saturation or pressure changes in the model. In fact this “test” of running the simulator for a few years without any production or injection should always be performed to ensure that the model is indeed in hydrostatic equilibrium.

There are basically three methods to initialize a model:

- 1) **Equilibrium** Initialization: This is the primary method to initialize a model, and is in fact, the only manner that should be used to set the model at equilibrium conditions, unless a restart run is being initialized. Here, OPM Flow uses the previously entered PVT data and the fluid saturation tables ([SWFN](#), [SGFN](#), etc.), combined with the [EQUIL](#) keyword in the [SOLUTION](#) section to compute the capillary and fluid pressure gradients, and hence the fluid saturations and fluid densities in each cell. The [EQUIL](#) keyword basically sets the pressure at a datum depth together with fluid contacts for an equilibrium region, from which all the required grid block properties can be calculated. Note that it is not uncommon for hydrostatic equilibrium not be achieved initially, for example due to different PVT regions being in communication, aquifer properties incorrectly defined, etc. Which is why it is important to run the simulator for a few years without any production or injection to ensure that the model is indeed in hydrostatic equilibrium, and to rectify any issues.
- 2) **Enumeration** Initialization: This is the option where one defines the pressures for all the grid cells together with the cells initial gas and water saturations, the oil saturation in this case is back calculated by the simulator using $S_o = 1.0 - S_g - S_w$. Thus, in three phase runs only two phase saturations need to be entered and for two phase runs only one saturation array needs to be provided. This type of initialization is rarely used in the industry due to the complexity of ensuring that the model is in gravity equilibrium; however, the facility has been used by some companies in the past, using in-house proprietary software, to generate suitable input to the simulator. Enumeration of course can also be used to initialize the model in non-equilibrium state, similar to the restart initialization described next.
- 3) **Restart** Initialization: This is a form of enumeration initialization but in this case the condition of gravity equilibrium is not applied; in other words the model is initialized at a point in time when after the model has produced or injected fluids. This type of initialization is very common, as it enables the run to “restart” from a previously run case. For example, once a model has been history matched, all the prediction cases can be “restarted” from the end of the final history match run, as oppose to running the cases from the start of the history match to the end of the predication cases. In some fields, the reservoir may not be in hydrostatic equilibrium, for example, due to a pressure differential across the field caused by a re-charging aquifer. In these rare cases the model is initialized using the hydrostatic equilibrium approach but then run until the present day conditions match the measured present day conditions by adjusting the aquifer influx. The actual history match/prediction runs are then restarted from the resulting present day restart file.

Equilibrium Initialization uses the PVT data and saturation tables entered in the [PROPS](#) section combined with data entered on the [EQUIL](#) keyword in the [SOLUTION](#) section. As capillary forces are generally small relative to viscous forces in field scale simulations, one may expect the errors introduced by use of the drainage capillary pressure curves to simulate an imbibition process to be small. Thus, it is common practice in field scale simulations to use the capillary pressure drainage curves in initializing a model, mainly as drainage curves are routinely measured in the laboratory, whereas imbibition curve are not. The exception of course is when hysteresis is applicable or when simulating small scale systems. If the hysteresis option has been activated via HYSTER variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section, then the simulator will use the appropriate imbibition capillary pressure data entered via the saturation table keywords ([SWFN](#), [SGFN](#), etc.) in determining the fluid saturation.

The initial vertical distribution of fluids will be governed by the fluid densities and capillary forces, with a gas zone, a gas-oil transition zone, an oil zone, an oil-water transition zone, and finally the water zone within a hydrostatic unit or region. Not all zones may be present in a hydrostatic unit, for example in dry gas reservoirs there will be only gas, gas-water transition, and water zones. This is fundamental to construct given the density of the in situ phases. However, there are rare exceptions to this principle. When there is an exception, the more likely explanation is the various fluids are not in the same hydrostatic unit.

In the methodology outline below, it has been assumed that the initial oil pressure has been specified at some depth, z_{init} within the oil zone. Note it is not necessary that the pressure is specified within the oil zone, but it makes describing the analysis somewhat simpler. Thus we have:

$$p = p_{init}^{oil} \text{ at } z = z_{init} \quad (10.1)$$

The oil pressure at any other point in the oil column can be calculated as shown below:

$$p_{oil} = p_{init}^{oil} + \gamma^{oil} (z - z_{init}) \quad (10.2)$$

where γ^{oil} is the oil pressure gradient and z is the given depth. We can then calculate the oil pressure at the Water-Oil Contact ("WOC"), that is p_{woc}^{oil} from:

$$p_{woc}^{oil} = p_{init}^{oil} + \gamma^{oil} (z_{woc} - z_{init}) \quad (10.3)$$

Employing the definition of water-oil capillary pressure, one can write an expression for the water pressure at the water-oil contact:

$$p_{woc}^{water} = p_{woc}^{oil} - p_{cwoc} \quad (10.4)$$

The water pressure at any other point in the reservoir can now be related to the water pressure at the water-oil contact as shown below:

$$p_{water} = p_{woc}^{water} + \gamma^{water} (z - z_{woc}) \quad (10.5)$$

where γ^{water} is the water pressure gradient and z is the given depth.

Similarly, for gas we have the oil phase pressure at the Gas-Oil Contact ("GOC").

$$p_{goc}^{oil} = p_{init}^{oil} + \gamma^{oil} (z_{goc} - z_{init}) \quad (10.6)$$

Again, we can use gas-oil capillary pressure to obtain the gas phase pressure at the GOC, that is p_{goc}^{gas} :

$$p_{goc}^{gas} = p_{goc}^{oil} + p_{cgoc} \quad (10.7)$$

and the gas phase pressure at any point in the reservoir is therefore determined from:

$$p_{gas} = p_{goc}^{gas} + \gamma^{gas} (z - z_{goc}) \quad (10.8)$$

where γ^{gas} is the gas pressure gradient.

Having calculated the phase pressures at all depths in the reservoir (all grid block cells in the model), we can then calculate the capillary pressures, as the difference between the oil and water pressures, and the oil and gas pressures, depending on the phases present, that is:

$$P_{cwo} = P_{oil} - P_{water} \quad (10.9)$$

$$P_{cgo} = P_{gas} - P_{oil} \quad (10.10)$$

Under the assumption that the reservoir is initially at capillary equilibrium, we now calculate the average water saturation for each grid block in the computational grid on the basis of the capillary pressure at the center of that grid block.

$$S_w = S_w \left(P_{COW_{mid-point\ of\ grid\ cell}} \right) \quad (10.11)$$

The calculation of initial gas saturations is exactly analogous to the above treatment of water saturations.

$$S_g = S_g \left(P_{COG_{mid-point\ of\ grid\ cell}} \right) \quad (10.12)$$

Note also that fluid saturations in a grid block should sum to one, that is, $S_g + S_o + S_w = 1.0$.

With grid blocks of sufficiently small size, the procedure described above will yield accurate saturation distributions for the static equilibrium situation. Unfortunately, in practice we are often forced to use grid blocks that are much larger than we would like because of computer time and storage considerations. One of the results of using such large blocks is that the saturation distributions and hence the oil- and gas-in-place volumes can sometimes be substantially in error.

This is easily demonstrated by imagining a case where the water-oil contact lies infinitesimally above the center of a 20-foot thick grid block. In this case, the procedure described above would cause the water saturation to be calculated as 1.0, whereas in actual fact it should be very nearly 0.5 ($1.0 + S_{wc}$). Hence, we would make an error approaching half the pore volume of the grid block in the oil and water in-place in that block. To overcome this we can sub-divide each grid block in the vertical direction, and calculate the saturations for the sub-blocks, with the number of sub-blocks defined via the EQLOPT3 option on the [EQLNUM](#) keyword. The simulator can then sum the sub-blocks to obtain the average grid block saturation and thus obtain a more accurate estimate of the volumes in-place. Note this is less of a concern these days due to fact that most models now use 0.5 to 1.0 metres (one to three feet) in the vertical resolution due to the improvement in computer efficiency.

There are variations to the above general methodology depending on the data entered, for example:

- 1) If the [SWATINIT](#) array, that stipulates the initial water saturation for each cell, has been entered in the [PROPS](#) section then the capillary pressure data entered on the saturation tables are re-scaled to honor the [SWATINIT](#) array whenever possible, and the re-scaled capillary pressure data are used in the equilibrium initialization calculations. Thus, each cell will have its own scaling of the capillary pressure that needs to be taken into account during the simulation.
- 2) For reservoirs with both oil and dissolved gas phases ([OIL](#) and [DISGAS](#) keywords in the [RUNSPEC](#) section), the dissolved gas ratio, Rs, or the saturation pressure Pb, may also be entered as a function of depth. See the [RSVD](#) and [PBVD](#) keywords in the [SOLUTION](#) section. If neither of these keywords have been entered then Rs is set to equal to the Rs value at the GOC. This also means that the datum depth on the [EQUIL](#) keyword in the [SOLUTION](#) section, must equal the GOC depth, in order for the saturated Rs to be calculated from the datum pressure and the associated PVT table.

- 3) Similarly, for reservoirs with both wet gas and vaporized oil ([GAS](#) and [VAPOIL](#) keywords in the [RUNSPEC](#) section), the dissolved vaporized oil (more commonly referred to as condensate) gas ratio, R_v , or the dew-point pressure P_d , may also be entered as a function of depth. See the [RVVD](#) and [PDVD](#) keywords in the [SOLUTION](#) section. If neither of these keywords have been entered then R_v is set to equal to the R_v value at the GOC. Again, this also means that the datum depth must set to the GOC depth in order for the saturated R_v to be calculated from the datum pressure and the associated PVT table.

In OPM Flow²⁹⁷ initial conditions are calculated in a similar manner by using the pressure at a reference depth, the depths of the water-oil and oil-gas contacts, and the capillary pressures at those contact depths. It then solves the following ordinary differential equations:

$$\begin{aligned}\frac{dp_{oil}}{dz} &= \rho_{oil}(z, p_{oil})g \\ \frac{dp_{gas}}{dz} &= \rho_{gas}(z, p_{gas})g \\ \frac{dp_{wat}}{dz} &= \rho_{wat}(z, p_{wat})g\end{aligned}\tag{10.13}$$

where

ρ_i	= the oil, gas, water density,
z	= the depth,
p_i	= the oil, gas, water pressure, and
g	= the gravity term.

The equations are solved numerically using a 4th-order Runge-Kutta²⁹⁸ and ²⁹⁹ method, with the order of the phases decided by the location of the datum depth. First, the simulator solves for the phase in whose zone the datum is located, using the datum pressure to bind the pressure solution. Then the simulator solves for phases corresponding to the zones above and below. To tie the phase pressures, the simulator uses the already computed phase pressure(s) evaluated at the zone contact depth and the input capillary pressures.

If the initial water saturations have been specified using the [SWATINIT](#) keyword then the same approach is used for the water saturation but the capillary pressure is scaled to match the specified initial water saturation. Each cell will thus have its own scaling of the capillary pressure that needs to be taken into account during the simulations. Initial fluid properties (GOR, formation volume factors, etc.) are calculated from the cell equilibrium pressure and the entered fluid property tables, as described earlier.

²⁹⁷ Rasmussen, A.F., Sandve, T.H., Bao, K., Lauser, A., Hove, J., Skaflestad, B., Klöfkorn, R., Blatt, M., Rustad, A.B., Sævareid, O. et al. [2019]. The Open Porous Media Flow Reservoir Simulator. <https://doi.org/10.1016/j.camwa.2020.05.014>.

²⁹⁸ Runge, C. D. T. (1895), "Über die numerische Auflösung von Differentialgleichungen", *Mathematische Annalen*, Springer, 46 (2): 167–178, doi:10.1007/BF01446807.

²⁹⁹ Kutta, M. (1901), "Beitrag zur näherungsweisen Integration totaler Differentialgleichungen", *Zeitschrift für Mathematik und Physik*, 46: 435–453.

10.2 DATA REQUIREMENTS

10.2.1 EQUILIBRIUM INITIALIZATION

For the standard Equilibrium Initialization the [EQUIL](#) keyword in the [SOLUTION](#) section is used to initialize the model. The keyword contains the datum depth and pressure, oil-water and gas-oil contacts, the capillary pressure values at the fluid contacts, as well as various equilibrium initialization options, including how the dissolve gas and vaporized oil should be calculated, and if a refined in-place volume calculation should be performed. There should be one record for each equilibration region with the number of regions defined by the NTEQUL parameter on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.

Note that if a fluid contact is not present in the reservoir, for example in two phase runs containing only gas and water, the water oil-water contact on the [EQUIL](#) keyword should be set to the gas-water contact. Also, in cases where there is no gas cap (or free gas) then gas-oil should be set to a value shallower than the top of the reservoir, and for three phases cases where there is initially no oil zone, as for a gas condensate field for example, the gas-oil contact should be set to the same depth as water-oil contact. Table 10.1 outlines the minimum set of keywords required by OPM Flow in order for the simulator to initialize the model For the standard Equilibrium Initialization option.

No.	SOLUTION Keyword	Equilibrium Initialization Description	Type Of Initialization
1	EQUIL	EQUIL – Define the Equilibration Initialization Data - this is the main keyword for the Equilibration initialization that sets the datum depth, the pressure and fluid contacts for an equilibrium region.	Equilibration
2	PBVD	PBVD – Equilibration Bubble-Point versus Depth Tables - used for Live Black-Oil equilibration regions to define the saturation pressure versus depth relationship.	Equilibration
3	PDVD	PDVD – Define Equilibration Dew-Point versus Depth Tables - used for Wet Gas equilibration regions to define the saturation pressure versus depth relationship.	Equilibration
4	RSVD	RSVD – Equilibration Dissolved Gas-Oil Ratio (Rs) versus Depth Tables - used for Live Black-Oil equilibration regions to define the gas-oil ratio versus depth relationship.	Equilibration
5	RVVD	RVVD – Equilibration Vaporized Oil-Gas Ratio (Rv) versus Depth Tables - use for Wet Gas equilibration regions to declare the condensate-gas ratio versus depth relationship.	Equilibration
6	SALTVD	SALTVD – Equilibration Salt Concentration versus Depth Tables - this is a standard commercial simulator keyword used with the Brine model, but may also be used with OPM Flow's Vaporized Water and Salt Precipitation models.	Equilibration
7	THPRES	THPRES – Define Equilibration Region Threshold Pressures	Equilibration Enumeration
		Vaporized Water Model (OPM Flow Specific Keywords)	
1	RVWVD	RVWVD – Equilibration Vaporized Water-Gas Ratio (Rvw) versus Depth Tables	Equilibration
2	SALTPVD	SALTPVD – Initial Precipitated Salt Volume Fraction versus Depth Tables	Equilibration

No.	SOLUTION Keyword	Equilibrium Initialization Description	Type Of Initialization
3	SALTVD	<i>SALTVD – Equilibration Salt Concentration versus Depth Tables</i> - this is a standard commercial simulator keyword used with the Brine model, but may also be used with OPM Flow's Vaporized Water and Salt Precipitation models.	Equilibration
Notes: 1) Only the Equilibration and Restart initialization type of initialization are used in practice. 2) For the Equilibration option the PBVD, PDVD, RSVD, and RSVVD keywords may be optional if the datum depth on the EQUIL keyword is set equal to the gas-oil contact ("GOC"), otherwise one of these keywords must be used to set the initial RS ("GOR") or RV ("GCR") values for the equilibrium region being defined. 3) For Live Black-Oil equilibrium regions if the datum depth is not equal to the GOC then one can either use the PBVD keyword to set the saturation pressure (bubble-point) versus depth, or the RSVD keyword to set the GOR versus depth. Similarly for Wet Gas equilibrium regions, one can either use the PDVD keyword to set the saturation pressure (dew-point) versus depth, or RVVD keyword to set the CGR versus depth. 4) The THPRES keyword is entirely optional but is commonly used to isolate various equilibrium regions.			

Table 10.1: Minimum Set of SOLUTION Equilibrium Keywords Required by OPM Flow

The minimum set of SOLUTION keywords is generally for three phases runs, that is where the OIL, DISGAS, GAS, VAPOIL, and/or WATER keywords in the RUNSPEC section have been invoked. If additional phases or options have been invoked in the RUNSPEC section, then additional SOLUTION keywords may be required in order to initialize the model. For example, if the BRINE keyword in the RUNSPEC section has been used to activate the BRINE model then the SALTVD keyword in the SOLUTION section may or may not be used to set the initial salt concentration versus depth relationship.

10.2.2 ENUMERATION INITIALIZATION

Enumeration Initialization is based on user supplying all the required data for the simulator to determine the pressures, fluid saturations, and fluid properties for each cell in the model. This is similar to entering the grid property array data in the GRID section, in that all the data is required for all the cells in the model.

To define the cell pressures for all the grid blocks, the PRESSURE keyword in the SOLUTION section should be employed. There is also the PRVD keyword in the SOLUTION section that can be used to define a pressure versus depth relationship to enable the simulator to calculate the cell pressures instead of entering the data via the PRESSURE keyword; however, this keyword is not implemented in OPM Flow.

For the fluid saturations, it is only necessary to enter the saturations sufficient to define the system, so for a two phase system it is only necessary to define one phase, normally the water phase, as the other phase will be calculated by the simulator. Similarly, for three phases only two phases must be entered with the water phase always normally entered, and the other phase dependent on the type of reservoir. The data is entered using the SGAS, SOIL, and SWAT keywords in the SOLUTION section.

Finally the initial fluid property data needs to be defined based on the fluid phases active in the model. For oil reservoirs either the RS keyword that defines a cell's initial gas-oil ratio, or the PBUB keyword that defines the oil's initial bubble point pressure for each cell are required. For gas reservoirs, either the RV keyword that defines vaporized oil-gas ratio (or condensate gas-oil ratio), or the PDEW keyword that defines the initial dew point pressure for each cell must be entered. Table 10.2 outlines the minimum set of keywords required by OPM Flow in order for the simulator to initialize the model. For the Enumeration Initialization option - all the keywords are in the SOLUTION section.

No.	SOLUTION Keyword	Enumeration Initialization Description	Type Of Initialization
1	PRESSURE	<i>PRESSURE – Define the Initial Equilibration Pressures for All Grid Blocks.</i> This keyword must be present for all Enumeration equilibration regions.	Enumeration
2	PBUB	<i>PBUB – Define the Initial Equilibration Bubble-Point Pressure for All Grid Blocks</i> - used for Live Black-Oil equilibration regions to define the saturation pressure for each grid cell.	Enumeration
3	PDEW	<i>PDEW – Define the Initial Equilibration Dew-Point Pressure for All Grid Blocks</i> - used for Wet Gas equilibration regions to set the saturation pressure for each cell.	Enumeration
4	RS	<i>RS – Define the Initial Equilibration GOR (Rs) for All Grid Blocks</i> - used for Live Black-Oil equilibration regions to define the gas-oil ratio for each grid blocks.	Enumeration
5	RV	<i>RV – Define the Initial Equilibration CGR (Rv) for All Grid Blocks</i> - use for Wet Gas equilibration regions to set the condensate-gas ratio for each cell.	Enumeration
6	SALT	<i>SALT – Define the Initial Equilibration Salt Concentration for All Grid Blocks</i> - used to set the initial salt concentration in the Brine model and OPM Flow's Salt Precipitation model.	Enumeration
7	SGAS	<i>SGAS – Define the Initial Equilibration Gas Saturation for All Grid Blocks</i> - if gas is present in the model then the keyword is used to set the initial gas saturation.	Enumeration
8	SOIL	<i>SOIL – Define the Initial Equilibration Oil Saturation for All Grid Blocks</i> - if oil is present in the model then the keyword is used to set the initial oil saturation.	Enumeration
9	SWAT	<i>SWAT – Define the Initial Equilibration Water Saturation for All Grid Blocks</i> - since water is normally always present in petroleum reservoirs, this keyword is always used to define the initial water saturation. For two-phase runs, oil-water or gas-water, then it is only necessary to use this keyword, as the second phase saturations are automatically calculated by the simulator.	Enumeration
10	THPRES	<i>THPRES – Define Equilibration Region Threshold Pressures</i>	Equilibration Enumeration
		Microbial Induced Calcite Precipitation And Biofilm Model Enumeration (OPM Flow Specific Keywords)	
1	SBIOF	<i>SBIOF – Define The Initial Equilibration Biofilm Volume Fraction For All Grid Blocks</i>	Enumeration
2	SCALC	<i>SCALC – Define The Initial Equilibration Calcite Volume Fraction For All Grid Blocks</i>	Enumeration
3	SMICR	<i>SMICR – Define The Initial Equilibration Microbial Concentration For All Grid Blocks</i>	Enumeration

No.	SOLUTION Keyword	Enumeration Initialization Description	Type Of Initialization
4	SOXYG	<i>SOXYG – Define The Initial Equilibration Oxygen Concentration For All Grid Blocks</i>	Enumeration
5	SUREA	<i>SUREA – Define The Initial Equilibration Urea Concentration For All Grid Blocks</i>	Enumeration
		Vaporized Water Model (OPM Flow Specific Keywords)	
1	RVW	<i>RVW – Define the Initial Equilibration Vaporized Water in Gas Ratio for All Grid Blocks</i> - used in both the Vaporized Water model and the Salt Precipitation model.	Enumeration
2	SALT	<i>SALT – Define the Initial Equilibration Salt Concentration for All Grid Blocks</i> - this is a standard commercial simulator keyword used with the Brine model, but may also be used with OPM Flow's Vaporized Water and Salt Precipitation models.	Enumeration
3	SALTP	<i>SALTP – Define the Initial Precipitated Salt Volume Fraction for All Grid Blocks.</i>	Enumeration
		Dissolved Gas In Water Model (OPM Flow Specific Keywords)	
1	RSW	<i>RSW – Define the Initial Equilibrium Solution Gas in Water Ratio for All Grid Blocks</i> - this is a commercial compositional simulator keyword, but may also be used with OPM Flow's Dissolved Gas in Water model.	Enumeration
1) In an Enumeration type of initialization only sufficient data to define the pressures and fluid distributions are required, for example, in a two phase gas-water run it is only necessary to use the PRESSURE, RV, and either SGAS or SWAT keywords to formulate the solution. 2) The THPRES keyword is entirely optional but is commonly used to isolate various equilibrium regions. 3) The Microbial Induced Calcite Precipitation ("MICP") model is OPM Flow's implementation of <i>MICP</i> used to investigate leakage remediation. Note currently the model assumes 100% water saturation and therefore it is not necessary to use the SWAT keyword. 4) SBIOF and SMICR can be set in the <i>BIOFILM</i> model.			

Table 10.2: Minimum Set of SOLUTION Enumeration Keywords Required by OPM Flow

Again, the minimum set of SOLUTION keywords is for general three phases runs, that is where the OIL, DISGAS, GAS, VAPOIL, and/or WATER keywords in the RUNSPEC section have been invoked. If additional phases or options have been invoked, then additional SOLUTION keywords may be required in order to initialize the model.

10.2.3 RESTART INITIALIZATION

Restart Initialization is performed by the simulator by loading a previously generated RESTART file and equilibrating the model accordingly. In nearly all cases the model will not satisfy the condition of gravity equilibrium; in other words the model is initialized at a point in time after the model has produced or injected fluids.

To write a restart record to the RESTART file, either the RPTSOL or RPTRST keyword in the SOLUTION section can be used to write the initial restart record at time equal to zero. To write restart records at various time points as the simulation progresses through time, then either the RPTRST or RPTSCHED

keywords may be used. Both the RPTSOL and RPTSCHED keywords enable just the basic restart record to be written to the RESTART file, which is generally sufficient in most cases. In comparison with the dedicated RPTRST keyword, RPTRST provides additional functionality on the data to be written to the RESTART file and is therefore the preferred keyword to use. For example to request the standard restart data be written out every month using the RPTRST keyword, one would use the following:

```
--
--      RESTART CONTROL BASIC = 4 (YEARLY) 5 (MONTHLY)
--
RPTRST
      BASIC=5
```

Subsequently one of the monthly restart points can be used to restart the run.

The keywords used to create restart points in various sections of the input deck and to initialize the model from a RESTART file are shown in Table 10.3.

No.	Keyword	Restart Creation Description	Keyword Section
1	RPTRST	RPTRST – Define Data to be Written to the RESTART File	SOLUTION
2	RPTSOL	RPTSOL – Define SOLUTION Section Reporting	
3	RPTRST	RPTRST – Define Data to be Written to the RESTART File	SCHEDULE
4	RPTSCHED	RPTSCHED – Define SCHEDULE Section Reporting	
No.	Keyword	Restart Initialization Description	Keyword Section
5	RESTART	RESTART – Restart Run From an Existing Restart File - this keyword is used to initialize the model from a previously run OPM Flow run or from the commercial simulator. Note that due to the complexities of the RESTART file, OPM Flow may not always be able to restart from the commercial simulators RESTART file.	SOLUTION
Notes: - OPM Flow can only restart runs from a RESTART file, and the commercial simulator's SAVE file format is not supported. - The THPRES keyword is not required as the data is included in the RESTART file. - SCHEDULE section keywords <u>not</u> stored in the RESTART and must therefore remain in the input deck if required include: - GLOBAL section keywords: COLUMNS, DEBUG, ECHO, EXTRAPMS, FORMFEED, INCLUDE, MESSAGES, NOECHO, NOWARN, and the OPTIONS keyword. - SCHEDULE section keywords: PIMULTAB, RPTSCHED, RPTRST, SCDPTAB, SCDATAB, TUNING, VFPPROD, and VFPINJ keywords.			

Table 10.3: SOLUTION Restart Initialization Keywords Required by OPM Flow

In the input deck that is being employed to restart the run, all the equilibration keywords (EQUIL, RSVD, etc.) or the enumeration equilibration keywords (PRESSURE, SGAS, SOIL, SWAT, etc.) in the SOLUTION section used to initialize the model should be deleted. Secondly, it is good practice to insert the SKIPREST keyword at the very beginning of the SCHEDULE section. The SKIPREST keyword causes the simulator to

only read in data it requires for restarting the run up to the RESTART point. This is because certain keywords always need to be present in a restart run in the SCHEDULE section, as the data is not stored on the RESTART file, for example the VFP tables (VFPPROD and VFPINJ keywords in the SCHEDULE section). The SKIPREST keyword automatically processes the input deck and reads the required data.

10.2.4 SECTION KEYWORDS

A complete list of SOLUTION keywords in alphabetic order is shown in Table 10.4 together with a generalized Topic column that classifies the functionality of the keyword. Note that not all keywords and features listed in Table 10.4 are implemented in OPM Flow. Cells not colored in the No. column indicate the keyword is supported by OPM Flow, cells colored gray indicate that the keyword is not applicable, and finally, cells colored in orange indicate keywords that are not currently supported by OPM Flow.

No.	RUNSPEC Keyword	Description	Topic
1	ADD	<i>ADD – Add a Constant to a Specified Array.</i>	Input
2	ADDREG	<i>ADDREG – Add a Constant to an Array based on a Region Number.</i>	Input
3	APIVD	<i>APIVD – Equilibration Oil API Gravity versus Depth Tables.</i>	API
4	AQANCONL	<i>AQANCONL – Define Analytical Connections to a LGR Grid.</i>	Aquifer
5	AQANNC	<i>AQANNC – Define Analytic Aquifer Non-Neighbor Connections.</i>	Aquifer
6	AQANTRC	<i>AQANTRC – Define Analytic Aquifer Initial Tracer Concentrations.</i>	Aquifer
7	AQUALIST	<i>AQUALIST – Define An Analytic Aquifer Name to Aquifer Numbers .</i>	Aquifer
8	AQUANCON	<i>AQUANCON – Define Analytical Connections to the Grid.</i>	Aquifer
9	AQUCHGAS	<i>AQUCHGAS – Define Constant Pressure Gas Analytical Aquifer Properties.</i>	Aquifer
10	AQUCHWAT	<i>AQUCHWAT – Define Constant Pressure Water Analytical Aquifer Properties.</i>	Aquifer
11	AQUCON	<i>AQUCON – Define Numerical Aquifer Connections to the Grid.</i>	Aquifer
12	AQUCT	<i>AQUCT – Define Carter-Tracy Analytical Aquifers.</i>	Aquifer
13	AQUFET	<i>AQUFET – Define Fetkovich Analytical Aquifer and Connections .</i>	Aquifer
14	AQUFETP	<i>AQUFETP – Define Fetkovich Analytical Aquifers .</i>	Aquifer
15	AQUFLUX	<i>AQUFLUX – Define Constant Flux Analytical Aquifer.</i>	Aquifer
16	BC	<i>BC – Define Boundary Conditions.</i> This is an OPM Flow specific keyword defines the boundary conditions for the model, and can be used to set boundary conditions for when external influx or efflux volumes are influencing the reservoir pressure and production history.	Equilibration
17	BOUNDARY	<i>BOUNDARY – Define a Boundary Box for Printing.</i>	Output
18	BOX	<i>BOX – Define a Range of Grid Blocks to Enter Property Data.</i>	Input
19	COLUMNS	<i>COLUMNS – Define Input File Column Margins.</i>	Input
20	COPY	<i>COPY – Copy Array Data to Another Array.</i>	Input
21	COPYREG	<i>COPYREG – Copy an Array to Another Array based on a Region Number.</i>	Input

No.	RUNSPEC Keyword	Description	Topic
22	DATUM	<i>DATUM – Define the Datum Depth for the Model.</i>	Output
23	DATUMR	<i>DATUMR – Define Datum Depths for the FIPNUM Regions.</i>	Output
24	DATUMRX	<i>DATUMRX – Define Datum Depths for the FIP Allocated Regions .</i>	Output
25	DEBUG	<i>DEBUG – Define the Debug Data to be Printed to File.</i>	Output
26	DYNAMICR	<i>DYNAMICR – Start of Dynamic Region Parameter Definition.</i>	Input
27	ECHO	<i>ECHO – Activate Echoing of User Input Files to the Print File.</i>	Output
28	END	<i>END – Define the End of the Input File.</i>	Input
29	ENDBOX	<i>ENDBOX – Define the End of the BOX Defined Grid.</i>	Input
30	ENDDYN	<i>ENDDYN – End of Dynamic Region Parameter Definition.</i>	Input
31	ENDFIN	<i>ENDFIN – End the Definition of a Local Grid Refinement.</i>	LGR
32	ENDINC	<i>ENDINC – Define the End of an Include File.</i>	Input
33	ENDSKIP	<i>ENDSKIP – Deactivate Skipping of Keywords and Input Data.</i>	Input
34	EQUALREG	<i>EQUALREG – Sets an Array to a Constant by Region Number.</i>	Input
35	EQUALS	<i>EQUALS – Sets a Specified Array to a Constant.</i>	Input
36	EQUIL	<i>EQUIL – Define the Equilibration Initialization Data.</i>	Equilibration
37	EXTRAPMS	<i>EXTRAPMS – Activate Extrapolation Warning Messages.</i>	Output
38	FILEUNIT	<i>FILEUNIT – Activate Unit Consistency Checking.</i>	Input
39	FORMFEED	<i>FORMFEED – Defined the Print File Form-Feed Character.</i>	Output
40	GASCONC	<i>GASCONC – Define the Initial Equilibration Coal Gas Concentration for All Grid Blocks.</i>	CBM
41	GASSATC	<i>GASSATC – Define the Initial Equilibration Saturated Coal Gas Concentration for All Grid Blocks.</i>	CBM
42	GCVD	<i>GCVD – Define Equilibration Coal Gas Concentration versus Depth Tables.</i>	CBM
43	GETDATA	<i>GETDATA – Load and Assign Data Array from INIT or RESTART Files.</i>	Input
44	GETGLOB	<i>GETGLOB – Activate Loading of Global Grid Restart Data Option.</i>	Input
45	GI	<i>GI – Define the Initial Equilibration Gi Values for All Grid Blocks.</i>	Enumeration
46	HMAQUCT	<i>HMAQUCT – History Match Carter-Tracy Aquifer Gradient Parameters.</i>	Gradient
47	HMAQUFET	<i>HMAQUFET – History Match Fetkovich Aquifer Gradient Parameters.</i>	Gradient
48	HMMLCTAQ	<i>HMMLCTAQ – History Match Carter-Tracy Aquifer Gradient Multipliers.</i>	Gradient
49	HMMLFTAQ	<i>HMMLFTAQ – History Match Fetkovich Aquifer Gradient Multipliers.</i>	Gradient

No.	RUNSPEC Keyword	Description	Topic
50	HMLTWCN	<i>HMLTWCN – History Match Well Connection and Skin Multipliers.</i>	Gradient
51	HMWELCON	<i>HMWELCON – History Match Well Connection and Skin Parameters.</i>	Gradient
52	IMPORT	<i>IMPORT – Import Grid File Data at the Current Position.</i>	Input
53	INCLUDE	<i>INCLUDE – Load Another Data File at the Current Position.</i>	Input
54	MESSAGE	<i>MESSAGE – Output User Message.</i> The MESSAGE keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files.	Output
55	MESSAGES	<i>MESSAGES – Define Message Print Limits and Stop Limits.</i> The MESSAGES keyword defines the print and stops levels for various messages.	Output
56	MULTIPLY	<i>MULTIPLY – Multiply a Specified Array by a Constant.</i>	Input
57	MULTIREG	<i>MULTIREG – Multiply an Array by a Constant based on a Region Number.</i>	Input
58	NOECHO	<i>NOECHO – Deactivate Echoing of User Input Files to the Print File.</i>	Output
59	NOHMD	<i>NOHMD – Deactivate History Match Gradient Derivative Calculations.</i>	Gradient
60	NOHMO	<i>NOHMO – Deactivate History Match Gradient Derivative Calculations (Alias).</i>	Gradient
61	NOWARN	<i>NOWARN – Deactivate Warning Messages.</i>	Output
62	OILAPI	<i>OILAPI – Define the Initial Equilibration Oil API for All Grid Blocks.</i>	API
63	OPERATE	<i>OPERATE – Define Mathematical Operations on Arrays.</i>	Input
64	OPERATER	<i>OPERATER – Define Mathematical Operations on Arrays by Region.</i>	Input
65	OUTSOL	<i>OUTSOL – Define Data to be Written to the RESTART File (Retired).</i>	Output
66	PBUB	<i>PBUB – Define the Initial Equilibration Bubble-Point Pressure for All Grid Blocks.</i>	Enumeration
67	PBVD	<i>PBVD – Equilibration Bubble-Point versus Depth Tables.</i>	Equilibration
68	PDEW	<i>PDEW – Define the Initial Equilibration Dew-Point Pressure for All Grid Blocks.</i>	Enumeration
69	PDVD	<i>PDVD – Define Equilibration Dew-Point versus Depth Tables.</i>	Equilibration
70	PRESSURE	<i>PRESSURE – Define the Initial Equilibration Pressures for All Grid Blocks.</i>	Enumeration
71	PRVD	<i>PRVD – Define the Initial Equilibration Pressures versus Depth.</i>	Enumeration
72	PYEND	<i>PYEND – End the Definition of a PYINPUT Section.</i> This keyword is an OPM Flow specific keyword for the simulator's scripting facility using the standard Python interpreter.	Action

No.	RUNSPEC Keyword	Description	Topic
73	PYINPUT	<i>PYINPUT – Define the Start of a PYINPUT Section.</i> This keyword is an OPM Flow specific keyword for the simulator's scripting facility using the standard Python interpreter.	Action
74	RAINFALL	<i>RAINFALL – Constant Flux Aquifer Rainfall Flux by Month.</i>	Aquifer
75	RBEDCONT	<i>RBEDCONT – Define River Grid Block Contact Area versus Depth.</i>	Rivers
76	REFINE	<i>REFINE – Start the Definition of a Local Grid Refinement.</i>	LGR
77	RESTART	<i>RESTART – Restart Run From an Existing Restart File.</i>	Restart
78	RIVERSYS	<i>RIVERSYS – Define River System (Branch Structure and Boundary Conditions).</i>	Rivers
79	RPTRST	<i>RPTRST – Define Data to be Written to the RESTART File.</i>	Output
80	RPTSOL	<i>RPTSOL – Define SOLUTION Section Reporting.</i>	Output
81	RS	<i>RS – Define the Initial Equilibration GOR (Rs) for All Grid Blocks.</i>	Enumeration
82	RSVD	<i>RSVD – Equilibration Dissolved Gas-Oil Ratio (Rs) versus Depth Tables.</i>	Equilibration
83	RSW	<i>RSW – Define the Initial Equilibrium Solution Gas in Water Ratio for All Grid Blocks.</i> This is a commercial compositional simulator keyword, but may also be used with OPM Flow's Dissolved Gas in Water model.	Enumeration
84	RTEMP	<i>RTEMP – Define the Initial Reservoir Temperature for the Model.</i>	Thermal
85	RTEMPA	<i>RTEMPA – Define the Initial Reservoir Temperature for the Model.</i>	Thermal
86	RTEMPVD	<i>RTEMPVD – Define the Initial Reservoir Temperature versus Depth Tables.</i>	Thermal
87	RV	<i>RV – Define the Initial Equilibration CGR (Rv) for All Grid Blocks.</i>	Enumeration
88	RVVD	<i>RVVD – Equilibration Vaporized Oil-Gas Ratio (Rv) versus Depth Tables.</i>	Equilibration
89	RVW	<i>RVW – Define the Initial Equilibration Vaporized Water in Gas Ratio for All Grid Blocks.</i> This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by the VAPWAT in the RUNSPEC section.	Water Vaporization
90	RVWVD	<i>RVWVD – Equilibration Vaporized Water-Gas Ratio (Rvw) versus Depth Tables</i> This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by the VAPWAT in the RUNSPEC section.	Water Vaporization
91	SALT	<i>SALT – Define the Initial Equilibration Salt Concentration for All Grid Blocks.</i>	Brine

No.	RUNSPEC Keyword	Description	Topic
		A standard commercial simulator keyword used with the Brine Model, but may also be used with OPM Flow's Vaporized Water and Salt Precipitation models.	
92	SALTP	<i>SALTP – Define the Initial Precipitated Salt Volume Fraction for All Grid Blocks.</i> This is an OPM Flow specific keyword for the simulator's Salt Precipitation Model that is activated by the PRECSALT keyword and declaring that vaporized water is present in the run via the VAPWAT in the RUNSPEC section.	Water Vaporization
93	SALTPVD	<i>SALTPVD – Initial Precipitated Salt Volume Fraction versus Depth Tables.</i> This is an OPM Flow specific keyword for the simulator's Salt Precipitation Model that is activated by the PRECSALT keyword and declaring that vaporized water is present in the run via the VAPWAT in the RUNSPEC section.	Water Vaporization
94	SALTREST	<i>SALTREST – Define the Restart Salt Concentration for All Grid Blocks.</i>	Brine
95	SALTVD	<i>SALTVD – Equilibration Salt Concentration versus Depth Tables.</i> A standard commercial simulator keyword used with the Brine model, but may also be used with OPM Flow's Vaporized Water and Salt Precipitation models.	Brine
96	SBIOF	<i>SBIOF – Define The Initial Equilibration Biofilm Volume Fraction For All Grid Blocks.</i>	MICP/Biofilm
97	SCALC	<i>SCALC – Define The Initial Equilibration Calcite Volume Fraction For All Grid Blocks.</i>	MICP
98	SCVD	<i>SCVD – Define Equilibration Coal Solvent Concentration versus Depth Tables.</i>	CBM
99	SFOAM	<i>SFOAM – Define the Initial Equilibration Foam Concentration for All Grid Blocks.</i>	Enumeration
100	SGAS	<i>SGAS – Define the Initial Equilibration Gas Saturation for All Grid Blocks.</i>	Enumeration
101	SKIP	<i>SKIP – Activate Skipping of All Keywords and Input Data.</i>	Input
102	SKIP100	<i>SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator.</i>	Input
103	SKIP300	<i>SKIP300 – Activate Skipping of Keywords by Compositional Simulator.</i>	Input
104	SMICR	<i>SMICR – Define The Initial Equilibration Microbial Concentration For All Grid Blocks.</i>	MICP/Biofilm
105	SOIL	<i>SOIL – Define the Initial Equilibration Oil Saturation for All Grid Blocks.</i>	Enumeration
106	SOLUTION	<i>SOLUTION – Define the Start of the SOLUTION Section of Keywords.</i>	Input
107	SOLVCONC	<i>SOLVCONC – Define the Initial Coal Solvent Concentration for All Grid Blocks.</i>	CBM

No.	RUNSPEC Keyword	Description	Topic
108	SOLVFRAC	<i>SOLVFRAC – Define the Initial Gas Solvent Fraction for All Grid Blocks</i>	Solvent
109	SOXYG	<i>SOXYG – Define The Initial Equilibration Oxygen Concentration For All Grid Blocks.</i>	MICP
110	SPOLY	<i>SPOLY – Define the Initial Equilibration Polymer Concentration for All Grid Blocks.</i>	Polymer
111	SPOLYMW	<i>SPOLYMW – Define The Initial Equilibration Polymer Molecular Weights For All Grid Blocks.</i> This is an OPM Flow specific keyword for the simulator's Polymer Molecular Weight Transport option that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure.	Polymer
112	SSOL	<i>SSOL – Define the Initial Equilibration Solvent Saturation for All Grid Blocks.</i>	Solvent
113	SUREA	<i>SUREA – Define The Initial Equilibration Urea Concentration For All Grid Blocks.</i>	MICP
114	SURF	<i>SURF – Define the Initial Equilibration Surfactant Concentration for All Grid Blocks.</i>	Surfactant
115	SWAT	<i>SWAT – Define the Initial Equilibration Water Saturation for All Grid Blocks.</i>	Enumeration
116	TBLK	<i>TBLK – Define Tracer Initial Grid Block Concentrations.</i>	Tracers
117	TEMPI	<i>TEMPI – Define the Initial Temperature Values for All Cells.</i>	Thermal
118	TEMPVD	<i>TEMPVD – Define the Initial Reservoir Temperature versus Depth Tables.</i>	Thermal
119	THPRES	<i>THPRES – Define Equilibration Region Threshold Pressures.</i>	Equilibration
120	TVDP	<i>TVDP – Define the Initial Equilibration Tracer Saturation versus Depth Functions.</i>	Tracers
121	VAPPARS	<i>VAPPARS – Oil Vaporization Parameters.</i>	Phases
122	VISDATES	<i>VISDATES – Define External Reservoir Geo-Mechanics VISAGE Stress Dates.</i>	Geo-Mechanics
123	VISOPTS	<i>VISOPTS – Define External Reservoir Geo-Mechanics VISAGE Options.</i>	Geo-Mechanics
124	WARN	<i>WARN – Activate Warning Messages.</i>	Output
Notes: - Note that not all keywords and features listed above are implemented in OPM Flow. Cells not colored in the No. column indicate the keyword is supported, cells colored gray indicate that the keyword is not applicable, and finally, cells colored in orange indicate keywords that are not supported by OPM Flow. - For the Topic column the definition of the entries is defined as follows:			

No.	RUNSPEC Keyword	Description	Topic
1)	Action:	Keywords associated with the ACTIONX or OPM Flow's Python facility.	
2)	API:	API option equilibration keywords.	
3)	Biofilm:	BIOFILM option equilibration keywords.	
4)	Brine:	BRINE option equilibration keywords.	
5)	CBM:	COAL BED METHANE equilibration keyword (not implemented in OPM Flow).	
6)	Enumeration:	Enumeration equilibration keywords.	
7)	Equilibration:	Equilibration keywords.	
8)	Geo-Mechanics:	GEO_MECHANICS equilibration keywords (this option is not available in OPM Flow).	
9)	Gradient:	GRADIENT equilibration keywords., used in history matching (this option is not available in OPM Flow).	
10)	Input:	Input control and options.	
11)	LGR:	LOCAL GRID REFINEMENT equilibration keywords (not implemented in OPM Flow).	
12)	MICP:	Microbial Induced Calcite Precipitation model equilibration keywords.	
13)	Output:	Output control and options.	
14)	Phases:	Equilibration keywords related to the active phases.	
15)	Polymer:	POLYMER equilibration keywords.	
16)	Restart:	Restart equilibration keywords.	
17)	Rivers:	Keywords associated with the RIVERS facility (not implemented in OPM Flow).	
18)	Solvent:	SOLVENT equilibration keywords.	
19)	Thermal:	THERMAL equilibration keywords.	
20)	Tracers:	TRACER equilibration keywords.	
21)	Water Vaporization:	WATER VAPORIZATION equilibration keywords.	

Table 10.4: Alphabetic List of SOLUTION Keywords

Note that a number of keywords can be classified under several topics, so the Topic column should only be used as a general reference guide.

Example

The first example is taken from the Norne field, in which the model is in SI units, and the NTEQUL varibale on the EQLDIMS keyword in the RUNSPEC section has been set to five for five equilibration regions in the model.

```
--
--          DATUM    DATUM    OWC    PCOW    GOC    PCGO    RS    RV    N
--          DEPTH    PRESS    DEPTH    ----    DEPTH    ----    OPT    OPT    OPT
EQUIL
    2582.0    268.56    2692.0    0.0    2582.0    0.0    1    0    0 / C+D:Garn
    2500.0    263.41    2585.5    0.0    2500.0    0.0    2    0    0 / G:  Garn
    2582.0    269.46    2618.0    0.0    2582.0    0.0    3    0    0 / E:  Garn
    2200.0    236.92    2400.0    0.0    2200.0    0.0    4    0    0 / G:Ile-Tilje
    2585.0    268.77    2693.3    0.0    2585.0    0.0    5    0    0 / C+D+E:Ile-Ti

--
--          DEPTH    RS
--                      sm3/sm3
--          -----
RSVD
    2582.0    120.25
    2597.0    110.00
    2660.7    106.77
    2697.0    106.77
--
--          -----
    2500.0    94.50
    2510.0    94.30
    2590.0    94.10
--
--          -----
    2582.0    120.25
    2597.0    110.00
    2660.7    106.77
    2697.0    106.77
--
--          -----
    2500.0    94.50
    2510.0    94.30
    2590.0    94.10
--
--          -----
    2585.9    120.29
    2599.9    110.00
    2663.6    106.77
    2699.9    106.77
--
--          EQLNUM    EQLNUM    THPRES
--          FROM      TO        VALUE
THPRES
    1          2          0.588031
    2          1          0.588031
    1          3          0.787619
    3          1          0.787619
    1          4          7.000830
    4          1          7.000830
/
```

Note that even though the datum depth are the gas-oil contact depths are the same the RSVD keyword has been used to defined the RS (GOR) versus depth relationship. Secondly, the THPRES keyword has been used to set a threshold pressure between the various equilibration regions to prevent flow between the regions until the threshold is exceeded.

The next example is taken from a field containing only dead³⁰⁰ oil with one equilibration region and Fetkovich³⁰¹ analytical aquifer connected on the flanks of the field.

```
--
--          DATUM    DATUM    OWC    PCOW    GOC    PCGO    RS    RV    N
--          DEPTH    PRESS    DEPTH    ----    DEPTH    ----    OPT    OPT    OPT
EQUIL
          3707      1609      3707      0        1*       1*       0      0      0 /
--
--          DATUM
--          DEPTH
DATUM
          3500                                     / DATUM DEPTH
--
--                                     FETKOVICH AQUIFER DESCRIPTION
--
--          ID    DATUM    AQF    AQF    AQF    AQF    AQF    SALT
--          NUM    DEPTH    PRESS    VOLM    COMP    PI      PVT    CONC
--
AQUFETP
          1    1130.    1*      1.0E+12 3.0E-5  500E3  1      0.0      /
/
--
--                                     ANALYTIC AQUIFER CONNECTION
--
--          ID    ----- BOX -----    CONNECT    AQF    AQF    ADJOIN
--          NUMBER I1  I2    J1  J2    K1  K2    FACE      INFLX  MULTI  CELLS
--
--                                     ANALYTIC AQUIFER CONNECTION
--
--          ID    ----- BOX -----    CONNECT    AQF    AQF    ADJOIN
--          NUMBER I1  I2    J1  J2    K1  K2    FACE      INFLX  MULTI  CELLS
AQUANCON
          1      57  57      28  36      46  58      'I+'    1*     1*     'NO' /
          1       1   1      14  14       1  29      'J-'    1*     1*     'NO' / NORTHERN
          1       2   2      12  12       1  29      'J-'    1*     1*     'NO' /
          1       3   3      10  10       1  29      'J-'    1*     1*     'NO' /
          1       4   5       9   9       1  29      'J-'    1*     1*     'NO' /
          1       6   6       8   8       1  29      'J-'    1*     1*     'NO' /
          1      14  14       4   4       1  29      'J-'    1*     1*     'NO' /
          1      15  15       2   2       1  29      'J-'    1*     1*     'NO' /
          1      16  77       1   1       1  29      'J-'    1*     1*     'NO' /
          1       1   1      14 136       1  29      'I-'    1*     1*     'NO' / WESTERN
          1      13  13     137 141       1  29      'I-'    1*     1*     'NO' /
          1      12  12     142 143       1  29      'I-'    1*     1*     'NO' /
          1      11  11     144 144       1  29      'I-'    1*     1*     'NO' /
          1      10  10     145 145       1  29      'I-'    1*     1*     'NO' /
          1       9   9     146 146       1  29      'I-'    1*     1*     'NO' /
          1       8   8     147 147       1  29      'I-'    1*     1*     'NO' /
```

Notice how the gas-oil contact has been defaulted as there is no gas in the model only oil and water. Alternatively, one could have located the gas-oil contact above the reservoir. Secondly the DATUM keyword has been used to reset the datum depth for pressure reporting.

³⁰⁰ “Dead” oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

³⁰¹ Fetkovich, M. J. “A Simplified Approach to Water Influx Calculations - Finite Aquifer Systems,” *Journal of Petroleum Technology*, (1971) 23, No. 7, 814-828.

10.3 KEYWORD DEFINITIONS

10.3.1 ADD – ADD A CONSTANT TO A SPECIFIED ARRAY

The [ADD](#) keyword adds a constant to a specified array or part of an array. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [ADD](#) keyword is being applied.

See [ADD – Add a Constant to a Specified Array](#) in the [GRID](#) section for a full description.

10.3.2 ADDREG – ADD A CONSTANT TO AN ARRAY BASED ON A REGION NUMBER

The [ADDREG](#) keyword adds a constant to a specified array or part of an array based on cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [ADDREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [ADDREG](#) keyword is being used.

See [ADDREG – Add a Constant to an Array based on a Region Number](#) in the [GRID](#) section for a full description.

10.3.3 APIVD – EQUILIBRATION OIL API GRAVITY VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [APIVD](#) keyword defines the oil [API](#) gravity versus depth tables for each equilibration region when [API](#) Tracking as been activated by the [API](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding API gravity values, API .			None
		feet	m	cm	
2	API	A columnar vector of real values that defines the API gravity at the corresponding DEPTH . The American Petroleum Institute (“API”) classifies oils based on an API gravity (γ_{API}), or degrees API (°API), the relationship between relative density (γ_o) of oil and API gravity (γ_{API}) is given by: $\gamma_{API} = \frac{141.5}{\gamma_o} - 131.5$			None
		°API	°API	°API	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 10.5: APIVD Keyword Description

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the bubble-point versus depth functions.

```
--
--      DEPTH      API
--      GRAVITY
--      -----
APIVD
      3000.0      41.10
      8000.0      41.10                               / API VS DEPTH EQUIL REGN 01
--      -----
      3000.0      41.10
      8000.0      38.50                               / API VS DEPTH EQUIL REGN 02
--      -----
      3000.0      41.10
      8000.0      38.50                               / API VS DEPTH EQUIL REGN 03
```

Here three tables are entered; the first table has a constant [API](#) gravity versus depth relationship for equilibration region number one and the other two equilibration regions have the [API](#) gravity varying with depth.

10.3.4 AQANCONL – DEFINE ANALYTICAL CONNECTIONS TO A LGR GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The AQANCON keyword defines how analytical aquifers are connected to a Local Grid Refinement ("LGR") grid, this includes the Carter-Tracy, Fetkovich and Constant Flux analytical aquifers, all of which are implemented in OPM Flow. Carter-Tracy analytical aquifers are characterized by the [AQUCT](#) keyword in the [GRID](#) section and Fetkovich analytical aquifers are defined by either the [AQUFET](#) or [AQUFETP](#) keywords in the [SOLUTION](#) section. Finally, the Constant Flux aquifer is defined by the [AQUFLUX](#) keyword in [SOLUTION](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	AQUID is a positive integer greater than or equal to one and less than the maximum number of analytical aquifers as defined by the NANAQU variable on the AQUDIMS keyword in the RUNSPEC section, that defines the aquifer to be connected to the grid.			None
2	LGRNAME	A character string of up to eight characters in length that defines the name of the LGR that will connect to an analytical aquifer AQUID. The LGR must have been previously defined by the either the CARFIN (Cartesian LGR grid) keyword, or the RADIN/RADIN4 (radial LGR grid) keyword in the GRID section.			None
3	I1	A positive integer that defines the LGR's lower bound of the cells in the I-direction to be connected to the aquifer, and must be greater than or equal to one and less than or equal to I2 and NX on the CARFIN keyword for Cartesian grids.			I
4	I2	A positive integer that defines the LGR's upper bound of the cells in the I-direction to be connected to the aquifer, and must be greater than or equal to I1 and less than or equal to NX on the CARFIN keyword for Cartesian grids.			NX
5	J1	A positive integer that defines the LGR's lower bound of the cells in the J-direction to be connected to the aquifer, and must be greater than or equal to one and less than or equal to J2 and NY on the CARFIN keyword for Cartesian grids.			I
6	J2	A positive integer that defines the LGR's upper bound of the cells in the J-direction to be connected to the aquifer, and must be greater than or equal to J1 and less than or equal to NY on the CARFIN keyword for Cartesian grids.			NY
7	K1	A positive integer that defines the LGR's lower bound of the cells in the K-direction to be to be connected to the aquifer, and must be greater than or equal to one and less than or equal to K2 and NZ on the CARFIN keyword for Cartesian grids.			I

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	K2	A positive integer that defines the LGR's upper bound of the cells in the K-direction to be connected to the aquifer, and must be greater than or equal to K1 and less than or equal to NZ on the <i>CARFIN</i> keyword for Cartesian grids.			NZ
9	AQUFACE	AQUFACE is a character string that sets the connection “face” of the cells declared by this record and should be set to one of the following: 1) X+, Y+, or Z+ for the positive direction, or X-, Y- or Z- for the negative direction transmissibilities. 2) I+, J+, or K+ for the positive direction, or I-, J- or K- for the negative direction transmissibilities.			None
10	<i>AQUFLUX</i>	<i>AQUFLUX</i> is a positive real value that sets the fraction of the total influx between the aquifer and the defined cells declared on this keyword. If defaulted the cell face for each cell is applied and if a values is declared then this values is applied to all cells declared by this record.			I*
		ft ²	m ²	cm ²	
11	AQUCOEF	AQUCOEF is a real positive values that scales the calculated connection between the aquifer and the cells declared on this record.			1.0
		dimensionless	dimensionless	dimensionless	
12	AQUOPT	AQUOPT is a character string that sets the cell face connection and should be set to one of the following: 1) YES:Aquifer connections <u>can adjoin</u> to active cells allowing for connections inside the reservoir grid. It is not recommended to use this option without thoroughly checking the connections in the model. 2) NO:Aquifer connections <u>cannot adjoin</u> to active cells preventing connections inside the reservoir grid. This is the recommended and the default value.			NO

Notes:
1) Where NX, NY and NZ are the dimensions of the LGRNAME as defined on the *CARFIN* keyword in the *GRID* section, or the NR, NTHEATA and NZ dimensions on the *RADFIN* keyword in the *GRID* section.
2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 10.6: AQANCONL Keyword Description

Note

If the [AQANCONL](#) keyword has been utilized in the run deck then OPM Flow will write the [AQUIFERA](#) array to the *.INIT file in order to visualize the aquifer connections in OPM ResInsight.

This is accomplished by setting the [AQUIFERA](#) value to 2(AQUID-1) for cells connected to aquifer [AQUID](#). If a cell is connected to multiple analytical aquifers then [AQUIFERA](#) is summed for all aquifers connected to a cell. Note that connecting cells to multiple aquifers is best avoided.

Example

The following example defines aquifer number one connected to the J- face of various cells in the LGROP001 [LGR](#), and a second basal aquifer connected to the K+ face.

```
--
--
--          LGR ANALYTIC AQUIFER CONNECTION
--
--      ID   LGR      ----- BOX -----  CONNECT  AQF   AQF   ADJOIN
--      NUM  NAME      I1  I2   J1  J2   K1  K2   FACE    INFLX  MULTI  CELLS
AQANCONL
      1     LGROP001    1  10    1  15    1  10    J-     1*    1.0    'NO' /
      1     LGROP001    1  10    1  15   12  18    J-     1*    1.0    'NO' /
      2     LGROP001    1  10    1  15   20  20    K+     1*    1.0    'NO' /
/
```

See the [AQUCT](#) keyword in the [GRID](#) section for a complete example on defining and connecting a Carter-Tracy aquifer to a simulation grid.

10.3.5
AQANNC – DEFINE ANALYTIC AQUIFER NON-NEIGHBOR CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

AQANNC defines the analytic aquifer non-neighbor connections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.6 AQANTRC – DEFINE ANALYTIC AQUIFER INITIAL TRACER CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [AQANTRC](#) keyword defines the initial tracer concentrations for analytic aquifers that have previously been defined by the AQCT keyword in the [GRID](#), [PROPS](#), or [SOLUTION](#) sections for Carter-Tracy analytical aquifers, or the AQFET and AQFETP keywords in the [SOLUTION](#) section for Fetkovich analytical aquifers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.7 AQUALIST – DEFINE AN ANALYTIC AQUIFER NAME TO AQUIFER NUMBERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [AQUALIST](#) keyword defines named lists of analytic aquifers identified by aquifer numbers for greater readability in the output.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.8 AQUANCON – DEFINE ANALYTICAL CONNECTIONS TO THE GRID

The [AQUANCON](#) keyword defines how analytical aquifers are connected to the simulation grid, this includes Carter-Tracy and Fetkovich analytical aquifers, both of which are implemented in OPM Flow.

See [AQUANCON – Define Analytical Connections to the Grid](#) in the [GRID](#) section for a full description.

10.3.9 AQUCHGAS – DEFINE CONSTANT PRESSURE GAS ANALYTICAL AQUIFER PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [AQUCHGAS](#) keyword defines the properties of constant pressure gas analytical aquifers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.10 AQUCHWAT – DEFINE CONSTANT PRESSURE WATER ANALYTICAL AQUIFER PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [AQUCHWAT](#) keyword defines the properties of constant pressure water analytical aquifers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.11 AQUICON – DEFINE NUMERICAL AQUIFER CONNECTIONS TO THE GRID

[AQUICON](#) keyword defines how numerical aquifers are connected to the simulation grid.

See [AQUICON – Define Numerical Aquifer Connections to the Grid](#) in the [GRID](#) section for a full description.

10.3.12 AQUCT – DEFINE CARTER-TRACY ANALYTICAL AQUIFERS

The [AQUCT](#) keyword defines a Carter Tracy aquifer, the properties of the aquifer, including the Carter-Tracy aquifer influence function associated with the aquifer, defined by the [AQUCTAB](#) keyword in the [PROPS](#) section.

See [AQUCT – Define Carter-Tracy Analytical Aquifers](#) in the [GRID](#) section for a full description

10.3.13 AQUFET – DEFINE FETKOVICH ANALYTICAL AQUIFER AND CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [AQUFET](#) keyword defines Fetkovich³⁰² analytical aquifers, the aquifer properties, together with the cell connections to the aquifer. Each row entry in the [AQUFETP](#) keyword defines one Fetkovich analytical aquifer and one cell face to be connected to the aquifer.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; however, see the [AQUFETP](#) keyword in the [SOLUTION](#) section and [AQUANCON](#) keyword in the [GRID](#) section, on how to define and connect Fetkovich analytical aquifers.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DATUM	DATUM is a single positive value that defines the Fetkovich reference datum depth for PRESS.			None
		feet	m	cm	
2	PRESS	PRESS is a single positive value that defines the aquifer pressure at DATUM. If PRESS is defaulted then the simulator will set the aquifer's initial reservoir pressure to be in equilibrium with the cells the aquifer is contacted to. Defaulting this parameter will avoid inconsistent equilibration pressures between the reservoir cells and the aquifer.			I*
		psia	barsa	atma	
3	PORV	A real positive value that defines the initial water volume of the aquifer.			None
		stb	sm ³	scc	
4	COMP	COMP is a real number defining the total compressibility (C _t) of the aquifer, that is the rock compressibility (C _f) plus the water compressibility (C _w) at the aquifer datum pressure (DATUM) and is defined as: $C_t = C_f + C_w$			None
		1/psia	1/barsa	1/atma	
5	PI	A real positive number that defines the aquifer productivity index based on the aquifer influx rate per unit pressure drop.			None
		stb/d/psia	sm ³ /barsa	scc/hr/atma	
6	PVTW	A positive integer that defines the aquifer's PVTW water property table.			I
7	I1	A positive integer that defines the lower bound of the cells in the I-direction to be connected to the aquifer and must be greater than or equal to one and less than or equal to I2 and NX.			I

³⁰² Fetkovich, M. J. "A Simplified Approach to Water Influx Calculations - Finite Aquifer Systems," *Journal of Petroleum Technology*, (1971) 23, No. 7, 814-828.

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	I2	A positive integer that defines the upper bound of the cells in the I-direction to be connected to the aquifer and must be greater than or equal to I1 and less than or equal to NX			NX
9	J1	A positive integer that defines the lower bound of the cells in the J-direction to be connected to the aquifer and must be greater than or equal to one and less than or equal to J2 and NY.			I
10	J2	A positive integer that defines the upper bound of the cells in the J-direction to be connected to the aquifer and must be greater than or equal to J1 and less than or equal to NY.			NY
11	K1	A positive integer that defines the lower bound of the cells in the K-direction to be to be connected to the aquifer and must be greater than or equal to one and less than or equal to K2 and NZ.			I
12	K2	A positive integer that defines the upper bound of the cells in the K-direction to be connected to the aquifer and must be greater than or equal to K1 and less than or equal to NZ.			NZ
13	AQUFACE	AQUFACE is a character string that sets the connection “face” of the cells declared by this record and should be set to one of the following: 1) X+, Y+, or Z+ for the positive direction, or X-, Y- or Z- for the negative direction transmissibilities. 2) I+, J+, or K+ for the positive direction, or I-, J- or K- for the negative direction transmissibilities.			None
14	SALTCON	SALTCON is a real positive number that defines the initial salt concentration in the aquifer. This variable is ignored by OPM Flow.			0.0
		lb/stb	kg/sm ³	gm/scc	

Notes:
1) Where NX, NY and NZ are the dimensions of the model as defined on the *DIMENS* keyword in the *RUNSPEC* section.
2) The keyword is followed by up to NANAQU records as defined on the *AQUDIMS* keyword in the *RUNSPEC* section
3) The keyword should be terminated by a “/”.

Table 10.7: AQUIFET Keyword Description

Note this keyword should only be used in equilibration and enumerated initialize runs, that is it should be omitted from [RESTART](#) runs.

Note

If the model is unstable then this may be due to an aquifer not being in equilibrium with the connecting reservoir blocks, for example the aquifer is connected to only hydrocarbon reservoir cells. Try commenting out the aquifer and see if this resolves the instabilities.

10.3.14 AQUIFETP – DEFINE FETKOVICH ANALYTICAL AQUIFERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [AQUIFETP](#) keyword defines Fetkovich³⁰³ analytical aquifers and the aquifer properties. Each row entry in the [AQUIFETP](#) keyword defines one Fetkovich analytical aquifer. In order to fully define this type of aquifer, the aquifer must be connected to the reservoir using the [AQUANCON](#) keyword in the [GRID](#) or [SOLUTION](#) sections.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	A positive integer greater than or equal to one and less than or equal to NANAQ on the AQUIDIMS keyword in the RUNSPEC section, that defines the Fetkovich aquifer number.			I
2	DATUM	DATUM is a single positive value that defines the Fetkovich reference datum depth for PRESS.			None
		feet	m	cm	
3	PRESS	PRESS is a single positive value that defines the aquifer pressure at DATUM . If PRESS is defaulted then the simulator will set the aquifer's initial reservoir pressure to be in equilibrium with the cells the aquifer is contacted to. Defaulting this parameter will avoid inconsistent equilibration pressures between the reservoir cells and the aquifer.			I*
		psia	barsa	atma	
4	PORV	A real positive value that defines the initial water volume of the aquifer.			None
		stb	sm ³	scc	
5	COMP	COMP is a real number defining the total compressibility (C _t) of the aquifer, that is the rock compressibility (C _f) plus the water compressibility (C _w) at the aquifer datum pressure (DATUM) and is defined as: $C_t = C_f + C_w$			None
		l/psia	l/barsa	l/atma	
6	PI	A real positive number that defines the aquifer productivity index based on the aquifer influx rate per unit pressure drop.			None
		stb/d/psia	sm ³ /d/barsa	scc/hr/atma	
7	PVTW	A positive integer that defines the aquifer's PVTW water property table.			I

³⁰³ Fetkovich, M. J. "A Simplified Approach to Water Influx Calculations - Finite Aquifer Systems," *Journal of Petroleum Technology*, (1971) 23, No. 7, 814-828.

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	SALTCON	SALTCON is a real positive number that defines the initial salt concentration in the aquifer, for when the simulator's Brine Model has been activated via the <i>BRINE</i> keyword in the <i>RUNSPEC</i> section. This variable is ignored by OPM Flow.			0.0
		lb/stb	kg/sm ³	gm/scc	
9	<i>TEMP</i>	<i>TEMP</i> is a real positive number that defines the initial temperature of the aquifer at <i>DATUM</i> for use with OPM Flow's thermal option. The <i>THERMAL</i> keyword in the <i>RUNSPEC</i> section must be activated to use this option.			I*
		°F	°C	°C	

Notes:

- 1) The keyword is followed by up to NANAQ records as defined on the *AQUDIMS* keyword in the *RUNSPEC* section
- 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”. Note the commercial simulator only requires a terminating “/” if the number of records on this keyword are less than NANAQ on the *AQUDIMS* keyword in the *RUNSPEC* section. If the number of records are equal to NANAQ and a terminating “/” has been entered then the commercial simulator will issue a warning message; however, the commercial simulator run will proceed as expected.

Table 10.8: AQUFETP Keyword Description

Note this keyword should only be used in equilibration and enumerated initialize runs, that is it should be omitted from [RESTART](#) runs.

Note

If the model is unstable then this may be due to an aquifer not being in equilibrium with the connecting reservoir blocks, for example if the aquifer is connected to some hydrocarbon reservoir cells. Try commenting out the aquifer and see if this resolves the instabilities, and if so amend the aquifer connections accordingly.

Example

Given the following grid and aquifer dimensions in the *RUNSPEC* section:

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      MAX      MAX      MAX
--      NDIVIX   NDIVIY   NDIVIZ
-- DIMENS
--      20       1       5
--
--      AQF      AQF      AQF      AQF      AQF      AQF      AQF      AQF
--      MXAQN    MXNAQC   NIFTBL  NRIFTB  NANAQU   NCAMAX  MXNALI  MXAAQL
-- AQUODIMS
--      1*       1*       5       100     1       1*       1*       1*
--

```

The Fetkovich analytical aquifer is defined in the *SOLUTION* sections as:

```

-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION --
--
--      FETKOVICH AQUIFER DESCRIPTION
--
--      ID  DATUM  AQF  AQF  AQF  AQF  AQF  AQF  SALT
--      NUM DEPTH PRESS VOLM COMP PI  PVT  CONC
--
--      AQUFETP
--      1  1130.  1*   1.0E+12 3.0E-5 500E3  1   0.0
--

```

And the connection of the aquifer is set in the *GRID* or the *SOLUTION* sections as:

```

--
--      ANALYTIC AQUIFER CONNECTION
--
--      ID  ----- BOX -----  CONNECT  AQF  AQF  ADJOIN
--      NUMBER I1 I2  J1 J2  K1 K2  FACE    INFLX MULTI CELLS
--
--      AQUANCON
--      1    1  1  1  1  1  1  J-    1.0  1.0  'NO'
--

```

Here one Fetkovich analytical aquifer is connected to a single cell (1, 1, 1) at the J- face (or Y- face) of the cell.

10.3.15 AQUFLUX – DEFINE CONSTANT FLUX ANALYTICAL AQUIFER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [AQUFLUX](#) keyword defines the properties of Constant Flux Analytical Aquifers, that allows for a constant water influx to the connected grid blocks. This type of aquifer is connected to the model using either the [AQUANCON](#) keyword for global cells, or the [AQUANCONL](#) keyword for cells belonging to a Local Grid Refinement ("LGR"), both the aforementioned keywords are in the [GRID](#) and [SOLUTION](#) sections. The keyword itself may be utilized in both the [SOLUTION](#) and [SCHEDULE](#) sections, with subsequent entries overwriting the previous entry.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	AQUID	A positive integer greater than or equal to one and less than or equal to NANAQ on the AQUIDIMS keyword in the RUNSPEC section, that defines the AQUFLUX aquifer number.			1
2	AQFLUX	A real positive value the defines the aquifer water influx rate, per unit area of the connected grid cell face.			None
		stb/day/ft2	sm ³ /day/m2	scc/hour/cm2	
3	SALTCON	SALTCON is a real positive number that defines the initial salt concentration in the aquifer, for when with simulator's Brine Model has been activated via the BRINE keyword in the RUNSPEC section. This variable is ignored by OPM Flow.			0.0
		lb/stb	kg/sm ³	gm/scc	
4	TEMP	TEMP is a real positive number that defines the initial temperature of the aquifer for use with OPM Flow's thermal option. The THERMAL keyword in the RUNSPEC section must be activated to use this option. If the THERMAL keyword is absent from the input deck, then the parameter is ignored.			1*
		°F	°C	°C	
5	PRESS	PRESS is a single positive value that defines the aquifer pressure for use with OPM Flow's thermal option. The THERMAL keyword in the RUNSPEC section must be activated to use this option. If the THERMAL keyword is absent from the input deck, then the parameter is ignored.			1*
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by up to NANAQ records as defined on the [AQUIDIMS](#) keyword in the [RUNSPEC](#) section
- 2) Each record is terminated by a "/" and the keyword should be terminated by a "/". Note the commercial simulator only requires a terminating "/" if the number of records on this keyword are less than NANAQ on the [AQUIDIMS](#) keyword in the [RUNSPEC](#) section. If the number of records are equal to NANAQ and a terminating "/" has been entered then the commercial simulator will issue a warning message; however, the commercial simulator run will proceed as expected.

Table 10.9: AQUFLUX Keyword Description

The water flow rate into a connected grid cell for this type of aquifer, is calculated from:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

$$q_w = \text{AQUFLUX}(\text{AQFLUX}) \times A_{(i,j,k)} \times \text{AQUANCON}(\text{AQUCOEF}) \tag{10.14}$$

Where:

AQUFLUX(AQFLUX)

= the AQFLUX parameter on the [AQUFLUX](#) keyword,

AQUANCON(AQUCOEF)

= the AQUCOEF parameter on the [AQUANCON](#) keyword

Note that $A_{(i,j,k)}$ is calculated from the connected cell geometry, and thus the [AQUANCON\(AQUFLUX\)](#) and the [AQUANCONL\(AQUFLUX\)](#) parameters are ignored for this type of aquifer.

The [AQUFLUX](#) keyword should only be used in runs that have been initialized by equilibration or with enumerated values, and should not be used in [RESTART](#) runs.

See also the [BCCON](#) keyword in the [GRID](#) section and the [BCPROP](#) keyword in the [SCHEDULE](#) section that can be used to specify boundary conditions.

Example

Given a total of five analytical aquifers as declared by the NANAQ parameter on the [AQUDIMS](#) keyword being set to five, of which three are Constant Flux Aquifers and two Carter-Tracy Aquifers, then:

```

--
--                                CONSTANT FLUX AQUIFER DESCRIPTION
--
--      ID      WATER      SALT      AQF      AQU
--      NUM      INFLUX     CONC     TEMP     PRES
--
AQUFLUX
    1    0.0004    1*        1*        1*
    2    0.0005    1*        1*        1*
    3    0.0003    1*        1*        1*
/
--
--                                CARTER-TRACY AQUIFER DESCRIPTION
--
--      ID      DATUM      AQF      AQF      AQF      AQF      AQF      AQF      AQF      INFL      PVT      AQU
--      NUM      DEPTH     PRESS     PERM     PORO     RCOMP     RE       DZ       ANGLE    NUM     TAB
--
AQUCT
    4    2000.0    269      100.0    0.30     3.0e-5    330     10.0    360.0    1      2      /
    5    2000.0    269      100.0    0.30     3.0e-5    330     10.0    360.0    1      2      /
/

```

Defines three Constant Flux Aquifers and three Carter-Tracy Aquifers, and the connection of the aquifers are set in the [GRID](#) or [SOLUTION](#) sections via:

```

--
--                                ANALYTIC AQUIFER CONNECTION
--
--      ID      ----- BOX -----      CONNECT      AQF      AQF      ADJOIN
--      NUMBER  I1   I2   J1   J2   K1   K2   FACE      INFLX    MULTI    CELLS
--
AQUANCON
    1          1   198   1   14   1   10   J-        1*      1.0      'NO'
    2          1   198   1   14   12  22   J-        1*      1.0      'NO'
    3          1   198   1   14   23  54   J-        1*      1.0      'NO'
    4          1   198   1   14   61  70   J-        1*      1.0      'NO'
    5          1   198  15  172  71  71   K+        1*      1.0      'NO'
/

```


10.3.16 BC – DEFINE BOUNDARY CONDITIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [BC](#) keyword defines the boundary conditions for the model, and can be used to set boundary conditions for when external influx or efflux volumes are influencing the reservoir pressure and production history. For example, when the average reservoir pressure remains constant throughout the production period due to water influx, or gas migration from an external source.

The [BC](#) keyword has been replaced by the [BCCON](#) keyword in the [GRID](#) section that defines the boundary condition connections and the [BCPROP](#) keyword in the [SCHEDULE](#) section that defines the boundary condition properties.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	I1	A positive integer that defines the lower bound of the grid in the I-direction for which the boundary conditions are to be applied, must be greater than or equal to 1 and less than or equal to I2 and NX.			I
2	I2	A positive integer that defines the upper bound of the grid in the I-direction for which the boundary conditions are to be applied, must be greater than or equal to I1 and less than or equal to NX			NX
3	J1	A positive integer that defines the lower bound of the grid in the J-direction for which the boundary conditions are to be applied, must be greater than or equal to 1 and less than or equal to J2 and NY.			I
4	J2	A positive integer that defines the upper bound of the grid in the J-direction for which the boundary conditions are to be applied, must be greater than or equal to J1 and less than or equal to NY.			NY
5	K1	A positive integer that defines the lower bound of the grid in the K-direction for which the boundary conditions are to be applied, must be greater than or equal to one and less than or equal to K2 and NZ.			I
6	K2	A positive integer that defines the upper bound of the grid in the K-direction for which the boundary conditions are to be applied, must be greater than or equal to K1 and less than or equal to NZ.			NZ
7	DIRECT	A character string that defines the direction to apply the boundary conditions, and should be set to one of the following X, Y, or Z for the positive direction, or X-, Y- or Z- for the negative direction.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	TYPE	A defined character string that defines the type of boundary condition to be applied, and should be set to one of the following character strings: 1) DIRICHLET: for a user defined boundary conditions. In this case, items (1) through (7) must be set, together with PHASE for the fluid type, and PRESS and TEMP for the constant pressure and temperature boundary conditions. - This option is currently not supported but will be available in the next release. 2) FREE: for the initial state of the boundary to be kept throughout the simulation, that is a constant boundary condition. For this option only items (1) through (7) need to be defined. 3) RATE: for the boundary to have a constant influx or efflux rate. Again items (1) through (7) are required, plus PHASE for the fluid type, and RATE to set the PHASE rate. Only the FREE and RATE options are currently supported; however the next release will support the DIRICHLET option.			None
9	PHASE	A defined character string that sets fluid type used in the boundary calculations, and should be set to one of the following character strings: 1) GAS: the gas phase will be used to control the boundary conditions for when the TYPE has been set to DIRICHLET or RATE. 2) OIL: the oil phase will be used to control the boundary conditions for when the TYPE has been set to DIRICHLET or RATE. 3) WATER or WAT: the water phase will be used to control the boundary conditions for when the TYPE has been set to DIRICHLET or RATE.			None
10	RATE	A real value that defines the constant surface oil, gas or water rate to be injected or withdrawn at the boundary, for when TYPE has been set to RATE. Note a negative value implies an injection rate, whereas, a positive value indicates a withdrawal rate.			0.0
		Liquid: stb/day Gas: Mscf/day	Liquid: sm ³ /day Gas: sm ³ /day	Liquid: scc/hr Gas: scc/hr	
11	PRESS	PRESS is a real positive value that defines the constant pressure boundary condition. PRESS should only be entered if TYPE has been set to DIRICHLET. If the pressure at the boundary is less than PRESS, then the fluid type declared via PHASE will flow across the boundary. The default value of I* will use the simulator's calculated value based on data entered via the EQUIL keyword in the SOLUTION section.			I*
		psia	barsa	atma	

No.	Name	Description			Default
		Field	Metric	Laboratory	
12	TEMP	TEMP is a real positive number that defines the constant temperature boundary condition. TEMP should only be entered if TYPE has been set to DIRICHLET. The default value of I* will use the simulator's calculated value based on data entered via one of the following reservoir temperature keywords: RTEMP, RTEMPA, RTEMPVD, TEMPI, or TEMPVD, in the SOLUTION section. Note that all of the aforementioned reservoir temperature keywords, except for TEMPI, may also be used in the PROPS section as well.			I*
		°F	°C	°C	

Notes:

- 1) Where NX, NY and NZ are the dimensions of the model as defined on the [DIMENS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record must be terminated by a “/” and the keyword is terminated by a “/”.

Table 10.10: BC Keyword Description

See also the [AQUFLUX – Define Constant Flux Analytical Aquifer](#) keyword that is supported by OPM Flow in both the [SOLUTION](#) and [SCHEDULE](#) sections, to define the aquifer influx with greater flexibility.

If the [BC](#) keyword is not present in the input deck, then the boundary conditions for the model are set to be no flow, which is the normal behavior in both OPM Flow and the commercial simulator.

Examples

The first example shows how to set a constant pressure boundary using TYPE equal to FREE:

```
--
--      DEFINE MODEL BOUNDARY CONDITIONS (OPM FLOW SOLUTION GRID KEYWORD)
--
--      ----- BOX -----      BC      BC      BC      BC      BC      BC
--      I1  I2   J1  J2   K1  K2  DIRC  TYPE      PHASE  RATE  PRESS  TEMP
BC
      1   1     1   1*   1   1*  X-   FREE      1*    1*    1*    1*   /
      1   1*    1   1    1   1*  Y    FREE
/
```

With this option it is only necessary to define the boundary cells and all the other parameters (PHASE, RATE, PRESS, and TEMP) can be defaulted, as they are ignored for when TYPE equals FREE.

The next example is based on NX, NY and NZ equal to 20, 1, 10 respectively, on the [DIMENS](#) keyword in the [RUNSPEC](#) section, and shows how different boundary types can be assigned to different parts of the model.

```
--
--      DEFINE MODEL BOUNDARY CONDITIONS (OPM FLOW SOLUTION GRID KEYWORD)
--
--      ----- BOX -----      BC      BC      BC      BC      BC      BC
--      I1  I2   J1  J2   K1  K2  DIRC  TYPE      PHASE  RATE  PRESS  TEMP
BC
      1   1     1   1    1   10  X-   RATE      GAS    1*    256.0  100.0 /
      20  20    1   1    1   10  X    FREE      4*
/
```

The last example shows how the DIRICHLET boundary condition option may be used:

```
--
--      DEFINE MODEL BOUNDARY CONDITIONS (OPM FLOW SOLUTION GRID KEYWORD)
--
--      ----- BOX ----- BC   BC      BC   BC   BC   BC
--      I1  I2   J1  J2   K1  K2  DIRC  TYPE      PHASE  RATE  PRESS  TEMP
BC
      1   1     1   1*   1   1*  X-   DIRICHLET  WAT   1*   256.0  100.0 /
      1   1*    1   1    1   1*  Y    DIRICHLET  WAT   1*   1*    100.0 /
/
```

Here, the first line sets both the pressure and temperature at the boundary, and the second line defaults the pressure entry, so that the simulator calculated initial boundary pressure will be used.

10.3.17 BOUNDARY – DEFINE A BOUNDARY BOX FOR PRINTING

The [BOUNDARY](#) keyword defines a rectangular grid for printing various arrays to the output print file (*.PRT); thus, avoiding printing all the elements in the selected array.

See [BOUNDARY – Define a Boundary Box for Printing](#) in the [GRID](#) section for a full description.

10.3.18 BOX – DEFINE A RANGE OF GRID BLOCKS TO ENTER PROPERTY DATA

[BOX](#) defines a range of grid blocks for which subsequent data is assigned for all the cells in the defined [BOX](#). Note that the [BOX](#) grid is reset by the keyword [ENDBOX](#) by resetting the current defined [BOX](#) to be the whole grid. The keyword can be used for any array and for all grid types.

See [BOX – Define a Range of Grid Blocks to Enter Property Data](#) in the [GRID](#) section for a full description.

10.3.19 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The [COLUMNS](#) keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the [GLOBAL](#) section for a full description.

10.3.20 COPY – COPY ARRAY DATA TO ANOTHER ARRAY

The [COPY](#) keyword copies an array (or part of an array) to another array or part of an array. The arrays can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [COPY](#) keyword is being used.

See [COPY – Copy Array Data to Another Array](#) in the [GRID](#) section for a full description.

10.3.21 COPYREG – COPY AN ARRAY TO ANOTHER ARRAY BASED ON A REGION NUMBER

The [COPYREG](#) keyword copies a specified array or part of an array based on cells with a specific region number to another array. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [COPYREG](#) keyword is read by the simulator. The property arrays can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [COPYREG](#) keyword is being used.

See [COPYREG – Copy an Array to Another Array based on a Region Number](#) in the [GRID](#) section for a full description.

10.3.22 DATUM – DEFINE THE DATUM DEPTH FOR THE MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DATUM](#) keyword defines the datum depth for the model. This allows for all grid block potentials (depth corrected pressures) to be calculated at a common depth.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DATUM	DATUM is a single positive value that defines the datum depth for the model.			None
		feet	m	cm	

Notes:

- I) The keyword is terminated by a “/”.

Table 10.11: DATUM Keyword Description

See also the [DATUMR](#) and [DATUMRX](#) keywords in the [SOLUTION](#) section that also define the datum depth for the model.

Example

```
--      DATUM
--      DEPTH
--      -----
DATUM      5000.0                               / DATUM DEPTH FOR REPORTING
```

The above example defines the datum for the model to be 5000.0

10.3.23 DATUMR – DEFINE DATUM DEPTHS FOR THE FIPNUM REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DATUMR](#) keyword defines the datum depth for each fluid in-place region ([FIPNUM](#)) declared in the model. This allows for all grid block potentials (depth corrected pressures) to be calculated at a common depth within a [FIPNUM](#) region.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DATUMR	DATUMR is a vector of positive values that defines the datum depth for each fluid in-place region.			None
		feet	m	cm	
Notes: 1) The keyword is followed by NTFIP values as defined on REGDIMS keyword in the RUNSPEC section. 2) The keyword is terminated by a “/”.					

Table 10.12: DATUMR Keyword Description

See also the [DATUM](#) and [DATUMRX](#) keywords in the [SOLUTION](#) section that also define the datum depth for the model.

Example

```
--
--      DATUM
--      DEPTH
--      -----
DATUMR      4800.0
              4900.0
              5000.0                               / DATUM DEPTH FOR REPORTING
```

The above example defines the datum depth for three [FIPNUM](#) regions, for when NTFIP has been set equal to three on the [REGDIMS](#) keyword in the [RUNSPEC](#) section.

10.3.24 DATUMRX – DEFINE DATUM DEPTHS FOR THE FIP ALLOCATED REGIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [DATUMRX](#) keyword defines the datum depth for each fluid in-place family region defined by the [FIP](#) keyword. This allows for all grid block potentials (depth corrected pressures) to be calculated at a common depth within a given [FIP](#) region. The [FIP](#) keyword in the [REGIONS](#) section allows one to define additional sets of fluid in-place regions to the standard [FIPNUM](#) keyword. For example, one could use [FIPNUM](#) to define the reservoir layers as fluid in-place regions and the [FIP](#) keyword to define the fluid in-place region for fault blocks.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	FIPNAME	A character string of up to five characters in length that defines the <i>FIP</i> family name for which the datum depth data is being defined. The default value of I* will set <i>DATUMR</i> to the standard <i>FIPNUM</i> region numbers.			I*
2	<i>DATUMR</i>	<i>DATUMR</i> is a vector of positive values that defines the datum depth for each fluid in-place family region. There must be one entry for each region in the <i>FIP</i> family name. A maximum of NTFIP values, as declared by the <i>REGDIMS</i> keyword in the <i>RUNSPEC</i> section, may be entered for each FIPNAME entry.			None
		feet	m	cm	
Notes: 1) The keyword is followed any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 10.13: [DATUMR](#) Keyword Description

See also the [FIP](#) keyword in the [REGIONS](#) section to define [FIP](#) family regions, and the [DATUM](#) and [DATUMR](#) keywords in the [SOLUTION](#) section that also define the datum depth for the model.

Example

```
--
--      FIP      DATUM
--      NAME     DEPTH
DATUMRX  'FLTBL'  5000.0  5000.0  5000.0  5000.0  / DATUM DEPTH FOR REPORTING
          'LICBL'  5000.0  5050.0                / DATUM DEPTH FOR REPORTING
/
```

The above example defines the datum depth for two [FIP](#) families, FLTBL and LICBL, with the datum set to a constant 5000.0 psia for FLTBL family and different values for each of the regions in the LICBL family of regions.

10.3.25 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the GLOBAL section for a full description.

10.3.26 DYNAMICR – START OF DYNAMIC REGION PARAMETER DEFINITION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DYNAMICR](#) keyword marks the start of a Dynamic Region section and defines the parameters used for Dynamic Regions that allows for property and reporting regions to vary as the run progresses, based on the parameters and logic defined by this keyword and section. A Dynamic Region section is terminated by the [ENDDYN](#) keyword in the [SOLUTION](#) or [SCHEDULE](#) sections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.27 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

10.3.28 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the GLOBAL section for a full description.

10.3.29 ENDBOX – DEFINE THE END OF THE BOX DEFINED GRID

This keyword marks the end of a previously defined [BOX](#) sub-grid as defined by a previously entered [BOX](#) keyword. The keyword resets the input grid to be the full grid as defined by the NX, NY, and NZ variables on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

See [ENDBOX – Define the End of the BOX Defined Grid](#) in the [GRID](#) section for a full description.

10.3.30 ENDDYN – END OF DYNAMIC REGION PARAMETER DEFINITION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ENDDYN](#) keyword marks the end of a Dynamic Region section that was started with the [DYNAMICR](#) keyword in the [SOLUTION](#) or [SCHEDULE](#) sections. Dynamic Regions allow for property and reporting regions to vary as the run progresses, based on the parameters and logic defined within the section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.31 ENDFIN – END THE DEFINITION OF A LOCAL GRID REFINEMENT

[ENDFIN](#) defines the end of a Cartesian or radial local grid refinement (“LGR”) definition and a [LGR](#) property definition data set.

See [ENDFIN – End the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

10.3.32 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the GLOBAL section for a full description.

10.3.33 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the GLOBAL section for a full description.

10.3.34 EQUALREG – SETS AN ARRAY TO A CONSTANT BY REGION NUMBER

The [EQUALREG](#) keyword sets a specified array to a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [EQUALREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [EQUALREG](#) keyword is being used.

See [EQUALREG – Sets an Array to a Constant by Region Number](#) in the [GRID](#) section for a full description.

10.3.35 EQUALS – SETS A SPECIFIED ARRAY TO A CONSTANT

The [EQUALS](#) keyword sets a specified array or part of an array to a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the [EQUALS](#) keyword is being used.

See [EQUALS – Sets a Specified Array to a Constant](#) in the [GRID](#) section for a full description.

10.3.36 EQUIL – DEFINE THE EQUILIBRATION INITIALIZATION DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

This keyword defines the parameters used to initialize the model for when equilibration is calculated by OPM Flow. This is the standard methodology to initialize a model, the non-standard formulation of entering the pressures and saturations for each grid cell is seldom employed in the industry. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DATUM	DATUM is a single positive value that defines the reference datum depth for PRESS.			0.0
		feet	m	cm	
2	PRESS	PRESS is a single positive value that defines the pressure at DATUM . If the DATUM depth lies above the GOC then PRESS is the pressure with respect to the gas phase. If the DATUM depth is below OWC then PRESS refers to the water phase pressure. Otherwise, PRESS refers to the oil phase pressure.			0.0
		psia	barsa	atma	
3	WATCONT	1) For three phase runs containing oil, gas and water WATCONT is the depth of the oil-water contact (OWC). 2) For two phase runs containing oil and water WATCONT is the depth of the oil-water contact (OWC). 3) For two phase runs containing gas and water WATCONT is the depth of the gas-water contact (GWC).			0.0
		feet	m	cm	
4	WATCAP	1) For three phase runs containing oil, gas and water WATCAP is the oil-water capillary pressure at the OWC. 2) For two phase runs containing oil and water WATCAP is the oil-water capillary pressure at the OWC. 3) For two phase runs containing gas and water WATCAP is the gas-water capillary pressure at the GWC			0.0
		psia	barsa	atma	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	GASCONT	1) For three phase runs containing oil, gas and water GASCONT is the depth of the gas-oil contact (GOC). 2) Note in cases where there is no gas cap (or free gas) then GASCONT should be set to a value shallower than the top of the reservoir. 3) In cases where there is initially no oil zone, as for a gas condensate field for example, the GASCONT should be set to the same depth as WATCONT. 4) For two phase runs containing oil and water, or gas and water, GASCONT is ignored.			0.0
		feet	m	cm	
6	GASCAP	1) For three phase runs containing oil, gas and water GASCAP is the gas-oil capillary pressure at the GWC. 2) For two phase runs containing oil and water, or gas and water, GASCAP is ignored.			0.0
		psia	barsa	atma	
7	EQLOPTI	EQLOPTI is an integer value that sets the initialization option for when dissolved gas is present in the run, as activated by the <i>DISGAS</i> keyword in the <i>RUNSPEC</i> section. 1) A positive value of EQLOPTI results in the gas-oil ratio being calculated from data entered on the <i>PBVD</i> (saturation pressure or bubble-point pressure versus depth table) or the <i>RSVD</i> keyword (gas-oil ratio versus depth table). If this option is selected, then either the <i>PBVD</i> or <i>RSVD</i> keywords must be present in the input deck. 2) Note that the allocation of multiple <i>PBVD</i> and <i>RSVD</i> tables to each grid cell is through the <i>EQLNUM</i> keyword and not the <i>PVTNUM</i> keyword. 3) A zero value of EQLOPTI results in the gas-oil ratio being set to the saturated gas-oil ratio at the GOC. In this case DATUM must be equal GASCONT and the <i>PBVD</i> and <i>RSVD</i> keywords may be omitted. 4) A negative value of EQLOPTI results in the same option for when EQLOPTI is zero. EQLOPTI is ignored if there is no dissolved gas in the run.			0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	EQLOPT2	EQLOPT2 is an integer value that sets the initialization option for when vaporized oil (condensate) is present in the run, as activated by the VAPOIL keyword in the RUNSPEC section. 1) A positive value of EQLOPT2 results in the condensate-gas ratio being calculated from data entered on the PDVD (saturation pressure or dew point pressure versus depth table) or the RVVD keyword (condensate-gas ratio versus depth table). If this option is selected, then either the PDVD or RVVD keywords must be present in the input deck 2) Note that the allocation of multiple PDVD and RVVD tables to each grid cell is through the EQLNUM keyword and not the PVTNUM keyword. 3) A zero value of EQLOPT2 results in the condensate-gas ratio being set to the saturated condensate-gas ratio at the GOC. In this case DATUM must be equal GASCONT and the PDVD and RVVD keywords may be omitted. 4) A negative value of EQLOPT2 results in the same option for when EQLOPT2 is zero. EQLOPT2 is ignored if there is no vaporized oil in the run.			0
		dimensionless	dimensionless	dimensionless	
9	EQLOPT3	EQLOPT3 is an integer value that sets the initialization accuracy options for the equilibration calculation. 1) A zero value of EQLOPT3 results in OPM Flow using the fluid saturations at the center of the grid block in the equilibration calculation. This results in a stable initialization at the expense of a potentially less accurate fluid in-place calculation, especially for large thick grid blocks with a fluid contact in the block. 2) A negative value of EQLOPT3 results in the simulator dividing each grid cell into $2 N + 1$ horizontal sub-blocks for the equilibration calculation. This results in an accurate fluid in-place calculation at the expense of initialization stability, that is there may be some movement of fluids when there is no production at the start of the run. Increasing the value of N increases the accuracy of the calculation, with the maximum value of N being set to 20 by OPM Flow. 3) A positive value of EQLOPT3 results in the same option for when EQLOPT3 is negative, except that tilted fault blocks are used in the calculation. Again, increasing the value of N increases the accuracy of the calculation, with the maximum value of N being set to 20 by OPM Flow. Note this option should be used with Irregular Corner-Point Grids. EQLOPT3 is ignored for Radial Grids. Only EQLOPT3 options (1) and (2), that is values less than and equal to zero, are supported by OPM Flow.			0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
10	EQLOPT4	A positive integer value greater than or equal one and less than or equal to three, that sets the initialization option in the commercial compositional simulator. EQLOPT4 should be defaulted with 1*, as it is not used by OPM Flow.			None
		dimensionless	dimensionless	dimensionless	
11	EQLOPT5	A positive integer value that if set to one forces PRESS to be used for the datum pressure in the commercial compositional simulator. EQLOPT5 should be defaulted with either 1*, as it is not used by OPM Flow.			None
		dimensionless	dimensionless	dimensionless	
12	EQLOPT6	EQLOPT6 is an integer value that sets the initialization option for when vaporized water is present in the run, as activated by the VAPWAT keyword in the RUNSPEC section. Note this is an OPM Flow specific parameter for use with simulator's Vaporized Water Model. 1) A positive value of EQLOPT6 results in the vaporized water-gas ratio being calculated from data entered on the RVWVD keyword (vaporized water-gas ratio versus depth table). If this option is selected, then the RVWVD keyword must be present in the input deck Note that the allocation of multiple RVWVD tables to each grid cell is through the EQLNUM keyword and not the PVTNUM keyword. 2) A zero value of EQLOPT2 results in the vaporized water-gas ratio being set to the saturated vaporized water-gas ratio at the GOC. In this case DATUM must be equal GASCONT and the RVWVD keyword may be omitted. 3) A negative value of EQLOPT6 results in the same option for when EQLOPT6 is zero. EQLOPT6 is ignored if there is no vaporized water in the run.			0
		dimensionless	dimensionless	dimensionless	

Notes:
1) The keyword is followed by NTEQUL records as declared on the EQLDIMS keyword in the [RUNSPEC](#) section.
2) Each record is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.14: EQUIL Keyword Description

Note

A common method to initialize a model is by using the [SWATINIT](#) property array to set the initial water saturation for each cell in the model. This property is normally exported from a static model, where Saturation Height Functions (“SHF”) have been used to describe the water saturation profile with depth. In the dynamic model capillary pressure functions are used to described the water profile versus depth.

Note that if the [SWATINIT](#) array has been used to initialize the model then the fine grid block initialization via the EQLOPT3 variable, should not normally be used, and should be defaulted or set equal to zero; otherwise, the resulting water saturation will not strictly honor the [SWATINIT](#) array.

See also the [PRESSURE](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords in the [SOLUTION](#) section to initialize the model using the non-standard formulation of entering the pressures and saturations for each grid cell.

Example

--												
--	DATUM	DATUM	OWC	PCOW	GOC	PCGO	RS	RV	N	E300	RVW	
--	DEPTH	PRESS	DEPTH	----	DEPTH	----	OPT	OPT	OPT	OPTS	OPT	
EQUIL												
	3650.0	1560.0	3712.0	0.00	1000.0	0.00	1	0	-5	2*	1*	/
	3650.0	1560.0	3741.0	0.00	1000.0	0.00	1	0	-5	2*	1*	/
	3650.0	1560.0	3741.0	0.00	1000.0	0.00	1	0	-5	2*	1*	/

The above example defines three equilibration records for when NTEQUL equals three on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section. Here there is no gas cap and the GOC has been set to a value above the reservoirs (1000.0), and the value of EQLOPT3 (-5) has been explicitly stated.

10.3.37 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

10.3.38 FILEUNIT – ACTIVATE UNIT CONSISTENCY VERIFICATION

The [FILEUNIT](#) keyword defines the units of the data set, and is used to verify that the units in the input deck and any associated include files are consistent. The keyword does not provide for the conversion between different sets of units.

See [FILEUNIT – Activate Unit Consistency Checking](#) in the [GRID](#) section for a full description.

10.3.39 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be placed at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a full description.

10.3.40 GASCONC – DEFINE THE INITIAL EQUILIBRATION COAL GAS CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GASCONC](#) keyword defines the initial equilibration coal gas concentration values for all matrix grid cells in the model and should be used in conjunction with the [GCVD](#) keyword in the [SOLUTION](#) section, to fully describe the initial state of the model. The keyword should only be used if the coal phase has been activated in the model via the [COAL](#) keyword in the [RUNSPEC](#) section. Note both [GASCONC](#) and [GCVD](#) are optional as the simulator will calculate the coal gas concentration based on the equilibrium concentration and the block pressure.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	GASCONC	GASCONC is an array of real positive numbers that define the initial equilibration coal gas concentration values to each matrix cell in the model. Repeat counts may be used, for example 20*75.0.			None
		Mscf/ft3	sm ³ /m ³	scc/cc	

Notes:

- 1) The number of entries should correspond to number of matrix cells in the model, that is (NX x NY x NZ) divided by two, as per the parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 10.15: GASCONC Keyword Description

See also the [GCVD](#) keyword in the [SOLUTION](#) section to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION COAL GAS CONCENTRATION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 6
--
GASCONC      1000*75.500      1000*65.500      1000*60.000      /
```

The above example defines the initial equilibration coal gas concentration values to be 75.500 for all the matrix cells in the first layer, 65.500 for all the cells in the second layer, and finally 60.000 for all the cells in the third layer.

10.3.41 GASSATC – DEFINE THE INITIAL EQUILIBRATION SATURATED COAL GAS CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GASSATC](#) keyword defines the initial equilibration saturated coal gas concentration values for all grid cells in the model. The keyword should only be used if the coal phase has been activated in the model via the [COAL](#) keyword in the [RUNSPEC](#) section. The keyword is used to re-scale the Langmuir isotherms entered via the [LANGMUIR](#) keyword in the [PROPS](#) section, in conjunction with a matrix grid blocks initial reservoir pressure. The keyword is optional, and if absent from the input file, the matrix grid block Langmuir isotherm is left unscaled.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GASSATC	GASSATC is an array of real positive numbers that define the initial equilibration saturated coal gas concentration values to each cell in the model. Repeat counts may be used, for example 20*75.0.			None
		Mscf/ft3	sm ³ /m ³	scc/cc	

Notes:

- 1) The number of entries should correspond to number of matrix cells in the model, that is (NX x NY x NZ) divided by two, as per the parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 10.16: GASSATC Keyword Description

See also the [GASCONC](#) and the [GCVD](#) keywords in the [SOLUTION](#) section to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION SAT COAL GAS CONCENTRATION ALL CELLS MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 6
--
GASSATC      1000*75.500      1000*65.500      1000*60.000      /
```

The above example defines the initial equilibration saturated coal gas concentration values to be 75.500 for all the matrix cells in the first layer, 65.500 for all the cells in the second layer, and finally 60.000 for all the cells in the third layer.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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10.3.42 GCVD – DEFINE EQUILIBRATION COAL GAS CONCENTRATION VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [GCVD](#) keyword defines the initial coal gas concentration versus depth tables for each equilibration region for when the coal phase has been activated in the run via the [COAL](#) keyword in the [RUNSPEC](#) section. The keyword may be used in conjunction with the [GASCONC](#) keyword in the [SOLUTION](#) section, to fully describe the initial state of the model. Note both [GASCONC](#) and [GCVD](#) are optional as the simulator will calculate the coal gas concentration based on the equilibrium concentration and the block pressure.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding coal gas concentration, GCVALS.			None
		feet	m	cm	
2	GCVALS	A columnar vector of real values that defines the coal gas concentration values at the corresponding DEPTH.			None
		Mscf/ft3	sm ³ /m ³	scc/cc	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the EQLDIMS keyword in the RUNSPEC section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the EQLDIMS keyword in the RUNSPEC section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.17: GCVD Keyword Description

See also the [GASCONC](#) and [GASSATC](#) keywords in the [SOLUTION](#) section.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the coal gas concentration versus depth functions.

```
--
--      DEPTH      GC
--      MSCF/FT
--      -----
GCVD
      100.0      75.5000
      1000.0     75.5000                      / GC VS DEPTH EQUIL REGN 01
--      -----
      100.0      65.5000
      1000.0     65.5000                      / GC VS DEPTH EQUIL REGN 02
--      -----
      100.0      60.0000
      1000.0     60.0000                      / GC VS DEPTH EQUIL REGN 03
```

Here three tables are entered with a constant coal gas concentration versus depth relationship for each equilibration region.

10.3.43 GETDATA – LOAD AND ASSIGN DATA ARRAY FROM INIT OR RESTART FILE

The [GETDATA](#) keyword loads a data array from a previously generated [INIT](#) or [RESTART](#) file and assigns the loaded array to either same array in the run or another array name.

See [GETDATA – Load and Assign Data Array from INIT or RESTART Files](#) in the [GRID](#) section for a full description

10.3.44
GETGLOB – ACTIVATE LOADING OF GLOBAL GRID RESTART DATA OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [GETGLOB](#), switches on the global grid read option for when the run is restarting from a [RESTART](#) file. Only the global grid will be loaded in the subsequent [RESTART](#) keyword and any Local Grid Refinements (“LGR”) on the [RESTART](#) file will be ignored.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

```

--
--      ACTIVATE LOADING OF GLOBAL GRID RESTART DATA OPTION
--
GETGLOB

```

The above example switches on the option to only load the global grid from the [RESTART](#) file.

10.3.45
GI – DEFINE THE INITIAL EQUILIBRATION GI VALUES FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GI](#) keyword defines the initial equilibration [GI](#) values for all grid cells in the model and should be used in conjunction with the other enumeration equilibration keywords; [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the [GI](#) Pseudo Compositional option has been activated in the model via the [GIMODEL](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

See also the [GIALL](#) keyword in the [PROPS](#) section that sets the [GI](#) values as a function of pressure, as well as setting the corresponding [RVGI](#), [RSGI](#), [BGGI](#) and [BOGI](#) values at the same time.

10.3.46
HMAQUCT – HISTORY MATCH CARTER-TRACY AQUIFER GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMAQUCT](#) keyword defines the history match analytical Carter-Tracy aquifer gradient parameters for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and analytical Carter-Tracy aquifers have been specified in the model via the [AQUCT](#) and connected to the grid using the [AQUANCON](#) or [AQANCONL](#) keywords. All keywords are in the [SOLUTION](#) section.

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of aquifers that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.47
HMAQUFET – HISTORY MATCH FETKOVICH AQUIFER GRADIENT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMAQUFET](#) keyword defines the history match analytical Fetkovich aquifer gradient parameters for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and analytical Fetkovich aquifers have been specified in the model via the [AQUFET](#) and/or the [AQUFETP](#) keywords and connected to the grid using [AQUANCON](#) or [AQANCONL](#) keywords. All keywords are in the [SOLUTION](#) section.

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of aquifers that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.48
HMMLCTAQ – HISTORY MATCH CARTER-TRACY AQUIFER GRADIENT MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMMLCTAQ](#) keyword defines the history match analytical Carter-Tracy aquifer gradient multipliers for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and analytical Carter-Tracy aquifers have been specified in the model via the [AQUCT](#) and connected to the grid using the [AQUANCON](#) or [AQANCONL](#) keywords. All keywords are in the [SOLUTION](#) section.

Multipliers can be declared for the Carter-Tracy aquifer permeability, aquifer angle of influence and the aquifer depth.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.49
HMMLFTAQ
– HISTORY MATCH FETKOVICH AQUIFER GRADIENT MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [HMAQUFET](#) keyword defines the history match analytical Fetkovich aquifer gradient multipliers for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section, and analytical Fetkovich aquifers have been specified in the model via the [AQUFET](#) and/or the [AQUFETP](#) keywords and connected to the grid using [AQUANCON](#) or [AQANCONL](#) keywords. All keywords are in the [SOLUTION](#) section.

Multipliers can be declared for the Fetkovich aquifer water volume, aquifer permeability, and the aquifer depth.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.50 HMMLTWCN – HISTORY MATCH WELL CONNECTION AND SKIN MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [HMMLTWCN](#), defines the history match gradient multipliers for well connection factors and connection skins, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. Wells must be specified using the WELPSECS keyword in the [SCHEDULE](#) section and their connections defined by the [COMPDAT](#) and/or [COMPDATL](#) keywords, also in the [SCHEDULE](#) section

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.51
HMWELCON
– HISTORY MATCH WELL CONNECTION AND SKIN PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [HMWELCON](#), defines the history match gradient parameters for well connection factors and connection skins, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. Wells must be specified using the WELPSECS keyword in the [SCHEDULE](#) section and their connections defined by the [COMPDAT](#) and/or [COMPDATL](#) keywords, also in the [SCHEDULE](#) section

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.52
IMPORT – IMPORT GRID FILE DATA AT THE CURRENT POSITION

The [IMPORT](#) keyword informs the simulator to continue reading input data from the specified [IMPORT](#) file. When the end of the [IMPORT](#) file is reached, input data is read from the next keyword in the current file. Normally [IMPORT](#) files are generated by grid pre-processing software and the keyword allows for both formatted and unformatted (binary) files to be loaded.

See [IMPORT – Import Grid File Data at the Current Position](#) in the [GRID](#) section for a full description.

10.3.53
INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDINC](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the [GLOBAL](#) section for a full description.

10.3.54
MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the [GLOBAL](#) section for a full description.

10.3.55
MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the [GLOBAL](#) section for a full description.

10.3.56
MULTIPLY – MULTIPLY A SPECIFIED ARRAY BY A CONSTANT

The [MULTIPLY](#) keyword multiplies a specified array or part of an array by a constant. The constant can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [MULTIPLY – Multiply a Specified Array by a Constant](#) in the [GRID](#) section for a full description.

10.3.57
MULTIREG – MULTIPLY AN ARRAY BY A CONSTANT BASED ON A REGION NUMBER

The [MULTIREG](#) keyword multiplies an array or part of an array by a constant for cells with a specific region number. The region number array can be [FLUXNUM](#), [MULTNUM](#) or [OPERNUM](#) and these arrays must be defined and be available before the [MULTIREG](#) keyword is read by the simulator. The constant can be real or integer depending on the property array type; however, the property arrays that can be operated on is dependent on which section the [MULTIREG](#) keyword is being used.

See [MULTIREG – Multiply an Array by a Constant based on a Region Number](#) in the [GRID](#) section for a full description.

10.3.58 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active. but can subsequently be switched off by the *NOECHO* activation keyword.

See *NOECHO – Deactivate Echoing of User Input Files to the Print File* in the GLOBAL section for a full description.

10.3.59 NOHMD – DEACTIVATE HISTORY MATCH GRADIENT DERIVATIVE CALCULATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [NOHMD](#) deactivates various history match gradient derivative calculations for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The keyword consists of a series of character strings that define which derivative should be switch off based on the keyword that requested the derivatives to be calculated, for example [HMFaults](#) keyword in the [GRID](#) section. If an empty list is entered then all the gradient derivative calculations previously requested are switch off. The keyword is useful for changing from history matching runs to predication cases, as the prediction cases will be more computationally efficient without the burden of the gradient derivative calculations.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.60
NOHMO
– DEACTIVATE HISTORY MATCH GRADIENT DERIVATIVE CALCULATIONS (ALIAS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [NOHMO](#) deactivates various history match gradient derivative calculations for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The keyword consists of a series of character strings that define which derivative should be switch off based on the keyword that requested the derivatives to be calculated, for example [HMFAULTS](#) keyword in the [GRID](#) section. If an empty list is entered then all the gradient derivative calculations previously requested are switch off. The keyword is useful for changing from history matching runs to predication cases, as the prediction cases will be more computationally efficient without the burden of the gradient derivative calculations.

The keyword is an alias for the [NOHMD](#) keyword in the [SOLUTION](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.61 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

10.3.62 OILAPI – DEFINE THE INITIAL EQUILIBRATION OIL API FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *OILAPI* keyword defines the initial equilibration oil *API* gravity pressures for all grid cells in the model, for when the Oil *API* Tracking option as been invoked by the *API* keyword in the *RUNSPEC* section. The keyword should be used in conjunction with the *PBUB*, *PDEW*, *PRESSURE*, *RS*, *RV*, *SGAS*, *SOIL* and *SWAT* keywords etc., to fully describe the initial state of the model.

The keyword is used by the Enumeration Initialization method to initialize the model, as opposed to the Equilibration Initialization method that utilizes the *EQUIL* keyword in the *SOLUTION* section. This is the non-standard formulation to initialize the model and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the *EQUIL* keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	OILAPI	OILAPI is an array of real positive numbers assigning the initial equilibration oil API gravity to each cell in the model. The American Petroleum Institute (“API”) classifies oils based on an API gravity (γ_{API}), or degrees API ($^{\circ}API$), the relationship between relative density (γ_o) of oil and API gravity (γ_{API}) is given by: $\gamma_{API} = \frac{141.5}{\gamma_o} - 131.5$ Repeat counts may be used, for example 20*38.5			None
		$^{\circ}API$	$^{\circ}API$	$^{\circ}API$	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.18: OILAPI Keyword Description

See also the *PBUB*, *PDEW*, *PRESSURE*, *RS*, *RV*, *SGAS*, *SOIL* and *SWAT* keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION OIL API FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
OILAPI      1000*40.2      1000*39.5      1000*38.2      /
```

The above example defines the initial equilibration oil *API* gravity to be 40.2 for all the cells in the first layer, 39.5 for all the cells in the second layer, and finally 38.2 for all the cells in the third layer.

10.3.63 OPERATE – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS

This keyword, [OPERATE](#), defines mathematical operations on property arrays ([NTG](#), [PORO](#) etc.) and optionally using another property array as input to the function. The keyword allows for various mathematical functions and their associated variables to be defined and applied to the selected array data. Input constants can be real or integer depending on the array type; however, the arrays that can be operated on is dependent on which section the keyword is being used.

See [OPERATE – Define Mathematical Operations on Arrays](#) in the [GRID](#) section for a full description.

10.3.64 OPERATER – DEFINE MATHEMATICAL OPERATIONS ON ARRAYS BY REGION

This keyword defines the mathematical operations on arrays for specific regions in the commercial simulator and is currently not supported by OPM Flow. However, similar functionality is provided by the [ADD](#) and [MULTIPLY](#) keywords.

See [OPERATER – Define Mathematical Operations on Arrays by Region](#) in the [GRID](#) section for a full description.

10.3.65

OUTSOL – DEFINE DATA TO BE WRITTEN TO THE RESTART FILE (RETIRED)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the data and frequency of the data to be written to the [RESTART](#) file at each requested restart point. The keyword has been replaced by the [RPTRST](#) keyword in the [SOLUTION](#) and [SCHEDULE](#) sections and is therefore considered retired.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

10.3.66 PBUB – DEFINE THE INITIAL EQUILIBRATION BUBBLE-POINT PRESSURE FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [PBUB](#) keyword defines the initial equilibration bubble-point saturation pressures values for all grid cells in the model and should be used in conjunction with the [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if dissolved gas has been activated in the model via the [DISGAS](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PBUB	PBUB is an array of real positive numbers assigning the initial equilibration bubble-point saturation pressure values to each cell in the model. Repeat counts may be used, for example 20*3500.0			None
		psia	barsa	atma	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.19: PBUB Keyword Description

See also the [PBVD](#), [PDEW](#), [PRESSURE](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION PSAT VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
PBUB      1000*3500.0      1000*3525.0      1000*0.3535.0      /
```

The above example defines the initial equilibration bubble-point saturation pressure values to be 3500.0 for all the cells in the first layer, 3525.0 for all the cells in the second layer, and finally 3535.0 for all the cells in the third layer.

10.3.67 PBVD – EQUILIBRATION BUBBLE-POINT VERSUS DEPTH TABLES

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Description

The *PBVD* keyword defines the bubble-point pressure versus depth tables for each equilibration region that should be used when there is dissolved gas in the model (*DISGAS* has been activated in the *RUNSPEC* section) and the *EQLOPT1* variable has been set to a positive integer on the *EQUIL* keyword in the *SOLUTION* section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	<i>DEPTH</i>	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding bubble-point values, PBVALS.			None
		feet	m	cm	
2	PBVALS	A columnar vector of real values that defines the oil bubble-point values at the corresponding <i>DEPTH</i> .			None
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the *EQLDIMS* keyword in the *RUNSPEC* section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the *EQLDIMS* keyword in the *RUNSPEC* section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 10.20: PBVD Keyword Description

Alternatively, the dissolved gas-oil ratio versus depth tables may be entered using the [RSVD](#) keyword in the [SOLUTION](#) section instead of this keyword. See also the [RSVD](#) and [EQUIL](#) keywords in the [SOLUTION](#) section.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the bubble-point versus depth functions.

DEPTH	PSAT PRESS	
3000.0	3000.0	
8000.0	3025.0	/ PSAT VS DEPTH EQUIL REGN 01
3000.0	3100.0	
8000.0	3125.0	/ PSAT VS DEPTH EQUIL REGN 02
3000.0	3200.0	
8000.0	3225.0	/ PSAT VS DEPTH EQUIL REGN 03

Here three tables are entered and each table is terminated by a “/” and there is no keyword terminating “/”.

10.3.68 PDEW – DEFINE THE INITIAL EQUILIBRATION DEW-POINT PRESSURE FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [PDEW](#) keyword defines the initial equilibration dew-point pressure values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if vaporized oil been activated in the model via the [VAPOIL](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	PDEW	PDEW is an array of real positive numbers assigning the initial equilibration dew-point pressure values to each cell in the model. Repeat counts may be used, for example 20*3525.0			None
		psia	barsa	atma	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.

Table 10.21: PDEW Keyword Description

See also the [PBUB](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION PSAT VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
PDEW      1000*3500.0      1000*3525.0      1000*0.3535.0      /
```

The above example defines the initial equilibration dew-point saturation pressure values to be 3500.0 for all the cells in the first layer, 3525.0 for all the cells in the second layer, and finally 3535.0 for all the cells in the third layer.

10.3.69 PDVD – DEFINE EQUILIBRATION DEW-POINT VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [PDVD](#) keyword defines the dew-point pressure versus depth tables for each equilibration region that should be used when there is vaporized oil in the model ([VAPOIL](#) has been activated in the [RUNSPEC](#) section) and the [EQLOPT2](#) variable has been set to a positive integer on the [EQUIL](#) keyword in the [SOLUTION](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding dew-point values, PDVALS.			None
		feet	m	cm	
2	PDVALS	A columnar vector of real values that defines the gas dew-point values at the corresponding DEPTH .			None
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.22: PDVD Keyword Description

Alternatively, the vaporized oil-gas ratio (condensate-gas ratio) versus depth tables may be entered using the [RVVD](#) keyword in the [SOLUTION](#) section instead of this keyword.

See also the [RVVD](#) and [EQUIL](#) keywords in the [SOLUTION](#) section.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the bubble-point versus depth functions.

```
--
--      DEPTH      PSAT
--      PRESS
--      -----
PDVD
      3000.0      2000.0
      8000.0      2025.0                               / PSAT VS DEPTH EQUIL REGN 01
--      -----
      3000.0      2100.0
      8000.0      2125.0                               / PSAT VS DEPTH EQUIL REGN 02
--      -----
      3000.0      2200.0
      8000.0      2225.0                               / PSAT VS DEPTH EQUIL REGN 03
```

Here three tables are entered and each table is terminated by a "/" and there is no keyword terminating "/".

10.3.70 PRESSURE – DEFINE THE INITIAL EQUILIBRATION PRESSURES FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [PRESSURE](#) keyword defines the initial equilibration pressures for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model.

The keyword is used by the Enumeration Initialization method to initialize the model, as opposed to the Equilibration Initialization method that utilizes the [EQUIL](#) keyword in the [SOLUTION](#) section. This is the non-standard formulation to initialize the model and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	PRESSURE	PRESSURE is an array of real positive numbers assigning the initial equilibration pressures to each cell in the model. Repeat counts may be used, for example 20*4200.0.			None
		psia	barsa	atma	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.

2)

The keyword is terminated by a “/”.

Table 10.23: PRESSURE Keyword Description

See also the [PBUB](#), [PDEW](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION PRESSURES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
PRESSURE      1000*4500.0    1000*4510.0    1000*4520.0      /
```

The above example defines the initial equilibration pressures to be 4500.0 for all the cells in the first layer, 4510.0 for all the cells in the second layer, and finally 4520.0 for all the cells in the third layer.

10.3.71 PRVD – DEFINE THE INITIAL EQUILIBRATION PRESSURES VERSUS DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [PRVD](#) keyword defines the initial reservoir pressure versus depth and should be used in conjunction with the [PBUB](#), [PDEW](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. [PRVD](#) is an alternative to the [PRESSURE](#) keyword in the [SOLUTION](#) section, that defines the initial equilibration pressures for all grid cells in the model

The keyword is used by the Enumeration Initialization method to initialize the model, as opposed to the Equilibration Initialization method that utilizes the [EQUIL](#) keyword in the [SOLUTION](#) section. This is the non-standard formulation to initialize the model and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding reservoir oil pressures values, PRESSURE .			None
		feet	m	cm	
2	PRESSURE	A columnar vector of real values that defines the initial equilibration oil pressure values at the corresponding DEPTH .			None
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.24: PRVD Keyword Description

See also the [PBUB](#), [PDEW](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to five on the *EQLDIMS* keyword in the *RUNSPEC* section, then the following example defines the initial oil reservoir pressure versus depth

--			
--	DEPTH	INIT	
--		PRESS	
--	-----	-----	
PRVD			
	3000.0	3000.0	
	4000.0	3345.0	
	5000.0	3690.0	
	7000.0	4700.0	
	7200.0	4769.0	/ POIL VS DEPTH EQUIL REGN 01
--	-----	-----	
	3000.0	3100.0	
	4000.0	3445.0	
	5000.0	3790.0	
	7000.0	4700.0	
	7200.0	4769.0	/ POIL VS DEPTH EQUIL REGN 02
--	-----	-----	
	3000.0	3150.0	
	4000.0	3495.0	
	5000.0	3840.0	
	7000.0	4700.0	
	7200.0	4769.0	/ POIL VS DEPTH EQUIL REGN 03

Here three tables are entered and each table is terminated by a "/" and there is no keyword terminating "/".

10.3.72 PYEND – END THE DEFINITION OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword on a separate single line.

See [PYEND – End the Definition of a PYINPUT Section](#) in the [GRID](#) section for a full description.

10.3.73 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator. [PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line.

See [PYINPUT – Define the Start of a PYINPUT Section](#) in the [GRID](#) section for a full description.

10.3.74 RAINFALL – CONSTANT FLUX AQUIFER RAINFALL FLUX BY MONTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This [RAINFALL](#) keyword defines the month by month rainfall flux for constant flux aquifers.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.75 RBEDCONT – DEFINE RIVER GRID BLOCK CONTACT AREA VERSUS DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [RBEDCONT](#) keyword defines the river grid block contact area versus depth tables, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.76 REFINE – START THE DEFINITION OF A LOCAL GRID REFINEMENT

The [REFINE](#) keyword defines the start of a Cartesian or radial local grid refinement (“LGR”) definition that sets the properties of the selected [LGR](#). The keyword is then followed by the property keywords associated with the section where the keyword is being invoked. For example, if the [REFINE](#) keyword is used in the [GRID](#) section then most of the keywords in that section can be used to set the grid properties for the [LGR](#).

See [REFINE – Start the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

10.3.77 RESTART – RESTART RUN FROM AN EXISTING RESTART FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [RESTART](#) keyword defines the parameters to restart the simulation from a previous run that has written a [RESTART](#) file out to disk. Only restarting from [RESTART](#) files is permitted by OPM Flow; restarting from [SAVE](#) files is not implemented.

No.	Name	Description	Default
1	RSNAME	The RSNAME variable is a character string that defines the root name of the RESTART file to be read into the current input deck.	None
2	RSNUM	A positive integer that defines the restart point on the RESTART file to be read and to be used to initialize the model. When OPM Flow writes a restart point a message is printed to the *.PRT file indicating the time step the restart was written out.	None
3	RSTYPE	A defined character sting set to SAVE to read the restart data from the SAVE file, otherwise defaulted to I* to read the data from the RESTART file. The SAVE file option is not supported by OPM Flow and should be defaulted with I*.	I*
4	RSFORMAT	A defined character string that defines the format of the SAVE file to be read if RSTYPE has been set to SAVE , and should be set to one of the following: 1) FORMATTED: If the file is formatted as ASCII i.e. a text file, as oppose to a binary file. The option can be abbreviated to just the letter F. 2) UNFORMATTED: If the file is in binary format, note this option can be abbreviated to just the letter U. This type of file is operating system dependent If the variable RSFORMAT omitted then the default is for binary file input. This option is not supported by OPM Flow and should be defaulted with I*.	U

Notes:

- 1) OPM Flow can only restart runs from a [RESTART](#) file, the commercial simulator's [SAVE](#) file format is not supported.
- 2) Note that due to the complexities of the [RESTART](#) file, OPM Flow may not always be able to restart from the commercial simulators [RESTART](#) file.
- 3) The keyword is terminated by a “/”.

Table 10.25: RESTART Keyword Description

The most direct way to start a restart run is to:

- 1) Copy the existing data file that created the [RESTART](#) file and give it a new name. For example if the [RESTART](#) file is from a case named NOR-OPM-A01.DATA, then the copied data file could be named NOR-OPM-A01-R1.DATA.
- 2) Edit the copied data file (NOR-OPM-A01-R1.DATA) and delete all equilibration keywords ([EQUIL](#), [RSVD](#), etc.) or the enumeration equilibration keywords ([PRESSURE](#), [SGAS](#), [SOIL](#), [SWAT](#), etc.) in the [SOLUTION](#) section used to initialize the model.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

- 3)

In the [SOLUTION](#) section of NOR-OPM-A01-RI.DATA file insert the [RESTART](#) keyword, using NOR-OPM-A01 as RSNAME and the required RSNUM value for the time step to restart from.
- 4)

In the [SCHEDULE](#) section of NOR-OPM-A01-RI.DATA file insert the [SKIPREST](#) keyword at the very beginning of the [SCHEDULE](#) section. The [SKIPREST](#) keyword causes the simulator to only read in data it requires for restarting the run up to the [RESTART](#) point (RSNUM). Note that certain keywords always need to be present in a restart run in the [SCHEDULE](#) section as the data is not stored on the [RESTART](#) file, for example the VFP tables ([VFPPROD](#) and [VFPINJ](#) keywords). The [SKIPREST](#) keyword automatically processes the input deck and reads the required data.
- 5)

In the [SCHEDULE](#) section of NOR-OPM-A01-RI.DATA file after the [RESTART](#) point make any required changes, save the file and run the NOR-OPM-A01-RI.DATA with OPM Flow.

See also [RPTRST](#), [RPTSCHED](#) and [SKIPREST](#) keywords.

Example

The example below defines a restart from the previously run NOR-OPM-A01 case at time step number 40.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      FLEXIBLE RESTART FROM PREVIOUS SIMULATION RUN
--
--      FILE          RESTART   RESTART   FILE
--      NAME          NUMBER    TYPE      FORMAT
RESTART
      'NOR-OPM-A01'   40        1*       1*      /
```

In addition in the [SCHEDULE](#) section the [SKIPREST](#) keyword should be used to correctly read in the schedule data up to the [RESTART](#) point.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      ACTIVATE SKIPREST OPTION TO AVOID MODIFYING SCHEDULE SECTION
--
SKIPREST
```

Note is advisable to place the [SKIPREST](#) keyword at the very beginning of the [SCHEDULE](#) section.

10.3.78
RIVERSYS – DEFINE RIVER SYSTEM (BRANCH STRUCTURE AND BOUNDARY CONDITIONS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

RIVERSYS defines a river system by specifying the branch structure of the river together with the branch's associated boundary conditions, for when the River option has been activated via the RIVRDIMS keyword in the RUNSPEC section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.79 RPTRST – DEFINE DATA TO BE WRITTEN TO THE RESTART FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the data to be written to the [RESTART](#) file and the frequency at which restart points will be created. In addition to the solution data arrays required to restart a run, the user may request extra data to be written to the restart file for visualization in OPM ResInsight.

The [RPTRST](#) keyword should be followed by a series of character strings that indicate the data to be written. In most cases the character string is the keyword used to load the data in the OPM Flow input deck, for example [KRG](#) for the gas relative permeability of each grid block at the requested times. It is anticipated that OPM Flow will support additional functionality as development progresses.

No.	Mnemonic	Description	Array Name
1	ALLPROPS	An alias for the BG, BO, BW, DEN, KRG , KRO , KRW , VGAS, VOIL and VWAT mnemonics combined that writes all the associated properties.	
2	BASIC	The BASIC mnemonic defines the frequency at which the restart data and the additional requested data is written to the RESTART file. The mnemonic is assigned a value, using the form BASIC=OPTION, where OPTION is an integer variable set to one of the following: 1) OPTION = 0 then output to the restart file is stopped. 2) OPTION = 1 then the restart files are written at every report time, but only the last output is kept. This minimizes the restart file size but limits the visualization in OPM ResInsight to the final conditions. 3) OPTION = 2 then the restart files are written at every report time until this switch is reset and all the restarts are kept. 4) OPTION = 3 then the restart files are written every n^{th} report time with the frequency determined by the mnemonic "FREQ=n". 5) OPTION = 4 then the restart files are written at the first report time of each year. If FREQ=n is set greater than one then restart files will be written every n^{th} year. 6) OPTION = 5 then the restart files are written at the first report time of each month. If FREQ=n is set greater than one then restart files will be written every n^{th} month. 7) OPTION = 6 then the restart files are written after every time step. This option is not supported by OPM Flow and the simulation will be stopped.	
3	BG	Gas formation volume factor.	I OVERBG
4	BO	Oil formation volume factor.	I OVERBO
5	BW	Water formation volume factor.	I OVERBW
6	CONV	Identify the worst n cells causing convergence problems based on the residuals of each equation (CONV_XXX), using the form CONV=n. In addition, the number of Newton iterations required by each cell in order to satisfy the solution change convergence criteria at the last time step (CONV_NEW).	CNV_GAS CNV_OIL CNV_WAT CONV_NEW

No.	Mnemonic	Description	Array Name
7	DEN	Fluid phase densities at reservoir conditions.	GAS_DEN OIL_DEN WAT_DEN
8	<i>FIP</i>	Volumes in place for each phase at surface conditions.	FIPGAS FIPOIL FIPWAT
9	FLAWS	Inter-block volumetric flows at surface conditions for each phase (<i>GAS</i> , <i>OIL</i> , <i>WAT</i>) in the I+, J+ and K+ directions. The Array Name column shows the arrays for just the <i>GAS</i> phase, the other phases will have similar arrays.	FLOGASI+ FLOGASJ+ FLOGASK+
10	FLAWS-	Inter-block volumetric flows at surface conditions for each phase (<i>GAS</i> , <i>OIL</i> , <i>WAT</i>) in the I-, J- and K- directions. The Array Name column shows the arrays for just the <i>GAS</i> phase, the other phases will have similar arrays.	FLOGASI- FLOGASJ- FLOGASK-
11	FLORES	Inter-block volumetric flows at reservoir conditions for each phase (<i>GAS</i> , <i>OIL</i> , <i>WAT</i>) in the I+, J+ and K+ directions. The Array Name column shows the arrays for just the <i>GAS</i> phase, the other phases will have similar arrays.	FLRGASI+ FLRGASJ+ FLRGASK+
12	FLORES-	Inter-block volumetric flows at reservoir conditions for each phase (<i>GAS</i> , <i>OIL</i> , <i>WAT</i>) in the I-, J- and K- directions. The Array Name column shows the arrays for just the <i>GAS</i> phase, the other phases will have similar arrays.	FLRGASI- FLRGASJ- FLRGASK-
13	FREQ	The FREQ mnemonic defines the frequency at which restart data is written when BASIC=3, 4 or 5, using the form FREQ=n.	
14	<i>KRG</i>	Gas relative permeability at the grid block's gas saturation.	GASKR
15	<i>KRO</i>	Oil relative permeability at the grid block's oil saturation.	OILKR
16	<i>KRW</i>	Water relative permeability at the grid block's water saturation.	WATKR
17	PBPD	Bubble point and dew point pressures.	<i>PBUB</i> <i>PDEW</i>
18	PCGW	Gas-water capillary pressure (gas-water simulations).	PCGW
19	PCOG	Gas-oil capillary pressure.	PCOG
20	PCOW	Oil-water capillary pressure.	PCOW
21	RESIDUAL	Component residuals after the last Newton iteration.	RES_GAS RES_OIL RES_WAT
22	RFIP	Volumes in place for each phase at reservoir conditions.	RFIPGAS RFIPOIL RFIPWAT

No.	Mnemonic	Description	Array Name
23	ROCKC	Rock compaction: Overburden pressure, minimum oil pressure, maximum water saturation, pore volume multiplier and transmissibility multiplier.	PRES_OVB PRESROCC SWMAX PORV_RC TMULT_RC
24	RSSAT	Saturated dissolved gas-oil ratio.	RSSAT
25	RSWSAT	Saturated dissolved gas-water ratio. This is an OPM Flow specific mnemonic that requires the command line option --enable-opm-rst-file=true is specified.	RSWSAT
26	RVSAT	Saturated vaporized oil-gas ratio.	RVSAT
27	RVWSAT	Saturated vaporized water-gas ratio. This is an OPM Flow specific mnemonic that requires the command line option --enable-opm-rst-file=true is specified.	RVWSAT
28	SFIP	Volumes in place for each phase at surface/separator conditions.	SFIPGAS SFIPOIL SFIPWAT
29	VGAS	Gas viscosity at reservoir conditions.	GAS_VISC
30	VISC	Fluid phase viscosities at reservoir conditions.	GAS_VISC OIL_VISC WAT_VISC
31	VOIL	Oil viscosity at reservoir conditions.	OIL_VISC
32	VWAT	Water viscosity at reservoir conditions.	WAT_VISC
Notes: 1) The keyword is terminated by a "/".			

Table 10.26: RPTRST Keyword Description

Note, older versions of the commercial simulator used integer values to control the output to the [RESTART](#) file, this type of format is not supported by OPM Flow and thus the integers should be converted to the mnemonic format as outlined in Table 10.26.

Note that OPM Flow automatically writes out all the data required to make a restart run as outlined in the table below:

No.	Mnemonic	Description	Array Name
I	BIOFILM	Biofilm volume fraction used when the BIOFILM or MICP option has been activated in the RUNSPEC section.	BIOFILM

No.	Mnemonic	Description	Array Name
2	CALCITE	Calcite volume fraction used when the <i>MICP</i> option has been activated in the <i>RUNSPEC</i> section.	CALCITE
3	<i>FOAM</i>	Foam concentration used when the <i>FOAM</i> option has been activated in the <i>RUNSPEC</i> section.	<i>FOAM</i>
4	MICROBES	Microbe concentration used when the <i>BIOFILM</i> or <i>MICP</i> option has been activated in the <i>RUNSPEC</i> section.	MICROBES
5	<i>MULTPV</i>	Pore volume multiplier output when <i>MULTPV</i> keyword is present in the input deck.	<i>MULTPV</i>
6	OXYGEN	Oxygen concentration used when the <i>MICP</i> option has been activated in the <i>RUNSPEC</i> section.	OXYGEN
7	<i>PERMFACT</i>	Permeability multiplication factor as a function of porosity change used when the <i>PRECSALT</i> option has been activated in the <i>RUNSPEC</i> section	<i>PERMFACT</i>
8	<i>POLYMER</i>	Polymer concentration used when the <i>POLYMER</i> option has been activated in the <i>RUNSPEC</i> section	<i>POLYMER</i>
9	PPCW	Modified oil-water capillary pressure end-points used when the <i>SWATINIT</i> keyword has been included in the input deck.	PPCW
10	<i>PRESSURE</i>	Pressure data for each grid block.	<i>PRESSURE</i>
11	<i>RS</i>	Dissolved gas-oil ratio for each grid block.	<i>RS</i>
12	<i>RSW</i>	Dissolved gas-water ratio for each grid block for use when the <i>DISGASW</i> option has been activated in the <i>RUNSPEC</i> section.	<i>RSW</i>
13	RSWSOL	Dissolved solvent-water ratio for each grid block for use when the <i>SOLVENT</i> and <i>DISGASW</i> options have been activated in the <i>RUNSPEC</i> section.	RSWSOL
14	<i>RV</i>	Vaporized oil-gas ratio for each grid block.	<i>RV</i>
15	<i>RVW</i>	Vaporized water-gas ratio for each grid block for use when the <i>VAPWAT</i> option has been activated in the <i>RUNSPEC</i> section.	<i>RVW</i>
16	<i>SALT</i>	Salt concentration for each grid block for use when the <i>BRINE</i> option has been activated in the <i>RUNSPEC</i> section.	<i>SALT</i>
17	<i>SALTP</i>	Precipitated salt volume fraction for each grid block for use when the <i>PRECSALT</i> option has been activated in the <i>RUNSPEC</i> section.	<i>SALTP</i>
18	<i>SGAS</i>	Gas saturation for each grid block.	<i>SGAS</i>
19	SGMAX	Maximum gas saturation reached used when gas-oil non-wetting phase relative permeability hysteresis has been activated.	SGMAX
20	SHMAX	Maximum wetting phase saturation reached used when gas-oil wetting phase relative permeability hysteresis has been activated.	SHMAX
21	<i>SOIL</i>	Oil saturation each grid block.	<i>SOIL</i>

No.	Mnemonic	Description	Array Name
22	SOMAX	Maximum oil saturation reached used when the VAPPARS keyword has been included in the input deck, or when oil-water non-wetting phase relative permeability hysteresis has been activated.	SOMAX
23	SOMIN	Minimum oil saturation reached used when gas-oil capillary pressure hysteresis has been activated.	SOMIN
24	SSOLVENT	Solvent saturation for each grid block used when the SOLVENT option has been activated in the RUNSPEC section.	SSOLVENT
25	SWAT	Water saturation for each grid block.	SWAT
26	SWHYI	Minimum water saturation reached used when oil-water capillary pressure hysteresis has been activated.	SWHYI
27	SWMAX	Maximum water saturation reached used when oil-water wetting phase relative permeability hysteresis has been activated.	SWMAX
28	TEMP	Temperature of each grid block used when the THERMAL option has been activated in the RUNSPEC section. OPM Flow also supports the output of temperatures for non-thermal simulations when the TEMP mnemonic is specified.	TEMP
29	UREA	Urea concentration used when the MICP option has been activated in the RUNSPEC section.	UREA
Notes: - Only items that are necessary to restart a run are written to the restart file, for example if the SOLVENT option has not been activated in the RUNSPEC section then the SSOLVENT array will not be written to the restart file. - The tracer distribution is output when the TRACERS option has been activated in the RUNSPEC section. The array name is the tracer name followed by either 'F' (free) or 'S' (solution).			

Table 10.27: Data Sets Automatically Written to the RESTART File

Note

Currently, OPM Flow distinguishes between those arrays which are required for restarting a run and those which are "merely" for visualization and analysis. However, this classification is imperfect and leads to potentially exporting arrays that are incompatible with the commercial simulator, such as the [TEMP](#) array in non-thermal simulation runs.

If one is willing to forego the ability to restart an OPM Flow simulation run, using the commercial simulator, e.g., simulating the historic period using OPM Flow and the prediction period using the commercial simulator, then additional arrays are available, including the [PBUB](#) and [PDEW](#) vectors generated by the PBDP option, by using the following command line option:

```
--enable-opm-rst-file=true
```

If the `--enable-opm-rst-file=true` command line option has been specified then the following OPM Flow specific data are also automatically output for visualization only as outlined in the table below:

No.	Mnemonic	Description	Array Name
1	XMFCO2	CO2 liquid-phase mole fractions written when the CO2STORE option has been activated in the RUNSPEC section.	XMFCO2
2	XMFH2	H2 liquid-phase mole fractions written when the H2STORE option has been activated in the RUNSPEC section.	XMFH2
3	YMFWAT	Water gas-phase mole fractions written when either the CO2STORE or H2STORE option has been activated in the RUNSPEC section.	YMFWAT
Notes: The above OPM Flow specific data are only written to the restart file if the --enable-opm-rst-file=true command line option has been specified and the specified options have been activated in the RUNSPEC section.			

Table 10.28: Data Sets Automatically Written to the RESTART File for Visualization Only

Examples

The first example request that the standard restart data be written out every month.

```
--
--      RESTART CONTROL BASIC = 4 (YEARLY) 5 (MONTHLY)
--
RPTRST
      BASIC=5
```

The next example requests that the standard restart data be written at every report time step until this switch is reset and all the restarts are kept. In addition to the standard the data the gas, oil and water relative permeability data will also be written out at each report time step.

```
--
--      RESTART CONTROL BASIC = 4 (YEARLY) 5 (MONTHLY)
--
RPTRST
      BASIC=2   KRG   KRO   KRW
```

10.3.80 RPTSOL – DEFINE SOLUTION SECTION REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the data in the [SOLUTION](#) section that is to be printed to the output print file in human readable format. The keyword has two distinct forms, the first of which consists of the keyword followed by a series of integers on the next line indicating the data to be printed (see the first example). This is the original formal in the commercial simulator and was subsequently superseded by the second format. The second format consists of the keyword followed by a series of character strings that indicate the data to be printed. In most cases the character string is the keyword used to load the data in the OPM Flow input deck, for example [PVDG](#) for the dry gas PVT tables. Its is anticipated that OPM Flow will eventually support the functionality of the second format only, the first format although recognized will be completely ignored.

OPM Flow provides only limited supported for this keyword and will ignore the unsupported options because they have no effect on the results.

No.	Name	Description	Default
1	DENO	Print the oil reservoir density array	N/A
2	EQUIL	Print the equilibration report.	N/A
3	FIP	Print the fluid in-place report. The parameter is assigned a value, OPTION, using the form <i>FIP</i> =OPTION, where OPTION is an integer variable set to: 1) OPTION = 1 then the report is for the field only. 2) OPTION = 2 then in addition to the field report, a report is produced for each <i>FIPNUM</i> region, as defined by the <i>FIPNUM</i> keyword in the <i>REGIONS</i> section. Note the commercial simulator also prints the flows to other regions as well as the flows from the wells. This additional reporting option has not been implemented in OPM Flow. 3) OPTION = 3 then in addition to the above, a balance report is also produced for fluid in-place regions defined by the <i>FIP</i> keyword in the <i>REGIONS</i> section.	<i>FIP</i> =2
4	FIPRESV	Print the reservoir volumes in-place report.	N/A
5	WELSPECS	<i>WELSPECS</i> switches on reporting of the well connections, wells and groups at each report time step. There are numerous reports associated with this option. Unlike the other reporting parameters that produce a report for each reporting time step, the <i>WELSPECS</i> report option only produces a report if an associated keyword has been activated at the current reporting time step. For example, if the reporting time steps are January, February, and March 2020, and the <i>RPTSCHED WELSPECS</i> option is activated in January, with wells OP01 and OP02 being declared via the <i>WELSPECS</i> and <i>COMPDAT</i> keywords, then a report will be printed for January for these two wells. If there are no further well activations until March, with well OP03 being declared, then there will be no report for February, and only well OP03 will reported at the March reporting time step.	N/A
Notes: 1) The keyword is terminated by a “/”.			

Table 10.29: RPTSOL Keyword Description

Note

Except for non-array like data, *FIP* etc., this keyword has the potential to produce very large print files that some text editors may have difficulty loading. A more efficient solution for array type data is to load the *.INIT and *.RESTART files into OPM ResInsight to view the data graphically, this also has the benefit of being able to filter the grid based on I, J, K ranges and grid properties.

Examples

The first example shows the original format of this keyword; although the keyword and format are recognized by OPM Flow, the format is ignored and is unlikely to be implemented in the simulator.

```
--
--      DEFINE SOLUTION SECTION REPORT OPTION (ORIGINAL FORMAT)
--
RPTSOL      1      2*0      1      3*1      /
```

The next example shows the second format of the keyword which may be supported in a future release of OPM Flow.

```
--
--      DEFINE SOLUTION SECTION REPORT OPTIONS
--
RPTSOL      FIP=2      FIPRESV      RESTART=3      /
```

10.3.81 RS – DEFINE THE INITIAL EQUILIBRATION GOR (RS) FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [RS](#) keyword defines the initial equilibration gas-oil ratio values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if dissolved gas has been activated in the model via the [DISGAS](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RS	RS is an array of real positive numbers assigning the initial equilibration gas-oil ratio values to each cell in the model. Repeat counts may be used, for example 20*1.30.			None
		Mscf/stb	sm ³ /sm ³	scc/scc	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.30: RS Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION GOR VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
RS      1000*1.3500   1000*1.3010   1000*1.3000   /
```

The above example defines the initial equilibration GOR values to be 1.3500 for all the cells in the first layer, 1.3010 for all the cells in the second layer, and finally 1.3000 for all the cells in the third layer.

10.3.82 RSVD – EQUILIBRATION DISSOLVED GAS-OIL RATIO (Rs) VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [RSVD](#) keyword defines the dissolved gas-oil ratio (Rs) versus depth tables for each equilibration region that should be used when there is dissolved gas in the model ([DISGAS](#) has been activated in the [RUNSPEC](#) section) and the EQLOPTI variable has been set to a positive integer on the [EQUIL](#) keyword in the [SOLUTION](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding dissolve gas-oil ratio values, RS .			None
		feet	m	cm	
2	RS	A columnar vector of real values that defines the dissolved gas-oil ratio values at the corresponding DEPTH .			None
		Mscf/stb	sm ³ /sm ³	scc/scc	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.31: RSVD Keyword Description

Alternatively, the oil bubble-point pressure versus depth tables may be entered using the [PBVD](#) keyword in the [SOLUTION](#) section instead of this keyword.

See also the [PBVD](#) and [EQUIL](#) keywords in the [SOLUTION](#) section.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the bubble-point versus depth functions.

```
--
--      DEPTH      RS
--      MSCF/STB
--      -----
RSVD
      3000.0      1.400
      8000.0      1.400                               / RS VS DEPTH EQUIL REGN 01
--      -----
      3000.0      1.400
      8000.0      1.400                               / RS VS DEPTH EQUIL REGN 02
--      -----
      3000.0      1.400
      8000.0      1.400                               / RS VS DEPTH EQUIL REGN 03
```

Here three tables are entered with a constant GOR versus depth relationship.

10.3.83 RSW – DEFINE THE INITIAL EQUILIBRIUM SOLUTION GAS IN WATER RATIO FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [RSW](#) keyword defines the initial equilibration solution gas in water ratio values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if both gas and water phases have been activated in the model via the [GAS](#) and [WATER](#) keywords, and the [DISGASW](#) keyword is also present activating OPM Flow's Dissolved Gas in Water Model. All the aforementioned keywords are in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RSW	RSW is an array of positive real numbers assigning the initial equilibration solution gas-water ratio values to each cell in the model. Repeat counts may be used, for example 20*1.30.			None
		Mscf/stb	sm ³ /sm ³	scc/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.32: RSW Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Note

This is an OPM Flow specific keyword for the simulator's Dissolved Gas in Water Model that is activated by declaring that dissolved gas in water is present in the run using the [DISGASW](#) keyword in the [RUNSPEC](#) section.

Use the command line option --enable-opm-rst-file=true to output the [RSW](#) data to the [RESTART](#) file.

Example

```
--
--      INITIAL EQUILIBRATION SOLUTION GAS IN WATER RATIO VALUES FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
RSW      1000*0.0000    1000* 0.0000    1000*1.3000    /
```

The above example defines the initial equilibration solution gas-water values to be 0.000 for all the cells in the first and second layers and 1.3000 for all the cells in the third layer.

10.3.84 RTEMP – DEFINE THE INITIAL RESERVOIR TEMPERATURE FOR THE MODEL

This keyword defines the initial reservoir temperature for the model. Note that the [RTEMP](#) keyword is an alias for [RTEMPA](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

See [RTEMP – Define the Initial Reservoir Temperature for the Model](#) in the [PROPS](#) section for a full description.

10.3.85 RTEMPA – DEFINE THE INITIAL RESERVOIR TEMPERATURE FOR THE MODEL

This keyword defines the initial reservoir temperature for the model. Note that the [RTEMPA](#) keyword is an alias for [RTEMP](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

See [RTEMPA – Define the Initial Reservoir Temperature for the Model](#) in the [PROPS](#) section for a full description.

10.3.86 RTEMPVD – DEFINE THE INITIAL RESERVOIR TEMPERATURE VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword defines the initial reservoir temperature versus depth tables for each equilibration region. Note that the [RTEMPVD](#) keyword is an alias for [TEMPVD](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

The initial reservoir temperature must be defined when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil temperature model, and the [THERMAL](#) keyword to activate the compositional thermal model.

The initial reservoir temperature should be defined when OPM Flow's CO₂ or H₂ storage option has been activated by the [CO2STORE](#) or [H2STORE](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth for corresponding reservoir temperature parameter TEMP .			None
		feet	m	cm	
2	TEMP	A columnar vector of real monotonically increasing down the column values that defines the corresponding reservoir temperature for the given depth.			None
		°F	°C	°C	

Notes:

- 1) The keyword is followed by NTEQUIL tables as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQIDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.33: RTEMPVD Keyword Description

See also the [RTEMP](#) keyword in the [PROPS](#) section for an alternative way to define a uniform initial reservoir temperature.

Note

The keyword is documented here in the [SOLUTION](#) section, the same as the commercial simulator, but it can also be used in the [PROPS](#) section by OPM Flow.

Example

```

--
--      INITIAL RESERVOIR TEMPERATURE VERSUS DEPTH TABLE
--
RTEMPVD
--      DEPTH      TEMPERATURE
--      FEET      DEG F
--      -----
--      1000.0      90.000
--      2000.0      100.000
--      3000.0      130.000
--      4000.0      160.000
--
--
--      / TABLE NO. 01
--
--      DEPTH      TEMPERATURE
--      FEET      DEG F
--      -----
--      1000.0      90.000
--      2000.0      100.000
--      3000.0      130.000
--      4000.0      160.000
--
--
--      / TABLE NO. 02
--
--      DEPTH      TEMPERATURE
--      FEET      DEG F
--      -----
--      1000.0      90.000
--      2000.0      100.000
--      3000.0      130.000
--      4000.0      160.000
--
--
--      / TABLE NO. 03

```

The above example defines three identical reservoir depth versus temperature tables for the three NTEQUIL regions defined on the *EQLDIMS* keyword in the *RUNSPEC* section.

10.3.87 RV – DEFINE THE INITIAL EQUILIBRATION CGR (RV) FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [RV](#) keyword defines the initial equilibration vaporized oil-gas ratio values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if vaporized oil been activated in the model via the [VAPOIL](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	RV	RV is an array of real positive numbers assigning the initial equilibration vaporized oil-gas ratio values to each cell in the model. Repeat counts may be used, for example 20*0.00720			None
		stb/Mscf	sm ³ /sm ³	scc/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.34: RV Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION CGR VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
RV      1000*0.00720    1000*0.00725    1000*0.00730      /
```

The above example defines the initial equilibration GOR values to be 0.00720 for all the cells in the first layer, 0.00725 for all the cells in the second layer, and finally 0.00730 for all the cells in the third layer.

10.3.88 RVVD – EQUILIBRATION VAPORIZED OIL-GAS RATIO (RV) VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [RVVD](#) keyword defines the vaporized oil-gas ratio (Rv) versus depth tables for each equilibration region that should be used when there is vaporized oil in the model ([VAPOIL](#) has been activated in the [RUNSPEC](#) section) and the EQLOPT2 variable has been set to a positive integer on the [EQUIL](#) keyword in the [SOLUTION](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding vaporized oil-gas ratio values, RV .			None
		feet	m	cm	
2	RV	A columnar vector of real values that defines the vaporized oil-gas ratio values at the corresponding DEPTH .			None
		stb/Mscf	sm ³ /sm ³	scc/scc	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.35: RVVD Keyword Description

Alternatively, the gas dew-point pressure versus depth tables may be entered using the [PDVD](#) keyword in the [SOLUTION](#) section instead of this keyword.

See also the [PDVD](#) and [EQUIL](#) keywords in the [SOLUTION](#) section.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the dew-point versus depth functions.

```
--
--      DEPTH      RV
--      STB/MSCF
--      -----
RVVD
      3000.0      0.00725
      8000.0      0.00725                               / RV VS DEPTH EQUIL REGN 01
--      -----
      3000.0      0.00730
      8000.0      0.00730                               / RV VS DEPTH EQUIL REGN 02
--      -----
      3000.0      0.00750
      8000.0      0.00750                               / RV VS DEPTH EQUIL REGN 03
```

Here three tables are entered with a constant CGR versus depth relationship for each equilibration region.

10.3.89 RVW – DEFINE THE INITIAL EQUILIBRATION VAPORIZED WATER IN GAS RATIO FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [RVW](#) keyword defines the initial equilibration vaporized water in gas ratio values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if both gas and water phases have been activated in the model via the [GAS](#) and [WATER](#) keywords, and the [VAPWAT](#) is also present activating OPM Flow's Vaporized Water Model. All the aforementioned keywords are in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	RVW	RVW is an array of real positive numbers assigning the initial equilibration gas-vaporized water ratio values to each cell in the model. Repeat counts may be used, for example 20*1.30.			None
		stb/Mscf	sm ³ /sm ³	rcc/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.36: RVW Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Note

This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by declaring that vaporized water is present in the run using the [VAPWAT](#) keyword in the [RUNSPEC](#) section. Use the command line option --enable-opm-rst-file=true to output the [RVW](#) data to the [RESTART](#) file.

Example

```
--
--      INITIAL EQUILIBRATION WATER VAPOR IN GAS RATIO VALUES FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
RVW      1000*0.0000    1000* 0.0000    1000*1.3000      /
```

The above example defines the initial equilibration gas-vaporized water values to be 0.000 for all the cells in the first and second layers and 1.3000 for all the cells in the third layer.

10.3.90 RVWVD – EQUILIBRATION VAPORIZED WATER-GAS RATIO (RVW) VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [RVWVD](#) keyword defines the vaporized water-gas ratio (Rvw) versus depth tables for each equilibration region that should be used when there is vaporized water in the model and the EQLOPT6 variable has been set to a positive integer on the [EQUIL](#) keyword in the [SOLUTION](#) section.

The keyword should only be used if both gas and water phases have been activated in the model via the [GAS](#) and [WATER](#) keywords, and the [VAPWAT](#) is also present activating OPM Flow's Vaporized Water Model. All the aforementioned keywords are in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding vaporized oil-gas ratio values, RVW			None
		feet	m	cm	
2	RVW	A columnar vector of real values that defines the vaporized water-gas ratio values, values at the corresponding DEPTH .			None
		stb/Mscf	sm ³ /sm ³	scc/scc	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.37: RVWVD Keyword Description

Alternatively, the vaporized water-gas ratio for each cell may be set via the [RVW](#) keyword in the [SOLUTION](#) section, if the non-standard method to initialize the model via enumeration is being employed.

See also the [EQUIL](#) keywords in the [SOLUTION](#) section.

Note

This is an OPM Flow specific keyword for the simulator's Water Vaporization Model that is activated by declaring that vaporized water is present in the run using the [VAPWAT](#) keyword in the [RUNSPEC](#) section. Use the command line option --enable-opm-rst-file=true to output the [RVW](#) data to the [RESTART](#) file.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the vaporized water-gas ratio versus depth functions.

```
--
--      DEPTH      RVW
--      STB/MSCF
--      -----
RVWVD
      3000.0      0.00000
      8000.0      0.00000                               / RVW VS DEPTH EQUIL REGN 01
--      -----
      3000.0      0.00000
      8000.0      0.00000                               / RVW VS DEPTH EQUIL REGN 02
--      -----
      3000.0      0.00100
      8000.0      0.00100                               / RVW VS DEPTH EQUIL REGN 03
```

The example shows three tables for three regions with constant [RVW](#) versus depth relationships for each equilibration region, with the first two tables having a zero vaporized water-gas ratio and the last region having a constant 0.001 stb/Mscf versus depth relationship.

10.3.91 SALT – DEFINE THE INITIAL EQUILIBRATION SALT CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SALT](#) keyword defines the initial equilibration salt concentration values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SGAS](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the salt (brine) phase has been activated in the model via the [BRINE](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SALT	SALT is an array of real positive numbers that are greater than or equal to zero assigning the initial equilibration salt concentration values to each cell in the model. Repeat counts may be used, for example 20*15.0.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.38: SALT Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION SALT CONCENTRATION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SALT      10000*0.0000      10000*0.0000      10000*15.000
/
```

The above example defines the initial equilibration salt concentration values to be 0.0000 for all the cells in the first and second layers and finally 15.000 for all the cells in the third layer.

10.3.92 SALTP – DEFINE THE INITIAL PRECIPITATED SALT VOLUME FRACTION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SALTP](#) keyword defines the initial equilibration precipitated salt volume fraction values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the salt (brine) phase has been activated in the model via the [BRINE](#) keyword, and the [PRECSALT](#) keyword to activate OPM Flow's Salt Precipitation Model. Both keywords are in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SALTP	SALTP is an array of real positive numbers that are greater than or equal to zero and less than or equal to one, that define the initial equilibration salt volume fraction values to each cell in the model. Repeat counts may be used, for example 20*0.15.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.39: SALTP Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Note

This is an OPM Flow specific keyword for the simulator's Salt Precipitation Model that is activated by the [PRECSALT](#) keyword and declaring that vaporized water is present in the run via the [VAPWAT](#) in the [RUNSPEC](#) section.

This keyword defines the initial precipitated salt volume fraction contained within the pore space. See [SALT](#) in the [SOLUTION](#) section that defines the initial salt concentration within the water phase.

Example

The example activates the standard Brine Tracking model using the [BRINE](#) keyword, OPM Flow's Salt Precipitation model using the [PRECSALT](#) keyword, and OPM Flow's vaporized water phase with the [VAPWAT](#) keyword; all three keywords are in the [RUNSPEC](#) section.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      MAX      MAX      RSVD      TVDP      TVDP
--      EQLNUM   DEPTH    NODES     TABLE  NODES
EQLDIMS
      3          1*       20        1*       1*                               /
--
--      ACTIVATE STANDARD BRINE MODEL
--
BRINE
--
--      ACTIVATE THE OPM FLOW SALT PRECIPITATION MODEL (OPM FLOW KEYWORD)
--
PRECSALT
--
--      VAPORIZED WATER IN WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
VAPWAT
```

Then in the [SOLUTION](#) section the [SALTP](#) keyword would be of the form:

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      DEFINE INITIAL PRECIPITATED SALT VOLUME FRACTION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SALTP
      1000*0.0000      1000*0.0000      1000*0.100                               /
```

Here the initial equilibration precipitated salt volume fraction values are set to 0.0000 for all the cells in the first and second layers and finally 0.1000 for all the cells in the third layer.

10.3.93 SALTPVD – INITIAL PRECIPITATED SALT VOLUME FRACTION VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [SALTPVD](#) keyword defines the initial precipitated salt volume fraction versus depth tables for each equilibration region for when OPM Flow's Salt Precipitation Model has been activated in the input deck via the [PRECSALT](#) keyword in the [RUNSPEC](#) section. The keyword defines the initial deposited salt as a volume fraction (Ss), that is solid salt saturation.

Note

This is an OPM Flow specific keyword for the simulator's Salt Precipitation Model that is activated by the [PRECSALT](#) keyword and declaring that vaporized water is present in the run via the [VAPWAT](#) in the [RUNSPEC](#) section. This is the initial precipitated salt volume fraction contained within the pore space, see [SALTVD](#) in the [SOLUTION](#) section that defines the initial salt concentration within the water phase.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth for corresponding salt volume fraction SALTPSAT.			None
		feet	m	cm	
2	SALTPSAT	A columnar vector of real values that defines the corresponding volume fraction of precipitated salt for the given depth. Note only the standard Brine Model is supported and therefore there should be only one columnar vector of SALTPSAT.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTEQUL tables as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQIDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no " terminator for the keyword.

Table 10.40: SALTPVD Keyword Description

Precipitated salt can be viewed in OPM ResInsight via the [SALTP](#) dynamic grid variable that represents the precipitated salt saturation.

Example

The example activates the standard Brine Tracking model using the [BRINE](#) keyword, OPM Flow’s Salt Precipitation model using the [PRECSALT](#) keyword, and OPM Flow’s vaporized water phase with the [VAPWAT](#) keyword; all three keywords are in the [RUNSPEC](#) section. The example also sets the number of equilibrium regions to three (NTEQUL set to three on the [EQLDIMS](#) keyword also in the [RUNSPEC](#)), that is:

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--          MAX      MAX      RSVD      TVDP      TVDP
--          EQLNUM    DEPTH    NODES    TABLE    NODES
EQLDIMS
--          3          1*       20        1*        1*
--
--          ACTIVATE STANDARD BRINE MODEL
--
BRINE
--
--          ACTIVATE THE OPM FLOW SALT PRECIPITATION MODEL (OPM FLOW KEYWORD)
--
PRECSALT
--
--          VAPORIZED WATER IN WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
VAPWAT
```

Then in the [SOLUTION](#) section the [SALTPVD](#) keyword would be of the form:

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--          PRECIPITATED SALT VOLUME FRACTION VERSUS DEPTH (OPM FLOW KEYWORD)
--
--          DEPTH      SALTPSAT
--          -----
SALTPVD
--          3000.0      0.000
--          8000.0      0.000
--
--          -----
--          3000.0      0.000
--          8000.0      0.000
--
--          -----
--          3000.0      0.000
--          8000.0      0.000
```

Here the initial precipitated salt volume fraction has been set to zero for all three equilibration regions.

10.3.94 SALTREST – DEFINE THE RESTART SALT CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SALTREST](#) keyword defines restart salt concentration values for all grid cells in the model and should be used in runs that are using the [RESTART](#) facility, where the initial run has not used the Low Salt or Brine options. This allows for initial runs that have used the standard water PVT properties via the [PVTW](#) keyword in the [PROPS](#) section, to be restarted with salt dependent water properties. The keyword should only be used if the salt (brine) phase has been activated in the current restart run (not the initial run) via the [BRINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SALTREST	SALTREST is an array of real positive numbers that are greater than or equal to zero assigning the restart salt concentration values to each cell in the model. Repeat counts may be used, for example 20*15.0.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.41: SALTREST Keyword Description

See also the [PVTWSALT](#) keyword in the [PROPS](#) section and the [RESTART](#) keyword in the [SOLUTION](#) section.

Example

```
--
--      DEFINE RESTART SALTREST VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SALTREST      1000*0.0000      1000*0.0000      1000*15.000      /
```

The above example defines the restart salt concentration values to be 0.0000 for all the cells in the first and second layers and finally 15.000 for all the cells in the third layer.

10.3.95 SALTVD – EQUILIBRATION SALT CONCENTRATION VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [SALTVD](#) keyword defines the initial salt concentration versus depth tables for each equilibration region for when the salt (brine) phase has been activated in the model via the [BRINE](#) keyword in the [RUNSPEC](#) section, and the [EQLOPTI](#) variable has been set to a positive integer on the [EQUIL](#) keyword in the [SOLUTION](#) section. Secondly, the keyword should also be used to set the initial salt concentration versus depth if OPM Flow's [PRECSALT](#) keyword in the [RUNSPEC](#) section has been used to activate the simulators Salt Precipitation model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth for corresponding salt concentrations SALTCON.			None
		feet	m	cm	
2	SALTCON	A columnar vector of real monotonically increasing down the column values that defines the corresponding salt concentration within the water phase for the given depth. There should be one columnar vector for each type of salt. For the standard Brine Model there is only one salt type and therefore there should be only one columnar vector of SALTCON. However, if the BRINE keyword has been invoked with the ECLMC keyword in the RUNSPEC section, then there should one columnar SALTCON vector for each declared salt type. <u>It is recommended to provide initial salt concentrations less then or equal to values provided by SALTSOL keyword in the PROPS section.</u>			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

- 1) The keyword is followed by NTEQUIL tables as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQIDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "I" terminator for the keyword.

Table 10.42: SALTVD Keyword Description

Note

This is the initial salt concentration contained within the water phase, see the [SALTPVD](#) keyword in the [SOLUTION](#) section that defines the initial salt volume fraction that has been precipitated into the pore space.

Examples

The first example activates the standard Brine Tracking model using the `BRINE` keyword in the `RUNSPEC` section and sets the number of equilibrium regions to three (NTEQUIL set to 3 on the `EQLDIMS` keyword also in the `RUNSPEC`), that is:

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      MAX      MAX      RSVD      TVDP      TVDP
--      EQLNUM    DEPTH    NODES    TABLE    NODES
EQLDIMS
--      3         1*      20        1*        1*
--
--      ACTIVATE STANDARD BRINE MODEL
--
BRINE
```

Then in the `SOLUTION` section the `SALTVD` keyword would be of the form:

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      DEPTH      SALT-1      SALT-2      SALT-3      SALT-4
--      -----      SALTCON      SALTCON      SALTCON      SALTCON
--
SALTVD
--      3000.0      1.200
--      8000.0      1.200
--
--      3000.0      1.300
--      8000.0      1.300
--
--      3000.0      1.400
--      8000.0      1.400
```

The next example shows how the `SALTVD` keyword is entered when both the `ECLMC` and `BRINE` keywords have activated the Multi-Component Brine model in the `RUNSPEC` section, that is:

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      MAX      MAX      RSVD      TVDP      TVDP
--      EQLNUM  DEPTH    NODES    TABLE  NODES
EQLDIMS
--      3        1*      20        1*      1*                               /
--
--      ACTIVATE MULTI-COMPONENT BRINE MODEL
--
ECLMC
--
--      DEFINE WATER PHASE MULTI-COMPONENT BRINE COMPONENTS
--
--      SALT1    SALT2    SALT3    SALT4    SALT5
BRINE
--      NACL     CACL     MGC03                               /
```

The above example activates the Multi-Component Brine model with three different water salinities for three equilibrium regions. In this case the resulting [SALTVD](#) keyword would be of the form:

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      DEPTH      SALT-01  SALT-02  SALT-03  SALT-04
--      -----  CONCENTR  CONCENTR  CONCENTR  CONCENTR
--
SALTVD
--      3000.0    1.200    0.540    0.020
--      7000.0    1.200    0.640    0.040                               / EQUIL REGN 01
--
--      3000.0    1.300    0.440    0.020
--      8000.0    1.300    0.540    0.040                               / EQUIL REGN 02
--
--      5000.0    1.400    0.640    0.002
--      8000.0    1.400    0.640    0.002                               / EQUIL REGN 03
```

In this case there are three data sets, one for each equilibrium region and three SALTCON columnar vectors, one for each salt type (NACL, CACL and MGC03) declared via the [BRINE](#) keyword in the [RUNSPEC](#) section.

Note that the Multi-Component Brine model is not available in OPM Flow.

10.3.96 SBIOF – DEFINE THE INITIAL EQUILIBRATION BIOFILM VOLUME FRACTION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SBIOF](#) keyword defines the initial equilibration biofilm volume fraction for all grid cells in the model. The keyword should only be used if either the [BIOFILM](#) or [MICP](#) model has been activated in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SBIOF	SBIOF is an array of real numbers that are greater than or equal to zero and less than or equal to one assigning the initial equilibration biofilm volume fraction values to each cell in the model. Repeat counts may be used, for example 20*0.0010.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) Physical values are less or equal to the grid cell porosity. For the MICP model, then the sum of SBIOF and SCALC has to be less or equal to the grid cell porosity.
- 3) The keyword is terminated by a “/”.

Table 10.43: SBIOF Keyword Description

For both [BIOFILM](#) and [MICP](#) models, see also the [SMICR](#) keyword, and for the [MICP](#) model, the [SCALC](#), [SOXYG](#), and [SUREA](#) keywords to define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION BIOFILM VOLUME FRACTION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SBIOF      10000*0.0000 10000*0.0000 10000*0.0010
/
```

The above example defines the initial equilibration biofilm volume fraction values to be 0.0000 for all the cells in the first and second layers and finally 0.0010 for all the cells in the third layer.

10.3.97 SCALC – DEFINE THE INITIAL EQUILIBRATION CALCITE VOLUME FRACTION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SCALC](#) keyword defines the initial equilibration calcite volume fraction for all grid cells in the model. The keyword should only be used if the [MICP](#) model has been activated in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SCALC	SCALC is an array of real numbers that are greater than or equal to zero and less than or equal to one assigning the initial equilibration calcite volume fraction values to each cell in the model. Repeat counts may be used, for example 20*0.0010.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The sum of SBIOW and SCALC has to be less or equal to the grid cell porosity.
- 3) The keyword is terminated by a “/”.

Table 10.44: SCALC Keyword Description

See also the [SBIOW](#), [SMICR](#), [SOXYG](#) and [SUREA](#) keywords to define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION CALCITE VOLUME FRACTION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SCALC      1000*0.0000 1000*0.0000 1000*0.0010      /
```

The above example defines the initial equilibration calcite volume fraction values to be 0.0000 for all the cells in the first and second layers and finally 0.0010 for all the cells in the third layer.

10.3.98 SCVD – DEFINE EQUILIBRATION COAL SOLVENT CONCENTRATION VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [SCVD](#) keyword defines the initial coal solvent concentration versus depth tables for each equilibration region for when the coal phase has been activated in the run via the [COAL](#) keyword in the [RUNSPEC](#) section. The keyword may be used in conjunction with the [SOLVCONC](#) keyword in the [SOLUTION](#) section, to fully describe the initial state of the model. Note both [SOLVCONC](#) and [SCVD](#) are optional as the simulator will calculate the coal gas concentration based on the equilibrium concentration and the block pressure.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding coal solvent concentration, SCVALS.			None
		feet	m	cm	
2	SCVALS	A columnar vector of real values that defines the coal solvent concentration values at the corresponding DEPTH.			None
		Mscf/ft3	sm ³ /m ³	scc/cc	

Notes:

- 1) The keyword is followed by NTEQUL records as declared on the EQLDIMS keyword in the RUNSPEC section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the EQLDIMS keyword in the RUNSPEC section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 10.45: SCVD Keyword Description

See also the [SOLVCONC](#), [GCVD](#), [GASCONC](#) and [GASSATC](#) keywords in the [SOLUTION](#) section.

Example

Given NTEQUL equals three and NDRXVD is greater than or equal to two on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the following example defines the coal solvent concentration versus depth functions.

```
--
--      DEPTH      SOLVC
--      MSCF/FT
--      -----
SCVD
      100.0      75.5000
      1000.0     75.5000                      / SC VS DEPTH EQUIL REGN 01
--      -----
      100.0      65.5000
      1000.0     65.5000                      / SC VS DEPTH EQUIL REGN 02
--      -----
      100.0      60.0000
      1000.0     60.0000                      / SC VS DEPTH EQUIL REGN 03
```

Here three tables are entered with a constant coal solvent concentration versus depth relationship for each equilibration region

10.3.99 SFOAM – DEFINE THE INITIAL EQUILIBRATION FOAM CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SFOAM](#) keyword defines the initial equilibration foam concentration values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SGAS](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the foam phase has been activated in the model via the [FOAM](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SFOAM	<i>SFOAM</i> is an array of real positive numbers that are greater than or equal to zero assigning the initial equilibration foam concentration values to each cell in the model. Units are dependent on the transport phase specified via the FOAMOPT I variable on the <i>FOAMOPTS</i> keyword in the <i>PROPS</i> section. FOAMOPT I should be set to either <i>GAS</i> or <i>WATER</i>. Repeat counts may be used, for example 20*0.5			None
		Gas: lb/Mscf Water: lb/stb	Gas: kg/sm ³ Water: kg/sm ³	Gas: gm/scc Water: gm/scc	

Notes:

1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2) The keyword is terminated by a “/”.

Table 10.46: SFOAM Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION FOAM VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SFOAM
      1000*0.0000      1000*0.0000      1000*0.500      /
```

The above example defines the initial equilibration foam concentration values to be 0.0000 for all the cells in the first and second layers and finally 0.500 for all the cells in the third layer.

10.3.100 SGAS – DEFINE THE INITIAL EQUILIBRATION GAS SATURATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SGAS](#) keyword defines the initial equilibration gas saturation values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the gas phase has been activated in the model via the [GAS](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SGAS	SGAS is an array of real positive numbers that are greater than or equal to zero and less than or equal to one assigning the initial equilibration gas saturation values to each cell in the model. Repeat counts may be used, for example 20*0.600.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.47: SGAS Keyword Description

Note for two phase runs it is only necessary to enter one saturation array of one of the phases present in the run ([SGAS](#), [SOIL](#), or [SWAT](#)), as the simulator will calculate the other phases by difference. Similarly for three phase runs it is only necessary to enter the array data for two of the phases, as the third saturation will again be calculated by the simulator.

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION GAS SAT VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SGAS      1000*0.7000      1000*0.6500      1000*0.6000      /
```

The above example defines the initial equilibration gas saturation values to be 0.7000 for all the cells in the first layer, 0.6500 for all the cells in the second layer, and finally 0.6000 for all the cells in the third layer.

10.3.101 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

10.3.102 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

10.3.103 SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

10.3.104 SMICR – DEFINE THE INITIAL EQUILIBRATION MICROBIAL CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [SMICR](#) keyword defines the initial equilibration microbial concentration values for all grid cells in the model. The keyword should only be used if either the [BIOFILM](#) or [MICP](#) model has been activated in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SMICR	SMICR is an array of real numbers that are greater than or equal to zero assigning the initial equilibration microbial concentration values to each cell in the model. Repeat counts may be used, for example 20*0.1500.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.48: SMICR Keyword Description

For both [BIOFILM](#) and [MICP](#) models, see also the [SBIOF](#) keyword, and for the [MICP](#) model, the [SCALC](#), [SOXYG](#), and [SUREA](#) keywords to define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION MICROBIAL CONCENTRATION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SMICR
      1000*0.0000 1000*0.0000 1000*0.1500 /
```

The above example defines the initial equilibration microbial concentration values to be 0.0000 for all the cells in the first and second layers and finally 0.1500 for all the cells in the third layer.

10.3.105 SOIL – DEFINE THE INITIAL EQUILIBRATION OIL SATURATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SOIL](#) keyword defines the initial equilibration oil saturation values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the oil phase has been activated in the model via the [OIL](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOIL	SOIL is an array of real positive numbers that are greater than or equal to zero and less than or equal to one assigning the initial equilibration oil saturation values to each cell in the model. Repeat counts may be used, for example 20*0.600.			None
		dimensionless	dimensionless	dimensionless	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.

2)

The keyword is terminated by a “/”.

Table 10.49: SOIL Keyword Description

Note for two phase runs it is only necessary to enter one saturation array of one of the phases present in the run ([SGAS](#), [SOIL](#), or [SWAT](#)), as the simulator will calculate the other phases by difference. Similarly for three phase runs it is only necessary to enter the array data for two of the phases, as the third saturation will again be calculated by the simulator.

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION OIL SAT VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SOIL      1000*0.7000      1000*0.6500      1000*0.6000      /
```

The above example defines the initial equilibration oil saturation values to be 0.7000 for all the cells in the first layer, 0.6500 for all the cells in the second layer, and finally 0.6000 for all the cells in the third layer.

10.3.106 SOLUTION – DEFINE THE START OF THE SOLUTION SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SOLUTION](#) activation keyword marks the end of the [REGIONS](#) section and the start of the [SOLUTION](#) section that defines the parameters used to initialize the model, by:

- 1) defining fluid contacts and pressures, or

2) defining pressures and fluid saturations for all cells in the model, or

3) by restarting from a previously OPM Flow completed run.

There is no data required for this keyword.

Example

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
```

The above example marks the end of the [REGIONS](#) section and the start of the [SOLUTION](#) section in the OPM Flow data input file.

10.3.107 SOLVCONC – DEFINE THE INITIAL COAL SOLVENT CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SOLVCONC](#) keyword defines the initial coal solvent concentration values for all matrix grid cells in the model and should be used in conjunction with the [SCVD](#) keyword in the [SOLUTION](#) section, to fully describe the initial state of the model. The keyword should only be used if the coal phase has been activated in the model via the [COAL](#) keyword in the [RUNSPEC](#) section. Note both [SOLVCONC](#) and [SCVD](#) are optional as the simulator will calculate the coal solvent concentration based on the equilibrium concentration and the block pressure.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SOLVCONC	SOLVCONC is an array of real positive numbers that define the initial equilibration coal solvent concentration values to each matrix cell in the model. Repeat counts may be used, for example 20*75.0.			None
		Mscf/ft3	sm ³ /m ³	scc/cc	

Notes:

- 1) The number of entries should correspond to number of matrix cells in the model, that is (NX x NY x NZ) divided by two, as per the parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a “/”.

Table 10.50: SOLVCONC Keyword Description

See also the [SCVD](#) keyword in the [SOLUTION](#) section to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION COAL SOLVENT CONCENTRATION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 6
--
SOLVCONC
      1000*75.500      1000*65.500      1000*60.000      /
```

The above example defines the initial coal solvent concentration values to be 75.500 for all the matrix cells in the first layer, 65.500 for all the cells in the second layer, and finally 60.000 for all the cells in the third layer.

10.3.108 SOLVFRAC – DEFINE THE INITIAL GAS SOLVENT FRACTION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [SOLVFRAC](#) keyword defines the initial solvent fraction within the gas phase values for all matrix grid cells in the model. The keyword should only be used if the coal phase has been activated in the model via the [COAL](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

[SOLVFRAC](#) is used with the [standard](#) equilibration method to initialize the model via the [EQUIL](#) keyword in the [RUNSPEC](#) section, as oppose to the non-standard enumeration method.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SOLVFRAC	SOLVFRAC is an array of real positive numbers that define the initial solvent fraction within the gas phase values for each matrix cell in the model. Repeat counts may be used, for example 20*0.075.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to number of matrix cells in the model, that is (NX x NY x NZ) divided by two, as per the parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.51: SOLVFRAC Keyword Description

See also the [EQUIL](#) keyword in the [SOLUTION](#) section to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION GAS SOLVENT FRACTION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 6
--
SOLVFRAC      1000*0.0250      1000*0.0350      1000*0.0500      /
```

The above example defines the initial gas solvent fraction values to be 0.250 for all the matrix cells in the first layer, 0.0350 for all the cells in the second layer, and finally 0.0500 for all the cells in the third layer.

10.3.109 SOXYG – DEFINE THE INITIAL EQUILIBRATION OXYGEN CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

The [SOXYG](#) keyword defines the initial equilibration oxygen concentration values for all grid cells in the model. The keyword should only be used if the [MICP](#) model has been activated in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SOXYG	SOXYG is an array of real numbers that are greater than or equal to zero assigning the initial equilibration oxygen concentration values to each cell in the model. Repeat counts may be used, for example 20*0.1500			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.52: SOXYG Keyword Description

See also the [SBI OF](#), [SCALC](#), [SMICR](#), and [SUREA](#) keywords to define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION OXYGEN CONCENTRATION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SOXYG      1000*0.0000 1000*0.0000 1000*0.1500      /
```

The above example defines the initial equilibration oxygen concentration values to be 0.0000 for all the cells in the first and second layers and finally 0.1500 for all the cells in the third layer.

10.3.110 SPOLY – DEFINE THE INITIAL EQUILIBRATION POLYMER CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SPOLY](#) keyword defines the initial equilibration polymer concentration values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SGAS](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the polymer phase has been activated in the model via the [POLYMER](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SPOLY	SPOLY is an array of real positive numbers that are greater than or equal to zero assigning the initial equilibration polymer concentration values to each cell in the model. Repeat counts may be used, for example 20*25.0.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.53: SPOLY Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION POLYMER VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SPOLY      1000*0.0000      1000*0.0000      1000*15.000      /
```

The above example defines the initial equilibration polymer concentration values to be 0.0000 for all the cells in the first and second layers and finally 15.000 for all the cells in the third layer.

10.3.111 SPOLYMW – DEFINE THE INITIAL EQUILIBRATION POLYMER MOLECULAR WEIGHTS FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SPOLYMW](#) keyword defines the initial equilibration polymer molecular weights for all grid cells in the model and should only be used with OPM Flow's Polymer Molecular Weight Transport option, together with the other standard equilibration keywords, in order to fully describe the initial state of the model.

This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

The model has been tested using metric units; however, using either field or laboratory units with the option should be considered experimental.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SPOLYMW	SPOLYMW is an array of real positive numbers that are greater than or equal to zero assigning the initial equilibration polymer molecular weights to each cell in the model. Repeat counts may be used, for example 20*5.0			0.0
		lb/lb-M	kg/kg-M	gm/gm-M	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the [DIMENS](#) keyword in the [RUNSPEC](#) section, unless the [BOX](#) keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the [BOX](#) statement.
- 2) The keyword is terminated by a "/".

Table 10.54: SPOLYMW Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      INITIAL EQUILIBRATION POLYMER MOLECULAR WEIGHTS FOR ALL CELLS
--
--      ARRAY      CONSTANT      ----- BOX -----
--                                     I1  I2   J1  J2   K1  K2
EQUALS
      SPOLYMW      0.0000      1*  1*   1*  1*   1   5 / LAYERS 1 TO 5
      SPOLYMW      5.0000      1*  1*   1*  1*   6   7 / LAYERS 6 TO 7
      SPOLYMW      0.0000      1*  1*   1*  1*   8  20 / LAYERS 8 TO 20
/
```

The above example defines the initial equilibration polymer molecular weights to be 0.0000 for all the cells, except for layers six to seven, where the polymer molecular weight is set to five for these cells.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

10.3.112 SSOL – DEFINE THE INITIAL EQUILIBRATION SOLVENT SATURATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	-----------------	-------------------------	--------------------------

Description

The [SSOL](#) keyword defines the initial equilibration solvent saturation values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the solvent phase has been activated in the model via the [SOLVENT](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SSOL	SSOL is an array of real positive numbers that are greater than or equal to zero and less than or equal to one assigning the initial equilibration solvent saturation values to each cell in the model. Repeat counts may be used, for example 20*0.000.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.55: SSOL Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#), and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION GAS SAT VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SSOL      1000*0.0000      1000*0.0000      1000*0.0000      /
```

The above example defines the initial equilibration solvent saturation values to be 0.0 for all the cells in the model.

10.3.113 SUREA – DEFINE THE INITIAL EQUILIBRATION UREA CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SUREA](#) keyword defines the initial equilibration urea concentration values for all grid cells in the model. The keyword should only be used if the [MICP](#) model has been activated in the [RUNSPEC](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SUREA	SUREA is an array of real numbers that are greater than or equal to zero assigning the initial equilibration urea concentration values to each cell in the model. Repeat counts may be used, for example 20*30.0.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.

2)

The keyword is terminated by a “/”.

Table 10.56: SUREA Keyword Description

See also the [SBI OF](#), [SCALC](#), [SMICR](#), and [SOXYG](#) keywords to define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION UREA CONCENTRATION FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SUREA      1000*0.0000 1000*0.0000 1000*20.0      /
```

The above example defines the initial equilibration urea concentration values to be 0.0000 for all the cells in the first and second layers and finally 20.0 for all the cells in the third layer.

10.3.114 SURF – DEFINE THE INITIAL EQUILIBRATION SURFACTANT CONCENTRATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SURF](#) keyword defines the initial equilibration surfactant concentration values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SGAS](#) and [SWAT](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the surfactant phase has been activated in the model via the [SURFACT](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SURF	SURF is an array of real positive numbers that are greater than or equal to zero assigning the initial equilibration surfactant concentration values to each cell in the model. Repeat counts may be used, for example 20*25.0.			None
		lb/stb	kg/sm ³	gm/scc	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.57: SURF Keyword Description

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#), [SOIL](#) and [SWAT](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION SURFACTANT VALUES FOR ALL CELLS
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SURF      1000*0.0000      1000*0.0000      1000*0.2500      /
```

The above example defines the initial equilibration surfactant concentration values to be 0.0000 for all the cells in the first and second layers and finally 0.2500 for all the cells in the third layer.

10.3.115 SWAT – DEFINE THE INITIAL EQUILIBRATION WATER SATURATION FOR ALL GRID BLOCKS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [SWAT](#) keyword defines the initial equilibration water saturation values for all grid cells in the model and should be used in conjunction with the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#) and [SOIL](#) keywords etc., to fully describe the initial state of the model. The keyword should only be used if the water phase has been activated in the model via the [WATER](#) keyword in the [RUNSPEC](#) section.

This is the non-standard method to initialize the model via enumeration and is seldom employed in the industry. The standard methodology is for OPM Flow to initialize a model using the parameters on the [EQUIL](#) keyword combined with other keywords to fully describe the initial state of the model. The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	SWAT	SWAT is an array of real positive numbers that are greater than or equal to zero and less than or equal to one assigning the initial equilibration water saturation values to each cell in the model. Repeat counts may be used, for example 20*0.300.			None
		dimensionless	dimensionless	dimensionless	

Notes:

1)

The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.

2)

The keyword is terminated by a “/”.

Table 10.58: SWAT Keyword Description

Note for two phase runs it is only necessary to enter one saturation array of one of the phases present in the run ([SGAS](#), [SOIL](#), or [SWAT](#)), as the simulator will calculate the other phases by difference. Similarly for three phase runs it is only necessary to enter the array data for two of the phases, as the third saturation will again be calculated by the simulator.

See also the [PBUB](#), [PDEW](#), [PRESSURE](#), [RS](#), [RV](#), [SGAS](#) and [SOIL](#) keywords to fully define the initial state of the model.

Example

```
--
--      DEFINE INITIAL EQUILIBRATION WAT SAT VALUES FOR ALL CELLS IN THE MODEL
--      BASED ON NX = 100, NY = 100 AND NZ = 3
--
SWAT      1000*0.2000      1000*0.2500      1000*0.4500      /
```

The above example defines the initial equilibration water saturation values to be 0.2000 for all the cells in the first layer, 0.2500 for all the cells in the second layer, and finally 0.4500 for all the cells in the third layer.

10.3.116 TBLK – DEFINE TRACER INITIAL GRID BLOCK CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

TBLK keyword defines the initial tracer concentration for all or selected cells in the model, for when the **TRACERS** keyword in the **RUNSPEC** section has declared the maximum number of tracers for each phase, and the **TRACER** keyword in the **PROPS** section has defined the tracer. This keyword is not in the standard keyword format due to the tracer name being concatenated to the keyword **TBLK** to fully define the tracer being initialized.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NAME	A character string of up to eight characters, consisting of <i>TBLK</i> as the first four characters followed by a four letter character string defining the tracer's name. The fifth character should either be the letter F or the letter S, that indicates the state of the tracer either to be free (F) or in solution (S). For example, TBLKFIGS (free) or TBLKSIGS (solution). The last three characters of NAME (the effective tracer name) must also match an entry on the <i>TRACER</i> keyword's NAME parameter, in the <i>PROPS</i> section. Note it is best to void names beginning with the letters F, S, and T as these names may create naming issues in post-processing software.			None
2	<i>TBLK</i>	<i>TBLK</i> is an array of real numbers greater than or equal to zero, that are assigned the tracer concentration values for each cell in the model or the current input <i>BOX</i>. Repeat counts may be used, for example 200*0.0. The units for the tracer, if required, are set on the <i>TRACER</i> keyword in the <i>PROPS</i> section. This should be the same as the PHASE in the model.			None
		Liquid: <i>TBLK</i> /stb Gas: <i>TBLK</i> /Mscf	Liquid: <i>TBLK</i> /sm ³ Gas: <i>TBLK</i> /sm ³	Liquid: <i>TBLK</i> /scc Gas: <i>TBLK</i> /scc	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the *DIMENS* keyword in the *RUNSPEC* section, unless the *BOX* keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the *BOX* statement.
- 2) The keyword is terminated by a “/”.

Table 10.59: TBLK Keyword Description

See also the **TRACERS** keyword in the **RUNSPEC** section to declared the maximum number of tracers for each phase, the **TRACER** keyword in the **PROPS** section to define the tracer, and the **WTRACER** keyword in the **SCHEDULE** section that defines the wells injecting the tracer.

Note

Currently, one cannot initialize tracers using the **EQUALS** keyword. Instead use the array format, that is the keyword followed by the required number of values, or the **TVDP** keyword in the **SOLUTION** section to set the initial tracer concentrations as a function of depth.

Example

The following `TRACERS` keyword in the `RUNSPEC` section declares the number of tracers in the model.

```
--
--      NUMBER AND TYPE OF TRACERS
--      NO OIL  NO WAT  NO GAS  NO ENV  DIFF    MAX    MIN    TRACER
--      TRACERS TRACERS TRACERS TRACERS CONTL  NONLIN NONLIN NONLIN
TRACERS
      0          0          1          0      'NODIFF' 1*      1*      1*      /
```

And the `TRACER` keyword in the `PROPS` section declares the tracer name and the phase for the tracer.

```
--
--      DEFINE TRACER NAMES
--
--      TRACER  TRACER
--      NAME    PHASE
--      -----
TRACER
      'IGS'      'GAS'                      / INJECTED GAS
/
```

Finally, the `TBLK` keyword in the `SOLUTION` section sets the initial tracer grid block concentrations in both the free and solution states.

```
--
--      INITIAL TRACER CONCENTRATIONS
--
TBLKFIGS
      1000*0.0                      / TRACER FIGS CONCENTRATIONS

TBLKSIGS
      1000*0.0                      / TRACER SIGS CONCENTRATIONS
```

Here the initial concentrations are set to zero.

Then in the `SCHEDULE` section one can us the `WTRACER` keyword to define the well injecting the tracer and the tracer concentration being injected,.

```
--
--      DEFINE CONCENTRATION OF TRACERS IN THE INJECTION STREAMS,
--      INJECTION TRACER CONCENTRATIONS NOT DEFINED USING THE WTRACER
--      KEYWORD ARE ASSUMED TO BE ZERO.
--
-- WELL  NAME    TRACER  TRACER  TRACER
-- NAME  TRACER  VALUE   CUM     GROUP
WTRACER
'GI01'  'IGS'    1.0                /
/
```

In this case, well GI01 is a gas injection well injecting gas with a tracer concentration of 1.0. The example shows how to track dry gas injection in a gas condensate reservoir, although, the example can be used for any type of gas injection.

10.3.117 TEMPI – DEFINE THE INITIAL TEMPERATURE VALUES FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	---------------------------------	-------------------------	--------------------------

Description

This keyword defines the initial reservoir temperature for each cell in the model. The keyword is used to explicitly define the initial reservoir temperature via the Enumeration Initialization method rather than defining a uniform initial temperature or defining temperature versus depth tables.

The initial reservoir temperature must be defined when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil temperature model, and the [THERMAL](#) keyword to activate the compositional thermal model.

The initial reservoir temperature should be defined when OPM Flow's CO₂ or H₂ storage option has been activated by the [CO2STORE](#) or [H2STORE](#) keyword in the [RUNSPEC](#) section.

The keyword can be used with all grid types.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TEMPI	TEMPI is an array of real positive numbers assigning the initial temperature to each cell in the model. Repeat counts may be used, for example 20*100.0.			None
		°F	°C	°C	

Notes:

- 1) The number of entries should correspond to the NX x NY x NZ parameters on the DIMENS keyword in the RUNSPEC section, unless the BOX keyword defines a sub area of the grid, in which case the total number of entries should correspond to the number of cells defined by the BOX statement.
- 2) The keyword is terminated by a “/”.

Table 10.60: TEMPI Keyword Description

See also the [RTEMP](#) keyword in the [PROPS](#) section and the [RTEMPVD](#) keyword in the [SOLUTION](#) section for alternative ways to initialize the model's initial temperature.

Example

```
--
--      DEFINE GRID BLOCK TEMPERATURE FOR ALL CELLS
--      (BASED ON NX x NY x NZ = 300)
--
TEMPI      100*212.0    100*215.0    100*220.0      /
```

The above example defines the initial temperature to be 212.0, 215.0, and 220.0 oF for the first, second and third layers in the model for all 300 cells, as defined by the [DIMENS](#) keyword in the [RUNSPEC](#) section.

10.3.118 TEMPVD – DEFINE THE INITIAL RESERVOIR TEMPERATURE VERSUS DEPTH TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the initial reservoir temperature versus depth tables for each equilibration region. Note that the [TEMPVD](#) keyword is an alias for [RTEMPVD](#), and that both keywords are supported by OPM Flow, in both the [PROPS](#) and [SOLUTION](#) sections, but are treated as being mutually exclusive.

The initial reservoir temperature must be defined when OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil temperature model, and the [THERMAL](#) keyword to activate the compositional thermal model.

The initial reservoir temperature should be defined when OPM Flow's CO2 or H2 storage option has been activated by the [CO2STORE](#) or [H2STORE](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth for corresponding reservoir temperature parameter RTEMP .			None
		feet	m	cm	
2	RTEMP	A columnar vector of real monotonically increasing down the column values that defines the corresponding reservoir temperature for the given depth.			None
		°F	°C	°C	

Notes:

- 1) The keyword is followed by NTEQUL tables as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NDRXVD rows as declared on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 10.61:TEMPVD Keyword Description

See also the [RTEMP](#) keyword in the [PROPS](#) section for an alternative way to define a uniform initial reservoir temperature.

Example

```
--
--      INITIAL RESERVOIR TEMPERATURE VERSUS DEPTH TABLE
--
TEMPVD
--      DEPTH          TEMPERATURE
--      FEET           DEG F
--      -----
--      1000.0         90.000
--      2000.0         100.000
--      3000.0         130.000
--      4000.0         160.000                                / TABLE NO. 01
--
--      -----
--      1000.0         90.000
--      2000.0         100.000
--      3000.0         130.000
--      4000.0         160.000                                / TABLE NO. 02
--
--      -----
--      1000.0         90.000
--      2000.0         100.000
--      3000.0         130.000
--      4000.0         160.000                                / TABLE NO. 03
```

The above example defines three identical reservoir depth versus temperature tables for the three NTEQUAL regions defined on the [EQLDIMS](#) keyword in the [RUNSPEC](#) section.

10.3.119 THPRES – DEFINE EQUILIBRATION REGION THRESHOLD PRESSURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **THPRES** defines the threshold pressure between various equilibration regions that have been defined by the **EQLNUM** keyword in the **REGIONS** section. The threshold pressure defines the potential difference between two regions which must be exceeded before flow can occur between the two regions. Once flow occurs the potential between the two regions is reduced by the threshold pressure.

This option must be activated by **THPRES** variable on **EQLOPTS** keyword in the **RUNSPEC** section in order to utilize this feature. Note that the irreversible option, as defined by **IRREVER** variable on **EQLOPTS** keyword in the **RUNSPEC** section, is not supported.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	EQLNUM1	EQLNUM1 is an a positive integer that is greater or equal to one and less than or equal to NTEQUL on the EQLDIMS keyword in the RUNSPEC section, that defines the “from” equilibration region number.			None
		dimensionless	dimensionless	dimensionless	
2	EQLNUM2	EQLNUM1 is an a positive integer that is greater or equal to one and less than or equal to NTEQUL on the EQLDIMS keyword in the RUNSPEC section, that defines the “to” equilibration region number.			None
		dimensionless	dimensionless	dimensionless	
3	THPRES	THPRES defines the threshold pressure from EQLNUM1 to EQLNUM2 and from EQLNUM2 to EQLNUM1. The default value of I* sets the threshold pressure to a value that initially prevents flow between the two equilibration regions. Any subsequent production or injection in either of the two equilibration regions will therefore result in flow between the two regions. Thus, this default initially isolates the two equilibration regions. If a equilibration region number pair has not been explicitly defined by this keyword the THPRES is set to zero, for no threshold pressure.			I*
		psia	barsa	atma	

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword is terminated by a “/”.

Table 10.62: THPRES Keyword Description

See also the **MULTREGT** keyword in the **GRID** section that uses the transmissibility between the **MULTNUM**, **FLUXNUM** or **OPERNUM** region arrays to control the flow between various regions within the model.

Note

Care should be taken that cells in different **EQLNUM** regions are not in communication, as this will result in an unstable initial equilibration.

Examples

Given NTEQUL is equal to six on the *EQLDIMS* keyword in the *RUNSPEC* section,

```
--
--      EQLNUM  EQLNUM  THPRES
--      FROM    TO      VALUE
THPRES
      1         2        0.588031      / REGN 1 TO REGN 2
      2         1        0.588031      / REGN 2 TO REGN 1
      1         3        0.787619      / REGN 1 TO REGN 3
      3         1        0.787619      / REGN 3 TO REGN 1
      1         4        7.000830      / REGN 1 TO REGN 4
      4         1        7.000830      / REGN 4 TO REGN 1
/
```

The above example defines the threshold pressures between equilibration regions one and two, one and three and one and four. As the threshold pressures between regions one and five and one and six (as well as other combinations), have not been explicitly set in the example, the threshold pressures for these combinations are set to zero.

However, as the irreversible option, as defined by IRREVER variable on *EQLOPTS* keyword in the *RUNSPEC* section, is not supported, then example can be simplified to:

```
--
--      EQLNUM  EQLNUM  THPRES
--      FROM    TO      VALUE
THPRES
      1         2        0.588031      / REGN 1 AND REGN 2
      1         3        0.787619      / REGN 1 AND REGN 3
      1         4        7.000830      / REGN 1 AND REGN 4
/
```

Again, as the threshold pressures between regions one and five and one and six (as well as other combinations), have not been explicitly set in the example, the threshold pressures for these combinations are set to zero.

10.3.120 TVDP – DEFINE THE INITIAL EQUILIBRATION TRACER SATURATION VERSUS DEPTH FUNCTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the initial equilibration tracer concentration versus depth functions for each grid cell in the model, for when the Tracer option has been enabled by the [TRACERS](#) keyword in the [RUNSPEC](#) section. The maximum number of tracers for each phase are declared on the [TRACERS](#) keyword in the [RUNSPEC](#) section. Unlike other keywords, the [TVDP](#) keyword must be concatenated with the name of the tracer declared by the [TRACER](#) keyword in the [PROPS](#) section as outlined in Table 10.63.

No.	Name	Description	Default
1	TVDP	A four letter character string equal to TVDP that is the root keyword name for this data set function.	None
2	STATE	A one letter character string defining the tracer's state, which is concatenated to TVDP to give the partial name of the keyword. STATE should be set to F ("free") or S ("solution"), with the latter related to a tracer that is dissolved in an associated phase. If the tracer defined by NAME is associated with the OIL , GAS or WATER phases then STATE should be set to F. If the tracer is associated with the DISGAS or VAPOIL phases then STATE should be set to S.	None
3	NAME	A three letter character string defining the tracer's name as defined by the TRACER keyword, which is concatenated to TVDP and STATE to given the full name of the keyword. For example, if the TRACER keyword has been used to define a tracer named SEA in the WAT phase, then the full keyword name would be TVDPFSEA. Whereas, if a tracer was defined as IGS for the GAS phase then it could be TVDPFIGS and/or TVDPSIGS depending if one was wishing to track the "free" or "solution" gas phase, or both. Note it is best to avoid names beginning with the letters F, S, and T as these names may create naming issues in post-processing software.	None

Table 10.63: [TVDP](#) Keyword Name Format

Following the declaration of the full keyword name, TVDPSTATENAME, the keyword is followed by the data as outlined below.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DEPTH	A columnar vector of real monotonically increasing down the column values that defines the depth values for the corresponding initial tracer saturations, TVDP			None
		feet	m	cm	
2	TVDPVAL	A columnar vector of real values, greater than or equal to zero, that defines the initial tracer concentration values at the corresponding DEPTH . If tracer units have been defined by the UNITS parameter on the TRACER keyword in the PROPS section, then the units of TVDPVAL are the ratio of UNITS divided by the TVDPVAL units given below. For example, if UNITS was defined as kg, then for metric units TVDPVAL units would be kg/sm ³ .			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
		Liquid: stb Gas: Mscf	Liquid: sm ³ Gas: sm ³	Liquid: scc Gas: scc	

Notes:

- 1) The keyword is followed by NTTRVD records as declared on the *EQLDIMS* keyword in the *RUNSPEC* section.
- 2) Each table must contain a minimum of two rows and a maximum of NSTRVD rows as declared on the *EQLDIMS* keyword in the *RUNSPEC* section.
- 3) Each table is terminated by a “/” and there is no “/” terminator for the keyword.

Table 10.64:TVDP Keyword Description

See also the *TBLK* keyword in the *SOLUTION* section that also sets the initial tracer concentration for each grid block.

Example

This example is taken from Norne model, in which there are seven tracers related to the water phase.

```

-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--          MAX      MAX      RSVD      TVDP      TVDP
--          EQLNUM    DEPTH    NODES    TABLE    NODES
EQLDIMS
--          9        1*       20        7         1*                /
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--          DEFINE TRACER NAMES
--
--          TRACER      TRACER
--          NAME         PHASE
--          - - - - -    - - - - -
TRACER
--          SEA         WAT                /
--          HTO         WAT                /
--          S36         WAT                /
--          2FB         WAT                /
--          4FB         WAT                /
--          DFB         WAT                /
--          TFB         WAT                /
/

```

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--          INITIAL EQUILIBRATION TRACER SATURATION VERSUS DEPTH
--
--          DEPTH      TRACER
--          -----
--          -----
TVDPFSEA
          1000.0      0.0
          5000.0      0.0
                                     / TRACER FSEA CONCENTRATION VS DEPTH

TVDPFHTO
          1000.0      0.0
          5000.0      0.0
                                     / TRACER FHTO CONCENTRATION VS DEPTH

TVDPFS36
          1000.0      0.0
          5000.0      0.0
                                     / TRACER FS36 CONCENTRATION VS DEPTH

TVDPF2FB
          1000.0      0.0
          5000.0      0.0
                                     / TRACER F2FB CONCENTRATION VS DEPTH

TVDPF4FB
          1000.0      0.0
          5000.0      0.0
                                     / TRACER F4FB CONCENTRATION VS DEPTH

TVDPFDFB
          1000.0      0.0
          5000.0      0.0
                                     / TRACER FDFB CONCENTRATION VS DEPTH

TVDPFTFB
          1000.0      0.0
          5000.0      0.0
                                     / TRACER FTFB CONCENTRATION VS DEPTH
```

Here we first define the number of tracers in the model via the [EQLDIMS](#) keyword in the [RUNSPEC](#) section, then the actual tracers themselves in the [PROPS](#) section using the [TRACER](#) keyword, and finally the initial tracer concentrations are all set to zero via the [TVDP](#) keyword in the [PROPS](#) section.

10.3.121 VAPPARS – OIL VAPORIZATION PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

VAPPARS defines the rate of oil vaporization in the presence of undersaturated gas and the rate at which the remaining oil gets “heavier” via the reduction in the solution gas-oil ratio (“Rs”). This keyword should only be used if the **OIL**, **GAS**, **DISGAS** and **VAPOIL** keywords in the **RUNSPEC** section have been invoked to allow oil, gas, dissolved gas and vaporized oil to be present in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	VAPPAR1	VAPPAR1 is a real positive dimensionless number that defines the rate at which oil vaporizes into the available undersaturated gas in a grid block. The default value of zero invokes the standard black-oil formulation in which all oil vaporizes into the available undersaturated phase in a grid cell. Increasing this parameter decreases the rate of vaporization. Typical values for VAPPAR1 range from zero to five.			0
		dimensionless	dimensionless	dimensionless	
2	VAPPAR2	VAPPAR2 is a real positive dimensionless number that defines the rate at which the Rs of the remaining oil in a grid cell decreases The default value of zero invokes the standard black-oil formulation in which the remaining oil's Rs does not change as the oil vaporizes into the available undersaturated gas in a grid cell. Increasing this parameter increases the difference between the remaining oil and the vaporized oil Rs values. Typical values for VAPPAR2 are less than one.			0
		dimensionless	dimensionless	dimensionless	

Notes:

1) The keyword is terminated by a “/”.

Table 10.65: VAPPARS Keyword Description

Note this keyword is normally used in history matching field performance to control the availability of the vaporized oil phase.

See also the **DRSDT** and **DRVDT** keywords in the **SCHEDULE** section that control the rate at which the solution gas-oil ratio and the vaporized oil-gas ratio increase within a grid block, respectively.

Examples

The first example sets the black-oil default parameters

```
--
--      OIL VAPORIZATION PARAMETERS
--
--      OIL-VAP   RS-INCS
--      VAPPAR1   VAPPAR2
VAPPARS
      0           0           /
```

And the second example decreases the rate at which the oil vaporizes into the available undersaturated gas and increases the difference between the grid block oil saturation Rs and the vaporized oil Rs within a grid cell.

```
--
--      OIL VAPORIZATION PARAMETERS
--
--      OIL-VAP   RS-INCS
--      VAPPAR1   VAPPAR2
VAPPARS
      1.5         0.150       /
```

Again, the keyword is normally used in history matching field performance to control the availability of the vaporized oil phase.

10.3.122
VISDATES
– DEFINE EXTERNAL RESERVOIR GEO-MECHANICS VISAGE STRESS DATES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [VISDATES](#) keyword defines External Reservoir Geo-Mechanics [VISAGE](#) option stress dates. The keyword should not be used in input decks as the associated data is generated by an external program.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.123
VISOPTS – DEFINE EXTERNAL RESERVOIR GEO-MECHANICS VISAGE OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [VISDATES](#) keyword defines External Reservoir Geo-Mechanics [VISAGE](#) option modeling options. The keyword should not be used in input decks as the associated data is generated by an external program.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

10.3.124 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

CHAPTER 11: SUMMARY SECTION

11.1 INTRODUCTION

The [SUMMARY](#) section defines the variables to be written to the summary files that are used to generate line graphs of properties such as oil flow rate versus time, grid block pressure versus time, etc. Unlike the other sections, the [SUMMARY](#) section has basically two types of keywords: keywords that request a variable to be written out to the summary file as described in section [11.2 Data Requirements](#), and the conventional type of keyword described in section [11.3 Keyword Definitions](#) that perform an action. There are literally hundreds of keywords associated with the former indicating the variety of data that can be written to the summary file. Whereas, for the latter there are less than twenty keywords.

Standard summary file output consists of three files as outlined in Table 11.1, with the raw requested data stored in [SUMMARY](#) index and [SUMMARY](#) data files, the [SUMMARY](#) RSM file is an ASCII file that is generated from the other two files by the simulator at the end of the run. There is also the option to output an OPM Flow specific Enhanced [SUMMARY](#) output file.

File Type	Data Type	Description	OPM Flow Status
ESMRY	Dynamic Vector Data	This is an OPM FLOW specific Enhanced SUMMARY (*.ESMRY) output file format that is optimized for fast loading of selected vectors by post-processing applications. The *.ESMRY file is re-written for every time step and does not have a performance impact on the simulation. Load times are in the order of 30 times faster compared with fetching selected vectors using the *.UNSMRY file format in post processing software. The file type is compatible with OPM ResInsight and the option is activated via the --enable-esmry=true command line option (true is the default value). Note that the FMOUT, MULTOUT, and UNIFOUT keywords have no impact on this file type.	OPM Flow Fully Supported.
RSM	Output File	The RSM file contains the variables requested to be written to the SUMMARY file via the keywords described in the 11.2 Data Requirements, in a tabulated ASCII output format. The format of the file enables the data to be easily loaded into the LibreOffice Calc program for further processing, as each vector represents one column. The report is written at the end of the simulation run by parsing the SUMMARY Index and SUMMARY Data files.	Fully Supported
SUMMARY Index	Dynamic Vector Index	The SUMMARY index file type specifies and defines the format and data type written to the SUMMARY Data file.	Fully Supported
SUMMARY Data	Dynamic Vector Data	The SUMMARY data file contains the variables requested to be written to the file via the keywords described in the 11.2 Data Requirements. The data consists of vectors that are used to generate line graphs of properties such as oil flow rate versus time, grid block pressure versus time, etc. The properties to be stored on the SUMMARY file are written to the summary file at the end of each successful time step. The data can be used to compare actual production data with the simulation derived results in post processing software like OPM ResInsight.	Fully Supported
Notes: 1) If no summary data is requested then the files will not be created. 2) The SUMMARY files can be loaded or written out in either unified or non-unified formats as well as in binary or ASCII formats.			

Table 11.1: OPM Flow Summary Output File Summary

As mentioned in Table 11.1 all files can be written out in either ASCII or binary formats and in addition the [SUMMARY](#) file can be loaded or written out in either unified or non-unified formats. The file type (ASCII or binary) and file structure format (unified or non-unified formats) is set via a set of keywords in the [RUNSPEC](#) section, as described in Table 11.2 for easy of reference.

Process	RUNSPEC Keyword	Description	Files
Input	FMTIN	The keyword defines the input files to be formatted as ASCII, i.e., text files, as opposed to binary files.	*.FSMSPEC *.FUNSMRY
	MULTIN	The keyword defines the input files to be non-unified multiple files, as opposed to unified files.	*.SMSPEC *.S0001
	UNIFIN	UNIFIN defines the input files to be unified files, as opposed to non-unified multiple files.	*.SMSPEC *.UNSMRY
Output	FMTOUT	The keyword sets all output files to be formatted as ASCII, i.e., text files, as opposed to binary files.	*.FSMSPEC *.FUNSMRY
	MULTOUT	MULTOUT defines the output files to be non-unified multiple files, as opposed to unified files.	*.SMSPEC *.S0001
	UNIFOUT	UNIFOUT defines the output files to be unified files, as opposed to non-unified multiple files.	*.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. 2) For unified files if the run terminates unexpectedly, or there is insufficient disk space, then the last reported output is not stored. The main advantage of unified files is that if a number of simulation cases reside in the same directory, the number of output files per case can be far fewer than using non-unified multiple files. There is no limit on the number of reporting steps that a unified file can store.			

Table 11.2: RUNSPEC Input and Output File Format Keywords

OPM Flow automatically generates the [SUMMARY](#) file names based on the input file name and the output options selected via the keywords in the [RUNSPEC](#) section as summarized in in Table 11.2. For example, starting OPM Flow using the following command from the terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA. Here CASENAME is the “root” of the filename and DATA is the extension of the filename. OPM Flow will generate the [SUMMARY](#) output files based on the CASENAME with the extension based on the type (ASCII or binary) and file structure format (unified or non-unified formats), as outlined in Table 11.3.

File Type	Unformatted Extension	Description	Formatted Extension
RSM	*.RSM	Output RSM file.	*.RSM
SUMMARY Index	*.SMSPEC	The SUMMARY index file for both unified and non-unified formats.	*.FSMSPEC
SUMMARY Data Non-Unified	*.Snnnn	The SUMMARY data files containing the variables requested to be written to the SUMMARY files. For example, the SUMMARY non-unified binary files would be: CASENAME.S0001 CASENAME.S0002 CASENAME.S0003 etc. CASENAME.SMSPEC	*.Annnn
SUMMARY Data Unified	*.UNSMRY	The SUMMARY data file containing the variables requested to be written to the SUMMARY file. For example, the SUMMARY data unified binary files: CASENAME.UNSMRY CASENAME.SMSPEC	*.FUNSMRY
Notes: 1) The above file naming convention is for Linux type operating systems, as OPM Flow is currently only officially supported for Linux distributions.			

Table 11.3: OPM Flow File Naming Conventions

The default behavior is to write out the requested variables at each time step. As this can lead to large files (or a large number of files for non-unified output), especially for full field simulation models, the [RPTONLY – Activate the Report Time Steps Only Option for the SUMMARY File](#) keyword allows one to write out the data only at a report time step instead.

The [SUMMARY](#) section is terminated by the [SCHEDULE](#) keyword.

11.2 DATA REQUIREMENTS

To generate a summary vector, the appropriate keyword must be entered in the [SUMMARY](#) section of the input data file; only variables explicitly requested will be written to the SUMMARY files. Normally the data is written to the summary file at the end of each successful time step, but as mentioned previously, this can be changed by the [RPTONLY](#) keyword to write out the data only at report time steps, thus reducing the overall size of the SUMMARY files.

The following sections describe the summary variable mnemonic syntax which defines the type of summary variable object (Aquifer, Field, Group, Well, etc.), the variable format for a given variable object, and the variable names for the various variable objects.

11.2.1 SUMMARY VARIABLE MNEMONIC SYNTAX

As mentioned earlier, there are literally hundreds of variables that can be written to the SUMMARY file, which can make the process of requesting the data rather complex. Fortunately in most cases, but not always, the variable names follow a four or five letter syntax that defines the variable mnemonic used to describe the data to be written out. Table 11.4 outlines the general syntax employed in deriving the variable mnemonic for some of the more commonly used summary variables.

Summary Variable Mnemonic Syntax									
First Character Object		Second Character Object Type		Third Character Flow Type		Four Character Value Type		Fifth Character Special	
A	Aquifers	A	Analytical Aquifer	Q	Influx	R			
		L ¹	Analytical Aquifer List	L	Liquid influx	T			
		N	Numerical Aquifer						
B	Block (Grid Cell)	O	Oil	P	Production	R	Rate	L	Liquid
F	Field	W	Water	I	Injection	T	Total (cumulative)	H	History
G	Group	G	Gas	F	Flow			G	Gas
R	Region	V	Volume at reservoir conditions						
W	Well	L	Liquid						
C	Well Connection	P ²	Pressure						
S	Well Segment	T	Tracer						

Summary Variable Mnemonic Syntax				
First Character Object	Second Character Object Type	Third Character Flow Type	Four Character Value Type	Fifth Character Special
Notes: 1) Analytical Aquifer Lists are currently not supported by OPM Flow; however the standard analytical and numerical aquifer vectors are supported to a varying degree. 2) For Field and Region pressures the mnemonic is PR, that is FPR for the field reservoir pressure and RPR for region average pressure. Whereas, a well's bottom-hole pressure is WBHP and its tubing head pressure is WTHP.				

Table 11.4: Summary Variable Mnemonic Syntax

So for example, FOPR would mean Field Oil Production Rate. Also if there is historical data associated with a variable, then the same variable mnemonic is used with the suffix H, that is for the historical field oil production rate the mnemonic would be FOPRH. Table 11.5 shows some examples of variable mnemonics that follow the general syntax.

Summary Variable Mnemonics Examples	
Name	Description
BOFT	Block Oil Flow Total
COFR	Connection Oil Flow Rate
FOPR	Field Oil Production Rate
GVPR	Group Volume Production Rate
RTIT	Region Tracer Injection Total
WWPT	Well Water Production Total

Table 11.5: Summary Variable Mnemonics Examples

And Table 11.6 shows some examples that do not follow the general syntax. Where applicable, each of these keywords would also have an equivalent mnemonic for historical input data available in history matching runs, that is: FWCTH, WBHPH, WTHPH, WWCTH, etc.

Summary Variable Mnemonics Examples Not Following the General Syntax	
Name	Description
BAPI	Block API
BOSAT	Block Oil Saturation
BPR	Block Pressure
FAQR	Field Aquifer Influx Rate
FPR	Field Average Pressure
FWCT	Field Water Cut

Summary Variable Mnemonics Examples Not Following the General Syntax	
Name	Description
WBHP	Well Bottom-Hole Pressure
WTHP	Well Tubing Head Pressure
WWCT	Well Water Cut

Table 11.6: Summary Variable Mnemonics Examples Not Following the General Syntax

11.2.2 SUMMARY VARIABLE FORMAT

In addition to the general mnemonic syntax, each object type (Field, Group, Region, Well, etc.) has additional syntax governing which specific object (group, well, etc.) should be written out to the SUMMARY file as explained in Table 11.7.

Summary Variable Format		
Summary Object	Format	Example
AQUIFER	Analytical aquifers are identified by the first two characters AA in the variable mnemonic and numerical aquifers are identified by the first two characters of AN. Aquifer variables for a given aquifer can be followed by a list of aquifer numbers and therefore a terminating “/” is required to end the list of aquifers. A blank list requests output for all the aquifers.	For instance to obtain the aquifer influx rate and cumulative influx for all analytical aquifers in the model, one would use: <pre>AAQR / AAQT /</pre> or just for numerical aquifers one, two, and three: <pre>ANQR 1 2 3 / ANQT 1 2 3 /</pre>
	Analytical aquifer lists defined via the AQUALIST keyword in the SOLUTION section are identified by the first two characters AL in the variable mnemonic. Aquifer variables for aquifer lists can be followed by any number of aquifer list names and therefore a terminating “/” is required to end the list of aquifers. A blank list requests output for all the aquifers.	For instance to obtain the total aquifer influx rate and cumulative influx for all analytical aquifers in aquifer lists EAST and WEST, one would use: <pre>ALQR 'EAST' 'WEST' / ALQT 'EAST' 'WEST' /</pre> <u>Note OPM Flow does not currently support Analytical Aquifer Lists.</u>
BLOCK (Grid Cells)	Block variables are followed by a list of cell (I, J, K) indices, with each line terminated by a “/”, and the list terminated with another terminating “/”.	To request the oil saturation for a series of grid cells one would use the following: <pre>BOSAT 1 1 1 / 10 10 3 / 15 15 10 / /</pre>
FIELD	Field variables take no additional parameters and therefore do not require a terminating “/”.	For example: <pre>FOPR FOPT FPR</pre> Would export the field oil production rate and total, plus the field average reservoir pressure to the SUMMARY file.

Summary Variable Format		
Summary Object	Format	Example
GROUP	Group variables can be followed by a list of group names enclosed in quotes and therefore a terminating “/” is required to end the list of groups. A blank list requests output for all groups.	As an example to get the oil production rates and totals for two groups representing two platforms one would use <pre>GOPR 'PLAT1' 'PLAT2' /</pre> GOPT 'PLAT1' 'PLAT2' / or <pre>GOPR / GOPT /</pre> if there are only two groups in the model.
REGION	Region variables for a given region can be followed by a list of region numbers and therefore a terminating “/” is required to end the list of regions. A blank list requests output for all the regions.	For instance to obtain the reservoir pressures and oil production totals for all regions: <pre>RPR / ROPT /</pre> or just for regions one, two, and three: <pre>RPR 1 2 3 / ROPT 1 2 3 /</pre>
	Region variables that are based on flows between two regions must be followed by a list of region number pairs, indicating the two regions. Each region pair is terminated by a “/”. In addition the variable mnemonic should also be terminated by a terminating “/”.	To obtain the oil and water flow total (or cumulative) between regions one and two, and also three and four, one would use: <pre>ROFT 1 2 / 3 4 / / RWFT 1 2 / 3 4 / /</pre>
WELL	Well variables can be followed by a list of well names enclosed in quotes and therefore a terminating “/” is required to end the list of wells. A blank list requests output for all wells.	As an example to get the oil production rates and totals for all oil wells beginning with the letters OP one would use <pre>WOPR 'OP*' /</pre> WOPT 'OP*' / To output the oil production rate for all wells, one may use: <pre>WOPR /</pre> Note, this would write out the oil production rate for all well types: producers and injectors, as well as wells declared as gas and water wells.

Summary Variable Format		
Summary Object	Format	Example
WELL COMPLETIONS	Well variables for well completion vectors³⁰⁴ generally consist of the well summary mnemonic name suffixed with the letter "L", for example WOPRL, for the completion oil production rate. The completion mnemonic is then followed by a list of well names enclosed in quotes and the completion number, each terminated by a "/". In addition the list is terminated by a terminating "/". A blank list for the completions results in all completions for a well being written out, and a blank list for the well will result in all well completions being written to the SUMMARY file. Care should be exercised when defaulting the list of wells and completions as there is the potential to generate large volumes of data.	For example, to get the completion oil production rates for well OP01 completions two, three, and four, and for completion two for well OP02 one would use: <pre> WOPRL 'OP01' 2 / 'OP01' 3 / 'OP01' 4 / 'OP02' 2 / / </pre> or to get the oil production rates for each of the completions for well OP01: <pre> WORPL 'OP01' / / </pre>
WELL CONNECTIONS	Connection variables are followed by a list of well names in quotes and connection I, J, K indices, each terminated by a "/". In addition the list is terminated by a terminating "/". A blank list for the connections results in all connections for a well being written out. And a blank list for the well will result in all well connections being written to the SUMMARY file. Again, care should be exercised when defaulting the list of wells and connections as there is the potential to generate large volumes of data.	To request the connection gas-oil ratio and water cut for wells OP01 and OP02 for several connections, one would use: <pre> CGOR 'OP01' 1 1 1 / 'OP01' 1 1 2 / 'OP02' 10 10 1 / / CWCT 'OP01' 1 1 1 / 'OP01' 1 1 2 / 'OP02' 10 10 1 / / </pre>

³⁰⁴
The term well connection is used to describe individual connections from the wellbore to the reservoir grid, as opposed to well completions. A well completion is used to describe a set of connections, for example, a well may consist of several completions with each completion consisting of multiple connections.

Summary Variable Format		
Summary Object	Format	Example
WELL CONNECTIONS (COMPLETIONS)	This type of data is the same as for the Well Completion data, except that the Well Connection series of mnemonics are used instead of the well mnemonics. Here the completion vectors consist of the connection mnemonic name suffixed with the letter "L", for example COPRL, for the completion oil production rate. The completion mnemonic is then followed by a list of well names enclosed in quotes and the connection I, J, K indices, each terminated by a "/". In addition, the list is terminated by a terminating "/". If any connection has been defined as being within a completion via the COMPLUMP keyword in the SCHEDULE section, then the completion data is written out instead of the connection data. A blank list for the completions results in all completions for a well being written out. However, the well name cannot be defaulted, unlike the other well and connection mnemonics. Care should be exercised when defaulting the list of completions as there is the potential to generate large volumes of data.	To request the gas-oil ratio and water cut for wells OP01 and OP02 for several completions identified by connections that have been lumped into those completions, one would use: <pre> CGORL 'OP01' 1 1 1 / 'OP01' 1 1 2 / 'OP02' 10 10 1 / / CWCTL 'OP01' 1 1 1 / 'OP01' 1 1 2 / 'OP02' 10 10 1 / / </pre>
WELL SEGMENTS	Well variables for multi-segment well segments can be followed by a list of well names enclosed in quotes and the segment number, each terminated by a "/". In addition, the list is terminated by a terminating "/". A blank list for the segments results in all segments for a well being written out. And a blank list for the well will result in all well segments being written to the SUMMARY file. Care should be exercised when defaulting the list of wells and segments as there is the potential to generate large volumes of data.	For example, to get the segment oil flow rates for well OP01 segments two, three, and four and for segment two for well OP02 one would use: <pre> SOFR 'OP01' 2 / 'OP01' 3 / 'OP01' 4 / 'OP02' 2 / / </pre> or to get all the segment oil flow rates for all the segments for well OP01: <pre> SOFR 'OP01' / / </pre>

Table 11.7: Summary Variable Format

11.2.3 AQUIFER VARIABLES

Table 11.8 outlines the aquifer summary variables based on the type of aquifer: analytical, analytical list and numerical. Note that the analytical list aquifer type is just an analytical aquifer “set” defined using the [AQUALIST](#) keyword in the SOLUTION section, that assigns an analytic aquifer list name to a set of aquifer numbers for greater readability in the output. This type of aquifer is not supported by OPM Flow and neither are numerical aquifers.

Aquifer Summary Variables						
Variable	Root	Field	Analytical Aquifer	Analytical Aquifer List	Numerical Aquifer	Comment
Aquifer Influx Rate (Water Aquifers)	QR	FAQR	AAQR	ALQR	ANQR	
Aquifer Influx Total (Water Aquifers)	QT	FAQT	AAQT	ALQT	ANQT	
Aquifer Influx Rate (Gas Aquifers)	QRG	FAQRG	AAQRG	ALQRG		
Aquifer Influx Total (Gas Aquifers)	QTG	FAQTG	AAQTG	ALQTG		
Aquifer Pressure	QP		AAQP		ANQP	Water pore volume weighted average
Aquifer Carter-Tracy Dimensionless Pressure	QPD		AAQPD			
Aquifer Carter-Tracy Dimensionless Time	QTD		AAQTD			
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available, for example, numerical aquifers are currently not supported by OPM Flow.						

Table 11.8: Aquifer Summary Variables

Example

The following example requests the field and analytical aquifer pressure, rate and cumulative water influxes for all analytical aquifers to be written to the SUMMARY file.

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--     ANALYTICAL AQUIFERS (FIELD)
--
FAQR
FAQT
--
--     ANALYTICAL AQUIFERS
--
AAQP
/
AAQR
/
AAQT
/
```

11.2.4 FIELD, GROUP, WELL, WELL CONNECTION, AND COMPLETION VARIABLES

The following table (Table 11.9) lists the field, group, well, well connection, and well completion production, injection, pressure, and productivity summary variables. Note that not all of these variables are available in OPM Flow; however, the simulator will issue a warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

See also section [11.2.6 Field, Region, and Block Variables](#) that outlines the field, region, and block production and injection data summary variables, and section [11.2.7 Field and Region Recovery Variables](#) that describes the summary variables for oil recovery efficiency and the oil recovery mechanism.

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Flow	Artificial Lift Quantity	ALQ			WALQ			
Flow	Artificial Lift and VFP Table Number	MVFP			WMVFP			Well VFP table number.
Flow	Efficiency Factor	EFF		GEFF	WEFF			
Flow	Efficiency Factor Product	EFFG			WEFFG			Efficiency factor of the well times the product of all the groups above the well.
Flow	Energy Production Rate	EPR	FEPR	GEPR	WEPR			
Flow	Energy Production Total	EPT	FEPT	GEPT	WEPT			
Flow	Gas Consumption Rate	GCR	FGCR	GGCR				
Flow	Gas Consumption Total	GCT	FGCT	GGCT				
Flow	Gas Flow Rate	GFR				CGFR	CGFRL	Positive for production and negative for injection.
Flow	Gas Flow Rate (Free)	GFRF				CGFRF		
Flow	Gas Flow Rate (Solution)	GFRS				CGFRS		
Flow	Gas Flow Rate (Upstream)	GFRU				CGFRU		Sum of connection gas flow rates upstream of, and including, this connection.
Flow	Gas Import Rate	GIMR	FGIMR	GGIMR				
Flow	Gas Import Total	GIMT	FGIMT	GGIMT				
Flow	Gas Injection Guide Rate	GIGR		GGIGR	WGIGR			
Flow	Gas Injection Rate	GIR	FGIR	GGIR	WGIR	CGIR	CGIRL	
Flow	Gas Injection Rate History	GIRH	FGIRH	GGIRH	WGIRH			
Flow	Gas Injection Rate Target/Limit	GIRT	FGIRT	GGIRT	WGIRT			

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Flow	Gas Injection Total	GIT	FGIT	GGIT	WGIT	CGIT	CGITL	
Flow	Gas Injection Total History	GITH	FGITH	GGITH	WGITH			
Flow	Gas Lift Rate	GLIR	FGLIR	GGLIR	WGLIR			Gas lift gas with group and field total rates based on well efficiency.
Flow	Gas Potential Injection Rate	GPI	FGPI	GGPI	WGPI	CGPI		Also WGIP for WGPI.
Flow	Gas Potential Injection Rate	GPI2	FGPI2	GGPI2	WGPI2			Including shut and stopped wells.
Flow	Gas Potential Production Rate	GPP	FGPP	GGPP	WGPP	CGPP		
Flow	Gas Potential Production Rate	GPP2	FGPP2	GGPP2	WGPP2			Including shut and stopped wells.
Flow	Gas Potential Production Rate (Free)	GPPF	FGPPF	GGPPF	WGPPF			
Flow	Gas Potential Production Rate (Free)	GPPF2	FGPPF2	GGPPF2	WGPPF2			Including shut and stopped wells.
Flow	Gas Potential Production Rate (Soln)	GPPS	FGPPS	GGPPS	WGPPS			
Flow	Gas Potential Production Rate (Soln)	GPPS2	FGPPS2	GGPPS2	WGPPS2			Including shut and stopped wells.
Flow	Gas Production Guide Rate	GPGR		GGPGR	WGPGR			
Flow	Gas Production Rate	GPR	FGPR	GGPR	WGPR	CGPR		Produced reservoir gas only, gas lift gas excluded.
Flow	Gas Production Rate (Free)	GPRF	FGPRF	GGPRF	WGPRF			
Flow	Gas Production Rate (Solution)	GPRS	FGPRS	GGPRS	WGPRS			
Flow	Gas Production Rate History	GPRH	FGPRH	GGPRH	WGPRH			
Flow	Gas Production Rate Target/Limit	GPRT	FGPRT	GGPRT	WGPRT			
Flow	Gas Production Total	GPT	FGPT	GGPT	WGPT	CGPT	CGPTL	
Flow	Gas Production Total (Free)	GPTF	FGPTF	GGPTF	WGPTF			
Flow	Gas Production Total (Solution)	GPTS	FGPTS	GGPTS	WGPTS			
Flow	Gas Production Total History	GPTH	FGPTH	GGPTH	WGPTH			
Flow	Gas Sales Rate	GSR	FGSR	GGSR				
Flow	Gas Sales Rate	SGR	FSGR	GSGR				Same as GSR.

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Flow	Gas Sales Total	GST	FGST	GGST				
Flow	Gas Sales Total	SGT	FSGT	GSGT				Same as GST.
Flow	Gas Voidage Injection Rate	GVIR			WGVIR			
Flow	Gas Voidage Production Rate	GVPR			WGVPR			
Flow	Liquid Flow Rate	LFR				CLFR	CLFRL	Positive for production and negative for injection.
Flow	Liquid Production Rate	LPR	FLPR	GLPR	WLPR	CLPR		
Flow	Liquid Production Rate History	LPRH	FLPRH	GLPRH	WLPRH			
Flow	Liquid Production Rate Target/Limit	LPRT	FLPRT	GLPRT	WLPRT			
Flow	Liquid Production Total	LPT	FLPT	GLPT	WLPT	CLPT	CLPTL	
Flow	Liquid Production Total History	LPTH	FLPTH	GLPTH	WLPTH			
Flow	Oil Flow Rate	OFR				COFR	COFRL	Positive for production and negative for injection.
Flow	Oil Flow Rate (Free)	OFRF				COFRF		
Flow	Oil Flow Rate (Solution)	OFRS				COFRS		
Flow	Oil Flow Rate (Upstream)	OFRU				COFRU		Sum of connection oil flow rates upstream of, and including this connection.
Flow	Oil Injection Guide Rate	OIGR		GOIGR	WOIGR			
Flow	Oil Injection Rate	OIR	FOIR	GOIR	WOIR	COIR	COIRL	
Flow	Oil Injection Rate History	OIRH	FOIRH	GOIRH	WOIRH			
Flow	Oil Injection Rate Target/Limit	OIRT	FOIRT	GOIRT	WOIRT			
Flow	Oil Injection Total	OIT	FOIT	GOIT	WOIT	COIT	COITL	
Flow	Oil Injection Total History	OITH	FOITH	GOITH	WOITH			
Flow	Oil Potential Injection Rate	OPI	FOPI	GOPI	WOPI			
Flow	Oil Potential Injection Rate	OPI2	FOPI2	GOPI2	WOPI2			Including shut and stopped wells.
Flow	Oil Potential Production Rate	OPP	FOPP	GOPP	WOPP	COPP		
Flow	Oil Potential Production Rate	OPP2	FOPP2	GOPP2	WOPP2			Including shut and stopped wells.
Flow	Oil Production Guide Rate	OPGR		GOPGR	WOPGR			
Flow	Oil Production Rate	OPR	FOPR	GOPR	WOPR	COPR		

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Flow	Oil Production Rate (Free)	OPRF	FOPRF	GOPRF	WOPRF			
Flow	Oil Production Rate (Solution)	OPRS	FOPRS	GOPRS	WOPRS			
Flow	Oil Production Rate History	OPRH	FOPRH	GOPRH	WOPRH			
Flow	Oil Production Rate Target/Limit	OPRT	FOPRT	GOPRT	WOPRT			
Flow	Oil Production Total	OPT	FOPT	GOPT	WOPT			
Flow	Oil Production Total (Free)	OPTF	FOPTF	GOPTF	WOPTF			
Flow	Oil Production Total (Solution)	OPTS	FOPTS	GOPTS	WOPTS			
Flow	Oil Production Total History	OPTH	FOPTH	GOPTH	WOPTH			
Flow	Res.Vol. Flow Rate	VFR				CVFR	CVFRL	Positive for production and negative for injection.
Flow	Res.Vol. Flow Rate (Free)	VFRF				CVFRF		
Flow	Res.Vol. Injection Guide Rate	VIGR		GVIGR	WVIGR			
Flow	Res.Vol. Injection Rate	VIR	FVIR	GVIR	WVIR	CVIR	CVIRL	
Flow	Res.Vol. Injection Rate Target/Limit	VIRT	FVIRT	GVIRT	WVIRT			
Flow	Res.Vol. Injection Total	VIT	FVIT	GVIT	WVIT	CVIT	CVITL	
Flow	Res.Vol. Potential Injection Rate	VPI	FVPI	GVPI	WVPI			
Flow	Res.Vol. Potential Injection Rate	VPI2	FVPI2	GVPI2	WVPI2			Including shut and stopped wells.
Flow	Res.Vol. Potential Production Rate	VPP	FVPP	GVPP	WVPP	CVPP		
Flow	Res.Vol. Potential Production Rate	VPP2	FVPP2	GVPP2	WVPP2			Including shut and stopped wells.
Flow	Res.Vol. Production Guide Rate	VPGR		GVPPGR	WVPPGR			
Flow	Res.Vol. Production Rate	VPR	FVPR	GVPR	WVPR	CVPR		
Flow	Res.Vol. Production Rate Target/Limit	VPRT	FVPRT	GVPRRT	WVPRRT			
Flow	Res.Vol. Production Total	VPT	FVPT	GVPT	WVPT	CVPT		
Flow	Water Flow Rate	WFR				CWFR	CWFRL	Positive for production and negative for injection.
Flow	Water Flow Rate (Upstream)	WFRU				CWFRU		Sum of connection water flow rates upstream of, and including, this connection.

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Flow	Water Injection Guide Rate	WIGR		GWIGR	WWIGR			
Flow	Water Injection Rate	WIR	FWIR	GWIR	WWIR	CWIR	CWIRL	
Flow	Water Injection Rate History	WIRH	FWIRH	GWIRH	WWIRH			
Flow	Water Injection Rate Target/Limit	WIRT	FWIRT	GWIRT	WWIRT			
Flow	Water Injection Total	WIT	FWIT	GWIT	WWIT	CWIT	CWITL	
Flow	Water Injection Total History	WITH	FWITH	GWITH	WWITH			
Flow	Water Potential Injection Rate	WPI	FWPI	GWPI	WWPI			Also WWIP for WWPI.
Flow	Water Potential Injection Rate	WPI2	FWPI2	GWPI2	WWPI2			Including shut and stopped wells.
Flow	Water Potential Production Rate	WPP	FWPP	GWPP	WWPP	CWPP		
Flow	Water Potential Production Rate	WPP2	FWPP2	GWPP2	WWPP2			Including shut and stopped wells.
Flow	Water Produce/Injected Ratio	WPIR	FWPIR	GWPIR	WWPIR			Reported as a percent.
Flow	Water Production Guide Rate	WPGR		GWPGR	WWPGR			
Flow	Water Production Rate	WPR	FWPR	GWPR	WWPR	CWPR		
Flow	Water Production Rate History	WPRH	FWPRH	GWPRH	WWPRH			
Flow	Water Production Rate Target/Limit	WPRT	FWPRT	GWPRT	WWPRT			
Flow	Water Production Total	WPT	FWPT	GWPT	WWPT	CWPT	CWPTL	
Flow	Water Production Total History	WPTH	FWPTH	GWPTH	WWPTH			
Flow	Water Voidage Injection Guide Rate	WVIR			WWVIR			
Flow	Water Voidage Injection Guide Total	WVIT			WWVIT			
Flow	Water Mass Injection Rate	AMIR	FAMIR	GAMIR	WAMIR	CAMIR	CAMIRL	
Flow	Water Mass Injection Total	AMIT	FAMIT	GAMIT	WAMIT	CAMIT	CAMITL	
Flow	Water Mass Production Rate	AMPR	FAMPR	GAMPR	WAMPR	CAMPR	CAMPRL	
Flow	Water Mass Production Total	AMPT	FAMPT	GAMPT	WAMPT	CAMPT	CAMPTL	
No.Wells	Number of abandoned injection wells	MWIA	FMWIA	GMWIA				
No.Wells	Number of abandoned production wells	MWPA	FMWPA	GMWPA				
No.Wells	Number of drilling events in total	MWDT	FMWDT	GMWDT				
No.Wells	Number of drilling events this timestep	MWDR	FMWDR	GMWDR				
No.Wells	Number of injection wells currently flowing	MWIN	FMWIN	GMWIN				

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
No. Wells	Number of injectors on group control	MWIG	FMWIG	GMWIG				
No. Wells	Number of injectors on own reservoir volume rate limit control	MWIV	FMWIV	GMWIV				
No. Wells	Number of injectors on own surface rate limit control	MWIS	FMWIS	GMWIS				
No. Wells	Number of injectors on pressure control	MWIP	FMWIP	GMWIP				
No. Wells	Number of producers controlled by own oil rate limit	MWPO	FMWPO	GMWPO				
No. Wells	Number of producers on group control	MWPG	FMWPG	GMWPG				
No. Wells	Number of producers on own reservoir volume rate limit control	MWPV	FMWPV	GMWPV				
No. Wells	Number of producers on own surface rate limit control	MWPS	FMWPS	GMWPS				
No. Wells	Number of producers on pressure control	MWPP	FMWPP	GMWPP				
No. Wells	Number of producers using artificial lift (with ALQ > 0.0)	MWPL	FMWPL	GMWPL				
No. Wells	Number of production wells currently flowing	MWPR	FMWPR	GMWPR				
No. Wells	Number of unused injection wells	MWIU	FMWIU	GMWIU				
No. Wells	Number of unused production wells	MWPU	FMWPU	GMWPU				
No. Wells	Number of workover events in total	MWWT	FMWWT	GMWWT				
No. Wells	Number of workover events this time step.	MWWO	FMWWO	GMWWO				
No. Wells	The number of connections capable of flowing in the well	MCON			WMCON			
No. Wells	Total number of injection wells	MWIT	FMWIT	GMWIT				
No. Wells	Total number of production wells	MWPT	FMWPT	GMWPT				
Pressure	Blocking Factor (GPP)	CDBF				CCDBF		
Pressure	Bottom-Hole Pressure	BHP			WBHP			
Pressure	Bottom-Hole Pressure (Five Point	BP5			WBP5			
Pressure	Bottom-Hole Pressure (Four Point)	BP4			WBP4			
Pressure	Bottom-Hole Pressure (Nine Point)	BP9			WBP9			
Pressure	Bottom-Hole Pressure (One Point)	BP			WBP			
Pressure	Bottom-Hole Pressure History	BHPH			WBHPH			
Pressure	Bottom-Hole Pressure Target/Limit	BHPT			WBHPT			

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Pressure	Connection Pressure	PR				CPR	CPRL	Completion pressure is an average pressure for the completion.
Pressure	Connection D-Factor	DFAC				CDFAC		
Pressure	Connection Transmissibility Factor	TFAC				CTFAC		
Pressure	Productivity (Gas Phase)	PIG			WPIG			
Pressure	Productivity (Liquid Phase)	PIL			WPIL			
Pressure	Productivity (Oil Phase)	PIO			WPIO			
Pressure	Productivity (Preferred Phase – Five Point)	PI5			WPI5			
Pressure	Productivity (Preferred Phase – Four Point)	PI4			WPI4			
Pressure	Productivity (Preferred Phase – Nine Point)	PI9			WPI9			
Pressure	Productivity (Preferred Phase – One Point)	PI1			WPI1			
Pressure	Productivity (Preferred Phase)	PI			WPI	CPI		
Pressure	Productivity (Water Phase)	PIW			WPIW			
Pressure	Tubing Head Pressure	THP			WTHP			
Pressure	Tubing Head Pressure History	THPH			WTHPH			
Ratio	Gas-Liquid Ratio	GLR	FGLR	GGLR	WGLR	CGLR	CGLRL	
Ratio	Gas-Liquid Ratio (Bottom-Hole)	BGLR			WBGLR			
Ratio	Gas-Liquid Ratio History	GLRH	FGLRH	GGLRH	WGLRH			
Ratio	Gas-Oil Ratio	GOR	FGOR	GGOR	WGOR	CGOR	CGORL	
Ratio	Gas-Oil Ratio History	GORH	FGORH	GGORH	WGORH			
Ratio	Oil-Gas Ratio	OGR	FOGR	GOCR	WOCR	COGR	COGRL	
Ratio	Oil-Gas Ratio History	OGRH	FOGRH	GOCRH	WOCRH			
Ratio	Water -Gas Ratio	WGR	FWGR	GWGR	WWGR	CWGR	CWGRL	
Ratio	Water Cut	WCT	FWCT	GWCT	WWCT	CWCT	CWCTL	
Ratio	Water Cut History	WCTH	FWCTH	GWCTH	WWCTH			
Ratio	Water-Gas Ratio History	WGRH	FWGRH	GWGRH	WWGRH			

Field, Group, Well, Well Connection, and Completion Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Comment
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.								

Table 11.9: Field, Group, Well, Well Connection, and Completion Summary Variables

11.2.5 FIELD, GROUP AND WELL CONTROL MODE VARIABLES

In addition to the production, injection, pressure, and productivity summary variables, there are also summary variables that output the control mode at which the field, groups and wells are being controlled. Table 11.10 summarizes the mnemonics for the field and groups together with a description and the values written out to the SUMMARY file.

Field and Group Control Mode Reporting						
Object	Field	Group	Field	Group	Field	Group
Mnemonic	FMCTP	GMCTP	FMCTW	GMCTW	FMCTG	GMCTG
Description	Production Group.		Water Injection Group.		Gas Injection Group.	
Values	Value	Mnemonic	Value	Mnemonic	Value	Mnemonic
	0.0	NONE	0.0	NONE	0.0	NONE
	1.0	ORAT	1.0	RATE	1.0	RATE
	2.0	WRAT	2.0	RESV	2.0	RESV
	3.0	GRAT	3.0	REIN	3.0	REIN
	4.0	LRAT	4.0	VREP	4.0	VREP
	5.0	RESV	5.0		5.0	
	6.0	PRBL	6.0		6.0	
	7.0	ENERGY	7.0		7.0	
	8.0	WGRA	8.0	WGRA	8.0	WGRA
	9.0	CRAT	9.0	Availability	9.0	Availability
	10.0	PBGS	10.0		10.0	
	11.0	PBWS	11.0		11.0	
	12.0	CRAT	12.0		12.0	
Notes:						
1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow.						
2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination.						
3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.						
4) For both production and injection groups the mode of control set to negative if the rate was set by a higher group.						
5) Note that the output to the SUMMARY file is the numeric values, whereas for the RSM file it is the mnemonics, that is: ORAT, WRAT, etc.						
6) See Table 11.11 for an explanation of the mnemonics used in Table 11.10.						

Table 11.10: Field and Group Control Mode Reporting

Similarly, Table 11.11 shows the control mode reporting variables for wells.

Well Control Mode Reporting						
Object	Well			Well		
Mnemonic	WSTAT			WMCTL		
Description	Well Status indicator.			Well Mode of Control indicator.		
Values	Value	Mnemonic	Description	Value	Mnemonic	Description
	1.0	PROD	Well is a producing well.	-1.0		Well is under group control.
	2.0	INJ	Well is an injection well.	0.0	SHUT or STOP	Well is shut-in or stopped.
	3.0	SHUT	Well is shut-in except those under priority control.	1.0	ORAT	Well is on oil rate control.
	4.0	STOP	Well is stopped except those under priority control.	2.0	WRAT	Well is on water rate control.
	5.0	PSHUT	Well is shut-in under priority control.	3.0	GRAT	Well is on gas rate control.
	6.0	PSTOP	Well is stopped under priority control	4.0	LRAT	Well is on liquid rate control.
				5.0	RESV	Well is on reservoir voidage rate control.
				6.0	THP	Well is on THP control.
				7.0	BHP	Well is on BHP control.
				9.0	CRAT	Not used
				11.0	GOR Penalty	
				12.0	Drawdown	Well is under drawdown control.
				13.0	NGL	Not used
				30.0	Availability	
				31.0	REIN	
				32.0	TMRA	
				33.0	WGRA	
				34.0	CVAL	
				41.0 - 51.0		

Well Control Mode Reporting		
Object	Well	Well
Mnemonic	WSTAT	WMCTL
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available. - Note that the output to the SUMMARY file is the numeric values, whereas for the RSM file it is the mnemonics, that is: PROD, INJ, etc.		

Table 11.11: Well Control Mode Reporting

11.2.6 FIELD, REGION, AND BLOCK VARIABLES

The next set of summary data is based on the block, field and region summary variables and the variables are tabulated in Table 11.9. Note when the data is for a block variable, like for example the oil saturation, this will be the actual value for the a grid cell (block), whereas, for the field and region items this will be an average value.

See also section [11.2.4 Field, Group, Well, Well Connection, and Completion Variables](#) that outlines the field production data summary variables, and section [11.2.7 Field and Region Recovery Variables](#) that describes the summary variables for oil recovery efficiency and the oil recovery mechanism.

Block, Field, and Region Summary Variables						
Type	Variable	Root	Field	Region	Block	Comment
Flow	Gas Flow (Inter-Block in I+ Direction)	FLOGI			BFLOGI	
Flow	Gas Flow (Inter-Block in J+ Direction)	FLOGJ			BFLOGJ	
Flow	Gas Flow (Inter-Block in K+ Direction)	FLOGK			BFLOGK	
Flow-Reg	Gas Flow Rate Inter-Region	GFR		RGFR		OPM Flow implementation.
Flow-Reg	Gas Flow Rate Inter-Region (Positive Contribution)	GFR+		RGFR+		
Flow-Reg	Gas Flow Rate Inter-Region (Negative Contribution)	GFR-		RGFR-		
Flow-Reg	Gas Flow Total Inter-Region (Liquid and Gas Phases)	GFT		RGFT		
Flow-Reg	Gas Flow Total Inter-Region (+VE Liquid and Gas Phases)	GFT+		RGFT+		OPM Flow Implementation
Flow-Reg	Gas Flow Total Inter-Region (-VE Liquid and Gas Phases)	GFT-		RGFT-		
Flow-Reg	Gas Flow Total Inter-Region (Gas Phase)	GFTG		RGFTG		
Flow-Reg	Gas Flow Total Inter-Region (Liquid Phase)	GFTL		RGFTL		
Flow	Gas Injection Rate	GIR		RGIR		
Flow	Gas Injection Total	GIT		RGIT		
Flow	Gas Potential	PPG	FPPG	RPPG	BPPG	
Flow	Gas Production Rate	GPR		RGPR		
Flow	Gas Production Rate (Free)	GPRF		RGPRF		
Flow	Gas Production Rate (Solution)	GPRS		RGPRS		
Flow	Gas Production Total	GPT		RGPT		Liquid and gas phases
Flow	Gas Production Total (Free)	GPTF		RGPTF		
Flow	Gas Production Total (Net From Region)	GP		RGP		Minus injected gas.
Flow	Gas Production Total (Solution)	GPTS		RGPTS		
Flow	Gas Velocity (Inter-Block in I+ Direction)	VELGI			BVELGI	
Flow	Gas Velocity (Inter-Block in J+ Direction)	VELGJ			BVELGJ	
Flow	Gas Velocity (Inter-Block in K+ Direction)	VELGK			BVELGK	

Block, Field, and Region Summary Variables						
Type	Variable	Root	Field	Region	Block	Comment
Flow	Oil Flow (Inter-Block in I+ Direction)	FLOOI			BFLOOI	
Flow	Oil Flow (Inter-Block in J+ Direction)	FLOOJ			BFLOOJ	
Flow	Oil Flow (Inter-Block in K+ Direction)	FLOOK			BFLOOK	
Flow-Reg	Oil Flow Rate Inter-Region	OFR		ROFR		OPM Flow implementation.
Flow-Reg	Oil Flow Rate Inter-Region (Positive Contribution)	OFR+		ROFR+		
Flow-Reg	Oil Flow Rate Inter-Region (Negative Contribution)	OFR-		ROFR-		
Flow-Reg	Oil Flow Total Inter-Region (Liquid and Gas Phases)	OFT		ROFT		
Flow-Reg	Oil Flow Total Inter-Region (+VE Liquid and Gas Phases)	OFT+		ROFT+		OPM Flow implementation.
Flow-Reg	Oil Flow Total Inter-Region (-VE Liquid and Gas Phases)	OFT-		ROFT-		
Flow-Reg	Oil Flow Total Inter-Region (Gas Phase)	OFTG		ROFTG		
Flow-Reg	Oil Flow Total Inter-Region (Liquid Phase)	OFTL		ROFTL		
Flow	Oil Injection Rate	OIR		ROIR		
Flow	Oil Injection Total	OIT		ROIT		
Flow	Oil Potential	PPO	FPPO	RPPO	BPPO	
Flow	Oil Production Rate	OPR		ROPR		
Flow	Oil Production Total	OPT		ROPT		
Flow	Oil Production Total (Net From Region)	OP		ROP		
Flow	Oil Velocity (Inter-Block in I+ Direction)	VELOI			BVELOI	
Flow	Oil Velocity (Inter-Block in J+ Direction)	VELOJ			BVELOJ	
Flow	Oil Velocity (Inter-Block in K+ Direction)	VELOK			BVELOK	
Flow	Water Flow (Inter-Block in I+ Direction)	FLOWI			BFLOWI	
Flow	Water Flow (Inter-Block in J+ Direction)	FLOWJ			BFLOWJ	
Flow	Water Flow (Inter-Block in K+ Direction)	FLOWK			BFLOWK	
Flow-Reg	Water Flow Rate Inter-Region	WFR		RWFR		OPM Flow implementation.
Flow-Reg	Water Flow Rate Inter-Region	WFR+		RWFR+		
Flow-Reg	Water Flow Rate Inter-Region	WFR-		RWFR-		
Flow-Reg	Water Flow Total Inter-Region	WFT		RWFT		
Flow-Reg	Water Flow Total Inter-Region	WFT+		RWFT+		OPM Flow implementation.
Flow-Reg	Water Flow Total Inter-Region	WFT-		RWFT-		
Flow	Water Injection Rate	WIR		RWIR		

Block, Field, and Region Summary Variables						
Type	Variable	Root	Field	Region	Block	Comment
Flow	Water Injection Total	WIT		RWIT		
Flow	Water Potential	PPW	FPPW	RPPW	BPPW	
Flow	Water Production Rate	WPR		RWPR		
Flow	Water Production Total	WPT		RWPT		
Flow	Water Production Total (Net From Region)	WP		RWP		
Flow	Water Velocity (Inter-Block in I+ Direction)	VELWI			BVELWI	
Flow	Water Velocity (Inter-Block in J+ Direction)	VELWJ			BVELWJ	
Flow	Water Velocity (Inter-Block in K+ Direction)	VELWK			BVELWK	
Mass	Water In-place	AMIP	FAMIP	RAMIP	BAMIP	
Pressure	Gas Phase Pressure	GPR			BGPR	
Pressure	Oil Phase Pressure	PR	FPR	RPR	BPR BPRESSUR	Field and Region HCPV weighted.
Pressure	Oil Phase Pressure	PRH	FPRH	RPRH		Field and Region HCPV weighted.
Pressure	Oil Phase Pressure (PV Weighted)	PRP	FPRP	RPRP		Field and Region PV weighted.
Pressure	Water Phase Pressure	WPR			BWPR	
Property	Block Hydrocarbon Phase State	STATE			BSTATE	Gas (1), Gas & Oil (2), and Oil (3).
Property	Bubble Point Pressure	PBUB			BPBUB	
Property	Capillary Pressure (Gas-Oil)	GPC			BGPC	
Property	Capillary Pressure (Water-Oil)	WPC			BWPC	
Property	Dew Point	PDEW			BPDEW	
Property	Gas P/Z	PRGZ	FPRGZ	RPRGZ		
Property	Gas Reservoir Density	GDEN	FGDEN	RGDEN	BGDEN	
Property	Gas Reservoir Density	DENG			BDENG	Same as BGDEN.
Property	Gas Saturation	GSAT	FGSAT	RGSAT	BGSAT	Also BSGAS.
Property	Gas Saturation (Drainage to Imbibition)	GSHY			BGSHY	Leaving gas saturation for gas capillary pressure hysteresis.
Property	Gas Saturation (Dynamically Trapped)	GTPD			BGTPD	WAG hysteresis only.
Property	Gas Saturation (Trapped Critical)	GSTRP			BGSTRP	For gas capillary pressure hysteresis.
Property	Gas Saturation (Trapped)	GTRP			BGTRP	WAG hysteresis only.
Property	Gas Viscosity	GVIS	FGVIS	RGVIS	BGVIS	

Block, Field, and Region Summary Variables						
Type	Variable	Root	Field	Region	Block	Comment
Property	Gas-Oil Ratio (Saturated)	RSSAT			BRSSAT	
Property	Initial Contact Corrected Potential	PPC	FPPC	RPPC	BPPC	
Property	Oil Reservoir Density	ODEN	FODEN	RODEN	BODEN	
Property	Oil Reservoir Density	DENO			BDENO	Same as BODEN.
Property	Oil Saturation	OSAT	FOSAT	ROSAT	BOSAT	Also BSOIL.
Property	Oil Viscosity	OVIS	FOVIS	ROVIS	BOVIS	Also BVOIL.
Property	Oil-Gas Ratio (Saturated)	RVSAT			BRVSAT	
Property	Relative Permeability (Gas in the R- Direction)	GKRR-			BGKRR-	
Property	Relative Permeability (Gas in the R+ Direction)	GKRR			BGKRR	
Property	Relative Permeability (Gas in the Theta- Direction)	GKRT-			BGKRT-	
Property	Relative Permeability (Gas in the Theta+ Direction)	GKRT			BGKRT	
Property	Relative Permeability (Gas in the X- Direction)	GKRX-			BGKRX-	Also BGKRI-.
Property	Relative Permeability (Gas in the X+ Direction)	GKRX			BGKRX	Also BGKRI.
Property	Relative Permeability (Gas in the Y- Direction)	GKRY-			BGKRY-	Also BGKRJ-.
Property	Relative Permeability (Gas in the Y+ Direction)	GKRY			BGKRY	Also BGKRJ.
Property	Relative Permeability (Gas in the Z- Direction)	GKRZ-			BGKRZ-	Also BGKRK-.
Property	Relative Permeability (Gas in the Z+ Direction)	GKRZ			BGKRZ	Also BGKRK.
Property	Relative Permeability (Gas)	KRG			BKRG	Also BGWR.
Property	Relative Permeability (Krw Reduction Factor)	RK			BRK	Due to Polymer.
Property	Relative Permeability (Oil Two Phase to Gas)	KROG			BKROG	
Property	Relative Permeability (Oil Two Phase to Water)	KROW			BKROW	
Property	Relative Permeability (Oil in the R- Direction)	OKRR-			BOKRR-	
Property	Relative Permeability (Oil in the R+ Direction)	OKRR			BOKRR	
Property	Relative Permeability (Oil in the Theta- Direction)	OKRT-			BOKRT-	
Property	Relative Permeability (Oil in the Theta+ Direction)	OKRT			BOKRT	
Property	Relative Permeability (Oil in the X- Direction)	OKRX-			BOKRX-	Also BOKRI-.
Property	Relative Permeability (Oil in the X+ Direction)	OKRX			BOKRX	Also BOKRI.
Property	Relative Permeability (Oil in the Y- Direction)	OKRY-			BOKRY-	Also BOKRJ-.
Property	Relative Permeability (Oil in the Y+ Direction)	OKRY			BOKRY	Also BOKRJ.
Property	Relative Permeability (Oil in the Z- Direction)	OKRZ-			BOKRZ-	Also BOKRK-.
Property	Relative Permeability (Oil in the Z+ Direction)	OKRZ			BOKRZ	Also BOKRK.

Block, Field, and Region Summary Variables						
Type	Variable	Root	Field	Region	Block	Comment
Property	Relative Permeability (Oil)	KRO			BKRO	Also BOKR.
Property	Relative Permeability (Water in the R- Direction)	WKRR-			BWKRR-	
Property	Relative Permeability (Water in the R+ Direction)	WKRR			BWKRR	
Property	Relative Permeability (Water in the Theta- Direction)	WKRT-			BWKRT-	
Property	Relative Permeability (Water in the Theta+ Direction)	WKRT			BWKRT	
Property	Relative Permeability (Water in the X- Direction)	WKRX-			BWKRX-	Also BWKRI-.
Property	Relative Permeability (Water in the X+ Direction)	WKRX			BWKRX	Also BWKRI.
Property	Relative Permeability (Water in the Y- Direction)	WKRY-			BWKRY-	Also BWKRJ-.
Property	Relative Permeability (Water in the Y+ Direction)	WKRY			BWKRY	Also BWKRJ.
Property	Relative Permeability (Water in the Z- Direction)	WKRZ-			BWKRZ-	Also BWKRK-.
Property	Relative Permeability (Water in the Z+ Direction)	WKRZ			BWKRZ	Also BWKRK.
Property	Relative Permeability (Water)	KRW			BKRW	Also BWKR.
Property	Water Reservoir Density	WDEN	FWDEN	RWDEN	BWDEN	
Property	Water Reservoir Density	DENW			BDENW	Same as BWDEN.
Property	Water Saturation	WSAT	FWSAT	RWSAT	BWSAT	Also BSWAT.
Property	Water Saturation (Drainage to Imbibition)	WSHY			BWSHY	Leaving water saturation for gas capillary pressure hysteresis.
Property	Water Saturation (Maximum Wetting)	WSMA			BWSMA	Capillary pressure hysteresis maximum water saturation.
Property	Water Viscosity	WVIS	FWVIS	RWVIS	BWVIS	
Ratio	Gas-Oil Ratio	RS	FRS	RRS	BRS	
Ratio	Oil-Gas Ratio	RV	FRV	RRV	BRV	
Volumes	Gas In-Place	GIP	FGIP	RGIP	BGIP	Liquid and gas phases.
Volumes	Gas In-Place (Gas Phase)	GIPG	FGIPG	RGIPG	BGIPG	
Volumes	Gas In-Place (Liquid Phase)	GIPL	FGIPL	RGIPL	BGIPL	Only for cases that have dissolved gas in the water phase.
Volumes	Gas In-Place (Reservoir Conditions)	GIPR	FGIPR			Compositional vector implemented in OPM Flow.
Volumes	Oil In-Place	OIP	FOIP	ROIP	BOIP	Liquid and gas phases.
Volumes	Oil In-Place (Gas Phase)	OIPG	FOIPG	ROIPG	BOIPG	
Volumes	Oil In-Place (Liquid Phase)	OIPL	FOIPL	ROIPL	BOIPL	

Block, Field, and Region Summary Variables						
Type	Variable	Root	Field	Region	Block	Comment
Volumes	Oil In-Place (Reservoir Conditions)	OIPR	FOIPR			Compositional vector implemented in OPM Flow.
Volumes	Pore Volume (Gas)	GPV	FGPV	RGPV	BGPV	
Volumes	Pore Volume (HCPV)	HPV	FHPV	RHPV	BHPV	
Volumes	Pore Volume (Oil)	OPV	FOPV	ROPV	BOPV	
Volumes	Pore Volume (Reservoir Conditions)	RPV	FRPV	RRPV	BRPV	Also BPORV.
Volumes	Pore Volume (Water)	WPV	FWPV	RWPV	BWPV	
Volumes	Rock Compaction (Trans. Multiplier X Direction)	PERMMDX			BPERMMDX	Also BPERMOD.
Volumes	Rock Compaction (Trans. Multiplier Y Direction)	PERMMDY			BPERMMDY	
Volumes	Rock Compaction (Trans. Multiplier Z Direction)	PERMMDZ			BPERMMDZ	
Volumes	Rock Compaction (Trans. Multiplier)	RTM	FRTM	RRTM	BRTM	
Volumes	Rock Compaction (Dual Porosity SIGMA Multiplier)	SIGMMOD			BSIGMMOD	
Volumes	Rock Compaction (PV Multiplier)	PORVMOD			BPORVMOD	
Volumes	Water In-Place	WIP	FWIP	RWIP	BWIP	
Volumes	Water In-Place (Reservoir Conditions)	WIIPR	FWIIPR			Compositional vector implemented in OPM Flow.
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.						

Table 11.12: Block, Field, and Region Summary Variables

11.2.7 FIELD AND REGION RECOVERY VARIABLES

Oil recovery factors (or efficiency) may be reported at both the field and the region levels using a variety of parameters as summarized in Table 11.13 and equations (11.1) to (11.6). In addition to the actual recovery factors the simulator can also report the oil volumes produced by various drive mechanisms as described in the table.

See also section [11.2.4 Field, Group, Well, Well Connection, and Completion Variables](#) that outlines the field production data summary variables, and section [11.2.6 Field, Region, and Block Variables](#) that outlines the field, region and block production and injection data summary variables.

Field and Region Summary Recovery Variables					
Type	Variable	Root	Field	Region	Comment
Recovery	Oil Recovery Efficiency (STOIIP)	OE	FOE	ROE	Based on STOIIP.
Recovery	Oil Recovery Efficiency (Wells)	OEW	FOEW	ROEW	Based on well production and STOIIP.
Recovery	Oil Recovery Efficiency (STOIIP Mobile with Respect to Water)	OEIW	FOEIW	ROEIW	Based on STOIIP and initial mobile oil saturate with respect to water.
Recovery	Oil Recovery Efficiency Wells Mobile with Respect to Water)	OEWW	FOEWW	ROEWW	Based on well production and initial mobile oil saturate with respect to water.
Recovery	Oil Recovery Efficiency (STOIIP Mobile with Respect to Gas)	OEIG	FOEIG	ROEIG	Based on STOIIP and initial mobile oil saturate with respect to gas.
Recovery	Oil Recovery Efficiency (Wells Mobile with Respect to Gas)	OEWG	FOEWG	ROEWG	Based on well production and initial mobile oil saturate with respect to gas.
Recovery	Oil Production Total (Free Gas Gas)	OTMF	FOTMF	ROTMF	Volumes reported at stock tank conditions.
Recovery	Oil Production Total (Gas Influx)	ORMG	FORMG	RORMG	
Recovery	Oil Production Total (Oil Expansion)	ORME	FORME	RORME	
Recovery	Oil Production Total (Rock Compaction)	ORMR	FORMR	RORMR	
Recovery	Oil Production Total (Solution Gas)	ORMS	FORMS	RORMS	
Recovery	Oil Production Total (Water Influx)	ORMW	FORMW	RORMW	
Notes:					
1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow.					
2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination.					
3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.					

Table 11.13: Field and Region Summary Recovery Variables

These summary variables are computed as:

$$OE = \frac{(STOIIP_{t=0} - STOIIP_{t=i})}{STOIIP_{t=0}} \quad (11.1)$$

$$OEIW = \frac{(Well\ Oil\ Production_{t=i})}{STOIPP_{t=0}}$$

(11.2)

$$OEIW = \frac{(STOIPP_{t=0} - STOIPP_{t=i})}{(Initial\ Mobile\ Oil\ Saturation\ with\ respect\ to\ Water_{t=0})}$$

(11.3)

$$OEWW = \frac{(Well\ Oil\ Production_{t=i})}{(Initial\ Mobile\ Oil\ Saturation\ with\ respect\ to\ Water_{t=0})}$$

(11.4)

$$OEIG = \frac{(STOIPP_{t=0} - STOIPP_{t=i})}{(Initial\ Mobile\ Oil\ Saturation\ with\ respect\ to\ Gas_{t=0})}$$

(11.5)

$$OEWG = \frac{(Well\ Oil\ Production_{t=i})}{(Initial\ Mobile\ Oil\ Saturation\ with\ respect\ to\ Gas_{t=0})}$$

(11.6)

In addition to all the aforementioned summary variables, there are variables associated with specific features that may have been enabled in the input deck, for instance, the Brine Model, or the Polymer Model. The next set of sections describe the variables associated with these specific options.

11.2.8 OPTION SPECIFIC VARIABLES – ALKALINE MODEL

This feature has not been implemented in OPM Flow.

11.2.9 OPTION SPECIFIC VARIABLES – API AND TRACER TRACKING

The summary variables in this section are for use for when either the API tracking model has been activated by the API keyword in the RUNSPEC section so that the various “oil types” are tracked in the model, or for when the Tracer Model has been requested via the TRACERS keyword in the RUNSPEC section, that defines the number of tracers in the model and the various passive tracer tracking options. Currently, only the Tracer model is implemented in OPM Flow.

API and Tracer Tracking Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block
Flow	Oil API	API	FAPI	GAPI	WAPI	CAPI	RAPI	BAPI
Flow	Tracer Flow Rate	TFR				CTFR		
Flow	Tracer Flow Total (Inter Region)	TFTT					RTFTT	
Flow	Tracer Flow Total (Free Inter Region)	TFTF					RTFTF	
Flow	Tracer Flow Total (Solution Inter Region)	TFTS					RTFTS	
Flow	Tracer Injection Concentration	TIC	FTIC	GTIC	WTIC	CTIC		
Flow	Tracer Injection Concentration (Free)	TICF	FTICF	GTICF	WTICF	CTICF		
Flow	Tracer Injection Concentration (Solution)	TICS	FTICS	GTICS	WTICS	CTICS		
Flow	Tracer Injection Rate	TIR	FTIR	GTIR	WTIR	CTIR		
Flow	Tracer Injection Rate (Free)	TIRF	FTIRF	GTIRF	WTIRF	CTIRF		
Flow	Tracer Injection Rate (Solution)	TIRS	FTIRS	GTIRS	WTIRS	CTIRS		
Flow	Tracer Injection Total	TIT	FTIT	GTIT	WTIT	CTIT		
Flow	Tracer Injection Total (Free)	TITF	FTITF	GTITF	WTITF	CTITF		
Flow	Tracer Injection Total (Solution)	TITS	FTITS	GTITS	WTITS	CTITS		
Flow	Tracer Production Concentration	TPC	FTPC	GTPC	WTPC	CTPC		
Flow	Tracer Production Concentration (Free)	TPCF	FTPCF	GTPCF	WTPCF	CTPCF		
Flow	Tracer Production Concentration (Solution)	TPCS	FTPCS	GTPCS	WTPCS	CTPCS		
Flow	Tracer Production Rate	TPR	FTPR	GTPR	WTPR	CTPR		
Flow	Tracer Production Rate (Free)	TPRF	FTPRF	GTPRF	WTPRF	CTPRF		
Flow	Tracer Production Rate (Solution)	TPRS	FTPRS	GTPRS	WTPRS	CTPRS		
Flow	Tracer Production Total	TPT	FTPT	GTPT	WTPT	CTPT		
Flow	Tracer Production Total (Free)	TPTF	FTPTF	GTPTF	WTPTF	CTPTF		
Flow	Tracer Production Total (Solution)	TPTS	FTPTS	GTPTS	WTPTS	CTPTS		
Volume	Tracer Concentration	TCN						BTCN

API and Tracer Tracking Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block
Volume	Tracer Concentration (Free)	TCNF						BTCNF
Volume	Tracer Concentration (Solution)	TCNS						BTCNS
Volume	Tracer In-Place	TIPT	FTIPT				RTIPT	BTIPT
Volume	Tracer In-Place (Free)	TIPF	FTIPF				RTIPF	BTIPF
Volume	Tracer In-Place (Solution)	TIPS	FTIPS				RTIPS	BTIPS
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.								

Table 11.14: API and Tracer Tracking Summary Variables

Note that for the tracer summary variables the mnemonic should be concatenated with the tracer name as defined by the [TRACER](#) keyword in the [PROPS](#) section. This keyword defines a series of passive tracers that are associated with a phase (oil, water, or gas) in the model, see the example for reference.

Example

In the PROPS section the TRACER keyword defines four passive tracers one for a gas injection well, one for tracking the dissolved gas, and two to track the injected water from two water injection wells.

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS
--
--      DEFINE TRACER NAMES
--
--      TRACER      TRACER
--      NAME        PHASE
--      -----
TRACER
      'IGS'         'GAS'           / GAS INJECTOR
      'DGS'         'GAS'           / DISOLVED GAS
      'IW1'         'WAT'           / WAT INJECTOR 1
      'IW2'         'WAT'           / WAT INJECTOR 2
/
```

If we wish to report the field total amount of the tracers in-place we would use the following in the SUMMARY section:

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY

FTIPTIGS
FTIPTDGS
FTIPTIW1
FTIPTIW2
```

Note that there is no terminating “/” for field summary variables.

11.2.10 OPTION SPECIFIC VARIABLES – ASPHALTENE MODEL

This feature has not been implemented in OPM Flow.

11.2.11 OPTION SPECIFIC VARIABLES – BRINE MODEL

The available Brine model summary keywords are dependent on which form of the Brine model has been invoked in the simulation input deck. In the [RUNSPEC](#) section the [BRINE](#) keyword activates the standard Brine Tracking model and optionally defines the water phase to have various salinities if the [ECLMC](#) keyword in the [RUNSPEC](#) section has been used to activate the Multi-Component Brine model. The Multi-Component Brine model allows for the water phase to have multiple water salinities.

Note that the Multi-Component Brine model is currently not supported by OPM Flow.

If OPM Flow's Salt Precipitation Model has been activated in the input deck via the [PRECSALT](#) keyword in the [RUNSPEC](#) section, then the Brine model summary keywords can also be employed with this option.

Brine and Multi-Component Brine Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Flow	Salt Flow Rate	SFR				CSFR			Brine and Salt Precipitation Models.
Flow	Salt Injection Rate	SIR	FSIR	GSIR	WSIR	CSIR			
Flow	Salt Injection Total	SIT	FSIT	GSIT	WSIT	CSIT			
Flow	Salt Production Rate	SPR	FSPR	GSPR	WSPR	CSPR			
Flow	Salt Production Total	SPT	FSPT	GSPT	WSPT	CSPT			
Property	Salt Corrected Water Viscosity	EMV_SAL						BEMV_SAL	
Volume	Salt Concentration (Block)	SCN						BSCN	
Volume	Salt Concentration (Injection)	SIC	FSIC	GSIC	WSIC	CSIC			
Volume	Salt Concentration (Production)	SPC	FSPC	GSPC	WSPC	CSPC			
Volume	Salt In-Place	SIP	FSIP				RSIP	BSIP	
Flow	Anion Concentration (Block)	TCNFANI						BTCNFANI	Multi-Component Brine Model.
Flow	Anion Flow Rate	TFRANI				CTFRANI			
Flow	Anion Injection Rate	TIRANI	FTIRANI	GTIRANI	WTIRANI				
Flow	Anion Injection Total	TITANI	FTITANI	GTITANI	WTITANI	CTITANI			
Flow	Anion Production Rate	TPRANI	FTPRANI	GTPRANI	WTPRANI				
Flow	Anion Production Total	TPTANI	FTPTANI	GTPTANI	WTPTANI	CTPTANI			
Flow	Cation Flow Rate	TFRCAT				CTFRCAT			
Flow	Cation Injection Rate	TIRCAT	FTIRCAT	GTIRCAT	WTIRCAT				
Flow	Cation Injection Total	TITCAT	FTITCAT	GTITCAT	WTITCAT	CTITCAT			
Flow	Cation Production Rate	TPRCAT	FTPRCAT	GTPRCAT	WTPRCAT				
Flow	Cation Production Total	TPTCAT	FTPTCAT	GTPTCAT	WTPTCAT	CTPTCAT			

Brine and Multi-Component Brine Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Property	Effective Salinity (Polymer)	ESALPLY						BESALPLY	
Property	Effective Salinity (Surfactant)	ESALSUR						BESALSUR	
Volume	Cation Concentration (Block)	TCNFCAT						BTCNFCAT	
Volume	Cation Concentration (Rock Associated)	TRADCAT						BTRADCAT	
Volume	Cation Concentration (Surfactant Associated)	TSADCAT						BTSADCAT	
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.									

Table 11.15: Brine and Multi-Component Brine Model Summary Variables

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

11.2.12 OPTION SPECIFIC VARIABLES - CO2STORE/H2STORE MODEL

The variables in this section are for OPM Flow's black-oil [CO2STORE/H2STORE](#) Model, and are based on the commercial simulator's compositional vectors (GCDI, GCDM, and WCD series), as the model is only available in the commercial simulator's compositional simulator.

CO2STORE/H2STORE Field, Group and Well Summary Variables							
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion
Flow	CO ₂ /H ₂ Mass Injection Rate	GMIR	FGMIR	GGMIR	WGMIR	CGMIR	CGMIRL
Flow	CO ₂ /H ₂ Mass Injection Total	GMIT	FGMIT	GGMIT	WGMIT	CGMIT	CGMITL
Flow	CO ₂ /H ₂ Mass Production Rate	GMPR	FGMPR	GGMPR	WGMPR	CGMPR	CGMPRL
Flow	CO ₂ /H ₂ Mass Production Total	GMPT	FGMPT	GGMPT	WGMPPT	CGMPT	CGMPTL
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.							

Table 11.16: CO2STORE/H2STORE Field, Group and Well Summary Variables

CO2STORE/H2STORE Field, Region, and Block Summary Variables					
Type	Variable	Root	Field	Region	Block
Mass	CO ₂ /H ₂ In-place	GMIP	FGMIP	RGMIP	BGMIP
Mass	CO ₂ /H ₂ In-place (Gas Phase)	GMGP	FGMGP	RGMGP	BGMGP
Mass	CO ₂ /H ₂ In-place (Liquid Phase)	GMDS	FGMDS	RGMDS	BGMDS
Mass	CO ₂ /H ₂ Effectively Residual Trapped ('Stranded') in the gas phase	GMST	FGMST	RGMST	BGMST
Mass	CO ₂ /H ₂ Effectively Residual Untrapped ('Unstranded') in the gas phase	GMUS	FGMUS	RGMUS	BGMUS
Mass	CO ₂ /H ₂ Maximum Potential Residual Trapped in the gas phase	GMTR	FGMTR	RGMTR	BGMTR
Mass	CO ₂ /H ₂ Maximum Potential Residual Untrapped in the gas phase	GMMO	FGMMO	RGMMO	BGMMO
Mass	CO ₂ /H ₂ Immobile in the gas phase (Non-wetting relative permeability equals zero)	GKTR	FGKTR	RGKTR	BGKTR
Mass	CO ₂ /H ₂ Mobile in the gas phase (Non-wetting relative permeability greater than zero)	GKMO	FGKMO	RGKMO	BGKMO
Moles	CO ₂ /H ₂ Immobile in the gas phase	GCDI	FGCDI	RGCDI	BGCDI
Moles	CO ₂ /H ₂ Mobile in the gas phase	GCDM	FGCDM	RGCDM	BGCDM

Moles	CO ₂ /H ₂ Immobile in the gas phase (Non-wetting relative permeability equals zero)	GKDI	FGKDI	RGKDI	BGKDI
Moles	CO ₂ /H ₂ Mobile in the gas phase (Non-wetting relative permeability greater than zero)	GKDM	FGKDM	RGKDM	BGKDM
Moles	CO ₂ /H ₂ Dissolved in the water phase	WCD	FWCD	RWCD	BWCD
Volume	Water in the gas phase	WIPG	FWIPG	RWIPG	BWIPG
Volume	Water in the water phase	WIPL	FWIPL	RWIPL	BWIPL

Notes:

- 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow.
- 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination.
- 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.

Table 11.17: CO2STORE/H2STORE Field, Region, and Block Summary Variables

See the OpenGoSim documentation (https://docs.opengosim.com/cirrus/theory/residual_and_stranded.html) for more details of how the trapped gas saturation and the stranded gas saturation contribute to the maximum potential residual trapping (GMTR) and the effective residual trapping (GMST) of CO₂ in saline aquifers when relative permeability hysteresis is modelled.

The 11th Society of Petroleum Engineers Comparative Solution Project (<http://spe.org/csp>) defines the “immobile free-phase CO₂” as “CO₂ at saturations for which the non-wetting phase relative permeability equals zero” (GKDI, GKTR).

Note that the GK* and GM* series of variables are OPM Flow specific variables.

Note that the WIPG and WIPL series of variables are OPM Flow specific variables.

11.2.13 OPTION SPECIFIC VARIABLES – COAL BED METHANE MODEL

The Coal Bed Methane model is not supported by OPM Flow.

11.2.14 OPTION SPECIFIC VARIABLES – CONDUCTIVE FAULTS MODEL

Conductive faults have not been implemented in OPM Flow.

11.2.15 OPTION SPECIFIC VARIABLES – ENVIRONMENTAL TRACERS

Environmental tracers are not supported by OPM Flow.

11.2.16 OPTION SPECIFIC VARIABLES – FORMATION DAMAGE MODEL

The variables in this section are for the OPM Flow's Formation Damage Model. The [WINJDAM](#) keyword sets up the filter cake properties for the specific water injection well in the [SCHEDULE](#) section.

Formation Damage Model Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block
Flow	Filtrate Injection Concentration	INJFC			WINFC			
Flow	Filtrate Volume Injection Rate	INJFVR			WINJFVR	CINJFVR		
Flow	Filtrate Volume Injection Total	INJFVT			WINJFVT	CINJFVT		
Property	Filter Cake Area Of Flow	FCAOF				CFCAOF		
Property	Filter Cake Permeability	FCPERM				CFCPERM		
Property	Filter Cake Porosity	FCPORO				CFCPORO		
Property	Filter Cake Radius	FCRAD				CFCRAD		
Property	Filter Cake Skin	FCSKIN				CFCSKIN		
Property	Filter Cake Width	FCWIDTH				CFCWIDTH		
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.								

Table 11.18: Formation Damage Model Summary Variables

Note that these are OPM Flow specific variables.

11.2.17 OPTION SPECIFIC VARIABLES – FOAM MODEL TRACERS

The Foam phase and model are activated via the **FOAM** keyword in the **RUNSPEC** section. Note in the commercial simulator the FOAM phase and model can be used in conjunction with the POLYMER and SURFACT phases; this is not the case for OPM Flow. OPM Flow's FOAM phase and model is a standalone implementation and cannot be used in conjunction with either the POLYMER or SURFACT phases.

Foam Model Tracers Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Flow	Foam Flow Rate	TFRFOA				CTFRFOA			
Flow	Foam Injection Rate	TIRFOA	FTIRFOA	GTIRFOA	WTIRFOA				
Flow	Foam Injection Total	TITFOA	FTITFOA	GTITFOA	WTITFOA	CTITFOA			
Flow	Foam Production Rate	TPRFOA	FTPRFOA	GTPRFOA	WTPRFOA				
Flow	Foam Production Total	TPTFOA	FTPTFOA	GTPTFOA	WTPTFOA	CTPTFOA			
Property	Foam Adsorption Total	TADSFOA	FTADSFOA			CTADSFOA	RTADSFOA	BTADSFOA	Block is current value.
Property	Foam Capillary Number	TCNMFOA						BTCNMFOA	
Property	Foam Decayed Half Life	THLFFOA						BTHLFFOA	
Property	Foam Decayed Tracer	TDCYFOA	FTDCYFOA				RTDCYFOA	BTDCYFOA	
Property	Foam Gas Mobility Factor	TMOBFOA	FTMOBFOA						Excluding shear.
Property	Foam in Solution	TIPTFOA	FTIPTFOA			CTIPTFOA	RTIPTFOA	BTIPTFOA	
Volume	Foam Concentration	TCNFFOA						BTCNFFOA	
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.									

Table 11.19: Foam Model Tracers Summary Variables

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

11.2.18 OPTION SPECIFIC VARIABLES – GAS FIELD OPERATIONS MODEL

The Gas Field Operations model has not been implemented in OPM Flow.

11.2.19 OPTION SPECIFIC VARIABLES – GAS LIFT OPTIMIZATION MODEL

For the Gas Lift Optimization model only a few SUMMARY vectors are available, both in OPM Flow and the commercial simulator, as depicted in Table 11.20. Gas lift optimization is activated by the LIFTOPT keyword in the SCHEDULE section as gas lift activation can vary through time depending on the flow characteristics of the wells.

Gas Lift Optimization Model Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block
Flow	Gas lift gas rate, based on ALQ	GLIR	FGLIR	GGLIR	WGLIR			
Flow	Incremental oil rate per unit of incremental gas lift gas quantity, that is the well's gradient as defined by equation (11.7) below	OGLR			WOGLR			
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.								

Table 11.20: Gas Lift Optimization Model Summary Variables

The incremental gas lift gas supply is allocated to a well based on the well's current incremental gradient using the following formulae:

$$Gradient = \frac{\beta_w \times \Delta Q_{Oil}}{\Delta Q_{GasLift} + \beta_g \times \Delta Q_{Gas}} \quad (11.7)$$

where

β_w	= a weighting factor for the preferential allocation of lift gas,
β_g	= the gas production rate weighting factor,
ΔQ_{Oil}	= the increment/decrement in oil production rate,
ΔQ_{Gas}	= the increment/decrement in gas production rate, and
$\Delta Q_{GasLift}$	= the increment/decrement in gas lift gas rate.

Normally β_g is defaulted to zero and in which case equation (11.7) reduces to:

$$Gradient = \frac{\beta_w \times \Delta Q_{Oil}}{\Delta Q_{GasLift}} \quad (11.8)$$

Note that if gas lift optimization has been activated in the model then the gas production SUMMARY variables (FGPR, GGPR, WGPR, FGPT, etc.) only contain the produced gas volumes, that is the reported values exclude the gas associated with gas lift gas.

Secondly, the well gas lift rates are adjusted by the well efficiencies in the calculating the group and field gas lift rates.

11.2.20 OPTION SPECIFIC VARIABLES – GAS CALORIFIC VALUE REPORTING

This feature has not been implemented in OPM Flow.

11.2.21 OPTION SPECIFIC VARIABLES – GI PSUEDO-COMPOSITIONAL MODEL

This feature has not been implemented in OPM Flow.

11.2.22 OPTION SPECIFIC VARIABLES – LGR AND COARSENING

Local Grid Refinement (“LGR”) and cell coarsening have not been implemented in OPM Flow.

11.2.23 OPTION SPECIFIC VARIABLES – BIOFILM AND MICP MODEL

The **MICP** keyword in the **RUNSPEC** section activates the Microbially Induced Calcite Precipitation (“MICP”) model for the run. The summary variables for this option are outlined in Table 11.21. Summary variables related to microbial and biofilm quantities can be also used with the **BIOFILM** model.

Biofilm and MICP Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Well Completion	Region	Block
Flow	Microbial Mass Injection Rate	MMIR	FMMIR	GMMIR	WMMIR	CMMIR	CMMIRL		
Flow	Oxygen Mass Injection Rate	MOIR	FMOIR	GMOIR	WMOIR	CMOIR	CMOIRL		
Flow	Urea Mass Injection Rate	MUIR	FMUIR	GMUIR	WMUIR	CMUIR	CMUIRL		
Flow	Microbial Mass Injection Total	MMIT	FMMIT	GMMIT	WMMIT	CMMIT	CMMITL		
Flow	Oxygen Mass Injection Total	MOIT	FMOIT	GMOIT	WMOIT	CMOIT	CMOITL		
Flow	Urea Mass Injection Total	MUIT	FMUIT	GMUIT	WMUIT	CMUIT	CMUITL		
Flow	Microbial Mass Production Rate	MMPR	FMMPR	GMMPR	WMMPR	CMMPR	CMMPRL		
Flow	Oxygen Mass Production Rate	MOPR	FMOPR	GMOPR	WMOPR	CMOPR	CMOPRL		
Flow	Urea Mass Production Rate	MUPR	FMUPR	GMUPR	WMUPR	CMUPR	CMUPRL		
Flow	Microbial Mass Production Total	MMPT	FMMPT	GMMPT	WMMPT	CMMPT	CMMPTL		
Flow	Oxygen Mass Production Total	MOPT	FMOPT	GMOPT	WMOPT	CMOPT	CMOPTL		
Flow	Urea Mass Production Total	MUPT	FMUPT	GMUPT	WMUPT	CMUPT	CMUPTL		
Mass	Microbes In-place	MMIP	FMMIP					RMMIP	BMMIP
Mass	Oxygen In-place	MOIP	FMOIP					RMOIP	BMOIP
Mass	Urea In-place	MUIP	FMUIP					RMUIP	BMUIP
Mass	Biofilm In-place	MBIP	FMBIP					RMBIP	BMBIP
Mass	Calcite In-place	MCIP	FMCIP					RMCIP	BMCIP
Notes: Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination.									

Table 11.21: Biofilm and MICP Model Summary Variables

11.2.24 OPTION SPECIFIC VARIABLES – MULTI-SEGMENT WELLS

The summary variables in this section are for use for when multi-segment wells are active in the model, and the variables only apply to the well segments requested for a multi-segment well, not to the well, well connection, or well completion.

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

Multi-Segment Wells Summary Variables				
Type	Variable	Root	Well Segment	Comment
Flow	Artificial Lift Quantity	ALQ	SALQ	See keyword WSEGTABL .
Flow	Brine Flow Rate	FR	SFR	Surfactant and Polymer model.
Flow	Gas Flow Rate	GFR	SGFR	
Flow	Gas Flow Rate (Free)	GFRF	SGFRF	
Flow	Gas Flow Rate (Solution)	GFRS	SGFRS	
Flow	Gas Flow Total (Absolute)	GFTA	SGFTA	
Flow	Gas Flow Total	GFT	SGFT	
Flow	Gas Flow Velocity	GFV	SGFV	
Flow	Gas Holdup Fraction	GHF	SGHF	
Flow	Gas Import Rate	GIMR	SGIMR	See keyword WSEGEXSS .
Flow	Gas Import Total	GIMT	SGIMT	See keyword WSEGEXSS .
Flow	Oil Absolute Flow Total	OFTA	SOFTA	
Flow	Oil Flow Rate	OFR	SOFR	
Flow	Oil Flow Rate (Free)	OFRF	SOFRF	
Flow	Oil Flow Rate (Solution)	OFRS	SOFRS	
Flow	Oil Flow Total	OFT	SOFT	
Flow	Oil Flow Velocity	OFV	SOFV	
Flow	Oil Holdup Fraction	OHF	SOHF	
Flow	Polymer Flow Rate	CFR	SCFR	Polymer model.
Flow	Tracer Flow Rate	TFR	STFR	Tracer model.
Flow	Water Flow Rate	WFR	SWFR	
Flow	Water Flow Total	WFT	SWFT	
Flow	Water Flow Total (Absolute)	WFTA	SWFTA	
Flow	Water Flow Velocity	WFRV	SWFRV	
Flow	Water Holdup Fraction	WHF	SWHF	

Multi-Segment Wells Summary Variables				
Type	Variable	Root	Well Segment	Comment
Flow	Water Import Rate	WIMR	SWIMR	See keyword WSEGEXSS .
Flow	Water Import Total	WIMT	SWIMT	See keyword WSEGEXSS .
Pressure	Pressure	PR	SPR	
Pressure	Pressure Drop	PRD	SPRD	
Pressure	Pressure Drop Component due to Friction	PRDF	SPRDF	
Pressure	Pressure Drop Component due to Hydrostatic head	PRDH	SPRDH	
Pressure	Pressure Drop due to Acceleration head	PRDA	SPRDA	
Pressure	Pressure Drop Frictional Multiplier	PRDM	SPRDM	See keyword WSEGMULT .
Property	API	API	SAPI	API model.
Property	Fluid Mixture Density (Excludes Exponents)	DENM	SDENM	Excludes exponents of flowing fractions.
Property	Fluid Mixture Density (Includes Exponents)	MDEN	SMDEN	Includes exponents of flowing fractions.
Property	Fluid Mixture Viscosity (Includes Exponents)	MVIS	SMVIS	Includes exponents of flowing fractions.
Property	Fluid Viscosity (Effective Mixture)	EMVIS	SEMVIS	Water/polymer fluid.
Property	Gas Density	GDEN	SGDEN	
Property	Gas Viscosity	GVIS	SGVIS	
Property	Gas-Liquid Drift Velocity (Vd)	GLVD	SGLVD	Drift flux slip model.
Property	Gas-Liquid Profile Parameter (C0)	GLPP	SGLPP	Drift flux slip model.
Property	Oil Density	ODEN	SODEN	
Property	Oil Viscosity	OVIS	SOVIS	
Property	Oil-Water Drift Velocity (Vd)	OWVD	SOWVD	Drift flux slip model.
Property	Oil-Water Profile Parameter (C0)	OWPP	SOWPP	Drift flux slip model.
Property	Pump Working Power	PPOW	SPPOW	See keyword WSEGPULL .
Property	Scale Deposition Diameter (Karst Conduit Calcite Dissolution)	FD	SFD	Scale Deposition model.
Property	Setting of Segment	FOPN	SFOPN	See keywords WSEGVALV , WSEGAICD , WSEGSICD , WSEGTABL , WSEGLABY , and WSEGFLIM .
Property	Strength of ICD on Segment	STR	SSTR	See keywords WSEGVALV , WSEGAICD , and WSEGSICD .
Property	Water Density (Pure Water)	WDEN	SWDEN	
Property	Water Viscosity (Pure Water)	WVIS	SWVIS	

Multi-Segment Wells Summary Variables				
Type	Variable	Root	Well Segment	Comment
Ratio	Gas-Oil Ratio	GOR	SGOR	
Ratio	Oil-Gas Ratio	OGR	SOGR	
Ratio	Water Cut	WCT	SWCT	
Ratio	Water Gas Ratio	WGR	SWGR	
Volume	Brine Concentration	SCN	SSCN	Surfactant and Polymer model
Volume	Polymer Concentration	CCN	SCCN	Polymer model
Volume	Tracer Concentration	TFC	STFC	Tracer model
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.				

Table 11.22: Multi-Segment Wells Summary Variables

Examples

For example, to get the segment oil flow rates for well OP01 segments two, three, and four and for segment two for OP02 one would use:

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY

SOFR
'OP01' 2 /
'OP01' 3 /
'OP01' 4 /
'OP02' 2 /
/

```

or to get all the oil flow rates for all the segments for well OP01:

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY

SOFR
'OP01' /
/

```

Care should be exercised when defaulting the list of wells and segments as there is the potential to generate large volumes of data.

11.2.25 OPTION SPECIFIC VARIABLES – NETWORK MODEL

For the Network models only a few SUMMARY vectors for groups and nodes are available, both in OPM Flow and the commercial simulator, as depicted in Table 11.23. There are two types of network option in the simulator: a Standard Network and an Extended Network option, the latter is activated by the NETWORK keyword in the RUNSPEC section. The summary vectors apply to both the Standard and Extended Network options.

Network Model Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block
Pressure	Group/Node pressure in a production network based on rates at end of timestep (OPM Flow specific). An alias for NPR.			GNETPR				
Pressure	Group/Node pressure in a production network.			GPR				
Pressure	Group /Node pressure in a gas injection network.			GPRG				
Pressure	Group/Node pressure in a water injection network.			GPRW				
Pressure	Pressure drop along the group's or node's outlet branch in a production network.			GPRB				
Pressure	Pressure drop along the group's or node's inlet branch in a gas injection network.			GPRBG				
Pressure	Pressure drop along the group's or node's inlet branch in a water injection network.			GPRBW				
Pressure	Group/Node pressure in a production network based on rates at end of timestep (OPM Flow specific).			NPR				
Flow	Artificial Lift Quantity ("ALQ") in the group's or node's outlet branch in a production network. Note no units are stated as ALQ units are variable, for example Hz for ESP or rate for gas lift gas.			GALQ				
Flow	Oil flow rate along the group's or node's outlet branch in a production network.			GOPRNB				
Flow	Water flow rate along the group's or node's outlet branch in a production network.			GWPRNB				
Flow	Gas flow rate along the group's or node's outlet branch in a production network.			GGPRNB				
Flow	Liquid flow rate along the group's or node's outlet branch in a production network.			GLPRNB				

Network Model Summary Variables								
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block
Flow	Water flow rate along the group's or node's inlet branch in water injection network.			GWIRNB				
Flow	Gas flow rate along the group's or node's inlet branch in gas injection network.			GGIRNB				
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.								

Table 11.23: Network Model Summary Variables

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

11.2.26 OPTION SPECIFIC VARIABLES – OPM FLOW SIMULATION PERFORMANCE

The following table (Table 11.24) lists the OPM Flow simulation performance summary variables that can be written to the SUMMARY file. Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

OPM Flow Simulation Performance		
Variable Description	Variable	Comment
Elapsed - Cumulative waiting time per process in seconds.	DTMWAIT	
Elapsed - Elapsed time in seconds.	ELAPSED	No data written to file.
Iterations - Number of linear iterations for each gradient calculation (Gradient Option).	HLINEARS	
Iterations - Total number of linear iterations for each gradient pressure (Gradient Option).	HSUMLINS	
Maximum Courant-Friedrichs-Lewy value for each time step.	MAXCFL	Compositional keyword.
Maximum variation in energy for each time step.	MAXDE	Compositional keyword.
Maximum variation in pressure for each time step.	MAXDPR	
Maximum variation in gas saturation for each time step.	MAXDSG	
Maximum variation in oil saturation for each time step.	MAXDSO	
Maximum variation in water saturation for each time step.	MAXDSW	
Maximum variation in temperature for each time step.	MAXDT	Compositional keyword.
Memory - Operating system reported maximum current memory usage across all processors.	MEMORYTS	
CPR Solver - Number of pressure iterations for each time step.	MLINEARP	
Iterations - Number of linear iterations for each time step.	MLINEARS	
Messages - Cumulative number of BUG messages.	MSUMBUG	
Messages - Cumulative number of COMMENT messages.	MSUMCOMM	
Messages - Cumulative number of ERROR messages.	MSUMERR	
CPR Solver - Cumulative number of pressure iterations.	MSUMLINP	
Iterations - Cumulative number of linear iterations.	MSUMLINS	
Messages - Cumulative number of MESSAGES messages.	MSUMMESS	
Iterations - Cumulative number of Newton iterations.	MSUMNEWT	

OPM Flow Simulation Performance		
Variable Description	Variable	Comment
Messages - Cumulative number of PROBLEM messages.	MSUMPROB	
Messages - Cumulative number of WARNING messages.	MSUMWARN	
Number of flash calculations for each time step	NBAKFL	Compositional keyword.
Memory - Peak usage of dynamically allocated memory reported by OPM Flow. For parallel cases this is the maximum across all processors.	NBYTOT	
CPR Solver - Average number of pressure iterations per linear iteration for each time step.	NCPRLINS	
Iterations – Cumulative average number of Newton iterations per cell in flash calculations..	NEWTFL	Compositional keyword.
Iterations - Number of Newton iterations used for each time step.	NEWTON	
CPR Solver - Average number of pressure iterations per Newton iteration per time step.	NLINEARP	
Iterations - Average number of linear iterations per Newton iteration for each time step.	NLINEARS	For runs with LGRs, LLINEARS will automatically be exported for each LGR.
Iterations - Maximum number of linear iterations in the Newton iterations per time step.	NLINSMAX	
Iterations - Minimum number of linear iterations in the Newton iterations per time step.	NLINSMIN	
Cumulative number of flash calculations.	NNUMFL	Compositional keyword.
Cumulative number of stability tests.	NNUMST	Compositional keyword.
Non-linear residual total.	NLRESSUM	Compositional keyword.
Non-linear residual maximum value.	NLRESMAX	Compositional keyword.
Number of time steps – taken.	NTS	Compositional keyword.
Number of time steps – pressure converged.	NTSPCL	Compositional keyword.
Number of time steps – molar density converged.	NTSMCL	Compositional keyword.
Number of time steps – energy density converged.	NTSECL	Compositional keyword.
Time Step – Criteria used to select the length of the time step, set to one of the following:	STEPTYPE	No data written to file. Note if the RUNSUM keyword has been activated in the SUMMARY section, then the mnemonics are written to the RSM file instead of integer values.
Value		
Criteria		
Definition		
1	INIT	Initial time step.
2	TRNC	Controlled via time truncation error.

OPM Flow Simulation Performance				
Variable Description			Variable	Comment
3	MINF	Minimum ratio between one time step and the next.		
4	MAXF	Maximum ratio between one time step and the next.		
5	MINS	Minimum permitted time step.		
6	MAXS	Maximum permitted time step.		
7	REPT	Report time step.		
8	HALF	Half of the time set to reach the next report time step.		
9	CHOP	Time step chopped due non-convergence of non-linear equations.		
10	SATM	Maximum saturation change.		
11	PCHP	Time step chopped due to maximum pressure change in IMPES formulation.		
12	DIFF	Difficult time step after a time step CHOP.		
13	LGRC	Determined by LGR fluid in-place error targets.		
14	SURF	Set by maximum expected grid block surfactant concentration change.		
15	NETW	Time step determined by network balancing controls.		
16	THRP	Time step controlled by maximum through put ratio.		
17	EMTH	Time step selected to match the end of a month.		
18	MAXP	Set by maximum expected grid block pressure change.		
19	WCYC	Determined by well cycling on/off time.		
20	MAST	Chosen by the master simulation in a reservoir coupling run.		
21	SLVR	Selected by a slave simulation to match a slave reporting time step, in a reservoir coupling run.		

OPM Flow Simulation Performance				
Variable Description			Variable	Comment
22	SLVC	Time step selected by a slave simulation due to a slave's expected flow rate change in a reservoir coupling run.		
23	MAXW	Maximum time step size after a well control event.		
24	EFF+	Selected by ZIPPY optimum time selection algorithm.		
25	EFF-			
26	NLTR			
27	EFFT			
28	DLYA			
29	ACTN			
30	RAIN			
CPU - Current CPU usage in seconds.			TCPU	It does not consider the time taken by inter-process communications in parallel runs, whereas ELAPSED does. Thus, for parallel jobs, ELAPSED is the most relevant time measurement.
CPU - CPU time per day (or hour in lab units depending on run units system).			TCPUDAY	
CPU – Cumulative CPU time for each gradient calculation (Gradient Option).			TCPUH	
CPU - Cumulative CPU time for all gradient calculations (Gradient Option).			TCPUHT	
CPU - Cumulative CPU time used in SCHEDULE section.			TCPUSCH	
CPU - CPU time per time step in seconds.			TCPUTS	
CPU - CPU time per time step for each gradient calculation (Gradient Option).			TCPUTSH	
CPU - CPU time per time step for all gradient calculations (Gradient Option).			TCPUTSHT	
CPU – Cumulative Tracer CPU time.			TCPUTR	Compositional keyword.
CPU – Cumulative Tracer implicit CPU time.			TCPUTRSV	Compositional keyword.
CPU – Cumulative Tracer Lagrangian tracer stramline CPU time.			TCPULGTR	Compositional keyword.

OPM Flow Simulation Performance		
Variable Description	Variable	Comment
CPU – Cumulative Tracer Lagrangian tracer solver CPU time.	TCPULGSV	Compositional keyword.
Elapsed - Elapsed time per day (or hour in lab units).	TELAPDAY	
Elapsed - Elapsed time per linear iteration in seconds.	TELAPLIN	No data written to file.
Elapsed - Elapsed time per time step in seconds.	TELAPTS	
Time Step - Length of time step.	TIMESTEP	
Iterations - Number of well Newton iterations taken in the last global Newton iteration. A negative value indicates that the well failed to converge.	WNEWTON	
Time Step - Predicted efficiency of the time step (developer use).	ZIPEFF	
Time Step - Predicted efficiency of the time step divided by the actual efficiency (developer use).	ZIPEFFC	
Notes: - Cells under the Variable column not colored indicate that the summary variable is available in OPM Flow. - Cells under the Variable column colored in gray indicate that the summary variable is recognized by the parser but the summary variable is not available. - Cells under the Variable column colored in orange indicate that the summary variable is not recognized by the parser and the summary variable is not available. These summary variables may cause the simulator to abort.		

Table 11.24: OPM Flow Simulator Performance Summary Variables

See also the PERFORMA and NMESSAGE keywords in section [11.3 Keyword Definitions](#) that write out a selection of the variables in Table 11.24.

11.2.27 OPTION SPECIFIC VARIABLES – POLYMER MODEL

The summary variables in this section are associated with Polymer phase and the polymer flood model. The feature is activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

Polymer Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Flow	Polymer Flow Rate	CFR				CCFR			
Flow	Polymer Flow Total (Inter-Region)	CFT					RCFT		
Flow	Polymer Injection Rate	CIR	FCIR	GCIR	WCIR	CCIR			
Flow	Polymer Injection Total	CIT	FCIT	GCIT	WCIT	CCIT			
Flow	Polymer Production Rate	CPR	FCPR	GCPR	WCPR	CCPR			
Flow	Polymer Production Total	CPT	FCPT	GCPT	WCPT	CCPT			
Flow	Salt Flow Rate	SFR				CSPR			
Flow	Salt Injection Rate	SIR	FSIR	GSIR	WSIR	CSIR			
Flow	Salt Injection Total	SIT	FSIT	GSIT	WSIT	CSIT			
Flow	Salt Production Rate	SPR	FSPR	GSPR	WSPR	CSPR			
Flow	Salt Production Total	SPT	FSPT	GSPT	WSPT	CSPT			
Property	Polymer Adsorbed (PLYTRRFA)	CABnnn						BCABnnn	The highest temperature band at which residual resistance factor was calculated, see keyword PLYTRRFA . The band number nnn can range from 001 to 999, but must be less than or equal to the argument on the PLYTRRFA keyword.
Property	Polymer Adsorption Total	CAD	FCAD				RCAD	BCAD	Block is concentration.
Property	Polymer Solution (Effective Viscosity)	EPVIS						BEPVIS	Also BVPOLY.
Property	Polymer Solution Shear Viscosity (Inter-Block I+ Direction)	SHWVISI						BSHWVISI	
Property	Polymer Solution Shear Viscosity (Inter-Block J+ Direction)	SHWVISJ						BSHWVISJ	

Polymer Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Property	Polymer Solution Shear Viscosity (Inter-Block K+ Direction)	SHWVISK						BSHWVISK	
Property	Polymer Thermal Degradation	CDCS						BCDCS	Total mass degraded in previous time step.
Property	Polymer Thermal Degradation Rate	CDCR						BCDCR	
Property	Polymer Thermal Degradation Rate (Adsorbed)	CDCA						BCDCA	
Property	Polymer Thermal Degradation Rate (Solution)	CDCP						BCDCP	
Property	Polymer Water (Effective Viscosity)	EMVIS						BEMVIS	Based on block center properties.
Property	Water Flow Rate Times Shear Multiplier (Inter-Block I+ Direction)	FLOW0I						BFLOW0I	Multiplied by the corresponding shear multiplier (PLYSHLOG and PLYSHEAR options only).
Property	Water Flow Rate Times Shear Multiplier (Inter-Block J+ Direction)	FLOW0J						BFLOW0J	
Property	Water Flow Rate Times Shear Multiplier (Inter-Block K+ Direction)	FLOW0K						BFLOW0K	
Property	Water Shear Rate After Shear Effects (Inter-Block I+ Direction)	SRTWI						BSRTWI	
Property	Water Shear Rate After Shear Effects (Inter-Block J+ Direction)	SRTWJ						BSRTWJ	
Property	Water Shear Rate After Shear Effects (Inter-Block K+ Direction)	SRTWK						BSRTWK	
Property	Water Shear Rate Prior to Shear Effects (Inter-Block I+ Direction)	SRTW0I						BSRTW0I	
Property	Water Shear Rate Prior to Shear Effects (Inter-Block J+ Direction)	SRTW0J						BSRTW0J	
Property	Water Shear Rate Prior to Shear Effects (Inter-Block K+ Direction)	SRTW0K						BSRTW0K	
Property	Water Velocity (Inter-Block I+ Direction)	VELW0I						BVELW0I	
Property	Water Velocity (Inter-Block J+ Direction)	VELW0J						BVELW0J	
Property	Water Velocity (Inter-Block K+ Direction)	VELW0K						BVELW0K	
Property	Water Viscosity (Effective)	EWV_POL						BEWV_POL	
Property	Water Viscosity Sheared Factor (Inter-Block I+ Direction)	PSHLZI						BPSHLZI	

Polymer Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Property	Water Viscosity Sheared Factor (Inter-Block J+ Direction)	PSHLZJ						BPSHLZJ	
Property	Water Viscosity Sheared Factor (Inter-Block K+ Direction)	PSHLZK						BPSHLZK	
Volume	Polymer Concentration (Block)	CCN						BCCN	
Volume	Polymer Concentration (Injection)	CIC	FCIC	GCIC	WCIC	CCIC			
Volume	Polymer Concentration (Production)	CPC	FCPC	GCPC	WCPC	CCPC			
Volume	Polymer Solution (In Solution)	CIP	FCIP				RCIP	BCIP	
Volume	Salt Cell Concentration	SCN						BSCN	
Volume	Salt In Place	SIP	FSIP				RSIP	BSIP	
Volume	Salt inter-region Flow Total	SFT					RSFT		
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.									

Table 11.25: Polymer Model Summary Variables

11.2.28 OPTION SPECIFIC VARIABLES – PSEUDO STEADY STATE

This feature has not been implemented in OPM Flow.

11.2.29 OPTION SPECIFIC VARIABLES – RESERVOIR COUPLING

This feature has not been implemented in OPM Flow.

11.2.30 OPTION SPECIFIC VARIABLES – SOLVENT MODEL

The **SOLVENT** keyword in the **RUNSPEC** section activates the solvent phase in the model and activates the four component solvent model for the run. In addition to this keyword, the oil, water, and gas phases should also be declared for the run using the **OIL**, **WATER**, and **GAS** keywords. The summary variables for this option are outlined in Table 11.26.

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

Solvent Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Flow	Solvent Flow Rate	NFR				CNFR			Field, group, and well gas rates and cumulative volumes include the solvent gas.
Flow	Solvent Flow Total (Inter-Region)	NFT					RNFT		
Flow	Solvent Injection Rate	NIR	FNIR	GNIR	WNIR				
Flow	Solvent Injection Total	NIT	FNIT	GNIT	WNIT	CNIT			
Flow	Solvent Production Rate	NPR	FNPR	GNPR	WNPR				
Flow	Solvent Production Total	NPT	FNPT	GNPT	WNPT	CNPT			
Property	Solvent Relative Permeability	NKR						BNKR	
Volume	Solvent In-Place	NIP	FNIP				RNIP	BNIP	
Volume	Solvent Saturation	NSAT						BNSAT	
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.									

Table 11.26: Solvent Model Summary Variables

11.2.31 OPTION SPECIFIC VARIABLES – SURFACTANT MODEL

This feature has not been implemented in OPM Flow.

11.2.32 OPTION SPECIFIC VARIABLES – THERMAL MODEL

The temperature option (*TEMP* keyword in the *RUNSPEC* section) and the thermal option (*THERMAL* keyword in the *RUNSPEC* section) are two separate modeling facilities in the commercial simulator; although some keywords can be used by both options. OPM Flow's thermal implementation is based on solving the energy equation fully coupled with the black-oil equations so the results are not directly equivalent to the commercial simulator's black-oil *TEMP* or compositional *THERMAL* formulations. The *THERMAL* keyword is used to invoke OPM Flow's thermal option. The summary variables for this option are listed in Table 11.16.

Note that not all these variables are available in OPM Flow; however, the simulator will issue warning messages if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

Thermal Model Summary Variables									
Type	Variable	Root	Field	Group	Well	Well Connection	Region	Block	Comment
Flow	Energy Injection Rate	TIRHEA	FTIRHEA	GTIRHEA	WTIRHEA				
Flow	Energy Injection Total	TITHEA	FTITHEA	GTITHEA	WTITHEA				
Flow	Energy Production Rate	TPRHEA	FTPRHEA	GTPRHEA	WTPRHEA				
Flow	Energy Production Total	TPTHEA	FTPTHEA	GTPTHEA	WTPTHEA				
Property	Temperature (Block)	TCNFHEA						BTCNFHEA	
Property	Temperature (Injection)	TICHEA	FTICHEA	GTICHEA	WTICHEA				
Property	Temperature (Production)	TPCHEA	FTPCHEA	GTPCHEA	WTPCHEA			BTPCHEA	
Volume	Energy In-Place Difference	TIPTHEA	FTIPTHEA				RTIPTHEA ⁴	BTIPTHEA	Difference in energy in-place from start of run to current time.

Notes:

- 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow.
- 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination.
- 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.
- 4) RTIPTHEA is currently not reported as unsupported by the simulator.

Table 11.27: Thermal Model Summary Variables

See the *THERMAL* keyword in the *RUNSPEC* section for additional information on this feature.

11.2.33 OPTION SPECIFIC VARIABLES – USER DEFINE QUANTITIES

The **UDQ** keyword in the **SCHEDULE** section defines variables and operations used to access the User Defined Quantities features in OPM Flow. UDQ variables can be constants, summary variables, as defined in the **SUMMARY** section, or a formula using various mathematical functions together with constants and summary variables.

The variable names defined by the **UDQ** keyword in the **SCHEDULE** section consist of a character string of up to eight characters in length, which stipulates the name of the user defined variable based on the type of variable. The variable type is defined by the first two characters of the variable name and must be set to one of the following:

- 1) CU: For variables that are associated with connections, for example summary variable COFR (Connection Oil Flow Rate).
- 2) FU: For variables that are associated with field data, for example summary variable FOPR (Field Oil Production Rate).
- 3) GU: For variables that are associated with groups, for example summary variable GLPR (Group Liquid Production Rate).
- 4) RU: For variables that are associated with regions, for example summary variable RPR (Region Pressure).
- 5) SU: For variables that are associated with multi-segment wells, for example summary variable SOFR (Segment Oil Flow Rate).
- 6) WU: For variables that are associated with wells, for example summary variable WWCT (Well Water Cut).
- 7) AU: For variables that are associated with aquifers, for example summary variable AAQP (Analytical Aquifer Pressure).
- 8) BU: For variables that are associated with blocks, for example summary variable BPR (Block oil phase Pressure).

The summary variable format for UDQ defined variables is presented in Table 11.17.

User Define Quantities Summary Variables							
Variable	Root	Field	Group	Well	Well Connection	Region	Well Segment
UDQ Variable	XXXXXX	FUXXXXXX	GUXXXXXX	WUXXXXXX	CUXXXXXX	RUXXXXXX	SUXXXXXX
Notes: 1) Where FUXXXXXX, GUXXXXXX, and so forth are the variable names defined by the UDQ keyword in the SCHEDULE section. 2) Output of aquifer (AU) and block (BU) summary variables is not supported.							

Table 11.28: User Define Quantities Summary Variables

The following example illustrates the use of this type of summary variable.

Examples

The example shows how to define some constant field variables used for calculating facilities corrected condensate yields in a wet gas model in the SCHEDULE section. The corrected condensate rate is stored as variable named FU_FNGLR.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
-- DEFINE START OF USER DEFINED QUANTITY SECTION
--
UDQ
--
-- OPERATOR VARIABLE  EXPRESSION
--
ASSIGN      FUNGLYLD  1.100000                      / Condensate Yield Correction
--
-- OPERATOR VARIABLE  EXPRESSION
--
DEFINE      FU_FNGLR  FGPR *(FOGR * FUNGLYLD) / Calculate Condensate Rate Field
UPDATE      FU_FNGLR  ON /
UNITS       FU_FNGLR  STBD /

/ DEFINE END OF USER DEFINED QUANTITY SECTION
```

In order to report the field corrected condensate values, one would declare FU_FNGLR in the SUMMARY section, as shown below.

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY

FU_FNGLR
```

Note that the normal summary variables syntax rules still apply to groups, regions, wells, etc. So for example, in addition, if the same calculations were performed on two wet gas producers, GP01 and GP02, then one would use:

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY

FU_FNGLR

WU_FNGLR
GP01
GP02
/
```

Assuming of course that WU_FNGLR was defined in the SCHEDULE section.

11.2.34 OPTION SPECIFIC VARIABLES – WELLBORE FRICTION MODEL

This feature has not been implemented in OPM Flow.

11.3 KEYWORD DEFINITIONS

11.3.1 ALL – EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	----------------	--------------------------

Description

This keyword activates the writing out of a standard set of summary production and injection data vectors for the field, group and well objects to the [SUMMARY](#) (*.SMSPEC and *.UNSMRY) and RSM (*.RSM) files. Table 11.29 lists the production, injection, pressure and volume summary variables written out by the [ALL](#) keyword, and Table 11.30 list the aquifer variables.

Standard Production, Injection, and Pressures Summary Variables					
Variable	Root	Field	Group	Well	Comment
Gas Injection Rate	GIR	FGIR	GGIR	WGIR	
Gas Injection Total	GIT	FGIT	GGIT	WGIT	
Gas Production Rate	GPR	FGPR	GGPR	WGPR	Produced reservoir gas only, gas lift gas is excluded.
Gas Production Total	GPT	FGPT	GGPT	WGPT	
Oil Injection Rate	OIR	FOIR	GOIR	WOIR	
Oil Injection Total	OIT	FOIT	GOIT	WOIT	
Oil Production Rate	OPR	FOPR	GOPR	WOPR	
Oil Production Total	OPT	FOPT	GOPT	WOPT	
Res.Vol. Injection Rate	VIR	FVIR	GVIR	WVIR	
Res.Vol. Injection Total	VIT	FVIT	GVIT	WVIT	
Res.Vol. Production Rate	VPR	FVPR	GVPR	WVPR	
Res.Vol. Production Total	VPT	FVPT	GVPT	WVPT	
Water Injection Rate	WIR	FWIR	GWIR	WWIR	
Water Injection Total	WIT	FWIT	GWIT	WWIT	
Water Production Rate	WPR	FWPR	GWPR	WWPR	
Water Production Total	WPT	FWPT	GWPT	WWPT	
Bottom-Hole Pressure	BHP			WBHP	
Productivity	PI			WPI	Preferred Phase
Tubing Head Pressure	THP			WTHP	
Gas-Oil Ratio	GOR	FGOR	GGOR	WGOR	
Water -Gas Ratio	WGR	FWGR	GWGR	WWGR	
Water Cut	WCT	FWCT	GWCT	WWCT	

Standard Production, Injection, and Pressures Summary Variables					
Pressures and Volumes					
Variable	Root	Field	Group	Well	Comment
Oil Phase Pressure	PR	FPR			Field and Region HCPV weighted
Gas In-Place	GIP	FGIP			Liquid and gas phases
Gas In-Place (Gas Phase)	GIPG	FGIPG			
Gas In-Place (Liquid Phase)	GIPL	FGIPL			Only for cases that have dissolved gas in the water phase
Oil In-Place	OIP	FOIP			Liquid and gas phases
Oil In-Place (Gas Phase)	OIPG	FOIPG			
Oil In-Place (Liquid Phase)	OIPL	FOIPL			
Water In-Place	WIP	FWIP			
Notes: - Cells colored in gray indicate that the summary variable is not available for this combination. - Cells colored orange under the Type column indicate that this set of summary variables are not available in OPM Flow because the underlying feature is not available in OPM Flow.					

Table I I.29: Standard Production, Injection, and Pressure Summary Variables

Standard Aquifer Summary Variables						
Variable	Root	Field	Analytical Aquifer	Analytical Aquifer List	Numerical Aquifer	Comment
Aquifer Influx Rate (Water Aquifers)	QR	FAQR	AAQR			
Aquifer Influx Total (Water Aquifers)	QT	FAQT	AAQT			
Aquifer Influx Rate (Gas Aquifers)	QRG	FAQRG	AAQRG			
Aquifer Influx Total (Gas Aquifers)	QTG	FAQTG	AAQTG			
Notes: - Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. - Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. - Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.						

Table I I.30: Standard Aquifer Summary Variables

There is no data required for this keyword and there is no terminating "/" for this keyword.

Examples

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--     EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--     ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--     ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

Note the [SEPARATE](#) keyword is not required for OPM Flow as this is the default behavior; however, it is probably good practice to include it if the same input decks are being run with commercial simulator.

11.3.2 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The [COLUMNS](#) keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the GLOBAL section for a full description.

11.3.3 DATE – ACTIVATE THE DATE OPTION FOR THE SUMMARY FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the writing out of the date of each time step to the SUMMARY file. Normally only the time in days and decimal years are written out to the SUMMARY file, activating the DATE option also results in the DATE being written out to the SUMMARY file as well. This option is normally used when the RUNSUM keyword in the SUMMARY section has been activated to produce a RSM file.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

The following example shows an example RSM file output when the DATE option has not been activated.

SUMMARY OF RUN NO-DATE-KEYWORD

TIME	YEARS	FPR	FOEW	FOPR	FOPT
DAYS	YEARS	PSIA		STB/DAY	STB
0	0	4467.125	0	0	0
1.000000	0.002738	4466.943	0.000239	3235.662	3235.662
31.00000	0.084873	4464.476	0.007407	3230.117	100256.4
60.00000	0.164271	4462.717	0.014291	3193.902	193421.5
91.00000	0.249144	4460.813	0.021523	3127.557	291306.3
121.0000	0.331280	4458.909	0.028362	3055.878	383879.7
152.0000	0.416153	4456.914	0.035262	2982.212	477271.4

And activating the SUMMARY file DATE option with:

```
--
--      ACTIVATE DATE SUMMARY FILE OPTION
--
DATE
```

Results in the following example RSM file output.

SUMMARY OF RUN WITH-DATE-KEYWORD

DATE	YEARS	DAY	MONTH	YEAR	FPR	FOEW	FOPR
	YEARS				PSIA		STB/DAY
1-JAN-98	0	19	10	1992	4467.125	0	0
2-JAN-98	0.002738	20	10	1992	4466.943	0.000239	3235.662
31-JAN-98	0.084873	21	10	1992	4464.476	0.007407	3230.117
28-FEB-98	0.164271	24	10	1992	4462.717	0.014291	3193.902
31-MAR-98	0.249144	28	10	1992	4460.813	0.021523	3127.557
30-APR-98	0.331280	3	11	1992	4458.909	0.028362	3055.878
31-MAY-98	0.416153	14	11	1992	4456.914	0.035262	2982.212

11.3.4
DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the GLOBAL section for a full description.

11.3.5
ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

11.3.6
END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the GLOBAL section for a full description.

11.3.7
ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the GLOBAL section for a full description.

11.3.8
ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the GLOBAL section for a full description.

11.3.9 EXCEL – ACTIVATE THE EXCEL OPTION FOR THE SUMMARY FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates the writing out of the RSM file data in a format that can easily be loaded into Microsoft's EXCEL spreadsheet program or LibreOffice's CALC spreadsheet program. The RSM file output is activated by the [RUNSUM](#) keyword in the [SUMMARY](#) section.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Examples

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--     ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--     ACTIVATE EXCEL SUMMARY FILE OPTION
--
EXCEL
```

The above example activates the [SUMMARY](#) file [EXCEL](#) option for directly loading the RSM file into either Microsoft's EXCEL or LibreOffice's CALC spreadsheet programs.

11.3.10 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

11.3.11 FMWSET – EXPORT WELL STATUS VECTORS FOR THE FIELD TO FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	----------------	--------------------------

Description

This keyword is similar to the [ALL](#) keyword in the [SUMMARY](#) section, in that it results in a group of summary variables to be written out to the [SUMMARY](#) file. In this case the keyword activates the writing out of a set of data vectors that give the production and injections status of all the wells in the model, as well as the number of wells in the drilling queue and the number of workover events occurring within a time step. Both instantaneous and cumulative well counts and events are written out as listed in Table 11.31.

There is no data required for this keyword.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Field and Group Well Status Summary Variables				
Variable	Root	Field	Group	Comment
Number of abandoned injection wells	MWIA	FMWIA	GMWIA	
Number of abandoned production wells	MWPA	FMWPA	GMWPA	
Number of drilling events in total	MWDT	FMWDT	GMWDT	
Number of drilling events this time step	MWDR	FMWDR	GMWDR	
Number of injection wells currently flowing	MWIN	FMWIN	GMWIN	
Number of injectors on group control	MWIG	FMWIG	GMWIG	
Number of injectors on own reservoir volume rate limit control	MWIV	FMWIV	GMWIV	
Number of injectors on own surface rate limit control	MWIS	FMWIS	GMWIS	
Number of injectors on pressure control	MWIP	FMWIP	GMWIP	
Number of producers controlled by own oil rate limit	MWPO	FMWPO	GMWPO	
Number of producers on group control	MWPG	FMWPG	GMWPG	
Number of producers on own reservoir volume rate limit control	MWPV	FMWPV	GMWPV	
Number of producers on own surface rate limit control	MWPS	FMWPS	GMWPS	
Number of producers on pressure control	MWPP	FMWPP	GMWPP	
Number of producers using artificial lift (with ALQ > 0.0)	MWPL	FMWPL	GMWPL	
Number of production wells currently flowing	MWPR	FMWPR	GMWPR	
Number of unused injection wells	MWIU	FMWIU	GMWIU	
Number of unused production wells	MWPU	FMWPU	GMWPU	
Number of workover events in total	MWWT	FMWWT	GMWWT	
Number of workover events this time step.	MWWO	FMWWO	GMWWO	
Total number of injection wells	MWIT	FMWIT	GMWIT	
Total number of production wells	MWPT	FMWPT	GMWPT	

Field and Group Well Status Summary Variables				
Variable	Root	Field	Group	Comment
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.				

Table 11.31: *FMWSET* - Standard Field and Group Well Status Summary Variables

See also the *GMWSET* keyword in the *SUMMARY* schedule that has similar functionality but at a group level.

Example

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT WELL STATUS VECTORS FOR THE FIELD TO FILE
--
FMWSET
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

The above example exports the field standard well status variables to the *SUMMARY* file.

11.3.12 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be place at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a dull description.

11.3.13 GMWSET – EXPORT WELL STATUS VECTORS BY GROUP TO FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	----------------	--------------------------

Description

This keyword is similar to the [ALL](#) keyword in the [SUMMARY](#) section, in that it results in a group of summary variables to be written out to the [SUMMARY](#) file. In this case the keyword activates the writing out of a set of data vectors that give the production and injections status of all the wells in the named groups, as well as the number of wells in the drilling queue and the number of workover events occurring within a time step for the requested groups. Both instantaneous and cumulative well counts and events for the groups are written out as tabulated in Table 11.32.

Note that [GMWSET](#) should be followed by a list of group names enclosed in quotes and therefore a terminating “/” is required to end the list of groups. A blank list requests output for all groups.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

Field and Group Well Status Summary Variables				
Variable	Root	Field	Group	Comment
Number of abandoned injection wells	MWIA	FMWIA	GMWIA	
Number of abandoned production wells	MWPA	FMWPA	GMWPA	
Number of drilling events in total	MWDT	FMWDT	GMWDT	
Number of drilling events this timestep	MWDR	FMWDR	GMWDR	
Number of injection wells currently flowing	MWIN	FMWIN	GMWIN	
Number of injectors on group control	MWIG	FMWIG	GMWIG	
Number of injectors on own reservoir volume rate limit control	MWIV	FMWIV	GMWIV	
Number of injectors on own surface rate limit control	MWIS	FMWIS	GMWIS	
Number of injectors on pressure control	MWIP	FMWIP	GMWIP	
Number of producers controlled by own oil rate limit	MWPO	FMWPO	GMWPO	
Number of producers on group control	MWPG	FMWPG	GMWPG	
Number of producers on own reservoir volume rate limit control	MWPV	FMWPV	GMWPV	
Number of producers on own surface rate limit control	MWPS	FMWPS	GMWPS	
Number of producers on pressure control	MWPP	FMWPP	GMWPP	
Number of producers using artificial lift (with ALQ > 0.0)	MWPL	FMWPL	GMWPL	
Number of production wells currently flowing	MWPR	FMWPR	GMWPR	
Number of unused injection wells	MWIU	FMWIU	GMWIU	
Number of unused production wells	MWPU	FMWPU	GMWPU	
Number of workover events in total	MWWT	FMWWT	GMWWT	
Number of workover events this time step.	MWWO	FMWWO	GMWWO	
Total number of injection wells	MWIT	FMWIT	GMWIT	
Total number of production wells	MWPT	FMWPT	GMWPT	

Field and Group Well Status Summary Variables				
Variable	Root	Field	Group	Comment
Notes: 1) Cells colored in gray with a mnemonic indicate that the summary variable is not available in OPM Flow. 2) Cells colored in gray with no mnemonic indicate that the summary variable is not available for this combination. 3) Cells colored orange show combinations that are not available in OPM Flow because the underlying feature is not available.				

Table 11.32: *GMWSET* - Standard Field and Group Well Status Summary Variables

See also the *FMWSET* keyword in the *SUMMARY* schedule that has similar functionality but at the field level.

Examples

The first example below exports all the group standard well status variables to the *SUMMARY* file.

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT WELL STATUS VECTORS FOR NAMED GROUPS TO FILE
--
GMWSET
/
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

The second example exports all the group standard well status variables for just the PLAT1 and PLT2 groups only to the *SUMMARY* file.

```
--
--      EXPORT WELL STATUS VECTORS FOR NAMED GROUPS TO FILE
--
GMWSET
'PLAT1' 'PLAT2'
/
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

11.3.14 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDINC](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the GLOBAL section for a full description.

11.3.15 MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the GLOBAL section for a full description.

11.3.16 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the GLOBAL section for a full description.

11.3.17 MONITOR – ACTIVATE OUTPUT OF THE MONITORING DATA AND FILE

The [MONITOR](#) keyword activates the writing out of the run time monitoring information used by post-processing graphics software to display run time information, for example the simulated production and injection rates and cumulative values. OPM Flow does not have this functionality.

See [MONITOR – Activate Output of the Monitoring Data and File](#) in the [RUNSPEC](#) section for a full description.

11.3.18 NARROW – ACTIVATE RUN SUMMARY NARROW COLUMN OUTPUT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *NARROW* keyword activates the Run Summary Narrow Column Output option, for when printed *SUMMARY* data has been requested by the *RUNSUM* keyword in the *SUMMARY* section. The option increases the number of columns “printed on the page”.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

11.3.19 NEWTON – ACTIVATE NEWTON ITERATION SUMMARY OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates the writing out of the Newton iteration vector (the number of non-linear iterations per time step) to the [SUMMARY](#) file, and the RSM file if the RSM file option has been requested by the [RUNSUM](#) keyword in the [SUMMARY](#) section,

There is no data required for this keyword and there is no terminating “/” for this keyword.

Although the keyword is recognized by OPM Flow only zeros are written to the [SUMMARY](#) file.

Example

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE      --
--      ACTIVATE NEWTON ITERATION SUMMARY OUTPUT
--
NEWTON
```

The above example actives the writing out of the Newton iteration vector to the [SUMMARY](#) file.

11.3.20 NMESSAGE – EXPORT CUMULATIVE MESSAGE SUMMARY VARIABLES TO FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the writing out of a standard set of summary OPM Flow simulation performance summary variables to the [SUMMARY](#) (*.SMSPEC and *.UNSMRY) and RSM (*.RSM) files, namely the number of messages written per message class. Table 11.33 lists the summary variables written out by the [NMESSAGE](#) keyword.

The keyword is recognized by OPM Flow but none of the variables are currently supported.

OPM Flow Simulator Performance Summary Variables		
Cumulative Message Variables		
Variable Description	Variable	Comment
Messages - Cumulative number of BUG messages.	MSUMBUG	Unsupported.
Messages - Cumulative number of COMMENT messages.	MSUMCOMM	Unsupported.
Messages - Cumulative number of ERROR messages.	MSUMERR	Unsupported.
Messages - Cumulative number of MESSAGES messages.	MSUMMESS	Unsupported.
Messages - Cumulative number of PROBLEM messages.	MSUMPROB	Unsupported.
Messages - Cumulative number of WARNING messages.	MSUMWARN	Unsupported.
Notes: - Cells under the Variable column not colored indicate that the summary variable is available in OPM Flow. - Cells under the Variable column colored in gray indicate that the summary variable is recognized by the parser but the summary variable is not available. - Cells under the Variable column colored in orange indicate that the summary variable is not recognized by the parser and the summary variable is not available. These summary variables may cause the simulator to abort.		

Table 11.33: Simulator Performance Summary Variables (Cumulative Messages)

Example

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT PERFORMANCE CUMULATIVE MESSAGE VARIABLE VECTORS TO FILE
--
NMESSAGE
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

Note the [SEPARATE](#) keyword is not required for OPM Flow as this is the default behavior; however, it is probably good practice to include it if the same input decks are being run with the commercial simulator.

11.3.21 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active. but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

11.3.22 NOMONITO – DEACTIVATE OUTPUT OF THE MONITORING DATA AND FILE

The [NOMONITO](#) keyword deactivates the writing out of the run time monitoring information used by post-processing graphics software to display run time information, for example the simulated production and injection rates and cumulative values. OPM Flow does not have this functionality.

See [NOMONITO – Deactivate Output of the Monitoring Data and File](#) in the [RUNSPEC](#) section for a full description.

11.3.23 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

11.3.24 OFM – ACTIVATE OFM FILE OUTPUT OF THE SUMMARY DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates the writing out of the [SUMMARY](#) file data in the Oil Field Manager (“OFM”) file format to enable the simulated data to be directly loaded into Oil Field Manager.

There is no data required for this keyword and there is no terminating “/” for this keyword.

See also the [EXCEL](#), [RPTONLY](#), [RUNSUM](#) and [SEPARATE](#) keywords in the [SUMMARY](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

11.3.25 PERFORMA – EXPORT STANDARD SIMULATOR PERFORMANCE SUMMARY VARIABLES TO FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	----------------	--------------------------

Description

The [PERFORMA](#) keyword activates the writing out of a standard set of OPM Flow simulation numerical performance summary variables to the [SUMMARY](#) (*.SMSPEC and *.UNSMRY) and RSM (*.RSM) files. Table 11.34 lists the summary variables written out by the keyword.

Note that not all these variables are available in OPM Flow; however, the simulator will issue a warning message if this is indeed the case. It is anticipated that the number of recognized summary variables will increase in future releases of OPM Flow.

OPM Flow Simulator Performance Summary Variables		
Numerical Performance Variables		
Variable Description	Variable	Comment
Elapsed - Elapsed time in seconds.	ELAPSED	No data written to file.
Iterations - Number linear iterations for each time step.	MLINEARS	
Iterations - Cumulative number of linear iterations.	MSUMLINS	
Iterations - Cumulative number of Newton iterations.	MSUMNEWT	
Iterations - Number of Newton iterations used for each time step.	NEWTON	
Iterations - Average number of linear iterations per Newton iteration for each time step.	NLINEARS	For runs with LGRs, LLINEARS will automatically be exported for each LGR .
Iterations - Maximum number of linear iterations in the Newton iterations per time step.	NLINSMAX	
Iterations - Minimum number of linear iterations in the Newton iterations per time step.	NLINSMIN	
Time Step – Criteria used to select the length of the time step. See section 11.2.26 Option Specific Variables – OPM Flow Simulation Performance for the definition of the STEPTYPE mnemonics.	STEPTYPE	No data written to file.
CPU - Current CPU usage in seconds.	TCPU	Does not consider the time taken by inter-process communications in parallel runs, whereas ELAPSED does. Thus, for parallel jobs, ELAPSED is the most relevant time measurement.
CPU - CPU time per day (or hour in lab units depending on run units system).	TCPUDAY	No data written to file.
CPU - CPU time per time step in seconds.	TCPUTS	No data written to file.

OPM Flow Simulator Performance Summary Variables		
Numerical Performance Variables		
Variable Description	Variable	Comment
Elapsed - Elapsed time per linear iteration in seconds.	TELAPLIN	No data written to file.
Time Step - Length of time step.	TIMESTEP	
Notes: - Cells under the Variable column not colored indicate that the summary variable is available in OPM Flow. - Cells under the Variable column colored in gray indicate that the summary variable is recognized by the parser but the summary variable is not available. - Cells under the Variable column colored in orange indicate that the summary variable is not recognized by the parser and the summary variable is not available. These summary variables may cause the simulator to abort.		

Table 11.34: Simulator Performance Summary Variables (Numerical Performance)

Note

If the summary vector data is unavailable then zeros are written at each time step to the [SUMMARY](#) and RSM file.

Example

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT NUMERICAL PERFORMANCE SUMMARY VARIABLES TO FILE
--
PERFORMA
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE

```

Note the [SEPARATE](#) keyword is not required for OPM Flow as this is the default behavior; however, it is probably good practice to include it if the same input decks are being run with commercial simulator.

11.3.26 PYEND – END THE DEFINITION OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword on a separate single line.

See [PYEND – End the Definition of a PYINPUT Section](#) in the [GRID](#) section for a full description.

11.3.27 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow's Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator. [PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line.

See [PYINPUT – Define the Start of a PYINPUT Section](#) in the [GRID](#) section for a full description.

11.3.28 RPTONLY – ACTIVATE THE REPORT TIME STEPS ONLY OPTION FOR THE SUMMARY FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword activates the writing out of the [SUMMARY](#) file and the RSM file data, if the RSM file option has been requested by the [RUNSUM](#) keyword in the [SUMMARY](#) section, at report time steps only. The default is for the data to be written out for all time steps to the [SUMMARY](#) files. This keyword reduces the file size at the expense of lower resolution in the time domain.

There is no data required for this keyword and there is no terminating “/” for this keyword.

The option can be deactivated by the [RPTONLYO](#) keyword in the [SUMMARY](#) section.

Example

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
--
--      ACTIVATE REPORT TIME STEPS ONLY SUMMARY FILE OPTION
--
RPTONLY
```

The above example activates the writing out of the [SUMMARY](#) file at report time steps only.

11.3.29 RPTONLYO – DEACTIVATE THE REPORT TIME STEPS ONLY OPTION FOR THE SUMMARY FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword deactivates the writing out of the [SUMMARY](#) file and the RSM file data, if the RSM file option has been requested by the [RUNSUM](#) keyword in the [SUMMARY](#) section, at report time steps only, and switches on writing out all the time steps to the files. This option is the default behavior for when [RPTONLY](#) has not been activated.

There is no data required for this keyword and there is no terminating “/” for this keyword.

The option can be deactivated by the [RPTONLY](#) keyword in the [SUMMARY](#) section that will switch on writing the data at every report time step instead of every time step.

Example

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
--
--      DEACTIVATE REPORT TIME STEPS ONLY SUMMARY FILE OPTION
--
RPTONLYO

```

The above example deactivates the writing out of the [SUMMARY](#) file at report time steps only, and switches on writing out all the time steps to the file.

11.3.30 RPTSMRY – ACTIVATE OR DEACTIVATE SUMMARY LIST REPORT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates or deactivates a listing of all the summary variables that are going to be written to the [SUMMARY](#) file and the RSM file, if the RSM file option has been requested by the [RUNSUM](#) keyword in the [SUMMARY](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description	Default
I	RPTSMRY	An integer value set to zero for no report, or one to produce the report.	0

Notes: I) The keyword is terminated by a “/”.			
---	--	--	--

Table 11.35: RPTSMRY Keyword Description

Example

```

-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--     EXPORT STANDARD SUMMARY VARIABLE VECTORS TO FILE
--
ALL
--
--     ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--     ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
--
--     ACTIVATE OR DEACTIVATE SUMMARY LIST REPORT
--
RPTSMRY
      1                               /

```

The example switches on the summary list report.

11.3.31 RUNSUM – ACTIVATE RSM FILE OUTPUT OF THE SUMMARY DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates the writing out of the [SUMMARY](#) file data in a columnar format to the PRT file. Normally the [SEPARATE](#) keyword in the [SUMMARY](#) section is invoked in the same run to direct the data stream to a separate RSM file for easy loading into other programs, for example, Microsoft's EXCEL or LibreOffice's CALC spreadsheet programs.

There is no data required for this keyword and there is no terminating “/” for this keyword.

See also the [EXCEL](#), [RPTONLY](#) and [SEPARATE](#) keywords in the [SUMMARY](#) section.

Example

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--      ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--      ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

Note unlike the commercial simulator, OPM Flow always writes out the data to a separate file.

11.3.32 SEPARATE – ACTIVATE THE SEPARATE RSM FILE OUTPUT OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates the writing out of the [SUMMARY](#) file date in a columnar format to the RSM file, if the [RUNSUM](#) keyword has also been activated in the [SUMMARY](#) section. Both the [SEPARATE](#) and the [RUNSUM](#) keywords need to be invoked. If the [SEPARATE](#) option is not activated then the RSM output is directed to the end of the PRT file. Normally the both the [SEPARATE](#) and [RUNSUM](#) keywords are invoked in the same run to enable easy loading of the data into Microsoft's EXCEL or LibreOffice's CALC spreadsheet programs.

There is no data required for this keyword and there is no terminating “/” for this keyword.

See also the [EXCEL](#), [RPTONLY](#) and [RUNSUM](#) keywords in the [SUMMARY](#) section.

Example

```
-- =====
--
-- SUMMARY SECTION
--
-- =====
SUMMARY
--
--     ACTIVATE COLUMNAR SUMMARY DATA REPORTING OPTION
--
RUNSUM
--
--     ACTIVATE SUMMARY DATA RSM FILE OUTPUT OPTION
--
SEPARATE
```

Note unlike the commercial simulator, OPM Flow always writes out the data to a separate file; however, it is probably good practice to include it if the same input decks are being run with commercial simulator.

11.3.33 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

11.3.34 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

This keyword turns on skipping of black-oil keywords until the [ENDSKIP](#) activation keyword is encountered. All black-oil keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by OPM Flow.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

11.3.35 SKIP300 – ACTIVATE SKIPPING OF COMPOSITIONAL KEYWORDS AND INPUT DATA

Turns on skipping of “compositional” keywords until the [ENDSKIP](#) activation keyword is encountered. All “compositional” keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by OPM Flow.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

11.3.36 SUMMARY – DEFINE THE START OF THE SUMMARY SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	----------------	--------------------------

Description

The [SUMMARY](#) activation keyword marks the end of the [SOLUTION](#) section and the start of the [SUMMARY](#) section that defines the variables to be written out to the [SUMMARY](#) file for reporting and plotting of grid block data, production data, etc.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

```
-- =====  
--  
-- SUMMARY SECTION  
--  
-- =====  
SUMMARY
```

The above example marks the end of the [SOLUTION](#) section and the start of the [SUMMARY](#) section in the OPM Flow data input file.

11.3.37 SUMTHIN – DEFINE SUMMARY DATA REPORTING TIME STEPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	----------------	--------------------------

Description

This keyword defines a time interval for writing out the [SUMMARY](#) data to the [SUMMARY](#) file and the RSM file, if the [RUNSUM](#) keyword has also been activated in the [SUMMARY](#) section. Only the data for the first time step in the time interval is written out and the other time steps are skipped until the next time interval, except for report time steps. Note that report time steps data are always written out regardless of the setting on this keyword. This enables the size of the [SUMMARY](#) files to be reduced depending on the size of the time interval. However, the keyword will produce irregular time steps reports of the [SUMMARY](#) data.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	SUMSTEP	SUMSTEP is a real positive number that defines the time interval for which the first time step of data will be written to the <i>SUMMARY</i> file (and the RSM file if RSM output has been activated). For example, if SUMSTEP is set to 30 days, and if the simulator takes time steps of 0, 5, 10, 16, 24, 30, 40, 45, 60, 90 days. Then the <i>SUMMARY</i> data will be written out at time steps 0, 30, 40 and 60 days.			None
		days	days	hours	
Notes: I) The keyword is terminated by a “/”.					

Table 11.36: SUMTHIN Keyword Description

See also the [RPTONLY](#) keyword in the [SUMMARY](#) section that forces the [SUMMARY](#) data to be only written out at report time steps, as oppose to all time steps or SUMSTEP time intervals.

Example

```
--
--      DEFINE SUMMARY DATA REPORTING TIME STEP INTERVAL
--
--      SUMSTEP
SUMTHIN      30.0      /
```

The above example defines the [SUMMARY](#) file time step interval to be 30 days for both field and metric units.

11.3.38 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

CHAPTER 12: SCHEDULE SECTION

12.1 INTRODUCTION

The [SCHEDULE](#) section is used to specify the production and injection targets and constraints for the entire model, advance the model through time, and stipulate any other data that depends on time. For example, drilling wells at a particular time, incorporating production and/or injection rate changes as the model progresses through time, and opening and shutting wells and well connections/completions at particular times during the history of the reservoir. Field development and operating strategy are also incorporated in the [SCHEDULE](#) section, for example the timing and effect of upgrading of the field facilities liquid and water handling capacities. All these actions must be specified in the [SCHEDULE](#) section at the time they occur, that is all the events in the [SCHEDULE](#) section are time dependent. This is different to the other sections of the input deck where the order of keywords is largely unimportant; in the [SCHEDULE](#) section it is essential because the order in which events take place in the field have to be preserved. For instance, the keyword defining wellhead locations via the [WELSPECS](#) keyword must precede the keyword defining the locations and properties of the well connections using the [COMPDAT](#) keyword. The rules on keyword order generally follow the same sequence as the order of events in a production plan, that is we drill wells first ([WELSPECS](#)), then complete them ([COMPDAT](#)), and put them on production ([WCONHIST](#), [WCONPROD](#), [WCONINJH](#), or [WCONINJE](#)).

In most cases a significant part of the contents of the [SCHEDULE](#) section is often imported from other applications. For instance, the Vertical Lift Performance (“VLP”) tables used to convert from Bottom-Hole Pressure (“BHP”) to Tubing Head Pressure (“THP”) and entered via the [VFPPROD](#) keyword, are always generated by an external Nodal Analysis application. These tables are then included in the input deck via the [INCLUDE](#) keyword. Note as the VFP tables are assigned to a well via the [WCONHIST](#), [WCONPROD](#), [WCONINJH](#), or [WCONINJE](#) keywords, then the tables need to have been previously entered prior to assigning them to a well, that is the data is order dependent. Well historical production data is also normally “included” from an external source, typically the production data base, and may have to be edited to ensure that the correct connections/completions are opened and shut at the appropriate time.

Note

The term well connection is used to describe individual connections from the wellbore to a reservoir grid cell, as opposed to a well completion. A well completion is used to describe a set of connections that are “grouped” into well completions using the [COMPLUMP](#) keyword in the [SCHEDULE](#) section. For instance, a well may consist of several completions with each completion consisting of multiple connections.

The reservoir management flexibility offered in the [SCHEDULE](#) section is indicated by the more than 300 keywords available in the section and thus makes quality assurance of the data challenging. This means that one should not assume that the data included from third party applications are error free and therefore suitable error checking should be performed.

The majority of the keywords are related to controlling how wells and groups are operated, by applying production/injection targets and constraints, economic criteria and associated actions if a criteria is violated, drilling of additional wells to meet production targets or constraints, as well as the relationship between the different wells and groups in the model. In addition, the [DATES](#), [TIME](#), and [TSTEP](#) keywords are used to progress the simulation through time and the [TUNING](#) series of keywords control the numerical parameters in defining numerical convergence of each time step. Note that OPM Flow can be instructed to read the [TUNING](#) keyword if the appropriate command line parameter has been activated (see section [2.2 Running OPM Flow From The Command Line](#)).

Note

When the [DATES](#), [TIME](#), and [TSTEP](#) keywords are used to advance the simulation through time, this “new” time step is known as a “report time step”, as either an action (open a well for example) or a report request will be made. This is different to a simulator’s automatic time step used to advance the simulator to the next “reporting time step”.

For instance, if the simulation is currently at January 1, 2020 and the next reporting time step is January 1, 2021, then the simulator may not be able to advance directly in one time step to January 1, 2021. Instead, a series of time steps of various duration will be automatically selected by the simulator based on the current set of numerical convergence controls until the required “report time step” is reached.

The next section summarizes the data requirements for the [SCHEDULE](#) section and offers some guidance on structuring the format of the keywords in order to increase readability of the input deck and to minimize potential input errors.

12.2 DATA REQUIREMENTS

Apart from the keywords that advance the simulator through time (the [DATES](#), [TIME](#), and [TSTEP](#) keywords), the minimum required data is associated with defining a well ([WELSPEC](#)), defining a well’s connection to the reservoir ([COMPDAT](#)), and the operating and production targets and constraints for the well ([WCONHIST](#), [WCONPROD](#), [WCONINJH](#), or [WCONINJE](#)). Well connections can be “grouped” into completions using the [COMPLUMP](#) keyword to represent actual physical well completions in the model. Wells can either operate independently or under group control. That is when a well is allocated to a group, then the group can dictate how the wells in the group under group control are operated. For example, a group may have production targets and constraints and all wells under group control within the group will be operated in such a manner as to satisfy the group’s targets and constraints. Note that wells can belong to a group but do not necessary have to be under group control. The top level group, level one, has the name FIELD and under this level can be wells, groups, and sub groups to the higher level groups. By default three group levels are defined that sets the wells as level three, reporting directly to defined groups at level two, and the level two groups reporting to the FIELD group at level one. If a different configuration is required, then the [GRUPTREE](#) keyword should be used to define the group hierarchy by defining a lower level group that reports directly to a higher level group.

The major well specification keywords are summarized in Table 12.1 for ease of reference.

Major Well Specification Keywords		
Purpose	Keywords	Description
Vertical Flow Tubing Performance Table Specification	VFPINJ	VFPINJ – Define Injection Vertical Flow Performance Tables . VFPINJ declares similar data to VFPPROD but for injection wells.
	VFPPROD	VFPPROD – Define Production Vertical Flow Performance Tables . The VFPPROD keyword defines production Vertical Flow Performance (“VFP”) tables that are used to determine the outflow or downstream pressure based on the inlet or upstream pressure and the phases flowing through the system. For a well this means the table relates the flowing bottom-hole pressure (“FBHP”) to the well’s tubing head pressure (“THP”) based on the oil, gas, and water rates (and any artificial lift quantities like gas lift gas), or phases ratios, flowing up the wellbore.

Major Well Specification Keywords		
Purpose	Keywords	Description
Well Completion Data	WELSPECS	WELSPECS – Define Well Specifications . The WELSPECS keyword defines the general well specification data for all well types, and must be used for all wells before any other well specification keywords are used in the input file. The keyword declares the name of the well, the initial group the well belongs to, the wellhead location and other key parameters.
	COMPDAT	COMPDAT – Define Well Connections to the Grid . COMPDAT defines how a well is connected to the reservoir by defining or modifying existing well connections.
	COMPLUMP	COMPLUMP – Assign Well Connections to Completions . The COMPLUMP keyword assigns connections, as defined by the COMPDAT keyword, to completion intervals. This “lumping” or “grouping” of the connections to various completion intervals allows automatic workovers and economic criteria to be applied to the completions (that is a set of connections) as opposed to the individual connections.
	COMPTRAJ	COMPTRAJ – Define Well Trajectory Connections to the Grid . COMPTRAJ keyword defines how a well that has been declared as a trajectory well, using the WELTRAJ keyword, is connected to the reservoir model by defining or modifying existing well perforation depths. The keyword can only be used for wells defined by the WELTRAJ keyword, This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.
Historical Production and Injection Data	WELTRAJ	WELTRAJ – Define Well Trajectory Data . WELTRAJ defines a trajectory well together with the well trajectory data, and is used in conjunction with the COMPTRAJ keyword to define the well connections to the simulation grid blocks. The keyword can only be used for trajectory wells that employ the COMPTRAJ keyword to define the connections to the grid, This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.
	WCONHIST	WCONHIST – Define Well Historical Production Rates and Pressures . WCONHIST defines production rates and pressures for wells that have been declared history matching wells by the use of this keyword. History matching wells are handled differently than ordinary wells that use the WCONPROD keyword for controlling their production targets and constraints.
Predictive Production and Injection Data	WCONINJH	WCONINJH – Well Historical Observed Injection Rates and Pressures . WCONINJH declares similar data as the WCONHIST keyword but for injection wells.
	WCONPROD	WCONPROD – Define Well Production Targets and Constraints . The WCONPROD keyword defines production targets and constraints for wells that have previously been defined by the WELSPECS keyword.
	WCONINJE	WCONINJE – Well Injection Targets and Constraints . WCONINJE declares similar data as the WCONPROD keyword but for injection wells.

Major Well Specification Keywords		
Purpose	Keywords	Description
Well Control	WELCNTL	<i>WELCNTL – Modify Well Control and Targets.</i> The WELCNTL keyword modifies a well's target control and value, both rates and pressures, for previously defined wells without having to define all the variables on the well control keywords: WCONPROD, WCONHIST, WCONINJE, or WCONINJH keywords. See also the WELTARG keyword below.
	WECON	<i>WECON – Well Economic Criteria for Production Wells.</i> WECON defines the economic criteria for production wells that have previously been defined by the WELSPECS and WCONPROD keywords.
	WEFAC	<i>WEFAC – Define Well Efficiency.</i> WEFAC defines a well's efficiency or up-time factor as opposed to setting the efficiency factors at the group level.
	WHISTCTL	<i>WHISTCTL – Define Well Historical Target Phase.</i> The WHISTCTL keyword changes the target control phase for wells declared as history match wells via the WCONHIST keyword. The target phase is set on the WCONHIST keyword and WHISTCTL overrides this value for all subsequent entries on the WCONHIST keyword.
	WELOPEN	<i>WELOPEN – Define Well and Well Connection Flowing Status.</i> WELOPEN defines the status of wells and the well connections, and is used to open and shut previously defined wells and well connections without having to re-specify all the data on the well control keywords: WCONPROD, WCONHIST, WCONINJE, or WCONINJH keywords.
	WELTARG	<i>WELTARG – Modify Well Target and Constraint Values.</i> The WELTARG keyword modifies the target and constraints values of both rates and pressures for previously defined wells without having to define all the variables on the well control keywords: WCONPROD, WCONHIST, WCONINJE, or WCONINJH keywords. See also the WELCNTL keyword above.
Well Productivity Adjustments (Normally used in History Matching)	WELPI	<i>WELPI – Define Well Productivity and Injectivity Indices.</i> WELPI is used to define a well's productivity or injectivity index and the values entered on this keyword for a given well will override any previously calculated values, or values previously entered using this keyword.
	WPIMULT	<i>WPIMULT – Define Well Connection Multipliers.</i> The WPIMULT keyword defines a well connection multiplier factor that scales the existing well connection values. The resulting effect is to scale the well's productivity at the reporting time step the keyword is entered.
Notes: 1) Cells colored orange under the Keywords column are currently not supported by OPM Flow.		

Table 12.1: Major Well Specification Keywords

Wells are initially allocated to groups via the WELSPECS keyword and groups have a similar set of keywords as for wells, as outlined in Table 12.2. However only a limited set of keywords have been implemented in OPM Flow compared with the commercial simulator's set of keywords.

Major Group Specification Keywords		
Purpose	Keywords	Description
Group Specification	GRUPTREE	<i>GRUPTREE – Define Group Tree Hierarchy.</i> GRUPTREE defines the hierarchy of groups that have been created by having wells assigned to them via the WELSPECS keyword. By default three group levels are defined that sets the wells as level three, reporting directly to defined groups at level two, and the level two groups reporting to the FIELD group at level one.
Predictive Production and Injection Data	GCONPROD	<i>GCONPROD – Group Production Targets and Constraints.</i> The GCONPROD keyword defines production targets and constraints for groups, including the top most group in the group hierarchy known as the FIELD group.
	GCONINJE	<i>GCONINJE – Group Injection Targets and Constraints.</i> GCONINJE declares similar data as for GCONPROD but for group injection targets and constraints.
Group Control	GECON	<i>GECON – Group Economic Criteria for Production Groups.</i> The GECON keyword defines economic criteria for production groups, including the field level group FIELD, that have previously been defined by the WELSPECS and GCONPROD keyword.
	GEFAC	<i>GEFAC – Define Group Efficiency.</i> GEFAC defines a group's efficiency or up-time factor as opposed to setting the efficiency factors for individual wells.
	GRUPTARG	<i>GRUPTARG – Modify Group Targets and Constraints Values.</i> GRUPTARG keyword modifies the production targets and constraints of both rates and pressures for previously defined groups without having to define all the variables on the group production control keywords: GCONPROD or GCONPRI keywords.
Notes: 1) Cells colored orange under the Keywords column are currently not supported by OPM Flow.		

Table 12.2: Major Group Specification Keywords

The final set of keywords define various controls for the model, the keywords available to advance the simulator through time, and the available reporting keywords, as illustrated in Table 12.3.

Schedule Advancement, Control, And Reporting Keywords		
Purpose	Keywords	Description
Control RS and RV Behavior	DRSDT	<i>DRSDT – Solution Gas (Rs) Maximum Rate of Increase Parameters.</i> DRSDT defines the maximum rate at which the solution gas-oil ratio (Rs) can be increased in a grid cell. The keyword is similar in functionality to the DRSDTR keyword, that defines the maximum rate at which Rs can be increased in a grid cell by region.
	DRSDTR	<i>DRSDTR – Solution Gas (Rs) Maximum Rate of Increase Parameters by Region.</i> DRSDTR defines the maximum rate at which the solution gas-oil ratio (Rs) can be increased in a grid cell for various regions in the model. The keyword is similar in functionality to the DRSDT keyword, that defines the maximum rate at which Rs can be increased in a grid cell for all cells in the model.

Schedule Advancement, Control, And Reporting Keywords		
Purpose	Keywords	Description
Schedule Advancement	DRVDT	<i>DRVDT – Vaporized Oil (Rv) Maximum Rate of Increase Parameters.</i> DRVDT defines the maximum rate at which the solution oil-gas ratio or condensate-gas ratio (Rv) can be increased in a grid cell. The keyword is similar in functionality to the DRVDTTR keyword, that defines the maximum rate at which Rv can be increased in a grid cell by region.
	DRVDTTR	<i>DRVDTTR – Vaporized Oil (Rv) Maximum Rate of Increase Parameters by Region.</i> DRVDTTR defines the maximum rate at which the solution oil-gas ratio or condensate-gas ratio (Rv) can be increased in a grid cell for various regions in the model. The keyword is similar in functionality to the DRVDT keyword, that defines the maximum rate at which Rv can be increased in a grid cell for all cells in the model.
	DATES	<i>DATES – Advance Simulation by Reporting Date.</i> DATES advances the simulation to a given report date after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the SCHEDULE section keywords, or further DATES keywords may be entered to advance the simulator to the next report date.
Schedule Advancement	TIME	<i>TIME – Advance Simulation by Cumulative Reporting Time.</i> TIME advances the simulation to a given cumulative report time after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the SCHEDULE section keywords, or further TIME keywords may be entered to advance the simulator to the next report time.
	TSTEP	<i>TSTEP – Advance Simulation by Reporting Time.</i> TSTEP keyword advances the simulation to a given report time after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the SCHEDULE section keywords, or further TSTEP keywords may be entered to advance the simulator to the next report time
Schedule Advancement Control	TUNING	<i>TUNING – Numerical Tuning Control.</i> TUNING defines the parameters used for controlling the commercial simulator's numerical convergence parameters for the global grid. The keyword is mostly ignored by OPM Flow; however, the simulator can be instructed to read the TUNING keyword if the appropriate command line parameter has been activated (see section 2.2 Running OPM Flow From The Command Line).
	NEXTSTEP	<i>NEXTSTEP – Maximum Next Time Step Size .</i> NEXTSTEP defines the maximum time step size the simulator should take for the next time step. This keyword can be used to reset the time step for when known large changes to the model are taking place that may result in time step chops.
Reporting	RPTSCHED	<i>RPTSCHED – Define SCHEDULE Section Reporting.</i> RPTSCHED keyword defines the data in the SCHEDULE section that is to be printed to the output print file in human readable format.

Schedule Advancement, Control,And Reporting Keywords		
Purpose	Keywords	Description
	RPTRST	<i>RPTRST – Define Data to be Written to the RESTART File.</i> RPTRST keyword defines the data and frequency of the data to be written to the RESTART file at each requested restart point. In addition to the solution data arrays required to restart a run, the user may request additional data to be written to the restart file for visualization in OPM ResInsight.
Notes: 1) Cells colored orange under the Keywords column are currently not supported by OPM Flow.		

Table 12.3: Schedule Advancement, Control, and Reporting Keywords

In terms of structuring the format of the keywords in the SCHEDULE section it is advisable to declared all the VLP tables, wells and groups at the start of SCHEDULE section as oppose to declaring the items as they are needed, or when they come on stream at various times during the simulation. This produces a cleaner input deck and tends to limit unforeseen errors, as all items are declared upfront and only the operational changes (opening wells, changing group and well targets, etc.) are needed. The example SCHEDULE section given on the following pages illustrates a typical SCHEDULE based on this philosophy.

The first segment of the example shows the start of the SCHEDULE section and the group definitions and controls. Here there are group controls only at the FIELD level and there are both production and injection rate targets, as well as water and liquid handling constraints applied on the FIELD level. These keywords are activated from the start of the simulation as set by the START keyword in the RUNSPEC section, in this case January 1, 2020.

```

-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
-- GROUP PRODUCTION AND INJECTION CONTROLS
--
--
-- DEFINE GROUP TREE HIERARCHY
--
-- LOWER      HIGHER
-- GROUP      GROUP
GRUPTREE
    FLTBLK1  FIELD
    FLTBLK2  FIELD
    FLTBLK3  FLTBK2
/
--
-- GROUP PRODUCTION CONTROLS
--
-- GRUP  CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME  MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF    WAT
GCONPROD
FIELD   GRAT  1*    8E3   125E3  10E3   1*    1*    1*    1*    1*
/

```



```
--
--          GROUP INJECTION TARGETS AND CONSTRAINTS
--
-- GRUP  FLUID  CNTL   SURF   RESV   REINJ   VOID   GRUP   GUIDE   GUIDE  GRUP  GRUP
-- NAME  TYPE   MODE   RATE   RATE   FRAC    FRAC   CNTL   RATE   DEF   REINJ  RESV
GCONINJE
FIELD   GAS    RATE   100E3   1*     1*      1.0    YES    1*     1*    1*     1*   /
/
```

The next segment covers the well specification and includes the loading of the VFP tables via include files, declaring the wells using the WELSPECS keyword, and connecting the wells to the reservoir using the COMPDAT keyword. In addition, the COMPLUMP keyword is invoked to assign the well connections to well completions.

In multi-stacked reservoirs it is a good idea to associate the same completion number with a given reservoir for all the wells. So for example completion number one is always associated with the Lower Talang Akar Formation Unit A, completion number two with the Lower Talang Akar Formation Unit B, and completion number three with the Upper Talang Akar Formation Unit C, etc. In this way one can easily identify which zone a well is producing from or completed in.

Finally, the WCONPROD keyword is used to define the operating conditions for the gas producers (GP01 and GP02) and to assign the VFP tables to the producing wells. Whereas the WCONINJE keyword performs a similar function for the single gas injector, GI01.

Notice also the well naming convention that easily identifies the type of well: the letter G for a gas well, O for an oil well, and W for a water well, and the function of the well, P for a producer and I for an injector.

```
-- -----
-- WELL SPECIFICATIONS AND COMPLETIONS
-- -----
--
--          LOAD INCLUDE FILES WITH VLP TABLES
--
INCLUDE
    'WEL-P50-VLP01.inc'      / VFP TABLE #1 5 1/2 inch tubing

INCLUDE
    'WEL-P50-VLP02.inc'      / VFP TABLE #2 4 1/2 inch tubing

--
--          WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP      PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME        I      J  DEPTH  FLUID  AREA   EQUANS  SHUT  FLOW  TABLE
WELSPECS
GI01    FIELD       14    13   1*     GAS    1*     GPP    SHUT   NO    1*     /
GP01    FLTBLK1     64    80   1*     GAS    1*     GPP    SHUT   NO    1*     /
GP02    FLTBLK1     24   110   1*     GAS    1*     GPP    SHUT   NO    1*     /
/

--
--          WELL CONNECTION DATA
--
-- WELL  --- LOCATION ---  OPEN  SAT  CONN  WELL  KH  SKIN  D  DIR
-- NAME  II JJ K1 K2  SHUT  TAB  FACT DIA  FACT FACT FACT PEN
COMPDAT
GI01    1* 1*  1 60  SHUT  1*  1*  0.708 1*  0.0  1*  'Z'  /
GP01    1* 1*  1 60  SHUT  1*  1*  0.708 1*  0.0  1*  'Z'  /
GP01    1* 1*  1 60  SHUT  1*  1*  0.708 1*  0.0  1*  'Z'  /
/

--
--          ASSIGN WELL CONNECTIONS TO COMPLETIONS
--
```

```
-- WELL    --- LOCATION    --- COMPL
-- NAME    II  JJ  K1  K2    NO.
COMPLUMP
GI01      14  13    1  20    1          / COMPLETION NO. 01
GI01      14  13   35  60    2          / COMPLETION NO. 02
GP01      64  80    1  20    1          / COMPLETION NO. 01
GP01      64  80   22  60    2          / COMPLETION NO. 02
GP02      24 110    1  20    1          / COMPLETION NO. 01
GP02      24 110   25  60    2          / COMPLETION NO. 02
/
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL    OPEN/  CNTL   OIL    WAT    GAS    LIQ    RES    BHP    THP    VFP    VFP
-- NAME    SHUT   MODE   RATE   RATE   RATE   RATE   RATE   PRES   PRES   TABLE  ALFQ
WCONPROD
GP01      SHUT   GRUP   1*    1*    70E3   1*    1*    500.0  120.0  1        /
GP02      SHUT   GRUP   1*    1*    70E3   1*    1*    500.0  120.0  2        /
/
--
--          WELL INJECTION CONTROLS
--
-- WELL    FLUID   OPEN/  CNTL   SURF   RESV   BHP    THP    VFP
-- NAME    TYPE    SHUT   MODE   RATE   RATE   PRSES   PRES   TABLE
WCONINJE
GI01      GAS     SHUT   RATE  125E3   1*    10E3   1*    1*          /
/
```

The final part of the well specification segment is shown below and sets the well efficiency for the wells via the WEFAC keyword, the producing wells economic limits via the WECON keyword, and then advances the simulation to January 25, 2020. Thus, there is no production up to January 25, 2020 as all the wells are shut-in.

```
--
--          WELL EFFICIENCY FACTORS
--
-- WELL    EFF    NETWK
-- NAME    FACT   OPTN
WEFAC
'GI*'     0.950          /
'GP*'     0.950          /
/
--
--          WELL ECONOMIC CRITERIA FOR PRODUCTION WELLS
-- WELL    MIN    MIN    MAX    MAX    MAX    CNTL   END    WELL
-- NAME    ORAT   GRAT   WCUT   GOR    WGR    MODE   RUN    NAME
WECON
'GP*'     1*     5.0E3  1*     1*     1*    'WELL' 'NO'          /
/
DATES
25 JAN    2020          /
/
```

On January 25, 2020 yearly restarts are requested (RPTRST) together with various printed reports (RPTSCHED), as well as the maximum time step size of 31 days being set via the TUNING keyword. More importantly, well GP01 is opened together with completion number one to put the well on production. Note that one needs to open both the well and the completion; hence, the two lines in the WEOPEN keyword.

```
-- -----
-- SCHEDULE SECTION FOR PHASE 1 DEVELOPMENT
-- -----
--
--          RESTART CONTROL BASIC = 4 (ALL=2, YEARLY=4, MONTHLY=5, TSTEP=6)
--
```

```
RPTRST
    BASIC=4
--
--      DEFINE SCHEDULE SECTION REPORT OPTION
--
RPTSCHED
    'WELLS=2'      'WEL SPECS'      'CPU=2'      'FIP=2'
--
--  DEFAULT TUNING PARAMETERS (SET MAX TSTEP = 31 days)
--
TUNING
1.0      31.0      0.1      0.15      3      0.3      0.1      1.25      0.75
/
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
--  WELL WELL  --LOCATION--  COMPLETION
--  NAME STAT   I   J   K  FIRST LAST
WELOPEN
GP01      OPEN
GP01      OPEN      0   0   0      1   1
/

DATES
1  FEB    2020  /
/
```

On February 1, 2020 further SCHEDULE keywords are processed. The first keyword is the RPTSCHED keyword that in this case switches off all reports printed to the *.PRT file, as only annual reports are required for this particular model.

Next the gas producer GP02 is put on production using two statements in the WELOPEN keyword, and the simulator is requested to advance to report time step of March 1, 2020.

```
--
--      DEFINE SCHEDULE SECTION REPORT OPTION
--
RPTSCHED
'NOTHING'
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
--  WELL WELL  --LOCATION--  COMPLETION
--  NAME STAT   I   J   K  FIRST LAST
WELOPEN
GP02      OPEN
GP02      OPEN      0   0   0      1   1
/

DATES
1  MAR    2020  /
/
```

Then finally on March 1, 2020 the gas injector is placed on stream and the simulation advances on a monthly report basis to the end of the year.

```
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
--  WELL WELL  --LOCATION--  COMPLETION
--  NAME STAT   I   J   K  FIRST LAST
WELOPEN
GI01      OPEN
GI01      OPEN      0   0   0      1   1
/
```

```
/
DATES
1 APR 2020 /
1 MAY 2020 /
1 JUN 2020 /
1 JUL 2020 /
1 AUG 2020 /
1 SEP 2020 /
1 OCT 2020 /
1 NOV 2020 /
1 DEC 2020 /
/
```

Next a detailed report is produced as of January 1, 2021, and reporting is then switch off and the simulation continues to the end of the run.

```
-- -----
-- SCHEDULE SECTION - 2021-01-01
-- -----
```

```
RPTSCHED
'WELLS=2'      'WELSPECS'      'CPU=2'      'FIP=2'      /
```

```
DATES
1 JAN 2021 /
/
```

```
RPTSCHED
'NOTHING'      /
```

```
DATES
1 FEB 2021 /
1 MAR 2021 /
1 APR 2021 /
...
/
```

12.2.1 HISTORY MATCHING RUNS

History matching is the process of matching actual historical production data versus the model's production performance. During history matching the engineer specifies the actual field measured production and injection rates for a reservoir which has been producing for a period of time. Although other reservoir characteristics such as permeability, layering structure, aquifer strength, and individual well performance are specified as normal in the remainder of the data file, they are only one realization out of possibly thousands of realizations. There is therefore significant uncertainty associated with the data set. Thus in most cases, the simulated past performance of the reservoir, that is the well rates calculated by OPM Flow up to the present day, will not in general correspond to the measured rates specified by the user. History matching then becomes an exercise in identifying those reservoir properties subject to the greatest uncertainty and adjusting them to bring the simulated and measured rates to an acceptable degree of agreement.

Prior to beginning the matching process the production data should be reviewed and quality checked. This step is needed mainly to:

- Allocate production data areally and vertically to the pertinent areas and zones, as well as ensuring that the correct well completions have been implemented into the model's schedule section. Particular care should be taken with multiple completed wells to ensure that the wells produce from the correct zones at the correct times. Wells that are completed across fault planes also need to be checked so that the completions are on the right side of the fault plane.
- Correct field production data to separator conditions consistent with model conditions, or vice versa.
- Correct reported pressures for simulation use (e.g., Peaceman correction), or use the weighted areal average well bottom-hole pressures from the simulator to match against actual data.

Note also that not all production volumes have the same level of quality assurance. Product streams that are sold are going to have a higher level of quality assurance because they have value, compared with production streams that are disposed off or re-injected, as they have no value. Thus, in an oil field selling the produced oil but flaring or re-injecting the produced gas, the oil volumes will be back allocated based on "ticketed sales", and thus the field volumes are reasonable certain. However, even here there are uncertainties associated with allocating production volumes back to the individual wells and well completions. As the gas volumes are not sold they have no value and thus have greater uncertainty than the oil volumes, especially when back allocating back to the wells.

History matching is an iterative process, in which steps are repeated a number of times with variations in reservoir characterization (permeability, pore volume, relative permeability modifications, etc.), until a reasonable match is obtained. There are no precise rules for conducting a history match but the methodology is well established. Generally one first matches pressures globally, then overall saturations (and well rates), and repeat until an overall match is achieved, as the process of matching pressures also effects the saturations and vice versa. Which is why this is an iterative process. This methodology is then repeated on a regional level until the process is complete and reasonable matches have been obtained at the field, region, and well levels.

Generally, group control, targets, and constraints are not utilized in the history matching part of a simulation study. Instead, the WCONHIST keyword is used to enter the well production (oil, gas, and water rates), pressure data (BHP and THP when available), and VFP table assignment, on a well by well basis and on discrete time intervals, normally monthly. WCONHIST allows one to set the "target" phase and rate the well will attempt to produce at, if possible, and to produce the other phases that come with the target rate. That is there are no constraints applied to the other phases, unlike when using the WCONPROD keyword. If well productivity has been matched then the target phase rates will be satisfied and the quality of the history match is determined by the accuracy of the other phases, together with the pressure match. Thus, in an oil field the target phase will normally be the oil phase and the oil rate, and the quality of the history would be determined by the produced gas and water volumes plus the pressure match at various levels (field, region, reservoir, and well). If water breakthrough has not occurred and the field is producing above the saturation

pressure then there is very little to match, as the gas production in this case is a direct function of the oil rate.

The WCONHIST keyword also has a RESV target phase where the target is set to the in situ reservoir volume rate which is calculated by the simulator using the oil, water, gas, and liquid rates declared on the keyword. This is useful in the initial part of the history matching study where one is trying to match the overall pressure behavior, as the option forces the simulator to produce the correct number of reservoir barrels (or m3), although the actual oil, gas, and water rates will most likely not match at this stage of the study. Note that it is not necessary to edit the WCONHIST keywords for a run to change the target phase, as the WHISTCNTL keyword can be used to do this from the time this keyword is invoked, thus avoiding changing the control mode on all subsequent WCONHIST keywords.

As mentioned above, the rates and volumes of production streams that are sold are more reliable than those that are being re-injected, flared, or otherwise disposed of, and therefore in an oil field the preferred target phase is oil and not liquid, which some engineers prefer to use.

The WCONINJH keyword has similar functionality as the WCONHIST keyword, but is used for history matching injection wells instead.

More recent developments in history matching incorporate Assisted History Matching (“AHM”) techniques [305](#), [306](#), [307](#), [308](#), [309](#), [310](#) and [311](#) in which various statistical techniques are applied in specialized software to match the field data by the user supplying the parameters to vary, the range of the parameters values, and the objective function to minimize. This type of software can also generate a range of “reasonable” history match models that can then be used to quantify the uncertainty range in the prediction phase.

³⁰⁵ *History Matching and Uncertainty Quantification: Multi-objective Particle Swarm Optimisation Approach (SPE 143067)*, L. Mohamed, M. Christie, V. Demyanov, Vienna, Austria, 23–26 May 2011.

³⁰⁶ *Field-Scale Assisted History Matching Using a Systematic, Massively Parallel Ensemble Kalman Smoother Procedure (SPE00182617)*, Binghui Lin, Paul I Crumpton, and Ali H. Dogru, Society of Petroleum Engineers (February 2017).

³⁰⁷ *Enhancing the Geological Models Consistency in Ensemble-Based History Matching - An Integrated Approach (SPE00186049)*, A. Perrone, F Pennadoro, A Tiani, E Della Rossa, and J. Saetrom, Society of Petroleum Engineers (May, 2017).

³⁰⁸ *Correlation-Based Adaptive Localization for Ensemble-Based History Matching - Applied to the Norne Field Case Study (SPE00191305)*, Xiaodong Luo, Rolf Lorentzen, Randi Valestrand, and Geir Evensen, Society of Petroleum Engineers (April, 2018).

³⁰⁹ *Rapid Forecast Calibration Using Nonlinear Simulation Regression with Localization (SPE-193845-MS)*, Jincong He, Wenyue Sun, and Xian-Huan Wen, SPE Reservoir Simulation Conference held in Galveston, Texas, USA, 10–11 April 2019.

³¹⁰ *Multilevel Strategies and Geological Parameterizations for History Matching Complex Reservoir Models (SPE-193895-MS)*, Yimin Liu and Louis J. Durlofsky, SPE Reservoir Simulation Conference held in Galveston, Texas, USA, 10–11 April 2019.

³¹¹ *Improved Estimation and Forecast Through Model Error Estimation – Norne Field Example (IPTC-19142-MS)*, Minjie Lu and Yan Chen, International Petroleum Technology Conference held in Beijing, China, 26 – 28 March 2019.

12.2.2 PREDICTION RUNS

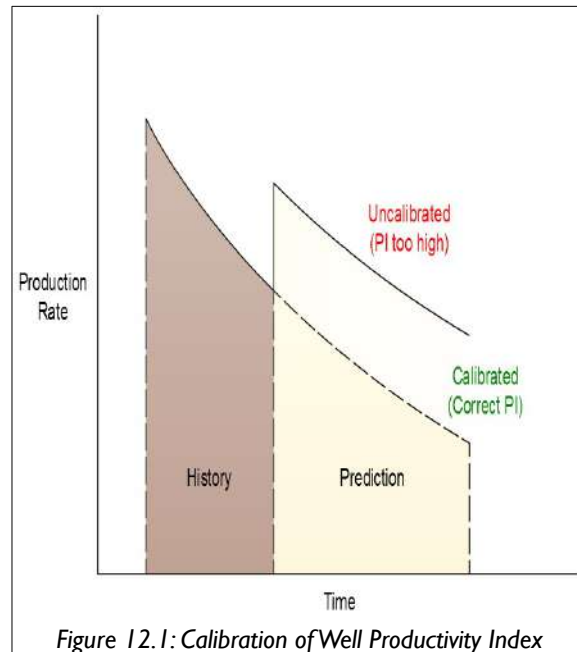
Prediction runs are used to optimize future production from the field and can take two forms:

- Prediction runs may follow directly from a history match case using the reservoir description estimated during the history matching phase. The history matching and prediction runs are frequently in separate data files, with the prediction cases “restarting” from the final history match run, as the history match restart file contains sufficient data to continue the simulation into the future. Therefore, the prediction data set contains the same reservoir description as the history match but is initialized using the restart file output from the history match case. In general, the base case scenario is the No Further Activity (“NFA”) scenario. This is just a continuation of the history match into the future using the well controls in force at the end of the history matching period. This is the simplest way to run the field since the surface facilities remain unchanged and no workovers take place. However, the switch from history matching mode to prediction mode should be treated with care in order to ensure a smooth transition between the two phases. Subsequent cases will then be based on the NFA scenario, and will attempt to optimize recovery by examining the impact of drilling infill wells, modifying facilities constraints, varying the reservoir management strategy, etc.
- Prediction cases may also be based on predicting production from structures that have not been put on production at the time of the study, that is there is no historical production available. In this case, the reservoir description is based on the best available data set and interpretation, although there may be various variations of the models to cover the range of uncertainties in the static earth modeling. Typically, Low, Best, and High models are generated to capture the range of uncertainty in the static modeling, although full uncertainty analysis is becoming more common as computational computer power becomes more available. Additional dynamic sensitivities are then run to determine the Low, Best, and High models that address both the static and dynamic uncertainties. These models then form the basis of the supplementary predictions using a variety of different production strategies that are used to improve the petroleum recovery and reduce production costs. For each prediction case, an aspect of the surface facilities or well management policy is changed and the case is compared to the base prediction. Most simulators, including OPM Flow, have a wide range of well, group and surface facility modeling features and it is likely that several will be invoked during recovery optimization to model events such as drilling infill wells, modifying facilities constraints, investigating various depletion strategies, etc.

In both scenarios more recent developments within the industry apply a full uncertainty analysis to the models, in which multiple cases are run under various parameter changes to derive a distribution of recoverable volumes, which in turn are used to select the Low, Best, and High scenarios. This can result in hundreds of cases being run and is computationally expensive.

As mentioned above, the switch from history matching mode to prediction mode should be treated with care in order to ensure a smooth transition between the two phases. History matching of a well’s BHP has always been somewhat problematic, as the field measured data consists of shutting-in the well for a specific period for the BHP to build backup to a well’s average drainage pressure (usually assumed to be the reservoir pressure in the surrounding area). History matching time steps are normally based on monthly time steps to coincide with the monthly back allocated field reported production volumes. Thus, to correctly match the measured Shut-in Bottom-Hole Pressure (“SBHP”) data in the model, one would have to shut-in the well in the model as per the field, and then open up the well back up at the end of the BHP survey. One would also have to re-adjust the well rates to match the cumulative volumes produced for the time the well was shut-in. These steps are impracticable and engineers originally handled the problem by assigning the wells a high productivity index, so that the difference between the Flowing Bottom-Hole Pressure (“FBHP”) and the SBHP was small.

This would lead to a mismatch in rates when switching from history matching mode to prediction mode as shown in Figure 12.1. Thus, the final step in history matching was to calibrate the well rates based on adjusting the Productivity Indices (“PI”) of the wells, in order to get a smooth transition between the two modes of operating the wells.



Calibration is achieved by running the model at the back pressures held against the wells at one or several time-steps. The well PIs and/or wellbore flow parameters are adjusted to duplicate field-observed rates. Although the calibration step was usually done after the history-matching step, there are times when it must be done beforehand (e.g., when modeling multi-layer flow into a well). This is because in stratified reservoirs the correct allocation of rates among layers is dependent on the absolute value of the well's PI (and not solely on the ratios of PIs of the various layers).

Although well PI matching is still performed as part of the general history matching methodology, it is no longer necessary to set a well's PI to a high value in order to mimic the field measured SBHP. This is because most simulators are now capable of reporting an "average" bottom-hole pressure around the well, including one, four, five, and nine grid cell averaging, as well as alternative averaging schemes. Thus, the engineer now matches the field measured SBHP with one of the simulator's average grid cell BHPs, normally either the five or nine grid cell average BHP values.

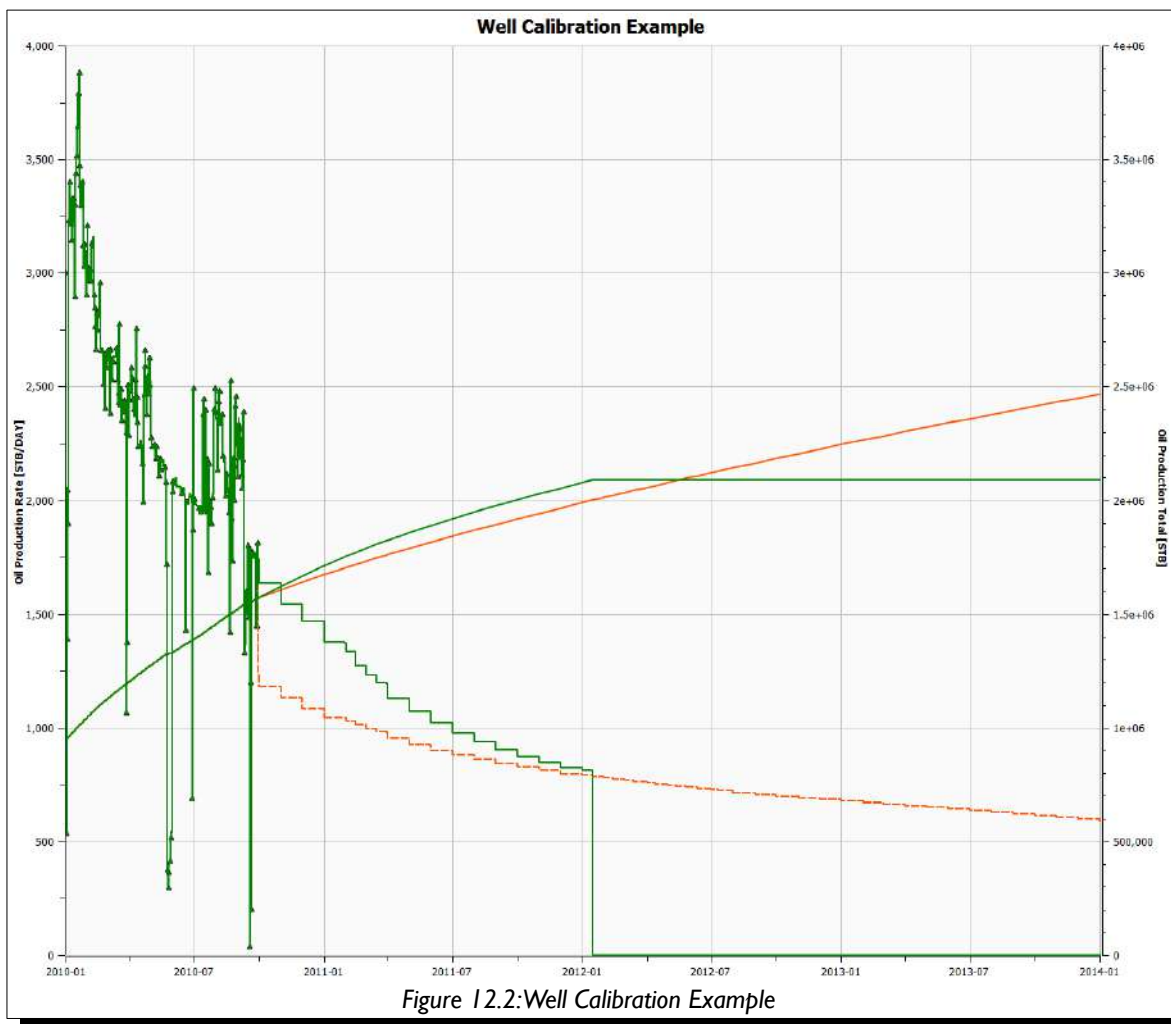
This leaves the question on what controls should be used when moving from history matching mode to prediction mode and ensuring a smooth transition between the two, and at the same time ensuring the resulting predictive forecast is reasonable. There are basically three approaches to this problem:

- 1) set the well's prediction constraint to the last observed rate,
- 2) set the well's prediction constraint to the average FBHP and adjusting a well's PI, and thirdly,
- 3) set the well's prediction constraint to the last FBHP or THP.

The next three sections discusses these options in some detail.

(I) Prediction Constraint Based on a Well's Last Observed Rate

In the history matching mode the wells are controlled by a rate target, so using the last observed well rate would seem a sensible approach. Thus, this approach uses the last observed well rates as well constraints in the prediction run. The typical approach is to constrain production based on a preferred phase (oil rate, gas rate, or liquid rate), and then set a reasonable flowing BHP value as a limit for when the well can no longer meet the rate target set. The main requirement for using this approach is therefore that the history-matched model reproduces the rates and flowing bottom-hole pressures fairly exactly.



However, there are some drawbacks to this approach as one can observe in Figure 12.2, which shows the oil rate and cumulative oil for a well, the symbols on the plot are the actual historical oil production rates. If the last period of production history were not matched successfully, a smooth transition from the history simulation to prediction phase will not be achieved, as shown by the orange lines in Figure 12.2.

A second drawback to this approach, is that the well and field production rate profiles will tend to have a flat trend for a few months (or even years) when the well can still deliver the rate target. In real field management such rate profiles can be optimistic if the specified boundary conditions are not representative of actual operating conditions in the field.

(2)
Prediction Constraint Based on a Well's Average FBHP and Adjusting a Well's PI

This approach has been adopted by some reservoir engineers as a result of the issues mentioned for the first approach.

Here, an average FBHP of each well, based on most recent production history, is used as a target to control the wells during the prediction phase. Usually, this will lead to abrupt changes in rate and pressure profiles during the transition. The next step is therefore to dampen these abrupt changes in the well rate profiles by applying well PI multipliers (at the well and/or completion level) until a smooth transition is obtained in the rate profiles, similar to that shown by the green lines after the history matching period in Figure 12.1.

The problem with this approach is that it is synonymous to re-calibrating a supposedly history-matched simulation model (with a given permeability distribution based on a certain geological characterization), and thus this approach is rather questionable and is not recommended.

(3)
Prediction Constraint Based on a Well's Last Flowing BHP or THP

This methodology is the recommended approach for switching between historical and prediction controls in a reservoir model. It involves automatically applying the last flowing bottom-hole pressure at the instant the transition takes place as the production target, and therefore setting the latest flow rates to be constraints. This ensures a smooth transition between history and prediction phases without having to resort to unreasonable changes to the model. In OPM Flow this is accomplished by using the WELTARG keyword to specify that all the wells should use the bottom-hole pressure as the target at the beginning of the restart run, and defaulting the actual value for the BHP. By defaulting the BHP value the simulator will use the current value of FBHP for a well as the constraint. Note that the WELTARG keyword only defines the variable to be changed, it does not change how a well is controlled. However, by setting the BHP to equal to the current FBHP this has a similar effect as changing the operating target.

Using this approach, all wells will continue flowing at the last flowing bottom-hole pressure from the previous time step, using this value of FBHP as the effective target while the last historical rates are kept as limits (if BHP is requested as shown in the example below). This approach achieves two important objectives, firstly it results in a smooth transition from the history-match phase to prediction phase in well and field production profiles And secondly, it gives a smooth decline in well production rate, as is expected of real wells in the field.

Although this discussion has been based on using the BHP as the effective control mechanism, one could also use THP, and to a lesser extent, liquid rate, instead.

The following example below shows how to apply the recommended method to obtain a smooth transition to the predictive phase using the BHP as the controlling mechanism.

Example

In the example, December 2019 is the last month of historical data and therefore the predication phase starts from January 1, 2020.

```
DATES
01 DEC 2019 /
/
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE ALFQ  PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.5E3  100.0  1550   10      1*    900.0 1*    /
OP02     OPEN   ORAT   10.3E3  500.0  1550   10      1*    700.0 1*    /
OP03     OPEN   ORAT   5.4E3  2100.0 1550   10      1*    500.0 1*    /
/
-----
-- 01 JAN 2020 START OF PREDICTION SCHEDULE SECTION
-----
DATES
01 JAN 2020 /
/
--
--          WELL PRODUCTION AND INJECTION TARGETS
--
-- WELL  WELL   TARGET
-- NAME  TARG   VALUE
WELTARG
OP*     BHP                                /
/
```

From December 1, 2019 to January 1, 2020 the three oil producing wells will be under history matching control with the oil rate being the target rate for the wells. On January 1, 2020 the wells will be converted to normal producers as the WELTARG keyword changes a well status from a history match well to a regular well. At the same time the WELTARG keyword sets all the wells to have their current FBHP as a constraint, by setting the target BHP values to the default values, which effectively makes the FBHP a target as it is now the “active” constraint.

The alternative keyword for implementing the same approach is the [WELCNTL](#) keyword, but unfortunately this is currently not implemented in OPM Flow.

12.2.3 WELL PRODUCTIVITY

Throughout this section the equations are presented in terms of oil production or liquid flow (i.e., oil and water). Note that similar equations exist for gas flow in terms of p^2 and pseudo pressure formulations ($\psi(p)$).

Under the assumption of (1) one-dimensional radial flow into the wellbore, (2) steady-state pressure behavior, and (3) single-phase flow, it can be shown that the flow rate of a well is given by:

$$q = (0.001127) (2\pi) \frac{kh}{\mu} \left[\frac{p_e - p_w}{\ln \left(\frac{r_e}{r_w} \right) + S} \right] \quad (12.1)$$

where

q	= production rate (rb/day),
p_e	= pressure at the drainage radius of the well (psia),
p_w	= bottom-hole well pressure (psia),
r_e	= drainage radius (feet),
r_w	= wellbore radius (feet),
S	= skin factor (dimensionless), and
kh	= effective permeability thickness (mD.ft).

This is extended to multi-phase flow by introduction of relative permeabilities and adjusted to surface conditions by use of the formation volume factor resulting in the following expression for oil production rate:

$$q_o = (0.001127) (2\pi) \frac{kk_{ro} h}{B_o \mu_o} \left[\frac{p_e - p_w}{\ln \left(\frac{r_e}{r_w} \right) + S} \right] \quad (12.2)$$

If the modified radial flow equation is written for each perforated interval and summed over all intervals, the resulting equation is the one used to calculate oil production rates in the simulator. One way to specify the flow capacities of wells is to enter r_e , r_w , S , and p_w values for the well, with the kh values for each perforated interval. The simulator then determines the other parameter values from the calculated pressure and saturation distributions.

Because the pressure at the drainage radius is not readily determined in the simulator, it substitutes p_b (a representative pressure of each perforation grid block) for p_e . That is readily accommodated by replacing the drainage radius by a probe radius as defined by Peaceman³¹² for a vertical well:

$$r_b = 0.28 \frac{\left[\sqrt{\frac{K_y}{K_x}} \Delta x^2 + \sqrt{\frac{K_x}{K_y}} \Delta y^2 \right]^{0.5}}{\left(\frac{K_y}{K_x} \right)^{0.25} + \left(\frac{K_x}{K_y} \right)^{0.25}} \quad (12.3)$$

where

K_x and K_y	= x- and y- direction absolute permeabilities, and
Δx and Δy	= grid block sizes (feet).

³¹² Peaceman, D.W. "Interpretation of Well-Block Pressures in Numerical Reservoir Simulation with Nonsquare Grid Blocks and Anisotropic Permeability." SPE 10528 presented at the Sixth SPE Symposium on Reservoir Simulation, New Orleans, (February 1982), pp. 553-569.

The simulator then uses the modified radial flow equation, written for each connection:

$$q_o = \frac{(0.001127) (2\pi) \left(\frac{k k_{ro} h}{B_o \mu_o} (p_b - p_w) \right)}{\ln \left(\frac{r_b}{r_w} \right) + S} \quad (12.4)$$

The connection term on the COMPDAT keyword in the SCHEDULE section, CONFACT, is simply the transmissibility portion of equation (12.4) for a given connection that is:

$$T_{xy} = \frac{(0.001127) (2\pi) (kh)}{\ln \left(\frac{r_b}{r_w} \right) + S} \quad (12.5)$$

And the oil phase mobility term is defined as:

$$M_o = \frac{k_{ro}}{B_o \mu_o} \quad (12.6)$$

Substituting (12.5) and (12.6) into (12.4) and summing all the connections for the well to obtain the oil rate for the well gives:

$$q_o = \sum_1^N (T_{xy} M_o (p_b - p_w)) \quad (12.7)$$

Note in the case of partially completed wells, the effective permeability thickness of a perforated interval may be substantially greater than the permeability thickness of the formation along the perforation.

Engineers commonly use the Productivity Index ("PI") as a parameter to defined an oil well's deliverability, as it is derived relatively easy from field measurements, that is:

$$q_o = PI (p_e - p_w) \quad (12.8)$$

or in terms of the productivity index:

$$PI = \frac{q_o}{p_e - p_w} \quad (12.9)$$

It is significant that this equation for the Productivity Index is written in terms of the pressure at the well drainage radius, p_e , rather than grid block pressure, p_b . It must be written this way since PI is a property that is determined by field production tests. One can also relate the above equation to the multilayered extension of the steady-state radial flow equation (12.7), to give the grid block Productivity Index:

$$PI = \frac{q_o}{p_b - p_w} = \sum_{n=1}^N (T_{xy} M_o)_n \quad (12.10)$$

One very important point to note is that the productivity index is, in general, not constant for a given well since both relative permeability and drainage radius can be time dependent. This makes PI an inappropriate quantity for computer prediction of production rates, but it is often the only type of data available on well performance. To overcome this limitation, most simulators accept well drainage area PI data as input, and then immediately convert this assuming steady state radial flow with a uniform mobility throughout the drainage area, resulting in the following equation.

$$PI = \sum_{n=1}^N \left(T_{xy} M_o \left(\frac{\ln \left(\frac{r_b}{r_w} \right) + S}{\ln \left(\frac{r_e}{r_w} \right) + S} \right) \right)_n \quad (12.11)$$

Since the drainage radius, r_e , is generally different from the grid block radius, r_b , the ratio of logarithms may be significant; hence, the use of the PI option normally requires specification of both r_e and r_b .

Equation (12.11) is used to calculate the PI for the well and is used to print the PI on the WELLS production report requested via the RPTSCHED keyword in the SCHEDULE section. Note that for gas wells with non-zero D-factors, the non-Darcy skin factor is added to the skin (S) term in equation (12.11) in both the numerator and the denominator. Secondly, r_e is often undetermined or unknown and it is common to default the DRADIUS parameter on the WELSPICS keyword, in the SCHEDULE section. If the well drainage radius (DRADIUS) is defaulted then equation (12.11) simplifies to:

$$PI = \sum_{n=1}^N (T_{xy} M_o)_n \quad (12.12)$$

In this case the PI should be considered a grid block productivity index and not a well drainage area productivity index. Note also that if DRADIUS is set to a negative number, then this results in a well's potential being written out to the WELLS production report instead of the productivity index. In this instance this is the rate based on only applying the BHP and THP constraints only, all other constraints are ignored.

12.3 KEYWORD DEFINITIONS

12.3.1 ACTION – DEFINE ACTION CONDITIONS AND COMMAND PROCESSING (FIELD)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *ACTION* keyword defines a series of conditions that invoke run time processing of *ACTION* functions and is similar to executing a run time script for conditions and variables at the field level.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; use the *ACTIONX* keyword instead.

12.3.2 ACTIONG – DEFINE ACTION CONDITIONS AND COMMAND PROCESSING (GROUPS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [ACTIONG](#) keyword defines a series of conditions that invoke run time processing of [ACTION](#) functions and is similar to executing a run time script for conditions and variables at the group level

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; use the [ACTIONX](#) keyword instead.

12.3.3 ACTIONR – DEFINE ACTION CONDITIONS AND COMMAND PROCESSING (REGIONS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [ACTIONR](#) keyword defines a series of conditions that invoke run time processing of [ACTION](#) functions and is similar to executing a run time script for conditions and variables at the region level.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; use the [ACTIONX](#) keyword instead.

12.3.4 ACTIONS – DEFINE ACTION CONDITIONS AND COMMAND PROCESSING (WELL SEGMENTS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ACTIONS](#) keyword defines a series of conditions that invoke run time processing of [ACTION](#) functions and is similar to executing a run time script for conditions and variables associated with well segments.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; use the [ACTIONX](#) keyword instead.

12.3.5 ACTIONW – DEFINE ACTION CONDITIONS AND COMMAND PROCESSING (WELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword starts the definition of an [ACTIONW](#) section that stipulates the Boolean conditions to test the nominated well parameters, and the resulting [SCHEDULE](#) keywords to be executed, if the Boolean condition evaluates to true. An [ACTIONW](#) Definition Section is terminated by an [ENDACTIO](#) keyword on a separate line. Here, the keyword defines a series of conditions applied to wells only, that invoke run time processing of [ACTION](#) functions, and is similar to executing a run time script for conditions and variables at the well level. The [ACTION](#) series of keywords ([ACTION](#), [ACTIONG](#), [ACTIONR](#), [ACTIONS](#) and [ACTIONW](#)) can apply Boolean conditional tests to variables at the field, group, region, well segment and well levels.

Note that one should use caution when using this facility as it may result in the simulator aborting, because the [ACTIONW](#) keyword enables the user to implement complex functionality and therefore it is advisable to start with simple expressions before adding the desired complexity.

See also the [PYACTION](#) keyword in the [SCHEDULE](#) that implements OPM Flow's Python scripting facility using the Python scripting language.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped; use the [ACTIONX](#) keyword instead.

No.	Name	Description	Default
ACTIONW		Defines the start of an ACTIONW Definition Section. This is followed on a new line by the ACTIONW record that stipulates the conditions for which the defined action will be executed, and then this is followed by the various operations to be performed if the condition is satisfied.	
I-1	ACTNAME	A character sting of up to eight characters in length that defines the name of this action definition. If ACTNAME has previously been used by any ACTION series keyword, then the previous ACTION definitions will be replaced by the definition declared by this ACTIONW Definition Section.	None
I-2	ACTWELL	A character string of up to eight characters in length that defines the well name for which the ACTIONW Definition Section is being defined. Note that the well name (ACTWELL) must have been declared previously using the WEL SPECS keyword in the SCHEDULE section, otherwise an error may occur.	None
I-3	ACTLHS	A character string of up to eight characters in length that stipulates a well SUMMARY variable on the left hand side of a Boolean conditional test. For example, WOPR that stands for the Well Oil Production Rate. In addition, ACTLHS can also be a well User Defined Quantity ("UDQ") defined by the UDQ – Declare User Define Quantities ("UDQ") keyword in the SCHEDULE section. In this case, the first two characters of ACTLHS must be set to WU and the UDQ variable must have previously been declared with the UDQ keyword. A complete list of well SUMMARY vectors that can be used with ACTIONW keyword by OPM Flow and the commercial black-oil simulator is summarized in Table I2.5.	Not Applicable

No.	Name	Description	Default
I-4	ACTTEST	A defined character string that defines the Boolean operator and must be set to one of the following Boolean conditionals: 1) >: Greater than. 2) <: Less than. 3) >=: Greater than or equal to. 4) <=: Less than or equal to. 5) =: Equals to. 6) !=: Not equal to Note that the OPM Flow implementation of ACTTEST has been enhanced to support all of the above conditions. Whereas, the commercial simulator only supports (1) and (2). Thus, one should be mindful of this fact if the input deck is to be run by both OPM Flow and the commercial simulator.	Not Applicable
I-5	ACTRHS	A numeric value that defines the constant on the right hand side of the Boolean conditional test. The set of wells for which the Boolean condition is true, is captured and can be used as a general "well list" with the symbol '?' in subsequent well keywords. For example, to shut wells with a gas production rate less than 5 MMscf/d, one would use: <pre> ACTIONW ACT01 'GP*' WGPR < 5000.0 10000 / WELOPEN '?' SHUT / / ... ENDACTIO </pre>	Not Applicable
I-6	ACTNSTEP	A positive integer that defines the maximum number times that the ACTNAME action is to be executed. The <i>ACTIONW</i> action is executed at the end of each time step if at least one well satisfies the Boolean condition until the action has been executed the specified number of times. The default value of one means that the definition will be executed only once. One can use a large value, for example 10,000 for the definition to be executed at every time step. Note that the counter only affects successful evaluations; i.e. if ACTNSTEP is set equal to one (the default), then the simulator will test the action at the end of every time step until it evaluates to true.	1

No.	Name	Description	Default
I-3	ACTINCR	ACTINCR is a real negative or positive value that specifies a value to increment ACTRHS every time the Boolean condition evaluates to true. For example, if ACTINCR is set to 500.0: <pre> ACTIONW ACT02 'GP*' WGPR < 5000.0 10000 500.0 / WELOPEN '?' SHUT / / ... ENDACTIO </pre> Then after the third time the Boolean conditional has evaluated to true, the condition would effectively be: <pre> ACTIONW ACT02 'GP*' WGPR < 6500.0 10000 500.0 / ... ENDACTIO </pre>	0.0
I-4	/	Record terminated by a "/"	Not Applicable
		The next section contains any number of standard SCHEDULE keywords that will be executed if the Boolean expression evaluates to true. For example, to reduce the tubing head pressure constraint when any of the oil producers' oil rate drop below 100 stb/d then one could use: <pre> ACTIONW ACT03 'OP*' WOPR < 100.0 10000 / -- -- FLOW WELLS THROUGH LOW PRESSURE SEPARATOR -- -- WELL WELL TARGET -- NAME TARG VALUE WELTARG 'OP*' THP 150 / / ENDACTIO </pre> In theory, most SCHEDULE keywords can be used in an ACTIONW Definition Section here, except for the time stepping keywords, i.e, TSTEP and DATES. See the ACTIONX keyword for a list of the SCHEDULE keywords that are known to work with the ACTION series of keywords.	Not Applicable
	ENDACTIO	Define the end of ACTIONW Definition Section.	Not Applicable
Notes: I) There is no terminating "/" for this keyword, instead the ENDACTIO keyword terminates the keyword.			

Table 12.4: ACTIONW Keyword Description

The well [SUMMARY](#) vectors that can be used with the [ACTIONW](#) keyword by OPM Flow and the commercial black-oil simulator are outlined in Table 12.5.

No.	Well SUMMARY Vector Description	SUMMARY Mnemonic
1	Bottom-Hole Pressure	WBHP
2	Gas Injection Rate	WGIR
3	Gas Injection Total	WGIT
4	Gas-Liquid Ratio	WGLR
5	Gas-Liquid Ratio (Bottom Hole)	WBGLR
6	Gas-Oil Ratio	WGOR
7	Gas Production Rate	WGPR
8	Gas Production Total	WGPT
9	Liquid Production Rate	WLPR
10	Liquid Production Rate	WLPT
11	Oil Injection Rate	WOIR
12	Oil Injection Total	WOIT
13	Oil Production Rate	WOPR
14	Oil Production Total	WOPT
15	Polymer Production Concentration	WCPC
16	Polymer Production Rate	WCPR
17	Salt Production Concentration	WSPC
18	Salt Production Rate	WSPR
19	Tracer Production Concentration (Tracer name should be added to WTPC)	WTPC
20	Tracer Production Rate (Tracer name should be added to WTPC)	WTPR
21	Tubing Head Pressure	WTHP
22	Voidage Injection Rate	WVIR
23	Voidage Injection Total	WVIT
24	Voidage Production Rate	WVPR
25	Voidage Production Total	WVPT
26	Water Cut	WWCT
27	Water Injection Rate	WWIR
28	Water Injection Total	WWIT

No.	Well <i>SUMMARY</i> Vector Description	<i>SUMMARY</i> Mnemonic
29	Water Production Rate	WWPR
30	Water Production Total	WWPT
31	Water-Gas Ratio	WWGR
32	User Defined Quantity	WUXXXXXX
Notes: - Cells under the No. column not colored show that the <i>SUMMARY</i> vector is functional within an <i>ACTIONW</i> block. - Cells under the No. column colored orange show that the <i>SUMMARY</i> vector is currently not supported within an <i>ACTIONW</i> block.		

Table 12.5: *ACTIONW* Supported Well *SUMMARY* Variables

See also the *ACTDIMS* and *UDADIMS* keywords in the *RUNSPEC* section to define the dimensions for the *ACTION* series of keywords and associated variables. In addition, the *EXIT* keyword in the *SCHEDULE* section that allows for terminating the simulation for when a condition within an *ACTIONW* definition is satisfied

Although most *SCHEDULE* section keywords should work with the *ACTIONW* keyword, Table 12.8 under the description of *ACTIONX – Define Action Conditions and Command Processing* shows the status of keywords that have been tested and known to work, together with keywords that are currently planned to be implemented.

As mentioned previously, the *UDQ* keyword stipulates the variables and operations used to access the User Defined Quantities features in OPM Flow. *UDQ* variables can be constants, *SUMMARY* variables, as defined in the *SUMMARY* section, or a formula using various mathematical functions together with constants and *SUMMARY* variables.

Note

Within an *ACTIONW* Definition Section any *UDQ* variables utilizing well variables, must have their associated wells previously fully defined in the commercial simulator; otherwise an error will occur. For example, if a well's GOR is being used as part of a *UDQ* definition, then the well must be fully characterized prior to declaring the *UDQ* definition.

This restriction does not apply to OPM Flow; however, it should be considered if the same deck is to be run with both simulators.

User Defined Quantities can also be used as User Defined Arguments (“UDA”) in the *SCHEDULE* section with various group, well, and connection keywords. In this case, the UDA variables are used to replace numerical values on these keywords by UDA variables that have been defined by the *UDQ* keyword. For example, if we wish to make the oil rate for certain wells to be a function of their water cut, then one can define the function using the *UDQ* keyword that results in a *UDQ* variable, WU_WOPR say, and then use WU_WOPR as a UDA variable on the *WCONPROD* keyword for the ORAT parameter. See Table 12.78 for a list of keywords that can be used with UDA variables in the *UDQ – Declare User Define Quantities (“UDQ”)* keyword section.

Examples

The first example uses the [ACTIONW](#) keyword to re-complete a gas injection well, GI01, when the well's bottom-hole pressure exceeds the fracture pressure of the formation, and to open up a structurally higher zone in the well. The keyword [NEXTSTEP](#) is used to set the next step to 0.1 days to avoid convergence issues due to a well event.

```
--
--      ACTIONW WELL COMMANDS
--
ACTIONW
--      ACTION      WELL      ACTION      ACTION      ACTION      MAX      INCREMENT
--      NAME        NAME        LHS        TEST        RHS        ACTIONS  RHS
--      ACTW-1      GI01      WBHP        >        6800.0
--
--      ACTION COMMANDS TO BE EXECUTED
--
--      NEXT      ALL
--      STEP      TIME
NEXTSTEP
--      0.1      'NO'
--
-- WELL PRODUCTION STATUS
--
--      WELL WELL      --LOCATION--  COMPLETION
--      NAME STAT      I      J      K      FIRST LAST
WEOPEN
GI01      SHUT
GI01      SHUT      0      0      0      2      2
GI01      OPEN
GI01      OPEN      0      0      0      1      1
/
ENDACTIO
```

The second example shows how to reduce a producing well's deliverability due to water production. Here, well OP01's productivity index is reduced by 2% when the well's liquid production total is greater than 100,000 stb. The reduction is repeated 300 times and each time the action is executed the ACTRHS constant is increased by 10,000 stb.

```
--
--      ACTIONW WELL COMMANDS
--
ACTIONW
--      ACTION      WELL      ACTION      ACTION      ACTION      MAX      ICREMENT
--      NAME        NAME        LHS        TEST        RHS        ACTIONS  RHS
--      ACTW-01      OP01      WLPT        >        100E3      300      10E3
--
--      DEFINE WELL CONNECTION MULTIPLIERS
--
--      WELL PI      --LOCATION--  COMPLETION
--      NAME MULT      I      J      K      FIRST LAST
WPIMULT
OP01      0.980      1*      1*      1*      1*      1*
/
ENDACTIO
```

The final example is similar to the previous example, and shows how to nest `ACTIONW` blocks and how to apply `ACTIONW` to all the oil producers when the Boolean condition are satisfied.

```
--
--      ACTIONW WELL COMMANDS
--
ACTIONW
--      ACTION      WELL      ACTION      ACTION      ACTION      MAX      ICREMENT
--      NAME        NAME      LHS        TEST       RHS        ACTIONS  RHS
--      ACTW-01A    'OP*'    WLPT      >         100E3      300      10E3    /

--
--      ACTIONW WELL COMMANDS
--
ACTIONW
--      ACTION      WELL      ACTION      ACTION      ACTION      MAX      ICREMENT
--      NAME        NAME      LHS        TEST       RHS        ACTIONS  RHS
--      ACTW-01B    'OP*'    WPI       >          5.0        1        0.0     /

--
--      DEFINE WELL CONNECTION MULTIPLIERS
--
-- WELL  PI      --LOCATION--  COMPLETION
-- NAME  MULT    I    J    K    FIRST LAST
WPIMULT
'? '    0.980    1*   1*   1*    1*    1*
/

ENDACTIO
ENDACTIO
```

In this case the outer `ACTIONW` checks if the liquid production for each `OP*` well is great than the calculated `ACTRHS` constant, if it is, then inner `ACTIONW` reduces all the selected wells' productivity indices by 2%, provided the well's productivity index is greater than five.

12.3.6 ACTIONX – DEFINE ACTION CONDITIONS AND COMMAND PROCESSING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [ACTIONX](#) keyword defines a series of conditions that invoke run time processing of [ACTION](#) functions and is similar to executing a run time script. This is the general purpose version of the [ACTION](#) series of keywords that can apply Boolean conditional tests to variables at the field, group, region, well segment and well levels. The [ACTION](#), [ACTIONG](#), [ACTIONR](#), [ACTIONS](#) and [ACTIONW](#) keywords are not implemented in OPM Flow and are unlikely to be so, as the [ACTIONX](#) keyword implements their functionality with greater flexibility.

This keyword starts the definition of an [ACTIONX](#) section that stipulates the Boolean conditions to test and the resulting [SCHEDULE](#) keywords to be executed if the Boolean condition evaluates to true. An [ACTIONX](#) Definition Section is terminated by an [ENDACTIO](#) keyword on a separate single line.

Although this keyword is read by OPM Flow and the [ACTION](#) and [UDQ](#) computational logic and calculations have been implemented, one should use caution when using this facility as it may result in OPM Flow aborting. This is because the [ACTIONX](#) keyword enables the user to implement complex functionality and therefore it is advisable to start with simple expressions before adding the desired complexity.

See also the [PYACTION](#) keyword in the [SCHEDULE](#) section that implements OPM Flow's Python scripting facility using the standard Python scripting language.

No.	Name	Description	Default
ACTIONX		Define the start of ACTIONX Definition Section. This is then followed on a new line by any number of ACTIONX records that define the conditions for which the defined action will be executed and the various operations to be performed if the conditions are satisfied.	
I-1	ACTNAME	ACTNAME is a character sting of up to eight characters in length, that defines the name of this action definition. If ACTNAME has previously been used by any ACTION series keyword, then the previous ACTION series definition will be replaced by the definition declared by this ACTIONX Definition Section.	
I-2	ACTNSTEP	ACTNSTEP is a positive integer that defines the number times that the ACTNAME definition is executed. ACTIONX definitions are activated at the end of a time step and this parameter is used to set how many time steps the ACTNAME definition will be invoked. The default value of one means that the definition will be executed only once. Use a large value, for example 10,000 for the definition to be executed at every time step. Note that the counter only affects successful evaluations; i.e. if ACTNSTEP is set equal to one (the default), then the simulator will test the action at the end of every time step until it evaluates to true.	1
I-3	ACTDELTA	ACTDELTA is a real positive value that defines the minimum duration of time after the conditions defined on the second record have been satisfied before the ACTIONX actions are executed. For example, if ACTDELTA is defaulted the actions will be executed at the end of the time step for which the conditions are met. If set to say 30, then a minimum of 30 days will pass before the actions are executed (assuming field or metric units).	0.0
		days days hours	
I-4	/	Record terminated by a "/"	Not Applicable

No.	Name	Description	Default
2-1	ACTLHS	ACTLHS is a series of character strings, each up to eight characters in length, that defines a constant, <i>UDQ</i> defined value, or a <i>SUMMARY</i> variable on the left hand side of a Boolean conditional test. The format for ACTLHS is dependent on the <i>SUMMARY</i> variable type: Aquifer, Block, Field, Group, Region, Time, Well, Well Connection, Well Local Grid Refinement Connection, or a Well Segment. In addition to <i>SUMMARY</i> variables, an <i>UDQ</i> defined value or a Constant variable can be used. The format for the various data types is given in Table 12.7.	Not Applicable
2-2	ACTTEST	ACTTEST is a defined character string that states the Boolean operator and must be set to one of the following Boolean conditionals: 1) >: Greater than. 2) <: Less than. 3) >=: Greater than or equal to. 4) <=: Less than or equal to. 5) =: Equal to. 6) !=: Not equal to. For example to test if the field's gas production rate is less than 600 MMscf/d then one would use: <pre> ACTIONX PHASE2 1 GGPR 'FIELD' < 600E3 / ... ENDACTIO </pre>	Not Applicable
2-3	ACTRHS	ACTRHS is a numeric value or a series of character strings, each up to eight characters in length, that defines a constant, an <i>UDQ</i> defined value, or a <i>SUMMARY</i> variable on the right hand side of a Boolean conditional test, as outlined in Table 12.7 (see also ACTLHS). In the case of well quantities the set of matching wells is captured and can be used as a general "well list" with the symbol '?' in subsequent well keywords. For example, to shut-in all oil producing wells ('OP*') with a water cut greater than 90% for every time the field water production rate exceeds 60,000 stb/d one would use: <pre> ACTIONX MXWATER 10000 GWPR 'FIELD' > 60E3 AND WWCT 'OP*' > 0.90 / -- WELL PRODUCTION STATUS -- -- WELL WELL --LOCATION-- COMPLETION -- NAME STAT I J K FIRST LAST WELOPEN '?' SHUT / ENDACTIO </pre>	Not Applicable

No.	Name	Description	Default
2-4	ANDOR	An optional defined character string that specifies a Boolean operator that must be set to either AND or OR if included on this record, that links this record with additional records of this type. For example, to test if the field's gas production rate is less than 600 MMscf/d after 2020 then one would use: <pre> ACTIONX PHASE2 1 GGPR 'FIELD' < 600E3 AND YEAR > 2020 / ... ENDACTIO </pre> This item should be left blank if not required.	Not Applicable
2.5	/	Termination of an ACTIONX Boolean condition record. Note that multiple numbers of records of this type can be entered with each record terminated by a “/”, as illustrated above.	Not Applicable
3-1	/	The Boolean condition section of the ACTIONX keyword is terminated by an empty line with a single “/”.	Not Applicable
		The next section contains any number of standard SCHEDULE keywords that will be executed if the Boolean expression evaluates to true. For example, to test if the field's gas production rate is less than 600 MMscf/d after 2020 and to open up additional wells if this occurs, then one would use: <pre> ACTIONX PHASE2 1 GGPR 'FIELD' < 600E3 AND YEAR > 2020 / -- WELL PRODUCTION STATUS -- -- WELL WELL --LOCATION-- COMPLETION -- NAME STAT I J K FIRST LAST WELOPEN GP10 OPEN GP11 OPEN / ENDACTIO </pre> In theory, most SCHEDULE keywords can be used in an ACTIONX Definition Section here, except for the time stepping keywords, i.e, TSTEP and DATES. See Table 12.8 for a list of the SCHEDULE keywords that are known to work with the ACTIONX keyword.	Not Applicable
	ENDACTIO	Define the end of ACTIONX Definition Section.	Not Applicable
Notes: l) There is no terminating “/” for this keyword, instead the ENDACTIO keyword terminates the keyword.			

Table 12.6: [ACTIONX](#) Keyword Description

The variable types and the associated definitions that are available for use with Boolean conditionals are outlined in Table 12.7.

Variable Type	Description
AQUIFER	AQUIFER variable consists of two parameters: 1) Aquifer SUMMARY variable; for example, Analytical Aquifer Influx Rate, AAQR. 2) Aquifer number consisting of a positive integer greater than zero that defines the aquifer to be used.
BLOCK	BLOCK variable consists of four parameters: 1) Block SUMMARY variable; for example Block Oil Saturation, BOSAT. 2) Block I location which should be a positive integer greater than or equal to one and less than or equal to NX. 3) Block J location which should be a positive integer greater than or equal to one and less than or equal to NY. 4) Block K location which should be a positive integer greater than or equal to one and less than or equal to NZ. The NX, NY, and NZ parameters are defined on the DIMENS keyword in the RUNSPEC section.
CONSTANTS	CONSTANTS can consist of one or optionally two parameters: 1) A numerical value defining the Constant. 2) An optional numerical value defining the Increment. If not specified then the Increment defaults to zero. The Increment is added to the Constant after each time the action is executed. The Increment can be positive or negative.
FIELD	The FIELD variable consists of any field SUMMARY variable; for example the Field average Pressure, as shown below: <pre> ACTIONX WIPHASE 1 / FPR < 2500 / / ... ENDACTIO </pre> The above would action a set of SCHEDULE keywords if the field average pressure fell below 2,500 psia for a run using FIELD units.
GROUP	GROUP variable definition consists of: 1) Group SUMMARY variable; for example, Group Oil Production Rate, GOPR. 2) Group Name which is a character string of up to eight characters in length that defines an existing group, note that the group named FIELD is the top most group. To enable an action for when the field's oil production rate drops below 20,000 stb/d then one could use. <pre> ACTIONX OILMIN 1 / GOPR 'FIELD' < 20.0E3 / / ... ENDACTIO </pre>

Variable Type	Description
REGION	REGION variable definition consists of: 1) Region SUMMARY variable; selected from one of the following only: RPR, RGSAT, ROSAT, RVVSAT, RGIP, ROIP, and RWIP. No other region summary fields are permitted in the expressions. 2) Fluid In-Place region number which is a positive integer greater than or equal to zero that defines the region number. The value should less than or equal to the NTFIP variable on either the REGDIMS or TABDIMS keywords in the RUNSPEC section. Note that a zero value indicates the whole model. 3) Fluid In-Place region family (not used by OPM Flow). For example, <pre> ACTIONX WIPHASE 1 / RPR 0 < 2500 / / ... ENDACTIO </pre> Would action a set of SCHEDULE keywords if the field average pressure fell below 2,500 psia for a run using FIELD units.
TIME	TIME variable definition consists of one parameter that can have three values: DAY for the current simulation day of the month, MNTH for the current simulation month, and YEAR for the current simulation year. Thus, to set an action for April 1, 2025 one would use: <pre> ACTIONX ACT01 1 / DAY = 1 AND / MNTH = 'APR' AND / YEAR = 2025 / / ... ENDACTIO </pre> Note that the value for the MNTH variable, 'APR', in the example, can also be entered without the quotes, that is: <pre> MNTH = APR AND / </pre> In addition, numerical values for MNTH, similar to DAY and YEAR, are also permitted and are converted to the nearest integer for comparison, so for example: <pre> ACTIONX ACT01 1 / DAY = 0.95 AND / MNTH = 4.40 AND / YEAR = 2024.9 / / ... ENDACTIO </pre> Would again result in the action taking place on April 1, 2025.

Variable Type	Description
WELL	WELL variable definition consists of: 1) Well SUMMARY variable; for example, Well Oil Production Rate, WOPR. 2) Well Name which is a character string of up to eight characters in length that defines the well name, which must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur. The Well Name may be defaulted (*), a well root (e.g 'OP*') or a well list (e.g. '*BLK-I') when used on the LHS of a condition. Note that the use of well lists is an OPM Flow specific feature. To reduce the tubing head pressure constraint for when any of the oil producers' oil rate drop below 100 stb/d then one could use. <pre> ACTIONX WOILMIN 1 / WOPR 'OP*' < 100.0 / / -- -- FLOW WELLS THROUGH LOW PRESSURE SEPARATOR -- -- WELL WELL TARGET -- NAME TARG VALUE WELTARG 'OP*' THP 150 / / ENDACTIO </pre>
WELL CONNECTION	WELL CONNECTION variable definition is comprised of: 1) Well connection SUMMARY variable; for example, Connection Oil Flow Rate, COFR. 2) Well Name which is a character string of up to eight characters in length that defines the well, which must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur. 3) I- Connection: A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction 4) J- Connection: A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction. 5) K- Connection: A positive integer greater than or equal to one and less than or equal to NZ that defines the connection location in the K-direction. The NX, NY, and NZ parameters are defined on the DIMENS keyword in the RUNSPEC section.

Variable Type	Description
WELL LOCAL GRID REFINEMENT CONNECTION	WELL LOCAL GRID REFINEMENT CONNECTION variable definition is comprised of: 1) Well local grid refinement connection SUMMARY variable; for example, the Local Grid Refinement Connection Oil Flow Rate, LCOFR. 2) Well Name which is a character string of up to eight characters in length that defines the well, which must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur. 3) Local Grid Refinement Name which is a character string of up to eight characters in length that defines the Local Grid Refinement ("LGR"), which must have been declared previously using the CARFIN or RADFIN keywords in the GRID section, otherwise an error may occur. 4) I- Connection:A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction within the LGR. 5) J-Connection:A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction within the LGR. 6) K- Connection:A positive integer greater than or equal to one and less than or equal to NZ that defines the connection location in the K-direction within the LGR. The NX, NY, and NZ parameters are defined on either the CARFIN or RADFIN keywords in the GRID section depending upon whether a Cartesian or radial local grid refinement is being utilized. Note Local Grid Refinements are currently not implemented in OPM Flow.
WELL SEGMENT	WELL SEGMENT variable definition consists of: 1) Well Segment SUMMARY variable; for example, Segment Oil Flow Rate, SOFR. 2) Multi-Segment Well Name which is a character string of up to eight characters in length that defines the well name which must have been declared previously using the WELSPECS and WELSEGS keywords in the SCHEDULE section, otherwise an error may occur. 3) Segment Number, which is a positive integer greater than or equal to two and less than or equal to MXSEGS on WSEGDIMS keyword in the RUNSPEC section that defines the segment. Note that the total number of wells should be defined via the WELLSDIMS keyword and the number of multi-segment wells should be declared on the WSEGDIMS keyword, both keywords are in the RUNSPEC section.

Table 12.7: ACTIONX Variable Definitions

See also the [ACTDIMS](#) and [UDADIMS](#) keyword in the [RUNSPEC](#) section to define the dimensions for the [ACTIONX](#) keyword and associated variables. In addition, the [EXIT](#) keyword in the [SCHEDULE](#) section that allows for terminating the simulation for when a condition within an [ACTIONX](#) definition is satisfied.

Although most [SCHEDULE](#) keywords should work with the [ACTIONX](#) keyword, Table 12.8 shows the status of keywords that have been tested and known to work, together with keywords that are currently planned to be implemented.

ACTIONX					
Schedule Section Keywords Status					
Action Keywords	Group Keywords	Well Keywords		Connection Keywords	Miscellaneous Keywords
ACTIONX	GCONINJE	WCONINJE	WPIMULT	COMPDAT	BOX
UDQ	GCONPROD	WCONPROD	WSEGVAVL	COMPLUMP	ECHO
	GCONSUMP	WECON	WTEST	COMPSEGS	ENDBOX
	GLIFTOPT	WEFAC	WTMULT		EXIT ⁵
	GRUPNET	WEOPEN			INCLUDE ⁷
	GRUPTARG	WELPI			MULTX
	GRUPTREE	WELSEGS			MULTX-
	GSATINJE	WELSPECS ⁶			MULTY
	GSATPROD	WELTARG			MULTY-
		WGRUPCON			MULTZ
		WINJMULT			MULTZ-
					NEXT
					NEXTSTEP
					NOECHO

Notes:

- Cells not colored show that the keyword has been tested and is functional within an ACTIONX block.
- Cells colored in gray indicate that the keyword has been tested in OPM Flow but the results are currently different to the commercial simulator.
- Cells colored orange show keywords that are currently available in OPM Flow but have not been tested; however, testing is ongoing.
- Cells colored red show keywords currently unavailable in OPM Flow because the underlying feature is not available; however, there are plans to implement this functionality.
- OPM Flow specific keyword.
- If a standard well is fully declared in an ACTIONX block which is then activated at a later date, and later the well is modified to be a multi-segment well using the WELSEGS and COMPSEGS keywords, then this will cause the simulator to abort with an assert failure.
- The INCLUDE keyword is permissible in an OPM Flow ACTIONX block, but this will cause an exception in the commercial simulator. Users must be cautious about the contents of the included file, as conflicts with the base DATA file (e.g., time progression settings) or keywords unsupported by ACTIONX may cause issues. Future updates to the parser may also alter or remove this functionality, so it's advisable not to rely on it for long-term compatibility.

Table 12.8: ACTIONX Schedule Section Keyword Status

As mentioned previously, the UDQ keyword stipulates the variables and operations used to access the User Defined Quantities features in OPM Flow. UDQ variables can be constants, SUMMARY variables, as defined in

the [SUMMARY](#) section, or a formula using various mathematical functions together with constants and [SUMMARY](#) variables.

Note

Within an [ACTIONX](#) Definition Section any [UDQ](#) variables utilizing group and well variables, must have their associated groups and wells previously fully defined in the commercial simulator; otherwise an error will occur. For example, if a well's GOR is being used as part of a [UDQ](#) definition, then the well must be fully characterized prior to declaring the [UDQ](#) definition.

This restriction does not apply to OPM Flow; however, it should be considered if the same deck is to be run with both simulators.

User Defined Quantities can also be used as User Defined Arguments (“UDA”) in the [SCHEDULE](#) section with various group, well, and connection keywords. In this case, the UDA variables are used to replace numerical values on these keywords by UDA variables that have been defined by the [UDQ](#) keyword. For example, if we wish to make the oil rate for certain wells to be a function of their water cut, then one can define the function using the [UDQ](#) keyword that results in a [UDQ](#) variable, WU_OPR say, and then use WU_OPR as a UDA variable on the [WCONPROD](#) keyword for the ORAT parameter. See Table 12.78 for a list of keywords that can be used with UDA variables in the [UDQ – Declare User Define Quantities \(“UDQ”\)](#) keyword section.

Examples

The first example uses the [UDQ](#) keyword to sort the oil wells from high water cut to low, via the WU_WLIST variable, and then use the [ACTIONX](#) keyword to shut-in the worst offending well when the field's water production is greater than 30,000 stb/d.

```
--
--      DEFINE START OF USER DEFINED QUANTITY SECTION
--
UDQ
--
--      OPERATOR VARIABLE  EXPRESSION
--
DEFINE  WU_WCUT    WWCT 'OP*'                / WELL WWCT LIST
DEFINE  WU_LIST    SORTD(WU_WCUT)            / WELL WWCT LIST SORTED
/                                              END OF UDQ SECTION
--
--      DEFINE START OF ACTIONX SECTION
--
ACTIONX
      WSHUTIN
10
      GWPR 'FIELD' > 30E3  AND
      WU_LIST 'OP*' = 1
/
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
--      WELL WELL      --LOCATION--  COMPLETION
--      NAME STAT      I    J    K    FIRST LAST
WELOPEN
'?'      SHUT
'?'      SHUT      0    0    0      0      0
/
```

ENDACTIO

Apart from checking that the field's water production rate is greater than 30,000 stb/d the Boolean conditional also checks that there is at least one well in the sorted well list. Notice also the use of '?' symbol as a substitution of the well name and that the ACTIONX WSHUTIN series of commands will be executed a total of ten times.

The second example checks to see if the field's gas rate is below 600 MMscf/d and if the simulation time is greater that July 1, 2030. If it is, then compression is installed by re-setting all the gas producing well's THP and BHP pressures to 450 psia and 300 psia respectively. In addition all gas wells currently shut-in are tested to see if they can be opened up under the new THP and BHP constraints.

```
--
--      START ACTIONX FIELD PHASE-3 AUTOMATIC COMPRESSION
--
ACTIONX
    PHASE-3      1
    GGPR  'FIELD' < 600E3 AND
    DAY    >= 1    AND
    MNTH   >= JUL  AND
    YEAR   >= 2030
/

--
--      INSTALL COMPRESSION AND RESET WELL THP AND BHPS
--
--  WELL      WELL      TARGET
--  NAME      TARG      VALUE
WELTARG
'GP*'      THP      450
'GP*'      BHP      300
/

--
--      TEST AND OPEN ALL WELLS UNDER COMPRESSION CONSTRAINTS
--
--  WELL      TEST      CLOSE      NO.      START
--  NAME      INTV      CHECK      CHECK      TIME
WTEST
'GP*'      1.0      PE      1      3
/

--
--      END OF ACTIONX FIELD PHASE-3 AUTOMATIC COMPRESSION DEFINITION
--
ENDACTIO
```

12.3.7 AITS – ACTIVATE INTELLIGENT TIME STEPPING

Turns on the commercial simulator's intelligent time stepping.

See [AITS – Activate Intelligent Time Stepping](#) in the [RUNSPEC](#) section.

12.3.8 AITSOFF – DEACTIVATE INTELLIGENT TIME STEPPING

Turns off the commercial simulator's intelligent time stepping.

See [AITSOFF – Deactivate Intelligent Time Stepping](#) in the [RUNSPEC](#) Section

12.3.9 APILIM – DEFINE API TRACKING GRID BLOCK LIMITS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [APILIM](#) keyword defines [API](#) Tracking grid block limits for when [API](#) Tracking has been activated via the [API](#) keyword in the [RUNSPEC](#) section. The keyword enables the simulator to monitor the grid blocks outside the limits defined on the keyword, as well as to optionally constrain the values within a given range.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.10

AQUCHGAS – DEFINE CONSTANT PRESSURE GAS ANALYTICAL AQUIFER PROPERTIES

The [AQUCHGAS](#) keyword defines the properties of constant pressure gas analytical aquifers.

See [AQUCHGAS – Define Constant Pressure Gas Analytical Aquifer Properties](#) in the [SOLUTION](#) section for a full description.

12.3.11

AQUCHWAT – DEFINE CONSTANT PRESSURE WATER ANALYTICAL AQUIFER PROPERTIES

The [AQUCHWAT](#) keyword defines the properties of constant pressure water analytical aquifers.

See [AQUCHWAT – Define Constant Pressure Water Analytical Aquifer Properties](#) in the [SOLUTION](#) section for a full description.

12.3.12

AQUCT – DEFINE CARTER-TRACY ANALYTICAL AQUIFERS

The [AQUCT](#) keyword defines Carter-Tracy³¹³ analytical aquifers, the properties of the aquifer, including the Carter-Tracy aquifer influence function associated with the aquifer, defined by the [AQUCTAB](#) keyword in the [PROPS](#) section.

See [AQUCT – Define Carter-Tracy Analytical Aquifers](#) in the [GRID](#) section.

³¹³

Carter, R. D. and Tracy, G.W. “An Improved Method for Calculating Water Influx,”Transactions of AIME (1960) 219, 215-417.

12.3.13

AQUCWFAC – MODIFY CONSTANT PRESSURE WATER ANALYTICAL AQUIFER PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [AQUCWFAC](#) keyword modifies the datum depth and pressure for all aquifers specified by the [AQUCHWAT](#) keyword in the [SOLUTION](#) or [SCHEDULE](#) sections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.14 AQUFETP – DEFINE FETKOVICH ANALYTICAL AQUIFERS

The [AQUFETP](#) keyword defines Fetkovich Analytical aquifers and the aquifer properties. Each row entry in the [AQUFETP](#) keyword defines one Fetkovich Analytical aquifer. In order to fully define this type of aquifer, the aquifer must be connected to the reservoir using the [AQUANCON](#) keyword in the [GRID](#) or [SOLUTION](#) sections.

See [AQUFETP – Define Fetkovich Analytical Aquifers](#) in the [SOLUTION](#) section for a full description.

12.3.15 AQUFLUX – DEFINE CONSTANT FLUX ANALYTICAL AQUIFER

The [AQUFLUX](#) keyword defines the properties of Constant Flux Analytical Aquifers, that allows for a constant water influx to the connected grid blocks

See [AQUFLUX – Define Constant Flux Analytical Aquifer](#) in the [SOLUTION](#) section for a full description.

12.3.16 BCPROP – DEFINE BOUNDARY CONDITIONS PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [BCPROP](#) keyword defines the type and properties of the boundary conditions.

Together the [BCCON](#) and [BCPROP](#) keywords define the boundary conditions for the model, and can be used to set boundary conditions for when external influx or efflux volumes are influencing the reservoir pressure and production history. For example, when the average reservoir pressure remains constant throughout the production period due to water influx, or gas migration from an external source.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	INDEX	A positive integer that identifies the boundary condition.			None
2	TYPE	A defined character string that defines the type of boundary condition to be applied, and should be set to one of the following character strings: 1) DIRICHLET: for user defined boundary conditions. In this case, COMPONENT for the fluid type, and PRESS and TEMP for the constant pressure and temperature boundary conditions must be specified. 2) FREE: for the initial state of the boundary to be kept throughout the simulation, that is a constant boundary condition. The remaining items should be defaulted. 3) RATE: for the boundary to have a constant influx or efflux rate. In this case, COMPONENT for the fluid type, and RATE to set the mass rate per unit area must be specified. 4) THERMAL: Constant temperature boundary condition (no-flow of mass). 5) NONE: No flow boundary condition.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	COMPONENT	A defined character string that sets fluid type used in the boundary calculations, and should be set to one of the following character strings: 1) GAS: the gas phase will be used to control the boundary conditions for when the TYPE has been set to DIRICHLET or RATE. 2) OIL: the oil phase will be used to control the boundary conditions for when the TYPE has been set to DIRICHLET or RATE. 3) WATER or WAT: the water phase will be used to control the boundary conditions for when the TYPE has been set to DIRICHLET or RATE. 4) SOLVENT: the solvent component will be used to control the boundary conditions for when the TYPE has been set to RATE. 5) POLYMER: the polymer component will be used to control the boundary conditions for when the TYPE has been set to RATE. 6) MICR: the microbial component will be used to control the boundary conditions for when the TYPE has been set to RATE. 7) OXYG: the oxygen component will be used to control the boundary conditions for when the TYPE has been set to RATE. 8) UREA: the urea component will be used to control the boundary conditions for when the TYPE has been set to RATE. 9) NONE: the fluid type is undefined. COMPONENT must be declared and not equal to NONE if TYPE has been set to either DIRICHLET or RATE. MICR requires the BIOFILM or MICP keyword, and OXYG or UREA requires the MICP keyword.			NONE
4	RATE	A real value that defines the constant mass rate per unit area of the specified COMPONENT to be injected or withdrawn at the boundary, when TYPE has been set to RATE. Note a negative value implies an influx rate, whereas, a positive value indicates an efflux.			0.0
		lb/day/ft ²	kg/day/m ²	gm/hour/cm ²	
5	PRESS	PRESS is a real positive value that defines the constant pressure boundary condition. PRESS should only be entered if TYPE has been set to DIRICHLET. If the pressure at the boundary is less than PRESS, then the fluid type declared via COMPONENT will flow across the boundary. The default value of I* will use the simulator's calculated value based on data entered via the EQUIL keyword in the SOLUTION section.			I*
		psia	barsa	atma	
6	TEMP	TEMP is a real positive number that defines the constant temperature boundary condition. TEMP should only be entered if TYPE has been set to DIRICHLET or THERMAL. The default value of I* will use the simulator's calculated value based on data entered via one of the following reservoir temperature keywords: RTEMP, RTEMPA, RTEMPVD, TEMPI, or TEMPVD, in the SOLUTION section. Note that all of the aforementioned reservoir temperature keywords, except for TEMPI, may also be used in the PROPS section as well.			I*
		°F	°C	°C	

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	MECHTYPE	A defined character string that defines the type of geo-mechanical boundary condition to be applied, and should be set to one of the following character strings: 1) FREE: Surface is free to move with no external stress. 2) FIXED: Surface is fixed in the x/y/z-directions if FIXEDX / FIXEDY / FIXEDZ > 0. The displacement in each direction defined by DISPX / DISPY / DISPZ. The external stress is defined by STRESSXX / STRESSYY / STRESSZZ. 3) NONE: No geo-mechanical boundary condition.			NONE
8	FIXEDX	A positive integer that identifies the whether the boundary is free or fixed in the x-direction. A value of 0 implies the boundary is free to move in the x-direction. A value > 0 implies the boundary is fixed in the x-direction with displacement defined by DISPX,			1
9	FIXEDY	A positive integer that identifies the whether the boundary is free or fixed in the y-direction. A value of 0 implies the boundary is free to move in the y-direction. A value > 0 implies the boundary is fixed in the y-direction with displacement defined by DISPY,			1
10	FIXEDZ	A positive integer that identifies the whether the boundary is free or fixed in the z-direction. A value of 0 implies the boundary is free to move in the z-direction. A value > 0 implies the boundary is fixed in the z-direction with displacement defined by DISPZ,			1
	STRESSXX	A real value that defines the diagonal component σ_{xx} of the stress tensor.			0
		psia	barsa	atma	
	STRESSYY	A real value that defines the diagonal component σ_{yy} of the stress tensor.			0
		psia	barsa	atma	
	STRESSZZ	A real value that defines the diagonal component σ_{zz} of the stress tensor.			0
		psia	barsa	atma	
	DISPX	A real value that defines the x-direction displacement boundary condition.			0
		feet	m	cm	
	DISPY	A real value that defines the y-direction displacement boundary condition.			0
		feet	m	cm	
	DISPZ	A real value that defines the z-direction displacement boundary condition.			0
		feet	m	cm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) The boundary condition connections must have been specified using the <i>BCCON</i> keyword. 2) Cells colored orange indicate parameters for the geo-mechanical boundary conditions that are only supported by the experimental geo-mechanical simulator. 3) Each record must be terminated by a “/” and the keyword is terminated by a “/”.					

Table 12.9: BCPROP Keyword Description

See also the [AQUFLUX](#) keyword that is supported by OPM Flow in both the [SOLUTION](#) and [SCHEDULE](#) sections, to define a constant flux analytical aquifer.

If the [BCCON](#) and [BCPROP](#) keywords are not present in the input deck, then the boundary conditions for the model are set to be no flow, which is the normal behavior in both OPM Flow and the commercial simulator.

The [BC](#) keyword has been replaced by the [BCCON](#) and [BCPROP](#) keywords.

Examples

The first example shows how to set a constant pressure boundary using TYPE equal to FREE:

```
--
--          DEFINE BOUNDARY CONDITIONS CONNECTION (OPM FLOW GRID KEYWORD)
--
--INDEX  ----- BOX ----- BC
--          I1  I2   J1  J2   K1  K2   DIRC
BCCON
1          1    1    1    1*   1    1*   X-   /
2          1    1*   1    1    1    1*   Y    /
/

--
--          DEFINE BOUNDARY CONDITIONS PROPERTIES (OPM FLOW SCHEDULE KEYWORD)
--
--INDEX  BC      BC      BC      BC      BC
--          TYPE    COMPT  RATE  PRESS  TEMP
BCPROP
1          FREE    1*      1*      1*      1*   /
2          FREE
/
```

With this option it is only necessary to define the boundary cells and all the other parameters (COMPONENT, RATE, PRESS, and TEMP) can be defaulted, as they are ignored when TYPE equals FREE.

The next example is based on NX, NY and NZ equal to 20, 1, 10 respectively, on the [DIMENS](#) keyword in the [RUNSPEC](#) section, and shows how different boundary types can be assigned to different parts of the model.

```
--
--          DEFINE BOUNDARY CONDITIONS CONNECTION (OPM FLOW GRID KEYWORD)
--
--INDEX  ----- BOX ----- BC
--          I1  I2   J1  J2   K1  K2   DIRC
BCCON
1          1    1    1    1    1    10   X-   /
2          20   20   1    1    1    10   X    /
/
```

```
--
--      DEFINE BOUNDARY CONDITIONS PROPERTIES (OPM FLOW SCHEDULE KEYWORD)
--
--INDEX BC      BC      BC      BC      BC
--      TYPE      COMPT  RATE  PRESS  TEMP
BCPROP
1      DIRICHLET  GAS    1*    256.0  100.0 /
2      FREE      4*          /
/
```

The last example shows how the DIRICHLET boundary condition option may be used:

```
--
--      DEFINE BOUNDARY CONDITIONS CONNECTION (OPM FLOW GRID KEYWORD)
--
--INDEX ----- BOX ----- BC
--      I1  I2   J1  J2   K1  K2  DIRC
BCCON
1      1   1    1   1*   1   1*  X-   /
2      1   1*   1   1    1   1*  Y    /
/

--
--      DEFINE BOUNDARY CONDITIONS PROPERTIES (OPM FLOW SCHEDULE KEYWORD)
--
--INDEX BC      BC      BC      BC      BC
--      TYPE      COMPT  RATE  PRESS  TEMP
BCPROP
1      DIRICHLET  WAT    1*    256.0  100.0 /
2      DIRICHLET  WAT    1*    1*      100.0 /
/
```

Here, the first line sets both the pressure and temperature at the boundary, and the second line defaults the pressure entry, so that the simulator calculated initial boundary pressure will be used.

12.3.17 BOUNDARY – DEFINE A BOUNDARY BOX FOR PRINTING

The [BOUNDARY](#) keyword defines a rectangular grid for printing various arrays to the output print file (*.PRT); thus, avoiding printing all the elements in the selected array.

See [BOUNDARY – Define a Boundary Box for Printing](#) in the [GRID](#) section for a full description.

12.3.18 BOX – DEFINE A RANGE OF GRID BLOCKS TO ENTER PROPERTY DATA

[BOX](#) defines a range of grid blocks for which subsequent data is assigned for all the cells in the defined [BOX](#). Note that the [BOX](#) grid is reset by the keyword [ENDBOX](#) by resetting the current defined [BOX](#) to be the whole grid. The keyword can be used for any array and for all grid types.

See [BOX – Define a Range of Grid Blocks to Enter Property Data](#) in the [GRID](#) section for a full description.

12.3.19 BRANPROP – DEFINE NETWORK BRANCH PROPERTIES FOR EXTENDED NETWORK OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[BRANPROP](#) defines network branch properties for the extended network option for when the Extended Network Model has been activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section. There are two types of network facilities in the simulator, the Standard Network model, which is defined with the [GRUPNET](#) keyword in the [SCHEDULE](#) section and the Extended Network Model defined by the [BRANPROP](#) and [NODEPROP](#) keywords, again in the [SCHEDULE](#) section.

For the Extended Network Model the group hierarchy can be different to that defined by the [GRUPTREE](#) keyword; however, the bottom most nodes in the network tree associated with wells, must be the same as that defined by the [GRUPTREE](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DOWNNODE	A character string of up to eight characters in length that defines the down stream node name for this branch, that is the node closest to the wells. Thus for a production network, this will be an inlet node as the wells are importing fluid into the branch node. Whereas for an injection node, this is an outlet node as injection fluid is being exported to the wells.			None
2	UPNODE	A character string of up to eight characters in length that defines the up stream node name for this branch, that is the node furthestmost from the wells. Thus for a production network, this will be an outlet node as the wells are exporting fluid from the branch node. Whereas for an injection node, this is an inlet node as the wells are importing the injection fluid.			None
3	VFPTAB	A positive integer greater than or equal to zero that defines the vertical lift performance table to be used for calculating the pressure behavior between the inlet (for production) or outlet (for injection) node (DOWNNODE) and the outlet (for production) or inlet (for injection) node (UPNODE). For a production network this must reference a table associated with the VFPPROD keyword, and for an injection network a table declared via the VFPINJ keyword. Both keywords are in the SCHEDULE section. If the pressure behavior between the two nodes is zero (no pressure loss in the network branch), then a value of 9999 should be entered for this variable. A value of zero for VFPTAB removes the branch from the extended network and the resulting flows are ignored in the network flow stream.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	ALQ-NODE	A real positive value that defines the artificial lift quantity to be used in conjunction with the VFPPROD assigned to the branch via the VPFTAB variable. VFPTAB vertical lift performance table and the artificial lift quantity ALQ-NODE are used with the branch fluid rates to calculate the pressure behavior through the branch. For a network this can be considered to be either a pump to pump fluid through the network or a compressor to compress gas to a higher export pressure. Basically,ALQ-NODE is used to reduce the pressure loss through the branch. Note that the units for ALQ-NODE are dependent on the associated variable on the VFPPROD keyword. Should be set to zero if ALQ-DEN is set to either DENO or DENG, or if the branch is associated with an automatic compressor.			0.0
5	ALQ-DEN	A defined character string that defines that ALQ-NODE variable represents either as a surface density or as an artificial lift quantity for a pump or a compressor, and should be set to one of the following: 1) DENO: ALQ-NODE represents the average surface oil density flowing through the branch. 2) DENG: ALQ-NODE represents the average surface gas density flowing through the branch. 3) NONE: ALQ-NODE is the artificial lift quantity as defined on the VFPPROD keyword to model a pump or a compressor. The VFPPROD keyword should be consistent with this variable, that is, if ALQ-DEN is set to DENO then the surface oil should be used as the ALQ variable on the VFPPROD keyword.			NONE

Notes:

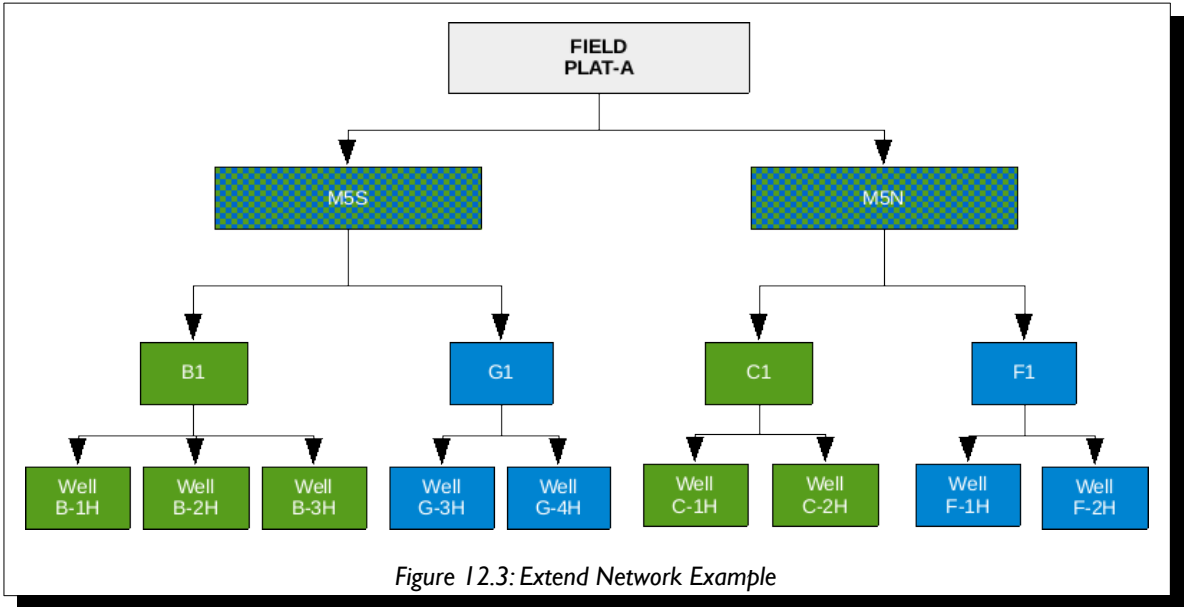
- 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.10: BRANPROP Keyword Description

See also the [NETWORK](#) keyword in the [RUNSPEC](#) section and the [NODEPROP](#) keyword in the [SCHEDULE](#) section.

Example

Given the following Extended Network model in Figure 12.3.



First the Extended Network model should be used invoked in the [RUNSPEC](#) section, and then the [BRANPROP](#) keyword should be used to define the branch network, and finally the [NODEPROP](#) keyword is used to describe the node properties.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--     ACTIVATE THE EXTENDED NETWORK OPTION AND DEFINE PARAMETERS
--
--     MAX.    MAX    NOT
--     NODE    LINK   USED
NETWORK
3           2       1*
/
.....
.....
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--     EXTENDED NETWORK BRANCH PROPERTIES
--
--     DOWN    UP      VFP    VFP
--     NODE    NODE    TABLE ALFQ
BRANPROP
B1          PLAT-A    5       1*
C1          PLAT-A    4       1*
/
```

--	EXTENDED NETWORK NODE PROPERTIES						
--							
--	NODE	NODE	CHOKE	GAS	CHOKE	SOURCE	NETWORK
--	NAME	PRESS	OPTN	LIFT	GROUP	SINK	TYPE
NODEPROP							
PLAT-A	21.0	NO	NO				/
B1	1*	NO	NO				/
C1	1*	NO	NO				/
/							

Here the main platform for the field, PLAT-A, has a fixed 21 barsa pressure applied as an operating constraint.

12.3.20 CALTRAC – DEFINE A GAS CALORFIC VALUE TRACER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [CALTRAC](#) keyword is used to assign a gas calorfic value to a tracer, for when the Tracer option has been invoked by the [TRACER](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

12.3.21
CECON – DEFINE WELL CONNECTIONS ECONOMIC LIMIT CRITERIA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

CECON sets the economic cut-off criteria for a well’s connection to the simulation grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.22 CECONT – DEFINE WELL CONNECTIONS TRACER ECONOMIC LIMIT CRITERIA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [CECONT](#), sets the tracer economic cut-off criteria for a well's connection to the simulation grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.23 COLUMNS – DEFINE INPUT FILE COLUMN MARGINS

The [COLUMNS](#) keyword defines the input file column margins; characters outside the margins are ignored by the input parser.

See [COLUMNS – Define Input File Column Margins](#) in the GLOBAL section for a full description.

12.3.24 COMPDAT – DEFINE WELL CONNECTIONS TO THE GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COMPDAT](#) keyword defines how a well is connected to the reservoir by defining or modifying existing well connections. Ideally the connections should be declared in the correct sequence, starting with the connection nearest the well head and then working along the wellbore towards the bottom or toe of the well, however this may not be possible or convenient, for example when connections are added or removed from a well during the simulation (see [COMPORD](#) in the [SCHEDULE](#) section for options regarding connection ordering).

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	I	A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction. If set to zero or defaulted with I* the location is taken from the wellhead location I-direction value on the WELSPECS keyword in the SCHEDULE section.			0
3	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction. If set to zero or defaulted with I* the location is taken from the wellhead location J-direction value on the WELSPECS keyword in the SCHEDULE section.			0
4	K1	A positive integer greater than or equal to one and less than or equal to NZ that defines the UPPER connection location in the K-direction.			None
5	K2	A positive integer greater than or equal to K1 and less than or equal to NZ that defines the LOWER connection location in the K-direction.			None
6	STATUS	A defined character string of length four that defines the connections' operational status, STATUS should be set to one of the following character strings: 1) OPEN: the connections are open to flow. 2) SHUT: the connections are closed to flow (shut-in). 3) AUTO: the connection are initially closed, but may be opened automatically if an economic limit is violated. Currently this option is not supported by OPM Flow.			OPEN

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	SATNUM	An integer greater than or equal to zero and less than NTSFUN as declared on the TABDIMS keyword in the RUNSPEC section, that defines the saturation table number to be used for flow between the reservoir grid block and the well connections. If SATNUM is set to zero or defaulted with I* then: 1) The saturation table allocated to the grid block that the connections are located within are used. 2) If the hysteresis option has been activated via the HYSER variable on the SATOPTS keyword in the RUNSPEC section, then both the imbibition and drainage saturation tables allocated to the grid block that the connections are located within are used. The imbibition table allocation can be changed by the COMPIMB keyword in the RUNSPEC section, provided it is entered after the COMPDAT keyword.			0
8	CONFACT	A real value greater than or equal to zero that defines the transmissibility connection factor between the well bore and the reservoir grid block. If set to zero or defaulted with I* then items (9) through (13) are used to calculate CONFACT.			Defined
		cPrb/day/psia 0	cPrm ³ /day/bars 0	cPrcc/hr/atm 0	
9	DW	A real positive value that defines the well bore diameter of the connections for the well. DW is used in calculating a well's productivity or injectivity index; however the value will be ignored in calculating the connections CONFACT value if CONFACT has been directly entered.			None
		feet	m	cm	
10	KH	A real value that defines the effective KH (permeability x length) for the connections. If less than or equal to zero, or defaulted by I*, then KH is calculated from the connected grid blocks. KH is ignored if CONFAC has been directly entered.			Calculated from connected grid blocks
		mD.ft	mD.m	mD.cm	
11	SKIN	A real value that defines the connections dimensionless skin factor. SKIN is used in calculating a well's productivity or injectivity index; however, the value will be ignored in calculating the connections CONFACT value if CONFAC has been directly entered.			0.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
12	DFACT	A real value that defines the non-Darcy D factor coefficient for gas wells. It is recommended that the value should be defaulted with 1* and the non-Darcy D factor coefficient for gas wells defined via the WDFAC keyword in the SCHEDULE section. If the value is greater than or equal to zero then the value is used to define the well D factor and the simulator evaluates the connection D factors internally. If the value is less than zero then its sign is reversed and then used directly to define the connection D factors. In this case any D factor values previously defined using the WDFAC keyword for other connections will be retained, whereas, any connection D factors previously defined for the entire well using the WDFACCOR keyword will be discarded.			0.0
		day/Mscf	day/m ³	hour/sc	
13	DIRECT	A defined character string of length one that defines the orientation of the connections and should be set to either X, Y, or Z. The direction of connections also determines the length of the connection used to calculate the connection factor if CONFACT has not been entered directly. The default value is for a vertical connection, that is DIRECT is defaulted to Z.			Z

Notes:

- 1) The keyword is followed by up to MXCONS records as declared on the [WELLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.11: COMPDAT Keyword Description

Multiple grid block connections can be defined on one record for vertical wells by assigning different values to K1 and K2, for deviated and horizontal wells this may not be possible and therefore each grid block connection must be separately defined by using one record per connection, setting K1 equal to K2 in each record.

See also the [WEL SPECS](#) keyword to define wells, the [COMPIMB](#) to reset the imbibition relative permeability table allocation, and the [COMPORD](#) to re-order the completions along the well trajectory. In addition, the [COMPLUMP](#) keyword groups well connections together to form well completions for a well. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Note

The term well connection is used to describe individual connections from the wellbore to the reservoir grid, as opposed to well completions. A well completion is used to describe a set of connections, for example, a well may consist of several completions with each completion consisting of multiple connections.

Example

The following example defines two vertical oil wells using the [WELSPECS](#) keyword and their associated connection data.

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  OPEN  CROSS  PRESS
-- NAME  NAME        I      J  DEPTH  FLUID   AREA  EQUANS SHUT  FLOW   TABLE
WELSPECS
OP01    PLATFORM    14    13   1*     OIL    1*     STD   SHUT   NO     1*  /
OP02    PLATFORM    35    96   1*     OIL    1*     STD   SHUT   NO     1*  /
/
--
--      WELL CONNECTION DATA
--
-- WELL  --- LOCATION ---  OPEN  SAT    CONN  WELL  KH    SKIN  D    DIR
-- NAME  II  JJ  K1  K2  SHUT  TAB   FACT  DIA  FACT  FACT  FACT  PEN
COMPDAT
OP01    1*  1*  20  56  OPEN  1*    1*    0.708 1*    0.0   1*    'Z' /
OP01    1*  1*  75 100  SHUT  1*    1*    0.708 1*    0.0   1*    'Z' /
OP02    35  96  75 100  OPEN  1*    1*    0.708 1*    0.0   1*    'Z' /
```

Well OP01 has two sets of connections; the first one connects grid cells (14, 13, 20) to (14, 13, 56) to the well and is open to flow and the second connecting grid cells (14, 13, 75) to (14, 13, 100) is shut. Well OP02 has only one open set of connections from cells (35, 96, 75) to cells (35, 96, 100).

12.3.25 COMPDATL – DEFINE WELL CONNECTIONS TO A LGR GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [COMPDATL](#) keyword defines how a well in a Local Grid Refinement (“LGR”) is connected to the reservoir by declaring the [LGR](#) and defining or modifying existing well connections. Ideally the connections should be declared in the correct sequence, starting with the connection nearest the well head and then working along the wellbore towards the bottom or toe of the well, however this may not be possible or convenient, for example when connections are added or removed from a well during the simulation (see the [COMPORD](#) keyword in the [SCHEDULE](#) section for options regarding connection ordering).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the well LGR connection data are being defined. Note that the well name (LGRNAME) must have been declared previously using the WELSPECL keyword in the SCHEDULE section, otherwise an error may occur. If defaulted with I* the LGR on the WELSPECL keyword will be utilized.			Defined
3	I	A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction. If set to zero or defaulted with I* the location is taken from the wellhead location I-direction value on the WELSPECS keyword in the SCHEDULE section.			0
4	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction. If set to zero or defaulted with I* the location is taken from the wellhead location J-direction value on the WELSPECS keyword in the SCHEDULE section.			0
5	K1	A positive integer greater than or equal to one and less than or equal to NZ that defines the UPPER connection location in the K-direction.			None
6	K2	A positive integer greater than or equal to K1 and less than or equal to NZ that defines the LOWER connection location in the K-direction.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	STATUS	A character string of length four that defines the connections' operational status, STATUS should be set to one of the following character strings: 1) OPEN: the connections are open to flow. 2) SHUT: the connections are closed to flow (shut-in). 3) AUTO: the connection are initially closed, but may be opened automatically if an economic limit is violated. Currently this option is not supported by OPM Flow			OPEN
8	SATNUM	An integer greater than or equal to zero and less than NTSFUN as declared on the TABDIMS keyword in the RUNSPEC, that defines the saturation table number to be used for flow between the reservoir grid block and the well connections. If SATNUM is set to zero or defaulted with I* then: 1) The saturation table allocated to the grid block that the connections are located within are used. 2) If the hysteresis option has been activated via the HYSTER variable on the SATOPTS keyword in the RUNSPEC section, then both the imbibition and drainage saturation tables allocated to the grid block that the connections are located within are used. The imbibition table allocation can be changed by the COMPIMB keyword in the RUNSPEC section, provided it is entered after the COMPDAT keyword.			0
9	CONFACT	A real value greater than or equal to zero that defines the transmissibility connection factor between the well bore and the reservoir grid block. If set to zero or defaulted with I* then items (9) through (13) are used to calculate CONFACT.			Defined
		cPrb/day/psia 0	cPrm ³ /day/bars 0	cPrcc/hr/atm 0	
10	DW	A real positive value that defines the well bore diameter of the connections for the well. DW is used in calculating a well's productivity or injectivity index; however the value will be ignored in calculating the connections CONFACT value if CONFAC has been directly entered.			None
		feet	m	cm	
11	KH	A real value that defines the effective KH (permeability x length) for the connections. If less than or equal to zero or defaulted by I* then KH is calculated from the connected grid blocks. KH is ignored if CONFAC has been directly entered.			Calculated from connected grid blocks
		mD.ft	mD.m	mD.cm	
12	SKIN	A real value that defines the connections dimensionless skin factor. SKIN is used in calculating a well's productivity or injectivity index; however, the value will be ignored in calculating the connections CONFACT value if CONFAC has been directly entered.			0.0

No.	Name	Description			Default
		Field	Metric	Laboratory	
		dimensionless	dimensionless	dimensionless	
13	DFACT	A real value that defines the non-Darcy D factor coefficient for gas wells. This value should be defaulted with I* and the non-Darcy D factor coefficient for gas wells defined via the WDFAC keyword in the SCHEDULE section. Currently this option is not supported by OPM Flow.			I*
		day/Mscf	day/m ³	hour/sc	
14	DIRECT	A one letter character string that defines the orientation of the connections and should be set to either X, Y, or Z. The direction of connections also determines the length of the connection used to calculate the connection factor if CONFAC has not been entered directly. The default value is for a vertical connection, that is DIRECT is defaulted to Z.			Z

Notes:

- 1) The keyword is followed by up to MXCONS records as declared on the [WELLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.12: COMPDATL Keyword Description

Multiple grid block connections can be defined on one record for vertical wells by assigning different values to K1 and K2, for deviated and horizontal wells this may not be possible and therefore each grid block connection must be separately defined by using one record per connection, setting K1 equal to K2 in each record.

See also the [WEL SPECS](#) keyword to define wells, the [COMPIMB](#) to reset the imbibition relative permeability table allocation, and the [COMPORD](#) to re-order the completions along the well trajectory. In addition, the [COMPLUMP](#) keyword groups well connections together to form well completions for a well. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Note

The term well connection is used to describe individual connections from the wellbore to the reservoir grid, as opposed to well completions. A well completion is used to describe a set of connections, for example, a well may consist of several completions with each completion consisting of multiple connections.

Example

The following example defines two vertical oil wells using the [WELSPECS](#) keyword and their associated connection data.

```
--
--      WELL LGR SPECIFICATION DATA
--
--      WELL GROUP  LGR    -LOCATION- BHP    PHASE DRAIN INFLOW SHUT CROSS PVT
--      NAME NAME    NAME      I      J  DEPTH FLUID AREA  EQUA.  IN   FLOW  TABLE
WELSPECL
      OP01  PLAT  OP01LGR  14   13   1*    OIL   1*    STD   SHUT  NO   1*  /
      OP02  PLAT  OP02LGR  28   96   1*    OIL   1*    STD   SHUT  NO   1*  /
/
--
--      WELL LGR CONNECTION DATA
--
--      WELL  LGR    ---LOCATION---  OPEN  SAT CONN  WELL  KH  SKIN  D  DIR
--      NAME  NAME      II  JJ  K1  K2  SHUT  TAB FACT DIA  FACT FACT FACT PEN
COMPDATL
      OP01  OP01LGR  1*   1*   20  56  OPEN  1*  1*  0.708  1*   1*   1*  Z  /
      OP01  OP01LGR  1*   1*   75 100  SHUT  1*  1*  0.708  1*   1*   1*  Z  /
      OP02  OP02LGR  35   96   75 100  OPEN  1*  1*  0.708  1*   1*   1*  Z  /
/
```

Well OP01 has two sets of connections; the first one connects grid cells (14, 13, 20) to (14, 13, 56) to the well and is open to flow and the second connecting grid cells (14, 13, 75) to (14, 13, 100) is shut. Well OP02 has only one open connection from cells (35, 96, 75) to cells (35, 96, 100).

12.3.26
COMPDATM – DEFINE WELL CONNECTIONS TO AN AMALGAMATED LGR GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COMPDATM](#) keyword is an alias for the [COMPDATL](#) keyword. [COMPDATM](#) defines how a well in an amalgamated Local Grid Refinement (“LGR”) is connected to the reservoir by declaring the [LGR](#) and defining or modifying existing well connections. Ideally the connections should be declared in the correct sequence, starting with the connection nearest the well head and then working along the wellbore towards the bottom or toe of the well, however this may not be possible or convenient, for example when connections are added or removed from a well during the simulation (see the [COMPORD](#) keyword in the [SCHEDULE](#) section for options regarding connection ordering).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.27 COMPFLSH – ASSIGN COMPDAT DIFFERENTIAL-FLASH LIBERATION RATIOS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	----------

Description

[COMPFLSH](#) is used to assign saturated PVT differential-flash liberation ratios to individual well completions for both the oil formation volume factor and the gas-oil ratio. Only the saturated oil properties are changed by this keyword, that is only data entered via the [PVCO](#) or [PVTO](#) keywords in the [PROPS](#) section will be used in the calculation. The other fluid PVT property keywords: [PVTW](#), [PVTG](#), [DENSITY](#), etc., are not used.

The differential liberation expansion (“DLE”) experiment is designed to approximate the depletion process of an oil reservoir, and thereby provide suitable PVT data for calculating reservoir performance. This is then coupled with surface faculties via the multi-stage separator flash data. The multi-stage separator test is performed on an oil sample primarily to provide a basis for converting differential liberation data from a residual oil to a stock-tank oil basis. Occasionally, several separator tests are run to help choose separator conditions that maximize stock-tank oil production. Usually two or three stages of separation are used, with the last stage being at atmospheric pressure and near-ambient temperature (60 to 80 °F).

Note, normally the data on the [PVCO](#) and [PVTO](#) keywords should have already taken the two datasets into account and therefore this keyword should not normally be used in practice, unless there is a specific requirement to adjust the oil PVT data on a well connection basis.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.28 COMPIMB – ASSIGN IMBIBITION SATURATION TABLES TO WELL CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [COMPIMB](#) keyword assigns imbibition saturation tables to well connections. The [COMPDAT](#) keyword in the [SCHEDULE](#) section also assigns imbibition saturation tables to connections, but in this case the table number is the same as for the drainage curve. If this is not the required assignment then the [COMPIMB](#) keyword can be used to reset the imbibition saturation table number. For this to be effective the [COMPIMB](#) keyword must precede the [COMPDAT](#) keyword, otherwise it will have no effect.

The [COMPIMB](#) keyword should only be used if the hysteresis option has been activated via the [HYSTER](#) variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	I	A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction. If set to zero or defaulted with I* then all connections in the I-direction that also satisfy J, K1 and K2 criteria are assigned the IMBNUM imbibition table number.			0
3	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction. If set to zero or defaulted with I* then all connections in the J-direction that also satisfy I, K1 and K2 criteria are assigned the IMBNUM imbibition table number.			0
4	K1	A positive integer greater than or equal to one and less than or equal to NZ that defines the UPPER connection location in the K-direction. If set to zero or defaulted with I* then the upper most connection in the well is used.			0
5	K2	A positive integer greater than or equal to K1 and less than or equal to NZ that defines the LOWER connection location in the K-direction. If set to zero or defaulted with I* then the lowest most connection in the well is used.			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	IMBNUM	An integer greater than or equal to zero and less than NTSFUN as declared on the TABDIMS keyword in the RUNSPEC, that defines the imbibition saturation table number to be used for flow between the reservoir grid block and the well connections. If IMBNUM is set to zero or defaulted with I* then the inhibition saturation table allocated to the grid block that the connections are located within is used. If I, J, K1, K2 are all set to zero or defaulted to I*, then IMBNUM is allocated to all connections in the well.			0
Notes: 1) The keyword is followed by up to MXCONS records as declared on the WELLDIMS keyword in the RUNSPEC section. 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.13: COMPIMB Keyword Description

Multiple grid block connections can be defined on one record for vertical wells by assigning different values to K1 and K2, for deviated and horizontal wells this may not be possible and therefore each grid block connection must be separately defines by setting K1 equal to K2.

See also the [COMPDAT](#) keyword in the [SCHEDULE](#) section.

Example

The following example defines the connections for two vertical oil wells using the [COMPDAT](#) keyword and then re-sets the imbibition saturation functions using the [COMPIMP](#) keyword.

```
--
--      WELL CONNECTION DATA
--
-- WELL  --- LOCATION --- OPEN  SAT   CONN  WELL  KH   SKIN  D   DIR
-- NAME  II  JJ  K1  K2  SHUT  TAB   FACT  DIA  FACT FACT  FACT  PEN
COMPDAT
OP01      1*  1*  20  56  OPEN   1     1*   0.708  1*   0.0   1*   'Z' /
OP01      1*  1*  75 100  SHUT   2     1*   0.708  1*   0.0   1*   'Z' /
OP02      35  96  75 100  OPEN   1     1*   0.708  1*   0.0   1*   'Z' /
--
-- ASSIGN IMBIBITION SATURATION TABLES TO CONNECTIONS
--
-- WELL      ---LOCATION---  SAT
-- NAME      II  JJ  K1  K2  TAB
COMPIMP
OP01          1*  1*  20  56   11
OP01          1*  1*  75 100   12
OP02          1*  1*  1*  1*   11
/
```

Well OP01 has two sets of [COMPIMP](#) records to overwrite the imbibition saturation tables, one for connections (14, 13, 20) to (14, 13, 56) resetting the imbibition saturation table number from one to 11 and one for connections (14, 13, 75) to (14, 13, 100) that resets the imbibition table number from 2 to 12. Well OP02 has only one connection from cells (35, 96, 75) to cells (35, 96, 100), so all the default values for I, J, K1, and K2 can be used to set the imbibition table numbers from 2 to 11. Note in all cases the drainage saturation table retains the value as specified by the [COMPDAT](#) keyword, that is one, two and one.

12.3.29 COMPINJK – ASSIGN INJECTION WELL RELATIVE PERMEABILITY VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [COMPINJK](#) keyword assigns injection well relative permeability values to well connections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.30 COMPLMPL – ASSIGN WELL LGR CONNECTIONS TO COMPLETIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **COMPLMPL** keyword assigns well connections in a **LGR**, as defined by the **COMPDATL** keyword in the **SCHEDULE** section, to completion intervals. This “lumping” of the connections to various completion intervals allows automatic workovers and economic criteria to be applied to the completions (that is a set of connections) as opposed to the connections. This allows for a more realistic approach for workovers operations.

For example, if the water cut criteria for working over a well was set to 95%, and the average grid block connection thickness was one meter, then once a well’s water cut reached 95% the worst offending one meter connection would be shut-in. If the well’s actual perforation interval was 10 meters and the 10 connections were lumped as one completion, then when the water cut limit of 95% is reach, the completion would be shut-in, that is all of the 10 connections within the completion would be shut-in.

As the keyword is used to lump connections into a completions, the simulator adds together the contribution from all connections in the completion and uses the total values to test the economic limits. Note that a connection can only belong to one completion. In addition, completions can be used instead of connections in the **WELOPEN** and **WPIMULT** keywords if the completions have been defined by **COMPLUMP** for a well.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the well LGR connection data are being defined. Note that the well name (LGRNAME) must have been declared previously using the WELSPECL keyword in the SCHEDULE section, otherwise an error may occur. If defaulted with I* the LGR on the WELSPECL keyword will be utilized.			Defined
3	I	A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction. If set to zero or defaulted with I* then all connections in the I-direction that also satisfy J, K1 and K2 criteria are assigned the ICOMP completion number.			0
4	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction. If set to zero or defaulted with I* then all connections in the J-direction that also satisfy I, K1 and K2 criteria are assigned the ICOMP completion number.			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	K1	A positive integer greater than or equal to one and less than or equal to NZ that defines the UPPER connection location in the K-direction. If set to zero or defaulted with I* then the upper most connection in the well is used.			0
6	K2	A positive integer greater than or equal to K1 and less than or equal to NZ that defines the LOWER connection location in the K-direction. If set to zero or defaulted with I* then the low most connection in the well is used.			0
7	ICOMP	An integer greater than or equal to one and less than or equal to MXCONS as defined on the WELLDIMS keyword in the RUNSPEC section, that defines the completion number of the currently defined set of connections. If I, J, K1, K2 are all set to zero or defaulted to I*, then all connections in the well have the same completion number of ICOMP.			None

Notes:

- 1) The keyword is followed by any number of records.
- 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.14: COMPLUPL Keyword Description

Multiple grid block connections can be defined on one record for vertical wells by assigning different values to K1 and K2, for deviated and horizontal wells this may not be possible and therefore each grid block connection must be separately defined by setting K1 equal to K2.

See also the [COMPDATL](#) keyword in the [SCHEDULE](#) section.

Example

The following example defines the connections for two vertical oil wells using the [COMPDATL](#) keyword and the re-allocation of the connections to completions intervals using the [COMPLMPL](#) keyword.

```

--
--      WELL CONNECTION DATA FOR LGR WELLS
--
--      WELL      LGR      --- LOCATION ---  OPEN  SAT   CONN   WELL   D      DIR
--      NAME      NAME      II  JJ  K1  K2   SHUT  TAB   FACT   DIA   FACT  PEN
COMPDATL
      OP01      OP01LGR    14  13  20  56   OPEN  1*    1*    0.708  3*    Z /
      OP01      OP01LGR    14  13  75 100   SHUT  1*    1*    0.708  3*    Z /
      OP02      OP02LGR    35  96  75 100   OPEN  1*    1*    0.708  3*    Z /
/
--
--      ASSIGN WELL CONNECTIONS TO COMPLETIONS
--
--      WELL      LGR      --- LOCATION ---  COMPL
--      NAME      NAME      II  JJ  K1  K2   NO.
COMPLMPL
      OP01      OP01LGR    1*  1*  20  56    1          / COMPLETION NO. 01
      OP01      OP01LGR    1*  1*  75 100    2          / COMPLETION NO. 02
      OP02      OP02LGR    1*  1*  75  85    1          / COMPLETION NO. 01
      OP02      OP21LGR    1*  1*  86 100    2          / COMPLETION NO. 02
/

```

Here the well OP01 connections (14, 13, 20) to (14, 13, 56) are assigned to completion number one and connections (14, 13, 75) to (14, 13, 100) are assigned to completion number two. Well OP02 has only one set of connection data from cells (35, 96, 75) to cells (35, 96, 100), but they have split into two separate completion intervals, with connections (35, 96, 75) to (35, 96, 85) assigned to completion interval number one and (35, 96, 86) to (35, 96, 100) to completion number two.

12.3.31 COMPLUMP – ASSIGN WELL CONNECTIONS TO COMPLETIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [COMPLUMP](#) keyword assigns connections, as defined by the [COMPDAT](#) keyword in the [SCHEDULE](#) section, to completion intervals. This “lumping” of the connections to various completion intervals allows automatic workovers and economic criteria to be applied to the completions (that is a set of connections) as opposed to the connections. This allows for a more realistic approach for workovers operations.

For example, if the water cut criteria for working over a well was set to 95%, and the average grid block connection thickness was one meter, then once a well’s water cut reached 95% the worst offending one meter connection would be shut-in. If the well’s actual perforation interval was 10 meters and the 10 connections were lumped as one completion, then when the water cut limit of 95% is reach, the completion would be shut-in, that is all of the 10 connections within the completion would be shut-in.

As the keyword is used to lump connections into a completions, the simulator adds together the contribution from all connections in the completion and uses the total values to test the economic limits. Note that a connection can only belong to one completion. In addition, completions can be used instead of connections in the [WELOPEN](#) and [WPIMULT](#) keywords if the completions have been defined by [COMPLUMP](#) for a well.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	I	A positive integer greater than or equal to zero and less than or equal to NX that defines the connection location in the I-direction. If set to zero or defaulted with I* then all connections in the I-direction that also satisfy J, K1 and K2 criteria are assigned the ICOMP completion number.			0
3	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction. If set to zero or defaulted with I* then all connections in the J-direction that also satisfy I, K1 and K2 criteria are assigned the ICOMP completion number.			0
4	K1	A positive integer greater than or equal to one and less than or equal to NZ that defines the UPPER connection location in the K-direction. If set to zero or defaulted with I* then the uppermost connection in the well is used.			0
5	K2	A positive integer greater than or equal to K1 and less than or equal to NZ that defines the LOWER connection location in the K-direction. If set to zero or defaulted with I* then the lowermost connection in the well is used.			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	ICOMP	An integer greater than or equal to one and less than or equal to MXCONS as defined on the WELLDIMS keyword in the RUNSPEC section, that defines the completion number of the currently defined set of connections. If I, J, K1, K2 are all set to zero or defaulted to 1*, then all connections in the well have the same completion number of ICOMP.			None
Notes: 1) The keyword is followed by any number of records. 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.15: COMPLUMP Keyword Description

Multiple grid block connections can be defined on one record for vertical wells by assigning different values to K1 and K2, for deviated and horizontal wells this may not be possible and therefore each grid block connection must be separately defined by setting K1 equal to K2. See also the [COMPDAT](#) keyword in the [SCHEDULE](#) section.

Example

The following example defines the connections for two vertical oil wells using the [COMPDAT](#) keyword and the re-allocation of the connections to completions intervals using the [COMPLUMP](#) keyword.

```
--
--      WELL CONNECTION DATA
--
-- WELL  --- LOCATION --- OPEN  SAT   CONN  WELL  KH   SKIN  D   DIR
-- NAME  II  JJ  K1  K2  SHUT  TAB   FACT  DIA  FACT  FACT  FACT  PEN
COMPDAT
OP01      14  13  20  56  OPEN  1*   1*   0.708  1*   0.0   1*   'Z' /
OP01      14  13  75 100  SHUT  1*   1*   0.708  1*   0.0   1*   'Z' /
OP02      35  96  75 100  OPEN  1*   1*   0.708  1*   0.0   1*   'Z' /
/
--
--      ASSIGN WELL CONNECTIONS TO COMPLETIONS
--
-- WELL  --- LOCATION --- COMPL
-- NAME  II  JJ  K1  K2  NO.
COMPLUMP
OP01      1*  1*  20  56  1                               / COMPLETION NO. 01
OP01      1*  1*  75 100  2                               / COMPLETION NO. 02
OP02      1*  1*  75  85  1                               / COMPLETION NO. 01
OP02      1*  1*  86 100  2                               / COMPLETION NO. 02
/
```

Here the well OP01 connections (14, 13, 20) to (14, 13, 56) are assigned to completion number one and connections (14, 13, 75) to (14, 13, 100) are assigned to completion number two. Well OP02 has only one set of connection data from cells (35, 96, 75) to cells (35, 96, 100), but they have been split into two separate completion intervals, with connections (35, 96, 75) to (35, 96, 85) assigned to completion interval number one and (35, 96, 86) to (35, 96, 100) assigned to completion number two.

12.3.32 COMPOFF – DEACTIVATE NETWORK AUTOMATIC COMPRESSORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [COMPOFF](#) keyword deactivates network automatic compressors defined via the [GASFCOMP](#) keyword in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.33 COMPORD – DEFINE WELL CONNECTION ORDERING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [COMPORD](#) keyword defines how the well connection data entered on the [COMPDAT](#) keyword in the [SCHEDULE](#) section are to be ordered for a well.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	<i>COMPORD</i>	A character string that defines the method for ordering the well connections given on the <i>COMPDAT</i> keyword, and should be set to <i>DEPTH</i> , <i>INPUT</i> , or <i>TRACK</i> . 1) <i>DEPTH</i> : The connections are ordered by a connection's true vertical depth from the shallowest to the deepest. If multiple connections are at the same depth then these connections are sub ordered by the sequence they were entered on the <i>COMPDAT</i> keyword. 2) <i>INPUT</i> : This option results in the connections being ordered in the same sequence as entered via the <i>COMPDAT</i> keyword. In this case the connections should be declared in the correct sequence, starting with the connection nearest the well head and then working along the wellbore towards the bottom or toe of the well. 3) <i>TRACK</i> : This option enables OPM Flow to trace the well connections through the grid to obtain the correct order for the connections. If the supplied <i>COMPDAT</i> indicates the well is vertical (via the <i>DIRECT</i> variable being equal to Z on the <i>COMPDAT</i> keyword) then the <i>DEPTH</i> option will be applied instead. All options are now supported by OPM Flow.			TRACK

Notes:

- 1) The keyword is followed by any number of records.
- 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.16: COMPORD Keyword Description

See also the [COMPDAT](#) keyword in the [SCHEDULE](#) section.

Note

If visual inspection of the well trajectories in the model indicate problematic or unrealistic well connections, the options on this keyword may be useful in correcting the issue.

Example

The following example defines the connections for two vertical oil wells using the `COMPDAT` keyword and the `COMPORD` to defined the connection ordering for the wells.

```
--
--      WELL CONNECTION DATA
--
-- WELL  --- LOCATION --- OPEN  SAT   CONN  WELL  KH   SKIN  D   DIR
-- NAME   II  JJ  K1  K2   SHUT  TAB   FACT  DIA  FACT FACT  FACT  PEN
COMPDAT
OP01      1*  1*  20  56   OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
OP01      1*  1*  75 100   SHUT  1*   1*   0.708 1*   0.0  1*   'Z' /
OP02      35  96  75 100   OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
--
--      DEFINE WELL CONNECTION ORDERING
--
-- WELL  COMPL
-- NAME  ORDER
COMPORD
OP01    DEPTH
OP02    DEPTH
/
```

The `DEPTH` option has been chosen because both wells are vertical. Also one could use the following format instead for the `COMPORD`:

```
--
--      DEFINE WELL CONNECTION ORDERING
--
-- WELL  COMPL
-- NAME  ORDER
COMPORD
*      DEPTH
/
```

as both wells should utilize the `DEPTH` option. This version would set all wells in the model to `DEPTH` connection ordering.

12.3.34 COMPRIV – DEFINE GRID CELL CONNECTIONS TO A RIVER

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [COMPRIV](#) keyword defines grid cell connections to a river, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.35 COMPRP – RE-SCALE FLUID SATURATIONS OF WELL CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The COPMPRP keyword re-scales the fluid saturations of a well's connection to the grid block.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.36 COMPRL – RE-SCALE FLUID SATURATIONS OF WELL LGR CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The COMPRL keyword re-scales the fluid saturations of a well’s connection to an [LGR](#) grid block, for when the Local Grid Refinement (“LGR”) option has been activated by the [LGR](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.37 COMPSEGL – DEFINE WELL CONNECTIONS FOR MULTI-SEGMENT WELLS IN A LGR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The COMSEGL keyword defines how a multi-segment well is connected to the reservoir by defining or modifying existing well connections in an [LGR](#). Note that well must have been previously define by the [WELSPECL](#) keyword in the [SCHEDULE](#) section and the well connections must have been previously defined via the [COMPDATL](#) keyword in the [SCHEDULE](#) section. The [COMPSEGL](#) keyword should be repeated for each multi-segment well in the model.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECL keyword in the SCHEDULE section, otherwise an error may occur.			None
1-2	/	Record terminated by a "/"			Not Applicable
2-1	LGRNAME	A character string of up to eight characters in length that defines the name of the local grid refinement for which the well is assigned to.			None
2-2	I	A positive integer greater than or equal to one and less than or equal to NX that defines the connection location in the I-direction.			None
2-3	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the connection location in the J-direction.			None
2-4	K	A positive integer greater than or equal to zero and less than or equal to NZ that defines the connection location in the K-direction.			None
2-5	IBRANCH	A positive integer greater than or equal to one and less than or equal to MXBRAN on WSEGDIMS keyword in the RUNSPEC section that defines the branch number of the defined I, J and K connection.			None
2.6	DEPTH1	DEPTH1 is a real positive value that defines the length of the tubing from the tubing head or wellhead at the surface to the <u>start</u> of the connection in the I, J, K cell.			None
		feet	m	cm	
2-7	DEPTH2	DEPTH2 is a real positive value that defines the length of the tubing from the tubing head or wellhead at the surface to the end of the connection in the I, J, K cell.			None
		feet	m	cm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-8	DIRECT	A one letter character string that defines the orientation of the connections and should be set to either X, Y, or Z. The direction of connections also determines the length of the connection The default value is for a vertical connection, that is DIRECT is defaulted to Z.			Z
2-9	IEND	IEND is positive or negative integer, that is not equal to zero that is set to one of the following: 1) a value between -NX and +NX that is not equal to zero that defines the last connection location in the I-direction, 2) a value between -NY and +NY that is not equal to zero that defines the last connection location in the J-direction, or 3) a value between -NZ and +NZ that is not equal to zero that defines the last connection location in the K-direction, that defines the end of the range of the connections depending on the value of DIRECT. For example, if DIRECT is equal to Y or J then the IEND will be associated with the J-direction. The value may be positive or negative but must be calculated to remain within the grid. For example for NY is set 100 on the DIMENS keyword in the RUNSPEC section and J on this record set to 50, then IEND most range between -49 to +50. Currently this option is not supported by OPM Flow.			None
2-10	DEPTH3	DEPTH3 is a real positive value that defines the datum depth for this set of connection, normally taken as the mid-point of the perforations associated with this set of connections. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-11	LENGTH	LENGTH is a real positive value that defines the length of the well for this set of completions that is used in thermal calculations.. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-12	ISEG	A real positive values equal to or greater than zero that defines the coordinate in the x-direction of the nodal point of this segment that is used for display purposes only. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-13	/	Record terminated by a "/"			Not Applicable

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) Each multi-segment wells must be defined by a separate <i>COMPSEGL</i> keyword that consists of two records, with entries 1-1 to 1-2 representing record one items and 2-1 to 2-13 representing record number two items in the “No.” column in this table. 2) Record number two of the keyword, items 2-1 to 2-13 is followed by up to MXCONS records as declared on the <i>WSEGDIMS</i> keyword in the <i>RUNSPEC</i> section. 3) Each of the records are terminated by a “/” and is explicitly shown in the above rows and the keyword should be terminated by a “/”.					

Table 12.17: COMPSEGL Keyword Description

The total number of wells and completions should be defined via the WELLSDIMS keyword and the number of multi-segment wells and completions should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section.

See also the [WELSPECL](#) keyword to define wells in an [LGR](#), the [COMPDATL](#) keyword to define the well connections for both ordinary wells and multi-segment wells with an [LGR](#), and the [COMPSEGS](#) keyword to define a multi-segment connections in the global grid. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the completions for two oil producing segment oil wells (OP01 and OP02) using the [COMPSEGS](#) keywords.

```
--      COMPLETION SEGMENT SPECIFICATION DATA
--
-- WELL
-- NAME
COMPSEGS
OP01
--      LGR  --LOCATION--  BRAN  TUBING  NODAL  DIR  LOC  MID  COMP  ISEG
--      NAME  II  JJ  K1  NO  LENGTH  DEPTH  PEN  I, J, K  PERFS  LENGTH
LGR01 10  10  1  1  2512.5  2525.0
LGR01 10  10  2  1  2525.0  2550.0
LGR01 10  10  3  1  2550.0  2575.0
LGR01 10  10  4  1  2575.0  2600.0
LGR01 10  10  5  1  2600.0  2625.0
LGR01 10  10  6  1  2625.0  2650.0
/
LGR01 9  10  2  2  2637.5  2837.5
LGR01 8  10  2  2  2837.5  3037.5
LGR01 7  10  2  2  3037.5  3237.5
LGR01 6  10  2  2  3237.5  3437.5
LGR01 5  10  2  2  3437.5  3637.5
/
```

Note that the [COMPDATL](#) keyword in the [SCHEDULE](#) section must also be defines for this well.

12.3.38 COMPSEGS – DEFINE WELL CONNECTIONS FOR MULTI-SEGMENT WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines how a multi-segment well is connected to the reservoir by allocating each well connection to a well segment. Note that the well must have previously been defined as a multi-segment well using the [WELSEGS](#) keyword in the [SCHEDULE](#) section and the well connections must have previously been defined via the [COMPDAT](#) keyword in the [SCHEDULE](#) section.

The [COMPSEGS](#) keyword should be repeated for each multi-segment well in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
I-2	/	Record terminated by a “/”			Not Applicable
2-1	I	A positive integer greater than or equal to one and less than or equal to NX that defines the connection location in the I-direction.			None
2-2	J	A positive integer greater than or equal to one and less than or equal to NY that defines the connection location in the J-direction.			None
2-3	K	A positive integer greater than or equal to one and less than or equal to NZ that defines the connection location in the K-direction.			None
2-4	IBRANCH	A positive integer greater than or equal to one and less than or equal to MXBRAN on WSEGDIMS keyword in the RUNSPEC section that defines the branch number of the defined I, J and K connection.			None
2-5	LENGTH1	A real positive value that defines the length of the tubing from the tubing reference point (for example Measure Depth Relative to Kelly Bushing, MDRKB) to the <u>start</u> of the connection in the I, J, K cell.			None
		feet	m	cm	
2-6	LENGTH2	A real positive value that defines the length of the tubing from the tubing reference point to the <u>end</u> of the connection in the I, J, K cell.			None
		feet	m	cm	
2-7	DIRECT	A one letter character string that defines the orientation of the connections and should be set to either X, Y or Z, or either I, J or K. The direction of connections may also determine the length of the connection. Currently this option is not supported by OPM Flow.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-8	IJKEND	A positive integer that is set to one of the following depending on the orientation of the connections (DIRECT): 1) X or I: A value between 1 and NX that defines the last connection location in the I-direction, 2) Y or J: A value between 1 and NY that defines the last connection location in the J-direction, or 3) Z or K: A value between 1 and NZ that defines the last connection location in the K-direction. For example, if DIRECT is equal to Y or J then the IJKEND will be associated with the J-direction. The value may be less than or greater than Y or J but must be between 1 and NY. Currently this option is not supported by OPM Flow.			None
2-9	DEPTH	A real positive value that defines the datum depth for this set of connections, normally taken as the mid-point of the perforations associated with this set of connections. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-10	THLEN	A real positive value that defines the length of the well in the connection cells for this set of completions that is used in thermal calculations. If this value is defaulted then the thickness of the cell in the direction of the well orientation (DIRECT) is used. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-11	ISEG	A real positive value that defines the segment number to assign to this range of connections. If this value is defaulted then each connection is assigned to the segment with the nodal point nearest to the centre of the connection. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-12	/	Record terminated by a "/"			Not Applicable

Notes:

- Each multi-segment wells must be defined by a separate **COMPSEGS** keyword that consists of two records, with entries 1-1 to 1-2 representing record one items and 2-1 to 2-12 representing record number two items in the "No." column in this table.
- Record number two of the keyword, items 2-1 to 2-12 can be repeated up to a maximum of MXCONS times as declared on the **WELLDIMS** keyword in the **RUNSPEC** section.
- Each of the records are terminated by a "/" and is explicitly shown in the above rows and the keyword should be terminated by a "/"

Table 12.18: COMPSEGS Keyword Description

The maximum number of wells and completions should be defined via the [WELLDIMS](#) keyword and the maximum number of multi-segment wells, segments and branches should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section.

See also the [WELSPECS](#) keyword to define wells, the [COMPDAT](#) keyword to define the well connections for both standard wells and multi-segment wells, and the [COMPSEGL](#) keyword to define multi-segment connections in a [LGR](#). All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the completions for two multi-segment oil production wells (OP01 and OP02) using the [COMPSEGS](#) keyword.

```
--          COMPLETION SEGMENT SPECIFICATION DATA
--
-- WELL
-- NAME
COMPSEGS
OP01
--          --LOCATION--  BRAN  START  END      DIR  LOC   MID   COMP   ISEG
--          II  JJ  K1   NO    LENGTH LENGTH PEN  I, J, K  PERFS  LENGTH
--          10  10  1    1     2512.5 2525.0
--          10  10  2    1     2525.0 2550.0
--          10  10  3    1     2550.0 2575.0
--          10  10  4    1     2575.0 2600.0
--          10  10  5    1     2600.0 2625.0
--          10  10  6    1     2625.0 2650.0
--
--          9  10  2    2     2637.5 2837.5
--          8  10  2    2     2837.5 3037.5
--          7  10  2    2     3037.5 3237.5
--          6  10  2    2     3237.5 3437.5
--          5  10  2    2     3437.5 3637.5
--
--
--          COMPLETION SEGMENT SPECIFICATION DATA
--
-- WELL
-- NAME
COMPSEGS
OP02
--          --LOCATION--  BRAN  START  END      DIR  LOC   MID   COMP   ISEG
--          II  JJ  K1   NO    LENGTH LENGTH PEN  I, J, K  PERFS  LENGTH
--          1  9  3    1     2662.5 2862.5
--          1  8  3    1     2862.5 3062.5
--          1  7  3    1     3062.5 3262.5
--          1  6  3    1     3262.5 3462.5
--          1  5  3    1     3462.5 3662.5
--
--          2  10  5    2     2712.5 2912.5
--          2  10  5    2     2912.5 3112.5
--          4  10  5    2     3112.5 3312.5
--          5  10  5    2     3312.5 3512.5
--          6  10  5    2     3512.5 3712.5
--
--          1  9  6    3     2737.5 2937.5
--          1  8  6    3     2937.5 3137.5
--          1  7  6    3     3137.5 3337.5
--          1  6  6    3     3337.5 3537.5
--          1  5  6    3     3537.5 3737.5
--
--
```

Note that the [COMPDAT](#) keyword in the [SCHEDULE](#) section must also be defined for these two wells.

12.3.39 COMPTRAJ – DEFINE WELL TRAJECTORY CONNECTIONS TO THE GRID

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COMPTRAJ](#) keyword defines how a well that has been declared as a trajectory well, using the [WELTRAJ](#) keyword in the [SCHEDULE](#) section, is connected to the reservoir model by defining or modifying existing well perforation depths. The keyword can only be used for wells defined by the [WELTRAJ](#) keyword, and [WELTRAJ](#) defined wells must use the [COMPTRAJ](#) keyword to define the connections to the grid, that is one cannot use [COMPDAT](#) for these type of wells.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	IBRANCH	A positive integer greater than or equal to one and less than or equal to MXBRAN on WSEGDIMS keyword in the RUNSPEC section that defines the branch number of a segment. All segments on the main stem must have IBRANCH set to one and lateral branches should have values between two and MXSEGS on the WSEGDIMS keyword in the RUNSPEC section. Only the default value of one is currently supported, that is only the main branch of a multi-segment well is supported, or a single trajectory for a conventional well.			1
3	TOP	A real positive value that defines the depth of the top of the perforation interval, and should be less than the value entered for the BOT parameter (the bottom perforation depth).			None
		feet	m	cm	
4	BOT	A real positive value that defines the depth of the base of the perforation interval, and should be greater than the value entered for the TOP parameter (the top perforation interval).			None
		feet	m	cm	
5	REF	REF is a defined character string that defines the reference depth type for TOP and BOT, and should be set to either: 1) MD for the depths referencing Measured Depth, or 2) TVD for depths referencing True Vertical Depth. Only measured depth is currently supported, that is MD.			MD

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	ICOMP	An integer greater than or equal to one and less than or equal to MXCONS as defined on the WELLDIMS keyword in the RUNSPEC section, that defines the completion number of the currently defined perforation interval (connection interval). If defaulted with I*, then ICOMP is set equal to one.			I
7	STATUS	A defined character string of length four that defines the connections' operational status within the perforation interval, STATUS should be set to one of the following character strings: 1) OPEN: the connections are open to flow. 2) SHUT: the connections are closed to flow (shut-in). 3) AUTO: the connection are initially closed, but may be opened automatically if an economic limit is violated. Currently this option is not supported by OPM Flow.			OPEN
8	SATNUM	An integer greater than or equal to zero and less than NTSFUN as declared on the TABDIMS keyword in the RUNSPEC , that defines the saturation table number to be used for flow between the reservoir grid block and the well connections. If SATNUM is set to zero or defaulted with I* then: 1) The saturation table allocated to the grid block that the connections are located within are used. 2) If the hysteresis option has been activated via the HYSSTER variable on the SATOPTS keyword in the RUNSPEC section, then both the imbibition and drainage saturation tables allocated to the grid block that the connections are located within are used. The imbibition table allocation can be changed by the COMPIMB keyword in the RUNSPEC section, provided it is entered after the COMPTRAJ keyword.			0
9	CONFACT	A real value greater than or equal to zero that defines the transmissibility connection factor between the well bore and the reservoir grid block. If set to zero or defaulted with I* then items (10) through (13) are used to calculate CONFACT.			Defined
		cP.rb/day/psia 0	cP.rm ³ /day/bars 0	cP.rcc/hr/atm 0	
10	DW	A real positive value that defines the well bore diameter of the connections for the well. DW is used in calculating a well's productivity or injectivity index; however the value will be ignored in calculating the connections CONFACT value if CONFAC has been directly entered.			None
		feet	m	cm	
11	KH	A real value that defines the effective KH (permeability x length) for the connections within the perforation interval. If less than or equal to zero, or defaulted by I*, then KH is calculated from the connected grid blocks. KH is ignored if CONFACT has been directly entered.			Calculated from connected grid blocks

No.	Name	Description			Default
		Field	Metric	Laboratory	
		mD.ft	mD.m	mD.cm	
12	SKIN	A real value that defines the connections dimensionless skin factor. SKIN is used in calculating a well's productivity or injectivity index; however, the value will be ignored in calculating the connections CONFACT value if CONFACT has been directly entered.			0.0
		dimensionless	dimensionless	dimensionless	
13	DFACT	A real value that defines the non-Darcy D factor coefficient for gas wells. This value should be defaulted with I* and the non-Darcy D factor coefficient for gas wells defined via the WDFAC keyword in the SCHEDULE section. Currently this option is not supported by OPM Flow.			I*
		day/Mscf	day/m ³	hour/sc	

Notes:

1)

Note unlike the [COMPDAT](#) keyword, there is no direction parameter, [COMPDAT](#)(DIRECT), for [COMPTRAJ](#), as the wellbore direction is implicit in the well trajectory defined by the [WELTRAJ](#) keyword in the [SCHEDULE](#) section.

2)

The keyword is followed by up to MXCONS records as declared on the [WELLDIMS](#) keyword in the [RUNSPEC](#) section.

3)

Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.19: COMPTRAJ Keyword Description

Using the [WELTRAJ](#) and [COMPTRAJ](#) keywords to define wells and how they are connected to grid, offers several advantages compared to the conventional approach based on the (I, J, K) co-ordinates of the grid. The approach allows for the wells to be independent of the grid, which is particularly useful when running ensemble cases, as the well connections are no longer required to be re-calculated for each ensemble case. In addition, quality control of the model is improved by using consistent perforation data in both the static and dynamic models.

Note

The term well connection is used to describe individual connections from the wellbore to the reservoir grid, as opposed to well completions. A well completion is used to describe a set of connections, for example, a well may consist of several completions with each completion consisting of multiple connections.

For wells defined using the [WELTRAJ](#) and [COMPTRAJ](#) keywords, the [WELTRAJ](#) keyword defines the trajectory of the well within the model, and the [COMPTRAJ](#) defines the perforation intervals in the well. A perforation interval will automatically generate various well connections to the grid, and in addition multiple perforation intervals may be grouped into a completion.

Example

The following example defines two trajectory wells oil wells, OP01 and OP02, using the [WELTRAJ](#) keyword, together with their perforations using the [COMPTRAJ](#) keyword.

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME        I      J  DEPTH  FLUID   AREA   EQUANS  SHUT  FLOW   TABLE
WELSPecs
OP01    PLATFORM    1*    1*    1*      OIL    1*      STD    SHUT    NO    1*    /
OP02    PLATFORM    1*    1*    1*      OIL    1*      STD    SHUT    NO    1*    /
/

--
--      WELL TRAJECTORY DATA
--
-- WELL  BRAN  XCORD          YCORD          TVDSS      MD
-- NAME  NO    DEPTH          DEPTH          DEPTH      DEPTH
-----
WELTRAJ
OP01    1*     2.805445e+06  3.602948e+06  -100.000000  0.0    /
OP01    1*     2.805445e+06  3.602948e+06   877.000000  977.0   /
OP01    1*     2.805445e+06  3.602948e+06   957.995024  1058.0  /
OP01    1*     2.805444e+06  3.602946e+06  1051.976081  1152.0  /
.....
OP02    1*     2.810828e+06  3.604507e+06   9371.792711  11418.0 /
OP02    1*     2.810885e+06  3.604525e+06   9443.657000  11511.0 /
OP02    1*     2.810952e+06  3.604546e+06   9531.966162  11624.0 /
OP02    1*     2.810973e+06  3.604553e+06   9560.411742  11660.0 /
/

--
--
--      WELL TRAJECTORY CONNECTION DATA
--
-- WELL  BRAN  -- PERFORATION -- COMPL  OPEN  SAT  CONN  WELL  KH  SKIN  D
-- NAME  NO.   TOP    BOT    REF  NO.   SHUT  TAB  FACT DIA  FACT FACT FACT
COMPTRAJ
OP01    1*     8230   8244   MD    1     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP01    1*     8352   8380   MD    1     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP01    1*     9070   9100   MD    1     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP01    1*     9220   9250   MD    2     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP01    1*     9266   9280   MD    2     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP01    1*     9693   9703   MD    3     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP01    1*     9940   9974   MD    3     SHUT  1*  1*    0.708  1*    0.0  1*  /
OP02    1*     9979   9985   TVD    1*    SHUT  1*  1*    0.708  1*    0.0  1*  /
OP02    1*    10173  10183  TVD    1*    SHUT  1*  1*    0.708  1*    0.0  1*  /
OP02    1*    10190  10204  TVD    1*    SHUT  1*  1*    0.708  1*    0.0  1*  /
OP02    1*    10327  10333  TVD    1*    SHUT  1*  1*    0.708  1*    0.0  1*  /
OP02    1*    10339  10345  TVD    1*    SHUT  1*  1*    0.708  1*    0.0  1*  /
OP02    1*    11528  11538  TVD    1*    SHUT  1*  1*    0.708  1*    0.0  1*  /
/
```

Well OP01 has eight perforation intervals, with the intervals one to three grouped into one completion, perforation intervals four to five grouped into completion number two, and finally the bottom three perforations are grouped into completion number three. In contrast, OP02 has six perforated intervals with their completion interval defaulted to one.

12.3.40 COMPVE – RE-DEFINE WELL CONNECTION DEPTHS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [COMPVE](#) keyword is used to re-define the well connection depths in the global grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.41 COMPVEL – RE-DEFINE WELL LGR CONNECTION DEPTHS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *COMPVEL* keyword is used to re-define the well connection depths in a Local Grid Refinement (“LGR”) grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.42 CPIFACT – DEFINE WELL CONNECTION TRANSMISSIBILITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *CPIFACT* keyword is used to define well connection transmissibility multipliers for well connections to the global grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.43 CPIFACTL – DEFINE WELL CONNECTION TRANSMISSIBILITY MULTIPLIERS IN A LGR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [CPIFACT](#) keyword is used to define well Local Grid Refinement (“LGR”) connection transmissibility multipliers for well connections to a [LGR](#), for when the [LGR](#) option has been invoked by the [LGR](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

12.3.44 CSKIN – RE-DEFINE WELL CONNECTION SKIN FACTORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [CSKIN](#), is used to re-define a well's connection skin factors and as such will result in the well's connection transmissibility factors being updated accordingly. The well connections must already be defined using the [COMPDAT](#) keyword in the [SCHEDULE](#) section before this keyword can be used.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection skin is being re-defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	I	A positive integer greater than or equal to one and less than or equal to NX that defines the connection location in the I-direction. If set to zero or defaulted with I* then connections in any I-direction location will be modified.			0
3	J	A positive integer greater than or equal to one and less than or equal to NY that defines the connection location in the J-direction. If set to zero or defaulted with I* then connections in any J-direction location will be modified.			0
4	KI	A positive integer greater than or equal to one and less than or equal to K2 and NZ that defines the UPPER connection location in the K-direction. If set to zero or defaulted with I* then this will be taken from the top connection of the well.			0
5	K2	A positive integer greater than or equal to KI and less than or equal to NZ that defines the LOWER connection location in the K-direction. If set to zero or defaulted with I* then this will be taken from the bottom connection of the well.			0
6	SKIN	A real value that defines the connections' dimensionless skin factors.			0.0
		dimensionless	dimensionless	dimensionless	
Notes:					
1) Each record is terminated by a "/" and the keyword should be terminated by a "I".					

Table 12.20: CSKIN Keyword Description

Example

The following example re-defines the skin factor for all well OP01 connections in layers 4 to 6.

```
--
--      CONNECTION SKIN DATA
--
-- WELL  --- LOCATION --- SKIN
-- NAME   II  JJ  K1  K2   FACT
CSKIN
OP01      1*  1*   4   6   2.0   /
/
```

12.3.45 DATES – ADVANCE SIMULATION BY REPORTING DATE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword advances the simulation to a given report date after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the [SCHEDULE](#) section keywords, or further [DATES](#) data sets or keywords may be entered to advance the simulator to the next report date.

If the [DATES](#) keyword is to be used during the simulation, then the [START](#) keyword in the [RUNSPEC](#) section must be declared to set the start date for the run.

No.	Name	Description	Default
1	DAY	A positive integer that defines the day of the month for the data set, the value should be greater than or equal to one and less than or equal to 31.	None
2	MONTH	Character string for the month for the data set and should be one of the following 'JAN', 'FEB', 'MAR', 'APR', 'MAY', 'JUN', 'JUL' (or 'JLY'), 'AUG', 'SEP', 'OCT', 'NOV', or 'DEC'	None
3	YEAR	A positive four digit integer value representing the year for the data set, which must be specified fully by four digits, that is 1986.	None
4	TIME	A numeric character string that defines the time for the data set in the form of: HH:MM:SS.SSSS The default value means in most cases this parameter can be defaulted. TIME is normally used when detailed DST matching is performed to enable the pressures and rates to be stated at specific dates and times.	00:00:00

Notes:

- 1) The keyword is followed by a number of data sets (or rows) representing one [DATE](#) record per row.
- 2) Each record (or row) is terminated by a “/” and the keyword is terminated by a “/”.

Table 12.21: DATES Keyword Description

Note that OPM Flow uses the standard Gregorian calendar and therefore leap years are accounted for in the [DATES](#) keyword. Thus, it is more accurate to use the [DATES](#) keyword to progress the simulator through time if one is matching actual production data.

See also the [TIME](#) and [TSTEP](#) keywords in the [SCHEDULE](#) section.

Whenever possible it is a good idea to always set the start date to be at the beginning of the year, as like most simulators, OPM Flow reports are always stated at the number of days from the start date (and sometimes at a given date). If the start date is at the beginning of the year, then calculating the actual date is relatively straight forward and simple.

Examples

Given a start date of January 1, 2020 set via the [START](#) keyword in the [RUNSPEC](#) section, the following example advances the simulator from the start date of January 1, 2020 to January 1, 2021, using quarterly reporting time steps.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE

-- -----
-- SCHEDULE SECTION - 2020-01-01
-- -----
RPTSCHED
    'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'
/

DATES
    2  JAN  2020  /
/

RPTSCHED
    'NOTHING'
/

DATES
    1  APR  2020  /
    1  JUL  2020  /
    1  OCT  2020  /
/

-- -----
-- SCHEDULE SECTION - 2021-01-01
-- -----
RPTSCHED
    'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'
/

DATES
    1  JAN  2021  /
/

RPTSCHED
    'NOTHING'
/

DATES
    1  APR  2021  /
    1  JUL  2021  /
    1  OCT  2021  /
/
```

The above example writes out a series of report at the start of the run and then advances the simulation one day to January 2, 2020 and switches off the reporting. The simulation then advances to April 1, July 1 and October 1, 2020 with no further changes to the run. After October 1, 2020 reporting is switched on again to enable a report on January 1, 2021, which is then subsequently switched off after the January 1, 2021 report time step has been completed.

Note if one wishes to terminate the run at the end of year (as opposes to the beginning of the year and get a final report for the year, then the next example demonstrates the keyword sequence to enable this.

```

-----
-- SCHEDULE SECTION - 2021-01-01
-----
RPTSCHED
    'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'
/

DATES
    2  JAN    2021  /
/

RPTSCHED
    'NOTHING'
/

DATES
    1  FEB    2021  /
    1  MAR    2021  /
    1  APR    2021  /
    1  MAY    2021  /
    1  JUN    2021  /
    1  JULY   2021  /
    1  AUG    2021  /
    1  SEP    2021  /
    1  OCT    2021  /
    1  NOV    2021  /
    1  DEC    2021  /
/
--
-- FINAL REPORT AND RESTART AT YEAR END
--
RPTSCHED
    'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'
/

RPTRST
    'BASIC=2'
/

DATES
    31 DEC    2021  /
/

```

In the above example monthly reporting time steps have been used instead of quarterly and report is requested after the December 1, 2021 time step and is therefore written out on December 31, 2021.

12.3.46 DCQDEFN – DEFINE GAS DCQ UNITS AS RATE OR ENERGY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DCQDEFN](#) keyword defines the DCQ units to be rate or energy (calorific value) when using the Gas Field Operation model and the Gas Calorific Value control option. The gas DCQ rates are controlled by the [GASYEAR](#), [GASPERIO](#), [GDCQ](#), [GASFTARG](#) or [GASFDECR](#) keywords in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.47 DEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE

This keyword defines the debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See [DEBUG – Define the Debug Data to be Printed to File](#) in the GLOBAL section for a full description.

12.3.48 DELAYACT – DEFINE DELAYED ACTION KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [DELAYACT](#) keyword defines a series of keywords that should be executed after an [ACTION](#) keyword has been triggered by the [ACTION](#), [ACTIONG](#), [ACTIONR](#), [ACTIONW](#), [ACTIONS](#), or [ACTIONX](#) keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.49 DIFFMMF – DEFINE DIFFUSIVITY MULTIPLIERS FOR MATRIX-FRACTURES

This keyword, [DIFFMMF](#), defines the diffusivity multipliers for matrix-fractures for when the Dual Porosity option has been activated by either the [DUALPORO](#) or [DUALPERM](#) keywords, or the Coal Bed Methane option is selected by the [COAL](#) keyword, and the Diffusivity option has been activated by the [DIFFUSE](#) keywords; all four keywords are in the [RUNSPEC](#) section.

See [DIFFMMF – Define Diffusivity Multipliers for Matrix-Fractures](#) in the [GRID](#) section for a full description.

12.3.50 DIMPES – DEFINE IMPES DYNAMIC SOLUTION PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [DIMPES](#), defines the Implicit in Pressure Explicit Saturation (“IMPES”) dynamic solution parameters and results in the simulator switching from the current solution formulation to the [IMPES](#) formulation. OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

Example

```
--
--      ACTIVATE THE IMPES SOLUTION OPTION
--
DIMPES
```

The above example switches on the [IMPES](#) solution option; however, this has no effect in OPM Flow input decks.

12.3.51
DIMPLICT
–
ACTIVATE
FULLY
IMPLICIT
DYNAMIC
SOLUTION
FORMULATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [DIMPLICT](#), activates the Fully Implicit Formulation and results in the simulator switching from the current solution formulation to the fully implicit formulation. OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness.

There is no data required for this keyword.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

Example

```
--
--      ACTIVATES THE FULLY IMPLICIT SOLUTION OPTION
--
DIMPLICT
```

The above example switches on the fully implicit solution option; however, this has no effect in OPM Flow input decks.

12.3.52 DRILPRI – DEFINE PRIORITIZED DRILLING QUEUE PRIORITY PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, *DRILPRI*, defines the prioritized drilling queue priority parameters used in the priority formulae for this type of drilling queue.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.53 DRSDT – SOLUTION GAS (Rs) MAXIMUM RATE OF INCREASE PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

DRSDT defines the maximum rate at which the solution gas-oil ratio (Rs) can be increased in a grid cell. The keyword is similar in functionality to the **DRSDTR** keyword, that defines the maximum rate at which Rs can be increased in a grid cell by region. Both keywords should only be used if the **OIL**, **GAS**, and **DISGAS** keywords in the **RUNSPEC** section have been invoked to allow oil, gas and dissolved gas to be present in the model. The keyword only affects the behavior of an increasing Rs, for example when gas is being injected into an oil reservoir, and is subject to the availability of free gas and the ability of the undersaturated oil to adsorb this gas.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DRSDT1	DRSDT1 is a real positive number that defines the maximum rate at which the solution gas-oil ratio is allowed to increase in a grid cell, that is the maximum rate the gas can dissolve into the available undersaturated oil. A value of zero means that Rs cannot increase and free gas cannot dissolve into the unsaturated oil in a grid cell. Alternatively a very large value of DRSDT1 allows Rs to increase rapidly until there is no free gas or the oil within the grid block is fully saturated. Note if the keyword is not present in the input deck then DRSDT1 is assumed to be a very large number resulting in complete re-solution of the gas into the available undersaturated oil.			None
		Mscf/stb/d	sm ³ /sm ³ /day	scc/scc/day	
2	DRSDT2	DRSDT2 is a defined character string that defines whether the DRSDT1 is applied to either all grid blocks or just those grid blocks containing free gas: 1) <i>ALL</i>: means the DRSDT1 maximum rate at which Rs is allowed to increase in a grid cell is applied to all grid blocks. 2) <i>FREE</i>: means the DRSDT1 maximum rate at which Rs is allowed to increase in a grid cell is applied to grid blocks only containing free gas. Note if the keyword is not present in the input deck then DRSDT2 is set to the default value of <i>ALL</i>.			<i>ALL</i>
Notes:					
1) The keyword is terminated by a “/”.					

Table 12.22: DRSDT Keyword Description

Note this keyword can be used in history matching field performance to control the availability of the movable gas phase.

See also the **VAPPARS** keyword in the **SOLUTION** section and the **DRSDTR**, **DRVDT** and **DRVDTTR** keywords in the **SCHEDULE** section that controls how vaporized oil is treated and the rate at which the dissolved phase ratio increases within a grid block.

Examples

The first example prevents the solution gas-oil ratio from increasing and applies this to all grid cells.

```
--
--      SOLUTION GAS (RS) MAXIMUM RATE OF INCREASE FOR MODEL
--
DRSDT
--      MAX RS      ALL/FREE
--      DRSDT1      DRSDT2
--      -----
--      0.000      ALL
```

And the second example below applies 0.0005 Mscf/stb/d as the maximum rate at which the solution gas-oil ratio is allowed to increase in a grid cell, and applies this to only cells containing free gas.

```
--
--      SOLUTION GAS (RS) MAXIMUM RATE OF INCREASE FOR MODEL
--
DRSDT
--      MAX RS      ALL/FREE
--      DRSDT1      DRSDT2
--      -----
--      0.0005      FREE
```

Again, the keyword parameters when applied are subject to the availability of free gas and the ability of the undersaturated oil to adsorb this gas.

12.3.54 DRSDTCON – CO2 CONVECTIVE DISSOLUTION PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [DRSDTCON](#) keyword defines the parameters that control the convective dissolution of carbon dioxide (CO₂) into the in-situ brine within a grid cell, as described by Mykkeltvedt *et al.*³¹⁴, based on an assumption of vertical equilibrium. The keyword internally causes the simulator to calculate the solution gas-oil ratio (R_s), normally controlled by the [DRSDTI](#) parameter on the [DRSDT](#) keyword in the [SCHEDULE](#) section, making the [DRSDT](#) keyword redundant.

The keyword should only be used if the [CO2STORE](#) keyword in the [RUNSPEC](#) section has been activated to model CO₂ storage via OPM Flow's CO₂-Brine PVT model. Note that the [CO2STORE](#) keyword must be used together with either: (1) the [GAS](#), [WATER](#) and [DISGASW](#) keywords (or alternatively the [GASWAT](#) and [DISGASW](#) keywords), or (2) the [GAS](#), [OIL](#) and [DISGAS](#) keywords in the [RUNSPEC](#) section. See the [CO2STORE](#) keyword for details.

The [DRSDTCON](#) keyword is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	CHI	A real positive value (χ) that defines the proportionality constant related to the maximum rate of increase of CO₂ solution gas-oil ratio (R_s) in the Linear regime. A value of zero means that convective dissolution of CO₂ into in-situ brine does not occur and free CO₂ cannot dissolve into the brine. Alternatively, a non-zero value of CHI allows convective dissolution of CO₂. Note if the CO2STORE keyword is present but the DRSDTCON keyword is absent from the input deck, then this results in instantaneous dissolution of CO₂ into the available undersaturated in-situ brine.			0.04
		dimensionless	dimensionless	dimensionless	
I	PSI	A real positive value (ψ) that defines the normalised CO₂ solution gas-oil ratio ($\tilde{X}$) at the transition between the Linear and the Steady-State regimes.			0.34
		dimensionless	dimensionless	dimensionless	
I	OMEGA	A real positive value (ω) that defines the maximum rate of increase in CO₂ solution gas-oil ratio (R_s) in the Steady-State regime.			3.0e-9
		1/s	1/s	1/s	
I	OPTION	A defined character string that specifies in which cells the convective dissolution rate limit is applied, and should be set to one of the following character strings: 1) ALL: the limit is applied to all cells. 2) FREE: the limit is only applied to cells with free gas.			ALL
Notes:					

³¹⁴ Mykkeltvedt, T.S., Sandve, T.H. & Gasda, S.E. New Sub-grid Model for Convective Mixing in Field-Scale CO₂ Storage Simulation. *Transp Porous Med* 152, 9 (2025). <https://doi.org/10.1007/s11242-024-02141-5>

l) The keyword is terminated by a “/”.

Table 12.23: DRSDTCON Keyword Description

Mykkeltvedt *et al.* describe four regimes that characterise the evolution of the CO₂ solution gas-oil ratio in a control volume:

- Initial Phase: Near instantaneous equilibrium partitioning and diffusion.
- Linear Phase: Fingers of CO₂-rich brine propagate downwards towards the base of the control volume.
- Steady-State Phase: Fingers have passed through the base of the control volume.
- Decline Phase: Gas phase CO₂ has been completely dissolved in the brine.

In the Initial phase a near instantaneous jump in dissolved CO₂ is assumed if the control volume spans the CO₂-brine interface (otherwise the solution gas-oil ratio is zero). The initial jump comes from the capillary transition zone, the height of which depends on the capillary pressure function. Capillary equilibrium happens almost immediately and therefore the CO₂ solution gas-oil ratio is allowed to increase at unlimited rate until capillary equilibrium is reached. Capillary equilibrium is assumed when:

$$R_s > R_{s,sat} S_o \quad (12.13)$$

Where:

R_s = solution gas-oil ratio,
 $R_{s,sat}$ = saturated solution gas-oil ratio,
 S_o = oil saturation.

This is followed by the Linear phase where the finger speed is assumed to be relatively constant and therefore it is reasonable to assume that the CO₂ gas-oil ratio will increase at a constant rate (Linear regime). The maximum dissolution rate, as per the [DRSDT](#) keyword in the [SCHEDULE](#) section during the Linear regime (F_{lin}) is given by:

$$F_{lin} = \chi \left(\frac{R_{s,sat} K_z \Delta \rho_c g}{\mu_o S_o D_z \phi} \right) \quad (12.14)$$

Where:

χ = the non-dimensional parameter CHI controlling the dissolution rate,
 $R_{s,sat}$ = the maximum gas-oil ratio at the CO₂ solubility limit in the brine,
 K_z = the vertical permeability,
 $\Delta \rho_c$ = the difference between the brine density saturated with CO₂ at $R_{s,sat}$ and the brine density without dissolved CO₂,
 g = the gravity constant,
 μ_o = the oil (brine) viscosity, and
 S_o = the oil (brine) saturation.

The third phase is characterised by a change in slope of the rate of dissolution as the system transitions from linear build-up to quasi-steady state (Steady-State regime). The parameter PSI defines the normalised gas-oil ratio at the transition from the Linear to the Steady-State regime, where the normalised gas-oil ratio is given by:

$$\tilde{X} = \frac{R_s - R_{s,sat} S_g}{R_{s,sat} (1 - S_g)} \quad (12.15)$$

The rate of dissolution during the Steady-State regime is defined by the parameter OMEGA.

The final Decline phase occurs when the CO₂ has been completely dissolved in the brine at which point the rate of dissolution is zero.

The values of CHI, PSI and OMEGA can either be estimated from fine-scale simulation, historical data, or from laboratory tests³¹⁵. Analysis of fine-scale simulation results by Mykkeltvedt *et al* indicates that 0.04 +/- 0.01 is a reasonable value for CHI. Elenius *et al.*³¹⁶ also state that 0.04 is a reasonable value for CHI for the Utsira formation.

Note

In the commercial simulator a constant or regional value for *DRSDT* can be given as an input parameter. This can be used to include the effect of convective mixing as shown by Thibeau and Dutin³¹⁷.

OPM Flow's approach differs from this in that the *DRSDT* value is computed internally and is dependent on both the static and dynamic cell properties.

Example

The example below is similar to that shown under the *CO2STORE* keyword in the *RUNSPEC* section. In the *RUNSPEC* section one declares that the carbon dioxide storage model is active for the run to account for both carbon dioxide and water phase solubility using OPM Flow's CO₂-Brine PVT model.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
-- -----
-- FLUID TYPES AND TRACER OPTIONS
-- -----
--
--      ACTIVATE CO2 STORAGE IN THE MODEL (OPM FLOW CO2 STORAGE KEYWORD)
--
CO2STORE
--
--      ACTIVATE GAS-WATER THE MODEL (OPM FLOW KEYWORD)
--
GASWAT
--
--      DISSOLVED GAS IN WATER IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
DISGASW
--
--      VAPORIZED WATER IN DRY/WET GAS IS PRESENT IN THE RUN (OPM FLOW KEYWORD)
--
VAPWAT
```

³¹⁵ Taheri, A., Torsæter, O., Lindeberg, E., Hadia, N. J., & Wessel-Berg, D. (2018). Qualitative and quantitative experimental study of convective mixing process during storage of CO₂ in heterogeneous saline aquifers. *International Journal of Greenhouse Gas Control*, 71, 212-226.

³¹⁶ Elenius, M. T., Nordbotten, J. M., & Kalisch, H. (2014). Convective mixing influenced by the capillary transition zone. *Computational Geosciences*, 18(3-4), 417-431

³¹⁷ Thibeau, S., & Dutin, A. (2011). Large scale CO₂ storage in unstructured aquifers: Modeling study of the ultimate CO₂ migration distance. *Energy Procedia*, 4, 4230-4237.

The second part of the example sets the maximum dissolution rate for convective CO₂ mixing via the [DRSDTCON](#) keyword in the [SCHEDULE](#) section using the base case parameters calculated by Mykkeltvedt et al from fine-scale simulations.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--          CO2 CONVECTIVE DISSOLUTION PARAMETERS
--
DRSDTCON
--          CHI          PSI          OMEGA      OPTION
          0.04          0.34          3.0E-09    ALL
```

See also the [CO2STORE](#) keyword in the [RUNSPEC](#) section for further information on OPM Flow's CO₂ storage facility.

12.3.55 DRSDTR – SOLUTION GAS (Rs) MAXIMUM RATE OF INCREASE PARAMETERS BY REGION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[DRSDTR](#) defines the maximum rate at which the solution gas-oil ratio (Rs) can be increased in a grid cell for various regions in the model. The keyword is similar in functionality to the [DRSDT](#) keyword, that defines the maximum rate at which Rs can be increased in a grid cell for all cells in the model. The number of [DRSDTR](#) vector data sets is defined by the NTPVT parameter on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and the allocation of the [DRSDTR](#) records to different grid blocks in the model is done via the [PVTNUM](#) keyword in the REGION section. One data set consists of one record or line which is terminated by a “/”.

[DRSDTR](#) should only be used if the [OIL](#), [GAS](#), and [DISGAS](#) keywords in the [RUNSPEC](#) section have been invoked to allow oil, gas and dissolved gas to be present in the model. The keyword only affects the behavior of an increasing Rs, for example when gas is being injected into an oil reservoir, and is subject to the availability of free gas and the ability of the undersaturated oil to adsorb this gas.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	DRSDT1	DRSDT1 is a real positive number that defines the maximum rate at which the solution gas-oil ratio is allowed to increase in a grid cell, that is the maximum rate the gas can dissolve into the available undersaturated oil. A value of zero means that Rs cannot increase and free gas cannot dissolve into the unsaturated oil in a grid cell. Alternatively a very large value of DRSDT1 allows Rs to increase rapidly until there is no free gas or the oil within the grid block is fully saturated. Note if the keyword is not present in the input deck then DRSDT1 is assumed to be a very large number resulting in complete re-solution of the gas into the available undersaturated oil.			None
		Mscf/stb/d	sm ³ /sm ³ /day	scc/scc/day	
2	DRSDT2	DRSDT2 is a defined character string that defines whether the DRSDT1 is applied to either all grid blocks or just those grid blocks containing free gas: 1) ALL: means the DRSDT1 maximum rate at which Rs is allowed to increase in a grid cell is applied to all grid blocks. 2) FREE: means the DRSDT1 maximum rate at which Rs is allowed to increase in a grid cell is applied to grid blocks only containing free gas. Note if the keyword is not present in the input deck then DRSDT2 is set to the default value of ALL.			ALL

Notes:

1) The keyword is followed by NTPVT records as declared on the **TABDIMS** keyword in the **RUNSPEC** section.

1) Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 12.24: DRSDTR Keyword Description

Note this keyword can be used in history matching field performance to control the availability of the movable gas phase.

See also the [VAPPARS](#) keyword in the [SOLUTION](#) section and the [DRSDT](#), [DRVDT](#) and [DRVDTT](#) keywords in the [SCHEDULE](#) section that controls how vaporized oil is treated and the rate at which the dissolved phase ratio increases within a grid block.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored but is documented here for completeness, as it is expected to be available in the next release of OPM Flow.

Examples

The first example prevents the solution gas-oil ratio from increasing and applies this to all regions for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to three.

```
--
--      SOLUTION GAS (RS) MAXIMUM RATE OF INCREASE BY REGION
--
DRSDTR
--      MAX RS      ALL/FREE
--      DRSDT1      DRSDT2
--      -----
--      0.0000      ALL                      /
--      0.0000      ALL                      /
--      0.0000      ALL                      /
```

The second example below prevents the solution gas-oil ratio from increasing and applies this to all grid cells in [PVTNUM](#) region one. For [PVTNUM](#) regions one and two the keyword applies 0.0005 Mscf/stb/d as the maximum rate at which the solution gas-oil ratio is allowed to increase in a grid cell, and applies this to only cells containing free gas.

```
--
--      SOLUTION GAS (RS) MAXIMUM RATE OF INCREASE BY REGION
--
DRSDTR
--      MAX RS      ALL/FREE
--      DRSDT1      DRSDT2
--      -----
--      0.0000      ALL                      /
--      0.0005      FREE                     /
--      0.0005      FREE                     /
```

Again, the keyword parameters when applied are subject to the availability of free gas and the ability of the undersaturated oil to adsorb this gas.

12.3.56 DRVDT – VAPORIZED OIL (RV) MAXIMUM RATE OF INCREASE PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

DRVDT defines the maximum rate at which the vaporized oil-gas ratio or condensate-gas ratio (Rv) can be increased in a grid cell. The keyword is similar in functionality to the **DRVDTI** keyword, that defines the maximum rate at which Rv can be increased in a grid cell by region. Both keywords should only be used if the **OIL**, **GAS**, and **VAPOIL** (condensate) keywords in the **RUNSPEC** section have been invoked to allow oil, gas and condensate to be present in the model. The keyword only affects the behavior of an increasing Rv, for example when gas is being injected into a gas condensate reservoir as part of a gas re-cycling scheme, and is subject to the availability of free oil (condensate) and the ability of the undersaturated gas to adsorb this condensate.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DRVDTI	DRVDTI is a real positive number that defines the maximum rate at which the vaporized oil-gas ratio is allowed to increase in a grid cell, that is the maximum rate at which the oil can dissolve into the available undersaturated gas. A value of zero means that Rv cannot increase and free oil cannot vaporize into the unsaturated gas in a grid cell. Alternatively a very large value of DRVDTI allows Rv to increase rapidly until there is no free oil or the gas within the grid block is fully saturated. Note if the keyword is not present in the input deck then DRVDTI is assumed to be a very large number resulting in complete re-vaporization of the oil into the available undersaturated gas.			None
		stb/Mscf/d	sm ³ /sm ³ /day	scc/scc/day	
Notes:					
I) The keyword is terminated by a “/”.					

Table 12.25: DRVDT Keyword Description

Note this keyword can be used in history matching field performance to control the availability of the movable gas phase. See also the **VAPPARS** keyword in the **SOLUTION** section and the **DRVDTI**, **DRSDT** and **DRSDTR** keywords in the **SCHEDULE** section that controls how solution gas is treated and the rate at which the vaporized phase ratio increases within a grid block.

Example

The example prevents the solution oil-gas ratio from increasing.

```
--
--      VAPORIZED OIL (RV) MAXIMUM RATE OF INCREASE FOR MODEL
--
DRVDT
--      MAX RV
--      DRVDT1
--      -----
--      0.000 /
```

Again, the keyword parameters when applied are subject to the availability of free oil and the ability of the undersaturated gas to adsorb this oil.

12.3.57 DRVDTR – VAPORIZED OIL (RV) MAXIMUM RATE OF INCREASE PARAMETERS BY REGION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[DRVDTR](#) defines the maximum rate at which the vaporized oil-gas ratio or condensate-gas ratio (Rv) can be increased in a grid cell for various regions in the model. The keyword is similar in functionality to the [DRVDT](#) keyword, that defines the maximum rate at which Rv can be increased in a grid cell for all cells in the model. The number of [DRVDTR](#) vector data sets is defined by the NTPVT parameter on the [TABDIMS](#) keyword in the [RUNSPEC](#) section and the allocation of the [DRVDTR](#) records to different grid blocks in the model is done via the [PVTNUM](#) keyword in the REGION section. One data set consists of one record or line which is terminated by a “/”.

This keyword should only be used if the [OIL](#), [GAS](#), and [VAPOIL](#) (condensate) keywords in the [RUNSPEC](#) section have been invoked to allow oil, gas and condensate to be present in the model. The keyword only affects the behavior of an increasing Rv, for example when gas is being injected into a gas condensate reservoir as part of a gas re-cycling scheme, and is subject to the availability of free oil (condensate) and the ability of the undersaturated gas to adsorb this condensate.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	DRVDTI	DRVDTI is a real positive number that defines the maximum rate at which the vaporized oil-gas ratio is allowed to increase in a grid cell, that is the maximum rate at which the oil can vaporize into the available undersaturated gas. A value of zero means that Rv cannot increase and free oil cannot vaporize into the unsaturated gas in a grid cell. Alternatively a very large value of DRVDTI allows Rv to increase rapidly until there is no free oil or the gas within the grid block is fully saturated. Note if the keyword is not present in the input deck then DRVDTI is assumed to be a very large number resulting in complete re-vaporization of the oil into the available undersaturated gas.			None
		stb/Mscf/d	sm ³ /sm ³ /day	scc/scc/day	

Notes:

- 1) The keyword is followed by NTPVT records as declared on the [TABDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record is terminated by a “/” and there is no “/” terminator for the keyword.

Table 12.26: DRVDTR Keyword Description

Note this keyword can be used in history matching field performance to control the availability of the movable gas phase.

See also the [VAPPARS](#) keyword in the [SOLUTION](#) section and [DRVDT](#), [DRSDT](#), and [DRSDTR](#) keywords in the [SCHEDULE](#) section that controls how dissolved gas is treated and the rate at which the vaporized phase ratio increases within a grid block.

Examples

The first example prevents the vaporized oil-gas ratio from increasing and applies this to all regions for when NTPVT on the [TABDIMS](#) keyword in the [RUNSPEC](#) section is set to three.

```
--
--      VAPORIZED OIL (RV) MAXIMUM RATE OF INCREASE PARAMETERS BY REGION
--
DRVDT1
--      MAX RV
--      DRVDT1
--      -----
--      0.000 /
--      0.000 /
--      0.000 /
```

The second example below prevents the vaporized oil-gas ratio from increasing and applies this to all grid cells in [PVTNUM](#) region one. For [PVTNUM](#) regions one and two the keyword applies 0.005 stb//Mscf/d as the maximum rate at which the vaporized oil-gas ratio is allowed to increase in a grid cell,

```
--
--      VAPORIZED OIL (RV) MAXIMUM RATE OF INCREASE PARAMETERS BY REGION
--
DRVDT1
--      MAX RV
--      DRVDT1
--      -----
--      0.0005 /
--      0.0005 /
--      0.0005 /
```

Again, the keyword parameters when applied are subject to the availability of free oil and the ability of the undersaturated gas to adsorb this oil.

12.3.58 DUMPCUPL – ACTIVATE OUTPUT TO THE RESERVOIR COUPLING FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [DUMPCUPL](#), activates output to the Reservoir Coupling file from the reservoir coupling file in the master run for when reservoir coupling is invoked by the [GRUPMAST](#) and [SLAVES](#) keywords in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.59 DYNAMICR – DEFINE DYNAMIC REGION PARAMETERS

The [DYNAMICR](#) keyword marks the start of a Dynamic Region section and defines the parameters used for Dynamic Regions that allows for property and reporting regions to vary as the run progresses, based on the parameters and logic defined by this keyword and section. A Dynamic Region section is terminated by the [ENDDYN](#) keyword in the [SOLUTION](#) or [SCHEDULE](#) sections.

See [DYNAMICR – Start of Dynamic Region Parameter Definition](#) in the [SOLUTION](#) section for a full description.

12.3.60 ECHO – ACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns on echoing of all the input files to the print file; note that this keyword is activated by default and can subsequently be switched off by the [NOECHO](#) activation keyword.

See [ECHO – Activate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

12.3.61 END – DEFINE THE END OF THE INPUT FILE

This keyword marks the end of the input file and can occur in any section. Any keywords and data after the [END](#) keyword are ignored.

See [END – Define the End of the Input File](#) in the GLOBAL section for a full description.

12.3.62 ENDACTIO – END THE DEFINITION OF ACTION COMMANDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *ENDACTIO* keyword defines the end of a series of conditions that invoke run time processing of the *ACTION* series of keywords, namely: *ACTION*, *ACTIONG*, *ACTIONR*, *ACTIONS*, *ACTIONW* and *ACTIONX*. Only the *ACTIONX* keyword is implemented in OPM Flow as this keyword implements the *ACTION*, *ACTIONG*, *ACTIONR*, *ACTIONS*, *ACTIONW* functionality with greater flexibility. See the *ACTIONX* keyword in the *SCHEDULE* section for a full description of the *ACTION* facility.

There is no data required for this keyword and there is no terminating “/” for this keyword.

Example

The example shows the use of the *ACTIONX* and *ENDACTIO* keywords to test if the field’s gas production rate is less than 600 MMscf/d after 2020 and to open up additional wells if this occurs.

```
--
-- START OF ACTIONX FIELD PHASE-2 DEVELOPMENT DEFINITION
--
ACTIONX
  PHASE2      1      /
  GGPR  'FIELD' < 600E3 AND  /
  YEAR > 2020      /
/
-- WELL PRODUCTION STATUS
--
-- WELL      WELL  --LOCATION--  COMPLETION
-- NAME      STAT   I   J   K  FIRST LAST
WELOPEN
GP10        OPEN                      /
GP11        OPEN                      /
/
--
-- END OF ACTIONX FIELD PHASE-2 DEVELOPMENT DEFINITION
--
ENDACTIO
```

12.3.63 ENDBOX – DEFINE THE END OF THE BOX DEFINED GRID

This keyword marks the end of a previously defined [BOX](#) sub-grid as defined by a previously entered [BOX](#) keyword. The keyword resets the input grid to be the full grid as defined by the NX, NY, and NZ variables on the [DIMENS](#) keyword in the [RUNSPEC](#) section.

See [ENDBOX – Define the End of the BOX Defined Grid](#) in the [GRID](#) section for a full description.

12.3.64 ENDDYN – END OF DYNAMIC REGION PARAMETER DEFINITION

The [ENDDYN](#) keyword marks the end of a Dynamic Region section that was started with the [DYNAMICR](#) keyword in the [SOLUTION](#) or [SCHEDULE](#) sections. Dynamic Regions allow for property and reporting regions to vary as the run progresses, based on the parameters and logic defined within the section.

See [ENDDYN – End of Dynamic Region Parameter Definition](#) in the [SOLUTION](#) for a full description.

12.3.65 ENDFIN – END THE DEFINITION OF A LOCAL GRID REFINEMENT

[ENDFIN](#) defines the end of a Cartesian or radial local grid refinement (“LGR”) definition and a [LGR](#) property definition data set.

See [ENDFIN – End the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

12.3.66 ENDINC – DEFINE THE END OF AN INCLUDE FILE

This keyword marks the end of an include file specified on the [INCLUDE](#) keyword. When the [ENDINC](#) keyword is encountered in the [INCLUDE](#) file, input data is read from the next keyword in the current file. Any keywords and data after the [ENDINC](#) keyword in the [INCLUDE](#) file are ignored.

See [ENDINC – Define the End of an Include File](#) in the [GLOBAL](#) section for a full description.

12.3.67 ENDSKIP – DEACTIVATE SKIPPING OF KEYWORDS AND INPUT DATA

The [ENDSKIP](#) keyword deactivates the skipping of keywords that was activated by the [SKIP](#), [SKIP100](#), or [SKIP300](#) keywords. Each [SKIP](#) keyword should be paired with an [ENDSKIP](#) keyword.

See [ENDSKIP – Deactivate Skipping of Keywords and Input Data](#) in the [GLOBAL](#) section for a full description.

12.3.68 EPSDBGS – WRITE END-POINT DEBUG DATA TO THE DEBUG FILE (MULTIPLE)

This keyword, [EPSDBGS](#), defines the end-point debug data for multiple grid blocks that should be written to the debug file (*.DBG) for when the End-Point Scaling option has been activated by the [ENDSCALE](#) keyword in the [RONSPEC](#) section.

See [EPSDBGS – Write End-Point Debug Data to the DEBUG File \(Multiple\)](#) in the [PROPS](#) section for a full description.

12.3.69 EPSDEBUG – WRITE END-POINT DEBUG DATA TO THE DEBUG FILE (INDIVIDUAL)

This keyword, [EPSDEBUG](#), defines the end-point debug data for individual grid blocks that should be written to the Debug file (*.DBG) for when the End-Point Scaling option has been activated by the [ENDSCALE](#) keyword in the [RUNSPEC](#) section.

See [EPSDEBUG – Write End-Point Debug Data to the DEBUG File \(Individual\)](#) in the [PROPS](#) section for a full description.

12.3.70 EXCAVATE – SET THE STATUS OF A GRID BLOCK TO ACTIVE OR EXCAVATE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [EXCAVATE](#), sets the status of global and [LGR](#) grid blocks to active or excavate. Excavated grid blocks have all the transmissibilities set to zero thus disabling flow between the surrounding grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.71 EXIT – EXIT SIMULATION FROM WITHIN AN ACTION SECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [EXIT](#) keyword is part of OPM Flow's [ACTION](#) facility that allows for terminating the simulation for when a condition within an [ACTIONX](#) definition is satisfied. Invoking the keyword within an [ACTIONX](#) definition will result in the simulation terminating with an exit status code. The [ACTION](#) facility allows the user to enter computational logic to the simulation run based on the how the simulation run is proceeding – see the [ACTIONX](#) keyword in the [SCHEDULE](#) section.

Note

This is an OPM Flow specific keyword for the simulator's [ACTION](#) facility and will therefore cause an error if used in the commercial simulator.

No.	Name	Description	Default
I	EXITCODE	An optional integer that sets the exit code printed to the *.PRT file, if not defined the default value of zero will be used.	0
Notes: I) The keyword is terminated by a "/".			

Table 12.27: EXIT Keyword Description

The [EXIT](#) keyword should only be used as part of an [ACTIONX](#) block, if found elsewhere in the input deck it will be ignored.

Examples

The first example uses the [ACTIONX](#) keyword to define a condition for when the Field Oil Production Rate ("FOPR") falls below 1,000 stb/d (or 1,000 m³) using the default value for the EXITCODE.

```
--
--      DEFINE START OF ACTIONX SECTION
--
ACTIONX
      'CHECK_FOPR' 100000      /
      FOPR < 1000             /
/
--
--      TERMINATE AND EXIT SIMULATION
--
EXIT
/

ENDACTIO
```

The next example terminates the simulation with EXITCODE one when the Field Pressure (“FPR”) falls below 200 psia (or 200 barsa).

```
--
--      DEFINE START OF ACTIONX SECTION
--
ACTIONX
    'CHECK_FPR'  100000      /
    FPR < 200      /
/
--
--      TERMINATE AND EXIT SIMULATION
--
EXIT
    1      /

ENDACTIO
```

Note is is probably good practice to always set the EXITCODE to be able to identify the reason for the simulation stopping.

12.3.72 EXTRAPMS – ACTIVATE EXTRAPOLATION WARNING MESSAGES

The [EXTRAPMS](#) keyword activates extrapolation warning messages for when OPM Flow extrapolates the PVT or VFP tables. Frequent extrapolation warning messages should be investigated and resolved as this would indicate possible incorrect data and may result in the simulator extrapolating to unrealistic values.

See [EXTRAPMS – Activate Extrapolation Warning Messages](#) in the GLOBAL section for a full description.

12.3.73 FBHPDEF – DEFINE WELL DEFAULT BHP TARGET AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [FBHPDEF](#), defines the default well BHP target for production wells and the default BHP constraint for injection wells.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TARGET_BHP	A real positive value that defines the default well BHP target for production wells.			1.01325 barsa
		psia	barsa	atma	
2	LIMIT_BHP	A real positive value that defines the default well BHP limit for injection wells.			6895 barsa
		psia	barsa	atma	

Notes:

- The values defined by the *FBHPDEF* keyword will be used when a well BHP target or limit is later defaulted using one of the well control keywords in the *SCHEDULE* section (e.g. *WCONPROD*, *WCONHIST*, *WCONINJE* and *WCONINJH*). Well BHP targets or limits defaulted prior to the *FBHPDEF* keyword will not be affected.
- The keyword should be terminated by a “/”.

Table 12.28: FBHPDEF Keyword Description

See also the [WELTARG](#) keyword in the [SCHEDULE](#) section that modifies a well’s target and constraint values.

Example

The following examples sets the default BHP target for production wells to 1.01325, and the default BHP limit for injection wells to 6895:

```
--
--      DEFAULT BHP TARGET AND CONSTRAINTS
--
FBHPDEF      1.01325  6895.  /
```

12.3.74 FILEUNIT – ACTIVATE UNIT CONSISTENCY VERIFICATION

The [FILEUNIT](#) keyword defines the units of the data set, and is used to verify that the units in the input deck and any associated include files are consistent. The keyword does not provide for the conversion between different sets of units.

See [FILEUNIT – Activate Unit Consistency Checking](#) in the [GRID](#) section for a full description.

12.3.75 FORMFEED – DEFINED THE PRINT FILE FORM-FEED CHARACTER

The [FORMFEED](#) keyword defines the form-feed character, or carriage control character, for the output print (*.PRT) run summary (*.RSM) files. The keyword should be placed at the very top of the input file.

See [FORMFEED – Defined the Print File Form-Feed Character](#) in the GLOBAL section for a full description.

12.3.76 GASBEGIN – DEFINE START OF ANNUAL SCHEDULING SECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GASBEGIN](#), defines the start of an Annual Scheduling section set of keywords used when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. An Annual Scheduling section starts with the [GASBEGIN](#) keyword and is terminated by the [GASEND](#) keyword, with keywords in between used to control and write reports at selected times between the start and end of a contract period. Only one Annual Scheduling section is activate at a time, that is, a subsequent Annual Scheduling section overwrites the previous set of entries.To clear the current Annual Schedule section enter the [GASBEGIN](#) keyword followed by the [GASEND](#) keyword word with no other keywords in between.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.77 GASEND – DEFINE END OF ANNUAL SCHEDULING SECTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [GASEND](#), defines the end of an Annual Scheduling section set of keywords used when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. An Annual Scheduling section starts with the [GASBEGIN](#) keyword and is terminated by the [GASEND](#) keyword, with keywords in between used to control and write reports at selected times between the start and end of a contract period. Only one Annual Scheduling section is activate at a time, that is, a subsequent Annual Scheduling section overwrites the previous set of entries.To clear the current Annual Schedule section enter the [GASBEGIN](#) keyword followed by the [GASEND](#) keyword word with no other keywords in between.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.78 GASFCOMP – DEFINE AUTOMATIC GAS COMPRESSORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GASFCOMP](#), defines automatic gas compressors for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section and the Standard Network option has been specified by the [GRUPTREE](#), [GRUPNET](#) and [GNETINJE](#) series of keywords in the [SCHEDULE](#) section. Automatic gas compressors are automatically switch on for a group if a group’s gas production target cannot be satisfied. In addition, if a group’s gas target is reduced then the automatic compressors are initially switch off to test that the reduced gas rate target can be met without compression, if not, compression is switched back on. Note that all automatic compressors are “switch on” when calculating a field’s gas deliverability.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.79 GASFDECR – DEFINE FIELD GAS SALES CONTRACT MONTHLY REDUCTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GASFDECR](#), defines the field’s monthly reduction in the gas sales contract quantity for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. Gas contracts are commonly based on a Daily Contract Quantity (“DCQ”) that determines the gas rate that the field should be produced at, which is normally expressed as a multiple of the DCQ, for example 1.33, and is often referred to as the “swing factor”. Some gas contracts also define a maximum DCQ (“Max DCQ”) and/or a minimum take or pay DCQ (“Min DCQ”), as well as seasonal demand characteristics. For example, gas rates may be set higher in the winter months in order to meet heating demand compared with summer months in colder climates, and the opposite in warmer climates where air conditioning demand is high.

Thus, the DCQ must be calculated first over a contract period, where the contract period is commonly contract years, or in some instances contract quarters. This is performed by the simulator using the current DCQ and checking to see if the (DCQ x Swing Factor) can be satisfied throughout the current contract period, if not the DCQ is re-calculated so that the (DCQ x Swing Factor) condition is satisfied. Once this condition is met, the second and final pass uses the calculated DCQ in conjunction with the monthly scaling profile data to set the monthly gas rate for the field:

$$Q_{month} = DCQ \times SWINGFAC_{month} \tag{12.16}$$

Where:

- Qmonth = the monthly gas production target
- DCQ = Daily Contract Quantity
- [SWINGFAC](#)_{month} = monthly rate scaling factor that takes into account seasonal demand, etc.

Here the [GASFDECR](#) keyword allows for a reduction in the calculated final pass monthly gas rates and thus equation (12.16) become:

$$Q_{month} = (DCQ \times SWINGFAC_{month}) - GASFDECR_{month} \tag{12.17}$$

Where:

- Qmonth = the monthly gas production target
- DCQ = Daily Contract Quantity
- [SWINGFAC](#)_{month} = monthly rate scaling factor that takes into account seasonal demand, etc.
- [GASFDECR](#)_{month} = monthly gas rate reduction.

Since the simulator must make two passes to calculate the final rates this will naturally decrease computational efficiency.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.80 GASFDELC – DEFINE GAS DELIVERABILITY CALCULATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GASFDELC](#) keyword defines how the field’s gas deliverability calculation should be performed for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.81 GASFTARG – DEFINE FIELD GAS SALES CONTRACT MONTHLY TARGET

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GASFTARG](#), defines the field's monthly gas sales contract quantity for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. Gas contracts are commonly based on a Daily Contract Quantity ("DCQ") that determines the gas rate that the field should be produced at, which is normally expressed as a multiple of the DCQ, for example 1.33, and is often referred to as the "swing factor". Some gas contracts also define a maximum DCQ ("Max DCQ") and/or a minimum take or pay DCQ ("Min DCQ"), as well as seasonal demand characteristics. For example, gas rates may be set higher in the winter months in order to meet heating demand compared with summer months in colder climates, and the opposite in warmer climates where air conditioning demand is high.

Thus, the DCQ must be calculated first over a contract period, where the contract period is commonly contract years, or in some instances contract quarters. This is performed by the simulator using the current DCQ and checking to see if the (DCQ x Swing Factor) can be satisfied throughout the current contract period, if not the DCQ is re-calculated so that the (DCQ x Swing Factor) condition is satisfied. Once this condition is met, the second and final pass uses the calculated DCQ in conjunction with the monthly scaling profile data to set the monthly gas rate for the field:

$$Q_{month} = DCQ \times SWINGFAC_{month} \quad (12.18)$$

Where:

Q_{month} = the monthly gas production target
 DCQ = Daily Contract Quantity
 $SWINGFAC_{month}$ = monthly rate scaling factor that takes into account seasonal demand, etc.

Here the [GASFTARG](#) keyword sets a minimum target rate in the calculated final pass monthly gas rates and thus equation (12.18) become:

$$Q_{month} = \text{Minimum} \left((DCQ \times SWINGFAC_{month}), GASFTARG_{month} \right) \quad (12.19)$$

Where:

Q_{month} = the monthly gas production target
 DCQ = Daily Contract Quantity
 $SWINGFAC_{month}$ = monthly rate scaling factor that takes into account seasonal demand, etc.
 $GASFTARG_{month}$ = minimum monthly gas rate target.

Since the simulator must make two passes to calculate the final rates this will naturally decrease computational efficiency.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.82 GASMONTH – DEFINE START OF ANNUAL SCHEDULING EVENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GASMONTH](#), states the month for which subsequent scheduling events take place within an Annual Schedule section for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. The keyword must lie in between the [GASBEGIN](#), that defines the start of an Annual Scheduling section and the [GASEND](#) keyword that ends the section. Optionally, the keyword can be used to write a report to the print file (*.PRT) at the requested month.

See also the [GASBEGIN](#) and [GASEND](#) keywords in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.83 GASPERIO – ADVANCE SIMULATION BY GAS CONTRACT PERIOD

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword advances the simulation over one or more gas contract periods for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. A contract period in this case is the period over which the Daily Contract Quantity is fixed, this can a be year or one or more months. If the contract period is a year then the [GASYEAR](#) keyword in the [SCHEDULE](#) section can be used instead of GASPERIOD.

[GASPERIO](#) is an alternative to the [DATES](#), [TIME](#) and [TSTEP](#) keywords in the [SCHEDULE](#) section that advances the simulation to a given report time after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the [SCHEDULE](#) section keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.84 GASYEAR – ADVANCE SIMULATION BY GAS CONTRACT YEAR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword advances the simulation over one or more gas contract years for when the Gas Field Operations option has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. A contract year in this case is the period over which the Daily Contract Quantity is fixed, this can a be year, this keyword or the [GASPERIO](#) keyword in the [SCHEDULE](#) section, or one or more months. If the contract period is over one or more months then the [GASPERIO](#) keyword in the [SCHEDULE](#) section can be used instead of [GASYEAR](#).

[GASYEAR](#) is an alternative to the [DATES](#), [TIME](#) and [TSTEP](#) keywords in the [SCHEDULE](#) section that advances the simulation to a given report time after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the [SCHEDULE](#) section keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.85 GCALECON – GROUP ECONOMIC CRITERIA FOR PRODUCTION CALORIFIC GROUPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCALECON](#) keyword defines economic criteria for production groups, including the field level group [FIELD](#), that have previously been defined by the [GCONPROD](#) keyword in the [SCHEDULE](#) section and have had their rate targets and constraints set by calorific value via the [GCONVAL](#) keyword in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Note that wells are allocated to a group when they are specified by the [WELSPECS](#) keyword and wells can also have economic controls. Wells under group control are therefore subject to the economic criteria set via the [GCONPROD](#) and [CECON](#) keywords in the [SCHEDULE](#) section and the controls specified by the [WECON](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	ENEVAL	A real positive value that defines the minimum economic surface energy production rate, below which an economic action of shutting in or stopping all the wells in the group, as requested by item (9) of the WELSPECS keyword. A value less than or equal to zero switches of this criteria.			0.0
		BTU/day	kJ/day	J/hour	
3	CALVAL	A real positive value that defines the minimum economic surface calorific value, below which an economic action of shutting in or stopping all the wells in the group, as requested by item (9) of the WELSPECS keyword. A value less than or equal to zero switches of this criteria,			0.0
		Btu/Mscf	kJ/sm ³	J/scc	
8	END	A defined character string that defines if the simulation should terminate if all the producing wells in the group, including the FIELD group, are shut or stopped. END should be set to one of the following character strings: 1) NO: no action is taken and the run continues. 2) YES: terminate the run at the next report time step.			NO

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.29: GCALECON Keyword Description

See also the [WELSPECS](#) keyword to define a wells shut-in or stop options, [GCONPROD](#), [GCONCAL](#), [GCONENG](#) for group controls, and [WECON](#) for setting a well's economic criteria. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the economic criteria for the field with a minimum economic surface energy production rate of 5 x 109 BTU/day and a minimum economic surface calorific value of 900 Btu/Mscf

```
--
--          GROUP ECONOMIC CRITERIA FOR PRODUCTION GROUPS UNDER CALORIFIC CONTROL
--
-- GRUP  ENERGY  CALORIFIC  END
-- NAME  RATE      VALUE      RUN
GCALECON
FIELD   5E9       900.0      'YES'
/
```

If the economic limits are violated then the run will stop at the next report time step.

12.3.86 GCONCAL – GROUP PRODUCTION CALORIFIC TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCONCAL](#) keyword defines calorific production targets and constraints for groups, including the top most group in the group hierarchy known as the FIELD group. Wells are allocated to groups when the wells are specified by the [WELLSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group's target calorific value is being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	CALVAL	A real positive value that defines the target surface calorific value of the produced gas. The default value of 1 x 1020 switches off calorific control for the group.			1.0 x 10 ²⁰
		Btu/Mscf	kJ/sm ³	J/hour	
3	ACTION	A defined character string that defines the action to be taken if the CALVAL is violated. ACTION should be set to one of the following character strings: 1) NONE: no action is taken and the group's produced calorific value will therefore no longer meet CALVAL. 2) RATE: scale back the gas rate of various wells under the group by the value of FACTOR until the calorific target (CALVAL) is satisfied. The corrective action takes places at the end of the time step in which the constraint is violated.			None
4	FACTOR	A real positive value that is less than or equal to one that defines the amount wells can be scaled back in order to satisfy CALVAL. Note this assumes that there are wells within the group that are producing with higher and lower calorific values, and the simulator is thus able to fine a combination of wells that satisfy the group's CALVAL target.			None
		dimensionless	dimensionless	dimensionless	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.30: GCONCAL Keyword Description

See also the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the

[GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection rate targets and constraints, the [WCONPROD](#) keyword to define a production well's targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the calorific production target for the field.

```
---
--          GROUP CALORIFIC PRODUCTION CONTROLS
--
--  GRUP  CALORIFIC  ACTION  CUT
--  NAME  VALUE              BACK
GCONCAL
FIELD    1010E3      RATE    0.95
/
```

Here the calorific production target has been set to 1,010 × 103 Btu/Mscf for the field and if the target cannot be met then the well rates are reduced by 0.95 at each iteration until the target is satisfied.

12.3.87 GCONENG – GROUP PRODUCTION ENERGY TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCONCAL](#) keyword defines energy production targets and constraints for groups, including the top most group in the group hierarchy known as the FIELD group. Wells are allocated to groups when the wells are specified by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group's target calorific value is being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the <i>GRUPTREE</i> keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	ENGVAL	A real positive value that defines the surface energy target for the group. The default value of 1 x 1020 switches off the energy target for the group.			1.0 x 10 ²⁰
		BTU/day	kJ/day	J/hour	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.31: GCONENG Keyword Description

See also the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection rate targets and constraints, the [WCONPROD](#) keyword to define a production well's targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the energy production target for the field.

```
--
--      GROUP ENERGY PRODUCTION CONTROLS
--
-- GRUP  ENERGY
-- NAME  VALUE
GCONENG
FIELD   1010E9
/
```

Here the energy production target has been set to 1,010 x 10⁹ Btu/day for the field.

12.3.88 GCONINJE – GROUP INJECTION TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCONINJE](#) keyword defines injection targets and constraints for groups, including the top most group in the group hierarchy known as the FIELD group. Wells are allocated to groups when the wells are specified by the [WEL SPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their injection rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the whole field. Note that the group hierarchy should be defined by the GRUPTREE keyword in the SCHEDULE , when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	TYPE	A defined character string that defines the type of injection fluid. TYPE should be set to one of the following character strings: 1) GAS : for a gas injection well. 2) OIL : for a water injection well. 3) WAT : for a water injection well.			None
3	TARGET	A defined character string that sets the target injection control for the group, all the other phases will therefore act as constraints. The simulator will attempt to meet the TARGET based on the phase rate stated in items (4) to (7) on this keyword. TARGET should be set to one of the following character strings: 1) NONE : the group has no target phase, but if entered, constraints are still defined and active. 2) FLD : this group is controlled from a higher level group, including the FIELD group. 3) RATE : the injection phase will be control by the surface fluid rate for the phase defined by the TYPE variable. For example, if TYPE has been set to WAT then this would mean the water injection rate as defined by item (4). 4) RESV : the target is set to the in situ reservoir volume rate as defined by item (5). 5) REIN : the target is set to the group's production of the phase defined by TYPE multiplied by the value on item (6). For example, if TYPE has been set to WAT then this would mean the group's water production multiplied by item (6). 6) VREP : the target is set to the group's voidage replacement ratio as defined by item (7).			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	RATE	A real positive value that defines the maximum surface injection rate target or constraint for the phase declared by the TYPE variable. This value may be specified using a User Defined Argument (UDA).			None
		Liquid stb/d Gas Mscf/d	Liquid sm ³ /day Gas sm ³ /day	Liquid scc/hour Gas scc/hour	
5	RESV	A real positive value that defines the maximum reservoir volume injection rate target or constraint. This value may be specified using a User Defined Argument (UDA). Note setting a value here other than the default means that TYPE, item (2) will be the supplement or “make up” phase.			None
		rtb/d	rm ³ /day	rcc/hour	
6	REIN	A real positive value that defines the target or constraint re-injection fraction for the produced phase defined by the TYPE variable. This value may be specified using a User Defined Argument (UDA). For example, if TYPE is equal to GAS and REIN is equal to 0.85, then 85% of the produced gas will be re-injected.			None
		dimensionless	dimensionless	dimensionless	
7	VREP	A real positive value that defines the target or constraint of the voidage replacement ratio based on all the produced fluids. This value may be specified using a User Defined Argument (UDA). For example, if TYPE is equal to WAT and VREP is equal to 1.00, then 100% of the produced reservoir volume will be re-inject as an equivalent water volume. Note setting a value here other than the default means that TYPE, item (2) will be the supplement phase.			None
		dimensionless	dimensionless	dimensionless	
8	GRPCNTL	A defined character string that determines if this group is subject to higher level group control. 1) YES: then this group is subject to a higher level group's control and the flow rates for this group will be adjusted accordingly. 2) NO: then this group is NOT subject to a higher level group's control and the flow rates for this group will only be controlled by the parameters for this group. This variable is ignored if GRPNAME is equal to FIELD .			YES

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	GRPGUIDE	A real positive value that defines a group's injection guide rate expressed as a dimensionless number. A group requires a value for GRPGUIDE only if it is required to produce a specified proportion of a higher level group's rate. Defaulting GRPGUIDE results in the subordinate groups and wells under guide control having their rates dictated by any higher level groups under guide rate control. In other words the GRPNAME is masked out. Setting GRPGUIDE to a real positive value and GUIPHASE to either RATE or RESV will result in a constant injection guide rate.			None
		dimensionless	dimensionless	dimensionless	
10	GUIPHASE	A defined character string that sets the guide phase to which the guide rate in item (9) applies. GUIPHASE should be set to one of the following character strings: 1) NETV: the guide phase is set to production rate minus any injected reservoir rate from the other phases. The net volume is calculated at the beginning of a time step and the option should only be used for groups under voidage replacement control. Note that using this option means that TYPE, item (2) will be the supplement or "make up" phase and the value entered for the group's guide rate (GRPGUIDE) will be ignored with this option. 2) RATE: the guide phase is set to the surface injection rate. Setting GUIPHASE to RATE and GRPGUIDE to a real positive value will result in a constant injection guide rate. 3) RESV: the guide phase is set to the in situ reservoir volume rate. Setting GUIPHASE to RESV and GRPGUIDE to a real positive value will result in a constant injection guide rate. 4) VOID: the guide rate is calculated at the beginning of each time step based on the group's net voidage rate. OPM Flow now supports all guide rate options. The default value of I* means that the group has no injection guide rate for this phase.			None
11	GRPREIN	A character string of up to eight characters in length that defines the group name whose production rate should be used for applying the REIN quantity to be injected into GRPNAME. This variable is used to re-inject the REIN production fraction from another group (GRPREIN) via this group (GRPNAME). If GRPREIN is defaulted then the re-injection quantity for GRPNAME will be based on the production from GRPNAME itself.			GRPNAME
12	GRPVREP	A character string of up to eight characters in length that defines the group name whose production rate should be used for applying the VREP quantity to be injected into GRPNAME. This variable is used to re-inject the VREP production fraction from another group (GRPVREP) via this group (GRPNAME). If GRPVREP is defaulted then the voidage quantity for GRPNAME will be based on the production from GRPNAME itself.			GRPNAME
13	WGASRATE	Wet gas injection rate used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the SCHEDULE section. User Defined Arguments are activated using the UDADIMS keyword in the RUNSPEC section, and defined by the UDQ keyword in the SCHEDULE section. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.32: GCONINJE Keyword Description

See also the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [GCONPROD](#) keyword to define a group's production targets and constraints, the [WCONPROD](#) keyword to define a production well's targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the injection targets and constraints for the field and two groups that are one level below the field group, since the [GRUPTREE](#) keyword has not been entered to define the group hierarchy.

```
--
--          GROUP INJECTION TARGETS AND CONSTRAINTS
--
-- GRUP  FLUID CNTL  SURF  RESV  REINJ  VOID  GRUP  GUIDE  GUIDE  GRUP  GRUP
-- NAME  TYPE  MODE  RATE  RATE  FRAC  FRAC  CNTL  RATE  DEF  REINJ  RESV
GCONINJE
FIELD    WAT   VREP   35E3  1*    1*    1*    NO   1*    1*    1*    1*  /
GRP01    WAT   VREP    1*    1*    1*    1.0  YES  1*    1*    1*    1*  /
GRP02    WAT   VREP    1*    1*    1*    1.0  YES  1*    1*    1*    1*  /
/
```

In this example, group GRP01 and GRP02 are injecting water via voidage replacement with a voidage replacement of one and are under the control on the field group, that imposes a 35,000 m3/day total water injection limit.

12.3.89
GCONPRI
–
GROUP PRODUCTION PRIORITY TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCONPRI](#) keyword defines production targets and constraints for groups, including the top most group in the group hierarchy known as the FIELD group, for when groups and their associated wells are operating under priority control as oppose to guide rate control. Priority control is activated by the [PRIORITY](#) keyword in the [SCHEDULE](#) section. Wells are allocated to groups when the wells are specified by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	ORAT	A real positive value that defines the maximum surface oil production rate constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	OILACT	A defined character string that defines the action to be taken if ORAT is exceeded. OILACT should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion in the worst offending well will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: close the worst offending well. 5) PLUG: plug back the worst offending well based on the plug back length and options defined on the WPLUG keyword in the SCHEDULE. 6) PRI: control the group production rate using the first priority formulae defined by the PRIORITY keyword in the SCHEDULE section. 7) PR2: control the group production rate using the second priority formulae defined by the PRIORITY keyword in the SCHEDULE section.			"NONE"
4	WRAT	A real positive value that defines the maximum surface water production rate constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/d	sm ³ /day	scc/hour	
5	WATACT	A defined character string that defines the action to be taken if WRAT is exceeded. WATACT should be set to a character string described by the OILACT parameter on this record.			None
6	GRAT	A real positive value that defines the maximum surface gas production rate constraint This value may be specified using a User Defined Argument (UDA).			None
		Mscf/d	sm ³ /day	scc/hour	
7	GASACT	A defined character string that defines the action to be taken if GRAT is exceeded. GASACT should be set to a character string described by the OILACT parameter on this record.			None
8	LRAT	A real positive value that defines the maximum surface liquid (oil plus water) production rate constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/d	sm ³ /day	scc/hour	
9	LIQACT	A defined character string that defines the action to be taken if LRAT is exceeded. LIQACT should be set to a character string described by the OILACT parameter on this record.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
10	RESV	A real positive value that defines the maximum reservoir volume production rate constraint. This value may be specified using a User Defined Argument (UDA).			None
		rb/d	rm ³ /day	rcc/hour	
11	RESVFRAC	A real positive value that defines a group's maximum production balancing fraction constraint. This value may be specified using a User Defined Argument (UDA).			None
		dimensionless	dimensionless	dimensionless	
12		Not used should be defaulted with I*.			I*
13		Not used should be defaulted with I*.			I*
14		Not used should be defaulted with I*.			I*
15		Not used should be defaulted with I*.			I*
16	LINCOMB	A real positive value that defines the linearly combined maximum surface target rate or constraint, as per the LINCOM keyword in the SCHEDULE section.			I*
17	LINACT	A defined character string that defines the action to be taken if LINCOMB is exceeded. LINACT should be set to a character string described by the OILACT parameter on this record.			None

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the [SCHEDULE](#) section. User Defined Arguments are activated using the [UDADIMS](#) keyword in the [RUNSPEC](#) section, and defined by the [UDQ](#) keyword in the [SCHEDULE](#) section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.33: GCONPRI Keyword Description

See also the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [GCONPROD](#) keyword to define a group's production targets and constraints, the [GCONINJE](#) keyword to define a group's injection targets and constraints, the [WCONPROD](#) keyword to define a production well's targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the production targets and constraints for the field and two groups that are one level below the field group, since the [GRUPTREE](#) keyword has not been entered to define the group hierarchy.

```
--
--          GROUP PRIORITY PRODUCTION CONTROLS
--
-- GRUP  ORAT  ORAT  WRAT  WRAT  GRAT  GRAT  LRAT  LRAT  RVOL  RVOL
-- NAME  LIMIT ACTN  LIMIT ACTN  LIMIT ACTN  LIMIT ACTN  LIMIT ACTN
GCONPRI
FIELD   40E3  PRI    30E3  +CON  125E3  PRI    1*    1*    1*    1*    /
```

GRP01	25E3	PRI	1*	1*	1*	1*	1*	1*	1*	1*	/
GRP02	25E3	PRI	1*	1*	1*	1*	1*	1*	1*	1*	/
/											

All groups are controlled by oil constraints, but only the field level has water and gas constraints to reflect the actual production facility constraints. The oil production constraint of 40,000, 25,000 and 25,000 stb/d are defined for the field, GRP01 and GRP02 groups, respectively. If the oil rate constraint is exceeded then the wells will be controlled using the priority formulae one, as defined on the [PRIORITY](#) keyword in the [SCHEDULE](#) section. Similarly for the field, for when the gas constraint is exceeded. Finally, if the field water constraint is surpassed then the worst offending connection and below in the worst offending well are shut.

12.3.90 GCONPROD – GROUP PRODUCTION TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCONPROD](#) keyword defines production targets and constraints for groups, including the top most group in the group hierarchy known as the FIELD group. Wells are allocated to groups when the wells are specified by the [WEL SPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword in the SCHEDULE section, when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	TARGET	A defined character string that specifies the production rate control mode for the group. The simulator will attempt to meet the TARGET rate as defined by remaining items on this keyword. TARGET should be set to one of the following character strings: 1) NONE: the group has no target rate, but if entered, constraints are still defined and active. 2) FLD: this group is controlled by a higher level group (including the FIELD group) subject to its guide rate as defined by items (9) and (10). 3) ORAT: the target is set to the surface oil production rate as defined by item (3). 4) WRAT: the target is set to the surface water production rate as defined by item (4). 5) GRAT: the target is set to the surface gas production rate as defined by item (5). 6) LRAT: the target is set to the surface liquid (oil plus water) production rate as defined by item (6). 7) RESV: the target is set to the in situ reservoir volume rate as defined by item (14). Note that the commercial simulator includes additional options (not listed above) that are not currently supported by OPM Flow.			None
3	ORAT	A real positive value that defines the maximum surface oil production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	WRAT	A real positive value that defines the maximum surface water production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/d	sm ³ /day	scc/hour	
5	GRAT	A real positive value that defines the maximum surface gas production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		Mscf/d	sm ³ /day	scc/hour	
6	LRAT	A real positive value that defines the maximum surface liquid (oil plus water) production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/d	sm ³ /day	scc/hour	
7	ACTION	A defined character string that specifies the action to be taken if the constraints in (3) to (6) are violated. ACTION should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion in the worst offending well will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: close the worst offending well. 5) PLUG: plug back the worst offending well. 6) RATE: control the group production rate to equal the upper limit. This effectively changes the TARGET to be the violated phase constraint. The corrective action takes places at the end of the time step in which the constraint is violated. Note that only the NONE, WELL and RATE options are currently supported by OPM Flow.			None
8	GRPCNTL	A defined character string that determines if this group is subject to higher level group control. 1) YES: then this group is subject to a higher level group's control and the flow rates for this group will be adjusted accordingly. 2) NO: then this group is NOT subject to a higher level group's control and the flow rates for this group will only be controlled by the parameters for this group. GRPCNTL will be ignored for the FIELD group.			None
9	GRPGUIDE	A real positive value that defines a group's production guide rate expressed as a dimensionless number. A group requires a value for GRPGUIDE only if it is required to produce a specified proportion of a higher level group's rate			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
		dimensionless	dimensionless	dimensionless	
10	GUIPHASE	A defined character string that sets the guide phase to which the guide rate in item (9) applies. GUIPHASE should be set to one of the following character strings: 1) OIL: the guide phase is set to oil. 2) WAT: the guide phase is set to water. 3) GAS: the guide phase is set to gas. 4) LIQ: the guide phase is set to liquid (oil plus water). 5) COMB: the guide phase is set to a linear combination of phases with the coefficients specified by the LINCOM keyword in the SCHEDULE section. 6) INJV: the guide rate is set to the group's reservoir volume injection rate at the start of each timestep. 7) POTN: the guide rate is set to the group's production potential at the start of each timestep. 8) FORM: the guide rate will be based on the formulae defined via the GUIDERAT keyword in the SCHEDULE section. The formulae enables subordinate groups and wells to decrease their contribution from wells producing too much gas or too much water. 9) ' ' or I* : the group is not under guide rate control, and therefore superordinate group production targets will be pro-rated directly down to its subordinate groups and wells. Note that only the OIL, WAT, GAS, LIQ and default options are currently supported by OPM Flow.			I*
11	ACTWAT	A defined character string that defines the action to be taken if the WRAT constraint, item (4), is violated. ACTWAT should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion in the worst offending well will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: close the worst offending well. 5) PLUG: plug back the worst offending well. This option is not implemented in OPM Flow. 6) RATE: control the group production rate to equal the upper limit. This effectively changes the TARGET to be the violated phase constraint. If defaulted then procedure defined by ACTION, item (7), is applied. The corrective action takes places at the end of the time step in which the constraint is violated. Note that only the NONE and RATE options are currently supported by OPM Flow.			I*

No.	Name	Description			Default
		Field	Metric	Laboratory	
12	ACTGAS	A defined character string that defines the action to be taken if the GRAT constraint, item (5), is violated. ACTGAS should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion in the worst offending well will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: close the worst offending well. 5) PLUG: plug back the worst offending well. 6) RATE: control the group production rate to equal the upper limit. This effectively changes the TARGET to be the violated phase constraint. If defaulted then procedure defined by ACTION, item (7), is applied. The corrective action takes places at the end of the time step in which the constraint is violated Note that only the NONE and RATE options are currently supported by OPM Flow.			I *
13	ACTLIQ	A defined character string that defines the action to be taken if the LRAT constraint, item (6), is violated. ACLIQT should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion in the worst offending well will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: close the worst offending well. 5) PLUG: plug back the worst offending well. This option is not implemented in OPM Flow. 6) RATE: control the group production rate to equal the upper limit. This effectively changes the TARGET to be the violated phase constraint. If defaulted then procedure defined by ACTION, item (7), is applied. The corrective action takes places at the end of the time step in which the constraint is violated. Note that only the NONE and RATE options are currently supported by OPM Flow.			I *
14	RESV	A real positive value that defines the maximum reservoir volume production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		rb/d	rm ³ /day	rcc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
15	RESVFRAC	A real positive value that defines the maximum reservoir volume production balancing fraction. Not used and should be defaulted with 1*.			1*
16	WGASRATE	Wet gas production rate used in the commercial compositional simulator. Not used and should be defaulted with 1*.			1*
17	CALRATE	Calorific production rate used in the commercial compositional simulator. Not used and should be defaulted with 1*.			1*
18	GASFRAC	Surface gas production fraction used in the commercial compositional simulator Not used and should be defaulted with 1*.			1*
19	WATFRAC	Surface water production fraction used in the commercial compositional simulator. Not used and should be defaulted with 1*.			1*
20	COMBRATE	Linearly combined production rate used in the commercial compositional simulator. Not used and should be defaulted with 1*.			1*
21	COMBPROC	Linearly combined procure for when exceeding COMBRATE, used in the commercial black-oil simulator. Not used and should be defaulted with 1*.			1*

Notes:

1)

User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the [SCHEDULE](#) section. User Defined Arguments are activated using the [UDADIMS](#) keyword in the [RUNSPEC](#) section, and defined by the [UDQ](#) keyword in the [SCHEDULE](#) section.

2)

The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.34: GCONPROD Keyword Description

See also the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [GCONINJE](#) keyword to define a group's injection targets and constraints, the [WCONPROD](#) keyword to define a production well's targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the production targets and constraints for the field and two groups that are one level below the field group, since the [GRUPTREE](#) keyword has not been entered to define the group hierarchy.

```
--
--          GROUP PRODUCTION CONTROLS
--
-- GRUP  CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME  MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF    WAT
GCONPROD
FIELD   ORAT  40E3   60E3   300E3  60E3   1*    1*    1*    1*    1*    /
```

GRP01	FLD	25E3	1*	1*	1*	1*	1*	1*	1*	1*	/
GRP02	FLD	25E3	1*	1*	1*	1*	1*	1*	1*	1*	/
/											

All groups are controlled by oil rate targets or constraints, but only the field level has water, gas and liquid constraints to reflect the actual production facility constraints. The wells under group control will be produced based on oil potential of each of the wells under group control, such that the field oil production target of 40,000 stb/d is honored and subject to the other phase fluid constraints. In addition, GRP01 and GRP02 oil rate values of 25,000 stb/d are constraints as these two groups are subject to the FIELD level targets and constraints.

12.3.91 GCONSALE – DEFINE GROUP SALES GAS PRODUCTION TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[GCONSALE](#) defines group sales gas production targets and constraints for when the gas production from an oil field group is exported under a Gas Sales Agreement (“GSA”) and the oil field group also has oil production targets and constraints.

Note that the keyword should not be used to control sales gas for a gas field group, as the gas injection rate is used to control the sales gas production with this keyword, that is:

$$\begin{aligned} \text{Gas Sales Rate} = & \text{Total Group Gas Production Rate} \\ & - \text{Group Gas Injection Rate} \\ & + \text{Total Group Gas Import Rate} \\ & - \text{Total Group Gas Consumption} \end{aligned} \quad (12.20)$$

Thus, surplus gas that cannot be sold is re-injected, which requires that there are active gas injectors in the model that are subordinate to groups with gas sales targets. Note that the surplus gas re-injection rates are automatically calculated by OPM Flow at each time step.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group gas sales target and constraints are being defined. The group named FIELD is the top most group and should be used to set the gas sales targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	GSALE	GSALE should either be set to: 1) A real positive value that defines the gas sales rate for the group, or, 2) A real negative value that switches off both gas sales and gas re-injection for the group. This value may be specified using a User Defined Argument (UDA). Note that if GSALE has been set to switch off both gas sales and gas re-injection, then the GCONINJE keyword in the SCHEDULE section may be used to re-enable gas re-injection again.			None
		Mscf/d	sm ³ /day	scc/hour	
3	GSALEMAX	A real positive value that must be greater than GSALE that defines the maximum allowed gas sales rate. If GSALE exceeds GSALEMAX then the action defined by the ACTION variable on this keyword is implemented at the end of the current time step. This value may be specified using a User Defined Argument (UDA).			1 x 10 ²⁰
		Mscf/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	GSALEMIN	A real positive value that must be less than GSALE that defines the minimum allowed gas sales rate. If GSALE is less than GSALEMIN then one of the following actions will be implemented at the end of the current time step: 1) If the group's maximum gas production rate constraint is constraining the gas rate, then reset the constraint to satisfy the group's minimum gas sales rate (GSALEMIN), else: 2) If the group has active subordinate dedicated gas producers, as defined by the <i>WGASPROD</i> keyword in the <i>SCHEDULE</i> section, then reset their gas target rates to satisfy GSALEMIN, else: 3) If there are subordinate dedicated gas producers for the group in a drilling queue, open the next dedicated well and set the well's gas rate to satisfy GSALEMIN. Note that only wells that are subordinate to the group and are not under gas rate control or group prioritization are considered for opening. 4) If there are no appropriate subordinate dedicated gas producers for the group in a drilling queue, open the next non-dedicated well and set the well's gas rate to satisfy GSALEMIN. Again, note that only wells that are subordinate to the group and are not under gas rate control or group prioritization are considered for opening. If none of the above actions can be implemented then the minimum gas sales rate will not be satisfied. This value may be specified using a User Defined Argument (UDA).			-1×10^{20}
5	<i>ACTION</i>	A defined character string that defines the action to be taken if the maximum gas sales rate, GSALEMAX, is violated. <i>ACTION</i> should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion in the worst offending well will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: close the worst offending well. 5) PLUG: plug back the worst offending well. This option is not implemented in OPM Flow. 6) RATE: the group's production rate target is reduced to equal GSALEMAX, after accounting for fuel gas (<i>GCONSUMP</i> keyword) and the current rate of re-injection. This will also place the group on gas production control. 7) <i>END</i>: stop the simulation at the end of the report step. The corrective action takes places at the end of the time step in which the constraint is violated.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the <i>SCHEDULE</i> section. User Defined Arguments are activated using the <i>UDADIMS</i> keyword in the <i>RUNSPEC</i> section, and defined by the <i>UDQ</i> keyword in the <i>SCHEDULE</i> section. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.35: GCONSALE Keyword Description

[GCONSALE](#) can also be used with the [GCONINJE](#) keyword in the [SCHEDULE](#) section in order to apply additional limits, for example by applying a maximum group injection rate. In this scenario, the TARGET variable on the [GCONINJE](#) keyword must be set to "REIN", and if desired, a re-injection fraction (REIN on the [GCONINJE](#) keyword), or any other constraint.

See also the [GCONSUMP](#) in the [SCHEDULE](#) section that defines the fuel gas requirements for groups.

Example

The following examples sets the field gas sales target rate:

```
--
--      GROUP GAS SALES FOR OIL FIELDS
--
-- GRUP  GAS      MAX      MIN      CNTL
-- NAME  SALES    RATE     RATE     ACTN
GCONSALE
FIELD   40E3     50E3     20E3     RATE
/
```

Here the field has a gas sales target of 40 MMscf/d, with a maximum rate of 50 MMscf/d and a minimum of 20 MMscf/d. If the maximum gas sales rate is exceeded then the group’s gas production rate target is reduced to equal GSALEMAX, after accounting for fuel gas and the current rate of re-injection. This will also place the group on gas production control.

12.3.92 GCONSUMP – DEFINE GROUP GAS CONSUMPTION AND GAS IMPORT TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[GCONSUMP](#) defines the group gas consumption rate either as an actual rate or as a percentage of the group's production. In both oil and gas fields produced gas is commonly used as fuel to support the processing and utility facilities needed to run the plant.

In addition to defining gas consumption, the keyword can also be used to define the group's gas import rate, if required. This is used to import gas into the model from other sources (fields, reservoirs etc.) that are not included in the current run. For example, if a several fields are supplying gas to a power plant (field A, B and C), but only one is being modeled in the current import deck (A), then production from the other two fields (B and C) can be incorporated into model in order to meet the plant demand. Note in this case the import gas rates from fields B and C are fixed, and therefore field A acts like a "swing" producer to match the gas demand target.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group gas consumption is being defined. The group named FIELD is the top most group and should be used to set the fuel consumption for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	GASFUEL	A real value that defines the gas consumption, that is the fuel gas consumed by the group, either defined as a volumetric rate or as a fraction of the group's gas production. The two options are implemented by: 1) Volumetric Rate Option: Setting GASFUEL to a value greater than zero will be interpreted as the actual fuel gas rate consumed by the group. 2) Fraction of Produced Gas: Setting GASFUEL to a negative value between minus one and zero will be interpreted as a fraction of the groups production rate. For example, if GASFUEL is entered as -0.05, would mean 5% of the group's produced gas will be consumed as fuel. This value may be specified using a User Defined Argument (UDA).			0.0
		Mscf/d	sm ³ /day	scc/hour	
3	GASIMP	A real positive value greater than zero that defines the amount of gas to be imported into the group. This value may be specified using a User Defined Argument (UDA). This option is currently not supported by OPM Flow			0.0
		stb/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	GASNODE	A character string of up to eight characters in length that defines the network node in the Extended Network Model, for which the fuel gas should be removed (GASFUEL) or the imported gas (GASIMP) assigned. This option is currently not supported by OPM Flow			None

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the *SCHEDULE* section. User Defined Arguments are activated using the *UDADIMS* keyword in the *RUNSPEC* section, and defined by the *UDQ* keyword in the *SCHEDULE* section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.36: GCONSUMP Keyword Description

If the group is acting under Group Gas Sales control via the [GCONSALE](#) keyword in the [SCHEDULE](#) section, then the sales gas is calculated by:

$$\begin{aligned} \text{Gas Sales Rate} = & \text{Total Group Gas Production Rate} \\ & - \text{Group Gas Injection Rate} \\ & + \text{Total Group Gas Import Rate} \\ & - \text{Total Group Gas Consumption} \end{aligned} \quad (12.21)$$

If the group is acting under Group Gas Re-Injection control via the [GCONINJE](#) keyword in the [SCHEDULE](#) section, then the group gas injection rate calculated by:

$$\begin{aligned} \text{Group Gas Injection Rate} = & \text{Group Gas Injection Rate} \times \text{Group Re-Injection Fraction} \\ & + \text{Total Group Gas Import Rate} \\ & - \text{Total Group Gas Consumption} \end{aligned} \quad (12.22)$$

Note

In oil fields with no gas compression typical values of fuel gas range from three to five percent.

Example

The first example sets the fuel gas consumption to 3.0 MMscf/d for the field.

```
--
--      GROUP GAS CONSUMPTION (FUEL) AND IMPORT
--
--  GRUP   GAS      GAS
--  NAME   FUEL     IMPORT
--  -----
GCONSUMP
FIELD    3.0E3 /
```

The second example sets group PLAT-WST's fuel consumption to be 5% of the platform's produced gas and group PLAT-EST's to a constant 1.0 MMscf/d.

```
--
--      GROUP GAS CONSUMPTION (FUEL) AND IMPORT
--
```

-- GRUP	GAS	GAS	
-- NAME	FUEL	IMPORT	
--	-----	-----	
GCONSUMP			
PLAT-WST	-0.050		/
PLAT-EST	1.0E3		/
/			

12.3.93 GCONTOL – DEFINE GROUP CONSTRAINT TOLERANCE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GCONTOL](#) keyword defines the tolerance and parameters used to control the accuracy of group targets and constraints, including the field's targets and constraints. The keyword sets the tolerance and number of Newton iterations for each time step so that the wells under group control can match the desired group targets and constraints.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPCNV	GRPCNV is a real positive value less than one that sets the group tolerance criteria used to define if convergence has been satisfied for group's under rate control. Here, GRPCNV is an acceptable fraction of the group's rate target. Thus, the a numerical quantity 0.01 means that values must be with 0.01 (or 1.0%) of the groups' production target. For groups under priority control, as per GCONPRI keyword in the SCHEDULE section, GRPCNV is ignored. Note that this criteria may not be satisfied if the number of Newton iterations used in updating the well targets, as set by the NUPCOL keyword in the RUNSPEC section, or the NUPCOL parameter on this keyword, is exceeded. In this case, and only if the well potentials allow, the well rates are re-calculated ignoring the NUPCOL limit. Group tolerance criteria can also be set by the GRPCNV parameter on the NETBALAN keyword in the SCHEDULE section. If both values of GRPCNV have been entered, then the minimum of the two is used. The default value means there is no convergence criteria.			1.0 × 1020
		dimensionless	dimensionless	dimensionless	
2	NUPCOL	A positive integer that defines the maximum number of Newton iterations used to update well targets within a time step. Wells under group control may suffer from some dependency with other wells in the same group that are under group control. This may cause some oscillation in the production and injection well rates within the group. In order to avoid this, after the number Newton iterations within a time step surpasses NUPCOL, the group well rates are frozen until the time step has converged. Reducing the potential of well rate oscillations within the time step may result in the group targets and limits not being exactly being met in this case. Increasing the value of NUPCOL will improve the accuracy of the group targets and limits at the expense of computational efficiency. NUPCOL can also be set by the NUPCOL keyword in the RUNSPEC section. A value entered on this keyword overwrites any previously entered values of NUPCOL, including the value set by the NUPCOL keyword. The default value of 1* invokes the last previously entered value of either by the NUPCOL keyword or this keyword.			1*

No.	Name	Description			Default
		Field	Metric	Laboratory	
		dimensionless	dimensionless	dimensionless	
3	GASCNV	In the commercial compositional simulator GASCNV is a real positive value less than one that sets the tolerance criteria used to define if convergence has been satisfied for well gas injection rates. GASCNV is also used in conjunction with well availability gas fraction quantities in the commercial compositional simulator The value is ignored by OPM Flow.			1.0 x 10-3
		dimensionless	dimensionless	dimensionless	
4	GASMXITE	In the commercial compositional simulator GASMXITE is a positive integer value that defines the maximum number of iterations for the gas injection computation. The value is ignored by OPM Flow.			5
		dimensionless	dimensionless	dimensionless	

Notes:

1) The keyword is terminated by a “/”.

Table 12.37: GCONTROL Keyword Description

See also the [NUPCOL](#) keyword in the [RUNSPEC](#) section, and the [NETBALAN](#) and [WLINTOL](#) keywords in the [SCHEDULE](#) section that control the network balancing convergence criteria and the tolerance parameters for wells, respectively.

Examples

The example sets the group target convergence criteria to 1% (0.01) with a maximum of four Newton iterations ([NUPCOL](#)).

```
--
--          GROUP CONSTRAINT TOLERANCE
--
--          GROUP  NEWTON  INJEC  NEWTON
--          TOL    ITERS   TOL    ITERS
--          -----
GCONTROL
          0.01    4        1*     1*          /
```

12.3.94 GCUTBACK – DEFINE GROUP CUTBACK LIMITS AND PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GCUTBACK](#), defines a production group’s cutback limits and parameters. See also the [WCUTBACK](#) keyword in the [SCHEDULE](#) section that provides similar functionality for wells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.95 GCUTBACT – DEFINE GROUP TRACER CUTBACK LIMITS AND PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GCUTBACT](#), defines a production group’s cutback limits and parameters based on the named produced tracer from the group. See also the [WCUTBACT](#) keyword in the [SCHEDULE](#) section that provides similar functionality for groups.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.96 GDCQ – DEFINE GROUP MULTIPLE DAILY CONTRACT QUANTITIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GDCQ](#) keyword defines the Daily Contract Quantities (“DCQ”) for when multiple group contracts are required when the Gas Field Operations model has been activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section, or the GWSINGF has been invoked to define multiple group contracts in the [SCHEDULE](#) section. The group contracts must first be defined by the [GSWINGF](#) keyword, followed by the GCDQ keyword, and then the [GASYEAR](#) or [GASPERIO](#) keywords. GCDQ may be repeated in the [SCHEDULE](#) section to reset group DCQs.

See also the [SWINGFAC](#) keyword that set a single group DCQ at the field level, as opposed to having multiple DCQ group contracts using the [GDCQ](#) keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.97 GDCQECON – GROUP ECONOMIC CRITERIA FOR DCQ PRODUCTION GROUPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GDCQECON](#) keyword defines economic criteria for DCQ production groups, including the field level group [FIELD](#), that have previously been defined by the [GCONPROD](#) keywords in the [SCHEDULE](#) section. Note that wells are allocated to a group when they are specified by the [WELSPECS](#) keyword and wells can also have economic controls. Wells under group control are therefore subject to the economic criteria set via the [GCONPROD](#) and [CECON](#) keywords in the [SCHEDULE](#) section and the controls specified by the [WECON](#) keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the <i>GRUPTREE</i> keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	DCQ	A real positive value that defines the minimum economic DCQ gas production rate, below which an economic action of shutting in or stopping all the wells in the group, as requested by item (9) of the <i>WELSPECS</i> keyword. Note if GRPNAME is equal to <i>FIELD</i> then the run will be terminated. A value less than or equal to zero switches of this criteria.			0.0
		Mscf/d	sm ³ /day	scc/hour	

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.38: GDCQECON Keyword Description

See also the [WELSPECS](#) keyword to define a wells shut-in or stop options, [GCONPROD](#) for group controls, and [WECON](#) for setting a well's economic criteria. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the minimum DCQ for the field to be 10 MMscf/d.

```
--
--      GROUP ECONOMIC CRITERIA FOR DCQ PRODUCTION GROUPS
--
--  GRUP  GAS
--  NAME  DCQ
GDCQECON
FIELD    10E3
/
```

12.3.98
GDRILPOT – DEFINE GROUP POTENTIAL RATES FOR AUTOMATIC DRILLING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GDRILPOT](#), defines the minimum group potential rate that will result in a well from the one of the automatic drilling queues, as defined by either the [QDRILL](#) or [WDRILPRI](#) keywords in the [SCHEDULE](#) section, to be drilled and placed on production. The advantage of using a group’s potential, as oppose to a minimum rate limit, is that setting the potential greater than the group’s minimum flow rate, will result in well being drilled in time to support the desired production rate.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.99 GECON – GROUP ECONOMIC CRITERIA FOR PRODUCTION GROUPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GECON](#) keyword defines economic criteria for production groups, including the top level FIELD group, that have been created by having wells assigned to them via the [WELSPECS](#) keyword in the [SCHEDULE](#) section.

Note that wells are allocated to a group when they are specified by the [WELSPECS](#) keyword and wells can also have well level economic controls. Wells under group control are therefore subject to the economic criteria set via the [GECON](#) and [WECON](#) keywords and the controls specified by the [GCONPROD](#) keyword in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	ORAT	A real positive value that defines the minimum economic surface oil production rate, below which an economic action of shutting in or stopping all the wells in the group, as requested by the AUTO variable item (9) of the WELSPECS keyword. This value may be specified using a User Defined Argument (UDA). A value less than or equal to zero switches off this criteria.			0.0
		stb/d	sm ³ /day	scc/hour	
3	GRAT	A real positive value that defines the minimum economic surface gas production rate, below which an economic action of shutting in or stopping all the wells in the group, as requested by the AUTO variable item (9) of the WELSPECS keyword. This value may be specified using a User Defined Argument (UDA). A value less than or equal to zero switches off this criteria,			0.0
		Mscf/d	sm ³ /day	scc/hour	
4	WCUT	A real positive value that defines the maximum economic surface water cut, above which an economic action will take place. This value may be specified using a User Defined Argument (UDA). Water cut is defined as: $f_w = \frac{q_w}{q_w + q_o}$, and the various actions that are available if the water cut limit is exceeded are described in item (7). A value less than or equal to zero switches off this criteria.			0.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	GOR	A real positive value that defines the maximum economic surface gas-oil ratio, above which an economic action will take place, as defined by item (7). This value may be specified using a User Defined Argument (UDA). A value less than or equal to zero switches off this criteria.			0.0
		Mscf/stb	sm ³ /sm ³	scc/scc	
6	WGR	A real positive value that defines the maximum economic surface water-gas ratio, above which an economic action will take place, as defined by item (7). This value may be specified using a User Defined Argument (UDA). A value less than or equal to zero switches off this criteria.			0.0
		stb/Mscf	sm ³ /sm ³	scc/scc	
7	ACTION	A defined character string that defines the action to be taken if the economic WCUT, GOR, or WGR limits are violated. ACTION should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending well. If connections have been grouped as completions then the worst offending completion will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it will be closed. 4) WELL: shut or stop the worst offending well as per the AUTO variable on the WEL SPECS keyword. 5) PLUG: plug back the worst offending well based on the plug back length and options defined on the WPLUG keyword in the SCHEDULE. The corrective action takes places at the end of the time step in which the constraint is violated. Only the NONE option is currently supported by the simulator.			None
8	END	A defined character string that defines if the simulation should terminate if all the producing wells in the group, including the FIELD group, are shut or stopped. END should be set to one of the following character strings: 1) NO: no action is taken and the run continues. 2) YES: terminate the run at the next report time step. Only the NO option is currently supported by the simulator.			NO
9	MXWELS	A positive integer defining the maximum number of producing and injecting wells for this this group and any subordinate groups. The default value of zero implies that there is no limit to the number of wells. This is not currently supported by the simulator and must be defaulted.			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes:					
1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the SCHEDULE section. User Defined Arguments are activated using the UDADIMS keyword in the RUNSPEC section, and defined by the UDQ keyword in the SCHEDULE section. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.39: GECON Keyword Description

See also the [WEL SPECS](#) keyword to define a well's shut-in or stop options, [GCONPROD](#) for group controls, and [WECON](#) for setting a well's economic criteria. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the economic criteria for the field with a minimum oil rate of 2,000 m³/day and a maximum water cut of 95%.

```
--
--      GROUP ECONOMIC CRITERIA FOR PRODUCTION GROUPS
--
-- GRUP  OIL    GAS    WCT    GOR    WGR    WORK  END  MAX
-- NAME  MIN    MIN    MAX    MAX    MAX    OVER  RUN  WELLS
GECON
FIELD   2E3    1*     0.95    1*     1*     CON  'YES'  1*  /
/
```

If the economic limits are violated then the run will stop at the next report time step.

12.3.100 GECONT – GROUP TRACER ECONOMIC CRITERIA FOR PRODUCTION GROUPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **GECONT** keyword defines tracer economic criteria for production groups, including the field level group **FIELD**, that have previously been defined by the **GCONPROD** keywords in the **SCHEDULE** section, for tracers define by the **TRACER** keyword in the **PROPS** section.

Note that wells are allocated to a group when they are specified by the **WELSPECS** keyword and wells can also have economic controls. Wells under group control are therefore subject to the economic criteria set via the **GCONPROD** and **CECON** keywords in the **SCHEDULE** section and the controls specified by the **WECON** keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
I-2	ACTION	A defined character string that defines the action to be taken if the economic WCUT , GOR , or WGR limits are violated. ACTION should be set to one of the following character strings: 1) NONE : no action is taken. 2) CON : close the worst offending connection in the worst offending. If connections have been grouped as completions then the worst offending completion will be closed. 3) +CON : close the worst offending connection and all below it in the worst offending well. If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL : shut or stop the well as per the AUTO variable on the WELSPECS keyword. 5) PLUG : plug back the worst offending well based on the plug back length and options defined on the WPLUG keyword in the SCHEDULE . The corrective action takes places at the end of the time step in which the constraint is violated.			None
I-3	END	A defined character string that defines if the simulation should terminate if all the producing wells in the group, including the FIELD group, are shut or stopped. END should be set to one of the following character strings: 1) NO : no action is taken and the run continues. 2) YES : terminate the run at the next report time step.			NO

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-4	MXWELS	A positive integer defining the maximum number of producing and injecting wells for this group and any subordinate groups. The default value of zero implies that there is no limit to the number of wells.			0
1-5	/	Record one terminated by a “/”			Not Applicable
2-1	NAME	A three letter character string defining the tracer’s name. Note it is best to void names beginning with the letters F, S, and T as these names may create naming issues in post-processing software.			None
2-2	MXTOTAL	A real positive value that defines the maximum total (free plus solution) tracer rate. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-3	MXFREET	A real positive value that defines the maximum total (free plus solution) tracer concentration. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-4	MXFREEQ	A real positive value that defines the maximum free tracer rate. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-5	MXCONC	A real positive value that defines the maximum free tracer concentration. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-6	MXSOLNQ	A real positive value that defines the maximum solution rate. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-7	MXSOLNC	A real positive value that defines the maximum solution concentration. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-8	/	Record two terminated by a “/”			Not Applicable
3-1	/	Group terminated by a “/”			Not Applicable

Notes:

- 1) **GECONT** keyword consists of two records, with entries 1-1 to 1-5 representing record one items and 2-1 to 2-8 representing record number two items, in the “No.” column in this table. A maximum of three type two records can be entered following a type one record.
- 2) Each type one and type two records are terminate by a “/” as indicated in the table, and a group data set is terminated by a further “/”, after which additional group data sets can be entered stating with a record of type one followed by type two.
- 3) The keyword the keyword should be terminated by an additional “/” after the group data set termination “/” character.

Table 12.40: GECONT Keyword Description

See also the **WEL SPECS** keyword to define a wells shut-in or stop options, **GCONPROD** for group controls, and **WECON** for setting a well’s economic criteria. All the aforementioned keywords are described in the **SCHEDULE** section.

Example

The following example defines the tracer economic criteria for the field and two groups, FLTBLK1 and FLTBLK,.

```

--
--          GROUP TRACER ECONOMIC CRITERIA FOR PRODUCTION GROUPS
--
--  GRUP  WORK   END   MAX
--  NAME  OVER   RUN   WELLS
GECONT
FIELD    +CON    'YES'  1*                               /  START OF GROUP
--
--          TRACER  TRACER  TRACER  TRACER  TRACER  TRACER  TRACER
--          NAME    TOTAL   TOTAL   FREE   FREE   SOLN   SOLN
--          RATE    CONCEN  RATE    CONCEN  RATE    CONCEN
--          PLY      1000.0
--          BRI      1000.0
--          TR1      1*      0.7500
--
FLTBLK1  +CON    'YES'  1*                               /  START OF GROUP
--
--          TRACER  TRACER  TRACER  TRACER  TRACER  TRACER  TRACER
--          NAME    TOTAL   TOTAL   FREE   FREE   SOLN   SOLN
--          RATE    CONCEN  RATE    CONCEN  RATE    CONCEN
--          PLY      800.0
--          BRI      800.0
--
FLTBLK2  +CON    'YES'  1*                               /  START OF GROUP
--
--          TRACER  TRACER  TRACER  TRACER  TRACER  TRACER  TRACER
--          NAME    TOTAL   TOTAL   FREE   FREE   SOLN   SOLN
--          RATE    CONCEN  RATE    CONCEN  RATE    CONCEN
--          PLY      800.0
--          BRI      800.0
--
--          /  END OF GROUP
--          /  END OF KEYWORD

```

If the economic limits are violated then the worst offending connection and all below it in the worst offending well will be closed, If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed.

12.3.101 GEFAC – DEFINE GROUP EFFICIENCY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

Defines a group's efficiency or up-time factor as opposed to setting the efficiency factors for individual wells.

Note that wells are allocated to a group when they are specified by the [WELSPECS](#) keyword and wells can also have efficiency factors.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group efficiency factor is being defined. The group named FIELD is the top most group and cannot have an efficiency factor set. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	FACTOR	A real positive value less than or equal to one that defines the efficiency factor for the group. If a group's down time is 5% then FACTOR should be set to 0.95 (1.0 – 0.05).			1.0
		dimensionless	dimensionless	dimensionless	
3	GRPNETWK	A defined character string that determines if the GRPNAME efficiency factor should be transferred to the equivalent Extended Network Model node, and should be set to either: 1) NO: The group's equivalent Extended Network Model node flow rates are not multiplied by the efficiency factor FACTOR. 2) YES: The group's equivalent Extended Network Model node flow rates are multiplied by the efficiency factor FACTOR This option is only applicable for the Extended Network Model. In the Standard Network Model groups' reported flow rates are always used in the calculation of pressure drops.			YES
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.41: GEFAC Keyword Description

See also the [WEFAC](#) and [NEFAC](#) keywords in the [SCHEDULE](#) section to define the efficiency factor for a well or a node.

Example

```
--
--      GROUP EFFICIENCY FACTORS
--
-- GRUP  EFF      NETWK
-- NAME  FACT      OPTN
--      -----
GEFAC
PLATFORM 0.950
SUBSEA1  0.860
/
```

In the above example the group PLATFORM has its up-time factor set to 95% and the subsea group SUBSEA1 has an up-time factor of 86%.

12.3.102 GLIFTLIM – GROUP ARTIFICIAL LIFT CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GLIFTLIM](#) keyword defines the maximum number of wells on artificial lift and the maximum amount of the artificial lift that is available for a group, including the top most group in the group hierarchy known as the FIELD group. Wells are allocated to groups when the wells are specified by the [WELSPERCS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group's artificial lift constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	MXLIFT	A real positive value that defines the total amount of artificial lift available for this group and any subordinate groups. The units for MXLIFT are the same as that defined by the ALQ parameter on the VFPPROD keyword in the SCHEDULE section. For example, if ALQ has been set to GRAT on the VFPPROD keyword, then MXLIFT would be the maximum amount of gas lift gas available for this group and any subordinate groups, and the units would Mscf, assuming FIELD units had been activated in the RUNSPEC section. The default value of zero implies that there is no limit applied to the group and its subordinate groups.			0
		See VFPPROD (ALQ)	See VFPPROD (ALQ)	See VFPPROD (ALQ)	
3	MXWELS	A positive integer defining the maximum number of producing wells on artificial lift for this group and any subordinate groups. The default value of zero implies that there is no limit to the number of wells.			0
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.42: GLIFTLIM Keyword Description

See also the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection rate targets and constraints, the [WCONPROD](#) keyword to define a production well's targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

Example

The following example defines the artificial lift constraints for the field, assuming all the wells are on gas lift.

```
---
--      GROUP ARTIFICIAL LIFT CONSTRAINTS
--
-- GRUP  MAX    MAX
-- NAME  ALQ    WELLS
GLIFTLIM
FIELD    20E3  20                                /
/
```

Here the maximum amount of gas lift gas for the field is set to 20.0 MMscf/f and a maximum of 20 wells can utilize gas lift at a time.

12.3.103 GLIFTOPT – DEFINE GROUP GAS OPTIMIZATION LIMITS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GLIFTOPT](#) keyword defines the maximum amount of gas lift gas available and the maximum amount of gas the group can produce, including the top most group in the group hierarchy known as the FIELD group, for when gas lift optimization has been activated via the [LIFTOPT](#) keyword in the [SCHEDULE](#) section. Note that the [LIFTOPT](#) keyword should precede the [GLIFTOPT](#) keyword in the [SCHEDULE](#) section in order to activate the gas lift optimization facility.

Wells are allocated to groups when the wells are specified by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells, including any well gas lift optimization parameters on the [WLIFTOPT](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group's gas lift optimization parameters are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	MXLIFT	A real value that defines the total amount of gas lift gas available for this group and any subordinate groups, multiplied by their respective efficiency factors. The units for MXLIFT are the same as that defined by the ALQ parameter on the VFPPROD keyword in the SCHEDULE section. In this case ALQ should be GRAT on the VFPPROD keyword, as MXLIFT applies to the maximum amount of gas lift gas available. The default value of zero, or a negative value implies that there is no limit applied to the group and its subordinate groups.			0
		Mscf/d	sm ³ /day	scc/hour	
3	MXGAS	A real value that defines the total amount of gas the group can process. This is the sum of the gas lift gas plus the produced gas for this group and any subordinate groups, multiplied by their respective efficiency factors. The default value of zero, or a negative value implies that there is no limit applied to the group and its subordinate groups			0
		Mscf/d	sm ³ /day	scc/hour	

Notes:
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.43: GLIFTOPT Keyword Description

See also the [LIFTOPT](#) keyword to activate gas lift optimization, the [WLIFTOPT](#) keyword to define the wells under gas lift optimization control, the [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection rate targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example first switches on gas lift optimization via the [LIFTOPT](#) keyword and then defines the artificial lift constraints for the field, assuming all the well are on gas lift, using the [GLIFTOPT](#) keyword.

```
--          ACTIVATE GAS LIFT OPTIMIZATION AND PARAMETERS
--
-- INCR      INCR      TSTEP      NEWTON
-- GAS       OIL       INTVAL     OPTN
LIFTOPT
12.5E3      5E-3      0.0        YES
/
--
--          GROUP GAS LIFT OPTIMIZATION CONSTRAINTS
--
-- GRUP      MAX              MAX
-- NAME      GAS ALQ          TOTAL GAS
GLIFTOPT
FIELD       200E3           1*
/
```

Here the [LIFTOPT](#) keyword defines the maximum incremental gas lift gas quantity to be 12.5 × 10³ m³, the minimum incremental oil gain per m³ of gas lift gas is set to 5.0 × 10⁻³ m³, the time step interval is set to zero to perform the gas optimization every time step, and finally the gas lift optimization will be performed [NUPCOL](#) Newton iterations for the time step.

The [GLIFTOPT](#) sets the maximum amount of gas lift gas for the field to 200,000 m³ and there is no maximum limit for the total maximum amount of gas that the group can process.

12.3.104 GNETDP – GROUP NETWORK PRESSURE AND RATE CONTROLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GNETDP](#) keyword sets a group’s minimum and maximum network pressure and rate controls for when the either the Standard Network or the Extended Network options have been activated, and the group is part of a network. The keywords allows for the pressure of the group to vary in order to satisfy the rate conditions declared by this keyword. The Standard Network option is invoked if the [GRUPTREE](#), [GRUPNET](#), [GNETINJE](#), [GNETPUMP](#), etc. series of keywords have been used in the [SCHEDULE](#) section. Whereas, the Extended Network option is activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section. Several keywords, including, [GNETDP](#), can be used by both network options.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.105 GNETINJE – DEFINE GROUP INJECTION NETWORK CONFIGURATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GNETINJE](#) keyword defines the configuration of a group injection network for when the either the Standard Network or the Extended Network options have been activated. The Standard Network option is invoked if the [GRUPTREE](#), [GRUPNET](#), [GNETINJE](#), [GNETPUMP](#), etc. series of keywords have been used in the [SCHEDULE](#) section. Whereas, the Extended Network option is activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section. Several keywords, including, [GNETINJE](#), can be used by both network options.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.106
GNETPUMP – STANDARD NETWORK AUTOMATIC COMPRESSOR AND PUMPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GNETPUMP](#) keyword defines the configuration of automatic compressors and pumps in a production Standard Network, for when the Standard Network option is invoked by the [GRUPTREE](#), [GRUPNET](#), [GNETINJE](#), [GNETPUMP](#), etc., series of keywords in the [SCHEDULE](#) section. Although several keywords can be used by both the Standard and Extended Network options, [GNETPUMP](#) can only be used with the Standard Network option. The equivalent keyword for the Extended Network option is the [NETCOMPA](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.107 GPMaint – Define Group Pressure Maintenance Targets and Controls

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GPMaint](#) keyword defines the groups under pressure maintenance control, the associated flow rate and pressure targets, and fluid in-place regions associated with pressure maintenance, as well as various pressure maintenance controls. [GPMaint](#) allows for various regions, as defined by the [FIPNUM](#) or [FIP](#) keywords in the [REGIONS](#) section, to have their average reservoir pressure maintained at a specified value.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group's associated FIPNUM region will have a targeted average reservoir pressure maintained. The group named FIELD is the top most group and can also be used to set pressure target for the whole field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	GRPCNTL	A defined character string of length four, that sets the production or injection control for the group, used to maintain the average pressure for the FIPNUM region. GRPCNTL should be set to one of the following character strings: 1) NONE: the group will no longer maintain the average reservoir pressure for the FIPNUM region by adjusting production and injection rates. 2) GINJ: GRPNAME's gas in situ reservoir volume injection rate (RESV) will be adjusted in order to maintain FIPNUM 's average reservoir pressure. 3) GINS: GRPNAME's gas surface rate (GRAT) will be used to maintain FIPNUM 's average reservoir pressure. 4) OINJ: GRPNAME's oil in situ reservoir volume injection rate (RESV) will be adjusted in order to maintain FIPNUM 's average reservoir pressure. 5) OINS: GRPNAME's oil surface rate (ORAT) will be used to maintain FIPNUM 's average reservoir pressure. 6) PROD: GRPNAME's total in situ reservoir volume production rate (RESV) will be adjusted in order to maintain FIPNUM 's average reservoir pressure. Note that group cannot be under prioritization group control, as per the GCONPRI keyword in the SCHEDULE section, with this option. 7) WINJ: GRPNAME's water in situ reservoir volume injection rate (RESV) will be adjusted in order to maintain FIPNUM 's average reservoir pressure. 8) WINS: GRPNAME's water surface rate (WRAT) will be used to maintain FIPNUM 's average reservoir pressure.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	<i>FIPNUM</i>	A positive integer value that defines the region for which pressure maintenance is to be applied by GRPNAME, where <i>FIPNUM</i> may be define as per the: 1) <i>FIPNUM</i> keyword in the <i>REGIONS</i> section. 2) The <i>FIP</i> keyword in the <i>REGIONS</i> section, used to define the fluid in-place name and the associated region numbers for each grid block. This option is used in conjunction with FIPNAME (item (4)) to define the name of fluid in-place region and <i>FIPNUM</i>, which in this case is the region number associated with FIPNAME. 3) Alternatively, a value of zero may be entered, in which case pressure maintenance is applied to the whole field. The maximum number of <i>FIPNUM</i> and <i>FIP</i> regions is set by the <i>REGDIMS</i>(NMFIPR) or the <i>TABDIMS</i>(NTFIP) keywords(variables) in the <i>RUNSPEC</i> section. If both <i>REGDIMS</i>(NMFIPR) and <i>TABDIMS</i>(NTFIP) have been defined then the maximum of the two is used.			None
4	FIPNAME	A character string of up to five characters in length, defining the fluid in-place's name, as defined by the <i>FIP</i> keyword in the <i>REGIONS</i> section. For example, if the <i>FIP</i> keyword has been used to define a <i>FIP</i> group or family, named FIPBLK-A, then FIPNAME would be set to BLK-A. The default value of I* means that <i>FIPNUM</i> on this keyword applies to the standard <i>FIPNUM</i> array and not to the <i>FIP</i> defined group regions.			I*
5	TARGET	A real positive value that defines the average hydrocarbon pore volume weighted reservoir pressure target for the region, for which GRPNAME should attempt to satisfy. Note that the average pressure report in the <i>RPTSCHED</i> series of reports is a pore volume weight average reservoir pressure, not the average hydrocarbon pore volume weighted reservoir pressure specified by TARGET on this keyword. Thus, there will be small differences between the two numbers.			None
		psia	barsa	atma	
6	ALPHA	A real positive value that defines the proportionality constant used to control the flow rates in order to maintain/reach the TARGET average hydrocarbon pore volume pressure for the region. See equation (12.23). Larger values of ALPHA accelerate the time for the region's pressure to satisfy the required target pressure (TARGET). However, ALPHA values that are too large may cause the region's pressure to oscillate. Thus, the value of ALPHA should be a value that gives the expected steady state flow rate (TARGET) of the group divided by a reasonable transient pressure error.			None
		Liquid: stb/d/psi Gas: Mscf/d/psi RESV: rb/d/psi	Liquid: sm ³ /day/bars Gas: sm ³ /day/bars RESV: rm ³ /day/bars	Liquid: scc/hour/atm Gas: scc/hour/atm RESV: rcc/hour/atm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	BETA	A real positive value that defines the time interval used to control the flow rates in order to maintain the TARGET average hydrocarbon pore volume pressure for the region. See equation (12.23). Larger values of BETA reduce the tendency for the regional pressure to oscillate, but consequently decrease the rate at which the pressure reaches its target value. This is because the pressure error is measured at the end of the previous time step, resulting in a “delay” in the response. Also, the larger the time steps the greater the propensity for the pressure to oscillate for given ALPHA and BETA. As a guide BETA should be set to a value that is at least as large as the maximum time step size (see the TUNING keyword in the SCHEDULE section and section 2.2 Running OPM Flow From The Command Line for setting the maximum time step size).			
		day	day	day	*

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.44: GPMaint Keyword Description

The [GPMaint](#) keyword utilizes control theory^{318, 319, 320} and ³²¹ to calculate the required injection and production rates to ensure some stability in the behavior of the average hydrocarbon pore volume reservoir pressure for a given region and the injection rates over time. For example, if the group’s flow control (GRPCNTL) is set to one of the in situ reservoir volume injection rates (GINJ, OINJ or WINJ), say Q , then the group’s flow injection rates for each time step is calculated by:

$$Q = Q_i + \alpha \left((P_{TARGET} - P_{i-1}) + \frac{\sum_{i=1}^{i-1} (P_{TARGET} - P_{i-1}) \times \Delta t_i}{\beta} \right) \quad (12.23)$$

Where:

- P_{TARGET} = the regions average hydrocarbon pore volume weighted reservoir pressure target, the TARGET parameter in Table 12.44.
- P_{i-1} = the resulting average hydrocarbon pore volume weighted reservoir pressure for the region from the preceding completed time step.
- Q = the new in situ reservoir volume rate.
- Q_i = the initial in situ reservoir volume rate at the time the keyword was activated in the input. Subsequent [GPMaint](#) keywords will reset Q_i .
- α = the proportionality constant used to control the flow rates, the ALPHA parameter in Table 12.44.
- β = the time interval parameter used to control the flow rates, the BETA parameter in Table 12.44.

³¹⁸ John Doyle, Bruce Francis, Allen Tannenbaum, *Feedback Control Theory*, Macmillan Publishing Co., 1990.

³¹⁹ Karl J. Åström; Richard M. Murray (2008). *Feedback Systems: An Introduction for Scientists and Engineers*. Princeton University Press. ISBN 978-0-691-13576-2.

³²⁰ Andrei, Neculai (2005)., *Modern Control Theory – A Historical Perspective*, Research Institute for Informatics, Center for Advanced Modeling and Optimization, 8-10, Avereșcu Avenue, Bucharest 1, Romania.

³²¹ Sontag, Eduardo (1998). *Mathematical Control Theory: Deterministic Finite Dimensional Systems. Second Edition*, Springer. ISBN 978-0-387-98489-6.

The $\sum_{i=1}^{i-1} (P_{TARGET} - P_{i-1}) \times \Delta t_i$ term in equation (12.23) is the cumulative sum of the average hydrocarbon pore volume weighted reservoir pressure error for the region, times the time step length, from when the last [GPMAINT](#) keyword was entered, up to the preceding completed time step.

The objective of equation (12.23) is to stabilize the injection/production rates for the group for when there are changes to the average hydrocarbon pore volume weighted reservoir pressure target set by the initial and subsequent [GPMAINT](#) keywords, or by any group constraints applied afterwards. In addition, if the pressure error is small ($P_{TARGET} - P_{i-1}$) then the equation should result in approximately the same rates as the previous time step, if the [GPMAINT](#) has been previously entered.

However, if the first time the keyword is entered coincides with the start of injection, then Qi in equation (12.23) by definition will be zero, and the injection rate will start from zero and increase as calculated by the equation. The ALPHA and BETA parameters in Table 12.44 can be used to control how the rate is built up to meet the desired average hydrocarbon pore volume weighted reservoir pressure target.

Examples

The first example uses the gas surface rate to maintain the fields' average hydrocarbon pore volume pressure at 225 barsa, with the ALPHA and BETA parameters set to 40.0 and 70.0 respectively.

```
--
--          GROUP PRESSURE MAINTENANCE TARGETS AND CONTROLS
--
-- GRUP  CNTL  FIPNUM  FIP      PRESS  ALPHA  BETA
-- NAME  MODE  REGION  FIPNAME TARGET  CONST  CONST
GPMAINT
FIELD    GINS    0      1*      225    40.0   70.0      /
/
```

The second example uses group's BLK-A gas surface rate to maintain the average hydrocarbon pore volume pressure at 200 barsa for [FIPNUM](#) regions one to four, with the ALPHA and BETA parameters set to 40.0 and 70.0 respectively.

```
--
--          GROUP PRESSURE MAINTENANCE TARGETS AND CONTROLS
--
-- GRUP  CNTL  FIPNUM  FIP      PRESS  ALPHA  BETA
-- NAME  MODE  REGION  FIPNAME TARGET  CONST  CONST
GPMAINT
BLK-A    GINS    1      1*      200    40.0   70.0      /
BLK-A    GINS    2      1*      200    40.0   70.0      /
BLK-A    GINS    3      1*      200    40.0   70.0      /
BLK-A    GINS    4      1*      200    40.0   70.0      /

BLK-B    WINJ    1      FLT-B    215    30.0   65.0      /
BLK-B    WINJ    2      FLT-B    215    30.0   65.0      /
BLK-B    WINJ    3      FLT-B    215    30.0   65.0      /
BLK-B    WINJ    4      FLT-B    215    30.0   65.0      /
/
```

For group BLK-B, water in situ reservoir volume injection rate (RESV) is used to maintain [FIP](#) group/family FLT-B's average reservoir pressure at 215 barsa for FIPFLT-B regions one to four. Here the ALPHA and BETA parameters set to 30.0 and 65.0 respectively.

12.3.108 GRADGRUP – DEFINE GROUP HISTORY MATCH GRADIENT FILE OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GRADGRUP](#) keyword defines the [SUMMARY](#) field and group vectors that should be written to the History Match Gradient output file, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.109 GRADRESV – DEFINE SOLUTION DERIVATIVE HISTORY MATCH GRADIENT OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GRADRESV](#) keyword defines the [SOLUTION](#) derivative arrays that should be written to the History Match Gradient output file, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.110 GRADRFT – DEFINE RFT DERIVATIVE HISTORY MATCH GRADIENT OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [GRADRFT](#) keyword defines the derivative well RFT data, the [SOLUTION](#) pressure and saturations at a well's connected grid block, that should be written to the History Match Gradient output file, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.111 GRADWELL – DEFINE WELL HISTORY MATCH GRADIENT FILE OUTPUT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GRADWELL](#) keyword defines the [SUMMARY](#) well vectors that should be written to the History Match Gradient output file, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.112 GRDREACH – DEFINE RIVER AND GRID BLOCK CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [GRDREACH](#) keyword defines the location of grid blocks connecting to a previously defined river, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.113 GRUPMAST – DEFINE MASTER AND SLAVE GROUPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [GRUPMAST](#) keyword defines master groups and their associated slave groups for when the Reservoir Coupling option as been activated by the [GRUPMAST](#) and [SLAVES](#) keywords in the [SCHEDULE](#) section. Reservoir coupling allows for independent reservoir simulation decks ([SLAVES](#)) to be controlled by a separate master run file. For example, if there are five separate reservoir models each representing one field, one of the four would be used as the master and the other four would be the subordinate [SLAVES](#).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.114 GRUPNET – DEFINE GROUP STANDARD NETWORK PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GRUPNET](#) keyword defines the standard group network parameters used to model the flow and pressures behavior within the network. The group hierarchy is defined by the [GRUPTREE](#) keyword and wells are assigned to groups using the [WELSPECS](#) keyword, both keywords are in the [SCHEDULE](#) section.

Group pressure values are optionally entered for each group in the network together with a vertical flow (lift) performance (“VFP”) table that determines the pipeline pressure behavior from the LOWER to the HIGHER group given the current flowing conditions; the group relationship is defined by the [GRUPTREE](#) keyword. The VFP table is entered via the [VFPPROD](#) keyword for production pipelines and [VFPINJ](#) for injection pipelines. These are the same keywords as used for well modeling, however, the manner in which they are generated by an external program is completely different when used for pipeline modeling.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the network parameters are being defined. The group named FIELD is the top most group and may be used as a GRPNAME.			None
2	PRES	A real value that defines the fixed pressure for this group when the group is a terminating group. If the group is not a terminating group then PRES should be defaulted with I* or set to a negative number.			I*
		psia	barsa	atma	
3	VFPTAB	A positive integer greater than or equal to zero that defines the VFPPROD or VFPINJ vertical lift performance table to be used for calculating the pipeline pressures connecting the LOWER and HIGHER groups in the network. Note that: 1) The default value of zero implies that there is no pipeline connecting the LOWER and HIGHER groups. 2) If PRES is set to a real positive number then VFPTAB should be set to zero as this implies that GRPNAME is a terminating group and therefore there is no pipeline connecting GRPNAME to a HIGHER group. 3) If PRES and VFPTAB are defaulted with I* or zero, then GRPNAME is not part of the network. 4) IF VFPTAB is set equal to 9999 then this implies that there is no pressure change between the LOWER and HIGHER group. If a non-zero value is entered then the vertical lift performance tables must be entered via the VFPPROD or VFPINJ keyword in the SCHEDULE section.			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	ALQ-PIPE	A real positive value that defines the artificial lift quantity to be used in conjunction with the vertical lift performance table (VFPPROD) assigned to the group via VPFTAB variable. The vertical lift performance table and the artificial lift quantity ALQ-WELL are used with the pipeline fluid rates to calculate the pipeline change pressure between the LOWER and HIGHER groups. Note that the units for ALQ-PIPE are dependent on the associated variable on the VFPPROD keyword and may represent a pump or a compressor depending how the VFPPROD table was generated by an external program.			0.0
5	OPTIONI	A defined character string that defines if a group's production target should be achieved by adjusting the tubing pressure of the wells within the group or by the adjusting the well rates by their guide rate. OPTIONI should be set to one of the following character strings: 1) YES: the group production target is achieved by adjusting the tubing pressure of the wells within the group, so that all wells flow at the same tubing head pressure. This is normally used for wells that flow into a common manifold, for example a sub-sea completion manifold. If a group is using this option and has a higher group with production targets or constraints, then this group should have it's guide rate set via the GCONPROD keyword in the SCHEDULE section, to ensure that the well's within this group operate at the same tubing head pressure. 2) NO: the group production target is achieved by adjusting the guide rates of the wells within the group. This is the standard method in matching group targets and may result with the wells within the having different tubing head pressures. Only groups containing wells can use OPTIONI equal to YES or NO, a group without wells should set OPTIONI to NO. Numerical convergence controls and iteration limits for wells using OPTIONI set equal to YES are defined via the NETBALAN keyword in the SCHEDULE section. Only the NO option is currently supported by the simulator.			NO

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	OPTION2	A defined character string that defines if well gas lift gas flows through the group's pipeline. OPTION2 should be set to one of the following character strings: 1) NO: no well gas lift gas is allowed to flow through the pipeline only produced reservoir gas is allowed to flow through the pipeline. 2) FLO: the well gas lift gas is added to the gas flow along the pipeline. The well gas lift gas is the sum of the calculated ALQ-WELL values for all the subordinate wells multiplied by their efficiency factors. 3) ALQ: the pipeline gas lift gas value (ALQ-PIPE) is the sum of the calculated ALQ-WELL values for all the subordinate wells multiplied by their efficiency factors. This means that the ALQ-PIPE gas lift gas value declared on item (4) is ignored. If either FLO or ALQ have been selected then artificial lift quantity for the pipeline (ALQ-PIPE) and the wells (ALQ-WELL) must be defined as gas lift gas on the VFPPROD tables. A well's specific gas lift gas quantity is set via the ALQ-WELL variable on the WCONPROD keyword in the SCHEDULE section. Only the NO and FLO options are currently supported by the simulator.			NO
7	OPTION3	A defined character string that defines if the ALQ-PIPE variable should be reset to an equivalent surface oil or gas density flowing along the pipeline. OPTION3 should be set to one of the following character strings: 1) DENO: set ALQ-PIPE to the average surface density of the oil flowing along the pipeline. 2) DENG: set ALQ-PIPE to the average surface density of the gas flowing along the pipeline. 3) NONE: no change to ALQ-PIPE. If either DENO or DENG have been selected then artificial lift quantity on the VFPPROD tables must be based on the same density parameter. These options are normally used when a mixture of oil or gas with different surface densities flows into the network. Only the NONE option is currently supported by the simulator.			NONE

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.45: GRUPNET Keyword Description

See also the [WEL SPECS](#) keyword to define wells, the [VFPPROD](#) and [VFPINJ](#) keywords that the define vertical lift performance table to be used for calculating the pipeline pressures connecting the LOWER and HIGHER group in the network. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example defines a network based on two groups

```
--
--      DEFINE GROUP STANDARD NETWORK PARAMETERS
--
-- GRUP  CNTL   VFP    PUMP   MANIFOLD  INCLUDE  ALQ
```

```
-- NAME      PRES      TABLE  POWER  GROUP      LIFT GAS  DENS
GRUPNET
PROD-A      1200.    1*
PROD-B      1*      1      1*      'YES'      1*      1*
/
```

The next example is more complex and is taken form the Norne model.

```
--
--          DEFINE GROUP STANDARD NETWORK PARAMETERS
--
-- GRUP     CNTL      VFP      PUMP      MANIFOLD  INCLUDE  ALQ
-- NAME     PRES      TABLE  POWER      GROUP      LIFT GAS  DENS
GRUPNET
FIELD      20.0      5*
PROD       20.0      5*
MANI-B2    1*      8      1*      NO      2*
MANI-B1    1*      8      1*      NO      2*
MANI-K1    1*     9999      4*
B1-DUMMY   1*     9999      4*
MANI-D1    1*      8      1*      NO      2*
MANI-D2    1*      8      1*      NO      2*
MANI-K2    1*     9999      4*
D2-DUMMY   1*     9999      4*
MANI-E1    1*      9      1*      NO      2*
MANI-E2    1*      9      4*
```

Here the *FIELD* controlling pressure is set at 20 barsa and the same limit is used for group PROD which sits directly under the FIELD group (see Figure 12.4).

12.3.115 GRUPRIG – GROUP DRILLING AND WORKOVER RIG SPECIFICATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Defines a groups drilling and workover specifications.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.116 GRUPLAV – DEFINE SLAVE GROUPS IN SLAVE RESERVOIRS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [GRUPLAV](#) keyword defines slave groups in a slave input deck and their associated master groups in the master run, for when the Reservoir Coupling option as been activated by the [GRUPMAST](#) and [SLAVES](#) keywords in the [SCHEDULE](#) section. This keyword is required for every slave input deck. Reservoir coupling allows for independent reservoir simulation decks ([SLAVES](#)) to be controlled by a separate master run file. For example, if there are five separate reservoir models each representing one field, one of the four would be used as the master and the other four would be the subordinate [SLAVES](#).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.117 GRUPTARG – MODIFY GROUP TARGETS AND CONSTRAINTS VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GRUPTARG](#) keyword modifies the target and constraints values of both rates and pressures for previously defined groups without having to define all the variables on the group control keywords: [GCONPROD](#) or [GCONPRI](#) keywords. Variables not changed by the [GRUPTARG](#) keyword remain the same as those previously entered via the group control keywords or previously entered [GRUPTARG](#) keywords. Note that the group must still be initially be fully defined using the [GCONPROD](#) or [GCONPRI](#) keywords. All the aforementioned keywords are described in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Note that wells are allocated to groups when the wells are specified by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the group name for which the group target and constraints are being defined. The group named FIELD is the top most group and should be used to set targets and constraints for the field. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy.			None
2	TARGET	A defined character string that sets the item to be changed for the group the value of the item is set by item (3). 1) ORAT: reset the surface oil production rate value as defined by item (3). 2) WRAT: reset the surface water production rate value as defined by item (3). 3) GRAT: reset the surface gas production rate value as defined by item (3). 4) LRAT: reset the surface liquid (oil plus water) production rate value as defined by (3). 5) RESV: reset the in situ reservoir volume rate value as defined by (3). 6) GUID: reset the guide rate value for wells operating under group control. Note TARGET only defines the variable to be changed, it does not change how a group is controlled. For example, if a group is operating on ORAT control, as defined by the previously entered GCONPROD keyword, entering TARGET equal to LRAT with a value, changes the liquid constraint but the group still remains on ORAT control. Use the GCONPROD or GCONPRI keywords in the SCHEDULE section to change the control mode of a well.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	VALUE	A real positive value that defines the value of the variable declared by TARGET			None
	Liquid	stb/d	sm ³ /day	scc/hour	
	Gas	Mscf/d	sm ³ /day	scc/hour	
	Res Vol	rb/d	rm ³ /day	rcc/hour	
	Pressure	psia	barsa	atma	

Notes:

1) The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.46: GRUPTARG Keyword Description

See also the [WELTARG](#) and [WELCNTL](#) keyword, in the [SCHEDULE](#) section that can be used to reset a well's control mode, as well as a well's target and constraints of both rates and pressures.

Example

The following example below shows the oil rates for the field at the start of the schedule section (January 1, 2000).

```

-----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-----
--
--          GROUP PRODUCTION CONTROLS
--
-- GRUP  CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME  MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF   WAT
GCONPROD
FIELD   ORAT  40E3   60E3   30E3   65E3   1*    1*    1*    1*    1*    /
/
DATES
01 FEB 2000 /
/
--
--          GROUP PRODUCTION AND INJECTION TARGETS
--
--          GROUP    GROUP    TARGET
--          NAME     TARG     VALUE
GRUPTARG
FIELD     ORAT     45E3
FIELD     LIQ      75E3
/
/

```

From January 1, 2000 to February 1, 2000 the field is on oil rate control and has a target oil rate of 40,000 stb/d, a maximum water handling capacity of 60,000 stb/d, a maximum liquid capacity of 65,000 stb/d, and a maximum gas constraint of 30 MMscf/d. After February 1, 2000 the field's target oil rate is increased to 45,000 stb/d and the maximum liquid constraint is increased to 75,000 stb/s; all the other parameters remain unchanged.

12.3.118 GRUPTREE – DEFINE GROUP TREE HIERARCHY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

GRUPTREE defines the group hierarchy of groups that have been created by having wells assigned to them via the **WEL SPECS** keyword in the **SCHEDULE** section. By default three group levels are defined that sets the wells as level three, reporting directly to defined groups at level two, and the level two groups reporting to the **FIELD** group at level one. If a different configuration is required then the **GRUPTREE** keyword should be used to define the group hierarchy by defining a lower level group that reports directly to a higher level group.

No.	Name	Description	Default
1	LOWER	A character string of up to eight characters in length that defines the group name which belongs to the HIGHER group. The group named FIELD is the top most group and should NOT be used as a group name for the LOWER group name. Undefined group relationships are automatically assigned to the FIELD group.	None
2	HIGHER	A character string of up to eight characters in length that defines the HIGHER group name that the LOWER group belongs to. The group named FIELD is the top most group and can be used as the HIGHER group name. Undefined group relationships are automatically assigned to the FIELD group.	None
Notes: 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".			

Table 12.47: GRUPTREE Keyword Description

A group hierarchy can have any number of levels but groups that have other groups as **LOWER** groups cannot also have wells for the **HIGHER** group. Thus, a group either contains wells or has **LOWER** groups

See also the **GCONPROD** and **GCONINJE** for defining group production and injection volumes, and the **WEL SPECS** keywords to allocate wells to groups. All the aforementioned keywords are described in the **SCHEDULE** section.

Examples

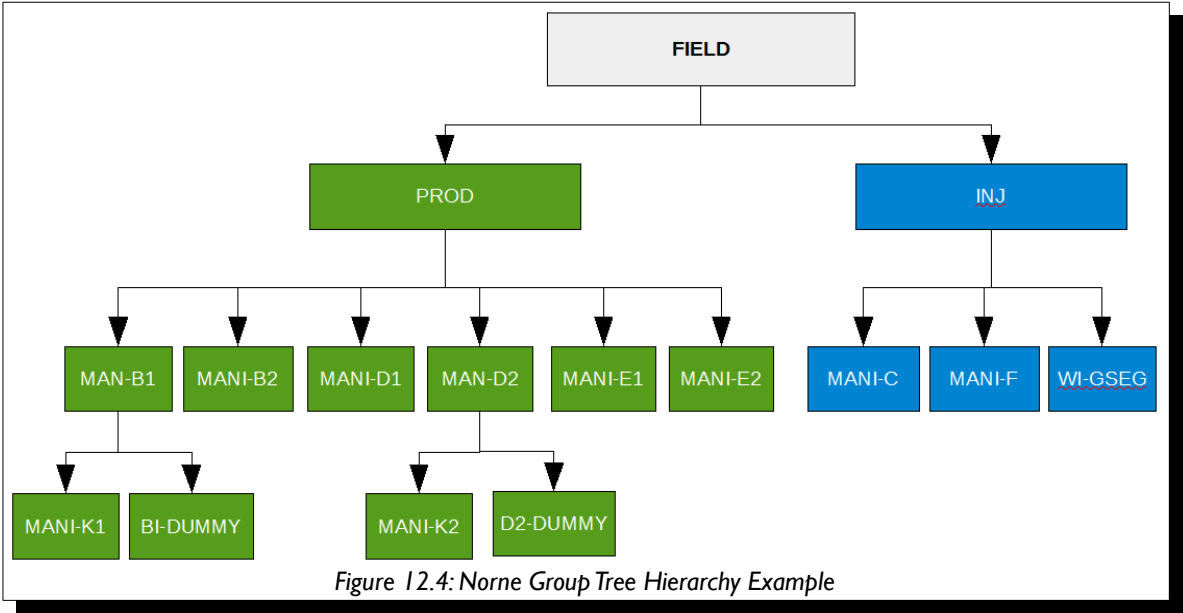
The first example defines **PLAT01** and **PLAT03** reporting to the **FIELD** level (default if these records are omitted) and **PLAT02** reporting to **PLAT01**.

```
--
--      DEFINE GROUP TREE HIERARCHY
--
--      LOWER      HIGHER
--      GROUP      GROUP
GRUPTREE
      PLAT01      FIELD
      PLAT02      PLAT01
      PLAT03      FIELD
/
```

The next example is more complex and is taken form the Norne model.

```
--
--      DEFINE GROUP TREE HIERARCHY
--
--      LOWER      HIGHER
--      GROUP      GROUP
GRUPTREE
    'INJE'        'FIELD'
    'PROD'        'FIELD'
    'MANI-B2'     'PROD'
    'MANI-B1'     'PROD'
    'MANI-D1'     'PROD'
    'MANI-D2'     'PROD'
    'MANI-E1'     'PROD'
    'MANI-E2'     'PROD'
    'MANI-K1'     'MANI-B1'
    'MANI-K2'     'MANI-D2'
    'MANI-C'      'INJE'
    'MANI-F'      'INJE'
    'WI-GSEG'     'INJE'
    'B1-DUMMY'    'MANI-B1'
    'D2-DUMMY'    'MANI-D2'
```

The group hierarchy for this example is shown below.



Here groups PROD, INJ, MAN-B1, and MAN-D2 report to higher level groups and the other remaining groups all have individual wells allocated to them instead.

12.3.119 GSATINJE – DEFINE GROUP SATELLITE INJECTION RATES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GSATINJE](#) keyword defines a satellite group’s oil, gas and water injection rates in the model. Satellite groups are not connected to the reservoir model and therefore have no wells or subordinate groups associated with them, they are nevertheless connected to other higher level groups and higher level groups within a network model (if activated). Thus, they provide a means to “add-in” outside injection and production to the model without modeling the “add-in” reservoirs.

The keyword is used to define injection rates into the model from other sources (fields, reservoirs etc.) that are not defined in the current run. For example, if Fields A, B and C are all producing through a common plant, but only Field A is being modeled in the current input deck, then injection and production from Fields B and C can be incorporated into the model in order to meet the plant injection rates for Field A. Note in this case the import rates from Fields B and C are fixed, and therefore Field A would act like a “swing” injector to match the overall injection requirement.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the satellite group name for which the group injection rates are being defined. The group named FIELD is the top most group and should not be used to define a satellite group. Note that the group hierarchy should be defined by the GRUPTREE keyword in the SCHEDULE section, when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy. Note that a satellite group cannot have subordinate groups or wells.			None
2	TYPE	A defined character string that specifies the type of injection fluid. TYPE should be set to one of the following character strings: 1) GAS: for gas injection. 2) OIL: for oil injection. 3) WAT: for water injection. If the satellite group injects more than one phase then the rates should be specified in separate records.			None
3	RATE	A positive real value that defines the surface injection rate for the phase declared by the TYPE variable.			0.0
		Liquid: stb/d Gas: Mscf/d	Liquid: sm ³ /day Gas: sm ³ /day	Liquid: scc/hour Gas: scc/hour	
4	RESV	A positive real value that defines the reservoir volume injection rate for the phase declared by the TYPE variable for the satellite group. Generally, RESV should be set to zero, unless one wishes to have the satellite reservoir volume rate added to the group's superordinate groups. If RESV is set to zero for all satellite groups and if the superordinate group has a reservoir volume injection target, a voidage replacement target, or a production balancing fraction target, then only the non-satellite group volumes will be used. This item is not supported by OPM Flow and should be defaulted (I*) or set to zero.			0.0
		rb/d	rm ³ /day	rcc/hour	
5	CALRATE	A positive real value that defines the calorific injection rate used in the commercial compositional simulator. This item is not supported by OPM Flow and should be defaulted (I*) or set to zero.			0.0

Notes:
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.48: GSATINJE Keyword Description

See also the [GSATPROD](#) and [GRUPTREE](#) keywords to define satellite production rates and the group hierarchy, respectively. For non-satellite groups see the [GCONINJE](#) and [GCONPROD](#) keywords. All the aforementioned keywords are in the [SCHEDULE](#) section.

Note

Once a group has been defined to be a satellite group, via the [GSATINJE](#) and [GSATPROD](#) keywords, then the equivalent modeled group keywords, [GCONINJE](#) and [GCONPROD](#) in the [SCHEDULE](#) section, cannot be used to set the operating conditions for satellite groups, only the [GSATINJE](#) and [GSATPROD](#) keywords may be used.

Example

The example below is based on an oil field, with the main field FLD-A being modeled in the input deck with two groups (FLD-A1 and FLD-A2), combined with one satellite field supplementing oil production and water injection (FLD-B).

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      DEFINE GROUP TREE HIERARCHY
--      LOWER      HIGHER
--      GROUP      GROUP
GRUPTREE
      FLD-A      FIELD
      FLD-A1     FLD-A
      FLD-A2     FLD-A
      FLD-B      FIELD
/
--
--      GROUP PRODUCTION CONTROLS
--
-- GRUP  CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME  MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF   WAT
GCONPROD
FIELD   ORAT   20E3   30E3   50E3   1*    1*    1*
FLD-A   FLD    20E3   30E3   50E3   1*    1*    1*
/
--
--      LOAD INCLUDE FILE - LOAD ALL WELLS AND VFP TABLES
--
INCLUDE
      'FLD-A-P50-WELLS.INC' /
--
--      GROUP INJECTION TARGETS AND CONSTRAINTS
--
-- GRUP  FLUID  CNTL    SURF    RESV    REINJ    VOID    GRUP  GUIDE  GUIDE  GRUP  GRUP
-- NAME  TYPE   MODE    RATE    RATE    FRAC     FRAC    CNTL  RATE   DEF   REINJ  RESV
GCONINJE
FIELD   WAT    REIN    35E3    1*      1*      1.0     NO    1*    1*    1*    1* /
/
--
--      SATELLITE GROUP PRODUCTION CONTROLS
--
-- GRUP  OIL    WAT    GAS    RESV    GAS    CALORIFIC
-- NAME  RATE   RATE   RATE   RATE    LIFT   RATE
GSATPROD
FLD-B   5E3     0.00   2E3    1*      1*
/
--
--      SATELLITE GROUP INJECTION CONTROLS
```

```
--
-- GRUP FLUID SURF RESV CALORIFIC
-- NAME TYPE RATE RATE RATEV
GSATINJE
FLD-B WAT 10E3 1* 1* /
/
--
-- ADVANCE SIMULATION BY REPORTING DATE
--
DATES
1 FEB 2021 /
/
--
-- SATELLITE GROUP PRODUCTION CONTROLS
--
-- GRUP OIL WAT GAS RESV GAS CALORIFIC
-- NAME RATE RATE RATE RATE LIFT RATE
GSATPROD
FLD-B 5E3 0.00 2E3 0.0 1* /
/
--
-- SATELLITE GROUP INJECTION CONTROLS
--
-- GRUP FLUID SURF RESV CALORIFIC
-- NAME TYPE RATE RATE RATEV
GSATINJE
FLD-B WAT 10E3 1* 1* /
/
```

Here FLD-B will supplement FLD-A's water re-injection rate by 10,000 stb/d, subject to a maximum field water injection rate of 35,000 stb/d on the [GCONINJE](#) keyword. This means that FLD-A's maximum water injection rate cannot exceed 25,000 stb/d. In terms of oil rates, [GCONPROD](#) defines a maximum oil rate of 20,000 stb/d of which 5,000 stb/d is from FLD-B. If any of the constraints on the [GCONINJE](#) and [GCONPROD](#) keywords are violated, then the appropriate action will occur only for FLD-A, in order to ensure the group constraints are honored.

12.3.120 GSATPROD – DEFINE GROUP SATELLITE PRODUCTION RATES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GSATPROD](#) keyword defines a satellite group's oil, water and gas production rates in the model. Satellite groups are not connected to the reservoir model and therefore have no wells or subordinate groups associated with them, they are nevertheless connected to other higher level groups and higher level groups within a network model (if activated). They provide a means to “add-in” outside injection and production to the model without modeling the “add-in” reservoirs.

This keyword is used to import additional fluid streams into the model from other sources (fields, reservoirs, etc.) that are not defined in the current run. For example, if Fields A, B and C are supplying gas to a power plant, but only Field A is being modeled in the current input deck, then production from Fields B and C can be incorporated into model in order to meet the plant demand. Note in this case the import gas rates from Fields B and C are fixed, and therefore Field A acts like a “swing” producer to match the gas demand target.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRPNAME	A character string of up to eight characters in length that defines the satellite group name for which the group production import rates are being defined. The group named FIELD is the top most group and should not be used with this keyword. Note that the group hierarchy should be defined by the GRUPTREE keyword in the SCHEDULE section when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy. <u>Note that a satellite group cannot have subordinate groups or wells.</u>			None
2	ORAT	A real value, greater than or equal to zero, that defines the satellite's surface oil production rate to be imported into the model. This value may be specified using a User Defined Argument (UDA).			0.0
		stb/d	sm ³ /day	scc/hour	
3	WRAT	A real value, greater than or equal to zero, that defines the satellite's surface water production rate to be imported into the model. This value may be specified using a User Defined Argument (UDA).			0.0
		stb/d	sm ³ /day	scc/hour	
4	GRAT	A real value, greater than or equal to zero, that defines the satellite's surface gas production rate to be imported into the model. This value may be specified using a User Defined Argument (UDA).			0.0
		Mscf/d	sm ³ /day	scc/hour	
5	RESV	A real value, greater than or equal to zero, that defines the satellite's reservoir volume production rate to be imported into the model. This item is not supported by OPM Flow and should be defaulted (1*) or set to zero.			0.0
		rb/d	rm ³ /day	rcc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	GASLIFT	A real value, greater than or equal to zero, that defines the satellite’s gas lift gas to be imported for this group. A value for GASLIFT is only required if externally supplied gas lift is being imported into the model, and: 1) The Network Model is active in the input deck via the NETWORK keyword in the RUNSPEC section. Here, GASLIFT is applied to the production network in order to calculate the pressure drop through the network, for branches that have: (1) OPTION2 set to FLO on the GRUPNET keyword in the SCHEDULE section, or (2) the GASLIFT parameter on the NODEPROP keyword in the SCHEDULE section set to YES, for when the Extended Network Model is being used. as per the BRANPROP and NODEPROP keywords, in the SCHEDULE section. 2) The Gas Lift Optimization Model is active in the deck via the LIFTOPT keyword in the SCHEDULE section. In this case GASLIFT is applied to GRPNAME’s superordinate group’s gas supply constraints based on the GLIFTOPT keyword in the SCHEDULE section. For clarity, GLIFTOPT constraint violations result in the groups being modeled having their rates adjusted, whereas satellite group rates are not affected. This item is not supported by OPM Flow and should be defaulted (1*) or set to zero.			
		Mscf/d	sm ³ /day	scc/hour	0.0
7	CALRATE	A real value greater than or equal to zero that defines the satellite’s calorific production rate. This item is not supported by OPM Flow and should be defaulted (1*) or set to zero.			0.0

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.49: GSATPROD Keyword Description

See also the [GSATINJE](#) and [GRUPTREE](#) keywords to define satellite injection rates and the group hierarchy, respectively. For non-satellite groups see the [GCONINJE](#) and [GCONPROD](#) keywords. All the aforementioned keywords are in the [SCHEDULE](#) section.

Note

Once a group has been defined to be a satellite group, via the [GSATINJE](#) and [GSATPROD](#) keywords, then the equivalent modeled group keywords, [GCONINJE](#) and [GCONPROD](#) in the [SCHEDULE](#) section, cannot be used to set the operating conditions for satellite groups, only the [GSATINJE](#) and [GSATPROD](#) keywords may be used.

Example

The example below is based on a dry gas field, with the main field FLD-A being modeled in the input deck with two groups (FLD-A1 and FLD-A2), combined with two satellite gas fields, FLD-B and FLD-C.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      DEFINE GROUP TREE HIERARCHY
--      LOWER      HIGHER
--      GROUP      GROUP
GRUPTREE
    FLD-A      FIELD
    FLD-A1     FLD-A
    FLD-A2     FLD-A
    FLD-B      FIELD
    FLD-C      FIELD
/
--
--      GROUP PRODUCTION CONTROLS
--
-- GRUP  CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME  MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF   WAT
GCONPROD
FIELD   GRAT  1*    1*    450E3  1*    1*    1*
FLD-A   FLD   1*    1*    450E3  1*    1*    1*
/
--
--      LOAD INCLUDE FILE - LOAD ALL WELLS AND VFP TABLES
--
INCLUDE
    'FLD-A-P50-WELLS.INC' /
--
--      SATELLITE GROUP PRODUCTION CONTROLS
--
-- GRUP  OIL    WAT    GAS    RESV    GAS    CALORIFIC
-- NAME  RATE   RATE   RATE   RATE    LIFT   RATE
GSATPROD
FLD-B   10.0    0.00   30E3   1*      1*
FLD-C   20.0    5.50   20E3   0.0     1*
/
--
--      ADVANCE SIMULATION BY REPORTING DATE
--
DATES
    1  FEB    2021  /
/
--
--      SATELLITE GROUP PRODUCTION CONTROLS
--
-- GRUP  OIL    WAT    GAS    RESV    GAS    CALORIFIC
-- NAME  RATE   RATE   RATE   RATE    LIFT   RATE
GSATPROD
FLD-B   10.0    0.00   30E3   0.0     1*
FLD-C   20.0    9.00   20E3   0.0     1*
/
```

Since the field gas rate is set to 450 MMscf/d and the satellite production is 50 MMscf/d, then FLD-A will produce only 400 MMscf/d and not the stipulated 450 MMscf/d on the [GCONPROD](#) keyword.

12.3.121 GSEPCOND – ASSIGN GROUP SEPARATORS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GSEPCOND](#) keyword assigns previously defined separators to a group. Group separators are specified by the [SEPVALS](#) keyword in the [SCHEDULE](#) section. The facility is used in black-oil modeling to re-scale the PVT data entered via the [PROPS](#) section, based on the saturation point oil formation volume factor (Bob) and the initial saturated gas-oil ratio (Rsi) entered on the SEVPALS keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.122 GSSCPTST – PERFORM SUSTAINABLE CAPACITY TEST

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [GSSCPTST](#) keyword instructs the simulator to perform a sustainable capacity test. This causes the model to be saved in it's current state via the [RESTART](#) file, and the test performed by running the simulation under the current conditions combine with the parameters on this keyword. After the test is perform, the simulator will restart from the point prior to the test by loading in the [RESTART](#) file. This type of testing is normally applied to gas fields for which the gas sales contracts stipulate that the gas sales rates are based on a sustainable capacity rate over a fixed period of time.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.123 GSWINGF – DEFINE GROUP MULTIPLE GAS CONTRACT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [GSWINGF](#), defines the gas contract parameters, swing factor and the monthly seasonal profile factor, for when there are multiple gas contracts being used in the model. The keyword is used with the Gas Field Operations option which is activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. Gas contracts are commonly based on a Daily Contract Quantity (“DCQ”) that determines the gas rate that the field should be produced at, which is normally expressed as a multiple of the DCQ, for example 1.33, and is often referred to as the “swing factor”. Some gas contracts also define a maximum DCQ (“Max DCQ”) and/or a minimum take or pay DCQ (“Min DCQ”), as well as seasonal demand characteristics. For example, gas rates may be set higher in the winter months in order to meet heating demand compared with summer months in colder climates, and the opposite in warmer climates where air conditioning demand is high.

Thus, the DCQ must be calculated first over a contract period, where the contract period is commonly contract years, or in some instances contract quarters. This is performed by the simulator using the current DCQ and checking to see if the (DCQ x Swing Factor) can be satisfied throughout the current contract period, if not the DCQ is re-calculated so that the (DCQ x Swing Factor) condition is satisfied. Once this condition is met, the second and final pass uses the calculated DCQ in conjunction with the monthly scaling profile data to set the monthly gas rate for the field:

$$Q_{month} = DCQ \times SWINGFAC_{month} \quad (12.24)$$

Where:

Q_{month} = the monthly gas production target
 DCQ = Daily Contract Quantity
 $SWINGFAC_{month}$ = monthly rate scaling factor that takes into account seasonal demand, etc.

Here the [GSWINGF](#) keyword allows for different gas contract parameters to be assign to different groups and is mutually exclusive to the [SWINGFAC](#) keyword in the [SCHEDULE](#) section, that sets the gas contract parameters for a single contract at the FIELD group level.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.124
GTADD – ADD A CONSTANT TO A GROUP TARGET OR CONSTRAINT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GTADD](#), adds a numerical constant to a group’s target or constraint value. The group must have been initially fully defined using the [GCONPROD](#) or [GCONPRI](#) keywords for producers or [GCONINJE](#) for injectors. Variables not changed by the [GTADD](#) keyword remain the same as those previously entered via the group control keywords or previously entered [GTADD](#) keywords. See also the [GRUPTARG](#) keyword that sets the values for a group’s target and constraints and the [GTMULT](#) keyword that multiplies a group target or constraint by a constant. All the aforementioned keywords are described in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.125 GTMULT – MULTIPLY GROUP TARGET OR CONSTRAINT BY A CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [GTMULT](#), multiplies a group’s target or constraint value by a numerical constant. The group must have been initially fully defined using the [GCONPROD](#) or [GCONPRI](#) keywords for producers or [GCONINJE](#) for injectors. Variables not changed by the [GTMULT](#) keyword remain the same as those previously entered via the group control keywords or previously entered [GTMULT](#) keywords. See also the [GRUPTARG](#) keyword that sets the values for a group’s target and constraints, and the [GTADD](#) keyword that adds a constant to a group’s target or constraint. All the aforementioned keywords are described in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.126 GUIDECAL – SCALE GUIDE RATES BASED ON GAS CALORIFIC VALUE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GUIDECAL](#) keyword defines a well or group’s guide rate as a function of their calorific values, for when the individual wells and groups are under guide rate control. Group and well guide rates that have not been directly defined are set equal to their production potentials at the start of each time step. In this case the [GUIDECAL](#) keyword can be used to specify the coefficients of a function that takes into account the calorific value of the produced gas, effectively scaling the guide rates based on the calorific value of the gas being produced.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.127 GUIDERAT – DEFINE GROUP GUIDE RATE FORMULA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines a general formulae used to define a group's and well's guide rate as a function of the their potential. The default behavior, that is when this keyword is not invoked, is to set the target control mode and rate via the [GCONPROD](#) keyword in the [SCHEDULE](#) section. In this case the target rate is distributed between the group's wells that are under group control using a well's guide rate. If a well's guide rate has not been defined, for example by this keyword, then the well potential of the group controlling phase at the beginning of the time step is used. For example, if the group target rate and phase is oil, then the well's under group control will have their oil rates determined by their oil rate potential³²². The [GUIDERAT](#) keyword substitutes the potential calculation with a more general formulae in the aforementioned distribution and allocation of the rates:

$$\text{Phase Guide Rate} = \frac{(\text{Potential}_{\text{Phase}})^A}{B + C(\text{Potential Ratio}_1)^D + E(\text{Potential Ratio}_2)^F} \quad (12.25)$$

Where:

- $\text{Potential}_{\text{Phase}}$ = the potential of the phase,
- A to F = constants defined on this keyword,
- Potential Ratio_1 = the potential phase ratio as defined by this keyword,
- Potential Ratio_2 = the potential phase ratio as defined by this keyword.

The formulae can be used to control high water cut or high GOR wells in an oil field, such as the offending wells are given progressively smaller guide rates as they water out or gas out.

Note that groups can only have potential guide rates if they are subordinate in another group and required to produce a proportion of the superior group's target rate. In this case the [GUIDERAT](#) keyword can optionally be applied by setting GUIPHASE variable on the [GCONPROD](#) keyword in the [SCHEDULE](#) section. Group potentials are the sum of the potentials of their subordinate open wells.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TSTEP	A real positive value that defines the minimum time interval to re-calculate the guide rates. The guide rates are calculated at the start of a time step and the default value of zero means that the guide rates are calculated for each time step. A non-zero value for TSTEP resets the minimum interval, for example setting TSTEP equal to 30 would mean the guide rates are calculate every 30 days, or to the nearest associated time step. Calculating guide rates every time step may cause issues due to the rate dependent behavior, for example gas cusping or water coning causing the well rates to oscillate. In this case using a non-zero value of TSTEP may eliminate this oscillating behavior.			0.0
		days	days	hours	

³²² Production and injection potentials are based on rates that are unrestricted. For wells this implies that well potential is calculated based on either the BHP or THP limit, which ever is the more constraining.

No.	Name	Description			Default																		
		Field	Metric	Laboratory																			
2	PHASE	A defined character string that sets the potential phase guide rate for the group and well, the resulting Phase Guide Rate in equation (12.25). PHASE should be set to one of the following character strings: 1) OIL: is set as the potential phase guide rate and in this case the Potential Ratio1 variable is the water-oil ratio ("WOR") and Potential Ratio2 refers to the gas-oil ratio ("GOR") in equation (12.25). 2) LIQ: is set as the potential phase guide rate and the Potential Ratio1 variable is the water cut ("WCT") and Potential Ratio2 refers to the gas-liquid ratio ("GLR") in equation (12.25). 3) GAS: is set as the potential phase guide rate with the Potential Ratio1 variable being the water cut ("WCT") and Potential Ratio2 referring to the oil-gas ratio ("OGR") in equation (12.25). 4) RES: here the potential phase guide rate is defined as the reservoir fluid volume rate and the Potential Ratio1 variable is the water-oil ratio ("WOR") and Potential Ratio2 refers to the gas-oil ratio ("GOR") in equation (12.25). 5) COMB: this option uses the linearly combined phase guide rate based on the values entered on the LINCOM keyword in the SCHEDULE section. Here the <i>Potential Ratio₁</i> variable is the water divided by linearly combined phase and <i>Potential Ratio₂</i> refers to the gas divided by linearly combined phase in equation (12.25). This option is not available in OPM Flow. 6) NONE: the Phase Guide Rate calculation is switched off and the well guide rates revert to the well potentials of their group target phase. For reference, the units for the various options is given below . <table><tr><td>WOR: dimensionless</td><td>dimensionless</td><td>dimensionless</td></tr><tr><td>WCT: dimensionless</td><td>dimensionless</td><td>dimensionless</td></tr><tr><td>WGR: stb/Mscf</td><td>dimensionless</td><td>dimensionless</td></tr><tr><td>GOR: Mscf/stb</td><td>dimensionless</td><td>dimensionless</td></tr><tr><td>GLR: Mscf/stb</td><td>dimensionless</td><td>dimensionless</td></tr><tr><td>OGR: stb/Mscf</td><td>dimensionless</td><td>dimensionless</td></tr></table>			WOR: dimensionless	dimensionless	dimensionless	WCT: dimensionless	dimensionless	dimensionless	WGR: stb/Mscf	dimensionless	dimensionless	GOR: Mscf/stb	dimensionless	dimensionless	GLR: Mscf/stb	dimensionless	dimensionless	OGR: stb/Mscf	dimensionless	dimensionless	None
WOR: dimensionless	dimensionless	dimensionless																					
WCT: dimensionless	dimensionless	dimensionless																					
WGR: stb/Mscf	dimensionless	dimensionless																					
GOR: Mscf/stb	dimensionless	dimensionless																					
GLR: Mscf/stb	dimensionless	dimensionless																					
OGR: stb/Mscf	dimensionless	dimensionless																					
3	A	A real value greater than or equal to -3 and less than or equal to 3, that defines coefficient A in equation (12.25).			0.0																		
4	B	B is a real positive value that defines coefficient B in equation (12.25).			0.0																		
5	C	C is a real value that defines coefficient C in equation (12.25).			0.0																		
6	D	D is a real value greater than or equal to -3 and less than or equal to 3, that defines coefficient D in equation (12.25).			0.0																		
7	E	E is a real value that defines coefficient E in equation (12.25).			0.0																		
8	F	F is a real value greater than or equal to -3 and less than or equal to 3, that defines coefficient F in equation (12.25).			0.0																		

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	GROPT01	A defined character string that determines if calculated phase guide rates should be allowed to increase (YES) or not (NO), and should be set to one of the following: 1) NO: The phase guide rates calculated from equation (12.25) are not allowed to increase above the current value, and will be reset to the current value if the calculated value does exceed the current value. 2) YES: The phase guide rates calculated from equation (12.25) are allowed to increase at each calculation. This may increase the propensity for oscillations in the rates if the water cut and GOR are rate dependent. Note only the default value is currently supported by OPM Flow.			YES
10	GROPT02	A real positive value greater than or equal to zero and less than or equal to one that “dampens” the calculated phase guide rate based on the following formula: $(Phase\ Guide\ Rate)_t^{new} = GROPT\ 02 \times (Phase\ Guide\ Rate)_t + (1 - GROPT\ 02) \times (Phase\ Guide\ Rate)_{(t-1)}$ The option is intended to have a similar effect as the GROPT01 NO option to reduce oscillations as a result of either the water cut or GOR being rate dependent. Values approaching one allows the calculated phase guide rates to change instantaneously with the phase potentials, whereas values approaching zero dampen the potential guide rates towards the previously calculated values, thereby reducing the potential for oscillating behavior.			1.0
11	GROPT03	A defined character string that determines if “free” gas potential rates for the Potential Ratio2 variable in equation (12.25) should be used (YES), or if “free and associated” gas should be used (NO), and should be set to one of the following: 1) NO: Use “free and associated” gas, that is total gas in the Potential Ratio2 variable. 2) YES: Only utilize “free” gas in the Potential Ratio2 variable.			NO
12	GROPT04	A real positive value that sets the minimum potential guide rate. If the calculated potential guide is below this value it will be reset to GROPT04. The option is meant to avoid groups and wells being ignored due to the calculated potential guide rates being minuscule.			1.0 x10-6
Notes:					
1) The keyword is terminated by a “/”.					

Table 12.50: GUIDERAT Keyword Description

Note that the [GUIDERAT](#) keyword only applies to production groups and wells. Injection groups and wells are still controlled by their potential guide rates.

Finally, as mentioned previously, if the [GUIDERAT](#) or [WGRUPCON](#) keywords are not present in the input deck then the group and well potential guide rates will be calculated using the well's potential rates. The [WGRUPCON](#) keyword in the [SCHEDULE](#) section can be used to set a constant potential guide rate for a well.

Note

[GUIDERAT](#) can be used to penalize wells producing excessive water by utilizing the C and D coefficients in equation (12.25), and to discriminate against wells that are gassing out by setting the E and F coefficients.

Note that the value range through which Potential Ratio1 and Potential Ratio2 vary is variable. For example, if Potential Ratio1 is water cut, then the value should be between zero and one, whereas for the water-oil ratio the value can vary between zero and infinity. The same applies to the units of Potential Ratio2 which are dependent on if the GOR, GLR or OGR ratio is used in the calculation.

One can use the C and E coefficients to scale these terms to the required relative magnitudes in the denominator and the D and F powers to influence how quickly the penalty increases with increasing water and gas fractions. High positive value for D and F coefficients will make production fall off rapidly as the water or gas fraction increases, while a negative values will favor producing these type of wells.

Note that the B coefficient should always be positive to prevent the denominator's going to zero.

Finally, if one wishes each well to produce in proportion to its potential when the water fraction and gas fraction are equal (the usual case), then the A coefficient should be set to one.

Examples

The first example sets the guide phase to oil and the resulting Phase Guide Rate based on oil potential based on setting the A and B coefficients to to one, that is:

$$\begin{aligned} \text{Phase Guide Rate} &= \frac{(\text{Potential}_{\text{Phase}})^A}{B + C(\text{Potential Ratio}_1)^D + E(\text{Potential Ratio}_2)^F} \\ &= \frac{\text{Oil Potential}^{1.0}}{1.0} \end{aligned} \quad (12.26)$$

with all the other parameters defaulted, except for the minimum time interval to re-calculate the guide rates which is set to 30 days.

```
--
--      SETS GUIDE RATES FOR GROUPS AND WELLS UNDER GUIDE RATE CONTROL
--
--      TIME  GUIDE  A    B    C    D    E    F    INCR  DAMP  FREE
--      STEP  PHASE  POW  CON  CON  POW  CON  POW  OPTN  OPTN  GAS
GUIDERAT
      30    'OIL'   1.0  1.0  1*  1*  1*  1*  1*   1*   1*   /
```

The next example sets the Phase Guide Rate to the reservoir fluid volume rate, with preference given to low GOR wells and with high GOR wells penalized, based on setting A and B to one, C and D to zero, E equal to 10 and F equal to two, that is:

$$\begin{aligned}
 \text{Phase Guide Rate} &= \frac{(Potential_{Phase})^A}{B + C(Potential Ratio_1)^D + E(Potential Ratio_2)^F} \\
 &= \frac{Reservoir Fluid Volume Potential^{1.0}}{1.0 + 10 \times (GOR)^2}
 \end{aligned}$$

(12.27)

with all the other parameters defaulted.

```

--
--          SETS GUIDE RATES FOR GROUPS AND WELLS UNDER GUIDE RATE CONTROL
--
--          TIME  GUIDE  A    B    C    D    E    F    INCR  DAMP  FREE
--          STEP  PHASE  POW  CON  CON  POW  CON  POW  OPTN  OPTN  GAS
GUIDERAT
1*      'RES'  1.0   1.0   1*   1*   10   2    1*    1*    1*    /

```

The [GUIDERAT](#) keyword is very flexible but can also lead to unexpected results, thus it is probably useful to perform some manual calculations outside of the simulator before implementing the selected scheme in the input deck.

12.3.128 GUPFREQ – INSTANTANEOUS GRADIENT OPTION UPDATE FREQUENCY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GUPFREQ](#) keyword sets the update frequency of the Instantaneous Gradient option for when this option has been activated by the [GDIMS](#) keyword in the [RUNSPEC](#) section. The Instantaneous Gradient option calculates derivatives of solution quantities at the current time step with respect to variations in the variables at the current time step. This is different to Gradient option that calculates the derivatives of solution quantities at the current time step with respect to variations in the variables at the initial time step, that is a time equal to zero. Consequently, the Instantaneous Gradient option can be switched on and off by the [GUPFREQ](#) keyword in the [SCHEDULE](#) section, whereas the Gradient option cannot.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.129 GWRTWCV – INSTANTANEOUS GRADIENT OPTION WELL VARIABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [GWRTWCV](#) keyword defines the wells and instantaneous gradient parameters to be calculated and exported as [SUMMARY](#) vectors to the summary file, for when the Instantaneous Gradient option has been activated by the [GDIMS](#) keyword in the [RUNSPEC](#) section. The Instantaneous Gradient option calculates derivatives of solution quantities at the current time step with respect to variations in the variables at the current time step. This is different to Gradient option that calculates the derivatives of solution quantities at the current time step with respect to variations in the variables at the initial time step, that is a time equal to zero. Consequently, the Instantaneous Gradient option can be switched on and off by the [GUPFREQ](#) keyword in the [SCHEDULE](#) section, whereas the Gradient option cannot.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.130
HMWPIMLT
–
HISTORY
MATCH
WELL
PRODUCTIVITY
INDEX
PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [HMWPIMLT](#), defines the history match gradient parameters for well productivity indices, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. Wells must be specified using the WELPSECS keyword in the [SCHEDULE](#) section and their connections defined by the [COMPDAT](#) and/or [COMPDATL](#) keywords, also in the [SCHEDULE](#) section

See also the [HMDIMS](#) keyword in the [RUNSPEC](#) section that specifies the dimensions for the gradient option, including the maximum number of gradient wellss that can be used with the History Match Gradient option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.131 INCLUDE – LOAD ANOTHER DATA FILE AT THE CURRENT POSITION

The [INCLUDE](#) keyword informs OPM Flow to continue reading input data from the specified [INCLUDE](#) file. When the end of the [INCLUDE](#) file is reached, or the [ENDING](#) keyword is encountered in the included file, input data is read from the next keyword in the current file.

See [INCLUDE – Load Another Data File at the Current Position](#) in the GLOBAL section for a full description.

12.3.132
LGRFREE – ACTIVATE LOCAL GRID REFINEMENT INDEPENDENT TIME STEPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LGRFREE](#) keyword activates the Local Grid Refinement (“LGR”) Independent Time Step option that allows the [LGR](#) to have solution time steps independent of the host grid for the stated [LGR](#), and for when LGRs have been declared by the [LGR](#) keyword in the [RUNSPEC](#) section, and defined by the [CARFIN](#) (Cartesian [LGR](#) grid) or [RADIN/RADIN4](#) (radial [LGR](#) grid) keywords in the [GRID](#) section. [LGR](#) independent solution time stepping can be deactivated by the [LGRLOCK](#) keyword in the [SCHEDULE](#) section.

Currently, OPM Flow does not support the local grid refinement feature and therefore this keyword is ignored by the simulator.

No.	Name	Description	Default
I	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which independent solution time stepping is to be activated. The LGR must have been previously defined by the CARFIN (Cartesian LGR grid) or RADIN/RADIN4 (radial LGR grid) keywords in the GRID section.	None
Notes: I) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.			

Table 12.51: LGRFREE Keyword Description

Example

The example below defines three oil LGRs([LGR](#)-OP01,-OP02, and -OP03) and all the gas well LGRs ([LGR](#)-GP*) that should use independent solution time steps.

```

--
--      ACTIVATE LOCAL GRID REFINEMENT INDEPENDENT TIME STEPS
--
--      LGRNAME
LGRFREE
      LGR-OP01
      LGR-OP02
      LGR-OP03
      LGR-GP*
/

```

12.3.133 LGRLOCK – DEACTIVATE LOCAL GRID REFINEMENT INDEPENDENT TIME STEPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [LGRLOCK](#) keyword deactivates the Local Grid Refinement (“LGR”) Independent Time Step option that allows the [LGR](#) to have solution time steps independent of the host grid for the stated [LGR](#), that is the [LGR](#) will now follow the global grid solution time steps. LGRs must have declared by the [LGR](#) keyword in the [RUNSPEC](#) section, and defined by the [CARFIN](#) (Cartesian [LGR](#) grid) or RADIN/RADIN4 (radial [LGR](#) grid) keywords in the [GRID](#) section. [LGR](#) independent solution time stepping can be activated by the [LGRFREE](#) keyword in the [SCHEDULE](#) section.

Currently, OPM Flow does not support the local grid refinement feature and therefore this keyword is ignored by the simulator.

No.	Name	Description	Default
I	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which independent solution time stepping is to be deactivated. The LGR must have been previously defined by the CARFIN (Cartesian LGR grid) or RADIN/RADIN4 (radial LGR grid) keywords in the GRID section.	None
Notes: I) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.			

Table 12.52: LGRLOCK Keyword Description

Example

The example below defines three oil LGRs(LGR-OP01,-OP02, and -OP03) and all the gas well LGRs (LGR-GP*) that should have their independent solution time steps deactivated.

```
--
--      DEACTIVATE LOCAL GRID REFINEMENT INDEPENDENT TIME STEPS
--
--      LGRNAME
LGRLOCK
      LGR-OP01
      LGR-OP02
      LGR-OP03
      LGR-GP*
/
```

12.3.134 LGROFF – DEACTIVATE A LOCAL GRID REFINEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **LGROFF** keyword deactivates a stated Local Grid Refinement (“LGR”) and optionally sets the minimum number of wells below which the **LGR** will be automatically deactivated. LGRs must have declared by the **LGR** keyword in the **RUNSPEC** section, and defined by the **CARFIN** (Cartesian **LGR** grid) or **RADIN/RADIN4** (radial **LGR** grid) keywords in the **GRID** section. LGRs can subsequently be activated by the **LGRON** keyword in the **SCHEDULE** section.

Currently, OPM Flow does not support the local grid refinement feature and therefore this keyword is ignored by the simulator.

No.	Name	Description	Default
1	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the LGR is being deactivated. The LGR must have been previously defined by the CARFIN (Cartesian LGR grid) or RADIN/RADIN4 (radial LGR grid) keywords in the GRID section.	None
2	MNWELLS	A positive integer greater than or equal to zero that defines the minimum number of active wells, below which the LGR will be automatically deactivated. The default value of zero implies that there is no limit to the number of wells and results in the LGR being unconditionally being deactivated.	0

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.53: LGROFF Keyword Description

Example

The example below unconditionally deactivates LGR-OP01, and sets the minimum number of active wells for deactivating LGR-OP02 and LGR-OP03 to one. For all the gas well LGRs (LGR-GP*) the minimum number of wells for deactivation is set to two.

```
--
--      DEACTIVATE LOCAL GRID REFINEMENTS
--
--      LGRNAME      MNWELLS
LGROFF
      LGR-OP01                      /
      LGR-OP02      1                /
      LGR-OP03      1                /
      LGR-GP*       2                /
/
```

12.3.135 LGRON – ACTIVATE A LOCAL GRID REFINEMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The **LGRON** keyword activates a stated Local Grid Refinement (“LGR”) and optionally sets the minimum number of wells above which the **LGR** will remain active. LGRs must have declared by the **LGR** keyword in the **RUNSPEC** section, and defined by the **CARFIN** (Cartesian **LGR** grid) or **RADIN/RADIN4** (radial **LGR** grid) keywords in the **GRID** section. LGRs can subsequently be deactivated by the **LGROFF** keyword in the **SCHEDULE** section.

Currently, OPM Flow does not support the local grid refinement feature and therefore this keyword is ignored by the simulator.

No.	Name	Description	Default
1	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the LGR is being activated. The LGR must have been previously defined by the CARFIN (Cartesian LGR grid) or RADIN/RADIN4 (radial LGR grid) keywords in the GRID section.	None
2	MNWELLS	A positive integer greater than or equal to zero that defines the minimum number of active wells, below which the LGR will be automatically deactivated. The default value of zero implies that there is no limit to the number of wells and results in the LGR being unconditionally being activated.	0
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.			

Table 12.54: LGRON Keyword Description

Example

The example below unconditionally activates LGR-OP01, and sets the minimum number of active wells for activating LGR-OP02 and LGR-OP03 to one. For all the gas well LGRs (LGR-GP*) the minimum number of wells for activating these LGRs is set to two.

```
--
--      ACTIVATE LOCAL GRID REFINEMENTS
--
--      LGRNAME      MNWELLS
LGRON
      LGR-OP01                      /
      LGR-OP02      1                /
      LGR-OP03      1                /
      LGR-GP*       2                /
/
```

12.3.136 LIFTOPT – ACTIVATE GAS LIFT OPTIMIZATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LIFTOPT](#) keyword activates the gas lift optimization option and defines the gas lift gas increment size, the minimum incremental oil improvement, as well as the timing of the calculations. Note that the [LIFTOPT](#) keyword should precede any [GLIFTOPT](#) and [WLIFTOPT](#) keywords in the [SCHEDULE](#) section in order to activate the gas lift optimization facility.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GASLIFT	A real positive number that defines the gas lift gas size increment that is used to increase the gas lift size quantity in steps. For example, if GASLIFT is set to 0.5 MMscf/d then gas lift gas will be allocated in step of 0.5 MMscf/d to each well during the optimization process. A zero or negative value switches off gas lift optimization.			None
		Mscf/d	sm ³ /day	scc/hour	
2	MINOIL	MINOIL is a real positive value that defines the minimum increase in oil rate for a given quantity of gas lift gas, for when gas lift gas should be applied to a well. Additional GASLIFT will only be assigned to a well if: $\frac{\Delta Q_{oil} \times OPTWGT}{GASLIFT} > MINOIL$ Where ΔQ_{oil} is the incremental oil and OPTWGT is the well's weighting factor defined by the OPTWGT variable on the WLIFTOPT keyword in the SCHEDULE section.			None
		stb/Mscf	sm ³ /sm ³	scc/scc	
3	TSTEP	TSTEP is a real positive value that defines the frequency of the gas lift optimization calculations, for example setting TSTEP equal to 30 days would result in the gas lift optimization calculation being performed approximately every 30 days. The default value of zero will result in the calculations being performed every time step. Note if the group or well is part of a production network then gas lift optimization is performed at the same time as the network is being balance, that is this parameter is ignored in this scenario. See the NETBALAN keyword in the SCHEDULE section to set the network balancing frequency in this case.			0.0
		days	days	hours	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	OPTLIFT	A defined character string that determines if the gas lift optimization iterations should be performed for the same number of Newton iterations within a time step as used to update well targets, or to just use the first Newton iteration only. The NUPCOL keyword in the RUNSPEC section determines the number of Newton iterations used to update well targets during a time step. OPTLIFT should be set to one of the following: 1) NO: In this case the gas lift optimization is only performed for the first Newton iteration for a time step and the distributed gas lift gas is then held constant for the groups and wells through resultant Newton iterations during the time step. This leads to better numerical performance, but may lead to the targets and constraints not being exactly satisfied if the reservoir conditions change during the time step calculations. 2) YES: Sets the number gas lift optimization iterations to the same number of Newton iterations within a time step as used to update well targets, as per the NUPCOL keyword in the RUNSPEC section. This results in greater target and constraint accuracy even if the reservoir conditions change during the time step calculations, but at the expense of numerical performance. Similar to the NO option, after the NUPCOL Newton iteration during the time step, the distributed gas lift gas is then held constant for the groups and wells in subsequent Newton iterations.			
Notes:					
1) The keyword is terminated by a “/”.					

Table 12.55: LIFTOPT Keyword Description

See also the [GLIFTOPT](#) keyword to define the group gas lift optimization controls and the [WLIFTOPT](#) keyword to define the wells under gas lift optimization control, both keywords are described in the [SCHEDULE](#) section. The [NUPCOL](#) keyword in the [RUNSPEC](#) section that determines the number of Newton iterations used to update well targets and gas lift optimization calculations during a time step may also be of interest.

Example

The following example activates gas lift optimization for the field and defines the optimization parameters.

```
--
--      ACTIVATE GAS LIFT OPTIMIZATION AND PARAMETERS
--
--      INCR   INCR   TSTEP   NEWTON
--      GAS    OIL    INTVAL   OPTN
LIFTOPT
      12.5E3  5E-3   0.0      YES    /
```

Here the maximum incremental gas lift gas quantity is set to 12.5×10^3 m³, the minimum incremental oil gain per m³ of gas lift gas is set to 5.0×10^{-3} m³, the time step interval is set to zero to perform the gas optimization every time step, and finally the gas lift optimization will be performed [NUPCOL](#) Newton iterations for the time step.

12.3.137 LINCOM – DEFINE LINEAR COMBINATION COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [LINCOM](#) keyword defines the oil, gas and water coefficients for the Linear Combination facility which allows for a linear combination of the aforementioned phase rates and volumes to be used as targets and constraints in controlling group and well production and injection data. See also the [LCUNIT](#) in the [PROPS](#) section that defines the units for linear combination equation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.138 MATCORR – ACTIVATE THE MATERIAL BALANCE CORRECTION OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The *MATCORR* keyword activates the Material Balance Correction option used to adjust the accumulated material balance error in the simulation.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.139 MESSAGE – OUTPUT USER MESSAGE

The [MESSAGE](#) keyword outputs a user message to the terminal, as well as to the print (*.PRT) and debug (*.DBG) files. Note this is different to the [MESSAGES](#) keyword, that defines OPM Flows message print limits and stop limits generated by the simulator.

See [MESSAGE – Output User Message](#) in the GLOBAL section for a full description.

12.3.140 MESSAGES – DEFINE MESSAGE PRINT LIMITS AND STOP LIMITS

The [MESSAGES](#) keyword defines the print and stops levels for various messages. The “print limits” set the maximum number of messages that will be printed, after which no more messages will be printed and the “stop limits” terminate the run when these limits are exceeded. There are six levels of message that increase in severity from informative all the way to programming errors, as outlined in Table 4.5.

See [MESSAGES – Define Message Print Limits and Stop Limits](#) in the GLOBAL section for a full description.

12.3.141 MESSOPTS – RESET SEVERITY LEVEL FOR FORCED TIME STEPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [MESSOPTS](#), resets the severity level for time steps that are forced to be accepted by the simulator. The normal severity level for this type of simulator generated message is PROBLEM and this can result in the run stopping depending on the parameters entered on the [MESSAGES](#) keyword. [MESSOPTS](#) can be used to reset the severity level to [MESSAGE](#), COMMENT, WARNING, or PROBLEM; for example, to avoid the run terminating due to too many PROBLEM messages.

Note that the [MESSAGES](#) keyword is a global keyword can therefore be used in any section; however, only the last instance of the keywords is active. The [MESSOPTS](#) keyword can only be used in the [SCHEDULE](#) section but can be used multiple times to change the severity level for forced time steps. Again, only the last occurrence of the keyword is active.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

12.3.142
MULSGGD – MULTIPLY MATRIX-FRACTURE COUPLING FOR OIL-GAS GRAVITY DRAINAGE FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [MULSGGD](#), defines a constant multiplier to modify the matrix-fracture coupling transmissibility for dual porosity and dual permeability models, for when the alternative matrix-fracture coupling transmissibilities for oil-gas gravity drainage has been selected. The alternative matrix-fracture coupling transmissibilities for oil-gas gravity drainage option is activated via the [SIGMAGD](#) or [SIGMAGDV](#) keywords in the [GRID](#) section, and the dual porosity or dual permeability models are activated by the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section, respectively.

This keyword applies the multiplier for all cells in model; whereas, the [MULSGGDV](#) keyword in the [SCHEDULE](#) section applies the multiplier to individual grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.143

MULSGGDV – MULTIPLY MATRIX-FRACTURE COUPLING FOR OIL-GAS GRAVITY DRAINAGE FOR INDIVIDUAL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [MULSGGDV](#), defines a constant multiplier to modify the matrix-fracture coupling transmissibility for dual porosity and dual permeability models, for when the alternative matrix-fracture coupling transmissibilities for oil-gas gravity drainage has been selected. The alternative matrix-fracture coupling transmissibilities for oil-gas gravity drainage option is activated via the [SIGMAGD](#) or [SIGMAGDV](#) keywords in the [GRID](#) section, and the dual porosity or dual permeability models are activated by the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section, respectively.

This keyword applies the multiplier for individual cells in model; whereas, the [MULSGGD](#) keyword in the [SCHEDULE](#) section applies the multiplier to all grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.144 MULTFLT – MULTIPLY THE TRANSMISSIBILITY OF A DEFINED FAULT BY A CONSTANT

MULTFLT enables the transmissibilities across defined faults, as declared by the **FAULTS** keyword, to be modified. They keyword allows for the re-scaling of the existing fault transmissibilities calculated by OPM Flow, for example setting a fault to be completely sealing by setting the multiplier to zero..

See [MULTFLT – Multiply the Transmissibility of a Defined Fault by a Constant](#) in the **GRID** section for a full description.

12.3.145 MULTPV – MULTIPLY CELL PORE VOLUMES BY A CONSTANT

MULTPV multiplies the pore volumes of a cell by a real positive constant for all the cells in the model via an array. An alternative to defining the complete array is to use the **BOX** keyword to define an area of the grid and then use the **MULTPV** keyword to set the multipliers just for the area defined by the **BOX** keyword (see the example).

See [MULTPV – Multiply Cell Pore Volumes by a Constant](#) in the **GRID** section for a full description.

12.3.146 MULTR – MULTIPLY CELL TRANSMISSIBILITY IN THE +R DIRECTION

MULTR multiplies the transmissibility between two cell faces in the +R direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I+1, J, K). The keyword should only be used with radial grids, as declared by the **RADIAL** keyword in the **RUNSPEC** section.

See [MULTR – Multiply Cell Transmissibility in the +R Direction](#) in the **GRID** section for a full description.

12.3.147 MULTR- – MULTIPLY CELL TRANSMISSIBILITY IN THE -R DIRECTION

MULTX- multiplies the transmissibility between two cell faces in the -X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I-1, J, K) and (I, J, K). The keyword should only be used with radial grids, as declared by the **RADIAL** keyword in the **RUNSPEC** section.

See [MULTR- – Multiply Cell Transmissibility in the -R Direction](#) in the **GRID** section for a full description.

12.3.148 MULTREGT – MULTIPLY TRANSMISSIBILITIES BETWEEN REGIONS

The **MULTREGT** keyword multiplies the transmissibility between two regions by a constant. The region number array can be **FLUXNUM**, **MULTNUM** or **OPERNUM** and these arrays must be defined and be available before the **MULTREGT** keyword is read by the simulator. The constant should be a real number.

See [MULTREGT – Multiply Transmissibilities Between Regions](#) in the **GRID** section for a full description.

12.3.149 MULTSIG – MULTIPLY MATRIX-FRACTURE COUPLING FOR ALL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [MULTSIG](#), defines a constant multiplier to modify the matrix-fracture coupling transmissibility for dual porosity and dual permeability models, for when the matrix-fracture coupling transmissibilities have been specified via the [SIGMAGD](#) or [SIGMAGDV](#) keywords in the [GRID](#) section, and the dual porosity or dual permeability models have been activated by the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section, respectively.

This keyword applies the multiplier for all cells in model; whereas, the [MULTSIGV](#) keyword in the [SCHEDULE](#) section applies the multiplier to individual grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.150
MULTSIGV – MULTIPLY MATRIX-FRACTURE COUPLING FOR INDIVIDUAL CELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [MULTSIGV](#), defines a constant multiplier to modify the matrix-fracture coupling transmissibility for dual porosity and dual permeability models, for when the matrix-fracture coupling transmissibilities have been specified via the [SIGMAGD](#) or [SIGMAGDV](#) keywords in the [GRID](#) section, and the dual porosity or dual permeability models have activated by the [DUALPORO](#) or [DUALPERM](#) keywords in the [RUNSPEC](#) section, respectively.

This keyword applies the multiplier for individual cells in model; whereas, the [MULTSIG](#) keyword in the [SCHEDULE](#) section applies the multiplier to all grid blocks.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.151 MULTTHT – MULTIPLY CELL TRANSMISSIBILITY IN THE +THETA DIRECTION

MULTTHT multiplies the transmissibility between two cell faces in the +Theta direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J+I, K). The keyword should only be used with radial grids, as declared by the *RADIAL* keyword in the *RUNSPEC* section.

See *MULTTHT – Multiply Cell Transmissibility in the +Theta Direction* in the *GRID* section for a full description.

12.3.152 MULTTHT- – MULTIPLY CELL TRANSMISSIBILITY IN THE -THETA DIRECTION

MULTTHT- multiplies the transmissibility between two cell faces in the -Theta direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J-I, K) and (I, J, K). The keyword should only be used with radial grids, as declared by the *RADIAL* keyword in the *RUNSPEC* section.

See *MULTTHT- – Multiply Cell Transmissibility in the -Theta Direction* in the *GRID* section for a full description.

12.3.153 MULTX – MULTIPLY CELL TRANSMISSIBILITY IN THE +X DIRECTION

MULTX multiplies the transmissibility between two cell faces in the +X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I+I, J, K).

See *MULTX – Multiply Cell Transmissibility in the +X Direction* in the *GRID* section for a full description.

12.3.154 MULTX- – MULTIPLY CELL TRANSMISSIBILITY IN THE -X DIRECTION

MULTX- multiplies the transmissibility between two cell faces in the -X direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I-I, J, K) and (I, J, K).

See *MULTX- – Multiply Cell Transmissibility in the -X Direction* in the *GRID* section for a full description.

12.3.155 MULTY – MULTIPLY CELL TRANSMISSIBILITY IN THE +Y DIRECTION

MULTY multiplies the transmissibility between two cell faces in the +Y direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J+I, K).

See *MULTY – Multiply Cell Transmissibility in the +Y Direction* in the *GRID* section for a full description.

12.3.156 MULTY- – MULTIPLY CELL TRANSMISSIBILITY IN THE -Y DIRECTION

MULTY- multiplies the transmissibility between two cell faces in the -Y direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J-I, K) and (I, J, K).

See *MULTY- – Multiply Cell Transmissibility in the -Y Direction* in the *GRID* section for a full description.

12.3.157 MULTZ – MULTIPLY CELL TRANSMISSIBILITY IN THE +Z DIRECTION

MULTZ multiplies the transmissibility between two cell faces in the +Z direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K) and (I, J, K+I).

See *MULTZ – Multiply Cell Transmissibility in the +Z Direction* in the *GRID* section for a full description.

12.3.158 MULTZ- – MULTIPLY CELL TRANSMISSIBILITY IN THE -Z DIRECTION

MULTZ- multiplies the transmissibility between two cell faces in the -Z direction for all the cells in the model via an array, that is the keyword sets the transmissibility multiplier of block (I, J, K) between the cells (I, J, K-I) and (I, J, K).

See *MULTZ- – Multiply Cell Transmissibility in the -Z Direction* in the *GRID* section for a full description.

12.3.159 NCONSUMP – NODE GAS CONSUMPTION (EXTENDED NETWORK)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [NCONSUMP](#) keyword defines an extended network node’s gas consumption rate, for when the Extended Network option has been activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section. The keyword can also be used to attribute the gas consumption to a previously defined group. See also the [GCONSUMP](#) keyword in the [SCHEDULE](#) section that overs more flexibility and can also be used with the Extended Network option.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.160 NEFAC – NODE EFFICIENCY FACTORS (EXTENDED NETWORK)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [NEFAC](#) keyword defines an extended network node's efficiency factor, for when the Extended Network option has been activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section. See also the [GEFAC](#) keyword in the [SCHEDULE](#) section that can also be used with the Extended Network option.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NODE	A character string of up to eight characters in length that defines the node name for which the node efficiency factor is being defined.			None
2	FACTOR	A real positive value that is less than or equal to one that defines the efficiency factor for the node. If a node's down time is 5% then FACTOR should be set to 0.95 (1.0 – 0.05).			1.0
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.56: NEFAC Keyword Description

See also the [WEFAC](#) keyword in the [SCHEDULE](#) section to define a well's' efficiency factor.

12.3.161 NETBALAN – NETWORK BALANCING PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

NETBALAN keyword causes the simulator to perform a network balancing operation at the next time step. In addition, the keyword defines the network balancing parameters used to control how network balancing is performed on a network, as well as the frequency of the network balancing calculations. If any of the parameters on the keyword are defaulted, then either the previously entered values are used, or if there are no previously entered values, then the keyword default values are employed.

No.	Name	Description			Default		
		Field	Metric	Laboratory			
1	NTSTEP	NSTEP is a real value that defines the criteria for the network balancing interval, used to define the frequency of the network balancing algorithm. NSTEP may be a negative number, a value of zero, or a positive number, as described below: 1) Negative Value: If NSTEP is any negative value then the time stepping interval for the network balancing is based on the number of Newton iterations used in updating the well targets, as set by the NUPCOL keyword in the RUNSPEC section, or the NUPCOL parameter on the GCONTOL keyword in the SCHEDULE section. If NUPCOL is defined on both keywords, then the GCONTOL value takes precedence. 2) Thus, any negative value of NSTEP means the network will be balanced for the first NUPCOL Newton iterations for each time step. 3) Consequently, if this option is used it may result in some oscillations as the THP of the wells are updated at each Newton network re-balance computation. 4) Zero Value: Here the network is balanced at the beginning of every time step, and should result in small network balancing errors for the time step, provided of course the reservoir conditions are fairly constant over the time step. This is the default value. 5) Positive Value: In this case NSTEP is a time interval between the previous network balance and the next schedule network balancing. For example, if NSTEP is set equal to 91.25 days (365/4), then network balancing will occur every quarter. Again, the balancing occurs at the beginning of the time step. <u>Note only negative or zero values are currently supported.</u>			0.0		
		days or dimensionless	days or dimensionless	hours or dimensionless			
		2	NCNV	NCNV is a real positive value that defines the nodal pressure convergence variance, used to define if convergence has been satisfied.			Defined
psia 1.45	barsa 0.10			atma 0.09869			

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	NMXITER	A positive integer value that defines the maximum number of network balancing operations that may be performed during a network balancing computation.			10
		dimensionless	dimensionless	dimensionless	
4	GRPCNV	GRPCNV is a real positive value less than one that sets the group tolerance criteria used to define if convergence has been satisfied for: 1) Wells in subsea completion manifold groups under rate control. In this case GRPCNV is applied to the THP values of the wells within the group. In the field, these type of wells must flow against a common manifold pressure, that is all the wells must flow at approximately the same THP, for a given group's flow rate. 2) Automatic chokes for group's under rate control. Here, GRPCNV is applied to the pressure drop across the automatic chokes. In both cases GRPCNV is an acceptable fraction of the group's rate target. Thus, the default value 0.01 means that network balance calculated rate must be within 0.01 (or 1.0%) of the groups' production target. Note that this criteria may not be satisfied if the number of Newton iterations used in updating the well targets, as set by the NUPCOL keyword in the RUNSPEC section, or the NUPCOL parameter on the GCONTROL keyword in the SCHEDULE section, is exceeded. Group convergence criteria can also be set by the GRPCNV parameter on the GCONTROL keyword in the SCHEDULE section. If both values of GRPCNV have been entered, then the minimum of the two is used.			0.01
		dimensionless	dimensionless	dimensionless	
5	THPMXITE	A positive integer value that defines the maximum number of well THP iterations for wells in subsea completion manifold groups under rate control. There is no equivalent iteration maximum for automatic chokes in groups under rate control, as the pressure drop across the chokes is automatically calculated in the network balancing calculation. Thus, NMXITER is used for these calculations.			10
		dimensionless	dimensionless	dimensionless	
6	NTRGERR	A real positive value that stipulates the maximum target branch error in a network balance calculation, at the end of the time step. NTRGERR is compared with the branch error (Error), where Error is the difference between the pressure drop along the branch from the previous network balance and the current calculation, again, at the end of the time step. Subject to existing targets and constraints, the simulator will attempt to adjust the time step size in order to roughly match NTRGERR. Note also that NTRGERR should be a value that is sufficient to allow convergence as defined by NNCNV and GRPCNV parameters. The default value of 1.0×1020 means that this parameter has no effect in the selection of the time step size.			1.0×1020
		psia	barsa	atma	

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	NMAXERR	A real positive value that stipulates the maximum branch error in a network balance calculation, at the end of the time step. Again, NMAXERR is compared against the branch error (Error), where Error is the difference between the pressure drop along the branch from the previous network balance and the current calculation, at the end of the time step. If Error is greater than NMAXERR then the simulator will enforce a time step chop. Time step chops are computationally expensive and should therefore be minimized. Hence, care should be used in setting this value, which should be significantly greater than NTRGERR. The default value of 1.0×1020 means that this parameter has no effect in the selection of the time step size.			1.0×1020
		psia	barsa	atma	
8	NTSMIN	NTSMIN is a real positive value that sets the minimum time step size for when NTRGERR and NMAXERR have values. Large network balancing errors (Error) can take place for various reasons, a common occurrence is for wells producing near their operating limit which may shut in during a network balance operation. If NTRGERR and NMAXERR have values, then this will result in the simulator enforcing time step chops to perhaps very small time steps. NTSMIN can therefore be used to set the minimum time step size under, and only, these circumstances. The default value, TSMINZ, is taken from TSMINZ parameter on the TUNING keyword in the SCHEDULE section.			TSMINZ
		days	days	hour	
Notes:					
I) The keyword is terminated by a “/”.					

Table 12.57: NETBALAN Keyword Description

Examples

The first example sets network balancing to occur at the beginning of each time step using a 1.0 psia convergence criteria, and a maximum of 10 iterations. All the other parameters are defaulted.

```
--
--      NETWORK BALANCE CONTROL OPTIONS
--
--      BAL  CONV  MAX  WTHP  WTHP  ERR  ERR  TSTEP
--      INTV  TOL  ITRS  TOL  ITS  TARG  MAX  MIN
NETBALAN
      0.0   1.0   10   1*   1*   1*   1*   1*   /
```


The next example sets network balancing to occur for the first *NUPCOL* Newton iterations for each time step. All the other parameters are defaulted.

```
--
--      NETWORK BALANCE CONTROL OPTIONS
--
--      BAL   CONV   MAX   WTHP   WTHP   ERR   ERR   TSTEP
--      INTV  TOL   ITRS  TOL   ITS   TARG  MAX   MIN
NETBALAN
      -1     1*     1*     1*     1*     1*     1*     1*      /
```

12.3.162
NETCOMPA
–
DEFINE
AUTOMATIC
COMPRESSORS
(EXTENDED NETWORK)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [NETCOMPA](#) keyword defines automatic compressors in an extended network, for when the Extended Network option has been activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.163 NEXT – MAXIMUM NEXT TIME STEP SIZE (ALIAS FOR NEXTSTEP)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the maximum time step size the simulator should take for the next time step. This keyword can be used to reset the time step for when known large changes to the model are taking place that may result in time step chops. For example, if the reporting time size is using monthly reporting steps via the [DATES](#) keyword in the [SCHEDULE](#) section, then if for example, a group of wells start production at a given date, then the [NEXT](#) keyword can be used to shorten the next step in order to avoid a time step chop.

Time steps chops are computationally expensive as the simulator cannot solve the current time step at the given tolerance, and therefore has to reduce the time step size. For example, if the previous completed time step was at day 365 and the current time step ending at 396 days cannot be solved, then the simulator will reduce the current time step to perhaps end at day 370, if this still cannot be solved then the time step will be chopped back again to perhaps to less than one day. Using the [NEXT](#) or [NEXTSTEP](#) keyword, the simulator is instructed to take a small time step in the anticipation that this will avoid time step chops and thus improve computational performance.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NSTEPI	NSTEPI is a real positive value that defines the maximum length of the next time step.			None
		days	days	hours	
2	NSTEP2	NSTEP2 is a character string that should be set to either NO or YES to state if the NSTEPI should be applied to future reporting time steps. 1) NO: Means that NSTEPI should not be applied to subsequent reporting time steps. 2) YES: Means that NSTEPI should be applied to subsequent reporting time steps. The default value of NO means that NSTEPI will only be applied once.			NO

Notes:

1) The keyword is terminated by a “/”.

Table 12.58: NEXT Keyword Description

See also the [DATES](#) and [TSTEP](#) keywords in the [RUNSPEC](#) section that are used to advance the simulation through time.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored. See section [2.2 Running OPM Flow From The Command Line](#) on how to control time stepping for OPM Flow.

Example

```
--      NEXT      ALL
--      STEP      TIME
--      ----      ----
NEXT
          1      'NO'      /
```

Here the next step size is set to one day and should only be used once.

12.3.164 NEXTSTEP – MAXIMUM NEXT TIME STEP SIZE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the maximum time step size the simulator should take for the next time step. This keyword can be used to limit the time step size when known large changes to the model are taking place that may otherwise result in time step chops. For example, if the simulator is using monthly reporting steps defined by the [DATES](#) keyword in the [SCHEDULE](#) section, then if a group of wells start production at a given date, then the [NEXTSTEP](#) keyword can be used to limit the next time step in order to avoid a time step chop.

Time steps chops are computationally expensive as the simulator cannot solve the current time step at the given tolerance, and therefore has to reduce the time step size. For example, if the previous completed time step was at day 365 and the current time step ending at 396 days cannot be solved, then the simulator will reduce the current time step to perhaps end at day 370, if this still cannot be solved then the time step will be chopped back again and so on, possibly reducing the time step to less than one day. Using the [NEXT](#) or [NEXTSTEP](#) keyword, the simulator is instructed to take a small time step in the anticipation that this will avoid time step chops and thus improve computational performance.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NSTEPI	A real positive value that defines the maximum length of the next time step.			None
		days	days	hours	
2	NSTEP2	A defined character string that should be set to either NO or YES to state if the NSTEPI should be applied to future reporting time steps. 1) NO: Means that NSTEPI should not be applied to subsequent reporting time steps. 2) YES: Means that NSTEPI should be applied to subsequent reporting time steps. The default value of NO means that NSTEPI will only be applied once.			NO

Notes:

1) The keyword is terminated by a “/”.

Table 12.59: NEXTSTEP Keyword Description

See also the [DATES](#) and [TSTEP](#) keywords in the [RUNSPEC](#) section that are used to advance the simulation through time.

See section [2.2 Running OPM Flow From The Command Line](#) on how to control time stepping for OPM Flow that allows for greater flexibility.

Examples

The first example shows the direct use of the `NEXTSTEP` keyword. Here the next time step size is limited to one day for the first time step following the next report time only.

```
--      NEXT      ALL
--      STEP      TIME
--      ----      ----
NEXTSTEP
      1          'NO'          /
```

The next example shows a more complete use of the keyword for when the field oil production has increased dramatically from 10,000 stb/d to 50,000 stb/d as indicated by the two `GCONPROD` keywords.

```
-- -----
-- SCHEDULE SECTION - 2021-01-01
-- -----
--
-- GROUP PRODUCTION CONTROLS
--
-- GRUP      CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME      MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF    WAT
GCONPROD
'FIELD'     'ORAT' 10E3   60E3   300E3  60E3   1*    1*    1*    1*    1*    /
/
RPTSCHED
'WELLS=2'   'WELSPECS' 'CPU=2' 'FIP=2'          /

DATES
  2  JAN    2021  /
/
RPTSCHED
'NOTHING'   /

DATES
  1  FEB    2021  /
  1  MAR    2021  /
/
-- GROUP PRODUCTION CONTROLS
--
-- GRUP      CNTL  OIL    WAT    GAS    LIQ    CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME      MODE  RATE   RATE   RATE   RATE   OPT   CNTL  RATE   DEF    WAT
GCONPROD
'FIELD'     'ORAT' 50E3   90E3   300E3  90E3   1*    1*    1*    1*    1*    /
/
--
--      NEXT      ALL
--      STEP      TIME
--      ----      ----
NEXTSTEP
      1          'NO'          /

DATES
  1  APR    2021  /
  1  MAY    2021  /
  1  JUN    2021  /
  1  JUL    2021  /
  1  AUG    2021  /
  1  SEP    2021  /
  1  OCT    2021  /
  1  NOV    2021  /
  1  DEC    2021  /
/
```

Given a start date of January 1, 2020 set via the [START](#) keyword in the [RUNSPEC](#) section, the above example shows the initial oil production of 10,000 stb/d starting in January 1, 2020, and continuing up to March 1, 2021. At the March 1, 2021 report time the field oil production rate is increased to 50,000 stb/d and the maximum next time step is set to one day. After the one day time step is completed (March 2, 2012), the simulator will progressively increase the time step size until a maximum of 31 days is reached. The 31 day maximum is a result of requesting monthly time steps via the [DATES](#) keyword. The intent of using the [NEXTSTEP](#) keyword in this case is to prevent time step chops occurring due to the “shock” to the system caused by the large increase in oil production.

12.3.165 NEXTSTPL – MAXIMUM NEXT TIME STEP SIZE (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the maximum time step size the simulator should take for the next time step for all Local Grid Refinements (“LGR”). This keyword can be used to reset the time step for when known large changes to the model are taking place that may result in time step chops. For example, if the reporting time size is using monthly reporting steps via the [DATES](#) keyword in the [SCHEDULE](#) section, then if for example, a group of wells start production at a given date, then the [NEXTSTPL](#) keyword can be used to shorten the next step in all the LGRs in order to avoid a time step chop.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NSTEPI	NSTEPI is a real positive value that defines the maximum length of the next time step.			None
		days	days	hours	
2	NSTEP2	NSTEP2 is a character string that should be set to either NO or YES to state if the NSTEPI should be applied to future reporting time steps. 1) NO: Means that NSTEPI should not be applied to subsequent reporting time steps. 2) YES: means that STEP1 should be applied to subsequent reporting time steps. The default value of NO means that NSTEPI will only be applied once.			NO

Notes:

1) The keyword is terminated by a “/”.

Table 12.60: NEXTSTPL Keyword Description

See also the [NEXT](#) and [NEXTSTEP](#) keywords in the [SCHEDULE](#) section that are used to control the global grid’s next time step.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped. See section [2.2 Running OPM Flow From The Command Line](#) on how to control time stepping for OPM Flow.

Example

```
--
--      NEXT      ALL
--      STEP      TIME
--      ----      ----
NEXTSTEP
      1          'NO'          /
```

Here the next step size for all LGRs is set to one day and should only be used once.

12.3.166 NODEPROP – DEFINE NETWORK NODE PROPERTIES FOR EXTENDED NETWORK

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the network node properties for the extended network option for when the Extended Network Model has been invoked by the [NETWORK](#) keyword in the [RUNSPEC](#) section. There are two types of network facilities in the simulator, the Standard Network model, which is defined with the [GRUPNET](#) keyword in the [SCHEDULE](#) section and the Extended Network Model defined by the [BRANPROP](#) and [NODEPROP](#) keywords, again in the [SCHEDULE](#) section.

For the Extended Network Model the group hierarchy can be different to that defined by the [GRUPTREE](#) keyword; however, the bottom most nodes in the network tree associated with wells, must be the same as that defined by the [GRUPTREE](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	NODE	A character string of up to eight characters in length that defines the node name for the data on this keyword record.			None
2	PRESS	A real value that sets the terminal fixed pressure for the node, this should be set to: 1) A real positive value to define the fixed pressure for the node if the node is a terminal node, otherwise: 2) PRESS should be set to the default value of I* or a real negative value if the node is not a terminal node.			I*
		psia	bars	atm	
3	CHOKE	CHOKE is a defined character string that sets if the downstream branch (or uptree branch, being the one closer towards the terminal node) from this node should have the capability to choke back the flow rate in order impose a flow constraint. Here the downstream branch is the node furthestmost from the wells. Thus for a production network, this will be an outlet branch as the wells are exporting fluid from the branch node. Whereas for an injection node, this is an inlet branch as the wells are importing the injection fluid.			NO
4	GASLIFT	A defined character string that sets if the associated subordinate well's produced gas lift gas should be included in the node's flow stream (YES), or not (NO). GASLIFT should be set to either: 1) NO: Do not include gas lift gas in the node's production. This means that only the produced gas from the node's subordinate wells will be included in the node's production stream. 2) YES: Include both gas lift gas and produced gas in the node's production. This means that all gas from the subordinate wells will be included in the node's production stream. If NODE does not have any subordinate wells or satellite groups (see the GSATPROD keyword in the SCHEDULE section) directly attached to the node, then GASLIFT should be defaulted (I*) or set to NO. The option is only valid for producing networks			NO

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	GROUP	A character string of up to eight characters in length that defines the group for which the automatic choke will be applied in order to match the group's target rate (the TARGET variable on the GCONPROD keyword in the SCHEDULE section). The target rate is matched by adjusting the pressure drops across the automatic choke. The default value of I* uses NODE (item one) as the group if it exists in the run. In addition, if NODE is just connected to subordinate wells then GROUP should also be defaulted.			I*
6	GRPNAME	GRPNAME defines the name of the source or sink group in the commercial compositional simulator; however, the variable is not used as sources and sinks nodes must have the same name as their comparable groups in the Extended Network Model in both OPM Flow and the commercial black-oil simulator.			I*
7	NETYPE	NETYPE defines the network type in the commercial compositional simulator: PROD,WINJ or GINJ. However, the variable is not used as only production networks are supported in the Extended Network Model in both OPM Flow and the commercial black-oil simulator.			I*
Notes: 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.61: NODEPROP Keyword Description

See also the [NETWORK](#) keyword in the [RUNSPEC](#) section and the [BRANPROP](#) in the [SCHEDULE](#) section.

Example

Given the following Extended Network model in Figure 12.5.

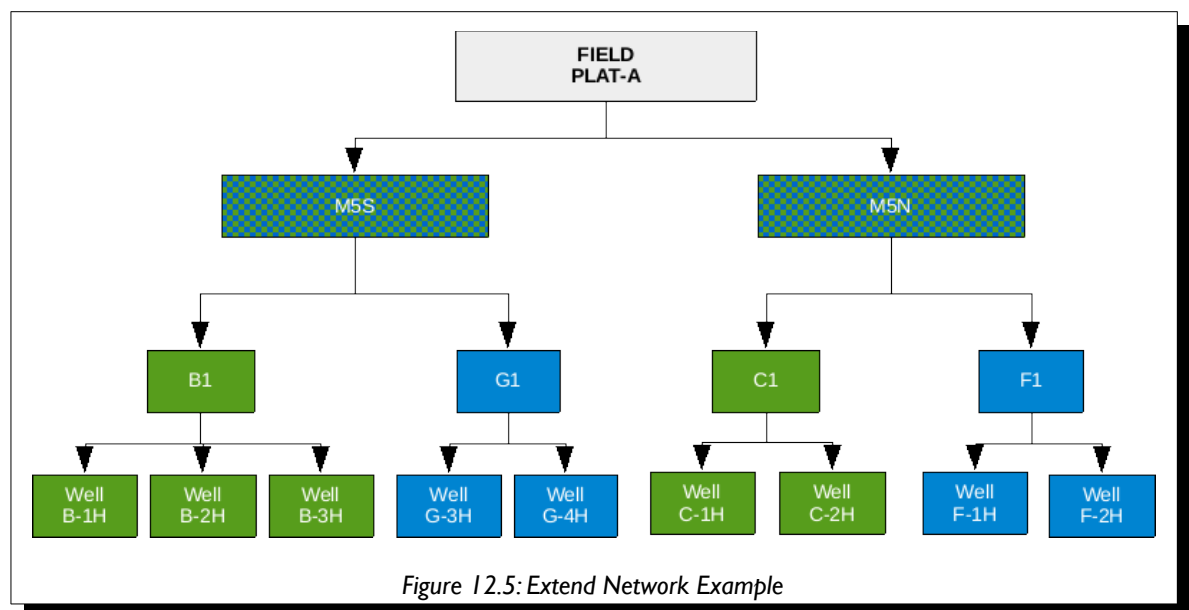


Figure 12.5: Extend Network Example

First the Extended Network model should be used invoked in the [RUNSPEC](#) section, and then the [BRANPROP](#) keyword should be used to define the branch network, and finally the [NODEPROP](#) keyword is used to describe the node properties with the network.

```
-- =====
--
-- RUNSPEC SECTION
--
-- =====
RUNSPEC
--
--      ACTIVATE THE EXTENDED NETWORK OPTION AND DEFINE PARAMETERS
--
--      MAX.      MAX      NOT
--      NODE      LINK      USED
NETWORK
      3          2          1*                               /
.....
.....
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      EXTENDED NETWORK BRANCH PROPERTIES
--
-- DOWN  UP      VFP      VFP
-- NODE  NODE      TABLE  ALFQ
BRANPROP
B1      PLAT-A    5        1*                               /
C1      PLAT-A    4        1*                               /
/
--
--      EXTENDED NETWORK NODE PROPERTIES
--
-- NODE  NODE  CHOKE  GAS   CHOKE  SOURCE  NETWORK
-- NAME  PRESS OPTN  LIFT  GROUP  SINK    TYPE
NODEPROP
PLAT-A  21.0   NO     NO
B1      1*    NO     NO
C1      1*    NO     NO
/
```

Here the main platform for the field, PLAT-A, has a fixed 21 barsa pressure applied as an operating constraint.

12.3.167 NOECHO – DEACTIVATE ECHOING OF USER INPUT FILES TO THE PRINT FILE

Turns off echoing of all the input files to the print file. Note by default echoing of the inputs files is active, but can subsequently be switched off by the [NOECHO](#) activation keyword.

See [NOECHO – Deactivate Echoing of User Input Files to the Print File](#) in the GLOBAL section for a full description.

12.3.168 NOHMD – DEACTIVATE HISTORY MATCH GRADIENT DERIVATIVE CALCULATIONS

The [NOHMD](#) deactivates various history match gradient derivative calculations for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The keyword consists of a series of character strings that define which derivative should be switch off based on the keyword that requested the derivatives to be calculated, for example [HMFAULTS](#) keyword in the [GRID](#) section. If an empty list is entered then all the gradient derivative calculations previously requested are switch off. The keyword is useful for changing from history matching runs to predication cases, as the prediction cases will be more computationally efficient without the burden of the gradient derivative calculations.

See [NOHMD – Deactivate History Match Gradient Derivative Calculations](#) in the [SOLUTION](#) section for a full description.

12.3.169 NOHMO – DEACTIVATE HISTORY MATCH GRADIENT DERIVATIVE CALCULATIONS (ALIAS)

The [NOHMO](#) deactivates various history match gradient derivative calculations for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section. The keyword consists of a series of character strings that define which derivative should be switch off based on the keyword that requested the derivatives to be calculated, for example [HMFAULTS](#) keyword in the [GRID](#) section. If an empty list is entered then all the gradient derivative calculations previously requested are switch off. The keyword is useful for changing from history matching runs to predication cases, as the prediction cases will be more computationally efficient without the burden of the gradient derivative calculations.

See [NOHMO – Deactivate History Match Gradient Derivative Calculations \(Alias\)](#) in the [SOLUTION](#) section for a full description.

12.3.170 NOSIM – ACTIVATE THE NO SIMULATION MODE FOR DATA FILE CHECKING

[NOSIM](#) switches the mode of OPM Flow to data input checking mode. In this mode the input file is read and all messages and print instructions are sent to the respective output files. The [SCHEDULE](#) section is read but the simulation is not performed.

See [NOSIM – Activate the No Simulation Mode for Data File Checking](#) in the [RUNSPEC](#) section for a full description

12.3.171 NOWARN – DEACTIVATE WARNING MESSAGES

Turns off warning messages to be printed to the print file; note that this keyword is deactivated by default and can subsequently be switched off by the [WARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#).

See [NOWARN – Deactivate Warning Messages](#) in the GLOBAL section for a full description.

12.3.172 NUPCOL – DEFINE THE NUMBER OF NEWTON ITERATIONS USED TO UPDATE WELL TARGETS

The [NUPCOL](#) keyword defines the maximum number of Newton iterations within a time step that may be used to update the well production and injection targets, after which the well targets will be frozen until the time step calculations have converged and the time step is complete.

See [NUPCOL – Define the Maximum Number of Newton Iterations Used to Update Well Targets](#) in the [RUNSPEC](#) section for a full description and also section [2.2 Running OPM Flow From The Command Line](#) on how to set various other numerical control parameters for OPM Flow.

12.3.173 NWATREM – NODE WATER REMOVAL (EXTENDED NETWORK)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [NWATREM](#) keyword defines an extended network node as a point where water is removed from the network, for when the Extended Network option has been activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section. The water to be removed can be specified as a rate or as a fraction of the total volume passing through the node.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.174 OUTSOL – DEFINE DATA TO BE WRITTEN TO THE RESTART FILE (RETIRED)

This keyword defines the data and frequency of the data to be written to the [RESTART](#) file at each requested restart point. The keyword has been replaced by the [RPTRST](#) keyword in the [SOLUTION](#) and [SCHEDULE](#) sections and is therefore considered retired.

See [OUTSOL – Define Data to be Written to the RESTART File \(Retired\)](#) in the [SOLUTION](#) section for a full description.

12.3.175 PICOND – DEFINE THE GENERALIZED PSEUDO PRESSURE PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *PICOND* keyword defines the Generalized Pseudo Pressure (“GPP”)³²³ and ³²⁴ parameters used in a gas condensate well connection inflow equations. GPP accounts for both the impact of condensate drop out and compressibility in the mobility inflow term . If the keyword is absent from the input deck then the default values are applied.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

323

Whitson, C. H. and Fevang, Ø. “Generalised Pseudopressure Well Treatment in Reservoir Simulation,” Presented at the IBC Technical Services Conference on Optimisation of Gas Condensate Fields, Aberdeen, UK (June 26-27, 1997)

324

Whitson, C. H. and Fevang, Ø. “Modeling Gas Condensate Well Deliverability,” paper SPE 30714, SPE Reservoir Engineering (1996) 11, No. 4, 221-230; also presented at the SPE Annual Technical Conference and Exhibition, Dallas, Texas, USA (October 22-25, 1995).

12.3.176 PIMULTAB – DEFINE WELL PRODUCTIVITY INDEX VERSUS WATER CUT TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[PIMULTAB](#) defines productivity index multiplier versus water cut tables that are used to scaled a well's connection factors based on connection's current producing water cut. The tables are used for modeling the productivity decline due to increasing water cut. Allocation of the tables to a well is via the [WPITAB](#) keyword in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WCUT	A real monotonically increasing positive columnar vector that defines the maximum surface water cut for the corresponding PIMULT vector. Water cut is defined as $f_w = \frac{q_w}{q_w + q_o}$.			None
		dimensionless	dimensionless	dimensionless	
2	PIMULT	A real positive decreasing columnar vector that defines the productivity index multiplier used to scale a well's connection factors, for the corresponding WCUT vector.			None
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) The keyword is followed by NTPIMT tables as stated on the [PIMTDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each table must contain a minimum of two rows and a maximum of NRPIMT rows as declared on the [PIMTDIMS](#) keyword in the [RUNSPEC](#) section.
- 3) Each table is terminated by a "/" and there is no "/" terminator for the keyword.

Table 12.62: PIMULTAB Keyword Description

See also the [WPITAB](#) keyword that allocates the tables to the wells, and also the [WPIMULT](#) keyword that scales a well's productivity index by a constant value, both of which are in the [SCHEDULE](#) section. This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Examples

Given NTPIMT equals two and NRPIMT equals four on [PIMTDIMS](#) keyword in the [RUNSPEC](#) section, then:

```
--      DEFINE WELL PRODUCTIVITY INDEX VERSUS WATER CUT TABLES
--
--      MAX      PI
--      WCUT     MULT
--      -----
PIMULTAB
      0.0000    1.0000
      0.2500    0.9500
      0.5000    0.8500
      0.7500    0.7500
--
```



```
--
      0.0000    1.0000
      0.2500    0.9500
      0.5000    0.8500
      0.7500    0.7500      /
```

The next example is summarized from the Norne model with NTPIMT equals one and NRPIMT equals to 51 on the *PIMTDIMS* keyword in the *RUNSPEC* section.

```
--
--      DEFINE WELL PRODUCTIVITY INDEX VERSUS WATER CUT TABLES
--      The following is the reviewed model in Aug-2006, low-high case
--      a=0.25, b=0.1; PIMULT=(1-a)/exp(fw/b)+a
--
--      MAX      PI
--      WCUT     MULT
--      -----
PIMULTAB
      0.000    1.0000
      0.025    0.8341
      0.050    0.7049
      0.075    0.6043
      0.100    0.5259
      0.125    0.4649
      0.150    0.4173
      0.175    0.3803
      0.200    0.3515
      0.225    0.3290
      0.250    0.3116
      0.275    0.2979
      0.300    0.2873
      0.325    0.2791
      0.350    0.2726
      0.375    0.2676
      0.400    0.2637
      0.425    0.2607
      0.450    0.2583
      0.475    0.2565
      0.500    0.2551
      0.525    0.2539
      0.550    0.2531
      0.575    0.2524
      0.600    0.2519
      0.625    0.2514
      0.650    0.2511
      0.675    0.2509
      0.700    0.2507
      0.725    0.2505
      0.750    0.2504
      0.775    0.2503
      0.800    0.2503
      0.825    0.2502
      0.850    0.2502
      0.875    0.2501
      0.900    0.2501
      0.925    0.2501
      0.950    0.2501
      0.975    0.2500
      1.000    0.2500 /
```

12.3.177 PLYADS – DEFINE POLYMER ROCK ADSORPTION TABLES

The [PLYADS](#) keyword defines the rock polymer adsorption tables for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. Alternatively, the functions can be entered via the [PLYADSS](#) keyword in the [PROPS](#) section for when salt sensitivity is to be considered.

See [PLYADS – Define Polymer Rock Adsorption Tables](#) in the [PROPS](#) section for a full description.

12.3.178 PLYDHFLF – DEFINE POLYMER THERMAL DEGRADATION HALF-LIFE TABLES

The [PLYDHFLF](#) keyword defines the polymer thermal degradation half-life with respect to temperature functions for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

See [PLYDHFLF – Define Polymer Thermal Degradation Half-Life Tables](#) in the [PROPS](#) section for a full description.

12.3.179 PLYMAX – DEFINE POLYMER-SALT VISCOSITY MIXING CONCENTRATIONS

The [PLYMAX](#) keyword defines maximum polymer and salt concentrations that are to be used in the mixing parameter calculation of the fluid component viscosities, for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

See [PLYMAX – Define Polymer-Salt Viscosity Mixing Concentrations](#) in the [PROPS](#) section for a full description.

12.3.180 PLYROCKM – MODIFY POLYMER-ROCK PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [PLYROCKM](#) keyword modifies rock properties entered via the [PLYCAMAX](#), [PLYKRRF](#), [PLYRMDEN](#), and [PLYROCK](#) keywords in the [PROPS](#) section, for when the Polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.181 PLYSHEAR – ACTIVATE AND DEFINE POLYMER SHEARING PARAMETERS

The [PLYSHEAR](#) keyword activates and defines the polymer shear thinning-thickening option for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

See [PLYSHEAR – Activate and Define Polymer Shearing Parameters](#) in the [PROPS](#) section for a full description.

12.3.182 PLYSHLOG – ACTIVATE AND DEFINE THE POLYMER SHEARING LOGARITHMIC PARAMETERS

This keyword activates and defines the parameters for the logarithm-based polymer shear thinning/thickening option.

See [PLYSHLOG – Activate and Define the Polymer Shearing Logarithmic Parameters](#) in the [PROPS](#) section for a full description.

12.3.183 PLYVISC – DEFINE POLYMER VISCOSITY SCALING FACT

PLYVISC defines the polymer viscosity scaling factors used to determine the relationship of pure water viscosity with respect to increasing polymer saturation within a grid block. The polymer option must be activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section in order to use this keyword.

See [PLYVISC – Define Polymer Viscosity Scaling Factors](#) in the [PROPS](#) section for a full description.

12.3.184 PLYVISCS – DEFINE POLYMER-SALT VISCOSITY SCALING FACTORS

PLYVISCS defines the polymer-salt viscosity scaling factor tables applied to pure water that are used to determine the viscosity of a polymer-salt mixture with respect to increasing polymer saturation within a grid block. The polymer option must be activated by the [POLYMER](#) keyword, as well as the brine phase declared by the [BRINE](#) keyword in the [RUNSPEC](#) section in order to use this keyword. However the [ECLM](#) keyword in the [RUNSPEC](#) must not be used with this keyword.

See [PLYVISCS – Define Polymer-Salt Viscosity Scaling Factors](#) in the [PROPS](#) section for a full description.

12.3.185 PLYVISCT – DEFINE POLYMER-TEMPERATURE VISCOSITY SCALING FACTORS

PLYVISCT defines the polymer-temperature viscosity scaling factor tables applied to pure water that are used to determine the viscosity of the polymer at a given temperature with respect to increasing polymer saturation within a grid block. Both the polymer option must be activated by the [POLYMER](#) keyword and the temperature option invoked by the [TEMP](#) keyword in the [RUNSPEC](#) section in order to use this keyword. However the [BRINE](#) keyword in the [RUNSPEC](#) must not be used with this keyword, that is the salt sensitivity options should be deactivated.

See [PLYVISCT – Define Polymer-Temperature Viscosity Scaling Factors](#) in the [PROPS](#) section for a full description.

12.3.186 PLYVSCST – DEFINE POLYMER-SALT-TEMPERATURE VISCOSITY SCALING FACTORS

[PLYVSCST](#) defines the polymer-salt-temperature viscosity scaling factor tables applied to pure water that are used to determine the viscosity of the polymer at a given salt concentration and for a given temperature, with respect to increasing polymer saturation within a grid block. Both the polymer option must be activated by the [POLYMER](#) keyword and the temperature option invoked by the [TEMP](#) keyword in the [RUNSPEC](#) section in order to use this keyword. In addition, the [BRINE](#) keyword in the [RUNSPEC](#) must also be invoked. The keyword is used in conjunction with the [SALTNODE](#) keyword to define the various salt concentrations and the [TEMPNODE](#) keyword to define the various reservoir temperatures. Both keywords are in the [PROPS](#) section.

See [PLYVSCST – Define Polymer-Salt-Temperature Viscosity Scaling Factors](#) in the [PROPS](#) section for a full description.

12.3.187 PRIORITY – ACTIVATE AND DEFINE WELL PRIORITIZATION COEFFICIENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	-----------------

Description

The **PRIORITY** keyword activates the Well Priority option and defines the coefficients in the well priority equation. Wells under group control are ranked based on their well potential in order to satisfy group controls. For example if a group's oil target is exceeded, then the group may shut-in the lease productive oil wells based on their well potential. The Priority option is an alternative form of ranking the wells based on the following equation:

$$\text{Priority} = \frac{a_1 + a_2 Q_{\text{oil}} + a_3 Q_{\text{water}} + a_4 Q_{\text{gas}}}{b_1 + b_2 Q_{\text{oil}} + b_3 Q_{\text{water}} + b_4 Q_{\text{gas}}} \quad (12.28)$$

Where:

Q_{oil}	= well oil potential
Q_{water}	= well water potential
Q_{gas}	= well gas potential
a_{1-4}	= priority coefficients supplied by this keyword
b_{1-4}	= priority coefficients supplied by this keyword

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TIME	A real positive integer that defines the minimum time interval between executing the well priority calculation. The calculation is performed at the beginning of the time step that exceeds the previous calculation (t_0) by a minimum of TIME, that is for when $t_n \geq (t_0 + \text{TIME})$. Note that the default value of zero means that the calculation is performed at each time step. As a consequence, this may result in some oscillation as well wells are switched on/off at subsequent time steps.			0
		days	days	hours	
2	A1	A real positive integer greater than or equal to zero that defines a_1 priority coefficient in equation (12.28).			0
3	A2	A real positive integer greater than or equal to zero that defines a_2 priority coefficient in equation (12.28).			0
4	A3	A real positive integer greater than or equal to zero that defines a_3 priority coefficient in equation (12.28).			0
5	A4	A real positive integer greater than or equal to zero that defines a_4 priority coefficient in equation (12.28).			0
6	B1	A real positive integer greater than or equal to zero that defines b_1 priority coefficient in equation (12.28).			0

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	B2	A real positive integer greater than or equal to zero that defines b_2 priority coefficient in equation (12.28).			0
8	B3	A real positive integer greater than or equal to zero that defines b_3 priority coefficient in equation (12.28).			0
9	B4	A real positive integer greater than or equal to zero that defines b_4 priority coefficient in equation (12.28).			0
Notes: 1) The keyword is terminated by a “/”.					

Table 12.63: PRIORITY Keyword Description

Example

```
--
--      SETS COEFFICIENTS FOR WELL PRIORITIZATION OPTION
--
--      TIME   A      B      C      D      E      F      G      H
--      STEP   Qo     Qw     Qg     Qo     Qw     Qg     Qo     Qw     Qg
PRIORITY
0.0    0.0    1.0    0.0    0.0    1.0    0.0    0.0    0.0 / High Oil Pot
--      0.0    0.0    1.0    1.0    0.0    0.0    0.0    1.0    0.0 / Low Water Cut Pot
--      0.0    0.0    1.0    0.0    0.0    0.0    0.0    0.0    1.0 / Low GOR Pot
```

The above example defines the well priority calculation to be based on a well's oil potential, with calculation to be performed at each time step. Note that the low water cut and low GOR options are given for reference but are commented out and therefore ignored by the simulator.

12.3.188 PRORDER – DEFINE A GROUP PRODUCTION RULES SEQUENCE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

PRORDER defines the order of group production rules to be implemented fore when a group’s target is not satisfied.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.189 PYACTION – DEFINE PYTHON BASED ACTION CONDITIONS AND COMMAND PROCESSING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	-----------------

Description

The [PYACTION](#) keyword is part of OPM Flow's Python scripting facility that loads a standard Python script file that can be used to define a series of conditions and actions as the simulation proceeds through time. The "included" Python script file is executed by the standard Python interpreter. Thus, OPM Flow's Python scripting facility offers greater flexibility compared to the commercial simulator's [ACTION](#) series of keywords ([ACTION](#), [ACTIONG](#), [ACTIONR](#), [ACTIONS](#), [ACTIONW](#) and [ACTIONX](#)) that can apply Boolean conditional tests to variables at the field, group, region, well segment and well levels. Note that OPM Flow has also implemented the commercial simulator's [ACTIONX](#) keyword, but not the [ACTION](#), [ACTIONG](#), [ACTIONR](#), [ACTIONS](#) and [ACTIONW](#) keywords, as the [ACTIONX](#) keyword implements their functionality with greater flexibility. For a python class documentation and for further documentation and examples, also refer to the [OPM Online Python Documentation](#).

This keyword starts the definition of a [PYACTION](#) section that specifies the name of the action and a string indicating the number of times the action should be run; this is then followed by the name of the file containing the Python script. Note that unlike an [ACTIONX](#) block that is terminated by the [ENDACTIO](#) keyword the [PYACTION](#) section has no terminating keyword.

Although this keyword is read by OPM Flow and the script processing has been implemented, one should use caution when using this facility as it may result in OPM Flow aborting. This is because the [PYACTION](#) facility allows the user to implement complex functionality and the implementation was new for the 2020-04 release. Users should therefore use caution when using this facility.

Note

This is an OPM Flow specific keyword for the simulator's scripting facility using the standard Python interpreter, as such it gives more flexibility than the commercial simulator's [ACTIONX](#) keyword, although OPM Flow also supports this as well.

The keyword should be considered experimental as details of the OPM Flow - Python interface might change for future releases. In particular, the current implementation is quite minimal; however, future releases are expected to add more entry points in the Schedule class which can be used to manipulate the reservoir model as the simulation progresses. Users are encouraged to make suggestions for new features in this regard.

The [PYACTION](#) keyword is a very powerful keyword and allows any piece of Python code to be included and run, including potentially malicious code. The important point is to scrutinize any [PYACTION](#) keyword in a deck you receive from other parties.

No.	Name	Description	Default
	PYACTION	PYACTION declares the start of a PYACTION Definition Section. This is then followed by one record that defines the name of the action and a string indicating the number of times the action should be run; this is then followed by a second record indicating the file containing the Python script.	Not Applicable

No.	Name	Description	Default
I-1	ACTNAME	ACTNAME is a character sting of any length enclose in quotes that defines the name of this action definition.	None
I-2	ACTNSTEP	ACTNSTEP is a defined character string that indicates the number of times the action should be performed, and should be set to one of the following: 1) SINGLE: Here the action script is run only once. 2) UNLIMITED: If this option is selected then the action is run at the end of every time step. 3) FIRST_TRUE: This option forces the action to be run at the end of every time step until the script returns a value of True. Note that the FIRST_TRUE option is only supported for back compatibility when using the deprecated run() function style Python scripts.	SINGLE
I-4	/	Record terminated by a "/"	Not Applicable
2-1	FILENAME	A character string enclosed in quotes that defines the Python module/script file to read in and to be processed by OPM Flow.	None
2-2	/	Record terminated by a "/"	Not Applicable
Notes: 1) There is no terminating "/" for this keyword.			

Table 12.64: PYACTION Keyword Description

The [PYACTION](#) keyword is a result of combining two programming languages, the interactive Python interpreter and OPM Flow's source code language C++. When combining two languages one extends and embeds one into the other. When extending Python with C++ the functionality implemented in C++ is made available to Python applications, when embedding Python in C++ one can call Python functions from within C++. The [PYACTION](#) keyword is based on embedding a Python interpreter in the C++ OPM Flow simulator, such that Python code is run from within the simulator.

Note that to use the [PYACTION](#) keyword OPM Flow must be built with embedded python enabled in opm-common/CMakeLists.txt.

```
option(OPM_ENABLE_EMBEDDED_PYTHON "Enable embedded python?" ON)
```

In order to fully benefit from the power of the [PYACTION](#) keyword one should familiarize oneself with the Python wrapper classes for the various OPM Flow C++ classes. These classes are essential to share the simulator state with the [PYACTION](#) script. That is, the function provides the interface between the two programming languages.

To learn more about these classes you can use the pydoc utility, for example, to learn more about the SummaryState class, in a Linux terminal one would issue the following bash command:

```
bash% pydoc opm.io.sim.SummaryState
```

The Python script file (FILENAME in the [PYACTION](#) keyword) should be a standard Python module³²⁵ that defines the [PYACTION](#) script and should consist of 100% pure Python. The [PYACTION](#) Python module (FILENAME) is imported during processing of the input deck and as such this implies:

- 1) Basic Python syntax checking is performed when the Python module (FILENAME) is read in.
- 2) It is verified that the module has the correct format.

The syntax of the Python module (FILENAME) is given in Table 12.65 together with a description of the Python module opm_embedded.

Python Script Definition		
<pre> # # OPM Flow PYACTION Module Script # import opm_embedded ecl_state = opm_embedded.current_ecl_state schedule = opm_embedded.current_schedule report_step = opm_embedded.current_report_step summary_state = opm_embedded.current_summary_state ... </pre>		
No.	Name	Description
1	ecl_state	The ecl_state variable is an instance of the opm.io.ecl_state.EclipseState class which is initialized with all the static information in the simulation, for example the 3D grid properties and the PVT and saturation tables. This object can be read by the Python script but the available data should not be modified as any updates will not be passed back to the simulator.

³²⁵ A Python module is a file containing Python definitions and statements. The file name is the module name with the suffix .py appended. Within a module, the module's name (as a string) is available as the value of the global variable `__name__`.

Python Script Definition		
<pre> # # OPM Flow PYACTION Module Script # import opm_embedded ecl_state = opm_embedded.current_ecl_state schedule = opm_embedded.current_schedule report_step = opm_embedded.current_report_step summary_state = opm_embedded.current_summary_state ... </pre>		
No.	Name	Description
2	schedule	The schedule variable is an instance of the <code>opm.io.sched.Schedule</code> class which has all the SCHEDULE information internalized (Schedule object). For a complete documentation of the available call points of <code>opm.io.schedule.Schedule</code>, we refer to https://opm.github.io/opm-python-documentation/master/common.html#opm.io.schedule.Schedule. The most notable ones are: <pre> schedule.open_well(well_name, report_step) schedule.shut_well(well_name, report_step) schedule.stop_well(well_name, report_step) </pre> Which can be used to open, shut or stop a well at a particular report step (or at the current report step if the <code>report_step</code> argument is not specified), and: <pre> schedule.insert_keywords(kw) </pre> Which can be used to insert various keywords where <code>kw</code> is a string defining the keyword and associated records. For the 2024.10 release this has been tested with the following keywords: BOX, COMPSEGS, ENDBOX, EXIT, FIELD, INCLUDE, GCONINJE, GCONPROD, GCONSUMP, GEFAC, GRUPTREE, METRIC, MULTX, MULTX-, MULTY, MULTY-, MULTZ, MULTZ-, NEXT, NEXTSTEP, WCONINJE, WCONPROD, WECON, WEFAC, WEOPEN, WELPI, WELTARG, WGRUPCON, WELSEGS, WELSPECS, WSEGVALV, WLIST, WPIMULT, WTEST, WTMULT This list is subject to change for future releases. The parsing strictness for the ACTIONX and PYACTION keywords can be set using the <code>--action-parsing-strictness</code> command line option. If the default option <code>--action-parsing-strictness="normal"</code> has been set then OPM Flow will only apply keywords that have been tested, whereas if the option <code>--action-parsing-strictness="low"</code> has been set then OPM Flow will attempt to apply all keywords (however the simulation results may be incorrect). Adding additional call points to the <code>opm.io.schedule.Schedule</code> instance in order to update the Schedule object in more ways is an obvious candidate for improving the PYACTION functionality.

Python Script Definition		
<pre># # OPM Flow PYACTION Module Script # import opm_embedded ecl_state = opm_embedded.current_ecl_state schedule = opm_embedded.current_schedule report_step = opm_embedded.current_report_step summary_state = opm_embedded.current_summary_state ...</pre>		
No.	Name	Description
3	report_step	The current report step.
4	summary_state	The summary_state variable is an instance of the <code>opm.io.sim.SummaryState</code> class, the purpose of this class is to serve as a container for SUMMARY variables that contain the simulation results; for example, the Field Oil Rate (FOPR) or a Well's Water CuT (WWCT). See the SUMMARY SECTION for a detailed description of the variables available. The summary_state variable will typically be the variable one will use to access the state of the simulation. For example, to check if the water cut in well OP01 exceeds 0.5 one would use the following statement: <pre>if summary_state.well_var("OP01", "WWCT") > 0.50: ...</pre> The summary_state variable can also be use to update variables including UDQ variables, i.e: <pre>summary_state.update_well_var("OP01", "WUXX", 0.25)</pre> The above assigns a value of 0.25 to the well UDQ variable WUXX for well OP01.

Table I 2.65: PYACTION Module Script Definition

See also the [PYINPUT](#) and [PYEND](#) keywords in the [GRID³²⁶](#) section which are also part of OPM Flow's Python scripting facility, that process standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords.

Examples

The first example checks if well OP01 has a water cut greater than 0.8 and if so then the well is shut in. In the input deck we would have:

```
--
--          START OF PYACTION SECTION
--
--          ACTNAME          ACTNSTEP
PYACTION
          'MAXWCUT'          'UNLIMITED'          /
          'pthon/script/MAXWCUT.py'              /
--
--          END OF PYACTION SECTION
--
```

And then in the Python module file 'pthon/script/MAXWCUT.py' one would have:

³²⁶ Note the [PYINPUT](#) and [PYEND](#) keywords can be used in the [GRID](#), [EDIT](#), [PROPS](#), [SOLUTION](#), [SUMMARY](#) and [SCHEDULE](#) sections, but are described in the [GRID](#) section.

```
#
# OPM Flow PYACTION Module Script
#
import opm_embedded
schedule = opm_embedded.current_schedule
summary_state = opm_embedded.current_summary_state

if (not 'setup_done' in locals()):
    executed = False
    setup_done = True

if (not executed and summary_state.well_var("OP1", "WWCT") > 0.80):
    message = "Well OP01 has been shut-in due to WWCT > 0.80\n"
    opm_embedded.OpmLog.info(message)
    schedule.shut_well("OP1")
    executed = True
```

Note how the 'executed' variable is used to ensure that the action is only executed once the first time the water cut is greater than 0.8.

The next example is based on the first example from the [ACTIONX](#) keyword ([ACTIONX – Define Action Conditions and Command Processing](#)). The Python script first checks if the field's water production is greater than 30,000 stb/d, and if not returns control back to the simulator. If the field water production is greater than 30,000 stb/d then the script uses a Python variable count to keep track of the number of times the script has been executed, and then sorts the wells from high water cut to low, via the wct_list variable, and then shuts in the worst offending well. If a well is shut-in the count variable is increased by one and control is passed back to the simulator. The script is executed as maximum of ten times.

```
--
--      START OF PYACTION SECTION
--
--      ACTNAME      ACTNSTEP
PYACTION      'WSHUTIN'      'UNLIMITED'      /
              'pthon/script/WSHUTIN.py'      /
--
--      END OF PYACTION SECTION
--
```

And then in the Python module file 'pthon/script/WSHUTIN.py' one would have:

```
#
# OPM Flow PYACTION Module Script
#
import opm_embedded

schedule = opm_embedded.current_schedule
summary_state = opm_embedded.current_summary_state

if (not 'setup_done' in locals()):
    execution_counter = 0
    setup_done = True

if (execution_counter < 10 and summary_state["FWPR"] >= 30000):
    #
    # Get Sorted Well List
    #
    wct_list = sorted( [ (well, summary_state.well_var(well, "WWCT"))
                        for well in summary_state.wells], reverse=True )
    #
    # Shut-in Well with Highest Water Cut
    #
    well, wwct = wct_list[0]
```

```

if wwct > 0:
    schedule.shut_well(well)
    execution_counter += 1

```

Note that by using `PYACTION` it makes sense to combine the `UDQ` variable into the `PYACTION` statement, like shown in the example above, although this is not necessary, as one could in principle use a normal `UDQ` statement and then access the variables in the `PYACTION` script using the `summary_state` variable.

The final example checks to see if the field's gas rate is below 600 MMscf/d and if the simulation time is greater than January 1, 2030. If it is, then compression is installed by re-setting all the gas producing well's THP and BHP pressures to 450 psia and 300 psia respectively. In addition all gas wells currently shut-in are tested to see if they can be opened up under the new THP and BHP constraints.

```

--
--      START OF PYACTION SECTION
--
--      ACTNAME      ACTNSTEP
PYACTION
    'WSHUTIN'      'UNLIMITED'      /
    'python/script/PHASE3.py'        /
--
--      END OF PYACTION SECTION
--

```

And then in the Python module file 'python/script/PHASE3.py' one would have:

```

#
# OPM Flow PYACTION Module Script
#
import datetime
import opm_embedded

schedule = opm_embedded.current_schedule
summary_state = opm_embedded.current_summary_state

if (not 'setup_done' in locals()):
    executed = False
    setup_done = True

if not executed:
    sim_time = schedule.start +
                datetime.timedelta( seconds = summary_state.elapsed() )

    if (summary_state["FGPR"] < 600000 and
        sim_time > datetime.datetime(2030, 1, 1)):
        #
        # Do WELTARG and WTEST action
        #
        ...
        executed = True

```

Note how the current simulation time (`sim_time`) is evaluated from the simulation start time (`schedule.start`) and elapsed time (`summary_state.elapsed`); and how it is compared with 00:00 on 1 January 2030 using the Python `datetime` module.

12.3.190 PYEND – END THE DEFINITION OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords. A [PYINPUT](#) Definition Section is terminated by a [PYEND](#) keyword on a separate single line.

See [PYEND – End the Definition of a PYINPUT Section](#) in the [GRID](#) section for a full description.

12.3.191 PYINPUT – DEFINE THE START OF A PYINPUT SECTION

The [PYINPUT](#) and [PYEND](#) keywords are a part of OPM Flow’s Python scripting facility that processes standard Python commands that can be used to manipulate and define the simulators input parameters during processing of the input deck. The main purpose of the facility is to script the construction of the various keywords used by the simulator. [PYINPUT](#) declares the start of a [PYINPUT](#) Definition Section on a single separate line, which is then followed by various standard Python commands, with one command per line.

See [PYINPUT – Define the Start of a PYINPUT Section](#) in the [GRID](#) section for a full description.

12.3.192 QDRILL – DEFINE SEQUENTIAL DRILLING QUEUE WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [QDRILL](#) keyword places previously defined wells in the Sequential Drilling Queue. Wells in this type of queue will be automatically drilled and completed in the sequence entered in order to satisfy group targets, as defined by the [GCONPROD](#), [GCONINJE](#) and [GCONSALE](#) keywords. or a group’s production potential as per the GDRILLPOT keyword. All the previously mentioned keywords are in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.193 RAINFALL – CONSTANT FLUX AQUIFER RAINFALL FLUX BY MONTH

This keyword, [RAINFALL](#), defines the month by month rainfall flux for constant flux aquifers.

See [RAINFALL – Constant Flux Aquifer Rainfall Flux by Month](#) in the [GRID](#) section for a full description.

12.3.194
RCMASTS – RESERVOIR COUPLING GROUP MINIMUM TIME STEP FOR FLOW RESTRICTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

RCMASTS is used when reservoir coupling is invoked by the GRUPMAST and SLAVES keywords in the SCHEDULE section. The keyword should be placed within the master file and it sets the minimum time step size for groups for when a group is being restricted by a group’s limiting flow rate fractional change (see the GRUPMAST keyword in the SCHEDULE section).

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.195 REACHES – DEFINE RIVER REACHES STRUCTURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *REACHES* keyword defines the reach structure of a previously characterized river system using the *RIVERSYS* keyword in the *SCHEDULE* section. The River option must be activated via the *RIVRDIMS* keyword in the *RUNSPEC* section in order to use these keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.196 READDATA – READ SCHEDULE DATA BASED ON CURRENT TIME STEP

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [READDATA](#) keyword enables the simulator to read [SCHEDULE](#) data files generated by external programs on the fly, that is as the run is progressing. The external program can “hold” the simulation by using a file lock, in order for the external program to evaluate the current simulation results, then write out a [SCHEDULE](#) data file for the next time step, and finally releasing the “hold” by deleting the file lock and continuing with newly written [SCHEDULE](#) data file. The mechanism can be repeated so that the external program is dictating how the simulation progresses through time,

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.197 REFINE – START THE DEFINITION OF A LOCAL GRID REFINEMENT

The [REFINE](#) keyword defines the start of a Cartesian or radial local grid refinement (“LGR”) definition that sets the properties of the selected [LGR](#). The keyword is then followed by the property keywords associated with the section where the keyword is being invoked. For example, if the [REFINE](#) keyword is used in the [GRID](#) section then most of the keywords in that section can be used to set the grid properties for the [LGR](#).

See [REFINE – Start the Definition of a Local Grid Refinement](#) in the [GRID](#) section for a full description.

12.3.198 RIVDEBUG – DEFINE THE DEBUG DATA TO BE PRINTED TO FILE (RIVERS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines the debug data associated with rivers to be written to the debug file (*.DBG), for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.199 RIVERSYS – DEFINE RIVER SYSTEM (BRANCH STRUCTURE AND BOUNDARY CONDITIONS)

[RIVERSYS](#) defines a river system by specifying the branch structure of the river together with the branch's associated boundary conditions, for when the River option has been activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section.

See [RIVERSYS – Define River System \(Branch Structure and Boundary Conditions\)](#) in the [SOLUTION](#) section for a full description.

12.3.200 RIVRPROP – MODIFY RIVER REACHES PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [RIVRPROP](#) keyword modifies the individual reaches in a river structure of a previously characterized river system using the [RIVERSYS](#) and the [REACHES](#) keywords in the [SCHEDULE](#) section. The River option must be activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section in order to use this keyword. [RIVRPROP](#) is an alternative and a more concise way to changing the individual reaches in a river structure than the [REACHES](#) keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.201 RIVSALT – DEFINE RIVER UPSTREAM FLOW SALT CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [RIVSALT](#) keyword defines the injected salt concentration in individual river branches in a previously characterized river system using the [RIVERSYS](#) and the [REACHES](#) keywords in the [SCHEDULE](#) section. The River option must be activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section in order to use this keyword. In addition, the Brine option must also be enabled via the [BRINE](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.202
RIVTRACE
–
DEFINE
RIVER
UPSTREAM
FLOW
TRACER
CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [RIVTRACE](#) keyword defines the injected tracer concentration in individual river branches in a previously characterized river system using the [RIVERSYS](#) and the [REACHES](#) keywords in the [SCHEDULE](#) section. The River option must be activated via the [RIVRDIMS](#) keyword in the [RUNSPEC](#) section in order to use this keyword. In addition, the Tracer option must also be enabled by the [TRACER](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.203 RPTHMG – DEFINE WELL HISTORY MATCH GRADIENT REPORTING (GROUPS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [RPTHMG](#), either enables or disables history match output reporting to the history match file (*.HMD) for the named group, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

12.3.204 RPTHMW – DEFINE WELL HISTORY MATCH GRADIENT REPORTING (WELLS)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [RPTHMW](#), either enables or disables history match output reporting to the history match file (*.HMD) for the named well, for when the History Match Gradient option has been activated by the [HMDIMS](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

12.3.205 RPTONLY – ACTIVATE THE REPORT TIME STEPS ONLY OPTION FOR THE SUMMARY FILE

This keyword activates the writing out of the [SUMMARY](#) file and RSM file data, if the RSM file option has been requested by the [RUNSUM](#) keyword in the [SUMMARY](#) section, at report time steps only. The default is for all time steps to be written out to the files. This keyword reduces the file size at the expense of lower resolution in the time domain. There is no data required for this keyword.

See [RPTONLY – Activate the Report Time Steps Only Option for the SUMMARY File](#) in the [SUMMARY](#) section for a full description.

12.3.206 RPTONLYO – DEACTIVATE THE REPORT TIME STEPS ONLY OPTION FOR THE SUMMARY FILE

This keyword deactivates the writing out of the [SUMMARY](#) file and RSM file data, if the RSM file option has been requested by the [RUNSUM](#) keyword in the [SUMMARY](#) section, at report time steps only, and switches on writing out all the time steps to the files. This option is the default behavior for when [RPTONLY](#) has not been activated. There is no data required for this keyword.

See [RPTONLYO – Deactivate the Report Time Steps Only Option for the SUMMARY File](#) in the [SUMMARY](#) section for a full description.

12.3.207 RPTRST – DEFINE DATA TO BE WRITTEN TO THE RESTART FILE

This keyword defines the data to be written to the [RESTART](#) file at each requested restart point.

See [RPTRST – Define Data to be Written to the RESTART File](#) in the [SOLUTION](#) section for a full description.

12.3.208 RPTSCHED – DEFINE SCHEDULE SECTION REPORTING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	-----------------

Description

This keyword defines the data in the [SCHEDULE](#) section that is to be printed to the output print file in human readable format. The keyword has two distinct forms, the first of which consists of the keyword followed by a series of integers on the next line indicating the data to be printed (see the first example). This is the original format in the commercial simulator and was subsequently superseded by the second format. The second format consists of the keyword followed by a series of character strings that indicate the data to be printed. In most cases the character string is the keyword used to define the data in the OPM Flow input deck, for example [WEL SPECS](#) to defined the basic well definitions.

No.	Name	Description	Default
1	FIP	Print the fluid in-place report. The parameter is assigned a value, OPTION , using the form FIP = OPTION , where OPTION is an integer variable set to: 1) OPTION = 1 then the report is for the field only. 2) OPTION = 2 then in addition to the field report, a report is produced for each FIPNUM region, as defined by the FIPNUM keyword in the REGIONS section. Note the commercial simulator also prints the flows to other regions as well as the flows from the wells. This additional reporting option has not been implemented in OPM Flow. 3) OPTION = 3 then in addition to the above, a balance report is also produced for fluid in-place regions defined by the FIP keyword in the REGIONS section.	FIP =2
2	FIPRESV	Print the reservoir volumes in-place report.	None
3	NOTHING	Switches off all printed reports in the SCHEDULE section.	N/A
4	SALT	Print grid block salt concentration values. Note this is an OPM Flow specific keyword.	N/A
5	RESTART	RESTART defines the frequency at which the restart data for restarting a run is written to the RESTART file. The parameter is assigned a value, OPTION , using the form RESTART = OPTION , where OPTION is an integer variable set to: 1) OPTION = 1 then the restart files are written at every report time, but only the last one in the run is kept. This minimizes the restart file size but only the final results are stored, limiting the visualization in OPM ResInsight. 2) OPTION = 2 then the phase inter-blocks are written to the restart files, in addition to the standard data. 3) OPTION = 3 then the fluid in-place and phase potentials are also written to the restart file. 4) OPTION = 6 then the restart files are written at every time step. See the RPTRST keyword in the SOLUTION section for a more flexible way to write out restart files.	

No.	Name	Description	Default
6	WELLS	The WELLS option turns on production and injection rate and cumulative volume reporting for produced and injected fluids. The parameter has several levels of reporting details set by the assigned OPTION value, using the form WELLS = OPTION, where OPTION is an integer variable set to: 1) OPTION = 1 report volumes at the well level. 2) OPTION = 2 report volumes at the well and the well connection levels. 3) OPTION = 3 report volumes for layer totals. 4) OPTION = 4 report volumes for layer totals and for wells. 5) OPTION = 5 report volumes for layer totals and for wells and well connections. Only OPTION equal to one is supported by OPM Flow.	WELLS=1
7	WELSPECS	WELSPECS switches on reporting of the well connections, wells and groups at each report time step. There are numerous reports associated with this option. Unlike the other reporting parameters that produce a report for each reporting time step, the WELSPECS report option only produces a report if an associated keyword has been activated at the current reporting time step. For example, if the reporting time steps are January, February, and March 2020, and the RPTSCHED WELSPECS option is activated in January, with wells OP01 and OP02 being declared via the WELSPECS and COMPDAT keywords, then a report will be printed for January for these two wells. If there are no further well activations until March, with well OP03 being declared, then there will be no report for February, and only well OP03 will be reported at the March reporting time step.	
Notes: 1) The keyword is terminated by a “/”.			

Table 12.66: RPTSCHED Keyword Description

Development is current progressing on developing reports in a similar format to the commercial simulator and this section will be updated as additional reports are added to OPM Flow's functionality.

Note

Unlike the other reporting keywords in the **RUNSPEC**, **GRID**, **EDIT**, **PROPS** and **SOLUTION** keywords, the requested reports on the this keyword remain in effect until they are switched off by this keyword, that is, the reports are written out every report time step until requested to stop. Use the 'NOTHING' parameter to switch off all reporting.

An example [FIP](#) report is shown in Figure 12.6 from the Norne field, note only the field and the first two region reports are shown.

=====									
: Field Totals :									
: PAV = 268.423 BARSA :									
: PORV = 673813759 RM3 :									
: Pressure is weighted by hydrocarbon pore volume :									
: Porv volumes are taken at reference conditions :									
=====									
	Liquid	Oil	SM3		Wat	SM3		Gas	SM3
		Vapour		Total	Total		Free	Dissolved	Total
:Currently in place	92852273	506017	93358290	480610201	11202115432	9841722029	21043837461		
:Originally in place	160196505	559893	160756399	402039147	9744843573	17311090896	27055934469		
=====									
: FIPNUM report region 1 :									
: PAV = 269.913 BARSA :									
: PORV = 44729956 RM3 :									
: Pressure is weighted by hydrocarbon pore volume :									
: Porv volumes are taken at reference conditions :									
=====									
	Liquid	Oil	SM3		Wat	SM3		Gas	SM3
		Vapour		Total	Total		Free	Dissolved	Total
:Currently in place	5398282	259439	5567722	12179015	5263878362	594212703	5858991065		
:Originally in place	5358346	311293	5669639	11497281	5426018016	591714035	6017732051		
=====									
: FIPNUM report region 2 :									
: PAV = 266.851 BARSA :									
: PORV = 14745585 RM3 :									
: Pressure is weighted by hydrocarbon pore volume :									
: Porv volumes are taken at reference conditions :									
=====									
	Liquid	Oil	SM3		Wat	SM3		Gas	SM3
		Vapour		Total	Total		Free	Dissolved	Total
:Currently in place	2744221	88023	2832244	3078782	1655632008	318616757	1974248764		
:Originally in place	3229038	88236	3317274	3087870	1535013525	362544609	1897558134		
=====									

Figure 12.6: RPTSCHED: FIP (Balance) Report

Figure 12.7 illustrates the reservoir volumes in-place report for the first two regions from the Norne field.

=====					
: RESERVOIR VOLUMES M3 :					
: REGION :	: TOTAL PORE :	: PORE VOLUME :	: PORE VOLUME :	: PORE VOLUME :	: PORE VOLUME :
:	: VOLUME :	: CONTAINING :	: CONTAINING :	: CONTAINING :	: CONTAINING :
:	:	: OIL :	: WATER :	: GAS :	: HYDRO-CARBON :
=====					
: 1:	44719142:	7048934:	11937133:	25733075:	32782009:
=====					
: 2:	14747585:	4266286:	3206400:	7274899:	11541185:
=====					

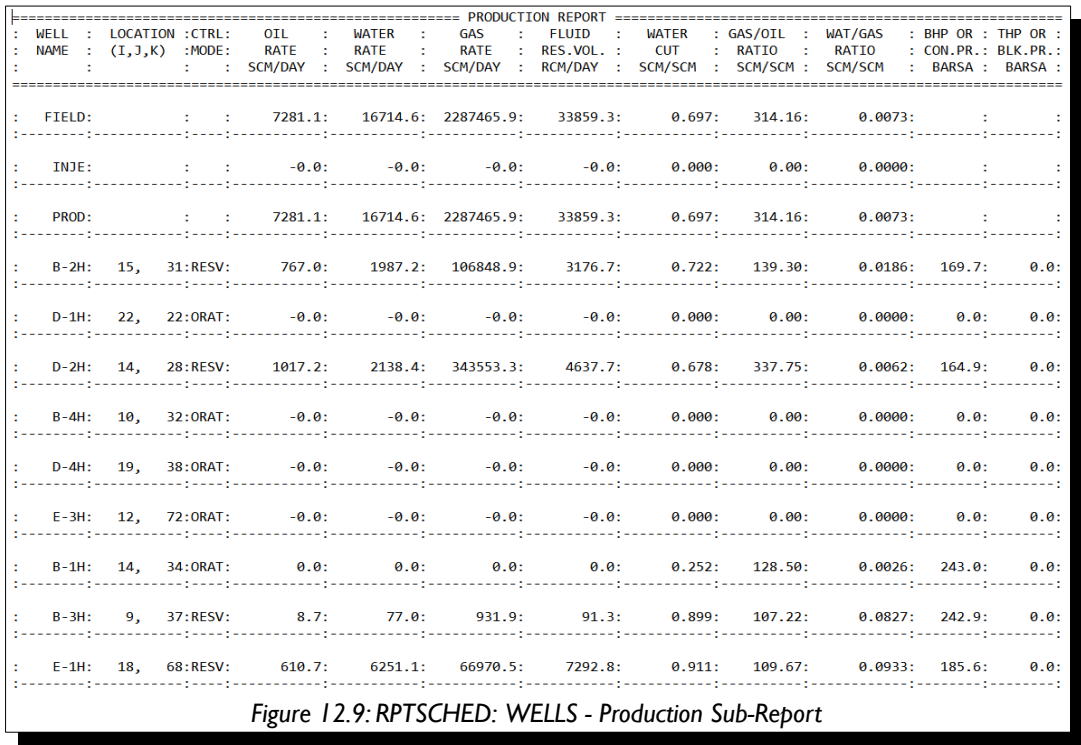
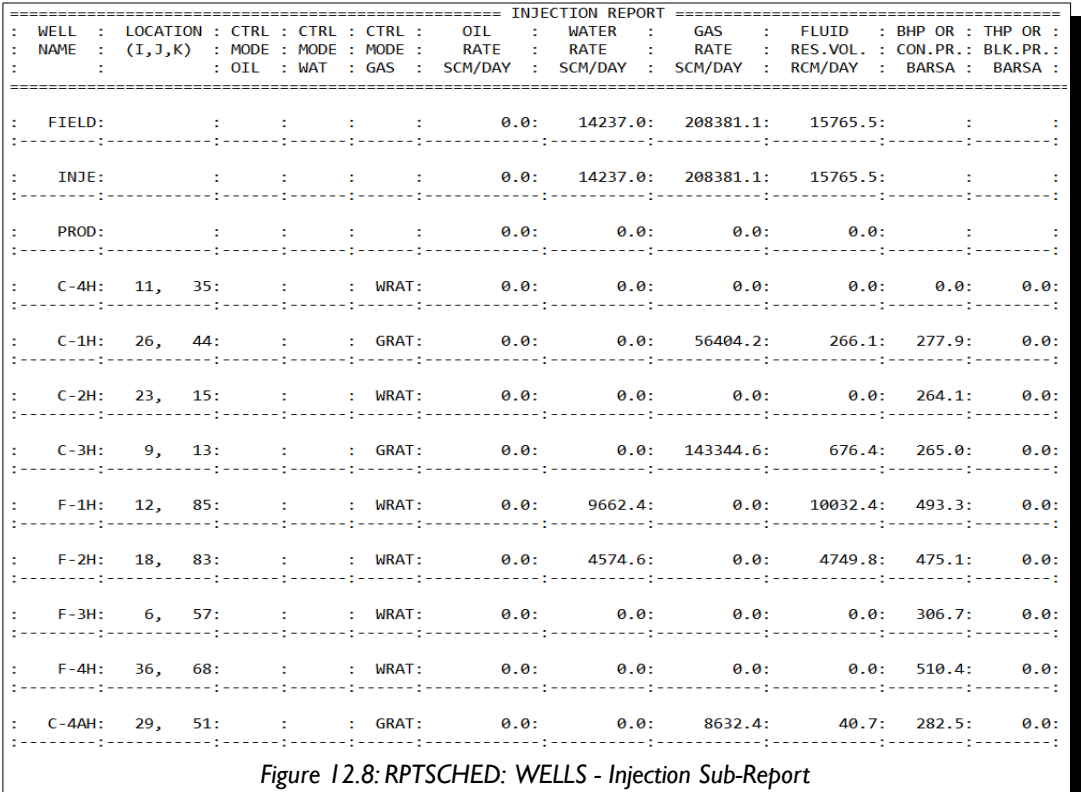
Figure 12.7: RPTSCHED: FIPRESV (Reservoir Volumes) Report

Note

Note that the “PORV” quantity in the [FIP](#) (Balance) Report, as shown in Figure 12.6, is reported at reference conditions, meaning there is no pressure dependence involved. However, the “TOTAL PORE VOLUME” values in the Reservoir Volumes Report (Figure 12.7) are pressure dependent pore volumes. Thus, for region one the “PORV” value is 44,729,956 rm3 (Figure 12.6) and the “TOTAL PORE VOLUME” (Figure 12.7) is 44,719,142 rm3.

This is the same as the commercial simulator.

The WELLS report consists of several sub-reports depending on the selected option for this report type. Figure 12.8 and Figure 12.9 show example Injection and Production sub-reports



The third and final report is the Cumulative Production and Injection sub-report, shown in Figure 12.10.

CUMULATIVE PRODUCTION/INJECTION REPORT												
WELL	LOCATION	WELL	CTRL	OIL	WATER	GAS	Prod	OIL	WATER	GAS	INJ	
NAME	(I,J,K)	TYPE	MODE	PROD	PROD	PROD	RES.VOL	INJ	INJ	INJ	RES.VOL	
				MSCM	MSCM	MMSCM	MRCM	MSCM	MSCM	MMSCM	MRCM	
FIELD:				67201.8:	23494.0:	14651446.1:	149177.9:	0.0:	102036.7:	8684539.9:	149925.7:	
INJE:				0.0:	178.7:	0.0:	185.7:	0.0:	102036.7:	8684539.9:	149925.7:	
PROD:				67201.8:	23315.3:	14651446.1:	148992.2:	0.0:	0.0:	0.0:	0.0:	
C-4H:	11, 35:	INJ:WRAT:		0.0:	0.0:	0.0:	0.0:	0.0:	5468.3:	2927677.5:	20664.3:	
B-2H:	15, 31:	PROD:RESV:		10803.4:	1766.2:	1840432.3:	19370.5:	0.0:	0.0:	0.0:	0.0:	
D-1H:	22, 22:	PROD:ORAT:		5497.5:	507.5:	908176.2:	9406.7:	0.0:	0.0:	0.0:	0.0:	
D-2H:	14, 28:	PROD:RESV:		7533.6:	1245.2:	2698602.4:	20280.4:	0.0:	0.0:	0.0:	0.0:	
B-4H:	10, 32:	PROD:ORAT:		1307.0:	0.6:	269357.3:	2383.2:	0.0:	0.0:	0.0:	0.0:	
D-4H:	19, 38:	PROD:ORAT:		3179.7:	84.5:	680216.1:	6036.1:	0.0:	0.0:	0.0:	0.0:	
C-1H:	26, 44:	INJ:GRAT:		0.0:	0.0:	0.0:	0.0:	0.0:	12795.6:	2186307.8:	24278.4:	
E-3H:	12, 72:	PROD:ORAT:		1687.3:	198.9:	179311.1:	2441.4:	0.0:	0.0:	0.0:	0.0:	
C-2H:	23, 15:	INJ:WRAT:		0.0:	0.0:	0.0:	0.0:	0.0:	21661.2:	0.0:	22504.3:	
B-1H:	14, 34:	PROD:ORAT:		3581.4:	476.1:	1413055.5:	10326.8:	0.0:	0.0:	0.0:	0.0:	

Figure 12.10: RPTSCHED: WELLS - Cumulative Production and Injection Sub-Report

Similarly as for the WELLS report, the [WEL SPECS](#) report consists of several sub-reports, including the Well Production Control report shown in Figure 12.11 for the Volve field.

WEL SPECS AT0.00 DAYS*FLOW

REPORT 031 DEC 2007*RUN AT 12:41 ON 12 SEP 2016

WELL SPECIFICATION DATA

WELL	GROUP	WELLHEAD	B.H. REF	PREF-	DRAINAGE	GAS	SHUT-IN	CROSS	PVT	WELL	FIP	WELL
NAME	NAME	LOCATION	DEPTH	ERRED	RADIUS	INFL	INSTRCT	FLOW	TAB	DENS	REG	D-FACTOR 1
		(I, J)	METRES	PHASE	METRES	EQU		ABLT		CALC		DAY/SM3
I-F-5	AAR	84, 41	2997.3	WAT	P.EQUIV.R:	STD	OPEN	YES	4	SEG	0	0
P-F-14	AAR	65, 48	2476.1	OIL	P.EQUIV.R:	STD	OPEN	YES	2	SEG	0	0
I-F-4	SRAR	73, 28	2931.0	WAT	P.EQUIV.R:	STD	OPEN	YES	3	SEG	0	0
I-F46	SRAR	73, 28	2931.0	GAS	P.EQUIV.R:	STD	OPEN	YES	3	SEG	0	0
P-F-12	SRAR	58, 37	2705.0	OIL	P.EQUIV.R:	STD	OPEN	YES	1	SEG	0	0
P-F-15C	SRAR	55, 28	2551.0	OIL	P.EQUIV.R:	STD	OPEN	YES	11	SEG	0	0
P-F-11B	SRAR	66, 42	2451.0	OIL	P.EQUIV.R:	STD	SHUT	YES	2	SEG	0	0
I-F-1B	P_NW	66, 58	2988.5	WAT	P.EQUIV.R:	STD	SHUT	YES	6	SEG	0	0
P-F-15D	AAR	63, 47	2844.9	OIL	P.EQUIV.R:	STD	SHUT	YES	2	SEG	0	0
P-F-1C	P_NW	57, 51	2889.1	OIL	P.EQUIV.R:	STD	SHUT	YES	10	SEG	0	0
PIL-N	P_NORTH	80, 54	3153.9	OIL	P.EQUIV.R:	STD	OPEN	YES	5	SEG	0	0
PIL-NW	P_NW	64, 59	2959.5	OIL	P.EQUIV.R:	STD	SHUT	YES	6	SEG	0	0

1: The WELL D-FACTOR is not implemented - and the report will always show the default value 0.

1: The WELL D-FACTOR is not implemented - and the report will always show the default value 0.

Figure 12.11: RPTSCHED: WEL SPECS - Production Well Control Sub-Report

The **COMPDAT** keyword data is listed on the Well Connection Data sub-report as depicted in Figure 12.12 for the I-F-5 well. Note that the data is repeated for all connections and for all wells declared at the reporting time step.

WELL CONNECTION DATA													
WELL NAME	GRID BLOCK	CMPL NO#	CENTRE DEPTH METRES	OPEN SHUT	SAT TAB	CONNECTION FACTOR* CPM3/D/B	INT DIAM METRES	K H VALUE MD.METRE	SKIN FACTOR	CONNECTION D-FACTOR 1 DAY/SM3	SATURATION SCALING DATA		
											SWMIN	SWMAX	SGMIN SGMAX 2
I-F-5	84, 41,	1	2997.3	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	2	2998.3	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	3	2999.4	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	4	3000.4	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	5	3001.5	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	6	3002.5	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	7	3003.6	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	8	3004.6	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	9	3005.6	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	10	3006.6	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	11	3007.7	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	12	3008.8	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	13	3009.8	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	14	3010.9	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	84, 41,	15	3012.0	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 41,	17	3035.9	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 41,	19	3038.6	SHUT	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 41,	20	3040.0	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	20	3050.7	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	21	3051.9	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	22	3053.0	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	23	3054.2	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	24	3055.4	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	25	3056.6	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	26	3058.1	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	27	3059.9	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	28	3061.7	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	29	3063.6	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				
I-F-5	85, 42,	30	3065.4	OPEN	1	0.000000	0.216000	0.000000:0.000000	0				

1: The well connection D-FACTOR is not implemented in opm and the report will always show 0.
2: The saturation scaling data has not been implemented in the report and will show blank

Figure 12.12: RPTSCHED: WELSPECS - Well Connection Data Sub-Report

For multi-segment wells both the Production Well Control and Well Connection sub-reports are printed as per Figure 12.11 and Figure 12.12, and in addition the equivalent multi-segment well data is printed as well, as shown in Figure 12.13 and Figure 12.14, as shown on the following page.

Figure 12.13 shows the Multi-Segment Well Segment Structure sub-report for a single multi-segment well, OP01. See the [WELSEGS](#) keyword [Example](#) in the [SCHEDULE](#) section to see how the OP01 well is defined.

MULTI-SEGMENT WELL: SEGMENT STRUCTURE														
WELLNAME AND SEG TYPE	SEG NO	BRN NO	MAIN INLET SEGMENT	OUTLET SEGMENT	SEGMENT LENGTH METRES	TOT LENGTH TO END METRES	DEPTH CHANGE METRES	T.V. AT END METRES	DEPTH:DIA OR F: SCALING:ABS ROUGHN: X-SECTN METRES : METRES : METRES	AREA M**2	VOLUME M3	P DROP MULT FACTOR		
OP01	1	1	2	0	2512.5			2512.5	0	0	0.000			
HFA	2		3	1	2537.5			2534.5	0.3000	0.000100	0.07068	1.767		
	3		4	2	2562.5			2560.5	0.3000	0.000100	0.07068	1.767		
	4		5	3	2587.5			2593.5	0.3000	0.000100	0.07068	1.767		
	5		6	4	2612.5			2614.5	0.3000	0.000100	0.07068	1.767		
	6		22	5	2637.5			2635.5	0.3000	0.000100	0.07068	1.767		
	7	2	8	2	2737.5			2538.5	0.2000	0.000100	0.03141	6.283		
	8		9	7	2937.5			2537.5	0.2000	0.000100	0.03141	6.283		
	9		10	8	3137.5			2539.5	0.2000	0.000100	0.03141	6.283		
	10		11	9	3337.5			2535.5	0.2000	0.000100	0.03141	6.283		
	11		0	10	3537.5			2536.5	0.2000	0.000100	0.03141	6.283		
	12	3	13	3	2762.5			2563.5	0.2000	0.000100	0.03141	6.283		
	13		14	12	2962.5			2562.5	0.1000	0.000100	0.00785	1.570		
	14		15	13	3162.5			2562.5	0.1000	0.000100	0.00785	1.570		
	15		16	14	3362.5			2564.5	0.1000	0.000100	0.00785	1.570		
	16		0	15	3562.5			2562.5	0.1000	0.000100	0.00785	1.570		
	17	4	18	5	2812.5			2613.5	0.2000	0.000100	0.03141	6.283		
	18		19	17	3012.5			2612.5	0.1000	0.000100	0.00785	1.570		
	19		20	18	3212.5			2612.5	0.1000	0.000100	0.00785	1.570		
	20		21	19	3412.5			2612.5	0.1000	0.000100	0.00785	1.570		
	21		0	20	3612.5			2613.5	0.1000	0.000100	0.00785	1.570		
	22	5	23	6	2837.5			2634.5	0.2000	0.000100	0.03141	6.283		
	23		24	22	3037.5			2637.5	0.2000	0.000100	0.03141	6.283		
	24		25	23	3237.5			2638.5	0.2000	0.000100	0.03141	6.283		
	25		26	24	3437.5			2639.5	0.1000	0.000100	0.00785	1.570		
	26		0	25	3637.5			2639.5	0.1000	0.000100	0.00785	1.570		

Figure 12.13: RPTSCHED: WELSPECS - Multi Segment Well Structure Sub-Report

And Figure 12.14 depicts Multi-Segment Well Connection Data sub-report for the same well.

MULTI-SEGMENT WELL: CONNECTION DATA											
WELL NAME	CONNECTION NUMBER	SEGMENT NUMBER	BRANCH ID	TUB LENGTH START PERFS METRES	TUB LENGTH END PERFS METRES	TUB LENGTH CENTR PERFS METRES	TUB LENGTH END SEGMT METRES	CONNECTION DEPTH METRES	SEGMENT DEPTH METRES	GRID BLOCK DEPTH METRES	
OP01	10	10	1	1			2512.5	2512.5	2512.5	2512.5	
	10	10	2	2			2537.5	2534.5	2534.5	2537.5	
	10	10	3	3			2562.5	2560.5	2560.5	2562.5	
	10	10	4	4			2587.5	2593.5	2593.5	2587.5	
	10	10	5	5			2612.5	2614.5	2614.5	2612.5	
	10	10	6	6			2637.5	2635.5	2635.5	2637.5	
	9	10	7	7			2737.5	2538.5	2538.5	2537.5	
	8	10	2	8			2937.5	2537.5	2537.5	2537.5	
	7	10	2	9			3137.5	2539.5	2539.5	2537.5	
	6	10	2	10			3337.5	2535.5	2535.5	2537.5	
	5	10	2	11			3537.5	2536.5	2536.5	2537.5	
	10	9	3	12			2762.5	2563.5	2563.5	2562.5	
	10	8	3	13			2962.5	2562.5	2562.5	2562.5	
	10	7	3	14			3162.5	2562.5	2562.5	2562.5	
	10	6	3	15			3362.5	2564.5	2564.5	2562.5	
	10	5	3	16			3562.5	2562.5	2562.5	2562.5	
	9	10	5	17			2812.5	2613.5	2613.5	2612.5	
	8	10	5	18			3012.5	2612.5	2612.5	2612.5	
	7	10	5	19			3212.5	2612.5	2612.5	2612.5	
	6	10	5	20			3412.5	2612.5	2612.5	2612.5	
	5	10	5	21			3612.5	2613.5	2613.5	2612.5	
	10	9	6	22			2837.5	2634.5	2634.5	2637.5	
	10	8	6	23			3037.5	2637.5	2637.5	2637.5	
	10	7	6	24			3237.5	2638.5	2638.5	2637.5	
	10	6	6	25			3437.5	2639.5	2639.5	2637.5	
	10	5	6	26			3637.5	2639.5	2639.5	2637.5	

Figure 12.14: RPTSCHED: WELSPECS - Multi Segment Well Connection Sub-Report

Example

The first example shows the original format of this keyword.

```
--
--      DEFINE SCHEDULE SECTION REPORT OPTION (ORIGINAL FORMAT)
--
RPTSCHED      1          2*0      1          3*1          /
```

The next example shows the second format of the keyword.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE

-- -----
-- SCHEDULE SECTION - 2000-01-01
-- -----
RPTSCHED      'WELLS=2'      'WELSPECs'      'CPU=2'      'FIP=2'      /

DATES
1 JAN 2000 /
/

RPTSCHED      'NOTHING'      /

DATES
1 FEB 2000 /
1 MAR 2000 /
1 APR 2000 /
1 MAY 2000 /
1 JUN 2000 /
1 JUL 2000 /
1 AUG 2000 /
1 SEP 2000 /
1 OCT 2000 /
1 NOV 2000 /
1 DEC 2000 /
/
```

In the above example monthly reporting time steps have been used with a [SCHEDULE](#) section report on the January 1, 2000; after which all reports are switched off for the subsequent reporting time steps.

12.3.209 SAVE – ACTIVATE OUTPUT OF A SAVE FILE FOR FAST RESTARTS

This keyword activates output of a [SAVE](#) file for fast restarts. There is no data required for this keyword.

See [SAVE – Activate Output of a SAVE File for Fast Restarts](#) in the [RUNSPEC](#) section for a full description.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

12.3.210 SCDATAB – WELL CONNECTION PI MULTIPLIERS VERSUS SCALE DEPOSIT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SCDATAB defines well connection Productivity Index (“PI”) reduction multipliers versus scale deposited per unit length of the perforated interval tables, for when the Scale Deposition option has been activated by declaring the dimensions of the scaling deposition tables using the SCDPDIMS keyword in the RUNSPEC section. The SCDATAB tables are allocated to individual wells using the WSCTAB keyword and the rate of scale accumulation around the well connections is given by the SCDPTAB keyword; both keywords are in the SCHEDULE section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.211 SCDETAB – WELL CONNECTION KARST AQUIFER PROPERTIES FOR SCALE DEPOSIT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SCDETAB defines well connection karst³²⁷ aquifer properties for modeling scale deposited by dissolution of calcite from the aquifer water, for when the Scale Deposition option has been activated by declaring the dimensions of the scaling deposition tables using the SCDPDIMS keyword in the RUNSPEC section. The SCDETAB tables are allocated to individual wells using the WSCTAB keyword in the SCHEDULE section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

327

Karst is a topography formed from the dissolution of soluble rocks such as limestone, dolomite, and gypsum. Karst aquifers are characterized by a network of conduits and caves, with the conduits and caves draining the pore space between the limestone grains (intergranular or primary porosity) and the fractures (secondary porosity) formed by joints, bedding planes, and faults.

12.3.212
SCDPTAB – WELL CONNECTION SCALE DEPOSITION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

SCDATAB defines the well connection scale deposition rate as a function of sea water flow rate, for when the Scale Deposition option has been activated by declaring the dimensions of the scaling deposition tables using the [SCDPDIMS](#) keyword in the [RUNSPEC](#) section. The [SCDATAB](#) tables are allocated to individual wells using the [WSCTAB](#) keyword and the sea water fraction is based on a water tracer entered via the [SCDPTRAC](#) keyword; both keywords are in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.213 SCDPTRAC – ALLOCATE SEA WATER TRACER FOR SCALE DEPOSITION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [SCDPTRAC](#) keyword is used to allocate an existing passive water tracer defined by the [TRACER](#) keyword in the [PROPS](#) section, to represent the sea water flowing into a well connection as a fraction of the total water influx. The keyword is used together with the [SCDPTAB](#) keyword in the [SCHEDULE](#) section to calculated the volume of scale deposited around the well connections.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	NAME	A three letter character string defining the tracer's name that has previously been defined by the <i>TRACER</i> keyword in the <i>PROPS</i> section			None
Notes: I) The keyword is terminated by a “/”.					

Table 12.67: SCDPTRAC Keyword Description

Example

In the [PROPS](#) section define a tracer in the water phase, for example:

```
--
--      DEFINE TRACER NAMES
--
--      TRACER      TRACER
--      NAME        PHASE
--      -----
TRACER      'SEA'      'WAT'                               / SEA WATER TRACER
/
```

Then in the [SCHEDULE](#) section allocate the previously defined water tracer as a sea water tracer to be used with the scale deposition facility, that is:

```
--
--      ALLOCATE SEA WATER TRACER FOR SCALE DEPOSITION
--
--      TRACER
--      NAME
--      -----
SCDPTRAC    'SEA'                               / SEA WATER TRACER
/
```

12.3.214 SCHEDULE – DEFINE THE START OF THE SCHEDULE SECTION OF KEYWORDS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *SCHEDULE* activation keyword marks the end of the *SUMMARY* section and the start of the *SCHEDULE* section that defines the group and well definitions, operating and economic constraints, as well as how OPM Flow should advance through time. Numerical controls are also defined in this section and all parameters can be varied through time.

There is no data required for this keyword.

Example

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
```

The above example marks the end of the *SUMMARY* section and the start of the *SCHEDULE* section in the OPM Flow data input file.

12.3.215
SEPVALS – DEFINE SEPARATOR OIL FORMATION VOLUME FACTOR AND GOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SEPVALS](#) keyword defines the initial and subsequent separator oil formation volume factor (Bo) and Gas Oil Ratio (“GOR” or Rs). The facility is used in black-oil modeling to re-scale the PVT data entered via the [PROPS](#) section, based on the saturation point oil formation volume factor (Bob) and the initial saturated gas-oil ratio (Rsi) entered on the [SEPVALS](#) keyword. The first occurrence of this keyword sets the initial conditions and must be followed by the [GSEPCOND](#) keyword that assigns previously defined separators to a group.

Note that the keyword can only be used in runs with oil and dissolve gas only, with no vaporized oil (condensate) in the gas phase.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.216 SHRATE – ACTIVATE AND DEFINE THE POLYMER SHEARING LOGARITHMIC PARAMETERS

This keyword activates the logarithm-based polymer shear thinning/thickening option and defines the shear rate constant. This keyword can only be used in conjunction with the [PLYSHLOG](#) in the [PROPS](#) section

See [SHRATE – Activate Log-based Polymer Shearing and Define the Shear Rate Constant](#) in the [PROPS](#) section for a full description.

12.3.217 SIMULATE – ACTIVATE THE SIMULATION MODE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[SIMULATE](#) switches the mode of the simulation to run simulation mode from the data input checking mode activated by the [NOSIM](#) keyword in the [SCHEDULE](#) section. Note that if [NOSIM](#) has been used in the [RUNSPEC](#) section then [SIMULATE](#) will have no effect.

There is no data required for this keyword and there is no terminating “/” for this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

Example

The example below switches OPM Flow to no simulation mode for data checking of the input deck.

```
--
--      ACTIVATE SIMULATION MODE TO RUN THE MODEL
--
SIMULATE
```

12.3.218 SKIP – ACTIVATE SKIPPING OF ALL KEYWORDS AND INPUT DATA

The [SKIP](#) keyword activates skipping of all keywords and input data until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP](#) and [ENDSKIP](#) keywords are ignored.

See [SKIP – Activate Skipping of All Keywords and Input Data](#) in the GLOBAL section for a full description.

12.3.219 SKIP100 – ACTIVATE SKIPPING OF BLACK-OIL KEYWORDS AND INPUT DATA

The [SKIP100](#) keyword activates skipping of all keywords and input data by the commercial black-oil simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP100](#) and [ENDSKIP](#) keywords are ignored by the commercial black-oil simulator. The [SKIP100](#) keyword is ignored by the commercial compositional simulator.

See [SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator](#) in the GLOBAL section for a full description.

12.3.220 SKIP300 – ACTIVATE SKIPPING OF KEYWORDS BY COMPOSITIONAL SIMULATOR

The [SKIP300](#) keyword activates skipping of all keywords and input data by the commercial compositional simulator until the [ENDSKIP](#) keyword is encountered. All keywords between the [SKIP300](#) and [ENDSKIP](#) keywords are ignored by the commercial compositional simulator. The [SKIP300](#) keyword is ignored by the commercial black-oil simulator.

See [SKIP300 – Activate Skipping of Keywords by Compositional Simulator](#) in the GLOBAL section for a full description.

12.3.221 SKIPREST – ACTIVATE SKIPPING OF RESTART SCHEDULE DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword turns on skipping of keywords up to the start of the restart point, as defined on the [RESTART](#) keyword in the [RUNSPEC](#) section. The [RESTART](#) keyword defines the parameters to restart the simulation from a previous run that has written a [RESTART](#) file out to disk. Activating the [SKIPREST](#) keyword causes the simulator to only read in data it requires for restarting the run up to the [RESTART](#) point (RSNUM on the [RESTART](#) keyword in the [SOLUTION](#) section). Note that certain keywords always need to be present in a restart run in the [SCHEDULE](#) section as the data is not stored on the [RESTART](#) file, for example the VFP tables ([VFPPROD](#) and [VFPINJ](#) keywords). The [SKIPREST](#) keyword automatically processes the input deck and reads the required data.

There is no data required for this keyword.

Example

The example below defines a restart from the previously run NOR-OPM-A01 case at time step number 40.

```
-- =====
--
-- SOLUTION SECTION
--
-- =====
SOLUTION
--
--      FLEXIBLE RESTART FROM PREVIOUS SIMULATION RUN
--
--      FILE          RESTART   RESTART   FILE
--      NAME          NUMBER    TYPE      FORMAT
RESTART
      'NOR-OPM-A01'    40        1*        1*      /
```

Then in the [SCHEDULE](#) section the [SKIPREST](#) keyword is used to correctly read in the schedule data up to the [RESTART](#) point.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--      ACTIVATE SKIPREST OPTION TO AVOID MODIFYING SCHEDULE SECTION
--
SKIPREST
```

12.3.222 SLAVES – DEFINE SLAVE RESERVOIR SIMULATION PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [SLAVES](#), defines the name of the slave reservoirs and their associated simulation input files, for when the Reservoir Coupling option has been declared active by the [GRUPMAST](#) and [SLAVES](#) keywords in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.223 SOURCE – DEFINE SOURCE TERM

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [SOURCE](#) keyword defines a mass and heat source term for a given pseudo component in the specified grid block.

One source term can be added to each grid block for each pseudo component active in the model. If the source term for a given pseudo component is later redefined in the same block this will replace the previous source term.

This is an OPM Flow specific keyword that is not supported by the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	I	A positive integer greater than zero and less than or equal to NX that defines the location of the source term in the I-direction.			None
2	J	A positive integer greater than zero and less than or equal to NY that defines the location of the source term in the J-direction.			None
3	K	A positive integer greater than zero and less than or equal to NZ that defines the location of the source term in the K-direction.			None
4	COMP	A defined character string that defines the pseudo component to be injected by the source term, COMP should be set to one of the following character strings: 1) OIL: the source term will inject oil. 2) GAS: the source term will inject gas. 3) WATER: the source term will inject water. 4) SOLVENT: the source term will inject solvent. 5) POLYMER: the source term will inject polymer. 6) MICR: the source term will inject microbes. 7) OXYG: the source term will inject oxygen. 8) UREA: the source term will inject urea. 9) NONE: no pseudo component specified. The SOLVENT or POLYMER option should only be specified if the solvent or polymer model has been activated by the SOLVENT or POLYMER keyword respectively in the RUNSPEC section. MICR, OXYG, or UREA requires the MICP keyword. A pseudo component must be specified if the mass injection rate is non-zero.			NONE
5	RATE	A real value that defines the source term's mass injection rate of the specified component COMP.			0.0
		lb/day	kg/day	gm/hour	
6	HRATE	A real value that defines the source term's heat injection rate ¹ . The heat injection rate should only be specified if the thermal model has been activated by the THERMAL keyword in the RUNSPEC section.			0.0

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

No.	Name	Description			Default
		Field	Metric	Laboratory	
		Btu/day	kJ/day	J/hour	
7	TEMP	A real value that defines the temperature of the injected component ¹ . The temperature of the source term should only be specified if the thermal model has been activated by the THERMAL keyword in the RUNSPEC section.			None
		°F	°C	°C	

Notes:

- 1) The heat injection rate HRATE and the temperature of the injected component [TEMP](#) cannot be specified in the same record, if one of these has been specified the other should be defaulted.
- 2) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.68: SOURCE Keyword Description

See also the [BCCON](#) keyword in the [GRID](#) section and the [BCPROP](#) keyword in the [SCHEDULE](#) section that can be used to specify boundary conditions.

Example

The first example adds two source terms to block (I=2, J=4, K=6) that inject gas at a mass flow rate of 25,000 lb/day and inject water at a mass flow rate of 1,000 lb/day respectively.

```
--
--      DEFINE SOURCE TERM
--
-- I      J      K      COMP      MASS      HEAT      TEMP
-- -----
SOURCE
  2      4      6      GAS      25000      1*        1*        /
  2      4      6      WATER     1000      1*        1*        /
/
```

The second example adds a source term to block (I=2, J=4, K=6) that injects gas at a mass flow rate of 25,000 lb/day and a heat injection rate of 1.25×10^6 Btu/day in a thermal black-oil simulation.

```
--
--      DEFINE SOURCE TERM
--
-- I      J      K      COMP      MASS      HEAT      TEMP
-- -----
SOURCE
  2      4      6      GAS      25000      1.25E6      1*        /
/
```

The third example adds a source term to block (I=2, J=4, K=6) that injects gas at a mass flow rate of 25,000 lb/day and a temperature of 200°F in a thermal black-oil simulation.

```
--
--      DEFINE SOURCE TERM
--
-- I      J      K      COMP      MASS      HEAT      TEMP
-- -----
SOURCE
  2      4      6      GAS      25000      1*        200        /
/
```

12.3.224 SUMTHIN – DEFINE SUMMARY DATA REPORTING TIME STEPS

This keyword defines a time interval for writing out the [SUMMARY](#) data to the [SUMMARY](#) file and the RSM file, if the [RUNSUM](#) keyword has been has also been activated in the [SUMMARY](#) section. Only the data for the first time step in the time interval is written out and the other time steps are skipped until the next time interval. This enable the size of the [SUMMARY](#) files to be reduced depending on the size of the time interval. However, the keyword will produce irregular time steps reports of the [SUMMARY](#) data.

See [SUMTHIN – Define SUMMARY Data Reporting Time Steps](#) in the [SUMMARY](#) section for a full description.

12.3.225 SURFVISC – SURFACTANT SOLUTION VISCOSITY VERSUS CONCENTRATION

SURFSVISC defines the surfactant viscosity relationship of solution water viscosity with respect to increasing surfactant concentration within a grid block. The surfactant option must be activated by the [SURFACT](#) keyword in the [RUNSPEC](#) section in order to use this keyword.

See [SURFVISC – Surfactant Solution Viscosity versus Concentration](#) in the [PROPS](#) section for a full description.

12.3.226 SWINGFAC – DEFINE FIELD GAS CONTRACT PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [SWINGFAC](#), defines the gas contract parameters, swing factor and the monthly seasonal profile factor, for when there is a single gas contract being used in the model. The keyword is used with the Gas Field Operations option which is activated by the [GASFIELD](#) keyword in the [RUNSPEC](#) section. Gas contracts are commonly based on a Daily Contract Quantity (“DCQ”) that determines the gas rate that the field should be produced at, which is normally expressed as a multiple of the DCQ, for example 1.33, and is often referred to as the “swing factor”. Some gas contracts also define a maximum DCQ (“Max DCQ”) and/or a minimum take or pay DCQ (“Min DCQ”), as well as seasonal demand characteristics. For example, gas rates may be set higher in the winter months in order to meet heating demand compared with summer months in colder climates, and the opposite in warmer climates where air conditioning demand is high.

Thus, the DCQ must be calculated first over a contract period, where the contract period is commonly contract years, or in some instances contract quarters. This is performed by the simulator using the current DCQ and checking to see if the (DCQ x Swing Factor) can be satisfied throughout the current contract period, if not the DCQ is re-calculated so that the (DCQ x Swing Factor) condition is satisfied. Once this condition is met, the second and final pass uses the calculated DCQ in conjunction with the monthly scaling profile data to set the monthly gas rate for the field:

$$Q_{month} = DCQ \times SWINGFAC_{month} \quad (12.29)$$

Where:

Q_{month} = the monthly gas production target
 DCQ = Daily Contract Quantity
 $SWINGFAC_{month}$ = monthly rate scaling factor that takes into account seasonal demand, etc.

Here the [SWINGFAC](#) keyword sets the gas contract parameters for a single contract at the FIELD group level and is mutually exclusive to [GSWINGF](#) keyword in the [SCHEDULE](#) section that allows for different gas contract parameters to be assign to different groups.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.227 TIGHTEN – TIGHTEN AND RELAX NUMERICAL CONTROLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [TIGHTEN](#) keyword tightens up or slackens the numerical controls for the linear, non-linear and material balance convergence targets and also tightens or relaxes the maximum values for the aforementioned parameters. The keyword should be used with caution as it may result in significantly increasing the run times.

Note that any subsequent use of the [TUNING](#) keyword in the [SCHEDULE](#) section will result in resetting the numerical controls. See also the [TIGHTENP](#) in the [SCHEDULE](#) section that allows for greater flexibility in modifying the numerical controls.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.228
TIGHTENP
–
TIGHTEN
AND
RELAX
NUMERICAL
CONTROLS
INDIVIDUALLY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [TIGHTENP](#) keyword is similar to the [TIGHTEN](#) keyword in the [SCHEDULE](#) section, in that it tightens up or slackens the numerical controls for the linear, non-linear and material balance convergence targets and also tightens or relaxes the maximum values for the aforementioned parameters. However, [TIGHTENP](#) allows for greater flexibility as there are four parameters on this keyword, as opposed to just one on the [TIGHTEN](#) keyword, that can be used to modify the numerical controls. The keyword should be used with caution as it may result in significantly increasing the run times.

Note that any subsequent use of the [TUNING](#) keyword in the [SCHEDULE](#) section will result in resetting the numerical controls. See also the [TIGHTEN](#) keyword in the [SCHEDULE](#) section that has more limited flexibility in modifying the numerical controls.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.229 TIME – ADVANCE SIMULATION BY CUMULATIVE REPORTING TIME

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword advances the simulation to a given cumulative report time after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the [SCHEDULE](#) section keywords, or further [TIME](#) keywords may be entered to advance the simulator to the next report time.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TIME	A vector of real positive numbers that define the cumulative length of the of report times.			None
		days	days	hours	
Notes: I) The keyword is followed by a vector of numbers separated by a space and the keyword is terminated by a “/”.					

Table I 2.69:TIME Keyword Description

See also the [DATES](#) and [TSTEP](#) keyword in the [SCHEDULE](#) section. Note that since OPM Flow uses the standard Gregorian calendar and therefore leap years are accounted for in the [DATES](#) keyword. Thus, it is more accurate to use the [DATES](#) keyword to progress the simulator through time if one is matching actual production data.

Whenever possible it is a good idea to always set the start date to be at the beginning of the year, as like most simulators, OPM Flow reports are always stated at the number of days from the start date (and sometimes at a given date). If the start date is at the beginning of the year, then calculating the actual date is relatively straight forward and simple.

Examples

The fist example shows how to advance the simulation three years using the [TIME](#) keyword, from the given start date of January 1, 2022 set via the [START](#) keyword in the [RUNSPEC](#) section.

```

-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE

-- -----
-- SCHEDULE SECTION - 2022-01-01
-- -----
--
-- ADVANCE SIMULATION BY REPORTING TIME
--
TIME
    365.25  730.50  1095.75
/

```

The second example shows the same advance but using the [TSTEP](#) keyword instead.

```
-- =====
--
-- SCHEDULE SECTION
-- =====
SCHEDULE

-- -----
-- SCHEDULE SECTION - 2022-01-01
-- -----
RPTSCHED
      'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'      /
--
-- ADVANCE SIMULATION BY REPORTING TIME
--
TSTEP
      3*365.25
/
```

Again, if the simulated production targets are actual production data or the results are going to be used in economic evaluations then the [DATES](#) keyword may be more useful in advancing the simulation via the reporting time steps, as the exact dates will be honored.

12.3.230 TSTEP – ADVANCE SIMULATION BY REPORTING TIME

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword advances the simulation to a given report time after which additional keywords may be entered to instruct OPM Flow to perform additional functions via the [SCHEDULE](#) section keywords, or further [TSTEP](#) keywords may be entered to advance the simulator to the next report time.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	TSTEP	A vector of real positive numbers that define the length of the time intervals to subsequent report steps. Repeat counts may be used, for example 10*365.25.			None
		days	days	hours	
Notes: I) The keyword is followed by a vector of numbers separated by a space and the keyword is terminated by a “/”.					

Table 12.70: TSTEP Keyword Description

See also the [DATES](#) and [TIME](#) keywords in the [SCHEDULE](#) section. Note that since OPM Flow uses the standard Gregorian calendar and therefore leap years are accounted for in the [DATES](#) keyword. Thus, it is more accurate to use the [DATES](#) keyword to progress the simulator through time if one is matching actual production data.

Whenever possible it is a good idea to always set the start date to be at the beginning of the year, as like most simulators, OPM Flow reports are always stated at the number of days from the start date (and sometimes at a given date). If the start date is at the beginning of the year, then calculating the actual date is relatively straight forward and simple.

Examples

The first example shows how to advance the simulation via the reporting time steps from the given start date of January 1, 2022 set via the [START](#) keyword in the [RUNSPEC](#) section, to the next year, without any actions or reporting taking place.

```

-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE

-- -----
-- SCHEDULE SECTION - 2022-01-01
-- -----
--
-- ADVANCE SIMULATION BY REPORTING TIME
--
-- JAN FEB MAR APR MAY JUN JULY AUG SEP OCT NOV DEC
TSTEP
    31  28  31  30  31  30  31  31  30  31  30  31
/

```

The second example is similar to the previous example but with quarterly reporting time steps used instead based on $\frac{365.25}{4}=91.3125$ days per quarter

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
-- SCHEDULE SECTION - 2022-01-01
--
RPTSCHED      'WELLS=2'      'WELSPECS'      'CPU=2'      FIP=2'      /

--
-- ADVANCE SIMULATION BY REPORTING TIME
--
-- QUARTERLY
TSTEP
4*91.3125
/
```

Again, if the simulated production targets are actual production data or the results are going to be used in economic evaluations then the [DATES](#) keyword may be more useful in advancing the simulation via the reporting time steps, as the exact dates will be honored.

12.3.231 TUNING – NUMERICAL TUNING CONTROL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [TUNING](#) keyword defines the parameters used to control the commercial simulator's time stepping and numerical convergence for the global grid. The keyword is similar to the [TUNINGDP](#) keyword in the [SCHEDULE](#) section that is optimized for high throughput runs. The keyword is mostly ignored by OPM Flow; however, the simulator can be instructed to read some parameters from the [TUNING](#) keyword if the appropriate command line parameter has been activated (see section [2.2 Running OPM Flow From The Command Line](#)).

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	TSINIT	TSINIT is a positive real value that defines the maximum length of the next time step. Note that whenever the keyword is used TSINIT is always set back to the default value of one, unless explicitly over written.			1.0
		days	days	hours	
I-2	TSMAXZ	TSMAXZ is a positive real value that defines the maximum length of time steps following the next time step (TSINIT).			365.0
		days	days	hours	
I-3	TSMINZ	TSMINZ is a positive real value that defines the minimum length of all time steps.			0.1
		days	days	hours	
I-4	TSMCHP	TSMCHP is a positive real value that sets the minimum time step length that can be chopped.			0.15
		days	days	hours	
I-5	TSFMAX	TSFMAX is a positive real value that specifies the maximum growth rate a time step can be increased by, subject to the maximum allowable time step size set by TSMAXZ. For example, if the current time step has converged at 10 days and TSFMAX is set to the default value, then the next time step will be 3.0 x 10 days, that is 30 days provided it is less than TSMAXZ.			3.0
		dimensionless	dimensionless	dimensionless	
I-6	TSFMIN	TSFMIN is a positive real value that specifies the minimum decay rate a time step can be decreased by, subject to the minimum allowable time step size set by TSMINZ. For example, if the current time step has not converged at 10 days and TSFMIN is set to the default value, then the next time step will be 0.3 x 10 days, that is the maximum of 3 days and TSMINZ.			0.3
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-7	TSFCNV	TSFCNV is a positive real value that specifies the decay rate a time step can be decreased by after a convergence failure.			0.1
		dimensionless	dimensionless	dimensionless	
I-8	TFDIFF	TFDIFF is a positive real value that sets the time step growth factor of the time step after a convergence failure. For example, if the chopped current convergent time step is 10 days and TFDIFF is set to the default value, then the time step will be increased to 1.25×10 days, that is the minimum of 12.5 days and TSMAXZ.			1.25
		dimensionless	dimensionless	dimensionless	
I-9	THRUPT	THRUPT is a positive real value that specifies the maximum throughput ratio over a time step.			1.0×10^{20}
		dimensionless	dimensionless	dimensionless	
I-10	TMAXWC	TMAXWC is a positive real value that defines maximum allowed time step after a well event; for example, when a well is opened or closed, etc.			None
		days	days	hours	
I-11	/	Record terminated by a "/"			Not Applicable
2-1	TRGTTE	TRGTTE is a positive real value that sets the target time truncation error.			0.1
		dimensionless	dimensionless	dimensionless	
2-2	TRGCNV	TRGCNV is a positive real value that defines the target non-linear convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
2-3	TRGMBE	TRGMBE is a positive real value that specifies the target material balance error.			1.0×10^{-7}
		dimensionless	dimensionless	dimensionless	
2-4	TRGLCV	TRGLCV is a positive real value that specifies the target linear convergence error.			0.0001
		dimensionless	dimensionless	dimensionless	
2-5	XXXTTE	XXXTTE is a positive real value that sets the maximum time truncation error.			10.0
		dimensionless	dimensionless	dimensionless	
2-6	XXXCNV	XXXCNV is a positive real value that defines the maximum non-linear convergence error.			0.01
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-7	XXXMBE	XXXMBE is a positive real value that specifies the maximum mass balance error, that is the tolerated mass balance error relative to total mass present.			1.0 × 10 ⁻⁶
		dimensionless	dimensionless	dimensionless	
2-8	XXXLCV	XXXLCV is a positive real values that sets the maximum linear convergence error.			0.0001
		dimensionless	dimensionless	dimensionless	
2-9	XXXWFL	XXXWFL is a positive real values that fixes the maximum well flow convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
2-10	TRGFIP	TRGFIP is a positive real value that stipulates the target fluid in-place error in Local Grid Refinements.			0.025
		dimensionless	dimensionless	dimensionless	
2-11	TRGSFT	TRGSFT is a positive real values that defines the target surfactant change when the Surfactant Model is active in the run.			None
		dimensionless	dimensionless	dimensionless	
2-12	THIONX	THIONX is a positive real value used to set the threshold for damping in the ion exchange calculation for when the Brine Model is active in the run.			0.01
		dimensionless	dimensionless	dimensionless	
2-13	TRWGHT	TRWGHT is an integer that stipulates the implicitness for active tracer updates within the Newton iterations, and should be set to: 0 : The calculation is explicit, that is fully decoupled. 1 : The calculation is implicit, that is fully coupled.			1
		dimensionless	dimensionless	dimensionless	
2-14	/	Record terminated by a "/"			Not Applicable
3-1	NEWTMX	NEWTMX is a positive integer greater or equal to NEWTMN that stipulates the maximum number of Newton iterations for a time step.			12
		dimensionless	dimensionless	dimensionless	
3-2	NEWTMN	NEWTMN is a positive integer that is less or equal to NEWTMX that defines the minimum number of Newton iterations for a time step.			1
		dimensionless	dimensionless	dimensionless	
3-3	LITMAX	LITMAX is a positive integer greater or equal to LITMIN that sets the maximum number of linear iterations within a Newton iteration.			25
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
3-4	LITMIN	LITMIN is a positive integer less or equal to LITMAX that sets the minimum number of linear iterations within a Newton iteration.			1
		dimensionless	dimensionless	dimensionless	
3-5	MXWSIT	MXWSIT is a positive integer that defines the maximum number of iterations within a well flow calculation.			8
		dimensionless	dimensionless	dimensionless	
3-6	MXWPIT	MXWPIT is a positive integer that stipulates the maximum number of iterations for solving the bottom-hole pressure for wells under tubing head pressure control within a well flow calculation.			8
		dimensionless	dimensionless	dimensionless	
3-7	DDPLIM	DDPLIM is a positive real value that stipulates the maximum pressure change at the last Newton iteration.			1.0 × 10 ⁶
		psia	barsa	atma	
3-8	DDSLIM	DDSLIM is a positive real value that sets the maximum saturation change at the last Newton iteration.			1.0 × 10 ⁶
		dimensionless	dimensionless	dimensionless	
3-9	TRGDPR	TRGDPR is a positive real value that defines the target maximum pressure change within a time step.			1.0 × 10 ⁶
		psia	barsa	atma	
3-10	XXXDPR	XXXDPR is a positive real value that stipulates the maximum tolerable pressure change within a time step.			None
		psia	barsa	atma	
3-11	MNWRFP	MNWRFP is a positive integer greater than one and less than NEWTMX that defines the minimum number of Newton iterations before invoking the bisection algorithm for when the polymer phase is active in the model via the <i>POLYMER</i> keyword in the <i>RUNSPEC</i> section.			4
		dimensionless	dimensionless	dimensionless	
3-12	/	Record terminated by a “/”			Not Applicable

Notes:

1) The keyword consists of three records, with items 1-1 to 1-11 representing record one items, 2-1 to 2-14 record two and 3-1 to 3-12 representing record number three. Each of the records are terminated by a “/” and is explicitly shown in the above rows.

2) There is no keyword terminating “/”.

Table 12.71: TUNING Keyword Description

Note that for record number two (items 2-1 to 2-13) the maximum values should always be greater than the associated target value; for example, XXXCNV should be greater than TRGCNV. Also note that the *TUNING*

keyword is stored on the restart files (see [RPTST – Define Data to be Written to the RESTART File](#)) enabling the parameters to be utilized in a restart run without re-specifying the keyword.

Example

```
--
--      DEFAULT TUNING PARAMETERS
--
TUNING      1.0      365.0      0.1      0.15      3      0.3      0.1      1.25  1E20  1*  /
                                                    /
                                                    /
```

The above example explicitly sets the default parameters for OPM Flow for when the appropriate command line parameter has been activated (see section [2.2 Running OPM Flow From The Command Line](#)) to instruct the simulator to read the first record of the [TUNING](#) keyword. Alternatively one could just use the following to accomplish the same thing.

```
TUNING
/
/
/
```

12.3.232 TUNINGDP – NUMERICAL TUNING CONTROL FOR HIGH THROUGHPUT CASES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Defines the parameters used for controlling the commercial simulator's numerical convergence parameters. The keyword is similar to the [TUNING](#) keyword in the [SCHEDULE](#) section, but the defaults on this keyword are optimized for high throughput runs. See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TRGLCV	TRGLCV is a real positive value that specifies the linear convergence error target. The default value is ten times greater than the default value on the TUNING keyword.			0.0001
		dimensionless	dimensionless	dimensionless	
2	XXXLCV	XXXLCV is a real positive values that sets the maximum linear convergence error. The default value is ten times greater than the default value on the TUNING keyword.			0.001
		dimensionless	dimensionless	dimensionless	
3	TRGDDP	TRGDDP a real positive value that stipulates the maximum pressure change during a Newton iteration that enables the solution to be accepted when the residual pressure is still outside its convergence criteria.			1.0
		psia	barsa	atma	
4	TRGDDS	TRGDDS a real positive value that sets the maximum saturation change during a Newton iteration that enables the solution to be accepted when the residual saturation is still outside its convergence criteria.			0.001
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) There is no keyword terminating “/”.

Table 12.72: TUNINGDP Keyword Description

Note that the [TUNING](#) keyword is stored on the restart files (see [RPTRST – Define Data to be Written to the RESTART File](#)) enabling the parameters to be utilized in a restart run without re-specifying the keyword.

Example

```
--  
--      DEFAULT TUNINGDP PARAMETERS  
--  
TUNINGDP  
/
```

The above example explicitly sets the default parameters.

12.3.233 TUNINGH – NUMERICAL TUNING CONTROL FOR HISTORY MATCH GRADIENT CALCULATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Defines the parameters used for controlling the commercial simulator's numerical convergence parameters. The keyword is similar to the [TUNING](#) keyword in the [SCHEDULE](#) section, but the defaults on this keyword are optimized for high throughput runs. See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	GRGLCV	GRGLCV is a real positive value that specifies the linear convergence error target.			0.0001
		dimensionless	dimensionless	dimensionless	
2	GXXLCV	GXXLCV is a real positive values that sets the maximum linear convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
3	GMSLCV	GMSLCV is a real positive value that specifies the linear convergence residual reduction.			1.0 x 10 ⁻²⁰
		dimensionless	dimensionless	dimensionless	
4	LGTMIN	LGTMIN is a positive integer less or equal to LGTMAX that sets the minimum number of linear iterations within a Newton iteration.			1
		dimensionless	dimensionless	dimensionless	
5	LGTMAX	LGTMAX is a positive integer greater or equal to LGTMIN that sets the maximum number of linear iterations within a Newton iteration.			25
		dimensionless	dimensionless	dimensionless	
Notes: 1) There is no keyword terminating “/”.					

Table 12.73: TUNINGH Keyword Description

Example

```
--
--      DEFAULT TUNINGH PARAMETERS
--
TUNINGH
/
```

The above example explicitly sets the default parameters.

12.3.234 TUNINGL – NUMERICAL TUNING CONTROL FOR ALL LGRs

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

TUNINGL defines the parameters used for controlling the commercial simulator's numerical convergence parameters for all Local Grid Refinements ("LGR"). The keyword is the same as the **TUNING** keyword in the **SCHEDULE** section that applies the tuning parameters to the global grid. See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	TSINIT	TSINIT is a real positive value that defines the maximum length of the next time step. Note that whenever the keyword is used TSINIT is always set back to the default value of one, unless explicitly over written.			1.0
		days	days	hours	
I-2	TSMAXZ	TSMAXZ is a real positive value that defines the maximum length of the next time step following TSINIT.			365.0
		days	days	hours	
I-3	TSMINZ	TSMINZ is a real positive value that defines the minimum length of all time steps.			0.1
		days	days	hours	
I-4	TSMCHP	TSMCHP is a real positive values that sets the minimum length of all chopped time steps.			0.15
		days	days	hours	
I-5	TSFMAX	TSFMAX is a real positive value that specifies the maximum growth rate a time step can be increased by, subject to the maximum allowable time step size set by TSMAXZ. For example, if the current time step has converged at 10 days and TSFMAX is set to the default value, then the next time step will be 3.0 x 10 days, that is 30 days provided it is less than TSMAXZ.			3.0
		dimensionless	dimensionless	dimensionless	
I-6	TSFMIN	TSFMIN is a real positive value that specifies the minimum decay rate a time step can be decreased by, subject to the minimum allowable time step size set by TSMINZ. For example, if the current time step has not converged at 10 days and TSFMAX is set to the default value, then the next time step will be 0.3 x 10 days, that is the maximum of 0.3 days and TSMINZ.			0.3
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-7	TSFCNV	TSFCNV real positive value that specifies the decay rate a time step can be decreased by after the number of target iterations has been exceeded.			0.1
		dimensionless	dimensionless	dimensionless	
I-8	TFDIFF	TFDIFF is a real positive value that sets the time step growth factor of the time step after a convergence failure. For example, if the chopped current convergent time step is 10 days and TFDIFF is set to the default value, then the time step will be increased to 1.25×10 days, that is the minimum of 11.25 days and TSMAXZ.			1.25
		dimensionless	dimensionless	dimensionless	
I-9	THRURPT	THRURPT is a real positive value that specifies the maximum throughput ratio over a time step.			1.0×10^{20}
		dimensionless	dimensionless	dimensionless	
I-10	TMAXWC	TMAXWC is a real double precision value that defines maximum allowed time step after a well event; for example, when a well is opened or closed, etc.			None
		days	days	hours	
I-11	/	Record terminated by a "/"			Not Applicable
2-1	TRGTTE	TRGTTE is a real positive value that sets the time truncation error target.			0.1
		dimensionless	dimensionless	dimensionless	
2-2	TRGCNV	TRGCNV a real positive value that defines the non-linear convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
2-3	TRGMBE	TRGMBE is a real positive value that specifies then target material balance error.			1.0×10^{-7}
		dimensionless	dimensionless	dimensionless	
2-4	TRGLCV	TRGLCV is a real positive value that specifies the linear convergence error target.			0.00001
		dimensionless	dimensionless	dimensionless	
2-5	XXXTTE	XXXTTE is a real positive value that sets the maximum time truncation error.			10.0
		dimensionless	dimensionless	dimensionless	
2-6	XXXCNV	XXXCNV is a real positive value that defines the maximum non-linear convergence error.			0.01
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-7	XXXMBE	XXXMBE is a real positive value that specifies the maximum mass balance error, that is the tolerated mass balance error relative to total mass present.			1.0 × 10 ⁻⁶
		dimensionless	dimensionless	dimensionless	
2-8	XXXLCV	XXXLCV is a real positive values that sets the maximum linear convergence error.			0.0001
		dimensionless	dimensionless	dimensionless	
2-9	XXXWFL	XXXWFL is a real positive values that fixes the maximum well flow convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
2-10	TRGFIP	TRGFIP is a real positive value that stipulates the target fluid in-place error in Local Grid Refinements.			0.025
		dimensionless	dimensionless	dimensionless	
2-11	TRGSFT	TRGSFT is a real positive values that defines the target surfactant change when the Surfactant Model is active in the run.			None
		dimensionless	dimensionless	dimensionless	
2-12	THIONX	THIONX is a positive real value used to set the threshold for damping in the ion exchange calculation for when the Brine Model is active in the run.			0.01
		dimensionless	dimensionless	dimensionless	
2-13	TRWGHT	TRWGHT is a positive integer that stipulates the implicitness for active tracer updates within the Newton iterations, and should be set to: 1 : The calculation is explicit, that is fully decoupled. 2 : The calculation is implicit, that is fully coupled.			1
		dimensionless	dimensionless	dimensionless	
2-14	/	Record terminated by a "/"			Not Applicable
3-1	NEWTMX	NEWTMX is a positive integer greater or equal to NEWTMN that stipulates the maximum number of Newton iterations for a time step.			12
		dimensionless	dimensionless	dimensionless	
3-2	NEWTMN	NEWTMN is a positive integer that is less or equal to NEWTMX that defines the minimum number of Newton iterations for a time step.			1
		dimensionless	dimensionless	dimensionless	
3-3	LITMAX	LITMAX is a positive integer greater or equal to LITMIN that sets the maximum number of linear iterations within a Newton iteration.			25
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
3-4	LITMIN	LITMIN is a positive integer less or equal to LITMAX that sets the minimum number of linear iterations within a Newton iteration.			1
		dimensionless	dimensionless	dimensionless	
3-5	MXWSIT	MXWSIT is a positive integer that defines the maximum number of iterations within a well flow calculation.			8
		dimensionless	dimensionless	dimensionless	
3-6	MXWPIT	MXWPIT is a positive integer that stipulates the maximum number of iterations for solving the bottom-hole pressure for wells under tubing head pressure control within a well flow calculation.			8
		dimensionless	dimensionless	dimensionless	
3-7	DDPLIM	DDPLIM a real positive value that stipulates the maximum pressure change at the last Newton iteration.			1.0 x 10 ⁻⁶
		psia	barsa	atma	
3-8	DDSLIM	DDSLIM a real positive value that sets the maximum saturation change at the last Newton iteration.			1.0 x 10 ⁻⁶
		dimensionless	dimensionless	dimensionless	
3-9	TRGDP	TRGDP is a real positive value that defines the target pressure change within a time step.			1.0 x 10 ⁻⁶
		psia	barsa	atma	
3-10	XXXDPR	XXXDPR is a real positive value that stipulates the maximum tolerable pressure change within a time step.			1.0 x 10 ⁻⁶
		psia	barsa	atma	
3-11	MNWRFP	MNWRFP is a positive integer greater than one and less than NEWTMX that defines the minimum number of Newton iterations before invoking the bisection algorithm for when the polymer phase is active in the model via the POLYMER keyword in the RUNSPEC section.			4
		dimensionless	dimensionless	dimensionless	
3-12	/	Record terminated by a "/"			Not Applicable

Notes:

- 1) The keyword consists of three records, with items 1-1 to 1-11 representing record one items, 2-1 to 2-14 record two and 3-1 to 3-12 representing record number three. Each of the records are terminated by a "/" and is explicitly shown in the above rows.
- 2) There is no keyword terminating "/".

Table 12.74: TUNINGL Keyword Description

Note that for record number two (items 2-1 to 2-13) the maximum values should always be greater than the associated target value; for example, XXXCNV should be greater than TRGCNV. Also note that the [TUNINGS](#) keyword is stored on the restart files (see [RPTST – Define Data to be Written to the RESTART File](#)) enabling the parameters to be utilized in a restart run without re-specifying the keyword.

See also the [TUNINGS](#) keyword in the [SCHEDULE](#) section that sets the tuning parameters for individual LGRs.

Example

```
-  
--          DEFAULT TUNINGL PARAMETERS  
--  
TUNINGL  
/  
/  
/
```

The above example explicitly sets the default parameters for all LGRs.

12.3.235 TUNINGS – NUMERICAL TUNING CONTROL FOR INDIVIDUAL LGRs

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

TUNINGS defines the parameters used for controlling the commercial simulator's numerical convergence parameters for individual Local Grid Refinements ("LGR"). The keyword is similar to the [TUNINGL](#) keyword in the [SCHEDULE](#) section that applies the tuning parameters to the all LGRs, except for an additional first record that includes the [LGR](#) name.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See section [2.2 Running OPM Flow From The Command Line](#) on how to invoke various numerical schemes via the OPM Flow command line interface.

No.	Name	Description			Default
		Field	Metric	Laboratory	
0-1	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the tuning data is being defined.			None
0-2	/	Record terminated by a "/"			Not Applicable
1-1	TSINIT	TSINIT is a real positive value that defines the maximum length of the next time step. Note that whenever the keyword is used TSINIT is always set back to the default value of one, unless explicitly over written.			1.0
		days	days	hours	
1-2	TSMAXZ	TSMAXZ is a real positive value that defines the maximum length of the next time step following TSINIT.			365.0
		days	days	hours	
1-3	TSMINZ	TSMINZ is a real positive value that defines the minimum length of all time steps.			0.1
		days	days	hours	
1-4	TSMCHP	TSMCHP is a real positive values that sets the minimum length of all chopped time steps.			0.15
		days	days	hours	
1-5	TSFMAX	TSFMAX is a real positive value that specifies the maximum growth rate a time step can be increased by, subject to the maximum allowable time step size set by TSMAXZ. For example, if the current time step has converged at 10 days and TSFMAX is set to the default value, then the next time step will be 3.0 x 10 days, that is 30 days provided it is less than TSMAXZ.			3.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-6	TSFMIN	TSFMIN is a real positive value that specifies the minimum decay rate a time step can be decreased by, subject to the minimum allowable time step size set by TSMINZ. For example, if the current time step has not converged at 10 days and TSFMAX is set to the default value, then the next time step will be 0.3×10 days, that is the maximum of 0.3 days and TSMINZ.			0.3
		dimensionless	dimensionless	dimensionless	
I-7	TSFCNV	TSFCNV real positive value that specifies the decay rate a time step can be decreased by after the number of target iterations has been exceeded.			0.1
		dimensionless	dimensionless	dimensionless	
I-8	TFDIFF	TFDIFFA is a real positive value that sets the time step growth factor of the time step after a convergence failure. For example, if the chopped current convergent time step is 10 days and TFDIFF is set to the default value, then the time step will be increased to 1.25×10 days, that is the minimum of 1.25 days and TSMAXZ.			1.25
		dimensionless	dimensionless	dimensionless	
I-9	THRURPT	THRURPT is a real positive value that specifies the maximum throughput ratio over a time step.			1.0×10^{20}
		dimensionless	dimensionless	dimensionless	
I-10	TMAXWC	TMAXWC is a real double precision value that defines maximum allowed time step after a well event; for example, when a well is opened or closed, etc.			None
		days	days	hours	
I-11	/	Record terminated by a "/"			Not Applicable
2-1	TRGTTE	TRGTTE is a real positive value that sets the time truncation error target.			0.1
		dimensionless	dimensionless	dimensionless	
2-2	TRGCNV	TRGCNV a real positive value that defines the non-linear convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
2-3	TRGMBE	TRGMBE is a real positive value that specifies then target material balance error.			1.0×10^{-7}
		dimensionless	dimensionless	dimensionless	
2-4	TRGLCV	TRGLCV is a real positive value that specifies the linear convergence error target.			0.00001
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-5	XXXTTE	XXXTTE is a real positive value that sets the maximum time truncation error.			10.0
		dimensionless	dimensionless	dimensionless	
2-6	XXXCNV	XXXCNV is a real positive value that defines the maximum non-linear convergence error.			0.01
		dimensionless	dimensionless	dimensionless	
2-7	XXXMBE	XXXMBE is a real positive value that specifies the maximum mass balance error, that is the tolerated mass balance error relative to total mass present.			1.0 x 10 ⁻⁶
		dimensionless	dimensionless	dimensionless	
2-8	XXXLCV	XXXLCV is a real positive values that sets the maximum linear convergence error.			0.0001
		dimensionless	dimensionless	dimensionless	
2-9	XXXWFL	XXXWFL is a real positive values that fixes the maximum well flow convergence error.			0.001
		dimensionless	dimensionless	dimensionless	
2-10	TRGFIP	TRGFIP is a real positive value that stipulates the target fluid in-place error in Local Grid Refinements.			0.025
		dimensionless	dimensionless	dimensionless	
2-11	TRGSFT	TRGSFT is a real positive values that defines the target surfactant change when the Surfactant Model is active in the run.			None
		dimensionless	dimensionless	dimensionless	
2-12	THIONX	THIONX is a positive real value used to set the threshold for damping in the ion exchange calculation for when the Brine Model is active in the run.			0.01
		dimensionless	dimensionless	dimensionless	
2-13	TRWGHT	TRWGHT is a positive integer that stipulates the implicitness for active tracer updates within the Newton iterations, and should be set to: 1 : The calculation is explicit, that is fully decoupled. 2 : The calculation is implicit, that is fully coupled.			1
		dimensionless	dimensionless	dimensionless	
2-14	/	Record terminated by a "/"			Not Applicable
3-1	NEWTMX	NEWTMX is a positive integer greater or equal to NEWTMN that stipulates the maximum number of Newton iterations for a time step.			12
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
3-2	NEWTMN	NEWTMN is a positive integer that is less or equal to NEWTMX that defines the minimum number of Newton iterations for a time step.			1
		dimensionless	dimensionless	dimensionless	
3-3	LITMAX	LITMAX is a positive integer greater or equal to LITMIN that sets the maximum number of linear iterations within a Newton iteration.			25
		dimensionless	dimensionless	dimensionless	
3-4	LITMIN	LITMIN is a positive integer less or equal to LITMAX that sets the minimum number of linear iterations within a Newton iteration.			1
		dimensionless	dimensionless	dimensionless	
3-5	MXWSIT	MXWSIT is a positive integer that defines the maximum number of iterations within a well flow calculation.			8
		dimensionless	dimensionless	dimensionless	
3-6	MXWPIT	MXWPIT is a positive integer that stipulates the maximum number of iterations for solving the bottom-hole pressure for wells under tubing head pressure control within a well flow calculation.			8
		dimensionless	dimensionless	dimensionless	
3-7	DDPLIM	DDPLIM a real positive value that stipulates the maximum pressure change at the last Newton iteration.			1.0 x 10 ⁻⁶
		psia	barsa	atma	
3-8	DDSLIM	DDSLIM a real positive value that sets the maximum saturation change at the last Newton iteration.			1.0 x 10 ⁻⁶
		dimensionless	dimensionless	dimensionless	
3-9	TRGDP	TRGDP is a real positive value that defines the target pressure change within a time step.			1.0 x 10 ⁻⁶
		psia	barsa	atma	
3-10	XXXDPR	XXXDPR is a real positive value that stipulates the maximum tolerable pressure change within a time step.			1.0 x 10 ⁻⁶
		psia	barsa	atma	
3-11	MNVWRF	MNVWRF is a positive integer greater than one and less than NEWTMX that defines the minimum number of Newton iterations before invoking the bisection algorithm for when the polymer phase is active in the model via the POLYMER keyword in the RUNSPEC section.			4
		dimensionless	dimensionless	dimensionless	
3-12	/	Record terminated by a "/"			Not Applicable

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) The keyword consists of three records, with items 1-1 to 1-11 representing record one items, 2-1 to 2-14 record two and 3-1 to 3-12 representing record number three. Each of the records are terminated by a “/” and is explicitly shown in the above rows. 2) There is no keyword terminating “/”.					

Table 12.75:TUNINGS Keyword Description

Note that for record number two (items 2-1 to 2-13) the maximum values should always be greater than the associated target value; for example, XXXCNV should be greater than TRGCNV. Also note that the [TUNING](#) keyword is stored on the restart files (see [RPTRST – Define Data to be Written to the RESTART File](#)) enabling the parameters to be utilized in a restart run without re-specifying the keyword.

See also the [TUNINGL](#) keyword in the [SCHEDULE](#) section that sets the tuning parameters for all LGRs.

Example

```
--
--          DEFAULT TUNINGS PARAMETERS
--
TUNINGS
OP01-LGR
/
/
/
```

The above example explicitly sets the default parameters for the [LGR](#) named OP01-LGR

12.3.236 UDQ – DECLARE USER DEFINE QUANTITIES (“UDQ”)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword starts the definition of a [UDQ](#) section that stipulates the variables and operations used to access the User Defined Quantities features in OPM Flow. [UDQ](#) variables can be constants, [SUMMARY](#) variables, as defined in the [SUMMARY](#) section, or a formula using various mathematical functions together with constants and [SUMMARY](#) variables. Available operations include the ASSIGN, DEFINE, UNITS and UPDATE commands that are sub-keywords to the [UDQ](#) section keyword.

User Defined Quantities can be output to the summary file and used as User Defined Arguments (“UDA”) in the [SCHEDULE](#) section with various group, well, and connection keywords.

Although this keyword is read by OPM Flow and the [ACTION](#) and [UDQ](#) computational logic and calculations have been implemented, one should use caution when using this facility as it may result in OPM Flow aborting.

No.	Name	Description	Default
UDQ		Define the start of UDQ Definition Section. This is then followed on a new line by any number of UDQ records that define the various operations to be performed using the ASSIGN, DEFINE UNITS and UPDATE sub-keywords for the OPERATOR.	
I	OPERATOR	OPERATOR is a defined character string that specifies the type of operation to perform, and should be one of the following: 1) ASSIGN: Assigns a constant numerical value to the UDQ defined by VARIABLE and sets the UPDATE status to OFF. 2) DEFINE: Defines a mathematical formula and assigns it to the UDQ defined by VARIABLE. The UDQ is initialized with the formula and the UPDATE status is set to ON. 3) UNITS: Sets the reporting units for the UDQ defined by VARIABLE. The units have no effect on the calculations. The UDQ must already have been defined prior to using this option. 4) UPDATE: Stipulates when the UDQ defined by VARIABLE should be evaluated.	

No.	Name	Description	Default
2	VARIABLE	VARIABLE is a character string of length eight that stipulates the name of the user defined variable that will processed by the OPERATOR command. The first two characters of VARIABLE must be set based on the type of variable being defined, that is: 1) CU: For variables that are associated with connections, for example SUMMARY variable COFR (Connection Oil Flow Rate). 2) FU: For variables that are associated with field data, for example SUMMARY variable FOPR (Field Oil Production Rate). 3) GU: For variables that are associated with groups, for example SUMMARY variable GLPR (Group Liquid Production Rate). 4) RU: For variables that are associated with regions, for example SUMMARY variable RPR (Region Pressure). 5) SU: For variables that are associated with multi-segment wells, for example SUMMARY variable SOFR (Segment Oil Flow Rate). 6) WU: For variables that are associated with wells, for example SUMMARY variable WWCT (Well Water Cut). 7) AU: For variables that are associated with aquifers, for example SUMMARY variable AAQP (Analytical Aquifer Pressure). 8) BU: For variables that are associated with blocks, for example SUMMARY variable BPR (Block oil phase Pressure). OPM Flow currently only supports field, group, segment and well variables (FU*, GU*, SU* and WU*).	
3	EXPRESSION	The data type for EXPRESSION is based on the OPERATOR option above, namely if OPERATOR is set to: 1) ASSIGN: EXPRESSION should be a numerical value. 2) DEFINE: EXPRESSION should be a mathematical expression. Allowable tokens for the expression include SUMMARY or UDQ variables, real numbers (can be integers or scientific notation), opening and closing brackets '(' and ')', and basic mathematical operators plus '+', minus '-', multiply '*', divide '/' and exponent '^'. 3) UNITS: EXPRESSION should be a character string (enclosed in quotes if it contains blanks) with a maximum length of eight characters, that declares the units that will used when reporting the UDQ defined by VARIABLE. 4) UPDATE: EXPRESSION should be a defined character string (ON, OFF or NEXT): ON to evaluate the UDQ defined by VARIABLE at all time steps, OFF to not evaluate the UDQ, or NEXT to evaluate the UDQ at the next time step.	
	/	Termination of a UDQ record. Note that multiple numbers of records can be entered within a UDQ section with each record terminated by a “/”.	
	/	Define the end of UDQ Definition Section	
Notes: 1) The keyword is terminated by a “/”.			

Table 12.76: UDQ Keyword Description

All the functions available for use in the DEFINE EXPRESSION in the commercial simulator are listed in Table 12.77 below.

UDQ - Description Of Functions			
Function	Precedence	Type	Description
(			Open bracket.
)			Close bracket.
[			Start of user defined table argument list.
]			End of user defined table argument list.
,			Separator for multi-dimensional user defined table argument list.
-	6	unary elemental	Negation of defined elements.
ABS()	6	unary elemental	Absolute value of defined elements.
AVEA()	6	unary scalar	Arithmetic average of defined elements.
AVEG()	6	unary scalar	Geometric average of defined elements.
AVEH()	6	unary scalar	Harmonic average of defined elements.
DEF()	6	unary elemental	Returns 1 if the element is defined, otherwise returns undefined.
EXP()	6	unary elemental	Exponential of defined elements.
IDV()	6	unary elemental	Returns 1 if the element is defined, otherwise returns 0.
LN()	6	unary elemental	Natural logarithm of defined elements.
LOG()	6	unary elemental	Logarithm base 10 of defined elements.
MAX()	6	unary scalar	Maximum of defined elements.
MIN()	6	unary scalar	Minimum of defined elements.
NORM1()	6	unary scalar	1 norm of defined elements.
NORM2()	6	unary scalar	2 norm of defined elements.
NORMI()	6	unary scalar	Infinity norm of defined elements.
NINT()	6	unary elemental	Nearest integer to defined elements.
PROD()	6	unary scalar	Product of defined elements.
RANDN()	6	unary elemental	Random sample from Normal distribution $N(0,1)$, where the seed is specified by the UDQPARAM keyword Item 1.
RANDU()	6	unary elemental	Random sample from Uniform distribution $U(-1,1)$, where the seed is specified by the UDQPARAM keyword Item 1.
RRNDN()	6	unary elemental	Random sample from Normal distribution $N(0,1)$, where the seed is determined from the current time. By default restart simulations will

UDQ - Description Of Functions			
Function	Precedence	Type	Description
			use the seed from the base simulation (see UDQDIMS keyword Item 11).
RRNDU()	6	unary elemental	Random sample from Uniform distribution U(-1,1), where the seed is determined from the current time. By default restart simulations will use the seed from the base simulation (see UDQDIMS keyword Item 11).
SORTA()	6	unary elemental	Position of the element in an ascending sort of defined elements.
SORTD()	6	unary elemental	Position of the element in an descending sort of defined elements.
SUM()	6	unary scalar	Sum of defined elements.
UNDEF()	6	unary elemental	Returns 1 if element is undefined, otherwise returns undefined.
TU*[]	6	unary elemental	Lookup value of the user defined table based on the arguments in the square brackets. All arguments must have the same type or be scalar. A scalar is returned if all the arguments are scalar, or the same type of UDQ set is returned as the UDQ set arguments.
<=	5	binary intersection	Returns 1 if the LHS is less than or equal to the RHS, otherwise return 0. The tolerance for equality is specified by the UDQPARAM keyword Item 4.
>=	5	binary intersection	Returns 1 if the LHS is greater than or equal to the RHS, otherwise return 0. The tolerance for equality is specified by the UDQPARAM keyword Item 4.
<	5	binary intersection	Returns 1 if the LHS is less than the RHS, otherwise return 0.
>	5	binary intersection	Returns 1 if the LHS is greater than the RHS, otherwise return 0.
==	5	binary intersection	Returns 1 if the LHS is equal to the RHS. The tolerance for equality is specified by the UDQPARAM keyword Item 4.
!=	5	binary intersection	Returns 1 if the LHS is not equal to the RHS. The tolerance for equality is specified by the UDQPARAM keyword Item 4.
^	4	binary intersection	Exponentiation.
*	3	binary intersection	Multiplication.
/	3	binary intersection	Division.
+	2	binary intersection	Addition.
-	2	binary intersection	Subtraction.
UADD	1	binary union	Returns sum of intersecting elements from the union of two UDQ sets.
UMAX	1	binary union	Returns maximum of intersecting elements from the union of two UDQ sets.

UDQ - Description Of Functions			
Function	Prece- dence	Type	Description
UMIN	I	binary union	Returns minimum of intersecting elements from the union of two UDQ sets.
UMUL	I	binary union	Returns product of intersecting elements from the union of two UDQ sets.
Notes: 1) Functions are evaluated in order of decreasing precedence. Binary functions of equal precedence are evaluated left to right. Unary functions of equal precedence are evaluated right to left. 2) Binary functions operate on either the union or the intersection of two UDQ sets. For intersections the result is only defined if both corresponding elements are defined. For unions the result is defined if either of the corresponding elements are defined. Both arguments of a binary function either need to be of the same type or one needs to be scalar. 3) Unary functions operate on a single argument and are either elemental or return a scalar value.			

Table 12.77: UDQ - Description of Functions

See also the [UDADIMS](#), [UDQDIMS](#) and [UDQPARAM](#) keywords in the [RUNSPEC](#) section to define the dimensions for the [UDQ](#) keyword and associated variables.

Note

Wells and/or groups needed in a UDQ DEFINE statement must be present at the point of definition. In particular, wells must have been introduced using the [WELSPECS](#) keyword prior to the UDQ keyword if any specific well name is used in the defining expression. Similarly, well lists must have been introduced and populated through the [WLIST](#) keyword before using the well list name in the UDQ definition. Finally, groups must be introduced through the [WELSPECS](#) or [GRUPTREE](#) keywords before using any specific group names in the defining expression.

User Defined Quantities can also be used as User Defined Arguments (“UDA”) in the [SCHEDULE](#) section with various group, well, and connection keywords. In this case, the UDA variables are used to replace numerical values on these keywords by UDA variables that have been defined by the [UDQ](#) keyword. For example, if we wish to make the oil rate for certain wells be a function of their water cut, then one can define the function using the [UDQ](#) keyword that results in a [UDQ](#) variable, WU_WCUT say, and then use WU_WCUT as a UDA variable on the [WCONPROD](#) keyword for the ORAT parameter. Table 12.78 lists the keywords that can be used with UDA variables.

UDQ - User Defined Argument Supported Keywords					
Schedule Section Keywords Status					
Number	Group Keywords	Well Keywords		Connection Keywords	Miscellaneous Keywords
1	GCONINJE	WALKALIN	WINJEDET	CECON	LINCOM
2	GCONPRI	WAPI	WINJFCNC	CPIFACT	
3	GCONPROD	WCONHIST	WINJTEMP	CPIFACTL	

UDQ - User Defined Argument Supported Keywords Schedule Section Keywords Status					
Number	Group Keywords	Well Keywords		Connection Keywords	Miscellaneous Keywords
4	GCONSALE	WCONINJE	WPOLYMER		
5	GCONSUMP	WCONPROD	WSALT		
6	GECON	WECON	WSOLVENT		
7	GRUPFUEL	WECONCMF	WSURFACT		
8	GRUPSALE	WELDRAW	WTADD		
9	GSATPROD	WELLSTRE	WTMULT		
10	GTADD	WELTARG	WTRACER		
11	GTMULT	WFOAM			
		Multi-Segment Well Keywords			
1		WSEGTABL	WSEGVALV		
Notes: - Cells not colored show that the keyword has been tested and is functional within an ACTIONX block using a User Defined Argument. - Cells colored in gray indicate that the keyword has not been tested in OPM Flow. - Cells colored orange show keywords currently unavailable in OPM Flow because the underlying feature is not available; either because the keyword is a compositional keyword in the commercial simulator, or the keyword has not been implemented in OPM Flow.					

Table 12.78: UDQ - User Defined Argument Supported Keywords

Note

Note that after the terminating "/" for the ASSIGN operator normally any comments can be entered; however, if there is "/" within the comment field, as per:

```
ASSIGN FUNGLYLD 1.196 / Condensate Yield (63.5 / 56.7) / (1.0 - 0.065)
```

then the simulator will abort. The work around is to manually place the comment characters "--" after the ASSIGN terminating "/", like so:

```
ASSIGN FUNGLYLD 1.196 / -- Condensate Yield (63.5 / 56.7) / (1.0 - 0.06)
```

Examples

The first example shows how to define some constant field variables used for calculating facilities corrected condensate and Liquefied Petroleum Gas³²⁸ ("LPG") yields in a wet gas model:

--

³²⁸ Liquefied Petroleum Gas or LPG consists mainly of propane, propylene, butane, and butylene in various mixtures. It is produced as a by-product of natural gas processing and petroleum refining. The components of LPG are gases at standard conditions.

```
-- DEFINE START OF USER DEFINED QUANTITY SECTION
--
UDQ
--
-- OPERATOR VARIABLE    EXPRESSION
--
ASSIGN      FUNGLYLD    1.100000          / -- Condensate Yield (stb/Mscf)
ASSIGN      FUNGLSHK    0.000000          /  Condensate Shrinkage Factor to Zero
ASSIGN      FULPGYLD    0.065775          / -- LPG Sep Gas Yield (stb/Mscf)
ASSIGN      FULPGSHK    0.080410          /   LPG           Shrinkage Factor
ASSIGN      FUFACSHK    0.000935          / Facilities Shrinkage Factor
ASSIGN      FUFULSHK    0.052924          / Fuel           Utilization
ASSIGN      FUDELTA     1E-10              / Value to avoid dividing by zero
errors

/  DEFINE END OF USER DEFINED QUANTITY SECTION
```

The next example is a continuation of this example by showing how one can calculate the adjusted field condensate and LPG rates. Note both examples could be merged into a single [UDQ](#) definition but have been stated separately for ease of reference.

```
--
-- DEFINE START OF USER DEFINED QUANTITY SECTION
--
UDQ
--
-- OPERATOR VARIABLE    EXPRESSION
--
DEFINE      FU_FNGLR    FGPR * (FOGR * FUNGLYLD) / Calculate Condensate Rate Field
UPDATE      FU_FNGLR    ON                        /
UNITS       FU_FNGLR    STBD                      /

DEFINE      FU_FLPGR    FU_FWGPR * FULPGYLD       / Calculate LPG Rate Field
UPDATE      FU_FLPGR    ON                        /
UNITS       FU_FLPGR    STBD                      /

/  DEFINE END OF USER DEFINED QUANTITY SECTION
```

In the above the **DEFINE** operator is use to define the equations to calculate the corrected condensate (**FU_FNGLR**) and LPG rates (**FU_FLPGR**) with the **UPDATE** operator set to **ON** so that the rates are calculate at every time step, and finally, the **UNITS** operator is used to set the units of the calculated rates.

The final example show the use of the [UDADIMS](#) and [UDQDIMS](#) keywords in the [RUNSPEC](#) section, followed by the keywords in the [SCHEDULE](#) section that define a [UDQ](#) definition that uses the **DEFINE** operator to calculate adjusted well rates based on an expression. The final set of keywords show how the [UDQ](#) defined variables are employed on the [WCONPROD](#) keyword to control the production constraints for several wells.

```

RUNSPEC SECTION KEYWORDS
-----
--
--      USER DEFINED ARGUMENT DIMENSIONS
--      NO.      NOT      TOTAL
--      ARGS    USED    UDQ
UDADIMS
--      10      1*      10
--
--      USER DEFINED ARGUMENT DIMENSIONS FACILITY
--      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      MAX      RAND
--      FUNCS    ITEMS    CONNS    FIELD    GROUP    REGS    SEGTM    WELL    AQUF    BLCKS    OPT
UDQDIMS
--      50      25      0      50      50      0      0      0      0      0      0      N /
```

And the [SCHEDULE](#) section part of the example is shown below.

```

                                SCHEDULE SECTION KEYWORDS
                                -----
--
-- DEFINE START OF USER DEFINED QUANTITY SECTION
--
UDQ
--
-- OPERATOR VARIABLE EXPRESSION
--
DEFINE      WUOPRL (WOPR OPL01 - 150) * 0.90 / OIL & LIQ CAPACITIES
DEFINE      WULPRL (WLPR OPL01 - 200) * 0.90 / at GEFAC = 0.8995
DEFINE      WUOPRU (WOPR OPU01 - 250) * 0.80 /
DEFINE      WULPRU (WLPR OPU01 - 300) * 0.80 /
--
UNITS      WUOPRL SM3/DAY          / DEFINE REPORTING UNITS
UNITS      WULPRL SM3/DAY          / FOR UDQ VARIABLES
UNITS      WUOPRU SM3/DAY          /
UNITS      WULPRU SM3/DAY          /
/ DEFINE END OF USER DEFINED QUANTITY SECTION
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL  OIL   WAT   GAS   LIQ   RES   BHP   THP   VFP   VFP
-- NAME  SHUT   MODE  RATE  RATE  RATE  RATE  RATE  PRES  PRES  TABLE ALFQ
WCONPROD
OP01     SHUT   GRUP   1*    1*    1*    1*    1*    200.0                /
OP02     SHUT   GRUP   1*    1*    1*    1*    1*    200.0                /
/
DATES
      1  FEB   2020  /
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL  OIL   WAT   GAS   LIQ   RES   BHP   THP   VFP   VFP
-- NAME  SHUT   MODE  RATE  RATE  RATE  RATE  RATE  PRES  PRES  TABLE ALFQ
WCONPROD
OP01     OPEN   GRUP   WUOPRL 1*    1*    WULPRL 1*    60.0                /
OP02     OPEN   GRUP   WUOPRL 1*    1*    WULPRL 1*    00.0                /
/
DATES
      1  MAR   2020  /
      1  APR   2020  /
      1  MAY   2020  /
      1  JUN   2020  /
      1  JUL   2020  /
      1  AUG   2020  /
      1  SEP   2020  /
/

```

12.3.237 UDT – DECLARE USER DEFINE TABLES (“UDT”)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [UDT](#) keyword defines a single multi-dimensional User Defined Table (“UDT”).

User Defined Quantities (“UDQ”) defined using the [UDQ](#) keyword in the [SCHEDULE](#) section can be assigned values by looking up a [UDT](#).

This keyword is partially supported by OPM Flow; only one-dimensional lookup tables are supported.

Each User Defined Table must be defined with a separate [UDT](#) keyword that consists of a number of records defining the table name and dimensions; the interpolation type and interpolation points for each dimension; and the table values for each set of interpolation points. The total number of records depends on the table dimensions and the number of interpolation points.

No.	Name	Description	Default
I-1	NAME	A character string of up to eight characters in length beginning with ‘TU’ that defines the table name.	None
I-2	NDIMS	An integer value between one and MXDIMS that defines the number of dimensions for the table. MXDIMS is the maximum number of UDT table dimensions defined using the UDTDIMS keyword in the RUNSPEC section. OPM Flow only supports one-dimensional tables (NDIMS = 1).	None
I-3	/	Record terminated by a “/”	Not Applicable
2-1	TYPE	A defined character string that defines the type of interpolation for this dimension and must one of the following: 1) ‘NV’ this dimension will use the nearest value. 2) ‘LC’ this dimension will use linear interpolation within the table and clamp to the first or last value respectively for extrapolation. 3) ‘LL’ this dimension will use linear interpolation within the table and linear extrapolation outside of the table. 4) ‘ID’ this dimension will use a string or identifier to select the interpolation point. If no matching interpolation point is found then the lookup will return the value given to undefined UDQ variables as specified by the DEFAULT value on the UDQPARAM keyword in the RUNSPEC section.	None
2-2	POINTS	A monotonically increasing vector of real values that defines the numerical value of the interpolation points for this dimension if the specified type of interpolation TYPE is either ‘NV’, ‘LC’ or ‘LL’. Otherwise, a vector of character strings of up to eight characters in length defining the ‘ID’.	None
2-2	/	One record for each of the NDIMS dimensions each terminated by a “/”	Not Applicable
3-1	VALUES	A vector of real values corresponding to each of the interpolations points POINTS in the first dimension.	None

No.	Name	Description	Default
3-2	/	Record (3-1) is included once if the number of dimensions NDIMS equals one. Otherwise, if the number of dimensions is greater than one then Record (3-1) is repeated for each of the interpolation POINTS in the second dimension. Each record is terminated by a "/".	Not Applicable
3-3	/	Item (3-2) is followed by an additional "/".	Not Applicable
3-4	/	If the number of dimensions NDIMS is greater than or equal to three then records (3-1) to (3-3) are repeated for each of the interpolation POINTS in the third dimension, and then followed by an additional "/".	Not Applicable
3-5	/	If the number of dimensions NDIMS is equal to four then records (3-1) to (3-4) are repeated for each of the interpolation POINTS in the fourth dimension, and then followed by an additional "/".	Not Applicable
3-6	/	The keyword is terminated by a "/".	Not Applicable
Notes: 1) Each User Defined Table must be entered with a separate <i>UDT</i> keyword. 2) The table values (VALUES) are ordered with the interpolation points for the first dimension cycling fastest. 3) Each of the records are terminated by a "/" and is explicitly shown in the above rows. 4) The keyword is terminated by a "/".			

Table 12.79: UDT Keyword Description

The format for a one-dimensional table with m interpolation points is outlined below:

```
UDT
(NAME) 1 / ----- NAME, NDIMS
--
-- Interpolation points for each dimension:
--
(TYPE1) a1 a2 ... am / ----- TYPE, POINTS
--
-- VALUES: f1 = f(ai)
--
f1 f2 ... fm / ----- end of 1D f(a1) ... f(am)
/ ----- end of 2D f(*)
/ ----- end of keyword
```

The format for a two-dimensional table with m and n interpolation points in the first and second dimensions respectively is outlined below:

```
UDT
(NAME) 2 / ----- NAME, NDIMS
--
-- Interpolation points for each dimension:
--
(TYPE1) a1 a2 ... am / ----- TYPE, POINTS
(TYPE2) b1 b2 ... bn /
--
-- VALUES: fij = f(ai,bj)
--
f11 f21 ... fm1 / ----- end of 1D f(a1,b1) ... f(am,b1)
```



```
f12 f22 ... fm2 /
...
f1n f2n ... fmn / ----- end of 1D f(a1,bn) ... f(am,bn)
/ ----- end of 2D f(*, b1) ... f(*, bn)
/ ----- end of keyword
```

The format for a three-dimensional table with m, n and p interpolation points in the first, second and third dimensions respectively is outlined below:

```
UDT
(NAME) 3 / ----- NAME, NDIMS
--
-- Interpolation points for each dimension:
--
(TYPE1) a1 a2 ... am / ----- TYPE, POINTS
(TYPE2) b1 b2 ... bn /
(TYPE3) c1 c2 ... cp /
--
-- VALUES: fijik = f(ai,bj,ck)
--
f111 f211 ... fm11 / ----- end of 1D f(a1,b1,c1) ... f(am,b1,c1)
f121 f221 ... fm21 /
...
f1n1 f2n1 ... fmn1 / ----- end of 1D f(a1,bn,c1) ... f(am,bn,c1)
/ ----- end of 2D f(*, b1,c1) ... f(*, bn,c1)
f112 f212 ... fm12 /
f122 f222 ... fm22 /
...
f1n2 f2n2 ... fmn2 /
/

...

f11p f21p ... fm1p /
f12p f22p ... fm2p /
...
f1np f2np ... fmpn / ----- end of 1D f(a1,bn,cp) ... f(am,bn,cp)
/ ----- end of 2D f(*, b1,cp) ... f(*, bn,cp)
/ ----- end of 3D f(*, *, c1) ... f(*, *, cp)
/ ----- end of keyword
```

The format for a four-dimensional table with m, n, p and q interpolation points in the first, second, third and forth dimensions respectively is outlined below:

```
UDT
(NAME) 4 / ----- NAME, NDIMS
--
-- Interpolation points for each dimension:
--
(TYPE1) a1 a2 ... am / ----- TYPE, POINTS
(TYPE2) b1 b2 ... bn /
(TYPE3) c1 c2 ... cp /
(TYPE4) d1 d2 ... dq /
--
-- VALUES: fijikl = f(ai,bj,ck,d1)
--
-- f(*, *, *, d1):
--
f1111 f2111 ... fm111 / ----- end of 1D f(a1,b1,c1,d1) ... f(am,b1,c1,d1)
f1211 f2211 ... fm211 /
...
f1n11 f2n11 ... fmn11 / ----- end of 1D f(a1,bn,c1,d1) ... f(am,bn,c1,d1)
/ ----- end of 2D f(*, b1,c1,d1) ... f(*, bn,c1,d1)
```

```

f1121 f2121 ... fm121 /
f1221 f2221 ... fm221 /
...
f1n21 f2n21 ... fmn21 /
/

...

f11p1 f21p1 ... fm1p1 /
f12p1 f22p1 ... fm2p1 /
...
f1np1 f2np1 ... fmn1p1 / ----- end of 1D f(a1,bn,cp,d1) ... f(am,bn,cp,d1)
/ ----- end of 2D f(*, b1,cp,d1) ... f(*, bn,cp,d1)
/ ----- end of 3D f(*, *, c1,d1) ... f(*, *, cp d1)
--
-- f(*, *, *,d2):
--
f1112 f2112 ... fm112 / ----- end of 1D f(a1,b1,c1,d2) ... f(am,b1,c1,d2)
f1212 f2212 ... fm212 /
...
f1n12 f2n12 ... fmn12 / ----- end of 1D f(a1,bn,c1,d2) ... f(am,bn,c1,d2)
/ ----- end of 2D f(*, b1,c1,d2) ... f(*, bn,c1,d2)
f1122 f2122 ... fm122 /
f1222 f2222 ... fm222 /
...
f1n22 f2n22 ... fmn22 /
/

...

f11p2 f21p2 ... fm1p2 /
f12p2 f22p2 ... fm2p2 /
...
f1np2 f2np2 ... fmn1p2 / ----- end of 1D f(a1,bn,cp,d2) ... f(am,bn,cp,d2)
/ ----- end of 2D f(*, b1,cp,d2) ... f(*, bn,cp,d2)
/ ----- end of 3D f(*, *, c1,d2) ... f(*, *, cp,d2)
--
-- f(*, *, *,d3):
--

...

--
-- f(*, *, *,dq):
--
f111q f211q ... fm11q / ----- end of 1D f(a1,b1,c1,dq) ... f(am,b1,c1,dq)
f121q f221q ... fm21q /
...
f1n1q f2n1q ... fmn1q / ----- end of 1D f(a1,bn,c1,dq) ... f(am,bn,c1,dq)
/ ----- end of 2D f(*, b1,c1,dq) ... f(*, bn,c1,dq)
f112q f212q ... fm12q /
f122q f222q ... fm22q /
...
f1n2q f2n2q ... fmn2q /
/

...
f11pq f21pq ... fm1pq /
f12pq f22pq ... fm2pq /
...
f1npq f2npq ... fmn1pq / ----- end of 1D f(a1,bn,cp,dq) ... f(am,bn,cp,dq)
/ ----- end of 2D f(*, b1,cp,dq) ... f(*, bn,cp,dq)
/ ----- end of 3D f(*, *, c1,dq) ... f(*, *, cp,dq)
/ ----- end of 4D f(*, *, *, d1) ... f(*, *, *, dq)

```

/ ----- end of keyword

Examples

The following example shows a one-dimension User Defined Table that will linearly interpolate between values of 100 and 180 for values of FOPR between 100 and 500 and will clamp to the end points for values of FOPR outside this range.

```
--
-- DECLARE USER DEFINED TABLE
--
UDT
-- NAME NDIMS
-- 'TU_FBHP' 1 /
-- TYPE POINTS
-- 'LC' 100.0 500.0 / -- FOPR values
-- VALUES
-- 100.0 180.0 / -- FBHP values
/
/
--
-- DECLARE USER DEFINED QUANTITIES
--
UDQ
ASSIGN FU_WBHP 0 /
DEFINE FU_WBHP0 FU_WBHP /
DEFINE FU_WBHP (TU_FBHP[FOPR] UMIN WBHP 'PROD') UMIN FU_WBHP0 /
/
--
-- WELL PRODUCTION TARGETS AND CONSTRAINTS
--
WCONPROD
'PROD' 'OPEN' ORAT 500.0 4* FU_WBHP /
/
```

In the example above, the [UDT](#) specifies how the bottom hole pressure limit should reduce as the field oil production rate decreases (note the interpolation point values must be monotone increasing). The User Defined Quantity FU_WBHP uses a lookup value from the [UDT](#) in the form TU_BHP[FOPR].

12.3.238 USECUPL – LOAD A RESERVOIR COUPLING FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [USECUPL](#) keyword causes the simulator to read a Reservoir Coupling file that has been previously created in a master run using the [DUMPCUPL](#) keyword in the [SCHEDULE](#) section, for when reservoir coupling is invoked by the [GRUPMAST](#) and [SLAVES](#) keywords in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.239 VAPPARS – OIL VAPORIZATION PARAMETERS

[VAPPARS](#) defines the rate of oil vaporization in the presence of undersaturated gas and the rate at which the remaining oil gets “heavier” via the reduction in the solution gas-oil ratio (“Rs”). This keyword should only be used if the [OIL](#), [GAS](#), [DISGAS](#) and [VAPOIL](#) keywords in the [RUNSPEC](#) section have been invoked to allow oil, gas, dissolved gas and vaporized oil to be present in the model.

In OPM Flow, [VAPPARS](#) can only be set once, and not changed multiple times in the [SCHEDULE](#) section.

See [VAPPARS – Oil Vaporization Parameters](#) in the [SOLUTION](#) section for a full description.

12.3.240 VFPCHK – DEFINE PRODUCTION VERTICAL FLOW PERFORMANCE BHP CHECK

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [VFPPROD](#) keyword defines production Vertical Flow Performance (“VFP”) tables that are used to determine the outflow or downstream pressure based on the inlet or upstream pressure and the phases flowing through the system. For a well this means the table relates the flowing bottom-hole pressure (“BHP”) to the well’s tubing head pressure (“THP”) based on the oil, gas and water rates (and any artificial lift quantities like gas lift gas), or phases ratios, flowing up the wellbore.

The VFP tables are generated by external programs and may result in some inconsistencies. A common inconsistency is that the curves of BHP versus flow rate at adjacent THP values cross, that is the BHP is increasing with decreasing THP. This will result in numerical convergence problems and should therefore be avoided; which is why the simulator checks for this particular inconsistency. However, there are cases when the external software generating the VFP table outputs “valid” high BHP values, for example, when the flow exceeds the erosion velocity limits, or the flow is supersonic, indicating a no flow condition. The VFPCHK keyword sets the BHP check pressure ([VFPCHK](#)) for subsequent [VFPPROD](#) tables, so that crossing BHP values above [VFPCHK](#) will be ignored.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	VFPCCHK	VFPCCHK is a real positive value that defines the BHP pressure above which crossingVFP curves will be ignored. Setting VFPCCHK to a large number like the default value number will cause all crossing curves to be checked.Also if the keyword is omitted from the input deck then the check is performed using the default value.			1.0 ¹⁰
		psia	barsa	atma	
Notes: I) The keyword is terminated by a “/”.					

Table 12.80: VFPCHK Keyword Description

Note

One reason for external programs generating crossing VFP curves is that the curves have been generated with too much resolution. For example, if the GOR entries has been generated with values of 100, 150, 200, 250, 300, 350, 400, 450 and 500, then use a geometric spacing instead to generated the VFP table, that is: 100, 300, 900. This will enable the simulator to interpolate the curves consistently and avoid crossing VFP curves.

Example

Here the example sets the maximum BHP to be 1.0×10^6 above which crossing VFP curves will be ignored.

```
--  
--      DEFINE PRODUCTION VFP CHECK MAX BHP  
--  
--      MXBHP  
VFPCHK      1.0E6  
/
```

12.3.241 VFPINJ – DEFINE INJECTION VERTICAL FLOW PERFORMANCE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [VFPINJ](#) keyword defines injection Vertical Flow Performance (“VFP”) tables that are used to determine the outflow or downstream pressure based on the inlet or upstream pressure and the phases being injected into the system. For a well this means the table relates the flowing bottom-hole pressure (“BHP”) to the well’s tubing head pressure (“THP”) based on the oil, gas or water injection rates. The table is also used to describe the pressure relationship when the network option is being used, although the Network option is not currently implemented in OPM Flow. In this case the table describes the pipeline pressure behavior from the HIGHER group (inlet node) to the LOWER group (outlet node) given the current flowing conditions (the group relationship is defined by the [GRUPTREE](#) keyword in [SCHEDULE](#) section).

The data for this keyword is generated by an external program and is normally included into the input deck using the [INCLUDE](#) keyword as described in section [CHAPTER 4: GLOBAL SECTION KEYWORDS](#), as the data can be quite voluminous.

Each [VFPINJ](#) table must be entered with a separate [VFPINJ](#) keyword that consists of four records, with 1-1, 1-2 representing record one items (1) and (2) and 2-1, 2-2 representing record number two items (1) and (2) and so on in the “No.” column in Table 12.81. Each record is terminated by a “/”. The fourth record must be repeated to give BHP data as a function of FLO for all THP values.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	VFPTAB	A positive integer greater than zero and less than or equal to the MXVFPTAB variable as defined on the VFPIDIMS keyword in the RUNSPEC section, that defines the vertical flow performance table number.			None
1-2	VFPREF	A real positive value that defines the reference depth used to generate this VFPINJ table data set. OPM Flow automatically corrects any difference between VFPREF and the BHPREF on the WELSPRES and WPAVDEP keywords in the SCHEDULE section, using the current hydrostatic head.			None
1-3	FLO	A defined character string that defines the injection phases, and should be set to one of the following character strings: 1) OIL : for injecting phase being oil. 2) GAS : for injecting phase being gas. 3) WAT : for injecting phase being water.			None
1-4	VFPTYPE	A defined character string that should be defaulted or set equal to THP.			THP
1-5	VFPUNITS	Units used for the BHP-DATA on this keyword. This variable is ignored by OPM Flow and should be defaulted with I*.			I*
		FIELD	METRIC	LAB	
1-6	VFPVALUE	A defined character string that should be defaulted or set equal to BHP. This variable is ignored by OPM Flow and should be defaulted with I*.			BHP

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-7	/	Record terminated by a “/”			Not Applicable
2-1	FLO-DATA	A real positive monotonically increasing vector that defines the numerical values of the injection phase declared by the FLO variable. The number of entries must greater than two and less than or equal to MXMFLO as defined on the <i>VFPIDIMS</i> keyword in the <i>RUNSPEC</i> section.			None
		Liquid: stb Gas: Mscf	Liquid: sm ³ Gas: sm ³	Liquid: scc Gas: scc	
2-2	/	Record terminated by a “/”			Not Applicable
3-1	THP-DATA	A real positive monotonically increasing vector that defines the numerical values of the tubing head pressure values. The number of entries must be greater than two and less than or equal to MXMTHP as defined on the <i>VFPIDIMS</i> keyword in the <i>RUNSPEC</i> section.			None
		psia	barsa	atma	
3-2	/	Record terminated by a “/”			Not Applicable
4-1	NTHP	This data record consists of an integer value that defines the index of THP values entered via the THP-DATA records on this keyword. For example, if THP-DATA is equal to 1000, 2000, 3000 and 3500 and NTHP is equal to three then NTHP refers to third entry, that is THP equal to 3000.			None
	BHP-DATA	NTHP is then followed by a real vector of BHP values for each FLO injection rate for the corresponding index value (NTHP) and is then terminated with a”/” The (4-1) record, which consists of both NTHP and BHP-DATA data, is then repeated, until all combinations of (NTHP and FLO) and the associate BHP data has been entered.			
		psia	barsa	atma	
4-2	/	Each Index (NTHP, BHP-DATA) data set is terminated by a “/”			Not Applicable

Notes:

- Each *VFPINJ* table must be entered with a separate VFINJ keyword that consists of four records, with items 1-1 to 1-7 representing record one items and 2-1 to 2-2 representing record number two items, etc., in the “No.” column in this table.
- Each of the records are terminated by a “/” and is explicitly shown in the above rows.
- There is no keyword terminating “/”.

Table 12.81: VFPINJ Keyword Description

See also the [WEL SPECS](#) keyword to define wells and the [WCONINJE](#) keyword that is used to allocate the [VF PINJ](#) tables to specific wells.

Note that one [VF PINJ](#) table can be allocated to one or more wells, provided the wells in question have a similar trajectory and similar flow characteristics, for example vertical water injection wells injecting into the same reservoir.

The [VFPPROD](#) keyword is used to enter VFP tables for production wells or to describe the pressure relationship when the network option is being used. In this case the table describes the pipeline pressure behavior from the LOWER group (inlet node) to the HIGHER group (outlet node) given the current production conditions.

All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example shows the [VF PINJ](#) table for a water injection well and is taken from the Norne OPM Flow model.

```

VF PINJ

-- Table      Datum Depth      Rate Type
-- -----
      12          2718.07          'WAT' /

-- 'WAT' units - SM3/DAY
      500.0  1263.2  2026.3  2789.5  3552.6
      4315.8  5078.9  5842.1  6605.3  7368.4
      8131.6  8894.7  9657.9 10421.1 11184.2
      11947.4 12710.5 13473.7 14236.8 15000.0 /

-- 'THP' units - BARSA
      21.01   63.24  105.46  147.68  189.90
      232.12  274.35  316.57  358.79  401.01 /

1  254.51  253.95  252.27  249.83  246.69
   242.88  238.42  233.32  227.59  221.22
   214.23  206.62  198.38  189.53  180.06
   169.97  159.26  147.95  136.00  123.46
/
2  297.02  296.49  294.82  292.39  289.26
   285.47  281.01  275.92  270.20  263.84
   256.87  249.28  241.05  232.22  222.76
   212.70  202.01  190.71  178.79  166.27
/

.....
.....

9  594.67  594.29  592.70  590.34  587.29
   583.57  579.16  574.17  568.55  562.25
   555.40  547.92  539.79  531.09  521.74
   511.82  501.25  490.13  478.34  466.01
/
10 637.19  636.83  635.26  632.91  629.86
    626.16  621.76  616.78  611.17  604.89
    598.05  590.59  582.47  573.79  564.45
    554.56  544.01  532.91  521.14  508.83
/

```

The example shows the first two and the last two records of the fourth kind, as the data is too voluminous to be included.

Note

The VFPTAB variable defines the table number of the [VFPINJ](#) data set; if more then one [VFPINJ](#) keyword is entered with the same VFPTAB number then the [VFPINJ](#) data set will be overwritten by the last [VFPINJ](#) keyword with the same VFPTAB number.

The same comment is also applicable to the [VFPPROD](#) keyword.

12.3.242 VFPPROD – DEFINE PRODUCTION VERTICAL FLOW PERFORMANCE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [VFPPROD](#) keyword defines production Vertical Flow Performance (“VFP”) tables that are used to determine the outflow or downstream pressure based on the inlet or upstream pressure and the phases flowing through the system. For a well this means the table relates the flowing bottom-hole pressure (“BHP”) to the well’s tubing head pressure (“THP”) based on the oil, gas and water rates (and any artificial lift quantities like gas lift gas), or phases ratios, flowing up the wellbore. The table is also used to describe the pressure relationship when the network option is being used. In this case the table describes the pipeline pressure behavior from the LOWER group (inlet node) to the HIGHER group (outlet node) given the current flowing conditions (the group relationship is defined by the [GRUPTREE](#) keyword in [SCHEDULE](#) section).

Each [VFPPROD](#) table must be entered with a separate [VFPPROD](#) keyword that consists of seven records, with 1-1, 1-2 representing record one items (1) and (2) and 2-1, 2-2 representing record number two items (1) and (2) in the “No.” column in Table 12.82. Each record is terminated by a “/”. The seventh record must be repeated to give BHP data as a function of FLO for all THP values.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-1	VFPTAB	A positive integer greater than zero and less than or equal to the MXVFPTAB variable as defined on the VFPPDIMS keyword in the RUNSPEC section, that defines the vertical lift performance table number.			None
1-2	VFPREF	A real positive value that defines the reference depth used to generate this VFPPROD table data set. OPM Flow automatically corrects any difference between VFPREF and the BHPREF on the WELSPECS and WPAVEDEP keywords in the SCHEDULE section, using the current hydrostatic head.			None
1-3	FLO	A defined character string that defines the flowing phases, and should be set to one of the following character strings: 1) GAS : for flowing phase being the gas rate. 2) OIL : for flowing phase being the oil rate. 3) LIQ : for flowing phase being the liquid (oil plus water) rate.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-4	WFR	A defined character string that defines the flowing water fraction and should be set to one of the following character strings: 1) WOR: for the water fraction being the water-oil ratio $\frac{q_w}{q_o}$ and should be used if FLO is set to 'OIL' or 'LIQ' 2) WCT: for the water fraction being the water cut $\frac{q_w}{q_o + q_w}$ and should be used if FLO is set to 'OIL' or 'LIQ' 3) WGR: for the water fraction being the water-gas ratio $\frac{q_w}{q_g}$ and should be used if FLO is set to 'GAS'.			None
I-5	GFR	A defined character string that defines the flowing gas fraction and should be set to one of the following character strings: 1) GOR: for the gas fraction being the gas-oil ratio $\frac{q_g}{q_o}$ and should be used if FLO is set to 'OIL' or 'LIQ' 2) GLR: for the gas fraction being the gas-liquid ratio $\frac{q_g}{q_o + q_w}$ and should be used if FLO is set to 'OIL' or 'LIQ' 3) OGR: for the gas fraction being the oil-gas ratio $\frac{q_o}{q_g}$ and should be used if FLO is set to 'GAS'.			None
I-6	VFPTYPE	A defined character string that should be defaulted or set equal to THP.			THP

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-7	ALQ	A defined character string that defines the artificial lift quantity and should be set to one of the following character strings: 1) GRAT: for the artificial lift quantity being the gas lift gas injection rate. 2) IGLR: for the artificial lift quantity being the gas lift gas, injection gas-liquid ratio. 3) TGLR: for the artificial lift quantity being the gas lift gas, injection total gas-liquid ratio. 4) COMP: for the artificial lift quantity being the compressor power, for a compressor. 5) PUMP: for the artificial lift quantity being the pump rating for a pump. 6) DENO: for oil surface density. 7) DENG: for gas surface density. 8) BEAN: for multi-segment wells choke parameter. 9) " ": for undefined, that is for no ALQ data. The ALQ parameter is just another variable used to interpolate the outflow pressure based on the inlet pressure, together with the phases flowing through the system. As such, ALQ can represent any parameter; however, the units should be consistent with that used to set the value. For example, if pump speed (Hz) is used to set a well's artificial lift quantity via the WCONPROD(ALQ-WELL) parameter in the SCHEDULE section, then the ALQ-DATA should represent pump speed in Hz. The default value of I* is " " or undefined, that covers the case when the ALQ variable is not entered, except for when gas lift is employed in the model. When gas lift is active then the default value for ALQ is set to GRAT provided: 1) Gas lift optimization is active via the LIFTOPT keyword, or 2) The GLIFTLIM keyword in the SCHEDULE section is used to constrain the total amount of gas lift used within a group, or 3) Gas lift flows are included in the network pressure loss calculations as determined by the GRUPNET(OPTION2), or the NODEPROP(GASLIFT) parameters in the SCHEDULE section. In addition, if any of the above is true then ALQ must be set to GRAT.			I*
I-8	VFPUNITS	Units used for the VFP-DATA on this keyword. This variable is ignored by OPM Flow and should be defaulted with I*.			I*
		FIELD	METRIC	LAB	
I-9	VFPVALUE	A defined character string that defines the type of data in the VFP-DATA vector. This should be set equal to BHP if the data contains the bottom-hole pressure data, or TEMP if the data is the Tubing Head Temperature (THT) data. This variable is ignored by OPM Flow and should be defaulted with I*.			BHP
I-10	/	Record terminated by a "/"			Not Applicable

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-1	FLO-DATA	A real positive monotonically increasing vector that defines the numerical values of the flowing phase declared by the FLO variable. The number of entries must greater than two and less than or equal to MXMFLO as defined on the VFPPDIMS keyword in the RUNSPEC section.			None
		Liquid: stb Gas: Mscf	Liquid: sm ³ Gas: sm ³	Liquid: scc Gas: scc	
2-2	/	Record terminated by a "/"			Not Applicable
3-1	THP-DATA	A real positive monotonically increasing vector that defines the numerical values of the tubing head pressure values. The number of entries must greater than two and less than or equal to MXMTHP as defined on the VFPPDIMS keyword in the RUNSPEC section.			None
		psia	barsa	atma	
3-2	/	Record terminated by a "/"			Not Applicable
4-1	WFR-DATA	A real positive monotonically increasing vector that defines the numerical values of the flowing water fraction declared by the WFR variable. The number of entries must greater than two and less than or equal to MXMWFR as defined on the VFPPDIMS keyword in the RUNSPEC section.			None
		WOR: dimensionless WCT: dimensionless WGR: stb/Mscf	dimensionless dimensionless dimensionless	dimensionless dimensionless dimensionless	
4-2	/	Record terminated by a "/"			Not Applicable
5-1	GFR-DATA	A real positive monotonically increasing vector that defines the numerical values of the flowing gas fraction declared by the GFR variable. The number of entries must greater than two and less than or equal to MXMGFR as defined on the VFPPDIMS keyword in the RUNSPEC section.			None
		GOR: Mscf/stb GLR: Mscf/stb OGR: stb/Mscf	dimensionless dimensionless dimensionless	dimensionless dimensionless dimensionless	
5-2	/	Record terminated by a "/"			Not Applicable

No.	Name	Description			Default
		Field	Metric	Laboratory	
6-1	ALQ-DATA	A real positive monotonically increasing vector that defines the numerical values of the artificial lift quantity declared by the ALQ variable. The number of entries must greater than two and less than or equal to MXMALQ as defined on the VFPPDIMS keyword in the RUNSPEC section.			None
		GRAT: Mscf/d IGLR: Mscf/stb TGLR: Mscf/d DENO: lb/ft ³ DENG: lb/ft	sm ³ /day dimensionless dimensionless kg/m ³ kg/m ³	scc/hour dimensionless dimensionless gm/cc gm/cc	
6-2	/	Record terminated by a "/"			Not Applicable
7-1	NTHP	This data record consists of a series of integer values that defines the index of THP, WFR, GFR, ALQ entered via the those records on this keyword. The first index, NTHP, is an integer value that defines the index of THP values entered via the THP-DATA records on this keyword. For example, if THP-DATA is equal to 100, 200, 300 and 350 and NTHP is equal to three then NTHP refers to third entry, that is THP equal to 300.			None
	NWFR	The second index, NWFR, is an integer value that defines the index of the water fraction values entered via the WFR-DATA records on this keyword. For example, if WFR-DATA is equal to 0.00, 0.25, 0.50 and 0.75 and NWFR is equal to two then NWFR refers to second entry, that is WFR equal to 0.25.			None
	NGFR	The third index, NGFR, is an integer value that defines the index of the gas fraction values entered via the GFR-DATA records on this keyword. For example, if GFR-DATA is equal to 100.0, 200.0, 500.0 and 750.0 and NGFR is equal to three then NGFR refers to third entry, that is GFR equal to 500.0.			None
	NALQ	The fourth and final index entry, NALQ, is an integer value that defines the index of artificial lift values via the ALQ-DATA records on this keyword. For example, if ALQ-DATA is equal to 50, 100, 200 and 300 and NALQ is equal to one then NALQ refers to first entry, that is ALQ equal to 50. The fourth index is then followed by the VFP-DATA, containing either the BHP or THT vector values.			None
	VFP-DATA	VFP-DATA is a real vector of either BHP or THT values for each FLO production rate for the corresponding index value (NTHP, NWFR, NGFR, NALQ) and is then terminated with a"/". The (7-1) record, which consists of the four indices and BHP (or THT) data, is then repeated until all combinations of (NTHP, NWFR, NGFR, NALQ) and the associate BHP (or THT) data has been entered. Note that the VFPVALUE parameter, determines the type of VFP-DATA, that is BHP for bottom-hole pressure, or THT for tubing head temperature.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
		psia	barsa	atma	
7-2	/	Each Index (NTHP, NWFR, NGFR, NALQ.VFP-DATA) data set is terminated by a “/”			Not Applicable

Notes:

- 1) Each *VFPPROD* table must be entered with a separate *VFPPROD* keyword that consists of seven records with entries 1-1 to 1-10 representing record one items and 2-1 to 2-2 representing record number two items, etc., in the “No.” column in this table.
- 2) Each of the records are terminated by a “/” and is explicitly shown in the above rows.
- 3) There is no keyword terminating “/”.

Table 12.82:VFPPROD Keyword Description

The data for this keyword is generated by an external program and is normally included into the input deck using the [INCLUDE](#) keyword as described in section [CHAPTER 4: GLOBAL SECTION KEYWORDS](#), as the data can be quite voluminous.

Note that for equivalent two phase runs:

- For example oil-water runs with only the [OIL](#) and [WATER](#) keywords in the [RUNSPEC](#) section, or runs that model dead oil³²⁹ with a constant solution gas-oil ratio value defined by the [RSCONST](#) keyword in the [PROPS](#) section, then the FLO parameter in Table 12.82 must be set to either [OIL](#) or LIQ, WFR to either WCT or WOR, and GFR to GOR.
- Gas-water models with only the [GAS](#) and [WATER](#) keywords in the [RUNSPEC](#) section, or models that only have dry gas³³⁰ with a constant condensate-gas ratio defined by the [RVCONST](#) keyword in the [PROPS](#) section, then the FLO parameter in Table 12.82 must be set to [GAS](#), WFR to WGR, and GFR to OGR.

Note

It is possible to have only the [OIL](#) and [WATER](#) keywords in the [RUNSPEC](#) section and to use gas lift for the wells, without declaring the [GAS](#) phase in the [RUNSPEC](#) section.

In this case, the FLO parameter in Table 12.82 must be set to either [OIL](#) or LIQ, WFR to either WCT or WOR, and GFR to GOR. In this case the ALQ parameter is optional, but if present must set to GRAT. If the ALQ and ALQ-DATA parameters are absent then the GFR-DATA will be used based on the flowing GOR plus the stipulated gas lift gas.

See also the [WELSPECS](#) keyword to define wells and the [WCONPROD](#) keyword that is used to allocate the [VFPPROD](#) tables to specific wells. Note that one [VFPPROD](#) table can be allocated to one or more wells, provided the wells in question have a similar trajectory and similar flow characteristics, for example vertical oil wells producing from the same reservoir, or different reservoirs with similar PVT properties.

³²⁹ "Dead" oil is oil that it contains no dissolved gas or a relatively thick oil or residue that has lost its volatile components.

³³⁰ Natural gas that occurs in the absence of condensate or liquid hydrocarbons, or gas that had condensable hydrocarbons removed, is called dry gas. It is primarily methane with some intermediates. The hydrocarbon mixture is solely gas in the reservoir and there is no liquid (condensate surface liquid) formed either in the reservoir or at surface. The term dry indicates that the gas does not contain heavier hydrocarbons to form liquids at the surface conditions. Dry gas typically has GOR's greater than 100,000 scf/stb or 18,000 Sm³/m³.

The [VFPINJ](#) keyword is used to enter VFP tables for injection wells or to describe the pressure relationship when the network option is being used. In this case the table describes the pipeline pressure behavior from the HIGHER group (inlet node) to the LOWER group (outlet node) given the current injection conditions.

All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example shows the [VFPPROD](#) table for a production gas well and is taken from the Norne OPM Flow model. Here WFR has been set to water-gas ratio and GFR has been set to the oil-gas ratio, and the ALQ value is defaulted.

```

VFPPROD

-- Table      Datum Depth      Rate Type      WFR Type      GFR Type
-----
--           5              2623.39          'GAS'          'WGR'          'OGR' /

-- 'GAS' units - SM3/DAY
50000.0 100000.0 200000.0 400000.0 800000.0
1200000.0 1600000.0 1999999.9 3000000.0 3999999.8
5000000.5 /

-- 'THP' units - BARSA
10.00 20.00 40.00 80.00 120.00
150.00 200.00 250.00 /

-- 'WGR' units - SM3/SM3
0 1e-9 1e-6 1e-5 0.0001
0.001 0.01 0.1 /

-- 'OGR' units - SM3/SM3
1e-7 1e-6 1e-5 0.0001 0.001
0.01 /

-- 'ALQ' units -
0 /

1 1 1 1 11.93 12.22 13.35 17.24 27.93
39.83 52.06 64.38 95.20 125.89
156.52
/
1 1 2 1 11.93 12.22 13.35 17.24 27.94
39.84 52.07 64.39 95.21 125.91
156.55
/

.....
.....

8 8 5 1 483.75 511.15 614.09 1044.78 2757.56
5592.55 9528.36 14567.24 32005.79 56375.24
87684
/
8 8 6 1 487.68 516.24 624.74 1075.40 2860.16
5803.92 9880.58 15093.76 33119.59 58297.57
90639
/

```

The example shows the first two and the last two records of type seven, as the data is too voluminous to be included.

The next example below shows an example oil producing well [VFPPROD](#), again taken from Norne OPM Flow model. Here WFR has been set to water cut and GFR has been set to the gas-oil ratio, and the ALQ value is defaulted.

VFPPROD

```
-- Table      Datum Depth      Rate Type      WFR Type      GFR Type      TAB Type
-- -----
--      37      2641.02      'LIQ'      'WCT'      'GOR'      /
```

-- Prosper files are corrected from RKB to MSL depth. lmarr

```
-- Table      Datum Depth      Rate Type      WFR Type      GFR Type      TAB Type
-- -----
--      37      2617.02      'LIQ'      'WCT'      'GOR'      /
```

```
-- 'LIQ' units - SM3/DAY
200.0  500.0  1000.0  1500.0  2000.0
2500.0 3000.0 3500.0 4000.0 4500.0
5000.0 5500.0 6000.0 6500.0 7000.0
7500.0 8000.0 10000.0 14000.0 /
```

```
-- 'THP' units - BARSA
21.01  51.01  61.01  81.01  101.01
121.01 141.01 161.01 181.01 201.01 /
```

```
-- 'WCT' units - FRACTION
0      0.1      0.2      0.3      0.4
0.5    0.6      0.7      0.8      1 /
```

```
-- 'GOR' units - SM3/SM3
90      100      150      200      500
1000    2000 /
```

```
-- 'ALQ' units -
0 /
```

```
1  1  1  1  160.82  136.70  119.79  115.86  117.38
      121.16  126.08  131.56  137.48  143.74
      150.29  157.07  164.02  171.07  178.13
      185.11  192.09  220.38  280.86
```

/

```
1  1  2  1  155.63  129.40  112.32  108.64  110.44
      114.74  120.15  126.09  132.47  139.05
      146.02  153.41  160.67  167.91  175.13
      182.34  189.55  218.81  281.02
```

/

```
10 10  6  1  439.30  437.95  437.53  437.79  438.39
      439.26  440.36  441.67  443.19  444.92
      446.85  448.99  451.32  453.85  456.58
      459.51  462.64  477.11  515.47
```

/

```
10 10  7  1  439.30  437.95  437.53  437.79  438.39
      439.26  440.36  441.67  443.19  444.92
      446.85  448.99  451.32  453.85  456.58
      459.51  462.64  477.11  515.47
```

/

The example shows the first two and the last two records of type seven, as the data is too voluminous to be included.

12.3.243 VFPTABL – DEFINE PRODUCTION VERTICAL FLOW PERFORMANCE ALQ INTERPOLATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [VFPTABL](#) keyword defines the interpolation method for production Vertical Flow Performance (“VFP”) tables for the Artificial Lift Quantity (“ALQ”). Production VFP data is entered via the [VFPPROD](#) keyword in the [SCHEDULE](#) section. By default the simulator interpolates all the variables in the VFP tables using linear interpolation, including the ALQ quantity. However, if the ALQ values represent gas lift, then linear interpolation may not be insufficient, as the gradient change between the tabulated ALQ values may result in sudden changes. This is particularly important in gas lift optimization studies where the available gas lift gas is being allocated to a group of wells in order to maximize oil production rates. To overcome this issue the [VFPTABL](#) keyword allows the ALQ values to be interpolated using cubic spline interpolation, and results in a smoother transition between the various ALQ entries.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	VFPTABL	VFPTABL is a defined positive integer that specifies the interpolation method to be used with the ALQ quantity in the VFP production tables, and should be set to one of the following: 1) Apply linear interpolation to all VFPPROD variables. 2) Apply linear interpolation to all VFPPROD variables, except for the ALQ variable, for which cubic spline interpolation should be used. If the keyword is absent from the input deck then linear interpolation will be used for all variables.			I
		dimensionless	dimensionless	dimensionless	

Notes: 1) The keyword is followed by a vector of numbers separated by a space and the keyword is terminated by a “/”.					
--	--	--	--	--	--

Table I 2.83:VFPTABL Keyword Description

Example

The example sets cubic spline interpolation for the ALQ quantity in the [VFPPROD](#) tables, with linear interpolation used for all the variables.

```
--
--      ALQ INTERPOLATION OPTION
--
--      OPTION
VFPTABL
      2
/
```

12.3.244 WAITBAL – WAIT ON NETWORK BALANCE BEFORE ALLOWING FURTHER ACTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword sets the network balance option for all networks when networks are active in the model. Basically, the keyword either activates the [PRORDER](#) and [GDRILPOT](#) stipulated actions before or after the network has been balanced

The network option is normally used to ensure that the tubing head pressure (“THP”) of a group of wells flowing into a common network node is consistent with a group’s flow rates, that is each well’s THP is flowing at the same THP and at the same time satisfying well and group targets and constraints. This is accomplished by calculating the well THP limits dynamically by balancing the flow rates and pressure losses in the network.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.245 WALKALIN – DEFINE WATER INJECTION ALKALINE CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WAKALIN keyword defines the water injection alkaline concentration for water injection wells for when the surfactant and/or polymer models have been activated by the [SURFACT](#), [SURFACTW](#), or the [POLYMER](#) keywords in the [RUNSPEC](#) section, combined with the [ALKALINE](#) keyword which is also in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.246 WALQCALC – DEFINE WELL VFP SURFACE ALQ PHASE DENSITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [WALQCALC](#) keyword defines the well VFP surface ALQ phase density use in the VFP table lookup and interpolation to be gas surface density, oil surface density, or neither. Note that the user should ensure that generated VFP tables have been generated consistent with the setting on this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.247 WAPI – DEFINE OIL WELL INJECTION API GRAVITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines an oil injection well's [API](#) gravity for when [API](#) tracking has been made active via the [API](#) keyword in the [RUNSPEC](#) section. The American Petroleum Institute ([API](#)) classifies oils based on an [API](#) gravity (γ_{API}), or degrees [API](#) (oAPI), the relationship between relative density (γ_o) of oil and [API](#) gravity (γ_{API}) is given by:

$$\gamma_{API} = \frac{141.5}{\gamma_o} - 131.5$$

(12.30)

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.248 WARN – ACTIVATE WARNING MESSAGES

Turns on warning messages to be printed to the print file (*.PRT); note that this keyword is activated by default and can subsequently be switched off by the [NOWARN](#) activation keyword. The warning messages may be turned on and off using keywords [WARN](#) and [NOWARN](#). OPM Flow always prints error messages.

See [WARN – Activate Warning Messages](#) in the GLOBAL section for a full description.

12.3.249 WBHGLR – DEFINE WELL BOTTOM-HOLE GLR CONSTRAINT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [WBHGLR](#), defines a well’s bottom-hole Gas Liquid Ratio (“GLR”) constraint, where the GLR is the is the ratio of the “free” gas rate and liquid rate at bottom-hole conditions. The reference depth for bottom-hole conditions is given by the BHPREF variable on the [WELSPECS](#) keyword in the [SCHEDULE](#) section.

Normally this type of well control is applied to pumping wells to avoid the well “pumping off”, that is when the liquid column above the pump is low, resulting in an increase in gas intake and an associated loss in pump efficiency.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.250 WBOREVOL – DEFINE EFFECTIVE WELLBORE STORAGE VOLUME

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WBOREVOL](#) defines a well’s effective wellbore storage volume. The primary purpose of the keyword is to enable matching of the wellbore storage effects in well tests and the corresponding pressure response observed in the test. Normally, as part of well test interpretation, the pressure, permeability, effective wellbore storage, etc., are derived from the analytical interpretation of the test. This keyword therefore allows the engineer to enter the analytical derived effective wellbore storage.

Wellbore storage, in terms of well testing, is an important variable when the well is shut-in at the surface, as the well continues to flow down-hole until the fluids obtain equilibrium. Most well tests are now conducted using specialized tools that shut-in the well down-hole, thus eliminating, or mostly eliminating, wellbore storage effects.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.251
WCALCVAL – DEFINE GAS WELL CALORIFIC VALUE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword defines a gas well’s calorific value for when the Gas Calorific Value option has been activated by specifying a target calorific value for a group via the [GCONCAL](#) keyword in the [SCHEDULE](#) section. If this option is invoked then the gas calorific value must be set either by this keyword for a well by well allocation of the calorific value, or by using the Tracer Tracking option (activated by the [TRACER](#) keyword in the [RUNSPEC](#) section) combined with [CALTRAC](#) keyword in the [SCHEDULE](#) section that defines the tracer for the calorific value.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.252 WCONHIST – DEFINE WELL HISTORICAL PRODUCTION RATES AND PRESSURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WCONHIST](#) keyword defines production rates and pressures for wells that have been declared history matching wells by the use of this keyword. History matching wells are handled differently than ordinary wells that use the [WCONPROD](#) keyword for controlling their production targets and constraints. However, the wells still need to be defined like ordinary production wells using the [WELSPECS](#) keyword in the [SCHEDULE](#) section.

Note that although wells can be allocated to a group when they are specified by the [WELSPECS](#) keyword, history matching wells cannot operate under group control. Field and group reporting is still consistent for all wells allocated to a group.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the wells observed production rates and pressures are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	STATUS	A defined character string that declares the status of the well. STATUS should be set to one of the following character strings: 1) OPEN: the well is open to flow and will attempt to produce the required production volumes. 2) STOP: the well is “stopped” at the surface and will not produce any fluids to surface; however, if there any open connections then flow may occur within the wellbore and between the open connections depending on a connection’s potential with respect to all the other connections. Inter-connection flow (cross flow) can be prevented by setting the XFLOW variable on the WELSPECS keyword to NO. In this case the well’s behavior will be similar to the SHUT option described below. 3) SHUT: the well is shut at the surface and downhole, this results in no flow at the surface and no cross flow downhole. Note a well’s STATUS should always be set either STOP or SHUT if the well’s production is to be set to zero. Just setting a well’s production rate to zero means that the well is open to flow with a zero rate.			OPEN

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	TARGET	A defined character string that sets the observed target production phase for the well, all the other phases are calculated unconstrained and used for reporting only. The simulator will attempt to meet the TARGET based on the phase rate stated in items (4) to (6) and (10) on this keyword. TARGET should be set to one of the following character strings: 1) ORAT: the target is set to the surface oil production rate as defined by item (4). 2) WRAT: the target is set to the surface water production rate as defined by item (5). 3) GRAT: the target is set to the surface gas production rate as defined by item (6). 4) LRAT: the target is set to the surface liquid (oil plus water) production rate and is calculated by the simulator using (4) and (5). 5) RESV: the target is set to the in situ reservoir volume rate and is calculated by the simulator using items (4), (5) and (6). 6) BHP: the target rate is set to the bottom-hole pressure as defined by item (10). Note the TARGET control mode may be reset using the WHISTCTL keyword in the SCHEDULE section, from the time the WHISTCTL is invoked, thus avoiding changing the control mode on all subsequent WCONHIST keywords.			None
4	ORAT	A real positive value that defines the observed surface oil production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			Defined
		stb/d 0.0	sm ³ /day 0.0	scc/hour 0.0	
5	WRAT	A real positive value that defines the observed surface water production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			Defined
		stb/d 0.0	sm ³ /day 0.0	scc/hour 0.0	
6	GRAT	A real positive value that defines the observed surface gas production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			Defined
		Mscf/d 0.0	sm ³ /day 0.0	scc/hour 0.0	

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	VFPTAB	A positive integer greater than or equal to zero that defines the vertical lift performance tables to be used for calculating the tubing head pressure for the well. If a non-zero value is entered then the vertical lift performance tables must be entered via the <i>VFPPROD</i> keyword in the <i>SCHEDULE</i> section and allocated to the well via this item. The default value of zero implies no vertical lift performance table initially. If this value is then reset to be greater than zero then the table will be used to calculate the well's tubing head pressure. Subsequently, the default is to use the previously declared table number.			0
8	ALQ-WELL	A real positive value that defines the artificial lift quantity to be used in conjunction with the <i>VFPPROD</i> assigned to the well via that keyword's VFPTAB variable. This value may be specified using a User Defined Argument (UDA). VFPTAB vertical lift performance table and the artificial lift quantity ALQ-WELL are used with the well fluid rates to calculate the well's tubing head pressures values from the bottom-hole pressure. Note that the units for ALQ-WELL is dependent on the associated variable on the <i>VFPPROD</i> keyword.			None
9	THP	A real positive value that defines the observed tubing head pressure. This value may be specified using a User Defined Argument (UDA). This parameter is only used for comparing the actual tubing head pressure given here with those calculated by the simulator; that is history matching wells can only be controlled by either the surface injection rate or their bottom-hole pressure.			Defined
		psia 0.0	barsa 0.0	atma 0.0	
10	BHP	A real positive value that defines the observed bottom-hole pressure. This value may be specified using a User Defined Argument (UDA).			Defined
		psia 0.0	barsa 0.0	atma 0.0	
11	WGRA	A real positive value that defines the observed wet gas rate in the commercial compositional simulator. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
12	NGL	A real positive value that defines the observed Natural Gas Liquid ("NGL") rate in the commercial compositional simulator. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0

Notes:

1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the *SCHEDULE* section. User Defined Arguments are activated using the *UDADIMS* keyword in the *RUNSPEC* section, and defined by the *UDQ* keyword in the *SCHEDULE* section.

2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.84:WCONHIST Keyword Description

See also the [WHISTCTL](#) keyword that can be used to reset the TARGET phase, the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection targets and constraints, and the [WCONPROD](#) keyword to define a production well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Note

One can use TARGET set to RESV in the initial history matching runs to get a “reasonable” pressure match, this ensures that the total reservoir withdrawals are correct, although the individual phase withdrawals will not match. Once a reasonable pressure match is achieved for the reservoir then one can reset TARGET to the sales phase, [OIL](#) or [GAS](#), and continue with the matching of all the phases.

In oil reservoirs some engineers prefer to use LIQ rather than [OIL](#) as the TARGET phase, although one should consider that as the water phase has no commercial value, the measurement accuracy is significantly less than the oil sales phase.

History matching wells are converted to ordinary wells by restating a well's control mode using either the [WCONPROD](#) or [WELTARG](#) keywords in the [SCHEDULE](#) section.

Examples

The following example below shows the observed production rates for the OP01 oil producer for the first quarter of 2000.

```

-----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-----
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.5E3  100.0  1550   10      1*    900.0  1*    /
/
DATES
01 FEB 2000 /
/
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.2E3  150.0  1520   1*      1*    875.0  3250.0 /
/
DATES
01 MAR 2000 /
/
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.0E3  200.0  1500   1*      1*    850.0  1*    /
/

```

From January 1, 2000 well OP01 is open and is on oil rate control, and produces 15,500 stb/d oil, with the observed rates of 100 stb/d of water and 15.5 MMscf/d of gas. The well uses [VFPPROD](#) vertical lift table

number 10 so that OPM Flow can calculate the tubing head pressures based on the fluids produced and the calculated pressures in the simulator.

The next example illustrates how to convert OP01 from a history match well to a normal production well at the start for the forecast run at August 1, 2017 using the [WELTARG](#) keyword.

DATES

01 AUG 2017 /

/

--

WELL PRODUCTION AND INJECTION TARGETS

--

-- WELL WELL TARGET

-- NAME TARG VALUE

WELTARG

OP01 THP 1*

/

/

Here by defaulting the bottom-hole pressure via 1* OPM Flow automatically applies the last bottom-hole pressure from the previous time step as the “constraining phase” together with the last historical rates as constraints. This ensures a smooth transition between history and prediction without having to resort to unreasonable changes to the model. This option is currently not implemented in OPM Flow but is expected to be incorporated in a future release.

12.3.253 WCONINJ – WELL INJECTION TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WCONINJ](#) is a legacy keyword that is no longer used in the commercial simulator and is not supported by OPM Flow. Instead well injection targets and constraints should be defined using the [WCONINJE](#) keyword in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.254 WCONINJE – WELL INJECTION TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WCONINJE](#) keyword defines injection targets and constraints for wells that have previously been defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Note that wells can be allocated to a group when they are specified by the [WELSPECS](#) keyword. Wells defined to be under group control will have their injection rates controlled by the group to which they belong, in addition to any well constraints defined for the wells using this keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well injection targets and constraints data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	TYPE	A defined character string that defines the type of injection well. TYPE should be set to one of the following character strings: 1) GAS: for a gas injection well. 2) OIL: for an oil injection well. 3) WAT: for a water injection well.			None
3	STATUS	A defined character string that declares the status of the well. STATUS should be set to one of the following character strings: 1) OPEN: the well is open for injection and will attempt to inject the required injection volumes. 2) STOP: the well is “stopped” at the surface and will not inject any fluids; however, if there are any open connections then flow may occur within the wellbore and between the open connections depending on a connection’s potential with respect to all the other connections. Inter-connection flow (cross flow) can be prevented by setting the XFLOW variable on the WELSPECS keyword to NO. In this case the well’s behavior will be similar to the SHUT option described below. 3) SHUT: the well is shut at the surface and downhole, this results in no injection and no cross flow downhole. 4) AUTO: the well is initially SHUT, but may be opened automatically if an economic limit is violated. This option is currently not supported by OPM Flow. Note a well’s STATUS should always be set either STOP or SHUT if the well’s injection is to be set to zero. Just setting a well’s injection rate to zero means that the well is open for injection with a zero rate, this will cause numerical issues especially for wells under THP control.			OPEN

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	TARGET	A defined character string that sets the target injection control mode for the well. TARGET should be set to one of the following character strings: 1) RATE: the injection phase will be controlled by the surface fluid rate for the given well type as defined by the TYPE variable. For example, if TYPE has been set to WAT then this would mean the surface water injection rate as defined by item (5). 2) RESV: the injection phase will be controlled by the in situ reservoir volume fluid rate for the given well type as defined by the TYPE variable. For example, if TYPE has been set to GAS then this would mean the gas reservoir volume injection rate as defined by item (6). 3) BHP: the target rate is set according to the bottom-hole pressure as defined by item (7). 4) THP: the target rate is set according to the tubing head pressure as defined by item (8). If this option is selected then the vertical lift performance tables must be entered via the VFPINJ keyword in the SCHEDULE section and allocated to the well via item (9). 5) GRUP: the well is under group control and injects its share of the group's target as set using the GCONINJE keyword in the SCHEDULE section.			None
5	RATE	A real positive value that defines the maximum surface injection rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		Liquid stb/d Gas Mscf/d	Liquid sm ³ /day Gas sm ³ /day	Liquid scc/hour Gas scc/hour	
6	RESV	A real positive value that defines the maximum reservoir volume injection rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		rb/d	rm ³ /day	rcc/hour	
7	BHP	A real positive value that defines the maximum bottom-hole pressure target or constraint. This value may be specified using a User Defined Argument (UDA). Note the default value basically means unlimited injection or no constraint and should therefore be avoided as the BHP will result in unrealistic well potentials as well as optimistic injection forecasts for the well.			Defined
		psia 10,000	barsa 6,895	atma 6,803	
8	THP	A real positive value that defines the maximum tubing head pressure target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		psia	barsa	atma	

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	VFPTAB	A positive integer greater than or equal to zero that defines the vertical lift performance tables to be used for calculating the tubing head pressure for the well. If a non-zero value is entered then the vertical lift performance tables must be entered via the <i>VFPINJ</i> keyword in the <i>SCHEDULE</i> section and allocated to the well via this item. The default value of zero implies no vertical lift performance tables and in this case TARGET cannot be set to THP and in addition item (10) should be defaulted or set to zero.			0
10	RSRVINJ	The dissolved gas-oil ratio in the injected oil, or the vaporized oil-gas ratio in the injected gas.			0.0
		Gas Injection: stb/Mscf Oil Injection: Mscf/stb	Gas Injection: sm ³ /sm ³ Oil Injection: sm ³ /sm ³	Gas Injection: scc/scc Oil Injection: scc/scc	
11	RSSTEAM	Thermal/Temperature gas-steam ratio for steam-gas injectors. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
12	OILFRAC	Surface oil fraction in a multi-phase injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
13	WATFRAC	Surface water fraction in a multi-phase injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
14	GASFRAC	Surface gas fraction in a multi-phase injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
15	OILSTEAM	Surface oil volume to steam volume ratio in a steam-oil injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
Notes: 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the <i>SCHEDULE</i> section. User Defined Arguments are activated using the <i>UDADIMS</i> keyword in the <i>RUNSPEC</i> section, and defined by the <i>UDQ</i> keyword in the <i>SCHEDULE</i> section. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table I 2.85:WCONINJE Keyword Description

See also the [GCONPROD](#) and the [GCONINJE](#) keywords to define a group's production and injection targets and constraints, and the [WCONPROD](#) keyword to define a production well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the injection targets and constraints for one gas injection well and one water injection well as follows:

```
--
--      WELL INJECTION CONTROLS
--
-- WELL  FLUID  OPEN/  CNTL  SURF  RESV  BHP  THP  VFP
-- NAME  TYPE   SHUT   MODE  RATE  RATE  PRSES PRES TABLE
WCONINJE
GI01     GAS    OPEN   GRUP  50E3  1*    1*    1*    1*  /
WI01     WAT    OPEN   RATE  25E3  1*    5000.  1*    1*  /
/
```

Well GI01 is a gas injection well directly under group control constrained by a maximum surface gas injection rate of 50 MMscf/d and well WI01 is an open water injection well with a surface water injection rate target of 25,000 stb/d, subject to a maximum bottom-hole pressure constraint 5,000 psia.

12.3.255 WCONINJH – WELL HISTORICAL OBSERVED INJECTION RATES AND PRESSURES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WCONINJH](#) keyword defines injection rates and pressures for wells that have been declared history matching wells by the use of this keyword. History matching wells are handled differently than ordinary wells that use the [WCONINJE](#) keyword for controlling their injection targets and constraints. However, the wells still need to be defined like ordinary injection wells using the [WELSPECS](#) keyword in the [SCHEDULE](#) section.

Note that although wells can be allocated to a group when they are specified by the [WELSPECS](#) keyword, history matching wells cannot operate under group control. Field and group reporting is still consistent for all wells allocated to a group, but history matching wells cannot be under group control.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the wells observed injection rates and pressures are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	TYPE	A defined character string that defines the type of injection well. TYPE should be set to one of the following character strings: 1) GAS : for a gas injection well. 2) OIL : for an oil injection well. 3) WAT : for a water injection well.			None
3	STATUS	A defined character string that declares the status of the well. STATUS should be set to one of the following character strings: 1) OPEN : the well is open for injection and will attempt to inject the observed injection volumes. 2) STOP : the well is “stopped” at the surface and will not inject fluids; however, if there any open connections then flow may occur within the wellbore and between the open connections depending on a connection’s potential with respect to all the other connections. Inter-connection flow (cross flow) can be prevented by setting the XFLOW variable on the WELSPECS keyword to NO. In this case the well’s behavior will be similar to the SHUT option described below. 3) SHUT : the well is shut at the surface and downhole, this results in no injection and no cross flow downhole. Note a well’s STATUS should always be set either STOP or SHUT if the well’s injection is to be set to zero. Just setting a well’s injection rate to zero means that the well is open to flow with a zero injection rate, this may cause numerical issues.			OPEN

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	RATE	A real positive value that defines the observed surface injection rate.			0.0
		Liquid stb/d Gas Mscf/d	Liquid sm ³ /day Gas sm ³ /day	Liquid scc/hour Gas scc/hour	
5	BHP	A real positive value that defines the observed bottom-hole pressure.			0.0
		psia	barsa	atma	
6	THP	A real positive value that defines the observed tubing head pressure. This parameter is only used for comparing the actual tubing head pressure given here with those calculated by the simulator; that is history matching wells can only be controlled by either the surface injection rate or their bottom-hole pressure.			None
		psia	barsa	atma	
7	VFPTAB	A positive integer greater than or equal to zero that defines the vertical lift performance tables to be used for calculating the tubing head pressure for the well. If a non-zero value is entered then the vertical lift performance tables must be entered via the VFPINJ keyword in the SCHEDULE section and allocated to the well via this item. The default value of zero implies no vertical lift performance table initially. If this value is then reset to be greater than zero then the table will be used to calculate the well's tubing head pressure. Subsequently, the default is to use the previously declared table number.			0
8	RSRVINJ	Dissolved gas fraction in injected oil or vaporized oil fraction in injected gas. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
9	OILFRAC	Surface oil fraction in a multi-phase injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
10	WATFRAC	Surface water fraction in a multi-phase injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
11	GASFRAC	Surface gas fraction in a multi-phase injector. The parameter is ignored by OPM Flow and should be defaulted or set to the default value of zero.			0.0
12	TARGET	A defined character string that sets the target injection control mode for the well. TARGET should be set to one of the following character strings: 1) RATE: the injection well will be controlled by the surface injection rate for the given well type as defined by the TYPE variable. For example, if TYPE has been set to WAT then this would mean the surface water injection rate as defined by item (4). 2) BHP: the injection well will be controlled by the bottom-hole pressure as defined by item (5).			RATE

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.86:WCONINJH Keyword Description

This keyword should be repeated at various time steps to fully describe the historical injection performance of the wells. For example, as most production and injection data is reconciled on a monthly basis, then monthly time steps covering the injection history of the wells should be used with [WCONINJH](#) keyword entered on a monthly basis.

History matching well are converted to ordinary wells by restating a well’s control mode using either the [WCONINJE](#) or [WELTARG](#) keywords in the [SCHEDULE](#) section.

Example

The following example below shows the observed gas rates for the GI01 gas injector for the first quarter of 2000.

```

-- -----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-- -----
--
--          WELL HISTORICAL INJECTION CONTROLS
--
-- -- WELL  FLUID  OPEN/  SURF  BHP   THP   VFP   NOT   CNTL
-- -- NAME  TYPE   SHUT   RATE  PRES  PRES  TABLE USED  MODE
WCONINJH
GI01      GAS    OPEN   15.5E3  1*    5462   12    4*    1*   /
/
DATES
01 FEB 2000 /
/
--
--          WELL HISTORICAL INJECTION CONTROLS
--
-- -- WELL  FLUID  OPEN/  SURF  BHP   THP   VFP   NOT   CNTL
-- -- NAME  TYPE   SHUT   RATE  PRES  PRES  TABLE USED  MODE
WCONINJH
GI01      GAS    OPEN   15.9E3  1*    5468   1*    4*    1*   /
/
DATES
01 MAR 2000 /
/

```

```
--
--      WELL HISTORICAL INJECTION CONTROLS
--
-- WELL  FLUID  OPEN/  SURF  BHP   THP   VFP   NOT   CNTL
-- NAME  TYPE   SHUT   RATE  PRES  PRES  TABLE USED  MODE
WCONINJH
GI01     GAS    OPEN  17.2E3  1*    5489   1*    4*    1*   /
/
```

Well GI01 is declared as a gas injection well under gas rate control as TARGET variable is defaulted to rate control by using 1* (the last entry on the record). In addition, the well uses vertical lift table [VFPINJ](#) number 12 (as shown at January 1, 2000) to calculate the tubing head pressures for the well. Note that it is not necessary to declare the [VFPINJ](#) table number if it remains the same for subsequent time steps and thus the default 1* is used to indicate the last entry should be used.

12.3.256 WCONINJP – DEFINE WELL INJECTION TARGETS AND CONSTRAINTS FOR PATTERN FLOOD WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WCONINJP](#), defines well injection targets and constraints for pattern flood wells. The keyword is similar to the [WCONINJE](#) keyword in the [SCHEDULE](#) section except that the injection control is applied to a group of wells defined by the first record of this keyword, combined with a second record that defines the wells in the pattern and their contribution to the pattern.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.257 WCONPROD – DEFINE WELL PRODUCTION TARGETS AND CONSTRAINTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	-----------------

Description

The [WCONPROD](#) keyword defines production targets and constraints for wells that have previously been defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Note that wells can be allocated to a group when they are specified by the [WELSPECS](#) keyword. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells using this keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well production targets and constraints data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	STATUS	A defined character string that declares the status of the well. STATUS should be set to one of the following character strings: 1) OPEN: the well is open to flow and will attempt to produce the required production volumes. 2) STOP: the well is “stopped” at the surface and will not produce any fluids to surface; however, if there any open connections then flow may occur within the wellbore and between the open connections depending on a connection’s potential with respect to all the other connections. Inter-connection flow (cross flow) can be prevented by setting the XFLOW variable on the WELSPECS keyword to NO. In this case the well’s behavior will be similar to the SHUT option described below. 3) SHUT: the well is shut at the surface and downhole, this results in no flow at the surface and no cross flow downhole. 4) AUTO: the well is initially SHUT, but may be opened automatically if an economic limit is violated. This option is currently not supported by OPM Flow. Note a well’s STATUS should always be set either STOP or SHUT if the well’s production is to be set to zero. Just setting a well’s production rate to zero means that the well is open to flow with a zero rate, this will cause numerical issues especially for wells under THP control.			OPEN

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	TARGET	A defined character string that sets the target production phase for the well, all the other phases will therefore act as constraints. The simulator will attempt to meet the TARGET based on the phase rate stated in items (4) to (10) on this keyword. TARGET should be set to one of the following character strings: 1) ORAT: the target is set to the surface oil production rate as defined by item (4). 2) WRAT: the target is set to the surface water production rate as defined by item (5). 3) GRAT: the target is set to the surface gas production rate as defined by item (6). 4) LRAT: the target is set to the surface liquid (oil plus water) production rate as defined by item (7). 5) RESV: the target is set to the in situ reservoir volume rate as defined by item (8). 6) BHP: the target rate is set to the bottom-hole pressure as defined by item (9). 7) THP: the target rate is set to the tubing head pressure as defined by item (10). If this option is selected then the vertical lift performance tables must be entered via the VFPPROD keyword in the SCHEDULE section and allocated to the well via item (11). 8) GRUP: the well is under group control and produces its share of the group's target as set using the GCONPROD keyword in the SCHEDULE section. Here the default tokens of <code>I*</code> or <code>"</code> may be used, and both default tokens behave in the same manner; however, the actual default value used when TARGET is defaulted is dependent on the data entered on this keyword as described below: 1) If the well's group parameters have not been defined via the GCONPROD keyword, in the SCHEDULE section, then TARGET is set to the first non-defaulted hydrocarbon rate (ORAT, GRAT, LRAT, or RESV). If all the rates are defaulted and a value for BHP is entered, then TARGET is set to BHP control. Similarly, if BHP is also defaulted, and THP has been entered, then TARGET is set to THP control. Finally, if all parameters are defaulted, then the default value for BHP, one atmosphere, will be used with BHP control. 2) If, <u>and only if</u>, the well's group parameters have been defined via the GCONPROD keyword, then TARGET is set equal to GRUP, and the remaining parameters on the WCONPROD keyword are used as well constraints. Note the default value of one atmosphere should be avoided as the BHP will result in unrealistic well potentials as well as optimistic production forecasts for the well.			I*
4	ORAT	A real positive value that defines the maximum surface oil production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/day	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	WRAT	A real positive value that defines the maximum surface water production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/day	sm ³ /day	scc/hour	
6	GRAT	A real positive value that defines the maximum surface gas production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		Mscf/day	sm ³ /day	scc/hour	
7	LRAT	A real positive value that defines the maximum surface liquid (oil plus water) production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		stb/day	sm ³ /day	scc/hour	
8	RESV	A real positive value that defines the maximum reservoir volume production rate target or constraint. This value may be specified using a User Defined Argument (UDA).			None
		rtb/day	rm ³ /day	rcc/hour	
9	BHP	A real positive value that defines the minimum bottom-hole pressure target or constraint. This value may be specified using a User Defined Argument (UDA). Note the default value of one atmosphere should be avoided as the BHP will result in unrealistic well potentials as well as optimistic production forecasts for the well.			Defined
		psia 14.70	barsa 1.01325	atma 1.0	
10	THP	A real positive value that defines the minimum tubing head pressure target or constraint. This value may be specified using a User Defined Argument (UDA). Note the default value of zero should be avoided if the well's control TARGET has been set to THP, as this will result in optimistic production forecasts for a well, since a well must flow against a back pressure imposed by the surface facilities.			Defined
		psia 0.0	barsa 0.0	atma 0.0	

No.	Name	Description			Default
		Field	Metric	Laboratory	
11	VFPTAB	A positive integer greater than or equal to zero that defines the vertical lift performance tables to be used for calculating the tubing head pressure for the well. If a non-zero value is entered then the vertical lift performance tables must be entered via the VFPPROD keyword in the SCHEDULE section and allocated to the well via this item. The default value of zero implies no vertical lift performance tables and in this case TARGET cannot be set to THP and in addition item (10) should be defaulted or set to zero.			0
12	ALQ-WELL	A real positive value that defines the artificial lift quantity to be used in conjunction with the VFPPROD assigned to the well via the VFPTAB variable. This value may be specified using a User Defined Argument (UDA). VFPTAB vertical lift performance table and the artificial lift quantity ALQ-WELL are used with the well fluid rates to calculate the well's tubing head pressures values from the bottom-hole pressure. Note that the units for ALQ-WELL are dependent on the associated variable on the VFPPROD keyword.			0.0
13	WGASRATE	Wet gas production rate used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
14	MOLARATE	Total molar rate used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
15	STEAMRAT	Thermal/Temperature steam rate (Cold Water Equivalent) for steam producers used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
16	DELTAP	Thermal/Temperature delta pressure offset for steam producers used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
17	DELTAT	Thermal/Temperature delta temperature offset for steam producer used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
18	CALRATE	Calorific production rate used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
19	COMBPROC	Linearly combined procedure for when exceeding COMBRATE, used in the commercial compositional simulator. Not used and should be defaulted with I*.			I*
20	NGL	A real positive value that defines the observed Natural Gas Liquid ("NGL") rate in the commercial compositional simulator. Not used and should be defaulted with I*.			I*

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the <i>SCHEDULE</i> section. User Defined Arguments are activated using the <i>UDADIMS</i> keyword in the <i>RUNSPEC</i> section, and defined by the <i>UDQ</i> keyword in the <i>SCHEDULE</i> section. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.87:WCONPROD Keyword Description

See also the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the production targets and constraints for five wells as follows:

```
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL  OIL   WAT   GAS   LIQ   RES   BHP   THP   VFP   VFP
-- NAME  SHUT   MODE  RATE  RATE  RATE  RATE  RATE  PRES  PRES  TABLE  ALFQ
WCONPROD
OP01    OPEN   GRUP   5E3   1*    1*    1*    1*    500.0
OP02    OPEN   GRUP  10E3   1*    1*    1*    1*    200.0  500.0  2      0.0  /
OP03    OPEN   GRUP  15E3   1*    1*    1*    1*    200.0  500.0  3     10.0  /
OP04    OPEN   ORAT  20E3   1*    1*    1*    1*    500.0
OP05    SHUT   GRUP  20E3   1*    1*    1*    1*    500.0
/
```

Well OP01 is open and is on group control, subject to a maximum oil rate constraint of 5,000 stb/d and a minimum bottom-hole pressure of 500 psia. OP02 is also open and on group control but it's maximum oil rate constraint has been set 10,000 stb/d, and is subject to a minimum bottom-hole pressure limit of 200 psia and a minimum tubing head pressure limit of 500 psia using [VFPPROD](#) vertical lift table number two. Well OP03 is very similar to OP02, but with a 15,000 stb/d maximum oil constraint and using [VFPPROD](#) vertical lift table number three with an artificial lift parameter of 10. The next well is not on group control. Well OP04 is open and has an oil rate target of 20,000 stb/d, subject to a minimum bottom-hole pressure of 500 psia. Finally, well OP05 is shut and will not be brought back on production despite being put under group control, as the well has been declared shut.

The next example defines the production targets and constraints for five wells, of which well OP01 is under group control as the well's group target and constraints have been set with the [GCONPROD](#) keyword, and wells OP02 to OP05 belong to groups that have not had their group constraints set by [GCONPROD](#).

```
--
--          GROUP PRODUCTION CONTROLS
--
-- GRUP  CNTL  OIL   WAT   GAS   LIQ   CNTL  GRUP  GUIDE  GUIDE  CNTL
-- NAME  MODE  RATE  RATE  RATE  RATE  OPT   CNTL  RATE  DEF   WAT
GCONPROD
GRP01   SAT   25E3   1*    1*    1*    1*    1*    1*    1*    1*    /
/
--
--          WELL SPECIFICATION DATA
--
-- WELL  GROUP  LOCATION  BHP  PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME    I    J  DEPTH  FLUID  AREA   EQUANS  SHUT  FLOW  TABLE
WELSPECS
OP01    SAT      14   13  1*    OIL   1*    P-P    SHUT  NO    1*    /
OP02    PLATFORM  64   80  1*    OIL   1*    GPP    SHUT  NO    1*    /
OP03    PLATFORM  24  110  1*    OIL   1*    STD    SHUT  NO    1*    /
OP04    PLATFORM  24  110  1*    OIL   1*    STD    SHUT  NO    1*    /
OP05    PLATFORM  24  110  1*    OIL   1*    STD    SHUT  NO    1*    /
/
```

--	WELL PRODUCTION WELL CONTROLS											
--	WELL	OPEN/	CNTL	OIL	WAT	GAS	LIQ	RES	BHP	THP	VFP	VFP
--	NAME	SHUT	MODE	RATE	RATE	RATE	RATE	RATE	PRES	PRES	TABLE	ALFQ
WCONPROD												
OP01	OPEN		1*	15E3	1*	1*	1*	1*	500.0			/
OP02	OPEN		1*	1*	5E3*	15E6	1*	1*	200.0	500.0	2	0.0 /
OP03	OPEN		1*	1*	1*	1*	1*	1*	1*	500.0	3	10.0 /
OP04	OPEN		'	1*	1*	1*	1*	1*	1*	1*	1*	1* /
OP05	SHUT		1*	1*	1*	1*	1*	1*	1*	1*	1*	1* /
/												

Here, well OP01's control mode, TARGET, will be set to group control (GRUP) and the well's production parameters will all act as constraints. For well OP02, the well's control mode is set to GRAT, and the other production parameters all act as constraints. Well OP03 control mode is set to THP and if the well's BHP value has been previously defined then that value will be used as a constraint; otherwise the default value of one atmosphere will be used instead. Finally, wells OP04 and OP05 will have their control mode set to BHP control. Note in this case, OP04's status has been set to OPEN; thus the well will produce at a target BHP of one atmosphere, which is unrealistic.

12.3.258 WCUTBACK – DEFINE WELL CUTBACK LIMITS AND PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WCUTBACK](#), defines a well’s cutback limits and parameters for both production and injection wells. See also the [GCUTBACK](#) keyword in the [SCHEDULE](#) section that provides similar functionality for groups.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.259 WCUTBACT – DEFINE WELL TRACER CUTBACK LIMITS AND PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WCUTBACT](#), defines a production well's cutback limits and parameters based on the named produced tracer from the well. See also the [GCUTBACT](#) keyword in the [SCHEDULE](#) section that provides similar functionality for groups.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.260 WCYCLE – DEFINE AUTOMATIC WELL OPENING AND CLOSING CYCLING PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WCYCLE](#) keyword defines automatic well opening and closing cycling parameters. These are used to model for example “Huff and Puff” cyclic steam injection in heavy oil reservoirs or Water-Alternating-Gas (“WAG”) processes in enhanced oil recovery modeling. The keyword defines specific time periods for automatically cycling wells on and off. For example in a WAG scheme the water injection wells would have one set of cycling parameters and the gas injection wells another, such that only one type of well is active at a time.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well automatic cycling parameters are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
4	ONTIME	A real positive value that defines the length of time for the on period. The well will be turned off at the beginning of the first time step after the on period has elapsed. If ONTIME is zero or negative then the well will not be turned on by the automatic well cycling.			0.0
		day	day	hour	
5	OFFTIME	A real positive value that defines the length of time for the off period. If the well was turned off by automatic cycling then it will be turned on at the beginning of the first time step after the off period has elapsed. If the well was turned off other than by automatic cycling then the well will remain turned off. If OFFTIME is zero or negative then the well will not be turned on by the automatic well cycling.			0.0
		day	day	hour	
6	STARTTIME	A real positive value that defines the length of time for the start-up period during which the well efficiency factor will be ramped up. The start-up period begins at the start of the on period. The well's efficiency factor is ramped up linearly from zero at the start of the on period to the value specified previously by the WEFAC keyword after the start-up period has elapsed. The well efficiency factor to apply during each time step is linearly interpolated based on the time at the end of the time step. If the time step size is small compared with STARTTIME then the rate will be ramped up gradually. Whereas, if the time step size is greater than STARTTIME then the rate will ramp up instantaneously potentially causing convergence problems.			0.0

No.	Name	Description			Default
		Field	Metric	Laboratory	
		day	day	hour	
7	MAXTS	A real positive value that defines the maximum time step length when the well is turned on by automatic cycling. This overrides the standard maximum time step length. This can help to reduce potential convergence problems caused when a high rate well is opened. If MAXTS is zero then the standard maximum time step length applies.			0.0
		day	day	hour	
13	CTRLTS	A defined character string that specifies whether the time steps should be controlled to coincide exactly with the automatic cycling times for this well. CTRLTS should be set to one of the following: 1) YES: time steps are controlled to coincide exactly with the automatic cycling times for this well. Note this will exactly match the specified on and off periods for this well. However, this is likely to reduce the time step size, especially if this option is set for a number automatic cycling wells that have different parameters. 2) NO: time steps are not controlled to coincide exactly with the automatic cycling times for this well. Note this is unlikely to match the specified on and off periods for this well unless they coincide with the report steps. The next period will start at the next time step after the current on/off period has elapsed. Therefore, the on/off periods are likely to be longer than specified.			NO
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.88:WCYCLE Keyword Description

Example

The following example defines the cycle lengths for a Water-Alternating-Gas injection scheme with a water-gas slug length ratio of 1:2, and initiates automatic cycling in the water injection well. A maximum time step of 5 days has been specified following the start of each injection period. The simulation then proceeds to the end of the first water injection cycle when automatic cycling is initiated in the gas injection well.

```
--
-- AUTOMATIC CYCLING OF WELLS ON AND OFF
--
WCYCLE
--WELL      ON      OFF      STARTUP  STARTUP  CTRL
--NAME      PERIOD  PERIOD  TIME     MAXTS    TIMESTEP
  I01W      30.0    60.0     1*       5.0     YES /
  I01G      60.0    30.0     1*       5.0     YES /
--
-- OPEN WELLS
--
WELOPEN
--WELL      OPEN
--NAME      SHUT
  I01W      OPEN /
  I01G      SHUT /
```

```
/
--
-- REPORT STEPS
--
TSTEP
  30.0      /
--
-- OPEN WELLS
--
WELOPEN
--WELL      OPEN
--NAME      SHUT
  I01G      OPEN /
/
```

Note that the wells initially need to be opened manually in order to start the automatic cycling. In addition, this needs to be done at the correct times to ensure that the water and gas injection wells are synchronised.

12.3.261 WDFAC – DEFINE GAS FLOW DEPENDENT SKIN FACTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WDFAC](#) keyword defines a gas well's D-factor (flow dependent skin factor), which is normally derived from well tests or calculated analytically based on the coefficient of inertial resistance, usually known as β , in Forchheimer's flow equation³³¹⁻³³² and ³³³.

Internally the simulator evaluates a D-factor for each well connection by assuming the D-factors are inversely proportional to the connection transmissibility factors. This is done at the time the well D-factor is specified. If a connection is later closed then the well D-factor is likely to change but the connection D-factors for each of the remaining open connections should not be affected.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well D-factor is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section.			None
2	DFACTOR	A real positive value greater than or equal to zero that defines the D-factor for the well.			0.0
		day/Mscf	day/m ³	hour/scc	

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.89:WDFAC Keyword Description

See also the [WDFACCOR](#) keyword in the [SCHEDULE](#) section that uses Dake's³³⁴ correlation to calculate the well D-factor; and the [COMPDAT](#) keyword in the [SCHEDULE](#) section that can be used to specify the connection D-factors directly.

Example

```
--
--          WELL D-FACTORS
--
-- WELL  D-
-- NAME  FACTOR
WDFAC
  GP01   1.0E-03 /
/
```

In the above example a well D-factor of 10^{-3} is specified for gas well GP01.

³³¹ Geertsma, J., 1974. Estimating the Coefficient of Inertial Resistance in Fluid Flow Through Porous Media. Soc.Pet.Eng.J., October: 445-450.

³³² Gewers, C.W.W. and Nichol, L.R., 1969. Gas Turbulence Factor in a Microvugular Carbonate. J.Can.Pet.Tech., April.

³³³ Wong, S.W., 1970. Effects of Liquid Saturation on Turbulence Factors for Gas Liquid Systems. J.Can.Pet.Tech., October

³³⁴ Dake, L.P. Fundamentals of Reservoir Engineering, Amsterdam, The Netherlands, Elsevier Science BV (1978) Chapter 8.6, pages 252-257.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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12.3.262 WDFACCOR – GAS FLOW DEPENDENT SKIN FACTOR (CORRELATION)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WDFACCOR keyword defines the parameters to calculate a gas well's connection D-factors (flow dependent skin factor) for each connection based on a correlation for the coefficient of inertial resistance, usually known as β , in Forchheimer's flow equation³³⁵⁻³³⁶⁻³³⁷ and ³³⁸. This keyword uses Dake's correlation to calculate the D-factor.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well D-factor correlation is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section.			None
2	A	A real value greater than or equal to zero that defines the coefficient A in the D-factor correlation.			0.0
		cP.day.ft ² /mD/Mscf	cP.day.m ² /mD/sm ³	cP.hour.cm ² /mD/scc	
3	B	A real value that defines the exponent B of the grid block permeability in the D-factor correlation.			0.0
		dimensionless	dimensionless	dimensionless	
4	C	A real value that defines the exponent C of the grid block porosity in the D-factor correlation.			0.0
		dimensionless	dimensionless	dimensionless	

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.90:WDFACCOR Keyword Description

The simulator evaluates the connection D-factors using the following expression based on Dake's correlation:

$$D = A K_e^B \phi^C \frac{K_e}{h} \frac{1}{r_w} \frac{\gamma_g}{\mu_g} \quad (12.31)$$

Where:

K_e = effective permeability of the grid block,
 ϕ = porosity of the grid block,

³³⁵ Dake, L.P. *Fundamentals of Reservoir Engineering*, Amsterdam, The Netherlands, Elsevier Science BV (1978) Chapter 8.6, pages 252-257.

³³⁶ Geertsma, J., 1974. Estimating the Coefficient of Inertial Resistance in Fluid Flow Through Porous Media. *Soc.Pet.Eng.J.*, October: 445-450.

³³⁷ Gewers, C.W.W. and Nichol, L.R., 1969. Gas Turbulence Factor in a Microvugular Carbonate. *J.Can.Pet.Tech.*, April.

³³⁸ Wong, S.W., 1970. Effects of Liquid Saturation on Turbulence Factors for Gas Liquid Systems. *J.Can.Pet.Tech.*, October

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

h	= connection length,
r_w	= wellbore radius,
γ_g	= gas gravity at surface conditions,
μ_g	= gas viscosity at bottom hole conditions.

Note that since μ_g is dependent on pressure the D-factor will also vary with pressure.

See also the [WDFAC](#) keyword in the [SCHEDULE](#) section that can be used to specify the well D-factor; and the [COMPDAT](#) keyword in the [SCHEDULE](#) section that can be used to specify the connection D-factors directly.

Examples

In the first example the connection D-factors for gas well GP01 are evaluated for correlation coefficients specified in field units based on the laboratory determined relationship between β and the absolute permeability presented by [Dake in Fig. 8.8](#) (variation of β with porosity has been neglected).

```
--
--      WELL D-FACTOR CORRELATIONS
--
-- WELL
-- NAME      A          B          C
WEFAC
  GP01      6.04E-05   -1.1045    0.0 /
/
```

In the second example the same correlation coefficients are specified in metric units.

```
--
--      WELL D-FACTOR CORRELATIONS
--
-- WELL
-- NAME      A          B          C
WEFAC
  GP01      1.20E-07   -1.1045    0.0 /
/
```

12.3.263 WDRILPRI – ADD WELLS TO THE DRILLING PRIORITY DRILLING QUEUE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WDRILPRI](#), adds wells to the Drilling Priority Drilling Queue and defines the well priority and drilling unit number or batch queue sequence for the well. The batch queue sequence number enables all wells with the same sequence number to drilled at the same time.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.264
WDRILRES – ACTIVATE PREVENTION OF MULTI-COMPLETIONS IN THE SAME CELL FOR QUEUED WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WDRILRES](#) keyword activates the prevention of multi-completions being completed in the same cell for wells in a drilling queue. Setting this option stops any well defined as a queued well via the [QDRILL](#) and [WDRILLPRI](#) keywords in the [SCHEDULE](#) section, or any wells set to automatic opening by setting the [STATUS](#) variable to [AUTO](#) on the [WCONPROD](#) keyword in the [RUNSPEC](#) section, from opening if there is an already existing active well connection to a cell.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.265 WDRILTIM – DEFINE DRILLING PARAMETERS FOR AUTOMATIC DRILLING OF NEW WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WDRILTIM](#) defines the automatic drilling parameters used to describe the numbers of days taken to drill a well, the drilling status of the well, and status of other wells when drilling an automatically drilled well.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.266 WECON – WELL ECONOMIC CRITERIA FOR PRODUCTION WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WECON](#) keyword defines the economic criteria for production wells that have previously been defined by the [WELSPECS](#) and [WCONPROD](#) keywords in the [SCHEDULE](#) section.

Note that wells can be allocated to a group when they are specified by the [WELSPECS](#) keyword and groups can also have economic controls. Wells under group control are therefore subject to the economic criteria set via the [GCONPROD](#) and [GECON](#) keywords in the [SCHEDULE](#) section and the economic criteria specified by the [WECON](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well economic criteria data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	ORAT	A real positive value that defines the minimum economic surface oil production rate, below which an economic action will take place, as outlined below: 1) If there are any remaining connections in the well with the STATUS variable set to AUTO on the COMPDAT keyword in the SCHEDULE section, then one of these connections (or completion) will be opened. 2) If there are no remaining connections in the well with the STATUS variable set to AUTO on the COMPDAT keyword, then the well will be shut or stopped as requested by item (9) of the WELSPECS keyword. Only option (2) is supported by OPM Flow as STATUS equals AUTO on the COMPDAT keyword is currently not supported by the simulator. Hence, the well be either shut or stopped. A value less than or equal to zero switches off this criterion. This value may be specified using a User Defined Argument (UDA).			0.0
		stb/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	GRAT	A real positive value that defines the minimum economic surface gas production rate, below which an economic action will take place, as outlined below: 1) If there are any remaining connections in the well with the STATUS variable set to AUTO on the COMPDAT keyword in the SCHEDULE section, then one of these connections (or completion) will be opened. 2) If there are no remaining connections in the well with the STATUS variable set to AUTO on the COMPDAT keyword, then the well will be shut or stopped as requested by item (9) of the WELSPECS keyword. Only option (2) is supported by OPM Flow as STATUS equals AUTO on the COMPDAT keyword is currently not supported by the simulator. Hence, the well be either shut or stopped. A value less than or equal to zero switches off this criterion. This value may be specified using a User Defined Argument (UDA).			0.0
		Mscf/d	sm ³ /day	scc/hour	
4	WCUT	A real positive value that defines the maximum economic surface water cut, above which an economic action will take place. This value may be specified using a User Defined Argument (UDA). Water cut is defined as: $f_w = \frac{q_w}{q_w + q_o}$, and the various actions that are available if the water cut limit is exceeded are described in item (7). A value less than or equal to zero switches off this criterion.			0.0
		dimensionless	dimensionless	dimensionless	
5	GOR	A real positive value that defines the maximum economic surface gas-oil ratio, above which an economic action will take place, as defined by item (7). A value less than or equal to zero switches off this criterion. This value may be specified using a User Defined Argument (UDA).			0.0
		Mscf/stb	sm ³ /sm ³	scc/scc	
6	WGR	A real positive value that defines the maximum economic surface water-gas ratio, above which an economic action will take place, as defined by item (7). A value less than or equal to zero switches off this criterion. This value may be specified using a User Defined Argument (UDA).			0.0
		stb/Mscf	sm ³ /sm ³	scc/scc	

No.	Name	Description			Default
		Field	Metric	Laboratory	
7	ACTION	A defined character string that defines the action to be taken if the economic WCUT, GOR, or WGR limits are violated. ACTION should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection. If connections have been grouped as completions then the worst offending completion will be closed. 3) +CON: close the worst offending connection and all below it. If connections have been grouped as completions then the worst offending completion and all below it will be closed. 4) WELL: shut or stop the well as per the AUTO variable on the WELSPECS keyword. The corrective action takes places at the end of the time step in which the constraint is violated. The +CON option is not currently supported by OPM Flow.			NONE
8	END	A defined character string that defines if the simulation should terminate if the well is shut or stopped. END should be set to one of the following character strings: 1) NO: no action is taken and the run continues. 2) YES: terminate the run at the next report time step. The YES option is not currently supported by OPM Flow.			NO
9	WELOPEN	A character string of up to eight characters in length that defines the name of the well, which will be opened when the current well (WELNAME) has been automatically closed/shut by the simulator. Wells closed manually do not invoke this option. The well name (WELOPEN) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
10	ELTOPT	A defined character string that defines the type of quantity used to test the economic limit, and should be set to one of the following character strings: 1) POTN: The Economic Limit Test ("ELT") is based on a well's potential rate. A well's potential rate is calculated using only the flowing bottom-hole pressure and tubing head pressure constraints, all other constraints are ignored, including group constraints. However, if the WELDRAW keyword in the SCHEDULE section has been used to limit a well's pressure draw down, then this will also be considered in the potential calculation, as it influences the bottom-hole pressure constraint. 2) RATE: The economic limit is based a well's surface production rate as oppose to the potential rate, this is the default behavior. Wells under group control via the GCONPROD keyword in the SCHEDULE section will be subject to the ELT, except for when the group's production target rate is set to zero. Wells under group control via the GCONPRI keyword in the SCHEDULE section will be subject to the ELT, except for when the group has curtailed a well's production.			RATE

No.	Name	Description			Default
		Field	Metric	Laboratory	
11	WCUT2	A real positive value that defines the secondary maximum economic surface water cut, above which an economic action will take place. A value less than or equal to zero switches off this criterion. This item is not supported by OPM Flow and should be defaulted (1*) or set to zero.			0.0
12	ACTION2	A defined character string that defines the action to be taken if the secondary economic WCUT2 limits is violated. This item is not supported by OPM Flow and should be defaulted (1*).			1*
13	GLR	A real positive value that defines the maximum economic surface gas-liquid ratio, above which an economic action will take place, as defined by item (7). A value less than or equal to zero switches off this criterion. This item is not supported by OPM Flow and should be defaulted (1*).			None
14	LRAT	A real positive value that defines the minimum liquid rate, below which the well (WELNAME) is shut-in. This item is not supported by OPM Flow and should be defaulted (1*).			0.0
15	TEMP	A real positive value that defines the maximum economic temperature in the commercial compositional Thermal/Temperature model for a well. This item is not supported by OPM Flow and should be defaulted (1*).			None
16	RESV	A real positive value that defines the minimum reservoir volume rate, below which the well (WELNAME) is shut-in. This item is not supported by OPM Flow and should be defaulted (1*).			0.0
Notes:					
1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the SCHEDULE section. User Defined Arguments are activated using the UDADIMS keyword in the RUNSPEC section, and defined by the UDQ keyword in the SCHEDULE section.					
2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.91: WECON Keyword Description

See also the [WEL SPECS](#) keyword to define a wells shut-in or stop options, [GCONPROD](#) for group controls, and [GECON](#) for setting a group's economic criteria. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines one oil well and one gas well using the [WEL SPECS](#) keyword, together with their economic criteria.

```
--
-- WELL SPECIFICATION DATA
--
-- WELL      GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  SHUT  CROSS  PRESS
-- NAME      NAME        I      J    DEPTH FLUID  AREA   EQUA.  IN    FLOW   TABLE
WEL SPECS
GP01        PLATFORM    14     13   1*     GAS   1*     GPP    SHUT   NO     1* /
OP01        PLATFORM    28     96   1*     OIL   1*     STD    SHUT   NO     1* /
```

```

/
--
-- WELL ECONOMIC CRITERIA FOR PRODUCTION WELLS
-- NAME MIN ORAT MIN GRAT MAX WCUT MAX GOR MAX WGR CNTL MODE END RUN
WECON
GP01 1* 5.0E3 1* 1* 1* 'WELL' 'NO' /
OP01 500 1* 0.95 15 1* 'WELL' 'YES' /
/
```

Well GP01 has a minimum economic gas rate of 5 MMscf/d and will shut-in if the gas rate falls below this rate, but the simulation will continue even if this occurs. Well OP01 has a minimum economic oil rate of 500 stb/d, a maximum water cut limit of 95%, and a maximum GOR of 15 Mscf/stb, if any any of these limits are violated the well will be shut-in and the run should be terminated at the next reporting time step, however the option to end the run is not currently supported by OPM Flow.

12.3.267 WECONINJ – WELL ECONOMIC CRITERIA FOR INJECTION WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WECONINJ](#) keyword defines economic criteria for injection wells that have previously been defined by the [WELSPECS](#) and [WCONINJE](#) keywords in the [SCHEDULE](#) section.

Note that wells can be allocated to a group when they are specified by the [WELSPECS](#) keyword and groups can also have economic controls. Wells under group control are therefore subject to the economic criteria set via the [GCONINJE](#) keyword in the [SCHEDULE](#) section and the controls specified by this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well economic injection criteria data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	MINVALUE	A real positive value that defines the minimum economic injection value, below which an economic action will take place, as defined by the AUTO parameter on the WELSPECS keyword (SHUT or STOP). Note that TYPE determines if the minimum value is applied to the well's actual injection rate or the well's potential. A value less than or equal to zero switches off this criterion.			0.0
		Liquid stb/d Gas Mscf/d	Liquid sm ³ /day Gas sm ³ /day	Liquid scc/hour Gas scc/hour	
3	TYPE	A defined character string that determines if MINVALUE is applied to a well's actual rate or potential, and should be set to one of the following: 1) RATE: In this case the MINVALUE is applied to a well's actual rate. 2) POTN: Here, MINVALUE is applied to the well's potential with only the BHP and THP constraints applied. The default value is RATE.			RATE
Notes: 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.92: WECONINJ Keyword Description

See also the [WELSPECS](#) keyword to define a wells shut-in or stop options and [GCONINJE](#) for group controls in the [SCHEDULE](#) section.

Example

The following example defines the economic injection parameters for all gas and water injection wells.

```
--
--      WELL ECONOMIC LIMIT DATA FOR INJECTION WELLS
--
-- WELL  MIN    RATE
-- NAME  VALUE  POTN
WECONINJ
GI*      2.0E3  RATE
WI*      5.0E3  POTN
/
```

Here all the gas injection wells have a minimum economic gas injection rate of 2 MMscf/d and the water injection wells have a minimum water potential rate of 5,000 stb/d. The AUTO parameter on the [WELSPECS](#) keyword will determine if the wells will be shut-in or stopped.

12.3.268 WECONT – WELL ECONOMIC TRACER CRITERIA FOR PRODUCTION WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The **WECONT** keyword defines the tracer economic criteria for production wells that have previously been defined by the **WELSPECS** and **WCONPROD** keywords in the **SCHEDULE** section, for tracers defined by the **TRACER** keyword in the **PROPS** section.

Note that wells can be allocated to a group when they are specified by the **WELSPECS** keyword and groups can also have economic controls. Wells under group control are therefore subject to the economic criteria set via the **GECONT** keyword in the **SCHEDULE** section and the controls specified by this keyword. Note that **GECONT** is not supported by OPM Flow in the current release.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well target and constraints are being defined.			None
I-2	ACTION	A defined character string that defines the action to be taken if the economic WCUT, GOR, or WGR limits are violated. ACTION should be set to one of the following character strings: 1) NONE: no action is taken. 2) CON: close the worst offending connection in the worst offending. If connections have been wellbore completions then the worst offending completion will be closed. 3) +CON: close the worst offending connection and all below it in the worst offending well. If connections have been wellbore completions then the worst offending completion and all below it in the worst offending well will be closed. 4) WELL: shut or stop the well as per the AUTO variable on the WELSPECS keyword. 5) PLUG: plug back the worst offending well based on the plug back length and options defined on the WPLUG keyword in the SCHEDULE. The corrective action takes place at the end of the time step in which the constraint is violated.			None
I-3	END	A defined character string that defines if the simulation should terminate if the well is shut or stopped. END should be set to one of the following character strings: 1) NO: no action is taken and the run continues. 2) YES: terminate the run at the next report time step.			NO
I-4	WELL	A character string of up to eight characters in length that defines the well name of a fully defined well that will be “opened” when the well WELNAME is shut-in or stopped.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
1-5	/	Record one terminated by a “/”			Not Applicable
2-1	NAME	A three letter character string defining the tracer’s name. Note it is best to void names beginning with the letters F, S, and T as these names may create naming issues in post-processing software.			None
2-2	MXTOTAL	A real positive value that defines the maximum total (free plus solution) tracer rate. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-3	MXFREET	A real positive value that defines the maximum total (free plus solution) tracer concentration. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-4	MXFREEQ	A real positive value that defines the maximum free tracer rate. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-5	MXCONC	A real positive value that defines the maximum free tracer concentration. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-6	MXSOLNQ	A real positive value that defines the maximum solution rate. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			None
2-7	MXSOLNC	A real positive value that defines the maximum solution concentration. Tracer units are per those defined by the carrying fluid, oil, gas, water, etc.			
2-8	/	Record two terminated by a “/”			Not Applicable
3-1	/	Well terminated by a “/”			Not Applicable

Notes:

- 1) [WECONT](#) keyword consists of two records, with entries 1-1 to 1-5 representing record one items and 2-1 to 2-8 representing record number two items, in the “No.” column in this table. A maximum of three type two records can be entered following a type one record.
- 2) Each type one and type two records are terminate by a “/” as indicated in the table, and a well data set is terminated by a further “/”, after which additional well data sets can be entered stating with a record of type one followed by type two.
- 3) The keyword the keyword should be terminated by an additional “/” after the well data set termination “/” character.

Table 12.93:WECONT Keyword Description

See also the [WEL SPECS](#) keyword to define a wells shut-in or stop options, and [WECON](#) for setting a well’s economic criteria. Both the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the tracer economic criteria for the field and two wells, OP01 and OP02.

```
--
--      WELL TRACER ECONOMIC CRITERIA FOR PRODUCTION WELLS
--
-- WELL  WORK  END  MAX
-- NAME OVER  RUN  WELLS
WECONT
OP01    +CON  'YES'  1*                               / START OF WELL
--
--      TRACER  TRACER  TRACER  TRACER  TRACER  TRACER  TRACER
--      NAME    TOTAL  TOTAL  FREE   FREE   SOLN   SOLN
--            RATE  CONCEN  RATE   CONCEN  RATE   CONCEN
--      PLY      800.0
--      BRI      800.0                               /
--                                           /
--                                           /
OP02    +CON  'YES'  1*                               / START OF WELL
--
--      TRACER  TRACER  TRACER  TRACER  TRACER  TRACER  TRACER
--      NAME    TOTAL  TOTAL  FREE   FREE   SOLN   SOLN
--            RATE  CONCEN  RATE   CONCEN  RATE   CONCEN
--      PLY      800.0
--      BRI      800.0                               /
--                                           / END OF WELL
--                                           / END OF KEYWORD
```

If the economic limits are violated then the worst offending connection and all below it in the worst offending well will be closed, If connections have been grouped as completions then the worst offending completion and all below it in the worst offending well will be closed

12.3.269 WEFAC – DEFINE WELL EFFICIENCY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Defines a well's efficiency or up-time as opposed to setting the efficiency at the group level.

Note that wells are allocated to a group when they are specified by the [WELSPECS](#) keyword and groups can also have efficiency factors.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well efficiency factor is being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	FACTOR	A real positive value greater than zero and less than or equal to one that defines the efficiency factor for the well. If a well's down time is 5% then FACTOR should be set to 0.95 (1.0 – 0.05). Note that well pressures and rates are calculated at their full flowing conditions but subject to any operating constraints, that is <u>without</u> the well efficiency being applied (FACTOR), in order to represent the actual flowing conditions in the field. The effective rates and volumes are calculated by applying FACTOR when summing individual well rates to their group level and higher, including summing to the top most group FIELD. In terms of a well's cumulative production, FACTOR is applied to the well rate times the time interval for the time step. This ensures that correct effective volume is withdrawn from (or injected to) the reservoir. This approach means that wells are effectively arbitrarily offline for a period during a time step, as opposed to all wells going offline concurrently. And thus the group and field rates and volumes are the effective rates and volumes for the field.			1.0
		dimensionless	dimensionless	dimensionless	
3	WELNETWK	A defined character string that determines if the WELNAME efficiency factor should be applied when calculating the flows and pressure losses in the Extended Network Model, and should be set to either: 1) NO: The well's equivalent Extended Network Model node flow rates are not multiplied by the efficiency factor FACTOR. 2) YES: The well's equivalent Extended Network Model node flow rates are multiplied by the efficiency factor FACTOR This option is only applicable for the Extended Network Model, as in the Standard Network Model groups flow rates are always used in the calculation of pressure drops (this is equivalent to the default option, YES).			YES

Notes:
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.94: WEFAC Keyword Description

See also the [GEFAC](#) keyword in the [SCHEDULE](#) section to set the efficiency at the group level, as opposed to applying the efficiency to individual wells.

Note

One can also apply plant efficiencies through the [GEFAC](#) keyword in the [SCHEDULE](#) section. If all the wells in a group are flowing through a facility that has an overall efficiency factor, then it is more appropriate to apply the efficiency factor at the group level. This of course does not preclude applying additional well efficiencies to individual wells. For example, subsea wells (wet trees) may have additional down time compared to platform wells (dry trees) even though both sets of well are flowing through the same platform. Another example would be gas lift wells and wells using electrical submersible pumps, as their artificial lift mechanisms.

Example

```
--
--      WELL EFFICIENCY FACTORS
--
-- WELL  EFF      NETWK
-- NAME  FACT      OPTN
WEFAC
'GP*   ' 0.950
'OP*   ' 0.862
/
```

In the above example the all the gas wells are are defined as having a well efficiency factor (up time) of 0.950 and all the oil wells have a lower efficiency factor of 0.862.

12.3.270 WELCNTL – MODIFY WELL CONTROL AND TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELCNTL](#) keyword modifies a well's target control and value, both rates and pressures, for previously defined wells without having to define all the variables on the well control keywords: [WCONPROD](#), [WCONHIST](#), [WCONINJE](#), or [WCONINJH](#) keywords. Variables not changed by the [WELCNTL](#) keyword remain the same as those previously entered via the well control keywords or previously entered [WELCNTL](#) keywords. Note that the well must still be initially be fully defined using the [WCONPROD](#) or [WCONINJE](#) keywords. All the aforementioned keywords are described in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well production rates and pressures data are being redefined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS and WCONPROD (or WCONINJE) keywords in the SCHEDULE section, otherwise an error may occur.			None
2	TARGET	A defined character string that sets the item to be changed for the well the value of the item is set by item (3). 1) ORAT: reset the surface oil production rate value as defined by item (3). 2) WRAT: reset the surface water production rate value as defined by item (3). 3) GRAT: reset the surface gas production rate value as defined by item (3). 4) LRAT: reset the surface liquid (oil plus water) production rate value as defined by (3). 5) RESV: reset the in situ reservoir volume rate value as defined by (3). 6) BHP: reset the bottom-hole pressure value as defined by item (3). 7) THP: reset the tubing head pressure value for the well as defined by item (3). 8) VFP: reset the vertical lift performance table number as defined by (3). 9) LIFT: reset the artificial lift quantity for use with vertical lift performance tables. 10) GUID: reset the guide rate value for wells operating under group control. Note TARGET redefines the target controlled for a well and the control value on item (4). For example, if a well is operating on ORAT control, as defined by the previously entered WCONPROD keyword, entering TARGET equal to LRAT with a value, sets the TARGET to liquid rate with the given value. That is the well will be targeting a liquid rate not the previously requested oil rate. Use the WELTARG keyword in the SCHEDULE section to change the target and constraint values for a well.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	VALUE	A real positive value that defines the value of the variable declared by TARGET			None
	Liquid	stb/d	sm ³ /day	scc/hour	
	Gas	Mscf/d	sm ³ /day	scc/hour	
	Res Vol	rb/d	rm ³ /day	rcc/hour	
	Pressure	psia	barsa	atma	
	VFP	dimensionless	dimensionless	dimensionless	
	LIFT	same as <i>VFPPROD</i> or <i>VFPINJ</i>	same as <i>VFPPROD</i> or <i>VFPINJ</i>	same as <i>VFPPROD</i> or <i>VFPINJ</i>	
Notes:					
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.95: WELCNTL Keyword Description

If a well is currently a history matching well, then *WELCNTL* can be used to change the well to a standard well. See also the *WELTARG* keyword, in the *SCHEDULE* section that can be used to reset a well's target and constraints of both rates and pressures.

Example

The following example below shows the oil rates for the OP01 oil producer at the start of the schedule section (January 1, 2000).

```

-----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-----
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL  OIL    WAT    GAS    LIQ    RES    BHP    THP    VFP    VFP
-- NAME SHUT   MODE  RATE   RATE   RATE   RATE   RATE   PRES  PRES  TABLE ALFQ
WCONPROD
OP01    OPEN   ORAT   3000   1*    1*    1*    1*    750.0  500.  9      1* /
/
DATES
01 FEB 2000 /
/
--
--          WELL CONTROL MODE AND OPERATING TARGET
--
-- WELL  WELL    TARGET
-- NAME CNTL    VALUE
WELCNTL
OP01    LRAT    5000
/

```

From January 1, 2000 to February 1, 2000 well OP01 is open and is on oil rate control and has a target oil rate of 3,000 stb/d and uses *VFPPROD* vertical lift table number 9 with a minimum tubing head pressure constraint of 500 psia. After February 1, 2000 the well is changed to liquid control with a target rate of 5,000 stb/d of liquid and all the other parameters remain unchanged.

12.3.271 WELDEBUG – DEFINE THE WELL DEBUG DATA TO BE PRINTED TO FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword defines the well debug data to be written to the debug file (*.DBG). This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.272 WELDRAW – DEFINE MAXIMUM DRAW DOWN FOR PRODUCTION WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [WELDRAW](#), defines the maximum draw down for production wells The keyword may be useful in wells that are subject to fines or sand production to limit the draw down between the sand face and the well in order to limit or avoid sand production.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.273 WELEVNT – DEFINE WELL WPWEM SUMMARY MNEMONIC OUTPUT VALUE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [WELEVNT](#), defines an integer value to be assigned to an individual well's WPWEM summary variable that is written to the [SUMMARY](#) file. The value is set to zero after the current time step.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.274 WELLSTRE – DEFINE INJECTION STREAM COMPOSITION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WELLSTRE](#) keyword defines a well stream together with the compositional component mole fractions, associated with the well stream, as such it should have the same number of entries as that declared via the [COMPS](#) keyword in the [RUNSPEC](#) section, and the [NCOMPS](#) keyword in the [PROPS](#) section. Once a gas well stream has been defined, it can be used with either the [WINJGAS](#) or [GINJGAS](#) keywords in the [SCHEDULE](#) section, to set the injected gas composition. Similarly, if an oil well stream has been defined by [WELLSTRE](#), then the well stream can be used with the [WINJOIL](#) keyword in the [SCHEDULE](#) section, to specify the injected oil composition.

The keyword should only be used if the [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section have also been activated for the gas-water two component model.

Note

This is an OPM Flow keyword used with OPM Flow's [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section, and should not be confused with the more general version of the [WELLSTRE](#) keyword used in the commercial compositional simulator.

Secondly, although OPM Flow parses the keyword, the simulator currently ignores the data for this keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	STREAM	STREAM is a character string of up to eight characters in length, representing the well stream name. The WELLDIMS (MXSTRMS) parameter in the RUNSPEC section determines the maximum number of well streams allowed in the model.			None
2	ZCOMP	A row vector, with each item representing a compositional component mole fraction for a given component. In addition, the sum of the compositional component mole fractions must sum to one, otherwise an error will occur. Note that the number ZCOMP values, should be the same as that entered via the NCOMPS keyword in the PROPS section, and the COMPS keyword in the RUNSPEC section. However, for a given ZCOMP component, the mole fraction may be defaulted with I*, in which case the mole fraction is set to zero. Secondly, ZCOMP may be terminated early, in this case the undefined mole fractions will be set to zero. Finally, only the default value of two components are currently supported by OPM Flow.			None
		mole fraction	mole fraction	mole fraction	

Notes:

- 1) The keyword is followed by up to MXSTRMS records as declared on the [WELLDIMS](#) keyword in the [RUNSPEC](#) section.
- 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.96:WELLSTRE Keyword Description

Example

The following example defines how to specify a two component formulation, together with defining the names of the composition components, to be used with the [CO2STORE](#) and [GASWAT](#) options.

```
-- =====
--
-- PROPS SECTION
--
-- =====
PROPS      --
--          CONFIRM NUMBER OF COMPOSITIONAL COMPONENTS (OPM FLOW KEYWORD)
--
NCOMPS
--          2 /
--
--          DEFINE COMPOSITIONAL COMPONENTS NAMES (OPM FLOW KEYWORD)
--
CNAMES
--          'CO2'
--          'H2O' /
```

The second part of the example, defines the well stream for the above two component CO₂ water system.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
--          WELL STREAM INJECTION COMPOSITION (OPM FLOW Keyword)
--
-- WELL      -- WELL STREAM COMPOSITIONAL COMPONENT      --
-- STREAM    --          MOLE FRACTIONS                    --
WELLSTRE
'CO2STREAM' 1.000  0.000 /
/
```

Here the well stream consists of 100% CO₂ and zero water.

12.3.275 WELMOVEL – MOVE GLOBAL WELL INTO AN LGR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [WELMOVEL](#), moves a previously defined gloabal well into a previously declared Local Grid Refinement (“LGR”), in a [RESTART](#) run. The keyword should only be used in [RESTART](#) runs.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
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12.3.276 WELOPEN – DEFINE WELL AND WELL CONNECTION FLOWING STATUS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELOPEN](#) keyword defines the status of wells and well connections, and is used to open and shut previously defined wells and well connections without having to re-specify all the data on the well control keywords: [WCONPROD](#), [WCONHIST](#), [WCONINJE](#), or [WCONINJH](#) keywords. Note that the well must still be initially fully defined using the [WCONPROD](#) or [WCONINJE](#) keywords. All the aforementioned keywords are described in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well or well connection status data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	STATUS	A defined character string of length four that defines the well or well connection operational status, STATUS should be set to one of the following character strings: 1) OPEN: the well or connections are open to flow. 2) SHUT: the well or connections are closed to flow (shut-in). 3) AUTO: the well or connections are initially closed, but may be opened automatically if an economic limit is violated.			OPEN
3	I	An integer less than or equal to NX that defines the connection location in the I-direction.			Negative
4	J	An integer less than or equal to NY that defines the connection location in the J-direction.			Negative
5	K	An integer less than or equal to NZ that defines the connection location in the K-direction.			Negative
6	C1	An integer less than or equal to the total number of completions that defines the first completion in the range.			Negative
7	C2	An integer less than or equal to the total number of completions that defines the last completion in the range.			Negative

Notes:

- 1) When connections are defined using the [COMPDAT](#) keyword they are assigned individual completion numbers in the order they are defined. The connections may then be lumped into completions using the [COMPLUMP](#) keyword. The range specified by C1 and C2 refers to all the completions in this range and all the connections contained within these completions.
- 2) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.97: WELOPEN Keyword Description

If variables I, J, K, C1 and C2 are all set to a negative number or defaulted with 1* then STATUS is applied to the well and the well connections remain unchanged.

If any of the variables I, J, K, C1 and C2 are set to zero or a positive value then STATUS is applied to the defined well connections and the well status remains unchanged. The defined well connections are those with the specified I, J, K locations and that are in the completion number range specified by C1 and C2. If a zero or negative value is specified for I, J or K then this matches any value for that index. If a zero or negative value is specified for C1 or C2 respectively then no lower or upper bound is applied to the completion range.

See also the [COMPDAT](#) keyword to define a well's connections, the [COMPLUMP](#) keyword to group well connections into well completions, the [WCONPROD](#) and [WCONINJE](#) keywords to define a well's production and injections targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example defines two vertical oil wells using the [WELSPECS](#) keyword and their associated connection data.

```
-- WELL SPECIFICATION DATA
--
-- WELL      GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  SHUT   CROSS  PRESS
-- NAME      NAME              I      J    DEPTH FLUID  AREA   EQUA.  IN     FLOW   TABLE
WELSPECS
OP01        PLATFORM      14     13   1*      OIL    1*      STD    OPEN   NO     1* /
OP02        PLATFORM      28     96   1*      OIL    1*      STD    OPEN   NO     1* /
/
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    LIQ    RES    BHP    THP    VFP    VFP
-- NAME  SHUT   MODE  RATE   RATE  RATE   RATE  RATE   PRES  PRES  TABLE  ALFQ
WCONPROD
' '*    SHUT   GRUP   1*     1*     1*     1*     1*     200.0                /
/
--
--          WELL CONNECTION DATA
--
-- WELL  --- LOCATION ---  OPEN  SAT  CONN  WELL  KH    SKIN  D    DIR
-- NAME  II  JJ  K1  K2  SHUT  TAB  FACT DIA  FACT FACT FACT  PEN
COMPDAT
OP01    1*  1*   1  10  SHUT  1*   1*   0.708 1*   0.0  1*   'Z' /
OP01    1*  1*  15  30  SHUT  1*   1*   0.708 1*   0.0  1*   'Z' /
OP01    1*  1*  35  90  SHUT  1*   1*   0.708 1*   0.0  1*   'Z' /
OP02    1*  1*   1  10  SHUT  1*   1*   0.708 1*   0.0  1*   'Z' /
/
```

```
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
--      WELL WELL      --LOCATION--  COMPLETION
--      NAME STAT      I   J   K   FIRST LAST
WELOPEN
OP01      OPEN                      /
OP01      OPEN      0   0   1      /
OP01      OPEN      0   0   2      /
OP01      OPEN      0   0   3      /
OP01      OPEN      0   0   4      /
OP02      OPEN                      /
OP02      OPEN      0   0   0      /
/
```

In this example the first record for each well in the [WELOPEN](#) keyword changes the well status from shut (as per the [WCONPROD](#) keyword) to open. Then for well OP01 well connections in layers one to four are opened for flow and all the connections for well OP02 are opened, that is the connections in layers one to ten.

The next example shows the use of the [COMPLUMP](#) keyword to group the well connections into well completions for wells OP01, OP02 and OP03, and then use the [WELOPEN](#) keyword to open the well and the well completions.

```
--
--      ASSIGN WELL CONNECTIONS TO COMPLETIONS
--
--      WELL      --- LOCATION ---  COMPL
--      NAME      II  JJ  K1  K2  NO.
COMPLUMP
OP01      0   0   1  10   1      / COMPLETION NO. 01
OP01      0   0  15  30   2      / COMPLETION NO. 02
OP01      0   0  35  90   3      / COMPLETION NO. 03
OP02      0   0  15  30   2      / COMPLETION NO. 02
OP02      0   0  35  90   3      / COMPLETION NO. 03
OP03      0   0   1  10   1      / COMPLETION NO. 01
OP03      0   0  35  90   3      / COMPLETION NO. 03
/
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
--      WELL WELL      --LOCATION--  COMPLETION
--      NAME STAT      I   J   K   FIRST LAST
WELOPEN
OP01      OPEN                      /
OP01      OPEN      0   0   0      3   3      /
OP02      OPEN                      /
OP02      OPEN      0   0   0      2   2      /
OP03      OPEN                      /
OP03      OPEN      0   0   0      0   0      /
/
```

Again, the first record of each well [WELOPEN](#) keyword changes the well status from shut (as per the [WCONPROD](#) keyword) to open. Then for well OP01 well completion number three is opened (connections 35 to 90), completion two (connections 15 to 30) for well OP02 and completion numbers one to three (all the connections) for well OP03. Note that there is no completion number two for OP03 so connections 15 to 30 will not be opened.

Note the last completion number for well OP03 was named completion number three, but it could have been named number two as well. The reason why it was named number three instead of two was because it

was assumed (for the example) that layers 35 to 90 represent a particular reservoir, and therefore allowing for the tracking of completions for individual reservoirs, as shown in the example.

This example shows how one can open all the wells and well completions for a given reservoir.

```
--
--      DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
-- WELL WELL  --LOCATION--  COMPLETION
-- NAME STAT   I   J   K  FIRST LAST
WELOPEN
' * '      OPEN                               /
' * '      OPEN      0   0   0      3      3      /
OP02      SHUT      0   0   0      0      0      /
OP02      OPEN      0   0   0      2      2      /
/
```

In this case wells OP01, OP02 and OP03 are opened via completion number three. All the connections for OP02 are then shut, and the connections associated with completion two are opened instead for this well.

12.3.277 WELOPENL – DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELOPENL](#) keyword defines the status of wells and well connection in Local Grid Refinement Grids (“LGR”) and is used to open and shut previously defined well and well connections without having to re-specify all the data on the well control keywords: [WCONPROD](#), [WCONHIST](#), [WCONINJE](#), or [WCONINJH](#) keywords. Note that the well must still be initially be fully defined using the [WCONPROD](#) or [WCONINJE](#) keywords. All the aforementioned keywords are described in the [SCHEDULE](#) section

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well and well connection status data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the well LGR connection data are being defined. Note that the well name (LGRNAME) must have been declared previously using the WELSPECL keyword in the SCHEDULE section, otherwise an error may occur. If defaulted with I* the LGR on the WELSPECL keyword will be utilized.			Defined
3	STATUS	A character string of length four that defines the well and a well's connections' operational status, STATUS should be set to one of the following character strings: 1) OPEN: the connections are open to flow. 2) SHUT: the connections are closed to flow (shut-in). 3) AUTO: the connection are initially closed, but may be opened automatically if an economic limit is violated.			OPEN
4	I	An integer less than or equal to NX that defines the connection location in the I-direction.			I*
5	J	An integer less than or equal to NY that defines the connection location in the J-direction.			I*
6	K	An integer less than or equal to NZ that defines the connection location in the K-direction.			I*
7	KI	An integer less than or equal to NZ that defines the UPPER completion location in the K-direction. Connections are lumped into completions via the COMPLUMP keyword, and KI refers to the first completion number, as defined by the COMPLUMP keyword, and all the connections contained within the KI completion.			I*

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	K2	An integer less than or equal to NZ that defines the LOWER completion location in the K-direction. Connections are lumped into completions via the COMPLUMP keyword, and K2 refers to the last completion number, as defined by the COMPLUMP keyword, and all the connections contained within the K2 completion.			I*

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.98:WEOPENL Keyword Description

If variables I, J K, K1 and K2 are all set to a negative number or defaulted with I* then STATUS is applied to the well and the well connection remain unchanged.

If variables I, J K, K1 and K2 are all set to zero or a positive value then STATUS is applied to the defined connections and the well status remains unchanged. The defined connections are those values with the I, J, K specified locations and connections within the completion number range specified by K1 and K2.

See also the [COMPDAT](#) keyword to define a well's connections, the [COMPLUMP](#) keyword to group well connections into well completions, the [WCONPROD](#) and [WCONINJE](#) keywords to define a well's production and injections targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example shows the use of the [COMPLMPL](#) keyword to group the well connections into well completions for well OP01 and then use the [WEOPEN](#) keyword to open the well and the well connections.

```
--
--      ASSIGN WELL LGR CONNECTIONS TO COMPLETIONS
--
--      WELL      LGR      ---LOCATION---      COMPL
--      NAME      NAME      II  JJ  K1  K2      NO.
COMPLMPL
      OP01      LGR1      26  58   1   3       1
      OP01      LGR1      26  58   4  10       2
      OP01      LGR1      26  58  11  12       3
/
```

--	WELL PRODUCTION STATUS FOR LGR WELLS									
--	WELL	LGR	WELL	--LOCATION--			COMPLETION			
--	NAME	NAME	STAT	I	J	K	FIRST	LAST		
WELOPENL										
OP01	LGR1	OPEN								/
OP01	LGR1	OPEN	0	0	0		1	2		/
OP01	LGR2	SHUT	0	0	0		3	3		/
/										

The first record of the [WELOPENL](#) keyword changes the well status from shut (as per the [WCONPROD](#) keyword) to open, in case it has been shut-in. Then well completion number one and two are opened (connections 1 to 10), and completion number three shut-in (connections 11 to 12).

12.3.278 WELPI – DEFINE WELL PRODUCTIVITY AND INJECTIVITY INDICES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELPI](#) keyword is used to define a well's productivity or injectivity index at the time the keyword is activated. Productivity and injectivity indices are a function of both bottom-hole pressure and mobility and thus will vary in time as the bottom-hole pressure and fluids produce by the well are changing. Thus, the values enter on this keyword for a given well will override any previously calculated values, or values previously entered, using this keyword. This keyword should only be invoked after a well's connection factors have been declared via the [COMPDAT](#) keyword in the [SCHEDULE](#) section.

No.	Name	Description			Default		
		Field	Metric	Laboratory			
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well productivity index is being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None		
2	<i>WELPI</i>	A real positive value that defines the productivity or injectivity of a well at the time the keyword is activated in the input deck. Subsequent use of the keyword for a given well will override all previously entered data, including the original productivity index calculated from the well's <i>COMPDAT</i> connection data. Note that: 1) Only the connections currently <u>opened</u> will have their connection factors re-scaled to honor the productivity and injectivity index. Connections not opened will remain unchanged. 2) Additional, previously defined connections added to the well at a later time, will not be influenced by a previous <i>WELPI</i> keyword. It may therefore be necessary to add another <i>WELPI</i> keyword at the time the new connections are opened, if desired. 3) However, re-defining connections via the <i>COMPDAT</i> keyword resets the calculated productivity and injectivity index for those re-defined connections in the well. Subsequent usage of the keyword for the same well may lead to unintended values; use the WPI series of variables in the <i>SUMMARY</i> section to output and verify that the values are consistent with the desired results.					
		Liquid stb/d/psia Gas Mscf/d/psia	Liquid sm ³ /day/bars Gas sm ³ /day/bars	Liquid scc/hour/atm Gas scc/hour/atm	1.0		

Notes:
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.99: WELPI Keyword Description

Note that only a well's currently opened connection factors, as entered via the [COMPDAT](#) keyword, are re-scaled to match the required productivity and injectivity index. See section [Well Productivity](#) on well productivity for further information on well productivity.

Note

One should not modify the productivity and injectivity index of gas wells that use:

- 1) Russell Goodrich³³⁹ pressure square inflow equation, as declared via 'R-G' or 'YES' for the INFLOW parameter on the **WELSPECS** keyword in the **SCHEDULE** section, or
- 2) the general dry gas pseudo pressure inflow equation, as declared via 'P-P' for the INFLOW parameter on the **WELSPECS** keyword in the **SCHEDULE** section, or
- 3) the generalized gas pseudo pressure inflow equation, as declared via 'GPP' for the INFLOW parameter on the **WELSPECS** keyword in the **SCHEDULE** section, or
- 4) those gas wells that use the non-Darcy D factor coefficient, as declared by the DFACT property on the **COMPDAT** and **WDFAC** keywords in the **SCHEDULE** section.

See also the **WPIMULT** keyword in the **SCHEDULE** section to set the productivity and injectivity index for a well by scaling the current index by a scaling factor.

Example

```
--
--      DEFINE WELL PRODUCTIVITY/INJECTIVITY INDEX
--
-- WELL  PI
-- NAME  MULT
WELPI
OP01    1.250
OP02    2.750
OP03    1.100
/
```

In the above example the oil wells are defined as having a well productivity indices of 1.250, 2.750 and 1.100 for the OP01, OP02 and OP03 wells, respectively.

³³⁹ Russell, D.G., Goodrich, J.H., Perry, G.E and Bruskotter, J.F "Methods of Predicting Gas Well Performance", Transactions of the ASME, Journal of Petroleum Technology (1966) 99-108.

12.3.279 WELPRI – ASSIGN WELL PRIORITY

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELPRI](#) keyword is used to re-assign a priority number to a well for when the [PRIORITY](#) keyword has been used in the [SCHEDULE](#) section. The [PRIORITY](#) keyword activates the Well Priority option and defines the coefficients in the well priority equation; [WELPRI](#) keyword can be used to over write these calculated priority numbers

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.280 WELSEGS – DEFINE MULTI-SEGMENT WELLS AND THEIR SEGMENT STRUCTURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELSEGS](#) keyword defines a well to be a multi-segment well and defines the well's segment structure. Note that the well must have previously been defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section and that the [WELSEGS](#) keyword should be repeated for each multi-segment well in the model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
I-2	TOPDEP	A real value that defines the depth of the nodal point of the top segment. This is used as the reference depth for reporting the bottom-hole pressure for the multi-segment well. If the keyword is entered multiple times for the same well, due to for example the well configuration changing through time, then it is only necessary to enter this data the first time the keyword is used for a well.			None
		feet	m	cm	
I-3	TOPLEN	A real positive value that defines the length of the tubing from the tubing reference point (for example Measured Depth Relative to Kelly Bushing, MDRKB) to the nodal point of the top segment. Tubing pressures between the nodal point of the top segment and the well head are not calculated by the multi-segment well option as these are taken into account by the VFP tables allocated to the well and entered via the VFPPROD and VFPINJ keywords in the SCHEDULE section. If TOPLEN is set to zero or defaulted then the tubing length is measured from the nodal point of the top segment.			0
		feet	m	cm	
I-4	WBORVOL	A real positive value that defines the effective wellbore volume for the top segment, that is from the tubing head or wellhead at the surface to the nodal point of the top segment. The default value of 1.0×10^{-5} results in minimal wellbore storage.			1.0E-5
		ft ³	m ³	cm ³	

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-5	TUBOPT	A defined character string that specifies the type of length and depth data entered for LENGTH and DEPTH on the second record and should be set to one of the following: 1) INC: Incremental changes in values of length and depth for each segment. 2) ABS: Absolute values of length and depth for each segment. As there is no default value for TUBOPT one of the above options must be explicitly defined.			None
I-6	PRESOPT	A defined character string that specifies the pressure drop calculation used for each well segment and should be set to one of the following: 1) HFA: Sets the pressure calculation to include the hydrostatic, friction and acceleration terms. 2) HF-: Sets the pressure calculation to include the hydrostatic and friction terms only. 3) H--: Sets the pressure calculation to include the hydrostatic pressure drop term only. The default value for PRESOPT of HFA sets the pressure calculation to include the hydrostatic, friction and acceleration terms.			HFA
I-7	FLOWOPT	A defined character string that specifies the type of multi-phase calculation used for each well segment and should be set to one of the following: 1) HO: Sets the multi-phase calculation to the homogeneous model, that is all phases flow at the same velocity. 2) DF-: Sets the multi-phase calculation to the Drift Flux Slip model. OPM Flow only supports the default value of HO.			HO
I-8	TOPX	A real positive value equal to or greater than zero that defines the coordinate in the x-direction of the nodal point of the top segment that is used for display purposes only. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
I-9	TOPY	A real positive value equal to or greater than zero that defines the coordinate in the y-direction of the nodal point of the top segment that is used for display purposes only. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
I-10	XAREATH	A real positive value equal to or greater than zero that defines the cross-sectional area of the pipe wall used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			None
		ft ²	m ²	cm ²	

No.	Name	Description			Default
		Field	Metric	Laboratory	
I-11	VHEATCAP	A real positive value equal to or greater than zero that defines the volumetric heat capacity of the pipe wall used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	
I-12	THCON	A real positive value equal to or greater than zero that defines the thermal conductivity of the pipe wall used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	
I-13	/	Record terminated by a “/”			Not Applicable
2-1	ISEG1	A positive integer greater than or equal to two and less than or equal to MXSEGS on WSEGDIMS keyword in the RUNSPEC section that defines the start of the segment range.			None
2-2	ISEG2	A positive integer greater than or equal to ISEG1 on this record and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section that defines the end of the segment range.			None
2-3	IBRANCH	A positive integer greater than or equal to one and less than or equal to MXBRAN on WSEGDIMS keyword in the RUNSPEC section that defines the branch number of a segment. All segments on the main stem must have IBRANCH set to one, and those on lateral branches should have IBRANCH set to between two and MXBRAN.			None
2-4	ISEG3	A positive integer greater than or equal to one and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section that defines the outlet segment for the segment at the start of the segment range (ISEG1).			None
2-5	LENGTH	A real positive value that: 1) If TUBOPT is set to INC then LENGTH is the incremental length of the tubing for each segment in the segment range. 2) If TUBOPT is set to ABS then LENGTH is the length of the tubing from the tubing reference point to the last segment in the range. The length between the nodal point of the last segment and the outlet segment is divided equally between each of the segments in the segment range.			None
		feet	m	cm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
2-6	DEPTH	A real positive value that: 1) If TUBOPT is set to INC then DEPTH is the incremental depth change of the tubing for each segment in the segment range. 2) If TUBOPT is set to ABS then DEPTH defines the depth of the tubing at the nodal point of the last segment in the segment range. The depths for other segments in the segment range are obtained by linear interpolation.			None
		feet	m	cm	
2-7	ID	A real positive value that defines the tubing internal diameter of each segment in the segment range.			None
		feet	m	cm	
2-8	EPSILON	A real positive value that defines the tubing absolute roughness of each segment in the segment range.			None
		feet	m	cm	
2-9	XAREA	A real positive value equal to or greater than zero that defines the cross-sectional area for fluid flow. Currently this option is not supported by OPM Flow.			None
		ft ²	m ²	cm ²	
2-10	VOLSEG	VOLSEG is a real positive value that defines the effective segment volume for the this segment. Currently this option is not supported by OPM Flow.			None
		ft ³	m ³	cm ³	
2-11	XCORDS	A real positive value that: 1) If TUBOPT is set to INC then XCORDS is the incremental change in the x-coordinate of the tubing for each segment in the segment range. 2) If TUBOPT is set to ABS then XCORDS defines the x-coordinate of the tubing at the nodal point of the last segment in the segment range. The x-coordinates for other segments in the segment range are obtained by linear interpolation. Currently this option is not supported by OPM Flow.			None
		feet	m	cm	
2-12	YCORDS	A real positive value that: 1) If TUBOPT is set to INC then YCORDS is the incremental change in the y-coordinate of the tubing for each segment in the segment range. 2) If TUBOPT is set to ABS then YCORDS defines the y-coordinate of the tubing at the nodal point of the last segment in the segment range. The y-coordinates for other segments in the segment range are obtained by linear interpolation. Currently this option is not supported by OPM Flow.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
		feet	m	cm	
2-13	XAREAS	A real positive value equal to or greater than zero that defines the cross-sectional area of the pipe wall for this segment, that is used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			None
		ft ²	m ²	cm ²	
2-14	VHEATSEG	A real positive value equal to or greater than zero that defines the volumetric heat capacity of the pipe wall for this segment, that is used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	
2-15	THCSEG	A real positive value equal to or greater than zero that defines the thermal conductivity of the pipe wall for this segment, that is used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			None
		Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K	
2-16	/	Record terminated by a “/”			Not Applicable

Notes:

- Each multi-segment wells must be defined by a separate [WELSEGS](#) keyword that consists of two records with entries 1-1 to 1-12 representing record one items and 2-1 to 2-15 representing record number two items in the “No.” column in this table.
- Record number two of the keyword, items 2-1 to 2-15, can be repeated a total of up to MXSEGS-1 times as declared on the [WSEGDIMS](#) keyword in the [RUNSPEC](#) section.
- Each of the records are terminated by a “/” and is explicitly shown in the above rows and the keyword should be terminated by a “/”.

Table 12.100: WELSEGS Keyword Description

The maximum number of wells should be defined via the [WELLDIMS](#) keyword and the maximum number of multi-segment wells should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section.

See also the [WELSPECS](#) keyword to define wells, the [COMPDAT](#) keyword to define the well completions for both ordinary wells and multi-segment wells, and the [COMPSEGS](#) keyword to define multi-segment well segment completions. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines one multi-segment oil production well (OP01) using the [WELSPECS](#), [WELSEGS](#), [COMPDAT](#) and [COMPSEGS](#) keywords, and one standard water injection well (WI01) using the [WELSPECS](#) and [COMPDAT](#) keywords.

--	WELL SPECIFICATION DATA											
--												
-- WELL	GROUP	LOCATION		BHP	PHASE	DRAIN	INFLOW	OPEN	CROSS	PVT		
-- NAME	NAME	I	J	DEPTH	FLUID	AREA	EQUANS	SHUT	FLOW	TABLE		
WEL SPECS												
OP01	PLATFORM	10	10	1*	OIL						/	
WI01	PLATFORM	1	1	1*	WATER						/	
/												
--												
--	WELL CONNECTION DATA											
--												
-- WELL	---	LOCATION		---	OPEN	SAT	CONN	WELL	KH	SKIN	D	DIR
-- NAME	II	JJ	K1	K2	SHUT	TAB	FACT	DIA	FACT	FACT	FACT	PEN
COMP DAT												
OP01	10	10	1	1	OPEN	1*	200.	0.5				/
OP01	10	10	2	2	OPEN	1*	200.	0.5				/
OP01	10	10	3	3	OPEN	1*	200.	0.4				/
OP01	10	10	4	4	OPEN	1*	200.	0.4				/
OP01	10	10	5	5	OPEN	1*	200.	0.4				/
OP01	10	10	6	6	OPEN	1*	200.	0.4				/
OP01	9	10	2	2	OPEN	1*	200.	0.4				/
OP01	8	10	2	2	OPEN	1*	200.	0.4				/
OP01	7	10	2	2	OPEN	1*	200.	0.4				/
OP01	6	10	2	2	OPEN	1*	200.	0.4				/
OP01	5	10	2	2	OPEN	1*	200.	0.4				/
OP01	10	9	3	3	OPEN	1*	200.	0.4				/
OP01	10	8	3	3	OPEN	1*	200.	0.4				/
OP01	10	7	3	3	OPEN	1*	200.	0.4				/
OP01	10	6	3	3	OPEN	1*	200.	0.4				/
OP01	10	5	3	3	OPEN	1*	200.	0.4				/
OP01	9	10	5	5	OPEN	1*	200.	0.4				/
OP01	8	10	5	5	OPEN	1*	200.	0.4				/
OP01	7	10	5	5	OPEN	1*	200.	0.4				/
OP01	6	10	5	5	OPEN	1*	200.	0.4				/
OP01	5	10	5	5	OPEN	1*	200.	0.4				/
OP01	10	9	6	6	OPEN	1*	200.	0.4				/
OP01	10	8	6	6	OPEN	1*	200.	0.4				/
OP01	10	7	6	6	OPEN	1*	200.	0.4				/
OP01	10	6	6	6	OPEN	1*	200.	0.4				/
OP01	10	5	6	6	OPEN	1*	200.	0.4				/
WI01	1	1	7	9	OPEN	1*	200.	0.5				/
/												
--												
--	WELL SEGMENT SPECIFICATION DATA											
--												
-- WELL	NODAL	LEN		WELL	DEPH	PRESS	FLOW					
-- NAME	DEPTH	TUBING		VOLM	OPTN	CALC	MODEL					
WEL SEGS												
OP01	2512.5	2512.5		1.0E-5	ABS	HFA	H0					
--												
--	SEG	SEG	BRAN	SEG	TUBING	NODAL	TUBE	TUBE	XSEC	VOL		
--	ISTR	IEND	NO	NO	LENGTH	DEPTH	ID	ROUGH	AREA	SEG		
	2	2	1	1	2537.5	2534.5	0.3	0.00010			/	
	3	3	1	2	2562.5	2560.5	0.3	0.00010			/	
	4	4	1	3	2587.5	2593.5	0.3	0.00010			/	
	5	5	1	4	2612.5	2614.5	0.3	0.00010			/	
	6	6	1	5	2637.5	2635.5	0.3	0.00010			/	

```

/
--
--      COMPLETION SEGMENT SPECIFICATION DATA
--
-- WELL
-- NAME
COMPSEGS
OP01
/
--
--      --LOCATION--  BRAN  START  END      DIR  LOC  MID  COMP  ISEG
--      II  JJ  K1  NO  LENGTH LENGTH  PEN  I, J, K  PERFS  LENGTH  NO.
--
10  10  1  1  2512.5  2525.0
10  10  2  1  2525.0  2550.0
10  10  3  1  2550.0  2575.0
10  10  4  1  2575.0  2600.0
10  10  5  1  2600.0  2625.0
10  10  6  1  2625.0  2650.0
/
9  10  2  2  2637.5  2837.5
8  10  2  2  2837.5  3037.5
7  10  2  2  3037.5  3237.5
6  10  2  2  3237.5  3437.5
5  10  2  2  3437.5  3637.5
/
10  9  3  3  2662.5  2862.5
10  8  3  3  2862.5  3062.5
10  7  3  3  3062.5  3262.5
10  6  3  3  3262.5  3462.5
10  5  3  3  3462.5  3662.5
/
9  10  5  4  2712.5  2912.5
8  10  5  4  2912.5  3112.5
7  10  5  4  3112.5  3312.5
6  10  5  4  3312.5  3512.5
5  10  5  4  3512.5  3712.5
/
10  9  6  5  2737.5  2937.5
10  8  6  5  2937.5  3137.5
10  7  6  5  3137.5  3337.5
/

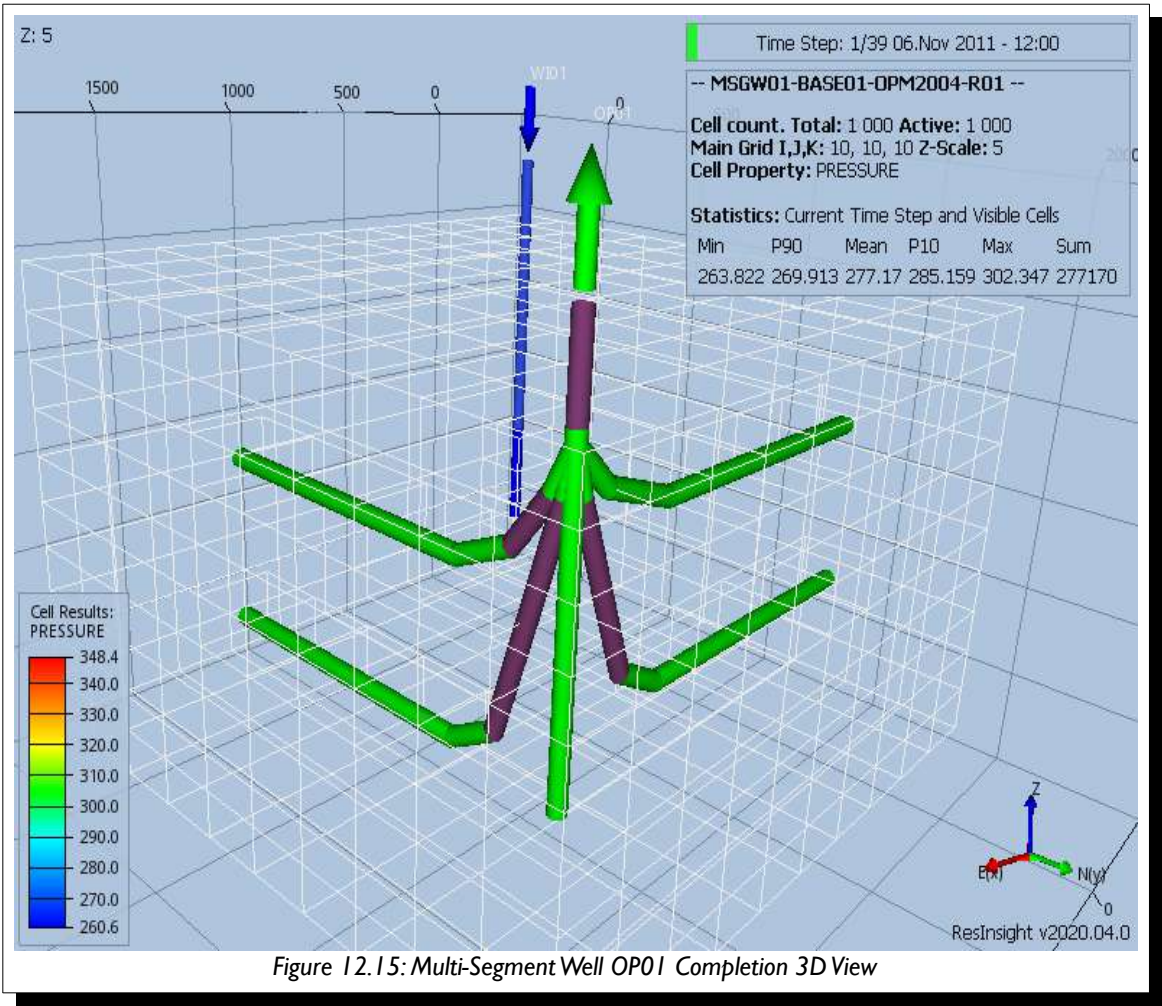
```

10	6	6	5	3337.5	3537.5	/
10	5	6	5	3537.5	3737.5	/

/

Note the use of both the [COMPDAT](#) and [COMPSEGS](#) keywords to fully define a multi-segment well's completion.

Finally, Figure 12.15 depicts the resulting well configuration for both wells, with the conventional water injection well shown in blue and the multi-segment oil producer shown in green.



12.3.281 WELSOMIN – DEFINE WELL CONNECTION MINIMUM OIL SATURATION FOR OPENING

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

WELSOMIN defines a minimum oil saturation for a well connection above which the connection will be opened automatically. If the grid block connection is below WELSOMIN then connection will not be automatically opened. Automatic opening of connection is controlled by the STATUS parameter on the COMPDAT keyword in the SCHEDULE section. Note that if the COMPLUMP keyword in the SCHEDULE section has been used to lump connections into completions then WELSOMIN is compared to the average oil saturation of the completion.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.282 WELSPECL – DEFINE WELL SPECIFICATIONS FOR LOCAL GRID REFINEMENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [WELSPECL](#) keyword defines the general well specification data for all well types and must be used for all wells contained within a Local Grid Refinement (“LGR”) instead of the [WELSPECS](#) keyword. [WELSPECL](#) must declare wells first before any other [LGR](#) well specification keywords are used in the input file. The keyword declares the name of well, the group the well belongs to, the [LGR](#) the well is incorporated into, the wellhead location and other key parameters.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well specification data is being defined.			None
2	GRPNAME	A character string of up to eight characters in length that defines the group name for which the well is assigned to. The group named FIELD is the top most group and thus GRPNAME cannot be set to FIELD, although this is allowed in the commercial compositional simulator but not the commercial black-oil simulator. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy. Secondly, groups defined by the GRUPTREE keyword cannot contain other groups <u>and</u> wells; that is, groups must either contain other groups or wells but not both. If necessary, wells can be re-allocated to a different group by re-entering a well's WELSPECS data together with a new value for GRPNAME.			None
3	LGRNAME	A character string of up to eight characters in length that defines the name of the local grid refinement for which the well is assigned to.			None
4	I	A positive integer greater than or equal to zero and less than or equal to NX on the CARFIN keyword for Cartesian grids, that defines the wellhead location for a vertical or deviated well, or the heel for a horizontal well in the I-direction within the LGR. For radial LGRs this parameter should be set to one.			None
5	J	A positive integer greater than or equal to zero and less than or equal to NY on the CARFIN keyword for Cartesian grids, that defines the wellhead location for a vertical or deviated well, or the heel for a horizontal well in the J-direction within the LGR. For radial LGRs this parameter should be set to one.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	BHPREF	A real value that defines the reference depth for reporting the bottom-hole pressure for the well. Ideally this value should be set to the midpoint of the perforations as defined by the COMPDATL keyword in the SCHEDULE section. If defaulted by I* or set to a value less than or equal to zero, then the mid-point of shallowest connection defined by the COMPDATL keyword will be used.			Mid-point of shallowest connection defined by the COMPDATL keyword
		feet	m	cm	
7	TYPE	A defined character that defines the “main” phase for the well, and should be set to one of the following character strings: 1) GAS: for a gas well. 2) OIL: for an oil well. 3) WAT: for a water injection well. 4) LIQ: for an oil well when the liquid productivity index is required for the well. This parameter defines the phase used to calculate a well’s productivity or injectivity index and the type of well, or a well’s connection, to close when a group’s production constraints, as defined on the GCONPROD keyword in the SCHEDULE section, have been violated. For example, if the well is declared as an oil well, then excessive gas and water connections will be subject to closure.			None
8	DRADIUS	A real value that defines the well drainage radius for the well used to calculate a well’s productivity or injectivity index. A default of zero results in the pressure equivalent radius of the grid blocks containing the well connections are used.			0.0
		feet	m	cm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
9	INFLOW	A defined character string that defines the inflow equation to be used for the well in calculating the well's flow rates. INFLOW should be set to one of the following character strings: 1) STD: the standard inflow equation will be used. This is normally used for wells that are primary oil or water wells. 2) NO: an alias for STD. 3) R-G: the Russell Goodrich³⁴⁰ pressure square inflow equation will be used. This option can be used for dry gas wells. 4) YES: an alias for R-G. 5) P-P: the general dry gas pseudo pressure inflow equation will be used. Normally used for dry gas wells. 6) GPP: the generalized gas pseudo pressure inflow equation used with wet gas wells, that is condensate gas wells. This inflow equation is based on the formulation of Whitson et al.³⁴¹ For oil and water wells the INFLOW should be set to STD, why for dry gas wells INFLOW can be set to either R-G or P-P; however, the P-P option is preferred for dry gas wells due to the more rigorous treatment of gas flow. For wet gas wells, that is gas condensate wells, INFLOW should be set to GPP. Only INFLOW equal to STD and NO are currently implemented in OPM Flow.			STD
10	AUTO	A defined character string that defines the automatic action to be taken if the economic WCUT, GOR, or WGR limits are violated and the well is to cease production. AUTO should be set to one of the following character strings: 1) STOP: the well is "stopped" at the surface and will not produce any fluids to surface; however, if there any open connections then flow may occur within the wellbore and between the open connections depending on a connection's potential with respect to all the other connections. Inter-connection flow (cross flow) can be prevented by setting the XFLOW variable to NO. In this case the well's behavior will be similar to the SHUT option described below. 2) SHUT: the well is shut at the surface and downhole, this results in no flow at the surface and no cross flow downhole. The corrective action takes places at the end of the time step in which the constraint is violated.			SHUT

³⁴⁰ Russell, D.G., Goodrich, J.H., Perry, G.E and Bruskotter, J.F "Methods of Predicting Gas Well Performance", Transactions of the ASME, Journal of Petroleum Technology (1966) 99-108.

³⁴¹ Whitson, C. H. and Fevang, Ø. "Generalised Pseudopressure Well Treatment in Reservoir Simulation," Presented at the IBC Technical Services Conference on Optimisation of Gas Condensate Fields, Aberdeen, UK (June 26-27, 1997).

No.	Name	Description			Default
		Field	Metric	Laboratory	
11	XFLOW	A defined character string that defines the if cross flow should occur within the wellbore, and should be set to either: 1) YES: to allow cross flow in the wellbore through well connections. 2) NO: to disallow cross flow within the wellbore, even if the flow potentials in the well connections would allow such flow to occur. In some cases numerical issues can occur if this variable is set to YES, and resetting it to NO may resolve the issue; however the results may not represent the physical process in this case.			YES
12	PVTNUM	A positive integer greater than or equal to zero that defines the PVT table used to calculate the wellbore fluid properties that define the relationship between reservoir and surface volume rates. The default value of zero sets <i>PVTNUM</i> to be the PVT table of the deepest connection in the well.			0
13	DENOPT	A defined character string that sets the type of density calculation used in calculating the wellbore hydrostatic head, and should be set to one of the following character strings: 1) SEG: sets the hydrostatic head density calculation to segmented. In this cases the density is calculated between neighboring well connections and the volumes flowing from the connections. This is the more accurate calculation if the fluid properties flowing from the well connections are variable. The density calculation itself is explicit, i.e. uses the flowing volumes of the last time step. 2) AVG: sets the hydrostatic head density calculation to the average density calculation. Here the density is considered uniform across a given reservoir and is dependent on total inflow rates of each phase and the well's bottom-hole pressure The default option of I* invokes the SEG option and is the only option implemented in OPM Flow.			SEG
14	FIPNUM	An integer value defines the <i>FIPNUM</i> region used to determine the reservoir conditions in calculating the well's reservoir volumes. If set to a negative integer value then the <i>FIPNUM</i> region of the deepest connection in the well will be used. If set to zero, the default value, then the average properties for the field will be used. If set to an integer value greater than zero, then the <i>FIPNUM</i> indicated by this value will be used.			0
15		Not used.			
16		Not used.			
17		Not used.			
18		Not used.			

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”. 2) Note that the total number of individual wells declared by the <i>WELSPECS</i> cannot exceed the valued entered via the MXWELS variable on the <i>WELLDIMS</i> keyword in the <i>RUNSPEC</i> section.					

Table 12.101: WELSPECL Keyword Description

See also the [COMPDATL](#) keyword to define a well's connections in a [LGR](#), the [WCONPROD](#) and [WCONINJE](#) keywords to define a well's production and injections targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines three wells using the [WELSPECL](#) keyword

```
--
-- WELL SPECIFICATION DATA FOR LGR WELLS
--
-- WELL   GROUP   LGR   LOCATION  BHP   PHASE  DRAIN  INFLOW  SHUT  CROSS  PVT
-- NAME   NAME    NAME   I      J    DEPTH FLUID  AREA   EQUA.  IN     FLOW  TABLE
WELSPECL
GI01     PLATFORM LGR01  14    13    1*    GAS    1*     P-P    SHUT   NO     1* /
GP01     PLATFORM LGR01  64    80    1*    GAS    1*     GPP    SHUT   NO     1* /
OP01     PLATFORM LGR02  24    10    1*    OIL    1*     STD    SHUT   NO     1* /
/
```

Here, well GI01 and GP01 are in the same [LGR](#) named LGR01 and OP01 is in a separate [LGR](#) named LGR02. GI01 is a dry gas injection well that uses the dry gas pseudo inflow equation, GP01 is a gas condensate well that uses the generalized gas pseudo pressure inflow equation, and finally, OP01 is an oil well that uses the standard inflow equation. All wells: will be shut if they are required to cease production, all wells disallow cross flow, and the hydrostatic head calculation is defaulted to the segment option for all wells.

12.3.283 WELSPECS – DEFINE WELL SPECIFICATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WELSPECS](#) keyword defines the general well specification data for all well types, and must be used for all wells before any other well specification keywords are used in the input file. The keyword declares the well, the name of the well, the group the well initial belongs to, the wellhead location and other key parameters.

OPM Flow also supports the use of the [WELSPECS](#) keyword to change the properties of one or more wells previously specified using the [WELSPECS](#) keyword. In particular, this supports changing the controlling group (GRPNAME) without affecting any other well properties such as the location of the well head (I, J) or the well reference depth (BHPREF), since defaulted properties are not changed in this case. However, the well reference depth can be re-defaulted by specifying a negative value for BHPREF. The wells to be changed can be specified using a well name, well name pattern or well list (WELNAME).

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well specification data are being defined.			None
2	GRPNAME	A character string of up to eight characters in length that defines the group name for which the well is assigned to. The group named FIELD is the top most group. GRPNAME can be set to FIELD although this is discouraged and a warning message will be issued. This is allowed in the commercial compositional simulator but not the commercial black-oil simulator. The FIELD group cannot contain both groups and wells. Note that the group hierarchy should be defined by the GRUPTREE keyword when there is more than one level of groups, otherwise all the groups will sit directly under the FIELD group in the group tree hierarchy. Secondly, groups defined by the GRUPTREE keyword cannot contain other groups and wells; that is, groups must either contain other groups or wells but not both. If necessary, wells can be re-allocated to a different group by re-entering a well's WELSPECS data together with a new value for GRPNAME.			None
3	I	A positive integer greater than or equal to zero and less than or equal to NX that defines the wellhead location for a vertical or deviated well, or the heel for a horizontal well in the I-direction. For wells being specified with the COMPTRAJ and WELTRAJ keywords in SCHEDULE section, that allow for an alternative manner to define the well connections to the simulation grid blocks, this parameter should be defaulted with I*. Since the simulator will calculate the wellhead location from the trajectory data on the WELTRAJ keyword. Note that the COMPTRAJ and WELTRAJ keywords are OPM Flow specific keywords, and will cause an error in the commercial simulator.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	J	A positive integer greater than or equal to zero and less than or equal to NY that defines the wellhead location for a vertical or deviated well, or the heel for a horizontal well in the J-direction. For wells being specified with the COMPTRAJ and WELTRAJ keywords in SCHEDULE section, that allows for an alternative manner to define the well connections to the simulation grid blocks, this parameter should be defaulted with 1*. Since the simulator will calculate the wellhead location from the trajectory data on the WELTRAJ keyword. Note that the COMPTRAJ and WELTRAJ keywords are OPM Flow specific keywords, and will cause an error in the commercial simulator.			None
5	BHPREF	A real value that defines the reference depth for reporting the bottom-hole pressure for the well. Ideally this value should be set to the midpoint of the perforations as defined by the COMPDAT keyword in the SCHEDULE section. If defaulted by 1* or set to a value less than or equal to zero, then the mid-point of shallowest connection defined by the COMPDAT keyword will be used.			Mid-point of shallowest connection defined by the COMPDAT keyword
		feet	m	cm	
6	TYPE	A defined character string that defines the “main” phase for the well, and should be set to one of the following character strings: 1) GAS: for a gas well. 2) OIL: for an oil well. 3) WAT: for a water injection well. 4) LIQ: for an oil well when the liquid productivity index is required for the well. This parameter defines the phase used to calculate a well’s productivity or injectivity index and the type of well, or a well’s connection, to close when a group’s production constraints, as defined on the GCONPROD keyword in the SCHEDULE section, have been violated. For example, if the well is declared as an oil well, then excessive gas and water connections will be subject to closure. Note OPM Flow only currently supports options one to three, that is option four (LIQ) is not supported. For producing wells this mostly matters if one plots the WPI summary vector (productivity index for well’s preferred phase). In the current treatment WPI will not have contributions from the water phase if the declared preferred phase is LIQ. For injecting wells WELSPECS’s preferred phase does not matter, since the preferred phase is (typically) reset to the injected phase via the WCONINJE and WCONINJH keywords.			None
7	DRADIUS	A real value that defines the well drainage radius for the well used to calculate a well’s productivity or injectivity index. A default of zero results in the pressure equivalent radius of the grid blocks containing the well connections are used.			0.0
		feet	m	cm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	INFLOW	A defined character string that defines the inflow equation to be used for the well in calculating the well's flow rates. INFLOW should be set to one of the following character strings: 1) STD: the standard inflow equation will be used. This is normally used for wells that are primary oil or water wells. 2) NO: an alias for STD. 3) R-G: the Russell Goodrich³⁴² pressure square inflow equation will be used. This option can be used for dry gas wells. 4) YES: an alias for R-G. 5) P-P: the general dry gas pseudo pressure inflow equation will be used. Normally used for dry gas wells. 6) GPP: the generalized gas pseudo pressure inflow equation used with wet gas wells, that is condensate gas wells. This inflow equation is based on the formulation of Whitson et al.³⁴³ For oil and water wells the INFLOW should be set to STD, whereas for dry gas wells INFLOW can be set to either R-G or P-P; however, the P-P option is preferred for dry gas wells due to the more rigorous treatment of gas flow. For wet gas wells, that is gas condensate wells, INFLOW should be set to GPP. Only INFLOW equal to STD and NO are currently implemented in OPM Flow.			STD
9	AUTO	A defined character string that specifies the action to be taken if the well is automatically closed/shut by the simulator, and should be set to one of the following: 1) STOP: the well is "stopped" at the surface and will not produce any fluids to surface; however, if there any open connections then flow may occur within the wellbore and between the open connections depending on a connection's potential with respect to all the other connections. Inter-connection flow (cross flow) can be prevented by setting the XFLOW variable to NO. In this case the well's behavior will be similar to the SHUT option described below. 2) SHUT: the well is shut at the surface and downhole, this results in no flow at the surface and no cross flow downhole. The corrective action takes places at the end of the time step in which the constraint is violated.			SHUT
10	XFLOW	A defined character string that defines the if cross flow should occur within the wellbore, and should be set to either: 1) YES: to allow cross flow in the wellbore through well connections. 2) NO: to disallow cross flow within the wellbore, even if the flow potentials in the well connections would allow such flow to occur. In some cases numerical issues can occur if this variable is set to YES, and resetting it to NO may resolve the issue; however, the results may not represent the physical down hole process in this case.			YES

³⁴² Russell, D.G., Goodrich, J.H., Perry, G.E and Bruskotter, J.F "Methods of Predicting Gas Well Performance", Transactions of the ASME, Journal of Petroleum Technology (1966) 99-108.

³⁴³ Whitson, C. H. and Fevang, Ø. "Generalised Pseudopressure Well Treatment in Reservoir Simulation," Presented at the IBC Technical Services Conference on Optimisation of Gas Condensate Fields, Aberdeen, UK (June 26-27, 1997).

No.	Name	Description			Default
		Field	Metric	Laboratory	
11	PVTNUM	A positive integer greater than or equal to zero that defines the PVT table used to calculate the wellbore fluid properties that define the relationship between reservoir and surface volume rates. The default value of zero sets PVTNUM to be the PVT table of the deepest connection in the well.			0
12	DENOPT	A defined character string that sets the type of density calculation used in calculating the wellbore hydrostatic head, and should be set to one of the following character strings: 1) SEG: sets the hydrostatic head density calculation to segmented. In this cases the density is calculated between neighboring well connections and the volumes flowing from the connections. This is the more accurate calculation if the fluid properties flowing from the well connections are variable. The density calculation itself is explicit, i.e. uses the flowing volumes of the last time step. 2) AVG: sets the hydrostatic head density calculation to the average density calculation. Here the density is considered uniform across a given reservoir and is dependent on total inflow rates of each phase and the well's bottom-hole pressure The default option of I* invokes the SEG option and is the only option implemented in OPM Flow.			SEG
13	FIPNUM	An integer value defines the FIPNUM region used to determine the reservoir conditions in calculating the well's reservoir volumes and is determined by: 1) If set to a negative integer value then the FIPNUM region of the deepest connection in the well will be used. 2) If set to zero, the default value, then the average properties for the field will be used. 3) If set to an integer value greater than zero, then the FIPNUM indicated by this value will be used.			0
14	STRMLIN1	Not used and should be defaulted with I*.			I*
15	STRMLIN2	Not used and should be defaulted with I*.			I*
16	TYPECOMP	Commercial compositional simulator well type model option that is not used and should be defaulted with either STD or I*.			STD
17	POLYTAB	A positive integer greater than or equal to zero that defines the polymer mixing table, as defined by the PLMIXPAR and PLYMAX keywords, to be used in calculating the well's well bore properties. The default value of zero means the table allocated via the PLMIXNUM array for the deepest connection in the well bore is utilized. Only the default value of zero is supported by OPM Flow.			0
Notes:					
1) The keyword is followed by any number records with each record terminated by a "/" and the keyword should be terminated by a "/".					
2) Note that the total number of individual wells declared by the WELSPECS cannot exceed the valued entered via the MXWELS variable on the WELLDIMS keyword in the RUNSPEC section.					

Table 12.102: WELSPECS Keyword Description

See also the [COMPDAT](#) keyword to define a well's connections, the [WCONPROD](#) and [WCONINJE](#) keywords to define a well's production and injection targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines three wells using the [WELSPECS](#) keyword

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME        I      J  DEPTH  FLUID   AREA   EQUANS SHUT  FLOW  TABLE
WELSPECS
GI01    PLATFORM    14    13   1*     GAS    1*     P-P    SHUT   NO    1*     /
GP01    PLATFORM    64    80   1*     GAS    1*     GPP    SHUT   NO    1*     /
OP01    PLATFORM    24   110   1*     OIL    1*     STD    SHUT   NO    1*     /
/
```

Here, well GI01 is a dry gas injection well that uses the dry gas pseudo inflow equation, GP01 is a gas condensate well that uses the generalized gas pseudo pressure inflow equation, and finally, OP01 is an oil well that uses the standard inflow equation. All wells will be shut if they are required to cease production, all wells disallow cross flow, and the hydrostatic head calculation is defaulted to the segment option for all wells.

If the same three wells are using the [COMPTRAJ](#) and [WELTRAJ](#) keywords to specify the connections to the simulation grid, then the [WELSPECS](#) keyword should be;

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME        I      J  DEPTH  FLUID   AREA   EQUANS SHUT  FLOW  TABLE
WELSPECS
GI01    PLATFORM    1*    1*   1*     GAS    1*     P-P    SHUT   NO    1*     /
GP01    PLATFORM    1*    1*   1*     GAS    1*     GPP    SHUT   NO    1*     /
OP01    PLATFORM    1*    1*   1*     OIL    1*     STD    SHUT   NO    1*     /
/
```

Notice how the well location parameters have been defaulted with I* in this case.

The following example illustrates how the [WELSPECS](#) keyword can be used to change the controlling group for oil production wells previously specified by the [WELSPECS](#) keyword that have an oil production rate less than 100.

```
--
--      ACTIONX BLOCK
--
ACTIONX
ACT01 1 /
WOPR 'OP*' < 100.0 /
/
WELSPECS
'?' 'LOWPRESS' /
/
ENDACTIO
```

12.3.284 WELTARG – MODIFY WELL TARGET AND CONSTRAINT VALUES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELTARG](#) keyword modifies the target and constraints values of both rates and pressures for previously defined wells without having to define all the variables on the well control keywords: [WCONPROD](#), [WCONHIST](#), [WCONINJE](#), or [WCONINJH](#). Variables not changed by the [WELTARG](#) keyword remain the same as those previously entered via the well control keywords or previously entered [WELTARG](#) keywords. Note that the well must still be initially fully defined using the [WCONPROD](#) or [WCONINJE](#) keywords. All the aforementioned keywords are described in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well production rates and pressures data are being redefined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS and WCONPROD (or WCONINJE) keywords in the SCHEDULE section, otherwise an error may occur.			None
2	TARGET	A defined character string that sets the item to be changed for the well the value of the item is set by item (3). 1) ORAT: reset the surface oil production rate value as defined by item (3). 2) WRAT: reset the surface water production rate value as defined by item (3). 3) GRAT: reset the surface gas production rate value as defined by item (3). 4) LRAT: reset the surface liquid (oil plus water) production rate value as defined by (3). 5) CRAT: reset the linearly combined calculated rate value as defined by (3). <u>This option is not supported.</u> 6) RESV: reset the in situ reservoir volume rate value as defined by (3). 7) BHP: reset the bottom-hole pressure value as defined by item (3). 8) THP: reset the tubing head pressure value for the well as defined by item (3). 9) VFP: reset the vertical lift performance table number as defined by (3). 10) LIFT: reset the artificial lift quantity for use with vertical lift performance tables. 11) GUID: reset the guide rate value for wells operating under group control. The commercial compositional simulator options: WGRA, NGL, CVAL, REIN, STRA, SATP and SATT are not applicable. Note that TARGET only defines the variable to be changed, it does not change how a well is controlled. For example, if a well is operating on ORAT control, as defined by the previously entered WCONPROD keyword, entering TARGET equal to LRAT with a value, changes the liquid constraint but the well still remains on ORAT control. Use the WELCNTL keyword in the SCHEDULE section to change the control mode of a well.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	VALUE	A real positive value that defines the value of the variable declared by TARGET This value may be specified using a User Defined Argument (UDA).			None
	Liquid	stb/d	sm ³ /day	scc/hour	
	Gas	Mscf/d	sm ³ /day	scc/hour	
	Res Vol	rb/d	rm ³ /day	rcc/hour	
	Pressure	psia	barsa	atma	
	VFP	dimensionless	dimensionless	dimensionless	
	LIFT	same as <i>VFPPROD</i> or <i>VFPINJ</i>	same as <i>VFPPROD</i> or <i>VFPINJ</i>	same as <i>VFPPROD</i> or <i>VFPINJ</i>	

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the *SCHEDULE* section. User Defined Arguments are activated using the *UDADIMS* keyword in the *RUNSPEC* section, and defined by the *UDQ* keyword in the *SCHEDULE* section.
- 2) The keyword is followed by any number records with each record terminated by a "/" and the keyword should be terminated by a "/"

Table 12.103:WELTARG Keyword Description

If a well is currently a history matching well, then *WELTARG* should only be used to change a well's bottom-hole pressure limit, vertical flow performance table number or the artificial lift quantity.

See also the *WELCNTL* keyword, in the *SCHEDULE* section that can be used to reset the control mode, as well as a well's target and constraints of both rates and pressures.

Example

The following example below shows the oil rates for the OP01 oil producer at the start of the schedule section (January 1, 2000).

```

-----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-----
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL  OIL   WAT   GAS  LIQ   RES   BHP   THP   VFP   VFP
-- NAME SHUT   MODE  RATE  RATE  RATE RATE  RATE  PRES  PRES  TABLE ALFQ
WCONPROD
OP01    OPEN   ORAT   3000  1*   1*   1*   1*   750.0  500.  9     1* /
/
DATES
01 FEB 2000 /
/
--
--          WELL PRODUCTION AND INJECTION TARGETS
--
-- WELL WELL  TARGET
-- NAME TARG  VALUE
WELTARG
OP01    ORAT   2000
/

```

/

From January 1, 2000 to February 1, 2000 well OP01 is open and is on oil rate control and has a target oil rate of 3,000 stb/d, and uses [VFPPROD](#) vertical lift table number 9 with a minimum tubing head pressure constraint of 500 psia. After February 1, 2000 the well's oil rate is reduced to 2,000 stb/d and all the other parameters remain unchanged.

12.3.285 WELTRAJ – DEFINE WELL TRAJECTORY DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WELTRAJ](#) keyword defines a trajectory well together with the well trajectory data (simplified directional survey data), and is used in conjunction with the [COMPTRAJ](#) keyword in the [SCHEDULE](#) section to define the well connections to the simulation grid blocks. The keyword can only be used for trajectory wells that employ the [COMPTRAJ](#) keyword to define the connections to the grid, that is, one cannot use [COMPDAT](#) keyword in the [SCHEDULE](#) section for declaring the connections to the grid for these type of wells.

Although [WELTRAJ](#) and [COMPTRAJ](#) keywords are sufficient to define the wellbore path and connections to the grid, it is still necessary to defined the general well specification parameters using the [WELSPECS](#) keyword in the [SCHEDULE](#) section. In this case, the wellhead location parameters, [WELSPECS](#)(I, J), should be defaulted with I*.

Note

This is an OPM Flow specific keyword and will therefore cause an error in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well trajectory data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur. Secondly, the wellhead location parameters on the WELSPECS keyword, WELSPECS(I, J), should be defaulted with I* for trajectory wells, as the well location will be calculated by the simulator.			None
2	IBRANCH	A positive integer greater than or equal to one and less than or equal to MXBRAN on WSEGDIMS keyword in the RUNSPEC section that defines the branch number of a segment. All segments on the main stem must have IBRANCH set to one and lateral branches should have values between two and MXSEGS on the WSEGDIMS keyword in the RUNSPEC section. Only the default value of one is currently supported, that is only the main branch of a multi-segment well is supported, or a single trajectory for a conventional well.			1
3	XCORD	A real positive value representing the X coordinate in three dimensional space of the well trajectory path. The Geodesy Reference System should be same as the static model used to generate the dynamic model.			None
		feet	m	cm	
4	YCORD	A real positive value representing the Y coordinate in three dimensional space of the well trajectory path. Again, the Geodesy Reference System should be same as the static model used to generate the dynamic model.			None
		feet	m	cm	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	TVD	A real negative or positive value that defines the True Vertical Depth along the wellbore path at XCORD and YCORD. Normally, TVD is referenced to the subsea elevation, that is TVDSS.			None
		feet	m	cm	
6	MD	A real positive value that defines the equivalent measured depth along the wellbore path at TVD. Normally, MD is referenced to a reference point on the drilling rig that drilled the well, for example, Relative to Kelly Bushing, MDRKB.			None
		feet	m	cm	

Notes:

- 1) For this keyword the Laboratory set of units are not supported.
- 2) The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.
- 3) Note that the total number of individual wells declared by the [WELSPECS](#) cannot exceed the valued entered via the MXWELS variable on the [WELLDIMS](#) keyword in the [RUNSPEC](#) section.

Table 12.104: WELTRAJ Keyword Description

Using the [WELTRAJ](#) and [COMPTRAJ](#) keywords to define wells and how they are connected to grid, offers several advantages compared to the conventional approach based on the (I, J, K) co-ordinates of the grid. The approach allows for the wells to be independent of the grid, which is particularly useful when running ensemble cases, as the well connections are no longer required to be re-calculated for each ensemble case. In addition, quality control of the model is improved by using consistent perforation data in both the static and dynamic models.

See the example on the following page.

Example

The following example defines two trajectory wells oil wells, OP01 and OP02, using the [WELSPECS](#) and [WELTRAJ](#) keywords, together with their perforations using the [COMPTRAJ](#) keyword.

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME        I      J  DEPTH  FLUID   AREA  EQUANS  SHUT  FLOW   TABLE
WELSPECS
OP01    PLATFORM    1*    1*    1*      OIL    1*      STD    SHUT    NO    1*    /
OP02    PLATFORM    1*    1*    1*      OIL    1*      STD    SHUT    NO    1*    /
/

--
--      WELL TRAJECTORY DATA
--
-- WELL  BRAN  XCORD      YCORD      TVDSS      MD
-- NAME  NO    DEPTH      DEPTH      DEPTH      DEPTH
-----
WELTRAJ
OP01    1*      2.805445e+06  3.602948e+06  -100.000000  0.0    /
OP01    1*      2.805445e+06  3.602948e+06  877.0000000  977.0    /
OP01    1*      2.805445e+06  3.602948e+06  957.9950240  1058.0   /
OP01    1*      2.805444e+06  3.602946e+06  1051.976081  1152.0   /
.....
OP02    1*      2.810828e+06  3.604507e+06  9371.792711  11418.0  /
OP02    1*      2.810885e+06  3.604525e+06  9443.657000  11511.0  /
OP02    1*      2.810952e+06  3.604546e+06  9531.966162  11624.0  /
OP02    1*      2.810973e+06  3.604553e+06  9560.411742  11660.0  /
/

--
--
--      WELL TRAJECTORY CONNECTION DATA
--
-- WELL  BRAN  -- PERFORATION -- COMPL  OPEN  SAT  CONN  WELL  KH  SKIN  D
-- NAME  NO.   TOP    BOT    REF  NO.   SHUT  TAB  FACT  DIA  FACT  FACT  FACT
COMPTRAJ
OP01    1*      8230   8244   MD    1     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP01    1*      8352   8380   MD    1     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP01    1*      9070   9100   MD    1     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP01    1*      9220   9250   MD    2     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP01    1*      9266   9280   MD    2     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP01    1*      9693   9703   MD    3     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP01    1*      9940   9974   MD    3     SHUT  1*  1*    0.708  1*    0.0   1*  /
OP02    1*      9979   9985   TVD    1*    SHUT  1*  1*    0.708  1*    0.0   1*  /
OP02    1*     10173  10183  TVD    1*    SHUT  1*  1*    0.708  1*    0.0   1*  /
OP02    1*     10190  10204  TVD    1*    SHUT  1*  1*    0.708  1*    0.0   1*  /
OP02    1*     10327  10333  TVD    1*    SHUT  1*  1*    0.708  1*    0.0   1*  /
OP02    1*     10339  10345  TVD    1*    SHUT  1*  1*    0.708  1*    0.0   1*  /
OP02    1*     11528  11538  TVD    1*    SHUT  1*  1*    0.708  1*    0.0   1*  /
```

Here, well OP01 has eight perforation intervals, with the intervals one to three grouped into one completion, perforation intervals four to five grouped into completion number two, and finally the bottom three perforations are grouped into completion number three. In contrast, OP02 has six perforated intervals with their completion interval defaulted to one.

12.3.286 WFOAM – DEFINE WELL FOAM INJECTION CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WFOAM](#) keyword defines an injection wells foam concentration. The foam option must be activated by the [FOAM](#) keyword in the [RUNSPEC](#) section in order to use this keyword. Note if a well's foam concentration is not set with this keyword then default value of zero is assigned to a well.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well injection foam concentration is being defined.			None
2	FOAMCON	A real positive value that defines the well's injection foam concentration. This value may be specified using a User Defined Argument (UDA). Units are dependent on the transport phase specified via the FOAMOPT I variable on the FOAMOPTS keyword in the PROPS section. FOAMOPT I should be set to either GAS or WATER . Currently OPM Flow only supports injecting foam via the GAS phase.			None
		Gas: lb/Mscf Water: lb/stb	Gas: kg/sm ³ Water: kg/sm ³	Gas: gm/scc Water: gm/scc	

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the [SCHEDULE](#) section. User Defined Arguments are activated using the [UDADIMS](#) keyword in the [RUNSPEC](#) section, and defined by the [UDQ](#) keyword in the [SCHEDULE](#) section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.105:WFOAM Keyword Description

See also the [FOAM](#) keyword in the [RUNSPEC](#) section, the [FOAMADS](#), [FOAMOPTS](#) and [FOAMROCK](#) keywords in the [PROPS](#) section.

Example

```
--
--      WELL INJECTION FOAM CONCENTRATION
--
-- WELL  FOAM
-- NAME  FOAMCON
WFOAM
GI01    0.020 /
GI02    0.020 /
GI03    0.020 /
/
```

Here three gas wells are given an injection foam concentration of 0.020 lb/Mscf, assuming field units.

12.3.287 WFRICSEG – CONVERT FRICTION WELL TO MULTI-SEGMENT WELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WFRICSEG](#) converts a previously defined friction well, as per the [WFRICTN](#) keyword in the [SCHEDULE](#) section, to a multi-segment well. The keyword thus acts as a replacement for the [WELSEGS](#) and [COMPSEGS](#) keywords for multi-segment wells. See also the [WFRICSG](#) keyword in the [SCHEDULE](#) section that performs similar functionality for wells in Local Grid Refinements.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.288
WFRICSGL – CONVERT FRICTION WELL TO MULTI-SEGMENT WELL (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WFRICSGL](#) converts a previously defined Local Grid Refinement (“LGR”) friction well, as per the [WFRICTNL](#) keyword in the [SCHEDULE](#) section, to a multi-segment [LGR](#) well. The keyword thus acts as a replacement for the [WELSEGS](#) and [COMPSEGL](#) keywords for [LGR](#) multi-segment wells. See also the [WFRICSEG](#) keyword in the [SCHEDULE](#) section that performs similar functionality for wells in the global grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.289 WFRICTN – DEFINE WELL AS A FRICTION WELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WFRICTN](#) keyword is used to declare a previously defined well as a friction well and to set the characteristics for this type of well including: tubing size, pipe roughness, and the connections to the grid. Wellbore friction is important in horizontal and multi-lateral wells where the pressure loss along the pipe can effect a well's performance. Note that unlike other [SCHEDULE](#) section well keywords, multiple wells cannot be entered with one [WFRICTN](#) keyword, that is, the keyword must be repeated for each well.

See also the [WFRICTNL](#) keyword in the [SCHEDULE](#) section that performs similar functionality for wells in Local Grid Refinements.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.290 WFRICTNL – DEFINE WELL AS A FRICTION WELL (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WFRICTNL](#) keyword is used to declare a previously defined Local Grid Refinement (“LGR”) well as a [LGR](#) friction well and to set the characteristics for this type of well including: tubing size, pipe roughness, and the connections to the grid. Wellbore friction is important in horizontal and multi-lateral wells where the pressure loss along the pipe can effect a well’s performance. Note that unlike other [SCHEDULE](#) section well keywords, multiple wells cannot be entered with one [WFRICTNL](#) keyword, that is, the keyword must be repeated for each well.

See also the [WFRICTN](#) keyword in the [SCHEDULE](#) section that performs similar functionality for wells in the global grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.291 WGASPROD – DEFINE SALE GAS WELL PRODUCTION TARGETS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WGASPROD](#) keyword declares wells to be Sales Gas producers and sets the incremental gas rate for a well and the maximum number of increments that this rate can be increased. Wells must have been previously been defined via the [WELSPECS](#) and [WCONPROD](#) keywords in the [SCHEDULE](#) section and are subject to any targets or constraints on [WCONPROD](#) keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.292 WGORPEN – DEFINE WELL GOR PENALTY PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WGORPEN](#) keyword defines a well’s Gas-Oil Ratio (“GOR”) penalty parameters used to calculate a well’s oil production target for the current month, as a function of the well’s previous month’s average GOR. The [WGORPEN](#) calculated oil rate overwrites any oil targets set by the [WCONPROD](#) and [WELTARG](#) keywords in the [SCHEDULE](#) section. In North American, it is common practice for the regulator to enforce GOR penalties, in order to control gas production in depletion drive oil reservoirs, with the stated intention to maximize oil recovery by limiting the energy loss from the reservoir by excessive gas production.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.293 WGRUPCON – DEFINE WELL GUIDE RATES FOR GROUP CONTROL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WGRUPCON](#) keyword defines a well's production or injection guide rate for when a well is under group control. The guide rate is used to determine a well's production target under group control in order to satisfy a group's targets and constraints, including any higher level related groups as well as the FIELD group.

Wells must have been previously defined and allocated to a group by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates controlled by the group to which they belong, in addition to any well constraints defined for the wells via the [WCONPROD](#) and [WCONINJE](#) keywords in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well production targets and constraints data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	STATUS	A defined character string that declares the status of the well to be under group control or not under group control. STATUS should be set to one of the following character strings: 1) YES: the well is under group control and its production behavior will be influenced by its assigned group, including connecting higher level groups as well as the FIELD group. 2) NO: the well is NOT under group control and its production behavior will only be influenced by its own targets and constraints. Note the default value of YES puts all wells under group control unless specified otherwise by the STATUS variable, or the TARGET variable on the WCONPROD and WCONINJE keywords in the SCHEDULE section.			YES
3	GUIDERAT	A dimensionless real number that determines the well's share of its group production (or injection) target rate. If GUIDERAT is a positive number then the guide rate for the well is fixed until modified by this keyword at a subsequent time. If TARGET variable on this keyword is not equal to the group's controlling phase, then the GUIDERAT is converted into the groups' controlling phase and is updated every time step. If GUIDERAT is less than or equal to zero then the well's guide rate is based on the well's potential (unrestricted flow) and the potential is calculated every time step.			-1.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	TARGET	A defined character string that sets the well's guide rate phase that the GUIDERAT value should be applied to. TARGET should be set to one of the following character strings: 1) OIL : the well's guide rate applies to the surface oil production rate. 2) WAT : the well's guide rate applies to the surface water production rate. 3) GAS : the well's guide rate applies to the surface gas production rate. 4) LIQ : the well's guide rate applies to the surface liquid (oil plus water) production rate. 5) RES : the well's guide rate applies to the in situ reservoir volume rate. 6) RAT : the well's guide rate applies to the surface rate of the injection phase. This should only be used if the well has been declared an injection well via the WCONINJE keyword in the SCHEDULE section. 7) COMB : the well's guide rate applies to the linearly combined calculated rate. This option is not supported by OPM Flow. TARGET may be defaulted if GUIDERAT has been defaulted, either by I* or a value less than or equal to zero.			None
5	SCALE	A real value that is used to multiple the GUIDERAT or the calculated well potentials to determine the final GUIDERAT for the well.			1.0
		dimensionless	dimensionless	dimensionless	

Notes:

1)

The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.106:WGRUPCON Keyword Description

See also the [GCONPROD](#) the [GCONINJE](#) keywords to define a group's production and injection targets and constraints, and the [WCONPROD](#) and [WCONINJE](#) keyword to define a well's production and injection characteristics. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the guides rates for all oil and gas producers and the gas injectors as follows:

```
--
--      DEFINE WELL GUIDES FOR GROUP CONTROL
--
-- WELL  GRUP  GUIDE  GUIDE  SCALE
-- NAME  CNTL  RATE   PHASE  FACT
WGRUPCON
'GI*'   YES   0       RAT    1.0      /
'GP*'   YES   0       GAS    1.0      /
'OP*'   NO    2       OIL    1.0      /
/
```

Both the gas producers ('GP*') and injectors ('GI*') are under group control with their guide rates based on their potentials. The gas injectors are controlled based on their potential surface gas injection rates and the gas producers on their potential surface gas production rates. In comparison, the oil wells (OP*) are controlled by their own targets and constraints.

12.3.294 WHEDREFD – DEFINE WELL HYDRAULIC HEAD REFERENCE DEPTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WHEDREFD](#) keyword sets the hydraulic head reference depth for reporting the hydraulic head pressure for the well, for wells that have previously been defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well hydraulic head reference depth data is being defined.			None
2	HYDREF	A real value that defines the hydraulic head reference depth for reporting the hydraulic head pressure for the well. HYDREF cannot be defaulted on the keyword; however if a well has not been set by this keyword HYDREF is set equal to the value on the <i>HYDRHEAD</i> keyword.			None
		feet	m	cm	

Notes:

1) The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.107: WHEDREFD Keyword Description

See also the [HYDRHEAD](#) keyword in the [PROPS](#) section.

Example

The following example defines three wells hydraulic head reference depths for reporting, using the [WHEDREFD](#) keyword

```
--
--      WELL HYDRAULIC HEAD REFERENCE DEPTH
--
-- WELL  HYDREF
-- NAME  DEPTH
WHEDREFD
OP01    150.0
OP02    175.0
OP03    150.0
/
```

Here, well OP01 and OP03 have their hydraulic head reference depths set to 150.0 ft and well OP02's hydraulic head reference depth is set to 175.0 ft.

12.3.295 WHISTCTL – DEFINE WELL HISTORICAL TARGET PHASE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WHISTCTL](#) keyword changes the target control for wells declared as history match wells via the [WCONHIST](#) keyword in the [SCHEDULE](#) section. The target phase is set on the [WCONHIST](#) keyword and [WHISTCTL](#) overrides this value for all subsequent entries on the [WCONHIST](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	TARGET	A defined character string that sets the observed target production phase for the well, all the other phases are calculated unconstrained and used for reporting only. The simulator will attempt to meet the TARGET based on the phase rate stated in items (4) to (6) and (10) on the WCONHIST keyword. TARGET should be set to one of the following character strings: 1) ORAT: the target is set to the surface oil production rate as defined by item (4) on the WCONHIST keyword. 2) WRAT: the target is set to the surface water production rate as defined by item (5) on the WCONHIST keyword. 3) GRAT: the target is set to the surface gas production rate as defined by item (6) on the WCONHIST keyword. 4) LRAT: the target is set to the surface liquid (oil plus water) production rate and is calculated by the simulator using (4) and (5) on the WCONHIST keyword. 5) RESV: the target is set to the in situ reservoir volume rate and is calculated by the simulator using items (4), (5) and (6) on the WCONHIST keyword. 6) BHP: the target rate is set to the bottom-hole pressure as defined by item (10) on the WCONHIST keyword. 7) NONE: revert back to the TARGET control mode on the WCONHIST keyword. The TARGET control mode defined on this keyword resets the TARGET control mode on the WCONHIST keyword in the SCHEDULE section, from the time the WHISTCTL is invoked, thus avoiding changing the control model on all subsequent WCONHIST keywords.			None
2	END	A defined character string that defines if the simulation should terminate if the well has switch to BHP control by the simulator, and should be set to one of the following character strings: 1) NO: no action is taken and the run continues. 2) YES: terminate the run at the next report time step. Wells set to BHP control via the WCONHIST or WHISTCTL keywords are ignored. Only END equal to NO is currently supported in OPM Flow.			NO

Notes:

- 1) The keyword is terminated by a “/”.

Table 12.108: WHISTCTL Keyword Description

History matching wells are handled differently then ordinary wells that use the [WCONPROD](#) keyword for controlling their production targets and constraints. However, the wells still need to be defined like ordinary production wells using the [WELSPECS](#) keyword in the [SCHEDULE](#) section. History matching wells are converted to ordinary wells by restating a well’s control mode using either the [WCONPROD](#) or [WELTARG](#) keywords in the [SCHEDULE](#) section.

See also the [WCONHIST](#) and [WCONINJH](#) keywords that are used to define the historical production and injection data, All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The example below shows the observed gas rates for the OP01 oil producer for the first quarter of 2000.

```
-----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-----
--
--      DEFINE WELL HISTORICAL TARGET PHASE
--
--      CNTL      BHP
--      MODE      STOP
WHISTCTL
--      RESV      NO
--
--
--      WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL  OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE  RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.5E3  100.0  1550   10      1*    900.0  1*
/
DATES
01 FEB 2000 /
/
--
--      WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL  OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE  RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.2E3  150.0  1520   1*      1*    875.0  3250.0 /
/
DATES
01 MAR 2000 /
/
--
--      WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL  OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE  RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.0E3  200.0  1500   1*      1*    850.0  1*
/
```

From January 1, 2000 the [WCONHIST](#) keyword defines well OP01, which is open and is on oil rate control, to produce 15,500 stb/d oil, with the observed rates of 100 stb/d of water and 15.5 MMscf/d of gas. However the WHISCTL keyword resets the target control to reservoir voidage from January 1, 2000 and onward. This is useful in initial history matching runs to get a “reasonable” pressure match, by ensuring that the total reservoir withdrawals are correct, although the individual phase withdrawals will not match. Once a reasonable pressure match is achieved for the reservoir then one can reset TARGET to the sales phase, [OIL](#) or [GAS](#), and continue with the matching of all the phases.

12.3.296 WHTEMP – DEFINE WELL TUBING HEAD TEMPERATURE PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WHTEMP](#), sets the parameters for the Tubing Head Temperature calculation, which can either be a constant value, or from a table lookup using a [VFPPROD](#) table, via the [VFPPROD](#) keyword in the [SCHEDULE](#) section, containing tubing head temperature data.

This keyword can only be used if the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well production targets and constraints data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	VFPTAB	A positive integer greater than or equal to zero that references the production vertical lift performance table (VFPPROD), containing the tubing head temperature data for the well. Note, a well must have both a VFPPROD pressure and a VFPPROD temperature table, if the tubing head temperatures are to be calculated. Alternatively, if a constant tubing head temperature for a production well is to be defined via the TEMP parameter, then VFPTAB should be defaulted with I* instead.			None
		dimensionless	dimensionless	dimensionless	
3	TEMP	A real positive value greater than zero that defines a constant tubing head temperature for a production well. In this case the VFPTAB parameter should be defaulted with I* if a constant tubing head temperature for a production well.			None
		°F	°C	°C	

Notes:

- 1) Note either VFPTAB or [TEMP](#) must be supplied, that is one can specify both.
- 2) The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.109: WHTEMP Keyword Description

See also the [VFPPROD](#) keyword in the [SCHEDULE](#) section.

Example

The following example defines three wells tubing head temperature parameters using the [WHTEMP](#) keyword

```
--
--      DEFINE WELL TUBING HEAD TEMPERATURE PARAMETERS
--
-- WELL  VFP      TUB
-- NAME  TABLE   TEMP
WHTEMP
OP01    5                               /
OP02    1*      150                      /
OP03    5                               /
/
```

Here, well OP01 and OP03 used [VFPPROD](#) table number five to calculate the tubing head temperature, and well OP02's uses a constant 150o tubing head temperature.

12.3.297 WINJCLN – CLEAN A FRACTION OF A DEPOSITED FILTER CAKE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WINJCLN](#) keyword signals that a filter cake should be completely or partially cleaned – this effectively multiplies the accumulated filter cake skin.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the filter cake properties are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	FCLNFRAC	A real positive value between 0 and 1 that defines the fraction of filter cake permeability (skin factor) to be removed. The accumulated filter cake skin factor for matching connections will be multiplied by (1 – FCLNFRAC), so the default value of 1 will completely clean the filter cake.			1
		dimensionless	dimensionless	dimensionless	
3	I	An integer that defines the matching connection location in the I-direction. If set to < 1 then all connections in the I-direction that also satisfy J and K criteria are selected.			-1
4	J	An integer that defines the matching connection location in the J-direction. If set to < 1 then all connections in the J-direction that also satisfy I and K criteria are selected.			-1
5	K	An integer that defines the matching connection location in the K-direction. If set to < 1 then all connections in the K-direction that also satisfy I and J criteria are selected.			-1

Notes:

1)

The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

2)

Repeated occurrences of this keyword will have a multiplicative effect.

Table 12.110: WINJCLN Keyword Description

See also the WINJDAN keyword to define the filter cake properties and the [WINJFCNC](#) keyword to define a well's injected filtrate concentration. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example signals the filter cake clean up for water injection wells using [WINJCLN](#):

```
--
--      WELL FILTER CAKE CLEAN UP
```

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

```
--  
-- WELL CLEAN --LOCATION--  
-- NAME FRAC II JJ KK  
WINJDAM  
INJ-A1 0.4 0 0 3 /  
INJ-A1 0.4 0 0 4 /  
INJ-B* 0.9 /  
/
```

In well INJ-A1 forty percent of the filter cake is cleaned up in well connections in layers 3 and 4 (filter cake skin is multiplied by 0.6). In wells matching INJ-B* ninety percent of the filter cake is cleaned up in all well connections (filter cake skin is multiplied by 0.1).

12.3.298 WINJDAM – DEFINE WELL FILTER CAKE PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WINJDAM](#) keyword defines filter cake properties for wells that have previously been defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. This keyword must be accompanied by the [WINJFCNC](#) keyword to define the injected filtrate concentration.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the filter cake properties are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	GEOMETRY	A defined character string that the filter cake geometry of the well. GEOMETRY must be set to one of the following character strings: 1) LINEAR : the filter cake model will assume a linear geometry. 2) RADIAL : the filter cake model will assume a radial geometry.			None
3	FCPERM	A real positive value that defines the filter cake permeability.			None
		mD	mD	mD	
4	FCPORO	A real positive value that defines the filter cake porosity.			0.3
		dimensionless	dimensionless	dimensionless	
5	FCRADIUS	Well radius to use in skin factor calculations. If FCRADIUS is defaulted then half the diameter from the COMPDAT keyword item 9 will be used if it is defined otherwise 0.5 feet will be used.			0.5 feet
		feet	m	cm	
6	FCAOF	A real positive value that defines the flow area for each connection. If FCAOF is defaulted then the value will be evaluated as $2\pi r_w h$, where r_w is equal to FCRADIUS, and h is evaluated using the permeability thickness kh value from the COMPDAT keyword item 10 and the permeability FCPERM item 3 above.			$2\pi r_w h$
		ft ²	m ²	cm ²	
7	I	An integer that defines the matching connection location in the I-direction. If set to < I then all connections in the I-direction that also satisfy J and K criteria are selected.			-I
8	J	An integer that defines the matching connection location in the J-direction. If set to < I then all connections in the J-direction that also satisfy I and K criteria are selected.			-I
9	K	An integer that defines the matching connection location in the K-direction. If set to < I then all connections in the K-direction that also satisfy I and J criteria are selected.			-I
Notes:					
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					
2) New records overwrite previously specified data for matching connections.					

Table 12.111: WINJDAM Keyword Description

See also the [WINJFCNC](#) keyword to define a well's injected filtrate concentration and the [WINJCLN](#) keyword to signal that a filter cake should be completely or partially cleaned. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the filter cake properties for a water injection well using [WINJDAM](#):

```
--
--      WELL FILTER CAKE PROPERTIES
--
-- WELL  LINEAR/  PERM   PORO   WELL  FLOW   --LOCATION--
-- NAME  RADIAL              RAD   AREA   II  JJ  KK
WINJDAM
INJ      LINEAR   10      0.32   /
INJ      RADIAL   1       0.25   1*    1*    0   0   3 /
/
```

Well INJ initially has a linear filter cake model for all completions with permeability of 10 mD, porosity of 0.32, and default well radius and flow area based on data from the [COMPDAT](#) keyword. This is then overwritten for completions in layer 3 with a radial filter cake model with permeability of 1 mD and porosity of 0.25 (again with default well radius and flow area).

12.3.299 WINJFCNC – DEFINE INJECTION WELL FILTRATE CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WINJFCNC](#) keyword defines the injected filtrate concentration for wells that have previously been defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the filter cake properties are being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	FCONCPPM	A real positive value that defines the volumetric concentration of filtrate in the injected water. This value may be specified using a User Defined Argument (UDA).			0
		ppm	ppm	ppm	

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the *SCHEDULE* section. User Defined Arguments are activated using the *UDADIMS* keyword in the *RUNSPEC* section, and defined by the *UDQ* keyword in the *SCHEDULE* section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".
- 3) New records overwrite previously specified data.

Table 12.112: WINJFCNC Keyword Description

See also the [WINJDAM](#) keyword to define the filter cake properties and the [WINJCLN](#) keyword to signal that a filter cake should be completely or partially cleaned. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the filtrate injection concentration for a water injection well using [WINJFCNC](#):

```
--
--      WELL FILTRATE INJECTION CONCENTRATION
--
-- WELL  FILTRATE
-- NAME  CONC
WINJFCNC
INJ-A1   10   /
INJ-B*   30   /
/
```

Well INJ-A1 has a filtrate concentration of 10 ppm in the injection water, and wells matching INJ-B* have a concentration of 30 ppm.

12.3.300 WINJGAS – DEFINE GAS INJECTION PROPERTIES FOR A WELL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WINJGAS](#) keyword defines the properties of the injection gas stream, for a given well. Once a gas well stream has been defined via the [WELLSTRE](#) keyword in the [RUNSPEC](#) section, it can be used with either the [WINJGAS](#) or [GINJGAS](#) keywords, to set the injected gas composition. Similarly, if an oil well stream has been defined by [WELLSTRE](#), then the well stream can be used with the [WINJOIL](#) keyword in the [SCHEDULE](#) section, to specify the injected oil composition. Note that, it is unnecessary to use [WINJGAS](#) for wells subordinate to a group having gas injection control, with the gas properties set by [GINJGAS](#) keyword in the [RUNSPEC](#) section. In this case the injection stream is defined by the [GINJGAS](#) keyword. However, if a gas injection well under group control uses the [WINJGAS](#) keyword, then this fluid, and not the group's fluid will be injected instead, at a rate controlled by the group.

The keyword should only be used if the [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section have also be activated for the gas-water two component model.

Note

This is an OPM Flow keyword used with OPM Flow's [CO2STORE](#) and [GASWAT](#) keywords in the [RUNSPEC](#) section, and should not be confused with the more general version of the [WINJGAS](#) keyword used in the commercial compositional simulator.

Secondly, although OPM Flow parses the keyword, the simulator currently ignores the data for this keyword.

No.	Name	Description	Default
1	WELNAME	A character string of up to eight characters in length that defines the injection well name, for which the gas injection properties are being specified. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.	None
2	STREAM	A defined character string that determines the properties of the injection gas, and as should be set to one of the following values: 1) GAS: The field separator gas composition is used as the injected gas. 2) GRUP: The well's group or upstream group injection fluid is used as the injected gas. 3) GV: This option enables the vapor production of a group, as declared via the SOURCE parameter on this record, to be used as the injected gas composition. 4) MIX: In this case, the gas injection composition is taken from either the WINJMIX or WINJORD keywords in the SCHEDULE section, with the name of injected fluid given by the SOURCE parameter on this record. 5) STREAM: Here the gas injection composition is given by the WELLSTRE keyword in the SCHEDULE section, with the name of injected fluid given by the SOURCE parameter on this record. 6) WV: This option enables the vapor production from a given well to be used as the injected gas, with the name of the well given by the SOURCE parameter on this record.	GRUP

No.	Name	Description	Default
		Only the STREAM option is supported by OPM Flow.	
3	SOURCE	A character string of up to eight characters in length, that defines the source of the gas injection stream, based on the value of STREAM . If STREAM equals GV , then source should be set to a group name. For STREAM equal to MIX or STREAM , then SOURCE should be set to the name of the gas stream, as defined by the WINJMIX , WINJORD , or WELLSTRE keywords.	None
4	MAKEUP	The name of the well stream used for the make-up gas, if make-up gas is required for WELNAME to match the injection target for the well. This option is not supported by OPM Flow.	None
5	STAGE	STAGE defines the separator stage from which the injection gas should be taken from. In this case, the vapor phase from any stage may be used, and the default value of zero uses the total vapor phase from the separator. This option is not supported by OPM Flow.	0

Notes:

- The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.113: WIN/GAS Keyword Description

Example

The following example defines how to specify a two component formulation, together with defining the names of the composition components, to be used with the *CO2STORE* and *GASWAT* options.

```

=====
--
--
-- PROPS SECTION
--
=====
PROPS      --
--          CONFIRM NUMBER OF COMPOSITIONAL COMPONENTS (OPM FLOW KEYWORD)
--
NCOMPS
          2
--
--          DEFINE COMPOSITIONAL COMPONENTS NAMES (OPM FLOW KEYWORD)
--
CNAMES
          'C02'
          'H2O'

```

The second part of the example, defines the well stream for the above two component CO₂ water system.

```
-- =====
--
-- SCHEDULE SECTION
--
-- =====
SCHEDULE
--
-- WELL STREAM INJECTION COMPOSITION (OPM FLOW Keyword)
--
-- WELL      -- WELL STREAM COMPOSITIONAL COMPONENT      --
-- STREAM    --              MOLE FRACTIONS              --
WELLSTRE
'C02STREAM'  1.000  0.000
/
--
-- WELL GAS INJECTION PROPERTIES
--
-- WELL  STREAM  SOURCE      MAKEUP  SEP
-- NAME  OPTION  DEPTH      GAS      STAGE
WINJGAS
GI01     STREAM  C02STREAM  1*      1* /
/
```

Here the well stream consists of 100% CO₂ and zero water, with well GI01 using the gas injection properties as defined by the [WELLSTRE](#) keyword and allocated via the [WINJGAS](#) keyword.

Finally, the gas injection rate is set via the [WCONINJE](#) keyword as shown below.

```
--
-- WELL INJECTION CONTROLS
--
-- WELL  FLUID  OPEN/  CNTL  SURF  RESV  BHP  THP  VFP
-- NAME  TYPE   SHUT   MODE  RATE  RATE  PRES PRES  TABLE
WCONINJE
GI01     GAS    OPEN   RATE  10E4  1*    300   1*   1*
/
```

Thus, gas injector GI01 will inject $10 \times 10^4 \text{ m}^3$ of CO₂ per day, assuming metric units.

12.3.301 WINJMULT – DEFINE WELL PRESSURE DEPENDENT INJECTIVITY MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WINJMULT](#) keyword defines pressure dependent injectivity multipliers for injection wells and can be used to approximate the increase or decrease in a well's injectivity due to hydraulic fracturing in water injection wells. Only injection wells are processed by this keyword, even if production wells have been entered by the keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	FRACPRES	FRACPRES is the fracture opening pressure (Pfractue) used in equation 12.32.			None
		psia	barsa	atma	
3	ALPHA	ALPHA is the multiplier gradient, α , in equation 12.32.			Defined
		l/psia 0.0	l/barsa 0.0	l/atma 0.0	
4	OPTION	A defined character string that determines how the data on this keyword is applied, and should be set to one of the following character strings: 1) CIRR: The injectivity multiplier as applied to the selected connections (I, J, K) is <u>irreversible</u>, and the pressure at the <u>connection's sand face</u> (Pwbhp) is used in equation 12.32, instead of the well's flowing bottom-hole pressure. This option means that even if the pressures in the wellbore, and therefore the sand face pressures, later declines, injectivity remains unchanged for all the connections. 2) CREV: The injectivity multiplier as applied to the selected connections (I, J, K) is <u>reversible</u>, and the pressure at the <u>connection's sand face</u> is used in equation 12.32, The reversibility of this option means that if the pressures in the wellbore, and therefore the sand face pressures, later declines, injectivity will also decline for <u>all</u> the connections. 3) WREV: The injectivity multiplier as applied to all connections in the well is reversible, and the wells' <u>flowing bottom-hole pressure</u> (P_{wbhp}) is used in equation 12.32, instead of the pressure at the connection's sand face. The connections stipulated by (I, J, K) are ignored. The reversibility of this option means that if the pressures in the wellbore, later declines, injectivity will also decline for all the connections.			WREV

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	I	An integer value less than or equal to NX that defines the connection location in the I-direction. If set to zero, a negative value, or defaulted with I* then all connections in the I-direction will be multiplied by ALPHA, depending on the selected OPTION value.			I*
6	J	An integer value less than or equal to NY that defines the connection location in the J-direction. If set to zero, a negative value, or defaulted with I* then all connections in the J-direction will be multiplied by ALPHA, depending on the selected OPTION value.			I*
7	K	An integer value less than or equal to NZ that defines the connection location in the K-direction. If set to zero, a negative value, or defaulted with I* then all connections in the K-direction will be multiplied by ALPHA, depending on the selected OPTION value.			I*
Notes: 1) The keyword is followed by any number of records. 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.114: WINJMULT Keyword Description

The methodology for applying the well pressure dependent injectivity multipliers is outlined in equation 12.32.

$$\begin{aligned} \text{Multiplier} &= 1.0 + \alpha (P_{WBHP} - P_{fracture}) && \text{for } P_{WBHP} > P_{fracture} \\ \text{Multiplier} &= 1.0 && \text{for } P_{WBHP} < P_{fracture} \end{aligned} \quad (12.32)$$

where:

- Multiplier = the resulting multiplier to be applied to the selected connections.
- α = the ALPHA multiplier gradient in Table 12.114.
- P_{WBHP} = either the well's current flowing bottom-hole pressure, or the selected individual connection's sand face pressure.
- $P_{fracture}$ = the effective fracture opening pressure, FRACPRES in Table 12.114.

The equation is applied every time there is a calculation to determine a well's flow rate, this results in the calculated mobility rates being scaled up by the Multiplier value in equation 12.32. Note also that since the scaling is performed on the connection fluid mobility values, then the reported connection transmissibilities in the print file etc. remain unchanged.

Note

If all the connection parameters (I, J, K) are defaulted, or OPTION is set equal to WREV, then the multiplier is applied to all connections in the well.

If any of the connection parameters (I, J, K) have positive values and OPTION is set equal to CIRR or CREV, then the multiplier is applied to the selected connections as determined by the (I, J, K) parameters.

Example

The example below show the `WINJMULT` keyword for three water injection wells.

```
--
--      DEFINE WELL CONNECTION MULTIPLIERS
--
-- WELL  FRAC  MULT   FRAC  --LOCATION--
-- NAME  PRES  VALUE  OPTN   I    J    K
WINJMULT
WI01    4200  0.0250  1*      1*   1*   1*      /
WI02    4250  0.0025  CIRR    1*   1*  145      /
WI02    4250  0.0025  CIRR    1*   1*  146      /
WI02    4250  0.0025  CIRR    1*   1*  147      /
WI03    4400  0.0055  CREV    1*   1*  160      /
WI03    4400  0.0055  CREV    1*   1*  165      /
/
```

The first well, WI01, uses the default value for OPTION, that is WREV, which means that the injectivity multiplier will be applied to all the connections in the well and the well's bottom-hole pressure is used in the calculation. In this case the process is reversible. The second well, WI02, applies the injectivity multiplier to all connections in layers 145 to 147 using the sand face pressures in the calculation, and the process is irreversible. Finally for well WI03, the multiplier is applied to all connections in layers 160 and 165 using the sand face pressures in the calculation, and in this case the process is reversible.

12.3.302 WINJTEMP – DEFINE INJECTION FLUID THERMAL PROPERTIES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WINJTEMP](#) defines the injection fluid thermal properties for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Only water and gas injection is supported.

This keyword can only be used if OPM Flow's thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well injection fluid thermal properties are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	STEAMQAL	STEAMQAL is a real positive value greater than or equal to zero and less than or equal to one that defines the steam quality of the injected fluid for the defined well. This parameter should be defaulted using I* as STEAMQAL is not used by OPM FLOW, as only water and gas injection is supported. This data is used by the commercial simulator's THERMAL option and is not supported by OPM Flow's THERMAL option.			I*
		dimensionless	dimensionless	dimensionless	
3	TEMP	TEMP is a real positive value that defines the temperature of the injected fluid for the defined well.			None
		°F	°C	°C	
4	PRES	PRES is a real positive value that defines the pressure of the injected fluid for the defined well.			None
		psia	barsa	atma	
5	ENTHALPY	ENTHALPY is a real positive value that defines the specific enthalpy of the injected fluid for the defined well. This is data is used by the commercial simulator's THERMAL option and is not supported by OPM Flow's THERMAL option.			None
		Btu/lbs-M	kJ/kg-M	J/gm-M	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.115: WINJTEMP Keyword Description

Example

The following example shows the [WINJTEMP](#) keyword for when OPM Flow’s temperature option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section.

```
--
--      INJECTION FLUID THERMAL PROPERTIES
--
-- WELL  STEAM  INJ    INJ    SPEC
-- NAME  QUAL   TEMP   PRES   ENTH
WINJTEMP
WI01    1*     68.0   220.0   1*      /
WI02    1*     70.0   230.0   1*      /
/
```

Here the water injection fluid’s temperature and pressure, in field units, for two water injections well are defined. Notice that both the steam quality and the specific enthalpy of the injected fluid for the defined wells are defaulted (or skipped), as OPM Flow’s [THERMAL](#) option does not support this data.

12.3.303 WLIFT – DEFINE WELL RE-TUBING, THP AND LIFT SWITCHING WORKOVER OPERATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WLIFT](#) defines the automatic workovers parameters for changing out wellbore tubing, changing the THP limit (for example switching from the high stage pressure separator to the low stage pressure separator), or changing the artificial lift parameters, for wells.

OPM Flow does not have this feature and hence this keyword is ignored by OPM Flow and the [WLIFT](#) keyword has no effect on the simulation.

12.3.304 WLIFTOPT – DEFINE WELL GAS LIFT OPTIMIZATION PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WLIFTOPT](#) defines which wells should use the Gas Lift Optimization facility in order to maximize oil production, as well as defining the associated gas lift optimization parameters for a given well. The keyword can also be used to switch off gas lift optimization for a well. Gas lift optimization is invoked via the [LIFTOPT](#) keyword in the [SCHEDULE](#) section. Note that the [LIFTOPT](#) keyword should precede the [WLIFTOPT](#) keyword in the [SCHEDULE](#) section in order to activate the gas lift optimization facility.

Wells are allocated to groups when the wells are specified by the [WELSPECS](#) keyword in the [SCHEDULE](#) section. Wells defined to be under group control will have their production rates and gas lift gas constraints ([GLIFTOPT](#) keyword in the [SCHEDULE](#) section) controlled by the group to which they belong, in addition to any well constraints defined for the wells, including the gas optimization parameters on the [WLIFTOPT](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well gas lift optimization parameters are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	OPTLIFT	A defined character string that sets if a well's gas lift gas rate should be calculated by the gas lift optimization facility or not, and should be set to: 1) NO: In this case the gas lift gas is a constant determined from the MXLIFT variable on this keyword, the ALQ-WELL variable on the WCONPROD keyword, or the TARGET and VALUE variables on the WELTARG keyword. 2) YES: Activates the gas lift optimization for the given well.			None
3	MXLIFT	A real value that defines the total amount of gas lift gas available for this well, multiplied by the well's efficiency factor. 1) If OPTLIFT is defined as NO then MXLIFT is considered a fixed gas lift gas rate. However, if MXLIFT is defaulted (I*) then MXLIFT is unchanged from the previously entered value. 2) If OPTLIFT equals YES and MXLIFT is defaulted (I*), then MXLIFT is taken from the largest value of the ALQ variable on the well's associated VFPPROD table. Note that the value entered here should be in the range entered in the VFPPROD table allocated to the well, otherwise errors may occur when optimizing the gas lift gas injection rate for the well.			I*
		Mscf/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	OPTWGT	OPTWGT (β_w) is real positive value that defines a weighting factor for allocating the available gas lift gas to a well. An increment of gas lift gas supply is allocated to a well based on the well's current incremental gradient multiplied by OPTWGT using the following formulae: $Gradient = \left(\frac{(\beta_w \times \Delta Q_{Oil})}{(\Delta Q_{GasLift} + \beta_g \times \Delta Q_{Gas})} \right)$ Where: β_w = is OPTWGT, the weighting factor for the preferential allocation of lift gas, β_g = is the gas production rate weighting factor, ΔQ_{Oil} = is the increment/decrement in oil production rate, ΔQ_{gas} = is the increment/decrement in gas production rate, and $\Delta Q_{GasLift}$ = is the increment/decrement in gas lift gas rate. Note by default β_g, the gas production rate weighting factor, is set to zero, and therefore the gradient equation simplifies to: $Gradient = \left(\frac{\beta_w \times \Delta Q_{Oil}}{\Delta Q_{GasLift}} \right)$ OPTWGT is ignored if OPTLIFT is equal to NO.			1
		dimensionless	dimensionless	dimensionless	
5	MINGAS	A real value that defines the minimum amount of gas lift gas available to the well, multiplied by the well's efficiency factor. The allocation of the gas lift gas is determined by: 1) If MINGAS is a positive value then this value is allocated to the well unless the well is unable flow with this quantity of gas lift gas. Alternatively, if the well is able to meet it's target rate without applying MINGAS, then the MINGAS rate is <u>not applied</u> to the well. 2) If MINGAS is a negative value, then the well is supplied with sufficient gas lift gas to allow the well to flow, subject to the maximum allowed gas lift quantity, as per MXLIFT variable on this keyword. The negative value itself is not used in any calculations. 3) If there is insufficient available gas lift gas, the wells are assigned values of MINGAS based on the decreasing order of their weighting factors as calculated per OPTWGT variable. 4) Wells belonging to groups that can meet their production targets without gas lift, will have their MINGAS values not applied, that is no gas lift is applied. The exception is that if OPTWGT has been set to a value greater than or equal to one, then the well will use the MINGAS value for it's gas lift gas, even if the group's target can be satisfied without gas lift. However, if both the well's group and the well can meet their production targets, then MINGAS will not be applied. This parameter is ignored if OPTLIFT is defined as NO.			0.0
		Mscf/d	sm ³ /day	scc/hour	

No.	Name	Description			Default
		Field	Metric	Laboratory	
6	OPTGAS	OPTGAS (β_g) is real positive value that defines the incremental gas weighting factor for allocating the available gas lift gas to a well. An increment of gas lift gas supply is allocated to a well based on the well's current incremental gradient as described by the definition of the OPTWGT (β_g) variable above, that is by the following formulae: $Gradient = \left(\frac{(\beta_w \times \Delta Q_{Oil})}{(\Delta Q_{GasLift} + \beta_g \times \Delta Q_{Gas})} \right)$ See OPTWGT for a definition of the variables in the equation. This parameter is ignored if OPTLIFT is defined as NO.			0.0
		dimensionless l	dimensionless	dimensionless	
		7	OPTLIMIT	A defined character string that defines if additional gas lift gas should be applied to the well, if the well's group gas target has been satisfied but the group's oil rate limit has not been achieved. 1) NO: Additional gas lift gas is not available for the given well. 2) YES: Additional gas lift gas is available for the given well. In cases where a well receiving additional gas lift may cause the well's group to exceed the group's gas target, normally the well will not be assigned the additional gas lift gas. However, if OPTLIMIT is set to YES, then this constraint is removed. This results in the gas lift optimization procedure continuing to maximize the oil rate, subject to available constraints. However, upon completion of the optimization process, applying the group controls may negate the gain from the gas lift optimization process. This parameter is ignored if OPTLIFT is defined as NO.	

Notes:
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.116: WLIFTOPT Keyword Description

See also the [LIFTOPT](#) keyword to activate gas lift optimization, the [GLIFTOPT](#) keyword to define the group gas lift optimization controls, [GRUPTREE](#) keyword to define the hierarchy of the groups below the FIELD level, the [WCONPROD](#) and [WCONINJE](#) keywords to define a well's production and injection rate targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example first switches on gas lift optimization via the `LIFTOPT` keyword and then defines the artificial lift constraints for PLAT-A, using the `GLIFTOPT` keyword and sets the well gas lift parameters using the `WLIFTOPT` keyword.

```
--      ACTIVATE GAS LIFT OPTIMIZATION AND PARAMETERS
--
--  INCR  INCR  TSTEP  NEWTON
--  GAS   OIL   INTVAL  OPTN
LIFTOPT
12.5E3   5E-3   0.0     YES                               /
/
--
--      GROUP GAS LIFT OPTIMIZATION CONSTRAINTS
--
--  GRUP  MAX      MAX
--  NAME  GAS ALQ   TOTAL GAS
GLIFTOPT
PLAT-A   200E3    1*                               /
/
--
--      WELL GAS LIFT OPTIMIZATION PARAMETERS
--
--  WELL  OPTN  MAX  WEIGHT  MIN  GAS  OPTN
--  NAME  LIFT  LIFT  FACTOR  LIFT  FACTOR  LIMIT
WLIFTOPT
OP01     YES   150E3  1.01   -1.0                               /
OP02     YES   150E3  1.01   -1.0                               /
OP03     YES   150E3  1.01   -1.0                               /
OP04     YES   150E3  1.01   -1.0                               /
OP05     YES   150E3  1.01   -1.0                               /
/
```

Here the `LIFTOPT` keyword defines the maximum incremental gas lift gas quantity to be 12.5×10^3 m3, the minimum incremental oil gain per m3 of gas lift gas is set to 5.0×10^{-3} m3, the time step interval is set to zero to perform the gas optimization every time step, and finally the gas lift optimization will be performed `NUPCOL` Newton iterations for the time step.

The `GLIFTOPT` keyword sets the maximum amount of gas lift gas for PLAT-A to 200,000 m3 and there is no maximum limit for the total maximum amount of gas that the group can process. In addition, `WLIFTOPT` sets all five wells to have gas lift gas optimization implemented with a maximum gas lift gas value of 150,000 m3 per well, with equal weighting factors and all wells are supplied with sufficient gas lift gas to allow the wells to flow, (minimum lift set to a negative value) subject to the maximum allowed gas lift quantity for the well (150,000 m3).

12.3.305 WLIMTOL – DEFINE WELL CONSTRAINT TOLERANCE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WLIMTOL keyword defines the tolerance to be used for various constraints applied to connections, completions (if connections have been lumped via the *COMPLUMP* keyword in the *SCHEDULE* section), wells, and groups, including the field group. See also the *GCONTOL* keyword in the *SCHEDULE* section that sets the tolerance parameters for groups.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.306 WLIST – DEFINE WELL LISTS (STATIC)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WLIST declares a group of wells to belong to a named static well list. Wells in a named well list are treated as a group of wells for which the standard well keywords can be applied. For example, instead of repeating a well keyword for each well, the keyword only needs to have the named well list instead, for the action to be applied to all wells in the named well list. In general any well keyword that allows well name roots as a well name, for example, PROD*, can use a named well list.

No.	Name	Description	Default
1	WLIST	A character string of up to eight characters in length, enclosed in quotes, that defines the well list name for the WELNAMES declared by this record. Note the first character must be asterisk ("*") and the second character must be a letter, for example, *PROD.	None
2	ACTION	A defined character string that determines how the WELNAMES should be handled with respect to the named well list (WLIST). ACTION should be set to one of the following: 1) ADD : Add the WELNAMES to an existing WLIST . 2) DEL : Delete WELNAMES from an existing WLIST . 3) MOV : WELNAMES from another existing named well list and ADD them to WLIST . 4) NEW : Define a new named well list and add the WELNAMES to WLIST .	
3-52	WELNAMES	A character string of up to eight characters in length that defines the well name that belongs to the named well list (WLIST). A total of 50 well names can be added to WLIST at a time. If additional wells are needed to added then use the ADD option of ACTION to add additional wells. Well names roots may all be used in WELNAMES as long as they are enclosed in quotes and end with an asterisk ("*"). In this case all wells that match the specification will be added to the list. For example, wells named OP01, OP02 and OP03, can be added as group by using "OP*" as the well name. Note that the well names must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur	
Notes: 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".			

Table 12.117: WLIST Keyword Description

Note that named well list production data in the **SUMMARY** file is well dependent, that is, if the wells belonging to a named well list is changed through time, the **SUMMARY** data will be based on the wells in the named well list group at the end of the run. Thus, if there are three wells in a named well list called *PROD1 at the beginning of a run; OP01, OP02 and OP03, and during the run OP03 is moved to a well named list

called *PROD2, then the [SUMMARY](#) data for *PROD1 will only contain the production data for wells OP01 and OP02 and *PROD2 will only contain the [SUMMARY](#) data for OP03 from the start to the end of the run.

Example

The following example defines two named well lists using the [WLIST](#) keyword.

```
--
--          WELL LIST SPECIFICATION
--
-- LIST      OPER      WELL NAME LIST
-- NAME
WLIST
'*BLK-1'    NEW      WEL-01M WEL-02M WEL-03M WEL-04M WEL-05M WEL-06M WEL-07M /
'*BLK-1'    ADD      WEL-08M WEL-09M WEL-10M WEL-11M WEL-12M WEL-13M WEL-14M /
'*BLK-1'    ADD      WEL-15M WEL-16M WEL-17M WEL-18M WEL-19M WEL-20M WEL-23M /
'*BLK-1'    ADD      WEL-24M WEL-25M WEL-26M WEL-28M /

'*BLK-2'    NEW      WEL-03U WEL-05U WEL-06U WEL-10U WEL-11U WEL-13U WEL-14U /
'*BLK-2'    ADD      WEL-15U WEL-16U WEL-17U WEL-18U WEL-19U WEL-25U WEL-27U /
/
DATES
      1 JAN 2020 /
/
--
--          DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
-- WELL WELL  --LOCATION-- COMPLETION
-- NAME STAT   I   J   K  FIRST LAST
WELOPEN
'*BLK-1' OPEN          0   0   0      0   0 /
'*BLK-1' OPEN          0   0   0      0   0 /
/
DATES
      1 JAN 2021 /
      1 JUL 2021 /
      1 OCT 2021 /
/
--
--          DEFINE WELL AND WELL CONNECTIONS FLOWING STATUS
--
-- WELL WELL  --LOCATION-- COMPLETION
-- NAME STAT   I   J   K  FIRST LAST
WELOPEN
'*BLK-2' OPEN          0   0   0      0   0 /
'*BLK-2' OPEN          0   0   0      0   0 /
/
```

In this example the wells in named well list “*BLK-1” are opened on January 1, 2020 and wells in named well list “*BLK-2” are opened October 1, 2021.

12.3.307
WLISTARG – MODIFY WELL LIST TARGET AND CONSTRAINT VALUES (STATIC)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WLISTARG](#) keyword modifies the target and constraint values of both rates and pressures for wells previously defined in a well list by the [WLIST](#) or [WLISTNAM](#) keywords. [WLISTARG](#) is similar to the [WELTARG](#) keyword in it that allows for modifying targets and constraints without having to define all the variables on the well control keywords: [WCONPROD](#), [WCONHIST](#), [WCONINJE](#), or [WCONINJH](#) keywords. Variables not changed by the [WLISTARG](#) keyword remain the same as those previously entered via the well control keywords or previously entered [WLISTARG](#) keywords. Note that the well must still be initially be fully defined using the [WCONPROD](#) or [WCONINJE](#) keywords. All the aforementioned keywords are described in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	WLIST	A character string of up to eight characters in length, enclosed in quotes, that defines the well list name declared by the WLIST keyword. Note the first character must be asterisk ("*") and the second character must be a letter, for example, *PROD.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
2	TARGET	A defined character string that sets the item to be changed for the well the value of the item is set by item (3). 1) ORAT: reset the surface oil production rate value as defined by item (3). 2) WRAT: reset the surface water production rate value as defined by item (3). 3) GRAT: reset the surface gas production rate value as defined by item (3). 4) LRAT: reset the surface liquid (oil plus water) production rate value as defined by (3). 5) RESV: reset the in situ reservoir volume rate value as defined by (3). 6) BHP: reset the bottom-hole pressure value as defined by item (3). 7) THP: reset the tubing head pressure value for the well as defined by item (3). 8) VFP: reset the vertical lift performance table number as defined by (3). 9) LIFT: reset the artificial lift quantity for use with vertical lift performance tables. 10) GUID: reset the guide rate value for wells operating under group control. Note TARGET only defines the variable to be changed, it does not change how a well is controlled. For example, if a well is operating on ORAT control, as defined by the previously entered WCONPROD keyword, entering TARGET equal to LRAT with a value, changes the liquid constraint but the well still remains on ORAT control. Use the WELCNTL keyword in the SCHEDULE section to change the control mode of a well.			None
	VALUE	A real positive vector of values that defines the value of the variable declared by TARGET for all the wells contained in WLIST . For example if there are four wells in WLIST then there must four real numbers for VALUE. The vector should be terminated by a “/” as indicated in the notes below.			None
3	Liquid Gas Res Vol Pressure VFP LIFT	stb/d Mscf/d rb/d psia dimensionless same as VFPPROD or VFPINJ	sm ³ /day sm3/day rm ³ /day barsa dimensionless same as VFPPROD or VFPINJ	scc/hour scc/hour rcc/hour atma dimensionless same as VFPPROD or VFPINJ	
Notes: 1) The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.118: WLISTARG Keyword Description

If a well is currently a history matching well, then WLISELTARG should only be used to change a wells bottom-hole pressure limit, vertical flow performance table number or the artificial lift quantity.

See also the [WELCNTL](#) keyword, in the [SCHEDULE](#) section that can be used to reset the control mode, as well as a well's target and constraints of both rates and pressures.

Example

The following example defines two named well lists using the [WLIST](#) keyword.

```
--
--      WELL LIST SPECIFICATION
--
-- LIST  OPER      WELL NAME LIST
-- NAME
WLIST
'*BLK-1' NEW      WEL-01M WEL-02M WEL-03M
/*BLK-2' NEW      WEL-03U WEL-05U WEL-06U WEL-10U
/
--
--      WELL PRODUCTION AND INJECTION TARGETS
--
-- WELL WELL  TARGET
-- NAME TARG  VALUE
WLISTARG
'*BLK-1' ORAT   2000.0  2000.00  2000.0
/*BLK-2' ORAT   3000.0  3500.00  4000.0  2000.0
/
```

The wells in the '*BLK-1' well list are all given an oil rate of 2,000 stb/d and wells in the '*BLK-2' well list are given rates of 3,000, 3,500, 4,000 and 2,000 stb/d.

12.3.308 WLISTNAM – DEFINE WELL LISTS (WLISTARG)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WLISTNAM](#) declares a group of wells to belong to a named [WLISTARG](#) well list for use with the [WLISTARG](#) keyword. Only the [WLISTARG](#) keyword can be used with this type of well list, and therefore it is better to use the [WLIST](#) keyword instead, that defines a static well list but offers more flexibility than a [WLISTNAM](#) well list.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description	Default
1	WLIST	A character string of up to eight characters in length, enclosed in quotes, that defines the well list name for the WELLNAMES declared by this record. Note the first character must be asterisk ("*") and the second character must be a letter, for example, *PROD.	None
2-51	WELNAMES	A character string of up to eight characters in length that defines the well name that belongs to the named well list (WLIST). A total of 50 well names can be added to WLISTNAM at a time. If the first well name in the list is the default value ("*1"), then the list is first cleared of all wells, before adding the subsequent wells in WELLNAMES. Well names roots may all be used in WELLNAMES as long as they are enclosed in quotes and end with an asterisk ("*"). In this case all wells that match the specification will be added to the list. For example, wells named OP01, OP02 and OP03, can be added as group by using "OP*" as the well name. Note that the well names must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.	1*

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.119: WLISTNAM Keyword Description

Note that named well list production data in the [SUMMARY](#) file is well dependent, that is, if the wells belonging to a named well list is changed through time, the [SUMMARY](#) data will be based on the wells in the named well list group at the end of the run. Thus, if there are three wells in a named well list called *PROD1 at the beginning of a run; OP01, OP02 and OP03, and during the run OP03 is moved to a well named list called *PROD2, then the [SUMMARY](#) data for *PROD1 will only contain the production data for wells OP01 and OP02 and *PROD2 will only contain the [SUMMARY](#) data for OP03 from the start to the end of the run.

Example

The following example defines two named well lists using the [WLISTNAM](#) keyword.

```
--
--          WELL LIST SPECIFICATION
--
-- LIST      WELL NAME LIST
-- NAME
WLISTNAM
'*BLK-1'    WEL-01M WEL-02M WEL-03M WEL-04M WEL-05M WEL-06M WEL-07M      /
'*BLK-1'    WEL-08M WEL-09M WEL-10M WEL-11M WEL-12M WEL-13M WEL-14M      /
'*BLK-1'    WEL-15M WEL-16M WEL-17M WEL-18M WEL-19M WEL-20M WEL-23M      /
'*BLK-1'    WEL-24M WEL-25M WEL-26M WEL-28M                                /

'*BLK-2'    1*      WEL-03U WEL-05U WEL-06U WEL-10U WEL-11U WEL-13U WEL-14U  /
'*BLK-2'    WEL-15U WEL-16U WEL-17U WEL-18U WEL-19U WEL-25U WEL-27U      /
/
```

Here well list '*BLK-1' contains 28 wells, that is wells WEL-01M to WEL-28M. For the '*BLK-2' well list all wells are first deleted due to the "1*" default value and then wells WEL-03U to WEL-27U are added to the list.

12.3.309 WMICP – DEFINE WATER INJECTION WELL'S MICROBIAL, OXYGEN, AND UREA CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WMICP](#) keyword defines a water injection well's microbial, growth, and cementation injection stream solutions, where the rate-limiting components are suspended microbes, oxygen, and urea concentrations respectively. These concentrations are used when the [MICP](#) keyword in the [RUNSPEC](#) section has been used to activate OPM Flow's Microbially Induced Calcite Precipitation model. See Landa-Marbán et al³⁴⁴ and ³⁴⁵ for a description of the model.

Note the keyword should only be used for wells declared as water injection wells via the [WCONINJE](#) keyword in the [SCHEDULE](#) section.

Note

This is an OPM Flow specific keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data is being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	MICRCON	MICRCON is a real positive value that defines the microbial concentration of the well's injection stream.			0.0
		lb/stb	kg/sm ³	gm/scc	
3	OXYGCON	A real positive value that defines the oxygen concentration of the well's injection stream.			0.0
		lb/stb	kg/sm ³	gm/scc	
4	UREACON	UREACON is a real positive value that defines the urea concentration of the well's injection stream.			0.0
		lb/stb	kg/sm ³	gm/scc	

Notes:

1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.120: WMICP Keyword Description

³⁴⁴ Landa-Marbán, D., Tveit, S., Kumar, K., Gasda, S.E., 2021. Practical approaches to study microbially induced calcite precipitation at the eld scale. *Int. J. Greenh. Gas Control* 106, 103256. <https://doi.org/10.1016/j.jggc.2021.103256>.

³⁴⁵ Landa-Marbán, D., Kumar, K., Tveit, S., Gasda, S.E., 2021. Numerical studies of CO₂ leakage remediation by micp-based plugging technology. In: Røkke, N.A. and Knuutila, H.K. (Eds) *Short Papers from the 11th International Trondheim CCS conference*, ISBN: 978-82-536-1714-5, 284-290.

Water injection wells that are not declared via this keyword have their concentrations defaulted to zero.

Example

```
--
--      DEFINE WATER INJECTION WELL MICROBIAL, OXYGEN, AND UREA CONCENTRATIONS
--
-- WELL  MICROBIAL  OXYGEN  UREA
-- NAME  MICRCON    OXYGCON  UREA CON
--      -----
WMICP
WI01    0.01
WI02    1*         0.04
WI03    1*         1*         60.0
WI04    1*         0.04         60.0
/
```

Here the microbial concentration for well WI01 is set to 0.01, the oxygen concentration for well WI02 is set to 0.04, the urea concentration for well WI03 is set to 60, and the oxygen and urea concentrations for well WI04 are set to 0.04 and 60 respectively.

12.3.310
WNCTRL
– DEFINE WELL CONTROL FOR NETWORK CONTROL OPTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The WNENTCNTL keyword sets a well’s control mode that should remain fixed after each network balancing calculation, for when the either the Standard Network or the Extended Network options have been activated, and the well is part of a network. The keyword allows for a well’s Tubing Head Pressure (“THP”), oil, gas, liquid, or water rate to be selected as fixed after each network balance calculation. Normally this should be the THP, and if the keyword is absent from the input deck then THP will be used as the default value. The Standard Network option is invoked if the [GRUPTREE](#), [GRUPNET](#), [GNETINJE](#), [GNETPUMP](#), etc. series of keywords have been used in the [SCHEDULE](#) section. Whereas, the Extended Network option is activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.311 WNETDP – DEFINE WELL THP TO NETWORK PRESSURE DROP

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WNETDP](#) keyword allows for a constant pressure drop between a well's Tubing Head Pressure ("THP") and the well's connecting network node, for when the either the Standard Network or the Extended Network options have been activated, and the well is part of a network. For production wells in a production network, [WNETDP](#) is added to the well's connecting network node pressure to arrive at the well's THP value. Whereas for injection wells in an injection network, [WNETDP](#) is subtracted from the well's connecting network node pressure to arrive at the well's THP value. The Standard Network option is invoked if the [GRUPTREE](#), [GRUPNET](#), [GNETINJE](#), [GNETPUMP](#), etc. series of keywords have been used in the [SCHEDULE](#) section. The Extended Network option is activated by the [NETWORK](#) keyword in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.312 WORKLIM – DEFINE WELL WORKOVER TIME

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WORKLIM sets the numbers of days taken to complete a workover.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.313
WORKTHP – DEFINE WELL WORKOVER OPTIONS FOR THP KILLED WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WORKTHP](#) keyword defines workover options for when a well dies, that is unable to produce at the current operating conditions, when under Tubing Head Pressure (“THP”) control. For example, if a well is producing to the high pressure separator and therefore has a high THP constraint, then the [WORKTHP](#) keyword can be used to switch the well to the lower pressure separator via re-setting the THP constraint.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.314 WPAVE – WELL BLOCK AVERAGE PRESSURE CALCULATION PARAMETERS FOR ALL WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPAVE](#) keyword defines the method and parameters for calculating a well's block average pressures for all wells in the model. The resulting average pressure can be written out to the [SUMMARY](#) and RSM files in order to compare with field observed data via the WBP, WBP4, WBP5 and WBP9 vectors in the [SUMMARY](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WPAVE1	A real dimensionless value that defines the weighting factor between the inner block and the surrounding blocks used in the calculation of the connection factor weighted average pressure. If WPAVE1 is greater than or equal to zero and less than or equal to one, then the average pressure for each well connection is calculated based on this weighting factor. A value of zero indicates only the surrounding blocks should be used in the calculation; and a value of one indicates only the inner blocks should be used. If WPAVE1 is less than zero, then the average pressure for each well connection is weighted based on the pore volumes of the inner and surrounding blocks.			0.5
2	WPAVE2	A real dimensionless value greater than or equal to zero and less than or equal to one, that defines the weighting factor between the connection factor weighted average pressures and the pore volume weighted average pressures. If WPAVE2 is equal to one, then the average pressures are calculated based only on the connection factor weighted average pressures. If WPAVE2 is equal to zero, then average pressures are calculated based only on the pore volumes weighted average pressures.			1.0
3	WPAVE3	A defined character string that determines how the hydrostatic head calculation is performed in correcting the pressures to the BHP reference depth on the WELSPECS or WPAVEDEP keywords in the SCHEDULE section. WPAVE3 should be set to one of the following character strings: 1) WELL: the hydrostatic head is calculated using the density of the fluid in the wellbore at the well connections. 2) RES: the hydrostatic head is calculated using the density of the fluid in the reservoir with well connections and averaged over the connections. 3) NONE: no hydrostatic correction is applied to the pressures.			WELL

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	WPAVE4	A defined character string that determines which connections should be used in the calculations, WPAVE4 should be set to one of the following character strings: 1) OPEN: only open connections and associated grid blocks should be used in the calculations. This option may result in pressure discontinuities if connections are opened and closed during the run. 2) ALL: all currently defined open and closed connections and associated grid blocks are used in the calculations. The pressure discontinuities issue mentioned above can be avoided with this option and defining all the well connections for a well at the beginning of the run. Only the OPEN option is currently supported by the simulator.			OPEN
Notes: 1) The keyword should be terminated by a “/”.					

Table 12.121: WPAVE Keyword Description

The keyword is not applicable and should not be used with radial and spider grid geometries.

See also the [WEL SPECS](#) keyword that defines a well and a well's bottom-hole pressure reference depth, the [WPAVE DEP](#) keyword that also defines a well's bottom-hole pressure reference depth, and the [COMP DAT](#) keyword to define a well's connections. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example defines the default well block average pressure calculation parameters

```
--
--      DEFINE WELL BLOCK AVERAGE PRESSURE CALCULATION PARAMETERS
--
--      INNER  PORV  WELL  OPEN
--      OUTER  CONN  RES   ALL
WPAVE
      0.5      1.0      WELL  ALL /
```

And the next example shows the parameters used in the Norne model.

```
--
--      DEFINE WELL BLOCK AVERAGE PRESSURE CALCULATION PARAMETERS
--
--      INNER  PORV  WELL  OPEN
--      OUTER  CONN  RES   ALL
WPAVE
      1*      0.0      WELL  ALL /
```

Here only pore volume weighting is used instead of connection weighting.

12.3.315 WPAVEDEP – DEFINE WELL REFERENCE DEPTH FOR PRESSURE CALCULATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPAVEDEP](#) keyword defines the reference depth to be used to calculate and report grid block average bottom-hole pressures for a well. This keyword can be used to override the values entered or defaulted on the [WELSPECS](#) keyword in the [SCHEDULE](#) section. The simulator corrects the grid block calculated pressures to a well's reference depth using the hydrostatic well of the producing fluids.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well and well connection status data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	BHPREF	A real value that defines the reference depth for reporting the bottom-hole pressure for the well. Ideally this value should be set to the midpoint of the perforations as defined by the COMPDAT keyword in the SCHEDULE section. If defaulted by 1* or set to a value less than or equal to zero, then the mid-point of shallowest connection defined by the COMPDAT keyword will be used.			Mid-point of shallowest connection defined by the COMPDAT keyword
		feet	m	cm	

Notes:

- 1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.122: WPAVEDEP Keyword Description

See also the [WELSPECS](#) keyword that defines a well, the [COMPDAT](#) keyword to define a well's connections, and the [WPAVE](#) for defining how the average bottom-hole pressure should be calculated. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Note

The keyword is normally used to reset a well's bottom-hole pressure depth to match the pressure gauge depth for when observed pressure is available, for example when conducting a history match for a well test, or when attempting to match static bottom-hole surveys conducted on a well.

Example

The following example illustrates how to set the bottom-hole reference depth for wells completed in different reservoirs that have different datum depths. Here it is assumed that all wells in a reservoir A have RES-A as part of their well name, and similarly for reservoirs B and C.

```
--
-- WELL SPECIFICATION DATA
--
-- WELL      GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  SHUT   CROSS  PRESS
-- NAME      NAME        I        J    DEPTH  FLUID   AREA    EQUA.   IN     FLOW   TABLE
WELSPECS
RES-AOP1 PLATFORM      14      13    1*      OIL     1*      STD     OPEN   NO     1* /
RES-AOP2 PLATFORM      17      16    1*      OIL     1*      STD     OPEN   NO     1* /
RES-AOP3 PLATFORM      21      19    1*      OIL     1*      STD     OPEN   NO     1* /
RES-BOP4 PLATFORM      28      96    1*      OIL     1*      STD     OPEN   NO     1* /
RES-BOP5 PLATFORM      34      89    1*      OIL     1*      STD     OPEN   NO     1* /
RES-COP6 PLATFORM     128      52    1*      OIL     1*      STD     OPEN   NO     1* /
RES-COP7 PLATFORM     134      56    1*      OIL     1*      STD     OPEN   NO     1* /
RES-COP8 PLATFORM     138      50    1*      OIL     1*      STD     OPEN   NO     1* /
RES-COP9 PLATFORM     120      52    1*      OIL     1*      STD     OPEN   NO     1* /
/
--
--          DEFINE WELL REFERENCE DEPTH FOR PRESSURE CALCULATIONS
--
-- WELL  REF
-- NAME  DEPTH
-- ----  -----
WPAVEDEP
'RES-A*' 3100.0
'RES-B*' 3300.0
'RES-C*' 5909.0
/
```

In the example the all wells dedicated to RES-A will have their bottom-hole reference depth set to 3,000 ft. TVDSS, RES-B wells to 3,300 ft. TVDSS and well RES-C wells to 5909 ft. TVDSS.

12.3.316 WPIMULT – DEFINE WELL CONNECTION MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPIMULT](#) keyword defines a well connection factor multiplier that scales the existing well connection factor values. The resulting effect is to scale the well's productivity at the reporting time step the keyword is entered.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well and well connection status data is being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	PIMULT	A real positive value that will be used to scale the well connection factors defined by I, J, K, C1 and C2 below.			1.0
3	I	An integer less than or equal to NX that defines the connection locations in the I-direction.			I*
4	J	An integer less than or equal to NY that defines the connection locations in the J-direction.			I*
5	K	An integer less than or equal to NZ that defines the connection locations in the K-direction.			I*
6	C1	An integer value that defines the first completion number in the range. Connections are lumped into completions via the <i>COMPLUMP</i> keyword, and C1 refers to the first completion number, as defined by the <i>COMPLUMP</i> keyword, and all the connections contained within the C1 completion.			I*
7	C2	An integer that defines the last completion in the range. Connections are lumped into completions via the <i>COMPLUMP</i> keyword, and C2 refers to the last completion number, as defined by the <i>COMPLUMP</i> keyword, and all the connections contained within the C2 completion.			I*
Notes:					
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.123: WPIMULT Keyword Description

If variables I, J, K, C1 and C2 are all negative values or defaulted with I*, then PIMULT is applied to all the well connections in the well.

If any of the variables I, J, K, C1 and C2 are set to zero (meaning any or all values), or a positive value then PIMULT is applied to the defined connections. The defined connections are those with the I, J, K variables in the specified location and a completion number in the range specified by C1 and C2.

Note, that PIMULT is applied at the time the [WPIMULT](#) keyword is entered and is cumulative if applied to the same well connections, provided there are intervening report time steps between consecutive [WPIMULT](#) keywords. Consequently, if there are no intervening report time steps between consecutive [WPIMULT](#) keywords utilizing the same well connections, then only the last set is applied.

See also the [WELPI](#) keyword to set a well's productivity or injectivity index at the time the keyword is activated, and also the [PIMULTAB](#) keyword that defines productivity index multiplier versus water cut tables that are used to scaled a well's connection factors based on a wells connection current producing water cut. Both keywords are documented in the [SCHEDULE](#) section.

Example

The following example defines three vertical oil wells using the [WEL SPECS](#) keyword and their associated connection data.

```
--
-- WELL SPECIFICATION DATA
--
-- WELL      GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  SHUT   CROSS  PRESS
-- NAME      NAME        I      J    DEPTH FLUID  AREA   EQUA.  IN     FLOW   TABLE
WEL SPECS
OP01        PLATFORM    14     13   1*      OIL    1*      STD    OPEN   NO     1*  /
OP02        PLATFORM    28     96   1*      OIL    1*      STD    OPEN   NO     1*  /
OP03        PLATFORM    128    56   1*      OIL    1*      STD    OPEN   NO     1*  /
/
--
-- WELL PRODUCTION WELL CONTROLS
--
-- WELL      OPEN/  CNTL   OIL    WAT    GAS    LIQ    RES    BHP    THP    VFP    VFP
-- NAME      SHUT   MODE   RATE   RATE   RATE   RATE   RATE   PRES  PRES  TABLE ALFQ
WCONPROD
'*'         SHUT   GRUP   1*     1*     1*     1*     1*     200.0                /
/
--
-- WELL CONNECTION DATA
--
-- WELL      --- LOCATION ---  OPEN  SAT  CONN  WELL  KH  SKIN  D  DIR
-- NAME      II  JJ  K1  K2  SHUT  TAB  FACT DIA  FACT FACT FACT PEN
COMPDAT
OP01        1*  1*   1  10  OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
OP01        1*  1*  15  30  OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
OP01        1*  1*  35  90  OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
OP02        1*  1*   1  10  OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
OP03        1*  1*  35  90  OPEN  1*   1*   0.708 1*   0.0  1*   'Z' /
/
--
-- ASSIGN WELL CONNECTIONS TO COMPLETIONS
--
-- WELL      --- LOCATION ---  COMPL
-- NAME      II  JJ  K1  K2  NO.
COMPLUMP
OP03        1*  1*  35  45   1                / COMPLETION NO. 01
OP03        1*  1*  50  90   2                / COMPLETION NO. 02
/
--
-- DEFINE WELL CONNECTION MULTIPLIERS
--
-- WELL      PI      --LOCATION--  COMPLETION
-- NAME      MULT      I    J    K  FIRST LAST
WPIMULT
OP01        1.250      1*  1*   1*   1*   1*
OP02        0.750      1*  1*  10   1*   1*
OP03        1.100      1*  1*   1*   1   2
```

/

In this example the [WPIMULT](#) scales the well productivity of well OP01 by 1.25, and scales all the well connection factors in layer 10 only by 0.75 for well OP02. For well OP03, [WPIMULT](#) scales all the connections in completions one and two by 1.1.

12.3.317 WPIMULTL – DEFINE WELL CONNECTION MULTIPLIERS (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPIMULTL](#) keyword defines a well connection factor multiplier that scales the existing well connection factor values, for a well in a Local Grid Refinement (“LGR”). The resulting effect is scale the well’s productivity at the reporting time step the keyword is entered.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well and well connection status data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECL keyword in the SCHEDULE section, otherwise an error may occur.			None
2	PIMULT	A real positive value that will be used to scale the well connection factors defined by I, J, K, C1 and C2 below.			1.0
3	LGRNAME	A character string of up to eight characters in length that defines the LGR name for which the well LGR connection multiplier factor (PIMULT) is being defined. Note that LGRNAME must have been declared previously using the WELSPECL keyword in the SCHEDULE section, otherwise an error may occur. If defaulted with I* the LGR on the WELSPECL keyword will be utilized.			Defined
3	I	An integer less than or equal to NX that defines the connection location in the I-direction.			I*
4	J	An integer less than or equal to NY that defines the connection location in the J-direction.			I*
5	K	An integer less than or equal to NZ that defines the connection location in the K-direction.			I*
6	C1	An integer value that defines the first completion in the range. Connections are lumped into completions via the COMPLUMP keyword, and C1 refers to the first completion number, as defined by the COMPLUMP keyword, and all the connections contained within the C1 completion			I*
7	C2	An integer value that defines the last completion in the range. Connections are lumped into completions via the COMPLUMP keyword, and C2 refers to the last completion number, as defined by the COMPLUMP keyword, and all the connections contained within the C2 completion			I*

No.	Name	Description			Default
		Field	Metric	Laboratory	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.124: WPIMULT Keyword Description

If variables I, J K, C1 and C2 are all defaulted with a negative value or I*, then PIMULT is applied to all the well connections in the well.

If variables I, J K, C1 and C2 are set to zero (meaning any or all values), or a positive value then PIMULT is applied to the defined connections. The defined connections are those with the I, J, K variables in the specified location and a completion number in the range specified by C1 and C2.

Note, that PIMULT is applied at the time the WPIMULTL keyword is entered and is cumulative if applied to the same well connections, provided there are intervening report time steps between consecutive WPIMULTL keywords. Consequently, if there are no intervening report time steps between consecutive WPIMULTL keywords utilizing the same well connections, then only the last set is applied.

See also the PIMULTAB keyword that defines productivity index multiplier versus water cut tables that are used to scaled a well’s connection factors based on a wells connection current producing water cut. The keyword is documented in the SCHEDULE section.

Example

The following example defines two vertical oil wells using the WELSPECL keyword and their associated connection data.

```
--
--      WELL LGR SPECIFICATION DATA
--
--      WELL GROUP  LGR   -LOCATION- BHP   PHASE DRAIN INFLOW SHUT  CROSS PVT
--      NAME NAME    NAME    I      J  DEPTH FLUID AREA  EQUA.  IN   FLOW  TABLE
WELSPECL
      OP01  PLAT  OP01LGR  14   13   1*    OIL   1*    STD   SHUT  NO   1*  /
      OP02  PLAT  OP02LGR  28   96   1*    OIL   1*    STD   SHUT  NO   1*  /
/
--
--      WELL LGR CONNECTION DATA
--
--      WELL  LGR   ---LOCATION---  OPEN  SAT CONN  WELL  KH  SKIN  D  DIR
--      NAME  NAME    II  JJ  K1  K2  SHUT  TAB FACT DIA  FACT FACT FACT PEN
COMPDATL
      OP01  OP01LGR  1*   1*   20  56  OPEN  1*  1*  0.708  1*  1*  1*  Z  /
      OP01  OP01LGR  1*   1*   75 100  SHUT  1*  1*  0.708  1*  1*  1*  Z  /
      OP02  OP02LGR  1*   1*   75 100  OPEN  1*  1*  0.708  1*  1*  1*  Z  /
      OP03  OP02LGR  1*   1*   75 100  OPEN  1*  1*  0.708  1*  1*  1*  Z  /
/
--
--      ASSIGN WELL CONNECTIONS TO COMPLETIONS
--
--      WELL  LGR   --- LOCATION ---  COMPL
--      NAME  NAME    II  JJ  K1  K2  NO.
COMPLMPL
      OP03    OP02LGR  1*   1*   75  85    1          / COMPLETION NO. 01
      OP03    OP21LGR  1*   1*   86 100    2          / COMPLETION NO. 02
/
```

```
--
--      DEFINE WELL CONNECTION MULTIPLIERS
--
-- WELL  PI      LGR      --LOCATION--  COMPLETION
-- NAME  MULT    NAME      I    J    K    FIRST LAST
WPIMULTL
OP01    1.250  OP01LGR    1*   1*   1*   1*   1*   /
OP02    0.750  OP01LGR    1*   1*  10   1*   1*   /
OP03    1.100  OP02LGR    1*   1*   1*   1    2    /
/
```

In this example the [WPIMULTL](#) keyword scales the well productivity of well OP01 by 1.25, scales all the well connection factors in layer 10 only by 0.75 for well OP02, and for OP03, scales all the connections in completions one and two by 1.100.

12.3.318 WPITAB – ASSIGN WELL PRODUCTIVITY INDEX VERSUS WATER CUT TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPITAB](#) keyword assigns the well productivity index multiplier versus water cut tables, that are used to scaled a well's connection factors based on the connection's current producing water cut, to a well. The tables are defined via the [PIMULTAB](#) keyword in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well productivity index multiplier versus water cut table, <i>PIMULTAB</i> , is being assigned. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	<i>PIMULTAB</i>	A positive integer value that defines the corresponding <i>PIMULTAB</i> table to be allocated to the well. A value less than or equal to zero means that no <i>PIMULTAB</i> table is allocated to the well			0
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.125:WPITAB Keyword Description

See also the [PIMULTAB](#) keyword that defines productivity index multiplier versus water cut tables and also the [WPIMULT](#) keyword that scales a well's productivity index by a constant value, both of which are in the [SCHEDULE](#) section.

Example

Given NTPIMT equals two on the [PIMTDIMS](#) keyword in the [RUNSPEC](#) section, then:

```
--
--      ASSIGN WELL PRODUCTIVITY INDEX VS WATER CUT TABLE
--
-- WELL  PI
-- NAME  TABLE
WPITAB
OP01    1                               /
OP02    1                               /
OP03    2                               /
/
```

Assigns [PIMULTAB](#) table one to wells OP01 and OP02 and table two to OP03.

12.3.319 WPLUG – DEFINE WELL PLUG BACK LENGTH

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Various keywords in the [SCHEDULE](#) section ([WECON](#), [GECON](#) etc.) allow for a well to be automatically plugged back if the well violates a constraint, that is to close existing perforations (well connections). For example if the water cut exceeds 90%, then plug back the well. The [WPLUG](#) keyword defines for automatic plug backs the length of the perforations (length of connections) to be closed each time an automatic plug back is performed, together with various options on how the workover should be performed, top down, bottom up, etc.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.320 WPMITAB – ASSIGN WELL POLYMER MOLECULAR MODEL INJECTION TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPMITAB](#) keyword assigns the well polymer molecular injection tables to water injection wells in OPM Flow's Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure. This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated. The keyword assigns the [PLYMWINJ](#) tables that are defined via the [PLYMWINJ](#) keyword in the [PROPS](#) section.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length, that defines the water injection well name, for which the well polymer molecular injection table, <i>PLYMWINJ</i> , is to be assigned. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	<i>PLYMWINJ</i>	A positive integer value that defines the corresponding <i>PLYMWINJ</i> table to be allocated to the water injection well. A value less than or equal to zero means that no PLYMWIN table is allocated to the well			0
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.126: WPMITAB Keyword Description

See also the [PLYMWINJ](#) keyword in the [PROPS](#) section, that describes the relationship of the injected polymer molecular weight as a function of polymer throughput and polymer velocity, for the simulator's Polymer Molecular Weight Transport option, as well as the [SKPRWAT](#), [SKPRPOLY](#), and [PLYVMH](#) keywords, also in the [PROPS](#) section, that are the additional keywords required for the Polymer Molecular Weight Transport option.

The [WSKPTAB](#) keyword in the [SCHEDULE](#) section may be used to assign the [SKPRWAT](#) and [SKPRPOL](#) tables to water injections wells, that enable the calculation of the wellbore skin pressure based on the fluids being injected.

Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

Given NTPMWINJ equals two on the *PINTDIMS* keyword in the *RUNSPEC* section, then:

```
--
--      ASSIGN WELL POLYMER MOLECULAR MODEL INJECTION TABLES
--
-- WELL  PLYMWINJ
-- NAME  TABLE
WPMITAB
WI01    1                               /
WI02    1                               /
WI03    2                               /
/
```

Assigns *PLYMWINJ* table one to wells WI01 and WI02 and table two to WI03.

12.3.321
WPOLYMER – DEFINE WATER INJECTION WELL POLYMER AND SALT CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WPOLYMER](#) keyword defines a water injection well's polymer and salt injection stream concentrations that are to be used for when the polymer and salt options have been activated by the [POLYMER](#) and [BRINE](#) keywords in the [RUNSPEC](#) section.

Note that if the Brine option has not be activated by the [BRINE](#) keyword in the [RUNSPEC](#) section, then the salt concentrations in the third column are ignored. Secondly, if the brine phase is declared but the polymer phase has not been made active, then the [WSALT](#) keyword in the [SCHEDULE](#) section can be used to set the salt concentration.

Currently the Brine option is not implemented in OPM Flow and therefore both the SALTCON and GRPSALT variables on this keyword are ignored.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well connection data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	POLCON	A real positive value that defines the polymer concentration of the well's injection stream. This value may be specified using a User Defined Argument (UDA).			None
		lb/stb	kg/sm ³	gm/scc	
3	SALTCON	A real positive value that defines the salt concentration of the well's injection stream. This value may be specified using a User Defined Argument (UDA). This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.			None
		lb/stb	kg/sm3	gm/scc	
4	GRPPOL	A character string of up to eight characters in length that defines the group name for which the group's produced polymer concentration should be used instead of the well's POLCON value stated on this keyword.			None
5	GRPSALT	A character string of up to eight characters in length that defines the group name for which the group's produced salt concentration should be used instead of the well's SALTCON value stated on this keyword. This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.			None

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the [SCHEDULE](#) section. User Defined Arguments are activated using the [UDADIMS](#) keyword in the [RUNSPEC](#) section, and defined by the [UDQ](#) keyword in the [SCHEDULE](#) section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.127:WPOLYMER Keyword Description

Water injection wells that are not declared via this keyword have their concentrations defaulted to zero.

See also the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the polymer and salt injection stream concentrations for three water injection wells for when the polymer option has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section.

```
--
--      DEFINE WATER INJECTION WELL POLYMER AND SALT CONCENTRATIONS
--
```

-- WELL	POLYMER	SALT	POLYMER	SALT
-- NAME	POLCON	SALTCON	GROUP	GROUP
--	-----	-----	-----	-----
WPOLYMER				
WI01	0.2500			/
WI02	1*	1*	GRPINJ1	/
WI03	0.2500	1*	GRPINJ1	/
/				

The polymer concentration for well WI01 is set to 0.25 and the stated polymer concentration for well WI02 will be ignored, as both WI02 and WI03 will re-inject the produced polymer from the GRPINJ1 group.

12.3.322 WPOLYRED – DEFINE WELL POLYMER-WATER VISCOSITY REDUCTION FACTOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WPOLYRED](#) keyword defines the polymer-water reduction factor for injection wells, for when the polymer phase has been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section. [WPOLYRED](#) should be set to a value greater than or equal to zero and less than or equal to one that determines the injection mixture’s viscosity. A value of zero indicates for pure water injection and a value of one will use the simulator’s valuated mixture viscosity. A value between zero and one will use an interpolated mixture viscosity.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.323 WREGROUP – AUTOMATIC RE-ASSIGNMENT OF WELLS TO GROUPS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WREGROUP defines the criteria to automatically re-assign wells to various other groups. This can be used, for example, to move wells on THP control flowing through a high pressure separator group to a low pressure separator group in order for the wells to be under different group controls for low pressure wells.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.324 WRFT – ACTIVATE WELL RFT REPORTING TO THE RFT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword activates reporting of a well's pressure and saturation profile versus depth for the connected grid blocks, to the RFT file for the requested wells at the time the keyword is activated. Data written out by OPM Flow is used to match the field measured data collected from a Repeat Formation Tester ("RFT") tool.

The RFT tool is an open hole device which is an updated version of the Formation Interval Tester ("FIT"), both of which are run on wire line. Both tools take multiple pressure readings (at various depths) thus enabling a pressure depth profile to be obtained from the formation, and, in addition, they can also take fluid samples from the formation. The latest tool available from Schlumberger is the Modular Formation Dynamics Tester ("MDT"), which, as its name suggests, is a modular tool that can be assembled in different configurations depending on what are the objectives for running the tool. Note other vendors have similar wire line logging tools with alternative names for the tools. Throughout this section the term RFT applies to all tools that measure a pressure profile versus depth (RFT/FIT/MDT etc.).

No.	Name	Description			Default
		Field	Metric	Laboratory	
I	WELNAME	A columnar vector of character strings of up to eight characters in length for each item, that defines the well name for which the RFT data should be written to the RFT file. Note that the WELNAME must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur. If the WELNAME is left blank then the data is written out for all wells at the time they are first opened to flow. If the WELNAME is given, then the RFT data for the well at the time step the keyword is invoked is written out.			None
Notes: 1) The keyword is followed by any number of records. 2) Each record is terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.128:WRFT Keyword Description

See also the [WRFTPLT](#) keyword in the [SCHEDULE](#) section that has more flexible reporting options.

Examples

The first example activates RFT reporting for all wells at the time a well is first opened to flow:

```
--
--      ACTIVATE WELL RFT REPORTING TO THE RFT FILE
--
-- WELL
-- NAME
WRFT
/
```

Ideally, this version of the keyword should be placed at the beginning of the [SCHEDULE](#) section to obtain the data for the wells in the run before they are opened up through time.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

The next example shows how to use the keyword to request the output for several wells at different reporting time steps.

```
-----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-----
DATES
15 JAN 2000 /
/
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.5E3  0.0    1550    10      1*    900.0  1*    /
OP02     SHUT
/
--
--          ACTIVATE WELL RFT REPORTING TO THE RFT FILE
--
-- WELL
-- NAME
WRFT
OP01
OP02
/
DATES
01 FEB 2000 /
/
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.5E3  0.0    1550    10      1*    900.0  1*    /
OP02     SHUT
/
--
--          ACTIVATE WELL RFT REPORTING TO THE RFT FILE
--
-- WELL
-- NAME
WRFT
OP01
OP02
/
DATES
01 MAR 2000 /
/
--
--          WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    VFP    VFP    THP    BHP
-- NAME  SHUT   MODE   RATE   RATE   RATE   TABLE  ALFQ   PRES  PRES
WCONHIST
OP01     OPEN   ORAT   15.5E3  0.0    1550    10      1*    900.0  1*    /
OP02     OPEN   ORAT   10.5E3  0.0    1000    10      1*    900.0  1*    /
/
```

In this example, both well's have their RFT written out on February 1 and March 1 2000.

12.3.325 WRFTPLT – ACTIVATE WELL RFT AND PLT REPORTING TO THE RFT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword activates reporting of a well's depth pressure and fluid rates profile to the RFT file for the requested wells at the time the keyword is activated. Data written out by the simulator is used to match the field measured data collected from both the Repeat Formation Tester ("RFT") tool and various Production Logging Tools ("PLT").

See the [WRFT](#) keyword in the [SCHEDULE](#) section for a brief description of the RFT data set. This keyword also activates the writing out of each well connection's fluid rates, connection factors and KH data, etc., as the PLT data. The PLT data is used to compare with measured data from wire line production logging tools.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A columnar vector of character strings of up to eight characters in length for each item, that defines the well name for which the RFT data should be written to the RFT file. Note that the WELNAME must have been declared previously using the WEL SPECS keyword in the SCHEDULE section, otherwise an error may occur. If the WELNAME is left blank then the data is written out for all wells at the time they are first opened to flow. If the WELNAME is given, then the RFT data for the well at the time step the keyword is invoked is written out.			None
2	RFT	A defined character string that sets the RFT data set output options and should be set to one of the following character strings. 1) NO: do not write RFT data for the well. 2) YES: write out the RFT data at the current reporting time step. 3) REPT: write out the RFT data at the current reporting time step and all subsequent reporting time steps. 4) TIMESTEP: write out the RFT data at the current reporting time step and all subsequent time steps. 5) FOPN: write out the RFT data at the current reporting time step for the well if it is opened, otherwise write the RFT data out the first time the named well is opened.			NO
3	PLT	A defined character string that sets the PLT data set output options and should be set to one of the following character strings. 1) NO: do not write PLT data for the well. 2) YES: write out the PLT data at the current reporting time step. 3) REPT: write out the PLT data at the current reporting time step and all subsequent reporting time steps. 4) TIMESTEP: write out the PLT data at the current reporting time step and all subsequent time steps.			NO

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	MULTISEG	A defined character string that sets the output options for multi-segment wells, that is the flow rates and pressures through each well segment, and should be set to one of the following character strings. 1) NO: do not write multi-segment well data for the well. 2) YES: write out the multi-segment well data at the current reporting time step. 3) REPT: write out the multi-segment well data at the current reporting time step and all subsequent reporting time steps. 4) TIMESTEP: write out the multi-segment well data at the current reporting time step and all subsequent time steps. Note the commercial simulator also uses MULTISEG to control the output of “rivers” for when the RIVERS Model has been enabled via the <i>RIVRDIMS</i> keyword in the <i>RUNSPEC</i> section. OPM Flow does not support the RIVERS Model.			NO

Notes:

- 1) The keyword is followed by any number of records terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.129: WRFTPLT Keyword Description

See also the [WRFT](#) keyword in the [SCHEDULE](#) section that has less flexible reporting options.

Examples

The first example activates RFT output at the current reporting time step for all the wells that are opened to flow, otherwise the RFT data is written out the first time a well is opened.

```
--
--      WELL RFT, PLT AND SEGMENT DATA
--
-- WELL  RFT   PLT   SEGMENT
-- NAME  DATA DATA DATA
WRFTPLT
' * '      FOPN                                     /
/
```

The next example writes out the RFT and PLT data for two wells at the current reporting time step.

```
--
--      WELL RFT, PLT AND SEGMENT DATA
--
-- WELL  RFT   PLT   SEGMENT
-- NAME  DATA DATA DATA
WRFTPLT
OP01    YES    YES                                     /
OP02    YES    YES                                     /
/
```

The final example is shown below:

```
--
--          WELL RFT, PLT AND SEGMENT DATA
--
-- WELL   RFT   PLT   SEGMENT
-- NAME   DATA DATA DATA
WRTPLT
OP01     REPT  NO
OP02     NO    YES
/
```

In this case the RFT data for well OP01 is written out at the current reporting time step and all subsequent reporting time steps. For well OP02, no RFT is written out but the PLT data is written out for the current report time step only.

12.3.326 WSALT – DEFINE WATER INJECTION WELL SALT CONCENTRATIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSALT](#) keyword defines a water injection well's salt injection stream concentration that is to be used for when the salt option has been activated by the [BRINE](#) keywords in the [RUNSPEC](#) section. Note that if the Polymer option has also been activated by the [POLYMER](#) keyword in the [RUNSPEC](#) section, then the [WPOLYMER](#) keyword in the [SCHEDULE](#) section should be used to enter both the polymer and salt concentrations.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the injection salt concentrations are being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	SALTCON	A real positive vector of values that defines the salt concentration of the well's injection stream and consists of: 1) If the Standard Brine model has been invoked by the <i>BRINE</i> keyword, then SALTCON consist of one value representing the injection salt concentration; 2) If OPM Flow's Water Vaporization and Salt Precipitation models have been activated by the <i>VAPWAT</i> and <i>PRECSALT</i> keywords in the <i>RUNSPEC</i> section, then SALTCON consist of one value representing the injection salt concentration; or, 3) If the Multi-Component Brine option has been activated by the <i>BRINE</i> and <i>ECLMC</i> keywords in the <i>RUNSPEC</i> section, then SALTCON consists of a vector of values representing the salt concentration of each brine within the injected brine mixture. Only options (1) and (2) are currently supported. This value may be specified using a User Defined Argument (UDA). Note if SALTCON is defaulted (I*) then the well's salt concentration will be equal to the well's group salt concentration.			I*
		lb/stb	kg/sm3	gm/scc	

Notes:

1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well and connection keywords in the *SCHEDULE* section. User Defined Arguments are activated using the *UDADIMS* keyword in the *RUNSPEC* section, and defined by the *UDQ* keyword in the *SCHEDULE* section.

2) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.

Table 12.130: WSALT Keyword Description

Water injection wells that are not declared via this keyword have their concentrations defaulted to zero.

See also the [GCONPROD](#) and [GCONINJE](#) keywords to define a group's production and injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example defines the salt injection stream concentration for three water injection wells for when the brine phase has been activated by the [BRINE](#) keyword in the [RUNSPEC](#) section.

```
--
--      DEFINE WATER INJECTION WELL SALT CONCENTRATIONS (STANDARD)
--
-- WELL  SALT-1      SALT-2      SALT-3      SALT-4
-- NAME  SALTCON    SALTCON    SALTCON    SALTCON
--      -----
WSALT
WI01    0.2500
WI02    1*
WI03    0.2500
/
```

The salt concentration for both well WI01 and WI03 is set to 0.25, and for well WI02 the salt concentration will be taken from the well's group salt concentration.

The next example is based on using the Multi-Component Brine option, that is the [BRINE](#) and [ECLMC](#) keywords have been used in the [RUNSPEC](#) section, and assuming three salts.

```
--
--      DEFINE WATER INJECTION WELL SALT CONCENTRATIONS (MULTIPLE)
--
-- WELL  SALT-1      SALT-2      SALT-3      SALT-4
-- NAME  SALTCON    SALTCON    SALTCON    SALTCON
--      -----
WSALT
WI01    0.1500      0.0500      0.0500
WI02    0.1500      0.0500      0.0500
WI03    0.2000      0.0500      0.0600
/
```

Here the salt concentrations for both well WI01 and WI02 are set to 0.1500, 0.0500, 0.0500 for the three salts and for well WI03 the salt concentrations are 0.2000, 0.0500 and 0.0600.

Note that OPM Flow does not currently support the Multi-Component brine model.

12.3.327 WSCCLEAN – WELL DEPOSITED SCALE ADJUSTMENT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSCCLEAN](#) keyword adjusts the amount of scale currently accumulated around a well's well connections for wells located in the global grid. For example, if a workover has been performed on a well to remove (or reduce) the deposited scale over the perforations, then this keyword can be used to implement the effects of the workover. Scale deposits reduce the productivity of well and this relationship is defined in the [SCDPTAB](#) and [SCDATAB](#) keywords in [SCHEDULE](#) section. The tables are allocated to a well via the [WSCTAB](#) keyword, which is also in the [SCHEDULE](#) section. Note that the Scale Deposition option must have been activated by declaring the dimensions of the scaling deposition tables using the [SCDPDIMS](#) keyword in the [RUNSPEC](#) section.

See also the WSSCLENL keyword in the [SCHEDULE](#) section that performs similar functionality for wells located in a Local Grid Refinement.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.328 WSCCLENL – WELL DEPOSITED SCALE ADJUSTMENT (LGR)

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSCCLENL](#) keyword adjusts the amount of scale currently accumulated around a well's well connections for wells located in a Local Grid Refinement ("LGR"). For example, if a workover has been performed on a well to remove (or reduce) the deposited scale over the perforations, then this keyword can be used to implement the effects of the workover. Scale deposits reduce the productivity of well and this relationship is defined in the [SCDPTAB](#) and [SCDATAB](#) keywords in [SCHEDULE](#) section. The tables are allocated to a well via the [WSCTAB](#) keyword, which is also in the [SCHEDULE](#) section. Note that the Scale Deposition option must have been activated by declaring the dimensions of the scaling deposition tables using the [SCDPDIMS](#) keyword in the [RUNSPEC](#) section.

See also the [WSCCLEAN](#) keyword in the [SCHEDULE](#) section that performs similar functionality for wells located in the global grid.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.329 WSCTAB – ASSIGN WELL SCALE DEPOSITION AND SCALE DAMAGE TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WSCTAB](#) assigns scale deposition and scale damage tables to a well, for when the Scale Deposition option has been activated by declaring the dimensions of the scaling deposition tables using the [SCDPDIMS](#) keyword in the [RUNSPEC](#) section. Scale deposits reduce the productivity of well and this relationship is defined in the [SCDPTAB](#) and [SCDATAB](#) keywords in the [SCHEDULE](#) section, and are allocated to a well by the [WSCTAB](#) keyword.

See also the [WPIMULT](#) keyword in the [SCHEDULE](#) section that adjusts a well's productivity index by a constant value.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.330 WSEGAICD – DEFINE MULTI-SEGMENT WELL AUTONOMOUS ICD CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WSEGAICD](#) keyword defines a multi-segment well segment to be an autonomous Inflow Control Device (“ICD”) as part of a completion for a multi-segment well. Note that the well must have been previously defined by the [WELSPECS](#) and [WELSEGS](#) keywords in the [SCHEDULE](#) section and that the data for the keyword should be repeated for each multi-segment completion that contains an autonomous ICD.

An ICD is a well completion component usually installed along the producing section of a well to minimize the unwanted water and gas breakthrough in an oil well, or early water production in a gas well, due to an uneven inflow profile over the completed interval. Permeability variations over the producing interval cause the high permeability zones to produce higher quantities of fluids than the lower permeability zones and this uneven producing fluid profile may result in bypassed hydrocarbons. Secondly, for horizontal wells, the pressure loss from the “toe” to the “heel” of the well again results in an uneven fluid profile over the producing interval. In order to rectify this ICDs can be installed so that the well fluids have to flow through an ICD before entering the tubing; thus, creating an additional “designed” pressure loss.

An autonomous ICD inhibits the production of high-mobility fluids such as water and gas since the pressure drop in each unit is dependent on fluid properties and mobility, the device automatically increases the pressure differential across zones with high water or gas saturations, thus choking back production from these zones.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using both the WELSPECS and WELSEGS keywords in the SCHEDULE section, otherwise an error may occur.			None
2	ISEG1	An integer greater than or equal to two and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section that defines the start of the segment range.			None
3	ISEG2	An integer greater than or equal to two and not less then ISEG1 on this record and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section, that defines the end of the segment range.			None
4	ICDSTREN	A positive real value that defines an empirical constant for the strength of the given ICD as determined from measurements using the calibrated fluid.			None
		$\text{psi}/((\text{lb}/\text{ft}^3)(\text{rft}^3/\text{day})^2)$	$\text{bars}/((\text{kg}/\text{m}^3)(\text{rm}^3/\text{day})^2)$	$\text{atm}/((\text{gm}/\text{cc})(\text{rcc}/\text{hr})^2)$	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	ICDLEN	A real value that defines the length of the ICD used in conjunction with NSCALFAC to calculate a scaling factor to be applied to the reservoir flow to adjust the flow through each ICD, that is: 1) If NSCALFAC equals zero: then the scale factor is equal to the length of the ICD (ICDLEN) divided by the length of the tubing section, that is the parent of the ICDs, then this allows for the case when the ICD segment may represent a number of ICDs in parallel. 2) If NSCALFAC equals one: then the scale factor is equal to the absolute value of ICDLEN. 3) If NSCALFAC equals two: then the scale factor is equal to the length of ICDLEN, divided by the total length of the completions which supply the ICD. NSCALFAC explicitly sets which of the above three options is used. If NSCALFAC is defaulted, then option 1) is used whenever ICDLEN is positive and option 2) when ICDLEN is negative.			Defined
		feet 39.37	m 12.00	cm 1,2000	
6	CALDEN	A positive real value that defines the density of the calibrating fluid at surface conditions.			Defined
		lb/ft ³ 62.416	kg/m ³ 1000.25	gm/cc 1.00025	
7	CALVISC	A positive real value that defines the viscosity of the calibrating fluid at surface conditions.			1.45
		cP	cP	cP	
8	EMLCRT	A positive real value that defines the “local water” in liquid fraction used to determine whether the “water-in-oil” or “oil-in-water” emulsion viscosity equation should be applied.			0.5
		dimensionless	dimensionless	dimensionless	
9	EMLTRANS	A positive real value that defines the width of the transition zone around EMLCRT and is used to ensure that the calculated viscosity forms a continuous function of water in liquid fraction. Within this region, the emulsion viscosity is a linear interpolation between the “water-in-oil” and “oil-in-water” viscosity values either side of the region.			0.05
		dimensionless	dimensionless	dimensionless	
10	EMLMAX	A positive real value that defines the maximum emulsion viscosity to continuous phase viscosity (oil or water) ratio.			5.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
11	NSCALFAC	An integer value greater than or equal to zero, that specifies the method to be used when applying the scaling factor and should be set to one of the following: 1) If NSCALFAC equals zero: then the scale factor is equal to the length of the ICD (ICDLEN) divided by the length of the tubing section, that is the parent of the ICDs, then this allows for the case when the ICD segment may represent a number of ICDs in parallel. 2) If NSCALFAC equals one: then the scale factor is equal to the absolute value of ICDLEN. 3) If NSCALFAC equals two: then the scale factor is equal to the length of ICDLEN, divided by the total length of the completions which supply the ICD. NSCALFAC explicitly sets which of the above three options is used. If NSCALFAC is negative, then option 1) is used whenever ICDLEN is positive and option 2) when ICDLEN is negative.			-1
		dimensionless	dimensionless	dimensionless	
12	CALRATE	A positive real value that defines the maximum surface flow rate for which the ICD was calibrated. Values calculated greater than CALRATE will use linear extrapolation.			None
		scf/d	sm ³ /day	scc/hour	
13	RATEXP	A real value greater than or equal to zero that defines the flow rate exponent in equation (12.33).			None
		dimensionless	dimensionless	dimensionless	
14	VISCEXP	A real value greater than or equal to zero that defines the viscosity exponent in equation (12.33).			None
		dimensionless	dimensionless	dimensionless	
15	STATUS	A defined character string that specifies the ICD's operational status, STATUS should be set to one of the following character strings: 1) OPEN: the ICD connections are open to flow. 2) SHUT: the ICD connections are closed to flow (shut-in).			OPEN
16	A1	A real value that defines the density mixture <i>OIL</i> flowing fraction exponent in equation (12.34)			1.0
17	A2	A real value that defines the density mixture <i>WATER</i> flowing fraction exponent in equation (12.34)			1.0
18	A3	A real value that defines the density mixture <i>GAS</i> flowing fraction exponent in equation (12.34)			1.0
19	B1	A real value that defines the viscosity mixture <i>OIL</i> flowing fraction exponent in equation (12.35)			1.0
20	B2	A real value that defines the viscosity mixture <i>WATER</i> flowing fraction exponent in equation (12.35)			1.0

No.	Name	Description			Default
		Field	Metric	Laboratory	
21	B3	A real value that defines the viscosity mixture GAS flowing fraction exponent in equation (12.35)			1.0
22	DENEXP	A real value greater than or equal to zero that defines the density exponent in equation (12.33).			None
		dimensionless	dimensionless	dimensionless	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.131: WSEGAICD Keyword Description

The total number of wells should be defined via the [WELLDIMS](#) keyword and the number of multi-segment wells should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section. In addition, the [WELSPCE](#) keyword should be used to define wells, the [COMPDAT](#) keyword to define the well completions for both standard wells and multi-segment wells, and the [COMPSEGS](#) keyword to define a multi-segment well segment completions. Finally, the [WSEGSICD](#) keyword can be used to define spiral ICD connections for the well. All the aforementioned keywords are described in the [SCHEDULE](#) section.

The equations used to calculate the pressure drop across the ICD are given below and illustrate how the pressure reduction is dependent on the density and viscosity of the fluid flowing through the device.

$$\Delta P = \left(\frac{\rho_{mixture}}{\rho_{calibrated}} \right)^{\rho_{exp}} \cdot \left(\frac{\mu_{calibrated}}{\mu_{mixture}} \right)^{\mu_{exp}} \cdot \rho_{mixture} \cdot \beta \cdot \left(\frac{q}{q_{calibrated}} \right)^{(Q_{exp} - 2)} \cdot q^2 \quad (12.33)$$

Where:

ΔP = the pressure drop across the device.
 $\rho_{mixture}$ = the density of the mixture, as per:

$$\rho_{mixture} = (\alpha_{oil}^{a_1} \cdot \rho_{oil}) + (\alpha_{wat}^{a_2} \cdot \rho_{wat}) + (\alpha_{gas}^{a_3} \cdot \rho_{gas}) \quad (12.34)$$

$\rho_{calibrated}$ = CALDEN, the density of the calibrating fluid at surface conditions.
 $\mu_{calibrated}$ = CALVISC, the viscosity of the calibrating fluid at surface conditions.
 $\mu_{mixture}$ = the viscosity of the mixture, as per:

$$\mu_{mixture} = (\alpha_{oil}^{b_1} \cdot \mu_{oil}) + (\alpha_{wat}^{b_2} \cdot \mu_{wat}) + (\alpha_{gas}^{b_3} \cdot \mu_{gas}) \quad (12.35)$$

β = ICDSTREN, the strength of the ICD as measured using the calibrated fluid.
 q = the local flow rate through the ICD at local conditions adjusted for scaling based on the ICDLEN and NSCALFAC parameters in Table 12.131.
 $q_{calibrated}$ = CALRATE, the maximum surface flow rate for which the ICD was calibrated.
 ρ_{exp} = DENEXP, the density exponent in Table 12.131.
 μ_{exp} = VISCEXP, the viscosity exponent in Table 12.131.
 Q_{exp} = RATEXP, the flow rate exponent in Table 12.131.

In equation (12.34) α_i represent the volume fractions of oil, water and gas at local conditions and a_i the density exponents of the three phases (A1, A2 and A3 in Table 12.131). Similarity for equation (12.35), α_i represent the volume fractions of oil, water and gas at local conditions and b_i the viscosity exponents of the three phases (B1, B2 and B3 in Table 12.131).

See also the [WSEGSICD](#) keyword in the [SCHEDULE](#) section for spiral ICDs, that operate in a similar fashion to autonomous ICDs.

Example

The following example defines a multi-segment oil production well (OP01) using the [WELSPECS](#), [WELSEGS](#) [COMPDAT](#) and [COMPSEGS](#) keywords, followed by the [WSEGAICD](#) keyword to define the autonomous inflow control devices for the well.

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP   LOCATION  BHP   PHASE  DRAIN  INFLOW  OPEN  CROSS PVT   DEN  FIP
-- NAME  NAME      I      J  DEPTH FLUID  AREA   EQUANS SHUT  FLOW  TABLE CAL  NUM
WELSPECS
OP01    PLATFORM 14      8  1*    OIL    0.0    STD    SHUT  YES   0      SEG  0 /
/
--
--      WELL CONNECTION DATA
--
-- WELL  --- LOCATION --- OPEN  SAT  CONN      WELL  KH  SKIN  D  DIR
-- NAME  II  JJ  K1  K2  SHUT  TAB  FACT      DIA  FACT FACT FACT PEN
COMPDAT
OP01    14   8   1   1  OPEN  1*   1.972960E+2 0.216 1*   0.00 1*  'Z' /
OP01    14   7   1   1  OPEN  1*   1.619450E+2 0.216 1*   0.00 1*  'Z' /
OP01    14   7   2   2  OPEN  1*   4.126449E+2 0.216 1*   0.00 1*  'Z' /
OP01    14   7   3   3  OPEN  1*   2.033290E+1 0.216 1*   0.00 1*  'Z' /
OP01    15   7   3   3  OPEN  1*   9.095613E+1 0.216 1*   0.00 1*  'Z' /
OP01    15   6   3   3  OPEN  1*   2.090607E+2 0.216 1*   0.00 1*  'Y' /
OP01    15   6   4   4  OPEN  1*   3.010669E+1 0.216 1*   0.00 1*  'Y' /
OP01    16   6   4   4  OPEN  1*   7.123814E+1 0.216 1*   0.00 1*  'Y' /
OP01    16   5   4   4  OPEN  1*   4.414386E+2 0.216 1*   0.00 1*  'Y' /
OP01    16   4   4   4  OPEN  1*   4.345126E+2 0.216 1*   0.00 1*  'Y' /
OP01    16   3   4   4  OPEN  1*   2.894573E+2 0.216 1*   0.00 1*  'Y' /
OP01    17   3   4   4  OPEN  1*   1.329523E+2 0.216 1*   0.00 1*  'Y' /
OP01    17   2   4   4  OPEN  1*   6.981252E+1 0.216 1*   0.00 1*  'Y' /
OP01    17   2   5   5  OPEN  1*   1.392382E+2 0.216 1*   0.00 1*  'Y' /
/
--
--      WELL SEGMENT SPECIFICATION DATA
--
-- WELL  NODAL      LEN  WELL  DEPH  PRESS  FLOW
-- NAME  DEPTH      TUBING VOLM  OPTN  CALC  MODEL
WELSEGS
OP01    2041.56259  0.0000  1*    INC  'HF - '  /
--
--      SEG  SEG  BRAN  SEG  TUBING  NODAL  TUBE  TUBE  XSEC  VOL
--      ISTR  IEND NO    NO    LENGTH DEPTH  ID    ROUGH AREA  SEG
--      2      2   1     1     9.56005 0.61168 0.152 0.0000100 /
--      3      3   1     2    17.40714 1.11376 0.152 0.0000100 /
--      4      4   1     3    41.24996 2.38255 0.152 0.0000100 /
--      5      5   1     4    38.35922 2.06899 0.152 0.0000100 /
--      6      6   1     5    27.13029 1.02248 0.152 0.0000100 /
--      7      7   1     6    73.45099 2.40389 0.152 0.0000100 /
--      8      8   1     7    54.95304 1.64832 0.152 0.0000100 /
--      9      9   1     8    12.37381 0.23781 0.152 0.0000100 /
--     10     10   1     9    62.61459 0.73403 0.152 0.0000100 /
```

	11	11	1	10	106.9805	1.37749	0.152	0.0000100		/
	12	12	1	11	88.42739	0.90931	0.152	0.0000100		/
	13	13	1	12	51.59899	0.27327	0.152	0.0000100		/
	14	14	1	13	24.75582	0.30814	0.152	0.0000100		/
	15	15	1	14	29.77334	0.49371	0.152	0.0000100		/
--										
-- PERFORATION VALVE SEGMENTS										
--										
--	SEG	SEG	BRAN	SEG	TUBING	NODAL	TUBE	TUBE	XSEC	VOL
--	ISTR	IEND	NO	NO	LENGTH	DEPTH	ID	ROUGH	AREA	SEG
	16	16	2	2	0.10000	0	0.152	0.0000100		/
	17	17	3	3	0.10000	0	0.152	0.0000100		/
	18	18	4	4	0.10000	0	0.152	0.0000100		/
	19	19	5	5	0.10000	0	0.152	0.0000100		/
	20	20	6	6	0.10000	0	0.152	0.0000100		/
	21	21	7	7	0.10000	0	0.152	0.0000100		/
	22	22	8	8	0.10000	0	0.152	0.0000100		/
	23	23	9	9	0.10000	0	0.152	0.0000100		/
	24	24	10	10	0.10000	0	0.152	0.0000100		/
	25	25	11	11	0.10000	0	0.152	0.0000100		/
	26	26	12	12	0.10000	0	0.152	0.0000100		/
	27	27	13	13	0.10000	0	0.152	0.0000100		/
	28	28	14	14	0.10000	0	0.152	0.0000100		/
	29	29	15	15	0.10000	0	0.152	0.0000100		/
/										
--										
-- MULTISEGMENT WELL COMPLETION SEGMENT SPECIFICATION DATA										
--										
-- WELL										
-- NAME										
COMPSEGS										
OP01										
/										
--										
--	--LOCATION--			BRAN	TUBING	NODAL	DIR	LOC	MID	COMP
--	II	JJ	K1	NO	LENGTH	DEPTH	PEN	I, J, K	PERFS	LENGTH
--										ISEG
--										NO.
	14	8	1	2	0.000000	0.10000				/
	14	7	1	3	0.000000	0.10000				/
	14	7	2	4	0.000000	0.10000				/
	14	7	3	5	0.000000	0.10000				/
	15	7	3	6	0.000000	0.10000				/
	15	6	3	7	0.000000	0.10000				/
	15	6	4	8	0.000000	0.10000				/
	16	6	4	9	0.000000	0.10000				/
	16	5	4	10	46.76168	46.86168				/
	16	4	4	11	0.000000	0.10000				/
	16	3	4	12	0.000000	0.10000				/
	17	3	4	13	0.000000	0.10000				/
	17	2	4	14	0.000000	0.10000				/
	17	2	5	15	42.50573	42.60573				/
/										
--										
-- MULTI-SEGMENT WELL ICD SEGMENT SPECIFICATION DATA										
--										
-- WELL	SEG	SEG	ICD	ICD	CAL	CAL	EML	EML	EML	SCAL
-- NAME	ISTR	IEND	STREN	LEN	DEN	VISC	CRIT	TRAN	MAX	FAC
										RAT
										EXP
										VISC
										OPEN
										CLOSE
WSEGAICD										
OP01	16	16	7.5e-5	-0.028975	1020	0.48	0.7	1*	1*	1*
OP01	17	17	7.5e-5	-0.023783	1020	0.48	0.7	1*	1*	1*
OP01	18	18	7.5e-5	-0.101240	1020	0.48	0.7	1*	1*	1*
OP01	19	19	7.5e-5	-0.015022	1020	0.48	0.7	1*	1*	1*
OP01	20	20	7.5e-5	-0.067205	1020	0.48	0.7	1*	1*	1*
OP01	21	21	7.5e-5	-0.155410	1020	0.48	0.7	1*	1*	1*
/										


```

OP01      22      22      7.5e-5 -0.011141 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
OP01      23      23      7.5e-5 -0.026362 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
OP01      24      24      7.5e-5 -0.163410 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
OP01      25      25      7.5e-5 -0.160830 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
OP01      26      26      7.5e-5 -0.107180 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
--
-- WELL SEG SEG ICD ICD CAL CAL EML EML EML SCAL CAL RATE VISC OPEN
-- NAME ISTR IEND STREN LEN DEN VISC CRIT TRAN MAX FAC RAT EXP EXP CLOSE
OP01      27      27      7.5e-5 -0.049206 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
OP01      28      28      7.5e-5 -0.025824 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
OP01      29      29      7.5e-5 -0.064414 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
/

```

The [WSEGAICD](#) keyword defines the autonomous ICD over segments 16 to 29 using the default value for NSCALFAC of -1, which sets the scale factor equal to the absolute value of ICDLEN, as all the ICDLEN values are negative. Note that the [WSEGAICD](#) should be repeated for each multi-segment well that has this type of ICD. So for example for two wells one would have:

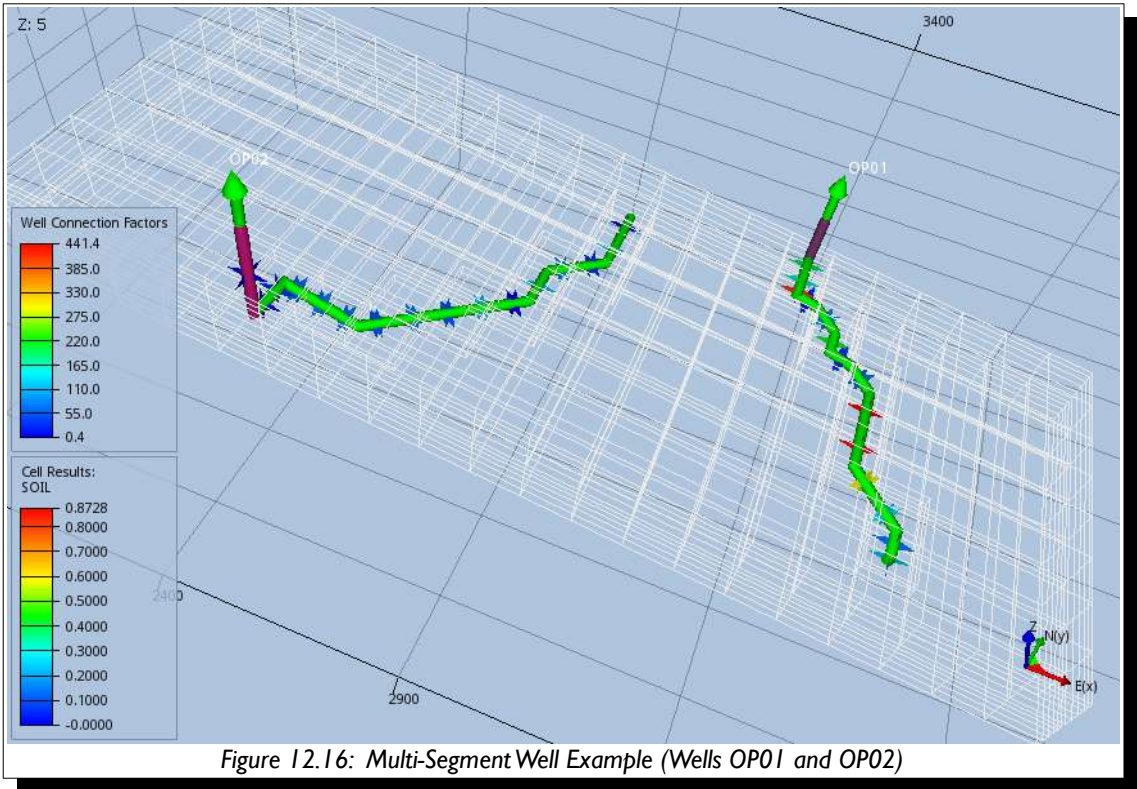
```

--
--          MULTI-SEGMENT WELL ICD SEGMENT SPECIFICATION DATA
--
-- WELL SEG SEG ICD ICD CAL CAL EML EML EML SCAL CAL RATE VISC OPEN
-- NAME ISTR IEND STREN LEN DEN VISC CRIT TRAN MAX FAC RAT EXP EXP CLOSE
WSEGAICD
OP01      16      16      7.5e-5 -0.028975 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
...
OP01      28      28      7.5e-5 -0.025824 1020 0.48 0.7 1* 1* 1* 1* 2.2 0.5 7* /
/

-- WELL SEG SEG ICD ICD CAL CAL EML EML EML SCAL CAL RATE VISC OPEN
-- NAME ISTR IEND STREN LEN DEN VISC CRIT TRAN MAX FAC RAT EXP EXP CLOSE
WSEGAICD
OP02      17      17      1.75E-4 .0049945 1020 0.48 1* 1* 1* 1* 1* 2.1 1.2 7* /
...
OP02      31      31      1.75E-4 .0038583 1020 0.48 1* 1* 1* 1* 1* 2.1 1.2 7* /
/

```

Figure 12.16 shows the layout of the two example wells.



12.3.331 WSEGDFIN – DEFINE MULTI-SEGMENT WELL DRIFT FLUX SLIP MODEL INPUT DATA

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The [WSEGDFIN](#) keyword defines a multi-segment well's drift flux slip model parameters. A slip model³⁴⁶ and ³⁴⁷ enables the different phases in the wellbore to flow at different velocities, for example gas will flow up the tubing at a higher velocity than oil and water. The option is activated by the FLOWOPT parameter on the [WELSEGS](#) keyword in the [SCHEDULE](#) section; however, the slip model flow calculation is not available in OPM Flow.

See also the WSEGFMMD keyword in the [SCHEDULE](#) section that sets which type of slip model should be used.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

³⁴⁶

Shi, H., Holmes, J.A., Durlofsky, L. J., Aziz, K., Diaz, L. R., Alkaya, B., and Oddie, G. “Drift-Flux Modeling of Two-Phase Flow in Wellbores,” paper SPE 84228, Society of Petroleum Engineers Journal (2005) 10, No. 1, 24-33; also presented as “Drift-Flux Modeling of Multiphase Flow in Wellbores,” at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA (October 5-8, 2003).

³⁴⁷

Shi, H., Holmes, J.A., Diaz, L. R., Durlofsky, L. J., and Aziz, K. “Drift-Flux Parameters for Three-Phase Steady-State Flow in Wellbores,” paper SPE 89836, Society of Petroleum Engineers Journal (2005) 10, No. 2, 130-137; also presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA (September 26-29, 2004).

12.3.332 WSEGDFMD – DEFINE MULTI-SEGMENT WELL DRIFT FLUX SLIP MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WSEGDFMD](#) keyword defines a multi-segment well's drift flux slip model definition that sets the type of slip model to be used. A slip model³⁴⁸ and ³⁴⁹ enables the different phases in the wellbore to flow at different velocities, for example gas will flow up the tubing at a higher velocity than oil and water. The option is activated by the FLOWOPT parameter on the [WELSEGS](#) keyword in the [SCHEDULE](#) section; however, the slip model flow calculation is not available in OPM Flow.

See also the [WSEGDFIN](#) keyword that sets the slip model's input parameters and the [WSEGDFPA](#) keyword that allows the model default values employed by [WSEGDFMD](#) to be modified..

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

³⁴⁸

Shi, H., Holmes, J.A., Durlofsky, L. J., Aziz, K., Diaz, L. R., Alkaya, B., and Oddie, G. “Drift-Flux Modeling of Two-Phase Flow in Wellbores,” paper SPE 84228, Society of Petroleum Engineers Journal (2005) 10, No. 1, 24-33; also presented as “Drift-Flux Modeling of Multiphase Flow in Wellbores,” at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA (October 5-8, 2003).

³⁴⁹

Shi, H., Holmes, J.A., Diaz, L. R., Durlofsky, L. J., and Aziz, K. “Drift-Flux Parameters for Three-Phase Steady-State Flow in Wellbores,” paper SPE 89836, Society of Petroleum Engineers Journal (2005) 10, No. 2, 130-137; also presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA (September 26-29, 2004).

12.3.333 WSEGDFPA – DEFINE MULTI-SEGMENT WELL DRIFT FLUX SLIP MODEL PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [WSEGDFPA](#), enables modification of a multi-segment well’s drift flux slip model default parameters used by the WSEGDFMA keyword in the [SCHEDULE](#) section to define the model.A slip model³⁵⁰ and ³⁵¹ enables the different phases in the wellbore to flow at different velocities, for example gas will flow up the tubing at a higher velocity than oil and water.The option is activated by the FLOWOPT parameter on the [WELSEGS](#) keyword in the [SCHEDULE](#) section; however, the slip model flow calculation is not available in OPM Flow.

See also the [WSEGDFIN](#) keyword that sets the slip model’s input parameters and the [WSEGDFMD](#) keyword that sets which drift flux slip model should be used. Note if the [WSEGDFPA](#) keyword is used than it must be placed after the [WSEGDFMD](#) keyword, but before the [WELSEGS](#) keyword that defines a multi-segment well. All the aforementioned keywords are in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

³⁵⁰ Shi, H., Holmes, J.A., Durlofsky, L. J., Aziz, K., Diaz, L. R., Alkaya, B., and Oddie, G. “Drift-Flux Modeling of Two-Phase Flow in Wellbores,” paper SPE 84228, Society of Petroleum Engineers Journal (2005) 10, No. 1, 24-33; also presented as “Drift-Flux Modeling of Multiphase Flow in Wellbores,” at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA(October 5-8, 2003).

³⁵¹ Shi, H., Holmes, J.A., Diaz, L. R., Durlofsky, L. J., and Aziz, K. “Drift-Flux Parameters for Three-Phase Steady-State Flow in Wellbores,” paper SPE 89836, Society of Petroleum Engineers Journal(2005) 10, No. 2, 130-137; also presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA (September 26-29, 2004).

12.3.334 WSEGEXSS – DEFINE MULTI-SEGMENT WELL IMPORT-EXPORT SEGMENT VOLUMES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [WSEGEXSS](#), enables the import or export of fluids from a segment in a multi-segment well. This can be used to, for example, model gas lift injection for oil wells under artificial lift, or to approximate the behavior of a down-hole separator. The import-export fluid volumes can either be expressed as rates or defined as a function of a segment's pressure value.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.335 WSEGFLIM – DEFINE MULTI-SEGMENT WELL ARTIFICIAL CHOKE CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

[WSEGFLIM](#) enables an artificial choke that chokes a given phase flow rate for a segment in a multi-segment well. This can be used, for example, to constraint unwanted production phase through a section of tubing, or to model a down-hole choke. The keyword provides coefficients that are applied to the frictional pressure drop across a multi-segment well's segment in order to inhibit production from that particular zone or segment. As such, the keyword does not actually model a down-hole choke; hence, the term artificial.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.336 WSEGMOD – DEFINE MULTI-SEGMENT WELL MODEL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

The [WSEGMOD](#) declares the multi-phase flow model to be used to calculate the pressure drop within an individual segment for multi-segment wells. The FLOWOPT parameter on the [WELSEGS](#) keyword in the [SCHEDULE](#) section sets the default multi-segment well model. FLOWOPT is a character string that can be set to HO that activates the homogeneous model, that is all phases flow at the same velocity, or DF that invokes the Drift Flux Slip model (note OPM Flow only supports the default value of HO for the homogeneous model). Here [WSEGMOD](#) can be used to set the flow model for a segment to either the homogeneous model or the Drift Flux Slip model, and addition a: VLP table allocated via the [WSEGTABL](#) keyword, or a specific model as defined by the [WSEGVALV](#), [WSEGFLIM](#) and [WSEGLABY](#) keywords. All the aforementioned keywords are in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.337 WSEGINIT – DEFINE MULTI-SEGMENT WELL INITIAL CONDITIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

Normally the simulator calculates the initial conditions for multi-segment wells, that is the pressure and fluid distributions in each segment. However, there are occasions when manually setting the pressures and phase distributions for each segment to investigate certain flow conditions may be useful. In this case the [WSEGINIT](#) keyword may be used to specify the initial conditions manually. Note that segments not initialized by this keyword will be automatically initialized by the simulator

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.338
WSEGITER
–
DEFINE
MULTI-SEGMENT
WELLS
ITERATION
PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

The *WSEGITER* keyword defines the multi-segment well solution iteration sequence and solution controls.

OPM Flow uses a different numerical scheme which makes this keyword redundant; hence, OPM Flow ignores this keyword. It is documented here for completeness. See section [2.2 Running OPM Flow From The Command Line](#) on how to set the numerical control parameters for OPM Flow.

12.3.339 WSEGLABY – DEFINE MULTI-SEGMENT WELL LABYRINTH ICD CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSEGSICD](#) keyword defines a multi-segment well segment to be a labyrinth Inflow Control Device. ("ICD") as part of a completion for a multi-segment well. Note that the well must have been previously defined by the [WELSPECS](#) and [WELSEGS](#) keywords in the [SCHEDULE](#) section and that the data for the keyword should be repeated for each multi-segment completion that contains a labyrinth ICD.

An ICD is a well completion component usually installed along the producing section of a well to minimize the unwanted water and gas breakthrough in an oil well, or early water production in a gas well, due to an uneven flow profile over the completed interval. Permeability variations over the producing interval cause the high permeability zones to produce higher quantities of fluids than the lower permeability zones and this uneven producing fluid profile may result in bypassed hydrocarbons. Secondly, for horizontal wells, the pressure loss from the "toe" to the "heel" of the well again results in an uneven fluid profile over the producing interval. In order to rectify this ICDs can be installed so that the well fluids have to flow through an ICD before entering the tubing; thus, creating an additional "designed" pressure loss.

A labyrinth ICD is a type of frictional ICD that adds an additional pressure loss by directing the fluid along a series of channels before entering the tubing. The channel flow path is designed in such a manner as to create the desired pressure loss for a given ICD. By placing various ICD's over the production interval one can design a completion that results in a more uniform producing fluid profile throughout the length of the producing interval. Although this type of ICD is not implemented in OPM Flow, it works in a similar fashion to how a spiral ICD works. Spiral ICDs are implemented in OPM Flow and the data is entered via the [WSEGSICD](#) keyword in the [SCHEDULE](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.340 WSEGLINK – DEFINE MULTI-SEGMENT WELL LOOPED FLOW PATHS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

This keyword, [WSEGLINK](#), specifies multi-segment well looped flow paths as part of a completion for a multi-segment well. A looped segment results in the nodes of the two individual segments that are looped (or connected) having the same solution pressures and oil, water and gas flowing rates.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.341
WSEGMULT – DEFINE MULTI-SEGMENT WELL FRICTIONAL PRESSURE LOSS MULTIPLIERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
---------	------	------	-------	---------	----------	---------	----------

Description

WSEGMULT supplies a set of constants used to modify (or scale) a multi-segment well's segment frictional pressure drop between connecting segments The constants enable either a constant pressure to be applied, or for the pressure drop to vary as a function of the Gas-Oil Ratio ("GOR") or the Water-Oil Ratio ("WOR"). The simulator calculated pressure drop is multiplied by the following resulting value:

$$Frictional\ Loss\ Multipler = min\left(x_1 + x_2(WOR)^{x_3} + x_4\left(\frac{GOR}{GOR_{min}}\right)^{x_5}, 1.0\right)$$

(12.36)

Where the constants x_1 to x_5 are defined by the values on this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.342 WSEGPORP – MODIFY MULTI-SEGMENT WELLS AND THEIR SEGMENT STRUCTURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSEGPORP](#) keyword allows for the editing of exiting multi-segment wells created by [WELSEGS](#) keyword in the [SCHEDULE](#) section without having to re-define all the information that is on the [WELSEGS](#) keyword. Note that the well must have been previously define by both the [WELSPECS](#) and [WELSEGS](#) keywords in the [SCHEDULE](#) section to use the [WSEGPORP](#) keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	ISEG1	A positive integer greater than or equal to two and less than or equal to MXSEGS on WSEGDIMS keyword in the RUNSPEC section that defines the start of a segment			None
3	ISEG2	A positive integer greater than or equal to two and less than or equal to ISEG1 on this record and MXSEGS on the WSEGDIMS keyword in the RUNSPEC section that defines the end of a segment.			None
4	ID	A real positive value that defines the tubing internal diameter of the segment for the well.			Previous Entered Value
		feet	m	cm	
5	EPSILON	A real positive value that defines the tubing absolute roughness of the segment for the well.			Previous Entered Value
		feet	m	cm	
6	XAREA	XAREA is real positive value equal to or greater than zero that defines the cross sectional area for fluid flow. Currently this option is not supported by OPM Flow.			Previous Entered Value
		ft ²	m ²	cm ²	
7	VOLSEG	VOLSEG is a real positive value that defines the effective segment volume for the this segment. Currently this option is not supported by OPM Flow.			Previous Entered Value
		ft3	m3	cm3	

No.	Name	Description			Default
		Field	Metric	Laboratory	
8	XAREAS	XAREAS is real positive value equal to or greater than zero that defines the cross sectional area of the pipe wall for this segment, that is used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			Previous Entered Value
	ft ²	m ²	cm ²		
9	VHEATCAP	VHEATCAP is real positive value equal to or greater than zero that defines the volumetric heat capacity of the pipe wall used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			Previous Entered Value
	Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K		
10	THCON	THCON is real positive value equal to or greater than zero that defines the thermal conductivity of the pipe wall used in thermal conductivity calculations for when the temperature calculation is activated by the TEMP keyword in the RUNSPEC section. Currently this option is not supported by OPM Flow.			Previous Entered Value
	Btu/ft/day/°R	kJ/m/day/K	J/cm/hr/K		
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.132:WSEGPROP Keyword Description

The total number of wells should be defined via the [WELLDIMS](#) keyword and the number of multi-segment wells should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section.

See also the [WELSPECs](#) keyword to define wells, the [COMPDAT](#) keyword to define the well completions for both ordinary wells and multi-segment wells, and the [COMPSEGS](#) keyword to define a multi-segment well segment completions. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example modifies two segments in well OP01 and one segment in well OP02.

```
--
--          WELL SEGMENT SPECIFICATION DATA
--
-- WELL  SEG  SEG  TUBE  TUBE  XSEC  VOL
-- NAME  ISTR  IEND  ID    ROUGH  AREA  SEG
WSEGP
OP01    12    14    0.3   0.00010
OP01    13    15    0.275 0.00010
OP02    14    14    0.275 0.00010
/
```

Note that the two multi-segment wells and their respective segments must have been previously defined by the [WELSEGS](#) keyword.

12.3.343
WSEGPULL
–
DEFINE
A
MULTI-SEGMENT
WELL
DOWN-HOLE
SEPARATOR
PUMP

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	--------------------------

Description

This keyword, [WSEGPULL](#), specifies a multi-segment well segment to be a pull-through pump for a down-hole water separator, defined by the [WSEGSEP](#) keyword in the [SCHEDULE](#) section, and defines the various parameters for this type of pump. Down-hole separators are used to separate water or free gas from the in situ fluid entering the wellbore in order to increase hydrocarbon recovery.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.344 WSEGSEP – DEFINE A MULTI-SEGMENT WELL DOWN-HOLE SEPARATOR

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WSEGSEP](#) specifies a multi-segment well segment to be a down-hole separator, that enables the separation of fluids down-hole. Down-hole separators are used to separate water or free gas from the in situ fluid entering the wellbore in order to increase hydrocarbon recovery. See also the [WSEGPULL](#) keyword in the [SCHEDULE](#) section that specifies a pull-through pump for a down-hole water separator.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.345 WSEGSICD – DEFINE MULTI-SEGMENT WELL SPIRAL ICD CONNECTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
-------------------------	----------------------	----------------------	-----------------------	-------------------------	--------------------------	-------------------------	-----------------

Description

The [WSEGSICD](#) keyword defines a multi-segment well segment to be a spiral Inflow Control Device (“ICD”) as part of a completion for a multi-segment well. Note that the well must have been previously defined by the [WELSPECS](#) and [WELSEGS](#) keywords in the [SCHEDULE](#) section and that the data for the keyword should be repeated for each multi-segment completion that contains a spiral ICD.

An ICD is a well completion component usually installed along the producing section of a well to minimize the unwanted water and gas breakthrough in an oil well, or early water production in a gas well, due to an uneven flow profile over the completed interval. Permeability variations over the producing interval cause the high permeability zones to produce higher quantities of fluids than the lower permeability zones and this uneven producing fluid profile may result in bypassed hydrocarbons. Secondly, for horizontal wells, the pressure loss from the “toe” to the “heel” of the well again results in an uneven fluid profile over the producing interval. In order to rectify this ICDs can be installed so that the well fluids have to flow through an ICD before entering the tubing; thus, creating an additional “designed” pressure loss.

A spiral ICD is a type of frictional ICD that adds an additional pressure loss by directing the fluid along a “channel” (spiral/helix) before entering the tubing. The channel flow path is designed in such a manner as to create the desired pressure loss for a given ICD. By placing various ICD’s over the production interval one can design a completion that results in a more uniform producing fluid profile throughout the length of the producing interval.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using both the WELSPECS and WELSEGS keywords in the SCHEDULE section, otherwise an error may occur.			None
2	ISEG1	A positive integer greater than or equal to two and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section that defines the start of a segment			None
3	ISEG2	A positive integer greater than or equal to two and not less then ISEG1 on this record and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section, that defines the end of a segment			None
4	ICDSTREN	A real positive value greater than zero that defines an empirical constant for the strength of the given ICD as determined from measurements using the calibrated fluid.			None
		psia(rft3/day)2	barsa/(rm3/day)2	atma/(rcc/hr)2	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	ICDLEN	A real value that defines the length of the ICD used in conjunction with NSCALFAC to calculate a scaling factor to be applied to the reservoir flow to adjust the flow through each ICD, that is: 1) If NSCALFAC equals zero: then the scale factor is equal to the length of the ICD (ICDLEN) divided by the length of the tubing section, that is the parent of the ICDs, then this allows for the case when the ICD segment may represent a number of ICDs in parallel. 2) If NSCALFAC equals one: then the scale factor is equal to the absolute value of ICDLEN. 3) If NSCALFAC equals two: then the scale factor is equal to the length of ICDLEN, divided by the total length of the completions which supply the ICD. NSCALFAC explicitly sets which of the above three options is used. If NSCALFAC is defaulted, then option 1) is used whenever ICDLEN is positive and option 2) when ICDLEN is negative.			Defined
		feet 39.37	m 12.00	cm 1,2000	
6	CALDEN	CALDEN is a real positive value greater than zero that defines the density of the calibrating fluid at surface conditions.			Defined
		lb/ft ³ 62.416	kg/m ³ 1000.25	gm/cc 1.00025	
7	CALVISC	CALVISC is a real positive value greater than zero that defines the viscosity of the calibrating fluid at surface conditions.			1.45
		cP	cP	cP	
8	EMLCRT	EMLCRT is a real positive value greater than zero that defines the “local water” in liquid fraction used to determine whether the “water-in-oil” or “oil-in-water” viscosity emulation equation should be applied.			0.5
		dimensionless	dimensionless	dimensionless	
9	EMLTRANS	EMLTRANS is a real positive value greater than zero that defines the width of the transition zone around EMLCRT and is used to ensure that the calculated viscosity forms a continuous function of water in liquid fraction. Within this region, the emulsion viscosity is a linear interpolation between the “water-in-oil” and “oil-in-water” viscosity values either side of the region.			0.05
		dimensionless	dimensionless	dimensionless	
10	EMLMAX	EMLMAX is a real positive value greater than zero that defines the maximum emulsion viscosity to continuous phase viscosity (oil or water) ratio.			5.0
		dimensionless	dimensionless	dimensionless	

No.	Name	Description			Default
		Field	Metric	Laboratory	
11	NSCALFAC	NSCALFAC is a positive integer value that is greater than or equal to zero, that sets the method to be used when applying the scaling factor and should be set to one of the following: 1) If NSCALFAC equals zero: then the scale factor is equal to the length of the ICD (ICDLEN) divided by the length of the tubing section, that is the parent of the ICDs, then this allows for the case when the ICD segment may represent a number of ICDs in parallel. 2) If NSCALFAC equals one: then the scale factor is equal to the absolute value of ICDLEN. 3) If NSCALFAC equals two: then the scale factor is equal to the length of ICDLEN, divided by the total length of the completions which supply the ICD. NSCALFAC explicitly sets which of the above three options is used. If NSCALFAC is defaulted, then option 1) is used whenever ICDLEN is positive and option 2) when ICDLEN is negative.			-1
12	CALRATE	A real positive value that defines the maximum surface flow rate for which the ICD was calibrated.			None
		scf/d	sm ³ /day	scc/hour	
13	STATUS	A defined character string of length four that defines the ICD's operational status, STATUS should be set to one of the following character strings: 1) OPEN: the ICD connections are open to flow. 2) SHUT: the ICD connections are closed to flow (shut-in).			OPEN
Notes:					
1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.133: WSEGSICD Keyword Description

The total number of wells should be defined via the [WELLDIMS](#) keyword and the number of multi-segment wells should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section. In addition, the [WEL SPECS](#) keyword should be used to define wells, the [COMP DAT](#) keyword to define the well completions for both ordinary wells and multi-segment wells, and the [COMP SEGS](#) keyword to define a multi-segment segment completions. Finally, the [WSEGSICD](#) keyword can then be use to define ICD connections for the well. All the aforementioned keywords are described in the [SCHEDULE](#) section.

The equations used to calculate the pressure drop across the ICD are given below and illustrate how the pressure reduction is dependent on the density and viscosity of the fluid flowing through the device.

$$\Delta P = \left(\frac{\rho_{\text{calibrated}}}{\rho_{\text{mixture}}} \cdot \frac{\mu_{\text{mixture}}}{\mu_{\text{calibrated}}} \right)^{\frac{1}{4}} \cdot \frac{\rho_{\text{mixture}}}{\rho_{\text{calibrated}}} \cdot \beta \cdot q^2 \quad (12.37)$$

Where:

ΔP = the pressure drop across the device.
 ρ_{mixture} = the density of the mixture, as per:

$$\rho_{\text{mixture}} = (\alpha_{\text{oil}} \cdot \rho_{\text{oil}}) + (\alpha_{\text{wat}} \cdot \rho_{\text{wat}}) + (\alpha_{\text{gas}} \cdot \rho_{\text{gas}}) \quad (12.38)$$

$\rho_{\text{calibrated}}$ = CALDEN, the density of the calibrating fluid at surface conditions.
 $\mu_{\text{calibrated}}$ = CALVISC, the viscosity of the calibrating fluid at surface conditions.
 μ_{mixture} = the viscosity of the mixture, as per:

$$\mu_{\text{mixture}} = (\alpha_{\text{oil}} + \alpha_{\text{wat}}) \cdot \mu_{\text{emulsion}} + (\alpha_{\text{gas}} \cdot \mu_{\text{gas}}) \quad (12.39)$$

μ_{emulsion} = the viscosity of the oil-water emulsion at local conditions.
 β = ICDSTREN, the strength of the ICD as measured using the calibrated fluid.
 q = the local flow rate through the ICD at local conditions adjusted for scaling based on ICDLEN and NSCALFAC parameters in Table 12.133.

In equation (12.38), α_i represents the volume fraction of oil, water and gas at local conditions and ρ_i the density of the three phases. Similarity for equation (12.39), α_i represents the volume fraction of oil, water and gas at local conditions and μ_i the viscosity of the three phases.

See also the [WSEGAICD](#) keyword in the [SCHEDULE](#) section for autonomous ICDs, that work in a similar fashion to how sprial ICDs work.

Example

The following example defines one producing well segment oil well (OP01) using the [WELSPECS](#), [WELSEGS](#) [COMPDAT](#) and [COMPSEGS](#) keywords, followed by the [WSEGSICD](#) keyword to define the spiral inflow control devices for the well.

```
--
--      WELL SPECIFICATION DATA
--
-- WELL  GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  OPEN  CROSS  PVT
-- NAME  NAME        I      J  DEPTH  FLUID   AREA   EQUANS  SHUT  FLOW  TABLE
WELSPECS
OP01    PLATFORM    10    10    1*     OIL
/
--
--      WELL CONNECTION DATA
--
-- WELL  --- LOCATION ---  OPEN  SAT  CONN  WELL  KH  SKIN  D  DIR
-- NAME  II  JJ  K1  K2  SHUT  TAB  FACT  DIA  FACT  FACT  FACT  PEN
COMPDAT
OP01    10  10    1    1  OPEN  1*   200.  0.5
OP01    10  10    2    2  OPEN  1*   200.  0.5
OP01    10  10    3    3  OPEN  1*   200.  0.4
OP01    10  10    4    4  OPEN  1*   200.  0.4
OP01    10  10    5    5  OPEN  1*   200.  0.4
```

OPM OPEN POROUS MEDIA

OPM FLOW REFERENCE MANUAL (2025-10)

Revision: Rev-0

OP01	10	10	6	6	OPEN	1*	200.	0.4	/
OP01	9	10	2	2	OPEN	1*	200.	0.4	/
OP01	8	10	2	2	OPEN	1*	200.	0.4	/
OP01	7	10	2	2	OPEN	1*	200.	0.4	/
OP01	6	10	2	2	OPEN	1*	200.	0.4	/
OP01	5	10	2	2	OPEN	1*	200.	0.4	/
OP01	10	9	3	3	OPEN	1*	200.	0.4	/
OP01	10	8	3	3	OPEN	1*	200.	0.4	/
OP01	10	7	3	3	OPEN	1*	200.	0.4	/
OP01	10	6	3	3	OPEN	1*	200.	0.4	/
OP01	10	5	3	3	OPEN	1*	200.	0.4	/
OP01	9	10	5	5	OPEN	1*	200.	0.4	/
OP01	8	10	5	5	OPEN	1*	200.	0.4	/
OP01	7	10	5	5	OPEN	1*	200.	0.4	/
OP01	6	10	5	5	OPEN	1*	200.	0.4	/
OP01	5	10	5	5	OPEN	1*	200.	0.4	/
OP01	10	9	6	6	OPEN	1*	200.	0.4	/
OP01	10	8	6	6	OPEN	1*	200.	0.4	/
OP01	10	7	6	6	OPEN	1*	200.	0.4	/
OP01	10	6	6	6	OPEN	1*	200.	0.4	/
OP01	10	5	6	6	OPEN	1*	200.	0.4	/

/

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WELL SEGMENT SPECIFICATION DATA

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WELL NAME	NODAL DEPTH	LEN TUBING	WELL VOLM	DEPH OPTN	PRESS CALC	FLOW MODEL				
OP01	2512.5	2512.5	1.0E-5	ABS	HFA	H0				/
	SEG ISTR	SEG IEND	BRAN NO	SEG NO	TUBING LENGTH	NODAL DEPTH	TUBE ID	TUBE ROUGH	XSEC AREA	VOL SEG
	2	2	1	1	2537.5	2534.5	0.3	0.00010		/
	3	3	1	2	2562.5	2560.5	0.3	0.00010		/
	4	4	1	3	2587.5	2593.5	0.3	0.00010		/
	5	5	1	4	2612.5	2614.5	0.3	0.00010		/
	6	6	1	5	2637.5	2635.5	0.3	0.00010		/
	7	7	2	2	2737.5	2538.5	0.2	0.00010		/
	8	8	2	7	2937.5	2537.5	0.2	0.00010		/
	9	9	2	8	3137.5	2539.5	0.2	0.00010		/
	10	10	2	9	3337.5	2535.5	0.2	0.00010		/
	11	11	2	10	3537.5	2536.5	0.2	0.00010		/
	12	12	3	3	2762.5	2563.5	0.2	0.00010		/
	13	13	3	12	2962.5	2562.5	0.1	0.00010		/
	14	14	3	13	3162.5	2562.5	0.1	0.00010		/
	15	15	3	14	3362.5	2564.5	0.1	0.00010		/
	16	16	3	15	3562.5	2562.5	0.1	0.00010		/
	17	17	4	5	2812.5	2613.5	0.2	0.00010		/
	18	18	4	17	3012.5	2612.5	0.1	0.00010		/
	19	19	4	18	3212.5	2612.5	0.1	0.00010		/
	20	20	4	19	3412.5	2612.5	0.1	0.00010		/
	21	21	4	20	3612.5	2613.5	0.1	0.00010		/
	22	22	5	6	2837.5	2634.5	0.2	0.00010		/
	23	23	5	22	3037.5	2637.5	0.2	0.00010		/
	24	24	5	23	3237.5	2638.5	0.2	0.00010		/

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

	25	25	5	24	3437.5	2639.5	0.1	0.00010	/				
	26	26	5	25	3637.5	2639.5	0.1	0.00010	/				
/													
--													
--	COMPLETION SEGMENT SPECIFICATION DATA												
--													
--	WELL												
--	NAME												
COMPSEGS													
OP01									/				
--													
--	--LOCATION--		BRAN	TUBING	NODAL	DIR	LOC	MID	COMP	ISEG			
--	II JJ K1		NO	LENGTH	DEPTH	PEN	I, J, K	PERFS	LENGTH	NO.			
	10 10 1		1	2512.5	2525.0						/		
	10 10 2		1	2525.0	2550.0						/		
	10 10 3		1	2550.0	2575.0						/		
	10 10 4		1	2575.0	2600.0						/		
	10 10 5		1	2600.0	2625.0						/		
	10 10 6		1	2625.0	2650.0						/		
	9 10 2		2	2637.5	2837.5						/		
	8 10 2		2	2837.5	3037.5						/		
	7 10 2		2	3037.5	3237.5						/		
	6 10 2		2	3237.5	3437.5						/		
	5 10 2		2	3437.5	3637.5						/		
	10 9 3		3	2662.5	2862.5						/		
	10 8 3		3	2862.5	3062.5						/		
	10 7 3		3	3062.5	3262.5						/		
	10 6 3		3	3262.5	3462.5						/		
	10 5 3		3	3462.5	3662.5						/		
	9 10 5		4	2712.5	2912.5						/		
	8 10 5		4	2912.5	3112.5						/		
	7 10 5		4	3112.5	3312.5						/		
	6 10 5		4	3312.5	3512.5						/		
	5 10 5		4	3512.5	3712.5						/		
	10 9 6		5	2737.5	2937.5						/		
	10 8 6		5	2937.5	3137.5						/		
	10 7 6		5	3137.5	3337.5						/		
	10 6 6		5	3337.5	3537.5						/		
	10 5 6		5	3537.5	3737.5						/		
--													
--	MULTI-SEGMENT WELL ICD SEGMENT SPECIFICATION DATA												
--													
--	WELL	SEG	SEG	ICD	ICD	CAL	CAL	EML	EML	EML	SCAL	CAL	OPEN
--	NAME	ISTR	IEND	STRNEN	LEN	DEN	VISC	CRIT	TRANS	MAX	FAC	RATE	CLOSE
WSEGSICD													
OP01	7	10	0.00025	1*	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
OP01	12	15	0.00025	1*	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
OP01	17	20	0.00025	1*	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
OP01	22	22	0.00025	1*	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
OP01	23	23	0.00025	1*	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
OP01	24	24	0.00025	1*	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
OP01	25	25	0.00050	10.0	1.0	0.45	0.50	0.05	5.0	2	1*	OPEN	/
/													

Branch number two uses one ICD for segments seven to ten, branch number three again uses only one ICD for segments 12 to 15 and similarly branch number three uses one ICD. The fifth and final branch has a total of four ICDs with the last interval having a 10 foot length. Since NSCALFAC equals two for the ICDs, then

the scale factor is equal to the length of ICDLEN, divided by the total length of the completions which supply the ICD. Where ICDLEN is defaulted, the default value of 39.37 ft will be used for field units and 12 m for SI units.

12.3.346 WSEGSOLV – DEFINE MULTI-SEGMENT WELL ITERATIVE LINEAR SOLVER PARAMETERS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSEGSOLV](#) keyword defines the numerical control parameters for the iterative linear solver for multi-segment well looped flow paths, as defined by the [WSEGLINK](#) keyword in the [SCHEDULE](#) section. A looped segment results in the nodes of the two individual segments that are looped (or connected) having the same solution pressures and oil, water and gas flowing rates.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.347
WSEGTABL
– ASSIGN MULTI-SEGMENT WELL VLP TABLES TO SEGMENTS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WSEGTABL](#) assigns previously defined Vertical Lift Performance (“VLP”) tables as specified by the [VFPPROD](#) keyword in the [SCHEDULE](#) section, to multi-segment well segments, as well as stipulating how the tables are to be applied.

The FLOWOPT parameter on the [WELSEGS](#) keyword in the [SCHEDULE](#) section sets the default multi-segment well model. FLOWOPT either activates the homogeneous model, that is all phases flow at the same velocity, or the Drift Flux Slip model. However, the [WSEGFMOD](#) keyword in the [SCHEDULE](#) section, can be used to set the flow model for a segment to either the homogeneous model or the Drift Flux Slip model, and addition a: VLP table allocated via the [WSEGTABL](#) keyword, or a specific model as defined by the [WSEGVAlV](#), [WSEGFLIM](#) and [WSEGLABY](#) keywords.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.348 WSEGVAlV – DEFINE MULTI-SEGMENT WELL SUB-CRITICAL VALVE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSEGVAlV](#) keyword defines a multi-segment well segment to be a sub-critical valve Inflow Control Device (“ICD”) as part of a completion for a multi-segment well. Note that the well must have been previously defined by the [WELSPECS](#) and [WELSEGS](#) keywords in the [SCHEDULE](#) section and that the data for the keyword should be repeated for each multi-segment completion that contains a sub-critical valve ICD.

An ICD is a well completion component usually installed along the producing section of a well to minimize the unwanted water and gas breakthrough in an oil well, or early water production in a gas well, due to an uneven flow profile over the completed interval. Permeability variations over the producing interval cause the high permeability zones to produce higher quantities of fluids than the lower permeability zones and this uneven producing fluid profile may result in bypassed hydrocarbons. Secondly, for horizontal wells, the pressure loss from the “toe” to the “heel” of the well again results in an uneven fluid profile over the producing interval. In order to rectify this ICDs can be installed so that the well fluids have to flow through an ICD before entering the tubing; thus, creating an additional “designed” pressure loss.

A sub-critical valve ICD is a type of frictional ICD that adds an additional pressure loss by directing the fluid through a constriction before entering the tubing. The pressure drop is a function of the fluid density, the volumetric flow rate, and the diameter of the constriction. By placing various ICD’s over the production interval one can design a completion that results in a more uniform producing fluid profile throughout the length of the producing interval.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which a multi-segment well is being defined. Note that the well name (WELNAME) must have been declared previously using both the WELSPECS and WELSEGS keywords in the SCHEDULE section, otherwise an error may occur.			None
2	ISEG I	A positive integer greater than or equal to two and less than or equal to MXSEGS on the WSEGDIMS keyword in the RUNSPEC section that defines the segment containing the sub-critical valve.			None
3	ICDCV	A real positive value greater than zero that defines the dimensionless flow coefficient for the valve (C_v). This a vendor specific value for a given vendor’s ICD.			None
		dimensionless	dimensionless	dimensionless	
4	AREAREST	A real positive value that defines the cross-sectional area of flow in the restricted section of the valve (A_v) and should have a minimum value of 1.0×10^{-10} . This value may be specified using a User Defined Argument (UDA). AREAREST is used to convert the segment volumetric flow rate into the flow velocity at the constriction (u_v).			None
		ft ²	m ²	cm ²	

No.	Name	Description			Default
		Field	Metric	Laboratory	
5	SEGLEN	A real positive value greater than or equal to zero that defines the additional pipe length for the frictional pressure drop (L). If set to zero then there is no additional pressure loss due to friction, whereas, if set to the default (1*), then the segment pipe length is calculated from the corresponding WELSEGS keyword.			Defined
		ft	m	cm	
6	ID	A real positive value that defines the pipe internal diameter of the segment used to calculate the pressure drop due to friction (D). The value is used to replace the segment pipe internal diameter defined on the WELSEGS keyword in record 2-7, also named ID. If ID is defaulted on this keyword then the equivalent value on the WELSEGS keyword will be used instead. Note for non-circular pipe segments use the equivalent diameter instead, that is: $\text{Equivalent ID} = \frac{4.0 \times (\text{Cross-Sectional Area})}{\text{Perimeter}}$			Defined
		feet	m	cm	
7	EPSILON	A real positive value that defines the pipe absolute roughness for this segment. The value is used to replace the segment pipe absolute roughness defined on the WELSEGS keyword in record 2-8, also named EPSILON. If EPSILON is defaulted on this keyword then the equivalent value on the WELSEGS keyword will be used instead.			Defined
		feet	m	cm	
8	AREAPIPE	A real positive value that defines the cross-sectional area of flow in the pipe (A_p), as opposed to the restricted section of the valve (A_r). AREAPIPE is used to convert the segment volumetric flow rate into the flow velocity through the pipe (u_p). The value is used to replace the segment pipe cross-sectional area of flow defined on the WELSEGS keyword in record 2-9, named XAREA. If AREAPIPE is defaulted on this keyword then the equivalent value on the WELSEGS keyword will be used instead.			Defined
		ft ²	m ²	cm ²	
9	STATUS	A character string of length four that defines the ICD's operational status, STATUS should be set to one of the following character strings: 1) OPEN: the ICD connections are open to flow. 2) SHUT: the ICD connections are closed to flow (shut-in).			OPEN
10	AREAMAX	A real positive value that defines the maximum cross-sectional area of flow in the restricted section of the valve (A_{max}). AREAMAX is used to convert the segment volumetric flow rate into the maximum flow velocity at the constriction (u_r). If defaulted then AREAPIPE will be used if defined, otherwise XAREA (item 2-9) on the WELSEGS keyword will be used.			Defined

No.	Name	Description			Default
		Field	Metric	Laboratory	
		ft ²	m ²	cm ²	
Notes: 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the SCHEDULE section. User Defined Arguments are activated using the UDADIMS keyword in the RUNSPEC section, and defined by the UDQ keyword in the SCHEDULE section. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.134: WSEGVAlV Keyword Description

The total number of wells should be defined via the [WELLDIMS](#) keyword and the number of multi-segment wells should be declared on the [WSEGDIMS](#) keyword, both keywords are in the [RUNSPEC](#) section. In addition, the [WELSPCS](#) keyword should be used to define wells, the [COMPDAT](#) keyword to define the well completions for both ordinary wells and multi-segment wells, and the [COMPSEGS](#) keyword to define a multi-segment segment completions. Finally, the [WSEGVAlV](#) keyword can then be use to define ICD connections for the well. All the aforementioned keywords are described in the [SCHEDULE](#) section.

OPM flow calculates the pressure drop across the ICD using a homogeneous sub-critical flow through a constriction model. The model consists of two variables, the pressure drop due to the constriction, $\delta P_{restriction}$, and the pressure drop due to friction, $\delta P_{friction}$, as shown in equation (12.40).

$$\delta P = \delta P_{restriction} + \delta P_{friction}$$

where

$$\delta P_{restriction} = C_1 \frac{\rho v_r^2}{2 C_v^2} \quad (12.40)$$

$$\delta P_{friction} = 2 C_2 f \frac{L}{D} v_p^2$$

Where:

- C_1 and C_2 = Unit conversion constants as listed in Table 12.135.
- C_v = Vendor supplied dimensionless flow coefficient for the valve.
- D = Diameter of the pipe.
- f = Fanning friction factor.
- L = Segment pipe length.
- ρ = Fluid mixture density.
- v_r = The flow velocity of the mixture through the constriction.
- v_p = The flow velocity of the mixture through the segment pipe.

Conversion Factor Constants and Variable Units					
No.	Name	Field	Metric	Laboratory	Property
1	C ₁	2.159 × 10 ⁻⁴	1.0 × 10 ⁻⁵	9.869 × 10 ⁻⁷	Constant
2	C ₂	2.892 × 10 ⁻¹⁴	1.340 × 10 ⁻¹⁵	7.615 × 10 ⁻¹⁴	
3	Density	lb/ft ³	kg/m ³	gm/cc	Units
4	Pressure	psia	bars	atm	
5	Velocity	ft/s	m/s	cm/s	
6	Volumetric Flow	ft ³ /day	m ³ /day	cc/hour	

Table 12.135: WELVALV Conversion Factor Constants and Units

In addition as both v_r and v_p are dependent on their respective cross-section areas then the volumetric flow (q) through the device requires that:

$$q = v_r A_r = v_p A_p \quad (12.41)$$

Substituting equation (12.41) for $\delta P_{restriction}$ in equation (12.40) one obtains:

$$\delta P_{restriction} = C_2 \frac{\rho q^2}{2 C_v^2 A_r^2} \quad (12.42)$$

Where:

- A_r = Cross-sectional area of the constriction.
- C_2 = Unit conversion constants as listed in Table 12.135.
- C_v = Vendor supplied dimensionless flow coefficient for the valve.
- ρ = Fluid mixture density.
- q = Volumetric flow rate.

The base strength of the device, K is defined using equation (12.42) as follows:

$$K = \frac{C_2}{2 C_v^2 A_r^2} \quad (12.43)$$

Note if K is greater than 0.1 then the device will be shut.

The setting of the device, that is how open the device is, is related to the restricted area and the maximum restricted area of the device as follows:

$$Setting\ of\ Device = \frac{A_r}{A_{max}} \quad (12.44)$$

Where:

- A_r = Cross-sectional area of the constriction.
- A_{max} = Maximum cross-sectional area of the constriction.

Example

The following example is based on one producing well segment oil well (OP01) using the [WELSPECS](#), [WELSEGS](#) [COMPDAT](#) and [COMPSEGS](#) keywords, as per the [WSEGS/ICD](#) keyword example ([Example](#)), and is therefore not repeated here.

```
--
--          MULTI-SEGMENT WELL ICD SEGMENT SPECIFICATION DATA
--
-- WELL   SEG  DEVICE  AREA   PIPE   PIPE   PIPE   PIPE   OPEN  MAX
-- NAME   NO   CV      REST   LENG   ID     EPSI   AREA   SHUT  AREA
WSEGVAlV
OP01      7  0.960    0.012  1*     1*     1*     1*     OPEN  1*  / DEFAULT VALUES
OP01     12  0.960    0.012  1*     1*     1*     1*     OPEN  1*  / TAKEN FROM
OP01     17  0.960    0.012  1*     1*     1*     1*     OPEN  1*  / WELSEGS KEYWORD

OP01     22  0.850    0.100  1*     1*     1*     1*     OPEN  1*  /
OP01     23  0.850    0.100  1*     1*     1*     1*     OPEN  1*  /
OP01     24  0.850    0.100  1*     1*     1*     1*     OPEN  1*  /
OP01     25  0.850    0.100  1*     1*     1*     1*     OPEN  1*  /
/
```

Here segments 7, 12 and 17 have the same type of sub-critical valves with their pipe properties taken from the [WELSEGS](#) keyword used to define well OP01 as a multi-segment well. Similarly, segments 22 to 25 have the same ICD properties, and again the pipe properties are taken from the [WELSEGS](#) keyword.

12.3.349 WSKPTAB – ASSIGN WELL POLYMER MOLECULAR MODEL SKIN TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WSKPTAB](#) keyword assigns the well polymer molecular water and polymer skin tables to water injection wells in OPM Flow's Polymer Molecular Weight Transport option, that uses the polymer molecular weight in calculating the polymer viscosity, as well as accounting for formation damage due to the water and polymer injection, by adjusting the wellbore skin pressure. This keyword should only be used if the [POLYMER](#) and [POLYMW](#) keywords in the [RUNSPEC](#) section are also activated. The keyword assigns the water [SKPRWAT](#) tables, that are defined via the [SKPRWAT](#) keyword in the [PROPS](#) section, that are used to calculate the wellbore skin pressure during water injection. As well as the polymer [SKPRPOLY](#) tables, that are defined via the [SKPRPOLY](#) keyword in the [PROPS](#) section, that are used to calculate the wellbore skin pressure during polymer injection.

Note

This is an OPM Flow specific keyword that employs an alternative polymer flood model based on a Polymer Molecular Weight Transport equation, that is not available in the commercial simulator.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length, that defines the water injection well name, for which the well water injection skin table, SKKPRWAT, and the polymer skin injection table, SKPRPOLY , are to be assigned. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	SKPRWAT	A positive integer value that defines the corresponding SKPRWAT table to be allocated to the water injection well. A value less than or equal to zero means that no SKPRWAT table is allocated to the well			0
3	SKPRPOLY	A positive integer value that defines the corresponding SKPRPOLY table to be allocated to the water injection well. A value less than or equal to zero means that no SKPRPOLY table is allocated to the well			0
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.136: WSKPTAB Keyword Description

See also the [SKPRWAT](#) and [SKPRPOLY](#) keywords in the [PROPS](#) section, that describe the relationship of the wellbore skin pressure with respect to the injected water and polymer throughput and fluid velocity, for the simulator's Polymer Molecular Weight Transport option. As well as the [PLYMWINJ](#) and [PLYVMH](#) keywords, also in the [PROPS](#) section, that are the additional keywords required for the Polymer Molecular Weight Transport option.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

The [WPMITAB](#) keyword in the [SCHEDULE](#) section may be used to assign the [PLYMWINJ](#) keyword in the [PROPS](#) section. Where [PLYMWINJ](#), describes the relationship of the injected polymer molecular weight as a function of polymer throughput and polymer velocity, for the simulator's Polymer Molecular Weight Transport option

Note that the standard polymer property data keywords: [PLYROCK](#), [PLYADS](#), [PLYMAX](#), etc., are still required to fully describe the polymer fluid.

Example

Given NTSKWAT equals two and NTSKPOLY equals three on the [PINTDIMS](#) keyword in the [RUNSPEC](#) section, then:

```
--
--      ASSIGN WELL POLYMER MOLECULAR MODEL SKIN TABLES
--
-- WELL   SKPRWAT   SKPRPOLY
-- NAME   TABLE   TABLE
WSKPTAB
WI01     1         1
WI02     1         3
WI03     2         2
/
```

Assigns [SKPRWAT](#) table one to wells WI01 and WI02 and table two to WI03, and [SKPRPOLY](#) tables one, two and three to wells WI01, WI03, and WI02, respectively.

12.3.350 WSOLVENT – DEFINE GAS INJECTION WELL SOLVENT FRACTION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WSOLVENT](#) defines a gas injection well's solvent fraction in the injection stream that is to be used when the Solvent option has been activated by the [SOLVENT](#) keyword in the [RUNSPEC](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name of a gas injection well for which the solvent fraction data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
4	SOLFRA	A real positive value greater than or equal to zero and less than or equal to one that defines the fraction of solvent in the gas well's injection stream. This value may be specified using a User Defined Argument (UDA).			None
		fraction	fraction	fraction	

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the [SCHEDULE](#) section. User Defined Arguments are activated using the [UDADIMS](#) keyword in the [RUNSPEC](#) section, and defined by the [UDQ](#) keyword in the [SCHEDULE](#) section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.137: WSOLVENT Keyword Description

Gas injection wells that are not declared via this keyword have their solvent fractions set to zero.

See also the [GCONINJE](#) keyword to define a group's injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the solvent fractions for three gas injection wells for when the solvent option has been activated by the [SOLVENT](#) keyword in the [RUNSPEC](#) section.

```
--
--      DEFINE GAS INJECTION WELL SOLVENT FRACTION
--
-- WELL  SOLVENT
-- NAME  FRACTION
--      -----
WSOLVENT
GI01    0.0000      /
GI02    0.5000      /
GI03    0.5000      /
/
```

The solvent fraction for the GI01 gas injector is set to zero and both GI02 and GI03 gas injectors have solvent fraction values of 0.5 for their injection streams.

12.3.351 WSURFACT – DEFINE WATER INJECTION WELL SURFACTANT CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

[WSURFACT](#) defines a water injection well's surfactant concentration in the injection stream that is to be used when the Surfactant phase has been activated by either the [SURFACT](#) or [SURFACTW](#) keywords in the [RUNSPEC](#) section.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name of a gas injection well for which the solvent fraction data is being defined. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> keyword in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	SURCON	A real positive value that defines the surfactant concentration of the well's injection stream.			None
		lb/stb	kg/sm ³	gm/scc	
Notes: 1) The keyword is followed by any number of records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.138: WSURFACT Keyword Description

Water injection wells that are not declared via this keyword have their surfactant concentrations set to zero.

See also the [GCONINJE](#) keyword to define a group's injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the surfactant concentrations for three water injection wells for when the surfactant phase option has been activated by either the [SURFACT](#) or [SURFACTW](#) keywords in the [RUNSPEC](#) section. Here the surfactant concentration has been set to 0.200 for all three wells.

```
--
--      DEFINE WATER INJECTION WELL SURFACTANT CONCENTRATION
--
-- WELL  SURFACT
-- NAME  SURCON
--      -----
WSURFACT
WI01    0.2000      /
WI02    0.2000      /
WI03    0.2000      /
/
```

12.3.352 WTADD – Add a Constant to a Well Target or Constraint

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WTADD](#), adds a constant to a previously define well's target or constraint, as stated on the [WCONPROD](#), [WCONINJE](#), or [WELTARG](#) keywords, but not for the history matching wells using the [WCONHIST](#) or [WCONINJH](#) keywords. All the aforementioned keywords are in the [SCHEDULE](#) section. The constant can be positive or negative.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.353 WTEMP – DEFINE AN INJECTION WELL’S FLUID TEMPERATURE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WTEMP](#) keyword defines the temperature of the injection fluid being injected by an injection well.

This keyword can only be used if OPM Flow’s thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section. Note this is different to the commercial simulator that uses the [TEMP](#) keyword in the [RUNSPEC](#) section to activate the black-oil thermal model.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for an injection well for which the injection well fluid's temperature data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	TEMP	A real positive value greater than zero that defines the temperature of the injected fluid.			None
		°F	°C	°C	
Notes: 1) Injection wells that are not declared via this keyword have their injection fluid temperatures set to zero degrees in the run's units. 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.139: WTEMP Keyword Description

See also the [GCONINJE](#) keyword to define a group’s injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well’s targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the injected fluid temperatures for three water injection wells for when the thermal option has been activated by the [THERMAL](#) keyword in the [RUNSPEC](#) section.

```
--
--      DEFINE INJECTION WELL FLUID TEMPERATURE
--
-- WELL  FLUID
-- NAME  TEMP.
--      -----
WTEMP
WI01    39.00      /
WI02    37.00      /
WI03    39.00      /
/
```

Here wells WI01 and WI03 inject water with a water temperature of 39 oF and well WI02’s injection water temperature is 37 oF.

12.3.354 WTEMPQ – OUTPUT WELL NAMES AND WELL LISTS TO THE PRINT FILE

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WTEMPQ](#) prints out a user defined selected list of currently defined wells and well lists to the print file (*.PRT). The keyword allows for sub-setting the well names etc., using the normal well and well list naming conventions. For example to list all wells beginning with the characters “OP” then one would use “OP*” as the well name on this keyword.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.355 WTEST – WELL TESTING CRITERIA FOR RE-OPENING CLOSED WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WTEST](#) keyword outlines the testing procedures to be applied to wells that are closed for various reasons to see if the wells are capable flowing under the current operating conditions. The keyword can be applied to single wells or groups of wells.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name of the well to be tested. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	TIME	A real value greater than zero that defines the minimum period of time that must elapse between well tests. The next well test is performed at the beginning of the next time step after the specified period of time has elapsed since the previous well test, for example if TIME is set equal to 365.25 (days), the test is performed approximately every year.			None
		days	days	hours	
3	TEST	A character string of up to five characters in length that defines a list of reasons why a well could have been closed. If a well was closed for one of the specified reasons then the well is tested to see if it can be put back on production. The characters that can be used to define TEST are as follows: 1) P: meaning the well was closed due to a bottom-hole or tubing head pressure limit, or other physical limit then the well is tested to see if it can flow, if it can, then it is put back on production, otherwise it remains closed. 2) E: meaning the well was closed due to a well or a well connection economic constraint then the well is tested to see if it can flow, if it can then it is put back on production, otherwise it remains closed. Here, if the well has at least one connection open then the well is opened. If a well has no open connections, then those connections that have been closed due to an economic limit test ("ELT"), are all individually tested and reopened if they meet the ELT criteria as defined by the WECON , CECON and WECONINJ keywords in the SCHEDULE section. And the well is opened if it has at least one open connection. Where the ELT applies to ORAT, GRAT, WCUT, GOR and WGR etc. 3) G: meaning the well was closed due to a group economic constraint, as per the GECON , GECONT , GCONCAL , GCONPRI , GCONPROD , and GCONSALE , keywords in the SCHEDULE section, then the well is tested to see if it can flow, if it can, then it is put back on production, otherwise it remains closed. 4) D: not supported by OPM Flow. 5) C: not supported by OPM Flow. The default value is an empty string " " that switches off testing.			" "

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------

No.	Name	Description			Default
		Field	Metric	Laboratory	
		Note that only the P, E and G options are currently supported in OPM Flow.			
4	NTIME	A positive integer greater than or equal to zero that define the number of times a well can be tested. The default value of zero means an infinite number of times.			0
5	START	A real positive value that defines the start up time used to prorated the rate at which the well is put back on production. If <i>START</i> is large compared to the time step size, then the well is brought on gradually, if it is less than the time step size, then the well is opened faster. The rate is prorated based on the following: $Q_i = Q_{final} \times \left(\frac{T_{End\ of\ Time\ Step} - T_{opened}}{START} \right)$ The default value of 0.0 means the well is opened immediately.			0.0
		days	days	hours	
Notes: 1) The keyword is followed by any number of records and each record is terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.140:WTEST Keyword Description

See also the *WELSPECS* keyword to define a wells shut-in or stop options, *WECON* for setting a well's economic criteria, *GCONPROD* and *GCONINJE* for group controls, and *GECON* for setting a group's economic criteria. All the aforementioned keywords are described in the *SCHEDULE* section.

Example

The following example defines test criteria for all gas wells ('GP*') and three oil wells (OP01, OP02, and OP03).

```
--
--
--          WELL TESTING CRITERIA FOR RE-OPENING CLOSED WELLS
--
-- WELL   TST   TST   NO.   STRT
-- NAME  INTV  TYPE  TSTS  TIME
-- ----  ----  ----  ----  ----
WTEST
'GP*'   365.25 P      5      0.0
OP01    30.0  PEG    0      0.0
OP02    30.0  PEG    0      0.0
OP03    30.0  PEG    0      0.0
/
```

All the gas wells are test annually if they have been shut-in due to a bottom-hole or tubing head pressure limit, are tested five times after they have been closed, and are opened up immediately. The oil wells are tested every 30 days if they have been closed due to a bottom-hole or tubing head pressure limit, a well economic limit or a group economic limit. All the oil wells are tested an infinite amount of times and are opened up immediately. Note that only the P, E and G options are currently supported in OPM Flow.

12.3.356
WTHPMAX – DEFINE A WELL’S MAXIMUM FLOWING THP FOR SHUT-IN

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

WTHPMAX stipulates a well’s maximum flowing Tubing Head Pressure (“THP”), above which the well will be shut-in. The facility is useful if the THP exceeds the wellhead maximum design pressure, which can occur if excessive gas invades the wellbore. In addition to setting the maximum THP, the keyword defines the criteria for re-testing the well to see if the THP has fallen below the maximum value.

This keyword is not supported by OPM Flow but would change the results if supported so the simulation will be stopped.

12.3.357 WTMULT – MULTIPLY A WELL TARGET OR CONSTRAINT BY A CONSTANT

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WTMULT](#), multiplies a defined well's target or constraint by a constant, for the target and constraints previously stipulated on the [WCONPROD](#), [WCONINJE](#), or [WELTARG](#) keywords, but not for the history matching wells using the [WCONHIST](#) or [WCONINJH](#) keywords. All the aforementioned keywords are in the [SCHEDULE](#) section. The constant should be positive value.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well target or constraint (CONTROL) is being adjusted by the multiplier (FACTOR). Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	CONTROL	A defined character string that declares the well target or constraint that will be adjusted by multiplying its current value by the FACTOR defined in (3). CONTROL should be set to one of the following character strings: 1) ORAT: the well's surface oil production rate will be adjusted. 2) WRAT: the well's surface water production rate will be adjusted. 3) GRAT: the well's surface gas production rate will be adjusted. 4) LRAT: the well's surface liquid (oil plus water) production rate will be adjusted. 5) CRAT: the well's linearly combined maximum surface rate, as per the LINCOM keyword in the SCHEDULE section, will be adjusted. 6) RESV: the well's in situ reservoir volume rate will be adjusted. 7) BHP: the well's bottom-hole pressure will be adjusted. 8) THP: the well's tubing head pressure will be adjusted. In this case a vertical performance table must have been previously assigned to the well via the WCONPROD or WCONINJE keywords. The tables are entered via the VFPINJ or the VFPPROD keywords. All the keywords are in the SCHEDULE section. 9) LIFT: the well's artificial lift quantity will be adjusted. Again, as for the THP, a vertical performance table must have been assigned to the well. 10) GUID: the well's guide rate will be adjusted. Only wells under group control and have a guide rate stipulated by the WGRUPCON keyword in the SCHEDULE section, may use this option. 11) CVAL: the well's calorific rate will be adjusted. 12) NGL: the well's natural gas liquid rate will be adjusted. Note that the CRAT, CVAL and the NGL options are not available in OPM Flow and should not be used.			None

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	FACTOR	A positive real value that defines the FACTOR that is used to multiply the CONTROL specified in (2). This value may be specified using a User Defined Argument (UDA).			None
		dimensionless	dimensionless	dimensionless	
4	NTIME	A positive integer greater than or equal to one that defines the number of report time steps for which the well target or constraint (CONTROL) is multiplied by the FACTOR. This is only applied when the FACTOR is specified by a User Defined Argument (UDA). The default value of one means that the multiplication is applied only at the current time step for the UDA variable. This option is not supported by OPM Flow.			1
		dimensionless	dimensionless	dimensionless	

Notes:

- 1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the [SCHEDULE](#) section. User Defined Arguments are activated using the [UDADIMS](#) keyword in the [RUNSPEC](#) section, and defined by the [UDQ](#) keyword in the [SCHEDULE](#) section.
- 2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".

Table 12.141:WTMULT Keyword Description

See also the [WELTARG](#) and [WELCNTL](#) keywords in the [SCHEDULE](#) section, that can be used to reset the well's target and constraints of both rates and pressures, as well as the well's control mode.

Example

The example shows three oil wells having the flow streams adjusted.

```
--
-- WELL TARGET/LIMIT MULTIPLIER
--
-- WELL      WELL  MULT   REPORT
-- NAME      CNTL  FACTOR  TIMES
WTMULT
OP01         ORAT   0.90           /
OP02         BHP   0.95           /
OP03         LIFT   1.25           /
/
```

Well OP01 has its current oil rate target multiplied by 0.90, well OP02 has its bottom-hole pressure constraint multiplied by 0.95, and well OP03 has its artificial lift quantity increased by 1.25 times.

12.3.358 WTRACER

– DEFINE AN INJECTION WELL’S TRACER CONCENTRATION

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WTRACER](#) keyword defines the tracer concentration of the injection fluid being injected by an injection well. This keyword should only be used if the Tracer option has been invoked by the [TRACERS](#) keyword in the [RUNSPEC](#) section and the tracers have been declared via the [TRACER](#) keyword in the [PROPS](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name of an injection well for which the tracer fraction data is being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	NAME	A three letter character string defining the tracer's name which has previously been defined via the TRACER keyword in the PROPS section. Note it is best to avoid names beginning with the letters F, S, and T as these names may create naming issues in post-processing software.			None
3	TRCON	A real positive value that defines the tracer concentration of the well's injection stream. This value may be specified using a User Defined Argument (UDA).			None
		Liquid: l/stb Gas: l/Mscf	Liquid: l/sm ³ Gas: l/sm ³	Liquid: l/scc Gas: l/scc	
4	TRCUM	A real positive value that defines the cumulative tracer concentration factor of the well's injection stream. This value may be specified using a User Defined Argument (UDA). <u>This feature is currently not supported by OPM Flow.</u>			None
		Liquid: l/stb Gas: l/Mscf	Liquid: l/sm ³ Gas: l/sm ³	Liquid: l/scc Gas: l/scc	
5	GRPNAME	A character string of up to eight characters in length that defines the group from which the produced tracer concentration should be used for the well's injection stream. GRPNAME must have been previously defined via the GCONPROD keyword in the SCHEDULE section, unless the FIELD group has been specified here. Note if GRPNAME is not defined then TRCON will be used for the tracer concentration of the well's injection stream. <u>This feature is currently not supported by OPM Flow.</u>			None
Notes:					
1) User Defined Quantities ("UDQ") can be used as User Defined Arguments ("UDA") for various group, well, and connection keywords in the SCHEDULE section. User Defined Arguments are activated using the UDADIMS keyword in the RUNSPEC section, and defined by the UDQ keyword in the SCHEDULE section.					
2) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.142: WTRACER Keyword Description

Injection wells that are not declared via this keyword have their tracer concentrations set to zero.

See also the [GCONINJE](#) keyword to define a group's injection targets and constraints, and the [WCONINJE](#) keyword to define an injection well's targets and constraints. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Example

The following example defines the tracer concentrations for two gas injectors and three water injection wells, with water injection well WVI02 having no 'WAT' tracer injected in the water phase.

```
--
--      DEFINE CONCENTRATION OF TRACERS IN THE INJECTION STREAMS,
--      INJECTION TRACER CONCENTRATIONS NOT DEFINED USING THE WTRACER
--      KEYWORD ARE ASSUMED TO BE ZERO.
--
-- WELL  NAME      TRACER  TRACER  TRACER
-- NAME  TRACER    CONC    CUM     GROUP
WTRACER
GI01    'GAS'      1.0                /
GI02    'GAS'      1.0                /

WI01    'WAT'      1.0                /
WI02    'WAT'      0.0                /
WI03    'WAT'      1.0                /
/
```

Note the terminating “/” for the keyword.

12.3.359 WVFDP – MODIFY WELL BHP OBTAINED FROM VFP TABLES

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WVFDP](#) keyword modifies a well's Bottom-Hole Pressure ("BHP") estimated by the simulator by interpolation of the Vertical Flow Performance ("VFP") tables. The production VFP tables are entered via the [VFPPROD](#) keyword and the injection tables by the [VFPINJ](#) keyword; both keywords are in the [SCHEDULE](#) section.

Note that simulator automatically adjusts the interpolated BHP to account for hydrostatic head using the density of the wellbore fluid and the difference between a well's BHP reference depth, as per the [BHPREF](#) parameter on the [WELSPECS](#) or [WELSPECL](#) keywords in the [SCHEDULE](#) section, and the [VFPREF](#) parameter reference depth on the [VFPPROD](#) and [VFPINJ](#) keywords. Thus, [WVFDP](#) applies an additional adjustment in order to match a well's flow rate to a given tubing head pressure, by adjusting the BHP.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which VFP interpolated BHP adjustment is to be applied. Note that the well name (WELNAME) must have been declared previously using the <i>WELSPECS</i> and <i>WCONPROD</i> (or <i>WCONINJE</i>) keywords in the <i>SCHEDULE</i> section, otherwise an error may occur.			None
2	DELTAP	A real positive or negative value that is added to the VFP interpolated BHP value (BHPVFP). A positive value of DELTAP increases the BHP and therefore makes a production well less productive; whereas, a negative value is subtracted from the BHP and therefore increases the productivity of a production well. Consequently, the opposite effect occurs for injection wells, that is, a positive value of BHPVFP increases the BHP and therefore increases an injection well's injectivity; whereas, a negative value is subtracted from the BHP and therefore decreases the injectivity of an injection well.			0.0
		psia	barsa	atma	
3	MULTP	MULTP is a real positive or negative value that scales the tubing pressure loss by the following equation; $BHP_{Adjusted} = THP + MULTP(BHP_{VFP} - THP)$ Thus, a MULTP value greater then 1.0 increases the BHP and therefore makes a production well less productive; whereas, a value less than 1.0 increases the productivity of a production well. Consequently, the opposite effect occurs for injection wells			1.0
		dimensionless	dimensionless	dimensionless	
Notes: 1) The keyword is followed by any number records with each record terminated by a “/” and the keyword should be terminated by a “/”.					

Table 12.143:WVFDP Keyword Description

Example

The following example below shows three oils operating under THP control.

```
-- -----
-- 01 JAN 2000 START OF SCHEDULE SECTION
-- -----
--
--          WELL PRODUCTION WELL CONTROLS
--
--
-- WELL  OPEN/  CNTL   OIL    WAT    GAS    LIQ    RES    BHP    THP    VFP    VFP
-- NAME SHUT   MODE   RATE   RATE   RATE   RATE   RATE   PRES  PRES  TABLE  ALFQ
WCONPROD
OP01    OPEN   THP     1*     1*     1*    5000    1*    750.0  500.   9      1* /
OP02    OPEN   THP     1*     1*     1*    5000    1*    750.0  500.   9      1* /
OP03    OPEN   THP     1*     1*     1*    5000    1*    750.0  500.   9      1* /
/
--
--          WELL VFP BHP-THP CORRECTION DATA
--
-- WELL  BHP      BHP
-- NAME DELTAP    MULTP
WVFPDIP
OP01    20.0      1*
OP01    -5.0      1*
OP01     0.0      1.10
/
```

Well OP01 has a delta pressure correction of 20 psia applied to it's BHP resulting in a reduction in the well's productivity for the given 500.0 psia THP operating target. For well OP02, the well's productivity is increased by subtracting 5.0 psia from the BHP.And finally for well OP03, the MULTP value of 1.10 decreases the well's productivity by increasing the pressure loss between the THP and BHP by 10%.

12.3.360 WVFPEXP – DEFINE WELL VFP INTERPOLATION OPTIONS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

This keyword, [WVFPEXP](#), defines how Vertical Flow Performance (“VFP”) tables are interpolated and can be used to resolve certain issues with wells operating under tubing head pressure control. For example, setting the VFP table to interpolate explicitly, that is using the previous time step results of the gas and water ratios for an oil well, may improve convergence. The default is to use implicit interpolation that uses the current time step values and may result in solution convergence oscillations in solving the linear equations. The [WCONPROD](#) keyword is used to allocate the [VFPPROD](#) tables to specific production wells. Note that one VFP table can be allocated to one or more wells; however, [WVFPEXP](#) is applied to a well’s allocated VFP table, not to all wells that use the same table, unless specifically requested. All the aforementioned keywords are in the [SCHEDULE](#) section.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well’s VFP interpolation options are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	IMPEXP	A defined character string of length three that defines the how the well’s VFPPROD tables are to be interpolated and should be set to one of the following character strings: 1) IMP: For this option the simulator interpolates a well’s VFP table implicitly, that is the latest gas fraction (GOR, GLR, etc.), water fraction (WCUT, WOR etc.), and ALQ is used to determine the required pressure. This is the most accurate option but may lead to oscillating behavior during the Newton/Linear iterations. This is the default behavior. 2) EXP: This options forces the simulator to use the previous time step values of a well’s gas fraction, water fraction, and ALQ to interpolate a well’s VFP table in order to determine the required pressure. This is less accurate than the IMP option but minimizes any oscillating behavior during the Newton/Linear iterations. If a well’s WCUT and GOR is varying as a function of BHP inside a Newton/Linear iteration (as for example when gas cusping or water coning is occurring), the implicit lookup of the VFP tables might indicate that the well has died, due to the interpolation/extrapolation. If a well dies using implicit WCUT and GOR VFP lookup then the simulator automatically switches to explicit (previous time step) VFP lookup to prevent the well from premature closure, and writes a message to the screen and print file stating the fact. Note the explicit treatment is just for the VFP lookup only. Using the IMPEXP option allows one to select wells that should use either the implicit or explicit lookup option, which may be useful in improving run time performance. Note that if the default value of IMP is used, the simulator will still switch to explicit VFP lookup to prevent a well from premature closure if the condition occurs.			IMP

No.	Name	Description			Default
		Field	Metric	Laboratory	
3	STATUS	A defined character string that defines if the well's operating condition should be checked to see if the VFP table lookup indicates if the well is operating on the flat or horizontal part of stabilized portion of the VLP table (curve), and should be set to one of the following character strings: 1) NO: For this option the check is not performed. 2) YES: In this case the check is performed. Hence, if a well under THP control is found to operating on the horizontal stabilized portion of the VLP curve, then the well is SHUT or STOPPED depending on the AUTO parameter on the WELSPECS keyword. This option is specific to the commercial simulator's VFPI program that can generate VFPF tables where the unstable part of the VFP curves to the left of the minimum is replaced with a horizontal line at the minimum BHP value. This option is not supported by OPM Flow.			NO
4	CONTROL	A defined character string that defines the behavior of rate control wells operating on the unstabilized portion of the VFP table, and should be set to one of the following character strings: 1) NO: This option does not prevent changes to a well's operating mode when the well is constrained to be operating on the unstabilized portion of the VFP table. This option may cause the well control to oscillate between rate and THP control during the Newton/Linear iterations. 2) YES1: For this option the well is prevented from changing from rate control to THP control provided it can produce at a higher rate under THP control. In this case the well will continue to produce at the required rate even though the calculated THP may be below the well's minimum THP value. Note however, the well will still be allowed to die if the IPR does not intersect the VFP curve. In the commercial simulator this option will also print a message the first time a well is prevented from changing its control mode, whereas OPM Flow always prints a message every time a well is prevented from changing its control mode. 3) YES2: This option behaves the same as the YES1 option, except a message is printed every time a well is prevented from changing its control mode. The default value of NO allows wells to die when they are constrained by a lower rate that forces them to operate on the unstable section of the VFP curve. This can occur if a well's rate is being set to match its group target rate or constraint limit. Using either the YES1 or YES2 option will help prevent the "hunting between control modes" issue or the well prematurely being shut-in.			NO
5	EXTRAP	The EXTRAP parameter is a defined character string in the commercial composition simulator that declares how the extrapolation of a well's gas fraction, water fraction, and ALQ values is to be conducted. This option is not supported by OPM Flow.			WG
Notes:					
1) The keyword is followed by any number of records with each record terminated by a "/" and the keyword should be terminated by a "/".					

Table 12.144: WVFPEXP Keyword Description

Example

The following example defines two oil wells using the [WELSPECS](#) and [WCONPROD](#) keywords, together with the [WVFPEXP](#) keyword that declares how the [VFPPROD](#) table lookup should be performed.

```
--
-- WELL SPECIFICATION DATA
--
-- WELL      GROUP      LOCATION  BHP    PHASE  DRAIN  INFLOW  SHUT   CROSS  PRESS
-- NAME      NAME        I      J    DEPTH FLUID  AREA   EQUA.  IN     FLOW   TABLE
WELSPECS
OP01        PLATFORM    14    13   1*      OIL    1*      STD    SHUT    NO     1*  /
OP02        PLATFORM    28    96   1*      OIL    1*      STD    SHUT    NO     1*  /
/
--
--          WELL PRODUCTION WELL CONTROLS
--
-- WELL  OPEN/  CNTL  OIL    WAT    GAS    LIQ    RES    BHP    THP    VFP    VFP
-- NAME  SHUT   MODE  RATE   RATE   RATE   RATE   RATE   PRES  PRES  TABLE ALFQ
WCONPROD
OP01     SHUT   GRUP   1*     1*     1*     1*     1*     500.0 100.0   1      /
OP02     SHUT   GRUP   1*     1*     1*     1*     1*     500.0 100.0   1      /
/
--
-- WELL OPTIONS FOR PROBLEMATIC THP CONTROLLED WELLS
--
-- WELL  IMP  CLSE  RATE  VFP
-- NAME  EXP  WELL  CNTL  EXT
WVFPEXP
'OP*'   '  1*   1*   YES1  1*
/
```

Here both wells are declared as initially shut on the [WELSPECS](#) keyword and use [VFPPROD](#) table number one as declared on the [WCONPROD](#) keyword. The [WVFPEXP](#) keyword declares all wells with a name beginning with OP to use implicit lookup and the wells are prevented from changing from rate control to THP control provided they can produce at a higher rate under THP control.

12.3.361 WWPAVE – WELL BLOCK AVERAGE PRESSURE CALCULATION PARAMETERS FOR INDIVIDUAL WELLS

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [WWPAVE](#) keyword defines the method and parameters for calculating a well's block average pressures for individual wells that can be written to the [SUMMARY](#) and RSM files via the [VWBP](#), [WBP4](#), [WBP5](#) and [WBP9](#) vectors in the [SUMMARY](#) section. The resulting average pressure can be written out to the summary file in order to be compared with field observed data. The keyword is similar to the [WPAVE](#) keyword in the [SCHEDULE](#) section that has similar functionality, but is applied to all wells in the model.

Note that [WWPAVE](#) will overwrite any parameters on the [WPAVE](#) keyword for a given well, and that [WWPAVE](#) can also be overwritten by any subsequent [WPAVE](#) keyword.

No.	Name	Description			Default
		Field	Metric	Laboratory	
1	WELNAME	A character string of up to eight characters in length that defines the well name for which the well average pressure calculation parameters are being defined. Note that the well name (WELNAME) must have been declared previously using the WELSPECS keyword in the SCHEDULE section, otherwise an error may occur.			None
2	WPAVE1	A real dimensionless value that defines the weighting factor between the inner block and the surrounding blocks used in the calculation of the connection factor weighted average pressure. If WPAVE1 is greater than or equal to zero and less than or equal to one, then the average pressure for each well connection is calculated based on this weighting factor. A value of zero indicates only the surrounding blocks should be used in the calculation; and a value of one indicates only the inner blocks should be used. If WPAVE1 is less than zero, then the average pressure for each well connection is weighted based on the pore volumes of the inner and surrounding blocks.			0.5
3	WPAVE2	A real dimensionless value greater than or equal to zero and less than or equal to one, that defines the weighting factor between the connection weighted average pressures and the pore volume weighted average pressures. If WPAVE2 is equal to one, then the average pressures are calculate based only using the connection factor calculated pressures. If WPAVE2 is equal to zero, then average pressures are calculate based on only using the pore volumes calculated pressures.			1.0

No.	Name	Description			Default
		Field	Metric	Laboratory	
4	WPAVE3	A defined character string that determines how the hydrostatic head calculation is performed in correcting the pressures to the BHP reference depth on the WELSPECS or WPAVEDEP keywords in the SCHEDULE section.WPAVE3 should be set to one of the following character strings: 1) WELL: the hydrostatic head is calculated using the density of the fluid in the wellbore at the well connections. 2) RES: he hydrostatic head is calculated using the density of the fluid in the reservoir with well connections and averaged over the connections. 3) NONE: no hydrostatic correction is applied to the pressures.			WELL
5	WPAVE4	A defined character string that determines which connections should be used in the calculations,WPAVE4 should be set to one of the following character strings: 1) OPEN: only open connections and associated grid blocks should be used in the calculations. This option may result in pressure discontinuities if connections are opened and closed during the run. 2) ALL : all currently defined open and closed connections and associated grid blocks are used in the calculations.The pressure discontinuities issue mentioned above can be avoided with this option and defining all the well connections for a well at the beginning of the run. Only the OPEN option is currently supported by the simulator.			OPEN

Notes:

1) The keyword should be terminated by a “/”.

Table 12.145: WPAVE Keyword Description

The keyword is not applicable and should not be used with radial and spider grid geometries.

See also the [WELSPECS](#) keyword that defines a well and a well's bottom-hole pressure reference depth, the [WPAVEDEP](#) keyword that also defines a well's bottom-hole pressure reference depth, and the [COMPDAT](#) keyword to define a well's connections. All the aforementioned keywords are described in the [SCHEDULE](#) section.

Examples

The following example defines the default well block average pressure calculation parameters for three oil wells: OP01, OP02 and OP03.

```
--
--      DEFINE WELL BLOCK AVERAGE PRESSURE CALCULATION PARAMETERS
--
-- WELL  INNER  PORV  WELL  OPEN
-- NAME  OUTER  CONN  RES   ALL
WPAVE
OP01    0.5    1.0    WELL  ALL
OP02    0.5    1.0    WELL  ALL
OP03    0.5    1.0    1*    1*
```

And the next example shows the parameters used in the Norne model.

```
--
--          DEFINE WELL BLOCK AVERAGE PRESSURE CALCULATION PARAMETERS
--
-- WELL  INNER  PORV  WELL  OPEN
-- NAME  OUTER  CONN  RES   ALL
WPAVE
OP01    1*      0.0   WELL  ALL
OP02    1*      0.0   WELL  ALL
OP03    1*      0.0   WELL  ALL
```

Here only pore volume weighting is used instead of connection weighting.

12.3.362 ZIPP2OFF – DEACTIVATE AUTOMATIC TIME STEP CONTROL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ZIPP2OFF](#) keyword deactivates the commercial simulator’s alternative automatic time step selection algorithm that assumes no prior knowledge of the problem, as opposed to the standard time step algorithm that is controlled via the [TUNING](#) keyword in the [SCHEDULE](#) section, combined with posterior knowledge gained from previous time steps.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See section [2.2 Running OPM Flow From The Command Line](#) on how to control time stepping for OPM Flow.

12.3.363 ZIPPY2 – ACTIVATE AUTOMATIC TIME STEP CONTROL

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Description

The [ZIPPY2](#) keyword activates the commercial simulator’s alternative automatic time step selection algorithm that assumes no prior knowledge of the problem, as opposed to the standard time step algorithm that is controlled via the TUNNING keyword in the [SCHEDULE](#) section, combined with posterior knowledge gained from previous time steps.

This keyword is not supported by OPM Flow but has no effect on the results so it will be ignored.

See section [2.2 Running OPM Flow From The Command Line](#) on how to control time stepping for OPM Flow.

APPENDIX A: KEYWORD INDEX - ALPHABETIC LISTING

A

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter A	Status
ACF – Define Acentric Factors	
ACTCO2S – Activity Model for CO2	
ACTDIMS – ACTION Keyword Dimensions	
ACTION – Define Action Conditions and Command Processing (Field)	
ACTIONG – Define Action Conditions and Command Processing (Groups)	
ACTIONR – Define Action Conditions and Command Processing (Regions)	
ACTIONS – Define Action Conditions and Command Processing (Well Segments)	
ACTIONW – Define Action Conditions and Command Processing (Wells)	
ACTIONX – Define Action Conditions and Command Processing	
ACTNUM – Set the Status of a Grid Block To Active or Inactive	
ACTPARAM – Define Action Facility Target and Tolerance Parameters	
ADD – Add a Constant to a Specified Array	
ADDREG – Add a Constant to an Array based on a Region Number	
ADDZCORN – Add a Constant to the ZCORN Depth Array	
ADSALNOD – Salt Concentration Based on SATNUM Array	
ADSORP – Define Generalized Langmuir Adsorption Function	
AITs – Activate Intelligent Time Stepping	
AITSOFF – Deactivate Intelligent Time Stepping	
ALKADS – Define Alkaline Adsorption Functions	
ALKALINE – Activate the Alkaline Phase and Model	
ALKROCK – Define Rock Alkaline Properties	
ALL – Export Standard Summary Variable Vectors to File	
ALPOLADS – Polymer Adsorption versus Alkaline Concentration Multipliers	
ALSURFAD – Surfactant Adsorption versus Alkaline Concentration Multipliers	
ALSURFST – Water-Oil Surface Tension versus Alkaline Concentration Multipliers	
AMALGAM – Define LGR Amalgamations	

A

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter A	Status
API – Activate API Tracking	
APIGROUP – Define API Tracking Number of Grouped Oil PVT Tables	
APILIM – Define API Tracking Grid Block Limits	
APIVD – Equilibration Oil API Gravity versus Depth Tables	
AQANCONL – Define Analytical Connections to a LGR Grid	
AQANNC – Define Analytic Aquifer Non-Neighbor Connections	
AQANTRC – Define Analytic Aquifer Initial Tracer Concentrations	
AQUALIST – Define An Analytic Aquifer Name to Aquifer Numbers	
AQUANCON – Define Analytical Connections to the Grid	
AQUCHGAS – Define Constant Pressure Gas Analytical Aquifer Properties	
AQUCHWAT – Define Constant Pressure Water Analytical Aquifer Properties	
AQUCON – Define Numerical Aquifer Connections to the Grid	
AQUCT – Define Carter-Tracy Analytical Aquifers	
AQUCWFAC – Modify Constant Pressure Water Analytical Aquifer Properties	
AQUDIMS – Define Aquifer Dimensions	
AQUFET – Define Fetkovich Analytical Aquifer and Connections	
AQUFETP – Define Fetkovich Analytical Aquifers	
AQUFLUX – Define Constant Flux Analytical Aquifer	
AQUNNC – Define Numerical Aquifer Non-Neighbor Connections	
AQUNUM – Define Numerical Aquifer Properties	
AQU TAB – Define Carter-Tracy Aquifer Influence Functions	
AUTOCOAR – Define Auto Refinement Grid Coarsen Area	
AUTOREF – Define Auto Refinement Options	

B

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter B	Status
BC – Define Boundary Conditions	OPM Flow
BCCON – Define Boundary Conditions Connections	OPM Flow
BCPROP – Define Boundary Conditions Properties	OPM Flow
BDENSITY – Define the Surface Brine Density for the Fluid	
BGGI – Define Gi Gas Formation Volume Factor Pressure Tables	
BIC – Binary Interaction Coefficients	
BIGMODEL – Activate Big Model Option (Retired)	
BIOFILM – Activate the Biofilm Model	OPM Flow
BIOFPARA – Define Properties for Models with Biofilms	OPM Flow
BLACKOIL – Activate Black-Oil Phases	
BOGI – Define Gi Oil Formation Volume Factor Pressure Tables	
BOUNDARY – Define a Boundary Box for Printing	
BOX – Define a Range of Grid Blocks to Enter Property Data	
BPARA – Activate Block Parallel License	
BPIDIMS – Define the Dimensions of the Interpolated Block Quantities	
BRANPROP – Define Network Branch Properties for Extended Network Option	
BRINE – Activate Brine Tracking Option	
BTOBALFA – Dual Porosity Matrix to Fracture Multiplier (All Cells)	
BTOBALFV – Dual Porosity Matrix to Fracture Multiplier (Individual Cells)	

C

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter C	Status
CALTRAC – Define a Gas Calorific Value Tracer	
CARFIN – Define a Cartesian Local Grid Refinement	
CART – Activate Cartesian Geometry	
CBMOPTS – Define Coal Bed Methane Options	
CECON – Define Well Connections Economic Limit Criteria	
CECONT – Define Well Connections Tracer Economic Limit Criteria	
CNAMES – Define Compositional Component Names	OPM Flow
CO2SOL – Activate Dissolved CO2 in the Water Phase	OPM Flow
CO2STORE – Activate the CO2 Storage Model	OPM Flow
COAL – Activate the Coal Phase (CBM Model)	
COALADS – Define Gas and Solvent Relative Adsorption Tables	
COALNUM – Define the Coal Region Numbers	
COALPP – Define Gas and Solvent Partial Pressure Adsorption Tables	
COARSEN – Define Grid Coarsening Cells	
COLLAPSE – Define Compressed Vertical Equilibrium Cells	
COLUMNS – Define Input File Column Margins	
COMPDAT – Define Well Connections to the Grid	
COMPDATL – Define Well Connections to a LGR Grid	
COMPDATM – Define Well Connections to an Amalgamated LGR Grid	
COMPFLSH – Assign COMPDAT Differential-Flash Liberation Ratios	
COMPIMB – Assign Imbibition Saturation Tables to Well Connections	
COMPINJK – Assign Injection well Relative Permeability Values	
COMPLMPL – Assign Well LGR Connections to Completions	
COMPLUMP – Assign Well Connections to Completions	
COMPOFF – Deactivate Network Automatic Compressors	
COMPORD – Define Well Connection Ordering	

C

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter C	Status
COMPRIV – Define Grid Cell Connections to a River	
COMPRP – Re-Scale Fluid Saturations of Well Connections	
COMPRPL – Re-Scale Fluid Saturations of Well LGR Connections	
COMPS – Activate Compositional Modeling Formulation	OPM Flow
COMPSEGL – Define Well Connections for Multi-Segment Wells in a LGR	
COMPSEGS – Define Well Connections for Multi-Segment Wells	
COMPTRAJ – Define Well Trajectory Connections to the Grid	OPM Flow
COMPVE – Re-Define Well Connection Depths	
COMPVEL – Re-Define Well LGR Connection Depths	
COORD – Define a Set of Coordinates Lines for a Reservoir Grid	
COORDSYS – Define Coordinate Grid Options	
COPY – Copy Array Data to Another Array	
COPYBOX – Copy Array Data Defined by a Box	
COPYREG – Copy an Array to Another Array based on a Region Number	
CPIFACT – Define Well Connection Transmissibility Multipliers	
CPIFACTL – Define Well Connection Transmissibility Multipliers in a LGR	
CPR – Activate Constrained Pressure Residual (“CPR”) Linear Solver	
CRITPERM – Define Minimum Permeability for Vertical Equilibrium Grid Cell Compression	
CSKIN – Re-Define Well Connection Skin Factors	

D

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter D	Status
<i>DATE – Activate the DATE Option for the SUMMARY File</i>	
<i>DATES – Advance Simulation by Reporting Date</i>	
<i>DATUM – Define the Datum Depth for the Model</i>	
<i>DATUMR – Define Datum Depths for the FIPNUM Regions</i>	
<i>DATUMRX – Define Datum Depths for the FIP Allocated Regions</i>	
<i>DCQDEFN – Define Gas DCQ Units as Rate or Energy</i>	
<i>DEBUG – Define the Debug Data to be Printed to File</i>	
<i>DELAYACT – Define Delayed Action Keywords</i>	
<i>DENAQA – Specify Ezrokhi Coefficients for Aqueous Density</i>	
<i>DENSITY – Define the Surface Oil, Water Gas Densities for the Fluids</i>	
<i>DEPTH – Edits the Depth at the Center of Each Cell</i>	
<i>DEPTHTAB – River Time and Depth Tables</i>	
<i>DIAGDISP – Activate Alternate Form of Tracer Dispersion</i>	
<i>DIFFAGAS – Define PVT Region Gas Component Diffusion Coefficients (Mass Fraction Formulation)</i>	OPM Flow
<i>DIFFAWAT – Define PVT Region Water Component Diffusion Coefficients (Mass Fraction Formulation)</i>	OPM Flow
<i>DIFFC – Define PVT Region Molecular Diffusion Tables</i>	
<i>DIFFCGAS – Define PVT Region Gas Component Diffusion Coefficients</i>	OPM Flow
<i>DIFFCOAL – Define Coal Bed Methane Gas Diffusion Data</i>	
<i>DIFFCWAT – Define PVT Region Water Component Diffusion Coefficients</i>	OPM Flow
<i>DIFFDP – Activate Dual Porosity Molecular Diffusion for Matrix-Fracture Flow Only</i>	
<i>DIFFMICP – Define PVT Region MICP Diffusion Coefficients (Mass Concentration Formulation)</i>	OPM Flow
<i>DIFFMMF – Define Diffusivity Multipliers for Matrix-Fractures</i>	
<i>DIFFMR – Define Grid Block Radial Direction Diffusivity Multipliers</i>	
<i>DIFFMR- – Define Grid Block Negative Radial Direction Diffusivity Multipliers</i>	
<i>DIFFMTHT – Define Grid Block Theta Direction Diffusivity Multipliers</i>	
<i>DIFFMTH- – Define Grid Block Negative Theta Direction Diffusivity Multipliers</i>	

D

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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DIFFMX – Define Grid Block X-Direction Diffusivity Multipliers	
DIFFMX- – Define Grid Block Negative X-Direction Diffusivity Multipliers	
DIFFMY – Define Grid Block Y-Direction Diffusivity Multipliers	
DIFFMY- – Define Grid Block Negative Y-Direction Diffusivity Multipliers	
DIFFMZ – Define Grid Block Z-Direction Diffusivity Multipliers	
DIFFMZ- – Define Grid Block Negative Z-Direction Diffusivity Multipliers	
DIFFR – Define Grid Block Radial Direction Diffusivity Values	
DIFFTHT – Define Grid Block Theta Direction Diffusivity Values	
DIFFUSE – Activate Molecular Diffusion Option	
DIFFX – Define Grid Block X-Direction Diffusivity Values	
DIFFY – Define Grid Block Y-Direction Diffusivity Values	
DIFFZ – Define Grid Block Z-Direction Diffusivity Values	
DIMENS – Define the Dimensions of the Model	
DIMPES – Define IMPES Dynamic Solution Parameters	
DIMPLICIT – Activate Fully Implicit Dynamic Solution Formulation	
DISGAS – Activate the Dissolved Gas Phase in the Model	
DISGASW – Activate Dissolved Gas in the Water Phase in the Model	OPM Flow
DISPDIMS – Define the Maximum Number of Dispersion Tables	
DISPERC – Define the Mechanical Dispersivity for All Cells	OPM Flow
DISPERSE – Define Dispersion Tables	
DOMAINS – Define the Parallel Domain Properties	
DPGRID – Activate The Matrix Cell to Fracture Cell Option	
DPKRMOD – Modify Matrix Oil Relative Permeability Data	
DPNUM – Define Dual and Single Porosity Grid Block Array	
DR – Define the Size of Grid Blocks in the R Direction for All Cells	
DRILPRI – Define Prioritized Drilling Queue Priority Parameters	
DRSDT – Solution Gas (Rs) Maximum Rate of Increase Parameters	

D

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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<i>DRSDTCON – CO2 Convective Dissolution Parameters</i>	OPM Flow
<i>DRSDTR – Solution Gas (Rs) Maximum Rate of Increase Parameters by Region</i>	
<i>DRV – Define the Size of Grid Blocks in the R Direction via a Vector</i>	
<i>DRVDT – Vaporized Oil (Rv) Maximum Rate of Increase Parameters</i>	
<i>DRVDT – Vaporized Oil (Rv) Maximum Rate of Increase Parameters by Region</i>	
<i>DSPDEINT – Activate Brine Tracer Dispersion Interpolation by Water Density</i>	
<i>DTHETA – Define the Size of Grid Blocks in the THETA Direction for All Cells</i>	
<i>DTHETAV – Sets the Size of Grid Blocks in THETA Direction via a Vector</i>	
<i>DUALPERM – Activate Dual Permeability Model</i>	
<i>DUALPORO – Activate Dual Porosity Model</i>	
<i>DUMPCUPL – Activate Output to the Reservoir Coupling File</i>	
<i>DUMPFLUX – Activate Writing Out of a Flux File</i>	
<i>DX – Define the Size of Grid Blocks in the X Direction for All Cells</i>	
<i>DXV – Define the Size of Grid Blocks in the X Direction via a Vector</i>	
<i>DY – Define the Size of Grid Blocks in the Y Direction for All Cells</i>	
<i>DYNAMICR – Start of Dynamic Region Parameter Definition</i>	
<i>DYNRDIMS – Define Dynamic Region Dimensions</i>	
<i>DYV – Define the Size of Grid Blocks in the Y Direction via a Vector</i>	
<i>DZ – Define the Size of Grid Blocks in the Z Direction for All Cells</i>	
<i>DZMATRIX – Matrix Block Height for Gravity Drainage Model For All Cells</i>	
<i>DZMTRX – Matrix Block Height for Gravity Drainage Model for the Grid</i>	
<i>DZMTRXV – Matrix Block Height for Gravity Drainage Model For All Cells</i>	
<i>DZNET – Define Grid Block Net Thickness for All Cells</i>	
<i>DZV – Define the Size of Grid Blocks in the Z Direction via a Vector</i>	

E

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter E	Status
ECHO – Activate Echoing of User Input Files to the Print File	
ECLMC – Activate Multi-Component Brine Model	
EDIT – Define the Start of the EDIT Section of Keywords	
EDITNNC – Scale Non-Neighbor Connections Between Cells Manually	
EDITNNCR – Reset Non-Neighbor Connections Between Cells Manually	
EHYSTR – Define Hysteresis Model and Parameters	
EHYSTRR – Define Hysteresis Model and Parameters via SATNUM	
END – Define the End of the Input File	
ENDACTIO – End the Definition of ACTION Commands	
ENDBOX – Define the End of the BOX Defined Grid	
ENDDDYN – End of Dynamic Region Parameter Definition	
ENDFIN – End the Definition of a Local Grid Refinement	
ENDINC – Define the End of an Include File	
ENDNUM – Define the End-Point Scaling Depth Region Numbers	
ENDSCALE – Activate Relative Permeability End-Point Scaling Option	
ENDSKIP – Deactivate Skipping of Keywords and Input Data	
ENKRVD – Define Relative Permeability End-Points versus Depth Functions	
ENKRVDX	
ENKRVDX-	
ENKRVDY	
ENKRVDY-	
ENKRVDZ	
ENKRVDZ-	
ENPCVD – Define Maximum Capillary Pressure versus Depth Functions	
ENPTVD – Define Relative Permeability Saturation End-Points versus Depth	
ENPTVDX	

E

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter E	Status
ENPTVDX-	
ENPTVDY	
ENPTVDY-	
ENPTVDZ	
ENPTVDZ-	
<i>ENSPCVD – Define Capillary Pressure End-Points versus Depth</i>	
<i>EOS – Specify Equation of State</i>	
<i>EOSNUM – Define the Equation of State Region Numbers</i>	
<i>EPSDBGS – Write End-Point Debug Data to the DEBUG File (Multiple)</i>	
<i>EPSDEBUG – Write End-Point Debug Data to the DEBUG File (Individual)</i>	
<i>EQLDIMS – Define the Equilibration Data Dimensions</i>	
<i>EQLNUM – Define the Equilibration Region Numbers</i>	
<i>EQLOPTS – Activate the Equilibration Options</i>	
<i>EQLZCORN – Modify the Depth of the Corner-Point Depth Array</i>	
<i>EQUALREG – Sets an Array to a Constant by Region Number</i>	
<i>EQUALS – Sets a Specified Array to a Constant</i>	
<i>EQUIL – Define the Equilibration Initialization Data</i>	
<i>ESSNODE – Define Salt Concentration Data for Water-Oil Surface Tension</i>	
<i>EXCAVATE – Set the Status of a Grid Block To Active or Excavate</i>	
<i>EXCEL – Activate the EXCEL Option for the SUMMARY File</i>	
<i>EXIT – Exit Simulation from within an Action Section</i>	OPM Flow
<i>EXTFIN – Define an External Unstructured Local Grid Refinement</i>	
<i>EXTHOST – Define Host Cells for External LGRs</i>	
<i>EXTRAPMS – Activate Extrapolation Warning Messages</i>	
<i>EXTREPGL – Define Host Cells for External Unstructured LGRs</i>	

F

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter F	Status
FAULTDIM – Define the Number of Fault Segments	
FAULTS – Define Faults in the Grid Geometry	
FBHPDEF – Define Well Default BHP Target and Constraints	
FHERCHBL – Define Herschel-Bulkley Data versus Polymer Concentration	
FIELD – Activate the Oil Field System of Units for the Model	
FILEUNIT – Activate Unit Consistency Checking	
FILLEPS – Activate Saturation End-Point Export to the INIT File	
FIP – Define the Fluid In-Place Names and Region and Numbers	
FIPNUM – Define the Fluid In-Place Region Numbers	
FIPOWG – Activate Oil, Gas, and Water FIP Zone Reporting	
FLUXNUM – Define the Flux Regions	
FLUXREG – Define Active Flux Regions	
FLUXTYPE – Defines the Flux Boundary Type	
FMTHMD – Activate The Format History Match Gradient File Option	
FMTIN – Activate The Format Input File Option	
FMTOUT – Activate The Format Output File Option	
FMWSET – Export Well Status Vectors for the Field to File	
FOAM – Activate the Foam Phase and Model	
FOAMADS – Define Foam Rock Adsorption Tables	
FOAMDCYO – Define Foam Decay versus Oil Saturation Tables	
FOAMDCYW – Define Foam Decay versus Water Saturation Tables	
FOAMFCN – Define Foam Gas Mobility Reduction versus Capillary Number	
FOAMFRM – Define Foam Gas Mobility Reduction versus Reference Mobility	
FOAMFSC – Define Foam Gas Mobility versus Surfactant Concentration Functions	
FOAMFSO – Define Foam Gas Mobility Reduction versus Oil Saturation	
FOAMFST – Define Foam Gas-Water Surface Tension versus Surfactant Concentration	

F

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter F	Status
FOAMFSW – Define Foam Gas Mobility Reduction versus Water Saturation	
FOAMMOB – Define Foam Gas Mobility versus Foam Concentration Tables	
FOAMMOBP – Define Foam Mobility Reduction versus Oil Pressure	
FOAMMOBS – Define Foam Mobility Reduction versus Shear	
FOAMOPTS – Define Foam Model Options	
FOAMROCK – Define Foam Rock Properties	
FORMFEED – Defined the Print File Form-Feed Character	
FRICTION – Activate Wellbore Friction Option	
FULLIMP – Activate Fully Implicit Solution Option	

G

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter G	Status
GAS – Activate the Gas Phase in the Model	
GASBEGIN – Define Start of Annual Scheduling Section	
GASCONC – Define the Initial Equilibration Coal Gas Concentration for All Grid Blocks	
GASDENT – Define Gas Density Temperature Coefficients	OPM Flow
GASEND – Define End of Annual Scheduling Section	
GASFCOMP – Define Automatic Gas Compressors	
GASFDECR – Define Field Gas Sales Contract Monthly Reduction	
GASFDEL – Define Gas Deliverability Calculation	
GASFIELD – Define Gas Field Operations Options	
GASFTARG – Define Field Gas Sales Contract Monthly Target	
GASJT – Define Gas Joule-Thomson Coefficient	OPM Flow
GASMONTH – Define Start of Annual Scheduling Event	
GASPERIO – Advance Simulation by Gas Contract Period	
GASSATC – Define the Initial Equilibration Saturated Coal Gas Concentration for All Grid Blocks	
GASVISCT – Define Gas Viscosity versus Temperature Functions	OPM Flow
GASWAT – Activate the Gas-Water Model Formulation	OPM Flow
GASYEAR – Advance Simulation by Gas Contract Year	
GCALECON – Group Economic Criteria for Production Calorific Groups	
GCONCAL – Group Production Calorific Targets	
GCONENG – Group Production Energy Targets	
GCONINJE – Group Injection Targets and Constraints	
GCONPRI – Group Production Priority Targets and Constraints	
GCONPROD – Group Production Targets and Constraints	
GCONSALE – Define Group Sales Gas Production Targets and Constraints	
GCONSUMP – Define Group Gas Consumption and Gas Import Targets	
GCONTOL – Define Group Constraint Tolerance	

G

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter G	Status
GCUTBACK – Define Group Cutback Limits and Parameters	
GCUTBACT – Define Group Tracer Cutback Limits and Parameters	
GCVD – Define Equilibration Coal Gas Concentration versus Depth Tables	
GDCQ – Define Group Multiple Daily Contract Quantities	
GDCQECON – Group Economic Criteria for DCQ Production Groups	
GDFILE – Load a Grid File	
GDIMS – Activate Instantaneous Gradient Option and Define Dimensions	
GDORIENT – Define Grid Orientation Parameters	
GDRILPOT – Define Group Potential Rates for Automatic Drilling	
GECON – Group Economic Criteria for Production Groups	
GECONT – Group Tracer Economic Criteria for Production Groups	
GEFAC – Define Group Efficiency	
GETDATA – Load and Assign Data Array from INIT or RESTART Files	
GETGLOB – Activate Loading of Global Grid Restart Data Option	
GI – Define the Initial Equilibration Gi Values for All Grid Blocks	
GIALL – Define Gi Values and PVT Properties versus Pressure	
GIMODEL – Activate Gi Pseudo Compositional Option	
GINODE – Define Gi Node Values	
GLIFTLIM – Group Artificial Lift Constraints	
GLIFTOPT – Define Group Gas Optimization Limits	
GMWSET – Export Well Status Vectors by Group to File	
GNETDP – Group Network Pressure and Rate Controls	
GNETINJE – Define Group Injection Network Configuration	
GNETPUMP – Standard Network Automatic Compressor and Pumps	
GPMAINT – Define Group Pressure Maintenance Targets and Controls	
GRADGRUP – Define Group History Match Gradient File Output	

G

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter G	Status
GRADRESV – Define Solution Derivative History Match Gradient Output	
GRADRFT – Define RFT Derivative History Match Gradient Output	
GRADWELL – Define Well History Match Gradient File Output	
GRAVCONS – Re-Define Gravity Constant	
GRAVDR – Activate Gravity Drainage and Imbibition for Dual Porosity Model	
GRAVDRB – Activate Vertical Discretized Gravity Drainage and Imbibition for Dual Porosity Model	
GRAVDRM – Activate Alternative Gravity Drainage and Imbibition for Dual Porosity Model	
GRAVITY – Define the Surface Oil, Water Gas Gravities for the Fluids	
GRDREACH – Define River and Grid Block Connections	
GRID – Define the Start of the GRID Section of Keywords	
GRIDFILE – Set the Grid File Output Options	
GRIDOPTS – Grid Processing Options	
GRIDUNIT – Define the Grid Units	
GRUPMAST – Define Master and Slave Groups	
GRUPNET – Define Group Standard Network Parameters	
GRUPRIG – Group Drilling and Workover Rig Specifications	
GRUPLAV – Define Slave Groups in Slave Reservoirs	
GRUPTARG – Modify Group Targets and Constraints Values	
GRUPTREE – Define Group Tree Hierarchy	
GSATINJE – Define Group Satellite Injection Rates	
GSATPROD – Define Group Satellite Production Rates	
GSEPCOND – Assign Group Separators	
GSF – Gas Saturation Function Tables (Gas-Water Systems)	OPM Flow
GSSCPTST – Perform Sustainable Capacity Test	
GSWINGF – Define Group Multiple Gas Contract Parameters	
GTADD – Add a Constant to a Group Target or Constraint	

G

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Alphabetic Listing Of Keywords Starting With The Letter G	Status
<i>GTMULT – Multiply Group Target or Constraint by a Constant</i>	
<i>GUIDECAL – Scale Guide Rates Based on Gas Calorific Value</i>	
<i>GUIDERAT – Define Group Guide Rate Formula</i>	
<i>GUPFREQ – Instantaneous Gradient Option Update Frequency</i>	
<i>GWRTWCV – Instantaneous Gradient Option Well Variables</i>	

H

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter H	Status
H2SOL – Activate Dissolved H2 in the Water Phase	OPM Flow
H2STORE – Activate the H2 Storage Model	OPM Flow
HALFTRAN – Activate Half Block Transmissibility Calculations	
HA – History Match End-Point Gradient Additive Modifier	
HAKRG	
HAKRGR	
HAKRO	
HAKRORG	
HAKRORW	
HAKRW	
HAKRWR	
HAPCG	
HAPCG	
HASGLPC	
HASOGCR	
HASOWCR	
HASWL	
HASWLPC	
HBNUM – Define Herschel-Bulkley Region Numbers	
HDISP – Define Tracer Mechanical Dispersivity Parameters	
HEATCR – Define Reservoir Rock Heat Capacity for All Cells	
HEATCRT – Define Reservoir Rock Heat Capacity Temperature Dependence for All Cells	
HM – History Match End-Point Gradient Multiplicative Modifier	
HMKRG	
HMKRGR	
HMKRO	

H

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter H	Status
HMKRORG	
HMKRORW	
HMKRW	
HMKRWR	
HMPCG	
HMPCG	
HMSGLPC	
HMSOGCR	
HMSOWCR	
HMSWL	
HMSWLPC	
<i>HM – History Match Region Gradient Parameters</i>	
HMKRG	
HMKRGR	
HMKRO	
HMKRORG	
HMKRORW	
HMKRW	
HMKRWR	
HMPERMx	
HMPERMy	
HMPERMz	
HMPORVM	
HMPRMXY	
HMSGCR	
HMSGL	

H

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter H	Status
HMSGLPC	
HMSIGMA	
HMSOGCR	
HMSOWCR	
HMSWCR	
HMSWL	
HMSWLPC	
HMTRANX	
HMTRANY	
HMTRANZ	
HMTRNXY	
HMAQUCT – History Match Carter-Tracy Aquifer Gradient Parameters	
HMAQUFET – History Match Fetkovich Aquifer Gradient Parameters	
HMAQUNUM – History Match Numerical Aquifer Gradient Parameters	
HMDIMS – Define History Match Gradient Parameter Dimensions	
HMFAULTS – History Match Fault Gradient Parameters	
HMMLAQUN – History Match Numerical Aquifer Gradient Multipliers	
HMMLCTAQ – History Match Carter-Tracy Aquifer Gradient Multipliers	
HMMLFTAQ – History Match Fetkovich Aquifer Gradient Multipliers	
HMMLT – History Match Grid Permeability Gradient Cumulative Multipliers	
HMMLTPR	
HMMLTPTH	
HMMLTPX	
HMMLTPXY	
HMMLTPY	
HMMLTPZ	

H

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter H	Status
HMMLTWCN – History Match Well Connection and Skin Multipliers	
HMMMREGT – History Match Region Transmissibility Gradient Cumulative Multipliers	
HMMROCK – History Match Rock Compressibility Gradient Cumulative Multipliers	
HMMROCKT – History Match Rock Compaction Gradient Cumulative Multipliers	
HMMULRGT – History Match Region Transmissibility Parameters	
HMMULT – History Match Grid Transmissibility & Pore Volume Gradient Cumulative Multipliers	
HMULTPV	
HMMULTR	
HMMULTTH	
HMMULTX	
HMMULTXY	
HMMULTY	
HMMULTZ	
HMMULTFT – History Match Fault Transmissibility Gradient Cumulative Multipliers	
HMMULTSG – History Match Dual porosity Sigma Gradient Cumulative Multipliers	
HMPROPS – History Match End-Point Section Start	
HMROCK – History Match Rock Compressibility Gradient Parameters	
HMROCKT – History Match Rock Compaction Gradient Parameters	
HMRREF – History Match Rock Table Reference Pressure Values	
HMWELCON – History Match Well Connection and Skin Parameters	
HMWPIMLT – History Match Well Productivity Index Parameters	
HRFIN – Define the Ratio of LGR Grid Blocks in the R-Direction	
HWKRO – End-Point Scaling of Grid Cell Kro(Swl) (High Salinity and Water Wet)	
HWKRORG – End-Point Scaling of Grid Cell Kro(Sgcr) (High Salinity and Water Wet)	
HWKRORW – End-Point Scaling of Grid Cell Kro(Swcr) (High Salinity and Water Wet)	
HWKRW – End-Point Scaling of Grid Cell Krw(Sw =1.0) (High Salinity and Water Wet)	

H

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter H	Status
HWKRWR – End-Point Scaling of Grid Cell KRWR(Sw=1.0) (High Salinity and Water Wet)	
HWPWCW – End-Point Scaling of Grid Cell Water Capillary Pressure (High Salinity and Water Wet)	
HWSNUM – Define the Saturation Table Region Numbers (High Salinity and Water Wet)	
HWSOGCR – End-Point Scaling Grid Cell SOGCR (High Salinity and Water Wet)	
HWSOWCR – End-Point Scaling Grid Cell SOWCR (High Salinity and Water Wet)	
HWSWCR – End-Point Scaling Grid Cell SWCR (High Salinity and Water Wet)	
HWSWL – End-Point Scaling Grid Cell SWL (High Salinity and Water Wet)	
HWSWLPC – End-Point Scaling Grid Cell SWLPC (High Salinity and Water Wet)	
HWSWU – End-Point Scaling Grid Cell SWU (High Salinity and Water Wet)	
HXFIN – Define the Ratio of LGR Grid Blocks in the X-Direction	
HYDRHEAD – Define Hydraulic Head Output Reference Data	
HYFIN – Define the Ratio of LGR Grid Blocks in the Y-Direction	
HYMOBGDR – Activate Carlson and Killough Alternative Drainage Hysteresis Option	
HYST – Activate the Hysteresis Option (Retired)	
HYSTCHCK – Activate Hysteresis Imbibition and Drainage End-Point Validation	
HZFIN – Define the Ratio of LGR Grid Blocks in the Z-Direction	

I

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter I	Status
<i>IHOST – Assign LGRs to Parallel Process Number</i>	
<i>IKRG – End-Point Scaling of Grid Cell Krg(Sgu) (Imbibition)</i>	
IKRGX	
IKRGX-	
IKRGY	
IKRGY-	
IKRGZ	
IKRGZ-	
<i>IKRGR – End-Point Scaling of Grid Cell Krgr(1-Sogcr) (Imbibition)</i>	
IKRGRX	
IKRGRX-	
IKRGRY	
IKRGRY-	
IKRGRZ	
IKRGRZ-	
<i>IKRO – End-Point Scaling of Grid Cell Kro(Swl) (Imbibition)</i>	
IKROX	
IKROX-	
IKROY	
IKROY-	
IKROZ	
KROZ-	
<i>IKRORG – End-Point Scaling of Grid Cell Kro(Sgcr) (Imbibition)</i>	
IKRORGX	
IKRORGX-	
IKRORGY	

I

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter I	Status
IKRORGY-	
IKRORGZ	
IKRORGZ-	
<i>IKRORW – End-Point Scaling of Grid Cell Kro(Swcr) (Imbibition)</i>	
IKRORWX	
IKRORWX-	
IKRORWY	
IKRORWY-	
IKRORWZ	
IKRORWZ-	
<i>IKRW – End-Point Scaling of Grid Cell Krw(Sw =1.0) (Imbibition)</i>	
IKRWX	
IKRWX-	
IKRWY	
IKRWY-	
IKRWZ	
IKRWZ-	
<i>IKRWR – End-Point Scaling of Grid Cell KRWR(Sowcr) (Imbibition)</i>	
IKRWRX	
IKRWRX-	
IKRWRY	
IKRWRY-	
IKWRVZ	
IKWRVZ-	
<i>IMBNUM – Define the Imbibition Saturation Table Region Numbers</i>	
IMBNUMX	

I

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter I	Status
IMBNUMX-	
IMBNUMY	
IMBNUMY-	
IMBNUMZ	
IMBNUMZ-	
<i>IMBNUMMF – Define the Imbibition Saturation Table Region Numbers (Matrix-Fracture)</i>	
<i>IMKRVD – Imbibition Relative Permeability End-Points versus Depth Functions</i>	
IMKRVDX	
IMKRVDX-	
IMKRVDY	
IMKRVDY-	
IMKRVDX	
IMKRVDZ-	
<i>IMPCVD – Imbibition Maximum Capillary Pressure versus Depth Functions</i>	
<i>IMPES – Activate Implicit Pressure Explicit Saturation Solution Option</i>	
<i>IMPLICIT – Activate Fully Implicit Solution Option</i>	
<i>IMPORT – Import Grid File Data at the Current Position</i>	
<i>IMPTVD – Imbibition Relative Permeability Saturation End-Points versus Depth</i>	
IMPTVDX	
IMPTVDX-	
IMPTVDY	
IMPTVDY-	
IMPTVDZ	
IMPTVDZ-	
<i>IMSPCVD – Imbibition Capillary Pressure Connate Saturations versus Depth</i>	

I

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter I	Status
INCLUDE – Load Another Data File at the Current Position	
INIT – Activate the INIT File Option	
INRAD – Define the Inner Radius of a Radial Grid	
INSPEC – Activate the INSPEC File Option	
INTPC – Activate Dual Porosity Integrated Capillary Pressure Option	
IONROCK – Define the Ion Exchange Capacity for All the Cells	
IONXROCK – Define Ion Exchange Constant by Saturation Table Regions	
IONXSURF – Define Surfactant Ion Exchange Constant by Saturation Table Regions	
IPCG – End-Point Scaling of Grid Cell Gas Capillary Pressure (Imbibition)	
IPCW – End-Point Scaling of Grid Cell Water Capillary Pressure (Imbibition)	
ISGCR – End-Point Scaling of Grid Cell Critical Gas Saturation (Imbibition)	
ISGCRX	
ISGCRX-	
ISGCRY	
ISGCRY-	
ISGCRZ	
ISGCRZ-	
ISGL – End-Point Scaling of Grid Cell Connate Gas Saturation (Imbibition)	
ISGLX	
ISGLX-	
ISGLY	
ISGLY-	
ISGLZ	
ISGLZ-	
ISGLPC – End-Point Scaling of Grid Cell Capillary Pressure Connate Gas Saturation (Imbibition)	
ISGU – End-Point Scaling of Grid Cell Maximum Gas Saturation (Imbibition)	

I

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter I	Status
ISGUX	
ISGUX-	
ISGUY	
ISGUY-	
ISGUZ	
ISGUZ-	
<i>ISOGCR – End-Point Scaling of Grid Cell Critical Oil Saturation to Gas (Imbibition)</i>	
ISOGCRX	
ISOGCRX-	
ISOGCRY	
ISOGCRY-	
ISOGCRZ	
ISOGCRZ-	
<i>ISOLNUM – Define the Independent Reservoir Regions</i>	
<i>ISOWCR – End-Point Scaling of Grid Cell Critical Oil Saturation to Water (Imbib.)</i>	
ISOWCRX	
ISOWCRX-	
ISOWCRY	
ISOWCRY-	
ISOWCRZ	
ISOWCRZ-	
<i>ISWCR – End-Point Scaling of Grid Cell Critical Water Saturation (Imbibition)</i>	
ISWCRX	
ISWCRX-	
ISWCRY	
ISWCRY-	

I

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter I	Status
ISWCRZ	
ISWCRZ-	
<i>ISWL – End-Point Scaling of Grid Cell Connate Water Saturation (Imbibition)</i>	
ISWLX	
ISWLX-	
ISWLY	
ISWLY-	
ISWLZ	
ISWLZ-	
<i>ISWLPC – End-Point Scaling Grid Cell Capillary Pressure Connate Water Saturations (Imbibition)</i>	
<i>ISWU – End-Point Scaling of Grid Cell Maximum Water Saturation (Imbibition)</i>	
ISWUX	
ISWUX-	
ISWUY	
ISWUY-	
ISWUZ	
ISWUZ-	

J

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter J	Status
JFUNC – Activate the Leverett J-function Option	
JFUNCR – Activate the Leverett J-function Saturation Table Option	

K

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter K	Status
<i>KRG – End-Point Scaling of Grid Cell Krg(Sgu) (Drainage)</i>	
KRGX	
KRGX-	
KRGY	
KRGY-	
KRGZ	
KRGZ-	
<i>KRGR – End-Point Scaling of Grid Cell Krgr(1-Sogcr) (Drainage)</i>	
KRGRX	
KRGRX-	
KRGY	
KRGY-	
KRGRZ	
KRGRZ-	
<i>KRNUM – Define the Directional Saturation Table Region Numbers</i>	
KRNUMR	
KRNUMR-	
KRNUMT	
KRNUMT-	
KRNUMX	
KRNUMX-	
KRNUMY	
KRNUMY-	
KRNUMZ	
KRNUMZ-	
<i>KRNUMMF – Define the Saturation Table Region Numbers (Matrix-Fracture)</i>	

K

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter K	Status
<i>KRO – End-Point Scaling of Grid Cell Kro(Swl) (Drainage)</i>	
KROX	
KROX-	
KROY	
KROY-	
KROZ	
KROZ-	
<i>KRORG – End-Point Scaling of Grid Cell Kro(Sgcr) (Drainage)</i>	
KRORGX	
KRORGX-	
KRORGY	
KRORGY-	
KRORGZ	
KRORGZ-	
<i>KRORW – End-Point Scaling of Grid Cell Kro(Swcr) (Drainage)</i>	
KRORWX	
KRORWX-	
KRORWY	
KRORWY-	
KRORWZ	
KRORWZ-	
<i>KRW – End-Point Scaling of Grid Cell Krw(Sw =1.0) (Drainage)</i>	
KRWX	
KRWX-	
KRWY	
KRWY-	

K

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter K	Status
KRWZ	
KRWZ-	
<i>KRWR – End-Point Scaling of Grid Cell KRWR(Sowcr) (Drainage)</i>	
KRWRX	
KRWRX-	
KRWRY	
KRWRY-	
KRWRZ	
KRWRZ-	

L

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter L	Status
<i>LAB – Activate the Laboratory System of Units for the Model</i>	
<i>LANGMPL – Define Langmuir Pressure Grid Cell Multiplier</i>	
<i>LANGMUIR – Langmuir Adsorption Isotherm Tables</i>	
<i>LANGSOLV – Langmuir Adsorption Isotherm Solvent Tables</i>	
<i>LCUNIT – Define Linear Combination Rate and Volume Units</i>	
<i>LGR – Define Local Grid Refinement Dimensions and Parameters</i>	
<i>LGRCOPY – Activate Local Grid Refinement Inheritance</i>	
<i>LGRFREE – Activate Local Grid Refinement Independent Time Steps</i>	
<i>LGRLOCK – Deactivate Local Grid Refinement Independent Time Steps</i>	
<i>LGROFF – Deactivate a Local Grid Refinement</i>	
<i>LGRON – Activate a Local Grid Refinement</i>	
<i>LICENSES – Define Required Licenses for Run</i>	
<i>LIFTOPT – Activate Gas Lift Optimization</i>	
<i>LINCOM – Define Linear Combination Coefficients</i>	
<i>LINKPERM – Assign Cell Permeabilities to Cell Faces</i>	
<i>LIVEOIL – Activate the Live Oil Phase (Oil with Free and Dissolved Gas)</i>	
<i>LKRO – End-Point Scaling of Grid Cell Kro(Swl) (Low Salinity and Oil Wet)</i>	
<i>LKRORG – End-Point Scaling of Grid Cell Kro(Sgcr) (Low Salinity and Oil Wet)</i>	
<i>LKRORW – End-Point Scaling of Grid Cell Kro(Swcr) (Low Salinity and Oil Wet)</i>	
<i>LKRW – End-Point Scaling of Grid Cell Krw(Sw =1.0) (Low Salinity and Oil Wet)</i>	
<i>LKRWR – End-Point Scaling of Grid Cell KRWR(Sowcr) (Low Salinity and Oil Wet)</i>	
<i>LOAD – Load a SAVE File for a Fast Restart</i>	
<i>LOWSALT – Activate the Low Salt Brine Phase in the Brine Model</i>	
<i>LPCW – End-Point Scaling of Grid Cell Water Capillary Pressure (Low Salinity and Oil Wet)</i>	
<i>LSALTFNC – Define Low Salt Weighting Factors versus Salt Concentration Functions</i>	
<i>LSLTWNUM – Define the Low Salt Water Wet Saturation Table Region Numbers</i>	

L

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter L	Status
LSNUM – Define the Low Salt Oil Wet Saturation Table Region Numbers	
LSOGCR – End-Point Scaling Grid Cell SOGCR (Low Salinity and Oil Wet)	
LSOWCR – End-Point Scaling Grid Cell SOWCR (Low Salinity and Oil Wet)	
LSWCR – End-Point Scaling Grid Cell SWCR (Low Salinity and Oil Wet)	
LSWL – End-Point Scaling Grid Cell SWL (Low Salinity and Oil Wet)	
LSWLPC – End-Point Scaling Grid Cell SWLPC (Low Salinity and Oil Wet)	
LSWU – End-Point Scaling Grid Cell SWU (Low Salinity and Oil Wet)	
LTOSIGMA – Dual Porosity Viscous Displacement Sigma Parameters	
LWKRO – End-Point Scaling of Grid Cell Kro(Swl) (Low Salinity and Water Wet)	
LWKROG – End-Point Scaling of Grid Cell Kro(Sgcr) (Low Salinity and Water Wet)	
LWKROW – End-Point Scaling of Grid Cell Kro(Swcr) (Low Salinity and Water Wet)	
LWKRW – End-Point Scaling of Grid Cell Krw(Sw =1.0) (Low Salinity and Water Wet)	
LWKRW – End-Point Scaling of Grid Cell KRWR(Sw=1.0) (Low Salinity and Water Wet)	
LWPCW – End-Point Scaling of Grid Cell Water Capillary Pressure (Low Salinity and Water Wet)	
LWSLTNUM – Define the Low Salt Oil Wet Saturation Table Region Numbers	
LWSNUM – Define the Low Salt Water Wet Saturation Table Region Numbers	
LWSOGCR – End-Point Scaling Grid Cell SOGCR (Low Salinity and Water Wet)	
LWSOWCR – End-Point Scaling Grid Cell SOWCR (Low Salinity and Water Wet)	
LWSWCR – End-Point Scaling Grid Cell SWCR (Low Salinity and Water Wet)	
LWSWL – End-Point Scaling Grid Cell SWL (Low Salinity and Water Wet)	
LWSWLPC – End-Point Scaling Grid Cell SWLPC (Low Salinity and Water Wet)	
LWSWU – End-Point Scaling Grid Cell SWU (Low Salinity and Water Wet)	
LX – Dual Porosity Viscous Displacement X Direction Matrix Size for All Cells	
LXFIN – Define Logarithmic LGR Grid Block Spacing in the X-Direction	
LY – Dual Porosity Viscous Displacement Y Direction Matrix Size for All Cells	
LYFIN – Define Logarithmic LGR Grid Block Spacing in the Y-Direction	

L

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Alphabetic Listing Of Keywords Starting With The Letter L	Status
<i>LZ – Dual Porosity Viscous Displacement Z Direction Matrix Size for All Cells</i>	
<i>LZFIN – Define Logarithmic LGR Grid Block Spacing in the Z-Direction</i>	

M

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter M	Status
MAPAXES – Define the Map Origin Input Data	
MAPUNITS – Define the Map Axes Units	
MASSFLOW – Define River Mass Flow versus Time Tables	
MATCORR – Activate the Material Balance Correction Option	
MAXVALUE – Sets a Maximum Value for an Array Element	
MEMORY – Define Allocated Memory (Retired)	
MESSAGE – Output User Message	
MESSAGES – Define Message Print Limits and Stop Limits	
MESSOPTS – Reset Severity Level for Forced Time Steps	
MESSSRVC – Activate or Deactivate Database Message File Output	
METRIC – Activate the Metric System of Units for the Model	
MICP – Activate the Microbially Induced Calcite Precipitation Model	OPM Flow
MICPPARA – Define Microbially Induced Calcite Precipitation Parameters (Retired)	OPM Flow
MINNCT – Set a Minimum Non-Neighbor Connection Transmissibility	
MINNPCOL – Define the Minimum Number of Newton Iterations Used to Update Well Targets	OPM Flow
MINPORV – Set a Minimum Grid Block Pore Volume Threshold for All Cells	
MINPV – Set a Minimum Grid Block Pore Volume Threshold for All Cells	
MINPVV – Set a Minimum Grid Block Pore Volume Threshold for Individual Cells	
MINVALUE – Set a Minimum Value for an Array Element	
MISC – Define Solvent Miscibility-Immiscibility Transform Functions	
MISCIBLE – Define Miscibility Todd-Longstaff Parameters	
MISCNUM – Define the Miscibility Region Numbers	
MLANG – Define Langmuir Maximum Gas Concentration for All Grid Cells	
MLANGSLV – Define Langmuir Maximum Solvent Concentration for All Grid Cells	
MONITOR – Activate Output of the Monitoring Data and File	
MPFANUM – Define Multi-Point Flux Discretization Regions	

M

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter M	Status
MPFNNC – Define Multi-Point Flux Non-Nighbor Connections	
MSFN – Miscible Normalized Relative Permeability Tables	
MSGFILE – Activate or Deactivate Message File Output	
MULSGGD – Multiply Matrix-Fracture Coupling for Oil-Gas Gravity Drainage for All Cells	
MULSGGDV – Multiply Matrix-Fracture Coupling for Oil-Gas Gravity Drainage for Individual Cells	
MULTFLT – Multiply the Transmissibility of a Defined Fault by a Constant	
MULTIN – Activate the Non-Unified Multiple Input File Option	
MULTIPLY – Multiply a Specified Array by a Constant	
MULTIREG – Multiply an Array by a Constant based on a Region Number	
MULTNUM – Define the Multiple Transmissibility Regions	
MULTOUT – Activate the Non-Unified Multiple Output File Option	
MULTOUTS – Activate Non-Unified Multiple Summary Output File Option	
MULTPV – Multiply Cell Pore Volumes by a Constant	
MULTR – Multiply Cell Transmissibility in the +R Direction	
MULTR- – Multiply Cell Transmissibility in the -R Direction	
MULTREAL – Activate Commercial Simulator's Multi-Realization License	
MULTREGD – Multiply Diffusivities Between Regions	
MULTREGH – Multiply Thermal Conductivities Between Regions	
MULTREGP – Multiply Pore Volumes Based On Region Number	
MULTREGT – Multiply Transmissibilities Between Regions	
MULTSIG – Multiply Matrix-Fracture Coupling for All Cells	
MULTSIGV – Multiply Matrix-Fracture Coupling for Individual Cells	
MULTTHT – Multiply Cell Transmissibility in the +Theta Direction	
MULTTHT- – Multiply Cell Transmissibility in the -Theta Direction	
MULTX – Multiply Cell Transmissibility in the +X Direction	
MULTX- – Multiply Cell Transmissibility in the -X Direction	

M

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Alphabetic Listing Of Keywords Starting With The Letter M	Status
<i>MULTY – Multiply Cell Transmissibility in the +Y Direction</i>	
<i>MULTY- – Multiply Cell Transmissibility in the -Y Direction</i>	
<i>MULTZ – Multiply Cell Transmissibility in the +Z Direction</i>	
<i>MULTZ- – Multiply Cell Transmissibility in the -Z Direction</i>	
<i>MW – Molecular Weights</i>	

N

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter N	Status
NARROW – Activate Run Summary Narrow Column Output Option	
NCOMPS – Confirm Number of Compositional Components	OPM Flow
NCONSUMP – Node Gas Consumption (Extended Network)	
NEFAC – Node Efficiency Factors (Extended Network)	
NETBALAN – Network Balancing Parameters	
NETCOMPA – Define Automatic Compressors (Extended Network)	
NETWORK – Activate the Extended Network Option and Define Parameters	
NEWTON – Activate Newton Iteration SUMMARY Output	
NEWTRAN – Activate Irregular Corner-Point Grid Transmissibilities	
NEXT – Maximum Next Time Step Size (Alias for NEXTSTEP)	
NEXTSTEP – Maximum Next Time Step Size	
NEXTSTPL – Maximum Next Time Step Size (LGR)	
NINENUM – Define the Nine-Point Discretization Region	
NINEPOIN – Activate the Nine-Point Discretization Option	
NMATOPTS – Define the Discretized Matrix Dual Porosity Parameters	
NMATRIX – Activate the Discretized Matrix Dual Porosity Option	
NMESSAGE – Export Cumulative Message Summary Variables to File	
NNC – Define Non-Neighbor Connections Between Cells Manually	
NNEWTF – Activate the Non-Newtonian Fluid Model	
NOCASC – Activate Linear Solver Tracer Algorithm	
NODEPROP – Define Network Node Properties for Extended Network	
NODPPM – Deactivate Fracture Porosity-Permeability Calculation	
NOECHO – Deactivate Echoing of User Input Files to the Print File	
NOGGF – Deactivate Output of Grid Geometry File	
NOHMD – Deactivate History Match Gradient Derivative Calculations	
NOHMO – Deactivate History Match Gradient Derivative Calculations (Alias)	

N

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter N	Status
NOHYST – Deactivate the Hysteresis Option	
NOINSPEC – Deactivate Output of the INIT Index File	
NOMONITO – Deactivate Output of the Monitoring Data and File	
NONNC – Deactivate Non-Neighbor Connections	
NORSSPEC – Deactivate Output of the RESTART Index File	
NOSIM – Activate the No Simulation Mode for Data File Checking	
NOWARN – Deactivate Warning Messages	
NOWARNEP – Deactivate End-Point Scaling Warning Messages	
NRSOUT – Defined Maximum Number of RESTART Elements	
NSTACK – Define the Stack Length for the Iterative Linear Solver	
NTG – Define the Net-to-Gross Ratio for All the Cells	
NUMRES – Define the Number of Reservoir Grids	
NUPCOL – Define the Maximum Number of Newton Iterations Used to Update Well Targets	
NWATREM – Node Water Removal (Extended Network)	
NXFIN – Define the Number of LGR Grid Blocks in the X-Direction	
NYFIN – Define the Number of LGR Grid Blocks in the Y-Direction	
NZFIN – Define the Number of LGR Grid Blocks in the Z-Direction	

O

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter O	Status
OFM – Activate OFM File Output of the SUMMARY Data	
OIL – Activate the Oil Phase in the Model	
OILAPI – Define the Initial Equilibration Oil API for All Grid Blocks	
OILDENT – Define Oil Density Temperature Coefficients	OPM Flow
OILJT – Define Oil Joule-Thomson Coefficient	OPM Flow
OILVISCT – Define Oil Viscosity versus Temperature Functions	OPM Flow
OLDTRAN – Activate Cartesian Regular Grid Transmissibilities	
OLDTRANR – Activate Radial Regular Grid Transmissibilities	
OPERATE – Define Mathematical Operations on Arrays	
OPERATER – Define Mathematical Operations on Arrays by Region	
OPERNUM – Define Regions for Mathematical Operations on Arrays	
OPTIONS – Activate Various Program Options	
OUTRAD – Define the Outer Radius of a Radial Grid	
OUTSOL – Define Data to be Written to the RESTART File (Retired)	
OVERBURD – Define Rock Overburden Pressure versus Depth Tables	

P

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter P	Status
PARALLEL – Define Parallel Run Configuration	
PARAOPTS – Define Parallel Run Options	
PARTTRAC – Activate and Define Partitioned Tracer Option	
PATHS – Define Filename Directory Path Aliases	
PBUB – Define the Initial Equilibration Bubble-Point Pressure for All Grid Blocks	
PBVD – Equilibration Bubble-Point versus Depth Tables	
PCFACT – Capillary Pressure Multiplication Factor as a Function of Porosity Change	OPM Flow
PCRIT – Critical Pressures	
PCG – End-Point Scaling of Grid Cell Maximum Gas Capillary Pressure (Drainage)	
PCG32D – Gas-Oil Capillary Pressure versus Oil and Water Saturation Tables	
PCW – End-Point Scaling of Grid Cell Water Capillary Pressure (Drainage)	
PCW32D – Water-Oil Capillary Pressure versus Oil and Gas Saturation Tables	
PDEW – Define the Initial Equilibration Dew-Point Pressure for All Grid Blocks	
PDVD – Define Equilibration Dew-Point versus Depth Tables	
PEBI – Activate and Defined PEBI Grid Options	
PECOEFS – Define Petro-Elastic Model Coefficients	
PEDIMS – Define Petro-Elastic Model Regions and Table Dimensions	
PEGTAB – Petro-Elastic Pressure Shear Modulus Table	
PEGTAB0	
PEGTAB1	
PEGTAB2	
PEGTAB3	
PEGTAB4	
PEGTAB5	
PEGTAB6	
PEGTAB7	

P

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter P	Status
PEKTAB – Petro-Elastic Pressure Bulk Modulus Table	
PEKTAB0	
PEKTAB1	
PEKTAB2	
PEKTAB3	
PEKTAB4	
PEKTAB5	
PEKTAB6	
PEKTAB7	
PENUM – Define the Petro-Elastic Region Numbers	
PERFORMA – Export Standard Simulator Performance Summary Variables to File	
PERMAVE – Define Average Transmissibility Coefficients	
PERMFACT – Permeability Multiplication Factor as a Function of Porosity Change	OPM Flow
PERMJFUN – Define Leverett J-Function Permeability for All Cells	
PERMR – Define the Permeability for Each Cell in the R Direction	
PERMTHT – Define the Permeability for Each Cell in the THETA Direction	
PERMX – Define the Permeability in the X Direction for All the Cells	
PERMY – Define the Permeability in the Y Direction for All the Cells	
PERMZ – Define the Permeability in the Z Direction for All the Cells	
PETGRID – Load a Generic Simulation Grid File	
PETOPTS – Define Petrel and Generic Simulation File Options	
PICOND – Define the Generalized Pseudo Pressure Parameters	
PIMTDIMS – Define Well Productivity Scaling Table Dimensions	
PIMULTAB – Define Well Productivity Index versus Water Cut Tables	
PINCH – Define Pinch-Out Layer Options	
PINCHNUM – Define Pinch-Out Regions for the PINCHREG Keyword	

P

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter P	Status
PINCHOUT – Define Pinch-Out Layers Option (Fixed)	
PINCHREG – Define Pinch-Out Region Options	
PINCHXY – Define Pinch-Out Areal Options	
PINTDIMS – Define Polymer Molecular Weight Model Table Dimensions	OPM Flow
PLMIXNUM – Define the Polymer Region Numbers	
PLMIXPAR – Define the Polymer Todd-Longstaff Mixing Parameters	
PLYADS – Define Polymer Rock Adsorption Tables	
PLYADSS – Define Polymer Rock Adsorption with Salt Dependence Tables	
PLYATEMP – Define Polymer Adsorption Table Temperature	
PLYCAMAX – Define Polymer-Rock Maximum Adsorption by Cell	
PLYDHFLF – Define Polymer Thermal Degradation Half-Life Tables	
PLYESAL – Define Polymer Effective Salinity Coefficient	
PLYKRRF – Define Polymer Rock Permeability Reduction by Cell	
PLYMAX – Define Polymer-Salt Viscosity Mixing Concentrations	
PLYMWINJ – Polymer Molecular Weight Model Throughput and Velocity Table	OPM Flow
PLYRMDEN – Define Polymer Model In Situ Rock Density	
PLYROCK – Define Polymer-Rock Properties	
PLYROCKM – Modify Polymer-Rock Properties	
PLYSHEAR – Activate and Define Polymer Shearing Parameters	
PLYSHLOG – Activate and Define the Polymer Shearing Logarithmic Parameters	
PLYTRRF – Define Polymer Rock Permeability Reduction versus Temperature	
PLYTRRFA – Define Polymer Rock Permeability Reduction versus Temperature Option	
PLYVISC – Define Polymer Viscosity Scaling Factors	
PLYVISCs – Define Polymer-Salt Viscosity Scaling Factors	
PLYVISCT – Define Polymer-Temperature Viscosity Scaling Factors	
PLYVMH – Polymer Molecular Weight Model Polymer Viscosity Constants	OPM Flow

P

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter P	Status
PLYVSCST – Define Polymer-Salt-Temperature Viscosity Scaling Factors	
PMAx – Maximum and Minimum Pressure for Total Compressibility Check	
PMISC – Define Miscibility versus Pressure Tables	
POLYMER – Activate the Polymer Phase in the Model	
POLYMW – Activate the Polymer Molecular Weight Transport Option	OPM Flow
PORO – Define the Porosity Values for All the Cells	
PORV – Define the Pore Volumes for All the Cells	
PPCWMAx – Define SWATINIT Calculated Capillary Pressure Constraints	
PRECSALT – Activate the OPM Flow Salt Precipitation Model	OPM Flow
PRESSURE – Define the Initial Equilibration Pressures for All Grid Blocks	
PRIORITY – Activate and Define Well Prioritization Coefficients	
PROPS – Define the Start of the PROPS Section of Keywords	
PRORDER – Define a Group Production Rules Sequence	
PRVD – Define the Initial Equilibration Pressures versus Depth	
PSTEADY – Activate Pseudo Steady State Flow Calculation Option	
PVCDO – Oil PVT Properties for Dead Oil (Constant Compressibility)	
PVCO – Oil PVT Properties for Live Oil	
PVDG – Gas PVT Properties for Dry Gas	
PVDO – Oil PVT Properties for Dead Oil	
PVDS – Solvent PVT Properties for the Solvent Model	
PVTG – Gas PVT Properties for Wet Gas with Vaporized Oil	
PVTGW – Gas PVT Properties for Dry Gas with Vaporized Water	OPM Flow
PVTGWO – Gas PVT Properties for Wet Gas with Vaporized Water and Oil	OPM Flow
PVTNUM – Define the PVT Regions	
PVTO – Oil PVT Properties for Live Oil	
PVTSOL – Oil PVT Properties for Live Oil versus CO2 Mass Fraction	OPM Flow

P

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter P	Status
PVTW – Define Water Fluid Properties for Various Regions	
PVTWSALT – Define Brine Water Fluid Properties for Various Regions	
PVZG – Gas PVT Properties for Dry Gas (Z-Factor)	
PYACTION – Define Python Based Action Conditions and Command Processing	OPM Flow
PYEND – End the Definition of a PYINPUT Section	OPM Flow
PYINPUT – Define the Start of a PYINPUT Section	OPM Flow

Q

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter Q	Status
QDRILL – Define Sequential Drilling Queue Wells	
QHRATING – Define River Mass Flow versus Depth Tables	
QMOBIL – Activate or Deactivate LGR End-Point Mobility Correction	

R

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter R	Status
RADFIN – Define a Radial Local Grid Refinement with One Column	
RADFIN4 – Define a Radial Local Grid Refinement with Four Columns	
RADIAL – Radial Grid Activation Option	
RAINFALL – Constant Flux Aquifer Rainfall Flux by Month	
RBEDCONT – Define River Grid Block Contact Area versus Depth	
RCMASTS – Reservoir Coupling Group Minimum Time Step for Flow Restriction	
REACHES – Define River Reaches Structure	
READDATA – Read Schedule Data Based on Current Time Step	
REFINE – Start the Definition of a Local Grid Refinement	
REGDIMS – Define the Maximum Number of Regions for a Region Array	
REGIONS – Define the Start of the REGIONS Section of Keywords	
RESIDNUM – Define Vertical Equilibrium Residual Flow Region Numbers	
RESTART – Restart Run From an Existing Restart File	
RESVNUM – Define Reservoir Coordinate Data Set	
RIVDEBUG – Define the Debug Data to be Printed to File (Rivers)	
RIVERSYS – Define River System (Branch Structure and Boundary Conditions)	
RIVRDIMS – Define the River Dimensions and Associated Data	
RIVRPROP – Modify River Reaches Properties	
RIVRXSEC – Define River Cross-Section versus Depth Parameters	
RIVSALT – Define River Upstream Flow Salt Concentrations	
RIVTRACE – Define River Upstream Flow Tracer Concentrations	
RKTRMDIR – Activate ROCKTAB Keyword Directional Transmissibility Multipliers	
ROCK – Define the Rock Compressibility for Various Regions	
ROCK2D – Pore Volume Compaction versus Pressure and Sw Tables	
ROCK2DTR – Transmissibility Compaction versus Pressure and Sw Tables	
ROCKCOMP – Activate Rock Compaction	

R

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter R	Status
ROCKFRAC – Define the Rock Volume to Bulk Volume Fraction for All the Cells	
ROCKNUM – Define Rock Compaction Table Region Numbers	
ROCKOPTS – Define Rock Compaction and Compressibility Options	
ROCKPAMA – Define Coal Palmer-Mansorri Rock Model Parameters	
ROCKTAB – Rock Compaction Tables	
ROCKTABH – Rock Compaction Hysteresis Tables	
ROCKTABW – Rock Compaction Tables (Water Induced)	
ROCKTHSG – Rock Compaction Hysteresis Tables (Dual Porosity)	
ROCKTSIG – Rock Compaction Tables (Dual Porosity)	
ROCKWNOD – Water Saturation Values for Compaction Pressure-Sw Tables	
RPTCPL – Activate Couple Simulation Reporting	
RPTGRID – Define GRID Section Reporting	
RPTGRIDL – Define GRID Section Reporting for LGRs	
RPTHMD – Define Well History Match Gradient Reporting Options	
RPTHMG – Define Well History Match Gradient Reporting (Groups)	
RPTHMW – Define Well History Match Gradient Reporting (Wells)	
RPTINIT – Define Output to the INIT File	
RPTISOL – Activate Isolated Reservoir Number Reporting	
RPTONLY – Activate the Report Time Steps Only Option for the SUMMARY File	
RPTONLYO – Deactivate the Report Time Steps Only Option for the SUMMARY File	
RPTPROPS – Define PROPS Section Reporting	
RPTREGS – Define REGIONS Section Reporting	
RPTRST – Define Data to be Written to the RESTART File	
RPTRUNSP – Activate RUNSPEC Reporting	
RPTSCHED – Define SCHEDULE Section Reporting	
RPTSMRY – Activate or Deactivate Summary List Report	

R

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter R	Status
RPTSOL – Define SOLUTION Section Reporting	
RS – Define the Initial Equilibration GOR (Rs) for All Grid Blocks	
RSCONST – Define Constant GOR (Rs) for All Dead Oil PVT Fluids	
RSCONSTT – Define Constant GOR (Rs) for Each Dead Oil PVT Fluid	
RSGI – Define Gas-Oil Ratio versus Pressure and Gi Tables	
RSSPEC – Activate Output of the RESTART Index File	
RSVD – Equilibration Dissolved Gas-Oil Ratio (Rs) versus Depth Tables	
RSW – Define the Initial Equilibrium Solution Gas in Water Ratio for All Grid Blocks	OPM Flow
RTEMP – Define the Initial Reservoir Temperature for the Model	
RTEMPA – Define the Initial Reservoir Temperature for the Model	
RTEMPVD – Define the Initial Reservoir Temperature versus Depth Tables	
RUNSPEC – Define the Start of the RUNSPEC Section of Keywords	
RUNSUM – Activate RSM File Output of the SUMMARY Data	
RV – Define the Initial Equilibration CGR (Rv) for All Grid Blocks	
RVCONST – Define Constant CGR (Rv) for All Dry Gas PVT Fluids	
RVCONSTT – Define Constant CGR (Rv) for Each Dry Gas PVT Fluid	
RVGI – Define Condensate-Gas Ratio versus Pressure and Gi Tables	
RVVD – Equilibration Vaporized Oil-Gas Ratio (Rv) versus Depth Tables	
RVW – Define the Initial Equilibration Vaporized Water in Gas Ratio for All Grid Blocks	OPM Flow
RVWVD – Equilibration Vaporized Water-Gas Ratio (Rvw) versus Depth Tables	OPM Flow
RWGSALT – Water Vaporization versus Pressure and Salt Concentration	OPM Flow

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
SALINITY – Define the Reservoir Salinity for All Cells	OPM Flow
SALT – Define the Initial Equilibration Salt Concentration for All Grid Blocks	
SALTNODE – Salt Concentration Based PVTNUM Array	
SALTMF – Define the Salt Liquid-Phase Mole Fraction for All Cells	OPM Flow
SALTP – Define the Initial Precipitated Salt Volume Fraction for All Grid Blocks	OPM Flow
SALTPVD – Initial Precipitated Salt Volume Fraction versus Depth Tables	OPM Flow
SALTREST – Define the Restart Salt Concentration for All Grid Blocks	
SALTSOL – Define the Salt Solubility Limit by Region	OPM Flow
SALTVD – Equilibration Salt Concentration versus Depth Tables	
SAMG – Activate Algebraic Multi-Grid Linear Solver	
SATNUM – Define the Saturation Table Region Numbers	
SATOPTS – Activate Relative Permeability Assignment Options	
SAVE – Activate Output of a SAVE File for Fast Restarts	
SBIOF – Define The Initial Equilibration Biofilm Volume Fraction For All Grid Blocks	OPM Flow
SCALC – Define The Initial Equilibration Calcite Volume Fraction For All Grid Blocks	OPM Flow
SCALECRS – Define End-Point Scaling Option	
SCALELIM – End-Point Scaling versus Depth Maximum Water Saturation	
SCDATAB – Well Connection PI Multipliers versus Scale Deposit	
SCDETAB – Well Connection Karst Aquifer Properties for Scale Deposit	
SCDPDIMS – Define Scale Deposition and Damage Table Dimensions	
SCDPTAB – Well Connection Scale Deposition Tables	
SCDPTRAC – Allocate Sea Water Tracer for Scale Deposition	
SCHEDULE – Define the Start of the SCHEDULE Section of Keywords	
SCVD – Define Equilibration Coal Solvent Concentration versus Depth Tables	
SDENSITY – Define the Miscible or Solvent Surface Gas Density	
SEPARATE – Activate the Separate RSM File Output Option	

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
SEPVALS – Define Separator Oil Formation Volume Factor and GOR	
SFOAM – Define the Initial Equilibration Foam Concentration for All Grid Blocks	
SGAS – Define the Initial Equilibration Gas Saturation for All Grid Blocks	
SGCR – End-Point Scaling Grid Cell Critical Gas Saturations	
SGCRX	
SGCRX-	
SGCRY	
SGCRY-	
SGCRZ	
SGCRZ-	
SGCWMIS – Miscible Critical Gas versus Water Saturation Functions	
SGF32D – Gas Saturation Tables versus Oil and Water Saturations	
SGFN – Gas Saturation Tables (Format Type 2)	
SGL – End-Point Scaling Grid Cell Connate Gas Saturations	
SGLX	
SGLX-	
SGLY	
SGLY-	
SGLZ	
SGLZ-	
SGLPC – End-Point Scaling Grid Cell Capillary Pressure Connate Gas Saturations	
SGOF – Gas-Oil Saturation Tables versus Gas (Format Type 1)	
SGOFLET – Gas-Oil LET Relative Permeability Functions	OPM Flow
SGU – End-Point Scaling Grid Cell Gas Saturation	
SGUX	
SGUX-	

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
SGUY	
SGUY-	
SGUZ	
SGUZ-	
SGWFLET – Gas-Water LET Relative Permeability Functions	OPM Flow
SGWFN – Gas-Water Saturation Tables (Format Type 2)	
SHRATE – Activate Log-based Polymer Shearing and Define the Shear Rate Constant	
SIGMA – Dual Porosity Matrix to Fracture Sigma (All Cells)	
SIGMAGD – Dual Porosity Matrix to Fracture Sigma for Gravity Drainage (All Cells)	
SIGMAGDV – Dual Porosity Matrix to Fracture Sigma Gravity Drainage (Individual Cells)	
SIGMAV – Dual Porosity Matrix to Fracture Sigma (Individual Cells)	
SIMULATE – Activate the Simulation Mode	
SKIP – Activate Skipping of All Keywords and Input Data	
SKIP100 – Activate Skipping of Keywords by Black-Oil Simulator	
SKIP300 – Activate Skipping of Keywords by Compositional Simulator	
SKIPREST – Activate Skipping of Restart Schedule Data	
SKPRPOLY – Polymer Molecular Weight Model Polymer Injection Skin Table	OPM Flow
SKPRWAT – Polymer Molecular Weight Model Water Injection Skin Table	OPM Flow
SKRO – End-Point Scaling of Grid Cell Kro(Swl) (Surfactant)	
SKRORG – End-Point Scaling of Grid Cell Kro(Sgcr) (Surfactant)	
SKRORW – End-Point Scaling of Grid Cell Kro(Swcr) (Surfactant)	
SKRW – End-Point Scaling of Grid Cell Krw(Sw =1.0) (Surfactant)	
SKRWR – End-Point Scaling of Grid Cell KRWR(Sowcr) (Surfactant)	
SLAVES – Define Slave Reservoir Simulation Parameters	
SLGOF – Gas-Oil Saturation Tables versus Gas (Format Type 1)	
SMICR – Define The Initial Equilibration Microbial Concentration For All Grid Blocks	OPM Flow

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
SMRYDIMS – Define Maximum Number of Summary Vectors to be Written	
SMULTX – Multiply Cell Transmissibility in the +X Direction (Auto-Refinement)	
SMULTY – Multiply Cell Transmissibility in the +Y Direction (Auto-Refinement)	
SMULTZ – Multiply Cell Transmissibility in the +Z Direction (Auto-Refinement)	
SOCRS – End-Point Scaling Grid Cell Miscible Critical Oil Saturation with Respect to Water	
SOF2 – Oil Saturation Tables with Respect to Gas or Water (Format Type 2)	
SOF3 – Oil Saturation Tables with Respect to Gas and Water (Format Type 2)	
SOF32D – Oil Saturation Tables with Respect to Water and Gas (Three Phase)	
SOGCR – End-Point Scaling Grid Cell Critical Oil Saturation with Respect to Gas	
SOGCRX	
SOGCRX-	
SOGCRY	
SOGCRY-	
SOGCRZ	
SOGCRZ-	
SOIL – Define the Initial Equilibration Oil Saturation for All Grid Blocks	
SOLUTION – Define the Start of the SOLUTION Section of Keywords	
SOLVCONC – Define the Initial Coal Solvent Concentration for All Grid Blocks	
SOLVDIMS – Define PEBI Grid Nested Factorization Solver Dimensions	
SOLVDIRS – Define Linear Solver Principal Directions	
SOLVENT – Activate the SOLVENT Phase in the Model	
SOLVFRAC – Define the Initial Gas Solvent Fraction for All Grid Blocks	
SOLVNUM – Define PEBI Grid Correspondence to Solver Order	
SOMGAS – STONE1 Model Minimum Oil Saturation versus Gas Saturation	
SOMWAT – STONE1 Model Minimum Oil Saturation versus Water Saturation	
SORWMIS – Miscible Residual Oil versus Water Saturation Functions	

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
SOURCE – Define Source Term	OPM Flow
SOWCR – End-Point Scaling Grid Cell Critical Oil Saturation with Respect to Water	
SOWCRX	
SOWCRX-	
SOWCRY	
SOWCRY-	
SOWCRZ	
SOWCRZ-	
SOXYG – Define The Initial Equilibration Oxygen Concentration For All Grid Blocks	OPM Flow
SPECGRID – Define the Dimensions of a Corner-Point Grid	
SPECHEAT – Define the Specific Heat of Oil, Water and Gas	
SPECROCK – Define the Specific Heat of the Reservoir Rock	
SPIDER – Spider Grid Activation Option	OPM Flow
SPOLY – Define the Initial Equilibration Polymer Concentration for All Grid Blocks	
SPOLYMW – Define The Initial Equilibration Polymer Molecular Weights For All Grid Blocks	OPM Flow
SSFN – Solvent and Gas Relative Permeability Tables	
SSGCR – End-Point Scaling Grid Cell Surfactant Critical Gas Saturations	
SSGL – End-Point Scaling Grid Cell Surfactant Connate Gas Saturations	
SSOGCR – End-Point Scaling Grid Cell Surfactant Critical Oil Saturation with Respect to Gas	
SSOL – Define the Initial Equilibration Solvent Saturation for All Grid Blocks	
SSOWCR – End-Point Scaling Grid Cell Surfactant Critical Oil Saturation with Respect to Water	
SSWCR – End-Point Scaling Grid Cell Critical Water Saturation	
SSWL – End-Point Scaling Grid Cell Surfactant Connate Water Saturation	
SSWU – End-Point Scaling Grid Cell Surfactant Maximum Water Saturation	
START – Simulation Start Date	
STOG – Define Capillary Pressure Oil-Gas Surface Tension versus Pressure	

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
STONE – Activate Stone's Second Three Phase Oil Relative Permeability Model (Alias for STONE2)	
STONE1 – Activate Stone's First Three Phase Oil Relative Permeability Model	
STONE1EX – Define Stone's First Three Phase Oil Relative Permeability Parameter	
STONE2 – Activate Stone's Second Three Phase Oil Relative Permeability Model	
STOW – Define Capillary Pressure Oil-Water Surface Tension versus Pressure	
STWG – Define Capillary Pressure Water-Gas Surface Tension versus Pressure	
SUMMARY – Define the Start of the SUMMARY Section of Keywords	
SUMTHIN – Define SUMMARY Data Reporting Time Steps	
SUREA – Define The Initial Equilibration Urea Concentration For All Grid Blocks	OPM Flow
SURF – Define the Initial Equilibration Surfactant Concentration for All Grid Blocks	
SURFACT – Activate the Surfactant Phase in the Model	
SURFACTW – Activate the Surfactant Phase with Wettability Changes in the Model	
SURFADDW – Defined Surfactant Adsorbed Concentration versus Wettability Fraction	
SURFADS – Define Surfactant Rock Adsorption Tables	
SURFCAPD – Capillary Number versus Miscibility Tables	
SURFESAL – Define Surfactant Effective Salinity Coefficient	
SURFNUM – Define the Surfactant Miscible Saturation Table Region Numbers	
SURFROCK – Define Surfactant-Rock Properties	
SURFST – Surfactant Water-Oil Surface Tension versus Surfactant Concentration	
SURFSTES – Surfactant Water-Oil Surface Tension versus Surfactant and Salt Concentrations	
SURFVISC – Surfactant Solution Viscosity versus Concentration	
SURFNUM – Define the Saturation Table Region Numbers (High Salinity and Water Wet)	
SWAT – Define the Initial Equilibration Water Saturation for All Grid Blocks	
SWATINIT – Define the Initial Water Saturation Array for Capillary Pressure Scaling	
SWCR – End-Point Scaling Grid Cell Critical Water Saturation	
SWCRX	

S

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter S	Status
SWCRX-	
SWCRY	
SWCRY-	
SWCRZ	
SWCRZ-	
<i>SWF32D – Water Saturation Tables with Respect to Oil and Gas (Three Phase)</i>	
<i>SWFN – Water Saturation Tables (Format Type 2)</i>	
<i>SWINGFAC – Define Field Gas Contract Parameters</i>	
<i>SWL – End-Point Scaling Grid Cell Connate Water Saturation</i>	
SWLX	
SWLX-	
SWLY	
SWLY-	
SWLZ	
SWLZ-	
<i>SWLPC – End-Point Scaling Grid Cell Capillary Pressure Connate Water Saturations</i>	
<i>SWOF – Water-Oil Saturation Tables (Format Type 1)</i>	
<i>SWU – End-Point Scaling Grid Cell Maximum Water Saturation</i>	
SWUX	
SWUX-	
SWUY	
SWUY-	
SWUZ	
SWUZ-	

T

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter T	Status
TABDIMS – Define the Number of Tables and the Table Dimensions	
TBLK – Define Tracer Initial Grid Block Concentrations	
TCRIT – Critical Temperatures	
TEMP – Activate the Temperature Modeling Option	
TEMPI – Define the Initial Temperature Values for All Cells	
TEMPNODE – Temperature Table for Polymer Solution Viscosity	
TEMPTVD – Activate Temperature Flux Limited Transport Option	
TEMPVD – Define the Initial Reservoir Temperature versus Depth Tables	
THCO2MIX – Specify Thermal Mixing Models	
THCGAS – Define Gas Phase Thermal Conductivity for All Cells	
THCOIL – Define Oil Phase Thermal Conductivity for All Cells	
THCONR – Define Rock and Fluid Thermal Conductivity for All Cells	
THCONSF – Define Gas Saturation Dependent Thermal Conductivity Scaling Factor for All Cells	
THCROCK – Define Reservoir Rock Thermal Conductivity for All Cells	
THCWATER – Define Water Phase Thermal Conductivity for All Cells	
THERMAL – Activate the Thermal Modeling Option	
THPRES – Define Equilibration Region Threshold Pressures	
THPRESFT – Define Fault Threshold Pressures	
TIGHTEN – Tighten and Relax Numerical Controls	
TIGHTENP – Tighten and Relax Numerical Controls Individually	
TIME – Advance Simulation by Cumulative Reporting Time	
TITLE – Define the Title for the Input Deck	
TLMIXPAR – Define the Miscible Todd-Longstaff Mixing Parameters	
TNUM – Define Passive Tracer Concentration Regions	
TOLCRIT – Define The Critical Saturation Tolerance	
TOPS – Define the Depth at the Center of the Top Face for Each Cell	

T

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter T	Status
TPAMEPS – Volumetric Strain versus Coal Gas Concentration Tables	
TPAMEPSS – Volumetric Strain versus Coal Solvent Concentration Tables	
TRACER – Define Passive Tracer Variables	
TRACERKM – Multi-Partitioned Tracer Option K(P) Tables	
TRACERKP – Standard Partitioned Tracer Option K(P) Tables	
TRACERS – Activate Tracer Options and Set Tracer Array Dimensions	
TRACITVD – Activate and Define Tracer Implicit Flux Limited Transport Option	
TRACTVD – Activate Tracer Explicit Flux Limited Transport Option	
TRADS – Environmental Tracer Adsorption Tables	
TRANGL – Define Non-Neighbor Connections Between Global and LGR Cells Manually	
TRANR – Define the Transmissibility in the +R Direction for All the Cells	
TRANHT – Define the Transmissibility in the +Theta Direction for All the Cells	
TRANX – Define the Transmissibility in the X Direction for All the Cells	
TRANX – Define the Transmissibility in the Y Direction for All the Cells	
TRANZ – Define the Transmissibility in the Z Direction for All the Cells	
TRDCY – Environmental Tracer Decay Tables	
TRDIF – Tracer Diffusion Tables	
TRDIS – Tracer Dispersion Table Number Allocation	
TRKPF – Define Partitioned Tracer Regions	
TRNHD – Activate Dispersion Non-Homogeneous Diffusion Option	
TRPLPORO – Activate the Triple Porosity Model Option	
TRROCK – Environmental Tracer-Rock Property Data	
TSTEP – Advance Simulation by Reporting Time	
TUNING – Numerical Tuning Control	
TUNINGDP – Numerical Tuning Control for High Throughput Cases	
TUNINGH – Numerical Tuning Control for History Match Gradient Calculations	

T

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Alphabetic Listing Of Keywords Starting With The Letter T	Status
<i>TUNINGL – Numerical Tuning Control for All LGRs</i>	
<i>TUNINGS – Numerical Tuning Control for Individual LGRs</i>	
<i>TVDP – Define the Initial Equilibration Tracer Saturation versus Depth Functions</i>	
<i>TZONE – End-Point Scaling Transition Zone Options</i>	

U

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter U	Status
<i>UDADIMS – Define the Dimensions of the User Defined Arguments</i>	
<i>UDQ – Declare User Define Quantities (“UDQ”)</i>	
<i>UDQDIMS – Define the Dimensions of the User Defined UDQ Feature</i>	
<i>UDQPARAM – Define Parameters for the User Defined Quantity Feature</i>	
<i>UDT – Declare User Define Tables (“UDT”)</i>	
<i>UDTDIMS – Define the Dimensions of the User Defined Tables</i>	
<i>UNCODHMD – Activate History Match Gradient Unencoded Output</i>	
<i>UNIFIN – Activate The Unified Input File Option</i>	
<i>UNIFOUT – Activate The Unified Output File Option</i>	
<i>UNIFOUTS – Activate The Unified Output Summary File Option</i>	
<i>UNIFSAVE – Activate The Unified Output Save File Option</i>	
<i>USECUPL – Load a Reservoir Coupling File</i>	
<i>USEFLUX – Activate Flux Boundary Model and Define Flux File</i>	
<i>USENOFLO – Activate Flux Boundary Model Without a Flux File</i>	

V

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter V	Status
<i>VAPOIL – Activate the Vaporized Oil in Wet Gas Phase in the Model</i>	
<i>VAPPARS – Oil Vaporization Parameters</i>	
<i>VAPWAT – Activate Vaporized Water in the Dry and Wet Gas Phases</i>	OPM Flow
<i>VCRIT – Critical Volumes</i>	
<i>VDFLOW – Velocity Dependent Flow Coefficient for Grid Block Flow (Grid)</i>	
<i>VDFLOWR – Velocity Dependent Flow Coefficient for Grid Block Flow (Region)</i>	
<i>VE – Activate Vertical Equilibrium Model (Global)</i>	
<i>VEDEBUG – Vertical Equilibrium Debug Data Output</i>	
<i>VEFIN – Activate Vertical Equilibrium Model (LGR)</i>	
<i>VEFRAC – Vertical Equilibrium Relative Permeability Fraction (Grid)</i>	
<i>VEFRACP – Vertical Equilibrium Capillary Pressure Fraction (Grid)</i>	
<i>VEFRACPV – Vertical Equilibrium Capillary Pressure Fraction (Cell)</i>	
<i>VEFRACV – Vertical Equilibrium Relative Permeability Fraction (Cell)</i>	
<i>VFPCHK – Define Production Vertical Flow Performance BHP Check</i>	
<i>VFPIDIMS – Injection Vertical Flow Performance Table Dimensions</i>	
<i>VFPINJ – Define Injection Vertical Flow Performance Tables</i>	
<i>VFPPDIMS – Production Vertical Flow Performance Table Dimensions</i>	
<i>VFPPROD – Define Production Vertical Flow Performance Tables</i>	
<i>VFPTABL – Define Production Vertical Flow Performance ALQ Interpolation</i>	
<i>VISAGE – Activate External Reservoir Geo-Mechanics VISAGE Option</i>	
<i>VISCAQA – Specify Ezrokhi Coefficients for Aqueous Viscosity</i>	
<i>VISCD – Activate Dual Porosity Viscous Displacement Option</i>	
<i>VISCREf – Define Viscosity-Temperature Reference Conditions</i>	
<i>VISDATES – Define External Reservoir Geo-Mechanics VISAGE Stress Dates</i>	
<i>VISOPTS – Define External Reservoir Geo-Mechanics VISAGE Options</i>	

W

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter W	Status
WAGHYSTR – Define Water-Alternating-Gas Hysteresis Parameters	W
WAITBAL – Wait On Network Balance Before Allowing Further Actions	
WALKALIN – Define Water Injection Alkaline Concentration	
WALQCALC – Define Well VFP Surface ALQ Phase Density	
WAPI – Define Oil Well Injection API Gravity	
WARN – Activate Warning Messages	
WATDENT – Define Water Density Temperature Coefficients	
WATER – Activate the Water Phase in the Model	
WATJT – Define Water Joule-Thomson Coefficient	OPM Flow
WATVISCT – Define Water Viscosity versus Temperature Functions	
WBHGLR – Define Well Bottom-Hole GLR Constraint	
WBOREVOL – Define Effective Wellbore Storage Volume	
WCALCVAL – Define Gas Well Calorific Value	
WCONHIST – Define Well Historical Production Rates and Pressures	
WCONINJ – Well Injection Targets and Constraints	
WCONINJE – Well Injection Targets and Constraints	
WCONINJH – Well Historical Observed Injection Rates and Pressures	
WCONINJP – Define Well Injection Targets and Constraints for Pattern Flood Wells	
WCONPROD – Define Well Production Targets and Constraints	
WCUTBACK – Define Well Cutback Limits and Parameters	
WCUTBACT – Define Well Tracer Cutback Limits and Parameters	
WCYCLE – Define Automatic Well Opening and Closing Cycling Parameters	
WDFAC – Define Gas Flow Dependent Skin Factor	
WDFACCOR – Gas Flow Dependent Skin Factor (Correlation)	
WDRILPRI – Add Wells to the Drilling Priority Drilling Queue	
WDRILRES – Activate Prevention of Multi-Completions in the Same Cell for Queued Wells	

W

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter W	Status
WDRILTIM – Define Drilling Parameters for Automatic Drilling of New Wells	
WECON – Well Economic Criteria for Production Wells	
WECONINJ – Well Economic Criteria for Injection Wells	
WECONT – Well Economic Tracer Criteria for Production Wells	
WEFAC – Define Well Efficiency	
WELCNTL – Modify Well Control and Targets	
WELDEBUG – Define the Well Debug Data to be Printed to File	
WELDRAW – Define Maximum Draw Down for Production Wells	
WELEVNT – Define Well WPWEM Summary Mnemonic Output Value	
WELLDIMS – Define the Wells and Group Dimensions	
WELLSTRE – Define Injection Stream Composition	OPM Flow
WELMOVEL – Move Global Well Into an LGR	
WEOPEN – Define Well and Well Connection Flowing Status	
WEOPENL – Define Well and Well Connections Flowing Status (LGR)	
WELPI – Define Well Productivity and Injectivity Indices	
WELPRI – Assign Well Priority	
WELSEGS – Define Multi-Segment Wells and Their Segment Structure	
WELSOMIN – Define Well Connection Minimum Oil Saturation for Opening	
WELSPECL – Define Well Specifications for Local Grid Refinements	
WELSPECS – Define Well Specifications	
WELTARG – Modify Well Target and Constraint Values	
WELTRAJ – Define Well Trajectory Data	OPM Flow
WFOAM – Define Well Foam Injection Concentrations	
WFRICSEG – Convert Friction Well to Multi-Segment Well	
WFRICSGE – Convert Friction Well to Multi-Segment Well (LGR)	
WFRICTN – Define Well as a Friction Well	

W

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter W	Status
WFRICTNL – Define Well as a Friction Well (LGR)	
WGASPROD – Define Sale Gas Well Production Targets	
WGORPEN – Define Well GOR Penalty Parameters	
WGRUPCON – Define Well Guide Rates for Group Control	
WH2NUM – Define WAG Hysteresis Saturation Table Region Numbers (Two Phase)	
WH3NUM – Define WAG Hysteresis Saturation Table Region Numbers (Three Phase)	
WHEDREFD – Define Well Hydraulic Head Reference Depth	
WHISTCTL – Define Well Historical Target Phase	
WHTEMP – Define Well Tubing Head Temperature Parameters	
WINJCLN – Clean a Fraction of a Deposited Filter Cake	
WINJDAM – Define Well Filter Cake Properties	
WINJFCNC – Define Injection Well Filtrate Concentration	
WINJGAS – Define Gas Injection Properties for a Well	OPM Flow
WINJMULT – Define Well Pressure Dependent Injectivity Multipliers	
WINJTEMP – Define Injection Fluid Thermal Properties	
WLIFT – Define Well Re-Tubing, THP and Lift Switching Workover Operations	
WLIFTOPT – Define Well Gas Lift Optimization Parameters	
WLIMTOL – Define Well Constraint Tolerance	
WLIST – Define Well Lists (Static)	
WLISTARG – Modify Well List Target and Constraint Values (Static)	
WLISTNAM – Define Well Lists (WLISTARG)	
WMICP – Define Water Injection Well's Microbial, Oxygen, And Urea Concentrations	OPM Flow
WNETCTRL – Define Well Control for Network Control Option	
WNETDP – Define Well THP to Network Pressure Drop	
WORKLIM – Define Well Workover Time	
WORKTHP – Define Well Workover Options for THP Killed Wells	

W

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter W	Status
WPAVE – Well Block Average Pressure Calculation Parameters for All Wells	
WPAVEDEP – Define Well Reference Depth for Pressure Calculations	
WPIMULT – Define Well Connection Multipliers	
WPIMULTL – Define Well Connection Multipliers (LGR)	
WPITAB – Assign Well Productivity Index versus Water Cut Tables	
WPLUG – Define Well Plug Back Length	
WPMITAB – Assign Well Polymer Molecular Model Injection Tables	OPM Flow
WPOLYMER – Define Water Injection Well Polymer and Salt Concentrations	
WPOLYRED – Define Well Polymer-Water Viscosity Reduction Factor	
WPOTCALC – Well Potential Calculation Options	
WREGROUP – Automatic Re-Assignment of Wells to Groups	
WRFT – Activate Well RFT Reporting to the RFT File	
WRFTPLT – Activate Well RFT and PLT Reporting to the RFT File	
WSALT – Define Water Injection Well Salt Concentrations	
WSCCLEAN – Well Deposited Scale Adjustment	
WSCCLENL – Well Deposited Scale Adjustment (LGR)	
WSCTAB – Assign Well Scale Deposition and Scale Damage Tables	
WSEGAICD – Define Multi-Segment Well Autonomous ICD Connections	
WSEGDFIN – Define Multi-Segment Well Drift Flux Slip Model Input Data	
WSEGDFMD – Define Multi-Segment Well Drift Flux Slip Model	
WSEGDFPA – Define Multi-Segment Well Drift Flux Slip Model Parameters	
WSEGDIMS – Define Multi-Segment Well Dimensions	
WSEGEXSS – Define Multi-Segment Well Import-Export Segment Volumes	
WSEGFLIM – Define Multi-Segment Well Artificial Choke Connections	
WSEGFMOD – Define Multi-Segment Well Model	
WSEGINIT – Define Multi-Segment Well Initial Conditions	

W

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter W	Status
WSEGITER – Define Multi-Segment Wells Iteration Parameters	
WSEGLABY – Define Multi-Segment Well Labyrinth ICD Connections	
WSEGLINK – Define Multi-Segment Well Looped Flow Paths	
WSEGMULT – Define Multi-Segment Well Frictional Pressure Loss Multipliers	
WSEGPROP – Modify Multi-Segment Wells and Their Segment Structure	
WSEGPULL – Define a Multi-Segment Well Down-Hole Separator Pump	
WSEGSEP – Define a Multi-Segment Well Down-Hole Separator	
WSEGSICD – Define Multi-Segment Well Spiral ICD Connections	
WSEGSOLV – Define Multi-Segment Well Iterative Linear Solver Parameters	
WSEGTABL – Assign Multi-Segment Well VLP Tables to Segments	
WSEGVALLV – Define Multi-Segment Well Sub-Critical Valve	
WSF – Water Saturation Function Tables (Gas-Water Systems)	OPM Flow
WSKPTAB – Assign Well Polymer Molecular Model Skin Tables	OPM Flow
WSOLVENT – Define Gas Injection Well Solvent Fraction	
WSURFACT – Define Water Injection Well Surfactant Concentration	
WTADD – Add a Constant to a Well Target or Constraint	
WTEMP – Define An Injection Well's Fluid Temperature	
WTEMPQ – Output Well Names and Well Lists to the Print File	
WTEST – Well Testing Criteria for Re-Opening Closed Wells	
WTHPMAX – Define a Well's Maximum Flowing THP for Shut-In	
WTMULT – Multiply a Well Target or Constraint by a Constant	
WTRACER – Define An Injection Well's Tracer Concentration	
WVFPDP – Modify Well BHP Obtained from VFP Tables	
WVFPEXP – Define Well VFP Interpolation Options	
WWPAVE – Well Block Average Pressure Calculation Parameters for Individual Wells	

X

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Alphabetic Listing Of Keywords Starting With The Letter X	Status
There are No Keywords Beginning with the Letter X	

Y

RUNSPEC	GRID	EDIT	PROPS	REGIONS	SOLUTION	SUMMARY	SCHEDULE
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Alphabetic Listing Of Keywords Starting With The Letter Y	Status
There are No Keywords Beginning with the Letter Y	

Z

<i>RUNSPEC</i>	<i>GRID</i>	<i>EDIT</i>	<i>PROPS</i>	<i>REGIONS</i>	<i>SOLUTION</i>	<i>SUMMARY</i>	<i>SCHEDULE</i>
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Alphabetic Listing Of Keywords Starting With The Letter Z	Status
<i>ZCORN – Define the Depth of Each Corner-Point of a Grid Block</i>	
<i>ZIPP2OFF – Deactivate Automatic Time Step Control</i>	
<i>ZIPPY2 – Activate Automatic Time Step Control</i>	
<i>ZMFVD – Define Compositional Components versus Depth</i>	OPM Flow

APPENDIX B: OPM FLOW RELEASE HISTORY

APPENDIX TABLE OF CONTENTS**APPENDIX B: OPM FLOW RELEASE HISTORY.....2357**

B.1 Release 2025-10.....	2360
B.1.1 New and Deprecated Command Line Options.....	2361
B.1.2 New Features.....	2363
B.1.3 Improvements.....	2365
B.1.4 Bug Fixes.....	2368
B.1.5 Known Issues.....	2370
B.2 Release 2025-04.....	2372
B.2.1 New and Deprecated Command Line Options.....	2373
B.2.2 New Features.....	2375
B.2.3 Improvements.....	2377
B.2.4 Bug Fixes.....	2382
B.2.5 Known Issues.....	2384
B.3 Release 2024-10.....	2387
B.3.1 New and Deprecated Command Line Options.....	2388
B.3.2 New Features.....	2391
B.3.3 Improvements.....	2392
B.3.4 Bug Fixes.....	2395
B.3.5 Known Issues.....	2399
B.4 Release 2024-04.....	2401
B.4.1 New and Deprecated Command Line Options.....	2402
B.4.2 New Features.....	2404
B.4.3 Improvements.....	2406
B.4.4 Bug Fixes.....	2412
B.4.5 Known Issues.....	2415
B.5 Release 2023-10.....	2418
B.5.1 New and Deprecated Command Line Options.....	2418
B.5.2 New Features.....	2421
B.5.3 Improvements.....	2422
B.5.4 Bug Fixes.....	2426
B.5.5 Known Issues.....	2428

B.1 RELEASE 2025-10

The 2025-10 release consists of some new features and various improvements and bug fixes. These include the following highlights:

- Added support for the biofilm model in two-phase gas-water systems.
- The peak memory allocation has been reduced when recreating the linear solver.
- Improved GPU support for blackoil fluid systems.
- Memory consumption has been reduced when the hysteresis model is not being used.
- Support has been added for link time optimization.
- Optimized update calls for CPR-type preconditioners.
- Memory usage is now reduced when energy, diffusion, or dispersion storage is not used.
- The intensive quantities cache now stores values for the current time step only and not the previous time step. This reduces the peak memory usage of the entire simulator by 10 %, without any performance degradation.
- Added a OPM Flow specific option to the Killough hysteresis model ([EHYSTR](#) keyword item 14) to correct the construction of the capillary pressure scanning curve.
- The new saturation function consistency checks have been enabled by default.
- Added support for running OPM Flow single phase (water) simulations from Python code using the Python bindings.
- Added support for running OPM Flow gas-water simulations, for example [H2STORE](#) or [CO2STORE](#) cases, from Python code using the Python bindings.
- Added support for additional [TUNING](#) keyword items: NEWTMX, NEWTMN, TRGMBE, TRGCNV, and XXXCNV.
- Improved support for satellite injection and production groups.
- A user defined argument can now be used for group reservoir volume production rate (RESV) controls when using the [GCONPROD](#) keyword.
- Restart file support has been added for the number of Newton iterations required by each cell to satisfy the solution change convergence criteria at the last time step (CONV_NEW).
- Support has been added for running simulations beyond 31-Dec-9999 (up to 292 million years).
- Added support for [WECON](#) keyword item 9, the name of the follow-on well to be open if the current well is shut.
- Added support for [WSEGAICD](#) keyword item 22, the density function exponent.
- Added UDA support for items 3 and 4 of the [WTRACER](#) keyword.
- Support has been added for the use of UDAs for ORAT, WRAT, and GRAT in the [GSATPROD](#) keyword.
- Full support has been added for running the Constrained Pressure Residual with Algebraic Multigrid (CPR-AMG) preconditioner entirely on the GPU for serial or single-GPU runs, using the Gpulstl framework.
- Support has been added for the Hypre preconditioner with GPU input types.
- The new GPU aware ISTL solver now has MPI support.

- An experimental Hybrid Newton method has been added that enhances nonlinear preconditioning using machine learning.
- The NLDD (Nonlinear Domain Decomposition) solver has reduced the time spent on local solves by skipping subdomains that haven't changed significantly between iterations.
- Partitioner zoltanwell (default) no longer ignores the command line argument --allow-distributed-wells. This makes its behaviour more like that of the other partitioners.
- Added support for water mass in place and water mass injection/production rate/total summary vectors.
- Added support for gas sales summary vectors.
- The well's preferred phase is now used to calculate the well productivity index.
- Improved robustness of well solves and well group behaviour.

B.1.1 NEW AND DEPRECATED COMMAND LINE OPTIONS

The major command line changes made for this release are summarized in Table B.1

OPM Flow 2025-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
1	--check-satfunc-consistency	Saturation function consistency checking is now activated by default. A boolean value. Whether or not to check saturation function consistency requirements.	true
2	--cpr-weights-thread-parallel	A boolean value. Enable OpenMP thread parallelization of CPR weight calculation. This can improve performance for large models but is disabled by default.	false
3	--debug-verbosity-level	An integer value. Set debug verbosity level globally. Default is 1, increasing values give additional output, and 0 disables most messages to the .DBG file.	1
4	--edge-conformal	A boolean value. Edge conformal cornerpoint processing.	false
5	--edge-weights-method	The edge-weighting method is now chosen by specifying a string rather than an integer. A character string. Choose edge-weighting strategy: 'uniform', 'transmissibility', or 'logtrans' (logarithm of transmissibility).	"transmissibility"
6	--gpu-aware-mpi	A boolean value. MPI communication use GPU aware MPI in the sense that it will use GPU direct communication. Setting this to true will require that the MPI implementation supports GPU direct communication. If you are unsure, set this to false.	false
7	--hy-ne-config-file	A character string. Use config files for Hybrid Newton.	"hybridNewtonConfig.json"

OPM Flow 2025-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
8	--linear-solver-accelerator	A character string. Choose the backend for the linear solver, either cpu or gpu.	"cpu"
9	--max-well-status-switch-for-wells	An integer value. Maximum number of status switching (stop<->open) for a well during a time-step.	99
10	--max-well-status-switch-in-inner-iter-wells	An integer value. Maximum number of status switching (stop<->open) for a well during inner iterations.	99
11	--milu-variant	The names of the possible variants have been converted to lower case. A character string. Specify which variant of the modified-ILU preconditioner ought to be used. Possible variants are: ilu (default, plain ILU), milu_1 (lump diagonal with dropped row entries), milu_2 (lump diagonal with the sum of the absolute values of the dropped row entries), milu_3 (if diagonal is positive add sum of dropped row entries, otherwise subtract them), milu_4 (if diagonal is positive add sum of dropped row entries, otherwise do nothing).	"ilu"
12	--nldd-relative-mobility-change-tol	A positive real value. Threshold for single cell relative mobility change in the NLDD solver.	0.1
13	--partition-method	The names of the partitioning methods have been converted to lower case. A character string. Choose partitioning method: 'simple', 'zoltan', 'metis', or 'zoltanwell' (Zoltan with all cells perforated by a well represented by a single vertex).	"zoltanwell"
14	--time-step-control-max-reduction-time-step	A positive real value (only applicable for the general 3rd order controller, using 'control-error-filtering' as time-step-control-tolerance-test-version). If the (proposed) relative change in time step size is larger than this parameter, the time step will be rejected.	0.1
15	--time-step-control-parameters	A character string (only applicable for the general 3rd order controller). Parameters for the general 3rd order controller. Should be given as 'beta_1;beta_2;beta_3;alpha_2;alpha_3'.	"0.125;0.25;0.125;0.75;0.25"
16	--time-step-control-reject-completed-step	A boolean value (only applicable for the general 3rd order controller). Include rejection of completed time steps if the relative change is larger than the time step control tolerance.	false

OPM Flow 2025-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
17	--time-step-control-tolerance-test-version	A character string (only applicable for the general 3rd order controller). Ways to decide if the time step should be rejected. Options: 'standard' and 'control-error-filtering'. The standard version compares relative change to tolerance directly to decide if the time step should be rejected, while the control-error-filtering version compares the relative change in time step size to the max-reduction-time-step parameter.	"standard"
18	--use-hy-ne	A boolean value. Wheter or not to use Hybrid Newton.	false
19	--verify-gpu-aware-mpi	A boolean value. Verify that the MPI implementation supports GPU aware MPI. If this is set to true *and* --gpu-aware-mpi=true, the simulation will fail if the MPI implementation does not support GPU aware MPI. Note that the verification is not exhaustive, and some configurations might not be verified, but will work in practice.	false
20	--well-group-constraints-max-iterations	An integer value. Maximum number of iterations in the well/group switching algorithm.	1
Notes: 1) Cells colored in green in the "No." column indicate a new command line option or a change to an existing option for this release. Cells colored green in the "Default" column indicate that the default value for an existing option has been changed.			

Table B.1: OPM Flow 2025-10 New and Deprecated Command Line Options

B.1.2 NEW FEATURES

In addition to the changes to the command line options the following new features have been added to the simulator:

- 1) Added support for the biofilm model in two-phase gas-water systems, which can be activated using the [BIOFILM](#) keyword in the [RUNSPEC](#) section in [CO2STORE](#) or [H2STORE](#) runs ([#6462](#)). The model includes biofilm as a solid phase and as microbes suspended in the water. This can be used in applications such as underground storage where dissolved hydrogen consumed by microbes leads to injectivity and hydrogen loss.
- 2) Added an optimized version of the preconditioner update calls, at the cost of apply times, to tailor the implementation more towards CPR-type preconditioners with smaller apply/update ratios ([#6310](#)).
- 3) Added support for edge conformal grid processing, activated using the --edge-conformal command line option ([#5806](#), [#814](#)). Edge conformal grids are very useful in the context of geo-mechanical fracturing processes.
- 4) The new saturation function consistency checks have been enabled by default ([#6252](#)).

- 5) Added support for running OPM Flow single phase (water) simulations from Python code using the Python bindings ([#6482](#)). This has many potential applications including coupling to external geochemistry and geomechanics solvers.
- 6) Added additional restart file items to maintain compatibility with user defined arguments ([#4439](#)).
- 7) Added support for specifying the maximum and minimum number of Newton iterations for a time step (NEWTMX and NEWTMN) on the [TUNING](#) keyword in the [SCHEDULE](#) section ([#6378](#)).
- 8) Added parser support for the [GSATINJE](#) keyword to specify satellite injection at the group level ([#4700](#)).
- 9) A check has been added to ensure that all wells and/or groups needed in a [UDQ](#) DEFINE statement are present at the point of definition ([#4619](#)). An error message will be generated for any missing well objects, for example:

```
Error: Unrecoverable errors while loading input:

Problem with UDQ
In CASE.DATA line 251
DEFINE cannot be evaluated:
Missing object in defining expression for FUBHPGI
-> No existing well matches the name XD-1H
```

- 10) Restart file support has been added for the number of Newton iterations required by each cell to satisfy the solution change convergence criteria at the last time step (CONV_NEW) ([#6374](#)). This output can be requested by including the CONV mnemonic on the [RPTREST](#) keyword. Note that simulations using NLDD are not currently supported.
- 11) Added support for [WSEGAICD](#) keyword item 22, the density function exponent ([#6473](#), [#4736](#)).
- 12) Experimental support has been added for the [WELSPECL](#) and [COMPDATL](#) keywords ([#6297](#)).
- 13) Added a command line option (`--linear-solver-accelerator=cpu or gpu`) to optionally carry out the linear solver operations on the GPU ([#6270](#)). By default all operations are carried out on the CPU. When using the gpu option, the linear solvers can be specified with the normal `--linear-solver` argument, which can point to a JSON file which can have the same format as the one for the CPU. Note that some options and preconditioners are currently limited to the CPU.
- 14) Full support has been added for running the Constrained Pressure Residual with Algebraic Multigrid (CPR-AMG) preconditioner entirely on the GPU for serial or single-GPU runs, using the Gpustl framework ([#6308](#), [Lye et al. \(2025\)](#)). It supports both AMGX and Hypre BoomerAMG solvers, allows consistent configuration across CPU and GPU, and introduces fast GPU-native implementations for CPR operations and quasiimpes weights. The setup minimizes memory usage and transfer overhead by sharing a single GPU matrix across solver components. Initial benchmarks show comparable or improved convergence versus CPU runs.

It is now possible to run the identical CPR-AMG preconditioner setup on both CPU and GPU (with different hardware specific AMG solvers) by using the command line option `linear-solver-accelerator=cpu` and `gpu` respectively.
- 15) An experimental Hybrid Newton method has been added that enhances nonlinear preconditioning using machine learning. Machine learning-based corrections have been integrated into the standard Newton solver of the black-oil simulator ([#6424](#), [Lechevallier, et al. \(2025\)](#)). One or more trained Hybrid Newton models can now be used to adjust the initial guess of the nonlinear solver at specific timesteps. Two new command line options have been added: setting `--use-hy-ne=true` activates the Hybrid Newton method, and `--hy-ne-config-file=<path>` specifies the path of the JSON configuration file that allows OPM Flow to correctly interpret how to precondition Newton's method.
- 16) The NLDD (Nonlinear Domain Decomposition) solver has reduced the time spent on local solves by skipping subdomains that haven't changed significantly between iterations ([#6346](#)). Mobility changes

indicate significant alterations in fluid flow characteristics, making it an effective indicator for determining when subdomain re-solving is necessary. The threshold for relative mobility change detection (default: 0.1) is specified using the `--nlld-relative-mobility-change-tol` command line option. By setting this parameter to 0, the mobility thresholding will be disabled.

- 17) Partitioner `zoltanwell` (default) no longer ignores the command line argument `--allow-distributed-wells` (#874). This makes its behaviour more like that of the other partitioners.
- 18) Added option to limit the relative change in the solution variables (#6009). If the relative change in the solution variables from one time step to the next becomes too large, it might be best to discard the computations even though the time step completed without numerical errors. The time step size would then be adjusted down before retrying the same time step. This option is only available when using the `general3rdorder` option for time-step-control. (See the new `--time-step-control-reject-completed-step` command line argument).
- 19) Added the option to use control error filtering as the tolerance test, instead of comparing relative change directly, to the general third order time step controller (#6280).
- 20) Added support for water mass in place summary vectors at the field, region and block level `[FRB]AMIP`, and water mass injection/production rates/totals at the field, group, well, connection, and completion levels `[FGWC]AM[IP][RT][L]` (#6181, #4569). Note that AM has been used to indicate water (aqueous) mass since WM, for example `GWMIR`, is already used to indicate water molar mass.
- 21) Satellite injection rates have been added to group level summary vectors (#4728).
- 22) A new application networkgraph has been created (#4535). This can be used to visualize a production network (extended network model) including wells.
- 23) Added the option to perform additional iterations in the well/group switching algorithm (#5933). See the command line argument `--well-group-constraints-max-iterations`, which has a default value of 1.

B.1.3 IMPROVEMENTS

Improvements to the simulator in this release include the following:

- 1) Both the linear solver command line options (e.g. `linear-solver=ilu`) and the parameters in JSON configuration files, that select the solvers and preconditioners, have been made case-insensitive by internally transforming all input into lowercase (#6388).
- 2) The following command line options have changed from taking integer arguments to taking string arguments (#6264). The defaults and behaviour have not been changed.
 - `--edge-weights-method` allowed values are: `uniform`, `transmissibility`, or `logtrans`. Replacing 0, 1, and 2.
 - `--partition-method` allowed values are: `simple`, `zoltan`, `metis`, or `zoltanwell`. Replacing 0-3.
- 3) The verbosity of the output from the time step controllers can be specified using the `--time-step-verbosity=BOOLEAN` command line argument (#6320). Setting this to true (default) increases the verbosity, or to false decreases the verbosity.
- 4) The code has been simplified eliminating ambiguous calls to `broadcast()` (#6180).
- 5) The compatibility of the interface with Dune-Grid has been improved (#892).
- 6) Reduced peak memory consumption when the command line option `--cpr-reuse-setup` is set to recreate the linear solver (which it does by default), see that option and `--cpr-reuse-interval` for details (#6345).
- 7) Improved GPU support for blackoil fluid systems (#6365). The `BlackoilIntensiveQuantities::update()` function can now be called without an `ElementContext`. The motivation is to make it a better fit for the GPU code, but it should also allow some more optimizations on the CPU, such as static dispatch for the "multiplexed" (runtime switched) functions of the fluid and material subsystems.

- 8) Memory consumption has been reduced when the hysteresis model is not being used ([#6510](#)). A new variant of the simulator (`flow_blackoil_nohyst`) has been added. This variant is automatically chosen whenever there is no hysteresis in the case and the black-oil variant (`flow_blackoil`) would otherwise be chosen.
- 9) Support has been added for link time optimization ([#4666](#)).
- 10) The code now avoids using an `ElementContext` just to evaluate `IntensiveQuantities` and demonstrates the use of compile-time dispatch to select a particular three-phase model avoiding run-time switching per cell ([#6367](#), [#4658](#)). This gives a 5-10% speedup for the property calculations.
- 11) Improved the performance of the Extended Summary file (ESMRY) output routine and fixed the RSTEP vector output ([#4677](#)).
- 12) Detailed instrumentation using Tracy has been enabled for individual subsystems of the simulator ([#4751](#)). This is used by developers to profile the application and improve performance, and not an end user feature.
- 13) Improved the efficiency of the formation volume factor and viscosity evaluation ([#4674](#), [#6386](#)). The "Props/update time" for Norne was reduced from 3.03s to 2.16s (subject to some variation).
- 14) Memory usage is now reduced when energy, diffusion, or dispersion storage is not used ([#6351](#)).
- 15) The efficiency of the CPR weight calculation has been improved ([#6457](#)).
- 16) The intensive quantities cache now stores values for the current time step only and not the previous time step ([#6357](#)). This reduces the peak memory usage of the entire simulator by 10 %, without any performance degradation.
- 17) The memory consumption during simulation has been reduced by 96 MB per million cells by removing unused capillary pressure storage ([#4650](#)).
- 18) There is a speed improvement in some cases where a single `PYACTION` block generates multiple keywords ([#4624](#)).
- 19) GPU matrix updates have been optimized by using direct memory transfer and pinned memory for Gpu-ISTL, providing significant performance improvements for GPU linear solver operations ([#6406](#)).
- 20) Improved error messages when invalid grid block ranges are defined using the `BOX` keyword ([#4600](#)).
- 21) Messages reporting that a well has converged in zero iterations (for example, 'Well P-1 converged in 0 inner iterations') will now only be output to the Debug (DBG) file if an elevated verbosity level has been specified by the user ([#6350](#)).
- 22) Added support for running OPM Flow gas-water simulations, for example `H2STORE` or `CO2STORE` cases, from Python code using the Python bindings ([#6075](#)).
- 23) The Python documentation for OPM Flow has been updated ([#4615](#), [OPM Python Documentation](#)).
- 24) Added effect of satellite surface injection rates, and updated satellite surface production rates ([#6454](#)).
- 25) Added support for additional `TUNING` keyword items: target non-linear convergence error (TRGCNV), maximum non-linear convergence error (XXXCNV), and target material balance error (TRGMBE) ([#6413](#)).
- 26) A user defined argument can now be used for group reservoir volume production rate (RESV) controls when using the `GCONPROD` keyword ([#4503](#)). In addition, restart support has been added for this control.
- 27) Support has been added for running simulations beyond 31-Dec-9999 (up to 292 million years) ([#6336](#)).
- 28) Support has been added for underscore characters in user defined region set names ([#4611](#)). These were previously rejected at the input level.

- 29) The implementation of the exponentiation operator "^" has been extended to support combinations of UDQ sets and UDQ scalars in the definition of UDQs (see the [UDQ](#) keyword) ([#4757](#)). Previously, this operator only supported UDQ sets of the same size.
- 30) Added support for [WECON](#) keyword item 9, the name of the follow-on well to be open if the current well is shut ([#6050](#), [#4507](#)).
- 31) Added UDA support for items 3 and 4 of the [WTRACER](#) keyword ([#6549](#), [#4800](#)).
- 32) If the [GSATPROD](#) keyword specifies a group that doesn't exist at this point in the schedule then it will be created ([#4689](#)). Groups created in this way will be parented directly to the FIELD group.
- 33) Support has been added for the use of UDAs for ORAT, WRAT, and GRAT in the [GSATPROD](#) keyword ([#6429](#), [#4701](#)).
- 34) Added the option to output [FIP](#) and RESV data to a separate CSV file ([#6329](#)). This is requested by including the CSVFIP mnemonic in the [RPTSOL](#) keyword.
- 35) A more user friendly message is now output if an attempt is made to restart a simulation from report step zero, which is not supported by OPM Flow ([#4638](#)).
- 36) Added support for the parallel Hydre preconditioner for the solution of scalar systems ([#6384](#)). This enables the use of Hydre for CPR, CPRW and possibly other systems written as scalar compressed row storage (CRS) matrices, for example, geomechanics or separate temperature equations.
- 37) Support has been added for the Hydre preconditioner with GPU input types by adding a unified, backend-agnostic HydreInterface that enables seamless integration with both CPU and GPU data types ([#6444](#)). Using the functionality from [#6384](#), this also makes it possible to use Hydre MPI parallel on the GPU. This is an important functionality for the broader CPR-AMG on GPU implementation in [#6308](#).
- 38) A new AmgxInterface class has been added that provides a unified, backend-agnostic interface to the AMGX functionality, enabling seamless usage with both CPU and GPU data types ([#6423](#)). This provides important functionality for the broader CPR-AMG on GPU implementation in [#6308](#) that is essential for achieving good GPU performance, as it ensures minimal data transfers.
- 39) A new sparse matrix implementation has been added that uses cuSPARSE's modern Generic API instead of the legacy API ([#6401](#)). This addition is essential for supporting block size 1 operations in the GPU CPR-AMG implementation in [#6308](#).
- 40) The CPR weight calculation has been parallelized using OpenMP threading to improve performance, particularly important for GPU configurations where the CPU is idle and the CPU-based weight calculation can become a significant proportion of linear solve time ([#6419](#)).
- 41) The new GPU aware ISTL solver now has MPI support ([#6379](#)). This setup only makes sense if you have multiple GPUs either on one node or spread across multiple nodes in a cluster. To use this simply run flow through MPI:

```
mpirun -np 2 flow --linear-solver-accelerator=gpu --matrix-add-well-contributions=true CASENAME
```

We additionally support using GPUDirect/GPU aware MPI/CUDA aware MPI should the cluster and MPI libraries used for compilation support this. To enable, use the `--gpu-aware-mpi=BOOLEAN` option. Using GPU aware MPI is expected to give a speedup.

Furthermore, currently limited to OpenMPI systems, we can verify if GPU aware MPI support is available. To enable or disable this check, use the `--verify-gpu-aware-mpi=BOOLEAN` option. Note that this check is not always conclusive, and for most systems it is recommended to turn it off and rather test correctness through other means.

- 42) The Gpu-ISTL code structure has been reorganised to reduce memory consumption and facilitate fewer host-to-device copies in the future ([#6241](#)).
- 43) Improved robustness by checking that the specified external partition file exists before trying to read it ([#6326](#)). If the external partition file does not exist, an error message will be reported.

- 44) Some types of run-time errors have been converted to compile-time errors, making it easier to avoid certain types of programming mistakes ([#4651](#)).
- 45) Added support for gas sales summary vectors [FG]GS[RT] and [FG]SG[RT] ([#4731](#)).
- 46) The well's preferred phase is now used to calculate the well productivity index (WPI, WPI4, WPI5, and WPI9) summary vectors ([#4770](#)). Previously, the oil phase was used.
- 47) The performance of the TracerModel has been improved by adding OpenMP parallelization over grid elements ([#6206](#)).
- 48) A new command line parameter (--max-well-status-switch-iter-wells) has been added to limit the maximum number of well switches between stopped and open during a time step ([#6456](#)). This parameter should be used with care as it could lead to a well being shut too early.
- 49) Added a command line option (--max-well-status-switch-in-inner-iter-wells=99) to specify the maximum number of status switches (between shut and open) in inner iterations for wells ([#6338](#)).
- 50) If the IPR-line lies above all (FLO, BHP) points in the lift table (VFP), the intersection is now determined by extrapolation to higher rates based on the last two points ([#6187](#)).
- 51) Only surface rates are allowed to be assigned to satellite production or injection groups ([#6480](#)). If higher level groups containing satellite groups have a reservoir volume rate (RESV) or voidage replacement (VREP) control, the treatment of the satellite rates is not well defined. A check has been added and an error message is now output if this is the case.
- 52) Previously well control changes weren't logged during local switching ([#6145](#)). Well control changes are now logged when the *NUPCOL* iteration threshold is reached.
- 53) The initialization procedure for production wells is now more robust, in particular it ensures that valid gas and water fractions are obtained for the VFP table lookup ([#6114](#)). Also, the gas and water fractions of a well are stored across iterations to better deal with situations such as stopped wells, where it is not possible to reconstruct gas and water fractions from the rates.
- 54) When detecting oscillations only the number of group switches after *NUPCOL* non-linear iterations are now considered ([#6358](#)). This makes the behaviour of groups similar to that of wells.

B.1.4 BUG FIXES

The following bug fixes have been incorporated into this release:

- 1) Corrected the application of region to region transmissibility multipliers (*MULTREGT* keyword) to NNCs created by pinched-out cells (*PINCH* keyword) in some specific cases ([#6498](#)).
- 2) Fixed a small bug for grid topology that did not affect OPM Flow, but did affect the geomechanics codes ([#913](#)).
- 3) Added a OPM Flow specific option to the Killough hysteresis model (*EHYSTR* keyword item I4) to correct the construction of the capillary pressure scanning curve ([#4676](#)). This option is currently inactive by default but this is likely to change in the future as the correction is believed to be a bug fix.
- 4) The old style *PYACTION* scripts are now assumed to have run successfully if they do not return a value ([#4725](#)). This fixes backwards compatibility.
- 5) The *TUNING* keyword now correctly sets the minimum length of all time steps Item I-3 TSMINZ instead of Item I-6 TSFMIN ([#6332](#)).
- 6) A bug has been fixed in the unit handling of the *OPERATE* keyword ([#4765](#)). This resolves an reported issue with the units when performing operations on permeability arrays ([#4597](#)).
- 7) Allow the *WELTARG* keyword to accept an empty well list ([#4654](#)). In this case no well targets will be changed and a warning message will be output.

- 8) A bug has been fixed where previously [DRSDT](#) was activated in restart runs even if the original simulation did not contain the [DRSDT](#) keyword ([#4655](#)).
- 9) Fixed the handling of the group production control ([GCONPROD](#)) keyword item 2 equals FLD ([#4635](#)). This is now correctly interpreted as inheriting the control from the group's parent(s).
- 10) If [WLIST](#) keyword add (ADD) or move (MOV) operations are performed on a non-existent well list then a new list will now be created ([#4768](#)). Also, if an attempt is made to delete (DEL) a non-existent well list then an error message is output.
- 11) OPM Flow now respects [GCONPROD](#) keyword items 11 to 13 properly ([#6331](#)). These are the actions to be taken if water, gas, and liquid rate constraints are violated. Previously the action would default to that specified by item 7 whichever of the constraints was violated.
- 12) The code now protects against possible division by zero when applying the [UDQ](#) comparison operators, which have a relative tolerance defined by the [UDQPARAMS](#) keyword item 4 ([#4758](#)).
- 13) [UDQ](#) values are now only updated once at the end of the next time step if the [UDQ](#) keyword UPDATE flag is set to NEXT ([#4767](#)). Previously, the [UDQ](#) values were also updated at the beginning of the next report step.
- 14) An error is now reported if the specified initial gas-oil contact is deeper than the initial oil-water contact specified on the [EQUIL](#) keyword in the [SOLUTION](#) section ([#4771](#)).
- 15) Fixed the summary (RSM) file generation failure in cases with formatted output ([#4653](#)). Also, corrected the generation of the SMSPEC file name used as the source when creating RSM file output in certain special cases that were previously not handled.
- 16) Fixed the time step counter in cases where the [SUMTHIN](#) or [RPTONLY](#) keywords are used ([#4644](#)).
- 17) Invalid region index pairs specified when requesting inter-region flow summary vectors (RxFT) are now ignored and a warning message is output ([#4704](#)).
- 18) Since the TIME vector in the summary file is stored as a single precision floating-point number, small time steps can give TIME values that are not strictly increasing especially at long elapsed times ([#4647](#)). This is known to cause problems for some post-processing tools. Summary output is now filtered to prevent this occurring.
- 19) The summary vector calculation of group and field level cumulative production volumes ([FG]xPT) have been amended to incorporate group level efficiency factors ([GEFAC](#)) for satellite production ([#4683](#)).
- 20) An incorrect reference solTracerRate has been fixed in TracerModel ([#6207](#)).
- 21) A well's preferred phase is now correctly restored following a restart ([#4740](#)). Previously, all producing wells would be restarted as oil producers, the common case, but this would be wrong for gas producers.
- 22) The last valid network pressure is now stored and used for initialization if the current iteration fails ([#6200](#)).
- 23) The new operability check introduced for wells under THP control has now also been introduced for wells not under THP control ([#6317](#)). This fixes convergence issues seen in a couple of test cases.
- 24) Well group targets are now stored: to make group targets consistent for local switching during well solves, to avoid using global vectors with name mapping from Wellstate, to only use singleWellState in the well switching algorithm, and to avoid recomputing well group targets ([#6311](#)).
- 25) Well status checks based on the well objects in the schedule have been replaced with the ones based on the well state ([#6051](#)).
- 26) A bug has been fixed in the LeftExtreme interpolation of 2D tables (where the y sample values are entered in ascending order) ([#4668](#)). Only small effects have been observed but could be significant in corner cases.

- 27) Allow restart runs for cases with binary (IMPORT'ed) input files ([#4708](#)).
- 28) Avoid very small denominator values when the product $R_s.R_v$ or $R_{sw}.R_{vw}$ is too large ([#6227](#)). This fixes the spikes seen in the gas injection potentials in a few regression tests.
- 29) Compilation issues have been resolved on architectures where long double maps to `__float128` with QuadMath ([#4756](#), [#6497](#)). This resolves issue [#4702](#).
- 30) Diagnostic messages for `DELAYED_EXIT1` will now be printed at the end of the parsing process ([#4613](#)).
- 31) Several bugs in the rocSPARSE backend for the GPU have been fixed including a memory leakage that caused a segmentation fault for a big model, and the rocSPARSE based CPR implementation has been updated to be inline with the OPM default CPR implementation ([#6339](#)).
- 32) The `baseName` has been separated from the file removal regular expression since it's a user-controlled string that might contain regular expression metacharacters like '+' or '-' ([#6293](#)). In rare corner cases, those characters might lead to matching too many or not enough files. The simulator now uses regular string matching for the `baseName`. Symbolic links and directories are omitted from consideration. Grid files are also omitted from file removal since it's possible for `CASE.DATA` to use `CASE.EGRID` as an input file (see the [GDFILE](#) keyword).

B.1.5 KNOWN ISSUES

Known issues in this release of the simulator include:

- 1) There is an error with the RSM header for summary vectors whose NUMS entry in the SMSPEC file is derived from more than a single number source (e.g., single region or segment ID). This applies to all block vectors (BGPV, BOPV, BWPV, etc.), connection level quantities (COPT, etc.), and inter-region flows such as ROFT etc ([#3078](#)). The work around is to plot the data in OPM ResInsight and right-click on the plot to view and copy the data.
- 2) As per previous releases of the radial model, the [COORDSYS](#) keyword item three must be set to COMP to complete the circle, this has not been implemented in this release. Also there appears to be a bug for full radial models when a well goes on BHP control that causes the well not to respect the BHP constraint, this eventually causes the well to die prematurely. See [#2640](#) for a discussion on the topic.
- 3) As in previous releases there are some issues with the [OPERATE](#) and [OPERATER](#) keywords associated with the input parsing; for various reasons a few of the fields require special case treatment in the grid processing, including (at least) [MULTZ](#), [PORV](#), and [ACTNUM](#), and for those keywords the [OPERATE/OPERATER](#) keyword does not work. The work around is to use the [MULTIPLY](#) keyword instead.
- 4) For the [UDQ](#) ASSIGN operator after the terminating "/" normally any comments can be entered; however, if there is "/" within the comment field, as per:

```
ASSIGN FUNGLYLD 1.196 / Condensate Yield (63.5/56.7)/(1.0 - 0.065)
```

then the simulator will abort. The work around is to manually place the comment characters "--" after the ASSIGN terminating "/", like so:

```
ASSIGN FUNGLYLD 1.196 / -- Condensate Yield (63.5/56.7)/(1.0 - 0.065)
```
- 5) At the moment, one cannot initialize tracers using the [EQUALS](#) keyword. Instead use the array format, that is the keyword followed by the required number of values, or the [TVDP](#) keyword in the [SOLUTION](#) section to set the initial tracer concentrations as a function of depth.
- 6) Currently, gas tracers cannot be used if the dissolved gas phase, as per the [DISGAS](#) keyword in the [RUNSPEC](#) section, is active in the model.

- 7) The summary vector RTIPTHEA, that defines the energy in-place between the initial and the current time for regions, is not supported unlike the FTIPTHEA and BTIPTHEA vectors. Secondly, the error message:

```
Warning: Problem with summary keyword RTIPTHEA
In RSM-THERMAL.data line 492
FIP region FIPHEA not defined in REGIONS section - RTIPTHEA ignored
```

is incorrect, as the message indicates that it is being treated like a named region, as per the *FIP* keyword, when it is actually a *SUMMARY* vector (#3870).

- 8) If there are cells that are very distorted, which can occur near fault planes, then the simulator may abort because it cannot calculate the pore volume of such cells. The work around is to re-generate the grid in the static model, taking care that the cells around the fault planes are more or less orthogonal (#2992 and #3770).
- 9) Currently the *OPERATER* keyword in the *EDIT* section does not work with the *DEPTH*, *TRANX*, *TRANY*, and *TRANZ* property arrays (#2994 and #748).
- 10) If a standard well is fully declared in an *ACTIONX* block which is then activated at a later date, and later the well is modified to be a multi-segment well using the *WELSEGS* and *COMPSEGS* keywords, then this will cause the simulator to abort with an assert failure. The solution to this issue is to not use this type of work flow in declaring wells (#2891 and #2895).
- 11) Although the *ACTIONX EXIT* command works as expected, it does not write out the requested RSM file at the end of the run. However, the other *SUMMARY* and *RESTART* files are written out (#2877).
- 12) There are small differences in the behavior of the *NEXTSTEP* keyword in the *RUNSPEC* section between OPM Flow and the commercial simulator that remain unresolved (#3745).
- 13) OPM cannot be built with dune-fem version 2.9 or later (#4934). Please use a previous version.
- 14) The simulator uses an irregular corner-point grid geometry with adjusted pore volumes to represent radial grids so it is not possible to create a full ring (360 degree disk) with only one cell in the theta direction (NY=1). The work around is to model a slice (say *DTHETA*=60 degrees). Note that as the angle increases larger pore volume adjustments are required (#4755).
- 15) In some cases with the network option the simulator can wrongly report that a well has no THP constraints, for example

```
GLIFT WTEST: Well S-3H does not have THP constraints
```

when THP constraints have been defined (#4887).

- 16) Dispersion in the gas phase leads to convergence issues for the 11th SPE CSP (<https://spe.org/csp>) models Version 11B and Version 11C (and is not fully tested). Therefore dispersion in the gas phase has been removed and a warning added (#5101, #859).

B.2 RELEASE 2025-04

The 2025-04 release consists of some new features and various improvements and bug fixes. These include the following highlights:

- A command line option `--check-satfnc-consistency` has been added that enables improved saturation function consistency checking.
- A new hybrid linear-radial filter cake modelling option `LINRAD` has been added to the `WINJDAM` keyword.
- The simulator now prints a message to the log files and standard output for all errors that occur while parsing the input deck, previously no message was reported for some errors before aborting.
- Gas consumption (`GCONSUMP` keyword) is now supported at multiple levels in the group hierarchy.
- The linear setup time has been reduced when using two-stage (CPR-like) preconditioners.
- Network solver sub-iterations have been added to avoid updating the gas lift, group controls and so on at every network update. This gives a significant speed up in cases including gas lift and network.
- The ability to pass standard command line parameters to the simulator from Python has been added.
- The handling of the `NUPCOL` keyword has been updated.
- To avoid group control oscillations the code now limits the maximum number of times a group can switch to the same control.
- Support has been added for using the Hypre library's BoomerAMG solver in serial runs, providing a high-performance algebraic multigrid solver option particularly beneficial for large-scale simulations and the possibility to run the algebraic multigrid (AMG) solver on the GPU.
- Support has been added for the `WCYCLE` keyword in the `SCHEDULE` section that defines automatic well opening and closing cycle parameters.
- Support has been added for using the AmgX library in serial runs. AmgX is a high-performance algebraic multigrid solver for NVIDIA GPUs.
- Added support for `ROCKOPTS` keyword item 2 equal to `STORE`. If `ROCKOPTS` item 2 is equal to `STORE`, the initial pressure is used in `ROCK` and `ROCKTAB` keywords.
- The simulator now uses the TPFA linearizer for Black Oil Thermal runs.
- The default load balancer has been changed to Zoltan with the grid represented by a graph. All cells potentially perforated by a well are represented by a single vertex, therefore the partitioning is unable to split a well across more than one process. This has resulted in a general improvement and has resolved convergence issues in some hard cases.
- The `WELPI` and `WPIMULT` keywords can now be used with the "insert keywords" function in `PYACTION` blocks.
- Added partial support for the `GSATPROD` keyword in the `SCHEDULE` section to define group satellite production rates.
- Added command line parameters for network solver sub-iterations and pressure update damping.
- Added the following command line arguments to specify load balancing parameters: add corners to partition (`--add-corners=false`), and numbers of layers overlap in parallel partition (`--num-overlap=1`).
- Support has been added for instantaneous flow rates in extended network models (`WEFAC` and `GEFAC` item 3). The `WEFAC` and `GEFAC` keywords are now fully supported.
- The permeability multiplication factor as a function of porosity change (`PERMFACT` keyword) can now be defined for each saturation table region.

- Support has been added for the [WCONHIST](#) and [WCONINJH](#) keywords in [ACTIONX](#) blocks.
- Added support for multi-segment well output to the PRT file for [RPTSCHED](#) mnemonic [WELLS=x](#).
- Support has been added for the [RSW](#) keyword in the [SOLUTION](#) section to explicitly initialise the dissolved gas-water ratio in each cell when dissolved gas in the water phase has been activated.
- Defaulted tables are now allowed in the [SWOF](#), [SGOF](#), [SWFN](#), [SGFN](#) and [SLGOF](#) keywords provided the first table has been defined. A defaulted table is set equal to the previous table.
- Added support for outputting saturation function tables defined using the [SLGOF](#) keyword to the initial data ([INIT](#)) file.
- The [WELTRAJ](#) and [COMPTRAJ](#) keywords now handle any [MAPAXES](#) specifications.
- Support has been added for using the special name selector '?' in well level [UDQ](#) assignments within an [ACTIONX](#) block.
- Support has been added for gas consumption and gas import summary vectors.
- Threshold pressures ([THPRES](#) keyword) are now restored from the [RESTART](#) file.
- Added support for the [WTMULT](#) and [WLIST](#) keywords in [PYACTION](#) blocks.
- Support has been added for the [GEFAC](#) keyword in [ACTIONX](#) blocks and in [PYACTION](#) code.
- The use of named regions is now supported in [ACTIONX](#) conditions.
- Support has been added for summary output of group and field production and injection targets.
- Support has been added for the [CO2STORE/H2STORE](#) specific block level CO₂/H₂ mass, moles and volumes in place summary variables; mass production rate (GMPR) and mass production total (GMPT) variables; and mass injection/production variables at the connection and completion level (for example, CGMIR and CGMIRL).
- The microbially induced calcite precipitation model (see the [MICP](#) keyword) has been updated to use the new lineariser, support dispersion, diffusion, sources, boundary conditions, and new summary variables. In addition, the new [BIOFPARA](#) keyword (which can be set per [SATNUM](#) region) is now used to set the parameters instead of the [MICPPARA](#) keyword.

B.2.1 NEW AND DEPRECATED COMMAND LINE OPTIONS

The major command line changes made for this release are summarized in Table B.2

OPM Flow 2025-04 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
1	--add-corners	A boolean value. Add corners to partition.	false
2	--allow-splitting-inactive-wells	A boolean value. Allow inactive (never non-shut) wells to be split across multiple domains.	true
3	--bda-device-id	Renamed as --gpu-device-id.	0
4	--check-satfunc-consistency	A boolean value. Whether or not to check saturation function consistency requirements.	false

OPM Flow 2025-04 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
5	--gpu-device-id	An integet value. Choose device ID for cusparseSolver or openclSolver, use 'nvidia-smi' or 'clinfo' to determine valid IDs.	0
6	--local-domains-partition-well-neighbor-levels	An integer value. Number of neighbor levels around wells to include in the same domain during NLDD partitioning.	1
7	--maximum-number-of-group-switches	An integer value. Maximum number of times a group can switch to the same control.	3
8	--network-max-iterations	Replaced by --network-max-outer-iterations.	200
9	--network-max-outer-iterations	An integer value. Maximum outer number of iterations in the network solver before giving up.	10
10	--network-max-pressure-update-in-bars	A real value. Maximum pressure update in the inner network pressure update iterations.	5
11	--network-max-strict-iterations	Replaced by --network-max-strict-outer-iterations.	100
12	--network-max-strict-outer-iterations	An integer value. Maximum outer iterations in network solver before relaxing tolerance.	10
13	--network-max-sub-iterations	An integer value. Maximum number of sub-iterations to update network pressures (within a single well/group control update).	20
14	--network-pressure-update-damping-factor	A real value. Damping factor in the inner network pressure update iterations.	0.1
15	--num-overlap	An integer value. Numbers of layers overlap in parallel partition.	1
16	--num-satfunc-consistency-sample-points	Maximum number of reported failures for each individual saturation function consistency check.	5
17	--nupcol-group-rate-tolerance	A real value. Tolerance for acceptable changes in VREP/RAIN group rates.	0.001
18	--partition-method	An integer value. Choose partitioning strategy: 0=simple, 1=Zoltan, 2=METIS, 3=Zoltan with all cells of well represented by one vertex.	3
19	--pre-solve-network	A boolean value. Pre solve and iterate the network model at start-up.	true
20	--slave	A boolean value. Specify if the simulation is a slave simulation in a master-slave simulation.	false

OPM Flow 2025-04 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
21	--threads-per-process	An integer value. The maximum number of threads to be instantiated per process ('-1' means 'automatic').	2
22	--time-step-control-safety-factor	A real value to be multiplied with the time step control tolerance to ensure that the target relative change is lower than the tolerance.	0.8
23	--zoltan-phg-edge-size-threshold	A real value. Low-level threshold fraction in the range [0,1] controlling which hypergraph edge to omit. Used if --zoltan-params="graph" or if --zoltan-params="hypergraph".	0.35
Notes: 1) Cells colored in green in the "No." column indicate a new command line option or a change to an existing option for this release. Cells colored green in the "Default" column indicate that the default value for an existing option has been changed. 2) Cells colored in orange in the "No." column indicate the command line option is available but is now "hidden" from the main help listing, --help. These options can be listed using the --help-all command line option, that lists all the command line options included in the release, including experimental, obsolete, hidden and deprecated options. 3) Cells colored in red in the "No." column indicate a deprecated command line option for this release.			

Table B.2: OPM Flow 2025-04 New and Deprecated Command Line Options

B.2.2 NEW FEATURES

In addition to the changes to the command line options the following new features have been added to the simulator:

- 1) A command line option --check-satfnc-consistency has been added that enables improved saturation function consistency checking. In addition, the new command line option --num-satfnc-consistency-sample-points allows the user to select the maximum number of reported failures for each individual consistency check. By default the simulator will report at most five failures for each check. If end-point scaling has been activated then the scaled curves are checked otherwise the unscaled curves are checked (#5596).
- 2) A new hybrid linear-radial filter cake modelling option LINRAD has been added to the [WINJDAM](#) keyword in the [SCHEDULE](#) section (#5717). The cake thickness is computed assuming linear geometry. This thickness is then used to compute a damage skin in radial geometry.
- 3) Running flow with the parameter --partition-method=3 will use Zoltan partitioner with the grid represented by a graph (#5749). All cells potentially perforated by a well are represented by a single vertex, therefore the partitioning is unable to split a well into several processes. This should lead to better partitioning results on problems with a complex well structure, where other partitioners might split wells over several processes after which some well cells would have to be remapped (when distributed wells are not allowed).
- 4) An experimental feature has been added that enables multi-segment wells to be distributed across several processes when using the command line argument --allow-distributed-wells=true (#5746).
- 5) Support has been added for using the Hypr library's BoomerAMG solver, providing a high-performance algebraic multigrid solver option particularly beneficial for large-scale simulations and the possibility to run the algebraic multigrid (AMG) solver on the GPU (#5762).

- 6) Support has been added for the [WCYCLE](#) keyword in the [SCHEDULE](#) section that defines automatic well opening and closing cycle parameters ([#5792](#)).
- 7) Support has been added for using the AmgX library in serial runs ([#5808](#)). AmgX is a high-performance algebraic multigrid solver for NVIDIA GPUs.
- 8) Added support for [ROCKOPTS](#) keyword item 2 equal to STORE ([#5785](#)). If [ROCKOPTS](#) item 2 is equal to STORE, the initial pressure is used in [ROCK](#) and [ROCKTAB](#) keywords.
- 9) A mixed precision option has been added to GPU DILU similar to that for GPU ILU0 ([#5674](#)).
- 10) A mixed precision option where off-diagonals are stored as float and diagonals are stored as double has been added to both GPU DILU and GPU ILU0 ([#5688](#)).
- 11) The [WELPI](#) and [WPIMULT](#) keywords can now be used with the "insert keywords" function in [PYACTION](#) blocks ([#5892](#)).
- 12) The following cubic equations of state can now be selected using the [EOS](#) keyword: Peng-Robinson (PR), modified Peng-Robinson (PRCORR), Redlich-Kwong (RK) and Soave-Redlich-Kwong (SRK) when using the experimental compositional simulator ([#5886](#)).
- 13) Added partial support for the [GSATPROD](#) keyword in the [SCHEDULE](#) section to define group satellite production rates ([#5894](#)).
- 14) Added command line parameters for network solver sub-iterations and pressure update damping ([#5918](#)). See the following command line arguments: maximum number of sub-iterations to update network pressures, within a single well/group control update (--network-max-sub-iterations), damping factor in the inner network pressure update iterations (--network-pressure-update-damping-factor), and maximum pressure update in the inner network pressure update iterations (--network-max-pressure-update-in-bars).
- 15) Added the following command line arguments to specify load balancing parameters: add corners to partition (--add-corners=false), and numbers of layers overlap in parallel partition (--num-overlap=1) ([#5942](#)).
- 16) The experimental compositional simulator has been expanded to three phases by the addition of a ("dummy") immiscible water phase ([#5851](#)).
- 17) Support has been added for instantaneous flow rates in extended network models ([WEFAC](#) and [GEFAC](#) item 3). The [WEFAC](#) and [GEFAC](#) keywords are now fully supported ([#5410](#)).
- 18) A third-order time step controller has been added that can be used as an alternative to the PID controller ([#5974](#)). The new controller can be specified by using the --time-step-control="general3rdorder" command line argument.
- 19) Added a command line parameter (--pre-solve-network=true) to make the pre-step network rebalance optional ([#6048](#)).
- 20) Added command line parameter (--zoltan-phg-edge-size-threshold=0.35) that controls which hypergraph edges to omit/discard by the Zoltan partitioner ([#6036](#)). This is to enable runtime experimentation with this aspect of the partitioning algorithm, the default value should not normally be altered.
- 21) The permeability multiplication factor as a function of porosity change ([PERMFACT](#) keyword) can now be defined for each saturation table region ([#6074](#)).
- 22) Added support for [RPTSCHED](#) mnemonic WELLS=2. In particular, this appends connection level results to the injection/production output sent to the print (PRT) file ([#6099](#)).
- 23) Support has been added for the [WCONHIST](#) and [WCONINJH](#) keywords in [ACTIONX](#) blocks ([#6092](#)).
- 24) Added support for multi-segment well output to the PRT file for [RPTSCHED](#) mnemonic WELLS=x ([#6110](#)).

- 25) Support has been added for the [RSW](#) keyword in the [SOLUTION](#) section to explicitly initialise the dissolved gas-water ratio in each cell when dissolved gas in the water phase has been activated by specifying the [DISGASW](#) keyword in the [RUNSPEC](#) section ([#6111](#)).
- 26) Added support for block summary vectors related to mass and volume in-place for [CO2STORE](#) and [H2STORE](#) runs including: BGMIP, BGMGP, BGMDs, BGMST, BGMUS, BGMTR, BGMMO, BGKTR, BGKMO, BGCDI, BGCDM, BGKDI, BGKDM, BWCD, BWIPG, BWIPL ([#6127](#)).
- 27) Added the option to merge each well with a layer(s) of cells surrounding it, in order to keep the well further from the subdomain boundary, when partitioning the grid ([#811](#)). To use this option, flow must be called with parameter `--zoltan-params=*.json` taking a json file. The json file must contain the keyword `EnvelopeWellLayers` with the number of layers (default 0). The feature is supported only by the partitioner using the `GraphOfGrid` (`--partition-method=3`).
- 28) Added an option to do serial partitioning with Zoltan using the graph representation of grid (`GraphOfGrid`) ([#821](#)). The parallel partitioning with `GraphOfGrid` is also available and is currently the default partitioner.
- 29) The `zoltanGoG` (Zoltan Graph of Grid) partitioner has replaced `zoltan` as the default partitioner (if there are no wells `zoltanGoG` is equivalent to `zoltan`) ([#840](#)).
- 30) Defaulted tables are now allowed in the [SWOF](#), [SGOF](#), [SWFN](#), [SGFN](#) and [SLGOF](#) keywords provided the first table has been defined ([#4308](#)). A defaulted table is set equal to the previous table.
- 31) Added support for 'LINRAD' as a valid option for item 2 (GEOMETRY) of the keyword [WINJDAM](#) ([#4320](#)). This option specifies a hybrid filter cake model where the cake thickness is computed assuming linear geometry (as with item 2 = [LINEAR](#)). This thickness is then used to compute a damage skin in radial geometry (as with item 2 = [RADIAL](#)).
- 32) Support has been added for gas consumption and gas import summary vectors `GGCR`, `GGIMR`, `GGCT`, `GGIMT`, `FGCR`, `FGIMR`, `FGCT` and `FGIMT` and for the corresponding restart input/output ([#4342](#)).
- 33) Support has been added for group name pattern matching in [UDQ](#) assignments ([#4433](#)).
- 34) Support has been added for summary output of group (and field) production and injection targets ([#4500](#)). The following summary vectors are now supported: `G[OWGLV]PRT`, `G[WGV]IRT`, `F[OWGLV]PRT`, and `F[WGV]IRT`.
- 35) Support has been added for the [CO2STORE/H2STORE](#) specific block level CO_2/H_2 mass, moles and volumes in place summary variables; mass production rate (`GMPR`) and mass production total (`GMPT`) variables; and mass injection/production variables at the connection and completion level (for example, `CGMIR` and `CGMIRL`) ([#4550](#)).
- 36) The microbially induced calcite precipitation model (see the [MICP](#) keyword) has been updated to use the new lineariser, support dispersion, diffusion, sources, boundary conditions, and new summary variables ([#4538](#)). In addition, the new [BIOFPARA](#) keyword (which can be set per [SATNUM](#) region) is now used to set the parameters instead of the [MICPPARA](#) keyword.

B.2.3 IMPROVEMENTS

Improvements to simulator in this release include the following:

- 1) When partitioning, the simulator now only accounts for wells that might be active at some point in time during the run ([#5609](#)).
- 2) Logging has been added for [PYACTION](#) blocks similar to the current logging for [ACTIONX](#) blocks ([#5699](#)).
- 3) The experimental compositional simulator now outputs `INIT`, `EGRID`, `SMSPEC`, `UNRST` and `UNSMRY` files ([#5687](#)).
- 4) The device name and compute capability are now output for every rank ([#5611](#)).

- 5) Since the code can now switch to/from GRUP control in the inner well iterations this is problematic because the group reduction rates (the combined rates of all wells that are not on group rate) are not updated, which in turn may result in excessive switching (#5695). As a temporary solution a new command line argument has been added (`--check-group-constrains-well-iterations=false`). Setting `--check-group-constrains-well-iterations=true` means that individual well constraints will not be checked during the inner well iterations if the well is on GRUP control.
- 6) The code no longer repeatedly recomputes the well BHP from the THP when evaluating well potentials for gaslift optimisation (#5696). This gives a 10 times speed up of the gaslift optimisation for many cases since the BHP is not repeatedly computed for ALQ values that don't converge.
- 7) The simulator now prints a message to the log files and standard output for all errors that occur while parsing the input deck, previously no message was reported for some errors before aborting (#5707).
- 8) If one well operating on a well constraint (typically THP or BHP) in a group struggles to converge it may lead to problems for the other wells in the group. In this case, the simulator now only shuts the unconverged well rather than all the wells in the group (#5735).
- 9) Previously, using the `WTEST` keyword on a gas lifted well typically resulted in the maximum artificial lift quantity (ALQ) being used after the well was reopened. The simulator now optimises the ALQ to give the maximum incremental gradient. This has reduced the large spikes in ALQ seen after reopening wells in the gas lift test cases (#5757).
- 10) Debug output from THP calculations previously output to the PRT file will now be output to the DBG file (#5761). The volume of output can be large especially for cases with gas lift.
- 11) Gas consumption (`GCONSUMP` keyword) is now supported at multiple levels in the group hierarchy (#5739).
- 12) The simulator now avoids duplicating the output of warnings and errors to the standard error stream (`stderr`) (#5765).
- 13) The linear setup time has been reduced when using two-stage (CPR-like) preconditioners. Previously, the second stage ILU0 preconditioner was unnecessarily recreated every Newton iteration. The simulator now only updates the preconditioner (#5758). This will be especially important when using GPU preconditioners with `CPR`, as these GPU preconditioners typically involve expensive GPU allocations, matrix analysis and autotuning.
- 14) Network solver sub-iterations have been added to avoid updating the gas lift, group controls and so on at every network update (#5767). This gives a significant speed up in cases including gas lift and network.
- 15) The ability to pass standard command line parameters to the simulator from Python has been added (#5760).
- 16) The handling of the `NUPCOL` keyword has been updated (#5724). Wells are no longer allowed to change to group control when the number of iterations is greater than `NUPCOL`. Voidage replacement and reinjection target rates are updated even if the number of iterations is greater than `NUPCOL` provided the change is greater than a specified relative tolerance (see `--nupcol-group-rate-tolerance`).
- 17) The fine-level smoothing parameters (`pre_smooth` and `post_smooth`) for the `CPR` AMG implementation in OPM have been exposed by adding them to the linear solver JSON configuration tree output to the DBG file (#5786).
- 18) To avoid group control oscillations the code now limits the maximum number of times a group can switch to the same control (#5753). The number of switches is only checked when the number of iterations is greater than `NUPCOL` (that is multiple switches are allowed whilst the network is iterating). See the command line argument `--maximum-number-of-group-switches`.
- 19) The code now ensures that tracer flows for cross-flowing injector connections are included (#5789).

- 20) The 16 day restriction on timesteps in prediction mode with THP constraints has been removed from the adaptive time stepping algorithm ([#2970](#)). The time-stepping, convergence and well models have all improved since this hard-coded limit was originally put in place.
- 21) Group guide rates are now allowed in cases where a group has an automatic choke in an extended network model ([#5754](#)).
- 22) The memory requirement has been reduced by 15 bytes per cell by reducing the size of enums ([#5841](#)).
- 23) The simulator now uses the Two Point Flux Approximation (TPFA) linearizer for Black Oil Thermal runs ([#5843](#)).
- 24) The default load balancer has been changed to Zoltan with the grid represented by a graph ([#5800](#)). All cells potentially perforated by a well are represented by a single vertex, therefore the partitioning is unable to split a well across more than one process. Also the weights of the faces will be added up due to the merging of cells. A cell that is not perforated by any well is still represented by one vertex with weight one. This has resulted in a general improvement and has resolved convergence issues in some hard cases.
- 25) If the partition has been loaded from a file containing an externally generated partition (`--external-partition="filename"`) and the output of binary files compatible with the commercial simulator has been deactivated (`--enable-ecl-output=false`), then the expensive routine `allocTrans()` is now skipped ([#5817](#)).
- 26) Repeated calculation of the cell centroids has been avoided by caching the values ([#5888](#)). This can significantly reduce the setup time for large models.
- 27) The default maximum number of threads per process has been changed from -1 ('automatic') to 2 (see command line argument `--threads-per-process=2`). If OpenMP is available the maximum number of threads will be defaulted to 2 unless the environment variable `OMP_NUM_THREADS` is set or the command line argument is used. Note that the `OMP_NUM_THREADS` takes precedence over the `--threads-per-process` command line argument ([#5856](#)).
- 28) Empty command line arguments are now ignored, this avoids error messages of the form "Parameter 'EclDeckFileName' specified multiple times as a command line parameter" ([#5679](#)).
- 29) In an extended network model the bottommost (leaf) nodes associated with wells must be the same as those defined by [GRUPTREE](#) keyword. However, the extended network model will now allow empty leaf nodes (with no wells) that do not exist in the [GRUPTREE](#) ([#5919](#), [#6010](#)).
- 30) Improved the logic for shutting failed wells ([#5900](#)). In particular, allow for failed wells to shut even though they haven't failed consistently in the last 3 attempts if the time step has already been reduced enough.
- 31) A check has been added to guard against high roughness values in the [WELSEGS](#) keyword (that can result in a singularity) whilst reading the input deck ([#5945](#)).
- 32) Information about wells being stopped or revived before the timestep has converged will now only be sent to the debug (DBG) file. However, changes to the well status when the timestep has converged will still be output to the terminal and the print (PRT) file ([#6012](#)).
- 33) The stagnation and oscillation code for multi-segment wells has been modified. If the well stagnates five times with full damping then the well is marked as unconverged and the simulator tries again at next Newton iteration ([#6020](#)).
- 34) The output of some NaN's has been prevented for ratios of rates, for example produced gas-oil ratio ([#6077](#)). If the numerator is zero then zero is now output instead of dividing it by the denominator that might also be zero.
- 35) The gaslift optimiser information messages are now output to the debug (DBG) file rather than the print (PRT) file to avoid exceeding the message limit ([#6082](#)).

- 36) Updates generated by [PYACTION](#) or [ACTIONX](#) blocks are now collected and applied together ([#6070](#)). Previously, the updates would have been applied one at a time, potentially causing unnecessary recalculations.
- 37) When testing a well (see [WTEST](#) keyword) belonging to a group that corresponds to a node in an extended network model (see [NETWORK](#) keyword) the well's dynamic THP constraint will now be set based on the network's nodal pressure ([#6116](#)).
- 38) Previously, the original fluid in place values in a restart run would be zero ([#6119](#)). If a restart file is available for report step zero then this will now be used to generate the original fluids in place.
- 39) The well rates used to avoid oscillations are now taken from the network iterations (see [NUPCOL](#) keyword) rather than using the previous well rates ([#6128](#)). This is expected to give more accurate results in particular when wells startup.
- 40) The maximum number of Newton iterations used to updated well targets (see [NUPCOL](#) keyword) is now applied correctly, rather than the specified value plus one ([#6100](#)). Also, gas lift is no longer optimized in the last iteration to allow the network to converge (unless the maximum number of iterations is less than two).
- 41) The value of the saturated dissolution factor is now stored giving a significant speed up in cases with convective mixing (see [DRSDTCON](#) keyword) ([#5774](#)).
- 42) Some bugs that were present in the Non-Linear Domain Decomposition (NLDD) reporting have been fixed, and the way statistics from the NLDD solver are collected, displayed, and analysed has been improved ([#6124](#)).
- 43) The well equation is no longer re-solved when computing the initial guess for BHP at the THP constraint if the well state is updated ([#6081](#)).
- 44) Non-neighbour connections generated due to pinched out cells (see [PINCH](#) keyword) have been added to the log output ([#798](#)).
- 45) The code now uses unsigned integers when processing the grid to allow larger grid sizes ([#818](#)).
- 46) Support for lowercase file extensions has been added to the file conversion utility (convertECL) ([#4274](#)).
- 47) Added support for outputting saturation function tables defined using the [SLGOF](#) keyword to the initial data ([INIT](#)) file ([#4307](#)).
- 48) Added GPU support for evaluation of CO₂ PVT functions ([#4222](#)).
- 49) The simulator now allows a non-zero artificial lift quantity (ALQ) to be used in the defining expression of a user defined quantity ([UDQ](#)) in the case where no vertical flow performance (VFP) tables have been provided ([#4324](#)).
- 50) The simulator now gives a more meaningful error message if the well connections defined in a [COMPDAT](#) keyword within an [ACTIONX](#) block are not part of the grid ([#4336](#)).
- 51) Added support for restart runs with the standard network model, where the network dimensions have not been defined by the [NETWORK](#) keyword in the [RUNSPEC](#) section ([#4337](#)).
- 52) Support has been added for using the special name selector '?' in well level [UDQ](#) assignments within an [ACTIONX](#) block ([#4343](#)). In particular, OPM Flow now supports [ACTIONX](#) blocks like

```

ACTIONX
  TESTW 25 1 /
  DAY > 10 AND /
  WMCTL '*PROD?' = 0 /
/

UDQ
  ASSIGN WU_TEST '?' 1 /
/

```


ENDACTIO

in which the selector '?' represents all wells from all well lists matching the well list template '*PROD?' for which the WMCTL is zero, meaning the well is shut.

- 53) Support has been added for using a leading backslash (\) to disambiguate a leading wildcard characters (* or ?) in a name pattern as being truly a wildcard, for example wells matching '*PROD' rather than the well list '*PRODS' (#4345).
- 54) Threshold pressures (*THPRES* keyword) are now restored from the *RESTART* file (#5743, #4347).
- 55) The code will now issue a warning message if the *THPRES* keyword is present in a restart run (and it will be ignored) (#4347). When performing a restart the solution section should be updated, and in particular the *THPRES* keyword should be removed. The *THPRES* values should be read from the restart file instead.
- 56) Added restart file support for field level user defined arguments (UDA) (#4357).
- 57) The default well injection temperature (see the *WTEMP* keyword in the *SCHEDULE* section) has been changed from standard conditions to zero degrees Celsius in runs using the temperature modelling option (see the *TEMP* keyword in the *RUNSPEC* section) (#4042).
- 58) The depth correction algorithm for the WBPn well block pressure summary vectors has been revised (#4376). The revised approach uses the common mixture density, but corrects each cell pressure individually based on that cell's centre depth before aggregating the per-connection contributions.
- 59) Added support for the *WTMULT* and *WLIST* keywords in *PYACTION* blocks (#4397, #4402).
- 60) Variable and function names in the Python Schedule class have been changed from "timestep" to "reportstep" where these refer to report steps rather than time steps (#4412).
- 61) Added documentation to Python classes: SummayState, Schedule, Connection, Deck, DeckKeyword, DeckItem, DeckRecord, UDValue, EclipseGrid, EclipseState, ESmry and EGrid (#4405, #4413, #4414, #4428, #4497).
- 62) Support has been added for the *GEFAC* keyword in *ACTIONX* blocks and in *PYACTION* code (#4408).
- 63) The use of named regions is now supported in *ACTIONX* conditions, for example 'RPR I RE3 < 215 /' (#4449).
- 64) Added support for writing the combined thermal conductivity of rock plus fluid array (*THCONR*) to the initial data (*INIT*) file (#4474).
- 65) If necessary the gas PVT tables entered using either the *PVDG* or *PVTG* keywords are now extended down to a pressure of 1 barsa (#3779). This change guards against potential negative formation volume factors (densities) and/or viscosities when extrapolating the original tables to pressures in the order of one atmosphere.
- 66) The surface densities of CO₂ and brine are now output to the terminal in the same units as the input deck for *CO2STORE/CO2SOL* runs (#4510).
- 67) The simulator now avoids calculating the vapour pressure when not required for the CO₂-Brine PVT model (#4515). Note that in Spycher and Pruess 2010 they use 31 °C which is inconsistent with the critical temperature (30.95 °C) used in OPM Flow.
- 68) The simulator now properly restores gas-lift optimisation parameters following a restart (#4517).
- 69) *NODEPROP* keyword Item 4 (GASLIFT) for leaf nodes in an extended network model is now saved to and restored from the restart file (#4499).

- 70) The [COMPSEGS](#) keyword now allows the distance to the start of the connection (Item 5) to be equal to the distance to the end of the connection (Item 6) to define very short completion intervals ([#4265](#)).
- 71) A warning is now issued if the depth change along a well segment is larger than the segment length ([#3976](#)).
- 72) Invalid keyword combinations encountered whilst parsing [ACTIONX](#) blocks are now ignored ([#4511](#)).
- 73) If a well is completed in a numerical aquifer cell then the connection properties are now taken from the numerical aquifer rather than the grid cell ([#4547](#)). Previously, this could give not-a-number (NaN) values for the permeability-thickness (kh) product and the connection length since the grid's permeability would be explicitly reset to zero in numerical aquifer cells. This impacts the [WELSPECS](#) report generated by the [RPTSCHED](#) keyword and the SCON restart file array.
- 74) Report step zero will now always be written to the restart file (unless this has been explicitly turned off) ([#4545](#)). This is used to generate initial fluid in place ([FIP](#)) reports in restart runs.
- 75) Support has been restored for the old-style "integer control" format for PRT file report generation using the [RPTSCHED](#) keyword ([#4526](#)).
- 76) The lookup of source terms defined by the [SOURCE](#) keyword is now much faster ([#4553](#)). Previously, the lookup could be very slow when using the [SOURCE](#) keyword to couple OPM Flow with another simulator where there could be a source term in every cell.

B.2.4 BUG FIXES

The following bug fixes and improvements have been incorporated into this release:

- 1) A bug that resulted in an out-of-bounds error when restarting parallel runs with tracers has been fixed ([#5665](#)). The tracer values are now the same in a parallel restart run as in the equivalent sequential restart run.
- 2) A memory bug that affected runs where the NLDD nonlinear solver was used and where there was an unconverged local NLDD domain, which had a multi-segment well on that domain, has been fixed ([#5675](#)).
- 3) The bottom hole temperature for stopped injection wells is now assumed to be equal to the reservoir temperature ([#5605](#)). Previously, the injection temperature was used for stopped injection wells.
- 4) A number of fixes have been applied to the gaslift code for cases with complex group controls with liquid or water constraints ([#5694](#)).
- 5) Fixed a bug where a local index was used instead of a global index when extracting well contributions to add to the matrix for NLDD with the GPU bridge activated ([#5720](#)).
- 6) The simulator no longer tries to compute the initial solution for inactive multi-segment wells split across processes ([#5751](#)).
- 7) Fixed a bug where the total group rate was used to check a higher level group constraint rather than just the portion subject to the higher level control ([#5729](#)).
- 8) Avoid updating the state of wells that are not under group control when updating and communicating group data ([#5748](#)).
- 9) A potential deadlock that could occur if the Unsymmetric MultiFront (UMFPACK) solver encountered a singular matrix has been fixed ([#5773](#)).
- 10) The simulator now always calculates the initial in place volumes since these may be used by various keywords in the [SCHEDULE](#) section even if [FIP](#) is not requested by the [RPTSOL](#) keyword ([#5784](#)).
- 11) The code now allows efficiency scaling factors to be queried in off-process wells where required ([#5816](#)). Previously, this could cause MPI to hang due to an exception thrown by `well_state[name]`.

- 12) When calculating the voidage replacement (VREP) injection rate for a given phase the simulator now accounts for possible injection rate controls on the other phases ([#5783](#)).
- 13) The RFT output has been fixed for shut wells (and in the case of partition method 3 for inactive wells) split across processors ([#5824](#)). Previously, zeroes would be output for some connections.
- 14) Several issues related to zero group net voidage rates (NETV) have been fixed ([#5822](#)).
- 15) The current well state is now used as an initial guess for the implicit well potential calculations ([#5929](#)). This avoids some occasional convergence issues.
- 16) Fixed the output of residuals for gas-water systems, previously the residuals were only output for the gas phase ([#6002](#)).
- 17) Summary output has been corrected for cumulative quantities (MSUMLINS and MSUMNEWT) at time steps between report steps ([#6039](#)). Previously, this was only output correctly at report steps.
- 18) Corrected application of THP constraints for wells subject to group constraints ([#6018](#)).
- 19) Fixed VTK output for dry runs (requested using `--enable-vtk-output=true` and `--enable-dry-run=true`) in cases without the oil phase present ([#6057](#)) or with the `DRSDT` keyword in the `SCHEDULE` section ([#6060](#)).
- 20) Fixed problem with group pressure maintenance targets and controls applied at the Field level using the `GPMaint` keyword ([#5916](#)).
- 21) A potential division by zero has been prevented that could occur when using the CPRW linear solver and a well has no active perforations on some rank ([#6053](#)).
- 22) Previously, a small non-zero oil saturation could be generated when the model was initialised by enumerated gas and water saturations due to the numerical precision. This caused an issue when setting up the primary variables. This issue has been resolved by setting initial oil saturations less than a small tolerance value to zero ([#6079](#)).
- 23) The list of well events is now cleared out once they have been processed at the start of a report step preventing them from potentially being executed again ([#6078](#)).
- 24) The simulator now re-iterates the network solver if the well group controls are changed, the gaslift quantity is changed, or the inner iterations did not converge ([#6062](#)).
- 25) The `MICP` model now works with the default linear solver settings ([#6109](#)).
- 26) Groups subject to higher level controls are now allowed to switch to the `FIELD` group controls ([#6135](#)).
- 27) Fixed an issue where the wrong phase index could be used for restart output if output was only requested for one phase ([#6134](#)).
- 28) The correct ALQ dimension is now restored when restarting a run ([#4271](#)).
- 29) Specifying UPDATE NEXT in an `ACTIONX` block now causes the `UDQ` to be updated on the next time step rather than the next report step ([#4325](#)).
- 30) A few issues with the `WELTRAJ` and `COMPTRAJ` keywords have been fixed ([#4279](#)). Any `MAPAXES` specifications were not handled, and cases with the origin not at zero failed. To solve this, the well coordinates are now transformed when they are read. There was a bug in `COMPTRAJ` where defaulting the saturation table column was not handled, leading to a segmentation fault. The code for converting the trajectory to cell indices only took the top and bottom coordinates into account, effectively assuming linear trajectories. Now intermediate points are added, allowing arbitrary trajectories.
- 31) All group constraints are now written to the restart file even if the group is controlled by a higher-level group ([#4346](#)). Previously, production groups controlled by a higher-level group could be given the wrong group control following a restart.

- 32) The minimum grid block pore volume thresholds for individual cells specified using the [MINPVV](#) keyword will now take precedence over the threshold set of all cells using the [MINPV](#) or [MINPORV](#) keyword ([#4360](#)). Previously, [MINPVV](#) would be ignored if either [MINPV](#) or [MINPORV](#) was used.
- 33) The simulator can now handle deck files with sub-second report steps with a resolution of 1ms ([#4374](#)).
- 34) A potential divide by zero error at report step zero has been avoided ([#4392](#)).
- 35) The [GRUPTARG](#), [GSATINJE](#), [GSATPROD](#), [GRUPNET](#) and TEST keywords have been removed from the list of supported keywords in [ACTIONX](#) blocks since they are not supported by OPM Flow ([#4400](#), [#4436](#)). Note that TEST is not a keyword.
- 36) Previously, there was a risk that the defining expression for a [UDQ](#) could overflow the 128 character limit when stored in the restart file ([#4422](#)). The simulator now stores a "fused" version of the defining expression that honours the character limit.
- 37) The code now ensures that any events that might happen during the first time step after the restart time are preserved ([#4417](#)).
- 38) Fixed the handling of the group (GRUP) control mode for wells in auto-choke groups in extended network models ([#4435](#)).
- 39) Fixed a bug where the simulator tried to restore [UDQ](#) unit strings as [UDQ](#) variables after a restart ([#4457](#)).
- 40) The code now uses the branch number to correctly determine if a segment of a multi-segment well is active at restart ([#4483](#)).
- 41) When a keyword is expected the simulator will now ignore any line where the first non-whitespace character is a slash ([#4462](#)). Previously, if the slash was followed by any non-whitespace characters on the same line then an exception would have been raised.
- 42) The simulator now updates the well efficiency factor immediately if the [WEFAC](#) keyword is run from an [ACTIONX](#) block during a report step rather than at the end of the report step ([#4495](#)).
- 43) The handling of daylight-saving time when importing the start date from a summary file (or enhanced summary file) using Python bindings has been fixed ([#4509](#)).

B.2.5 KNOWN ISSUES

Known issues in this release of the simulator include:

- 1) There is an error with the RSM header for summary vectors whose NUMS entry in the SMSPEC file is derived from more than a single number source (e.g., single region or segment ID). This applies to all block vectors (BGPV, BOPV, BWPV, etc.), connection level quantities (COPT, etc.), and inter-region flows such as ROFT etc ([#3078](#)). The work around is to plot the data in OPM ResInsight and right-click on the plot to view and copy the data.
- 2) As per previous releases of the radial model, the [COORDSYS](#) keyword item three must be set to COMP to complete the circle, this has not been implemented in this release. Also there appears to be a bug for full radial models when a well goes on BHP control that causes the well not to respect the BHP constraint, this eventually causes the well to die prematurely. See [#2640](#) for a discussion on the topic.
- 3) As in previous releases there are some issues with the [OPERATE](#) and [OPERATER](#) keywords associated with the input parsing; for various reasons a few of the fields require special case treatment in the grid processing, including (at least) [MULTZ](#), [PORV](#) and [ACTNUM](#), and for those keywords the [OPERATE/OPERATER](#) keyword does not work. The work around is to use the [MULTIPLY](#) keyword instead.
- 4) For the [UDQ](#) ASSIGN operator after the terminating "/" normally any comments can be entered; however, if there is "/" within the comment field, as per:

ASSIGN FUNGLYLD 1.196 / Condensate Yield (63.5/56.7)/(1.0 - 0.065)

then the simulator will abort. The work around is to manually place the comment characters "--" after the ASSIGN terminating "/", like so:

ASSIGN FUNGLYLD 1.196 / -- Condensate Yield (63.5/56.7)/(1.0 - 0.06)

- 5) At the moment, one cannot initialize tracers using the [EQUALS](#) keyword. Instead use the array format, that is the keyword followed by the required number of values, or the [TVDP](#) keyword in the [SOLUTION](#) section to set the initial tracer concentrations as a function of depth.
- 6) Currently, gas tracers cannot be used if the dissolved gas phase, as per the [DISGAS](#) keyword in the [RUNSPEC](#) section, is active in the model.
- 7) The summary vector RTIPTHEA, that defines the energy in-place between the initial and the current time for regions, is not supported unlike the FTIPTHEA and BTIPTHEA vectors. Secondly, the error message:

```
Warning: Problem with summary keyword RTIPTHEA
In RSM-THERMAL.data line 492
FIP region FIPHEA not defined in REGIONS section - RTIPTHEA ignored
```

is incorrect, as the message indicates that it is being treated like a named region, as per the [FIP](#) keyword, when it is actually a [SUMMARY](#) vector ([#3870](#)).

- 8) If there are cells that are very distorted, which can occur near fault planes, then the simulator may abort because it cannot calculate the pore volume of such cells. The work around is to re-generate the grid in the static model, taking care that the cells around the fault planes are more or less orthogonal ([#2992](#) and [#3770](#)).
- 9) Currently the [OPERATER](#) keyword in the [EDIT](#) section does not work with the [DEPTH](#), [TRANX](#), [TRANX](#) and [TRANZ](#) property arrays ([#2994](#) and [#748](#)).
- 10) If a standard well is fully declared in an [ACTIONX](#) block which is then activated at a later date, and later the well is modified to be a multi-segment well using the [WELSEGS](#) and [COMPSEGS](#) keywords, then this will cause the simulator to abort with an assert failure. The solution to this issue is to not use this type of work flow in declaring wells ([#2891](#) and [#2895](#)).
- 11) Although the [ACTIONX EXIT](#) command works as expected, it does not write out the requested RSM file at the end of the run. However, the other [SUMMARY](#) and [RESTART](#) files are written out ([#2877](#)).
- 12) Although the [GCONSUMP](#) keyword in the [SCHEDULE](#) section is fully implemented as documented, it is not possible to verify the output as the associated [SUMMARY](#) vectors are not written out, that is the [SUMMARY](#) sales gas vectors FGSR, FGST, GGSR and GGST, and fuel vectors FGCR, FGCT, GGCR, and GGCT have not been implemented ([#2679](#)).
- 13) There are small differences in the behavior of the [NEXTSTEP](#) keyword in the [RUNSPEC](#) section between OPM Flow and the commercial simulator that remain unresolved ([#3745](#)).
- 14) There is a unit handling issue associated with [OPERATE](#) keyword. If the [OPERATE\(X\)](#) parameter has units, as for example [PERMX](#), then the conversion is always done in SI units, despite the input deck declaring the deck to be fields units, as per [FIELD](#) keyword in the [RUNSPEC](#) section. Note that OPM Flow performs all of its calculations internally in SI and performs unit handling only when inputting the *.DATA file and when outputting result files. Thanks to [Irijkels](#) for reporting the issue. See [#4597](#) for details.
- 15) There is an issue associated with restarting from a restart file with the solution gas (Rs) maximum rate of increase, as defined by the [DRSDT](#) keyword in the [SCHEDULE](#), that has been set to zero. This is because, the simulator does not save/restore this setting in simulator's restart files, which means that simulator misses the essential value zero upon restarting the case. As a work-around one can use the option `--sched-restart=true`, when running the restart case. This will initialize the restarted simulation based on information from the complete [SCHEDULE](#) section, instead of just the parts that

we're going to simulate and the rest from the restart file. Thanks to [goncalvesmachadoc](#) for reporting the issue. See [#4272](#) for details.

- 16) OPM cannot be built with dune-fem version 2.9 or later ([#4934](#)). Please use a previous version.
- 17) The simulator uses an irregular corner-point grid geometry with adjusted pore volumes to represent radial grids so it is not possible to create a full ring (360 degree disk) with only one cell in the theta direction (NY=1). The work around is to model a slice (say [DTHETA=60](#) degrees). Note that as the angle increases larger pore volume adjustments are required ([#4755](#)).
- 18) In some cases with the network option the simulator can wrongly report that a well has no THP constraints, for example

```
GLIFT WTEST: Well S-3H does not have THP constraints
```

when THP constraints have been defined ([#4887](#)).

- 19) Dispersion in the gas phase leads to convergence issues for the 11th SPE CSP (<https://spe.org/csp>) models Version 11B and Version 11C (and is not fully tested). Therefore dispersion in the gas phase has been temporarily removed and a warning added ([#5101](#), [#859](#)).

B.3 RELEASE 2024-10

The 2024-10 release consists of some new features and various improvements and bug fixes. These include the following highlights:

- Linear solver default parameters have been updated to improve accuracy and performance including.
- Add summary output vectors for stranded CO₂ in the gas phase (GMST) and maximum potential residual trapping (GMTR).
- Added PSI and OMEGA parameters to *DRSDTCON* to control the regime and rate of convective mixing. Now supports *GAS/WATER* systems in addition to *OIL/GAS* systems.
- Support has been added for MPI communication being performed directly between GPUs instead of being sent via the CPU when using a multi-gpu linear solver.
- Added support for *WTMULT* to multiply a well target or constraint by a constant.
- Simulator is now able to make better usage of the Damaris middleware if enabled (<https://project.inria.fr/damaris/>).
- Extended the tracer model to include partitioning of tracers into free and solution states whenever dissolved gas/vaporized oil exist.
- Added support for automatic chokes (*NODEPROP* item 3 CHOKE) in extended network models.
- Added support for Killough's hysteresis model for both wetting and non-wetting phase hysteresis in a water wet system (*EHYSTR* item 2 HYSTMOD = 4).
- Added the option to specify the linear solver for subdomains when using the NLDD nonlinear solver.
- Added foundations for future experiments using the four-byte float type instead of the eight-byte double type as the primary array element in order to save memory.
- Support added for RFIP and SFIP mnemonics for the *RPTRST* keyword.
- Support added for mnemonic WELLS=N for the *RPTSCHED* keyword.
- Previously the THP was set to zero if a well was stopped or set to a zero rate (well or group). The THP will now be calculated provided a VFP table is active.
- Corrected transmissibilities calculated for pinched out cells when using *PINCH* option 4 equals *ALL*.
- Support added for the *EQUALREG* keyword.
- Added support for using METIS in addition to Zoltan to partition corner point grids.
- Proper *INIT* file table output for the saturation functions in LET format.
- Add support for outputting *MULTPV* to the Restart file.
- Add support for additional keywords in *PYACTION* blocks.
- Added support for *DENAQA* and *VISCAQA* keywords when using *CO2STORE*. Also added support for *SALTMF* keyword.
- Support has been expanded for saving/restoring scaling factors for segment level devices (SICDs, AICDs and Valves) to/from the Restart file.
- Added support for output of well level target or constraint summary vectors.
- Added support for field, group and well gas mass injection rate and injection total summary vectors for use with *CO2STORE*.
- Proper handling of the *WHISTCTL* in restarted simulations.

- Expanded consistency checks for array operations such as [ADD](#), [COPY](#), and [MULTIPLY](#), as well as the per-region counterparts.
- Performance improvements in well name matching on platforms which support the Posix fnmatch() function.
- Initial and experimental support for compositional simulations—currently supporting 2 to 7 components.
- Fixed two-phase gas/water and gas/oil model initialisation (equilibration).

B.3.1 NEW AND DEPRECATED COMMAND LINE OPTIONS

The major command line changes made for this release are summarized in Table B.2

OPM Flow 2024-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
1	--action-parsing-strictness	Added a new command line parameter that specifies the parsing strictness for ACTIONX and PYACTION blocks (#5547 , #4167). A defined character string. Set strictness of parsing process for ActionX and PyAction. Available options are normal (do not apply keywords that have not been tested for ActionX or PyAction) and low (try to apply all keywords, beware: the simulation outcome might be incorrect).	"normal"
2	--check-group-constraints-inner-well-iterations	A boolean value. Allow checking of group constraints during inner well iterations.	true
3	--convergence-monitoring	A new convergence monitoring feature (#5590) has been introduced to improve the robustness and efficiency of the simulator based on Moyner et al., 2024 (https://doi.org/10.3997/2214-4609.202437057). A boolean value. Enable convergence monitoring.	false
4	--convergence-monitoring-cut-off	An integer value. Cut off limit for convergence monitoring.	6
5	--convergence-monitoring-decay-factor	A real value. Decay factor for convergence monitoring.	0.75
6	--enable-drift-compensation	Linear solver default parameters have been updated to improve accuracy and performance including: material balance tolerance reduced to 1.0e-7, local well solve control switching is turned on, use implicit IPR is turned on, and enable drift compensation is turned off (#5157). A boolean value. Enable partial compensation of systematic mass losses via the source term of the next time step.	false

OPM Flow 2024-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
7	--imbalance-tol	A real value. Tolerable imbalance of the load balancing.	1.1
8	--inj-mult-damp-mult	Added damping to the injectivity multipliers defined by the WINJMULT keyword (#5631). A real value. Injection multiplier dampening factor (dampening multiplied by this each time oscillation is detected).	0.9
9	--inj-mult-min-damp-factor	A real value. Minimum injection multiplier dampening factor (maximum dampening level).	0.05
10	--inj-mult-osc-threshold	A real value. Injection multiplier oscillation threshold (used for multiplier dampening).	0.1
11	--input-skip-mode	Added a command line argument to control the way OPM Flow treats SKIP100 and SKIP300 keywords (#5329). By default OPM Flow behaves like the commercial black-oil simulator. A defined character string. Set compatibility mode for the SKIP100/SKIP300 keywords. Options are: 100 (skip SKIP100 ..ENDSKIP, keep SKIP300 ..ENDSKIP), 300 (skip SKIP300 ..ENDSKIP, keep SKIP100 ..ENDSKIP) or all (skip both SKIP100 ..ENDSKIP and SKIP300 ..ENDSKIP).	"100"
12	--linear-solver	A defined character string. Configuration of solver. Valid options are: cprw (default), ilu0, dilu, cpr (an alias for cprw), cpr_quasiimpes, cpr_trueimpes, cpr_trueimpesanalytic, amg or hybrid (experimental). Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json'.	"cprw"
13	--local-well-solve-control-switching	A boolean value. Allow control switching during local well solutions.	true
14	--metis-params	A character string. Configuration of Metis partitioner. You can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See http://glaros.dtc.umn.edu/gkhome/fetch/sw/memis/manual.pdf for available METIS options.	"default"
15	--min-strict-cnv-iter	An integer value. Minimum number of Newton iterations before relaxed tolerances can be used for the CNV convergence criterion.	-1

OPM Flow 2024-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
16	--nlld-local-linear-solver	Added the option to specify the linear solver for subdomains when using the NLDD nonlinear solver (#5553). Previously, the simulator was hardcoded to use the ILU0 solver with the default maximum number of iterations and default reduction tolerance. A defined character string. Configuration of NLDD local linear solver. Valid options are: ilu0 (default), dilu, cpr_quasiimpes and amg. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json'.	"ilu0"
17	--nlld-local-linear-solver-max-iter	An integer value. The maximum number of iterations of the NLDD local linear solver.	200
18	--nlld-local-linear-solver-reduction	A real value. The minimum reduction of the residual which the NLDD local linear solver must achieve.	0.01
19	--partition-method	Added support for using METIS in addition to Zoltan to partition corner point grids (#725). An integer value. Choose partitioning strategy: 0=simple, 1=Zoltan, 2=METIS.	1
20	--tolerance-cnv-energy	Added separate energy convergence tolerance and energy balance tolerance (#5321). Previously the tolerances for mass were used. A real value. Local energy convergence tolerance (maximum of local energy errors).	0.01
21	--tolerance-cnv-energy-relaxed	A real value. Relaxed local energy convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0
22	--tolerance-energy-balance	A real value. Tolerated energy balance error relative to (scaled) total energy present.	1e-07
23	--tolerance-energy-balance-relaxed	A real value. Relaxed tolerated energy balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
24	--tolerance-mb	A real value. Tolerated mass balance error relative to total mass present.	1e-07
25	--use-implicit-ipr	A boolean value. Compute implicit IPR for stability checks and stable solution search.	true

OPM Flow 2024-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
26	--print-properties	Support for this command line option has been removed (#5514). An integer value. Print the values of the compile time properties at the start of the simulation.	2
27	--zoltan-imbalance-tol	A real value. Tolerable imbalance of the loadbalancing provided by Zoltan. This command line option has been deprecated; --imbalance-tol should be used instead.	1.1
Notes: - Cells colored in green in the "No." column indicate a new command line option or a change to an existing option for this release. Cells colored green in the "Default" column indicate that the default value for an existing option has been changed. - Cells colored in orange in the "No." column indicate the command line option is available but is now "hidden" from the main help listing, --help. These options can be listed using the --help-all command line option, that lists all the command line options included in the release, including experimental, obsolete, hidden and deprecated options. - Cells colored in red in the "No." column indicate a deprecated command line option for this release.			

Table B.3: OPM Flow 2024-10 New and Deprecated Command Line Options

B.3.2 NEW FEATURES

In addition to the changes to the command line options the following new features have been added to the simulator:

- 1) Add summary output vectors for stranded CO₂ in the gas phase (GMST) and maximum potential residual trapping (GMTR) ([#5344](#), https://docs.opengosim.com/theory/residual_and_stranded/).
- 2) Added support for [WTMULT](#) to multiply a well target or constraint by a constant ([#5411](#)).
- 3) Simulator is now able to make better usage of the Damaris middleware if enabled (<https://project.inria.fr/damaris/>). Added the ability to pass multiple variables to Damaris using the DamarisWriter class ([#5352](#)). A command line option has also been added to limit the variables to be passed using --damaris-limit-variables=<CSV list>.
- 4) Restart files can now be output every time step if the command line option --enable-write-all-solutions=true ([#5426](#)). The numbering of the restart and summary files will follow the time step index and not the report index. This will not be compatible with restart, but it is sometimes useful to investigate the results at every time step. Note that this doesn't add support for [RPTRST](#) BASIC=6.
- 5) Added support for automatic chokes ([NODEPROP](#) item 3 CHOKE) in extended network models ([#4935](#)).
- 6) Added support for using [CPR](#) on AMD GPU using the rocSPARSE library ([#5408](#)).
- 7) Added support for outputting the RSWSAT and RVWSAT arrays to the Restart file ([#5483](#)).
- 8) Added PSI and OMEGA parameters to [DRSDTCN](#) to control the regime and rate of convective mixing ([#5491](#)). Now supports [GAS/WATER](#) systems in addition to [OIL/GAS](#) systems.
- 9) Added an OPM implementation of the ILU0 preconditioner in CUDA similar to that implemented in CuDILU ([#5441](#)).

- 10) Added support for Killough's hysteresis model for both wetting and non-wetting phase hysteresis in a water wet system ([EHYSTR](#) item 2 HYSTMOD = 4) (#5273).
- 11) Extended implementation of [DRSDTCON](#) to apply to multiple PVT regions (#5527). Also added support for deactivation of individual [DRSDT](#), [DRVDT](#) and [DRSDTCON](#) regions.
- 12) Added foundations for future experiments using the four-byte float type instead of the eight-byte double type as the primary array element in order to save memory (#5560).
- 13) Support added for RFIP and SFIP mnemonics for the [RPTRST](#) keyword to request output of fluid in place volumes at surface and reservoir conditions to the Restart file (#5621). Note that [FIP](#) is an alias for SFIP.
- 14) Support added for mnemonic WELLS=N for the [RPTSCHEd](#) keyword to request output of well reports to the Print file (#5626).
- 15) Support added for the [EQUALREG](#) keyword to specify property arrays by region (#5648).
- 16) Added a failure flag to the TaskletRunner allowing errors in a tasklet to be caught and logged (#909).
- 17) Add support for outputting [MULTPV](#) to the Restart file if it is specified in the [GRID](#) or [EDIT](#) section (#4021).
- 18) Add support for [WELSPECs](#), [COMPSEGS](#), [WELSEGS](#), [WSEGVAlV](#), [WECON](#), [WTEST](#), [WGRUPCON](#), [GCONSUMP](#), [GRUPTREE](#) and [EXIT](#) keywords in [PYACTION](#) blocks (#4059, #4061, #4069, #4070, #4071).
- 19) Added support for [CNAMES](#) to specify the component names (#5405, #4091).
- 20) Added support for [DENAQA](#) and [VISCAQA](#) keywords to specify parameters for calculating brine density and viscosity using Ezrokhi's method when using [CO2STORE](#) (#4091). Also added support for [SALTMF](#) to allow the initial salt concentration to be input as a mole fraction.
- 21) Extended the tracer model to include partitioning of tracers into free and solution states whenever dissolved gas/vaporized oil exist (#5268, #4001). Essentially the tracer linear system has been extended with equations for free and solution tracers with simple mass transfer coupling between them. Both free and solution tracers are now output to the Restart file.
- 22) Added support for output of well level target or constraint summary vectors: WBHPT, W[OGLVW]PRT, and W[OGVW]IRT where O, G, L, V and W represent oil, gas, liquid, reservoir volume and water respectively (#4093).
- 23) Initial support for enumerated compositional initialisation (#4110).
- 24) Added support for [FGW]GMI[RT] field, group and well gas mass injection rate and injection total summary vectors for use with [CO2STORE](#) (#4162).
- 25) A mixing model has been added to the thermal black-oil simulator (#4014). The internal energy of each phase now also includes the energy for the dissolved component. The meaning of the [TEMP](#) keyword has been changed. In the Thermal model ([THERMAL](#)) enthalpy equals internal energy plus work. Whereas, the Temperature model ([TEMP](#)) now assumes enthalpy equals internal energy.
- 26) Added support for the [COPY](#), [OPERATE](#), [ADDREG](#), [COPYREG](#), [EQUALREG](#), and [MULTIREG](#) keywords for data with global storage (#4236, #4237, #4238).
- 27) Extended API to allow a source term to be added to a cell (#4241). In particular this allows a source term to be generated by Python code without needing to pass in a DeckRecord.
- 28) Added parsing support for ZMF keyword (#4200).
- 29) The convertECL utility has been updated to support conversion of [LGR](#) output between the unformatted (.LGR) and formatted (.FLGR) file types (#4249).

B.3.3 IMPROVEMENTS

Improvements to simulator in this release include the following:

- 1) Support has been added for MPI communication being performed directly between GPUs instead of being sent via the CPU when using a multi-gnu linear solver (#5145).
- 2) Python bindings can now control whether `MPI_Init()` and `MPI_Finalize()` will be called when creating an `Opm::Main` object (#5325). This allows the Python unit test files to run more than a single unit test.
- 3) The fraction of pore-volume whose CNV targets are violated as a new per-iteration quantity is now included in the INFOITER file (`--output-extra-convergence-info=iteration`), with the column header `CnvErrPvFrac` (#5302).
- 4) Support has been added for using hipify on the CUDA code to allow for GPU support on both AMD and Nvidia architectures (#5253).
- 5) The simulator now checks if relaxed tolerances are tighter than the strict tolerances and if so adjusts the relaxed tolerances (#5337).
- 6) The code now resets the group pressure maintenance state to avoid oscillations when the target pressure is reached and the pressure maintenance rate is set to zero (#5350).
- 7) Optimized the memory operations time by overlapping the creation (which includes a host memory copy) of the Jacobian matrix used in the block Jacobi ILU with copying the matrix data to the GPU (#5256).
- 8) SummaryState Objects are now Aware of Undefined `UDQ` Values (#5334).
- 9) Previously the code would only start handling oscillations when two phases had oscillating residuals. The code will now start handling oscillations when just one phase is oscillating (#5398).
- 10) Added Newton update damping for the situation where the solution for a domain oscillates (#5342).
- 11) Reduced execution time for parallel AMG/CPR simulations. Allowed use of AMG with command line option `--matrix-add-well-contributions=false` (#5182).
- 12) The code has been updated to only issue warnings the first time the transmissibilities are calculated thus removing spurious and duplicate warnings that can occur when using load balancing (#5351).
- 13) Nearly all exceptions thrown when computing wells will not abort the simulator but result in a timestep chop. Hence those should not be counted as errors and be reported in the PRT file. The simulator now only reports problems for exceptions thrown when updating the well potentials (#5423).
- 14) The well economic (`WECON`) minimum rates are no longer applied if the well is subject to a zero group rate target or constraint (#5424).
- 15) Directly output the SGMAX, SHMAX, SOMAX, SOMIN and SWHYI arrays rather than the KRNSW_GO, KRNSW_OW, PCSWM_GO, PCSWM_OW and SOMAX arrays to the Restart file for cases with relative permeability and/or capillary pressure hysteresis (#5402).
- 16) Sphinx documentation (<https://opm.github.io/opm-python-documentation/master/index.html>) has been added for the `opm.simulators.BlackOilSimulator` Python module (#5243).
- 17) The simulator now more dynamically picks the thread block size when using CUDA giving up to a 50% speed up on linear iterations (#5433).
- 18) If OpenMP is available then the simulator will now default to two threads per process unless the `OMP_NUM_THREADS` environment variable is set or the command line option `--threads-per-process` is set. A warning will be issued if both the environment variable and the command line option are set, but the environment variable will take precedence (#5454).
- 19) Visualization Toolkit (VTK) files can now be output for dry runs by using the `--enable-vtk-output=I --enable-dry-run=I` command line options (#5499).

- 20) Linear solver default parameters have been updated to improve accuracy and performance including: material balance tolerance reduced to 1.0e-7, local well solve control switching is turned on, use implicit IPR is turned on, and enable drift compensation is turned off (#5157).
- 21) A more helpful message is now output if the `PETGRID` keyword is used to try to load a generic simulation grid (*.GSG) file (#5541): "OPM Flow cannot read GSG format grid input, re-export case from Petrel using GRDECL instead."
- 22) The check for active wells as a condition for drift compensation has been removed (#5110). For CO₂ injection simulation the drift compensation is still beneficial even when wells are no longer active.
- 23) The command line option `--cpr-reuse-setup` now only applies to the `CPR` preconditioner (#5445).
- 24) The simulator now uses a mobility weighted average of cell level densities for the connection level mixture densities in no-flow producing wells (#5479). This mobility weighted average approach gives a more monotone pressure build up.
- 25) Output cell counts and pore volume fractions to INFOITER file for cells that: (1) satisfy the strict convergence tolerance, (2) satisfy the relaxed but not the strict convergence tolerance, (3) do not satisfy the relaxed convergence tolerance (#5338).
- 26) Consistency checks have been added for oil, gas and water phase saturation functions (#5570, #5571, #5572).
- 27) The simulator now avoids computing the default number of domains based on a specific target number of cells per domain if the user chooses a specific number of NLDD domains (#5579).
- 28) The simulator now avoids doing unnecessary updates in the numerical aquifer calculations (#5632).
- 29) Previously the THP was set to zero if a well was stopped or set to a zero rate (well or group). The THP will now be calculated provided a VFP table is active (#5634).
- 30) Consistency checks have been added for three-point saturation scaling `SCALECRS=YES` (#5573, #5640). These checks collectively guarantee a mobile displacing oil saturation in the two phase gas-oil and oil-water systems.
- 31) The dependencies on the OPM PyPi packages have been updated to add support for newer versions of Python, versions 3.6 to 3.12 are now supported (#5644).
- 32) A new strategy for preventing wells from oscillating between being stopped and revived has been introduced (#5659). The cycle of oscillations are broken by requiring that for a well to remain stopped after it has been stopped in the local solve, that it should have had zero rate when it entered the local solve, meaning that both the global and local solve agree that the well should be stopped.
- 33) It is no longer necessary to use add well contributions to matrix for NLDD. Also, the CPRW preconditioner is now supported for NLDD (#5341, #5676).
- 34) The number of pinch-out connections generated will now be reported to the print file (#767).
- 35) An error message will now be generated if the `COMPSEGS` keyword is used before the multi-segment well is defined using the `WELSEGS` keyword (#4031).
- 36) The reservoir volume rate (RESV) constraint for a history matching producer is evaluated by converting the observed oil, water and gas surface rates to the current reservoir conditions hence its value is not fixed. However, to test if the well is subject to a zero reservoir volume rate constraint it is sufficient to test if all of the observed rates are zero (#4039).
- 37) The path of the Restart file (either absolute or relative to the .DATA file) as specified by the `RESTART` keyword is now saved for writing to the SMSPEC file (#4056).
- 38) Restart file paths of between 73 and 132 characters in length are now also written to the SMSPEC file (#4057).
- 39) Support has been added for formatted FMSPEC files in the summary utility (#4080).

- 40) Improved the reading of possibly corrupt summary files (#4079).
- 41) The `ACTNUM` array is now updated immediately after scanning the `GRID` and `EDIT` sections (#4082). This avoids problems where cells are deactivated by setting `PORO` equal to zero while at the same time setting regions numbers for these cells to invalid values (which is not currently supported).
- 42) The compatibility with restart files has been improved specifically reading the well closure reason for history matching wells, and not using a guide rate model for wells with a target type of NONE (#4090).
- 43) The `INIT` file output code's assumption that tabulated saturation functions have at most NSSFUN (`TABDIMS` keyword item 3) rows per saturation region has been relaxed (#4066). This allows larger saturation function tables to be output to the `INIT` file without an error being generated.
- 44) Added support for restarting cases with group injection controlled voidage replacement (#4108).
- 45) Support has been expanded for saving/restoring scaling factors for segment level devices (SICDs, AICDs and Valves) to/from the Restart file (#3978).
- 46) Avoid unnecessarily checking if the fluid is saturated in `CO2STORE` and `H2STORE` cases (#4125).
- 47) If the injection fluid temperature has not been set for an injection well then it now defaults to the cell temperature in the upper most completion (#5490, #4155, #4152).
- 48) A warning message will now be generated and the well connection will be skipped if the well connection is in a cell that will be deactivated due a pore volume cut off specified by the `MINPV` or `MINPVV` keyword (#4174). This also avoids a crash that would have occurred if any such well was re-completed at a later stage in the simulation.
- 49) Restart support has been added for `DRSDT` (#4166).
- 50) The code now uses the failure flag of the tasklet runner to detect a failure while writing output and throw an error if this happens (#5478, #4178).
- 51) Added ESMRY methods to get a list of dates for summary results and obtain the units for a summary vector have been exported to Python (#4184).
- 52) Add support for converting between formatted (.FGRID) and unformatted (.GRID) grid files to the convertECL utility (#4208).
- 53) An error message will now be generated if an attempt is made to copy an undefined array (for example using the `COPY` keyword). Similarly, operations on an array (for example the `ADD` keyword) other than assignment (for example using the `EQUALS` keyword) will now generate an error message if the array being operated on has not been previously been defined in the input deck (#4109, #4235). However, operations are allowed on multiplier arrays (`MULTX`, `MULTY`, etc.) that have not been explicitly defined in the input deck but have a default value of one (#4235).
- 54) Performance improvements in well name matching on platforms which support the Posix fnmatch() function (#4243).

B.3.4 BUG FIXES

The following bug fixes and improvements have been incorporated into this release:

- 1) Avoid sending and receiving from the same Buffer to fix errors with MPICH on Red Hat Linux (#5297).
- 2) Redistribute gaslift for groups with an ALQ limit (#5296). Previously, gaslift was only redistributed for groups limited by rate.
- 3) Fixed output of CO₂ mass in place (#5324).
- 4) Initialise the reservoir temperature in `CO2STORE` restart runs using the initial reservoir temperature if no temperatures are stored in the restart file, as is now the case for isothermal runs (#5331).

- 5) Mark the option 'BASIC=0' as allowed for the [RPTST](#) keyword ([#5335](#)).
- 6) Corrected definitions of GMDS and GMGP summary vectors ([#5292](#)).
- 7) Fixed a typo that would cause problems when compiling using CUDA-aware MPI ([#5368](#)).
- 8) When checking for zero target rates, also check wells under group control where required ([#5232](#)).
- 9) Avoid possible use of out-of-scope temporary ([#5393](#)).
- 10) Fixed an issue where a [UDQ](#) assignment would not be performed if it was triggered on the last time step of report step ([#5379](#)).
- 11) Three small fixes: Make sure that the same explicit/implicit fraction is used when computing THP(BHP) and when computing BHP(THP). Reduced risk of switching control in case of extreme VFP-extrapolation. Disallow local well control switching when the maximum number of well switches has been reached ([#5395](#)).
- 12) Fix the modification of transmissibility arrays (for example, using the [MULTIPLY](#) keyword in the [EDIT](#) section to modify [TRANX](#)) when running in parallel ([#5414](#)).
- 13) Avoid recursively deleting results in all subfolders if the command line argument --output-dir is empty ([#5427](#)).
- 14) Avoid mass creation when using [DRSDT](#) or [DRSDTCON](#) if hysteresis was deactivated ([#5407](#)).
- 15) If convergence fails due to NaN then report this as a problem rather than an error, then chop the time step and continue ([#5434](#)).
- 16) Support was previously added for [GCONPROD](#) items I1 to I3 but generally did not take effect. This has been now been fixed ([#5432](#)).
- 17) Check that a well exists before closing it due to a group action ([#5462](#)).
- 18) Switch to using size_t instead of int for buffer position to deal with buffers exceeding size intmax in MPI calls ([#5473](#)).
- 19) Avoid generating an exception from std::stoi() if the environment variable OPM_NUM_THREADS is not set ([#5504](#), [#5516](#)).
- 20) Ensure default ALQ is set for all production wells on all processes. Fixes an MPI bug for cases where a well switches from producer to injector at the restart step ([#5534](#)).
- 21) Avoids an exception that would be thrown if one or more wells are shut in a [CO2STORE](#) run ([#5546](#)).
- 22) Avoid deleting derivatives in the case of zero pressure or zero flux that would have resulted in an incorrect matrix ([#5236](#)).
- 23) Fixed conditional expression in [DRVDT](#) initialisation code ([#5565](#)).
- 24) Set the correct number of target iterations in the simple iteration timestep controller ([#5476](#)).
- 25) Ensure well and accumulated group efficiency factors are applied to the connection energy rates ([#5323](#)).
- 26) Fixed calculation of injection well BHP for [CO2STORE](#) and [H2STORE](#) runs ([#5602](#)).
- 27) Fixed output of well temperature ([#5305](#)).
- 28) Avoid adding gas lift gas of non-open wells to the network ([#5601](#)). Also allow non-zero artificial lift quantities (ALQ) for extended network models.
- 29) Fixed an indexing error in the PID controller formula as used in the time stepping algorithm ([#4855](#)).
- 30) SFC (Space-Filling Curve) reordering for cells in AluGrid is now the default choice to optimise data access patterns for adaptive mesh refinements/coarsening ([#5630](#)).

- 31) Actually use fixed ordering when constructing matrices in the AMG hierarchy to get reproducible results in parallel runs ([#5646](#)).
- 32) Corrected transmissibilities calculated for pinched out cells when using [PINCH](#) option 4 equals [ALL](#) ([#5577](#)).
- 33) Call method with the correct precision in the rocSPARSE solver ([#5655](#)).
- 34) The lookup index for tracer concentrations when restarting parallel runs has been fixed ([#5670](#)).
- 35) Fixed a memory bug that only occurred when using the NLDD non-linear solver when there was an unconverged local NLDD domain, which had a multi-segment well on that domain ([#5676](#)).
- 36) Fixed a bug which prevented restarts for gas-water cases ([#5681](#)).
- 37) Solution data is now marked as invalid at the end of a time step in the experimental black-oil simulator (flowexp_blackoil). Previously, many quantities were not evaluated or updated and output to the Restart file ([#5685](#)).
- 38) Initialise the water saturation to one if the primary variable is switched from water saturation to gas-water ratio ([#908](#)).
- 39) Add scoped and named variable to avoid using out-of-scope temporary ([#724](#)).
- 40) Make methods that handle well connections aware of possible future connections that might be opened in [ACTIONX](#) ([#742](#)).
- 41) Updated the molar mass of n-Decane ("C10") defined in the code from 0.0142 to 0.142 kg/gmol ([#4037](#)).
- 42) Change the dimension for critical temperature ([TCRIT](#)) from temperature to absolute temperature ([#4041](#)).
- 43) Previously three modules were exported to Python for use with [PYACTION](#), now only opmcommon_python is exported ([#4029](#)).
- 44) Avoid computing a well THP if the VFP table is defaulted or explicitly set to zero ([#4058](#)).
- 45) Proper [INIT](#) file table output for the saturation functions in LET format ([#4051](#)).
- 46) Fix deadlock when outputting [MULTPV](#) to [INIT](#) file in parallel ([#4068](#)).
- 47) The value specified by the [UDQPARAM](#) keyword for undefined UDQs is now used as a fallback default for undefined or missing UDQs, in particular well, group or segment UDQs ([#4054](#)).
- 48) Fixed comparison and serialization from Runspec object where multiple members were ignored ([#4086](#)).
- 49) Fix output of LET curves by taking into account the connate saturations ([#4085](#)).
- 50) Trigger an update of the well targets when running [WTMULT](#) in an [ACTIONX](#) block ([#4095](#)).
- 51) Fix salinity input in [CO2STORE](#) module and allow [CO2STORE](#) to be used with multiple PVT regions ([#4099](#)).
- 52) Wells modified by user-defined arguments (UDA) of a [WSEGVALLV](#) keyword will be updated during time steps not only report steps ([#4121](#)).
- 53) Use gas-water ratio ([RSW](#)) rather than saturated gas-water ratio (RSWSAT) when calculating water density ([#4124](#)).
- 54) Hysteresis parameter get and set methods now work directly on min/max saturations rather than relative permeability and capillary pressure values to avoid conversion in the output layer ([#4089](#)). Also, imbibition and drainage curves were mixed up, and connate water saturation was incorrectly added to the drainage wetting phase critical saturation.

- 55) Correctly iterate through file deck container while erasing. This would fail if for example the last keyword of a block is erased. Also the iteration endpoint was not updated correctly ([#4122](#)).
- 56) Avoid referencing object until its index is known to be good ([#4127](#)). This is the minimal change needed to allow the use of field level user defined quantities to define group input/production targets. However, such models cannot be reliably restarted.
- 57) Fix warning message when resetting control modes other than 'RATE','BHP' to 'RATE' for historical injection wells ([#4138](#)).
- 58) Avoid duplicating objects on pointer serialization/de-serialization ([#4141](#)).
- 59) Avoid division by zero when evaluating corner point coordinates if coordinate lines ([COORD](#)) have equal top and bottom depths ([#4145](#)).
- 60) Ensure that the unit conversion offset is not discarded when parsing Temperature units ([#4155](#)). Also the units for the Joule-Thomson coefficient have been changed from Temperature/Pressure to AbsoluteTemperature/Pressure on the [GASJT](#), [OILJT](#) and [WATJT](#) keywords, and the units for the thermal expansion ratio for a cell in mechanics models have been changed from I/Temperature to I/AbsoluteTemperature on the THERMEXR keyword.
- 61) The summary tool now prints the help message and returns failure if you run it with no parameters (or no parameters after the options). Previously, the tool would generate a segmentation fault ([#4156](#)).
- 62) Wetting phase hysteresis model has been fixed. The formulation as in Killough 1976 is followed more faithfully, in particular, the non-wetting phase saturation is now used to determine the turning points ([#4160](#)).
- 63) Use correct bhp limit for production wells under historic control when restarting a run ([#4112](#)).
- 64) Fix assignment of selected well variables using the [UDQ](#) keyword, and improve handling of quoted well names ([#4177](#)).
- 65) Proper handling of the [WHISTCTL](#) in restarted simulations ([#4119](#)).
- 66) The serializer no longer passes null pointers to the packers, and a workaround for Dune 2.6 has been removed ([#4192](#)).
- 67) Flag indicating whether the parameters for the logarithm-based polymer shear thinning/thickening option include a reference temperature is now assigned to the correct member of the [PLYSHLOG](#) table object ([#4195](#)).
- 68) Fixed serialization and output of Ezrokhi density coefficients for parallel runs ([#4209](#)).
- 69) Fix limit on maximum number of iterations in flash calculation ([#4214](#)).
- 70) Added missing throw statements for logic errors ([#4203](#)).
- 71) If the VFP table number is defaulted for history wells using [WCONHIST](#) or [WCONINJH](#) then the previously defined value now will be used ([#4228](#)).
- 72) Previously, a warning 'well is not connected to the grid' is output if any connection is not connected to the grid. Now a warning message is only output if the well is completely unconnected from the grid after the entire [COMPDAT](#) record has been processed ([#4223](#)).
- 73) Added missing UDA Dimension for group production ([GCONPROD](#)) gas rate target or constraint (GRAT). Without this a restarted simulation with UDAs for GCONPROD's GRAT target or constraint with fail to load ([#4248](#)).
- 74) Fixed a bug where well and group rates that can be specified by UDAs were converted to SI units twice ([#4269](#)).
- 75) The code now correctly identifies when a new well object is encountered ([#4278](#)). Previously, the simulator would produce inconsistent results for polymer cases run in parallel.

B.3.5 KNOWN ISSUES

Known issues in this release of the simulator include:

- 1) There is an error with the RSM header for summary vectors whose NUMS entry in the SMSPEC file is derived from more than a single number source (e.g., single region or segment ID). This applies to all block vectors (BGPV, BOPV, BWPV, etc.), connection level quantities (COPT, etc.), and inter-region flows such as ROFT etc ([#3078](#)). The work around is to plot the data in OPM ResInsight and right-click on the plot to view and copy the data.
- 2) As per previous releases of the radial model, the [COORDSYS](#) keyword item three must be set to COMP to complete the circle, this has not been implemented in this release. Also there appears to be a bug for full radial models when a well goes on BHP control that causes the well not to respect the BHP constraint, this eventually causes the well to die prematurely. See [#2640](#) for a discussion on the topic.
- 3) As in previous releases there are some issues with the [OPERATE](#) and [OPERATER](#) keywords associated with the input parsing; for various reasons a few of the fields require special case treatment in the grid processing, including (at least) [MULTZ](#), [PORV](#) and [ACTNUM](#), and for those keywords the [OPERATE/OPERATER](#) keyword do not work. The work around is to use the [MULTIPLY](#) keyword instead.
- 4) For the [UDQ](#) ASSIGN operator after the terminating "/" normally any comments can be entered; however, if there is "/" within the comment field, as per:

```
ASSIGN FUNGLYLD 1.196 / Condensate Yield (63.5/56.7)/(1.0 - 0.065)
```

then the simulator will abort. The work around is to manually place the comment characters "--" after the ASSIGN terminating "/", like so:

```
ASSIGN FUNGLYLD 1.196 / -- Condensate Yield (63.5/56.7)/(1.0 - 0.065)
```

- 5) At the moment, one cannot initialize tracers using the [EQUALS](#) keyword. Instead use the array format, that is the keyword followed by the required number of values, or the [TVDP](#) keyword in the [SOLUTION](#) section to set the initial tracer concentrations as a function of depth.
- 6) Currently, gas tracers cannot be used if the dissolved gas phase, as per the [DISGAS](#) keyword in the [RUNSPEC](#) section, is active in the model.
- 7) The summary vector RTIPTHEA, that defines the energy in-place between the initial and the current time for regions, is not supported unlike the FTIPTHEA and BTIPTHEA vectors. Secondly, the error message:

```
Warning: Problem with summary keyword RTIPTHEA
In RSM-THERMAL.data line 492
FIP region FIPHEA not defined in REGIONS section - RTIPTHEA ignored
```

is incorrect, as the message indicates that it is being treated like a named region, as per the [FIP](#) keyword, when it is actually a [SUMMARY](#) vector ([#3870](#)).

- 8) If there are cells that are very distorted, which can occur near fault planes, then the simulator may abort because it cannot calculate the pore volume of such cells. The work around is to re-generate the grid in the static model, taking care that the cells around the fault planes are more or less orthogonal ([#2992](#) and [#3770](#)).
- 9) Currently the [OPERATER](#) keyword in the [EDIT](#) section does not work with the [DEPTH](#), [TRANX](#), [TRANX](#) and [TRANZ](#) property arrays ([#2994](#) and [#748](#)).
- 10) If a standard well is fully declared in an [ACTIONX](#) block which is then activated at a later date, and later the well is modified to be a multi-segment well using the [WELSEGS](#) and [COMPSEGS](#) keywords, then this will cause the simulator to abort with an assert failure. The solution to this issue is to not use this type of work flow in declaring wells ([#2891](#) and [#2895](#)).

- 11) Although the [ACTIONX EXIT](#) command works as expected, it does not write out the requested RSM file at the end of the run. However, the other [SUMMARY](#) and [RESTART](#) files are written out ([#2877](#)).
- 12) Although the [GCONSUMP](#) keyword in the [SCHEDULE](#) section is fully implemented as documented, it is not possible to verify the output as the associated [SUMMARY](#) vectors are not written out, that is the [SUMMARY](#) sales gas vectors FGSR, FGST, GGSR and GGST, and fuel vectors FGCR, FGCT, GGCR, and GGCT have not been implemented ([#2679](#)).
- 13) There are small differences in the behavior of the [NEXTSTEP](#) keyword in the [RUNSPEC](#) section between OPM Flow and the commercial simulator that remain unresolved ([#3745](#)).
- 14) **There is a unit handling issue associated with [OPERATE](#) keyword. If the [OPERATE\(X\)](#) parameter has units, as for example [PERMX](#), then the conversion is always done in SI units, despite the input deck declaring the deck to be fields units, as per [FIELD](#) keyword in the [RUNSPEC](#) section. Note that OPM Flow performs all of its calculations internally in SI and performs unit handling only when inputting the *.DATA file and when outputting result files. Thanks to [Irijkels](#) for reporting the issue. See [#4597](#) for details.**
- 15) There is an issue associated with restarting from a restart file with the solution gas (Rs) maximum rate of increase, as defined by the [DRSDT](#) keyword in the [SCHEDULE](#), that has been set to zero. This is because, the simulator does not save/restore this setting in simulator's restart files, which means that simulator misses the essential value zero upon restarting the case. As a work-around one can use the option `-sched-restart=true`, when running the restart case. This will initialize the restarted simulation based on information from the complete [SCHEDULE](#) section, instead of just the parts that we're going to simulate and the rest from the restart file. Thanks to [goncalvesmachadoc](#) for reporting the issue. See [#4272](#) for details.
- 16) OPM cannot be built with dune-fem version 2.9 or later ([#4934](#)). Please use a previous version.
- 17) The simulator uses an irregular corner-point grid geometry with adjusted pore volumes to represent radial grids so it is not possible to create a full ring (360 degree disk) with only one cell in the theta direction (NY=1). The work around is to model a slice (say [DTHETA](#)=60 degrees). Note that as the angle increases larger pore volume adjustments are required ([#4755](#)).
- 18) In some cases with the network option the simulator can wrongly report that a well has no THP constraints, for example

GLIFT WTEST: Well S-3H does not have THP constraints

 when THP constraints have been defined ([#4887](#)).
- 19) Dispersion in the gas phase leads to convergence issues for the 11th SPE CSP (<https://spe.org/csp>) models Version 11B and Version 11C (and is not fully tested). Therefore dispersion in the gas phase has been temporarily removed and a warning added ([#5101](#), [#859](#)).

B.4 RELEASE 2024-04

The 2024-04 release consists of some new features and various improvements and bug fixes. The new features and improvements include the following highlights:

- Added support for the [CO2SOL](#) and [H2SOL](#) keywords for modeling CO₂ or H₂ injection in hydrocarbon reservoirs. These are similar to the existing [CO2STORE](#) and [H2STORE](#) keywords for modeling CO₂ or H₂ injection in saline aquifers.
- The option has been added to output the liquid phase H₂ mole fraction and vapor phase water mole fraction to the [RESTART](#) file when using the [H2STORE](#) option ([RPTRST](#) keyword mnemonics XMFH2 and YMFWAT).
- Added CO₂ mass in place summary vectors for the field and region level when using [CO2STORE](#), including trapped masses based on the definition of immobile gas in the 11th SPE Comparative Solution Project.
- Added support for the [SOURCE](#) keyword for modeling a simple source term.
- Support has been added for [WDFAC](#) and [WDFACCOR](#) keywords in the [SCHEDULE](#) section. The option of setting connection specific D factors (Item 12 in [COMPDAT](#)) is now also supported.
- Added support for setting [GCONPROD](#) keyword item 7 ([ACTION](#)) equal to WELL. Added support for reason equals G (group) to the [WTEST](#) keyword item 3 (TEST).
- Added support for the [DIFFAGAS](#) and [DIFFAWAT](#) keywords to define respectively the gas phase and water phase diffusion coefficients based on a mass fraction formulation.
- Support has been added for the [FBHPDEF](#) keyword in the [SCHEDULE](#) section to define the default well BHP target for production wells and the default BHP constraint for injection wells.
- Support has been added for the [SKIP](#), [SKIP100](#), [SKIP300](#) and [ENDSKIP](#) keywords.
- Support has been added for the [NONNC](#) keyword to deactivate non-neighbor connections.
- Added partial support for [DATUMR](#) and [DATUMRX](#) keywords in the [SOLUTION](#) section for use with block level summary keywords (BPP*).
- Limited support has been added for User Defined Tables ([UDT](#)).
- Added support for using region-level summary vectors in the defining expressions of field-level user defined quantities (UDQs).
- Support has been added for output of addition summary vectors, initialization arrays and restart arrays.
- A new brine-CO₂ mutual solubility model has been implemented from Spycher & Pruess, Transp. Porous Med., 2010. This model improves the accuracy of the solubility calculations in [CO2STORE](#) at temperatures above 100 degrees Celsius. In addition, a new keyword, [ACTCO2S](#), has been added to choose between activity models.
- Added support for the [PCFACT](#) keyword in the [PROPS](#) sections to define the capillary pressure multiplication factor as a function of porosity change; used in conjunction with OPM Flow's Salt Precipitation model.
- Added support for the [THCO2MIX](#) keyword in the [PROPS](#) section to specify the thermal mixing models for salt in water, CO₂ dissolved in brine, and water vaporized in CO₂.
- Added support for using well lists ([WLIST](#)) in [ACTIONX](#) blocks with the following keywords: [COMPDAT](#), [COMPLUMP](#), [WCONINJE](#), [WCONPROD](#), [WECON](#), [WEFAC](#), [WELOPEN](#), [WELPI](#), [WELSPECS](#), [WELTARG](#), [WGRUPCON](#), [WPIMULT](#), [WSEGVAVL](#), [WTEST](#) and [WTMULT](#).
- Added support for an expanded set of summary vectors to be used in defining expressions for UDQs and in the condition blocks of the [ACTIONX](#) keyword.

- Removed the need for a run-function in [PYACTION](#) blocks. EclipseState, Schedule, ReportStep and SummaryState are now available as attributes of the module opm_embedded. The code maintains backwards compatibility with the previous usage.
- Tooltips can now be enabled when writing code in a Python IDE by typing "import opm_embedded".
- An error will now be reported and the simulation stopped when the [PYACTION](#) keyword is used if flow was built without embedded Python.
- Repaired the [PYACTION](#) functions open_well(), shut_well() and stop_well(), which open/shut/stop a well at the specified report step, and added the option to use these functions at the current report step (if no report step is specified).
- Added [PYACTION](#) function insert_keyword() to insert a keyword at the specified later report step or at the current report step (if no report step is specified). This functionality is available for the keywords: [FIELD](#), [ENDBOX](#), [GCONINJE](#), [GCONPROD](#), [METRIC](#), [MULTX](#), [MULTX-](#), [MULTY](#), [MULTY-](#), [MULTZ](#), [MULTZ-](#), [NEXT](#), [NEXTSTEP](#), [WCONINJE](#), [WCONPROD](#), [WEFAC](#), [WEOPEN](#) and [WELTARG](#).

B.4.1 NEW AND DEPRECATED COMMAND LINE OPTIONS

The major command line changes made for this release are summarized in Table B.4

OPM Flow 2024-04 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
1	--accelerator-mode	Added rocspase option. A defined character string that selects the linear solver; usage: '--accelerator-mode=[none cusparse opencl amgcl rocalution rocspase]'. 	"none"
2	--debug-emit-cell-partition	Added new command line option. A boolean value that determines whether (true) or not (false) to emit cell partitions as a debugging aid (#4938). 	false
3	--explicit-rock-compaction	Added new command line option (#5089). A boolean value. Use pressure from end of the last time step when evaluating rock compaction (true) or not (false). 	false
4	--external-partition	Added new experimental command line option (#4882). A quoted character string. Name of file from which to load an externally generated partitioning of the model's active cells for MPI distribution purposes. If empty, the built-in partitioning method will be employed. 	""
5	--linear-solver	Added the dilu and cpr_trueimpesanalytic options (#5119 , #5002 , #4899). A defined character string that selects the configuration of solver. Valid options are: ilu0, dilu, cprw, cpr (an alias for cprw), cpr_quasiimpes, cpr_trueimpes, cpr_trueimpesanalytic, amg or hybrid (experimental). Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json'. 	"ilu0"

OPM Flow 2024-04 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
6	--local-domains-ordering-measure	Replaced the “pressure” option with the “maxpressure” and “averagepressure” options. A defined character string that selects the subdomain ordering measure. Allowed values are 'maxpressure', 'averagepressure' and 'residual'.	"maxpressure"
7	--local-solve-approach	Changed the default value from “jacobi” to “gauss-seidel”. A defined character string to choose the local solve approach. Valid choices are jacobi and gauss-seidel.	"gauss-seidel"
8	--local-tolerance-scaling-cnv	Changed the default value from 0.01 to 0.1. A positive real number. Set to a lower than 1.0 to use stricter convergence tolerance for local solves.	0.1
9	--min-strict-mb-iter	Added new command line option (#5173). An integer value defining the number of Newton iterations before relaxed tolerances can be used for the MB convergence criterion. Default -1 means that the relaxed tolerance is used when maximum number of Newton iterations are reached.	-1
10	--newton-max-error	Changed the default value from 1e+10 to 1e+100. A positive real value. The maximum error tolerated by the Newton method to which does not cause an abort.	1e+100
11	--newton-min-iterations	Changed the default value from 1 to 2. A positive integer value. The minimum number of Newton iterations per time step.	2
12	--pressure-max	Added new command line option. A positive real value. Maximum absolute pressure.	1e+99
13	--pressure-min	Added new command line option. A real value. Minimum absolute pressure.	-1e+99
14	--pressure-scale	Added new command line option. A positive real value. Scaling of pressure primary variable.	1
15	--tolerance-mb-relaxed	Added new command line option (#5173). A positive real value. Relaxed tolerated mass balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
16	--use-implicit-ipr	Added new command line option. A boolean value. Compute implicit IPR for stability checks and stable solution search.	false
17	--ecl-enable-drift-compensation	Renamed to --enable-drift-compensation	true

OPM Flow 2024-04 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
18	--newton-target-iterations	Changed the default value from 6 to 10. A positive integer value. The 'optimum' number of Newton iterations per time step. This is an experimental option that is not used by the standard version of OPM Flow; the --time-step-control-target-newton-iterations option should be used instead.	10
19	--vtk-write-ecl-tracer-concentration	Renamed to --vtk-write-tracer-concentration	false
20	--ecl-enable-tuning	Deprecated (#5213)	
21	--ecl-newton-relaxed-tolerance	Deprecated	
22	--ecl-newton-relaxed-volume-fraction	Deprecated	
23	--ecl-newton-strict-iterations	Deprecated	
24	--ecl-newton-sum-tolerance	Deprecated	
25	--ecl-newton-sum-tolerance-exponent	Deprecated	
Notes: - Cells colored in green in the "No." column indicate a new command line option or a change to an existing option for this release. - Cells colored in orange in the "No." column indicate the command line option is available but is now "hidden" from the main help listing, --help. These options can be listed using the --help-all command line option, that lists all the command line options included in the release, including experimental, obsolete, hidden and deprecated options. - Cells colored in red in the "No." column indicate a deprecated command line option for this release.			

Table B.4: OPM Flow 2024-04 New and Deprecated Command Line Options

B.4.2 NEW FEATURES

In addition to the above the following new features have been added to the simulator:

- The Forchheimer term has been added as an additional skin term that depends (explicitly) on the rate ([#4832](#)). Support has been added for [WDFAC](#) and [WDFACCOR](#) keywords in the [SCHEDULE](#) section. These keywords can be used to model a Forchheimer term in the well model either by specifying the D factor directly or by using the Dake model ([#3661](#)). The option of setting connection specific D factors (Item 12 in [COMPDAT](#)) is now also supported ([#4971](#)).
- Support has been added for the [FBHPDEF](#) keyword in the [SCHEDULE](#) section to define the default well BHP target for production wells and the default BHP constraint for injection wells ([#4969](#), [#3747](#)).
- Added partial support for fluid in place region ([FIP](#)) output to the [RPTSOL](#) and [RPTSCHED](#) keywords ([#4982](#), [#4978](#), [#4980](#), [#3756](#)). The simulator supports mnemonic [FIP](#) = 1 or 2, and partially supports [FIP](#) = 3.
- Added support for mechanical dispersion as a compile time option ([#4960](#), [#3737](#), [#847](#)). This adds support for the linear dispersion model described in the 11th SPE CSP benchmark [document](#) (Eq. 2.3).

- 5) Support has been added for specifying `WCONPROD` keyword item 12 (ALQ) with a user defined argument (UDA) (#5033).
- 6) Support has been added for specifying `WSEGVAVL` keyword item 4 (AREAREST) with a user defined argument (UDA) (#4873).
- 7) Support has been added for the `SKIP`, `SKIP100`, `SKIP300` and `ENDSKIP` keywords (#5085, #3817). Note that OPM Flow will skip data between the `SKIP300` and `ENDSKIP` keywords whereas the commercial black-oil simulator will ignore the `SKIP300` keyword.
- 8) Added support for defining a fluid mass (and enthalpy) source term within a grid cell using the new OPM Flow specific `SOURCE` keyword in the `SCHEDULE` section (#5050, #3835, #1106, #5107, #3864).
- 9) Support has been added for the `NONNC` keyword in the `RUNSPEC` section to deactivate non-neighbor connections (#5134).
- 10) Added support for the `DIFFAGAS` and `DIFFAWAT` keywords in the `PROPS` section to define respectively the gas phase and water phase diffusion coefficients for each pseudo component and for each PVT region. The `DIFFAGAS` and `DIFFAWAT` keywords are equivalent to `DIFFCGAS` and `DIFFCWAT` keywords, but assume a mass fraction formulation of diffusion rather than the default mole fraction formulation (#5143, #3878).
- 11) Added partial support for `DATUMR` and `DATUMRX` keywords in the `SOLUTION` section for use with block level summary keywords (BPP*) (#5241, #5240, #3958).
- 12) Added support for using region-level summary vectors in the defining expressions of field-level UDQs (#5250, #5247, #3977, #3980). For example:

```
UDQ
  DEFINE FURE2 (ROIP_RE2 2)+(ROIP_RE2 4)+(ROIP_RE2 5) /
/
```

- 13) Added support for `GCONPROD` item 7 (`ACTION`) equal to WELL. Added support for reason `WTEST` item 3 (TEST) equals G (group) (#5172, #3894).
- 14) Limited support has been added for User Defined Tables (see the `UDTDIMS` keyword in the `RUNSPEC` section and `UDT` keyword in the `SCHEDULE` section), one-dimensional tables are now supported (#4940, #3728). The maximum number of dimensions in any given User Defined Table is specified by item 4 (MXDIMS) of the `UDTDIMS` keyword.
- 15) Added support for setting a no flow and constant temperature boundary condition for the I Ith SPE CSP benchmark cases B/C (#865).
- 16) Added support for updating individual well properties for one or more wells (based on a well name, well name pattern or well list) using the `WELSPECS` keyword in the `SCHEDULE` section. In particular, this revised logic enables the controlling group to be changed without affecting any other well properties such as the location of the well head or the well reference depth (#3703). As an example, this new `WELSPECS` behavior enables the following usage:

```
ACTIONX
  A 1 /
  WOPR 'P*' < 123.4 /
/
WELSPECS
  '?' 'LOWPRESS' /
/
ENDACTIO
```

to move all wells matching the pattern 'P*', and with a low oil production rate, to the group 'LOWPRESS'. This could, in turn, apply a different set of group-level production controls to those wells.

- 17) A new brine-CO₂ mutual solubility model has been implemented from Spycher & Pruess³⁵². This model improves the accuracy of the solubility calculations in [CO2STORE](#) at temperatures above 100 degrees Celsius. In addition, a new keyword, [ACTCO2S](#), has been added to choose between activity models. The original [CO2STORE](#) solubility and salt activity models are currently the default behavior (i.e., [ACTCO2S](#) keyword model number 3) ([#3694](#)).
- 18) Added the [CO2SOL](#) and [H2SOL](#) keywords in the [RUNSPEC](#) section which activate either dissolved carbon dioxide (CO₂) or hydrogen (H₂) in the water phase, where CO₂ or H₂ is represented by the [SOLVENT](#) pseudo component, using the simulator's CO₂-Brine or H₂-Brine PVT model. The [CO2SOL](#) and [H2SOL](#) keywords can be used when modeling CO₂ or H₂ injection in depleted hydrocarbon reservoirs ([#3764](#), [#851](#), [#4991](#)).
- 19) Added support for the [PCFACT](#) keyword in the [PROPS](#) sections to define the capillary pressure multiplication factor as a function of porosity change ([#3857](#)). Currently the keyword is used in conjunction with OPM Flow's Salt Precipitation model.
- 20) Added methods to the Python module `opm.simulators.BlackOilSimulator` to access primary variables and fluid state variables ([#4950](#)).
- 21) Added docstrings for the Python bindings. Currently, the docstrings can be used to get context help in editors or in IPython. However, the plan is to also use these docstrings to generate sphinx documentation for the Python bindings ([#5242](#)).
- 22) Added [PYACTION](#) functions `insert_keyword(const std::string& deck_string)` and `insert_keyword(const std::string& deck_string, std::size_t report_step)` to insert a keyword. A keyword can be inserted at a later report step or at the current report step; inserting a keyword at a past report step or at a report step exceeding the total number of report steps throws an error ([#3994](#)).
- 23) Added support for the [THCO2MIX](#) keyword in the [PROPS](#) section to specify the thermal mixing models for salt in water, CO₂ dissolved in brine, and water vaporized in CO₂ ([#4012](#)).

B.4.3 IMPROVEMENTS

Improvements include:

- 1) Previously, the simulator would perform unnecessary network iterations if all wells belonging to the network were in history mode. The simulator has been updated to avoid this and only compute the network pressures for reporting purposes ([#4881](#)).
- 2) Previously, if no well THP constraint had been specified then the THP would be set to zero. The simulator has been updated so that the THP would only be set to zero if no valid VFP table has been provided. If a VFP table has been provided but no THP constraint has been specified then the THP value is calculated for reporting purposes. In addition, if the well is part of a network then the THP value is required for a constraint check ([#4930](#)).
- 3) Updated code to output more information and avoid harsh termination due to assert in `StandardWellPrimaryVariables` ([#4946](#)).
- 4) The simulator has been updated to reduce well connection transmissibility factor due to salt precipitation ([#4919](#)).
- 5) The simulator has been updated to compute the reservoir volume rates based on the average pressure and temperature in the well for injectors in [THERMAL](#) cases ([#4826](#)).
- 6) The code has been updated to ensure there is a non-zero initial guess for well rates, and that non-zero water/gas fractions are available from the previous time step for the (explicit) VFP table

³⁵² Spycher, N., Pruess, K. A Phase-Partitioning Model for CO₂-Brine Mixtures at Elevated Temperatures and Pressures: Application to CO₂-Enhanced Geothermal Systems. *Transp Porous Med* 82, 173–196 (2010).

interpolation. Previously, if this was not the case then the well was likely to converge to the zero rate and/or zero fractions case and would be deemed to be inoperable ([#4952](#)).

- 7) The simulator has been updated to allow the creating or opening of a well as a result of an [ACTIONX](#) keyword block in the middle rather than at the end of a report step ([#4749](#)).
- 8) Previously, numerical problems were reported as errors in the print file (*.PRT), before continuing the simulation. This resulted in successful runs being incorrectly reported as having errors. The code has been updated to report them as problems rather than errors ([#4977](#), [#4957](#)).
- 9) Rerouting is now allowed in the standard and extended network models ([#3695](#), [#4945](#)).
- 10) Added support for networks with multiple fixed-pressure nodes ([#3702](#), [#4915](#)).
- 11) Added extra information about well convergence failures to the INFOITER file (if this file has been requested, the default is for this file not to be output) ([#4975](#)).
- 12) The simulator will now log a problem if asked to continue with the run after a convergence failure of the non-linear solver has occurred, by specifying the command line option `--solver-continue-on-convergence-failure=true` ([#4979](#)).
- 13) If parsing strictness is set to low with the command line option `--parsing-strictness=low` and no matching wells are found by various keywords in the [SCHEDULE](#) section then a warning will be issued and the run will continue ([#4905](#), [#3698](#)).
- 14) Previously, numerical problems during well testing would be caught at a higher level, causing time step chops in serial mode and possibly MPI communication errors in parallel. With this change wells that experience such problems during testing will simply not open, and the MPI errors are no longer present ([#5032](#)).
- 15) A guard has been added against not a number (NaN) from certain expressions (e.g. evaluating a power of a negative number) in the valve models ([#5041](#)).
- 16) If Hierarchical Data Format (HDF5) input/output is requested but no support is available then an exception is thrown early instead of logging later an error when save/load is to be performed ([#5042](#)).
- 17) Functionality has been added for computing an implicit Inflow Performance Relationship (IPR). Moreover several functions are now utilizing this for stability checking, estimating the operability of stopped/unconverged wells and searching for solutions (VFP/IPR intersections) in problematic cases. In addition, direct computation of potentials is included. For the new code to take effect, currently simulations must be run with both options `--local-well-solve-control-switching=true` and `--use-implicit-ipr=true` ([#4986](#)).
- 18) The code has been updated to apply MULTX, [MULTY](#) and [MULTZ](#) transmissibility multipliers to boundary transmissibilities ([#5047](#)).
- 19) The name of the destination file is now included in warning messages about unhandled report keywords ([#5048](#)).
- 20) Reformatted production and injection reports in the Print file (*.PRT) by removing dashed lines between records ([#5049](#)).
- 21) Previously information about parallel runs was only output to the standard out. To help with debugging and replicating cases (e.g. in case of crashes) without saved standard output, the information about MPI processes and OMP threads is now also sent to the PRT file ([#5031](#)).
- 22) The test of whether the gas phase at the well connection is saturated or not has been made more robust by simply comparing the oil-gas ratio with saturated oil-gas ratio at the connection pressure ([#5075](#)).
- 23) Output of reservoir volumes to the Print file (*.PRT) has been added for custom fluid in place ([FIP](#)) regions. In addition, the compatibility of the output with the commercial simulator has been improved ([#5068](#)).

- 24) Previously the flow reports in the PRT file were restricted to wells active on the current MPI rank. The simulator now outputs the flow reports for all wells active at the current report step (#5090).
- 25) The output of Newton progress has been improved for the NLDD option (#5104).
- 26) Added parallel well domain and error logging for NLDD (#5116).
- 27) Currently the NLDD option requires the command line option `--matrix-add-well-contributions=true`. An error message will be now be generated if NLDD is used without this option set (#5120).
- 28) Improved handling of wells with zero rate where cross-flow is turned on and off (#5139, #3880).
- 29) The simulator has been updated to ignore the `COORDSYS` keyword in the `GRID` section (#5161). Note that the simulator will stop if the number of reservoir grids to process, set using the `NUMRES` keyword in the `GRID` section, is greater than one.
- 30) Improved error message to be more informative when an exception occurs in `computeWellPotentials()` (#5170).
- 31) The simulator now removes possible old simulation output files from the specified output directory before logging starts (#5168).
- 32) A number of minor improvements have been made to the time reporting output including the addition of setup time (#5174).

```

===== End of simulation =====
Number of MPI processes:      1
Threads per MPI process:     2
Setup time:                  0.29 s
  Deck input:                 0.21 s
Number of timesteps:         123
Simulation time:              0.98 s
  Assembly time:              0.31 s (Wasted: 0.0 s; 0.0%)
    Well assembly:            0.04 s (Wasted: 0.0 s; 0.0%)
  Linear solve time:          0.23 s (Wasted: 0.0 s; 0.0%)
    Linear setup:              0.06 s (Wasted: 0.0 s; 0.0%)
  Props/update time:          0.19 s (Wasted: 0.0 s; 0.0%)
  Pre/post step:              0.14 s (Wasted: 0.0 s; 0.0%)
  Output write time:          0.07 s
Overall Linearizations:        483      (Wasted: 0; 0.0%)
Overall Newton Iterations:    360      (Wasted: 0; 0.0%)
Overall Linear Iterations:    1311     (Wasted: 0; 0.0%)

```

- 33) The code now avoids unnecessarily looking up the artificial lift quantity (ALQ) for injection wells (#5221).
- 34) Improved formatting of small numbers in adaptive time stepping output (#5237).
- 35) With multi-segment wells, the segment pressure can be lower than the BHP, so it is not desirable to use the BHP lower limit to limit the segment pressures. The simulator has been updated so that the BHP lower limit is only used to limit the BHP; a lower limit of zero is imposed on the segment pressure (#5234).
- 36) The simulator has been updated to allow for the possibility that not all well connections are in active cells for the purpose of calculating the WBPn summary vectors. This situation might arise for example when "re-parsing" portions of the `SCHEDULE` section following the successful triggering of an `ACTIONX` block (#5258, #3988).
- 37) Avoid repeatedly getting a parameter value in a performance-critical section of the code, which was causing a significant performance loss for the Norne model, by getting the parameter once in the constructor instead (#5318).

- 38) Avoid expensive call to `FaceDir::FromIntersectionIndex()` in a performance-critical section of the code, this fixes a significant performance regression on the order of 5% or more in the flux calculations that was introduced earlier (#898).
- 39) Added support for using well lists (`WLIST`) in `ACTIONX` blocks with the following keywords: `COMPDAT`, `COMPLUMP`, `WCONINJE`, `WCONPROD`, `WECON`, `WEFAC`, `WELOPEN`, `WELPI`, `WELSPECS`, `WELTARG`, `WGRUPCON`, `WPIMULT`, `WSEGVAVL`, `WTEST` and `WTMULT` (#3741).
- 40) Output computed surface densities for CO₂ and Brine to the PRT file if the `CO2STORE` option is enabled (#3730).
- 41) The simulator will now output the time step in units of hours when using laboratory (`LAB`) units. Previously units of days would have been used for all unit systems (#3829).
- 42) Improved the handling of errors related to the grid keywords. Missing or ambiguous keywords are detected and the error messages list the valid options (#3849, #3551).
- 43) Added support for pattern matching in the `MULTFLT` keyword (#3851).
- 44) Improved the error reporting for `SALTSOL` tables when a column is missing (#3850).
- 45) Added support for setting the same region for the source and target in the `MULTREGT` keyword. In this case the multipliers will be applied to all connections within the specified region as well as to connections between the specified region and all other regions (#3845).
- 46) The code has been updated so that for mixed wettability systems it no longer scales the negative part of the oil-water capillary pressure curve when using the `SWATINIT` keyword (#3720). This behavior is controlled by an option in the commercial simulator.
- 47) The `CO2STORE` option could lead to slow runs if a lot of extrapolation was done resulting in numerous warning messages. These messages have been turned off unless the debug option has been set. Note that extrapolation of tables can be avoided by limiting the temperatures and pressures using the `-temperature-min`, `-temperature-max`, `-pressure-min` and `-pressure-max` command line options (#3858).
- 48) The simulator now outputs a meaningful error message in the case where it fails to open a potentially corrupt file (#3870).
- 49) Improved parsing of sections in the input deck and support for optional sections `EDIT`, `REGIONS` and `SUMMARY` (#3863).
- 50) Enforce consistency checks on keywords specifying the standard and extended network models respectively (#3885).
- 51) The simulator has been updated to check whether the standard network model is active when initializing a network balancing operation. Previously, the simulator would only check for the extended network model. This was an issue because the pressure convergence tolerance is set to zero unless a network model is active (#3874).
- 52) Add support for operations (e.g. `EQUALS`) on the `MINPVV` array (#3897).
- 53) Added support for numbers in user defined region set names, for example `FIPAB1` and `FIPAB2` (#3979).
- 54) Allow the use of the `FIELD` group for selected group level keywords when handling UDQs and especially `ACTIONX` condition blocks, for example `GGOR 'FIELD' > 123.4` (#3993).
- 55) Allow an expanded set of summary vectors to be used in defining expressions for UDQs and in the condition blocks of the `ACTIONX` keyword. In principle, this change enables using all known summary vector categories as part of the defining expressions in a `UDQ`, but additional testing is needed before claiming to fully support such usage (#3968).
- 56) The simulator will now report an error if the `PYACTION` keyword is used when flow has been built without embedded Python (#3998). Previously, only a warning message would have been reported.

- 57) Removed the need for a run-function in the Python code in [PYACTION](#). The EclipseState, Schedule, ReportStep and SummaryState have been made available as attributes of the module opm_embedded instead. This enables tooltips for writing code in a Python IDE, after importing the library as "import opm_embedded", typing "opm_embedded." will show the tooltips. The code will still be backwards compatible ([#3986](#)).
- 58) Added check that when a non-default THP constraint has been specified for a well that a valid VFP table has been defined ([#4005](#)).
- 59) Previously, when temperature results were not available 0 K would be converted to the units system specified in the deck and output to the [SUMMARY](#) file (for example -273.15 or -459.67 depending on the unit system). The simulator will now output a zero temperature in the specified units system when temperature results are not available ([#4013](#)).
- 60) Added [PYACTION](#) functions to open, close or shut a well without specifying the report step ([#4019](#)).
- 61) Added an error message if an attempt is made to use the [COMPSEGS](#) keyword to define the well segment connections before the structure of the multi-segment well has been defined using the [WELSEGS](#) keyword ([#4031](#)).
- 62) The code now determines if a history matching producer with a reservoir volume rate (RESV) target is subject to a zero rate constraint by checking if the specified historic rates for oil, gas and water are all zero ([#4039](#)).

In addition, the following new [SUMMARY](#) keywords are now recognized as described by the comments in Table B.5.

No.	Summary Keyword	Comment
1	NPR, GNETPR	Added support for output of network pressures based on rates at end of the time step (#3701 , #4914). Note that GNETPR is an alias for NPR included since NPR can cause problems for certain post-processors.
2	CGFRF, CGFRS, COFRF and COFRS	Added support for output of well connection free gas, solution gas, liquid oil and vaporized oil flow rate vectors (#5010 , #3782).
3	CDFAC	Added support for output of well connection D factor (#4832).
4	FGMIP, FGMGP, FGMDS, FGMTR and FGMMO	Added CO ₂ mass in place summary vectors for the field when using the CO2STORE option: total mass in place, mass in gas phase, mass dissolved in water phase, mass in trapped gas phase and mass in mobile gas phase (#5114 , #3868).
5	RGMIP, RGMGP, RGMDS, RGMTR and RGMMO	Added CO ₂ mass in place summary vectors for FIP regions when using the CO2STORE option: total mass in place, mass in gas phase, mass dissolved in water phase, mass in trapped gas phase and mass in mobile gas phase (#5114 , #3868).
6	BGIP, BGIPL	Added support for block total gas in place (BGIP) and block gas in place in the water phase (BGIPL) for gas-water systems (#5188).
7	BPPO, BPPG and BPPW	Added support for output of block level datum depth corrected phase pressures (#5240 , #3959 , #3958).
8	FGKDI, FGKDM, FGKTR and FGKMO	The 11th Society of Petroleum Engineers Comparative Solution Project (http://spe.org/csp) defines the "immobile free-phase CO ₂ " as "CO ₂ at saturations for which the non-wetting phase relative permeability equals zero". Field and Region level summary vectors have been added when using the

	RGKDI, RGKDM, RGKTR and RGKMO	CO2STORE option based on this definition as follows (#5281 , #4010): Gas Moles in the Immobile Gas Phase (Non-wetting relative permeability equals zero), Gas Moles in the Mobile Gas Phase (Non-wetting relative permeability greater than zero), Gas Mass in the Immobile Gas Phase, and Gas Mass in the Mobile Gas Phase.
--	-------------------------------------	--

Table B.5: New [SUMMARY](#) Keywords for the 2024-04 Release

Support has been added for the following initialization output as described by the comments in Table B.6.

No.	Mnemonic	Comment
1	MULTX , MULTY and MULTZ MULTX- , MULTY- and MULTZ-	Added support for outputting transmissibility multiplier arrays to the INIT file. The MULTX , MULTY and MULTZ arrays are always written. If the first item of the GRIDOPTS keyword is NO then only the MULTX- , MULTY- and MULTZ- arrays that have been specified in the deck will be written. If the first item of the GRIDOPTS keyword is YES then all transmissibility multiplier arrays are written (#3997).
2	SWATINIT	The simulator will now output the SWATINIT array to INIT file if it exists in the input deck (#3832).

Table B.6: New Initialization Output for the 2024-04 Release

Supported has been added for the following restart output as described by the comments in Table B.7.

No.	Mnemonic	Comment
1	RESIDUAL	Added option to output the residuals at the end of the time step. To activate this output the command line option <code>--enable-opm-rst-file=true</code> should be specified and the mnemonic RESIDUAL should be added to the RPTRST keyword (#4937).
2	XMFH2,YMFWAT	The option has been added to output the liquid phase H ₂ mole fraction and vapor phase water mole fraction (XMFH2 and YMFWAT) when using H2STORE . To activate this output the command line option <code>--enable-opm-rst-file=true</code> should be specified (#4964). Previously the mnemonics XMFCO2 and YMFWAT were only available for CO2STORE .
3	RSWSOL	Support has been added for output of dissolved solvent to water ratio RSWSOL (#4991 , #3764 , #851).
4	FLows-, FLOres-	Support has been added for output of negative direction inter-block flows at surface conditions (FLows-) and at reservoir conditions (FLOres-) (#5038 , #3823 , #854).
5	RPORV	Support has been added for output of pore volumes at reservoir conditions using the RPORV mnemonic of the RPTRST keyword (#5092).
6	CONV	Added partial support for output of cells causing convergence problems for the residuals of the oil, water, gas, polymer, brine and solvent equations to the restart file using the CONV mnemonic of the RPTRST keyword (#5054 , #5112).
7	PCGW	Added support for outputting gas-water capillary pressure (Pcgw) to the restart file using the PCGW mnemonic of the RPTRST keyword (#5126).
8	FIPOIL, FIPWAT, FIPGAS, RFIPOIL,	Added support for output of fluid in place arrays for oil, gas and water to the RESTART file. The SFIPOIL, SFIPWAT and SFIPGAS are the surface conditions volumes for oil, water and gas respectively (these are aliases for FIPOIL,

	RFIPWAT, RFIPGAS	FIPWAT and FIPGAS). The RFIPOIL, RFIPWAT and RFIPGAS are the reservoir conditions volumes for oil, water and gas respectively (#5224 , #3952).
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Table B.7: New Restart Output for the 2024-04 Release

B.4.4 BUG FIXES

The following bug fixes and improvements have been incorporated into this release.

- 1) The simulator has been updated to use the `applyRestartSwatInit()` member function instead of directly mutating a data member of the `EpsInfo` structure. This way we defer the PCOW re-scaling for [SWATINIT](#) to a context with a more complete view of the changes needed to convey the information to all components involved. With this change the simulator is mostly able to reproduce the base run in a restart run from the end of the historic period for a real field case that has strong [SWATINIT](#) scaling of the oil-water capillary pressure function ([#4944](#), [#3731](#)).
- 2) The code was updated to include a new event to address an issue with a reported case where [WELOPEN](#) failed to open a well previously shut due to an economic limit ([#4924](#), [#3707](#)).
- 3) The logger is now initialized before parsing ([#5007](#)). Previously, uninitialized values would have been used when the parser logged some errors.
- 4) Fixed an issue where [BCPROP](#) would be ignored if it was not set in the first time step. This update makes it possible to set [BCPROP](#) later in the schedule ([#4912](#), [#833](#)).
- 5) Previously, a well with shut-in instruction "shut" that changes to non-operable during an iteration will give the message well xxx gets SHUT during iteration, but no further actions are triggered. This is due to the flag `changed_to_stopped_this_step_` not being set. The simulator has been updated so that a non-operable well will give the message well xxx gets STOPPED during iteration (irrespective of it's shut-in instructions), and at the end of the time-step get closed according to it's shut/stop instruction (with accompanied message) ([#5014](#), [#4902](#)).
- 6) Fixed an issue where there was an inconsistency between well BHP and perforation pressures, and between well rate and perforation rate ([#5015](#)).
- 7) Fixed a bug in the Gauss-Seidel NLDD approach that would actually do a Jacobi initialization for the very first iteration of a local domain solve. We add the `maxpressure` option for Gauss-Seidel NLDD domain ordering, and make it the new default ordering, and make Gauss-Seidel the default approach. Finally, we make the default number of domains equal to one domain per 1000 cells, instead of just a single domain, and tweak the default convergence tolerance scaling for local solves. These changes are intended to lower the bar to testing the `-nonlinear-solver=nldd` option. You still need to use `-matrix-add-well-contributions=true` to run with NLDD, otherwise the defaults should be sufficient for reasonable behavior ([#5021](#)).
- 8) Previously, combining `-load-step` with different `-output-dir` triggered a warning. The code has been updated to not issue a warning since these options can safely be used together ([#5024](#)).
- 9) Previously, when singular matrices occurred when solving multi-segment wells an exception would be thrown only on one process resulting in MPI errors about truncated messages. The code has now been updated so that all MPI processes will throw, terminate the non-linear solver and cut the time step to continue the simulation ([#5025](#), [#5036](#)).
- 10) Fixed the sign of the production rate for group pressure maintenance ([#5026](#)).
- 11) Get a complete list of fluid in place (FIP) regions from `FieldPropsManager` rather than from `SummaryConfig`, which only holds regions with summary keywords ([#5034](#), [#3813](#)).
- 12) Previously an exception thrown by `WetGasPVT::saturationPressure` due to failure to converge within 20 iterations was only seen on one MPI rank and the others continued. The code now communicates the problem and throws on all MPI processes. This results in the time step being cut as a result and simulation continuing ([#5046](#)).

- 13) Ensure the input temperature is used in the density calculations in the initialization ([#5058](#)).
- 14) Avoid modifying outputDir, leave empty if defaulted ([#5073](#)).
- 15) The code has been updated to only output FLOWS and FLORES for interior cells. This fixes an issue with parallel output of FLOWS and FLORES ([#5043](#)).
- 16) The setting of solvent values has been made conditional on the [SOLVENT](#) being active ([#5086](#)).
- 17) Previously when assembling the acceleration pressure drop terms for multi-segment wells certain derivatives for reverse flow were skipped. The code has been rewritten to fix this issue and improves convergence in some cases ([#5040](#)).
- 18) Code has been updated to use the gas pressure rather than the oil pressure in gas-water simulations when modifying rock properties due to compaction ([#5081](#)).
- 19) The code has been updated to honor the default minimum number of Newtons iterations. Previously the simulator would perform a minimum of two Newton iterations even though the default was one ([#5094](#)). Note that also the default has been changed from one to two.
- 20) The code has been updated to use the arithmetic average of compaction transmissibility multipliers rather than the upstream value ([#5088](#), [#858](#)).
- 21) A fix has been provided for a crash when cleaning up the Damaris mesh offset variable due to the shared memory buffer size being off by one ([#5109](#)).
- 22) The simulator now supports at least NTFIP ([REGDIMS](#) keyword item 1) distinct regions for inter-region flow to avoid a possible segmentation failure ([#5117](#)).
- 23) A couple of changes have been made to fix network convergence issues. The lower bound on the network pressure update has been removed as this could lead to convergence issues. In addition, the convergence tolerance has been tightened by a factor of 10 for wells that are controlled under a dynamic THP-limit since the default (standard) well tolerance was not sufficiently strict to get network convergence in some cases ([#5128](#)).
- 24) Previously the solvent production rate was not included in the total gas production rate when applying the gas production rate limit (GRAT) in the well control keyword [WCONPROD](#). The gas production rate limit is now applied to the total gas production rate including both the gas and solvent pseudo components ([#5140](#)).
- 25) A bug revealed by ([#5124](#)) in the local well control switching option where previously stopped wells would re-open during the local solve has been fixed ([#5129](#)).
- 26) Previously, [NEXTSTEP](#) was ignored if the user did not run OPM Flow explicitly setting the command line parameter `--enable-tuning=true`. Now [NEXTSTEP](#) will be applied regardless of which parameter value was set ([#5215](#)).
- 27) The handling of the [NEXTSTEP](#) keyword in an [ACTIONX](#) block has now been fixed ([#5222](#)).
- 28) The code has been updated to ensure that the simulator does not advance beyond the current report step and that the next time step is not too small if the [NEXTSTEP](#) keyword is used in an [ACTIONX](#) block ([#5259](#)).
- 29) Fixed an issue where previously the low parsing strictness command line option was not being respected when reprocessing the schedule section after the application of an action ([#5220](#)).
- 30) Fixed a crash in the Python module due to duplicate symbols that depended on the order of the object libraries in the linker command during the build ([#5235](#)).
- 31) Added missing convergence metrics to the domain convergence report. Previously, for NLDD local solves the metrics were always zero and non-converged local solves were accepted ([#5249](#)).
- 32) The code has been updated to call loadRestartData on all processes even if there are no local wells since group data is still required ([#5252](#)).

- 33) Corrected formatting of debug message in MultisegmentWellSegments.cpp to avoid throwing exception in libfmt (#5261).
- 34) Fixed a potential out-of-bounds reference that could occur if there are no wells (#5264).
- 35) Fixed a segmentation fault due to taking a reference to a temporary variable in the WBP calculation service (#5267).
- 36) Fixed a bug introduced when the ROCm dependency on OpenCL was removed (#4883), which disabled the option to run the rocSPARSE and cuSPARSE backends with the optimized version of the block Jacobi incomplete lower-upper (ILU) preconditioner (#5255).
- 37) Fixed dangling references in Logger for test_LogOutputHelper that can result in a segmentation fault that has been seen on the ppc64le architecture (#5263).
- 38) The code now makes sure that the fast AMG preconditioner is reconstructed when performing an update (#5284).
- 39) Fixed an issue where the send and receive buffers were the same which caused an MPICH error on MPICH based Red Hat Enterprise Linux 7/8 builds (#5297).
- 40) Fixed a bug which caused an error whenever item 4 (PINCHCAL) of the *PINCH* keyword was set to *ALL*. The case where one of the pinched out cells has zero z-direction permeability (or z-direction transmissibility multiplier) is now supported. However, the *ALL* option is still not fully supported (#707, #701).
- 41) Fixed an integer overflow in the initial processing of grids with more than 134 million cells that would cause a segmentation fault (#710).
- 42) The simulator now correctly allows for solvent flux when the dissolved gas in water option has been activated by the *DISGASW* keyword in the *RUNSPEC* section (#866).
- 43) The 'meaning' variables are now initialized to avoid potential out-of-bounds dereferences (#880).
- 44) Fixed a bug where setting the saturation table number (*SATNUM*) in the *COMPDAT* keyword (item 7) to a value other than the cell *SATNUM* resulted in a segmentation fault in two-phase cases. The simulator now checks which two-phase approach is active and sets the correct relative permeability parameters accordingly (#3739).
- 45) Fixed an issue where the simulator interpreted the month 'FEB' in an *ACTIONX* condition as a field-level summary vector and wrote a value of zero rather two (for February) to the restart file. This resulted in an unhelpful error message when later trying to perform a restart (#3826).
- 46) The simulator now assumes that the reservoir density and internal energy of CO₂ and H₂ are not affected by their water content (#3806). This avoids the use of calculations based on the ideal gas assumptions implemented in simpleHuDuanH2O.hpp. The formulation is now as in the 11th SPE CSP.
- 47) The simulator previously assumed that all UDQs loaded from the restart file had an UPDATE status of ON (#3808). The simulator now inspects the actual update status and uses that to configure the *UDQ* when restarting the simulation (#3833).
- 48) Previously the simulator conflated history control with the *WHISTCTL* keyword setting, thus failing to respect any control mode setting in the *WCONHIST* keyword in the case of a simulation restart in the historic period. The *WHISTCTL* keyword setting has been decoupled from that of the control mode setting specified in the *WCONHIST* keyword thereby supporting both modes when a simulation is restarted in the historic period. This, in turn, fixes a simulation restart on a real field (#3854).
- 49) Fixed output of the group hierarchy chart to the PRT and DBG files as requested by the *WEL SPECS* mnemonic of the *RPTSCHED* keyword (#3856).
- 50) Corrected the units for region level CO₂ summary vectors RGCDI, RGCDM and RWCD to be moles instead of surface volumes (#3866).
- 51) Fixed support for the not equals operators '.ne.' and '!=' in *ACTIONX* keywords (#3893).

- 52) Account for well and/or group efficiency factors when evaluating rates for non-default region sets, for example ROPR_ABC and ROPT_ABC (#3955).
- 53) The code has been updated so that TSINIT (item 1) of the [TUNING](#) keyword only applies to the next time step (#3953).
- 54) Fixed the handling of the [MULTX](#), [MULTX-](#), [MULTY](#), [MULTY-](#), [MULTZ](#), [MULTZ-](#) keywords and [EQUALS](#) [MULT*](#) keywords in the [EDIT](#) section. Previously multiple occurrences of these keywords in the [EDIT](#) section would have been applied cumulatively. Now the [EDIT](#) section has its own set of multipliers which are applied to the ones in the [GRID](#) section when the end of the [EDIT](#) section is reached (#4002).
- 55) Added a check that a keyword is valid for use with the [PYACTION](#) keyword before it is inserted into the deck (#4008).
- 56) Fixed the initialization of gas-water simulations with a transition zone (#5293).

B.4.5 KNOWN ISSUES

- 1) There is an error with the RSM header for summary vectors whose NUMS entry in the SMSPEC file is derived from more than a single number source (e.g., single region or segment ID). This applies to all block vectors (BGPV, BOPV, BWPV, etc.), connection level quantities (COPT, etc.), and inter-region flows such as ROFT etc (#3078). The work around is to plot the data in OPM ResInsight and right-click on the plot to view and copy the data.
- 2) The [GDFILE](#) keyword in the [GRID](#) section loads a grid file in various formats, with the FMTOPT parameter setting the format type of the file. If the variable FMTOPT is omitted then the default is for binary file input for the commercial simulator; whereas, OPM Flow derives FMTOPT from the file extension (*.EGRID or *.FEGRID), making FMTOPT superfluous. However, if the extension is lower case then OPM Flow may incorrectly determine the file type. The work around is change the extension to upper case.
- 3) As per previous releases of the radial model, the [COORDSYS](#) keyword item three must be set to COMP to complete the circle, this has not been implemented in this release. Also there appears to be a bug for full radial models when a well goes on BHP control that causes the well not to respect the BHP constraint, this eventually causes the well to die prematurely. See #2640 for a discussion on the topic.
- 4) As in previous releases there are some issues with the [OPERATE](#) and [OPERATER](#) keywords associated with the input parsing; for various reasons a few of the fields require special case treatment in the grid processing, including (at least) [MULTZ](#), [PORV](#) and [ACTNUM](#), and for those keywords the [OPERATE/OPERATER](#) keyword does not work. The work around is to use the [MULTIPLY](#) keyword instead.
- 5) For the [UDQ](#) ASSIGN operator after the terminating "/" normally any comments can be entered; however, if there is "/" within the comment field, as per:

```
ASSIGN FUNGLYLD 1.196 / Condensate Yield (63.5/56.7)/(1.0 - 0.065)
```

then the simulator will abort. The work around is to manually place the comment characters "--" after the ASSIGN terminating "/", like so:

```
ASSIGN FUNGLYLD 1.196 / -- Condensate Yield (63.5/56.7)/(1.0 - 0.06)
```

- 6) At the moment, one cannot initialize tracers using the [EQUALS](#) keyword. Instead use the array format, that is the keyword followed by the required number of values, or the [TVDP](#) keyword in the [SOLUTION](#) section to set the initial tracer concentrations as a function of depth.
- 7) Currently, gas tracers cannot be used if the dissolved gas phase, as per the [DISGAS](#) keyword in the [RUNSPEC](#) section, is active in the model.

- 8) The summary vector RTIPTHEA, that defines the energy in-place between the initial and the current time for regions, is not supported unlike the FTIPTHEA and BTIPTHEA vectors. Secondly, the error message:

```
Warning: Problem with summary keyword RTIPTHEA
In RSM-THERMAL.data line 492
FIP region FIPHEA not defined in REGIONS section - RTIPTHEA ignored
```

is incorrect, as the message indicates that it is being treated like a named region, as per the *FIP* keyword, when it is actually a *SUMMARY* vector (#3870).

- 9) If there are cells that are very distorted, which can occur near fault planes, then the simulator may abort because it cannot calculate the pore volume of such cells. The work around is to re-generate the grid in the static model, taking care that the cells around the fault planes are more or less orthogonal (#2992 and #3770).
- 10) Currently the *OPERATER* keyword in the *EDIT* section does not work with the *DEPTH*, *TRANX*, *TRANY* and *TRANZ* property arrays (#2994 and #748).
- 11) If a standard well is fully declared in an *ACTIONX* block which is then activated at a later date, and later the well is modified to be a multi-segment well using the *WELSEGS* and *COMPSEGS* keywords, then this will cause the simulator to abort with an assert failure. The solution to this issue is to not use this type of work flow in declaring wells (#2891 and #2895).
- 12) Although the *ACTIONX EXIT* command works as expected, it does not write out the requested RSM file at the end of the run. However, the other *SUMMARY* and *RESTART* files are written out (#2877).
- 13) Although the *GCONSUMP* keyword in the *SCHEDULE* section is fully implemented as documented, it is not possible to verify the output as the associated *SUMMARY* vectors are not written out, that is the *SUMMARY* sales gas vectors FGSR, FGST, GGSR and GGST, and fuel vectors FGCR, FGCT, GGCR, and GGCT have not been implemented (#2679).
- 14) There are small differences in the behavior of the *NEXTSTEP* keyword in the *RUNSPEC* section between OPM Flow and the commercial simulator that remain unresolved (#3745).
- 15) **There is a unit handling issue associated with *OPERATE* keyword. If the *OPERATE(X)* parameter has units, as for example *PERMX*, then the conversion is always done in SI units, despite the input deck declaring the deck to be fields units, as per *FIELD* keyword in the *RUNSPEC* section. Note that OPM Flow performs all of its calculations internally in SI and performs unit handling only when inputting the *.DATA file and when outputting result files. Thanks to *Irijkels* for reporting the issue. See #4597 for details.**
- 16) There is an issue associated with restarting from a restart file with the solution gas (Rs) maximum rate of increase, as defined by the *DRSDT* keyword in the *SCHEDULE*, that has been set to zero. This is because, the simulator does not save/restore this setting in simulator's restart files, which means that simulator misses the essential value zero upon restarting the case. As a work-around one can use the option *-sched-restart=true*, when running the restart case. This will initialize the restarted simulation based on information from the complete *SCHEDULE* section, instead of just the parts that we're going to simulate and the rest from the restart file. Thanks to *goncalvesmachadoc* for reporting the issue. See #4272 for details.
- 17) OPM cannot be built with dune-fem version 2.9 or later (#4934). Please use a previous version.
- 18) The simulator uses an irregular corner-point grid geometry with adjusted pore volumes to represent radial grids so it is not possible to create a full ring (360 degree disk) with only one cell in the theta direction (NY=1). The work around is to model a slice (say *DTHETA*=60 degrees). Note that as the angle increases larger pore volume adjustments are required (#4755).
- 19) In some cases with the network option the simulator can wrongly report that a well has no THP constraints, for example

```
GLIFT WTEST: Well S-3H does not have THP constraints
```

when THP constraints have been defined ([#4887](#)).

- 20) Dispersion in the gas phase leads to convergence issues for the 11th SPE CSP (<https://spe.org/csp>) models Version 11B and Version 11C (and is not fully tested). Therefore dispersion in the gas phase has been temporarily removed and a warning added ([#5101](#), [#859](#)).

B.5 RELEASE 2023-10

We dedicate this OPM 2023-10 release to our colleague and friend David Baxendale. David passed away in late June 2023 after a short severe illness. Our thoughts are with his wife and son. The OPM community is very thankful for all his contributions to OPM and the fruitful discussions with him about issues with and advancements of the simulator.

David started contributing to OPM in 2016 as [OPMUSER](#) on github and continued his good work until his very last days. We owe the OPM Reference manual to him. He started this heroic effort in 2017 and it now has thousands of pages. We, his colleagues and friends, are and will be surely missing him with his reservoir engineering expertise and know-how, his enthusiasm, and humor.

The 2023-10 release consists of some new features and various improvements and bug fixes. Our main target was to support more keywords used for relevant fields and reduce differences between OPM flow and the commercial simulator. These improvements include:

- Added support for temperature ([THERMAL](#)) plus salt precipitation ([PRECSALT](#)) modeling in gas-water-brine ([GAS-WATER-BRINE](#)) systems.
- Added support for modeling dissolved gas in water ([DISGASW](#)) and vaporized water in the gas phase ([VAPWAT](#)) in the thermal-gas-water simulator.
- Support for modeling [FOAM](#) combined with [SOLVENT](#).
- Partial support for [WAGHYSTR](#) keyword (Water-Alternating-Gas hysteresis).
- Improvements to many user-facing error messages.
- More graceful exits for problems in parallel runs.
- Temperature is output to the [RESTART](#) file if requested via [RPTRST](#).
- Added support for [WBP](#), [WBP4](#), [WBP5](#) and [WBP9](#) in the [SUMMARY](#) section to output well block averaged pressures for open completions.
- Added support for initializing constant flux aquifers from a restart.
- Faster two-point flux-approximation introduced in the last release is now also used for linearizing gas-oil cases with energy (with diffusion) and gas-oil diffusion.

B.5.1 NEW AND DEPRECATED COMMAND LINE OPTIONS

The major command line changes made for this release are summarized in Table B.8

OPM Flow 2023-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
1	--linear-solver	Added option to use the experimental hybrid solver configuration. Valid options are: <code>ilu0</code> (default), <code>cprw</code> , <code>cpr</code> (an alias for <code>cprw</code>), <code>cpr_quasiimpes</code> , <code>cpr_trueimpes</code> , <code>amg</code> or <code>hybrid</code> (experimental). Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with <code>'.json.'</code>	<code>ilu0</code>
2	--linear-solver-print-json-definition	Added option to write the JSON definition of the linear solver setup to the DBG file.	<code>true</code>

OPM Flow 2023-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
3	--linear-solver-reduction	The minimum reduction of the residual which the linear solver must achieve.	0.01
4	--local-domain-ordering-measure	Added parameter to specify the domain ordering measure as either residual or pressure when using the NLDD solver (#4738).	pressure
5	--local-domains-partitioning-imbalance	Subdomain partitioning imbalance tolerance for the NLDD solver. 1.03 is 3 percent imbalance.	1.03
6	--local-domains-partitioning-method	Subdomain partitioning method for the NLDD solver. Allowed values are 'zoltan', 'simple', and the name of a partition file ending with '.partition'.	zoltan
7	--local-solve-approach	Choose local solve approach for the NLDD solver. Valid choices are jacobi and gauss-seidel.	jacobi
8	--local-tolerance-scaling-cnv	Set lower than 1.0 to use stricter convergence tolerance for local solves when using the NLDD solver.	0.01
9	--local-tolerance-scaling-mb	Set lower than 1.0 to use stricter convergence tolerance for local solves when using the NLDD solver.	1
10	--local-well-solve-control-switching	Added trigger to enable (true) or disable (false) well control/status switching during local well equation solves when using the NLDD solver (#4895).	false
11	--max-local-solve-iterations	Max iterations for local solves with NLDD nonlinear solver.	20
12	--maximum-water-saturation	Maximum water saturation.	1
13	--network-max-iterations	Maximum number of iterations in the network solver before giving up.	200
14	--network-max-strict-iterations	Maximum iterations in network solver before relaxing tolerance.	100
15	--nldd-num-initial-newton-iter	Added option to specify number of global non-linear (Newton) iterations performed by the NLDD solver before starting the local non-linear iterations. The default value of 1 preserves the current behavior (#4922).	1
16	--nonlinear-solver	Choose nonlinear solver. Valid choices are newton or nldd.	newton

OPM Flow 2023-10 New And Deprecated Command Line Options			
No.	Variable Name	Description	Default
17	--num-local-domains	Number of local domains for NLDD nonlinear solver. Note this is an experimental feature in the current release 2023.10 that is expected to be more complete and tested by the 2024.10 release.	0
18	--save-step	Added new options to keep only the last step. Save serialized state to .OPMRST file. Either a specific report step, "all" to save all report steps or "x" to save every x'th step. Use negative values of "x" to keep only the last written step, or "last" to save every step, keeping only the last (#4807).	""
19	--water-only-threshold	Cells with water saturation above or equal to this value are considered one-phase water only.	1
20	--load-file	FileName for .OPMRST file used to load serialized state. If empty (""), CASENAME.OPMRST is used.	""
21	--num-pressure-points-equil	Number of pressure points (in each direction) in tables used for equilibration (#4718).	2000
22	--save-file	FileName for .OPMRST file used for saving serialized state. If empty, CASENAME.OPMRST is used.	""
23	--use-average-density-ms-wells	Approximate segment densities by averaging over segment and its outlet.	false
24	--ecl-max-time-step-size-after-well-event	Maximum time step size after a well event (seconds). Default is equivalent to 1 year. Deprecated.	3.15576e+07
25	--ecl-restart-shrink-factor	Factor by which the time step is reduced after convergence failure. Deprecated	3

Notes:

- 1) Cells colored in green in the "No." column indicate a new command line option or a change to an existing option for this release.
- 2) Cells colored in orange in the "No." column indicate the command line option is available but is now "hidden" from the main help listing, --help. These options can be listed using the --help-all command line option, that lists all the command line options included in the release, including experimental, obsolete, hidden and deprecated options.
- 3) Cells colored in red in the "No." column indicate a deprecated command line option for this release.

Table B.8: OPM Flow 2023-10 New and Deprecated Command Line Options

B.5.2 NEW FEATURES

In addition to the above the following new features have been added to the simulator:

- 1) When the general well specification data is defined by the [WELSPECS](#) keyword in the [SCHEDULE](#) section the well can now be assigned directly to the [FIELD](#) group in item 2 GRPNAME (#3485 and #4608). Previously there was a restriction preventing wells from being parented directly to [FIELD](#). Although wells are now allowed to be parented directly to [FIELD](#), this is discouraged and a warning message will be issued. Mixing wells and groups as children of a single group is still forbidden.
- 2) Added support for dissolved gas in water ([DISGASW](#)) in the gas-water simulator with salt precipitation ([PRECSALT](#)) and vaporized water in the gas phase ([VAPWAT](#)) (#4623). However, input of gas solubility in water is not currently supported so this option is only useful in combination with [CO2STORE](#).
- 3) Added support for modeling dissolved gas in water ([DISGASW](#)) and vaporized water in the gas phase ([VAPWAT](#)) in the thermal-gas-water simulator (#4661).
- 4) Added support for temperature ([THERMAL](#)) plus salt precipitation ([PRECSALT](#)) modeling in gas-water-brine (GAS-WATER-BRINE) systems as specified in the [RUNSPEC](#) section (#4650). This has to date only been tested in combination with [CO2STORE](#). Note that generally, if the [PRECSALT](#) keyword has been activated in the input deck then the [VAPWAT](#) keyword should also be activated.
- 5) Added support for modeling foam ([FOAM](#)) plus solvent ([SOLVENT](#)) in the simulator (#4654, #3523 and #805). In addition, gas or water is now allowed as the transport phase for the foam as specified by [FOAMOPTS](#) item 1 (previously only gas was supported as the foam transport phase).
- 6) Added support for [WVFPDP](#) in the [SCHEDULE](#) section (#4620 and #3504). The [WVFPDP](#) keyword modifies a well's Bottom-Hole Pressure ("BHP") estimated by the simulator by interpolation of the Vertical Flow Performance ("VFP") tables.
- 7) Added support for [GCONPROD](#) item 7 [ACTION](#) equal to NONE in the [SCHEDULE](#) section, which specifies that no action is to be taken if the oil, water, gas or liquid rate constraints are violated. Previously only the RATE option was supported by the simulator (#4658).
- 8) Added partial support for [WAGHYSTR](#) keyword in the [PROPS](#) section (#4710 and #3542). This keyword defines the parameters for the Water-Alternating-Gas ("WAG") hysteresis option, when the hysteresis option has been activated by the [HYSTER](#) variable on the [SATOPTS](#) keyword in the [RUNSPEC](#) section. Only gas phase hysteresis is currently supported by the [WAGHYSTR](#) keyword. The residual oil modification fraction, which would only be active when the [STONE1](#) three-phase oil relative permeability model is used, is not currently supported.
- 9) Support added for [PPCWMAX](#) to limit the maximum capillary pressure scaling when initializing the model using [SWATINIT](#), and the option to modify the connate water saturation to match the input water saturation if the capillary pressure is exceeded (#4707 and #3570).
- 10) Added support for [WINJMULT](#) in the [SCHEDULE](#) section to define pressure dependent injectivity multipliers for injection wells and can be used to approximate the change in injectivity due to hydraulic fracturing (#4686).
- 11) Added partial support for [WPAVE](#) and [WWPAVE](#) defining the method and parameters for calculating a well's block average pressures for either all wells or specific wells (#4695, #4694 and #4693). Calculation of block averages pressures is currently only supported for OPEN completions (not for ALL completions) as specified in [WPAVE](#) item 4 and [WWPAVE](#) item 5.
- 12) Aquifer keywords have been enabled when [H2STORE](#) is specified in the [RUNSPEC](#) section with brine modeled by the [OIL](#) phase (#4791). This is similar to the behavior when [CO2STORE](#) is specified.
- 13) Added a model for formation damage due to suspended solids in the injection water (#4346 and #3313). New OPM specific well keywords [WINJDAM](#), [WINJFCNC](#) and [WINJCLN](#) have been added to the [SCHEDULE](#) section. [WINJDAM](#) sets up the filter cake properties for the specified water injection

well, [WINJFCNC](#) defines the filtrate concentration in the injection water, and [WINJCLN](#) specifies that a fraction of the filter cake has been cleaned up.

- 14) User defined arguments (UDA) have been enabled for the [WINJFCNC](#) keyword item 2 (FCONCPPM) the injection concentration parameter ([#4763](#) and [#3614](#)).
- 15) Added partial support for [GRUPNET](#), which defines the standard group network parameters used to model the flow and pressure behavior within the network ([#4760](#), [#3609](#), [#4815](#) and [#3651](#)). [GRUPNET](#) item 5 OPTION1 equals NO only is supported (defines how the groups production target should be achieved). Item 6 OPTION2 equals NO or FLO is supported but ALQ is not supported (defines how gas lift gas flows through the group's pipeline). Item 7 OPTION3 equals NONE only is supported (defines if the ALQ-PIPE variable should be reset).
- 16) Added partial support for [GECON](#) ([#4819](#) and [#3657](#)). Item 7 WORKOVER only supports NONE, workover procedures are not currently implemented. Item 8 ENDRUN only supports NO, end run is not currently implemented. Item 9 MXWELLS is not supported and must be defaulted.
- 17) Support has been added for the [CSKIN](#) keyword in the [SCHEDULE](#) section to modify the connection skin factor ([#4871](#) and [#3681](#)).
- 18) Support for [WSF](#) and [GSF](#) keywords in the [PROPS](#) section has been added for [H2STORE](#) runs (as for [CO2STORE](#)) ([#4528](#)). The [WSF](#) and [GSF](#) keywords define the water relative permeability data versus water saturation tables and the gas relative permeability data versus gas saturation tables for when only gas and water are present in the input deck. These keywords can only be used with either the [CO2STORE](#) or [H2STORE](#) models.
- 19) Support has been added for setting the [WGRUPCON](#) keyword Item 2 STATUS in the [SCHEDULE](#) section equal to NO ([#4575](#)).
- 20) Support has been added for setting the [PINCH](#) item 2 PINCHOPT parameter equal to NOGAP in the [GRID](#) section. This parameter is now fully supported by the simulator. Previously only the default GAP option was supported ([#4603](#)).
- 21) Added support for initializing constant flux aquifers from a restart ([#4520](#) and [#4519](#)). In particular, form constant flux aquifer objects from the restart step's collection if available and properly initialize their total produced volume.
- 22) Added support for gas-water-solvent systems ([#4568](#) and [#4548](#)).
- 23) Added partial support for setting items 11, 12 and 13 of [GCONPROD](#) in the [SCHEDULE](#) section: ACTWAT, ACTGAS and ACTLIQ respectively define the action to be taken if the water rate (VRAT), gas rate (GRAT) and liquid rate (LRAT) constraints defined by [GCONPROD](#) are violated. Supported options are now NONE or RATE. Options CON, +CON, WELL, and PLUG are not currently supported ([#4748](#)).
- 24) Added support for Non-linear Domain Decomposition (NLDD) as the non-linear solver.
- 25) The option has been added to allow well control or well status to be updated during the iteration process during local solve for the well equations ([#4895](#)). In this way the converged well is given the correct control/status for the current reservoir state. A command line argument `-local-well-solve-control-switching=true` has been added to trigger the use of the function (this is *false* by default).

B.5.3 IMPROVEMENTS

Improvements include:

- 1) The correct version of flow is now selected for two-phase gas-water systems with either water vaporized in the gas phase ([VAPWAT](#)) and/or gas dissolved in the water phase ([DISGASW](#)) as defined in the [RUNSPEC](#) section ([#4592](#)). Previously the wrong version of flow was selected for gas-water systems with water vaporized in the gas phase (but no gas dissolved in the water phase).

- 2) The boundary conditions keyword `BC` has been deprecated and been split into two parts: `BCCON` in the `GRID` section where the block indices and direction are set, and `BCPROP` in the `SCHEDULE` section where the type and value of the boundary condition are set (#3482, #802 and #949).
- 3) The maximum number of allowable `EQLNUM` regions has been increased from 255 to 65525 (#4726).
- 4) A stricter convergence tolerance has been imposed on standard (not multi-segment) wells with a zero rate target. This change improves the results of a reported case with zero reservoir volume rate (RESV) control and potentially improves the material balance of the whole system (#4572).
- 5) When testing a gas lift well under THP control, if the well does not converge with the maximum artificial lift quantity (ALQ) then the simulator will now try to reduce ALQ in increments to check if the well equations converge with a smaller ALQ (#4579).
- 6) The bisection algorithm has been updated to keep track of the size of its search interval. The algorithm will now stop iterating if the minimum interval size is reached (#4617).
- 7) The simulator has been modified to ensure that THP constraints are only applied to prediction wells and not history match wells (#4615).
- 8) The simulator has been updated to avoid requesting the ALQ value for injectors, the ALQ value is only requested for producers (#4648).
- 9) The simulator has been modified to output temperature information to the restart file if `TEMP` is specified in the `RPTRST` keyword or if `THERMAL` is set in the `RUNSPEC` section even if the command line option `-enable-opm-rst-file=false` is set (#4646). The command line option `-enable-opm-rst-file` is a Boolean value that controls the output of OPM specific data sets to the commercial simulator's restart file to enable restart of OPM Flow runs by OPM Flow (`true`), or not to output the data (`false`).
- 10) Update the calculation of immobile gas saturation to take into the account trapped saturation (calculated when the hysteresis model is used) in the calculation of mobile and immobile fluid volumes in place (#4642 and #3517). The hysteresis model is activated by the specifying the `HYSTER` parameter of the `SATOPTS` keyword in the `RUNSPEC` section.
- 11) An error message will now be generated if invalid region numbers `SATNUM`, `PVTNUM`, `IMBNUM` or `EQLNUM` are input (#4705). Valid region numbers are positive integers less than or equal to the maximum number of regions specified in the `RUNSPEC` section.
- 12) Added `setVapPars()` at the start of the report step to properly handle oil vaporization parameters (`VAPPARS`) updates in the `SCHEDULE` section (#4677).
- 13) If vaporised water (`VAPWAT`) is present in the model but the initial equilibrium vaporized water in gas ratio (`RVW`) for an equilibration region is not explicitly defined by a value versus depth table (`RVWVD`), and the datum depth is not at either the gas-oil contact (if oil is present) or the gas-water contact (if oil is not present) then the values of `RVW` will be initialised to zero (#4647 and #4688). This makes it easier to include `VAPWAT` in `CO2STORE` cases.
- 14) The mass balance limit (`XXXMBE`) from the keyword `TUNING` is now used if it is explicitly specified (not defaulted) and the command line argument `-enable-tuning=true` is used (#4621 and #3522).
- 15) The simulator now gives a more informative error message if the input grid has no active cells at all. Often this points to an error in the input data and this change might help in finding the problem (#4735).
- 16) A more user-friendly error message is now reported when the time step is cut too often or too much. (#4746).
- 17) Perform a more graceful exit instead of `MPI_Abort` for expected exceptions in parallel runs. Instead of unconditionally issuing `MPI_Abort` if a fatal exception is encountered, the simulator will try to test whether all processes have experienced this exception and if this is the case just terminate normally with an exit code that signals an error. `MPI_Abort` is still used if not all processes get an exception as this is the only way to ensure that the program aborts (#4750). This approach also works around issues in some MPI implementations that might not correctly return the error.

- 18) Hydrostatic and acceleration pressure losses have been included for well segments representing an inflow control device (ICD) (#4824). Valves may be placed in long segments with significant depth differences (for example, at branch inlets), where the hydrostatic contribution in particular may be significant.
- 19) An error message is now reported if a negative oil or water saturation is passed to the *RV* and *RVW* initialization routine (#4675).
- 20) When approximately zero well rates are encountered during iterations (pre-convergence), gas/water fractions become highly inaccurate which in turn may lead to the solver getting stuck or the well getting shut prematurely. The code has been updated to switch to explicit VFP table lookup whenever the rate drops below the lowest value in the table. The logic around explicit lookup is also updated so that this approach also works for just opened wells (#4669).
- 21) Information about linear system sizes is now output to the DBG file (#4734 and #4754).
- 22) For stopped or zero-rate-target wells the *alternative_well_rate_init* procedure previously returned unscaled well-rates resulting from a zero-bhp condition. This could lead to convergence failures for network balancing since the initialized rates could be off by orders of magnitude. The code now skips this procedure for stopped or zero-rate-target wells reducing the risk of convergence failures (#4817).
- 23) The two-point flux-approximation (TPFA) has been added as a linearizer for gas-oil cases with energy (with diffusion) and gas-oil diffusion (#4825 and #4816). All energy cases now have diffusion enabled.
- 24) Added *RESTART* file output for the geomechanical module (#4803 and #4588).
- 25) Relaxation factors slightly outside the interval [0, 1] in the standard well model are reset to the interval limits, while keeping the assertion for factors further outside the interval to possibly pick up failure cases (#4862).
- 26) Connections between reservoir cells and numerical aquifer cells, or between numerical aquifer cells when multiple such cells define a single numerical aquifer, are now always treated as NNCs for output purposes (#4821).
- 27) Regional transmissibility multipliers such as those entered in the *MULTREGT* keyword are now applied to explicit input non-neighbor connections (*NNC*). This now implements all known connection behaviors for inter-region connections. In the commercial simulator, if the region numbers specified in *MULTREGT* items 1 and 2 are equal and positive then the transmissibilities within the region as well as any transmissibilities connecting any other regions are multiplied by the given value. The option to specify transmissibility multipliers in this way using *MULTREGT* is not yet supported (#4822 and #4821).
- 28) The name of the missing OPM restart file is reported in the error message when it cannot be located (#4870).

```
Error locating serialized restart file PATH/FILENAME.OPMRST
```
- 29) Added PRT file output when group economic criteria for production groups *GECON* is activated (#4866).
- 30) Reduced repetitive output of network information when running in parallel (#4879).
- 31) The lower limit for bottom hole pressure (BHP) in Newton updates has been reduced slightly from 1 bar. This allows for cases that might have a defaulted BHP constraint of 1 bar (#4877).
- 32) The region set name matching algorithm has been changed to using unique prefixes. This enables the simulator to recognize that the region set name FIPUNI should match the user defined region name FIPUNIT (#4868).

- 33) Debug output sent to the DBG file is no longer also echoed to the console as well ([#4955](#) and [#4941](#)).
- 34) Updated the derivatives and also the Jacobian matrix for the multisegment well pressure equations ([#4640](#)).
- 35) The intensive quantities for `timelidx==1` corresponding to the start of the current timestep are now updated if we do not use the storage cache, or if we cannot recycle the first iteration storage ([#4662](#)).
- 36) Improved message issued when shutting a well because it fails to converge ([#4741](#)).
- 37) Damaris was updated in version 1.8.0 to support the HDF5 `H5Sselect_elements()` capability to rewrite data in memory to another order on disk. This allows (MPI decomposed) local simulation data to be written back to its original global position on disk. To support this a new element was added to the Damaris XML `<variable...>` type, named "select-file", along with some other options (not required by OPM Flow). This keyword is added to the basic in-built Damaris XML file of OPM Flow ([#4830](#)).

In addition, the following new [SUMMARY](#) keywords are now recognized as described by the comments in Table B.9.

No.	Summary Keyword	Comment
1	WBP, WBP4, WBP5 and WBP9	Added support for SUMMARY vectors to output well block averaged pressures for open completions (#4695 , #4694 and #4693).
2	WINJFVR, WINJFVT and WINJFC	Added support for SUMMARY keywords relating to filtration injection modelling (#4790 , #3628 and #3626). Added injection well SUMMARY vectors for filtrate volume injection rate (volume rate), filtrate volume injection total (volume), and filtrate injection concentration (ppm).
3	CINJFVR, CINJFVT, CFCSKIN, CFCWIDTH, CFCPERM, CFCPORO, CFCRAD and CFCAOF	Added support for SUMMARY keywords relating to filtration injection modelling (#4790 , #3628 and #3626). Added injection well connection SUMMARY vectors for filtrate volume injection rate (volume rate), filtrate volume injection total (volume), skin factor due to filtration (dimensionless), thickness of filter cake (Length), permeability of filter cake (MD), porosity of filter cake (dimensionless), well bore radius used in the filtration modeling (Length), and well bore area of flow used in the filtration modeling (Area).
4	SxDEN, SDENM and SMDEN	Added support for SUMMARY vectors to output fluid densities at the block and well segment levels (#4744 , #3594 and #3593). Added well segment SUMMARY vectors for phase density of phase x (segment conditions), fluid mixture density weighted by phase flowing fractions, and fluid mixture density weighted by phase flowing fractions with exponents. Phase x is one of O (oil), G (gas), or W (water).
5	BxDEN and BDENx	Added support for SUMMARY vectors to output fluid densities at the block and well segment levels (#4744 , #3594 and #3593). Added block SUMMARY vectors for phase density of phase x. Phase x is one of O (oil), G (gas), or W (water).
6	BFLOWI, BFLOWJ, and BFLOWK	Added support for SUMMARY vectors BFLOWI, BFLOWJ and BFLOWK to request block oil/gas/water flow rates at surface conditions (#4867 , #3675 and #827). This does not require FLOWS to be specified in <code>RPTRST</code> .

Table B.9: New [SUMMARY](#) Keywords for the 2023-10 Release

B.5.4 BUG FIXES

The following bug fixes and improvements have been incorporated into this release.

- 1) Fixed the issue of incorrectly using the gas-oil contact depth for initialization of vaporized water (*RVW*) in the special case of a two-phase gas-water system. The gas-water contact depth is now used in this case ([#4688](#) and [#4647](#)).
- 2) Fixed a bug in the group pressure maintenance (*GPMAINT*) code ([#4664](#)). Previously, if negative group pressure maintenance rates were calculated, they would first be used to incorrectly update the *GPMAINT* error integral, before the rates were later set to zero. Now the simulator only calculates rates for injectors if the pressure is below the regional pressure target (or for producers if the pressure is above the regional pressure target).
- 3) Bug fixes related to the handling of gas dissolved in water ratio (*RSW*) and water vaporized in gas ratio (*RVW*) in the standard well model ([#4591](#)).
- 4) The max size of the next time step is now only used when specified in *TUNING* or *NEXTSTEP* ([#4660](#)).
- 5) If the data file contains the *CPR* keyword then the "CPR" preconditioner should be used unless it is overridden by a command line argument. Previously this was incorrectly overridden by the command line argument `--linear-solver-max-iter`, the correct command line argument `--linear-solver` is now used ([#4700](#)).
- 6) The user is now able to specify any number of threads, this prevents only one thread being used on some hardware where the number of processors would always be reported as one when using MPI and OpenMP (irrespective of the actual number). The behaviour is now: (a) if nothing is specified then 2 threads are used, (b) if *OMP_NUM_THREADS* is specified then this number is used regardless of the hardware and the command line argument `--threads-per-process` is ignored, and (c) if `--threads-per-process` is used and *OMP_NUM_THREADS* is not set then the number specified on the command line is used ([#4709](#)). There is a check to make sure that the number of threads used by the linearizer is consistent ([#811](#)).
- 7) Fixed a bug related to indexing in the temperature boundary condition ([#4761](#)).
- 8) Gas dissolved in oil (*RS*) and oil vaporized in gas (*RV*) ratio initialized using value versus depth tables (*RSVD*, *RVVD*) should be limited by their respective saturated values. Fixed a bug where this limit was not applied at depths outside the range of depths in the *RSVD* or *RVVD* table ([#4723](#)).
- 9) For producers where all perforations have zero rates the perforation mixture fraction is approximated using the (inverse formation volume factor times mobility) ratio, and weight the perforation rates using the well transmissibility. The perforation mixture fraction was previously approximated using only the mobility ratio ([#4681](#)).
- 10) A case under investigation has some cells containing only oil and water (zero gas saturation), which have zero relative permeability (and mobility) for all the three phases. This caused problems in part of the code which assumed the total mobility was non-zero. For perforations having zero mobility for all the phases, the simulator now uses a small value to generate small perforation rates for those perforations, at the same time, the simulator can use these rates to recover the mixing ratios for those perforations ([#4682](#) and [#4681](#)).
- 11) The correct formation volume factors are now used in the case of zero phase rates resulting in zero *RS*, *RV*, *RSW* or *RVW*. Previously the saturated formation volume factor was used in these cases ([#4590](#)).
- 12) The reservoir volume rate constraint (RESV) in *GCONPROD* was not honoured. This has been fixed ([#4687](#)).
- 13) Previously, in the case of zero threshold pressure and zero pressure difference, the code would set the pressure difference explicitly to zero. This would also set any derivatives to zero, which could

disconnect the corresponding matrix rows. The code no longer sets the pressure difference to zero when not necessary (#4701).

- 14) Code has been updated to avoid dividing by zero when scaling the well rates in `updateWellStateRates()` (#4715 and #4649).
- 15) The code has been updated to avoid round off errors in phase saturations leading to the initialization of dissolved or vaporized fluid ratios (for example saturated `RS`) with saturated values instead of taking values from the input ratio versus depth table (for example `RSVD`) (#4720).
- 16) Code has been updated to avoid writing beyond array limits (#4753).
- 17) The well closure reason is now set to `GROUP` rather than `ECONOMIC` if a group economic constraint (`GECON`) is reached. This prevents the closed well being reopened with `WTEST` item 3 `TEST` equal to 'E' (#4854).
- 18) The simulator now sets the well THP to be zero in the `WellState` only if there is no VFP table associated with the well (#4932). Previously, this would be done if the well had no target THP or THP limit.
- 19) The code now checks whether `LIFTOPT` is active first to avoid unnecessarily running routines relating to gas lift optimization (#4956).
- 20) MPI is now initialized before creating the communicator and a couple of fields are now set to `ROOT_ONLY` in `opmrst_inspect()` (#4601).
- 21) The `updateWaterMobilityWithPolymer()` method updates values, it does not rewrite them. Hence we have to feed it the scalar values not zeros. Scalar values are now used (#4655).
- 22) The restriction on the length of the next timestep following an event will now be applied following production or injection updates (`WCONPROD`, `WCONINJE` keywords) (#4781). The maximum length of the next timestep following an event can be set using either the `TUNING` keyword item 10 (`TMAXWC`) with the command line argument `-enable-tuning=true`, or by using the command line argument `-time-step-after-event-in-days=x` (where `x` is the number of days).
- 23) Added perforation data (`perf_data`) comparison in the equality operator for `SingleWellState` (#4783).
- 24) Added check that rock compaction table indices (`ROCKNUM`) are within the bounds of the tables defined by `ROCKTAB` (#4788).
- 25) Code modified to avoid a possible segmentation fault in the cleanup routine if the simulator has not been set up (#4794).
- 26) Fixed a bug where the code could attempt to access entries in a zero element array due to not dereferencing a pointer to inspect array elements when deciding whether to apply `TRANX`, `TRANY`, etc. keywords (#4801).
- 27) Code has been modified to prevent slightly negative oil fractions in well segments occurring due to round-off. This could have lead to failure if the segment represented for example an autonomous Inflow Control Device (ICD) valve in a multi-segment well (#4834).
- 28) Fixed a bug with the `DIRICHLET` option in the boundary condition property definition (`BCPROP`) not working in gas-water runs. Additionally, fixed a bug where `BCPROP` had to be defined at every report step (#4835).
- 29) If a well was `SHUT` due to economic or physical reasons the well-state quantities were mostly set to zero (including the BHP). The code has been modified to set the BHP to the BHP limit when its value has not been initialized when under well target rate control (#4841). This helps to prevent the Newton update from stagnating under certain circumstances (for example, a reported issue introduced by #4772).
- 30) Fixed a bug which occurred when using non-neighbor connections (`NNC`) in thermal simulations (#4900).

- 31) Fixed computation of temperature for distributed wells ([#4888](#)). This was not an issue for non-distributed wells.
- 32) Code has been modified to always allocate buffers for storing flows if BFLOWI, BFLOWJ or BFLOWK is requested. This fixes a segmentation fault that occurred if these keywords were requested without [RPTRST](#) in the [SOLUTION](#) section ([#4904](#)).
- 33) Code has been modified so that there will be no vertical connection between cells if the layers in between are inactive or collapsed unless the [PINCH](#) keyword has been specified ([#4901](#)).
- 34) Fixed a bug which occurred in parallel runs where all processes other than rank zero would keep iterating beyond the maximum number of allowed iterations ([#4909](#)).
- 35) Fixed a bug where regional multipliers ([MULTREGT](#)) were applied twice to non-neighbor connections entered in [EDITNNC](#) ([#4921](#)).
- 36) Fixed a bug where if `-enable-vtk-output=true` and `-enable-write-all-solutions=false` (it is false by default), then Visualization Toolkit (VTK) files were written for all solutions. The code has been modified so that VTK files are only written for report steps (unless `-enable-write-all-solutions=true`) ([#4916](#)).
- 37) After DUNE version 2.7, the method `getCollectiveCommunication()` was changed to `getCommunication()`. The `getCollectiveCommunication()` method issues the deprecation warning and assumes all process take part in the communication. However, this might not be the case, namely, only a subset might be used. This fix consists of getting the communication object used by the grid as this uses only processes that are actually computing and not only outputting ([#4962](#)).
- 38) Fixed computation of dissolved gas in water ratio (Rsw) used in calculation of wellbore density for standard wells ([#4976](#)).

B.5.5 KNOWN ISSUES

- 1) There is an error with the RSM header for summary vectors whose NUMS entry in the SMSPEC file is derived from more than a single number source (e.g., single region or segment ID). This applies to all block vectors (BGPV, BOPV, BWPV, etc.), connection level quantities (COPT, etc.), and inter-region flows such as ROFT etc ([#3078](#)). The work around is to plot the data in OPM ResInsight and right-click on the plot to view and copy the data.
- 2) OPM Flow does not support using LIQ as a well's preferred phase with the [WELSPECS](#) keyword, that is [WELSPECS\(TYPE\)](#) equals LIQ. This is a long-standing bug/omission in the simulator stemming from a somewhat naive internal notion of phases so we don't have an entry for a liquid phase, only for the distinct oil and water phases. For producing wells this mostly matters if you plot the WPI summary vector (productivity index for well's preferred phase). In the current treatment WPI will not have contributions from the water phase if the declared preferred phase is LIQ. For injecting wells WELSPECS's preferred phase doesn't really matter at all since the preferred phase is (typically) reset to the injected phase in [WCONINJE/WCONINJH](#) anyway. See issue [#3075](#).
- 3) If the simulator finds well connection being declared as connections via the [COMPDAT](#) keyword in the [SCHEDULE](#) section, then it writes out a warning message:

```
Warning: Problem with COMPDAT keyword
In SIM_PEE_Basic_Simulator.DATA line 535
The cell (0,6,4) in well INJ1 is not active and the connection will be ignored

Warning: Problem with COMPDAT keyword
In SIM_PEE_Basic_Simulator.DATA line 535
The cell (16,0,4) in well INJ2 is not active and the connection will be ignored
```

However, the reported cell references are offset by minus one, meaning the correct warning messages should be:


```
Warning: Problem with COMPDAT keyword
In SIM_PEE_Basic_Simulator.DATA line 535
The cell (1,7,5) in well INJ1 is not active and the connection will be ignored
```

```
Warning: Problem with COMPDAT keyword
In SIM_PEE_Basic_Simulator.DATA line 535
The cell (17,1,5) in well INJ2 is not active and the connection will be ignored
```

See issue [#3167](#) for details.

- 4) The [GDFILE](#) keyword in the [GRID](#) section loads a grid file in various formats, with the FMTOPT parameter setting the format type of the file. If the variable FMTOPT is omitted then the default is for binary file input for the commercial simulator; whereas, OPM Flow derives FMTOPT from the file extension (*.EGRID or *.FEGRID), making FMTOPT superfluous. However, if the extension is lower case then OPM Flow may incorrectly determine the file type. The work around is change the extension to upper case.
- 5) As per previous releases of the radial model, the [COORDSYS](#) keyword item three must be set to COMP to complete the circle, this has not been implemented in this release. Also there appears to be a bug for full radial models when a well goes on BHP control that causes the well not to respect the BHP constraint, this eventually causes the well to die prematurely. See [#2640](#) for a discussion on the topic.
- 6) As in previous releases there are some issues with the [OPERATE](#) and [OPERATER](#) keywords associated with the input parsing; for various reasons a few of the fields require special case treatment in the grid processing, including (at least) [MULTZ](#), [PORV](#) and [ACTNUM](#), and for those keywords the [OPERATE/OPERATER](#) keyword does not work. The work around is to use the [MULTIPLY](#) keyword instead.
- 7) For the [UDQ](#) ASSIGN operator after the terminating "/" normally any comments can be entered; however, if there is "/" within the comment field, as per:

```
ASSIGN FUNGLYLD 1.196 / Condensate Yield (63.5/56.7)/(1.0 - 0.065)
```

then the simulator will abort. The work around is to manually place the comment characters "--" after the ASSIGN terminating "/", like so:

```
ASSIGN FUNGLYLD 1.196 / -- Condensate Yield (63.5/56.7)/(1.0 - 0.06)
```

- 8) At the moment, one cannot initialize tracers using the [EQUALS](#) keyword. Instead use the array format, that is the keyword followed by the required number of values, or the [TVDP](#) keyword in the [SOLUTION](#) section to set the initial tracer concentrations as a function of depth.
- 9) Currently, gas tracers cannot be used if the dissolved gas phase, as per the [DISGAS](#) keyword in the RUNSPEC section, is active in the model.
- 10) The summary vector RTIPTHEA, that defines the energy in-place between the initial and the current time for regions, is not supported unlike the FTIPTHEA and BTIPTHEA vectors. Secondly, the error message:

```
Warning: Problem with summary keyword RTIPTHEA
In RSM-THERMAL.data line 492
FIP region FIPHEA not defined in REGIONS section - RTIPTHEA ignored
```

is incorrect, as the message indicates that it is being treated like a named region, as per the [FIP](#) keyword, when it is actually a [SUMMARY](#) vector ([#3870](#)).

- 11) If there are cells that are very distorted, which can occur near fault planes, then the simulator may abort because it cannot calculate the pore volume of such cells. The work around is to re-generate the grid in the static model, taking care that the cells around the fault planes are more or less orthogonal ([#2992](#) and [#3770](#)).

- 12) Currently the `OPERATER` keyword in the `EDIT` section does not work with the `DEPTH`, `TRANX`, `TRANX` and `TRANZ` property arrays (#2994 and #748).
- 13) If a standard well is fully declared in an `ACTIONX` block which is then activated at a later date, and later the well is modified to be a multi-segment well using the `WELSEGS` and `COMPSEGS` keywords, then this will cause the simulator to abort with an assert failure. The solution to this issue is to not use this type of work flow in declaring wells (#2891 and #2895).
- 14) Although the `ACTIONX EXIT` command works as expected, it does not write out the requested RSM file at the end of the run. However, the other `SUMMARY` and `RESTART` files are written out (#2877).
- 15) Although the `GCONSUMP` keyword in the `SCHEDULE` section is fully implemented as documented, it is not possible to verify the output as the associated `SUMMARY` vectors are not written out, that is the `SUMMARY` sales gas vectors FGSR, FGST, GGSR and GGST, and fuel vectors FGCR, FGCT, GGCR, and GGCT have not been implemented (#2679).
- 16) There are small differences in the behavior of the `NEXTSTEP` keyword in the `RUNSPEC` section between OPM Flow and the commercial simulator that remain unresolved (#3745).
- 17) **There is a unit handling issue associated with `OPERATE` keyword. If the `OPERATE(X)` parameter has units, as for example `PERMX`, then the conversion is always done in SI units, despite the input deck declaring the deck to be fields units, as per `FIELD` keyword in the `RUNSPEC` section. Note that OPM Flow performs all of its calculations internally in SI and performs unit handling only when inputting the *.DATA file and when outputting result files. Thanks to [Irijkels](#) for reporting the issue. See #4597 for details.**
- 18) There is an issue associated with restarting from a restart file with the solution gas (Rs) maximum rate of increase, as defined by the `DRSDT` keyword in the `SCHEDULE`, that has been set to zero. This is because, the simulator does not save/restore this setting in simulator's restart files, which means that simulator misses the essential value zero upon restarting the case. As a work-around one can use the option `--sched-restart=true`, when running the restart case. This will initialize the restarted simulation based on information from the complete `SCHEDULE` section, instead of just the parts that we're going to simulate and the rest from the restart file. Thanks to [goncalvesmachadoc](#) for reporting the issue. See #4272 for details.
- 19) OPM cannot be built with dune-fem version 2.9 or later. Please use a previous version.
- 20) The simulator uses an irregular corner-point grid geometry with adjusted pore volumes to represent radial grids so it is not possible to create a full ring (360 degree disk) with only one cell in the theta direction (NY=1). The work around is to model a slice (say `DTHETA=60` degrees). Note that as the angle increases larger pore volume adjustments are required (#4755).
- 21) In principle the `PYACTION` code can control anything, however the simulator generally deals poorly with wells not explicitly controlled by the input file (for example opening or closing wells). It is recommended to utilize the normal `ACTIONX` machinery for well controls by constructing an `ACTIONX` block in the Python code and then passing that back to the Schedule object (#4810 and #4813).
- 22) If the `--save-step` command line option is used to request the serialized state is saved to an .OPMRST file and HDF5 support is missing then an error is generated at the report step where the save was requested. This error should be generated during startup to save time and resources (#4812).
- 23) In some cases with the network option the simulator can wrongly report that a well has no THP constraints, for example

```
GLIFT WTEST: Well S-3H does not have THP constraints
```

when THP constraints have been defined (#4887).

APPENDIX C: OPMRUN – FLOW JOB SCHEDULER

APPENDIX TABLE OF CONTENTS

APPENDIX C: OPMRUN - FLOW JOB SCHEDULER.....2431

C.1 Introduction.....	2435
C.2 OPMRUN Functionality.....	2437
C.2.1 Add Job and Select Run Type.....	2438
C.2.2 Edit Job Data and Parameter File.....	2440
C.2.3 Load Previously Saved Queue.....	2442
C.2.4 Reset Job Queue Parameters.....	2443
C.2.5 Run Jobs in Queue with Various Options.....	2444
C.2.6 Menu Options.....	2447
File Menu Options.....	2447
Edit Menu Options.....	2447
View Menu Options.....	2450
Tools Menu Options.....	2450
C.3 OPMRUN Tools.....	2451
C.3.1 OPMRUN Tools: Job File Compression Utility for Saving Space and Archiving.....	2451
C.3.2 OPMRUN Tools: Simulator Input/Keywords.....	2452
Keywords: Menu Options.....	2454
Keywords: File Imports.....	2456
Keywords: Section Standard Set of Keywords.....	2457
Keywords: SUMMARY Section Variables.....	2458
Keywords: SCHEDULE Section Keywords and Date Schedule.....	2459
Keywords: DATA (Sets) Option.....	2460
Keywords: MODEL Option.....	2461
Keywords: USER Templates Option.....	2462
C.3.3 OPMRUN Tools: Simulator Input/Production Schedule.....	2463
C.3.4 OPMRUN Tools: Simulator Input/Sensitivities.....	2466
C.3.5 OPMRUN Tools: Simulator Input/Well Specification.....	2472
C.3.6 OPMRUN Tools: ResInsight.....	2475
C.3.7 OPMRUN Tools: Well Trajectory Conversion.....	2475

APPENDIX INDEX OF FIGURES

Figure C.1: OPMRUN - Initial Display.....	2437
Figure C.2: OPMRUN - Display Elements.....	2438
Figure C.3: OPMRUN - Add Job Dialog Box.....	2438
Figure C.4: OPMRUN - Add Job and Select Run Type.....	2439
Figure C.5: OPMRUN - Edit Job Data and Parameter File.....	2440
Figure C.6: OPMRUN - Edit Job Parameter File.....	2441
Figure C.7: OPMRUN - Load Job Queue Dialog Box and Queue Display.....	2442
Figure C.8: OPMRUN - Reset Job Queue Parameters.....	2443
Figure C.9: OPMRUN - Reset Job Queue Parameters Dialog.....	2443
Figure C.10: OPMRUN - Run Jobs in the Queue.....	2444
Figure C.11: OPMRUN - Select Run Option Dialog.....	2444
Figure C.12: OPMRUN - Log File Creation.....	2445
Figure C.13: OPMRUN - Session Log.....	2446
Figure C.14: OPMRUN - Edit Data File.....	2447
Figure C.15: OPMRUN - Set Default Parameter Set.....	2448
Figure C.16: OPMRUN - Set Application Options.....	2449
Figure C.17: OPMRUN - Set Project Directories.....	2449
Figure C.18: OPMRUN Tools -Compression Utility.....	2451
Figure C.19: OPMRUN Tools - Compression Utility Application.....	2451
Figure C.20: OPMRUN Tools - Keywords Application.....	2452
Figure C.21: OPMRUN Tools - Keywords HEADER Comment Example.....	2453
Figure C.22: OPMRUN Tools - Keywords File Menu.....	2454
Figure C.23: OPMRUN Tools - Keywords Edit Menu.....	2454
Figure C.24: OPMRUN Tools - Keywords Generate Menu.....	2454
Figure C.25: OPMRUN Tools - Keywords Help.....	2454
Figure C.26: OPMRUN Tools - Keywords Generator Help.....	2455
Figure C.27: OPMRUN Tools - Keywords Velocity Template Help.....	2455
Figure C.28: OPMRUN Tools - Keywords File Imports.....	2456
Figure C.29: OPMRUN Tools - Keywords Section Standard Set of Keywords Example.....	2457
Figure C.30: OPMRUN Tools - Keywords Summary Section Keywords.....	2458
Figure C.31: OPMRUN Tools - Keywords Schedule Section and Date Schedule.....	2459
Figure C.32: OPMRUN Tools - Keywords Data (Sets) Option.....	2460
Figure C.33: OPMRUN Tools - Keywords Model Option.....	2461
Figure C.34: OPMRUN Tools - Keywords User Templates Option.....	2462
Figure C.35: OPMRUN Tools - Production Schedule Input File Example.....	2463
Figure C.36: OPMRUN Tools - Production Schedule Application.....	2464
Figure C.37: OPMRUN Tools - Production Schedule Output Example.....	2465
Figure C.38: OPMRUN Tools - Sensitivity Application.....	2466
Figure C.39: OPMRUN Tools - Sensitivity Base Case Data File Example.....	2467
Figure C.40: OPMRUN Tools - Sensitivity Load Factors Example.....	2468
Figure C.41: OPMRUN Tools - Sensitivity Factor Editing Example.....	2469
Figure C.42: OPMRUN Tools - Sensitivity Scenario Option Selection Example.....	2470
Figure C.43: OPMRUN Tools - Sensitivity Case Generation and OPMRUN Queue File.....	2471
Figure C.44: OPMRUN Tools - Well Specification OPM ResInsight Well Completion File Format.....	2472
Figure C.45: OPMRUN Tools - Well Specification OPM ResInsight Reservoir Layer File.....	2473
Figure C.46: OPMRUN Tools - Well Specification Application.....	2474
Figure C.47: OPMRUN Tools - Well Specification WELSPECS and COMPDAT Output Example.....	2474
Figure C.48: OPMRUN Tools - Well Specification OPM ResInsight Perforation File Output.....	2475
Figure C.49: OPMRUN Tools - Well Trajectory Conversion Input File.....	2475
Figure C.50: OPMRUN Tools - Well Trajectory Application.....	2476
Figure C.51: OPMRUN Tools - Well Trajectory Output Trajectory File.....	2477

C.1 INTRODUCTION

OPMRUN is graphical user interface to Flow that has similar functionality to the commercial simulator's ECLRUN program. Target audience are Reservoir Engineers in a production environment. Developers and experienced Linux users will already have compatible work flows. The application performs the following:

- Allows editing and management of OPM Flow's run time parameters. Default parameters are automatically loaded from OPM Flow, and the user can reset the default parameter set either from a parameter or PRT file. Editing of a job's parameter file is also available.
- Runs under Linux and Windows 10. For Windows 10 OPM Flow is run via the Windows Subsystem for Linux ("WSL").
- Allows simulation jobs to be queued and run in either foreground (under OPMRUN), or in background in a xterm terminal session in Linux or WSL under Windows 10. Jobs in the queue can be set to run in *NOSIM* mode or RUN mode.
- Foreground jobs can be killed from OPMRUN, with the option of killing all the jobs in the queue.
- Queues can be edited, saved and loaded.

Various additional simulation input generation and conversion utilities are available including:

- Compressing a job to save space (DATA, and all OPM Flow output files) and uncompressing previously compressed jobs,
- Keywords, a keyword generator based on the Apache Velocity Template Language ("VTL"). The templates can therefore also be used with any editor that supports VTL, jEdit for example. There is one template per keyword, with the formatting the same as the OPM Flow manual. Over 450 templates are currently implemented. One can also customize the existing templates as well as creating User defined templates. The keywords are examples, one still has to edit the resulting deck with the actual required data, but the format with comments should make this a straight forward process.
- A Production Schedule application that takes a comma delimited CSV file containing historical production and injection data and converts the data to an OPM Flow *SCHEDULE* file using the *WCONHIST* series of keywords. Currently only production data is supported.
- Sensitivities application that generates sensitivity cases based on a "Base" case file. The Base file contains "Factors" (variable names), \$X01, \$X02, etc., that are substituted with user defined values using the data entered and the type of Sensitivity Scenario selected.
- A Well Specification application that uses the standard well export files from OPM ResInsight to reformat the data in a more user-friendly manner for the *WELSPECS* and *COMPDAT* keywords. Optionally, the application can generate the *COMPLUMP* keyword based on the OPM ResInsight layers file, with one completion per defined reservoir layer.
- Calling OPM ResInsight and loading the currently selected job into OPM ResInsight for viewing.
- A Well Trajectory Conversion application that converts a Schlumberger Petrel exported well trajectory file into an OPM ResInsight file, containing all the wells.

The software can be downloaded from the following link:

<https://github.com/OPM/opm-utilities/tree/master/opmrunk>

OPMRUN is written in Python 3 and tested under various Ubuntu distributions. Note that only Python 3 is supported and tested, Python2 support has been deprecated.

The program requires the following standard module Python libraries:

- datetime, getpass, importlib, os, numpy, pkg_resources, pandas, pathlib, platform, psutil, sys, re, subprocess, and tkinter as tk.

In addition, the following non-standard Python modules are required:

- airspeed, notify-py, pyDOE2, and PySimpleGUI.

For some Linux systems the relevant package manager may have to be used to install *tkinter* as *tk*; whereas for Windows 10 users the tkinter package is pre-installed with Python.

C.2 OPMRUN FUNCTIONALITY

OPMRUN enables the editing and management of OPM Flow's run time parameters, setting up job queues to run a series of simulation jobs sequentially, as well as the management of the job queues. Figure C.1 shows the initial display.

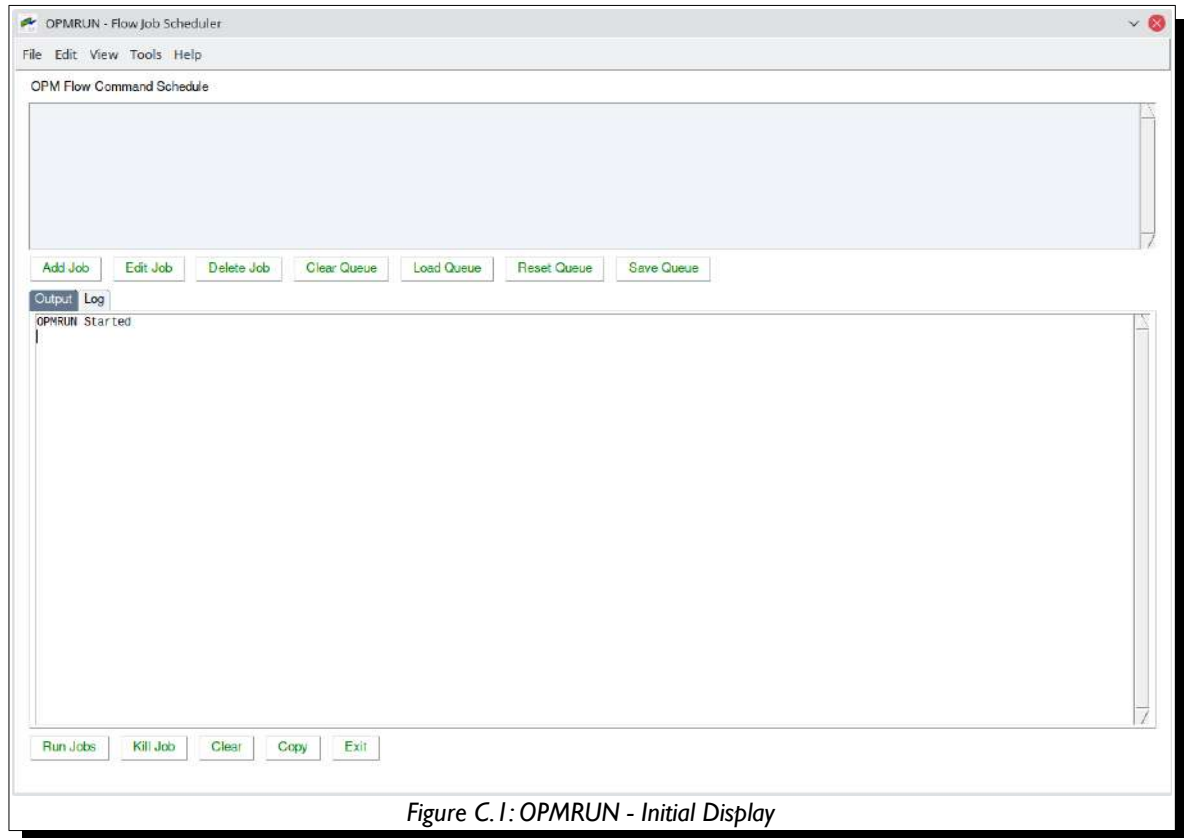


Figure C.1: OPMRUN - Initial Display

Upon launch the program runs OPM Flow to get a list of command line parameters from the current version of OPM Flow. These default parameters can be edited for each case, or alternative default parameter sets can be loaded from an existing parameter file from another job, or a *.PRT file from a completed simulation.

As can be seen in Figure C.2 the program has upper and lower display elements. The upper element shows a list of simulation jobs that are in the job queue and the lower element consists of two elements, one for the OPM Flow Output (the terminal output from OPM Flow) and a second element (OPM Run Log) that is a session log of the jobs run by OPMRUN. Clicking the OPM Flow Output and OPM Run Log tabs switches the display on the lowered element between two display types.

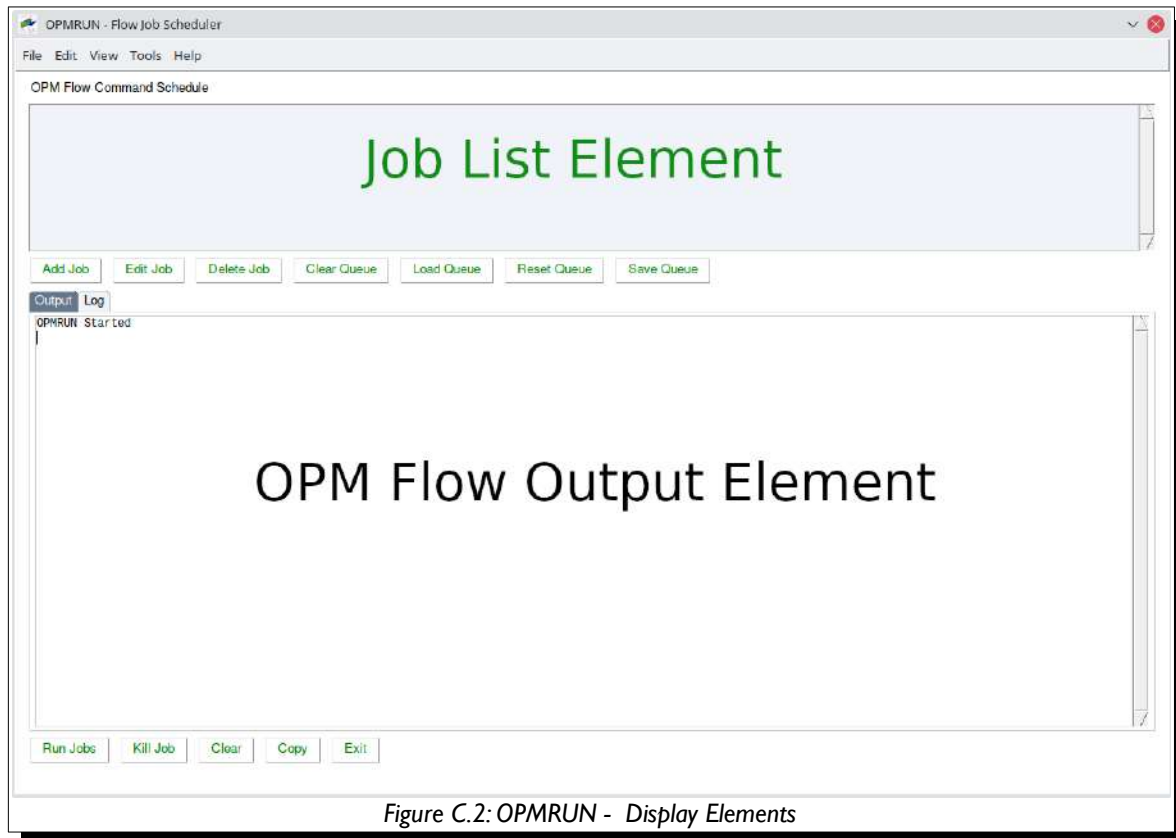


Figure C.2: OPMRUN - Display Elements

C.2.1 ADD JOB AND SELECT RUN TYPE

To add jobs to the queue use the Add Job button or load an existing job queue using the Load Queue button. Jobs can be edited or deleted from the queue using the Edit Job and Delete Job buttons, and a series of jobs can be saved as a job queue by using the Save Queue button. The Clear Queue button deletes all jobs from the queue.

Pressing the *Add Job* button will display the following dialogue box:

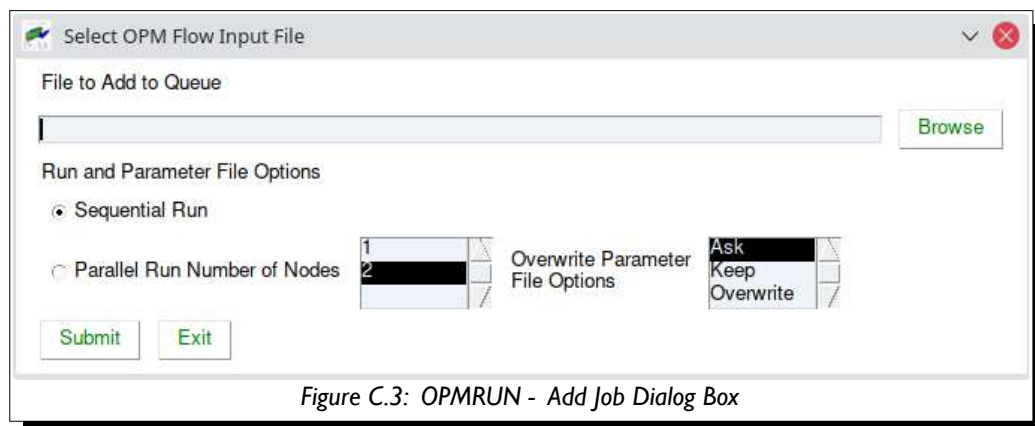


Figure C.3: OPMRUN - Add Job Dialog Box

Use the *Browse* button to select the input file to add to the queue, then select the Run Parameters for this input file, then press the *Submit* button to add the input file to the job queue (Figure C.4).

The Overwrite Parameter File Options allow for different default treatments of existing *.PARAM files, which is particularly useful when adding multiple jobs at the same time.

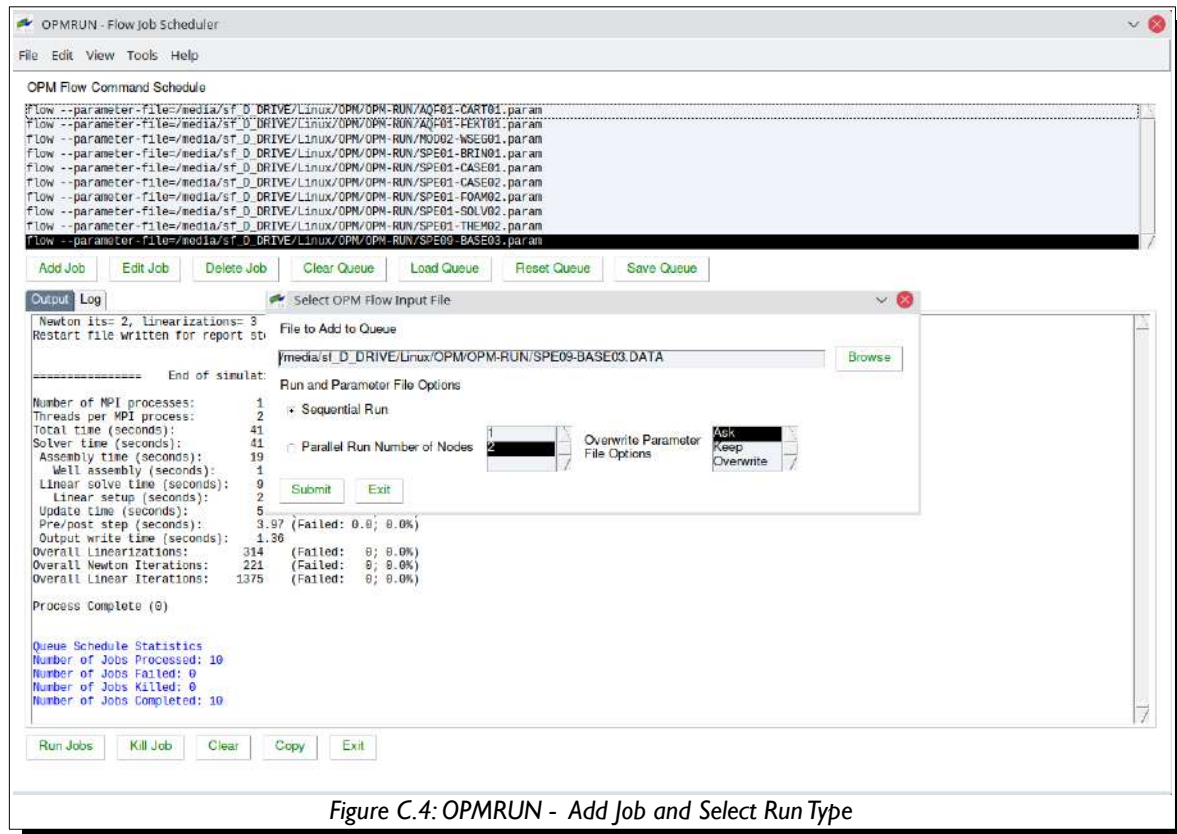


Figure C.4: OPMRUN - Add Job and Select Run Type

The reason for this is because different versions of OPM Flow have different parameter sets and if a newer version of OPM Flow runs with a previous version's *.PARAM file then the simulator will stop with an error message for the various invalid parameters for the current version of the simulator.

C.2.2 EDIT JOB DATA AND PARAMETER FILE

Jobs in the queue can be edited by selecting the *Edit Job* button that will display two options (Figure C.5): one to edit the input file using the defined editor and the second to edit the OPM Flow Parameter File.

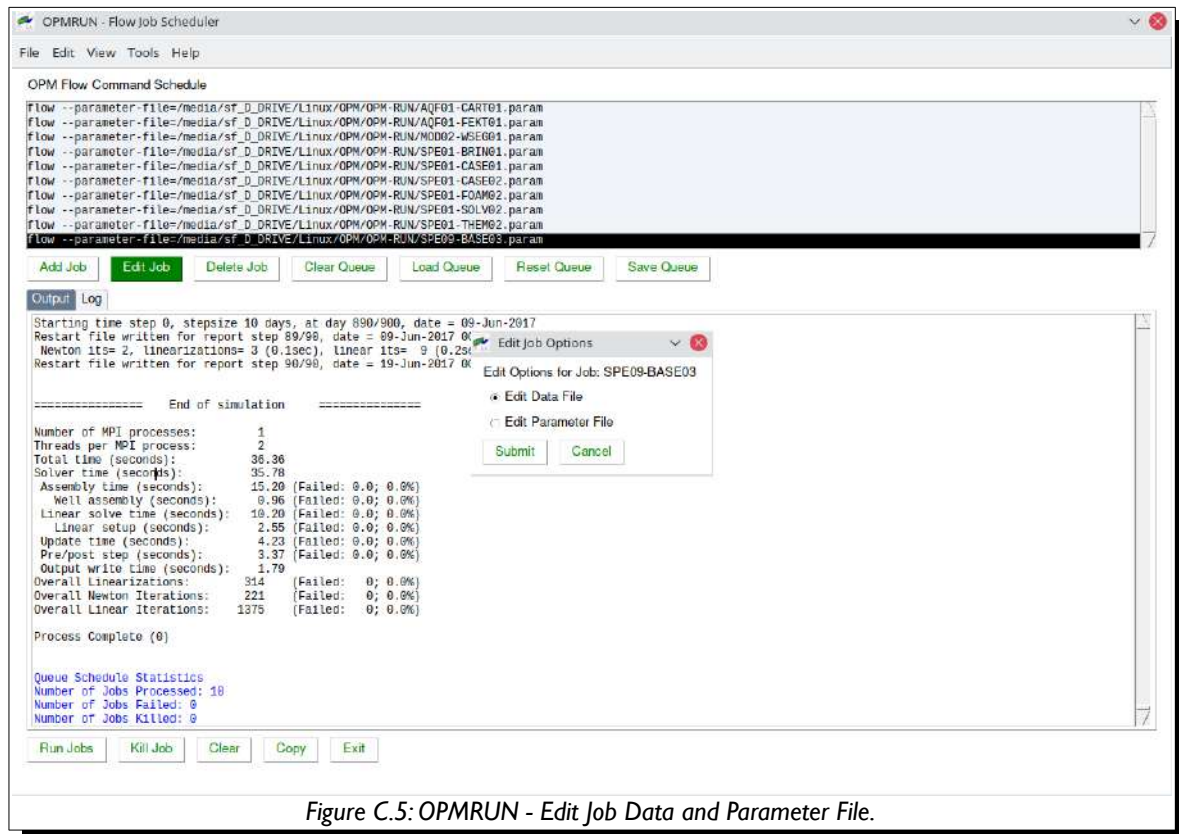
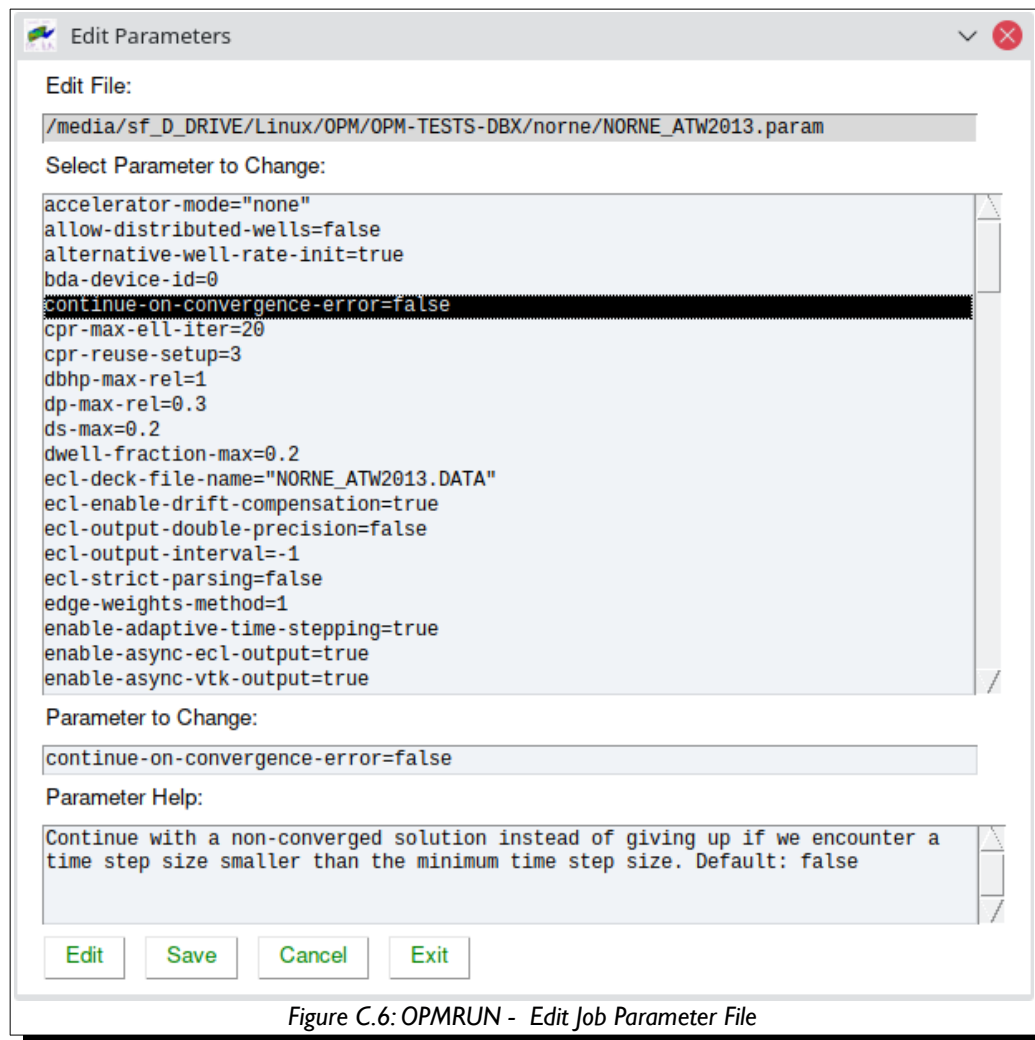


Figure C.5: OPMRUN - Edit Job Data and Parameter File.

If the second option is selected OPMRUN will display a dialog box that shows a list of the OPM Flow command line parameters together with the parameter help information (Figure C.7). Selecting a parameter from the list and selecting the Edit button will display the setting for the selected parameter (alternatively one can double click the required entry). One can then edit the parameter as required. Use the Save button to save the change and use the Exit button to save all the changes to the parameter file. The Cancel button will cancel all changes to the parameter file.



Alternatively one can use the:

- 24) Edit OPM Flow Parameter menu option to edit the parameter file for a job.
- 25) List OPM Flow Parameters menu option to list the commands in the parameter file for a job.
- 26) Set OPM Flow Default Parameters to set the default parameters for all subsequent jobs added to the queue. This option allows the user to load a default set of parameters from (1) OPM Flow, (2) an OPM Flow Parameter File, or (3) an OPM Flow print file (*.PRT).

One can also right-click on a job and select one of the available options.

C.2.3 LOAD PREVIOUSLY SAVED QUEUE

To load a previously saved job queue, press the *Load Queue* button this will display a dialog box allowing the user to select a queue file (*.que), after pressing the *OK* button the jobs will be displayed in the Job List Element as illustrated in Figure C.7.

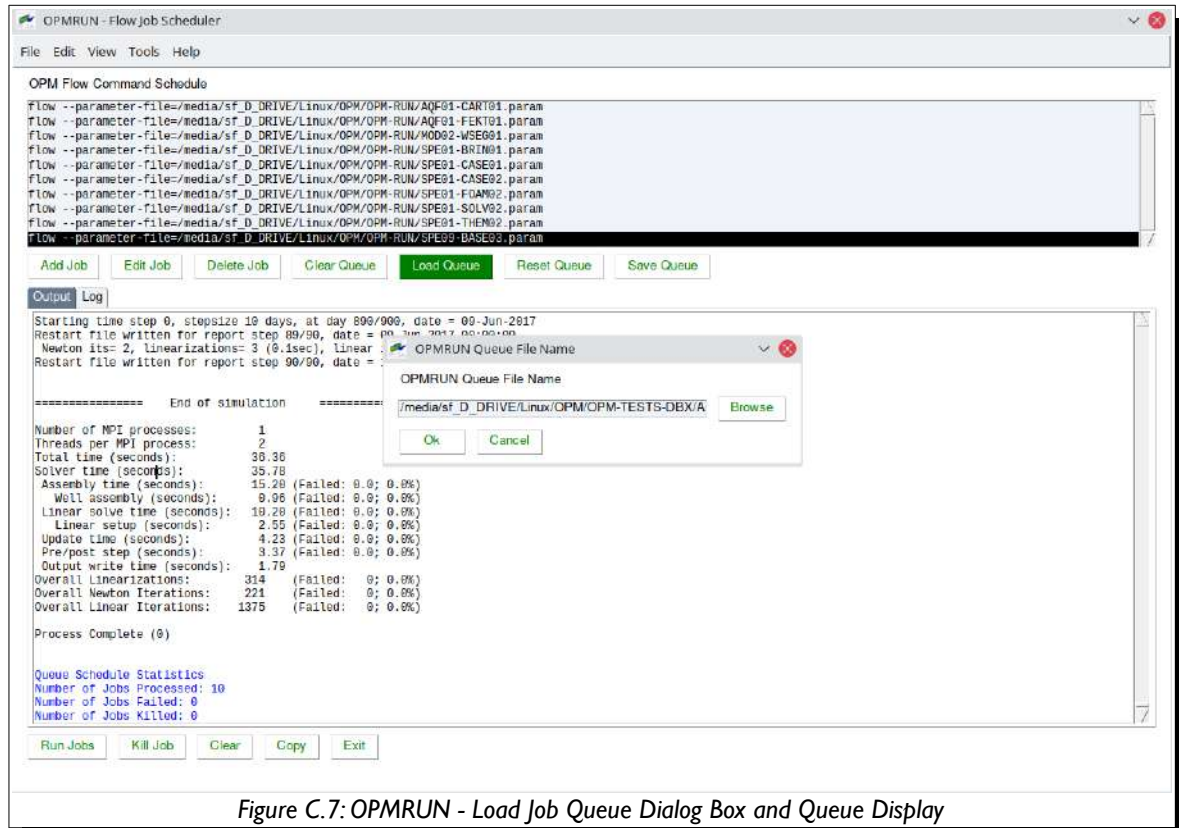


Figure C.7: OPMRUN - Load Job Queue Dialog Box and Queue Display

Queue files allow for various jobs to be load efficiently, especially for ensemble and sensitivity cases and may contain a large number of cases.

Note

When running under Windows 10 the Job names will follow the Windows 10 file naming convention and OPMRUN automatically handles the file names for running the jobs under the Window Subsystem for Linux. However, care is needed for any “included” file in the input deck. In this case the [PATHS – Define Filename Directory Path Aliases](#) keyword in the [RUNSPEC](#) section may be of use.

C.2.4 RESET JOB QUEUE PARAMETERS

Reset Job Queue Parameters allows jobs run under Windows 10 WSL to be renamed for running under Linux, and changing jobs from serial to parallel and vice versa.

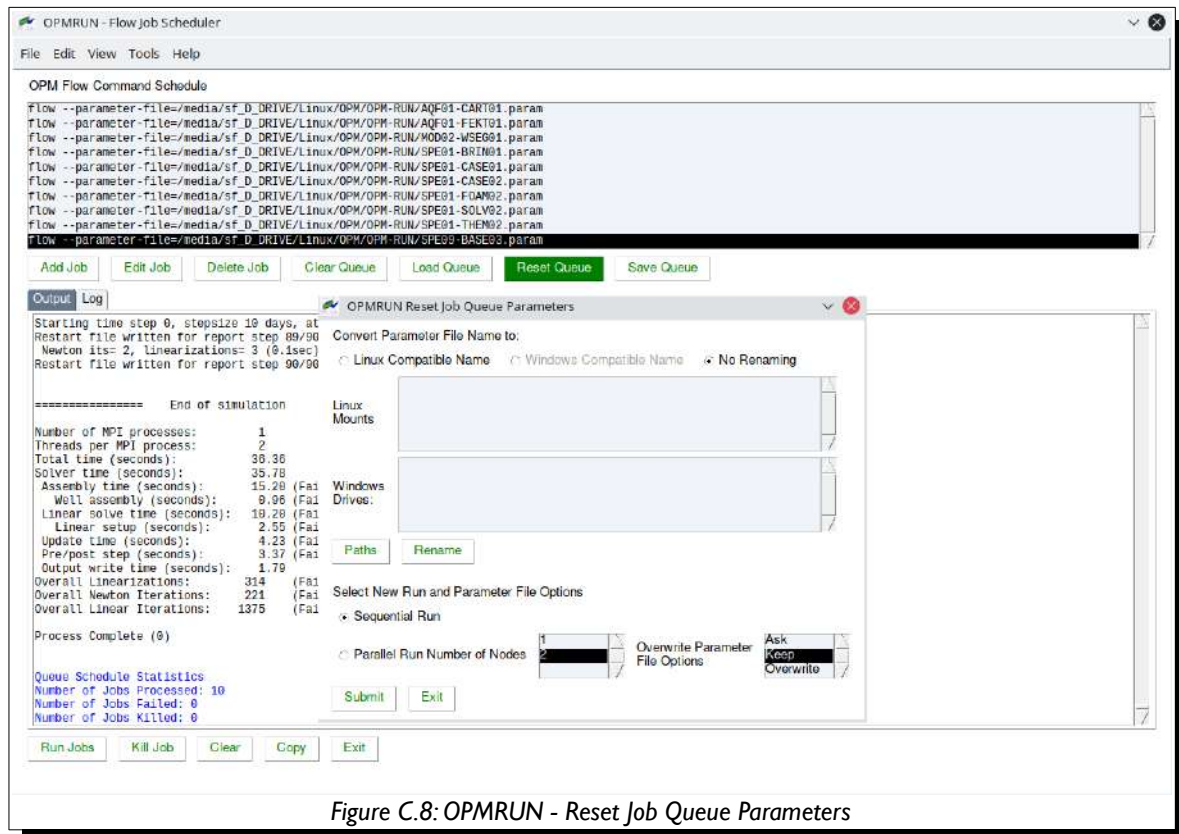


Figure C.8: OPMRUN - Reset Job Queue Parameters

The application will attempt to list or Linux mount points and Windows drives depending on the host operating system, once the two systems mount and drive points have been selected then the files in the queue will be renamed from the previous host system to the current host system. This is only performed for the *.DATA files, included files in the input deck are currently not converted.

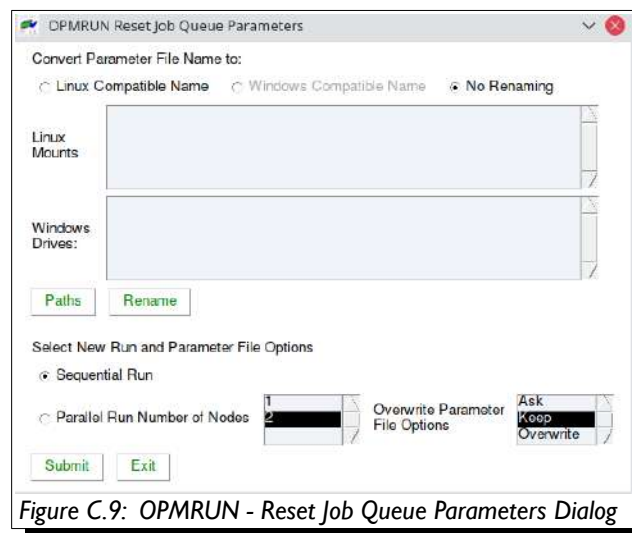


Figure C.9: OPMRUN - Reset Job Queue Parameters Dialog

As previously mentioned, one can also change the queue run time parameters from serial to parallel or vice versa.

C.2.5 RUN JOBS IN QUEUE WITH VARIOUS OPTIONS

Selecting the Run Jobs button displays the Select Run Option dialog box shown in Figure C.10 and Figure C.11.

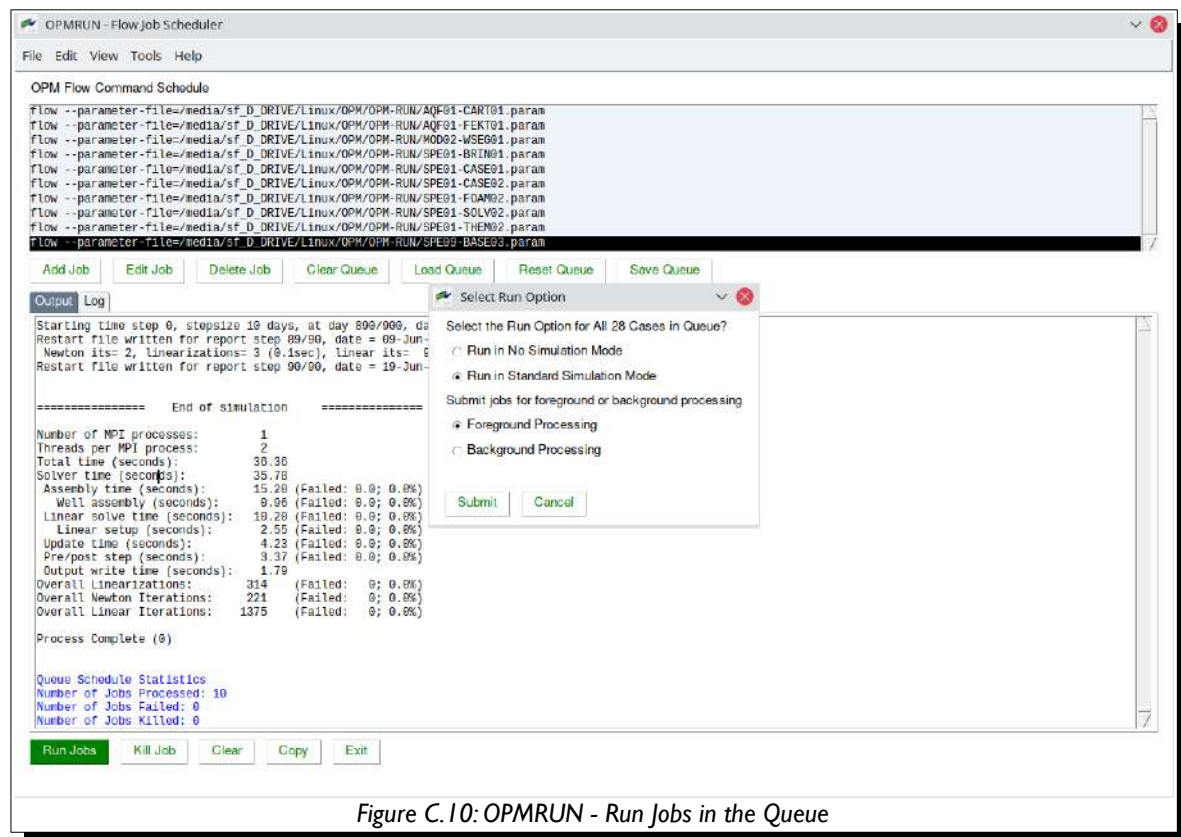


Figure C.10: OPMRUN - Run Jobs in the Queue

On the Select Run Dialog, the Run in No Simulation Mode option is equivalent to setting the [NOSIM](#) option in the input deck for all jobs in the queue (see section [5.3.101 NOSIM – Activate the No Simulation Mode for Data File Checking](#) and the `--enable-dry-run` command line parameter in Table 2.1 in section [2.2 Running OPM Flow From The Command Line](#)). This allows for checking all the jobs at once.

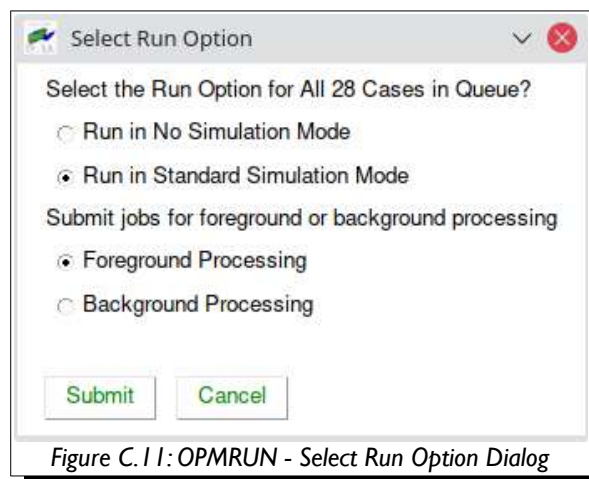


Figure C.11: OPMRUN - Select Run Option Dialog

Selecting Run in Standard Simulation Mode will run all the jobs in the queue sequentially, with the OPM Flow terminal output directed to OPM Flow Output Element, as shown in Figure C.12. The terminal output is also directed to a *.LOG file as well, similar to what the commercial simulator does.

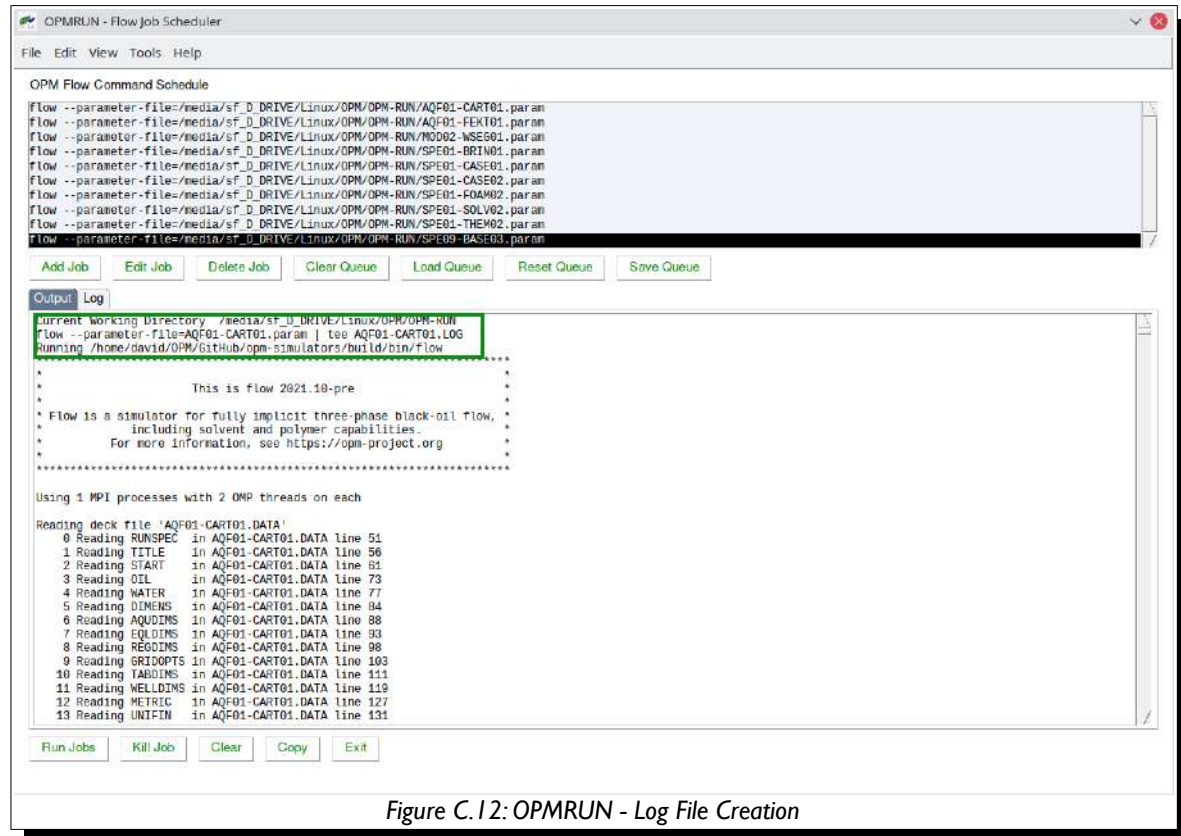
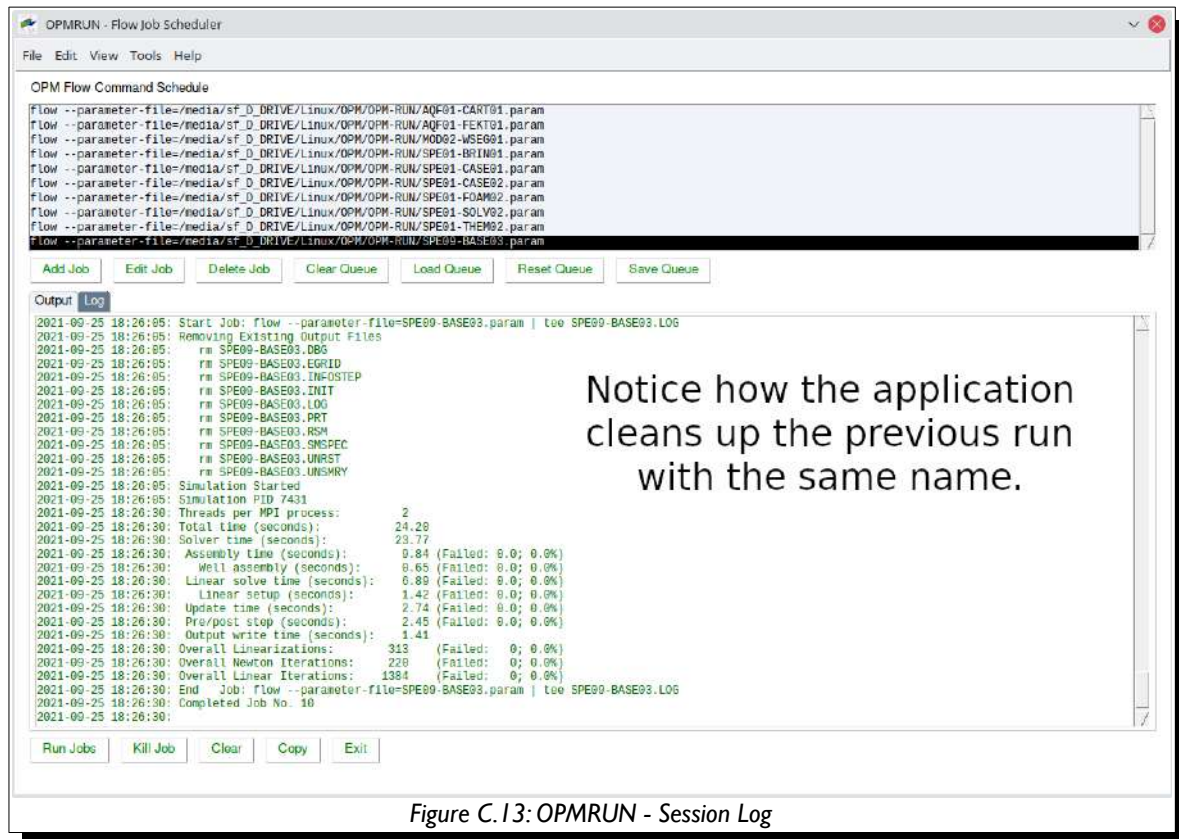


Figure C.12: OPMRUN - Log File Creation

Clicking the OPM Run Log tab displays the OPMRUN's session log file that records the time and date of the major events that have occurred, including the start and end times of each run. Notice also how OPMRUN deletes all the existing output files for a given job, if they exist, before running OPM Flow, as well as creating a Schedule Log for tracking progress (Figure C.13).



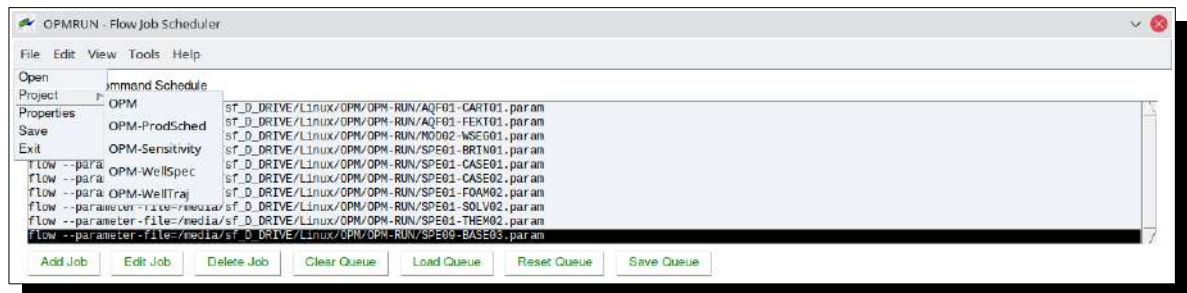
The Kill button will ask the user if the current running job should be killed, and if the job is to be killed, the application will prompt as to whether or not all the jobs in the queue should be killed.

The Clear button clears the OPM Flow Output Element from the currently displayed tab (Output or Log) and the Copy button copies the data to the clipboard.

C.2.6 MENU OPTIONS

File Menu Options

Enables open and saving the job queue, switching projects and listing OPMRUN's user properties.



Edit Menu Options

Lets one add jobs, add jobs recursively (all jobs in the selected directory and below), edit the data file (Figure C.14) and the parameter file for the selected job,(see section C.2.2 Edit Job Data and Parameter File) edit, list and set the default parameters for running jobs that will be added to the queue, set OPMRUN options, and set the project's project directories.

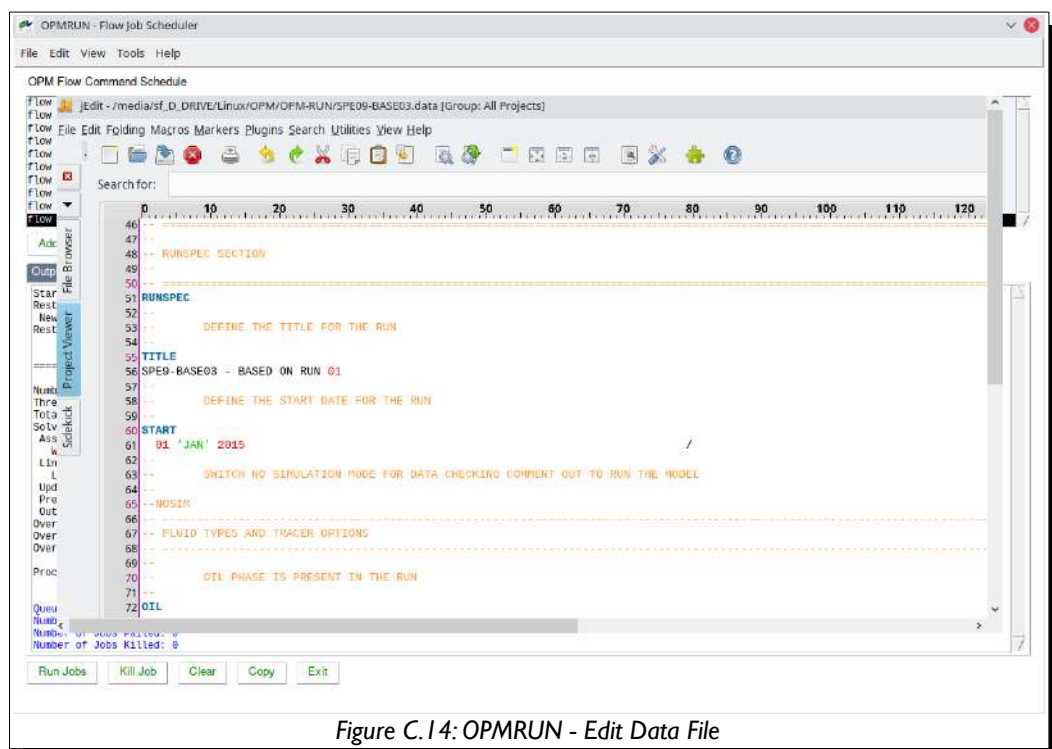
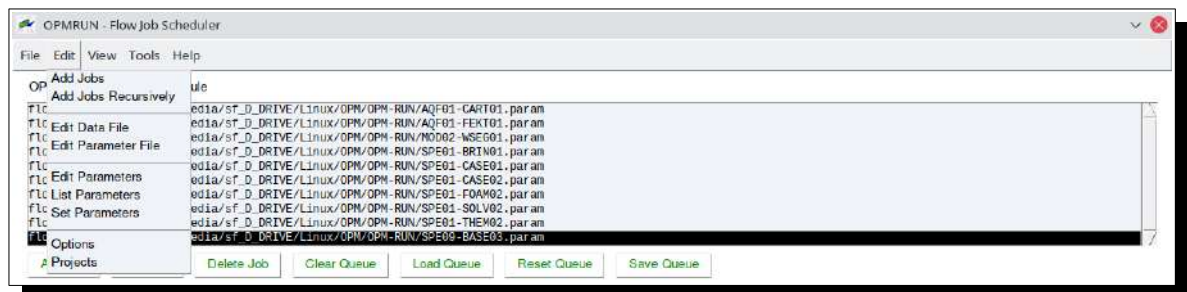


Figure C.14: OPMRUN - Edit Data File

The options are also available by right-clicking a job in the **Job List Element**

The Edit Parameters, List Parameters and Set Parameters relate to the default parameter set, not the parameter set for a particular job. When a new job is added to the queue the application checks if a *.PARAM file exists for the job, if not then the default parameter set is used for the job. Editing the default parameter set is the same as editing a job parameter set (see section C.2.2 Edit Job Data and Parameter File for more information).

One can also define the default parameter set by using the OPM Flow default values, which is the set created when OPMRUN is first initialized. In addition, one can load an existing parameter set from an existing *.PARAM file or from an OPM Flow *.PRT file (Figure C.15).

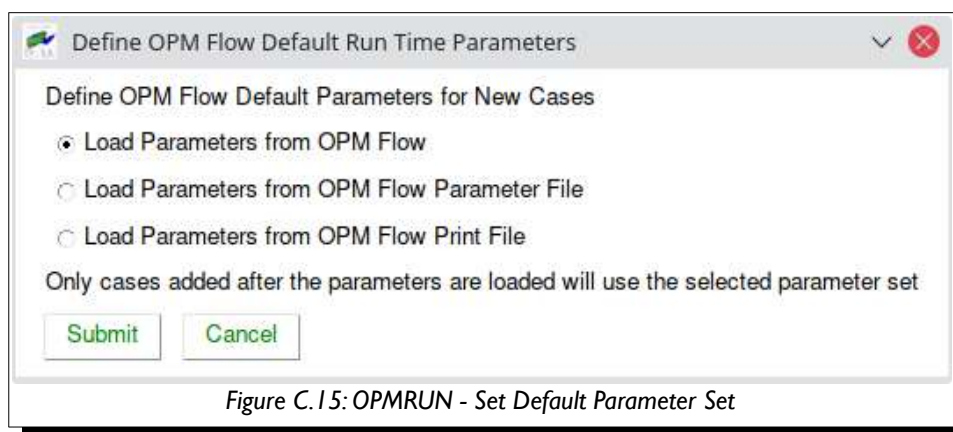


Figure C.15: OPMRUN - Set Default Parameter Set

OPMRUN has several configuration options that can be set via the Edit/Options menu option as illustrated in Figure C.16. These include setting:

- The location of the OPM Flow manual.
- The Keyword Generator Template Directory, one of tools supplied with OPMRUN, see section C.3.2 OPMRUN Tools: Simulator Input/Keywords for further information on this application.
- The location of the ResInsight program, for loading the results of a simulation run for viewing.
- The editor command to used to edit the input deck and view the resulting simulator output files.
- The terminal console to be used for background jobs. WSL ("Windows Subsystem for Linux") should be selected if OPMRUN is running under Windows 10 to enable jobs to be submitted to the installed Linux distribution.
- The User Information series of fields are used by various supplied tools and in some templates used by the Keyword Generator application (see C.3.2 OPMRUN Tools: Simulator Input/Keywords for more information on this application). Note if a "User Information" field is not defined then the template variable will be output instead – this can easily be deleted in the application.
- One can also define the main OPMRUN windows configuration parameters define: input (Job List Element) and output panel's (OPM Flow Output Element) size, font and font size.

Edit Options

OPM Flow Manual Location

OPM Keyword Generator Template Directory

ResInsight Command

Editor Command for Editing Input Files

Terminal Command for Running in Background Mode

User Information for Reporting

Author

Company Name

Address Line 1

Address Line 2

Email Address

Main Window Configuration Setting

Input Element Width

Input Element Height

Output Element Width

Output Element Height

Output Element Font

Output Element Font Size

Figure C.16: OPMRUN – Set Application Options

In addition to the aforementioned options, the Edit/Projects menu item enables the setting of project directories that allows the user to set a default directory for loading and saving files within OPMRUN and the auxiliary applications (Figure C.17).

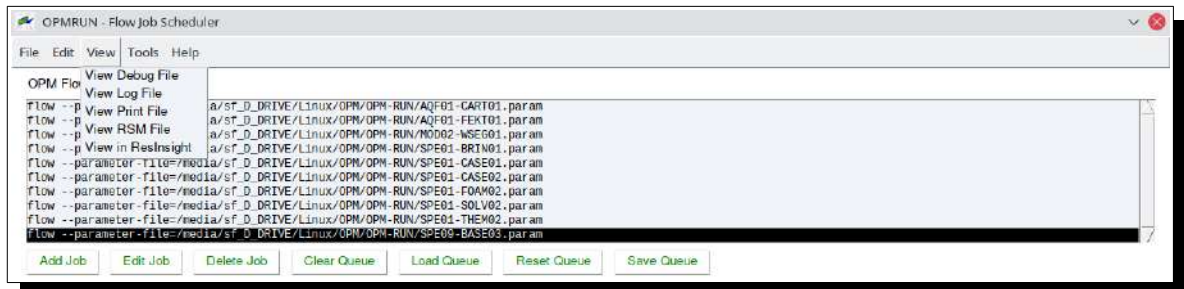
Edit Projects

No.	Project Name	Project Directory	
1.	OPM	/media/sf_D_DRIVE/Linux/OPM	Browse
2.	OPM-ProdSched	/media/sf_D_DRIVE/Linux/OPM/GitHub/opm-utilities/opmrun/.examples/opm_prodsched	Browse
3.	OPM-Sensitivity	/media/sf_D_DRIVE/Linux/OPM/GitHub/opm-utilities/opmrun/.examples/opm_sensitivity	Browse
4.	OPM-WellSpec	/media/sf_D_DRIVE/Linux/OPM/GitHub/opm-utilities/opmrun/.examples/opm_wellspec	Browse
5.	OPM-WellTraj	/media/sf_D_DRIVE/Linux/OPM/GitHub/opm-utilities/opmrun/.examples/opm_welltraj	Browse

Figure C.17: OPMRUN – Set Project Directories

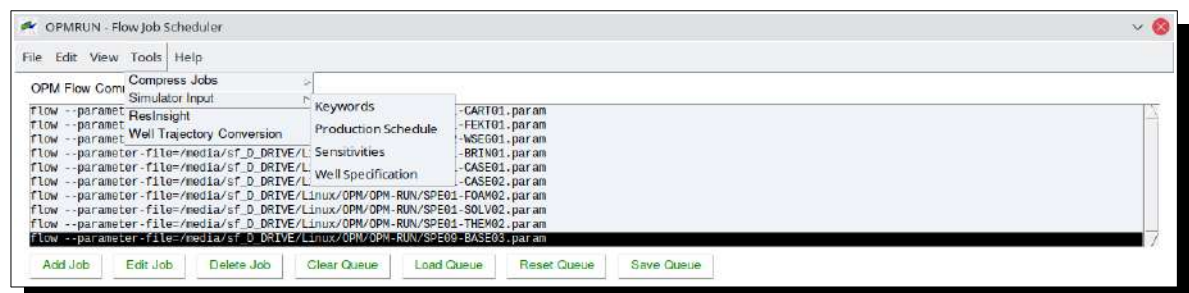
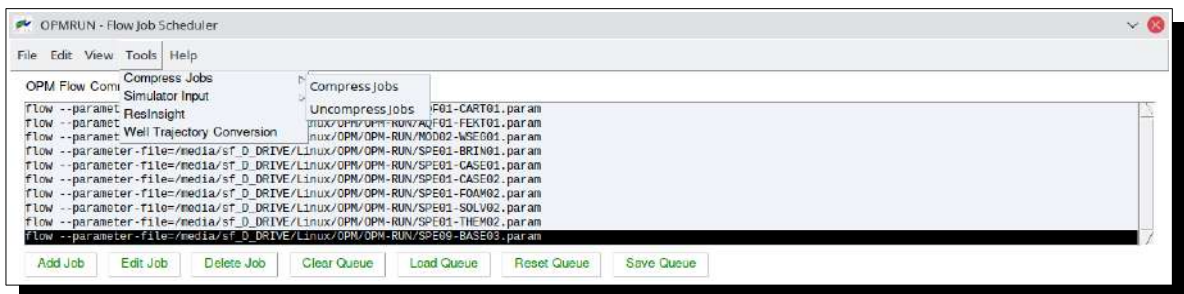
View Menu Options.

Allows the user to view the results of an OPM Flow simulation run using the default editor. The options are also available by right-clicking a job in the *Job List Element*.



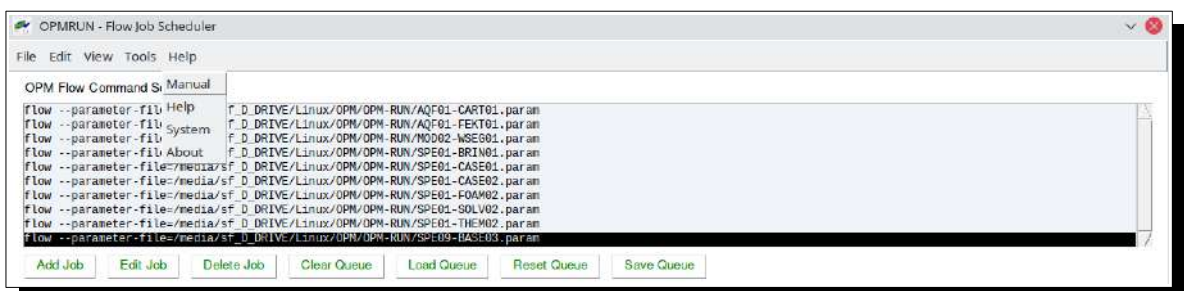
Tools Menu Options

Contains various tool that may be useful in building a simulation model.



See the section C.3 OPMRUN Tools for further details on the available tools.

Help Menu Options



Use the Edit / Options menu option to select the location of the OPM Flow Manual.

C.3 OPMRUN TOOLS

C.3.1 OPMRUN TOOLS: JOB FILE COMPRESSION UTILITY FOR SAVING SPACE AND ARCHIVING.

Simulation input and output files can be extremely large, especially for large full field models. Running multiple cases in these circumstances can easily use up all available disk space, especially if multiple users are running multiple cases. The Tools/Compression Jobs option allows the user to compress a series of jobs into individual zip files (one zip file per job), as well as uncompressing previously zip job files.

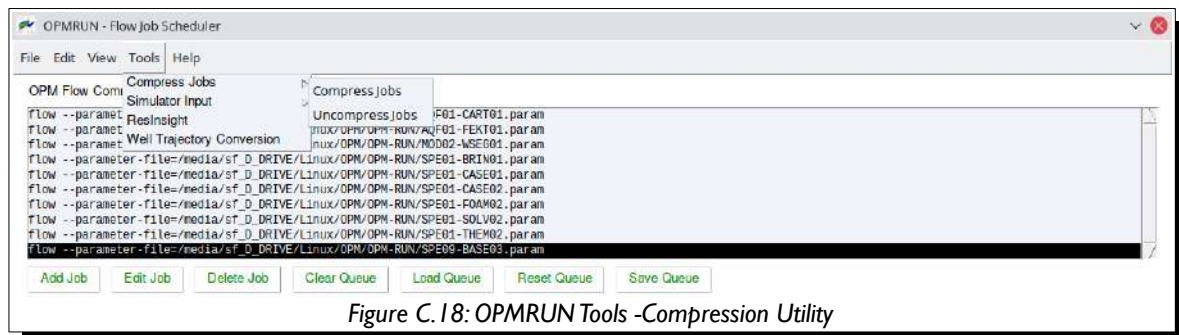


Figure C.18: OPMRUN Tools -Compression Utility

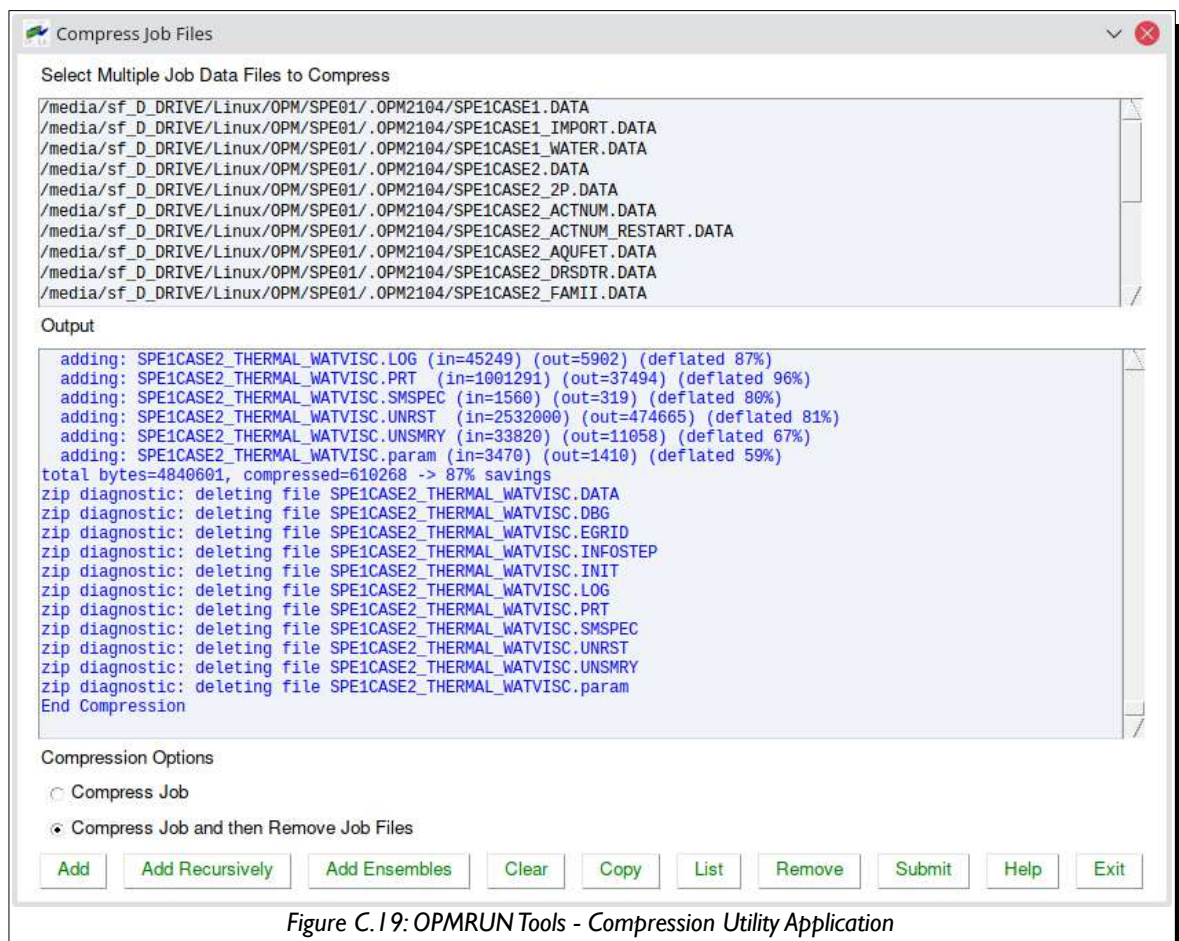


Figure C.19: OPMRUN Tools - Compression Utility Application

Note there is a similar application for uncompressing zip files and that the tool uses the Linux zip and unzip programs both on Linux host systems and Windows 10 systems using WSL.

C.3.2 OPMRUN TOOLS: SIMULATOR INPUT/KEYWORDS

The OPMRUN Keyword generator is an application that generates OPM Flow keywords that can be cut and pasted into any editor or saved to a separate file to form the basis of a new input deck (*.DATA files). The application utilizes templates based on the Apache Velocity Template Language (“VTL”), a commonly used template language used by software engineers. The templates can therefore also be used with any editor that supports VTL, jEdit for example, a popular open source Java based editor and PyCharm, a Python integrated development environment used in computer programming, specifically for the Python language.

There is one template per keyword, with formatting of the keywords being the same as the OPM Flow manual. Currently there are over 450 templates implemented and the intention is for additional keywords to be added as the their usage is implemented within the simulator and documented within the manual. The application allows one to customize the existing templates as well as creating user defined templates by including the templates in the template directory and following the VTL language syntax. One still has to edit the resulting keywords to match the data require, but the structure and comments are provided by the application.

The application is accessed via the Tool/Simulator Input/Keywords menu item and the tool can generate specific keywords, as well as complete sections (Figure C.20)

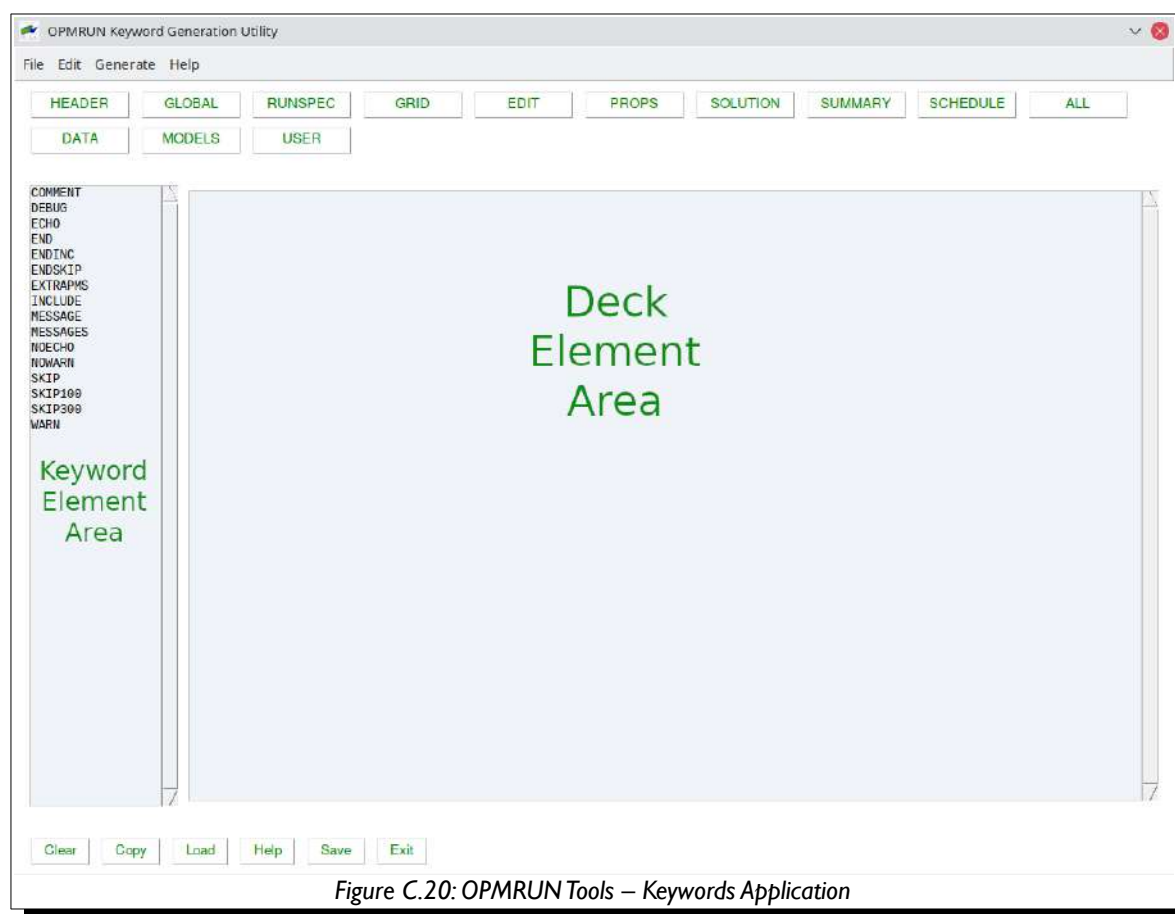
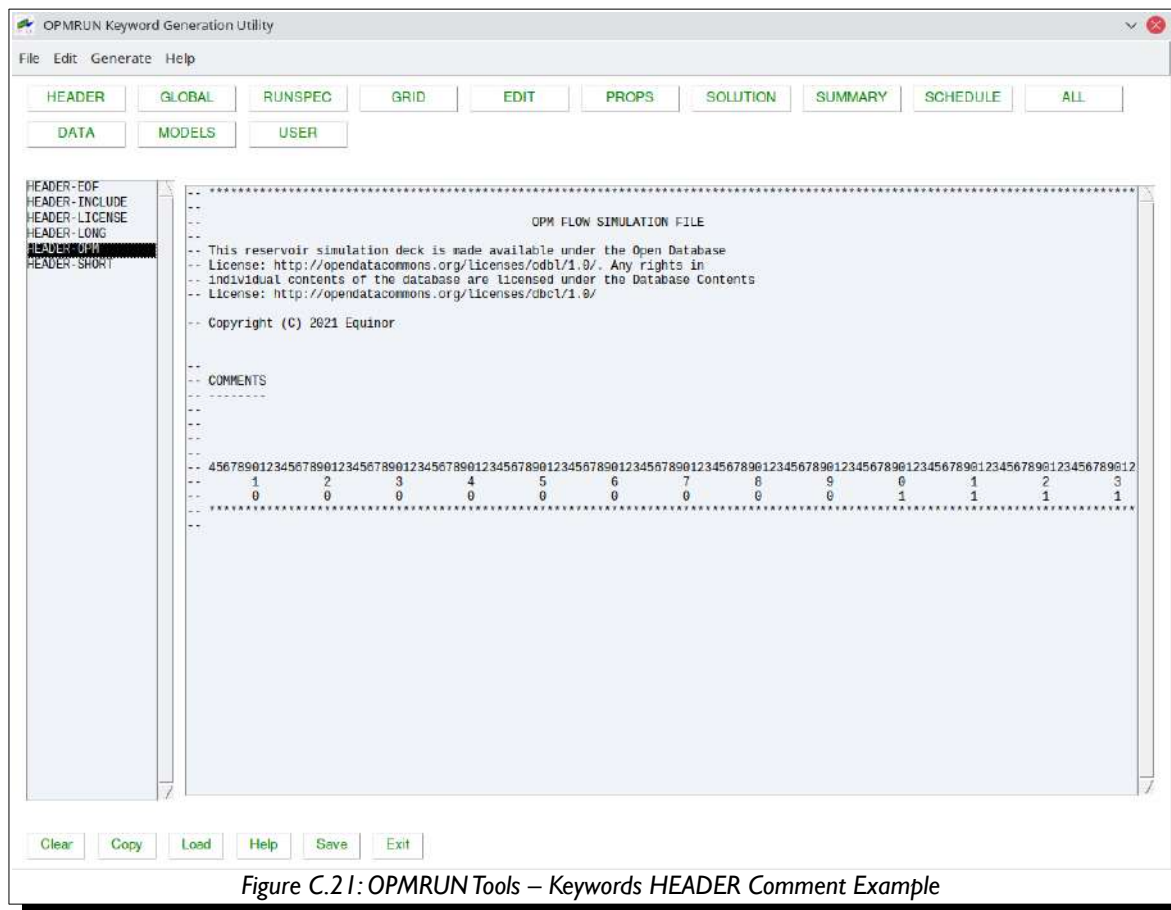


Figure C.20: OPMRUN Tools – Keywords Application

The application consists of several elements, a conventional menu system at the top, a *Deck Element Area* that will contain the resulting generated keywords, a *Keyword Element Area* for the user to select the keyword, data, models or user templates, and finally a series of buttons, *HEADER*, *GLOBAL*, etc., that are used to select the keywords in a OPM Flow section, specific data sets, models or user defined templates. The selection will appear in the *Keyword Element Area*.

Clicking on an item in the *Keyword Element Area* will generate the data for the item in the *Deck Element Area*, as shown below for the OPM Flow copyright header in Figure C.21.



The *Deck Element Area* is editable by simply clicking anywhere in the element and making changes. Use the *Clear* button to clear the *Deck Element Area* display, the *Copy* button to copy the *Deck Element Area* data to the clipboard, and the *Save* button to save the data to a file. The *Load* button allows one to load an existing file into the *Deck Element Area* for additional editing.

Note that the *HEADER* section is not an OPM Flow section, but a list of various comment block headers used to make the deck more readable.

Keywords: Menu Options

The various menu options include the File Menu



Figure C.22: OPMRUN Tools – Keywords File Menu

Where the *Open* and *Save* options load and save a file, and the *Properties* displays OPMRUN's properties.



Figure C.23: OPMRUN Tools – Keywords Edit Menu

The Edit Menu provides some basic standard editing facilities



Figure C.24: OPMRUN Tools – Keywords Generate Menu

Next, the Generate Menu options allows one to generate a complete section of keywords, as described below. These options are equivalent to selecting the equivalent section keyword (*RUNSPEC*, *GRID*, etc.) in the *Keyword Element Area*.



Figure C.25: OPMRUN Tools – Keywords Help

Finally, the Help Menu option display the Keyword Help information (Figure C.26):

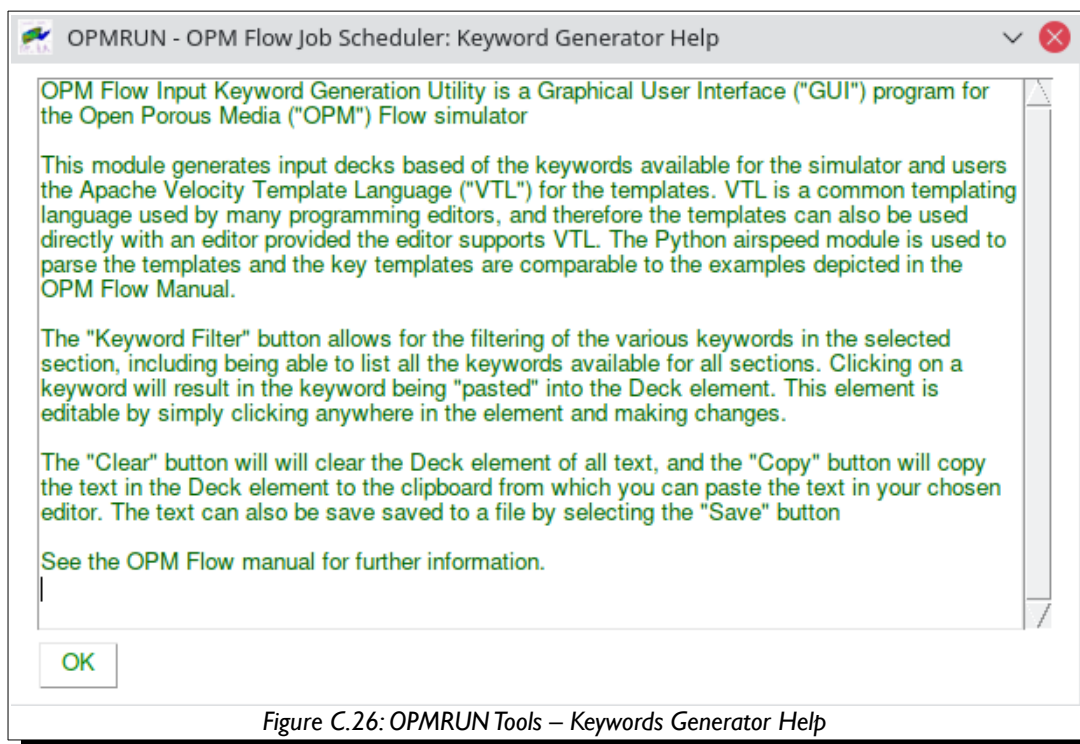


Figure C.26: OPMRUN Tools – Keywords Generator Help

And the Velocity Template Help (Figure C.27).

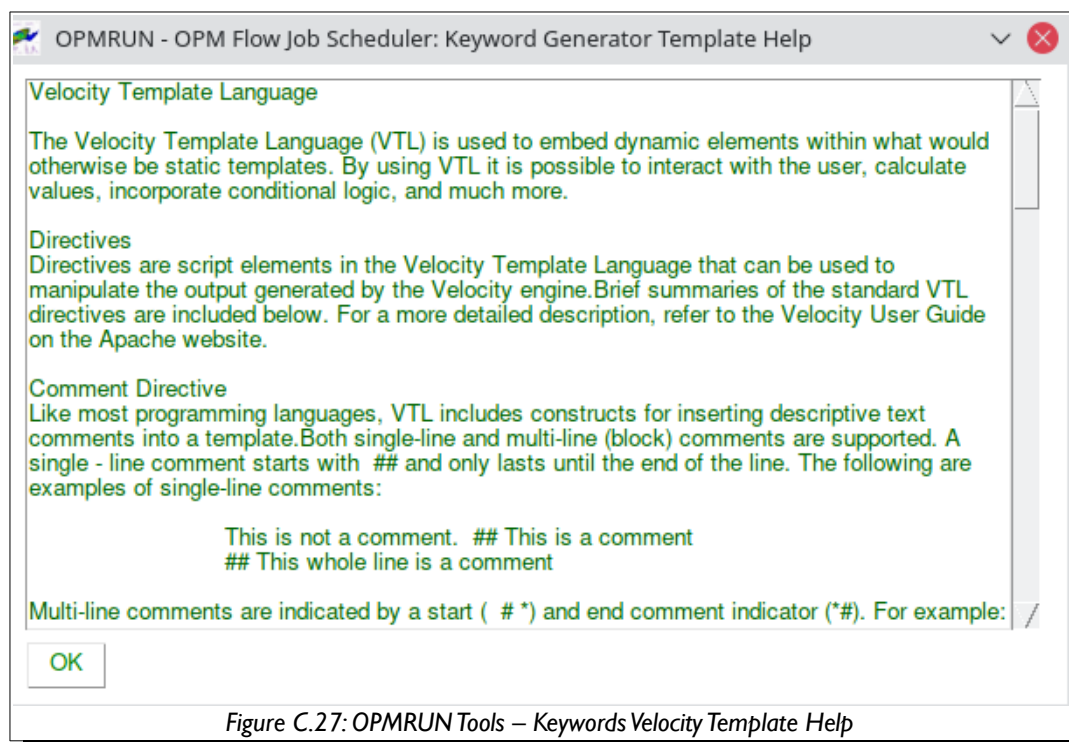


Figure C.27: OPMRUN Tools – Keywords Velocity Template Help

As mentioned previously, the tool uses the Apache Velocity Template Language ("VTL") for the templates, and therefore the templates can also be used directly with an editor, provided the editor supports VTL.

Keywords: File Imports

If a keyword requires a file, for example, the *INCLUDE* and *LOAD* keywords, then a dialog box is presented to enable the file to be selected. The application will also allow one to select the file name format, after the file has been selected.

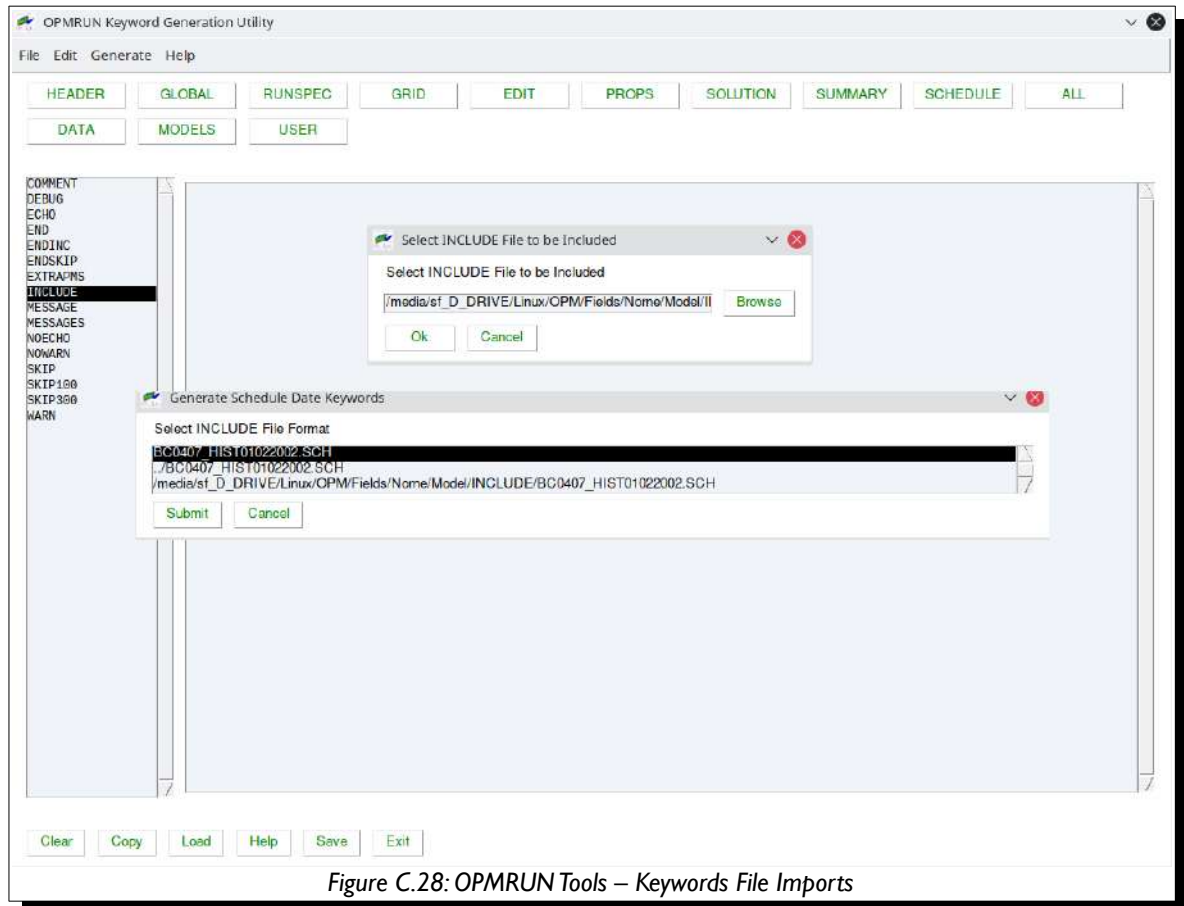


Figure C.28: OPMRUN Tools – Keywords File Imports

Note that COMMENT template is not an actual keyword, but a comment block to make the deck more readable for the user.

Keywords: Section Standard Set of Keywords

Selecting a Generate Menu option or a Section keyword ([RUNSPEC](#), [GRID](#), [EDIT](#), [PROPS](#), [SOLUTION](#), [SUMMARY](#), and [SCHEDULE](#)) in the *Keyword Element Area* will give an option to generate a representative set of keywords for that section, as per the [RUNSPEC](#) example in Figure C.29.

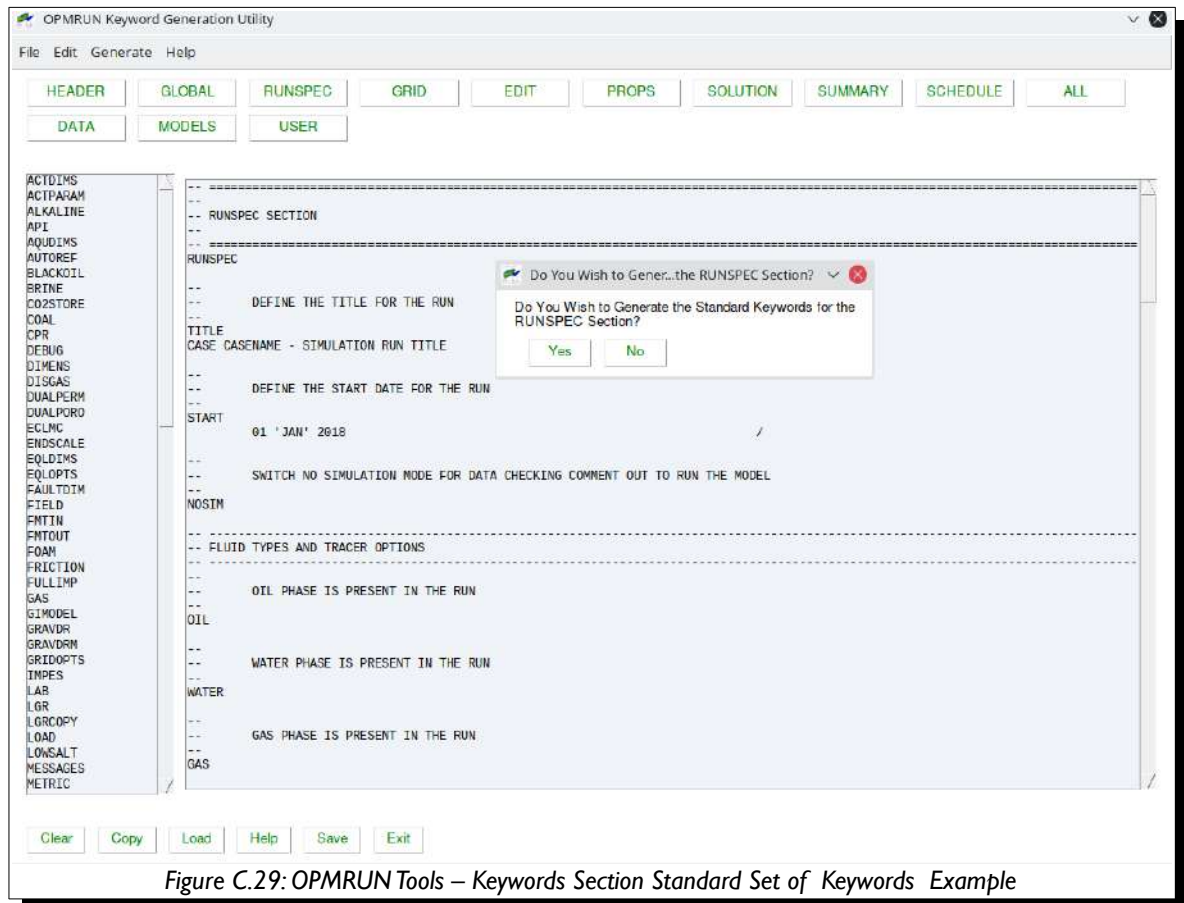


Figure C.29: OPMRUN Tools – Keywords Section Standard Set of Keywords Example

One can therefore generate a complete input deck in a matter of minutes; however, you still have to edit the generated input deck with your actual data.

Keywords: SUMMARY Section Variables

For the [SUMMARY](#) section keyword, one can also generate various sets of summary variables based on the options being used in the model. Note that not all the variables are currently available in OPM Flow, but additional variables are added at each release.

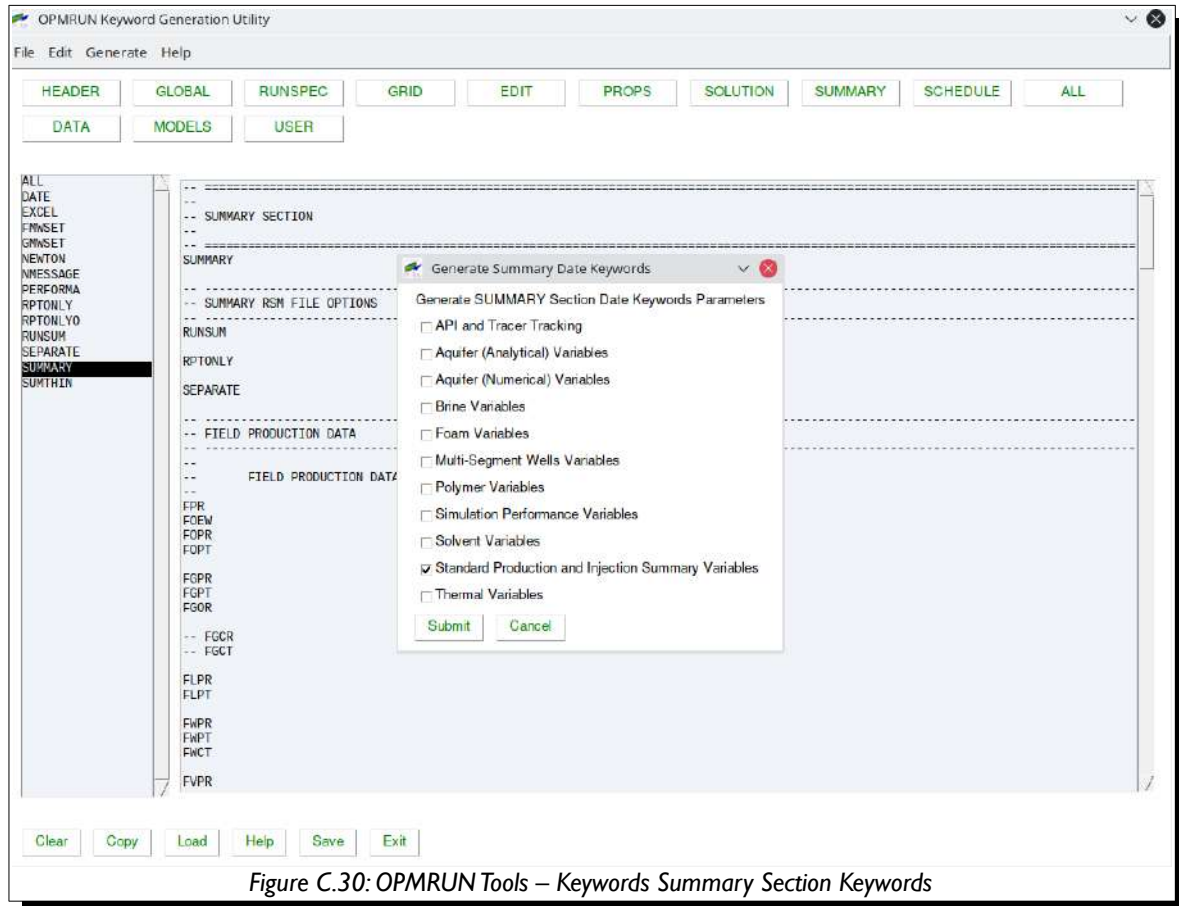


Figure C.30: OPMRUN Tools – Keywords Summary Section Keywords

For [SUMMARY](#) variables not recognized by OPM Flow, the simulator will issue a warning message and ignore those variables not implemented.

Keywords: SCHEDULE Section Keywords and Date Schedule

For the *SCHEDULE* Section keyword, one can also generate a representative set of *SCHEDULE* keywords, plus a date schedule from a start year to an end year, using Annual, Quarterly, or Monthly time steps.

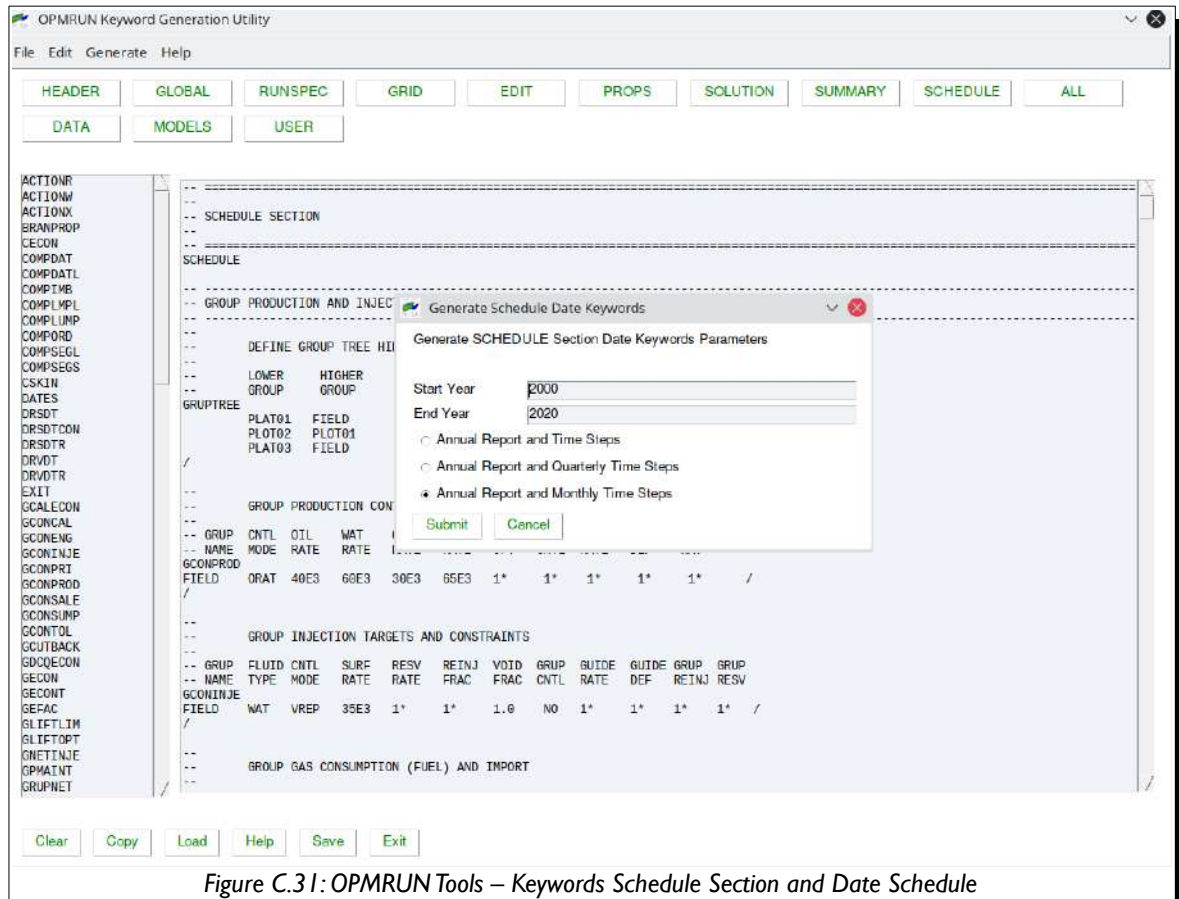


Figure C.31: OPMRUN Tools – Keywords Schedule Section and Date Schedule

This option also writes a standard report using the *RPTSCHED* keyword at the beginning of each year which is subsequently switch off for the intermediate Quarterly and Monthly time steps. A final report is written at the end of the run.

Keywords: DATA (Sets) Option

There is also a *DATA* option which is not an OPM Flow section, but a series of data sets, as shown in Figure C.32.

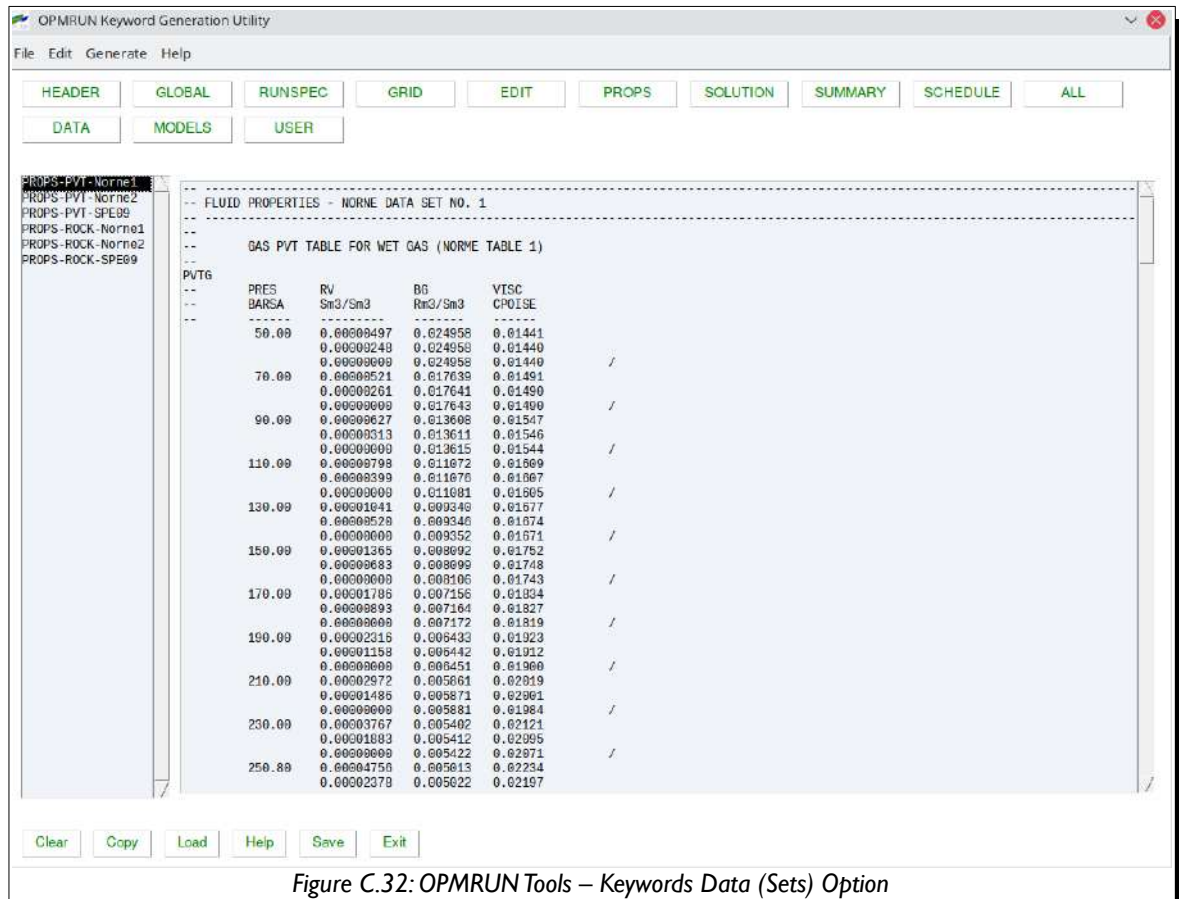


Figure C.32: OPMRUN Tools – Keywords Data (Sets) Option

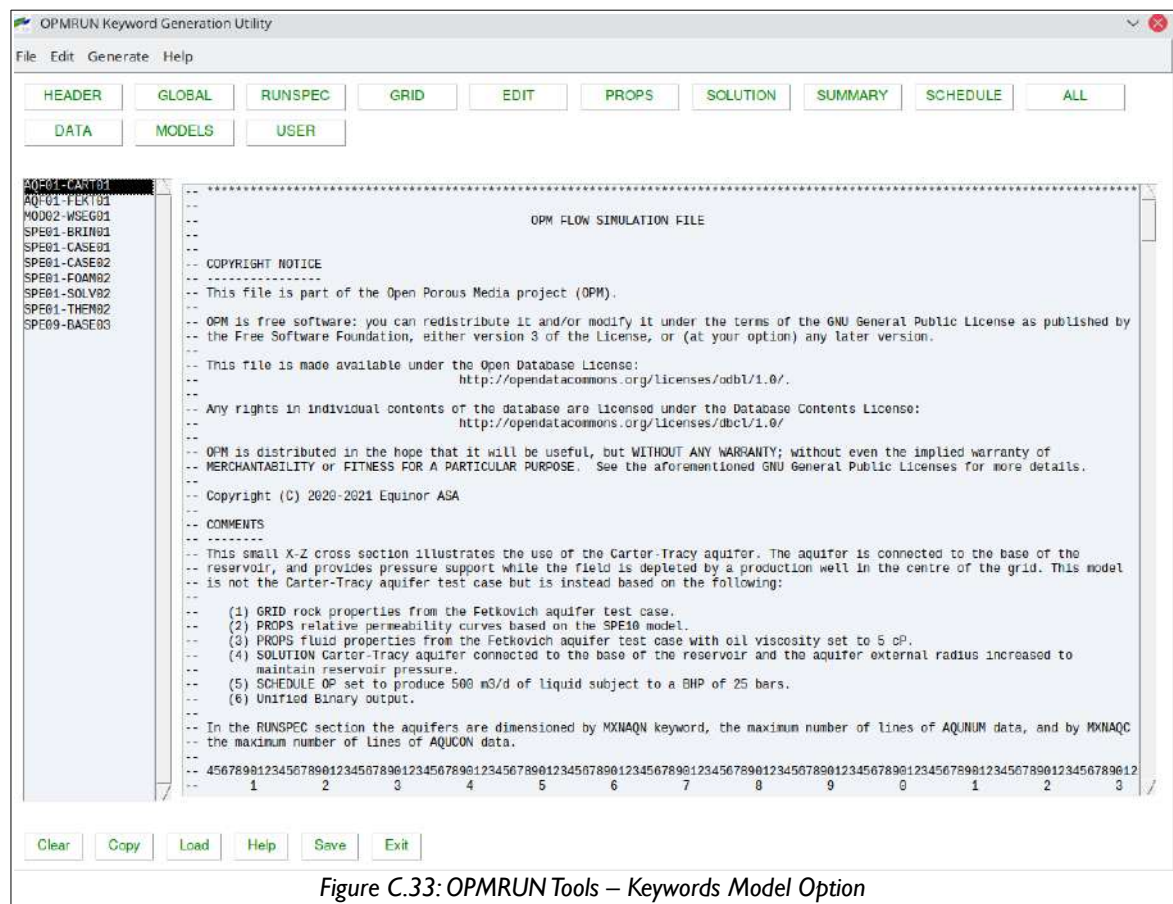
The data sets are complete examples for a given type of data used in OPM Flow, for example a PVT data set for a wet gas reservoir, or three phase relative permeability data set. The data sets are intended to be used as a guide for generating ones own keyword input, or for building models for testing.

The intention is to expand the collection of data sets over time as more data becomes available.

Keywords: MODEL Option

Like the DATA option, the MODEL option is not an OPM Flow section, but is instead a collection of working models, as illustrated in Figure C.33.

The purpose of the models is to illustrate the functionality of various features implemented in OPM Flow and to act as guide for users in building their own models.



Additional models will be added when available.

Keywords: USER Templates Option

Finally, the *USER* option is where users can store their own templates. *USER* templates with the “vm” extension will automatically be listed by the *USER* button. To use this feature, after selecting a keyword, right clicking on the keyword allows one to load the actual template for the keyword. One can then edit the template and save the changes back to the same template or another template using the Save button.

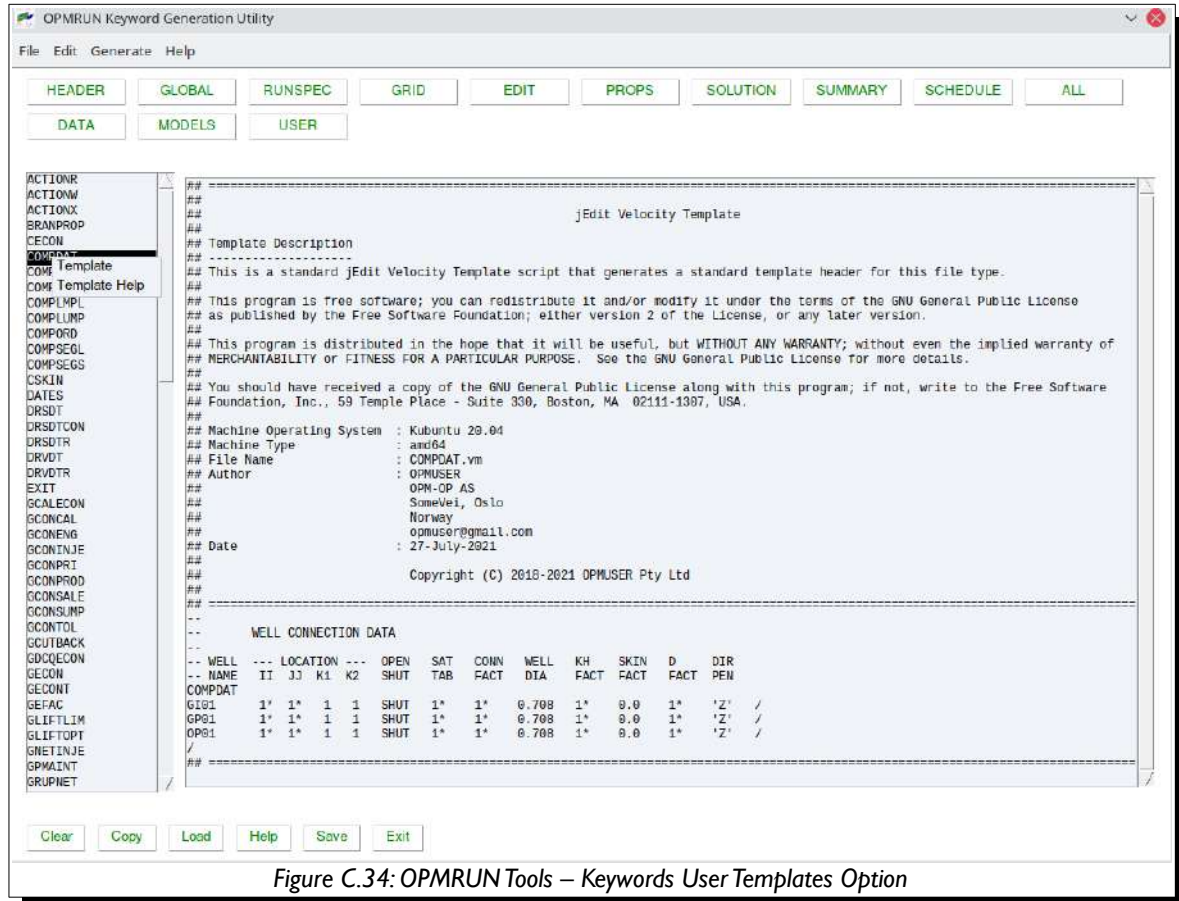


Figure C.34: OPMRUN Tools – Keywords User Templates Option

The Template Help option displays a brief introduction to VTL for further reference.

C.3.3 OPMRUN TOOLS: SIMULATOR INPUT/PRODUCTION SCHEDULE

The *Tools/Simulator Input/Production Schedule* application takes a comma delimited CSV file containing historical production and injection data and converts the data to an OPM Flow *SCHEDULE* file using the *WCONHIST* series of keywords. An example input file is shown below:

	A	B	C	D	E	F	G	H
1	Xy.Wellcompl	Date	Prd_dly.Time	Prd_dly.Cond	Prd_dly.Water	Prd_dly.Gas	Prd_dly.whp	Prd_dly.wht
2	GP-P1-01	03/01/2020	24	25	16	10528	551	133
3	GP-P1-01	03/02/2020	24	25	16	10507	551	133
4	GP-P1-01	03/03/2020	24	25	16	10513	550	131
5	GP-P1-01	03/04/2020	24	25	16	10506	549	132
6	GP-P1-01	03/05/2020	24	25	16	10503	550	133
7	GP-P1-01	03/06/2020	24	25	16	10506	549	132
8	GP-P1-01	03/07/2020	24	25	16	10416	556	131
9	GP-P1-01	03/08/2020	24	24	15	10054	595	132
10	GP-P1-01	03/09/2020	24	25	15	10180	584	131
11	GP-P1-01	03/10/2020	24	24	16	10493	549	132
12	GP-P1-01	03/11/2020	24	25	15	10493	549	133
13	GP-P1-01	03/12/2020	24	25	16	10476	550	133
14	GP-P1-01	03/13/2020	24	25	16	10243	607	132
15	GP-P1-01	03/14/2020	24	25	15	10276	596	130
16	GP-P1-01	03/15/2020	24	24	16	10414	548	131
17	GP-P1-01	03/16/2020	24	25	15	10400	549	132
18	GP-P1-01	03/17/2020	24	26	17	10471	601	130
19	GP-P1-01	03/18/2020	24	29	15	10819	647	131
20	GP-P1-01	03/19/2020	24	26	16	10348	653	132
21	GP-P1-01	03/20/2020	24	27	17	10843	626	132
22	GP-P1-01	03/21/2020	24	28	16	10929	649	131
23	GP-P1-01	03/22/2020	24	26	16	10901	648	131
24	GP-P1-01	03/23/2020	24	24	17	10693	575	133

Figure C.35: OPMRUN Tools – Production Schedule Input File Example

The first row in the input file is a header row that declares the data type for a column, the example shows typical Oil Field Manager (“OFM”) header variable names, but various variable names can be used to define the data type.

The tool can convert daily production data to a: daily production schedule, monthly average, or monthly on-stream average production schedule, as shown below:

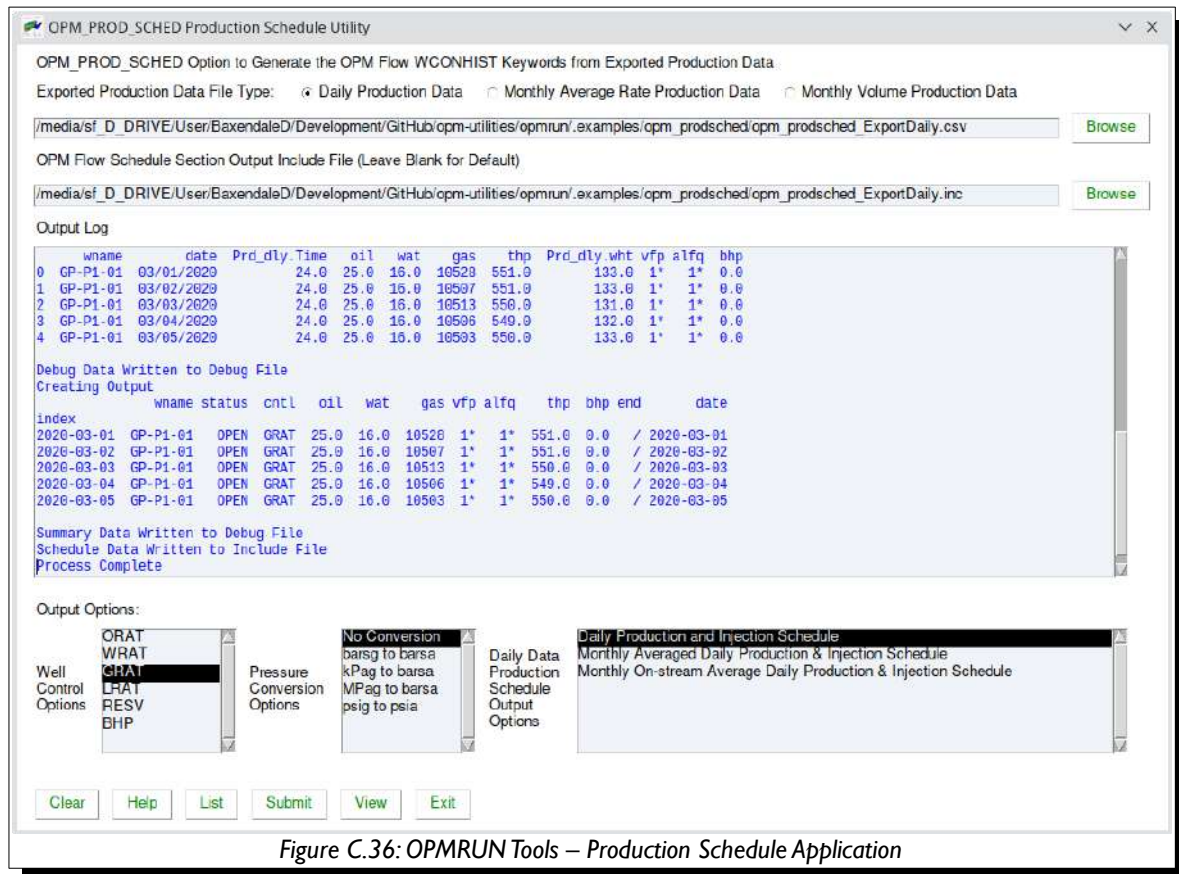


Figure C.36: OPMRUN Tools – Production Schedule Application

Notice that the application checks various variable names for the column headers. For example for the BHP data, the column names can be: bhp, bottom-hole pressure, BHP, or BOTTOM-HOLE [PRESSURE](#).

A sample of the generated output file is shown in Figure C.37.

```

NOECHO
--
DATES
01 Mar 2020 /
/
--
-- WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL OPEN CNTL OIL WAT GAS VFP VFP THP BHP
-- NAME SHUT MODE RATE RATE RATE TABLE ALFQ PRES PRES
WCONHIST
GP-P1-01 OPEN GRAT 25.0 16.0 10528 1* 1* 551.0 0.0 / 2020-03-01
GP-P1-02 OPEN GRAT 21.0 21.0 13680 1* 1* 557.0 0.0 / 2020-03-01
GP-P1-03 OPEN GRAT 39.0 34.0 16432 1* 1* 625.0 0.0 / 2020-03-01
GP-P1-05 OPEN GRAT 187.0 27.0 51043 1* 1* 1907.0 0.0 / 2020-03-01
GP-P1-06 OPEN GRAT 41.0 17.0 9427 1* 1* 619.0 0.0 / 2020-03-01
GP-P2-02 OPEN GRAT 0.0 0.0 13365 1* 1* 690.0 0.0 / 2020-03-01
GP-P2-03 OPEN GRAT 0.0 0.0 10486 1* 1* 663.0 0.0 / 2020-03-01
GP-P2-04 OPEN GRAT 0.0 0.0 8009 1* 1* 646.0 0.0 / 2020-03-01
/
DATES
02 Mar 2020 /
/
--
-- WELL HISTORICAL PRODUCTION CONTROLS
--
-- WELL OPEN CNTL OIL WAT GAS VFP VFP THP BHP
-- NAME SHUT MODE RATE RATE RATE TABLE ALFQ PRES PRES
WCONHIST
GP-P1-01 OPEN GRAT 25.0 16.0 10507 1* 1* 551.0 0.0 / 2020-03-02
GP-P1-02 OPEN GRAT 21.0 21.0 13653 1* 1* 556.0 0.0 / 2020-03-02
GP-P1-03 OPEN GRAT 39.0 35.0 16400 1* 1* 624.0 0.0 / 2020-03-02
GP-P1-05 OPEN GRAT 186.0 25.0 51054 1* 1* 1905.0 0.0 / 2020-03-02
GP-P1-06 OPEN GRAT 41.0 17.0 9409 1* 1* 618.0 0.0 / 2020-03-02
GP-P2-02 OPEN GRAT 0.0 0.0 13485 1* 1* 681.0 0.0 / 2020-03-02
GP-P2-03 OPEN GRAT 0.0 0.0 10494 1* 1* 659.0 0.0 / 2020-03-02
GP-P2-04 OPEN GRAT 0.0 0.0 8024 1* 1* 641.0 0.0 / 2020-03-02
/
DATES
03 Mar 2020 /
/

```

Figure C.37: OPMRUN Tools – Production Schedule Output Example

Note

Note the current release only support production data via the **WCONHIST** keyword, injection data via **WCONINJH** keyword is not supported.

C.3.4 OPMRUN TOOLS: SIMULATOR INPUT/SENSITIVITIES

The *Tools/Simulator Input/Sensitivities* option generates sensitivity cases based on a "Base" case file. The Base file contains "Factors" (variable names), \$X01, \$X02, etc., that are substituted with user defined values using the data entered and the type of Sensitivity Scenario selected. Thus, the first step is to configure the Base file in a text editor by replacing actual values by the variable names, previously mentioned.

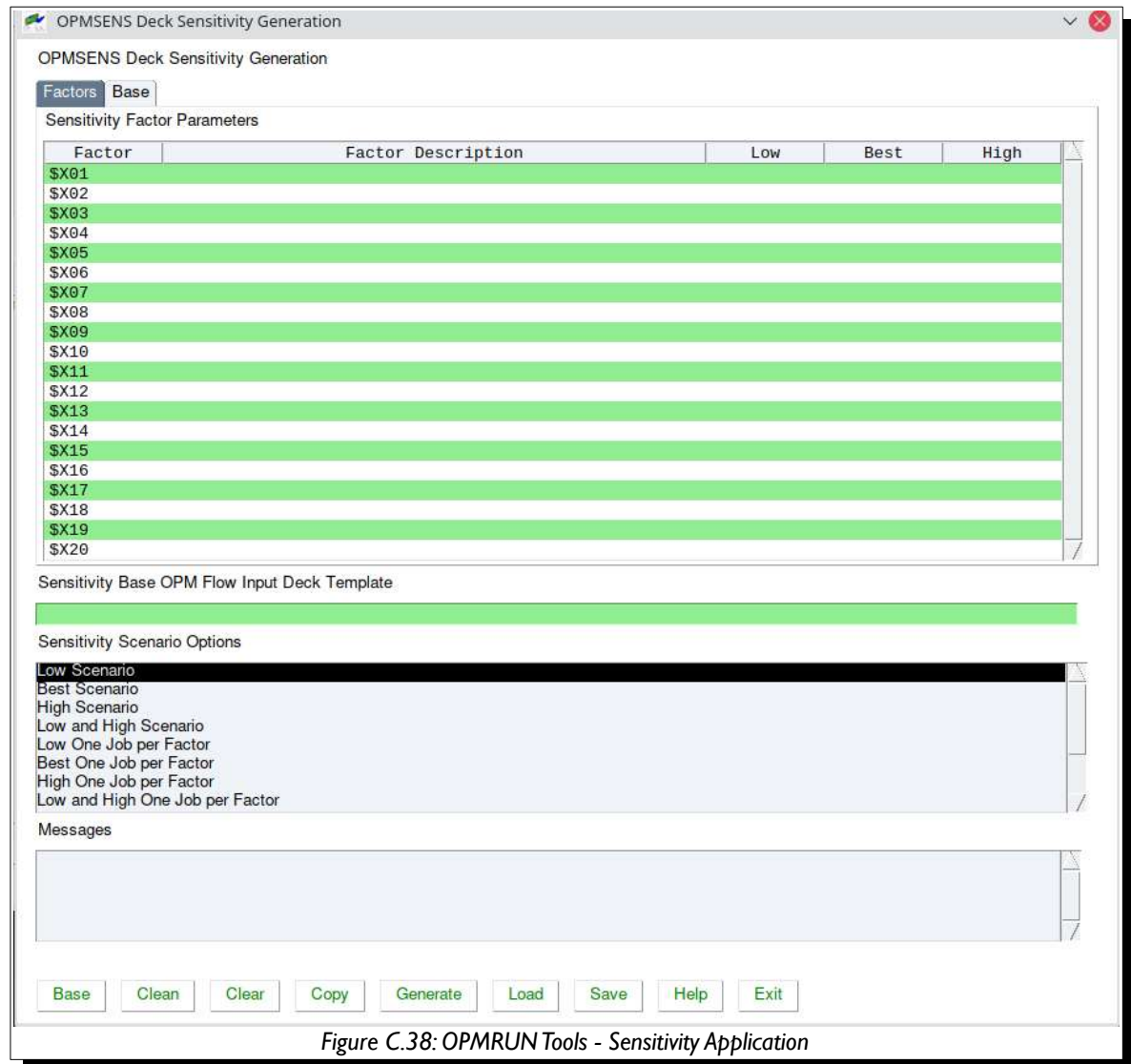


Figure C.38: OPMRUN Tools - Sensitivity Application

After editing the Base file, the next step is to load the Base file into the application using the *Base* button, which will prompt the user for the file to load and then display the file in the *Base* tab, as shown in Figure C.39



Figure C.39: OPMRUN Tools – Sensitivity Base Case Data File Example

Limited editing of the *Base* file is supported on the above screen.

The next step is to define the "Factors" and the factor values. A total of 20 factors are available and each factor consist of a Low, Best and High estimates. Note it is not necessary to enter all three estimates, if one wishes just to generate a limited sensitivity case. For example, if one wishes to only run a Low Scenario sensitivity then it is only necessary to enter data for the Low factor values.

Previously saved factor data can be loaded via the *Load* button, as shown below:

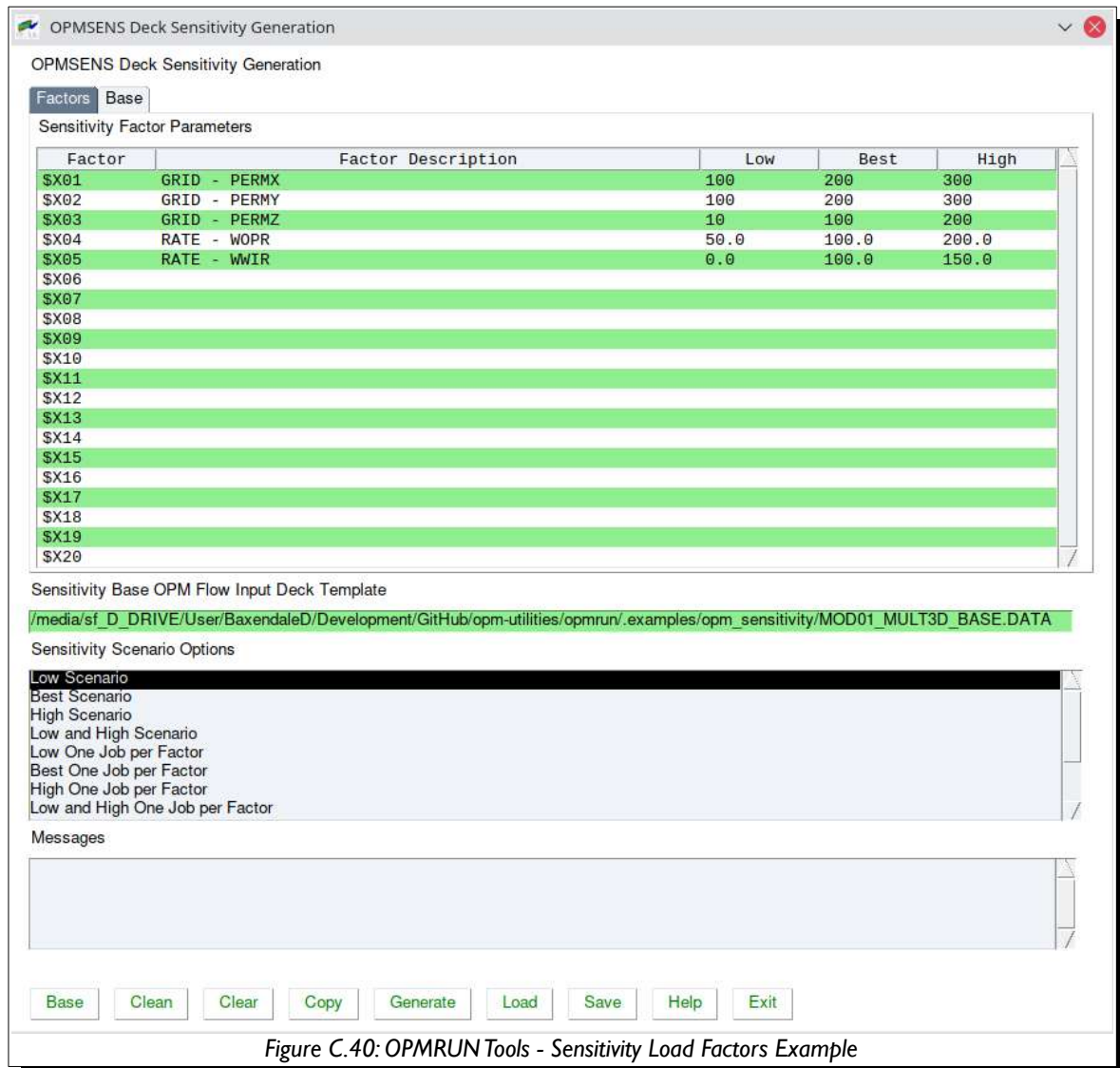


Figure C.40: OPMRUN Tools - Sensitivity Load Factors Example

Selecting a Factor Description row allows one to define a description for the factor variable, so for \$X01 in the above figure the description is *GRID - PERMX*. When selecting a Factor Description, a popup dialog will be displayed to enter the data, and if one right-clicks on the popup's Factor Description field one can select a description for one of the pre-defined descriptions as illustrated in the next figure.

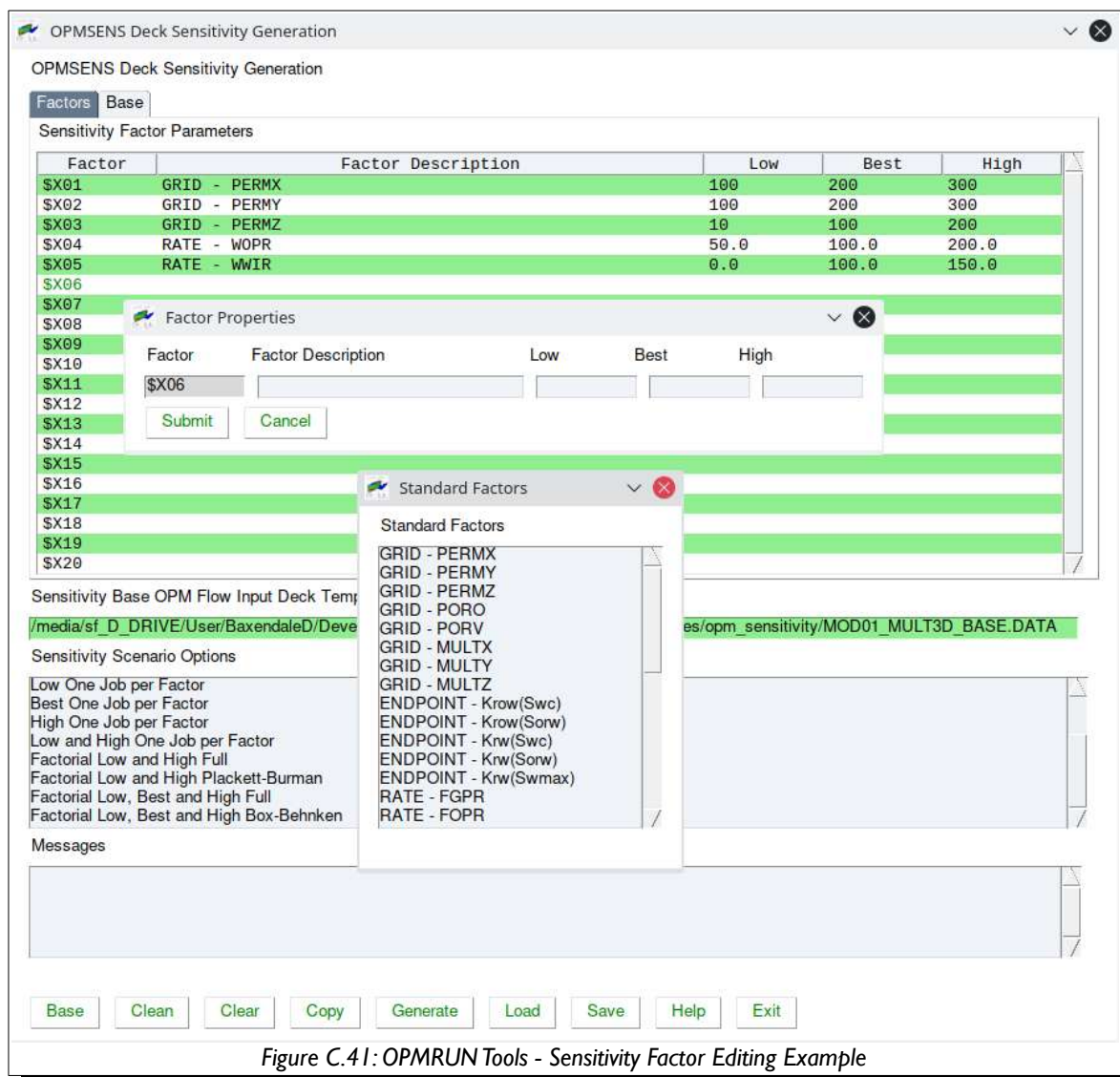


Figure C.41: OPMRUN Tools - Sensitivity Factor Editing Example

After the Sensitivity Factors have been entered one can then select the Sensitivity Scenario that one wishes to use generate the sensitivity cases. In the example in figure Figure C.42 the *Factorial: Low, Best and High Box-Behnken* DOE (Design of Experiments) has been selected. Selecting the *Generate* button, runs a series of checks, and if there are no errors the program will inquire if you wish to generate the set of cases (Figure C.42).

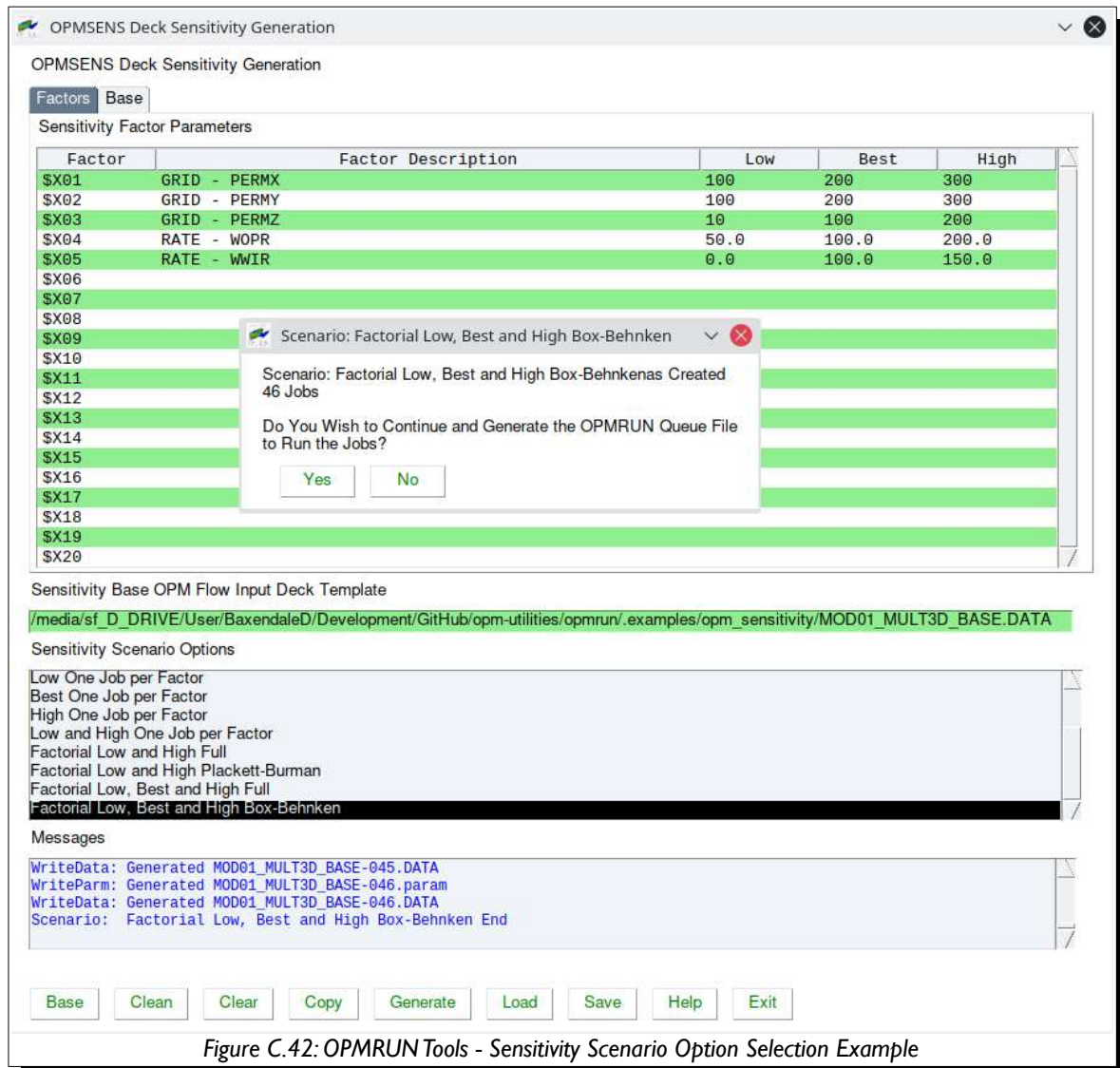


Figure C.42: OPMRUN Tools - Sensitivity Scenario Option Selection Example

If the Yes option is selected then the cases will be generated and the application will ask for the name of OPMRUN Queue file to write the jobs to, as depicted in

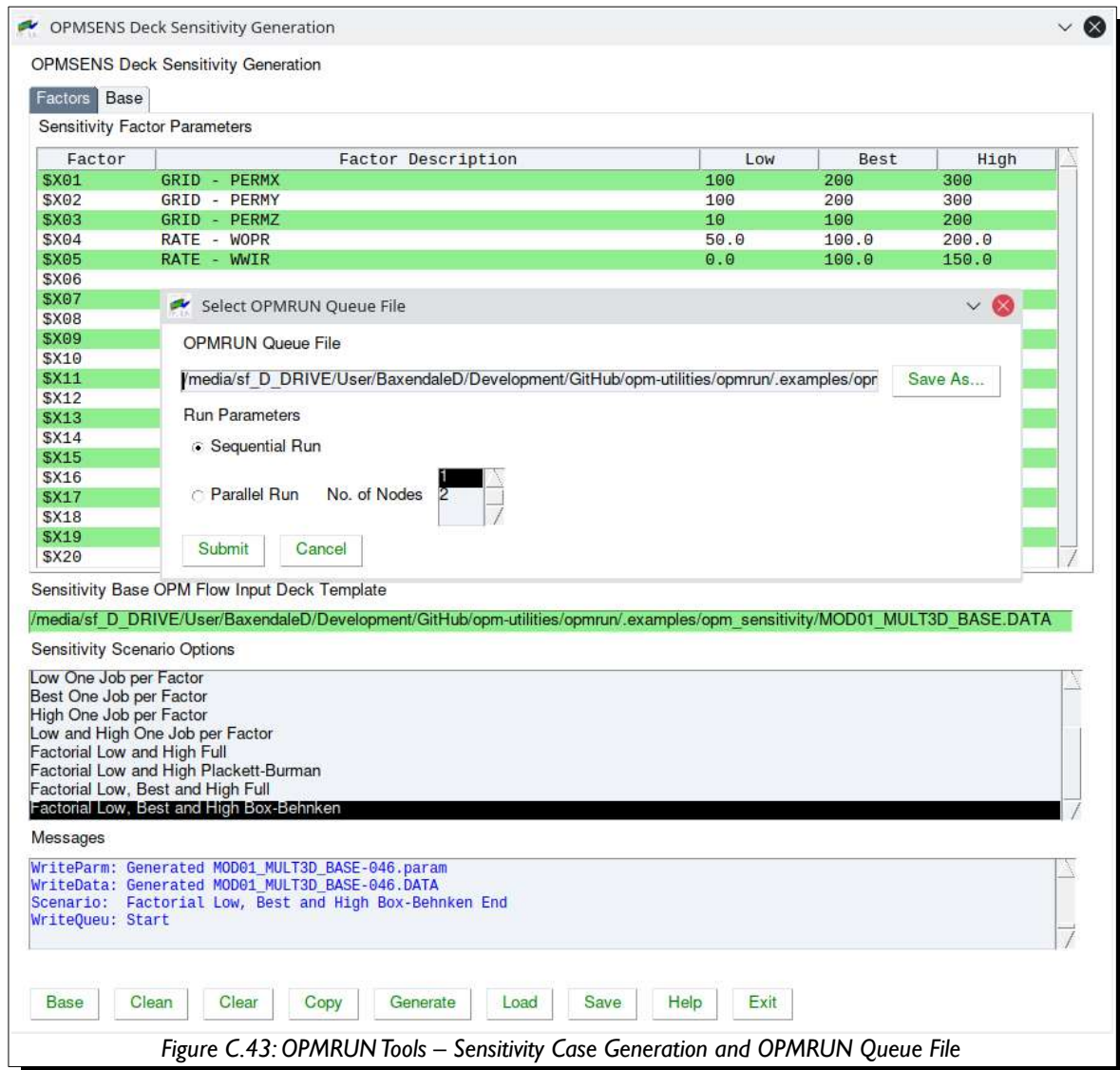


Figure C.43: OPMRUN Tools – Sensitivity Case Generation and OPMRUN Queue File

This allows the user to load the queue file into OPMRUN and to run all the jobs.

C.3.5 OPMRUN TOOLS: SIMULATOR INPUT/Well SPECIFICATION

This tool, *Tools/Simulator Input/Well Specification*, uses the standard well export files from OPM ResInsight to reformat the data in a more user-friendly manner for the [WELSPECS](#) and [COMPDAT](#) keywords. Optionally, the application can generate the [COMPLUMP](#) keyword based on the OPM ResInsight layers file, with one completion per defined reservoir layer.

An example OPM ResInsight Exported Well Completion File Format(*.exp) is shown in Figure C.44

-- WELL	GROUP			BHP	PHASE	DRAIN	INFLOW	OPEN	CROSS	PVT	HYDS	FIP
-- NAME	NAME	I	J	DEPTH	FLUID	AREA	EQUANS	SHUT	FLOW	TABLE	DENS	REGN
WELSPECS												
GP-P1-06	1*	82	139	1*	OIL	0.0	STD	STOP	YES	0	SEG	0 /
GP-P1-05	1*	72	89	1*	OIL	0.0	STD	STOP	YES	0	SEG	0 /
GP-P1-01	1*	73	102	1*	OIL	0.0	STD	STOP	YES	0	SEG	0 /
GP-P1-02	1*	79	112	1*	OIL	0.0	STD	STOP	YES	0	SEG	0 /
GP-P2-04	1*	74	79	1*	OIL	0.0	STD	STOP	YES	0	SEG	0 /
/												
-- WELL					OPEN	SAT	CONN					
-- NAME	I	J	K1	K2	SHUT	TAB	FACT	WELL	KH		SKIN	D
COMPDAT												
---- Completions for completion type Perforation ----												
-- Perforation Completion : MD In: 10115.8 - MD Out: 10125.6 Transmissibility: 0.0855562												
GP-P1-01	73	102	49	49	OPEN	1*	8.555621E-02	0.70900	8.712182E+01	2.00000	1*	'Z' /
-- Perforation Completion : MD In: 10125.6 - MD Out: 10136.6 Transmissibility: 0.153124												
GP-P1-01	73	102	50	50	OPEN	1*	1.531236E-01	0.70900	1.559257E+02	2.00000	1*	'Z' /
-- Perforation Completion : MD In: 10136.6 - MD Out: 10147.6 Transmissibility: 0.153119												
GP-P1-01	73	102	51	51	OPEN	1*	1.531194E-01	0.70900	1.559215E+02	2.00000	1*	'Z' /
-- Perforation Completion : MD In: 10147.6 - MD Out: 10154.2 Transmissibility: 0.141482												
GP-P1-01	73	102	52	52	OPEN	1*	1.414819E-01	0.70900	1.440710E+02	2.00000	1*	'Z' /
-- Perforation Completion : MD In: 10162.4 - MD Out: 10169.6 Transmissibility: 0.0383522												
GP-P1-01	73	102	53	53	OPEN	1*	3.835216E-02	0.70900	6.450968E+01	6.70000	1*	'Z' /
-- Perforation Completion : MD In: 10169.6 - MD Out: 10180.6 Transmissibility: 0.0356561												
GP-P1-01	73	102	54	54	OPEN	1*	3.565613E-02	0.70900	5.997487E+01	6.70000	1*	'Z' /
-- Perforation Completion : MD In: 10180.6 - MD Out: 10191.7 Transmissibility: 0.0030597												
GP-P1-01	73	102	55	55	OPEN	1*	3.059698E-03	0.70900	5.146520E+00	6.70000	1*	'Z' /
-- Perforation Completion : MD In: 10191.7 - MD Out: 10202.7 Transmissibility: 0.0116847												
GP-P1-01	73	102	56	56	OPEN	1*	1.168469E-02	0.70900	1.965406E+01	6.70000	1*	'Z' /
-- Perforation Completion : MD In: 10202.7 - MD Out: 10213.7 Transmissibility: 0.00139438												
GP-P1-01	73	102	57	57	OPEN	1*	1.394380E-03	0.70900	2.345397E+00	6.70000	1*	'Z' /
-- Perforation Completion : MD In: 10213.7 - MD Out: 10224.7 Transmissibility: 0.0584769												
GP-P1-01	73	102	58	58	OPEN	1*	5.847693E-02	0.70900	9.836029E+01	6.70000	1*	'Z' /

Figure C.44: OPMRUN Tools - Well Specification OPM ResInsight Well Completion File Format

And an OPM ResInsight Imported Formation Layer File (.Lyr) example is illustrated in Figure C.45

```
--
-- ResInsight Formation Names
-- Zone          Top   Base
-- Name          Layer Layer
--
'Pre-A'          1 - 2
'Shale'          3 - 3
'A Upper'        4 - 5
'Shale'          6 - 6
'A Lower'        7 - 8
'A Lower Pay'    9 - 14
'Shale'          15 - 15
'B Upper'        16 - 18
'Shale'          19 - 19
'B Lower'        20 - 26
'Shale'          27 - 27
'C Upper'        28 - 30
'C Upper Inte'   31 - 36
'C Upper Lobe'   37 - 40
'Shale'          41 - 41
'C Lower'        42 - 47
'Shale'          48 - 48
'D Upper'        49 - 59
'D Middle'       60 - 72
'D Lower'        73 - 80
'Shale'          81 - 81
'D Lower Lobe 1'  82 - 87
'Shale'          88 - 88
'D Lower Lobe 2'  89 - 90
'Shale'          91 - 91
'E Sand'         92 - 97
'Shale'          98 - 98
```

*Figure C.45: OPMRUN Tools - Well
Specification OPM ResInsight Reservoir Layer
File*

The application also can generate a well a OPM ResInsight perforation file with the formation names for cross-checking the perforations.

The application user interface is shown in Figure C.46. Note that in Figure C.46 the *Output Header* options are used for comments only, no unit conversion is performed.

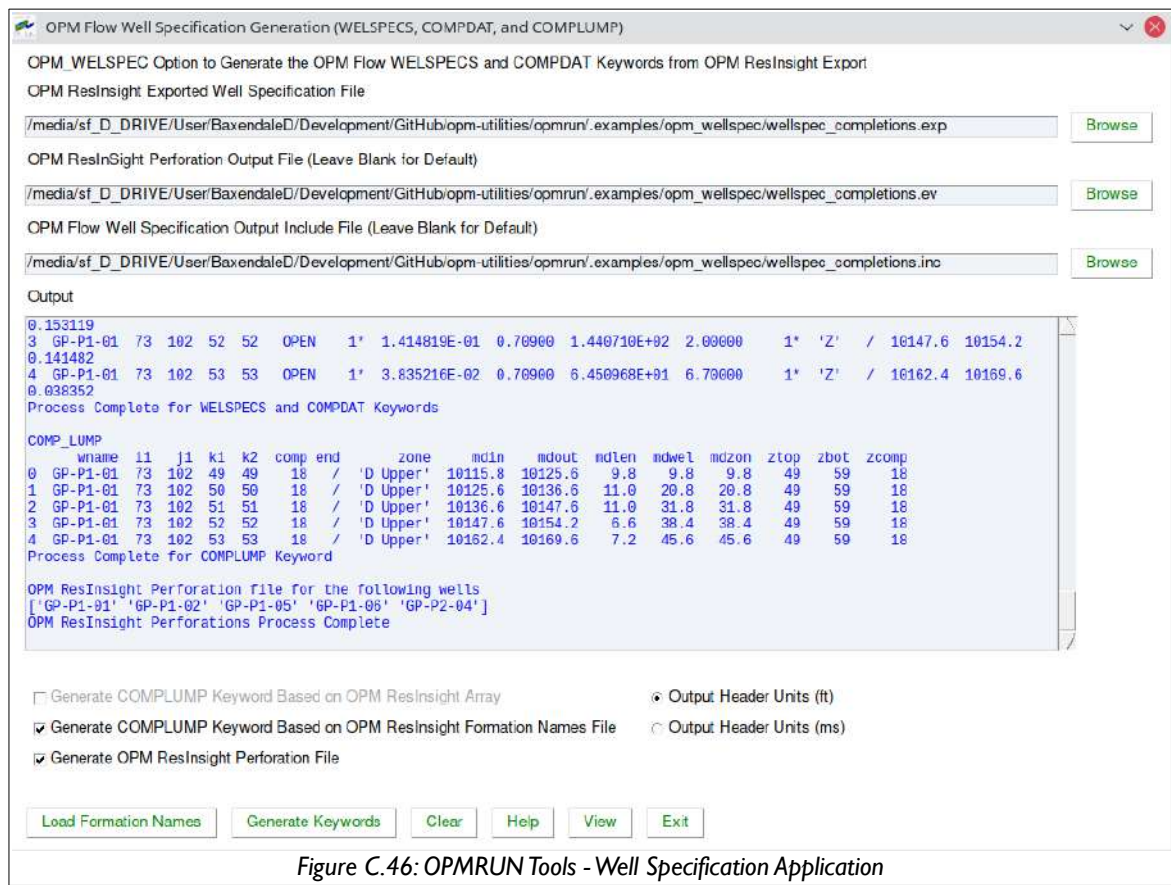


Figure C.46: OPMRUN Tools - Well Specification Application

In terms of output, the next figure shows the resulting well completion file to be used with OPM Flow, showing the [WELSPECS](#) and [COMPDAT](#) keywords (the [COMPLUMP](#) keyword is not shown in this example)

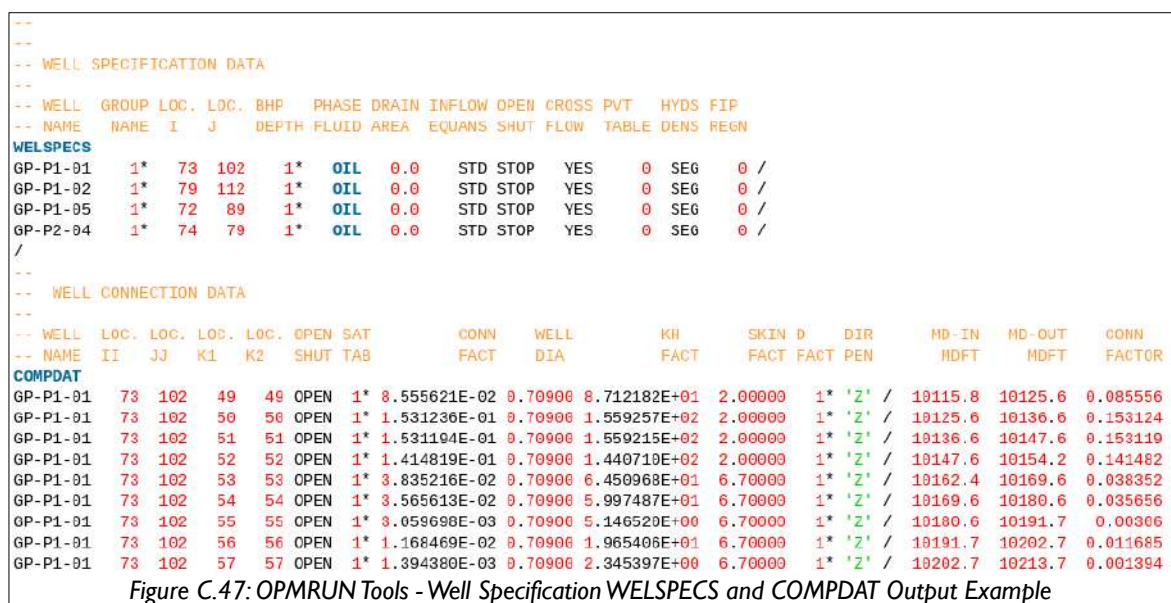


Figure C.47: OPMRUN Tools - Well Specification WELSPECS and COMPDAT Output Example

The final figure for this tool shows the resulting generated OPM ResInsight perforation file.

```
--
UNITS FIELD
--
--
-- DATE PERFORATION TOP BOT WELL SKIN ZONE
-- MDFT MDFT DIAM FACT
WELLNAME GP-P1-01
"SOH" perforation 10115.8 10125.6 0.70900 2.00000 -- 'D Upper'
"SOH" perforation 10125.6 10136.6 0.70900 2.00000 -- 'D Upper'
"SOH" perforation 10136.6 10147.6 0.70900 2.00000 -- 'D Upper'
"SOH" perforation 10147.6 10154.2 0.70900 2.00000 -- 'D Upper'
"SOH" perforation 10162.4 10169.6 0.70900 6.70000 -- 'D Upper'
"SOH" perforation 10169.6 10180.6 0.70900 6.70000 -- 'D Upper'
"SOH" perforation 10180.6 10191.7 0.70900 6.70000 -- 'D Upper'
"SOH" perforation 10191.7 10202.7 0.70900 6.70000 -- 'D Upper'
"SOH" perforation 10202.7 10213.7 0.70900 6.70000 -- 'D Upper'
"SOH" perforation 10213.7 10224.7 0.70900 6.70000 -- 'D Upper'
"SOH" perforation 10224.7 10228.7 0.70900 6.70000 -- 'D Upper'
```

Figure C.48: OPMRUN Tools - Well Specification OPM ResInsight Perforation File Output

C.3.6 OPMRUN TOOLS: RESINSIGHT

This option, *Tools/ResInsight*, loads the currently selected job into OPM ResInsight for viewing, this done via a Python sub-process call in OPMRUN, rather than using OPM ResInsight's Python [API](#).

C.3.7 OPMRUN TOOLS: WELL TRAJECTORY CONVERSION

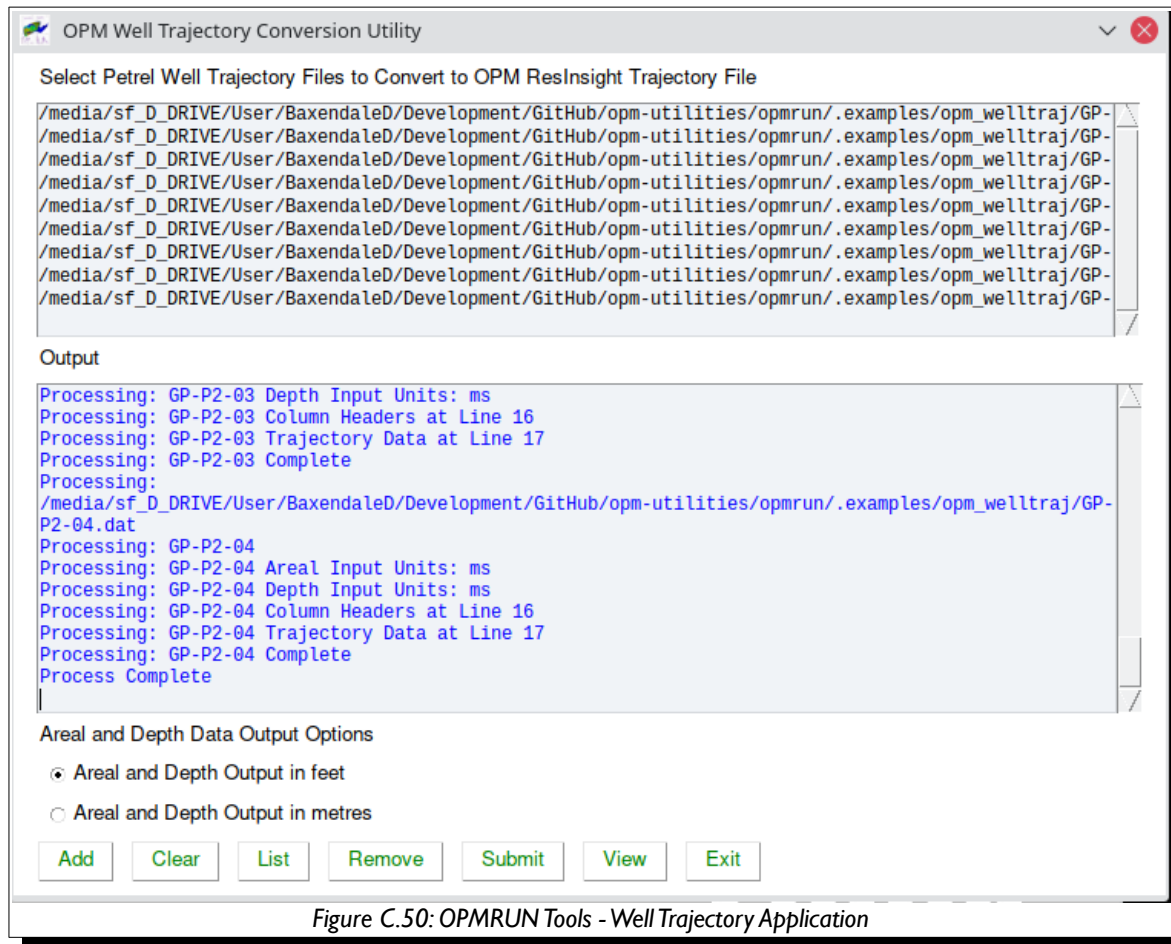
OPM ResInsight can read well trajectories in a given format into the program, the *Tools/Well Trajectory Conversion* option converts a Schlumberger Petrel exported well trajectory file, as shown Figure C.49, into a OPM ResInsight well trajectory file containing all the wells.

```
# WELL TRACE FROM PETREL
# WELL NAME: GP-P1-01
# DEFINITIVE SURVEY: Explicit survey 1
# WELL HEAD X-COORDINATE: 294774.82000000 (m)
# WELL HEAD Y-COORDINATE: 2623340.30000000 (m)
# WELL DATUM (KB, Kelly bushing, from MSL): 13.60000000 (m)
# WELL TYPE: GAS
# MD AND TVD ARE REFERENCED (=0) AT WELL DATUM AND INCREASE DOWNWARDS
# ANGLES ARE GIVEN IN DEGREES
# XYZ TRACE IS GIVEN IN COORDINATE SYSTEM [Petrel]
# AZIM_TN: azimuth in True North
# AZIM_GN: azimuth in Grid North
# DX DY ARE GIVEN IN GRID NORTH IN m-UNITS
# DEPTH (Z, tvd_z) GIVEN IN m-UNITS
#=====
# MD X Y Z TVD DX DY AZIM_TN INCL DLS AZIM_GN
#=====
0.000000000 294774.82000 2623340.3000 13.600000000 0.000000000 -0.000000001 -0.000000002 359.19018141 0.000000000 0.000000000 0.000000000
224.20000000 294774.88839 2623341.0993 -210.5980866 224.1980866 0.0683874065 0.7993438398 4.0801814067 0.4100000000 0.0548617306 4.8900000000
252.80000000 294774.98621 2623341.2814 -239.1972995 252.7972995 0.1662053919 0.9814199335 47.230181407 0.4800000000 0.3500120164 48.0400000000
282.20000000 294775.19584 2623341.4957 -268.5957535 282.1957535 0.3758445395 1.1957111340 41.010181407 0.6900000000 0.2235585568 41.8200000000
311.10000000 294775.59178 2623341.9055 -297.4898516 311.0898516 0.7717802449 1.6054957099 44.170181407 1.5700000000 0.9154355591 44.9800000000
339.70000000 294776.51776 2623342.6035 -326.0654115 339.6654115 1.6977563415 2.3034936697 56.230181407 3.1000000000 1.6768888626 57.0400000000
368.60000000 294778.20113 2623343.6423 -354.8966452 368.4966452 3.3811279249 3.3422523835 58.350181407 4.7500000000 1.7191224028 59.1600000000
397.50000000 294780.74744 2623345.2481 -383.6372139 397.2372139 5.9274355074 4.9480945668 56.030181407 7.2100000000 2.5654105761 56.8400000000
426.10000000 294784.30376 2623347.6883 -411.9070216 425.5070216 9.4837615496 7.3882779982 53.810181407 10.1400000000 3.0928640683 54.6200000000
454.90000000 294788.99742 2623351.1132 -440.1115661 453.7115660 14.177422776 10.813246789 52.500181407 13.1400000000 3.1369063547 53.3100000000
483.90000000 294794.78585 2623355.5409 -468.1780891 481.7780891 19.965846340 15.240878427 51.180181407 15.9700000000 2.9474446939 51.9900000000
```

Figure C.49: OPMRUN Tools - Well Trajectory Conversion Input File

The utility allows for the multiple wells to be converted at once and for conversion of units.

Note in some areas of the world it is not uncommon for the areal units to be in UTM and the depth to be in feet. This configuration is also handled by the application.



An example output file is shown in Figure C.5I


```
-- WELL TRACE FROM PETREL
-- WELL NAME: GP-P1-01
-- DEFINITIVE SURVEY: Explicit survey 1
-- WELL HEAD X-COORDINATE: 294774.82000000 (m)
-- WELL HEAD Y-COORDINATE: 2623340.30000000 (m)
-- WELL DATUM (KB, Kelly bushing, from MSL): 13.60000000 (m)
-- WELL TYPE: GAS
-- MD AND TVD ARE REFERENCED (=0) AT WELL DATUM AND INCREASE DOWNWARDS
-- ANGLES ARE GIVEN IN DEGREES
-- XYZ TRACE IS GIVEN IN COORDINATE SYSTEM [Petrel]
-- AZIM_TN: azimuth in True North
-- AZIM_GN: azimuth in Grid North
-- DX DY ARE GIVEN IN GRID NORTH IN m-UNITS
-- DEPTH (Z, tvd_z) GIVEN IN m-UNITS
--
-- APPLIED AREAL CONVERSION FACTOR: 3.28083989501312
-- APPLIED DEPTH CONVERSION FACTOR: 3.28083989501312
--
WELLNAME:GP-P1-01
--          XCORD          YCORD          TVDSS          MD
--          DEPTH          DEPTH
--          ft            ft            ft            ft
-----
  9.671090e+05  8.606760e+06   -44.62         0.00
  9.671092e+05  8.606762e+06    690.94        735.56
  9.671095e+05  8.606763e+06    784.77        829.40
  9.671102e+05  8.606763e+06    881.22        925.85
  9.671115e+05  8.606765e+06    976.02       1020.67
  9.671146e+05  8.606767e+06   1069.77       1114.50
  9.671201e+05  8.606770e+06   1164.36       1209.32
  9.671284e+05  8.606776e+06   1258.65       1304.13
  9.671401e+05  8.606784e+06   1351.40       1397.97
  9.671555e+05  8.606795e+06   1443.94       1492.45
  9.671745e+05  8.606810e+06   1536.02       1587.60
  9.671965e+05  8.606827e+06   1626.60       1682.41
  9.672208e+05  8.606846e+06   1715.81       1776.90
  9.672483e+05  8.606868e+06   1803.86       1871.72
```

Figure C.5 I: OPMRUN Tools - Well Trajectory Output Trajectory File.

APPENDIX D: PYTHON INTERFACE TO OPM FLOW

APPENDIX TABLE OF CONTENTS

APPENDIX D: PYTHON INTERFACE TO OPM FLOW.....	2479
D.1 Introduction.....	2482
D.2 Working with the Input Deck.....	2482
D.2.1 The Parser Object.....	2482
D.2.2 Loading an Input Deck.....	2483
D.2.3 Units.....	2483
D.2.4 Accessing the Input Deck.....	2484
D.2.5 Special Parsing.....	2485
D.2.6 Creating EclipseState and Grid Objects.....	2486
D.2.7 Creating a Schedule.....	2487
D.3 Working with the Result Files.....	2487
D.3.1 Loading a Grid File.....	2487
D.3.2 Loading a Summary Case of Results.....	2487
D.3.3 Loading a Restart File.....	2488

D.1 INTRODUCTION

Some of the C++ code in OPM Flow has been “wrapped” in Python which means that one can invoke the simulator’s C++ code from one’s own Python programs. At the moment the wrappers for the input layer and the code for working with the result files is quite complete and usable. For a python class documentation and for further documentation and examples, please refer to the [OPM Online Python Documentation](#). There is also the option to interact with running simulations from within Python using the [PYACTION](#) keyword.

The goal is for the Python code to be structured like a native Python API, but some design decisions are certainly influenced by the underlying C++ implementation, and to get a deeper understanding of what is possible and how to achieve specialized tasks one might need to consult the simulator’s C++ code.

Note

Observe that the Python bindings described here are not very mature. The API might change in future releases. It is in general quite simple to expose new functionality to Python.

D.2 WORKING WITH THE INPUT DECK

The reading of the input deck in OPM Flow is a two step process, first a data structure called a “Deck” is created - the “Deck” is essentially a collection of keywords where all elements have been converted to the correct type and default values have been injected into the keyword. The “Deck” is a quite low level data structure, and the actual simulation is based on higher level data structures where the origin in “keywords” from a *.DATA file is no longer so apparent. The most notable high level objects are the “EclipseState” and “Schedule” classes. Large parts of this functionality is available from Python.

D.2.1 THE PARSER OBJECT

The initial building block in the parsing process is the “Parser” object which knows how to interpret the input keywords and create data structures. Virtually all scripts working with the input files will start with creating a Parser object:

```
#!/usr/bin/env python3
from opm.io.parser import Parser
# Create a parser object which can be used to parse a string or input files
parser = Parser()
```

In most cases one will just create a default parser object and be done with it, but it is possible to both add your own keyword definitions to the parser and alternatively create a custom parser which only accepts a subset of keywords. These features will be demonstrated briefly in section [Special Parsing](#).

D.2.2 LOADING AN INPUT DECK

The “Deck” data structure is essentially a list of keywords which have been loaded as “DeckKeyword” instances. The Deck can either be loaded from an input data file or from a string. The following example shows how to load two [SCHEDULE](#) section keywords: [WELSPECS](#) and [COMPDAT](#):

```
#!/usr/bin/env python3
from opm.io.parser import Parser
deck_string = """
WELSPECS
  'W1'  'G1'   19   4      1*  WATER  1*   1*   SHUT  1*  1*   1*   /
/
COMPDAT
  'W1'   11   3   1   5   OPEN  1*   1*   0.216  1*   0   1*   Z   1* /
/
"""
parser = Parser()
deck = parser.parseString( deck_string )
```

After running this small script the deck variable will contain the two keywords [WELSPECS](#) and [COMPDAT](#). For these keywords the “*” have been replaced with the correct default value and the items have been converted to the correct type. In addition, all numerical values have been converted to SI units, more about units can be found in section [Units](#). Alternatively you can use the `Parser.parseFile()` method to parse an entire input file. In this case [INCLUDE](#) and [IMPORT](#) keywords will be resolved and everything will be coalesced into one large “DeckKeyword” data structure:

```
#!/usr/bin/env python3
import os.path
import sys
from opm.io.parser import Parser
data_file = sys.argv[1]
print(f"Loading deck from {data_file}")
parser = Parser()
deck = parser.parseFile( data_file )
```

The code above loads a *.DATA file given by calling the Python script. For example, if the name of the input deck, that may contain a complete input deck or just a selection of keywords, is called NORNE.DATA, and the name of the above script is LoadDeck.py, then:

```
python LoadDeck NORNE.DATA
```

would load the NORNE.DATA into the simulator.

D.2.3 UNITS

Internally in OPM Flow all quantities are managed as SI units, whereas input decks use the unit systems [METRIC](#), [FIELD](#), or [LAB](#) (laboratory). The [METRIC](#) unit system is comparable to SI units, the most notable difference is that time is expressed in days instead of seconds, the [FIELD](#) unit system is based on historical units like barrels of oil (stb) and feet for length.

The important point is that when you access the elements in the Deck the values you will get are in SI units, whereas the input you provided has been in one of other set of units. Consider the example:

```
#!/usr/bin/env python3
```

```
from opm.io.parser import Parser
deck_string = """
WCONPROD
  'W1'  'OPEN'  'ORAT'  86400 /
/
"""
parser = Parser()
deck = parser.parseString( deck_string )
kw = deck[0]
record0 = kw[0]
orat = record0["ORAT"].value
well = record0["WELL"].value
print("==> The Oil rate in well:{} is {} m3/s".format(orat, well))
```

When this script is run it will print:

```
==> The Oil rate in well:W1 is 1.0 m3/s
```

because the input value of 86,400 m3/day is internally converted to 1.0 m3/second.

The default input unit system is *METRIC*, that means that if the input is to be interpreted in either *FIELD* or *LAB* units, then one needs to add a keyword to declare the units before entering the data. Thus, to interpret the *WCONPROD* keyword's oil rate as stb/d in the previous example, the *FIELD* keyword needs to be added to the example, that is:

```
#!/usr/bin/env python3
from opm.io.parser import Parser
deck_string = """

FIELD

WCONPROD
  'W1'  'OPEN'  'ORAT'  86400 /
/
"""
```

D.2.4 ACCESSING THE INPUT DECK

After you have loaded a Deck you can query it with normal Python functions:

```
# Check if deck has keyword:
if "GRID" in deck:
    print("Deck contains 'GRID' keyword")
else:
    print("Deck does not have 'GRID' keyword")

# Loop through all the keywords:
for kw in deck:
    print("kw: {}".format(kw.name))
```

To get a keyword from the deck you can access it with index, name, or a combination of name and occurrence index:

```
# Get keyword 10
kw10 = deck[10]

# Get the last DATES keyword
last_dates = deck["DATES"]

# Get the third WELSPECS keyword
welspecs3 = deck[("WELSPECS", 3)]
```

A deck keyword is composed of “records”, which are again composed of “items”. The example below shows how one can iterate through the wells in a *WELSPECS* keyword:

```
kw = deck["WELSPECS"]
print(f"Keyword has {len(kw)} records")
for record in kw:
    well = record["WELL"].get_str()
    group = record[1].get_str()

    print(f"Well {num}: {well} is part of group{group}")
```

D.2.5 SPECIAL PARSING

The keywords that OPM Flow recognizes are configured via JSON files which are embedded in the source distribution. The JSON keywords for the simulator are in the following public GitHub repository:

<https://github.com/OPM/opm-common/tree/master/opm/input/eclipse/share/keywords>

By default the Parser class will recognize all the keywords which are known to OPM Flow, but one can create your own custom parser with only a subset of keywords:

```
from opm.io.parser import Builtin
# Create a special parser which only recognizes the corner point grid
# keywords
parser = Parser(add_default = False)
builtin = Builtin()
parser.add_keyword( builtin.COORD )
parser.add_keyword( builtin.ZCORN )
parser.add_keyword( builtin.ACTNUM )
```

The opposite is also possible, here is how one can add a special keyword GCLOSE which could be used to close all the wells in the group:

```
# Create a fictitious home mode keyword GCLOSE:
GCLOSE = {"name" : "GCLOSE",
          "sections" : ["SCHEDULE"],
          "items" : [
              {"name" : "GROUP", "value_type" : "STRING"}
          ]}

parser = Parser()
# Add the keyword description via a JSON string
parser.add_keyword(json.dumps(GCLOSE))
```

By default the parser used in OPM Flow is quite strict - a tad stricter than the one used by the commercial simulator. However, one can configure how the parser should react to certain error conditions. By default the parser will fail with an exception if an *INCLUDE* file is not found, but one can for instance change this behavior to ignore that error condition, for example:

```
parse_context = ParseContext([('PARSE_MISSING_INCLUDE',
                                opm.io.action.ignore)])

parser = Parser()
deck = parser.parseFile(self.norne_fname, parse_context)
```


D.2.6 CREATING ECLIPSESTATE AND GRID OBJECTS

The “Deck” is a quite raw structure and not very user friendly. Before actually used in the simulator the properties from the “Deck” are coalesced to a form more easily usable format. Consider, for example the *PORO* array configured below where the configuration is given in three steps:

```
-- GRID Dimensions 10 x 10 x 3
DIMENS
  10 10 3 /
--
-- Set A Global Value 0.10 For All Of The Grid
--
PORO
  300*0.10 /
--
-- Set A Value Of 0.15 For All The Cells In The Middle Layer
--
BOX
  1 10 1 10 2 2 /
PORO
  100*0.15 /
ENDBOX
--
-- Scale The Bottom Layer By A Factor OF Two
--
MULTIPLY
  PORO 2 1 10 1 10 3 3 /
/
```

In the deck representation this will be a collection of five different keywords: {"PORO", "BOX", "PORO", "ENDBOX", "MULTIPLY"}, whereas in the “EclipseState” this will be one property "PORO" where all the modifications have been completed. In addition to the grid properties like "PERMX" and "PORO" the “EclipseState” object contains numerous other objects like the fault properties, PVT tables, and more.

```
#!/usr/bin/env python3
import sys
from opm.io.parser import Parser
from opm.io.ecl_state import EclipseState
parser = Parser()
data_file = sys.argv[1]
deck = parser.parseFile( data_file )
es = EclipseState( deck )
#
# Get the FieldPropsManager from the EclipseState. The FieldPropsManager
# is then used to query and look up grid properties like PERMX and PORO.
# Observe that the properties you get from the FieldPropsManager only have
# the active cells only.
#
fp = es.field_props()
poro = fp["PORO"]
satnum = fp["SATNUM"]
#
# Get the input grid from the EclipseState:
#
grid = es.grid()
# Get a manager for all the tables
tables = es.tables()
```

D.2.7 CREATING A SCHEDULE

The Schedule object contains all the dynamic information in the model, in particular that includes all information about wells and groups. The Schedule object is constructed from the “Deck” and the “EclipseState” objects.:

```
#!/usr/bin/env python3
import sys
from opm.io.parser import Parser
from opm.io.ecl_state import EclipseState
parser = Parser()
data_file = sys.argv[1]
deck = parser.parseFile( data_file )
es = EclipseState( deck )
schedule = Schedule(deck, es)
print(f"Schedule file has {len(schedule)} report steps")
print("List of wells")
for well in schedule.well_names():
    print(well)
```

The example above prints the number of report time steps and all the well names in the [SCHEDULE](#) section.

D.3 WORKING WITH THE RESULT FILES

In addition to the functionality to load and inspect the input deck, OPM Flow also has a Python interface to load and inspect the various result files, [SUMMARY](#) and [RESTART](#) files. There is also some functionality to create files with the correct formatting.

D.3.1 LOADING A GRID FILE

Using the class “EclipseGrid” one can load a grid representation from a *.EGRID file on disk:

```
#!/usr/bin/env python3
import sys
from opm.io.ecl import EGrid
grid_file = sys.argv[1]
grid = EGrid(grid_file)
```

The return value from “EclipseState.ecl_grid()” is also of this type.

D.3.2 LOADING A SUMMARY CASE OF RESULTS

Using the class “ESMry” one can load the Extended [SUMMARY](#) file for a simulations that contains all the [SUMMARY](#) vectors.

```
#!/usr/bin/env python3
import sys
from opm.io.ecl import ESMry
case = sys.argv[1]
summary = Esmry(case)
```

D.3.3 LOADING A RESTART FILE

Finally, one load the solution arrays stored in the *.RESTART file using the Erst class, as shown below.

```
#!/usr/bin/env python
import sys
from opm.io.ecl import ERst

rst_file = sys.argv[1]
rst = ERst(rst_file)
```

Note that a restart file can be large, so care must be taken on how these type of files are loaded into Python.

APPENDIX E: RUNNING PREVIOUS RELEASES OF OPM FLOW

APPENDIX TABLE OF CONTENTS

APPENDIX E: RUNNING PREVIOUS RELEASES OF OPM FLOW.....2489

E.1 Running OPM Flow 2025-04.....

E.2 Running OPM Flow 2024-10.....

E.3 Running OPM Flow 2024-04.....

E.4 Running OPM Flow 2023-10.....

2492

2514

2535

2555

E.1 RUNNING OPM FLOW 2025-04

The command line syntax for running OPM Flow is:

```
flow [OPTIONS] CASENAME
```

and typing the following command from your terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA.

OPM Flow accepts command line options to control various aspects of the simulator's run time behavior, as described in the table below. To give a command line option use "key=value" syntax, with no spaces around the equals sign. It is also possible to put multiple options together in a parameter file. To do so, put one option on each line of the file with ".param" as the extension and pass that filename as a command line parameter to OPM Flow using:

```
flow --parameter-file=CASENAME.param
```

to instruct OPM Flow to read the parameter file.

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
1	-h or --help	A character string that causes OPM Flow to print a help message that gives a brief description of the available command line parameters. This now only prints the active user facing command line options.	N/A
2	--help-all	Prints all the command line options included in the release, including obsolete, hidden and deprecated options.	N/A
3	--accelerator-mode	A defined character string that selects the linear solver, usage: '--accelerator-mode=[none cusparse opencl amgcl rocalution rocsparse]'. 	"none"
4	--action-parsing-strictness	A defined character string. Set strictness of parsing process for ActionX and PyAction. Available options are normal (do not apply keywords that have not been tested for ActionX or PyAction) and low (try to apply all keywords, beware: the simulation outcome might be incorrect).	"normal"
5	--add-corners	A boolean value. Add corners to partition.	false
6	--allow-distributed-wells	Allow the perforations of a well to be distributed to interior of multiple processes.	false
7	--allow-splitting-inactive-wells	A boolean value. Allow inactive (never non-	true

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		shut) wells to be split across multiple domains.	
8	--alternative-well-rate-init	Use alternative well rate initialization procedure.	true
9	--bda-device-id	Renamed as --gpu-device-id.	0
10	--check-group-constraints-inner-well-iterations	A boolean value. Allow checking of group constraints during inner well iterations.	true
11	--check-satfunc-consistency	A boolean value. Whether or not to check saturation function consistency requirements.	false
12	--continue-on-convergence-error	Continue with a non-converged solution instead of giving up if we encounter a time step size smaller than the minimum time step size.	false
13	--convergence-monitoring	A boolean value. Enable convergence monitoring.	false
14	--convergence-monitoring-cut-off	An integer value. Cut off limit for convergence monitoring.	6
15	--convergence-monitoring-decay-factor	A real value. Decay factor for convergence monitoring.	0.75
16	--cpr-reuse-interval	A positive integer that sets the reuse preconditioner interval. Used when --cpr-reuse-setup is set to 4, then the preconditioner will be fully recreated instead of reused every N linear solve, where N is this parameter.	30
17	--cpr-reuse-setup	A positive integer that defines if the CPR solver should re-use the AMG setup. Valid options are: 0 : Recreate the preconditioner for every linear solve. 1 : Recreate once every time step. 2 : Recreate if last linear solve took more than 10 iterations. 3 : Never recreate. 4 : Recreate every N linear solves, where N is the parameter --cpr-reuse-interval. Changed the default value from three to four.	4
18	--dbph-max-rel	A real positive value that sets the maximum relative change of the bottom-hole pressure in a single iteration.	1.0
19	--debug-emit-cell-partition	A boolean value that determines whether	false

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		(true) or not (false) to emit cell partitions as a debugging aid.	
20	--dp-max-rel	A real positive double precision value that sets the maximum allowed relative change of pressure per iteration.	0.3
21	--ds-max	A real positive double precision value that sets the maximum allowed change in saturation per iteration.	0.2
22	--dwell-fraction-max	A real positive value that sets the maximum allowed change in well's volume fraction per iteration.	0.2
23	--ecl-deck-file-name	A character string that defines the name of the OPM Flow input file which contains the simulator's ECLIPSE formatted input deck to be simulated. Note that there must be no spaces in the filename. If there are spaces, which can typically occur on Windows based systems, then the filename should be enclosed in quotes; thus instead of FILE NAME use "FILE NAME".	""
24	--ecl-output-double-precision	A boolean value that switches on (true) or off (false) double precision in restart files. Useful for 'perfect' restarts.	false
25	--ecl-output-interval	An integer value defining the number of report steps that ought to be skipped between two writes of restart files results.	-1
26	--edge-weights-method	A defined positive integer value that defines the edge-weighting strategy: 0 : For uniform, 1 : for trans, or 2 : for log(trans).	1
27	--enable-adaptive-time-stepping	A boolean value that turns on (true) or off (false) adaptive time stepping. If set to false the report time steps provided by the input deck are used.	true
28	--enable-async-ecl-output	A boolean value that sets the output mode to be asynchronous (true), letting the simulator continue computing the next time step while writing results to restart and summary files in the commercial simulator's format on a separate thread.	true
29	--enable-async-vtk-output	A boolean value that sets the output mode to be asynchronous (true), letting the simulator continue computing the next time step while writing results to the VTK output files on a separate thread.	true

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
30	--enable-drift-compensation	A boolean value that enables (true) or disables (false) partial compensation of systematic mass losses via the source term of the next time step.	false
31	--enable-dry-run	A boolean value that specifies if the simulation should actually run (true), or just check the input deck (false). This option is equivalent to activating the <i>NOSIM</i> keyword in the <i>RUNSPEC</i> section of the input deck (see section 5.3.101 <i>NOSIM – Activate the No Simulation Mode for Data File Checking</i>). The default value of “auto” will use whatever is stipulated in the input deck via the <i>NOSIM</i> keyword.	“auto”
32	--enable-ecl-output	A boolean value that specifies if the binary output files (restart, summary files, etc.) should be written in the commercial simulator’s format (true), or OPM Flow’s format (false).	true
33	--enable-esmry	A boolean value that switches on (true) or off (false) the output of <i>SUMMARY</i> vectors to the ESMRY file for fast loading of summary data.	false
34	--enable-gravity	A boolean value that switches on (true) or off (false) the use of the gravity correction for the pressure gradients.	true
35	--enable-grid-adaptation	A boolean value that enables (true) or disables (false) adaptive grid refinement/coarsening. Note that Eclipse grid output will be disabled if adaptive grid refinement/coarsening is enabled.	false
36	--enable-intensive-quantity-cache	A boolean value that switches on (true) or off (false) the caching of intensive quantities.	true
37	--enable-logging-fallout-warning	A boolean value that sets the developer option to see whether logging was on non-root processors (true). If set to true output will be appended to the *.DBG or *.PRT files.	false
38	--enable-opm-rst-file	A boolean value set to true or false to write OPM specific data sets to the commercial simulators restart file to enable restart of OPM Flow runs by OPM Flow (true), or not to write the data (false).	false
39	--enable-storage-cache	A boolean value that turns on (true) or off (false) storing previous storage terms and avoid re-calculating them.	true

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
40	--enable-terminal-output	A boolean value that turns on (true) or off (false) high-level information about the simulation's progress to the terminal	true
41	--enable-thermodynamic-hints	A boolean value that enables (true) or disables (false) thermodynamic hints.	false
42	--enable-tuning	A boolean value that instructs OPM Flow to read the time stepping and convergence parameters from the <i>TUNING</i> keyword in the input deck, if set to true. Note that only some of the <i>TUNING</i> keyword parameters are processed.	false
43	--enable-vtk-output	A boolean value that turns on (true) or off (false) a global switch for writing VTK files.	false
44	--enable-well-operability-check	A boolean value that enables (true) checking of a well's operating status, or disables (false) the checking.	true
45	--enable-well-operability-check-iter	A boolean value that enables (true) checking of a well's operating status during iterations, or disables (false) the checking during iterations.	false
46	--enable-write-all-solutions	A boolean value that turns on (true) or off (false) the writing of all solutions to disk instead of only the ones for the report steps.	false
47	--end-time	A real value. The simulation time at which the simulation is finished [s].	1e+100
48	--explicit-rock-compaction	A boolean value. Use pressure from end of the last time step when evaluating rock compaction (true) or not (false).	false
49	--external-partition	A quoted character string. Name of file from which to load an externally generated partitioning of the model's active cells for MPI distribution purposes. If empty, the built-in partitioning method will be employed.	""
50	--force-disable-fluid-in-place-output	A boolean value that instructs OPM Flow not to print the Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
51	--force-disable-resv-fluid-in-place-output	A boolean value that instructs OPM Flow to not to print the Reservoir Volume Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
52	--full-time-step-initially	A boolean value that instructs OPM Flow to always attempt to finish a report step using a single time step (true) or not (false).	false
53	--gpu-device-id	An integer value. Choose device ID for cusparseSolver or openclSolver, use 'nvidia-smi' or 'clinfo' to determine valid IDs.	0
54	--ignore-keywords	A character string that defines a list of keywords which should be ignored by OPM Flow. The keyword string should be enclosed in quotes and each keyword separated by a colon, that is ':	""
55	--ilu-fillin-level	A positive integer value that sets the fill in level for the ILU preconditioner.	0
56	--ilu-redblack	A boolean value that instructs OPM Flow to use red-black partitioning for the ILU preconditioner.	false
57	--ilu-relaxation	A real positive double precision value that sets the relaxation factor of the linear solver's ILU preconditioner.	0.9
58	--ilu-reorder-spheres	A boolean value that specifies OPM Flow to reorder the entries of the matrix in the red-black ILU preconditioner in spheres starting at an edge (true) or not (false). If false the original ordering is preserved in each color. Otherwise try to ensure D4 ordering (in a 2D structured grid, the diagonal elements are consecutive).	false
59	--imbalance-tol	A real value. Tolerable imbalance of the load balancing.	1.1
60	--initial-time-step-in-days	A real double precision value that sets the size of initial time step in days.	1.0
61	--initial-time-step-size	A real double precision value that sets the size of initial time step in seconds.	86400
62	--inj-mult-damp-mult	A real value. Injection multiplier dampening factor (dampening multiplied by this each time oscillation is detected).	0.9
63	--inj-mult-min-damp-factor	A real value. Minimum injection multiplier dampening factor (maximum dampening level).	0.05
64	--inj-mult-osc-threshold	A real value. Injection multiplier oscillation threshold (used for multiplier dampening).	0.1
65	--input-skip-mode	A defined character string. Set compatibility mode for the SKIP100/SKIP300 keywords. Options are: 100 (skip SKIP100 ..ENDSKIP, keep SKIP300 ..ENDSKIP), 300 (skip SKIP300 ..ENDSKIP, keep	"100"

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		SKIP100 ..ENDSKIP) or all (skip both and SKIP100 ..ENDSKIP SKIP300 ..ENDSKIP).	
66	--linear-solver	A defined quoted character string that sets the configuration of the linear solver, valid values are: 1) cprw^{353 354 355} (default), 2) ilu0, 3) dilu, 4) cpr (an alias for cprw), 5) cpr_quasiimpes, 6) cpr_trueimpes, 7) cpr_trueimpesanalytic, 8) amg³⁵⁶, 9) hybrid (experimental), or 10) a file name that has the extension ".json", that contains the linear solver configuration parameters. Option (1) extends the existing Constrained Pressure Residual ("CPR") preconditioner to include wells. This option can also be invoked via the CPR keyword in the RUNSPEC section; however, the command line parameter takes precedence. For option (10) one enters a character string enclosed in quotes that defines the filename of a JSON configuration file for the flexible linear solver system. In this case the file extension should be '.json'. Note that the *.PRT file contains the "Property tree for the linear solver" listing, which is the JSON specification of the current case, and can be used to configure a user specific linear solver JSON file.	"cprw"
67	--linear-solver-ignore-convergence-failure	A boolean value that if set to true convergences failures in the linear solver are ignored. This option should be used with care, as the results might be unreliable.	false

³⁵³ Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

³⁵⁴ R. Scheichl, M. Roland, J. Wendebourg, Decoupling and block preconditioning for sedimentary basin simulations, *Computational Geosciences* 7 (2003) 295-318.

³⁵⁵ Klemetsdal, Ø.S., Møyner, O. & Lie, K.A. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. *Comput Geosci* 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

³⁵⁶ M. Blatt, A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients, Ph.D. thesis, Ruprecht-Karls-Universität Heidelberg (2010).

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
68	--linear-solver-max-iter	A positive integer value that defines the maximum number of linear iterations.	200
69	--linear-solver-print-json-definition	Write the JSON definition of the linear solver setup to the DBG file.	true
70	--linear-solver-reduction	A real positive double precision value that sets the minimum reduction of the residual for the linear solver. The linear solver convergences when the residual is reduced sufficiently. The simulator now overrides the default reduction to be 0.005 instead of 0.01, if the linear solver has been set to one of the cpr options, in a similar manner as how the default maximum number of linear iterations for the cpr and cprw options are changed to 20 instead of 200, unless specified by the command line option by the user.	0.01
71	--linear-solver-restart	A positive integer value that sets the number of iterations after which GMRES is restarted.	40
72	--linear-solver-verbosity	A positive integer value that defines the output from linear solver: 0 : No extra output, 1 : output per solution iteration, or 2 : output per iteration.	0
73	--load-file	A character string enclosed in quotes that defines the FileName for the .OPMRST file used to load the serialized state. If empty, CASENAME.OPMRST is used.	""
74	--load-step	An integer value that determines if the simulator should load the serialized state from OPM Flow's version of the restart file, and should be set to either a specific report step, or 0 to load the last stored report step. The default value of -1 does not load the data from the OPM Flow specific restart file. OPM Flow's version of the restart file, is written using the --save-step=N option.	-1
75	--local-domains-ordering-measure	A character string enclosed in quotes that defines the subdomain ordering measure. Allowed values are 'maxpressure', 'averagepressure' and 'residual'.	"maxpressure"
76	--local-domains-partition-well-neighbor-levels	An integer value. Number of neighbor levels around wells to include in the same domain during NLDD partitioning.	1

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
77	--local-domains-partitioning-imbalance	A real positive double precision value that sets the subdomain partitioning imbalance tolerance. 1.03 is 3 percent imbalance.	1.03
78	--local-domains-partitioning-method	A character string enclosed in quotes that defines the subdomain partitioning method. Allowed values are 'zoltan', 'simple', and the name of a partition file ending with '.partition'.	"zoltan"
79	--local-solve-approach	A character string enclosed in quotes that defines the local solve approach. Valid choices are 'jacobi' and 'gauss-seidel'.	"gauss-seidel"
80	--local-tolerance-scaling-cnv	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	0.1
81	--local-tolerance-scaling-mb	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	1
82	--local-well-solve-control-switching	A boolean value that allows (true) or disallows (false) control switching during local well solutions.	true
83	--matrix-add-well-contributions	A boolean value that if set to true explicitly specifies the influences of wells between cells in the Jacobian and preconditioner matrices.	false
84	--max-inner-iter-ms-wells	A positive integer value that defines the maximum number of inner iterations for multi-segment wells.	100
85	--max-inner-iter-wells	A positive integer value that defines the maximum number of inner iterations for standard wells.	50
86	--max-local-solve-iterations	A positive integer value that defines the maximum number of iterations for local solves with NLDD nonlinear solver.	20
87	--max-newton-iterations-with-inner-well-iterations	A positive integer that specifies the maximum Newton iterations with inner well iterations.	8
88	--max-pressure-change-ms-wells	A real positive value that defines the maximum relative pressure change for a single iteration of the multi-segment well model.	1e+06
89	--max-residual-allowed	A real positive value that sets the absolute maximum tolerance for residuals without cutting the time step size.	1e+07
90	--max-single-precision-days	A real positive value that sets the maximum	20.0

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		time step size where single precision floating point arithmetic can be used for solving the linear systems of equations.	
91	--max-temperature-change	A real positive value that stipulates the maximum absolute change of temperature in a single iteration.	5
92	--max-time-step-divisions	A positive integer that specifies the maximum number of divisions by two of the timestep size before the simulation bails out.	10
93	--max-time-step-size	A real positive value that defines the maximum size to which all time steps are limited to in seconds.	inf
94	--max-welleq-iter	A positive integer that defines the maximum number of iterations to determine the solution to the well equations.	30
95	--maximum-number-of-group-switches	An integer value. Maximum number of times a group can switch to the same control.	3
96	--maximum-number-of-well-switches	A positive integer values that stipulates the maximum number of times a well can switch to the same control.	3
97	--maximum-water-saturation	A real positive value that defines the maximum water saturation.	1.0
98	--metis-params	A character string. Configuration of Metis partitioner. You can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See http://glaros.dtc.umn.edu/gkhome/fetch/sw/metis/manual.pdf for available METIS options.	"default"
99	--milu-variant	A defined character string that specifies which variant of the modified ILU preconditioner ought to be used. Possible variants are: 1) ILU (default, plain ILU), 2) MILU_1 (lump diagonal with dropped row entries), 3) MILU_2 (lump diagonal with the sum of the absolute values of the dropped row entries), 4) MILU_3 (if diagonal is positive add sum of dropped row entries. Otherwise subtract them), 5) MILU_4 (if diagonal is positive add sum of dropped row entries. Otherwise do nothing).	"ILU"

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
100	--min-strict-cnv-iter	A positive integer that sets the minimum number of Newton iterations before relaxed tolerances are used for the CNV convergence criterion.	-1
101	--min-strict-mb-iter	An integer value defining the number of Newton iterations before relaxed tolerances can be used for the MB convergence criterion. Default -1 means that the relaxed tolerance is used when maximum number of Newton iterations are reached.	-1
102	--min-time-step-based-on-newton-iterations	A real positive value that sets the minimum time step size (in days for field and metric unit and hours for lab unit) can be reduced to based on Newton iteration counts.	0.0
103	--min-time-step-before-shutting-problematic-wells-in-days	A real positive value that sets the minimum time step size in days for which problematic wells are not shut. Time steps below this value will result in problematic wells being shut-in by the simulator.	0.01
104	--min-time-step-size	A real positive value that sets the minimum size to which all time steps are limited to in seconds.	0.0
105	--network-max-iterations	Replaced by --network-max-outer-iterations.	200
106	--network-max-outer-iterations	An integer value. Maximum outer number of iterations in the network solver before giving up.	10
107	--network-max-pressure-update-in-bars	A real value. Maximum pressure update in the inner network pressure update iterations.	5
108	--network-max-strict-iterations	Replaced by --network-max-strict-outer-iterations.	100
109	--network-max-strict-outer-iterations	An integer value. Maximum outer iterations in network solver before relaxing tolerance.	10
110	--network-max-sub-iterations	An integer value. Maximum number of sub-iterations to update network pressures (within a single well/group control update).	20
111	--newton-max-error	A real positive value that sets the maximum error tolerated by the Newton method to which does not cause an abort.	1e+100
112	--newton-max-iterations	A positive integer that defines the maximum number of Newton iterations per time step used by the simulator.	20
113	--newton-max-relax	A real positive value that sets the maximum	0.5

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		relaxation factor of a Newton iteration used by the simulator.	
I 14	--newton-min-iterations	A positive integer that sets the minimum number of Newton iterations per time step used by the simulator.	2
I 15	--network-pressure-update-damping-factor	A real value. Damping factor in the inner network pressure update iterations.	0.1
I 16	--newton-relaxation-type	A character string. The type of relaxation used by the Newton method. Valid options are: dampen or sor.	"dampen"
I 17	--newton-target-iterations	A positive integer that sets the 'optimum' number of Newton iterations per time step.	10
I 18	--newton-tolerance	A real positive value that sets the maximum raw error tolerated by the Newtonmethod for considering a solution to be converged.	0.01
I 19	--newton-verbose	A boolean value that specifies whether the Newton method should inform the user about its progress (true) or not (false).	true
I 20	--newton-write-convergence	A boolean value that specifies whether to write the convergence behaviour of the Newton method to a VTK file (true) or not (false).	false
I 21	--nldd-local-linear-solver	A defined character string. Configuration of NLDD local linear solver. Valid options are: ilu0 (default), dilu, cpr_quasiimpes and amg. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json'.	"ilu0"
I 22	--nldd-local-linear-solver-max-iter	An integer value. The maximum number of iterations of the NLDD local linear solver.	200
I 23	--nldd-local-linear-solver-reduction	A real value. The minimum reduction of the residual which the NLDD local linear solver must achieve.	0.01
I 24	--nldd-num-initial-newton-iter	A positive integer that sets the number of initial global Newton iterations when running the NLDD nonlinear solver.	1
I 25	--nonlinear-solver	A character string that specifies the nonlinear solver. Valid choices are newton or nldd.	"newton"
I 26	--num-local-domains	A positive integer that sets the number of local domains for NLDD nonlinear solver. Note this is an experimental feature in the current release 2023.10 that is expected to	0

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		be more complete and tested by the 2024.10 release.	
127	--num-overlap	An integer value. Numbers of layers overlap in parallel partition.	1
128	--num-pressure-points-equil	A positive integer that sets the number of pressure points (in each direction) in tables used for equilibration.	2000
129	--num-satfunc-consistency-sample-points	An integer value. Maximum number of reported failures for each individual saturation function consistency check.	5
130	--nupcol-group-rate-tolerance	A real value. Tolerance for acceptable changes in VREP/RAIN group rates.	0.001
131	--opencl-ilu-parallel	A boolean value that if set to true then parallelize the ILU decomposition and application on GPU, or not (false).	true
132	--opencl-platform-id	A positive integer that specifies the platform identification ("ID") for the openSolver. Use the "clinfo" command to determine valid IDs.	0
133	--output-dir	A character string that defines the directory to which OPM Flow is to write the commercial simulator compatible output files (restart, summary files etc.). The default value results in the files be written to the same directory as the input file.	""
134	--output-extra-convergence-info	A defined character string that specifies whether to provide additional convergence output to separate files for diagnostic purposes. The available options are: 1) "none" results in no extra output and overrides all other options. 2) "steps" writes out convergence information per time step, to a CASENAME.INFOSTEP file. The file is useful for identifying numerical issues. 3) "iterations" writes out non-linear convergence metrics, i.e., the MB and CNV values, per phase, for each non-linear iteration in each time step, to a CASENAME.INFOITER file. Options can be combined with commas, e.g. "steps,iterations" for multiple outputs. The default value of "none" prevents the two files from being written out, for better compatibility with the commercial simulator.	"none"
135	--output-interval	A positive integer that specifies the number	1

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		of report steps between two consecutive writes of restart data.	
136	--output-mode	A defined character string that defines the output to the *.PRT and *.DBG files: 1) "none": No output to the files. 2) "log" or false: Output logging information only. 3) "all" or true: Output everything. For example to just output logging information use: <pre>--output-mode="log".</pre> or <pre>--output-mode=false</pre>	"all"
137	--owner-cells-first	A boolean value that determines if the cells owned by a rank should be ordered before ghost/overlapping the cells (true), or not (false).	true
138	--parameter-file	A character string that defines the name of a parameter file which contains the simulator's set of run-time parameters, as listed in this table.	""
139	--parsing-strictness	Set strictness of parsing process. Available options are: 1) "normal": stop for critical errors, and for unsupported keywords that would change the simulator results if supported. 2) "high": stop for all errors, that is even for unsupported keywords that do not effect the results, for example <i>ECHO</i> and <i>NOECHO</i>. 3) "low": same as normal, except do not stop due to unsupported keywords that would change the simulator results if supported, and even if marked critical. Default: "normal"	"normal"
140	--partition-method	An integer value. Choose partitioning strategy: 0=simple, 1=Zoltan, 2=METIS, 3=Zoltan with all cells of well represented by one vertex.	3
141	--pre-solve-network	A boolean value. Pre solve and iterate the network model at start-up.	true
142	--predetermined-time-steps-file	A file with a list of predetermined time step sizes (one time step per line).	""
143	--pressure-max	A positive real value. Maximum absolute pressure.	1e+99

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
144	--pressure-min	A real value. Minimum absolute pressure.	-1e+99
145	--pressure-scale	A positive real value. Scaling of pressure primary variable.	1
146	--pri-var-oscilation-threshold	A real positive value that defines the threshold value for the primary variable switching conditions after its meaning has switched to hinder oscillations.	1e-05
147	--print-parameters	A positive integer value that request that the run time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
148	--project-saturations	A boolean value that determines if all the fluid saturations should be scaled to ensure the values are in the interval (0, 1), including runs that use solvents (true), or not (false).	false
149	--regularization-factor-wells	A real positive value that defines the "regularization factor" for wells.	100
150	--relaxed-linear-solver-reduction	A real positive value that defines the minimum reduction of the residual which the linear solver need to achieve for the solution to be accepted.	0.01
151	--relaxed-max-pv-fraction	A real positive value that defines The fraction of the pore volume of the reservoir where the volumetric error (CNV) might be violated during strict Newton iterations.	0.03
152	--relaxed-pressure-tol-msw	A real positive value that sets the relaxation tolerance for the multi-segment well pressure solution in Pascals.	10000
153	--relaxed-well-flow-tol	A real positive value that sets the relaxation tolerance for the well flow residual in reservoir cubic metres (rm3).	0.001
154	--restart-time	A real positive value that sets the simulation time at which a restart should be attempted [s].	-1e+35
155	--restart-writing-interval	A positive integer value that sets the frequency at which time steps are serialized to disk.	16777215
156	--save-file	A character string that specifies the FileName for .OPMRST file used for saving serialized state. If empty, CASENAME.OPMRST is used.	""

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
157	--save-step	A character string that determines if the simulator should save the serialized state of the OPM Flow simulator at one or more report steps to a special *.OPMRST file. This is in addition to the normal restart files written, and consumes significantly more space than the normal restart files, but restarting OPM Flow from this file using the --load-step option deviates less from the original run, compared to restarting using the normal restart features. The files produced are not compatible with other simulators, and also will not be compatible between different releases of OPM Flow The parameter should be set to one of the following: 1) "all" to save all report steps, 2) "x" to save every x'th step, or 3) "x" to save a specif time step. The default value of "" does not write anything to the OPM Flow specific restart file.	""
158	--scale-linear-system	A boolean value that switches on (true) or off (false) the scaling of linear system of equations according to the equation scale and primary variable types.	false
159	--sched-restart	A boolean value that determines for a restart run if the case should initialize wells and groups from the historical SCHEDULE section (true), or from the well and group data on the restart file (false). Note that the commercial simulator always uses data from the restart file (false).	false
160	--serial-partitioning	A boolean value that determines if partitioning for parallel runs on a single process (true), or not (false).	false
161	--shut-unsolvable-wells	A boolean value that determines if the simulator should shut unsolvable wells (true), or not (false).	true
162	--slave	A boolean value. Specify if the simulation is a slave simulation in a master-slave simulation. Note that reservoir coupling is not currently supported. This is a developer option that should not be used.	false
163	--solve-welleq-initially	A boolean value that determines if the simulator should fully solve the well equations before each iteration of the reservoir model (true), or not (false).	true

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		Note that the well equations are always added to the full system and solved until convergence.	
164	--solver-continue-on-convergence-failure	A boolean value that stipulates if the simulator should continue (true) instead of stopping (false) when the minimum solver time step is reached.	false
165	--solver-growth-factor	A real positive value that specifies the growth factor a time step can be increased by when recovering from one or more time step chops, subject to the maximum allowable time step size set by the --solver-max-time-step-in-days parameter. For example, if the current time step has converged at 5 days after a time step chop, and --solver-growth-factor is set to the default value of 2.0, then the next time step will be $2.0 * 5$ days, that is at 10 days.	2.0
166	--solver-max-growth	A real positive value that specifies the maximum growth factor a time step can be increased by after a report time step, subject to the maximum allowable time step size set by the --solver-max-time-step-in-days parameter. Thus, if the current time step has converged at 5 days after at a report step, and --solver-max-growth is set to the default value of 3.0, then the next time step will be $3.0 * 5$ days, that is at 15 days.	3.0
167	--solver-max-restarts	A positive integer that defines the number of allowed consecutive restarts (or time step chops) before the simulation is terminated.	10
168	--solver-max-time-step-in-days	A real positive double precision value that specifies the maximum allowed time step size in days.	365
169	--solver-min-time-step	A real positive double precision value that specifies the minimum size of a time step in days for field and metric units, and hours for laboratory units If a time step cannot converge without getting cut below this time step size the simulator will stop.	1e-12
170	--solver-restart-factor	A real positive double precision value that sets the time step chop factor of the time step after a convergence failure. For example, if the current non-convergent time step is 30 days and --solver-restart-factor is set to the default value, then the time step will be repeated using $0.33 * 30$	0.33

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		days as the time step, that is 9.9 days.	
171	--solver-verbosity	A positive integer that specifies the "chattiness" of the non-linear solver.	1
172	--strict-inner-iter-wells	A positive integer that specifies the number of inner well iterations with strict tolerance.	40
173	--strict-outer-iter-wells	A positive integer that specifies the number of Newton iterations for which wells are checked with strict tolerance.	6
174	--temperature-max	A real positive value that sets the maximum absolute temperature.	1e+09
175	--temperature-min	A real positive value that sets the minimum absolute temperature.	0
176	--threads-per-process	A positive integer value that stipulates the maximum number of threads to be instantiated per process ('-1' means 'automatic').	2
177	--time-step-after-event-in-days	A real double precision value that sets the maximum allowed time step after an event; for example, when a well is open or closed etc. The default value of -1 means that events to do effect the time stepping.	-1
178	--time-step-control	A defined character string that defines the time stepping control algorithm and is set to one of the following: 1) pid: PID controller based adaptive time step control as suggested by Turek and Kuzmin ³⁵⁷ . 2) pid+iteration: Use PID and linear iteration numbers to guide the time step. 3) pid+newtoniteration: Use PID and non-linear iterations numbers to guide the time step. 4) Hardcoded: Use time steps supplied by user via the time-step-control-filename parameter.	"pid+newtoniteration"
179	--time-step-control-decay-damping-factor	A real positive value that specifies the decay rate a time step can be decreased by after the number <i>target iterations</i> has been exceeded	1.0
180	--time-step-control-decay-rate	A real positive value that specifies the decay rate a time step size can be decreased by	0.75

³⁵⁷ Algebraic Flux Correction III. Incompressible Flow Problems. Uni Dortmund, Turek and Kuzmin, January, 2006 (DOI: 10.1007/3-540-27206-2_8)

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		after the number of target iterations has been exceeded	
181	--time-step-control-file-name	A character string enclosed in quotes that specifies a filename where time steps are specified. The default is the character string timesteps. For instance time steps can be generated by the ecl_summary application in libecl as per the following UNIX command line: <pre>path_to_libecl_applications/ ecl_summary DECK TIME > filename</pre> Where: DECK is the name of the data deck you want to get the time steps from, TIME tells the application to return the timing for the run, and "filename" is the name of the file the times are piped to.	"timesteps"
182	--time-step-control-growth-damping-factor	A real positive value that specifies the growth rate of the time step increase when the number of target iterations are undercut.	3.2
183	--time-step-control-growth-rate	A real positive value that specifies the growth rate of the time step size when the number of target iterations are undercut.	1.25
184	--time-step-control-safety-factor	A real value to be multiplied with the time step control tolerance to ensure that the target relative change is lower than the tolerance.	0.8
185	--time-step-control-target-iterations	A positive integer that defines the number of linear iterations which the time step control scheme should aim for (if applicable).	30
186	--time-step-control-target-newton-iterations	A positive integer that specifies the number of Newton iterations which the time step control scheme should aim for (if applicable).	8
187	--time-step-control-tolerance	A real double precision value that sets the tolerance for PID (only used with the pid and pid+ options defined by the --time-step-control option).	0.1
188	--time-step-verbosity	A positive integer that specifies the "chattiness" during the time integration.	1
189	--tolerance-cnv	A real positive double precision value that specifies the maximum non-linear tolerance	0.01

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		error. This is the local convergence tolerance (maximum of local saturation errors).	
190	--tolerance-cnv-energy	A real value. Local energy convergence tolerance (maximum of local energy errors).	0.01
191	--tolerance-cnv-energy-relaxed	A real value. Relaxed local energy convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0
192	--tolerance-cnv-relaxed	A real positive value that defines the relaxed local convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0
193	--tolerance-energy-balance	A real value. Tolerated energy balance error relative to (scaled) total energy present.	1e-07
194	--tolerance-energy-balance-relaxed	A real value. Relaxed tolerated energy balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
195	--tolerance-mb	A real positive double precision value that sets the maximum mass balance error; that is the tolerated mass balance error relative to total mass present.	1e-07
196	--tolerance-mb-relaxed	A positive real value. Relaxed tolerated mass balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
197	--tolerance-pressure-ms-wells	A real positive double precision value that specifies the tolerance for the pressure equations for multi-segment wells.	1000
198	--tolerance-well-control	A real positive double precision value that sets the maximum tolerance for the well control equations.	1e-07
199	--tolerance-wells	A real positive double precision value that defines the maximum non-linear error for the well equations.	1e-04
200	--update-equations-scaling	A boolean value that switches on (true) or off (false) the updating of the scaling factors for mass balance equations during the simulation.	false
201	--use-average-density-ms-wells	A boolean value that specifies whether to approximate segment densities by averaging over the segment and its outlet (true) or not (false).	false
202	--use-gmres	A boolean value that when set to true	false

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
		OPM Flow will use Generalized Minimal Residual ("GMRES") ³⁵⁸ and ³⁵⁹ solver instead of Biconjugate Gradient Stabilized ("BiCGSTAB") ³⁶⁰ and ³⁶¹ as the linear solver within the Newton iterations.	
203	--use-implicit-ipr	A boolean value. Compute implicit IPR for stability checks and stable solution search.	true
204	--use-multisegment-well	A boolean value that when set to true the simulator will use the well model for multi-segment wells instead of the one for single-segment wells.	true
205	--use-update-stabilization	A boolean value that switches on (true) or off (false) the stabilized Newton option, that attempts to detect and correct oscillations or stagnation during the Newton iterations. This option may improve convergence for some cases.	true
206	--water-only-threshold	A real positive value that defines the saturation threshold, for which cells with water saturations above or equal to this threshold are considered one-phase water only.	1.0
207	--zoltan-imbalance-tol	Replaced by --imbalance-tol.	1.1
208	--zoltan-params	A character string that specifies the configuration of the Zoltan partitioner. Valid options are: graph, hypergraph or scotch. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See https://sandialabs.github.io/Zoltan/ug_html/ug.html for available Zoltan options.	"graph"
209	--zoltan-phg-edge-size-threshold	A real value. Low-level threshold fraction in the range [0,1] controlling which hypergraph edge to omit. Used if --zoltan-params="graph" or if --zoltan-params="hypergraph".	0.35
Notes: 1) Cells colored green in the No. column indicate new command line parameters for this release. Similarly for			

³⁵⁸ Y. Saad, A flexible inner-outer preconditioned GMRES algorithm, *SIAM J. Sci. Statist. Comput.*, 14, (1993).

³⁵⁹ Y. Saad and M.H. Schultz, GMRES: A generalized minimum residual algorithm for solving nonsymmetric linear systems, *SIAM J. Sci. Statist. Comput.*, 7 (1986), pp. 856-869.

³⁶⁰ Van der Vorst, H. A. (1992). "Bi-CGSTAB: A Fast and Smoothly Converging Variant of Bi-CG for the Solution of Nonsymmetric Linear Systems". *SIAM J. Sci. Stat. Comput.* 13 (2): 631–644. doi:10.1137/0913035. hdl:10338.dmlcz/104566

³⁶¹ Sleijpen, G. L. G.; Fokkema, D. R. (November 1993). "BiCGstab(l) for linear equations involving unsymmetric matrices with complex spectrum" (PDF). *Electronic Transactions on Numerical Analysis*. Kent, OH: Kent State University. 1: 11–32. ISSN 1068-9613.

OPM Flow 2025-04 Command Line Options			
No.	Variable Name	Description	Default
the Default column, cells colored green indicate the default value has changed from the previous release. 2) Items shaded in gray are considered to be developer options that should be used with caution as the values associated with these command line parameters are subject to change. Cells colored orange in the No. column indicate new developer options for this release. 3) Cells colored red in the No. column indicate command line parameters that have either been deprecated or removed from this release. 4) VTK Graphics Command Line Parameters³⁶² set of commands are no longer listed via the basic help command line command: <pre>flow --help</pre> instead use: <pre>flow --help-all</pre> to get a list of supported command line Parameters. 5) The --enable-vtk-output option above, if set to “true” will write out the data as stipulated by the compile options. One may need to compile the source code to obtain the desired output. 6) As per all UNIX and LINUX based systems the input is case sensitive. 7) If a parameter file is used to set one of the above parameters, the leading “--” should not be incorporated in the parameter file.			

Table E.1: OPM Flow 2025-04 Command Line Options

³⁶² Virtualization ToolKit (“VTK”) files used to visualize the 3D results from the model using ParaView, an open-source, multi-platform general purpose data analysis and visualization application (<https://www.paraview.org/overview/>). For the commercial simulator’s compatible output files OPM’s ResInsight 3D visualization package can be used instead and is specifically tailored for OPM Flow and the commercial simulator.

E.2 RUNNING OPM FLOW 2024-10

The command line syntax for running OPM Flow is:

```
flow [OPTIONS] CASENAME
```

and typing the following command from your terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA.

OPM Flow accepts command line options to control various aspects of the simulator's run time behavior, as described in the table below. To give a command line option use "key=value" syntax, with no spaces around the equals sign. It is also possible to put multiple options together in a parameter file. To do so, put one option on each line of the file with ".param" as the extension and pass that filename as a command line parameter to OPM Flow using:

```
flow --parameter-file=CASENAME.param
```

to instruct OPM Flow to read the parameter file.

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
1	-h or --help	A character string that causes OPM Flow to print a help message that gives a brief description of the available command line parameters. This now only prints the active user facing command line options.	N/A
2	--help-all	Prints all the command line options included in the release, including obsolete, hidden and deprecated options.	N/A
3	--accelerator-mode	A defined character string that selects the linear solver, usage: '--accelerator-mode=[none cusparse opencl amgcl rocalution rocsparse]'. 	"none"
4	--action-parsing-strictness	A defined character string. Set strictness of parsing process for ActionX and PyAction. Available options are normal (do not apply keywords that have not been tested for ActionX or PyAction) and low (try to apply all keywords, beware: the simulation outcome might be incorrect).	"normal"
5	--allow-distributed-wells	Allow the perforations of a well to be distributed to interior of multiple processes.	false
6	--alternative-well-rate-init	Use alternative well rate initialization procedure.	true

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
7	--bda-device-id	Choose device ID for cusparseSolver or openclSolver; use 'nvidia-smi' or 'clinfo' commands to determine valid IDs.	0
8	--check-group-constraints-inner-well-iterations	A boolean value. Allow checking of group constraints during inner well iterations.	true
9	--continue-on-convergence-error	Continue with a non-converged solution instead of giving up if we encounter a time step size smaller than the minimum time step size.	false
10	--convergence-monitoring	A boolean value. Enable convergence monitoring.	false
11	--convergence-monitoring-cut-off	An integer value. Cut off limit for convergence monitoring.	6
12	--convergence-monitoring-decay-factor	A real value. Decay factor for convergence monitoring.	0.75
13	--cpr-reuse-interval	A positive integer that sets the reuse preconditioner interval. Used when --cpr-reuse-setup is set to 4, then the preconditioner will be fully recreated instead of reused every N linear solve, where N is this parameter.	30
14	--cpr-reuse-setup	A positive integer that defines if the CPR solver should re-use the AMG setup. Valid options are: 0 : Recreate the preconditioner for every linear solve. 1 : Recreate once every time step. 2 : Recreate if last linear solve took more than 10 iterations. 3 : Never recreate. 4 : Recreate every N linear solves, where N is the parameter --cpr-reuse-interval. Changed the default value from three to four.	4
15	--dbph-max-rel	A real positive value that sets the maximum relative change of the bottom-hole pressure in a single iteration.	1.0
16	--debug-emit-cell-partition	A boolean value that determines whether (true) or not (false) to emit cell partitions as a debugging aid.	false
17	--dp-max-rel	A real positive double precision value that sets the maximum allowed relative change of pressure per iteration.	0.3
18	--ds-max	A real positive double precision value that sets the maximum allowed change in	0.2

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		saturation per iteration.	
19	--dwell-fraction-max	A real positive value that sets the maximum allowed change in well's volume fraction per iteration.	0.2
20	--ecl-deck-file-name	A character string that defines the name of the OPM Flow input file which contains the simulator's ECLIPSE formatted input deck to be simulated. Note that there must be no spaces in the filename. If there are spaces, which can typically occur on Windows based systems, then the filename should be enclosed in quotes; thus instead of FILE NAME use "FILE NAME".	""
21	--ecl-output-double-precision	A boolean value that switches on (true) or off (false) double precision in restart files. Useful for 'perfect' restarts.	false
22	--ecl-output-interval	An integer value defining the number of report steps that ought to be skipped between two writes of restart files results.	-1
23	--edge-weights-method	A defined positive integer value that defines the edge-weighting strategy: 0 : For uniform, 1 : for trans, or 2 : for log(trans).	1
24	--enable-adaptive-time-stepping	A boolean value that turns on (true) or off (false) adaptive time stepping. If set to false the report time steps provided by the input deck are used.	true
25	--enable-async-ecl-output	A boolean value that sets the output mode to be asynchronous (true), letting the simulator continue computing the next time step while writing results to restart and summary files in the commercial simulator's format on a separate thread.	true
26	--enable-async-vtk-output	A boolean value that sets the output mode to be asynchronous (true), letting the simulator continue computing the next time step while writing results to the VTK output files on a separate thread.	true
27	--enable-drift-compensation	A boolean value that enables (true) or disables (false) partial compensation of systematic mass losses via the source term of the next time step.	false
28	--enable-dry-run	A boolean value that specifies if the simulation should actually run (true), or just check the input deck (false).	"auto"

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		This option is equivalent to activating the NOSIM keyword in the RUNSPEC section of the input deck (see section 5.3.101 NOSIM – Activate the No Simulation Mode for Data File Checking). The default value of “auto” will use whatever is stipulated in the input deck via the NOSIM keyword.	
29	--enable-ecl-output	A boolean value that specifies if the binary output files (restart, summary files, etc.) should be written in the commercial simulator's format (true), or OPM Flow's format (false).	true
30	--enable-esmry	A boolean value that switches on (true) or off (false) the output of SUMMARY vectors to the ESMRY file for fast loading of summary data.	false
31	--enable-gravity	A boolean value that switches on (true) or off (false) the use of the gravity correction for the pressure gradients.	true
32	--enable-grid-adaptation	A boolean value that enables (true) or disables (false) adaptive grid refinement/coarsening. Note that Eclipse grid output will be disabled if adaptive grid refinement/coarsening is enabled.	false
33	--enable-intensive-quantity-cache	A boolean value that switches on (true) or off (false) the caching of intensive quantities.	true
34	--enable-logging-fallout-warning	A boolean value that sets the developer option to see whether logging was on non-root processors (true). If set to true output will be appended to the *.DBG or *.PRT files.	false
35	--enable-opm-rst-file	A boolean value set to true or false to write OPM specific data sets to the commercial simulators restart file to enable restart of OPM Flow runs by OPM Flow (true), or not to write the data (false).	false
36	--enable-storage-cache	A boolean value that turns on (true) or off (false) storing previous storage terms and avoid re-calculating them.	true
37	--enable-terminal-output	A boolean value that turns on (true) or off (false) high-level information about the simulation's progress to the terminal	true
38	--enable-thermodynamic-hints	A boolean value that enables (true) or disables (false) thermodynamic hints.	false
39	--enable-tuning	A boolean value that instructs OPM Flow to read the time stepping parameters from	false

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		the <i>TUNING – Numerical Tuning Control</i> keyword in the input deck, if set to true. Note that only the first record of the <i>TUNING</i> keyword is processed.	
40	--enable-vtk-output	A boolean value that turns on (true) or off (false) a global switch for writing VTK files.	false
41	--enable-well-operability-check	A boolean value that enables (true) checking of a well's operating status, or disables (false) the checking.	true
42	--enable-well-operability-check-iter	A boolean value that enables (true) checking of a well's operating status during iterations, or disables (false) the checking during iterations.	false
43	--enable-write-all-solutions	A boolean value that turns on (true) or off (false) the writing of all solutions to disk instead of only the ones for the report steps.	false
44	--end-time	A real value. The simulation time at which the simulation is finished [s].	1e+100
45	--explicit-rock-compaction	A boolean value. Use pressure from end of the last time step when evaluating rock compaction (true) or not (false).	false
46	--external-partition	A quoted character string. Name of file from which to load an externally generated partitioning of the model's active cells for MPI distribution purposes. If empty, the built-in partitioning method will be employed.	""
47	--force-disable-fluid-in-place-output	A boolean value that instructs OPM Flow not to print the Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
48	--force-disable-resv-fluid-in-place-output	A boolean value that instructs OPM Flow to not to print the Reservoir Volume Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
49	--full-time-step-initially	A boolean value that instructs OPM Flow to always attempt to finish a report step using a single time step (true) or not (false).	false
50	--ignore-keywords	A character string that defines a list of keywords which should be ignored by OPM Flow. The keyword string should be enclosed in quotes and each keyword separated by a colon, that is ':'.	""

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
51	--ilu-fillin-level	A positive integer value that sets the fill in level for the ILU preconditioner.	0
52	--ilu-redblack	A boolean value that instructs OPM Flow to use red-black partitioning for the ILU preconditioner.	false
53	--ilu-relaxation	A real positive double precision value that sets the relaxation factor of the linear solver's ILU preconditioner.	0.9
54	--ilu-reorder-spheres	A boolean value that specifies OPM Flow to reorder the entries of the matrix in the red-black ILU preconditioner in spheres starting at an edge (true) or not (false). If false the original ordering is preserved in each color. Otherwise try to ensure D4 ordering (in a 2D structured grid, the diagonal elements are consecutive).	false
55	--imbalance-tol	A real value. Tolerable imbalance of the load balancing.	1.1
56	--initial-time-step-in-days	A real double precision value that sets the size of initial time step in days.	1.0
57	--initial-time-step-size	A real double precision value that sets the size of initial time step in seconds.	86400
58	--inj-mult-damp-mult	A real value. Injection multiplier dampening factor (dampening multiplied by this each time oscillation is detected).	0.9
59	--inj-mult-min-damp-factor	A real value. Minimum injection multiplier dampening factor (maximum dampening level).	0.05
60	--inj-mult-osc-threshold	A real value. Injection multiplier oscillation threshold (used for multiplier dampening).	0.1
61	--input-skip-mode	A defined character string. Set compatibility mode for the SKIP100 / SKIP300 keywords. Options are: 100 (skip SKIP100 ..ENDSKIP, keep SKIP300 ..ENDSKIP), 300 (skip SKIP300 ..ENDSKIP, keep SKIP100 ..ENDSKIP) or all (skip both SKIP100 ..ENDSKIP and SKIP300 ..ENDSKIP).	"100"
62	--linear-solver	A defined quoted character string that sets the configuration of the linear solver; valid values are: 1) cprw ^{363 364 365} (default),	"cprw"

³⁶³ Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

³⁶⁴ R. Scheichl, M. Roland, J. Wendebourg, Decoupling and block preconditioning for sedimentary basin simulations, Computational Geosciences 7 (2003) 295-318.

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		2) ilu0, 3) dilu, 4) cpr (an alias for cprw), 5) cpr_quasiimpes, 6) cpr_trueimpes, 7) cpr_trueimpesanalytic, 8) amg³⁶⁶, 9) hybrid (experimental), or 10) a file name that has the extension ".json", that contains the linear solver configuration parameters. Option (1) extends the existing Constrained Pressure Residual ("CPR") preconditioner to include wells. This option can also be invoked via the <i>CPR</i> keyword in the <i>RUNSPEC</i> section; however, the command line parameter takes precedence. For option (10) one enters a character string enclosed in quotes that defines the filename of a JSON configuration file for the flexible linear solver system. In this case the file extension should be '.json'. Note that the *.PRT file contains the "Property tree for the linear solver" listing, which is the JSON specification of the current case, and can be used to configure a user specific linear solver JSON file.	
63	--linear-solver-ignore-convergence-failure	A boolean value that if set to true convergences failures in the linear solver are ignored. This option should be used with care, as the results might be unreliable.	false
64	--linear-solver-max-iter	A positive integer value that defines the maximum number of linear iterations.	200
65	--linear-solver-print-json-definition	Write the JSON definition of the linear solver setup to the DBG file.	true
66	--linear-solver-reduction	A real positive double precision value that sets the minimum reduction of the residual for the linear solver. The linear solver convergences when the residual is reduced sufficiently. The simulator now overrides the default reduction to be 0.005 instead of 0.01, if the linear solver has been set to one of the cpr	0.01

³⁶⁵ Klemetsdal, Ø.S., Møyner, O. & Lie, KA. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. *Comput Geosci* 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

³⁶⁶ M. Blatt, A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients, Ph.D. thesis, Ruprecht-Karls-Universität at Heidelberg (2010).

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		options, in a similar manner as how the default maximum number of linear iterations for the cpr and cprw options are changed to 20 instead of 200, unless specified by the command line option by the user.	
67	--linear-solver-restart	A positive integer value that sets the number of iterations after which GMRES is restarted.	40
68	--linear-solver-verbosity	A positive integer value that defines the output from linear solver: 0 : No extra output, 1 : output per solution iteration, or 2 : output per iteration.	0
69	--load-file	A character string enclosed in quotes that defines the FileName for the .OPMRST file used to load the serialized state. If empty, CASENAME.OPMRST is used.	""
70	--load-step	An integer value that determines if the simulator should load the serialized state from OPM Flow's version of the restart file, and should be set to either a specific report step, or 0 to load the last stored report step. The default value of -1 does not load the data from the OPM Flow specific restart file. OPM Flow's version of the restart file, is written using the --save-step=N option.	-1
71	--local-domains-ordering-measure	A character string enclosed in quotes that defines the subdomain ordering measure. Allowed values are 'maxpressure', 'averagepressure' and 'residual'.	"maxpressure"
72	--local-domains-partitioning-imbalance	A real positive double precision value that sets the subdomain partitioning imbalance tolerance. 1.03 is 3 percent imbalance.	1.03
73	--local-domains-partitioning-method	A character string enclosed in quotes that defines the subdomain partitioning method. Allowed values are 'zoltan', 'simple', and the name of a partition file ending with '.partition'.	"zoltan"
74	--local-solve-approach	A character string enclosed in quotes that defines the local solve approach. Valid choices are 'jacobi' and 'gauss-seidel'.	"gauss-seidel"
75	--local-tolerance-scaling-cnv	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	0.1

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
76	--local-tolerance-scaling-mb	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	1
77	--local-well-solve-control-switching	A boolean value that allows (true) or disallows (false) control switching during local well solutions.	true
78	--matrix-add-well-contributions	A boolean value that if set to true explicitly specifies the influences of wells between cells in the Jacobian and preconditioner matrices.	false
79	--max-inner-iter-ms-wells	A positive integer value that defines the maximum number of inner iterations for multi-segment wells.	100
80	--max-inner-iter-wells	A positive integer value that defines the maximum number of inner iterations for standard wells.	50
81	--max-local-solve-iterations	A positive integer value that defines the maximum number of iterations for local solves with NLDD nonlinear solver.	20
82	--max-newton-iterations-with-inner-well-iterations	A positive integer that specifies the maximum Newton iterations with inner well iterations.	8
83	--max-pressure-change-ms-wells	A real positive value that defines the maximum relative pressure change for a single iteration of the multi-segment well model.	1e+06
84	--max-residual-allowed	A real positive value that sets the absolute maximum tolerance for residuals without cutting the time step size.	1e+07
85	--max-single-precision-days	A real positive value that sets the maximum time step size where single precision floating point arithmetic can be used for solving the linear systems of equations.	20.0
86	--max-temperature-change	A real positive value that stipulates the maximum absolute change of temperature in a single iteration.	5
87	--max-time-step-divisions	A positive integer that specifies the maximum number of divisions by two of the timestep size before the simulation bails out.	10
88	--max-time-step-size	A real positive value that defines the maximum size to which all time steps are limited to in seconds.	inf
89	--max-welleq-iter	A positive integer that defines the maximum number of iterations to	30

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		determine the solution to the well equations.	
90	--maximum-number-of-well-switches	A positive integer values that stipulates the maximum number of times a well can switch to the same control.	3
91	--maximum-water-saturation	A real positive value that defines the maximum water saturation.	1.0
92	--metis-params	A character string. Configuration of Metis partitioner. You can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See http://glaros.dtc.umn.edu/gkhome/fetch/sw/metis/manual.pdf for available METIS options.	"default"
93	--milu-variant	A defined character string that specifies which variant of the modified ILU preconditioner ought to be used. Possible variants are: 1) ILU (default, plain ILU), 2) MILU_1 (lump diagonal with dropped row entries), 3) MILU_2 (lump diagonal with the sum of the absolute values of the dropped row entries), 4) MILU_3 (if diagonal is positive add sum of dropped row entries. Otherwise subtract them), 5) MILU_4 (if diagonal is positive add sum of dropped row entries. Otherwise do nothing).	"ILU"
94	--min-strict-cnv-iter	A positive integer that sets the minimum number of Newton iterations before relaxed tolerances are used for the CNV convergence criterion.	-1
95	--min-strict-mb-iter	An integer value defining the number of Newton iterations before relaxed tolerances can be used for the MB convergence criterion. Default -1 means that the relaxed tolerance is used when maximum number of Newton iterations are reached.	-1
96	--min-time-step-based-on-newton-iterations	A real positive value that sets the minimum time step size (in days for field and metric unit and hours for lab unit) can be reduced to based on Newton iteration counts.	0.0
97	--min-time-step-before-shutting-problematic-wells-in-days	A real positive value that sets the minimum time step size in days for which problematic wells are not shut. Time steps below this value will result in problematic wells being	0.01

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		shut-in by the simulator.	
98	--min-time-step-size	A real positive value that sets the minimum size to which all time steps are limited to in seconds.	0.0
99	--network-max-iterations	A positive integer that sets the maximum number of iterations in the network solver before giving up.	200
100	--network-max-strict-iterations	A positive integer that sets the maximum iterations in network solver before relaxing tolerance.	100
101	--newton-max-error	A real positive value that sets the maximum error tolerated by the Newton method to which does not cause an abort.	1e+100
102	--newton-max-iterations	A positive integer that defines the maximum number of Newton iterations per time step used by the simulator.	20
103	--newton-max-relax	A real positive value that sets the maximum relaxation factor of a Newton iteration used by the simulator.	0.5
104	--newton-min-iterations	A positive integer that sets the minimum number of Newton iterations per time step used by the simulator.	2
105	--newton-relaxation-type	A character string. The type of relaxation used by the Newton method. Valid options are: dampen or sor.	"dampen"
106	--newton-target-iterations	A positive integer that sets the 'optimum' number of Newton iterations per time step.	10
107	--newton-tolerance	A real positive value that sets the maximum raw error tolerated by the Newtonmethod for considering a solution to be converged.	0.01
108	--newton-verbose	A boolean value that specifies whether the Newton method should inform the user about its progress (true) or not (false).	true
109	--newton-write-convergence	A boolean value that specifies whether to write the convergence behaviour of the Newton method to a VTK file (true) or not (false).	false
110	--nlld-local-linear-solver	A defined character string. Configuration of NLDD local linear solver. Valid options are: ilu0 (default), dilu, cpr_quasiimpes and amg. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json'.	"ilu0"

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
I 11	--nlld-local-linear-solver-max-iter	An integer value. The maximum number of iterations of the NLDD local linear solver.	200
I 12	--nlld-local-linear-solver-reduction	A real value. The minimum reduction of the residual which the NLDD local linear solver must achieve.	0.01
I 13	--nlld-num-initial-newton-iter	A positive integer that sets the number of initial global Newton iterations when running the NLDD nonlinear solver.	1
I 14	--nonlinear-solver	A character string that specifies the nonlinear solver. Valid choices are newton or nlld.	"newton"
I 15	--num-local-domains	A positive integer that sets the number of local domains for NLDD nonlinear solver. Note this is an experimental feature in the current release 2023.10 that is expected to be more complete and tested by the 2024.10 release.	0
I 16	--num-pressure-points-equil	A positive integer that sets the number of pressure points (in each direction) in tables used for equilibration.	2000
I 17	--opencl-ilu-parallel	A boolean value that if set to true then parallelize the ILU decomposition and application on GPU, or not (false).	true
I 18	--opencl-platform-id	A positive integer that specifies the platform identification ("ID") for the openSolver. Use the "clinfo" command to determine valid IDs.	0
I 19	--output-dir	A character string that defines the directory to which OPM Flow is to write the commercial simulator compatible output files (restart, summary files etc.). The default value results in the files be written to the same directory as the input file.	""
I 20	--output-extra-convergence-info	A defined character string that specifies whether to provide additional convergence output to separate files for diagnostic purposes. The available options are: 1) "none" results in no extra output and overrides all other options. 2) "steps" writes out convergence information per time step, to a CASENAME.INFOSTEP file. The file is useful for identifying numerical issues. 3) "iterations" writes out non-linear convergence metrics, i.e., the MB and CNV values, per phase, for each non-linear iteration in each time step, to a	"none"

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		CASENAME.INFOITER file. Options can be combined with commas, e.g."steps,iterations" for multiple outputs. The default value of "none" prevents the two files from being written out, for better compatibility with the commercial simulator.	
121	--output-interval	A positive integer that specifies the number of report steps between two consecutive writes of restart data.	1
122	--output-mode	A defined character string that defines the output to the *.PRT and *.DBG files: 1) "none": No output to the files. 2) "log" or false: Output logging information only. 3) "all" or true: Output everything. For example to just output logging information use: --output-mode="log". or --output-mode=false	"all"
123	--owner-cells-first	A boolean value that determines if the cells owned by a rank should be ordered before ghost/overlapping the cells (true), or not (false).	true
124	--parameter-file	A character string that defines the name of a parameter file which contains the simulator's set of run-time parameters, as listed in this table.	""
125	--parsing-strictness	Set strictness of parsing process. Available options are: 1) "normal": stop for critical errors, and for unsupported keywords that would change the simulator results if supported. 2) "high": stop for all errors, that is even for unsupported keywords that do not effect the results, for example <i>ECHO</i> and <i>NOECHO</i>. 3) "low": same as normal, except do not stop due to unsupported keywords that would change the simulator results if supported, and even if marked critical. Default: "normal"	"normal"
126	--partition-method	An integer value. Choose partitioning strategy: 0=simple, 1=Zoltan, 2=METIS.	1

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
127	--predetermined-time-steps-file	A file with a list of predetermined time step sizes (one time step per line).	""
128	--pressure-max	A positive real value. Maximum absolute pressure.	1e+99
129	--pressure-min	A real value. Minimum absolute pressure.	-1e+99
130	--pressure-scale	A positive real value. Scaling of pressure primary variable.	1
131	--pri-var-oscillation-threshold	A real positive value that defines the threshold value for the primary variable switching conditions after its meaning has switched to hinder oscillations.	1e-05
132	--print-parameters	A positive integer value that request that the run time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
133	--print-properties	A positive integer value that request that the compile time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
134	--project-saturations	A boolean value that determines if all the fluid saturations should be scaled to ensure the values are in the interval (0, 1), including runs that use solvents (true), or not (false).	false
135	--regularization-factor-wells	A real positive value that defines the "regularization factor" for wells.	100
136	--relaxed-linear-solver-reduction	A real positive value that defines the minimum reduction of the residual which the linear solver need to achieve for the solution to be accepted.	0.01
137	--relaxed-max-pv-fraction	A real positive value that defines The fraction of the pore volume of the reservoir where the volumetric error (CNV) might be violated during strict Newton iterations.	0.03
138	--relaxed-pressure-tol-msw	A real positive value that sets the relaxation tolerance for the multi-segment well <u>pressure</u> solution in Pascals.	10000
139	--relaxed-well-flow-tol	A real positive value that sets the relaxation tolerance for the well flow	0.001

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		residual in reservoir cubic metres (rm3).	
140	--restart-time	A real positive value that sets the simulation time at which a restart should be attempted [s].	-1e+35
141	--restart-writing-interval	A positive integer value that sets the frequency at which time steps are serialized to disk.	16777215
142	--save-file	A character string that specifies the FileName for .OPMRST file used for saving serialized state. If empty, CASENAME.OPMRST is used.	""
143	--save-step	A character string that determines if the simulator should save the serialized state of the OPM Flow simulator at one or more report steps to a special *.OPMRST file. This is in addition to the normal restart files written, and consumes significantly more space than the normal restart files, but restarting OPM Flow from this file using the <i>--load-step</i> option deviates less from the original run, compared to restarting using the normal restart features. The files produced are not compatible with other simulators, and also will not be compatible between different releases of OPM Flow The parameter should be set to one of the following: 1) "all" to save all report steps, 2) "x" to save every x'th step, or 3) "x" to save a specif time step. The default value of "" does not write anything to the OPM Flow specific restart file.	""
144	--scale-linear-system	A boolean value that switches on (true) or off (false) the scaling of linear system of equations according to the equation scale and primary variable types.	false
145	--sched-restart	A boolean value that determines for a restart run if the case should initialize wells and groups from the historical SCHEDULE section (true), or from the well and group data on the restart file (false). Note that the commercial simulator always uses data from the restart file (false).	false
146	--serial-partitioning	A boolean value that determines if partitioning for parallel runs on a single process (true), or not (false).	false

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
147	--shut-unsolvable-wells	A boolean value that determines if the simulator should shut unsolvable wells (true), or not (false).	true
148	--solve-welleq-initially	A boolean value that determines if the simulator should fully solve the well equations before each iteration of the reservoir model (true), or not (false). Note that the well equations are always added to the full system and solved until convergence.	true
149	--solver-continue-on-convergence-failure	A boolean value that stipulates if the simulator should continue (true) instead of stopping (false) when the minimum solver time step is reached.	false
150	--solver-growth-factor	A real positive value that specifies the growth factor a time step can be increased by when recovering from one or more time step chops, subject to the maximum allowable time step size set by the <code>--solver-max-time-step-in-days</code> parameter. For example, if the current time step has converged at 5 days after a time step chop, and <code>--solver-growth-factor</code> is set to the default value of 2.0, then the next time step will be $2.0 * 5$ days, that is at 10 days.	2.0
151	--solver-max-growth	A real positive value that specifies the maximum growth factor a time step can be increased by after a report time step, subject to the maximum allowable time step size set by the <code>--solver-max-time-step-in-days</code> parameter. Thus, if the current time step has converged at 5 days after at a report step, and <code>--solver-max-growth</code> is set to the default value of 3.0, then the next time step will be $3.0 * 5$ days, that is at 15 days.	3.0
152	--solver-max-restarts	A positive integer that defines the number of allowed consecutive restarts (or time step chops) before the simulation is terminated.	10
153	--solver-max-time-step-in-days	A real positive double precision value that specifies the maximum allowed time step size in days.	365
154	--solver-min-time-step	A real positive double precision value that specifies the minimum size of a time step in days for field and metric units, and hours for laboratory units If a time step cannot converge without getting cut below this time step size the simulator will stop.	1e-12

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
155	--solver-restart-factor	A real positive double precision value that sets the time step chop factor of the time step after a convergence failure. For example, if the current non-convergent time step is 30 days and --solver-restart-factor is set to the default value, then the time step will be repeated using $0.33 * 30$ days as the time step, that is 9.9 days.	0.33
156	--solver-verbosity	A positive integer that specifies the "chattiness" of the non-linear solver.	1
157	--strict-inner-iter-wells	A positive integer that specifies the number of inner well iterations with strict tolerance.	40
158	--strict-outer-iter-wells	A positive integer that specifies the number of Newton iterations for which wells are checked with strict tolerance.	6
159	--temperature-max	A real positive value that sets the maximum absolute temperature.	1e+09
160	--temperature-min	A real positive value that sets the minimum absolute temperature.	0
161	--threads-per-process	A positive integer value that stipulates the maximum number of threads to be instantiated per process ('-1' means 'automatic').	-1
162	--time-step-after-event-in-days	A real double precision value that sets the maximum allowed time step after an event; for example, when a well is open or closed etc. The default value of -1 means that events to do effect the time stepping.	-1
163	--time-step-control	A defined character string that defines the time stepping control algorithm and is set to one of the following: 1) pid: PID controller based adaptive time step control as suggested by Turek and Kuzmin ³⁶⁷ . 2) pid+iteration: Use PID and linear iteration numbers to guide the time step. 3) pid+newtoniteration: Use PID and non-linear iterations numbers to guide the time step. 4) Hardcoded: Use time steps supplied by user via the time-step-control-filename parameter.	"pid+newtoniteration"

³⁶⁷ Algebraic Flux Correction III. Incompressible Flow Problems. Uni Dortmund, Turek and Kuzmin, January, 2006 (DOI: 10.1007/3-540-27206-2_8)

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
164	--time-step-control-decay-damping-factor	A real positive value that specifies the decay rate a time step can be decreased by after the number target iterations has been exceeded	1.0
165	--time-step-control-decay-rate	A real positive value that specifies the decay rate a time step size can be decreased by after the number of target iterations has been exceeded	0.75
166	--time-step-control-file-name	A character string enclosed in quotes that specifies a filename where time steps are specified. The default is the character string timesteps. For instance time steps can be generated by the ecl_summary application in libecl as per the following UNIX command line: <pre>path_to_libecl_applications/ ecl_summary DECK TIME > filename</pre> Where: - DECK is the name of the data deck you want to get the time steps from, - TIME tells the application to return the timing for the run, and "filename" is the name of the file the times are piped to.	"timesteps"
167	--time-step-control-growth-damping-factor	A real positive value that specifies the growth rate of the <u>time step</u> increase when the number of target iterations are undercut.	3.2
168	--time-step-control-growth-rate	A real positive value that specifies the growth rate of the <u>time step size</u> when the number of target iterations are undercut.	1.25
169	--time-step-control-target-iterations	A positive integer that defines the number of linear iterations which the time step control scheme should aim for (if applicable).	30
170	--time-step-control-target-newton-iterations	A positive integer that specifies the number of Newton iterations which the time step control scheme should aim for (if applicable).	8
171	--time-step-control-tolerance	A real double precision value that sets the tolerance for PID (only used with the pid and pid+ options defined by the --time-step-control option).	0.1
172	--time-step-verbosity	A positive integer that specifies the "chattiness" during the time integration.	1

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
173	--tolerance-cnv	A real positive double precision value that specifies the maximum non-linear tolerance error. This is the local convergence tolerance (maximum of local saturation errors).	0.01
174	--tolerance-cnv-energy	A real value. Local energy convergence tolerance (maximum of local energy errors).	0.01
175	--tolerance-cnv-energy-relaxed	A real value. Relaxed local energy convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0
176	--tolerance-cnv-relaxed	A real positive value that defines the relaxed local convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0
177	--tolerance-energy-balance	A real value. Tolerated energy balance error relative to (scaled) total energy present.	1e-07
178	--tolerance-energy-balance-relaxed	A real value. Relaxed tolerated energy balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
179	--tolerance-mb	A real positive double precision value that sets the maximum mass balance error, that is the tolerated mass balance error relative to total mass present.	1e-07
180	--tolerance-mb-relaxed	A positive real value. Relaxed tolerated mass balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
180	--tolerance-pressure-ms-wells	A real positive double precision value that specifies the tolerance for the pressure equations for multi-segment wells.	1000
181	--tolerance-well-control	A real positive double precision value that sets the maximum tolerance for the well control equations.	1e-07
182	--tolerance-wells	A real positive double precision value that defines the maximum non-linear error for the well equations.	1e-04
183	-update-equations-scaling	A boolean value that switches on (true) or off (false) the updating of the scaling factors for mass balance equations during the simulation.	false
184	--use-average-density-ms-wells	A boolean value that specifies whether to approximate segment densities by	false

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
		averaging over the segment and its outlet (true) or not (false).	
185	--use-gmres	A boolean value that when set to true OPM Flow will use Generalized Minimal Residual ("GMRES") ³⁶⁸ and ³⁶⁹ solver instead of Biconjugate Gradient Stabilized ("BiCGSTAB") ³⁷⁰ and ³⁷¹ as the linear solver within the Newton iterations.	false
186	--use-implicit-ipr	A boolean value. Compute implicit IPR for stability checks and stable solution search.	true
187	--use-multisegment-well	A boolean value that when set to true the simulator will use the well model for multi-segment wells instead of the one for single-segment wells.	true
188	--use-update-stabilization	A boolean value that switches on (true) or off (false) the stabilized Newton option, that attempts to detect and correct oscillations or stagnation during the Newton iterations. This option may improve convergence for some cases.	true
189	--water-only-threshold	A real positive value that defines the saturation threshold, for which cells with water saturations above or equal to this threshold are considered one-phase water only.	1.0
190	--zoltan-imbalance-tol	A real value. Tolerable imbalance of the loadbalancing provided by Zoltan. This command line option has been deprecated; --imbalance-tol should be used instead.	1.1
191	--zoltan-params	A character string that specifies the configuration of the Zoltan partitioner. Valid options are: graph, hypergraph or scotch. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See https://sandialabs.github.io/Zoltan/ug_html/ug.html for available Zoltan options.	"graph"
Notes:			

³⁶⁸ Y. Saad, A flexible inner-outer preconditioned GMRES algorithm, *SIAM J. Sci. Statist. Comput.*, 14, (1993).

³⁶⁹ Y. Saad and M.H. Schultz, GMRES: A generalized minimum residual algorithm for solving nonsymmetric linear systems, *SIAM J. Sci. Statist. Comput.*, 7 (1986), pp. 856{869.

³⁷⁰ Van der Vorst, H. A. (1992). "Bi-CGSTAB: A Fast and Smoothly Converging Variant of Bi-CG for the Solution of Nonsymmetric Linear Systems". *SIAM J. Sci. Stat. Comput.* 13 (2): 631–644. doi:10.1137/0913035. hdl:10338.dmlcz/104566

³⁷¹ Sleijpen, G. L. G.; Fokkema, D. R. (November 1993). "BiCGstab(l) for linear equations involving unsymmetric matrices with complex spectrum" (PDF). *Electronic Transactions on Numerical Analysis*. Kent, OH: Kent State University. 1: 11–32. ISSN 1068-9613.

OPM Flow 2024-10 Command Line Options			
No.	Variable Name	Description	Default
1)		Cells colored green in the No. column are new command line parameters for this release. Similarly for the Default column, cells colored green indicate the default value has changed from the previous release.	
2)		Items shaded in gray are considered to be developer options that should be used with caution as the values associated with these command line parameters are subject to change.	
3)		VTK Graphics Command Line Parameters ³⁷² set of commands are no longer listed via the basic help command line command: flow --help instead use: flow --help-all to get a list of supported command line Parameters.	
4)		The --enable-vtk-output option above, if set to "true" will write out the data as stipulated by the compile options. One may need to compile the source code to obtain the desired output.	
5)		As per all UNIX and LINUX based system the input is case dependent.	
6)		If a parameter file is used to set one of the above parameters, the leading "--" should not be incorporated in the parameter file.	

Table E.2: OPM Flow 2024-10 Command Line Options

³⁷²
Virtualization ToolKit ("VTK") files used to visualize the 3D results from the model using ParaView, an open-source, multi-platform general purpose data analysis and visualization application (<https://www.paraview.org/overview/>). For the commercial simulator's compatible output files OPM's ResInsight 3D visualization package can be used instead and is specifically tailored for OPM Flow and the commercial simulator.

E.3 RUNNING OPM FLOW 2024-04

The command line syntax for running OPM Flow is:

```
flow [OPTIONS] CASENAME
```

and typing the following command from your terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA.

OPM Flow accepts command line options to control various aspects of the simulator's run time behavior, as described in the table below. To give a command line option use "key=value" syntax, with no spaces around the equals sign. It is also possible to put multiple options together in a parameter file. To do so, put one option on each line of the file with ".param" as the extension and pass that filename as a command line parameter to OPM Flow using:

```
flow --parameter-file=CASENAME.param
```

to instruct OPM Flow to read the parameter file.

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
1	-h or --help	A character string that causes OPM Flow to print a help message that gives a brief description of the available command line parameters. This now only prints the active user facing command line options.	N/A
2	--help-all	Prints all the command line options included in the release, including obsolete, hidden and deprecated options.	N/A
3	--accelerator-mode	A defined character string that selects the linear solver, usage: '--accelerator-mode=[none cusparse opencl amgcl rocalution rocsparse]'. 	"none"
4	--allow-distributed-wells	Allow the perforations of a well to be distributed to interior of multiple processes.	false
5	--alternative-well-rate-init	Use alternative well rate initialization procedure.	true
6	--bda-device-id	Choose device ID for cusparseSolver or openclSolver, use 'nvidia-smi' or 'clinfo' commands to determine valid IDs.	0
7	--continue-on-convergence-error	Continue with a non-converged solution instead of giving up if we encounter a time step size smaller than the minimum time step size.	false

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
8	--cpr-reuse-interval	A positive integer that sets the reuse preconditioner interval. Used when <i>--cpr-reuse-setup</i> is set to 4, then the preconditioner will be fully recreated instead of reused every N linear solve, where N is this parameter.	30
9	--cpr-reuse-setup	A positive integer that defines if the <i>CPR</i> solver should re-use the AMG setup. Valid options are: 0 : Recreate the preconditioner for every linear solve. 1 : Recreate once every time step. 2 : Recreate if last linear solve took more than 10 iterations. 3 : Never recreate. 4 : Recreate every N linear solves, where N is the parameter <i>--cpr-reuse-interval</i> . Changed the default value from three to four.	4
10	--dbph-max-rel	A real positive value that sets the maximum relative change of the bottom-hole pressure in a single iteration.	1.0
11	--dp-max-rel	A real positive double precision value that sets the maximum allowed relative change of pressure per iteration.	0.3
12	--ds-max	A real positive double precision value that sets the maximum allowed change in saturation per iteration.	0.2
13	--dwell-fraction-max	A real positive value that sets the maximum allowed change in well's volume fraction per iteration.	0.2
14	--debug-emit-cell-partition	A boolean value that determines whether (true) or not (false) to emit cell partitions as a debugging aid.	false
15	--ecl-deck-file-name	A character string that defines the name of the OPM Flow input file which contains the simulator's ECLIPSE formatted input deck to be simulated. Note that there must be no spaces in the filename. If there are spaces, which can typically occur on Windows based systems, then the filename should be enclosed in quotes; thus instead of FILE NAME use "FILE NAME".	""
16	--enable-drift-compensation	A boolean value that enables (true) or disables (false) partial compensation of systematic mass losses via the source term	true

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		of the next time step.	
17	--ecl-enable-tuning	Honor some aspects of the TUNING keyword from the ECL deck.	false
18	--ecl-newton-relaxed-tolerance	The maximum error which the volumetric residual may exhibit if it is in a 'relaxed' region during a strict iteration.	1e+09
19	--ecl-newton-relaxed-volume-fraction	The fraction of the pore volume of the reservoir where the volumetric error may be violated during strict Newton iterations.	0.03
20	--ecl-newton-strict-iterations	The number of Newton iterations where the volumetric error is considered.	8
21	--ecl-newton-sum-tolerance	The maximum error tolerated by the Newton method for considering a solution to be converged.	0.0001
22	--ecl-newton-sum-tolerance-exponent	The exponent used to scale the sum tolerance by the total pore volume of the reservoir.	0.333333
23	--ecl-output-double-precision	A boolean value that switches on (true) or off (false) double precision in restart files. Useful for 'perfect' restarts.	false
24	--ecl-output-interval	An integer value defining the number of report steps that ought to be skipped between two writes of restart files results.	-1
25	--edge-weights-method	A defined positive integer value that defines the edge-weighting strategy: 0 : For uniform, 1 : for trans, or 2 : for log(trans).	1
26	--enable-adaptive-time-stepping	A boolean value that turns on (true) or off (false) adaptive time stepping. If set to false the report time steps provided by the input deck are used.	true
27	--enable-async-ecl-output	A boolean value that sets the output mode to be asynchronous (true), letting the simulator continue computing the next time step while writing results to restart and summary files in the commercial simulator's format on a separate thread.	true
28	--enable-async-vtk-output	A boolean value that sets the output mode to be asynchronous (true), letting the simulator continue computing the next time step while writing results to the VTK output files on a separate thread.	true
29	--enable-dry-run	A boolean value that specifies if the simulation should actually run (true), or just	"auto"

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		check the input deck (false). This option is equivalent to activating the NOSIM keyword in the RUNSPEC section of the input deck (see section 5.3.101 NOSIM – Activate the No Simulation Mode for Data File Checking). The default value of “auto” will use whatever is stipulated in the input deck via the NOSIM keyword.	
30	--enable-ecl-output	A boolean value that specifies if the binary output files (restart, summary files, etc.) should be written in the commercial simulator's format (true), or OPM Flow's format (false).	true
31	--enable-esmry	A boolean value that switches on (true) or off (false) the output of SUMMARY vectors to the ESMRY file for fast loading of summary data.	false
32	--enable-gravity	A boolean value that switches on (true) or off (false) the use of the gravity correction for the pressure gradients.	true
33	--enable-grid-adaptation	A boolean value that enables (true) or disables (false) adaptive grid refinement/coarsening. Note that Eclipse grid output will be disabled if adaptive grid refinement/coarsening is enabled.	false
34	--enable-intensive-quantity-cache	A boolean value that switches on (true) or off (false) the caching of intensive quantities.	true
35	--enable-logging-fallout-warning	A boolean value that sets the developer option to see whether logging was on non-root processors (true). If set to true output will be appended to the *.DBG or *.PRT files.	false
36	--enable-opm-rst-file	A boolean value set to true or false to write OPM specific data sets to the commercial simulators restart file to enable restart of OPM Flow runs by OPM Flow (true), or not to write the data (false).	false
37	--enable-storage-cache	A boolean value that turns on (true) or off (false) storing previous storage terms and avoid re-calculating them.	true
38	--enable-terminal-output	A boolean value that turns on (true) or off (false) high-level information about the simulation's progress to the terminal	true
39	--enable-thermodynamic-hints	A boolean value that enables (true) or disables (false) thermodynamic hints.	false

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
40	--enable-tuning	A boolean value that instructs OPM Flow to read the time stepping parameters from the <i>TUNING – Numerical Tuning Control</i> keyword in the input deck, if set to true. Note that only the first record of the <i>TUNING</i> keyword is processed.	false
41	--enable-vtk-output	A boolean value that turns on (true) or off (false) a global switch for writing VTK files.	false
42	--enable-well-operability-check	A boolean value that enables (true) checking of a well's operating status, or disables (false) the checking.	true
43	--enable-well-operability-check-iter	A boolean value that enables (true) checking of a well's operating status during iterations, or disables (false) the checking during iterations.	false
44	--enable-write-all-solutions	A boolean value that turns on (true) or off (false) the writing of all solutions to disk instead of only the ones for the report steps.	false
45	--end-time	The simulation time at which the simulation is finished [s].	1e+100
46	--explicit-rock-compaction	A boolean value. Use pressure from end of the last time step when evaluating rock compaction (true) or not (false).	false
47	--external-partition	A quoted character string. Name of file from which to load an externally generated partitioning of the model's active cells for MPI distribution purposes. If empty, the built-in partitioning method will be employed.	""
48	--force-disable-fluid-in-place-output	A boolean value that instructs OPM Flow not to print the Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
49	--force-disable-resv-fluid-in-place-output	A boolean value that instructs OPM Flow to not to print the Reservoir Volume Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
50	--full-time-step-initially	A boolean value that instructs OPM Flow to always attempt to finish a report step using a single time step (true) or not (false).	false
51	--ignore-keywords	A character string that defines a list of keywords which should be ignored by OPM Flow. The keyword string should be	""

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		enclosed in quotes and each keyword separated by a colon, that is ':'.	
52	--ilu-fillin-level	A positive integer value that sets the fill in level for the ILU preconditioner.	0
53	--ilu-redblack	A boolean value that instructs OPM Flow to use red-black partitioning for the ILU preconditioner.	false
54	--ilu-relaxation	A real positive double precision value that sets the relaxation factor of the linear solver's ILU preconditioner.	0.9
55	--ilu-reorder-spheres	A boolean value that specifies OPM Flow to reorder the entries of the matrix in the red-black ILU preconditioner in spheres starting at an edge (true) or not (false). If false the original ordering is preserved in each color. Otherwise try to ensure D4 ordering (in a 2D structured grid, the diagonal elements are consecutive).	false
56	--initial-time-step-in-days	A real double precision value that sets the size of initial time step in days.	1.0
57	--initial-time-step-size	A real double precision value that sets the size of initial time step in seconds.	86400
58	--linear-solver	A defined quoted character string that sets the configuration of the linear solver; valid values are: 1) ilu0 (default), 2) dilu, 3) cprw or cpr (an alias for cprw), ^{373 374} and ³⁷⁵ 4) cpr_quasiimpes, 5) cpr_trueimpes, 6) cpr_trueimpesanalytic, 7) amg, ³⁷⁶ 8) hybrid (experimental), or 9) a file name that has the extension ".json", that contains the linear solver configuration parameters.	"ilu0"

³⁷³ Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

³⁷⁴ R. Scheichl, M. Roland, J. Wendebourg, Decoupling and block preconditioning for sedimentary basin simulations, Computational Geosciences 7 (2003) 295{318.

³⁷⁵ Klemetsdal, Ø.S., Møyner, O. & Lie, KA. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. Comput Geosci 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

³⁷⁶ M. Blatt, A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients, Ph.D. thesis, Ruprecht-Karls-Universit t Heidelberg (2010).

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		Option (3) extends the existing Constrained Pressure Residual ("CPR") preconditioner to include wells. This option can also be invoked via the CPR keyword in the RUNSPEC section; however, the command line parameter takes precedence. For option (9) one enters a character string enclosed in quotes that defines the filename of a JSON configuration file for the flexible linear solver system. In this case the file extension should be '.json'. Note that the *.PRT file contains the "Property tree for the linear solver" listing, which is the JSON specification of the current case, and can be used to configure a user specific linear solver JSON file. The option "cpr" now is an alias for "cprw" instead of "cpr_trueimpes".	
59	--linear-solver-ignore-convergence-failure	A boolean value that if set to true convergences failures in the linear solver are ignored. This option should be used with care, as the results might be unreliable.	false
60	--linear-solver-max-iter	A positive integer value that defines the maximum number of linear iterations.	200
61	--linear-solver-print-json-definition	Write the JSON definition of the linear solver setup to the DBG file.	true
62	--linear-solver-reduction	A real positive double precision value that sets the minimum reduction of the residual for the linear solver. The linear solver convergences when the residual is reduced sufficiently. The simulator now overrides the default reduction to be 0.005 instead of 0.01, if the linear solver has been set to one of the cpr options, in a similar manner as how the default maximum number of linear iterations for the cpr and cprw options are changed to 20 instead of 200, unless specified by the command line option by the user.	0.01
63	--linear-solver-restart	A positive integer value that sets the number of iterations after which GMRES is restarted.	40
64	--linear-solver-verbosity	A positive integer value that defines the output from linear solver: 0 : No extra output, 1 : output per solution iteration, or 2 : output per iteration.	0

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
65	--load-file	A character string enclosed in quotes that defines the FileName for the .OPMRST file used to load the serialized state. If empty, CASENAME.OPMRST is used.	""
66	--load-step	An integer value that determines if the simulator should load the serialized state from OPM Flow's version of the restart file, and should be set to either a specific report step, or 0 to load the last stored report step. The default value of -1 does not load the data from the OPM Flow specific restart file. OPM Flow's version of the restart file, is written using the --save-step=N option.	-1
67	--local-domains-ordering-measure	A character string enclosed in quotes that defines the subdomain ordering measure. Allowed values are 'maxpressure', 'averagepressure' and 'residual'.	"maxpressure"
68	--local-domains-partitioning-imbalance	A real positive double precision value that sets the subdomain partitioning imbalance tolerance. 1.03 is 3 percent imbalance.	1.03
69	--local-domains-partitioning-method	A character string enclosed in quotes that defines the subdomain partitioning method. Allowed values are 'zoltan', 'simple', and the name of a partition file ending with '.partition'.	"zoltan"
70	--local-solve-approach	A character string enclosed in quotes that defines the local solve approach. Valid choices are 'jacobi' and 'gauss-seidel'.	"gauss-seidel"
71	--local-tolerance-scaling-cnv	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	0.1
72	--local-tolerance-scaling-mb	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	1
73	--local-well-solve-control-switching	A boolean value that allows (true) or disallows (false) control switching during local well solutions.	false
74	--matrix-add-well-contributions	A boolean value that if set to true explicitly specifies the influences of wells between cells in the Jacobian and preconditioner matrices.	false
75	--max-inner-iter-ms-wells	A positive integer value that defines the maximum number of inner iterations for multi-segment wells.	100

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
76	--max-inner-iter-wells	A positive integer value that defines the maximum number of inner iterations for standard wells.	50
77	--max-local-solve-iterations	A positive integer value that defines the maximum number of iterations for local solves with NLDD nonlinear solver.	20
78	--max-newton-iterations-with-inner-well-iterations	A positive integer that specifies the maximum Newton iterations with inner well iterations.	8
79	--max-pressure-change-ms-wells	A real positive value that defines the maximum relative pressure change for a single iteration of the multi-segment well model.	1.0×10^6
80	--max-residual-allowed	A real positive value that sets the absolute maximum tolerance for residuals without cutting the time step size.	1.0×10^7
81	--max-single-precision-days	A real positive value that sets the maximum time step size where single precision floating point arithmetic can be used for solving the linear systems of equations.	20.0
82	--max-temperature-change	A real positive value that stipulates the maximum absolute change of temperature in a single iteration.	5
83	--max-time-step-divisions	A positive integer that specifies the maximum number of divisions by two of the timestep size before the simulation bails out.	10
84	--max-time-step-size	A real positive value that defines the maximum size to which all time steps are limited to in seconds.	inf
85	--max-welleq-iter	A positive integer that defines the maximum number of iterations to determine the solution to the well equations.	30
86	--maximum-number-of-well-switches	A positive integer values that stipulates the maximum number of times a well can switch to the same control.	3
87	--maximum-water-saturation	A real positive value that defines the maximum water saturation.	1.0
88	--milu-variant	A defined character string that specifies which variant of the modified ILU preconditioner ought to be used. Possible variants are: 1) ILU (default, plain ILU), 2) MILU_I (lump diagonal with dropped row entries),	"ILU"

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		3) MILU_2 (lump diagonal with the sum of the absolute values of the dropped row entries), 4) MILU_3 (if diagonal is positive add sum of dropped row entries. Otherwise subtract them), 5) MILU_4 (if diagonal is positive add sum of dropped row entries. Otherwise do nothing).	
89	--min-strict-cnv-ite	A positive integer that sets the minimum number of Newton iterations before relaxed tolerances are used for the CNV convergence criterion.	0
90	--min-strict-mb-iter	An integer value defining the number of Newton iterations before relaxed tolerances can be used for the MB convergence criterion. Default -1 means that the relaxed tolerance is used when maximum number of Newton iterations are reached.	-1
91	--min-time-step-based-on-newton-iterations	A real positive value that sets the minimum time step size (in days for field and metric unit and hours for lab unit) can be reduced to based on Newton iteration counts.	0.0
92	--min-time-step-before-shutting-problematic-wells-in-days	A real positive value that sets the minimum time step size in days for which problematic wells are not shut. Time steps below this value will result in problematic wells being shut-in by the simulator.	0.01
93	--min-time-step-size	A real positive value that sets the minimum size to which all time steps are limited to in seconds.	0.0
94	--network-max-iterations	A positive integer that sets the maximum number of iterations in the network solver before giving up.	200
95	--network-max-strict-iterations	A positive integer that sets the maximum iterations in network solver before relaxing tolerance.	100
96	--newton-max-error	A real positive value that sets the maximum error tolerated by the Newton method to which does not cause an abort.	1e+100
97	--newton-max-iterations	A positive integer that defines the maximum number of Newton iterations per time step used by the simulator.	20
98	--newton-max-relax	A real positive value that sets the maximum relaxation factor of a Newton iteration used by the simulator.	0.5

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
99	--newton-min-iterations	A positive integer that sets the minimum number of Newton iterations per time step used by the simulator.	2
100	--newton-relaxation-type	A character string. The type of relaxation used by the Newton method. Valid options are: dampen or sor.	"dampen"
101	--newton-target-iterations	A positive integer that sets the 'optimum' number of Newton iterations per time step.	10
102	--newton-tolerance	A real positive value that sets the maximum raw error tolerated by the Newtonmethod for considering a solution to be converged.	0.01
103	--newton-verbose	A boolean value that specifies whether the Newton method should inform the user about its progress (true) or not (false).	true
104	--newton-write-convergence	A boolean value that specifies whether to write the convergence behaviour of the Newton method to a VTK file (true) or not (false).	false
105	--nldd-num-initial-newton-iter	A positive integer that sets the number of initial global Newton iterations when running the NLDD nonlinear solver.	1
106	--nonlinear-solver	A character string that specifies the nonlinear solver. Valid choices are newton or nldd.	"newton"
107	--num-local-domains	A positive integer that sets the number of local domains for NLDD nonlinear solver. Note this is an experimental feature in the current release 2023.10 that is expected to be more complete and tested by the 2024.10 release.	0
108	--num-pressure-points-equil	A positive integer that sets the number of pressure points (in each direction) in tables used for equilibration.	2000
109	--opencl-ilu-parallel	A boolean value that if set to true then parallelize the ILU decomposition and application on GPU, or not (false).	true
110	--opencl-platform-id	A positive integer that specifies the platform identification ("ID") for the openSolver. Use the "clinfo" command to determine valid IDs.	0
111	--output-dir	A character string that defines the directory to which OPM Flow is to write the commercial simulator compatible output files (restart, summary files etc.).	""

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		The default value results in the files be written to the same directory as the input file.	
I 12	--output-extra-convergence-info	A defined character string that specifies whether to provide additional convergence output to separate files for diagnostic purposes. The available options are: 1) "none" results in no extra output and overrides all other options. 2) "steps" writes out convergence information per time step, to a CASENAME.INFOSTEP file. The file is useful for identifying numerical issues. 3) "iterations" writes out non-linear convergence metrics, i.e., the MB and CNV values, per phase, for each non-linear iteration in each time step, to a CASENAME.INFOITER file. Options can be combined with commas, e.g. "steps,iterations" for multiple outputs. The default value of "none" prevents the two files from being written out, for better compatibility with the commercial simulator.	"none"
I 13	--output-interval	A positive integer that specifies the number of report steps between two consecutive writes of restart data.	1
I 14	--output-mode	A defined character string that defines the output to the *.PRT and *.DBG files: 1) "none": No output to the files. 2) "log" or false: Output logging information only. 3) "all" or true: Output everything. For example to just output logging information use: <pre>--output-mode="log".</pre> or <pre>--output-mode=false</pre>	"all"
I 15	--owner-cells-first	A boolean value that determines if the cells owned by a rank should be ordered before ghost/overlapping the cells (true), or not (false).	true
I 16	--parameter-file	A character string that defines the name of a parameter file which contains the simulator's set of run-time parameters, as listed in this table.	""
I 17	--parsing-strictness	Set strictness of parsing process. Available options are:	"normal"

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		1) "normal": stop for critical errors, and for unsupported keywords that would change the simulator results if supported. 2) "high": stop for all errors, that is even for unsupported keywords that do not effect the results, for example <i>ECHO</i> and <i>NOECHO</i> . 3) "low": same as normal, except do not stop due to unsupported keywords that would change the simulator results if supported, and even if marked critical. Default: "normal"	
118	--predetermined-time-steps-file	A file with a list of predetermined time step sizes (one time step per line).	""
119	--pressure-max	A positive real value. Maximum absolute pressure.	1e+99
120	--pressure-min	A real value. Minimum absolute pressure.	-1e+99
121	--pressure-scale	A positive real value. Scaling of pressure primary variable.	1
122	--pri-var-oscillation-threshold	A real positive value that defines the threshold value for the primary variable switching conditions after its meaning has switched to hinder oscillations.	1.0×10^{-5}
123	--print-parameters	A positive integer value that request that the run time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
124	--print-properties	A positive integer value that request that the compile time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
125	--project-saturations	A boolean value that determines if all the fluid saturations should be scaled to ensure the values are in the interval (0, 1), including runs that use solvents (true), or not (false).	false
126	--regularization-factor-wells	A real positive value that defines the "regularization factor" for wells.	100

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
127	--relaxed-linear-solver-reduction	A real positive value that defines the minimum reduction of the residual which the linear solver need to achieve for the solution to be accepted.	0.01
128	--relaxed-max-pv-fraction	A real positive value that defines The fraction of the pore volume of the reservoir where the volumetric error (CNV) might be violated during strict Newton iterations.	0.03
129	--relaxed-pressure-tol-msw	A real positive value that sets the relaxation tolerance for the multi-segment well <u>pressure</u> solution in Pascals.	10000
130	--relaxed-well-flow-tol	A real positive value that sets the relaxation tolerance for the well flow residual in reservoir cubic metres (rm3).	0.001
131	--restart-time	A real positive value that sets the simulation time at which a restart should be attempted [s].	-1e+35
132	--restart-writing-interval	A positive integer value that sets the frequency at which time steps are serialized to disk.	16777215
133	--save-file	A character string that specifies the FileName for .OPMRST file used for saving serialized state. If empty, CASENAME.OPMRST is used.	""
134	--save-step	A character string that determines if the simulator should save the serialized state of the OPM Flow simulator at one or more report steps to a special *.OPMRST file. This is in addition to the normal restart files written, and consumes significantly more space than the normal restart files, but restarting OPM Flow from this file using the <code>--load-step</code> option deviates less from the original run, compared to restarting using the normal restart features. The files produced are not compatible with other simulators, and also will not be compatible between different releases of OPM Flow The parameter should be set to one of the following: 1) "all" to save all report steps, 2) "x" to save every x'th step, or 3) "x" to save a specif time step. The default value of "" does not write anything to the OPM Flow specific restart file.	""
135	--scale-linear-system	A boolean value that switches on (true) or	false

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		off (false) the scaling of linear system of equations according to the equation scale and primary variable types.	
136	--sched-restart	A boolean value that determines for a restart run if the case should initialize wells and groups from the historical <i>SCHEDULE</i> section (true), or from the well and group data on the restart file (false). Note that the commercial simulator always uses data from the restart file (false).	false
137	--serial-partitioning	A boolean value that determines if partitioning for parallel runs on a single process (true), or not (false).	false
138	--shut-unsolvable-wells	A boolean value that determines if the simulator should shut unsolvable wells (true), or not (false).	true
139	--solve-welleq-initially	A boolean value that determines if the simulator should fully solve the well equations before each iteration of the reservoir model (true), or not (false). Note that the well equations are always added to the full system and solved until convergence.	true
140	--solver-continue-on-convergence-failure	A boolean value that stipulates if the simulator should continue (true) instead of stopping (false) when the minimum solver time step is reached.	false
141	--solver-growth-factor	A real positive value that specifies the growth factor a time step can be increased by when recovering from one or more time step chops, subject to the maximum allowable time step size set by the <i>--solver-max-time-step-in-days</i> parameter. For example, if the current time step has converged at 5 days after a time step chop, and <i>--solver-growth-factor</i> is set to the default value of 2.0, then the next time step will be $2.0 * 5$ days, that is at 10 days.	2.0
142	--solver-max-growth	A real positive value that specifies the maximum growth factor a time step can be increased by after a report time step, subject to the maximum allowable time step size set by the <i>--solver-max-time-step-in-days</i> parameter. Thus, if the current time step has converged at 5 days after at a report step, and <i>--solver-max-growth</i> is set to the default value of 3.0, then the next time step will be $3.0 * 5$ days, that is at 15 days.	3.0

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
143	--solver-max-restarts	A positive integer that defines the number of allowed consecutive restarts (or time step chops) before the simulation is terminated.	10
144	--solver-max-time-step-in-days	A real positive double precision value that specifies the maximum allowed time step size in days.	365
145	--solver-min-time-step	A real positive double precision value that specifies the minimum size of a time step in days for field and metric units, and hours for laboratory units If a time step cannot converge without getting cut below this time step size the simulator will stop.	1.0×10^{-12}
146	--solver-restart-factor	A real positive double precision value that sets the time step chop factor of the time step after a convergence failure. For example, if the current non-convergent time step is 30 days and --solver-restart-factor is set to the default value, then the time step will be repeated using $0.33 * 30$ days as the time step, that is 9.9 days.	0.33
147	--solver-verbosity	A positive integer that specifies the "chattiness" of the non-linear solver.	1
148	--strict-inner-iter-wells	A positive integer that specifies the number of inner well iterations with strict tolerance.	40
149	--strict-outer-iter-wells	A positive integer that specifies the number of Newton iterations for which wells are checked with strict tolerance.	6
150	--temperature-max	A real positive value that sets the maximum absolute temperature.	1.0×10^9
151	--temperature-min	A real positive value that sets the minimum absolute temperature.	0
152	--threads-per-process	A positive integer value that stipulates the maximum number of threads to be instantiated per process ('-1' means 'automatic').	-1
153	--tolerance-mb-relaxed	A positive real value. Relaxed tolerated mass balance error that applies for iterations after the iterations with the strict tolerance.	1e-06
154	--time-step-after-event-in-days	A real double precision value that sets the maximum allowed time step after an event; for example, when a well is open or closed etc. The default value of -1 means that events	-1

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		to do effect the time stepping.	
155	--time-step-control	A defined character string that defines the time stepping control algorithm and is set to one of the following: 1) pid: PID controller based adaptive time step control as suggested by Turek and Kuzmin³⁷⁷. 2) pid+iteration: Use PID and linear iteration numbers to guide the time step. 3) pid+newtoniteration: Use PID and non-linear iterations numbers to guide the time step. 4) Hardcoded: Use time steps supplied by user via the time-step-control-filename parameter.	"pid+newtoniteration"
156	--time-step-control-decay-damping-factor	A real positive value that specifies the decay rate a time step can be decreased by after the number target iterations has been exceeded	1.0
157	--time-step-control-decay-rate	A real positive value that specifies the decay rate a time step size can be decreased by after the number of target iterations has been exceeded	0.75
158	--time-step-control-file-name	A character string enclosed in quotes that specifies a filename where time steps are specified. The default is the character string timesteps. For instance time steps can be generated by the ecl_summary application in libecl as per the following UNIX command line: <pre>path_to_libecl_applications/ ecl_summary DECK TIME > filename</pre> Where: DECK is the name of the data deck you want to get the time steps from, TIME tells the application to return the timing for the run, and "filename" is the name of the file the times are piped to.	"timesteps"
159	--time-step-control-growth-damping-factor	A real positive value that specifies the growth rate of the time step increase when the number of target iterations are undercut.	3.2

³⁷⁷ Algebraic Flux Correction III. Incompressible Flow Problems. Uni Dortmund, Turek and Kuzmin, January, 2006 (DOI: 10.1007/3-540-27206-2_8)

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
160	--time-step-control-growth-rate	A real positive value that specifies the growth rate of the <u>time step size</u> when the number of target iterations are undercut.	1.25
161	--time-step-control-target-iterations	A positive integer that defines the number of linear iterations which the time step control scheme should aim for (if applicable).	30
162	--time-step-control-target-newton-iterations	A positive integer that specifies the number of Newton iterations which the time step control scheme should aim for (if applicable).	8
163	--time-step-control-tolerance	A real double precision value that sets the tolerance for PID (only used with the pid and pid+ options defined by the --time-step-control option).	0.1
164	--time-step-verbosity	A positive integer that specifies the "chattiness" during the time integration.	1
165	--tolerance-cnv	A real positive double precision value that specifies the maximum non-linear tolerance error. This is the local convergence tolerance (maximum of local saturation errors).	0.01
166	--tolerance-cnv-relaxed	A real positive value that defines the relaxed local convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0
167	--tolerance-mb	A real positive double precision value that sets the maximum mass balance error; that is the tolerated mass balance error relative to total mass present.	1.0×10^{-6}
168	--tolerance-pressure-ms-wells	A real positive double precision value that specifies the tolerance for the pressure equations for multi-segment wells.	1000
169	--tolerance-well-control	A real positive double precision value that sets the maximum tolerance for the well control equations.	1.0×10^{-7}
170	--tolerance-wells	A real positive double precision value that defines the maximum non-linear error for the well equations.	1.0×10^{-4}
171	-update-equations-scaling	A boolean value that switches on (true) or off (false) the updating of the scaling factors for mass balance equations during the simulation.	false
172	--use-average-density-ms-wells	A boolean value that specifies whether to	false

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
		approximate segment densities by averaging over the segment and its outlet (true) or not (false).	
173	--use-gmres	A boolean value that when set to true OPM Flow will use Generalized Minimal Residual ("GMRES") ³⁷⁸ and ³⁷⁹ solver instead of Biconjugate Gradient Stabilized ("BiCGSTAB") ³⁸⁰ and ³⁸¹ as the linear solver within the Newton iterations.	false
174	--use-implicit-ipr	A boolean value. Compute implicit IPR for stability checks and stable solution search.	false
175	--use-multisegment-well	A boolean value that when set to true the simulator will use the well model for multi-segment wells instead of the one for single-segment wells.	true
176	--use-update-stabilization	A boolean value that switches on (true) or off (false) the stabilized Newton option, that attempts to detect and correct oscillations or stagnation during the Newton iterations. This option may improve convergence for some cases.	true
177	--water-only-threshold	A real positive value that defines the saturation threshold, for which cells with water saturations above or equal to this threshold are considered one-phase water only.	1.0
178	--zoltan-imbalance-tol	A real positive value that defines the tolerable imbalance of the load balancing provided by Zoltan package.	1.1
179	--zoltan-params	A character string that specifies the configuration of the Zoltan partitioner. Valid options are: graph, hypergraph or scotch. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See https://sandialabs.github.io/Zoltan/ug_html/ug.html for available Zoltan options.	"graph"
Notes:			

³⁷⁸ Y. Saad, A flexible inner-outer preconditioned GMRES algorithm, *SIAM J. Sci. Statist. Comput.*, 14, (1993).

³⁷⁹ Y. Saad and M.H. Schultz, GMRES: A generalized minimum residual algorithm for solving nonsymmetric linear systems, *SIAM J. Sci. Statist. Comput.*, 7 (1986), pp. 856{869.

³⁸⁰ Van der Vorst, H. A. (1992). "Bi-CGSTAB: A Fast and Smoothly Converging Variant of Bi-CG for the Solution of Nonsymmetric Linear Systems". *SIAM J. Sci. Stat. Comput.* 13 (2): 631–644. doi:10.1137/0913035. hdl:10338.dmlcz/104566

³⁸¹ Sleijpen, G. L. G.; Fokkema, D. R. (November 1993). "BiCGstab(l) for linear equations involving unsymmetric matrices with complex spectrum" (PDF). *Electronic Transactions on Numerical Analysis*. Kent, OH: Kent State University. 1: 11–32. ISSN 1068-9613.

OPM Flow 2024-04 Command Line Options			
No.	Variable Name	Description	Default
1)		Cells colored green in the No. column are new command line parameters for this release. Similarly for the Default column, cells colored green indicate the default value has changed from the previous release.	
2)		Items shaded in gray are considered to be developer options that should be used with caution as the values associated with these command line parameters are subject to change.	
3)		VTK Graphics Command Line Parameters ³⁸² set of commands are no longer listed via the basic help command line command: flow --help instead use: flow --help-all to get a list of supported command line Parameters.	
3)		The --enable-vtk-output option above, if set to "true" will write out the data as stipulated by the compile options. One may need to compile the source code to obtain the desired output.	
4)		As per all UNIX and LINUX based system the input is case dependent.	
5)		If a parameter file is used to set one of the above parameters, the leading "--" should not be incorporated in the parameter file.	

Table E.3: OPM Flow 2024-04 Command Line Options

³⁸² Virtualization ToolKit ("VTK") files used to visualize the 3D results from the model using ParaView, an open-source, multi-platform general purpose data analysis and visualization application (<https://www.paraview.org/overview/>). For the commercial simulator's compatible output files OPM's ResInsight 3D visualization package can be used instead and is specifically tailored for OPM Flow and the commercial simulator.

E.4 RUNNING OPM FLOW 2023-10

The command line syntax for running OPM Flow is:

```
flow [OPTIONS] CASENAME
```

and typing the following command from your terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA.

OPM Flow accepts command line options to control various aspects of the simulator's run time behavior, as described in the table below. To give a command line option use "key=value" syntax, with no spaces around the equals sign. It is also possible to put multiple options together in a parameter file. To do so, put one option on each line of the file with ".param" as the extension and pass that filename as a command line parameter to OPM Flow using:

```
flow --parameter-file=CASENAME.param
```

to instruct OPM Flow to read the parameter file.

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
1	-h or --help	A character string that causes OPM Flow to print a help message that gives a brief description of the available command line parameters. This now only prints the active user facing command line options.	N/A
2	--help-all	Prints all the command line options included in the release, including obsolete, hidden and deprecated options.	N/A
3	--accelerator-mode	Choose a linear solver, usage: '--accelerator-mode=[none cusparse opencl amgcl rocalution]'.	"none"
4	--allow-distributed-wells	Allow the perforations of a well to be distributed to interior of multiple processes.	false
5	--alternative-well-rate-init	Use alternative well rate initialization procedure.	true
6	--bda-device-id	Choose device ID for cusparseSolver or openclSolver, use 'nvidia-smi' or 'clinfo' commands to determine valid IDs.	0
7	--continue-on-convergence-error	Continue with a non-converged solution instead of giving up if we encounter a time step size smaller than the minimum time step size.	false

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
8	--cpr-reuse-interval	A positive integer that sets the reuse preconditioner interval. Used when --cpr-reuse-setup is set to 4, then the preconditioner will be fully recreated instead of reused every N linear solve, where N is this parameter.	30
9	--cpr-reuse-setup	A positive integer that defines if the CPR solver should re-use the AMG setup. Valid options are: 0 : Recreate the preconditioner for every linear solve. 1 : Recreate once every time step. 2 : Recreate if last linear solve took more than 10 iterations. 3 : Never recreate. 4 : Recreate every N linear solves, where N is the parameter --cpr-reuse-interval. Changed the default value from three to four.	4
10	--dbph-max-rel	A real positive value that sets the maximum relative change of the bottom-hole pressure in a single iteration.	1.0
11	--dp-max-rel	A real positive double precision value that sets the maximum allowed relative change of pressure per iteration.	0.3
12	--ds-max	A real positive double precision value that sets the maximum allowed change in saturation per iteration.	0.2
13	--dwell-fraction-max	A real positive value that sets the maximum allowed change in well's volume fraction per iteration.	0.2
14	--ecl-deck-file-name	A character string that defines the name of the OPM Flow input file which contains the simulator's ECLIPSE formatted input deck to be simulated. Note that there must be no spaces in the filename. If there are spaces, which can typically occur on Windows based systems, then the filename should be enclosed in quotes; thus instead of FILE NAME use "FILE NAME".	""
15	--ecl-enable-drift-compensation	A boolean value that enables (true) or disables (false) partial compensation of systematic mass losses via the source term of the next time step.	true
16	--ecl-enable-tuning	Honor some aspects of the TUNING keyword from the ECL deck.	false

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
17	--ecl-newton-relaxed-tolerance	The maximum error which the volumetric residual may exhibit if it is in a 'relaxed' region during a strict iteration.	1e+09
18	--ecl-newton-relaxed-volume-fraction	The fraction of the pore volume of the reservoir where the volumetric error may be violated during strict Newton iterations.	0.03
19	--ecl-newton-strict-iterations	The number of Newton iterations where the volumetric error is considered.	8
20	--ecl-newton-sum-tolerance	The maximum error tolerated by the Newton method for considering a solution to be converged.	0.0001
21	--ecl-newton-sum-tolerance-exponent	The exponent used to scale the sum tolerance by the total pore volume of the reservoir.	0.333333
22	--ecl-output-double-precision	A boolean value that switches on (true) or off (false) double precision in restart files. Useful for 'perfect' restarts.	false
23	--ecl-output-interval	An integer value defining the number of report steps that ought to be skipped between two writes of restart files results.	-1
24	--edge-weights-method	A defined positive integer value that defines the edge-weighting strategy: 0 : For uniform, 1 : for trans, or 2 : for log(trans).	1
25	--enable-adaptive-time-stepping	A boolean value that turns on (true) or off (false) adaptive time stepping. If set to false the report time steps provided by the input deck are used.	true
26	--enable-async-ecl-output	A boolean value that sets the output mode to be asynchronously (true), letting the simulator continue computing the next time step while writing results to restart and summary files in the commercial simulators format.	true
27	--enable-async-vtk-output	A boolean value that sets the output mode to be asynchronously (true), letting the simulator continue computing the next time step while writing results to the VTK output files.	true
28	--enable-dry-run	A boolean value that specifies if the simulation should actually run, (true) or just check the input deck (false). This option is equivalent to activating the NOSIM keyword in the RUNSPEC section of the input deck (see section 5.3.101	"auto"

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		<i>NOSIM – Activate the No Simulation Mode for Data File Checking).</i> The default value of “auto” will use whatever is stipulated in the input deck via the <i>NOSIM</i> keyword.	
29	--enable-ecl-output	A boolean value that specifies if the binary output files (restart, summary files, etc.) should be written in the commercial simulator's format (true), or OPM Flow's format (false).	true
30	--enable-esmry	A boolean value that switches on (true) or off (false) the output <i>SUMMARY</i> vectors to the ESMRY file for fast loading of summary data.	false
31	--enable-gravity	A boolean value that switches on (true) or off (false) the use of the gravity correction for the pressure gradients.	true
32	--enable-grid-adaptation	A boolean value that enables (true) or disables (false) adaptive grid refinement/coarsening. Note that Eclipse grid output will be disabled if adaptive grid refinement/coarsening is enabled.	false
33	--enable-intensive-quantity-cache	A boolean value that switches on (true) or off (false) the caching of intensive quantities.	true
34	--enable-logging-fallout-warning	A boolean value that sets the developer option to see whether logging was on non-root processors (true). If set to true output will be appended to the *.DBG or *.PRT files.	false
35	--enable-opm-rst-file	A boolean value set to true or false to write OPM specific data sets to the commercial simulators restart file to enable restart of an OPM Flow runs (true) by OPM Flow, or not to write the data (false).	false
36	--enable-storage-cache	A boolean value that turns on (true) or off (false) storing previous storage terms and avoid re-calculating them.	true
37	--enable-terminal-output	A boolean value that turns on (true) or off (false) high-level information about the simulation's progress to the terminal	true
38	--enable-thermodynamic-hints	A boolean value that enables (true) or disables (false) thermodynamic hints.	false
39	--enable-tuning	A boolean value that instructs OPM Flow to read the time stepping parameters from the <i>TUNING – Numerical Tuning Control</i> keyword in the input deck, if set to true.	false

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		Note that only the first record of the TUNING keyword is processed.	
40	--enable-vtk-output	A boolean value that turns on (true) or off (false) a global switch for writing VTK files.	false
41	--enable-well-operability-check	A boolean value that enables (true) checking of a well's operating status, or disables (false) the checking.	true
42	--enable-well-operability-check-iter	A boolean value that enables (true) checking of a well's operating status during iterations, or disables (false) the checking during iterations.	false
43	--enable-write-all-solutions	A boolean value that turns on (true) or off (false) the writing of all solutions to disk instead of only the ones for the report steps.	false
44	--end-time	The simulation time at which the simulation is finished [s].	1e+100
45	--force-disable-fluid-in-place-output	A boolean value that instructs OPM Flow not to print the Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
46	--force-disable-resv-fluid-in-place-output	A boolean value that instructs OPM Flow to not to print the Reservoir Volume Fluid In-Place report after each report time step (true) or not (false). Note this parameter will override the print request in the input deck.	false
47	--full-time-step-initially	A boolean value that instructs OPM Flow to always attempt to finish a report step using a single time step (true) or not (false).	false
48	--ignore-keywords	A character string that defines a list of keywords which should be ignored by OPM Flow. The keyword string should be enclosed in quotes and each keyword separated by a colon, that is ':'.	""
49	--ilu-fillin-level	A positive integer value that sets the fill in level for the ILU preconditioner.	0
50	--ilu-redblack	A boolean value that instructs OPM Flow to use red-black partitioning for the ILU preconditioner.	false
51	--ilu-relaxation	A real positive double precision value that sets the relaxation factor of the linear solver's ILU preconditioner.	0.9
52	--ilu-reorder-spheres	A boolean value that specifies OPM Flow	false

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		to reorder the entries of the matrix in the red-black ILU preconditioner in spheres starting at an edge (true) or not (false). If false the original ordering is preserved in each color. Otherwise try to ensure D4 ordering (in a 2D structured grid, the diagonal elements are consecutive).	
53	--initial-time-step-in-days	A real double precision value that sets the size of initial time step in days.	1.0
54	--initial-time-step-size	A real double precision value that sets the size of initial time step in seconds.	86400
55	--linear-solver	A defined quoted character string that sets the configuration of the linear solver; valid values are: 1) ilu0 (default), 2) cprw or cpr (an alias for cprw),^{383 384} and ³⁸⁵ 3) cpr_quasiimpes, 4) cpr_trueimpes, 5) amg,³⁸⁶ 6) hybrid (experimental), or 7) a file name that has the extension ".json", that contains the linear solver configuration parameters. Option (2) extends the existing Constrained Pressure Residual ("CPR") preconditioner to include wells. This option can also be invoked via the <i>CPR</i> keyword in the <i>RUNSPEC</i> section; however, the command line parameter takes precedence. For option (7) one enters a character string enclosed in quotes that defines the filename of a JSON configuration file for the flexible linear solver system. In this case the file extension should be '.json'. Note that the *.PRT file contains the "Property tree for the linear solver" listing, which is the JSON specification of the current case, and can be used to configure a user specific linear solver JSON file.	"ilu0"

³⁸³ Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).

³⁸⁴ R. Scheichl, M. Roland, J. Wendebourg, Decoupling and block preconditioning for sedimentary basin simulations, *Computational Geosciences* 7 (2003) 295{318.

³⁸⁵ Klemetsdal, Ø.S., Møyner, O. & Lie, K.A. Accelerating multiscale simulation of complex geomodels by use of dynamically adapted basis functions. *Comput Geosci* 24, 459–476 (2020). <https://doi.org/10.1007/s10596-019-9827-z>.

³⁸⁶ M. Blatt, A parallel algebraic multigrid method for elliptic problems with highly discontinuous coefficients, Ph.D. thesis, Ruprecht-Karls-Universität Heidelberg (2010).

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		The option "cpr" now is an alias for "cprw" instead of "cpr_trueimpes".	
56	--linear-solver-ignore-convergence-failure	A boolean value that if set to true convergences failures in the linear solver are ignored. This option should be used with care, as the results might be unreliable.	false
57	--linear-solver-max-iter	A positive integer value that defines the maximum number of linear iterations.	200
58	--linear-solver-print-json-definition	Write the JSON definition of the linear solver setup to the DBG file.	true
59	--linear-solver-reduction	A real positive double precision value that sets the minimum reduction of the residual for the linear solver. The linear solver convergences when the residual is reduced sufficiently. The simulator now overrides the default reduction to be 0.005 instead of 0.01, if the linear solver has been set to one of the cpr options, in a similar manner as how the default maximum number of linear iterations for the cpr and cprw options are changed to 20 instead of 200, unless specified by the command line option by the user.	0.01
60	--linear-solver-restart	A positive integer value that sets the number of iterations after which GMRES is restarted.	40
61	--linear-solver-verbosity	A positive integer value that defines the output from linear solver: 0 : No extra output, 1 : output per solution iteration, or 2 : output per iteration.	0
62	--load-file	A character string enclosed in quotes that defines the FileName for the .OPMRST file used to load the serialized state. If empty, CASENAME.OPMRST is used.	""
63	--load-step	An integer value that determines if the simulator should load the serialized state from OPM Flow's version of the restart file, and should be set to either a specific report step, or 0 to load the last stored report step. The default value of -1 does not load the data from the OPM Flow specific restart	-1

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		file. OPM Flow's version of the restart file, is written using the <code>--save-step=N</code> option.	
64	<code>--local-domains-ordering-measure</code>	A character string enclosed in quotes that defines the subdomain ordering measure. Allowed values are 'pressure' and 'residual'.	"pressure"
65	<code>--local-domains-partitioning-imbalance</code>	A real positive double precision value that sets the subdomain partitioning imbalance tolerance. 1.03 is 3 percent imbalance.	1.03
66	<code>--local-domains-partitioning-method</code>	A character string enclosed in quotes that defines the subdomain partitioning method. Allowed values are 'zoltan', 'simple', and the name of a partition file ending with '.partition'.	"zoltan"
67	<code>--local-solve-approach</code>	A character string enclosed in quotes that defines the local solve approach. Valid choices are 'jacobi' and 'gauss-seidel'.	"jacobi"
68	<code>--local-tolerance-scaling-cnv</code>	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	0.01
69	<code>--local-tolerance-scaling-mb</code>	A real positive double precision value that sets the convergence tolerance for local solves. Set to lower than 1.0 to use stricter convergence tolerance for local solves.	1
70	<code>--local-well-solve-control-switching</code>	A boolean value that allows (true) or disallows (false) control switching during local well solutions.	false
71	<code>--matrix-add-well-contributions</code>	A boolean value that if set to true explicitly specifies the influences of wells between cells in the Jacobian and preconditioner matrices.	false
72	<code>--max-inner-iter-ms-wells</code>	A positive integer value that defines the maximum number of inner iterations for multi-segment wells.	100
73	<code>--max-inner-iter-wells</code>	A positive integer value that defines the maximum number of inner iterations for standard wells.	50
74	<code>--max-local-solve-iterations</code>	A positive integer value that defines the maximum number of iterations for local solves with NLDD nonlinear solver.	20
75	<code>--max-newton-iterations-with-inner-well-iterations</code>	A positive integer that specifies the maximum Newton iterations with inner well iterations.	8
76	<code>--max-pressure-change-ms-wells</code>	A real positive value that defines the maximum relative pressure change for a	1.0×10^6

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		single iteration of the multi-segment well model.	
77	--max-residual-allowed	A real positive value that sets the absolute maximum tolerance for residuals without cutting the time step size.	1.0×10^7
78	--max-single-precision-days	A real positive value that sets the maximum time step size where single precision floating point arithmetic can be used for solving the linear systems of equations.	20.0
79	--max-temperature-change	A real positive value that stipulates the maximum absolute change of temperature in a single iteration.	5
80	--max-time-step-divisions	A positive integer that specifies the maximum number of divisions by two of the timestep size before the simulation bails out.	10
81	--max-time-step-size	A real positive value that defines the maximum size to which all time steps are limited to in seconds.	inf
82	--max-welleq-iter	A positive integer that defines the maximum number of iterations to determine the solution to the well equations.	30
83	--maximum-number-of-well-switches	A positive integer values that stipulates the maximum number of times a well can switch to the same control.	3
84	--maximum-water-saturation	A real positive value that defines the maximum water saturation.	1.0
85	--milu-variant	A defined character string that specifies which variant of the modified ILU preconditioner ought to be used. Possible variants are: 1) ILU (default, plain ILU), 2) MILU_1 (lump diagonal with dropped row entries), 3) MILU_2 (lump diagonal with the sum of the absolute values of the dropped row entries), 4) MILU_3 (if diagonal is positive add sum of dropped row entries. Otherwise subtract them), 5) MILU_4 (if diagonal is positive add sum of dropped row entries. Otherwise do nothing). The default is "ILU"	"ILU"
86	--min-strict-cnvt-ite	A positive integer that sets the minimum number of Newton iterations before	0

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		relaxed tolerances are used for the CNV convergence criterion.	
87	--min-time-step-based-on-newton-iterations	A real positive value that sets the minimum time step size (in days for field and metric unit and hours for lab unit) can be reduced to based on Newton iteration counts.	0.0
88	--min-time-step-before-shutting-problematic-wells-in-days	A real positive value that sets the minimum time step size in days for which problematic wells are not shut. Time steps below this value will result in problematic wells being shut-in by the simulator.	0.01
89	--min-time-step-size	A real positive value that sets the minimum size to which all time steps are limited to in seconds.	0.0
90	--network-max-iterations	A positive integer that sets the maximum number of iterations in the network solver before giving up.	200
91	--network-max-strict-iterations	A positive integer that sets the maximum iterations in network solver before relaxing tolerance.	100
92	--newton-max-error	A real positive value that sets the maximum error tolerated by the Newton method to which does not cause an abort.	1e+10
93	--newton-max-iterations	A positive integer that defines the maximum number of Newton iterations per time step used by the simulator.	20
94	--newton-max-relax	A real positive value that sets the maximum relaxation factor of a Newton iteration used by the simulator.	0.5
95	--newton-min-iterations	A positive integer that sets the minimum number of Newton iterations per time step used by the simulator. The default value of one ensures that at least one Newton iteration is performed after the previous time step.	1
96	--newton-relaxation-type	A character string. The type of relaxation used by the Newton method. Valid options are: dampen or sor.	"dampen"
97	--newton-target-iterations	A positive integer that sets the 'optimum' number of Newton iterations per time step.	6
98	--newton-tolerance	A real positive value that sets the maximum raw error tolerated by the Newton method for considering a solution to be converged.	0.01
99	--newton-verbose	A boolean value that specifies whether the	true

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		Newton method should inform the user about its progress (true) or not (false).	
100	--newton-write-convergence	A boolean value that specifies whether to write the convergence behaviour of the Newton method to a VTK file (true) or not (false).	false
101	--nlld-num-initial-newton-iter	A positive integer that sets the number of initial global Newton iterations when running the NLDD nonlinear solver.	1
102	--nonlinear-solver	A character string that specifies the nonlinear solver. Valid choices are newton or nlld.	"newton"
103	--num-local-domains	A positive integer that sets the number of local domains for NLDD nonlinear solver. Note this is an experimental feature in the current release 2023.10 that is expected to be more complete and tested by the 2024.10 release.	0
104	--num-pressure-points-equil	A positive integer that sets the number of pressure points (in each direction) in tables used for equilibration.	2000
105	--opencl-ilu-parallel	A boolean value that if set to true then parallelize the ILU decomposition and application on GPU, or not (false).	true
106	--opencl-platform-id	A positive integer that specifies the platform identification ("ID") for the openSolver. Use the "clinfo" command to determine valid IDs.	0
107	--output-dir	A character string that defines the directory to which OPM Flow is to write the commercial simulator compatible output files (restart, summary files etc.). The default value results in the files be written to the same directory as the input file.	""
108	--output-extra-convergence-info	A defined character string that specifies whether to provide additional convergence output to separate files for diagnostic purposes. The available options are: 1) "none" results in no extra output and overrides all other options. 2) "steps" writes out convergence information per time step, to a CASENAME.INFOSTEP file. The file is useful for identifying numerical issues. 3) "iterations" writes out non-linear convergence metrics, i.e., the MB and CNV values, per phase, for each non-	"none"

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		linear iteration in each time step, to a CASENAME.INFOITER file. Options can be combined with commas, e.g."steps,iterations" for multiple outputs. The default value of "none" prevents the two files from being written out, for better compatibility with the commercial simulator.	
109	--output-interval	A positive integer that specifies the number of report steps between two consecutive writes of restart data.	1
110	--output-mode	A defined character string that defines the output to the *.PRT and *.DBG files: 1) "none": No output to the files. 2) "log" or false: Output logging information only. 3) "all" or true: Output everything. For example to just output logging information use: --output-mode="log". or --output-mode=false	"all"
111	--owner-cells-first	A boolean value that determines if the cells owned by a rank should be ordered before ghost/overlapping the cells (true), or not (false).	true
112	--parameter-file	A character string that defines the name of a parameter file which contains the simulator's set of run-time parameters, as listed in this table.	""
113	--parsing-strictness	Set strictness of parsing process. Available options are: 1) "normal": stop for critical errors, and for unsupported keywords that would change the simulator results if supported. 2) "high": stop for all errors, that is even for unsupported keywords that do not effect the results, for example <i>ECHO</i> and <i>NOECHO</i> . 3) "low": same as normal, except do not stop due to unsupported keywords that would change the simulator results if supported, and even if marked critical. Default: "normal"	"normal"
114	--predetermined-time-steps-file	A file with a list of predetermined time step sizes (one time step per line).	""

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
I 15	--pri-var-oscillation-threshold	A real positive value that defines the threshold value for the primary variable switching conditions after its meaning has switched to hinder oscillations.	1.0×10^{-5}
I 16	--print-parameters	A positive integer value that request that the run time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
I 17	--print-properties	A positive integer value that request that the compile time parameters be printed at the start of the run: 0 : No output to the files. 1 : Output *.DBG file. 2 : Output to *.DBG and *.PRT files (default).	2
I 18	--project-saturations	A boolean value that determines if all the fluid saturations should be scaled to ensure the values are in the interval (0, 1), including runs that use solvents (true), or not (false).	false
I 19	--regularization-factor-wells	A real positive value that defines the "regularization factor" for wells.	100
I 20	--relaxed-linear-solver-reduction	A real positive value that defines the minimum reduction of the residual which the linear solver need to achieve for the solution to be accepted.	0.01
I 21	--relaxed-max-pv-fraction	A real positive value that defines The fraction of the pore volume of the reservoir where the volumetric error (CNV) might be violated during strict Newton iterations.	0.03
I 22	--relaxed-pressure-tol-msw	A real positive value that sets the relaxation tolerance for the multi-segment well <u>pressure</u> solution in Pascals.	10000
I 23	--relaxed-well-flow-tol	A real positive value that sets the relaxation tolerance for the well flow residual in reservoir cubic metres (rm3).	0.001
I 24	--restart-time	A real positive value that sets the simulation time at which a restart should be attempted [s].	-1e+35
I 25	--restart-writing-interval	A positive integer value that sets the frequency at which time steps are serialized to disk.	16777215
I 26	--save-file	A character string that specifies the	""

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		FileName for .OPMRST file used for saving serialized state. If empty, CASENAME.OPMRST is used.	
127	--save-step	A character string that determines if the simulator should save the serialized state of the OPM Flow simulator at one or more report steps to a special *.OPMRST file. This is in addition to the normal restart files written, and consumes significantly more space than the normal restart files, but restarting OPM Flow from this file using the <i>--load-step</i> option deviates less from the original run, compared to restarting using the normal restart features. The files produced are not compatible with other simulators, and also will not be compatible between different releases of OPM Flow The parameter should be set to one of the following: 1) "all" to save all report steps, 2) "x" to save every x'th step, or 3) "x" to save a specif time step. The default value of "" does not write anything to the OPM Flow specific restart file.	""
128	--scale-linear-system	A boolean value that switches on (true) or off (false) the scaling of linear system of equations according to the equation scale and primary variable types.	false
129	--sched-restart	A boolean value that determines for a restart run if the case should initialize wells and groups from the historical SCHEDULE section (true), or from the well and group data on the restart file (false). Note that the commercial simulator always uses data from the restart file (false).	false
130	--serial-partitioning	A boolean value that determines if partitioning for parallel runs on a single process (true), or not (false).	false
131	--shut-unsolvable-wells	A boolean value that determines if the simulator should shut unsolvable wells (true), or not (false).	true
132	--solve-welleq-initially	A boolean value that determines if the simulator should fully solve the well equations before each iteration of the reservoir model (true), or not (false). Note that the well equations are always added to the full system and solved until convergence.	true

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
133	--solver-continue-on-convergence-failure	A boolean value that stipulates if the simulator should continue (true) instead of stopping (false) when the minimum solver time step is reached.	false
134	--solver-growth-factor	A real positive value that specifies the growth factor a time step can be increased by when recovering from one or more time step chops, subject to the maximum allowable time step size set by the <code>--solver-max-time-step-in-days</code> parameter. For example, if the current time step has converged at 5 days after a time step chop, and <code>--solver-growth-factor</code> is set to the default value of 2.0, then the next time step will be $2.0 * 5$ days, that is at 10 days.	2.0
135	--solver-max-growth	A real positive value that specifies the maximum growth factor a time step can be increased by after a report time step, subject to the maximum allowable time step size set by the <code>--solver-max-time-step-in-days</code> parameter. Thus, if the current time step has converged at 5 days after at a report step, and <code>--solver-max-growth</code> is set to the default value of 3.0, then the next time step will be $3.0 * 5$ days, that is at 15 days.	3.0
136	--solver-max-restarts	A positive integer that defines the number of allowed consecutive restarts (or time step chops) before the simulation is terminated.	10
137	--solver-max-time-step-in-days	A real positive double precision value that specifies the maximum allowed time step size in days.	365
138	--solver-min-time-step	A real positive double precision value that specifies the minimum size of a time step in days for field and metric units, and hours for laboratory units If a time step cannot converge without getting cut below this time step size the simulator will stop.	1.0×10^{-12}
139	--solver-restart-factor	A real positive double precision value that sets the time step chop factor of the time step after a convergence failure. For example, if the current non-convergent time step is 30 days and <code>--solver-restart-factor</code> is set to the default value, then the time step will be repeated using $0.33 * 30$ days as the time step, that is 9.9 days.	0.33
140	--solver-verbosity	A positive integer that specifies the "chattness" of the non-linear solver.	1

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
I41	--strict-inner-iter-wells	A positive integer that specifies the number of inner well iterations with strict tolerance.	40
I42	--strict-outer-iter-wells	A positive integer that specifies the number of Newton iterations for which wells are checked with strict tolerance.	6
I43	--temperature-max	A real positive value that sets the maximum absolute temperature.	1.0×10^9
I44	--temperature-min	A real positive value that sets the minimum absolute temperature.	0
I45	--threads-per-process	A positive integer value that stipulates the maximum number of threads to be instantiated per process ('-1' means 'automatic').	-1
I46	--time-step-after-event-in-days	A real double precision value that sets the maximum allowed time step after an event; for example, when a well is open or closed etc. The default value of -1 means that events to do effect the time stepping.	-1
I47	--time-step-control	A defined character string that defines the time stepping control algorithm and is set to one of the following: 1) pid: PID controller based adaptive time step control as suggested by Turek and Kuzmin ³⁸⁷ . 2) pid+iteration: Use PID and linear iteration numbers to guide the time step. 3) pid+newtoniteration: Use PID and non-linear iterations numbers to guide the time step. 4) Hardcoded: Use time steps supplied by user via the time-step-control-filename parameter.	"pid+newtoniteration"
I48	--time-step-control-decay-damping-factor	A real positive value that specifies the decay rate a time step can be decreased by after the number target iterations has been exceeded	1.0
I49	--time-step-control-decay-rate	A real positive value that specifies the decay rate a time step size can be decreased by after the number of target iterations has been exceeded	0.75
I50	--time-step-control-file-name	A character string enclosed in quotes that specifies a filename where time steps are	"timesteps"

³⁸⁷ Algebraic Flux Correction III. Incompressible Flow Problems. Uni Dortmund, Turek and Kuzmin, January, 2006 (DOI: 10.1007/3-540-27206-2_8)

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		specified. The default is the character string timesteps. For instance time steps can be generated by the ecl_summary application in libecl as per the following UNIX command line: <pre>path_to_libecl_applications/ ecl_summary DECK TIME > filename</pre> Where: DECK is the name of the data deck you want to get the time steps from, TIME tells the application to return the timing for the run, and "filename" is the name of the file the times are piped to.	
151	--time-step-control-growth-damping-factor	A real positive value that specifies the growth rate of the <u>time step</u> increase when the number of target iterations are undercut.	3.2
152	--time-step-control-growth-rate	A real positive value that specifies the growth rate of the <u>time step size</u> when the number of target iterations are undercut.	1.25
153	--time-step-control-target-iterations	A positive integer that defines the number of linear iterations which the time step control scheme should aim for (if applicable).	30
154	--time-step-control-target-newton-iterations	A positive integer that specifies the number of Newton iterations which the time step control scheme should aim for (if applicable).	8
155	--time-step-control-tolerance	A real double precision value that sets the tolerance for PID (only used with the pid and pid+ options defined by the --time-step-control option).	0.1
156	--time-step-verbosity	A positive integer that specifies the "chattiness" during the time integration.	1
157	--tolerance-cnv	A real positive double precision value that specifies the maximum non-linear tolerance error. This is the local convergence tolerance (maximum of local saturation errors).	0.01
158	--tolerance-cnv-relaxed	A real positive value that defines the relaxed local convergence tolerance that applies for iterations after the iterations with the strict tolerance.	1.0

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
159	--tolerance-mb	A real positive double precision value that sets the maximum mass balance error, that is the tolerated mass balance error relative to total mass present.	1.0×10^{-6}
160	--tolerance-pressure-ms-wells	A real positive double precision value that specifies the tolerance for the pressure equations for multi-segment wells.	1000
161	--tolerance-well-control	A real positive double precision value that sets the maximum tolerance for the well control equations.	1.0×10^{-7}
162	--tolerance-wells	A real positive double precision value that defines the maximum non-linear error for the well equations.	1.0×10^{-4}
163	-update-equations-scaling	A boolean value that switches on (true) or off (false) the updating of the scaling factors for mass balance equations during the simulation.	false
164	--use-average-density-ms-wells	A boolean value that specifies whether to approximate segment densities by averaging over the segment and its outlet (true) or not (false).	false
165	--use-gmres	A boolean value that when set to true OPM Flow will use Generalized Minimal Residual ("GMRES") ³⁸⁸ and ³⁸⁹ solver instead of Biconjugate Gradient Stabilized ("BiCGSTAB") ³⁹⁰ and ³⁹¹ as the linear solver within the Newton iterations.	false
166	--use-multisegment-well	A boolean value that when set to true the simulator will use the well model for multi-segment wells instead of the one for single-segment wells.	true
167	--use-update-stabilization	A boolean value that switches on (true) or off (false) the stabilized Newton option, that attempts to detect and correct oscillations or stagnation during the Newton iterations. This option may improve convergence for some cases.	true
168	--water-only-threshold	A real positive value that defines the saturation threshold, for which cells with	1.0

³⁸⁸ Y. Saad, A flexible inner-outer preconditioned GMRES algorithm, *SIAM J. Sci. Statist. Comput.*, 14, (1993).

³⁸⁹ Y. Saad and M.H. Schultz, GMRES: A generalized minimum residual algorithm for solving nonsymmetric linear systems, *SIAM J. Sci. Statist. Comput.*, 7 (1986), pp. 856-869.

³⁹⁰ Van der Vorst, H. A. (1992). "Bi-CGSTAB: A Fast and Smoothly Converging Variant of Bi-CG for the Solution of Nonsymmetric Linear Systems". *SIAM J. Sci. Stat. Comput.* 13 (2): 631–644. doi:10.1137/0913035. hdl:10338.dmlcz/104566

³⁹¹ Sleijpen, G. L. G.; Fokkema, D. R. (November 1993). "BiCGstab(l) for linear equations involving unsymmetric matrices with complex spectrum" (PDF). *Electronic Transactions on Numerical Analysis*. Kent, OH: Kent State University. 1: 11–32. ISSN 1068-9613.

OPM Flow 2023-10 Command Line Options			
No.	Variable Name	Description	Default
		water saturations above or equal to this threshold are considered one-phase water only.	
169	--zoltan-imbalance-tol	A real positive value that defines the tolerable imbalance of the load balancing provided by Zoltan package.	1.1
170	--zoltan-params	A character string that specifies the configuration of the Zoltan partitioner. Valid options are: graph, hypergraph or scotch. Alternatively, you can request a configuration to be read from a JSON file by giving the filename here, ending with '.json.' See https://sandialabs.github.io/Zoltan/ug_html/ug.html for available Zoltan options.	"graph"
Notes: - Cells colored green in the No. column are new command line parameters for this release. Similarly for the Default column, cells colored green indicate the default value has changed from the previous release. - Items shaded in gray are considered to be developer options that should be used with caution as the values associated with these command line parameters are subject to change. - VTK Graphics Command Line Parameters³⁹² set of commands are no longer listed via the basic help command line command: <code>flow --help</code> instead use: <code>flow --help-all</code> to get a list of supported command line Parameters. - The --enable-vtk-output option above, if set to "true" will write out the data as stipulated by the compile options. One may need to compile the source code to obtain the desired output. - As per all UNIX and LINUX based system the input is case dependent. - If a parameter file is used to set one of the above parameters, the leading "--" should not be incorporated in the parameter file.			

Table E.4: OPM Flow 2023-10 Command Line Options

³⁹² Virtualization ToolKit ("VTK") files used to visualize the 3D results from the model using ParaView, an open-source, multi-platform general purpose data analysis and visualization application (<https://www.paraview.org/overview/>). For the commercial simulator's compatible output files OPM's ResInsight 3D visualization package can be used instead and is specifically tailored for OPM Flow and the commercial simulator.

APPENDIX F: OPM FLOW OUTPUT FILE FORMATS

APPENDIX TABLE OF CONTENTS

APPENDIX F: OPM FLOW OUTPUT FILE FORMATS.....2575

F.1 Overview.....	2580
F.1.1 File Types.....	2580
F.1.2 File Naming Conventions.....	2585
F.1.3 Unformatted File Format Considerations.....	2587
F.1.4 File Output Format.....	2588
F.2 EGRID - Model Structural Data for Irregular Corner-Point Grids File.....	2591
F.2.1 EGRID Header Keywords.....	2593
F.2.2 EGRID Global Irregular Corner Point Grid Keywords.....	2597
F.2.3 EGRID LGR Irregular Corner Point Grid Keywords.....	2601
F.2.4 EGRID Non-Neighbor Connections for Irregular Corner Point Grid Keywords.....	2604
F.3 EGRID - Model Structural Data for Unstructured Grids File.....	2607
F.4 EGRID - Model Structural Data for Mixed Grids File.....	2607
F.5 GRID - Model Structural Data File.....	2607
F.5.1 RESTART Data - LGR Grid Header Keywords.....	2607
F.6 INIT - Model Initialization and Static Data File.....	2610
F.6.1 INIT Index File.....	2610
F.6.2 INIT Data File Specification.....	2611
F.6.3 INIT Data - Static Data Header Keywords.....	2614
F.6.4 INIT Data - Static Grid Array Data Keywords.....	2619
F.6.5 INIT Data - LGR Static Data Header Keywords.....	2622
F.6.6 INIT Data - LGR Grid Array Data Keywords.....	2625
F.6.7 INIT Data - Static Tabular Data Header Keyword.....	2627
F.6.8 INIT Data - Static Tabular and Constant Data Array Keywords.....	2630
F.7 LOG File - OPM Flow Terminal Output.....	2632
F.7.1 LOG File - Header.....	2632
F.7.2 LOG File - Reading Deck.....	2633
F.7.3 LOG File - Processing Data.....	2633
F.7.4 LOG File - Saturation Function Diagnostics.....	2634
F.7.5 LOG File - Main Simulation Loop.....	2635
F.7.6 LOG File - End of Simulation.....	2636
F.8 RESTART Files - Dynamic Data (Solution, Groups, Wells, Connections, Aquifers, etc.).....	2637
F.8.1 RESTART Index Files.....	2638
F.8.2 RESTART Data File Specification.....	2639
F.8.3 RESTART Data - Header Keywords.....	2641
F.8.4 RESTART Data - Group, Well, and Connection Data Keywords.....	2653
F.8.5 RESTART Data - UDQ and ACTIONX Keywords.....	2673
F.8.6 RESTART Data - Aquifer Data Keywords.....	2680
F.8.7 RESTART Data - Hidden Keyword.....	2685
F.8.8 RESTART Data - Solution Data keyword.....	2686
F.8.9 RESTART Data - LGR Solution Data Keywords.....	2694
F.9 RFT Files - Pressure, Saturation and Production Log Data.....	2697
F.9.1 RFT Data File Specification.....	2697
F.9.2 RFT Data File - Time and Date keywords.....	2698
F.9.3 RFT Data File - Well and Connection Data Keyword.....	2699
F.9.4 RFT Data File - RFT Solution Data Keyword.....	2701
F.9.5 RFT Data File - Solution PLT Data Keywords.....	2703
F.9.6 RFT Data File - Solution Multi-Segment Well Keywords.....	2703
F.9.7 RFT Data File - Solution River Keywords.....	2703
F.10 SUMMARY Files - Time Based Vector Data.....	2704
F.10.1 Enhanced SUMMARY Data File Keywords.....	2704
F.10.2 SUMMARY Index File keywords.....	2707
F.10.3 SUMMARY Data File Keywords.....	2712

F.11 SAVE Files - Initialization and Solution Data.....2714

APPENDIX INDEX OF TABLES

Table F.1: OPM Flow Output File Types Summary.....	2583
Table F.2: RUNSPEC Input and Output File Format Keywords.....	2584
Table F.3: OPM Flow File Naming Conventions.....	2586
Table F.4: Big-Endian and Little-Endian Representation.....	2587
Table F.5: Output File Keyword Format.....	2588
Table F.6: INTEHEAD Keyword - IWEL Example.....	2590
Table F.7: EGRID - Model Structural Data for Irregular Corner-Point Grids.....	2593
Table F.8: EGRID Header Keywords.....	2595
Table F.9: EGRID Global Irregular Corner Point Grid Keywords.....	2599
Table F.10: EGRID LGR Irregular Corner Point Grid Keywords.....	2602
Table F.11: EGRID NNC Keywords for Irregular Corner Point Grids.....	2605
Table F.12: RESTART Data - LGR Grid Header Keywords.....	2608
Table F.13: INIT Data File Structure.....	2613
Table F.14: INIT Data - Static Data Header Keywords.....	2617
Table F.15: INIT Data - Static Grid Array Data Keywords.....	2621
Table F.16: INIT Data - LGR Static Data Header Keywords.....	2623
Table F.17: INIT Data - LGR Grid Array Data Keywords.....	2625
Table F.18: INIT Data - Static Tabular Data Header Keyword.....	2629
Table F.19: INIT Data - Static Tabular and Constant Data Array Keywords.....	2630
Table F.20: RESTART Data File Structure.....	2640
Table F.21: RESTART Data - Header Keywords.....	2651
Table F.22: RESTART Data - Group, Well, and Connection Keywords (Groups).....	2658
Table F.23: RESTART Data - Group, Well, and Connection Keywords (Multi-Segment Wells).....	2661
Table F.24: RESTART Data - Group, Well, and Connection Keywords (Wells).....	2667
Table F.25: RESTART Data - Group, Well, and Connection Keywords (Connections).....	2669
Table F.26: RESTART Data - Group, Well, and Connection Keywords (Tracers).....	2670
Table F.27: RESTART Data - Group, Well, and Connection Keywords (Network).....	2671
Table F.28: RESTART Data - UDQ Keywords.....	2675
Table F.29: RESTART Data - ACTIONX Keywords.....	2678
Table F.30: RESTART Data - Analytical Aquifer Data Keywords.....	2682
Table F.31: RESTART Data - Numerical Aquifer Data Keywords.....	2683
Table F.32: RESTART Data - Hidden Keyword.....	2685
Table F.33: RESTART Data - Solution Data keyword.....	2686
Table F.34: RESTART Data - Solution Data Keyword for Tracer Concentration Name.....	2687
Table F.35: RESTART Data - Solution Data Keywords.....	2692
Table F.36: RESTART Data - LGR Solution Data Keywords.....	2694
Table F.37: RFT Data File Format.....	2698
Table F.38: RFT Data File - Time and Date Keywords.....	2698
Table F.39: RFT Data File - Well and Connection Data Keyword.....	2700
Table F.40: RFT Data File - RFT Solution Data Keywords.....	2701
Table F.41: Enhanced SUMMARY File Keywords.....	2706
Table F.42: SUMMARY Index File Keywords (Global).....	2709
Table F.43: SUMMARY Index File Keywords (LGR).....	2710
Table F.44: SUMMARY Index File Keywords (Commercial Simulator).....	2710
Table F.45: SUMMARY Data File Keywords.....	2712
Table F.46: SUMMARY Data File Keywords (LGR).....	2713

F.1 OVERVIEW

This section of the manual describes the various output files generated by OPM Flow and attempts to outline the format of the various files, in order for third party software to read and write these type of files. From the 2019-04 release a substantial effort has been undertaken to make OPM Flow's output files compatible with the commercial simulator's output files. This compatibility enables OPM Flow to "restart" from the commercial simulator's generated [RESTART](#) files as well as the commercial simulator to "restart" from the OPM Flow's [RESTART](#) files. Although, this might not be the case for all models, as restarting a simulation model from another simulators restart file is complicated and may not be possible in all cases.

Where applicable, files written by OPM Flow can be loaded into OPM ResInsight post processing software for further analysis and for displaying the results. Please see section [2.2 Running OPM Flow From The Command Line](#) for the various command line options for setting the output format type.

F.1.1 FILE TYPES

OPM Flow, similar to the commercial simulator, writes out various files, some of which are used by post processing software (OPM ResInsight) and some that are directly used by the user, for example the *.PRT file that contains various reports. Table F.1 summaries the various file formats and the status of the file formats currently supported by OPM Flow.

File Type	Data Type	Description	OPM Flow Status
DATA	Input Data	DATA files contain the input data in ASCII format used to run OPM Flow.	Fully Supported, as outlined in the manual
DBG	Debug Data	This file contains ASCII developer debug output specific to OPM Flow, that is there no compatibility with the commercial simulator's DBG file.	OPM Flow Specific
EGRID	Structure Data	EGRID files contain the structural information for the model via the COORD , ZCORN , etc. keywords, and employs the extensible GRID format, which is more compact than the GRID formatted files and is the only format supported by OPM Flow. The output for this file type must be requested via the GRIDFILE – Set the Grid File Output Options in the GRID section. The data is used in post-processing software to view the grid and is not in itself used by OPM Flow.	Fully Supported
GRID	Structure Data	This file type contains the structural information for the model via the COORD , ZCORN , etc. keywords, and consists of two data formats: 1) The standard GRID file format that contains only the active global cells. 2) The extended GRID file format that contains (1) plus local grid refinements and local coarsening grid specifications. Neither of these two formats are currently supported by OPM Flow.	Not Supported
INIT Index	Static Property Index	The INIT index file type specifies and defines the format and data type written to the INIT Data file. This file is not required by OPM Flow or OPM ResInsight to be able to read and write the INIT data file.	Not Supported and Not Required

File Type	Data Type	Description	OPM Flow Status
<i>INIT</i> Data	Static Property Data	This file type contains static model properties, that is, 1) Grid property data: porosity (<i>PORO</i> array), permeability (<i>PERMX</i>, <i>PERMY</i>, and <i>PERMZ</i>), net-to-gross (<i>NTG</i>), etc. 2) Region allocation array data: <i>EQLNUM</i>, <i>FIPNUM</i>, <i>PVTNUM</i>, <i>SATNUM</i>, etc. 3) Fluid and rock property static data, including end-point scaling arrays: <i>SGCR</i>, <i>SGL</i>, <i>SOWCR</i>, <i>PCW</i>, etc. 4) Fluid and rock property tables: <i>PVDG</i>, <i>PVTO</i>, <i>PCW</i>, <i>SGFN</i>, <i>SWFN</i>, etc. In order to request this type of output see the <i>INIT – Activate the INIT File Option</i> in the <i>GRID</i> section. Activating this keyword results in both the <i>INIT</i> static property index and data files being written out by the commercial simulator. OPM Flow only writes out this file as the index file is not required by OPM ResInsight. The data is used in post-processing software to view the grid property data and is not in itself used by OPM Flow.	Fully Supported
LOG	Output File	The LOG file for the commercial simulator contains an ASCII copy of the output from the terminal. The file itself is not generated by the commercial simulator but by an auxiliary program, ECLRUN, that is used to call and execute all the commercial simulators. OPM Flow does not generate a LOG file; however, the file can be generated by using the Linux <i>tee</i> command, for example: <pre>flow CASENAME tee CASENAME.LOG</pre> Which will copy the terminal output to the CASENAME.LOG file. Alternatively, one can use the OPMRUN, the Graphical User Interface (“GUI”) program for OPM Flow, that provides similar functionality to the commercial simulator’s auxiliary program.	OPM Flow Specific
PRT	Output File	This file contains ASCII user output reports specific to OPM Flow, although there is a strong desire to make the OPM Flow reports similar to those that are produced by the commercial simulator.	OPM Flow Specific
<i>RESTART</i> Index	Dynamic Solution Index	The <i>RESTART</i> index file type specifies and defines the format and data type written to the <i>RESTART</i> Data file. This file is not required by OPM Flow or OPM ResInsight to be able to read and write the <i>RESTART</i> data file.	Not Supported and Not Required

File Type	Data Type	Description	OPM Flow Status
RESTART Data	Dynamic Solution Data	This file type contains dynamic solution data for each active cell in the model at the user requested time steps, that is, 1) Grid solution pressure and saturation data (PRESSURE, SGAS, SOIL, and SWAT). Note that Ternary saturations are normally computed by the post-processing software. 2) Grid solution fluid property data (GAS_DEN, GAS_VISC, OIL-DENS, etc.). The RPTRST and the RPTSCHED keywords in the SCHEDULE defines the data and frequency of the data to be written to the RESTART file at each requested restart point. Activating this keyword results in both the RESTART index and data files being written out by the commercial simulator; OPM Flow only writes this file. In addition to the solution data, this file type also includes group and well configuration data (number of connections, connections open and closed, etc.) This information is required to ensure that the correct group and well configuration is available at a given time in order to be able to restart the simulation at a restart point. The data is used to visualize the simulation results of the model in two and three dimensional space using post-processing software. The file is also used by OPM Flow to “restart” from a previous simulation case.	Fully Supported
RFT	Dynamic Wellbore Vector Data	Data written to the RFT file consists of wellbore vector data, for example, pressure and saturation versus wellbore depth at various time steps. The data written out is not restricted to just Repeat Formation Tester (“RFT”) data, but can contain any Production Logging Tool (“PLT”) data made available in the simulator. Note that only the RFT data set is currently supported by OPM Flow. The keywords WRFT and WRFTPLT in the SCHEDULE section are used to define the wells, data, and time step at which the data should be written to the RFT file. The data is used in post-processing software to compare the actual wire line logging data with the simulation derived results. The file is not in itself used by OPM Flow.	Fully Supported (RFT)
RSM	Output File	The RSM file contains the variables requested to be written to the SUMMARY file via the keywords described in the SUMMARY SECTION, in a tabulated ASCII output format. The format of the file enables the data to be easily loaded into the LibreOffice Calc program for further processing, as each vector represents one column. The report is written at the end of the simulation run by parsing the SUMMARY Index and SUMMARY Data files.	Fully Supported
SAVE	Static and Dynamic Data	SAVE files are basically a combination of both the INIT and RESTART files, except for the static initialization data that also includes the processed data, e.g., PORV, TRANX, TRANX, TRANZ, etc. This enables “fast restarts” as the pore volumes and transmissibility arrays do not have to be re-calculated in a “restart” run. This file type is currently not supported by OPM Flow.	Not Supported
SUMMARY ESMRY	Dynamic Vector Data	This is an OPM FLOW specific Enhanced SUMMARY (*.ESMRY) output file format that is optimized for fast loading of selected vectors by post-processing applications. The *.ESMRY file is re-written for every time step and does not have a performance impact on the simulation. Load times are in the order of 30 times faster compared with fetching selected vectors using the *.UNSMRY file format in post processing software.	Fully Supported

File Type	Data Type	Description	OPM Flow Status
<i>SUMMARY</i> Index	Dynamic Vector Index	The <i>SUMMARY</i> index file type specifies and defines the format and data type written to the <i>SUMMARY</i> Data file.	Fully Supported
<i>SUMMARY</i> Data	Dynamic Vector Data	The <i>SUMMARY</i> data file contains the variables requested to be written to the file via the keywords described in the <i>SUMMARY SECTION</i>. The data consists of vectors that are used to generate line graphs of properties such as oil flow rate versus time, grid plot pressure versus time, etc. The properties to be stored on the <i>SUMMARY</i> file are written to the summary file at the end of each successful time step. The data can be used to compare actual production data with the simulation derived results in post processing software. Note that although the file format is fully supported, not all of the <i>SUMMARY</i> variables are available.	Fully Supported
Notes: 1) All files can be written out in either ASCII or binary formats, except for DBG, LOG, PRT, and RSM files that are always written in ASCII format. 2) In addition, <i>SUMMARY</i> and <i>RESTART</i> files can be loaded or written out in either unified or non-unified formats. 3) File Type cells colored in gray indicate the output may be different to the commercial simulators equivalent file type. File Type cells colored in orange represent file types that are not currently supported by OPM Flow.			

Table F.1: OPM Flow Output File Types Summary

As mentioned in Table F.1 all files can be written out in either ASCII or binary formats. In addition, the *SUMMARY* and *RESTART* files can be loaded or written out in either unified or non-unified formats. The file type (ASCII or binary) and file structure format (unified or non-unified formats) are set via various keywords in the *RUNSPEC* section, as outlined in Table F.2.

Process	<i>RUNSPEC</i> Keyword	Description	Files
Input	<i>FMTIN</i>	The keyword defines the input files to be formatted as ASCII, i.e., text files, as oppose to binary files. The input deck file is always of this type. The option relates to the OPM Flow derived files that are used as input, for example when restarting from another case. <u>If the keyword is omitted then the default is for binary file input.</u>	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTIN</i>	A character string that defines the input files to be non-unified multiple files, as opposed to unified files. In this case, one file is read in per reporting time step, as opposed to all time steps reports being read from one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X0001 *.SMSPEC *.S0001

Process	<i>RUNSPEC</i> Keyword	Description	Files
	<i>UNIFIN</i>	<i>UNIFIN</i> defines the input files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. For this option a single summary file and a single restart file will be read. <u>If the keyword is omitted then the default is for one file per report time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Output	<i>FMTOUT</i>	The keyword sets all output files to be formatted as ASCII i.e. text files, as opposed to binary files. The *.DBG, *.LOG, *.PRT, and *.RSM files are always of this type. The option relates to the OPM Flow output files only. In this case the files will be portable across operating systems, but will also be very large in terms of hard disk space. For this reason it is recommend that the default option is used so that only binary files are written out. <u>If the keyword is omitted then the default is for binary file input.</u>	*.FEGRID *.FINSPEC *.FINIT *.FRSSPEC *.FUNRST *.FSMSPEC *.FUNSMRY
	<i>MULTOUT</i>	A character string that defines the output files to be non-unified multiple files, as opposed to unified files. In this case, one file is written for each reporting time step, as opposed to all time steps reports being written in one file. <u>If the keyword is omitted then the default is for one file per reporting time step.</u>	*.RSSPEC *.X0001 *.SMSPEC *.S0001
	<i>UNIFOUT</i>	<i>UNIFOUT</i> defines the output files to be unified files, as opposed to non-unified multiple files. A unified file is a single file containing output for each reporting time step. Here a single summary file and a single restart file will be generated, as opposed to one file per report time step. <u>If the keyword is omitted then the default is for one file per report time step.</u>	*.RSSPEC *.UNRST *.SMSPEC *.UNSMRY
Notes: 1) A binary file is computer-readable but not human-readable. All executable programs are stored in binary files, as are most numeric data files. In contrast, text files are stored in a form (usually ASCII) that is human-readable. 2) For unified files if the run terminates unexpectedly, or there is sufficient disk space, then the last report output is not stored. The main advantage of unified files is that if a number of simulation cases reside in one directory, the number of output files per case is minimum compared to using non-unified multiple files. There is no limit on the number of reporting steps that a unified file can store.			

Table F.2: RUNSPEC Input and Output File Format Keywords

F.1.2 FILE NAMING CONVENTIONS

OPM Flow automatically generates the file names based on the input file name and the output options selected via the keywords in the [RUNSPEC](#) summarized in Table F.2. For example, the command line syntax for running OPM Flow is:

```
flow [OPTIONS] CASENAME
```

and typing the following command from the terminal:

```
flow CASENAME
```

will start the simulator and run the case specified by CASENAME.DATA. Here CASENAME is the “root” of the filename and DATA is the extension of the filename. OPM Flow will generate the output files based on the CASENAME with the extension based on the type (ASCII or binary) and file structure format (unified or non-unified formats), as outlined in Table F.3. Note that if CASENAME includes a relative or absolute path to the data file, the run directory will be set so that all output files are generated in the directory holding the data file.

File Type	Unformatted Extension	Description	Formatted Extension
DATA		Input data file.	*.DATA
DBG		Output debug file.	*.DBG
EGRID	*.EGRID	EGRID files containing the structural grid data and is the only format supported by OPM Flow. This output for this file type must be requested via the GRIDFILE – Set the Grid File Output Options in the GRID section.	*.FEGRID
GRID	*.GRID	Grid file in either the standard GRID file format or the extended GRID file format Neither of these two formats are currently supported by OPM Flow.	*.FGRID
INIT Index	*.INSPEC	The INIT index file for both unified and non-unified formats	*.FINSPEC
INIT	*.INIT	This file type contains static model properties.	*.FINIT
LOG		Output log file that contains a copy of the output from the terminal. The file itself is not generated by the commercial simulator but by an auxiliary program, ECLRUN, that is used to call and execute all the commercial simulators.	*.LOG
PRT		Output print file.	*.PRT
RESTART Index	*.RSSPEC	The RESTART index file for both unified and non-unified formats. This file is not required by OPM Flow or OPM ResInsight to be able to read and write the RESTART data file.	*.FRSSPEC

File Type	Unformatted Extension	Description	Formatted Extension
<i>RESTART</i> Data Non-Unified	*.Xnnnn	The <i>RESTART</i> data files containing the solution arrays requested to be written to the <i>RESTART</i> files. For example: CASENAME.X0001 CASENAME.X0002 CASENAME.X0003 etc. CASENAME.RSSPEC	*.Fnnnn
<i>RESTART</i> Data Unified	*.UNRST	The <i>RESTART</i> data file containing the solution arrays requested to be written to the <i>RESTART</i> file. For example: CASENAME.UNRST CASENAME.RSSPEC	*.FUNRST
RFT	*.RFT	The RFT data file containing wellbore vector data requested to be written to the RFT file. For example: CASENAME.RFT	*.FRFT
RSM		Output RSM file.	*.RSM
<i>SAVE</i>	*.SAVE	The <i>SAVE</i> file type is currently not supported by OPM Flow.	*.FSAVE
<i>SUMMARY</i> ESMRY	*.ESMRY	OPM Flow Enhanced <i>SUMMARY</i> (*.ESMRY) output file format that is optimized for fast loading of selected vectors by post-processing applications. Note that the <i>FMTOUT</i> , <i>MULTOUT</i> , and <i>UNIFOUT</i> keywords have no impact on this file type.	
<i>SUMMARY</i> Index	*.SMSPEC	The <i>SUMMARY</i> index file for both unified and non-unified formats.	*.FSMSPEC
<i>SUMMARY</i> Data Non-Unified	*.Snnnn	The <i>SUMMARY</i> data files containing the variables requested to be written to the <i>SUMMARY</i> files. For example: CASENAME.S0001 CASENAME.S0002 CASENAME.S0003 etc. CASENAME.SMSPEC	*.Annnn
<i>SUMMARY</i> Data Unified	*.UNSMRY	The <i>SUMMARY</i> data file containing the variables requested to be written to the <i>SUMMARY</i> file. For example: CASENAME.UNSMRY CASENAME.SMSPEC	*.FUNSMRY
Notes: 1) The above file naming convention is for Linux type operating systems, as OPM Flow is currently only officially supported for Linux distributions. 2) File Type cells colored in gray indicate the output may be different to the commercial simulators equivalent file type. File Type cells colored in orange represent file types that are not supported by OPM Flow. 3) Formatted and Unformatted Extension cells colored in gray indicate the format is not available or not applicable.			

Table F.3: OPM Flow File Naming Conventions

F.1.3 UNFORMATTED FILE FORMAT CONSIDERATIONS

Originally the commercial simulator was written in FORTRAN 77 and the current version is still written in a version of FORTRAN. FORTRAN unformatted file output is dependent on the FORTRAN compiler used to generated the executable program. Typically an unformatted record consists of a four byte prefix outlining the length of the record in bytes, then the actual data record, followed by suffix containing the length of the record in bytes. Most but not all compilers use four bytes. This aids in reading records, for example, the four byte record size at the end of the record assists with a backspace operation. If the record size is greater than two Gigabytes (2^{30} bytes), the record is divide into sub-records. In this case the sign bit of the prefix inform that the record is continued by a sub-record or not and the sign bit of the suffix indicate whether or not there is a preceding sub record.

OPM Flow unformatted files are written using the big-endian mode, that is the most significant value in the sequence is stored at the lowest storage address, that is first. This is opposite to the little-endian mode, where the least significant value in the sequence is stored first. For example, consider the number 1025 (2 to the tenth power plus one) stored in a four byte integer:

00000000 00000000 00000100 00000001

Address	Big-Endian Representation Of 1025	Little-Endian Representation Of 1025
00	00000000	00000001
01	00000000	00000100
02	00000100	00000000
03	00000001	00000000

Table F.4: Big-Endian and Little-Endian Representation

OPM Flow is written in C++ using the open source GNU C++ compiler and therefore developers using C++ need to ensure that unformatted files are read or written using the correct big-endian mode.

F.1.4 FILE OUTPUT FORMAT

All the files consist of a series of “header” keywords that defined various parameters based on their format type: integer, real, Boolean, double precision, and character. For example for the *INIT* and *RESTART* files the header keywords are:

- 1) SEQNUM: Report time step keyword (*RESTART* file only).
- 2) INTEHEAD: Integer (four byte) keyword that contains constant values for example, the dimensions of the model, as well as the location and length of data contained in the subsequent property keywords.
- 3) LOGIHEAD: Boolean or logical (four byte) keyword that consists of a set of either T (true) or F (false) character strings, for example, if dissolved gas is present in the model then set to T for true or F for false.
- 4) DOUBHEAD: Real double precision keyword (eight byte) that contains real values, for example, the current time step and *TUNING* keyword parameters. Note that the integer parameters on the *TUNING* keyword are stored in the INTEHEAD keyword data array.

These are then followed by a series of “property” keywords that outline the various additional data. For the *RESTART* files the header keywords are repeated for each time step a *RESTART* record is written. There are also additional header keywords for Local Grid Refinements (“LGR”) that are repeated for each *LGR* and for time step.

All the keywords follow the same format, including the property keywords, as outlined in Table F.5, which shows a portion of the INTEHEAD header keyword definition for the *RESTART* file.

No.	Description	Output File Keyword Format			Status Or Value
		Keyword	No. of Entries	Data Type	
	INTEHEAD	Global grid header keyword for the <i>RESTART</i> Data file that defines the integer variables for this time step. NIHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model. Undefined values should be set to zero.			Required
1-1	Format	INTEHEAD	NIHEAD	INTE	
2-1	Data	An encoded integer corresponding to the time the file was created. For files not originating from the commercial simulator, this value may be set to zero.			ISNUM
2-2	Data	Simulator version, should be set to zero.			VERSION
2-3	Data	Unit system used in the simulation, set to 1 for metric, 2 for field, and 3 for laboratory			
2-4	Data	Undefined.			0
Notes: 1) Rows shaded in gray indicate the start of a keyword. 2) Cells colored red in the No. column indicate that this item is either “Not Used” or “Undefined”.					

Table F.5: Output File Keyword Format

The row colored gray in the table specifies the start of a keyword definition and shows the format for the keyword (INTEHEAD) together with a description of the keyword. The next row, item 1-1 in the No. column, outlines the format of the keyword, that is the keyword name (INTEHEAD), the number of data entries for the keyword (NIHEAD), and the type of entries (INTE for integers). The next set of rows specify the individual data entries, for example, item 2-3 in the No. column is the third integer value that declares the

unit system for the file. Note that Cells colored red in the No. column indicate that this item is either “Not Used” or “Undefined”.

Note

Under the No. column in Table F.5 the count base is one (offset one), as per FORTRAN; however, in C++ the base is zero (offset zero) and therefore C++ programmers must subtract one from this column to obtain the correct reference.

This reference adjustment should be applied to all references in this Appendix.

OPM Flow is written in C++, then developers should use offset zero in reviewing or modifying the code.

The Data Type variable in row I-I in Table F.5 should have one of the following values:

- 1) CHAR for eight character words enclosed in single quotes (ASCII character decimal 39) for formatted output.
- 2) DOUB for double precision eight byte real values.
- 3) INTE for standard four byte integers.
- 4) LOGI for standard four byte Boolean (logical) values.
- 5) MESS keyword that has no associated data.
- 6) REAL for single precision four byte reals values.

All keywords follow the same format including the property array like *PORO* and *PRESSURE*. So for example, the INTEHEAD keyword would be like this in a typical formatted *RESTART* file:

```
'INTEHEAD'      249 'INTE '
-955283513      200400      2      -2345      -2345      -2345
      -2345      -2345      10      10      1      100
      10      -2345      7      -2345      0      1
      0      2      2      0      0      0
      110      108      109      3      97      93
      -2345      -2345      19      38      53      -2345
      97      93      146      5      0      1
      15      24      8      5      2      4
```

And the pressure and water saturation arrays would be of the same form in a typical formatted *RESTART* file:

```
'PRESSURE'      44431 'REAL '
0.26889725E+03  0.26838983E+03  0.26826810E+03  0.26820352E+03
0.26814493E+03  0.26809378E+03  0.26807767E+03  0.26806363E+03
0.26802625E+03  0.26798474E+03  0.26795001E+03  0.26791434E+03
0.26787915E+03  0.26783920E+03  0.26777118E+03  0.26769208E+03

'SWAT'          44431 'REAL '
0.10500000E+00  0.10500000E+00  0.10500000E+00  0.10500000E+00
0.10500000E+00  0.10500000E+00  0.10500000E+00  0.14000000E+00
0.14000000E+00  0.14500000E+00  0.14500000E+00  0.14500000E+00
0.14500000E+00  0.14500000E+00  0.16000000E+00  0.16000000E+00
```

Secondly, the No. Entries is the number of values following the keyword and as such cannot be used to determine the size of an array that is not associated with the grid, say for example the number of active

connections in a well. Instead, the array size is given in another property keyword. In this case, the number of elements for each well in the IWEL property array is given in the INTEHEAD keyword as shown in Table F.6 (see Table F.21 for a full description of the [RESTART](#) INTEHEAD keyword).

No.	Keyword Name	RESTART Data - Header Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	INTEHEAD	Global grid header keyword for the RESTART Data file.			Required
1-1	Format	INTEHEAD	NIHEAD	INTE	
2-17	Data	Number of wells in the model.			NWELLS
2-25	Data	Number of values per well in the IWEL array (default value is 97).			NIWELZ

Table F.6: INTEHEAD Keyword - IWEL Example

Now if INTEHEAD(17) was equal to three and INTEHEAD(5) was equal to five then the IWEL property array for the three wells would have the data arranged as shown in Figure F.1.

IWEL Property Array Data Layout INTEHEAD(17) = 3 and INTEHEAD(25)= 5														
Well-1					Well-2					Well-3				
Item 1	Item 2	Item 3	Item 4	Item 5	Item 1	Item 2	Item 3	Item 4	Item 5	Item 1	Item 2	Item 3	Item 4	Item 5
Pos 1	Pos 2	Pos 3	Pos 4	Pos 5	Pos 6	Pos 7	Pos 8	Pos 9	Pos 10	Pos 11	Pos 12	Pos 13	Pos 14	Pos 15

Figure F.1: IWEL Property Keyword Example

Here the Items are the values affiliated with a specific well, and Pos(ition) is the actual position in the IWEL property array, shown as gray boxes in Figure F.1.

Similarly for well connections and completions. In this case the maximum number of connections for all the wells is given in INTEHEAD(18) element and a well's current number of connections is given via the IWEL(5) element in the IWEL property array (see Table F.27 for a detailed description of the IWEL keyword).

Now if the number of wells equal two (INTEHEAD(17)) and INTEHEAD(18) is equal to four, that is the maximum number of connections is set to four, together with IWEL(5) equal to three for the first well and four for the second well, the ICON property array for the two wells would have the data arranged as shown in Figure F.2.

ICON Property Array Data Layout

INTEHEAD(17) = 2 and INTEHEAD(18) = 4

Well-1 with IWEL(5) = 3 and Well-2 with IWEL(5) = 4

ICON Array																										
Well-1												Well-2														
Conn-1			Conn-2			Conn-3			Conn-4			Conn-1			Conn-2			Conn-3			Conn-4					
I	J	K	I	J	K	I	J	K	0	0	0	I	J	K	I	J	K	I	J	K	I	J	K	I	J	K
Position																										
1	2	3	4	5	6	7	8	9	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2		
									0	1	2	3	4	5	6	7	8	9	0	1	2	3	4			

Figure F.2: ICON Property Keyword Example

Here the connection co-ordinates are given by (I, J, K) and the actual position in the ICON array is shown in the gray colored boxes. Note that Well-1 has the required maximum number of connections but the fourth connection has the default values of zero, as the IWEL(5) value for this well only has three active connections.

Note also that the number of data elements per connection in the ICON vector is given by NICONZ which is given by INTEHEAD(33) and that the size of the ICON vector is NICONZ x NCWMAX x NWELLS, where NCWMAX is set by INTEHEAD (18) and NWELLS by INTEHEAD(17). In the example, NICONZ was assumed to be three for illustrative purposes, while the actual value is a constant 25 in the most recent versions of the commercial simulator. These arbitrary constants, like NICONZ, use to determine vector lengths, appear to be constant for various versions of the commercial simulator.

The following sections outline the format of the various individual files supported by OPM Flow, except for the DBG, LOG, PRT, and RSM ASCII format files.

F.2 EGRID - MODEL STRUCTURAL DATA FOR IRREGULAR CORNER-POINT GRIDS FILE

EGRID files contain the structural information for the model via the [COORD](#), [ZCORN](#), etc. keywords, and employs the extensible [GRID](#) format, which is more compact than the [GRID](#) formatted files and is the only format supported by OPM Flow.

The output for this file type must be requested via the [GRIDFILE – Set the Grid File Output Options](#) in the [GRID](#) section, as shown below:

```
--
--      GRID FILE OUTPUT OPTIONS
--      GRID      EGRID
--      OPTN      OPTN
GRIDFILE
      0          1
```

The above example defines that no [GRID](#) file will be written out and that the extensible [GRID](#) (that is the EGRID geometry format) file will be produced. This is the only configuration that OPM Flow supports.

The [FMTIN](#) and [FMTOUT](#) keywords in the [RUNSPEC](#) section defines the input and output files to be formatted as ASCII i.e. text files, (*.FEGRID), as oppose to unformatted binary files (*.EGRID). If these keywords are omitted then the default is for unformatted binary file input.

The EGRID file structure consists of a series header keywords that define the header keyword data type, the number of data entries, and then the associated header keyword data. There are a total of four major keyword header types for this file type: (1) File Header, (2) Global Grid Header, (3) Local Grid Refinement Header³⁹³ (repeated for each local grid refinement in the model), and (4) Non-Neighbor Connection Header.

In addition to the structural data, this file type also contains information describing the geometry relationship between the global grid and any Local Grid Refinements (“LGR”) that are present in the model, together with the Non-Neighbor Connections (“NNC”) that may be present due to structural discontinuities. The overall structure of this file is outlined in Table F.7

Reference Section	Global Section Keywords	Global Data Keywords	LGR Section Keywords	LGR Data Keywords	Notes
F2.1	FILEHEAD MAPUNITS MAPAXES GRIDUNIT GDORIENT				EGRID header keyword that defines various Global Grid properties.
F2.2	GRIDHEAD	COORD ZCORN etc.			EGRID global irregular corner point grid keyword that defines the grid geometry.
F2.2	ENDGRID				Marks the end of the EGRID global irregular corner point grid keyword.
F2.3			LGR LGRPARNT LGRSGRID		EGRID LGR irregular corner point grid keyword that defines the LGR name and optionally the nested or parent LGRs associated with this LGR.
F2.3			GRIDHEAD	COORD ZCORN etc.	EGRID LGR irregular corner point grid keyword that defines the grid geometry.
F2.3			ENDGRID		Marks the end of the EGRID LGR irregular corner point grid keyword.
F2.3			ENDLGR		Sets the end of current LGR section.
F2.3			LGR LGRPARNT LGRSGRID		EGRID LGR irregular corner point grid keyword that defines the LGR name and optionally the nested or parent LGRs associated with this LGR
F2.3			GRIDHEAD	COORD ZCORN etc.	EGRID LGR irregular corner point grid keyword that defines the grid geometry.
F2.3			ENDGRID		End of the EGRID LGR grid section.
F2.3			ENDLGR		Sets the end of current LGR section.

³⁹³ Several keywords are used for both global and LGR grids, for example the GRIDHEAD, COORD, ZCORN keywords.

Reference Section	Global Section Keywords	Global Data Keywords	LGR Section Keywords	LGR Data Keywords	Notes
F.2.4	NNCHEAD	NNC1 NNC2			EGRID Global Non-Neighbor Connections for Irregular Corner Point Grid keywords.
F.2.4			NNCHEAD	NNC1 NNC2 NNCL NNCG	First <i>LGR</i> EGRID global and local non-neighbor connections for irregular corner point grid keywords.
F.2.4			NNCHEAD	NNC1 NNC2 NNCL NNCG	Second <i>LGR</i> EGRID global and local non-neighbor connections for irregular corner point grid keywords.
F.2.4			NNCHEADA	NNA1 NNA2	EGRID amalgamated and local non-neighbor connections for irregular corner point grid keywords.
Notes: 1) Cells shaded in gray indicate the combination is not applicable and cells colored orange indicate that this item is not currently supported by OPM Flow.					

Table F.7: EGRID - Model Structural Data for Irregular Corner-Point Grids

Each major header keyword is subdivided into a series of sub-header keywords that define a particular data set. The general format for a header keyword is the keyword (enclosed in single quotes), followed by the number of data entries (an integer value), followed by the data type which is set to CHAR, INTE, or REAL (enclosed in quotes). This will then be followed by the data for the keyword.

The following sections describe the format of the various keywords and the associated data sets.

F.2.1 EGRID HEADER KEYWORDS

The FILEHEAD keyword defines the key attributes of the file via a series of integer constants, this is then followed by the *MAPUNITS*, *MAPAXES*, *GRIDUNIT*, and *GDORIENT* keywords and their data sets. Note that some keywords are optional and are not used by OPM Flow per se, but are used by pre- and post processing software. Secondly, some features are not supported by OPM Flow, for example local grid refinements, and thus OPM will not write out the data associated with these features.

Table F.8 outlines the structure of the EGRID Header Keywords and their affiliated data.

No.	Keyword Name	EGRID Header Keyword Table F.8			Status Or Value
		Keyword	No. of Entries	Data Type	
	FILEHEAD	This keyword marks the start of the EGRID file and defines various parameters associated with this file type.			Required
1-1	Format	FILEHEAD	100	INTE	
2-1	Data	Version number for this file type, for example 3.			3

No.	Keyword Name	EGRID Header Keyword Table F.8			Status Or Value
		Keyword	No. of Entries	Data Type	
2-2	Data	The year this version of the file format was released, for example 2004.			2004
2-3	Data	Not used.			
2-4	Data	Version number of earliest this file format is supported, normally set to 0.			0
2-5	Data	Type of grid, set to 0 for Irregular Corner-Point Grids, 1 for Unstructured Grids, or 2 for a mixture of Irregular Corner-Point and Unstructured Grids (Mixed Grids).			0
2-6	Data	Type of rock model, set to 0 for a single porosity/permeability rock model, 1 for dual porosity rock model, or 3 for a dual permeability rock model.			0
2-7	Data	Original grid format, set to 0 for unknown, 1 for Irregular Corner-Point Grids, or 2 for Cartesian Regular Grids.			
2-8 to 2-100	Data	Not used			0
	MAPUNITS	This keyword defines the MAPUNITS.			Optional
1-1	Format	MAPUNITS	1	CHAR	
2-1	Data	The units of the map data, normally set to FEET or METRES.			
	MAPAXES	The MAPAXES keyword defines the grid position relative to the map coordinates. The six values in the data fields define the relative map (x, y) coordinates for three locations to enable conversion from grid to map coordinate in pre- and post processing software.			Optional
1-1	Format	MAPAXES	6	REAL	
2-1	Data	X-coordinate on the y-axis located at the end of the y-axis.			
2-2	Data	Y-coordinate on the y-axis located at the end of the y-axis.			
2-3	Data	X-coordinate at the origin.			
2-4	Data	Y-coordinate at the origin.			
2-5	Data	X-coordinate on the x-axis located at the end of the x-axis.			
2-6	Data	Y-coordinate on the x-axis located at the end of the x-axis.			
	GRIDUNIT	The GRIDUNIT keyword defines units of the COORD and ZCORN arrays and the grid coordinate space			Required
1-2	Format	GRIDUNIT	2	CHAR	
2-1	Data	The first entry defines units of the COORD and ZCORN arrays and should be normally set to FEET or METRES (enclosed in single quotes).			
2-2	Data	The second entry defines the grid coordinate space (MAP or MAPFT if COORD and ZCORN are defined in map coordinates, or blank if in grid coordinates (' ')).			

No.	Keyword Name	EGRID Header Keyword Table F.8			Status Or Value
		Keyword	No. of Entries	Data Type	
	<i>GDORIENT</i>	Grid orientation keyword defines the property ordering in the I, J, and K dimensions, the direction of the Z-direction, and the “handedness of the grid.			Optional
1-1	Format	<i>GDORIENT</i>	5	CHAR	
2-1	Data	Define the property ordering in the I dimension (set to either INC for increasing or DEC for decreasing).			
2-2	Data	Define the property ordering in the J dimension (set to either INC for increasing or DEC for decreasing).			
2-3	Data	Define the property ordering in the K dimension (set to either INC for increasing or DEC for decreasing).			
2-4	Data	Declare the direction of the Z-direction, should be set to either UP or DOWN.			
2-5	Data	Set the “handedness of the grid., should be set to either LEFT or RIGHT.			
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.					
3) For formatted output all character variables, including the keywords should be enclosed in single quotes, for example the FILEHEAD keyword should be written out as ‘FILEHEAD’.					
4) <i>GDORIENT</i> keyword in the <i>GRID</i> section is not supported by OPM Flow.					

Table F.8: EGRID Header Keywords

Example: EGRID Header Keywords

The following example shows a typical formatted file EGRID header keyword data set taken from the Norne model.

```
'FILEHEAD'      100 'INTE'
                  3      2004      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
                  0      0      0      0      0      0
'MAPAXES '      6 'REAL'
  0.00000000E+00  0.10000000E+03  0.00000000E+00  0.00000000E+00
  0.10000000E+03  0.00000000E+00
'GRIDUNIT'      2 'CHAR'
'METRES' ' '      ' '      ' '      ' '      ' '      ' '
```

Note that the *GDORIENT* keyword is not written out.

F.2.2 EGRID GLOBAL IRREGULAR CORNER POINT GRID KEYWORDS

Following the EGRID Header Keyword data set, the EGRID Global Irregular Corner Point Grid Keywords should be read or written to the EGRID file. This data set includes the actual structural data for the model, that is the [COORD](#) and [ZCORN](#) keyword data, as well as the [ACTNUM](#) keyword data that defines if a given cell is active (set to 1), or inactive (set to zero). The structure for this type data is defined in Table F.9.

No.	Keyword Name	EGRID Global Irregular Corner Point Grid Keywords Table F.9			Status Or Value
		Keyword	No. of Entries	Data Type	
	GRIDHEAD	Global grid keyword that defines the start of this keyword type.			Required
1-1	Format	GRIDHEAD	100	INTE	
2-1	Data	Type of grid, set to 0 for a mixture of Irregular Corner-Point and Unstructured Grids, 1 for Irregular Corner-Point Grids, and 2 for Unstructured Grids. Only the default value of one is supported by OPM Flow.			1
2-2	Data	The number of grid blocks in the x-direction (NX).			
2-3	Data	The number of grid blocks in the y-direction (NY).			
2-4	Data	The number of grid blocks in the z-direction (NZ).			
2-5	Data	The grid reference number set to zero for the global grid or a value greater than zero to represent a LGR . For example if there are five LGR 's, then this complete data set is repeated six times, once for the global grid, and five times for the five LGR s., with the first LGR having a reference number of 1 and the last LGR having a reference number of 5.			
2-6 to 2-24	Data	Not used.			0
2-25	Data	NUMRES the number of reservoirs, that is the number of COORD data sets to be processed by OPM Flow. This should be set to one as the only value supported by OPM Flow			1
2-26	Data	NSEG the number of coordinate line segments, should be set to 1.			1
2-27	Data	Cartesian/Radial grid indicator set to 0 for Cartesian grids and greater than zero for radial grids. Currently OPM Flow does not support radial geometries so this value should always be set to zero.			0
2-28	Data	The LGR location of the lower I-index LGR in the global grid. This value should be set to zero if there is no LGR grids.			0
2-29	Data	The LGR location of the lower J-index LGR in the global grid. This value should be set to zero if there is no LGR grids.			0
2-30	Data	The LGR location of the lower K-index LGR in the global grid. This value should be set to zero if there is no LGR grids.			0
2-31	Data	The LGR location of the upper I-index LGR in the global grid. This value should be set to zero if there is no LGR grids.			0
2-32	Data	The LGR location of the upper J-index LGR in the global grid. This value should be set to zero if there is no LGR grids.			0

No.	Keyword Name	EGRID Global Irregular Corner Point Grid Keywords			Status Or Value
		Table F.9			
		Keyword	No. of Entries	Data Type	
2-33	Data	The <i>LGR</i> location of the uper K-index <i>LGR</i> in the global grid. This value should be set to zero if there is no <i>LGR</i> grids.			0
2-34 to 2-100	Data	Not used by OPM Flow.			0
	BOXORIG	Not supported.			Optional
1-1	Format	BOXORIG	3	INTE	
2-1 to 2-3	Data	Ignored by OPM Flow.			
	<i>COORD</i>	Property keyword that defines the start of the <i>COORD</i> data set. The <i>COORD</i> data defines a set of coordinate lines or pillars for a reservoir grid via an array. The number of data values for this keyword should be 6 x (NX+1) x (NY+1) x <i>NUMRES</i> . Where <i>NUMRES</i> represents the number of separate coordinate lines (or reservoirs). In OPM Flow <i>NUMRES</i> can only be set to one.			Required
1-1	Format	<i>COORD</i>	6 x (NX+1) x (NY+1)	REAL	
2-1	Data	<i>COORD</i> data in the same format as the <i>COORD</i> keyword described in the <i>GRID</i> section of the manual.			
	<i>COORDSYS</i>	Coordinate definitions for each <i>NUMRES</i> reservoir. The number of data values for this keyword should be 6 x <i>NUMRES</i> . Where <i>NUMRES</i> represents the number of separate coordinate lines (or reservoirs). In OPM Flow <i>NUMRES</i> can only be set to one.			Optional
1-1	Format	<i>COORDSYS</i>	6	INTE	
2-1 to 2-6	Data	Ignored by OPM Flow.			
	<i>ZCORN</i>	Property keyword that defines the start of the <i>ZCORN</i> data set. <i>ZCORN</i> defines the depth of each corner point of a grid block on the pillars defining the reservoir grid. The number of data values for this keyword should be 8 x NX x NY x NZ.			Required
1-2	Format	<i>ZCORN</i>	8 x NX x NY x NZ	REAL	
1-3	Type	Type of data in the Data fields.			
2-1	Data	<i>ZCORN</i> data in the same format as the <i>ZCORN</i> keyword described in the <i>GRID</i> section of the manual.			

No.	Keyword Name	EGRID Global Irregular Corner Point Grid Keywords Table F.9			Status Or Value
		Keyword	No. of Entries	Data Type	
	ACTNUM	Property keyword that defines the start of the ACTNUM data set. ACTNUM specifies which grid blocks are active or inactive. A value of 1 indicates the block is active and a value of 0 indicates the block is inactive. Although this data set status is set to Optional, it is normally always written out by pre-processing software. The number of data values for this keyword should be NX x NY x NZ integer values.			Optional
I-1	Format	ACTNUM	NX x NY x NZ	INTE	
2-1	Data	ACTNUM data in the same format as the ACTNUM keyword described in the GRID section of the manual.			
	CORSNUM	Property keyword that defines the start of the CORSNUM data set for when the grid has been coarsened. The number of data values for this keyword should be NX x NY x NZ integer values. Grid coarsening is currently not supported in OPM Flow.			Optional
I-1	Format	CORSNUM	NX x NY x NZ	INTE	
2-1	Data	CORSNUM data set.			
I-1	ENDGRID	The ENDGRID keyword marks the end of the these set of keywords. Always set to 0 to indicate that there is no data section following this keyword.			Required
I-2	Format	ENDGRID	0	INTE	0
Notes: 1) Rows shaded in gray indicate the start of a keyword. 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either "Not Used" or "Undefined". 3) For formatted output all character variables, including the keywords, should be enclosed in single quotes, for example the GRIDHEAD keyword should be written out as 'GRIDHEAD'.					

Table F.9: EGRID Global Irregular Corner Point Grid Keywords

F.2.3 EGRID LGR IRREGULAR CORNER POINT GRID KEYWORDS

The data structure for Local Grid Refinement (“LGR”) Irregular Corner Point Grids is similar to the global grid format described in the previous section. Additional data that defines the *LGR* properties (*LGR* Name for example) are included in this definition and the *LGR* keywords are repeated for each *LGR* in the model. The keyword description is outlined in Table F.10.

Note that currently OPM Flow does not support *LGR* grids and therefore this series of keywords cannot be used by OPM Flow.

No.	Keyword Name	EGRID <i>LGR</i> Irregular Corner Point Grid Keywords Table F.10 (Repeated For Each <i>LGR</i> In The Model)			Status Or Value
		Keyword	No. of Entries	Data Type	
	<i>LGR</i>	<i>LGR</i> keyword defines the start of the <i>LGR</i> section of keywords and is required when LGRs are in the model.			Required
1-1	Format	<i>LGR</i>	1	CHAR	
2-1	Data	Name of the <i>LGR</i>			
	LGRPARNT	Start of LGRPARNT keyword that defines the parent <i>LGR</i> for when nested <i>LGR</i> have been used.			Optional
1-1	Format	LGRPARNT	1	CHAR	
2-1	Data	Name of parent <i>LGR</i>			
	LGRSGRID	Start of the LGRSGRID keyword for LGRs parented from a sub-grid.			Optional
1-1	Format	LGRSGRID	1	CHAR	
2-1	Data	Name of parent <i>LGR</i>			
	GRIDHEAD	<i>LGR</i> grid keyword that defines the start of this keyword type for the <i>LGR</i> . The format is the same as the Global grid keyword - see Table F.9.			Required
	BOXORIG	Not supported. The format is the same as the Global grid keyword - see Table F.9.			Optional
	<i>COORD</i>	Property keyword that defines the start of the <i>LGR COORD</i> data set. The <i>COORD</i> data defines a set of coordinate lines or pillars for <i>LGR</i> grid via an array. The format is the same as the Global grid keyword - see Table F.9.			Required
	<i>COORDSYS</i>	Coordinate definitions for the <i>LGR</i> . The format is the same as the Global grid keyword - see Table F.9.			Optional
	<i>ZCORN</i>	Property keyword that defines the start of the <i>LGR ZCORN</i> data set. The format is the same as the Global grid keyword - see Table F.9.			Required
	<i>ACTNUM</i>	Property keyword that defines the start of the <i>LGR ACTNUM</i> data set. The format is the same as the Global grid keyword - see Table F.9.			Optional

No.	Keyword Name	EGRID <i>LGR</i> Irregular Corner Point Grid Keywords Table F.10 (Repeated For Each <i>LGR</i> In The Model)			Status Or Value
		Keyword	No. of Entries	Data Type	
	HOSTNUM	Property keyword that defines the start of the HOSTNUM data set, which is only applicable to <i>LGR</i> grids. Number of data values for this keyword should be NX x NY x NZ integer values.			Required
1-1	Format	HOSTNUM	NX x NY x NZ	INTE	
2-1	Data	HOSTNUM data set.			
	ENDGRID	The ENDGRID keyword marks the end of the <i>LGR</i> Grid Header section. Always set to 0 to indicate that there is no Data section for this keyword.			Required
1-1	Format	ENDGRID	0	INTE	0
	ENDLGR	ENDLGR keyword marks the end of the <i>LGR</i> Header section. Always set to 0 to indicate that there is no Data section for this keyword.			Required
1-1	Format	ENDLGR	0	INTE	0
Notes: 1) Rows shaded in gray indicate the start of a keyword. 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow. 3) For formatted output all character variables, including the keywords, should be enclosed in single quotes, for example the <i>LGR</i> keyword should be written out as 'LGR'.					

Table F.10: EGRID *LGR* Irregular Corner Point Grid Keywords

Note that this data set is repeated for each *LGR* in the model as demonstrated in the following example.

Example: EGRID *LGR* Irregular Corner Point Grid Keywords

The following example shows a typical formatted data set for this series of keywords with two Cartesian *LGR* grids, named LGR-1 and LGR-2 define via the following *CARFIN* keyword statements:

```
--
--      CARFIN LGR GRID COMMANDS
--
--      LGR      ----- FINE GRID -----      -- CARFIN GRID --  MAX      HOST
--      NAME      I1  I2  J1  J2  K1  K2      NX    NY    NZ    WELLS  NAME
CARFIN
    'LGR-1'      2  2  2  2  1  1      2    2    2    1    GLOBAL /
CARFIN LGR GRID PARAMETERS

--
--      CARFIN LGR GRID COMMANDS
--
--      LGR      ----- FINE GRID -----      -- CARFIN GRID --  MAX      HOST
--      NAME      I1  I2  J1  J2  K1  K2      NX    NY    NZ    WELLS  NAME
CARFIN
    'LGR-2'      9  9  9  9  1  1      2    2    2    1    GLOBAL /
CARFIN LGR GRID PARAMETERS

ENDFIN
```

The resulting keywords are as follows:

```

'LGR' 1 'CHAR'
'LGR-1'
'LGRPARNT' 1 'CHAR'
'
'GRIDHEAD' 100 'INTE'
1 2 2 2 1 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
1 1 0 2 2 1
.....
0 0 0 0
'COORD' 54 'REAL'
0.10000000E+04 0.10000000E+04 0.00000000E+00 0.10000000E+04
0.10000000E+04 0.10000000E+04 0.15000000E+04 0.10000000E+04
0.00000000E+00 0.15000000E+04 0.10000000E+04 0.10000000E+04
0.20000000E+04 0.10000000E+04 0.00000000E+00 0.20000000E+04
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.15000000E+04
.....
0.20000000E+04 0.10000000E+04
'ZCORN' 64 'REAL'
0.83250000E+04 0.83250000E+04 0.83250000E+04 0.83250000E+04
0.83250000E+04 0.83250000E+04 0.83250000E+04 0.83250000E+04
0.83250000E+04 0.83250000E+04 0.83250000E+04 0.83250000E+04
0.83350000E+04 0.83350000E+04 0.83350000E+04 0.83350000E+04
.....
0.83450000E+04 0.83450000E+04 0.83450000E+04 0.83450000E+04
'ACTNUM' 8 'INTE'
1 1 1 1 1 1
1 1
'HOSTNUM' 8 'INTE'
12 12 12 12 12 12
12
'ENDGRID' 0 'INTE'
'ENDLGR' 0 'INTE'
'LGR' 1 'CHAR'
'LGR-2'
'LGRPARNT' 1 'CHAR'
'
'GRIDHEAD' 100 'INTE'
1 2 2 2 2 0
.....
0 0 0 0
'COORD' 54 'REAL'
0.80000000E+04 0.80000000E+04 0.00000000E+00 0.80000000E+04
0.80000000E+04 0.10000000E+04 0.85000000E+04 0.80000000E+04
.....
0.90000000E+04 0.10000000E+04
'ZCORN' 64 'REAL'
0.83250000E+04 0.83250000E+04 0.83250000E+04 0.83250000E+04
.....
0.83450000E+04 0.83450000E+04 0.83450000E+04 0.83450000E+04
'ACTNUM' 8 'INTE'
1 1 1 1 1 1
1 1
'HOSTNUM' 8 'INTE'
89 89 89 89 89 89
89
'ENDGRID' 0 'INTE'
'ENDLGR' 0 'INTE'

```

End of Example

F.2.4 EGRID Non-Neighbor Connections for Irregular Corner Point Grid Keywords

The keywords in this section are used to define Non-Neighbor Connections (“NNC”) for both the global grid and any *LGR*’s included in the model. This data set is only written out if there are NNCs in the model and can be skipped if this is the case; however, apart from very simple models, nearly all full field models will have NNCs as a result of modeling the structural faults in the model. Likewise, the keywords affiliate with LGRs are also skipped if there are no LGRs in the model.

Note that actual *NNC* transmissibility data for the NNCs (*TRANX*, *TRANY*, and *TRANZ*) are stored in the *INIT* Data File and not in the EGRID File.

The keyword definitions are tabulated in Table F.11.

No.	Keyword Name	EGRID Non-Neighbor Connections Keywords For Irregular Corner Point Grids			Status Or Value
		Table F.11			
		Keyword	No. of Entries	Data Type	
	NNCHEAD	This keyword marks the start of the non-neighbor keywords and defines the number of non-neighbor connections.			Required
1-1	Format	NNCHEAD	10	INTE	
2-1	Data	The number of non-neighbor connections (NUMNNC).			
2-2	Data	Grid identifier set to zero for the global grid, or a value greater than zero to represent a <i>LGR</i> .			
2-3 to 2-10	Data	Not used.			
	NNC1	NNC1 defines the cell numbers for the UPSTREAM non-neighbor connections, where NUMNNC is the number of non-neighbor connections.			Required
1-1	Format	NNC1	NUMNNC	INTE	
2-1	Data	NUMNNC integer values defining the cell numbers for the UPSTREAM non-neighbor connections.			
	NNC2	NNC2 defines the cell numbers for the DOWNSTREAM non-neighbor connections, where NUMNNC is the number of non-neighbor connections.			Required
1-1	Format	NNC2	NUMNNC	INTE	
2-1	Data	NUMNNC integer values defining the cell numbers for the DOWNSTREAM non-neighbor connections.			
	NNCL	This keyword defines the local cells connected to the global grid, where NCONGL is the number of entries.			Optional
1-1	Format	NNCL	NCONCL	INTE	
2-1	Data	NCONCL integer values defining the local cells connected to the global grid.			

No.	Keyword Name	EGRID Non-Neighbor Connections Keywords For Irregular Corner Point Grids			Status Or Value
		Table F.11			
		Keyword	No. of Entries	Data Type	
	NNCG	This keyword defines the global cells connected to the current local grid being defined by this section of keywords, where NCONGL is the number of entries.			Optional
1-1	Format	NNCG	NCONGL	INTE	
2-1	Data	NCONGL integer values defining the global cells connected to the current local grid.			
	NNCHEADA	This keyword defines the connection between two LGRs that have been amalgamated.			Optional
1-1	Format	NNCHEADA	10	INTE	
2-1	Data	LGR index of the first <i>LGR</i> in the amalgamation (ILOC1).			
2-2	Data	LGR index of the second <i>LGR</i> in the amalgamation (ILOC2).			
2-3 to 2-10		Not used.			
	NNA1	This keyword defines the ILOC1 cells connected in the amalgamation, where NUMNCA is the number of entries.			Optional
1-1	Format	NNA1	NUMNCA	INTE	
2-1	Data	NUMNCA integer values defining the ILOC1 cells connected in the amalgamation.			
	NNA2	This keyword defines the ILOC2 cells connected in the amalgamation, where NUMNCA is the number of entries.			
1-1	Format	NNA2	NUMNCA	INTE	
2-1	Data	NUMNCA integer values defining the ILOC2 cells connected in the amalgamation.			

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.
- 3) For formatted output all character variables, including the keywords, should be enclosed in single quotes, for example the NNCHHEAD keyword should be written out as ‘NNCHHEAD’.
- 4) Local Grid Refinements are currently not supported by OPM Flow and neither therefore is *LGR* amalgamation of LGRs.

Table F.11: EGRID NNC Keywords for Irregular Corner Point Grids

Note that if *LGR*'s are present in the model then the keywords relating to LGRs are repeated for each *LGR* in the model.

Example: EGRID Grid Keywords for Non-Neighbor Connections for Irregular Corner Point Grids

The example shows the NNCs for the global grid and two LGRs.

'NNCHEAD '	10	'INTE '			
11363	0	0	0	0	0
0	0	0	0		
'NNC1 '	11363	'INTE '			
5717	5717	21173	26325	31477	36629
41781	46933	52085	57237	62389	67541
72693	77845	82997	88149	93301	103605
.....					
'NNC2 '	11363	'INTE '			
566	10870	26326	31478	36630	41782
46934	52086	57238	62390	67542	72694
77846	82998	88150	93302	98454	98454
103606	612	5764	26372	21220	31524
.....					
'NNCHEAD '	10	'INTE '			
0	1	0	0	0	0
0	0	0	0		
'NNC1 '	0	'INTE '			
'NNC2 '	0	'INTE '			
'NNCL '	16	'INTE '			
1	5	3	7	1	5
2	6	2	6	4	8
3	7	4	8		
'NNCG '	16	'INTE '			
11	11	11	11	2	2
2	2	13	13	13	13
22	22	22	22		
'NNCHEAD '	10	'INTE '			
0	2	0	0	0	0
0	0	0	0		
'NNC1 '	0	'INTE '			
'NNC2 '	0	'INTE '			
'NNCL '	16	'INTE '			
1	5	3	7	1	5
2	6	2	6	4	8
3	7	4	8		
'NNCG '	16	'INTE '			
88	88	88	88	79	79
79	79	90	90	90	90
99	99	99	99		

End of Example

F.3 EGRID - MODEL STRUCTURAL DATA FOR UNSTRUCTURED GRIDS FILE

This file format is currently not supported by OPM Flow.

F.4 EGRID - MODEL STRUCTURAL DATA FOR MIXED GRIDS FILE

This file format is currently not supported by OPM Flow.

F.5 GRID - MODEL STRUCTURAL DATA FILE

This file format is currently not supported by OPM Flow.

F.5.1 RESTART DATA - LGR GRID HEADER KEYWORDS

This set of keywords define an *LGR*'s properties, *LGR* Name for example, and always proceeds an *LGR* data set for a given *LGR*; that is the keywords are repeated for each *LGR* data set in the model and each time step. Note that a given *LGR* data set is terminated by the ENDLGR keyword and the SEQNUM keyword terminates the input for a time step, which switches the input back to the global grid properties. The keyword description is outlined in Table F.12.

Note that currently OPM Flow does not support *LGR* grids and therefore this set of keywords cannot be used by OPM Flow.

No.	Keyword Name	RESTART Data - LGR Grid Header Keywords Table F.12 (Repeated For Each LGR In The Model)			Status Or Value
		Keyword	No. Of Entries	Data Type	
	<i>LGR</i>	<i>LGR</i> grid header keyword that defines the name of the <i>LGR</i> .			Always Required
1-1	Keyword	<i>LGR</i>	1	CHAR	
2-1	Data	Name of the <i>LGR</i> .			
	LGRHEAD1	<i>LGR</i> that defines the integer variables for this time step.			Always Required
1-1	Format	LGRHEAD1	45	INTE	
2-1 to 2-45	Data	Undefined.			
	LGRHEADQ	<i>LGR</i> grid keyword that defines the logical variables (T for true and F for false) for this time step.			Always Required
1-1	Format	LGRHEADQ	5	LOGI	
2-1 to 2-5	Data	Undefined.			

No.	Keyword Name	RESTART Data - LGR Grid Header Keywords Table F.12 (Repeated For Each LGR In The Model)			Status Or Value
		Keyword	No. Of Entries	Data Type	
	LGRHEAD	LGR grid keyword that defines the double precision REAL variables for this time step.			Always Required
1-1	Format	LGRHEAD	5	DOUB	5
2-1 to 2-5	Data	Undefined.			
	ENDLGR	The ENDLGR keyword marks the end of an LGR section for a given LGR.			Always Required
1-1	Format	ENDLGR	1	MESG	
	Data	LGR number.			

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.
- 3) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the LGR keyword should be written out as ‘LGR ’.

Table F.12: RESTART Data - LGR Grid Header Keywords

This keyword type is repeated for each LGR in the model with the last LGR data set terminated by the LGRSGONE keyword to mark the end of the LGR input.

Note

The LGR, LGRHEAD, LGRHEADQ, and LGRHEADD always proceed LGR data sets and the ENDLGR keyword terminates the end of a data section of LGR data.

Example: RESTART Data - LGR Grid Header Keywords

The following example shows a typical formatted *LGR* grid header data set for a single *LGR* grid named LGR-I.

```
'LGR      '          1 'CHAR'  
'LGR-1    '  
'LGRHEADI'         45 'INTE'  
      1      100      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
      -2345      -2345      -2345      -2345      -2345      -2345  
'LGRHEADQ'          5 'LOGI'  
F F F F F  
'LGRHEADD'          5 'DOUB'  
  0.000000000000000D+00  -0.10000000200409D+21  -0.10000000200409D+21  
 -0.10000000200409D+21  -0.10000000200409D+21
```

End of Example

F.6 INIT – MODEL INITIALIZATION AND STATIC DATA FILE

The *INIT* files contain the static data specified in the *GRID*, *PROPS*, and *REGIONS* sections. For example, the *PORO*, *PERMX*, and *NTG* arrays from the *GRID* section and the resulting *PORV*, *TRANX*, *TRANY*, and *TRANZ* arrays. The data is used in post-processing software, for example OPM ResInsight, to visualize the static grid properties and the tabular PVT and relative permeability data, and is not used per se by the simulator.

The data is written to file if the *INIT* keyword in the *GRID* section has been activated (*INIT – Activate the INIT File Option*). Similar to the EGRID file, the *INIT* file can either be written out in formatted form as ASCII i.e. text files, if the *FMTOUT* keyword has been activated, or binary format if the *FMTOUT* keyword has not been activated. Normally, this option is always activated by the user and when activated the binary form of the file is used.

There are two file types associated with the *INIT* data:

- 1) an index file (*INIT* Index) that contains a list of variables, the array type (character, integer, or real) and the size of the arrays that are written out to the initialization data file and,
- 2) the initialization data file (*INIT* Data) that contains the actual data written out at the beginning of the run once the model has been initialized.

The commercial simulator writes out both files, whereas OPM Flow only writes out the *INIT* Data file, as the data can be read by most post-processing software, including OPM ResInsight, using only the *INIT* Data file as input.

The following sections outline the format for these two file types.

F.6.1 INIT INDEX FILE

This file type is not supported or required by OPM Flow or OPM ResInsight to read or write the static data written to the *INIT* Data file.

F.6.2 INIT DATA FILE SPECIFICATION

This file type contains the global and any *LGR* grid property data present in the model³⁹⁴, for example porosity (*PORO*) and permeability data arrays (*PERMX*, *PERMY*, and *PERMZ*) for properties allocated to each grid cell, as well as the fluid and rock property functions tabular data. The overall structure of this file is similar to the *RESTART* file with the individual keywords being structured as shown in Table F.13.

Reference Section	Global Section Keywords	Global Data Keywords	<i>LGR</i> Section Keywords	<i>LGR</i> Data Keywords	Notes
F.6.3	INTEHEAD LOGIHEAD DOUBHEAD				Global grid header keywords that defines the integer, logical, and double precision variables for this header keyword.
F.6.4	Property Keywords	<i>PORV</i> <i>PORO</i> etc.			Global grid property keywords that defines the global grid property data defined in the <i>GRID</i> and Edit sections. For example, <i>PORV</i> , <i>NTG</i> , <i>PERMX</i> , etc.)
F.6.5			<i>LGR</i> LGRHEAD LGRHEADQ LGRHEADD		Defines the start of an <i>LGR</i> section of keywords and the <i>LGR</i> header keywords.
F.6.5			INTEHEAD LOGIHEAD DOUBHEAD		<i>LGR</i> grid header keywords defining the same data as the global keywords, except for the given <i>LGR</i> . This set of keywords are only written once for each <i>LGR</i> .
F.6.4			Property Keywords	<i>PORV</i> <i>PORO</i> etc.	<i>LGR</i> property keywords; same format as the global grid keywords.
F.6.5			<i>LGR</i> LGRHEAD LGRHEADQ LGRHEADD		The <i>LGR</i> header keywords.
F.6.5			INTEHEAD LOGIHEAD DOUBHEAD		The <i>LGR</i> header keywords which are only written once for each <i>LGR</i> .
F.6.4			Property Keywords	<i>PORV</i> <i>PORO</i> etc.	<i>LGR</i> property keywords.
			etc.		Repeated for each <i>LGR</i> in the model.
F.6.5			LGRSGONE		Marks the end of <i>LGR</i> section of keywords.

³⁹⁴ Several keywords are used for both global and *LGR* grids, for example the *INTEHEAD*, *LOGIHEAD*, *DOUBHEAD*, *PROPERTY*, and *REGION* keywords and arrays.

Reference	Section	Global Section Keywords	Global Data Keywords	LGR Section Keywords	LGR Data Keywords	Notes
F.6.7		TABDIMS				TABDIMS defines the tables dimensions, for the subsequent TAB keywords.
F.6.8		TAB				Rock and fluid tabular property data keywords.
F.6.8		CON				CON keyword defines the constant PVT dead oil data (the data associated with RSCONST , RSCONSTT , RVCONST , and RVCONSTT keywords).
F.6.4	Region Keywords	PVTNUM SATNUM etc.				Region property data for the global grid as defined in the REGION section (PVTNUM , SATNUM , etc.). The keyword is repeated to account for all region property data.
F.6.5				LGR LGRHEAD LGRHEADQ LGRHEADD		The LGR header keywords.
F.6.4			Region Keywords	PVTNUM SATNUM etc.		LGR region property data for the LGR grid.
F.6.5				LGR LGRHEAD LGRHEADQ LGRHEADD		The LGR header keywords.
F.6.4			Region Keywords	PVTNUM SATNUM etc.		LGR region property data for the LGR grid.
			etc.			Repeated for each LGR in the model.
F.6.5				LGRSGONE		Marks the end of LGR section of keywords.
F.6.4	End-Point Keywords	SWATINIT SWL etc.				Global grid saturation and end-point data keywords. The keyword uses the same keywords outlined in the GRID and PROPS section, for example, the SWL , SWATINIT , KRG , PCW , etc. arrays
F.6.5				LGR LGRHEAD LGRHEADQ LGRHEADD		The LGR header keywords.
F.6.4			End-Point Keywords	SWATINIT SWL etc.		LGR grid saturation and end-point data keywords.

Reference Section	Global Section Keywords	Global Data Keywords	LGR Section Keywords	LGR Data Keywords	Notes
F.6.5			LGR LGRHEAD LGRHEADQ LGRHEADD		The LGR header keywords.
F.6.4			End-Point Keywords	SWATINIT SWL etc.	LGR grid saturation and end-point data keywords.
			etc.		
F.6.5			LGRSGONE		Marks the end of LGR section of keywords.
F.6.4	NNC Keywords	TRANNC HEATNNC etc.			Non-neighbor connection property data ("NNC") for the global grid. The keyword is repeated to account for various NNC arrays.
F.6.5			LGR LGRHEAD LGRHEADQ LGRHEADD		The LGR header keywords.
F.6.4			NNC Keywords	TRANNC TRANGL etc.	LGR NNC property data.
F.6.5			LGR LGRHEAD LGRHEADQ LGRHEADD		The LGR header keywords.
F.6.4			NNC Keywords	TRANNC TRANGL etc.	LGR NNC property data.
			etc.		Repeated for each LGR in the model.
F.6.5			LGRSGONE		Marks the end of LGR section of keywords.
Notes: 1) Cells shaded in gray indicate the combination is not applicable and cells colored orange indicate that this item is not currently supported by OPM Flow.					

Table F.13: INIT Data File Structure

F.6.3 INIT DATA – STATIC DATA HEADER KEYWORDS

The *INIT* Data – Static Data Header Keyword section is the first data set that should be read or written to the *INIT* Data file. This data set includes INTEHEAD, LOGIHEAD, and DOUBHEAD keywords that define versus parameters used in subsequent keywords in the *INIT* Data file. This is then followed by a series of global grid PROPERTY data keywords that define the various global grid property arrays for each grid cell in the model, as well as the static tabular data (relative permeability tables and PVT tables), etc.

The structure for this set of keywords is defined in Table F.14.

No.	Keyword Name	<i>INIT</i> Data – Static Data Header Keywords Table F.14			Status Or Value
		Keyword	No. of Entries	Data Type	
	INTEHEAD	Global static header keyword for the <i>INIT</i> Data file that defines the start of <i>INIT</i> static data set, and contains the integer variables for this header keyword. NIHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model.			Required
1-1	Format	INTEHEAD	NIHEAD	INTE	
2-1		Undefined.			0
2-2		Undefined.			0
2-3	Data	Unit system used in the simulation, set to 1 for metric, 2 for field, and 3 for laboratory.			
2-4 to 2-8		Undefined.			0
2-9	Data	<i>DIMENS</i> key word in the <i>RUNSPEC</i> section - the number of grid blocks in the x-direction.			NX
2-10	Data	<i>DIMENS</i> key word in the <i>RUNSPEC</i> section - the number of grid blocks in the y-direction.			NY
2-11	Data	<i>DIMENS</i> key word in the <i>RUNSPEC</i> section - the number of grid blocks in the z-direction.			NZ
2-12	Data	Number of global active cells in the model.			NACTIV
2-13	Data	Not used.			0
2-14	Data	Type of grid, set to 0 for Irregular Corner-Point Grids, 1 for Unstructured Grids, or 2 for a mixture of Irregular Corner-Point, Unstructured Grids (Mixed Grids) and 3 for Cartesian grids.			I
2-15	Data	Type of phases in the model, set to 1 for oil, 2 for water, 3 for oil-water, 4 for gas, 5 for oil-gas, 6 for gas-water and 7 for oil-water-gas.			PHASE
2-16 to 2-64		Undefined.			0
2-65	Data	First part of start date of the run, DAY, the value should be greater than or equal to one and less than or equal to 31 for the calendar days.			DAY

No.	Keyword Name	<i>INIT</i> Data – Static Data Header Keywords Table F.14			Status Or Value
		Keyword	No. of Entries	Data Type	
2-66	Data	Second part of start date of the run, MONTH, the value should be greater than or equal to one and less than or equal to 12 for the calendar month.			MONTH
2-77	Data	Third part of start date of run, YEAR, a positive four digit integer value of the start year, which must be specified fully by four digits, that is 1986.			YEAR
2-68 to 2-94		Undefined.			0
2-95	Data	Simulator code used to generate the file, set to 100 for Schlumberger's ECLIPSE 100, 300 for Schlumberger's ECLIPSE 300, 500 for Schlumberger's ECLIPSE 300 Thermal, 700 for Schlumberger's INTERSECT simulator, 800 for Schlumberger's FrontSim simulator, or a negative value for other simulators. OPM Flow uses a value of 100.			I PROG
2-96 to 206		Undefined.			0
2-207	Data	The first part of the current simulation time in the form HH:MM:SS, that is HH for hours. The value should be greater than or equal to zero and less than or equal to 23.			I HOURZ
2-208	Data	The second part of the current simulation time in the form HH:MM:SS, that is MM for minutes. The value should be greater than or equal to zero and less than or equal to 59.			I MINTS
2-209 to 410		Undefined.			0
411	Data	The third part of the current simulation time in the form HH:MM:SS, that is SS for seconds. The value should be greater than or equal to zero and less than or equal to 59.			I SECND
	LOGIHEAD	Global static data keyword for the <i>INIT</i> Data file that defines the logical variables (T for true and F for false) for this keyword. NLHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model. Undefined items should be set to F (False).			Required
I-1	Format	LOGIHEAD	NLHEAD	LOGI	
2-1	Data	Dissolved gas is present in the model indicator, set to T if present in the model else set to F.			<i>DISGAS</i>
2-2	Data	Vaporized oil is present in the model indicator, set to T if present in the model else set to F.			<i>VAPOIL</i>
2-3	Data	<i>SATOPTS</i> in the <i>RUNSPEC</i> section - directional relative permeability curves are active in the model.			DIRECT

No.	Keyword Name	<i>INIT</i> Data – Static Data Header Keywords Table F.14			Status Or Value
		Keyword	No. of Entries	Data Type	
2-4	Data	<i>SATOPTS</i> keyword in the <i>RUNSPEC</i> section - reversible relative permeability (black-oil) are active in the model.			IRREVERS
		Radial grid geometry has been used for a compositional model indicator, set to T or F. Note that currently OPM Flow does not support radial grids and therefore this item should be set to F.			<i>RADIAL</i>
2.5	Data	Radial grid geometry has been used for a black-oil model indicator, set to T or F. Note that currently OPM Flow does not support radial grids and therefore this item should be set to F.			<i>RADIAL</i>
		Reversible relative permeability (compositional) are active in the model.			IRREVERS
2.6		Undefined.			F
2-7	Data	The hysteresis option has been activated in the model indicator, set to T or F.			HYSTER
2-8 to 2-14	Data	Undefined.			F
2-15	Data	The dual porosity option has been activated in the model indicator, set to T or F. Note that currently OPM Flow does not support dual porosity grids and therefore this item should be set to F.			<i>DUALPORO</i>
2-16		Undefined.			F
2-17	Data	<i>ENDSCALE</i> keyword in the <i>RUNSPEC</i> section - enable end-point scaling indicator.			<i>ENDSCALE</i>
2-18	Data	<i>ENDSCALE</i> keyword in the <i>RUNSPEC</i> section - directional end-point scaling indicator.			DIRECT
2-19	Data	<i>ENDSCALE</i> keyword in the <i>RUNSPEC</i> section - reversible end-point scaling indicator.			IRREVERS
2-20	Data	<i>SCALECRS</i> keyword in the <i>PROPS</i> section - alternative three point end-point scaling indicator.			<i>SCALECRS</i>
2-21 to 2-35		Undefined.			F
2-36	Data	Miscible displacement has been activated in the model indicator, set to T or F. See the <i>MISCIBLE – Define Miscibility Todd-Longstaff Parameters</i> keyword in the <i>RUNSPEC</i> section for further information.			
2-37 to 2-55		Undefined.			F

No.	Keyword Name	INIT Data – Static Data Header Keywords			Status Or Value
		Table F.14			
		Keyword	No. of Entries	Data Type	
2-56	Data	The scale water capillary pressure at maximum water saturation (that is minimum pressure) has been activated in the model, set to T or F.			
2-57	Data	The scale water capillary pressure at maximum water saturation (that is minimum pressure) has been activated in the model, set to T or F.			
2-58 to 2-127		Undefined.			0
2-128	Data	Coal Bed Methane option has been activated in the compositional model indicator; set to T or F. Note that currently OPM Flow does not support the Coal Bed Methane option and therefore this item should be set to F.			F
	DOUBHEAD	Global static data keyword for the INIT Data file that defines the double precision REAL variables for this keyword. NDHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model.			Required
I-I	Format	DOUBHEAD	NDHEAD	DOUB	
2-I to 2-185		Undefined.			0.0

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.
- 3) For formatted output all character variables, including the keywords should be enclosed in single quotes, for example the SEQNUM should be written out as ‘SEQNUM’.

Table F.14: *INIT* Data – Static Data Header Keywords

Example: INIT Data – Static Header Keywords

The following example shows a typical formatted *INIT* Data header set of keywords.

```

'INTEHEAD'          249 'INTE'
-955283513          200400           2          -2345          -2345          -2345
      -2345          -2345          10           10           1          100
      10           -2345           7          -2345           0           1
      0              2              2              0              0           0
      110           108           109           3           97           93
      -2345          -2345          19           38           53          -2345
      97            93           146           5              0           1
      15            24            8            5              2           4
      0              0              0              0              0           0
      0              0              0              0              0           0
      0              0              0              0             19          10
      1982           0              0              0              1           0
      0              0              0              0              1          10
      0              0             12             1             25           1
.....
'LOGIHEAD'          79 'LOGI'
T F F T F F F F T F F F F F F F T F T F F F F F F
F F F F F F F F F F F F F F F F F F F F F F F
F F F F F F F F F F F F F F F F F F F F F F
F F F F
'DOUBHEAD'          185 'DOUB'
0.00000000000000D+00 0.10000000000000D+01 0.36500000000000D+03
0.100000000149012D+00 0.150000000596046D+00 0.30000000000000D+01
0.300000001192093D+00 0.100000000149012D+00 0.100000000149012D+00
-0.100000000200409D+21 0.10000000000000D+01 0.10000000000000D+01
0.10000000000000D+01 0.00000000000000D+00 0.00000000000000D+00
0.10000000000000D+01 0.10000000000000D-03
.....

```

End of Example

F.6.4 INIT DATA – STATIC GRID ARRAY DATA KEYWORDS

This series of keywords define various static global and *LGR* grid property arrays for each grid cell in the model. The grid array keywords use the same grid property keywords described in the *GRID* section of the manual. For example, the *DEPTH*, *PORO*, *PERMX*, *PERMY*, *PERMZ*, *NTG*, *TOPS* arrays etc. In addition, the keywords also include the calculated pore volumes (*PORV*) and transmissibility arrays (*TRANX*, *TRANY*, and *TRANZ*) after processing the *GRID* and *EDIT* sections. Note the first property should always be the *PORV* array for both global and *LGR* grid types.

This format is also used to define the global grid *REGION*, *NNC*, and *ENPOINTS* section arrays, with the keyword name corresponding to the array name. For example, for the *REGION* keywords the same keywords outlined in the *REGION* section of the manual; that is *EQLNUM*, *FIPNUM*, *PVTNUM*, *ROCKNUM*, and *SATNUM* are utilized as the keywords in this section.

The structure for this type of data and the keywords are defined in Table F.15.

No.	Keyword Name	INIT Data – Static Grid Array Data Keywords Table F.15			Status Or Value
		Keyword	No. of Entries	Data Type	
	PROPERTY	The PROPERTY keyword defines the start of a grid property array section of the <i>INIT</i> file. PROPERTY is the keyword name of the array being written out, as outlined in the <i>GRID</i> section. For example, <i>DEPTH</i>, <i>PORO</i>, <i>PERMX</i>, <i>PERMY</i>, <i>PERMZ</i>, <i>NTG</i>, <i>TOPS</i> keywords and the calculated <i>PORV</i>, <i>TRANX</i>, <i>TRANY</i>, and <i>TRANZ</i> arrays. The PROPERTY keyword (items 1-1 and 2-1) is repeated for each grid property. The <i>PORV</i> property must be the first data set for this keyword type and is written out in full for each cell in the model (that is NXYZ = NX x NY x NZ values). Note the <i>PORV</i> array values for inactive cells are set to a constant value of zero. Subsequent grid property arrays can be in any order and only the active cells are written out (NACTIV).			Required
1-1	Format	PROPERTY	NXYZ or NACTIV	REAL	
2-1	Data	PROPERTY data set.			
LGR Property Data for Each LGR Keyword and Data					
Table and Constant Keywords (<i>TABDIMS</i> , <i>TAB</i> , and <i>CON</i>)					
	REGION	The REGION keyword defines the start of the grid region array section. REGION is the keyword name of the region array being written out, as outlined in the <i>REGIONS</i> section. For example, <i>EQLNUM</i>, <i>FIPNUM</i>, <i>PVTNUM</i>, <i>ROCKNUM</i>, <i>SATNUM</i>, etc. For analytical aquifers in the model, the AQUIFERA variable name is used to indicate which cells are connected to a given aquifer. If a cell is connected to an aquifer number N, then the cell value will be set to 2(N-1) in the array. The REGION keyword (items 1-1 and 2-1) is repeated for each global region array and the arrays can be written out in any order.			Required
1-1	Format	REGION	NACTIV	INTE	
2-1	Data	REGION data set.			

No.	Keyword Name	INIT Data – Static Grid Array Data Keywords			Status Or Value
		Table F.15			
		Keyword	No. of Entries	Data Type	
LGR Region Keywords and Data for Each Grid					
	ENDPOINT	The ENDPOINT keywords define the start of a grid's saturation and end-point data arrays. ENDPOINT is the keyword name of the saturation (SWL, SWATINIT, etc.) array or end-point array (PCW, KRORW, etc.) being written out, as described in the GRID and PROPS sections. The type of data for the keyword should be set to REAL for arrays with real values (SWL, SWATINIT, etc) or INTE for integer arrays (ENDNUM, etc). ENDPOINT arrays can be in any order and only the active cells are written out (NACTIV). This keyword (items 1-1 and 2-1) is repeated for each saturation and end-point array being written out. The data written out is dependent on the various options declared in the RUNSPEC section as well as the data declared in the GRID and PROPS sections.			Required
1-1	Keyword	ENDPOINTS	NACTIV	INTE or REAL	
2-1	Data	ENDPOINT data set.			
LGR Saturation End-Point Keywords and Data for Each LGR					
	NNC	The NNC keyword defines the start of a grid's non-neighbor connection array data. NNC is the keyword name of the non-neighbor connection array being written out, and should be set to one of the following keywords: 1) TRANNNC: the non-neighbor connection transmissibilities within the grid. Always required even if there are no NNC, in which case NUMNCC is set to zero. 2) DIFFNNC: the non-neighbor connection diffusivities within the grid. Only required if the Diffusivity option has been activated by the DIFFUSE keyword in the RUNSPEC section. This option is currently not supported by OPM Flow 3) HEATNNC: the non-neighbor connection thermal transmissibilities within the grid. Only required if OPM Flow's THERMAL option has been activated by the THERMAL keyword in the RUNSPEC section. The following two additional NNC keywords are required for when the data for an LGR is being defined: 1) TRANCC: the non-neighbor connection transmissibilities within the LGR (always required for an LGR). 2) TRANGL: the non-neighbor connections transmissibilities between the LGR and the global grid (always required for an LGR). The number of data values for this keyword data set should set to the number of non-neighbor connection to be read or written out (NUMNNC) for the TRANNC, DIFFNNC, HEATNCC, TRANCC, and TRANGL arrays. The NNC keyword (items 1-1 and 2-1) are repeated for each non-neighbor connection array being written out.			Required
1-1	Format	NNC	NUMNCC	REAL	
2-1	Data	NNC data set.			
LGR Non-Neighbor Connection Keywords for Each LGR					

No.	Keyword Name	INIT Data – Static Grid Array Data Keywords Table F.15			Status Or Value
		Keyword	No. of Entries	Data Type	
Notes: 1) Rows shaded in gray indicate the start of a keyword and rows colored orange indicate the location of LGR data if LGRs are present in the model. 2) For formatted output all character variables, including the keyword should be enclosed in single quotes, for example the PORV and PROPERTY fields should be written out as 'PORV ' and 'DEPTH ', etc.					

Table F.15: INIT Data – Static Grid Array Data Keywords

Example: INIT Data – Static Grid Array Data Keywords

The following example shows a typical formatted global grid property data keyword.

```
'PORV ' 100 'REAL '
0.10686456E+07 0.10686456E+07 0.10686456E+07 0.10686456E+07
0.10686456E+07 0.10686456E+07 0.10686456E+07 0.10686456E+07
.....
0.10686456E+07 0.10686456E+07 0.10686456E+07 0.10686456E+07
0.10686456E+07 0.10686456E+07 0.10686456E+07 0.10686456E+07
'DX ' 100 'REAL '
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
.....
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
'DY ' 100 'REAL '
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
.....
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
0.10000000E+04 0.10000000E+04 0.10000000E+04 0.10000000E+04
'DZ ' 100 'REAL '
0.20000000E+02 0.20000000E+02 0.20000000E+02 0.20000000E+02
0.20000000E+02 0.20000000E+02 0.20000000E+02 0.20000000E+02
.....
0.20000000E+02 0.20000000E+02 0.20000000E+02 0.20000000E+02
0.20000000E+02 0.20000000E+02 0.20000000E+02 0.20000000E+02
'PERMX ' 100 'REAL '
0.50000000E+03 0.50000000E+03 0.50000000E+03 0.50000000E+03
.....
0.50000000E+03 0.50000000E+03 0.50000000E+03 0.50000000E+03
0.50000000E+03 0.50000000E+03 0.50000000E+03 0.50000000E+03
```

TABULAR DATA

```
'PVTNUM ' 100 'INTE '
1 1 1 1 1
1 1 1 1 1
.....
1 1 1 1 1
'SATNUM ' 100 'INTE '
1 1 1 1 1
1 1 1 1 2
1 1 2 2 2
.....
'EQLNUM ' 100 'INTE '
1 1 1 1 1
1 1 1 1 1
```

End of Example

F.6.5 INIT DATA - LGR STATIC DATA HEADER KEYWORDS

The file structure for this series of keywords consists of two sets of keywords, the first set is repeated each time an *LGR* data set is written out and the second set, which is the same as global static header keywords in section INIT Data – Static Data Header Keywords, is only written once. The first set of keywords defines an *LGR*'s properties, *LGR* Name for example, and always proceeds an *LGR* data set for a given *LGR*; that is this series of keywords are repeated for each *LGR* data set in the model. This is then followed by the second set of static data header keywords, or the actual array or table keywords. Termination of the *LGR* data set occurs after the last *LGR* data set has been written out using the LGRSGONE keyword, which switches input back to the global grid properties. Table F.16 outlines the structure of the *INIT* Data - *LGR* Static Header Keywords and their affiliated data.

Note that currently OPM Flow does not support *LGR* grids and therefore this set of keywords cannot be used by OPM Flow.

No.	Keyword Name	<i>INIT</i> Data - <i>LGR</i> Static Header Keywords Table F.16 (Repeated Each Time An <i>LGR</i> Each Data Set Is Written Out)			Status Or Value
		Keyword	No. of Entries	Data Type	
	<i>LGR</i>	<i>LGR</i> static data header keyword for the <i>INIT</i> Data file that defines the name of the <i>LGR</i> .			Always Required
1-1	Keyword	<i>LGR</i>	1	CHAR	
2-1	Data	Name of the <i>LGR</i> .			
	LGRHEAD1	<i>LGR</i> that defines the integer variables for this keyword.			Always Required
1-1	Format	LGRHEAD1	45	INTE	
2-1 to 2-45	Data	Undefined.			
	LGRHEADQ	<i>LGR</i> grid keyword for the <i>INIT</i> Data file that defines the logical variables (T for true and F for false) for this keyword.			Always Required
1-1	Format	LGRHEADQ	5	LOGI	
2-1 to 2-5	Data	Undefined.			
	LGRHEADD	<i>LGR</i> grid keyword for the <i>INIT</i> Data file that defines the double precision REAL variables for this keyword.			Always Required
1-1	Format	LGRHEADD	5	DOUB	5
2-1 to 2-5	Data	Undefined.			

No.	Keyword Name	INIT Data - LGR Static Header Keywords Table F.16 (Repeated Each Time An LGR Each Data Set Is Written Out)			Status Or Value
		Keyword	No. of Entries	Data Type	
	LGRSGONE	The LGRSGONE keyword marks the end of the LGR section. Number of entries is always set to zero. There is no data set associated with this keyword.			Always Required
I-I	Format	LGRSGONE	0	MESG	
	INTEHEAD	The keywords are the same as the INIT Data Static Data Header keywords but the data pertains to the current LGR. See Table F.14 for a detailed description of this keyword type.			Required Only Once
	LOGIHEAD	See Table F.14 for a detailed description of this keyword type.			Required Only Once
	DOUBHEAD	See Table F.14 for a detailed description of this keyword type.			Required Only Once

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.
- 3) For formatted output all character variables, including the keyword name should be enclosed in single quotes, for example the LGR keyword should be written out as ‘LGR ’.

Table F.16: *INIT* Data - *LGR* Static Data Header Keywords

This series of keywords are repeated for each *LGR* in the model followed by the pertinent data and in the order outlined in Table F.13 and with the last *LGR* data set terminated by the LGRSGONE keyword to mark the end of the *LGR* input section.

Note

The *LGR*, LGRHEAD1, LGRHEADQ, and LGRHEADD keywords always proceeds and *LGR* data sets and the LGRSGONE keyword terminates the end of the *LGR* data section, which causes the data to revert back to the global grid data.

Example: INIT Data - LGR Static Header Keywords

The following example shows a typical formatted *INIT* Data - *LGR* Grid Header Keyword data set for a single *LGR* grid named LGR-I. As this is the first time *LGR* data is being written out both sets of *LGR* header keywords are written out followed by the property array data.

```

'LGR'          1 'CHAR'
'LGR-1'
'LGRHEAD1'     45 'INTE'
      1      100      -2345      -2345      -2345      -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345
'LGRHEADQ'     5 'LOGI'
  F F F F F
'LGRHEADD'     5 'DOUB'
  0.000000000000000D+00 -0.10000000200409D+21 -0.10000000200409D+21
 -0.10000000200409D+21 -0.10000000200409D+21
.....

'INTEHEAD'     249 'INTE'
-955283513      200400      2      -2345      -2345      -2345
    -2345    -2345      2      2      2      8
.....
      1      6      1      10      1      10
      1      1      1      0      30      3
      18     10      9
'LOGIHEAD'     79 'LOGI'
  T F F T F F F F T F F F F F F F T F T F F F F F F
  F F F F F F F F F F F F F F F F F F F F F F F F
  F F F F F F F F F F F F F F F F F F F F F F F
  F F F F
'DOUBHEAD'     185 'DOUB'
  0.000000000000000D+00  0.100000000000000D+01  0.365000000000000D+03
  0.100000000149012D+00  0.150000000596046D+00  0.300000000000000D+01
.....
  0.100000000000000D+01  0.000000000000000D+00  0.000000000000000D+00
  0.100000000000000D+01  0.100000000000000D-03
.....

'PORV'         8 'REAL'
  0.13358070E+06  0.13358070E+06  0.13358070E+06  0.13358070E+06
  0.13358070E+06  0.13358070E+06  0.13358070E+06  0.13358070E+06
'DX'           8 'REAL'
  0.50000000E+03  0.50000000E+03  0.50000000E+03  0.50000000E+03
  0.50000000E+03  0.50000000E+03  0.50000000E+03  0.50000000E+03
'DY'           8 'REAL'
  0.50000000E+03  0.50000000E+03  0.50000000E+03  0.50000000E+03
  0.50000000E+03  0.50000000E+03  0.50000000E+03  0.50000000E+03
'DZ'           8 'REAL'
  0.10000000E+02  0.10000000E+02  0.10000000E+02  0.10000000E+02
  0.10000000E+02  0.10000000E+02  0.10000000E+02  0.10000000E+02
'PERMX'        8 'REAL'
  0.50000000E+03  0.50000000E+03  0.50000000E+03  0.50000000E+03
  0.50000000E+03  0.50000000E+03  0.50000000E+03  0.50000000E+03
.....

```

End of Example

F.6.6 INIT DATA - LGR GRID ARRAY DATA KEYWORDS

The file structure for this set of keywords is identical to the Static Grid Array Data Keywords described in Table F.15, except that *LGR* header keywords prefix the actual data keywords. Both the *LGR* header and data keywords are repeated for each *LGR* in the model, and the last *LGR* data set is terminated by the LGRSGONE keyword to mark the end of the *LGR* input section. The keyword description for this type of data set is outlined in Table F.17.

Note that currently OPM Flow does not support *LGR* grids and therefore this series of keywords cannot be used by OPM Flow.

No.	Keyword Name	<i>INIT</i> Data - <i>LGR</i> Grid Array Data Keywords Table F.17 (Repeated For Each <i>LGR</i> In The Model)	Status Or Value
I-1	<i>LGR</i>	<i>LGR</i> Header Keywords (see Table F.16).	Required
	LGRHEAD1		
	LGRHEADQ		
	LGRHEADD		
I-2	INTEHEAD	<i>LGR</i> Header Keywords (see Table F.16).	Required Only Once
	LOGIHEAD		
	DOUBHEAD		
I-3	PROPERTY REGION <i>NNC</i> or ENPOINIT	<i>LGR</i> array data set for the current <i>LGR</i> – same format as the <i>INIT</i> Data - Grid Array Data Keywords (see Table F.15).	Required
I-4	LGRSGONE	<i>LGR</i> Header Section termination keyword (see Table F.16).	Required
Notes: 1) Rows shaded in gray indicate the start of a keyword. 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow. 3) For formatted output all character variables, including the keyword should be enclosed in single quotes, for example the <i>LGR</i> keyword should be written out as 'LGR '.			

Table F.17: *INIT* Data - *LGR* Grid Array Data Keywords

This keyword data set is repeated for each *LGR* in the model with the last *LGR* data set terminated by the LGRSGONE keyword to mark the end of the *LGR* data section.

Example: INIT Data - LGR Static Grid Array Data Keywords

The following example shows a typical formatted *INIT* Data - *LGR* Static Grid Array Data Keywords for the region data set with two *LGR* grids named LGR-1 and LGR-2.

```
'LGR      ' 1 'CHAR'
'LGR-1    '
'LGRHEAD1' 45 'INTE'
      1      100      -2345      -2345      -2345      -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345
'LGRHEADQ' 5 'LOGI'
F F F F F
'LGRHEADD' 5 'DOUB'
  0.000000000000000D+00 -0.100000000200409D+21 -0.100000000200409D+21
 -0.100000000200409D+21 -0.100000000200409D+21
'PVTNUM   ' 8 'INTE'
      1      1      1      1      1      1
      1      1
'SATNUM   ' 8 'INTE'
      1      1      1      1      2      2
      2      2
'EQLNUM   ' 8 'INTE'
      1      1      1      1      1      1
      1      1
'FIPNUM   ' 8 'INTE'
      1      1      1      1      1      1
      1      1
'LGR      ' 1 'CHAR'
'LGR-2    '
'LGRHEAD1' 45 'INTE'
      2      100      -2345      -2345      -2345      -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345    -2345    -2345    -2345
    -2345    -2345    -2345
'LGRHEADQ' 5 'LOGI'
F F F F F
'LGRHEADD' 5 'DOUB'
  0.000000000000000D+00 -0.100000000200409D+21 -0.100000000200409D+21
 -0.100000000200409D+21 -0.100000000200409D+21
'PVTNUM   ' 8 'INTE'
      1      1      1      1      1      1
      1      1
'SATNUM   ' 8 'INTE'
      1      2      1      2      2      1
      2      1
'EQLNUM   ' 8 'INTE'
      1      1      1      1      1      1
      1      1
'FIPNUM   ' 8 'INTE'
      1      1      1      1      1      1
      1      1
'LGRSGONE' 0 'MESS'
```

End of Example

F.6.7 INIT DATA – STATIC TABULAR DATA HEADER KEYWORD

The *INIT* Data – Static Tabular Data Header Keyword precedes the TAB keyword and consists of just one keyword followed by a series of integer values that define the dimensions of the data in the TAB keyword. *TABDIMS* is then followed by the TAB keyword that contains the actual tabular data in one continuous array.

The structure for this keyword is defined in Table F.18.

No.	Keyword Name	<i>INIT</i> Data – Static Tabular Data Header Keyword Table F.18			Status Or Value
		Keyword	No. of Entries	Data Type	
	<i>TABDIMS</i>	Tabular header keyword for the <i>INIT</i> Data file that defines the start of the tabular data and contains the base addresses and dimension of subsequent tabular array keyword TAB. Missing values should be defaulted to one.			Required
1-1	Format	<i>TABDIMS</i>	100	INTE	
2-1	Data	Total size of TAB data array, value should be the same as that on the TAB keyword.			NTABDA
2-2	Data	Location of the first entry in the TAB array for the rock property table.			IBROCK
2-3	Data	Maximum number of rock property tables.			NTROCK
2-4	Data	Location of the first entry in the TAB array of rock compaction data.			IBROCC
2-5	Data	Maximum number of pressure nodes in rock compaction data table.			NPROCC
2-6	Data	Maximum number of rock compaction data tables.			NTROCC
2-7	Data	Location of the first entry in the TAB array for the oil PVT property tables.			IBPVTO
2-8	Data	Location of the first entry in the TAB array for the oil pressure nodes.			JBPVTO
2-9	Data	Maximum number of composition nodes in oil PVT tables.			NRPVTO
2-10	Data	Maximum number of pressure nodes in oil PVT tables.			NPPVTO
2-11	Data	Maximum number of oil PVT property tables.			NTPVTO
2-12	Data	Location of the first entry in the TAB array for water PVT property tables.			IBPVTW
2-13	Data	Maximum number of water property tables.			NTPVTW
2-14	Data	Location of the first entry in the TAB array for gas PVT property tables.			IBPVTG
2-15	Data	Location of the first entry in the TAB array for the gas pressure nodes.			JBPVTG
2-16	Data	Maximum number of composition nodes in the gas PVT tables.			NRPVTG
2-17	Data	Maximum number of pressure nodes in the gas PVT tables.			NPPVTG
2-18	Data	Maximum number of gas PVT property tables.			NTPVTG
2-19	Data	Location of the first entry in the TAB array for surface density tables.			IBDENS
2-20	Data	Maximum number of surface densities tables.			NTDENS
2-21	Data	Location of the first entry in the TAB array for water saturation tables.			IBSWFN

No.	Keyword Name	INIT Data – Static Tabular Data Header Keyword Table F.18			Status Or Value
		Keyword	No. of Entries	Data Type	
2-22	Data	Maximum number of saturation nodes in water saturation tables.			NSSWFN
2-23	Data	Maximum number of water saturation tables.			NTSWFN
2-24	Data	Location of the first entry in the TAB array for gas saturation tables.			IBSGFN
2-25	Data	Maximum number of saturation nodes in gas saturation tables.			NSSGFN
2-26	Data	Maximum number of gas saturation tables.			NTSGFN
2-27	Data	Location of the first entry in the TAB array for oil saturation tables.			IBSOFN
2-28	Data	Location of the first entry in the TAB array for array of connate water saturations.			IBSWCO
2-29	Data	Maximum number of saturation nodes in the oil saturation tables.			NSSOFN
2-30	Data	Maximum number of oil saturation tables.			NTSOFN
2-31 to 40		Undefined.			I
2-41	Data	Location of the first entry in the TAB array to the <i>VE</i> table data.			IBVETB
2-42	Data	Maximum number of items in a <i>VE</i> table.			NSVETB
2-43	Data	Maximum number of <i>VE</i> tables.			NTVETB
2-44	Data	Location of the first entry in the TAB array to the threshold pressure array.			IBTHPR
2.45	Data	Location of the first entry in the TAB array to <i>SCALELIM</i> data			IBSLIM
2.46	Data	Maximum number of end-point versus depth nodes.			NSENDP
2.47	Data	Maximum number of end-point versus depth tables.			NTENDP
2.48	Data	Location of the first entry in the TAB array to the reference temperature <i>RTEMP</i> used in <i>PVZG</i> keyword in the PROPD section.			IBRTEM
2.49	Data	Location of the first entry in the TAB array to the value specified by the <i>TOLCRIT</i> keyword. Note that this is a single value applied to all the saturation tables.			IBCTOL
2.50		Undefined.			I
2-51	Data	Location of the first entry in the TAB array to <i>LANGMUIR</i> table.			IBLANG
2-52	Data	Maximum number of columns in <i>LANGMUIR</i> tables.			NCLANG
2-53	Data	Maximum number of rows in the <i>LANGMUIR</i> tables.			NSLANG
2-54	Data	Maximum number of <i>LANGMUIR</i> tables			NTLANG
2-55	Data	Location of the first entry in the TAB array to a <i>LANGSOLV</i> table.			IBLANG2

No.	Keyword Name	INIT Data – Static Tabular Data Header Keyword Table F.18			Status Or Value
		Keyword	No. of Entries	Data Type	
2-56	Data	Location of the first entry in the TAB array to a COALPP table.			IBCADP
2-57	Data	Location of the first entry in the TAB array to a COALADS table.			IBCADS
2-58	Data	Location of the first entry in the TAB array to a ROCKPAMA table.			IBROCP
2-59	Data	Maximum number of ROCKPAMA tables.			NTRPMA
2-60 to 2-100		Undefined.			I

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.
- 3) For formatted output all character variables, including the keywords should be enclosed in single quotes, for example the [PORV](#) and PROPERTY fields should be written out as 'PORV ' and 'DEPTH ', etc.

Table F.18: INIT Data – Static Tabular Data Header Keyword

Example: INIT Data – Static Tabular Data Header Keyword

The following example shows a typical formatted *INIT* Data – Static Tabular Data Header Keyword and the affiliated data.

```
'TABDIMS '      100 'INTE '
77006           1           2           1           60           1
17             36017         60           60           2       36799
2              37139        73139         60           60           2
5               2          36809          33           2       73259
33              2          36137        36797          66           2
73589            5          73644          20           5       74044
20              5           0           0       74444          25
2              74494        75835          20           1       77003
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
0               0           0           0           0           0
```

End of Example

F.6.8 INIT DATA – STATIC TABULAR AND CONSTANT DATA ARRAY KEYWORDS

The *INIT* Data – Static Tabular and Constant Data Array Keyword section contains the actual tabular data in one continuous array, followed by the constant data array. The tabular array data is parse according to the phases present in the model and the location and dimension data given by the *TABDIMS* keyword.

The structure for this set of keywords is defined in Table F.19.

No.	Keyword Or Variable Name	INIT Data – Static Tabular And Constant Data Array Keyword Table F.19			Status Or Value
		Keyword	No. of Entries	Data Type	
	TAB	Tabular data array keyword for the <i>INIT</i> Data file that defines the start of the tabular data array. The total number of entries is given by the NTABDA and the value should be the same as declared on the <i>TABDIMS</i> keyword.			Required
1-1	Format	<i>TABDIMS</i>	NTABDA	DOUB	
2-1	Data	Tabular data array. The data should be written out in double precision and missing data should be set to less than or equal to -10.0 x 1020 or greater than or equal to 10.0 x 1020.			
	CON	Constant data array that contains the constant dead oil, dead vaporized oil, and saturation pressure PVT entered via the <i>RSCONST/RSCONSTT</i> and <i>RVCONST/RVCONSTT</i> keywords in the <i>PROPS</i> section.			Required
1-1	Format	CON	NPVT x 3	DOUB	
2-1	Data	The data consists of three values for each PVT data set (NPVT), <i>RS</i> , <i>RV</i> , and the saturation pressure of the active phase in the model (bubble-point or dew point).			
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) Note both TAB and CON keywords should be enclosed in single quotation marks, that is ‘TAB’ and ‘CON’.					

Table F.19: *INIT* Data – Static Tabular and Constant Data Array Keywords

The phases present in the model are given on the INTEHEAD keyword (Table F.14) in location 2-15 where the type of phases in the model are set via an integer: 1 for oil, 2 for water, 3 for oil-water, 4 for gas, 5 for oil-gas, 6 for gas-water, and 7 for oil-water-gas. This data is used to access the phase dependent data (PVT, saturation functions tables, etc.) combine with the location (or base address) and the dimension data given on the *TABDIMS* keyword in Table F.18.

Note that data given in Table F.18 refers to the maximum values and therefore missing data should be set to the default value.

Example: INIT Data – Static Tabular and Constant Data Array Keywords

The following example shows a typical formatted data set for this type set of keywords and the affiliated data. Here the CON array has two PVT tables but the values are all set to zero indicating that dead oil and vaporized dead oil are not active phases in the model.

```
'TAB'          '73576' 'DOUB'
  0.497000000000000D-05  0.248000000000000D-05  0.000000000000000D+00
 -0.200000000000000D+21 -0.200000000000000D+21 -0.200000000000000D+21
 -0.200000000000000D+21 -0.200000000000000D+21 -0.200000000000000D+21
.....
 -0.200000000000000D+21 -0.200000000000000D+21 -0.200000000000000D+21
  0.521000000000000D-05  0.261000000000000D-05  0.000000000000000D+00
 -0.200000000000000D+21 -0.200000000000000D+21 -0.200000000000000D+21
'CON'          '6' 'DOUB'
  0.000000000000000D+00  0.000000000000000D+00  0.000000000000000D+00
  0.000000000000000D+00  0.000000000000000D+00  0.000000000000000D+00
.....
```

End of Example

F.7 LOG FILE – OPM FLOW TERMINAL OUTPUT

The LOG file for the commercial simulator contains an ASCII copy of the output from the terminal. The file itself is not generated by the commercial simulator but by an auxiliary program, ECLRUN, that is used to call and execute all the commercial simulators.

OPM Flow does not generate a LOG file; however, the file can be generated by using the Linux tee command, for example:

```
flow CASENAME | tee CASENAME.LOG
```

Which will copy the terminal output to the CASENAME.LOG file. Alternatively, one can use OPMRUN, the Graphical User Interface (“GUI”) program for OPM Flow, that provides similar functionality to the commercial simulator’s auxiliary program.

The OPM Flow terminal output differs from that generated by the commercial simulator. This section outlines the format of the OPM Flow specific Terminal Output (LOG File) illustrated with examples.

Note that the OPM Flow terminal output is generally also sent in the Print (PRT) file. In addition to the terminal output, the Print file also includes:

- the name of the machine the simulation was run on,
- the operating system,
- the start time for the simulation,
- a list of the command line arguments specified when running OPM Flow,
- the property tree for the linear solver,
- any output reports requested by the user (for example, using the [RPTPROPS](#) keyword in the [PROPS](#) section, the [RPTSOL](#) keyword in the [SOLUTION](#) section, the [RPTSCHED](#) keyword in the [SCHEDULE](#) section, and so forth), and finally
- an error summary giving the total number of Warnings, Problems, Errors, and so forth reported during the simulation.

Also note that the contents of the LOG file will vary depending on the model being simulated, the command line parameters, and the numerical performance of the simulator. The main simulation loop for example may contain additional messages other than those illustrated including: when the simulator changes the control mode for a well, when a time step fails to converge, and so forth.

F.7.1 LOG FILE - HEADER

At the start of the LOG file OPM Flow writes a header giving the software version followed by information about the number of MPI processes if applicable. The following example is taken from the LOG file generated for the Norne model ([NORNE_ATW2013](#)):

```
*****
*
*                               This is flow 2024.04                               *
*
* Flow is a simulator for fully implicit three-phase black-oil flow, *
* including solvent and polymer capabilities. *
* For more information, see https://opm-project.org *
*
*****
```

Using 8 MPI processes with 2 OMP threads on each

In this example OPM Flow version 2024.04 has been run using eight MPI processes and each MPI process may itself use 2 OpenMP threads.

F.7.2 LOG FILE – READING DECK

Next OPM Flow parses the input deck including any files referenced with the [INCLUDE](#) keyword. As each keyword is parsed and its data records are read, the simulator reports the keyword name, the file it is located in, and the corresponding line number. Any warning messages about unsupported keywords or unsupported keyword items found in the deck are then reported. See the following shortened example:

```
Reading deck file 'NORNE_ATW2013.data'
  0 Reading RUNSPEC in NORNE_ATW2013.data line 9
  1 Reading DIMENS in NORNE_ATW2013.data line 11
...
1342 Reading DATES in /.../norne/INCLUDE/BC0407_HIST01122006.SCH line 9565
#[34;1m
Warning: Unsupported keywords or keyword items:

NSTACK: keyword not supported
In file: NORNE_ATW2013.data, line 102
...
RPTRST: invalid value '0' for item 1
In file: /.../norne/INCLUDE/BC0407_HIST01122006.SCH, line 9245
RPTRST(RPTRST): invalid option or unsupported integer control format#[0m
```

In this example OPM Flow reports that the [NSTACK](#) keyword in the main data file is not currently supported, and that an invalid option has been specified for item 1 on the [RPTRST](#) keyword in an include file.

Note that OPM Flow outputs text formatting to the terminal to provide color highlighting of warning messages etc. (for example, “#[34;1m”). This formatting may not be displayed in a text editor.

F.7.3 LOG FILE – PROCESSING DATA

The input that was read in is now processed. First the grid structure and the fault model is created, and in this example the tracers are initialized:

```
Creating corner-point grid from keywords COORD, ZCORN and others

Initializing tracers from TRACER in NORNE_ATW2013.data line 268
...

Loading faults from FAULTS in /.../norne/INCLUDE/FAULT/FAULT_JUN_05.INC line 9

Applying MULTFLT in /.../norne/INCLUDE/FAULT/FAULTMULT_AUG-2006.INC line 8
...
```

In the above example, the fault structure is read in from an include file (FAULT_JUN_05.INC), and then the corresponding fault multipliers are read from different include files (FAULTMULT_AUG-2006.INC, etc.).

Next the dynamic information from the [SCHEDULE](#) section is processed to prepare the report steps:

```
Processing dynamic information from
NORNE_ATW2013.data line 449
Initializing report step 0/247 at 1997-11-06 0 DAYS line 449
Processing keyword DRSDT at line 452
...
Reading from: /.../norne/INCLUDE/VFP/NEW_D2_GAS_0.00003.VFP line 62
Processing keyword VFPPROD at line 62
#[34;1m
Warning: VFPPROD table 5 has 7 non-monotonic points of BHP(THP)
In /.../norne/INCLUDE/VFP/NEW_D2_GAS_0.00003.VFP line 62
This may cause convergence issues due to switching between BHP and THP control.
#[0m
...
Complete report step 1 (8 DAYS) at 1997-11-14 (0 DAYS)

Initializing report step 2/247 at 1997-11-14 (0 DAYS) line 204
```

```
Processing keyword WCONHIST at line 208
Processing keyword WCONINJE at line 212
Complete report step 2 (17 DAYS) at 1997-12-01 (8 DAYS)
...
```

In the above example where key values have been highlighted in yellow, the simulation will start the first report step (out of a total of 247) at the simulation start time of 6 November 1997; and will complete it after advancing the simulation by 8 days to 14 November 1997.

Whilst, initializing the first report step, lift table (*VFPPROD* table 5) is processed and OPM Flow reports that it has found 7 non-monotonic points of BHP versus THP, and warns that this may cause convergence issues. At this point the user may decide to adjust the table or continue with the unmodified table.

The simulation will then continue with report step 2 of 247 starting on 14 November 1997; and will complete report step 2 after advancing the simulation by a further 17 days to 1 December 1997 (elapsed time 25 days).

During report step 2 the *WCONHIST* and *WCONINJE* keywords have been processed. These modify the well production and injection controls effective from the start of the report step.

Note that OPM Flow only reports the initialization of the first five report steps.

Note at this point the dynamic data is only being processed to set up the report step; and the simulation has not yet started.

The grid is then processed after which the simulator reports the number of active cells in the model along with their total pore volume. The number of cells removed from the model due to the cut-off values applied and the resulting reduction in total pore volume are also reported. For example:

```
Processing grid
Total number of active cells: 44431 / total pore volume: 673248728 RM3
Removed 496 cells with a pore volume of 100450 RM3 (0.015 %) due to MINPV/MINPVV
```

For parallel runs information about the load balancing between the processes is then reported:

Load balancing distributes 44431 active cells on 8 processes as follows:

rank	owned cells	overlap cells	total cells
0	5821	3382	9203
1	6104	1771	7875
...			
sum	44431	13568	57999

Load balancing distributed the wells as follows:

well name	ranks with perforated cells
B-1AH	4
B-2H	1
...	

Finally, warning messages about unsupported keywords are reported, for example:

```
#[34;1m
Warning: Unhandled summary keyword DATE/DAY
In /.../norve/INCLUDE/SUMMARY/summary.data line 9#[0m
...
```

F.7.4 LOG FILE – SATURATION FUNCTION DIAGNOSTICS

Having processed the input data the simulator reports diagnostic information about the saturation functions including: the fluid system, the saturation function family, and the number of saturation function regions. For example:

=====Saturation Functions Diagnostics=====

System: Black-oil system.
Relative permeability input format: Saturation Family I.
Number of saturation regions: 2

In the above example a black-oil system has been specified, saturation function family I has been used, and there are two saturation function regions.

F.7.5 LOG FILE – MAIN SIMULATION LOOP

The simulator now enters the main simulation loop and reports the progress of the simulation as it advances from the start date to each of the report times specified in the input deck, for example:

===== Starting main simulation loop =====

Report step 0/247 at day 0/3312, date = 06-Nov-1997
Using Newton nonlinear solver.

Starting time step 0, stepsize 1 days, at day 0/8, date = 06-Nov-1997
Restart file written for report step 0/247, date = 06-Nov-1997 00:00:00
Newton its= 3, linearizations= 4 (0.1sec), linear its= 15 (0.1sec)

Starting time step 1, stepsize 3 days, at day 1/8, date = 07-Nov-1997
Newton its= 2, linearizations= 3 (0.0sec), linear its= 21 (0.1sec)

Starting time step 2, stepsize 4 days, at day 4/8, date = 10-Nov-1997
Newton its= 2, linearizations= 3 (0.0sec), linear its= 33 (0.1sec)

Report step 1/247 at day 8/3312, date = 14-Nov-1997

...

Report step 246/247 at day 3298/3312, date = 17-Nov-2006

Starting time step 0, stepsize 14 days, at day 3298/3312, date = 17-Nov-2006
Newton its= 3, linearizations= 4 (0.0sec), linear its= 70 (0.2sec)

In the example above, the simulation starts the first report step number 0 (out of a total of 247 report steps) on 6 November 1997 an elapsed time of 0 days (of a total of 3312 days).

The simulator reports that it is using the Newton nonlinear solver.

A restart file is written at the start of the first report step number 0 (out of a total of 247 report steps) on 6 November 1997 at 00:00:00.

The simulator then attempts to advance the simulation by a first time step of 1 day starting from 6 November 1997 an elapsed time of 0 days (towards the next report time at an elapsed time of 8 days). The time step is successfully completed after three Newton iterations, four linearizations (taking 0.1s) and 15 linear iterations (taking 0.1s).

The simulator then attempts to advance the simulation by a second time step of 3 days starting on 7 November 1997 an elapsed time of 1 day (towards the next report time at an elapsed time of 8 days). The time step is successfully completed after two Newton iterations, three linearizations (taking 0.0s), and 21 linear iterations (taking 0.1s). Note that since the previous time step successfully converged the time step size has been increased by OPM Flow's time stepping algorithm.

The simulator then attempts to advance the simulation by a third time step of 4 days starting on 10 November 1997 an elapsed time of 4 days (towards the next report time at an elapsed time of 8 days). The time step is successfully completed after two Newton iterations, three linearizations (taking 0.0s), and 33 linear iterations (taking 0.1s). Note the time step size has been constrained to advance the simulation to the next report time.

The simulation has now reached the start of report step 1 (of 247) on 14 November 1997 an elapsed time of 8 days (out of a total of 3312 days).

The above example now skips the intermediate report steps and jumps to the end of the simulation.

The simulation starts the final report step number 246 (of 247) on 17 November 2006 an elapsed time of 3298 days (out of a total of 3312 days).

The simulator then attempts to advance the simulation by a single time step of 14 days starting on 17 November 2006 an elapsed time of 3298 days (towards the final report time at an elapsed time of 3312 days). The time step is successfully completed after three Newton iterations, four linearizations (taking 0.0s), and 70 linear iterations (taking 0.2s). The simulation ends on 1 December 2006.

F.7.6 LOG FILE – END OF SIMULATION

The final section of the LOG file reports the numerical performance of the simulation including the number of processors, the simulation time, and the total numbers of linearizations, Newton iterations, and linear iterations, for example:

```
=====      End of simulation      =====
Number of MPI processes:           8
Threads per MPI process:          2
Setup time:                        8.40 s
  Deck input:                      5.18 s
Number of timesteps:              369
Simulation time:                  240.47 s
  Assembly time:                   22.44 s (Wasted: 0.0 s; 0.0%)
    Well assembly:                  2.74 s (Wasted: 0.0 s; 0.0%)
  Linear solve time:               83.22 s (Wasted: 0.0 s; 0.0%)
    Linear setup:                   8.16 s (Wasted: 0.0 s; 0.0%)
  Props/update time:               21.69 s (Wasted: 0.0 s; 0.0%)
  Pre/post step:                   35.75 s (Wasted: 0.0 s; 0.0%)
  Output write time:               69.24 s
Overall Linearizations:            1903      (Wasted:      0; 0.0%)
Overall Newton Iterations:        1534      (Wasted:      0; 0.0%)
Overall Linear Iterations:        22713     (Wasted:      0; 0.0%)
```

In the above example, the simulation took 8.4s to setup followed by 240.47s of simulation time during which it performed at total of 369 time steps, which required a total of 1534 Newton iterations. The summary includes a breakdown of the simulation time; of which the linear solve time was the largest component in this example.

The total linear solve time was 83.22s of which 0.0s (or 0.0% of the total) was wasted due to convergence failures (since there were no convergence failures in this example). Similarly, there were a total of 1,534 Newton iterations of which zero were wasted due to convergence failures.

F.8 RESTART FILES – DYNAMIC DATA (SOLUTION, GROUPS, WELLS, CONNECTIONS, AQUIFERS, ETC.)

RESTART files contain the solution data arrays for each active cell in the model, and represents a “snapshot” of the reservoir state at a reporting time step, for example the pressure (**PRESSURE**, **PPCW**, and **PPCG**), fluid saturations (**SGAS**, **SOIL**, and **SWAT**), and fluid state (**RS** and **RV**) for all active cells. The **PPCW** and **PPCG** arrays are the water and capillary solution arrays and the **RS** and **RV** arrays are the gas-oil and vaporized-oil ratio (condensate-gas ratio) solution arrays. The data written to the **RESTART** file is dependent on the fluid phases and options invoked in the model, as well as any additional user requested solution data. The default arrays written out are always sufficient to enable the simulator to restart from a previous run³⁹⁵. This was the original purpose of the **RESTART** file, that is saving the reservoir state in such a manner, is to enable a “restart” of the simulation.

RESTART files also contain the necessary information to do restart runs for Groups, Wells, Connections, Aquifer data, etc. This information enables restart runs based on Schedule information read from the Restart file only, using the **SKIPREST** keyword in the **SCHEDULE** section.

As of the OPM Flow 2019-04 release the simulator can “restart” from the commercial simulator’s generated **RESTART** files as well as the commercial simulator being to “restart” from the OPM Flow’s generated **RESTART** files. See section 2.2 *Running OPM Flow From The Command Line* for additional information.

As **RESTART** files contain a complete description of the reservoir state, they are also used in post processing software to visualize the reservoir solutions through time, for example by OPM ResInsight. And as the functionality of the both OPM Flow and the commercial simulator have expanded over time additional solution arrays have been added to the **RESTART** file to enable “restarts” (**POLYMER** – polymer saturations), as well as to write out additional user specific solution arrays (**FIPOIL** - fluid in-place oil array), that are not necessary for a “restart”, but are consider useful in understanding the reservoir performance.

The frequency and type of data written to the **RESTART** file is controlled via the **RPTSCHED** and **RPTRST** keywords in the **SCHEDULE**, with the latter having greater functionality and flexibility. For example, to request that the standard restart data be written out every month using the **RPTRST** keyword:

```
--
--      RESTART CONTROL BASIC = 4 (YEARLY) 5 (MONTHLY)
--
RPTRST
      BASIC=5 /
```

The next example requests that the standard restart data be written at every report time step until this switch is reset and all the restarts are kept. In addition to the standard data, the gas, oil, and water relative permeability data will also be written out at each report time step.

```
--
--      RESTART CONTROL BASIC = 4 (YEARLY) 5 (MONTHLY)
--
RPTRST
      BASIC=2   KRG   KRO   KRW /
```

RESTART solution data is written out via two separate files; the **RESTART** Index file that specifies and defines the format and data type written to the **RESTART** Data file, and the **RESTART** Data file that contains the solution data for each active cell in the model for the requested report time steps. The commercial simulator writes out both files, where as OPM Flow only writes out the **RESTART** Data file, as the data can be read by most post-processing software, including OPM ResInsight, using only the **RESTART** Data file as input. Secondly, both the commercial simulator and OPM Flow can be “restarted” using just the **RESTART** Data file.

The following sections outline the format for these two file types.

³⁹⁵ For example in a three phase (oil, gas, and water) runs only two phase saturations are needed for a “restart” run, as the third phase can be calculated in by summing two phases and subtracting from one.

F.8.1 RESTART INDEX FILES

This file type is not supported or required by OPM Flow or OPM ResInsight to read or write restart files.

F.8.2 RESTART DATA FILE SPECIFICATION

This file type contains the global and [LGR](#) grid solution data property data³⁹⁶ ([PRESSURE](#), [SGAS](#), [SOIL](#), and [SWAT](#)) and the group, well, and connection data³⁹⁷. The overall structure of this file is similar to the [INIT](#) file, and the individual data keywords are structured as presented in Table F.20.

Reference Section	Global Section Keywords	Global Data Keywords	LGR Data Keywords	Global Section Keywords	Notes
F.8.3	SEQNUM				Global grid header keyword that defines the start of a RESTART global grid time step data set.
F.8.3	INTEHEAD LOGIHEAD DOUBHEAD				Global grid header keyword that defines the integer, logical, and double precision variables for this header keyword.
F.8.4	IGRP SGRP etc.				Group, well, and connection data status keywords for this reporting time step.
F.8.6	IAAQ SAAQ ect.				Aquifer definition arrays.
F.8.7	HIDDEN				Defines HIDDEN solution data, that is a list of solution arrays NOT to be read by post processing software.
F.8.8	STARTSOL				This keyword marks the start of the solution variable section for the global grid.
F.8.8		PRESSURE SWAT etc.			Solution array section (PRESSURE , SGAS , SOIL , SWAT , etc.) for the Global grid.
F.8.8	LGRNAMES				A list of LGRs in this report step.
F.8.8	ENDSOL				Marks the end of the solution variable section for the global grid.
F.5.1			LGR LGRHEAD I LGRHEAD Q LGRHEAD D		LGR grid header keyword for that defines the start of a RESTART LGR data set.
F.8.3			INTEHEAD LOGIHEAD DOUBHEAD		LGR header keyword and defines the integer, logical, and double precision variables for this header keyword.

³⁹⁶ Several keywords are used for both global and [LGR](#) grids, for example the [INTEHEAD](#), [LOGIHEAD](#), and [DOUBHEAD](#) keywords and [SOLUTION](#) arrays.

³⁹⁷ The group, well, and connection production data is not stored on the [RESTART](#) file, but is instead stored on the [SUMMARY](#) file. The [RESTART](#) file contains the group, well, and connection configurations as they change versus time. For example, the status of a well ([OPEN](#) or [SHUT](#)), or the connections open to flow in a well.

Reference Section	Global Section Keywords	Global Data Keywords	LGR Data Keywords	Global Section Keywords	Notes
F8.4			IGRP SGRP etc.		Group, well, and connection data status keywords for this reporting time step.
F8.7			HIDDEN		Defines HIDDEN solution data section.
F8.8			STARTSOL		Start of LGR solution array section.
F8.8			Solution Keywords	PRESSURE SWAT etc.	LGR solution array section – same as global grid.
F8.8			ENDSOL		End of the solution variable section for the current LGR grid.
F5.1			ENDLGR		End of current LGR section.
F5.1			LGR LGRHEAD1 LGRHEADQ LGRHEADD		LGR grid header keywords.
F8.4			IGRP SGRP etc.		Group, well, and connection data status keywords for this reporting time step.
F8.7			HIDDEN		Defines HIDDEN solution data section.
F8.8			STARTSOL		Start of solution array section.
F8.8			Solution Keywords	PRESSURE SWAT etc.	LGR solution array section.
F8.8			ENDSOL		End of the solution variable section.
F5.1			ENDLGR		End of current LGR section.
F8.3	SEQNUM				Global grid header keyword that defines the start of a RESTART global grid time step data set.
Notes: 1) Cells shaded in gray indicate the combination is not applicable and cells colored orange indicate that this item is not currently supported by OPM Flow.					

Table F.20: RESTART Data File Structure

The individual keywords are described in the following sections.

F.8.3 RESTART DATA - HEADER KEYWORDS

This set of keywords³⁹⁸ are the first set of keywords that should be read or written to the *RESTART* file. This data set includes INTEHEAD, LOGIHEAD, and DOUBHEAD keywords that define versus parameters used in subsequent keywords in the *RESTART* file for the global and *LGR* grids.

The structure for this type of keyword is defined in Table F.21.

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
	SEQNUM	Global grid header keyword for the <i>RESTART</i> Data file that defines the start of a <i>RESTART</i> global grid time step data set for unified restart files only.			Optional
1-1	Format	SEQNUM	1	INTE	
2-1	Data	Report time step number.			0
	INTEHEAD	Global grid header keyword for the <i>RESTART</i> Data file that defines the integer variables for this time step. NIHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model. Undefined values should be set to zero.			Required
1-1	Format	INTEHEAD	NIHEAD	INTE	
2-1	Data	An encoded integer corresponding to the time the file was created. For files not originating from the commercial simulator, this value may be set to zero.			ISNUM
2-2	Data	Simulator version, should be set to zero.			VERSION
2-3	Data	Unit system used in the simulation, set to 1 for metric, 2 for field, and 3 for laboratory			
2-4 to 2-8	Data	Undefined.			0
2-9	Data	<i>DIMENS</i> key word in the <i>RUNSPEC</i> section - the number of grid blocks in the x-direction.			NX
2-10	Data	<i>DIMENS</i> key word in the <i>RUNSPEC</i> section - the number of grid blocks in the y-direction.			NY
2-11	Data	<i>DIMENS</i> key word in the <i>RUNSPEC</i> section - the number of grid blocks in the z-direction.			NZ
2-12	Data	Number of global active cells in the model .			NACTIV
2-13	Data	Not used.			0
2-14	Data	Not used.			0
2-15	Data	Type of phases in the model, set to 1 for oil, 2 for water, 3 for oil-water, 4 for gas, 5 for oil-gas, 6 for gas-water, and 7 for oil-water-gas.			PHASE
2-16	Data	Undefined.			0
2-17	Data	Number of wells in the model.			NWELLS

³⁹⁸ The *RESTART* Data – Header keywords are used both for global and *LGR* grids with the global grid data for the global grid and the individual *LGR* data for the *LGR* grids. The *LGR* data is preceded by a series of *LGR* header keywords and terminated by an *LGR* termination keyword.

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-18	Data	Maximum number of grid block connections per well.			NCWMAX
2-19	Data	Actual number of groups.			NGRP
2-20	Data	Maximum number of wells belonging to a group in the model.			NWGMAX
2-21	Data	Maximum number of groups for this model.			NGMAXZ
2-22 to 2-24	Data	Not used.			
2-25	Data	Number of values per well in the IWEL array (default value is 97).			NIWELZ
2-26	Data	Number of values per well in the SWEL array.			NSWELZ
2-27	Data	Number of values per well in the XWEL array.			NXWELZ
2-28	Data	Number of eight character words per well in the ZWEL array.			NZWELZ
2-29 to 2-32	Data	Not used.			
2-33	Data	Number of values per completion in ICON array (default 19).			NICONZ
2-34	Data	Number of values per completion in SCON array.			NSCONZ
2-35	Data	Number of values per completion in XCON array.			NXCONZ
2-36		Not used.			0
2-37	Data	Number of values per group in IGRP array.			NIGRPZ
2-38	Data	Number of values per group in SGRP array.			NISGRPZ
2-39	Data	Number of values per group in XGRP array.			NIXGRPZ
2-40	Data	Number of values per group in ZGRP array.			NIZGRPZ
2-41		Not used.			
2-42	Data	Maximum number of analytical aquifer connections.			NCAMAX
2-43	Data	Number of values per aquifer in the IAAQ array.			NIAAQZ
2-44	Data	Number of values per aquifer in the SAAQ array.			NSAAQZ
2-45	Data	Number of values per aquifer in the XAAQ array.			NXAAQZ
2-46	Data	Number of values per aquifer connection in the ICAQ array.			NICAQZ
2-47	Data	Number of values per aquifer connection in the SCAQ array.			NSCAQZ
2-48	Data	Number of values per aquifer connection in the ACAQ array.			NXCAQZ
2-49 to 2-51		Not used.			
2-52	Data	Index indicating if group control is used or not (1 – for <i>GCONPROD</i> group control, 2 for <i>GCONINJE</i> control, or 0 if no group control)			NGCTRL

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-53 to 2-58		Undefined.			
2-59	Data	Parameter to determine the nominated phase for the guide rate.			NGRNPH
2-60 to 2-64		Undefined.			
2-65	Data	Calendar day of report step, the value should be greater than or equal to one and less than or equal to 31 for the calendar days.			DAY
2-66	Data	Calendar month of report step, the value should be greater than or equal to one and less than or equal to 12 for the calendar month.			MONTH
2-67	Data	Calendar year of report step, a positive four digit integer value of the start year, which must be specified fully by four digits, that is 2019.			YEAR
2-68	Data	Numerical solver step, that is the number of solver steps the simulator has performed so far.			NUM_SOLVER_STEP
2-69	Data	Report step.			REPORT_STEP
2-70 to 2-71		Undefined.			0
2-72	Data	Index for <i>WHISTCTL</i> keyword.			WHISTC
2-73 to 2-80		Undefined.			
2-81	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum number of Newton iterations for a time step.			NEWTMX
2-82	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the minimum number of Newton iterations for a time step			NEWTMIN
2-83	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum number of linear iterations within a Newton iteration.			LITMAX
2-84	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the minimum number of linear iterations within a Newton iteration.			LITMIN
2-85		Undefined.			
2-86		Undefined.			
2-87	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum number of iterations within a well flow calculation.			MXWSIT
2-88	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum number of iterations for solving the bottom-hole pressure for wells under tubing head pressure control within a well flow calculation.			MXWPIT
2-89		Undefined.			

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-90	Data	<i>REGDIMS</i> and <i>TABDIMS</i> keywords in the <i>RUNSPEC</i> section. The maximum number of regions in the <i>FIPNUM</i> region array. Note that this parameter may also be set on the <i>TABDIMS</i> keyword as well. If NTFIP is set in both places then the maximum value should be used.			NTFIP
2-91 to 2-94		Undefined.			
2-95	Data	Simulator code used to generate the file, set to 100 for Schlumberger's ECLIPSE 100, 300 for Schlumberger's ECLIPSE 300, 500 for Schlumberger's ECLIPSE 300 Thermal, 700 for Schlumberger's INTERSECT simulator, 800 for Schlumberger's FrontSim simulator, or a negative value for other simulators. OPM Flow uses a value of 100.			IPROG
2-96 to 2-99		Undefined.			0
2-100	Data	<i>REGDIMS</i> keyword in the <i>RUNSPEC</i> section. The total maximum number of regions. The <i>FIPNUM</i> regions are defined by (2-90). If additional sets of fluid in-place regions are used, as per the FIPxxx series of fluid in-place region keywords, then NMFIPR is the sum of all <i>FIP</i> regions.			NMFIPR
2-101 to 2-131					
2-132	Data	<i>NETWORK</i> keyword in the <i>RUNSPEC</i> section - maximum number of nodes in an extended network model or zero if the extended network option has not been activated.			NODMAX
2-133	Data	<i>NETWORK</i> keyword in the <i>RUNSPEC</i> section - maximum number of branches in an extended network model or zero if the extended network option has not been activated.			NBRMAX
2-134	Data	Number of items per branch in the IBRAN array when the extended network model is active, or zero if the extended network option has not been activated.			NIBRAN
2-135	Data	Number of items per branch in the RBRAN array when the extended network model is active, or zero if the extended network option has not been activated.			NRBRAN
2-136	Data	Number of items per node in the INODE array when the extended network model is active, or zero if the extended network option has not been activated.			NINODE
2-137	Data	Number of items per node in the RNODE array when the extended network model is active, or zero if the extended network option has not been activated.			NRNODE
2-138	Data	Number of items per node in the ZNODE array when the extended network model is active, or zero if the extended network option has not been activated.			NZNODE

No.	Keyword Name	RESTART Data - Header Keywords			Status Or Value
		Table F.21			
		Keyword	No. of Entries	Data Type	
2-139	Data	Number of items in the INOBR array.			NINOBR
2-140 to 2-156		Undefined.			0
2-157	Data	The number of actions in the data set.			NO_ACT
2-158	Data	Maximum number of lines of schedule data for ACTION keyword - including ENDACTIO.			MAX_LINES
2-159	Data	Maximum number of eight character strings per input line of Action data (rounded up from input deck).			MAXSPRLINE
2-160 to 2-162		Undefined.			
2-163	Data	Maximum number of aquifer connections actually used in the model.			NGCAUS
2-164	Data	Maximum number of wells in the model.			NWMAXZ
2-165 to 2-174		Undefined.			0
2-175	Data	Number of multi-segmented wells defined with the WELSEGS keyword in the SCHEDULE section, for when multi-segment wells have been activated, or zero otherwise.			NSEGWL
2-176	Data	WSEGDIMS keyword in the RUNSPEC section - maximum number of multi-segment wells, for when multi-segment wells have been activated, or zero otherwise.			NSWLMX
2-177	Data	WSEGDIMS keyword in the RUNSPEC section - maximum number of segments per multi-segment well, for when multi-segment wells have been activated, or zero otherwise.			NSEGMX
2-178	Data	WSEGDIMS keyword in the RUNSPEC section - maximum number of branches per multi-segment well, including the main branch, for when multi-segment wells have been activated, or zero otherwise.			NLBRMX
2-179	Data	Number of entries per segment in the multi-segment well ISEG array, for when multi-segment wells have been activated, or zero otherwise.			NISEGZ
2-180	Data	Number of entries per segment in the multi-segment well RSEG array, for when multi-segment wells have been activated, or zero otherwise.			NRSEGZ
		The number of data items per well segment in the RSEG restart vector is dependent on the number of active phases (oil, gas, water) in the simulation run. In particular, the following relations hold:			
		Scenario	Phases	NRSEGZ	
		Single Phase	Oil, Gas, or Water	126	
		Two Phase	Oil/Gas, Oil/Water, Gas/Water	134	
		Three Phase	Oil/Gas/Water	146	

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-181	Data	Number of entries per segment in the multi-segment well ILBR array, for when multi-segment wells have been activated, or zero otherwise.			NILBRZ
2-181 to 2-206		Undefined.			0
2-207	Data	The first part of the current simulation time in the form HH:MM:SS, that is HH for hours. The value should be greater than or equal to zero and less than or equal to 23.			IHOURZ
2-208	Data	The second part of the current simulation time in the form HH:MM:SS, that is MM for minutes. The value should be greater than or equal to zero and less than or equal to 59.			IMINTS
2-209 to 2-223		Undefined.			0
2-224	Data	Number of INTEGER lines in the <i>AQUNUM</i> data set.			NILAQN
2-225	Data	Number of REAL lines in the <i>AQUNUM</i> data set.			NIRAQN
2-226		Undefined.			0
2-227	Data	Number of entries in the <i>AQUNUM</i> data set.			NUMAQN
2-228 to 2-234		Undefined.			0
2-235	Data	Number of items in the ICOT array.			NICOTZ
2-236	Data	Number of items in the XCOT array.			NXCOTZ
2-237	Data	Number of items in the IWET array.			NIWETZ
2-238	Data	Number of items in the XWET array.			NXWETZ
2-239	Data	Number of items in the IGRT array.			NIGRTZ
2-240	Data	Number of items in the XGRT array.			NXGRTZ
2-241	Data	Number of tracers in the model plus two.			NSTRA2
2-242 to 2-245		Undefined.			0
2-246	Data	Maximum number of conditions per action.			MAXACTC
2-253	Data	Maximum number of analytical aquifers in the model.			MAAQID
2-254 to 2-262		Undefined.			0
2-263	Data	Number of Field <i>UDQ</i> data (parameters).			NOFUDQS
2-264	Data	Number of Group <i>UDQ</i> data (parameters).			NOGUDQS

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-265		Undefined.			
2-266		Undefined.			
2-267	Data	Number of Well <i>UDQ</i> data (parameters).			NOWUDQS
2-268	Data	Integer seed value for the RAND.			UDQPAR_I
2-269		Undefined.			
2-270		Undefined.			
2-271	Data	Maximum number of Chord segment links per multi-segment well for when multi-segment wells have been activated, or zero otherwise.			NCRDMX
2-272 to 2-290		Undefined.			0
2-291		Number of integer IUADs.			NOIUADS
2-292		Number of integer IUAPs.			NOIUAPS
2-293 to 2-296		Undefined.			
2-297	Data				RSEED
2-298 to 2-410		Undefined.			
2-411	Data	The third part of the current simulation time in the form HH:MM:SS, that is SS for seconds. The value should be greater than or equal to zero and less than or equal to 59.			ISECND
	LOGIHEAD	Global grid keyword for the <i>RESTART</i> Data file that defines the logical variables (T for true and F for false) for this time step. NLHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model. Undefined items should be set to F (False).			Required
1-1	Format	LOGIHEAD	NLHEAD	LOGI	
2-1	Data	Dissolved gas is present in the model indicator, set to T if present in the model else set to F.			<i>DISGAS</i>
2-2	Data	Vaporized oil is present in the model indicator, set to T if present in the model else set to F.			<i>VAPOIL</i>
2-3	Data	<i>SATOPTS</i> in the <i>RUNSPEC</i> section - directional relative permeability curves are active in the model.			DIRECT
2-4	Data	<i>SATOPTS</i> keyword in the <i>RUNSPEC</i> section - reversible relative permeability curves (black-oil) are active in the model.			IRREVERS

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
		Radial grid geometry has been used for a compositional model indicator, set to T or F. Note that currently OPM Flow does not support radial grids and therefore this item should be set to F.			<i>RADIAL</i>
2-5	Data	Radial grid geometry has been used for a black-oil model indicator, set to T or F. Note that currently OPM Flow does not support radial grids and therefore this item should be set to F.			<i>RADIAL</i>
		Reversible relative permeability (compositional) are active in the model.			IRREVERS
2-6		Undefined.			
2-7	Data	Enable hysteresis indicator.			HYSER
2.8 to 2-14		Undefined.			F
2-15	Data	The dual porosity option has been activated in the model indicator, set to T or F. Note that currently OPM Flow does not support dual porosity grids and therefore this item should be set to F.			<i>DUALPORO</i>
2-16		Undefined.			F
2-17	Data	<i>ENDSCALE</i> keyword in the <i>RUNSPEC</i> section - enable end-point scaling indicator.			<i>ENDSCALE</i>
2-18	Data	<i>ENDSCALE</i> keyword in the <i>RUNSPEC</i> section - directional end-point scaling indicator.			DIRECT
2-19	Data	<i>ENDSCALE</i> keyword in the <i>RUNSPEC</i> section - reversible end-point scaling indicator.			IRREVERS
2-20	Data	<i>SCALECRS</i> keyword in the <i>PROPS</i> section - alternative three point end-point scaling indicator.			<i>SCALECRS</i>
2-21 to 2-30		Undefined.			F
2-31	Data	Coal Bed Methane option in the black-oil model indicator, set to T or F. Note that currently OPM Flow does not support the Coal Bed Methane option and therefore this item should be set to F.			F
2-32 to 2-38		Undefined.			F
2-39	Data	Constant oil compressibility (<i>PVCDO</i>) indicator.			ConstCo
2-40 to 2-75		Undefined.			
2-76	Data	Multi-segment wells are present in the model indicator.			HasMSWells

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-77 to 2-127		Undefined.			
2-128		Coal Bed Methane option in the compositional model indicator, set to T or F. Note that currently OPM Flow does not support the Coal Bed Methane option and therefore this item should be set to F.			F
	DOUBHEAD	Global grid keyword for the <i>RESTART</i> Data file that defines the double precision REAL variables for this time step. NDHEAD is the number of entries associated with this keyword, and is dependent on the features being used in the model. Undefined values should be set to zero.			Required
1-1	Format	DOUBHEAD	NDHEAD	DOUB	
2-1	Data	The simulation REPORT <i>TIME</i> STEP, expressed in days for all units, except for the <i>LAB</i> system of units which should be expressed in hours.			
2-2	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - maximum length of next time step.			TSINIT
2-3	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - maximum length of time step after next.			TSMAXZ
2-4	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - minimum length of all time steps.			TSMINZ
2-5	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - minimum length of all chopped time steps.			TSMCHP
2-6	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - maximum growth rate a time step can be increased by, subject to the maximum allowable time step size set by TSMAXZ.			TSFMAX
2-7	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - minimum decay rate a time step can be decreased by, subject to the minimum allowable time step size set by TSMINZ.			TSFMIN
2-8	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the decay rate a time step can be decreased by after the number of target iterations has been exceeded.			TSFCNV
2-9	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the time truncation error target.			TRGTTE
2-10	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the non-linear convergence error.			TRGCNV
2-11	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the target material balance error.			TRGMBE
2-12	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the linear convergence error target.			TRGLCV
2-13 to 2-16		Undefined.			

No.	Keyword Name	<i>RESTART</i> Data - Header Keywords Table F.21			Status Or Value
		Keyword	No. of Entries	Data Type	
2-17	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum time truncation error.			XXXTTE
2-18	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum non-linear convergence error.			XXXCNV
2-19	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum mass balance error.			XXXMBE
2-20	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum linear convergence error.			XXXLCV
2-21	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum well flow convergence error.			XXXWFL
2-22 to 2-82		Undefined.			
2-83	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - target pressure change within a time step.			TRGDPR
2-84	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - time step growth factor of the time step after a convergence failure.			TFDIFF
2-85	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum pressure change at the last Newton iteration.			DDPLIM
2-86	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum saturation change at the last Newton iteration.			DDSLIM
2-87		Undefined.			
2-88	Data	<i>GUIDERAT</i> keyword in the <i>SCHEDULE</i> section - guide rate parameter A.			GRPAR_A
2-89	Data	<i>GUIDERAT</i> keyword in the <i>SCHEDULE</i> section - guide rate parameter B.			GRPAR_B
2-90	Data	<i>GUIDERAT</i> keyword in the <i>SCHEDULE</i> section - guide rate parameter C.			GRPAR_C
2-91	Data	<i>GUIDERAT</i> keyword in the <i>SCHEDULE</i> section - guide rate parameter D.			GRPAR_D
2-92	Data	<i>GUIDERAT</i> keyword in the <i>SCHEDULE</i> section - guide rate parameter E.			GRPAR_E
2-93	Data	<i>GUIDERAT</i> keyword in the <i>SCHEDULE</i> section - guide rate parameter F.			GRPAR_F
2-94 to 2-97		Undefined.			
2-98	Data	Guide rate parameter delay interval.			GRPAR_INT
2-99		Undefined.			
2-100	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the maximum throughput ratio over a time step.			THRURPT
2-101	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - maximum tolerable pressure change within a time step.			XXXDPR
2-102	Data	<i>TUNING</i> keyword in the <i>SCHEDULE</i> section - the target fluid in-place error in Local Grid Refinements.			TRGFIP

No.	Keyword Name	RESTART Data - Header Keywords			Status Or Value
		Table F.21			
		Keyword	No. of Entries	Data Type	
2-103	Data	TUNING keyword in the SCHEDULE section - target surfactant change when the Surfactant Model is active in the run.			TRGSFT
2-104 to 2-144		Undefined.			
2-145	Data	Guide rate parameter damping factor.			GRPAR_DAMP
2-146 to 2-161		Undefined.			
2-161	Data	The simulation START time, expressed in days for all units, except for the LAB system of units which should expressed in hours.			
2-162	Data	Cumulative time from the start of the simulation, expressed in days for all units, except for the LAB system of units which should expressed in hours. That is the sum of values (2-1) and (2-161).			
2-163 to 2-212		Undefined.			0
2-213	Data	UDQPARAM item number 2 (Permitted range (+/-) of user-defined quantities).			UDQPAR_2
2-214	Data	UDQPARAM item number 3 (Value given to undefined elements when outputting data).			UDQPAR_3
2-215	Data	UDQPARAM item number 4 (fractional equality tolerance used in == <= etc. functions).			UDQPAR_4
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) Cells colored red in the No. column indicate that this item is either “Not Used” or “Undefined”.					
3) For formatted output all character variables, including the keyword should be enclosed in single quotes, for example the DOUBHEAD keyword should be written out as ‘DOUBHEAD’.					

Table F.21: *RESTART* Data - Header Keywords

Example: RESTART Data - Header Keywords

The following example shows a typical formatted *RESTART* Data - Header keywords.

```

'INTEHEAD'          249 'INTE'
-955283513          200400          2          -2345          -2345          -2345
-2345              -2345          10          10          1          100
10                 -2345          7          -2345          0          1
0                  2            2            0            0            0
110                108          109          3            97          93
-2345              -2345          19          38          53          -2345
97                 93           146          5            0            1
15                 24            8            5            2            4
0                  0            0            0            0            0
0                  0            0            0            0            0
0                  0            0            0            19          10
1982               0            0            0            1            0
0                  0            0            0            1            10
0                  0            12          1            25          1
-2345              -2345          8            8            5            1
2                  1            2            0            100          2
7                  0           -11          1            0            1
0                  0            0           10          10          1
100                1            1            1            1          10
10                 1            19          1            0            0
0                  0            0            1            1            1
0                  0            0            0            0            0
0                  14          10           10          16          1
1                  1            1            1            2            1
1                  1            1            1            1          31
108                0            0            0            0            0
0                  50          10           4            5            9
0                  2            8            8            12          1
25                 1          -32767       -32767       -32767       -32767
0                  1            1            1            18          86
5                  1            1            1            1          18
86                 -32767       -32767          0            0          109
53                 146          8            0            19          110
0                  0            1            1            86          0
0                  0            0            0            0            1
0                  0            0            0            1            0
0                  1            0           -1           12          0
0                  10          13          1            0            0
0                  0            2            0            0          3600
1                  6            1           10           1            10
1                  1            1            0           30          3
18                 10          9
'LOGIHEAD'          79 'LOGI'
T F F T F F F F T F F F F F F F T F T F F F F F F
F F F F F F F F F F F F F F F F F F F F F F F
F F F F F F F F F F F F F F F F F F F F F F
F F F F
'DOUBHEAD'          185 'DOUB'
0.000000000000000D+00 0.100000000000000D+01 0.365000000000000D+03
0.100000000149012D+00 0.150000000596046D+00 0.300000000000000D+01
0.300000001192093D+00 0.100000000149012D+00 0.100000000149012D+00
.....
-0.100000000200409D+21 0.100000000000000D+01 0.100000000000000D+01
0.100000000000000D+01 0.000000000000000D+00 0.000000000000000D+00
0.100000000000000D+01 0.100000000000000D-03

```

End of Example

F.8.4 RESTART DATA – GROUP, WELL, AND CONNECTION DATA KEYWORDS

The Restart Data – Well and Group Completion Keywords³⁹⁹ define various parameters associated with groups, wells, and connection keywords in the [RESTART](#) file for both global and [LGR](#) grids.

The structure for this type of keyword is based on the data type as summarized below:

- 1) The group data set keywords are specified in Table F.22 and consists of the IGRP, SGRP, XGRP, and ZGRP keywords.
- 2) Table F.23 contains the multi-segment well keywords that consist of the ISEG, RSEG, ILBS, ILBR, and ICRD keywords.
- 3) For wells the keywords are IWEL, SWEL, XWEL, and ZWEL and Table F.24 describes these keywords.
- 4) Well connection keywords are specified in Table F.25 and consist of the ICON, SCON, and XCON keywords.
- 5) Tracer keywords are listed in Table F.26 and consists of the ICOT, XCOT, IWET, XWET, IGRT, and XGRT keywords.
- 6) Finally, the network keywords INODE, IBRAN, INOBR, RNODE, RBRAN, and ZNODE are specified in Table F.27.

Note that if multi-segment wells, tracers, and networks are absent from the simulation deck then this data will not be written to the [RESTART](#) Data file.

The following table outlines the keywords for groups.

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.22 - Group Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	IGRP	Keyword defines the integer Group, well, and connection data for this time step. Undefined values should be set to zero.			Required
1-1	Format	IGRP	NSGRPZ x NGMAXZ	INTE	
2-1 to 2- NWG MAX	Data	Integer group data array IGRP(NIGRPZ, NGMAXZ) where NIGRPZ and NGMAXZ are defined on the INTEHEAD keyword. Entries 1 to NWGMAX define the index of each well in the group, if this is a well group, or the index of each child group in this group if it is a node group. Note that undefined items in this array may be set to zero.			
2 - (NWG MAX + 1)		The number of wells or child groups belonging to this group.			

³⁹⁹ The [RESTART](#) Data – Group, Well and Connection keywords are used both for global and [LGR](#) grids with groups, well, and connection data for the global grid, and wells and connection data for wells contained in the individual [LGR](#)s for the [LGR](#) grids. The [LGR](#) data is preceded by a series of [LGR](#) header keywords and terminated by an [LGR](#) termination keyword.

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.22 - Group Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2 - (NWG MAX + 2)		A parameter indicating production group control mode, set to: 3 : NONE or <i>FIELD</i> . 4 : Oil rate. 5 : Water rate. 6 : Gas rate. 7 : Liquid rate. 8 : Reservoir volume rate.			
2 - (NWG MAX + 6)		A parameter that defines if the group is controlled by its own targets/limits, or is open to respond to higher level control, or is controlled by a higher level group (see details in code). 0 : If group is injection group.			
2 - (NWG MAX + 7)		A parameter that is dependent on the higher level group control active and guide rate defined for the group. 0 : No group guide rate, or the group is the <i>FIELD</i> group. 1 : Unknown. 2 : Unknown. 3 : Unknown. 4 : Unknown. 5 : Unknown. 6 : Unknown. 7 : Unknown. 8 : Group guide rate defined and higher level guide rate control.			
2 - (NWG MAX + 8)		A parameter that defines which rate reduction option is used for a production group. If rate reduction is used, the value is generally four (4). 0 : <i>FIELD</i> group.			
2 - (NWG MAX + 11)		A parameter describing the production control item defined via item two, the TARGET variable, on the <i>GCONPROD</i> keyword: 0 : NONE or <i>FIELD</i> . 1 : Oil rate. 2 : Water rate. 3 : Gas rate. 4 : Liquid rate. 5 : Reservoir volume rate.			
2 - (NWG MAX + 17)		A parameter describing the water injection control defined via item three, the TARGET variable, on the <i>GCONINJE</i> keyword: 0 : NONE or <i>FIELD</i> . 1 : Rate. 2 : Reservoir volume rate. 3 : Re-injection fraction. 4 : Voidage replacement.			

No.	Keyword Name	<i>RESTART</i> Data – Group, Well, And Connection Keywords Table F.22 - Group Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2 - (NWG MAX + 18)		A parameter for a water injection group that defines if the group is controlled by its own targets/limits, or is open to respond to higher level control, or is controlled by a higher level group (see details in code).			
2 - (NWG MAX + 22)		A parameter dependent on the gas injection control defined via item three, the TARGET variable, on the <i>GCONINJE</i> keyword: 0 : NONE or <i>FIELD</i> . 1 : Rate. 2 : Reservoir volume rate. 3 : Re-injection fraction. 4 : Voidage replacement.			
2 - (NWG MAX + 23)		A parameter for a gas injection group that defines if the group is controlled by its own targets/limits, or is open to respond to higher level control, or is controlled by a higher level group (see details in code).			
2 - (NWG MAX + 27)		A variable that defines the group type with: 0 : for a well group. 1 : for a node group. 2 : for a satellite group. 3 : for a slave group.			
2 - (NWG MAX + 28)		The level of the group, with 0 representing the Field group.			
2 - (NWG MAX + 29)		The index of the parent group.			
2 - (NWG MAX + 34)		The sum of the number of active production and injection wells for the group.			
2 - (NWG MAX + 89)		Group sequence number according to the sequence in the input deck, equals NGMAXZ for the <i>FIELD</i> group.			
	SGRP	Keyword defines the real group data array SGRP(NIGRPZ, NGMAXZ) where NIGRPZ and NGMAXZ are defined on the INTEHEAD keyword.			Required
I-I	Format	SGRP	NSGRPZ x NGMAXZ	REAL	
2-1 to 2-6		Undefined.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.22 - Group Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-7	Data	Group's oil production target/limit.			
2-8	Data	Group's water production target/limit.			
2-9	Data	Group's gas production target/limit.			
2-10	Data	Group's liquid production target/limit.			
2-11	Data	Group's oil surface volume injection rate target/limit.			
2-12	Data	Group's oil reservoir volume injection rate target/limit.			
2-13	Data	Group's oil re-injection fraction target/limit.			
2-14	Data	Group's oil voidage injection fraction target/limit.			
2-15		Undefined.			
2-16	Data	Group's water surface volume injection rate target/limit.			
2-17	Data	Group's water reservoir volume injection rate target/limit.			
2-18	Data	Group's water re-injection fraction target/limit.			
2-19	Data	Group's water voidage injection fraction target/limit.			
2-20		Undefined.			
2-21	Data	Group's gas surface volume injection rate target/limit.			
2-22	Data	Group's gas reservoir volume injection rate target/limit.			
2-23	Data	Group's gas re-injection fraction target/limit.			
2-24	Data	Group's gas voidage injection fraction target/limit.			
	XGRP	Keyword defines the double precision group data array XGRP(NIGRPZ, NGMAXZ) where NIGRPZ and NGMAXZ are defined on the INTEHEAD keyword.			Required
1-1	Format	XGRP	NSGRPZ x NGMAXZ	DOUB	
2-1	Data	Group's oil production rate.			
2-2	Data	Group's water production rate.			
2-3	Data	Group's gas production rate.			
2-4	Data	Group's liquid production rate.			
2-5		Undefined.			
2-6	Data	Group's water injection rate.			
2-7	Data	Group's gas injection rate.			
2-8		Undefined.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.22 - Group Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-9	Data	Group's producing water cut.			
2-10	Data	Group's producing gas/oil ratio.			
2-11	Data	Group's total cumulative oil production.			
2-12	Data	Group's total cumulative water production.			
2-13	Data	Group's total cumulative gas production.			
2-14	Data	Group's total cumulative reservoir voidage production.			
2-15	Data	Undefined.			
2-16	Data	Group's total cumulative water injection.			
2-17	Data	Group's total cumulative gas injection.			
2-18	Data	Group's total cumulative reservoir volume injection.			
2-19		Undefined.			
2-20		Undefined.			
2-21		Undefined.			
2-22		Undefined.			
2-23	Data	Group's oil production potential.			
2-24	Data	Group's water production potential.			
2-25 to 2-86		Undefined.			
2-86	Data	Group's producer guide rate for oil.			
2-87	Data	Group's producer guide rate for water.			
2-88	Data	Group's producer guide rate for gas.			
2-89	Data	Group's producer guide rate for reservoir voidage volume.			
2-90	Data	Group's injection guide rate for oil.			
2-91		Undefined.			
2-92	Data	Group's injection guide rate for water.			
2-93	Data	Second copy of group's injection guide rate for water. Not fully characterized.			
2-94	Data	Groups injection guide rate for gas.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.22 - Group Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-95 to 2-127		Undefined.			
2-128	Data	Second copy of group's producer guide rate for oil. Not fully characterized.			
2-129	Data	Second copy of group's producer guide rate for water. Not fully characterized.			
2-130	Data	Second copy of group's producer guide rate for gas. Not fully characterized.			
2-131	Data	Second copy of group's producer guide rate for reservoir voidage volume. Not fully characterized.			
2-132 to 2-135		Undefined.			
2-136	Data	Group's total cumulative oil production (observed/historical rates).			
2-137		Undefined.			
2-138		Undefined.			
2-139		Undefined.			
2-140	Data	Group's total cumulative water production (observed/historical rates).			
2-141	Data	Group's total cumulative water injection (observed/historical rates).			
2-142		Undefined.			
2-143		Undefined.			
2-144	Data	Group's total cumulative gas production (observed/historical rates).			
2-145	Data	Group's total cumulative gas injection (observed/historical rates).			
	ZGRP	Keyword defines the character GROUP data for this keyword.			Required
1-1	Format	ZGRP	NSGRPZ x NGMAXZ	CHAR	
2-1	Data	Character group data array ZGRP(NIGRPZ, NGMAXZ) where NIGRPZ and NGMAXZ are defined on the INTEHEAD keyword.			

Table F.22: RESTART Data – Group, Well, and Connection Keywords (Groups)

The next table (Table F.23) contains the multi-segment well keywords that consist of the ISEG, RSEG, ILBS, ILBR, and ICRD keywords.

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.23 - Multi-Segment Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	ISEG	Integer multi-segment well data array ISEG(NISEGZ, NSEGMX, NSWLMX) where NISEGZ, NSEGMX, and NSWLMX are defined on the INTEHEAD keyword.			Optional
I-1	Format	ISEG	NISEGZ x NSEGMX x NSWLMX	INTE	
2-1	Data	Segment number (one-based). Special ordering – see code for details.			
2-2	Data	Outlet segment (one-based). Defines the multi-segment outlet segment number starting with 0 for the segment nearest wellhead (NISEGZ = 2).			
2-3	Data	Inflow segment current branch (one-based).			
2-4	Data	Branch number (one-based). Defines the multi-segment branch for this segment number 1 for main stem and 0 if not active (NISEGZ = 4).			
2-5 to 2-11		Undefined.			
2-12	Data	Segment type.			
2-123 to 2-18		Undefined.			
2-19	Data	ICD scaling mode.			
2-20	Data	ICD open/shut flag.			
	RSEG	Double precision multi-segment well data array RSEG(NISEGZ, NSEGMX, NSWLMX) where NISEGZ, NSEGMX, and NSWLMX are defined on the INTEHEAD keyword.			Optional
I-1	Format	RSEG	NISEGZ x NSEGMX x NSWLMX	DOUB	
2-1	Data	Segment's distance to outlet.			
2-2	Data	Segment's depth differential to outlet.			
2-3	Data	Internal diameter of segment.			
2-4	Data	Roughness parameter of segment.			
2-5	Data	Cross-sectional area of segment.			
2-6	Data	Physical volume of segment.			
2-7	Data	Segment's distance to BHP reference node.			
2-8	Data	Segment's depth differential to BHP reference node.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.23 - Multi-Segment Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-9	Data	Normalized total segment flow rate.			
2-10	Data	Normalized water flow rate fraction.			
2-11	Data	Normalized gas flow rate fraction.			
2-12	Data	Segment pressure.			
2-13 to 2-30		Undefined.			
2-31	Data	Very close to normalized water flow rate fraction.			
2-32 to 2-39		Undefined.			
2-40	Data	Undefined.			
2-41	Data	Length of valve.			
2-42	Data	Cross-sectional area of valve.			
2-43	Data	Valve's dimensionless flow coefficient.			
2-44	Data	Maximal cross-sectional valve area.			
2-45 to 2-86		Undefined.			
2-87	Data	Device base strength.			
2-88		Undefined.			
2-89	Data	Calibrated fluid density.			
2-90	Data	Calibrated fluid viscosity.			
2-91	Data	Critical water fraction.			
2-92	Data	Transition region width.			
2-93	Data	Maximum emulsion ratio.			
2-94 to 2-97		Undefined.			
2-98	Data	Maximum valid flow rate.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.23 - Multi-Segment Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-99 to 2-102		Undefined.			
2-103	Data	ICD length.			
2-104	Data	Valve area fraction.			
2-105 to 2-111		Undefined.			
	ILBS	Integer multi-segment well data array for lateral branches ILBR(NLBRMX, NSWLMX) where NLBRMX and NSWLMX are defined on the INTEHEAD keyword.			Optional
1-1	Format	ILBS	NLBRMX x NSWLMX	INTE	
2-1 to no_branches	Data	The segment number of the first segment in the branch for branch number two and upwards.			
	ILBR	Integer multi-segment well data array for chords ILBR(NILBRZ, NLBRMX, NSWLMX) where NILBRZ, NLBRMX, and NSWLMX are defined on the INTEHEAD keyword.			Optional
1-1	Format	ILBR	NILBRZ x NLBRMX x NSWLMX	INTE	
2-1	Data	Outlet segment number.			
2-2	Data	Number of segments in the branch.			
2-3	Data	First segment in branch.			
2-4	Data	Last segment in branch.			
2-5	Data	Branch number minus one.			
	ICRD	Integer multi-segment well data array for chords ICRD(2, NCRDMX, NSWLMX) where NCRDMX, and NSWLMX are defined on the INTEHEAD keyword.			Optional
1-1	Format	ILBR	2 x NCRDMX x NSWLMX	INTE	
2-1	Data	ICRD (2, NCRDMX, NSWLMX) array.			

Table F.23: RESTART Data – Group, Well, and Connection Keywords (Multi-Segment Wells)

The wells keywords are IWEL, SWEL, XWEL, and ZWEL and Table F.24 below describes these keywords in detail.

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.24 - Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	IWEL	Well integer data array IWEL(NIWELZ, NWELLS) where NIWELX and NWELLS are defined on the INTEHEAD keyword. Undefined values should be set to zero.			
I-1	Format	IWEL	NIWELZ × NWELLS	INTE	Required
2-1	Data	Defines the location of the wellhead in the grid for the I dimension (NIWELZ = 1).			
2-2	Data	Defines the location of the wellhead in the grid for the J dimension (NIWELZ = 2).			
2-3	Data	Layer ID (one-based) of top/first connection.			
2-4	Data	Layer ID (one-based) of bottom/last connection.			
2-5	Data	Number of active cells connected to a well.			
2-6	Data	Index (one-based) of well's current group.			
2-7	Data	Defines the well type, set 1 for a production well, 2 for an oil injection well, 3 for a water injection well, or 4 for a gas injection well (NIWELZ = 7).			
2-8	Data	Well's active target control mode (constraint).			
2-9		Undefined.			
2-10		Undefined.			
2-11	Data	Defines the well status, set to greater than zero for producing and less than or equal to zero for being shut (NIWELZ = 11).			
2-12	Data	ID (one-based) of well's current VFP table.			
2-13		Undefined.			
2-14		Undefined.			
2-15		Undefined.			
2-16	Data	Well's requested control mode from simulation deck (<i>WCONINJE</i> , <i>WCONPROD</i>).			
2-17 to 2-22		Undefined.			
2-23	Data	Cross flow indicator.			
2-24 to 2-42		Undefined.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.24 - Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-43	Data	Defines the LGR index for a well with local completions in a LGR (NIWELZ = 43).			
2-44 to 2-48		Undefined.			
2-49	Data	Defines the well friction indicator, set to non-zero for horizontal wells (NIWELZ = 49).			
2-50	Data	Well's requested control mode from simulation deck (WCONHIST , WCONINJH).			
2-51 to 2-70		Undefined.			
2-71	Data	Multi-segment well ID; value of one for regular wells, or zero for multi-segment wells.			
2-72	Data	Number of well segments: value of one for regular wells, or the number of segments for multi-segment wells.			
2-73 to 2-98		Undefined.			
2-99	Data	Well's completion ordering scheme, set to: 0 : Connections ordered along well track (increasing MD) 1 : Connections ordered by increasing true vertical depth. Not really supported in OPM Flow. 2 : Connections listed in order of appearance in simulation model's COMPDAT keyword.			
	SWEL	Well real data array SWEL(NSWELZ, NWELLS) where NSWELX and NWELLS are defined on the INTEHEAD record.			
I-I	Format	SWEL	NSWELZ x NWELLS	REAL	Required
2-1	Data	Well's current oil rate production target.			
2-2	Data	Well's current water rate production target.			
2-3	Data	Well's current gas rate production target.			
2-4	Data	Well's current liquid rate production target.			
2-5	Data	Well's current reservoir voidage rate production target.			
2-6	Data	Well's tubing head pressure target.			
2-7	Data	Well's bottom hole pressure target.			
2-8		Undefined.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.24 - Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-9		Undefined.			
2-10		Well's reference depth for BHP.			
2-11		Well's artificial lift quantity.			
2-12 to 2-17		Undefined.			
2-18		Well's drainage radius - item seven from the WELSPECS keyword.			
2-19 to 2-20		Undefined.			
2-25		Item two from WEFAC ; this value is repeated at two locations.			
2-26 to 2-31		Undefined.			
2-32		Item two from WEFAC keyword.			
2-33		Undefined.			
2-34		Well's historical/observed liquid rate target/limit.			
2-35 to 2-54		Undefined.			
2-55		Well's historical/observed gas rate target/limit.			
2-56		Well's historical/observed bottom hole pressure target/limit.			
	XWEL	Well's injection and production data array XWEL(NXWELZ, NWELLS).			
I-1	Format	XWEL	NSWELZ x NWELLS	DOUB	Required
2-1	Data	Well's oil production rate.			
2-2	Data	Well's water production rate.			
2-3	Data	Well's gas production rate.			
2-4	Data	Well's liquid production rate.			
2-5	Data	Well's reservoir voidage production rate.			
2-6	Data	Well's tubing head pressure.			
2-7	Data	Well's flowing/producing bottom hole pressure.			
2-8	Data	Well's producing water cut.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.24 - Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-9	Data	Well's producing gas/oil ratio.			
2-10 to 2-18		Undefined.			
2-19	Data	Well's total cumulative oil production.			
2-20	Data	Well's total cumulative water production.			
2-21	Data	Well's total cumulative gas production.			
2-22	Data	Well's total cumulative reservoir voidage production.			
2-23		Undefined.			
2-24	Data	Well's total cumulative water injection.			
2-25	Data	Well's total cumulative gas injection.			
2-26	Data	Well's total cumulative reservoir volume injection.			
2-27 to 2-34		Undefined.			
2-35	Data	Well's producing gas formation volume factor.			
2-36		Undefined.			
2-37		Undefined.			
2-38		Undefined.			
2-39 to 2-41		Undefined.			
2-42	Data	Well's current BHP Target/Limit.			
2-43 to 2-48		Undefined.			
2-49	Data	Well's "primary" guide rate (oil for producers, preferred phase for injectors).			
2-50	Data	Well's producer guide rate for water.			
2-51	Data	Well's producer guide rate for gas.			
2-52 to 2-68	Data	Undefined.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.24 - Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-69	Data	Well's producer guide rate for reservoir voidage volume.			
2-70 to 2-75		Undefined.			
2-76	Data	Well's total cumulative oil production (observed/historical rates).			
2-77	Data	Well's total cumulative water production observed/historical rates).			
2-78	Data	Well's total cumulative gas production (observed(historical rates).			
2-79 to 2-81		Undefined.			
2-82	Data	Well's total cumulative water injection (observed/historical rates).			
2-83	Data	Well's total cumulative gas injection (observed/historical rates).			
2-84 to 2-91		Undefined.			
2-92		Second copy of well's primary guide rate. Not fully characterized.			
2-93	Data	Second copy of well's producer guide rate for water. Not fully characterized.			
2-94	Data	Second copy of well's producer guide rate for gas. Not fully characterized.			
2-95	Data	Second copy of well's producer guide rate for reservoir voidage. Not fully characterized.			
2-96 to 2-122		Undefined.			
2-123	Data	Well's voidage production rate.			
2-124	Data	Well's voidage production rate.			
	ZWEL	Well character data array ZWEL(NXWELZ, NWELLS) where NXWELX and NWELLS are defined on the INTEHEAD keyword. All strings are eight characters in length.			
1-1	Format	ZWEL	NSWELZ x NWELLS	CHAR	Required
2-1	Data	Defines the well name consisting of eight characters (NXWELZ = 1).			
2-2	Data	Defines the name of the Well List the well belongs to, set to "blank" for the well does not belong to Well List (NXWELZ = 2), that is the ACTIONX name.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.24 - Well Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-3	Data	Defines the end of time step action for the well (NXWELZ = 3).			

Table F.24: RESTART Data – Group, Well, and Connection Keywords (Wells)

The well connection keywords are specified in Table F.25 and consist of the ICON, SCON, and XCON keywords.

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.25 - Connection Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	ICON	Connection integer data array ICON(NICONZ, NCWMAX, NWELLS) where NICONZ, NCWMAX, and NWELLS are defined on the INTEHEAD keyword. Data keywords are required for each completion in the well and the number of data items, N, should be NICONZ x NCWMAX x NWELLS. Undefined values should be set to zero.			
1-1	Format	ICON	N	INTE	
2-1	Data	Defines the well connect index (NICONZ = 1).			0
2-2	Data	Defines the location of connection in the grid for the I dimension (NICONZ = 2).			0
2-3	Data	Defines the location of connection in the grid for the J dimension (NICONZ = 3).			0
2-4	Data	Defines the location of connection in the grid for the K dimension (NICONZ = 4).			0
2-5		Undefined.			0
2-6	Data	Defines the status of the connection, set to less than or equal to zero for being shut or greater than zero for being open (NICONZ = 5).			0
2-7		Drainage saturation function table number.			0
2-8		Undefined.			0
2-9		Undefined.			0
2-10		Imbibition saturation function table number.			0
2-11		Undefined.			0
2-12		Undefined.			0
2-13		Completion ID (1-based).			0
2-14	Data	Defines the direction of connection, set to one for the x-direction, two for the y-direction, or three for the z-direction. The default value of zero indicates the z-direction (NICONZ = 14).			0

No.	Keyword Name	<i>RESTART</i> Data – Group, Well, And Connection Keywords Table F.25 - Connection Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-15	Data	Defines, for multi-segment wells, the segment number the connect belongs to, for conventional wells the default value of zero should be used (NICONZ = 15).			
	SCON	Connection real data array SCON(NSCONZ, NCWMAX, NWELLS) where NSCONZ, NCWMAX, and NWELLS are defined on the INTEHEAD keyword. Data are required for each completion in the well and the number of data items, N, should be NSCONZ x NCWMAX x NWELLS.			Required
1-1	Format	ICON	N	REAL	
2-1	Data	Connection transmissibility factor NSCONZ = 1 in the SCON array.			
2-2	Data	Connection center depth.			
2-3	Data	Connection diameter.			
2-4	Data	Connection KH, that is the net thickness times the permeability, NSCONZ = 4 in the SCON array.			
2-5	Data	Skin factor - item 'SKIN' from <i>COMPDAT</i> keyword in the <i>SCHEDULE</i> section.			
2-6 to 2-11		Undefined.			
2-12	Data	Connection transmissibility factor.			
2-13 to 2-20		Undefined.			
2-21	Data	Distance to end of connection in segment.			
2-22	Data	Distance to start of connection in segment.			
2-23 to 2-29		Undefined.			
2-30	Data	Undefined.			
2-31	Data	Undefined.			
2-32 to 2-40		Undefined.			
2-41	Data	Connection factor defined in input deck (set to one), or zero for undefined in input deck.			

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.25 - Connection Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	XCON	Connection double precision data array XCON(NXCONZ, NCWMAX, NWELLS) where NXCONZ, NCWMAX, and NWELLS are defined on the INTEHEAD keyword. The number of data items, N, should be NXCONZ x NCWMAX x NWELLS			Required
1-1	Format	ICON	N	DOUB	
2-1	Data	Oil surface rate.			
2-2	Data	Water surface rate.			
2-3	Data	Gas surface rate.			
2-4 to 2-34		Undefined.			
2-35		Connection pressure value.			
2-36 to 2-49		Undefined.			
2-50		Reservoir voidage rate.			

Table F.25: RESTART Data – Group, Well, and Connection Keywords (Connections)

The tracer keywords are listed in Table F.26 and consists of the ICOT, XCOT, IWET, XWET, IGRT, and XGRT keywords. Note that this series of keywords are only available if tracers are present in the run.

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords Table F.26 - Tracer Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	ICOT	Tracer connection integer data array ICOT(NICOTZ, NSTRA2, NCWMAX, NWELLS) where NICOTZ, NSTRA2, NCWMAX, and NWELLS are defined on the INTEHEAD keyword. The number of data items, N, should be NICONZ x NSTRA2 x NCWMAX x NWELLS.			Optional
1-1	Format	ICOT	N	INTE	
2-1	Data	ICOT(NICOTZ, NSTRA2, NCWMAX, NWELLS) array.			
	XCOT	Tracer connection double precision data array ICOT(NICOTZ, NSTRA2, NCWMAX, NWELLS) where NICOTZ, NSTRA2, NCWMAX, and NWELLS are defined on the INTEHEAD keyword. The number of data items, N, should be NICONZ x NSTRA2 x NCWMAX x NWELLS.			Optional
1-1	Format	XCOT	N	DOUB	
2-1	Data	XCOT(NICOTZ, NSTRA2, NCWMAX, NWELLS) array.			

No.	Keyword Name	<i>RESTART</i> Data – Group, Well, And Connection Keywords Table F.26 - Tracer Keywords			Status Or Value
	IWET	Tracer well integer data array IWET(NIWETZ, NSTRA2, NWMAXZ, NWELLS) where NIWETZ, NSTRA2, and NWMAXZ are defined on the INTEHEAD keyword. The number of items, N, should be NIWETNZ x NSTRA2 x NWMAXZ.			Optional
1-1	Format	IWET	N	INTE	
2-1	Data	IWET(NIWETZ, NSTRA2, NWMAXZ, NWELLS) array.			
	XWET	Tracer connection double precision data array XWET(NXWETZ, NSTRA2, NWMAXZ) where NXWETZ, NSTRA2, and NWMAXZ are defined on the INTEHEAD keyword. The number of items, N, should be NXWETNZ x NSTRA2 x NWMAXZ.			Optional
1-1	Format	XWET	N	DOUB	
2-1	Data	XWET(NXWETZ, NSTRA2, NWMAXZ) array.			
	IGRT	Tracer well integer data array IGRT(NIGRTZ, NSTRA2, NWMAXZ, NWELLS) where NIGRTZ, NSTRA2, and NWMAXZ are defined on the INTEHEAD keyword. The number of elements, N, should be NIGRTNZ x NSTRA2 x NWMAXZ.			Optional
1-1	Format	IGRT	N	INTE	
2-1	Data	IGRT(NIGRTZ, NSTRA2, NWMAXZ, NWELLS) array.			
	XGRT	Tracer connection double precision data array XGRT(NXGRTZ, NSTRA2, NWMAXZ) where NXGRTZ, NSTRA2, and NWMAXZ are defined in the INTEHEAD keyword. The number of data items, N, should be NXGRTNZ x NSTRA2 x NWMAXZ.			Optional
1-1	Format	XGRT	N	DOUB	
2-1	Data	XWET(NXWETZ, NSTRA2, NWMAXZ) array.			

Table F.26: *RESTART* Data – Group, Well, and Connection Keywords (Tracers)

Finally, the network keywords INODE, IBRAN, INOBR, RNODE, RBRAN, and ZNODE are specified in Table F.27. Again, this series of keywords are only available if networks are present in the run.

No.	Keyword Name	<i>RESTART</i> Data – Group, Well, And Connection Keywords Table F.27 - Network Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	INODE	Network node integer data array INODE(NINODE, NODMAX) where NINODE and NODMAX are defined in the INTEHEAD keyword.			Optional
1-1	Format	INODE	NINODE x NODMAX	INTE	
2-1	Data	INODE(NINODE, NODMAX) array			
	IBRAN	Network node branch integer data array IBRAN(NIBRAN, NBRMAX) where NIBRAN and NBRMAX are defined in the INTEHEAD keyword.			Optional
1-1	Format	IBRAN	NIBRAN x NBRMAX	INTE	

No.	Keyword Name	RESTART Data – Group, Well, And Connection Keywords			Status Or Value
		Table F.27 - Network Keywords			
2-I	Data	IBRAN(NIBRAN, NBRMAX)			
	INOBR	Network node-branch integer data array INOBR(NIOBR) where NIOBR are defined in the INTEHEAD keyword.			Optional
I-I		INOBR	NIOBR	INTE	
2-I		INOBR(NIOBR) array.			
	RNODE	Network node double precision data array RNODE(NRNODE, NODMAX) where NRNODE and NODMAX are defined in the INTEHEAD keyword.			Optional
I-I	Format	RNODE	NRNODE x NODMAX	DOUB	
2-I	Data	RNODE(NRNODE, NODMAX) array.			
	RBRAN	Network node branch integer data array RBRAN(NRBRAN, NBRMAX) where NRBRAN and NBRMAX are defined in the INTEHEAD keyword.			Optional
I-I	Format	RBRAN	NRBRAN x NBRMAX	DOUB	
2-I	Data	RBRAN(NRBRAN, NBRMAX) array.			
	ZNODE	Network node character data array ZNODE(NZNODE, NODMAX) where NZNODE and NODMAX are defined in the INTEHEAD keyword.			Optional
I-I	Format	ZNODE	NZNODE x NODMAX	CHAR	
2-I	Data	ZNODE(NZNODE, NODMAX) array.			
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) Cells colored red in the No. column indicate that this item is either “Not Used” or “Undefined”.					
3) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the ZNODE keyword should be written out as ‘ZNODE ’.					

Table F.27: RESTART Data – Group, Well, and Connection Keywords (Network)

Example: RESTART Data – Group, Well, and Connection Keywords

The following example shows a typical formatted **RESTART** Data – Group, Well, and Connection keyword data set, showing the IGRP, SGRP, XGRP, and ZGRP keywords.

[illegible]

End of Example

F.8.5 RESTART DATA – UDQ AND ACTIONX KEYWORDS

The restart data for the **UDQ** and **ACTIONX** keywords are written out in a set of **UDQ** and **ACTIONX** arrays. If there are no **UDQ** or **ACTIONX** data, the corresponding arrays are not written to the restart file and only the relevant arrays are written out.

The structure for this type of keyword is based on the data type as summarized below:

- 1) The **UDQ** data set keywords are specified in Table F.28 and consists of the IUDQ, IUAD, ZUDN, ZUDL, IGPH, IUAP, DUDW, DUDG, and DUDF keywords.
- 2) The **ACTIONX** data set keywords are specified in Table F.29 and consists of the keywords IACT, SACT, ZACT, ZLACT, ZACN, IACN, and SACN.

No.	Keyword Name	RESTART Data – UDQ And ACTIONX Keywords Table F.28 - UDQ Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	IUDQ	Keyword defines the integer properties as per the UDQ variable.			Required
I-I	Format	IUDQ	No. UDQ 's x 3	INTE	
2-I	Data	0 : ASSIGN UDQ 1 : Undefined. 2 : DEFINE UDQ			
2-2	Data	-4 : ASSIGN UDQ (-1) times the number of operators in expression: DEFINE UDQ (see code for details)			
2-3	Data	Sequence number of UDQ pr type (F, G, etc.)			
	IUAD	This keyword defines the various controls for wells, groups, etc.			Required
I-I	Format	IUAD	No. IUAD's x 5	INTE	
2-I	Data	Set to: 1) 200000 + 19 for GCONPROD and ORAT 2) 300000 + 19 for GCONPROD and WRAT 3) 400000 + 19 for GCONPROD and GRAT 4) 500000 + 19 for GCONPROD and LRAT 5) 300000 + 4 for WCONPROD + ORAT 6) 400000 + 4 for WCONPROD + WRAT 7) 500000 + 4 for WCONPROD + GRAT 8) 600000 + 4 for WCONPROD + LRAT 9) 300000 + 3 for WCONINJE + ORAT 10) 400000 + 3 for WCONINJE + RATE (surface rate) 11) 500000 + 3 for WCONINJE + RESV (reservoir vol rate) 12) 1000000 + 27 for CECON + minimum oil rate			
2-2	Data	Sequence number of UDQ used (from input sequence) for the actual constraint/target.			

No.	Keyword Name	RESTART Data – UDQ And ACTIONX Keywords Table F.28 - UDQ Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-3	Data	Set to one (1).			
2-4	Data	Number of times the UDQ variable is used.			
2-5	Data	The sequence number for the first use of the actual UDQ, equal to one (1) plus the (sum over the earlier UDQ's of the number of uses of the respective UDQ's).			
	IUAP	The keyword defines the sequence number of the actual well, group, etc. for each UDQ use in WCONPROD, GCONPROD, GCONINJE, etc.			Required
1-1	Format	IUAP	No. IUAP's	INTE	
2-1	Data	Equals the well, group, etc. sequence number for the UDQ-Variable used in actual target, limit.			
	IGPH	This keyword defines an integer number per group, and for the FIELD which is the last group in the list.			Required
1-1	Format	IGPH	Max Number Groups in the Field plus one.	INTE	
2-1	Data	The value of the integer denotes the phase (or component) for injection group control (GCONINJE): 0 : No injection control. 1 : Oil phase injection control. 2 : Water phase injection. 3 : Gas injection.			
	ZUDN	Keyword pairs of data for each UDQ.			Required
1-1	Format	ZUDN	No. UDQ's x 2	CHAR	
2-1	Data	UDQ name (maximum of eight characters per UDQ).			
2-2	Data	UDQ's units (maximum of eight characters per UDQ).			
	ZUDL	This keyword contains the EXPRESSION variable (item three) on the UDQ keyword in the input deck "Data for operation".			Required
1-1	Format	ZUDL	No. UDQ's x 16	CHAR	
2-1 to 2-16	Data	A character string that defines the EXPRESSION variable, "Data for operation", split into strings of eight characters each.			
	DUDW	Keyword containing the values of all the well UDQ's.			Required
1-1	Format	DUDW	No. Well UDQs x Max No. of Wells.	DOUBLE	
2-1 to no of well UDQ's	Data	The value of the UDQ for the actual well (repeated for all wells in the well input sequence).			

No.	Keyword Name	RESTART Data – UDQ And ACTIONX Keywords Table F.28 - UDQ Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	DUDG	Keyword containing the values of all the group UDQ's.			Required
I-I	Format	DUDG	No. Group_UDQs x Max No. of Groups	DOUBLE	
2-I to no of group UDQ's	Data	The value of UDQ for the actual group (repeated for all groups in the group input sequence).			
	DUDF	Keyword containing the values of all the field UDQ's.			Required
I-I	Format	DUDF	No. Field UDQs	DOUBLE	
2-I to no of field UDQ's	Data	Value of UDQ for field UDQs.			

Table F.28: RESTART Data – UDQ Keywords

No.	Keyword Name	RESTART Data – UDQ And ACTIONX Keywords Table F.29 - ACTIONX Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	IACT	The keyword defines the integer properties for the ACTIONX variable.			Required
I-I	Format	IACT	No. of Actions x 9	INTE	
2-I	Data	0 : Undefined.			
2-2	Data	Number of lines of SCHEDULE data.			
2-3	Data	1 : For condition and previous condition equal to AND, and combinations OR/AND 2 : For all conditions and previous conditions equal to OR.			
2-4	Data	7 : Undefined.			
2-5	Data	0 : Undefined.			
2-6	Data	The number of times the action is triggered.			
2-7	Data	0 : Undefined.			
2-8	Data	0 : Undefined.			
2-9	Data	The number of conditions for the action.			
	IACN	Keyword defines various parameters for			Required
I-I	Format	IACN	No.Actions x 26 x Max No. Conditions per Action.	INTE	

No.	Keyword Name	<i>RESTART</i> Data – <i>UDQ</i> And <i>ACTIONX</i> Keywords Table F.29 - <i>ACTIONX</i> Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-1 to 2-10	Data	0 : Undefined.			
2-11	Data	The type of quantity for the condition: 1 : For a field quantity (number of flowing producing wells). 2 : For a well quantity 3 : For a (node) group quantity. 9 : For a well group quantity 10 : For the DAY. 11 : For the MONTH. 12 : For the YEAR.			
2-12	Data	Quantity type for the Right Hand Side ("RHS") quantity: 1 : For field variables. 2 : For well variables? 3 : For group variables. 8 : For constant values.			
2-13	Data	Index for relational operator (<, =, >): 0 : For LHS quantity greater RHS quantity. 1 : For LHS quantity less than or equal to RHS quantity.			
2-14	Data	Relates to operator: 1 : AND 2 : OR			
2-17	Data	Defines the operator used in <i>ACTIONX</i> for defined quantities: 1 : For the > operator. 2 : For the < operator. 3 : For the >= operator. 4 : For the <= operator. 5 : For the = operator.			

No.	Keyword Name	RESTART Data – UDAQ And ACTIONX Keywords Table F.29 - ACTIONX Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-18	Data	The value relates to the operator for the triggering condition and if the right hand quantity is a constant or not. There is one value for each condition and the value is determined as follows: 1) For the first condition in an ACTIONX statement $iactn[17] = 0$ 2) For the second, third etc. conditions $iactn[17]$ is determined using the following logic: - If the previous condition has a constant rhs $\Rightarrow iactn[17] = 0$. - If previous rhs is of type {W,G,F} and - If previous boolean operator is AND $\Rightarrow iactn[17] = 1$ - If previous boolean operator is OR $\Rightarrow iactn[17] = 0$ Note as mentioned previously, the No. column in this table and all tables in APPENDIX F, the count is based on one (offset one); however, in C++ the base is zero (offset zero) and therefore C++ programmers must subtract one from this column to obtain the correct reference. Hence, the $iactn[17]$ reference is the correct reference for C++.			
	SACT	Keyword defines various parameters for an ACTIONX keyword			Required
1-1	Format	SACT	No.Actions x 5	REAL	
2-1 to 2-5	Data	0 : Undefined.			
	ZACT	Keyword defines the name of an action			Required
1-1	Format	ZACT	No.Actions x 4	CHAR	
2-1	Data	The name of an ACTION (maximum eight characters).			
	ZLACT	Keyword containing the actions defined for an ACTIONX keyword.			Required
1-1	Format	ZLACT	No_Actions x Max No. of Input Lines x Max No. Characters per Input Line.	CHAR	
2-1 to 2-?	Data	Contains the actions defined for an ACTIONX keyword.			
	ZACN	Keyword containing the EXPRESSION parameter (item 3) on the UDAQ keyword, the "Data for operation".			Required
1-1	Format	ZACN	No_Actions x Max No. of Conditions x 13	CHAR	
2-1	Data	Left hand side quantity if (DAY, MONTH or YEAR).			
2-2	Data	Right hand side quantity if (Well or Group).			
2-3	Data	Operator.			
2-4	Data	Well name – left hand quantity, if it is a well quantity.			
2-5	Data	Well name – right hand quantity, if it is a well quantity.			

No.	Keyword Name	<i>RESTART</i> Data – <i>UDQ</i> And <i>ACTIONX</i> Keywords Table F.29 - <i>ACTIONX</i> Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-6	Data	Group name – left hand quantity, if it is a group quantity.			
2-7	Data	Group name – right hand quantity, if it is a group quantity.			
	SACN				Required
1-1	Format	SACN	No. of Actions x Max No. Conditions x 16	DOUBLE	
2-1	Data	0 : Undefined.			
2-2	Data	0 : Undefined.			
2-3	Data	Value of condition: 1 : If Month otherwise zero.			
2-4	Data	0 : Undefined.			
2-5	Data	0 : Well or Field data, or 1 : Group data.			
2-6	Data	Value of right hand side.			
2-7	Data	Value of left hand side.			
2-8	Data	Value of right hand side.			
2-9	Data	Value of left hand side.			
2-10	Data	Value of right hand side.			

 Table F.29: *RESTART* Data – *ACTIONX* Keywords

Example: RESTART Data – Group, Well, and Connection Keywords

The following example shows a typical formatted *RESTART* Data – *UDQ* – data.

ZUDN		22	'CHAR'						
'WUWI3			'WUPR1			'WUPR3		'WUPR4	
	'WUPR5			'GUOPR1			'GULPR1		
'GUOPR2			'GULPR2			'GUOPR3		'GULPR3	
'ZUDL		176	'CHAR'						
('GOPR	'L	'OWER'	*0.	.931 *0.	'50 *1)				
				'1/(WWCT	'OP*')				
						SORTA(WU	PR1)		
								'1/(WWCT	
'OPU*')									
	SORTA(WU	PR4)							
UPPER *0	.15) * 0	.899		(2500 -	GOPR LOW	ER *0.03	-	GOPR	
ER *0.03	- GOPR	UPPER *0	.15) * 0	.899		(4500 -	GOPR LOW		
(2500 -	GOPR LOW	ER *0.03	-	GOPR	UPPER *0	.15) * 0	.920		
.15) * 0	.920	(4500 -	GOPR LOW	ER *0.03	-	GOPR	UPPER *0		
- GOPR	UPPER *0	.15) * 0	.880		(2500 -	GOPR LOW	ER *0.03		
GOPR LOW	ER *0.03	-	GOPR	UPPER *0	.15) * 0	.880		(4500 -	
IUDQ		33	'INTE'						
	2	-4		1	2	-1		2	
	2	-1		3	2	-1		4	
	2	-1		5	2	-4		1	
	2	-4		2	2	-4		3	
	2	-4		4	2	-4		5	
	2	-4		6					

End of Example

F.8.6 RESTART DATA – AQUIFER DATA KEYWORDS

This set of keywords are used to define the various aquifer arrays written out by OPM Flow. If there are no aquifers in the model then this set of keywords are skipped. If an analytical or numerical model aquifer is present in the model, a complete set of keywords, for a given aquifer type, should be written out. For example, if a numerical model is defined in the model then the IAQN and RAQN keywords should be written to the file.

The structure for this set of keywords is dependent on the aquifer type, for analytical aquifers the format is outlined in Table F.30 and numerical aquifers are characterized in Table F.31.

No.	Keyword Name	RESTART Data – Aquifer Data Keywords Table F.30 - Analytical Aquifer Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	IAAQ	Analytical aquifers integer data array IAAQ(NIAAQZ, MAAQID) where NIAAQZ and MAAQIDR are defined in the INTEHEAD keyword.			Optional
1-1	Format	IAAQ	NIAAQZ x MAAQID	INTE	
2-1	Data	Aquifer connection number.			
2-2	Data	Water property table.			PVTNUM
2-3 to 2-9		Undefined.			
2-10	Data	Analytical aquifer type, set to zero for Fetkovich aquifers and one for CarterTracy aquifers.			
2-11	Data	Type Related 2.			
	SAAQ	Analytical aquifers real data array SAAQ(NSAAQZ, MAAQID) where NSAAQZ and MAAQIDR are defined in the INTEHEAD keyword.			Optional
1-1	Format	SAAQ	NSAAQZ x MAAQID	REAL	
2-1	Data	Total Compressibility.			COMP
2-2	Data	Fetkovich analytical aquifer pore volume.			PORV
		Carter-Tracy aquifer external radius.			RE
2-3	Data	Fetkovich analytical aquifer productivity index.			PI
		Carter-Tracy aquifer permeability.			PERM
2-4	Data	Fetkovich analytical aquifer time constant.			
		Carter-Tracy aquifer porosity.			PORO
2-5	Data	Analytical aquifer pressure.			PRESS
2-6	Data	Analytical aquifer datum depth.			DATUM
2-7	Data	Carter-Tracy aquifer thickness.			DZ
2-8	Data	Carter-Tracy aquifer influence angle.			ANGLE
2-9	Data	Carter-Tracy aquifer mass density of water.			

No.	Keyword Name	<i>RESTART</i> Data – Aquifer Data Keywords Table F.30 - Analytical Aquifer Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-10	Data	Carter-Tracy aquifer water viscosity.			
	XAAQ	Analytical aquifers double precision data array XAAQ(NXAAQZ, MAAQID) where NXAAQZ and MAAQIDR are defined in the INTEHEAD keyword.			Optional
1-1	Format	XAAQ	NXAAQZ x MAAQID	DOUB	
2-1	Data	Aquifer flow rate (AAQR: N).			
2-2	Data	Dynamic aquifer pressure (AAQP: N).			
2-3	Data	Liquid volume produced from the aquifer into the reservoir (AAQT: N).			
2-4	Data	Total aquifer influx coefficient across all aquifer connections.			
2-5	Data	Reciprocal time constant for Carter-Tracy aquifer.			
2-6	Data	Influx constant "beta" for Carter-Tracy aquifer.			
2.7	Data	Unknown.			
2-8	Data	Unknown.			
2-9	Data	Dimensionless time for Carter-Tracy aquifer (AAQTD: N)			
2-10	Data	Dimensionless pressure for Carter-Tracy aquifer (AAQPD: N)			
	IAQL	Analytical aquifers integer aquifer list data array IAQL(NIAQLX, MXNALI, MXAAQL) where NIAQLX, MXNALI, and MXAAQL are defined in the INTEHEAD keyword.			Optional
1-1	Format	IAQL	NIAQLX x MXNALI x MXAAQL	INTE	
2-1	Data	IAQL(NIAQLX, MXNALI, MXAAQL) array.			
	ZAQL	Analytical aquifers character aquifer list data array ZAQL(NZAQLX, MXNALI) where NZAQLX, and MXNALI are defined in the INTEHEAD keyword.			Optional
1-1	Format	ZAQL	NZAQLX x MXNALI	CHAR	
2-1	Data	ZAQL(NZAQLX, MXNALI) array.			
	ICAQ	Analytical aquifers integer aquifer connection data array MAAQID arrays, each of size ICAQ(NICAQZ, NGCAUS) where NICAQZ, NGCAUS, and MAAQID are defined in the INTEHEAD keyword.			Optional
1-1	Format	ICAQ	NICAQZ x NGCAUS x MAAQID	INTE	
2-1	Data	Reservoir connecting cell in the I-direction.			I
2-2	Data	Reservoir connecting cell in the J-direction.			J
2-3	Data	Reservoir connecting cell in the K-direction.			K

No.	Keyword Name	RESTART Data – Aquifer Data Keywords Table F.30 - Analytical Aquifer Keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
2-4	Data	Connecting cell status, with active equal to one otherwise zero.			
2-5	Data	Connecting “face” of the reservoir cell, where I- = 1, I+ = 2, J- = 3, J+ = 4, K- = 5, and K+ = 6.			AQUFACE
	SCAQ	Analytical aquifers real aquifer connection data array MAAQID arrays, each of size SCAQ(NSCAQZ, NGCAUS) where NSCAQZ, NGCAUS, and MAAQID are defined in the INTEHEAD keyword.			Optional
1-1	Format	SCAQ	NICAQZ x NGCAUS x MAAQID	REAL	
2-1	Data	Connection's fraction of total aquifer influx coefficient.			
2-2	Data	Connection's effective face area divided by aquifer's total influx coefficient.			
	ACAQ	Analytical aquifers double precision aquifer connection data array MAAQID arrays, each of size ACAQ(NACAQZ, NGCAUS) where NACAQZ, NGCAUS, and MAAQID are defined in the INTEHEAD keyword.			Optional
1-1	Format	ACAQ	NACAQZ x NGCAUS x MAAQID	DOUB	
2-1	Data	ACAQ(NSCAQZ, NGCAUS, MAAQID) array.			
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) For formatted output all character variables, including the keywords should be enclosed in single quotes, for example the ACAQ keyword should be written out as 'ACAQ '.					

Table F.30: RESTART Data – Analytical Aquifer Data Keywords

And for numerical aquifers the format is outlined in Table F.31.

No.	Keyword Name	RESTART Data – Aquifer Data Keywords Table F.31 - Numerical Aquifer keywords			Status Or Value
		Keyword	No. of Entries	Data Type	
	IAQN	Numerical aquifers integer data array IAQN(NIIAQN, NUMAQN) where NIIAQN and NUMAQN are defined in the INTEHEAD keyword.			Optional
1-1	Format	IAAQ	NIIAQN x NUMAQN	INTE	
2-1	Data	Aquifer identification number.			AQUID
2-2	Data	Aquifer coordinate in the I-direction			I
2-3	Data	Aquifer coordinate in the J-direction			J
2-4	Data	Aquifer coordinate in the K-direction			K
2-5	Data	Aquifer Water Property Table.			PVTNUM
2-6	Data	Aquifer SATNUM table.			SATNUM
	RAQN	Numerical aquifers double precision data array RAQN(NIRAQN, NUMAQN) where NIIAQN and NUMAQN are defined in the INTEHEAD keyword.			Optional
1-1	Format	RAAQ	NIRAQN x NUMAQN	REAL	
2-1	Data	Aquifer inflow area.			AREA
2-2	Data	Aquifer length.			LENGTH
2-3	Data	Aquifer porosity			PORO
2-4	Data	Aquifer permeability.			PERM
2-5	Data	Aquifer datum depth.			DATUM
2-6	Data	Aquifer pressure.			PRESS
2-7	Data	Unknown			
2-8	Data	Unknown			
2-9	Data	Unknown			
2-10	Data	Total aquifer pore-volume.			
2-11	Data	Aquifer inflow rate (ANQR: N).			
2-12	Data	Total liquid volume produced from aquifer (AQNT: N).			
2-13	Data	Dynamic aquifer pressure (ANQP: N).			
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the RAQN keyword should be written out as 'RAQN'.					

Table F.31: RESTART Data – Numerical Aquifer Data Keywords

Example: RESTART Data – Aquifer Data Keywords

The following example shows a typical formatted *RESTART* Data – Aquifer Data keyword data set for an analytical aquifer.

```

'IAAQ      '      15 'INTE'
      3          1          0          0          0          0
      0          0          0          1          1          1
      0          0          0
'SAAQ      '      24 'REAL'
0.29999999E-04 0.24000000E+04 0.20000000E+02 0.10000000E+00
0.49619409E+04 0.54000000E+04 0.14000000E+03 0.13333334E-01
0.63151001E+02 0.31000000E+00 0.00000000E+00 0.43846342E+00
0.00000000E+00 0.00000000E+00 0.00000000E+00 0.00000000E+00
0.00000000E+00 0.00000000E+00 0.00000000E+00 0.00000000E+00
0.00000000E+00 0.00000000E+00 0.00000000E+00 0.00000000E+00
'XAAQ      '      8 'DOUB'
0.63857101611063D+03 0.49619408179004D+04 0.26831697328212D+05
0.150000000000000D+06 0.23624684055200D-01 0.36097145428688D+02
0.000000000000000D+00 0.000000000000000D+00
'ICAQNUM   '      1 'INTE'
      1
'ICAQ      '      15 'INTE'
      5          1          1          13          2          5
      1          2          14          2          5          1
      3          15          2
'SCAQNUM   '      1 'INTE'
      1
'SCAQ      '      6 'REAL'
0.33333334E+00 0.33333334E+00 0.33333334E+00 0.33333334E+00
0.33333334E+00 0.33333334E+00

```

End of Example

F.8.7 RESTART DATA – HIDDEN KEYWORD

This keyword⁴⁰⁰ defines solution arrays that have been exported by OPM Flow that should not be processed by post-processing software.

The structure for this type of keyword is defined in Table F.32.

No.	Keyword Name	RESTART Data – Hidden Keyword Table F.32			Status Or Value
		Keyword	No. of Entries	Data Type	
	HIDDEN	The keyword defines the solution arrays that have not been exported by the simulator, where N is the number of solution variable names that have been “hidden”, each enclosed in single quotes and of eight characters in length.			Required
I-I	Format	HIDDEN	N	CHAR	
2-I	Data	Character list of solution arrays to be hidden from post processing software.			

Notes:

- 1) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the HIDDEN keyword should be written out as 'HIDDEN '.

Table F.32: RESTART Data – Hidden Keyword

Example: RESTART Data – Hidden Keyword

The following example shows a typical formatted hidden data set, showing 51 Solution arrays that should be hidden.

```
'HIDDEN '      51 'CHAR'
'ISTHGGI ' 'SWMAX ' 'SWHY1 ' 'SWHY2 ' 'SWHY3 ' 'ISTHW ' 'SOMAX '
'SGHY1 ' 'SGHY2 ' 'SGHY3 ' 'ISTHG ' 'SGMAX ' 'SHMAX ' 'WATQUIES'
'GASQUIES' 'PRESROCC' 'SPDOIL ' 'SPDWATER' 'SPDGAS ' 'SWINIT ' 'SGINIT '
'POLYMAX ' 'EIP0 ' 'EIP ' 'TINIT ' 'IOCN ' 'SGMIN ' 'SGSTRD '
'SWSTRD ' 'RISTRD ' 'SGSTRI ' 'SWSTRI ' 'RISTRI ' 'SGTRPD ' 'SGSTRWD '
'SGSTRWI ' 'SWMIN ' 'SGSTRWS ' 'RPTHMW ' 'PPCW ' 'GRADWELL' 'GRADCONN'
'SWHY4 ' 'SGHY4 ' 'GRADGRUP' 'GRADRESV' 'GRADFIP ' 'GRADSOL ' 'GRADEXP '
'IAQN ' 'RAQN '
```

End of Example

⁴⁰⁰ The RESTART Data – Hidden keyword format is used both for global and LGR grids.

F.8.8 RESTART DATA - SOLUTION DATA KEYWORD

The solution data keywords⁴⁰¹ define the solution for each active cell for both global and *LGR* grids. This is then followed by a series of global data for a global entry or *LGR* data for a *LGR* entry.

The structure for this type of keyword is defined in Table F.33 for standard solution data and Table F.34 for tracers. A list of all solution names is tabulated in Table F.35.

No.	Keyword Name	<i>RESTART</i> Data - Solution Data Keyword Table F.33			Status Or Value
		Keyword	No. of Entries	Data Type	
	STARTSOL	This keyword marks the start of the solution variable section for both global and local grids. Note that there is no data associated with this keyword.			Required
1-1	Format	STARTSOL	0	MESG	
	LGRNAMES	This keyword defines a list of <i>LGR</i> names for the reporting time step, where N is the number of <i>LGR</i> names.			
1-1	Format	LGRNAMES	N	CHAR	
2-1	Data	A character list of <i>LGR</i> names with each name enclosed in single quotes and each having a length of eight characters.			
	<i>SOLUTION</i>	The <i>SOLUTION</i> keyword defines the start of a <i>SOLUTION</i> array data set, where <i>SOLUTION</i> is the name of the array being written out, as outlined in Table F.35. For example, <i>PRESSURE</i> , <i>SWAT</i> , <i>SGAS</i> , <i>RS</i> , <i>RV</i> etc. The <i>SOLUTION</i> keyword (items 1-1 and 2-1) is repeated for each <i>SOLUTION</i> array being written out. <i>SOLUTION</i> arrays can be in any order and only the active cells are written out (NACTIV).			Required
1-1	Format	<i>SOLUTION</i>	NACTIV	REAL	
2-1	Data	<i>SOLUTION</i> data set.			
	ENDSOL	This keyword marks the end of the solution variable section for both global and local grids. Note that there is no data associated with this keyword.			Required
1-1	Format	ENDSOL	0	MESG	
Notes: 1) Rows shaded in gray indicate the start of a keyword. 2) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the <i>SOLUTION</i> array for <i>PRESSURE</i> should be written out as 'PRESSURE'.					

Table F.33: RESTART Data - Solution Data keyword

For *SOLUTION* tracer concentration arrays the ZTRACER keyword must precede the *SOLUTION* tracer data array – see Table F.34 for a description of ZTRACER keyword.

⁴⁰¹ The *RESTART* Data - Solution Data keywords are used both for global and *LGR* grids *SOLUTION* arrays. The *LGR* data is preceded by a series of *LGR* head keywords and terminated by an *LGR* termination keyword.

No.	Keyword Name	RESTART Data – Solution Data Keyword For Tracer Concentration Name			Status Or Value
		Table F.34			
		Keyword	No. of Entries	Data Type	
	ZTRACER	This keyword marks the start of the solution variable section for both global and local grids. The <i>TRACER</i> in the Data field is the name of the tracer concentration being processed.			Optional
1-1	Format	ZTRACER	2	CHAR	
2-1	Data	<i>TRACER</i>	‘ ‘		
	<i>TRACER</i>	<i>TRACER</i> is the name of the tracer concentration that is contained on the next keyword. Tracer arrays can be in any order and only the active cells are written out (NACTIV).			
1-1	Format	<i>TRACER</i>	NACTIV	REAL	
2-1	Data	Tracer solution data array.			
Notes:					
1) The ZTRACER and the <i>TRACER</i> keywords are repeated for each tracer.					
2) For formatted output all character variables, including the keyword, should be enclosed in single quotes, including the tracer name.					

Table F.34: *RESTART* Data – Solution Data Keyword for Tracer Concentration Name

The *SOLUTION* keywords are listed in Table F.35.

Model Type		Model Option	Solution Data Keywords	Solution Variable Description	OPM Output Status
Black-Oil	Compositional				
		STANDARD	IOVERBO	Reciprocal of oil formation volume factor.	
		STANDARD	IOVERBW	Reciprocal of water formation volume factor.	
		STANDARD	CNV_DGAS	Worst cells depending on the gas saturation increment / Rv increment / Rs increment.	
		STANDARD	CNV_DPRE	Worst cells depending on the pressure increment.	
		STANDARD	CNV_DWAT	Worst cells depending on the water saturation increment.	
		STANDARD	CNV_GAS	Worst cells depending on the residual of gas equation.	
		STANDARD	CNV_OIL	Worst cells depending on the residual of the oil equation.	
		STANDARD	CNV_WAT	Worst cells depending on the residual of the water equation.	

Model Type		Model Option	Solution Data Keywords	Solution Variable Description	OPM Output Status
Black-Oil	Compositional				
		STANDARD	CONV_NEW	Number of Newtons required by each cell in order to satisfy the solution change convergence criteria at the last time step.	
		STANDARD	CONV_PRU	Worst cells depending on the pressure update.	
		STANDARD	CONV_VBR	Worst cells depending on the volume balance residual.	
		STANDARD	DRAINAGE	Drainage region numbers.	
		STANDARD	DRAINMIN	Drainage sink indicator.	
		STANDARD	FIPGAS	Gas fluid-in-place.	
		STANDARD	FIPOIL	Oil fluid-in-place.	
		STANDARD	FIPWAT	Water fluid-in-place.	
		STANDARD	GAS_DEN	Gas density.	
		STANDARD	GAS_POTN	Gas potential.	
		STANDARD	GAS_PRES	Gas phase pressure.	
		STANDARD	GAS_VISC	Gas viscosity.	
		STANDARD	IOVERBG	Reciprocal of gas formation volume factor.	
		STANDARD	ISTHG	Gas capillary pressure state.	
		STANDARD	ISTHW	Water capillary pressure state.	
		STANDARD	OIL_DEN	Oil density.	
		STANDARD	OIL_POTN	Oil potential.	
		STANDARD	OIL_VISC	Oil viscosity.	
		STANDARD	<i>PBUB</i>	Bubble point pressure.	
		STANDARD	PCOG	Oil-gas capillary pressure.	
		STANDARD	PCOW	Oil-water capillary pressure.	
		STANDARD	<i>PDEW</i>	Dew point pressure.	
		STANDARD	<i>PORV</i>	Pore volume at surface conditions.	
		STANDARD	<i>PRESSURE</i>	Pressure.	
		STANDARD	RFIPGAS	Gas fluid-in-place at reservoir conditions.	
		STANDARD	RFIPOIL	Oil fluid-in-place at reservoir conditions.	

Model Type		Model Option	Solution Data Keywords	Solution Variable Description	OPM Output Status
Black-Oil	Compositional				
		STANDARD	RFIPWAT	Water fluid-in-place at reservoir conditions.	
		STANDARD	RPORV	Pore volume at reservoir conditions.	
		STANDARD	RS	Gas-oil ratio.	
		STANDARD	RSSAT	Saturated gas-oil ratio.	
		STANDARD	RV	Oil-gas ratio.	
		STANDARD	RVSAT	Saturated oil-gas ratio.	
		STANDARD	SFIPGAS	Gas fluid-in-place at surface/separator conditions.	
		STANDARD	SFIPOIL	Oil fluid-in-place at surface/separator conditions.	
		STANDARD	SFIPWAT	Water fluid-in-place at surface/separator conditions.	
		STANDARD	SGAS	Gas saturation.	
		STANDARD	SGMAX	Maximum gas saturation.	
		STANDARD	SGMIN	Minimum gas saturation.	
		STANDARD	SOIL	Oil saturation.	
		STANDARD	SOMAX	Maximum oil saturation.	
		STANDARD	SSOL	Solvent saturation.	
		STANDARD	STATES	Gas-oil state indicator.	
		STANDARD	SWAT	Water saturation.	
		STANDARD	SWMAX	Maximum water saturation.	
		STANDARD	SWMIN	Minimum water saturation.	
		STANDARD	WAT_DEN	Water density.	
		STANDARD	WAT_POTN	Water potential.	
		STANDARD	WAT_PRES	Water phase pressure.	
		STANDARD	WAT_VISC	Water viscosity.	
		STANDARD	XMF	Liquid mole fractions.	
		STANDARD	YMF	Vapor mole fractions.	
		STANDARD	ZMF	Total mole fractions.	
		ALKALINE	ALKADS	Alkaline adsorption.	

Model Type		Model Option	Solution Data Keywords	Solution Variable Description	OPM Output Status
Black-Oil	Compositional				
		ALKALINE	ALKALINE	Alkaline concentration.	
		ALKALINE	ALKMAX	Alkaline maximum historic concentration.	
		ALKALINE	PADMAX	Alkaline polymer maximum historic adsorption.	
		ALKALINE	PLADALK	Alkaline polymer adsorption multipliers.	
		ALKALINE	SFADALK	Alkaline surfactant adsorption multipliers.	
		ALKALINE	STMALK	Alkaline water/oil surface tension multipliers.	
		API	SDENO	Oil surface densities.	
		API	OILAPI	Oil API values.	
		AQUIFER	HYDH	Hydraulic head aquifer.	
		AQUIFER	HYDHFWD	Fresh water hydraulic head aquifer.	
		BRINE	CNV_DSAL	Worst cells depending on the brine concentration increment.	
		BRINE	CNV_SAL	Worst cells depending on the residual of the brine equation (BRINEoption when used with polymer).	
		BRINE	ESALSUR	Effective salinity for surfactant.	
		BRINE	SALT	Brine concentration.	
		COAL	COALGAS	Coal gas concentration for coal bed methane option.	
		COAL	COALSOLV	Solvent concentration for coal bed methane option.	
		COAL	GASSATC	Initial coal gas saturated content for coal bed methane option.	
		COAL	MLANG	Langmuir scaling factors for coal bed methane option.	
		COAL	MLANGSLV	Langmuir scaling factors for solvent in coal bed methane option.	
		ENDSCALE	GASKR	Gas relative permeability.	
		ENDSCALE	OILKR	Oil relative permeability.	
		ENDSCALE	WATKR	Water relative permeability.	
		EXCAVATE	EXCAVNUM	Excavation status identifier.	
		EXCAVATE	TRANEXX/Y/Z	Transmissibilities in excavation runs.	

Model Type		Model Option	Solution Data Keywords	Solution Variable Description	OPM Output Status
Black-Oil	Compositional				
		FOAM	FOAM	Foam concentration.	
		FOAM	FOAM_HL	Foam half-life.	
		FOAM	FOAMADS	Foam adsorption.	
		FOAM	FOAMCNM	Foam capillary numbers.	
		FOAM	FOAMDCY	Foam decay.	
		FOAM	FOAMMAX	Foam maximum historic concentration.	
		FOAM	FOAMMOB	Foam mobility multiplier.	
		GIMODEL	CNV_DGGI	Worst cells depending on the dry injection gas parameter (Gi) increment.	
		GIMODEL	CNV_GGI	Worst cells depending on the residual of the Gi Pseudo-Compositional model.	
		GIMODEL	GGI	Gi injected gas ratio.	
		GIMODEL	RFIPGGI	Dry injection gas in place at reservoir conditions.	
		GIMODEL	SFIPGGI	Dry injection gas in place at surface conditions.	
		POLYMER	CABINnnn	This is for use with the PLYTRRFA keyword only.	
		POLYMER	CNV_DPLY	Worst cells depending on the polymer concentration increment.	
		POLYMER	CNV_PLY	Worst cells depending on the residual of the polymer equation.	
		POLYMER	PADS	Adsorbed polymer concentrations.	
		POLYMER	POLYMAX	Maximum historic polymer concentration.	
		POLYMER	POLYMER	Polymer concentrations.	
		POLYMER	SFIPPLY,RFIPPLY	Polymer-in-place.	
		POLYMER	SFIPSAL,RFIPSAL	Salt-in-place (BRINE option when used with polymer).	
		ROCKCOMP	PRESROCC	Rock pressure values used for rock compaction model.	
		SOLVENT	CNV_DSOL	Worst cells depending on the solvent concentration increment.	
		SOLVENT	CNV_SOL	Worst cells depending on the residual of the solvent equation.	

Model Type		Model Option	Solution Data Keywords	Solution Variable Description	OPM Output Status
Black-Oil	Compositional				
		<i>SOLVENT</i>	RFIPSOL	Solvent-in-place at reservoir conditions.	
		<i>SOLVENT</i>	SFIPSOL	Solvent-in-place at surface conditions.	
		<i>SURFACT</i>	CATROCK	Divalent cation concentration associated with rock.	
		<i>SURFACT</i>	CATSURF	Divalent cation concentration associated with surfactant.	
		<i>SURFACT</i>	<i>SURFACT</i>	Surface interactions.	
		<i>SURFACT</i>	<i>SURFADS</i>	Adsorbed surfactant concentrations.	
		<i>SURFACT</i>	SURFCNM	Surfactant capillary numbers.	
		<i>SURFACT</i>	SURFMAX	Maximum surfactant concentrations.	
		<i>SURFACT</i>	<i>SURFST</i>	Surface tension in surfactant runs.	
		<i>TEMP</i>	<i>TEMP</i>	Temperature.	
OPM		<i>THERMAL</i>	<i>TEMP</i>	Temperature.	
		<i>VE</i>	EOGC	Maximum oil-gas contact.	
		<i>VE</i>	EOWC	Minimum oil-water contact.	
		<i>VE</i>	GWC	Gas-water contact.	
		<i>VE</i>	OGC	Oil-gas contact.	
		<i>VE</i>	OWC	Oil-water contact.	
		<i>VE</i>	POT_CORR	Initial contact corrected potential.	

Notes:

- 1) The Model Type columns indicate if the solution keyword applies to a black-oil model, a compositional model or both. Since OPM Flow is a black-oil model the compositional keywords are not valid, as shown by the orange cells under the Compositional column.
- 2) The Model Option column states the model option the solution keyword is associated with and a green cell in the Model Type column indicates that OPM Flow has this option.
- 3) Finally, the OPM Output Status column indicates if the current version of OPM Flow writes out the data associated with the solution keyword, with a green cell indicating it does and orange it does not.

Table F.35: RESTART Data - Solution Data Keywords

Example: RESTART Data – Solution Data Keyword

The following example shows a typical formatted global grid data set for this keyword set.

```

'STARTSOL'          0 'MESS'
'PRESSURE'          44431 'REAL'
  0.26889725E+03    0.26838983E+03    0.26826810E+03    0.26820352E+03
  0.26814493E+03    0.26809378E+03    0.26807767E+03    0.26806363E+03
  0.26802625E+03    0.26798474E+03    0.26795001E+03    0.26791434E+03
  0.26787915E+03    0.26783920E+03    0.26777118E+03    0.26769208E+03
.....
'SWAT'              44431 'REAL'
  0.10500000E+00    0.10500000E+00    0.10500000E+00    0.10500000E+00
  0.10500000E+00    0.10500000E+00    0.10500000E+00    0.14000000E+00
  0.14000000E+00    0.14500000E+00    0.14500000E+00    0.14500000E+00
  0.14500000E+00    0.14500000E+00    0.16000000E+00    0.16000000E+00
.....
'SGAS'              44431 'REAL'
  0.00000000E+00    0.89499998E+00    0.89499998E+00    0.89499998E+00
  0.89499998E+00    0.89499998E+00    0.89499998E+00    0.86000001E+00
  0.86000001E+00    0.85500002E+00    0.85500002E+00    0.85500002E+00
.....
'RS'                44431 'REAL'
  0.11697504E+03    0.12030303E+03    0.12023456E+03    0.12019825E+03
  0.12016528E+03    0.12013654E+03    0.12012746E+03    0.12011956E+03
  0.12009855E+03    0.12007520E+03    0.12005566E+03    0.12003561E+03
  0.12001582E+03    0.11999335E+03    0.11995509E+03    0.11991060E+03
  0.11987761E+03    0.11985194E+03    0.11983020E+03    0.11981972E+03
  0.11980798E+03    0.11977385E+03    0.11970672E+03    0.11963727E+03
.....
'RV'                44431 'REAL'
  0.57879315E-04    0.57552861E-04    0.57483696E-04    0.57447018E-04
  0.57413723E-04    0.57384681E-04    0.57375513E-04    0.57367535E-04
  0.57346311E-04    0.57322733E-04    0.57302990E-04    0.57282738E-04
  0.57262747E-04    0.57240050E-04    0.57201407E-04    0.57156467E-04
  0.57123143E-04    0.57097215E-04    0.57075260E-04    0.57064670E-04
  0.57052810E-04    0.57018340E-04    0.56950528E-04    0.56880377E-04
.....
'TRFIELD'           28 'DOUB'
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
  0.00000000000000D+00    0.00000000000000D+00    0.00000000000000D+00
.....
'ZTRACER'           2 'CHAR'
'SEAF'              ' '
'SEAF'              44431 'REAL'
  0.00000000E+00    0.00000000E+00    0.00000000E+00    0.00000000E+00
  0.00000000E+00    0.00000000E+00    0.00000000E+00    0.00000000E+00
  0.00000000E+00    0.00000000E+00    0.00000000E+00    0.00000000E+00
  0.00000000E+00    0.00000000E+00    0.00000000E+00    0.00000000E+00
  0.00000000E+00    0.00000000E+00    0.00000000E+00    0.00000000E+00
  0.00000000E+00    0.00000000E+00    0.00000000E+00    0.00000000E+00
.....

```

End of Example

F.8.9 RESTART DATA - LGR SOLUTION DATA KEYWORDS

The file structure for this series of keywords is similar to the global grid [RESTART](#) Data keywords described previously. However, additional data that defines the [LGR](#) properties ([LGR](#) Name for example) are included in this keyword definition and the [LGR](#) keyword is repeated for each [LGR](#) in the model. The keyword description is outlined in Table F.36.

Note that currently OPM Flow does not support [LGR](#) grids and therefore this keyword format cannot be used by OPM Flow.

No.	Keyword Name	RESTART Data - LGR Grid Property Data Keyword Table F.36 (Repeated For Each LGR In The Model)	Status Or Value
	LGR		
	LGRHEAD1	LGR Grid Header keyword. See section F.5.1 RESTART Data - LGR Grid Header Keywords.	Required
	LGRHEADQ		
	LGRHEADD		
	INTEHEAD	Header keyword. See section F.8.3 RESTART Data - Header Keywords.	
	LOGIHEAD		
	DOUBHEAD		
	IGRP SGRP etc.	Group, well, and connection data status keywords for this reporting time step. See section F.8.4 RESTART Data – Group, Well, and Connection Data Keywords.	Required
	IAAQ SAAQ ect.	Aquifer definition arrays, only applicable if an aquifer is connected to an LGR . See section F.8.6 RESTART Data – Aquifer Data Keywords.	Optional
	HIDDEN	HIDDEN solution data, that is a list of solution arrays NOT to be read by post processing software. See section F.8.7 RESTART Data – Hidden Keyword.	Required
	SOLUTION	LGR SOLUTION array section (PRESSURE , SGAS , SOILS , WAT , etc.). See section F.8.8 RESTART Data - Solution Data keyword.	Required
	ENDSOL	LGR SOLUTION section termination keyword. See section F.8.8 RESTART Data - Solution Data keyword.	Required
	ENDLGR	LGR grid header section termination keywords. See section F.5.1 RESTART Data - LGR Grid Header Keywords.	Required

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow.
- 3) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the [LGR](#) keyword should be written out as 'LGR'.

Table F.36: RESTART Data – LGR Solution Data Keywords

This set of keywords are repeated for each [LGR](#) in the model with the last [LGR](#) data set terminated by the ENDSOL and ENDLGR keywords to mark the end of the [LGR](#) input - see the example.

Example: RESTART Data - LGR Solution Data Keywords

The following example shows a typical formatted data set with two [LGR](#) grids named LGR-1 and LGR-2.

```

'LGR      '      1 'CHAR'
'LGR-1    '
'LGRHEAD1'      45 'INTE'
      1      100      -2345      -2345      -2345      -2345
.....
      -2345      -2345      -2345
'LGRHEADQ'      5 'LOGI'
  F  F  F  F  F
'LGRHEADD'      5 'DOUB'
  0.00000000000000D+00  -0.10000000200409D+21  -0.10000000200409D+21
 -0.10000000200409D+21  -0.10000000200409D+21
'INTEHEAD'      249 'INTE'
.....
'LOGIHEAD'      79 'LOGI'
.....
'DOUBHEAD'      185 'DOUB'
.....
'IGRP      '      358 'INTE'
      0      0      0      0      0      0
.....
      0      0      0      0
'SGRP      '      186 'REAL'
  0.000000000E+00  0.000000000E+00  0.000000000E+00  0.000000000E+00
.....
  0.000000000E+00  0.100000000E+01
'XGRP      '      348 'DOUB'
  0.00000000000000D+00  0.00000000000000D+00  0.00000000000000D+00
.....
  0.00000000000000D+00  0.00000000000000D+00  0.00000000000000D+00
'ZGRP      '      10 'CHAR'
'      '      '      '      '      '      '      '      '      '      '
'      '      '      '      '      '      '      '      '      '      '
'ICOT      '      0 'INTE'
'XCOT      '      0 'DOUB'
'IWET      '      0 'INTE'
'XWET      '      0 'DOUB'
'IGRT      '      18 'INTE'
.....
'XGRT      '      180 'DOUB'
  0.00000000000000D+00  0.00000000000000D+00  0.00000000000000D+00
.....
  0.10000000200409D+21  0.10000000200409D+21  0.10000000200409D+21
'DLYTIM      '      30 'DOUB'
  0.00000000000000D+00  0.00000000000000D+00  0.00000000000000D+00
.....
  0.00000000000000D+00  0.00000000000000D+00  0.00000000000000D+00
'HIDDEN      '      51 'CHAR'
'ISTHGGI      ' 'SWMAX      ' 'SWHY1      ' 'SWHY2      ' 'SWHY3      ' 'ISTHW      ' 'SOMAX
'
'IAQN      ' 'RAQN      '
'STARTSOL      '      0 'MESS'
'PRESSURE      '      9 'REAL'
  0.26823886E+03  0.26824997E+03  0.26826108E+03  0.26822879E+03
.....
'SWAT      '      9 'REAL'
  0.10500000E+00  0.10500000E+00  0.10500000E+00  0.10500000E+00
.....

```

'SGAS'	9	'REAL'	0.89499998E+00	0.89499998E+00	0.89499998E+00	0.89499998E+00
'RS'	9	'REAL'	0.12021812E+03	0.12022437E+03	0.12023062E+03	0.12021246E+03
'RV'	9	'REAL'	0.57467099E-04	0.57473411E-04	0.57479720E-04	0.57461380E-04
'ENDSOL'	0	'MESS'				
'ENDLGR'	1	'INTE'				
'LGR'	1	'CHAR'				
'LGR-2'						
'LGRHEAD1'	45	'INTE'				
'LGRHEADQ'	5	'LOGI'				
'LGRHEADD'	5	'DOUB'				
'INTEHEAD'	249	'INTE'				
'LOGIHEAD'	79	'LOGI'				
'DOUBHEAD'	185	'DOUB'				
'IGRP'	358	'INTE'				
'SGRP'	186	'REAL'				
'XGRP'	348	'DOUB'				
'ZGRP'	10	'CHAR'				
'ICOT'	0	'INTE'				
'XCOT'	0	'DOUB'				
'IWET'	0	'INTE'				
'XWET'	0	'DOUB'				
'IGRT'	18	'INTE'				
'XGRT'	180	'DOUB'				
'DLYTIM'	30	'DOUB'				
'HIDDEN'	51	'CHAR'				
'STARTSOL'	0	'MESS'				
'PRESSURE'	9	'REAL'				
'SWAT'	9	'REAL'				
'SGAS'	9	'REAL'				
'RS'	9	'REAL'				
'RV'	9	'REAL'				
'ENDSOL'	0	'MESS'				
'ENDLGR'	1	'INTE'				
'SEQNUM'	1	'INTE'				

End of Example

F.9 RFT FILES – PRESSURE, SATURATION AND PRODUCTION LOG DATA

The Repeat Formation Tester (“RFT”) file contains a well’s depth pressure profile at the time the data is requested using the [WRFT](#) or [WRFTPLT](#) keywords in the [SCHEDULE](#) section. The data is meant to be used to compare the simulator’s pressure profile with the field measured data collected from one of the open hole wire line logging tools: Formation Interval Tester (“FIT”), Repeat Formation Tester (“RFT”) tool, Modular Dynamics Tester (“MDT”), or similar tool. The keywords also export the fluid saturations for the well connections at the same time.

In addition to the pressure and saturation data, the [WRFTPLT](#) keyword can also active the writing out of each well connection’s fluid rates, connection factors and KH data, etc., collectively known as the Production Logging Tool (“PLT”) data. The PLT data is used to compare with measured data from wire line production logging tools. Currently, output of the PLT data is not supported by OPM Flow.

F.9.1 RFT DATA FILE SPECIFICATION

The file structure for the RFT Data File consists of various keywords: a Time and Date keyword, a Well and Connection keyword followed by the requested optional keywords of RFT, PLT, Multi-Segment Well, and River keywords, that are written out depending on the data requested. A brief description of the keywords is given in Table F.37. A complete RFT data set is written to file for each well and time step as requested on the [WRFT](#) and [WRFTPLT](#) keywords.

Reference Section	Keyword Name	Keyword Type	Description
F.9.2	TIME	Time and Date Date	This data set contains the simulated time and date information that the date was written out.
	DATE		
F.9.3	WELLETC	Well and Connection Data	The Well and Connection keywords outlines the type and units being written out, the type of well and the well connections at the simulated time the data is written out to the RFT file.
	CONIPOS		
	CONJPOS		
	CONKPOS		
	HOSTGRID		
F.9.4	DEPTH	RFT Solution Data	RFT keyword data consists of the pressure and saturations for each declared connection in the well at the simulated time the data is written out to file.
	PRESSURE		
	SWAT		
	SGAS		
	CPLY		
	CPLAD		
	CBRI		

Reference Section	Keyword Name	Keyword Type	Description
F9.5		PLT Solution Data	PLT series of keywords contain the production logging data which consists of the mid-point depth of each connection, the tubing length for the connection from the bottom-hole reference depth, the pressures and fluid rates for the connections. In addition, this data also includes the well pressures, fluid rates, connection transmissibility and KH vales, for the given simulated time. Currently this data set is not supported.
F9.6		Multi-Segment Well Data	The Multi-Segment Well data set consists of pressure and flow rate data for segments and branches, currently this data set is not supported.
F9.7		River Data	Data pertaining to the Rivers option, this option is not supported by OPM Flow.
Notes: 1) Reference Section cells colored orange indicate that this item is not supported by OPM Flow.			

Table F.37: RFT Data File Format

F.9.2 RFT DATA FILE – TIME AND DATE KEYWORDS

The time and date keywords define the start of an RFT data set and declares the time and date for the subsequent property data written to file. The structure for these two keywords is summarized in Table F.38.

No.	Keyword Name	RFT Data File – Time And Date Keywords			Status Or Value
		Table F.38			
		Keyword	No. of Entries	Data Type	
	TIME	This keyword marks the start of of an RFT file data set.			Required
1-1	Format	TIME	1	REAL	
2-1	Data	Simulation time the data is being written to file.			
	DATE	Three integer values representing the date of the data being written out.			Required
1-1	Format	DATE	3	INTE	
2-1	Data	Calendar day and should range between one and 31.			DAY
2-2	Data	Calendar month and should range between one and 12.			MONTH
2-3	Data	Calendar year in four digits, that is 2020.			YEAR
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) For formatted output all character variables, including the keyword should be enclosed in single quotes, for example the TIME should be written out as 'TIME'.					

Table F.38: RFT Data File – Time and Date Keywords

The *DATE* keyword is immediate followed by the well and connection keywords.

F.9.3 RFT DATA FILE – WELL AND CONNECTION DATA KEYWORD

The well and connection data set of keywords specify the well and the well connection data as well as the system of units for the pressure and flow data being written to the RFT file. The structure for this set of keywords is outlined in Table F.39 and consists of WELLET, CONIPOS, CONJPOS, CONKPOS and HOSTGRID keywords.

No.	Keyword Name	RFT Data File – Well And Connection Data Keyword Table F.39			Status Or Value
		Keyword	No. of Entries	Data Type	
	WELLET	This keyword marks the start of the well and connection data set and comprises of 16 character variables all enclosed in single quotes.			Required
1-1	Format	WELLET	16	CHAR	
2-1	Data	Time units.			
2-2	Data	Well name.			
2-3	Data	Local Grid Refinement (“LGR”) name or left blank if the well is not in an LGR . LGRs are not currently supported by OPM Flow.			Blank
2-4	Data	Depth units.			
2-5	Data	Pressure units.			
2-6	Data	Type of data being written out in subsequent keywords and should be set to R for RFT data, P for PLT data, or S for Segment data. The Segment data option is not currently supported in OPM Flow.			
2-7	Data	Well type data set to STANDARD for a standard well, FRICTION for a wellbore friction well, and MULTSEG for a multi-segment well. Only the STANDARD well option is currently supported by OPM Flow.			
2-8	Data	Liquid flow rate units.			
2-9	Data	Gas flow rate units.			
2-10	Data	Local volumetric flow rate units.			
2-11	Data	Flow velocity units.			
2-12	Data	Not used and should be a blank character string.			Blank
2-13	Data	Liquid and gas viscosity units.			
2-14	Data	Polymer and Brine concentration units.			
2-15	Data	Polymer and Brine flow rate units.			
2-16	Data	Adsorbed Polymer concentration units.			
	CONIPOS	This keyword defines the grid index in the x direction (I direction) for all NCON connections in the well associated with the RFT data set.			Required
1-1	Format	CONIPOS	NCON	INTE	
2-1	Data	Grid index in the I direction for each connection.			
	CONJPOS	This keyword defines the grid index in the y direction (J direction) for all NCON connections in the well associated with the RFT data set.			Required

No.	Keyword Name	RFT Data File – Well And Connection Data Keyword			Status Or Value
		Table F.39			
		Keyword	No. of Entries	Data Type	
1-1	Format	CONJPOS	NCON	INTE	
2-1	Data	Grid index in the J direction for each connection.			
	CONKPOS	This keyword defines the grid index in the z direction (K direction) for all NCON connections in the well associated with the RFT data set.			Required
1-1	Format	CONKPOS	NCON	INTE	
2-1	Data	Grid index in the K direction for each connection.			
	HOSTGRID	Defines the <i>LGR</i> name for each of the NCON connections. <i>LGR</i> 's have not been implemented in OPM Flow and therefore this keyword should contain NCON blank character strings.			Blank
1-1	Format	HOSTGRID	NCON	CHAR	Blank
2-1	Data	Connections host <i>LGR</i> grid name.			
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					
2) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the WELLETC should be written out as 'WELLETC '.					

Table F.39: RFT Data File – Well and Connection Data Keyword

This series of keywords are immediate followed by the RFT solution keywords: *DEPTH*, *PRESSURE*, *SWAT*, and *SGAS*.

F.9.4 RFT DATA FILE – RFT SOLUTION DATA KEYWORD

The RFT solution keywords specify the actual solution data (pressure, saturation etc.) for each connection in the well, as described in Table F.40.

No.	Keyword Name	RESTART Data - Solution Data Keyword Table F.40			Status Or Value
		Keyword	No. of Entries	Data Type	
1-1	DEPTH	Connection grid block depth.			Required
1-1	Format	DEPTH	NCON	REAL	
2-1	Data	Depth data for each connection.			
	PRESSURE	Connection grid block pressure at the requested time.			Required
1-1	Format	PRESSURE	NCON	REAL	
2-1	Data	Pressure data for each connection.			
	SWAT	Connection grid block water saturation at the requested time.			Required
1-1	Format	SWAT	NCON	REAL	
2-1	Data	Water saturation data for each connection.			
2-4	SGAS	Connection grid block gas saturation at the requested time.			Required
1-1	Format	SGAS	NCON	REAL	
2-1	Data	Gas saturation data for each connection.			
	CPLY	Connection grid block polymer concentration at the requested time.			Optional
1-1	Format	CPLY	NCON	REAL	
2-1	Data	Polymer concentration data for each connection.			
	CPLAD	Connection grid block adsorbed polymer concentration at the requested time.			Optional
1-1	Format	CPLAD	NCON	REAL	
2-1	Data	Adsorbed polymer concentration data for each connection.			
	CBRI	Connection grid block brine concentration at the requested time.			Optional
1-1	Format	CBRI	NCON	REAL	
2-1	Data	Brine concentration data for each connection.			

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow.
- 3) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the *PRESSURE* keyword should be written out as 'PRESSURE'.

Table F.40: RFT Data File – RFT Solution Data Keywords

F.9.5 RFT DATA FILE – SOLUTION PLT DATA KEYWORDS

This data set is currently not supported.

F.9.6 RFT DATA FILE – SOLUTION MULTI-SEGMENT WELL KEYWORDS

This data set is currently not supported.

F.9.7 RFT DATA FILE – SOLUTION RIVER KEYWORDS

This data set is currently not supported.

F.10 SUMMARY FILES – TIME BASED VECTOR DATA

The *SUMMARY* files contain the variables requested to be written to the summary files via the keywords declared in the *SUMMARY* section. The data are used to generate line graphs of properties such as oil flow rate versus time, grid block pressure versus time, etc. See sections [11.2 Data Requirements](#) and [11.3 Keyword Definitions](#) that declare the variables to be written out to the *SUMMARY* files. The default behavior is write out the requested variables at each time step. As this can lead to large files, especially for full field simulation models, the *RPTONLY – Activate the Report Time Steps Only Option for the SUMMARY File* keyword allows one to write out the data only at a report time step instead.

There are three types of *SUMMARY* files, the first *SUMMARY* file is the OPM Flow specific Enhanced *SUMMARY* (*.ESMRY) output file that is optimized for fast loading of selected vectors by post-processing applications, including OPM ResInsight. This file type is standalone, that is there is no separate index file.

The other two files types associated with the time based vector data, is an index file (*SUMMARY* Index) that contains a list of variables and objects (wells, groups, connections etc.) that are to be written out to summary data file, and the summary data file (*SUMMARY* Data) that contains the actual data written out at the required time steps. The commercial simulator uses the term “MINISTEP” to describe the time the data is written, and as mentioned above, “MINISTEP” may be either a time step or a reporting time step.

F.10.1 ENHANCED SUMMARY DATA FILE KEYWORDS

Again, this is an OPM FLOW specific Enhanced *SUMMARY* (*.ESMRY) output file format that is optimized for fast loading of selected vectors by post-processing applications. The *.ESMRY file is re-written for every time step and load times are in the order of 30 times faster compared with fetching selected vectors using the *.UNSMRY file format in post processing software. The file type is compatible with OPM ResInsight and the option is activated via the enable-esmry=true command line option.

The structure for this series of keywords is outlined in Table F.41 and depends on the data requested to be written out to the *SUMMARY* files.

No.	Keyword Name	Enhanced <i>SUMMARY</i> File Keywords Table F.41			Status Or Value
		Keyword	No. of Entries	Data Type	
	<i>START</i>	Three integer values representing the start date of the simulation., and four integers representing the hour, minutes, seconds and micro seconds the start date.			Required
1-1	Format	<i>START</i>	7	INTE	
2-1	Data	First part of start date of the run, DAY, the value should be greater than or equal to one and less than or equal to 31 for the calendar days.			DAY
2-2	Data	Second part of start date of the run, MONTH, the value should be greater than or equal to one and less than or equal to 12 for the calendar month.			MONTH
2-3	Data	Third part of start date of run, YEAR, a positive four digit integer value of the start year, which must be specified fully by four digits, for example 1986.			YEAR
2-4	Data	The first part of the start time in the form HH:MM:SS:MS, that is HH for hours. The value should be greater than or equal to zero and less than or equal to 23.			IHOURLZ
2-5	Data	The second part of the start time in the form HH:MM:SS:MS, that is MM for minutes. The value should be greater than or equal to zero and less than or equal to 59.			IMINTS

No.	Keyword Name	Enhanced SUMMARY File Keywords Table F.4I			Status Or Value
		Keyword	No. of Entries	Data Type	
2-6	Data	The third part of the start time in the form HH:MM:SS:MS, that is SS for seconds. The value should be greater than or equal to zero and less than or equal to 59.			ISECND
2-7	Data	The fourth part of the start time in the form HH:MM:SS:MS, that is MS for milliseconds. The value should be greater than or equal to zero and less than or equal to 999.			
	RESTART	RESTART is only present for restarted runs and contains the root file name (absolute or relative path) to the case which this run was restarted from.			Optional
1-1	Format	RESTART	I	C0nn	
2-1	Data	The name of file, enclosed in quotes, which contains the name of the case the current run was re-started from. The length of the character string is set by the Data Type, C0nn, for example, C125 would mean that the RESTART filename character length is 125 characters.			
	RSTNUM	The RSTNUM keyword is only present for restart runs and defines the report time step this case was restarted from.			Optional
1-1	Format	RSTNUM	I	INTE	
2-1	Data	Restart report time step. number.			
	KEYCHECK	A list of all the summary variable mnemonics written to the file at each time step., where NLIST is the number of vectors and C0nn is the constant fixed length of each variable character string. For example: <pre>'KEYCHECK' 6 'C010' 'TIME ' 'YEARS ' 'FOPR ' 'GGOR:PROD ' 'TCPU ' 'WOPR:PROD1'</pre> Where each vector name consists of 10 characters. For vectors associated with objects, groups, wells etc., the vector name is followed by a ":" and then the object name as shown in the above example.			Required
1-1	Format	KEYCHECK	NLIST	C0nn	
2-1	Data	Each mnemonic string should be enclosed in quotes and have a character length of C0nn, for example 'C010' in the above example.			
	UNITS	Units			Required
1-1	Format	UNITS	NLIST	CHAR	
2-1	Data	Units for the associated summary variable mnemonics.			
	RSTEP	An array that holds one integer value for each time step. An integer value of one indicates time steps that are also report time steps and a zero indicates time steps only.			Required
1-1	Format	RSTEP	NSTEP	INTE	
2-1	Data	One value for each time step and the values should be either zero or one.			

No.	Keyword Name	Enhanced <i>SUMMARY</i> File Keywords			Status Or Value
		Table F.4I			
		Keyword	No. of Entries	Data Type	
	<i>TSTEP</i>	An array holding one integer value for each time step with the values representing the time step indices. Note that the values starts from zero, such that the Data will have values of the form 0, 1, 2, ..., NSTEP - 1, that is unless the <i>RPTONLY</i> keyword has been used in the <i>SUMMARY</i> section to write only the report time steps to the file. Values should range between 0 to NSTEP - 1			Required
I-I	Format	<i>TSTEP</i>	NSTEP	INTE	
2-I	Data	One value for each time step. with the value ranging from zero to NSTEP - 1.			
	VX	One data array (length = NSTEP) for each summary vector specified in array KEYCHECK. Naming of these arrays should be V0,V1,V2, ... etc.			Required
I-I	Format	VX	NSTEP	REAL	
2-I	Data	One real value for each time step for one specific summary vector VX vector, where X ranges from zero to NLIST – 1, for the NLIST variables declared via the KEYCHECK keyword.			

Notes:

- 1) Keywords are order dependent.
- 2) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the *START* and KEYCHECK keywords should be written out as 'START ' and 'KEYCHECK'.

Table F.4I: Enhanced *SUMMARY* File Keywords

Example: Enhanced *SUMMARY* File Keywords

The following example shows a typical formatted Enhanced *SUMMARY* file set of keywords.

```
'START'      1      7 'INTE'
              0      11      2018      0      0      0
'KEYCHECK'   6 'C010'
'TIME'       'YEARS'      'FOPR'      'GGOR:PROD' 'TCPU'      'WOPR:PROD1'
'UNITS'      6 'CHAR'
'DAYS'       'SM3/DAY' 'SM3/SM3' 'SECONDS' 'SM3/DAY'
'RSTEP'      4 'INTE'
              1      1      0      1
'TSTEP'      4 'INTE'
              0      1      2      3
'V0'         4 'REAL'
              1.0000000E+00 4.0000000E+00 9.0000000E+00 1.4000000E+01
'V1'         4 'REAL'
              2.7378509E-03 1.0951404E-02 2.4640657E-02 3.8329910E-02
'V2'         4 'REAL'
              0.0000000E+00 0.0000000E+00 0.0000000E+00 0.0000000E+00
'V3'         4 'REAL'
              0.0000000E+00 0.0000000E+00 0.0000000E+00 0.0000000E+00
'V4'         4 'REAL'
              6.0681427E-01 6.5710169E-01 7.0169014E-01 7.6383120E-01
'V5'         4 'REAL'
              0.0000000E+00 0.0000000E+00 0.0000000E+00 0.0000000E+00
```

The example shows the file holding six summary vector and four time steps, with the first, second and fourth time steps being report time steps.

F.10.2 SUMMARY INDEX FILE KEYWORDS

The structure for this series of keywords is outlined in Table F.42, Table F.43, and Table F.44 and depends on the data requested to be written out to the [SUMMARY](#) Data File.

No.	Keyword Name	<i>SUMMARY</i> Index File Keywords (Global) Table F.42			Status Or Value
		Keyword	No. of Entries	Data Type	
	INTEHEAD	The integer header keyword contains two values indicating the units system and the simulator used to generate the file.			Required
I-1	Format	INTEHEAD	2	INTE	
2-1	Data	Unit system used in the simulation, set to 1 for metric, 2 for field, and 3 for laboratory			
2-2	Data	Simulator code used to generate the file, set to: 100 for Schlumberger's ECLIPSE 100, 300 for Schlumberger's ECLIPSE 300, 500 for Schlumberger's ECLIPSE 300 Thermal, 700 for Schlumberger's INTERSECT simulator, 800 for Schlumberger's FrontSim simulator, or a negative value for other simulators. OPM Flow uses a value of 100.			100
	<i>RESTART</i>	Contains the root file name of the case which this run was restarted from, if this was a restart run, other wise blank.			Required
I-1	Format	<i>RESTART</i>	9	CHAR	
2-1	Data	Maximum of 72 characters divided into eight character words. For example, if the data file is restarted from NOR01-PRED01-OPM2004-R00, then this data will be of the form: 'NOR01-PR' 'ED01-OPM' '2010-R00' '			
	<i>DIMENS</i>	This keyword contains the number of vectors and objects written to the <i>SUMMARY</i> Data file at each time step.			Required
I-1	Format	<i>DIMENS</i>	6	REAL	
2-1	Data	NLIST is the number of values stored at each time step on the <i>SUMMARY</i> Data File.			
2-2	Data	The number of grid blocks in the x-direction (NX).			
2-3	Data	The number of grid blocks in the y-direction (NY).			
2-4	Data	The number of grid blocks in the z-direction (NZ).			
2-5	Data	Not Used.			0
2-6	Data	Report time step of the case which this run was restated from, if this was a restart run.			0
	KEYWORDS	A list of all the summary variable mnemonics written to the <i>SUMMARY</i> Data file at each time step.			Required
I-1	Format	KEYWORDS	NLIST	CHAR	
2-1	Data	Each mnemonic string should be enclosed in quotes.			

No.	Keyword Name	SUMMARY Index File Keywords (Global) Table F.42			Status Or Value
		Keyword	No. of Entries	Data Type	
	WGNAMES	A list of all well and group objects associated with summary variable mnemonics			Required
1-1	Format	WGNAMES	NLIST	CHAR	
2-1	Data	Each well and group name should be enclosed in quotes			
	NUMS	Integer grid cell or region number associated with the summary variable mnemonics.			Required
1-1	Format	NUMS	NLIST	INTE	
2-1	Data	Integer values.			
	LENGTHS	Horizontal well length.			Optional
1-1	Format	LENGTHS	NLIST	REAL	
2-1	Data	The distance from the bottom-hole reference depth to the completion depth for the associated summary variable mnemonics for horizontal wells			
	LENUNITS	Horizontal well length units.			Optional
1-1	Format	LENUNITS	I	CHAR	
2-1	Data	Horizontal well length units.			
	MEASRMNT	Measurements associated with each summary variable mnemonic.			Optional
1-1	Format	MEASRMNT	NBLOCK x NLIST	CHAR	
2-1	Data	NBLOCK is equal to the number of elements provided by this keyword divided by NLIST.			
	UNITS	Units			Required
1-1	Format	UNITS	NLIST	CHAR	
2-1	Data	Units for the associated summary variable mnemonics.			
	STARTDAT	Three integer values representing the date and four integers representing the time of the data being written out.			Required
1-1	Format	STARTDAT	3	INTE	
2-1	Data	Calendar day and should range between one and 31.			DAY
2-2	Data	Calendar month should range between one and 12.			MONTH
2-3	Data	Calendar year in four digits, that is 2020.			YEAR
2-4	Data	Hour.			HOUR
2-5	Data	Minutes.			MINUTE
2-6	Data	Seconds in micro seconds, 0-59:999:999			SECOND

No.	Keyword Name	SUMMARY Index File Keywords (Global)			Status Or Value
		Table F.42			
		Keyword	No. of Entries	Data Type	
Notes:					
1) Keywords up to and including the NUMS keyword are order dependent, after the NUMS keyword the keywords may be in any order.					
1) Rows shaded in gray indicate the start of a keyword.					
2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow and those colored red are either “Not Used” or “Undefined”.					
3) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the INTEHEAD and DIMENS keywords should be written out as ‘INTEHEAD’ and ‘DIMENS’.					

Table F.42: SUMMARY Index File Keywords (Global)

In addition, Table F.43 summarizes the SUMMARY Index File keywords associated with Local Grid Refinements.

No.	Keyword Name	SUMMARY Index File Keywords (LGR) Table F.43			Status Or Value
		Keyword	No. of Entries	Data Type	
	LGRS	Local Grid Refinement names associated summary variable mnemonic for LGR grids.			Optional
1-1	Format	LGRS	NLIST	CHAR	
2-1	Data	Name of the LGRs.			
	NUMLX	This keyword defines the LGR grid index in the x direction (I direction) for the associated summary variable mnemonic for LGR grids.			Optional
1-1	Format	NUMLX	NLIST	INTE	
2-1	Data	LGR I location of local grid block data or the connection data.			
	NUMLY	This keyword defines the LGR grid index in the y direction (J direction) for the associated summary variable mnemonic for LGR grids.			Optional
1-1	Format	NUMLY	NLIST	INTE	
2-1	Data	LGR J location of local grid block data or the connection data.			
	NUMLZ	This keyword defines the LGR grid index in the z direction (K direction) for the associated summary variable mnemonic for LGR grids.			Optional
1-1	Format	NUMLX	NLIST	INTE	
2-1	Data	LGR K location of local grid block data or the connection data.			
	LGRNAMES	The names of all the LGRs in this case with the number of of LGRs in the case given by NLGR.			Optional
1-1	Format	LGRNAMES	NLGR	CHAR	
2-1	Data	List of LGR names.			
	LGRVEC	Integer list of the number of summary variable mnemonics for each LGR.			Optional

No.	Keyword Name	SUMMARY Index File Keywords (LGR)			Status Or Value
		Table F.43			
		Keyword	No. of Entries	Data Type	
1-1	Format	LGRVEC	NLGR	INTE	
2-1	Data	Number of summary variable mnemonics for each LGR.			
	LGRTIMES	A list of the total number of local time steps for each LGR.			
1-1	Format	LGRTIMES	NLGR	INTE	
2-1	Data	Number of local time steps for each LGR.			

Notes:

- 1) The LGR keywords may be in any order but must come after the NUMS keyword in Table F.42.
- 2) Rows shaded in gray indicate the start of a keyword.
- 3) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow
- 4) For formatted output all character variables, including the keyword, should be enclosed in single quotes, for example the LGRTIMES keyword should be written out as 'LGRTIMES'.

Table F.43: SUMMARY Index File Keywords (LGR)

Finally, Table F.44 summarizes the SUMMARY Index File keywords specific to the commercial simulator and are therefore not described in any detail.

No.	Keyword Name	<i>SUMMARY</i> Index File Keywords (Commercial Simulator) Table F.44	Status Or Value
	RUNTIMEI	Run time monitoring – not supported.	
	RUNTIMED	Run time monitoring – not supported.	
	STEPRESN	Run time monitoring – not supported.	
	XCOORD	<i>OFM</i> Data – not supported.	
	YCOORD	<i>OFM</i> – not supported.	
	TIMESTMP	Not supported.	

Notes:

- 1) Rows shaded in gray indicate the start of a keyword.
- 2) Cells colored orange in the No. column indicate that this item is not supported by OPM Flow.
- 3) For formatted output all character variables, including the keyword should be enclosed in single quotes.

Table F.44: SUMMARY Index File Keywords (Commercial Simulator)

Example: SUMMARY Index File Keywords

The following example shows a typical formatted *SUMMARY* Index file set of keywords.

```

'INTEHEAD'          2 'INTE'
      1          100
'RESTART'          9 'CHAR'
'          '
'          '
'DIMENS'            6 'INTE'
      164          10          10          10          0          0
'KEYWORDS'          164 'CHAR'
'TIME'      'YEARS'  'FGIP'  'FGIPG'  'FGIPL'  'FGIR'  'FGIT'
'FGOR'      'FGPR'  'FGPT'  'FOIP'  'FOIPG'  'FOIPL'  'FOIR'
'FOIT'      'FOPR'  'FOPT'  'FPR'   'FVIR'  'FVIT'  'FVPR'
'FVPT'      'FWCT'  'FWIP'  'FWIR'  'FWIT'  'FWPR'  'FWPT'
'GGIR'      'GGIT'  'GGOR'  'GGPR'  'GGPT'  'GOIR'  'GOIT'
'GOPR'      'GOPT'  'GVIR'  'GVIT'  'GVPR'  'GVPT'  'GWCT'
'GWIR'      'GWIT'  'GWPR'  'GWPT'  'SPR'   'SPR'   'SPR'
'SPR'       'SPR'   'SPR'   'SPR'   'SPR'   'SPR'   'SPR'

.....
'WWCT'      'WWCT'  'WWIR'  'WWIR'  'WWIT'  'WWIT'  'WWPR'
'WWPR'      'WWPT'  'WWPT'
'WGNAMEs'    164 'CHAR'
'++++++'    '++++++' '++++++' '++++++' '++++++' '++++++' '++++++'
'++++++'    '++++++' '++++++' '++++++' '++++++' '++++++' '++++++'
'++++++'    '++++++' '++++++' '++++++' '++++++' '++++++' '++++++'
'++++++'    '++++++' '++++++' '++++++' '++++++' '++++++' '++++++'
'PLATFORM'  'PLATFORM' 'PLATFORM' 'PLATFORM' 'PLATFORM' 'PLATFORM' 'PLATFORM'
'PLATFORM'  'PLATFORM' 'PLATFORM' 'PLATFORM' 'PLATFORM' 'PLATFORM' 'PLATFORM'
'PLATFORM'  'PLATFORM' 'PLATFORM' 'PLATFORM' 'OP01'  'OP01'  'OP01'
'OP01'      'OP01'  'OP01'  'OP01'  'OP01'  'OP01'  'OP01'
'OP01'      'OP01'  'OP01'  'OP01'  'OP01'  'OP01'  'OP01'
'OP01'      'OP01'  'OP01'  'OP01'  'OP01'  'OP01'  'OP01'
'OP01'      'OP01'  'OP01'  'OP01'  'OP01'  'OP01'  'OP01'
'OP01'      'OP01'  'OP01'  'OP01'  'OP01'  'OP01'  'OP01'

.....

'WI01'      'OP01'  'WI01'
'NUMS'      164 'INTE'
      0          0          0          0          0          0
      0          0          0          0          0          0
      0          0          0          0          0          0
      0          0          0          0          0          0
      0          0          0          0          0          0
      0          0          0          0          0          0
      0          0          0          0          1          2
      3          4          5          6          7          8
      9          10         11         12         13         14

.....
      0          0
'UNITS'      164 'CHAR'
'DAYS'      'YEARS'  'SM3'   'SM3'   'SM3'   'SM3/DAY' 'SM3'
'SM3/SM3'   'SM3/DAY' 'SM3'   'SM3'   'SM3'   'SM3'   'SM3/DAY'
'SM3'       'SM3/DAY' 'SM3'   'BARSA'  'RM3/DAY' 'RM3'   'RM3/DAY'
'RM3'       'SM3/SM3' 'SM3'   'SM3/DAY' 'SM3'   'SM3/DAY' 'SM3'

.....
'SM3/DAY'   'SM3'   'SM3'
'STARTDAT'  6 'INTE'
      1          1          2020          0          0          0

```

End of Example

F.10.3 SUMMARY DATA FILE KEYWORDS

The structure for this series of keywords is relatively simple compared to the [SUMMARY](#) Index file, in that it only consists of four keyword types that are written out at the requested time steps, as outlined in Table F.45.

No.	Keyword Name	SUMMARY Data File Keywords Table F.45			Status Or Value
		Keyword	No. of Entries	Data Type	
	SEQNUM	This keyword contains a sequence data value corresponding to an encoded integer representing the time the file was created. Although required the data is not used. Secondly, this keyword is written at each report step only and not for each inter leaving mini step, that is the file format is of the following format: SEQNUM MINISTEP PARAMS MINISTEP PARAMS MINISTEP PARAMS INISTEP PARAMS SEQNUM MINISTEP PARAMS MINISTEP PARAMS Report step n <			

Table F.45: SUMMARY Data File Keywords

For LGRs, the global cell data is written to the standard [SUMMARY](#) Data file containing an additional keyword that states the [LGR](#) name, as depicted in Table F.46.

No.	Keyword Name	SUMMARY Data File Keywords (LGR) Table F.46			Status Or Value
		Keyword	No. of Entries	Data Type	
	SEQHDR	As per Table F.45			Required
	LGRNAME	Name of Local Grid Refinement.			Required
	Format	LGRNAME	I	CHAR	
	Data	LGR name enclosed in single quotes.			
	MINSTEP	As per Table F.45.			Required
	PARAMS	As per Table F.45.			Required
Notes:					
1) Rows shaded in gray indicate the start of a keyword.					

Table F.46: [SUMMARY](#) Data File Keywords ([LGR](#))

The data for the [LGR](#) time steps, as opposed to the global time step, are written out to a separate [SUMMARY](#) Data File, with the suffix of [LGR](#) or FLG for fixed format output. In this case, the MINSTEP keyword states the “local” time step and the keyword is repeated for each local grid in the model.

Example: [SUMMARY](#) Data Data File keyword

The following example shows a typical formatted [SUMMARY](#) Data File keyword data set for one complete report time step and part of a second report time step.

```
'SEQHDR'      1 'INTE'
1
'MINISTEP'    1 'INTE'
0
'PARAMS'      164 'REAL'
0.50000000E+00 0.13689254E-02 0.18381236E+11 0.51459348E+10
0.13235301E+11 0.00000000E+00 0.00000000E+00 0.37649313E+03
0.95391960E+07 0.47695980E+07 0.10947282E+09 0.30000488E+06
0.10917282E+09 0.00000000E+00 0.00000000E+00 0.25336973E+05
0.12668486E+05 0.27468176E+03 0.15571577E+05 0.77857886E+04
0.63867320E+05 0.31933660E+05 0.12643164E-02 0.60474832E+08
0.15000000E+05 0.75000000E+04 0.32074501E+02 0.16037251E+02
0.00000000E+00 0.00000000E+00 0.37649313E+03 0.95391960E+07
0.47695980E+07 0.00000000E+00 0.00000000E+00 0.25336973E+05
0.12668486E+05 0.15571577E+05 0.77857886E+04 0.63867320E+05
0.31933660E+05 0.12643164E-02 0.15000000E+05 0.75000000E+04
0.32074501E+02 0.16037251E+02 0.26000000E+03 0.26172531E+03
0.26362421E+03 0.26596732E+03 0.26747589E+03 0.26896045E+03
.....
'SEQHDR'      1 'INTE'
2
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1
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0.13232370E+11 0.00000000E+00 0.00000000E+00 0.32637939E+03
.....
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End of Example

F.11 SAVE FILES - INITIALIZATION AND SOLUTION DATA

This file format is currently not supported by OPM Flow.

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The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the qualitative approach adopted, which involves in-depth interviews and focus group discussions. The researchers aimed to explore the experiences and perceptions of the participants, rather than testing a specific hypothesis.

The third part of the paper presents the findings of the study. It discusses the various themes that emerged from the data, such as the role of family in education and the influence of community norms. The researchers found that there were significant differences in the way that different cultural groups viewed education and learning.

The final part of the paper discusses the implications of the findings for practice. It suggests that educators and policymakers should take into account the cultural context of their students when designing educational programs. This could involve providing additional support for students from disadvantaged backgrounds or adapting teaching methods to better suit different learning styles.